

# Lexicographic Orders of Well-Orderings of the Surreal Numbers

Set approximations, the singular birthday transition, set-length words,  
class orders, and higher-order structure

Research report merged from twenty-eight independent manuscripts  
prepared for Vladimir Reshetnikov (sources 08, 09, 11 and 12 of batch 80;  
sources 13–18 of batch 81; sources 19–30 of batch 83; source 31 of batch 89;  
sources 32–35 of batch 96; source 36 of batch 97)

Sources dated 2 October 2026 (source 11: 3 October 2026, UTC), 3 October 2026 (sources  
13–30), 4 October 2026 (sources 31–34; source 31 UTC), 5 October 2026 (source 35) and  
October 2026 (source 36); merged 2–5 October 2026

## Abstract

The phrase *all well-orders of the surreal numbers* has several inequivalent meanings. Well-orders of a fixed birthday fragment form a set; well-orders of arbitrary set-sized subsets form an ordinary class; well-orders of the whole surreal line are themselves classes, and their totality belongs to a higher logical level. We develop these settings and distinguish set-like from unrestricted class well-orders.

For an infinite set-sized alphabet  $X$ , the existing ProveIt theory gives  $|\mathcal{W}(X)| = 2^{|X|}$  and the exact ordinal embedding spectrum  $\beta < (|X|)^+$ . We apply it to surreal birthday fragments. For arbitrary set-length surreal words, we give an explicit sign-sequence embedding into **No**, with birthday bounds, and identify the one-sided cuts created by limit-length words.

For the whole surreal line, we define lexicographic comparison directly on labeled class well-orders, without assuming ordinal enumerations or a class of class order types. Every such well-order has a canonical initial segment of type Ord; its remaining tail determines a convex fiber. In the set-like fragment, bounded-support permutations of one global enumeration form an ordinary class isomorphic to **No**, and interpolate every uniformly set-indexed strict cut of the entire fragment. The fragment contains every class linear order, admits binary class-word coding, and cannot have an exhaustive uniform class of set-codes. The unrestricted order has adjacent pairs, all arising from finite residual alphabets. An omitted-range construction exhibits a proper-class cut filled by a non-set-like well-order. In a full inaccessible-universe interpretation, the unrestricted order does not embed into its set-like fragment.

The proofs use ZF, ZFC, or GBC, as specified, with a further uniform nonembedding theorem in  $\text{KM} + \text{GC}$ . We give universe-relative and proof-assistant formulations, and propose further research on the higher-order structure.

[merge] This report merges four manuscripts written independently on 2 October 2026 on the same question and delivered together in batch 80: source 11 (the base, whose abstract is printed above), and sources 12, 09 and 08. They prove the same core up to four times. Each shared theorem is printed once; the other sources' versions are recorded in numbered notes, and their genuinely different arguments are kept as marked routes. Source 12 adds an exact dichotomy at birthday cutoffs: for an infinite cardinal  $\kappa$  and  $\mu = 2^{<\kappa}$ , the minimal-length enumeration order  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  is equimorphic with the binary cube  $\mathbb{B}_{\kappa \cdot \kappa}$  (ordinal product) exactly when  $\kappa$  is a singular strong limit, and with  $\mathbb{B}_\mu$  otherwise, with  $\kappa = \beth_\omega$  as an unconditional example. It also adds a second direct comparison of all class well-orders, by their largest common labelled prefix,

and an exact external identification of both global orders in a full-class model at an inaccessible cardinal. Source 09 adds bounded reflection of set-indexed families, an explicit unfilled (Ord, Ord) configuration, and the embeddability in **No** of every uniformly class-coded family. Source 08 adds a separator of exactly prescribed birthday, the failure of finite support, and relation-code lexicography of all class well-orders. Source 11's choiceless proof answers source 12's question on the weak-choice strength; the direct comparisons of sources 11 and 12 answer source 08's question on comparing non-set-like class well-orders; and source 12's dichotomy answers in part a question of the repository's report on the reals. The whole set-sized general-alphabet layer re-proves that report and is printed with pointers to it, never as new.

[merge] (3 October 2026.) Six further manuscripts of batch 81, sources 13–18, continue this report and are merged into it as Parts VI–IX. They compute the density, cellularity, characters, gap spectrum and gap multiplicities of the set-sized strata at birthday cutoffs and the coding lengths of the strata  $\mu \cdot \gamma + n$ ; classify the termination conventions of word orders; classify every proper class cut of the set-coded core and recognize the cuts of global enumerations, which answers source 11's question on represented cuts, with automorphism orbits, class-extension absoluteness and support thresholds; decompose every class well-order canonically into Ord-blocks, with every class well-order as a block-index order and explicit condensation histories; settle source 08's question on full virtual products; bound the comprehension needed for the uniform nonembedding by  $\Sigma_1^1$ ; and classify the full order of every countable  $\omega$ -standard model as a factorial shuffle. Results proved by several of them are printed once, with every proof credited; their re-proofs of this report are notes.

[merge] Twelve further manuscripts of batch 83 (sources 19–30, all of 3 October 2026) continue the report as Parts X–XIV. On set-sized strata they determine the embedded binary-cube spectrum of every fixed enumeration type of **No**<sub>< $\kappa$</sub>  and its coding length except at singular non-strong-limit residual cardinals, the weight and point characters of every fixed-length stratum, and the topology of the minimal slice: Baire space with real-line completion at the countable cutoff, a generalized Baire space at regular sizes, and, at uncountable sizes of countable cofinality, a space that is not Borel in its injection completion and has no complete metric; the maximal and self-normalizing bounded subgroup and the canonical group topology above the prefix topology. Across birthday cutoffs they show that the native alphabet **No**<sub>< $\theta$</sub>  remembers  $\theta$  exactly, glue coherent cutoff profiles to class well-orders and detect set-likeness on a club, and treat noncardinal cutoffs and coherent embeddings. For word orders they factor increasing words at the terminator and reduce the centered increasing order to  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ , and compare relation-table scans. For the core they classify branch configurations under core-and-cylinder automorphisms, show that exhaustiveness is invisible even to the cylinders, give the exact least ordinal support bound, and recover an available class realization from its labelled global enumerations. For unrestricted orders they prove a local coding dichotomy, describe the countable class models and their common complete theory, and show over GBC that canonical terminating condensation histories are equivalent to strong comparability of class well-orders. Many of these results are proved by several of the twelve independently; each is printed once with every source credited.

[merge] One manuscript of batch 89 (source 31, dated 4 October 2026) is printed as Part XV. Over Zermelo set theory with Choice, a selector required to choose only from nonempty countable sets, with Separation extended to it, is not conservative; the proof combines a bounded uniformization theorem with Glazer's published countermodels, used through an interface whose exact match with them has not been reviewed. In the full-class rank models  $(V_\lambda, \mathcal{P}(V_\lambda))$  at every limit height  $\lambda > \omega$ , global choice and birthday-monotone set-like well-orders of the sign carrier always exist, while ordinal enumeration of the sign carrier, birthday-monotone enumeration, universal set coding (equivalently, ordinal abstraction) and full Class Replacement have the distinct exact thresholds: a cardinal height  $\kappa$  with  $2^{<\kappa} = \kappa$ , a strong-limit cardinal height, a beth fixed point  $\kappa = \beth_{\kappa}$ , and a strongly inaccessible height; Class Replacement on one set holds exactly when its size is below the cofinality of the height. This answers in part source 28's question on separating Class Replacement from uniform set choice, and extends source 12's inaccessible full-class model to every limit height.

[merge] Four manuscripts of batch 96 (sources 32–35, dated 4 and 5 October 2026) are merged

as Part XVI. Two of them prove independently that strong comparability of class well-orders and the existence of canonical condensation histories, which source 29 showed equivalent over GBC, are equivalent over GB without set or global choice: the polynomial orders  $P(\Gamma)$  are class well-ordered by a choice-free minimization, and a class linear order is well-ordered as soon as its nonempty set subsets have least elements. The Part develops the arithmetic of class well-orders beyond Ord (sums, products, finite-support powers, division by  $\Omega = (\text{Ord}, <)$ ), a hereditary term order  $\mathbf{E} \cong \text{Ord}^{[\mathbf{E}]}$  that is the least such fixed point and has external type  $\varepsilon_{\kappa+1}$  over an inaccessible  $\kappa$ , and a fixed-point completion of every class well-order initially embedded in its polynomial order, giving a further equivalent of comparability, here proved over GB as well. No uniformly coded family of class well-orders is cofinal under embedding; for each definability level a single order escapes all orders of that level. Over  $V_\kappa$  the definable order types form an interval  $[0, \Theta)$  with  $\kappa < \Theta < \kappa^+$ , of countable cofinality under a named global well-order, closed under ordinal arithmetic with  $\kappa^\Theta = \Theta$ , and a truth predicate represents  $\Theta$  exactly.

[merge] A fifth manuscript on the same request (source 36, batch 97) is printed whole as Part XVII. Besides independent proofs of Part XVI's localization, power, hereditary, diagonal, definability-ceiling and truth-presentation theorems, it computes sums and products of the points of the hereditary order by explicit normal-form algorithms, characterizes when a finite-support power  $A^{[B]}$  is set-like, extends the epsilon-term functor to class well-orders, and shows that over every transitive model of ZFC of height  $\kappa$  the definable order types form an interval whose supremum  $\mathfrak{H}(M)$  is a fixed point of  $\xi \mapsto \varepsilon_\xi$  above  $\zeta_{\kappa+1}$ , and that over  $V_\kappa$  the ceilings are unbounded in  $\kappa^+$  as the named global enumeration varies.

**Status.** This is a mathematical research report with proofs, not a claim of machine verification or bibliographic priority. Existing general set-alphabet results are credited to the repository. The further constructions are proved here and should be evaluated by their arguments. The repository audit was targeted; no Lean build or remote change was made.

[merge] **Status of the merged report.** AI-assisted and unrefereed; not formalized. Sources 12 and 09 call themselves AI-assisted in their title pages; sources 08 and 11 do not use the word (author lines: 08 and 09 “Research manuscript prepared for Vladimir Reshetnikov”, 11 “Research report”, 12 “Research report prepared for Vladimir Reshetnikov”). None of the four claims historical priority, peer review or machine verification. Their finite checks verify finite instances only, never a transfinite or class-theoretic statement. No statement of this report is formalized; the formal status of the inputs it uses is in [section 1.5](#).

[merge] **Batch 81 (3 October 2026).** Sources 13–18 are research manuscripts with written proofs. Sources 13, 15, 17 and 18 call themselves AI-assisted, source 14 unrefereed, and source 16 says that its checks are not external peer review. None claims priority, peer review or machine verification, and none of their statements is formalized ([sections 1.4 and 1.8](#)).

[merge] **Batch 83 (3 October 2026).** Sources 19–30 are research manuscripts with written proofs. Sources 19–26, 28 and 29 call themselves AI-assisted, source 30 reports AI-assisted proof review, and source 27 claims no priority or external peer review. None claims historical priority, peer review or a machine-checked formalization, and none is formalized here ([section 1.9](#)).

[merge] **Batch 89 (3 October 2026).** Source 31 is a research article with written proofs (“Research article prepared for Vladimir Reshetnikov”). It claims no machine verification and no historical priority, uses Glazer’s countermodel theorem as an explicitly cited input, and names expert review of its localized theorem as the next step; it is not formalized here ([section 1.10](#)).

[merge] **Batch 96 (4 October 2026).** Sources 32 and 33 are “Research manuscript[s] prepared for Vladimir Reshetnikov” and call themselves AI-assisted and unrefereed; sources 34

and 35 are “Research report[s] prepared for Vladimir Reshetnikov”, “not a refereed publication” and “unrefereed”, with “independent mathematical checks during preparation”. None claims historical priority, peer review or a machine-checked formalization, and none is formalized here ([section 1.11](#)).

[merge] **Batch 97 (4 October 2026)**. Source 36 is a research manuscript with “conventional written proofs with independent review during preparation”; it has no author line and does not call itself AI-assisted. It claims no publication priority, no resolution of a recognized open problem, no referee and no proof-assistant certification, and it is not formalized here ([section 1.12](#)).

## Contents

|     |                                                                           |     |
|-----|---------------------------------------------------------------------------|-----|
| 1   | About this report                                                         | 13  |
| 2   | The objects being studied                                                 | 45  |
| 3   | Relationship to ProveIt and the literature                                | 46  |
| 4   | The other sources' introductions                                          | 47  |
| 5   | Surreal order facts and foundational conventions                          | 60  |
| I   | Sets of surreals                                                          | 69  |
| 6   | Well-order codes on a fixed set alphabet                                  | 69  |
| 7   | Birthday cutoffs and their set-sized orders                               | 76  |
| 8   | Birthday cutoffs and the singular transition (source 12)                  | 79  |
| II  | Set-length words                                                          | 81  |
| 9   | Set-length surreal words embed explicitly in the surreal line             | 81  |
| 10  | Local characters and set suprema of word orders                           | 82  |
| 11  | The complete obstruction to filling a set cut                             | 83  |
| 12  | Saturation at a regular stage                                             | 86  |
| III | Global choice and all class well-orders                                   | 87  |
| 13  | Global choice and well-ordering the surreal class                         | 87  |
| 14  | Lexicographic comparison without class ordinals                           | 92  |
| 15  | The canonical initial set-like part                                       | 95  |
| 16  | All jumps arise from finite residual permutations                         | 96  |
| 17  | Non-set-like well-orders and alternative coding (source 08)               | 99  |
| 18  | Numerically well-ordered subclasses and other interpretations (source 12) | 100 |
| IV  | Set-like global well-orders                                               | 101 |
| 19  | The lexicographic structure of global enumerations                        | 101 |

|     |                                                                                                           |     |
|-----|-----------------------------------------------------------------------------------------------------------|-----|
| V   | Foundations, models and formalization                                                                     | 118 |
| 20  | Formalizing the subject without changing its meaning                                                      | 118 |
| 21  | A full-class model that connects the two size levels (source 12)                                          | 122 |
| 22  | A concrete universe enlargement: all objects become sets (source 09)                                      | 124 |
| 23  | Universes, class models, and paradox avoidance (source 08)                                                | 125 |
| 24  | A formalization architecture with explicit size boundaries (source 12)                                    | 127 |
| 25  | Formalization strategies and paradox avoidance (source 09)                                                | 129 |
| 26  | A Lean formalization plan tied to ProveIt (source 08)                                                     | 131 |
| VI  | Set-sized strata: topology, cut spectra and block types                                                   | 133 |
| 27  | Density and cellularity at every fixed type (source 16)                                                   | 134 |
| 28  | Countable incompleteness and global characters (source 16)                                                | 138 |
| 29  | The cut spectrum of the alphabet $\text{No}_{<\kappa}$ (source 17)                                        | 139 |
| 30  | The minimal slice $\mathcal{W}_\mu(\text{No}_{<\kappa})$ (source 17; routes of sources 15, 16 and 18)     | 142 |
| 31  | Longer strata: every cardinal-block type (source 15)                                                      | 156 |
| 32  | Set-sized baselines and warnings in sources 13–18                                                         | 161 |
| 33  | What this Part answers                                                                                    | 165 |
| VII | Words, termination conventions and other comparison orders                                                | 166 |
| 34  | The six sources' maps of interpretations                                                                  | 166 |
| 35  | Shorter-first words: three classes, one prefix order (source 16; routes of sources 13, 14 and 18)         | 177 |
| 36  | Termination conventions: the criterion and all isomorphism types (source 16; routes of sources 13 and 18) | 182 |
| 37  | Centered increasing words and the entry zero (source 18)                                                  | 189 |
| 38  | Other comparison orders (source 15; remarks of sources 13, 14, 16, 17 and 18)                             | 195 |
| 39  | What this Part answers                                                                                    | 201 |

|      |                                                                                                    |     |
|------|----------------------------------------------------------------------------------------------------|-----|
| VIII | Cuts of the set-coded core: recognition, automorphisms, support and class realizations             | 201 |
| 40   | Foundations and the audited starting point (source 14)                                             | 202 |
| 41   | The set-coded skeleton and its cylinders (source 14)                                               | 204 |
| 42   | The core as re-proved in sources 13, 15, 16, 17 and 18                                             | 206 |
| 43   | A trichotomy for class cuts of the skeleton (source 14)                                            | 216 |
| 44   | Characters, saturation, and principal cuts (source 14)                                             | 218 |
| 45   | Recognizing exhaustive cuts and preserving old cuts (source 14)                                    | 220 |
| 46   | Five further proofs of the represented-cut theorem (sources 16, 15, 13, 17 and 18)                 | 221 |
| 47   | Automorphisms of the core (source 14; sources 17, 15 and 13)                                       | 237 |
| 48   | Class extensions (source 14; source 17)                                                            | 243 |
| 49   | Support-restricted suborders and the exact reserve (source 14; source 17)                          | 245 |
| 50   | What this Part answers                                                                             | 249 |
| IX   | Unrestricted class well-orders: canonical blocks, condensation, products, comprehension and models | 250 |
| 51   | Order facts and direct comparison as re-proved in sources 15, 16, 18, 17 and 13                    | 251 |
| 52   | The canonical block decomposition (source 15; routes of sources 16 and 18)                         | 263 |
| 53   | Condensation histories and the strength of iterated condensation (source 15; sources 16 and 18)    | 271 |
| 54   | Finite components and trace fibres (source 17; sources 18 and 14)                                  | 274 |
| 55   | Asymmetric products and the role of semantics (source 13)                                          | 278 |
| 56   | Coding, diagonalization, and a sharper comprehension bound (source 16; source 14)                  | 281 |
| 57   | Larger universes, class realizations and paradox avoidance (sources 16, 14, 13, 15 and 18)         | 284 |
| 58   | An exact external classification in countable models (source 18)                                   | 293 |
| 59   | Formalization plans of sources 13–18                                                               | 297 |
| 60   | What this Part answers                                                                             | 308 |

|     |                                                                                                                           |     |
|-----|---------------------------------------------------------------------------------------------------------------------------|-----|
| X   | Set-sized strata, topology and permutation groups                                                                         | 309 |
| 61  | The twelve batch-83 sources: scope and interpretations                                                                    | 311 |
| 62  | Thick alphabets, prefix cylinders and the prefix topology (source 24; sources 26, 28, 21, 22 and 19)                      | 349 |
| 63  | Fixed-length strata: residual cylinders, weight and point characters (source 22)                                          | 354 |
| 64  | Fixed types with infinite residual tails: the exact binary-cube spectrum (source 19; route of source 30)                  | 361 |
| 65  | Completions of the minimal slice (sources 24, 21 and 19)                                                                  | 371 |
| 66  | A generalized Baire model at every regular size (source 24)                                                               | 378 |
| 67  | The countable cutoff (source 24; routes of sources 19, 20, 21, 26, 29 and 30)                                             | 380 |
| 68  | Countable strata of length $\omega + n$ (sources 29 and 30)                                                               | 390 |
| 69  | Singular cardinalities: continuity, complete metrizability and non-Borel boundaries (sources 24 and 26)                   | 407 |
| 70  | Permutation groups at singular cardinalities: maximality, self-normalization and the canonical group envelope (source 28) | 421 |
| 71  | The class analogue: global prefix completion and its limits (source 24; sources 26 and 28)                                | 434 |
| 72  | What this Part answers                                                                                                    | 445 |
| XI  | Across birthday cutoffs: restrictions, coherent limits and birthday spectra                                               | 446 |
| 73  | Exactly when restriction preserves the native order (source 19)                                                           | 448 |
| 74  | Birthday limits: gluing, stabilization, and two different completions (source 20)                                         | 450 |
| 75  | Coherent embeddings within a cardinality band (source 21)                                                                 | 457 |
| 76  | The exact binary spectrum of a birthday alphabet (source 29)                                                              | 459 |
| 77  | Noncardinal birthday cutoffs (source 21)                                                                                  | 463 |
| 78  | All enumeration lengths at a noninitial birthday bound (source 29)                                                        | 466 |
| 79  | What this Part answers                                                                                                    | 472 |
| XII | Words, termination markers and comparison codes                                                                           | 474 |



|                                                                                           |                                                                                                                       |     |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----|
| 80                                                                                        | The four sources' maps of interpretations                                                                             | 474 |
| 81                                                                                        | Termination markers over a saturated alphabet (source 23; second routes to Part VII)                                  | 484 |
| 82                                                                                        | Increasing words: no saturating marker, a product at the terminator, and the zero conventions (sources 23, 27 and 29) | 490 |
| 83                                                                                        | Numerically well-ordered proper-class carriers (source 29)                                                            | 501 |
| 84                                                                                        | Relation-table lexicography: spectra, scans, extensions and density (sources 27 and 30)                               | 504 |
| 85                                                                                        | What this Part answers                                                                                                | 515 |
| XIII The core and its cylinders: symmetries, decorated completions and class realizations |                                                                                                                       | 516 |
| 86                                                                                        | The core as re-proved in sources 25, 22, 21 and 29                                                                    | 518 |
| 87                                                                                        | Unranked cylinders recover prefix heights (source 22)                                                                 | 530 |
| 88                                                                                        | What unranked cylinders remember (source 25)                                                                          | 533 |
| 89                                                                                        | The one-point profile extension theorem (source 25)                                                                   | 535 |
| 90                                                                                        | Exact configuration homogeneity (source 25)                                                                           | 536 |
| 91                                                                                        | Exhaustiveness is invisible to unlabelled cylinders (source 25)                                                       | 538 |
| 92                                                                                        | Extension to the full enumeration order (source 25)                                                                   | 539 |
| 93                                                                                        | Cross-node label coherence and rigidity (source 25)                                                                   | 540 |
| 94                                                                                        | Regular set-sized models and an exact group size (source 25)                                                          | 542 |
| 95                                                                                        | Cofinality obstructions and singular lengths (source 25)                                                              | 544 |
| 96                                                                                        | Ordered prefix trees without the core (sources 27 and 29)                                                             | 545 |
| 97                                                                                        | Automorphisms of the core and the failure of extension (sources 20 and 21)                                            | 554 |
| 98                                                                                        | Residual data and the completion of the raw order (source 19)                                                         | 566 |
| 99                                                                                        | Why residual classification is not automatically self-similar (source 19)                                             | 570 |
| 100                                                                                       | Persistent external cuts absent from the class realization (source 19)                                                | 573 |
| 101                                                                                       | The exact least ordinal support bound (source 27)                                                                     | 574 |

|       |                                                                                                         |     |
|-------|---------------------------------------------------------------------------------------------------------|-----|
| 102   | Class realizations: what labelled and marked enumeration spaces remember<br>(sources 22, 27, 21 and 30) | 577 |
| 103   | Unrestricted class well-orders and other readings (source 25)                                           | 592 |
| 104   | A size and recursion audit (source 25)                                                                  | 594 |
| 105   | What this Part answers                                                                                  | 596 |
| XIV   | Unrestricted class well-orders II: local size, condensation histories and<br>external models            | 597 |
| 106   | Direct comparison and set-interval condensation once more (source 23)                                   | 599 |
| 107   | Local size of intervals in the unrestricted order (sources 20 and 27)                                   | 601 |
| 108   | Complete condensation histories (sources 20, 23 and 29)                                                 | 609 |
| 109   | Countable class models: two external order types and one theory (sources 19,<br>20, 21 and 30)          | 625 |
| 110   | Foundations: full rank semantics and well-foundedness (source 27)                                       | 645 |
| 111   | Foundations, formalization plans and ledgers of sources 19–30                                           | 648 |
| 112   | What this Part answers                                                                                  | 716 |
| XV    | Countable global choice and surreal coding over Zermelo set theory                                      | 718 |
| XV.1  | The problem and the contributions                                                                       | 719 |
| XV.2  | Axioms, classes, and bounded constructions                                                              | 720 |
| XV.3  | Uniform presentations from restricted selection                                                         | 721 |
| XV.4  | Incidence coding without Collection                                                                     | 723 |
| XV.5  | Countable global selection is nonconservative                                                           | 725 |
| XV.6  | Full-class rank models and the sign carrier                                                             | 727 |
| XV.7  | Ordinal enumeration and birthday order                                                                  | 728 |
| XV.8  | Universal set coding and ordinal abstraction                                                            | 729 |
| XV.9  | The exact domain size for Class Replacement                                                             | 730 |
| XV.10 | The sizes of internal proper classes                                                                    | 732 |
| XV.11 | Explicit separating models                                                                              | 732 |

|        |                                                                                                               |     |
|--------|---------------------------------------------------------------------------------------------------------------|-----|
| XV.12  | Consequences for surreal foundations and formalization                                                        | 734 |
| XV.13  | Axiom ledger and proof dependencies                                                                           | 735 |
| XV.14  | Source audit and research status                                                                              | 735 |
| XV.15  | What this Part answers                                                                                        | 736 |
| XVI    | Definable class well-orders beyond Ord: arithmetic, histories over GB, fixed points and definability ceilings | 737 |
| XVI.1  | The question and the contribution                                                                             | 738 |
| XVI.2  | Foundations: classes, definability, and well-foundedness                                                      | 745 |
| XVI.3  | The boundary at $\Omega$ and elementary arithmetic                                                            | 750 |
| XVI.4  | Finite-support exponentiation without Global Choice                                                           | 757 |
| XVI.5  | The canonical-history theorem over GB                                                                         | 767 |
| XVI.6  | Hereditary normal forms and a first fixed point                                                               | 782 |
| XVI.7  | Uniform catalogs and truth jumps                                                                              | 799 |
| XVI.8  | External spectra and the effect of definability                                                               | 811 |
| XVI.9  | How far the constructions reach, and what requires more axioms                                                | 825 |
| XVI.10 | Rigorous formalization routes                                                                                 | 829 |
| XVI.11 | Ledgers, checks, and source audits of sources 32–35                                                           | 839 |
| XVI.12 | What this Part answers                                                                                        | 845 |
| XVII   | Finite-support arithmetic and definability horizons beyond Ord                                                | 848 |
| XVII.1 | The question and its three different answers                                                                  | 849 |
| XVII.2 | Class well-orders and the set boundary                                                                        | 852 |
| XVII.3 | Finite-support ordinal arithmetic                                                                             | 854 |
| XVII.4 | Hereditary $\Omega$ -normal forms                                                                             | 858 |
| XVII.5 | Arithmetic of the represented initial segments                                                                | 861 |
| XVII.6 | Epsilon notation and exact external spectra                                                                   | 863 |
| XVII.7 | Uniform notation systems and definability horizons                                                            | 871 |

|                |                                                        |             |
|----------------|--------------------------------------------------------|-------------|
| <b>XVII.8</b>  | <b>Rigorous formalization choices</b>                  | <b>880</b>  |
| <b>XVII.9</b>  | <b>Further research questions and topics</b>           | <b>884</b>  |
| <b>XVII.10</b> | <b>Proof status, source audit, and reproducibility</b> | <b>884</b>  |
| <b>XVII.11</b> | <b>What this Part answers</b>                          | <b>887</b>  |
|                | <b>Questions, conclusions and status</b>               | <b>890</b>  |
| <b>113</b>     | <b>Research questions</b>                              | <b>890</b>  |
| <b>114</b>     | <b>The sources' conclusions</b>                        | <b>968</b>  |
| <b>115</b>     | <b>Results, assumptions, and verification status</b>   | <b>971</b>  |
| <b>A</b>       | <b>Source crosswalk and reproducibility</b>            | <b>978</b>  |
| <b>B</b>       | <b>The finite checks as shipped</b>                    | <b>986</b>  |
| <b>C</b>       | <b>The other sources' title pages and abstracts</b>    | <b>989</b>  |
| <b>D</b>       | <b>Crosswalk of the sources' numbered statements</b>   | <b>1006</b> |

# 1 About this report

## 1.1 Four manuscripts, one report

[merge] (Added 3 October 2026: six further manuscripts of batch 81 are merged as sources 13–18 and described in [section 1.8](#). This subsection describes the four batch-80 sources.) [merge] All four manuscripts arrived in commit 4e270aa46 and were placed in ccc046989 (batch 80, cluster K3). Their batch-80 manuscript numbers are their source numbers here.

| Source       | Manuscript (archive)                                                                                                                                                                                                                                  | Pin and date line                                                                                              | Labels and files                        |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 11<br>(base) | <i>Lexicographic Orders of Well-Orderings of the Surreal Numbers: Set Approximations, Class Orders, and Higher-Order Structure</i> ; 2,807 lines, 35-page PDF<br>(surreal_well_orders (1).zip)                                                        | ae7c1e6aa (2 October 2026, 18:03:53 PDT = 3 October 2026, 01:03:53 UTC); dated “3 October 2026” (the UTC date) | prefix swo::; files<br>11-raw-orders-*  |
| 12           | <i>Well-Orders of the Surreal Numbers and Their Lexicographic Geometry: Class-sized common prefixes, exact embedding spectra, a singular-cardinal transition, and size-safe formalization</i> ; 1,689 lines, 24-page PDF<br>(surreal_well_orders.zip) | 63bb8b1d9 (2 October 2026, 18:01:32 PDT); dated 2 October 2026                                                 | prefix swo:st::; files<br>12-singular-* |
| 09           | <i>Well-orders of the Surreal Numbers: Lexicographic universality, a surreal dense core, and the boundary between sets and classes</i> ; 1,677 lines, 24-page PDF<br>(Surreal_Well_Orders_Research.zip)                                               | 63bb8b1d9; dated 2 October 2026                                                                                | prefix swo:ec::; files<br>09-core-*     |
| 08           | <i>Lexicographic Orders of Surreal Well-Orderings: Set-sized spectra, global enumeration orders, and a surreal-sized dense skeleton</i> ; 2,022 lines, 26-page PDF<br>(Surreal_Well_Orders_Research (1).zip)                                          | ae7c1e6aa; dated 2 October 2026                                                                                | prefix swo:sk::; files<br>08-skeleton-* |

The two pairs of archives sharing a name (08 and 09, 11 and 12) are not editions of one another. Every pair of the four texts has word 8-gram overlap of 0.6–1.1 % (preamble boilerplate), no member file is shared, the pins differ within each pair, and the result sets differ. They are four independent manuscripts on one spine; none supersedes another, and all four are merged (union).

Source 11 is the base. Its global-choice equivalence is proved over GB without set choice, the weakest base theory of the four ([theorem 13.1](#)); it covers set alphabets, set-length words, and set-like and unrestricted class well-orders; and it alone attributes the general set-alphabet theorems to the repository’s report on the reals, `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/lexicographic-well-orderings-of-reals/` (cited below as the reals report, labels lwo:).

## 1.2 How the four texts are printed

[merge] Source 11’s text is printed in full and in its own order. The other sources’ material is inserted where it belongs.

- Their introductions ([section 4](#)) and foundational sections are printed whole, as sections headed “Source  $N$ ”.

- A result that a printed result already proves by the same argument is recorded in a numbered *Note*: the note gives the source’s statement, its number in that source, and what differs (hypotheses, strength, extra remarks). Prose of the source that makes a point not made elsewhere is kept in the note.
- A genuinely different proof is printed in full as a marked route (“Source 12’s route”, “second proof”). A result proved only by one source is printed in full where it belongs, with its source named in the heading or the first sentence.
- Every label of every source survives, with the source’s prefix (**swo:** for 11, **swo:st:** for 12, **swo:ec:** for 09, **swo:sk:** for 08). Numbered statements without a label in their source carry **swo:sub:nnumber**, for example **swo:st:n10.4** for source 12’s Corollary 10.4. [Appendix D](#) maps every numbered statement of the four sources to its place here.
- Text written for the merge is marked **[merge]**. Dated notes of 2 October 2026 record where one source answers another ([section 1.7](#)).

Statements and sections are numbered by their place in this report; source numbers appear in headings, notes and [appendix D](#). Apart from the renamings of [section 1.3](#) and the corrections listed in [section 1.6](#), the sources’ text is verbatim. Statements of the reals report and of other repository reports are cited by label name, as the sources do; those reports are not reproduced. The same rules govern sources 13–18 ([section 1.8](#)); their introductions are in [section 4.5](#).

### 1.3 One notation for four sources

**[merge]** The sources’ symbols collide. The following convention is fixed before any of them is used; every renaming is listed, and the tempting false reading is printed where confusion is likely.

| Here                                       | Meaning, and the sources’ symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GB, GBC                                    | GB has a ZF set part, elementary class comprehension and class replacement, and <i>no</i> choice principle, as in the foundations report ( <b>found:sub:gbconvention</b> ) and the collection’s notation guide; GBC = GB + GC. Sources 09 and 12 analyse the necessity of choice “over GB plus set-level AC” (09, <a href="#">proposition 13.7</a> ; 12, <a href="#">theorem 13.2</a> ); that theory is written GB + AC in the merge text. Source 11 works over GB itself. <i>False reading</i> : the letters GB in sources 09 and 12 do not include set choice here. |
| bd( $x$ )                                  | Birthday, the length of the sign sequence (08, 09; the foundations and birthday-cutoff reports). Source 11’s $\ell(x)$ and source 12’s $b(s)$ are renamed bd( $x$ ), bd( $s$ ). In sources 08 and 09 the letter $b$ is the baseline enumeration, never a birthday.                                                                                                                                                                                                                                                                                                    |
| $\mathbf{No}_{<\kappa}$                    | $\{x \in \mathbf{No} : \text{bd}(x) < \kappa\}$ , the birthday cutoff (foundations report $\mathbf{No}_{<\alpha}$ ). Renamed: source 08’s $X_\lambda$ , $X_\kappa$ ; source 09’s $X_\theta$ (its $\kappa_\theta =  \mathbf{No}_{<\theta} $ ); source 11’s $X_\kappa$ ; source 12’s $S_\kappa$ . The reals report writes $S_{<\theta}$ . <i>False reading</i> : source 12’s $S_\kappa$ is $\mathbf{No}_{<\kappa} = S_{<\kappa}$ , not $S_{<\kappa^+}$ .                                                                                                                |
| $\ell_2(L)$                                | Binary coding length $\min\{\gamma : L \hookrightarrow \mathbb{B}_\gamma\}$ , as in the reals report ( <b>lwo:def:binary-length</b> ). Source 12 prints the same symbol $\ell_2$ and calls it the “binary coding rank” (its macro was named <b>\rk</b> ); its prose is kept. rank( $x$ ) is the von Neumann rank of a set (source 11’s <b>\rk</b> ), used only in the proof of <a href="#">theorem 13.1</a> . <i>False reading</i> : the “rank” of source 12 is $\ell_2$ , not rank.                                                                                  |
| $\mathcal{W}(X)$ , $\mathcal{W}_\alpha(X)$ | All exhaustive well-order enumerations of a set $X$ (lengths below $ X ^+$ ), and those of length exactly $\alpha$ , compared at the first unequal entry in the given order of $X$ (reals report’s $\mathcal{W}$ , $\mathcal{W}_\lambda$ ). The minimal slice is $\mathcal{W}_{ X }(X)$ . Source 11’s $\mathcal{P}_\lambda(X)$ is renamed $\mathcal{W}_\lambda(X)$ ; source 12’s script $\mathscr{W}$ is printed $\mathcal{W}$ .                                                                                                                                      |

| Here                                                              | Meaning, and the sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{I}(X), \mathcal{I}, \mathcal{T}$                        | $\mathcal{I}(X)$ : injective words of length below $ X ^+$ over a set $X$ (source 11).<br>$\mathcal{I} = \mathcal{I}(\mathbf{No})$ : the class of all injective set-length surreal words, proper prefixes first (source 11's $\mathcal{I}$ ; source 08's $\mathcal{I}(\mathbf{No})$ ; source 09's $\mathcal{T}$ , renamed). $\mathcal{T}$ : all set-length words, not necessarily injective (source 11's $\mathcal{T}$ ). <i>False reading</i> : source 09's $\mathcal{T}$ was the injective class. The reals report's $\mathcal{I}$ (injections $\kappa \rightarrow \mathbb{R}$ ) is unrelated.                                                                                                                                                                                                                                                                                     |
| $\mathfrak{W}_{sl}(\mathbf{No}), \mathfrak{W}_{all}(\mathbf{No})$ | Set-like global well-orders of $\mathbf{No}$ , equivalently class bijections $\text{Ord} \rightarrow \mathbf{No}$ ; and all class well-orders of $\mathbf{No}$ . Both are predicates on classes (“metanotation”), never classes of classes. Source 12's $\mathcal{E}(\mathbf{No})$ , written $\text{SL}(\mathbf{No})$ in its macros, and its $\mathcal{W}_{cl}(\mathbf{No})$ ; source 09's $\mathcal{E}$ ; source 08's “virtual” $\mathcal{E}$ are renamed. $\mathfrak{J}$ is source 11's collection of class injections $\text{Ord} \rightarrow \mathbf{No}$ .                                                                                                                                                                                                                                                                                                                      |
| $\mathbb{B}_\gamma, \mathbb{B}_{\text{Ord}}$                      | $\mathbb{B}_\gamma = (\{0, 1\}^\gamma, <_{\text{lex}})$ (reals report, 08, 12). $\mathbb{B}_{\text{Ord}}$ : class binary words on $\text{Ord}$ compared at their least disagreement (source 08's $\mathbb{B}_{\text{Ord}}$ ; source 12's $\mathfrak{B}$ , renamed; source 11 writes $\{0, 1\}^{\text{Ord}}$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\simeq_e, \hookrightarrow$                                       | Equimorphism (embeddings both ways; not isomorphism) and order embedding. Source 11's $\approx_{\text{emb}}$ is printed $\simeq_e$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $\mathcal{P}_{bd}(e_0), \mathcal{P}_s(b)$                         | The <i>core</i> : enumerations that agree with a fixed baseline enumeration outside a set of positions. Source 11 codes them by normalized permutations $\sigma : \lambda \rightarrow \lambda$ ( $\mathcal{P}_{bd}(e_0)$ , <a href="#">definition 19.3</a> ); sources 09 and 08 by fixed-point-free bijections $\sigma : S \rightarrow S$ of a set $S$ of ordinals (source 09's $\mathcal{C}_b$ , source 08's $\mathcal{P}_b$ , both written $\mathcal{P}_s(b)$ here; source 08's code carrier $\mathcal{P}$ is $\mathcal{P}_s$ , and its power-set $\mathcal{P}(V_\kappa)$ keeps the letter $\mathcal{P}$ ). The two codings have the same evaluations: restrict the identity extension of a support code to $\lambda = \sup\{\xi + 1 : \xi \in S\}$ , and conversely discard fixed points. With the same baseline, every order statement about one is a statement about the other. |
| $e_0, B, b$                                                       | The fixed baseline class bijection $\text{Ord} \rightarrow \mathbf{No}$ of sources 11, 12, and 09 and 08 respectively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| $d(F, G), \Delta(E, F), \delta(e, f)$                             | The least ordinal at which two class functions on $\text{Ord}$ differ (sources 11, 09, 08).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| $\eta_{\text{Ord}}, \eta_\kappa$                                  | The set-cut property, and its version for cuts of fewer than $\kappa$ elements, empty sides allowed (11, <a href="#">definition 5.1</a> ; 08, <a href="#">definition 5.8</a> ); source 09 writes “set-cut property”. The subscript is a cardinal bound, not an $\aleph$ -index.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Termination                                                       | Sign sequences compare with $- < \text{termination} < +$ (numerical order); words compare with proper prefixes first. Exhaustive enumerations of one set are prefix-free, so no convention is needed there.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| (S08. $n$ )                                                       | Source 08's displayed equations (1)–(13) keep their numbers as (S08.1)–(S08.13). (S08.10), an inequality inside a proof recorded in <a href="#">note 19.13</a> , is described there, not displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### 1.3.1 Addendum of 3 October 2026: notation for sources 13–18

[merge] The six batch-81 sources ([section 1.8](#)) reuse the symbols above with other meanings, and several reuse this report's macro names for other objects. The convention below is fixed before any of their material is printed; it applies to Parts VI–IX and changes nothing in Parts I–V. Every renaming is listed; the tempting false reading is printed where confusion is likely.

| Here                                                   | Meaning, and the batch-81 sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\kappa, \mu, r, q$                                    | $\kappa$ is an infinite cardinal used as a birthday cutoff, $\mu = 2^{<\kappa} =  \mathbf{No}_{<\kappa} $ , $r = \text{cf } \kappa$ and $q = \text{cf } \mu$ , as in source 17. Renamed: source 18's $\nu, \tau$ (printed $r, q$ ; in source 18's general principle, $\nu$ stays the point character of an abstract order $D$ ); source 14's $\tau = \text{cf } \kappa$ (printed $r$ ); source 15's $\delta = \text{cf } \kappa$ in its saturation section (printed $r$ ). Source 16 writes $\kappa$ for the cardinality of a general alphabet and $\theta$ for the cutoff; in its text these are printed $\mu$ and $\kappa$ . Sources 13–18's $X_\kappa$ and source 16's $X_\theta$ are printed $\mathbf{No}_{<\kappa}$ . <i>False reading:</i> $r$ is a cofinality, never a coding length (next row). |
| $X,  X  = \mu$                                         | A general alphabet: an infinite linearly ordered set, of cardinality $\mu$ (source 18's convention; source 16's $\kappa$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $\rho_\kappa, \mathbb{B}_\sigma \times_{\text{lex}} m$ | Source 15's $r_\kappa$ ( $\kappa \cdot \kappa$ at a singular strong-limit cutoff, $\mu$ otherwise; by <a href="#">theorem 8.3</a> it is $\ell_2(\mathcal{W}_\mu(\mathbf{No}_{<\kappa}))$ ), renamed to keep $r = \text{cf } \kappa$ ; and source 15's $D_{\sigma,m} = \mathbb{B}_\sigma \times_{\text{lex}} \{0, \dots, m-1\}$ , the binary cube with a finite chain of $m$ points as last coordinate, renamed because source 15's core is $\mathcal{D}$ .                                                                                                                                                                                                                                                                                                                                              |
| d, cell, w                                             | Source 18's letter $\rho$ for a regular cardinal is printed $\theta$ , as in source 17. Density, cellularity (a supremum, not necessarily attained) and weight of an order topology, as in the reals report; sources 15–18 write $d, c, w$ . Cellularity is never written $c$ . Source 15's $d(a, b)$ (a first-difference coordinate) and source 18's $d_* = \mu^{<\mu}$ keep their letters.                                                                                                                                                                                                                                                                                                                                                                                                            |
| ci, characters                                         | ci is coinitality. The left and right characters of a point $x$ are $\text{cf}(L_{<x})$ and $\text{ci}(L_{>x})$ (sources 17, 18; source 15 writes $\text{cf}^-(f)$ , $\text{ci}^+(f)$ ). A point character, the global cofinality of the order, and the character of a gap are three different invariants (source 18's warning).                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| GapChar( $L$ ), Reg                                    | The set of characters ( $\text{cf } A, \text{ci } B$ ) of the gaps $(A, B)$ of $L$ , a gap having nonempty sides with no endpoints (principal cuts and the external ends are excluded). Source 17's $\Sigma(L)$ and source 18's $\text{GapSpec}(L)$ are printed with the reals report's symbol GapChar; the report <b>surreal-fields-across-universes</b> writes GapSpec. Reg is the class of infinite regular cardinals (source 17's $\text{Reg}_\infty$ ; the reals report writes $\text{Reg}(<\kappa)$ for those below $\kappa$ ).                                                                                                                                                                                                                                                                   |
| $\eta_\nu$                                             | Source 17: for regular $\nu$ , every cut $A < B$ with $ A ,  B  < \nu$ is filled; sources 15 and 18: every cut with $ A  +  B  < \nu$ (resp. $ A \cup B  < \nu$ ) is filled, for any infinite cardinal $\nu$ . Empty sides are allowed throughout. For regular $\nu$ the three definitions agree with this report's $\eta_\kappa$ ( <a href="#">section 1.3</a> ), and every threshold statement printed concerns the regular cardinal $r$ .                                                                                                                                                                                                                                                                                                                                                            |
| $P, [p], C_p$ ; straddling                             | $P = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ is the minimal slice (sources 17, 18; source 15 writes $L$ ). For an injective prefix $p$ , $[p]$ (source 17) and $C_p$ (sources 16, 18, and source 11 in Part IV) are the same cylinder. A cylinder that meets both sides of a cut <i>straddles</i> it; this is the word of sources 16, 17 and 18 and is used in all batch-81 Parts. Sources 13 and 14 say that a prefix <i>crosses</i> a cut, source 15 that it is <i>persistent</i> ; these words are explained where printed.                                                                                                                                                                                                                                                                          |
| $\mathcal{T}, \mathcal{I}, \mathcal{T}^{\text{inc}}$   | All set-length words, injective set-length words, numerically increasing set-length words. Source 16's $\mathcal{T}, \mathcal{T}^{\text{inj}}, \mathcal{T}^{\text{inc}}$ , whose macro <code>\Words</code> names <i>all</i> words while this report's <code>\Words</code> names the injective ones; source 13's $\mathcal{T}, \mathcal{I}$ agree with this report. Source 18's centered classes $\mathcal{C}, \mathcal{C}_{\text{inj}}, \mathcal{C}_{\text{inc}}$ are printed $\mathcal{T}_c, \mathcal{I}_c, \mathcal{T}_c^{\text{inc}}$ (its $\mathcal{C}$ would clash with $\mathcal{C}_b$ ). Source 16's termination types $T_{\min}, T_\kappa$ are printed $\mathbf{T}_{\min}, \mathbf{T}_\kappa$ . <i>False reading:</i> source 16's $\mathcal{T}$ is not the injective class.                     |



| Here                                                                            | Meaning, and the batch-81 sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathfrak{W}_{\text{sl}}(\mathbf{No}), \mathfrak{W}_{\text{all}}(\mathbf{No})$ | As in <a href="#">section 1.3</a> . Renamed: $\mathfrak{E}$ (sources 13, 14, 16, 17) for set-like global orders (source 15 already writes $\mathfrak{W}_{\text{sl}}$ ); source 14's $\mathfrak{W}$ (macro <code>\Hwell</code> ), source 16's $\mathfrak{H}$ and source 17's $\mathfrak{W}$ (macro <code>\HW0</code> ) for all class well-orders (sources 13 and 15 already write $\mathfrak{W}_{\text{all}}$ ). Source 18's macros <code>\Hsl</code> , <code>\Hall</code> print script $\mathcal{H}_{\text{sl}}$ , $\mathcal{H}_{\text{all}}$ there and $\mathfrak{W}_{\text{sl}}$ , $\mathfrak{W}_{\text{all}}$ here.                                                                                                                                                                                                                                                                                                  |
| $\mathfrak{J}$                                                                  | Class injections $\text{Ord} \rightarrow \mathbf{No}$ , as source 11's $\mathfrak{J}$ (sources 13, 14, 17 write $\mathfrak{J}$ with the macro <code>\J</code> ; source 16 writes $\mathfrak{J}$ with the macro <code>\I</code> ; source 15's $\mathfrak{J}$ ). <i>False reading</i> : source 16's <code>\I</code> is not this report's $\mathcal{I} = \mathcal{I}$ , the class of injective set-length words.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $\mathcal{P}_{\text{bd}}(b), \mathcal{P}_{\text{s}}(b)$                         | The core, as in <a href="#">section 1.3</a> , in the source's own coding: normalized permutations $\mathcal{P}_{\text{bd}}(b)$ for sources 15 ( $\mathcal{D}$ ) and 16 ( $\mathcal{P}_b$ ); fixed-point-free support codes $\mathcal{P}_{\text{s}}(b)$ for sources 13 ( $\mathcal{D}$ ), 14 ( $\mathcal{P}_b$ ), 17 ( $\mathcal{C}_b$ ) and 18 ( $\mathcal{C}_b$ , macro <code>\Pbd</code> ). <i>False reading</i> : source 18's <code>\Pbd</code> is the support coding, not this report's $\mathcal{P}_{\text{bd}}(e_0)$ ; the two codings have the same evaluations.                                                                                                                                                                                                                                                                                                                                                 |
| $\mathcal{P}_{<\kappa}(b)$                                                      | Source 14's support-cardinality layer: codes whose support has cardinality less than $\kappa$ . <i>False reading</i> : it is not $\mathcal{P}_{\text{bd}}$ , whose subscript means bounded (set-sized) support; source 14 itself warns that a cardinality filtration and an ordinal-height filtration must not share the word "bounded".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $\simeq_e, <_{\text{lex}}$                                                      | Equimorphism is $\simeq_e$ for all six sources (source 16's $\simeq_{\text{emb}}$ ; source 18's macro <code>\equim</code> prints $\rightleftharpoons$ there and $\simeq_e$ here). Source 14's macro <code>\lex</code> prints the relation $<_{\text{lex}}$ ; this report's <code>\lex</code> is the subscript $\text{lex}$ , so source 14's relation is printed $<_{\text{lex}}$ . Source 18's bold $\mathbf{B}_\gamma$ is printed $\mathbb{B}_\gamma$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Cuts of the core                                                                | A normalized class cut $(L, U)$ has a lower side $L$ without greatest element; the cut of an injective branch $F$ is written $L_F$ (source 16's $L_F$ , source 17's $L_f$ ; source 14's $D_F, D_p^\pm, D_{p,H}$ and source 13's $(L, U)$ are explained where printed). In Part VIII, source 13's code cylinder $\mathcal{D}_p$ is printed $\mathcal{P}_{\text{s}}(b)_p$ , and source 17's $\mathcal{C}_p$ and source 18's $\mathcal{C}_b(p)$ are printed $\mathcal{C}_p$ ; in source 14's support section its decreasing sequence $(r_i)$ is printed $(u_i)$ and the regular cardinal of its definition of $\eta$ -orders $\theta$ , since $r = \text{cf } \kappa$ there.                                                                                                                                                                                                                                               |
| $\varepsilon, \text{end}, \dagger$                                              | A formal termination symbol adjoined to the alphabet of a word order; it is never a surreal number. Source 16's $\varepsilon$ sits at a cut $L_\varepsilon < \varepsilon < R_\varepsilon$ of $\mathbf{No}$ , and its $0^-, 0^+$ sit immediately below and above zero; source 14 writes <code>end</code> ; source 18's centered $\dagger$ takes the place of zero, which is then not a letter, and its $\dagger_-, \dagger_+$ sit immediately below and above an available zero. Source 13's central termination pads a word with zeros, zero again not being a letter. Source 18 writes $\ell(p)$ for the length of a word $p$ and $\mathcal{J}$ for the increasing words, printed $\mathcal{T}^{\text{inc}}$ . <i>False reading</i> : the Conway sign order $- < \text{end} < +$ is not prefix-first dictionary order.                                                                                                 |
| $(\text{SNN}.n)$ , labels                                                       | Displayed equations of source $NN$ keep their numbers as $(\text{SNN}.n)$ , as for source 08. Labels: <code>swo:fs</code> : (13, replacing its <code>swf</code> :), <code>swo:tc</code> : (14), <code>swo:cb</code> : (15, keeping its internal <code>cce</code> : and <code>new</code> :, since dropping them would merge two of its labels), <code>swo:ds</code> : (16), <code>swo:gs</code> : (17), <code>swo:sh</code> : (18, keeping its internal <code>wv</code> :); unlabelled numbered statements get <code>swo:sub:nnumber</code> ; labels written for the merge in Part VI begin <code>swo:vi</code> :, in Part VII <code>swo:vii</code> :, in Part VIII <code>swo:viii</code> :, in Part IX <code>swo:ix</code> :. Two labels of source 13 printed in Part VI carry its old prefix as well, <code>swo:fs:swf:sec:fixed</code> and <code>swo:fs:swf:thm:fixed</code> ; like every label they are not renamed. |

| Here                               | Meaning, and the batch-81 sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Blocks, ranks, $\kappa$ in Part IX | Source 15's $B_W$ , $C_b$ (least representatives, block of $b$ ), source 16's $J_R$ , $\alpha_R$ and source 18's $B_R$ , $\theta_R$ (representatives of the proper blocks, type of the final set block) are kept; $\rho_R(x)$ (sources 16 and 18) is a rank inside a block, not $\rho_\kappa$ . <i>False readings:</i> in source 15's block section $b$ is not the baseline and $C_b$ is not a cylinder; source 14's $J_R$ is an initial branch, not source 16's $J_R$ . In Part IX's sections on larger universes $\kappa$ is strongly inaccessible (source 14's $\Omega$ , sources 17's and 18's $\lambda$ ). Source 18's plain $H_{sl}$ , $H_{all}$ are printed $\mathfrak{W}_{sl}$ , $\mathfrak{W}_{all}$ . |

### 1.3.2 Addendum of 3 October 2026: notation for sources 19–30

[merge] The twelve batch-83 sources (section 1.9) reuse the symbols of sections 1.3 and 1.3.1 with other meanings, give one object up to five names, and reuse this report's macro names (`\Words`, `\BB`, `\JJ`, `\End`, `\I`) for other objects. The convention below is fixed before any of their material is printed; it applies to Parts X–XIV and to the batch-83 sections of the front and back matter, and changes nothing earlier. The conventions of section 1.3.1 continue to hold. Every renaming is listed; the tempting false reading is printed where confusion is likely. Each printed section names the renamings it uses.

| Here                                                          | Meaning, and the batch-83 sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\kappa$ , $\mu$ , $r$ , $q$                                  | As in section 1.3.1: $\kappa$ is an infinite cardinal used as a birthday cutoff, $\mu = 2^{<\kappa} =  \mathbf{No}_{<\kappa} $ , $r = \text{cf } \kappa$ , $q = \text{cf } \mu$ ; for a general alphabet $ X  = \mu$ . Renamed: source 26 writes $\kappa$ for $ X $ and $\theta$ for the cutoff, printed $\mu$ and $\kappa$ throughout (so its $\mu = 2^{<\theta}$ reads $\mu = 2^{<\kappa}$ , and its $H_\kappa$ reads $H_\mu$ ); source 27 writes $\kappa$ for $ X $ and $\theta$ for a cardinal cutoff in its scan and convex-copy sections, printed $\mu$ and $\kappa$ ; source 23 writes $\kappa =  X $ and $\lambda$ for the cutoff in its spectra section, printed $\mu$ and $\kappa$ ; source 29's fixed set of cardinality $\kappa$ in its baseline section is printed with $\mu$ ; source 24's $r = \text{cf } \mu$ and source 28's $\delta = \text{cf } \mu$ are printed $q$ . Sources 19–30's $X_\kappa$ , $X_\theta$ and source 27's $S_{<\theta}$ are printed $\mathbf{No}_{<\kappa}$ , $\mathbf{No}_{<\theta}$ . <i>False reading:</i> source 26's and source 27's $\kappa$ is not a birthday cutoff. |
| $\theta$ noncardinal; $\mu = 2^\kappa$ there                  | At a <i>noncardinal</i> cutoff (an infinite ordinal $\theta$ that is not a cardinal: source 21's noncardinal section, source 29's birthday-spectrum and noncardinal sections) the sources write $\kappa =  \theta $ and $\mu =  \mathbf{No}_{<\theta}  = 2^\kappa$ ; this is kept, with $\theta$ for the cutoff. <i>False reading:</i> there $\mu = 2^\kappa$ , not $2^{<\kappa}$ , and $\kappa$ is $ \theta $ , not the cutoff. In Part XI $\theta$ is any cutoff ordinal and $\mathbf{No}_{<\alpha} = \{x : \text{bd}(x) < \alpha\}$ for every ordinal $\alpha$ (source 20's $X_\alpha$ , source 29's $X_\theta$ ); elsewhere $\theta$ keeps its meaning in section 1.3.1 (a regular cardinal).                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\rho_\kappa$ ; $\text{wt}$ , $\varrho_{lo}$ , $\varrho_{up}$ | $\rho_\kappa$ is $\kappa \cdot \kappa$ at a singular strong-limit cutoff and $\mu$ otherwise (section 1.3.1). Source 19's and source 29's $r_\kappa$ are printed $\rho_\kappa$ (source 30 already writes $\rho_\kappa$ ). The residual-layer weights ( $\lambda \cdot \lambda$ for a singular strong-limit $\lambda$ , else $\lambda$ ), source 19's and source 29's $r(\lambda)$ , $u_\kappa(\lambda)$ and source 30's $r(\nu)$ , are printed $\text{wt}(\lambda)$ , $\text{wt}_\kappa^+(\lambda)$ , $\text{wt}(\nu)$ (Parts X and XI). The weighted lower and upper lengths $R$ , $U$ of sources 19 and 29 are printed $\varrho_{lo}$ , $\varrho_{up}$ , since $R$ , $S$ name raw class orders. Their $D_{\rho,m}$ (also source 30's) is printed $\mathbb{B}_\rho \times_{\text{lex}} m$ , as source 15's $D_{\sigma,m}$ was; source 21's $\text{br}$ and source 29's $\ell_2$ are this report's $\ell_2$ . <i>False reading:</i> $\varrho_{lo}$ is an ordinal, never a relation, and $r$ is still $\text{cf } \kappa$ .                                                                                           |

| Here                                                                                                                                                                                   | Meaning, and the batch-83 sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{P}_{\text{bd}}(b), \mathcal{P}_{\text{s}}(b)$ (the core)                                                                                                                     | The ordinary class of set codes of bounded-support perturbations of a baseline $b : \text{Ord} \rightarrow \mathbf{No}$ , ordered by evaluation, in the source's own coding (section 1.3): normalized permutations of an ordinal, printed $\mathcal{P}_{\text{bd}}(b)$ , for sources 19 ( $\mathcal{C}$ ), 20, 24, 26 and 28 ( $\mathcal{C}_b$ ), 21 ( $S_b = \mathcal{P}_b$ ) and 29 ( $P_e$ , printed $\mathcal{P}_{\text{bd}}(e)$ , its baseline being $e$ ); fixed-point-free support codes, printed $\mathcal{P}_{\text{s}}(b)$ , for sources 22, 25 and 27 ( $\mathcal{D}_b$ ) and 30 ( $\mathcal{B}_b$ ). The two codings have the same evaluations. <i>False reading</i> : source 25's $\mathcal{C}_b$ , printed $\mathbf{C}_b$ (next row), is not the core.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $\mathbf{C}_b$                                                                                                                                                                         | Source 25's two-sorted reduct: the core with its order, the injective set-length words as nodes with the prefix order, and the incidence of codes in node cylinders; no label relation. Source 25's script $\mathcal{C}_b$ (macro $\backslash \mathbf{C}$ ), printed $\mathbf{C}_b$ ; its set version at a regular $\kappa$ is $\mathbf{C}_b^\kappa$ , with $\mathcal{P}_{\text{s}}^\kappa(b)$ , $\mathcal{I}^\kappa$ , $\mathfrak{J}^\kappa$ for its $\mathcal{D}_{b,\kappa}$ , $\mathcal{T}_\kappa$ , $\mathfrak{J}_\kappa$ , and source 25's direction sign $\varepsilon(f, g)$ is printed $\text{sgn}(f, g)$ . <i>False reading</i> : sources 20, 24 and 26 write the calligraphic $\mathcal{C}_b$ for the core itself.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| $\mathcal{T}, \mathbf{T}_{\min}, \mathbf{T}_{\min}^+$                                                                                                                                  | $\mathcal{T}$ is the class of all set-length surreal words, with repetitions (section 1.3); source 21's $\mathcal{T}$ (macro $\backslash \text{Words}$ ) is printed $\mathcal{T}$ . Its shorter-first order, the type $\mathbf{T}_{\min}$ of source 16's termination family (theorem 36.2), is written $\mathbf{T}_{\min}$ , and $\mathbf{T}_{\min}^+$ is its nonempty part: source 27's $\mathcal{B} = \text{Words}_\perp(\mathbf{No})$ (macro $\backslash \text{Bword}$ ) and $D = \mathcal{B} \setminus \{\emptyset\}$ , source 29's bold $\mathbf{T}$ and $\mathbf{P}$ (macro $\backslash \text{Pnon}$ ). <i>False reading</i> : source 21's and source 27's macro $\backslash \text{Words}$ is not this report's $\mathcal{I}$ ; source 27's $\mathcal{B}$ is a word order, not a cube $\mathbb{B}$ , not source 28's ideal $\mathcal{B}_\mu$ , not source 24's $\mathfrak{B}$ (macro $\backslash \text{BB}$ , unused).                                                                                                                                                                                                                                                                                                                              |
| $\mathcal{I}$ , the prefix tree                                                                                                                                                        | Injective set-length surreal words (section 1.3), and, with the prefix order $\sqsubseteq$ , the ordered prefix tree. Renamed: the node classes $\mathcal{T}$ of sources 22, 25 and 27 (macros $\backslash \mathbf{T}$ , $\backslash \text{Tree}$ ); source 26's $\mathcal{T}_{\text{inj}}(\mathbf{No})$ ; source 29's script $\mathcal{I}$ is already $\mathcal{I}$ . <i>False reading</i> : $\mathcal{T}$ is $\mathcal{T}$ in this report, never the injective tree.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\varepsilon; \mathcal{T}_\varepsilon, \mathcal{I}_\varepsilon, \mathcal{T}_\varepsilon^{\text{inc}}$                                                                                  | A formal marker at a cut $A = L \sqcup R$ of an alphabet, as source 16's $\varepsilon$ (section 1.3.1), and the all, injective and strictly increasing set-length words over $A$ compared with it. Renamed: source 23's $\mathbf{e}$ , $\mathcal{W}_\mathbf{e}$ , $\mathcal{I}_\mathbf{e}$ , $\mathcal{U}_\mathbf{e}$ ; source 27's $\dagger$ at a cut and its $\text{Inc}_C$ ; source 29's $\dagger$ at a cut (macro $\backslash \text{End}$ ) and its $\text{Inc}_\dagger$ . The symbol $\dagger$ stays reserved for source 18's centered terminator in the place of zero, with $\dagger_-$ , $\dagger_+$ ; in the Conway sign order source 29's $\backslash \text{End}$ is printed <b>end</b> ( $- < \text{end} < +$ ). <i>False reading</i> : source 29's $\backslash \text{End}$ is not this report's <b>end</b> = <b>end</b> .                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $\mathcal{T}_{\text{sf}}^{\text{inc}}, \mathcal{T}_{\text{lf}}^{\text{inc}}; \mathcal{T}_c^{\text{inc}}, \mathcal{T}_{\dagger\pm}^{\text{inc}}, \mathcal{T}_{\text{pad}}^{\text{inc}}$ | Increasing words over an alphabet with proper prefixes first or last (source 27's $\text{Inc}_\perp$ , $\text{Inc}_\dagger$ ; source 29's $\text{Inc}_{\text{sf}}$ , $\text{Inc}_{\text{lf}}$ ); over $\mathbf{No}$ the first is $\mathcal{T}^{\text{inc}}$ , and source 29's increasing tree $\mathcal{J}$ is $\mathcal{T}^{\text{inc}}$ with $\sqsubseteq$ . The centered increasing words with zero excluded (source 18's $\mathcal{T}_c^{\text{inc}}$ ), with zero retained and $\dagger_-$ or $\dagger_+$ , and the zero-padding quotient: source 27's $J, J_-, J_+, Q_0$ and source 29's $\mathcal{C}, \mathcal{C}_-, \mathcal{C}_+, \mathcal{C}_{\text{pad}}$ . Source 27's residual maps $J(S, T; a, c)$ are unrelated and keep their letter. <i>False reading</i> : source 29's $\mathcal{C}$ is neither a core nor a cylinder.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\mathcal{W}_\mu(X), \mathcal{I}_\mu(X), \mathcal{I}_{<\mu}(X), \mathcal{I}_\mu^{\text{b}}$                                                                                            | For $ X  = \mu$ infinite: the minimal slice $\text{Bij}(\mu, X)$ ; the injections $\text{Inj}(\mu, X)$ , which form the length- $\mu$ layer of this report's $\mathcal{I}(X)$ ; the short prefixes $\bigcup_{\alpha < \mu} \text{Inj}(\alpha, X)$ ; and the forward-bounded injections of $\mu$ (bounded sets have bounded images). Renamed: the $\mathcal{P}_\mu(X), \mathcal{I}_\mu(X)$ of sources 21, 24, 26 and 28 (source 26's $\mathcal{P}_\kappa, \mathcal{I}_\kappa$ ), source 24's $\mathcal{T}_\mu(X)$ , source 26's $\mathcal{B}^+$ and source 28's $\mathcal{I}_\mu^{\text{b}}$ . Source 28's $S_\mu = \text{Sym}(\mu)$ and its subgroup $H_\mu$ (source 24's $H_\mu$ , source 26's $H_\kappa$ ) keep their letters; its one-sided loci $B_\mu^+, B_\mu^-$ are printed $H_\mu^+, H_\mu^-$ , so that $H_\mu = H_\mu^+ \cap H_\mu^-$ . Where these sources write $\mathcal{P}$ for the slice itself it is printed $P$ . <i>False reading</i> : $\mathcal{P}_\mu(X)$ is neither $\mathcal{P}_{\text{bd}}$ nor $\mathcal{P}_{\text{s}}$ , and $\mathcal{I}_\mu(X)$ is only one layer of $\mathcal{I}(X)$ (all injective words of length below $\mu^+$ ); source 11's $\mathcal{P}_\lambda(X)$ was renamed $\mathcal{W}_\lambda(X)$ (section 1.3). |

| Here                                                                                                                                                 | Meaning, and the batch-83 sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| $\mathcal{J}_b, \mathcal{J}$ , topologies                                                                                                            | Source 28's bounded ideal $\mathcal{B}_\mu$ (macro <code>\BB</code> ), printed $\mathcal{J}_b$ , and its admissible coordinate ideals $\mathcal{J}$ (macro <code>\JJ</code> ), with topologies $\tau_{\mathcal{J}}, \tau_b, \tau_{<\mu}, \tau_{\leftrightarrow}$ ; source 29's $\tau_{\text{lex}}, \tau_{\text{coord}}$ . <i>False reading:</i> $\mathcal{J}$ and $\mathcal{J}_b$ are not $\mathfrak{J} = \mathfrak{J}$ , and this report's <code>\BB</code> is the cube $\mathbb{B}$ ; source 26's <code>\JJ</code> is $\mathfrak{J}$ , printed $\mathfrak{J}$ , and source 24's $\mathfrak{J}$ (its <code>\JJ</code> ) is $\mathfrak{J}$ too. $\tau(h)$ is still the reserve of a partial injection (theorem 49.7), as in source 27.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\mathfrak{W}_{\text{sl}}, \mathfrak{W}_{\text{all}}, \mathfrak{J}; \mathfrak{W}_{\text{all}}^{\mathcal{M}}, \mathfrak{W}_{\text{sl}}^{\mathcal{M}}$ | As in sections 1.3 and 1.3.1. Renamed: $\mathfrak{E}$ (sources 22, 23, 24, 25, 26, 28), $\mathfrak{W}$ (sources 23, 25: <code>\Raw</code> ; 26: <code>\HH</code> ), source 19's script $\mathcal{H}_{\text{sl}}, \mathcal{H}_{\text{all}}$ and source 30's macros <code>\SL, \ALL</code> . The external sets of admitted class well-orders and of set-like ones of a set-sized two-sorted model $\mathcal{M}$ , as in Part IX: source 19's $\mathcal{H}_{\text{all}}^{\mathcal{M}}, \mathcal{H}_{\text{sl}}^{\mathcal{M}}$ , source 20's $H^{\mathcal{M}}$ , source 30's $\mathcal{W}^{\mathfrak{M}}$ (printed $\mathfrak{W}_{\text{all}}^{\mathfrak{M}}$ ); source 21's $L, L^{\mathcal{M}}$ are kept in its text, where they are defined. The external set-like order of a realization $(M, \mathcal{X})$ , the $H_{\mathcal{X}}$ of sources 21 and 27, keeps that name; source 30's $\mathcal{E}(\mathcal{C})$ (the graphs of the admitted set-like enumerations) and $\mathcal{K}_c(\mathcal{C})$ keep theirs. Realizations are written $\mathcal{X} \subseteq \mathcal{Y}$ (sources 21, 22, 27) or $\mathcal{C} \subseteq \mathcal{D}$ (source 30). <i>False reading:</i> source 30's $\mathcal{C}, \mathcal{D}$ are class collections, not cores, and $H_{\mathcal{X}}$ is not a fibre $H_q$ or a history $H_i$ . |
| $I_\infty, \mathcal{F}, \sigma(\mathcal{A})$                                                                                                         | $I_\infty = \omega + \sum_{q \in \mathbb{Q}} \mathbb{Z} + \omega^*$ , the external type of an internally finite interval of nonstandard length in a countable model (source 20's symbol; source 19's $K$ and source 21's $H$ renamed); $\mathcal{F} = \{n! : n \in \omega\}$ (source 20; source 19's $\mathcal{D}$ ); and $\sigma(\mathcal{A})$ , the dense shuffle of a set $\mathcal{A}$ of colours, as in Part IX (all three write $\sigma$ ). <i>False reading:</i> $I_\infty$ has a first and a last element; $\sigma(\mathcal{A})$ is not a core code $\sigma$ or $E_\sigma$ (Part VIII).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| $H_q; H, H_i, D_W(A), p_i, c_i, \text{NF}_H$                                                                                                         | Source 19's $H_q$ is the admitted raw fibre over a core cut $q$ . Condensation histories (sources 20 and 29): the least-representative condensation operator $D_W(A)$ , history slices $H_i$ (source 20's $H_x$ ), floor maps $p_i$ , digits $c_i$ , the finite normal form $\text{NF}_H$ . Source 23 writes $D(H)$ for one condensation of a class well-order $H$ and $E_t$ for its stages (unions at limits). Source 30's orientation predicate $H$ is its own. <i>False reading:</i> source 23's $H$ is a class well-order, not a history; $H_\mu$ is a group.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $C(R, S), Y(R, S), \equiv_{\text{set}}, K_R; E, T_R, n(R)$                                                                                           | Source 20's local-size section: the common-prefix carrier (its $P(R, S)$ , printed $C(R, S)$ as in Part III), the residual $Y(R, S) = \mathbf{No} \setminus C(R, S)$ , “the residual is a set”, and $K_R$ , the largest final set segment (source 16's $A_R$ ). The internal-finite-residual equivalence $E$ and the maximal finite terminal tail $T_R$ with its length $n(R) = n_R$ (sources 18–21); source 19's $E(x, y), S(x)$ are first-order formulas of the pure order; source 21's $C_R, P_R$ are its own. <i>False reading:</i> source 18's $K_R$ (Part IX) is an initial set-like part of $B_R$ , and source 20's $A_R$ is printed $\text{Min}_R$ (next row).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\pi_Y, \pi_{\beta\alpha}, \text{lr}_\alpha(x), \text{Min}_R$                                                                                        | Restriction maps $R \mapsto R \upharpoonright Y$ (source 19's $r_Y$ ) and $\mathcal{W}(\mathbf{No}_{<\beta}) \rightarrow \mathcal{W}(\mathbf{No}_{<\alpha})$ (source 20's $\rho_{\beta\alpha}$ ); source 20's local ranks $r_\alpha(x)$ and its class $A_R$ of strong-limit cutoffs $\kappa$ at which $\text{otp}(R \upharpoonright \mathbf{No}_{<\kappa}) = \kappa$ . Source 20's $X_\alpha$ is $\mathbf{No}_{<\alpha}$ for every ordinal $\alpha$ . <i>False reading:</i> $\text{Min}_R$ is not source 16's $A_R$ , now $K_R$ , and not source 30's $A_\alpha = e[\alpha]$ ; $\rho_\kappa$ keeps its meaning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| $P_{\omega+n}, E_n, S_n, \text{Tab}_n$                                                                                                               | For a countable dense alphabet $D$ (in particular $\mathbf{No}_{<\omega}$ or $\mathbb{Q}$ ): $P_{\omega+n} = \text{Bij}(\omega + n, D)$ and $E_n$ , the $\omega$ -injections omitting exactly $n$ labels. Source 29's notation is kept; source 30's $P_n$ is printed $P_{\omega+n}$ , its alphabet $X$ is $D$ , its set $D_n$ of nonisolated points is $S_n$ (source 29's $S$ ), and its relation tables $\mathcal{T}_n$ of type $\omega + n$ are printed $\text{Tab}_n$ (here $\mathcal{T} = \mathcal{T}$ ), with its maps $\Phi_n, r_n$ ; source 29's real completion $C$ is printed $\hat{I}$ . <i>False reading:</i> source 30's $P_2$ is $P_{\omega+2}$ , not $\text{Bij}(2, D)$ , and $\text{Tab}_n$ is not $\mathcal{T}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Here                                                                   | Meaning, and the batch-83 sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $s, \vartheta, t_R, N(p), \mathcal{L}_s(X)$                            | A relation-table scan $s : \vartheta \rightarrow X \times X$ , the table bits of $R$ , realized prefix cylinders and the linear-order tables: source 27's $q : \theta \rightarrow X \times X$ , $b_R, C(p), \mathcal{L}_q$ (and source 29's scan $s$ ). Source 27's density and cellularity $d, c$ are printed $d$ , cell, its $\mathbb{B}_\kappa$ is $\mathbb{B}_\mu$ , and its gap characters $\lambda, \rho$ are printed $\lambda_L, \lambda_R$ . <i>False reading:</i> $s$ is not this report's $q = \text{cf } \mu$ , and $t_R$ is not a baseline.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| $\text{Inv}(W), g; U(A), \prec_{\text{sf}}, \prec_{\text{lf}}$         | Source 30's inversion class $I(W)$ and the rank $\rho$ of a forced relation of a partial table, printed $\text{Inv}(W)$ (this report's $\mathcal{I}$ is the injective words) and $g$ ; its inverse ranks $\text{rk}_R$ are printed $r_R$ , as in Part VII. Source 29's next-label class $U(A)$ above a numerical class carrier (Part VII's $U(q)$ for set words) and its shorter- and longer-first carrier orders $\prec_{\text{sf}}, \prec_{\text{lf}}$ ( $\prec_{\text{lf}}$ is source 15's $<_\chi$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $P(\Gamma); \text{CTH}, \text{CWO}$                                    | Finite-support functions from a class well-order $\Gamma$ to nonzero ordinals, compared at the greatest disagreement, as Part IX prints source 15's $P(J)$ ; source 23's $P(\Gamma)$ (macro <code>\Po1</code> ) and source 29's $\mathcal{P}(\Gamma)$ are printed $P(\Gamma)$ . <b>CTH:</b> every class well-order $W$ has its canonical terminating history on $W + 1$ ; <b>CWO:</b> strong comparability of class well-orders (source 29). <i>False reading:</i> $P(\Gamma)$ is not a power class; $\mathcal{P}(V_\kappa)$ remains the power set.                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathbb{B}, \simeq_e, \times_{\text{lex}}, \text{Ord as a character}$ | As before: the cube $\mathbb{B}_\gamma$ (source 19's bold <b>B</b> , source 27's <code>\Binary</code> ), equimorphism $\simeq_e$ (source 27's $\rightleftharpoons$ , source 30's <code>\eqemb</code> ), $\times_{\text{lex}}$ (source 29's $\otimes_{\text{lex}}$ ). In a character pair, <b>Ord</b> means “no set-sized cofinal (coinitial) family”, never a cardinal (source 23's <b>On</b> , source 20's $\Omega$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Labels                                                                 | Sub-prefixes: <b>swo:rs:</b> (19), <b>swo:lm:</b> (20), <b>swo:fu:</b> (21), <b>swo:rg:</b> (22), <b>swo:tm:</b> (23), <b>swo:pc:</b> (24), <b>swo:ph:</b> (25), <b>swo:nb:</b> (26), <b>swo:sc:</b> (27), <b>swo:ge:</b> (28), <b>swo:bw:</b> (29), <b>swo:rt:</b> (30). Each delivered label $L$ becomes <b>swo:sub:</b> $L$ , internal prefixes included: source 21's and source 29's <b>gl:</b> become <b>swo:fu:gl:</b> and <b>swo:bw:gl:</b> and are unrelated to source 11's <b>swo:gl:</b> ; source 30's <b>fnd:</b> labels and its <b>fnd:prop:pairs</b> , <b>fnd:thm:faithful</b> (names also used by sources 21 and 22) are distinct under <b>swo:rt:</b> . Unlabelled numbered statements get <b>swo:sub:nnumber</b> ; displayed equations keep their numbers as $(\text{SNN}.n)$ . Labels written for the merge begin <b>swo:x:</b> , <b>swo:xi:</b> , <b>swo:xii:</b> , <b>swo:xiii:</b> , <b>swo:xiv:</b> in Parts X–XIV and <b>swo:b83:</b> in the front and back matter. |

### 1.3.3 Addendum of 3 October 2026: notation for source 31

[merge] Source 31 (section 1.10) works in first-order Zermelo set theory and in full-class rank models, and uses symbols that this report uses otherwise. The convention below applies to Part XV, to source 31's questions (section 113.26) and to its abstract (Appendix C); it changes nothing earlier, and the conventions of sections 1.3.1 and 1.3.2 continue to hold. Every renaming is listed; the tempting false reading is printed where confusion is likely.

| Here          | Meaning, and source 31's symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Z, ZC, ZC_F$ | Zermelo set theory in Glazer's convention, as source 31 fixes it: Extensionality, Empty Set, Pairing, Union, Power Set, Infinity and the Separation schema, with no Replacement, Collection, Foundation or Choice; $ZC = Z + \text{Choice}$ , and $ZC_F$ adds Foundation. Source 31's macro <code>\Z</code> is printed with <code>\Zer</code> (same symbol). <i>False reading:</i> $Z$ is a theory, not the integers; $ZC$ has neither classes nor Replacement, unlike <b>GB</b> (section 1.3). |

| Here                                                            | Meaning, and source 31's symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $c, \text{Sep}(c), \text{GC}_{\leq \omega}, \text{GC}_{\leq D}$ | A total unary operation symbol $c$ ; $\text{Sep}(c)$ is Separation for every formula that mentions $c$ ; $\text{GC}_{\leq \omega}$ requires $c(X) \in X$ only for nonempty $X \rightarrow \omega$ , and $\text{GC}_{\leq D}$ only for nonempty $D$ -small $X$ . <i>False reading:</i> $\text{GC}_{\leq \omega}$ is a restricted selector over $\text{ZC}$ , not this report's global choice $\text{GC}$ over $\text{GB}$ ; in the column "GC" of the table in <a href="#">section XV.11</a> , $\text{GC}$ is full global choice.                                                                                                                                                                                                                                                                                                                     |
| $X \rightarrow D$                                               | There is an injection of $X$ into $D$ , given by a set graph (by a class, where the text says "class injection"); source 31's $\hookrightarrow$ (macro <code>\inj</code> ), renamed. <i>False reading:</i> $\hookrightarrow$ remains an order embedding throughout this report; $\rightarrow$ is no order embedding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $D, \text{Small}_D, \mathcal{C}_D, \prec_D$                     | A fixed well-ordered set $(D, <_D)$ ; $\text{Small}_D(X)$ iff $X \rightarrow D$ ; the code set $\mathcal{C}_D = \mathcal{P}(D) \times \mathcal{P}(D \times D)$ and a fixed well-order $\prec_D$ of it. <i>False reading:</i> $D$ is not Part X's countable dense alphabet, and $\mathcal{C}_D$ is not a core (sources 20, 24 and 26 wrote $\mathcal{C}_b$ for the core, printed $\mathcal{P}_{\text{bd}}(b)$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $\text{Thr}_X, W_X, i_X, e_X$                                   | The set of threads on $X$ (source 31's $\mathcal{T}_X$ , renamed because $\mathcal{T} = \mathcal{T}$ is the class of all set-length words here); the selector-compatible well-order of <a href="#">lemma XV.3.1</a> ; the uniform injections into $D$ and the countable enumerations of <a href="#">theorem XV.3.3</a> and <a href="#">corollary XV.3.5</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $H_a, Q, \Theta, T, \mathbb{T}$                                 | In <a href="#">sections XV.4</a> and <a href="#">XV.5</a> : the incidence trace $H_a \subseteq I \times D$ , the partial surjection $Q$ , Glazer's obstruction sentence $\Theta$ and its target set $T$ , and the base theory $\mathbb{T}$ ( $\text{ZC}$ or $\text{ZC}_F$ ). <i>False reading:</i> $H_a$ is not a fibre $H_q$ , a history slice $H_i$ or the group $H_\mu$ ; the sans-serif theory $\mathbb{T}$ is not the order type $\mathbf{T}_{\min}$ .                                                                                                                                                                                                                                                                                                                                                                                          |
| $\mathcal{M}_\lambda, \lambda, \kappa$                          | $\mathcal{M}_\lambda = (V_\lambda, \mathcal{P}(V_\lambda))$ for a limit ordinal $\lambda > \omega$ , with every external subset of $V_\lambda$ as a class (source 31's macro <code>\M</code> , printed with <code>\RM</code> , same symbol); at a strongly inaccessible $\kappa$ it is source 12's $\mathcal{M}$ of <a href="#">section 21</a> . The height $\lambda$ is also the birthday cutoff of $S_\lambda$ , and $\kappa =  \lambda $ , as in the noncardinal convention of <a href="#">section 1.3.2</a> with $\lambda$ for $\theta$ . <i>False reading:</i> $\kappa$ is the cutoff only when $\lambda$ is a cardinal; $\mathcal{M}_\lambda$ is not one of the arbitrary set-sized GBC models $\mathcal{M}$ of Parts IX and XIV, and it satisfies Class Replacement only at strongly inaccessible heights ( <a href="#">theorem XV.9.3</a> ). |
| $S_\lambda, \text{bd}$                                          | The sign carrier $S_\lambda = \bigcup_{\alpha < \lambda} {}^\alpha 2$ and $\text{bd}(s) = \text{dom}(s)$ . Reading 0, 1 as $-, +$ , $S_\kappa = \mathbf{No}_{< \kappa}$ as a set of sign sequences, and $ S_\kappa  = 2^{< \kappa} = \mu$ ( <a href="#">lemma XV.7.1</a> ). Kept, not renamed $\mathbf{No}_{< \lambda}$ , because source 31 asserts no field operations on $S_\lambda$ at arbitrary cutoffs (Part XI's $\mathbf{No}_{< \theta}$ is the same set, used as an ordered alphabet). <i>False reading:</i> $S_\lambda$ is neither source 28's $S_\mu = \text{Sym}(\mu)$ nor Part X's set $S_n$ . Source 12's $S_\kappa$ , printed $\mathbf{No}_{< \kappa}$ in Parts I–V, is the same set as source 31's $S_\kappa$ .                                                                                                                       |
| $\text{CR}(a), \text{CR}(\text{Ord}), \text{CR}, \text{OA}$     | Class Replacement for the admitted class functions on the internal set $a$ , on every internal ordinal, and on every internal set; ordinal abstraction (every internal set well-order is isomorphic to an internal von Neumann ordinal by an internal set graph).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Numbers, labels                                                 | The sections of Part XV are numbered XV.1–XV.15, and source 31's statement $k.m$ is printed as XV. $k.m$ ; its equations (1)–(10) are tagged (S31.1)–(S31.10); its research questions 1–12 are printed in <a href="#">section 113.26</a> . Labels: <code>swo:gcz:</code> before each delivered label; unlabelled numbered statements <code>swo:gcz:nnumber</code> ( <code>swo:gcz:n3.4</code> ); its questions <code>swo:gcz:n1–swo:gcz:n12</code> , numbered consecutively as source 28's <code>swo:ge:nk</code> ; labels written for the merge <code>swo:gcz:part</code> , <code>swo:gcz:sub:...</code> , <code>swo:gcz:sec:status</code> , <code>swo:gcz:sec:qstatus</code> and <code>swo:gcz:app:...</code> . Source 31 defines a macro <code>\rk</code> that it never uses; its <code>\restr</code> and <code>\code</code> are this report's.   |



### 1.3.4 Addendum of 4 October 2026: notation for sources 32–35

[merge] Sources 32–35 (section 1.11) work with class well-orders as presentations, in GB, GBC and full or definable class models over  $V_\kappa$ , and use several symbols that this report and each other use otherwise. The convention below applies to Part XVI, to their questions (section 113.28) and to their abstracts (Appendix C); it changes nothing earlier, and the conventions of sections 1.3.1 to 1.3.3 continue to hold. Every renaming is listed; the tempting false reading is printed where confusion is likely.

| Here                                                                        | Meaning, and the sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Omega$                                                                    | The presentation $(\text{Ord}, <)$ of the class of ordinals, used as a factor, base or exponent (sources 32 and 35 write $\backslash\text{Om}$ , source 33 $\backslash\text{On}$ ). <i>False reading:</i> $\Omega$ is not a set ordinal of the same universe and not a surreal number; it is not source 14's inaccessible $\Omega$ (printed $\kappa$ in Part IX) nor source 20's $\Omega$ in a character pair (printed $\text{Ord}$ ). In the external models of Section XVI.8 it has external type $\kappa$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $A^{[B]}, A^{[B]}, P(\Gamma)$                                               | The finite-support power: finite partial functions from $B$ to $A$ minus its least point, compared at the greatest disagreement; $P(\Gamma) = \Omega^{[\Gamma]}$ is the polynomial order of Parts IX and XIV (source 15's $P(J)$ , source 23's $P(\Gamma)$ , source 29's $P(\Gamma)$ ). Source 34 writes $\backslash\text{Pow}\{A\}\{B\}$ (printed with $\backslash\text{fsp}$ , because $\backslash\text{Pow}$ is this report's power set) and $P(\Gamma)$ as $\backslash\text{PWO}$ ; source 35 writes $A^{[B]}$ and $P(\Gamma)$ , printed $P(\Gamma)$ ; sources 32 and 33 write $A^B$ and $P_\Gamma$ , printed $A^{[B]}$ and $P(\Gamma)$ wherever the operands are class orders. For set ordinals $A^{[B]}$ is ordinary ordinal exponentiation, and $\Omega^n$ for finite $n$ abbreviates $\Omega^{[n]} = \Omega \cdots \Omega$ . <i>False reading:</i> $A^{[B]}$ is not the class of all functions $B \rightarrow A$ and not cardinal exponentiation; in a hereditary term display $\Omega^t a$ the superscript is a term, not a power. A superscript $M$ , as in $\text{Ord}^M$ , is relativization to a model. |
| $\mathcal{E}, \mathcal{E}(A), \mathcal{E}[\kappa], \mathcal{E}^{\leq n}$    | The class of hereditary finite base- $\Omega$ terms (source 34's $\backslash\text{Eps} = \mathcal{E}$ , source 35's $\mathcal{E}$ ), its version over a base class well-order $A$ (source 34's $H(A)$ , source 32's $\mathcal{H}(A)$ ), the terms with coefficients below $\kappa$ , and the terms of height at most $n$ (source 34's $E_n$ ). <i>False reading:</i> $\mathcal{E}$ is not source 09's $\mathcal{E}_\kappa$ of Part V or the $\mathcal{E}$ of Section 1.3, and $\mathcal{E}(A)$ is not a history; $\mathcal{E}[\kappa]$ is not an initial segment of $\mathcal{E}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\mathbb{T}, T_n$                                                           | Source 33's tower $T_0 = \Omega$ , $T_{n+1} = \Omega^{[T_n]}$ and its direct limit, which source 33 writes $E$ . <i>False reading:</i> $\mathbb{T}$ is an order isomorphic to $\mathcal{E}$ (section XVI.6.8), not a truth predicate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $\hookrightarrow, \nrightarrow, \preceq_{\text{init}}, \prec_{\text{init}}$ | An order embedding exists; none exists; an embedding with initial image exists; one with proper initial image exists. Sources 33 and 34 write $\preceq_{\text{emb}}$ , source 32 $\preceq_e$ and $\preceq_i$ , sources 33–35 $\preceq_{\text{init}}$ ; all are printed as here. <i>False reading:</i> $\hookrightarrow$ needs no initial image and does not give a comparison of lengths without CWO.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $H, H_i, D_W, p_i, c_i, \text{NF}_H, \text{CWO}, \text{CTH}, \text{ETR}$    | As in Part XIV (section 108): histories, their slices, the condensation operator, floors, digits, normal forms and the three principles. $E_i, E_t, E_c, E_b, F_b$ are the stage equivalences of a history or a tower (sources 35, 32, 33). <i>False reading:</i> source 32's $D_W$ meant the domain of $W$ and is printed $\text{dom } W$ ; its and source 33's $D, D_n$ are diagonal orders, not $D_W$ ; source 34's $\Delta$ is its diagonal sum.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $G, T, T_A, A$                                                              | A named set-like global well-order; a full satisfaction class (source 34's $T$ for $(V, \in, A, G)$ , source 33's $T$ for $(V_\kappa, \in, G)$ , source 32's $T_A$ for $(V_\kappa, \in, G, A)$ ); a named class predicate. <i>False reading:</i> $G$ is not source 23's exponent class $G$ of Part XIV, and $T$ is not $\mathbf{T}_{\text{min}}$ . Source 35 also writes $T$ for the class of terms of its lemma XVI.4.8 and for the reflection tree in Section XVI.8, printed as delivered; in neither place is it a truth predicate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Here                                                                                           | Meaning, and the sources' symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Theta_L, \delta_A, \delta_G, \delta_G^0, \delta_0, \delta_{\text{ord}}, \delta_{\text{set}}$ | External suprema of definable order types: source 34's $\Theta_L$ for a finite language $L$ with set parameters (and $\Theta_{A,G}$ ), source 32's $\delta_A$ for $\{\in, G, A\}$ , source 33's $\delta_G$ and parameter-free $\delta_G^0$ for $\{\in, G\}$ , source 35's $\delta_0, \delta_{\text{ord}}, \delta_{\text{set}}$ for $\{\in\}$ with no, ordinal or set parameters; over $V_\kappa$ these are one invariant (the [merge] paragraph at the end of source 34's text in Section XVI.8). Source 34's remainder $\delta$ in $\Omega \cdot Q + \delta$ is Part IX's $\theta_R$ . <i>False reading:</i> $\Theta_L$ is not Glazer's obstruction sentence $\Theta$ of Part XV nor the inaccessible $\Theta$ of <a href="#">section 71</a> ; none of these suprema exists inside the model. |
| $\mathcal{S}_{\text{par}}^L, \mathcal{S}_0^L, S_A, S_0, S_{\text{ord}}, S_{\text{set}}$        | External sets of definable order types (sources 34, 32, 35). <i>False reading:</i> not the sign carrier $S_\lambda$ of Part XV or $\text{Sym}(\mu)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| FPE, $C(A, j), \iota, \Phi$                                                                    | Source 35's fixed-point extension principle and completion. <i>False reading:</i> $\Phi$ is also source 33's digit map (Section XVI.5); the two are different maps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| WO, Sat, Valid                                                                                 | Formula-level well-order and satisfaction predicates (sources 32–35, written $\backslash\text{WO}, \backslash\text{Sat}$ ). <i>False reading:</i> WO is not this report's $\mathcal{W} = \mathcal{W}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Numbers, labels                                                                                | The sections of Part XVI are numbered XVI.1–XVI.12; source 34's equation $(k)$ is tagged (S34. $k$ ), source 35's (S35. $k$ ), source 32's (S32. $k$ ), source 33's ( $s.k$ ) (S33. $s.k$ ). Labels: <b>swo:bo:</b> (34), <b>swo:fc:</b> (35), <b>swo:dc:</b> (32) and <b>swo:dh:</b> (33) before each delivered label; unlabelled numbered statements <b>nnumber</b> with that prefix ( <b>swo:fc:n2.1</b> ); source 34's agenda items <b>swo:bo:R1–R12</b> ; labels written for the merge <b>swo:part:beyondord</b> and <b>swo:xvi:...</b>                                                                                                                                                                                                                                                   |

### 1.3.5 Addendum of 4 October 2026: notation for source 36

[merge] Source 36 ([section 1.12](#)) uses the objects of Part XVI under other names, and some of its letters mean other things here. The convention below applies to Part XVII, to its questions ([section 113.30](#)) and to its abstract ([Appendix C](#)); the conventions of [section 1.3.4](#) and the earlier addenda continue to hold. Every renaming is listed; the tempting false reading is printed where confusion is likely.

| Here                                                          | Meaning, and source 36's symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Omega, \Omega_u$                                            | $\Omega$ is the presentation $(\text{Ord}, \in)$ , as in Part XVI (source 36's bold $\Omega$ , macro $\backslash\text{Om}$ ). In Section XVII.8 the light-face $\Omega_u$ is the order type of $\text{Ordinal}_u$ , an ordinal of the next universe level, printed as delivered. <i>False reading:</i> $\Omega_u$ is not the presentation $\Omega$ .                                                                                                                                                                                                                                                                          |
| $A^{[B]}, \Omega^t \alpha$                                    | The finite-support power (source 36's $A^B$ ), printed $A^{[B]}$ wherever the operands are class orders, as in Part XVI; $\Omega^n$ for finite $n$ abbreviates $\Omega^{[n]}$ , and for set ordinals $(a^b, \kappa^\xi, \omega^\alpha)$ the power is ordinary ordinal exponentiation. In a term display $\Omega^{t_0} \alpha_0 + \dots$ or a point display $\Omega^{b_0} \alpha_0 + \dots$ the superscript is a term or an exponent point, not a power, as source 36 explains. <i>False reading:</i> $0^{[0]} = 1$ and $0^{[B]} = 0$ are conventions, not instances of <a href="#">definition XVII.3.1</a> .                  |
| $\mathbf{E}, \mathbf{E}^{\leq n+1}, w_n, \bar{\alpha}, \ t\ $ | The hereditary term order, source 36's sans-serif $\mathbf{T}_\Omega$ (macro $\backslash\text{treeord}$ ), which is source 34's $\mathbf{E}$ ; its height stages, source 36's $T_n$ (height at most $n+1$ ), printed $\mathbf{E}^{\leq n+1}$ as in source 34; the tower codes $w_0 = [(\bar{1}, 1)]$ , $w_{n+1} = [(w_n, 1)]$ , which are source 34's terms $\tau_{n+1}$ ; the constant $\bar{\alpha} = [(0, \alpha)]$ ; and $\ t\  = \mathbf{E}_{<t}$ . <i>False reading:</i> $\mathbf{E}^{\leq n+1}$ is a subclass of $\mathbf{E}$ , isomorphic to but not equal to source 33's separate order $T_n$ ; $w_n$ is not a word. |



| Here                                                                                                            | Meaning, and source 36's symbols                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $f \boxplus g, f \boxtimes g, F_\beta$                                                                          | Normal-form addition and multiplication of points of $\mathbf{E}$ ( <a href="#">theorems XVII.5.1 and XVII.5.2</a> ); the addition rule also applies to points of $\Omega^{[B]}$ . The terms $F_\beta$ are the coefficient multiples ( <a href="#">S36.5.1</a> ). <i>False reading</i> : not the natural (Hessenberg) sum and product; $\ f \boxplus g\  \cong \ f\  + \ g\ $ is ordinal addition of presentations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $\text{Ep}(B), e_b, \llbracket t \rrbracket$                                                                    | The epsilon-term functor (source 36's sans-serif $\mathbf{E}$ , macro <code>\epsop</code> ), its constants and its set-case evaluation. Renamed because $\mathbf{E} = \mathbf{E}$ is the hereditary order here. <i>False reading</i> : $\text{Ep}$ is not $\mathbf{E}$ , not source 09's $\mathcal{E}_\kappa$ or the $\mathcal{E}$ of Section 1.3, and not a stage equivalence $E_i$ ; $\text{Ep}(\Omega) \cong \Omega$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| $\varepsilon_\xi, \zeta_\xi$                                                                                    | The enumerations of the fixed points of $\alpha \mapsto \omega^\alpha$ (with $\varepsilon_0$ the first) and of $\xi \mapsto \varepsilon_\xi$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $\mathfrak{H}(M), \mathfrak{H}_0(M), \mathfrak{H}_p(M), \mathfrak{H}_G, \mathfrak{H}_{G,T}, \mathcal{H}_\kappa$ | Definability horizons, source 36's $H(M), H_0(M), H_p(M), H_G, H_{G,T}$ : external suprema of definable order types with set parameters, without them, with one fixed tuple $p$ , and in the languages $\{\in, G\}$ and $\{\in, G, T\}$ ; $\mathcal{H}_\kappa$ is the set of the $\mathfrak{H}_G$ (as delivered). Renamed because $H, H_i$ are histories here. They are the invariants of Part XVI: $\mathfrak{H}(M) = \Theta_{\{\in\}}$ and $\mathfrak{H}_G = \Theta_{\{\in, G\}}$ in source 34's notation, $\mathfrak{H}_G = \delta_G$ (source 33) and $= \delta_A$ for trivial $A$ (source 32); over $V_\kappa$ , $\mathfrak{H}(V_\kappa)$ is source 35's $\delta_{\text{set}}$ . <i>False reading</i> : not source 16's $\mathfrak{H}$ (printed $\mathfrak{W}_{\text{all}}$ ); $H_{\kappa+}$ in <a href="#">proposition XVII.8.1</a> is the class of sets hereditarily of size at most $\kappa$ ; $\mathcal{H}_\kappa$ is not source 32's $\mathcal{H}(A)$ , printed $\mathbf{E}(A)$ . |
| $\text{Spec}(M), \text{Spec}_G, \text{Spec}_G^0, \text{Spec}_{\text{full}}$                                     | External sets of definable (or all) class order types, source 34's $\mathcal{S}_{\text{par}}^L$ and $\mathcal{S}_0^L$ , source 32's $S_A$ , source 35's $S_{\text{set}}$ . <i>False reading</i> : external sets of ordinals, not classes of the model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| $D_n, \mathcal{D}, \delta_n, h_n, A_{n,u}, E_G, W, \tau_n, T$                                                   | Diagonal orders (as source 33's $D_n$ ) and their external types; the complexity heights $h_n$ ; the validated candidates $A_{n,u}$ ; the truth-promotion order $E_G$ ; the zeta order $W$ of <a href="#">theorem XVII.6.7</a> ; the ordinals $\tau_n$ in the proof of <a href="#">theorem XVII.6.6</a> ; the full satisfaction predicate $T$ for $(V_\kappa, \in, G)$ , as source 33's $T$ . <i>False reading</i> : $h_n$ is not source 20's $h_\kappa$ ; $E_G$ is not a stage equivalence; $W$ is not a generic class well-order; $\tau_n$ there are ordinals, not source 34's terms; $D_n$ is not the operator $D_W$ ; $T$ is a truth predicate, not a term class (source 36's $T_\Omega$ and $T_n$ are printed $\mathbf{E}$ and $\mathbf{E}^{\leq n+1}$ ).                                                                                                                                                                                                                             |
| $G, G_A, \pi$                                                                                                   | A named bijection $\kappa \rightarrow V_\kappa$ (source 32's $G$ ; it induces source 33's set-like global well-order); the swap-coded bijection of <a href="#">lemma XVII.7.10</a> ; the canonical pairing of ordinals. <i>False reading</i> : $G_A$ codes a subset $A \subseteq \kappa$ ; it is not source 32's pair $(G, A)$ of predicates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| $\preccurlyeq_{\text{init}}, \prec_{\text{init}}, L_{<x}$                                                       | Source 36's $\preccurlyeq_{\text{init}}, \prec_{\text{init}}$ and <code>\cut</code> . Source 36 also defines $\preccurlyeq_{\text{emb}}$ and an operator $\text{WO}$ that it never uses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Numbers, labels                                                                                                 | The sections of Part XVII are numbered XVII.1–XVII.11; source 36's equation $(k.m)$ is tagged (S36. $k.m$ ) and its $(*)$ (S36. $*$ ); the credit notes of the merge are XVII. $k.A$ , XVII. $k.B$ , ... Labels: <b>swo:hn:</b> before each delivered label (source 36's prefixes <b>intro:</b> , <b>ar:</b> , <b>ns:</b> , <b>hz:</b> , <b>form:</b> , <b>agenda:</b> , <b>audit:</b> ); its unlabelled Definition 6.1 <b>swo:hn:n6.1</b> ; its research questions <b>swo:hn:q1–swo:hn:q10</b> ; labels written for the merge <b>swo:part:horizons</b> and <b>swo:xvii:....</b>                                                                                                                                                                                                                                                                                                                                                                                                           |

## 1.4 What the report claims, and what it does not

[merge] The report claims the written proofs printed below, each in the foundation its statement names (ZF, ZFC, GB, GB + AC, GBC, KM + GC, or ZFC with a strongly inaccessible cardinal). The results new against the repository are: the singular transition at birthday cutoffs ([theorem 8.3](#)); the choiceless global-choice equivalence ([theorem 13.1](#)); the direct comparison of all class well-orders

(theorem 14.2, theorem 14.7), its canonical initial part and convex fibres, and its adjacency criterion (theorems 15.1 and 16.2 and proposition 15.3); the core of bounded-support perturbations and its interpolation theorem (theorem 19.10 and corollary 19.15); binary-class equimorphism and the diagonal non-coding theorems (theorems 19.32, 19.37 and 19.38); omitted ranges (theorems 19.42 and 19.43 and corollary 19.46); bounded reflection (theorem 19.30); the failure of finite support (proposition 19.31); relation-code lexicography (proposition 17.1); explicit word coding and the word-cut classification (theorems 9.1 and 11.1); and the universe models and non-embedding theorems (theorems 20.1, 20.2, 21.1 and 22.1). None of these is claimed to be historically first.

It does *not* claim the following; every limitation stated by any source is kept.

- *No novelty for re-proofs.* The set-sized general-alphabet layer (prefix-freeness, cardinality  $2^{|X|}$ , universality, no  $|X|^+$ -chain, ordinal and cube spectra, coding length  $|X|^+$ , extrema, adjacency,  $n!$  blocks, strict cubes) re-proves the reals report for every infinite linear alphabet and is printed with pointers to it (section 6.1). Universality and homogeneity of **No** are classical (Ehrlich [15]; Hamkins [20]); all four sources say so. The coding of sets behind the choice equivalence is, in source 09's words, a standard well-founded-coding argument; the connection is discussed by Hamkins [18, 19].
- *Source 11* (its Status paragraph, its delivery README's "Scope and verification" and its audit): no machine verification, no Lean build, no new Lean file; no bibliographic priority for constructions proved in the report; research questions are proposed directions, not a certified list of published open problems; the repository audit was targeted and repository-wide correctness was not assessed; the finite checks do not verify transfinite proofs, cardinal arithmetic or class comprehension. Its README states: "The mathematical proofs underwent a separate proof review." That is the delivery's own statement; no such review is recorded in the repository, and this report does not rely on it.
- *Source 12* (title page, its Section 1.2, and 12-singular-RESEARCH\_STATUS.md, "What is not claimed"): AI-assisted, unrefereed; historical priority not claimed; no previously catalogued open problem is claimed settled, nor that its exact formulations are new; no new Lean or Rocq verification; the Python checks are not substitutes for the proofs; the bounded theorem is not a construction of a full class model at a singular cardinal; the full collection of class well-orders is not treated as a set or ordinary class at the same foundational level; no universal semantic decoder for formula codes is assumed; its nine research questions are proposed continuations; its source search was not sufficient to establish originality.
- *Source 09* (title page, its Appendix A, and 09-core-RESEARCH\_STATUS.md, "Important boundaries"): AI-assisted, not peer-reviewed or Lean-verified, no historical priority; classical ingredients and repository results are distinguished from its constructions; no class whose members are proper classes is formed; the global-enumeration representation covers set-like class well-orders only, and no complete classification of the relation-code comparison is claimed; the core realizes uniform set-sized cuts, not all proper-class cuts (an explicit counterexample is supplied); for fixed alphabets the absence of  $\kappa^+$  and its reverse is not claimed to be a sufficient embeddability criterion; automorphisms and higher-level cuts are not classified; its research directions are proposals; literature and repository searches were targeted, not exhaustive.
- *Source 08* (its Status box and delivery README): explicit derivations, not a claim of historical priority; not independently refereed and not checked in Lean; no new Lean formalization; repository declarations are separated from proposed extensions; repository sources were read but not rebuilt; the finite tests do not establish any transfinite or class-theoretic theorem; every use of "class" is relative to the stated foundational model.

- *Proposed Lean modules are proposals.* The module names in the Lean plans of [sections 20 and 24 to 26](#) do not exist in the repository; the existing declarations they cite do ([section 1.5](#)).
- *Review status.* The repository’s Hilbert’s-tenth research tree holds a written-proof review of the four archives (`Computability/HilbertTenthProblem/Papers/research-wip/naive-stream-queue/review_batch80_surreal.md`, commit `c338541e3`). It found their principal arguments sound under the stated foundations, with one scoped correction to source 08, which is applied here ([section 5.6](#)), and it reran the four finite checks. A second note there authenticates the placement (`review_surreal_placement_ccc046989.md`, commit `3aa123856`). That review is pinned to the delivered archives and does not cover the text written for this merge; it is not an entry of the surreal collection’s review record, where this report’s review is pending. ([merge] Added 3 October 2026: the same tree has since reviewed the merged text as well. Its synthesis review (`review_surreal_synthesis_0be9b9134.md`, commit `280583d2f`) read every passage written for the merge and found no mathematical defect under the stated hypotheses, but two overstatements, both corrected here in place: the choice recursion of [theorem 13.1](#) had been called a “coherent” sequence of well-orders in [remark 13.4](#) and [remark 113.15](#), and the README had understated the largest delivered file. Its preservation audit (`review_surreal_transfer_0be9b9134.md`, commit `afb706742`) accounts for all 166 formal statements of the four sources (124 printed with normalized-exact bodies, 41 as duplicate-result notes, one question inside an answered remark). Neither is an entry of the collection’s review record either.)

[merge] *Addendum of 3 October 2026: sources 13–18.* The batch-81 addition claims the written proofs printed in Parts VI–IX, each in the foundation its statement names. Its results that are new against the repository are those called new in [sections 33, 39, 50 and 60](#); its re-proofs of this report, of the reals report and of the report `surreal-fields-across-universes` are notes. None is claimed to be historically first, and results proved by several batch-81 sources are credited to all of them. Every limitation stated by the six sources is kept:

- *Source 13 (13-products-RESEARCH\_STATUS.md):* AI-assisted and unrefereed; no peer review or priority claimed; a targeted audit of this report at `0ecd158fd`, no Lean build, no new Lean proof. Its six “critical scope conditions”: cut reconstruction concerns normalized cuts and all set-ordinal levels, exhaustion being a separate coverage condition; the classification is relative to a supplied coded core and prefix system, which the pure order does not canonically recover; the product nonembedding is proved for arbitrary external maps in the full inaccessible universe and for fixed-formula uniform maps in `KM + GC`, and is not advertised as a theorem of bare `GBC` about a third sort; externally countable models admit external embeddings that need not be internally uniform; central termination reserves zero as a marker; ordinary classes of set codes, class relations and a higher collection of all class relations are never identified. Its finite checks do not verify ordinal limits, class comprehension, global choice or transfinite recursion.
- *Source 14 (14-trichotomy-RESEARCH\_STATUS.md):* `GBC` with a fixed baseline; the inaccessible model is an illustration, not a hypothesis; historical priority not established; no Lean theorem compiled. Its scope cautions: the trichotomy uses named prefix cylinders; the orbit theorem is a coarser classification; equimorphism is not isomorphism; a principal normalized cut has its point on the upper side; cardinal support bounds are not ordinal location bounds; the class-extension theorem fixes the sets and the baseline; raw trace fibres forget tails, and singleton fibres can be non-set-like; a coherent injective branch can omit labels; a virtual quotient is not a collection of class equivalence classes. Its status file says that the inspected

Lean simplicity theorem “is NOT an independent arbitrary-small-cut existence theorem”; that is true of `SignSequenceSimplicity.lean`, while the existence theorem is `small_cut_fillers` in `SignSequenceCut.lean` (section 1.5).

- *Source 15* (its delivery README, not shipped): AI-assisted and unrefereed; its extensions “received independent mathematical review within the task”, with no claim of priority or formal certification; the fixed-type classification covers  $\mu \cdot \gamma + n$  with finite  $n$  only.
- *Source 16* (`16-strata-SOURCE_STATUS.txt`): a targeted audit at 491ed8411; “The proofs underwent independent mathematical checks during preparation”, the delivery’s own statement, which “is not external peer review, a proof of historical novelty, or verification by a proof assistant”; no Lean build; its  $\Sigma_1^1$  theorem is a sufficient bound, with no reversal.
- *Source 17* (`17-gaps-research_status.md`): AI-assisted, no referee review, no compiled Lean; proposed module names only; a targeted audit; no priority. Its unresolved scope: no isomorphism classification of the minimal slice; nonminimal types and noncardinal cutoffs not classified; the automorphism group, arbitrary baseline changes, higher residual blocks and their recursion strength open; the condensed unrestricted order not identified with the set-like order or with **No**; the class-extension theorem not a reconstruction of the class realization. Its reserve formula minimizes the number of extra moved positions, not an ordinal support bound or a birthday.
- *Source 18* (its delivery README, not shipped): “conventional proofs and independent AI-assisted proof review”; no literature priority, peer review or machine-checked formalization; no Lean build or imported-axiom audit; its status table is printed whole (section 115.8).

The repository’s Hilbert’s-tenth research tree, which reviewed the batch-80 sources, holds no review of sources 13–18 at the time of this write.

[merge] *Addendum of 3 October 2026: sources 19–30.* The batch-83 addition claims the written proofs printed in Parts X–XIV, each in the foundation its statement names. Its results that are new against the repository are those called new in sections 72, 79, 85, 105 and 112; its re-proofs of this report, of sources 13–18 and of the reals report are notes or marked routes, and a result proved by several batch-83 sources is credited to all of them (section 1.9). None is claimed to be historically first. Every limitation stated by the twelve sources is kept in the sections that print them; the principal ones:

- *Source 19* (`19-residual-source_status.txt`): its proofs were checked “independently during preparation”, which “is not external peer review”; no Lean build. The least coding length of a fixed type is exact only when its two weighted bounds agree; at singular non-strong-limit residual cardinals at most  $\kappa$  it stays open (research question 113.94). Its common complete theory of all set-sized GBC models (theorem 109.9) is the strongest unreviewed claim of the addition.
- *Source 20* (its title page and delivery README, not shipped): AI-assisted; its results are claims “with internal mathematical review, not machine-checked theorems or claims of priority in the literature”; of its condensation-history theorems: “No necessity or reverse-mathematical equivalence is asserted”.
- *Source 21* (`21-obstructions-SOURCE_AUDIT.txt`, `21-obstructions-LEAN_INTERFACE_AUDIT.txt`): unrefereed and unformalized; “No exhaustive novelty search or historical priority claim is made”; its countable-model results “must not be silently transferred to arbitrary uncountable class realizations”.
- *Source 22* (`22-unranked-RESEARCH_STATUS.md`, `22-unranked-SOURCE_AUDIT.md`): AI-

assisted and unrefereed; no priority; its density and cellularity column re-proves source 16 (only weights and characters are new, [note 63.5](#)); its checks do not test cardinal arithmetic, cofinalities or class comprehension.

- *Source 23* (23-termination-RESEARCH\_STATUS.md, 23-termination-SOURCE\_AUDIT.md): no Lean proof generated; GBC + ETR is a sufficient framework, “not claimed necessary”; its termination criterion is a second route to source 16’s theorem.
- *Source 24* (24-completions-SOURCE\_AUDIT.md): “No claim of historical priority, peer review, or Lean kernel certification is made”; its homeomorphisms are not order isomorphisms.
- *Source 25* (25-symmetries-RESEARCH\_STATUS.md, 25-symmetries-SOURCE\_AUDIT.md): AI-assisted; not claimed refereed or historically first; its invisibility theorem concerns unlabelled cylinders and does not make exhaustiveness undefinable from label equality.
- *Source 26* (26-nonborel-RESEARCH\_STATUS.md): “a relative contribution statement, not a bibliographic-priority claim”; unrefereed; no Lean build; it treats minimal lengths and ordinary Borel sets.
- *Source 27* (27-scans-SOURCE\_AUDIT.txt): no claim of historical priority or external peer review; its least support bound optimizes ordinal height only, not birthdays.
- *Source 28* (28-envelope-SOURCE\_AUDIT.md): unrefereed; no priority, no Lean kernel verification; its character bound concerns the coarsest group refinement, and its headline metrizable question is answered by source 26 ([note 70.13](#)).
- *Source 29* (29-birthdays-PROOF\_STATUS.txt, 29-birthdays-SOURCE\_AUDIT.txt): AI-assisted, internal review only, “not external refereeing or proof-assistant certification”; it does *not* establish  $\text{CWO} \Rightarrow \text{ETR}$ .
- *Source 30* (30-tables-proof\_review.txt, 30-tables-source\_audit.txt): conventional proofs with AI-assisted proof review, which is not “a substitute for external peer review”; its classification under GCH is conditional.

No review of sources 19–30 or of Parts X–XIV exists in the repository at the time of this write.

[merge] *Addendum of 3 October 2026: source 31.* The batch-89 addition claims the written proofs printed in Part XV, in the settings its statements name: first-order  $\text{ZC}$  or  $\text{ZC}_F$  with a selector and expanded Separation ([sections XV.2 to XV.5](#)), and an external ZFC metatheory for the full-class rank models ([sections XV.6 to XV.11](#)). Its results that are new against the repository are listed in [section XV.15](#); none is claimed to be historically first. Every limitation it states is kept where it is printed; the principal ones (its Scope paragraph, [Appendix C](#); [sections XV.12 and XV.14](#); its delivery README, not shipped):

- Conventional mathematical proofs; no Lean or Rocq verification and no machine-checked formalization; its formalization architecture ([section XV.12](#)) is a proposal.
- The nonconservativity theorem ([theorem XV.5.1](#)) uses Glazer’s countermodel theorem [70, Propositions 3.8 and 4.11] as an explicitly cited input; the countermodels are not reconstructed. Source 31 states that input as an abstract interface ([section XV.5.1](#)); whether Glazer’s obstruction has exactly that shape has not been checked in this repository, and source 31 says that expert review of the localized statement “would be the next scholarly step”. The theorem and its corollaries are printed as claimed, with that dependency.
- The rank-model classification is proved in an external ZFC metatheory and concerns full classes; the “fourth row” (universal set coding, equivalently ordinal abstraction) is an equivalence within this family of full-class rank models, not over every weak class theory; no closure of  $S_\lambda$



under the surreal field operations is asserted at arbitrary cutoffs; nothing is asserted about first-order Replacement at singular heights.

- No claim that the repository’s supported-prefix extension, small-net rigidity or inverse-reconstruction schemes are each equivalent to a row of its table; no solution of the surreal transfer problem; no classification for definable-class realizations; no proof of the finite-input conservativity boundary.
- Targeted literature and repository searches only: “this is not a proof of priority”. Restricted global-choice principles have prior literature (Enayat’s global finite choice [71]); the bounded proof of the well-ordering theorem is classical in method; ordinal abstraction and the example  $V_{\omega_1}$  (Hamkins, Rin), set-like class orders without Replacement (Carneiro), the global-choice equivalences over GB (Hamkins), well-ordered Replacement (Freire and Hamkins) and beth fixed points (MathOverflow) are credited.

No review of source 31 or of Part XV exists in the repository at the time of this write; the Hilbert’s-tenth research tree has routed its archive without a correctness finding (section 1.10).

[merge] *Addendum of 4 October 2026: sources 32–35.* The batch-96 addition claims the written proofs printed in Part XVI, in the settings its statements name: GB with ZF sets for sources 34 and 35’s choice-free results, GBC where a statement says so, GBC with an assumed full satisfaction class for the truth jumps, and an external ZFC metatheory with a strongly inaccessible  $\kappa$  (or a transitive  $\beta$ -model) for the spectra. Its results that are new against the repository are listed in section XVI.12; none is claimed to be historically first. Every limitation the sources state is kept where it is printed; the principal ones:

- Neither source 34 nor source 35 proves CWO or CTH in GB; both leave open whether CWO implies ETR (“a dated literature-status statement, not a proof that no resolution exists”, source 34). The reduction to the surreal carrier keeps global choice. Source 34’s lemma XVI.2.2 is specific to linear orders and does not identify the absence of descending sequences with well-foundedness without choice.
- Source 32’s termination theorem is conditional on the existence of the hierarchy, and ETR is “sufficient, not proved necessary”; source 33’s digit theorem consumes a supplied tower. (Both are re-proofs of source 20’s theorem, remark XVI.1.1.)
- The spectra are external and assume an inaccessible  $\kappa$  with  $V_\kappa$ , or a transitive  $\beta$ -model; they do not transfer to arbitrary countable, non- $\beta$  or nonstandard models. Countable cofinality of a ceiling does not make it countable or internally represented; the inequalities among source 35’s three suprema are not claimed strict.
- The truth jumps assume a full satisfaction class; the orders  $J_\gamma$  are “not canonical absolute class ordinals”; source 32’s theorem does not compute  $\delta_{T_A}$ .
- The bounded-complexity diagonals are schemata in a standard  $n$ , not one definable sequence; source 35 claims no separation of two consecutive levels.
- Hereditary ordinal notation and class epsilon notation are not claimed as inventions (Weiermann, Freund, Rathjen through Jeon are credited); the fixed-point completion is new against the sources each author examined.
- The finite checks of sources 32, 33 and 35 check coding conventions only (“NOT machine proofs”); no Lean or Rocq development exists, and the module plans are proposals.
- Targeted repository and literature searches only; source 33 did not read the 860-page report line by line, source 32 read three files.

The Hilbert’s-tenth research tree’s bounded intake of the four archives checked source 34’s choice-free power theorem and set-subset criterion and found no defect, without auditing the rest of the GB equivalence or the other proofs; no review covers Part XVI as written ([section XVI.12](#)). (Dated note, 4 October 2026: an independent check of the four statements written for Part XVI, [theorem XVI.6.21](#) and [remarks XVI.5.24](#), [XVI.5.29](#) and [XVI.5.31](#), is recorded in [section XVI.12](#); it found them valid after editorial corrections.) A later bounded publication review and the empty-domain follow-up are also recorded there; Review remark H5 qualifies the earlier passes and records the remaining cases repaired in this text. [\[merge\]](#) *Addendum of 4 October 2026: source 36*. The batch-97 addition claims the written proofs printed in Part XVII, in the settings its sections name: GB with ZF sets and no choice for the arithmetic, notation and transfer results; an external ZFC metatheory with an arbitrary transitive set model of ZFC for the calibrations and spectra; and an inaccessible  $\kappa$  with a named enumeration  $G$  of  $V_\kappa$  (and a full truth predicate for the promotion) for the horizon theorems. What is new against Parts I–XVI is listed in [section XVII.11](#); nothing is claimed to be historically first, and source 36’s horizon theorems, bounded-complexity diagonal and truth promotion are theorems of sources 32–34 (credit notes in Part XVII). Every limitation source 36 states is kept where it is printed; the principal ones:

- “Publication priority and a breakthrough resolution of a recognized open problem are not asserted”; long class orders, their arithmetic and the epsilon notation functor “have established antecedents”; the converse from comparability to ETR is not established.
- The arithmetic is relative to ordinal coefficients; no normalization of general powers inside E is proved; the transfer theorem does not apply to an arbitrary normal ordinal function.
- The spectra count externally well-founded orders; an arbitrary transitive model need not recognize external ill-foundedness; the horizon theorem at  $V_\kappa$  uses the inaccessibility of  $\kappa$  and a named  $G$ , and “parameter-free” excludes only set parameters;  $\text{cf}(\mathfrak{H}(M)) = \omega$  is not claimed for unrestricted parameters over an arbitrary  $M$ ; the diagonal formulas are an external schema, not one evaluator; the size of the truth jump is not computed; the admissible bound’s equality cases are not claimed as new.
- Conventional written proofs only; no referee, no Lean or Rocq; the five proposed modules are design targets; the demonstrator checks natural coefficients only and “does not prove any class-theoretic theorem”.
- A targeted repository audit (README, this report’s README and `29-birthdays-PROOF_STATUS.txt`, the ListCoding README); this report’s article was not read end to end, and the audit is “not an exhaustive duplicate search”.

The Hilbert’s-tenth research tree’s bounded intake of the archive (commit `39bb02ece`) found no defect in the proofs it read. A later bounded publication review of `a21a42d8c` checks selected arithmetic and effective-interface proofs and corrects three editorial scope claims; its coverage and remaining exclusions are in [section XVII.11](#).

## 1.5 Formal status

[\[merge\]](#) No statement of this report is formalized, and no `swo:` label has a Lean mapping in the collection’s formalization ledger (`Algebra/SurrealNumbers/docs/FORMALIZATION.md`). Placement in the surreal collection, beside the Lean foundations of `Algebra/SurrealNumbers/Surreal/Foundations/`, confers no formal status.

Four inputs that the sources use *are* proved in Lean, in the universe-relative form recorded in that ledger, where a “set” of surreals is a family indexed by a `Small.{u}` type and the sign carrier



lives in `Type (u+1)`:

- properness and bounded birthdays (found:prop:proper; [proposition 5.4](#), first three clauses): `SignSequence`, `birthdays_bounded`, `small_bounded`, `not_small` in `SignSequence.lean`;
- filling of small cuts (found:sub:cutdata): `small_cut_fillers` in `SignSequenceCut.lean` and `exists_separator_of_small_sets` in `SignSequenceCutOperation.lean`;
- a separator of exactly a prescribed birthday above all option birthdays, [theorem 5.6](#): `exists_cut_separator_of_birthday_lt` in `SignSequenceCut.lean`, by the same route as source 08’s proof (the complete lexicographic lattice of Boolean words of one length, which is [lemma 5.5](#), there taken from Mathlib);
- the inconsistency of unrestricted cuts (found:prop:allcuts): `noUnrestrictedCuts` in `SizeObstructions.lean`.

The other declarations the sources cite exist at the current revision: `ordinalOrderEmbedding`, `small_strict_upper_bounds`, `simpler_wellFounded`, `existsUnique_simplest_separator`, `cut_realizes`, `cut_isPrefix`, `birthday_cut_le`, `birthday_cut_le_iSup`, `cut_congr`, `SmallCutData`, `HasSmallCutFillers`, `not_small_of_small_cut_fillers`, `lt_iff`, and the others in their tables. None of them states a result of this report.

[merge] (3 October 2026.) The same holds for sources 13–18: none ships Lean code, no statement of theirs is formalized, the declarations they cite exist at the current revision (including `le_iff_options_lt` and `lt_iff_exists_option` in `SignSequenceComparison.lean`, cited by source 18), and the modules they propose do not exist ([note 59.1](#)).

[merge] (3 October 2026, batch 83.) The same holds for sources 19–30: none ships Lean code or compiled a Lean file, no statement of theirs is formalized, and the module names in their formalization plans ([section 111](#)) are proposals: no module of those names exists. The declarations those plans name as existing were checked against `Algebra/SurrealNumbers/Surreal/Foundations/` when the plans were printed; none of them states a result of this report. Source 21’s audit of the declarations it uses is shipped as `21-obstructions-LEAN_INTERFACE_AUDIT.txt`.

[merge] (3 October 2026, batch 89.) Source 31 ships no Lean or Rocq code and cites no Lean declaration; no statement of Part XV is formalized. Its formalization architecture ([section XV.12](#)) proposes the domain-size criterion [lemma XV.9.2](#) as a first machine-checked milestone, and its [research question 113.243](#) a verified bounded uniformization theorem; no such development exists.

[merge] (4 October 2026, batch 96.) Sources 32–35 ship no Lean or Rocq code; no statement of Part XVI is formalized. Source 32 cites `SetTheory/ZF/Lean/ZF/Zf.lean` (its declarations `Sep_form`, `Repl_form`, `ZFax`, `ZFprov`) as a foundation for a future extension, and Mathlib’s `Ordinal.univ` and `Ordinal.type_lt_ordinal`; source 33 proposes a directory `SetTheory/ClassWellOrders/`, which does not exist. Their first proposed milestones are the choice-free power theorem and the capped normal-form reconstruction ([section 113.28](#)). [merge] (4 October 2026, batch 97.) Source 36 ships no Lean or Rocq code; no statement of Part XVII is formalized. It proposes five modules (`ClassOrderPresentation`, `FiniteSupportPower`, `OmegaNormalForm`, `EpsilonTerms`, `DefinabilityHorizon`), none of which exists, and cites `Logic/PeanoArithmetic/ListCoding/REA DME.md` and Mathlib’s `Mathlib.SetTheory.Ordinal.CantorNormalForm` as engineering patterns, not as theorems about classes. Its natural-coefficient demonstrator (`code/36-horizons-notation_demo.py`) is a finite test, not a formalization.

## 1.6 Where the merge had to choose

[merge]

- *Base and order.* Source 11, for the reasons in [section 1.1](#); its section order is kept, and member material follows the base result it duplicates or extends.
- *Strongest statement printed as the statement.* The choice equivalence is source 11's (over GB; sources 12 and 09 assume set choice, and source 09 treats set-like orders only). The diagonal theorem is printed twice: source 11's ([theorem 19.37](#), rows of set codes) and source 12's ([theorem 19.38](#), every uniform family of class well-orders, with a set-like witness); sources 09's and 08's are notes (source 08's indexes by Ord only). The birthday-cutoff statement is source 12's dichotomy; source 08's regular case is a note.
- *Different routes kept in full:* source 09's stabilization proof of the ordinal spectrum; source 12's route to the set-sized spectra through binary coding length; the second and third proofs that a set can be avoided ([lemmas 5.3](#) and [5.10](#)); source 12's and 09's proofs of the choice equivalence over GB + AC; source 12's largest common labelled prefix; source 12's residual decomposition and adjacency criterion; source 12's concatenation proof of the binary-class embedding; source 12's prefix extension through the complement of the prefix; source 09's support coding of the core; source 09's explicit unfilled configuration; source 09's general universality and homogeneity theorems; source 12's birthday-bounded universality.
- *Kept whole:* each member's introduction, foundational section, universe and model sections, Lean plan, research questions, conclusion, ledger and reproducibility record.
- *Corrections to source text.* The renamings of [section 1.3](#); one mistyped reference in source 11 (a proposition cited as "Theorem", in [section 12](#)); source 08's predecessor citation ([section 4.3](#) and the note after it); source 12's question on the weak-choice strength, printed as an answered remark ([remark 113.15](#)); source 08's question 11 annotated as answered; one sentence of source 08 on well-foundedness replaced by the correction of the Hilbert's-tenth review ([section 5.6](#) and the note before it). Statements that name delivery files (build scripts, `finite_checks.py`, `surreal_well_orders.tex`) are kept and annotated with the shipped names ([appendix B](#)).

## 1.7 Where one source answers another

[merge] Dated notes, 2 October 2026.

- Source 12's question on the weak-choice strength* ([remark 113.15](#), and its [remark 13.3](#)): answered by source 11. Over GB without set choice, a class well-order of **No** exists exactly when global choice holds ([theorem 13.1](#)); see [remark 13.4](#). The same theorem shows that the set-choice hypothesis of source 09's [proposition 13.7](#) is unnecessary. Source 09's direction on the exact axiomatic strength of *other* universality and homogeneity assertions ([section 113.3](#)) is unaffected.
- Source 08's question 11, comparison of non-set-like class well-orders* ([section 113.4](#)): answered by sources 11 and 12. [Theorems 14.2](#) and [14.7](#) compare all class well-orders of a labelled class directly, with elementary comprehension, without ETR and without comparing abstract class order types; this also supplies the single comparison that source 08's [section 17](#) says does not follow by aligning domains (see the note after it). Source 09's relation-code direction concerns a different comparison and stays open.
- Source 08's question 3, singular birthdays and code lengths:* answered in part by source 12's [theorem 8.3](#): at a singular cutoff the least binary coding length of the fixed-length enumeration order is  $\kappa \cdot \kappa$  when  $\kappa$  is a strong limit and  $2^{<\kappa}$  otherwise. Which cuts fail at singular cutoffs stays open.

- (d) *Source 08’s question 2, invariants at regular cutoffs*: partly answered by source 11’s [proposition 12.1](#) (the minimal slice is  $\eta_\kappa$ , with every global and local cofinality and coinitality  $\kappa$ ); density, cellularity and the cut cofinalities beyond these stay open.
- (e) *The reals report’s question “Other ground orders”* (its Section “Further research questions”, `lwo:sec:research`, which names “set-sized suborders of surreal numbers”): answered in part by [theorem 8.3](#) for  $X = \mathbf{No}_{<\kappa}$  and the minimal stratum ( $n = 0$ ):  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\mu$ , except at singular strong-limit  $\kappa$ , where it is  $\simeq_e \mathbb{B}_{\kappa \cdot \kappa}$  and not  $\simeq_e \mathbb{B}_\kappa$ . It stays open for  $n > 0$ , for longer strata, and for other alphabets (source 12’s [research question 113.11](#), source 08’s question 4).
- (f) *Hamkins’s question whether universality of the surreal line is a weak global choice principle [21]*: not settled here, as source 12 says after its [remark 113.15](#).

[merge] Dated note, 3 October 2026 (batch 81). Items (c) and (d) are completed and item (e) is extended by Part VI: the cut spectrum at singular cutoffs, the density, cellularity and cut cofinalities at regular cutoffs, and the longer strata  $\mu \cdot \gamma + n$  of  $\mathbf{No}_{<\kappa}$  ([section 33](#)).

[merge] Dated note, 3 October 2026 (batch 83). Sources 19–30 answer, fully or in part, source 12’s [research questions 113.11](#), [113.12](#), [113.18](#) and [113.19](#), source 11’s [research question 113.3](#), source 08’s question 9 and source 09’s fifth direction, open items of Parts VII and VIII, and questions of sources 13–18 and of each other; the dated status notes are in [section 113.25](#).

[merge] Dated note, 3 October 2026 (batch 89). Source 31 answers source 28’s [research question 113.209](#) in part and extends source 12’s full-class model [theorem 21.1](#) from inaccessible heights to every limit height ([sections XV.15](#) and [113.27](#)). Hamkins’s question in item (f) is not settled by it.

[merge] Dated note, 4 October 2026 (batch 96). Sources 34 and 35 answer over GB the strength of complete condensation histories, which [note 108.4](#), item (b) of [section 112](#) and source 20’s [research question 113.111](#) left open without global choice; together with sources 32 and 33 they answer source 31’s [research question 113.239](#) in part ([sections XVI.12](#) and [113.29](#)). Hamkins and Woodin’s Question 7 is not settled.

[merge] Dated note, 4 October 2026 (batch 97). Source 36 answers source 33’s [research question 113.286](#) and the addition and multiplication part of source 34’s [research question 113.275](#), and advances source 32’s [research question 113.284](#), source 33’s [research question 113.285](#) and source 34’s [research questions 113.272](#) and [113.283](#) ([sections XVII.11](#) and [113.31](#)). Hamkins and Woodin’s Question 7 is not settled.

## 1.8 The batch-81 addition: six further manuscripts

[merge] (3 October 2026.) Six manuscripts of batch 81, all dated 3 October 2026 and all continuing this report, are merged into it as one addition, sources 13–18. Sources 13–17 arrived in commit `9a4ce14e9` and were placed in `f0cd7032d` (batch 81L); source 18 arrived in `bd599ac06` and was placed in `c4720e1b2` (batch 81L2). Their local numbers continue this report’s sequence; the batch-81 manuscript number is in the second column.

| Src. | B81 | Manuscript (archive), pages                                                                                                  | Pin       | Labels, files                                       |
|------|-----|------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------|
| 13   | 01  | <i>Lexicographic Orders of Surreal Well-Orderings: Further Structure</i> ; 33 pp.<br>(Surreal_Well_Orders_Further_Study.zip) | 0ecd158fd | <code>swo:fs::</code><br><code>13-products-*</code> |

| Src. | B81 | Manuscript (archive), pages                                                                                                                                                                                               | Pin       | Labels, files               |
|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------|
| 14   | 02  | <i>Lexicographic Well-Orders of the Surreals II: cut classification, support thresholds, and class-model absoluteness</i> ; 33 pp. (lexicographic_surreal_well_orders_II.zip)                                             | 0ecd158fd | swo:tc;;<br>14-trichotomy-* |
| 15   | 03  | <i>Lexicographic Orders of Surreal Well-Orders: interpretations, canonical blocks, represented cuts, and fixed-type spectra</i> ; 37 pp. (surreal_lexicographic_orders.zip)                                               | 0ecd158fd | swo:cb;;<br>15-blocks-*     |
| 16   | 04  | <i>Lexicographic Orders of Surreal Well-Orders: Cuts, Canonical Blocks, and the Limits of Set-Sized Coding</i> ; 33 pp. (surreal_well_orders_further.zip)                                                                 | 491ed8411 | swo:ds;;<br>16-strata-*     |
| 17   | 05  | <i>Lexicographic Well-Orders of the Surreal Numbers, II: exact cut spectra, branch recognition, optimal support reserves, and automorphism-extension obstructions</i> ; 29 pp. (surreal_well_orders_further_study(1).zip) | 0ecd158fd | swo:gs;;<br>17-gaps-*       |
| 18   | 06  | <i>Lexicographic Well-Orders of the Surreal Numbers: Cuts, canonical class blocks, and changes of interpretation</i> ; 53 pp. (surreal_lexicographic_orders_further_study.zip)                                            | 83befe707 | swo:sh;;<br>18-shuffle-*    |

The six are independent texts, not editions of one another. Every pair shares 0.6–1.4% of word 8-grams (preamble, this report’s title and address, bibliography), no member file is shared, and the theorem sets differ. Sources 14 and 17 share only the inner directory name `surreal_well_orders_II` and a “II” in the title; source 18’s archive name extends source 15’s, but its pin, structure and results differ. None supersedes another; all six are merged (union). Sources 13–17 read this report at commit `0be9b9134`, before the corrections of `0f52c0dca`; source 18 read it at `83befe707`, after them, and before sources 13–17 arrived, so it never saw them.

They overlap heavily with each other and with this report. The represented-cut theorem that answers source 11’s [research question 113.1](#) is proved in all six by one argument; all six re-prove the core layer of Part IV (fresh separators, completion, interpolation, the core isomorphic to **No**, the diagonal theorem, direct comparison); and the set-sized layer re-proves the reals report. What is new is a second wave of results on questions of this report: the topology, cut spectra and coding lengths of the set-sized strata at birthday cutoffs; the termination classification of word orders; the trichotomy, characters, automorphism orbits, absoluteness and support thresholds for cuts of the set-coded core; the canonical block decomposition of class well-orders with explicit condensation histories; product (non)absorption; a  $\Sigma_1^1$  comprehension bound; and the countable models.

The addition is printed by theme, not by source, because the sources overlap: Part VI ([part VI](#)) prints the set-sized strata; Part VII ([part VII](#)) prints the six sources’ maps of interpretations, the word orders, the termination conventions and the other comparison orders; Part VIII ([part VIII](#)) prints the class cuts of the core: their classification and recognition, automorphisms, class extensions and support; Part IX ([part IX](#)) prints the unrestricted class well-orders: the canonical block decomposition and condensation histories, finite components and trace fibres, products, a  $\Sigma_1^1$  comprehension bound, larger universes and countable models, and the sources’ formalization plans. The sources’ introductions are printed in [section 4.5](#), their questions in [section 113.5](#), and their conclusions, ledgers, reproducibility records, title pages and statement crosswalks after those of the batch-80 sources. The rules of [section 1.2](#) apply unchanged: each shared result is printed once,

with every source that proves it named; the other versions are numbered notes, genuinely different proofs are marked routes, and every label of every source survives with the prefixes of [section 1.3.1](#). Re-proofs of results of this report or of other repository reports are printed as notes with pointers, never with a novelty claim.

*Status.* Sources 13, 15 and 17 call themselves AI-assisted and unrefereed, source 14 unrefereed ([merge] corrected 3 October 2026: source 14 does not call itself AI-assisted); source 18 calls itself “an AI-assisted research manuscript with mathematical proof review”, and its validation “consists of conventional proofs and independent AI-assisted proof review”; source 16 says its checks are “not external peer review”. None claims historical priority, peer review or a machine-checked formalization, and none is formalized here. The status files of sources 13, 14, 16 and 17 are shipped in this directory under their prefixed names.

*Where the last write of the addition had to choose* (3 October 2026). The six introductions are printed in [section 4](#), with the batch-80 introductions; the foundations sections, the universe sections and the Lean plans of sources 13–18 are printed at the end of Part IX rather than in Part V, so that no existing section is renumbered. Corrections made to source text: “In GBC” added to source 15’s [proposition 51.10](#); source 13’s “new point” about split products annotated as already contained in source 11’s proofs ([note 55.6](#)); source 15’s sentence that adding classes can destroy internal well-foundedness withdrawn for GBC realizations ([note 57.5](#)); source 17’s question table and source 14’s “Four main results” annotated ([section 4.5](#)); citations of this report replaced by cross-references, with the sources’ numbers “at the pin”. Source equations are tagged ( $SNN.n$ ).

## 1.9 The batch-83 addition: twelve further manuscripts

[merge] (3 October 2026.) Twelve manuscripts of batch 83, all dated 3 October 2026 and all continuing this report, are merged into it as one addition, sources 19–30. They arrived in five commits: b63f0c852 (sources 19–24), 5ad44b1ed (source 25), 0d7b2441d (source 26), 79049d58c (sources 27 and 28) and 51b0a69d7 (sources 29 and 30), and were placed together in 2e06337d5 (batch 83L). Their local numbers continue this report’s sequence; the batch-83 manuscript number is in the second column. Batch-83 manuscripts 07–10 are Research Reports placed elsewhere.

| Src. | B83 | Manuscript; archive and inner directory; pages                                                                                                                                                                  | Pin; arrival            | Labels; files                                                            | Parts               |
|------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------|---------------------|
| 19   | 01  | <i>Lexicographic Orders of Surreal Well-Orders: Residual spectra, decorated completions, and model invariants</i> ; <code>surreal_lexicographic_orders_continuation.zip</code> , same inner directory; 45 pp.   | d5bd4a67b;<br>b63f0c852 | <code>swo:rs::19-r</code><br><code>esidual-*</code>                      | X, XI,<br>XIII, XIV |
| 20   | 02  | <i>Lexicographic Orders of Surreal Well-Orders: Coherent Limits, Nonextendible Symmetries, and Countable Models</i> ; <code>surreal_orders_limits_symmetries_models.zip</code> , same; 50 pp.                   | d5bd4a67b;<br>b63f0c852 | <code>swo:lm::</code><br><code>20-limits-*</code>                        | XI, XIII,<br>XIV    |
| 21   | 03  | <i>Lexicographic Orders of Surreal Well-Orders: Further Structure, Automorphism Obstructions, and Model Dependence</i> ; <code>surreal_well_orders_further_structur</code><br><code>e.zip</code> , same; 44 pp. | 109aaca15;<br>b63f0c852 | <code>swo:fu::</code><br><code>21-obstructi</code><br><code>ons-*</code> | X, XI,<br>XIII, XIV |

| Src. | B83 | Manuscript; archive and inner directory;<br>pages                                                                                                                                                                                                                         | Pin; arrival            | Labels; files             | Parts              |
|------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------|--------------------|
| 22   | 04  | <i>Lexicographic Orders of Surreal Well-Orderings: Residual geometry, the finite-tail transition, unranked cylinders, and reconstruction of class realizations;</i><br>Surreal_Lexicographic_Orders_Residual_Geometry.zip, inner<br>Surreal_Lexicographic_Orders/; 33 pp. | d5bd4a67b;<br>b63f0c852 | swo:rg:: 22-unranked-*    | X, XIII            |
| 23   | 05  | <i>Lexicographic Surreal Well-Orders: Termination markers, increasing-word obstructions, and transfinite condensation;</i><br>Surreal_Lexicographic_Termination_Condensation.zip, inner<br>surreal_lexicographic_further/; 30 pp.                                         | 109aaca15;<br>b63f0c852 | swo:tm:: 23-termination-* | XII, XIV           |
| 24   | 06  | <i>Lexicographic Well-Orders of the Surreals: Prefix completions, generalized Baire models, and the set-class boundary;</i><br>Surreal_Lexicographic_Orders_Prefix_Completions.zip, inner<br>surreal_lex_prefix_completions/; 32 pp.                                      | 109aaca15;<br>b63f0c852 | swo:pc:: 24-completions-* | X                  |
| 25   | 11  | <i>Lexicographic Surreal Well-Orders: Prefix homogeneity, invisible exhaustiveness, and label rigidity;</i> Surreal_Well_Orders_Prefix_Symmetries.zip, inner<br>surreal_prefix_symmetries/; 33 pp.                                                                        | 7ff7736ec;<br>5ad44b1ed | swo:ph:: 25-symmetries-*  | XIII               |
| 26   | 12  | <i>Lexicographic Well-Orders of the Surreals: Non-Borel boundaries, closed universality, and safe foundations;</i><br>Surreal_Lexicographic_Boundaries.zip, same; 34 pp.                                                                                                  | 7ff7736ec;<br>0d7b2441d | swo:nb:: 26-nonborel-*    | X                  |
| 27   | 13  | <i>Lexicographic Orders of Surreal Well-Orders: Further Structure from Encodings, Prefix Trees, and Support Bounds;</i><br>surreal_lexicographic_orders.zip, same; 34 pp.                                                                                                 | 7ff7736ec;<br>79049d58c | swo:sc:: 27-scans-*       | XII, XIII, XIV     |
| 28   | 14  | <i>Lexicographic Well-Orders of the Surreals: Singular permutation geometry and the canonical group envelope;</i> Surreal_Lexicographic_Group_Envelope.zip, same; 32 pp.                                                                                                  | 7ff7736ec;<br>79049d58c | swo:ge:: 28-envelope-*    | X                  |
| 29   | 15  | <i>Lexicographic Orders of Surreal Well-Orders: Exact Birthday Spectra, Word Products, Topological Transitions, and Class Comparability;</i><br>surreal_orders_new_structures.zip, same; 57 pp.                                                                           | 7ff7736ec;<br>51b0a69d7 | swo:bw:: 29-birthdays-*   | XI, XII, XIII, XIV |
| 30   | 16  | <i>Lexicographic Well-Orders of the Surreals: Residual Codes, Topological Transitions, and Class Reconstruction;</i><br>surreal_well_orders.zip, inner<br>surreal_well_orders/; 47 pp.                                                                                    | 7ff7736ec;<br>51b0a69d7 | swo:rt:: 30-tables-*      | X, XII, XIII, XIV  |



All twelve also contribute to the head of Part X (their scope sections, [section 61](#)), the close of Part XIV (their foundations, formalization plans, ledgers and verification records, [section 111](#)), the research questions ([section 113.12](#)) and the appendix of abstracts. The pins are the revisions of this report that each source read: `d5bd4a67b` and `109aaca15` have Parts I–V only; `7ff7736ec` has Parts I–VIII, with Part IX announced but not written. Several sources also read unplaced manuscripts, now sources of this report: source 19 read sources 15, 16 and 18; source 20 read source 18; sources 22 and 23 read sources 13 and 17; source 24 read sources 13, 14 and 17; source 25 read sources 13, 17, 22 and 23; source 26 read source 24; source 28 read sources 13 and 24; source 29 read sources 19, 20 and 21 (its manifest records their hashes). Sources 21, 27 and 30 cite none of them.

The twelve are independent texts, not editions of one another or of earlier sources. Every pair shares at most 3.3 % of word 8-grams (source 28 with source 24; source 22 with source 25 2.4 %; all other pairs, and each source against this report and sources 13–18, at most 2.2 %); no member file is shared, byte-identical to another delivered file or present in the repository before `2e06337d5`. Three name coincidences are not editions: source 27’s archive and main file have the names of source 15’s (`surreal_lexicographic_orders.zip`; different title, pin and theorems, 0.6 % overlap); source 30’s archive has the name of source 12’s, and its inner directory and main file those of sources 11 and 08 (`surreal_well_orders.zip`; at most 0.5 % overlap); source 23’s inner directory is source 13’s (`surreal_lexicographic_further/`). None supersedes another; all twelve are merged (union).

They overlap heavily, with each other and with this report. The countable-cutoff theorem (the minimal slice of a countable dense alphabet is homeomorphic to Baire space and has the real line as Dedekind completion, with continuum many omitted points in every interval) is proved by sources 19, 20, 21, 24, 29 and 30, and its complete metrizability again by source 26; the represented-cut theorem of Part VIII again by sources 20, 21 and 22; the automorphism-extension criterion of Part VIII by sources 20, 21 and 25; the termination criterion of Part VII by source 23; the canonical blocks and condensation histories of Part IX by sources 23 and 29 (the blocks also by sources 19 and 21); the residual-code classification by sources 19 and 30; the countable-strata topology by sources 29 and 30; the faithful reconstruction of a class realization by sources 22, 27 and 30; the centered-word product by sources 27 and 29. Each is printed once, with every source that proves it named; the other versions are numbered notes, genuinely different proofs are marked routes. What is new is a third wave of results on questions of this report and of sources 13–18: the binary-cube spectrum and coding length of every fixed type (source 19); weights and characters of every fixed-length stratum (source 22); the exact birthday spectrum of  $\mathbf{No}_{<\theta}$  and the noncardinal cutoffs (sources 29 and 21); the topology of the minimal slice at every size and the non-Borel boundary (sources 24, 26 and 28); coherent limits and club detection across cutoffs (source 20); the increasing-word factorization and scan-dependent tables (sources 27, 29 and 30); configuration homogeneity, invisible exhaustiveness and label rigidity of the core (source 25); the least ordinal support bound (source 27); reconstruction of class realizations (sources 22, 27 and 30); the local coding dichotomy, countable class models and their common theory (sources 19, 20 and 21); and the equivalence of canonical condensation histories with class comparability (source 29).

*How Parts X–XIV are organized.* The addition is printed by theme, not by source, after Part IX. Part X prints the set-sized strata, their topology and the permutation groups of the minimal slice (sources 19, 21, 22, 24, 26, 28, 30). Part XI goes across birthday cutoffs: restrictions, coherent limits, noncardinal cutoffs and coherent embeddings (sources 19, 20, 21, 29). Part XII prints words, termination markers and comparison codes (sources 23, 27, 29, 30). Part XIII prints the core and its cylinders: automorphisms, decorated completions, support and class realizations (sources 25, 22, 20, 21, 19, 27, 29, 30). Part XIV prints unrestricted class well-orders: local size, condensation histories and external models (sources 20, 19, 21, 23, 27, 29, 30), and closes with the twelve



sources' foundations, formalization plans, ledgers and verification records ([section 111](#)). The rules of [section 1.2](#) and of [section 1.8](#) apply unchanged; re-proofs of results of this report, of sources 13–18 or of other repository reports are printed as notes with pointers, never with a novelty claim. Where a source says that this report lacks something it now has (most read it before Parts VI–IX existed), the sentence is kept as pinned provenance with a dated [merge] correction. Source 28's headline question, complete metrizable of the full singular permutation space, is answered negatively in ZFC by source 26, which source 28 did not see; it is printed with that status. The questions of this report and of sources 13–18 that the addition answers fully or in part are recorded in [section 113.25](#).

*Status.* Sources 19–26, 28 and 29 call themselves AI-assisted on their title pages; sources 23–26 and 28 add “unrefereed”, sources 19 and 29 “not a refereed publication” (“paper”), and sources 20, 21 and 29 that their proofs had internal review only. Source 27 makes “no claim of historical priority or external peer review”; source 30 reports “independent AI-assisted proof review” and no machine-checked formalization. None claims historical priority or a machine-checked formalization, none built the repository or compiled a new Lean file, and none is formalized here; their finite checks verify finite mechanisms only. Their status files, audits, finite checks and records are shipped in this directory under their prefixed names ([section 111](#)).

*Where the batch-83 merge had to choose.*

- *One theme, one Part.* Each mathematical section of the twelve sources is printed once, in the Part of its theme; a result already printed (in Parts I–IX or by another batch-83 source) becomes a numbered note that quotes the statement and keeps the source's label, followed by the source's proof where it differs. The maps of interpretations of sources 23, 27, 29 and 30 are printed in Part XII, beside the word orders and relation tables they introduce, and not in [section 61](#); source 29's section on global well-orders and represented cuts is printed in Part XIII ([section 86.5](#)).
- *The core.* The two codings of Part I are kept apart by source: normalized permutations of an ordinal are printed  $\mathcal{P}_{\text{bd}}(b)$  (sources 19, 20, 21, 24, 26, 28 and 29), fixed-point-free support codes  $\mathcal{P}_s(b)$  (sources 22, 25, 27 and 30). Source 20's  $\mathcal{C}_b$  is defined by normalized permutations  $\pi : \lambda \rightarrow \lambda$  with  $\lambda = \sup\{\xi + 1 : \pi(\xi) \neq \xi\}$ , so it is printed  $\mathcal{P}_{\text{bd}}(b)$  in every Part (a draft of Part XI had  $\mathcal{P}_s(b)$ ; the evaluations are the same).
- *Residual weights.* Source 19's and source 29's weights  $r(\lambda)$ ,  $u_\kappa(\lambda)$  and weighted lengths  $R$ ,  $U$  are printed  $\text{wt}(\lambda)$ ,  $\text{wt}_\kappa^+(\lambda)$ ,  $\varrho_{\text{lo}}$ ,  $\varrho_{\text{up}}$  in Parts X and XI alike, and source 30's  $r(\nu)$  is  $\text{wt}(\nu)$  ([section 1.3.2](#)).
- *Placement of the shared front and back matter.* The twelve sources' opening sections head Part X ([section 61](#)); their foundations, formalization plans, ledgers and verification records close Part XIV ([section 111](#)); their questions follow those of sources 13–18 ([section 113.12](#)); their title pages and abstracts close [appendix C](#); their numbered statements are mapped in [appendix D](#). No existing Part, section or label was renumbered or renamed; the sections of the questions part and the appendices move by the number of new sections.
- *Provenance correction.* The placement commit 2e06337d5 records source 25's archive `Surreal_1_Well_Orders_Prefix_Symmetries.zip` as arriving in 0d7b2441d; it arrived in 5ad44b1ed, as the table above says. The staged bytes are unaffected.

## 1.10 The batch-89 addition: source 31

[merge] (3 October 2026.) One manuscript of batch 89, its manuscript 01, continues this report and is printed as Part XV ([part XV](#)), source 31. It arrived in commit 7c0f2d9f9 and was placed in 23adb85f9. It shipped only its manuscript, its PDF and a delivery README, so the placement

staged no file here; 31 is its number in this report’s local sequence (the highest prefix in this directory is 30–). Batch-89 manuscripts 02–05 are placed in other reports.

| Src. | B89 | Manuscript; archive and inner directory;<br>pages                                                                                                                                                                                 | Pin; arrival            | Labels; files  | Parts |
|------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|-------|
| 31   | 01  | <i>Countable Global Choice and Surreal Coding over Zermelo Set Theory: A localized nonconservativity theorem and exact rank-model spectra</i> ;<br>global_choice_surreal_coding.zip,<br>same inner directory; 1,547 lines, 21 pp. | 2b7b388ba;<br>7c0f2d9f9 | swo:gcz:: none | XV    |

The pin 2b7b388ba (“Catalogue batches 81 to 86”) holds this article as blob 05807d292, the text written by the batch-83 write 07ee2b30c, with Parts I–XIV, unchanged until this write; source 31 names that blob, so the numbers it cites (for example Theorem 13.1, swo:cf:choice) are the current ones. Its title page reads “Research article prepared for Vladimir Reshetnikov” and is dated 4 October 2026, the UTC date of its arrival (3 October 2026, 21:36 PDT). It is not an edition of any earlier source: it shares no file with them, its subject (Zermelo set theory with a selector; full-class rank models at every limit height) is new to this report, 0.6% of its word 8-grams occur in this report (bibliography and standard phrasing), and it shares at most 0.33% with any other batch-89 manuscript. Two of those bear on it: manuscript 03, printed as Part VI of **birthday-cutoffs-and-hereditary-sets**, shows that  $V_{\omega+\omega}$  fails Replacement for definable families, the definable special case of [lemma XV.9.2](#) at a height of cofinality  $\omega$  (its Proposition 30.1, hset:zr:prop:rankmodel, which also refutes there its bounding scheme NIB inside  $(-1, 0)$ ; its Remark 24.7, hset:zr:rem:swo, records the inclusion; labels added 4 October 2026, after that write was committed); and manuscript 05, the report **naming-elementary-embeddings** of the research-report collection, studies Replacement and Collection schemes in a language expanded by named maps, a theme related to the expanded Separation of [proposition XV.5.5](#), with different theorems.

*How Part XV is printed.* Source 31 is the only source of its material, so nothing is merged across sources: Part XV prints its Sections 1–12 whole and in order as Sections XV.1–XV.12, then its Appendices A and B (axiom ledger and source audit) as Sections XV.13 and XV.14, and a status section XV.15. Its research questions are printed in the questions part ([section 113.26](#)), its title page, abstract and scope statement in [Appendix C](#), its delivery record in [appendix A.12](#), and its numbered statements are mapped in [appendix D](#). Where it re-proves or extends a result of Parts I–XIV, a [merge] paragraph after the statement says so; there are two re-proofs (the count of [lemma XV.7.1](#) at a cardinal cutoff, and the inaccessible case of [theorem XV.9.3](#)).

*Status.* A research article with written proofs; it claims no machine verification and no historical priority, uses Glazer’s countermodel theorem as an explicitly cited input, and names expert review of its localized theorem as the next step ([section XV.14](#)). It is not formalized here. The Hilbert’s-tenth research tree routed its archive with the other topology and surreal arrivals of that day (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review\_topology\_surreal\_arrivals\_20261003.md, commit 6aadaa4f2): it hashed the archive, read bounded line ranges and ran nothing, and found that source 31 yields no Diophantine construction; that note records scope, not a proof review, and has no correctness finding.

*Where the batch-89 write had to choose.*

- *Numbering.* The sections of Part XV are numbered XV.1–XV.15 instead of continuing from 112, so that the questions part keeps its Sections 113–115 and every statement, question and

appendix keeps the number it had before this write (the batch-81 and batch-83 writes moved the closing sections; this one does not). Source 31’s statement  $k.m$  is  $XV.k.m$ , and the [merge] paragraphs inside its sections add no numbered statement.

- *Placement.* Source 31’s questions follow those of sources 19–30 (section 113.26), with the dated notes of section 113.27; its abstract closes appendix C; its ledger and source audit close Part XV, as the batch-83 ledgers close Part XIV.
- *Citations of this report.* Source 31’s references to `swo:cf:choice` and `swo:ge:n10` keep its words and gain cross-references; “the inspected ProveIt report” is annotated; its bibliography entry for this report is [69].
- *Renamings* (section 1.3.3): threads  $\text{Thr}_X$  for source 31’s  $\mathcal{T}_X$ ; injections  $X \mapsto D$  for its  $X \hookrightarrow D$ . No other symbol changes.
- *Bibliography.* Its entry for Hamkins’s 2014 post on transfinite recursion is this report’s [63], not [26] (2017), which source 31’s key names; its entry for Hamkins’s post on global choice is [19]; its other entries carry the prefix `gcz`.
- *The Glazer interface* is printed as source 31 states it, with a [merge] paragraph (section XV.5.1) saying that it is unreviewed. No source text was corrected and no statement strengthened.

### 1.11 The batch-96 addition: sources 32–35

[merge] (4 October 2026.) Four manuscripts of batch 96, its manuscripts 08, 09, 10 and 11, continue this report and are merged as Part XVI (part XVI). They arrived together in commit `e3839ad2c` and were placed in `111c38012`. In this report’s local sequence they are sources 32–35 (the highest prefix in this directory was 30–; 31 is source 31, which staged nothing), with file prefixes `32-ceilings-`, `33-heights-`, `34-choicefree-` and `35-completion-` and label prefixes `swo:dc:`, `swo:dh:`, `swo:bo:` and `swo:fc:`. The other batch-96 manuscripts are placed in other reports.

| Src. | B96 | Manuscript; archive and inner directory; size                                                                                                                                                                          | Pin; arrival                                                                  | Labels; files            | Parts         |
|------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------|---------------|
| 32   | 08  | <i>Beyond Ord: Definable Class Well-Orders. Explicit arithmetic, condensation, definability ceilings, and truth jumps; Beyond_Ord_Research.zip</i> , same inner directory; 982 lines, 27 pp.                           | <code>commit 8f7d4a5c8</code><br>(its “root tree”);<br><code>e3839ad2c</code> | <code>swo:dc:</code> ; 4 | XVI           |
| 33   | 09  | <i>Beyond Ord: Definable Class Well-Orders, Arithmetic, Condensation, and Definability Heights; Beyond_Ord_Class_Well_Orders.zip</i> , same inner directory; 1,981 lines, 30 pp.                                       | none (two blobs);<br><code>e3839ad2c</code>                                   | <code>swo:dh:</code> ; 6 | XVI           |
| 34   | 10  | <i>Beyond Ord: Definable Class Well-Orders, Canonical Arithmetic, and the Limits of Notation; beyond_ord.zip</i> , same inner directory; 2,314 lines, 32 pp.                                                           | <code>e1d2f3048</code> ;<br><code>e3839ad2c</code>                            | <code>swo:bo:</code> ; 1 | XVI<br>(base) |
| 35   | 11  | <i>Definable Class Well-Orders Beyond Ord: Arithmetic, canonical notation, fixed-point completion, and the limits of uniform definability; class_orders_beyond_ord.zip</i> , same inner directory; 3,132 lines, 40 pp. | <code>f158f27b4</code> ;<br><code>e3839ad2c</code>                            | <code>swo:fc:</code> ; 2 | XVI           |

*Pins.* Source 34 read this report at `e1d2f3048` and source 35 at `f158f27b4` (both 4 October 2026, 16:34 and 16:37 PDT). Source 32 records a “root tree” `8f7d4a5c8`, which is a commit (16:32 PDT, tree `08039c482`), and two blobs, source 23’s status file and `SetTheory/ZF/Lean/ZF/Zf.lean`. Source 33 records no commit, only the blobs of `Algebra/SurrealNumbers/docs/README.md` (introduced in `c14a866bc`) and of this report’s README (introduced in `e50dde15b`). At every one of these revisions, and until this write, `article.tex` is blob `72eee5cc8`, the text with Parts I–XV, and the README is blob `d2e24b58`; so all four read the same report, and the numbers they cite (Theorems 52.3, 53.4, 108.16 and 108.19 among them) are the current ones. Sources 32, 33 and 34 are dated 4 October 2026; source 35 is dated 5 October 2026, after its pin and its arrival (4 October 2026, 17:22 PDT), and source 33’s files carry 5 October timestamps; the dates are printed as delivered. Title pages: sources 32 and 33 “Research manuscript prepared for Vladimir Reshetnikov”, sources 34 and 35 “Research report prepared for Vladimir Reshetnikov”.

*Not editions.* The four are independent answers to one request: no file is shared, and shared word 8-grams are at most 1.64% (sources 34 and 35) and at most 1.62% of any source against this report (preambles, bibliography, quoted phrasing of this report). Their main theorems overlap heavily, as Part XVI’s Notes record.

*How Part XVI is printed.* Source 34 is the base: on the shared theorems it has the weakest hypotheses (GB for the history equivalence, as source 35; GB for the hereditary order and its universal property, where sources 32, 33 and 35 use GBC; any transitive  $\beta$ -model for the definable ceiling, where sources 32 and 33 use  $V_\kappa$ ), and its set-subset criterion lowers the other sources’ GBC steps. Its Sections 1–10 are Sections XVI.1–XVI.10, each followed by the corresponding sections of sources 35, 32 and 33 as subsections; the four sources’ appendices and ledgers form Section XVI.11, and XVI.12 is the status section. The research questions of all four are merged by theme in [section 113.28](#), with dated notes in [section 113.29](#); their title pages and abstracts are in [Appendix C](#), their delivery records in [appendix A.13](#), and their numbered statements are mapped in [appendix D](#).

*Status.* Sources 32 and 33 call themselves AI-assisted and unrefereed; sources 34 and 35 report “independent mathematical checks during preparation” and are not refereed. All four give written proofs, claim no historical priority and no machine-checked formalization, and are not formalized here. The Hilbert’s-tenth research tree reviewed the archives in a bounded intake (commit `fcc4098ec`), with a placement audit (`b16786caf`) and a related computability note (`c6082d075`); what they checked and what they did not is recorded in [section XVI.12](#).

*Literature cited only by source 35* (status at this write). Checked online on 4 October 2026: Frittaion and Genovesi’s *Axiom Beta Implies Elementary Transfinite Recursion* is arXiv:2603.13913, version 3 of 14 July 2026, with the stated result for its weak theory **C**; Jeon and Walsh’s *Generalized ordinal analysis and reflection principles in set theory* was first published online in the Journal of Symbolic Logic on 28 April 2026 (arXiv:2312.12859, version of 20 December 2023); Jeon’s note states as its Proposition 1.5 the class order  $E_0 = \varepsilon_{\text{Ord}+1}$  with a reference to Rathjen’s Definition 4.1 and Lemma 4.4; version 2 (10 March 2021) of the class-forcing paper records that Gitman and Hamkins pointed out a flaw in the published proof of their Theorem 9 (clopen determinacy implies ETR), the implication remaining correct, as source 35 says; and Gitman and Hamkins prove the GB + DC form of the well-foundedness test in the discussion before their Definition 6 (in their Section 4, as source 34 cites it). Not checked: Glazer’s MathOverflow question 270535 and its answers by Hamkins and Taranovsky (the site refused automated access), the cited results of Williams’s dissertation, and Rathjen’s chapter itself, which source 35 also did not see.

*Bibliographic notes of the sources.* Entries of this report are reused where the sources cite the same work: [22] (source 34 adds “Contemporary Mathematics 690, 2017, pp. 121–143”), [27] (“the full text checked was arXiv version 1”, source 34; “Theorems 4–6, Observation 8, Theorem 9 and Question 7”), [24] (version 2, source 35), [35], [30], [26] and [28]. New entries carry the prefix **xvi**.

Where the batch-96 write had to choose.

- *Base and arrangement*: source 34, for the reasons above; the other sources' sections follow the base section they continue, whole where possible (source 35's Sections 3 and 9, source 32's Section 3 and source 33's Section 3 are split between two base sections).
- *Duplicates*: a statement proved again by the same argument is a Note; three source equations referred to elsewhere are printed inside their Notes, (S33.7.1), (S33.7.2) and (S35.15). Proofs that differ are printed as routes: source 35's power theorem, leastness and successor obstruction, source 32's power theorem and termination, source 33's definability height.
- *Numbering*: Sections XVI.1–XVI.12, so that no earlier number changes; the questions are one subsection, 113.28, with dated notes in 113.29 (the placement had projected one subsection per source, 113.28–113.31 and 113.32; the four lists are merged by theme instead).
- *Renamings* (section 1.3.4):  $A^{[B]}$  for source 34's  $\backslash\text{Pow}$ ;  $A^{[B]}$  and  $P(\Gamma)$  for sources 32 and 33's  $A^B$  and  $P_\Gamma$ ;  $E$ ,  $E(A)$ ,  $E^{\leq n}$  for  $\mathcal{E}$ ,  $H(A)$ ,  $\mathcal{H}(A)$  and source 34's  $E_n$ ;  $\mathbb{T}$  for source 33's direct limit  $E$ ;  $\text{dom } W$  for source 32's  $D_W$ ;  $\hookrightarrow$ ,  $\nrightarrow$ ,  $\preceq_{\text{init}}$  for the sources' embedding symbols; source 34's agenda items as numbered questions R1–R12.
- *Stale and answered claims*, kept on record with their proofs: source 32's termination priority (remark XVI.1.1); source 33's Remark 5.5 and question 12.7 (remark XVI.5.29); source 32's question 12.1 and source 33's question 12.6 (section 113.28).
- *Added by this write*: the identification of towers with histories (remark XVI.5.24), the consequence of source 33's converse (remark XVI.5.31), and the choice-free completion theorem (theorem XVI.6.21), which answers source 35's question 10.6 from source 34's criterion; none is a source's statement, and none is reviewed. (Dated note, 4 October 2026: all four have since been checked independently and found valid after editorial corrections, which are made and recorded in dated notes at each; section XVI.12.)
- *A correction to the placement record*: the placement dossier read Mathlib's `Ordinal.univ` and `Ordinal.type_lt_ordinal` as deprecated; only `Ordinal.univ_id` is, as source 32 says (Section XVI.10).
- No source text was corrected and no statement strengthened; the delivered texts are printed with the renamings and label prefixes above, with headings replaced by merge headings that carry the source's label.

## 1.12 The batch-97 addition: source 36

[merge] (4 October 2026.) One manuscript of batch 97, its manuscript 01, continues this report and is printed as Part XVII (part XVII), source 36. It arrived in commit d7cf7d554 (4 October 2026, 18:06 PDT), fifteen minutes after the placement 111c38012 of sources 32–35, and was placed in 6571ee1af (batch 97A). In this report's local sequence it is source 36 (the highest prefix in this directory was 35–), with file prefix 36-horizons- and label prefix swo:hn:. Batch-97 manuscripts 02–07 are placed in other reports.



| Src. | B97 | Manuscript; archive and inner directory; size                                                                                                                                                                                                                      | Pin; arrival            | Labels; files | Parts |
|------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------|-------|
| 36   | 01  | <i>Beyond Ord: Definable Class Well-Orders, Finite-Support Arithmetic, and Definability Horizons. A constructive notation calculus with exact model-relative bounds;</i><br>Beyond_Ord_Research_Package.zip,<br>inner directory Beyond_Ord; 2,950 lines,<br>40 pp. | e1d2f3048;<br>d7cf7d554 | swo:hn:: 5    | XVII  |

*Pin.* Source 36 read this report at e1d2f3048 (4 October 2026, 16:34:43 PDT, as its audit says in UTC), the pin of source 34, where `article.tex` is blob 72eee5cc8 (3,289,057 bytes, Parts I–XV) and the README blob d2e24b586; its audit adds that a code-search index first offered the older commit 555ee36fb. It read this report’s README, 29-birthdays-PROOF\_STATUS.txt, the repository README and Logic/PeanoArithmetic/ListCoding/README.md, not the article itself; it did not see Part XVI. Its title page has no author line (“An investigation prompted by the ProveIt repository”, October 2026; PDF author “Research manuscript on definable class well-orders”); its audit is dated 4–5 October 2026.

*Not an edition.* Source 36 shares no file with sources 32–35 (the Hilbert’s-tenth review compared every member byte for byte), and its shared word 8-grams are 0.41 %, 0.36 %, 0.72 % and 0.78 % against sources 32, 33, 34 and 35 and 0.90 % against this report with Part XVI (preambles, bibliography, standard phrasing). It is an independent fifth answer to the request of sources 32–35, and its main theorems overlap theirs heavily, as the credit notes of Part XVII record.

*How Part XVII is printed.* Part XVI was written and committed (62b16914e) before source 36 could join its merge, and inserting source 36’s material into it would renumber its Sections XVI.11–XVI.12 or put results after its status section. Source 36 is therefore printed whole as its own Part, as source 31 is in Part XV: its Sections 1–8 as Sections XVII.1–XVII.8, its Section 9 as Section XVII.9 with its questions in [section 113.30](#), its Appendix A as Section XVII.10, and a status section XVII.11. Its title page and abstract are in [Appendix C](#), its delivery record in [appendix A.14](#), and its numbered statements are mapped in [appendix D](#).

*Status.* Conventional written proofs with “independent review during preparation” (its words); no referee, no proof assistant, no priority claim. Not formalized here. The Hilbert’s-tenth research tree reviewed the archive in a bounded intake before placement (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_beyond_ord_package_d7cf7d554.md`, commit 39bb02ece); its scope is recorded in [section XVII.11](#).

*Literature cited only by source 36*, not checked by this write: Reshetnikov’s Mathematics Stack Exchange question 4072813 [95], the MathOverflow discussions 116590 and 379630 [100, 101], Freire and Williams’s Theorem 2.1.1 [96], Marcone and Montalbán’s Definition 2.3 [97], Traytel’s Isabelle theory [98] and the Mathlib documentation of Cantor normal forms [99]. Entries of this report are reused where source 36 cites the same work: [84] (version 4), [22] (source 36 adds Contemporary Mathematics 690, 2017, 121–143), [35] (its Question 2.35 and Lemma 3.17), [27] (Theorems 4–6 and Question 7), [24] and [19] (“especially the rank-first construction”). New entries carry the prefix xvii.

*Where the batch-97 write had to choose.*

- *A Part of its own*, printed whole, for the reason above; the placement 6571ee1af made the same decision.



- *Numbering*: Sections XVII.1–XVII.11 and statements XVII. $k.m$  equal to source 36’s  $k.m$ ; the credit notes are lettered (XVII. $k.A$ , ...), so that no source number moves; the questions are one subsection, 113.30, with dated notes in 113.31.
- *Credit, not correction*: the fifteen credit notes name the statements of Parts I–XVI that source 36 proves again; its novelty sentences (Section XVII.1.5, its title page, the status table of Section XVII.10) are printed as delivered with dated credit notes; its least-fixed-point theorem is printed as delivered, and the stronger form its proof gives is stated in [note XVII.4.B](#) with credit to source 34. No source text was changed and no source statement strengthened.
- *Renamings* ([section 1.3.5](#)):  $A^{[B]}$  for class-order powers  $A^B$ ;  $\mathbf{E}$  and  $\mathbf{E}^{\leq n+1}$  for  $\mathbf{T}_\Omega$  and  $T_n$ ;  $\mathbf{E}_\rho$  for the epsilon-term functor  $\mathbf{E}$ ;  $\mathfrak{H}$  for the horizons  $H$ ;  $\preccurlyeq_{\text{init}}$ ,  $\prec_{\text{init}}$ ,  $L_{<x}$  for its embedding and cut macros. The starred equation is cited by its tag in one sentence.
- *Unproved claims*: none of source 36’s theorems was found false, and none is unproved. Two sentences of its question commentary state expectations without proof (“Further finitary notation operators may force additional closure conditions”; “An audited coding might yield a small constant shift”); they are printed as delivered, marked as conjectures in [section 113.30](#).
- *Corrections to Part XVI*: this write also corrects four passages written on 4 October 2026 for Part XVI, after an independent check of them ([theorem XVI.6.21](#) and [remarks XVI.5.24](#), [XVI.5.29](#) and [XVI.5.31](#)); each carries a dated note.

## 2 The objects being studied

There are two orders in this problem. Each object under consideration is itself a well-order of some collection of surreal numbers. We then compare these objects lexicographically using the *numerical* order on their surreal labels. This second order need not be a well-order. Indeed, the orders below contain abundant descending sequences.

Write  $L \hookrightarrow M$  for an order embedding and  $L \simeq_e M$  for embeddings in both directions. Equimorphism does not imply isomorphism.

| Object                                   | Elements                                                                                                             | Logical size and interpretation                                  |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| $\mathcal{W}(X)$                         | Ordinal-indexed bijections onto a fixed set $X \subseteq \mathbf{No}$                                                | A set; ordinary ZFC order theory                                 |
| $\mathcal{I}(\mathbf{No})$               | Injective set-length ordinal words in $\mathbf{No}$                                                                  | A class of sets; well-orders of arbitrary set-sized subsets      |
| $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  | Set-like class well-orders of all $\mathbf{No}$ , equivalently class bijections $\text{Ord} \rightarrow \mathbf{No}$ | A higher-order totality; use class variables or uniform families |
| $\mathfrak{W}_{\text{all}}(\mathbf{No})$ | All class well-orders of all $\mathbf{No}$ , including non-set-like ones                                             | A higher-order totality; direct labeled comparison is needed     |

The fraktur notation is deliberate. It does not declare a class with proper classes as members. Statements about  $\mathfrak{W}_{\text{sl}}$  and  $\mathfrak{W}_{\text{all}}$  are formulas with class variables, or statements relative to a specified larger universe. An embedding of an ordinary class order into one of these totalities is a uniform family: one class relation encodes the object assigned to each set-valued index.

**Principal answers.** For a fixed infinite alphabet  $X$ , the exact ordinal ceiling is  $(|X|)^+$ . Passing to all set-length surreal words removes that ceiling: there is an explicit embedding of the entire class into  $\mathbf{No}$ , and the singleton words embed  $\mathbf{No}$  in the other direction. The two orders are nevertheless not isomorphic, because limit-length words have set-sized left cofinality.

Under global choice, the set-like global fragment contains a uniformly indexed copy of every class linear order. This includes every set ordinal,  $\text{Ord}$ , reverse  $\text{Ord}$ ,  $\text{Ord} + \text{Ord}$ , and the whole numerical surreal order. Yet the entire fragment cannot be exhausted by one uniform class of set-codes. Class universality of a subfamily is compatible with a larger totality of class objects.

The unrestricted global order is different again. It has adjacent pairs and convex residual fibers. Every raw well-order begins with a canonical  $\text{Ord}$ -sequence; if that sequence omits labels, those labels form a later well-ordered tail. This decomposition permits a precise study of *all* well-orders without treating them all as set-like. It also supplies a nonembedding obstruction: the unrestricted order has too many disjoint two-point fibers to embed into the set-like order in the full containing-universe model. A related uniform definability obstruction is provable in  $\text{KM} + \text{GC}$ .

### 3 Relationship to ProveIt and the literature

We inspected the repository at the immutable revision

`ae7c1e6aae3dc58c9c638b35646865ad6c10387d`.

The closest precursor is the report on lexicographic well-orderings of the reals [1], in the subdirectory `lexicographic-well-orderings-of-reals/` of the repository's ordinal-order reports. Its general alphabet theorems already establish the cardinality, universality, and ordinal cutoff for every infinite linearly ordered set  $X$ . Their application to birthday fragments is a transfer of existing results, not a new generic alphabet theorem.

That report also proves, with  $\mathfrak{c} = 2^{\aleph_0}$ ,

$$\mathbf{No}_{<\theta} \hookrightarrow \mathcal{W}(\mathbb{R}) \quad (\theta < \mathfrak{c}^+), \quad \mathbf{No}_{<\mathfrak{c}^+} \not\hookrightarrow \mathcal{W}(\mathbb{R}).$$

The negative assertion follows from the all-plus chain of length  $\mathfrak{c}^+$ . It warns against taking an incoherent union of stagewise embeddings.

The repository's foundational report [3] supplies the relevant size discipline: individual surreals are set-coded, bounded fragments are sets, arbitrary class cuts need not be fillable, and quantifying over class functions does not create a class of those functions. Its local set-valued recursion arguments are directly useful. A nearby vector-space report uses the same distinction to avoid an unnecessary ETR hypothesis [5].

The sign-sequence presentation is standard [12, 14]. Absolute universality and homogeneity of the surreal order are part of the established theory discussed by Ehrlich [15]. We supply the short recursion arguments needed here.

**The Kanovei–Shelah connection.** There is a methodological connection, but the objects are different. Kanovei and Shelah's actual index consists of maps

$$a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N})$$

whose ranges are ultrafilters. They compare these maps lexicographically using the binary order on  $\mathcal{P}(\mathbb{N})$  [16, Section 1]. The maps need not be injective and do not enumerate the entire real line. Their construction illustrates how a definable linear order on many codes can replace a definably selected well-order. The present problem shares that idea while introducing a different level of class-size complexity.

**Contribution and attribution.** The central developments here are explicit surreal-word compression, direct comparison of unrestricted class well-orders, their canonical initial-Ord decomposition, the bounded-support interpolation theorem, and the relation between omitted ranges and non-set-like tails. No claim about real-specific countability, separability, or gap spectra is transferred to surreal alphabets without proof.

## 4 The other sources' introductions

[merge] Each of the other manuscripts opens with its own statement of the problem, of its relation to ProveIt, and of the Kanovei–Shelah connection. These introductions are printed whole, one subsection each, followed by notes where the merge corrects or qualifies them. Their summaries of results are not repeated claims: every result they announce is printed, once, in the body.

### 4.1 Source 12: scope, predecessors, and the results at a glance

The phrase “well-orders of surreal numbers” hides several distinct objects. The usual numerical order of **No** is not a well-order. One may instead put an unrelated well-order on the entire surreal carrier, or consider numerical well-orders on subclasses, or work with an ordinary set of bounded-birthday surreals. This article treats all three viewpoints, but the principal object is the lexicographic comparison of *exhaustive well-orders of a fixed carrier*.

The supplied ProveIt repository contains a substantial predecessor, *Lexicographic Orders of the Well-Orderings of the Reals*, merging three reports and distinguishing the full enumeration order from its minimal-length slice [1]. The inspected README explicitly marks that report as unrefereed and not formalized. Its real-number spectra, binary coding, finite-tail geometry, and warnings about isomorphism versus equimorphism motivate the present work. The repository also contains a Lean development of simplest elements of convex sets of sign sequences [10]. These are useful foundations, not a formal verification of the new theorems below.

We pinned the audit to repository commit

63bb8b1d99359e7188abf0a33fb152e121858150.

The audit was targeted, not an exhaustive review of the repository. All essential arguments used here are written out, so no unreviewed research claim from the repository is required as a black box.

#### 4.1.1 Three different orders

For an infinite set-sized linear order  $X$  of cardinality  $\mu$ , define

$$\mathcal{W}_\alpha(X) = \{e : \alpha \longrightarrow X : e \text{ is bijective}\}, \quad \mathcal{W}(X) = \bigcup_{\mu \leq \alpha < \mu^+} \mathcal{W}_\alpha(X).$$

They are ordered by their first unequal entry, using the given order of  $X$ , not the well-order being represented.

For the proper class **No**, let  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  denote, as a *predicate on classes*, the class bijections  $E : \text{Ord} \rightarrow \mathbf{No}$ . These represent exactly the set-like global well-orders. Let  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  denote the predicate of being an arbitrary class well-order relation on **No**. Neither displayed notation asserts the existence of a class having these class-sized objects as members.

The comparison on  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  uses a least ordinal of disagreement. The comparison on  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  uses a largest common *labelled* initial segment. The latter can be a proper class, and this is essential rather than a cosmetic extension.

### 4.1.2 What is proved, and what is not being claimed

The principal developments are the elementary common-prefix theorem ([theorem 14.7](#)), the global-choice equivalence ([theorem 13.2](#)), the non-enumerability theorem ([theorem 19.38](#)), and the singular transition ([theorem 8.3](#)). The full and minimal bounded spectra are proved in [sections 6.2](#) and [8](#). The difference between the dense set-like order and finite-tail geometry in the full order is proved in [sections 16.1](#) and [19.2](#).

These are mathematical claims with supplied proofs. We do not claim that they settle a previously catalogued open problem, that their exact formulations are historically new, or that they have been independently refereed. The proposed research questions at the end are precisely that: questions suggested by the results, not an assertion of a complete literature census.

### 4.1.3 The Kanovei–Shelah connection

Kanovei and Shelah order maps  $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters, using a fixed lexicographic order on  $\mathcal{P}(\mathbb{N})$  [[16](#), §1]. Their maps need not be bijections, and their index is not the literal collection of all well-orders of  $\mathbb{R}$ , still less all class well-orders of **No**. The useful common idea is to compare representations at their first difference instead of choosing one canonical representative. Their index construction remains set-sized. Moving to all global surreal well-orders creates an additional size level that their construction does not remove.

[merge] *On source 12's introduction.* Source 12 read the reals report's README and the Lean file `SignSequenceSimplicity.lean`, not the reals report's article; it therefore re-proved the set-sized patterns it needed. They are printed here as a marked route with pointers to the reals report ([sections 6.1](#) and [6.2](#)). The reals report's article has not changed since source 12's pin. Source 12's principal developments are printed as [theorem 14.7](#) (a second route to [theorem 14.2](#)), [theorem 13.2](#) (a second route to [theorem 13.1](#), over  $\text{GB} + \text{AC}$ ), [theorem 19.38](#) (printed beside [theorem 19.37](#)) and [theorem 8.3](#) (new).

## 4.2 Source 09: the question and the principal conclusions

A surreal number is a set-sized object, but the totality **No** of surreal numbers is a proper class. Consequently, taking “all well-orders” changes logical level when one passes from  $\mathbb{R}$  to **No**. The main task is not merely to replace the real alphabet by a larger alphabet. It is to specify which objects are being ordered and which collections of those objects are available in the foundational theory.

Throughout, the word *embedding* means an order-preserving and order-reflecting injection. It need not preserve cuts, suprema, the order topology, birthdays, simplicity, or field operations. An order is *equimorphic* with another when each embeds into the other; this is weaker than isomorphism.

### 4.2.1 Three different carriers

For a set  $X \subseteq \mathbf{No}$ , let  $\mathcal{W}(X)$  consist of all bijections  $e : \alpha \rightarrow X$ , with  $\alpha$  an ordinal. These are exactly the exhaustive enumerations associated with well-orders on the fixed labeled set  $X$ . For different  $e, f$ , comparison takes place at the first unequal value. Because both are exhaustive, neither can be a proper initial segment of the other.

Let  $\mathcal{I}$  instead contain every set-length injective word  $p : \alpha \rightarrow \mathbf{No}$ , allowing its range to vary. Here proper-prefix pairs do occur, so an end-of-word convention is necessary. We use the convention that a proper prefix comes first. This is ordinary dictionary order, *not* the convention used to compare surreal sign sequences.

Finally, let  $\mathfrak{W}_{\text{sl}}$  be informal notation for the global class bijections  $E : \text{Ord} \rightarrow \mathbf{No}$ . Each represents a set-like global well-order. In GBC,  $\mathfrak{W}_{\text{sl}}$  is notation for a condition on class variables, not a class whose members are these class functions. A precise, ordinary class substitute is the set-coded core  $\mathcal{P}_s(b)$  of eventual perturbations of a fixed  $b : \text{Ord} \rightarrow \mathbf{No}$ .

**Main global conclusion.** In GBC, for every fixed global enumeration  $b$ ,

$$(\mathcal{P}_s(b), <_{\text{lex}}) \cong (\mathbf{No}, <).$$

The core is order-dense among all global enumerations and even supplies interpolants for every uniformly set-indexed separated cut involving arbitrary global enumerations. Nevertheless, there is no uniform class-indexed enumeration of *all* global enumerations, and certain proper-class-sized separated families have no interpolant at all.

#### 4.2.2 How this extends the repository’s real-line investigation

The inspected ProveIt snapshot contains a substantial merged report on lexicographic well-orderings of the reals [1]. That report studies ordinal spectra, binary coding, finite tails, local characters, and gaps. Its set-sized perspective is useful here, but its continuum bounds cannot simply be applied to the entire surreal class. We reprove the set-sized arguments needed below for an arbitrary linearly ordered set, without importing unverified transfinite claims from that report.

The genuinely global development is organized around four concrete constructions: bounded-support extension of an injective prefix; set-indexed lexicographic interpolation; an eventually-equal surreal core; and diagonalization at independent pairs of positions. These constructions are the main contribution of this manuscript. Whether particular formulations have appeared previously would require a more exhaustive literature search; they are not presented as established historical firsts.

Kanovei and Shelah’s nonstandard-model construction provides motivation for lexicographically ordering redundant codes. Their actual index is not the collection of all real well-orders: it consists of maps from the continuum cardinal to  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters, ordered at their first disagreement [16, §1]. This distinction remains essential in any proposed surreal analogue.

[merge] *On source 09’s introduction.* Source 09 read the reals report’s README and the Lean files `SignSequence.lean` and `SignSequenceCutOperation.lean`. Its set-sized arguments, which it “reproves ... without importing unverified transfinite claims”, are printed as notes and one marked route (section 6.3). The four constructions it calls its main contribution were found independently by the other sources on the same day: bounded-support prefix completion also by sources 11 and 08 (lemma 19.4 and note 19.8); set-indexed interpolation by sources 11, 12 and 08 (theorem 19.10 and notes 19.11 and 19.13); the eventually-equal core by sources 11 and 08 (corollary 19.15 and note 19.17); and the pair-switch diagonalization by sources 11, 12 and 08 (theorems 19.37 and 19.38 and note 19.41). No priority among the sources is implied by the order of printing. The box’s last sentence is theorem 19.42 and theorem 19.43.

#### 4.3 Source 08: the question, disambiguated

The phrase “well-orders of surreal numbers” has two independent ambiguities. First, is the carrier a set of surreal numbers or the whole proper class? Second, is a well-order required to be the restriction of the numerical order, or is it a new ordering of the labelled carrier? A third choice enters when the well-orders themselves are compared: which data are read lexicographically?

This paper uses the following principal convention. An arbitrary well-order on a labelled carrier is represented by its exhaustive increasing enumeration *in that well-order*. Two enumerations are then compared at their first unequal entry, using the *numerical* order of their surreal values. The enumerations themselves need not be numerically increasing. This distinction is indispensable.

For example, the two lists  $(0, 1, -1)$  and  $(0, -1, 1)$  represent two well-orders of the same three-element set. The second list is smaller in our lexicographic order, since  $-1 < 1$ . Both underlying relations have ordinal type 3. Passing to ordinal isomorphism types would identify them and destroy the object being studied.

#### 4.3.1 Five different carriers

| Object                      | Elements or parameters                                     | Foundational size                                       |
|-----------------------------|------------------------------------------------------------|---------------------------------------------------------|
| $\mathcal{W}(X)$            | Exhaustive enumerations of one fixed set $X$               | A set in ZFC                                            |
| $\mathcal{I}(\mathbf{No})$  | Injective ordinal-length, set-sized words in $\mathbf{No}$ | A class of sets                                         |
| $\mathfrak{M}_{sl}$         | Class bijections $\text{Ord} \rightarrow \mathbf{No}$      | A virtual collection of classes                         |
| $\mathcal{P}_s(b)$          | Set codes for set-supported changes of a fixed $b$         | A class of sets                                         |
| Arbitrary class well-orders | Class relations, possibly not set-like                     | A virtual collection; another comparison rule is needed |

A *virtual collection* is notation for a condition on class variables, not a new object whose members are proper classes. Thus “ $e \in \mathfrak{M}_{sl}$ ” abbreviates “ $e$  is a class bijection from  $\text{Ord}$  onto  $\mathbf{No}$ .” All statements involving several such parameters can be translated into the two-sorted language of class set theory. Whenever a family of classes is used, it is represented by one uniform class relation. This convention is made precise in [section 19.1](#).

#### 4.3.2 Main conclusions

The central distinction is between a very large ambient order and its class-sized dense skeleton. Fix a global enumeration  $b : \text{Ord} \rightarrow \mathbf{No}$ . Write  $\mathcal{P}_s(b)$  for canonical set codes for permutations  $\sigma$  of  $\text{Ord}$  that equal the identity outside a set, ordered by the numerical lexicographic comparison of  $b \circ \sigma$ . We prove

$$(\mathcal{P}_s(b), <_b) \cong (\mathbf{No}, <). \quad (\text{S08.1})$$

Moreover, if uniformly given set-sized families  $L, R$  of arbitrary global enumerations satisfy  $L <_{\text{lex}} R$ , then some  $b \circ \sigma$  strictly separates them. The separator is in the same fixed skeleton regardless of the supports, or lack of supports, of the input enumerations.

This yields a useful answer to the embedding question. Every *set* linear order, every ordinal, and indeed every *class linear order whose elements are sets* embeds into  $\mathcal{P}_s(b)$ . In particular,  $\text{Ord}$ ,  $\text{Ord} + 1$ ,  $\text{Ord} + \text{Ord}$ , their reversals,  $\mathbf{No}$ , and each set-indexed lexicographic power of  $\mathbf{No}$  embed. This does not say that the full virtual order  $\mathfrak{M}_{sl}$  is itself a class order or is isomorphic to  $\mathbf{No}$ .

At a fixed set-sized carrier  $X$ , the answer is different. If  $|X| = \kappa$  is infinite, then

$$|\mathcal{W}(X)| = 2^\kappa, \quad \beta \hookrightarrow \mathcal{W}(X) \iff \beta < \kappa^+. \quad (\text{S08.2})$$

Thus making the carrier larger changes the allowed ordinal spectrum; there is no legitimate substitution of a “cardinal  $|\mathbf{No}|$ ” into [\(S08.2\)](#).



### 4.3.3 Relationship to existing work

The set-sized part extends the exhaustive-enumeration viewpoint of the prior working manuscript on the continuum [2]. Its results are reproved here in a form that does not assume separability of the alphabet. The standard surreal background is the sign-sequence and cut-filling viewpoint associated with Conway, Gonshor, and Alling [11, 12, 13]; Ehrlich’s work places the surreal continuum in a broader universality framework [15].

The immediate formalization base is the inspected ProveIt sign-sequence construction and its small-cut theorem [7, 8]. We do not need the repository’s field arithmetic or stronger research claims. The resemblance to the Kanovei–Shelah definable hyperreal construction is real but limited: their actual index consists of fixed-length maps whose ranges are ultrafilters, not literally all well-orders of the reals [16]. We return to this distinction in [section 4.4](#).

[merge] *On source 08’s predecessor.* The “prior working manuscript on the continuum” that source 08 cites [2] is batch-76 manuscript 02, *Lexicographic Orders of Well-Orderings of the Continuum*, which is now the base source 02 of the repository’s reals report (placed in 2a04b60f2, written in 718bfa3b5 and 603f6ed31). Source 08 did not read that report. Its set-sized results re-prove the reals report’s general-alphabet theorems, which hold for every infinite linear alphabet, separable or not; they are recorded with pointers in [sections 6.1](#) and [6.4](#). The display (S08.1) is [note 19.17](#), and (S08.2) is [notes 6.21](#) and [6.23](#).

## 4.4 Source 08: the Kanovei–Shelah connection, carefully stated

Kanovei and Shelah’s definable nonstandard model uses an index  $A$  of maps

$$a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N})$$

whose ranges are ultrafilters. They compare these maps lexicographically using the binary characteristic-function order on subsets of  $\mathbb{N}$ . Repetitions are allowed; the index is not a well-order [16]. It is therefore not literally the order of all bijective well-order enumerations of  $\mathbb{R}$ .

The relevant common idea is that ordinal-indexed codes can be linearly compared at their first unequal value even when no definable choice of one representative from each coded object is supplied. Our paired codes, terminal codes, and virtual relations explore that order-theoretic idea in the surreal setting. They do not provide a class ultrapower or a transfer theorem.

An extension to a class-sized ultrapower would need separate decisions about admissible predicates, function families, equivalence-class coding, truth or satisfaction, and the exact size of saturation. Neither a dense index order nor a surreal-sized skeleton settles these questions. In particular, the set-cut interpolation theorem is an order-theoretic theorem, not a claim of saturation for every language or every class-sized family of types.

[merge] The Kanovei–Shelah distinction is made in the same terms by all four sources (source 11 in [section 3](#), source 12 in [section 4.1](#), source 09 in [section 4.2](#), source 08 above) and by the reals report itself, which analyses the actual index (`lwo:eq:KS`, `lwo:sec:KS-topology`, `lwo:rk:sec:KS`). The uniform constructions it suggests are research questions here ([research question 113.10](#); source 09’s and source 08’s last directions).

## 4.5 The introductions of sources 13–18

[merge] (3 October 2026.) The opening sections of the six batch-81 sources are printed whole in the following subsections, in source order, followed by the Kanovei–Shelah remarks of sources 14 and 17; the other four batch-81 sources discuss the Kanovei–Shelah index in their maps of interpretations

(Part VII, [section 34](#)). Every result they announce is printed once in Parts VI–IX. Their equations are tagged (SNN.*n*) as elsewhere. Corrections and qualifications:

- *Stale descriptions of this report.* All six describe this report at their pins, as merged from four manuscripts; sources 13 and 15 (and source 14 in its audit section) give its length as 115 pages. It now merges ten manuscripts in nine Parts. Their citations of this report are printed as cross-references, with section and theorem numbers “at the pin” where the source gives them.
- *One theorem announced six times.* Each source presents the represented-cut criterion among its main results; it is one theorem, proved independently in all six ([note 46.1](#)). Source 16’s “all ordinal-class blocks at once” and source 18’s ([S18.3](#)) are likewise one theorem with source 15’s ([note 52.1](#)), and source 18’s density display ([S18.2](#)) is source 16’s theorem ([theorem 27.2](#)). Source 18, which could not know sources 13–17, calls these results “advances” over the repository in its table; relative to the merged report they are independent proofs of results of the sibling sources, and only the items listed as new in [sections 33, 39, 50 and 60](#) carry that word here.
- *Source 14’s “Four main results”* lists five; the heading is kept.
- *Source 17’s question table* maps its reserve theorem to source 11’s [research question 113.6](#), which asks for birthday bounds of word embeddings; the reserve theorem bears on source 09’s direction 6 and source 08’s question 13 instead ([note 49.11](#)). The table also omits source 08’s questions 2, 3, 6 and 12, on which its results bear ([sections 33 and 50](#)).
- *Parameters.* Source 16’s overview writes  $\kappa$  for the size of an alphabet and source 18’s writes  $\nu, \tau, \rho$ ; they are printed  $\mu$  and  $r, q, \theta$  as in [section 1.3.1](#). Source 18’s display defining  $X_\kappa = \mathbf{No}_{<\kappa}$  is omitted with its symbol.

#### 4.6 Source 13: scope, provenance, and the main advances

The phrase “the lexicographic ordering of all well-orders of surreal numbers” does not determine a single object. There are three independent choices: what is being well-ordered, how that well-order is represented, and in what universe the collection of representations is allowed to exist. A fourth choice appears when representations can have different lengths: the position of a termination symbol.

The ProveIt repository already contains a substantial report on these issues, assembled from four independent manuscripts. Its maintained `article.pdf` is described as 115 pages, and the report explicitly distinguishes written proofs from machine-checked results (this report and its README at the pin; [\[merge\]](#) the page count and the number of sources have grown with the batch-81 addition, and the report is still not formalized). We use the snapshot `0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8`. The earlier report supplies the fixed-alphabet theory, the surreal-sized dense core, the direct comparison of raw class well-orders, and several nonembedding and coding arguments. These are credited below and, where needed for the new proofs, re-established in a compact form.

The present investigation addresses four of that report’s proposed continuations rather than relabeling its existing results as discoveries. The term “new” below means an extension developed in this manuscript relative to the inspected material; it is not a bibliographic priority claim.

**Main conclusions.** A normalized cut comes from an injective Ord-word exactly when it crosses a unique prefix cylinder at every set-ordinal length. It comes from a global enumeration exactly when these prefixes eventually include every label. Nonbranch cuts stop either at a successor step, where they separate whole next-letter cylinders, or at a limit step, where they lie on the boundary of one cylinder.

For the virtual order  $\mathfrak{W}_{\text{sl}}$  of all class bijections  $\text{Ord} \rightarrow \mathbf{No}$ ,

$$\mathfrak{W}_{\text{sl}} \cong \mathbf{No} \times_{\text{lex}} \mathfrak{W}_{\text{sl}},$$

but  $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \not\rightarrow \mathfrak{W}_{\text{sl}}$  in the full inaccessible model and in the specified uniform  $\text{KM} + \text{GC}$  sense. This is not a contradiction:  $\mathbf{No}$  is an ordinary class of sets, whereas the full  $\mathfrak{W}_{\text{sl}}$  is a domain of class objects.

An order automorphism can send a prefix cylinder to a non-cylinder. Consequently the numerical linear order on enumerations does not, by itself, canonically specify the prefix tree. Central termination, on the other hand, repairs the small one-sided characters of ordinary dictionary words and produces an order isomorphic to  $\mathbf{No}$ .

#### 4.6.1 What is proved, and what remains conditional

The cut-reconstruction, product-absorption, recoding, and central-word results are proved in GBC. The product obstruction is proved in two separate interpretations: ambient ZFC with a strongly inaccessible universe, and a fixed-formula uniformity schema in  $\text{KM} + \text{GC}$ . The latter proof identifies its impredicative comprehension step. We do not claim that the same argument works over bare GBC.

This separation also prevents a false “absolute” assertion about the full global order. In an externally countable model, the available global enumerations form an externally countable dense order without endpoints, hence a copy of the rationals. Arbitrary external maps and internally uniform constructions therefore answer different embedding questions.

#### 4.6.2 Reading paths

Section 34.1–32.2 survey the interpretations and size constraints. Section 42.2–46.4 establish the dense-core and cut machinery. Section 55–36.4 contain the other main extensions. Sections 57.3 and 59.1 specify foundational and proof-assistant interfaces. The final research agenda distinguishes questions resolved here from stronger problems still to be investigated.

### 4.7 Source 14: the continuation and its main conclusions

The phrase “the lexicographic ordering of all well-orders of surreal numbers” does not specify one mathematical object. It leaves open what is being well-ordered, how it is represented, how termination is compared, and which logical universe supplies the word “all.” These choices are not cosmetic: they change cardinality, adjacency, completeness, and even the logical sort of the objects under study.

The present article continues the merged ProveIt report (this report and its README), inspected at commit 0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8. That report combines four earlier manuscripts and already proves a large body of results. In particular, its Section 19 ([merge] Part IV here) supplies the dense set-supported skeleton, and its research agenda explicitly asks which cuts of that core represent exhaustive branches. Repeating the skeleton isomorphism or the set-sized ordinal spectrum would not answer that question.

Our main ambient theory is  $\text{GBC} = \text{GB} + \text{GC}$ , with Power Set and ordinary set Foundation. The letters  $\mathfrak{W}_{\text{sl}}$ ,  $\mathfrak{J}$ , and  $\mathfrak{W}_{\text{all}}$  denote predicates on classes, not classes whose members are proper classes. Fix a class bijection

$$b : \text{Ord} \longrightarrow \mathbf{No}.$$

The ordinary class  $\mathcal{P}_s(b)$  consists of set codes for permutations of  $\mathbf{Ord}$  with set support, evaluated against  $b$ . The virtual order  $\mathfrak{W}_{sl}$  consists of all admitted class bijections  $\mathbf{Ord} \rightarrow \mathbf{No}$ .

**Four main results.** ([merge] Five are listed below.)

**Cut classification and recognition.** Every proper normalized class cut of  $\mathcal{P}_s(b)$  is uniquely a cylinder edge, a numerical gap after a set prefix, or an  $\mathbf{Ord}$ -long injective branch. Such a cut is the trace of an element of  $\mathfrak{W}_{sl}$  exactly when its crossing prefixes occur at every ordinal length and together contain every surreal (Theorem 43.3 and 45.1).

**No newly filled old cuts.** For two GBC class realizations with the same sets and the same  $b$ , an old skeleton cut is realized by a global enumeration in the larger realization if and only if it already was in the smaller one (Theorem 48.1).

**Exact support threshold.** For every infinite set cardinal  $\kappa$ ,

$$\mathcal{P}_{<\kappa}(b) \text{ is } \eta_{cf \kappa} \text{ and is not } \eta_{(cf \kappa)^+}.$$

A witnessing omitted configuration has a minimum possible support cost of exactly  $\kappa$  in the full core (Theorem 49.3).

**What unrestricted tails hide.** Two labelled class well-orders have the same trace on  $\mathcal{P}_s(b)$  exactly when they have the same canonical initial  $\mathbf{Ord}$ -enumeration. Above a fixed injective branch  $F$ , the fiber is precisely the family of well-orders of  $\mathbf{No} \setminus \text{ran } F$  (Theorem 54.6).

**Bare-order cut orbits.** Character pairs completely determine the automorphism orbits of normalized cuts, yet exhaustive representability is not preserved by those automorphisms (Theorem 47.2).

The results are complementary. The first determines the missing boundary points; the second studies dependence on the admitted classes; the third measures the amount of permutation support needed for interpolation; and the fourth distinguishes the full relation order from its set-like part. No one of these conclusions identifies the full collection of class well-orders with an ordinary surreal-sized class.

#### 4.8 Source 15: scope, notation, and the audit boundary

The useful starting point is to specify what the entries of the proposed order actually are. A well-order relation on a fixed set, an enumeration of that set, an isomorphism type, a relation table, and a naturally increasing subclass of  $\mathbf{No}$  are different kinds of data. Some determine one another, but their natural-looking lexicographic comparisons need not agree.

The ProveIt report *Lexicographic Orders of Well-Orderings of the Surreal Numbers* combines four earlier manuscripts into a 115-page document ([merge] at that revision). We inspected its source, guide, the related real-alphabet report, and the sign-sequence foundation at revision 0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8. The surreal report's source blob is f20837b76f8d981cb1dce3da79f633e280edad41; the same source was present at the earlier search-index revision 491ed8411cccf7bd57916eb7666bbba762e92f96. This article uses the newer revision as its audit boundary (this report; [1, 7]).

The prior report already establishes general set-sized cardinal and ordinal spectra, direct comparison of labelled class well-orders without abstract class ordinals, the global-choice equivalence, a dense bounded-support core isomorphic to  $\mathbf{No}$ , an omitted-range construction, and the singular minimal-slice dichotomy. We preserve those results as background. The main continuation consists of the block decomposition in Section 52.1, the cut classification in Section 46.2, and the longer

fixed-type calculation in [Section 31](#). The interpretation obstructions in [Section 38](#) explain why these results cannot simply be transferred to every other coding.

**Foundational convention.** GB means Gödel–Bernays set theory with Power Set and with no choice axiom built into the notation.  $\text{GBC} = \text{GB} + \text{GC}$ , where global choice provides a class choice function on nonempty sets. Set-sized results are in ZFC unless stated otherwise. A class well-order is a class strict linear order in which every nonempty available subclass has a least element. This explicit convention avoids silently confusing internal class well-foundedness with well-foundedness in an external metatheory. The standard equivalent formulations under global choice are discussed in [Section 57.4](#).

**Order notation.** An embedding preserves and reflects strict order.  $L \simeq_e M$  means that there are embeddings in both directions; it does not mean that  $L$  and  $M$  are isomorphic. The product  $L \times_{\text{lex}} M$  compares the  $L$  coordinate first. Ordinary ordinal multiplication keeps its usual convention:  $\alpha \cdot \beta$  is  $\beta$  consecutive copies of  $\alpha$ . We write  $\mathbb{B}_\theta = (2^\theta, <_{\text{lex}})$ , where  $0 < 1$ . The binary coding length  $\ell_2(L)$  is the least ordinal  $\theta$  for which  $L \hookrightarrow \mathbb{B}_\theta$ , when it exists. Birthday is written  $\text{bd}(x)$ .

**An essential size distinction.** Symbols such as  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  and  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  name predicates on class variables, with a comparison formula. They are not assertions that an ordinary GB class has other proper classes as its elements. A statement about a uniform family means one class relation whose rows are indexed by sets, possibly by a proper class of set indices. A larger universe may instead make the whole collection into an external set. The two readings are related but must be kept explicit.

#### 4.9 Source 16: main conclusions and relationship to the earlier report

There is a coherent and quite rich theory here, but it is essential to say which objects are being compared. A well-order on surreal labels need not agree with their numerical order. An ordinal-indexed enumeration of the whole surreal class is necessarily a class function. An arbitrary class well-order need not possess such an enumeration. Finally, even when each well-order is a legitimate class, a collection having those classes as its elements is not an ordinary class of the same two-sorted theory.

The ProveIt manuscript (this report, read at revision 491ed8411) already establishes the basic distinctions, exact ordinal embedding bounds for set alphabets, the surreal-sized bounded-support core, direct comparison of non-set-like orders, and a full-universe nonembedding result. Its merged source contains four earlier manuscripts and explicitly leaves several sharper questions. The present article develops those questions rather than treating the earlier conclusions as new. The principal additions are as follows.

1. **Topology at every enumeration length.** For an infinite linearly ordered set  $X$  of cardinality  $\mu$  and an ordinal  $\alpha$  with  $|\alpha| = \mu$ ,

$$\text{d}(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha \in \{\mu, \mu + 1, \mu + 2\}, \\ 2^\mu, & \text{otherwise.} \end{cases}$$

For the full order, both invariants equal  $2^\mu$ . Every infinite stratum, and the full order, has a countable witness to failure of conditional Dedekind completeness.

2. **Regular-stage gap spectrum.** If  $X$  is an  $\eta_\mu$  order of size  $\mu$ , for regular infinite  $\mu$ , the unfilled two-sided cuts of  $\mathcal{W}_\mu(X)$  have exactly the characters  $(\mu, \lambda)$  and  $(\lambda, \mu)$ , where  $\lambda$  ranges over infinite regular cardinals at most  $\mu$ .
3. **Represented cuts.** A lower cut of the bounded-support core represents an injective Ord-branch exactly when it is straddled by a prefix cylinder of every set ordinal length. The branch is exhaustive exactly when those prefixes collectively contain every label. A failed test has a first successor or limit obstruction; arbitrary non-set-like well-orders do not remove either obstruction.
4. **All ordinal-class blocks at once.** Every class well-order  $R$  admits the canonical decomposition

$$(X, R) \cong \left( \sum_{j \in J_R} \text{Ord} \right) + \alpha_R,$$

where the summands are precisely the equivalence classes for set-sized closed intervals, and  $\alpha_R$  is the optional final set block. The block index can itself be any class well-order. The construction is elementary in GB, and directly defines the entire family of residual tails.

5. **Word conventions and formal strength.** For the shorter-prefix-first convention, arbitrary, injective, and numerically increasing set-length words are isomorphic in GBC by maps preserving lengths and prefixes. Other placements of a termination symbol can instead make the whole unrestricted or injective word order isomorphic to **No**. For elementary relationally specified transformations, a uniform nonembedding theorem needs only  $\Sigma_1^1$  class comprehension over GBC, rather than all of KM.

These are extensions relative to the inspected report, not announcements that previously catalogued problems in the literature have been solved. The proofs below are intended to be independently readable. When a longer existing theorem is only summarized, its provenance and exact scope are stated. We separate mathematical proof, finite diagnostic checks, and formal proof throughout.

## 4.10 Source 17: relation to the repository and statement of the advance

### 4.10.1 Pinned source and scope

The main input is the merged ProveIt report *Lexicographic Orders of Well-Orderings of the Surreal Numbers* (this report), at repository revision

0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8.

The relevant path is

Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex.

Its blob identifier is f20837b76f8d981cb1dce3da79f633e280edad41. The accompanying README describes a union of four independent manuscripts, not four successive editions. It also records the unrefereed and unformalized status of that report. We use the merged text rather than silently selecting one of the earlier conventions.

The source already proves the basic set-alphabet embedding results, the direct comparison of unrestricted labelled class well-orders, bounded-support interpolation, the existence of a surreal-sized dense core, a diagonal no-coding theorem, and a singular-cardinal transition in binary coding length. We do not present these as new. In particular, binary equimorphism is not an order-isomorphism classification, and it does not by itself determine local characters or the cut spectrum.



#### 4.10.2 Questions addressed

The following correspondence uses the source’s stable LaTeX labels, so it does not depend on pagination.

| Source question or topic                           | Result here                                                                                         | Limitation                                                          |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <code>swo:n16.7</code> ; <code>swo:st:n14.3</code> | Exact point characters, cut spectrum, density, cellularity, weight, saturation of the minimal slice | Not an isomorphism classification; not every nonminimal type        |
| <code>swo:n16.1</code> ; source 09, direction 3    | Cylinder completeness and coverage characterize represented cuts                                    | Uses the labelled prefix structure, not just the bare core order    |
| <code>swo:n16.6</code> ; source 09, direction 6    | Optimal additional-support cardinal for prefix completion                                           | Not an optimal birthday bound for every word code                   |
| <code>swo:n16.4</code> ; source 09, direction 4    | Exact automorphism extension criterion and a nonextendible core automorphism                        | No classification of the entire automorphism group                  |
| <code>swo:n16.8</code> ; <code>swo:st:n14.8</code> | Same-set class extensions are detected by newly represented cuts                                    | Does not reconstruct the entire class realization up to isomorphism |
| <code>swo:st:n14.7</code>                          | Maximal finite adjacency blocks and density of their quotient                                       | No full quotient isomorphism classification                         |

These are extensions of specific research questions posed in the repository, not assertions that the questions have an established history as major open problems. Every result used from the source is either restated with a proof or clearly identified as background.

#### 4.11 Source 18: main results and the scope of the continuation

The current **ProveIt** report on surreal well-orders (this report, read at revision `83befe707`) merges four earlier manuscripts. It already distinguishes set and class interpretations, proves the set-sized ordinal spectrum, analyzes binary coding at singular birthdays, compares arbitrary class well-orders directly, characterizes adjacency, and constructs a dense bounded-support core in the set-like global order. Repeating these conclusions would not answer the request for further study. We use them as a starting point, prove the elementary tools needed below, and address the outstanding questions on cuts, block decompositions, termination conventions, and model dependence.

Here is the principal bounded result. Throughout the article,  $\kappa$  is an infinite initial cardinal and

$$\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}, \quad r = \text{cf } \kappa, \quad q = \text{cf } \mu.$$

Let  $P = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  consist of all bijections  $\mu \rightarrow \mathbf{No}_{<\kappa}$ , ordered by their first unequal values. We prove

$$|P| = 2^\mu, \quad d(P) = \text{cell}(P) = \mu^{<\mu}, \quad \text{cf}(P) = \text{ci}(P) = r,$$

and every point has left and right character  $q$ . The order fills all cuts specified by fewer than  $r$  points. Its proper Dedekind gaps have exactly the character pairs

$$\{(r, r), (q, q)\} \cup \{(\theta, r), (r, \theta) : \theta < \mu \text{ is an infinite regular cardinal}\}. \quad (\text{S18.1})$$

This is a gap spectrum, not an isomorphism classification. Orders with the same spectrum can still have different global arrangements of cuts. There are exactly  $2^\mu$  proper gaps of character  $(q, q)$ , and exactly  $\mu^{<\mu}$  of every other character in the displayed spectrum. Thus the Dedekind completion has cardinality  $2^\mu$ .

For every infinite ordered alphabet  $X$  of size  $\mu$ , a separate fixed-type result gives

$$d(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha = \mu, \mu + 1, \mu + 2, \\ 2^\mu, & \mu + 3 \leq \alpha < \mu^+. \end{cases} \quad (\text{S18.2})$$

The third terminal position is the first that supplies an interior point of a finite permutation fiber. This is an order-topological transition, and its numerical values need not differ when  $\mu^{<\mu} = 2^\mu$ .

On the class side, let  $R$  be any supplied well-order of a proper class. Declare two labels equivalent when the closed interval between them is a set. Every block has an actual least label, which represents it without making a class an element. In GB this gives

$$R \cong (B_R \times \text{Ord})_{\text{lex}} + \theta_R, \quad (\text{S18.3})$$

with all proper blocks of type  $\text{Ord}$  and at most one final set block of type  $\theta_R$ . The block index  $B_R$  can be any class well-order; it can even be isomorphic to  $R$ . Thus simultaneous decomposition is elementary, while repeated quotienting need not lower isomorphism type.

A class cut of the bounded-support core comes from a raw global well-order precisely when, at every ordinal length, one prefix cylinder meets both sides. These prefixes uniquely reconstruct an injective  $\text{Ord}$ -word by elementary comprehension. The cut has a set-like representative exactly when that word is onto. A cut that would require the first two entries to be the same surreal is represented by no raw well-order. Allowing arbitrary class well-orders therefore does not fill all class cuts, even though it fills the omitted-range branch cuts.

A further theorem concerns an external, countable model of the class theory. If the model is  $\omega$ -standard, the full admitted order  $\mathfrak{W}_{\text{all}}$  is a dense shuffle of finite chains of sizes

$$1, 2, 6, 24, 120, \dots$$

Its finite-condensation quotient is the rational order with every factorial size appearing densely as a color. The admitted set-like order  $\mathfrak{W}_{\text{sl}}$  is externally isomorphic to  $\mathbb{Q}$ , but it is not definable in the pure full order, even with finitely many parameters. These are external statements; they do not supply an internally coded embedding of all global orders into a class of sets.

The final new group of results concerns set-length words. Ordinary dictionary comparison puts a proper prefix first. Under GBC, arbitrary, injective, and strictly numerically increasing words with this convention are isomorphic as prefix trees and as linear orders. Centered termination over  $\mathbf{No} \setminus \{0\}$  instead makes arbitrary and injective words isomorphic to  $\mathbf{No}$ . Increasing words retain a precisely identified limit-prefix obstruction. Keeping the actual digit 0 immediately above a distinct terminator creates adjacent pairs. Hence “lexicographic” must include a termination rule.

#### 4.11.1 What is established here

| Topic               | Repository starting point                          | Advance in this article                                                                               |
|---------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Birthday cutoffs    | Coding rank and selected regular-stage results     | Exact density, cellularity, point characters, saturation, and proper-gap spectrum with multiplicities |
| Fixed ordinal types | Finite-tail geometry; longer strata proposed       | Exact density/cellularity transition at $\mu + 3$                                                     |
| Class blocks        | Initial set-like part; iteration strength proposed | Elementary simultaneous decomposition, full quotient spectrum, self-similarity                        |
| Class cuts          | Dense core and omitted-range cuts                  | Exact prefix criterion, elementary reconstruction, and first-failure classification                   |
| Countable models    | Internal/external distinction                      | Factorial-shuffle classification and pure-order nondefinability                                       |
| Word conventions    | Prefix-first words and their limit obstructions    | Tree isomorphisms; centered conventions and exact increasing-word obstruction                         |

“Advance” means an extension of the inspected repository text. We make no assertion that these formulations have never appeared elsewhere, and do not describe the repository’s proposed directions as a certified list of open problems in the published literature. The proofs below are independent of the repository’s claimed Lean theorems except where a formalization reuse point is explicitly discussed.

#### 4.12 Source 14: the precise Kanovei–Shelah connection

Kanovei and Shelah’s definable nonstandard-model construction uses a lexicographically ordered family of maps from the continuum cardinal to  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters; the maps may repeat values [16]. That index set is not literally the set of all well-orders of  $\mathbb{R}$ . Its relevance is methodological: a linear order on an entire family of presentations can organize a construction without selecting one privileged presentation. Importing that method into a surreal setting must still state the indexing sort, support restriction, and evaluation relation. The diagonal theorem prevents replacing all global enumerations by one exhaustive ordinary class of set presentations.

#### 4.13 Source 17: the Kanovei–Shelah connection, with its size boundary

The definable nonstandard-real construction is a useful motivation for ordering many presentations rather than selecting one preferred presentation. Its actual index should be stated accurately. In Section 1 of Kanovei–Shelah [16], the index consists of maps from the continuum cardinal into  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters; the maps need not be injective. The authors order this *set* of maps by first difference, using the binary lexicographic order of subsets of  $\mathbb{N}$ , and then use finite-support functions. It is therefore not literally the order of all well-order relations on  $\mathbb{R}$ .

The methodological analogy remains valuable: a specified linear order of codes can organize a construction without a definable choice of one special code. In the present setting,  $\mathcal{W}_\mu(X)$  is a legitimate set index, and a uniform subfamily of global enumerations can be indexed by an ordinary class of set codes. Neither observation makes the full higher-order predicate  $\mathfrak{W}_{\text{sl}}$  into a set index. Any proposed surreal analogue must state where its functions, supports, evaluation maps, and

ultrafilters live. The diagonal obstruction of [Theorem 42.16](#) rules out one tempting shortcut, while leaving many bounded or explicitly uniform constructions available.

[merge] The Kanovei–Shelah distinction is thus made by all ten sources, in the same terms ([section 4.4](#) and the note after it).

## 5 Surreal order facts and foundational conventions

### 5.1 Signs, birthdays, and set cuts

A surreal number is represented by a sign sequence

$$s : \alpha \longrightarrow \{-, +\}$$

of set-ordinal length  $\alpha$ , its birthday  $\text{bd}(s)$ . Numerical comparison at the first disagreement uses

$$- < \text{termination} < +.$$

This differs from dictionary order with every proper prefix smaller than every extension. A sign sequence can be on either side of an extension, depending on its next sign.

For an ordinal  $\theta$ , let

$$\mathbf{No}_{<\theta} = \bigcup_{\alpha < \theta} \{-, +\}^\alpha.$$

It is a set. The full  $\mathbf{No}$  is proper-class-sized, because all-plus sequences embed all ordinals. Every set of surreals has bounded birthdays by Replacement.

**Definition 5.1.** A class linear order  $L$  has the *set-cut property*, called  $\eta_{\text{Ord}}$  here, if for all sets  $A, B \subseteq L$  with  $A < B$  there is  $x$  with  $A < x < B$ . Empty sides are allowed. For a cardinal  $\kappa$ , the set-sized notation  $\eta_\kappa$  requires the same property when  $|A \cup B| < \kappa$ .

The surreal cut construction supplies this property for  $\mathbf{No}$  [[14](#), Section 2]. A separator can be chosen canonically as the unique simplest separator. This statement concerns *sets* of options; applying it to the whole class against the empty right side would be inconsistent.

**Lemma 5.2** (Deleting a set). *If  $S \subseteq \mathbf{No}$  is a set, then  $\mathbf{No} \setminus S$  has the set-cut property. Every nonempty open interval of  $\mathbf{No}$  is a proper class.*

*Proof.* Let  $A < B$  be sets in  $\mathbf{No} \setminus S$ . First choose  $x$  with  $A < x < B$ , and put

$$S_- = \{s \in S : s \leq x\}, \quad S_+ = \{s \in S : x < s\}.$$

Then  $A \cup S_- < B \cup S_+$ . A separator of this new set cut avoids both parts of  $S$ . The argument allows empty sides. If an interval  $(a, b)$  were a set  $S$ , applying the result to  $\{a\} < \{b\}$  in  $\mathbf{No} \setminus S$  would be impossible.  $\square$

Choosing both separators by simplicity makes the deletion argument uniform in the set parameters without global choice.

[merge] *Third route to the first clause of [lemma 5.2](#) (source 12, Lemma 6.2).* Source 12 avoids a forbidden set by using that open intervals of  $\mathbf{No}$  are proper classes.

**Lemma 5.3** (Avoiding a forbidden set). *If  $A < B$  are sets of surreals and  $D$  is a set, then some separator  $z$  of  $A, B$  lies outside  $D$ .*

*Proof.* Choose  $u$  between  $A$  and  $B$ , and then  $v$  between  $\{u\}$  and  $B$ . Every point of  $(u, v)$  separates  $A, B$ , and that interval is a proper class. It cannot be contained in  $D$ . With a fixed global well-order, choose the first acceptable point, making the operation uniform.  $\square$

[merge] Source 08 opens with its own toolkit, printed next. Its [proposition 5.4](#) restates `found:prop:proper` (Lean-proved in universe-relative form; [section 1.5](#)) and adds that neither the numerical order nor “birthday first” well-orders **No**, which source 12 also proves ([section 13.1](#)). Its [theorem 5.6](#) sharpens the birthday bound of the cut theorem to a separator of *exactly* any prescribed birthday above the option birthdays; this is proved in Lean as `exists_cut_separator_of_birthday_lt`, by the same complete-lattice argument ([lemma 5.5](#)). Its [corollary 5.7](#) contains the first clause of [lemma 5.2](#) with a prescribed birthday, and its [definition 5.8](#) is [definition 5.1](#).

## 5.2 Source 08: surreal order and size, the minimal toolkit

We take a surreal number to be a sign sequence

$$s : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \text{Ord}.$$

Its birthday is  $\text{bd}(s) = \alpha$ . To compare two sequences, extend them by a termination symbol 0 and compare at the first disagreement using  $- < 0 < +$ . This is the numerical order. The simplicity relation is proper initial-segment extension, and is not the numerical order.

**Proposition 5.4.** *Every birthday-bounded collection of surreals is a set, every set of surreals has bounded birthdays, and **No** is a proper class. Neither the numerical order nor “birthday first, then numerical order” is a well-order of **No**.*

*Proof.* For a fixed ordinal  $\lambda$ , the sign sequences of length below  $\lambda$  form the set  $\bigcup_{\alpha < \lambda} \{-, +\}^\alpha$ . Replacement bounds the birthdays of any given set. Conversely, the all-plus sequences give a distinct surreal for every ordinal, so a set containing all surreals would bound all ordinals. The negative integers already descend numerically. For the second claim, let  $s_n$  have birthday  $\omega$ , with  $n$  initial minus signs followed by plus signs forever. At coordinate  $n$  one has  $s_{n+1}(n) = - < + = s_n(n)$ , so  $s_0 > s_1 > s_2 > \dots$  within a single birthday level.  $\square$

### 5.2.1 A birthday-controlled separator theorem

We record a proof of the exact order-theoretic fact used later. It also explains the repository’s bounded-word implementation [8].

**Lemma 5.5.** *For every set ordinal  $\beta$ , the binary lexicographic order  $\mathbb{B}_\beta = (\{0, 1\}^\beta, <_{\text{lex}})$  is a complete lattice.*

*Proof.* The least and greatest elements are the constant 0 and 1 words. Given a set  $A$  of words, form  $w$  recursively. At coordinate  $\xi$ , put  $w(\xi) = 1$  exactly when some  $a \in A$  agrees with the already constructed prefix  $w \upharpoonright \xi$  and has  $a(\xi) = 1$ ; otherwise put 0. If some  $a \in A$  exceeded  $w$ , its first disagreement would contradict the rule for a 0 coordinate. If  $v < w$ , then at their first disagreement  $\xi$  we have  $v(\xi) = 0$ ,  $w(\xi) = 1$ , and a witness  $a \in A$  agreeing with  $w$  through that 1 coordinate. Thus  $v < a$  and  $v$  is not an upper bound. This proves that  $w = \sup A$ , including  $A = \emptyset$ . Infima follow by reversing all bits.  $\square$

**Theorem 5.6** (Birthday-controlled cut filling). *Let  $A, B$  be sets of surreals with  $A < B$ , meaning  $a < b$  for all  $a \in A$ ,  $b \in B$ . If  $\beta$  is an ordinal strictly greater than every birthday in  $A \cup B$ , then a surreal  $z$  of birthday exactly  $\beta$  satisfies  $A < z < B$ . Either side may be empty.*

*Proof.* For a shorter word  $x$ , let  $u_\beta(x)$  be the word of length  $\beta$  obtained by appending a plus to  $x$  and then filling all remaining coordinates with minuses. It is the least full-length word strictly above  $x$ . Dually, let  $v_\beta(x)$  be  $x$  followed by a minus and then pluses; it is the greatest full-length word strictly below  $x$ . These assertions follow by inspecting the first differing sign: an extension above  $x$  must choose plus when  $x$  terminates, and the least possible continuation then uses only minuses.

If  $x < y$  and both have length below  $\beta$ , then  $u_\beta(x) \leq v_\beta(y)$ . Indeed  $u_\beta(x) \leq y$  by the same first-disagreement check; equality is impossible because their birthdays differ. Applying the dual greatest-word property gives the claim. By lemma 5.5, full-length sign words have a supremum, and

$$z = \sup\{u_\beta(a) : a \in A\}$$

satisfies  $a < u_\beta(a) \leq z$  for every  $a \in A$  and  $z \leq v_\beta(b) < b$  for every  $b \in B$ . All words in this complete lattice have birthday exactly  $\beta$ . If  $A$  is empty, its supremum is the all-minus word. The vacuous case  $\beta = 0$  causes no problem.  $\square$

**Corollary 5.7** (Fresh separators). *If  $A < B$  are sets and  $S$  is a set of forbidden surreals, a separator  $z \notin S$  exists. It can be required to have any specified birthday strictly exceeding all birthdays in  $A \cup B \cup S$ .*

**Definition 5.8.** A class linear order is an  $\eta_{\text{Ord}}$ -order if every pair of sets  $A < B$  has a strict separator, including empty sides. For an infinite cardinal  $\kappa$ , an  $\eta_\kappa$ -order satisfies the same condition when  $|A \cup B| < \kappa$ .

The notation  $\eta_\kappa$  here uses a cardinal bound directly; it should not be confused with conventions in which the subscript is the index  $\alpha$  in  $\aleph_\alpha$ . By theorem 5.6, **No** is an  $\eta_{\text{Ord}}$ -order. Every interval is a proper class: if  $(a, b)$  were a set, a separator of  $\{a\} \cup (a, b)$  and  $\{b\}$  would be a new member of the same interval above all its members. No set is cofinal or coinital in **No**.

### 5.3 Which recursion principle is used?

Here GB has a ZF set part, elementary class comprehension, and class replacement; choice is stated separately. We use  $\text{GBC} = \text{GB} + \text{GC}$ , which includes set choice. Cardinality calculations and cardinal-valued cofinality statements throughout the article are read in ZFC, or in the set part of GBC. The explicitly definable sign code and the direct set-cut constructions retain their stated choice-free strength.

Ordinary recursion along Ord, with a set output at every stage and a rule using the set of earlier values and fixed class parameters, is available in GB. On each set ordinal perform the usual recursion with those parameters. Uniqueness makes the local solutions agree; elementary comprehension assembles their graph.

This is different from arbitrary elementary transfinite recursion, ETR, whose stages may themselves be classes. The latter supports iterated truth predicates and is stronger than GBC [22]. The prefix searches, partial permutations, and back-and-forth maps used here have set-sized stages.

[merge] The other sources fix their foundational conventions in their own sections, printed whole next. They agree with source 11's conventions, with one difference of base theory: when it analyses choice, source 12 works over  $\text{GB} + \text{AC}$  (section 1.3).



## 5.4 Source 12: the surreal carrier and the foundational conventions

### 5.4.1 Sign sequences

We use the sign-sequence presentation

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha.$$

Each individual sequence is a set. Its length  $\text{bd}(s) = \text{lh}(s)$  is its birthday. At the first difference, signs and termination are compared by

$$- < \text{end} < +.$$

This is the usual numerical order, not the ordinary “shorter comes first” dictionary convention. Simplicity is the separate relation of being a proper sign prefix. The standard surreal construction and its simplicity structure are treated by Gonshor and Ehrlich [12, 15].

We shall use the following standard feature, also obtainable directly from the construction: whenever  $A, B$  are *sets* of surreals with  $A < B$ , there is a surreal  $z$  with  $A < z < B$ . There is a unique such separator of minimum birthday. Moreover, if every member of  $A \cup B$  has birthday less than an ordinal  $\gamma$ , a separator can be constructed with birthday at most  $\gamma$ . This is interpolation, not Dedekind completeness: no least upper bound is being asserted.

Every set of surreals has a birthday bound, by Replacement. Every nonempty numerical interval contains a proper class of surreals. To see the latter without arithmetic, start with  $a < b$  and recursively choose  $x_\alpha$  between  $\{a\} \cup \{x_\xi : \xi < \alpha\}$  and  $\{b\}$ . The previous values form a set at each stage. Thus one obtains distinct points for all ordinals  $\alpha$ .

### 5.4.2 Well-orders and set-likeness

A class well-order on a class  $X$  is a strict linear class relation  $R$  for which every nonempty subclass has an  $R$ -least element. In GBC this agrees with the usual formulations using nonempty set subsets or absence of descending  $\omega$ -sequences; the relationship is discussed in [27, §3]. We use strict relations throughout.

It is *set-like* if

$$\text{Pred}_R(x) = \{y \in X : yRx\}$$

is a set for every  $x$ . A proper-class set-like well-order is isomorphic to  $\text{Ord}$ : map  $x$  to the ordinal type of its set of predecessors. The range is an initial segment of  $\text{Ord}$ , and cannot be a set ordinal when  $X$  is proper. This construction uses only set-sized order types.

Not all class well-orders are set-like. The ordered sums  $\text{Ord} + 1$ ,  $\text{Ord} + \text{Ord}$ , and  $\text{Ord} + \text{Ord} + 1$  are examples. Their notation means ordered sums of class orders, not new von Neumann ordinals.

### 5.4.3 Foundational levels

Our set-sized results are theorems of ZFC. Our global results are in GBC, meaning Gödel–Bernays set theory with global choice. When analyzing the necessity of choice we work over GB plus ordinary, set-level AC. The full collection of global well-orders is handled by formulas with class variables, not by pretending that a proper class is an element of another GBC class.

Kelley–Morse theory allows stronger comprehension, but its classes still have sets, not proper classes, as members. Moving from GBC to KM does not by itself create a powerclass object of all subclasses of  $\mathbf{No}$ . One can instead use a larger set-theoretic universe, a set model with a specified collection of classes, or an explicitly higher-order theory. These are distinct interpretations and will be kept distinct.

#### 5.4.4 Why an ordinal-length recursion here need not invoke ETR

A recursion on  $\text{Ord}$  that produces one *set* at each stage and uses the set of previous values is available by the ordinary recursion theorem with class parameters in GB. In contrast, ETR permits a genuinely class-valued recursive stage, represented by a class of pairs of a stage and an arbitrary set. Even a truth hierarchy brings stronger principles into play [27, §2]. Our forth constructions, block lengths, and canonical enumeration of a class with set-sized initial segments are set-valued recursions. We do not invoke a universal satisfaction predicate or unrestricted class-valued recursion.

[merge] Like source 12, source 09 works over  $\text{GB} + \text{AC}$  when it analyses choice. Its lemma 5.10 is a second route to the first clause of lemma 5.2: it adds the forbidden points inside the cut to the left side, where source 11 first splits the forbidden set at a separator. Its bound (1) is  $\text{hset:eq:cut-bound}$  of the birthday-cutoff report [4], and its definition 5.9 is definition 5.1.

### 5.5 Source 09: foundational conventions and the surreal order

#### 5.5.1 Sets, classes, and the meaning of a family

All set-sized results are in ZFC. Global results are in GBC, here meaning Gödel–Bernays class theory with global choice. Classes have sets as members. A proper class cannot be an element of another class. An expression such as “the set of class functions  $E_i$ ” is therefore replaced by a *uniform family*: a class relation

$$F \subseteq I \times \text{Ord} \times \mathbf{No}$$

whose  $i$ th section is the graph of  $E_i$ . The index  $I$  may be a set or a class of sets. If  $I$  is a set, this is a uniformly set-indexed family, not a literal set of proper classes.

This distinction does not prevent quantification over individual class variables. For example, “for every two class bijections  $E, F$ , there is a least ordinal where they differ” is a legitimate GBC assertion. What is unavailable is a class having all such bijections as members. Rangel and Mariano explicitly distinguish the set-subsets of a class from a nonexistent class of all its subclasses [17, §1.1.1].

Global choice will be used in its standard consequence that a proper class can be enumerated by  $\text{Ord}$  with set-sized initial segments. One way to obtain this consequence is to well-order each rank layer using a global choice function, concatenate the layers, and collapse the resulting set-like well-order. Restricting such an order to any proper subclass again gives order type  $\text{Ord}$ .

#### 5.5.2 Sign sequences and the cut property

Use the sign-sequence realization

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha.$$

Each displayed element is a set function. Its domain is its *birthday*, denoted  $\text{bd}(x)$ . Numerical order compares the first different sign using

$$- < \text{termination} < +.$$

It is different both from ordering by birthday and from the dictionary order on arbitrary surreal-valued words used later. The all-plus word of length  $\alpha$  represents the ordinal  $\alpha$ .

The standard simplicity/cut theorem says that, for sets  $A, B \subseteq \mathbf{No}$  with  $A < B$ , there is a unique simplest surreal strictly between them. Here  $A < B$  means  $a < b$  for every  $a \in A, b \in B$ , with empty sides allowed. We write this simplest separator as  $\{A \mid B\}$ . A useful birthday bound is

$$\text{bd}\{A \mid B\} \leq \sup\{\text{bd}(x) + 1 : x \in A \cup B\}. \quad (1)$$

These are standard sign-sequence facts [12]; the inspected repository also contains explicit small-cut and birthday-bound statements on its sign carrier [9].

In particular, every set of surreal numbers has strict upper and lower bounds. No set is cofinal or cointial in **No**, and no set is the whole of **No**. These assertions do not authorize a cut constructor with arbitrary proper-class option families.

**Definition 5.9.** A class linear order  $D$  has the *set-cut property* if every pair of sets  $A, B \subseteq D$  satisfying  $A < B$  has an element  $d \in D$  with  $A < d < B$ . Empty sides are allowed.

The surreal order has the set-cut property. In terminology used in the literature this is an absolute or class analogue of an  $\eta$ -order. We use the explicit definition rather than relying on potentially ambiguous subscripts such as  $\eta_{\text{Ord}}$ .

**Lemma 5.10** (Avoiding a prescribed set). *If  $D$  has the set-cut property and  $U \subseteq D$  is a set, then  $D \setminus U$  also has the set-cut property.*

*Proof.* Let  $A, B \subseteq D \setminus U$  be sets with  $A < B$ , and put

$$U_0 = \{u \in U : A < u < B\}.$$

Then  $A \cup U_0 < B$ . Choose  $x$  with  $A \cup U_0 < x < B$ . This  $x$  is in the original open cut. If it belonged to  $U$ , it would belong to  $U_0$ , giving  $x < x$ . Thus  $x \notin U$ .  $\square$

### 5.5.3 The precise recursion strength used here

We repeatedly construct a sequence of *sets* indexed by ordinals, where the next value is a uniquely specified set obtained from the set-sized previous history and fixed class parameters. This recursion is available in GBC: set recursion with class parameters constructs all set-length approximations, uniqueness makes them compatible, and elementary class comprehension joins them.

This is not the principle ETR, whose stages can themselves be proper classes. The latter is genuinely stronger than GBC and must not be silently invoked by saying “perform class recursion” [24, §2]. Every global recursion in the proofs below is explicitly of the former, set-valued kind.

[merge] Source 08’s foundational section, printed next except for its Theorem 8.2, which is recorded in [note 5.12](#). One sentence is corrected, as proposed by the review of the four archives in the repository’s Hilbert’s-tenth research tree (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_batch80_surreal.md`, commit `c338541e3`, with the patch `surreal_research1_wellfounded_scope.patch` there). Source 08 wrote: “For non-set-like relations, the model’s class collection can matter to the well-order assertion.” Under GBC, internal well-foundedness of a fixed class relation is equivalent to the absence of a set-coded descending  $\omega$ -sequence ([27, Section 3]; source 11 makes the same distinction in [section 20](#)), so two class expansions with the same sets agree about it for a relation they share; what can change is which relations are available, and internal well-foundedness differs from external well-foundedness. The printed sentence is the patch’s text. No statement of source 08 depends on the original sentence.

## 5.6 Source 08: class foundations and set-like well-orders

From this section onward, the default theory is GBC: Gödel–Bernays class set theory with Global Choice. Sets are the objects that may be members; classes are collections of sets. A class function maps set arguments to set values. Its image of a set is a set by class replacement. Elementary class comprehension allows formulas with class parameters but only set quantifiers in the defining formula.

### 5.6.1 Why Global Choice is named explicitly

Global Choice supplies a set-like global well-order of the universe. Restricting it to any class gives a set-like well-order, and every proper class can then be enumerated by  $\text{Ord}$ . We use this to choose one baseline  $b : \text{Ord} \rightarrow \mathbf{No}$ , to enumerate class carriers during back-and-forth, and to select witnesses uniformly. Ordinary set Choice alone does not justify these global choices.

Many local results need less. Pairwise lexicographic comparison, the extension of one specified set prefix relative to a given  $b$ , and the set-cut argument use only class replacement, elementary comprehension, set Choice, and the displayed class parameters. Keeping  $b$  as an explicit parameter is therefore useful in weaker foundations.

**Definition 5.11.** A class well-order  $\prec$  is *set-like* when, for every element  $x$ , its predecessor class  $\{y : y \prec x\}$  is a set. By a class well-order here we mean a linear class relation such that every nonempty subclass available in the model has a least element.

For a set-like relation, it suffices to require well-foundedness on sets: a nonempty subclass has a witness  $x$ , and one may search the set of predecessors of  $x$  together with  $x$  for its least member. Over GBC, well-foundedness of any fixed class relation is equivalent to the absence of a set-coded descending  $\omega$ -sequence. Keeping the sets and the relation fixed while enlarging the class collection therefore does not change its internal well-foundedness. What may change is which class relations are available; internal and external well-foundedness must also be distinguished. [merge] (This replaces one sentence of source 08; see the note before this section.)

*Note 5.12* (Source 08, Theorem 8.2). *Every set-like well-order of a proper class has a unique order isomorphism from  $\text{Ord}$ . Conversely, every class bijection  $e : \text{Ord} \rightarrow \mathbf{No}$  induces a set-like well-order of  $\mathbf{No}$ .* This is the last paragraph of the proof of [theorem 13.1](#) and source 09’s [proposition 13.6](#), proved by the same rank map: the predecessors of  $x$  form a well-ordered set of ordinal type  $r(x)$ ; the range of  $r$  is downward closed, and it cannot be a set ordinal, since the inverse of  $r$  would then map a set ordinal onto a proper class, contradicting class replacement. Conversely the predecessors of  $e(\alpha)$  form the set  $e[\alpha]$ .

Thus all set-like proper-class well-orders have the same abstract order type. The interesting lexicographic structure lies in their labels, not in distinct ordinal types. It is incorrect either to introduce a set ordinal equal to “the order type of all ordinals” or to quotient away the enumerations and expect the lexicographic information to survive.

### 5.6.2 The recursion principle actually used

All transfinite constructions below have *set-valued* stages and set-sized histories at each ordinal stage. In GBC, fix the class parameters and a rule assigning a unique set to each set history. The ordinary set recursion theorem with those parameters constructs a solution through each set ordinal. Uniqueness makes these initial solutions compatible, and elementary comprehension forms their union as a class. This is the set-valued recursion needed here.

It is not the principle of elementary transfinite recursion with *class-valued* stages. For example, constructing a new truth class at each stage is a substantially different task. The distinctions among GBC, GBC + ETR, and KM are genuine; the class-recursion and class forcing literature documents their strength [23, 25]. No equivalence with those stronger principles is claimed for the constructions in this paper.

## 5.7 Universality and uniqueness of the saturated class order

**Proposition 5.13.** *In ZFC, every set linear order embeds into  $\mathbf{No}$ . In GBC, every class linear order embeds uniformly into  $\mathbf{No}$ . Any two proper-class  $\eta_{\text{Ord}}$  orders are isomorphic in GBC.*

*Proof.* Enumerate the source in a cardinal-length well-order, or in type Ord for a proper class, using global choice. At stage  $\alpha$ , the earlier images form a set. Separate the images of earlier points below the new point from those above it, and choose a separator. This extends an order embedding. Each partial map is a set.

For uniqueness, globally enumerate both class orders and alternately insert the first unused point from each enumeration. The appropriate set cut in the other order is fillable. Global choice selects a separator when none is distinguished. The union of the partial isomorphisms is a surjective class order isomorphism.  $\square$

The same construction, started from a specified set-sized partial isomorphism, gives set homogeneity. These are the order-theoretic universality and homogeneity features of  $\mathbf{No}$  [15]. The assertion that every class order embeds is made with global choice; it is not used as an unproved converse to global choice.

[merge] *Source 12's version (its Lemmas 6.5 and 6.6).* It exposes the size of the forth recursion, with a birthday bound for set orders, and proves the class isomorphism  $\mathbf{No} \setminus A \cong \mathbf{No}$  that source 11 uses in [proposition 19.25](#).

**Lemma 5.14** (Surreal universality, with the size of the recursion exposed). *In GBC, every class linear order admits a class order-embedding into  $\mathbf{No}$ . For a set order enumerated in type  $\lambda$ , one can arrange that the image of its  $\xi$ th enumerated element has birthday at most  $\xi$ , for each  $\xi < \lambda$ .*

*Proof.* Use a global-choice enumeration of the carrier, either by an ordinal or by Ord. When the next element is considered, its previously embedded predecessors and successors form two sets  $A < B$ . Send it to the simplest surreal separator. It is distinct from all previous images and preserves order. Every stage has set-sized history. Induction gives the birthday bound, because the earlier birthdays are less than the current stage. This is the standard forth argument behind surreal universality [15, 20].  $\square$

**Lemma 5.15** (Deleting a set does not change the numerical class order). *In GBC,  $\mathbf{No} \setminus A$  is order-isomorphic to  $\mathbf{No}$  for every set  $A$ . Every nonempty open numerical interval is also order-isomorphic to  $\mathbf{No}$ .*

*Proof.* Each target has interpolation for arbitrary separated sets, by [lemma 5.3](#), adding interval bounds when needed. Each is a proper class and has an Ord-enumeration by global choice. Carry out a back-and-forth between the two enumerations. At each ordinal stage, extend the partial isomorphism to the next unused domain point and then to the next unused target point. The previous partial map is a set, and the required cut can be filled while avoiding its set-sized range. At the end the class map is onto and order-preserving.  $\square$

[merge] *Source 09's route.* Source 09 proves universality into any class order with the set-cut property, uniqueness together with set-homogeneity, and the isomorphisms  $\mathbf{No} \setminus U \cong \mathbf{No}$  and  $(a, b) \cong \mathbf{No}$ ; its proofs are the forth and back-and-forth recursions of [proposition 5.13](#), stated for a general target.

## 5.8 Source 09: the classical universality mechanism, with proofs

The following facts are classical features of the surreal order; we include the arguments because the same mechanism will identify our lexicographic core. The class-universal and set-homogeneous viewpoint appears, for example, in [17].

**Theorem 5.16** (Class universality). *In GBC, every class linear order embeds into  $\mathbf{No}$ . More generally, every class linear order embeds into any class linear order with the set-cut property.*

*Proof.* For a set-sized domain, choose a well-ordering and enumerate it by an ordinal. For a proper-class domain  $L$ , use global choice to fix a class bijection  $a : \text{Ord} \rightarrow L$ . At stage  $\alpha$ , assume the values  $h(a(\xi))$  for  $\xi < \alpha$  have been selected. Form the two sets

$$A_\alpha = \{h(a(\xi)) : \xi < \alpha, a(\xi) <_L a(\alpha)\}, \quad B_\alpha = \{h(a(\xi)) : \xi < \alpha, a(\alpha) <_L a(\xi)\}.$$

Earlier choices preserve order, so  $A_\alpha < B_\alpha$ . Choose a separator in the target, using its set-cut property and a fixed global well-order for a definite choice. The new value is different from every earlier value and has exactly the required comparisons. At each stage the partial map is a set. The resulting class function is the embedding.  $\square$

**Theorem 5.17** (Uniqueness and set-homogeneity). *In GBC, any two class linear orders with the set-cut property are isomorphic. Moreover, an order isomorphism between set-sized suborders of these orders extends to a class order isomorphism.*

*Proof.* Neither order is a set: applying the set-cut property with the entire carrier on the left would otherwise be contradictory. Fix Ord-length enumerations of both carriers. Start with the given set partial isomorphism. At each ordinal stage, first put the least not-yet-mapped element of the first enumeration into the domain. Its required image is constrained by two sets of already mapped elements; the set-cut property supplies an image. Next put the least omitted element of the second enumeration into the range by the symmetric construction. Take unions at limit stages. Every initial partial map is a set, so the recursion is of the permitted kind. The final map is total and onto.  $\square$

**Corollary 5.18.** *For every set  $U \subseteq \mathbf{No}$ ,  $\mathbf{No} \setminus U \cong \mathbf{No}$ . Every nonempty open interval of  $\mathbf{No}$  is likewise isomorphic to  $\mathbf{No}$ .*

*Proof.* Use lemma 5.10 and theorem 5.17. For an interval  $(a, b)$ , add  $a$  to every lower constraint set and  $b$  to every upper constraint set.  $\square$

These isomorphisms concern numerical order alone and need not preserve simplicity or arithmetic.

[merge] Source 08's Section 12 ("The skeleton is a surreal line") contains two further statements of the classical mechanism and the skeleton theorem (note 19.17).

*Note 5.19* (Source 08, Theorem 12.1, "Uniqueness of a class saturated line"). *In GBC, any two proper-class  $\eta_{\text{Ord}}$ -orders whose elements are sets are order-isomorphic.* This is the last clause of proposition 5.13 and theorem 5.17, by the same back-and-forth. Source 08 spells out that the separator chosen in the forward step cannot be a previous image, that a fixed set-like global well-order chooses each separator uniformly among sets, and that the least-unmatched rule handles every element: after enough stages all elements of a given bounded index have been included.



*Note 5.20* (Source 08, Theorem 12.3, “Universality for class linear orders”). *Every class linear order  $L$  with set-valued elements admits a class order embedding into  $\mathbf{No}$ , and hence into  $\mathcal{P}_s(b)$ .* This is the second clause of [proposition 5.13](#) and [theorem 5.16](#), composed with [note 19.17](#). Source 08 adds: “The input order need not be set-like. A point with a proper class of predecessors is harmless because only the previously *processed* elements, rather than all its predecessors, enter a stage of the construction. This explains why  $\text{Ord} + 1$  and more complicated non-set-like class well-orders can embed into the skeleton even though each *enumeration represented by* a skeleton point induces a set-like well-order on  $\mathbf{No}$ .”

## Part I

# Sets of surreals

## 6 Well-order codes on a fixed set alphabet

Let  $X$  be an infinite linearly ordered set, and put  $\lambda = |X|$ . We work in ZFC throughout this section. A well-order on the whole carrier  $X$  has a unique increasing enumeration  $f : \alpha \rightarrow X$ , where increasing refers to that well-order, not to the native order of  $X$ . Its domain satisfies  $|\alpha| = \lambda$ , and hence  $\alpha < \lambda^+$ . Write  $\mathcal{W}(X)$  for these bijective enumeration codes, and introduce two related sets:

$$\mathcal{I}(X) = \bigcup_{\alpha < \lambda^+} \{f : \alpha \hookrightarrow X\}, \quad \mathcal{W}_\lambda(X) = \{f : \lambda \rightarrow X : f \text{ is bijective}\}.$$

Thus  $\mathcal{I}(X)$  codes well-orders of arbitrary subsets of  $X$ , including the empty subset, whereas  $\mathcal{W}(X)$  always uses the whole carrier. All three sets inherit the lexicographic order defined earlier. A proper prefix is smaller than its extension in  $\mathcal{I}(X)$ . Two distinct members of  $\mathcal{W}(X)$  cannot be proper prefixes of one another: the shorter enumeration would already have used every element of  $X$ .

The repository’s real-well-ordering report already proves the exhaustive and cardinal-length statements for arbitrary infinite alphabets [1]. We record those results together with the injective-word variant, and include a common proof to keep the size distinctions explicit. Neither density nor completeness of the native order is needed.

**Theorem 6.1** (Cardinality and exact ordinal spectrum). *Each of  $\mathcal{W}_\lambda(X)$ ,  $\mathcal{W}(X)$ , and  $\mathcal{I}(X)$  has cardinality  $2^\lambda$ , contains a copy of every linear order of cardinality at most  $\lambda$ , and contains neither  $\lambda^+$  nor  $(\lambda^+)^*$ . Consequently the ordinals embeddable in any of them are exactly the ordinals  $\alpha < \lambda^+$ .*

*Proof.* There are at most  $\lambda^+$  permitted domains. For each domain there are at most  $\lambda^\lambda = 2^\lambda$  functions, so  $|\mathcal{I}(X)| \leq \lambda^+ \cdot 2^\lambda = 2^\lambda$ . For the reverse inequality, partition  $X$  into pairs  $\{a_\xi, b_\xi\}$ ,  $\xi < \lambda$ , with  $a_\xi < b_\xi$ . The lexicographic order on  $\lambda \times \{0, 1\}$ , taking  $\xi$  first, has order type  $2 \cdot \lambda = \lambda$ . For  $\varepsilon \in 2^\lambda$ , list the  $\xi$ -th pair as  $(a_\xi, b_\xi)$  if  $\varepsilon(\xi) = 0$ , and as  $(b_\xi, a_\xi)$  otherwise. The resulting map

$$2^\lambda \rightarrow \mathcal{W}_\lambda(X)$$

is an order embedding when  $2^\lambda$  has lexicographic order. This proves all three cardinality assertions.

Let  $L$  be any linear order of cardinality at most  $\lambda$ , and fix an injection  $\iota : L \hookrightarrow \lambda$ . Associate to  $t \in L$  the binary function  $c_t$  whose value at  $\iota(s)$  is 1 precisely when  $s < t$ , and whose remaining coordinates are 0. If  $t < u$ , then  $c_t \leq c_u$  coordinatewise, and the inequality is strict at  $\iota(t)$ . Their first difference therefore has the correct direction. The preceding pair construction embeds  $L$  into  $\mathcal{W}_\lambda(X)$ .

It remains to exclude longer monotone sequences. Suppose that  $(f_\xi)_{\xi < \lambda^+}$  is strictly increasing or strictly decreasing in  $\mathcal{I}(X)$ . We recursively construct eventual common prefixes  $p_\alpha$  of length  $\alpha$ , for  $\alpha < \lambda^+$ . At a limit stage, take the union of the earlier prefixes. Regularity of  $\lambda^+$  ensures that a tail of the original sequence extends that union. At a successor step, at most one member of this tail equals  $p_\alpha$ ; discard it and all earlier indices. The values at coordinate  $\alpha$  are now monotone. A monotone sequence indexed by a tail of  $\lambda^+$ , taking at most  $\lambda$  values, is eventually constant: the first occurrence indices of all its values have a supremum below  $\lambda^+$ . Append this eventual value to  $p_\alpha$ . Every constructed prefix is injective because it occurs in members of the original sequence. The union of the prefixes through  $\lambda^+$  would therefore be an injection  $\lambda^+ \hookrightarrow X$ , a contradiction. Finally, an ordinal at least  $\lambda^+$  contains  $\lambda^+$ , whereas every smaller ordinal has cardinality at most  $\lambda$ .  $\square$

The distinction between the models also affects their elementary order topology. The order  $\mathcal{W}_\lambda(X)$  is dense. Indeed, let  $f <_{\text{lex}} g$ , with first difference at  $\xi$ . If the tail of  $f$  after  $\xi$  has an ascent, interchanging the two entries produces a point strictly between  $f$  and  $g$ . If the tail of  $g$  has a descent, the corresponding interchange works in the other direction. If neither exists, the first remaining tail is strictly decreasing and the second strictly increasing. Their common set of entries, after removing the two entries at  $\xi$ , is infinite and would be both well-ordered and reverse-well-ordered. This is impossible.

By contrast,  $\mathcal{W}(X)$  has adjacent pairs. Choose  $a < b$  in  $X$ , enumerate  $X \setminus \{a, b\}$  as a prefix  $p$ , and compare  $p \frown \langle a, b \rangle$  with  $p \frown \langle b, a \rangle$ . Any intervening enumeration would have to share  $p$ , after which only these two orders remain. Thus a cardinal-length convention changes the resulting linear order even though it does not change its cardinality or ordinal embedding spectrum.

**Proposition 6.2** (Exactly where jumps occur). *Suppose  $f <_{\text{lex}} g$  in  $\mathcal{W}(X)$ , with common prefix  $p$  up to their first difference. Put  $Y = X \setminus \text{ran}(p)$ ,  $a = f(\text{dom}(p))$ , and  $b = g(\text{dom}(p))$ , so  $a < b$ . The pair is adjacent precisely when  $Y$  is finite,  $a, b$  are consecutive in its native order, the tail of  $f$  after  $a$  lists  $Y \setminus \{a\}$  decreasingly, and the tail of  $g$  after  $b$  lists  $Y \setminus \{b\}$  increasingly.*

*Proof.* If an unused element lies strictly between  $a$  and  $b$ , start an enumeration with that element after  $p$  and complete it; this lies between  $f$  and  $g$ . Likewise, an ascent after  $a$ , or a descent after  $b$ , can be interchanged to create an intervening enumeration. Adjacency thus forces the two remaining tails to be decreasing and increasing, respectively. Their common native suborder  $Y \setminus \{a, b\}$  is then both well-ordered and reverse-well-ordered. It must be finite: an infinite well-order contains its first  $\omega$  elements, a subset with no greatest element. This proves necessity. Conversely, an intervening enumeration must share  $p$ . There is no possible next entry strictly between  $a$  and  $b$ . Among enumerations beginning with  $a$ , the displayed decreasing continuation is the largest; among those beginning with  $b$ , the increasing continuation is the smallest. Neither branch admits an intervening point.  $\square$

This description also identifies the finite components generated by adjacency. Write the domain of an enumeration uniquely as  $\theta = \delta + n$ , with  $\delta$  a limit ordinal and  $n < \omega$ . Its component consists of the  $n!$  permutations of the final  $n$  entries, keeping the prefix of length  $\delta$  fixed. In particular, words of limit length have no immediate neighbor. A jump cannot cross from one such component to another, because its common-prefix remainder would be infinite.

Endpoints can be classified just as concretely. The least element of  $\mathcal{W}(X)$ , if one exists, must always choose the least unused native element: otherwise interchanging that entry with a later smaller one decreases the enumeration. Hence a least element exists exactly when the native order on  $X$  is a well-order, and is its increasing enumeration. The greatest-element statement is the reverse assertion. For  $\mathcal{W}_\lambda(X)$ , the same argument shows that a least element exists exactly when the native

order type is the initial ordinal  $\lambda$ , not merely an arbitrary ordinal of cardinality  $\lambda$ . For example, an alphabet of native type  $\lambda + 1$  has a least full enumeration, but no least  $\lambda$ -length permutation.

## 6.1 Pointers to the reals report

[merge] The theorems of this section hold for every infinite linearly ordered set and are theorems of the repository’s reals report, which proves them in that generality. Source 11 says so (section 3); its repository audit (11-raw-orders-repository\_audit.md) adds that their application to birthday fragments “should be attributed as applications of existing repository results, not marketed as new general theorems”. All four sources re-prove them: source 08 from batch-76 manuscript 02 (now the reals report’s base source), sources 09 and 12 deliberately, and source 11 to keep the size distinctions explicit. They are printed once, here, with these pointers; no novelty is claimed for any of them.

| Statement                                                                                                                                                        | Here                                                                       | Reals report                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Prefix-freeness of exhaustive enumerations                                                                                                                       | before theorem 6.1; lemma 6.3; note 6.11; note 6.18                        | lwo:lem:prefix-free                                                          |
| Cardinality $2^\lambda$ ; pair swaps embed $\mathbb{B}_\lambda$ in $\mathcal{W}_\lambda(X)$ and $\mathbb{B}_\theta$ in $\mathcal{W}(X)$ for $\theta < \lambda^+$ | theorem 6.1; lemma 6.6 and theorem 6.8; note 6.12; notes 6.19 and 6.21     | lwo:prop:cardinality,<br>lwo:lem:all-small-cubes                             |
| Every order of cardinality $\leq \lambda$ embeds (cut coding), even in every fixed-type slice                                                                    | theorem 6.1; lemma 6.5; note 6.12; note 6.20                               | lwo:lem:cut-code,<br>lwo:thm:ordinal-spectrum(a),<br>lwo:sp:thm:universality |
| No monotone $\lambda^+$ -sequence; ordinal spectrum $\beta < \lambda^+$                                                                                          | theorem 6.1; lemma 6.7; lemma 6.13 and theorem 6.14; notes 6.22 and 6.23   | lwo:lem:eventual,<br>lwo:thm:no-long,<br>lwo:thm:ordinal-spectrum            |
| Strict hierarchy of binary cubes                                                                                                                                 | lemma 6.4                                                                  | lwo:thm:cube-strict                                                          |
| Cube spectrum $\theta < \lambda^+$ and coding length $\ell_2(\mathcal{W}(X)) = \lambda^+$                                                                        | theorem 6.8; notes 6.23 and 6.24                                           | lwo:thm:full-length                                                          |
| Least and greatest enumerations                                                                                                                                  | paragraph after proposition 6.2; note 6.16                                 | lwo:lem:extrema                                                              |
| Adjacent pairs; finite blocks of $n!$ points                                                                                                                     | proposition 6.2 and the paragraph after it; note 6.17; notes 6.25 and 6.26 | lwo:thm:adjacency,<br>lwo:thm:blocks                                         |
| Absence of $\lambda^+$ is not a sufficient embeddability criterion                                                                                               | remark 6.9; remark 6.15                                                    | the reals report’s non-claims                                                |

The arguments are the reals report’s: pair swaps, characteristic-function cut codes, stabilization of a common prefix along a long monotone sequence, and cylinder fusion for strict cubes (the reals report credits the strict cube hierarchy to Corollary cor:strict of the point-separating game report and prints its fusion proof as a second route). Two of the sources’ routes are printed in full: source 12’s Section 9, because the singular transition of section 8 uses its lemmas, and source 09’s stabilization lemma for a regular cardinal (lemma 6.13). Source 08’s statements are recorded in notes (section 6.4).

## 6.2 Source 12’s route: a general set-sized spectrum theorem

We temporarily forget the surreal structure. Let  $X$  be any infinite linearly ordered set, and write  $|X| = \mu$ , an initial cardinal. The spectra below depend only on the cardinality, not on density,

separability, or field operations of  $X$ .

**Lemma 6.3** (Prefix-freeness of exhaustive enumerations). *Two distinct exhaustive injective ordinal enumerations of  $X$  are never proper prefixes of one another. Hence first-difference lexicography is unambiguous on  $\mathcal{W}(X)$ .*

*Proof.* An exhaustive prefix has already used every element of  $X$ . An injective proper extension would have to use a new element, which is impossible.  $\square$

For any ordinal  $\gamma$ , let  $\mathbb{B}_\gamma = (2^\gamma, <_{\text{lex}})$ . For linear orders  $L, M$ , write  $L \hookrightarrow M$  for an order-embedding and  $L \simeq_e M$  for embeddings in both directions. Define

$$\ell_2(L) = \min\{\gamma : L \hookrightarrow \mathbb{B}_\gamma\}$$

when such a binary coding ordinal exists. This is an ordinal-valued rank, not a cardinal invariant.

**Lemma 6.4** (Strict binary hierarchy). *For ordinals  $\alpha, \beta$ ,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

*In particular  $\ell_2(\mathbb{B}_\alpha) = \alpha$ .*

*Proof.* Zero-padding proves the forward construction when  $\alpha \leq \beta$ . It suffices to rule out an embedding  $f : \mathbb{B}_{\beta+1} \rightarrow \mathbb{B}_\beta$ .

Recursively construct an input prefix  $s_\xi$  of length  $\xi$  and an output prefix  $t_\xi$  of length  $\xi$ , so that every input extending  $s_\xi$  is sent to an output extending  $t_\xi$ . At a successor stage, split the input cylinder into its two next-bit halves and split the output cylinder likewise. The two input halves have strictly ordered images. They cannot both have images meeting both output halves: an image point in the upper output half from the lower input half would exceed an image point in the lower output half from the upper input half. Thus one input half has image contained in a single output half. Choose those halves to continue.

At a limit take unions; the corresponding cylinders are nonempty and the containment persists. At stage  $\beta$ , two inputs remain, differing in their last bit, but their images have the same full  $\beta$ -long output. This contradicts injectivity. For  $\alpha > \beta$ , padding embeds  $\mathbb{B}_{\beta+1}$  into  $\mathbb{B}_\alpha$ , giving the general impossibility.  $\square$

**Lemma 6.5** (Characteristic-cut coding). *Every linear order of cardinality at most  $\mu$  embeds in  $\mathbb{B}_\mu$ .*

*Proof.* Enumerate its carrier by distinct coordinates  $d_\xi$ , padding unused coordinates if necessary. Send  $x$  to the bits  $\mathbf{1}[d_\xi < x]$ . For  $x < y$ , every differing coordinate changes from 0 to 1, and the coordinate representing  $x$  does change. Thus the least difference gives the correct order.  $\square$

**Lemma 6.6** (Pair orientation coding).  $\mathbb{B}_\mu \hookrightarrow \mathcal{W}_\mu(X)$ . *More generally,  $\mathbb{B}_\gamma \hookrightarrow \mathcal{W}(X)$  for every  $\gamma < \mu^+$ .*

*Proof.* Partition  $X$  into  $\mu$  unordered pairs, sort each pair by  $X$ 's order, and orient the pair according to its bit. The ordinal sum of  $\mu$  two-element blocks has type  $2 \cdot \mu = \mu$ .

For a general  $\gamma < \mu^+$ , choose  $|\gamma|$  disjoint pairs and retain a disjoint reserve of size  $\mu$ . Such a choice is possible even when  $|\gamma| = \mu$ . List the pairs in type  $\gamma$ , then well-order all remaining labels in type  $\mu$ . The resulting enumeration has length  $2 \cdot \gamma + \mu < \mu^+$ . Its comparison is decided by the first oriented pair at which the inputs differ.  $\square$

**Lemma 6.7** (No  $\mu^+$ -long monotone chain). *Neither  $\mu^+$  nor its reverse embeds into  $\mathcal{W}(X)$ .*

*Proof.* Suppose  $(e_i)_{i < \mu^+}$  is strictly increasing. We recursively stabilize its entries on final segments of the index set. Assume that a final segment agrees on a prefix  $p$  of length  $\delta < \mu^+$ . No enumeration in a further infinite final segment can end there, by [lemma 6.3](#). Their  $\delta$ th entries form a nondecreasing  $\mu^+$ -sequence in a set of size  $\mu$ .

Such a sequence is eventually constant: choose one first occurrence for each value in its range; their at most  $\mu$  indices have supremum below  $\mu^+$  by regularity. Past that supremum no larger value can first appear. At a limit stage  $\delta < \mu^+$ , take the supremum of the earlier stabilization indices, still below  $\mu^+$ , and the union of the earlier prefixes.

This would produce an injective sequence of labels from  $X$  of length  $\mu^+$ : every set-sized earlier part is shared by a final segment of injective enumerations. That contradicts  $|X| = \mu$ . Replace nondecreasing by nonincreasing for the reverse order.  $\square$

**Theorem 6.8** (Cardinality and exact spectra). *For every infinite linearly ordered set  $X$  of cardinality  $\mu$ ,*

$$\begin{aligned} |\mathcal{W}(X)| &= |\mathcal{W}_\mu(X)| = 2^\mu, \\ \beta &\hookrightarrow \mathcal{W}(X) \iff \beta < \mu^+, \\ \beta &\hookrightarrow \mathcal{W}_\mu(X) \iff \beta < \mu^+, \\ \mathbb{B}_\gamma &\hookrightarrow \mathcal{W}(X) \iff \gamma < \mu^+, \\ \ell_2(\mathcal{W}(X)) &= \mu^+. \end{aligned}$$

*The ordinal statements also hold for reversed ordinals, and every linear order of cardinality at most  $\mu$  embeds even into  $\mathcal{W}_\mu(X)$ .*

*Proof.* Pair coding gives  $2^\mu$  elements, while all well-orders are relations on  $X$ , giving the upper bound  $2^{|X \times X|} = 2^\mu$ . The lower ordinal bounds and universality follow from [lemmas 6.5](#) and [6.6](#); [lemma 6.7](#) gives the upper bounds. The same characteristic coding applies to reversed ordinals.

For the cube spectrum use [lemma 6.6](#). If  $\gamma \geq \mu^+$ ,  $\mathbb{B}_\gamma$  contains an increasing chain of type  $\mu^+$ : use the sequences that are 1 before a moving cut and 0 afterward. Thus it cannot embed into  $\mathcal{W}(X)$ .

Embed  $X$  into  $\mathbb{B}_\mu$  using [lemma 6.5](#). Encode an exhaustive enumeration of length  $\alpha < \mu^+$  by concatenating its  $\mu$ -long entry codes, then pad with zeroes to length  $\mu^+$ . The unpadded length  $\mu \cdot \alpha$  has cardinality at most  $\mu$ , so is less than  $\mu^+$ . Distinct enumerations disagree before either ends, and their encoded first disagreement preserves the order. This proves  $\mathcal{W}(X) \hookrightarrow \mathbb{B}_{\mu^+}$ .

If it embedded in  $\mathbb{B}_\rho$  for  $\rho < \mu^+$ , then the copy of  $\mathbb{B}_{\rho+1}$  supplied by [lemma 6.6](#) would contradict [lemma 6.4](#). Hence the rank is exactly  $\mu^+$ .  $\square$

*Remark 6.9* (What the rank does not say). Although  $\ell_2(\mathcal{W}(X)) = \mu^+$ , the cube  $\mathbb{B}_{\mu^+}$  does *not* embed back into  $\mathcal{W}(X)$ . An order can first admit a binary encoding at an ordinal without containing the full cube of that ordinal. Nor is the absence of  $\mu^+$  and its reverse claimed sufficient for an arbitrary order to embed in  $\mathcal{W}(X)$ .

### 6.3 Source 09's route

[merge] Source 09's Section 4 works for any linearly ordered set. Its first statements are recorded in notes; its stabilization lemma and its proof of the ordinal spectrum are a second route and are printed in full.

*Note 6.10* (Source 09, Definition 4.1).  $\mathcal{W}_\alpha(X) = \text{Bij}(\alpha, X)$  and  $\mathcal{W}(X)$  is the union over all ordinals  $\alpha$ ; if  $|X| = \kappa$  is infinite, the possible lengths are exactly  $\kappa \leq \alpha < \kappa^+$ . This is the  $\mathcal{W}(X)$  of [section 6](#) and of the reals report.

*Note 6.11* (Source 09, Lemma 4.2, “Exhaustiveness removes the prefix ambiguity”). *Distinct members of  $\mathcal{W}(X)$  have a first unequal coordinate strictly below both of their lengths, so  $e <_{\text{lex}} f$  at that coordinate is a strict linear order (for transitivity compare the two first-disagreement ordinals of  $e < f$  and  $f < g$ ). This is `lwo:lem:prefix-free` and [lemma 6.3](#), by the same argument.*

*Note 6.12* (Source 09, Theorem 4.3, “Cardinality and universality”). *For every infinite linearly ordered set  $X$  of cardinality  $\kappa$ ,  $|\mathcal{W}_\kappa(X)| = |\mathcal{W}(X)| = 2^\kappa$ ; the lexicographic cube  $2^\kappa$  embeds into  $\mathcal{W}_\kappa(X)$ ; consequently every linear order of size at most  $\kappa$  embeds into  $\mathcal{W}_\kappa(X)$  and hence into  $\mathcal{W}(X)$ . This is [theorem 6.1](#), `lwo:prop:cardinality` and `lwo:thm:ordinal-spectrum(a)`, with the same pair-orientation and characteristic-function codes. Source 09 stresses that  $2 \cdot \kappa = \kappa$  is ordinal multiplication.*

## Second route (source 09): eventual stabilization at a regular cardinal.

**Lemma 6.13** (Eventual stabilization). *Let  $\mu$  be a regular infinite cardinal and let  $X$  have cardinality less than  $\mu$ . Every nondecreasing map  $f : \mu \rightarrow X$  is eventually constant.*

*Proof.* Some fiber must be unbounded: otherwise the union of fewer than  $\mu$  bounded fibers would be bounded. Once a value in such a fiber is reached, monotonicity and the unboundedness of the fiber prevent any subsequent strict increase.  $\square$

**Theorem 6.14** (Exact ordinal spectrum). *For  $|X| = \kappa$  infinite and any ordinal  $\beta$ ,*

$$\beta \hookrightarrow \mathcal{W}(X) \iff \beta < \kappa^+.$$

*The same equivalence holds for  $\beta^*$ , the reverse order of  $\beta$ .*

*Proof.* The positive direction follows from [note 6.12](#). For the negative direction put  $\mu = \kappa^+$  and suppose that  $(e_\xi)_{\xi < \mu}$  is strictly increasing in  $\mathcal{W}(X)$ . We construct an injective common eventual prefix of length  $\mu$ , which is impossible in the alphabet  $X$ .

Assume a prefix  $p$  of length  $\delta < \mu$  is shared by all  $e_\xi$  beyond some index  $t_\delta < \mu$ . At limits this holds by taking the supremum of the earlier cutoffs; regularity ensures that the new cutoff remains below  $\mu$ . If  $p$  exhausts  $X$ , every enumeration on this tail is exactly  $p$ , contrary to strict increase. Otherwise each tail enumeration has an entry at coordinate  $\delta$ . As the words share all earlier coordinates, these entries are nondecreasing in  $\xi$ . By [lemma 6.13](#) they eventually have one common value. Add it to  $p$ . Injectivity of the enumerations makes this a new value, not used earlier in  $p$ .

Continuing for every  $\delta < \mu$  gives  $\mu$  distinct elements of  $X$ , a contradiction. An embedding of any  $\beta \geq \mu$  would restrict to one of  $\mu$ . Reversing the alphabet proves the negative statement for  $\mu^*$  and its longer extensions.  $\square$

*Remark 6.15.* The absence of  $\kappa^+$  and its reverse is not asserted to characterize all linear orders embeddable in  $\mathcal{W}(X)$ . Indeed  $\mathcal{W}(X)$  already contains the binary cube of cardinality  $2^\kappa$ , so “size at most  $\kappa$ ” is a universality guarantee, not a necessary cardinality condition for an individual order to embed.

*Note 6.16* (Source 09, Proposition 4.7).  *$\mathcal{W}(X)$  has a least element exactly when the alphabet order on  $X$  is a well-order, and a greatest element exactly when its reverse is.* This is the paragraph on endpoints after [proposition 6.2](#) and `lwo:lem:extrema`, by the same argument (a least enumeration must take the least unused element at every position).



*Note 6.17* (Source 09, Proposition 4.8, “Finite residuals are exactly the adjacency mechanism”). *If  $e < f$  in  $\mathcal{W}(X)$  have common prefix  $p$  before their first disagreement and  $R = X \setminus \text{ran}(p)$ , they are adjacent precisely when  $R$  is finite and their remaining words are consecutive permutations of  $R$  in finite lexicographic order.* This is [proposition 6.2](#) and `lwo:thm:adjacency`, by the same argument through [note 6.16](#).

Even for a dense alphabet  $X$ , the full space  $\mathcal{W}(X)$  can therefore have adjacent points. This finite-tail phenomenon disappears for global enumerations: deleting a set prefix from **No** never leaves a finite remainder.

## 6.4 Source 08’s set-sized sections

[merge] Source 08’s Sections 3 (“All well-orderings of a fixed set”), 4 (“The exact ordinal and binary-cube spectra”) and 5 (“Adjacent pairs and finite tails”) re-prove the reals report’s general theorems by the same arguments. They are recorded here with source 08’s own remarks. Throughout,  $(X, <)$  is an infinite linearly ordered set of cardinality  $\kappa$ , its initial ordinal.

*Note 6.18* (Source 08, display (3) and Lemma 3.1, “No prefix ambiguity”). Source 08 defines

$$\mathcal{W}_\alpha(X) = \{e : \alpha \rightarrow X : e \text{ is a bijection}\}, \quad \mathcal{W}(X) = \bigcup_{\kappa \leq \alpha < \kappa^+} \mathcal{W}_\alpha(X) \quad (\text{S08.3})$$

and proves: *two distinct members  $e, f$  of  $\mathcal{W}(X)$  disagree before either one ends; consequently*

$$e <_{\text{lex}} f \iff e(\delta) < f(\delta), \quad \delta = \min\{\xi < \min(\text{dom } e, \text{dom } f) : e(\xi) \neq f(\xi)\} \quad (\text{S08.4})$$

*defines a linear order without an end-of-word convention.* This is `lwo:lem:prefix-free`. Source 08 adds: “Sorting by length *before* reading entries gives a different order. For instance, choosing one word of each possible length would produce a chain of type  $\kappa^+$  in that alternative.” [Note 6.22](#) shows this is impossible in [\(S08.4\)](#).

*Note 6.19* (Source 08, Lemma 3.2, “Pair coding”). *For every ordinal  $\theta < \kappa^+$  there is an order embedding  $\mathbb{B}_\theta \hookrightarrow \mathcal{W}(X)$ ; in particular  $\mathbb{B}_\kappa \hookrightarrow \mathcal{W}_\kappa(X)$ .* This is `lwo:lem:all-small-cubes`, `lwo:prop:cardinality` and [lemma 6.6](#), with the same blocks  $(a_\xi, c_\xi)$  and  $(c_\xi, a_\xi)$  followed by a fixed enumeration of the unused elements.

*Note 6.20* (Source 08, Lemma 3.3). *Every linear order  $L$  of cardinality at most  $\kappa$  embeds into  $\mathbb{B}_\kappa$ , by the bits  $\mathbf{1}[d < x]$  at distinct coordinates.* This is `lwo:lem:cut-code` and [lemma 6.5](#).

*Note 6.21* (Source 08, Theorem 3.4).  *$\mathcal{W}(X)$  has cardinality  $2^\kappa$  and contains every linear order of cardinality at most  $\kappa$ ; the embedding can already be made into  $\mathcal{W}_\kappa(X)$ .* This is `lwo:prop:cardinality` and `lwo:thm:ordinal-spectrum(a)`. Source 08 notes that neither the continuum hypothesis nor any regularity of  $\kappa$  occurs, so the theorem applies to arbitrary birthday cutoffs and to numerically scattered as well as dense sets of surreals.

*Note 6.22* (Source 08, Theorem 4.1, “Prefix stabilization”). *There is no strictly increasing or strictly decreasing sequence of length  $\kappa^+$  in  $\mathcal{W}(X)$ .* This is `lwo:thm:no-long`; the proof is the stabilization of [lemma 6.13](#) and [theorem 6.14](#): a monotone  $\kappa^+$ -sequence with at most  $\kappa$  values has an unbounded fibre in the regular cardinal  $\kappa^+$  and is constant from its first occurrence on.

*Note 6.23* (Source 08, Corollary 4.2, “Exact spectra”). *For every ordinal  $\beta$  and set ordinal  $\theta$ :  $\beta \hookrightarrow \mathcal{W}(X)$ ,  $\beta^{\text{op}} \hookrightarrow \mathcal{W}(X)$  and  $\mathbb{B}_\theta \hookrightarrow \mathcal{W}(X)$  each hold exactly when the ordinal is below  $\kappa^+$ .* This is `lwo:thm:ordinal-spectrum` and the cube clause of `lwo:thm:full-length`, as in [theorem 6.8](#); the chain  $f_\eta = \mathbf{1}[\xi < \eta]$  of type  $\kappa^+$  in  $\mathbb{B}_{\kappa^+}$  excludes the larger cubes.

*Note 6.24* (Source 08, Proposition 4.3, “A uniform binary upper coding”). *There is an order embedding  $\mathcal{W}(X) \hookrightarrow \mathbb{B}_{\kappa^+}$ , by concatenating cut codes of the entries and padding with zeros. This is the upper bound of `lwo:thm:full-length`, as in [theorem 6.8](#). Source 08’s remark: these results locate  $\mathcal{W}(X)$  between a long hierarchy of binary cubes without identifying it with one; the absence of a  $\kappa^+$ -chain is compatible with cardinality  $2^\kappa$  and does not mean that all suborders are small.*

*Note 6.25* (Source 08, Theorem 5.1, “Adjacency classification”). *Let  $e < f$  in  $\mathcal{W}(X)$  first disagree at  $\delta$ , with common prefix  $p$ , remaining alphabet  $Y = X \setminus \text{ran } p$ ,  $a = e(\delta)$  and  $c = f(\delta)$ . They are adjacent exactly when (i)  $Y$  is finite, (ii)  $a < c$  are consecutive in  $Y$ , and (iii) after  $a$  the word  $e$  lists  $Y \setminus \{a\}$  decreasingly and after  $c$  the word  $f$  lists  $Y \setminus \{c\}$  increasingly. This is [proposition 6.2](#) and `lwo:thm:adjacency`, by the same argument.*

*Note 6.26* (Source 08, Corollary 5.2, “Finite condensation blocks”). *If the length of  $e$  is  $\lambda + n$  with  $\lambda$  zero or a limit and  $n < \omega$ , its finite-condensation block consists of the words with the same prefix of length  $\lambda$  and all  $n!$  permutations of the remaining  $n$  values; a limit-length word has a singleton block. This is the paragraph after [proposition 6.2](#) and `lwo:thm:blocks`. Source 08 adds: “The phrase ‘singleton block’ does not assert that the point is isolated: it means that it has no finite chain of jumps to another point.”*

## 7 Birthday cutoffs and their set-sized orders

Use the sign presentation of surreal numbers:  $x \in \mathbf{No}$  is a function from an ordinal  $\text{bd}(x)$  to  $\{-, +\}$ . At the first difference, the comparison is  $- < \text{termination} < +$ . This presentation and the fundamental set-cut theorem are treated in Gonshor and in the modern account of Bagayoko and van der Hoeven [12, 14]. For an infinite cardinal  $\kappa$ , define

$$\mathbf{No}_{<\kappa} = \{x \in \mathbf{No} : \text{bd}(x) < \kappa\}.$$

This is a set. Counting sign sequences gives

$$|\mathbf{No}_{<\kappa}| = \sum_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}.$$

Here  $2^{<\kappa}$  denotes the supremum of  $2^\mu$  over cardinals  $\mu < \kappa$ ; it is at least  $\kappa$ .

**Corollary 7.1.** *Put  $\lambda = 2^{<\kappa}$ . Then  $\mathcal{W}(\mathbf{No}_{<\kappa})$  has cardinality  $2^\lambda$ , is universal for linear orders of cardinality at most  $\lambda$ , and admits exactly the ordinals below  $\lambda^+$ . The same conclusions hold for its cardinal-length permutation suborder and for the injective-word order  $\mathcal{I}(\mathbf{No}_{<\kappa})$ .*

The parameter controlling this spectrum is the cardinality  $\lambda$  of the alphabet. It need not equal the birthday cutoff  $\kappa$ . For  $\kappa = \omega$ , the alphabet consists of the dyadic rationals. Its well-ordering order has cardinality  $2^{\aleph_0}$ , contains every countable linear order, and does not contain  $\omega_1$ . For  $\kappa = \omega_1$ , the alphabet consists of the surreals with countable sign expansions and has cardinality  $\mathfrak{c} = 2^{\aleph_0}$ . Its well-ordering order consequently has cardinality  $2^\mathfrak{c}$  and ordinal spectrum  $\alpha < \mathfrak{c}^+$ , without any use of the continuum hypothesis. When  $2^{<\kappa} = \kappa$ , the spectrum simplifies to  $\alpha < \kappa^+$ .

**Lemma 7.2.** *If  $\kappa$  is regular infinite,  $\mathbf{No}_{<\kappa}$  fills every cut  $A < B$  with  $|A \cup B| < \kappa$ . Such a cut can still be filled after excluding any prescribed subset of cardinality less than  $\kappa$ .*

*Proof.* The birthdays of the members of  $A \cup B$  have supremum below  $\kappa$ , by regularity. The simplest surreal between them has birthday at most the supremum of their birthdays plus one, and hence still belongs to  $\mathbf{No}_{<\kappa}$ . For the exclusion statement, let  $S$  be the forbidden set, and let  $S_0$  consist of its members that already lie strictly between  $A$  and  $B$ . Fill the cut  $A \cup S_0 < B$ . Its filler lies in the original interval but strictly above every forbidden point in that interval.  $\square$

We call this property  $\eta_\kappa$ , always allowing either side of a cut to be empty. The cardinal parameter in this notation is explicit; conventions indexing eta orders by ordinal subscripts vary in the literature. This elementary form of saturation is all that the stage arguments below require.

Regularity is necessary for this particular cutoff conclusion. If  $\kappa$  is singular, a cofinal sequence of ordinals below  $\kappa$ , of length  $\text{cf}(\kappa) < \kappa$ , is unbounded in  $\mathbf{No}_{<\kappa}$ . Indeed, a surreal of birthday  $\beta$  is at most the ordinal  $\beta$ , the all-positive sign expansion of that length. Every candidate upper bound is therefore exceeded by some member of the cofinal sequence. The cut consisting of this sequence on the left and the empty set on the right has no filler in  $\mathbf{No}_{<\kappa}$ . This failure concerns the bounded stage; it does not weaken the set-cut theorem for the full surreal class.

[merge] Source 09's Section 5 adds the stage sizes for every ordinal  $\theta$ , not only for cardinals, the density and cofinality of  $\mathbf{No}_{<\theta}$ , and a table. The cardinal clause of its [proposition 7.3](#) is source 11's count above, and (2) is [corollary 7.1](#).

## 7.1 Source 09: birthday-bounded versions of the problem

Set

$$\mathbf{No}_{<\theta} = \{x \in \mathbf{No} : \text{bd}(x) < \theta\}, \quad \kappa_\theta = |\mathbf{No}_{<\theta}|.$$

Each  $\mathbf{No}_{<\theta}$  is a set, and every set of surreal numbers is contained in one such  $\mathbf{No}_{<\theta}$ . The latter assertion follows by taking the supremum of successor birthdays.

**Proposition 7.3.** *For every ordinal  $\theta$ ,*

$$|\mathbf{No}_{<\theta}| = \sum_{\alpha < \theta} 2^{|\alpha|}.$$

*In particular, for an infinite cardinal  $\tau$ ,  $|\mathbf{No}_{<\tau}| = 2^{<\tau} := \sup_{\lambda < \tau} 2^\lambda$ . If  $\theta$  is an infinite limit ordinal,  $\mathbf{No}_{<\theta}$  is dense without endpoints and has cofinality and coinitality  $\text{cf}(\theta)$ .*

*Proof.* There are exactly  $2^{|\alpha|}$  sign words of length  $\alpha$ , and words of different lengths are distinct. For an infinite cardinal  $\tau$ , the supremum of these cardinalities is at least  $\tau$ , so multiplying it by  $\tau$  does not enlarge it.

For  $x < y$  in an infinite limit stage, the simplest cut separator has birthday at most  $\max(\text{bd}(x), \text{bd}(y)) + 1 < \theta$ . Larger positive and negative ordinals of birthday below  $\theta$  rule out endpoints. The ordinals below  $\theta$  are cofinal, and their negatives are coinital. Conversely, a subset whose birthdays are bounded below  $\theta$  is bounded in  $\mathbf{No}_{<\theta}$  by an ordinal below  $\theta$ . A cofinal subset must therefore have birthdays cofinal in  $\theta$ , establishing the lower cofinality bound.  $\square$

The fixed-set results now give the exact ordinal spectrum

$$\beta \hookrightarrow \mathcal{W}(\mathbf{No}_{<\theta}) \iff \beta < \kappa_\theta^+ \quad (\kappa_\theta \text{ infinite}). \quad (2)$$

This is a cardinal bound attached to the entire alphabet, not merely to the cofinality of its numerical order.

| Stage                                                | Alphabet size                 | Size of its well-ordering space | Ordinal spectrum         |
|------------------------------------------------------|-------------------------------|---------------------------------|--------------------------|
| $\mathbf{No}_{<\omega}$                              | $\aleph_0$                    | $2^{\aleph_0}$                  | $\beta < \omega_1$       |
| $\mathbf{No}_{<\omega_1}$                            | $2^{\aleph_0} = \mathfrak{c}$ | $2^{\mathfrak{c}}$              | $\beta < \mathfrak{c}^+$ |
| $\mathbf{No}_{<\tau}$ , $\tau$ strongly inaccessible | $\tau$                        | $2^\tau$                        | $\beta < \tau^+$         |

The second row does not identify  $\mathfrak{c}^+$  with  $\omega_2$  without an additional continuum hypothesis. At finite stages,  $|\mathbf{No}_{<n}| = 2^n - 1$ , so the number of well-orders is  $(2^n - 1)!$ .

[merge] Source 08's Section 6 begins with the same count, as (S08.5), and then treats the regular case; its statements are recorded in the notes after this extract.

## 7.2 Source 08: birthday cutoffs and the regular-cardinal model

For an ordinal  $\lambda$ , write

$$\mathbf{No}_{<\lambda} = \{x \in \mathbf{No} : \text{bd}(x) < \lambda\}.$$

Its exact cardinality is

$$|\mathbf{No}_{<\lambda}| = \sum_{\alpha < \lambda} 2^{|\alpha|}. \quad (\text{S08.5})$$

This follows directly from the disjoint birthday levels in the sign-sequence representation. If this cardinal is infinite, all the results of section 6.4 apply to  $\mathcal{W}(\mathbf{No}_{<\lambda})$ . No algebraic closure property of the cutoff is needed.

A particularly transparent case occurs at a regular infinite cardinal  $\kappa$  satisfying  $2^{<\kappa} = \kappa$ . This includes  $\kappa = \omega$  and every strongly inaccessible cardinal; it also includes successor cardinals under appropriate cardinal arithmetic hypotheses.

*Note 7.4* (Source 08, Proposition 6.1). *If  $\kappa$  is regular infinite and  $2^{<\kappa} = \kappa$ , then  $|\mathbf{No}_{<\kappa}| = \kappa$  and the numerical order of  $\mathbf{No}_{<\kappa}$  is an  $\eta_\kappa$ -order.* The second clause is lemma 7.2 (which needs only regularity), proved by the same birthday bound; it is the classical saturation of  $\mathbf{No}_{<\kappa}$  recorded in the birthday-cutoff report (`hset:rem:saturation`, through `hset:eq:cut-bound`) [4].

*Note 7.5* (Source 08, Lemma 6.2). *Replace each minus by 00 and each plus by 11 and append the terminal block 01; the resulting binary word  $d(s)$  of length  $2 \cdot \text{bd}(s) + 2$  is prefix-free and  $s < t$  exactly when  $d(s) <_{\text{lex}} d(t)$ .* This is source 12's lemma 19.33, by the same comparison  $00 < 01 < 11$ .

*Note 7.6* (Source 08, Theorem 6.3, “The initial-length cutoff order”). *If  $\kappa$  is regular infinite and  $2^{<\kappa} = \kappa$ , then*

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\kappa, \quad |\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa; \quad (\text{S08.6})$$

*moreover  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$  is an  $\eta_\kappa$ -order, and its ordinal and reverse-ordinal spectra consist exactly of the ordinals below  $\kappa^+$ .* The equimorphism is case 2 of source 12's theorem 8.3, proved the same way (pair coding one way, the unpadding code note 7.5 with partial sums below  $\kappa$  by regularity the other way). The  $\eta_\kappa$  clause is source 11's proposition 12.1 for  $X = \mathbf{No}_{<\kappa}$ ; source 08's set-sized cut argument is the same common-coordinate search, with fresh values from theorem 5.6 and a completion of length  $\kappa$ , and it refers to note 19.13 for the detailed invariant. The spectra are theorem 6.1.

*Note 7.7* (Source 08, Remark 6.4). Equimorphism in (S08.6) is not isomorphism: the binary cube has endpoints and adjacent pairs, whereas  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$  is dense without endpoints. Order embeddings need not preserve jumps or be continuous for the order topologies.

The regularity assumption has specific jobs: it bounds option birthdays and first-disagreement coordinates, and controls partial sums of code lengths. At a singular cutoff these steps may fail. The general cardinality and ordinal-spectrum theorems remain valid, but the  $\eta_\kappa$  and fixed-length equimorphism conclusions are not asserted without their hypotheses.

[merge] The singular case that this paragraph leaves open is settled for the coding length by source 12's dichotomy (section 8): at a singular strong limit the fixed-length order is  $\simeq_e \mathbb{B}_{\kappa \cdot \kappa}$ , not  $\mathbb{B}_\kappa$ , and it is  $\simeq_e \mathbb{B}_{2^{<\kappa}}$  when  $2^{<\kappa} > \kappa$  (section 1.7, item (c)). Source 11's remark at the end of section 7 gives the failure of  $\eta_\kappa$  at a singular cutoff.

[merge] The next section is source 12’s Section 10. Its dichotomy is new against the repository; it answers in part the reals report’s question “Other ground orders” (section 1.7). It uses source 12’s self-delimiting code (lemma 19.33, printed in part IV) and its lemmas of section 6.2. Here  $S_\kappa$  of source 12 is printed  $\mathbf{No}_{<\kappa}$ , and its “binary coding rank” is the binary coding length  $\ell_2$  of the reals report.

## 8 Birthday cutoffs and the singular transition (source 12)

Let  $\kappa$  be an infinite initial cardinal and put

$$\mathbf{No}_{<\kappa} = \{s \in \mathbf{No} : \text{bd}(s) < \kappa\}, \quad \mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa} = \sup_{\lambda < \kappa} 2^{|\lambda|}.$$

The cardinality identity follows by counting all levels. The supremum is at least  $\kappa$ , and summing  $\kappa$  levels of size at most that supremum does not increase it. The reverse inequality is supplied by the levels themselves. We do not assume  $\mathbf{No}_{<\kappa}$  is a field; none of the following proofs uses arithmetic closure.

### 8.1 A scheduler that preserves the first varying coordinate

**Lemma 8.1** (Row scheduler). *Suppose a set order  $X$  of infinite cardinality  $\kappa$  has disjoint subsets  $(X_i)_{i < \kappa}$ , each containing a copy of  $X$  and each of cardinality  $\kappa$ . Suppose their complement is either empty or has cardinality  $\kappa$ . Then*

$$X_{\text{lex}}^\kappa \hookrightarrow \mathcal{W}_\kappa(X).$$

*Proof.* Fix embeddings  $\phi_i : X \rightarrow X_i$  and a baseline enumeration of each  $X_i$  in length  $\kappa$ . For  $x \in X$ , transpose the label  $\phi_i(x)$  with the first label of that baseline, giving an enumeration  $r_{i,x}$  of  $X_i$  whose first value is  $\phi_i(x)$ . Its dependence on  $x$  is fixed and uniform. If the complement is nonempty, add a fixed filler row indexed before all variable rows. The row index order still has type  $1 + \kappa = \kappa$ .

Well-order row-stage pairs  $(i, j) \in \kappa \times \kappa$  first by  $\max(i, j)$  and then lexicographically within a level. This schedule has type  $\kappa$ : the predecessors of any pair have cardinality less than  $\kappa$ , since both coordinates there are bounded below  $\kappa$ , while the entire set of pairs has size  $\kappa$ . The first appearance of row  $i$  is  $(i, 0)$ , and these first appearances occur in increasing row order.

For an input  $(x_i)_{i < \kappa}$ , output the  $j$ th entry of  $r_{i,x_i}$  when pair  $(i, j)$  is scheduled. The output is an exhaustive bijection in type  $\kappa$ . If two inputs first differ in row  $i$ , every earlier row agrees, no later row has appeared before  $(i, 0)$ , and the first output discrepancy is  $\phi_i(x_i)$  versus  $\phi_i(y_i)$ . Its sign is exactly the sign of the input comparison.  $\square$

**Lemma 8.2** (Surreal cone decomposition). *If  $|\mathbf{No}_{<\kappa}| = \kappa$ , then*

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \simeq_e (\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa.$$

*Proof.* One embedding is the inclusion of bijective sequences among all sequences. For the other, partition the sign sequences with a minus into the cones with prefixes

$$p_i = +^i \frown -, \quad i < \kappa.$$

They are disjoint, and prefixing by  $p_i$  gives a numerical order isomorphism from  $\mathbf{No}_{<\kappa}$  onto that cone. For  $i, \alpha < \kappa$ ,  $i + 1 + \alpha < \kappa$ ; this uses that  $\kappa$  is an initial cardinal and is valid even when it is singular. Removing the initial prefix also leaves a sign sequence of length less than  $\kappa$ . The complement consists of all-plus sequences, one for each length below  $\kappa$ , so has cardinality  $\kappa$ . Apply lemma 8.1.  $\square$

**Theorem 8.3** (Exact minimal-slice dichotomy). *For  $\kappa$  and  $\mu = 2^{<\kappa}$  as above,*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \mu = \kappa \text{ and } \text{cf } \kappa < \kappa, \\ \mathbb{B}_\mu, & \text{otherwise.} \end{cases}$$

*Consequently its binary coding rank is respectively  $\kappa \cdot \kappa$  or  $\mu$ . The exceptional case is exactly the case of a singular strong-limit cardinal  $\kappa$ .*

*Proof.* We separate the three mechanisms.

*Case 1:  $\mu > \kappa$ .* By lemma 19.33, each member of  $\mathbf{No}_{<\kappa}$  has a prefix-free numerical binary code of length less than  $\kappa$ . Pad each such code to length  $\kappa$ . Concatenating the entry codes of a  $\mu$ -long enumeration produces a binary word of length  $\kappa \cdot \mu = \mu$ . The equality holds because  $\mu$  is an initial cardinal strictly larger than  $\kappa$ : every proper partial product  $\kappa \cdot \xi$ ,  $\xi < \mu$ , has cardinality below  $\mu$ . Thus  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \hookrightarrow \mathbb{B}_\mu$ . Pair coding gives the reverse embedding.

*Case 2:  $\mu = \kappa$  and  $\kappa$  is regular.* Use the unpadded, self-delimiting code  $c(s)$ . In a sequence of  $\kappa$  entries, the sum of code lengths before any stage  $\xi < \kappa$  is less than  $\kappa$ , by regularity. The full sum is  $\kappa$ , since every length is positive. Consequently concatenation embeds even  $(\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa$  into  $\mathbb{B}_\kappa$ . Pair coding again supplies the reverse embedding for  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ .

*Case 3:  $\mu = \kappa$  and  $\delta = \text{cf } \kappa < \kappa$ .* The padded single-entry code embeds  $\mathbf{No}_{<\kappa}$  into  $\mathbb{B}_\kappa$ , whence

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \hookrightarrow (\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa \hookrightarrow \mathbb{B}_{\kappa \cdot \kappa}.$$

Choose increasing nonzero ordinals  $(\lambda_j)_{j < \delta}$  cofinal in  $\kappa$ . Their ordinal sum is  $\kappa$ . As  $\delta \cdot \kappa = \kappa$ , partition the  $\kappa$  input coordinates into  $\kappa$  consecutive blocks of length  $\delta$ . In each block, restrict its  $j$ th coordinate to sign sequences of exactly length  $\lambda_j$ . The lexicographic product of those coordinates is the binary cube of the sum of their lengths. Thus, over all the blocks, this restricted product is

$$\mathbb{B}_{\sum_{\xi < \kappa} \sum_{j < \delta} \lambda_j} = \mathbb{B}_{\kappa \cdot \kappa}.$$

It embeds into  $(\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa$ , which embeds into  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$  by lemma 8.2. This gives the claimed equimorphism. The rank assertions follow from lemma 6.4.

Finally, a singular strong limit satisfies  $2^{<\kappa} = \kappa$ . Conversely, suppose  $\kappa$  is singular and  $2^{<\kappa} = \kappa$ . If  $2^\lambda = \kappa$  for some  $\lambda < \kappa$ , choose  $\rho < \kappa$  with  $\rho \geq \lambda, \text{cf } \kappa$ . Then  $\kappa \leq 2^\rho \leq 2^{<\kappa} = \kappa$ , so  $2^\rho = \kappa$ , contradicting König's inequality  $\text{cf}(2^\rho) > \rho$ . Thus  $2^\lambda < \kappa$  for every  $\lambda < \kappa$ .  $\square$

**Corollary 8.4** (A complete cube test for the minimal slice). *Let  $\rho = \kappa \cdot \kappa$  in the exceptional case of theorem 8.3, and let  $\rho = \mu$  otherwise. Then*

$$\mathbb{B}_\gamma \hookrightarrow \mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \iff \gamma \leq \rho.$$

*In every case its ordinal embedding spectrum is still precisely  $\{\beta : \beta < \mu^+\}$ .*

**Example 8.5** (An unconditional singular example). Let  $\kappa = \beth_\omega = \sup_{n < \omega} \beth_n$ . It is a singular strong-limit cardinal of cofinality  $\omega$ , independently of GCH. Thus

$$\ell_2(\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})) = \kappa \cdot \kappa > \kappa, \quad \ell_2(\mathcal{W}(\mathbf{No}_{<\kappa})) = \kappa^+.$$

All three displayed ordinals must be distinguished. The first two have the same cardinality, but are different coding lengths; the successor cardinal is larger still.

**Example 8.6** (Regular cutoffs). For  $\kappa = \omega$ ,  $\mathbf{No}_{<\kappa}$  is the order of finite-birthday surreals, and the minimal slice is equimorphic with  $\mathbb{B}_\omega$ . For  $\kappa = \omega_1$ , its cardinality is  $2^{\aleph_0}$ , and the minimal slice is equimorphic with  $\mathbb{B}_{2^{\aleph_0}}$ , whether or not CH holds. At a strongly inaccessible  $\kappa$ , it is equimorphic with  $\mathbb{B}_\kappa$ .



## 8.2 Why the singular phenomenon is genuinely ordinal

A set of  $\kappa$  coordinates does not license arbitrary reordering of those coordinates in a lexicographic product. In the singular case, a short cofinal block of coordinate lengths can already consume ordinal length  $\kappa$ ; repeating that event  $\kappa$  times consumes  $\kappa \cdot \kappa$ . Replacing this ordinal sum by its cardinality would incorrectly collapse the rank back to  $\kappa$ . The strict-cube lemma shows that the lost information is mathematically detectable.

The row scheduler addresses a separate possible error: an arbitrary bijection  $\kappa \times \kappa \rightarrow \kappa$  need not preserve the first varying row. Bounding by the maximum coordinate ensures that all earlier observable output is independent of later varying rows. This is why the product-to-permutation embedding is valid even at singular  $\kappa$ .

[merge] The last sentence of [corollary 8.4](#) is [corollary 7.1](#) for the minimal slice. The dichotomy computes binary coding length only; source 11's question on the exact cut spectrum and point characters of  $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa})$  without  $2^{<\kappa} = \kappa$  ([research question 113.7](#)) stays open.

[merge] Dated note, 3 October 2026 (batch 81): that question is now answered by sources 17, 15 and 18 in Part VI: point characters  $(q, q)$  with  $q = \text{cf } \mu$ , the exact gap spectrum and its multiplicities, and the saturation threshold  $\eta_{\text{cf } \kappa}$  of  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ , for every infinite cardinal  $\kappa$  ([sections 30](#) and [33](#)). Source 15 extends the dichotomy itself to the strata  $\mu \cdot \gamma + n$  ([theorem 31.2](#)).

## Part II

# Set-length words

## 9 Set-length surreal words embed explicitly in the surreal line

Let  $\mathcal{T}$  be the class of all set-length words  $f : \alpha \rightarrow \mathbf{No}$ , and let  $\mathcal{I}$  be its injective subclass. The latter codes well-orders of set-sized subsets of  $\mathbf{No}$ . Both are ordered lexicographically with proper prefixes smaller. Their members are sets, so these are ordinary definable classes in  $\mathbf{GB}$ , not collections whose members are proper classes. They should not be confused with all global well-orders of the entire surreal class.

In particular, a word cannot exhaust  $\mathbf{No}$ : its range is a set by Replacement. Passing from fixed alphabets to  $\mathcal{I}$  consequently changes which carrier is being well-ordered from one point of the lexicographic order to another. This is a mathematically natural extension, but it answers a different question from ordering all class well-orders on a single proper-class carrier. Making this distinction before forming any collection of codes is what prevents a hidden ascent from classes to collections of classes.

The following construction uses neither a global enumeration of  $\mathbf{No}$  nor Global Choice. It uses only ordinal concatenation of sign expansions, whose general properties are discussed in [\[14, Section 3.2\]](#).

**Theorem 9.1** (Explicit sign coding). *There is an explicitly defined order embedding  $\mathcal{E} : \mathcal{T} \rightarrow \mathbf{No}$ . It maps the empty word to 0, maps every nonempty word to a positive surreal, and restricts to an order embedding of  $\mathcal{I}$ . Singleton words give embeddings in the opposite direction.*

*Proof.* For a sign expansion  $s$ , let  $D(s)$  duplicate each sign: replace  $-$  by  $(-, -)$ , and  $+$  by  $(+, +)$ . For  $x \in \mathbf{No}$ , put

$$C(x) = (+) \frown D(s_x) \frown (+, -).$$

The length here is the ordinal  $1 + 2 \cdot \text{bd}(x) + 2$ . If  $s_x, s_y$  first differ at a sign, their doubled blocks have the same comparison. If  $s_x$  is a proper prefix of  $s_y$ , the terminal block  $(+, -)$  of  $C(x)$  is compared with the next doubled block of  $C(y)$ . We have

$$(-, -) < (+, -) < (+, +).$$

This is precisely the comparison of a missing sign with a negative or positive continuation. Thus  $C$  preserves the surreal order. Moreover, two distinct values of  $C$  cannot be prefixes of one another: the same first differing block supplies an actual differing sign before either code ends.

For  $f : \alpha \rightarrow \mathbf{No}$ , concatenate these blocks in domain order:

$$\mathcal{E}(f) = \bigcap_{i < \alpha} C(f(i)).$$

Replacement and ordinal summation make this a set-length sign expansion. If two words first differ at  $i$ , their preceding blocks coincide and their blocks at  $i$  have a first differing sign. Prefix freedom prevents that difference from being displaced into a subsequent block. The comparison therefore agrees with the comparison of the original entries. If  $f$  is a proper prefix of  $g$ , the expansion  $\mathcal{E}(g)$  continues  $\mathcal{E}(f)$  with the initial  $+$  of the next block, so  $\mathcal{E}(f) < \mathcal{E}(g)$ . This also proves the assertions about empty and nonempty words. Finally,  $x \mapsto \langle x \rangle$  embeds  $\mathbf{No}$  into both word classes.  $\square$

Accordingly,  $\mathcal{T}$ ,  $\mathcal{I}$ , and  $\mathbf{No}$  are pairwise equimorphic as class linear orders. This conclusion asserts embeddings both ways, not isomorphism. Section 10 exhibits an obstruction to isomorphism that persists after the empty word is removed.

**Proposition 9.2** (Birthday bounds). *The coding has the following bounds.*

1. If  $\kappa$  is regular infinite,  $\alpha < \kappa$ , and  $f(i) \in \mathbf{No}_{<\kappa}$  for every  $i < \alpha$ , then  $\mathcal{E}(f) \in \mathbf{No}_{<\kappa}$ .
2. For any infinite cardinal  $\kappa$ , put  $\lambda = 2^{<\kappa}$ . If  $f$  is an arbitrary injective word over  $\mathbf{No}_{<\kappa}$ , then  $\mathcal{E}(f) \in \mathbf{No}_{<\lambda^+}$ .

*Proof.* The birthday of the image is

$$\text{bd}(\mathcal{E}(f)) = \sum_{i < \alpha} (1 + 2 \cdot \text{bd}(f(i)) + 2),$$

with ordinal addition and multiplication. For the first assertion, this sums fewer than  $\kappa$  ordinals below  $\kappa$ , and regularity keeps the sum below  $\kappa$ . For the second, injectivity gives  $|\alpha| \leq \lambda$ ; each block has cardinality at most  $\lambda$ . The whole sign expansion has cardinality at most  $\lambda$ , so its ordinal length is below  $\lambda^+$ .  $\square$

The first bound need not hold at a singular cutoff. Choose distinct ordinals  $\gamma_i < \kappa$ ,  $i < \text{cf}(\kappa)$ , cofinal in  $\kappa$ , and regard them as surreal entries. The word has length less than  $\kappa$ , but the lengths of its code blocks are unbounded in  $\kappa$ , so its encoded birthday is at least  $\kappa$ . Also, injectivity is essential to the unrestricted second bound: constant words over a one-element alphabet can have every ordinal length.

## 10 Local characters and set suprema of word orders

In this section  $\mathcal{W}$  denotes either  $\mathcal{T}$  or  $\mathcal{I}$ . Deleting the set of entries of an existing prefix causes no difficulty when choosing a new coordinate: apply Lemma 5.2. We say that a segment has no set cofinal subset rather than assigning it a cardinal cofinality that does not exist.

**Theorem 10.1** (Exact local character). *Let  $f \in \mathcal{W}$  have domain  $\alpha$ .*

1. *If  $\alpha$  is a nonzero limit ordinal, then  $\text{cf}(\mathcal{W}_{<f}) = \text{cf}(\alpha)$ .*
2. *If  $\alpha$  is a successor ordinal, then  $\mathcal{W}_{<f}$  has no set cofinal subset.*
3. *For every  $\alpha$ , the segment  $\mathcal{W}_{>f}$  has no set coinitial subset.*

*The empty word is the minimum of  $\mathcal{W}$ .*

*Proof.* Suppose first that  $\alpha$  is a nonzero limit. Its proper prefixes  $f \upharpoonright \beta$ ,  $\beta < \alpha$ , are cofinal below  $f$ . A word  $g < f$  is either a proper prefix, or differs from  $f$  first at some coordinate  $\xi$ ; in the latter case it lies below every prefix long enough to include coordinate  $\xi$ . A cofinal sequence of indices in  $\alpha$  therefore supplies a cofinal subset of size  $\text{cf}(\alpha)$ . Conversely, to each  $g < f$  assign a prefix length strictly large enough that the corresponding prefix exceeds  $g$ . Fewer than  $\text{cf}(\alpha)$  such lengths are bounded below  $\alpha$ , so the corresponding words cannot be cofinal.

Next let  $\alpha = \beta + 1$ , write  $p = f \upharpoonright \beta$ , and take a set  $A < f$ . For each member  $a$  properly extending  $p$ , its next coordinate satisfies  $a(\beta) < f(\beta)$ . Choose an unused surreal  $d$  strictly above all these coordinates and strictly below  $f(\beta)$ . Then  $h = p \frown \langle d \rangle$  is below  $f$ . It exceeds every member of  $A$  extending  $p$ ; the remaining members are at most  $p$ , so they also lie below  $h$ . Thus  $A$  is not cofinal.

Finally, take a set  $B > f$ . Collect  $b(\alpha)$  for those  $b \in B$  properly extending  $f$ , and choose an unused  $d$  below all these coordinates. The word  $h = f \frown \langle d \rangle$  lies above  $f$ . It lies below the members of  $B$  extending  $f$ . Every other member of  $B$  first differs earlier with a larger coordinate, and consequently exceeds every extension of  $f$ , including  $h$ . Hence  $f < h < B$ .  $\square$

**Corollary 10.2.** *Neither  $\mathcal{T}$  nor  $\mathcal{I}$ , with or without its empty word, is order-isomorphic to  $\mathbf{No}$ . Each has nontrivial set suprema, but a set with no minimum cannot have an infimum in either word order.*

*Proof.* Take an injective word  $f$  of length  $\omega$ . Its nonempty finite prefixes increase to  $f$ , and  $f$  is their supremum by the preceding theorem. No surreal can be the supremum of a set lying strictly below it: the surreal set-cut property would insert a smaller upper bound. This distinguishes the orders even after deleting their minimum. If  $f$  were the infimum of a set  $B$  not containing  $f$ , part (3) would produce  $h$  with  $f < h < B$ , contradicting the definition of infimum.  $\square$

These local obstructions coexist with very rich intervals. Given  $f <_{\text{lex}} g$  with a first differing coordinate, vary that coordinate through a surreal interval strictly between the two values and disjoint from the set of preceding entries. If  $f$  is a proper prefix of  $g$ , vary a new coordinate below the next entry of  $g$  instead. The surreal set-cut property provides such an interval after excluding the preceding entries. Every surreal interval  $(u, v)$  is order-isomorphic to  $\mathbf{No}$ , for example by

$$t \mapsto u + \frac{v - u}{2} \left( 1 + \frac{t}{1 + |t|} \right).$$

Thus every open word interval contains a copy of  $\mathbf{No}$ , and Theorem 9.1 then places a copy of the entire word order inside it. Interval universality therefore does not imply that set cuts are filled.

## 11 The complete obstruction to filling a set cut

The preceding prefix example is the only nontrivial obstruction. Here and below,  $A < B$  means that every member of  $A$  is strictly below every member of  $B$ ; either set may be empty.

**Theorem 11.1** (Word-cut classification). *A set cut  $A < B$  in  $\mathcal{W}$  has no interpolant if and only if one of the following holds:*

1.  *$B$  contains the empty word;*
2.  *$B$  has a minimum  $b$  of nonzero limit length, and  $A$  is cofinal in  $\mathcal{W}_{<b}$ .*

*If neither obstruction occurs, an interpolant can be chosen to have nonzero successor length.*

*Proof.* Either displayed obstruction plainly prevents an interpolant. For the converse, perform the following common-prefix search. Start with  $p$  empty. Maintain these invariants: members of  $A$  not extending  $p$  lie below  $p$ ; members of  $B$  not extending  $p$  lie above every extension of  $p$ . Here extending allows equality.

If  $p \in B$ , stop. Otherwise, with  $\delta = \text{dom}(p)$ , form the sets of next coordinates

$$A(p) = \{a(\delta) : a \in A, p \text{ is a proper prefix of } a\}, \quad B(p) = \{b(\delta) : b \in B, p \text{ is a proper prefix of } b\}.$$

Every member of  $A(p)$  is at most every member of  $B(p)$ , since the corresponding words satisfy  $a < b$ . If these coordinate sets are disjoint, all these inequalities are strict. Choose a surreal  $c$  strictly between them, outside  $\text{ran}(p)$ . Then  $p \frown \langle c \rangle$  interpolates the original cut by the invariants, and the construction is complete.

If the two coordinate sets meet, their intersection consists of a unique element  $c$ : two distinct common elements would violate the preceding inequalities. Replace  $p$  by  $p \frown \langle c \rangle$ . The invariants persist because the discarded  $A$ -branches have smaller next coordinates and the discarded  $B$ -branches have larger ones. In the injective-word case,  $c$  is unused, since it is witnessed by an injective word extending the previous prefix. At a limit stage take the union of the prefixes. It is a set word, is injective when required, and satisfies the same invariants. A member of  $B$  cannot have become a proper prefix of this union: the search would already have stopped upon reaching that member.

This procedure requires only set recursion. Choose an ordinal strictly above the domains of all words in  $A \cup B$ . The search cannot continue taking common next coordinates up to that length, because there would be no words supplying those coordinates. It therefore terminates. The remaining possibility is termination at  $p \in B$ . The invariants show that  $p = \min B$ . If its length is zero, the first obstruction holds.

Termination cannot occur at a successor length  $\delta + 1$ . The last appended coordinate was chosen from the intersection, so some member of  $A$  extends this successor prefix. Such a word is at least  $p$ , contrary to  $A < B$ . Hence a nonzero terminal length  $\alpha$  is a limit. For every  $\beta < \alpha$ , a member of  $A$  witnessed the common coordinate at stage  $\beta$ ; it extends  $p \upharpoonright (\beta + 1)$  and lies below  $p$ . These witnesses exceed prefixes of arbitrarily large length. Since the proper prefixes are cofinal below  $p$ ,  $A$  is cofinal below  $p$ . This is exactly the second obstruction.  $\square$

**Corollary 11.2** (Successor words). *The suborder of  $\mathcal{W}$  consisting of words of nonzero successor length fills every set cut. In GBC, it is order-isomorphic to **No**.*

*Proof.* Neither obstruction in Theorem 11.1 can occur when the upper set consists of successor words. The construction returns another successor word, proving the set-cut property. Under Global Choice, enumerate each of the two proper classes by a set-like class well-order and perform back-and-forth. At every ordinal stage the previously matched elements form a set; their images specify a set cut, and an unused filler extends the partial order isomorphism. The union is an isomorphism. This is the class analogue of the usual eta-order back-and-forth, with its choice assumption made explicit; compare the class framework in [15].  $\square$

The distinction is precise. Deleting the empty word alone does not remove the prefix-supremum obstruction. Restricting to successor lengths does remove it. The explicit embeddings of Theorem 9.1 require no Global Choice, while the back-and-forth identification just given uses it.

[merge] Source 09’s class of set-length injective words is  $\mathcal{I}$  (its  $\mathcal{T}$ ). Its proposition 11.3 obtains the equimorphism with **No** from class universality in GBC; source 11’s theorem 9.1 gives explicit embeddings in ZF, and corollary 10.2 the non-isomorphism.

## 11.1 Source 09: allowing the underlying set to vary

### 11.1.1 The set-word class

Every well-order on a set of surreals has a unique injective exhaustive enumeration of its range. Thus all such labeled set well-orders are represented by the genuine proper class

$$\mathcal{I} = \{p : p \text{ is an injective function } \alpha \rightarrow \mathbf{No} \text{ for some ordinal } \alpha\}.$$

Every member of  $\mathcal{I}$  is a set. The collection is proper because it contains every singleton word  $(x)$  with  $x \in \mathbf{No}$ .

For  $p, q \in \mathcal{I}$ , compare at the first unequal entry when one exists; otherwise the proper prefix is smaller. This is a strict linear order. One can verify transitivity by padding words with a formal termination symbol below every surreal, but this symbol is *not itself a surreal number*. It is simply part of the definition of word order.

**Proposition 11.3.** *The dictionary order on  $\mathcal{I}$  is dense, has the empty word as its least element, and has no greatest element. In GBC it is equimorphic with **No**, but it is not isomorphic to **No**, even after removing the empty word.*

*Proof.* If  $p < q$  differ at coordinate  $\delta$ , their common prefix has set range. By lemma 5.10, choose a fresh surreal between the two entries at  $\delta$  and append it to this prefix. The resulting word lies strictly between  $p$  and  $q$ . If  $p$  is a proper prefix of  $q$ , append to  $p$  a fresh value smaller than  $q(\text{dom } p)$ ; this also gives an intermediate word. Every word has a proper injective extension, so there is no greatest word.

The singleton map  $x \mapsto (x)$  embeds **No** into  $\mathcal{I}$ . Conversely,  $\mathcal{I}$  is a class linear order and embeds into **No** by theorem 5.16. The final nonisomorphism assertion follows from the next proposition.  $\square$

*Note 11.4* (Source 09, Proposition 6.2, “Limit prefixes create small one-sided character”). *If  $p \in \mathcal{I}$  has nonzero limit length  $\lambda$ , its proper prefixes are cofinal in the open initial segment below  $p$ , the cofinality of that segment is exactly  $\text{cf}(\lambda)$ , and  $p = \sup_{\mathcal{T}}\{p \restriction \xi : \xi < \lambda\}$ .* This is part (1) of theorem 10.1 and the first step of the proof of corollary 10.2, by the same argument.

In **No**, the set-cut property forbids a set cofinal immediately below any fixed point. Thus the difference in note 11.4 is intrinsic, not just an artifact of the empty word. An embedding into **No** does not contradict this: the image need not preserve suprema in the ambient order.

### 11.1.2 Labeled objects versus order types

There is only one increasing exhaustive enumeration of a set already well-ordered by the numerical surreal order. In contrast,  $\mathcal{W}(X)$  allows all well-orders on the labeled set  $X$ . If one instead quotients all set well-orders by abstract order isomorphism, only their ordinal types remain. Lexicographic comparison of their numerical entries does not descend to that quotient: two distinct two-letter words can have identical abstract order type and different positions in dictionary order. The distinctions between labels, inherited order, and order type must be retained.

[merge] Source 08’s Section 7 (“Varying set-sized subsets: a different class order”) uses the same class  $\mathcal{I}(\mathbf{No}) = \mathcal{I}$  with proper prefixes first.

*Note 11.5* (Source 08, Proposition 7.1). *The prefix-small lexicographic order on  $\mathcal{I}(\mathbf{No})$  is linear; its empty word is a minimum, its nonempty part is dense without endpoints, and it contains a copy of  $\mathbf{No}$  through one-entry words. Nevertheless its nonempty part is not an  $\eta_{\text{Ord}}$ -order: for an injective word  $p$  of length  $\omega$ , the set of its nonempty finite prefixes and  $\{p\}$  have no separator. This is [corollary 10.2](#) with the obstruction (2) of [theorem 11.1](#), by the same argument. Its last paragraph is printed next.*

More generally, for a word  $p$  of nonzero limit length  $\lambda$ , its proper prefixes are cofinal below it, and the left cofinality at  $p$  is  $\text{cf}(\lambda)$ . For the lower bound, a family of fewer than  $\text{cf}(\lambda)$  words below  $p$  has its first disagreement or termination coordinates bounded below  $\lambda$ , and so is bounded by a proper prefix. Thus even very large numerical alphabets do not erase cofinalities created by termination conventions.

If “well-orders of subsets” is instead required to mean *numerically increasing* enumerations, it defines a further subclass of  $\mathcal{I}(\mathbf{No})$ . The singleton embedding of  $\mathbf{No}$  still works, and the limit-prefix obstruction persists using an increasing  $p$ . We do not identify that subclass with the arbitrary-enumeration class. Likewise a numerically well-ordered *proper subclass* of  $\mathbf{No}$  need not be set-like: a bounded increasing copy of  $\text{Ord}$  followed by its upper endpoint has type  $\text{Ord} + 1$ . Such copies also follow from the universality theorem in [section 5.8](#).

## 12 Saturation at a regular stage

For regular infinite  $\kappa$ , restrict to successor-length injective words of length below  $\kappa$ , with entries in  $\mathbf{No}_{<\kappa}$ . This order is  $\eta_\kappa$ . Indeed, a cut with fewer than  $\kappa$  words has all its word lengths bounded below  $\kappa$ . The common-prefix search in the preceding proof consequently stops below  $\kappa$ , and [Lemma 7.2](#) supplies the last coordinate while excluding the fewer than  $\kappa$  entries already used. [Proposition 9.2](#) embeds this order into  $\mathbf{No}_{<\kappa}$ ; singleton words embed  $\mathbf{No}_{<\kappa}$  back. When  $2^{<\kappa} = \kappa$ , both orders have cardinality  $\kappa$ , and the ordinary back-and-forth of length  $\kappa$  makes them isomorphic.

A related saturation statement applies to full permutations.

**Proposition 12.1.** *Let  $X$  be an  $\eta_\kappa$ -order of cardinality  $\kappa$ , where  $\kappa$  is regular infinite. Then  $\mathcal{W}_\kappa(X)$  is  $\eta_\kappa$ . Its global and local cofinalities and coinitalities are all exactly  $\kappa$ .*

*Proof.* For a cut  $A < B$  of total cardinality below  $\kappa$ , apply the common-coordinate search to its permutations. If both sides are nonempty, the first-disagreement indices of the pairs in  $A \times B$  have supremum below  $\kappa$ . The search cannot continue sharing a coordinate past this bound; if either active side becomes empty, it stops sooner. The common prefix therefore has length below  $\kappa$ . The  $\eta_\kappa$  property chooses an unused coordinate strictly between the active coordinate sets. Complete the resulting prefix to a  $\kappa$ -permutation: its remaining entries and remaining positions both have cardinality  $\kappa$ , and its remaining interval of positions has order type  $\kappa$ . The completed permutation fills the cut.

Saturation excludes cofinal or coinital subsets of size below  $\kappa$ , both globally and on either side of a point. Conversely, [Theorem 6.1](#) excludes monotone sequences of length  $\kappa^+$ . If one of these segments had no cofinal subset of size at most  $\kappa$ , recursively choosing a point above all earlier choices would produce such an increasing sequence. The decreasing argument is identical. Thus each of the stated characters equals  $\kappa$ .  $\square$



In particular this proposition applies to  $\mathbf{No}_{<\kappa}$  when  $2^{<\kappa} = \kappa$ . The resulting permutation order has cardinality  $2^\kappa$ , so it is substantially larger than its alphabet while retaining the same local characters. It is not  $\eta_{\kappa^+}$ : that stronger cut property would construct an increasing  $\kappa^+$ -sequence, contrary to Theorem 6.1.

## Part III

# Global choice and all class well-orders

## 13 Global choice and well-ordering the surreal class

The distinction between a set of surreal numbers and the entire surreal class changes the existence question. Here GB has the sets of ZF, without a choice axiom, and GC asserts the existence of a class function choosing an element of every nonempty set. A class well-order is a strict linear class relation for which every nonempty available subclass has a least element. Its underlying objects are still sets. The relation itself can be a proper class, and its predecessor classes need not be sets. These are the size distinctions emphasized in the repository's foundations discussion [3].

Throughout this section,  $\mathfrak{W}_{\text{all}}$ ,  $\mathfrak{W}_{\text{sl}}$ , and  $\mathcal{W}(X)$  when  $X$  is a proper class are metanotation for the indicated collections of class relations. We quantify over their members as class variables; we do not assert that those collections are GB classes. When  $X$  is a set,  $\mathcal{W}(X)$  retains its ordinary set meaning from the preceding section. All constructions below are uniform formulas in the stated class parameters. In particular, restricting such a construction to an already given uniformly indexed family produces an ordinary class relation on the indexing domain.

Concretely, a family indexed by a class  $I$  is supplied by one class  $\mathcal{R} \subseteq I \times \mathbf{No} \times \mathbf{No}$ , whose slice at  $i$  defines the relation  $R_i$ . Writing this family does not make  $R_i$  a set or a member of  $I$ : the index  $i$  is a set, and the slice is a class defined with that parameter. This distinction permits uniform families of proper class relations while preserving the usual GB typing rules. No exhaustive indexing family is assumed in this section.

**Theorem 13.1** (Existence and global choice). *Over GB, the following statements are equivalent:*

1. GC holds;
2. there is a class well-order of  $\mathbf{No}$ ;
3. there is a set-like class well-order of  $\mathbf{No}$ ;
4. there is a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ .

Here set-like means that the predecessors of every element form a set.

*Proof.* We give the implication from an arbitrary well-order to global choice in detail. It also explains why set choice need not be assumed in the base theory. Let  $W$  be a class well-order of the sign-sequence presentation of  $\mathbf{No}$ . By ordinary set-valued transfinite recursion, with the fixed class parameter  $W$ , construct a set relation  $w_\alpha$  well-ordering  $V_\alpha$ , for every ordinal  $\alpha$ .

Set  $w_0 = \emptyset$ . Suppose  $w_\beta$  has been constructed. There is a unique ordinal  $\theta$  and a unique order isomorphism

$$e : \theta \longrightarrow (V_\beta, w_\beta).$$

For  $A \subseteq V_\beta$ , let  $s_A$  be the sign sequence of length  $\theta$  whose value at  $\xi$  is  $+$  exactly when  $e(\xi) \in A$ . The map  $A \mapsto s_A$  is a bijection between  $\mathcal{P}(V_\beta)$  and the set of all sign sequences of length  $\theta$ . Define

$$A w_{\beta+1} B \iff s_A W s_B.$$

This is a set relation by separation with a class parameter, and it well-orders  $V_{\beta+1} = \mathcal{P}(V_\beta)$ . No selection of an enumeration is involved: the enumeration of an already specified set well-order is unique.

At a nonzero limit  $\lambda$ , order  $V_\lambda$  first by rank, and within rank  $\gamma$  by the restriction of  $w_{\gamma+1}$ :

$$x w_\lambda y \iff \text{rank}(x) < \text{rank}(y) \text{ or } (\text{rank}(x) = \text{rank}(y) = \gamma \text{ and } x w_{\gamma+1} y).$$

Since  $\lambda$  is a limit, every required  $\gamma + 1$  is below  $\lambda$ . A nonempty subset of  $V_\lambda$  has a least occurring rank; the specified well-order on that rank then supplies its least member. Thus the recursion continues. Its stages are sets, even though their union over all ordinals is a class. This is precisely the ordinary GB recursion theorem, rather than a recursion whose individual values are proper classes.

For a nonempty set  $A$ , every element of  $A$  belongs to  $V_{\text{rank}(A)}$ . Choose the  $w_{\text{rank}(A)}$ -least element of  $A$ . The recursion and this prescription define a single class choice function, establishing GC.

Conversely, global choice implies choice for sets. For each ordinal  $\alpha$ , the set of well-orders of the birthday layer  $S_\alpha = \{-, +\}^\alpha$  is nonempty. Apply the global choice function to this set, and order **No** first by birthday and then by the selected order within the birthday layer. This is a class well-order. Every predecessor class is contained in  $\bigcup_{\beta \leq \alpha} S_\beta$  for some  $\alpha$ , hence is a set. This proves the set-like version directly.

Finally, for any set-like well-order  $R$  of the proper class **No**, assign to  $x$  the ordinal order type of its predecessor set. These ranks form an initial subclass of Ord. That subclass cannot be a set: otherwise the inverse rank map, by class replacement on a set domain, would make **No** a set. It is therefore all of Ord, and the inverse rank map is the required bijection. Conversely, a class bijection from Ord transports its well-order to a set-like well-order of **No**.  $\square$

The connection between surreal enumerations and global choice is discussed by Hamkins [18]; the general equivalences for global choice are collected in [19]. The explicit rank recursion above proves the assertion over the stated choiceless GB base. It does not assert that a global choice class is definable without parameters in the first-order language of set theory. Definability is an additional requirement.

[merge] (3 October 2026, batch 89.) Source 31 (Part XV) shows where the base theory enters. The full-class rank models  $\mathcal{M}_\lambda = (V_\lambda, \mathcal{P}(V_\lambda))$ , for limit  $\lambda > \omega$ , satisfy ZC with Foundation, elementary class comprehension and global choice, but Class Replacement only at strongly inaccessible  $\lambda$  (proposition XV.6.1 and theorem XV.9.3). In every one of them the sign carrier  $S_\lambda$  has a set-like, even birthday-monotone, class well-order (proposition XV.6.2), while a class bijection between Ord and  $S_\lambda$  exists exactly when  $\lambda$  is a cardinal  $\kappa$  with  $2^{<\kappa} = \kappa$  (theorem XV.7.2). At  $\lambda = \omega + \omega$ , for example, clauses 1–3 above hold for  $S_\lambda$  and clause 4 fails (section XV.11): the equivalence uses the class Replacement of GB. Source 31's theorem XV.5.1 localizes global choice over Zermelo set theory instead: selection from countable sets alone, with Separation for the selector, is not conservative over ZC, by Glazer's countermodels through an interface not reviewed here (section XV.5.1).

[merge] *Second proof over GB + AC (source 12).* Source 12 proves the equivalence by coding transitive closures on ordinals; set choice is used only to code each  $\text{TC}(\{x\})$  on an ordinal. Its equivalence includes non-set-like well-orders, as source 11's does. Its subsection on birthday order follows its theorem.

### 13.1 Source 12's route over GB with set choice

**Theorem 13.2** (Choice equivalence). *Over GB plus set-level AC, the following are equivalent:*

- (i) *global choice;*

- (ii) a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ ;
- (iii) a set-like class well-order of  $\mathbf{No}$ ;
- (iv) a class well-order of  $\mathbf{No}$ , not necessarily set-like.

*Proof.* Global choice supplies a set-like global well-order of the universe, and hence of the proper class  $\mathbf{No}$ . A proper set-like well-order has type  $\text{Ord}$ , giving (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii). A bijection in (ii) pulls back ordinal order, giving (iii), and (iii) $\Rightarrow$ (iv) is immediate.

We prove the substantive reverse implication by coding sets. Suppose  $R$  well-orders  $\mathbf{No}$ . Let  $x$  be a nonempty set and let  $T_x = \text{TC}(\{x\})$ ; it contains  $x$  and all its elements. By set choice, there is a least initial ordinal  $\lambda$  equinumerous with  $T_x$ . There is a fixed definable ordinal  $\tau_\lambda$  large enough to code a binary relation on  $\lambda$ , together with one distinguished index, by a binary sequence of length  $\tau_\lambda$ . For example, concatenate a  $\lambda \cdot \lambda$ -long relation table and a  $\lambda$ -long singleton marker.

Consider the *set* of codes for well-founded extensional relations on  $\lambda$  whose Mostowski collapse has range  $T_x$ , and whose marked index collapses to  $x$ . This set is nonempty. Convert 0, 1 to  $-$ ,  $+$ , view the codes as surreals, and take its  $R$ -least member. Decode it, and choose the element of  $x$  represented by the least ordinal index that belongs to the marked node in the coded membership relation.

The result is an element of  $x$ , uniquely determined by  $x$  and the class parameter  $R$ . Elementary class comprehension gives the graph of this choice operation on all nonempty sets. It is a global choice function. Notice that only a set of codes for each  $x$  was minimized: set-likeness of  $R$  was not needed.  $\square$

*Remark 13.3* (The explicit set-choice hypothesis). The converse proof uses set choice to code  $T_x$  on an ordinal. We do not claim the same equivalence over GB with no set-level choice. A purported stronger choice equivalence must address that missing coding step.

### 13.1.1 Birthday is not a canonical global well-order

Ordering first by birthday and then by numerical value does not work. For  $n < \omega$ , let  $s_n$  be the length- $\omega$  sign sequence consisting of  $n$  minuses followed by pluses forever. Then

$$\text{bd}(s_n) = \omega, \quad s_{n+1} < s_n.$$

The whole descending sequence lies in one birthday level. Simplicity itself is well-founded but is only a partial order; two incompatible sign sequences need not be comparable by simplicity.

With global choice one may choose a well-order on every birthday level and concatenate them in birthday order. The result is set-like because all earlier levels, and every initial part of the current level, form a set. The noncanonical choices have not disappeared: they are exactly the kind of information accounted for by [theorem 13.2](#).

*Remark 13.4* ([merge]; source 12's question answered). Dated note, 2 October 2026. Source 12's [remark 13.3](#) declines the choiceless case, and its research question on the exact weak-choice strength ([remark 113.15](#)) asks which principle is equivalent to a class well-order of the sign-sequence surreals over GB without set choice. Source 11's [theorem 13.1](#) answers it: global choice itself. The “missing coding step” that [remark 13.3](#) names is avoided, not supplied: the well-orders  $w_\alpha$  of  $V_\alpha$  are built by set-valued recursion from the given well-order of the sign sequences of length  $\theta = \text{otp}(V_\beta, w_\beta)$ , a subset  $A \subseteq V_\beta$  being coded by the sign sequence  $s_A$  of length exactly  $\theta$ , so no ordinal coding of a transitive closure has to be chosen. The proof thus produces a uniformly defined sequence of well-orders of the sets  $V_\alpha$ , in particular of the power sets of ordinals, which is the comparison the question suggests. The same mechanism, at the level of sets, is the ordinal-graph coding of the birthday-cutoff report

(`hset:thm:choice`, whose range over ZF is the class of sets with well-orderable transitive closure) [4]; that theorem concerns a different statement and is unchanged. Hamkins’s question whether surreal universality implies global choice [21] is not affected. (Corrected in place on 3 October 2026, following the Hilbert’s-tenth programme’s review `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_surreal_synthesis_0be9b9134.md` (commit 280583d2f): this note had called the sequence “coherent”. The orders  $w_\alpha$  are uniformly defined, but a later one need not restrict to an earlier one. If  $W$  puts  $-$  before  $+$  among the sign sequences of length one and  $+-$  before  $--$  among those of length two, then  $w_2$  puts  $\emptyset$  before  $\{\emptyset\}$  (codes  $-$  and  $+$ ), while  $w_3$  puts  $\{\emptyset\}$  before  $\emptyset$  (codes  $+-$  and  $--$ ). The proof of [theorem 13.1](#) needs only the uniform definition: it selects the  $w_{\text{rank}(A)}$ -least element of each nonempty set  $A$ . Ordering all sets by rank and then, within rank  $\gamma$ , by  $w_{\gamma+1}$ , as at the limit stages, does give a class well-order whose restrictions to the sets  $V_\alpha$  cohere; the proof does not use it.)

[merge] Dated note, 4 October 2026 (batch 89): a set-level, choiceless calibration. Part VII of the birthday-cutoff report [4] (batch-89 manuscript 04; AI-assisted, unrefereed, not formalized) proves over ZF that a set injects into **No** iff it injects into the power set of an ordinal (its Theorem 42.1, `hset:kw:thm:coding`), and that embedding the free real rational-function fields on all sets into **No** is exactly the first Kinna–Wagner principle, strictly weaker than choice (its Theorem 43.5, `hset:kw:thm:equivalences`). For a *prescribed* order it proves only that well-orderable linear orders embed (`hset:kw:prop:wellordered`), characterizes order embeddability by convex resolutions (`hset:kw:thm:resolution`), and exhibits, in a ZFA permutation model, an ordered field all of whose finitely generated coordinate subfields embed but which has no field homomorphism into the internal **No** (`hset:kw:thm:orderedcounter`). Hamkins’s question whether surreal universality implies global choice [21], about the class **No** over GB, is therefore still not affected.

[merge] *Third proof over GB + AC, for set-like orders (source 09)*. Source 09’s [proposition 13.6](#) is the last paragraph of the proof of [theorem 13.1](#) (and source 08’s [note 5.12](#)); its [proposition 13.7](#) recovers global choice from a class bijection  $\text{Ord} \rightarrow \mathbf{No}$  through a definable class surjection of **No** onto  $V$ , using set choice only to prove surjectivity. By [theorem 13.1](#) that hypothesis is unnecessary ([remark 13.4](#)). Its last subsection leads to source 08’s relation codes ([section 17](#)).

## 13.2 Source 09: global well-orders and global enumerations

**Definition 13.5.** A class well-order  $\prec$  on a class  $D$  is *set-like* when the predecessors of every  $d \in D$  form a set. We take “class well-order” to mean that every nonempty subclass has a least element.

**Proposition 13.6.** *Set-like class well-orders on **No** correspond bijectively to class bijections  $E : \text{Ord} \rightarrow \mathbf{No}$ .*

*Proof.* Given  $E$ , transfer the ordinal order to its values. Every predecessor set is the image of a set ordinal, so this order is set-like and is a class well-order.

Conversely, for a set-like well-order, assign to  $x$  the ordinal order type of its set of predecessors. This assignment is an order isomorphism onto a downward-closed class of ordinals. Its range cannot be a set: otherwise its inverse, a class function on a set, would have set range **No**. A proper downward-closed class of ordinals is all of  $\text{Ord}$ . Invert the assignment.  $\square$

Global choice supplies at least one such bijection; fix one and call it  $b$ . None of these enumerations is increasing in the numerical order on **No**. The numerical order is dense and has descending sequences, so it is not a well-order.

**Proposition 13.7** (The baseline already entails global choice). *Over GB with ordinary choice for sets, the following are equivalent: global choice; a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ ; and a set-like class well-order of  $\mathbf{No}$ .*

*Proof.* Global choice gives the bijection as explained in [section 5.5](#), and [proposition 13.6](#) identifies the last two conditions without using global choice. It remains to recover global choice from  $b$ .

There is an elementary definable class surjection  $d : \mathbf{No} \rightarrow V$ , using only ordinary set choice to prove surjectivity. Here is the coding argument. A set code consists of a nonzero ordinal  $\gamma$ , a well-founded extensional relation  $R \subseteq \gamma \times \gamma$ , and a distinguished root  $r < \gamma$ . Its decoded value is the image of  $r$  under the Mostowski collapse of  $(\gamma, R)$ . Every set  $x$  has such a code: well-order a transitive set containing  $x$ , transport its membership relation to an ordinal, and mark the index corresponding to  $x$ .

These triples have a uniform injective encoding by sign sequences. Write  $\gamma$  plus signs, then one minus sign as a header. Append a binary word of length  $\gamma \cdot \gamma$  recording  $R$ , with  $(u, v)$  assigned to position  $\gamma \cdot v + u$ , and then a word of length  $\gamma$  whose unique plus sign marks  $r$ . The first minus sign recovers  $\gamma$ ; the prescribed block lengths recover  $R$  and  $r$ . Decode valid codes by their set Mostowski collapses and send invalid codes to  $\emptyset$ . Everything in this decoder concerns set-sized relations, so it defines a class function by elementary comprehension, not a class truth predicate.

The composite  $d \circ b : \text{Ord} \rightarrow V$  is a class surjection. Assign to each set  $x$  the least ordinal  $\alpha$  with  $d(b(\alpha)) = x$ . These indices define a set-like global well-order of  $V$ : predecessors of  $x$  occur among the set of decoded values before its index. Taking the least member of each nonempty set in that well-order gives a global choice function.  $\square$

Thus existence of the global baseline is not a choice-free substitute for global choice. The equivalence is a standard well-founded-coding argument, included to make the axiomatic cost explicit. It says nothing similar about merely well-ordering one birthday-bounded set.

### 13.2.1 Why non-set-like global well-orders are different

Not every class well-order is set-like. For example,  $\text{Ord} + 1$  and  $\text{Ord} + \text{Ord}$  are class well-orders; the final point in the first example, and every point in the second block of the second example, has a proper class of predecessors. Global choice permits transfer of these orders to the underlying class  $\mathbf{No}$ .

Such orders have no exhaustive enumeration by ordinal positions alone. They are deliberately excluded from  $\mathfrak{W}_{\text{sl}}$ . Even two presentations of orders with a full first block of type  $\text{Ord}$  can agree throughout that block and differ later; there is then no ordinal position at which their full presentations first differ.

There is a different, fully explicit way to compare all class well-order *relations*, including non-set-like ones: fix a class bijection  $c : \text{Ord} \rightarrow \mathbf{No} \times \mathbf{No}$  and compare the binary characteristic classes of the relations at their first differing  $c$ -coordinate. This is a legitimate comparison formula on class variables. However, it depends on the auxiliary coordinate enumeration and is not the exhaustive-enumeration lexicographic order analyzed here. Its collection of objects remains of the same higher logical level.

[merge] The relation-code comparison of source 09's last paragraph is source 08's [proposition 17.1](#), which computes its equimorphism type. Source 09's research direction on it ([section 113.3](#)) asks for more: a dense set-coded core and the representation of its cuts.

## 14 Lexicographic comparison without class ordinals

The set-sized enumeration method studied for the real numbers [1] suggests comparing the first unequal labels. Arbitrary class well-orders require a formulation that also permits a proper class of common earlier labels. No ordinal enumeration of that entire common prefix is needed.

For a strict class well-order  $R$  on a fixed class  $X$ , write

$$P_R(x) = \{y \in X : yRx\}.$$

A common labeled prefix of  $R, S$  is a subclass initial in both orders on which their restrictions agree. The adjective labeled matters: agreement here is equality of the actual elements and relations, rather than the existence of an abstract order isomorphism.

**Lemma 14.1** (Unique first disagreement). *If  $R \neq S$  are class well-orders of the same class  $X$ , there are unique distinct  $a, b \in X$  such that*

$$P_R(a) = P_S(b) = C, \quad R \upharpoonright C = S \upharpoonright C.$$

*The subclass  $C$  is their maximal common labeled prefix.*

*Proof.* The class  $D = \{x : P_R(x) \neq P_S(x)\}$  is nonempty, since predecessor classes determine a strict order. Let  $a$  be its  $R$ -least element and put  $C = P_R(a)$ . For every  $y \in C$ ,  $P_R(y) = P_S(y)$ . In particular,  $ySa$ : otherwise  $aSy$ , which would imply  $aRy$ , contradicting  $yRa$ . Thus  $C \subseteq P_S(a)$ . Also, if  $zSy$  with  $y \in C$ , then  $zRy$ , so  $z \in C$ . Therefore  $C$  is  $S$ -initial as well as  $R$ -initial, and the restrictions of the orders to it agree.

The inclusion  $C \subseteq P_S(a)$  is strict because  $a \in D$ . Let  $b$  be the  $S$ -least element outside  $C$ . The strict inclusion shows that  $bSa$ , hence  $b \neq a$ . Initiality and the minimality of  $b$  give  $P_S(b) = C$ .

For uniqueness, suppose  $a', b', C'$  satisfy the displayed conditions with  $a' \neq b'$ . Every element of  $C'$  has identical predecessor classes in the two orders. But  $b'Sa'$ , so the predecessor classes of  $a'$  differ. Consequently  $a'$  is exactly the  $R$ -least element of  $D$ ; hence  $a' = a$ ,  $C' = C$ , and  $b' = b$ . Any common prefix properly extending  $C$  would have to include incompatible next labels, so  $C$  is maximal.  $\square$

Fix now a native linear order  $<_X$  on the labels. For distinct  $R, S$ , define  $R <_{\text{lex}} S$  when the labels supplied by Lemma 14.1 satisfy  $a <_X b$ .

**Theorem 14.2** (Raw lexicographic order). *This rule is a strict linear order on the class well-orders of  $X$ , understood as a definable relation on class variables. In particular it defines  $\mathfrak{W}_{\text{all}}$  for  $X = \mathbf{No}$  with its native numerical order.*

*Proof.* Existence and uniqueness of first disagreement give totality and asymmetry; the relation is irreflexive by definition. To verify transitivity, suppose  $R <_{\text{lex}} S <_{\text{lex}} T$ . Write  $(a, b)$  for the first unequal labels of  $R, S$ , and  $(c, d)$  for those of  $S, T$ . Their common prefixes are  $C = P_S(b)$  and  $D = P_S(c)$ , respectively, and  $a <_X b$ ,  $c <_X d$ .

If  $bSc$ , then  $C \subsetneq D$ . The orders  $S, T$  agree through the label  $b$ , so the first unequal labels of  $R, T$  are  $a, b$ , with common prefix  $C$ . Thus  $R <_{\text{lex}} T$ . If  $cSb$ , then  $D \subsetneq C$ ; now  $R, S$  agree through  $c$ , and the first unequal labels of  $R, T$  are  $c, d$ , again giving the conclusion. If  $b = c$ , the prefixes coincide and the next labels of  $R, T$  are  $a, d$ . Since  $a <_X b = c <_X d$ , the conclusion follows in this case as well.  $\square$



This proof compares labeled relations directly. It does not compare arbitrary class well-orders up to isomorphism, and makes no appeal to a collection of class ordinal types. The defining conditions use only set quantifiers with the given class relations as parameters: equality of predecessor classes means a universal statement about their set elements.

For example, the predicate that  $a, b$  are the first unequal labels can be written without quantifying over a common-prefix class:

$$\begin{aligned} a \neq b \quad \wedge \quad & \forall z (zRa \leftrightarrow zSb) \\ & \wedge \quad \forall x \forall y ((xRa \wedge yRa) \longrightarrow (xRy \leftrightarrow xSy)). \end{aligned}$$

Adding  $a <_X b$  and existentially quantifying the two labels gives the lexicographic predicate. Consequently its restriction to the indexed family  $\mathcal{R}$  described above is a GB class of pairs of indices. If the indexing contains duplicate orders, this relation orders the distinct relations, and becomes a strict linear order on indices whenever the indexing is injective. Passing instead to abstract isomorphism types would discard exactly the label information on which the construction depends.

**Proposition 14.3** (Convex prefix cones). *Fix a well-ordered subclass  $(C, P)$  of  $X$ . The collection of all class well-orders of  $X$  extending  $(C, P)$  as an initial labeled segment is convex in the raw lexicographic order.*

*Proof.* Suppose  $R, S$  extend the prefix and  $R <_{\text{lex}} T <_{\text{lex}} S$ . Let  $D$  be the first-disagreement prefix of  $R, T$ . The subclasses  $C, D$  are initial in  $R$ , and therefore inclusion-comparable. If  $C \subseteq D$ , then  $T$  extends  $(C, P)$ , as required. If  $D \subsetneq C$ , the next  $R$ -label lies in  $C$ ; it is also the next  $S$ -label after  $D$ . Accordingly,  $T$  compares with  $R$  and  $S$  in the same direction, contradicting the strict inequalities. This reasoning permits  $C$  and  $D$  to be proper classes.  $\square$

[merge] *Second route (source 12): the largest common labelled prefix.* Source 12's Section 3 proves lemma 14.1 and theorem 14.2 differently. Instead of the class  $D$  of points whose predecessor classes differ, it defines the largest common labelled prefix  $C(R, S)$  by an elementary formula and proves directly that it is initial in both orders and contains every common labelled initial segment. Source 11's shipped check compares its predecessor rule with finite lexicographic order on all 266,272 pairs of permutations of at most six points (data/11-raw-orders-finite\_checks\_results.txt); source 12's checks the common-prefix formula on 15,017 finite pairs (data/12-singular-finite\_checks.json).

## 14.1 Source 12's route: the largest common labelled prefix

The central observation is that comparing two *labelled* well-orders is easier than comparing arbitrary abstract class order types. Fix a class  $X$  equipped with an ambient strict linear order  $<$ , and let  $R, S$  be class well-orders of the same carrier  $X$ .

**Definition 14.4** (Common labelled prefix). Define  $C(R, S) \subseteq X$  by putting  $x \in C(R, S)$  if

$$\begin{aligned} \text{Pred}_R(x) &= \text{Pred}_S(x), \\ \forall u, v \in \text{Pred}_R(x), \quad & uRv \iff uSv. \end{aligned}$$

Equality of the predecessor classes is expanded as a quantification over sets  $u \in X$ . Thus this is an elementary formula with class parameters  $R, S$ ; it is not a quantification over all possible common prefixes.

**Lemma 14.5** (Maximal-prefix lemma).  *$C(R, S)$  is an initial segment of both well-orders, their restrictions to it agree, and it contains every common labelled initial segment. If  $R \neq S$ , it is proper, and the least elements*

$$a = \min_R(X \setminus C(R, S)), \quad b = \min_S(X \setminus C(R, S))$$

*exist and are distinct.*

*Proof.* Suppose  $x \in C(R, S)$  and  $yRx$ . Equality of the predecessor classes puts  $y$  before  $x$  in  $S$  as well. For  $zRy$ , transitivity gives  $zRx$ ; agreement of  $R, S$  below  $x$  then gives  $zSy$ . Conversely,  $zSy$  implies  $zSx$ , hence  $zRx$ , and agreement below  $x$  yields  $zRy$ . The orders agree below  $y$  because they agree below  $x$ . Therefore  $y \in C(R, S)$ . The symmetric argument proves initiality in  $S$ .

If  $u, v \in C(R, S)$  and  $uRv$ , equality of the predecessor classes of  $v$  gives  $uSv$ . Thus the restrictions agree. If  $D$  is any common labelled initial segment and  $x \in D$ , both predecessor classes of  $x$  lie in  $D$  and agree, as do the restricted relations. Hence  $D \subseteq C(R, S)$ . If  $C(R, S) = X$ , equality of every predecessor class implies  $R = S$ .

For  $R \neq S$ , class well-foundedness supplies  $a, b$ . By initiality and minimality,  $\text{Pred}_R(a) = C(R, S)$  and  $\text{Pred}_S(b) = C(R, S)$ . If  $a = b$ , these predecessor classes agree and the two relations agree on them. The defining formula would put  $a$  in  $C(R, S)$ , a contradiction.  $\square$

**Definition 14.6** (Full lexicographic comparison). For distinct  $R, S$ , put  $R <_{\text{lex}} S$  precisely when  $a < b$ , with  $a, b$  from lemma 14.5. Identical relations are equal in this order.

**Theorem 14.7** (Full class lexicographic order). *In GBC, the preceding comparison is a strict linear ordering of the predicate of all class well-orders of a fixed linearly ordered class  $X$ . Its definition uses only elementary class comprehension. In particular, ETR and the comparability of abstract class well-order types are not required.*

*Proof.* Irreflexivity and trichotomy follow from the maximal-prefix lemma and trichotomy in  $X$ . For transitivity suppose  $R <_{\text{lex}} S <_{\text{lex}} T$ . The classes  $C(R, S)$  and  $C(S, T)$  are initial segments of the same well-order  $S$ , so they are comparable by inclusion.

If  $C(R, S)$  is a proper subset of  $C(S, T)$ , the first  $S$ -element  $b$  after  $C(R, S)$  belongs to  $C(S, T)$ . Thus  $T$  agrees with  $S$  through that element. The first disagreement of  $R, T$  is exactly the earlier  $R, S$  disagreement, with the same two labels  $a < b$ . Consequently  $R <_{\text{lex}} T$ . The opposite proper-inclusion case uses the earlier  $S, T$  disagreement and is identical in form.

If the two common prefixes are equal to  $C$ , let  $a, b, c$  be the first labels after  $C$  in  $R, S, T$ . The hypotheses give  $a < b < c$ . Hence  $R, T$  also first disagree after  $C$ , and  $R <_{\text{lex}} T$ . Every construction of  $C$ , and every assertion about its next elements, expands into set quantifiers with the given class relations as parameters.  $\square$

*Remark 14.8* (Why this does not prove class order-type comparability). In general one asks whether two arbitrary class well-orders admit an initial-segment isomorphism. ETR is a sufficient hypothesis for that principle [27, Theorem 6]. Here no such isomorphism is constructed. Only their *identical labelled* portions are compared. As soon as the labels disagree, the fixed ambient order decides the answer. Confusing these two tasks would impose an unnecessary stronger axiom.

**Example 14.9** (A disagreement after every ordinary ordinal). Fix  $a < b$  in **No**. Well-order **No**  $\setminus \{a, b\}$  in type Ord. Append first  $a, b$ , and in a second order append  $b, a$ . The orders agree throughout their initial Ord-block and differ only after it. No ordinary ordinal indexes that first disagreement, but theorem 14.7 compares them immediately: their common labelled prefix is **No**  $\setminus \{a, b\}$ .

**Corollary 14.10** (Agreement with enumeration lexicography). *For set-sized exhaustive ordinal enumerations, and for class bijections  $\text{Ord} \rightarrow X$ , the common-prefix definition agrees with first-difference lexicography.*

*Proof.* Before the first unequal entry, exactly the same labels occur in exactly the same order. Their ranges form the common labelled initial segment. No later label can belong to that segment by initiality and the distinct next labels. In the  $\text{Ord}$  case the disagreement is a nonempty class of ordinals; restricting it below any one witness gives its least member.  $\square$

[merge] Source 12’s [remark 14.8](#) is source 11’s remark after [theorem 14.2](#) and in [section 20](#): comparability of abstract class well-orders by initial-segment isomorphisms follows from ETR [27], and neither labelled comparison needs it. Its [example 14.9](#) is source 11’s [example 16.3](#), and its [corollary 14.10](#) is the corresponding sentence after [definition 19.1](#).

## 15 The canonical initial set-like part

Every class well-order of  $\mathbf{No}$  has a canonical initial segment of type  $\text{Ord}$ , even when its entire order is not set-like. This gives a structural description of all well-orders using ordinary class functions.

**Theorem 15.1** (Canonical initial part). *For  $R \in \mathfrak{W}_{\text{all}}$ , let*

$$I_R = \{x \in \mathbf{No} : P_R(x) \text{ is a set}\}.$$

*Then  $I_R$  is a proper initial subclass of the domain or is the entire domain; in either case it is a proper class. Its induced order is set-like and has a unique order enumeration*

$$F_R : \text{Ord} \longrightarrow I_R.$$

*The order  $R$  is the concatenation of this enumeration with its induced well-order on  $\mathbf{No} \setminus I_R$ .*

*Proof.* If  $x \in I_R$  and  $yRx$ , then  $P_R(y) \subseteq P_R(x)$ , so separation makes  $P_R(y)$  a set. Thus  $I_R$  is initial. Suppose it were a set. Since  $\mathbf{No}$  is proper, there would be an  $R$ -least element  $x$  outside it. Every predecessor of  $x$  would then belong to  $I_R$ , and initiality would place every member of  $I_R$  before  $x$ . Hence  $P_R(x) = I_R$ , a contradiction.

The induced order on  $I_R$  has the same predecessor sets as  $R$ . The ordinal-rank map therefore identifies it with an initial subclass of  $\text{Ord}$ . As in [Theorem 13.1](#), a set-sized range would make  $I_R$  a set by replacement. Its range is consequently all of  $\text{Ord}$ . The inverse map is unique, and initiality gives the asserted concatenation.  $\square$

**Example 15.2** (Two proper blocks). Under GBC, fix  $e : \text{Ord} \rightarrow \mathbf{No}$  bijective. The disjoint classes  $A = \{e(2\alpha) : \alpha \in \text{Ord}\}$  and  $B = \{e(2\alpha + 1) : \alpha \in \text{Ord}\}$  partition  $\mathbf{No}$ , where the product is ordinal multiplication  $2 \cdot \alpha$ . Order each block by its displayed ordinal index and place all of  $A$  before all of  $B$ . The resulting well-order has class type  $\text{Ord} + \text{Ord}$ . Its canonical initial part is exactly  $A$ : a point of  $B$  has the whole proper class  $A$  among its predecessors. Hence an  $\text{Ord}$ -enumeration of the canonical part is not, in general, an enumeration of the entire well-order. Both blocks consist of ordinary surreal labels; their size affects their positions in the chosen well-order, not the status of individual surreal numbers as sets.

Let  $\mathfrak{J}$  denote the metacollection of injective class functions  $F : \text{Ord} \rightarrow \mathbf{No}$ , with lexicographic comparison at their first ordinal of disagreement. Such a first disagreement exists for distinct functions: after choosing any witness, minimize within the set of earlier ordinals. Put  $Y_F = \mathbf{No} \setminus \text{ran}(F)$ .

**Proposition 15.3** (Convex fibers). *Assume GBC. Every  $F \in \mathfrak{J}$  occurs as  $F_R$  for some  $R \in \mathfrak{W}_{\text{all}}$ . The fiber over  $F$  is a convex copy of  $\mathcal{W}(Y_F)$ . Distinct fibers are ordered by the first unequal values of their functions  $F$ .*

*Proof.* The range of an injection from Ord is proper, since a set could not contain an injective image of all ordinals. Choose a well-order  $Q$  of  $Y_F$ , using the restriction of any fixed global well-order, and put the sequence  $F$  before  $Q$ . Each  $F(\alpha)$  has exactly the set  $F^{\alpha}$  of predecessors. Each residual point, however, has the proper class  $\text{ran}(F)$  among its predecessors. Thus the resulting order has canonical function exactly  $F$ .

All orders with that function extend the same initial labeled order on  $\text{ran}(F)$ , and conversely every extension has that canonical function. Proposition 14.3 proves convexity. Restriction to  $Y_F$  and concatenation with  $F$  are inverse operations; first disagreement inside the residual orders proves that they preserve lexicographic comparison. Finally, if  $F \neq G$ , their first ordinal disagreement occurs before either residual order begins. It therefore determines the comparison of every member of one fiber with every member of the other.  $\square$

We may summarize this result as the lexicographic-sum description

$$\mathfrak{W}_{\text{all}} \cong \sum_{F \in (\mathfrak{J}, <_{\text{lex}})} \mathcal{W}(Y_F).$$

This is uniform metanotation for the preceding theorem, not the assertion that its indexing hyperclass or all its fibers have been collected into a GB class. The set-like fragment  $\mathfrak{W}_{\text{sl}}$  consists exactly of the fibers with  $Y_F = \emptyset$ , or equivalently the exhaustive enumerations. For finite  $Y_F$  of size  $m$ , its fiber is a convex block of  $m!$  orders, with the usual lexicographic order on permutations.

*Remark 15.4* (Singleton fibers and iteration). A singleton fiber need not be set-like. If  $F$  omits exactly one surreal  $z$ , its unique completion puts  $z$  after the entire Ord-sequence; it has class order type  $\text{Ord} + 1$ . Conversely, every exhaustive  $F$  gives a singleton set-like fiber. Extracting the initial part once also does not justify transfinite iteration of this operation through class-sized tails. Such an iteration would require a separate specification of its indexing order, its objects, and its recursion principle; none is asserted here.

## 16 All jumps arise from finite residual permutations

The full order has jumps despite the extreme density of the native surreal line. A proper common prefix can already have used every label between the two labels at which comparison is decided.

**Lemma 16.1** (Extreme enumerations). *For a class  $Z$  with a native linear order, a class well-order is lexicographically greatest among all well-orders of  $Z$  exactly when it is the native reverse order. It is lexicographically least exactly when it is the native forward order.*

*Proof.* If a well-order places  $y$  before  $z$  with  $y < z$ , transpose these two labels throughout the order. The result is another class well-order. Its first changed label is  $z$  in place of  $y$ , so it is lexicographically greater. A greatest order must therefore be strictly decreasing in the native order, which determines the entire relation. Conversely, in the reverse native order the next label at any first disagreement is the greatest remaining label, so every different well-order is smaller. The least-order assertion follows by reversing the native order. The argument includes empty and singleton domains.  $\square$

The lemma is an existence criterion as well as a characterization: a greatest enumeration exists only when the native reverse order itself is well-founded. It is therefore not permissible to use “list

the remaining labels in decreasing order” as an unconditional completion rule. For example, on a copy of the natural numbers with its usual native order, the increasing enumeration is least, but there is no greatest well-order. Finite residual domains have both extrema, and the adjacency theorem shows that precisely such domains can support a jump after a common prefix.

**Theorem 16.2** (Exact adjacency criterion). *Let  $R <_{\text{lex}} S$  be class well-orders of  $X$ . Let  $C$  be their first-disagreement prefix, let  $a < b$  be their respective next labels, and put  $Y = X \setminus C$ . They are adjacent if and only if all three conditions hold:*

1. *there is no  $y \in Y$  with  $a < y < b$ ;*
2. *the  $R$ -tail on  $Y \setminus \{a\}$  is the native reverse order;*
3. *the  $S$ -tail on  $Y \setminus \{b\}$  is the native forward order.*

*These conditions force  $Y$  to be finite. Thus adjacency is exactly ordinary finite permutation adjacency after a possibly proper class common prefix.*

*Proof.* Suppose first that the pair is adjacent. A label  $y \in Y$  strictly between  $a, b$  could be placed immediately after  $C$ , followed by the restriction of  $R$  to the other remaining labels. This would give a well-order strictly between  $R, S$ ; thus condition 1 holds.

Any lexicographically greater replacement of the  $R$ -tail after  $a$  would also give an intermediate order: its next label after  $C$  would still be  $a < b$ . Hence this tail is greatest, and Lemma 16.1 gives condition 2. A lexicographically smaller replacement of the  $S$ -tail after  $b$  similarly gives condition 3. These modifications use restrictions and finite transpositions of existing class relations; they require no new choice principle.

The overlap  $Z = Y \setminus \{a, b\}$  is now well-ordered both forward and backward in the native order. If  $Z$  were not finite, repeatedly take its least element after deleting the finitely many already selected. Ordinary recursion on  $\omega$  and replacement produce a set of strictly increasing selected elements. That set has no greatest member, contradicting reverse well-ordering. Thus  $Z$ , and hence  $Y$ , is finite. This argument does not assume that  $Z$  was a set in advance.

Conversely, suppose the three conditions hold and let  $R <_{\text{lex}} T <_{\text{lex}} S$ . Convexity of the prefix cone forces  $T$  to extend  $C$ . Let  $t \in Y$  be its next label. A label below  $a$  makes  $T < R$ , and a label above  $b$  makes  $T > S$ ; condition 1 excludes intermediate labels. If  $t = a$ , its remaining tail cannot exceed the greatest tail of  $R$ , so  $T \leq R$ . If  $t = b$ , its tail cannot precede the least tail of  $S$ , so  $T \geq S$ . Every possibility contradicts strict betweenness.  $\square$

**Example 16.3** (An adjacent pair of type  $\text{Ord} + 2$ ). Assume GBC, choose surreal labels  $a < b$ , and give  $C = \mathbf{No} \setminus \{a, b\}$  a set-like well-order  $P$ . Consider

$$R = P + (a, b), \quad S = P + (b, a).$$

Each has class order type  $\text{Ord} + 2$ , and  $R, S$  are adjacent. They have the same canonical initial enumeration, and constitute its entire two-point fiber. The notation  $\text{Ord} + 2$  describes a class concatenation; it is not a set ordinal.

For  $X = \mathbf{No}$ , every adjacent pair has a cofinite, hence proper class, common prefix. Neither endpoint can be set-like. Moreover, if two set-like orders are given, their common prefix is a set. The deletion property of the surreal line leaves an unused label strictly between their next labels, and that set prefix can be completed to a set-like well-order. For completeness, this extension does not require an additional recursion principle. Retain the prescribed set prefix, put the new intermediate label next, and then append the restriction of a fixed set-like global well-order of  $\mathbf{No}$  to the unused labels. A predecessor class in the appended part is contained in the union of the prescribed set

prefix and a predecessor set of the fixed order, together with the one inserted label. It is therefore a set. Every label occurs once, and the concatenation is a well-order, so the completion belongs to  $\mathfrak{W}_{\text{sl}}$ . Thus the induced order  $\mathfrak{W}_{\text{sl}}$  is dense, while  $\mathfrak{W}_{\text{all}}$  has the finite terminal blocks classified above. Allowing all well-orders changes the local order structure in a concrete and unavoidable way.

[merge] *Second route (source 12): residual decomposition and adjacency.* Source 12's Section 8 proves [theorem 16.2](#) in the form “the residual class is finite and the residual permutations are consecutive”; the two criteria are equivalent, since a tail that is both well-ordered and reverse well-ordered is finite. Its [proposition 16.5](#) is the description of one prefix cone given by [propositions 14.3](#) and [15.3](#). Its [corollary 16.7](#) adds that interior points of the finite blocks with  $n \geq 3$  are isolated in the order topology and that the set-like suborder is not dense in the full order, and its last paragraph adds the order-reversing involution induced by negation.

### 16.1 Source 12's route: unrestricted class well-orders and finite-tail geometry

The full order  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  contains  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  as an ordered subcollection, but it is not dense. The effect is not a failure of density of the surreal alphabet; it is exhaustion of almost all labels by a proper-class prefix.

**Definition 16.4** (A prefix cylinder in the full order). Let  $C \subsetneq X$  carry a fixed well-order  $P$ . Consider all class well-orders of  $X$  that have  $(C, P)$  as a labelled initial segment. Write this cylinder as  $[P]_X$  and put  $D = X \setminus C$ .

**Proposition 16.5** (Residual-order decomposition).  *$[P]_X$  is convex and is order-isomorphic to the lexicographic order of all well-orders of the residual carrier  $D$ , with its inherited ambient order. If  $D$  is a set, this is the ordinary set order  $\mathcal{W}(D)$ , with the same notation extended to finite carriers. If  $|D| = n < \omega$ , the cylinder has exactly  $n!$  elements.*

*Proof.* Restriction to  $D$  and ordered concatenation with  $P$  are inverse operations. Concatenation of two class well-orders is a class well-order: a nonempty subclass has its minimum in the first part if it meets that part, and otherwise in the second. The maximal common prefix after  $P$  is exactly the common prefix of the tails, so comparisons are preserved.

If a third order differs before completing  $P$ , its first discrepancy with every extension of  $P$  is the same and places it on the same side of all of them. This proves convexity. The last assertion counts permutations of a finite carrier.  $\square$

**Theorem 16.6** (Exact adjacency criterion). *Let  $R <_{\text{lex}} S$  be two class well-orders of  $X$ , and let  $C = C(R, S)$ ,  $D = X \setminus C$ . They are adjacent in the full order if and only if  $D$  is finite and their residual permutations are consecutive in the ordinary finite lexicographic order.*

*Proof.* Let  $a < b$  be their first residual labels. If some  $d \in D$  satisfies  $a < d < b$ , begin the residual order with  $d$  and well-order the remaining labels by a restriction of  $R$ . This gives an order strictly between. So an adjacent pair can have no such  $d$ .

If the tail of  $R$  after  $a$  has two labels  $u$  before  $v$  with  $u < v$  in the ambient order, interchange the labels  $u, v$  at their two positions. The new well-order is larger than  $R$ , and still smaller than  $S$  because its first residual label remains  $a$ . Thus adjacency forces the  $R$  tail to be numerically decreasing. Dually, the tail of  $S$  after  $b$  must be numerically increasing.

On  $D \setminus \{a, b\}$ , both the ambient numerical order and its reverse are therefore well-orders. Such a carrier is finite: an infinite well-order contains an increasing  $\omega$ -sequence, contradicting well-foundedness of its reverse. Hence  $D$  is finite.



Conversely, if  $D$  is finite, convexity in [proposition 16.5](#) says that every order between  $R, S$  must be another permutation of  $D$  after  $C$ . There is none exactly when the two permutations are consecutive.  $\square$

**Corollary 16.7** (Proper-class prefixes are responsible). *Adjacent pairs in  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  share a proper-class prefix. There are convex blocks of every factorial size  $n!$ . For  $n \geq 3$ , interior elements of such blocks are isolated in the order topology. The set-like suborder is not dense in the full order.*

*Proof.* A set prefix cannot exhaust all but finitely many surreals. Conversely, well-order  $\mathbf{No} \setminus D$  in type Ord for any finite  $D$ , and append all its permutations. For  $|D| \geq 3$ , choose two nonconsecutive permutations in the resulting finite convex block. Their nonempty open interval contains only orders with a proper-class predecessor segment, so it contains no set-like order.  $\square$

The full order still has no endpoints and no set-indexed cofinal or cointial family, by the first-entry argument of [note 19.28](#). It contains every class linear order, by [note 19.21](#). But neither global density nor interpolation for all set-indexed cuts survives: an adjacent pair already gives a two-sided cut that cannot be filled.

Pointwise surreal negation gives an order-reversing involution on both global orders: transport a well-order by the bijection  $x \mapsto -x$ . This preserves enumeration type and set-likeness while reversing the ambient comparison of the first unequal labels.

[merge] Source 08’s Section 14 proposes two alternatives to the labelled comparison of [section 14](#): a fixed larger index order, and relation-code lexicography.

## 17 Non-set-like well-orders and alternative coding (source 08)

The restriction to set-like well-orders is substantive, not cosmetic. Choose distinct surreals  $a < c$ , and enumerate  $\mathbf{No} \setminus \{a, c\}$  by Ord. Append  $a, c$  at the end in one well-order and  $c, a$  in another. These are legitimate class well-orders of type Ord + 2, but they are not set-like: the final points have a proper class of predecessors.

The two enumerations agree at every *set ordinal* position. Their first disagreement occurs only after a proper-class initial segment, at the first appended point. Thus ([S08.7](#)), whose index is Ord, does not compare them. Treating “Ord” as one more set ordinal would conceal precisely the error.

### 17.1 A fixed larger index order

One can fix a class well-order  $J$ , such as Ord + 2, and compare class bijections  $J \rightarrow \mathbf{No}$  at their first difference, using the order on  $J$  and the numerical order on  $\mathbf{No}$ . For the indicated concrete  $J$  this is entirely rigorous. For a general  $J$ , the existence of least members for the disagreement subclasses must be part of the specified class well-order property.

What does not follow is a canonical single comparison for all bijections having all possible non-set-like class well-order domains. Unlike set ordinals, the domains do not automatically come with a common set-ordinal coordinate system. A comparison theorem for all such domains, or a recursive alignment of their proper-class initial segments, needs an explicit formulation and an axiomatic audit. We do not assume an unproved class analogue of every set well-order theorem.

### 17.2 Relation-code lexicography works, but changes the question

There is an alternative that does not align enumeration domains. Fix a class bijection

$$q : \text{Ord} \longrightarrow \mathbf{No} \times \mathbf{No}.$$

Encode each strict class relation  $R$  on  $\mathbf{No}$  by  $\chi_R(\xi) = 1$  iff  $q(\xi) \in R$ . Compare these characteristic functions lexicographically. This gives a comparison of *all* class well-order relations, set-like or not. It depends on  $q$ , and is not generally the numerical lexicographic comparison of their exhaustive enumerations.

**Proposition 17.1.** *For every fixed  $q$ , the virtual order of all class well-order relations under relation-code comparison is equimorphic with  $\mathbb{B}_{\text{Ord}}$ . The lower embedding can be chosen to use only set-like well-order relations.*

*Proof.* The upper embedding is the characteristic-function code itself. For the lower embedding, partition  $\mathbf{No}$  into disjoint two-element blocks using a baseline  $b$ . For each block  $\{a, c\}$ , let  $m$  be the smaller of the two positions where  $q$  lists  $(a, c)$  and  $(c, a)$ . The positions  $m$  are distinct and form a proper subclass of  $\text{Ord}$ ; they have a unique increasing enumeration by  $\text{Ord}$ . Order the blocks by this enumeration.

Within each block, choose its orientation so that the relation value at its first listed directed pair is the prescribed input bit. Order all elements in earlier blocks before all elements in later blocks. This is a set-like well-order, consisting of an  $\text{Ord}$ -sequence of two-element blocks. Reversing a block changes only the two directed relations internal to that block, not any relations between blocks. At the first differing input bit, its associated  $m$  is therefore the first changed coordinate of the characteristic function. The code preserves order. All selections and enumerations are uniform relative to  $q, b$ .  $\square$

This supplies two coherent meanings of “lexicographic ordering of all well-orders”: numerical enumeration comparison on the set-like part, and relation-code comparison on all class well-orders. They have uniform embeddings into one another through the binary class order, but their definitions, canonicity properties, and possible topological invariants should not be conflated.

[merge] Dated note, 2 October 2026. Source 08 says (its Section 14.1) that a canonical single comparison for all bijections from all non-set-like class well-order domains does not follow without an explicit alignment of proper-class initial segments, and asks (its question 11) for a comparison principle for arbitrary class well-orders and its recursion requirements. Sources 11 and 12 supply one without aligning domains: compare the two relations on their largest common labelled prefix, then compare the next labels ([theorems 14.2](#) and [14.7](#)); its definition uses elementary comprehension only, not ETR and not comparability of abstract class order types. For the pair of type  $\text{Ord} + 2$  displayed at the start of this section, the comparison is decided at the first appended point ([example 16.3](#), [example 14.9](#)). Source 08’s warning stands for the alignment of abstract domains, and relation-code lexicography remains a different order ([section 18](#) makes the same point on three points).

## 18 Numerically well-ordered subclasses and other interpretations (source 12)

The positive surreal ordinals form a numerically increasing class copy of  $\text{Ord}$ , but this is not a well-order of all of  $\mathbf{No}$ . More generally, [lemma 5.14](#) implies that every class well-order is isomorphic to a numerically well-ordered subclass of  $\mathbf{No}$ . That subclass need not be set-like in its inherited numerical order.

For an explicit example, the class

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

has numerical order type  $\text{Ord} + 1$ . The arithmetic here is surreal arithmetic on the embedded ordinal values; the reciprocal reverses positive order and the minus sign reverses it back. The final point has a proper class of predecessors. Putting a positive ordinal block after this negative block gives a numerical copy of  $\text{Ord} + \text{Ord}$ .

For a set  $A \subseteq \mathbf{No}$ , numerical well-ordering has the usual ordinal order type. Its birthday bound places it inside some  $\mathbf{No}_{<\kappa}$ . The class of all such set-sized well-ordered subsets is a legitimate class of sets. Its members are not exhaustive enumerations of a fixed carrier, however, so one must specify what happens when one enumeration is a prefix of another. The “shorter first” convention and the surreal  $- < \mathbf{end} < +$  convention are different choices; neither can be silently borrowed from the prefix-free exhaustive setting.

Finally, ordering binary relation tables rather than enumerations is another legitimate but different construction. If one fixes a set-like well-order of  $\mathbf{No} \times \mathbf{No}$ , compares the characteristic functions of relations at the first unequal table entry, and restricts to well-orders, one gets a table-dependent order. A first changed relation bit is not generally the first changed enumerated label. Even on three points the two lexicographic orders can differ. The present article uses labelled initial segments, not relation-table lexicography.

[merge] Source 12’s shipped check records a three-point example on which relation-table lexicography and enumeration lexicography differ (`data/12-singular-finite_checks.json`, key `relation_table_vs_enumeration_example`).

## Part IV

# Set-like global well-orders

## 19 The lexicographic structure of global enumerations

The set-like case has a particularly strong structure theorem. Although the full collection of global enumerations is a higher-order object, one ordinary class of set codes already meets every cut specified by set-indexed families. That class, with its induced order, is isomorphic to the surreal line. The distinction between a set-indexed family and an ordinal-indexed class family is essential: an explicit omitted-range construction shows exactly where the interpolation theorem stops.

Throughout this section we work in GBC and fix a class bijection

$$e_0 : \text{Ord} \longrightarrow \mathbf{No}.$$

The existence of such a baseline is supplied by global choice; the relevant equivalences are discussed in [19]. We use the surreal set-cut property, its preservation under deleting a set (Lemma 5.2), and the universality and uniqueness statements of Proposition 5.13. These are order-theoretic properties; the field operations on  $\mathbf{No}$  play no role in the main construction.

### 19.1 Class parameters and uniform families

**Definition 19.1.** For this section,  $\mathfrak{J}$  denotes the collection of class injections  $F : \text{Ord} \rightarrow \mathbf{No}$ , and  $\mathfrak{W}_{\text{sl}}$  denotes the subcollection of class bijections. For distinct  $F, G \in \mathfrak{J}$ , put

$$d(F, G) = \min\{\alpha \in \text{Ord} : F(\alpha) \neq G(\alpha)\}, \quad F <_{\text{lex}} G \iff F(d(F, G)) < G(d(F, G)).$$

A *uniformly set-indexed family*  $(F_i)_{i \in I}$  is given by a set  $I$  and one class relation whose  $i$ th row is the graph of  $F_i$ .

These notations are interpreted relationally in two-sorted class theory. An individual  $F$  is a class, not a possible element of a GB class. Thus “for all  $F \in \mathfrak{W}_{\text{sl}}$ ” abbreviates quantification over class functions with the stated property. Likewise, a uniformly indexed family is a single class of triples  $(i, \alpha, x)$ , not a literal set containing proper classes. The first-difference definition is legitimate: choose any coordinate of disagreement and minimize within the set of preceding ordinals. The usual first-difference proof gives a strict linear comparison on these objects.

Uniformity matters whenever a set bound is taken. For example, if two uniformly set-indexed families have different rows across every pair, Class Replacement makes their cross first-difference ordinals a set. No principle of choosing an arbitrary family of classes is being assumed. Finite lists of given class parameters are automatically uniform, and the families of set-coded objects constructed below have a fixed evaluation formula, so their set-indexed subfamilies are uniform as well.

[merge] Sources 09 and 08 define the same comparison of global enumerations (source 09 in [section 19.3](#); source 08 in its Section 9, “The virtual order of global enumerations”).

*Note 19.2* (Source 08, display (7) and Proposition 9.1). For two different global enumerations  $e, f$ ,

$$\delta(e, f) = \min\{\alpha \in \text{Ord} : e(\alpha) \neq f(\alpha)\}, \quad e <_{\text{lex}} f \iff e(\delta(e, f)) < f(\delta(e, f)), \quad (\text{S08.7})$$

and *this comparison satisfies trichotomy and transitivity; it defines a linear comparison in the two-sorted sense*. This is [definition 19.1](#) and the paragraph after it, by the same argument. Source 08’s Section 9.1 makes source 11’s point about uniform families: a family indexed by  $J$  is one class relation  $T \subseteq J \times \text{Ord} \times \mathbf{No}$ ; for a set  $J$ , class replacement bounds all cross first-disagreement coordinates; “merely saying ‘choose one class for every member of a set’ would not supply the uniform relation, and could hide an additional class-choice principle”; with repeated rows one works with the comparison preorder or an injective subfamily.

## 19.2 A class of bounded-support codes

**Definition 19.3.** A *normalized bounded-support code* is a set permutation  $\sigma : \lambda \rightarrow \lambda$  of an ordinal  $\lambda$  satisfying

$$\lambda = \sup\{\xi + 1 : \xi < \lambda \text{ and } \sigma(\xi) \neq \xi\}.$$

The empty permutation is allowed. Extend  $\sigma$  to a class permutation  $\hat{\sigma}$  of  $\text{Ord}$  by the identity outside  $\lambda$ , and evaluate the code as

$$E_\sigma = e_0 \circ \hat{\sigma}.$$

Write  $\mathcal{P}_{\text{bd}} = \mathcal{P}_{\text{bd}}(e_0)$  for the ordinary class of these codes, ordered by the lexicographic comparison of their evaluations.

Normalization removes the otherwise harmless duplication obtained by adding fixed points to the end of a code. Every class permutation with bounded support has exactly one normalized code: restrict it to the supremum of one plus its moved coordinates. Its support is invariant under the permutation, so this restriction is a permutation of the indicated ordinal. Consequently distinct normalized codes evaluate to distinct enumerations, and the induced order on  $\mathcal{P}_{\text{bd}}$  is a genuine class linear order. The class is proper, since the transpositions of 0 with arbitrary nonzero ordinals have distinct codes.

Here and below, identifying a code with its evaluated enumeration is a notational convenience. It does not convert the evaluated proper-class graph into a set. In particular, the assertion that  $\mathcal{P}_{\text{bd}}$  is an ordinary class refers to the codes, while assertions comparing its members with arbitrary members of  $\mathfrak{W}_{\text{sl}}$  use the common evaluation relation.

**Lemma 19.4** (Completion of an arbitrary set prefix). *Every injective set function  $p : \delta \rightarrow \mathbf{No}$ , with  $\delta$  an ordinal, is the initial segment of an enumeration represented in  $\mathcal{P}_{\text{bd}}$ . The completion can be chosen uniformly in  $p$ .*

*Proof.* Let  $q = e_0^{-1} \circ p$ . This is a set injection from  $\delta$  into  $\text{Ord}$ . Choose an infinite initial cardinal  $\gamma$  strictly larger than  $\delta$  and than  $\sup \text{ran}(q)$ . Both complements

$$\gamma \setminus \delta \quad \text{and} \quad \gamma \setminus \text{ran}(q)$$

have cardinality  $\gamma$ , since fewer than  $\gamma$  elements were removed. A bijection between the complements therefore extends  $q$  to a permutation of  $\gamma$ . Extend that permutation by the identity above  $\gamma$ , and normalize its code. Its evaluation agrees with  $p$  at every coordinate below  $\delta$ . To make the construction uniform, fix the least permissible  $\gamma$  and then the least complement bijection under a fixed global well-order. These choices select sets; they do not require a choice principle for families of proper classes.  $\square$

The lemma is stronger than merely extending  $p$  to some global well-order. Even an arbitrarily long set-sized prefix can be realized by changing the baseline only on a bounded set of coordinates. The bound need not be small: it must contain the baseline indices of all values prescribed by  $p$ .

**Source 12’s prefix extension.** [merge] Source 12’s Section 6 (“Set-like global well-orders: interpolation and universality”) proves the results of this section by the same arguments, except for its prefix extension, printed next, which is a different construction: it appends the restriction of the baseline order to the complement of the prefix, so its completion need not have bounded support. Source 12’s interpolation theorem, local characters and local universality are recorded in notes below; its avoidance lemma and its universality lemmas are printed in [section 5](#).

We now restrict to  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ , the class-bijection predicate. All statements about set- or class-indexed collections of its members mean uniform families, unless a larger ambient universe is explicitly specified.

**Lemma 19.5** (Extending a set prefix). *Every injective set sequence  $p : \alpha \rightarrow \mathbf{No}$  extends to a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ . The extension can be chosen uniformly from  $p$  and one fixed baseline bijection  $B$ .*

*Proof.* Let  $A = \text{ran}(p)$ . Restrict the baseline order to  $\mathbf{No} \setminus A$ . It is a proper-class set-like well-order, so it has its canonical increasing enumeration in type  $\text{Ord}$ . Append that enumeration to  $p$ . The class order  $\alpha + \text{Ord}$  is canonically isomorphic to  $\text{Ord}$ . The construction is set-valued recursion with  $B$  as a parameter.  $\square$

[merge] *Source 09’s coding of the core.* Source 09 codes the core by fixed-point-free support permutations ([section 1.3](#)). Its prefix completion [lemma 19.6](#) has the proof of [lemma 19.4](#), and its [remark 19.7](#) warns why a larger cardinal is needed.

## 19.3 Source 09’s coding: canonical support codes and prefix completion

### 19.3.1 Lexicographic comparison of global enumerations

If  $E, F : \text{Ord} \rightarrow \mathbf{No}$  are distinct, choose an ordinal  $\gamma$  where they differ. Their disagreement class intersected with  $\gamma + 1$  is a nonempty set and has a least member. Thus

$$\Delta(E, F) = \min\{\alpha : E(\alpha) \neq F(\alpha)\}$$

is a set ordinal. Define

$$E <_{\text{lex}} F \iff E(\Delta(E, F)) < F(\Delta(E, F)).$$

The same first-disagreement argument as before proves trichotomy and transitivity. This proof quantifies over individual classes, not over a new class whose members would be enumerations.

### 19.3.2 Canonical set codes for eventual perturbations

A *support code* is a set function  $\sigma : S \rightarrow S$ , where  $S \subseteq \text{Ord}$  is a set,  $\sigma$  is bijective, and  $\sigma(\alpha) \neq \alpha$  for every  $\alpha \in S$ . The empty code is allowed. Extend it by the identity outside  $S$ :

$$\hat{\sigma}(\alpha) = \begin{cases} \sigma(\alpha), & \alpha \in S, \\ \alpha, & \alpha \notin S. \end{cases}$$

The code determines a global enumeration

$$E_\sigma = b \circ \hat{\sigma}. \quad (3)$$

Let  $\mathcal{P}_s(b)$  be the class of these codes, ordered by the corresponding  $E_\sigma$ . Because fixed points are omitted, distinct codes yield distinct enumerations. The carrier of  $\mathcal{P}_s(b)$  genuinely consists of sets.

A class enumeration  $E$  occurs this way exactly when it agrees with  $b$  at every sufficiently large ordinal. Indeed any set of ordinals is bounded. Conversely, if  $E$  equals  $b$  beyond a bound, then  $b^{-1} \circ E$  has set support, and that support is invariant under the permutation and its inverse. This gives its unique support code.

**Lemma 19.6** (Bounded-support prefix completion). *Every injective set word  $p : \alpha \rightarrow \mathbf{No}$  is a prefix of some  $E_\sigma$  with  $\sigma \in \mathcal{P}_s(b)$ . More precisely, choose any infinite initial cardinal  $\nu$  strictly above  $\alpha$  and every ordinal in  $b^{-1}[\text{ran } p]$ . Then  $\sigma$  can be chosen with support contained in  $\nu$ .*

*Proof.* Let  $r = b^{-1} \circ p : \alpha \rightarrow \nu$ . Since  $\alpha < \nu$ , both  $\nu \setminus \alpha$  and  $\nu \setminus \text{ran } r$  have cardinality  $\nu$ . Choose a bijection between them. Together with  $r$ , this gives a permutation  $\pi : \nu \rightarrow \nu$ . Extend  $\pi$  by the identity outside  $\nu$  and remove fixed points from its set code. Then  $b \circ \pi \upharpoonright \alpha = p$ .  $\square$

*Remark 19.7* (Why extra unused positions matter). A partial injection need not extend to a permutation on the union of its domain and range. For example,  $n \mapsto n + 1$  on  $\omega$  cannot extend to a permutation of  $\omega$  while retaining every prescribed value. The proof avoids this defect by moving to a larger cardinal  $\nu$ , where both complements have the same cardinality. Merely saying “complete the partial permutation” would leave a genuine gap.

For each ordinal  $\gamma$ , permutations of  $\gamma$  extended by the identity form a set-sized suborder of the core. Increasing  $\gamma$  gives compatible embeddings. Therefore

$$\mathcal{P}_s(b) = \bigcup_{\gamma \in \text{Ord}} \{\text{support codes of permutations of } \gamma\}.$$

Unlike naive unions of  $\mathcal{W}(\mathbf{No}_{<\theta})$ , this is a coherent direct system: the same global enumeration is retained when the support bound is increased.

[merge] Source 08’s Section 10 (“Set-supported permutations and prefix extension”) uses source 09’s support codes, writes  $\mathcal{P}_s$  for their class and  $\mathcal{P}_s(b)$  for the class ordered by

$$s <_b t \iff b \circ \hat{s} <_{\text{lex}} b \circ \hat{t}, \quad (\text{S08.8})$$

an ordinary class relation since codes are sets, and calls  $O_b = \{b \circ \hat{s} : s \in \mathcal{P}_s\}$  the virtual orbit of  $b$ .



*Note 19.8* (Source 08, Lemma 10.1, “Prefix extension inside one orbit”). *Every injective set word  $p : \alpha \rightarrow \mathbf{No}$  is an initial segment of  $b \circ \widehat{s}$  for some  $s \in \mathcal{P}_s$ .* This is [lemma 19.4](#) and [lemma 19.6](#), by the same proof through an initial cardinal  $\theta$  above  $\alpha$  and  $\sup \text{ran}(b^{-1} \circ p)$ , where both complements have cardinality  $\theta$ . Source 08 notes that set choice supplies the complement bijection; that a support below any such  $\theta$  suffices; that an arbitrary partial permutation of a fixed infinite set need not extend to a permutation of that set; and that under global choice the extension can be selected uniformly as the least valid code in a fixed set-like global well-order, “a choice of *sets*, not an invocation of choice among arbitrary classes”.

*Note 19.9* (Source 08, Corollary 10.2). *There is a uniform order embedding  $j_b : \mathbf{No} \hookrightarrow \mathcal{P}_s(b)$  for which  $b \circ \widehat{j_b(x)}$  has first entry  $x$ ; its image is cofinal and cointial in  $\mathfrak{W}_{\text{sl}}$  in the uniform-family sense.* The code is the transposition of 0 and  $b^{-1}(x)$ . This is [proposition 19.20](#) (source 11’s  $E_x$ ) and the first part of source 12’s [note 19.28](#). Source 08: the images of the positive and of the negative ordinals are cofinal and cointial, so there are uniform class-indexed cofinal families although there are no set-indexed ones; “cofinality must not be confused with exhaustive parametrization”.

## 19.4 Interpolation of every uniformly set-indexed cut

**Theorem 19.10** (Prefix interpolation). *Let  $(A_i)_{i \in I}$  and  $(B_j)_{j \in J}$  be uniformly set-indexed families in  $\mathfrak{J}$ , with*

$$A_i <_{\text{lex}} B_j \quad (i \in I, j \in J).$$

*Either index set may be empty. There is an injective set prefix  $p$  such that every member of  $\mathfrak{J}$  extending  $p$  is strictly above every  $A_i$  and strictly below every  $B_j$ . In particular, a member of  $\mathcal{P}_{\text{bd}}$  lies between the two families.*

*Proof.* If both index sets are nonempty, Class Replacement gives a set of ordinals  $d(A_i, B_j)$ . Choose  $\beta$  so that

$$d(A_i, B_j) < \beta \quad (i \in I, j \in J).$$

If one side is empty, put  $\beta = 0$ . We build an injective prefix  $p$ , keeping track of the rows that still agree with it. Call these rows *active*. The invariant is that every discarded  $A_i$  is below every extension of  $p$ , and every discarded  $B_j$  is above every extension of  $p$ .

Suppose  $p$  has length  $\alpha$  and the construction has not stopped. Let

$$L = \{A_i(\alpha) : A_i \text{ is active}\}, \quad R = \{B_j(\alpha) : B_j \text{ is active}\}.$$

These are sets. Since active cross pairs agree before  $\alpha$ , the hypothesis implies  $\ell \leq r$  whenever  $\ell \in L$  and  $r \in R$ . Thus  $L \cap R$  has at most one member: two distinct common values, taken in reverse order as a left and a right value, would contradict this inequality.

If  $L \cap R$  is empty, all cross inequalities are strict. By [Lemma 5.2](#), choose

$$z \in \mathbf{No} \setminus \text{ran}(p), \quad L < z < R.$$

This notation includes an empty left or right side; if both are empty, choose any fresh value. Append  $z$  and stop. Every previously active left row is now strictly below every extension of the new prefix, and every previously active right row is strictly above it. The invariant handles the already discarded rows, proving the desired conclusion.

If  $L \cap R = \{z\}$ , append  $z$  and retain as active exactly the rows whose next value is  $z$ . The new value is fresh because each surviving row is injective. A discarded left value is strictly below  $z$ , since  $z \in R$ ; a discarded right value is strictly above  $z$ , since  $z \in L$ . Hence the invariant is preserved.

At a limit stage take the union of the preceding prefixes; its length is a set ordinal and it remains injective.

The procedure cannot continue with both sides active through a prefix of length  $\beta$ . Such active rows would agree below  $\beta$ , contrary to the bound on every cross first difference. Therefore it stops no later than the next step, giving a set prefix. Lemma 19.4 supplies a bounded-support completion. Notice that surjectivity of the input rows was never used; this is why the theorem was stated for  $\mathfrak{J}$ .  $\square$

All recursion histories in this proof are sets. Even when the input rows are proper classes, their restrictions to a set ordinal are sets, and their next values form sets because the index sets are sets. The proof therefore uses ordinary set-valued transfinite recursion with class parameters, not ETR. The same observation applies to the back-and-forth argument below.

*Note 19.11* (Source 12, Theorem 6.3, “Set-indexed interpolation”). *If  $I, J$  are sets and  $(E_i)_{i \in I}, (F_j)_{j \in J}$  are uniform families in  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  with  $E_i <_{\text{lex}} F_j$  for all  $i, j$ , there is  $G \in \mathfrak{W}_{\text{sl}}(\mathbf{No})$  with  $E_i <_{\text{lex}} G <_{\text{lex}} F_j$ ; empty sides are allowed, so  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  is dense without endpoints in the indexed-family sense. This is theorem 19.10, by the same forced-prefix construction; source 12 completes the prefix by lemma 19.5, so its interpolant need not lie in the bounded-support core.*

[merge] Sources 09 and 08 prove the same theorem with their support codes (source 09’s Section 9, “The interpolation theorem”, and source 08’s Section 11, “The set-cut interpolation theorem”).

*Note 19.12* (Source 09, Theorem 9.1, “Uniform set-indexed interpolation”). *For uniformly set-indexed families  $(E_i) <_{\text{lex}} (F_j)$  of global enumerations ( $I, J$  possibly empty) and every fixed baseline  $b$ , there is  $\sigma \in \mathcal{P}_s(b)$  with  $E_i <_{\text{lex}} E_\sigma <_{\text{lex}} F_j$ . This is theorem 19.10, by the same construction; source 09 bounds the forced prefix by  $\lambda + 1$ ,  $\lambda = \sup\{\Delta(E_i, F_j) + 1\}$ , and notes that the theorem “permits constraints from outside the chosen core, and the interpolant can be required to come from any specified baseline’s eventual-agreement class”.*

*Note 19.13* (Source 08, Theorem 11.1, “Orbit interpolation”). *For fixed  $b$  and uniform families  $(l_i)_{i \in I}, (r_j)_{j \in J}$  of global enumerations,  $I, J$  sets, with  $l_i <_{\text{lex}} r_j$ , there is  $s \in \mathcal{P}_s$  with*

$$l_i <_{\text{lex}} b \circ \hat{s} <_{\text{lex}} r_j \quad (i \in I, j \in J); \quad (\text{S08.9})$$

*either family may be empty.* This is theorem 19.10, by the same active-row invariant (its display (S08.10) is the inequality  $a \leq b$  between active next entries). Source 08 stresses that the separator lies in one fixed orbit “regardless of the supports, or lack of supports, of the input enumerations”.

*Note 19.14* (Source 08, Corollary 11.2). *The virtual order  $\mathfrak{W}_{\text{sl}}$  is dense without endpoints and has no uniformly set-indexed cofinal or coinital family; every orbit  $O_b$  is dense in  $\mathfrak{W}_{\text{sl}}$  and realizes every uniformly set-sized separated cut; the class order  $\mathcal{P}_s(b)$  is an  $\eta_{\text{Ord}}$ -order. This is corollary 19.15 and proposition 19.27, by specializing the interpolation theorem.*

**Corollary 19.15.** *The class order  $\mathcal{P}_{\text{bd}}$  is an  $\eta_{\text{Ord}}$  order and is order-isomorphic to  $(\mathbf{No}, <)$ . Moreover, it meets every strict cut in  $\mathfrak{W}_{\text{sl}}$  specified by uniformly set-indexed families.*

*Proof.* A set of codes is uniformly evaluable, so Theorem 19.10 applies to every separated pair of sets in  $\mathcal{P}_{\text{bd}}$ . Empty sides give the absence of endpoints, and singleton sides give density. We already know that  $\mathcal{P}_{\text{bd}}$  is a proper class. Proposition 5.13 therefore identifies it with  $\mathbf{No}$ .

Explicitly, global choice enumerates each of the two classes by ordinals. Alternately extend a set-sized partial isomorphism to cover the next element of one enumeration and then the next element of the other. The lower and upper images at either step are sets, so the cut property supplies the new point. The union of this ordinary set-valued recursion is an isomorphism. The last assertion is the direct specialization of Theorem 19.10 to bijective input rows.  $\square$

This is an isomorphism of the class of bounded-support codes with the surreal line. It is not an isomorphism between  $\mathbf{No}$  and the full higher-order collection  $\mathfrak{W}_{\text{sl}}$ . Different baselines produce classes with the same abstract order type, but the argument does not single out a canonical isomorphism.

*Note 19.16* (Source 09, Corollary 9.2, “The surreal core theorem”). *In GBC,  $\mathcal{P}_s(b) \cong \mathbf{No}$ . In particular  $\mathcal{P}_s(b)$  is dense without endpoints, every set-sized partial order isomorphism between its suborders extends to an automorphism, and every class linear order embeds into it.* This is [corollary 19.15](#) (with the coding of [section 1.3](#)), with homogeneity from [theorem 5.17](#). Source 09 adds that no set family is cofinal or coinital among all global enumerations, or immediately on either side of a fixed global enumeration, while proper-class cofinal and coinital families exist, with first entries running through cofinal positive and coinital negative ordinals.

*Note 19.17* (Source 08, Theorem 12.2, “Surreal-sized skeleton”). *For every fixed global enumeration  $b$ ,  $\mathcal{P}_s(b) \cong \mathbf{No}$ ; the code order is a proper class, is dense in  $\mathfrak{W}_{\text{sl}}$  through its represented enumerations, and realizes every uniform set cut of  $\mathfrak{W}_{\text{sl}}$ .* This is [corollary 19.15](#), by the same uniqueness theorem ([note 5.19](#)); properness comes from [note 19.9](#). Source 08 adds: “The isomorphism is an isomorphism of *linear orders*, not of fields, of simplicity trees, of birthday filtrations, or of permutation groups. No canonical isomorphism independent of Global Choice and  $b$  is asserted. One could transport a field structure along a selected isomorphism, but it would be additional, choice-dependent structure, not an arithmetic operation naturally defined by lexicographic comparison of enumerations.”

#### 19.4.1 Eventual agreement and interleaved orbits (source 08)

Two enumerations  $e, f$  differ on a set of coordinates exactly when  $e = b \circ \hat{s}$ ,  $f = b \circ \hat{t}$  for some common orbit, equivalently when  $e^{-1} \circ f$  has set support. Any set of ordinal coordinates is bounded, so this is also eventual equality: there is an ordinal  $\alpha$  such that  $e(\xi) = f(\xi)$  for every  $\xi \geq \alpha$ . The equivalence between the support conditions follows from injectivity of the baseline enumeration and class replacement.

Distinct orbits are disjoint as collections of enumerations, but every one is dense in  $\mathfrak{W}_{\text{sl}}$ . Hence these equivalence classes are not convex, and the quotient by eventual equality does not inherit a linear order by simply comparing representatives. This is a concrete obstruction, not merely a size warning.

**Corollary 19.18** (The class of realizations of a set cut). *Under the hypotheses of Theorem 19.10, the ordinary class*

$$K = \{p \in \mathcal{P}_{\text{bd}} : (\forall i \in I \ A_i <_{\text{lex}} p) \text{ and } (\forall j \in J \ p <_{\text{lex}} B_j)\}$$

*is itself order-isomorphic to  $\mathbf{No}$ .*

*Proof.* It is an ordinary class by comprehension with the two uniform input relations as parameters. Suppose  $C, D \subseteq K$  are sets with  $C < D$ . Add the evaluated codes of  $C$  to the lower input family and those of  $D$  to the upper input family. All cross inequalities remain strict: members of  $K$  already lie between the original sides. Theorem 19.10 supplies a member of  $K$  between  $C$  and  $D$ . Thus  $K$  has the set-cut property, including empty sides. It cannot be a set, since applying that property with lower side all of  $K$  would give a member above itself. The uniqueness statement of Proposition 5.13 now gives the isomorphism.  $\square$

Thus the interpolation theorem produces a full proper-class order of available choices, even when the input constraints are numerous and occur at arbitrarily complicated set-ordinal coordinates. In particular, no set of codes can be order-dense in  $\mathfrak{W}_{\text{sl}}$ . Given such a set, choose a point strictly above

all its evaluations, and then a second, larger point. The open interval between the two contains no code from the proposed dense set. The same argument applies within  $K$ : a set cannot be dense even in the class of realizations of one specified set cut. The ordinary class  $\mathcal{P}_{\text{bd}}$  is therefore a natural scale for density here, while an exhaustive class of set codes will be ruled out by a separate diagonal argument.

*Note 19.19* (Source 09, Corollary 9.3, “Dense copies in every interval”). *For global enumerations  $E < F$ , the class  $\{\sigma \in \mathcal{P}_s(b) : E < E_\sigma < F\}$  is isomorphic to  $\mathbf{No}$ ; the core meets every open interval of the global order and supplies a copy of every class linear order inside it.* This is [corollary 19.18](#) with  $I = \{E\}$  and  $J = \{F\}$ , by the same argument.

## 19.5 Universality, cylinders, and local cofinality

**Proposition 19.20.** *The surreal line embeds explicitly into  $\mathcal{P}_{\text{bd}}$ . Consequently every ordinary class linear order embeds into  $\mathcal{P}_{\text{bd}}$  and hence into  $\mathfrak{W}_{\text{sl}}$ , in the uniform evaluation sense.*

*Proof.* For  $x \in \mathbf{No}$ , transpose the values  $e_0(0)$  and  $x$  in  $e_0$ , and denote the resulting enumeration by  $E_x$ . Its coordinate permutation has support contained in  $\{0, e_0^{-1}(x)\}$ , so it has a code in  $\mathcal{P}_{\text{bd}}$ , and

$$E_x(0) = x, \quad x < y \iff E_x <_{\text{lex}} E_y.$$

For the general assertion, compose this embedding with the class-order universality of  $\mathbf{No}$  from [Proposition 5.13](#).  $\square$

Thus all set ordinals, the class order  $\text{Ord}$ , its reverse, and class orders such as  $\text{Ord} + \text{Ord}$  occur as suborders of the lexicographic comparison. This statement concerns the order *between* enumerations. Each individual enumeration in  $\mathfrak{W}_{\text{sl}}$  still presents a well-order of type  $\text{Ord}$ ; its lexicographic environment need not be well-ordered.

*Note 19.21* (Source 12, Theorem 6.7, “Local class universality and cylinders”). *Every class linear order has a uniform order-embedding into  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  and into every nonempty open interval of  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ ; for a set-sized injective prefix  $p$ , the cylinder  $[p]$  is convex and isomorphic to  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  by uniform transformations.* This is [proposition 19.20](#) and [proposition 19.25](#), by the same constructions ( $x \mapsto E_x$  composed with [lemma 5.14](#); a cylinder through  $\mathbf{No} \setminus \text{ran}(p) \cong \mathbf{No}$ , [lemma 5.15](#)). Source 12 lists the examples “all set ordinals, their reverses,  $\text{Ord} + 1$ ,  $\text{Ord} + \text{Ord}$ , all set-sized lexicographic powers of  $\mathbf{No}$  formed as class orders” and adds that “every class linear order” still means an order whose elements are sets; it does not include an unrestricted higher collection such as all class well-orders themselves.

*Note 19.22* (Source 09, Theorem 10.2). *Every set ordinal, every reverse set ordinal and every class linear order embeds as a uniformly class-indexed family of global enumerations; all these embeddings can be taken in  $\mathcal{P}_s(b)$ , or within any specified nonempty global open interval.* This is [proposition 19.20](#) with [corollary 19.18](#). Source 09 lists  $\text{Ord}$ ,  $\text{Ord}^*$ ,  $\text{Ord} + 1$ ,  $\text{Ord} + \text{Ord}$ , set-sized lexicographic powers of  $\mathbf{No}$  and every class linear order on a class of set codes, and stresses that “the class order *being embedded* need not be set-like, although each well-order represented by a target global enumeration is set-like. These are two entirely different orders.” Its direct embedding  $x \mapsto T_x \circ b$  (exchange  $b(0)$  and  $x$ ) is source 11’s  $E_x$ ; it “does not by itself give the surjectivity of the isomorphism”.

*Note 19.23* (Source 08, Corollary 12.4). *The skeleton contains copies of every ordinal and its reverse, every class well-order,  $\text{Ord} + 1$ ,  $\text{Ord} + \text{Ord}$ ,  $\mathbf{No} \times_{\text{lex}} \mathbf{No}$ , and  $(\mathbf{No}^\alpha, <_{\text{lex}})$  for every set ordinal  $\alpha$ ; every injectively parametrized uniform class family of global enumerations, with its induced lexicographic order, also embeds into  $\mathcal{P}_s(b)$ .* The first part is [proposition 19.20](#); the last clause is

source 09's [proposition 19.24](#), printed next. Source 08: this concerns a uniform class-indexed family, not all class enumerations simultaneously, and replacing the set ordinal  $\alpha$  by  $\text{Ord}$  in  $\mathbf{No}^\alpha$  changes the size level of the elements and is not covered.

**Proposition 19.24** (Representable suborders do not exceed surreal universality). *Every uniformly class-indexed family of global enumerations, modulo equal rows, has a class-coded linear order that embeds into  $\mathbf{No}$ .*

*Proof.* For a family with class index  $I$ , choose the least representative of each equal-row class in a fixed set-like global well-order of  $I$ . Equality of rows is the first-order condition  $\forall \alpha E_i(\alpha) = E_j(\alpha)$ , with the family as a class parameter. The representatives form a class of sets. Their first-disagreement order is a class linear order, to which [theorem 5.16](#) applies.  $\square$

This does not say that the full higher-level totality  $\mathfrak{W}_{\text{sl}}$  embeds into  $\mathbf{No}$ ; it says that every *uniformly class-coded portion* does.

**Proposition 19.25** (Prefix cylinders). *For an injective set prefix  $p$ , let*

$$C_p = \{E \in \mathfrak{W}_{\text{sl}} : E \upharpoonright \text{dom}(p) = p\}.$$

*The cylinder  $C_p$  is convex and order-isomorphic to  $\mathfrak{W}_{\text{sl}}$ , with the isomorphism interpreted as a uniform transformation of class parameters. Every nonempty open interval in  $\mathfrak{W}_{\text{sl}}$ , and every interval specified by a uniformly set-indexed strict cut, contains a whole such cylinder.*

*Proof.* An enumeration that first departs from  $p$  at a coordinate of  $p$  lies on the same side of all its extensions. This proves convexity. Put  $S = \text{ran}(p)$  and  $\delta = \text{dom}(p)$ . By [Lemma 5.2](#) and [Proposition 5.13](#), fix an order-isomorphism  $h : \mathbf{No} \rightarrow \mathbf{No} \setminus S$ . The ordinal tail  $[\delta, \text{Ord})$  has order type  $\text{Ord}$ , with its canonical increasing parametrization  $\xi \mapsto \delta + \xi$ . Given  $E \in \mathfrak{W}_{\text{sl}}$ , prepend  $p$  to the enumeration  $h \circ E$  on this tail. The result is a bijection onto  $\mathbf{No}$ , since its prefix covers  $S$  and its tail covers exactly the complement. Conversely, restrict an element of  $C_p$  to its tail and apply  $h^{-1}$ . These transformations are inverse and preserve first-difference comparisons. The final assertion follows from [Theorem 19.10](#).  $\square$

In particular, every such interval contains a copy of the entire comparison structure  $\mathfrak{W}_{\text{sl}}$ , not merely a copy of an arbitrarily large set order. This self-similarity does not invoke a class whose members are the cylinders; for each set prefix the displayed construction gives the required transformation, using the indicated class parameters.

*Note 19.26* (Source 09, [Proposition 10.4](#), “Cylinder self-similarity”). *Each cylinder  $[p]$  (the global enumerations extending an injective set word  $p$ ; convex, and nonempty by [lemma 19.6](#)) is isomorphic to the full global comparison order by an explicit invertible transformation of class enumerations preserving lexicographic order and having exactly  $[p]$  as its image. This is [proposition 19.25](#), by the same construction  $H_E \upharpoonright \alpha = p$ ,  $H_E(\alpha + \xi) = h(E(\xi))$  with  $h : \mathbf{No} \cong \mathbf{No} \setminus \text{ran } p$  ([corollary 5.18](#)). Source 09's paragraph on two baselines follows; source 08 makes the same observation in its subsection printed after [corollary 19.15](#).*

For two baselines  $b, c$ , their eventual-agreement collections are either identical or disjoint. They are identical exactly when  $b, c$  agree beyond some set ordinal. Otherwise no global enumeration can be eventually equal to both. Nevertheless, *each* such collection has a genuine set-coded presentation isomorphic to  $\mathbf{No}$  and is dense in the entire global comparison order. Eventual agreement is therefore not a convex relation, and its quotient does not inherit a lexicographic linear order by comparing arbitrary representatives.



**Proposition 19.27** (Approximants at every point). *For each  $F \in \mathfrak{J}$  there are uniformly ordinal-indexed families  $(a_\alpha)$  and  $(b_\alpha)$  of codes in  $\mathcal{P}_{\text{bd}}$  such that, whenever  $\alpha < \beta$ ,*

$$a_\alpha <_{\text{lex}} a_\beta <_{\text{lex}} F <_{\text{lex}} b_\beta <_{\text{lex}} b_\alpha.$$

*The lower family is cofinal among the global enumerations below  $F$ , and the upper family is coinital among those above  $F$ . Neither side admits a uniformly set-indexed cofinal or coinital family, respectively.*

*Proof.* Choose a fresh value below  $F(\alpha)$ , append it to  $F \upharpoonright \alpha$ , and use Lemma 19.4 to obtain  $a_\alpha$ . Choose a fresh value above  $F(\alpha)$  to obtain  $b_\alpha$ . Lemma 5.2 supplies the fresh values, and the fixed global well-order makes all choices uniform. For  $\alpha < \beta$ , the comparison between  $a_\alpha$  and  $a_\beta$  is decided at  $\alpha$ , giving the asserted inequality; the other comparisons are identical arguments.

If  $E <_{\text{lex}} F$ , choose  $\alpha > d(E, F)$ . Then  $a_\alpha$  agrees with  $F$  at the decisive coordinate, so  $E <_{\text{lex}} a_\alpha <_{\text{lex}} F$ . This proves cofinality, and the upper assertion is symmetric. Finally, given any uniformly set-indexed family below  $F$ , Theorem 19.10 applied with upper side  $\{F\}$  supplies a larger lower approximant in  $\mathcal{P}_{\text{bd}}$ . Hence the original family was not cofinal. The dual argument proves the coinitality assertion.  $\square$

For  $F \in \mathfrak{W}_{\text{sl}}$ , these are the local cofinality and coinitality statements at an actual point of the order. It is appropriate to say that ordinal-length class families suffice and set-sized families do not; assigning an ordinary set cardinal to these proper-class cofinalities would obscure the distinction.

*Note 19.28* (Source 12, Proposition 6.4, “Cofinality and local character”). *No set-indexed uniform family is cofinal or coinital in  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ ; there are increasing cofinal and decreasing coinital families indexed by Ord; at every point neither side has a set-indexed cofinal, respectively coinital, family, but both have Ord-indexed ones.* This is proposition 19.27 together with the remark that first entries of a set-indexed family are bounded, by the same argument; the cofinal family is  $(E_\alpha)_{\alpha \in \text{Ord}}$  for the positive ordinals  $\alpha$ .

**Corollary 19.29** (Set saturation is not Dedekind completeness). *A nonempty uniformly set-indexed family in  $\mathfrak{W}_{\text{sl}}$  with no greatest member has no supremum in  $\mathfrak{W}_{\text{sl}}$ . Dually, a family with no least member has no infimum. Both statements also hold in the ordinary class order  $\mathcal{P}_{\text{bd}}$ .*

*Proof.* Suppose that  $E$  is the supremum of a family  $(A_i)$  with no greatest member. Every  $A_i$  is then strictly below  $E$ . By Theorem 19.10, there is a bounded-support enumeration  $p$  with  $A_i < p < E$  for every  $i$ . This makes  $p$  a smaller upper bound, a contradiction. The infimum argument reverses the inequalities, and the same bounded-support interpolant proves the assertions internal to  $\mathcal{P}_{\text{bd}}$ .  $\square$

For example, take the enumerations  $E_n$  from the transposition construction, so that  $E_n(0) = n$  for  $n \in \mathbb{N}$ . They form an increasing sequence of bounded-support codes. The enumeration whose first value is the surreal ordinal  $\omega$  is an upper bound, but there is no least upper bound in either  $\mathcal{P}_{\text{bd}}$  or  $\mathfrak{W}_{\text{sl}}$ . The obstruction is not a failure to find bounds: Theorem 19.10, with one input side empty, bounds every uniformly set-indexed family above and below. Rather, whenever a proposed bound is strictly beyond every member, there remains room to move the bound closer. This distinction will also matter when the full order of Dedekind cuts is embedded below. An order embedding preserves comparisons, but does not automatically preserve the role of a point as the least upper bound of a specified family.

[merge] Source 08’s Section 11.2, “Saturation is not Dedekind completeness”, proves corollary 19.29 by the same argument (a strictly smaller upper bound from note 19.13) and adds that the same assertion holds in  $\mathcal{P}_s(b)$  and in  $\mathbf{No}$ : “Set-cut saturation supplies abundant *strict* separators; it is not the Dedekind completeness familiar from the real line.”

[merge] Bounded reflection is proved only by source 09.



## 19.6 Source 09: bounded reflection

### 19.6.1 Every set family reflects into one bounded permutation stage

**Theorem 19.30** (Bounded reflection). *Let  $(E_i)_{i \in I}$  be a uniformly set-indexed family of pairwise distinct global enumerations. For any baseline  $b$ , there are a set ordinal  $\lambda$ , an infinite initial cardinal  $\nu$ , and support codes  $\sigma_i \subseteq \nu \times \nu$  such that*

$$E_{\sigma_i} \upharpoonright \lambda = E_i \upharpoonright \lambda, \quad E_i < E_j \iff E_{\sigma_i} < E_{\sigma_j}.$$

The family order also embeds into  $\mathcal{W}_\kappa(\mathbf{No}_{<\theta})$  for some birthday bound  $\theta$ , where  $\kappa = |\mathbf{No}_{<\theta}|$ .

*Proof.* Choose  $\lambda \geq 1$  strictly greater than every first-disagreement ordinal of a pair of distinct rows. This is possible by Replacement because  $I \times I$  is a set. The set

$$U = \bigcup_{i \in I} \text{ran}(E_i \upharpoonright \lambda)$$

is a set of surreals. Choose one infinite initial cardinal  $\nu$  strictly above  $\lambda$  and every ordinal in  $b^{-1}[U]$ . Apply lemma 19.6 to every row prefix at this same bound. All comparisons were resolved before  $\lambda$ , so they are preserved. Choice for this set family of completions suffices.

For the birthday version, take  $\theta$  large enough that  $U \subseteq \mathbf{No}_{<\theta}$  and its cardinality  $\kappa$  is strictly larger than the ordinal  $\lambda$ . Each prefix then removes fewer than  $\kappa$  letters from  $\mathbf{No}_{<\theta}$ . Enumerate the remainder in type  $\kappa$  and append it. The total ordinal length is  $\lambda + \kappa = \kappa$ . The resulting exhaustive words preserve all first disagreements.  $\square$

Thus no counterexample involving a *uniform set family* needs unboundedly many coordinates in an essential way. But the bound depends on the family, and there is no single set bound working for every family. For each  $\kappa$ , the global core contains  $\kappa^+$ , whereas theorem 6.14 rules it out of any fixed alphabet of size  $\kappa$ .

[merge] The failure of finite support is proved only by source 08; it shows why the core must allow set support.

## 19.7 Why finite support is insufficient (source 08)

Let  $F_b$  be the suborder of  $\mathcal{P}_s(b)$  consisting of finite support codes. The explicit transpositions in note 19.9 show that  $F_b$  already contains a copy of  $\mathbf{No}$  and hence every class linear order. Nevertheless it does not have the saturation or ambient density of the full set-supported skeleton.

**Proposition 19.31.** *The order  $F_b$  is not an  $\eta_{\text{Ord}}$ -order, and its represented enumerations are not dense in  $\mathfrak{W}_{\text{sl}}$ .*

*Proof.* Put  $p(n) = b(n+1)$  for  $n < \omega$ . This injective prefix differs from  $b$  at every natural-number coordinate. Every finite injective prefix extends to a finite perturbation of  $b$ : after applying  $b^{-1}$ , its partial injection extends to a permutation of the finite union of its domain and range, because the two remaining finite complements have equal size.

For each  $n$ , choose fresh values  $a_n < p(n) < c_n$  outside  $\text{ran}(p \upharpoonright n)$ , and extend  $(p \upharpoonright n) \frown (a_n)$  and  $(p \upharpoonright n) \frown (c_n)$  to finite perturbations  $l_n, r_n$  of  $b$ . These choices can be made uniformly. For all  $n, m$  one has  $l_n < r_m$ : at the earlier of their altered coordinates one sees either  $a_n < p(n)$  or  $p(m) < c_m$ , and when  $n = m$  one sees  $a_n < c_n$ .

A separator of these two countable families must agree with  $p$  at every finite coordinate. Otherwise at its first disagreement  $k$ , the rows  $l_m, r_m$  for  $m > k$  both agree with  $p$  there, forcing the candidate

below a left row or above a right row. Such a separator moves every natural-number coordinate of  $b$  and cannot have finite support. This proves failure of set-cut saturation.

Finally choose  $u < v$  in  $\mathfrak{W}_{\text{sl}}$  with common prefix  $p$  of length  $\omega$  and distinct fresh entries immediately afterward, using [note 19.8](#). Every enumeration between them must share that prefix, so none is a finite perturbation of  $b$ . Hence  $F_b$  is not dense in  $\mathfrak{W}_{\text{sl}}$ .  $\square$

Thus containing every ordinary class order is weaker than being a surreal line or being a dense skeleton of the virtual ambient order. Set support, not merely finite support, is what permits arbitrary set-length prefixes to be imposed.

## 19.8 Binary class words, cuts, and diagonalization

**Theorem 19.32.** *The lexicographic orders  $\mathfrak{W}_{\text{sl}}$  and  $\{0,1\}^{\text{Ord}}$  are order-bi-embeddable, where a binary word of length  $\text{Ord}$  means a class characteristic function. Both embeddings have uniform relational definitions in GBC.*

*Proof.* Partition the coordinate ordinals into successive pairs  $2 \cdot \xi, 2 \cdot \xi + 1$ . Here the multiplication is ordinal multiplication with 2 on the left; every ordinal has a unique such pair position. Sort the corresponding baseline values as

$$u_\xi = \min\{e_0(2 \cdot \xi), e_0(2 \cdot \xi + 1)\}, \quad v_\xi = \max\{e_0(2 \cdot \xi), e_0(2 \cdot \xi + 1)\}.$$

For a class  $D \subseteq \text{Ord}$ , put these two values in the order  $(u_\xi, v_\xi)$  if  $\xi \notin D$ , and in the order  $(v_\xi, u_\xi)$  if  $\xi \in D$ . Denote the resulting enumeration by  $T_D$ . Each pair retains exactly its original two values, so  $T_D$  is a bijection onto  $\mathbf{No}$ . At the least differing bit of two characteristic functions, 0 chooses the smaller first value and 1 the larger one. Thus  $D \mapsto T_D$  preserves and reflects lexicographic comparison.

For the converse, fix an ordinal enumeration  $(p_\xi)_{\xi \in \text{Ord}}$  of the set codes in  $\mathcal{P}_{\text{bd}}$ . Associate with  $E \in \mathfrak{W}_{\text{sl}}$  the class

$$D_E = \{\xi \in \text{Ord} : p_\xi <_{\text{lex}} E\}.$$

If  $E <_{\text{lex}} F$ , then  $D_E \subseteq D_F$ , and the inclusion is strict: [Corollary 19.15](#) supplies a code strictly between  $E$  and  $F$ . At the least ordinal belonging to  $D_F \setminus D_E$ , the characteristic functions therefore have values 0 and 1, respectively. This proves the second order embedding. Both constructions use elementary class comprehension with the input word or enumeration as a class parameter.  $\square$

Bi-embeddability must not be replaced by isomorphism. Binary class words have a least and a greatest word and have adjacent pairs: after a set prefix, the word with next bit 0 and all subsequent bits 1 immediately precedes the word with next bit 1 and all subsequent bits 0. By contrast,  $\mathfrak{W}_{\text{sl}}$  is dense and has no endpoints.

[merge] *Second route (source 12).* Source 12's Section 7 embeds  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  into binary class words by concatenating self-delimiting codes of the entries, instead of source 11's characteristic function of a dense class; the other embedding is source 11's pair swap.

## 19.9 Source 12's route to the binary-class representation

Let  $\mathbb{B}_{\text{Ord}}$  denote the predicate of class subsets of  $\text{Ord}$ , regarded as binary functions on  $\text{Ord}$  and compared at their least disagreement. This is again notation for a predicate and relation on classes, not a powerclass object.

**Lemma 19.33** (A self-delimiting numerical code). *Replace every minus in a sign sequence by 00, every plus by 11, and append the terminal block 01. The resulting binary ordinal word  $c(s)$  has length  $2 \cdot \text{bd}(s) + 2$ . Distinct codewords are prefix-incomparable, and their first-difference comparison agrees with numerical comparison of the surreals.*

*Proof.* The three two-bit blocks have order  $00 < 01 < 11$ , exactly matching  $- < \text{end} < +$ . At a sign disagreement the first different block decides. When one sign sequence ends, its terminal block differs from the next sign block of the other. Hence neither complete codeword can be a prefix of another, and the numerical comparison is preserved.  $\square$

**Theorem 19.34** (Binary-class equimorphism). *In the sense of uniform class transformations,*

$$\mathfrak{W}_{\text{sl}}(\mathbf{No}) \simeq_e \mathbb{B}_{\text{Ord}}.$$

*Proof.* For  $E : \text{Ord} \rightarrow \mathbf{No}$ , concatenate  $c(E(\alpha))$  over all ordinals. The cumulative length before each stage is a set ordinal. All blocks are nonempty, so the cumulative lengths are unbounded in  $\text{Ord}$  and the result is a binary class on all of  $\text{Ord}$ . If  $E, F$  first disagree at  $\alpha$ , their earlier block lengths and bits agree, and the code of that entry decides the same inequality by lemma 19.33.

For the converse, use the disjoint pairs  $a_\alpha < b_\alpha$  in the proof of theorem 19.38. For a binary class  $A$ , list the pair in ascending order if its bit is 0, and descending order if its bit is 1. Put pair blocks in ordinal order. The first different bit gives the first different pair orientation and the correct lexical inequality. All labels occur exactly once.  $\square$

This is not an isomorphism assertion. The binary-class order has least and greatest elements and adjacent pairs, such as a zero followed by all ones versus a one followed by all zeros. By note 19.11,  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  has no endpoints or adjacent pairs. The distinction between equimorphism and isomorphism is already visible at this level.

[merge] Source 08’s Section 13 (“Binary class sequences and the size obstruction”) defines an embedding between virtual collections as a *uniform operator*: “a formula, with displayed class parameters, assigning a unique output class to every input class, with class existence proved in the base theory. It is not a new class whose members are ordered pairs of proper classes.”

*Note 19.35* (Source 08, Theorem 13.1, “Virtual binary equimorphism”). *Relative to a baseline  $b : \text{Ord} \rightarrow \mathbf{No}$ , there are uniform order embeddings*

$$\mathbb{B}_{\text{Ord}} \hookrightarrow \mathfrak{W}_{\text{sl}} \hookrightarrow \mathbb{B}_{\text{Ord}}; \tag{S08.11}$$

*thus these virtual orders are equimorphic in the uniform-operator sense; they are not order-isomorphic.* This is theorem 19.32. The first embedding is source 11’s pair swap; the second is source 12’s concatenation (theorem 19.34), with the code of note 7.5 and block endpoints  $D_0 = 0$ ,  $D_{\xi+1} = D_\xi + (2 \cdot \text{bd}(e(\xi)) + 2)$ ,  $D_\lambda = \sup_{\xi < \lambda} D_\xi$  defined by set-valued recursion. Source 08 adds that the jump “ $p \cap 0$  followed by constant 1” below “ $p \cap 1$  followed by constant 0” in  $\mathbb{B}_{\text{Ord}}$  ceases to be adjacent in the image, because  $\mathfrak{W}_{\text{sl}}$  is dense: “equimorphism does not preserve order topology”.

**Corollary 19.36.** *The higher-order order of all class initial segments of  $\mathbf{No}$ , ordered by inclusion, embeds into  $\mathfrak{W}_{\text{sl}}$ . In particular, this holds for any specified convention of Dedekind cuts. Conversely, each  $E \in \mathfrak{W}_{\text{sl}}$  determines the initial segment  $\{p \in \mathcal{P}_{\text{bd}} : p <_{\text{lex}} E\}$ , and these initial segments separate distinct enumerations.*

*Proof.* For a class initial segment  $C \subseteq \mathbf{No}$ , form  $e_0^{-1}[C] \subseteq \text{Ord}$  and apply the pair-swap embedding of Theorem 19.32. Proper inclusion of initial segments gives proper inclusion of their inverse images, which implies the correct lex comparison of their characteristic functions. The converse is the dense-class construction in the second half of that theorem.  $\square$

This corollary does not form a GB class of all Dedekind cuts of  $\mathbf{No}$ . Each individual cut is an allowed class parameter; the rule associates an enumeration to it and compares any two such results. An encompassing object belongs to a higher type or to an explicitly larger ambient universe. Nor does the corollary assert that the natural inclusion of  $\mathcal{P}_{\text{bd}}$  in  $\mathfrak{W}_{\text{sl}}$  is a Dedekind completion: an order embedding of cuts need not preserve their intended suprema or their original positions relative to the embedded line.

There is a useful description of the natural cuts produced by an enumeration outside the bounded-support family. If  $E \in \mathfrak{W}_{\text{sl}}$  has no code in  $\mathcal{P}_{\text{bd}}$ , then

$$L_E = \{p \in \mathcal{P}_{\text{bd}} : p <_{\text{lex}} E\}, \quad U_E = \{p \in \mathcal{P}_{\text{bd}} : E <_{\text{lex}} p\}$$

partition  $\mathcal{P}_{\text{bd}}$  into two nonempty classes. Proposition 19.27 shows that  $L_E$  has no set-sized cofinal subset and  $U_E$  has no set-sized coinital subset, while ordinal-indexed class families suffice on both sides. This cut is unfilled in  $\mathcal{P}_{\text{bd}}$ : a putative filling code and  $E$  would be separated by another bounded-support code. Transporting the cut through an isomorphism  $\mathcal{P}_{\text{bd}} \cong \mathbf{No}$  therefore exhibits a surreal class cut with neither side controlled by a cofinal or coinital set. This is a description of individual additional enumerations relative to the fixed dense class, rather than a classification of all surreal class cuts.

**Theorem 19.37** (No exhaustive uniform class of codes). *There is no ordinary class of set codes with a single class evaluation relation whose evaluated rows include every member of  $\mathfrak{W}_{\text{sl}}$ .*

*Proof.* If such a coding existed, global choice would enumerate its codes by ordinals, with repetitions if the class of codes were a set. Write the resulting row relation as  $\text{Eval}(\xi, \alpha, x)$ . Use the sorted pairs  $u_\xi < v_\xi$  from Theorem 19.32, and define the class

$$D = \{\xi : \text{Eval}(\xi, 2 \cdot \xi, u_\xi)\}.$$

The associated pair-swap enumeration  $T_D$  differs from row  $\xi$  on the pair  $(2 \cdot \xi, u_\xi)$ . If that pair belongs to the old row, it does not belong to the graph of  $T_D$ , whose value there is  $v_\xi$ . If it does not belong to the old row, it belongs to the new graph, whose value there is  $u_\xi$ . Thus  $T_D$  is a member of  $\mathfrak{W}_{\text{sl}}$  omitted by the proposed exhaustive list. The argument even permits invalid or nonfunctional rows in the evaluator.  $\square$

The theorem explains why class universality of  $\mathbf{No}$  cannot simply be applied to the full collection  $\mathfrak{W}_{\text{sl}}$ . It also pinpoints the limitation of coding only bounded modifications, or any other one uniformly evaluable family of set codes: such a family may be order-dense and may realize all set-indexed cuts, while still omitting global enumerations by diagonalization.

[merge] *Strengthening (source 12)*. Source 12's Section 5 proves that no uniform family of class well-orders indexed by a class of sets contains every class well-order of  $\mathbf{No}$ , and that the missing one can be chosen set-like; its corollary 19.39 is theorem 19.37. Its subsection on definable classes explains why enumerating formulas does not evade the theorem.

## 19.10 Source 12's strengthening: why the collection of all global well-orders is not a class

The formal obstacle is not that individual class relations are illegal. They are legitimate class variables. The obstacle is attempting to make them the elements of an ordinary class of the same theory.

A useful legal replacement is a *uniform family*. For example,  $F \subseteq I \times \text{Ord} \times \mathbf{No}$  is a class of triples of sets, and its section  $F_i$  can be a class bijection for every  $i \in I$ . Similarly a class of triples

$(i, x, y)$  can code a family of class order relations. The index class  $I$  may be proper. These are single classes, not functions whose values are class objects.

**Theorem 19.38** (No exhaustive class-coded family). *In GBC, no uniform family of class well-orders of  $\mathbf{No}$  indexed by a class of sets contains every class well-order of  $\mathbf{No}$ . In fact, for every such family there is a set-like class well-order not among its sections.*

*Proof.* Global choice permits the family to be reindexed by an ordinal or by  $\text{Ord}$ . Extend an ordinal-indexed family to an  $\text{Ord}$ -indexed one by repeating an arbitrary default well-order; an empty family is trivial. Write the resulting relations as  $R_\alpha$ .

Fix a bijection  $B : \text{Ord} \rightarrow \mathbf{No}$  and partition its range into the pairs

$$\{B(2 \cdot \alpha), B(2 \cdot \alpha + 1)\} = \{a_\alpha, b_\alpha\}, \quad a_\alpha < b_\alpha.$$

Here multiplication is ordinal multiplication. The blocks of length two partition  $\text{Ord}$ . Define a new enumeration  $E$  by placing the two labels of pair  $\alpha$  in the order *opposite* to their relative order in  $R_\alpha$ . Keep the pair blocks in increasing  $\alpha$  order.

This is a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ , defined elementarily from the family and  $B$ . Its induced well-order disagrees with  $R_\alpha$  on the relative order of  $a_\alpha, b_\alpha$ . Thus it is not any section of the family.  $\square$

**Corollary 19.39** (No uniform set-code solution). *There is no class  $D$  of set codes together with a uniform class decoder whose decoded relations are exactly all class well-orders of  $\mathbf{No}$ .*

This diagonal theorem applies equally to an attempted exhaustive coding of all class subsets of  $\text{Ord}$  through the pair construction. It does not say that an externally specified set model has too many classes to form an external set. It says that the corresponding exhaustive decoder is not available *internally at the same size level*.

### 19.10.1 Definable classes and the truth-predicate trap

Formula codes and set parameters are sets, but the semantic map “formula plus parameters  $\mapsto$  its extension in the universe” is not a uniformly available truth predicate for the universe. For each fixed formula one may define a class. One may also work externally with all definable classes of a chosen model. Neither maneuver supplies an internal universal decoder defeating [theorem 19.38](#). The same caution applies to quantifying over all possible Lean universe levels: universe polymorphism is not an internal type of all universes.

[merge] Sources 09 and 08 prove the diagonal theorem for families of global enumerations (source 09’s Section 12, “Why all global well-orders cannot be uniformly coded by sets”).

*Note 19.40* (Source 09, Theorem 12.1, “Uniform no-coding theorem”). *There is no class of set indices  $I$  and no uniform class relation  $F \subseteq I \times \text{Ord} \times \mathbf{No}$  whose rows are exactly, or even include, all global enumerations of  $\mathbf{No}$ . This is [theorem 19.37](#), by the same diagonal class  $D = \{\alpha : (c(\alpha), 2 \cdot \alpha, a_\alpha^-) \in F\}$  along a class surjection  $c : \text{Ord} \rightarrow I$ , valid with nonfunctional rows. Source 09’s preceding paragraph defines the switches  $E_A$  that list the pair  $\{a_\alpha^-, a_\alpha^+\} = \{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\}$  increasingly for  $\alpha \notin A$  and decreasingly for  $\alpha \in A$ ; they embed the binary-class comparison into the global one, “the proper-class counterpart of the pair construction” of [note 6.12](#), as in the first half of [theorem 19.32](#).*

*Note 19.41* (Source 08, Theorem 13.2, “No class-uniform exhaustive list”). *If a class relation  $T \subseteq \text{Ord} \times \text{Ord} \times \mathbf{No}$  uniformly gives a global enumeration  $T_\alpha$  for each ordinal  $\alpha$ , there is a global enumeration  $e^*$  different from every  $T_\alpha$ ; consequently no uniform family indexed by an ordinary set*

or class of sets exhausts  $\mathfrak{W}_{\text{sl}}$ . This is [theorem 19.37](#), by the same switch at the even coordinate  $2 \cdot \alpha$ ; families indexed by sets or proper classes are reindexed by  $\text{Ord}$ . Source 08 adds: “the class of set-supported perturbation codes does not parametrize the full collection. The isomorphism  $\mathcal{P}_s(b) \cong \mathbf{No}$  is compatible with the much larger size of  $\mathfrak{W}_{\text{sl}}$ : each orbit is a dense copy of the surreal line, not all global enumerations”, and “an assertion about an embedding of the entire virtual order into a class must first specify a higher-type notion of mapping; it cannot be inferred from ordinary class universality.”

### 19.10.2 What the theorem does and does not say

The theorem rules out a uniform master list by set parameters, including one indexed by a proper class of such parameters. It does not rule out large useful coded subfamilies such as  $\mathcal{P}_s(b)$ ; indeed the latter is dense and class-universal.

It also does not mean that class functions cannot be discussed in **GBC**, nor that stronger foundations are inconsistent. They can be discussed one or several at a time, in uniform families, or as elements of a higher-level collection in a suitably stratified theory. What fails is compression of *all* of them into one same-level, set-indexed evaluation relation.

Enumerating formula strings does not evade this issue. A putative uniform evaluation of all formulas over the set universe would require an appropriate truth predicate; it is not obtained from syntax alone. Definitions with additional class parameters also need not remain in the original parameter collection. One must distinguish a metatheoretic list of formulas, individual definable classes, and an internal uniform family of their extensions.

## 19.11 Omitted ranges and the transition to raw well-orders

**Theorem 19.42** (An omitted-range cut). *Fix  $r \in \mathbf{No}$  and a class bijection  $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r\}$ . There are increasing and decreasing ordinal-indexed families  $(a_\alpha)$  and  $(b_\alpha)$  in  $\mathcal{P}_{\text{bd}}$  such that*

$$a_\alpha <_{\text{lex}} b_\beta \quad (\alpha, \beta \in \text{Ord}),$$

*but no member of  $\mathfrak{W}_{\text{sl}}$  lies strictly between both entire families. The raw well-order that lists  $F$  and then  $r$  has type  $\text{Ord} + 1$  and is the unique member of  $\mathfrak{W}_{\text{all}}$  between these families.*

*Proof.* Use the approximants from [Proposition 19.27](#). They satisfy  $a_\alpha <_{\text{lex}} F <_{\text{lex}} b_\beta$  for every pair of indices. If a global bijection  $E$  lay between both families and differed from  $F$  first at  $\delta$ , an approximant with index greater than  $\delta$  would place  $E$  on the wrong side. Therefore  $E$  would equal  $F$  at every ordinal, which contradicts its surjectivity.

Let  $R$  be the raw order obtained by appending  $r$  after the enumeration  $F$ . Its set-like initial part is precisely  $F$ : every point listed by  $F$  has set-sized predecessors, whereas  $r$  has a proper class of predecessors. Its comparisons with the approximants are decided at their prescribed ordinal first differences, so it lies between both families. Conversely, the same first-difference argument forces the core  $F_S$  of any raw order  $S$  between the families to equal  $F$ . By the core decomposition of [Theorem 15.1](#), its residual order must be the unique order on the singleton  $\{r\}$ . Thus  $S = R$ .  $\square$

The lower and upper families here are indexed by all ordinals, not by sets. Accordingly, the missing interpolant does not contradict [Theorem 19.10](#). It gives a concrete boundary between the set-cut saturation of the set-like comparison and the additional points provided by non-set-like global well-orders.

[merge] *Source 09’s route.* Source 09’s Section 11 gives a concrete instance of [theorem 19.42](#), with the nonexhaustive word  $\xi \mapsto \xi$  and the fresh values  $-1$  and  $\alpha + 1$ ; its [remark 19.44](#) is the general mechanism, which is source 11’s theorem for an arbitrary omitted range.



### 19.12 Source 09's route: a proper-class cut with no interpolant

The interpolation theorem has an exact and visible size limitation. The following construction uses only ordinal-valued prefixes and the single nonordinal value  $-1$ .

**Theorem 19.43** (An explicit unfilled  $(\text{Ord}, \text{Ord})$  configuration). *For any baseline  $b$ , there are uniformly class-indexed families  $(L_\alpha)_{\alpha \in \text{Ord}}$  and  $(U_\alpha)_{\alpha \in \text{Ord}}$  lying in  $\mathcal{P}_s(b)$  such that*

$$\alpha < \beta \implies L_\alpha < L_\beta < U_\beta < U_\alpha, \quad L_\alpha < U_\beta \quad \text{for all } \alpha, \beta,$$

*but there is no global enumeration  $E$  with  $L_\alpha < E < U_\beta$  for every  $\alpha, \beta$ .*

*Proof.* Let  $g : \text{Ord} \rightarrow \mathbf{No}$  be  $g(\xi) = \xi$ . This is an injective class word but not an exhaustive enumeration of  $\mathbf{No}$ . For each  $\alpha$ , use [lemma 19.6](#) to choose core completions of the prefixes

$$(g \upharpoonright \alpha) \frown (-1), \quad (g \upharpoonright \alpha) \frown (\alpha + 1),$$

calling them  $L_\alpha$  and  $U_\alpha$ , respectively. A fixed global well-order on the set support codes makes these choices uniform. The appended letters are fresh. For  $\alpha < \beta$ , comparison at coordinate  $\alpha$  gives  $L_\alpha < L_\beta$  and  $U_\beta < U_\alpha$ . Every lower word is below every upper word, by comparison at the smaller divergence coordinate.

Suppose a global enumeration  $E$  lay between all these bounds. If  $E(\delta) \neq \delta$  for some ordinal, take the least such  $\delta$ . If  $E(\delta) < \delta$ , the lower word  $L_{\delta+1}$  agrees with  $g$  through coordinate  $\delta$  and therefore lies above  $E$ , a contradiction. If  $E(\delta) > \delta$ , the upper word  $U_{\delta+1}$  lies below  $E$ , also a contradiction. Hence  $E(\delta) = \delta$  for every ordinal  $\delta$ , making  $E = g$ . But  $g$  omits, for example,  $-1$ , and is not surjective onto  $\mathbf{No}$ .  $\square$

Here the notation  $(\text{Ord}, \text{Ord})$  describes an increasing lower and a decreasing upper cofinal presentation, not a pair of set cardinal cofinalities. More explicitly, compare every global enumeration to the nonexhaustive word  $g$ . Every enumeration differs from  $g$  at a set ordinal and lies on exactly one side. The lower family in the theorem is cofinal on the lower side, and the upper family is coinital on the upper side. Neither side has an endpoint. The missing point would have to be the nonexhaustive branch  $g$  itself.

Every set subfamily of these constraints has an interpolant by [note 19.12](#). The failure is therefore genuinely at the proper-class level, not a missed set-sized limit. In particular, the space of all global enumerations is not a class-cut completion of its surreal core.

*Remark 19.44* (A general missing-branch mechanism). The same proof works for any injective class word  $g : \text{Ord} \rightarrow \mathbf{No}$  which is not surjective. At each set stage choose fresh values below and above  $g(\alpha)$  outside the earlier range, and complete the two prefixes. Any interpolant would agree with  $g$  at every coordinate and would therefore fail surjectivity. The surreal set-cut property guarantees the required fresh values.

**Example 19.45** (Two omitted points produce adjacent raw orders). Let  $r < s$  and choose  $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r, s\}$  bijectively. The approximants to  $F$  have exactly two raw interpolants: append  $r, s$  in that order, or append  $s, r$ . Both raw orders have type  $\text{Ord} + 2$ , and the first is lexicographically smaller. They are adjacent in  $\mathfrak{W}_{\text{all}}$ . Indeed, a raw order between them cannot have a different core, since an ordinal first difference would put it on the same side of both. With core  $F$  fixed, its residual order is a well-order of a two-element set, and these are exactly the two displayed possibilities. Thus the density theorem for  $\mathfrak{W}_{\text{sl}}$  does not extend to all raw well-orders.

**Corollary 19.46** (Exactly which raw set cuts admit bounded completions). *Let  $(A_i)_{i \in I} <_{\text{lex}} (B_j)_{j \in J}$  be a uniformly set-indexed strict cut in  $\mathfrak{W}_{\text{all}}$ . A member of  $\mathcal{P}_{\text{bd}}$  lies between the two sides if and only if*

$$F_{A_i} \neq F_{B_j} \quad (i \in I, j \in J),$$

where  $F_R$  is the canonical ordinal enumeration of the set-like initial part of a raw well-order  $R$ . The condition is vacuous when a side is empty.

*Proof.* If the cross cores are different, their comparisons are decided at ordinal first differences and have the same direction as the given raw comparisons. Apply Theorem 19.10 to these uniform families in  $\mathfrak{J}$ . Every comparison of the resulting completion with a raw input is already decided within its core, giving the required raw inequalities.

Conversely, suppose a cross pair shares a core  $F$ . This core cannot be surjective: a surjective core determines the entire raw order, contradicting the strict comparison of the pair. Any  $E \in \mathfrak{W}_{\text{sl}}$  therefore differs from  $F$  at an ordinal coordinate. That first difference places  $E$  on the same side of both raw orders, so  $E$  cannot lie between them. This excludes a bounded-support completion in particular.  $\square$

The criterion isolates the obstruction without appealing merely to size. Distinct ordinal cores can be separated by one bounded change of the baseline, even for a whole set-indexed cut. Two opposite sides with the same core require information in the residual non-set-like order, beyond every ordinal coordinate available to a global set-like enumeration.

## Part V

# Foundations, models and formalization

## 20 Formalizing the subject without changing its meaning

### 20.1 A bounded development in ordinary set theory

For any ordinal  $\theta$ , use the set  $X = \mathbf{No}_{<\theta}$ . The well-order relations on  $X$  form a subset of  $\mathcal{P}(X \times X)$ . Their lexicographic comparison can be defined either by unique ordinal enumeration or by the predecessor formula. This is ordinary ZFC mathematics. No inaccessible cardinal, class theory, or new logical sort is required.

This approach is particularly good for proving the general alphabet theorem, investigating cardinal spectra, and checking how a proposed construction changes birthdays. One should distinguish  $\mathcal{W}(X)$ , which permits every possible ordinal type of cardinality  $|X|$ , from its minimal-length stratum. Their cardinalities agree, but their jumps and finer order structure need not.

A family of results for every  $\theta$  does not automatically give a coherent global map. To take a union, the maps must agree on overlaps. The repository's bounded-surreal nonembedding example makes this restriction concrete.

### 20.2 Second-order class theory

In GB or GBC, the formula

$$\text{ClassWO}(R) \wedge \text{ClassWO}(S) \implies R = S \vee \text{Lex}(R, S) \vee \text{Lex}(S, R)$$

is meaningful with class variables. One can prove trichotomy, transitivity, existence theorems, and uniform-family embedding theorems in this language. It is unnecessary to form a class whose members are  $R, S$ .

When a family is indexed by a class  $I$  of sets, one class relation  $T \subseteq I \times \mathbf{No} \times \mathbf{No}$  supplies its rows. If the rows are pairwise distinct, substituting them into the elementary predecessor formula produces a class linear order on  $I$ . With repeated rows, one instead has the corresponding comparison on codes modulo equal evaluations. This is the correct format for most class-sized examples and applications.

In contrast, an expression such as

$$\{R : R \text{ is a class well-order of } \mathbf{No}\}$$

does not define an ordinary GB class of members. Our fraktur notation abbreviates a domain of class quantification. The diagonal theorem shows why a single uniform set-code replacement cannot recover the full domain either.

### 20.3 Why KM does not erase the extra level

KM strengthens comprehension by allowing class quantifiers, but its classes still have *sets* as members. It therefore does not turn the totality of class well-orders into an ordinary class. A genuine object containing arbitrary classes as members requires another sort, or a containing universe in which those classes become sets.

This distinction matters when one wants an arbitrary topology of higher-order open collections, a measure on all global well-orders, or a power object of their totality. Such notions require an explicit ambient theory. Hyperclass forcing provides one example of higher-level foundational practice [28]; it is not a reason to silently ignore the level distinction in bare KM.

There is also a separate recursion issue. Extracting the canonical initial Ord-part of one raw well-order is an elementary class construction. Iteratively deleting proper-class tails through an arbitrary transfinite sequence would have class-valued stages and needs its own analysis. General comparability of abstract class well-orders by initial-segment isomorphisms is another stronger topic; ETR implies it [27]. None of that machinery is needed for the labeled comparison proved in this article.

### 20.4 An explicit containing-universe model

A useful concrete model assumes a strongly inaccessible cardinal  $\kappa$  in an ambient universe and takes

$$U = V_\kappa, \quad \mathcal{C} = \mathcal{P}(V_\kappa).$$

This is a sufficient setting for interpreting a full class universe. The internal surreal class becomes the ambient set

$$\mathbf{No}^U = \mathbf{No}_{<\kappa}, \quad |\mathbf{No}^U| = \kappa.$$

All its ambient well-orders form a set of cardinality  $2^\kappa$ . Those that are internally set-like have ambient order type  $\kappa$ ; unrestricted well-orders may have any ambient ordinal type of cardinality  $\kappa$ .

To see the set-like assertion, every initial segment of a  $\kappa$ -enumeration has fewer than  $\kappa$  members, with ranks bounded below  $\kappa$ ; regularity puts it in  $U$ . Conversely, an ordinal type strictly exceeding  $\kappa$  has an initial segment of cardinality  $\kappa$ , which cannot be a set of  $U$ . A smaller type cannot enumerate the entire carrier.

The bounded-support class becomes an ambient order of cardinality  $\kappa$ , while the full  $\kappa$ -permutation order has cardinality  $2^\kappa$ . Thus a  $\kappa$ -sized dense suborder can coexist with a larger order of all enumerations. The exact ambient ordinal cutoff remains  $\kappa^+$ .

Every ordinal of  $U$  is below  $\kappa$ . There is no contradiction between internally embedding all ordinals and having an ambient successor-cardinal barrier. Inaccessibility is an additional assumption for this full universe example; it is not required for an individual birthday fragment.

## 20.5 A strict nonembedding result

The containing-universe interpretation makes a further comparison decidable by ordinary set-order invariants.

**Theorem 20.1** (Nonembedding in a full universe). *In the preceding full interpretation  $U = V_\kappa$ ,  $\mathcal{C} = \mathcal{P}(V_\kappa)$ , with ambient ZFC and  $\kappa$  strongly inaccessible,*

$$\mathfrak{W}_{\text{sl}}(\mathbf{No}^U) \hookrightarrow \mathfrak{W}_{\text{all}}(\mathbf{No}^U), \quad \mathfrak{W}_{\text{all}}(\mathbf{No}^U) \not\hookrightarrow \mathfrak{W}_{\text{sl}}(\mathbf{No}^U).$$

*These are assertions about the actual ambient sets of internal class well-orders and arbitrary ambient order embeddings.*

*Proof.* The first embedding is inclusion. Put  $X = \mathbf{No}^U$ . The target of the proposed reverse embedding is the dense order of all bijections  $\kappa \rightarrow X$ . Its bounded-support suborder  $D = \mathcal{P}_{\text{bd}}^U$  is dense and has ambient cardinality  $\kappa$ . For the upper bound, codes with domain  $\lambda < \kappa$  are subsets of  $\lambda \times \lambda$ ; strong inaccessibility bounds their number below  $\kappa$ , and there are  $\kappa$  possible domains. Transpositions give the matching lower bound.

Fix labels  $a < b$  in  $X$ . There are  $2^\kappa$  bijections  $F : \kappa \rightarrow X \setminus \{a, b\}$ , by the fixed-alphabet cardinality theorem. Each determines a convex adjacent pair

$$R_F = F + (a, b) <_{\text{lex}} S_F = F + (b, a)$$

in the unrestricted order. Distinct  $F$ 's determine distinct, disjoint convex fibers; one entire pair precedes the other.

If  $\Phi : \mathfrak{W}_{\text{all}}(X) \hookrightarrow \mathfrak{W}_{\text{sl}}(X)$  existed, the intervals

$$(\Phi(R_F), \Phi(S_F))$$

would be  $2^\kappa$  pairwise disjoint nonempty open intervals in the target. Each contains a point of  $D$ . Choosing the least such point under a fixed well-order of  $D$  would inject  $2^\kappa$  into  $\kappa$ , a contradiction.  $\square$

Thus equal cardinality  $2^\kappa$  and equal ordinal embedding cutoff  $\kappa^+$  do not imply equimorphism. The useful invariant here is the size of a family of disjoint open intervals relative to a dense suborder. The repository already uses related separability and cellularity arguments for the real alphabet [1, Remark `lwo:sp:rem:conditional`]; the argument above supplies the precise surreal-universe version.

There is also a direct class-theoretic obstruction, provided the inverse evaluation relation can be formed by impredicative comprehension.

**Theorem 20.2** (No definable uniform embedding in  $\text{KM} + \text{GC}$ ). *For every fixed formula, with fixed set and class parameters, that defines a total transformation  $R \mapsto \Phi(R)$  from  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  to  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ , the theory  $\text{KM} + \text{GC}$  proves that this transformation is not an order embedding. The transformation is specified relationally: the defining formula assigns a unique output class graph to each input class well-order. This is a schema for fixed formulas, not quantification over a third sort of arbitrary transformations.*

*Proof.* Suppose such an order embedding were defined. Fix  $a < b$  and a global well-order of the set codes in  $\mathcal{P}_{\text{bd}}$ . For each class bijection  $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{a, b\}$ , form the adjacent raw pair  $R_F, S_F$  from the preceding proof. Density of  $\mathcal{P}_{\text{bd}}$  in  $\mathfrak{W}_{\text{sl}}$  supplies codes strictly between  $\Phi(R_F)$  and  $\Phi(S_F)$ . Let  $q_F$  be the globally least such code. Comprehension in KM makes this a legitimate least-element definition, even if the formula specifying  $\Phi$  contains class quantifiers. Disjointness of the image intervals implies

$$q_F = q_G \implies F = G.$$

Now use KM comprehension to form the ordinary class relation

$$\text{Eval}(q, \alpha, x) \iff \exists \text{ class } F [F : \text{Ord} \xrightarrow{\sim} \mathbf{No} \setminus \{a, b\} \wedge q = q_F \wedge F(\alpha) = x].$$

Its rows are empty or the graphs of unique such bijections, and every such bijection occurs. Compose all rows with one fixed class bijection  $\mathbf{No} \setminus \{a, b\} \rightarrow \mathbf{No}$ . The resulting single class evaluator, with the ordinary set codes  $q \in \mathcal{P}_{\text{bd}}$ , contains every global enumeration of  $\mathbf{No}$ . This contradicts Theorem 19.37.  $\square$

The inverse evaluator is the decisive extra comprehension step: its definition quantifies over an arbitrary class  $F$ . The preceding proof does not claim that elementary GB comprehension suffices. Nor does it exclude arbitrary external maps between the available-class collections of a nonfull model. In particular, for an externally countable model these collections can both be countable, and its externally dense set-like order can admit an external copy of its unrestricted order. Such a map need not be representable by any uniform definition in the model. This is why the stated interpretation of an embedding matters.

## 20.6 Internal and external well-foundedness

For a model  $(M, \mathcal{X})$ , an internally well-founded class relation need not be externally well-founded. An external descending sequence might not be represented by a set in  $M$ . Thus comparing the internal theory with an external collection of all relations requires care.

Over GBC, well-foundedness of a class relation can be characterized by the absence of a set-coded descending  $\omega$ -sequence [27, Section 3]. Keeping the sets and an existing relation fixed while changing only the available classes does not arbitrarily change that relation's internal well-foundedness in suitable models. What can change is which class relations are available at all. External correctness can be ensured by an appropriate well-founded full universe interpretation, as above.

## 20.7 A practical Lean architecture

The repository already has a universe-indexed sign carrier [7]. Its type structure can be displayed schematically as:

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : forall i, signAt i != 0 <-> i < birthday
```

This is mathematical pseudocode using ASCII logical notation, not a new executable Lean declaration. The key point is the level separation: the full carrier is one universe above the types indexing its small sign lengths.

The following existing declarations were inspected:

| Module or declaration                                                   | Role in the proposed development                            |
|-------------------------------------------------------------------------|-------------------------------------------------------------|
| <code>SignSequence.ordinalOrderEmbedding</code>                         | The all-plus ordinal embedding                              |
| <code>birthdays_bounded</code> , <code>small_bounded</code>             | Control of small parameter families and bounded fragments   |
| <code>SignSequence.not_small</code>                                     | The carrier cannot be treated as small at its ordinal level |
| <code>SignSequenceCutOperation.cut</code> and <code>cut_realizes</code> | Actual separators for small cut data                        |
| <code>SizeObstructions.noUnrestrictedCuts</code>                        | The obstruction to unrestricted cut filling                 |

These are source-level observations, not results of a fresh build. The current repository contains later arithmetic modules as well; historical proposal notes should not be mistaken for the present implementation status.

A sensible formalization proceeds in four parts.

1. **Set carriers.** For  $X : \text{Type } u$ , represent well-orders as relations on  $X$  together with their laws. Prove the predecessor-based comparison and its equivalence with ordinal enumeration. This avoids transporting across ordinal representatives at every comparison.
2. **Ordinal words.** Define words with small ordinal domains, an injective subtype, ordinal concatenation, and the sign code. Formalize prefix-freeness and first disagreement before proving the cardinal spectrum.
3. **Relative set-likeness.** If  $X$  lives in  $\text{Type } (u+1)$ , express the condition that each predecessor subtype is `Small.{u}`. A well-order of this larger  $X$  has its unrestricted order type at the larger ordinal universe. A same-level enumeration must not be inferred without the relative smallness hypothesis.
4. **Uniform families.** Keep an explicit index type and an evaluation relation. The bounded-support codes are sets of ordinal pairs at the selected level. Formulate the interpolation theorem only for small index families and the no-code theorem at its correct larger level.

Lean universes and Mathlib ordinals provide precisely the needed separation [29, 30]. Universe polymorphism is a schema of well-typed constructions, not a term denoting the union of all universes. This article supplies no new Lean file and makes no machine-verification claim.

[merge] Source 12's Section 11 identifies both global orders in the full-class model  $(V_\kappa, \mathcal{P}(V_\kappa))$  of [theorem 20.1](#). Its clause (iv) gives a second proof of that theorem's non-embedding: the full order  $\mathcal{W}(\mathbf{No}_{<\kappa})$  has binary coding length  $\kappa^+$  ([theorem 6.8](#)), whereas the set-like order  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$  has coding length  $\kappa$  ([theorem 8.3](#), regular case), and coding length cannot decrease under embeddings ([lemma 6.4](#)). Source 11's proof counts  $2^\kappa$  disjoint intervals against a dense suborder of size  $\kappa$ .

## 21 A full-class model that connects the two size levels (source 12)

Assume in an ambient ZFC universe that  $\kappa$  is strongly inaccessible. Consider the two-sorted structure

$$\mathcal{M} = (V_\kappa, \mathcal{P}(V_\kappa)),$$

with all external subclasses of  $V_\kappa$  available as its classes. This is a familiar full-class interpretation of strong class set theory: regularity and the strong-limit property ensure the required closure of set operations and class Replacement. It satisfies KM and global choice. We use this as an explicit relative model, not as a claim that inaccessible cardinals are necessary for the earlier GBC theorems.



**Theorem 21.1** (Exact external identification). *For the full-class interpretation above:*

- (i) *the internal surreal carrier is externally  $\mathbf{No}_{<\kappa}$ , and its internal ordinals are the external ordinals below  $\kappa$ ;*
- (ii) *the internal set-like global surreal well-orders, viewed as an external set order, are exactly  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ ;*
- (iii) *all internal global surreal well-orders, viewed externally, form exactly  $\mathcal{W}(\mathbf{No}_{<\kappa})$ ;*
- (iv) *both external orders have cardinality  $2^\kappa$  and ordinal spectrum  $\beta < \kappa^+$ , but their binary coding ranks are respectively  $\kappa$  and  $\kappa^+$ . Therefore the full order does not embed into the set-like order.*

*Proof.* A sign sequence belongs to  $V_\kappa$  precisely when its ordinal domain has length less than  $\kappa$ . Inaccessibility gives  $|\mathbf{No}_{<\kappa}| = \kappa$ .

Every external relation on  $\mathbf{No}_{<\kappa}$  is a permitted internal class: each ordered pair of its labels is an element of  $V_\kappa$ . External well-foundedness implies internal well-foundedness. Conversely, if a relation is externally ill-founded, ambient choice supplies a descending  $\omega$ -sequence. Its countably many labels have ranks bounded below  $\kappa$  by regularity, so the sequence is an element of  $V_\kappa$  and witnesses internal ill-foundedness. Thus the well-orders agree.

An internal set has external cardinality less than  $\kappa$ . Conversely, a subset of  $\mathbf{No}_{<\kappa}$  of cardinality less than  $\kappa$  has bounded ranks and is an element of  $V_\kappa$ . Therefore internal set-likeness means that every proper predecessor segment has cardinality less than  $\kappa$ . An external well-order of  $\kappa$  labels has that property exactly when its order type is  $\kappa$ , proving (ii). Arbitrary external well-orders on those labels have types exactly the ordinals of cardinality  $\kappa$ , proving (iii). Comparisons agree by [theorem 14.7](#). Finally apply [theorems 6.8](#) and [8.3](#) and the monotonicity of coding rank under embeddings.  $\square$

## 21.1 Why this does not contradict surreal universality

Internally, each class linear order has elements that are sets of  $V_\kappa$  and hence externally at most  $\kappa$  elements. The external collection of all class well-orders has  $2^\kappa$  elements and is not an internal class carrier. Thus it is outside the scope of the internal class-universality theorem.

If  $\lambda > \kappa$  is another inaccessible cardinal, the entire external set  $\mathcal{W}(\mathbf{No}_{<\kappa})$  can be studied as an ordinary set in  $V_\lambda$ . It then embeds into the larger surreal carrier there by set-order universality. Its labels have moved up one foundational level; no same-level universal object has been created.

The singular cutoff  $\beth_\omega$  in [section 8](#) must not be used as though  $V_{\beth_\omega}$  were this model. That rank lacks the regularity used in Replacement and in the well-foundedness argument. The singular theorem is an ordinary ZFC theorem about a bounded set, not a full-class model construction at a singular height.

## 21.2 Henkin classes, countable models, and relativity

If the class collection is some  $\mathcal{X} \subsetneq \mathcal{P}(M)$ , only those well-orders whose relations lie in  $\mathcal{X}$  are admitted. External cardinality and saturation can then differ from the full-class calculation. A countable model may have an externally countable collection of admitted classes and external countable cofinal families, while internally having no *coded set-indexed* cofinal family. The missing external families are simply not internal objects.

For nonstandard or non- $\beta$  class models, one must additionally separate internal and external well-foundedness. Our exact identification in [theorem 21.1](#) used transitivity, full classes, and regularity; it was not asserted for arbitrary models. The general subtleties of class well-foundedness are discussed in [\[27, §3\]](#).

[merge] Clause (ii) of [theorem 21.1](#) is source 11’s statement that the internally set-like well-orders have ambient type  $\kappa$ , and clause (iii) its statement that unrestricted ones have every ambient type of cardinality  $\kappa$  ([section 20](#)). Source 12’s remark that the singular cutoff  $\beth_\omega$  must not be used as though  $V_{\beth_\omega}$  were such a model is the counterpart of source 11’s remark that inaccessibility is needed for the full universe example but not for an individual birthday fragment.

[merge] (3 October 2026, batch 89.) Source 31 (Part XV) extends this model from inaccessible  $\kappa$  to the full-class rank structures  $\mathcal{M}_\lambda = (V_\lambda, \mathcal{P}(V_\lambda))$  at every limit height  $\lambda > \omega$ , with the sign carrier  $S_\lambda = \bigcup_{\alpha < \lambda} {}^\alpha 2$  (so  $S_\kappa = \mathbf{No}_{<\kappa}$  as a set of sign sequences; no field operations are asserted at other cutoffs). Global choice and birthday-monotone set-like class well-orders of  $S_\lambda$  exist at every such height ([propositions XV.6.1](#) and [XV.6.2](#)); an Ord-enumeration of  $S_\lambda$  exists exactly when  $\lambda = \kappa$  is a cardinal with  $2^{<\kappa} = \kappa$  ([theorem XV.7.2](#)), a birthday-monotone one exactly when  $\lambda$  is a strong-limit cardinal ([theorem XV.7.4](#)), universal set coding and ordinal abstraction exactly when  $\lambda = \kappa = \beth_\kappa$  ([theorems XV.8.2](#) and [XV.8.4](#)); Class Replacement on a set  $a$  holds exactly when  $|a| < \text{cf } \lambda$ , so full Class Replacement exactly when  $\lambda$  is strongly inaccessible ([lemma XV.9.2](#) and [theorem XV.9.3](#)). The inaccessibility assumed above is therefore exactly what full Class Replacement needs in this family, and the warning that  $V_{\beth_\omega}$  is not such a model is made exact: there Class Replacement fails already on the domain  $\omega$ , although  $S_{\beth_\omega}$  has an ordinal birthday enumeration ([section XV.11](#)). Source 31 does not identify the external orders of clauses (ii)–(iv) at other heights; the external types of the set-like well-orders of  $S_\lambda$  are its [research question 113.241](#).

[merge] Source 09’s inaccessible model is the same setting. Its clause (ii) is the bounded form of [corollary 19.15](#), its clause (iii) of [theorem 19.10](#), and its clause (iv) the bounded form of [theorem 19.43](#); its  $\mathcal{E}_\kappa$  is  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ , the set-like order of [theorem 21.1\(ii\)](#).

## 22 A concrete universe enlargement: all objects become sets (source 09)

A useful consistency-safe visualization of the size jump is available in ordinary set theory with a strongly inaccessible cardinal. This assumption is for the illustration, not for the earlier ZFC or GBC results.

Fix a strongly inaccessible cardinal  $\kappa$ , put  $X = \mathbf{No}_{<\kappa}$ , and choose a bijection  $b : \kappa \rightarrow X$ . Externally,  $X$  is a set of cardinality  $\kappa$ . In the smaller universe  $V_\kappa$ , it plays the role of the entire surreal class. Every element of  $X$  and every set-length prefix of length below  $\kappa$  belong to  $V_\kappa$ .

Define the external set

$$\mathcal{E}_\kappa = \text{Bij}(\kappa, X)$$

and let  $\mathcal{P}_s^{<\kappa}(b)$  contain its enumerations eventually equal to  $b$  below  $\kappa$ , equivalently the permutations of  $\kappa$  with bounded support. Regularity makes bounded support equivalent here to support of cardinality less than  $\kappa$ .

**Theorem 22.1** (Inaccessible set-sized model). *In this setting:*

- (i)  $|\mathcal{P}_s^{<\kappa}(b)| = \kappa$  and  $|\mathcal{E}_\kappa| = 2^\kappa$ .
- (ii)  $\mathcal{P}_s^{<\kappa}(b) \cong X$ , as linear orders, and the core is order-dense in  $\mathcal{E}_\kappa$ .
- (iii) Every separated pair of families in  $\mathcal{E}_\kappa$  of combined size less than  $\kappa$  has an interpolant in the core.
- (iv) There is a separated  $(\kappa, \kappa)$ -configuration in the core with no interpolant even in  $\mathcal{E}_\kappa$ .

*Proof.* By [proposition 7.3](#),  $|X| = 2^{<\kappa} = \kappa$ . There are fewer than  $\kappa$  permutations on any fixed  $\alpha < \kappa$ ,

because  $\kappa$  is strong limit. Their union over  $\kappa$  stages has size at most  $\kappa$ , and the swaps of position zero with other positions give the reverse bound. The cardinality of  $\mathcal{E}_\kappa$  is  $2^\kappa$  by [note 6.12](#).

For fewer than  $\kappa$  lower and upper constraints, regularity bounds all cross first-disagreement coordinates below  $\kappa$ . The forced-prefix construction in [note 19.12](#) therefore has length below  $\kappa$ . Its chosen separators remain in  $X$ : the option families and previously used values have size below  $\kappa$ , their birthdays are bounded below  $\kappa$ , and (1) supplies a separator of birthday below  $\kappa$ . Prefix completion can be carried out at a larger cardinal still below  $\kappa$ . This proves (iii), and in particular density and the  $< \kappa$ -cut property of the core.

The same birthday argument gives the  $< \kappa$ -cut property of  $X$ . A back-and-forth of length  $\kappa$ , with partial maps of size less than  $\kappa$  at every stage, gives the isomorphism in (ii). Finally repeat [theorem 19.43](#) for  $\alpha < \kappa$  and the nonexhaustive word  $\alpha \mapsto \alpha$  into  $X$ . Both  $-1$  and each required successor ordinal belong to  $X$ , and the same comparison argument proves (iv).  $\square$

This model explains the apparently conflicting assertions that every class-coded portion embeds into the surreal line and yet all enumerations are larger. A class of set codes over  $V_\kappa$  has external size at most  $\kappa$ ; the set  $\mathcal{E}_\kappa$  has external size  $2^\kappa$ . It is an object of the larger universe, not such a class of codes in the smaller universe. External arbitrary families of  $\kappa$  or more class objects are not automatically small-universe sets.

Even externally,  $\mathcal{E}_\kappa$  only represents well-orders of length  $\kappa$ . All well-orders on the external set  $X$  also include lengths  $\alpha$  with  $\kappa < \alpha < \kappa^+$ . Thus a universe jump must record not just the carrier but which ordinal lengths are admitted.

[merge] Source 08's Section 15 discusses the same frameworks and the same full-class model; its (S08.13) is [theorem 21.1\(ii\)](#).

## 23 Universes, class models, and paradox avoidance (source 08)

The preceding proofs can be studied in several foundations, provided the meaning of “all” is kept explicit.

### 23.1 ZFC with definable classes

In ordinary ZFC, **No** is given by a formula defining sign sequences. Statements about a specified finite list of definable class functions can be translated into statements about their formulas. Set-sized cutoffs and all theorems about  $\mathcal{W}(X)$  are ordinary set theorems. Global conclusions can be formulated as schemes relative to specified definable enumerations or in a theory supplying a definable global well-order.

However, “take the set of formulas defining class well-orders and interpret them all in  $V$ ” is not a free replacement for a class collection. Formula codes are sets, but uniform unrestricted truth for all formulas of the universe is a separate issue. One cannot silently insert a full satisfaction predicate for  $V$  into the base theory. Uniformity and definability claims must state their truth resources.

### 23.2 Two-sorted GBC and stronger class theories

In a model  $(M, \mathcal{C})$  of GBC, the virtual collection  $\mathfrak{W}_{\text{sl}}$  means the bijections whose graphs belong to  $\mathcal{C}$ . The cut families in [note 19.13](#) are represented by one relation in  $\mathcal{C}$ , indexed by an  $M$ -set. The skeleton consists of the  $M$ -set support codes, interpreted relative to  $b \in \mathcal{C}$ . All class unions and selections asserted by the proofs are made inside this model.

Passing from GBC to KM strengthens class comprehension, but it still does not make proper classes eligible as members of ordinary classes. Stronger comprehension is not an additional membership sort. A literal object containing every class enumeration belongs to an explicitly higher-order theory or to an external metatheory.

The literature on class recursion gives useful warnings here: even natural uniform truth or forcing relations may require principles stronger than GBC [23, 25]. Our elementary lexicographic operators do not themselves supply such truth relations.

### 23.3 A concrete full-class model at an inaccessible cardinal

Let  $\kappa$  be strongly inaccessible in an external set universe, and consider the full-class model

$$(V_\kappa, \mathcal{P}(V_\kappa)). \quad (\text{S08.12})$$

The second occurrence of  $\mathcal{P}_s$  here is the ordinary power-set notation, not the permutation-code class. Internally,  $\text{Ord}^{V_\kappa}$  is a proper class; externally it is the set  $\kappa$ . Likewise the internal surreal class is externally  $\mathbf{No}_{<\kappa}$  and has cardinality  $\kappa$ .

The full collection of internal global enumerations is, externally,

$$\{e : \kappa \longrightarrow \mathbf{No}_{<\kappa} : e \text{ is a bijection}\}, \quad (\text{S08.13})$$

a set of cardinality  $2^\kappa$ . An internal set-supported perturbation has support of size below  $\kappa$ , equivalently bounded support in  $\kappa$ . There are exactly  $\kappa$  canonical codes: at each bound there are fewer than  $\kappa$  possibilities by strong inaccessibility, and there are  $\kappa$  bounds; the first-entry embedding gives the matching lower bound. Thus externally the skeleton has size  $\kappa$  and the full enumeration order size  $2^\kappa$ .

This is a direct set-theoretic model of the size distinction. The internal isomorphism  $\mathcal{P}_s(b) \cong \mathbf{No}$  becomes an external isomorphism between two  $\kappa$ -sized  $\eta_\kappa$ -orders. Internally “every set cut” means externally every cut specified by fewer than  $\kappa$  elements, not every external set cut.

Also, (S08.13) is the *fixed-length* order  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ . It is not the external order of all well-orderings of the set  $\mathbf{No}_{<\kappa}$ , which also allows lengths  $\kappa + 1$ ,  $\kappa + \omega$ , and all other ordinals of cardinality  $\kappa$ . Those longer enumerations may have internally proper-class initial segments and are not internal set-like well-orders. This is an important bridge between section 7.2 and the global theory.

Strong inaccessibility is used here to obtain a convenient full model and its size calculation. It is not an additional hypothesis of the GBC theorems: it describes one external way to realize their set/class distinction.

### 23.4 Restricted classes and external viewpoints

For a general class model, enlarging  $\mathcal{C}$  may add new global enumerations and new binary class words while leaving the sets of  $M$  unchanged. Thus the virtual ambient orders depend on the chosen class collection. The explicit formulas for pair coding, prefix extension, and diagonalization still work with their stated parameters in each model, but one must not identify an internal “all” with an external one.

Similarly, an internal well-foundedness assertion in a nonstandard or nontransitive model need not mean external well-foundedness. Concrete full-class models such as (S08.12) avoid some of these ambiguities; unrestricted absoluteness claims do not follow from our proofs.

### 23.5 What prevents the paradoxes

The safeguards are structural. A surreal is a set sign sequence, not a sequence of length  $\text{Ord}$ . Its birthday is a set ordinal. A support code is a set, while its identity extension is a class function. A uniform family is one class relation with set-valued indices, not a set containing proper classes. The full collection of class enumerations is virtual or lives at a higher type. Finally, “every set cut is realized” never means that a separator is required between two arbitrary proper classes.

For instance, a surreal above every ordinal would contradict [proposition 5.4](#); the left options in that purported cut are a proper class, so [theorem 5.6](#) does not apply. Likewise a list of every set-supported code could be a class enumeration, but an exhaustive class-uniform list of all global enumerations is ruled out by [note 19.41](#). Neither obstruction is bypassed by changing notation.

[merge] The three remaining sections are the other sources’ formalization proposals. The modules they name are proposals and do not exist; every existing declaration they cite exists at the current revision, and none of them states a result of this report ([section 1.5](#)).

## 24 A formalization architecture with explicit size boundaries (source 12)

### 24.1 Start with a universe-polymorphic set theorem

The most efficient verified starting point is independent of surreal arithmetic. For a type  $X$  with a linear order, represent a well-order on  $X$  by its binary relation and a proof of well-foundedness and linearity. Define the common prefix by equality of predecessor sets and equality of the restricted relations, exactly as in [section 14.1](#). Prove maximality, distinct next labels, and transitivity of comparison. This gives a generic theorem reusable for reals, surreal cutoffs, and other ordered alphabets.

For a fixed ordinal domain, represent an enumeration as an equivalence between an ordinal’s carrier and  $X$ . For all well-orders of a small  $X$ , using relations on  $X$  often avoids unnecessarily quantifying over every ordinal in a larger universe. The ordinal enumeration can be recovered later from the well-order type. This is a representation choice, not an identification of all ordinals with a small type.

### 24.2 Use the existing sign-sequence foundations selectively

The inspected repository file

`Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean`

contains, among others,

```
isPrefix_of_minimum_birthday
existsUnique_prefix_of_ordConnected
existsUnique_minimum_birthday_of_ordConnected
existsUnique_simplest_separator
```

[10]. Its comments explicitly distinguish uniqueness of a simplest separator from existence of separators for arbitrary small cuts. The last theorem assumes a separator exists. It must not be imported as though it independently established all cut-filling instances.

A suitable interface for the order-theoretic development needs a birthday map, a numerical linear order, sign concatenation and truncation, the first-difference comparison, and small-cut existence with a birthday bound. Field operations can be omitted except in illustrative examples.

### 24.3 Suggested proof modules

| Module                         | Principal obligations                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <code>CommonPrefix</code>      | The elementary definition; initiality; maximality; next-label inequality; generic full lexical linearity.            |
| <code>ExhaustiveLex</code>     | Prefix-freeness; equivalence with ordinal first-difference comparison; transport between relations and enumerations. |
| <code>BinaryCubes</code>       | Cylinder fusion; strict hierarchy; characteristic-cut embeddings; precise ordinal concatenation.                     |
| <code>PermutationCoding</code> | Pair orientation, product scheduler, transpositions, and exhaustive output.                                          |
| <code>BirthdayCutoffs</code>   | Size of $\mathbf{No}_{<\kappa}$ ; prefix-free numerical code; cone partition; regular and singular sums.             |
| <code>ClassFamilies</code>     | A formal set/class model, sections of class relations, baseline enumeration, and diagonal non-enumerability.         |
| <code>UniverseModels</code>    | A chosen inaccessible-rank interpretation; internal/external well-foundedness and identification of the two orders.  |

The finite common-prefix theorem should be proved directly for relations, not by first developing all of ordinal arithmetic. The scheduler proof should record both surjectivity and the first-row-appearance invariant; a bare cardinal pairing equivalence is insufficient. The singular theorem must use ordinal sums, never a cardinal cast that silently loses order.

### 24.4 Universe accounting

Schematically, a sign-sequence type built from small ordinal lengths has one universe level for its ordinal index and potentially a higher level for the total carrier. One should retain explicit levels rather than promise a concrete declaration before checking the chosen ordinal API. A bounded carrier with a specified small birthday bound is the cleanest first implementation.

Three operations must remain visibly different: taking a subtype of one fixed surreal carrier; lifting an existing carrier to a larger universe; and constructing all surreals of the larger universe. The third adds new birthdays and is not merely a type lift. Likewise, a type of relations on a fixed carrier can exist at a higher level without becoming a same-level class of all class relations.

For global theorems, two safe routes are available. One may formalize GB or GBC models and express theorems about their class predicates. Or one may formalize the bounded analogue inside a hierarchy of explicit universes and separately prove its interpretation in a set/class model. Neither route requires an inconsistent type of all types.

### 24.5 Axioms, kernels, and honest validation

A full formalization should record the uses of classical logic, set choice, global choice, and any ambient model-existence assumptions. The common-prefix comparison should not acquire an ETR assumption merely because abstract class order comparison sometimes uses one. The inaccessible-rank example is conditional on that ambient cardinal, whereas the bounded singular theorem needs no inaccessible cardinal.

The present package includes exact executable tests of finite shadows: sign coding, pair coding, the scheduler, the common-prefix formula, and finite adjacency. These are useful implementation



checks and counterexample detectors. They are not kernel-checked proofs of any infinite or class theorem, and no supplied file is advertised as such.

[merge] Source 09's frameworks and Lean dependency plan.

## 25 Formalization strategies and paradox avoidance (source 09)

### 25.1 Four suitable frameworks

**Birthday-bounded ZFC.** For the most immediately formalizable results, fix  $\mathbf{No}_{<\theta}$  and a length bound. Everything is then a set: relations, permutations, words, embeddings, and counterexamples. The exact ordinal spectrum and finite residual adjacency results belong here. No inaccessible cardinal is needed.

**Two-sorted GBC.** Use set variables for surreal numbers, support codes, ordinals, and bounded prefixes; use class variables for global enumerations. Treat  $E <_{\text{lex}} F$  as a formula on class variables. State family theorems with an explicit uniform ternary relation. The core is a genuine class linear order. This framework proves all the global results above without a class-of-classes object.

**A larger universe or explicit higher sorts.** To make the entire collection of smaller-universe global enumerations an object, put it in a larger universe, or use separate sorts for sets, classes, and collections of classes. Membership and comprehension must be stratified. The inaccessible model in [section 22](#) demonstrates one controlled interpretation. A higher-sort theory requires its own axioms; merely writing a new symbol for all classes does not specify one.

**Universe-polymorphic dependent type theory.** A type-theoretic formalization can construct a carrier relative to a universe, then form its function types at an appropriate larger level. It must not infer that those functions are indexed by the original small ordinals or that every large subtype is small. Universe polymorphism handles repeated constructions; it does not identify the universe levels.

### 25.2 Kelley–Morse does not create classes of classes

KM strengthens class comprehension, but its ordinary two-sorted interpretation still has sets as the members of classes. It does not turn proper classes into elements of other classes. Similarly, adding ETR resolves certain class-valued recursion problems but does not invalidate [note 19.40](#). These are separate issues: recursion strength, comprehension strength, and object level.

For non-set-like class well-orders, recursion along predecessors may need class-valued histories. One should therefore not reuse our set-valued recursion justification without checking the new hypotheses. The literature on ETR supplies the appropriate distinctions [\[24, 26\]](#).

### 25.3 Concrete reuse of the ProveIt sign-sequence foundation

At the pinned snapshot, the foundational module declares

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : ...
```

The ASCII arrow above is only a typographical transcription. The actual module supplies a linear order via zero-extended signs and contains the following relevant named declarations [7]:

| Declaration                        | Role in this project                                   |
|------------------------------------|--------------------------------------------------------|
| <code>lt_iff</code>                | First-disagreement characterization of sign order.     |
| <code>ordinalOrderEmbedding</code> | Increasing embedding of small ordinals.                |
| <code>birthdays_bounded</code>     | Birthday bounds for explicitly small families.         |
| <code>small_bounded</code>         | Smallness of bounded sign fragments.                   |
| <code>not_small</code>             | The entire carrier does not fit in the lower universe. |

The companion cut-operation file contains `cut_realizes` and `birthday_cut_le_iSup`, together with the separator theorem `exists_separator_of_small_sets` [9]. Its documentation explicitly distinguishes this order/cut development from supplying field operations. The present proofs need only the order/cut development.

## 25.4 A proposed Lean dependency plan

A practical implementation should proceed in the following dependency order, keeping the formal claims smaller than the whole manuscript at first.

1. **Fixed-alphabet enumeration order.** Represent an enumeration by its ordinal domain and an equivalence onto a fixed small alphabet. Prove prefix-freeness before defining variable-domain first disagreement. Then prove pair-switch coding, cardinality bounds, and the  $\kappa^+$  stabilization obstruction.
2. **Small injective prefixes.** Prove the extension lemma for a partial injection into a sufficiently large ordinal. Keep the equal cardinality of both complements as an explicit lemma. This part is independent of surreals.
3. **Abstract small-cut alphabets.** Parameterize the avoidance and forced-prefix interpolation lemmas by a linear order with separators for explicitly small option families. Instantiate with the existing sign carrier only afterwards.
4. **Coded support core.** Use a small support, a permutation on it, and a no-fixed-point condition, or use bounded permutations followed by a proved equivalence relation and canonical normalization. Supply a baseline equivalence as an explicit parameter.
5. **Large-family interface.** Represent rows by a single function or relation, retain its universe levels, and prove bounded reflection for small index types. State separately which large index orders possess an ordinal enumeration with small initial segments.
6. **Back-and-forth and the diagonal.** Formalize the set-valued recursion invariant. Only after its smallness obligations are discharged should the abstract core isomorphism and no-coding statements be instantiated.

In a universe-indexed version, a proposed baseline has the shape  $\text{Ordinal}.\{u\} \simeq \text{SignSequence}.\{u\}$ . Its existence is not inferred here merely because both types live in  $\text{Type}(u + 1)$ ; equal universe level does not imply equipotence. Likewise, the type of *all* functions between these carriers can be formed in the ambient type theory without becoming a small-universe class of set codes. This is exactly the distinction exhibited by [theorem 22.1](#).

## 25.5 A checklist of invalid shortcuts

The following shortcuts would change the mathematical problem or introduce an unjustified assumption:

- treating **No** as a set in its original universe, or treating all global enumerations as members of an ordinary class;
- using an arbitrary class option family in the surreal cut constructor, or replacing set-indexed interpolation by class-indexed interpolation;
- assuming every class well-order has type `Ord`, or confusing the well-order represented by an enumeration with the lexicographic order of enumerations;
- completing a partial injection without checking complement cardinalities, or using a set of individually given class witnesses without a uniform family relation;
- transferring a proof of equimorphism to isomorphism, or assuming an order embedding preserves ambient suprema;
- attributing the new global core and diagonal theorems to existing Lean declarations merely because the requisite sign-cut primitives are present.

[merge] Source 08’s Lean plan. The declarations of its table exist; source 08 correctly identifies `exists_cut_separator_of_birthday_lt` as the formal landing point of [theorem 5.6](#) and [corollary 5.7](#) ([section 1.5](#)).

## 26 A Lean formalization plan tied to ProveIt (source 08)

The repository was inspected at commit

ae7c1e6aae3dc58c9c638b35646865ad6c10387d.

The relevant project is `Algebra/SurrealNumbers`. Its README explicitly distinguishes research drafts from machine-checked results and describes an axiom audit [6]. This manuscript uses that distinction rather than treating all repository prose as a formal theorem.

### 26.1 Inspected declarations

The source `Surreal/Foundations/SignSequence.lean` defines a universe-indexed carrier of the form

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff :
    forall i, Iff (Not (signAt i = 0)) (i < birthday)
```

This display is an ASCII rendering of the inspected declaration, not a new compiled file. Its zero extension uses the ordered values minus, termination, and plus. The source gives the numerical linear order via Mathlib’s lexicographic function `order` and proves `lt_iff`, `ordinalOrderEmbedding`, `birthdays_bounded`, `small_strict_upper_bounds`, `not_small`, and `small_bounded` [7].

Two declarations in `SignSequenceCut.lean` are particularly relevant: `exists_cut_separator_of_birthday_lt` and `small_cut_fillers`. The construction uses full-length binary words, uses their complete lexicographic order, and produces a separator of exactly a prescribed birthday above

all option birthdays [8]. This is the formal landing point for [theorem 5.6](#) and [corollary 5.7](#); it is stronger than merely having a dense linear order.

| Inspected ingredient                             | Role in a formalization                                                        |
|--------------------------------------------------|--------------------------------------------------------------------------------|
| <code>SignSequence.lt_iff</code>                 | Express every comparison by its least differing coordinate                     |
| <code>birthdays_bounded</code>                   | Bound options indexed by a lower-universe small type                           |
| <code>not_small</code>                           | Exclude collapsing the whole carrier into the birthday universe                |
| <code>small_bounded</code>                       | Represent a fixed birthday fragment by a genuinely small type                  |
| <code>exists_cut_separator_of_birthday_lt</code> | Choose a separator outside a small forbidden family by increasing its birthday |
| <code>small_cut_fillers</code>                   | Discharge the ordered-cut interface used by the abstract arguments             |

## 26.2 Smallness is not the same as having a type

In Lean, `SignSequence.{u}` and `Ordinal.{u}` live in `Type (u + 1)`. They are not small relative to universe `u`. A `Set SignSequence` is a predicate on the whole larger carrier; it need not be a small family of surreals. Thus a formal cut theorem must carry a condition such as `Small.{u} I` on its index type, not merely declare its options to be a `Set`.

The same applies to permutations. A predicate-defined support in the larger universe is not automatically an admissible set support at the smaller level. The orbit should use explicit small support codes or an equivalent `Small.{u}` hypothesis. Otherwise the crucial prefix bound and the intended distinction between skeleton and full enumeration space are lost.

One safe interface keeps an equivalence

```
b : Equiv Ordinal.{u} SignSequence.{u}
```

as an explicit parameter. The present manuscript does not claim that the repository already constructs this equivalence with all the required relative-smallness properties. An implementation must either prove that baseline theorem in its universe framework or retain the baseline as a stated assumption.

## 26.3 Suggested module boundaries

The following are proposed modules, not existing files or completed formalizations.

1. `WellOrders/Exhaustive.lean`: exhaustive enumerations of a fixed carrier, no-prefix ambiguity, first-disagreement order, and relations versus enumerations.
2. `WellOrders/PairCoding.lean`: binary pair blocks, small-order universality, cardinality, prefix stabilization, and the exact ordinal spectrum.
3. `WellOrders/FiniteTails.lean`: cylinder extrema, adjacency, and finite-condensation blocks.
4. `WellOrders/PrefixExtension.lean`: small partial injections, choice of a larger support bound, complement bijections, and canonical moved-point codes.
5. `WellOrders/SetCuts.lean`: the active-row invariant, the bound on cross-disagreement coordinates, stopping, and strict separation before completion.
6. `WellOrders/BoundedPermutations.lean`: the code carrier, its small-cut interface, and the relative-universe form of the surreal-skeleton theorem.
7. `WellOrders/ClassBinaryCoding.lean`: terminal block codes, ordinal concatenation, and uniform diagonalization in the chosen higher-type representation.

The first three modules can be developed without surreal arithmetic or class metatheory. The next three isolate the delicate smallness and extension arguments. The final module should be delayed until the meaning of a uniform operator on larger function types is fixed.

## 26.4 Two complementary implementation routes

A *cutoff-first* implementation proves theorems for  $\mathbf{No}_{<\kappa}$  and  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$  with explicit regularity and cardinal hypotheses. All carriers are ordinary types and the formal statements closely match [note 7.6](#). This is the simplest route to substantial checked mathematics.

A *universe-relative* implementation abstracts over a large ordered carrier, a lower-universe small-cut operation, and a baseline equivalence. It formalizes the local orbit argument relative to that notion of smallness. Care is needed when proving back-and-forth: an arbitrary well-order of a large type may have initial segments too large for the small-cut operation. One must use an enumeration whose proper initial segments are small, rather than invoke an unrestricted large-type choice order.

A third option is a deep formalization of a two-sorted class theory or of a specified model  $(M, \mathcal{C})$ . This is the best route for comparing axiomatic strengths and internal/external semantics, but is not necessary to certify the set-sized combinatorics.

## 26.5 Trust boundary

The cited repository files were read, not rebuilt in this work. No new Lean module accompanies this manuscript, and no new kernel-checking claim is made. A future implementation should pin the repository and Mathlib revisions, compile in dependency order, reject unfinished proofs, and inspect the resulting axiom dependencies. Finite tests of codes, even exhaustive ones, cannot certify the class interpolation or back-and-forth arguments.

[merge] The four plans agree on a first milestone that is independent of surreal arithmetic: the predecessor or common-prefix comparison of well-orders on a small carrier, prefix-freeness and the successor-cardinal obstruction, then bounded-support prefix completion and small-family interpolation, reusing the existing small-cut interface. This is a proposal recorded for the collection's formalization work; no such module exists.

## Part VI

# Set-sized strata: topology, cut spectra and block types

[merge] (3 October 2026.) This Part opens the batch-81 addition of six further manuscripts, sources 13–18 ([section 1.8](#)); its notation is fixed in [section 1.3.1](#). It prints everything the six sources prove about set-sized enumeration orders: the density and cellularity of every stratum  $\mathcal{W}_\alpha(X)$  of every infinite linear alphabet ([section 27](#)), conditional incompleteness and global characters ([section 28](#)), the cut spectrum of the alphabet  $\mathbf{No}_{<\kappa}$  ([section 29](#)), the characters, fusion theorem, exact gap spectrum and gap multiplicities of the minimal slice  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  ([section 30](#)), the coding lengths, cube spectra and density of the longer strata  $\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})$  ([section 31](#)), and the six sources' set-sized baselines and warnings ([section 32](#)). [Section 33](#) records which questions of this report and of the reals report the Part answers.

The bases are chosen by generality, as in Parts I–V: source 16 for density, cellularity and incompleteness (every infinite alphabet, every stratum), source 17 for the alphabet and the minimal slice (every infinite cardinal cutoff, with no hypothesis on  $2^{<\kappa}$ ), source 15 for the longer strata (the only source treating them). Source 18, written before sources 13–17 had arrived and without knowledge of them, proves most of the minimal-slice results a third time; its density, characters and interpolation proofs differ in detail and are printed as routes, its gap-spectrum proof repeats source 17's and is recorded in notes, and its general permutation principle and exact gap multiplicities are new and printed in full. The whole Part is set theory in ZFC; no class is involved.

Throughout the Part,  $\kappa$  is an infinite cardinal used as a birthday cutoff,  $\mu = 2^{<\kappa} = |\mathbf{No}_{<\kappa}|$ ,  $r = \text{cf } \kappa$  and  $q = \text{cf } \mu$ ; a general alphabet  $X$  is an infinite linearly ordered set of cardinality  $\mu$ . Source 16 wrote  $\kappa$  for  $|X|$  and  $\theta$  for the cutoff, and source 18 wrote  $\nu, \tau$  for  $r, q$ ; their text is printed in the convention above (section 1.3.1).

## 27 Density and cellularity at every fixed type (source 16)

[merge] Source 16's Section 5, printed whole. Its  $\kappa$  (the cardinality of  $X$ ) is printed  $\mu$ , its cutoff  $\theta$  is printed  $\kappa$ , and its  $d, c$  are printed  $d, \text{cell}$  (section 1.3.1); its equations keep their numbers as (S16.n). Source 18 proves the same theorem independently, with a different lemma and a different lower-bound family (section 27.1); the reals report proves the case  $X = \mathbb{R}$  (note 27.5).

For a set linear order  $L$  with its *order topology*, let  $d(L)$  be the least cardinality of a dense subset. Let  $\text{cell}(L)$  be the supremum of the cardinalities of families of pairwise disjoint nonempty open sets. Always  $\text{cell}(L) \leq d(L) \leq |L|$ . A prefix cylinder is the collection of enumerations extending one fixed injective prefix. Cylinders are convex, but need not themselves be open. Our lower bounds use actual open intervals lying inside cylinders; our upper bounds use a family that meets every nonempty open set.

**Lemma 27.1** (No jump between distinct short-prefix branches). *If  $e < f$  in  $\mathcal{W}_\alpha(X)$  first differ at a coordinate  $\delta < \mu$ , then they are not adjacent.*

*Proof.* Write  $p = e \restriction \delta = f \restriction \delta$  and  $a = e(\delta) < b = f(\delta)$ . If there is no point between  $e$  and  $f$ , then  $e$  must be the maximum of its  $p \frown a$  cylinder, and  $f$  the minimum of its  $p \frown b$  cylinder. Any ascent in the remaining suffix of  $e$  can be swapped to produce a larger member of its cylinder. Thus that suffix is numerically strictly decreasing. Similarly, the suffix of  $f$  must be strictly increasing. Their common remaining label set is  $X \setminus (\text{ran}(p) \cup \{a, b\})$ , of cardinality  $\mu$ . Its numerical order would be both a well-order and a reverse well-order. An infinite linear order cannot have both properties. This contradiction proves the claim.  $\square$

**Theorem 27.2** (Exact density and cellularity). *Let  $|X| = \mu$  be infinite and  $|\alpha| = \mu$ . Then*

$$d(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha = \mu, \mu + 1, \text{ or } \mu + 2, \\ 2^\mu, & \text{all other } \alpha. \end{cases} \quad (\text{S16.3})$$

*Moreover  $d(\mathcal{W}(X)) = \text{cell}(\mathcal{W}(X)) = 2^\mu$ . No regularity assumption is needed.*

*Proof.* For any  $\delta < \mu$  there are  $\mu^{|\delta|}$  injective prefixes of length  $\delta$ , and each extends to an admissible enumeration. One way to see the lower count is to choose, for each coordinate, its label in a separate  $\mu$ -sized cell of a partition of  $X$ ; the choices are automatically injective. Every such cylinder contains



at least three points and hence a nonempty ambient open interval. Distinct cylinders at one level are disjoint and convex. Consequently

$$\text{cell}(\mathcal{W}_\alpha(X)) \geq \sup_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu}.$$

Suppose  $\alpha = \mu + n$ ,  $n \leq 2$ . Group enumerations by their common prefix of length  $\mu$ . A nonempty group has exactly  $n!$  points, coming from the permutations of the omitted  $n$  labels. Thus every group has at most two points. By [Lemma 27.1](#), distinct groups have dense quotient order: if two groups were consecutive, the last point of the first and first point of the second would be adjacent and differ before  $\mu$ .

It follows that every nonempty open set contains a cylinder specified before  $\mu$ , except possibly a singleton consisting of an isolated global endpoint. Indeed, select a group strictly between the relevant boundary groups, choose a member of it, and bound its first disagreement with the boundaries by a coordinate below  $\mu$ . The resulting prefix cylinder lies inside the open set. Select one completion for each injective prefix of length less than  $\mu$  and add any isolated global endpoints. This gives a dense set of size at most

$$\sum_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu}.$$

Together with the lower bound this proves the first line of [\(S16.3\)](#), including singular  $\mu$ .

If  $\alpha = \mu + n$  for finite  $n \geq 3$ , fix  $F \subseteq X$  of size  $n$ . There are  $2^\mu$  bijections  $p : \mu \rightarrow X \setminus F$ . The completion cylinder for each is a finite convex chain of  $n! \geq 6$  points. An interior point has both immediate neighbors in that chain; the interval between those neighbors is its singleton, an open set in the entire order. Different prefixes give  $2^\mu$  disjoint open singletons.

In every remaining case  $\alpha \geq \mu + \omega$ . Let  $\tau$  be the order type of the interval  $[\mu, \alpha)$ , so  $\alpha = \mu + \tau$  and  $|\tau|$  is infinite. Choose  $Y \subseteq X$  with  $|Y| = |\tau|$  and  $|X \setminus Y| = \mu$ . Each of the  $2^\mu$  bijections  $p : \mu \rightarrow X \setminus Y$  gives a convex cylinder of completions with residual order  $\mathcal{W}_\tau(Y)$ . Every such cylinder has at least three points, so supplies a nonempty ambient open interval. These intervals are pairwise disjoint. The cardinal upper bound  $|\mathcal{W}_\alpha(X)| \leq 2^\mu$  finishes the proof for all strata.

For  $\mathcal{W}(X)$  use the same construction with a fixed residual set of three labels. The resulting six-point cylinders are convex even when all enumeration lengths are admitted. Thus  $\text{cell}(\mathcal{W}(X)) \geq 2^\mu$ , and the cardinality bound gives equality throughout.  $\square$

*Remark 27.3* (Why two terminal labels differ from three). There can be  $2^\mu$  adjacent pairs in  $\mathcal{W}_{\mu+2}(X)$  while its density is only  $\mu^{<\mu}$ . An adjacent pair does not contain an open interval. Three or more terminal labels give a permutation block with an interior point, and that point is isolated. This is why the topological transition occurs at  $\mu + 3$ , rather than  $\mu + 1$  or  $\mu + 2$ . If  $\mu^{<\mu} = 2^\mu$ , the two displayed values coincide; the structural distinction still holds.

**Corollary 27.4** (Birthday-stage topology). *For every infinite cardinal  $\kappa$ , put  $\mu = 2^{<\kappa}$ . Formula [\(S16.3\)](#) applies to  $\mathbf{No}_{<\kappa}$ , whose cardinality is  $\mu$ . If  $\kappa$  is regular and  $2^{<\kappa} = \kappa$ , the first three strata have density and cellularity  $\kappa$ , whereas all later strata and the full order have density and cellularity  $2^\kappa$ .*

*Proof.* Only the last simplification needs comment. Regularity and  $2^{<\kappa} = \kappa$  imply  $\kappa^{<\kappa} = \kappa$ : for  $\lambda < \kappa$ , every function  $\lambda \rightarrow \kappa$  has bounded range, and for  $\rho < \kappa$  one has  $\rho^\lambda \leq 2^{\rho^\lambda} \leq \kappa$ . Taking the union of  $\kappa$  bounds proves the assertion.  $\square$

*Note 27.5* ([merge]; the reals report's case). For  $X = \mathbb{R}$ , the reals report proves  $d(\mathcal{W}_\alpha) = \text{cell}(\mathcal{W}_\alpha) = 2^{<\mathfrak{c}}$  for  $\alpha \in \{\mathfrak{c}, \mathfrak{c} + 1, \mathfrak{c} + 2\}$  and  $2^\mathfrak{c}$  for  $\mathfrak{c} + 3 \leq \alpha < \mathfrak{c}^+$  (lwo:prop:all-strata-density), with the

adjacent pairs at  $\mathfrak{c} + 2$  and the isolated points from  $\mathfrak{c} + 3$  on (`lwo:prop:finite-topology`) and the minimal slice in `lwo:prop:fixed-topology`. Since  $\mathfrak{c}^\lambda = 2^\lambda$  for infinite  $\lambda < \mathfrak{c}$ , one has  $\mathfrak{c}^{<\mathfrak{c}} = 2^{<\mathfrak{c}}$ , and [theorem 27.2](#) with  $X = \mathbb{R}$  is exactly that result, by the same two mechanisms (short-prefix cylinders, and the finite terminal permutation block). Source 16 does not cite these statements; it cites the reals report only for the set baseline ([note 32.1](#)). New in source 16 is the generality: every infinite linear alphabet, of any cardinality including singular ones and  $\aleph_0$ , with no density or saturation assumption on  $X$ .

*Note 27.6* ([`merge`]; the birthday-cutoff instances, and why three formulas agree). For  $X = \mathbf{No}_{<\kappa}$ , [corollary 27.4](#) gives  $d = \text{cell} = \mu^{<\mu}$  on  $\mathcal{W}_\mu$ ,  $\mathcal{W}_{\mu+1}$ ,  $\mathcal{W}_{\mu+2}$  and  $2^\mu$  on every longer stratum. Three other sources compute instances: source 17 for the minimal slice ( $d = w = \text{cell} = \mu^{<\mu}$ , [theorem 30.2](#)), source 18 for every stratum of every alphabet ([theorems 27.8](#) and [27.9](#), printed next), and source 15 for the block types  $\mathcal{W}_{\mu \cdot \gamma + n}$ , with the value  $2^{<\mu}$  when  $\rho_\kappa = \mu$ ,  $\gamma = 1$  and  $n \leq 2$ , and  $2^\mu$  otherwise ([corollary 31.7](#)). The formulas agree. If  $\mu > \kappa$  then  $\mu \leq 2^\kappa$ , so for  $\lambda < \mu$  one has  $\mu^\lambda \leq 2^{\kappa \cdot \lambda} = 2^{\max(\kappa, \lambda)} \leq 2^{<\mu}$ , whence  $\mu^{<\mu} = 2^{<\mu}$ . If  $\mu = \kappa$  is regular then  $2^{<\kappa} = \kappa$  and  $\kappa^{<\kappa} = \kappa$  (proof of [corollary 27.4](#)). If  $\mu = \kappa$  is singular,  $\kappa$  is a strong limit,  $\rho_\kappa = \kappa \cdot \kappa$  falls under source 15's second case, and  $\kappa^{<\kappa} = 2^\kappa$  ([corollary 30.26](#)). In every case the four sources give the same cardinal.

## 27.1 Source 18's route: exact cardinal invariants of bounded enumeration orders

[`merge`] Source 18's Section 4 up to its residual-alphabet subsection, printed whole as a second route to [theorem 27.2](#). Source 18 presents these theorems as advances; within batch 81 source 16 proves them too, and source 18 could not know it (its pin precedes the arrival of sources 13–17). Source 18's status table says that its proof review “corrected the arbitrary-alphabet density construction to include possible isolated global endpoints”; source 16's proof treats the same endpoints (“except possibly a singleton consisting of an isolated global endpoint”). Its remaining subsections are printed in [section 30.8](#), its residual-alphabet lemmas in [notes 29.8](#) and [29.9](#).

Throughout this section,  $X$  is an infinite linearly ordered set of cardinality  $\mu$ , identified with its initial ordinal when it is used as an index. If  $|\alpha| = \mu$ , write

$$\mathcal{W}_\alpha(X) = \{e : \alpha \longrightarrow X : e \text{ is bijective}\},$$

ordered by the first unequal coordinate. A proper-prefix comparison never occurs between two exhaustive enumerations of the same carrier. We use  $d(L)$  for the least cardinality of a topologically dense subset of the order topology, and  $\text{cell}(L)$  for the supremum of the cardinalities of pairwise disjoint families of nonempty open sets. The lower-bound families constructed below attain the asserted cellularities, so no convention concerning whether the supremum is attained affects the answers. All assertions in this section are theorems of ZFC.

### 27.1.1 Prefix cylinders and the minimal enumeration type

For an injective prefix  $p : \delta \rightarrow X$ ,  $\delta \leq \alpha$ , its cylinder is

$$C_p = \{e \in \mathcal{W}_\alpha(X) : e \restriction \delta = p\}.$$

Every cylinder is convex. If  $\delta < \mu$ , the cylinder is nonempty: both the unused alphabet and the remaining index set have cardinality  $\mu$ , so a bijection between them completes the prefix. Such a cylinder has at least three elements, obtained, for example, by permuting three unused labels in a fixed completion. It therefore contains a nonempty open interval of the ambient enumeration order. The entire cylinder need not be open; the proof uses only the open interval it contains.

**Lemma 27.7** (The dense order of cofinite prefixes). *For  $n < \omega$ , let*

$$Q_n(X) = \{p : \mu \rightarrow X : p \text{ is injective and } |X \setminus \text{ran } p| = n\}.$$

*With first-difference comparison,  $Q_n(X)$  is a dense linear order. Consequently  $\mathcal{W}_\mu(X) = Q_0(X)$  is dense.*

*Proof.* Take  $p < p'$  and let  $\delta < \mu$  be their first disagreement. If there are  $\delta < \beta < \gamma < \mu$  with  $p(\beta) < p(\gamma)$ , interchanging these two entries gives  $p < s < p'$ , with  $s \in Q_n(X)$ . Similarly, a descent  $p'(\beta) > p'(\gamma)$  after  $\delta$  gives  $p < s < p'$  by an interchange in  $p'$ . If neither operation is available, the tail of  $p$  after  $\delta$  is strictly decreasing and the tail of  $p'$  after  $\delta$  is strictly increasing. Their ranges have an infinite common subset: they omit from  $X$  only their common prefix, of size less than  $\mu$ , two first-disagreement labels, and two finite sets of omitted labels. On that common subset the numerical order would be both a well-order and a reverse well-order. This is impossible for an infinite set. Indeed, an infinite well-order contains an initial suborder of type  $\omega$ , which has no largest element. Thus one of the interchanges is available.  $\square$

**Theorem 27.8** (Exact density and cellularity at the minimal type). *For every infinite linearly ordered alphabet  $X$  of cardinality  $\mu$ ,*

$$d(\mathcal{W}_\mu(X)) = \text{cell}(\mathcal{W}_\mu(X)) = \mu^{<\mu}.$$

*No regularity assumption on  $\mu$ , and no density assumption on  $X$ , is needed.*

*Proof.* There are  $\mu^{<\mu}$  injective prefixes of length less than  $\mu$ . For the upper bound, choose one completion for each such prefix. Given  $e < f$ , density supplies  $g$  strictly between them. Choose  $\delta < \mu$  larger than both first-disagreement coordinates of  $(e, g)$  and  $(g, f)$ . Then  $C_{g|\delta} \subseteq (e, f)$ , so the selected completions are dense. Since a dense linear order with at least two points has no isolated endpoint, meeting all nonempty open intervals also gives topological density. This proves  $d \leq \mu^{<\mu}$ , and a dense set meets every member of a disjoint nonempty open family, giving  $c \leq d$ .

For the lower bound, choose distinct labels  $a_\delta \in X$  for  $\delta < \mu$ . For each  $\delta$ , take every injective prefix of length  $1 + \delta$  whose first label is  $a_\delta$ . There are  $\mu^{|\delta|}$  such prefixes. To justify this count for infinite  $\delta$ , functions  $\delta \rightarrow \mu$  can be encoded as injective maps  $\delta \rightarrow \delta \times \mu \cong \mu$  by recording the coordinate alongside its value; the reverse inequality is immediate. The finite cases are equally direct. Prefixes in different chosen levels have different first labels, while two distinct prefixes within a level have equal lengths and disagree. Their cylinders are therefore pairwise disjoint. Choose one nonempty open interval in each cylinder. The resulting disjoint family has cardinality

$$\sum_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu},$$

because  $1 + \delta < \mu$  for every  $\delta < \mu$ . This proves the lower bound and also proves that the stated cellularity is attained.  $\square$

### 27.1.2 A sharp transition at the third terminal coordinate

**Theorem 27.9** (Density and cellularity for every fixed type). *If  $|X| = \mu$  is infinite and  $\mu \leq \alpha < \mu^+$ , then*

$$d(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha \in \{\mu, \mu + 1, \mu + 2\}, \\ 2^\mu, & \mu + 3 \leq \alpha < \mu^+. \end{cases}$$

*These formulas are valid for arbitrary numerical orders on  $X$ .*

*Proof.* First take  $\alpha = \mu + n$  with  $n \leq 2$ . Restriction to the first  $\mu$  coordinates exhibits  $\mathcal{W}_{\mu+n}(X)$  as a lexicographic sum over  $Q_n(X)$  of finite fibers of size  $n!$ ; the omitted  $n$  labels are the only possible final labels. Each fiber has at most two points. By Lemma 27.7, the quotient  $Q_n(X)$  is dense. Thus every nonempty open interval in  $\mathcal{W}_{\mu+n}(X)$  contains a point whose  $\mu$ -prefix lies strictly between the prefixes of its two bounds: if the bounds had the same prefix, their fiber of at most two points would have empty open interval. A prefix of this intermediate point of length less than  $\mu$ , taken beyond both relevant first disagreements, has its entire cylinder inside the interval. Choose one completion of every prefix of length less than  $\mu$ , and also include the global least and greatest points whenever they exist. The resulting family is topologically dense: it meets every nonempty open interval, while the added endpoints handle a possible isolated endpoint in a two-point terminal fiber. Adding at most two points does not change its cardinality bound  $\mu^{<\mu}$ . The antichain of cylinders from the preceding proof gives the matching lower bound. This argument explicitly deals with the possible jumps when  $n = 2$ ; it does not assume that every cylinder is open, or that every interval contains a point separated from both bounds merely by choosing an arbitrary intermediate point.

Now suppose  $\alpha \geq \mu + 3$ , and write  $\alpha = \mu + \beta$ , where  $\beta$  is the order type of its tail after the first  $\mu$  positions. Choose  $T \subseteq X$  with  $|T| = |\beta|$  and  $|X \setminus T| = \mu$ . If  $|\beta| = \mu$ , use a partition of  $X$  into two  $\mu$ -element sets; otherwise remove a set of size  $|\beta|$ . There are  $2^\mu$  bijections  $p : \mu \rightarrow X \setminus T$ . This standard count follows either from the cardinality theorem for well-order enumerations or directly by orienting  $\mu$  disjoint pairs of labels. For each such  $p$ , the cylinder  $C_p$  consists of all continuations that biject  $\beta$  onto  $T$ . Since  $\beta \geq 3$ , it contains at least three points and hence a nonempty open interval. The cylinders are pairwise disjoint because the prefixes are distinct and have the same length. We obtain  $2^\mu$  disjoint nonempty open intervals. On the other hand,  $|\mathcal{W}_\alpha(X)| \leq \mu^\mu = 2^\mu$ , so no density or cellularity can exceed  $2^\mu$ .  $\square$

*Remark 27.10.* For  $\mu^{<\mu} < 2^\mu$ , the values for  $\mu + 2$  and  $\mu + 3$  differ. Thus merely adding one terminal coordinate can change both density and cellularity. For singular strong-limit  $\mu$ , both expressions are already  $2^\mu$ , so the geometric distinction remains but the cardinal invariants coincide. The formula is a new consequence beyond the baseline's finite-residual adjacency classification; it is not a new claim about the already-established ordinal embedding spectrum.

## 28 Countable incompleteness and global characters (source 16)

[merge] Source 16's Section 6 up to its subsection on regular saturated stages, which is printed in section 30.4.

Conditional Dedekind completeness means that every nonempty bounded-above set has a supremum, equivalently every nonempty bounded-below set has an infimum. It should not be confused with filling all small strict cuts, with order density, or with the completeness of a separately adjoined Dedekind completion.

**Theorem 28.1** (Every infinite stratum is incomplete). *For every infinite linearly ordered set  $X$  and every ordinal  $\alpha$  with  $|\alpha| = |X|$ ,  $\mathcal{W}_\alpha(X)$  is not conditionally Dedekind complete. A countable bounded monotone family witnesses this failure. The same conclusion holds for  $\mathcal{W}(X)$ .*

*Proof.* Every infinite linear order contains an increasing or decreasing  $\omega$ -sequence. For completeness, choose a countable set of distinct points and color each pair of its indices according to the numerical comparison; the infinite Ramsey theorem gives such a sequence. First assume  $a_0 < a_1 < a_2 < \dots$  in  $X$ , and put  $p(n) = a_{2n}$ .

For each  $n \geq 0$  choose  $U_n$  extending the finite prefix

$$(a_0, a_2, \dots, a_{2n-2}, a_{2n+1}),$$

and for  $n \geq 1$  choose  $L_n$  extending

$$(a_0, a_2, \dots, a_{2n-2}, a_{2n-1}).$$

All these prefixes extend to the prescribed type  $\alpha$ . First disagreement gives  $L_n < L_{n+1}$ ,  $U_{n+1} < U_n$ , and  $L_n < U_m$  for all  $n, m$ . Hence  $(U_n)$  is bounded below.

Let  $g$  be any lower bound. If it first differs from the infinite prefix  $p$  at a finite coordinate  $j$ , it must differ downward: an upward difference would give  $g > U_{j+1}$ . Consequently  $g < L_n$  for every  $n > j$ , so it is not the greatest lower bound.

Otherwise  $g$  agrees with all of  $p$ . Its remaining suffix contains all odd labels  $a_{2n+1}$ . That suffix cannot be numerically strictly decreasing: the positions of these increasing labels would form an infinite descending sequence of ordinals. It therefore has an ascent. Swap the two labels of such an ascent. The new enumeration has the same type, is strictly above  $g$ , and still agrees with  $p$  at every finite coordinate. It remains below every  $U_n$ . Again  $g$  is not greatest. If no completion of  $p$  with the specified type exists, this second case simply never occurs.

Thus  $(U_n)$  has no infimum. Reversing the numerical alphabet treats a decreasing  $\omega$ -sequence and yields the dual failure of a supremum. The argument for the full order is identical: swapping an ascent preserves whatever ordinal type the candidate bound happens to have.  $\square$

**Proposition 28.2** (Global characters from the first label). *If  $X$  has no greatest element, then*

$$\text{cf}(\mathcal{W}_\alpha(X)) = \text{cf}(\mathcal{W}(X)) = \text{cf}(X)$$

*for every admissible infinite  $\alpha$ . There is a dual statement for coinitality when  $X$  has no least element. In particular, for every infinite cardinal cutoff  $\kappa$ , all the fixed strata and the full order on  $\mathbf{No}_{<\kappa}$  have global cofinality and coinitality  $\text{cf}(\kappa)$ .*

*Proof.* Complete a cofinal family of possible first labels to enumerations. It is cofinal in the lexicographic order. Conversely, the first labels of a cofinal family of enumerations must be cofinal in  $X$ , or a larger first label would put an enumeration above the entire family. For  $\mathbf{No}_{<\kappa}$ , the ordinals below  $\kappa$  are numerically cofinal and the negatives coinital. Any family with bounded birthdays is bounded numerically, so the least possible size is  $\text{cf}(\kappa)$ .  $\square$

*Note 28.3* ([merge]; global characters in sources 15, 17 and 18). For the minimal slice of  $\mathbf{No}_{<\kappa}$ , the global part of [proposition 28.2](#) ( $\text{cf} = \text{ci} = r$ ) is also proved by source 17 ([theorem 30.2](#), with the same value for every cylinder in [lemma 30.1](#)), by source 15 ([note 30.7](#)) and by source 18 ([theorem 30.24](#), also for  $\mathcal{W}_{\mu+n}$ , and [note 30.13](#) for cylinders), each by the same first-label projection. Source 16's proposition is the general form: it holds for every stratum and for the full order  $\mathcal{W}(X)$  of any alphabet without a greatest element. Source 16's subsection "Regular saturated stages", the last part of its Section 6, is printed in [section 30.4](#) with the other gap spectra.

## 29 The cut spectrum of the alphabet $\mathbf{No}_{<\kappa}$ (source 17)

[merge] Source 17's Section 4, printed whole. Its  $\Sigma(L)$  is printed  $\text{GapChar}(L)$  (the reals report's symbol), and its displayed equations keep their numbers as (S17. $n$ ). At regular uncountable  $\kappa$  its main theorem is already in the repository ([note 29.7](#)). Sources 15 and 18 prove the alphabet lemmas again; their statements follow in [notes 29.8 to 29.10](#).

For the remainder of the set-sized analysis, fix an infinite cardinal  $\kappa$ , regarded as an initial ordinal, and write

$$X = \mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}, \quad r = \text{cf } \kappa, \quad q = \text{cf } \mu. \quad (\text{S17.3})$$

Here  $\mathbf{No}_{<\kappa}$  consists of all sign sequences of ordinal length below  $\kappa$ , ordered by first difference with  $- < \text{termination} < +$ . We write  $\text{Reg}$  for the infinite regular cardinals. No arithmetic closure of  $X$  is assumed. In particular, no field hypothesis is being inferred from the mere fact that  $\kappa$  is a cardinal.

**Definition 29.1** (Characters and genuine gaps). For a point  $x$  of a dense order  $L$  without endpoints, its left and right characters are  $\text{cf}(L_{<x})$  and  $\text{ci}(L_{>x})$ . A *gap* is a partition  $L = A \sqcup B$  into nonempty initial and final parts, with  $A < B$ , such that  $A$  has no greatest element and  $B$  no least element. Its character is  $(\text{cf } A, \text{ci } B)$ ;  $\text{GapChar}(L)$  denotes the set of these pairs. Thus a split immediately before or after an actual point is not a gap. Cofinalities of orders with a greatest element would be 1, but do not occur in  $\text{GapChar}(L)$ .

## 29.1 Size, intervals, and deletion

**Lemma 29.2.** *The order  $X$  has cardinality  $\mu$ . Every nonempty open interval and every nonempty open ray in  $X$  has cardinality  $\mu$ . Consequently, deletion of fewer than  $\mu$  labels leaves an order-dense, cofinal and coinitial suborder, with  $\mu$  labels still in each nonempty interval and ray.*

*Proof.* There are  $2^{|\alpha|}$  sign sequences of length  $\alpha$ , so  $|X| = \sum_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}$ ; the supremum on the right is at least  $\kappa$  and absorbs the number of summands. Given  $a < b$ , choose a sign sequence  $z$  strictly between them, of length below  $\kappa$ . A sufficiently long extension of  $z$  fixes its comparisons with  $a, b$  before any subsequent signs. Hence an entire sign cone lies inside  $(a, b)$ . If its fixed prefix has length  $\delta < \kappa$ , then all extensions of lengths  $\delta + \gamma$ ,  $\gamma < \kappa$ , remain below  $\kappa$ : an infinite initial ordinal is closed under sums of two smaller ordinals. This cone has  $\mu$  elements. The same construction applies to rays. Removing fewer than  $\mu$  elements cannot empty a set of cardinality  $\mu$  and leaves its cardinality unchanged, even when  $\mu$  is singular.  $\square$

**Lemma 29.3.** *The cofinality and coinitiality of  $X$  are  $r$ , and every point of  $X$  has left and right character  $r$ . The same assertions hold in  $Y = X \setminus Z$  whenever  $|Z| < \mu$ .*

*Proof.* The ordinal surreals below  $\kappa$  are cofinal, and a cofinal sequence of ordinals of length  $r$  gives an upper bound  $r$ . A family of fewer than  $r$  sign sequences has bounded birthdays below  $\kappa$  and is bounded above and below by ordinal surreals, proving the reverse bound.

At  $x$ , the sequences

$$x \frown - \frown +^\gamma \quad \text{and} \quad x \frown + \frown -^\gamma \quad (\gamma < \kappa)$$

are respectively cofinal below and coinitial above  $x$ ; a cofinal choice of  $\gamma$ 's of length  $r$  suffices. Conversely, fewer than  $r$  points have bounded birthdays. A longer sign extension of  $x$  lies between  $x$  and all the points on the appropriate side. Hence the character is not smaller than  $r$ .

For  $Y$ , replace each member of a cofinal or coinitial sequence by a remaining label in an appropriate interval. Such labels exist by Lemma 29.2. Any smaller sequence cofinal in  $Y$  would also be cofinal in the relevant interval or ray of  $X$ , by density.  $\square$

## 29.2 The alphabet's own gaps

The alphabet is not Dedekind complete. Even at a regular uncountable cutoff, its gaps need not have two equal characters. Distinguishing this fact from its uniform *point* characters is essential.

**Theorem 29.4** (Cut spectrum of the birthday alphabet).

$$\text{GapChar}(X) = \{(r, r)\} \cup \{(\theta, r), (r, \theta) : \theta \in \text{Reg}, \theta < \kappa\}.$$

*If  $|Z| < \mu$ , the same formula holds for  $X \setminus Z$ .*



*Proof.* Let  $(A, B)$  be a gap of  $X$ . Define a sign sequence  $s$  of length  $\kappa$  by set recursion: at stage  $\alpha$ , the already defined prefix  $s \upharpoonright \alpha$  is in  $X$ , so put  $s(\alpha) = +$  if that prefix belongs to  $A$ , and  $s(\alpha) = -$  if it belongs to  $B$ . Put

$$I_+ = \{\alpha < \kappa : s(\alpha) = +\}, \quad I_- = \{\alpha < \kappa : s(\alpha) = -\}.$$

The nodes  $s \upharpoonright \alpha$  for  $\alpha \in I_+$  increase and are cofinal in  $A$ ; the nodes for  $\alpha \in I_-$  decrease and are cointial in  $B$ . To verify cofinality, compare any  $x \in A$  with the path  $s$ . If  $x$  is a path prefix it is one of the displayed lower nodes. Otherwise the first disagreement cannot place  $x$  to the right of the chosen path: at a minus choice the corresponding node is in  $B$ . It therefore places  $x$  below a lower node. The dual argument handles  $B$ .

Neither index set has a greatest element, since the cut is a gap. At least one is unbounded in  $\kappa$  and has cofinality  $r$ . An unbounded other side also has cofinality  $r$ ; a bounded other side has an infinite regular cofinality  $\theta < \kappa$ . This proves the upper inclusion. Conversely, the path  $+^\theta \frown -^\kappa$ , with the evident length  $\kappa$  interpretation, has lower and upper index cofinalities  $(\theta, r)$ . Reversing signs gives  $(r, \theta)$ . A path with both signs cofinal in  $\kappa$  gives  $(r, r)$ . In each case the induced partition has no endpoint, as the associated node sequences have no last member.

Now let  $Y = X \setminus Z$  be as stated. Density of  $Y$  implies that a gap of  $Y$  corresponds either to a gap of  $X$  or to a single removed point of  $X$ : there cannot be two  $X$ -points between the two sides, because a  $Y$ -point would lie between them. At an  $X$ -gap, density preserves both cofinalities by choosing intermediate  $Y$ -points. At a removed point both characters are  $r$  by Lemma 29.3. Conversely, every  $X$ -gap survives restriction to  $Y$ , with the same characters. This proves the deletion assertion.  $\square$

**Example 29.5** (A warning about equal-character guesses). For  $\kappa > \omega$ , the path which first has  $\omega$  plus signs and then only minus signs determines an  $(\omega, r)$  gap. It lies at a boundary of a limit sign cone. The cone itself contains many surreal numbers; none is its missing boundary point. Thus uniform point character  $r$  does not mean that every gap has character  $(r, r)$ .

**Lemma 29.6.** *One has  $\mu \geq \kappa$  and  $q \geq r$ .*

*Proof.* The first assertion follows from the count in Lemma 29.2. If the increasing cardinal sequence  $2^\nu$ ,  $\nu < \kappa$ , never reaches its supremum  $\mu$ , a purported cofinal family in  $\mu$  of size below  $r$  is bounded by some  $2^\nu$ : choose an index bound for each of its members and then use  $\text{cf } \kappa = r$  to bound all those indices. Thus  $\text{cf } \mu \geq r$ . If the supremum is attained, the powers are constantly  $\mu$  for all sufficiently large cardinals  $\nu < \kappa$ . König's inequality  $\text{cf}(2^\nu) > \nu$  then gives  $\text{cf } \mu \geq \kappa$ . For completeness, that inequality follows from  $\rho^{\text{cf } \rho} > \rho$  for infinite cardinals  $\rho$ , applied to  $\rho = 2^\nu$ : if  $\text{cf } \rho \leq \nu$ , then  $\rho^{\text{cf } \rho} \leq (2^\nu)^\nu = 2^\nu$ , a contradiction.  $\square$

*Note 29.7* ([merge]; the repository's case of theorem 29.4). At a regular uncountable cutoff  $\kappa$ , theorem 29.4 is the case  $M = N$  of the cutoff gap decomposition of the report `surreal-fields-a-cross-universes` (`univ:gs:lem:boundary`), quoted in `birthday-cutoffs-and-hereditary-sets` (`hset:rem:cutoffgaps`): with no fresh signs, the gap pairs of  $\mathbf{No}_{<\kappa}$  are  $(\theta, \kappa)$  and  $(\kappa, \theta)$  for every infinite regular  $\theta \leq \kappa$ , realized by the same sign paths  $+^\theta \frown -^\kappa$ . Source 17 does not cite these reports. New in source 17 are the singular cutoffs, where the bounded side ranges over the regular  $\theta < \kappa$  while the unbounded side has character  $r = \text{cf } \kappa < \kappa$ ; the countable cutoff  $\kappa = \omega$ , where only  $(\omega, \omega)$  occurs; and the deletion clause.

*Note 29.8* (Source 18, Lemmas 4.5 and 5.3, and source 15, Lemma 9.1). Source 18's Lemma 4.5 ("Large intervals and unchanged cut sides after deletion"): *every nonempty numerical open interval in  $\mathbf{No}_{<\kappa}$  has cardinality  $\mu$ ; if  $U \subseteq \mathbf{No}_{<\kappa}$  has cardinality less than  $\mu$ , then  $\mathbf{No}_{<\kappa} \setminus U$  meets every*

such interval; in particular (1)  $\text{cf}(\mathbf{No}_{<\kappa}) = \text{ci}(\mathbf{No}_{<\kappa}) = r$ , (2) immediately below or above any point the cofinality or coinitality is  $r$ , and (3) every nonprincipal side of a cut retains its cofinality or coinitality after deletion of  $U$ , as do the two open sides at a point, even a deleted one (an endpoint retained in the residual alphabet still gives a principal side of character 1). Its Lemma 5.3 (“Small deletions preserve order density”) repeats this for  $R = X \setminus U$  and adds that *every proper gap of  $R$  has character in  $\text{GapChar}(X) \cup \{(r, r)\}$* . Source 15’s Lemma 9.1 (“Full-size numerical intervals”): *every nonempty open interval of  $\mathbf{No}_{<\kappa}$ , and every nonempty open ray, contains a sign-prefix cone order-isomorphic to  $\mathbf{No}_{<\kappa}$ ; in particular each has cardinality  $\mu$* . These are [lemmas 29.2 and 29.3](#) and the deletion clause of [theorem 29.4](#), by the same sign-cone argument. Source 15 makes the cone explicit in three cases (neither endpoint a prefix of the other: take  $a \frown +$ ;  $a$  a proper prefix of  $b$ : take  $b \frown -$ ;  $b$  a proper prefix of  $a$ : take  $a \frown +$ ), and source 18 obtains the cone from the birthday-controlled separator theorem ([section 5.2](#)) and adds an explicit approach to a point from below by  $x \frown - \frown +^\gamma$ ,  $\gamma$  cofinal in  $\kappa$ , and from above by the sign-reversed words.

*Note 29.9* (Source 15, Lemma 9.3, and source 18, Lemma 4.6). Both sources state more than [lemma 29.6](#):  $q \geq r$ ; if the supremum  $2^{<\kappa}$  is not attained then  $q = r$ , and if it is attained,  $2^\lambda = \mu$  for some cardinal  $\lambda < \kappa$ , then  $q \geq \kappa$ . The proofs are source 17’s: cofinal exponents give  $q \leq r$  in the unattained case, a shorter cofinal family would have bounded exponent witnesses, and in the attained case König’s inequality  $\text{cf}(2^\lambda) > \lambda$  for all  $\lambda$  between the attaining exponent and  $\kappa$  gives  $q \geq \kappa$ .

*Note 29.10* (Source 18, Lemma 5.2, “Alphabet gap spectrum”). For  $X = \mathbf{No}_{<\kappa}$ ,  $\text{GapChar}(X) = \{(r, r)\} \cup \{(\theta, r), (r, \theta) : \theta < \kappa \text{ an infinite regular cardinal}\}$ ; every point has left and right character  $r$ , and the global cofinality and coinitality are also  $r$ . This is [theorem 29.4](#) with [lemma 29.3](#), by the same recursion  $s(\alpha) = +$  if and only if  $s \restriction \alpha \in A$ . Source 18 adds the warning that the lemma “corrects a tempting but false simplification: unfilled alphabet cuts need not all have character  $(r, r)$ ”; at  $\kappa = \omega_1$  the cut below all positive infinite elements, represented by  $+^\omega -^{\omega_1}$ , has character  $(\omega, \omega_1)$ , and “the asymmetric alphabet gaps will, however, already be covered by the longer-prefix construction in  $P$ ”. This is the point of [example 29.5](#).

### 30 The minimal slice $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ (source 17; routes of sources 15, 16 and 18)

[merge] Source 17’s Sections 5 and 6 are printed in their order, with two insertions: source 18’s general permutation principle ([section 30.3](#)), stated before source 17’s exact gap spectrum because that theorem and source 16’s are instances of it, and source 18’s multiplicities ([section 30.5](#)). The second and third routes to the characters and the saturation threshold (sources 15 and 18) follow source 17’s saturation corollary. Source 17 writes  $[p]$  for the cylinder of an injective prefix  $p$ ; sources 16 and 18 write  $C_p$  for the same set. All three call a cylinder meeting both sides of a cut *straddling*.

#### 30.1 Exact local and topological invariants of the minimal slice (source 17)

Put  $P = \mathcal{W}_\mu(X)$ . For an injective prefix  $p : \alpha \rightarrow X$  with  $\alpha < \mu$ , let

$$[p] = \{f \in P : f \restriction \alpha = p\}, \quad Y_p = X \setminus \text{ran } p.$$

The set  $Y_p$  has cardinality  $\mu$ . Every such prefix has a completion: the remaining ordinal interval  $[\alpha, \mu)$  has order type  $\mu$ , so a bijection from it to  $Y_p$  suffices. This elementary extension fact would be false for arbitrary prescribed prefixes of length  $\mu$ .

**Lemma 30.1** (The cylinder base). *Every  $[p]$  is nonempty, convex, and clopen in the order topology of  $P$ . The cylinders form a base. No cylinder has an endpoint. Moreover,*

$$\text{cf}[p] = \text{ci}[p] = r.$$

*Proof.* Convexity is the first-difference property. Given  $f \in [p]$ , change its entry at coordinate  $\alpha$  to a fresh smaller or larger label and then complete. Lemma 29.2 provides such labels after deletion of  $f[\alpha]$ . This produces bounds on either side of  $f$  within  $[p]$ , proving openness and absence of endpoints. All length- $\alpha$  cylinders partition  $P$ , so the complement is also open. For  $a < f < b$ , a prefix of  $f$  extending beyond its first differences with  $a, b$  has its entire cylinder inside  $(a, b)$ . This proves the base assertion.

The first unused coordinate maps the cylinder onto  $Y_p$  in lexicographic blocks. Choosing one completion for cofinally many first unused labels shows  $\text{cf}[p] \leq \text{cf } Y_p = r$ . Conversely, the first unused entries of a cofinal family in  $[p]$  must be cofinal in  $Y_p$ , since otherwise a larger entry starts a cylinder above the entire family. Cointinality is dual.  $\square$

**Theorem 30.2** (Characters and size invariants). *Every point  $f \in P$  has character  $(q, q)$ , and*

$$\text{cf } P = \text{ci } P = r, \quad \text{d}(P) = \text{w}(P) = \text{cell}(P) = \mu^{<\mu}. \quad (\text{S17.4})$$

Here  $\text{d}$  is density,  $\text{w}$  is topological weight, and  $\text{cell}$  is cellularity as the supremum of cardinalities of pairwise disjoint families of nonempty open sets. In particular,  $P$  is zero-dimensional. The formula for  $\text{cell}$  does not assert that its supremum is always attained by one family.

*Proof.* For each coordinate  $\delta < \mu$ , choose  $f_\delta^- < f < f_\delta^+$  whose first difference from  $f$  is exactly  $\delta$ . Such choices are possible by the preceding lemma. Along a cofinal set of coordinates of order type  $q$ , the lower choices are increasing and cofinal below  $f$ , and the upper choices are decreasing and cointial above it. Conversely, for a family of fewer than  $q$  elements below  $f$ , the first-difference coordinates are bounded below  $\mu$ . A lower approximation differing later lies above them all. Thus neither point character is below  $q$ . The global cofinalities are Lemma 30.1 at the root.

The number of prefixes is at most  $\sum_{\alpha < \mu} \mu^{|\alpha|} = \mu^{<\mu}$ , giving that upper bound for weight, density, and cellularity. At each fixed length  $\alpha$ , the cylinders are disjoint nonempty open sets. For  $1 \leq \lambda < \mu$ , the number of injective  $\lambda$ -words over  $X$  is  $\mu^\lambda$ : a fixed injection  $j : \lambda \times X \rightarrow X$  sends any function  $h$  to the injection  $\xi \mapsto j(\xi, h(\xi))$ . Hence both density and cellularity are at least  $\mu^\lambda$  for every such  $\lambda$ . Taking suprema gives the reverse bound. Finally  $d \leq w$  and  $c \leq w$  complete the equalities. The clopen base gives zero-dimensionality.  $\square$

*Remark 30.3.* The point character is controlled by  $\text{cf } \mu$ , whereas the cofinality of a whole cylinder is controlled by  $\text{cf } \kappa$ . These can differ. Confusing the two would lose the central feature of the cut classification below. Hereditary density also equals  $\mu^{<\mu}$ , since every subspace has weight at most  $\text{w}(P)$  and  $P$  itself attains this density.

*Note 30.4* (Source 17, Proposition 5.4, “Set-alphabet background, with a direct proof”). *The order  $P$  has cardinality  $2^\mu$ . Every linear order of cardinality at most  $\mu$  embeds into every nonempty open interval of  $P$ . An ordinal  $\beta$  embeds in  $P$  if and only if  $\beta < \mu^+$ ; the same bound holds for reverse well-ordered suborders.* Source 17 calls these “already part of the source’s theory”. They are the set baseline (theorem 6.1; reals report `lwo:prop:cardinality`, `lwo:thm:ordinal-spectrum`, `lwo:sp:thm:universality`) applied inside a cylinder, whose residual alphabet has size  $\mu$ , together with lemma 30.1; the proof is the pair-orientation, initial-segment and coordinate-stabilization argument of section 6.

### 30.1.1 Fusion with an exhaustiveness requirement

**Theorem 30.5** (A dense-open fusion theorem). *The intersection of fewer than  $q$  open sets is open. The intersection of at most  $q$  dense open subsets of  $P$  is dense. In particular, when  $q = \omega$ ,  $P$  is a Baire space.*

*Proof.* At a point in fewer than  $q$  open sets, choose a cylinder neighborhood for each. Their prefix lengths have supremum below  $\mu$ , so the longest union prefix gives a cylinder inside the intersection.

Let  $(U_i)_{i < q}$  be dense open sets and start with any nonempty cylinder. Write  $X = \bigcup_{i < q} X_i$ , where the  $X_i$  are increasing and each has cardinality below  $\mu$ . Recursively refine the current prefix to a cylinder contained in  $U_i$ , using density and the cylinder base, and then append every member of  $X_i$  not yet used, in a set well-order of length below  $\mu$ . At stages below  $q$ , the accumulated length is still below  $\mu$ , including at limit stages. At the end the union has length exactly  $\mu$  and range  $X$ : its length is at most  $\mu$ , and it has used  $\mu$  distinct labels. It is therefore an element of  $P$ , belongs to every  $U_i$ , and extends the starting prefix. Fewer dense open sets can be padded by copies of  $P$ .  $\square$

The coverage step is essential. Bare prefix fusion of length  $\mu$  can produce an injective word omitting a label, which is not a point of  $P$ . The theorem does not claim a complete metrizable or a general isomorphism classification.

*Note 30.6* ([merge]; repository analogues and the regular size-matched case). For regular  $\kappa$  with  $2^{<\kappa} = \kappa$  one has  $\mu = r = q = \kappa$ , and the character part of [theorem 30.2](#) is source 11's [proposition 12.1](#) (every local and global character  $\kappa$ , and  $\eta_\kappa$ ). For  $X = \mathbb{R}$  the reals report proves the analogues: point characters  $\text{cf } \mathfrak{c}$  and global characters  $\omega$  (`lwo:prop:fixed-topology`), weight  $2^{<\mathfrak{c}}$  (`lwo:sp:prop:minimalweight`), and that fewer than  $\text{cf } \mathfrak{c}$  open sets have open intersection (`lwo:rk:cor:Pspace`). Sources 15, 17 and 18 do not cite these. New in sources 15, 17 and 18 is the alphabet  $\mathbf{No}_{<\kappa}$  at every infinite cardinal  $\kappa$ , in particular the separation of the point character  $q = \text{cf } \mu$  from the global character  $r = \text{cf } \kappa$ ; new in source 17 is the second half of [theorem 30.5](#), where every fusion sequence must also exhaust the alphabet.

*Note 30.7* (Source 15, Theorem 9.2, “Local and global cofinalities”). Source 15 writes  $L = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  for  $P$ , and  $\text{cf}^-(f)$ ,  $\text{ci}^+(f)$  for the cofinality of  $L_{<f}$  and the coinitality of  $L_{>f}$ . For every  $f \in L$ ,  $\text{cf}^-(f) = \text{ci}^+(f) = \text{cf } \mu$ ; the global cofinality and coinitality are  $\text{cf } \kappa$ ; consequently the ordinary local topological character of every point of  $L$  is  $\text{cf } \mu$ . This is the character part of [theorem 30.2](#), by the same argument (a swap of the entry at  $\eta$  with a later occurrence of a fresh smaller or larger label gives  $f_\eta^\mp$  with first difference exactly  $\eta$ ; cofinally many  $\eta$  suffice, fewer than  $\text{cf } \mu$  first differences are bounded). Source 15 adds the local base of size  $\text{cf } \mu$  formed by pairs from the two sides, and the examples: “at the cutoff  $\kappa = \omega_1$  the alphabet cardinal is  $\mu = 2^{\aleph_0}$ . The global cofinality is always  $\omega_1$ , whereas the local character is  $\text{cf}(2^{\aleph_0})$ . Under cardinal arithmetic making these cardinals different, the distinction survives at every point of the permutation order. At the singular strong-limit cutoff  $\kappa = \beth_\omega$ , every point has countable local character, even though the density and cellularity of the minimal slice are  $2^\kappa$ .”

## 30.2 The complete nonprincipal cut spectrum of the minimal slice (source 17)

### 30.2.1 Three ways that a cut can occur

For a gap  $(A, B)$  of  $P$ , say that a cylinder  $[p]$  *straddles* the cut when it meets both  $A$  and  $B$ . There is at most one straddling cylinder at any fixed length, because the cylinders at that length are disjoint convex blocks.

**Lemma 30.8** (Cut trichotomy). *Every gap of  $P$  is of exactly one of the following structural forms, using the first stage at which the construction stops.*

1. **A coordinate cut.** *Some straddling  $[p]$ , of length  $\alpha < \mu$ , has no straddling child. Its children are partitioned by a proper cut of the residual alphabet  $Y_p$ .*
2. **A limit-prefix boundary.** *At a nonzero limit  $\delta < \mu$ , all shorter prefixes on a coherent path straddle, but the union cylinder  $[p]$  lies wholly on one side. The cut is exactly the left or right boundary of  $[p]$ .*
3. **An omitted-range branch.** *A coherent prefix straddles at every length below  $\mu$ . Their union is an injection  $f : \mu \rightarrow X$  which is not surjective, and the gap is comparison with  $f$ .*

*Proof.* Begin at the empty prefix, whose cylinder straddles. At a successor stage, continue into the unique straddling child when it exists; otherwise every child belongs wholly to one side, giving case 1. At a limit  $\delta < \mu$ , the union is an injective prefix and has a nonempty cylinder. If that cylinder does not straddle, it is wholly on one side. Every word strictly below it differs at an earlier coordinate, and is below a lower member of an earlier straddling cylinder, so belongs to  $A$ . The dual argument holds above the cylinder. This proves exactly the boundary assertion in case 2.

If the process continues through all lengths below  $\mu$ , let  $f$  be the union injection. For any  $g \in P$  different from  $f$ , its first difference determines its cut side: a later straddling cylinder gives witnesses which force the orientation. If  $f$  were surjective it would itself belong to  $P$ . Whichever side contains it would then have an endpoint, contrary to the gap hypothesis. Thus  $f$  is not surjective, and all words in  $P$  are classified by their comparison with it. These alternatives cover the set recursion through  $\mu$ , and are disjoint by the first stopping stage.  $\square$

**Lemma 30.9** (Characters in the three cases). *In case 1, the gap character belongs to  $\text{GapChar}(X) \cup \{(r, r)\}$ . In case 2 it is respectively (cf  $\delta, r$ ) or  $(r, \text{cf } \delta)$ . In case 3 it is  $(q, q)$ .*

*Proof.* In case 1 partition the residual alphabet into nonempty parts  $U < V$ , according as a child lies on the lower or upper side. If  $U$  has no largest label, the lower character is  $\text{cf } U$ ; choose one point in cofinally many children. If  $U$  has largest label  $u$ , the last lower child is cofinal on the lower side and has cofinality  $r$ . The upper side has the dual description. When neither side has an endpoint, [Theorem 29.4](#) applies to  $Y_p$ . When one has an endpoint, the other has character  $r$  by [Lemma 29.3](#); the endpoint side is replaced by an entire child of character  $r$ .

At the left boundary of a limit cylinder  $[p]$ , lower approximations obtained by a smaller fresh entry at coordinates cofinal in  $\delta$  are cofinal below the cylinder. No family of fewer than  $\text{cf } \delta$  can be cofinal, since its first disagreements with  $p$  would be bounded below  $\delta$ . The upper side has  $[p]$  as a cointial subset, and the cylinder has cointiality  $r$ . The right boundary is dual.

For an omitted-range branch, approximations from either side at cofinally many coordinates give upper bounds  $q$  for both characters. Fewer than  $q$  disagreements are bounded, so a later approximation is closer to the branch. This proves equality.  $\square$

### 30.3 A general permutation-order principle (source 18)

The tree argument is not specific to surreal numbers. It separates the contribution of the alphabet from the new cuts introduced by long prefixes and omitted branches.

**Theorem 30.10** (Gap spectrum under permutation expansion). *Let  $D$  be a dense linear order without endpoints of infinite cardinality  $\mu$ . Assume that every nonempty open interval has cardinality*



$\mu$ , every point has the same two-sided character  $\nu$ , and both global cofinalities are also  $\nu$ . Let  $\text{Perm}_\mu(D)$  be the lexicographic order on bijections  $\mu \rightarrow D$ , and put  $q = \text{cf } \mu$ . Then

$$\begin{aligned} \text{GapChar}(\text{Perm}_\mu(D)) = & \text{GapChar}(D) \cup \{(\nu, \nu), (q, q)\} \\ & \cup \{(\theta, \nu), (\nu, \theta) : \theta < \mu \text{ is an infinite regular cardinal}\}. \end{aligned}$$

*Proof.* Small deletions leave an order dense in  $D$ , so the residual-cut analysis and the straddling-cylinder trichotomy apply verbatim. These give the upper bound. All the explicitly added pairs are attained by the same prefix-boundary and omitted-branch constructions. Finally, each gap of  $D$  lifts to a gap of the permutation order by placing a permutation on the left or right according to its first digit. With neither digit side principal, the two characters are unchanged. This gives the reverse inclusion.  $\square$

[merge] *The steps the proof leaves to the reader.* Source 18 writes “apply verbatim”; the two facts it uses are written out here. Put  $P = \text{Perm}_\mu(D)$ .

*Small deletions add only  $(\nu, \nu)$ .* Let  $U \subseteq D$  with  $|U| < \mu$  and  $R = D \setminus U$ . Every interval of  $D$  has  $\mu$  points, so  $R$  meets every interval of  $D$ ; in particular  $R$  is dense, has no endpoints, and has global cofinalities  $\nu$  and point characters  $(\nu, \nu)$ , because cofinal and cointial families transfer between  $D$  and  $R$  through points of the other order. Let  $(A, B)$  be a gap of  $R$ . Two points of  $D$  cannot lie between  $A$  and  $B$ , since a point of  $R$  would lie between them. If no point of  $D$  lies between them, the downward closure of  $A$  and the upward closure of  $B$  in  $D$  form a gap of  $D$  with the same characters. If one point  $x$  of  $D$  lies between them, then  $x \in U$ ,  $A$  is cofinal in  $D_{<x}$  and  $B$  cointial in  $D_{>x}$ , and the character is the point character  $(\nu, \nu)$  of  $x$ . Hence  $\text{GapChar}(R) \subseteq \text{GapChar}(D) \cup \{(\nu, \nu)\}$ .

*The three cases.* Let  $(A, B)$  be a gap of  $P$ . The prefixes  $p$  of length below  $\mu$  whose cylinders  $C_p$  meet both sides form a chain under extension, as in lemma 30.8. If there is a longest one,  $p$ , the residual alphabet  $R_p = D \setminus \text{ran } p$  (with  $|\text{ran } p| < \mu$ ) is cut into the digits whose child cylinders lie in  $A$  and those whose child cylinders lie in  $B$ ; both parts are nonempty and,  $R_p$  being dense, they do not both have endpoints. If neither has one, the character of  $(A, B)$  is the character of this gap of  $R_p$ , which lies in  $\text{GapChar}(D) \cup \{(\nu, \nu)\}$  by the first step:  $A \cap C_p$  is cofinal in  $A$  and is the lexicographic sum of the lower child cylinders. If the lower part has a last digit  $u$ ,  $A$  is cofinally the child cylinder  $C_{p \frown u}$ , whose cofinality is that of its residual alphabet,  $\nu$ , while  $B$  approaches  $u$  from above in  $R_p$ , with cointiality  $\nu$ ; the dual case is the same, so the character is  $(\nu, \nu)$ . If the straddling lengths have a limit supremum  $\delta < \mu$ , the union  $p$  has a nonempty cylinder lying on one side, the gap is a boundary of  $C_p$ , and its character is  $(\text{cf } \delta, \nu)$  or  $(\nu, \text{cf } \delta)$ : the approximations with a smaller (larger) unused digit at coordinates cofinal in  $\delta$  exist because rays of  $D$  contain intervals of size  $\mu$ , and  $C_p$  has global characters  $\nu$ . If the lengths are cofinal in  $\mu$ , the union is a nonsurjective injection  $\mu \rightarrow D$  and the character is  $(q, q)$ , exactly as in case 3 of lemma 30.9. This proves the upper bound.

*Attainment.* For a regular  $\theta < \mu$ , the two boundaries of the cylinder of an injective prefix of length  $\theta$  have characters  $(\theta, \nu)$  and  $(\nu, \theta)$ ; the lower boundary of a one-digit cylinder  $C_{\langle x \rangle}$  has character  $(\nu, \nu)$ , since its lower side approaches the digit  $x$  from below and  $C_{\langle x \rangle}$  has cointiality  $\nu$ ; a bijection of  $\mu$  onto  $D \setminus \{x\}$  gives  $(q, q)$ ; and a gap of  $D$  lifts at the first digit, as source 18 says.  $\square$

**Theorem 30.11** (Exact gap spectrum). *For every infinite cardinal  $\kappa$ , with the parameters in (S17.3),*

$$\text{GapChar}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \{(r, r), (q, q)\} \cup \{(\theta, r), (r, \theta) : \theta \in \text{Reg}, \theta < \mu\}. \quad (\text{S17.5})$$

*Proof.* The upper inclusion follows from the two preceding lemmas and  $\kappa \leq \mu$ : alphabet gap pairs involving a regular cardinal below  $\kappa$  already belong to the displayed set.



For  $(r, r)$ , fix a label  $x \in X$  and place on the lower side every word whose first entry is at most  $x$ , and on the upper side every word whose first entry exceeds  $x$ . The last lower child has cofinality  $r$ , and the upper first-entry ray has coinitality  $r$ . Neither side has an endpoint.

For any infinite regular  $\theta < \mu$ , choose an injective prefix of length  $\theta$ . Its two boundary cuts have characters  $(\theta, r)$  and  $(r, \theta)$  by the limit-boundary calculation. The fresh smaller and larger labels used there exist at every coordinate, so both sides of both cuts are nonempty.

Finally choose a bijection  $f : \mu \rightarrow X \setminus \{x\}$  for some label  $x$ . Each proper prefix is completable in  $P$ , but  $f$  itself is not in  $P$ . Its comparison cut is a gap of character  $(q, q)$ .  $\square$

*Note 30.12* ([merge]; [theorem 30.11](#) as an instance).  $D = \mathbf{No}_{<\kappa}$  satisfies the hypotheses of [theorem 30.10](#) with  $\nu = r$ : [lemmas 29.2](#) and [29.3](#) give the interval size and the uniform point and global characters. With  $\text{GapChar}(\mathbf{No}_{<\kappa})$  from [theorem 29.4](#), whose asymmetric pairs  $(\theta < \kappa)$  are among those with  $\theta < \mu$ , the principle returns exactly ([S17.5](#)). Source 17's proof above is the direct one.

*Note 30.13* (Source 18, Theorem 5.1, “Exact gap spectrum”, and Lemma 5.4). Source 18's Section 5 “settles the cut-spectrum question for the minimal-length permutation order at every infinite cardinal birthday cutoff”. It writes  $P$  for the bijections  $\mu \rightarrow \mathbf{No}_{<\kappa}$ , defines a *proper Dedekind gap* as a partition  $P = A \sqcup B$  into nonempty sets with  $A < B$ ,  $A$  without greatest and  $B$  without least element, of character (cf  $A$ , ci  $B$ ), and writes  $\text{GapChar}(P)$  for the set of characters (“this convention excludes principal cuts and the two external ends”). Its Theorem 5.1 is ([S17.5](#)) together with: *every displayed pair is attained; (1) every point of  $P$  has left and right character  $q$ ; (2) the global cofinality and coinitality are  $r$ ; (3) the principal cuts have character  $(1, q)$  or  $(q, 1)$ ; (4) if empty sides are assigned cofinality 0, the external ends have character  $(0, r)$  and  $(r, 0)$* . Its Lemma 5.4: *every cylinder  $C_p$  has cofinality and coinitality  $r$  and no endpoints; if  $p$  has nonzero limit length  $\delta < \mu$ , the lower boundary of  $C_p$  is a proper gap of character (cf  $\delta, r$ ) and the upper boundary one of character  $(r, \text{cf } \delta)$* . These are [lemma 30.1](#), case 2 of [lemma 30.9](#) and [theorems 30.2](#) and [30.11](#), and source 18's proof is source 17's: the straddling prefixes form a chain under extension, and its three cases (a longest straddling prefix, a limit supremum  $\delta < \mu$ , lengths cofinal in  $\mu$ ) are the three cases of [lemma 30.8](#), in the same order, with the same character computations. Source 18 attains  $(r, r)$  by the boundary of a successor-length cylinder  $C_{\langle x \rangle}$  (the exterior side approaches the digit  $x$ ) rather than by source 17's first-entry split; “every infinite regular  $\theta < \mu$  can be used as  $\delta$ ” in Lemma 5.4, so the asymmetric pairs are attained “independently of which of them already occur as gaps in the alphabet”.

### 30.4 Source 16's route: regular saturated stages over any saturated alphabet

[merge] The last subsection of source 16's Section 6. It is printed here, after source 17's spectrum, because it computes the same spectrum for the regular size-matched case and generalizes it to every regular saturated alphabet of the same size ([note 30.15](#)).

Suppose  $\mu$  is regular infinite and  $X$  is an  $\eta_\mu$  order of size  $\mu$ . The earlier report proves that  $\mathcal{W}_\mu(X)$  is itself  $\eta_\mu$ , with every local and global cofinality and coinitality  $\mu$  ([proposition 12.1](#)). Here is the mechanism. If one side is empty, choose a fresh first label beyond all first labels on the other side and complete it. For two nonempty families of fewer than  $\mu$  lower and upper enumerations, bound their cross first disagreements below  $\mu$ . Follow their forced common coordinates until the active next labels have strictly separated lower and upper sets. Choose a fresh interpolating label using saturation, and complete the resulting prefix. Fewer than  $\mu$  used labels can be avoided by a further small cut. The procedure must stop before the bound is passed. Small cut filling gives the lower character bound; [Note 32.1](#) gives the upper bound.

**Theorem 30.14** (Complete gap-character spectrum). *Under the preceding hypotheses, the cofinality pairs of proper unfilled Dedekind cuts in  $\mathcal{W}_\mu(X)$  are exactly*

$$(\mu, \lambda) \quad \text{and} \quad (\lambda, \mu), \quad \omega \leq \lambda \leq \mu, \quad \lambda \text{ regular.} \quad (\text{S16.4})$$

*Both sides of a cut here are nonempty, and neither has an endpoint.*

*Proof.* The two characters are infinite regular cardinals. Neither exceeds  $\mu$ , because a larger character would produce a monotone sequence of type  $\mu^+$ , contrary to [Note 32.1](#). They cannot both be below  $\mu$ : cofinal and coinital sets of those sizes would form a strict small cut, and saturation would insert a point in the supposed unfilled cut. This proves necessity.

For an infinite regular  $\lambda < \mu$ , fix an injective prefix  $p$  of length  $\lambda$ . Its completion cylinder  $C_p$  is convex and has no endpoints; its cofinality and coinitality are  $\mu$ , by the same saturation argument. The cut just above the entire cylinder has lower character  $\mu$ . Enumerations above the cylinder first differ upward from  $p$  at some coordinate below  $\lambda$ . Choose such upward deviations at a cofinal sequence of coordinates in  $\lambda$ ; their completions form a coinital family of size  $\lambda$  on that upper side. Fewer than  $\lambda$  deviation coordinates are bounded, and a later upward deviation lies below all of them. Thus the upper character is exactly  $\lambda$ . The cylinder has no maximum and the upper side has no minimum, so this cut is unfilled. The lower edge gives  $(\lambda, \mu)$ .

For  $(\mu, \mu)$  fix  $a \in X$  and split the order according to first label  $\leq a$  or  $> a$ . The lower side is cofinally the cylinder with first label  $a$ , of character  $\mu$ . The upper side has coinitality  $\mu$ , inherited from the native segment above  $a$ . Both sides have no endpoint. This realizes the final case.  $\square$

For regular  $\kappa$  satisfying  $2^{<\kappa} = \kappa$ , these conclusions apply to  $\mathbf{No}_{<\kappa}$ . They answer the earlier report's regular-stage density, cellularity, and cut-character questions under precisely those hypotheses. Singular stages and regular stages of larger cardinality require separate analysis; the theorem does not erase the singular coding phenomenon in [\(S16.2\)](#).

*Note 30.15* ([merge]; source 16's theorem as an instance). An  $\eta_\mu$  order  $X$  of regular cardinality  $\mu$  satisfies the hypotheses of [theorem 30.10](#) with  $\nu = \text{cf } \mu = \mu$ : it is dense without endpoints, its intervals have size  $\mu$ , and every point and both ends have character  $\mu$ , by saturation and  $|X| = \mu$ . An  $\eta_\mu$  order has no gap with both characters below  $\mu$ , so the principle gives the pairs  $(\mu, \theta)$ ,  $(\theta, \mu)$  with  $\theta \leq \mu$  regular, which is [\(S16.4\)](#). For  $X = \mathbf{No}_{<\kappa}$  with  $\kappa$  regular and  $2^{<\kappa} = \kappa$ , [theorem 30.14](#) and [example 30.20](#) coincide. Source 17 is more general in the cutoff (any  $\kappa$ , no hypothesis on  $2^{<\kappa}$ ), source 16 in the alphabet (any regular saturated alphabet). For  $X = \mathbb{R}$  the reals report's `two:rk:thm:Pgaps` is the corresponding spectrum,  $\{(\omega, \theta), (\theta, \omega) : \theta < \mathfrak{c} \text{ regular}\} \cup \{(\text{cf } \mathfrak{c}, \text{cf } \mathfrak{c})\}$ ; it is source 17's formula with  $r = \omega$ ,  $q = \text{cf } \mathfrak{c}$ ,  $\mu = \mathfrak{c}$  and the alphabet spectrum  $\text{GapChar}(\mathbb{R}) = \emptyset$  (the formula's  $(r, r)$  comes from a principal digit split, and  $(\omega, \omega)$  is the case  $\theta = \omega$ ). Neither source 16 nor source 17 cites it.

### 30.5 How many gaps have each character? (source 18)

[merge] “The same classification” and “the three cases” below are the straddling-cylinder classification of the gaps of  $P$ : a longest straddling prefix, a limit supremum of straddling lengths below  $\mu$ , and lengths cofinal in  $\mu$ , which are cases 1, 2 and 3 of [lemma 30.8](#) ([note 30.13](#)). Source 18's  $d_*$  is the density  $\mu^{<\mu}$  of  $P$  ([theorem 30.2](#)); its prefix variable  $q$  is printed  $p'$ , since  $q = \text{cf } \mu$  here. The theorem is the minimal-slice analogue of the reals report's count of  $2^{\mathfrak{c}}$  gaps of  $\mathcal{W}$  (`two:sp:thm:gapcount`), which source 18 does not cite.

The same classification gives exact multiplicities, not just a list of the possible character pairs. Write

$$d_* = \mu^{<\mu} = \sup_{\lambda < \mu} \mu^{|\lambda|}.$$

This is also the cardinality of the tree of injective prefixes of length less than  $\mu$ .

**Theorem 30.16** (Exact multiplicities). *The minimal permutation order  $P$  has exactly  $2^\mu$  proper Dedekind gaps. For a pair  $(a, b)$  in the spectrum of [Note 30.13](#), let  $G_{a,b}$  be the set of proper gaps of that character. Then*

$$|G_{a,b}| = \begin{cases} 2^\mu, & (a, b) = (q, q), \\ d_*, & (a, b) \neq (q, q). \end{cases}$$

*In particular, the Dedekind completion of  $P$ , with or without its two external ends, has cardinality  $2^\mu$ .*

*Proof.* First, every full nonsurjective injection  $f : \mu \rightarrow X$  gives a proper gap of character  $(q, q)$ . Fix  $x_* \in X$  and consider injections into  $X \setminus \{x_*\}$ . There are  $2^\mu$  such injections: for a lower bound partition this set into  $\mu$  disjoint pairs and choose one entry of each pair according to a binary word; the upper bound is  $\mu^\mu = 2^\mu$ .

Distinct injections give distinct gaps. If  $f < g$  first differ at  $\alpha$ , choose an unused digit strictly between  $f(\alpha)$  and  $g(\alpha)$  and extend the common prefix followed by that digit to a bijection. This is possible because the intervening interval has cardinality  $\mu$ , while fewer than  $\mu$  digits have been used. The resulting member of  $P$  lies strictly between  $f$  and  $g$ . There are therefore at least  $2^\mu$  gaps of character  $(q, q)$ .

For an upper bound on all gaps, the tree of short prefixes has cardinality  $d_* \leq 2^\mu$ , an alphabet of size  $\mu$  has at most  $2^\mu$  cuts, and there are at most  $\mu^\mu = 2^\mu$  full injections. The three cases of the straddling-cylinder classification therefore give at most  $2^\mu$  gaps altogether. This proves both the total number and the multiplicity of  $(q, q)$ .

Next fix an infinite regular  $\theta < \mu$ . There are exactly  $d_*$  prefixes whose lengths have cofinality  $\theta$ . For the nontrivial lower bound, take any infinite cardinal  $\lambda$  with  $\theta \leq \lambda < \mu$  and put

$$\delta = \lambda \cdot \theta.$$

Then  $\delta < \mu$ ,  $|\delta| = \lambda$ , and  $\text{cf } \delta = \theta$ . There are  $\mu^\lambda$  injective words of this length. One way to see the lower bound is to partition  $X$  into  $\lambda$  disjoint sets, each of cardinality  $\mu$ , and choose the digit at each coordinate from its assigned set. Taking the supremum over  $\lambda$  gives  $d_*$ . The opposite bound is the cardinality of the whole prefix tree.

Distinct prefixes give distinct lower-boundary cuts. For incomparable prefixes, an unused digit strictly between their first disagreeing digits yields a permutation between their cylinders; the interval has size  $\mu$  and fewer than  $\mu$  digits have been used. If  $p'$  properly extends  $p$ , choose at the first additional coordinate an unused digit smaller than the corresponding digit of  $p'$ . This gives a point of  $C_p$  below all of  $C_{p'}$ ; a larger choice gives a point above all of  $C_{p'}$ . Thus the lower boundaries, and also the upper boundaries, are distinct in this case as well. Consequently there are at least  $d_*$  gaps of each character  $(\theta, r)$  and  $(r, \theta)$ .

It remains to bound the number of gaps of any character other than  $(q, q)$ . Full-length omitted branches contribute none of these. Limit-cylinder boundaries contribute at most  $d_*$  gaps. For the maximal-prefix case distinguish two possibilities.

If  $q = r$ , the character being counted is not  $(r, r)$ . In the alphabet sign-path description, a gap not of character  $(r, r)$  has one sign set bounded below  $\kappa$ , so its path is eventually constant. There are at most  $\sum_{\delta < \kappa} 2^{|\delta|} = \mu$  such paths. Deleting a small set only adds  $(r, r)$  gaps, and a principal digit cut likewise lifts to that character. Thus each maximal prefix contributes at most  $\mu$  gaps of the character under consideration, and all short prefixes together contribute at most  $d_* \cdot \mu = d_*$ .

If  $q \neq r$ , then  $\mu > \kappa$ : equality would give  $\text{cf } \mu = \text{cf } \kappa$ . Every alphabet cut is represented by a length- $\kappa$  sign word, so there are at most  $2^\kappa$  of them. Cuts created by deletions and retained points add at most  $\mu$ . Since  $\kappa < \mu$ ,

$$2^\kappa \leq \mu^\kappa \leq d_*.$$

Again all maximal prefixes together contribute at most  $d_*$  gaps.

Every allowed character other than  $(q, q)$  has the form  $(\theta, r)$  or  $(r, \theta)$  for a regular  $\theta < \mu$ . The only apparent exception would be  $(r, r)$  with  $r = \mu$ , but then  $\kappa = \mu$  is regular and  $q = r$ , so that pair is precisely the already excluded  $(q, q)$ . The prefix-boundary lower bound therefore applies to every remaining character, completing the multiplicity calculation.

Finally  $|P| = 2^\mu$ , and its completion adds one point for each proper gap (and optionally its two ends). The completion cardinality is consequently  $2^\mu$ .  $\square$

### 30.5.1 Consequences and distinctions

The exact formula makes several distinctions transparent.

1. If  $\mu = \kappa$  is regular, then  $r = q = \kappa$  and the spectrum is

$$\{(\kappa, \kappa)\} \cup \{(\theta, \kappa), (\kappa, \theta) : \theta < \kappa \text{ infinite regular}\}.$$

In particular, the order's  $\eta_\kappa$  property does not prohibit asymmetric gaps with one small side.

2. At a singular strong-limit cutoff,  $\mu = \kappa$  and  $r = q = \text{cf } \kappa$ . The asymmetric pairs range over every infinite regular  $\theta < \kappa$ , including regular cardinals much larger than  $\text{cf } \kappa$ .
3. If  $\kappa = \omega_1$  and  $2^{\aleph_0} = \aleph_2$ , the proper gap spectrum consists exactly of

$$(\omega, \omega_1), \quad (\omega_1, \omega), \quad (\omega_1, \omega_1), \quad (\omega_2, \omega_2).$$

Every point has character  $(\omega_2, \omega_2)$ , while both global cofinalities are  $\omega_1$ . Thus local character, global character, and gap character cannot be conflated.

4. If  $\kappa = \omega$ , then  $\mu = r = q = \omega$  and every proper gap has character  $(\omega, \omega)$ .

The set of possible character pairs depends only on  $\mu$ ,  $r$ , and  $q$ ; its proof retains substantially more information. Every gap has a precise structural cause: a split between next digits, a boundary of a limit-length cylinder, or a nonsurjective branch of full length. Coinciding character pairs need not have the same structural cause. Nor does equality of these spectra by itself establish order isomorphism.

Natural further problems include classifying the order of the gaps of each character, studying which of the three mechanisms can be recognized intrinsically from the pure linear order, and extending the analysis to strata of nonminimal enumeration length. In the last problem a prefix can leave a small residual alphabet near the end of the domain, so the uniform large-interval argument used here no longer applies without a separate analysis of the ordinal tail.

## 30.6 Saturation and illustrative cases (source 17)

**Definition 30.17.** For an infinite regular cardinal  $\tau$ , an order is an  $\eta_\tau$  order when every pair of separated sets  $A < B$ , each of size below  $\tau$ , has a strict interpolant. Empty sides are allowed.

**Corollary 30.18** (Exact saturation threshold). *The order  $P$  is an  $\eta_r$  order, and is not an  $\eta_{r+}$  order.*

*Proof.* Suppose  $A < B$  have sizes below  $r$  and no strict interpolant. If one side is empty, the other would be cofinal or coinital in  $P$ , contradicting (S17.4). If both are nonempty, let

$$D = \{x : \exists a \in A (x \leq a)\}, \quad U = \{x : \exists b \in B (b \leq x)\}.$$

They are disjoint, and no strict interpolant means that they cover  $P$ . If  $D$  has a greatest element  $x$ , then  $B$  is coinital immediately above  $x$ , contradicting right point character  $q \geq r$ . If  $U$  has a least element, the dual contradiction uses  $A$ . If neither has an endpoint,  $(D, U)$  is a gap with both characters below  $r$ , forbidden by (S17.5). An  $(r, r)$  gap exists; cofinal and coinital witnesses of size  $r$  show failure of  $\eta_{r+}$ .  $\square$

**Example 30.19** (The countable birthday cutoff). At  $\kappa = \omega$ ,  $X$  is the order of finite sign sequences,  $\mu = r = q = \omega$ , and  $P$  has countable weight and density, character  $(\omega, \omega)$  at every point, and only  $(\omega, \omega)$  gaps. It is a Baire, zero-dimensional order. These facts do not identify it with the real line, whose order topology has a completely different connectedness structure.

**Example 30.20** (Regular size-matched cutoffs). If  $\kappa$  is regular and  $2^{<\kappa} = \kappa$ , then  $\mu = r = q = \kappa$ . The gap spectrum is

$$\{(\kappa, \kappa)\} \cup \{(\theta, \kappa), (\kappa, \theta) : \theta \in \text{Reg}, \theta < \kappa\}.$$

The order is  $\eta_\kappa$  but not  $\eta_{\kappa+}$ . The small character on one side of a limit-prefix boundary does not violate  $\eta_\kappa$ , because the other side requires  $\kappa$  witnesses.

**Example 30.21** (A singular strong-limit cutoff). At  $\kappa = \beth_\omega$ , one has  $\mu = \kappa$  and  $r = q = \omega$ . Every point has countable character, but for every infinite regular  $\theta < \kappa$  there are gaps of character  $(\theta, \omega)$  and  $(\omega, \theta)$ . Weight, density and cellularity are  $\kappa^{<\kappa} = 2^\kappa$ . This complements rather than replaces the source's  $\kappa \cdot \kappa$  binary-coding-length theorem (theorem 8.3).

[merge] (3 October 2026.) Source 17 wrote “are  $\kappa^{<\kappa}$ , with no extra cardinal-arithmetic simplification asserted”. At a singular strong limit  $\kappa$  one has  $2^\kappa \leq \kappa^{\text{cf } \kappa} \leq \kappa^{<\kappa} \leq 2^\kappa$ , so the value is  $2^\kappa$ ; source 18 proves this identity in corollary 30.26, and source 15 obtains the same value  $2^\kappa$  from its block-type theorem (corollary 31.7).

*Remark 30.22* (What has not been classified). The preceding formulas concern the minimal type  $\mu$ . They do not extend by replacing  $\mu$  with an arbitrary ordinal  $\alpha$  of cardinality  $\mu$ : a prefix of length below  $\alpha$  can already use  $\mu$  labels, leaving a radically smaller residual alphabet. Nor do equal values of these invariants imply order-isomorphism. Both issues remain in the research program at the end of this article.

### 30.7 Source 15’s route to the saturation threshold

[merge] Source 15’s Section 9, “Exact local characters at every cardinal cutoff”. Its Lemma 9.1 is recorded in note 29.8, its Theorem 9.2 in note 30.7 and its Lemma 9.3 in note 29.9; its saturation theorem is proved directly, by a prefix search, not from the gap spectrum as in corollary 30.18, and is printed in full. Its  $\delta = \text{cf } \kappa$  is printed  $r$ . Its first sentence describes this report at source 15’s pin; the point characters it mentions are now theorem 30.2.

The repository leaves the point characters of  $\mathcal{W}_{2^{<\kappa}}(\mathbf{No}_{<\kappa})$  open when  $2^{<\kappa}$  need not equal  $\kappa$ . They can be computed exactly. The local invariant is controlled by the enumeration length, whereas the global invariant is controlled by the numerical alphabet.

Throughout this section let

$$X = \mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}, \quad r = \text{cf } \kappa, \quad L = \mathcal{W}_\mu(X),$$

where  $\kappa$  is an infinite initial cardinal. For  $f \in L$ , write  $\text{cf}^-(f)$  for the cofinality of the open initial segment  $L_{<f}$ , and  $\text{ci}^+(f)$  for the coinitality of the open final segment  $L_{>f}$ .

We use the convention that an order is  $\eta_\nu$  if every cut  $A < B$  with  $|A| + |B| < \nu$  has a strict interpolant, including the cases where one side is empty. This does not assert that an arbitrary cut with larger sides is filled.

**Theorem 30.23** (Exact saturation threshold for the minimal slice). *The order  $L = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  is  $\eta_{\text{cf } \kappa}$ . For every infinite cardinal  $\nu > \text{cf } \kappa$ , it is not  $\eta_\nu$ . Equivalently, among infinite cardinal cut-size parameters,*

$$L \text{ is } \eta_\nu \iff \nu \leq \text{cf } \kappa.$$

*In particular, at every regular  $\kappa$ , the minimal slice is  $\eta_\kappa$  without the additional assumption  $2^{<\kappa} = \kappa$ .*

*Proof.* Let  $A < B$  be subsets of  $L$  with  $|A| + |B| < r$ . Any family of fewer than  $r$  surreal labels has birthdays bounded below  $\kappa$ . Consequently, for any strict numerical cut formed by such labels, there are two points  $u < v$  of  $X$  strictly between its sides. This follows from the usual surreal cut construction applied twice; all resulting birthdays stay below  $\kappa$ . The same statement holds with an empty side. By [Note 29.8](#), the interval  $(u, v)$  has cardinality  $\mu$ . Thus a numerical cut of this size admits a separator outside any prescribed set of fewer than  $\mu$  labels.

If one of  $A, B$  is empty, choose the first entry strictly above all first entries on the left or below all first entries on the right, and complete to a permutation. Suppose both are nonempty. For each  $(a, b) \in A \times B$ , let  $d(a, b) < \mu$  be its first-difference coordinate. By [Note 29.9](#),

$$\beta = \sup\{d(a, b) + 1 : a \in A, b \in B\} < \mu.$$

Build an injective prefix  $p$ , keeping as active those rows of  $A$  and  $B$  that still agree with  $p$ . At a stage  $\eta$ , every active left value at coordinate  $\eta$  is at most every active right value. If a value occurs on both sides, it is unique; append that value to  $p$  and keep the rows taking it. Every discarded left row is already below all extensions of the new prefix, and every discarded right row is already above them. The appended value is fresh because it belongs to active injective rows.

If the active value sets have no common value, they form a strict numerical cut. Choose a separator outside the range of  $p$  using the preceding full-size-interval argument, append it, and stop. If an active side becomes empty, use the same argument for the resulting one-sided cut. At a limit stage take the union of the prefixes and retain the rows that extend it. The search cannot still have active rows on both sides at stage  $\beta$ : such a pair would agree at every coordinate below  $\beta$ , contradicting the definition of  $\beta$ . Therefore it stops with a prefix of length less than  $\mu$ .

That prefix has used fewer than  $\mu$  labels. Its remaining labels and remaining positions both have cardinality  $\mu$ , and an exhaustive completion has order type  $\mu$ . Every such completion lies strictly above  $A$  and below  $B$ , proving  $\eta_r$ .

Finally, [Note 30.7](#) gives a global cofinal subset of size  $r$ . Together with an empty right side it is a cut of total size below every cardinal  $\nu > r$ , but it has no upper interpolant. Hence no such  $\eta_\nu$  property holds.  $\square$

These results settle the point-character part, and the maximal uniform cut-size threshold, of the earlier minimal-slice question. They do not compute its full two-sided unrealized-cut spectrum. In particular, saturation below  $\text{cf } \kappa$  and failure above that threshold do not specify which individual larger pairs of regular cofinalities occur as gaps.

[merge] (3 October 2026.) The “full two-sided unrealized-cut spectrum” that source 15 leaves open is source 17’s [theorem 30.11](#) (and source 18’s [note 30.13](#)); its multiplicities are [theorem 30.16](#).



### 30.8 Source 18's route and extension: characters, interpolation and two regimes

[merge] The rest of source 18's Section 4, printed whole. Its point characters and interpolation theorem are source 17's [theorem 30.2](#) and [corollary 30.18](#) and source 15's [theorem 30.23](#), proved again with a different interpolation argument (bounded dropout coordinates); it extends both to the types  $\mu + n$  and to the finite-cofinite prefix orders  $Q_n$  ([lemma 27.7](#)), and it exhibits a bounded  $(r, r)$  obstruction explicitly. Its corollary on two regimes merges with source 17's examples [examples 30.20](#) and [30.21](#); its second part supplies the value  $\kappa^{<\kappa} = 2^\kappa$  used in [example 30.21](#).

#### 30.8.1 Exact point characters and global cofinalities

For a point  $e$  in a linear order, its left character means the cofinality of  $(-\infty, e)$ , and its right character means the coinitality of  $(e, \infty)$ . An immediate predecessor or successor therefore gives character 1 on the respective side.

**Theorem 30.24** (Complete local and global characters at the minimal type). *Every point of  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  has left and right character  $q = \text{cf}(\mu)$ , whereas*

$$\text{cf}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \text{ci}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = r.$$

*More generally,  $Q_n(\mathbf{No}_{<\kappa})$  has point characters  $(q, q)$  for every finite  $n$ . Accordingly, in  $\mathcal{W}_{\mu+n}(\mathbf{No}_{<\kappa})$ :*

- *if  $n = 0$  or 1, every point has characters  $(q, q)$ ;*
- *if  $n \geq 2$ , the first and last points of each finite  $n!$ -point fiber have characters  $(q, 1)$  and  $(1, q)$  respectively;*
- *all interior points of a finite fiber have characters  $(1, 1)$ .*

*Interior points exist exactly when  $n \geq 3$ . The global cofinality and coinitality remain  $r$  for every finite  $n$ .*

*Proof.* Fix  $p \in Q_n(\mathbf{No}_{<\kappa})$  and  $\delta < \mu$ . There are  $\mu$  labels numerically below  $p(\delta)$ , and equally many above it. The labels used before  $\delta$ , together with the finitely many labels omitted from  $p$ , exclude fewer than  $\mu$  labels. Choose a smaller available label which therefore occurs later in the  $\mu$ -prefix, and transpose its occurrence with position  $\delta$ . This gives  $p_\delta^- < p$  with first difference exactly  $\delta$ . An analogous transposition gives  $p_\delta^+ > p$ .

As  $\delta$  runs along an increasing cofinal sequence in  $\mu$ , the points  $p_\delta^-$  increase cofinally toward  $p$ , while  $p_\delta^+$  decrease coinitially toward it. This proves both upper bounds  $q$ . Conversely, fewer than  $q$  points below  $p$  have first-disagreement coordinates bounded below  $\mu$ . Changing  $p$  at a later coordinate gives a point above all of them and below  $p$ . They cannot be cofinal. The dual argument gives the right lower bound. This proves the character statement for  $Q_n$ .

The map  $e \mapsto e(0)$  is onto  $\mathbf{No}_{<\kappa}$  and respects comparisons whenever the first coordinates differ. A family cofinal in the enumeration order must therefore have first coordinates cofinal in  $\mathbf{No}_{<\kappa}$ ; conversely, choose one enumeration over every member of a numerical cofinal sequence. This proves the global cofinality formula, and the coinitality formula is dual. Finally,  $\mathcal{W}_{\mu+n}(\mathbf{No}_{<\kappa})$  is the lexicographic sum over  $Q_n$  of finite ordered fibers of size  $n!$ . Immediate neighbors inside a fiber give character 1, while approach to the outside of a fiber is controlled by the character  $q$  of the quotient point.  $\square$

### 30.8.2 Sharp interpolation at every infinite cardinal cutoff

We say that  $L$  is  $\eta_r$  if for all sets  $A, B \subseteq L$  with  $|A \cup B| < r$  and  $a < b$  for every  $a \in A, b \in B$ , there is  $z \in L$  satisfying  $A < z < B$ . Empty sides are allowed. The sharpness result below also uses a bounded cut with both sides nonempty, so it does not depend on the empty-side convention.

**Theorem 30.25** (Exact guaranteed interpolation). *For every infinite initial cardinal  $\kappa$ , with notation as above,*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \text{ is } \eta_r, \quad r = \text{cf}(\kappa).$$

*It has a bounded unfilled cut of character  $(r, r)$ , so it is not  $\eta_{r+}$ . The same interpolation and sharpness assertions hold for  $\mathcal{W}_{\mu+1}(\mathbf{No}_{<\kappa})$ . For  $n \geq 2$ , adjacent points in the terminal fibers already preclude  $\eta_\omega$ .*

*Proof of interpolation.* First  $\mathbf{No}_{<\kappa}$  is  $\eta_r$ . A family of fewer than  $r$  numerical options has birthdays bounded below  $\kappa$ , so the set-cut theorem supplies a separator with birthday less than  $\kappa$ . Applying it twice supplies a nonempty interval of possible separators, still between the prescribed options. By [Note 29.8](#), such an interval has  $\mu$  labels.

Let  $A < B$  be sets of input enumerations with  $|A \cup B| < r$ . We construct an injective common prefix  $p$  until a completion separating the inputs becomes possible. At a prefix of length  $\delta < \mu$ , the active inputs are

$$A_p = \{a \in A : a \restriction \delta = p\}, \quad B_p = \{b \in B : b \restriction \delta = p\}.$$

The invariant is that all discarded lower inputs lie below every completion of  $p$ , while all discarded upper inputs lie above every completion of  $p$ . Let  $L$  and  $U$  be the sets of next entries of the active lower and upper inputs. Then every  $l \in L$  is at most every  $u \in U$ .

If  $L \cap U = \emptyset$ , the two next-entry sets are strictly separated (with empty sides permitted). Choose an interval of numerical separators between them. It has  $\mu$  labels, whereas  $\text{ran } p$  has size less than  $\mu$ . Choose an unused separator  $z$  in it. Every completion of  $p \frown (z)$  lies strictly between all inputs, and an exhaustive completion of the required type exists by cardinality. The construction stops.

Otherwise,  $L \cap U$  consists of exactly one label  $z$ : two different common labels would contradict  $L \leq U$ . This label is unused, since an active input is injective. Append it to  $p$ . Inputs with a smaller lower entry or a larger upper entry are discarded in the correct direction, maintaining the invariant. At limits take the union of the prefixes.

Suppose the construction never stopped before  $\mu$ . The union is an injective word  $p$  of length  $\mu$ . Every input whose  $\mu$ -prefix differs from  $p$  drops out at its first disagreement. There are fewer than  $r \leq q = \text{cf}(\mu)$  inputs, so these dropout coordinates are bounded below  $\mu$ . At most one input can have  $\mu$ -prefix exactly  $p$ : for type  $\mu$  the whole enumeration is determined, and for type  $\mu + 1$  its single omitted label determines the last entry. Since  $A < B$ , that one input cannot lie on both sides. Beyond the bound on the dropout coordinates, at least one active side is therefore empty, forcing the stopping case. This contradiction proves that the construction stops at a coordinate below  $\mu$ .  $\square$

*Proof of the bounded obstruction.* Choose a sign word  $s$  of length exactly  $\kappa$  such that both its plus positions and minus positions are cofinal in  $\kappa$ , and whose first two signs are  $+, -$ . For example, use the parity of the finite tail of each ordinal coordinate. Then  $0 < s < 1$ , while  $s \notin \mathbf{No}_{<\kappa}$ . Prefixes  $s \restriction \gamma$  at cofinally many positions with  $s(\gamma) = +$  lie below  $s$  and are cofinal below its cut in  $\mathbf{No}_{<\kappa}$ ; prefixes at cofinally many minus positions are coinital above it. Choose such two cofinal collections indexed by  $r$ .

Every  $x \in \mathbf{No}_{<\kappa}$  has its comparison with  $s$  decided before  $\kappa$ . A sufficiently long chosen prefix therefore passes  $x$  on the appropriate side. No element of  $\mathbf{No}_{<\kappa}$  separates the two prefix families. Neither side can have a smaller cofinal family: fewer than  $r$  birthdays are bounded below  $\kappa$ , and a longer appropriate prefix passes every label in that family. Thus this is a bounded numerical gap of character  $(r, r)$ .

Pull the cut back under the first-coordinate map  $e \mapsto e(0)$ . Choose arbitrary completions for the two numerical cofinal families. A hypothetical separating enumeration would need its first entry to lie in the missing numerical cut, which is impossible. Conversely those completions are cofinal and cointial on their respective sides, and the first-coordinate projection proves the matching lower bounds. Enumerations starting with 0 and 1 bound the whole configuration. This supplies the claimed bounded  $(r, r)$  obstruction.  $\square$

### 30.8.3 Regular and singular consequences

**Corollary 30.26** (Two especially transparent regimes). *1. If  $\kappa$  is regular and  $2^{<\kappa} = \kappa$ , then*

$$|\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa, \quad \text{d} = \text{cell} = \kappa,$$

*and every local character and global cofinality or cointiality is  $\kappa$ . The order is  $\eta_\kappa$  and has a bounded unfilled  $(\kappa, \kappa)$ -cut.*

*2. If  $\kappa$  is a singular strong-limit cardinal, then*

$$|\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa, \quad \text{d} = \text{cell} = 2^\kappa,$$

*while every point character and both global cofinalities are  $\text{cf}(\kappa)$ . Its sharp interpolation threshold is  $\eta_{\text{cf}(\kappa)}$ .*

*Proof.* In the regular case,  $\kappa^{<\kappa} = \kappa$ . For a fixed  $\lambda < \kappa$ , every function  $\lambda \rightarrow \kappa$  has bounded range, and the possibilities for each range bound are at most  $2^{<\kappa} = \kappa$ ; summing over  $\kappa$  range bounds gives  $\kappa^\lambda = \kappa$  for nonzero  $\lambda < \kappa$ .

In the singular strong-limit case,  $2^{<\kappa} = \kappa$ . Let  $r = \text{cf}(\kappa)$  and take a cofinal sequence  $(\kappa_\xi)_{\xi < r}$  below  $\kappa$ . A subset of  $\kappa$  is determined by its intersections with the  $\kappa_\xi$ . Each intersection has fewer than  $\kappa$  possibilities by the strong-limit hypothesis, so

$$2^\kappa \leq \kappa^r \leq \kappa^{<\kappa} \leq \kappa^\kappa = 2^\kappa.$$

Thus  $\kappa^{<\kappa} = 2^\kappa$ . The other conclusions now follow from the preceding theorems and the baseline cardinality theorem.  $\square$

**What these results settle, and what they do not.** The baseline explicitly left density, cellularity and much of the singular-cutoff interpolation analysis as continuation questions. The preceding theorems determine density and cellularity for every fixed enumeration type and every infinite alphabet, and give exact local/global characters and sharp guaranteed interpolation at every infinite initial birthday cutoff for the minimal slice. They do not identify the order up to isomorphism. In particular, equality of these invariants is not an isomorphism theorem, and the old exact ordinal spectrum remains logically separate from the new topological information. The full spectrum of unfilled cut characters requires an additional analysis of boundaries of transfinite prefix cylinders.

### 31 Longer strata: every cardinal-block type (source 15)

[merge] Source 15's Section 8, printed whole: the only batch-81 treatment of strata longer than the minimal one. Its  $r_\kappa$  is printed  $\rho_\kappa$  (by [theorem 8.3](#),  $\rho_\kappa = \ell_2(\mathcal{W}_\mu(\mathbf{No}_{<\kappa}))$ ), so that  $r$  keeps the meaning of  $\kappa$ ; its  $D_{\sigma,m}$  is printed  $\mathbb{B}_\sigma \times_{\text{lex}} m$ , so that no letter clashes with its core  $\mathcal{D}$  ([section 1.3.1](#)). The section extends source 12's [theorem 8.3](#) ( $\gamma = 1$ ,  $n = 0$ ) along the targets of [research question 113.11](#). Its entry code ([lemma 31.1](#)) is source 12's [lemma 19.33](#) and source 16's mechanism in (S16.2); its cube clause is [lemma 6.4](#) (reals report `two:thm:cube-strict`); its row scheduler ([lemma 31.3](#)) is source 12's [lemma 8.1](#) run blockwise along  $\gamma$ ; its finite-deletion lemma ([lemma 31.4](#)) is [lemma 8.2](#) with a finite set removed; its finite-capacity lemma is, by its own account, the finite-fiber fusion argument of the reals report. Source 15 states its finite-deletion lemma for singular strong-limit  $\kappa$ ; the hypothesis is not used in its proof.

The earlier minimal-slice theorem ([section 8](#)) extends to all enumeration types whose remainder on division by the alphabet cardinal is finite. This includes the explicitly proposed next cases  $\mu \cdot 2$  and  $\mu \cdot \gamma + n$ . It does not classify an additional infinite remainder smaller than  $\mu$ . All products and sums of ordinals in this section are ordinal operations.

Let  $\kappa$  be an infinite initial cardinal, and put

$$X = \mathbf{No}_{<\kappa}, \quad \mu = |X| = 2^{<\kappa}, \quad \rho_\kappa = \begin{cases} \kappa \cdot \kappa, & \kappa \text{ is a singular strong-limit cardinal,} \\ \mu, & \text{otherwise.} \end{cases}$$

For an ordinal  $\alpha$  with  $|\alpha| = \mu$ , write  $\mathcal{W}_\alpha(X)$  for the bijections  $\alpha \rightarrow X$ , compared by their first unequal entry in the numerical order of  $X$ . For a positive integer  $m$ , write

$$\mathbb{B}_\sigma \times_{\text{lex}} m = \mathbb{B}_\sigma \times_{\text{lex}} \{0, \dots, m-1\}, \quad \mathbb{B}_\sigma = (2^\sigma, <_{\text{lex}}),$$

the finite chain  $m = \{0, \dots, m-1\}$  being the last coordinate.

**Lemma 31.1** (An explicit entry code and the strict cube hierarchy). *Replace each minus sign by 00, each plus by 11, and terminate with 01. The resulting code  $c(s)$  is prefix-free and preserves numerical surreal order. Its length is  $2 \cdot \text{bd}(s) + 2$ , less than  $\kappa$  for  $s \in \mathbf{No}_{<\kappa}$ . Furthermore,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

*Proof.* The three blocks  $00 < 01 < 11$  encode precisely minus, termination, and plus. Distinct completed codes cannot be proper prefixes: at the first termination of one of them the other either terminates too or has a different block. Padding a completed entry therefore does not alter its comparison with another entry.

For the cube assertion, extension by zeros gives the embedding when  $\alpha \leq \beta$ . Suppose  $\alpha > \beta$  and an embedding in the other direction exists. Successively read target bits, retaining at each step a full source child cylinder whose image lies on one side of the next target-bit split. Such a child exists: if the lower child does not lie wholly below the split, every point of the upper child lies above it. At limits take unions of the chosen prefixes. After  $\beta$  steps, all points extending one source prefix of length  $\beta$  have the same whole target word. There are at least two source extensions since  $\beta < \alpha$ , contradicting injectivity. This is the same cylinder fusion mechanism used in the prior binary-length calculations.  $\square$

When  $\mu = \kappa$  is singular,  $\kappa$  is a strong limit. Otherwise  $2^\rho = \kappa$  for some  $\rho < \kappa$  would also give  $2^\lambda = \kappa$  at  $\lambda = \max(\rho, \text{cf } \kappa) < \kappa$ , contradicting König's inequality  $\text{cf}(2^\lambda) > \lambda$ . Thus the three cases used below are exhaustive.

**Theorem 31.2** (Cardinal blocks and a finite terminal permutation). *For every  $0 < \gamma < \mu^+$  and every  $n < \omega$ ,*

$$\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \simeq_e (\mathbb{B}_{\rho_\kappa \cdot \gamma} \times_{\text{lex}} n!).$$

*In particular, if  $\kappa$  is a singular strong limit, then*

$$\mathcal{W}_{\kappa \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\kappa \cdot \kappa \cdot \gamma} \times_{\text{lex}} \{0, \dots, n! - 1\}.$$

*The statement concerns equimorphism; it does not identify these permutation orders with the generally nondense binary products.*

We give the scheduler detail before proving the theorem, since a cardinality-preserving reindexing alone would not preserve lexicographic order.

**Lemma 31.3** (Blockwise row scheduling). *Let  $Y$  be a linearly ordered set of infinite cardinality  $\kappa$ , let  $0 < \gamma < \kappa^+$ , and put  $\lambda = \kappa \cdot \gamma$ . Suppose that  $Y$  contains pairwise disjoint subsets  $(Y_i)_{i < \lambda}$ , each of cardinality  $\kappa$ , with an order embedding  $\phi_i : X \rightarrow Y_i$ , and that  $Z = Y \setminus \bigcup_{i < \lambda} Y_i$  is either empty or has cardinality  $\kappa$ . Then*

$$X_{\text{lex}}^\lambda \hookrightarrow \mathcal{W}_\lambda(Y).$$

*Proof.* For each  $i < \lambda$ , fix a bijection  $b_i : \kappa \rightarrow Y_i$ . For  $x \in X$ , transpose  $\phi_i(x)$  with  $b_i(0)$  in that enumeration; denote the result by  $b_{i,x}$ . Its first value is  $\phi_i(x)$ , and for equal  $x$  the whole row is the same.

Split the variable rows into the consecutive blocks  $\{\kappa \cdot \beta + \xi : \xi < \kappa\}$ ,  $\beta < \gamma$ . Within one block schedule row–column pairs  $(\xi, j) \in \kappa^2$  first by  $\max(\xi, j)$  and then lexicographically inside each level. Every pair has fewer than  $\kappa$  predecessors: all earlier pairs lie in a square bounded below  $\kappa$ , and such a square has cardinality less than  $\kappa$ , even when  $\kappa$  is singular. The schedule therefore has order type  $\kappa$ . The first occurrence of row  $\xi$  is  $(\xi, 0)$ , and these first occurrences have increasing row indices.

If  $Z \neq \emptyset$ , include its fixed  $\kappa$ -long enumeration as a filler row before the variable rows in the first block. The row-index order is  $1 + \kappa = \kappa$ , so the same construction applies. The filler row has no input parameter. Every later block uses only its  $\kappa$  variable rows.

For  $x = (x_i)_{i < \lambda}$ , output the entries of  $b_{i,x_i}$  by these successive block schedules. Each block has length  $\kappa$  and there are  $\gamma$  blocks, so the output has length exactly  $\lambda$ . The disjoint rows and the filler exhaust  $Y$  exactly once.

If inputs  $x <_{\text{lex}} x'$  first differ at row  $i = \kappa \cdot \beta + \xi$ , earlier blocks agree completely. In block  $\beta$ , every output before the first entry of row  $i$  comes from a lower row or from the fixed filler. Those outputs agree. The first unequal output is therefore  $\phi_i(x_i) < \phi_i(x'_i)$  at the first entry of row  $i$ . This proves both injectivity and order preservation. No regularity of  $\gamma$  or  $\kappa$  was used.  $\square$

**Lemma 31.4** (Finite deletion does not obstruct the singular scheduler). *If  $\kappa$  is a singular strong limit,  $X = \mathbf{No}_{<\kappa}$ ,  $F \subseteq X$  is finite, and  $\lambda = \kappa \cdot \gamma$  for  $0 < \gamma < \kappa^+$ , then*

$$X_{\text{lex}}^\lambda \hookrightarrow \mathcal{W}_\lambda(X \setminus F).$$

*Proof.* For  $\xi < \kappa$ , let  $C_\xi$  be the sign-sequence cone with prefix  $+\xi \frown -$ . The cones are disjoint, and prefixing gives an order isomorphism  $X \rightarrow C_\xi$ . Indeed,  $\xi + 1 + \eta < \kappa$  for  $\xi, \eta < \kappa$ , because  $\kappa$  is an initial cardinal; deletion of the prefix gives the inverse. Each cone has cardinality  $\kappa$ . At most finitely many cones meet  $F$ . Select  $\kappa$  unaffected cones and index them by  $\lambda$ , which has cardinality  $\kappa$ .

The complement of those selected cones in  $X \setminus F$  has cardinality  $\kappa$ : it contains all but finitely many of the all-plus sequences, one for each length below  $\kappa$ , and is a subset of  $X$ . The preceding lemma now applies, with that complement as the filler. In particular, no unexplained bijection of a grid with  $\lambda$  is being used in place of a lexicographic scheduler.  $\square$

*Proof of Theorem 31.2.* Write  $\lambda = \mu \cdot \gamma$  and  $\sigma = \rho_\kappa \cdot \gamma$ . We first construct the upper embedding. Use the repository's prefix-free, numerical-order-preserving binary code  $c(s)$  for each  $s \in X$ , with positive length less than  $\kappa$ .

If  $\mu > \kappa$ , pad each individual entry code to length  $\kappa$ . Prefix-freeness ensures that this preserves all entry comparisons. A block of  $\mu$  entries then consumes ordinal length  $\kappa \cdot \mu = \mu$ . Applying this separately to the  $\gamma$  successive blocks gives a prefix code of length  $\mu \cdot \gamma = \sigma$ .

If  $\mu = \kappa$  and  $\kappa$  is regular, concatenate the unpadded codes in each  $\kappa$ -entry block. Each proper partial sum of their lengths is below  $\kappa$  by regularity, while the full sum is  $\kappa$  because all code lengths are positive. Each block therefore consumes exactly  $\kappa$ , again giving length  $\sigma$ .

In the remaining case  $\mu = \kappa$  is singular, it is a strong limit. Pad each entry code to length  $\kappa$ . Each block now consumes  $\kappa \cdot \kappa$ , and the complete  $\lambda$ -entry prefix has code length  $\kappa \cdot \kappa \cdot \gamma = \sigma$ .

In all cases, a prefix of length  $\lambda$  from a bijection  $\lambda + n \rightarrow X$  has a binary code of length exactly  $\sigma$ . The remaining  $n$  labels form a finite set determined by that prefix. Append the rank of their terminal permutation in its increasing lexicographic list of  $n!$  permutations. Different prefixes compare inside the binary code; equal prefixes have the same remaining labels and compare by the finite rank. This gives  $\mathcal{W}_{\lambda+n}(X) \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} n!)$ .

For the reverse embedding in either nonexceptional case, reserve a fixed  $n$ -element set  $F \subseteq X$  and partition  $X \setminus F$  into pairs indexed by  $\lambda$ . Such a partition exists because  $|\lambda| = \mu$ . List each pair in its increasing or decreasing orientation according to the corresponding bit of a member of  $\mathbb{B}_\lambda$ . Since  $\lambda$  is a nonzero limit ordinal,  $2 \cdot \lambda = \lambda$ . Append the requested one of the  $n!$  permutations of  $F$ . Here  $\sigma = \lambda$ , giving the required embedding.

It remains to obtain the longer cube in the singular strong-limit case. Let  $\delta = \text{cf } \kappa < \kappa$ , and choose nonzero ordinals  $(\tau_j)_{j < \delta}$  cofinal in  $\kappa$ , with  $\sum_{j < \delta} \tau_j = \kappa$ . The identities

$$\delta \cdot \kappa = \kappa, \quad \delta \cdot \lambda = \delta \cdot (\kappa \cdot \gamma) = \lambda$$

allow the  $\lambda$  input coordinates of  $X^\lambda$  to be grouped into  $\lambda$  consecutive blocks of  $\delta$  coordinates. In each block, restrict its  $j$ th coordinate to surreals whose sign sequences have length exactly  $\tau_j$ . That coordinate is a copy of  $\mathbb{B}_{\tau_j}$ , so the entire restricted product is

$$\mathbb{B}_{\sum_{i < \lambda} \sum_{j < \delta} \tau_j} = \mathbb{B}_{\kappa \cdot \lambda} = \mathbb{B}_{\kappa \cdot \kappa \cdot \gamma} = \mathbb{B}_\sigma.$$

By Lemma 31.4, this product embeds into  $\mathcal{W}_\lambda(X \setminus F)$ . Append any of the  $n!$  permutations of  $F$ , obtaining  $(\mathbb{B}_\sigma \times_{\text{lex}} n!) \hookrightarrow \mathcal{W}_{\lambda+n}(X)$ .  $\square$

### 31.1 Exact coding lengths, embedded cubes, and comparisons

The next capacity argument is the finite-fiber fusion argument already used in the real-alphabet report. It is included to make the numerical consequences of the extension explicit.

**Lemma 31.5** (Finite fibers over a binary cube). *Let  $\sigma, \tau$  be nonzero limit ordinals and let  $m, n$  be positive integers. Then*

$$(\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow (\mathbb{B}_\tau \times_{\text{lex}} n) \iff \sigma < \tau \text{ or } (\sigma = \tau \text{ and } m \leq n).$$

Moreover,

$$\ell_2(\mathbb{B}_\sigma \times_{\text{lex}} m) = \sigma + \lceil \log_2 m \rceil, \quad \mathbb{B}_\theta \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} m) \iff \theta \leq \sigma + \lceil \log_2 m \rceil.$$



*Proof.* For equal binary lengths, suppose  $f : (\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} n)$ . Recursively choose a source binary prefix  $s_\xi$  and target binary prefix  $t_\xi$ , both of length  $\xi$ , such that

$$f(C_{s_\xi} \times \{0, \dots, m-1\}) \subseteq C_{t_\xi} \times \{0, \dots, n-1\}.$$

At a successor stage the target's next bit cuts the image into two ordered parts. Of the source cylinder's two full children, at least one lies wholly on one side of that cut: if its lower child is not wholly on the lower side, its upper child is wholly on the upper side. Keep that child and the corresponding target bit. At a limit stage take unions of prefixes. The invariant persists, and the source cylinder remains nonempty. At stage  $\sigma$ , all  $m$  points over one source binary sequence lie in one target fiber of size  $n$ . Thus  $m \leq n$ . The converse is immediate.

Coding the finite chain of size  $m$  by  $\lceil \log_2 m \rceil$  bits gives  $(\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow \mathbb{B}_{\sigma + \lceil \log_2 m \rceil}$ . No cube shorter than  $\sigma$  can contain  $(\mathbb{B}_\sigma \times_{\text{lex}} m)$ , since it already contains  $\mathbb{B}_\sigma$  and the binary cube hierarchy is strict. A target  $\mathbb{B}_{\sigma+j} = (\mathbb{B}_\sigma \times_{\text{lex}} 2^j)$  with  $j < \omega$  can contain it only if  $m \leq 2^j$ , by the same fusion argument. This proves the coding-length formula.

Likewise,  $\mathbb{B}_{\sigma+j}$  embeds into  $(\mathbb{B}_\sigma \times_{\text{lex}} m)$  exactly when  $2^j \leq m$ . Once the first forbidden finite extension has been excluded, every larger cube is excluded because it contains that extension. This proves the complete cube spectrum.

Finally, if  $\sigma < \tau$ , then  $\sigma + \lceil \log_2 m \rceil < \tau$  because  $\tau$  is a limit. Code into that shorter cube and then into the target. If  $\tau < \sigma$ , the target codes into a cube of length  $\tau + \lceil \log_2 n \rceil < \sigma$ , whereas the source contains  $\mathbb{B}_\sigma$ ; strictness excludes an embedding. This proves the comparison criterion.  $\square$

**Corollary 31.6** (The complete invariant on these block types). *Under the hypotheses of Theorem 31.2, put  $\sigma = \rho_\kappa \cdot \gamma$ . Then*

$$\ell_2(\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})) = \sigma + \lceil \log_2(n!) \rceil,$$

and

$$\mathbb{B}_\theta \hookrightarrow \mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \iff \theta \leq \sigma + \lceil \log_2(n!) \rceil.$$

For  $0 < \gamma, \eta < \mu^+$  and  $m, n < \omega$ ,

$$\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \hookrightarrow \mathcal{W}_{\mu \cdot \eta + m}(\mathbf{No}_{<\kappa}) \iff \gamma < \eta \text{ or } (\gamma = \eta \text{ and } n! \leq m!).$$

*Proof.* Apply the finite-capacity lemma and use strict monotonicity in the right argument of  $\gamma \mapsto \rho_\kappa \cdot \gamma$  for a fixed nonzero left factor. Here each  $\rho_\kappa \cdot \gamma$  is a nonzero limit ordinal.  $\square$

For an unconditional singular example, take  $\kappa = \beth_\omega$ . The types  $\kappa \cdot 2$ ,  $\kappa \cdot 2 + 3$ , and  $\kappa^2$  have binary coding lengths respectively

$$\kappa^2 \cdot 2, \quad \kappa^2 \cdot 2 + 3, \quad \kappa^3.$$

The type  $\kappa \cdot 2 + 3$  contains the cube of length  $\kappa^2 \cdot 2 + 2$  but not the cube of its own coding length. These distinctions are invisible to cardinality and to the ordinal embedding spectrum, which remain  $2^\kappa$  and  $\{\beta : \beta < \kappa^+\}$  throughout.

### 31.2 Density and cellularity, including the singular minimal slice

Write  $d(L)$  for the least cardinality of a dense subset in the order topology, and  $\text{cell}(L)$  for the supremum of cardinalities of families of pairwise disjoint nonempty open sets. An additional consequence of the coding theorem is an exact answer for these two invariants on the same block types. In particular, the singular minimal slice has the largest possible cellularity, a fact not visible from its unchanged ordinal embedding spectrum.

**Corollary 31.7** (Density and cellularity of the block types). *For  $L = \mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})$  as above,*

$$d(L) = \text{cell}(L) = \begin{cases} 2^{<\mu}, & \rho_\kappa = \mu, \gamma = 1, n \leq 2, \\ 2^\mu, & \text{otherwise.} \end{cases}$$

*Consequently*

$$d(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = c(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \begin{cases} 2^\kappa, & \kappa \text{ is a singular strong limit,} \\ 2^{<\mu}, & \text{otherwise.} \end{cases}$$

*Proof.* First consider  $\rho_\kappa = \mu$ ,  $\gamma = 1$ , and  $n \leq 2$ . The finite factor  $n!$  has size at most two. There is an embedding  $f : L \rightarrow (\mathbb{B}_\mu \times_{\text{lex}} n!)$ . No point of  $L$  is isolated. Indeed,  $L$  has no endpoints because the native surreal alphabet has neither a least nor a greatest element; swapping the first entry with a smaller or larger label moves either way without changing the enumeration type. The adjacency criterion for exhaustive permutations applies inside a fixed type: if the unused remainder at a first difference is infinite, an ascent in the lower tail or a descent in the upper tail gives an intermediate permutation of the same type. Adjacency is therefore confined to the finite terminal permutation component. For  $n \leq 2$  each such component has at most two points, so no point has both an immediate predecessor and an immediate successor. Thus no point is isolated, and every nonempty open set contains at least five points.

For every binary prefix  $s$  of length less than  $\mu$  whose cylinder meets the first-coordinate projection of  $f[L]$ , choose one element  $x_s \in L$  whose image has that prefix. There are at most  $\sum_{\xi < \mu} 2^{|\xi|} = 2^{<\mu}$  such choices. These choices are dense. To see this, take five increasingly ordered points  $x_0 < \dots < x_4$  in any given nonempty open interval. Each binary fiber has at most two points, so the binary first coordinate of  $f(x_2)$  is strictly between those of  $f(x_0)$  and  $f(x_4)$ . Choose a prefix long enough to distinguish it from both outer coordinates. Its whole binary cylinder lies strictly between them, and the chosen  $x_s$  lies in the original open interval. Hence  $d(L) \leq 2^{<\mu}$ .

Conversely  $L$  contains  $\mathbb{B}_\mu$ . For each  $\xi < \mu$ , its  $2^{|\xi|}$  cylinders of length  $\xi$  each contain three ordered points. The ambient open intervals in  $L$  between the images of the outer two points are nonempty, since they contain the image of the middle point, and they are pairwise disjoint. Therefore  $\text{cell}(L) \geq 2^{<\mu}$ , proving equality in the first case since always  $\text{cell}(L) \leq d(L)$ .

In every remaining case, put  $\sigma = \rho_\kappa \cdot \gamma$ . If  $\sigma > \mu$ , the embedded cube  $\mathbb{B}_\sigma$  has  $2^\mu$  distinct cylinders of prefix length  $\mu$ , each with at least three points because  $\sigma$  is a limit ordinal greater than  $\mu$ . The preceding interval construction gives  $\text{cell}(L) \geq 2^\mu$ . If  $\sigma = \mu$  but  $n \geq 3$ , the embedded order  $(\mathbb{B}_\mu \times_{\text{lex}} n!)$  has a fiber with at least three points over each of its  $2^\mu$  binary sequences. The intervals between the images of its first and last fiber points are again pairwise disjoint and nonempty. Thus  $\text{cell}(L) \geq 2^\mu$  here as well. Finally  $d(L) \leq |L| = 2^\mu$ , giving the claimed equalities.  $\square$

### 31.3 The boundary of the result

Write a general admissible type as  $\alpha = \mu \cdot \gamma + \zeta$ ,  $\zeta < \mu$ . The theorem settles the case that  $\zeta$  is finite. When  $\zeta$  is infinite, the preceding long prefix leaves a residual numerical suborder of cardinality  $|\zeta|$ ; that residual order varies with the prefix. Neither the size nor the coding behavior of this smaller alphabet can simply be replaced by those of  $\mathbf{No}_{<\kappa}$ . An exact treatment must control the possible residual suborders and the code lengths available uniformly for their permutations. Thus the result gives a precise affirmative continuation of the proposed cardinal-block program while leaving the smaller infinite-tail classification explicitly open.

*Remark 31.8* ([merge]; the reals report's display at  $X = \mathbf{No}_{<\kappa}$ ). The reals report asks under which hypotheses on  $X$  the display  $\mathcal{W}_{\lambda+n}(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$  still holds (its question ‘‘Other ground

orders”, `two:sec:research`); its batch-80 note records that [theorem 8.3](#) gives the case  $n = 0$ ,  $\lambda = \mu$  for  $X = \mathbf{No}_{<\kappa}$ , except at a singular strong limit. For  $\lambda = \mu \cdot \gamma$ ,  $0 < \gamma < \mu^+$ , and every  $n < \omega$ , [theorem 31.2](#) gives  $\mathcal{W}_{\lambda+n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\rho_\kappa \cdot \gamma} \times_{\text{lex}} n!$ , and by [lemma 31.5](#) two such products with nonzero limit lengths are equimorphic only when the lengths are equal. So the display holds whenever  $\kappa$  is not a singular strong limit ( $\rho_\kappa = \mu$ ). At a singular strong limit ( $\mu = \kappa$ ,  $\rho_\kappa = \kappa \cdot \kappa$ ) it holds exactly when  $\kappa \cdot \kappa \cdot \gamma = \kappa \cdot \gamma$ , that is, when  $\gamma = \kappa^\omega \cdot \delta$  for some ordinal  $\delta$  (then  $\kappa \cdot \gamma = \gamma$ , by Cantor normal form to base  $\kappa$ ); it fails for  $\gamma = 1, 2, \kappa, \dots$ . The display is therefore not simply false at singular strong limits. Strata  $\lambda$  that are not multiples of  $\mu$  (the first is  $\mu + \omega$ ), cutoffs that are not cardinals, and other alphabets remain open, as source 15 says in [section 31.3](#).

## 32 Set-sized baselines and warnings in sources 13–18

[merge] Every batch-81 source restates the set-sized baseline of the reals report and of Part I ([section 6](#) and [theorem 6.1](#)): cardinality  $2^\mu$ , universality for orders of size at most  $\mu$ , and the exact ordinal spectrum  $\beta < \mu^+$ . They are recorded here once, with the sources’ own provenance statements, followed by the set-level warnings of sources 13, 14 and 17. None of these statements is new.

*Note 32.1* (Source 16, Theorem 4.1, “Basic set-alphabet results”). Source 16’s Section 4 fixes “an infinite linearly ordered set  $X$  of cardinality  $\mu$ ” and records that exhaustive enumerations are prefix-incomparable and that an injective prefix of length less than  $\mu$  extends to a bijection of every ordinal type  $\alpha$  with  $|\alpha| = \mu$ . *The orders  $\mathcal{W}(X)$  and  $\mathcal{W}_\mu(X)$  have cardinality  $2^\mu$ . Every linear order of cardinality at most  $\mu$  embeds into  $\mathcal{W}_\mu(X)$ . For every ordinal  $\beta$ ,  $\beta \hookrightarrow \mathcal{W}(X)$  if and only if  $\beta < \mu^+$ , and the same statement holds with  $\mathcal{W}_\mu(X)$  in place of  $\mathcal{W}(X)$ , and for the reverse of  $\beta$ .* Its proof is the pair-orientation, initial-segment and coordinate-stabilization proof of [section 6](#). Source 16’s status line: “Theorem 4.1 is background from the ProveIt reals and surreal reports; it is reproved here to make the later arguments self-contained. In particular, ‘all ordinals embed’ is false at every fixed set stage.”

*Birthday cutoffs and a singular distinction* (source 16’s Subsection 4.1). For  $\mathbf{No}_{<\kappa}$ ,  $\kappa$  an infinite cardinal, substitute  $\mu = 2^{<\kappa}$  for the alphabet cardinal in [Note 32.1](#). The ordinal embedding cutoff is  $\mu^+$ , not automatically  $\kappa^+$ . The earlier report also proves a sharper binary-coding distinction:

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} 2_{\text{lex}}^{\kappa \cdot \kappa}, & \kappa \text{ is a singular strong-limit cardinal,} \\ 2_{\text{lex}}^\mu, & \text{otherwise.} \end{cases} \quad (\text{S16.2})$$

Here  $\kappa \cdot \kappa$  is ordinal multiplication. This statement is equimorphism, meaning embeddings in both directions, not isomorphism.

The upper coding mechanism replaces a negative sign by 00, a positive sign by 11, and termination by 01. These codes are prefix-free and preserve numerical comparison. At a regular stage, a sum of fewer than  $\kappa$  code lengths remains below  $\kappa$ . At a singular strong-limit stage it can already reach  $\kappa$  before all  $\kappa$  coordinates have been processed; padding every entry to length  $\kappa$  gives total length  $\kappa \cdot \kappa$ . The matching lower bound in the report uses a cofinal sequence of sign lengths and an order-preserving row scheduler ([theorem 8.3](#)). We do not reclassify the binary coding ranks of all longer strata here. The topological classification below, by contrast, covers every admissible enumeration length and every infinite alphabet.

*Note 32.2* (Source 18, Theorem 3.1, “Set-sized universality and the exact ordinal spectrum”). Source 18’s Section 3 defines  $\mathcal{W}(X)$  and  $\mathcal{W}_\alpha(X)$  as here, notes that two exhaustive enumerations are never proper prefixes of one another, so that no termination convention is needed, and that “labels are

essential: two two-point well-orders can be isomorphic as abstract orders and still have different first numerical labels”. *Let  $X$  be an infinite linear order of cardinality  $\mu$ . Then  $|\mathcal{W}(X)| = |\mathcal{W}_\mu(X)| = 2^\mu$ . Every linear order of cardinality at most  $\mu$  embeds into  $\mathcal{W}_\mu(X)$ . An ordinal  $\beta$  embeds into  $\mathcal{W}(X)$  if and only if  $\beta < \mu^+$ . The same exact spectrum holds for reverse ordinals and for every nonempty stratum  $\mathcal{W}_\alpha(X)$ .* The proof is that of [section 6](#); source 18 adds that the stabilization argument “uses the regularity of the successor cardinal  $\mu^+$ , not regularity of  $\mu$ ”, and that cardinality  $2^\mu$  is a poor guide, since such an order contains no increasing sequence of type  $\mu^+$ . Its subsection “What the existing binary-rank result adds” restates [theorem 8.3](#) (binary coding length  $\kappa \cdot \kappa$  at a singular strong-limit cutoff,  $\mu$  otherwise), calls it “background rather than a dependency of our new cut-spectrum proof”, and adds: “Replacing these ordinal sums by cardinalities would erase a genuine distinction. Our results below add topological information not determined by this coding length.”

*Note 32.3* (Source 15, Lemma 3.2 and Theorem 3.3, “Set-sized baseline, from the prior reports”). Source 15’s Lemma 3.2 (“Exhaustiveness removes prefix ambiguity”): *distinct exhaustive bijective enumerations of the same set have a first unequal coordinate in both domains; neither is a proper prefix of the other.* Its Theorem 3.3 is the baseline of [note 32.1](#) for every nonempty fixed-type slice, proved as in [section 6](#), with the fixed type  $\alpha = \lambda + n$  handled by reserving  $n$  final labels and orienting pairs indexed by  $\lambda$ . Source 15 says: “This proof uses no special property of real or surreal arithmetic. It is reproduced to fix the dependence of the later theorems; its content is already in the audited reports. If  $X$  is finite of size  $n$ , the whole order is the finite chain of its  $n!$  permutations. The empty carrier has one empty well-order.” Its subsections on birthday cutoffs and on set-length words, which close its Section 3, follow. Its Lemma 3.1 (fresh separators) belongs to the class layer and is printed with the other batch-81 restatements of [lemma 5.2](#) ([lemma 51.2](#)).

### 32.1 Source 15’s birthday cutoffs and set-length words

[merge] The last two subsections of source 15’s Section 3; its  $X_\kappa$  is printed  $\mathbf{No}_{<\kappa}$ , and its forward reference to its Section 8 points to [section 31](#).

#### 32.1.1 Birthday cutoffs

For an infinite initial cardinal  $\kappa$ , let

$$X_\kappa = \mathbf{No}_{<\kappa}, \quad \mu = |X_\kappa| = 2^{<\kappa}.$$

Counting sign levels proves the cardinality formula: the supremum of their cardinalities is at least  $\kappa$ , and the union of  $\kappa$  levels does not exceed it. The preceding theorem gives

$$|\mathcal{W}(X_\kappa)| = 2^{2^{<\kappa}}, \quad \beta \hookrightarrow \mathcal{W}(X_\kappa) \iff \beta < (2^{<\kappa})^+.$$

At regular  $\kappa$ , a cut using fewer than  $\kappa$  labels has birthdays bounded below  $\kappa$ , and the usual cut construction stays in the cutoff. Singular  $\kappa$  need not have this property. The fixed-type calculation in [Section 31](#) makes the distinction visible even when the cardinalities and ordinal spectra coincide.

#### 32.1.2 Why all set-length words are a different order

If  $X$  varies, the class of all pairs  $(X, R)$  with  $X$  a set of surreals and  $R$  a well-order is a legitimate class of sets. Its ordinal enumerations are injective set-length words. Proper prefixes now occur. For example  $(0)$  is a prefix of  $(0, 1)$ . One may put shorter words first, shorter words last, or use a

termination symbol placed inside the alphabet. These comparisons agree on exhaustive enumerations of a fixed carrier but need not agree on variable carriers.

The prior report gives an explicit self-delimiting sign code embedding all set-length surreal words, with shorter-first order, into **No**. Its code also exhibits why this order need not be isomorphic to **No**: a word of nonzero limit length is the supremum of its shorter prefixes, whereas a surreal point has no set-sized cofinal family below it. Restricting to successor-length words removes that particular obstruction and yields a class order isomorphic to **No** in GBC (this report’s [part II](#), source 11’s Sections 9–11 there). An embedding theorem therefore must not be strengthened to an isomorphism theorem merely because both carriers are proper classes.

### 32.2 Source 13’s fixed-set section

[merge] Source 13’s Section 4, “The fixed-set theory and birthday fragments”, printed whole; its theorem, a restatement credited by source 13 itself, is printed as a note, and its  $X_\kappa$  is printed **No**<sub><κ</sub>.

The set-sized layer is important both in its own right and as a safe formalization starting point. We record its already-established results with their provenance rather than repeating the earlier report in full.

*Note 32.4* (Source 13, Theorem 4.1, “Fixed-alphabet results from the repository”). Let  $X$  be an infinite linearly ordered set,  $|X| = \mu$ , and compare exhaustive enumerations using its label order. Then

$$|\mathcal{W}(X)| = 2^\mu, \quad \beta \hookrightarrow \mathcal{W}(X) \iff \beta < \mu^+,$$

and the same spectrum holds for reversed ordinals. Every linear order of cardinality at most  $\mu$  embeds in  $\mathcal{W}(X)$ .

*Provenance and proof mechanisms* (source 13). These are the general-alphabet results of the repository’s real well-ordering report, carried into Section 6 of the surreal report ([1]; [section 6](#)). The cardinal lower bound codes independent binary choices by pair reversals in an enumeration; the upper bound counts well-order relations on  $X$ . Universality uses binary codes for initial segments of a small order. The ordinal upper bound is a prefix-stabilization obstruction, stronger than a cardinality bound. We invoke that established written theorem here; this paragraph is not a replacement proof of its stabilization argument.

For an infinite cardinal  $\kappa$ , let

$$\mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}.$$

Counting signs gives  $|\mathbf{No}_{<\kappa}| = \mu$ : there are  $2^{|\alpha|}$  sign sequences of length  $\alpha$ , and the union over  $\alpha < \kappa$  has that supremum cardinality. Thus the ordinal spectrum of  $\mathcal{W}(\mathbf{No}_{<\kappa})$  is exactly the ordinals below  $(2^{<\kappa})^+$ . This does not say that only ordinals below the birthday cutoff embed.

#### 32.2.1 A warning about the minimal-length layer

Write  $\mathbb{B}_\theta = (\{0, 1\}^\theta, <_{\text{lex}})$ , and write  $L \simeq_e M$  for mutual order embeddability. The merged surreal report proves the following sharper distinction ([theorem 8.3](#), Theorem 8.3 of the report at its pin):

$$\mathcal{W}_\mu(X_\kappa) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \kappa \text{ singular strong limit,} \\ \mathbb{B}_\mu, & \text{otherwise,} \end{cases} \quad \mu = 2^{<\kappa}.$$

The multiplication in  $\kappa \cdot \kappa$  is *ordinal* multiplication. At a singular strong limit,  $\mu = \kappa$ , but the lengths  $\kappa$  and  $\kappa \cdot \kappa$  need not have the same lexicographic effect. This is an existing result, not an advance of the present manuscript.

There is no automatic passage from equimorphism to isomorphism, from a minimal-length stratum to all strata, or from independently chosen cutoff embeddings to a coherent global map. The last passage requires agreement on overlaps. These distinctions become especially visible in the product obstruction below.

### 32.2.2 Set-sized approximations to the new cut theorem

For an arbitrary set alphabet, prefix cylinders can terminate by exhaustion and can have extrema. The crossing arguments of source 13’s Section 8 (its reconstruction of cuts from prefix cylinders) still provide a useful tree description, but its clean all-ordinal formulation depends on two genuinely global facts: every set prefix has a completion, and after deleting its range there are fresh labels on both sides of any remaining set cut. A bounded alphabet need not have either property at its boundary. Therefore the new global theorem is not asserted by simply replacing **No** by every birthday fragment in its statement.

*Note 32.5* (Source 14, Theorem 2.1, “Set-carrier baseline, inherited”). *In ZFC, if  $|X| = \mu$  is infinite, then  $|\mathcal{W}(X)| = 2^\mu$  and  $\beta \hookrightarrow \mathcal{W}(X)$  if and only if  $\beta < \mu^+$ , for ordinals  $\beta$ . Every linear order of cardinality at most  $\mu$  embeds in  $\mathcal{W}(X)$ . Adjacent pairs are exactly the consecutive permutations of a finite residual alphabet after a common prefix.* This is [section 6](#) with source 11’s adjacency theorem (reals report `two:thm:adjacency`). Source 14 prints it as “inherited from the repository, not a new result of this continuation” and gives no proof: “The complete proof is in the predecessor report; we do not substitute counting for that argument.” Its set-level remark follows.

[merge] The rest of source 14’s Subsection 2.2, “Fixed carriers and the inherited spectra”.

The size assertion is plausible for two different reasons: a relation on  $X$  has at most  $2^\mu$  possibilities, while independently choosing the orientation of  $\mu$  disjoint pairs gives  $2^\mu$  distinct well-orders. The ordinal upper bound is a lexicographic obstruction, not a cardinality bound:  $2^\mu$  can be much larger than  $\mu^+$ . The complete proof is in the predecessor report; we do not substitute counting for that argument.

For an infinite cardinal  $\kappa$ , sign sequences give

$$|\mathbf{No}_{<\kappa}| = 2^{<\kappa}.$$

Thus [Note 32.5](#) applies with  $\mu = 2^{<\kappa}$ . The repository also proves a more refined minimal-slice dichotomy: with  $\mathbb{B}_\delta = (\{0, 1\}^\delta, <_{\text{lex}})$ ,

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \kappa \text{ is singular strong limit,} \\ \mathbb{B}_\mu, & \text{otherwise.} \end{cases}$$

Here  $\cdot$  is *ordinal* multiplication and  $\simeq_e$  means mutual order embeddability, not isomorphism. This inherited result should not be confused with the support-cardinal theorem proved below: birthday bounds restrict the alphabet, whereas support bounds restrict how much of a fixed global enumeration is changed.

### 32.3 Source 17’s labelled well-orders of a fixed set

[merge] Source 17’s Subsection 2.1, from its map of interpretations. Let  $(X, <)$  be a set with a specified linear order. A well-ordering  $R$  of  $X$  has a unique exhaustive enumeration  $e_R : \alpha_R \rightarrow X$ , where  $\alpha_R$  is its ordinal type. For two distinct well-orders, neither exhaustive enumeration is a proper



prefix of the other: after listing every label once there is no fresh label to append. Therefore a first unequal coordinate exists, and we define

$$R <_{\text{lex}} S \iff e_R(\delta) < e_S(\delta), \quad \delta = \min\{\xi : e_R(\xi) \neq e_S(\xi)\}.$$

Write  $\mathcal{W}(X)$  for all such well-orders, and  $\mathcal{W}_\alpha(X)$  for the suborder with enumeration type exactly  $\alpha$ . Equivalently,  $\mathcal{W}_\alpha(X)$  consists of bijections  $\alpha \rightarrow X$ . The notation  $\mathcal{W}_\mu(X)$  below always uses the initial ordinal of the cardinal  $\mu = |X|$ .

The source establishes, for infinite  $|X| = \mu$ ,

$$|\mathcal{W}(X)| = 2^\mu, \quad \beta \text{ embeds in } \mathcal{W}(X) \iff \beta < \mu^+,$$

and universality for linear orders of cardinality at most  $\mu$  ([theorem 6.1](#); [1]). These are labelled-order results. Quotienting well-orders by abstract order-isomorphism removes exactly the labels on which the comparison depends.

### 33 What this Part answers

[merge] Dated notes, 3 October 2026. Each item names the question, where it is asked, and the result that answers it. Where the question index of [section 113](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 11's question on the cut spectrum and point characters of the minimal slice without  $2^{<\kappa} = \kappa$*  ([research question 113.7](#)): **answered**. Point characters  $(q, q)$ , global characters  $r$ , density, weight and cellularity  $\mu^{<\mu}$  ([theorem 30.2](#)), the exact gap spectrum ([theorem 30.11](#)), its multiplicities ([theorem 30.16](#)), and the saturation threshold  $\eta_r$  ([corollary 30.18](#) and [theorems 30.23](#) and [30.25](#)).
- (b) *Source 12's question on the invariants of  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$*  ([research question 113.13](#)): answered except for the isomorphism type and homogeneity, which no source determines (source 17's [remark 30.22](#); source 18 says “equality of these invariants is not an isomorphism theorem”).
- (c) *Source 08's question 2, invariants at regular cutoffs* ([section 113.4](#)): **answered**. Density and cellularity of every stratum ([theorem 27.2](#) and [corollary 27.4](#)), point characters and cut cofinalities ([theorem 30.14](#) and [example 30.20](#)). This completes item (d) of [section 1.7](#).
- (d) *Source 08's question 3, which small cuts fail at singular cutoffs*: answered in character terms by [theorems 29.4](#) and [30.11](#) (source 17 does not claim this question); its coding half was already answered by [theorem 8.3](#) (item (c) of [section 1.7](#)).
- (e) *Source 12's question on all fixed enumeration types* ([research question 113.11](#)) and *source 08's question 4*: answered for  $\alpha = \mu \cdot \gamma + n$ ,  $0 < \gamma < \mu^+$ ,  $n < \omega$ , the first targets the question suggests ([theorem 31.2](#) and [corollary 31.6](#)); open for an infinite remainder  $\zeta < \mu$  ([section 31.3](#)). The density of every stratum is [theorem 27.2](#).
- (f) *Source 09's direction 7, what survives away from inaccessible truncations* ([section 113.3](#)): answered for the minimal slice; it is  $\eta_r$  and not  $\eta_{r+}$ ,  $r = \text{cf } \kappa$  ([corollary 30.18](#) and [theorem 30.23](#)).
- (g) *The reals report's question “Other ground orders”* (`lwo:sec:research`; item (e) of [section 1.7](#)): answered further for  $X = \mathbf{No}_{<\kappa}$ , every stratum  $\lambda = \mu \cdot \gamma$  and every  $n$  ([remark 31.8](#)); open for strata that are not of this form, for noncardinal cutoffs and for other alphabets.
- (h) *The reals report's questions on other alphabets* (`lwo:rk:q:alphabets`, `lwo:sp:q:alphabets`): the density and cellularity of every stratum and the incompleteness of every stratum are settled for every infinite linear alphabet ([theorems 27.2](#) and [28.1](#)); the gap spectrum is settled

for the minimal slice of every alphabet satisfying the hypotheses of [theorem 30.10](#), which bears on the “general cut-spine transfer theorem” that `lwo:sp:q:alphabets` asks for; the gap spectrum of the full order  $\mathcal{W}(X)$  stays open.

Source 16 also says that its regular-stage results “answer the earlier report’s regular-stage density, cellularity, and cut-character questions under precisely those hypotheses” (end of [section 30.4](#)), and source 15 that its block theorem gives “a precise affirmative continuation of the proposed cardinal-block program” ([section 31.3](#)); items (c) and (e) record the same.

[merge] Dated note, 3 October 2026 (batch 83). Items (b), (e), (f) and (g) are answered further by Parts X and XI: the cube spectrum of every fixed type, with its infinite remainders (so the open case of [section 31.3](#) is covered), the topology of the minimal slice, and noncardinal cutoffs ([sections 72 and 79](#); [section 113.25](#) (a)–(c)).

## Part VII

# Words, termination conventions and other comparison orders

[merge] (3 October 2026.) The second Part of the batch-81 addition ([section 1.8](#); notation in [section 1.3.1](#)). Part II ([part II](#)) compared set-length words with proper prefixes first; the six batch-81 sources ask what changes when the comparison convention changes. This Part prints the six sources’ maps of the possible interpretations ([section 34](#)); the isomorphism of the unrestricted, injective and numerically increasing shorter-first word orders ([section 35](#)); the criterion for every placement of a formal terminator and the list of all isomorphism types it produces ([section 36](#)); the exact cut classification of centered increasing words and of the conventions for the entry zero ([section 37](#)); and comparison orders other than numerical enumeration lexicography: inverse ranks, relation tables, restrictions and normal forms ([section 38](#)). [Section 39](#) records which questions the Part answers.

The bases are chosen by generality, as in Part VI: source 16 for the word isomorphism and the termination classification (every placement of a terminator, unrestricted and injective words), source 18 for centered increasing words and the zero conventions (the only source treating them), source 15 for the other comparison orders (the only source proving theorems about them). Source 13’s central termination theorem and source 18’s centered theorem are the same special case of source 16’s criterion, proved independently, with the same letter code; they are printed as routes. Source 18 was written before sources 13–17 arrived. Re-proofs of Part II are printed as notes. Source 16 uses the macro names of this report for other objects (its `\Words` is the class of *all* words, its `\I` the class injections); here they are printed  $\mathcal{T}$  and  $\mathfrak{J}$ , and  $\mathcal{I}$  keeps its meaning of injective words ([section 1.3.1](#)). Labels written for the merge in this Part begin `swo:vii:`.

## 34 The six sources’ maps of interpretations

[merge] Each batch-81 source opens with a map of what “all well-orders of the surreal numbers” can mean, as sources 12, 09 and 08 did ([section 4](#)). The six maps are printed here, each whole, because their rows on varying carriers, termination, increasing subsets, relation codes and abstract types are the subject of this Part; the subsections on a fixed set carrier that sources 14 and 17 put inside their maps are printed in Part VI ([note 32.5](#) and [section 32.3](#)). The six agree with one another and with the batch-80 sources; [note 34.2](#) lists what they share.

### 34.1 Source 13: the interpretations that must be separated

#### 34.1.1 A taxonomy by carrier and representation

The following table fixes the principal meanings. In every row, numerical comparison of surreal labels is distinct from the auxiliary well-order being represented.

| Interpretation                      | Representation                                                                                       | Logical size and caveat                                                                               |
|-------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Fixed set $X \subseteq \mathbf{No}$ | Well-order relations on $X$ ,<br>equivalently bijections $\alpha \rightarrow X$ from<br>set ordinals | A set; every possible well-order<br>type on $X$ is permitted.                                         |
| Fixed length on $X$                 | Bijections $\lambda \rightarrow X$                                                                   | A single stratum; $ \lambda  =  X $ is<br>necessary.                                                  |
| Arbitrary set subsets               | Injective words $p : \alpha \rightarrow \mathbf{No}$                                                 | An ordinary class of sets;<br>termination must be specified.                                          |
| Words with<br>repetitions           | Arbitrary $p : \alpha \rightarrow \mathbf{No}$                                                       | An ordinary class, but not a<br>class of well-orderings of the<br>ranges.                             |
| Inherited numerical<br>order        | Subsets on which the restriction of $<$<br>is a well-order                                           | Not arbitrary relabelings;<br>increasing enumerations only.                                           |
| Set-like whole-class<br>orders      | Class bijections $E : \text{Ord} \rightarrow \mathbf{No}$                                            | The virtual domain $\mathfrak{W}_{\text{sl}}$ ; its<br>elements are classes.                          |
| Unrestricted<br>whole-class orders  | Class well-order relations on $\mathbf{No}$                                                          | The virtual domain $\mathfrak{W}_{\text{all}}$ ;<br>proper-class predecessor<br>segments are allowed. |
| Unlabeled order types               | Quotients by order-isomorphism                                                                       | Label-sensitive lexicographic<br>comparison does not descend.                                         |
| Relation-code order                 | Characteristic bits of a relation, read<br>in a chosen order of pairs                                | A different, coding-dependent<br>lexicographic order.                                                 |

This covers the standard semantic axes, not every artificial encoding one could invent. Birthday-first comparison, a fixed global well-order of the labels in place of their numerical order, or comparison of unquotiented Conway expressions each changes the problem again.

#### 34.1.2 Fixed carriers have no end-marker ambiguity

For a set  $X$ , put

$$\mathcal{W}(X) = \{e : \alpha \xrightarrow{\sim} X : \alpha \text{ an ordinal}\}, \quad \mathcal{W}_\lambda(X) = \{e \in \mathcal{W}(X) : \text{dom}(e) = \lambda\}.$$

These are sets: if  $|X| = \mu$  is infinite, the possible domains are among the ordinals below  $\mu^+$  of cardinality  $\mu$ .

**Lemma 34.1** (Prefix-freeness of exhaustive enumerations). *Distinct members of  $\mathcal{W}(X)$  are never proper prefixes of one another. Their lexicographic comparison is determined at a least common coordinate where their values differ.*

*Proof.* A prefix that already enumerates all of  $X$  cannot be extended injectively by another member of  $X$ . If two different enumerations agreed on their common domain, the shorter would be such a prefix. Otherwise their set of disagreement coordinates has a least member.  $\square$

An auxiliary well-order  $R$  on  $X$  has one ordinal type and one increasing enumeration in that order. This is a bijection between relation representations and the displayed exhaustive words. Lexicographic comparison uses the *numerical* order of their values, not the order  $R$  itself.

### 34.1.3 Varying the range introduces a genuine convention

An injective word  $p : \alpha \rightarrow \mathbf{No}$  enumerates a well-order of its range. Once ranges may differ,  $p$  can be a proper prefix of another word. Under *end-first* dictionary comparison,  $p < p \frown x$  for every fresh  $x$ . Under end-last comparison the inequality is reversed. A terminal symbol placed inside an enlarged alphabet gives a third possibility.

Deleting repeated letters from arbitrary words is not a harmless change of representation. For example, with  $a < b$ , the two words  $(a, a, b)$  and  $(a, b)$  have different dictionary positions but the same first-occurrence enumeration. No quotient order is supplied merely by performing this deletion.

### 34.1.4 Inherited well-orders and unlabeled quotients

The numerical order of all surreals is not a well-order: it contains, for example, the descending sequence  $0, -1, -2, \dots$ . Its ordinal subclass, represented by all-plus signs, is a proper-class increasing well-order. Every set ordinal is likewise realized as a numerically well-ordered subset of  $\mathbf{No}$ .

The class of *set-sized* numerically increasing words is an ordinary class of set codes. Under end-first comparison it has an empty minimum and the limit-prefix phenomenon proved in [Section 35.2](#). It therefore differs from the surreal line despite containing a copy of  $\mathbf{No}$  through singleton words. The general class-universality argument below embeds it back into  $\mathbf{No}$ : equimorphism is not isomorphism.

If labels are forgotten altogether, lexicographic comparison is usually lost. On a two-element labeled carrier the two enumerations have the same ordinal type but opposite lexicographic positions. More dramatically, every individual set-like global well-order of  $\mathbf{No}$  has type  $\text{Ord}$ ; its quotient by abstract order-isomorphism has just one element, although  $\mathfrak{W}_{\text{sl}}$  itself is extremely rich.

### 34.1.5 The Kanovei–Shelah comparison is related, not identical

Kanovei and Shelah’s definable hyperreal construction uses a linearly ordered family of maps from the continuum into  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters. The maps are compared lexicographically; injectivity is not the defining requirement [16]. This is an important motivation for ordered collections of enumeration codes. It should not be identified without argument with the exhaustive well-orderings of one fixed labeled carrier. In particular, replacing its index by “all surreal well-orders” would change both the coding and the logical size level.

## 34.2 Source 14: which interpretation is intended?

### 34.2.1 A map of the natural meanings

In the table below, “ordinary class” means a collection of sets defined by a class predicate. “Virtual” means that each candidate is itself a class and the theory quantifies over candidates without forming their collection as another class.

| Interpretation                      | Objects and comparison                                                                                                   | Size and main warning                                                                              |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Fixed set $X \subseteq \mathbf{No}$ | All well-order relations on $X$ , compared by their exhaustive labelled ordinal enumerations.                            | A set; arbitrary well-order type of cardinality $ X $ is allowed.                                  |
| Fixed length $\alpha$               | Bijections $\alpha \rightarrow X$ , numerical first-disagreement comparison.                                             | A set; one slice of the preceding order.                                                           |
| Birthday fragment                   | $X = \mathbf{No}_{<\theta}$ , with either of the preceding choices.                                                      | A set; the ordinal cutoff and cardinality play different roles.                                    |
| Variable set domain                 | Pairs $(A, R)$ with $A \subseteq \mathbf{No}$ a set and $R$ a well-order of $A$ , or injective set-length surreal words. | An ordinary proper class; a termination convention is indispensable.                               |
| Unrestricted words                  | All functions $\alpha \rightarrow \mathbf{No}$ for set ordinals $\alpha$ , allowing repeated values.                     | An ordinary proper class; these are not enumerations of well-ordered labelled subsets.             |
| Increasing subsets                  | Subsets or subclasses well-ordered by the <i>numerical</i> order inherited from $\mathbf{No}$ .                          | A restriction on labels, not an arbitrary imposed well-order.                                      |
| Set-like global order               | Class bijections $\text{Ord} \rightarrow \mathbf{No}$ .                                                                  | Virtual order $\mathfrak{W}_{\text{sl}}$ ; each proper initial segment is a set.                   |
| Arbitrary global order              | Class well-order relations on all of $\mathbf{No}$ , including non-set-like ones.                                        | Virtual order $\mathfrak{W}_{\text{all}}$ ; direct labelled common-prefix comparison is available. |
| Relation-code lex                   | Characteristic functions of relations, using a fixed well-ordered schedule of ordered pairs.                             | A different order; depends on the schedule and on the bit convention.                              |
| Order types only                    | Isomorphism classes after forgetting surreal labels.                                                                     | Enumeration lex does not descend. Set-like proper class orders all have type $\text{Ord}$ .        |

This list covers the main structural ambiguities, rather than pretending that arbitrary encodings cannot create additional variants. Every subsequent theorem names the interpretation to which it applies.

[merge] Source 14’s Subsection 2.2 (“Fixed carriers and the inherited spectra”) is printed in Part VI (note 32.5); its Subsections 2.3–2.5 follow.

### 34.2.2 Variable domains and the terminal symbol

An imposed well-order on a set  $A \subseteq \mathbf{No}$  is represented uniquely by an injective word  $p : \lambda \rightarrow \mathbf{No}$ . For varying  $A$ , a word may be a proper prefix of another. Three reasonable choices are: *prefix first*, using  $\text{end} < x$  for every numerical letter; *prefix last*, using  $x < \text{end}$ ; and a marker inserted at a specified cut in the numerical alphabet. A fourth order is length-first lexicography. They do not agree, and none becomes a well-order simply because the word domains are ordinals: the one-letter words already contain the numerical order of  $\mathbf{No}$ .

The familiar sign-sequence order of the surreals uses  $- < \mathbf{end} < +$ . It is not prefix-first dictionary order. A useful self-delimiting numerical code replaces  $-$  by 00,  $+$  by 11, and appends 01. Comparisons are preserved because

$$00 < 01 < 11.$$

The terminal block makes distinct numerical codewords prefix-incomparable. Concatenating such codes is useful for embeddings, but does not turn arbitrary repeated-letter words into exhaustive enumerations. The predecessor report treats the resulting compression and the special cuts at limit word lengths (sections 9 to 11; Sections 9–11 of this report at the pin).

### 34.2.3 Numerically increasing well-orders

The numerical order of **No** is not a well-order; for example,  $0 > -1 > -2 > \dots$ . Hence no numerical order-preserving enumeration can well-order all of **No**. This is different from imposing another well-order on the same carrier.

Proper subclasses can nevertheless be numerically well-ordered and non-set-like. For instance,

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

together with numerical order has type  $\text{Ord} + 1$ . The ordinal arithmetic in the denominator is interpreted by the corresponding surreal ordinal followed by surreal addition; the displayed values are strictly increasing and all precede 0. Thus a numerically bounded subclass can have a point with a proper class of predecessors. Set-likeness is not a consequence of numerical boundedness.

### 34.2.4 Forgetting labels changes the problem

For a fixed infinite set of cardinality  $\mu$ , its possible abstract well-order types are the ordinals  $\alpha$  satisfying  $\mu \leq \alpha < \mu^+$ . Ordering these types by ordinal size gives a well-order of type  $\mu^+$ , which cannot embed in  $\mathcal{W}(X)$  by Note 32.5. Therefore the order on abstract types is not the induced quotient of enumeration lexicography.

More directly, different enumerations of the same type need not occupy a convex subset: choose first labels  $x < y < z$  and complete the first and third to type  $\mu$ , the middle to type  $\mu + 1$ . All three completions exist, and their lexical positions are decided by those first labels. For a proper class carrier every set-like well-order has abstract type  $\text{Ord}$ , so quotienting that fragment by abstract type collapses it to a single type. Almost all of the lexicographic information was in the labels, not in the abstract order type.

## 34.3 Source 15: a map of the interpretations

Every concrete version is determined by six choices: the carrier; whether orders are labelled or identified up to isomorphism; how an order is coded; which order compares the code entries; which well-order scans the entries; and the universe in which all codes are collected. Prefix conventions add a seventh choice when the coded sequences have variable domains and need not exhaust a common carrier.



| Interpretation                | Objects and comparison                                                                      | Main distinction                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Fixed carrier, all types      | All ordinal bijections $e : \alpha \rightarrow X$ , compared at first unequal labels in $X$ | A set in ZFC; exhaustive codes are prefix-free.                                    |
| Fixed enumeration type        | Only $e : \alpha \rightarrow X$ for one prescribed $\alpha$                                 | Same possible ordinal suborders need not mean the same binary coding length.       |
| Variable set carrier          | Injective set-length words, or all well-ordered sets of surreal labels                      | Proper prefixes occur; termination must be ordered explicitly.                     |
| Natural well-ordered subsets  | Subsets $A \subseteq \mathbf{No}$ with the inherited numerical order well-founded           | No arbitrary rearrangement is part of the data.                                    |
| Set-like class orders         | Bijections $\text{Ord} \rightarrow \mathbf{No}$ ; first ordinal difference                  | All predecessors are sets; a predicate on class functions.                         |
| All labelled class orders     | Arbitrary class well-orders on $\mathbf{No}$ ; common labelled prefix                       | May differ only after a proper class of identical earlier labels.                  |
| Fixed class index type        | Bijections $J \rightarrow \mathbf{No}$ for one given class well-order $J$                   | A chosen type such as $\text{Ord} + \text{Ord}$ restricts the domain.              |
| Inverse-rank lexicography     | Rank values of labels scanned in a fixed baseline well-order                                | Uses ordinal values rather than numerical surreal labels; has different endpoints. |
| Relation-table lexicography   | Characteristic bits of $R \subseteq \mathbf{No}^2$ in a fixed well-ordered scan             | Depends on scan and bit orientation; differs already on three points.              |
| Isomorphism types             | Quotient well-orders by abstract order isomorphism                                          | Native enumeration lex does not descend to this quotient.                          |
| Birthday/simplicity variants  | Birthday ranks, sign strings, or simplicity predecessors                                    | Birthday ties need a separate well-order; simplicity is only partial.              |
| Definable or computable codes | A specified syntactic class with an evaluator                                               | Restricts the available objects; syntax alone does not supply universal truth.     |
| Higher-universe totality      | All relevant relations become sets in a larger universe                                     | Results depend on the class realization and external well-foundedness.             |

This table is exhaustive at the level of the structural choices just listed, not a claim to enumerate every artificially possible encoding. Once arbitrary encodings and arbitrary scans are allowed, one can manufacture many further linearizations. The mathematically meaningful task is to say which properties are intrinsic to the chosen order and which depend on its presentation.

### 34.3.1 The historical Kanovei–Shelah construction

The original construction uses

$$A = \{a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N}) : \text{ran}(a) \text{ is an ultrafilter on } \mathbb{N}\},$$

ordered lexicographically in the initial ordinal  $\mathfrak{c}$  coordinates, with  $\mathcal{P}(\mathbb{N})$  itself ordered through characteristic functions. Maps may repeat entries, and different maps may represent the same

ultrafilter. This is an important precedent for ordering presentations without selecting a preferred presentation. It is not literally an order of all well-orders of  $\mathbb{R}$  or of **No** [16, Section 1].

### 34.4 Source 16: a systematic map of the interpretations

Three choices account for most natural readings: the carrier to be well-ordered, whether numerical labels are retained, and the coding used for comparison. A fourth choice, the treatment of termination, becomes necessary when carriers or word lengths vary. There cannot be a literal list of every possible encoding: composing any encoding with an arbitrary permutation can manufacture another order. The following table covers the principal mathematical interpretations and explains how the remaining variants arise.

| Interpretation                                   | Objects and comparison                                                                    | Size and main issue                                                                                        |
|--------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Fixed set of labels<br>$X \subseteq \mathbf{No}$ | Bijections $\alpha \rightarrow X$ ; first unequal entry compared numerically              | A set in ZFC; the order on enumerations is usually far from a well-order.                                  |
| Fixed enumeration type                           | Only bijections $\alpha \rightarrow X$ , with $\alpha$ fixed                              | Changing $\alpha$ changes topology and may change embedding invariants.                                    |
| Variable set carrier                             | Injective set-length words in <b>No</b>                                                   | An ordinary class of set codes; proper-prefix comparison must be specified.                                |
| Numerically well-ordered subsets                 | Increasing enumerations in the numerical order                                            | No arbitrary auxiliary well-order remains on each carrier; proper class subsets can still be non-set-like. |
| Set-like global well-orders                      | Class bijections $\text{Ord} \rightarrow \mathbf{No}$                                     | A predicate on class variables; not an ordinary class of such graphs.                                      |
| Arbitrary global class well-orders               | Class relations on all of <b>No</b> ; compare after the common labelled prefix            | Common prefixes can be proper classes; ordinal-coordinate comparison alone is insufficient.                |
| Relation-table encoding                          | Binary membership tables for order relations, read along an auxiliary well-order of pairs | Gives a different, coding-dependent lexicographic order.                                                   |
| Abstract order types                             | Forget surreal labels and identify isomorphic well-orders                                 | Numerical first-disagreement comparison no longer descends to the quotient.                                |
| Larger-universe interpretation                   | Regard internal classes as sets in an explicit containing universe                        | Makes entire collections available, while changing the meaning of “set”, “class”, and “all”.               |

#### 34.4.1 Fixed labels versus abstract types

For a linearly ordered set  $(X, <)$ , write

$$\mathcal{W}_\alpha(X) = \{e : \alpha \xrightarrow{\sim} X\}, \quad \mathcal{W}(X) = \bigcup_{|\alpha|=|X|} \mathcal{W}_\alpha(X).$$

A bijection is identified with the well-order it induces on  $X$ . Distinct exhaustive enumerations cannot be proper initial segments of one another: the shorter enumeration has already used every

label, leaving no fresh next entry for the longer one. Therefore every distinct pair has a first unequal entry, with no termination convention needed.

If labels are discarded, the order types are simply the ordinals  $\alpha$  of cardinality  $|X|$ . Their ordinary ordinal comparison is meaningful, but it is not numerical lexicography. Even two permutations of a finite set have the same abstract order type and different lexicographic positions. For arbitrary class well-orders, a single class of all abstract “class ordinals” is not supplied by GB or KM. Direct labelled comparison will avoid this issue entirely.

### 34.4.2 Relation tables genuinely change the answer

Choose an auxiliary well-order  $j : \delta \rightarrow X \times X$  and encode a relation  $R$  by the binary word  $\chi_R(j(\xi))$ . The earliest differing bit defines a linear comparison, but its answer depends on  $j$  and the bit convention. For example, on  $X = \{0, 1, 2\}$  the enumeration  $(0, 2, 1)$  precedes  $(1, 0, 2)$  numerically. In row-major relation-table order, their first differing entry is the pair  $(0, 1)$ : its bits are respectively 1 and 0. The table comparison therefore gives the opposite inequality. A class version needs a specified class ordering of the pairs and appropriately stratified quantification; the word “lexicographic” alone does not choose one.

### 34.4.3 The Kanovei–Shelah connection

The motivating analogy should not be mistaken for an identification. Kanovei and Shelah use the index

$$A = \{a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N}) : \text{ran}(a) \text{ is an ultrafilter on } \mathbb{N}\},$$

ordered at the first differing coordinate by binary characteristic-function order on  $\mathcal{P}(\mathbb{N})$ . Repetitions are permitted, and the displayed definition admits ultrafilters generally. The point is a definable order on redundant codes, used in an ultrapower construction. It is not literally the space of all bijective enumerations or all well-orders of  $\mathbb{R}$  [16, §2 in the published version; §1 in the preprint].

## 34.5 Source 17: what can “all well-orders of surreal numbers” mean?

We use the sign-sequence presentation and numerical order of surreal numbers; standard background is given in [11, 12]. A well-ordering relation and the numerical order of its labels are distinct. An enumeration of a well-order need not be numerically increasing. This single distinction already produces two different research programs.

[merge] Source 17’s Subsection 2.1 (“Labelled well-orders of a fixed set”) is printed in Part VI (section 32.3). Its displayed word code keeps its number as (S17.1).

### 34.5.1 Set-length words and varying supports

Every well-order of a set-sized subset of **No** is represented by an injective word  $w : \alpha \rightarrow \mathbf{No}$ . Now one word can be a proper prefix of another. We use the *shorter-prefix-first* convention unless specified otherwise. The resulting carrier is an ordinary class of sets. Allowing repeated entries gives a larger class of words, but such a word no longer represents a well-order of its range without an additional occurrence labelling.

There is a useful explicit compression, also present in the source. Represent  $x \in \mathbf{No}$  by its sign sequence. Replace a minus sign by  $--$ , a plus sign by  $++$ , and append the terminal block  $-+$ . Call the resulting prefix-free code  $c(x)$ . The comparison

$$-- < -+ < ++$$

encodes the surreal convention “minus < termination < plus”. For any set-length word put

$$\Phi(w) = \bigcap_{i < \alpha} (+ \cap c(w(i))). \quad (\text{S17.1})$$

At a first differing entry the prefix-free coefficient codes determine the comparison; at a proper word prefix the next sign is +, so the longer word is larger. Hence  $\Phi$  is an order embedding into  $\mathbf{No}$ , with birthday

$$\text{bd } \Phi(w) = \sum_{i < \alpha} (1 + 2 \cdot \text{bd}(w(i)) + 2).$$

Products and sums in this formula are ordinal operations. All lengths are sets. The empty word is least, so the entire word order is not isomorphic to the endpoint-free surreal line, despite embeddings in both directions. Using  $-$  instead of the leading  $+$  in each entry block gives the longer-prefix-first convention and makes the empty word greatest.

Length-first comparison, often called shortlex, is another legitimate linear order, but is not this dictionary order. It should not be silently substituted in an embedding or topology theorem.

### 34.5.2 Numerically well-ordered subsets

A different interpretation asks for subsets  $A \subseteq \mathbf{No}$  whose *inherited numerical order* is a well-order. Their set-sized increasing enumerations form a suborder of the injective word order. They include every singleton, so their lexicographic order contains a copy of  $(\mathbf{No}, <)$  and is not a well-order. They inherit the embedding (S17.1). Every set-sized increasing word admits a longer increasing extension, because a set of surreal numbers has a strict upper bound.

There is no numerically increasing exhaustive well-order of the whole surreal line: the sequence  $0, -1, -2, \dots$  already witnesses the failure of well-foundedness of the numerical order. This does not prohibit an arbitrary labelled class well-order of  $\mathbf{No}$ .

Proper-class *subsets* with their inherited numerical order must also be distinguished from the entire numerical line. They can be non-set-like well-orders. For example,

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

is increasingly well-ordered in type  $\text{Ord} + 1$ . The reciprocal is surreal field reciprocal, and  $\alpha$  denotes its ordinal surreal value. The set of predecessors of 0 in this class order is proper. A collection of all such class subsets again lives at the higher logical level. For varying supports, proper-prefix comparisons require an explicit termination convention, just as for set words. Our main raw comparison keeps the carrier fixed and therefore has no termination ambiguity.

### 34.5.3 Global enumerations and unrestricted relations

A *set-like* class well-order is one in which each element has a set of predecessors. A set-like well-order of the proper class  $\mathbf{No}$  has order type  $\text{Ord}$ , and is equivalently a class bijection

$$E : \text{Ord} \longrightarrow \mathbf{No}.$$

Write  $\mathfrak{W}_{\text{sl}}$  for this predicate on class relations, and  $\mathfrak{J}$  for the analogous predicate of injective, not necessarily exhaustive,  $\text{Ord}$ -words. These symbols are not names of ordinary GB classes whose elements are class functions.

Write  $\mathfrak{W}_{\text{all}}$  for arbitrary class well-orders of  $\mathbf{No}$ . Orders of type  $\text{Ord} + 1$ ,  $\text{Ord} + \text{Ord}$ , and many other non-set-like types belong here. For example, enumerate  $\mathbf{No} \setminus \{a, b\}$  first and append  $a, b$ .

Interchanging those last two labels creates a first disagreement after a proper-class prefix, not at a set ordinal.

One can also fix a particular class well-order  $J$  and compare bijections  $J \rightarrow \mathbf{No}$  at their least differing  $J$ -position. This is legitimate when least elements of nonempty admitted subclasses of  $J$  are part of the hypothesis. It is a fixed-domain alternative, not a universal set ordinal index for every class well-order.

### 34.5.4 Relation codes, isomorphism types, and ordinal-valued rankings

Fix a class bijection  $d : \text{Ord} \rightarrow \mathbf{No} \times \mathbf{No}$ . A class relation  $R$  has binary code  $\xi \mapsto \mathbf{1}_R(d(\xi))$ , which can be compared lexicographically. This yields a valid order on the admitted well-order relations, but depends on  $d$  and the bit convention. It is generally not the native labelled-enumeration order. Even on  $\{0, 1\}$ , listing coordinate  $(0, 1)$  first and using  $0 < 1$  compares the two well-orders in the reverse of their first-label comparison.

Abstract isomorphism types form yet another subject. For set well-orders they are ordinals, naturally ordered by initial-segment embeddability. The native lexicographic order does not descend to this quotient: the two enumerations  $(a, b)$  and  $(b, a)$  have the same abstract type but opposite label order when  $a < b$ . A rank function, a birthday-first code, or a chosen well-order of relation codes introduces additional structure rather than recovering a canonical lexicographic comparison from the abstract types.

## 34.6 Source 18: interpretations and foundational conventions

### 34.6.1 A map of the natural interpretations

| Object                                           | What a point represents                                                                           | Size and comparison                                                                       |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| $\mathcal{W}(X)$ , $X \subset \mathbf{No}$ a set | A labelled well-order of the fixed carrier $X$ , in any possible ordinal type                     | A set; compare unique exhaustive ordinal enumerations                                     |
| $\mathcal{W}_\alpha(X)$                          | A well-order whose type is the specified ordinal $\alpha$                                         | A stratum of $\mathcal{W}(X)$ ; nonempty exactly when $ \alpha  =  X $                    |
| Injective set words                              | A well-order of a variable set-sized subset of $\mathbf{No}$                                      | A genuine proper class of sets; a termination convention is needed                        |
| Increasing set words                             | A subset well-ordered by the numerical surreal order, with its unique increasing enumeration      | A subclass of the previous word class; no arbitrary reordering of a fixed subset          |
| $\mathfrak{W}_{\text{sl}}(\mathbf{No})$          | A set-like global well-order, equivalently a class bijection $\text{Ord} \rightarrow \mathbf{No}$ | A predicate on class relations; distinct enumerations have a set ordinal first difference |
| $\mathfrak{W}_{\text{all}}(\mathbf{No})$         | Any class well-order of the entire surreal carrier                                                | A predicate on class relations; the common labelled prefix can be proper                  |
| Inherited class well-orders                      | A subclass on which numerical $<$ is a well-order                                                 | A different restriction: $\mathbf{No}$ itself is not numerically well-ordered             |
| Order types modulo isomorphism                   | Labels are forgotten                                                                              | Numerical lexicography does not descend to this quotient                                  |

| Object                        | What a point represents                                                         | Size and comparison                                                             |
|-------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Characteristic-relation codes | A well-order relation encoded by its membership bits in a chosen order of pairs | A valid auxiliary lex order, usually different from enumeration comparison      |
| Containing-universe objects   | All orders admitted by a specified smaller set/class universe                   | Sets in an external universe; “all” is relative to the chosen class realization |

There are indefinitely many artificial encodings. The table covers the natural mathematical choices by varying carrier, enumeration type, admissible order, label retention, comparison rule, and size level. An arbitrary encoding cannot be expected to preserve the results for another row.

[merge] Source 18’s Subsections 2.2 and 2.3 (sign sequences with its lemma on surreal order tools, and set and class theories) are foundations, not interpretations, and are not printed in this Part. Its Subsection 2.4 follows.

### 34.6.2 Why the Kanovei–Shelah connection needs qualification

Kanovei and Shelah’s published construction [16, §2] orders maps  $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters, using characteristic-function lexicography on  $\mathcal{P}(\mathbb{N})$ . Repetitions are allowed in those maps and in the ultrafilter presentation. Their index is therefore not literally  $\mathcal{W}(\mathbb{R})$  or a permutation order of the reals. The useful connection is the method of linearly ordering many codes without choosing one privileged code. The present article studies different, precisely specified orders inspired by that method. This correction was already explicit in the repository’s reals report [1].

*Note 34.2* ([merge]; what the six maps share). All six separate the same carriers as the batch-80 sources (sections 2 and 4.3): a fixed set with all types or one type, varying set carriers (injective words, with a termination convention), unrestricted words, numerically well-ordered subsets, set-like and arbitrary class well-orders, relation codes and abstract types.

- *Termination.* Sources 13, 14, 16 and 17 name the three placements of a terminator (first, last, inside the alphabet) and length-first order as a fourth choice; source 14’s warning that the Conway sign order uses  $- < \text{end} < +$  and is not prefix-first dictionary order is the convention printed in section 5.
- *Increasing subclasses.* The example  $\{-1/(\alpha + 1) : \alpha \in \text{Ord}\} \cup \{0\}$  of numerical type  $\text{Ord} + 1$  is source 12’s (section 18); sources 14, 15, 16, 17 and 18 repeat it.
- *Relation codes.* The two- and three-point examples of sources 14, 16, 17 and 18 make the point of source 12’s shipped example and of proposition 17.1; source 15 proves theorems about these orders (theorem 38.2 and corollary 38.3).
- *Abstract types.* That enumeration lexicography does not descend to abstract order types (sources 13, 14, 15, 16, 17) is printed with source 15’s non-convexity example in section 38; source 14’s argument that the type order  $\mu^+$  cannot embed uses note 32.5.
- *Kanovei–Shelah.* The Kanovei–Shelah subsections of sources 13, 15, 16 and 18 printed here (sources 14 and 17 discuss it elsewhere in their texts) make source 08’s point (section 4.4): the index there is a lexicographically ordered family of redundant codes, not a space of well-orders. Source 16 cites Section 2 of the published version and Section 1 of the preprint; source 15 cites “Section 1”.
- *Prefix-freeness.* Source 13’s lemma 34.1 is the prefix-freeness of exhaustive enumerations used in section 6.



### 35 Shorter-first words: three classes, one prefix order (source 16; routes of sources 13, 14 and 18)

[merge] Source 16's Section 7, printed whole; its  $\mathcal{T}$ ,  $\mathcal{T}^{\text{inj}}$ ,  $\mathcal{T}^{\text{inc}}$  are printed  $\mathcal{T}$ ,  $\mathcal{I}$ ,  $\mathcal{T}^{\text{inc}}$  (section 1.3.1), and its equations keep their numbers as (S16. $n$ ). Its references to its own fresh-separator lemma, universality proposition, set-cut display and raw-comparison section are printed as references to lemma 5.2, proposition 5.13, and definition 5.1 and theorem 14.2, which state the same facts. The word orders of Part II (part II) are the cases  $\mathcal{T}$  and  $\mathcal{I}$ ; what this section adds is the increasing class  $\mathcal{T}^{\text{inc}}$  and the isomorphism of the three (theorem 35.1). Source 18 proves the isomorphism independently, with the uniformity of the back-and-forth made explicit (section 35.5); source 13's and source 18's word sections and source 14's explicit codes follow as notes and routes.

Let  $\mathcal{T}$  be the class of all set functions  $f : \alpha \rightarrow \mathbf{No}$  with  $\alpha$  an ordinal. Let  $\mathcal{I}$  be its injective subclass, and  $\mathcal{T}^{\text{inc}}$  its numerically strictly increasing subclass. Members of all three are sets. In  $\mathcal{I}$  the word imposes a well-order on its image; in  $\mathcal{T}^{\text{inc}}$  that well-order is the restriction of the numerical order. Initially give all three the convention that a proper prefix is smaller than its extension.

The empty word is then a minimum. More substantially, if  $f$  has nonzero limit length  $\lambda$ , its proper prefixes are cofinal below it, giving

$$\text{cf}(\mathcal{T}_{<f}) = \text{cf}(\lambda). \quad (\text{S16.5})$$

At successor-length words there is no set cofinal family below the point, and above any word there is no set coinitial family. For the unrestricted and injective cases these facts appear in the earlier report; they follow by choosing a fresh next letter between the set of active coordinates and the coordinate of the given word. Equation (S16.5) already prevents an isomorphism with  $\mathbf{No}$ , even if the empty word is removed: a surreal cannot be the supremum of a set lying strictly below it, by the set-cut property (definition 5.1).

**Theorem 35.1** (Three interpretations with the same prefix order). *In GBC,*

$$\mathcal{T} \cong \mathcal{I} \cong \mathcal{T}^{\text{inc}}$$

*under shorter-prefix-first comparison. There are isomorphisms preserving word length, the prefix relation, unions of set-length prefix chains, and every singleton word. The same assertion holds for the convention that proper prefixes are larger than their extensions.*

*Proof.* For an admissible output prefix  $q$ , its possible next letters form, respectively,

$$\mathbf{No}, \quad \mathbf{No} \setminus \text{ran}(q), \quad U(q) = \{x \in \mathbf{No} : \forall y \in \text{ran}(q) (y < x)\}.$$

Each is a proper class with the set-cut property. For the second use the fresh-separator lemma (lemma 5.2). For the third, given sets  $A < B$  in  $U(q)$ , fill the cut  $A \cup \text{ran}(q) < B$ ; this also proves the property with an empty side. Uniform back-and-forth as in class universality (proposition 5.13) gives, for every set parameter  $q$ , an order isomorphism  $h_q$  from  $\mathbf{No}$  onto its available-letter class. Take  $h_\emptyset$  to be the identity. A single class relation  $H(q, x, y)$  defines all these maps: a pair belongs when it appears in the uniquely specified set-length back-and-forth history. No choice of a proper class for each index is hidden here.

For a free word  $f : \alpha \rightarrow \mathbf{No}$ , define its output by set recursion:

$$q(i) = h_{q \upharpoonright i}(f(i)) \quad (i < \alpha).$$

The output is injective or increasing according to the target rule. Conversely, an admissible output  $q$  uniquely recovers  $f(i) = h_{q|i}^{-1}(q(i))$ . These operations are inverse and preserve prefixes and lengths. At a first differing coordinate, both inputs use the same strictly increasing local map, so their comparison is preserved. The endpoint termination rule depends only on the prefix relation, which is also preserved. The recursive definition makes compatibility with unions immediate, and the choice at the empty prefix fixes singletons.  $\square$

This is an isomorphism theorem, not a literal identification of the objects. Longer labels generally change. In particular, the theorem does not turn an arbitrary chosen well-order of a subset into the numerical order on the *same* subset. It shows that the bare lexicographic order does not recover that conceptual distinction.

### 35.1 Increasing proper classes are a further case

The preceding theorem concerns set-length words. A numerically well-ordered proper subclass can itself be non-set-like. For example, the increasing class

$$\left\{ -\frac{1}{\xi + 1} : \xi \in \text{Ord} \right\} \cup \{0\}$$

has order type  $\text{Ord} + 1$ ; the point 0 has a proper class of predecessors. The simpler subclass of nonnegative ordinal surreals has type  $\text{Ord}$ . In GBC, class universality ([proposition 5.13](#)) embeds every class well-order into the numerical surreal line, so every such abstract order can occur on a numerically well-ordered subclass. No increasing enumeration can exhaust **No**: it has a first value and omits every smaller surreal.

If distinct proper class carriers are allowed, comparison must also say what happens when one whole labelled class order is an initial segment of another. The direct common-prefix method of Part III ([theorem 14.2](#)), with an explicitly placed termination symbol, supplies the corresponding variant. It is a different problem from comparing well-orders on the same fixed carrier.

*Note 35.2* ([merge]; what transfers from Part II). Display (S16.5) and the two sentences after it are [theorem 10.1](#) (sources 11 and 09: [note 11.4](#)). Because the isomorphisms of [theorem 35.1](#) preserve length and the prefix relation, every statement of Part II about  $\mathcal{T}$  or  $\mathcal{I}$  that is phrased by lengths and prefixes holds in GBC for  $\mathcal{T}^{\text{inc}}$ : the exact local characters ([theorem 10.1](#)), the word-cut classification ([theorem 11.1](#)) and the successor-word corollary ([corollary 11.2](#)). Source 18 makes this remark for the cut classification ([section 35.3](#)); its direct proof of the characters of  $\mathcal{T}^{\text{inc}}$  needs no global choice ([note 35.5](#)). Part II's explicit code ([theorem 9.1](#)) restricts to  $\mathcal{T}^{\text{inc}}$  without choice. The example of source 16's Subsection 7.1 is source 12's ([section 18](#)).

### 35.2 Source 13: set-length words and their first obstruction

[merge] Source 13's Section 5, printed whole. Its proposition is [theorem 10.1](#) (1) of Part II (and source 09's [note 11.4](#)) and is printed as a note.

Let  $\mathcal{T}$  be the ordinary class of all set-length words over **No**, and let  $\mathcal{I} \subseteq \mathcal{T}$  be the injective words. The latter is exactly the class of well-orderings of arbitrary set-sized subsets, represented by their increasing enumeration in the auxiliary order. In this section the terminal symbol precedes every label.

*Note 35.3* (Source 13, Proposition 5.1, "Limit-prefix character"). *Suppose  $p$  has nonzero limit length  $\lambda$ . In either  $\mathcal{T}$  or  $\mathcal{I}$  when  $p$  is injective, the proper prefixes of  $p$  are cofinal below  $p$ . The cofinality of the strict lower side at  $p$  is exactly  $\text{cf}(\lambda)$ .* This is [theorem 10.1](#) (1), with the same proof.

*Source 13's proof.* If  $q < p$ , then either  $q$  is a proper prefix of  $p$ , or its comparison with  $p$  is decided by a smaller letter at some coordinate  $\xi < \lambda$ . In either case a sufficiently long proper prefix of  $p$  lies above  $q$  and below  $p$ . Restricting to a cofinal sequence of lengths gives the upper bound  $\text{cf}(\lambda)$ .

For the converse, assign to each  $q < p$  a bound  $\xi_q < \lambda$  beyond which proper prefixes of  $p$  lie above  $q$ . A family of cardinality less than  $\text{cf}(\lambda)$  has a common such bound. A later proper prefix then witnesses its failure to be cofinal.  $\square$

Every point of **No** has no set-sized cofinal family strictly below it: a proposed family together with that point is a set cut and has a strict interpolant. Thus the displayed word orders are not isomorphic to **No**, even after the empty minimum is deleted. They nevertheless embed into **No** by class universality ([proposition 5.13](#)), and **No** embeds back through singleton words. The earlier report gives explicit sign-coding embeddings and a more detailed end-first analysis ([sections 9 to 11](#); Sections 9–11 of this report at the pin).

The same limit-prefix proof applies to numerically increasing set words, since every proper prefix remains increasing. By reversing the position of termination, one obtains the corresponding small *right-hand* character. [Section 36.4](#) proves that a central termination convention has a fundamentally different outcome.

### 35.3 Source 18: set-length dictionary words and numerical increase

Let  $\mathcal{T}$  be the class of all functions  $f : \alpha \rightarrow \mathbf{No}$  with  $\alpha$  a set ordinal, and let  $\mathcal{I}$  and  $\mathcal{T}^{\text{inc}}$  be its injective and strictly numerically increasing subclasses. These are ordinary classes: their members are set graphs. In  $\mathcal{I}$  a point represents a well-order of its own range; in  $\mathcal{T}^{\text{inc}}$  that well-order is the inherited numerical order. No point exhausts **No**.

First use ordinary dictionary comparison: compare first unequal entries, and put a proper prefix before every proper extension. This convention is sometimes described by a formal terminator below all surreal letters. Such a terminator is not a surreal number.

*Note 35.4* (Source 18, Proposition 6.1, “Explicit order coding; repository result”). *There is an explicit order embedding  $\mathcal{T} \hookrightarrow \mathbf{No}$ , requiring no Global Choice. Singleton words embed **No** into each of  $\mathcal{T}, \mathcal{I}, \mathcal{T}^{\text{inc}}$ . Thus all three word orders and **No** are equimorphic, but the word orders are not isomorphic to **No**.* This is [theorem 9.1](#), with the same letter code  $K(x) = C(x) = (+) \cap D(s_x) \cap (+, -)$ ; source 18 labels it a repository result.

*Source 18's proof.* Duplicate each sign in the sign expansion  $s_x$  and write the result as  $D(s_x)$ . The code

$$K(x) = (+) \cap D(s_x) \cap (+, -)$$

is prefix-free and numerically order-preserving. A terminated source word is correctly compared with a longer one because its delimiter satisfies  $(-, -) < (+, -) < (+, +)$ . Concatenate the  $K(f(i))$  in ordinal domain order. Replacement and ordinal summation give a set sign word. At the first unequal entry prefix-freeness preserves the comparison; at the end of a proper prefix the next block begins with  $+$ , so the shorter word is smaller. The code length is exactly

$$\sum_{i < \alpha} (1 + 2 \cdot \text{bd}(f(i)) + 2).$$

This proves the embedding. The nonisomorphism follows from the next proposition and holds even after removing the empty word.  $\square$

*Note 35.5* (Source 18, Proposition 6.2, “The prefix-supremum obstruction”). *The empty word is the minimum of each dictionary word order. For any word  $f$  of nonzero limit length  $\alpha$ , its proper prefixes are cofinal below it, and the left character of  $f$  is  $\text{cf } \alpha$ . At every successor-length word the left segment has no set cofinal subset. At every word the right segment has no set coinital subset. These assertions hold in  $\mathcal{T}$ ,  $\mathcal{I}$ , and  $\mathcal{T}^{\text{inc}}$ . For  $\mathcal{T}$  and  $\mathcal{I}$  this is [theorem 10.1](#). The case  $\mathcal{T}^{\text{inc}}$  also follows in GBC from [theorem 35.1](#), which preserves lengths and prefixes; source 18’s proof below needs no global choice.*

*Source 18’s proof.* At a limit length, a smaller word is either a proper prefix or differs at a coordinate below  $\alpha$ . A sufficiently long proper prefix exceeds it. A cofinal set of prefix lengths therefore gives the upper bound of  $\text{cf } \alpha$ ; fewer than that many comparison coordinates are bounded and cannot be cofinal. At successor length, interpolate at the last entry above all relevant lower next values and below that entry. In the injective case avoid earlier labels; in the increasing case also stay above all earlier labels. These are set constraints and the surreal cut property satisfies them. To defeat a proposed set coinital family above  $f$ , append a new value below all of its relevant next values. In the increasing case impose the additional lower bound  $\text{ran } f$ . Every family member extending  $f$  already satisfies that bound, so a separator still exists.  $\square$

In particular, an increasing word of length  $\omega$  is the supremum of its finite prefixes. No surreal number is the supremum of a set strictly below it, since the surreal set-cut property inserts a smaller upper bound. This invariant distinguishes the word orders from **No**; mutual embeddings do not preserve suprema in the ambient order.

The repository proves a sharper dictionary cut criterion for arbitrary and injective words ([theorem 11.1](#)): a set cut  $A < B$  fails to have an interpolant exactly when  $B$  contains the empty word, or when  $B$  has a minimum of nonzero limit length and  $A$  is cofinal below that minimum. A common-prefix search proves this: unequal next-value sets can be separated, while a forced common entry is appended; the search is bounded by the input word lengths. Its only terminal failure is a forced upper minimum reached at a limit. Restricting to nonzero successor-length words removes both failures and gives a copy of **No** in GBC. The prefix-tree isomorphism proved below transfers this criterion to increasing dictionary words as well.

The following sections give the genuinely different outcomes obtained by placing termination near zero. They treat zero exclusion, zero padding, and a distinct terminator separately; these choices cannot be conflated.

### 35.4 Source 14: an explicit word embedding with a termination convention

[merge] The first subsection of source 14’s Section 11; it gives a second explicit code for the prefix-first word order (a second route to [theorem 9.1](#)) and a code for the prefix-last order. The rest of that section (binary class words, the no-coding theorem and the Kanovei–Shelah remark) is not printed in this Part; its relation-code subsection is in [section 38.8](#). The ordinary class of all set-length surreal words is much smaller in logical sort than the virtual collection  $\mathfrak{W}_{\text{sl}}$ . An explicit coding makes its embeddability into **No** transparent and illustrates why the terminal symbol matters. The underlying compression idea is inherited from the predecessor ([section 9](#); Section 9 of this report at the pin); we spell out a version that also permits repeated letters.

For a surreal  $x$ , let  $c(x)$  be its binary sign code: replace each minus by 00, each plus by 11, and append 01. The three blocks encode  $- < \text{end} < +$ . Different complete codes are prefix-incomparable, and their first-difference order is the numerical order of their labels. For a word  $p : \lambda \rightarrow \mathbf{No}$ , concatenate the codes  $c(p(\alpha))$  in ordinal order. Now replace each binary 0 by 01 and

each binary 1 by 11, and append 00 to terminate the entire word. Finally read 0, 1 as surreal signs  $-$ ,  $+$ .

**Proposition 35.6** (Source 14, Proposition 11.1, “Prefix-first word coding”). *This construction is an order embedding of the class of all set-length surreal words, with prefix-first lexicography, into  $\mathbf{No}$ . Its output birthday is*

$$\sum_{\alpha < \lambda} (4 \cdot \text{bd}(p(\alpha)) + 4) + 2,$$

where every sum and multiplication is ordinal arithmetic.

*Proof.* All lengths and concatenations are sets. At the first differing label, the prefix-free numerical codes determine the correct binary comparison; the outer replacement preserves it. If one whole word is a proper prefix of the other, its outer terminal block 00 is compared with one of the data blocks 01, 11, making the shorter word smaller. Complete outer codes are prefix-incomparable, so their comparison as surreal sign sequences never reaches an ambiguity about termination. The length of a label code is  $2 \cdot \text{bd}(x) + 2$  and outer replacement doubles that length on the left. Summing those lengths and adding the last two signs gives the displayed formula.  $\square$

For prefix-last order, use outer data blocks 00, 10 and terminal block 11 instead. The same proof applies. The embeddings do not claim that these word orders are isomorphic to the entire surreal line: for example the empty word is the least point in the prefix-first order. Nor do they turn an Ord-length class word into a set-length surreal. The birthday formula is explicitly a *set* ordinal expression.

### 35.5 Source 18’s route: shorter-first increasing words are not a new order type in GBC

[merge] The first part of source 18’s Section 10: a second proof of [theorem 35.1](#), written without knowledge of source 16. Its proof makes explicit that the family of local isomorphisms is one class given by one elementary formula from a set-like global well-order, and its statement adds that zero may be excluded from any of the alphabets. The corollary of that section, on cones of centered increasing words, is printed with them ([corollary 37.9](#)).

The preceding centered results should not be transferred to the shorter-first convention. For shorter-first words a stronger identification is possible, despite the constraints on coordinates.

**Theorem 35.7** (A prefix-preserving tree isomorphism). *In GBC, the classes of arbitrary, injective, and strictly increasing set-length surreal words, all with shorter prefixes first, are pairwise order-isomorphic. Isomorphisms can be chosen to preserve every word length and the prefix relation. The conclusion also holds if zero is excluded from any of the alphabets.*

*Proof.* For a permitted target prefix  $p$ , its admissible next-letter class is respectively

$$\mathbf{No}, \quad \mathbf{No} \setminus \text{ran}(p), \quad D_p = \{x \in \mathbf{No} : \text{ran}(p) < x\}.$$

If zero is excluded, delete it as well. Each class is proper and fills every set cut. For the increasing case, simply add  $\text{ran}(p)$  to the lower side of the proposed cut; every upper-side letter already exceeds that range. For the injective case use set avoidance.

Fix a set-like global well-order  $\prec$ . There is a single uniformly defined family of order isomorphisms

$$\theta_p : \mathbf{No} \longrightarrow D_p,$$

where  $D_p$  now denotes the relevant admissible class in any of the cases. Here is a construction that makes the uniformity and the choice strength explicit. Restrict  $\prec$  to  $\mathbf{No}$  and to  $D_p$ ; these give ordinal enumerations, definably in the set parameter  $p$ . Run the deterministic back-and-forth that, at each stage, treats the least unmatched source point and then the least unmatched target point, always choosing the  $\prec$ -least unused filler of the required cut. Every partial map and every cut at an ordinal stage is a set. For each set ordinal, ordinary set recursion defines the unique initial construction, with  $p$  and  $\prec$  as parameters. The assertion that a pair occurs in some such initial construction is one elementary formula with these parameters. Elementary class comprehension therefore gives all the  $\theta_p$ 's as one class relation.

In particular, we have not selected an arbitrary class isomorphism independently for each of a proper class of prefixes. No extra class-choice principle, and no recursion with proper-class-valued stages, has been used.

For an arbitrary source word  $f : \alpha \rightarrow \mathbf{No}$ , construct a target word by set recursion. Given its prefix  $p$  at position  $\xi$ , put the next target entry equal to  $\theta_p(f(\xi))$ ; take unions at limits. The output is a permitted word of the same length. Conversely, from any permitted target word  $g$  recover

$$f(\xi) = \theta_{g|_\xi}^{-1}(g(\xi)).$$

These procedures are inverse. They preserve prefixes, and the first unequal entries are transformed by the same increasing map  $\theta_p$ . Thus they preserve the shorter-first lexicographic order.  $\square$

This theorem sharpens equimorphism: under the displayed choice assumption the three shorter-first word orders are actually the same order type. It does not identify that type with  $\mathbf{No}$ . Their positive-length limit prefixes still have the familiar small left cofinalities and nontrivial set suprema.

### 36 Termination conventions: the criterion and all isomorphism types (source 16; routes of sources 13 and 18)

[merge] Source 16's Section 8, printed whole, as the base of this section: it decides, for every placement of a formal terminator in the alphabet  $\mathbf{No}$ , whether the unrestricted and the injective word orders fill every set cut ([theorem 36.1](#)), and lists every isomorphism type that occurs ([theorem 36.2](#)). Its termination types  $\mathcal{T}_{\min}$ ,  $T_\kappa$  are printed  $\mathbf{T}_{\min}$ ,  $\mathbf{T}_\kappa$ ; its references to its own universality proposition and set-cut display are printed as references to [proposition 5.13](#) and [definition 5.1](#). Source 13 ([section 36.4](#)) and source 18 ([section 36.5](#)) prove the central case independently, with the same letter code ([note 36.3](#)).

Choose an initial/final partition  $\mathbf{No} = L_\varepsilon + R_\varepsilon$  and adjoin a formal symbol  $\varepsilon$ , distinct from every surreal, with

$$L_\varepsilon < \varepsilon < R_\varepsilon.$$

Compare words at the first differing coordinate, treating a terminated word as having the symbol  $\varepsilon$  there. This includes prefix-first and prefix-last as the two extreme placements. The Conway convention for sign expansions,  $- < \varepsilon < +$ , uses a two-letter alphabet; it must not be confused with prefix-first dictionary order over all surreals.

**Theorem 36.1** (Termination criterion). *Let  $\mathcal{V}$  be either  $\mathcal{T}$  or  $\mathcal{I}$ . Its termination order has a strict interpolant for every separated pair of sets, including empty sides, if and only if:*

1.  $L_\varepsilon$  and  $R_\varepsilon$  are nonempty;
2.  $L_\varepsilon$  has a greatest member or has no set cofinal subset;



3.  $R_\varepsilon$  has a least member or has no set coinital subset.

The equivalence uses GB and the surreal set-cut property. In GBC these conditions are also equivalent to this word order being isomorphic to **No**.

*Proof.* If either side of the threshold is empty, the empty word is an endpoint. If  $L_\varepsilon$  has no maximum but a set cofinal subset  $C$ , the singleton words  $(c)$ ,  $c \in C$ , lie below the empty word and have no strict interpolant below it. Any proposed interpolant has a first letter  $x \in L_\varepsilon$ , and some  $c \in C$  is larger than  $x$ , placing  $(c)$  above it. The right-side argument is dual. This proves necessity.

For sufficiency, start with a separated set cut  $A < B$  of words. Follow a common prefix  $p$ , keeping active precisely the input words extending it, including equality. Every discarded lower word is below every extension of  $p$ , and every discarded upper word is above every such extension. At the next coordinate let  $U, V$  be the sets of active lower and upper symbols, using  $\varepsilon$  for a word ending at  $p$ . Every  $u \in U$  is at most every  $v \in V$ .

If  $U$  and  $V$  meet, their common member is unique. It cannot be  $\varepsilon$ , which would put the same word  $p$  on both sides. Append the common surreal letter and continue. In the injective case a witness among the active words ensures freshness. Take unions at limits. The search cannot reach an ordinal larger than every input word length, so eventually  $U < V$  are disjoint.

If neither set contains  $\varepsilon$ , choose a fresh surreal strictly between them and append it. Suppose  $\varepsilon \in U$ , so  $V \subseteq R_\varepsilon$ . If a numerical member of  $U$  already lies in  $R_\varepsilon$ , ordinary fresh separation works. Otherwise all numerical lower symbols lie in  $L_\varepsilon$ . If  $R_\varepsilon$  has no set coinital subset, choose  $r \in R_\varepsilon$  below the whole set  $V$ , and then a fresh letter strictly between  $r$  and  $V$ . If  $R_\varepsilon$  has a least member  $r$  not in  $V$ , the same interval above  $r$  works.

The remaining case is that  $r = \min R_\varepsilon$  belongs to  $V$ . Append  $r$ , which is fresh in the injective case because an active upper word supplies it. All lower comparisons are now decided. For the upper words continuing through  $r$ , collect their next numerical letters and choose a fresh  $d$  below this set and below some fixed member of  $L_\varepsilon$ . The extension  $p \frown (r, d)$  is below all those upper words, including any terminating just after  $r$ , and above every lower word. Thus two letters handle the possible adjacency between the termination symbol and a numerical letter. The case  $\varepsilon \in V$  is the reversed construction using  $\max L_\varepsilon$  when needed.

This proves interpolation using only a bounded set recursion. The word carrier is proper, since it contains all singleton surreals. In GBC, class universality ([proposition 5.13](#)) gives the claimed isomorphism with **No**.  $\square$

### 36.1 Concrete consequences and explicit encodings

Write  $0^-$  for the formal symbol immediately below numerical zero, and  $0^+$  for the one immediately above zero. All labels, including zero, remain available. For  $0^-$  the sides are  $x < 0$  and  $x \geq 0$ ; for  $0^+$  they are  $x \leq 0$  and  $x > 0$ . [Theorem 36.1](#) gives

$$\mathcal{T}_{0^-} \cong \mathcal{I}_{0^-} \cong \mathbf{No}, \quad \mathcal{T}_{0^+} \cong \mathcal{I}_{0^+} \cong \mathbf{No} \quad \text{in GBC.} \quad (\text{S16.6})$$

This includes the empty word and all limit-length words.

There are also explicit embeddings that do not use global choice. For a sign sequence  $s_x$ , let  $D(s_x)$  duplicate each sign, and put

$$C_-(x) = D(s_x) \frown (+, -), \quad C_+(x) = D(s_x) \frown (-, +).$$

Both letter codes are prefix-free and preserve numerical comparison, because either terminal block lies strictly between  $(-, -)$  and  $(+, +)$ . The first sign of  $C_-(x)$  is negative exactly when  $x < 0$ , and

that of  $C_+(x)$  exactly when  $x \leq 0$ . Concatenating the codes of the entries therefore embeds the corresponding termination order into **No**. The encoded birthday is

$$\sum_{i < \alpha} (2 \cdot \text{bd}(f(i)) + 2). \quad (\text{S16.7})$$

The operations are ordinal operations; the 2 is on the left. At regular infinite birthday cutoffs, words of length below the cutoff whose entries are below it remain below it under this encoding. Singular cutoffs do not satisfy that closure assertion.

For prefix-first words, add an initial + to each letter code. This makes every continued block begin above termination and again gives an explicit embedding into **No**. Singleton words embed **No** back. Thus equimorphism persists even where local characters prevent isomorphism.

An interior threshold alone does not guarantee saturation. Place it between the surreals bounded above by some natural number and those larger than every natural number. The singleton natural-number words are cofinal below the empty word, so the termination order fails the criterion. Furthermore, (S16.6) does not extend to increasing words: a positive increasing word of length  $\omega$  still has its finite prefixes cofinal below it. Its permissible continuation letters are all positive. Hence the centered order on  $\mathcal{T}^{\text{inc}}$  is not  $\eta_{\text{Ord}}$ . The isomorphism of Theorem 35.1 specifically uses an endpoint termination convention.

### 36.2 Complete isomorphism classification of the termination family

There is a stronger classification for unrestricted and injective words. For an infinite regular cardinal  $\kappa$ , place the formal terminator at the threshold

$$L_\kappa = \{x : \exists \alpha < \kappa (x < \alpha)\}, \quad R_\kappa = \{x : \forall \alpha < \kappa (\alpha < x)\},$$

and call the resulting unrestricted word order  $\mathbf{T}_\kappa$ . The two classes partition **No**: the regular infinite cardinal  $\kappa$  is a limit ordinal. Its left side has cofinality  $\kappa$ , while its right side has no set coinital family, since the set of ordinals below  $\kappa$  can be separated from any proposed such family by the set-cut property (definition 5.1).

**Theorem 36.2** (All termination types). *In GBC, the isomorphism types obtained from every formal termination cut, for either unrestricted or injective set words, are exactly*

$$\mathbf{T}_{\min}, \quad \mathbf{T}_{\min}^{\text{op}}, \quad \mathbf{No}, \quad \mathbf{T}_\kappa, \quad \mathbf{T}_\kappa^{\text{op}} \quad (\kappa \text{ infinite regular}). \quad (\text{S16.8})$$

*There are no repetitions in this list. Here  $\mathbf{T}_{\min}$  uses shorter-prefix-first comparison. At every point of  $\mathbf{T}_\kappa$  the left cofinality is  $\kappa$  and there is no set coinital family on the right; its dual has the reversed profile.*

*Proof.* The endpoint placements give the first two types, by Theorem 35.1 and numerical negation. For an interior placement, Theorem 36.1 gives **No** unless one threshold side has no endpoint and a set cofinal or coinital family. Both sides cannot fail this way: the two sets would contradict the set-cut property (definition 5.1). The one small character is an infinite regular cardinal, say  $\kappa$ .

For a maximum-free left threshold of cofinality  $\kappa$ , extensions  $p \frown (c)$  with  $c$  in a cofinal subset of the left threshold are cofinal below any word  $p$ . In the injective case one may move these letters through intervening nonempty intervals to avoid  $\text{ran}(p)$ . Conversely, the next letters of a set cofinal below  $p$ , among its extending members, must be cofinal in that threshold. This proves the left character  $\kappa$ . The right threshold has no set coinital family, and the same next-letter argument excludes one above  $p$ . If a threshold side has an endpoint instead, two-letter extensions as in

**Theorem 36.1** exclude small local character on that side. These observations also distinguish the profiles in the list.

Two maximum-free threshold cuts with left cofinality  $\kappa$  are conjugate by a numerical automorphism of **No**. Choose increasing cofinal sequences of length  $\kappa$  on their left sides, match them, and extend this set partial isomorphism by back-and-forth. Each left side is determined as the initial closure of its sequence, so the automorphism carries the whole threshold cut to the other one. Applying it entrywise identifies the unrestricted word orders, and likewise the injective ones. The right-sided case is dual.

To identify unrestricted and injective words in this remaining case, fix the cofinal left sequence. For every set  $S$ , choose uniformly a cofinal sequence of the same type in  $L_\varepsilon \setminus S$ , using fresh points in intervening surreal intervals. Start back-and-forth with the matching of these two set sequences. It gives a uniform isomorphism  $h_S : \mathbf{No} \rightarrow \mathbf{No} \setminus S$  preserving both threshold sides. Taking  $h_\emptyset$  as the identity, the recursion

$$q(i) = h_{\text{ran}(q \upharpoonright i)}(f(i))$$

is a length- and prefix-preserving isomorphism from free words to injective words, now preserving comparison with the middle terminator as well. All preparatory sequences and stage histories are sets. The saturated cases were already identified with **No**.

Finally endpoints distinguish the first two types. The surreal type has no small character on either side of any point, whereas the  $\mathbf{T}_\kappa$  types have the displayed one-sided character. Those characters distinguish the different regular cardinals and their duals. This proves exhaustiveness and absence of repetitions.  $\square$

Every termination order in this list is nevertheless equimorphic with **No**, even without global choice. For an arbitrary threshold, the increasing map  $\phi(x) = x/(1 + |x|)$  can be rescaled separately on its two sides into  $(-1, 0)$  and  $(0, 1)$ . This preserves all letter comparisons and puts termination at zero. The duplicated-sign codes above then give an explicit word embedding into **No**; singleton words supply an embedding in the opposite direction. Thus equimorphism leaves a proper class of different isomorphism types undistinguished. This classification excludes numerically increasing words with an interior terminator, whose behavior depends on the prefix as explained above.

### 36.3 Length-first ordering is another independent choice

In shortlex order one compares the ordinal lengths first, and compares entries only when those lengths agree. It is the ordered sum of the fixed-length layers over **Ord**, rather than one of the termination orders above. Its empty word is a minimum, and two words can be ordered by their lengths even when they disagree numerically at their first entry. This variant is rigorously definable on set codes but should be named explicitly; none of the global-exhaustive results implicitly uses it.

*Note 36.3* ([merge]; one letter code, five spellings). With  $D(s)$  the sign sequence  $s$  with every sign doubled, the letter codes of the six sources and of Part II are the following; each is prefix-free and order-preserving because its terminal block lies strictly between  $(-, -)$  and  $(+, +)$ .

- $D(s_x) \cap (+, -)$ : source 16's  $C_-$  (S16.7), source 13's  $c(x)$  (proposition 36.5) and source 18's centered  $C(x)$  (proposition 36.7); birthday  $2 \cdot \text{bd}(x) + 2$  per letter, as in (S16.7). Source 16's  $C_+$  is  $D(s_x) \cap (-, +)$ .
- $(+) \cap D(s_x) \cap (+, -)$ : Part II's prefix-first code (theorem 9.1), source 16's “add an initial +” and source 18's  $K(x)$  (note 35.4); birthday  $1 + 2 \cdot \text{bd}(x) + 2$  per letter. Source 17's code (S17.1) is  $(+) \cap D(s_x) \cap (-, +)$ , with the same birthday.

- 00, 11, 01 for  $-$ ,  $+$  and termination, read as signs: source 14's inner code ([proposition 35.6](#) and its Subsection 2.3) and source 15's entry code ([lemma 31.1](#)); this is source 16's  $C_+$ . Source 14 doubles it once more for the outer word, which gives  $4 \cdot \text{bd}(x) + 4$  per letter and 2 for the end.

Source 16 does not cite Part II for its prefix-first code. No source claims that any of these codes is optimal; source 11's question on optimal birthday bounds ([research question 113.6](#)) stays open ([section 39](#)).

### 36.4 Source 13's route: central termination restores a surreal word order

[merge] Source 13's Section 12, printed whole as a route to the central case of [theorem 36.1](#). Its order  $\mathcal{T}_0$  is the termination order of the alphabet  $\mathbf{No} \setminus \{0\}$  with the terminator in the place of the reserved zero, i.e. with threshold sides the negative and the positive surreals; neither side has an endpoint or a set cofinal (resp. coinital) subset, so [theorem 36.1](#), transported along an isomorphism  $\mathbf{No} \setminus \{0\} \cong \mathbf{No}$  in GBC, gives [theorem 36.4](#). The proofs are the same common-prefix search. Its code ([proposition 36.5](#)) is source 16's code  $C_-$  (S16.7), with the same birthday. Source 13's own caveat stands: it does not classify all placements of a marker; source 16 does ([theorem 36.2](#)). Its references to its fresh-separator lemma, class universality and core interpolation are printed as references to [lemma 5.2](#), [proposition 5.13](#), and [theorem 19.10](#).

The end-first convention of [Section 35.2](#) treats every extension as larger than its prefix, irrespective of the sign of the new label. We now keep a genuine middle termination value. To avoid collapsing distinct words, that value must not also be an ordinary letter.

#### 36.4.1 The precise alternative

Let  $A = \mathbf{No} \setminus \{0\}$ . A word  $p : \alpha \rightarrow A$  is padded by the surreal number 0 at every coordinate at or beyond  $\alpha$ . Compare two different words at the first difference of these padded functions, using numerical comparison in  $\mathbf{No}$ . Call this *central termination*, and denote the arbitrary-word and injective-word orders by  $\mathcal{T}_0$  and  $\mathcal{I}_0$ , respectively.

For example,

$$(-1) < () < (1), \quad p \frown (-1) < p < p \frown (1),$$

whenever the displayed appended letters are allowed. The empty word is now an interior point, not an endpoint. For injective words, fresh negative and positive letters can always be used in place of  $-1, 1$ .

Distinct words really are distinguished by padding: at the end of a proper prefix, the longer word has a nonzero letter. If 0 were also allowed as a letter,  $p$  and  $p \frown (0)$  would have identical padded functions. The resulting rule would fail even to define an antisymmetric order on raw words. Reserving the marker is essential.

**Theorem 36.4** (Central-word isomorphism). *In GBC,*

$$\mathcal{T}_0 \cong \mathbf{No}, \quad \mathcal{I}_0 \cong \mathbf{No}.$$

*Both isomorphisms concern ordinary classes of set codes. They do not identify the word structure, concatenation, or birthdays with surreal arithmetic.*

*Proof.* It suffices to prove set-cut saturation. Let a set of lower words be strictly below a set of upper words. Choose an ordinal  $\delta$  strictly above all their lengths. We use an active-prefix argument similar to the interpolation theorem of the core (source 13's Theorem 7.2, [theorem 19.10](#)), but now an active word that ends exactly at the current prefix contributes the value 0.

At a current nonzero-letter prefix  $p$  of length  $\beta$ , an active bound agrees with  $p$  and has length at least  $\beta$ . Let  $L_\beta$  and  $U_\beta$  be its next letters, or 0 for a bound ending there. Every cross lower value is at most every upper value. If there is a shared value  $t$ , it is unique. It cannot be 0: a lower word and an upper word both ending at  $p$  would be identical, contradicting the strict cut. Append  $t$  and retain the active rows with next value  $t$ . In the injective case  $t$  is fresh, because it belongs to an active injective word extending  $p$ .

If no value is shared, choose a strict separator  $x$  between the two sets, avoiding 0, and also avoiding  $\text{ran}(p)$  in the injective case. The fresh-separator lemma (lemma 5.2) supplies it. The *terminated* word  $p \frown x$  now lies strictly between every original bound. Every comparison is already decided at this new coordinate or an earlier one, so its termination afterwards is immaterial.

At limits, take the union of the earlier prefixes. The procedure cannot continue indefinitely: after all input lengths have been passed, no bound can remain active. More specifically, a shared nonzero letter extends some active word and so cannot occur at a stage  $\beta \geq \delta$ . The construction stops at a set stage. It also works with an empty side or with both sides empty. This proves set-cut saturation of each order. Their elements are sets; apply class universality (proposition 5.13) to obtain the two isomorphisms.  $\square$

This answers the termination-convention direction for an important interior convention. It is not a classification of all possible placements of a marker in every class alphabet. By translating labels, one may reserve any specified surreal  $c$  instead of 0. Since  $\mathbf{No} \setminus \{0\} \cong \mathbf{No}$ , the construction can also be transported to an abstract alphabet order-isomorphic to the full surreal line. Such a transport is not the identity on numerical labels and should not be hidden in the notation.

### 36.4.2 An explicit sign-code embedding

The isomorphism proof uses class back-and-forth. There is also a direct order embedding, useful for calculations and finite tests.

Given a surreal letter  $x$  with sign sequence  $s_x$  of length  $\alpha$ , replace each  $-$  by  $--$  and each  $+$  by  $++$ , then append the pair  $+ -$ . Write the resulting sign block as

$$c(x) = \text{double}(s_x) \frown (+, -).$$

The family of these blocks is prefix-free. If two original sign sequences first differ by actual signs, their doubled versions decide the comparison in the same direction. If one sequence ends first, its terminal pair  $+ -$  lies lexicographically between the two possible continuation pairs  $--$  and  $++$ . Thus the same statement holds when the original comparison used termination.

**Proposition 36.5** (Direct central-word coding). *For a set word  $p : \lambda \rightarrow \mathbf{No} \setminus \{0\}$ , concatenate the blocks  $c(p(i))$  in order and regard the result as a surreal sign sequence. This defines an order embedding  $C : \mathcal{T}_0 \hookrightarrow \mathbf{No}$ , and hence also an embedding of  $\mathcal{I}_0$ , with*

$$\text{bd}(C(p)) = \sum_{i < \lambda} (2 \cdot \text{bd}(p(i)) + 2).$$

*All sums and products in this formula are ordinal operations.*

*Proof.* Every block has positive set-ordinal length and the concatenation has set-ordinal length, so it is a surreal. At a first unequal letter, prefix-freeness and order preservation of the block code decide the same comparison before either block is exhausted.

If one outer word is a proper prefix of the other, its code terminates at the beginning of the next block. Because the next letter is nonzero, the first sign of its block has the sign of that

letter. Comparison of termination with this sign has exactly the central termination direction. Thus all comparisons are strict and preserved. The birthday formula is the definition of ordinal concatenation.  $\square$

No claim of surjectivity is made for this explicit map. The order isomorphism in [Theorem 36.4](#) is a separate conclusion. The code is a central-termination adaptation of the prefix-free sign-block methods used in the earlier set-word analysis of Part II ([theorem 9.1](#)); the accompanying finite tests check it by independent dyadic evaluation.

### 36.4.3 Why the obstruction disappears

At a limit-length central word  $p$ , the proper prefixes are not uniformly below it: their comparison depends on the next nonzero letter of  $p$ . More importantly, a set of lower approximants can be separated from  $p$  by a fresh next value at the decisive active stage. The set-cut proof supplies this separator regardless of a particular sign pattern. Hence the small cofinality invariant of [Note 35.3](#) no longer obstructs an isomorphism with **No**.

The two constructions therefore illustrate a general warning: lexicographic comparison of variable-length words is not a minor technical variation of fixed-domain lexicography. A termination convention can change endpoints, point characters, and the entire isomorphism type.

## 36.5 Source 18's route: centered termination changes the order

[merge] Source 18's Section 7, printed whole as a third proof of the central case. Its classes  $\mathcal{C}$ ,  $\mathcal{C}_{\text{inj}}$ ,  $\mathcal{C}_{\text{inc}}$  are printed  $\mathcal{T}_c$ ,  $\mathcal{I}_c$ ,  $\mathcal{T}_c^{\text{inc}}$  ([section 1.3.1](#)). Its centered order is source 13's  $\mathcal{T}_0$  (the terminator compared as zero with the next entry), and its code  $C(x)$  is source 13's  $c(x)$  and source 16's  $C_-(x)$ . Nothing in it is new against sources 13 and 16 except the remark that singleton words also embed **No** into  $\mathcal{T}_c^{\text{inc}}$ , which the next section uses. Source 18 was written before sources 13–17 arrived.

For a set word  $p$ , write  $\ell(p)$  for its ordinal length and  $p \smallfrown \langle x \rangle$  for ordinal concatenation. The surreal set-cut property used throughout this section is the following: if  $L < R$  are sets of surreals, possibly empty, there is a surreal strictly between them. The interpolant can also be required to avoid any prescribed set. Indeed, first find a nonempty surreal interval between the two sides, and then use the set-cut property inside that interval to escape the forbidden set. No global choice is needed for these interpolation statements.

Put  $D = \mathbf{No} \setminus \{0\}$ . Let  $\mathcal{T}_c$  be all set-length words over  $D$ , and let  $\mathcal{I}_c$  and  $\mathcal{T}_c^{\text{inc}}$  be its injective and strictly numerically increasing subclasses. Include the empty word in all three classes. Compare unequal words at their first unequal entry. If one word is a proper prefix of the other, compare its termination symbol  $\dagger$  with the next entry of the longer word, using

$$x < \dagger \quad (x < 0), \quad \dagger < x \quad (x > 0).$$

Thus  $\dagger$  occupies the missing numerical position of zero. This is a strict linear order, called the centered order below. The definition is not the shorter-first convention.

**Theorem 36.6** (Centered arbitrary and injective words). *Both  $\mathcal{T}_c$  and  $\mathcal{I}_c$  fill every set cut. Consequently, in GBC, both are order-isomorphic to **No**. They have no endpoints; neither side of any point has a set cofinal or a set coinital subset, respectively.*

*Proof.* For a prefix  $p$ , its cone consists of all words extending  $p$ , including  $p$  itself. A cone is convex. A word outside it is uniformly below the whole cone or uniformly above the whole cone: the comparison has already been decided before  $p$  ends.



Given a set cut  $A < B$ , start with the empty prefix. Maintain the invariant that discarded members of  $A$  are below its whole cone, and discarded members of  $B$  are above its whole cone. At a current prefix  $p$  of length  $\delta$ , assign next tokens to the active words: an actual next entry to a proper extension, and the numerical token 0 to  $p$  itself. Let  $L_p, R_p$  be the two sets of tokens. Every member of  $L_p$  is at most every member of  $R_p$ . Their intersection therefore has at most one member.

If the token sets are disjoint, choose

$$L_p < c < R_p, \quad c \notin \{0\} \cup \text{ran}(p).$$

Then  $p \smallfrown \langle c \rangle$  interpolates the original cut. Avoiding the range is needed only for the injective version. If the token sets meet, their common token cannot be zero: zero would mean that the same word  $p$  belonged to both sides. Append their common nonzero entry and continue. The new prefix is injective when required, since it is an actual prefix of a witnessing injective word. At a limit stage take the union. The cone invariant persists.

Choose an ordinal strictly above every length occurring in  $A \cup B$ . The search cannot keep finding common actual entries that long. It therefore terminates with a successor-length interpolant. Empty sides cause no difficulty. Taking one side to be a singleton gives the asserted local character properties.

In GBC, restrict a fixed set-like global well-order to each of the two proper classes and to **No**. An ordinal back-and-forth constructs an isomorphism: at each stage its partial map is a set, and the new point is chosen in the cut imposed by that set. Freshness follows from the set-avoidance version of interpolation. This is a recursion with set-sized partial maps, not an appeal to choice for arbitrary families of proper classes.  $\square$

The contrast with shorter-first words is substantive. At a limit word, the centered convention still permits extensions on both sides. These extensions prevent the proper prefixes from creating a one-sided set-cofinal obstruction.

**Proposition 36.7** (An explicit centered sign embedding). *There is an explicit order embedding of  $\mathcal{T}_c$  into **No**, without global choice. Conversely, **No** embeds into each of  $\mathcal{T}_c$ ,  $\mathcal{I}_c$ , and  $\mathcal{T}_c^{\text{inc}}$ .*

*Proof.* For a nonzero surreal  $x$ , let  $s_x$  be its nonempty sign expansion. Duplicate each sign, writing  $D(s_x)$ , and put

$$C(x) = D(s_x) \smallfrown (+, -).$$

This is a prefix-free order-preserving code. When one original sign expansion ends, the delimiter  $(+, -)$  is compared with the next doubled block; the inequalities

$$(-, -) < (+, -) < (+, +)$$

give exactly the surreal comparison at termination. The first sign of  $C(x)$  is the sign of  $x$ . For a word  $f$ , concatenate the blocks  $C(f(i))$  in ordinal domain order. Replacement makes the resulting sign expansion a set. First disagreements are preserved by prefix freedom; proper-prefix comparisons are preserved by the first sign of the next block. This proves the first assertion. For the reverse embedding send 0 to the empty word and send  $x \neq 0$  to  $\langle x \rangle$ .  $\square$

## 37 Centered increasing words and the entry zero (source 18)

[merge] Source 18's Sections 8, 9 and 11 and the corollary of its Section 10, printed whole; no other source treats these orders. Source 16 observed the first obstruction (end of its Subsection 8.1: a

positive increasing word of length  $\omega$  is the supremum of its finite prefixes, so the centered order on  $\mathcal{T}^{\text{inc}}$  is not  $\eta_{\text{Ord}}$  and asked for the classification of increasing words with an interior terminator (its question “Centered increasing words”); source 18 gives it for the threshold at zero (theorem 37.2), and settles the three ways of treating an actual entry zero (propositions 37.4 and 37.5 and corollary 37.7). Its proofs use the common-prefix search of theorem 11.1 with the termination token evaluated as zero.

### 37.1 Centered increasing words: an exact cut classification

Strict numerical increase changes which continuations remain available. An all-negative prefix admits fresh negative and positive continuations. Once a positive entry has occurred, every valid continuation is positive. This distinction gives the entire obstruction below.

**Theorem 37.1** (Local characters of centered increasing words). *Let  $f \in \mathcal{T}_c^{\text{inc}}$  have length  $\alpha$ . If  $\alpha$  is a nonzero limit and  $f$  has a positive entry, then*

$$\text{cf}((\mathcal{T}_c^{\text{inc}})_{<f}) = \text{cf}(\alpha).$$

*At every other point the open initial segment has no set cofinal subset. At every point the open final segment has no set coinital subset. The order has no endpoints and is dense.*

*Proof.* If  $f$  has a positive entry and has limit length, its sufficiently long proper prefixes form an increasing cofinal family below it. Every smaller word either terminates before  $f$ , or first differs at a smaller entry before  $\alpha$ ; it cannot properly extend  $f$ , since all such extensions are larger. It is consequently below every sufficiently long prefix. A cofinal sequence of prefix lengths proves the upper bound  $\text{cf}(\alpha)$ . Conversely, for each smaller word choose a prefix exceeding it. Fewer than  $\text{cf}(\alpha)$  chosen lengths are bounded, so that many words cannot be cofinal.

Suppose next that every entry of  $f$  is negative; this includes the empty word. Given a set  $A < f$ , collect the next entries of those members of  $A$  that properly extend  $f$ . They are all negative. Choose a negative  $d$  above this set and above  $\text{ran}(f)$ . The set-cut property permits this even when the range of  $f$  has no largest member. Then

$$A < f \frown \langle d \rangle < f.$$

Members of  $A$  that do not extend  $f$  first differ earlier with a smaller entry. They are also below this new word.

If  $f = p \frown \langle x \rangle$  and  $x > 0$ , a smaller word sharing  $p$  either stops there or has next entry below  $x$ . For a set of such words, choose  $d < x$  above all their next entries, above zero, and above  $\text{ran}(p)$ . The word  $p \frown \langle d \rangle$  is a strictly larger lower bound for  $f$  than any member of the given set. This handles the remaining successor case.

Finally, given a set  $B > f$ , the next entries of members that extend  $f$  are positive. Choose a positive  $d$ , above the range of  $f$ , but below all those entries. Then

$$f < f \frown \langle d \rangle < B.$$

All other members of  $B$  are above the whole cone of  $f$ . This proves the right-hand assertion and, with empty sets, the absence of endpoints.

For density, if two words first differ, choose a new coordinate strictly between their different entries. It is automatically above the common prefix. If the comparison is a proper-prefix comparison with next entry  $x > 0$ , choose a coordinate strictly between the set  $\{0\} \cup \text{ran}(p)$  and  $x$ . For a next entry  $x < 0$ , choose one strictly between  $\{x\} \cup \text{ran}(p)$  and zero.  $\square$

**Theorem 37.2** (All set-cut obstructions). *A set cut  $A < B$  in  $\mathcal{T}_c^{\text{inc}}$  has no interpolant if and only if  $B$  has a minimum  $b$  such that*

1.  $\ell(b)$  is a nonzero limit ordinal;
2.  $b$  has a positive entry;
3.  $A$  is cofinal in the open initial segment below  $b$ .

*In every other case an interpolant of nonzero successor length can be chosen.*

*Proof.* The stated obstruction plainly prevents interpolation. For the converse use the common-prefix search from [Theorem 36.6](#), evaluating the termination token as zero. Common actual coordinates can still be appended, since each is witnessed by an increasing word.

At a stage where the next-token sets are disjoint, a permissible new coordinate must satisfy

$$L_p \cup \text{ran}(p) < c < R_p, \quad c \neq 0.$$

Every actual token on the right exceeds the whole range of  $p$ . Thus these are consistent strict surreal constraints unless zero is a right token and  $p$  already has a positive entry. When zero is a right token and all entries of  $p$  are negative, the constraints are still consistent and a negative interpolating coordinate can be chosen. Empty right sides are harmless.

The sole exceptional event is therefore reaching  $p \in B$  after  $p$  has acquired a positive entry. Every proper extension of  $p$  is larger, so the active lower side is then empty and  $p = \min B$ . This event cannot occur at a successor length: the last common coordinate had a witness in  $A$  extending that successor prefix, which would be at least  $p$ , contradicting  $A < B$ . It also cannot occur at length zero.

It can occur at a limit length  $\alpha$ . For every sufficiently large  $\xi < \alpha$ , a lower-side witness supplied the coordinate  $p(\xi)$ . That witness extends  $p \upharpoonright (\xi + 1)$ , is at least this prefix, and remains below  $p$ . The sufficiently long prefixes are cofinal below  $p$  by [Theorem 37.1](#); hence  $A$  is cofinal there. As before, a bound on all input word lengths ensures that the search terminates using only set recursion.  $\square$

**Corollary 37.3.** *The centered increasing-word order is equimorphic with  $\mathbf{No}$ , but is not isomorphic to  $\mathbf{No}$ , even in GBC. Its successor-length suborder fills every set cut and is isomorphic to  $\mathbf{No}$  in GBC. The larger suborder obtained by adjoining the empty word and every all-negative limit word also fills every set cut and has the same isomorphism conclusion.*

*Proof.* Use the explicit embeddings of [Proposition 36.7](#). The increasing word  $\langle 1, 2, 3, \dots \rangle$  is the supremum of its finite prefixes in this word order. Such a nontrivial set supremum cannot occur in  $\mathbf{No}$ , by its set-cut property. For either stated suborder, an upper set cannot contain the exceptional minimum in [Theorem 37.2](#), and the proof returns a successor word in the suborder itself. Apply class back-and-forth in GBC.  $\square$

Every nonempty open interval in the centered increasing-word order contains an order copy of  $\mathbf{No}$ . The density proof can choose a whole surreal interval of permissible new coordinates, rather than one coordinate. This supplies the copy. Thus rich interval universality coexists with the precise limit-word obstructions.

### 37.2 What if the entry zero is allowed?

Saying only that termination lies “between negatives and positives” does not settle its relation to an actual entry equal to zero. There are at least three different mathematical constructions.

### 37.2.1 Zero padding and its quotient

One may pad each set word by zeros at every later ordinal coordinate. First-difference comparison then gives a total preorder, not a strict linear order on the original words. For instance the empty word and  $\langle 0 \rangle$  become equal. In general put

$$\sigma(f) = \sup\{\xi + 1 : \xi < \ell(f), f(\xi) \neq 0\}, \quad \rho(f) = f \upharpoonright \sigma(f).$$

Two padded words are equivalent exactly when their reduced words  $\rho(f)$  agree. A class of canonical set representatives is therefore available: trim the entire terminal zero tail. A limit-length reduced word has nonzero entries cofinal in its length. Reduction preserves injectivity and strict numerical increase.

**Proposition 37.4** (The zero-padding quotient). *For all set words, and separately for injective set words, the zero-padding quotient fills every set cut and is isomorphic to **No** in GBC. For strictly increasing set words, the quotient has exactly the cut obstruction in [Theorem 37.2](#), applied to the reduced representatives. Its exceptional points are precisely the reduced limit words having a positive entry.*

*Proof.* Work with reduced representatives. A set of input representatives has lengths bounded by a set ordinal  $\mu$ . Pad them by zeros through coordinate  $\mu$  and perform the numerical common-prefix search. This search now allows a common coordinate equal to zero.

In the injective case, the prefix produced by the search remains injective. A common nonzero coordinate is an actual coordinate of every witness. If zero is common, at least one witness has an actual zero at that coordinate: otherwise a witness on each side would have terminated already and, agreeing at all previous coordinates, would represent the same padded word. An actual witness certifies that the enlarged prefix is injective. At a separation step choose a nonzero coordinate outside the range of the prefix, using the surreal set-cut property. This returns a reduced injective word. At coordinate  $\mu$ , two distinct input classes cannot both remain active, so termination is guaranteed. This proves the first claim.

For increasing words the same witness argument makes every common prefix strictly increasing. At a separating step add its range to the lower numerical constraints. The only potential failure is an upper padded-zero token together with a nonnegative entry in the current prefix.

If the prefix has a positive entry, an increasing extension cannot consist of an extra padded zero. The active upper word must therefore be exactly this reduced prefix. As in the proof of [Theorem 37.2](#), a successor stopping stage contradicts the preceding lower-side witness, while a limit stopping stage gives the stated cofinal-prefix obstruction.

The remaining apparent failure would be a prefix  $p = q \smallfrown \langle 0 \rangle$ , where  $q$  is all-negative, and an active upper representative equal to  $q$ . This stage cannot be reached. The common zero that produced it would require a lower witness. If that witness has terminated, it represents  $q$  itself; if it has an actual zero, its reduced increasing continuation has a positive entry and lies above  $q$ . Both possibilities contradict the cut. There are no further failures. The converse obstruction is again supplied by cofinal proper prefixes of a positive limit word.  $\square$

For labeled well-orders the quotient is a change of subject that must be stated explicitly. It identifies distinct codes with different underlying sets: an all-negative increasing word and that word followed by zero are different labeled well-orders, although the padding convention makes them equivalent.

### 37.2.2 A distinct terminator immediately below or above zero

Alternatively, retain every word as a distinct object and use one of the two enlarged alphabets

$$\mathbf{No}_{<0} < \dagger_- < 0 < \mathbf{No}_{>0}, \quad \mathbf{No}_{<0} < 0 < \dagger_+ < \mathbf{No}_{>0}.$$

The indicated neighboring symbols are adjacent in the enlarged alphabet. These are genuine linear orders on the original words.

**Proposition 37.5.** *For arbitrary set words, and separately for injective set words, both distinct-terminator conventions fill every set cut. Each resulting class order is isomorphic to  $\mathbf{No}$  in GBC.*

*Proof.* Use the common-prefix search with tokens in the enlarged alphabet. At a strict token separation an unused surreal can normally be inserted. The only exception is its displayed adjacent pair.

For  $\dagger_- < 0$ , the exceptional separation has the current prefix  $p$  on the lower side, and upper active words with next entry zero. No lower word has next entry zero, since that would have caused a common-coordinate step. Append zero and then append a sufficiently small negative, unused coordinate. The resulting word is above the original lower side and below every upper word in the zero cone, including its root if that root occurs. For  $0 < \dagger_+$ , use the reverse construction: append zero and then a sufficiently large positive, unused coordinate. The existence of a zero cone certifies that zero has not already been used when injectivity is required. All choices avoid the remaining set of used entries. This resolves the only additional cases, and the length bound still makes the search a set recursion.  $\square$

Increasing words behave differently. The following general fact isolates the reason, independently of any surreal coding.

**Lemma 37.6** (Adjacency test for increasing words). *Let  $D$  be a linearly ordered alphabet with a distinct termination symbol  $\dagger$ . Suppose every allowed increasing set word has a further admissible entry above  $\dagger$ . In the corresponding order of all increasing set words,  $f < g$  are adjacent if and only if*

$$g = f \smallfrown \langle c \rangle, \quad c = \min\{x \in D : \text{ran}(f) < x, \dagger < x\}.$$

*The minimum in this assertion must exist.*

*Proof.* If  $f$  is not a prefix of  $g$ , append a positive-side admissible entry to  $f$ . The new word is above  $f$ , and its comparison with  $g$  was already settled before the appended entry; hence it remains below  $g$ . Such a pair is not adjacent.

If  $f$  is a proper prefix of  $g$ , its first new entry  $c$  must be above  $\dagger$ . All subsequent entries exceed  $c$ , so if  $g$  has any further entries, the one-letter extension  $f \smallfrown \langle c \rangle$  lies strictly between. For this one-letter extension to be adjacent, no admissible entry strictly between  $\dagger$  and  $c$  can exist. Conversely, if  $c$  is the stated minimum, every word in between would have to extend  $f$ ; its first new entry could neither be below  $c$  nor above it, and a proper continuation after  $c$  would be larger than  $g$ . No such word exists.  $\square$

**Corollary 37.7** (Exactly where the zero jump appears). *For increasing surreal words with  $\dagger_-$ , the adjacent pairs are exactly*

$$p < p \smallfrown \langle 0 \rangle \quad \text{with every entry of } p \text{ negative.}$$

*The empty prefix is allowed. With  $\dagger_+$ , or with zero excluded and centered termination, there are no adjacent pairs.*

*Proof.* Every set of surreal entries has a strictly larger positive surreal, so the lemma applies. With  $\dagger_-$ , an all-negative prefix has zero as its least positive-side admissible entry. Once a nonnegative entry is present, the admissible upper class has no least member, by the surreal set-cut property. With  $\dagger_+$ , its positive-side admissible class consists of positive surreals above the prefix and has no least member. The same argument applies when zero is excluded.  $\square$

Thus a convention that differs only in the placement of actual zero can create adjacent pairs even among finite increasing words. The absence of an eligible letter between termination and the zero entry is decisive: increasing words cannot use the negative two-letter detour from [Proposition 37.5](#).

For completeness, the same common-prefix argument gives the full set-cut classification for increasing words with the distinct terminators. With  $\dagger_+$ , only the positive limit-word obstruction of [Theorem 37.2](#) remains. With  $\dagger_-$ , one must additionally allow the adjacent-pair obstruction

$$\max A = p, \quad \min B = p \smallfrown \langle 0 \rangle, \quad \text{ran}(p) < 0.$$

Indeed, at the exceptional token gap the word  $p \smallfrown \langle 0 \rangle$  itself interpolates unless it is the upper minimum and  $p$  is the lower maximum. All other separating-coordinate steps work as before. A positive limit stopping stage is unchanged. For  $\dagger_+$ , the zero-then-positive detour is increasing and always resolves the adjacent token gap.

*Note 37.8* ([merge]; two of the zero conventions are source 16's). For arbitrary and injective words, the conventions  $\dagger_-$  and  $\dagger_+$  of [proposition 37.5](#) are source 16's  $0^-$  and  $0^+$ , and that proposition is (S16.6), a case of [theorem 36.1](#); both proofs resolve the adjacent pair by the same two-letter detour. New in source 18 are the zero-padding quotient ([proposition 37.4](#)) and, for increasing words, the exact adjacent pairs ([lemma 37.6](#) and [corollary 37.7](#)).

### 37.3 Cones of centered increasing words (source 18, Corollary 10.2)

**Corollary 37.9** (Two kinds of centered increasing cones). *Work in GBC, with zero excluded and centered termination. The cone above an all-negative increasing prefix  $p$ , including its root  $p$ , is isomorphic to the entire centered increasing-word order  $\mathcal{T}_c^{\text{inc}}$ . If  $p$  has a positive entry, its cone is isomorphic to the shorter-first arbitrary-word order, including that order's minimum. These isomorphisms can preserve the rooted prefix tree and the lengths of tails after the root.*

*Proof.* At an all-negative prefix, the two admissible classes

$$\{x : \text{ran}(p) < x < 0\}, \quad \{x : 0 < x\}$$

are both proper and fill every set cut relative to themselves. Once a positive coordinate is chosen, the negative admissible class disappears, and the positive admissible class continues to fill all set cuts. Use the uniform construction in the proof of [Theorem 35.7](#) separately on the negative and positive parts whenever both are present. The source and target therefore pass from the all-negative case to the positive case at the same coordinate, and both entry and termination comparisons are preserved. This proves the first assertion. In the positive case every later entry is above termination, so the rooted cone uses precisely shorter-first comparison. The one-part version of the same construction proves the second assertion.  $\square$

### 37.4 Regular set stages

The cut arguments also give a set-sized version. Let  $\kappa$  be regular infinite, restrict entries to  $\mathbf{No}_{<\kappa}$ , and restrict word lengths to be below  $\kappa$ . With zero excluded and centered termination, the arbitrary



and injective word orders are  $\eta_\kappa$ . Their increasing-word successor suborders are  $\eta_\kappa$  as well. The proof of every relevant cut statement uses fewer than  $\kappa$  input words, has its prefix length bounded below  $\kappa$ , and adds fewer than  $\kappa$  coordinate constraints or forbidden entries. Regularity and the  $\eta_\kappa$  property of  $\mathbf{No}_{<\kappa}$  supply the last coordinate. If  $2^{<\kappa} = \kappa$ , these orders have cardinality  $\kappa$  and are isomorphic to  $\mathbf{No}_{<\kappa}$  by ordinary set back-and-forth.

The complete increasing-word order retains every positive limit-length obstruction that is present below the cutoff. When  $\kappa = \omega$  no nonzero limit word length is allowed, so this particular obstruction disappears. These statements should not be extrapolated to singular cutoffs or to unrestricted word lengths over a fixed set alphabet: a prefix can then exhaust the available one-sided room. For example, over the nonzero dyadic rationals, an increasing negative word  $\langle -1, -1/2, -1/4, \dots \rangle$  has no further negative entry above its whole range, whereas every set of negative surreals does have such a continuation in the full surreal class.

*Note 37.10* ([merge]; regular stages in Part II). For shorter-first words the regular-stage statements are Part II's [section 12](#) and [proposition 9.2](#) (1); source 18's paragraph gives the centered analogues. Its warning about singular cutoffs is the one printed after [proposition 9.2](#).

## 38 Other comparison orders (source 15; remarks of sources 13, 14, 16, 17 and 18)

[merge] Source 15's Section 10, printed whole as the base: three lexicographies on three labels, inverse-rank lexicography ([theorem 38.1](#)), relation tables and the transfer of their endpoints ([theorem 38.2](#) and [corollary 38.3](#)), the failure of restriction-monotone linearizations ([theorem 38.5](#)), the isomorphism quotient, the normal-form embedding ([proposition 38.6](#)) and the birthday and simplicity remarks. Its macros  $\mathfrak{W}_{\text{sl}}$ ,  $\mathfrak{W}_{\text{all}}$ ,  $\mathcal{D}$  print as in this report; its rk, bl are this report's rank and  $\ell_2$ . The relation-code remarks of the other sources follow in [section 38.8](#).

### 38.1 Three lexicographies disagree on three labels

On  $0 < 1 < 2$ , an enumeration is a permutation word. Its inverse-rank vector lists the positions of labels 0, 1, 2. For table lexicography scan the directed pairs in this order:

$$(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1),$$

and use bit 1 when the first label precedes the second.

| Enumeration | Inverse ranks | Relation table |
|-------------|---------------|----------------|
| 012         | 012           | 110100         |
| 021         | 021           | 110001         |
| 102         | 102           | 011100         |
| 120         | 201           | 001110         |
| 201         | 120           | 100011         |
| 210         | 210           | 001011         |

The increasing orders on the same six objects are respectively

enumeration: 012, 021, 102, 120, 201, 210,  
inverse rank: 012, 021, 102, 201, 120, 210,  
table: 210, 120, 102, 201, 021, 012.

The table comparison is not just the reverse of enumeration comparison. As abstract finite orders, all three are of course the same six-point chain. The difference here concerns which labelled object precedes which; the next results distinguish abstract infinite order types as well.

### 38.2 Inverse-rank lexicography

Fix a baseline  $q : \text{Ord} \rightarrow \mathbf{No}$ . For a set-like well-order  $R$ , let  $r_R : \mathbf{No} \rightarrow \text{Ord}$  be its rank map. Compare the permutations

$$p_R = r_R \circ q : \text{Ord} \longrightarrow \text{Ord}$$

lexicographically, using the ordinal order on the values. Conversely, every class permutation  $p$  corresponds to the enumeration  $q \circ p^{-1}$ . This comparison uses ordinal ranks of fixed labels, rather than numerical labels at fixed positions. For a non-set-like order an ordinal rank does not exist at every label, so this version needs a further coding choice before it can be applied to all raw orders.

**Theorem 38.1** (Endpoint and character of inverse ranks). *The inverse-rank order of set-like global well-orders has a least element, the baseline order, and no greatest element. It is dense. For every nonzero limit ordinal  $\lambda$ , it has a point  $M_\lambda$  with left cofinality  $\text{cf}(\lambda)$ . No point has a set coinital family strictly above it. In particular this order is not isomorphic to native enumeration lexicography.*

*Proof.* The identity permutation is least. If  $p$  first differs from the identity at  $\alpha$ , all smaller ordinals have already appeared, so  $p(\alpha) > \alpha$ . On every tail of a permutation there is an ascending pair: otherwise a countable subsequence would descend through the ordinals. Swapping that ascending pair makes the permutation lexicographically larger. If  $p < q$ , perform such a swap strictly after their first disagreement; the result lies strictly between them. This also proves no greatest point. For a set family of points above  $p$ , bound all their first disagreements with  $p$  and perform an ascending swap later; the result is above  $p$  and below the entire family.

Let  $M_\alpha$  begin with  $\alpha$  and then list the remaining ordinals in increasing order. This is a permutation of type  $\text{Ord}$  and is the least permutation with first value  $\alpha$ , since it always chooses the least unused ordinal thereafter. For limit  $\lambda$ ,

$$M_\lambda = \sup_{\alpha < \lambda} M_\alpha.$$

Indeed, a point below  $M_\lambda$  has first value below  $\lambda$ ; some larger  $\alpha < \lambda$  therefore gives a point  $M_\alpha$  above it. A cofinal family below  $M_\lambda$  must have first values cofinal in  $\lambda$ , proving the exact character. Native enumeration  $\text{lex}$  has no least element and no set cofinal family below any point.  $\square$

For a set carrier  $X$  of cardinality  $\mu$ , scanning the labels by a bijection  $q : \mu \rightarrow X$  gives a least inverse-rank code among all well-orders, again the  $q$  order. On a fixed-type slice of type  $\alpha$ , a least inverse-rank code exists exactly when  $\alpha = \mu$ : a least rank vector can have no inversion, since swapping an inverted pair lowers it, and an increasing bijection of its ranks forces precisely that type. Thus even fixed-type endpoint behavior depends on the representation.

### 38.3 Relation tables and a theorem about all their endpoints

Choose a well-ordered scan of  $X \times X$ . For a proper class carrier assume the scan is set-like, so every proper initial part of the scan is a set. The table of a strict relation  $R$  is its characteristic function, compared with  $0 < 1$ . The native numerical order on  $\mathbf{No}^2$  cannot itself serve as a scan with a universal first-difference principle.

**Theorem 38.2** (Endpoint transfer for relation tables). *For a set-like scan, a well-order table is least among all well-order tables if and only if it is least among all strict linear-order tables. The dual statement holds for greatest elements. Both assertions also hold with the class of set-like well-orders in place of all well-orders.*

*Proof.* Suppose  $R$  is a claimed minimum and a linear order  $T$  has a smaller table. At the first differing entry  $(x, y)$ , one has  $xRy$  and  $yTx$ . The earlier bits impose a set  $P$  of directed order constraints: a true off-diagonal bit gives its indicated direction, and a false one gives the reversed direction. Both  $R$  and  $T$  satisfy these constraints.

Let  $D$  be the downward closure of  $y$  under finite  $P$ -paths, including  $y$ . It is a set, contained in  $\{y\}$  and the vertices of the earlier scan. Since  $T$  satisfies  $P$  and  $yTx$ , one has  $x \notin D$ . Move  $D$  to the front of  $R$ , preserving the old order within  $D$  and within its complement. The resulting relation  $R'$  is a well-order. It satisfies every earlier constraint because  $D$  is  $P$ -downward closed. It puts  $y$  before  $x$ , so its first changed table bit is smaller, a contradiction. If  $R$  was set-like,  $R'$  is still set-like: new predecessor classes are contained in an old predecessor set together with  $D$ . The proof for a maximum moves the appropriate opposite endpoint at the first larger bit. The reverse implications are immediate.  $\square$

**Corollary 38.3** (Exactly three possible endpoint patterns). *On an infinite carrier, table lexicography of well-orders has at most one endpoint. By choosing the scan, each of the following can occur: a minimum only, a maximum only, or neither. The same patterns occur for set-like class well-orders of **No**.*

*Proof.* Reversing a linear order complements all off-diagonal bits, so it reverses the table comparison. If a minimum and maximum well-order both existed, [Theorem 38.2](#) would make them endpoints among all linear orders, and hence reversals of one another. An order and its reversal cannot both be well-orders on an infinite carrier: the first  $\omega$  points of one form a descending sequence in the other.

For any given linear order  $T$ , scan each unordered pair in a two-entry block, putting its  $T$ -false direction first and its  $T$ -true direction second. Scan the blocks set-likely. Then  $T$  is the least linear-order table: a different order changes the first entry of its earliest changed block from 0 to 1. Reversing the two entries makes  $T$  greatest. Take  $T$  to be a well-order for the one-endpoint cases, and a linear order such that neither it nor its reverse is well-founded for the no-endpoint case. Such an order exists on every infinite carrier; on **No** its native order suffices. Endpoint transfer finishes the argument.  $\square$

A scan built around a non-set-like  $R$  makes  $R$  the minimum among all well-orders but leaves no minimum among set-like well-orders. The greatest linear-order table is then  $R^{\text{op}}$ , not a well-order, so the set-like subcollection has neither endpoint. Therefore a relation table may erase the natural numerical comparison while retaining or losing particular well-order types according to the chosen scan.

**Example 38.4** (Failure of a numerical pair scan). On the integer labels, compare the enumerations

$$0, -1, 1, -2, 2, -3, 3, \dots, \quad 0, 1, -1, 2, -2, 3, -3, \dots,$$

then append the same set-like enumeration of the remaining surreals. The resulting set-like orders disagree on the directed pairs  $(-n, n)$  and  $(n, -n)$ ,  $n \geq 1$ . Their numerical-lexicographic disagreement set in **No**<sup>2</sup> has no least pair, because its first coordinates include arbitrarily negative integers. A genuine well-ordered scan avoids this problem; the numerical order alone does not.

### 38.4 Restrictions, extensions, and relabelling

Restriction to a smaller carrier can reverse enumeration lexicography:  $021 < 102$  on  $0 < 1 < 2$ , but on  $\{1, 2\}$  the restrictions are  $21 > 12$ . The obstruction is stronger than a bad choice of lexicographic convention.

**Theorem 38.5** (No total linearization respects every restriction). *There is no total order on the six well-orders of a three-element carrier making all three restrictions to two-element carriers monotone, no matter which of the two orders is chosen on each two-element well-order space.*

*Proof.* Encode each pair restriction by 0 or 1 according to its chosen order. The three restrictions determine the original relation, so the six well-orders map injectively to  $\{0, 1\}^3$ . If all restrictions were monotone, the six objects in increasing order would give a coordinatewise increasing chain of six distinct Boolean vectors. Every strict step raises the sum of the coordinates, whose possible values are 0, 1, 2, 3. Such a chain has at most four points, a contradiction.  $\square$

Thus arbitrary restrictions do not form a monotone system automatically. The theorem does not rule out a carefully chosen nested filtration or restricted family. In the other direction, fixing a well-order  $Q$  of  $X \setminus Y$  makes  $R \mapsto R + Q$  an order embedding from the native well-order space of  $Y$  into that of  $X$ . First disagreement occurs before the fixed tail. Prepending a common labelled prefix works as well.

No linearization can be invariant under every relabelling of a carrier with at least two labels. A transposition acts as a nontrivial involution on its well-orders, since well-orders are rigid. But an increasing involution of a linear order must be the identity:  $x < f(x)$  would imply  $f(x) < f(f(x)) = x$ . Some auxiliary label structure is indispensable. Increasing automorphisms of the numerical surreal line preserve native enumeration lex; a table scan or baseline must instead be transported along with the labels.

### 38.5 The quotient by order isomorphism is a new construction

On the numerical carrier  $\mathbb{N}$ , let

$$R = (0, 1, 2, \dots), \quad S = (1, 2, 3, \dots, 0), \quad T = (2, 0, 1, 3, 4, \dots).$$

Their types are  $\omega, \omega + 1, \omega$ , respectively, and their first labels give  $R < S < T$ . Thus the equivalence class of type  $\omega$  is not convex. A linear-order quotient requires convex classes, so labelled lexicography does not descend to abstract order types.

One may separately order the possible set types by ordinal comparison. For  $|X| = \mu$  infinite they are precisely  $\mu \leq \alpha < \mu^+$ ; in that comparison their order type is  $\mu^+$ . The original labelled order embeds no ordinal  $\mu^+$ , by [Note 32.3](#). This is a particularly sharp sign that the quotient-type order is a different object. For set-like well-orders of all **No**, quotienting by class-order isomorphism collapses everything to the one type **Ord**. Arbitrary non-set-like class types require their own collection and comparison framework; raw labelled lexicography provides neither automatically.

### 38.6 Numerically well-ordered subsets and a normal-form embedding

Let  $A, B \subseteq \mathbf{No}$  be sets or classes well-ordered by the inherited numerical order. Their union is well-ordered: a nonempty subclass has its least member among the minima of its intersections with  $A$  and  $B$ . Consequently  $\min(A \triangle B)$  exists whenever  $A \neq B$ . This makes an intrinsic characteristic comparison possible without a global scan of the entire numerical line:

$$A <_{\chi} B \iff \min(A \triangle B) \in A.$$

It uses membership bits with  $1 < 0$ . Transitivity is first-difference transitivity on the well-ordered union of three carriers. For increasing enumerations it agrees with numerical first difference and *longer prefixes first*. The empty carrier is greatest. The shorter-first convention agrees on genuine unequal next labels but reverses proper-prefix pairs and makes the empty carrier least.

**Proposition 38.6** (Coefficient-one normal forms realize set supports). *For a set-sized naturally well-ordered subset  $A \subseteq \mathbf{No}$ , the map*

$$H(A) = - \sum_{a \in A} \omega^{-a}$$

*is a well-defined surreal, and  $A <_{\chi} B$  if and only if  $H(A) < H(B)$ .*

*Proof.* The exponent support  $\{-a : a \in A\}$  is reverse well-ordered and is a set, so the displayed Conway normal form is legitimate. At the least  $a$  in the symmetric difference, the largest differing exponent is  $-a$ . The set containing  $a$  has coefficient  $-1$  there rather than  $0$ , and therefore gives the smaller surreal. Uniqueness of normal form gives injectivity [11, 12].  $\square$

This particular construction does not admit proper-class support: the normal form of one surreal has set support. Naturally well-ordered proper subclasses nevertheless exist with every class well-order type by surreal universality. For example

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

has natural type  $\text{Ord} + 1$ . Its final point has a proper class of earlier points. The reciprocal and addition in this example are surreal operations on the embedded ordinal labels.

At a set limit-length increasing word, shorter-first order makes the word the supremum of its proper prefixes; longer-first order makes it their infimum. The associated one-sided cofinality is the cofinality of the length. These features expose the termination convention in the order itself, even when both conventions admit embeddings into  $\mathbf{No}$ .

### 38.7 Birthday and simplicity do not secretly provide a scan

Simplicity, interpreted as initial sign-sequence extension, is a partial tree order. The one-sign strings  $+$  and  $-$  are incomparable by it. Numerical sign comparison instead uses the middle termination symbol  $- < \text{undefined} < +$  and gives the surreal line, not a well-order.

Even birthday-first ordering with numerical tie-breaking fails. The strings

$$s_n = -^n \frown +^\omega \quad (n < \omega)$$

all have birthday  $\omega$ , since  $n + \omega = \omega$ , and satisfy  $s_{n+1} < s_n$ . Thus the birthday- $\omega$  level already contains a descending sequence under numerical tie-breaking. A genuine birthday-compatible global well-order requires well-ordering each level with additional coherent choice data. That choice must be stated; it is not supplied by birthdays or simplicity alone.

*Note 38.7* ([merge]; repository counterparts of source 15's Section 10). Relation-table lexicography is source 08's relation-code order ([proposition 17.1](#): equimorphic with  $\mathbb{B}_{\text{Ord}}$  for every coordinate enumeration) and source 12's remark with its three-point check ([section 18](#)). New in source 15 are the endpoint transfer ([theorem 38.2](#)) and the three endpoint patterns ([corollary 38.3](#)), which show that the endpoints of a relation-code order are artifacts of the scan; inverse-rank lexicography ([theorem 38.1](#)); the restriction theorem ([theorem 38.5](#)); and the normal-form embedding ([proposition 38.6](#)). The birthday tie-break example is source 12's ("Birthday is not a canonical global well-order", after [theorem 13.2](#)), and source 18 repeats it ([section 38.8](#)). The non-convexity of type classes is made by sources 13, 14, 16 and 17 in their maps ([section 34](#)).

### 38.8 Relation-code remarks of sources 13, 14 and 18

[merge] Three further remarks on relation codes, each printed whole: source 13's (from its foundations section), source 14's (from its Section 11) and source 18's (from its section on global well-orders). Source 16's is in its map (section 34.4), source 17's in its map (section 34.5). They make one point, that relation-code lexicography depends on the scan and the bit convention and is not enumeration lexicography; they prove no theorem about it beyond proposition 17.1 and section 38.

#### 38.8.1 Source 13: relation-bit lexicography is a different order

Fix a set-like global enumeration of the ordered pairs of surreal labels. Read the characteristic bits of a raw relation in this order and compare at the first different bit. This is a legitimate code-dependent class lexicography. It is not in general the labeled common-prefix comparison of theorem 14.2: the first edge in a chosen code order need not reveal the first disagreement of the enumerated labels.

The preceding report analyzes this relation-code alternative and its binary-word equimorphism (proposition 17.1; Section 17 of this report at the pin). A choice of pair enumeration is indispensable. Replacing the label order by birthday order or comparing raw Conway construction trees would impose still other auxiliary data. Such choices can be studied rigorously, but no one of them should be advertised as *the* lexicographic order without identifying the coding convention.

#### 38.8.2 Source 14: relation-code lexicography really is another order

One may instead fix a class bijection scheduling  $\mathbf{No} \times \mathbf{No}$  by  $\mathbf{Ord}$ , and compare characteristic functions of well-order relations at their first differing scheduled pair. This defines a linear comparison on the admitted relations but depends on the schedule and on the convention  $0 < 1$ .

The difference is visible with three labels. Let numerical order be  $0 < 1 < 2$ , and let the enumerations be  $e = (0, 1, 2)$  and  $f = (1, 0, 2)$ . Enumeration lex gives  $e < f$ . If the relation-coordinate schedule begins with  $(0, 1)$ , their strict-relation bits there are 1 and 0, so the relation-code comparison gives  $e > f$ . Starting instead with  $(1, 0)$  reverses those bits and that conclusion. This counterexample does not involve any proper-class complication. Results proved for numerical first-disagreement order cannot simply be transported to relation codes.

#### 38.8.3 Source 18: inherited numerical well-orders and relation codes

Numerical  $<$  on  $\mathbf{No}$  is not a well-order; the negative integers already descend. Nevertheless, many subclasses are numerically well-ordered. The ordinals give type  $\mathbf{Ord}$ , and

$$\{-1/(\alpha + 1) : \alpha \in \mathbf{Ord}\} \cup \{0\}$$

has numerical order type  $\mathbf{Ord} + 1$ . Under Global Choice, class universality gives a numerically ordered copy of every supplied class well-order. This concerns the range of an embedding in the surreal line, not a well-order of every surreal label.

One can instead order well-order *relations* by characteristic bits. Choose a set-like class enumeration of  $X \times X$ , and compare the bits of two relations at their first unequal pair. The resulting comparison is well-defined, but is a different auxiliary order. Even on  $\{0, 1\}$  the enumerations  $(0, 1) < (1, 0)$  compare in the opposite direction if the first tested pair is  $(0, 1)$  and  $0 < 1$  for relation bits. The enumeration rule tests first labels, while the relation rule tests a chosen predicate membership. Its dependence on the chosen pair order should be recorded explicitly.



Finally, ordering birthday first and then ordering sign expansions lexicographically does not produce a global well-order. The fixed birthday- $\omega$  slice already contains descending sequences. A genuine birthday-first global well-order must use an auxiliary well-order within each birthday level, not its numerical sign order.

### 39 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 33](#); where the question index of [section 113](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 09's direction "Other termination conventions for set words"* ([section 113.3](#)): **answered** for unrestricted and injective words. Exactly the placements of a formal terminator satisfying the three conditions of [theorem 36.1](#) give the set-cut property, hence (in GBC) an order isomorphic to **No**; every placement gives one of the types of [theorem 36.2](#); the distinguished internal value is [theorems 36.4](#) and [36.6](#), and the prefix-free codes are listed in [note 36.3](#). For increasing words with the terminator at zero see (b).
- (b) *Source 08's question 14, increasing-subset and alternate-termination orders* ([section 113.4](#)): **answered** for prefix-small, prefix-large and centered conventions on unrestricted and injective words ([theorems 36.1](#) and [36.2](#)), for shorter-first increasing words (isomorphic to the other two classes in GBC, [theorem 35.1](#)), and for centered increasing words up to their isomorphism type: local characters, all set-cut obstructions, equimorphism without isomorphism ([theorems 37.1](#) and [37.2](#) and [corollary 37.3](#)). Open: the isomorphism type of the centered increasing order (source 18's own question) and increasing words with an interior terminator at a threshold other than zero.
- (c) *Source 13's question "A classification of termination cuts"* (for a class alphabet isomorphic to **No**): answered by source 16 ([theorems 36.1](#) and [36.2](#)), transported along an isomorphism of the alphabet with **No** in GBC.
- (d) *Source 16's question "Centered increasing words"*: answered by source 18 for the threshold at zero (with zero excluded, padded, or adjacent to a distinct terminator: [section 37](#)); open for other interior thresholds.
- (e) *Source 08's question 9 and source 09's direction "Non-set-like well-order relations"* (relation-code orders; [sections 113.3](#) and [113.4](#)): answered in part. Which endpoints a relation-code order of well-orders has is an artifact of the scan: a minimum only, a maximum only, or neither, each by a suitable scan, and never both on an infinite carrier ([theorem 38.2](#) and [corollary 38.3](#)). The isomorphism question for two scans stays open.
- (f) *Source 11's question on optimal birthday bounds* ([research question 113.6](#)): not answered. Sources 13, 14, 16, 17 and 18 give explicit codes with birthdays  $\sum(2 \cdot \text{bd} + 2)$ ,  $\sum(4 \cdot \text{bd} + 4) + 2$  and  $\sum(1 + 2 \cdot \text{bd} + 2)$  ([note 36.3](#)); none is claimed optimal.

Sources 16 and 18 also ask how much of the word isomorphisms survives without a global well-order; source 18's proof ([section 35.5](#)) shows that one set-like global well-order suffices as a parameter, so the question is whether that parameter can be removed. It is open.

[merge] Dated note, 3 October 2026 (batch 83). Items (b), (d) and (e) are answered in part by Part XII: the centered increasing order is  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ , no marker saturates increasing words, and two scans of one carrier can give different orders with the same spectrum, while a dense set-coded core exists for every set-like scan ([section 85](#); [section 113.25](#) (j), (k)).

## Part VIII

# Cuts of the set-coded core: recognition, automorphisms, support and class realizations

[merge] (3 October 2026.) The third Part of the batch-81 addition ([section 1.8](#); notation in [section 1.3.1](#)). Part IV ([part IV](#)) built the core, the ordinary class of set codes for bounded-support perturbations of one baseline enumeration  $b$ , proved that it interpolates every uniformly set-indexed cut, and asked which of its class cuts come from global enumerations ([research question 113.1](#); source 09’s third direction in [section 113.3](#)). All six batch-81 sources answer that question, by one argument. This Part prints the answer and what the sources prove around it: source 14’s foundations for the class-level results ([section 40](#)); the core and its cylinders ([section 41](#)), with the five other sources’ re-proofs of Part IV ([section 42](#)); the classification of every proper class cut of the core into a cylinder edge, a numerical gap or a branch ([section 43](#)) and its characters ([section 44](#)); the recognition of the cuts of global enumerations ([section 45](#)) and the five further proofs of it ([section 46](#)); automorphisms of the core ([section 47](#)); class extensions ([section 48](#)); and support-restricted suborders and the exact reserve of a partial injection ([section 49](#)). [Section 50](#) records which questions the Part answers.

The base is source 14 throughout, by generality: it alone classifies *every* proper class cut with unique data, gives the complete orbit invariant, the absoluteness theorem and the support hierarchy. Source 17 is the base for the exact reserve cardinal, the automorphism extension criterion and the clause that every proper class extension adds a represented cut; source 15 adds the finest first-failure partition (a forbidden repeated label) and source 13 the non-rigidity of the cylinder family. The represented-cut theorem itself is printed once as source 14’s [theorems 43.3](#) and [45.1](#), and the other five proofs follow as routes, each whole, because each source builds different examples and corollaries on its own proof ([note 46.1](#)). Source 18 was written before sources 13–17 arrived.

Notation ([section 1.3.1](#)): the core is printed  $\mathcal{P}_s(b)$  in the support coding of sources 13, 14, 17 and 18 (their  $\mathcal{D}$ ,  $\mathcal{P}_b$ ,  $\mathcal{C}_b$ ,  $\mathcal{C}_b$ ; source 18’s macro  $\backslash\text{Pbd}$ ), and  $\mathcal{P}_{bd}(b)$  in the normalized-permutation coding of sources 15 and 16 (their  $\mathcal{D}$  and  $\mathcal{P}_b$ ); the two codings have the same evaluations. Cylinders of codes are printed  $\mathcal{C}_p$  (source 13’s  $\mathcal{D}_p$  is printed  $\mathcal{P}_s(b)_p$ , source 17’s  $\mathcal{C}_p$  and source 18’s  $\mathcal{C}_b(p)$  are printed  $\mathcal{C}_p$ ). Source 14’s  $\mathfrak{E}$ ,  $\mathfrak{J}$ ,  $\mathfrak{W}$  are printed  $\mathfrak{W}_{sl}$ ,  $\mathfrak{J}$ ,  $\mathfrak{W}_{all}$ , its relation macro  $\backslash\text{lex}$  as  $<_{lex}$ ; in its support section its  $\tau = \text{cf } \kappa$  is printed  $r$ , its decreasing sequence  $(r_i)$  is printed  $(u_i)$ , and the generic regular cardinal of its definition of  $\eta_\tau$ -orders is printed  $\theta$ . A cylinder meeting both sides of a cut *crosses* it in sources 13, 14 and 15 (source 15 also calls such cuts *persistent*) and *straddles* it in sources 16, 17 and 18; the two words mean the same. All six use  $\text{Ord}$  as a character symbol for “a uniform  $\text{Ord}$ -indexed cofinal family exists and no set-sized one does”, never as a cardinal. Labels written for the merge in this Part begin `swo:viii:`.

## 40 Foundations and the audited starting point (source 14)

### 40.1 Sets, classes, and uniform families

We use sign sequences as canonical set representatives of surreals. Each surreal has a set ordinal birthday. This avoids treating a proper class of equivalent game forms as an individual set.

In GBC, a class  $E \subseteq \text{Ord} \times \mathbf{No}$  may be the graph of a global enumeration. It cannot then be an element of a set or an ordinary class. Expressions such as

$$E \in \mathfrak{W}_{sl} = \text{Bij}(\text{Ord}, \mathbf{No})$$

are convenient predicate notation. They mean that the class variable  $E$  satisfies the indicated formula. They are not membership assertions in a powerclass object. The same convention applies to the virtual collection  $\mathfrak{C}(\mathcal{P}_s(b))$  of class cuts.

A uniformly set-indexed family  $(E_i)_{i \in I}$  is one class relation  $U \subseteq I \times \text{Ord} \times \mathbf{No}$  whose  $i$ -th section is  $E_i$ , with  $I$  a set. Replacement with the parameter  $U$  produces a set of first disagreement ordinals or a set of evaluated prefix values. Merely writing “a set of proper-class functions” does not provide such a family.

A class well-order is a class linear order for which every nonempty subclass has a least member. Under the choice assumptions used here, the standard set-subset well-foundedness formulation is equivalent; see [27, Section 3]. Internal well-foundedness in an arbitrary model is not asserted to be external well-foundedness.

## 40.2 Why a baseline is an assumption, not a notation

Global Choice supplies a set-like class well-order and hence a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . Conversely, the predecessor report proves over GB without even assuming set Choice that a class well-order of the sign-sequence surreals implies Global Choice (theorem 13.1; Section 13 of this report at the pin). The proof uniformly well-orders the power sets  $V_{\beta+1}$  ([merge] the delivered text says “coherently”; the orders need not extend one another, as corrected in this report on 3 October 2026, commit 0f52c0dca, after source 14’s pin) and constructs rank-by-rank presentations; it is stronger than the observation that each already well-orderable set embeds into  $\mathbf{No}$ . We use that reported equivalence as background, rather than claim to re-prove its choiceless direction here.

There is no paradox in using  $\text{Ord}$  as the domain of  $b$ : it is a proper class domain. There would be a paradox in calling it a set ordinal or applying a set power-set operation to it inside the same universe.

## 40.3 Elementary comprehension versus class-valued recursion

The new constructions use formulas with set quantifiers and previously given class parameters. For example, “a set prefix has a core completion on each side of this cut” quantifies over set codes, not over classes. The union of all such coherent prefixes is consequently an ordinary class obtained by elementary comprehension.

Set-valued recursion of length  $\text{Ord}$  with set histories at each stage can be implemented by coherent set-length recursions in GBC. This must not be conflated with recursions that choose an entire class at each stage, or with recursions along arbitrary non-set-like class well-orders. The literature proves that ETR implies abstract class-well-order comparability [27, Theorem 6]; our labelled comparison and prefix reconstruction are different operations and do not invoke that implication in reverse.

## 40.4 Repository audit and status of the inputs

The inspected report lives at

```
Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex.
```

Its README records a 115-page merged report, four source manuscripts, and a pending collection-level review; it explicitly says the report is not formalized (its README at the pin; [merge] the page count and the number of sources have grown with this addition, and the report is still not formalized). We inspected the introduction, set-supported and binary-coding sections, universe

discussion, and research questions relevant to the present work. This is a targeted audit, not a certification of the entire repository.

We also inspected the actual Lean file

Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean.

It proves uniqueness and prefix properties of simplest separators *assuming existence*. In particular, `existsUnique_simplest_separator` takes an existence hypothesis. It is not a theorem supplying all small-cut separators by itself [10]. This distinction matters for the proposed formalization. [merge] The pinned tree already supplies the existence half and composes the two: `small_cut_fillers` in `SignSequenceCut.lean` [8] proves that small cuts have fillers, and `cut` in `SignSequenceCutOperation.lean` [9] is defined from `existsUnique_simplest_separator` applied to it (section 1.5). Source 14’s recommendation to compose them is therefore already carried out.

The present main theorems have written proofs below. Their novelty claim is only that they extend the inspected predecessor results; historical priority has not been established by the literature check.

## 41 The set-coded skeleton and its cylinders (source 14)

For the remainder of the class-level development we work in GBC and fix a class bijection  $b : \mathbf{Ord} \rightarrow \mathbf{No}$ . All occurrences of a class parameter mean a class belonging to the chosen class realization.

### 41.1 Canonical support codes

A *support code* is a set bijection  $\sigma : S \rightarrow S$  with no fixed points, where  $S$  is a set of ordinals. The empty code is permitted. Extend it by the identity to a class permutation  $\hat{\sigma}$  of  $\mathbf{Ord}$ , and set

$$E_\sigma = b \circ \hat{\sigma}, \quad \text{supp}(\sigma) = S.$$

Let  $\mathcal{P}_s(b)$  be the ordinary class of these codes, ordered by  $\sigma < \tau$  if  $E_\sigma <_{\text{lex}} E_\tau$ . The codes are canonical: two of them have the same extension exactly when they are equal. Every class permutation with set support is represented, because its support is invariant under the permutation and its restriction has no fixed points. A set of ordinals is bounded, so “set support” and “bounded support” are equivalent here. This does not mean finite support.

Let  $\mathfrak{W}_{\text{sl}}$  stand for the virtual collection of class bijections  $\mathbf{Ord} \rightarrow \mathbf{No}$ , and  $\mathfrak{J}$  for that of class injections. A statement quantifying over  $\mathfrak{W}_{\text{sl}}$  or  $\mathfrak{J}$  is a statement with class variables; these symbols do not name additional class objects. Two distinct members of  $\mathfrak{J}$  have a least differing ordinal. Their common initial segment has set length, so first-difference comparison is unambiguous.

**Lemma 41.1** (Fresh separators). *If  $A, B, U$  are sets of surreals and  $A < B$ , there is a surreal  $x \notin U$  such that  $A < x < B$ . Consequently  $\mathbf{No} \setminus U$  has no endpoints and fills all cuts specified by sets of its elements.*

*Proof.* The ordinary surreal set-cut theorem supplies a separator ([15]; lemma 5.2). In fact there are more than  $|U|$  separators: recursively for  $|U|^+$  many steps choose a separator above the earlier choices and  $A$ , and below  $B$ . Every stage uses set options, and the resulting choices are distinct. At least one is outside  $U$ . The same argument permits either option set to be empty. This uses set recursion and the usual surreal cut existence theorem, not a purported cut with proper-class options.  $\square$

**Lemma 41.2** (Completion of a set prefix). *Every injective set word  $p : \lambda \rightarrow \mathbf{No}$  extends to some  $E_\sigma$  with  $\sigma \in \mathcal{P}_s(b)$ .*

*Proof.* Choose an infinite initial cardinal  $\nu$  strictly greater than  $\lambda$  and all the ordinals in  $b^{-1}[\text{ran } p]$ . The map  $\alpha \mapsto b^{-1}(p(\alpha))$  is a partial injection from  $\nu$  to itself with domain of cardinality  $< \nu$  and equally small range. Both complements have cardinality  $\nu$ . Choose a bijection between them to extend the partial map to a permutation of  $\nu$ , and fix every ordinal outside  $\nu$ . Restricting the resulting permutation to its moved points gives the required canonical code.  $\square$

**Definition 41.3** (Cylinder and residual alphabet). For an injective set prefix  $p : \lambda \rightarrow \mathbf{No}$ , put

$$C_p = \{\sigma \in \mathcal{P}_s(b) : E_\sigma \upharpoonright \lambda = p\}, \quad A_p = \mathbf{No} \setminus \text{ran } p.$$

The empty prefix has cylinder  $C_\emptyset = \mathcal{P}_s(b)$ .

**Proposition 41.4** (Cylinder geometry). *Every  $C_p$  is a nonempty convex proper class. It has neither an endpoint nor a set-sized cofinal or cointial subset. Its immediate subdivisions are*

$$C_p = \bigcup_{x \in A_p} C_{p \smallfrown (x)},$$

*in increasing order of  $x$ . The union is an elementary class statement, not a set-indexed union. Cylinders of two prefixes are either nested or disjoint and strictly ordered.*

*Proof.* Completion proves nonemptiness. If a word between two extensions of  $p$  first differed from  $p$  before  $\lambda$ , it would lie on the same side of both extensions, proving convexity. The child cylinders partition  $C_p$  by the value at coordinate  $\lambda$ , and first difference puts them in numerical order. [Lemma 41.1](#) and completion give a child below or above any specified set of child labels. Thus a set of members of  $C_p$  is neither cofinal nor cointial: take their next labels, choose a fresh label above or below all of them, and complete. In particular  $C_p$  cannot itself be a set. Finally, incomparable prefixes first differ at a set coordinate, which orders their entire cylinders; comparable prefixes give nested cylinders.  $\square$

**Lemma 41.5** (Density around injective branches). *If  $F, G \in \mathfrak{J}$  and  $F <_{\text{lex}} G$ , there is a code  $\sigma \in \mathcal{P}_s(b)$  with  $F <_{\text{lex}} E_\sigma <_{\text{lex}} G$ . For every  $F \in \mathfrak{J}$  and every set ordinal  $\alpha$ , there are codes below and above  $F$  agreeing with  $F$  before  $\alpha$  and differing from it at  $\alpha$ .*

*Proof.* For the first assertion use the first disagreement  $\delta$ , choose  $x$  strictly between  $F(\delta)$  and  $G(\delta)$  outside their common prefix range, and complete  $(F \upharpoonright \delta) \smallfrown (x)$ . For the second, choose fresh values below and above  $F(\alpha)$  and complete the two prefixes. A fixed global well-order of set codes makes all these choices uniform when a uniform family is needed.  $\square$

These elementary cylinder facts, rather than a powerclass completion, are the main engine of the next result. The predecessor proves the stronger statement that  $\mathcal{P}_s(b)$  interpolates every uniformly set-indexed strict cut of  $\mathfrak{W}_{\text{sl}}$  ([theorem 19.10](#); Section 19 of this report at the pin). We will also recover the ordinary set-cut property of  $\mathcal{P}_s(b)$  from our cut classification and its character calculation.

*Note 41.6* ([merge]; Part IV counterparts of source 14's Section 4). Source 14's support codes are source 09's coding  $\mathcal{P}_s(b)$  ([section 1.3](#)); its [lemma 41.1](#) is [lemma 5.2](#), its [lemma 41.2](#) is [lemma 19.4](#) with the cardinality check of [remark 19.7](#), its [proposition 41.4](#) is the code-level form of [proposition 19.25](#) (sources 11 and 09), and the two-sided approximants of [lemma 41.5](#) are those of [proposition 19.27](#).

Source 14 credits the interpolation theorem (theorem 19.10) and does not re-prove it; the elementary cylinder facts are what its classification uses. The class  $T_D$  of crossing prefixes of a cut, defined in the next section, is the class-scale form of the reals report's cut spine (`lwo:sp:lem:cutspine`, for  $\mathcal{W}(\mathbb{R})$ ); source 14 does not cite it.

## 42 The core as re-proved in sources 13, 15, 16, 17 and 18

[merge] Each of the other five sources re-proves the core layer of Part IV before its recognition theorem, and each credits it to this report. Their sections are printed here whole, in source order, as re-proofs of the following results of Part IV; none is new against the repository.

- *Fresh separators*: lemma 5.2 (source 13's lemma 42.1, source 17's lemma 42.12).
- *Class universality, uniqueness and uniform complements*  $h_S : \mathbf{No} \cong \mathbf{No} \setminus S$ : proposition 5.13 and theorem 5.17 (source 13's lemma 42.2, source 17's corollary 42.15). Source 16's termination section (section 36) uses the same family  $h_S$ .
- *Prefix completion*: lemma 19.4 with remark 19.7 (lemmas 42.3, 42.8, 42.9 and 42.17; source 17 obtains it from its reserve theorem, corollary 49.10).
- *Interpolation and the core isomorphic to  $\mathbf{No}$* : theorem 19.10 and corollary 19.15 (theorems 42.4, 42.10, 42.13 and 42.18 and corollaries 42.5, 42.11 and 42.14; source 15 cites them).
- *Approximants at a branch*: proposition 19.27 (lemma 42.6).
- *The diagonal theorem*: theorems 19.37 and 19.38 (theorems 42.7, 42.16 and 42.19).
- *Canonical first block and residual fibres* of a raw order: theorem 15.1, proposition 15.3, and remark 15.4 (source 16's Subsection 11.1 below, with displays (S16.12) and (S16.13)).
- *Global choice is an assumption*: theorem 13.1 (end of source 13's Section 3).

The proofs differ only in bookkeeping. Sources 13, 15, 16 and 18 complete a prefix inside an infinite initial cardinal, as Part IV does; source 17 completes it with the least possible reserve (section 49). Source 13's warning after lemma 42.3 is the point of remark 19.7. Source 16's Subsection 11.1 refers to its block decomposition, printed in Part IX (section 52.2); source 17's Section 8 and source 18's Section 14 are printed together with their remarks on what density does not give.

### 42.1 Source 13: surreal cuts and the size of the universe

#### 42.1.1 Signs, birthdays, and small cut filling

We use the sign-sequence presentation of the surreal numbers. A surreal  $x$  is a sequence of signs  $-$  and  $+$  indexed by a set ordinal  $\text{bd}(x)$ . Extend the sequence by 0 after its domain; comparison is at the first unequal coordinate using  $- < 0 < +$ . Here 0 in the sign alphabet is a termination value, not an extra sign before the birthday. This is also the representation used by the inspected Lean carrier [7]. General context for the universal ordered-continuum viewpoint is given in [15].

For sets  $A, B \subseteq \mathbf{No}$  with  $A < B$ , a surreal lies strictly between them. More precisely, if an ordinal  $\gamma$  strictly exceeds all their birthdays, there is such a separator of birthday exactly  $\gamma$ . The stronger statement is implemented at the source level by `exists_cut_separator_of_birthday_lt` [8]. We use it in the following form.

**Lemma 42.1** (Fresh set-cut separator). *Let  $A, B, S$  be sets of surreals and suppose  $A < B$ . There is an  $x \notin S$  with  $A < x < B$ . Empty sides are allowed.*



*Proof.* Choose  $\gamma$  strictly larger than every birthday occurring in  $A \cup B \cup S$ . A separator of birthday  $\gamma$  cannot belong to  $S$ .  $\square$

The adjective “set” cannot be removed. There is no surreal above every ordinal: any surreal is smaller than a sufficiently long all-plus sign sequence. Likewise, a proper-class sign sequence is not another surreal at a larger birthday.

#### 42.1.2 Set-cut orders and class back-and-forth

Call an ordinary class linear order *set-cut saturated* when every pair of sets  $A < B$  has a strict interpolant, including when a side is empty. This is sometimes expressed by an  $\eta_{\text{Ord}}$  terminology; we use the explicit condition rather than assigning a set cardinal to it.

**Lemma 42.2** (Class universality and uniqueness in GBC). *Every ordinary class linear order embeds into  $\mathbf{No}$ . Any two nonempty set-cut saturated ordinary class orders are order-isomorphic. In particular, for every set  $S \subseteq \mathbf{No}$ ,*

$$\mathbf{No} \setminus S \cong \mathbf{No}.$$

*The complement isomorphisms can be chosen as a single uniform class family  $h_S : \mathbf{No} \rightarrow \mathbf{No} \setminus S$ , indexed by sets  $S$ .*

*Proof.* Global choice supplies a set-like enumeration of any proper class of sets; for a set carrier use a set well-order. At a stage of the usual order-embedding recursion, the images already chosen form a set. Their lower and upper parts determine a set cut in  $\mathbf{No}$ , and a fresh interpolant gives the next image. The recursion has a set partial map at each set-ordinal stage.

For uniqueness, alternate the two enumerations. A forth step puts the next unused source point into the target cut determined by the current partial isomorphism; a back step does the same for the inverse. Strict interpolation keeps each map injective. Every point on both sides is visited, so the union is an isomorphism. A set-cut saturated nonempty order cannot be a set, since using all its points as a lower side would require a point above itself.

[Lemma 42.1](#) says that  $\mathbf{No} \setminus S$  has the same set-cut property. To make its isomorphism uniform, fix the global well-order and always choose its least eligible witness. For each set parameter  $S$  and each set stage, the partial history is a set satisfying a fixed recursive specification. Its existence and uniqueness are established by set-length recursion. Elementary class comprehension, quantifying over such set histories, gives one relation  $h(S, x, y)$ . No choice of a class witness for each  $S$  is being assumed.  $\square$

This is not an appeal to elementary transfinite recursion with *class-valued* stages. Our stages are sets, even though their total union is a class. Keeping that distinction explicit will matter in [source 13's](#) sections on asymmetric products and on foundational interpretations ([\[merge\]](#) its Sections 10 and 13, printed in Part IX as [sections 55](#) and [57.3](#)).

#### 42.1.3 Global choice is a real assumption

The earlier report proves, over GB without set choice, that the existence of a class well-order of  $\mathbf{No}$  is equivalent to global choice ([theorem 13.1](#); Theorem 13.1 at the pin). Thus a global enumeration is not a choice-free consequence of the numerical order of surreals. We work in GBC for the main global constructions and fix a baseline bijection

$$b : \text{Ord} \xrightarrow{\sim} \mathbf{No}.$$

The particular  $b$  affects the chosen dense core but not the definition of the full enumeration comparison.

## 42.2 Source 13: the global enumeration order and its dense core

### 42.2.1 Uniformity and the code carrier

Let  $\mathfrak{J}$  denote the virtual domain of injective class functions  $F : \text{Ord} \rightarrow \mathbf{No}$ , and let  $\mathfrak{W}_{sl} \subseteq \mathfrak{J}$  require surjectivity. Their lexicographic comparison is at the least disagreement ordinal. A nonempty class of ordinals has a least member: find one member and take the least member below it. Thus this comparison is elementary in its class parameters.

A *uniformly set-indexed family*  $(F_i)_{i \in I}$  means a set  $I$  and a single class relation coding all values  $F_i(\alpha)$ . A phrase such as “choose a class  $F_i$  for each  $i$ ” is not a substitute for this uniform family. We consistently use the latter.

Fix  $b : \text{Ord} \simeq \mathbf{No}$ . A set-supported permutation  $\sigma$  of  $\text{Ord}$  is the identity except on a set of ordinals. Code it by the set graph of its moved points, omitting all fixed points. Such normalized codes form an ordinary class  $\mathcal{P}_s(b)$ , and a code  $d$  evaluates to

$$E_d = b \circ \widehat{d}.$$

Equivalently, a code is a set bijection  $d : S \rightarrow S$ , where  $S$  is a set of ordinals and  $d(\xi) \neq \xi$  for every  $\xi \in S$ ; the empty code is allowed. This is an elementary condition on sets, so defining  $\mathcal{P}_s(b)$  does not require quantification over a collection of class permutations. Different normalized codes have different evaluations. Order  $\mathcal{P}_s(b)$  by these evaluations. The support of each code is bounded by a set ordinal, but there is no common set bound for all codes.

For an injective set word  $p$ , define

$$C_p = \{E \in \mathfrak{W}_{sl} : E \upharpoonright \text{dom}(p) = p\}, \quad \mathcal{P}_s(b)_p = \{d \in \mathcal{P}_s(b) : E_d \upharpoonright \text{dom}(p) = p\}.$$

$C_p$  is virtual notation;  $\mathcal{P}_s(b)_p$  is an ordinary class of set codes. Both are convex in their respective orders.

### 42.2.2 Completion with the missing cardinality check

**Lemma 42.3** (Set-prefix completion). *Every injective set prefix  $p : \alpha \rightarrow \mathbf{No}$  has a completion in  $\mathcal{P}_s(b)$ . The completion can be chosen uniformly from  $p$ .*

*Proof.* The class inverse of  $b$ , restricted to the set range of  $p$ , gives a set injection  $f = b^{-1} \circ p : \alpha \rightarrow \text{Ord}$ . Choose an infinite initial cardinal  $\delta$  strictly above  $\alpha$  and every member of  $\text{ran}(f)$ . Both  $\delta \setminus \alpha$  and  $\delta \setminus \text{ran}(f)$  then have cardinality  $\delta$ . A bijection between these complements extends  $f$  to a permutation of  $\delta$ . Extend it by the identity outside  $\delta$  and retain only its moved points as a code. Global choice makes the choices uniform.  $\square$

It would be incorrect to assert that every partial injection of a set into itself extends to a permutation without checking the complements. For instance, a bijection from a proper subset onto the entire carrier has no such extension. Enlarging to the cardinal  $\delta$  is exactly what prevents this obstruction.

### 42.2.3 The active-prefix interpolation argument

The next theorem is the earlier dense-core interpolation result ([theorem 19.10](#); Section 19 at the pin). We include the algorithm because both cut classification and the central-word theorem depend on its logic.

**Theorem 42.4** (Uniform set-cut interpolation). *Let  $(A_i)_{i \in I}$  and  $(B_j)_{j \in J}$  be uniform set-indexed families in  $\mathfrak{J}$ , with  $A_i < B_j$  for every cross pair. There is an injective set prefix  $p$  such that all its global completions lie strictly between the two families. In particular, a code in  $\mathcal{P}_s(b)$  interpolates the cut.*

*Proof.* For a cross pair let  $d(i, j)$  be its least disagreement. Class replacement gives a set of these ordinals. Put

$$\theta = \sup\{d(i, j) + 1 : i \in I, j \in J\},$$

with supremum 0 for an empty set. We construct an injective common prefix, keeping as *active* the rows that agree with it so far. Every inactive lower row is already strictly below all completions; every inactive upper row is already strictly above them.

At stage  $\beta$ , let  $L_\beta$  and  $U_\beta$  be the sets of next letters of the active lower and upper rows. Every  $l \in L_\beta$  is at most every  $u \in U_\beta$ , since otherwise the corresponding cross pair would be reversed. There are two cases.

If  $L_\beta \cap U_\beta$  is nonempty, it is a singleton  $\{t\}$ . Append  $t$ , which is fresh because the active rows are injective. Keep only the rows whose next letter is  $t$ . A dropped lower row has next letter less than  $t$ , and a dropped upper row has next letter greater than  $t$ , so the invariant is maintained.

If the intersection is empty, then  $L_\beta < U_\beta$ , allowing empty sides. Choose a fresh strict separator  $x$  outside the earlier range by Lemma 42.1, and append it. Every remaining active comparison is now strict. This prefix is the required  $p$ .

At a limit stage take the union of the earlier set prefixes and keep exactly the rows agreeing with it. The construction cannot continue with a shared letter through stage  $\theta$ : surviving active rows on both sides would agree beyond their first disagreement. It therefore stops by that stage. The prefix has length at most  $\theta + 1$ . Apply Lemma 42.3 to obtain a code.  $\square$

The histories in this proof have set length and set entries. Nothing requires collecting a set of class-valued choices, taking a surreal cut with proper-class sides, or invoking a class-valued recursion principle.

**Corollary 42.5** (The core and its cylinders).  *$\mathcal{P}_s(b) \cong \mathbf{No}$ , and every  $\mathcal{P}_s(b)_p \cong \mathbf{No}$ . The core is order-dense among global enumerations and among distinct injective class words. Every interval specified by a uniform set cut in  $\mathfrak{W}_{sl}$  contains a whole cylinder  $C_p$ .*

*Proof.* The theorem says that  $\mathcal{P}_s(b)$  fills all its set cuts, so Lemma 42.2 applies. To keep a prescribed prefix  $p$ , run the same active-row construction after  $p$ , forbidding  $\text{ran}(p)$  as well as the later used values; all forbidden values at a stage still form a set. This proves the set-cut property of  $\mathcal{P}_s(b)_p$ , including empty sides. The other assertions follow directly from the strict cylinder conclusion of the theorem.  $\square$

#### 42.2.4 Approximating one branch

**Lemma 42.6** (Two-sided ordinal approximants). *For  $F \in \mathfrak{J}$  there are uniform sequences of codes  $a_\alpha, b_\alpha$  indexed by Ord such that, for  $\alpha < \beta$ ,*

$$E_{a_\alpha} < E_{a_\beta} < F < E_{b_\beta} < E_{b_\alpha}.$$

*They are respectively cofinal and coinital among code evaluations strictly below and above  $F$ . Neither punctured side has a set-sized cofinal or coinital family.*

*Proof.* Complete  $F \upharpoonright \alpha$  followed by a fresh letter below  $F(\alpha)$  to obtain  $a_\alpha$ , and use a fresh letter above for  $b_\alpha$ . The inequalities are decided at the earlier index. A code differing from  $F$  at  $\xi$  lies on the appropriate far side of every approximant with index greater than  $\xi$ . Uniform choices come from the least completion code in the fixed global well-order. Finally, Theorem 42.4, with  $F$  on one side, puts a code strictly beyond any proposed set family on the other side.  $\square$

#### 42.2.5 The diagonal limitation on all set codes

**Theorem 42.7** (No exhaustive uniform evaluator). *No single class relation with set indices can have, among its rows, every class bijection  $\text{Ord} \rightarrow \mathbf{No}$ .*

*Proof.* Extend an alleged evaluator to all set indices by giving unused indices empty rows. Fix a class bijection  $\alpha \mapsto c_\alpha$  from  $\text{Ord}$  to the universe of sets. Partition  $\text{Ord}$  into pairs

$$(2 \cdot \alpha, 2 \cdot \alpha + 1).$$

The multiplication is ordinal multiplication with 2 on the left; these adjacent pairs partition all ordinals.

On pair  $\alpha$ , use the two values  $b(2 \cdot \alpha), b(2 \cdot \alpha + 1)$ . Put the second value first if the row  $c_\alpha$  reports  $b(2 \cdot \alpha)$  at coordinate  $2 \cdot \alpha$ ; otherwise put the first value first. Put the other value second. This elementary class construction defines a bijection  $E$ . Whenever the row  $c_\alpha$  is the graph of a function, it differs from  $E$  at that even coordinate. Thus every valid row misses  $E$ .  $\square$

The theorem does not say that  $\mathcal{P}_s(b)$  is too small to be dense. It says that density and exhaustive representation are different notions. It also does not prohibit defining some enumerations from set parameters; it prohibits one uniform evaluator from covering *all* of them.

### 42.3 Source 15: the ordinary class of bounded-support codes

Fix a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . A set-supported permutation of  $\text{Ord}$  equals the identity outside some ordinal. Code it by a permutation  $\pi : \gamma \rightarrow \gamma$  and extend it by the identity. Normalize  $\gamma = \sup\{\xi + 1 : \pi(\xi) \neq \xi\}$ , using  $\gamma = 0$  and the empty code for the identity. This makes codes unique. Let  $\mathcal{P}_{\text{bd}}(b)$  be the ordinary class of these set codes, with evaluation

$$E_d(\xi) = b(\pi_d(\xi)).$$

The values of an evaluation are sets; the whole evaluation is a class function uniformly given by a class relation.

**Lemma 42.8** (Every injective set prefix has a core completion). *Every injective set word  $p : \alpha \rightarrow \mathbf{No}$  extends to some  $E_d$ ,  $d \in \mathcal{P}_{\text{bd}}(b)$ , and a completion can be chosen uniformly.*

*Proof.* The map  $q = b^{-1} \circ p$  is a set injection into  $\text{Ord}$ . Choose an infinite initial cardinal  $\nu$  greater than  $\alpha$  and every member of  $\text{ran}(q)$ . The complements of  $\alpha$  and of  $\text{ran}(q)$  in  $\nu$  both have cardinality  $\nu$ . A bijection between them extends  $q$  to a permutation of  $\nu$ . Extend by the identity and normalize. The least permissible  $\nu$  and a fixed global well-order choose a particular complement bijection, so the construction is uniform.  $\square$

We use the following established core theorem from the prior report:  $\mathcal{P}_{\text{bd}}(b)$ , with the induced native lexicographic order, fills every set cut and is class-order-isomorphic to  $\mathbf{No}$  (theorem 19.10 and corollary 19.15; Theorem 19.10 and Corollary 19.15 at the pin, which source 15 cites as “Theorems

19.10 and 19.15"). Its proof combines prefix completion with fresh separators and then applies the ordinary class back-and-forth theorem. This is a statement about a class of sets, so it falls within surreal universality.

For an injective set word  $p : \alpha \rightarrow \mathbf{No}$ , its cylinder is

$$C_p = \{d \in \mathcal{P}_{\text{bd}}(b) : E_d \upharpoonright \alpha = p\}.$$

Cylinders are nonempty and convex. Those of a fixed length partition  $\mathcal{P}_{\text{bd}}(b)$  and are strictly ordered by their different prefixes. Each cylinder has no set cofinal or cointial subset: given a set of its codes, choose a fresh label above or below all their values at coordinate  $\alpha$ , append it to  $p$ , and complete it using [Lemma 42.8](#).

#### 42.4 Source 16: the bounded-support core of global enumerations

Work in GBC and fix a baseline bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . A normalized bounded-support code is a set permutation  $\sigma : \lambda \rightarrow \lambda$  of an ordinal satisfying

$$\lambda = \sup\{\xi + 1 : \sigma(\xi) \neq \xi\}.$$

The empty permutation is permitted. Extend  $\sigma$  by the identity above  $\lambda$ , obtaining  $\hat{\sigma}$ , and evaluate the code as  $E_\sigma = b \circ \hat{\sigma}$ . Let  $\mathcal{P}_{\text{bd}}(b)$  be the ordinary class of these set codes, ordered by comparison of their evaluations. Normalization makes the coding injective. The evaluations are class functions, but their codes are sets; the distinction is essential.

**Lemma 42.9** (Uniform completion of every set prefix). *Every injective set function  $p : \delta \rightarrow \mathbf{No}$  extends to an evaluation  $E_q$  with  $q \in \mathcal{P}_{\text{bd}}(b)$ . A completion can be selected uniformly in  $p$ .*

*Proof.* The function  $b^{-1} \circ p$  is a set injection into  $\text{Ord}$ . Choose an infinite initial cardinal  $\gamma$  strictly above  $\delta$  and all the ordinals in its range. The unused positions and unused values below  $\gamma$  both have cardinality  $\gamma$ . A bijection of their complements extends this injection to a permutation of  $\gamma$ . Extend by the identity and normalize. Choose the least suitable set bijection in a fixed global well-order to make the construction uniform.  $\square$

**Theorem 42.10** (Uniform set-cut interpolation). *Let  $(F_i)_{i \in I}$  and  $(G_j)_{j \in J}$  be uniformly set-indexed families in  $\mathfrak{J}$  satisfying  $F_i < G_j$  for all cross pairs. Either side may be empty. Some injective set prefix  $p$  has every continuation in  $\mathfrak{J}$  strictly between the families. In particular, a member of  $\mathcal{P}_{\text{bd}}(b)$  interpolates them.*

*Proof.* When both sides are nonempty, their cross first-disagreement ordinals form a set by Class Replacement. Bound them strictly below  $\beta$ ; otherwise put  $\beta = 0$ . Follow a prefix  $p$ , calling an input row active if it agrees with the whole prefix. Discarded lower rows are below every continuation and discarded upper rows above every continuation.

At coordinate  $\delta = \text{dom}(p)$ , let  $A$  and  $B$  be the sets of next values of the active lower and upper rows. Every value of  $A$  is at most every value of  $B$ . If they are disjoint, use the fresh-separator lemma ([lemma 5.2](#)) to append an unused value strictly between them and stop. This also handles empty sides. If they meet, their intersection is a singleton; append that forced value and retain only the rows using it. A surviving injective row guarantees freshness. At limits take unions of the set histories.

The process cannot retain both a lower and an upper row through length  $\beta$ , since such rows would agree beyond their bounded first disagreement. Thus it stops at a set stage, with all comparisons decided. [Lemma 42.9](#) provides the bounded-support completion. Surjectivity was not used for any input row.  $\square$

**Corollary 42.11** (A surreal-sized dense core). *The ordinary class order  $\mathcal{P}_{\text{bd}}(b)$  satisfies the set-cut property and is isomorphic to  $\mathbf{No}$ . It is dense in both  $\mathfrak{J}$  and  $\mathfrak{W}_{\text{sl}}$  in the sense of their definable comparisons. It has no set cofinal or coinitial subset. The interpolation theorem works for every chosen baseline, including constraints whose rows do not belong to that baseline's core.*

*Proof.* Apply Theorem 42.10 to evaluations of sets of codes. The core is proper, since transpositions of position zero with arbitrary ordinal positions give distinct codes. Class universality and uniqueness (proposition 5.13 and theorem 5.17) then gives the isomorphism. The remaining assertions are instances of the same interpolation theorem.  $\square$

**Status.** The completion and interpolation theorems, and the core isomorphism, are results of the earlier report (Part IV: lemma 19.4, theorem 19.10, and corollary 19.15). They are proved here because they are the inputs to the new recognition theorem below. Neither  $\mathcal{P}_{\text{bd}}(b) \cong \mathbf{No}$  nor its density says that all global enumerations can be coded by sets.

#### 42.4.1 Canonical first cores and residual fibers

For  $R \in \mathfrak{W}_{\text{all}}$ , the first block of source 16's canonical block decomposition (its Theorem 10.1, [merge] theorem 52.6) is

$$I_R = \{x \in \mathbf{No} : \{y : yRx\} \text{ is a set}\}.$$

It is always a proper class and has a unique enumeration  $F_R : \text{Ord} \rightarrow I_R$ . If  $I_R$  were a set, the least point outside it would have exactly that set of predecessors, a contradiction.

Conversely every  $F \in \mathfrak{J}$  occurs as such a first core: enumerate its range by  $F$  and then append any class well-order of its complement. The points in the range have set predecessors; every residual point has that whole proper range before it. Restriction and concatenation therefore identify the fiber over  $F$  with

$$\mathcal{W}(Y_F), \quad Y_F = \mathbf{No} \setminus \text{ran}(F), \quad (\text{S16.12})$$

where for a proper  $Y_F$  the notation denotes the raw class comparison. Each fiber is convex. Distinct fibers are ordered by the first ordinal disagreement of their cores. Symbolically,

$$\mathfrak{W}_{\text{all}} \cong \sum_{F \in \mathfrak{J}} \mathcal{W}(Y_F). \quad (\text{S16.13})$$

This is a summary of uniform relational constructions, not a same-level class sum whose index elements are proper classes. The set-like order  $\mathfrak{W}_{\text{sl}}$  consists precisely of the exhaustive, empty-residual fibers. A singleton fiber need not be exhaustive: omitting one label produces a unique raw completion of type  $\text{Ord} + 1$ .

#### 42.5 Source 17: the ordinary class core inside the global comparison

From now on work in GBC and fix  $b : \text{Ord} \rightarrow \mathbf{No}$ . Let  $\mathfrak{J}$  be the predicate of injective class functions  $\text{Ord} \rightarrow \mathbf{No}$ , and let  $\mathfrak{W}_{\text{sl}}$  be its surjective subpredicate. Comparisons are made at the least ordinal of first disagreement. These are predicates on class variables, not classes whose members are proper classes.



### 42.5.1 Canonical codes and interpolation

A *support code* is a set bijection  $\sigma : S \rightarrow S$ , where  $S \subseteq \text{Ord}$  is a set and  $\sigma(\xi) \neq \xi$  for every  $\xi \in S$ . The empty code is allowed. Extend it by the identity off  $S$  and put  $E_\sigma = b \circ \hat{\sigma}$ . Let  $\mathcal{P}_s(b)$  be the ordinary class of these codes, ordered by the associated enumerations. The fixed-point-free condition makes the coding unique. A global enumeration has such a code exactly when it agrees with  $b$  at every sufficiently large ordinal. By [Corollary 49.10](#), every injective set prefix occurs in this core.

We recall the central interpolation theorem of the source, with its proof because the later recognition results depend on its full strength for *injective, possibly nonexhaustive* rows.

**Lemma 42.12** (Avoiding a set of labels). *If  $L, R, Z$  are sets of surreal numbers with  $L < R$ , there is  $z \notin Z$  with  $L < z < R$ . Either side of the cut may be empty.*

*Proof.* The surreal set-cut property supplies a strict separator of any two separated sets. Repeat that construction  $|Z|^+$  times, each time adding all earlier chosen separators to the lower side. Every stage has a set history and gives a new separator below all of  $R$ . The resulting set has more elements than  $Z$ , so one separator is not forbidden. This is set-valued recursion with class parameters, not class-valued recursion.  $\square$

**Theorem 42.13** (Core interpolation; source result). *Let  $(A_i)_{i \in I}$  and  $(B_j)_{j \in J}$  be uniformly set-indexed families in  $\mathfrak{J}$ , with  $A_i <_{\text{lex}} B_j$  for every  $i, j$ . There is an injective set prefix  $p$  such that every row in  $\mathfrak{J}$  extending  $p$  lies strictly between both families. In particular there is such a row with a code in  $\mathcal{P}_s(b)$ . Empty index sets are permitted.*

*Proof.* When both sides are nonempty, Class Replacement collects their cross first-difference coordinates into a set. Choose an ordinal  $\beta$  strictly above all these coordinates. If one side is empty take  $\beta = 0$ .

Build an injective prefix, retaining as *active* the rows agreeing with it. At stage  $\alpha$  form the sets of active next labels

$$L_\alpha = \{A_i(\alpha) : A_i \text{ active}\}, \quad R_\alpha = \{B_j(\alpha) : B_j \text{ active}\}.$$

Every lower label is at most every upper label. Their intersection has at most one member. If it is empty, choose a fresh strict separator by [Lemma 42.12](#), append it, and stop. All active inequalities are now strict. If the intersection is  $\{z\}$ , append  $z$  and keep only rows with next value  $z$ . This value is fresh because a surviving row is injective. A discarded lower row is below every later extension; a discarded upper row is above every later extension. This is the invariant throughout the construction. At limits take unions of the set histories.

Both sides cannot still be active after a common prefix of length  $\beta$ , since an active cross pair would disagree only at or beyond  $\beta$ . Thus the procedure stops by the next step. Complete the final prefix in  $\mathcal{P}_s(b)$  using [Corollary 49.10](#). All recursions and collections in this proof have set-sized stages and values.  $\square$

**Corollary 42.14** (A dense surreal order core; source result). *The class order  $\mathcal{P}_s(b)$  is isomorphic to the numerical order of  $\mathbf{No}$ . It is order-dense in the global comparisons of both  $\mathfrak{W}_{\text{sl}}$  and  $\mathfrak{J}$ : between any two distinct rows there is a coded row. Every uniformly set-indexed separated cut of those comparisons has a core interpolant.*

*Proof.* The last statements are [Theorem 42.13](#). In particular  $\mathcal{P}_s(b)$  has the same set-cut interpolation property as  $\mathbf{No}$ . Under Global Choice enumerate each of these proper classes in a set-like well-order. Construct a back-and-forth partial order-isomorphism along  $\text{Ord}$ . At each stage its domain and range

are sets, so the required cut is a set cut; interpolation provides a fresh image or preimage. Global choice selects a least permissible choice. Every history at a set ordinal is a set, so elementary class comprehension together with set-valued transfinite recursion gives the full class map. Alternating forth and back ensures exhaustiveness on both sides.  $\square$

This is an isomorphism of *linear orders*, not of surreal fields or of structures with the simplicity relation. The additional structure studied in [14] cannot be inferred from numerical order alone. Nor is the core exhaustive in  $\mathfrak{W}_{\text{sl}}$ .

**Corollary 42.15** (Universality for ordinary class linear orders). *Every given linear order on a class of sets embeds by a class map into  $\mathcal{P}_s(b)$ , hence into  $\mathbf{No}$ ,  $\mathfrak{W}_{\text{sl}}$ , and  $\mathfrak{W}_{\text{all}}$  in their stated senses.*

*Proof.* Global choice supplies a set-like enumeration of the given class (or a set enumeration if its carrier is a set). At each stage, previous images form a set. Partition them by whether their source labels are below or above the next source label; interpolation in  $\mathcal{P}_s(b)$  supplies a fresh image. This forth-only recursion gives the required embedding. The core inclusion and Corollary 42.14 give the other targets. This is the source’s class-universality argument.  $\square$

The hypothesis “class of sets” cannot be dropped and replaced with “all class well-orders”. The latter is the higher-order predicate  $\mathfrak{W}_{\text{all}}$ , not a carrier to which this recursion theorem applies. This distinction is compatible with the no-coding theorem that follows.

#### 42.5.2 Why a dense class of codes is not a universal list

**Theorem 42.16** (Uniform diagonal obstruction; source result). *No ordinary class of set indices with one uniform evaluation relation lists all class well-orders of  $\mathbf{No}$ . A missing order can always be chosen set-like.*

*Proof.* Global choice permits reindexing a proposed nonempty list by  $\text{Ord}$ ; a set-sized list can be extended by repetitions. Write its rows as  $R_\alpha$ . Partition the baseline labels into the pairs

$$\{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\} = \{a_\alpha, c_\alpha\}, \quad a_\alpha < c_\alpha.$$

The products here are ordinal products with 2 on the left, so these successive two-element blocks partition  $\text{Ord}$ . In block  $\alpha$ , list the two labels in the order opposite to their relative order under  $R_\alpha$ . Keep the blocks in increasing  $\alpha$  order. Elementary comprehension defines a class bijection  $E : \text{Ord} \rightarrow \mathbf{No}$ . Its induced well-order disagrees with  $R_\alpha$  on that pair, for every  $\alpha$ . The empty-list case is immediate.  $\square$

Each input row and the diagonal output are legitimate classes. The forbidden object is a same-level exhaustive decoder, not an individual large relation. Strengthening comprehension does not make a class into a permissible element of another class of the same two-sorted theory. An external set universe, a higher sort, or a stated uniform operator provides a different, legitimate interpretation.

#### 42.6 Source 18: the set-like global order and its ordinary class core

Assume GBC and fix a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . The set-like global order can be studied through its ordinal enumerations: if  $e \neq f$ , the nonempty class  $\{\alpha : e(\alpha) \neq f(\alpha)\}$  has a least ordinal. Numerical comparison there agrees with source 18’s Theorem 12.3 ([merge] theorem 51.16; theorem 14.7).

### 42.6.1 Canonical bounded-support codes

A permutation  $\pi$  of  $\mathbf{Ord}$  with set-sized support can be encoded by the set graph of its nonfixed pairs; it is the identity elsewhere. This representation is unique. Let  $\mathcal{P}_s(b)$  be the genuine class of all such set codes, ordered by their evaluated global enumerations  $b \circ \pi$ . The notation distinguishes the set code from the class function that it evaluates.

**Lemma 42.17** (Completion of every set prefix). *Every injective set word  $p : \delta \rightarrow \mathbf{No}$  extends an evaluated member of  $\mathcal{P}_s(b)$ .*

*Proof.* The  $b$ -indices of the labels in  $\text{ran } p$  form a set. Choose an infinite initial cardinal  $\lambda$  greater than  $\delta$  and all these indices. The first  $\lambda$  values of  $b$  contain  $\text{ran } p$ . List  $p$  first, followed by a bijection of the unused values of this initial block in length  $\lambda$ . Since  $\delta + \lambda = \lambda$ , the result permutes that entire block. Leave all later positions unchanged. This is a bounded-support permutation extending  $p$ .  $\square$

**Theorem 42.18** (Uniform set-cut interpolation). *Let  $(e_i)_{i \in I}$  and  $(f_j)_{j \in J}$  be uniformly presented set-indexed families of global ordinal enumerations, and suppose  $e_i < f_j$  for all  $i, j$ . There is  $g \in \mathcal{P}_s(b)$  strictly between them. In particular,  $\mathcal{P}_s(b)$  fills every set cut and is order-isomorphic to  $\mathbf{No}$  in GBC. It is order dense in  $\mathfrak{W}_{\text{sl}}$ .*

*Proof.* When both families are nonempty, the first disagreements of all pairs  $(e_i, f_j)$  form a set of ordinals by Class Replacement. Take a bound on them. Run a common-prefix search: keep only the family members extending the current prefix; numerical next values on the lower side are at most those on the upper side. If the sets of next values meet, they meet in a unique value, which is appended. If they do not meet, choose a fresh surreal strictly between them, append it, and stop. At a limit stage take the union of the preceding set prefixes.

The search cannot continue through the bound on all pairwise first disagreements while both active sides remain nonempty. When a side becomes empty, the surreal set-cut property also supplies a fresh next value. If one original side is empty, the search stops at the first step. Each inactive lower member is below the whole current cone, and each inactive upper member is above it. Thus the resulting successor prefix puts every completion strictly between the given families. Apply [Lemma 42.17](#).

For the isomorphism, perform set-valued back-and-forth along fixed set-like enumerations of the two proper classes  $\mathcal{P}_s(b)$  and  $\mathbf{No}$ . At each stage only a set of previously matched points must be respected, and interpolation supplies the next point. Taking singleton families in the first assertion proves density in  $\mathfrak{W}_{\text{sl}}$ .  $\square$

The conclusion is stronger than pairwise density and weaker than filling arbitrary proper-class cuts. For example, choose an injective  $\mathbf{Ord}$  word that enumerates  $\mathbf{No} \setminus \{x\}$ . It determines a cut of  $\mathcal{P}_s(b)$  but cannot itself be a global enumeration of  $\mathbf{No}$ . The next section makes the distinction exact.

Class universality of  $\mathbf{No}$  now implies that every supplied class linear order embeds into  $\mathfrak{W}_{\text{sl}}$ , through  $\mathcal{P}_s(b)$ . The same is true inside any nonempty open interval of  $\mathfrak{W}_{\text{sl}}$ : a sufficiently long set prefix and a free numerical interval place a copy of the same core inside it. This is a statement about uniform class-indexed subfamilies. It does not say that  $\mathfrak{W}_{\text{sl}}$  itself is an ordinary class.

### 42.6.2 An elementary obstruction to exhaustive class coding

**Theorem 42.19** (No ordinary class of codes for all global well-orders). *There is no class-indexed uniform evaluator whose values include every class well-order of  $\mathbf{No}$ . The same holds if only set-like orders are required as values.*

*Proof.* Under Global Choice any class of set codes can be listed in an Ord-indexed family, repeating values if the code class is a set. Suppose  $(R_\alpha)_{\alpha \in \text{Ord}}$  were exhaustive. Split the fixed baseline  $b$  into consecutive pairs

$$\{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\}, \quad \alpha \in \text{Ord}.$$

In pair  $\alpha$ , choose the orientation opposite to the relative order of these two labels in  $R_\alpha$ . Keep the pair blocks in their original ordinal order. This is a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ , hence a set-like well-order. It differs from  $R_\alpha$  on the relative order of its designated pair, for every  $\alpha$ , contradicting exhaustiveness. The orientation class is defined elementarily from the proposed evaluator.  $\square$

This is the relevant Cantor-style diagonal. It blocks a uniform “all codes are sets” shortcut. Individual global well-orders can still be parameters, and any specified uniform family can be studied. Kelley–Morse comprehension strengthens which classes exist; it does not make proper classes into members of other ordinary classes.

## 43 A trichotomy for class cuts of the skeleton (source 14)

### 43.1 Normalization and the three constructions

A *proper normalized cut* of  $\mathcal{P}_s(b)$  is a class  $D \subseteq \mathcal{P}_s(b)$  which is nonempty, proper, downward closed, and has no greatest member. Its complement is allowed to have a least member. Thus the point  $\sigma$  is represented by the lower set  $\{\tau : \tau < \sigma\}$ , not by the lower set that includes  $\sigma$ . This convention is essential when we calculate characters.

We write  $D_p^-$  and  $D_p^+$  for the two edges of a nonempty-prefix cylinder:

$$\begin{aligned} D_p^- &= \{\sigma : (\forall \tau \in C_p) \sigma < \tau\}, \\ D_p^+ &= \{\sigma : (\exists \tau \in C_p) \sigma < \tau\}. \end{aligned}$$

Since  $C_p$  has no endpoints,  $D_p^-$  consists of the codes below the whole cylinder, whereas  $D_p^+$  consists of those codes together with all of  $C_p$ . Both are proper normalized cuts. The empty-prefix edges are the improper endpoint cuts and are excluded.

For a set prefix  $p$  and a nonempty proper initial class  $H \subset A_p$  with no greatest member and with  $A_p \setminus H$  having no least member, define a *numerical-gap cut*

$$D_{p,H} = D_p^- \cup \{\sigma \in C_p : E_\sigma(\lambda) \in H\}, \quad \lambda = \text{dom } p.$$

Here  $D_\emptyset^- = \emptyset$  by convention. The nonprincipality conditions prevent this notation from duplicating the two edges of a child cylinder.

Finally, for  $F \in \mathfrak{J}$  define the *branch trace*

$$D_F = \{\sigma \in \mathcal{P}_s(b) : E_\sigma <_{\text{lex}} F\}.$$

[Lemma 41.5](#) shows that it is a proper normalized cut. The branch need not be exhaustive. If  $F = E_\sigma$ , the cut is principal in the normalized sense; otherwise it has no point of the skeleton at its boundary.

### 43.2 Crossing cylinders and reconstruction

**Definition 43.1.** A prefix  $p$  *crosses*  $D$  if its cylinder meets both sides:

$$\text{Cross}_D(p) \iff (\exists \sigma \in C_p \cap D)(\exists \tau \in C_p \setminus D).$$

Let  $T_D$  be the class of crossing set prefixes.

All displayed witnesses are sets. Therefore  $T_D$  is obtained by *elementary* class comprehension from  $D$  and  $b$ .

**Lemma 43.2** (The crossing-chain lemma). *For every proper normalized cut  $D$ , the class  $T_D$  contains the empty prefix, is downward closed under prefix restriction, and is a chain under prefix extension. It contains at most one prefix of any given length. Its union is an injective function with domain either a set ordinal or all of Ord.*

*Proof.* Properness gives the empty prefix, and any ancestor of a crossing cylinder also crosses. If incomparable prefixes  $p, q$  both crossed, order their disjoint cylinders as  $C_p < C_q$ . An excluded point of  $C_p$  would then lie below an included point of  $C_q$ , contradicting downward closure. Thus they are comparable, which also proves uniqueness at each length. Their union is a function and is injective because every two of its coordinates occur together in a sufficiently long member of the chain. Its domain is an initial subclass of Ord. Such a subclass is all of Ord or a set ordinal. In the latter case, class Replacement on that set domain makes the union a set function.  $\square$

**Theorem 43.3** (Boundary trichotomy). *Every proper normalized class cut  $D$  of  $\mathcal{P}_s(b)$  has exactly one of the following forms:*

- (i) a cylinder edge  $D_p^-$  or  $D_p^+$  for a nonempty injective set prefix  $p$ ;
- (ii) a numerical-gap cut  $D_{p,H}$  with the nonprincipality conditions specified above;
- (iii) a branch trace  $D_F$  for an injective class function  $F : \text{Ord} \rightarrow \mathbf{No}$ .

*The data within the appropriate form, including the edge direction, are unique. The alternatives and their data can be recovered from the crossing-prefix class by elementary formulas with class parameters.*

*Proof.* We distinguish whether the lengths occurring in  $T_D$  are unbounded.

*Unbounded case.* Let  $F = \bigcup T_D$ . Its domain is Ord. For a code  $\sigma$  with  $E_\sigma \neq F$ , let  $\delta$  be their first disagreement. Take a crossing prefix of length greater than  $\delta$ . If  $E_\sigma <_{\text{lex}} F$ , then  $\sigma$  is below that entire cylinder and in particular below one of its included points, so  $\sigma \in D$ . If  $F <_{\text{lex}} E_\sigma$ , it is above an excluded point of that cylinder and cannot belong to  $D$ . If  $E_\sigma = F$ , membership in  $D$  would make  $\sigma$  its greatest member, since all larger codes have just been excluded. This is prohibited by normalization. Therefore  $D = D_F$ .

*Bounded case with no longest crossing prefix.* The union  $p$  has nonzero limit length  $\lambda$ . It does not cross, so either  $C_p \subseteq D$  or  $C_p \cap D = \emptyset$ . For every code outside  $C_p$ , its first disagreement with  $p$  occurs before  $\lambda$ ; a longer crossing prefix then decides its membership exactly as in the unbounded case. Hence  $D = D_p^+$  in the first alternative and  $D = D_p^-$  in the second.

*Bounded case with a longest crossing prefix  $p$ .* No child  $C_{p \frown (x)}$  crosses. Each child is consequently wholly included or wholly excluded. Let

$$H = \{x \in A_p : C_{p \frown (x)} \subseteq D\}.$$

It is an initial class of  $A_p$ . Because  $C_p$  crosses and the child cylinders partition it,  $H$  is nonempty and proper. Points outside  $C_p$  are below all of it or above all of it; the crossing witnesses show that

exactly those below belong to  $D$ . Thus  $D$  has the formula used to define  $D_{p,H}$ . If  $H$  has a greatest member  $x$ , that formula is  $D_{p\smallfrown(x)}^+$ . If  $A_p \setminus H$  has a least member  $x$ , it is  $D_{p\smallfrown(x)}^-$ . Both possibilities cannot occur, since  $A_p$  is dense. If neither occurs, we have precisely alternative (ii).

For uniqueness, in alternative (iii) the crossing prefixes are exactly all set restrictions of  $F$ ; in alternative (ii) they are exactly  $p$  and its restrictions; and at an edge they are exactly the *proper* restrictions of its prefix. If the last chain has no longest member, its union recovers the limit prefix and membership of its cylinder recovers the direction. If it has a longest member, the included child labels have a maximum or the excluded child labels have a minimum; that endpoint recovers the additional label and the direction. For a numerical gap neither endpoint exists. These descriptions distinguish the alternatives, and also recover every parameter uniquely.  $\square$

*Remark 43.4* (What the theorem does not construct). It does not collect all class cuts into an ordinary class, nor insert an Ord-length sign sequence as a surreal number. The theorem is a statement about an arbitrary supplied class  $D$ . Its elementary formulas can be used uniformly in models, but the virtual family of every such  $D$  still occupies a higher logical level.

### 43.3 Examples distinguishing the alternatives

For the one-letter prefix  $(0)$ , the cut  $D_{(0)}^-$  lies just below all core enumerations beginning with 0. There is no numerical “largest first entry below 0.” It is a cylinder boundary, not a branch trace. At the empty prefix, take the numerical gap

$$H = \{x \in \mathbf{No} : (\exists n \in \mathbb{N}) x < n\}.$$

Its complement is the class of surreals at least as large as every natural number. It has no least member: from such a surreal  $y$ , the number  $y - 1$  is still in the complement and is smaller. The class  $H$  has no greatest member. This produces a cut of type (ii), with a countable cofinal left side but no set-sized coinital right side. Finally  $F(\alpha) = \alpha$  is an injective branch of type (iii) which omits, for example,  $-1$ . Its trace is coherent at every set length but is not realized by any member of  $\mathfrak{W}_{\text{sl}}$ . Exhaustiveness is not implied by coherence.

*Note 43.5* ([merge]; the three examples in the repository). The first-label cut at 0 (with its upper-edge twin) is printed also by sources 15 and 16 (example 46.10 and source 16’s table after theorem 46.3); the numerical gap  $\{x : \exists n x < n\}$  also by sources 15 and 17 (examples 46.11 and 46.23). The branch  $F(\alpha) = \alpha$ , which omits  $-1$ , is source 09’s theorem 19.43 (printed in Part IV), which source 14 does not cite; source 13 recovers it as example 46.17, crediting this report, and source 15 prints it again (after example 46.11). Its one-label variant, a bijection  $\text{Ord} \rightarrow \mathbf{No} \setminus \{r\}$ , is source 11’s theorem 19.42, printed again by sources 16 and 17 (source 16’s table, example 46.25), which do not cite it. The limit boundary below the cylinder of  $p(n) = n$  ( $n < \omega$ ), which source 14 does not use, is printed by sources 13 (end of its Section 9), 15 (example 46.11), 16 (its table) and 17 (example 46.24, for any injective  $p$  of length  $\omega$ ).

## 44 Characters, saturation, and principal cuts (source 14)

We use  $\text{Ord}$  as a *character symbol*: it means that a uniform ordinal-indexed cofinal or coinital family exists, while no set-sized one exists. It is not an ordinary cardinal. For any nonempty class linear order with no set-sized cofinal subset, global choice permits an increasing cofinal Ord-sequence: at each set stage dominate the earlier choices and the elements seen so far in a global enumeration. If no such upper choice existed, the already considered set would be cofinal. This is set-valued



recursion. The dual argument handles coinitality. A set-sized character is the least cardinality of a cofinal or coinital subset; if there is a corresponding endpoint it is 1.

**Theorem 44.1** (Cut-character calculation). *Write the character of a normalized cut as (cf  $D$ , ci( $\mathcal{P}_s(b) \setminus D$ )). For the alternatives of Theorem 43.3, the following formulas hold.*

| <i>Cut</i>                                      | <i>Character</i>                   |
|-------------------------------------------------|------------------------------------|
| $D_F$ , with $F = E_\sigma$ for some core code  | (Ord, 1)                           |
| $D_F$ , with $F$ not represented by a core code | (Ord, Ord)                         |
| $D_p^-$ , dom $p = \lambda$ a nonzero limit     | (cf $\lambda$ , Ord)               |
| $D_p^+$ , dom $p = \lambda$ a nonzero limit     | (Ord, cf $\lambda$ )               |
| $D_p^-$ or $D_p^+$ , dom $p$ a successor        | (Ord, Ord)                         |
| $D_{p,H}$                                       | (cf $H$ , ci( $A_p \setminus H$ )) |

For the last row both characters cannot be set-sized. At every actual point of  $\mathcal{P}_s(b)$ , both strict local characters are Ord.

*Proof.* For a branch  $F$ , complete a prefix agreeing with  $F$  before  $\alpha$  and taking a fresh smaller entry at  $\alpha$ ; call the result  $a_\alpha$ . Similarly obtain  $b_\alpha$  with a larger entry. If  $\alpha < \beta$ , first difference gives

$$a_\alpha < a_\beta <_{\text{lex}} F <_{\text{lex}} b_\beta < b_\alpha.$$

Every code below or above  $F$  is surpassed on the appropriate side by one of these approximants, once  $\alpha$  exceeds its first disagreement. A set of codes differing from  $F$  has a set of first disagreements and hence a common ordinal bound; a later approximant defeats its alleged cofinality or coinitality. If  $F$  itself is a core point, the complement of  $D_F$  has that point as its least member. This gives (Ord, 1), not (Ord, Ord); only the strict right side has character Ord.

At the lower edge of a limit-length prefix  $p$ , take a cofinal sequence of coordinates  $\alpha < \lambda$  and complete a word that first differs from  $p$  there by a smaller entry. These codes are cofinal below the cylinder. Conversely the first-disagreement coordinates of any cofinal family must be cofinal in  $\lambda$ : a code which follows  $p$  longer would otherwise be above the entire family. This proves the character cf  $\lambda$ . On the other side the cylinder has no set-sized coinital subset by Proposition 41.4, and lies below all other points of the complement. Its character is Ord. The upper-edge calculation is dual.

For a successor-length prefix  $p = q \frown (x)$ , any set of points below  $C_p$  that first differ before the last coordinate can be surpassed by one following  $q$ . Among those which follow  $q$ , their entries at the last coordinate form a set below  $x$ . Lemma 41.1 provides a fresh entry between that set and  $x$ . Completion again surpasses the whole family while remaining below  $C_p$ . Thus the lower side of the lower edge has no set-sized cofinal subset. The opposite side follows from the cylinder property. The upper edge is the dual case.

For a numerical gap, project included points in  $C_p$  to their next-coordinate labels in  $H$ . A cofinal set of labels, with one completed word above each chosen label when necessary, produces a cofinal family of codes. Conversely a cofinal family of codes must have cofinal projected labels: otherwise a child cylinder with a larger label in  $H$  surpasses it. The absence of a greatest member of  $H$  removes any dependence on the internal character of one child. The dual argument proves the right character. If both were sets, cofinal labels on the left and coinital labels on the right would constitute a set cut of  $A_p$  with no separator, contradicting Lemma 41.1. Finally the branch argument at a core point already proves the assertion about strict local characters.  $\square$

**Corollary 44.2** (Recovery of the core saturation theorem). *Every separated pair of sets of codes  $A, B \subseteq \mathcal{P}_s(b)$  has a strict separator in  $\mathcal{P}_s(b)$ . In particular the core is an  $\eta_{\text{Ord}}$ -order: it fills all set-sized strict cuts, including those with an empty side.*

*Proof.* Empty-side bounds follow by choosing a first label above or below all first labels in the specified set. Suppose both sides are nonempty and no separator exists. If  $A$  has a greatest point  $a$ , then  $B$  would be coinital in the strict right side of  $a$ , contradicting its local character. The dual argument applies if  $B$  has a least point. Otherwise let  $D = \{\sigma : (\exists a \in A) \sigma < a\}$ . It is a proper normalized cut with  $A$  cofinal. Absence of a separator makes  $B$  coinital in its complement: any point there below all of  $B$  would be strictly above all of  $A$ . The character table has no nonprincipal cut with two set-sized characters, a contradiction.  $\square$

**Corollary 44.3** (The known surreal order type). *There is a class order isomorphism  $\mathcal{P}_s(b) \cong \mathbf{No}$  in GBC.*

*Proof.* Both are proper class linear orders filling all set cuts. Fix set-like global enumerations and perform back-and-forth along  $\text{Ord}$ . At a set stage the existing partial map has set domain and range; the next point's lower and upper constraints are sets and admit a fresh image. The union is an elementary definable class bijection preserving order. This is the standard argument, also used by the predecessor; the isomorphism statement itself is inherited, not claimed as a new result.  $\square$

The character calculation explains why the core is not Dedekind complete. For instance the increasing core points whose first entries are natural numbers have upper bounds but no supremum. Filling *strict* set cuts supplies another point between all those entries and any proposed upper bound. The countable/non-set-sized numerical-gap character above records this failure rather than contradicting saturation.

*Note 44.4* ([merge]; which rows are new). The branch rows of [theorem 44.1](#) are source 11's [proposition 19.27](#) and source 09's [theorem 19.43](#) (( $\text{Ord}, \text{Ord}$ ) at a non-core branch); [corollary 44.2](#) is [theorem 19.10](#) for sets of codes and [corollary 44.3](#) is [corollary 19.15](#), both stated by source 14 as recoveries. New are the characters of the edge and numerical-gap cuts and the statement that no nonprincipal cut has two set-sized characters. The same table is proved, case by case, by source 13 ([proposition 46.19](#)), source 15 ([proposition 46.12](#)), source 16 (the paragraph after its table) and source 17 for non-core branches ([proposition 46.26](#)); those statements are printed with their sources' recognition theorems in [section 46](#).

## 45 Recognizing exhaustive cuts and preserving old cuts (source 14)

**Theorem 45.1** (Exhaustive-cut recognition). *Let  $D$  be a proper normalized class cut of  $\mathcal{P}_s(b)$ . There exists  $E \in \mathfrak{W}_{\text{sl}}$  with  $D = D_E$  if and only if both conditions hold:*

$$\forall \alpha \in \text{Ord} \exists p [\text{dom } p = \alpha \wedge \text{Cross}_D(p)], \quad (\text{C})$$

$$\forall x \in \mathbf{No} \exists p [\text{Cross}_D(p) \wedge x \in \text{ran } p]. \quad (\text{X})$$

*Such an enumeration is unique, and its graph is given by*

$$E(\alpha) = x \iff \exists p [\text{Cross}_D(p) \wedge \alpha \in \text{dom } p \wedge p(\alpha) = x].$$

*All quantifiers in the criterion and reconstruction range over sets; only  $D$  and  $b$  are class parameters.*

*Proof.* For an exhaustive  $E$ , every restriction  $E \upharpoonright \alpha$  crosses its trace by the smaller and larger completions in [Lemma 41.5](#). Every surreal occurs in one of these restrictions, giving (C) and (X). Conversely (C) and [Lemma 43.2](#) produce an injective union  $F$  on all of  $\text{Ord}$ . Condition (X) makes

it surjective. The unbounded case of [Theorem 43.3](#) gives  $D = D_F$ . The displayed graph formula is exactly the union of the crossing chain; its functionality and uniqueness follow from the chain lemma.  $\square$

**A logical simplification.** For this proper-class alphabet, (X) by itself already implies (C). Indeed, the union of the crossing chain then has range all of  $\mathbf{No}$ ; if its domain were a set ordinal, class Replacement would make that range a set. Its domain is therefore Ord, and prefix closure gives (C). We retain the two conditions in the theorem to distinguish coherent nonexhaustive branches from failures that stop at a set stage; we do not claim that the conditions are logically independent.

This answers the predecessor’s question “Which cuts of the core represent exhaustive branches?” ([research question 113.1](#); source 09’s third direction in [section 113.3](#)) *in the prefix-enriched structure*. A bare linear-order isomorphism of the core does not come equipped with its prefix cylinders or the label evaluation. The theorem does not show that those enrichments are definable from the order relation alone. [Corollary 47.3](#) below proves that an automorphism-invariant bare-order criterion cannot recognize exhaustiveness: some core automorphisms carry exhaustive traces to nonbranch cylinder edges.

The criterion also separates two notions of completion. An injective branch always gives a cut coherent through every set stage. It gives a global enumeration only if it eventually visits every label. No condition at a single bounded stage can enforce the latter assertion. For the branch  $F(\alpha) = \alpha$ , (C) holds and (X) fails at  $x = -1$ . The two cylinder-edge alternatives fail (C) at a bounded length; a numerical-gap alternative also fails (C), because there is no crossing child after its last crossing prefix.

[merge] The second half of source 14’s Section 7, on preserving old cuts under class extensions (its [theorem 48.1](#), the corollary and the remark after it), is printed in [section 48](#).

## 46 Five further proofs of the represented-cut theorem (sources 16, 15, 13, 17 and 18)

*Note 46.1* ([merge]; one theorem, six proofs). The theorem that a normalized class cut of the core is the cut of an injective class word exactly when a crossing (straddling) prefix exists at every set length, that the word is then unique and elementary in the cut, and that it is exhaustive exactly when the ranges of these prefixes cover  $\mathbf{No}$ , is proved independently in all six batch-81 sources: source 14’s [theorem 43.3\(iii\)](#) and [theorem 45.1](#), source 16’s [theorem 46.2](#), source 15’s [theorem 46.6](#) with [corollary 46.8](#), source 13’s [theorem 46.15](#), source 17’s [theorem 46.21](#) and source 18’s [theorem 46.28](#) with [corollary 46.31](#). Each source presents it as its own answer to source 11’s [research question 113.1](#); it answers that question ([section 50](#)), and no source has priority over another. The six proofs are the same argument: uniqueness of a crossing prefix at each length, coherence under restriction, the elementary union, and the first disagreement with a code. They are printed as routes because each source builds different examples and corollaries on its own proof.

The analysis of a failing cut is likewise proved five times. Source 14 partitions the proper cuts by their form (edge, numerical gap, branch: [theorem 43.3](#)); sources 13, 15, 16 and 18 partition the failing cuts by their first failed length (successor or limit: [theorems 46.3](#), [46.9](#), [46.18](#) and [46.32](#)). The two partitions are equivalent: a limit failure is an edge of a limit-length cylinder, and a successor failure is an edge of a child cylinder or a numerical gap. Source 15’s successor case is the finest, separating a *forbidden repeated label* (the unused alphabet split at a label already used by the prefix, [example 46.11](#), source 16’s second table row and source 18’s [example 46.33](#)) from a numerical gap

with no surreal separator; in source 14's terms both are numerical-gap cuts  $D_{p,H}$ , the first with  $A_p \setminus H$  and  $H$  separated by a point of  $\text{ran } p$ .

Source 18 states its theorem “in GB with the fixed baseline”. A baseline  $b : \text{Ord} \rightarrow \mathbf{No}$  already yields global choice over GB (theorem 13.1), so the hypothesis is not weaker than GBC; the theorem's own content is that no class choice and no ETR are used beyond the baseline. Sources 13, 14, 15, 16 and 17 say GBC.

#### 46.1 Source 16: exact recognition of cuts represented by branches

Fix the core  $\mathcal{P}_{\text{bd}}(b)$  and its evaluation. For an injective set prefix  $p : \alpha \rightarrow \mathbf{No}$  write

$$C_p = \{q \in \mathcal{P}_{\text{bd}}(b) : E_q \upharpoonright \alpha = p\}.$$

These cylinders are nonempty and convex, and at each fixed length they form an ordered partition. Each cylinder has no endpoint: change the next coordinate of one completion to a fresh smaller or larger surreal and apply Lemma 42.9.

A *lower cut* here means an available class  $\emptyset \neq L \subsetneq \mathcal{P}_{\text{bd}}(b)$  that is initial and has no greatest member. Put  $U = \mathcal{P}_{\text{bd}}(b) \setminus L$ . A prefix *straddles* the cut if  $C_p \cap L$  and  $C_p \cap U$  are both nonempty. This is a property with only set quantifiers and the class parameters  $L, b$ .

**Theorem 46.2** (Prefix-cylinder recognition). *The following are equivalent:*

1. *There is an injective class function  $F : \text{Ord} \rightarrow \mathbf{No}$  such that*

$$L = \{q \in \mathcal{P}_{\text{bd}}(b) : E_q <_{\text{lex}} F\} =: L_F.$$

2. *At every ordinal length  $\alpha$  some injective prefix straddles the cut.*

*When these conditions hold, the prefix at each length is unique and is exactly  $F \upharpoonright \alpha$ . The branch is exhaustive if and only if every surreal occurs in the range of some straddling prefix.*

*Proof.* For a given  $F$ , choose fresh values below and above  $F(\alpha)$  and complete the corresponding one-letter extensions of  $F \upharpoonright \alpha$ . They lie on opposite sides of  $F$ , so its prefix straddles.

Two different prefixes at the same length cannot both straddle an initial cut. Their cylinders are entirely ordered one before the other, and an upper witness in the earlier cylinder would be below a lower witness in the later one. This contradicts initiality. Restrictions of straddling prefixes still straddle. Hence the prefixes  $p_\alpha$  supplied by condition 2 are unique and coherent.

Define the branch directly, by elementary comprehension:

$$F(\alpha) = x \iff \exists p [p \text{ straddles, } \text{dom}(p) = \alpha + 1, p(\alpha) = x]. \quad (\text{S16.14})$$

Here the prefix and the witnesses from both sides are sets. Coherence proves that this is a total injective class function and that  $F \upharpoonright \alpha = p_\alpha$ . This is not a class-valued recursion.

If  $E_q < F$  first differ at  $\delta$ , every point of  $C_{p_{\delta+1}}$  is above  $q$ . One such point is in  $L$ , so  $q \in L$ . If  $F < E_q$ , an upper witness in the same cylinder lies below  $q$ , so  $q \in U$ . Finally, if  $E_q = F$ , placing  $q$  in  $L$  would make it the greatest member of  $L$  by the preceding two implications. Thus it is in  $U$ , and  $L = L_F$ . The prefixes determine  $F$  uniquely, and their range union gives exactly the stated exhaustion criterion.  $\square$

The no-greatest convention matters only at a represented core point: without it, both  $\{q : E_q < F\}$  and  $\{q : E_q \leq F\}$  would straddle every level when  $F = E_q$  for a code. The two improper cuts fail already at the empty prefix and represent no injection.

**Theorem 46.3** (Complete classification of the first failure). *If a lower cut is not represented, let  $\lambda$  be the least length with no straddling prefix. Then  $\lambda > 0$ , and exactly one of the following occurs.*

1. **Successor failure.** *If  $\lambda = \alpha + 1$ , let  $p$  be the unique straddling prefix of length  $\alpha$ . The unused labels  $\mathbf{No} \setminus \text{ran}(p)$  split into nonempty initial and final classes  $A + B$ , with*

$$C_{p \smallfrown (x)} \subseteq L \ (x \in A), \quad C_{p \smallfrown (x)} \subseteq U \ (x \in B).$$

*All codes below  $C_p$  are in  $L$  and all codes above it in  $U$ .*

2. **Limit failure.** *If  $\lambda$  is a nonzero limit, the coherent earlier prefixes have an injective set union  $p$  of length  $\lambda$ . The cut is exactly the lower or upper edge of its cylinder:*

$$L = C_p^- \quad \text{or} \quad L = C_p^- \cup C_p, \quad C_p^- := \{q : \forall c \in C_p \ (q < c)\}.$$

*Proof.* Minimize below any ordinal witnessing failure; this uses no ordering of classes. The empty prefix straddles. At a successor, every nonempty child cylinder is wholly on one side, and the children partition the parent cylinder, which straddles. Both parts of the label split are therefore nonempty, and their order follows from that of the children. The outside-code assertion follows by comparison with a lower and an upper witness in the parent.

At a limit, Class Replacement applied to the unique prefixes indexed by the set  $\lambda$  makes their union a set. Coherence ensures injectivity, and completion ensures a nonempty cylinder. Since the length fails, that cylinder is wholly on one side. Any code outside it first differs from  $p$  at  $\delta < \lambda$ ; the shorter prefix of length  $\delta + 1$  still straddles. Its witnesses force the code to the lower side if it differs downward and to the upper side if upward. Including or excluding the entire limit cylinder gives exactly its two edges.  $\square$

A successor failure need not be a numerical gap with both sides lacking endpoints. It can be an edge of a whole child cylinder, or a cut for which the numerically suitable next label has already been used. The theorem distinguishes these from a failure of coherence at a limit length.

#### 46.1.1 Explicit cuts and their meaning

The following table gives concrete, fully defined examples.

| Lower side                                                                | First failure | Interpretation                                                                           |
|---------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------|
| $\{q : E_q(0) \leq 0\}$                                                   | 1             | Upper edge of the entire zero-first cylinder.                                            |
| $\{q : E_q(0) < 0\} \cup \{q : E_q(0) = 0, E_q(1) < 0\}$                  | 2             | The apparent second separator would repeat the already used zero.                        |
| $C_p^-$ or $C_p^- \cup C_p$ , with $p(n) = n$ for $n < \omega$            | $\omega$      | Every finite prefix straddles, but the whole countable-prefix cylinder lies on one side. |
| $L_F$ for $F : \text{Ord} \xrightarrow{\sim} \mathbf{No} \setminus \{r\}$ | None          | A genuine injective branch, failing only the exhaustion test.                            |

All displayed lower sides have no greatest member. For a nonzero limit prefix length  $\lambda$ , its lower cylinder edge has lower cofinality exactly  $\text{cf}(\lambda)$  and no set coinital family on its upper side. To see this, make a fresh downward deviation at cofinally many positions of  $p$ . The completions increase

cofinally below  $C_p$ ; any cofinal family must have unbounded deviation coordinates. On the upper side, the initial portion  $C_p$  has no set cointial family, since one can choose a fresh next value below any set of next values. The upper edge has the dual characters.

**Corollary 46.4** (Exactly what raw tails add). *For a raw order  $R \in \mathfrak{W}_{\text{all}}$ , its strict lower cut in  $\mathcal{P}_{\text{bd}}(b)$  is  $L_{F_R}$ . Thus the cut image of all raw orders consists exactly of the cuts passing the straddling test. Over the cut of  $F$ , the fiber is the convex order  $\mathcal{W}(\mathbf{No} \setminus \text{ran } F)$ . Neither successor nor limit first-failure cuts are filled by any raw class well-order.*

*Proof.* A bounded-support enumeration unequal to  $F_R$  first disagrees at a set ordinal, before any non-set-like tail begins. Its comparison with  $R$  is therefore its comparison with  $F_R$ . Equality is possible only when  $F_R$  is exhaustive, in which case it determines  $R$ . Conversely every injection occurs as a first core, by (S16.12). Distinct cores give different cuts by Theorem 46.2, proving exactly the asserted image and fibers.  $\square$

This resolves the earlier report’s represented-cut question using the prefix tree with its numerical labels. It does not claim that the tree is recoverable from the bare order on  $\mathcal{P}_{\text{bd}}(b)$ . Since  $\mathcal{P}_{\text{bd}}(b) \cong \mathbf{No}$ , forgetting those additional predicates can lose precisely the information used by the recognition test.

*Note 46.5* ([merge]; on source 16’s last paragraph). Source 16’s sentence that this “resolves the earlier report’s represented-cut question” is true of the theorem, which all six batch-81 sources prove (note 46.1). Its corollary 46.4 is the statement about raw orders that sources 13, 15, 17 and 18 make in corollary 46.16, corollary 46.8, proposition 46.27 and corollary 46.30; the fibre description in it is source 12’s proposition 15.3, and the image of all raw orders, exactly the cuts passing the straddling test, is the new part. Part IX prints source 14’s trace-fibre theorem (theorem 54.6).

## 46.2 Source 15: a prefix criterion for all represented core cuts

[merge] Source 15’s Section 7, from its Subsection 7.2 (Subsection 7.1, the class of codes, is printed in section 42.3; Subsection 7.5, the automorphism theorem, in section 47).

### 46.2.1 The persistent-cylinder theorem

A normalized class cut of  $\mathcal{P}_{\text{bd}}(b)$  is a nonempty proper initial subclass  $L$  with no greatest member; write  $R = \mathcal{P}_{\text{bd}}(b) \setminus L$ . The upper side may have a least element. A cylinder *crosses* the cut if it meets both sides. For an injective class word  $F : \text{Ord} \rightarrow \mathbf{No}$ , put

$$L_F = \{d \in \mathcal{P}_{\text{bd}}(b) : E_d <_{\text{lex}} F\}.$$

The comparison is defined at the first ordinal difference, unless the functions agree, in which case the strict inequality is false.

**Theorem 46.6** (Exact criterion for representation). *Extension proved in source 15. In GBC, for a normalized class cut  $L$  the following are equivalent:*

1.  $L = L_F$  for an injective class word  $F : \text{Ord} \rightarrow \mathbf{No}$ ;
2. for every ordinal  $\alpha$ , some cylinder  $C_p$  with  $\text{dom}(p) = \alpha$  crosses the cut.

*The representing word is unique. Its restrictions are exactly the unique crossing prefixes. Its construction uses elementary comprehension, not ETR.*



*Proof.* At most one cylinder at a given level crosses. If  $C_p < C_q$  both crossed, a point of  $R$  in  $C_p$  would precede a point of  $L$  in  $C_q$ , contradicting initiality. A restriction of a crossing prefix also crosses. Thus the unique prefixes  $p_\alpha$  in (2) are coherent.

Define  $F(\alpha) = x$  when the crossing prefix of length  $\alpha + 1$  has last value  $x$ . This is an elementary formula: prefixes and codes are sets, and crossing is expressed by two existential set quantifiers with the cut and evaluation as class parameters. Coherence shows that  $F \upharpoonright \alpha = p_\alpha$ ; injectivity holds on every such restriction and hence globally.

If  $E_d \neq F$ , let  $\delta$  be the first difference. The cylinder  $C_{p_{\delta+1}}$  crosses the cut. If  $E_d(\delta) < F(\delta)$ , the whole cylinder is above  $d$ , and its lower-side member forces  $d \in L$ . If  $E_d(\delta) > F(\delta)$ , its upper-side member lies below  $d$ , forcing  $d \in R$ . If  $E_d = F$ , the code  $d$  cannot lie in  $L$ , since it would then be its greatest member. This proves (1).

Conversely, at every  $\alpha$  choose fresh labels  $a < F(\alpha) < c$ , neither in  $\text{ran}(F \upharpoonright \alpha)$ . Completions of  $(F \upharpoonright \alpha) \cap a$  and  $(F \upharpoonright \alpha) \cap c$  lie on opposite sides of  $F$ . Thus  $C_{F \upharpoonright \alpha}$  crosses. These same perturbations show that  $L_F$  is nonempty, proper, and has no greatest member. Uniqueness follows from recovery of every prefix.  $\square$

*Remark 46.7* (The normalization is necessary). Without the no-greatest condition, persistence determines a unique  $F$  but allows either side of a principal point:

$$L_F \subseteq L \subseteq L_F \cup \{d : E_d = F\}.$$

Only a word already represented by a core code permits the two choices. Both  $\{d : E_d < F\}$  and  $\{d : E_d \leq F\}$  cross every level, but the second has a greatest member. Terminal cuts below have neither endpoint.

**Corollary 46.8** (Exhaustiveness and the kernel of the raw cut map). *A persistent cut is represented by a set-like global well-order exactly when*

$$\forall x \in \mathbf{No} \ \exists \alpha \quad x \in \text{ran}(p_\alpha).$$

*Every persistent cut is represented by an arbitrary raw class well-order. Two raw orders determine the same strict lower cut on  $\mathcal{P}_{\text{bd}}(b)$  exactly when their initial words  $F_R$  agree. The common fibre is precisely  $\mathcal{W}(\mathbf{No} \setminus \text{ran}(F_R))$ , interpreted at the appropriate size level.*

*Proof.* The displayed condition says exactly that the recovered word is onto. Every injective word has a raw completion by source 15's Proposition 4.2 (prefix convexity and initial-word fibres; [merge] proposition 51.10; it is proposition 15.3). A raw order and its initial word compare with a core evaluation at the same first ordinal disagreement. If a core evaluation equals that word, its exhaustiveness forces the raw order to have no residual tail. Thus its lower cut is  $L_{F_R}$ . Apply uniqueness in the theorem and the fibre description of the same proposition.  $\square$

This answers the prior report's represented-cut question by a necessary and sufficient condition on all set-length prefix levels. Missing labels are not a failure of persistence. They distinguish the exhaustive words inside the persistent branches. Any ordering information after the first proper block is invisible to comparisons with the bounded-support core.

#### 46.2.2 An exhaustive description of first failure

**Theorem 46.9** (Terminal-cut classification). *Extension proved in source 15. If a normalized class cut fails the criterion, let  $\theta > 0$  be the least length without a crossing cylinder. Exactly one of the following descriptions applies.*

1. **Successor failure.** If  $\theta = \alpha + 1$ , there is a unique crossing prefix  $p : \alpha \rightarrow \mathbf{No}$  and a nonempty initial/final partition

$$A + B = \mathbf{No} \setminus \text{ran}(p), \quad A < B,$$

such that each child  $C_{p \frown x}$  lies wholly in  $L$  for  $x \in A$  and wholly in  $R$  for  $x \in B$ . The four mutually exclusive forms are: an upper boundary of a child cylinder when  $A$  has a greatest element; a lower boundary when  $B$  has a least element; a unique forbidden repeated label separating  $A$  and  $B$  when neither endpoint exists but a surreal separator exists; or a numerical class gap with no surreal separator.

2. **Limit failure.** If  $\theta$  is limit, the coherent union  $p = \bigcup_{\alpha < \theta} p_\alpha$  is an injective set word. The cut is exactly the lower or upper boundary of its nonempty cylinder:

$$L = \{d : E_d \upharpoonright \theta < p\} \quad \text{or} \quad L = \{d : E_d \upharpoonright \theta \leq p\}.$$

*Proof.* The empty cylinder crosses, so  $\theta \neq 0$ . At a successor, child cylinders partition the crossing parent and none crosses. Each is nonempty, giving the asserted partition of allowed labels. Both parts are nonempty because the parent crosses. Their order follows from the order of the cylinders and initiality of  $L$ . All codes below the parent lie in  $L$ , and all those above it lie in  $R$ .

The numerical residual alphabet is dense by the fresh-separator lemma (source 15's Lemma 3.1, lemma 51.2; lemma 5.2), so  $A$  cannot have a greatest element while  $B$  has a least one. If neither endpoint exists and  $s$  strictly separates them,  $s$  cannot be an allowed label: membership in  $A$  or  $B$  would contradict strict separation. Hence  $s \in \text{ran}(p)$ . Two different separators would enclose a fresh allowed label, which is impossible. This proves all four subcases.

At a limit, replacement over the set ordinal  $\theta$  collects the coherent prefixes, and their union is an injective set word. Its cylinder is nonempty by prefix completion but does not cross. A code outside it differs from  $p$  at a coordinate  $\delta < \theta$ ; the crossing cylinder of length  $\delta + 1$  then fixes its side. The only remaining choice is which side contains all of  $C_p$ , giving precisely the two formulas. There is no incoherent-branch case: uniqueness at the earlier levels already ruled it out.  $\square$

**Example 46.10** (A cut not filled even by arbitrary class well-orders). The initial class  $L = \{d : E_d(0) < 0\}$  has no greatest member. Its empty cylinder crosses, but each length-one cylinder lies on one side. It is the lower boundary of  $C_{(0)}$ . Therefore no injective Ord-word and no raw class well-order represents this cut. The upper boundary  $\{d : E_d(0) \leq 0\}$  is likewise unrepresented.

**Example 46.11** (Repeated label, numerical gap, and limit boundary). The cut defined by

$$E_d(0) < 0 \quad \text{or} \quad (E_d(0) = 0 \wedge E_d(1) < 0)$$

first fails at length two: the putative continuation of  $(0)$  would repeat 0. For a numerical-gap example, use  $A = \{x : \exists n \in \omega \ x < n\}$  and its complementary final class of positive infinite surreals, and put  $d \in L$  when  $E_d(0) \in A$ . No surreal strictly separates these label classes; neither relevant endpoint exists. Finally, if  $p(n) = n$  for  $n < \omega$ , the lower and upper boundaries of  $C_p$  first fail at length  $\omega$  even though every finite prefix still crosses.

For a persistent but nonexhaustive example, take  $F(\alpha) = \alpha$ . Every prefix level crosses, but the range omits the negative surreals. Appending a well-order of the omitted range gives many raw completions with exactly the same lower core cut. If  $F$  instead omits just one label, it has one raw completion of type  $\text{Ord} + 1$ . These are sharply different from the terminal cuts in Example 46.10.

### 46.2.3 Local characters of terminal cuts

Here a “set cofinal family” is an actual set of codes cofinal in an ordinary class side of a cut. We say explicitly when none exists; Ord is not being used as a set cardinal.

**Proposition 46.12** (Cofinality read from first failure). *At a first limit failure  $\theta$ , the lower-boundary case has  $\text{cf}(L) = \text{cf}(\theta)$ , while  $R$  has no set coinitial family. The upper-boundary case has the dual properties. At a successor failure with label partition  $A + B$ , if  $A$  has no greatest label,  $L$  has exactly the same least set cofinality as  $A$ , or no set cofinality if  $A$  has none. If  $A$  has a greatest label,  $L$  has no set cofinal family. The dual statements hold for  $B$  and  $R$ .*

*Proof.* At a limit, choose cofinally many coordinates  $\beta < \theta$  and complete prefixes agreeing with  $p$  before  $\beta$  but smaller at  $\beta$ . These form an increasing cofinal family on the lower side. Any cofinal family must have first-divergence coordinates unbounded in  $\theta$ , since a later lower perturbation dominates every family with bounded divergence coordinates. This proves equality. The upper side begins with the whole cylinder  $C_p$ , which has no set coinitial family; all remaining upper-side points are above it.

At a successor with no greatest label in  $A$ , a cofinal set of labels can be lifted to completions, and conversely the next labels of any cofinal family eventually inside the parent cylinder must be cofinal in  $A$ . If  $A$  has a greatest label, its final child cylinder has no set cofinal family and is final on the lower side. The dual argument proves the remaining assertions.  $\square$

No gap of  $\mathcal{P}_{\text{bd}}(b)$  can have both a set cofinal lower side and a set coinitial upper side: the core’s set-cut filling theorem would put a point between such two sets. The character calculation is compatible with this restriction, while identifying many genuinely different class cuts.

### 46.2.4 What a completion would have to mean

The normalized class cuts of  $\mathcal{P}_{\text{bd}}(b)$ , ordered by inclusion, give the natural Dedekind-cut interpretation at the next size level. For a uniform family of such cuts whose union is nonempty and proper, the union is again a normalized class cut: its defining membership formula quantifies over set indices. This is a precise available-family completeness statement, not a same-level class containing all its subclasses.

The map from raw orders to these cuts collapses exactly the residual fibres of [Corollary 46.8](#). Therefore adjoining all raw orders is neither a faithful Dedekind completion of the core nor a realization of every core cut. It both retains multiplicity over persistent cuts and omits all terminal cuts. Any order embedding into a dense completion that fixes the core would have to identify distinct raw orders lying in one fibre, which is impossible. Arbitrary embeddings that move the core are a different question.

## 46.3 Source 13: exact reconstruction of cuts from prefix cylinders

We now address the earlier report’s question asking for a necessary and sufficient cylinder criterion for represented cuts ([research question 113.1](#)). The criterion is relative to the chosen core and prefix system; [Section 47.4](#) will show why it is not a statement about the pure linear order alone.

### 46.3.1 Normalization and crossing

**Definition 46.13.** A *normalized proper cut* of  $\mathcal{P}_s(b)$  is a pair of ordinary classes  $(L, U)$  such that  $L \sqcup U = \mathcal{P}_s(b)$ , both sides are nonempty, every member of  $L$  precedes every member of  $U$ , and  $L$  has no greatest member. The upper side may have a least member.

An injective set prefix  $p$  crosses this cut when

$$\mathcal{P}_s(b)_p \cap L \neq \emptyset, \quad \mathcal{P}_s(b)_p \cap U \neq \emptyset.$$

The normalization is essential. Immediately below and immediately above a code  $d$ , two different cuts would otherwise reconstruct the same branch  $E_d$ . We choose the lower trace convention, so the cut at  $d$  puts  $d$  in the upper side.

**Lemma 46.14** (Uniqueness and coherence of crossing prefixes). *At a fixed ordinal length there is at most one crossing prefix. Restrictions of a crossing prefix cross the same cut. Consequently crossing prefixes of different lengths are coherent.*

*Proof.* Distinct same-length prefixes first differ at a coordinate, placing one entire cylinder below the other. If both cylinders crossed, an upper-side point in the first would precede a lower-side point in the second, impossible. Restriction enlarges a cylinder, so preserves crossing. Apply same-length uniqueness to the restriction of a longer crossing prefix.  $\square$

For  $F \in \mathfrak{J}$ , its lower trace is the class

$$\text{tr}(F) = \{d \in \mathcal{P}_s(b) : E_d < F\}.$$

All comparisons use a least set-ordinal disagreement, except equality, which simply excludes a code from the lower trace.

**Theorem 46.15** (Exact branch representation). *In GBC, a normalized proper cut  $(L, U)$  of  $\mathcal{P}_s(b)$  is the lower trace of a unique  $F \in \mathfrak{J}$  if and only if*

$$\text{for every ordinal } \alpha \text{ there exists a crossing prefix of length } \alpha. \quad (\text{S13.1})$$

*The represented word is exhaustive, hence belongs to  $\mathfrak{W}_{\text{sl}}$ , if and only if it also satisfies*

$$\text{for every } x \in \mathbf{No} \text{ some crossing prefix has } x \text{ in its range.} \quad (\text{S13.2})$$

*Proof.* Suppose first that  $L = \text{tr}(F)$ . For each ordinal  $\alpha$ , the prefix  $F \upharpoonright \alpha$  has code completions below and above  $F$ , by choosing a fresh next value on the desired side. Thus it crosses. It is the unique crossing prefix of that length. Equation (S13.2) is then precisely surjectivity of  $F$ .

Conversely, let  $p_\alpha$  be the unique crossing prefix of length  $\alpha$ . Coherence gives  $p_\alpha = p_\beta \upharpoonright \alpha$  whenever  $\alpha < \beta$ . Define

$$F(\alpha) = p_{\alpha+1}(\alpha).$$

This is an elementary class definition quantifying over the set code  $p_{\alpha+1}$ . It does not ask for a class-valued choice or a class-valued recursive sequence of prefixes. Every two coordinates belong to one longer injective prefix, so  $F$  is injective.

We check the trace equality. If  $E_d < F$ , let  $\xi$  be the first disagreement and choose a crossing prefix of length greater than  $\xi$ . Every code in its cylinder is above  $d$ . One of those codes is in  $L$ , so downward closure gives  $d \in L$ . If  $E_d > F$ , the symmetric argument with an upper-side code gives  $d \in U$ .

The only possible remaining case is  $E_d = F$ . Every other lower-side code has smaller evaluation, by the two arguments just given. If  $d$  were in  $L$ , it would therefore be its greatest member, contradicting normalization. Thus  $L = \text{tr}(F)$ . Distinct words have a code strictly between them by Corollary 42.5, so their traces are distinct. Finally, the union of the prefix ranges is exactly  $\text{ran}(F)$ , which proves the exhaustion criterion.  $\square$

### 46.3.2 What raw well-orders add, and what they cannot add

**Corollary 46.16** (Trace image and fibers of raw orders). *For  $R \in \mathfrak{W}_{\text{all}}$ , define  $L_R = \{d \in \mathcal{P}_s(b) : E_d < R\}$ . The normalized proper cuts represented in this way are exactly those satisfying (S13.1). For such a cut the unique reconstructed word is  $F_R$ . Its raw realization fiber is precisely the order of all well-orders of  $\mathbf{No} \setminus \text{ran}(F_R)$ . The fiber contains a set-like realization exactly when the coverage condition holds, and then it is a singleton.*

*Proof.* Every comparison with a code is decided inside the ordinal head unless the code equals the entire raw order; equality can occur only when the head is exhaustive. Hence  $L_R = \text{tr}(F_R)$ . Conversely, append a well-order of the omitted range to the branch reconstructed by Theorem 46.15. Source 13's Proposition 6.2 ("Canonical head; prior structural lemma", [merge] proposition 51.8; it is theorem 15.1 and proposition 15.3) gives every realization and proves the fiber statement.  $\square$

If the omitted range has  $n$  elements, there are exactly  $n!$  raw realizations. A single omitted label gives one additional point of type  $\text{Ord} + 1$ ; two omitted labels give two adjacent realizations of type  $\text{Ord} + 2$ . These examples were already constructed in the earlier report. The new criterion determines the complete trace image: adding arbitrary non-set-like tails still does not fill a cut that fails the all-level crossing test.

**Example 46.17** (A branch that fails only exhaustion). Let  $F(\alpha) = \alpha$ , viewed as the all-plus surreal of length  $\alpha$ . Its lower trace crosses every prefix length, but its prefixes never contain  $-1$ . It has raw realizations obtained by well-ordering the nonordinal surreals after its head, and no realization in  $\mathfrak{W}_{\text{sl}}$ . This recovers the earlier omitted-range example through the exact criterion rather than through a single selected gap.

## 46.4 Source 13: a complete first-failure classification

The representation theorem reduces the problem to a question about crossing prefixes. The next theorem describes every way the test can fail. In particular, failure at a limit is not an unexplained compactness assumption: it has an exact geometric form.

**Theorem 46.18** (Branch, horizontal cut, or limit boundary). *Let  $(L, U)$  be a normalized proper cut of  $\mathcal{P}_s(b)$ . Exactly one of the following descriptions applies.*

- (i) *There is a crossing prefix at every ordinal length. The cut is the trace of a unique injective class word.*
- (ii) *There is a least failed length  $\lambda = \beta + 1$ . Let  $p$  be the crossing prefix of length  $\beta$ . There is a nonempty cut  $A < B$  partitioning  $Y = \mathbf{No} \setminus \text{ran}(p)$  such that, within  $\mathcal{P}_s(b)_p$ ,*

$$L \cap \mathcal{P}_s(b)_p = \bigcup_{x \in A} \mathcal{P}_s(b)_{p \smallfrown x}, \quad U \cap \mathcal{P}_s(b)_p = \bigcup_{x \in B} \mathcal{P}_s(b)_{p \smallfrown x}.$$

*Thus the cut separates whole next-letter cylinders. We call it horizontal at  $p$ .*

- (iii) *There is a least failed length  $\lambda$  that is a nonzero limit. The union  $p$  of the shorter crossing prefixes has length  $\lambda$ , and the cut is exactly the lower or upper boundary of  $\mathcal{P}_s(b)_p$ .*

*All descriptions are expressible with ordinary class parameters in GBC.*

*Proof.* The empty prefix crosses every proper cut. If some length fails, the class of failed ordinals has a least member  $\lambda > 0$ .

For  $\lambda = \beta + 1$ , uniqueness gives the crossing  $p$  of length  $\beta$ . Every child cylinder is nonempty by prefix completion, but none crosses the cut. Each is therefore entirely on one side. Put  $x \in A$  or  $x \in B$  according to that side. These are ordinary classes: the predicate “every code in this child is in  $L$ ” quantifies only over set codes. Their union is  $Y$ , they are disjoint, and the label order of the children shows  $A < B$ . Both sides are nonempty because the parent crosses. Every code outside the parent lies below all of it or above all of it, so its placement in the cut is forced.

For limit  $\lambda$ , coherence gives an injective set word  $p = \bigcup_{\alpha < \lambda} p_\alpha$ . Its cylinder is nonempty but cannot cross. If a code does not extend  $p$ , it first disagrees at some  $\xi < \lambda$ . A shorter crossing prefix of length greater than  $\xi$  forces its side: a lower disagreement puts it in  $L$ , and an upper disagreement puts it in  $U$ . Thus every code below  $\mathcal{P}_s(b)_p$  is in  $L$ , and every code above it is in  $U$ . The entire  $\mathcal{P}_s(b)_p$  lies on one side, giving precisely one of its two boundaries.

If there is no failed set ordinal, [Theorem 46.15](#) applies. The three cases are disjoint by their first failed length.  $\square$

There is no fourth case called “failure at Ord”. An injective branch that omits labels still crosses every set level; its failure is exhaustion, not prefix existence. Treating Ord as one more set stage would conflate the two tests.

#### 46.4.1 One-sided characters of the three cases

For a class order, say that a side has character Ord when it has no set-sized cofinal (respectively coinital) family but does have a uniform increasing (respectively decreasing) Ord-indexed family. This is a notation for a pair of properties, not a set cardinal. When a set-sized family suffices, use the usual least cofinal cardinal; an endpoint has character 1.

Every ordinary class order without a set-cofinal subset has an Ord-indexed increasing cofinal sequence in GBC: enumerate its points by the global well-order and at each stage choose a point above both the earlier set of choices and the next enumerated point. Absence of a set-cofinal subset guarantees this choice. The dual construction handles coinitality.

**Proposition 46.19** (Characters at first failure). *The cut descriptions have the following character consequences.*

- (i) *A branch trace has character (Ord, Ord) when its word is not a code evaluation. If  $F = E_d$ , its normalized trace has character (Ord, 1); the punctured upper side again has character Ord.*
- (ii) *A lower boundary of a limit-length cylinder  $\mathcal{P}_s(b)_p$ , of length  $\lambda$ , has character (cf  $\lambda$ , Ord). Its upper boundary has character (Ord, cf  $\lambda$ ).*
- (iii) *For a horizontal cut associated with  $A < B$ , the lower character is cf( $A$ ) if  $A$  has no greatest label, and is Ord if  $A$  has a greatest label. The upper character is ci( $B$ ) if  $B$  has no least label, and is Ord if it has a least label.*

*Proof.* The branch statement is [Lemma 42.6](#), with the possible equal code placed in the upper side.

At a lower limit boundary, choose prefixes agreeing with  $p$  up to  $\xi$  and then taking a fresh lower letter. Completions at lengths  $\xi$  cofinal in  $\lambda$  are cofinal below the whole cylinder. Conversely, fewer than cf  $\lambda$  many codes below it have their first-disagreement coordinates bounded below  $\lambda$ , so a later such completion lies above them all. The right side has  $\mathcal{P}_s(b)_p$  as a coinital subclass, and that cylinder has no set-coinital family by [Corollary 42.5](#). The upper-boundary proof is dual.

For a horizontal cut whose lower label side has no greatest member, choose one code in each child along a cofinal family of labels. This family is cofinal among all lower codes. Conversely, projecting a cofinal family of lower codes in the parent onto its next labels is cofinal in  $A$ . Codes



below the parent cannot improve this bound. If  $A$  has a greatest label  $a$ , the whole terminal child  $\mathcal{P}_s(b)_{p \smallfrown a}$  is cofinal on the lower side. Its character is  $\text{Ord}$ , not 1. The upper assertion is the reversed argument.  $\square$

#### 46.4.2 Equal characters do not imply representability

**Example 46.20** (A horizontal  $(\text{Ord}, \text{Ord})$  cut). Let  $s : \text{Ord} \rightarrow \{-, +\}$  be the proper-class sign string whose sign is  $+$  when the finite remainder of its ordinal index is even and  $-$  when it is odd. Both signs occur cofinally. This is *not* a surreal number.

Compare each set-length surreal sign sequence with  $s$  at its first difference, regarding its termination as 0. Every surreal differs from  $s$  by the time of its termination. Define

$$A = \{x \in \mathbf{No} : x < s\}, \quad B = \{x \in \mathbf{No} : s < x\}.$$

These are nonempty classes partitioning the label order. Neither side has an endpoint or a set-sized cofinal or coinital family. Indeed, the birthdays of any proposed set family are bounded by some  $\gamma$ . Truncating  $s$  at a later  $+$  position gives a larger lower label than all lower members of the family; truncating at a later  $-$  position gives a smaller upper label. The corresponding ordinal sequences of truncations supply cofinal and coinital class families.

Now partition the core by its first label:

$$L = \{d : E_d(0) \in A\}, \quad U = \{d : E_d(0) \in B\}.$$

This cut has character  $(\text{Ord}, \text{Ord})$  but no crossing prefix of length 1. It is horizontal at the root, and is not the trace of any injective class word or of any raw class well-order.

Thus neither saturation below set size nor the ordered pair of local characters identifies the represented cuts. The labeled prefix criterion contains strictly more information.

#### 46.4.3 An explicit limit-boundary example

Let  $p(n) = n$  for  $n < \omega$ . The cut just below  $\mathcal{P}_s(b)_p$  is crossed by every finite initial prefix of  $p$ , but not by any prefix of length  $\omega$ . At each finite level, some lower code leaves the ordinal prefix later, while an upper code extends all of  $p$ . The cut has character  $(\omega, \text{Ord})$  and is not represented by a raw global well-order. Its obstruction is a set-length limit boundary rather than an omitted proper-class range. This is the distinction that an informal “keep taking longer prefixes” argument would overlook.

### 46.5 Source 17: recognizing exactly which core cuts are global enumerations

The preceding source results place  $\mathfrak{W}_{\text{sl}}$  in a larger collection of cuts of  $\mathcal{P}_s(b)$ , but do not by themselves identify its image. We now give a necessary-and-sufficient test in terms of the displayed prefix cylinders. This addresses source question [swo:n16.1](#) ([research question 113.1](#)) and its parallel in source 09 (its third direction in [section 113.3](#)). It is a *labelled-tree* recognition theorem; a pure-order recognition theorem would be stronger and, as shown below, cannot hold in the naive invariant form.

### 46.5.1 The straddling test

For an injective set prefix  $p : \alpha \rightarrow \mathbf{No}$ , define the ordinary class

$$C_p = \{c \in \mathcal{P}_s(b) : E_c \upharpoonright \alpha = p\}.$$

Let  $L$  be a proper nonempty lower subclass of  $\mathcal{P}_s(b)$  with no greatest element, and put  $U = \mathcal{P}_s(b) \setminus L$ . We allow  $U$  to have a least element. Thus the convention is the *open lower cut*, including the one immediately before an actual core point. Say that  $p$  straddles  $L$  when  $C_p$  meets both  $L$  and  $U$ .

**Theorem 46.21** (Recognition of injective branches). *The following conditions are equivalent.*

1. *There is an injective class word  $f : \text{Ord} \rightarrow \mathbf{No}$  such that*

$$L = L_f := \{c \in \mathcal{P}_s(b) : E_c <_{\text{lex}} f\}.$$

2. *For every set ordinal  $\alpha$ , there is a straddling injective prefix of length  $\alpha$ .*

*When these conditions hold, both the straddling prefix at each length and the word  $f$  are unique. Moreover  $f$  is surjective if and only if*

$$\forall x \in \mathbf{No} \exists p (p \text{ straddles } L \wedge x \in \text{ran } p). \quad (\text{S17.8})$$

*The reconstruction uses elementary class comprehension over GBC, not ETR.*

*Proof.* Distinct cylinders of the same length are disjoint convex blocks in lexicographic order. At most one can meet both sides of a cut: if an earlier block met  $U$ , every later block would be in  $U$ .

Suppose first that  $f$  is given. For any  $\alpha$ , take its prefix  $p = f \upharpoonright \alpha$ . A fresh smaller next label than  $f(\alpha)$ , and a fresh larger one, yield core completions respectively below and above  $f$ . Thus  $p$  straddles. These completions also show that  $L_f$  has no greatest element and is proper and nonempty.

Conversely, let  $p_\alpha$  denote the unique straddling prefix at length  $\alpha$ . Restrictions of straddling prefixes straddle, so uniqueness implies  $p_\beta \upharpoonright \alpha = p_\alpha$  for  $\alpha < \beta$ . Define a class graph by

$$f(\alpha) = x \iff \exists p (p \text{ is a straddling prefix of length } \alpha + 1 \wedge p(\alpha) = x).$$

The quantifier over  $p$  ranges over sets. Straddling itself quantifies over the set codes in  $\mathcal{P}_s(b)$ , with  $L$  and  $b$  as class parameters. Thus this is elementary comprehension. Coherence makes  $f$  a total function; its set restrictions are injective, so it is injective.

If  $E_c \neq f$ , let  $\delta$  be their first disagreement. The cylinder of  $p_{\delta+1}$  straddles. If  $E_c(\delta) < f(\delta)$ , then  $c$  is below every code in that cylinder, including a code in  $L$ , so  $c \in L$ . The reversed inequality places  $c$  in  $U$ . If  $E_c = f$ , putting  $c$  in  $L$  would make it the greatest member of  $L$ , already excluded. Hence  $L = L_f$  exactly, including the principal-cut case. Two different words would be separated by a core code, so the representing word is unique. Finally  $\text{ran } f$  is the union of the ranges of its straddling prefixes; this proves the coverage criterion.  $\square$

*Remark 46.22* (Why every ordinal length and the endpoint convention matter). Checking only successor extensions along finite histories is inadequate. A cut can track a coherent prefix through all finite levels and then lie at the boundary of its length- $\omega$  cylinder. There is then no straddling cylinder at that limit. In contrast, [Theorem 46.21](#) requires a straddler at *every* set ordinal. Also, a closed principal lower cut would have the same straddling prefixes as the corresponding open one. The hypothesis that  $L$  has no greatest element removes that ambiguity.

### 46.5.2 Three independent failures

**Example 46.23** (A coordinate cut that does not follow a branch). Let  $A \subseteq \mathbf{No}$  consist of those  $x$  with  $x < n$  for some natural number  $n$ , and let  $B = \mathbf{No} \setminus A$ . Both are nonempty,  $A$  has no largest element, and  $B$  has no least element: a positive infinite member of  $B$  has a smaller positive infinite half. Partition core codes by whether their first label lies in  $A$  or  $B$ . The root cylinder straddles, but no length-one cylinder does. This is a genuine cut of the core, but not comparison with an injective Ord-word.

**Example 46.24** (A limit boundary that defeats finite-depth testing). Fix an injective prefix  $p : \omega \rightarrow \mathbf{No}$ . Take the cut immediately before  $C_p$ . At every finite length  $n$ , the cylinder for  $p \upharpoonright n$  meets both cut sides, because a fresh lower label at a later finite coordinate gives a code strictly below  $C_p$ . At length  $\omega$ , the entire cylinder lies on the upper side and no cylinder straddles. This is precisely the limit obstruction in the set-sized cut trichotomy, now at an ordinary class of set codes.

**Example 46.25** (A coherent branch which is not exhaustive). Fix  $r_0 \in \mathbf{No}$  and a bijection  $f : \text{Ord} \rightarrow \mathbf{No} \setminus \{r_0\}$ . The cut  $L_f$  passes the straddling test at every ordinal length, but fails (S17.8) for  $r_0$ . It is not represented by any member of  $\mathfrak{W}_{\text{sl}}$ , since the representing injective word is unique.

These examples separate numerical-cut failure, limit coherence, and exhaustiveness. The test is not merely the assertion that a representation exists restated in different notation: its quantifiers over prefixes and core witnesses are over *sets*, and the representing proper-class function is recovered by one elementary formula.

**Proposition 46.26** (Characters of branch cuts outside the core). *If  $f \in \mathfrak{J}$  has no code in  $\mathcal{P}_s(b)$ , both sides of its core cut have no set-sized cofinal or coinitial subset, respectively. Both admit uniform Ord-indexed approximation families. This is often abbreviated by saying that the cut has character  $(\text{Ord}, \text{Ord})$ ; that notation does not make Ord into a cardinal or set.*

*Proof.* For each  $\alpha$ , extend  $f \upharpoonright \alpha$  by a smaller or larger fresh label and complete in the core, making the choices uniformly using global choice. As  $\alpha$  increases, these codes approach the cut from their respective sides. For any set of codes below  $f$ , Class Replacement bounds all their first disagreements with  $f$  by a set ordinal. A later lower approximation lies above them all. The upper argument is identical. There is no equality case because  $f$  has no core code.  $\square$

### 46.5.3 What the same cut remembers about unrestricted orders

For  $f \in \mathfrak{J}$ , put  $D_f = \mathbf{No} \setminus \text{ran } f$ . Append to the well-order listed by  $f$  any class well-order of  $D_f$ , choosing one using global choice when necessary. The result is a raw class well-order of  $\mathbf{No}$ . Every appended label has a proper class of predecessors; the initial set-like part is exactly  $f$ .

**Proposition 46.27** (Raw lifts and loss of residual information). *All raw well-orders with canonical initial enumeration  $f$  induce the same comparison cut  $L_f$  on  $\mathcal{P}_s(b)$ . Their convex fibre is the lexicographic order of all well-orders of  $D_f$ . Consequently the fibre is a singleton when  $|D_f| = 0$  or 1, and has  $n!$  members when  $|D_f| = n < \omega$ . For larger complements it retains the full residual well-order problem.*

*Proof.* Unless a core enumeration equals  $f$ , its comparison is decided at a set ordinal before any appended label. Equality is possible only when  $f$  is exhaustive, in which case no tail remains. Restriction to  $D_f$  and concatenation with the fixed initial order are inverse order-preserving operations. The source proves this convex-fibre description for all raw orders (proposition 15.3); the argument here identifies its interaction with the reconstructed cut.  $\square$

Thus recognition of an injective branch determines the raw order's first Ord-part but not its residual order. The core is dense in  $\mathfrak{W}_{\text{sl}}$  and  $\mathfrak{J}$ , not in the interiors of these nontrivial raw fibres.

#### 46.6 Source 18: exactly which class cuts of a core are represented

Fix a baseline bijection  $b : \text{Ord} \rightarrow \mathbf{No}$  and let  $\mathcal{P}_s(b)$  be its ordinary class of normalized bounded-support permutation codes. Write  $E_c$  for the enumeration evaluated from a code  $c \in \mathcal{P}_s(b)$ . All such evaluations are supplied by one class relation. For an injective set prefix  $p : \delta \rightarrow \mathbf{No}$ , put

$$C_p = \{c \in \mathcal{P}_s(b) : E_c \upharpoonright \delta = p\}.$$

The core-completion lemma ensures that every such cylinder is nonempty. It is convex. At a fixed length  $\delta$ , these cylinders partition  $\mathcal{P}_s(b)$  and are linearly ordered without overlap.

Throughout this section, a cut is an ordinary class  $L \subseteq \mathcal{P}_s(b)$  that is nonempty, proper, initial, and has no greatest element. This convention uses the strict lower side of a represented point. In particular, it excludes the closed principal cut at a code. Say that  $p$  *straddles*  $L$  if

$$C_p \cap L \neq \emptyset, \quad C_p \setminus L \neq \emptyset.$$

This is an elementary predicate of the set  $p$ , with class parameters  $b, L$ . Its witnesses are set codes, not arbitrary classes.

**Theorem 46.28** (The prefix criterion for represented class cuts). *In GB with the fixed baseline above, the following are equivalent.*

1. *There is an injective class function  $f : \text{Ord} \rightarrow \mathbf{No}$  such that*

$$L = \{c \in \mathcal{P}_s(b) : E_c <_{\text{lex}} f\}.$$

2. *For every ordinal  $\delta$  there is an injective prefix  $p : \delta \rightarrow \mathbf{No}$  that straddles  $L$ .*

*When they hold,  $f$  is unique. At every length  $\delta$ , the unique straddling prefix is  $f \upharpoonright \delta$ . The function  $f$  is defined by an elementary formula from  $b, L$ ; no class choice or ETR is needed to assemble it.*

*Proof.* At most one prefix of a given length can straddle. Indeed, if the cylinder of  $p$  lies below that of  $q$ , an element outside  $L$  in the first cylinder lies below an element of  $L$  in the second, contradicting initiality.

Assume the second condition and denote the unique prefix at length  $\delta$  by  $p_\delta$ . If  $\gamma < \delta$ , the restriction  $p_\delta \upharpoonright \gamma$  also straddles, since its cylinder contains the straddling cylinder of  $p_\delta$ . Uniqueness gives

$$p_\delta \upharpoonright \gamma = p_\gamma.$$

Define

$$f(\alpha) = p_{\alpha+1}(\alpha).$$

The graph is elementary because each  $p_\delta$  is a uniquely specified set. Coherence proves that this is an injective class function and that  $f \upharpoonright \delta = p_\delta$ .

Take a core code  $c$  with  $E_c \neq f$ , and let  $\alpha$  be the first coordinate of disagreement. If  $E_c(\alpha) < f(\alpha)$ , then  $c$  lies below every code in  $C_{p_{\alpha+1}}$ . That cylinder meets  $L$ , so  $c \in L$  by initiality. If  $E_c(\alpha) > f(\alpha)$ , then  $c$  lies above every code in the same cylinder. Its member outside  $L$  shows that  $c \notin L$ . Finally, if  $E_c = f$  and  $c \in L$ , the absence of a greatest element of  $L$  supplies  $d \in L$  with  $E_d > f$ , contradicting the previous case. Hence the displayed cut identity holds in every case.

Conversely, let  $L$  be the strict lower cut of an injective class word  $f$ . For any ordinal  $\delta$ , its prefix  $p = f \upharpoonright \delta$  is a set. Choose fresh surreal labels

$$u < f(\delta) < v, \quad u, v \notin \text{ran}(p).$$

Such labels exist because a set cannot exhaust either numerical interval next to a given surreal. Complete  $p$  followed by  $u$ , and  $p$  followed by  $v$ , to core enumerations. Both completions belong to  $C_p$ , and they lie respectively below and above  $f$ . Thus every prefix straddles. Its uniqueness at each length also proves the uniqueness of  $f$ .  $\square$

For completeness, the strict lower cut of every injective class word is indeed nonempty, proper, and without a greatest element. The first two assertions follow by choosing a smaller or larger first label. If  $E_c < f$  first differs at  $\alpha$ , complete  $f \upharpoonright (\alpha + 1)$  with a fresh next label smaller than  $f(\alpha + 1)$ . The resulting core row lies strictly between  $E_c$  and  $f$ .

*Remark 46.29* (Why the strict-cut convention matters). If  $f = E_c$  is itself a core row, both its strict lower cut and its closed lower cut have a straddling cylinder at every length. The closed cut has greatest element  $c$ . The no-greatest-element hypothesis is therefore necessary for the stated strict-cut criterion; it is not an incidental regularity assumption.

#### 46.6.1 Set-like representation and all raw representations

For a raw class well-order  $R$  of  $\mathbf{No}$ , let  $F_R : \text{Ord} \rightarrow \mathbf{No}$  be the unique enumeration of its canonical initial set-like part. It is injective, and the remaining labels, if any, occur after its entire ordinal-class prefix. The earlier common-prefix theorem shows that a core row and  $R$  are compared at their first ordinal disagreement with  $F_R$ , unless they are identical.

**Corollary 46.30** (Surjectivity is the exact remaining obstruction). *A cut  $L$  is represented by a raw class well-order of  $\mathbf{No}$  if and only if it satisfies the prefix criterion of [Theorem 46.28](#). Let  $f$  be the unique class word recovered from such a cut. Then:*

1.  *$L$  is represented by a set-like global well-order if and only if  $f$  is surjective onto  $\mathbf{No}$ .*
2. *All its raw representatives are exactly the well-orders obtained by putting the sequence  $f$  first and then well-ordering  $Y_f = \mathbf{No} \setminus \text{ran}(f)$ .*
3. *These representatives form the previously described convex fiber isomorphic, in the typed uniform sense, to  $\mathcal{W}(Y_f)$ .*

*Proof.* For any raw representative  $R$ , its cut on the core is the strict lower cut of  $F_R$ . [Theorem 46.28](#) therefore applies and gives  $f = F_R$ . Conversely, append the restriction of the baseline well-order to  $Y_f$  after  $f$ . The range of  $f$  is proper, and its points have precisely set-many earlier  $f$ -values; every appended point has the whole proper range before it. Thus the canonical initial function is exactly  $f$ , and the resulting raw order has cut  $L$ .

A set-like representative has no residual labels, so its canonical function is a bijection. Conversely a bijective  $f$  directly defines a set-like representative. Uniqueness of the recovered class word excludes a different surjective word representing the same cut. The last two statements are the canonical-fiber description.  $\square$

Thus allowing arbitrary raw well-orders fills the omitted-range branch cuts, but it still does not fill every class cut of the core. The first-failure theorem below identifies exactly where an unsuccessful prefix criterion breaks down.

**Corollary 46.31** (Elementary recognition in uniform families). *Relative to the baseline, raw representability of a cut is an elementary property of its class parameter. Set-like representability is also elementary: in addition to a straddling prefix at every length, require*

$$\forall x \in \mathbf{No} \ \exists \delta \in \text{Ord} \ \exists p[p : \delta \rightarrow \mathbf{No} \text{ straddles } L \wedge x \in \text{ran}(p)].$$

*Consequently a uniformly class-indexed family of cuts has an elementary class of raw-represented indices, an elementary class of set-like-represented indices, and a uniform relation recovering the injective words of all raw-represented members.*

*Proof.* All displayed variables are sets. The coverage condition is equivalent to surjectivity of the uniquely recovered word. Insert the family index into the same formulas and apply elementary class comprehension. The input cut remains a class parameter; this does not provide set codes for all cuts or all class words.  $\square$

#### 46.6.2 The first failure: a numerical split or a limit boundary

**Theorem 46.32** (First-failure classification). *Suppose  $L$  fails the prefix criterion, and let  $\delta$  be the least length with no straddling prefix. Then  $\delta > 0$  and exactly one of the following descriptions applies.*

1. **Successor failure.** *If  $\delta = \eta + 1$ , let  $p$  be the unique straddling prefix of length  $\eta$ . The unused alphabet  $Y = \mathbf{No} \setminus \text{ran}(p)$  has a nontrivial class cut  $Y = A \sqcup B$ , with  $A < B$ , such that*

$$C_{p \smallfrown \langle x \rangle} \subseteq \begin{cases} L, & x \in A, \\ \mathcal{P}_s(b) \setminus L, & x \in B. \end{cases}$$

*Outside  $C_p$ , the codes below that cylinder lie in  $L$  and those above it do not.*

2. **Limit failure.** *If  $\delta$  is a limit ordinal, the shorter unique straddling prefixes have a coherent union  $p : \delta \rightarrow \mathbf{No}$ , which is an injective set prefix. Its nonempty cylinder lies wholly in  $L$  or wholly outside  $L$ . The cut is respectively the upper or lower boundary of that cylinder: codes strictly below the cylinder belong to  $L$ , and codes strictly above it do not.*

*Proof.* The empty prefix straddles because  $L$  and its complement are nonempty, so  $\delta > 0$ . Unique shorter straddling prefixes are coherent by the proof of [Theorem 46.28](#).

In the successor case, every next-letter cylinder is nonempty and fails to straddle, so it lies wholly on one side. The two sides of the parent cylinder supply a letter of each kind. Initiality orders the resulting classes  $A < B$ . The straddling parent also forces every code below it into  $L$  and every code above it outside  $L$ .

In the limit case, Class Replacement on the set ordinal  $\delta$  collects the uniquely defined shorter prefixes into a set, so their union  $p$  is a set. Its cylinder is nonempty by core completion, and does not straddle by the defining failure at  $\delta$ . A code outside it first differs from  $p$  at some  $\alpha < \delta$ . The straddling cylinder at length  $\alpha + 1 < \delta$  then decides its side exactly as in the main proof. All codes below the length- $\delta$  cylinder lie in  $L$ , all those above it lie outside, and the entire cylinder itself lies on one side.  $\square$

The successor split is a class cut in the *unused* numerical alphabet. It can be principal at a remaining label, can occur at a label deleted by the prefix, or can be an unfilled class cut of the surreal numerical order. Surreal set-cut interpolation does not imply that an arbitrary class cut has a surreal boundary. The limit case records the independent possibility of reaching an entire set prefix while the cut lies at one of its cylinder's boundaries. Neither failure type is represented even by a raw class well-order, by [Corollary 46.30](#).



**Example 46.33** (A deleted-label cut caused by repetition). Fix  $x \in \mathbf{No}$  and put

$$L_x = \{c \in \mathcal{P}_s(b) : E_c(0) < x \text{ or } (E_c(0) = x \text{ and } E_c(1) < x)\}.$$

This is a nonempty proper initial class without a greatest element. The unique straddling prefix of length 1 is  $\langle x \rangle$ . There is no straddling prefix of length 2: its first entry other than  $x$  already decides the side, and when its first entry is  $x$ , its second entry must be different from  $x$  by injectivity. The first failure is therefore at length 2, where the unused alphabet is split at the deleted label  $x$ .

The cut is not represented by any injective ordinal-class word, and hence by no raw class well-order of  $\mathbf{No}$ . A would-be word would have to begin with  $x, x$ . This is an explicit omitted class cut distinct from the omitted-range branch cuts: enlarging the order from set-like to unrestricted raw well-orders does not repair it. If repeated labels are allowed in class words, however, the constant word  $x, x, x, \dots$  represents this cut. Every injective core row differs from that word by coordinate 1 at the latest.

*Remark 46.34* (Class character does not decide representability). Both sides of  $L_x$  have no set-sized cofinal or coinital witness, respectively, although they have uniform witnesses indexed by  $\text{Ord}$ . To see this, complete the prefixes

$$\left\langle x, x - \frac{1}{\alpha + 1} \right\rangle, \quad \left\langle x, x + \frac{1}{\alpha + 1} \right\rangle \quad (\alpha \in \text{Ord})$$

uniformly to core rows. Since ordinals are cofinal in the numerical surreal order, the first family is cofinal in  $L_x$  and the second is coinital in its complement. Any proposed set-sized witness can be improved at its second coordinate by surreal set interpolation. Thus this unrepresented cut has the same  $(\text{Ord}, \text{Ord})$  class character as omitted-range branch cuts that do have raw representatives. This notation records the stated existence and nonexistence of uniform witnesses; it does not assign a set cardinal to a proper class.

### 46.6.3 Foundational accounting

All cylinders, cuts, and recovered graphs in this section are ordinary classes of sets. Prefixes are set functions; core rows are accessed through set codes. Uniqueness assembles coherent prefixes without a class-choice principle. Replacement is used only on set-sized domains, including the union at a limit failure. The argument never constructs a class containing all class words, never invokes a truth predicate for arbitrary formulas, and never uses ETR. The fixed baseline already supplies the choice strength needed for the previously proved completion and extension lemmas.

## 47 Automorphisms of the core (source 14; sources 17, 15 and 13)

[merge] Four sources show that the bare order of the core, or of the full order of global enumerations, does not remember what the recognition theorem uses. Source 14 classifies the automorphism orbits of class cuts by their characters (complete invariant) and derives that exhaustiveness is not invariant; source 17 gives an exact criterion for an automorphism of the core to extend to the global enumerations, the non-extension theorem and the positive family of numerical relabellings; source 15 gives a weaker form of source 14's corollary; source 13 proves a different theorem, that an automorphism of the full order  $\mathfrak{W}_{s1}$  can move a cylinder to a non-cylinder. Source 14 is printed first.

### 47.1 Source 14: automorphism orbits and the limits of bare-order recognition

The prefix classification has a complementary classification in the language of order alone. It is coarser: different prefix-boundary types can belong to the same orbit under class order automorphisms.

**Lemma 47.1** (Set homogeneity of the core). *Every increasing bijection between two set-sized suborders of  $\mathcal{P}_s(b)$  extends to a class order automorphism of  $\mathcal{P}_s(b)$ .*

*Proof.* Begin with the given partial map and carry out the back-and-forth construction used in [Corollary 44.3](#). At any set stage its domain and range are sets, even though they may already contain the entire initial partial map. The lower and upper images of a new point form a set cut. A strict separator automatically avoids all previously used image points, since each of those is on one of the two sides. The inverse step is symmetric. Taking the coherent union gives a surjective class order embedding extending the original map.  $\square$

**Theorem 47.2** (Complete character invariant for cut orbits). *Two proper normalized class cuts of  $\mathcal{P}_s(b)$  lie in the same orbit under class order automorphisms if and only if they have the same character pair. The possible pairs are exactly*

$$(\rho, \text{Ord}), \quad (\text{Ord}, \rho), \quad (\text{Ord}, 1), \quad (\text{Ord}, \text{Ord}),$$

where  $\rho$  ranges over the infinite regular set cardinals.

*Proof.* A class automorphism and its inverse carry sets to sets and preserve cofinality and coinitality, so equality of characters is necessary. A normalized lower cut has no maximum, so any set-sized cofinality is an infinite regular cardinal. The upper complement either has a least point, with character 1, has infinite regular set coinitality, or has character Ord. Both characters cannot be set-sized by the core set-cut property. These observations exhaust the displayed possibilities. They all occur: lower and upper edges of a prefix of length  $\rho$  realize the first two, a core point realizes the third, and a successor-prefix cylinder edge realizes the fourth.

Suppose first that  $D, D'$  have the same set-sized left cofinality  $\rho$ . Choose strictly increasing cofinal sequences of type  $\rho$  in them. Matching the sequences gives a set partial isomorphism. [Lemma 47.1](#) extends it to an automorphism. Since a normalized lower set is the strict downward closure of any cofinal sequence with no last member, that automorphism carries  $D$  onto  $D'$ . For equal right coinitalities use coinital sequences in their complements and the same argument with upward closures. This includes right character 1, where one simply matches the two least points of the complements.

It remains to consider two cuts with character (Ord, Ord). Each of their four sides is itself a proper class  $\eta_{\text{Ord}}$ -order. For example a set cut inside  $D$  with a nonempty upper side is filled by core saturation, with the interpolant still below an element of  $D$ ; a set with no upper constraints has an upper bound inside  $D$  because  $D$  has no set-sized cofinal subset. Lower bounds remain available from the ambient core. The corresponding argument for the complement uses its absence of a set-sized coinital subset. None of these sides has an endpoint. Back-and-forth therefore gives one class isomorphism  $D \rightarrow D'$  and another between their complements. Their union is an order automorphism of the core, since all points on a lower side precede all points on its upper side.  $\square$

**Corollary 47.3** (Exhaustiveness is not a bare-order invariant). *There are a global exhaustive enumeration  $E$  and a class order automorphism  $g$  of  $\mathcal{P}_s(b)$  such that  $g[D_E]$  is not the trace of any unrestricted class well-order on  $\mathbf{No}$ . In particular, exhaustive representability and even branch representability cannot be recognized by a criterion invariant under all class automorphisms of the bare core order.*

*Proof.* Let  $E$  swap every consecutive ordinal pair of the baseline:

$$E(2 \cdot \alpha) = b(2 \cdot \alpha + 1), \quad E(2 \cdot \alpha + 1) = b(2 \cdot \alpha).$$

It is exhaustive, but it is not represented by a core code, because it moves a proper class of coordinates. Hence  $D_E$  has character (Ord, Ord). The lower edge of the one-letter cylinder  $C_{(b(0))}$  has the same character. The orbit theorem supplies  $g$  carrying the former cut to the latter. A cylinder edge is not a branch trace by [Theorem 43.3](#), and is not a raw-order trace by source 14's [Theorem 10.2](#) (trace fibres; [\[merge\] theorem 54.6](#), and [proposition 15.3](#) for the fibre description). This proves the assertion.  $\square$

Thus the reference to named cylinders in the recognition theorem is not merely a technical limitation of its proof. A genuine symmetry of the bare linear order can destroy the property being recognized. Principal exhaustive cuts are distinguished by their right character 1, but non-core exhaustive cuts are not distinguished from other (Ord, Ord) boundaries. The stabilizer of the exhaustive-cut collection inside the full automorphism group is therefore a proper subgroup, deserving its own study.

*Note 47.4* ([\[merge\]](#); overlaps). [Lemma 47.1](#) is the set-homogeneity of an order filling all set cuts ([theorem 5.17](#), source 09's remark on set-sized partial isomorphisms in its direction 4). [Corollary 47.3](#) and source 17's [theorem 47.7](#) use the same exhaustive enumeration (the baseline with every consecutive pair swapped); source 14 sends its cut to a cylinder edge, source 17 to the cut of a nonexhaustive injective word, and source 15's [theorem 47.10](#) to a first-label cut, so all three show that neither represented cuts nor exhaustive cuts are invariant under automorphisms of the bare core. Source 15's proof that each side of such a cut is a copy of **No** is the argument of source 17's [lemma 47.6](#) and of the last paragraph of the proof of [theorem 47.2](#).

## 47.2 Source 17: automorphisms, an exact extension test and a negative result

Let  $\mathcal{R}_b(L)$  be the predicate saying that  $L$  is a proper nonempty open lower cut of  $\mathcal{P}_s(b)$  which satisfies both the straddling and coverage conditions of [Theorem 46.21](#). Thus  $\mathcal{R}_b$  describes exactly the cuts  $L_E$  with  $E \in \mathfrak{W}_{sl}$ . It is a predicate on individual class parameters, not an ordinary class of cuts.

For a class order-automorphism  $h : \mathcal{P}_s(b) \rightarrow \mathcal{P}_s(b)$ , its action on cuts is  $h_*L = h[L]$ . An extension of  $h$  to  $\mathfrak{W}_{sl}$  means a uniform operator sending each global enumeration to a global enumeration, preserving comparisons and having a uniform inverse, whose action on core enumerations is  $h$ . This convention states the higher-order typing of the assertion.

**Theorem 47.5** (Extension criterion). *A class order-automorphism  $h$  of  $\mathcal{P}_s(b)$  extends to an automorphism of  $\mathfrak{W}_{sl}$  if and only if*

$$\mathcal{R}_b(L) \iff \mathcal{R}_b(h_*L) \quad \text{for every admissible lower class cut } L. \quad (\text{S17.9})$$

*When it exists, the extension is unique and is reconstructed uniformly by [Theorem 46.21](#). No new class-valued recursion is required.*

*Proof.* If  $H$  extends  $h$ , order preservation gives

$$L_{H(E)} = h_*L_E.$$

Applying the same fact to  $H^{-1}$  proves preservation of the represented cut predicate in both directions.

Conversely assume (S17.9). For each  $E$ , form  $L_E$ , take its image under  $h$ , and reconstruct its unique exhaustive word by Theorem 46.21. Call this word  $H(E)$ . The entire construction is elementary from  $E, h, b$ . Distinct rows have distinct cuts, and  $E < F$  if and only if  $L_E \subsetneq L_F$ , because the core is dense between them. Hence  $H$  preserves and reflects order. The same recipe using  $h^{-1}$  is its inverse. For a code  $c$ , the image of the strict lower segment below  $c$  is the strict lower segment below  $h(c)$ , so  $H$  agrees with  $h$  on the core. Finally an extension's value is determined by its comparisons with every core code, proving uniqueness.  $\square$

The criterion gives a precise role to the represented-cut predicate. It is not automatic that an abstract automorphism of the dense core preserves it. In fact it can fail in a strong way.

**Lemma 47.6** (Homogeneity of large two-sided core gaps). *Let  $(L, U)$  be a gap of  $\mathcal{P}_s(b)$  such that  $L$  has no set-sized cofinal subset and  $U$  no set-sized coinital subset. Each of the class orders  $L$  and  $U$  is isomorphic to  $\mathbf{No}$ . Any two such gaps are conjugate by a class order-automorphism of  $\mathcal{P}_s(b)$ .*

*Proof.* Consider a separated pair of sets  $A < B$  in  $L$ . If  $B$  is nonempty, Theorem 42.13 gives a core interpolant below every member of  $B$ , hence lying in  $L$ . If  $B$  is empty, the absence of a set-sized cofinal subset of  $L$  supplies a member of  $L$  strictly above all of  $A$ ; this includes the case  $A = \emptyset$  after choosing any member. Therefore  $L$  has the full set-cut interpolation property. It is proper: a set  $L$  would be a set-sized cofinal subset of itself. The dual proof gives the same property and properness for  $U$ .

The class back-and-forth proof of Corollary 42.14 now yields  $L \cong \mathbf{No} \cong U$ . Given two gaps, choose order-isomorphisms between their lower sides and between their upper sides. The union of these two class maps is an order-automorphism of the core carrying the first gap to the second. This uses only two class maps at a time, not an exhaustive choice of maps for a higher collection of gaps.  $\square$

**Theorem 47.7** (Some core automorphisms do not extend). *In GBC there is a class order-automorphism of  $\mathcal{P}_s(b)$  which does not extend to an automorphism of  $\mathfrak{W}_{sl}$ . More precisely, there is one taking the cut of an exhaustive enumeration to the cut of a nonexhaustive injective word.*

*Proof.* Construct an exhaustive enumeration  $E$  by transposing the two entries in every successive pair of positions of  $b$ . It differs from  $b$  on a proper class of positions, so it has no core code. Choose also a bijection  $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r_0\}$  for one fixed label  $r_0$ . It cannot have a core code because it is not exhaustive.

By Proposition 46.26, both  $L_E$  and  $L_F$  are genuine two-sided core gaps with no small cofinal or coinital side. Lemma 47.6 supplies an automorphism  $h$  with  $h_*L_E = L_F$ . The first cut satisfies  $\mathcal{R}_b$ , whereas the second fails coverage. Theorem 47.5 rules out an extension. Alternatively, any proposed image  $H(E) \in \mathfrak{W}_{sl}$  would have cut  $L_F$ , whose unique representing injection is the nonexhaustive word  $F$ , a contradiction.  $\square$

**Corollary 47.8** (A limit to pure-order recognition). *Exhaustiveness of a branch cut is not invariant under all automorphisms of the bare class order  $\mathcal{P}_s(b)$ . In particular it cannot be characterized by a parameter-free, automorphism-invariant property of that order and its cuts alone. The two-sided character  $(\text{Ord}, \text{Ord})$  does not suffice.*

This does not preclude recognition using the baseline, labelled prefix tree, or suitable additional predicates. Theorem 46.21 provides exactly such a positive result. Nor does it rule out more subtle reconstruction after adding a distinguished represented-cut predicate. It shows why order-isomorphism of dense cores, by itself, cannot be used to extend maps of the entire higher-order comparison.

### 47.2.1 A natural positive family

**Proposition 47.9** (Relabelling by numerical automorphisms). *If  $\varphi : \mathbf{No} \rightarrow \mathbf{No}$  is an increasing class bijection, then  $E \mapsto \varphi \circ E$  is an automorphism of  $\mathfrak{W}_{\text{sl}}$ , and transporting relations by  $\varphi$  is an automorphism of the unrestricted comparison  $\mathfrak{W}_{\text{all}}$ . It carries the core associated with  $b$  onto the core associated with  $\varphi \circ b$ .*

*Proof.* The first differing labels are transformed by an increasing bijection, so their comparison is unchanged. Surjectivity, injectivity, and the class well-order property are preserved under relabelling. Finally  $\varphi \circ (b \circ \hat{\sigma}) = (\varphi \circ b) \circ \hat{\sigma}$ , with the same set support code.  $\square$

This is a sufficient family of extendible maps, not a classification of all of them. Arbitrary permutations of the label class need not preserve lexicographic order. A general isomorphism between two cores whose baselines differ on a proper class of positions need not extend either; [Theorem 47.5](#) identifies the additional condition that must be checked.

### 47.3 Source 15: the bare order does not determine represented cuts

**Theorem 47.10** (An automorphism destroys prefix representability). *Extension proved in source 15. In GBC, there is a class order automorphism of  $\mathcal{P}_{\text{bd}}(b)$  taking a cut represented by an injective Ord-word to a cut represented by no such word, and therefore by no raw class well-order.*

*Proof.* Take a nonexhaustive word  $F$ . Neither side of  $(L_F, R_F)$  has a set cofinality toward the cut: a set of first-disagreement coordinates can be bounded, and a perturbation agreeing with  $F$  beyond that bound improves every proposed bound. Each side consequently fills all its own set cuts. For a cut with constraints on both sides, use the core interpolation theorem between bounding points in that side. For an empty bounding side, use the absence of a set cofinal or cointial family toward the removed boundary, together with the unboundedness away from it. Thus each side is a copy of  $\mathbf{No}$ .

The class order  $\mathbf{No} + \mathbf{No}$  also fills every set cut. If constraints straddle the summands, choose a point in the first summand above the set of lower constraints; cases with both bounds in one summand reduce to that summand. The cases of constraints extending into the second summand are handled there. Class uniqueness gives  $\mathbf{No} + \mathbf{No} \cong \mathbf{No}$ . Hence there is a numerical partition  $\mathbf{No} = A + B$  with both sides isomorphic to  $\mathbf{No}$ .

Let  $L' = \{d : E_d(0) \in A\}$  and  $R' = \{d : E_d(0) \in B\}$ . This cut has no crossing cylinder at length one, so it is unrepresented. Neither side has a set cofinality toward the cut, as is seen already in its first labels. The preceding interpolation argument shows that  $L'$  and  $R'$  are also copies of  $\mathbf{No}$ . Choose class isomorphisms  $L_F \rightarrow L'$  and  $R_F \rightarrow R'$  and take their union. It is the required automorphism. All these maps have sets as inputs and outputs.  $\square$

Thus even knowing the abstract class order  $\mathcal{P}_{\text{bd}}(b) \cong \mathbf{No}$  is insufficient to identify the enumeration cuts. One must retain predicates for the cylinders, the evaluation map, or equivalent structure. This supplies a negative answer to a natural reconstruction question raised in the prior report. The theorem does not claim that every automorphism of the core extends to the higher collection of all global enumerations.

[merge] Source 15 states [theorem 47.10](#) without naming its foundation; its proof chooses class isomorphisms of the sides of two cuts with  $\mathbf{No}$ , so it is a theorem of GBC, and the words “In GBC” are added at the merge. Its Proposition 4.2 ([proposition 51.10](#)) needs GBC for the same reason. Its closing sentence, a “negative answer to a natural reconstruction question raised in the prior report”, answers [research question 113.4](#) in part only ([section 50](#)).

#### 47.4 Source 13: the pure order does not recover the prefix cylinders

The earlier report asks whether prefix structure can be recovered from the pure order ([research question 113.4](#)). We give a negative answer for the family of all set-prefix cylinders, using only a finite change in how information is grouped.

**Lemma 47.11** (Finite tuples as a surreal class order). *For each positive integer  $n$ , the class  $P_n$  of injective  $n$ -tuples of surreals, ordered lexicographically, is isomorphic to  $\mathbf{No}$ .*

*Proof.* First, finite lexicographic powers  $\mathbf{No}^n$  fill every set cut. At the first coordinate, the lower and upper projected values satisfy all cross weak inequalities. If no cross value is shared, choose a strict separator for that coordinate and fill the remaining coordinates arbitrarily. If a value is shared, it is the unique common one; keep that coordinate and use the active rows in the remaining coordinates. Continuing this way cannot reach two identical opposing tuples, because the original cut was strict. This proves the set-cut property. [Lemma 42.2](#) gives  $\mathbf{No}^n \cong \mathbf{No}$ .

There is a uniform order isomorphism  $\mathbf{No}^n \rightarrow P_n$ : at coordinate  $i$ , map the input value increasingly into the complement of the previously output finite set, using  $h_S$ . Recover the input by the inverse complement map coordinate by coordinate. First disagreements are preserved. Compose the two isomorphisms.  $\square$

Fix an order isomorphism  $k : \mathbf{No} \rightarrow P_2$ . Write  $k(x) = (u_x, v_x)$ , where  $u_x \neq v_x$ . There is a uniform increasing bijection

$$g_x : \mathbf{No} \setminus \{x\} \longrightarrow \mathbf{No} \setminus \{u_x, v_x\}, \quad g_x = h_{\{u_x, v_x\}} \circ h_{\{x\}}^{-1}.$$

**Theorem 47.12** (Cylinder non-rigidity). *In GBC there is a uniform order automorphism of  $\mathfrak{W}_{\text{sl}}$  that maps some prefix cylinder to a class of enumerations that is not any prefix cylinder. Thus the family of all prefix cylinders is not invariant under automorphisms of the pure linear order.*

*Proof.* Define  $H(E)$  by replacing the first label  $x = E(0)$  with the pair  $(u_x, v_x)$ , and translating the remaining labels by  $g_x$ :

$$H(E)(0) = u_x, \quad H(E)(1) = v_x, \quad H(E)(2 + \alpha) = g_x(E(1 + \alpha)).$$

This is a class bijection onto  $\mathbf{No}$ . The first two output labels determine  $x$  by  $k^{-1}$ ; the inverse of  $g_x$  recovers the tail. Therefore  $H$  is a bijection on the virtual domain. Different first input labels are compared by the order isomorphism  $k$ , and equal first labels leave an increasing tail transformation. Hence  $H$  is an order automorphism, with

$$H(C_{(x)}) = C_{k(x)}.$$

Fix a label  $u$ . Then

$$H^{-1}(C_{(u)}) = \bigcup_{x \in A_u} C_{(x)}, \quad A_u = \{x : \text{the first coordinate of } k(x) \text{ is } u\}.$$

The class  $A_u$  contains more than one label, because  $P_2$  contains  $(u, v)$  for every  $v \neq u$ . It is nonempty and proper, since pairs with another first coordinate also exist. The displayed inverse image therefore contains enumerations with different first labels, but is not the whole domain. It cannot be a nonempty-prefix cylinder, all of whose members have the same first label, and it cannot be the empty-prefix cylinder, which is the whole domain. Apply the automorphism  $H^{-1}$  to  $C_{(u)}$ .  $\square$



#### 47.4.1 What the counterexample does not say

The theorem rules out a parameter-free order-theoretic characterization whose members are exactly the prefix cylinders. It also rules out recovering their original tree canonically from the pure linear order. It does not say that no useful interval structure is recoverable, that every order automorphism arises by block recoding, or that the chosen core  $\mathcal{P}_s(b)$  is preserved by  $H$ .

In particular, the expanded structure containing the core, its set-support filtration, birthday data, or an explicit prefix predicate may have a much smaller automorphism group. The cut criterion of [Theorem 46.15](#) is correctly a criterion in such an expanded prefix presentation. Its non-invariance under forgetting that presentation is a substantive feature, not a foundational flaw.

#### 47.4.2 Finite block depth is not intrinsic either

Using  $P_n$  instead of  $P_2$  replaces one first label by any fixed positive finite number of distinct labels, with a complement isomorphism applied to the tail. The same construction gives an order automorphism that sends first-level cylinders to selected  $n$ th-level cylinders. Thus even finite prefix depth has no absolute meaning in the pure order, despite being indispensable to a particular lexicographic presentation.

[merge] Source 13's opening sentence refers to [research question 113.4](#), which its theorem answers in part. [Lemma 47.11](#) uses the family  $h_S$  of [lemma 42.2](#).

### 48 Class extensions (source 14; source 17)

[merge] Sources 14 and 17 prove the same theorem: for two class realizations with the same sets and a common baseline, an old core cut is the cut of a global enumeration in the larger realization exactly when it already is in the smaller one, and the enumeration is then old. The proof is the elementarity of the recognition criterion. Source 17 adds that every proper class extension adds a new global enumeration and a new represented cut ([theorem 48.4](#)). Source 14's text, from the second half of its Section 7, is printed first.

**Theorem 48.1** (No new realization of an old trace cut). *Suppose  $(M, \mathcal{C})$  and  $(M, \mathcal{D})$  are models of GBC with the same set structure,  $\mathcal{C} \subseteq \mathcal{D}$ , and a common baseline enumeration  $b \in \mathcal{C}$ . If a proper normalized core cut  $D$  belongs to  $\mathcal{C}$ , then*

$$(M, \mathcal{C}) \models \exists E \in \mathfrak{W}_{\text{sl}} (D = D_E) \iff (M, \mathcal{D}) \models \exists E \in \mathfrak{W}_{\text{sl}} (D = D_E).$$

*Moreover, when it exists, the realizing enumeration in the larger model is already a class of the smaller model.*

*Proof.* The canonical support codes, the cylinders, and their membership relations are defined from the same sets and  $b$ . Every formula in (C), (X), and the reconstruction formula has set quantifiers and the old class parameters  $D, b$  only. Its truth is therefore the same in the two structures. If the larger model contains a realizing enumeration, the recognition theorem gives (C) and (X) there and hence in the smaller model. Elementary comprehension in the smaller model constructs the displayed graph from  $D, b$ . It is the same graph as the larger model's enumeration by uniqueness. The converse is immediate.  $\square$

**Corollary 48.2.** *An enumeration newly available after such a class extension has a trace cut not belonging to the old class collection. An old nonexhaustive branch cut, cylinder edge, or numerical-gap cut is not turned into the trace of a global enumeration by enlarging only the class part.*

*Remark 48.3* (Scope of absoluteness). The theorem holds because realizability of this particular kind of cut has an elementary, uniquely reconstructing criterion. It is not an assertion that all class-existence questions are absolute, that the virtual enumeration order is unchanged, or that arbitrary internal well-foundedness is externally absolute. The set structure and the baseline must remain fixed. New classes can, and in suitable realizations do, supply new enumerations and new cuts simultaneously.

#### 48.1 Source 17: what changes when the available classes change?

Let  $(M, \mathcal{X}) \subseteq (M, \mathcal{Y})$  be two transitive two-sorted models of GBC with the same sets and  $\mathcal{X} \subseteq \mathcal{Y}$ . Assume the baseline  $b$  belongs to  $\mathcal{X}$ . The assertions below can also be stated internally as preservation statements for a specified same-set extension; the transitive formulation avoids irrelevant ambiguities about external well-foundedness.

**Theorem 48.4** (New enumerations are exactly new represented core cuts). *The set-code carrier  $\mathcal{P}_s(b)$ , its order relation, and its labelled set-prefix cylinders are unchanged between the two models. For any  $E \in \mathfrak{M}_{sl}^{\mathcal{Y}}$ ,*

$$E \in \mathcal{X} \iff L_E \in \mathcal{X}.$$

*Consequently every genuinely larger class realization  $\mathcal{X} \subsetneq \mathcal{Y}$  adds a global enumeration and a represented cut of the unchanged core.*

*Proof.* Support codes and prefixes are sets, and the two models have exactly the same sets. Their evaluations use the same baseline class. All relations in the definition of a cylinder are therefore the same.

If  $E \in \mathcal{X}$ , elementary comprehension in the smaller model gives  $L_E \in \mathcal{X}$ . Conversely, suppose  $L_E \in \mathcal{X}$ . At every ordinal length, the assertion that a set prefix straddles this cut has the same truth value in the two models: its witnesses are set codes and its quantifiers range over their common sets. Thus the smaller model can reconstruct the unique branch by the elementary formula in [Theorem 46.21](#). Its graph must be  $E$ , so  $E \in \mathcal{X}$ . Coverage is also an elementary assertion with the same set domain.

For the final assertion, choose  $A \in \mathcal{Y} \setminus \mathcal{X}$ . A global well-order already in  $\mathcal{X}$  gives a class bijection  $g : \text{Ord}^M \rightarrow M$ . The pullback  $D = g^{-1}[A]$  is a subclass of  $\text{Ord}^M$  in  $\mathcal{Y}$  and not in  $\mathcal{X}$ , since  $A$  is recovered from it by the same old  $g$ . Encode  $D$  by listing each sorted baseline pair in increasing or decreasing order according as its index is absent from or present in  $D$ . This yields  $E_D \in \mathfrak{M}_{sl}^{\mathcal{Y}}$ . Pair orientation recovers  $D$ , so  $E_D$  is not in  $\mathcal{X}$ . Its core cut is new by the first part of the theorem.  $\square$

For pre-existing global enumerations, their comparisons and their surjectivity are unchanged. A bijection from the common ordinal class still induces a well-order: any nonempty subclass of its index class has a least ordinal, witnessed below any one of its members. Thus the old set-like comparison sits inside the new one with its order intact, even though the represented-cut collection strictly expands.

This result answers a precise part of the source's class-realization question. It does not reconstruct the entire class collection from the *unlabelled external order type* of its admitted enumerations. The proof uses the baseline and its fixed decoder. That stronger reconstruction problem remains open here.

## 49 Support-restricted suborders and the exact reserve (source 14; source 17)

[merge] Source 14 measures how much support an interpolant needs; source 17 measures how much support beyond the forced positions a completion of a partial injection needs. Source 14's Section 9 is printed first, then source 17's Section 7.

### 49.1 Source 14: a sharp hierarchy of support-restricted orders

Fix an infinite set cardinal  $\kappa$  and define

$$\mathcal{P}_{<\kappa}(b) = \{\sigma \in \mathcal{P}_s(b) : |\text{supp}(\sigma)| < \kappa\}.$$

For an infinite regular cardinal  $\theta$ , an  $\eta_\theta$ -order is a linear order in which every separated pair  $A < B$  with  $|A \cup B| < \theta$  has a strict separator, including when one side is empty. This convention makes  $\eta_\omega$  mean dense without endpoints. The class-sized orders here need not have cardinality  $\theta$ .

The predecessor distinguishes finite support from arbitrary set support ([proposition 19.31](#); Section 19 of this report at the pin). We now give an exact intermediate hierarchy, including singular thresholds. The key is to compress the support of a separator without preserving its irrelevant later coordinates.

**Lemma 49.1** (Small-support extension of a partial injection). *Let  $f : D \rightarrow \text{Ord}$  be a set injection. It extends to a class permutation of  $\text{Ord}$  with finite support if  $D$  is finite, and with support of cardinality at most  $|D|$  if  $D$  is infinite. One may require that any specified set of already prescribed fixed positions, disjoint from  $D \cup f[D]$ , stay fixed.*

*Proof.* Put  $T_0 = D \cup f[D]$ . For finite  $D$ , the complements  $T_0 \setminus D$  and  $T_0 \setminus f[D]$  have the same finite cardinality. Choose a bijection between them and combine it with  $f$ ; fix everything outside  $T_0$ . For infinite  $D$  of cardinality  $\nu$ , adjoin a fresh set  $B$  of  $\nu$  ordinals disjoint from  $T_0$  and from the specified fixed positions. With  $T = T_0 \cup B$ , both  $T \setminus D$  and  $T \setminus f[D]$  have cardinality  $\nu$ . A bijection between them extends  $f$  to a permutation of  $T$ , again extended by the identity. The resulting support has cardinality at most  $\nu$ .  $\square$

**Lemma 49.2** (Support compression). *Let  $A, B$  be separated sets of core codes and let*

$$S = \bigcup \{\text{supp}(\sigma) : \sigma \in A \cup B\}.$$

*There is a strict separator whose support is finite if  $S$  is finite, and has cardinality at most  $|S|$  if  $S$  is infinite.*

*Proof.* By [Corollary 44.2](#), choose a separator  $Q \in \mathcal{P}_s(b)$ . If its support is contained in  $S$ , nothing remains to prove. Otherwise let

$$\gamma = \min(\text{supp}(Q) \setminus S).$$

Every input permutation fixes  $\gamma$ , while  $\hat{Q}$  moves it. Thus the first disagreement between  $E_Q$  and every input enumeration occurs no later than  $\gamma$ . Any permutation agreeing with  $\hat{Q}$  through  $\gamma$  will preserve all those comparisons.

Let  $D = \{\alpha \leq \gamma : \hat{Q}(\alpha) \neq \alpha\}$ . By minimality of  $\gamma$ ,  $D \subseteq S \cup \{\gamma\}$ . Keep the partial injection  $\hat{Q} \upharpoonright D$ . None of its images is a prescribed fixed position  $\beta \leq \gamma$  outside  $D$ : otherwise  $\hat{Q}(\beta) = \beta = \hat{Q}(\alpha)$  for a distinct  $\alpha \in D$ , contradicting injectivity. Apply [Lemma 49.1](#) while retaining those fixed positions. For infinite  $D$  its buffer can explicitly be chosen above  $\gamma$ . The resulting permutation agrees with  $\hat{Q}$  through  $\gamma$ , hence is still a strict separator. Its support is finite when  $S$  is finite, and is at most  $|S|$  when  $S$  is infinite, since  $|D| \leq |S \cup \{\gamma\}|$ .  $\square$

**Theorem 49.3** (Exact support saturation and an attained obstruction). *For every infinite set cardinal  $\kappa$ , put  $r = \text{cf } \kappa$ . Then  $\mathcal{P}_{<\kappa}(b)$  is an  $\eta_r$ -order but not an  $\eta_{r+}$ -order. More precisely, there are increasing  $(l_i)_{i<r}$  and decreasing  $(u_i)_{i<r}$  in  $\mathcal{P}_{<\kappa}(b)$  such that  $l_i < u_j$  for all  $i, j$ , and every core separator moves at least  $\kappa$  coordinates. A separator of support cardinality exactly  $\kappa$  exists in  $\mathcal{P}_s(b)$ .*

*Proof.* For the positive assertion take  $A, B \subseteq \mathcal{P}_{<\kappa}(b)$  with  $|A \cup B| < r$ . Fewer than  $\text{cf } \kappa$  sets, each of cardinality  $< \kappa$ , have union of cardinality  $< \kappa$ . Their union of supports therefore satisfies  $|S| < \kappa$ . Support compression gives an interpolant with support  $< \kappa$ . For  $\kappa = \omega$  both the family and all its supports are finite, so the finite clause applies.

For sharpness choose strictly increasing ordinals  $(\lambda_i)_{i<r}$  cofinal in  $\kappa$ . Consider the injective prefix

$$p : \kappa \rightarrow \mathbf{No}, \quad p(\alpha) = b(\alpha + 1).$$

Every infinite initial cardinal is a limit ordinal, so this formula uses coordinates below  $\kappa$ . It moves every coordinate of its domain relative to  $b$ . For each  $i$ , choose fresh labels

$$a_i < p(\lambda_i) < c_i, \quad a_i, c_i \notin \text{ran}(p \upharpoonright \lambda_i),$$

and extend the two prefixes

$$(p \upharpoonright \lambda_i) \frown (a_i), \quad (p \upharpoonright \lambda_i) \frown (c_i)$$

to enumerations with codes  $l_i, u_i$ . Their partial ordinal maps have domains of cardinality  $|\lambda_i + 1| < \kappa$ ; by Lemma 49.1 they can be completed with support  $< \kappa$ , even when a chosen label has a very large baseline index. Set choice suffices to make the  $r$  choices simultaneously.

For  $i < j$ , comparison at  $\lambda_i$  gives  $l_i < l_j$  and  $u_j < u_i$ . The same first-coordinate argument gives  $l_i < u_j$  for every  $i, j$ . Suppose any global enumeration  $E$  separated the two families and first differed from  $p$  at some  $\alpha < \kappa$ . Choose  $i$  with  $\lambda_i > \alpha$ . Both  $E_{l_i}$  and  $E_{u_i}$  agree with  $p$  through  $\alpha$ . According to its differing value there,  $E$  would lie below both or above both, a contradiction. Hence any separator satisfies  $E \upharpoonright \kappa = p$ . Relative to  $b$  it must move all  $\kappa$  coordinates of that prefix. In particular no member of  $\mathcal{P}_{<\kappa}(b)$  separates the families.

Finally extend  $p$  itself by Lemma 49.1. Its domain has cardinality  $\kappa$ , so the extension has support at most  $\kappa$ ; the forced moves show the support is exactly  $\kappa$ . It separates the two families by comparison at  $\lambda_i$ . The obstruction has combined cardinality  $r$ , hence violates  $\eta_{r+}$ , and its minimum support cost is attained.  $\square$

#### 49.1.1 Consequences and examples

For a regular  $\kappa$  the answer is exactly  $\eta_\kappa$  but not  $\eta_{\kappa+}$ . In particular finite support is dense without endpoints but can fail a countable cut; countable support fills every countable strict cut but can fail a cut of size  $\omega_1$ . For  $\kappa = \aleph_\omega$ , the cofinality is  $\omega$ , so support of every cardinality below  $\aleph_\omega$  still does not guarantee countable-cut filling. The same construction takes supports cofinal in the threshold and forces their full union at the limit.

This sharpness statement is stronger than merely exhibiting some gap: it quantifies how much extra support any solution must use. It also explains why replacing a singular support bound by its cardinal size in a saturation argument is invalid. The closure property used in the positive direction is closure under unions of fewer than  $\text{cf } \kappa$  supports, not fewer than  $\kappa$  supports.

**Proposition 49.4** (Universality without high saturation). *Every  $\mathcal{P}_{<\kappa}(b)$  contains an order-isomorphic copy of  $\mathbf{No}$ , already among the permutations with support at most two. Thus, in GBC, it contains every ordinary class linear order up to embedding. Nevertheless no fixed  $\mathcal{P}_{<\kappa}(b)$  is isomorphic to  $\mathbf{No}$ , and it is not dense in  $\mathfrak{W}_{\text{sl}}$ .*

*Proof.* For  $x \in \mathbf{No}$ , transpose the baseline positions 0 and  $b^{-1}(x)$ , using the identity if these coincide. Call the code  $t_x$ . Then  $E_{t_x}(0) = x$ , so  $x < y$  implies  $t_x < t_y$ . The familiar class universality of  $\mathbf{No}$  follows by inserting points of any ordinary class linear order, in a fixed set-like global enumeration, between the images of their earlier lower and upper neighbors; every stage has only set many constraints. Composing gives the asserted embeddings.

**Theorem 49.3** supplies a set cut not filled by  $\mathcal{P}_{<\kappa}(b)$ , whereas  $\mathbf{No}$  fills all set cuts, so the two orders are not isomorphic. For ambient nondensity, use the prefix  $p$  of length  $\kappa$  in that theorem. Choose two distinct fresh next labels and complete both prefixes to core enumerations. Every enumeration between them shares  $p$  and hence moves at least  $\kappa$  baseline coordinates. The nonempty open interval they determine misses  $\mathcal{P}_{<\kappa}(b)$  entirely.  $\square$

*Remark 49.5* (Cardinality versus location of support). The theorem restricts  $|\text{supp } \sigma|$ , not the ordinal height of the support. A transposition with an arbitrarily late baseline coordinate still has support two. Replacing the cardinality condition by  $\text{supp } \sigma \subseteq \gamma$  for one fixed ordinal  $\gamma$  gives a set-sized permutation space with quite different universality. These two filtrations should never share an undifferentiated “bounded” label.

*Note 49.6* ([merge]; the finite case is source 08’s). At  $\kappa = \omega$ , **theorem 49.3** is source 08’s **proposition 19.31** (printed in Part IV), with the same witness prefix  $p(n) = b(n+1)$  and the same fresh labels; source 14 cites the finite-support subsection but not the proposition. New in source 14 are the positive half ( $\mathcal{P}_{<\kappa}(b)$  is  $\eta_r$ ), the threshold at every infinite  $\kappa$ , including singular  $\kappa$ , and the attained support cost  $\kappa$  of a separator. **Proposition 49.4** extends source 08’s remark that the finite-support codes contain a copy of  $\mathbf{No}$  (**note 19.9**).

## 49.2 Source 17: the optimal extra support needed to extend a partial injection

The source extends a prescribed prefix inside a sufficiently large cardinal block. That argument is safe but leaves a more elementary optimization question: how many *additional moved positions* are actually necessary? The following answer applies to arbitrary set-sized partial injections, not just prefixes.

**Theorem 49.7** (Exact reserve cardinal). *Let  $A \subseteq \text{Ord}$  be a set and let  $h : A \rightarrow \text{Ord}$  be injective. Define*

$$D = \{\xi \in A : h(\xi) \neq \xi\}, \quad R = h[D], \quad S = D \cup R, \quad u = |R \setminus D|, \quad v = |D \setminus R|.$$

*Among class permutations  $\pi : \text{Ord} \rightarrow \text{Ord}$  with set support which extend  $h$ , the minimum possible value of  $|\text{supp}(\pi) \setminus S|$  is*

$$\tau(h) = \begin{cases} 0, & u = v, \\ \max(\aleph_0, u, v), & u \neq v. \end{cases} \quad (\text{S17.6})$$

*Moreover every point of  $S$  must move. An extension attaining the bound can be constructed by a bijection between two explicitly specified sets. The theorem uses only set choice and elementary operations with the class of ordinals; with a fixed global choice parameter the construction can be made uniform.*

*Proof.* The fixed points  $A \setminus D$  are disjoint from  $R$ , by injectivity of  $h$ . Every  $d \in D$  must move. If  $r \in R \setminus D$ , it already has a preimage  $d \neq r$  under the prescribed map, so an extending permutation cannot fix  $r$ . Thus every point of  $S$  must move.

Suppose  $\pi$  is an extension and put  $T = \text{supp}(\pi) \setminus S$ . No point of  $T$  is in  $A \setminus D$ , whose values are prescribed to be fixed. The support of a permutation is invariant under the permutation and

its inverse. After the prescribed bijection  $h : D \rightarrow R$  has been removed,  $\pi$  must therefore give a bijection

$$(R \setminus D) \cup T \longrightarrow (D \setminus R) \cup T. \quad (\text{S17.7})$$

In particular  $u + |T| = v + |T|$ . If  $u \neq v$  are finite, finite cancellation rules out finite  $T$ , and  $\aleph_0$  suffices. If at least one of  $u, v$  is infinite, unequal cardinals can be equalized by addition of  $|T|$  only when  $|T| \geq \max(u, v)$ . This follows from the formula for addition of infinite cardinals as their maximum. These observations prove the lower bound in (S17.6).

Conversely, choose a fresh set of ordinals  $T$  of cardinality  $\tau(h)$ , disjoint from  $A \cup R$ , and choose a bijection in (S17.7). Combine it with  $h : D \rightarrow R$ , and take the identity outside  $S \cup T$ . The result is a permutation extending  $h$ , with support contained in  $S \cup T$ . All points of  $S$  move, as proved above. If  $\tau(h) > 0$ , the lower bound forces exactly  $\tau(h)$  further points to move; if  $\tau(h) = 0$ , no reserve was used. This proves attainment, including cases where the chosen bijection happens to fix some points of  $T$ .  $\square$

**Example 49.8** (Why a finite reserve can never repair a one-sided shift). Prescribe  $h(n) = n + 1$  for every  $n < \omega$ . Then  $D = \omega$ ,  $R = \omega \setminus \{0\}$ , and  $(u, v) = (0, 1)$ . No permutation on  $\omega$  extends  $h$ . More strongly, no extension can use only finitely many additional moved positions outside  $\omega$ . Exactly countably many are necessary and sufficient. One concrete completion chooses fresh  $t_0, t_1, \dots$ , sends  $t_0$  to 0, and sends  $t_{n+1}$  to  $t_n$ ; all other ordinals are fixed.

**Example 49.9** (Finite partial injections require no reserve). When  $D$  is finite,  $|D| = |R|$  implies  $|R \setminus D| = |D \setminus R|$ . Hence every finite partial injection extends to a permutation supported exactly on  $D \cup R$ . The distinction between this fact and the preceding infinite example is precisely the failure of finite cancellation for infinite sets; finite testing alone cannot find the infinite obstruction.

**Corollary 49.10** (Application to prescribed surreal prefixes). *Fix a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . If  $p : \alpha \rightarrow \mathbf{No}$  is an injective set prefix, apply Theorem 49.7 to  $h = b^{-1} \circ p$ . The resulting  $b \circ \pi$  extends  $p$  and agrees with  $b$  outside a set. Formula (S17.6) gives the exact minimum number of moved coordinates beyond the necessarily moved set  $D \cup h[D]$ .*

This is a cardinal optimum relative to a given partial injection and baseline. It does not assert a minimum *ordinal* support bound, a minimum birthday of a code, or a canonical extension independent of choice. Those are different optimization problems. The source’s larger-cardinal completion remains a useful simple implementation; the reserve formula says exactly when its extra room is genuinely necessary.

*Note 49.11* ([merge]; the reserve theorem in this report). Theorem 49.7 sharpens source 09’s remark 19.7: the example  $n \mapsto n + 1$  there is source 17’s example 49.8, and the theorem says exactly when the larger cardinal of lemma 19.4 is needed and how many extra positions suffice. The two support theorems are consistent: lemma 49.1 bounds the support of an extension of a set injection  $f : D \rightarrow \text{Ord}$  by  $|D|$  when  $D$  is infinite and keeps it finite when  $D$  is finite, while theorem 49.7 computes the least number of moved positions outside the forced set: 0 for finite partial injections (example 49.9), and otherwise 0 or  $\max(\aleph_0, u, v) \leq |D|$ . Source 17’s question table maps the reserve theorem to source 11’s research question 113.6; that question asks for birthday bounds, and the theorem bears instead on source 09’s direction “Sharp birthday and support bounds” and source 08’s question 13 (section 50).



## 50 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 33](#); where the question index of [section 113](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 11's research question 113.1 and source 09's direction "Which cuts of the core represent exhaustive branches?"* ([section 113.3](#)): **answered**, in the labelled-tree form the question asks for. A normalized class cut of the core is the cut of an injective class word exactly when a crossing prefix exists at every set length; the word is unique and elementary in the cut; it is exhaustive exactly when the crossing prefixes cover **No** ([theorem 45.1](#); six proofs, [note 46.1](#)). Every other proper cut is a cylinder edge or a numerical gap after a set prefix, with unique data ([theorem 43.3](#)); the missing labels, the forbidden repeated labels and the cuts escaping every coherent branch that [research question 113.1's](#) commentary asks to separate are exactly the coverage failure, source 15's repeated-label case and the edge and gap cuts. A recognition from the bare order is impossible (item (c)).
- (b) *Source 08's question 8, higher-size cuts and completion* ([section 113.4](#)): answered for the class cuts of the core relative to its prefix presentation: the trichotomy and characters ([theorems 43.3](#) and [44.1](#)), the image of the raw orders ([corollary 46.4](#)), and source 15's remark that adjoining the raw orders is neither a Dedekind completion of the core nor a realization of every core cut ([section 46.2](#)). Cuts given by higher collections of classes stay open.
- (c) *Source 11's research question 113.4, source 09's direction "Automorphisms beyond set-sized partial maps" and source 08's question 6*: answered in part, negatively. The automorphism orbits of class cuts of the core are exactly the four character classes ([theorem 47.2](#)); neither represented nor exhaustive cuts are invariant under automorphisms of the bare core ([corollary 47.3](#) and [theorems 47.7](#) and [47.10](#)); the prefix cylinders of  $\mathfrak{W}_{\text{sl}}$  are not invariant under automorphisms of  $\mathfrak{W}_{\text{sl}}$ , nor is finite prefix depth ([theorem 47.12](#)). An automorphism of the core extends to  $\mathfrak{W}_{\text{sl}}$  exactly when it preserves represented cuts ([theorem 47.5](#)). Open: the stabilizer of the exhaustive cuts, and which extra predicates make the reconstruction unique (source 14's and source 17's own questions).
- (d) *Source 11's research question 113.5, baselines*: its suggested restricted first case is settled: a baseline change by an increasing class bijection of **No** is an automorphism of  $\mathfrak{W}_{\text{sl}}$  and of  $\mathfrak{W}_{\text{all}}$  carrying one core onto the other ([proposition 47.9](#), source 08's remark in its question 6). The general case is open.
- (e) *Source 11's research question 113.8, source 12's research question 113.18 and source 08's question 12, class realizations*: answered in part. Over a fixed set structure and baseline, the represented-cut status of an old core cut is absolute, its enumeration is old, and every proper extension of the class part adds a global enumeration and a represented cut ([theorems 48.1](#) and [48.4](#)). What the unlabelled external order type remembers is open here (Part IX prints source 18's countable models, [section 58](#), and [section 60](#) (e) records what they answer).
- (f) *Source 09's direction "Sharp birthday and support bounds" and source 08's question 13*: answered in part for support. The least support cardinal of an interpolant in  $\mathcal{P}_{<\kappa}(b)$  is governed by  $\text{cf } \kappa$  ([theorem 49.3](#)), and the least reserve of a completion of a partial injection is 0 or  $\max(\aleph_0, u, v)$  ([theorem 49.7](#)). Ordinal support bounds, birthday bounds and the forced-prefix length remain open, as source 17 says.
- (g) *Source 11's research question 113.6* is not affected by the reserve theorem ([note 49.11](#)).

[merge] Dated note, 3 October 2026 (batch 83). Items (c)–(f) are answered further by Parts XIII and XIV: unlabelled cylinders do not recognize exhaustiveness while equality of labels across

extension edges does, the admitted labelled enumerations recover a class realization, and the least ordinal support is computed (sections 105 and 112; section 113.25 (e), (h), (i)).

## Part IX

# Unrestricted class well-orders: canonical blocks, condensation, products, comprehension and models

[merge] (3 October 2026.) The fourth and last Part of the batch-81 addition (section 1.8; notation in section 1.3.1). Part III (part III) compared arbitrary class well-orders of **No** directly, extracted the canonical initial set-like part of each, and asked how far that extraction can be iterated (research question 113.3); Part V (part V) compared the two global orders in larger universes. This Part prints what sources 13–18 prove beyond those Parts, and every remaining section of theirs on foundations, universes and formalization:

- their restatements of the order facts and of the direct comparison of class well-orders (section 51), all re-proofs of Parts I and III;
- the canonical decomposition of a class well-order into proper blocks of type **Ord** and one optional final set block, and the theorem that every class well-order is a block-index order (section 52), proved by sources 15, 16 and 18;
- explicit condensation histories of every class length and what they say about recursion strength (section 53, source 15);
- the maximal finite components of the unrestricted order and the trace fibres of the core (section 54, sources 17 and 14);
- asymmetric products (section 55, source 13);
- binary class words, the diagonal theorem and a  $\Sigma_1^1$  comprehension bound for uniform nonembedding (section 56, sources 16 and 14);
- larger universes and class realizations (section 57, sources 16, 14, 13, 15 and 18);
- the external classification of the full order in countable  $\omega$ -standard models (section 58, source 18);
- the six formalization plans (section 59).

Section 60 records which questions the Part answers. The sources' foundations sections and Lean plans are printed here, at the end of the addition, rather than in Part V, so that no existing section is renumbered.

*Notation in this Part* (in addition to section 1.3.1). The block notations are the sources' own and differ: source 15 writes  $B_W$  for the class of least representatives of the blocks of a class well-order  $(X, W)$  and  $C_b = \pi_W^{-1}(b)$  for the block with least representative  $b$ ; source 16 writes  $J_R$  for the least representatives of the proper blocks of  $R$  and  $\alpha_R$  for the type of the final set block; source 18 writes  $B_R$  and  $\theta_R$ . *False readings:* in source 15's block section  $b$  is a point of the order, not the baseline enumeration, and  $C_b$  is a block, not a prefix cylinder  $C_p$  of the core; source 14's  $J_R$  (section 54) is the canonical initial branch  $\mathbf{Ord} \rightarrow \mathbf{No}$  of  $R$  (source 15's  $F_R$ , source 13's  $F_R$ ), not source 16's class of block representatives. Sources 16 and 18 write  $\rho_R(x)$  for the rank of  $x$  inside its block; it is unrelated to source 15's coding length  $\rho_\kappa$ . In the sections on larger universes  $\kappa$  denotes a

strongly inaccessible cardinal, as in Part V (source 14 writes  $\Omega$ , sources 17 and 18 write  $\lambda$ ); it is not a birthday cutoff there. Source 18 writes  $H_{\text{sl}}$ ,  $H_{\text{all}}$  with a plain  $H$  in its later sections; they are printed  $\mathfrak{W}_{\text{sl}}$ ,  $\mathfrak{W}_{\text{all}}$ , as its macros are. Source 15's finite ordinal polynomials  $P(\omega)$  and source 18's class  $\mathcal{P}$  of finite-support functions  $\omega \rightarrow \text{Ord}$  are the same order ([note 53.6](#)). Labels written for the merge in this Part begin `swo:ix:`.

## 51 Order facts and direct comparison as re-proved in sources 15, 16, 18, 17 and 13

*Note 51.1* ([[merge](#)]; Part I and Part III counterparts). Before their class-level results, sources 13, 15, 16, 17 and 18 re-prove the surreal order facts and the direct comparison of class well-orders that Parts I and III already contain; each credits this report (source 18 labels its comparison theorems “repository result”). Their sections are printed whole below, in the order 15, 16, 18, 17, 13 for the order facts and 13, 15, 16, 18 for the comparison; none is new against the repository.

- *Fresh separators*: [lemma 5.2](#) (source 15's [lemma 51.2](#), source 16's [lemma 51.3](#), parts (1)–(2) of source 18's [lemma 51.5](#)). Parts (3)–(4) of source 18's lemma are facts about the alphabet  $\mathbf{No}_{<\kappa}$  proved in Part VI ([theorem 29.4](#)).
- *Universality and uniqueness of  $\mathbf{No}$* : [proposition 5.13](#) and [theorem 5.17](#) (source 16's [proposition 51.4](#); the paragraph after source 15's [lemma 51.2](#)).
- *Choice*: [theorem 13.1](#) (source 16's [proposition 51.11](#), source 18's [theorem 51.14](#), the first subsection of source 15's [section 51.7](#); source 17 cites it in its foundations). Sources 15, 16 and 18 prove it over GB without set choice, as [theorem 13.1](#) does; source 16 says that the recursion well-orders the hierarchy “uniformly” and source 18 that its orders “need not extend one another”, and none of the three repeats the “coherent” overstatement corrected in this report on 3 October 2026.
- *Direct comparison by the largest common labelled prefix*: [theorems 14.2](#) and [14.7](#) (source 13's [proposition 51.7](#), source 15's [theorem 51.9](#), source 16's [lemma 51.12](#) with (S16.9), source 17's [proposition 51.6](#) with (S17.2), source 18's [lemma 51.15](#) and [theorem 51.16](#) with (S18.4); source 14's version opens [section 54](#)).
- *Canonical initial part and convex fibres*: [theorem 15.1](#), [proposition 15.3](#), and [remark 15.4](#) (source 13's [proposition 51.8](#), source 15's [proposition 51.10](#); source 14's [lemma 54.5](#) in [section 54](#)).
- *Adjacency*: [theorems 16.2](#) and [16.6](#) (source 16's [proposition 51.13](#), source 18's [theorem 51.17](#); source 17's [theorem 54.2](#) in [section 54](#)).

Source 15 states [proposition 51.10](#) without naming its foundation. Its convexity half is a theorem of GB, but the realization of every injective word as an initial word appends “any fixed well-order” of the omitted range, which needs global choice; the words “In GBC” are added at the merge (source 13's [proposition 51.8](#) likewise uses “global choice” for this step). Source 15's remark after it (“If  $F$  omits  $m$  labels, its fibre has  $m!$  elements”) is [remark 15.4](#) and [proposition 15.3](#), as it says. Source 13's Section 6 and source 16's Section 9 continue in the cut sections of Part VIII, which cite them as [proposition 51.8](#) and Theorem 10.1 of source 16 ([theorem 52.6](#)).

### 51.1 Source 15: sign sequences, properness, and cut filling

We realize a surreal as a sign sequence  $s : \alpha \rightarrow \{-, +\}$  of ordinal length  $\alpha$ . At a first difference the comparison is  $- < \text{undefined} < +$ ; the birthday is the length. The all-plus sequences give an increasing copy of  $\text{Ord}$ , so **No** is a proper class. Every set of surreals has a set bound on its birthdays, by replacement.

The numerical surreal order fills every set cut: if  $L, R$  are sets and  $L < R$ , some  $x$  satisfies  $L < x < R$ , with either side allowed to be empty. This is the familiar Conway construction. It implies density and no endpoints, but it does not assert filling for arbitrary class cuts. The numerical order and the simplicity relation are distinct: extending a sign string makes it more complicated in simplicity, but may increase or decrease its numerical value [11, 12, 14].

**Lemma 51.2** (Fresh separators). *If  $L < R$  are sets of surreals and  $S$  is a set, there is  $x \notin S$  with  $L < x < R$ . In particular, deleting a set from **No** leaves a dense order without endpoints.*

*Proof.* If every separator belonged to  $S$ , the class  $I = \{x : L < x < R\}$  would be a set, by separation from  $S$ . It is nonempty. Apply set-cut filling to  $L \cup I < R$ : every member of  $I$  is less than every member of  $R$ . A resulting separator belongs to  $I$  and exceeds all of  $I$ , a contradiction. Thus the interval has a point outside  $S$ . The same proof allows empty bounding sides.  $\square$

In GBC, a forth construction embeds every ordinary class linear order in **No**: enumerate its underlying class by  $\text{Ord}$ , and at each stage separate the sets of earlier lower and upper images. A back-and-forth construction shows uniqueness, up to class order isomorphism, of a proper class order with this set-cut property. Each stage has a set of earlier values; these are ordinary set-valued ordinal recursions. This is the classical universality and absolute-linear-continuum phenomenon [15, 20]. It does not apply to a collection whose elements are themselves class functions.

### 51.2 Source 16: foundations and surreal order facts

#### 51.2.1 Conventions about sets, classes, and choice

Here GB has ZF as its set theory and no choice principle built into its notation. We write  $\text{GBC} = \text{GB} + \text{GC}$ ; global choice implies ordinary set choice. The set-sized theorems are proved in ZFC. A *class well-order* is a strict class linear order for which every available nonempty subclass has a least element. A relation is *set-like* if each predecessor class is a set. A proper set-like class well-order is uniquely isomorphic to  $\text{Ord}$ : send each point to the ordinal type of its predecessor set.

An individual class function  $F : \text{Ord} \rightarrow \mathbf{No}$  is legitimate. The notation “ $F \in \mathfrak{W}_{\text{sl}}$ ” below abbreviates that this class function is a bijection; it does not make  $F$  an element of a GB class. Likewise, a uniform family  $(F_i)_{i \in I}$  means one class of triples  $(i, \alpha, x)$  whose rows are the graphs in question. Its index  $I$  may be a set or a class of sets. One cannot silently choose a class for each set index and assume that the rows assemble into an available uniform family.

#### 51.2.2 Sign sequences, small cuts, and bounded stages

We use the sign presentation

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha,$$

with numerical comparison at the first different or missing sign, using  $- < \text{missing} < +$ . The birthday  $\text{bd}(x)$  is the length of the sign sequence. Every set of surreals has bounded birthdays, whereas all birthdays occur. Thus **No** is proper. The basic separation property is:

$$A, B \text{ sets, } A < B \implies \exists x \in \mathbf{No} (A < x < B), \quad (\text{S16.1})$$

with either side allowed to be empty. These are standard facts about the surreal order [11, 12, 14]; the corresponding smallness-sensitive primitives in ProveIt are audited in Section 59.4.

**Lemma 51.3** (Fresh separators). *If  $A < B$  are sets of surreals and  $S$  is any set, some  $x \notin S$  satisfies  $A < x < B$ . Every nonempty numerical interval is a proper class. Deleting a set from **No** preserves the set-cut property.*

*Proof.* First choose  $u, v$  with  $A < u < v < B$  by two applications of (S16.1). The interval  $(u, v)$  cannot be a set: if it were, choose an element above all of  $(u, v)$  but below  $v$ , a contradiction. Hence the set  $S$  cannot exhaust this interval. Any remaining point works. The same argument gives the assertions about intervals and deletion.  $\square$

For infinite cardinal  $\kappa$ , put

$$\mathbf{No}_{<\kappa} = \{x : \text{bd}(x) < \kappa\}, \quad \mu = 2^{<\kappa}.$$

Then  $\mathbf{No}_{<\kappa}$  is a set of size  $\mu$ . For regular infinite  $\kappa$  it is an  $\eta_\kappa$  order: every separated pair of sets with total cardinality less than  $\kappa$  has an interpolant in the same birthday stage. Bound the option birthdays below  $\kappa$  using regularity, and use the usual birthday-controlled separator. At singular  $\kappa$  this argument fails because a small set can have birthdays cofinal in  $\kappa$ .

### 51.2.3 Which uses of recursion are elementary?

A recursion producing a set at each ordinal stage, with a set history before each stage and fixed class parameters, is available in GB. One constructs each set-length approximation by ordinary recursion and joins the compatible approximations by elementary comprehension. This applies to the forth and back-and-forth constructions used below.

By contrast, a recursion whose stages are classes is an instance of elementary transfinite recursion when its defining formula has only set quantifiers. Such a recursion may exceed GBC even along  $\omega$ ; truth recursion is a standard example. ETR supplies abstract class well-order comparability by initial-segment isomorphisms, but we neither identify comparability with ETR nor assert a necessity result for our particular constructions [27, 22]. Size of the stage and history, rather than the mere appearance of the symbol **Ord**, is decisive.

**Proposition 51.4** (Universality and uniqueness). *In GBC every class linear order embeds into **No**. Any two proper class linear orders satisfying (S16.1) are isomorphic, provided they have set-like global enumerations. The construction can be made uniform in set parameters when the carrier and order predicates are uniform.*

*Proof.* Enumerate the source by an ordinal or by **Ord** using global choice. At each stage insert its next point between the images of its earlier lower and upper points. These are sets. For an isomorphism alternate forth and back steps through fixed global enumerations of the two orders, inserting the next unused point on the appropriate side at each step. Each partial map is a set. Choose the least permissible point under fixed global well-orders to make every stage unique. Its graph is the union of the set-length approximations. With an extra set parameter carried throughout, the same defining formula gives a uniform class relation for the whole parameterized family. This is the classical mechanism of surreal universality and homogeneity [15, 33].  $\square$

### 51.3 Source 18: surreals as sign sequences

Represent a surreal number by a function  $s_x : \text{bd}(x) \rightarrow \{-, +\}$ , where  $\text{bd}(x)$  is an ordinal. Numerical order is comparison at the first unequal sign, counting an absent sign as strictly between  $-$  and  $+$ . Sign extension is the simplicity relation, which is a tree order, not a total order. This presentation and its substructures are developed in the standard surreal literature [12, 14].

We shall repeatedly use the following elementary consequences.

**Lemma 51.5** (Surreal order tools). *1. If  $L, R$  are sets of surreals and  $L < R$ , some surreal lies strictly between them. Either set may be empty.*

*2. The separator can avoid any prescribed set of forbidden values.*

*3. For infinite initial  $\kappa$ , every nonempty open interval of  $\mathbf{No}_{<\kappa}$  contains a sign-prefix cone isomorphic to  $\mathbf{No}_{<\kappa}$ ; hence it has cardinality  $\mu = 2^{<\kappa}$ .*

*4.  $\mathbf{No}_{<\kappa}$  fills every cut with fewer than  $r = \text{cf } \kappa$  input values. Its cofinality, coinitality, and both characters at every point are  $r$ .*

*Proof.* For (1), choose an ordinal strictly above all birthdays in  $L \cup R$ . Comparison in a sufficiently long full binary level supplies a sign sequence separating the two sets; equivalently use the usual simplest surreal cut construction. To get (2), the separator interval contains a proper class of values: an interval is order-isomorphic to  $\mathbf{No}$ , or one can extend a sign prefix by arbitrary ordinal lengths. A set of forbidden values cannot exhaust it.

For (3), first choose a sufficiently long but still  $< \kappa$  prefix whose whole cone lies inside the given interval. Prefix concatenation preserves numerical order. If  $\beta, \gamma < \kappa$ , then  $\beta + \gamma < \kappa$ , since this sum still has cardinality below the initial cardinal  $\kappa$ . This works at singular cardinals too. Thus concatenating the chosen prefix with every sign word of length  $< \kappa$  gives the required copy. Counting levels gives  $|\mathbf{No}_{<\kappa}| = \sup_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}$ .

For (4), fewer than  $\text{cf } \kappa$  birthdays are bounded below  $\kappa$ , so the construction in (1) stays below  $\kappa$ . Ordinal labels along a cofinal sequence in  $\kappa$  give a cofinal sequence; their negatives give a coinital one. No smaller family is cofinal, since its birthdays are bounded. At a fixed point, append a negative sign and then arbitrarily long positive tails to approach from below; use the dual construction above. These give character at most  $r$ . The small-cut property gives the reverse inequality.  $\square$

Part (2) is also available in the bounded setting when the forbidden set has size  $< \mu$  and a whole interval of separators is available. This distinction matters: a used prefix in a minimal permutation can have length much greater than  $\kappa$ , but it still uses fewer than  $\mu$  labels.

### 51.4 Source 18: set and class theories

Set-sized results are proved in ZFC. In class arguments, GB denotes Gödel–Bernays theory over ZF with elementary class comprehension and Class Replacement. GBC adds Global Choice. A class is a collection of sets, never an element of another ordinary class. We regard a class well-order  $R$  as a strict class linear order for which every nonempty available subclass has an  $R$ -least element. This definition is internal to the specified two-sorted model.

A class order is *set-like* if every predecessor class is a set. A proper set-like class well-order is isomorphic to  $\text{Ord}$ : map a point to the ordinal type of its predecessors. The range is an initial subclass of  $\text{Ord}$ , and if it were a set, Class Replacement would make the whole carrier a set. Conversely, an  $\text{Ord}$  enumeration is set-like.

All uses of class back-and-forth below choose one new set label at each ordinal stage. The preceding history at a stage is a set, and the next choice is determined using a fixed global well-order. This is ordinary set-valued recursion with class parameters. We do not infer that arbitrary



class-valued recursions follow from GBC. Elementary transfinite recursion has substantially greater strength in general [24].

An expression such as  $\mathfrak{M}_{\text{all}}$  is initially notation for a domain of class variables, not a class whose members are class relations. A uniform family  $(R_i)_{i \in I}$  means one class relation  $F(i, x, y)$  with set labels  $i, x, y$ . A set-indexed family of class relations is not silently replaced by a set of classes.

## 51.5 Source 17: foundations and direct comparison without a class of classes

### 51.5.1 Four safe levels

Our set theorems use ZFC. Our global theorems use GBC, meaning Gödel–Bernays set theory with Global Choice; GB alone includes no choice principle. This agrees with the conventions in the source and in Hamkins’s account of Global Choice [19].

A class well-order here is a linear class relation for which every nonempty admitted subclass has a least element. Set-likeness is an additional property, not part of that definition. Which subclasses are admitted is specified by the surrounding class theory or model.

There are four distinct safe ways to interpret a higher-order-looking statement. One may work with ordinary sets such as  $\mathcal{W}_\mu(X)$ ; with an ordinary class of *set codes* such as  $\mathcal{P}_s(b)$ ; with finitely many class variables or a uniform class relation coding a family; or externally in a larger universe in which the objects from the smaller universe are sets. None permits forming an unrestricted same-level set or class of all proper classes.

A uniform family indexed by a class  $I$  of sets is a single relation  $F \subseteq I \times \text{Ord} \times \mathbf{No}$ , with each row  $F_i$  an enumeration. A uniform transformation of enumerations is a formula which defines an output class from an input class and for which output existence and uniqueness are proved. It is not a GB function with proper classes as arguments stored in its graph.

### 51.5.2 The direct common-prefix formula

For a strict class well-order  $R$  let  $P_R(x) = \{y : yRx\}$ . Given two class well-orders  $R, S$  of the same labelled carrier  $X$ , define

$$C(R, S) = \{x : P_R(x) = P_S(x) \text{ and } R \upharpoonright P_R(x) = S \upharpoonright P_S(x)\}. \quad (\text{S17.2})$$

Equality and restriction here expand to quantifiers over elements of  $X$ ; no class quantifier is hidden in the defining formula.

**Proposition 51.6** (Direct labelled comparison; background with proof). *The class  $C(R, S)$  is the largest common labelled initial segment of  $R, S$ . If  $R \neq S$ , its complement is nonempty and the two least residual labels  $a = \min_R(X \setminus C)$  and  $b = \min_S(X \setminus C)$  are different. Declaring  $R <_{\text{lex}} S$  exactly when  $a < b$  gives a strict linear comparison of all admitted class well-orders of  $X$ .*

*Proof.* If  $x \in C$  and  $yRx$ , the equality of predecessor classes puts  $y$  before  $x$  in both orders; agreement below  $x$  then gives equal predecessor classes and equal predecessor orders for  $y$ . Thus  $y \in C$ . The same argument works with  $S$ , and proves agreement on  $C$ . Every common labelled initial segment satisfies the defining condition, so is contained in  $C$ . If  $C = X$ , the relations agree. Otherwise well-foundedness provides the least residual elements. If  $a = b$ , both its predecessor classes are  $C$  and the orders there agree, contradicting  $a \notin C$ .

For transitivity compare  $C(R, S)$  and  $C(S, T)$  as initial segments of  $S$ . Initial segments of a linear order are comparable by inclusion. If one is properly shorter, its first disagreement is the first disagreement relevant to comparing  $R, T$ , and has the same orientation. If the two prefixes

are equal, the three next labels  $a < b < c$  give  $a < c$ . Trichotomy and asymmetry are immediate. This is an elementary proof; it does not compare arbitrary abstract class order types by a recursive isomorphism construction.  $\square$

This result is already in the merged report (theorems 14.2 and 14.7). Its inclusion is important: it eliminates a false dilemma between restricting to Ord-enumerations and introducing a stronger class recursion theory merely to define the comparison.

### 51.5.3 Choice, recursion, and external semantics

Global Choice supplies a baseline bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . The source also proves the converse strength statement: existence of any class well-order of the sign-sequence surreals implies Global Choice over choiceless GB (theorem 13.1). That converse is not needed in our proofs; we explicitly assume GBC for the global results.

Recursion whose value at each ordinal stage is a set is available with class parameters in GBC: set recursion constructs compatible histories on each set ordinal, and elementary comprehension collects them. This is not the assertion that arbitrary class-valued recursions have solutions. The latter territory is governed by principles such as ETR, whose strength is substantially more than elementary comprehension [22]. We will reconstruct branches by a uniqueness formula, so that even the appearance of such a class-valued recursion is avoided.

In a transitive two-sorted model  $(M, \mathcal{X})$ , “all classes” means  $\mathcal{X}$ . An external observer may have more subclasses of  $M$ . With a strongly inaccessible cardinal  $\lambda$  in an ambient universe,  $(V_\lambda, \mathcal{P}(V_\lambda))$  provides a useful full-class interpretation. Externally, its set-like surreal enumerations are the  $\lambda$ -length permutations of  $\mathbf{No}_{<\lambda}$ , while its unrestricted well-orders allow all external ordinal types of cardinality  $\lambda$  (theorem 21.1). This is a conditional model construction, not an assertion that ZFC proves an inaccessible cardinal exists.

Finally, neither passing to KM nor enumerating formula strings gives an internal universal class decoder. The latter would need a uniform truth predicate, not merely syntax. Section 42.5 gives an independent diagonal explanation for the failure of exhaustive same-level coding.

## 51.6 Source 13: direct comparison of unrestricted class well-orders

The class  $\mathbf{No}$  itself is fixed throughout this section. A class well-order  $R$  is a strict linear order for which every nonempty subclass has an  $R$ -least member. A set-like class order additionally has a set of predecessors at every point. The totality  $\mathfrak{W}_{\text{all}}$  is notation for the domain of such class parameters, not an ordinary class whose members are relations.

### 51.6.1 The largest common labeled prefix

The following construction is from the earlier report (theorem 14.2; Section 14 at the pin); its explicit formula is useful because it does not demand an enumeration of a non-set-like order by Ord.

**Proposition 51.7** (Elementary labeled comparison). *Two distinct class well-orders  $R, S$  of  $\mathbf{No}$  have a largest common labeled initial part  $G$ . There are distinct labels  $r, s$  that are the respective first points after  $G$ . Defining*

$$R <_{\text{lex}} S \iff r < s$$

*gives a strict linear comparison on  $\mathfrak{W}_{\text{all}}$  by an elementary formula with  $R, S$  as class parameters. No class-well-order comparability principle is required.*

*Proof.* Define  $G$  to consist of those  $x$  such that  $\text{Pred}_R(x) = \text{Pred}_S(x)$  and the two relations agree on that common predecessor class. Equality and agreement here abbreviate formulas quantifying over set elements. They do not form the predecessor classes as elements of a larger collection.

If  $x \in G$ , all its predecessors belong to  $G$ . The restrictions of  $R$  and  $S$  to  $G$  agree, and any common labeled initial part is contained in  $G$ . If  $G$  were the whole carrier, the relations would agree. Otherwise take the least points  $r, s$  of the complements in the two orders. Their predecessor classes are both  $G$ . They cannot coincide, since their common value would then be a member of  $G$ .

Trichotomy follows from numerical trichotomy of  $r, s$ . For transitivity, consider  $R < S < T$ . The common prefixes of  $(R, S)$  and  $(S, T)$  are both initial parts of  $S$ , hence one contains the other. If one is properly shorter, its first disagreement also decides the comparison of  $R, T$ . If they are equal, the three next labels satisfy  $r < s < t$ , which gives  $R < T$ . This proves transitivity without assigning an ordinal length to a proper-class prefix.  $\square$

This is not comparison of abstract class order types by initial-segment isomorphisms. For that different problem, elementary transfinite recursion is a known sufficient principle [27, Theorem 6]. Our formula compares identical labels on one carrier, rather than constructing an isomorphism between unrelated carriers.

### 51.6.2 The ordinal head and the residual fiber

**Proposition 51.8** (Canonical head; prior structural lemma). *For  $R \in \mathfrak{W}_{\text{all}}$ , the points with a set of  $R$ -predecessors form a proper-class initial part  $H_R$ , uniquely enumerated by a class injection*

$$F_R : \text{Ord} \longrightarrow \mathbf{No}.$$

*Conversely, for every injective class word  $F : \text{Ord} \rightarrow \mathbf{No}$ , appending any class well-order of  $Y = \mathbf{No} \setminus \text{ran}(F)$  produces a raw order whose head is  $F$ . All raw orders with this head form a convex fiber, with their internal comparison equal to the labeled lexicographic comparison of well-orders of  $Y$ .*

*Proof.*  $H_R$  is initial. Were it a set, the least point outside it would have exactly  $H_R$  as its predecessor class and would belong to it, a contradiction. Ranking each point by the ordinal type of its set of predecessors gives the unique enumeration: its ranks form a proper initial subclass of  $\text{Ord}$  only if they are a set, so they are all of  $\text{Ord}$ .

After an  $\text{Ord}$ -head is listed, every residual point has a proper class of predecessors, whereas each head point has a set of predecessors. This proves the converse. Distinct heads disagree at a set ordinal, and that disagreement places one entire fiber below the other. Equal heads leave only the tail comparison, proving convexity. A well-order on  $Y$  is available from global choice, also when  $Y$  is proper.  $\square$

This is the earlier report's decomposition (theorem 15.1 and proposition 15.3; Section 15 at the pin). We shall identify exactly which cuts of a fixed dense core these fibers represent, thereby supplying an additional characterization not contained in the decomposition alone.

## 51.7 Source 15: native lexicography on class well-orders

### 51.7.1 Existence, set-likeness, and the first proper block

A class relation  $R$  on  $\mathbf{No}$  is *set-like* when  $\text{Pred}_R(x) = \{y : yRx\}$  is a set for every  $x$ . An already-given set-like class well-order of a proper class has a unique order isomorphism with  $\text{Ord}$ : assign to each

point the ordinal type of its predecessor set. The image is initial in  $\text{Ord}$  and cannot be a set, since the inverse map would make the whole carrier a set. This collapse needs GB, not a separate global choice assumption [19].

Existence is different. Over GB, the existence of any class well-order of  $\mathbf{No}$ , even a non-set-like one, is equivalent to global choice (theorem 13.1; Theorem 13.1 at the pin). One direction restricts a set-like global well-order of the universe. For the other, a well-order of  $\mathbf{No}$  orders the sign codes for every  $\mathcal{P}(\alpha)$ . Starting with an ordinal enumeration of a rank stage, transport this order to its power set; at limits combine the already constructed orders by rank. This set-valued ordinal recursion produces well-orders of all rank stages and hence a global well-order. This explains why merely assuming a convenient global baseline already carries substantive choice information.

An arbitrary class well-order can have type  $\text{Ord} + 1$ ,  $\text{Ord} + \text{Ord}$ , or much more complicated class types. The final element of an  $\text{Ord} + 1$  order has a proper class of predecessors. It is still an ordinary surreal label; the new size occurs in its position in the chosen order.

### 51.7.2 A first disagreement after a proper class of agreement

If  $R, S$  are class well-orders of the same labelled carrier  $X$ , put  $\text{Pred}_R(x) = \{y \in X : yRx\}$ . A common labelled prefix is initial in both orders and carries the same restricted relation. The comparison below is the prior report's elementary construction (theorem 14.2; Section 14 at the pin); we include its proof because it makes the foundational advantage precise.

**Theorem 51.9** (First unequal labels without class ordinals). *If  $R \neq S$ , there are unique distinct  $a, b$  such that*

$$\text{Pred}_R(a) = \text{Pred}_S(b) = C, \quad R \restriction C = S \restriction C.$$

*The class  $C$  is their maximal common labelled prefix. If the labels carry a fixed linear order  $<$ , declaring  $R <_{\text{lex}} S$  when  $a < b$  defines a strict linear comparison on the class-well-order predicate. Its formula has only set quantifiers with  $R, S$  as class parameters.*

*Proof.* Let  $a$  be the  $R$ -least point whose predecessor classes in  $R$  and  $S$  differ, and put  $C = \text{Pred}_R(a)$ . For  $y \in C$ , these predecessor classes agree. Such  $y$  must also precede  $a$  in  $S$ : otherwise  $aSy$  would imply  $aRy$ . If  $zSy$  for  $y \in C$ , agreement of predecessor classes gives  $zRy$ , so  $C$  is also  $S$ -initial. The restricted orders agree, and  $C \subsetneq \text{Pred}_S(a)$ . Let  $b$  be the  $S$ -least point outside  $C$ . Then  $bSa$ ,  $b \neq a$ , and  $\text{Pred}_S(b) = C$ . Any other first-disagreement pair must have its first member equal to the  $R$ -least point with changed predecessors, proving uniqueness.

For transitivity, suppose  $R < S < T$ . Their two common prefixes with the middle order  $S$  are initial segments of that same order, hence comparable. If one is shorter, its first difference also decides  $R$  versus  $T$ ; if they are equal, the next labels satisfy  $a < b < c$ , which decides the comparison. Totality and asymmetry follow from unique first disagreement. Finally the formula can be written directly as

$$\exists a, b [a < b \wedge \forall z(zRa \leftrightarrow zSb) \wedge \forall u, v((uRa \wedge vRa) \Rightarrow (uRv \leftrightarrow uSv))].$$

No quantification over a collection of prefix classes is required. □

For example, enumerate  $\mathbf{No} \setminus \{a, b\}$  in type  $\text{Ord}$ , where  $a < b$ , and append either  $a, b$  or  $b, a$ . Their first unequal labels occur after every ordinal stage of the common enumeration. The displayed formula still compares them. It is not necessary to invent a set ordinal “beyond  $\text{Ord}$ ” to name their first difference.

For one given class index order  $J$ , bijections  $J \rightarrow \mathbf{No}$  may instead be compared at the  $J$ -least unequal coordinate. This agrees with the raw comparison, but restricts the allowed order type. Two

different index orders should be compared through the induced labelled relations unless an explicit alignment has been supplied.

**Proposition 51.10** (Prefix convexity and initial-word fibres). *In GBC, orders extending a common labelled prefix form a convex family. For an arbitrary class well-order  $R$  of  $\mathbf{No}$ , let*

$$I_R = \{x : \text{Pred}_R(x) \text{ is a set}\}, \quad F_R : \text{Ord} \longrightarrow I_R$$

*be its unique increasing enumeration. Then  $I_R$  is a proper class. Every injective class word  $F : \text{Ord} \rightarrow \mathbf{No}$  occurs as  $F_R$ , and its fibre is a convex copy of the well-order space of  $Y_F = \mathbf{No} \setminus \text{ran}(F)$ . Distinct fibres compare at the first unequal coordinate of their words.*

*Proof.* A comparison that first diverges inside a common prefix puts the competing order on the same side of every extension, proving convexity. The class  $I_R$  is initial and set-like. If it were a set, the least point outside it would have precisely  $I_R$  as its predecessors, a contradiction. Its enumeration is therefore by all ordinals.

Given  $F$ , order its range by its ordinal indices and append any fixed well-order of  $Y_F$ . Every point of the range has set predecessors, and every residual point has the proper range of  $F$  among its predecessors. Thus the resulting initial word is exactly  $F$ . Restriction to  $Y_F$  and concatenation with  $F$  give inverse order-preserving maps on the fibre. For unequal words, first ordinal difference precedes both residual tails.  $\square$

If  $F$  omits  $m$  labels, its fibre has  $m!$  elements. An exhaustive word and a word omitting just one label both have singleton fibres; only the first is set-like. The unrestricted space has adjacent finite residual permutation blocks, whereas the set-like space is dense and has no endpoints. These distinctions are already established in the prior report.

## 51.8 Source 16: global choice and direct comparison of arbitrary class well-orders

Write  $\mathfrak{W}_{\text{sl}}$  for the predicate of class bijections  $\text{Ord} \rightarrow \mathbf{No}$ , and  $\mathfrak{W}_{\text{all}}$  for that of all class well-order relations on  $\mathbf{No}$ . These are notations for assertions about class variables. We will also use  $\mathfrak{J}$  for class injections  $\text{Ord} \rightarrow \mathbf{No}$ , allowing nonexhaustive ranges.

**Proposition 51.11** (Existence has a precise choice strength). *Over GB, the following are equivalent: global choice; a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ ; a set-like class well-order of  $\mathbf{No}$ ; and an arbitrary class well-order of  $\mathbf{No}$ .*

*Proof.* Global choice gives a set-like global well-order of the universe. Its restriction to the proper class  $\mathbf{No}$  has order type  $\text{Ord}$ , yielding the bijection. A bijection induces a set-like well-order, which is an arbitrary class well-order as well.

For the reverse implication suppose  $R$  well-orders  $\mathbf{No}$ . Every  $\mathcal{P}(\delta)$  injects into  $\mathbf{No}$  by sign sequences of length  $\delta$ , so  $R$  gives it a well-order. One can now well-order the hierarchy  $V_\beta$  uniformly by set-valued recursion, starting with  $w_0 = \emptyset$ . At a successor, use the existing well-order  $w_\beta$  of  $V_\beta$  to identify it with an ordinal  $\delta$ , transfer subsets to  $\mathcal{P}(\delta)$ , and use the restriction of  $R$  on their sign codes. At a nonzero limit  $\lambda$ , order the disjoint rank layers by their ranks, using  $w_{\gamma+1} \upharpoonright (V_{\gamma+1} \setminus V_\gamma)$  on layer  $\gamma$ , where  $\gamma + 1 < \lambda$ . Each history is a set; no choice of an unspecified family of well-orders is used. Finally order all sets first by rank and then by the resulting well-order on that rank layer. This is a set-like global well-order of the universe and supplies global choice. This expands the choice-equivalence argument recorded in the prior report ([theorem 13.1](#)).  $\square$

For distinct  $F, G \in \mathfrak{J}$ , their first unequal ordinal coordinate exists: minimize the disagreement class below any one witness. Compare  $F(\delta)$  and  $G(\delta)$  numerically. This is a total, transitive lexicographic comparison. For general  $R, S \in \mathfrak{W}_{\text{all}}$ , however, a first disagreement may occur after a proper class of common labels.

**Lemma 51.12** (A common labelled prefix without class ordinals). *Let  $R \neq S$  be class well-orders of the same class  $X$ . There are unique distinct labels  $a, b$  such that*

$$\{x : xRa\} = \{x : xSb\} = C, \quad R \restriction C = S \restriction C.$$

*The class  $C$  is their largest common labelled initial segment.*

*Proof.* Let  $a$  be  $R$ -least among the labels whose predecessor classes in  $R$  and  $S$  differ. For every  $xRa$  the two predecessor classes of  $x$  agree. Consequently  $C = \{x : xRa\}$  is also  $S$ -initial and the relations agree on it. Moreover  $C$  is a proper subclass of the  $S$ -predecessors of  $a$ . Let  $b$  be the  $S$ -least label outside  $C$ . Then  $bSa$ ,  $b \neq a$ , and its  $S$ -predecessor class is  $C$ . In any pair satisfying the displayed conditions, every earlier label has identical predecessor classes, while the next  $R$ -label does not. It must therefore be  $a$ , and then both  $C$  and  $b$  are determined.  $\square$

Define  $R <_{\text{lex}} S$  if the corresponding next labels satisfy  $a < b$  in the numerical or other fixed native order of  $X$ . In a triple  $R, S, T$ , the common prefixes of  $(R, S)$  and  $(S, T)$  are initial segments of  $S$  and are inclusion-comparable. Whichever ends first decides the comparison of  $R$  and  $T$ ; if they end together, transitivity of the next numerical labels decides. Thus the definition is a strict linear comparison.

The first-pair condition can be expressed using only set quantifiers:

$$\begin{aligned} a \in X \wedge b \in X \wedge a \neq b \quad \wedge \quad & \forall z (zRa \leftrightarrow zSb) \\ & \wedge \quad \forall x \forall y ((xRa \wedge yRa) \Rightarrow (xRy \leftrightarrow xSy)). \end{aligned} \tag{S16.9}$$

This is why no abstract comparison of class order types and no ETR is required. On an injectively indexed uniform family of class well-orders, it gives an ordinary class linear order on the set-valued indices.

**Proposition 51.13** (Convexity and finite residual jumps). *Fix any labelled common prefix, possibly a proper class. Its completions form a convex suborder of  $\mathfrak{W}_{\text{all}}$ . Two distinct raw orders are adjacent exactly when the labels remaining after their largest common prefix form a finite set and their residual permutations are consecutive.*

*Proof.* An order that disagrees before a fixed common prefix ends compares in the same direction with all its completions, proving convexity. If  $R < S$  were adjacent and the remaining set or class of labels were infinite,  $R$  would have to be maximal in its next-letter cylinder and  $S$  minimal in its next-letter cylinder. As in Lemma 27.1, swapping an ascent or descent forces the respective suffixes to be numerically decreasing and increasing. If  $Y$  is the residual carrier and  $a, b$  the first unequal labels, the numerical order on  $Y \setminus \{a, b\}$  is consequently both a well-order and a reverse well-order. Such a class is finite: otherwise repeatedly taking its least remaining member produces a set-coded increasing  $\omega$ -sequence, contradicting reverse well-foundedness. Thus  $Y$  is finite, and convexity reduces adjacency exactly to that of its finite permutation list. Conversely consecutive residual permutations have no intervening completion, and convexity excludes an intervening order outside the prefix cylinder.  $\square$



## 51.9 Source 18: global well-orders: existence and elementary comparison

### 51.9.1 Existence is exactly a global choice issue

No set-length function exhausts  $\mathbf{No}$ , since its range is a set. A well-order of the whole surreal carrier must therefore be a class relation. It need not be set-like:  $\text{Ord} + 1$ ,  $\text{Ord} \cdot 2$ , and more complicated class concatenations can be transported to the carrier under Global Choice. Here these expressions denote explicitly constructed class orders, not set ordinals.

**Theorem 51.14** (Existence of a global well-order; repository result). *Over GB with ZF as its set theory, the following are equivalent: Global Choice; the existence of a set-like global well-order of  $\mathbf{No}$ ; the existence of any class well-order of  $\mathbf{No}$ .*

*Proof.* Global Choice gives a set-like global well-order of the set universe and hence of  $\mathbf{No}$ . The second statement implies the third. For the reverse implication, let  $R$  be a class well-order of the sign-sequence surreals. For every ordinal  $\theta$ , the length- $\theta$  binary sign sequences form a set, and the restriction of  $R$  well-orders that set. It thereby well-orders  $\mathcal{P}(\theta)$ , uniformly in  $\theta$ .

Construct a well-order of each  $V_\alpha$  by set-valued recursion with parameter  $R$ . Start with the empty set. Given a well-order of  $V_\alpha$ , use its unique ordinal enumeration to identify subsets of  $V_\alpha$  with subsets of an ordinal; pull back the already specified well-order of that power set. At a limit stage, order the disjoint rank layers first by their ordinal rank and then by the relevant previously constructed well-order. These orders need not extend one another; uniform existence is what the argument requires. Finally order the whole universe first by rank and then by the chosen well-order of each rank layer. Its predecessor classes are sets, so it is a set-like global well-order and supplies Global Choice. This is the choiceless argument used in the repository ([theorem 13.1](#)).  $\square$

The theorem is distinct from asking whether universality of the *numerical* surreal order for all class linear orders implies Global Choice. Our arguments use Global Choice for that universality; they do not establish its converse. Under Global Choice, the usual forth construction embeds a class linear order in  $\mathbf{No}$ : enumerate its carrier in type  $\text{Ord}$ , and at each stage place the next point between the already embedded lower and upper sets. Hamkins gives a research-paper account of this class universality [[33](#), §5].

### 51.9.2 Comparing orders after a proper-class common prefix

Let  $R, S$  be class well-orders on the same linearly ordered class  $X$ . The numerical order on  $X$  need not coincide with either well-order. Write  $\text{Pred}_R(x) = \{y : yRx\}$ . Define  $C(R, S)$  to consist of the labels  $x$  such that

$$\text{Pred}_R(x) = \text{Pred}_S(x) \quad \text{and} \quad R \upharpoonright \text{Pred}_R(x) = S \upharpoonright \text{Pred}_R(x). \quad (\text{S18.4})$$

Both equalities expand to formulas quantifying only over set labels, with the two relations as class parameters.

**Lemma 51.15** (Largest common labelled initial segment).  *$C(R, S)$  is an initial segment of both orders, on which they agree. It contains every common labelled initial segment. If  $R \neq S$ , it is proper, and the first remaining labels  $a$  in  $R$  and  $b$  in  $S$  satisfy*

$$\text{Pred}_R(a) = C(R, S) = \text{Pred}_S(b), \quad a \neq b.$$

*Proof.* If  $x$  satisfies (S18.4), each of its predecessors does too: the two orders agree below  $x$ , and therefore assign identical predecessor classes below any such label. Thus  $C$  is initial in both, and agreement follows by testing a pair below its larger label. Every common labelled initial segment satisfies the displayed conditions. If  $C = X$ , the two relations agree everywhere. Otherwise their well-order property supplies least labels outside  $C$ . Their predecessor classes are exactly  $C$ . If those labels were equal, that label would satisfy (S18.4), a contradiction.  $\square$

**Theorem 51.16** (Elementary full lexicographic order). *For distinct  $R, S$ , declare  $R <_{\text{lex}} S$  if  $a < b$  in the numerical order, with  $a, b$  as in Lemma 51.15. This is a strict linear order on the available well-orders of  $X$ . It is elementary in the two class parameters. It requires neither a system of abstract class ordinals nor a principle comparing arbitrary class order types.*

*Proof.* Trichotomy and asymmetry are immediate. For transitivity, compare the common prefixes for  $R, S$  and for  $S, T$ . They are initial segments of the same order  $S$ , hence nested. If the first prefix ends earlier,  $S$  and  $T$  still agree at its next label, so the comparison of  $R, T$  is the comparison of  $R, S$ . The opposite strict containment gives the comparison of  $S, T$ . If the prefixes agree, the three next labels are numerically ordered and transitivity follows in  $X$ .

For explicit elementary definability, use the formula defining  $C$ , assert that the predecessors of  $a$  under  $R$  and of  $b$  under  $S$  are exactly the labels satisfying that formula, and assert  $a < b$ . No existential quantification over an additional class is needed.  $\square$

The theorem extends the usual first-ordinal-difference rule. In the case of distinct set-like orders,  $C$  is a set and has an ordinal length. In the full case, it may already have type  $\text{Ord}$ ,  $\text{Ord} \cdot 2$ , or a more complicated type. The comparison still exists because the two orders use the same labels. Abstractly isomorphic but differently labelled prefixes are not declared equal.

### 51.9.3 Residual cylinders and finite adjacency

Fix a labelled well-order  $Q$  of  $C \subsetneq X$ . Its cylinder is the family of all global orders beginning with  $Q$ . Put  $Y = X \setminus C$ . Restriction to  $Y$  and concatenation with  $Q$  are inverse, order-preserving operations. The cylinder is convex: any order that diverges before finishing  $Q$  lies on the same side of every extension of  $Q$ .

This gives the following useful size-sensitive reading of a cylinder. If  $|Y| = n < \omega$ , its tails form the  $n!$  permutations of  $Y$ . If  $Y$  is an infinite set of size  $\lambda$ , its tails form  $\mathcal{W}(Y)$ , of cardinality  $2^\lambda$  and ordinal spectrum  $\beta < \lambda^+$ . If  $Y$  is proper, its tails form a predicate on class relations. Under GBC this last family contains a uniformly presented copy of every class linear order. Indeed, characteristic functions of initial segments embed any class linear order into binary  $\text{Ord}$  words; independent pair swaps in a fixed enumeration of  $Y$  then embed those words into set-like well-orders of  $Y$ .

Even when  $Y$  is a set, the corresponding full orders of  $X$  are class relations. The statement is a set-parameterized isomorphism with their tails, not the assertion that those class relations themselves form an ordinary set.

**Theorem 51.17** (Adjacency is finite residual adjacency). *Let  $R < S$ , with common prefix  $C$ , next labels  $a < b$ , and residual carrier  $Y = X \setminus C$ . Then  $R, S$  are adjacent if and only if  $Y$  is finite and their residual permutations are consecutive.*

*Proof.* If  $R, S$  are adjacent, no remaining numerical label lies strictly between  $a, b$ , or it could be put next to produce an intermediate order. The tail of  $R$  after  $a$  must be lexicographically greatest: otherwise a greater replacement remains below  $S$ . A greatest well-order tail must list its labels in decreasing numerical order, since any numerical ascent can be transposed to increase the code.

Dually, the tail of  $S$  after  $b$  must be numerically increasing. On  $Y \setminus \{a, b\}$ , numerical order and its reverse would therefore both be well-orders. Such a carrier is finite: otherwise successively taking the least remaining element produces an increasing  $\omega$  sequence with no greatest member, contradicting reverse well-ordering. Conversely, convexity of the cylinder reduces a finite residual carrier exactly to finite permutation adjacency.  $\square$

For example, choose  $a < b$  in  $\mathbf{No}$  and give  $\mathbf{No} \setminus \{a, b\}$  a set-like order  $Q$ . The two orders  $Q + (a, b)$  and  $Q + (b, a)$  are adjacent and have type  $\text{Ord} + 2$ . Neither is set-like. The set-like global order has no adjacent pairs, because a set prefix leaves an entire proper class of unused numerical separators. It is consequently not dense in the full global order: intervals inside a six-point terminal permutation fiber contain only non-set-like orders.

## 52 The canonical block decomposition (source 15; routes of sources 16 and 18)

*Note 52.1* ([merge]; one theorem, three proofs). Three batch-81 sources decompose an arbitrary class well-order at once by the same relation:  $x \sim y$  exactly when the closed interval between  $x$  and  $y$  is a set. Each proves that  $\sim$  is an elementary convex equivalence, that every block is set-like of type  $\text{Ord}$  or a set ordinal, that at most one block is a set and it is final, and that the whole order is canonically the lexicographic sum of  $\text{Ord}$  over the class of proper-block representatives, followed by the optional final set block; each realizes every nonempty class well-order as a block-index order in GBC. The statements are source 15's [theorems 52.3](#) and [52.5](#) and [corollary 52.4](#), source 16's [theorems 52.6](#) and [52.8](#) with (S16.11), and source 18's [lemma 52.10](#), [theorem 52.11](#) with (S18.3), and [theorem 52.14](#). Source 15 is printed first because it also writes out the coordinate map  $x \mapsto (\pi_W(x), r_W(x))$  and the realization of every block profile; the other two follow as routes, whole, because each adds statements of its own: source 16 the largest set final segment ([corollary 52.7](#)) and the uniform family of residual tails; source 18 the functoriality of the decomposition ([proposition 52.12](#)), the cofinality of  $R$  read off its blocks ([corollary 52.13](#)), uniform removal of the first  $\alpha$  blocks by one closed definition, and a class well-order isomorphic to its own block quotient ([proposition 52.15](#)). The first block is source 11's canonical initial part ([theorem 15.1](#)), as all three say. The sources do not cite one another (source 18 was written before sources 15 and 16 arrived), and no source has priority. All three state that the decomposition is elementary in GB, without global choice or ETR, and that GBC is used only to transport an abstract order to  $\mathbf{No}$ . The block decomposition answers the block-index clause of [research question 113.3](#) ([section 60](#)).

### 52.1 Source 15: canonical condensation into proper-class blocks

The preceding extraction of the initial set-like part can be extended to *all* blocks at once. This resolves an ambiguity in the question of how much recursion is required: constructing the canonical first-level decomposition is elementary; enumerating its blocks by a prescribed index order, or iterating the decomposition on an arbitrary input, is a different request.

Interval condensation is a standard construction in order theory [34]. Here the size ideal consists of sets, not of countable subsets. Thus the construction belongs to class theory and depends on which collections are sets in the underlying model. Its comparison with familiar countable-interval condensation is structural, not an identification of the two notions.

Work first in GB, with no choice assumption, and let  $(X, W)$  be a class well-order. We use the

formulation that every nonempty available subclass of  $X$  has a least element. Put

$$[x, y]_W = \{z \in X : x \leq_W z \leq_W y\} \quad (x \leq_W y).$$

**Definition 52.2** (Condensation by set intervals). For  $x, y \in X$ , write  $x \sim_W y$  when the closed interval between them is a set. Let

$$B_W = \{b \in X : \text{there is no } aWb \text{ with } a \sim_W b\}.$$

The *condensation* of  $W$  is the induced order on  $B_W$ .

This definition is elementary. For example, “ $[x, y]_W$  is a set” can be written

$$\exists u \forall z (z \in u \longleftrightarrow (z \in X \wedge x \leq_W z \wedge z \leq_W y)),$$

where  $u, z$  are set variables and  $X, W$  are class parameters. Neither the equivalence classes nor a collection of those classes is being used as an element of another class. Their least representatives are sets and form the legitimate class  $B_W$ .

**Theorem 52.3** (Elementary canonical block decomposition). *In GB, the following hold.*

1. *The relation  $\sim_W$  is a convex equivalence relation. Every equivalence class has a least element, and the class function  $\pi_W : X \rightarrow B_W$  sending an element to that least representative is elementary and unique.*
2. *The order  $(B_W, W)$  is a class well-order. Each fiber  $C_b = \pi_W^{-1}(b)$  is set-like in its induced order. It therefore has the order type of a nonzero set ordinal if it is a set, and has order type  $\text{Ord}$  if it is a proper class.*
3. *Every nonfinal fiber is a proper class. Consequently at most one fiber is a set, and such a fiber must be final.*
4. *There is a uniform class order isomorphism*

$$x \longmapsto (\pi_W(x), r_W(x)), \quad r_W(x) = \text{otp}([ \pi_W(x), x ]_W),$$

*onto a lexicographically ordered subclass of  $B_W \times \text{Ord}$ . Every nonfinal vertical section is all of  $\text{Ord}$ ; a final section is either all of  $\text{Ord}$  or a nonzero set ordinal.*

*No elementary-transfinite-recursion principle is required.*

*Proof.* Reflexivity and symmetry are immediate. In a linear order the interval between  $x$  and  $z$  is contained in the union of the intervals between  $x, y$  and between  $y, z$ . A subclass of a set is a set by Separation with class parameters; hence  $x \sim_W y$  and  $y \sim_W z$  imply  $x \sim_W z$ . If  $x \leq_W z \leq_W y$  and  $x \sim_W y$ , the intervals from  $x$  to  $z$  and from  $z$  to  $y$  are subsets of  $[x, y]_W$ , proving convexity.

For any  $x$ , its equivalence class is nonempty and elementary in the parameters, so it has a  $W$ -least element  $b$ . Such  $b$  belongs to  $B_W$ , and conversely every member of  $B_W$  is the least member of its equivalence class. The graph of  $\pi_W$  is therefore given by one elementary formula. Every nonempty available subclass of  $B_W$  is also a subclass of  $X$ , and its  $W$ -least member belongs to it. This proves the quotient’s well-order property without any theorem comparing abstract class well-orders.

Fix  $b \in B_W$  and  $x \in C_b$ . The predecessors of  $x$  in  $C_b$  are exactly  $[b, x)_W$ , a set. Thus  $C_b$  is set-like. Its set-sized predecessor ranks form an initial subclass of  $\text{Ord}$ . If this initial subclass were a set ordinal while  $C_b$  were proper, the inverse rank map and class Replacement would make  $C_b$  a set.

Therefore the rank range is all of Ord precisely in the proper-class case. In the set case it is the ordinary ordinal order type of  $C_b$ .

Suppose now that  $C_b$  is a set and is not final. Let  $d$  be the least point after this fiber. More explicitly, choose the least member of the nonempty class

$$\{z \in X : bWz \text{ and } z \not\prec_W b\}.$$

Convexity and minimality give  $[b, d)_W = C_b$ . Hence  $[b, d]_W = C_b \cup \{d\}$  is a set, contradicting  $d \not\prec_W b$ . This proves the final-fiber assertion.

For each  $x$ , the set well-order  $[\pi_W(x), x)_W$  has a unique ordinal order type. Its order-isomorphism witnesses are sets, so  $r_W$  is defined by elementary comprehension. Elements in different fibers are compared by their representatives; elements in one fiber are compared by their ranks. The displayed map is therefore a class order isomorphism. Its image is defined directly as the class of pairs that occur, so no symbol denoting a “class ordinal” has to be treated as a set-valued section length.  $\square$

The first fiber is exactly the familiar initial part consisting of elements with set-sized predecessor classes. If  $X$  is proper, the following equivalence is particularly useful:

$$W \text{ is set-like} \iff |B_W| = 1.$$

This is an invariant of the individual class well-order. It does not claim that set-likeness is definable in the *bare lexicographic order of all labeled well-orders*; that reconstruction question remains separate.

**Corollary 52.4** (Allowed block profiles). *Every proper-class well-order is canonically a sum of Ord-blocks indexed by a class well-order, with the sole possible exception of a final block of nonzero set-ordinal type. The index order need not be set-like, or even have type Ord.*

The word “canonically” refers to the least-representative quotient and the rank map above. It does not assert that arbitrary class index orders have canonical set codes or admit an ordinal classification inside GB.

**Theorem 52.5** (Every class well-order occurs as a block index). *Assume GBC. Let  $J$  be any nonempty class well-order, with a set or proper-class carrier.*

1. *There is a class well-order of **No** whose canonical condensation is order-isomorphic to  $J$  and whose every block has type Ord.*
2. *For every nonzero set ordinal  $\alpha$ , there is one whose block index is  $J + 1$ , whose blocks indexed by  $J$  have type Ord, and whose final block has type  $\alpha$ .*

*More generally these realize every profile allowed by Theorem 52.3, subject only to the whole carrier being a proper class.*

*Proof.* Order  $J \times \text{Ord}$  lexicographically, comparing the  $J$  coordinate first. This is a class well-order: a nonempty subclass has a least first coordinate and then a least ordinal coordinate. It is proper because  $J$  is nonempty. The interval between two points of the same row is a set. Between points of different rows lies a final segment of the earlier Ord-row, which is proper. Consequently the rows are exactly the equivalence classes, with representatives  $(j, 0)$ .

Global Choice supplies a class bijection between this proper carrier and **No**. Transfer the relation along it. A class bijection takes sets to sets by class Replacement, and the same is true of its inverse; it therefore preserves the set/proper-class distinction used in the condensation definition. This proves the first assertion.

For the second, append the set ordinal  $\alpha$  to  $J \times_{\text{lex}} \text{Ord}$ . This final tail is one new equivalence class. It does not merge with any preceding row because an interval crossing into it contains a proper final segment of an Ord-row. Transfer again to **No**.

For a desired profile with final set block, remove the final index from its proposed index order and apply the second construction. The remaining order must be nonempty: otherwise the whole carrier would be a set. Profiles with no set block use the first construction.  $\square$

For instance, the formal class-order expression  $\text{Ord} \cdot (\text{Ord} + 1)$  has an  $\text{Ord} + 1$  block index. Its final block occurs after a proper class of complete Ord-blocks. No procedure that lists only set-ordinally many initial blocks reaches it. Here the product denotes the explicitly described ordered sum; it is not an ordinal object larger than every ordinal.

## 52.2 Source 16: simultaneous canonical decomposition into ordinal-class blocks

The earlier report extracts the first set-like part of a class well-order and asks how to organize the successive residual parts. A direct interval definition answers this question without carrying out a class-valued recursion. In this section GB suffices; assume that the given relation already is a class well-order in the sense of [Section 51.2](#).

For  $x, y \in X$ , let  $[x, y]_R$  mean the closed interval between them, regardless of which comes first. Put

$$x \sim_R y \iff [x, y]_R \text{ is a set.} \quad (\text{S16.10})$$

Formally the right side is  $\exists s \forall z (z \in s \leftrightarrow z \in [x, y]_R)$ , with  $s$  a set variable.

**Theorem 52.6** (Canonical block decomposition). *The relation  $\sim_R$  is a convex equivalence relation. Every block has a least member and is set-like in its inherited order. Each block is therefore either a set well-order of ordinal type or a proper class well-order of type Ord. There is at most one set block, and it is terminal.*

Let  $J_R$  be the class of least representatives of the proper blocks, ordered by  $R$ , and let  $\alpha_R$  be the ordinal type of the terminal set block, or zero when it is absent. There is a canonical class order isomorphism

$$(X, R) \cong \left( \sum_{j \in (J_R, R)} \text{Ord} \right) + \alpha_R. \quad (\text{S16.11})$$

All the data and the isomorphism are definable by elementary comprehension. The sum is the ordinary class  $J_R \times \text{Ord}$  ordered lexicographically, not a class whose members are classes.

*Proof.* The interval between  $x$  and  $z$  is contained in the union of the intervals between  $x, y$  and between  $y, z$ . Finite union and separation therefore prove transitivity; the other equivalence properties and convexity follow immediately.

Each block is an available nonempty subclass and has an  $R$ -least point  $m(x)$ . The formula saying that  $j \sim_R x$  and no  $R$ -earlier point is equivalent to  $x$  defines  $m$  elementarily. Thus representatives form a subclass of  $X$ , avoiding a quotient carrier made of proper classes. If  $j = m(x)$ , the predecessors of  $x$  within its block are exactly the set  $[j, x)_R$ . Define

$$\rho_R(x) = \text{otp}([j, x)_R, R).$$

This uses only a set well-order and its unique ordinal type. The range of  $\rho_R$  on the block is an initial subclass of Ord. If that range were a set, Class Replacement applied to its inverse would make the block a set. Hence a proper block has rank range all of Ord; a set block has an ordinal rank range.



Suppose a set block  $B$  were not terminal. Let  $c$  be the least point strictly above all of  $B$ . There is nothing between  $B$  and  $c$ . For  $b \in B$ , the interval  $[b, c]_R$  is contained in  $B \cup \{c\}$ , a set. This contradicts maximality of the equivalence block. Thus a set block is terminal, and there is at most one.

On proper blocks,  $x \mapsto (m(x), \rho_R(x))$  is the desired isomorphism; on the optional final block use its ordinal rank. All assertions defining these graphs quantify over sets with  $R$  as a fixed class parameter, so elementary comprehension suffices.  $\square$

### 52.2.1 Every residual tail is now available uniformly

For  $j \in J_R$  put

$$T_j = \{x : j \leq_R x\}, \quad B_j = \{x : m(x) = j\}.$$

The family of tails is one class relation of pairs  $(j, x)$ . Within  $T_j$ , a point  $x$  has a set of predecessors exactly when  $[j, x)_R$  is a set, equivalently when  $x \in B_j$ . Hence  $B_j$  is precisely the canonical initial set-like part of  $T_j$ . The tail  $T_j$  is what remains after all earlier blocks have been deleted. This supplies the whole successive residual family even when  $J_R$  is non-set-like.

**Corollary 52.7** (Largest set final segment). *The class*

$$A_R = \{x \in X : \{y : x \leq_R y\} \text{ is a set}\}$$

*is always a set. It is the largest set-sized final segment of  $R$ , and is exactly its optional terminal block.*

*Proof.* If  $A_R$  is nonempty, let  $a$  be its least point. Every later point has a tail contained in the set tail above  $a$ , so  $A_R$  is precisely that tail, a set. Every set final segment is contained in  $A_R$  by definition. Conversely, a block containing a point of  $A_R$  is the union of a set of block predecessors and a subset of its set tail, and so is a set block. [Theorem 52.6](#) identifies it.  $\square$

Deleting  $A_R$  leaves a labelled prefix consisting of all the proper blocks and having no nonempty set-sized final segment. If that prefix is fixed, its completions form a convex copy of  $\mathcal{W}(A_R)$ . Appending any well-order on  $A_R$  leaves it the largest set final segment, since a tail meeting an earlier Ord block would already be proper. This refines the earlier first-block decomposition: after every proper block is removed, the residual carrier is always a set.

**Theorem 52.8** (All block-index orders occur). *Every nonempty class well-order  $J$ , whether its carrier is a set or a proper class, occurs as the index of the proper blocks of a class well-order. An arbitrary set ordinal can be prescribed as its final block. In GBC the resulting order can be carried by surreal labels.*

*Proof.* Order  $J \times \text{Ord}$  with the  $J$  coordinate primary. A nonempty subclass has a least occupied  $J$  coordinate and then a least ordinal in that section, so this is a class well-order. Intervals inside one section are sets; an interval crossing two sections contains a proper tail of Ord. Its blocks are exactly the sections. Appending a set ordinal gives the stipulated final block. The product is proper, and global choice supplies a class bijection of its carrier with **No**, along which the relation can be transported.  $\square$

If  $J_R$  is a set of type  $\gamma$ , formula (S16.11) means  $\text{Ord} \cdot \gamma + \alpha_R$ . If  $J_R$  is proper and set-like, it means  $\text{Ord} \cdot \text{Ord} + \alpha_R$ . The example  $\text{Ord} \cdot (\text{Ord} + 1) + \alpha$  has block index  $\text{Ord} + 1$ : an Ord-indexed list of its initial blocks would miss its last full block, whereas (S16.10) finds it directly.

### 52.2.2 What this does and does not require from ETR

The construction resolves successive residual extraction for one original order. Repeatedly condensing the *quotient order itself* is another operation. If  $C_R(D)$  denotes the least representatives of set-interval blocks in the induced order on  $D \subseteq X$ , a tower such as

$$D_0 = X, \quad D_{\beta+1} = C_R(D_\beta), \quad D_\lambda = \bigcap_{\beta < \lambda} D_\beta$$

is an elementary class-valued recursion. Each specified finite iteration is elementary; appropriate ETR supplies a uniform transfinite tower. No necessity or exact reversal is established here. Also, since the block-index order can be an arbitrary class well-order, [Theorem 52.6](#) does not solve abstract class well-order comparability or give a universal class ordinal.

### 52.3 Source 18: a simultaneous decomposition into ordinal-class blocks

The repository asks how much class recursion is needed to iterate the extraction of the initial set-like part of a class well-order. A direct definition organizes all of these blocks simultaneously. It avoids a recursion whose successive states are proper classes.

Work in GB and let  $R$  be a class well-order on a proper class  $X$ . Here “class well-order” means that every nonempty available subclass has an  $R$ -least element. For  $x \leq_R y$ , write

$$[x, y]_R = \{z \in X : x \leq_R z \leq_R y\}.$$

The assertion that this interval is a set is the elementary formula

$$\exists a \forall z [z \in a \longleftrightarrow (z \in X \wedge x \leq_R z \leq_R y)].$$

In particular, no quantification over candidate classes is needed.

**Definition 52.9** (Set-interval equivalence). Put  $x \sim_R y$  when the closed  $R$ -interval with endpoints  $x, y$  is a set. Let  $m_R(x)$  be the  $R$ -least element of the equivalence class of  $x$ . A block is represented by this least label, never by placing a proper class inside another class.

**Lemma 52.10** (Block structure). *The relation  $\sim_R$  is an elementary, convex equivalence relation. Every one of its blocks is set-like in its induced order. Every nonfinal block is a proper class. Consequently there is at most one set-sized block, and if present it is final.*

*Proof.* Reflexivity and symmetry are immediate. For transitivity, the interval between two of three points is contained in the union of the intervals from those points to the third; separation and the union of two sets give the result. Convexity follows by taking subsets of an interval.

Fix a block  $C$ . Its least element  $b$  exists by the assumed class well-ordering. If  $x \in C$ , its predecessors within  $C$  form the set  $[b, x]_R \setminus \{x\}$ . Thus the block is set-like. Suppose now that  $C$  is a set and is not final. There is a least point  $d$  after all of  $C$ . Convexity and this minimality imply  $[b, d]_R = C \cup \{d\}$ . This is a set, so  $d \sim_R b$ , contradicting  $d \notin C$ . The final assertion follows because two distinct final blocks are impossible.  $\square$

Let  $B_R$  be the class of least labels of the proper-class blocks, ordered by  $R$ . If there is a final set block, let  $\theta_R$  be its ordinal order type; otherwise put  $\theta_R = 0$ . Since  $X$  is proper,  $B_R$  is nonempty. It may be a set or a proper class.

**Theorem 52.11** (Canonical division by the ordinal class). *In GB, every class well-order  $R$  on a proper class has a canonical class isomorphism*

$$(X, R) \cong (B_R \times \text{Ord})_{\text{lex}} + \theta_R,$$

where the product compares the  $B_R$  coordinate first. All proper blocks have type  $\text{Ord}$ . The block map, the block index order, and the isomorphism are elementary and uniform in  $R$ . No global choice or elementary transfinite recursion principle is used.

*Proof.* A proper, set-like class well-order has type  $\text{Ord}$ : assign to each point the ordinal type of its set of predecessors. The ranks form an initial subclass of the ordinals; if that subclass were a set, the inverse rank map and class replacement would make the domain a set. Hence the rank range is all of  $\text{Ord}$ .

Apply this observation to each proper block. More explicitly, define

$$\rho_R(x) = \text{otp}(\{z : m_R(x) \leq_R z <_R x\}, R).$$

The displayed predecessor interval is a set. Its restricted relation and its unique ordinal order type are sets as well. Thus the graph of  $\rho_R$  is elementary with class parameter  $R$ , and  $(m_R(x), \rho_R(x))$  gives the desired coordinate in a proper block. The same rank map identifies a final set block with  $\theta_R$ . Convexity of the blocks proves order preservation and exhaustiveness.

The class  $B_R$  consists of actual elements of  $X$ , characterized by an elementary formula. Class well-foundedness restricts to it. None of the construction forms a collection having the blocks themselves as elements.  $\square$

**Proposition 52.12** (Uniqueness and functoriality). *A class order isomorphism preserves set-interval equivalence, induces an isomorphism of the proper-block index orders, and preserves the final ordinal  $\theta_R$ . Hence the pair  $(B_R, \theta_R)$  is unique up to class order isomorphism of its first coordinate. The construction works simultaneously for any uniformly coded set- or class-indexed family of input relations.*

*Proof.* A class isomorphism and its inverse send sets to sets by class replacement. They therefore preserve whether an interval is a set, and send block minima to block minima. Ordinal order types of set tails are unique. Conversely an isomorphism of block indices and equality of the final ordinals combine, using the displayed coordinates, to produce an isomorphism of the original orders. Uniformity follows by inserting the family index into the same elementary formulas.  $\square$

The first proper block is exactly the repository's canonical initial set-like part  $I_R$ . Thus the theorem is a simultaneous form of the successive residual decomposition. It does not prove comparability of arbitrary abstract class well-orders: the remaining index order  $B_R$  can be as complicated as any class well-order.

**Corollary 52.13** (Set-sized cofinality from the blocks). *If  $\theta_R > 0$ , the least size of a set cofinal in  $R$  is  $\text{cf}(\theta_R)$ . If  $\theta_R = 0$  and  $B_R$  has a greatest element, no set is cofinal in  $R$ . If  $\theta_R = 0$  and  $B_R$  has no greatest element,  $R$  has a set cofinal subset exactly when  $B_R$  does, and the least sizes agree.*

*Proof.* A nonempty final set tail determines cofinality. A final proper block has type  $\text{Ord}$ , so every set of points in that block is bounded, while earlier blocks cannot help with cofinality. In the remaining case, projecting a set cofinal subset to block minima gives a set cofinal in  $B_R$ . Conversely a set cofinal in  $B_R$  provides a cofinal set of least block labels in  $R$ : because  $B_R$  has no maximum, a cofinal family includes a block strictly beyond the block of any prescribed point. Class replacement preserves the relevant size inequalities.  $\square$

In a full containing-universe interpretation based on an inaccessible  $\kappa$ , the theorem agrees with ordinary ordinal division. The internal ordinal class is externally  $\kappa$ . An internal class interval is an internal set exactly when its external order type is less than  $\kappa$ : regularity bounds the ranks of fewer than  $\kappa$  labels, and internal sets have size less than  $\kappa$ . Thus a well-order of external ordinal type  $\gamma$ , with  $|\gamma| = \kappa$ , decomposes as

$$\gamma = \kappa \cdot \beta + \rho, \quad \rho < \kappa.$$

The elementary internal argument does not require the existence of such a containing universe.

**Theorem 52.14** (Exact spectrum of block index orders). *In GBC, every nonempty class well-order  $\Gamma$  on a class of set labels occurs as  $B_R$ , up to isomorphism, for a class well-order  $R$  of No. Any prescribed set ordinal can simultaneously be realized as the final remainder  $\theta_R$ .*

*Proof.* Form  $(\Gamma \times \text{Ord})_{\text{lex}} + \theta$ . It is a class well-order: a nonempty subclass either meets the product, in which case choose its least first coordinate and then its least ordinal coordinate, or lies in the final ordinal. Within one product fiber, closed intervals are sets. Between distinct fibers, an interval contains a proper final part of an Ord fiber and is proper. The proper blocks are exactly the fibers, and the final  $\theta$ , when nonzero, is one set block. Global choice supplies a class bijection from this proper carrier to No; transport the order.  $\square$

### 52.3.1 Uniform removal through ordinary ordinal stages

The direct decomposition also answers a narrower recursion question. Let  $K_R$  be the initial set-like part of  $B_R$  (all of  $B_R$  if it is a set), and let  $r_R(b)$  be the ordinal type of the predecessors of  $b$  in  $B_R$ , for  $b \in K_R$ . For every set ordinal  $\alpha$ , put

$$U_\alpha = \{x \in X : m_R(x) \in K_R \wedge r_R(m_R(x)) < \alpha\}.$$

The family  $(U_\alpha)_{\alpha \in \text{Ord}}$  is one elementary class relation. Its complement carries the residual order after removal of the first  $\alpha$  proper blocks. At a successor stage, if a proper residual carrier remains, its initial Ord block is removed; at a limit stage the removed domain is the union of the earlier removed domains. If  $B_R$  is a set, removal stops after its ordinal type, leaving  $\theta_R$ .

Thus even this uniform ordinal-indexed instance of apparently class-valued recursion is already available in GB through a closed definition. This is a statement about this particular operation, not a claim that arbitrary class-valued recursions are provable in GB.

### 52.3.2 A class well-order isomorphic to its own block quotient

Let  $\mathcal{P}$  consist of all functions  $f : \omega \rightarrow \text{Ord}$  with finite support. For distinct  $f, g$ , put

$$f \triangleleft g \iff f(n) < g(n), \quad n = \max\{i : f(i) \neq g(i)\}.$$

**Proposition 52.15** (Self-similar block quotient). *The order  $(\mathcal{P}, \triangleleft)$  is a class well-order. Its set-interval blocks consist precisely of functions that agree at every coordinate  $i \geq 1$ . Every block has type Ord, and its block index order is isomorphic to  $(\mathcal{P}, \triangleleft)$  itself. In GBC the example can be transported to a class well-order of No.*

*Proof.* For a nonempty class of functions, the zero function is least if present. Otherwise choose its least possible largest nonzero support index  $d$ . Among functions with that index, minimize the coefficient at  $d$ , then at  $d - 1$ , and so on through 0. A nonempty class of ordinals has a least element,

and there are finitely many steps. The result is least among functions of degree  $d$ , and every larger degree is greater. This proves class well-foundedness.

If  $f, g$  agree above coordinate 0, every function between them has that same tail, and its zeroth coordinate lies in the set ordinal interval between  $f(0)$  and  $g(0)$ . Their interval is a set. If their tails differ and  $f \triangleleft g$ , keep the tail of  $f$  fixed and let the zeroth coefficient run through all ordinals greater than  $f(0)$ . The resulting proper class lies strictly between  $f$  and  $g$ . This proves the description of blocks.

The least label in a block has zeroth coefficient 0. Deleting that coordinate identifies the class of block minima, with its inherited order, with all of  $\mathcal{P}$ . Varying the zeroth coefficient identifies each block with Ord. Finally apply global choice to transfer the proper carrier to No.  $\square$

This example is a rigorous class analogue of the finite-polynomial presentation informally denoted by  $\text{Ord}^\omega$ . The notation is not needed in the construction. It shows that passage to the block quotient need not reduce the order to a simpler isomorphism type. It does not assert nontermination of every possible limit-stage condensation convention; a limit operation must be defined separately.

### 52.3.3 What remains open after the direct decomposition

The repository's question about uniformly organizing successive initial Ord blocks has an elementary answer above, including an exact unrestricted spectrum of block index orders. A distinct question remains: what is the strength of uniformly iterating the *quotient operation itself* through arbitrary class well-orders, with an explicitly specified limit convention? General ETR is a sufficient framework for suitable elementary class recursions, but the present arguments establish neither its necessity for this specific operation nor the strength of alternative limit conventions. The self-similar example rules out a proof based merely on a claimed strict decrease of quotient isomorphism types.

## 53 Condensation histories and the strength of iterated condensation (source 15; sources 16 and 18)

[merge] Source 15's Section 6 constructs, for every class well-order  $J$ , a class well-order whose iterated condensation runs through exactly  $J$  proper stages before reaching a singleton, by one elementary formula. Sources 16 and 18 discuss the same recursion question in the last subsections of their block sections (printed in [section 52](#)); the three discussions agree: finitely many condensations, fixed externally, are elementary; a uniform transfinite tower of condensations is a class-valued recursion for which a suitable ETR suffices; no source claims that ETR is necessary.

### 53.1 Source 15: explicit condensation histories of arbitrary class length

The block theorem admits an additional extension. Every class well-order can be realized as the order of the nontrivial stages of an explicit condensation history. The construction has a direct elementary formula and therefore does not require a general recursion axiom, even when some stages have proper-class many predecessors.

**Definition 53.1** (Finite ordinal polynomials over a class order). For a nonempty class well-order  $(J, <_J)$ , let  $P(J)$  consist of the *set codes* for finite partial functions

$$f : J \rightarrow \text{Ord} \setminus \{0\}.$$

Values off the finite domain are interpreted as zero. The empty function is denoted 0. For distinct  $f, g$ , let  $j$  be the greatest member of their finite disagreement set and put

$$f <_{P(J)} g \iff f(j) < g(j).$$

For completeness, put  $P(\emptyset) = \{0\}$ . These are finite set codes, not proper-class functions used as elements of a class. The greatest disagreement exists because its domain is finite; there is no last-coordinate assumption on  $J$  itself.

**Lemma 53.2** (The polynomial order is a class well-order). *In GBC,  $P(J)$  is a proper class well-order.*

*Proof.* The order laws reduce to ordinary finite lexicographic comparison on the union of the finitely many supports of the polynomials under comparison. If  $j_0$  is the least point of  $J$ , varying the positive ordinal coefficient at  $j_0$  shows that  $P(J)$  is proper.

We first rule out a descending  $\omega$ -sequence  $(f_n)$ . Its union of supports is a set  $A \subseteq J$ ; the restriction of  $<_J$  to  $A$  is a set well-order. Let  $r : A \rightarrow \eta$  be its ordinal rank map. Choose an additively indecomposable ordinal  $\Lambda$  strictly above every coefficient of every  $f_n$ , for example a sufficiently large power of  $\omega$ . For each  $f_n$ , form the ordinary ordinal

$$v(f_n) = \sum_{j \in \text{supp}(f_n)}^{\text{decreasing } <_J} \Lambda^{r(j)} \cdot f_n(j).$$

This is a finite sum. If two polynomials first differ from the top at  $j$ , all their lower terms are less than  $\Lambda^{r(j)}$ . Indeed, each lower exponent is  $\sigma < r(j)$ , each coefficient is less than  $\Lambda$ , and  $\Lambda^\sigma c < \Lambda^{\sigma+1} \leq \Lambda^{r(j)}$ ; finite sums remain below the latter additive principal ordinal. When  $r(j) = 0$  there are no lower terms. Thus the different coefficients at  $j$  decide the ordinal comparison, and  $v$  preserves it strictly. The sequence  $(v(f_n))$  would be descending in Ord, a contradiction.

For completeness, Global Choice converts a nonempty class with no least member into a descending  $\omega$ -sequence. Given  $f$  in such a class, consider the nonempty class of its smaller members. Its members of least set-theoretic rank form a nonempty set. Apply the fixed global choice function to that set to choose the next member. This is a class function, and ordinary set-valued recursion through  $\omega$  constructs the descending sequence. Hence every nonempty available subclass of  $P(J)$  has a least member.  $\square$

**Lemma 53.3** (One condensation removes the least exponent). *Let  $j_0 = \min J$ . In  $P(J)$ , two polynomials lie in the same set-interval block exactly when they agree at every  $j >_J j_0$ . The least representative of that block has coefficient zero at  $j_0$ . Consequently the canonical condensation is the evident copy of  $P(J \setminus \{j_0\})$ .*

*Proof.* If the polynomials agree above  $j_0$ , every polynomial between them has those same upper coefficients. Its coefficient at  $j_0$  lies in a bounded ordinal interval, so the closed interval is a set.

Conversely let  $f < g$  disagree at a coordinate above  $j_0$ . Keep all coefficients of  $f$  above  $j_0$  fixed and replace its coefficient at  $j_0$  by  $f(j_0) + \beta + 1$ , for  $\beta \in \text{Ord}$ . The resulting polynomials lie strictly above  $f$  and strictly below  $g$ : the greatest disagreement with  $g$  is unchanged. They form a proper class, so  $[f, g]$  is not a set. Finally zero is the least ordinal coefficient, which identifies the least block representative.  $\square$

**Theorem 53.4** (Every class well-order is an explicit condensation length). *In GBC, for every nonempty class well-order  $J$  there is a class well-order of **No** with an elementary, uniformly coded condensation history indexed by  $J + 1$ , such that every stage indexed by  $J$  is proper and the appended final stage is a singleton.*



*Proof.* It suffices to use  $P(J)$  and transfer the result to **No** by a class bijection. For  $j \in J$  define the retained representatives

$$H_j = \{f \in P(J) : \text{supp}(f) \subseteq J_{\geq j}\}, \quad H_* = \{0\},$$

where  $*$  is one new final index. Code the indices with disjoint tags if necessary. The class of pairs  $(j, f)$  satisfying this condition is defined by an elementary formula, including when  $J_{< j}$  is proper.

At the least index,  $H_j = P(J)$ . If  $k$  is the immediate successor of  $j$ , [Lemma 53.3](#) applied to the tail  $J_{\geq j}$  shows that the condensation of the induced order on  $H_j$  is exactly  $H_k$ . If  $j$  is last, that same argument gives  $H_*$ .

At a nonzero limit index  $j$ , a polynomial belongs to every  $H_k$  for  $k <_J j$  if and only if none of its support lies below  $j$ . For if  $t <_J j$  lies in the support, choose  $k$  with  $t <_J k <_J j$ ; then the polynomial is absent from  $H_k$ . Thus

$$H_j = \bigcap_{k <_J j} H_k.$$

This intersection is legitimate elementary notation: membership is the set-quantifier statement  $\forall k (k <_J j \Rightarrow (k, f) \in H)$ . If  $J$  has no greatest element, the same argument shows that the intersection over all  $j \in J$  is  $\{0\} = H_*$ .

For every  $j \in J$ , arbitrary ordinal coefficients supported at  $j$  belong to  $H_j$ , so  $H_j$  is proper. If  $j <_J k$ , the polynomial with coefficient one at  $j$  belongs to  $H_j \setminus H_k$ . The history is therefore strictly decreasing at distinct indices and first becomes a singleton at its appended final stage. All formulas, including the full class of stages, have been supplied explicitly. No invocation of elementary transfinite recursion has occurred.  $\square$

Examples include finite histories, a history of length  $\omega$ , one of length  $\text{Ord}$ , and one of length  $\text{Ord} + \text{Ord}$ . These refer to the order of the nontrivial stages, rather than to additional set ordinals. In particular there is no universal set-ordinal bound on the depth of the proper-class block hierarchy.

*Remark 53.5* (A proper quotient can have the same abstract order type). For  $J = \omega$ , deleting the least exponent leaves a tail isomorphic to  $\omega$ . Hence  $P(\omega)$  is order-isomorphic to its own canonical condensation, even though the retained representatives form a proper subclass. The explicit history nevertheless reaches the singleton at stage  $\omega$ . Likewise every set-ordinal tail of  $\text{Ord}$  is isomorphic to  $\text{Ord}$ , while the history for  $P(\text{Ord})$  becomes a singleton only at its appended class-limit stage. Consequently one cannot prove termination by asserting that each condensation strictly decreases the abstract order type.

### 53.1.1 Exactly what this says about recursion strength

For an arbitrary input  $W$ , every *externally fixed finite* number of successive condensations can be constructed in GB by composing elementary definitions. This finite theorem scheme should not silently be replaced by the assertion that one class uniformly codes all finite iterations. The standard principle  $\text{ETR}_\omega$  is a sufficient hypothesis for that uniform history; its final intersection can then be formed by elementary comprehension. Similarly, an appropriate elementary-recursion principle is a sufficient hypothesis for an arbitrary prescribed class-length history on a general input.

No lower bound or equivalence with  $\text{ETR}_\omega$  is claimed for this particular condensation operator. Proving that such a history exists for every  $W$  in GB, or calibrating its additional strength, remains a sharper reverse-mathematical question. The explicit polynomial histories show why a general warning about class-valued recursion must not be converted into an assertion that every individual long construction needs ETR. Here one elementary formula already defines every stage, even along arbitrary given class well-orders.

### 53.1.2 A full-universe consistency check

Suppose an ambient universe contains a strongly inaccessible cardinal  $\kappa$ , and use the full two-sorted model  $(V_\kappa, \mathcal{P}(V_\kappa))$ . For a subclass of  $V_\kappa$ , being an internal set is equivalent to having ambient cardinality less than  $\kappa$ . One direction uses the strong-limit property; the other uses regularity to bound the ranks of fewer than  $\kappa$  many elements. Thus the preceding condensation becomes condensation by intervals of cardinality less than  $\kappa$ .

If the ambient ordinal type of an internal class well-order is  $\theta$ , write its ordinary left ordinal division by  $\kappa$  as

$$\theta = \kappa \cdot \gamma + \delta, \quad \delta < \kappa.$$

Its canonical blocks are the successive  $\kappa$ -segments, followed by one  $\delta$ -segment if  $\delta > 0$ . The quotient has type  $\gamma$  if  $\delta = 0$  and  $\gamma + 1$  otherwise. This is exactly the canonical block theorem as seen from the containing universe. An internal Ord-block is an ambient  $\kappa$ -block, rather than an ambient proper class.

This check also identifies the essential relativity: changing which collections count as sets changes the condensation ideal. The construction is intrinsic to  $(X, W)$  together with the chosen set/class boundary; it is not absolute between arbitrary universes.

*Note 53.6* ([merge]; the self-similar example of sources 15 and 18). Source 15's [remark 53.5](#) and source 18's [proposition 52.15](#) describe the same class well-order: source 15's  $P(\omega)$ , the finite partial functions  $\omega \rightarrow \text{Ord} \setminus \{0\}$  compared at the greatest coordinate of disagreement, is source 18's  $(\mathcal{P}, \triangleleft)$ , the finite-support functions  $\omega \rightarrow \text{Ord}$  compared at the largest coordinate of disagreement. Both show that it is isomorphic to its own block quotient, so that one condensation step need not lower the isomorphism type; source 18 proves it in full, source 15 deduces it from [lemma 53.3](#). Their full-universe checks agree as well: source 15's subsection above, source 18's remark after [corollary 52.13](#), and source 18's [section 57.5](#) all identify the internal block decomposition in  $(V_\kappa, \mathcal{P}(V_\kappa))$ ,  $\kappa$  inaccessible, with left ordinal division by  $\kappa$ .

## 54 Finite components and trace fibres (source 17; sources 18 and 14)

*Note 54.1* ([merge]; what is repeated and what is new). Source 17's [theorem 54.2](#) is source 11's [theorem 16.2](#) and source 12's [theorem 16.6](#), which it calls a “source result”. Its [theorem 54.4](#) identifies the maximal finite-adjacency components of the unrestricted order: they are the  $n!$  permutations of the maximal finite final segment. Finite permutation blocks after a common prefix are already source 12's [corollary 16.7](#) and, for set alphabets, source 08's [note 6.26](#); new in [theorem 54.4](#) are the maximality of these blocks on the class carrier, the uniqueness of the factorization  $R = P + \tau$ , and the density of the quotient. Source 18 proves the same component theorem independently ([lemmas 58.1](#) and [58.2](#) and [proposition 58.3](#), printed with its countable models in [section 58](#)), and adds that every factorial block size occurs densely ([lemma 58.5](#)).

Source 14's Section 10, printed second, recalls the direct comparison and the canonical initial branch ([lemma 54.5](#); source 11's [theorem 15.1](#), source 13's [proposition 51.8](#), source 15's [proposition 51.10](#)) and proves that the trace of an unrestricted order on the core is the cut of its initial branch ([theorem 54.6](#)). Its [example 54.7](#) (singleton fibres need not be set-like;  $m!$  elements over  $m$  omitted labels) is source 11's [remark 15.4](#) and [proposition 15.3](#), which source 14 does not cite, and its [corollary 54.8](#) is the statement about raw orders printed in Part VIII by sources 13, 15, 16, 17 and 18 ([note 46.5](#)). New in source 14's section is the exact description of the trace fibres over the core as the convex families  $F + Q$ .

### 54.1 Source 17: unrestricted orders and their finite condensation

The restriction to set-like orders is not cosmetic. After a class prefix has exhausted almost every label, the residual alphabet can be finite, and numerical density of **No** is no help. We recall the source's exact adjacency theorem and then identify the maximal components generated by adjacency.

**Theorem 54.2** (Adjacency after a common prefix; source result). *Let  $R <_{\text{lex}} S$  be well-orders of the same labelled class  $X$ , and let  $C$  be their largest common labelled prefix. Write  $a < b$  for the next labels and  $Y = X \setminus C$ . The orders are adjacent if and only if  $Y$  is finite and their permutations of  $Y$  after  $C$  are consecutive in the finite lexicographic order.*

*Proof.* A remaining label strictly between  $a$  and  $b$  would start an intermediate order, followed by a restriction of one given well-order. Also, the tail of  $R$  after  $a$  must be a lexicographically greatest well-order of its remaining labels; otherwise replacing that tail gives an intermediate order. Such a greatest order exists exactly when the native decreasing order is itself a well-order: any pair listed in native increasing order can be transposed to make the enumeration larger. Dually, the  $S$ -tail after  $b$  must be the native increasing order.

Thus  $Y \setminus \{a, b\}$  is well-ordered both in native increasing and native decreasing order. It is finite. Indeed, otherwise successively choosing increasing elements by their well-order creates a strictly increasing  $\omega$ -sequence, a set with no greatest element, contradicting reverse well-ordering. Hence  $Y$  is finite, and no intermediate residual permutation exists precisely when the two permutations are consecutive. The converse follows directly from convexity of the common-prefix cylinder. These are the source's arguments (theorems 16.2 and 16.6).  $\square$

**Definition 54.3** (Finite-adjacency components). Two orders are in the same component when a finite chain of adjacent pairs connects them. This is a uniform relation on the well-order predicates, or an ordinary equivalence relation in an explicitly larger set-universe interpretation. We do not form a same-level class of proper-class equivalence classes.

**Theorem 54.4** (Exact maximal finite components). *Every well-order  $R$  of the whole surreal carrier has a unique factorization*

$$R = P + \tau,$$

*where  $\tau$  is a finite final segment of length  $n$ , and  $P$  has no greatest element. The integer  $n$  may be zero. The finite-adjacency component of  $R$  consists exactly of all  $n!$  permutations of these same  $n$  final labels following the fixed prefix  $P$ .*

*After identifying each such component, the resulting higher-order linear comparison is dense and has no endpoints. A nonsingleton component intrinsically determines its terminal length  $n \geq 2$  from its size  $n!$ . Singleton components do not distinguish terminal lengths 0 and 1.*

*Proof.* Repeatedly delete the greatest point for as long as one exists. This process must stop after finitely many deletions: otherwise set-valued recursion on  $\omega$  produces a strictly descending  $\omega$ -sequence in  $R$ , contradicting the well-order property. The remaining prefix is still a proper class, since only finitely many labels were removed, and has no greatest element. This establishes existence and uniqueness of the maximal finite final segment.

All its  $n!$  permutations form a finite convex block, by common-prefix convexity. Consecutive members are adjacent by Theorem 54.2, so they are in one component. Suppose an order in this block has an adjacent neighbor. Their common prefix is cofinite. It must contain all of  $P$ : if it stopped strictly inside  $P$ , its nonempty residual final part of  $P$  would be finite and would have a greatest element, making  $P$  have a greatest element. That is impossible. Thus an adjacent neighbor

can change only the finite final labels and remains in the displayed block. No adjacency chain leaves it.

All components are finite and consequently have first and last elements. If two distinct components were adjacent in the quotient, the last member of the earlier one and first member of the later one would be adjacent in the original order, joining the components. Hence the quotient is dense. The original comparison has no endpoints: replace its first label by any smaller or larger label, then append the restriction of its well-order to the remaining labels. Since each component is finite, the quotient has no endpoints either. Strict increase of  $n!$  for  $n \geq 2$  proves the intrinsic size assertion.  $\square$

This is a description of the *maximal adjacency components*, not just of some finite cylinders. It does not identify the quotient with  $\mathfrak{W}_{\text{sl}}$  or with **No**. The many non-set-like convex fibres from [Proposition 46.27](#) can retain substantial structure after all finite components have been collapsed.

## 54.2 Source 14: unrestricted class well-orders and the trace quotient

The enumeration picture only covers *set-like* well-orders of the whole surreal class. For arbitrary class well-orders, some elements may have proper-class predecessor collections, so an ordinal-valued position need not exist for every label. We first recall how to compare such orders without inventing those positions.

### 54.2.1 Elementary comparison of labelled relations

Let  $R, S$  be strict class well-orders on the same labelled carrier **No**. For a label  $x$ , write  $\text{Pred}_R(x) = \{y : yRx\}$ . Define  $C$  by

$$x \in C \iff \text{Pred}_R(x) = \text{Pred}_S(x) \text{ and } R \upharpoonright \text{Pred}_R(x) = S \upharpoonright \text{Pred}_S(x).$$

Restrictions in this formula concern binary relations on the indicated predecessor classes. Equality expands into quantifiers over set labels; no class variable is bound. It follows directly that  $C$  is initial in both orders and their restrictions to  $C$  agree. If they differ,  $C$  is proper. Let  $a$  be the  $R$ -least label outside  $C$  and  $c$  the  $S$ -least such label. Their existence uses that the relations are class well-orders. They are distinct: if they coincided, that label too would have identical predecessors and identical predecessor relations, and would belong to  $C$ . Set

$$R <_{\text{lex}} S \iff a < c \text{ in numerical surreal order.}$$

This is the direct comparison developed in the predecessor ([theorems 14.2, 15.1 and 16.2](#); Sections 14–16 at the pin). For completeness, its transitivity has a particularly small proof. If  $R <_{\text{lex}} S <_{\text{lex}} T$ , the common prefixes for  $(R, S)$  and  $(S, T)$  are initial segments of the same well-order  $S$ , hence nested. If one is shorter, its first discrepancy already compares  $R$  with  $T$  in the asserted direction; the third order still follows  $S$  there. If the two prefixes agree, the three next labels satisfy  $a < b < c$  in numerical order. This gives  $R <_{\text{lex}} T$ . Totality and asymmetry follow from the distinct next labels. When both orders are set-like, this definition agrees with enumeration comparison.

There is no step comparing two abstract class order types by a newly constructed class isomorphism. The labels allow comparison by literal agreement of predecessors. That is the reason no appeal to ETR is needed here.

### 54.2.2 The canonical initial branch

**Lemma 54.5** (Initial set-like branch). *Every  $R \in \mathfrak{M}_{\text{all}}$  has a canonical injective class function  $J_R : \text{Ord} \rightarrow \mathbf{No}$  whose range is an initial segment of  $R$ . It is characterized by*

$$J_R(\alpha) = x \iff \text{Pred}_R(x) \text{ is a set and } \text{otp}(\text{Pred}_R(x), R) = \alpha.$$

*Its range is exactly the labels with set-sized predecessor collections. The relation  $R$  is set-like if and only if  $J_R$  is exhaustive.*

*Proof.* For each set ordinal  $\alpha$ , recursively take the least unused label for  $\alpha + 1$  stages. At every intermediate stage the used labels are a set, whereas the carrier is proper, so the remaining class is nonempty and has an  $R$ -least element. Its predecessor set is precisely the labels already chosen. These set-length recursions are unique and compatible. Their elementary characterization in the display gives a class function on  $\text{Ord}$ . Conversely any label with set predecessors occurs at the order type of that predecessor set. Initiality and the remaining assertions follow immediately.  $\square$

For  $F \in \mathfrak{J}$ , put  $X_F = \mathbf{No} \setminus \text{ran } F$ . Concatenate the order given by  $F$  with any class well-order  $Q$  of  $X_F$ , writing  $F + Q$  for the resulting labelled relation. Such a  $Q$  exists, for example by restricting the baseline well-order to  $X_F$ . The concatenation is a class well-order: a nonempty subclass meeting the first block has a least ordinal position there, and otherwise has a least element in the second block. Its initial branch is exactly  $F$ , because every label in the second block has the proper class  $\text{ran } F$  among its predecessors. Thus every injective branch, exhaustive or not, arises as  $J_R$ .

### 54.2.3 What the skeleton can and cannot distinguish

Define the trace of an unrestricted relation on the core by

$$\text{Tr}_b(R) = \{\sigma \in \mathcal{P}_s(b) : E_\sigma <_{\text{lex}} R\},$$

where a core enumeration is identified with its induced set-like well-order relation.

**Theorem 54.6** (Trace fibers and their exact loss of information). *For every unrestricted labelled class well-order  $R$ ,*

$$\text{Tr}_b(R) = D_{J_R}.$$

*Consequently*

$$\text{Tr}_b(R) = \text{Tr}_b(S) \iff J_R = J_S.$$

*For a fixed  $F \in \mathfrak{J}$ , its trace fiber is the convex family*

$$\{F + Q : Q \text{ is a class well-order of } X_F\}.$$

*The quotient by equality of core traces is therefore identified, in the virtual sense, with the lexicographic order  $\mathfrak{J}$ . Its image in the normalized cuts is exactly the branch alternative of [Theorem 43.3](#), not the entire cut collection.*

*Proof.* If  $E_\sigma$  differs from  $J_R$  at  $\alpha$ , the labelled orders agree before that ordinal and their next labels are  $E_\sigma(\alpha)$  and  $J_R(\alpha)$ . Both comparisons therefore have the same sign. If they agree everywhere,  $J_R = E_\sigma$  is exhaustive, so  $R$  is exactly that set-like order and neither strict comparison holds. This proves the trace formula.

If two initial branches differ, [Lemma 41.5](#) gives a core code between them, which belongs to the larger trace but not the smaller one. Thus trace equality forces branch equality. The concatenation

description above proves the asserted fiber and its surjectivity over all  $F$ . Distinct branches are compared at a set coordinate before either tail, so every fiber is convex and the order of fibers agrees with the lexicographic order of  $\mathfrak{J}$ . Finally the trichotomy separates branch traces from cylinder and numerical edges.  $\square$

**Example 54.7** (Singleton fibers need not be set-like). If  $X_F$  is empty, the fiber is the singleton exhaustive order  $F$ . If  $X_F = \{x\}$ , it is again a singleton, but its order is the non-set-like concatenation of an Ord-long initial block with the last label  $x$ . Its trace is not realized by any exhaustive enumeration. Thus “singleton trace fiber” is not a criterion for being set-like. For finite  $|X_F| = m$ , the fiber has exactly  $m!$  elements, ordered as the finite permutations of that residual alphabet. Larger residual alphabets retain their full well-ordering problem inside the fiber.

This gives a useful diagram of interpretations:

$$\mathfrak{W}_{\text{sl}} \subsetneq \mathfrak{J} \longrightarrow \{\text{normalized core cuts}\}, \quad \mathfrak{W}_{\text{all}} \twoheadrightarrow \mathfrak{J},$$

where the first arrow into cuts is injective and the last map has the fibers just described. The braces and arrows here abbreviate virtual relations on classes. They do not form a same-sort diagram of class objects in GBC.

The quotient is not the usual finite condensation of a linear order. It can collapse an infinite, even higher-order, family of tails sharing a proper-class initial branch. Conversely finite adjacent blocks elsewhere in the full order are governed by the finite residual-alphabet analysis of the predecessor. Equality of traces is precisely the observational equivalence given by *all set-supported test enumerations*.

**Corollary 54.8** (Raw orders do not fill every core boundary). *A cylinder-edge or numerical-gap cut of  $\mathcal{P}_s(b)$  is not the trace of any unrestricted class well-order of the full surreal carrier. Allowing non-set-like tails adds realizations of omitted-range branch cuts, but cannot supply all missing core cuts.*

The distinction is sharper than saying that the set-like order is incomplete. Some missing boundaries correspond to an honest Ord-long initial branch with unvisited labels and are supplied by an unrestricted tail. Other boundaries stop at a set stage and are not supplied by any such tail.

## 55 Asymmetric products and the role of semantics (source 13)

### 55.1 Absorbing a class-sized left factor

An isomorphism involving  $\mathfrak{W}_{\text{sl}}$  is understood first as a pair of inverse uniform transformations of class parameters. For example, a map on  $\mathbf{No} \times \mathfrak{W}_{\text{sl}}$  takes a set label and a class enumeration and returns a class enumeration by a fixed definition. It is not asserted to be an ordinary class of pairs whose entries include arbitrary classes.

**Theorem 55.1** (Left-factor absorption). *In GBC there are uniform inverse order isomorphisms*

$$\mathfrak{W}_{\text{sl}} \cong \mathbf{No} \times_{\text{lex}} \mathfrak{W}_{\text{sl}}.$$

*Consequently, for every ordinary class linear order  $K$ , there is a uniform embedding*

$$K \times_{\text{lex}} \mathfrak{W}_{\text{sl}} \hookrightarrow \mathfrak{W}_{\text{sl}}.$$



*Proof.* Use the uniform complement isomorphisms of [Lemma 42.2](#). Given  $(x, E)$ , define  $\Phi(x, E)$  by

$$\Phi(x, E)(0) = x, \quad \Phi(x, E)(1 + \alpha) = h_{\{x\}}(E(\alpha)) \quad (\alpha \in \text{Ord}).$$

The map  $\alpha \mapsto 1 + \alpha$  is the increasing parametrization of the ordinal tail after its first point. The displayed function is a bijection, since its tail covers exactly  $\mathbf{No} \setminus \{x\}$ .

Recover  $x$  as the first value and apply  $h_{\{x\}}^{-1}$  to the tail to obtain the inverse. Different first labels are compared immediately; with the same first label, the increasing complement isomorphism preserves the first tail disagreement. Thus  $\Phi$  is an order isomorphism. Embed  $K$  into  $\mathbf{No}$  by [Lemma 42.2](#) and compose with  $\Phi$ .  $\square$

This includes every ordinary class-indexed lexicographic sum of copies of  $\mathfrak{W}_{\text{sl}}$ , including index orders such as  $\text{Ord}$ ,  $\text{Ord}^*$  and  $\text{Ord} + \text{Ord}$ . It does *not* include the full  $\mathfrak{W}_{\text{sl}}$  as the index order: its members are classes, not set codes.

## 55.2 A dense-core obstruction to splitting every point

**Lemma 55.2** (Split-order density bound). *Let  $M$  be a dense set-sized linear order and let  $D \subseteq M$  be order-dense. If  $K \times_{\text{lex}} 2$  embeds into  $M$ , then  $|K| \leq |D|$  in ZFC.*

*Proof.* For each  $k$ , the images of  $(k, 0)$  and  $(k, 1)$  enclose a nonempty open interval of  $M$ . The intervals for different  $k$  are disjoint. Choose a point of  $D$  in each, using the least one in a fixed well-order of  $D$ . This injects  $K$  into  $D$ .  $\square$

The source pairs may be adjacent; that does not invalidate the argument. An order embedding need not preserve adjacency, and the target's density puts a point between their images. The proof is an interval counting argument rather than a count of all points in the target.

**Theorem 55.3** (No full split in an inaccessible universe). *Work in ambient ZFC and let  $\kappa$  be strongly inaccessible. Interpret internal sets as  $V_\kappa$  and classes as all ambient subsets of  $V_\kappa$ . For the resulting global enumeration order,*

$$\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \not\hookrightarrow \mathfrak{W}_{\text{sl}}.$$

*In particular,  $\mathfrak{W}_{\text{sl}}^2 \not\hookrightarrow \mathfrak{W}_{\text{sl}}$  and  $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} \mathbf{No}^{V_\kappa} \not\hookrightarrow \mathfrak{W}_{\text{sl}}$ , although  $\mathbf{No}^{V_\kappa} \times_{\text{lex}} \mathfrak{W}_{\text{sl}} \cong \mathfrak{W}_{\text{sl}}$ .*

*Proof.* Externally, the internal surreal carrier is  $X = \mathbf{No}_{<\kappa}$  of cardinality  $\kappa$ , and the available set-like global enumerations are exactly all bijections  $\kappa \rightarrow X$ . Every restriction to an internal set has ranks bounded below  $\kappa$  by regularity, so these functions are legitimate full-model classes. The cardinality of this order is  $2^\kappa$ : permutations give the upper bound, and independent pair reversals the lower bound.

The core  $D$  of internal set-support codes has ambient cardinality  $\kappa$ . For the upper bound, all its codes belong to  $V_\kappa$ , whose ambient cardinality is  $\kappa$ ; transpositions give the lower bound. The internal interpolation theorem makes  $D$  order-dense in the external enumeration order. This order is dense. Applying [Lemma 55.2](#) would force  $2^\kappa \leq \kappa$ , contrary to Cantor's theorem.

Choose two distinct points in the second factor of either proposed larger product. This gives a suborder isomorphic to  $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2$ , so both nonembeddings follow. The reverse-order placement of  $\mathbf{No}^{V_\kappa}$  is covered by [Theorem 55.1](#).  $\square$

The earlier report uses the same dense-core mechanism to rule out an embedding of all raw well-orders into the set-like fragment ([theorem 20.1](#)). The new point is that merely splitting *each existing set-like point into two* already creates the obstruction. It is unnecessary to introduce

the entire raw well-order domain. [merge] That proof already uses only the adjacent raw pairs  $F + (a, b) < F + (b, a)$ , with  $F$  ranging over the bijections of  $\kappa$  onto  $X \setminus \{a, b\}$ ; these pairs form a copy of  $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2$  (the set-like orders of  $X \setminus \{a, b\}$  form a copy of  $\mathfrak{W}_{\text{sl}}$ ). So the proof of [theorem 20.1](#) already establishes [theorem 55.3](#); source 13's theorem isolates that conclusion and states it for  $\mathfrak{W}_{\text{sl}}$  itself, and its “new point” is a new statement, not a new mechanism.

### 55.3 The uniform Kelley–Morse version

**Theorem 55.4** (No definable uniform split embedding). *For each fixed formula, with fixed set and class parameters, that defines a total transformation from  $\mathfrak{W}_{\text{sl}} \times 2$  to  $\mathfrak{W}_{\text{sl}}$ , the theory  $\text{KM} + \text{GC}$  proves that the transformation is not an order embedding. The transformation may be specified relationally by a unique output class for each input. This is a schema about formulas, not quantification over arbitrary third-sort functions.*

*Proof.* Suppose a formula defines an embedding  $\Psi$ . For every  $E \in \mathfrak{W}_{\text{sl}}$ , choose the globally least code  $q_E \in \mathcal{P}_s(b)$  satisfying

$$\Psi(E, 0) < E_{q_E} < \Psi(E, 1).$$

The interval is nonempty and meets the core. The definition of its eligible code class can use the formula for  $\Psi$  and its output classes;  $\text{KM}$  permits the necessary comprehension. For distinct  $E, F$ , the image intervals are disjoint. Hence

$$q_E = q_F \implies E = F.$$

Now form the ordinary class evaluator

$$T(q, \alpha, x) \iff \exists \text{ class } E [E \in \mathfrak{W}_{\text{sl}} \wedge q = q_E \wedge E(\alpha) = x].$$

This is the decisive impredicative comprehension step. Each nonempty row is the graph of a unique enumeration, and every enumeration occurs in some row. It contradicts [Theorem 42.7](#).  $\square$

The same conclusion applies to uniform embeddings of  $\mathfrak{W}_{\text{sl}}^2$  or  $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} \mathbf{No}$  into  $\mathfrak{W}_{\text{sl}}$ , by restricting the second coordinate to two fixed choices. The argument adapts the prior report's impredicative inverse-evaluator proof for raw orders ([theorem 20.2](#)). Elementary  $\text{GB}$  comprehension alone does not justify the displayed existential class quantifier in  $T$ , and no claim to that effect is made. [merge] Applied to the same pairs  $F + (a, b) < F + (b, a)$ , the proof of [theorem 20.2](#) is the proof of [theorem 55.4](#).

### 55.4 Externally countable models reverse the conclusion

**Proposition 55.5** (External countable collapse of the order). *Let  $(M, \mathcal{C})$  be an externally countable model of  $\text{GBC}$ . The external order of its available global enumerations is isomorphic to  $(\mathbb{Q}, <)$ . Consequently, with arbitrary external maps,*

$$\mathfrak{W}_{\text{sl}}^2 \cong \mathfrak{W}_{\text{sl}}, \quad \mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \hookrightarrow \mathfrak{W}_{\text{sl}}.$$

*Proof.* The collection of available class objects is externally countable. The internal theorems asserting density and absence of endpoints provide actual witnesses among those objects, so its induced external order is a countable dense linear order without endpoints. The classical back-and-forth for such orders identifies it with  $\mathbb{Q}$ . The lexicographic square of a countable dense order without endpoints is again countable, dense, and without endpoints. The two-point split is a countable linear order, which embeds into  $\mathbb{Q}$  by the elementary countable embedding construction.  $\square$

Even if the model also satisfies  $\text{KM} + \text{GC}$ , such an external embedding need not be represented by an internally uniform definition. Likewise, an external countable cofinal sequence need not be an internal set-coded sequence. This explains exactly why the proposition and [Theorem 55.4](#) do not conflict.

The question “does the global order contain its lexicographic square?” therefore has no unqualified model-independent answer until the meaning of the collection and of an allowed map has been fixed.

*Note 55.6* ([merge]; credits, and source 08’s question on products). [Theorem 55.1](#) is proved with the uniform complement isomorphisms  $h_S$  of [lemma 42.2](#) (source 11’s [proposition 5.13](#), source 09’s [theorem 5.17](#)). Its map  $\Phi(x, \cdot)$  is the isomorphism of  $\mathfrak{W}_{\text{sl}}$  onto the first-label cylinder  $C_{(x)}$  constructed in [proposition 19.25](#) and [note 19.26](#), which source 13 does not cite; the absorption theorem assembles these cylinder isomorphisms over all first labels  $x$ . As the merge notes inside the section say, the proofs of [theorems 55.3](#) and [55.4](#) are the proofs of source 11’s [theorems 20.1](#) and [20.2](#) applied to the pairs they already use, and [proposition 55.5](#) is the observation source 11 makes after [theorem 20.2](#) (“for an externally countable model these collections can both be countable”) turned into an isomorphism with  $\mathbb{Q}$ ; source 18’s [theorem 58.6](#) reproves its second clause and classifies the full order ([section 58](#)). Together they answer source 08’s question 7 on full virtual lexicographic products ([section 60](#)); the index of [section 113](#) had it open on 2 October 2026, although the negative half already followed from source 11’s proofs. Source 13’s critical scope conditions apply: the product nonembedding is not claimed as a theorem of bare GBC about a third sort of maps (its status file, item 3).

## 56 Coding, diagonalization, and a sharper comprehension bound (source 16; source 14)

### 56.1 Binary class words and representable suborders

Let  $\mathbb{B}_{\text{Ord}}$  denote binary class functions on  $\text{Ord}$ , with first-disagreement comparison. In the sense of uniform transformations,

$$\mathfrak{W}_{\text{sl}} \simeq_e \mathfrak{J} \simeq_e \mathbb{B}_{\text{Ord}}. \quad (\text{S16.15})$$

For one direction, partition the baseline values into ordinal-indexed pairs and list each pair increasingly or decreasingly according to the input bit. Here the coordinate pairs are  $\{2 \cdot \xi, 2 \cdot \xi + 1\}$ , with ordinal multiplication. This embeds all binary classes into exhaustive enumerations. For the other, globally enumerate the set codes of  $\mathcal{P}_{\text{bd}}(b)$  as  $(q_\xi)_{\xi \in \text{Ord}}$  and send  $F \in \mathfrak{J}$  to

$$D_F = \{\xi : E_{q_\xi} < F\}.$$

If  $F < G$ , density gives a core point strictly between them. Thus  $D_F \subsetneq D_G$ , whose characteristic words have the same first difference direction. Inclusion  $\mathfrak{W}_{\text{sl}} \subseteq \mathfrak{J}$  finishes [\(S16.15\)](#). This extends the earlier report’s coding argument to the injective comparison as well.

Equipmorphism is not isomorphism. Binary class words have least and greatest words and adjacent pairs, while  $\mathfrak{W}_{\text{sl}}$  and  $\mathfrak{J}$  are dense and have no endpoints. Likewise, any one injectively and uniformly indexed class family in  $\mathfrak{W}_{\text{sl}}$  is an ordinary class linear order on its indices, and hence embeds into **No** in GBC. This does not supply an embedding of the entire virtual collection of global enumerations into **No**. There is no exhaustive uniform family at the same level.

**Theorem 56.1** (No exhaustive uniform class of set codes). *In GBC, no class of set indices with one uniform evaluation relation contains every class well-order of **No** among its rows. In fact a set-like well-order is missing from every such family.*

*Proof.* Use global choice to reindex the rows by Ord, repeating a default row if the original family was a nonempty set. For each ordinal  $\xi$ , sort the two labels  $b(2 \cdot \xi), b(2 \cdot \xi + 1)$  as  $a_\xi < c_\xi$ . Define a new enumeration by keeping the pair blocks in ordinal order. Within pair  $\xi$ , place  $c_\xi$  before  $a_\xi$  if  $(a_\xi, c_\xi)$  belongs to relation row  $\xi$ , and otherwise place  $a_\xi$  before  $c_\xi$ . This local rule is elementary even for empty or malformed rows. The new enumeration is exhaustive and set-like; it differs from every well-order row  $\xi$  on that pair. The empty-family case is immediate.  $\square$

Formula strings do not evade this theorem. Each fixed formula defines its own class when comprehension applies, but uniformly evaluating all formulas over the entire universe would require a suitable truth predicate. Neither a metatheoretic list of formulas nor a proper class of their finite syntactic codes supplies such an evaluator.

## 56.2 A sufficient bound below full Kelley–Morse

The prior report proves in KM plus global choice that no fixed definable total transformation from  $\mathfrak{W}_{\text{all}}$  to  $\mathfrak{W}_{\text{sl}}$  is an order embedding, allowing arbitrary formulas defining the transformation. For elementary formulas the following sharper sufficient bound holds.

**Theorem 56.2** (Elementary transformations and  $\Sigma_1^1$  comprehension). *Work in  $\text{GBC} + \Sigma_1^1\text{-CA}$ . Let  $\vartheta(R, E)$  be a fixed formula with only set quantifiers, with fixed set and class parameters allowed. Suppose it specifies a unique class output  $E \in \mathfrak{W}_{\text{sl}}$  for every  $R \in \mathfrak{W}_{\text{all}}$ . The specified transformation is not an order embedding.*

*Proof.* Assume an embedding is specified. Fix numerical labels  $a < c$ . For each class bijection  $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{a, c\}$  form

$$R_F = F + (a, c), \quad S_F = F + (c, a).$$

These are adjacent raw orders, and different  $F$  give disjoint, ordered pairs. Their alleged images  $E_0 < E_1$  have a bounded-support code between them. Let  $q_F$  be the least such code in a fixed global well-order of  $\mathcal{P}_{\text{bd}}(b)$ . The image intervals are disjoint, so  $q_F = q_G$  implies  $F = G$ .

The required inverse-evaluation predicate has precisely the form

$$\begin{aligned} \text{Eval}(q, \xi, x) \iff \exists F \exists E_0 \exists E_1 [ & \text{Bij}(F) \wedge \vartheta(R_F, E_0) \wedge \vartheta(S_F, E_1) \\ & \wedge \text{LeastBetween}(q, E_0, E_1) \wedge F(\xi) = x ]. \end{aligned} \quad (\text{S16.16})$$

The three displayed existential variables range over classes; the matrix has only set quantifiers. The constructed raw orders can be substituted directly into the elementary formula  $\vartheta$ , and “least code between” quantifies only over set codes. Thus  $\Sigma_1^1$  class comprehension forms this evaluator.

Its rows are empty or unique bijection graphs, and every such  $F$  occurs as a row. Compose each nonempty row with a fixed bijection  $\mathbf{No} \setminus \{a, c\} \rightarrow \mathbf{No}$ , and fill empty rows with a default. This is an exhaustive uniformly class-indexed family of global enumerations. Replace each bijection row by its induced well-order relation, an elementary conversion, and apply [Theorem 56.1](#) to obtain the contradiction.  $\square$

This proves a sufficient comprehension bound, not a reversal. It does not show that  $\Sigma_1^1$  comprehension is necessary, nor that the same proof works for arbitrary  $\vartheta$  with unrestricted class quantifier alternations. The syntax of the claimed transformation is part of the theorem. The distinction between elementary and impredicative class definitions is consistent with the known hierarchy around ETR and KM [\[27, 22, 24\]](#).

### 56.3 Source 14: binary class words and global permutations

A characteristic function  $d : \text{Ord} \rightarrow \{0, 1\}$  can be turned into a global enumeration by pair orientations. Partition the ordinals into pairs  $2 \cdot \alpha, 2 \cdot \alpha + 1$ ; ordinal multiplication has 2 on the left. Sort the corresponding baseline labels as  $u_\alpha < v_\alpha$ . Put the pair in order  $(u_\alpha, v_\alpha)$  when  $d(\alpha) = 0$  and  $(v_\alpha, u_\alpha)$  when  $d(\alpha) = 1$ . Every baseline label is used exactly once. The first differing bit produces the first differing pair in the correct lexical direction.

Conversely, enumerate the core codes by a fixed class bijection  $\text{Ord} \rightarrow \mathcal{P}_s(b)$ . The characteristic function of  $D_E$  in this schedule is an order embedding of  $\mathfrak{W}_{\text{sl}}$  into binary class words: if  $E < F$ , then  $D_E \subsetneq D_F$  by core density, and the first changed bit is 0 versus 1. Thus the known equimorphism

$$\mathfrak{W}_{\text{sl}} \simeq_e (\{0, 1\}^{\text{Ord}}, <_{\text{lex}})$$

is entirely uniform. It is not an isomorphism. The binary class-word order has endpoints and adjacent pairs of the form  $p \frown (0) \frown (1, 1, \dots)$  and  $p \frown (1) \frown (0, 0, \dots)$ ;  $\mathfrak{W}_{\text{sl}}$  is dense without endpoints. These are class-word constructions, not sets of every binary class.

**Theorem 56.3** (No exhaustive uniform class of codes, inherited). *There is no uniformly Ord-indexed family containing every member of  $\mathfrak{W}_{\text{sl}}$ . More generally no ordinary class of set codes, with one uniform evaluation relation, exhausts all global enumerations.*

*Proof.* Suppose  $(E_\alpha)_{\alpha \in \text{Ord}}$  were such a family. In the pair at  $2 \cdot \alpha, 2 \cdot \alpha + 1$ , choose the orientation whose first entry differs from  $E_\alpha(2 \cdot \alpha)$ . At least one of the two orientations works, since the pair labels are distinct. The resulting class function is a bijection, as it only reorders disjoint baseline pairs, and it differs from every row  $E_\alpha$  at the designated coordinate. This is an elementary diagonal construction.

An ordinary proper class of set codes admits a set-like global enumeration in GBC; a set of codes can be padded to an Ord-indexed family. Either kind of exhaustive uniform evaluation would therefore contradict the first assertion. This is the predecessor's diagonal obstruction in an explicit pair form.  $\square$

A skeleton is therefore a dense *test family*, not a list of all objects. The new cut recognition theorem explains what extra information a class cut of that test family can encode; it does not defeat the diagonal theorem by relabelling cuts as sets.

*Note 56.4* ([merge]; repetitions, and the new comprehension bound). The binary-class equimorphism (S16.15) and source 14's display are source 11's [theorem 19.32](#) (with source 08's [note 19.35](#) and source 12's [theorem 19.34](#)); source 16 adds the injective words  $\mathfrak{J}$  between the two ends, by the same core-density argument. The diagonal theorems [theorem 56.1](#), [theorem 56.3](#) and source 15's [theorem 57.4](#) (printed with source 15's foundations in [section 57](#)) are source 11's [theorem 19.37](#) and source 12's [theorem 19.38](#), as are source 13's [theorem 42.7](#), source 17's [theorem 42.16](#) and source 18's [theorem 42.19](#) printed in Part VIII; all credit this report. New is source 16's [theorem 56.2](#): for transformations defined by a formula with only set quantifiers, GBC with  $\Sigma_1^1$  class comprehension suffices to refute an order embedding of  $\mathfrak{W}_{\text{all}}$  into  $\mathfrak{W}_{\text{sl}}$ , where source 11's [theorem 20.2](#) used KM + GC and arbitrary formulas. It is a sufficiency result; source 16 asks for the reversal ([research question 113.63](#)). Its proof uses the adjacent raw pairs only through the disjointness of their image intervals; with the pairs  $(E, 0) < (E, 1)$  of  $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2$  in place of  $R_F < S_F$  the same  $\Sigma_1^1$  evaluator can be formed, which bears on source 13's [research question 113.24](#). (This adaptation is an observation of the merge, not a theorem of either source.)

## 57 Larger universes, class realizations and paradox avoidance (sources 16, 14, 13, 15 and 18)

*Note 57.1* ([merge]; Part V counterparts). Five sources discuss the full-class model  $(V_\kappa, \mathcal{P}(V_\kappa))$  at an inaccessible  $\kappa$  and the relativity of class notions. Source 16's [theorem 57.2](#) is source 12's exact external identification [theorem 21.1](#) (both identify  $\mathfrak{W}_{\text{sl}}$  with  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$  and  $\mathfrak{W}_{\text{all}}$  with  $\mathcal{W}(\mathbf{No}_{<\kappa})$ ), and its [theorem 57.3](#) is source 11's [theorem 20.1](#) with one sharpening: already  $\mathcal{W}_{\kappa+2}(X)$  does not embed into  $\mathcal{W}_\kappa(X)$ , although by [theorem 27.2](#) both have density and cellularity  $\kappa$ . Source 18 proves the same nonembedding by counting cellularity, display (S18.5). Source 14's model subsection recalls source 12's and source 09's calculations ([theorems 21.1](#) and [22.1](#)), and source 15's full-universe paragraph repeats the coding-length contrast of [theorem 21.1](#). The rest of these sections are foundational discussions in the manner of Part V ([section 23](#)); they are printed whole, in the order 16, 14, 13, 15, 18, because each states its own list of invalid inferences and its own boundary of comprehension.

### 57.1 Source 16: an explicit larger-universe interpretation

To treat the entire collections as actual sets, choose an ambient ZFC universe and assume that  $\kappa$  is strongly inaccessible. Consider the two-sorted structure

$$\mathcal{M} = (V_\kappa, \mathcal{P}(V_\kappa)).$$

Its sets are the elements of  $V_\kappa$ , and its available classes are *all* external subsets of  $V_\kappa$ . It satisfies KM plus global choice. Inaccessibility controls replacement: the image of an internal set under any available class function has size less than  $\kappa$  and bounded rank. Full class comprehension is immediate from the external power set. Such full models are a standard way to make the next logical level explicit [28].

**Theorem 57.2** (Exact external identification). *In this full model the internal surreal class is externally  $X = \mathbf{No}_{<\kappa}$ , of cardinality  $\kappa$ . Internally set-like well-orders are exactly the externally well-ordered relations of type  $\kappa$  on  $X$ . All internal class well-orders correspond exactly to the full external set  $\mathcal{W}(X)$ . Thus the two comparisons become*

$$\mathfrak{W}_{\text{sl}}^{\mathcal{M}} = \mathcal{W}_\kappa(X), \quad \mathfrak{W}_{\text{all}}^{\mathcal{M}} = \mathcal{W}(X). \quad (\text{S16.17})$$

*Proof.* All sign sequences of length below  $\kappa$  belong to  $V_\kappa$ , and these are exactly the internal sign sequences. Strong inaccessibility gives  $|X| = 2^{<\kappa} = \kappa$ . Every external binary relation on  $X$  is an available internal class. Well-foundedness agrees in this full interpretation; in particular any external descending  $\omega$ -sequence has bounded rank below  $\kappa$  and is an internal set witnessing failure.

A subset of  $X$  is an internal set exactly when it has external size less than  $\kappa$ . One direction follows from the strong-limit property of  $V_\kappa$ ; the other follows from regularity bounding the ranks of fewer than  $\kappa$  elements. Consequently an external well-order of cardinality  $\kappa$  is internally set-like exactly when all its proper initial segments have cardinality less than  $\kappa$ , which is exactly external order type  $\kappa$ . All other external well-orders are still admitted classes, yielding the unrestricted identification.  $\square$

**Theorem 57.3** (Strict embedding separation in the full model). *In the preceding external interpretation,*

$$\mathfrak{W}_{\text{sl}}^{\mathcal{M}} \hookrightarrow \mathfrak{W}_{\text{all}}^{\mathcal{M}}, \quad \mathfrak{W}_{\text{all}}^{\mathcal{M}} \not\hookrightarrow \mathfrak{W}_{\text{sl}}^{\mathcal{M}}.$$

*Indeed even  $\mathcal{W}_{\kappa+2}(X)$  does not embed into  $\mathcal{W}_\kappa(X)$ , although both have density and cellularity  $\kappa$ .*



*Proof.* Inclusion gives the forward embedding. The target  $\mathcal{W}_\kappa(X)$  is dense and has a dense subset of size  $\kappa$ . For fixed  $a < c$  there are  $2^\kappa$  prefixes  $F : \kappa \rightarrow X \setminus \{a, c\}$ . Their two completions  $F + (a, c) < F + (c, a)$  are disjoint adjacent pairs in  $\mathcal{W}_{\kappa+2}(X)$ . An order embedding into the dense target would turn these pairs into  $2^\kappa$  disjoint nonempty open intervals. Each interval must meet the  $\kappa$ -sized dense subset, contradicting Cantor's theorem.  $\square$

This rederives the earlier report's full-model nonembedding, while identifying why equal cardinality, ordinal spectra, density, and cellularity can still fail to imply equimorphism. The adjacent pairs provide an additional obstruction when the target is dense.

### 57.1.1 Henkin models and internal well-foundedness

For a general model  $(M, \mathcal{C})$ , one has only the admitted classes  $\mathcal{C} \subseteq \mathcal{P}(M)$ , not automatically every external subset. External families of these classes need not be coded by any one internal family relation. If the model is externally countable, both virtual collections can be externally countable, and the external dense set-like order can admit an external copy of the raw order. Such an embedding need not be internally definable or uniformly represented, and so does not contradict [Theorem 56.2](#) or the internal diagonal theorem.

There is also a precise correction to a common ambiguity. Over GBC, for a fixed class relation, well-foundedness on nonempty sets, well-foundedness on all available nonempty subclasses, and absence of a set-coded descending  $\omega$ -sequence are equivalent. Therefore two class expansions with the same sets agree about internal well-foundedness of any relation they share. Expanding the class collection can add relations; it does not change that property of a shared relation. Nevertheless, even a transitive class model can contain a relation it regards as well-founded that is externally ill-founded; one must distinguish a general model from a  $\beta$ -model [27, §3].

The full inaccessible model removes this ambiguity by its explicit fullness and rank closure. A singular birthday cutoff is a legitimate set alphabet for [Note 32.1](#); it is not an automatic replacement for  $V_\kappa$  in this model-theoretic construction.

## 57.2 Source 14: universe-relative models and paradox avoidance

### 57.2.1 A full-class model with externally set-sized objects

For a concrete interpretation, assume in an external set universe a strongly inaccessible cardinal  $\Omega$  and consider

$$(V_\Omega, \mathcal{P}(V_\Omega)).$$

Internally the ordinal and surreal collections are proper classes; externally they are the sets  $\Omega$  and  $X = \mathbf{No}_{<\Omega}$ . Strong inaccessibility gives  $|X| = \Omega$ . A baseline is externally a bijection  $b : \Omega \rightarrow X$ . The core consists of permutations with support of size  $< \Omega$ , equivalently bounded support in  $\Omega$ , and has external cardinality  $\Omega$ . The external global-enumeration order is

$$\text{Bij}(\Omega, X), \quad |\text{Bij}(\Omega, X)| = 2^\Omega.$$

These are the predecessor's full-model calculations ([theorems 20.1](#), [21.1](#) and [22.1](#); the universe-model sections at the pin). The lower bound for the second cardinality is pair orientation; the upper bound counts graphs. For the core, each bounded stage has fewer than  $\Omega$  permutations by the strong-limit property, while first-entry transpositions supply  $\Omega$  distinct codes.

Internally “every set cut” means externally cuts with fewer than  $\Omega$  parameters. Thus the core and  $X$  are externally  $\Omega$ -sized  $\eta_\Omega$ -orders. The full enumeration order has strictly more points while

the core remains dense in it. This is an ordinary set-theoretic example of a dense order and a larger order of its selected cuts; no paradox occurs.

The new trichotomy becomes an external classification of cuts using prefixes of length  $< \Omega$ . A branch extending through all internal ordinals has external length  $\Omega$ . It is not an internally set prefix to which one may apply a small-cut theorem. The coverage condition is precisely surjectivity onto  $X$ .

All external well-orders on  $X$  also allow order types  $\alpha$  with  $\Omega < \alpha < \Omega^+$ . They correspond to internal non-set-like orders, not to additional bijections  $\Omega \rightarrow X$ . Consequently “increase the universe so everything is a set” does not license replacing one interpretation by another without recording the allowed ordinal lengths.

### 57.2.2 ZFC schemes, GBC, KM, and higher sorts

In ordinary ZFC, sign sequences and their set-sized fragments can be handled directly; global statements can be treated as schemes for specified definable class parameters. A definable baseline must be supplied where needed. Formula codes alone do not solve the uniformity problem: evaluating every possible class definition in the whole universe requires truth resources not furnished by a list of formulas.

In two-sorted GBC, the class parameters and set-coded skeleton are literal objects of their respective sorts. The classification theorem has the form “for every class cut  $D$ , one of the stated alternatives holds.” It is not a third-sort object containing all those cuts. Stronger comprehension in KM does not change the rule that proper classes cannot be members of ordinary classes. A literal collection of class orders or cuts requires a specified additional sort or an external universe. Relative consistency assumptions for any such enlargement must be stated separately.

In a universe-polymorphic type theory, one can form a type of relations on a fixed small carrier. A larger-universe type can hold all such relations without identifying it with the original carrier. Lifting a carrier changes its ambient type level, not its mathematical elements; constructing surreals with larger allowed ordinal birthdays adds new elements. These are different operations. A proof assistant should not hide the distinction under coercions or ambiguous names.

### 57.2.3 Which assertions depend on the class realization?

Changing the available classes while retaining the same sets leaves each existing support code and each existing prefix unchanged. It can add new global enumerations, binary class words, and cut classes. [Theorem 48.1](#) is a more precise invariant than merely saying that first disagreements are unchanged: it prohibits a new enumeration from realizing an *old* cut. At the same time the external cardinality and order type of the entire admitted enumeration collection can change. The results are compatible, because a new point brings a new trace parameter.

For unrestricted orders, having an initial branch reconstructible from an old trace does not similarly force an arbitrary new *tail* to be old. The trace deliberately forgets that tail. This illustrates why one must not generalize the absoluteness theorem from set-like realizations to uniqueness of raw realizations.

### 57.2.4 A short paradox audit

There is no set ordinal containing all ordinals, no surreal whose sign sequence has domain Ord, and no application here of the surreal cut constructor to arbitrary proper-class option sets. All recursive histories used for the main constructions at a set stage are sets. Class predicates such as “is a global enumeration” are never turned into same-sort membership objects. Equal abstract class

types are not silently represented by a canonical set ordinal. The baseline choice is explicit. Finally, class-model absoluteness is proved from formulas with the stated old parameters, not assumed for all higher-order properties. These restrictions are what allow the positive theorems, rather than obstacles to developing the subject.

### 57.3 Source 13: foundational interpretations without paradoxes

#### 57.3.1 Ordinary set theory: the cutoff-first route

For any set ordinal  $\theta$ , the sign sequences of birthday below  $\theta$  form a set. Relations on this set form another set, and the subset satisfying well-order axioms is available by separation. All fixed-carrier order theory therefore lives in ordinary ZFC; no inaccessible cardinal or extra class sort is needed.

When a proof uses a family of all birthday cutoffs, the family is a class description rather than a set containing the entire surreal line. A compatible directed system requires explicitly commuting transition maps. Merely proving existence of an embedding at each cutoff does not make the union of independently chosen maps a function.

For set-length words, a word is itself a set of ordered pairs. The class of all such words is a legitimate class of sets. It is not the same logical object as the totality of class-length words.

#### 57.3.2 Two-sorted class theory: quantify, do not collect

In GB or GBC, class variables can stand for relations on **No**. One may assert

$$\forall R \forall S [\text{ClassWO}(R) \wedge \text{ClassWO}(S) \implies R = S \vee R <_{\text{lex}} S \vee S <_{\text{lex}} R].$$

There is no need to form an object whose members are  $R$  and  $S$ . The fraktur notations  $\mathfrak{W}_{\text{sl}}, \mathfrak{J}, \mathfrak{W}_{\text{all}}$  used here abbreviate precisely such restricted class quantification.

A family indexed by an ordinary class  $K$  of sets is supplied by one class relation  $T(k, \alpha, x)$ . First-disagreement comparison of two rows is elementary in  $T$ . If repeated rows occur, a fixed set-like global well-order selects the least set code in each equal-row class; row equality itself quantifies only over set coordinates and values. The distinct-row representatives then form an ordinary class linear order, which embeds into **No**.

This explains how every uniformly class-coded portion of  $\mathfrak{W}_{\text{sl}}$  can embed into **No** while an exhaustive uniform code family is impossible. The local statement is about a particular class evaluator. It does not quantify a class of all evaluators or identify all class functions with their set codes.

#### 57.3.3 The exact comprehension boundary of the new proofs

The crossing predicate

$$\text{Cross}(L, p) \iff (\exists d \in \mathcal{P}_s(b)_p \ d \in L) \wedge (\exists d \in \mathcal{P}_s(b)_p \ d \notin L)$$

quantifies over set codes. “For every ordinal length a crossing prefix exists” is therefore elementary with  $L$  as a class parameter. The uniqueness lemma makes  $F(\alpha) = p_{\alpha+1}(\alpha)$  an elementary class definition. This is the reason cut reconstruction does not need ETR or impredicative comprehension.

The uniform complement isomorphisms, support completions, and active prefixes similarly use set histories. In contrast, the inverse evaluator in [Theorem 55.4](#) asks whether *some class enumeration* produced a specified code. That is a genuine existential class quantifier in a class comprehension formula. The proof uses KM at exactly this point.

General elementary transfinite recursion allows class-valued stages along a class well-order. It is distinct from the set-valued recursion used above. Its relevance to abstract class-well-order comparison is discussed in [27]; none of our direct labeled comparison arguments relies on the stronger comparison theorem.

#### 57.3.4 Kelley–Morse does not turn classes into members

KM permits class quantifiers in comprehension formulas, but its classes still have sets as members. Thus even KM does not justify an ordinary power class whose elements are all class well-orders of **No**. A third sort, a containing universe, or a fixed syntactic schema is necessary for arbitrary maps, topologies, or families whose elements are such classes.

Hyperclass forcing is an example of work that explicitly addresses this higher level, with additional model and coding hypotheses [28]. Its existence as a theory should not be read as permission to suppress the level distinction in an argument stated only over GB or KM.

#### 57.3.5 Full universes versus selected class realizations

The full interpretation  $(V_\kappa, \mathcal{P}(V_\kappa))$  at a strongly inaccessible  $\kappa$  is a convenient sufficient model for our external cardinal arguments. The available internal surreals are externally the set **No** $_{<\kappa}$ , and internal set-like enumerations have external length  $\kappa$ . A raw well-order may have any external ordinal type of cardinality  $\kappa$ ; types beyond  $\kappa$  create proper-class predecessor segments internally.

For a nonfull model  $(M, \mathcal{C})$ , only classes in  $\mathcal{C}$  are available. Changing  $\mathcal{C}$  can change the available branches, global enumerations, cuts, and external maps without changing the underlying set universe. The cut theorem remains an internal statement about the cuts actually present. If an outer class realization adds a normalized cut satisfying the crossing criterion, its unique branch becomes available there by elementary comprehension from that cut. Thus missing classes, not merely missing set elements, are part of the geometry.

Internal well-foundedness must also be kept separate from external well-foundedness. Over the relevant global-choice class theories, class well-foundedness can be tested by the absence of a set-coded descending  $\omega$ -sequence; an external descending sequence need not be so coded in the model [27, Section 3]. This is why an arbitrary internally well-ordered class is not automatically a well-order in a surrounding metatheory. Our full inaccessible model avoids that particular ambiguity, while Proposition 55.5 deliberately studies an external order of available objects.

[merge] Source 13’s subsection “Relation-bit lexicography is a different order” comes here in its text; it is printed in section 38.

#### 57.3.6 A concise list of invalid inferences

The most tempting false steps are logically different. A proper-class ordinal word is not a surreal; a predicate on surreals is not necessarily a small set; a set of individual class witnesses is not automatically a uniform family; a class well-order need not be set-like; an order embedding need not preserve suprema, cuts, or adjacency; and equal cardinality, equimorphism, and isomorphism are three distinct claims. Finally, the isomorphism type of an external available-object order need not describe the internally definable transformations on that order. Each of the main theorems above uses at least one of these distinctions essentially.

### 57.4 Source 15: formalization without paradoxes

#### 57.4.1 Four sound frameworks and their different meanings

| Framework                               | What it directly represents                                                                        | Important boundary                                                            |
|-----------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| ZFC, fixed sets                         | Relations on a set $X$ , all ordinal enumeration types, binary cubes, birthday cutoffs             | No set of all surreals; a definable-class claim is a scheme or a formula.     |
| Two-sorted GB/GBC                       | Individual class orders, uniform families, elementary raw comparison and set-interval condensation | Classes contain sets, not other proper classes.                               |
| KM and recursion extensions             | Stronger class comprehension; appropriate class-valued recursions                                  | Stronger comprehension does not add a third sort automatically.               |
| Containing universe or stratified types | A lower universe's classes become higher-universe objects                                          | Size and well-foundedness are relative; lifting does not identify all levels. |

### 57.4.2 Why Burali–Forti and unrestricted cuts do not arise

In the sign presentation, each birthday is a set ordinal, while birthdays are unbounded in the class of ordinals. The order type of an arbitrary class well-order is not asserted to be a set ordinal. An expression such as  $\text{Ord} + \text{Ord}$  abbreviates a relation on a class of tagged sets. A collection of all these relations is treated either by class-variable predicates or in a genuinely larger universe.

Likewise, the cut principle permits sets  $L < R$ . Substituting all of **No** on the left and the empty class on the right would demand a surreal greater than every surreal and is outside that principle. The represented cut theorem explicitly admits unfilled class cuts, so it does not smuggle in such an unrestricted principle.

**Theorem 57.4** (No exhaustive uniform class of set codes). *In GBC no class of set codes with one uniform decoder can list every class well-order of **No**. An omitted order can always be chosen set-like.*

*Proof.* Reindex any proposed family by  $\text{Ord}$ , repeating a default row if the indexing class is a set. Let its rows be  $R_\alpha$ . Partition a baseline enumeration of **No** into consecutive two-label blocks, using positions  $2 \cdot \alpha, 2 \cdot \alpha + 1$ . In block  $\alpha$ , place its two labels in the order opposite to their relative order in  $R_\alpha$ . Listing these blocks constructs a class bijection  $\text{Ord} \rightarrow \text{No}$ , and its well-order differs from each row on that row's designated pair. Its definition uses only the one family relation and the baseline as class parameters.  $\square$

This is the prior report's diagonal obstruction, restated to locate the size boundary (theorems 19.37 and 19.38; Theorems 19.37–19.38 at the pin). An ordinary class of codes may be dense and realize every set cut without listing every global enumeration. Formula strings do not evade the theorem: a uniform semantic evaluation of all formulas of the universe would require an appropriate truth predicate. Individual formulas, an external collection of definable classes of a model, and an internal universal evaluator are different resources.

### 57.4.3 Exactly where recursion principles enter

In GBC, for a class strict linear order, the following internal conditions are equivalent: every nonempty subclass has a least element; every nonempty set subset has a least element; and there is no descending  $\omega$ -sequence in the model. If a subclass has no least element, choose below each of its

members a witness of least possible set rank, then use global choice within that rank. Ordinary recursion with this class parameter constructs a descending  $\omega$ -sequence, whose range is a set by class replacement. The other implications are the usual set well-foundedness arguments. This explains the role of choice in the equivalence; the article's primary definition remains the subclass minimum condition [27, Section 2].

Set-valued recursion along Ord can be assembled from its unique set initial solutions using elementary comprehension. This covers the ordinary surreal forth construction and prefix completions. The raw comparison formula, simultaneous canonical block quotient, and persistent-cut reconstruction do not even need a class-valued recursive construction.

A general recursion whose stage value is itself a proper class is a stronger request. The principle ETR supplies elementary transfinite recursion along class well-orders. It suffices for comparison of arbitrary abstract class well-orders by initial-segment isomorphisms [27, Theorem 6]. That is a different problem from comparing identical labelled prefixes. The cited work poses converse-strength questions; we do not turn those questions into a claimed independence theorem or assert that their historical open status is settled as of the date of this article. For general class recursion and truth-predicate strength see [22, 35].

For condensation, a fixed finite number of iterations is available by substituting elementary formulas. A uniform history on every arbitrary input is a separate statement; ETR of the appropriate length is sufficient, but this article proves no necessity. In the polynomial examples the entire history already has an elementary formula, including at stages with proper-class many predecessors. A warning about general recursion must not be used to reject an explicitly defined construction.

#### 57.4.4 Full-universe and Henkin interpretations

Let a containing universe have an inaccessible  $\kappa$ , and consider  $(V_\kappa, \mathcal{P}(V_\kappa))$ . Internally, the surreals are  $\mathbf{No}_{<\kappa}$  and the ordinals are the ambient  $\kappa$ . Every external subclass of the set universe is available as a class. Internal class well-foundedness therefore agrees with external well-foundedness for these relations.

The internal set-like enumeration space becomes the ambient  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ , while the full raw space becomes all ambient well-orders of the carrier. Both have cardinality  $2^\kappa$ , and both embed exactly the ordinals below  $\kappa^+$ , but their finer structure differs. The set-like space is equimorphic with  $\mathbb{B}_\kappa$ ; the unrestricted one has unbounded binary coding lengths below  $\kappa^+$  and cannot embed into that one cube. This is a genuine nonembedding theorem in a specified external model, not a same-level cardinality argument about a class of classes (theorems 20.1, 21.1 and 22.1; Sections 20–22 at the pin).

For a general two-sorted model  $(M, \mathcal{C})$ , only the classes in  $\mathcal{C}$  may occur as relations or witnesses. A relation can be internally well-founded while an external descending sequence is absent from  $\mathcal{C}$ . A countable model may have externally countably many classes without possessing an internal exhaustive decoder. Adding classes can add new global enumerations and can destroy the internal well-foundedness of an old relation by adding a descending witness. The elementary first-disagreement formula on previously available valid orders remains the same, but the ambient totality and existence assertions may change.

### 57.5 Source 18: containing universes, class realizations, and paradox avoidance

The mathematical distinctions above are also distinctions between formal objects. A type, a set, a definable class, an available class in a two-sorted model, and a collection of available classes



have different membership and quantifier rules. A rigorous development should choose one of the following settings for each theorem.

### 57.5.1 Ordinary ZFC and bounded birthday stages

All results about  $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$  are ordinary set theorems. There is no need for class theory, an inaccessible cardinal, or a global well-order to construct these spaces. Choice well-orders the set alphabet, and all enumerations and cuts live in ordinary power sets. In particular, the exact gap spectrum does not rest on the existence of an external model or on an unformalized “class ordinal.”

One may also work in ZFC with definable classes as metanotation. A particular formula defines a particular global relation, but one must not silently quantify in the object theory over all formulas and their semantic truth sets. A uniform truth predicate for the whole set universe is unavailable. A scheme covering each fixed defining formula is different from a single theorem about all available classes.

### 57.5.2 GBC and named class parameters

GBC is a natural setting for supplied global orders, uniform families, the elementary common-prefix comparison, the block decomposition, and the represented-cut criterion. These constructions use actual set labels to represent their quotients or prefixes. A block minimum is a set; the proper block it represents is not an element.

The status of a well-order is relative to the class realization. In a model  $\mathcal{M} = (M, \mathcal{X})$ , “every nonempty subclass” means every appropriate member of  $\mathcal{X}$ . An external descending sequence might not be represented in  $M$  or  $\mathcal{X}$ . Consequently external countability or external ill-foundedness cannot simply be substituted for internal set size or internal well-foundedness. The factorial-shuffle theorem carefully uses external countability and, for its finite blocks,  $\omega$ -standardness; it does not assume that all internal well-orders are externally well-founded.

### 57.5.3 A full-class model at an inaccessible cardinal

For a concrete comparison with the external countable setting, let  $\lambda$  be strongly inaccessible in an ambient ZFC universe and consider the full class model

$$(V_\lambda, \mathcal{P}(V_\lambda)).$$

Its internal surreal carrier is the ambient set of sign sequences of length below  $\lambda$ , of ambient cardinality  $\lambda$ . Every ambient subset of the carrier is available as a class. Class well-foundedness therefore means actual external well-foundedness. An internal predecessor class is a set precisely when its ambient cardinality is below  $\lambda$ : regularity bounds the ranks of a small set of elements of  $V_\lambda$ , and inaccessibility gives the converse cardinal bound.

It follows that, from the ambient viewpoint,

$$\mathfrak{W}_{\text{sl}} = \mathcal{W}_\lambda(\mathbf{No}_{<\lambda}), \quad \mathfrak{W}_{\text{all}} = \mathcal{W}(\mathbf{No}_{<\lambda}).$$

The former has cellularity  $\lambda$ , since  $\lambda^{<\lambda} = \lambda$ . The latter has cellularity  $2^\lambda$ : fix a residual set of three labels and vary the  $\lambda$ -long prefix on its complement. Each of  $2^\lambda$  disjoint cylinders contains a nonempty open interval. Hence

$$\mathfrak{W}_{\text{all}} \not\rightarrow \mathfrak{W}_{\text{sl}} \quad \text{in this full-class interpretation.} \tag{S18.5}$$

An order embedding would send the endpoints and an interior point of each such interval to  $2^\lambda$  disjoint nonempty intervals in the target, contradicting its cellularity.

There is no conflict with the external countable-model embedding in the preceding section. The two class realizations have different external sizes and admit different external families. There is also no conflict with surreal universality:  $\mathfrak{W}_{\text{all}}$  is an external set of size  $2^\lambda$ , not an internal class of set labels to which the internal universality theorem applies.

The canonical block theorem supplies a further consistency check. An ambient order of the carrier has ordinal type  $\gamma$  with  $|\gamma| = \lambda$ . Its division

$$\gamma = \lambda \cdot \beta + \rho, \quad \rho < \lambda,$$

is exactly the external interpretation of the internal decomposition into Ord blocks and a final set tail. Thus the class theorem recovers ordinary ordinal division in this model without treating Ord as an internal set.

#### 57.5.4 KM, stronger recursions, and genuine higher sorts

KM permits impredicative class comprehension. It remains a two-sorted theory: classes are not members of classes. To make the whole domain of global well-orders an object one needs, for example, an explicitly specified higher sort, an external containing universe, or a carefully defined unrolling. These are substantive changes of framework, not notation that erases the diagonal in [Theorem 42.19](#).

The use of ETR should be equally explicit. General class-valued recursions, including iterated truth constructions, exceed GBC [24]. ETR is a sufficient framework for suitable abstract class well-order comparisons and recursive quotient constructions [35]. In contrast, three procedures in this article have elementary closed definitions or set-valued histories: comparison of shared labels, simultaneous set-interval decomposition, and reconstruction of a branch from unique set prefixes. None requires ETR. We do not infer an exact reverse-mathematical equivalence for an unspecified iteration of the quotient operator.

#### 57.5.5 A concise list of invalid identifications

The following substitutions would change a theorem's meaning or invalidate its proof.

1. Replacing Ord by a set ordinal while still claiming to enumerate all surreals of the same universe.
2. Treating all class relations as members of a class merely because each individual relation is available.
3. Replacing an arbitrary class well-order by an Ord enumeration; this discards its later proper blocks and set tail.
4. Replacing a proper-class cut by a set cut when applying surreal interpolation.
5. Using cardinal multiplication where the proof concerns ordinal lexicographic coding length.
6. Assuming that an order embedding preserves suprema or density in the ambient order.
7. Treating a set-indexed uniform family of classes as an ordinary set of class objects.
8. Extending numerical lexicography to isomorphism classes after discarding the labels on which it depends.

These are mathematical type distinctions, not additional consistency axioms imposed on the bounded results.

*Note 57.5* ([merge]; internal well-foundedness under class extensions). The last paragraph of source 15’s subsection on full-universe and Henkin interpretations (in [section 57.4](#)) says that adding classes “can destroy the internal well-foundedness of an old relation by adding a descending witness”. For class realizations that are models of GBC with the same sets this is not so, and source 16’s subsection on Henkin models proves the opposite: over GBC a class relation is well-founded on all admitted subclasses exactly when no set-coded descending  $\omega$ -sequence exists (source 15 proves the same equivalence in its subsection “Exactly where recursion principles enter”), and a descending  $\omega$ -sequence admitted by a larger realization is a set of the common set part, by class Replacement, so it was already present. Hence two GBC realizations with the same sets agree on the internal well-foundedness of every relation they share. Source 15’s sentence can hold only for a larger class collection in which that equivalence fails, for example one that is not a model of global choice; as a statement about GBC realizations it is withdrawn here. The same correction was made to source 08’s foundations after the Hilbert’s-tenth review of batch 80 ([section 5.6](#) and the note before it). Source 15’s [research question 113.56](#) asks which statements “require well-foundedness preservation”; by the argument above, none does among GBC realizations with the same sets.

## 58 An exact external classification in countable models (source 18)

The distinction between an internal class and an external collection of classes leads to a surprisingly uniform external order type. The statement below concerns the raw labelled comparison of all admitted class well-orders, including those that are not set-like. It does not concern their abstract order types. It strengthens the observation that an external countable collection of admitted orders can embed into the external rational line.

For a linear order  $L$ , write  $x \sim_{\text{fin}} y$  when the closed interval between  $x$  and  $y$  is finite. These equivalence classes are the *finite-condensation blocks*. For a nonempty countable collection  $\mathcal{A}$  of linear order types, its *shuffle*  $\sigma(\mathcal{A})$  is obtained by densely colouring  $\mathbb{Q}$  with one colour for each member of  $\mathcal{A}$ , then replacing a point by the order type of its colour. A colour-preserving back-and-forth proves that the resulting order type is independent of the chosen dense colouring. These are standard order-theoretic constructions; see, for example, Dimitrov–Harizanov–Morozov–Shafer–Soskova–Vatev, *On cohesive powers of linear orders*, Definitions 3.3 and 6.2, [36]. The identification with the well-order space below is derived here.

### 58.1 Terminal tails and finite-condensation blocks

The next arguments can first be carried out internally in GBC. Let  $X$  be an infinite linearly ordered class, and consider the predicate of all class well-orders of  $X$ , with the raw labelled lexicographic comparison. It is enough that the required well-orders and constructions exist; for  $X = \mathbf{No}$ , global choice supplies them.

**Lemma 58.1** (The maximal finite terminal tail). *Every class well-order  $R$  on an infinite carrier has a unique finite terminal segment  $T_R$  such that  $R$  restricted to  $X \setminus T_R$  has no greatest element. Write  $n(R) = |T_R|$ . Every finite terminal segment of  $R$  is contained in  $T_R$ .*

*Proof.* If  $R$  has no greatest element, take  $T_R = \emptyset$ . Otherwise start at the greatest element and repeatedly take its immediate predecessor, for as long as one exists. If this process did not terminate at a finite stage, ordinary set-valued recursion with parameter  $R$  would produce a descending  $\omega$ -sequence. Class Replacement makes its range a set, contradicting well-foundedness. At termination

the selected finite chain is terminal and the remaining prefix has no greatest element. Any finite terminal segment is obtained by finitely many of the same uniquely determined predecessor steps, proving maximality and uniqueness.  $\square$

**Lemma 58.2** (Adjacency forces a finite residual carrier). *Suppose  $R < S$  are adjacent in the lexicographic order of all class well-orders of  $X$ . Let  $C$  be their common labelled prefix, and let  $a = \min_R(X \setminus C) < \min_S(X \setminus C) = b$  in the ambient label order. Then  $Y = X \setminus C$  is finite.*

*Proof.* Keep the prefix  $C$ , then the next label  $a$ , fixed. The  $R$ -tail must be the greatest available well-order in that prefix cylinder: any greater tail gives an order strictly between  $R$  and  $S$ . At each subsequent position, therefore, its label must be the greatest remaining label. Otherwise choose a larger remaining label and well-order the remaining carrier to obtain a greater tail. Consequently the reverse ambient label order on  $Y \setminus \{a\}$  is a class well-order.

The symmetric argument, fixing  $C$  then  $b$ , shows that the ambient label order on  $Y \setminus \{b\}$  is a class well-order. On  $Z = Y \setminus \{a, b\}$ , both an order and its reverse are therefore well-orders. This forces  $Z$  to be finite. Indeed, if  $Z$  were infinite, repeatedly choosing the least unused point in one of those well-orders would produce an ascending  $\omega$ -sequence in that order, hence a descending  $\omega$ -sequence in the other. Thus  $Y$  is finite.  $\square$

**Proposition 58.3** (Factorial blocks). *The finite-condensation block containing  $R$  consists exactly of the orders obtained by keeping  $R$  fixed on  $X \setminus T_R$  and permuting the  $n(R)$  labels in  $T_R$ . It is a finite chain of size  $n(R)!$ .*

*Proof.* The common-prefix cylinder with residual carrier  $T_R$  is convex and contains exactly its  $n(R)!$  permutations. In lexicographic order they form a finite chain, so they lie in one finite-condensation block.

Conversely, any adjacent order  $S$  has a finite residual carrier at the first disagreement, by the preceding lemma. That residual is a finite terminal segment in  $R$ , hence lies in  $T_R$ . Thus an adjacent step cannot leave the displayed permutation cylinder. A finite interval is a finite chain of adjacent steps; no member outside this cylinder can be at finite distance from  $R$ . This proves the assertion.  $\square$

The cases  $n = 0$  and  $n = 1$  both give a singleton block. The maximal finite-tail length distinguishes these as labelled well-orders, but their block sizes do not distinguish them in the pure linear order.

## 58.2 Every factorial block size occurs densely

**Lemma 58.4** (An interval across different blocks contains an infinite residual cylinder). *If  $R < S$  belong to different finite-condensation blocks, there is a labelled prefix cylinder lying entirely inside  $(R, S)$  whose residual carrier is infinite. The prefix may be a proper class.*

*Proof.* Let  $C$  be the common prefix,  $Y = X \setminus C$ , and let  $a < b$  be the first unequal labels. The class  $Y$  is infinite, since a finite residual would put the two orders in one finite permutation cylinder.

Consider the following two possible changes. In  $R$ , at a position  $x \in Y \setminus \{a\}$  outside  $T_R$ , keep the whole earlier prefix and replace  $x$  by a larger unused label. In  $S$ , at a position  $x \in Y \setminus \{b\}$  outside  $T_S$ , keep the whole earlier prefix and replace  $x$  by a smaller unused label. A change of the first kind gives a prefix cylinder above  $R$  and below  $S$ , because the earlier first label  $a$  remains fixed; a change of the second kind gives the same conclusion with the first label  $b$  fixed. The unused

carrier before such a change is infinite, since the changed position lies outside the maximal finite terminal tail. Removing one label leaves the new cylinder with infinite residual carrier.

At least one such change must be possible. Otherwise let

$$Z = Y \setminus (\{a, b\} \cup T_R \cup T_S).$$

This class is infinite. In  $R$ , every  $x \in Z$  must be numerically greater than each of its  $R$ -successors in  $Z$ , since any larger unused successor would permit the first change. Thus  $R \restriction Z$  is the reverse ambient label order. In  $S$ , every  $x \in Z$  must be numerically smaller than each of its  $S$ -successors in  $Z$ , since a smaller one would permit the second change. Thus  $S \restriction Z$  is the ambient order. Both restrictions are class well-orders, contradicting the infinitude of  $Z$  as in the preceding proof. The orders in this cylinder also lie in blocks different from both endpoint blocks. For an upward change, its first disagreement with  $R$  occurs while the residual is infinite, and its first disagreement with  $S$  occurs at the original infinite residual  $Y$ ; the downward case is symmetric.  $\square$

**Lemma 58.5** (Prescribing the finite tail). *For every  $n \in \omega$ , every interval joining different finite-condensation blocks contains an order  $W$  with  $n(W) = n$ .*

*Proof.* Take the cylinder from Lemma 58.4, with prefix  $P$  and infinite residual carrier  $Y$ . Choose  $n$  distinct labels  $F \subseteq Y$ . Well-order  $Y \setminus F$  without a greatest element: if it is a set, use its infinite initial cardinal as order type; if it is a proper class, use a set-like well-order of type Ord. Both are available in GBC. Append these labels after  $P$ , and then append the  $n$  labels of  $F$  in any order. The resulting well-order belongs to the cylinder, and its maximal finite terminal tail is exactly  $F$ .  $\square$

These arguments are parameterwise constructions of individual classes. They do not assert that the collection of all orders, or its quotient, is an internal class whose elements are classes.

### 58.3 The external classification

Let  $\mathcal{M} = (M, \mathcal{K})$  be an externally countable two-sorted model of GBC. Write  $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$  and  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  for the external sets of its admitted set-like and arbitrary class well-orders of its internal surreal carrier. Comparisons are those computed in  $\mathcal{M}$ .

**Theorem 58.6** (Countable-model factorial-shuffle theorem). *If  $\mathcal{M}$  is  $\omega$ -standard, then, externally,*

$$\mathfrak{W}_{\text{all}}^{\mathcal{M}} \cong \sigma(\{1, 2, 6, 24, 120, \dots\}) = \sigma(\{n! : n \in \omega\}).$$

*Its finite-condensation quotient is a copy of  $\mathbb{Q}$ , and each factorial block size occurs densely in that quotient. In particular, the pure external order type is independent of the particular countable  $\omega$ -standard model and its admitted collection of classes.*

*For every externally countable model of GBC, even without  $\omega$ -standardness,*

$$\mathfrak{W}_{\text{sl}}^{\mathcal{M}} \cong \mathbb{Q}.$$

*Proof.* Internal linearity gives an actual external linear order on each admitted collection. The collections are countable because the entire class sort is externally countable. Internal nonemptiness and variation of the first surreal label make them infinite and give no endpoints.

In an  $\omega$ -standard model, the number  $n(R)$  of Lemma 58.1 is a standard natural number. Its  $n(R)!$ -element permutation cylinder is therefore genuinely externally finite. Proposition 58.3 identifies all external finite-condensation blocks. Its converse is also external: an external finite interval is a

standard finite chain of adjacent steps, and adjacency has its literal external meaning because the class sort ranges over precisely the admitted orders.

The quotient is countable, has no endpoints, and is dense by [Lemma 58.5](#). The same lemma makes every block-size colour dense. The countable colour-preserving back-and-forth identifies it with the densely coloured rational order used in the definition of the shuffle. Each finite block is a rigid chain, so this quotient isomorphism lifts uniquely within each block to the claimed order isomorphism.

For the set-like orders, their internal lexicographic order is dense: at the first unequal ordinal position, the common prefix is a set; choose an unused surreal strictly between the next two labels and complete the prefix to a global enumeration. The surreal cut-filling property permits avoidance of the set of used labels. There are no endpoints by varying the first label. Thus its external order is a countable dense linear order without endpoints, hence isomorphic to  $\mathbb{Q}$ . This part does not use  $\omega$ -standardness.  $\square$

*Remark 58.7* (An explicit computable presentation). For  $q \in \mathbb{Q}$ , let  $d(q) > 0$  be its denominator in lowest terms, and set  $c(q) = (1 + v_2(d(q)))!$ . Then the lexicographically ordered set

$$\{(q, i) : q \in \mathbb{Q}, 0 \leq i < c(q)\}$$

is an explicit computable copy of the factorial shuffle. For each fixed  $k \geq 0$ , reduced rationals with denominator of 2-adic valuation  $k$  are dense: choose a sufficiently large odd denominator factor and an odd numerator inside the desired interval. Cancelling common odd factors does not change that valuation. This computable presentation does not supply a computable decoder for the model's class relations.

*Remark 58.8* (The hypothesis cannot simply be dropped). If  $\mathcal{M}$  is  $\omega$ -nonstandard, choose a nonstandard  $N \in \omega^{\mathcal{M}}$  and append an internally  $N$ -element tail to a global well-order without a last element. The corresponding internal  $N!$ -element permutation cylinder is externally an infinite discrete interval with endpoints. Starting at its first element and taking the immediate successor any standard finite number of times produces infinitely many points in one external finite-condensation block. Consequently the external blocks need not be finite, and the displayed factorial-shuffle classification fails. The proof requires standardness of  $\omega$ , not external well-foundedness of every admitted class relation; no  $\beta$ -model assumption is needed.

**Corollary 58.9** (External embeddings and intrinsic automorphism orbits). *In an externally countable  $\omega$ -standard model, the unrestricted order is not externally isomorphic to its set-like suborder: it has adjacent pairs. Nevertheless it externally embeds into that suborder, since every countable linear order embeds into  $\mathbb{Q}$ .*

*The automorphism group of the unrestricted external order is the automorphism group of the rational quotient coloured by factorial block size. Its point orbits are indexed by a factorial block size and a position within that block.*

*Proof.* Only the automorphism assertion needs comment. Finite-distance blocks and their sizes are preserved by every order automorphism. Every colour-preserving quotient automorphism lifts uniquely because finite chains are rigid. Conversely, the colour-preserving back-and-forth is transitive on the quotient points of each fixed colour. Hence two points lie in the same orbit exactly when their block sizes and within-block positions agree.  $\square$

**Corollary 58.10** (Set-likeness is not definable from the external order). *In the countable  $\omega$ -standard setting, the distinguished subset  $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$  is not first-order definable, even with a finite tuple of parameters, in the pure linear order  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$ .*



*Proof.* Set-like orders have no last element and hence lie in singleton blocks. Orders of type  $\text{Ord} + 1$  also lie in singleton blocks, since their maximal finite terminal tail has size one, but they are not set-like. The preceding automorphism description can exchange any two singleton-block points.

To allow finitely many parameters, take a surreal label smaller than the first labels of all the parameter orders. There are both a set-like order and an order of type  $\text{Ord} + 1$  starting with that label. They lie below all the parameters and occupy singleton blocks. A colour-preserving quotient back-and-forth sends one to the other while fixing each parameter block, and its lift fixes the parameter points. No formula with those parameters can distinguish the two points.  $\square$

The natural inclusion of  $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$  in  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  is not dense. For example, fix a proper class prefix of type  $\text{Ord}$  leaving three labels, and vary the final three labels. The resulting six-point cylinder has a nonempty open interval containing no set-like order. More substantially, leaving an infinite set of labels gives whole intervals between different finite-condensation blocks that contain no set-like order. Thus the pure order classification does not classify the expanded pair  $(\mathfrak{W}_{\text{all}}^{\mathcal{M}}, \mathfrak{W}_{\text{sl}}^{\mathcal{M}})$ , or the external evaluation relation connecting these orders to their surreal labels. Those richer structures remain appropriate further research targets.

*Note 58.11* ([merge]; relation to sources 13 and 17 and to Part V). The second clause of [theorem 58.6](#) ( $\mathfrak{W}_{\text{sl}} \cong \mathbb{Q}$  externally in every externally countable model) is source 13’s [proposition 55.5](#), with the same proof; source 11 already observed, after [theorem 20.2](#), that in such a model the external set-like order can admit an external copy of the unrestricted order, which source 18’s [corollary 58.9](#) makes precise. [Lemmas 58.1](#) and [58.2](#) and [proposition 58.3](#) are a second proof of source 17’s [theorem 54.4](#). New in this section are the factorial-shuffle classification of the full order in countable  $\omega$ -standard models, its computable presentation, the failure without  $\omega$ -standardness, the description of the automorphism orbits, and the non-definability of set-likeness in the pure order ([corollary 58.10](#)).

## 59 Formalization plans of sources 13–18

*Note 59.1* ([merge]; status of the six plans). None of sources 13–18 ships Lean code, and no statement of theirs is formalized ([section 1.5](#)). Their plans are printed whole, in source order. Every existing declaration they cite exists at the current revision, including `SignSequence`, `SignSequence.lt_iff`, `birthdays_bounded`, `small_bounded`, `not_small`, `ordinalOrderEmbedding`, `small_cut_fillers`, `exists_cut_separator_of_birthday_lt`, `cut`, `cut_realizes`, `birthday_cut_le_iSup`, `exists_separator_of_small_sets`, `existsUnique_simplest_separator`, `SmallCutData`, `noUnrestrictedCuts`, and, cited by source 18, `le_iff_options_lt` and `lt_iff_exists_option` in `SignSequenceComparison.lean`. The modules they propose do not exist: source 14’s nine (`LabelledCommonPrefix ... ClassRealizations`), source 17’s eight files under `WellOrders/`, `ClassOrders/` and `Surreal/`, and source 16’s `WellOrderOn X`; sources 13, 15 and 18 describe layers without file names. Source 14’s remark that the existence half must be imported separately is already carried out in the tree (the merge note inside its plan).

### 59.1 Source 13: a formalization plan tied to the inspected Lean code

#### 59.1.1 What already exists at the pinned snapshot

The inspected source `Surreal/Foundations/SignSequence.lean` defines [7]:

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
```

```
signAt_ne_zero_iff :
  forall i, signAt i != 0 <-> i < birthday
```

This display is an ASCII schematic rendering of the actual declaration, not a new compiled Lean file. The source uses Lean’s native logical notation. Its numerical linear order is lifted from the lexicographic order of zero-extended sign functions.

| Inspected declaration                            | Role in this development                                                                                           |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>SignSequence.lt_iff</code>                 | First-disagreement characterization of numerical comparison.                                                       |
| <code>ordinalOrderEmbedding</code>               | The all-plus embedding of the ordinal carrier into the sign carrier.                                               |
| <code>birthdays_bounded</code>                   | A strict birthday bound for a family with an explicit lower-universe smallness instance.                           |
| <code>not_small</code>                           | The whole sign carrier is not small in the universe indexing its birthdays.                                        |
| <code>small_bounded</code>                       | Each fixed birthday-bounded fragment is small relative to that lower universe.                                     |
| <code>exists_cut_separator_of_birthday_lt</code> | A separator of exactly a prescribed birthday above all option birthdays; the fresh separator lemma’s direct input. |
| <code>small_cut_fillers</code>                   | The constructed carrier realizes every small strict cut.                                                           |

The last two are in `SignSequenceCut.lean` [8]. They are not merely postulated field axioms: the inspected source constructs them from fixed-length Boolean lexicographic words. We inspected the files, not a fresh build of the project. No theorem in this manuscript is thereby automatically a kernel-checked theorem.

### 59.1.2 Relative smallness must stay visible

`SignSequence.{u}` and `Ordinal.{u}` both live in `Type (u + 1)`, while bounded sign sequences and initial segments are small relative to `u`. A `Set SignSequence` is merely a predicate on the larger carrier. It need not be an admissible small family of options. The formal counterpart of “set cut” therefore requires an index type with `Small.{u}`, not just the word `Set` in a declaration.

The same applies to supports and prefix lengths. A support that is a predicate on a larger type is not automatically a small support. A code should carry a small graph or an explicit smallness proof. Otherwise the completion bound, the dense-core distinction, and the diagonal interpretation can silently change meaning.

A universe-relative approach may take a baseline equivalence

$$b : \text{Ordinal}.\{u\} \simeq \text{SignSequence}.\{u\}$$

as an explicit parameter. Its existence is not inferred merely because the types occupy the same universe. A formal project must either construct the required equivalence, with the intended small-initial- segment properties, or retain it as a stated assumption.

### 59.1.3 Suggested modules and their proof obligations

The following names are proposals, not claims about existing files.

**Exhaustive orders and finite predicates.** A first module should define exhaustive ordinal enumerations on a small carrier, prove prefix-freeness, and compare the relation-based and enumeration-based first-disagreement definitions. These proofs do not require surreal arithmetic or class theory. The finite checker included here provides examples, not a substitute for this module.

**Prefix codes and completion.** Represent a moved-point graph canonically and prove evaluation injective on codes. Prove the complement-cardinality extension lemma before using any prefix-completion operation. Formulate cylinders by restriction equality and prove convexity and same-level disjointness.

**Small-family interpolation.** Use an explicit small index type for the two families. Establish a bound on cross first-disagreement coordinates, then implement the active-row invariant with a set-sized history. The stopping lemma, not a generic unbounded recursion tactic, is the critical finite-length formal obligation at this stage.

**Cut traces and failure.** Define crossing by two existential code witnesses. Same-level uniqueness and restriction coherence are purely order-theoretic. Under all-level existence, reconstruct the value at  $\alpha$  from the unique prefix of length  $\alpha + 1$ . Prove trace equality with the no-greatest-lower normalization stated explicitly. For first failure, split successor from limit ordinals and prove the boundary formula rather than assuming that crossing commutes with limit intersection.

**Products and recoding.** First formalize [Lemma 55.2](#) for ordinary set types and an order-dense subset. It has no class-theoretic ambiguity and should be a small, reusable result. The uniform complement-map interface then supports prefix insertion and finite-block automorphisms. The KM inverse evaluator belongs in a model or syntax layer, not in an unqualified theorem about arbitrary Lean functions on a larger type.

**Central words.** Represent termination by an option type or by a reserved symbol with an explicit proof that it is not a letter. For injective words, retain injectivity as an invariant when selecting the shared next letter. The explicit sign-block embedding can be formalized independently of the class back-and-forth isomorphism.

#### 59.1.4 Three complementary formalization routes

The *cutoff-first route* proves the fixed-set and density-bound lemmas inside ordinary types. It requires the least foundational infrastructure and makes cardinal hypotheses explicit.

The *universe-relative route* abstracts a large ordered carrier with a lower-universe small-cut operation, a set-like enumeration, and support codes. It can certify the local combinatorics and many of the new constructions relative to those interfaces. It should not be described as a proof of every GBC or KM metatheorem merely because it uses two universes.

The *deep model route* formalizes a two-sorted class theory or a specified model  $(M, \mathcal{C})$ . This is the route for the exact comprehension schema, the external countable contrast, and the full-model interpretation. Taking all predicates on an arbitrary carrier as “classes” does not by itself establish the intended replacement or universe axioms; the model assumptions must be proved or supplied.

### 59.1.5 Verification boundary

A completed formalization should pin both the repository revision and its dependency revisions, compile the new modules, and audit theorem axioms and unfinished proofs. None of those actions is claimed here. The deliverable is a proof manuscript, a reproducible LaTeX build, and finite consistency checks. Existing Lean inputs are identified by actual declarations and source hashes in the audit appendix, so that a future formalization has concrete starting points.

## 59.2 Source 14: a realistic formalization plan

The new proofs admit a modular formalization in which most of the work is ordinary order theory. A full class-theoretic metamodel should not be the first dependency. The finite checks shipped with this article are useful regression tests, but do not discharge any infinite or class-level theorem.

### 59.2.1 A generic set-level kernel

Start with a type  $X$  carrying a linear order and a chosen ordinal or well-ordered index type. An exhaustive presentation is a bijection from that index type to  $X$ . For arbitrary well-order types on one fixed carrier, relations are often easier than a dependent sum over every ordinal: a relation together with linearity and well-foundedness proofs is already a set-sized object at the carrier's type level.

Prove the labelled common-prefix comparison before importing surreal arithmetic. Its main obligations are initiality, agreement of the restricted relations, existence and inequality of the two next labels, and transitivity via nested initial segments. For fixed-domain words, prove the equivalent least-disagreement formulation. Finite relation matrices and permutation words give independently checkable examples.

A second module handles partial-injection extension. Its specification must retain *agreement on a prescribed prefix* and the *support cardinality bound*; bare existence of an extension is not enough for the compression theorem. The finite case and the infinite buffer case should be separate lemmas, since equality of infinite cardinalities of domain and range alone does not imply equality of their complement cardinalities.

### 59.2.2 An abstract saturated alphabet interface

The core theory needs an ordered alphabet with set-cut filling and fresh separators, a global baseline in the class version, and set-prefix completion. Numerical surreal field operations are not required. A bounded preliminary development may use a regular-cardinal saturated set order and prefix lengths below that cardinal, with all relevant smallness hypotheses explicit. It proves the crossing-chain and bounded versions of the trichotomy without formalizing surreal addition.

For actual sign sequences, the pinned repository already supplies simplicity lemmas. The inspected declaration `existsUnique_simplest_separator` requires a proof that a separator exists. The appropriate architecture imports an independent small-cut existence theorem and then applies this uniqueness result; it does not erase the hypothesis to manufacture saturation. [merge] The pinned tree already has this architecture: `small_cut_fillers` in `SignSequenceCut.lean` proves that small cuts have fillers, and `cut` in `SignSequenceCutOperation.lean` is defined from `existsUnique_simplest_separator` applied to it (section 1.5; the same remark in source 14's audit section, section 40).

### 59.2.3 Proposed modules and theorem dependencies

| Proposed module      | Principal obligations                                                                          |
|----------------------|------------------------------------------------------------------------------------------------|
| LabelledCommonPrefix | Initiality, next labels, lexical linearity; comparison with ordinal enumerations.              |
| SupportExtension     | Finite extension; infinite buffer; preservation of fixed positions; cardinal bounds.           |
| CoreCylinders        | Canonical support codes, prefix completion, convex cylinders, ordered children.                |
| CrossingPrefixes     | Elementary crossing predicate; chain and uniqueness by length; union graph.                    |
| CutTrichotomy        | Unbounded branch; bounded limit edge; terminal numerical gap; uniqueness.                      |
| ExhaustiveTrace      | Coherence and coverage; graph reconstruction; trace injectivity.                               |
| SupportSaturation    | Support compression; union bound at cf $\kappa$ ; sharp forced-prefix configuration.           |
| RawTraceFibers       | Canonical set-like initial branch; tail concatenation; equality and convexity of trace fibers. |
| ClassRealizations    | Same-set structures and elementary absoluteness with old parameters.                           |

These are proposed modules, not files claimed to exist or compile. A dependency path is

completion  $\longrightarrow$  cylinders  $\longrightarrow$  crossing chain  $\longrightarrow$  trichotomy  $\longrightarrow$  recognition.

Support compression can instead import the predecessor's verified-in-writing core interpolation theorem while that path is being formalized. The character theorem and support threshold then provide two largely independent audits of the prefix machinery.

### 59.2.4 Class-level interfaces and avoiding a hidden third sort

A formal metatheory of GBC can express cuts and enumerations as class variables and use elementary comprehension explicitly. An alternative is to formalize the external inaccessible model, where all relevant collections are sets in a larger type universe. These routes have different assumptions and should not be conflated.

The central reconstruction specification can be written schematically as

```

Crosses(D, p) :=
  exists a b, InCylinder(a,p) and InCylinder(b,p)
    and D(a) and not D(b)

RecoveredGraph(D, alpha, x) :=
  exists p, Crosses(D,p) and alpha in domain(p) and p(alpha)=x

Coherent(D) := forall alpha, exists p,
  domain(p)=alpha and Crosses(D,p)

Covers(D) := forall x, exists p, Crosses(D,p) and x in range(p)

```

This is logical pseudocode, not Lean syntax and not compiled code. Its purpose is to display the quantifiers:  $a, b, p, \alpha, x$  are set objects. No line chooses a class as an element of a class family. The proof of functionality precedes any use of the recovered graph as a function, and the coverage theorem is a separate proof of surjectivity.

### 59.2.5 What must be checked before a kernel-verification claim

A completed formalization must pin its Lean and Mathlib revisions, compile all files, expose any axioms beyond the chosen foundations, and connect the generic ordered alphabet to the actual surreal carrier. It must also verify that internal ordinals, universe lifts, and prefix domains match the statements. None of those steps has been completed for the new theorems in this package; the package contains conventional written proofs and executable finite diagnostics only.

## 59.3 Source 15: a concrete Lean architecture

The inspected ProveIt declaration is

$$\text{SignSequence.}\{u\} : \text{Type } (u+1).$$

Its birthday belongs to  $\text{Ordinal.}\{u\}$ . The file provides `ordinalOrderEmbedding`, `birthdays_bounded`, `small_bounded`, `not_small`, and a well-founded simplicity relation, among other declarations [7]. The essential statement is relative smallness: the entire sign carrier is not  $u$ -small, even though it is a legitimate type one universe higher. The official Lean universe rules support this distinction [29]. This article has inspected those declarations but has not rebuilt the repository or certified any new theorem in Lean.

A practical development should use the following dependencies.

1. **Generic set layer.** Define exhaustive ordinal enumerations of an arbitrary linearly ordered carrier; prove prefix-freeness, first-difference comparison, pair orientation, and the successor-cardinal obstruction. Surreal arithmetic is unnecessary.
2. **Bounded surreal layer.** Reuse sign order and birthdays. Formalize the 00/11 code with terminal 01, ordinal concatenation, the max-coordinate scheduler, and the three coding-length cases. Track ordinal products explicitly instead of replacing them by cardinalities.
3. **Relative-smallness layer.** For a relation on a higher-universe carrier, express the assertion that an interval or predecessor type is  $u$ -small. Implement the least-representative condensation and its rank map. This models the chosen set/class boundary without a type of all types.
4. **Prefix and cut layer.** Define one uniform evaluation relation on small support codes. State crossing using quantification over small prefixes and codes, and reconstruct a persistent branch. Quantify over the cut as a parameter at the permitted universe level.
5. **Model layer.** To claim a theorem about GB, GBC, or KM itself, formalize a model or an axiomatic interface for that theory. Universe polymorphism alone is not a proof that a chosen type-theoretic encoding satisfies precisely those set-theoretic axioms.

Each layer should record its universe parameters, smallness assumptions, choice use, and available recursion principle. A relation on the entire higher-universe sign type with ordinary Lean well-foundedness may encode external well-foundedness stronger than a Henkin model's internal class condition. These should be separate definitions, linked only by proved translation theorems.



## 59.4 Source 16: a concrete formalization plan tied to ProveIt

### 59.4.1 Inspected primitives and actual status

The following declarations were inspected at the pinned repository revision. They are existing foundations, not formal proofs of the new results of this article. The earlier well-order report explicitly states that no theorem with its `swo:` labels has a Lean mapping.

| Existing file                                               | Relevant declarations or interface                                                                                                             | Reuse here                                                                                         |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <code>SignSequence.lean</code>                              | <code>SignSequence</code> , <code>birthday</code> ,<br><code>birthdays_bounded</code> , <code>small_bounded</code> ,<br><code>not_small</code> | Carrier, numerical order,<br>bounded birthday stages,<br>and properness relative to a<br>universe. |
| <code>SignSequenceCut.lean</code>                           | <code>small_cut_fillers</code> , <code>exists_cut_sep</code><br><code>arator_of_birthday_lt</code>                                             | Small-cut realization and<br>prescribed upper birthday<br>control.                                 |
| <code>SignSequenceCutOperati</code><br><code>on.lean</code> | <code>cut</code> , <code>cut_realizes</code> ,<br><code>birthday_cut_le_iSup</code> ,<br><code>exists_separator_of_small_sets</code>           | Canonical separators with<br>explicit smallness<br>assumptions.                                    |
| <code>SignSequenceSimplicity</code><br><code>.lean</code>   | <code>existsUnique_simplest_separator</code><br>and minimum-birthday results                                                                   | Deterministic choices and<br>control of sign-prefix cuts.                                          |
| <code>SizeObstructions.lean</code>                          | <code>SmallCutData</code> , <code>noUnrestrictedCuts</code>                                                                                    | Keep small inputs distinct<br>from an unrestricted<br>self-cut.                                    |

All paths in the table lie below `Algebra/SurrealNumbers/Surreal/Foundations/`; the pinned sources are recorded in the bibliography [7, 8, 9, 31]. No Lean build was performed for this article, and no proposed module name below should be read as an existing theorem declaration.

### 59.4.2 Two formalization routes

The most economical first route is ordinary type-theoretic order theory. Use a bounded birthday carrier known to be small, transport it with `Shrink` if convenient, and prove the set-alphabet theorems. Store the numerical label order separately from the well-order being encoded. A proposed `WellOrderOn X` can store the binary relation, strict linearity, and `WellFounded` evidence; an enumeration stores an ordinal domain with a bijection to  $X$ . Prove the relation/enumeration correspondence, prefix-incomparability, and first-disagreement comparison before taking any quotient. Quotienting first by abstract isomorphism would destroy the required labels.

The second route explicitly formalizes a model of class theory. Introduce separate carriers of set and admitted-class codes, membership, formula semantics, and internally admitted family relations. State elementary comprehension, replacement, and any GC or ETR assumptions at that level. This route is appropriate for exact axiom-strength claims and Henkin models. It is heavier than proving the bounded combinatorial theorems, but it prevents an accidental use of all ambient predicates as internal classes.

### 59.4.3 Universe levels and logical levels are different

ProveIt's actual carrier is

$$\text{SignSequence.}\{u\} : \text{Type } (u + 1), \quad \text{birthday} : \text{Ordinal.}\{u\}.$$

Its small-cut interfaces explicitly require `Small.{u}` for the index families. A bounded birthday subtype is small at that level, while the full sign carrier is not.

One must not say that every power set automatically forces the next syntactic Lean universe. If  $S : \text{Type } (u + 1)$ , then both `Set S` and `S → Prop` also inhabit `Type (u + 1)`. Being well-typed does not make them  $u$ -small, nor does it identify them with the class collection of an internal GBC model. Native quantification over all such predicates is an ambient full-type interpretation. Likewise, native `WellFounded R` asserts ambient well-foundedness, which is not automatically a model's internal assertion [29, 38, 30].

For a direct universe-relative global development, keep a baseline  $b : \text{Ord}^u \simeq \mathbf{No}^u$  as an explicit parameter until its existence is separately established. Encode supported permutations by genuinely small data, and represent families of their evaluated graphs uniformly. This gives meaningful theorems without pretending that every ambient type is an internal set or every function on class predicates is a same-level class function.

#### 59.4.4 Suggested milestones

1. **Finite and set-alphabet layer.** Formalize first disagreement, prefix convexity, pair coding, and the stabilization bound. This isolates the ordinal-spectrum theorem from all surreal-specific facts.
2. **Topological layer.** Formalize the no-cross-fiber-jump lemma, prefix families meeting every open set, and disjoint finite-tail cylinders. Then prove the countable incompleteness construction and the regular-stage gap spectrum.
3. **Small-support layer.** Formalize normalization, completion of a small prefix, and the bounded common-prefix interpolation search. Reuse the existing small-cut separator rather than postulating a new unrestricted cut operation.
4. **Cut recognition.** Define straddling on set codes, prove uniqueness and coherence at each length, and implement the direct formula (S16.14). Only then package the exhaustive branch criterion and first-failure classification.
5. **Class semantics layer.** In an explicit model interface, formalize set-interval equivalence, least representatives, local rank, and the terminal set. These are elementary constructions; a general class-recursion engine is unnecessary for them.
6. **Higher comprehension.** State the inverse evaluator with its exact syntactic complexity before invoking a comprehension axiom. Prove the elementary diagonal theorem independently, and use it as the final contradiction in the nonembedding theorem.

These milestones deliberately separate ordinary mathematics about a large type from metatheorems about a two-sorted model. No new consistency axiom is needed for the bounded results, and the inaccessible model is an optional explicit interpretation rather than a prerequisite for the whole subject.

### 59.5 Source 17: formalization routes and a practical verification plan

#### 59.5.1 Start with the alphabet interface, not surreal arithmetic

The new set-sized results depend on a small interface. A formal development can begin with a linear order  $X$  of infinite cardinality  $\mu$ , with  $\mu$  elements in every nonempty open interval and ray. It should record its ray and point character  $r$  and the gap spectra of deletions of fewer than  $\mu$  elements. Prefix completion, clopen cylinders, divergence bounds, the cut trichotomy, and fusion can then

be proved abstractly. The birthday sign alphabet is one implementation of this interface. This organization prevents arithmetic lemmas about surreal addition or multiplication from obscuring a proof that uses only order and signs.

The exact gap theorem has four independently testable mathematical interfaces: a gap is a two-sided partition without endpoints; a straddling prefix is a set witness; the first stopping stage is a successor, a proper limit, or full length; and each resulting character is computed separately. Separating them makes it difficult to accidentally substitute point character  $q$  for cylinder character  $r$ .

The support theorem is another independent target. Its statement needs only a set partial injection, the necessarily moved domain and range, cardinal addition, and a complement bijection. No surreal construction appears until the final composition with a baseline.

### 59.5.2 Proposed modules and proof obligations

The following names describe a proposed architecture, not files claimed to exist in the repository or in this package.

| Proposed module                                 | Principal obligation                                                                   |
|-------------------------------------------------|----------------------------------------------------------------------------------------|
| <code>WellOrders/CommonPrefix.lean</code>       | Elementary predecessor formula, maximality, first residual labels, transitivity        |
| <code>WellOrders/PartialSupport.lean</code>     | Mandatory support, deficiency balance, exact reserve cardinal and extension            |
| <code>WellOrders/PrefixTopology.lean</code>     | Completion, convex clopen cylinders, point versus cylinder character, base cardinality |
| <code>WellOrders/CutTrichotomy.lean</code>      | Unique straddling path, first failure, limit boundary, nonexhaustive branch            |
| <code>Surreal/CutoffGapSpectrum.lean</code>     | Sign-path classification of alphabet gaps and stability under small deletion           |
| <code>WellOrders/Fusion.lean</code>             | Dense-open refinement with explicit coverage of the alphabet                           |
| <code>ClassOrders/BranchRecognition.lean</code> | A chosen model or size interface; elementary reconstruction from unique set witnesses  |
| <code>ClassOrders/CoreAutomorphisms.lean</code> | Represented-cut preservation, uniqueness of extension, nonextension witness            |

For the topology modules, one should formalize the definition of cellularity as a supremum before claiming an exact cardinal value. A supremum of disjoint-family sizes need not be attained. Likewise the gap spectrum should not include point cuts of character 1, and the recognition module should not quietly switch between open and closed principal lower cuts.

### 59.5.3 Universe-relative type theory

Lean’s universes offer a practical way to keep a fixed smaller universe separate from a larger carrier. A type, a type of relations on it, and a universe of all smaller types must have their levels checked explicitly. Function types do not automatically jump one level merely because they represent a mathematically large collection: their levels are computed from the domain and codomain. Universe polymorphism supplies a family of declarations, not an internal type of all universes; and universe lifting is explicit rather than an automatic cumulative coercion [29].

For this project, choose a lower notion of “small” and an ordinal index which well-orders the intended large carrier with small proper initial segments. State the cut operation only for lower-small families. A back-and-forth argument along an arbitrary choice well-order of the large type is unsafe:

its proper initial segments may be too large for the assumed cut operation. The correct interface is the small-history property used in the proof of [Corollary 42.14](#).

A set-sized cutoff implementation avoids this difficulty entirely. Every object is an ordinary type with explicit cardinal bounds, and a formal proof can be instantiated at many cardinals. A universe-relative version then transports the argument, without equating an ordinal type from one universe with all ordinals from every universe.

#### 59.5.4 A deep model-theoretic alternative

To study exactly which comprehension principles suffice, formalize a specified two-sorted model  $(M, \mathcal{K})$  and its elementary comprehension and replacement axioms. Each theorem about global rows is then a theorem about admissible class relations of that model. This is the most direct route for [Theorem 48.4](#) and for questions comparing GBC, ETR, and KM.

The two routes should not be conflated. A successful theorem about types in a fixed universe does not, without an interpretation theorem, prove a conservativity or axiomatic-strength statement about class theory. Conversely, a GBC proof with explicit class variables is not automatically a compiled theorem about the repository’s current sign-sequence carrier.

#### 59.5.5 Verification boundary

No Lean code accompanies this article and no kernel-verification claim is made. The source’s foundation modules were not rebuilt. A future formalization should pin both the repository and Mathlib revisions, compile the modules, reject unfinished proofs, and audit theorem axiom dependencies. The companion Python checks test only finite mechanisms. They cannot decide the existence of class maps, prove cardinal cofinality formulas, or certify the infinite support obstruction.

### 59.6 Source 18: a concrete formalization plan for ProveIt

#### 59.6.1 What the inspected source actually provides

The source audit used the pinned revision `83befe707f2840c2b53e0606701d8a2b28598e47`. The relevant files are in `Algebra/SurrealNumbers/Surreal/Foundations/`. The following declarations were inspected directly in the source files [\[32\]](#), rather than inferred from a report’s title.

| Source file                              | Useful declarations and their role                                                                                                                                                                                                     |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>SignSequence.lean</code>           | <code>SignSequence : Type (u + 1)</code> , numerical linear order, <code>lt_iff</code> , <code>birthdays_bounded</code> , <code>small_bounded</code> , <code>not_small</code> , prefixes, truncations, and the ordinal order embedding |
| <code>SignSequenceCut.lean</code>        | <code>exists_cut_separator_of_birthday_lt</code> and <code>small_cut_fillers</code> ; the bounded and small-family separator interface                                                                                                 |
| <code>SignSequenceSimplicity.lean</code> | <code>existsUnique_simplest_separator</code> and minimum-birthday uniqueness; deterministic choice of separators where appropriate                                                                                                     |
| <code>SignSequenceComparison.lean</code> | <code>le_iff_options_lt</code> , <code>lt_iff_exists_option</code> , and canonical-cut comparison                                                                                                                                      |

The declaration `SignSequence.{u}` living in `Type (u+1)` is crucial. It contains the surreals with  $u$ -small birthdays; the theorem `not_small` says that this whole type is not itself  $u$ -small. The existence of a Lean type is therefore not the same as being a set at the intended smaller size level. Lean’s universe hierarchy enforces these distinctions [\[29\]](#). A Tarski–Grothendieck implementation

is another well-developed option; Pāk and Kaliszyk’s Mizar formalization illustrates a concrete universe-based route [37].

This article does not assert that the new spectrum, condensation, or shuffle theorems are already formalized in these files. No Lean build or complete imported-axiom audit was performed for this article. Reading a proof file is not equivalent to running its elaborator and kernel in the pinned environment.

### 59.6.2 The first set-level modules

The most reusable initial target is independent of surreal arithmetic. Represent a labelled well-order by a relation on a fixed carrier and its well-order proof, or by a bijection from an ordinal domain. Prove the equivalence of these representations and the absence of proper prefixes for exhaustive enumerations. Keep “increasing by numerical order” as an extra predicate on words, not as part of the definition of a well-order code.

The suggested sequence is:

1. **Ordinal words.** First disagreement, concatenation, prefix cones, alternative terminators, and the finite permutation adjacency criterion.
2. **Set spectra.** Pair coding, eventual coordinate stabilization, and the  $\mu^+$  obstruction. The central induction is on a regular successor cardinal, with the domain and range cardinalities stated separately.
3. **Permutation topology.** Order topology, dense sets, disjoint interval families,  $Q_n(X)$ , and the density/cellularity transition. Include global endpoints in dense-set constructions for arbitrary alphabets.
4. **Surreal cutoff interface.** Large sign cones, preservation of cut characters after deleting fewer than  $\mu$  labels, and  $\text{cf}(2^{<\kappa}) \geq \text{cf } \kappa$ .
5. **Cut classification.** Encode the chain of straddling prefixes and prove its three cases. The multiplicity theorem then uses explicit injections and the cardinalities of the prefix tree and full branch set.

This order gives a checkable sequence of finite dependency boundaries. The proof of the gap spectrum depends only on order, ordinal words, cardinal arithmetic, and the cutoff interface. It does not require surreal addition, multiplication, or real closedness.

### 59.6.3 The class-level modules

There are two sensible implementations.

*Semantic set/class models.* Define a first-order set universe and a collection of admitted class predicates, with the GB axioms made explicit. A class well-order is a relation in that collection with the internal well-order property. Then prove the elementary common-prefix theorem, set-interval decomposition, uniform branch criterion, and factorial-block facts as statements about that model. An external countability hypothesis subsequently gives the shuffle classification. This route preserves the exact meaning of internal set size and available classes.

*Universe-relative carriers.* Fix the smallness level  $u$ , work with a larger carrier type, and formulate set-likeness using  $u$ -small predecessor types. Relations on that carrier live in an appropriate larger universe. This is convenient for a concrete containing-universe development, but theorems must say which subsets are admitted and which smallness level “set” means. Merely forming the type of all relations in the next universe does not identify it with a particular GBC class realization.

In either route, give quotient blocks actual representatives. For set-interval condensation use the least label of a block; for the external finite-condensation quotient use the usual set quotient in the ambient universe. Their types and their purposes differ.

#### 59.6.4 Suggested proof obligations and trust reporting

For a mechanized milestone, require the following explicit outputs: the precise theorem statement with universe levels; a compiling source file against the pinned toolchain; an imported-axiom report; and a short explanation of the semantic correspondence with the article. Finite examples can test a comparator implementation, but they do not verify the  $\mu^+$  argument, Global Choice equivalence, or any claim about proper classes. The current deliverable supplies conventional proofs and source provenance, rather than presenting finite computations as certificates for infinitary assertions.

## 60 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 33](#); where the question index of [section 113](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 11's research question 113.3 and source 12's research question 113.14, iterated residual decomposition and proper-class blocks*: the block-index clause is **answered**. Every class well-order decomposes at once, elementarily in GB, into proper blocks of type Ord indexed by a class well-order, followed by at most one final set block; every nonempty class well-order occurs as the index, with any final set ordinal ([note 52.1](#)); explicit condensation histories of every class length exist ([theorem 53.4](#)). In particular all the subpredicates  $\text{Ord} \cdot \gamma + \alpha$  of [research question 113.14](#) are nonempty and are recognized by their block profile. Open: the strength needed to iterate the condensation of an arbitrary input uniformly ([research question 113.48](#), [research question 113.64](#), [research question 113.86](#)), and, for [research question 113.14](#), the embedding and nonembedding theorems between those subpredicates and the set-like order, of which only the full-model separation  $\mathcal{W}_{\kappa+2} \not\leftrightarrow \mathcal{W}_{\kappa}$  ([theorem 57.3](#)) is proved.
- (b) *Source 11's research question 113.2, weakest comprehension for the uniform nonembedding*: answered in part. For transformations defined by formulas with only set quantifiers, GBC with  $\Sigma_1^1$  class comprehension suffices ([theorem 56.2](#)); whether it is necessary, and whether GBC alone suffices, is open ([research question 113.63](#)). The question's bounded variant, comparing density and cellularity of all well-orders of  $\mathbf{No}_{<\kappa}$  with the minimal stratum, is answered by [theorem 27.2](#) (Part VI).
- (c) *Source 08's question 7, full virtual lexicographic products* ([section 113.4](#)): **answered** for the interpretations it names. In GBC,  $\mathcal{W}_{\text{sl}} \cong \mathbf{No} \times_{\text{lex}} \mathcal{W}_{\text{sl}}$  and every ordinary class-indexed lexicographic sum of copies of  $\mathcal{W}_{\text{sl}}$  embeds into  $\mathcal{W}_{\text{sl}}$  ([theorem 55.1](#)); in the full inaccessible universe  $\mathcal{W}_{\text{sl}} \times_{\text{lex}} 2 \not\leftrightarrow \mathcal{W}_{\text{sl}}$ , hence  $\mathcal{W}_{\text{sl}} \times_{\text{lex}} \mathcal{W}_{\text{sl}} \not\leftrightarrow \mathcal{W}_{\text{sl}}$  ([theorem 55.3](#)); no fixed definable uniform embedding exists in  $\text{KM} + \text{GC}$  ([theorem 55.4](#)); in externally countable models  $\mathcal{W}_{\text{sl}} \times_{\text{lex}} \mathcal{W}_{\text{sl}} \cong \mathcal{W}_{\text{sl}}$  externally ([proposition 55.5](#)). The negative half already followed from the proofs of [theorems 20.1](#) and [20.2](#) ([note 55.6](#)). A finer absorption spectrum (source 13's [research question 113.25](#)) is open.
- (d) *Source 12's research question 113.17, automorphisms and finite condensation*: answered in part. The maximal finite components of the unrestricted order are the  $n!$  permutations of the maximal finite final segment, and the condensed order is dense without endpoints ([theorem 54.4](#), [proposition 58.3](#)); in every countable  $\omega$ -standard model the full order is externally the factorial



shuffle and its automorphism orbits are known ([theorem 58.6](#), [corollary 58.9](#)). The internal classification of the condensed order (source 17’s [research question 113.77](#)) is open.

- (e) *Source 11’s [research question 113.8](#), source 12’s [research question 113.18](#) and source 08’s question 12, class realizations*: answered further. In addition to [section 50](#) (e): the pure external order of all admitted class well-orders is the same factorial shuffle in every countable  $\omega$ -standard model, so it records nothing about the class realization there, and set-likeness is not definable in it even with parameters ([theorem 58.6](#) and [corollary 58.10](#)); among GBC realizations with the same sets, internal well-foundedness of a shared relation never changes ([note 57.5](#)). The expanded structures with the set-like predicate or the label evaluation are open ([research question 113.88](#)).
- (f) *Source 11’s [research question 113.4](#), recovering prefix structure from the bare order*: a further negative part. Set-likeness is not first-order definable, even with finitely many parameters, from the pure order of all admitted class well-orders of a countable  $\omega$ -standard model ([corollary 58.10](#)); within one class well-order it is recognized by the number of blocks (source 15:  $W$  is set-like exactly when it has one block). With [section 50](#) (c), the question’s negative half is settled; the positive question, which extra predicates suffice, stays open.
- (g) *Comprehension questions* (source 12’s [research question 113.16](#), source 09’s direction 2, source 08’s question 10): bear. The direct comparison, the block decomposition, the represented-cut reconstruction and the uniform removal of blocks all have elementary definitions in GB or GBC (sources 15, 16 and 18); no reversal is proved.
- (h) *Source 11’s [research question 113.10](#), source 09’s direction 9 and source 08’s question 15 (Kanovei–Shelah)*: not affected; sources 14 and 17 restate the size boundary ([section 4](#)).

The questions of sources 13–18 that a sibling source answers are marked in [section 113](#).

[merge] Dated note, 3 October 2026 (batch 83). Items (a), (d), (e) and (f) are answered further by Parts XIII and XIV: over GBC, canonical terminating condensation histories exist exactly when class well-orders are strongly comparable (source 29), the countable class models and the common theory of their external orders are described, and the extra predicates that recognize exhaustive branches are identified ([sections 105 and 112](#); [section 113.25](#) (d), (e), (g), (h)).

## Part X

# Set-sized strata, topology and permutation groups

[merge] (3 October 2026.) The first Part of the batch-83 addition, sources 19–30 (provenance in [section 1.9](#), notation in [section 1.3.2](#)). Part VI ([part VI](#)) determined the density, cellularity, point characters and gap spectra of the set-sized strata and the coding lengths of the types  $\mu \cdot \gamma + n$ ; it left open the topology of these spaces beyond their cardinal invariants, the strata with an infinite remainder below  $\mu$ , and the isomorphism type and homogeneity of the minimal slice ([section 33](#), items (b) and (e)). Nine batch-83 sources continue exactly there. This Part prints what they prove about set-sized enumeration orders as ordered and topological spaces, about the permutation groups acting on them, and about the class analogue of prefix completion:

- thick alphabets, prefix cylinders and the prefix topology, with the sources’ re-proofs of Parts I and VI as notes ([section 62](#));

- residual cylinders, the weight and the point characters of every fixed-length stratum (source 22, [section 63](#));
- the exact binary-cube spectrum of every fixed enumeration type, and its coding sandwich (source 19, with source 30's independent route, [section 64](#));
- three completions of the minimal slice: forward prefix completion (source 24), the closure of the minimal stratum in the full order (source 21) and normal forms for the Dedekind completion (source 19) ([section 65](#));
- a generalized Baire model of the minimal slice at every regular size (source 24, [section 66](#));
- the countable cutoff, printed once from source 24, with the other six proofs and source 21's homogeneity theorem ([section 67](#));
- the countable strata of types  $\omega + n$  (sources 29 and 30, [section 68](#));
- singular cardinalities: continuity of the permutation operations, the exact complete-metrizability classification, non-Borel boundaries and closed universality (sources 24 and 26, [section 69](#));
- maximality, self-normalization and the canonical group envelope of the boundedness-preserving permutation group (source 28, [section 70](#));
- the class analogue: global prefix completion, small-net rigidity and inverse stabilization (source 24, with sources 26 and 28, [section 71](#)).

[Section 72](#) records which questions of this report, of its batch-81 sources and of the batch-83 sources the Part answers.

The bases are chosen by generality and by what each source alone proves, as in Parts VI–VIII: source 24 for the prefix topology and the three completions (every infinite alphabet), source 26 for complete metrizable and Borel complexity (every infinite cardinal), source 28 for the permutation groups (every admissible coordinate ideal), source 19 for the infinite residual tails (the only source that determines the cube spectrum in every case), source 22 for weights and characters, and source 29 for the countable strata (it alone compares the order topology with the coordinate topology at every countable length). Source 26 answers source 24's complete-metrizability question; source 28 read source 24 but not source 26, and its own leading question is the one source 26 answers ([note 70.13](#)). Sources 29 and 30 were written without knowledge of each other and prove the same countable-strata theorems; the countable-cutoff theorem is proved seven times in the batch ([note 67.1](#)). Source 30's residual-code sections reprove source 19's [section 64](#) independently.

*Conventions.* Throughout,  $\kappa$  is an infinite cardinal used as a birthday cutoff,  $\mu = 2^{<\kappa} = |\mathbf{No}_{<\kappa}|$ ,  $r = \text{cf } \kappa$  and  $q = \text{cf } \mu$ , as in [section 1.3.1](#); a general alphabet  $X$  is an infinite linearly ordered set of cardinality  $\mu$ , identified with its initial ordinal when it is used as a domain. The following renamings are made in this Part and listed in the batch-83 notation addendum ([section 1.3.2](#)).

- The minimal slice  $\text{Bij}(\mu, X)$  is  $\mathcal{W}_\mu(X)$ . Sources 24, 26 and 28 write  $\mathcal{P}_\mu(X)$ ; *false reading*: this  $\mathcal{P}$  is neither the core  $\mathcal{P}_{\text{bd}}$ ,  $\mathcal{P}_s$  nor source 11's  $\mathcal{P}_\lambda$ , which was already renamed  $\mathcal{W}_\lambda$  ([section 1.3](#)). Where these sources write  $\mathcal{P}$  for the slice itself it is printed  $P$ , as in Part VI.
- $\mathcal{I}_\mu(X) = \text{Inj}(\mu, X)$ , the injective words of length exactly  $\mu$ , is the length- $\mu$  layer of this report's  $\mathcal{I}(X)$ ;  $\mathcal{I}_{<\mu}(X)$  is the set of injective words of length below  $\mu$  (source 24's  $\mathcal{T}_\mu(X)$ , renamed because  $\mathcal{T} = \mathcal{T}$  is the class of all words). Sources 24, 26, 28 and 21 write  $\mathcal{I}_\mu(X)$ , which agrees.
- Source 26 writes  $\kappa$  for  $|X|$  and  $\theta$  for the birthday cutoff; they are printed  $\mu$  and  $\kappa$ . Its  $\mathcal{B}^+$  and source 28's  $\mathcal{I}_\mu^b$  (the boundedness-preserving injections) are one object, printed  $\mathcal{I}_\mu^b$ .
- Source 28's bounded ideal  $\mathcal{B}_\mu$  is printed  $\mathcal{J}_b$  (its  $\mathcal{B}$  would read as this report's cubes  $\mathbb{B}$ ), and

its one-sided loci  $B_\mu^+$ ,  $B_\mu^-$  are printed  $H_\mu^+$ ,  $H_\mu^-$ , so that  $H_\mu = H_\mu^+ \cap H_\mu^-$ ; its admissible ideals  $\mathcal{J}$  keep their letter (*false reading*:  $\mathcal{J}$  is not  $\mathfrak{J} = \mathfrak{J}$ ). The cofinality cf  $\mu$ , written  $\delta$  by source 28 and  $r$  in one proof of source 24, is printed  $q$ .

- The cores of sources 24, 26 and 28 ( $\mathcal{C}_b$ ) code a set-supported permutation by its restriction to the least ordinal bounding its support; they are printed  $\mathcal{P}_{\text{bd}}(b)$ . Their  $\mathfrak{E}$  is  $\mathfrak{W}_{\text{sl}}$  and source 24's  $\mathfrak{J}$ , source 26's  $\mathfrak{J}$  are  $\mathfrak{J}$ .
- Source 19's  $r_\kappa$  is  $\rho_\kappa$  and its  $D_{\rho,m}$  is  $\mathbb{B}_\rho \times_{\text{lex}} m$ , as for source 15 (section 1.3.1); its residual weights  $r(\lambda)$ ,  $u_\kappa(\lambda)$  and their sums  $R$ ,  $U$  are printed  $\text{wt}(\lambda)$ ,  $\text{wt}_\kappa^+(\lambda)$ ,  $\varrho_{\text{lo}}$ ,  $\varrho_{\text{up}}$ , as source 29's are in Part XI, because  $r = \text{cf } \kappa$  and  $R$  is a raw class order elsewhere. Source 30's weight  $r(\nu)$  is printed  $\text{wt}(\nu)$ .
- In the countable strata (section 68) the countable dense alphabet is  $D$  and  $P_\alpha = \mathcal{W}_\alpha(D)$ ; source 30's  $P_n$  (length  $\omega + n$ ) is printed  $P_{\omega+n}$ , which is source 29's index; source 30's alphabet  $X$ , its set  $D_n$  of nonisolated points and its relation-table space  $\mathcal{T}_n$  are printed  $D$ ,  $S_n$  and  $\text{Tab}_n$ ; source 29's real completion  $C$  is printed  $\hat{I}$ .

Set-sized results are theorems of ZFC; the class-level results of section 71 are stated in GBC as schemes about supplied class parameters. Labels written for the merge in this Part begin `swo:x::`.

*Status.* All nine sources describe themselves as research manuscripts with written proofs that are neither refereed nor machine-checked, and none claims historical priority. Source 24: “an AI-assisted, unrefered proof manuscript . . . not by a new Lean verification or a claim of historical priority. Finite regression checks are included but do not prove the transfinite theorems.” Source 26: “AI-assisted and unrefered . . . no new Lean verification or historical-priority claim is made.” Source 28: “AI-assisted, unrefered mathematical manuscript . . . not by a claim of historical priority or a new Lean kernel verification. Finite regression checks accompany the text but do not prove its infinitary statements.” Sources 19–22, 29 and 30 say the same in their status sections; source 30 calls its proofs “conventional mathematical arguments with independent AI-assisted proof review”. Nothing in this Part is formalized.

## 61 The twelve batch-83 sources: scope and interpretations

[merge] (3 October 2026.) Each of the twelve batch-83 sources (section 1.9) opens with a statement of what it continues and claims, and with a map of what “the lexicographic order of all well-orders of the surreal numbers” can mean. These opening sections are printed here, whole and in source order, one subsection per source section, before the mathematics of Parts X–XIV (the maps of sources 23, 27, 29 and 30 are printed in Part XII, beside the word orders and relation tables they concern), as the batch-80 sources' introductions were (section 4) and the batch-81 sources' maps were (section 34). They keep every scope statement, priority caveat and non-claim of their sources. The sources read this report at three pins (d5bd4a67b: sources 19, 20, 22; 109aaca15: sources 21, 23, 24; 7ff7736ec: sources 25–30), the first two before Parts VI–VIII were written; where a sentence says that this report lacks something it now has, the sentence is kept as pinned provenance and a dated [merge] footnote or remark gives the present state. The maps agree with one another and with the earlier sources; note 61.5 lists what they repeat. Notation follows section 1.3.2; each subsection names the symbols it renames.

### 61.1 Source 19: scope, sources, and the meaning of further study

[merge] Source 19's Sections 1 and 2, printed whole. Its  $X_\kappa$  is printed  $\mathbf{No}_{<\kappa}$ , its  $r_\kappa$  is printed  $\rho_\kappa$  (this report's  $r$  is cf  $\kappa$ , section 1.3.1), and its weighted lower and upper lengths  $R$ ,  $U$  are printed  $\varrho_{\text{lo}}$ ,

$\mathcal{Q}_{\text{up}}$ , since  $R$  names raw class orders here (section 1.3.2).

The repository already establishes a substantial foundation for this problem. Its merged surreal report [39] treats fixed set alphabets, birthday cutoffs, set-length words, set-like global enumerations, and arbitrary class well-orders. Its elementary largest-common-labelled-prefix comparison is particularly useful: it compares raw class relations directly, even when their common prefix is a proper class. It therefore avoids any preliminary assignment of “class ordinals.”

Three same-day continuation manuscripts [44, 43, 46] go further.<sup>1</sup> Among other things, they analyze bounded gap characters, simultaneous decomposition into set-interval blocks, representability of core cuts, fixed types with finite tails, and the external order in countable  $\omega$ -standard models. These are treated as prior work here. In particular, the exact identification of a persistent cut fiber with a residual well-order space is a starting point, not a new claim of this article.

The present continuation asks what those results leave unresolved. The principal advances are a stronger analysis of infinite residual types; completion formulas retaining the residual fibers; extension criteria for automorphisms; the nonstandard countable-model case and a common elementary theory; a full restriction-map theorem; and a normal form for points in the bounded Dedekind completion. The last sections explain how each result can be formalized without identifying objects from different size levels.

Throughout, “new” means developed in this continuation relative to the sources just identified. No claim of historical priority in the literature is made. A proof supplied here is a conventional mathematical proof. Existing Lean lemmas are named only when their source declarations were inspected; no new Lean development is asserted.

### 61.1.1 A precise historical connection

The motivation from Kanovei and Shelah should be stated accurately. Their construction [16, §2]<sup>2</sup> lexicographically orders maps from the continuum cardinal into  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters. The maps need not be injective. It is not, literally, the set of all well-order relations on  $\mathbb{R}$ . The relevant common mechanism is ordinal-indexed first-disagreement comparison of labelled codes. Passing to surreal labels introduces proper-class and representation questions absent from that set-sized index construction.

### 61.1.2 Notation and conventions

For a linear order  $X$  and an ordinal  $\alpha$ ,

$$\mathcal{W}_\alpha(X) = \{f : \alpha \longrightarrow X : f \text{ is bijective}\}, \quad \mathbb{B}_\alpha = (\{0, 1\}^\alpha, <_{\text{lex}}).$$

We write  $L \hookrightarrow M$  for an order embedding and  $L \simeq_e M$  for embeddings in both directions. Equimorphism is weaker than isomorphism. All sums, products, and inequalities between ordinal indices are ordinal operations unless explicitly called cardinal. Thus  $\kappa \cdot \kappa$  and  $\kappa$  have the same infinite cardinality and different ordinal meanings.

For an infinite initial cardinal  $\kappa$ , put

$$\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}, \quad \rho_\kappa = \begin{cases} \kappa \cdot \kappa, & \kappa \text{ singular and strong limit,} \\ \mu, & \text{otherwise.} \end{cases} \quad (\text{S19.1})$$

<sup>1</sup>[merge] These are sources 16, 15 and 18 of this report, printed in Parts VI–IX; source 19 read them as unplaced manuscripts and saw this report at d5bd4a67b, with Parts I–V only.

<sup>2</sup>[merge] Section 2 of the published paper; the index is defined in Section 1, page 2, of the preprint, which the other sources and this report’s bibliography cite.

A strong limit cardinal  $\lambda$  satisfies  $2^\rho < \lambda$  for every cardinal  $\rho < \lambda$ . The exceptional ordinal  $\rho_\kappa$  is the repository's singular birthday transition ([theorem 8.3](#)).

We use **GB** for Gödel–Bernays set theory with ZF in its set sort and no choice assumption built into the notation. We use **GBC** for **GB** with Global Choice. Purely set-sized results are proved in ZFC. Results involving all admitted class relations as an actual domain are stated externally over a specified set-sized two-sorted model. This distinction is repeated wherever it affects a theorem.

### 61.1.3 The main conclusions at a glance

The complete fixed-type cube spectrum is stated in [Theorem 64.11](#). Divide the residual ordinal into finitely many decreasing cardinal layers,

$$\alpha = \mu\gamma + \sum_{i < k} \lambda_i \eta_i + n, \quad \mu > \lambda_0 > \cdots > \lambda_{k-1}, \quad n < \omega.$$

Give a singular strong-limit layer weight  $\lambda_i \cdot \lambda_i$  and every other residual layer weight  $\lambda_i$ , and add those weighted lengths to  $\rho_\kappa \gamma$  to obtain  $\varrho_{\text{lo}}$ . Then

$$\mathbb{B}_\theta \hookrightarrow \mathcal{W}_\alpha(\mathbf{No}_{<\kappa}) \iff \theta \leq \varrho_{\text{lo}} + \lfloor \log_2(n!) \rfloor.$$

When the explicitly computed coding upper bound agrees with  $\varrho_{\text{lo}}$ , the least binary coding length is  $\varrho_{\text{lo}} + \lfloor \log_2(n!) \rfloor$ . This agreement holds for every enumeration type when  $\mu \leq \aleph_\omega$ .

The global reconstruction theorem, [Theorem 98.2](#), works externally over any set-sized **GBC** model. It replaces each cut of the core by the completed interval of its admitted raw-order fiber, or by one point when the fiber is empty. Retaining this decoration is exactly what is needed to decide extension of core automorphisms ([Theorem 98.5](#)). The at-most-two trace ambiguity for general residual alphabets is isolated in [Proposition 99.2](#). [Proposition 99.3](#) characterizes its occurrence by a set deletion exposing numerical initial and final segments of types  $\text{Ord}$  and  $\text{Ord}^*$ .

For externally countable models, [Theorem 109.1](#) gives the two pure order types

$$\sigma(\{1, 2, 6, 24, \dots\}) \quad \text{and} \quad \sigma(\{1, 2, 6, 24, \dots\} \cup \{K\}), \quad K = \omega + \sum_{q \in \mathbb{Q}} \mathbb{Z} + \omega^*.$$

The second occurs in the  $\omega$ -nonstandard case. The two orders are nonisomorphic but satisfy the same first-order sentences; indeed the same complete theory arises from every externally set-sized **GBC** model ([Theorem 109.9](#)).

Two more concrete conclusions are the exact restriction classification ([Theorem 73.1](#)) and the first-cutoff topology ([Theorem 67.13](#)): its permutation order is homeomorphic to Baire space and has real-line completion, while omitting continuum many completion points in every interval.

## 61.2 The interpretations that must be distinguished

The ambiguity has several independent sources. One may fix a set of surreal labels or use the entire proper class; fix an enumeration type or vary it; keep labels or quotient by isomorphism; compare labels in their numerical order or compare relation bits in a background schedule; and regard a collection externally or attempt to represent it internally. There are arbitrarily many artificial encodings, so no finite list can exhaust every possible meaning. The following table covers the natural mathematical interpretations and makes their differences explicit.

| Interpretation                | Objects                                                                         | Comparison and size                                                                         |
|-------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Fixed set, fixed type         | Bijections $\alpha \rightarrow X$ , with $X \subset \mathbf{No}$ a set          | First unequal numerical labels; a set                                                       |
| Fixed set, all types          | All well-orders of the same $X$                                                 | Compare their unique ordinal enumerations; a set                                            |
| Variable set support          | Injective ordinal-length surreal words                                          | Must specify proper-prefix rule; a class of sets                                            |
| Induced numerical well-orders | Increasing ordinal-length words, or subsets well-ordered by the numerical order | A special family of labelled words; termination still matters                               |
| Set-like global orders        | Class well-orders of all $\mathbf{No}$ with set predecessors                    | Equivalent in GBC to bijections $\text{Ord} \rightarrow \mathbf{No}$ ; predicate on classes |
| Fixed class index             | Exhaustive enumerations indexed by a specified class well-order $J$             | Comparisons depend on $J$ ; may go beyond $\text{Ord}$                                      |
| All global orders             | Arbitrary class well-orders of the entire carrier                               | Common labelled initial segment; predicate on classes                                       |
| Unlabelled order types        | Well-orders modulo order isomorphism                                            | Set case reduces to ordinals; label lexicography does not descend                           |
| Rank or relation codes        | Inverse rank functions, or characteristic functions of order relations          | A different lexicographic order, dependent on a background coordinate well-order            |

### 61.2.1 The numerical order is not a global well-order

The surreal line is densely ordered and has no endpoints. It contains descending sequences such as  $0, -1, -2, \dots$ . Consequently the phrase “a well-order of the surreal numbers” cannot mean the numerical ordering of the whole class. It means an additional relation imposed on the labels, or a restriction to a numerically well-ordered subclass. Every set ordinal embeds numerically into  $\mathbf{No}$ , but  $\mathbf{No}$  itself is not an ordinal.

For set-supported numerical well-orders, each subset has exactly one increasing enumeration. Allowing different well-orders of the same subset gives many distinct words, one unique ordinal enumeration per relation. A prefix such as  $\langle 2, -1 \rangle$  is a valid injective well-order code and cannot be an increasing numerical code.

For a proper subclass, even numerical well-ordering need not be set-like. For instance,

$$a_\alpha = 1 - \frac{1}{\alpha + 2} \quad (\alpha \in \text{Ord})$$

is a strictly increasing proper-class sequence below 1. Adjoining 1 yields a numerically well-ordered class of type  $\text{Ord} + 1$ ; the new last point has a proper class of predecessors. The arithmetic is ordinary surreal field arithmetic, and  $\alpha$  denotes its ordinal surreal representative.

### 61.2.2 What happens when labels are forgotten

For a set  $X$  of infinite size  $\mu$ , the isomorphism types of well-orders of  $X$  are exactly the ordinals  $\alpha$  with  $\mu \leq \alpha < \mu^+$ . Their natural comparison is the ordinary ordinal order. It has little to do with the lexicographic order of their labelled representatives. Two different permutations of a fixed ordinal type are isomorphic as well-orders while occurring in different positions in the label order. In fact the isomorphism classes need not be convex. On the natural-number labels,

$$R = (0, 1, 2, \dots), \quad S = (1, 2, 3, \dots, 0), \quad T = (2, 0, 1, 3, 4, \dots)$$



satisfy  $R < S < T$  and have types  $\omega, \omega + 1, \omega$ . Thus numerical lexicography does not induce an ordered quotient by unlabelled order type.

For arbitrary class well-orders the situation is more delicate: the existence of an isomorphism or comparison between two abstract class orders is a separate class-theoretic problem. The elementary labelled comparison below never requires that problem to be solved.

### 61.2.3 Termination is part of the definition

When supports vary, one word can properly extend another. The shorter-first convention declares  $p < p \frown q$  for nonempty  $q$ . A centered convention introduces a termination token  $\dagger$  and compares it with the next actual label. With 0 excluded from the alphabet, one can use

$$\mathbf{No}_{<0} < \dagger < \mathbf{No}_{>0}.$$

For increasing words this can reverse the comparison of a word with one of its extensions: appending a negative allowed label places the extension below its prefix, while a positive label places it above.

The previous continuations prove that arbitrary and injective centered set-word orders are isomorphic to **No** in GBC, whereas increasing words have additional limit-prefix obstructions [46] ([merge] printed in Part VII: sections 36.5 and 37). Under shorter-first comparison, arbitrary, injective, and increasing words are mutually isomorphic as ordered prefix trees in GBC. Their order is still not **No**: if  $p$  has nonzero limit length, its proper prefixes have supremum  $p$ . A set of surreals in the full numerical line has no least strict upper bound, by the set-cut property.

Keeping an actual zero and placing a distinct terminator immediately below it creates adjacent increasing words  $p < p \frown \langle 0 \rangle$  when every entry of  $p$  is negative. Identifying a terminator with zero instead identifies distinct words unless a quotient convention is imposed. These are actual mathematical changes, not different notations for one order.

### 61.2.4 Inverse ranks and relation-bit codes

For a set carrier  $X$ , fix a well-order  $j$  as a coordinate schedule. A well-order  $R$  of  $X$  can be encoded by the ordinal rank of each label, and these ranks can be compared in  $j$ -order. This inverse-rank comparison need not agree with the comparison of the forward enumerations. Alternatively, well-order the coordinate set  $X \times X$  and compare the bits  $1_{xRy}$ . The result is again a linear order on codes, but it depends on the chosen pair schedule and on which bit is called smaller. Such encodings remain useful for implementation, provided their order is not silently identified with the numerical label order.

Ordinal-valued inverse ranks extend to set-like global well-orders, but are not available for arbitrary proper-class predecessor segments. Relation-bit coding remains available for supplied raw class orders when a genuine class well-order scans the pair coordinates. The code is then a class of bits. This changes none of the size boundaries: a class code is not an element of the set universe.

## 61.3 Source 20: what this continuation adds

[merge] Source 20's Sections 1 and 2, printed whole. Its bounded-support core  $\mathcal{C}_b$  (normalized permutation codes) is printed  $\mathcal{P}_{\text{bd}}(b)$ , and its  $\mathfrak{W}_{\text{sl}}, \mathfrak{W}_{\text{all}}$  are this report's  $\mathfrak{W}_{\text{sl}}, \mathfrak{W}_{\text{all}}$ .

The ProveIt report [39] already gives an extensive account of the fixed-set spectrum, the role of Global Choice, direct comparison of arbitrary class well-orders, and the bounded-support core. The subsequent continuation [46]<sup>3</sup> adds cardinal invariants and gap spectra, simultaneous decomposition

<sup>3</sup>[merge] Source 18 of this report, printed in Parts VI–IX. Source 20 saw this report at d5bd4a67b, with Parts I–V only, and none of sources 13–17.

into proper-class blocks, a criterion for represented cuts, and a countable  $\omega$ -standard model theorem. Those results are the starting point here. We do not treat their proposed research questions as a certified catalogue of unsolved problems in the published literature.

The main advances concern information that is lost when a rich enumeration structure is replaced by its underlying linear order. They also explain which limiting procedures genuinely recover global objects. The distinction is productive: the same family of bounded spaces supports a directed system recovering a surreal-sized core and a coherent restriction system recovering every admitted class well-order.

| Direction              | Earlier starting point                      | Further result here                                                                             |
|------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------|
| Birthday compatibility | Arbitrary tail choices need not compose     | Coherent gluing, finite changes, club detection of set-likeness, and compatible tail extensions |
| Core automorphisms     | Set homogeneity; extension problem proposed | Cut-orbit classification and automorphisms that have no global extension                        |
| Size of raw intervals  | Some finite or set residual fibers          | Exact local set/class-codeability dichotomy                                                     |
| Condensation           | One simultaneous block decomposition        | Uniform histories under ETR and a termination bound                                             |
| Countable models       | Factorial shuffle when $\omega$ is standard | Nonstandard shuffle, external finite blocks, elementary equivalence, and decidability           |
| Countable topology     | Density and gap counts                      | Permutation topology, Baire space, and the real-line completion                                 |

The contributions are intentionally stated at their actual foundational level. A theorem about a class of set codes is not automatically a theorem about a collection whose individual members are proper classes. An external classification of a countable model does not create an internal enumeration of its class objects.

## 61.4 The interpretations, separated

### 61.4.1 The choices that determine the question

A complete formulation specifies six things: the carrier, whether its labels are retained, whether its given numerical order must be respected, the allowed enumeration types, the comparison convention, and the universe in which the collection is formed. Varying these choices produces the following natural interpretations. There are infinitely many artificial coding conventions; “all interpretations” here means an account of these mathematically distinct choices, not an enumeration of every possible arbitrary encoding.

| Object                                               | Meaning of one point                                     | Size and comparison                                         |
|------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------|
| $\mathcal{W}(X)$ for a set $X \subseteq \mathbf{No}$ | An arbitrary labelled well-order of the same carrier $X$ | A set; compare exhaustive ordinal enumerations              |
| $\mathcal{W}_\alpha(X)$                              | Such an order with specified ordinal type $\alpha$       | A set stratum, nonempty exactly when $ \alpha  =  X $       |
| Injective set words                                  | A well-order of a variable set of surreal labels         | A proper class of sets; a rule for termination is necessary |

| Object                                   | Meaning of one point                                                     | Size and comparison                                                                   |
|------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Numerically increasing set words         | A set well-ordered by the numerical surreal order                        | No arbitrary permutation of the chosen subset                                         |
| $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  | A set-like well-order of the whole surreal class                         | A predicate on classes; equivalently a bijection $\text{Ord} \rightarrow \mathbf{No}$ |
| $\mathfrak{W}_{\text{all}}(\mathbf{No})$ | Any class well-order of the whole surreal class                          | A predicate on classes; common prefixes may be proper classes                         |
| Variable class domains                   | A class well-order on an arbitrary subclass of $\mathbf{No}$             | A class-variable predicate; one whole order may be a proper prefix of another         |
| Numerical class well-orders              | A subclass well-ordered by the inherited numerical order                 | Can have proper predecessor classes; not restricted to set-like orders                |
| Isomorphism types                        | Labels are forgotten                                                     | Numerical lexicography does not descend to the quotient                               |
| Relation-table words                     | Membership bits of the order relation, scanned in a fixed order of pairs | A valid different lexicographic comparison                                            |
| Objects of a smaller universe            | All orders available in a specified model or universe                    | Sets externally; the answer depends on the chosen notion of “available”               |

#### 61.4.2 Fixed labels and forgotten labels

For a set  $X$  with its numerical order, a well-order has a unique increasing enumeration relative to that well-order,  $e : \alpha \rightarrow X$ . The word “increasing” here refers to the new well-order, not to the numerical order used to compare the entries of two enumerations. If two exhaustive enumerations of the same  $X$  are distinct, neither is a proper prefix of the other. Thus their first differing entries determine their comparison without a termination rule.

Forgetting the labels changes the problem completely. On  $\{0, 1\}$  the orders enumerated by  $(0, 1)$  and  $(1, 0)$  are isomorphic, yet the first is lexicographically smaller. Consequently lexicography is not invariant under arbitrary relabelling and gives no induced order on well-order isomorphism types. If one instead orders set well-order types by their ordinals, the resulting order is simply the usual ordinal order; that is a different construction.

Nor does “all well-orders” mean that the lexicographic order itself is a well-order. Even on an infinite fixed alphabet it contains descending chains. Numerical negation of every label gives an order reversal between the spaces for  $X$  and  $-X$ ; when  $X = -X$ , this is a self-duality of the enumeration order.

#### 61.4.3 Inherited numerical well-orders

The numerical order of  $\mathbf{No}$  is not well-founded. For example, the negative integers form a descending sequence. Its well-ordered subclasses are therefore a restriction on the domain, not choices of a new order on the entire domain.

They need not be set-like. For each ordinal  $\alpha$ , let  $s_\alpha$  be the surreal whose sign sequence has length  $\alpha + 1$ , begins with a minus sign, and has only plus signs afterward. These labels strictly increase with  $\alpha$  and remain below 0. Their class followed by 0 has numerical order type  $\text{Ord} + 1$ . The

last point has a proper class of predecessors. More generally, under Global Choice the universality of the surreal line realizes every supplied class linear order as a numerical suborder, so inherited numerical class well-orders realize all supplied class well-order types as well. This is an existence statement, not a canonical choice of a representative for every type.

#### 61.4.4 Variable domains and termination

For set words, the usual dictionary convention puts a proper prefix before every proper extension. This convention has an empty least word and permits small cofinal sequences at limit-length words: the proper prefixes of a word of length  $\omega$  increase to that word. Hence this word order is not isomorphic to the set-cut-saturated surreal line.

For a variable-domain order the carrier is part of the object; an empty strict relation alone would not distinguish an empty domain from a one-point domain. The same prefix-first convention works for variable class domains. Use the largest common labelled initial segment. If both remainders are nonempty, compare their least labels numerically; if just one is empty, put the exhausted order first. A proper-class order of type  $\text{Ord}$  can then be a proper prefix of one of type  $\text{Ord} + 1$ . This possibility is absent when both orders exhaust the same carrier. The elementary common-prefix argument below proves transitivity in this variable-domain setting as well, with an exhausted branch treated as a least terminator.

An alternative is to place a terminator at a numerical cut of the digit alphabet. In particular, a terminator between the negative and positive nonzero surreal digits behaves like the undefined sign in a surreal sign sequence. The earlier continuation proves that arbitrary and injective set words with this centered convention are isomorphic to **No** in GBC, while increasing words retain a limit-prefix obstruction [46] ([merge] sections 36.5 and 37). If the digit 0 is also retained immediately above a distinct terminator, the empty suffix and the suffix (0) become adjacent. Endpoint and gap conclusions therefore require the convention to be stated.

Reversing the significance of coordinates introduces another issue. Two infinite words need not have a last differing position, so a proposed “compare the last difference” rule is not a total order without a support restriction. Comparing order types before labels also changes the order: it is a type-priority sum of strata, rather than native enumeration lexicography.

#### 61.4.5 The Kanovei–Shelah connection

Kanovei and Shelah use a definable lexicographic order of maps  $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters; repetitions are allowed. The entries are compared by their characteristic functions. This is not the collection of all well-orders of the reals, and their construction does not identify it with one of the orders above [16, §1]. The shared idea is to retain many ordered codes instead of selecting one privileged representative. Transferring that idea to class-sized indices requires a separate account of uniformity and support.

### 61.5 Source 21: the problem and the further results

[merge] Source 21’s Sections 1 and 2, printed whole. Its  $\mathcal{T}$  (macro `\Words`), the class of *all* set-length words, is printed  $\mathcal{T}$ ; this report’s  $\mathcal{I}$  is the injective class. Its internal labels `g1:` stay under its own prefix (`swo:fu:g1:`) and are unrelated to source 11’s `swo:g1:`.

The surreal numbers supply an unusually effective testing ground for lexicographic constructions. Their numerical order can interpolate every set-sized strict cut, yet the collection of all surreals is a proper class. Thus a finite or set-sized argument may remain perfectly valid while an apparently harmless change from “each” to “all” crosses a logical size boundary. The order of presentations

introduces a second source of complexity: labels, enumeration lengths, and terminal tails can survive in a lexicographic order even when the presented well-orders are abstractly isomorphic.

The pinned ProveIt treatment [39] is a substantial synthesis of four earlier manuscripts. It proves general set-alphabet spectra, surreal birthday-stage results, word codings, a direct comparison of class well-orders, dense bounded-support cores, and size obstructions. Its open-question list asks about prefix recovery, automorphisms, noncardinal cutoffs, coherent embeddings, and changes of the class realization. The present paper concentrates on these directions.<sup>4</sup> Canonical block decompositions and represented-cut criteria also arose in the preceding research; we include the versions needed for the new arguments rather than treating every restatement as a new result.

The main conclusions are these.

1. **An exact obstruction to extending core automorphisms.** The bare order of a surreal-sized dense core does not distinguish the large cuts arising from exhaustive global enumerations, those arising from nonexhaustive injective branches, and certain cuts arising from no class well-order at all. An automorphism can interchange these kinds. Consequently some core automorphisms admit no extension to an order embedding from the set-like space into the full class-well-order space. Preservation of exhaustive represented cuts is the exact criterion for automorphism extension within the set-like space.
2. **A concrete failure of prefix recovery.** For  $P = \mathcal{W}_\omega(\mathbb{Q})$ , all nonempty open intervals and open rays have order type  $P$ . Every finite increasing partial map extends to an automorphism, but the family of positive-length prefix cylinders is not preserved by all these automorphisms. No such cylinder is definable from the pure order using finitely many parameters.
3. **A precise topology behind the countable example.** The order topology of  $P$  is its topology of finite agreement. It is homeomorphic to  $\omega^\omega$ , but its order is not that of the irrationals. There are continuum many missing cuts even inside each prefix cylinder, caused both by numerical cuts and by failure to enumerate every label.
4. **Fibers and closure for set-sized alphabets.** Every well-order of an infinite  $X$  starts with an injective word of length  $|X|$ . Fixing this word leaves exactly the lexicographic order of all well-orders of the unused labels. Under a natural large-interval hypothesis, the closure of the minimum-length stratum consists of that stratum together with the existing endpoints of these fibers.
5. **Further bounded and model-theoretic information.** We compute basic invariants at noncardinal birthday cutoffs, construct coherent embeddings over bounded chains of equal cardinality, and separate standard finite permutation blocks from the externally infinite blocks hidden in nonstandard models. The abstract countable set-like order is always  $\mathbb{Q}$ , while the distinguished core retains information that this bare order forgets.

The conclusions are deliberately sensitive to interpretation. A dense class isomorphic to **No** does not turn the entire family of class enumerations into an ordinary class. A homeomorphism does not imply an order isomorphism. An internally finite terminal block need not be externally finite. These distinctions are mathematical content, not merely precautions about notation.

<sup>4</sup>[merge] Source 21 read this report at 109aaca15, before Parts VI–VIII were written. Part VIII has since answered the prefix-recovery, automorphism and class-realization questions in part (section 50); noncardinal cutoffs (research question 113.12) and coherent embeddings (research question 113.19) were still open there, and source 21's Sections 5 and 6 answer them in part (section 113.25).

### 61.5.1 The relation to Kanovei–Shelah

The motivating analogy should be stated accurately. In Kanovei–Shelah’s construction [16, §1], the index set is

$$A = \{a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N}) : \text{ran}(a) \text{ is an ultrafilter on } \mathbb{N}\}.$$

They order these maps by first disagreement, using the characteristic function order on  $\mathcal{P}(\mathbb{N})$ . The maps can repeat values. This is not literally the set of all well-orders of  $\mathbb{R}$ , and their resulting index order is not itself a well-order. What carries over to our problem is the usefulness of a linear order on a sufficiently rich family of presentations. Their definable hyperreal construction is not a theorem identifying any of the surreal spaces studied below.

## 61.6 Interpretations and the logical size boundary

### 61.6.1 Surreals as sign sequences

We use the sign representation

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha.$$

The union is class notation. Each individual sign sequence has a set ordinal as its length, called its birthday  $\text{bd}(x)$ . At the first place where two sequences differ, termination is compared by

$$- < \text{undefined} < +.$$

This is the numerical order of the surreals, not the rule that a shorter word always comes first. The sign construction and its relation to the simplicity tree are classical [55, 12].

For an ordinal  $\theta$  put

$$\mathbf{No}_{<\theta} = \{x : \text{bd}(x) < \theta\}$$

(source 21’s  $X_\theta$ ). This is a set. If  $\kappa$  is an infinite initial ordinal, then  $|\mathbf{No}_{<\kappa}| = 2^{<\kappa}$ , where  $2^{<\kappa} = \sup_{\lambda < \kappa} 2^{|\lambda|}$ . Each set of surreals has bounded birthdays. Conversely, allowing all birthdays produces a proper class, since the all-plus sign sequences already give a copy of  $\text{Ord}$ .

The cut property used in this paper is:

$$A, B \text{ sets, } A < B \implies \exists x \in \mathbf{No} (A < x < B). \quad (\text{S21.1})$$

Empty sides are allowed. This is often called the  $\eta_{\text{Ord}}$  property or the absolute linear continuum property [15]. It does not assert interpolation for arbitrary class-sized sides. For example, there is no surreal greater than every ordinal.

### 61.6.2 The main natural readings

No finite list exhausts every artificial coding convention. The following table covers the natural readings that change the mathematical problem. Independent choices in different rows can also be combined.

| Reading                | Objects being ordered                                             | Size and principal distinction                                                            |
|------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Fixed carrier          | All well-order relations on a fixed set $X \subseteq \mathbf{No}$ | A set $\mathcal{W}(X) \subseteq \mathcal{P}(X^2)$ ; labels matter.                        |
| Fixed enumeration type | Bijections $\alpha \rightarrow X$ , where $ \alpha  =  X $        | A stratum $\mathcal{W}_\alpha(X)$ ; different $\alpha$ need not have the same invariants. |



| Reading                         | Objects being ordered                                                                                | Size and principal distinction                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Varying set carrier             | Well-orders of arbitrary set-sized subsets of $\mathbf{No}$                                          | An ordinary class of set-coded objects; equivalent to injective set-length words after choosing a termination rule.               |
| Arbitrary words                 | Functions $\alpha \rightarrow \mathbf{No}$ , possibly with repetitions                               | An ordinary class; repetitions make these words different from well-order presentations of their ranges.                          |
| Numerically increasing subsets  | Sets or classes well-ordered by the inherited numerical $<$                                          | An additional monotonicity restriction; an auxiliary well-order is no longer being chosen.                                        |
| Set-like global well-orders     | Class bijections $\text{Ord} \rightarrow \mathbf{No}$ , or their induced relations                   | Each object is a class. Their totality is a predicate on classes, denoted $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ .               |
| Unrestricted global well-orders | Class well-order relations on all of $\mathbf{No}$ , with possibly proper-class predecessor segments | A predicate on classes, denoted $\mathfrak{W}_{\text{all}}(\mathbf{No})$ ; ordinal enumeration need not exist.                    |
| Relation-table lexicography     | Characteristic functions of well-order relations, read in a separately chosen order of $X^2$         | A different order on the same relations; depends on the coordinate code.                                                          |
| Unlabelled order types          | Isomorphism classes of the preceding well-orders                                                     | Numerical first disagreement does not descend to this quotient. For set well-orders, the usual comparison gives ordinals instead. |
| Universe-relative collection    | All well-orders of the surreals small in a fixed universe                                            | A larger-universe set or type; smallness must remain fixed throughout a theorem.                                                  |

### 61.6.3 Set presentations and termination

For a set  $X$  with a fixed linear order  $<$ , a well-order relation  $R$  on  $X$  has a unique increasing enumeration  $e_R : \alpha_R \rightarrow X$  in the order  $R$ . Here “increasing” refers to  $R$ , not to the numerical order on  $X$ . For two different relations on the same  $X$ , their enumerations cannot be proper extensions of one another: the shorter enumeration would already have used all of  $X$ . Thus they have a first disagreement, and

$$R <_{\text{lex}} S \iff e_R(\delta) < e_S(\delta) \text{ at their first disagreement } \delta.$$

No termination convention is needed in this fixed-carrier situation.

For varying carriers, a word can be a proper prefix of another. Three different choices should be kept separate: shorter-first lexicography, longer-first lexicography, and a termination symbol at a specified cut of the alphabet. Shortlex, which first compares lengths and only then uses lexicography, is another order altogether. The usual surreal sign order uses a termination symbol between  $-$  and  $+$ ; that special three-way convention should not be imported silently into words whose letters are arbitrary surreals. The repository gives explicit prefix-safe embeddings of set-length words into  $\mathbf{No}$  and analyzes the one-sided limit-word cuts [39, Sections 9–11] ([merge] Part II, [part II](#)).

**Example 61.1** (The unlabelled quotient is incompatible with lexicography). Let  $X = \mathbb{Q}$ . Choose minimum-length enumerations beginning with 0 and 2, and an enumeration of type  $\omega + 1$  beginning with 1. In lexicographic order the third lies between the first two. The two outer well-orders are isomorphic, while the middle well-order has a different type. Therefore equality of abstract order

type is not a convex equivalence relation. There is no induced linear order on order types obtained by comparing arbitrary representatives in numerical lexicography.

The inherited numerical order on all of  $\mathbf{No}$  is not a well-order, as the sequence  $0, -1, -2, \dots$  witnesses. Nevertheless, each set ordinal occurs as the type of a numerically increasing subset of  $\mathbf{No}$ , using ordinal surreals. The proper class  $\text{Ord} \subset \mathbf{No}$  is a numerically well-ordered class. These facts concern one inherited order, not the many auxiliary well-orders of the whole numerical line.

Under GBC, every given class well-order has a numerical copy in  $\mathbf{No}$ , by the class-order universality proved below. Such a copy need not be set-like. For example, using surreal field operations,

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

has numerical type  $\text{Ord} + 1$ : the negative points increase with  $\alpha$ , and the last point has a proper class of predecessors.

One can also fix a particular class well-order  $J$  and compare class bijections  $J \rightarrow \mathbf{No}$  at the  $J$ -least disagreement. This is a legitimate fixed-domain stratum, for example with  $J = \text{Ord} + 2$ . It does not supply a common ordinal coordinate system for all possible class domains. The labelled-relation comparison in [note 86.10](#) avoids that need.

Relation-table coding is different even on two labels. With  $0 < 1$ , the word  $(0, 1)$  is lexicographically before  $(1, 0)$ . But if the first off-diagonal table coordinate is  $(0, 1)$  and  $0 < 1$  is the bit order, the first relation has bit 1 and the second bit 0, reversing this comparison. At the class level a fixed enumeration of  $\mathbf{No}^2$  similarly makes first-bit comparison legitimate, but it remains an auxiliary-coordinate order. Its binary-class equimorphism result in the repository does not identify it with the labelled-prefix order [[39](#), Section 17] ([[merge](#)] [proposition 17.1](#)).

#### 61.6.4 Sets, classes, and predicates on classes

Our class theory convention is explicit. GB has the usual set and class axioms with elementary class comprehension and class replacement, but no choice principle by definition.  $\text{GBC} = \text{GB} + \text{GC}$ , where global choice includes a set-like global well-order of the set universe. Set-only results are proved in ZFC. The class constructions below use GBC unless a weaker hypothesis is stated.

In these theories classes contain sets, not proper classes. A class relation  $R \subseteq \mathbf{No}^2$  is a legitimate single class. The notation “the class of all such  $R$ ” is not legitimate as an ordinary class. Instead, write a formula  $\text{CWO}(R)$  with a class variable, and write a formula  $\text{Lex}(R, S)$  for comparison. A theorem such as

$$\forall R, S (\text{CWO}(R) \wedge \text{CWO}(S) \longrightarrow \text{exactly one comparison holds})$$

is a second-order theorem. It requires no object having all those classes as elements. We use the symbols  $\mathfrak{W}_{\text{sl}}$  and  $\mathfrak{W}_{\text{all}}$  as readable shorthand for this predicate-level language.

An ordinary class family of global enumerations must have an explicit uniform code, for instance a class  $E \subseteq I \times \text{Ord} \times \mathbf{No}$  whose sections  $E_i$  are enumerations and whose indices  $i$  are sets. Quantifying over such a coded family is different from quantifying over every externally conceivable family of classes. Every interpolation and back-and-forth statement below respects this distinction.

#### 61.6.5 What stronger foundations do and do not provide

KM permits class quantifiers in class comprehension. It still does not make proper classes into elements of an ordinary class. A genuine collection of all classes requires a third sort, a suitable

hyperclass theory, or a larger surrounding set universe. Higher-order comprehension and choice principles then have to be stated for that additional level. Global choice chooses sets from nonempty classes; it is not an unrestricted choice principle for selecting one class from each family of classes. This remains a real distinction even over KM [66].

Likewise, an ordinal-stage construction does not automatically require ETR. If the state at each stage is a set and the previous stages form a set, ordinary set-valued recursion suffices in GBC. In contrast,  $\text{ETR}_{\text{Ord}}$  permits class-valued slices at each ordinal stage and is a substantive strengthening [24]. Our set-sized partial isomorphisms and prefix reconstructions belong to the former situation. We do not assert a uniform isomorphism-comparison theorem for arbitrary unlabelled class well-orders from this observation.

The universe-relative alternative is equally precise. Fix a universe  $\mathcal{U}$  of small sets and form surreals whose birthdays and cut families are  $\mathcal{U}$ -small. The encompassing surreal object belongs to a larger universe. The HoTT construction states this shift explicitly [56, §11.6]. In an inaccessible-universe model,  $V_\Omega$  plays the role of the small universe and its classes can be treated externally as subsets of  $V_\Omega$ . This is a model for a specified interpretation, not a proof that every abstract GBC model has its full external power set as its class part.

## 61.7 Source 22: scope and relation to the earlier work

[merge] Source 22’s Sections 1 and 2, printed whole. Its core  $\mathcal{D}_b$  (support codes) is printed  $\mathcal{P}_s(b)$ , its  $\mathfrak{C}$  is  $\mathfrak{W}_{\text{sl}}$ , and its prefix tree  $\mathcal{T}$  of injective set-length words is printed  $\mathcal{I}$  (here  $\mathcal{T} = \mathcal{T}$  is the class of all words). The maintained ProveIt report [39, 40] already develops fixed set alphabets, birthday fragments, set-length words, global-choice issues, set-like and unrestricted class well-orders, and a set-coded dense core of global enumerations. Its README explicitly distinguishes mathematical manuscripts from formalized declarations. We inspected the revision

d5bd4a67b41b89688a079997574a94bcd855bb83.

The maintained report is a union of four manuscripts, not a sequence of superseding editions. Its current guide describes a 116-page compiled text.<sup>5</sup>

We also consulted two previous continuation manuscripts supplied through the user’s Library: *Lexicographic Well-Orders of the Surreal Numbers, II* [45], and *Lexicographic Orders of Surreal Well-Orderings: Cut reconstruction, asymmetric products, prefix non-rigidity, and size-safe foundations* [41]. Their results are not presented here as new.<sup>6</sup> In particular, the first gives minimal-stratum cut spectra and point characters; both develop labelled cylinder recognition of global branches. They explicitly ask about nonminimal enumeration types, whether unranked cylinders suffice, and how much of a class realization can be reconstructed.

The present work answers substantial versions of those three questions. “Extension” means relative to the inspected manuscripts, not a claim of priority in the literature.

**The principal advances.** The cardinal topological invariants of *all* fixed-length strata are classified, with a sharp transition at two and three extra positions. Local characters are computed for  $\mu + n$  and  $\mu + \omega$ . Prefix heights are recovered from cylinder inclusion rather than supplied as additional data. A labelled trace structure faithfully recovers the available class

<sup>5</sup>[merge] True at the pin d5bd4a67b, which has Parts I–V. The report now merges thirty sources; its README is updated with this addition.

<sup>6</sup>[merge] They are sources 17 and 13 of this report (printed in Parts VI–IX). Source 22’s audit says that they “are not assumed to have been merged into the pinned current repository”; that was true at d5bd4a67b and is no longer so.

predicates over a fixed set universe.

### 61.7.1 What is deliberately not asserted

We do not classify all nonprincipal cuts at every nonminimal length, the full automorphism group of the global enumeration order, or the order type of every unrestricted class-well-order quotient. The reconstruction theorem retains labels and fixed coding parameters; it does not recover a class model from the bare linear order alone. No unrestricted class-cut filling principle is assumed. No existence assertion about all proper-class well-orders is smuggled into an ordinary set or an ordinary class.

### 61.7.2 Reading paths

For the new set-sized mathematics, read [note 62.15](#) and [sections 63.1, 63.2 and 63.4](#). For the global reconstruction results, read [sections 86.3, 87 and 102.1](#). [Sections 61.8, 111.10 and 111.11](#) specify the interpretation and formalization boundaries. Every new result is proved within its stated framework.

## 61.8 The interpretations that must be kept separate

### 61.8.1 A numerical order is not a chosen well-order

The numerical order on **No** is a dense linear order, not a well-order. For example,  $0, -1, -2, \dots$  is descending. A well-order  $R$  of a labelled set  $X \subseteq \mathbf{No}$  is generally unrelated to the numerical order. Its unique exhaustive  $R$ -increasing enumeration

$$e_R : \alpha \longrightarrow X$$

need not be numerically increasing. To compare  $R$  and  $S$  lexicographically, we compare  $e_R$  and  $e_S$  at their first unequal label, using the *numerical* order of those labels.

For a fixed carrier  $X$ , distinct exhaustive enumerations cannot be proper prefixes of one another: a proper prefix that already exhausts  $X$  leaves no unused label for the longer enumeration. Thus the first-disagreement comparison is independent of an end-marker convention in this case.

| Interpretation                   | Mathematical representation                                | Size and comparison issue                                                                    |
|----------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| A fixed set, fixed order type    | $\mathcal{W}_\alpha(X) = \{e : \alpha \simeq X\}$          | A set; numerical first disagreement.                                                         |
| A fixed set, all order types     | All well-order relations on $X$                            | A set; varying exhaustive enumeration lengths cause no prefix ambiguity.                     |
| Varying set-sized carriers       | Injective ordinal-length words in <b>No</b>                | An ordinary proper class; a termination convention is needed.                                |
| Numerically well-ordered subsets | Strictly increasing words, or class well-ordered suborders | Not arbitrary labelled well-orders; variable carriers again require a convention.            |
| Set-like global well-orders      | Class bijections $\text{Ord} \rightarrow \mathbf{No}$      | A predicate on class variables, not an ordinary class of objects.                            |
| Unrestricted global well-orders  | Class relations on <b>No</b>                               | Proper-class predecessor segments may occur; compare common labelled initial parts directly. |
| Unlabelled order types           | Quotient by order isomorphism                              | Lexicography of labels does not descend to this quotient.                                    |
| Relation-code or shortlex orders | Bit strings, or length first and then labels               | Different orders depending on the coding or primary key.                                     |

### 61.8.2 Words, repetitions, and terminal markers

All set-length words, injective words, and strictly increasing words are three different carriers. A word with repeated entries is not an exhaustive enumeration of a well-order on its range. It can instead describe a well-ordered family with repeated labels.

If lengths vary, an end-first convention declares a proper prefix smaller than its proper extensions. The empty word is then a least element. End-last gives a greatest element. A central terminal marker, placed between negative and positive labels after reserving 0, produces a different order. The earlier continuation [41] proves surreal-line isomorphism results for this central convention under its stated class foundations ([merge] section 36.4). They must not be transferred to end-first lexicography.

Even when the empty word is deleted, the convention matters. Under end-first comparison, a word  $p$  of nonzero limit length  $\lambda$  can be approached from below by its proper prefixes. Any word below  $p$  either is such a prefix or has a first smaller label, and then lies below a sufficiently long prefix. Hence its left character is  $\text{cf } \lambda$ . The full surreal line has no set-sized cofinal family below a point: the set cut between such a family and that point would be fillable. This is an explicit obstruction to identifying these word orders.

Restricting to numerically increasing words does not restore well-foundedness of the lexicographic order: singleton words already contain a copy of **No**. A numerically increasing enumeration of the *whole* surreal class cannot exist, since it would provide a numerically least surreal. Proper subclasses can, of course, be numerically well-ordered; the ordinal surreals are a basic example.

### 61.8.3 Quotients and alternative codes

If labels are forgotten, set well-orders are represented by ordinal order types. All set-like global well-orders of the whole surreal class have order type  $\text{Ord}$ , so their unlabelled quotient has one type, even though their labelled lexicographic comparison is highly nontrivial. An order on relations obtained by reading membership bits according to a chosen well-order of pairs is another legitimate construction. Its first differing bit need not correspond to the first differing element in the two exhaustive enumerations.

These distinctions are structural, not merely notational. There are indefinitely many possible coding conventions, but the table separates the independent choices responsible for the principal mathematical interpretations.

### 61.8.4 The Kanovei–Shelah connection

Kanovei and Shelah’s definable nonstandard-model construction [16] uses a lexicographically ordered set of maps from the continuum cardinal to  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters. The maps need not be injective, and the index set is not itself a well-order. Thus “the lexicographic order of all well-orders of the reals” is not a literal description of that construction. The relevant methodological connection is to retain presentations and order them definably, rather than select one presentation of each object. Their construction is set-sized; a proposed surreal analogue must state separately where its functions and index families live.

## 61.9 Source 23: scope, provenance, and the new results

[merge] Source 23’s Section 1, printed whole; its taxonomy (Section 2) is printed in Part XII with the word orders (section 80.1). Its marker  $\mathbf{e}$  is printed  $\varepsilon$ , as source 16’s marker at a cut (Part XII), and its word classes  $\mathcal{W}_{\mathbf{e}}(A)$ ,  $\mathcal{I}_{\mathbf{e}}(A)$ ,  $\mathcal{U}_{\mathbf{e}}(A)$  (all, injective and strictly increasing set-length words)

are printed  $\mathcal{T}_\varepsilon(A)$ ,  $\mathcal{I}_\varepsilon(A)$ ,  $\mathcal{T}_\varepsilon^{\text{inc}}(A)$ ; its polynomial order  $P(\Gamma)$  is printed  $P(\Gamma)$ , as Part IX prints source 15's  $P(J)$ ; its  $\mathfrak{E}$  and  $\mathfrak{W}$  are  $\mathfrak{W}_{\text{sl}}$  and  $\mathfrak{W}_{\text{all}}$ ; its character symbol  $\text{On}$  is printed  $\text{Ord}$ , as in Part VIII.

The numerical order on the surreal numbers is a dense class linear order, not a well-order. A *well-order of surreals* is therefore usually an additional ordering of some carrier of surreal labels. It can also mean a subset already well-ordered by the numerical order. These readings lead to different mathematics. Even after choosing one, the words “all” and “lexicographic” leave significant choices unresolved.

The ProveIt repository's merged report [39] treats set approximations, the singular birthday transition, set-length words, direct comparison of raw class well-orders, and the bounded-support core of the set-like global order. Its guide distinguishes written arguments from checked prerequisites; it does not claim a Lean formalization of that report. We inspected the guide and the relevant sign-sequence interfaces at revision 109aaca1505c12d70ae169bdc2011f40bd476f7d. The branch moved during the investigation; this is a pinned source, not a claim about the latest branch tip.

The subsequent manuscript [41] asks, in its Question 15.9, for a classification of termination cuts, including markers adjacent to existing labels. Its Question 15.11 asks how far residual decomposition can be iterated safely. Those two questions motivate the main developments below.<sup>7</sup> Earlier continuation material also studies cut recognition, support bounds, and automorphism extension [45]; we do not present those results again as new.

### 61.9.1 What this article adds

Write  $A$  for a class order having the small-cut property of **No**, and let  $A = L \sqcup R$  be a class cut determining the position of a new termination symbol  $\varepsilon$ . All letters in  $L$  precede  $\varepsilon$ , and  $\varepsilon$  precedes all letters in  $R$ . Both arbitrary and injective words are allowed to have any set ordinal length, including zero.

**Termination criterion.** The word order is set-cut saturated if and only if  $L, R$  are nonempty and

$L$  has a greatest element, or no set is cofinal in  $L$ ,  
 $R$  has a least element, or no set is coinital in  $R$ .

This holds for arbitrary words and for injective words separately. Under global choice, either saturated order is isomorphic to **No**.

[merge] This is source 16's [theorem 36.1](#) for an abstract alphabet with the set-cut property; the condensation hierarchy below is source 15's (Part IX).

The endpoint exceptions in this statement are essential. An immediate neighbor of  $\varepsilon$  can be replaced by a whole continuation cylinder, which has room above and below every set of its elements. A mere “the augmented alphabet must itself be saturated” criterion is false. Theorem 81.12 supplies exact point characters, not merely a sufficiency argument.

Strictly increasing words behave differently. Once an increasing word has entered the region above the marker, it cannot extend downward relative to its termination point. A countable

<sup>7</sup>[merge] The manuscript cited is source 13 of this report. Its question 15.9 was answered by source 16 before source 23 was placed ([theorems 36.1 and 36.2](#), Part VII); source 23's termination criterion is therefore a second route to that theorem, for an abstract set-cut-saturated alphabet. Source 23 read this report at 109aaca15, before Parts VI–VIII were written.



increasing word then yields an explicit unfilled countable cut. Theorem 82.1 proves this for every fixed marker placement.

The second development concerns the order *being encoded*, not merely its code. For a raw class well-order  $H$ , identify two points when the closed interval between them is a set. Every equivalence class has a least element, so the quotient is representable by an ordinary class of set objects. For a given class well-order  $\Gamma$ , let  $P(\Gamma)$  consist of finitely supported functions from  $\Gamma$  to the ordinals, compared at their greatest coordinate of disagreement. The functions are represented by finite graphs; we never turn a proper-class total function into a set.

**Prescribed condensation hierarchy.** For  $t \in \Gamma$ , put

$$p \equiv_t q \iff p(\gamma) = q(\gamma) \text{ for all } \gamma \geq_\Gamma t.$$

Adjoin a final stage  $\top$  with universal equivalence. This is exactly the set-interval condensation hierarchy of  $P(\Gamma)$ : successor stages condense set intervals and limit stages are unions. Every stage before  $\top$  is nontrivial. The whole family is given by one elementary definition in GBC.

For example, deleting the first  $n$  coordinates gives a quotient of  $P(\omega)$  isomorphic to itself for every finite  $n$ , yet the limit stage  $\omega$  collapses it to a point. For  $\Gamma = \text{Ord}$ , every set stage has the original quotient order type, but the union of all those stages collapses. Thus repeated isomorphism of successive quotients does not imply that a limit stage has stabilized.

The lexicographic relevance is concrete: between any two distinct set-like exhaustive enumerations of **No**, there is a raw well-order isomorphic to  $P(\Gamma)$  for any prescribed nonempty  $\Gamma$  (Theorem 108.23). This asserts arbitrary *internal well-order complexity* in those intervals. It does not assert that the complexity is recoverable from the bare outer linear order.

### 61.9.2 Conventions about size and proofs

Our default class theory is  $\text{GBC} = \text{GB} + \text{GC}$ , with ordinary set Foundation and Power Set. The name GB itself is not used to imply choice. All members of classes are sets. An “ordinary class” below is a class of set objects. A “virtual order” is a predicate on class variables with a definable comparison; it is not a same-level class whose members are proper classes.

A class well-order means a class linear order in which every nonempty admitted subclass has a least element. We use the global-choice consequences needed for class enumerations and for producing a descending sequence from a nonempty subclass with no minimum. No arbitrary class-valued recursion is silently included in GBC.

The symbol  $\text{Ord}$  in a character pair means “no set-sized cofinal family” or its dual. It is not an additional cardinal. Under global choice, the relevant nonempty class orders have cofinal or cointial sequences indexed by  $\text{Ord}$ , so this notation also records their class-sized character.

### 61.10 Source 24: the continuation and its main results

[merge] Source 24’s Sections 1 and 2, printed whole. Its  $\mathcal{P}_\mu(X)$  and  $\mathcal{I}_\mu(X)$  are printed  $\mathcal{W}_\mu(X)$  and  $\mathcal{L}_\mu(X)$  (the length- $\mu$  layer of this report’s  $\mathcal{I}(X)$ ; this  $\mathcal{P}$  is not  $\mathcal{P}_s$ ), its core  $\mathcal{C}_b$  (normalized permutation codes) is printed  $\mathcal{P}_{\text{bd}}(b)$ , and its  $\mathfrak{E}, \mathfrak{J}$  are  $\mathfrak{W}_{\text{sl}}, \mathfrak{J}$ .

There is no unique object denoted by “the lexicographic ordering of all well-orders of the surreal numbers.” Even after the size level is fixed, one must choose whether labels matter, whether the carrier is fixed, whether the enumeration length is prescribed, and how termination is compared



with the next label. At the class level, there is the additional distinction between a well-order whose every proper initial segment is a set and a well-order that has a proper-class initial segment.

The merged ProveIt report [39] develops the foundational taxonomy, embedding spectra, finite-tail adjacency, and a set-coded dense core of the global enumeration order. Three subsequent research drafts [45, 41, 42]<sup>8</sup> study cut reconstruction, automorphisms, asymmetric products, support thresholds, and model dependence. The present article does not count these as new results. Instead, it addresses a different question: *what exactly converges when longer and longer initial parts of an enumeration stabilize?*

For a set  $X$  of infinite cardinality  $\mu$ , regarded as an initial ordinal, write

$$\mathcal{W}_\mu(X) = \text{Bij}(\mu, X), \quad \mathcal{I}_\mu(X) = \text{Inj}(\mu, X).$$

The first is the minimal-length slice of the labelled well-ordering space. The second allows a well-order of a proper subset whose cardinality is still  $\mu$ . Both carry the topology generated by initial-segment cylinders. For the surreal alphabets treated below, this is also their lexicographic order topology.

### 61.10.1 A precise statement of the advances

**Theorem 61.2** (Main results, with hypotheses explicit). *Let  $|X| = \mu$  be infinite.*

- (i) *The Hausdorff completion of  $\mathcal{W}_\mu(X)$  for agreement on bounded initial segments is  $\mathcal{I}_\mu(X)$ . Surjectivity is the intersection of the  $\mu$  open dense requirements that each label occur.*
- (ii) *If  $\mu$  is regular and  $\lambda = \mu^{<\mu}$ , then*

$$\mathcal{W}_\mu(X) \cong_{\text{top}} \lambda^\mu,$$

*where the right side has the bounded topology. This is a homeomorphism, not an asserted order isomorphism or a forward-uniform isomorphism.*

- (iii) *If  $X$  is countable and densely ordered without endpoints, the lexicographic order on  $\mathcal{W}_\omega(X)$  has a complete compatible metric and is homeomorphic to  $\omega^\omega$ . Its Dedekind completion is  $\mathbb{R}$ , and its complement there is a dense meager  $F_\sigma$  set of cardinality  $2^{\aleph_0}$ . In particular, this order is not isomorphic to  $\mathbb{R} \setminus \mathbb{Q}$ .*
- (iv) *In the bounded-prefix topology on  $\text{Sym}(\mu)$ , right translation by  $\sigma$  is continuous exactly when  $\sigma$  takes bounded subsets of  $\mu$  to bounded subsets. Inversion is continuous at  $\pi$  exactly when  $\pi^{-1}$  has that property. These conditions hold automatically for regular  $\mu$  and can fail for singular  $\mu$ .*
- (v) *In GBC, every uniformly set-indexed prefix-Cauchy net of global enumerations is eventually constant. The coherent class-length prefix completion, formulated as a scheme about individual class parameters, consists of injective Ord-words. Requiring inverse stabilization as well yields exhaustive enumerations.*

*For (i), (ii), and (iv), the topology depends only on the alphabet as a set. To identify it with a lexicographic order topology at uncountable sizes, we use the thickness hypothesis in Definition 62.1.*

The proof of (ii) is constructive in the order-theoretic sense: a node is extended until the least still-unused label, relative to a fixed baseline enumeration, first appears. This produces a full tree with  $\mu^{<\mu}$  successors at every node. Regularity keeps all intermediate prefixes short. The very step that makes this construction work identifies its singular-cardinal obstruction.

<sup>8</sup>[merge] Sources 17, 13 and 14 of this report.

Part (iii) answers the complete-metrizability question in [45] for the countable-size case. It does *not* settle that question for uncountable singular  $\mu$  with  $\text{cf } \mu = \omega$ . Those spaces are metrizable, but the incompleteness of their most obvious metric is not a proof that every compatible metric is incomplete.<sup>9</sup>

### 61.10.2 How to read the article

Sections 61.11–62.1 provide the interpretations and reusable elementary order theory. Sections 65.1–69.1 contain the principal set-sized results and their proofs. Sections 71.1–71.2 treat the class problem, including non-set-like orders and universe-sensitive interpretations. The formalization plan and research agenda are in Sections 111.23 and 113.18.

Set-sized arguments use ZFC. For class arguments, GB means Gödel–Bernays theory without a choice convention silently built into the abbreviation, and GBC means GB with Global Choice. A statement about a family of classes always specifies a uniform coding relation; it never means that proper classes have become elements of an ordinary class.

## 61.11 The interpretations that must remain distinct

### 61.11.1 Fixed carriers, labels, and enumeration length

Let  $X$  be a linearly ordered set. A *labelled well-ordering of  $X$*  is a well-order relation  $R$  on that very carrier. It has a unique exhaustive enumeration

$$f_R : \alpha_R \longrightarrow X$$

that is an order isomorphism from the ordinal  $\alpha_R$  to  $(X, R)$ . Its enumeration is not usually increasing in the given numerical order of  $X$ . The external well-order  $R$  says when a label is listed; the original order of  $X$  says which first differing label is smaller.

If  $R \neq S$ , neither exhaustive enumeration can be a proper initial segment of the other. Once one has listed every element of  $X$ , an injective continuation on the same carrier is impossible. Thus there is a first index  $\delta$  at which they differ, and

$$R <_{\text{lex}} S \iff f_R(\delta) <_X f_S(\delta).$$

There is no termination convention to choose in this fixed-carrier comparison.

The full order  $\mathcal{W}(X)$  allows all ordinal lengths  $\alpha$  of cardinality  $|X|$ . Its fixed-length slice is  $\mathcal{W}_\alpha(X) = \text{Bij}(\alpha, X)$ . The minimal slice  $\mathcal{W}_\mu(X)$ , where  $\mu = |X|$  is initial, is especially uniform: after every prefix of length less than  $\mu$ , the remaining domain still has order type  $\mu$  and the remaining labels still have cardinality  $\mu$ . Several conclusions below fail without this qualification. In particular, the full order can have adjacent pairs produced by finite residual permutation tails [39].

### 61.11.2 Varying set carriers and the end marker

An arbitrary set-sized labelled well-order of a set of surreals is represented by an injective word  $p : \alpha \rightarrow \mathbf{No}$ , with both  $\alpha$  and the word's graph sets. Two such words can stand in the proper-prefix relation. Three natural conventions are genuinely different:

<sup>9</sup>[merge] Source 26, which read source 24, settles it negatively in ZFC: for uncountable  $\mu$  of countable cofinality  $\mathcal{W}_\mu(X)$  is not Borel in  $\mathcal{I}_\mu(X)$ , hence not completely metrizable (theorem 69.18; note 69.7).

| Convention          | Comparison at termination                            | Immediate consequence                                  |
|---------------------|------------------------------------------------------|--------------------------------------------------------|
| End first           | Termination precedes every actual label.             | A proper prefix is smaller; the empty word is least.   |
| End last            | Termination follows every actual label.              | A proper prefix is larger; the empty word is greatest. |
| Central termination | Termination lies at a specified cut of the alphabet. | Extending a word can move either left or right.        |

For example, on a nonzero ordered alphabet with negative and positive letters, a termination symbol placed at zero gives

$$p \frown (-1) < p < p \frown (1).$$

The sign-sequence presentation of a surreal uses precisely a central convention on the *sign* alphabet:  $- < \text{termination} < +$ . This should not be confused with end-first dictionary comparison of *surreal-valued* words. Length-first comparison, or shortlex, is yet another order. Allowing repeated labels produces an arbitrary word space rather than a space of labelled well-orders.

### 61.11.3 Numerically increasing well-orders

One may instead ask that a subset of **No** be well-ordered by the *inherited numerical order*. Its increasing enumeration is then a special injective word. This is a far smaller and more constrained problem than arbitrarily well-ordering its labels. The whole numerical order of **No** is not a well-order: it contains  $0 > -1 > -2 > \dots$ .

At the class level, even a numerically increasing well-ordered subclass need not be set-like. Ordinal surreals give a copy of Ord. The ordered-field identity

$$\frac{v}{v+1} - \frac{u}{u+1} = \frac{v-u}{(u+1)(v+1)} > 0 \quad (0 \leq u < v)$$

shows that  $\{\alpha/(\alpha+1) : \alpha \in \text{Ord}\}$  is an increasing, numerically bounded copy of Ord. Appending the surreal 1 gives a well-ordered subclass of type  $\text{Ord} + 1$ . The final point has a proper-class predecessor collection. Numerical boundedness is therefore not a substitute for set-likeness.

### 61.11.4 The whole surreal class

A *set-like* class well-order is one in which each element has only a set of predecessors. A set-like well-order on a proper class has a canonical enumeration of type Ord: the index of an element is the ordinal order type of its predecessor set. In GBC, fix a baseline class bijection

$$b : \text{Ord} \longrightarrow \mathbf{No}.$$

We write  $\mathfrak{W}_{\text{sl}}$  for the *virtual domain* of all class bijections  $f : \text{Ord} \rightarrow \mathbf{No}$ , ordered by numerical first disagreement. Each  $f$  is a class;  $\mathfrak{W}_{\text{sl}}$  is not declared to be a class whose members are those classes.

An *arbitrary* class well-order of **No** can have type  $\text{Ord} + 1$ ,  $\text{Ord} + \text{Ord}$ , or more complicated class-order types. These are not covered by  $\mathfrak{W}_{\text{sl}}$ . Their lexicographic comparison requires a largest common labelled initial segment that may itself be a proper class; Section 71.2 supplies an elementary formula.

### 61.11.5 Codes and quotients are different constructions

Fixing a class enumeration of  $\mathbf{No} \times \mathbf{No}$  permits comparison of the characteristic functions of two well-order relations at the first differing *relation bit*. This relation-code order is legitimate, but it depends on that chosen enumeration and need not agree with numerical first-disagreement comparison of the exhaustive label lists.

Quotienting by unlabelled order isomorphism is more drastic. All set-like global well-orders have the same type  $\mathbf{Ord}$ , so their quotient has one element. The rich order  $\mathfrak{W}_{\mathbf{sl}}$  is a structure of labelled presentations, not a structure of those unlabelled types. Numerical lexicographic order generally does not descend to the quotient.

The same caution applies to the Kanovei-Shelah construction of a definable nonstandard model [16]. Their published index set consists of functions from the continuum cardinal into  $\mathcal{P}(\mathbf{N})$  whose ranges are ultrafilters, ordered lexicographically. The functions need not be injective. This motivates definable fixed-length indexing, but it is not literally the present order of all well-order relations on  $\mathbb{R}$  or  $\mathbf{No}$ .

## 61.12 Source 25: the continuation and its main conclusions

[merge] Source 25's Sections 1 and 2, printed whole. Its core  $\mathcal{D}_b$  (support codes) is printed  $\mathcal{P}_s(b)$ , its node class  $\mathcal{T}$  of injective set-length words is  $\mathcal{I}$ , and its two-sorted reduct  $\mathcal{C}_b$  is printed  $\mathbf{C}_b$ , as in Part XIII (sources 20, 24 and 26 write  $\mathcal{C}_b$  for the core itself).

The phrase “the lexicographic ordering of all well-orders of surreal numbers” leaves several independent choices unspecified. The word *well-order* may describe an additional relation on a fixed carrier or the numerical order inherited by an increasing subset. The carrier may be a set or the whole proper class  $\mathbf{No}$ . A representation may be an enumeration, a relation code, or an unlabelled order type. Finally, the word *all* must be interpreted at an explicit logical size level. These distinctions are mathematical, not merely notational.

The maintained ProveIt report [39, 40] develops these layers and the set-supported dense core of the global enumeration order. Subsequent continuation manuscripts investigate cut reconstruction, prefix completion, termination conventions, residual geometry, and condensation [41, 50, 51].<sup>10</sup> In particular, the residual-geometry continuation recovers prefix heights from unranked cylinder inclusion but explicitly leaves open whether the label information used to recognize exhaustiveness can be discarded. The present article answers that question negatively, in a stronger form that still preserves the entire core and all its cylinders.

Our new arguments are concentrated in Sections 89–95. The supporting material is re-established where it enters those proofs. The preliminary fixed-set spectrum and raw-class comparison are included to make clear which interpretation each conclusion concerns; they are not presented as new discoveries.

### 61.12.1 The principal objects

Fix, in GBC, a class bijection

$$b : \mathbf{Ord} \longrightarrow \mathbf{No}.$$

<sup>10</sup>[merge] Sources 13, 22 and 23 of this report; source 25 also read source 17. It saw this report at 7ff7736ec, with Parts I–VIII.

Let  $\mathcal{P}_s(b)$  be the ordinary class of normalized *set codes* for set-supported permutations of this enumeration. Its evaluated branches are exhaustive, but the codes themselves are sets. Let

$$\mathcal{I} = \bigcup_{\alpha \in \text{Ord}} \text{Inj}(\alpha, \mathbf{No})$$

be the ordinary class of set-length injective words. A node  $p \in \mathcal{I}$  determines a cylinder

$$C_p = \{d \in \mathcal{P}_s(b) : p \sqsubseteq d\}.$$

Here and below a core code is identified notationally with its evaluated class word; the coding distinction is maintained in the proofs.

Write  $\mathfrak{J}$  for the *virtual domain* of admitted class injections  $\text{Ord} \rightarrow \mathbf{No}$ , and  $\mathfrak{W}_{\text{sl}}$  for the subdomain of bijections. These are predicates on class variables, not ordinary classes whose members are themselves classes. An automorphism of the core-and-cylinder structure acts on its ordinary classes of set codes and induces an elementary transformation of any supplied class branch.

### 61.12.2 The main result in one statement

**Theorem 61.3** (Configuration classification; precise version in Section 90). *Let  $(f_i)_{i \in I}$  and  $(g_i)_{i \in I}$  be uniformly set-indexed families of pairwise distinct members of  $\mathfrak{J}$ . There is an order automorphism of  $\mathcal{P}_s(b)$  preserving the entire unranked cylinder family and carrying every  $f_i$  to  $g_i$  on the induced branch action if and only if, for all applicable indices,*

$$\begin{aligned} f_i \in \mathcal{P}_s(b) &\iff g_i \in \mathcal{P}_s(b), \\ \delta(f_i, f_j) &= \delta(g_i, g_j), \\ f_i <_{\text{lex}} f_j &\iff g_i <_{\text{lex}} g_j. \end{aligned}$$

Here  $\delta(f, g)$  is the least ordinal at which distinct branches disagree.

Several consequences are worth distinguishing. An arbitrary noncore exhaustive branch and an arbitrary nonexhaustive injective branch lie in the same orbit. Thus exhaustiveness is not invariant under the unlabelled cylinder structure. This also produces cylinder-preserving core automorphisms with no order-automorphism extension to  $\mathfrak{W}_{\text{sl}}$ . On the other hand, preserving the equality of labels across different extension edges forces a single alphabet automorphism. Requiring that automorphism to preserve  $\mathcal{P}_s(b)$  forces it to be the identity.

These are statements about particular reducts and expansions. They do not say that the original, fully labelled definition of a global enumeration is ambiguous. They identify exactly which information was lost when labels were forgotten.

### 61.12.3 What is not claimed

The word “new” means an extension relative to the inspected repository and continuation material, not established historical priority. We do not classify all automorphisms of the *bare* order  $\mathfrak{W}_{\text{sl}}$ , nor all automorphisms of  $\mathcal{P}_s(b)$  after forgetting cylinders. We do not transfer the regular-cardinal theorem to arbitrary singular birthday fragments. We do not infer a proper-class theorem from finite experiments. Every such stronger question is separated from the proofs below.

### 61.13 A taxonomy of the interpretations

#### 61.13.1 Fixed carriers and canonical ordinal enumerations

For a linearly ordered set  $X$ , a well-order  $R$  on the labelled carrier  $X$  has a unique increasing enumeration with respect to  $R$ ,

$$e_R : \alpha_R \longrightarrow X.$$

The word “increasing” here refers to  $R$ , not to the pre-existing numerical order on  $X$ . For distinct well-orders on the same carrier, neither enumeration can be a proper prefix of the other: an exhaustive proper prefix leaves no new label available for an injective extension. Consequently there is a first disagreement, and one may define

$$R <_{\text{lex}} S \iff e_R(\delta) <_X e_S(\delta).$$

Different ordinal lengths are allowed; no termination convention is needed for two exhaustive words on the same carrier.

We use  $\mathcal{W}(X)$  for all such well-orders and  $\mathcal{W}_\alpha(X)$  for those of enumeration type  $\alpha$ . The latter is nonempty exactly when  $|\alpha| = |X|$ . In particular, “all well-orders on  $X$ ” and “all bijections  $|X| \rightarrow X$ ” need not be the same domain.

#### 61.13.2 Varying the carrier or allowing repetitions

An arbitrary injective set-length word represents a well-order on its range. If the ranges can differ, a word can be a proper prefix of another. An extra convention is then indispensable. Placing termination before every letter gives end-first dictionary order. Placing it after every letter gives the opposite prefix convention. Placing it inside an alphabet cut gives a third family of orders. Equal words are, of course, compared as equal.

Allowing repetitions produces all ordinal-length words, but these are no longer enumerations of well-orders on their ranges. They are still useful as comparison and coding spaces. The recent termination manuscript [51] studies how different termination cuts affect set-cut saturation ([merge] as source 16 does, section 36). Its results do not make those conventions interchangeable.

Shortlex orders compare length before letters and change the construction again. A right-to-left or “last disagreement” comparison is generally undefined on infinite words: the disagreement positions need not have a largest element. Sign expansions of individual surreal numbers have their own central termination convention,  $- < 0 < +$ ; that order must not be confused with an end-first order on words whose letters are themselves surreals.

#### 61.13.3 Numerically increasing words

A set  $A \subseteq \mathbf{No}$  can already be well-ordered by the numerical order. Its increasing enumeration is then an injective, numerically increasing ordinal word. This is a much smaller domain than arbitrary labelled well-orders on  $A$ : for a given  $A$  there is only one inherited numerical order.

For example, the finite list  $(3, -1, 2)$  specifies a perfectly good additional well-order of three surreal labels, but it is not numerically increasing. Our principal homogeneous structure contains all injective prefixes; restricting it to increasing prefixes removes the freedom that makes the one-point extension argument work. Section 103 returns to this distinction for class-sized increasing subsets.



#### 61.13.4 Global set-like and unrestricted class well-orders

A class well-order is *set-like* when every proper initial segment below an element is a set. On the proper class **No**, a set-like class well-order has type **Ord** and is equivalent to a class bijection  $\mathbf{Ord} \rightarrow \mathbf{No}$ . This is the interpretation represented by  $\mathfrak{W}_{sl}$ .

An unrestricted class well-order can have proper-class initial segments. A copy of  $\mathbf{Ord} + \mathbf{Ord}$  illustrates the difference. Such a well-order has no exhaustive ordinal-class enumeration of type **Ord**, so it is not an element of  $\mathfrak{W}_{sl}$ . It can nevertheless be compared directly with another labelled class well-order through their largest common labelled initial part; the definition is recalled in Section 103.

#### 61.13.5 Relation codes and unlabelled order types

A relation can instead be encoded as a binary characteristic word using a chosen well-order of the ordered pairs of labels. The resulting lexicographic order depends on that auxiliary coding and need not agree with enumeration lexicography. It is a valid but different construction.

Quotienting by order isomorphism discards labels. For a fixed infinite set  $X$  of cardinality  $\mu$ , the possible unlabelled types are the ordinals  $\alpha$  with  $|\alpha| = \mu$ . Ordering these types by ordinal comparison is not the labelled lexicographic order. For global set-like well-orders, the unlabelled type is always **Ord**, so this quotient has only one type. Arbitrary class well-order types require additional care: a class of proper-class isomorphism classes cannot simply be formed as an ordinary class in GBC.

| Interpretation      | Objects compared                                        | Logical size in the stated setting               |
|---------------------|---------------------------------------------------------|--------------------------------------------------|
| Fixed set $X$       | Well-order relations or exhaustive ordinal words        | An ordinary set in ZFC                           |
| Variable set ranges | Injective set-length surreal words                      | An ordinary class of sets                        |
| Repeated words      | All set-length surreal words                            | An ordinary class, but not all are well-orders   |
| Inherited order     | Numerically increasing words or subsets                 | Set/class distinction still applies              |
| Global set-like     | Class bijections $\mathbf{Ord} \rightarrow \mathbf{No}$ | A predicate on class variables                   |
| Global unrestricted | Class well-order relations on <b>No</b>                 | A predicate on class variables                   |
| Relation-code order | Binary codes under a selected pair order                | Set or higher-level domain, depending on carrier |
| Unlabelled types    | Isomorphism types rather than labels                    | A different quotient problem                     |

#### 61.13.6 The Kanovei–Shelah motivation, accurately scoped

The construction in Kanovei and Shelah [16, Section 1] uses lexicographically ordered maps  $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters; those maps may repeat values. It is not literally the order of all well-order relations on  $\mathbb{R}$ , and it is not a construction involving the proper class of all surreal labels. Its relevant methodological idea is to order presentations even when selecting one distinguished presentation would be undesirable. We retain that motivation without conflating the index spaces.

### 61.14 Source 26: the continuation and its principal results

[merge] Source 26’s Sections 1 and 2, printed whole. Source 26 writes  $\kappa$  for the cardinality of the alphabet and  $\theta$  for a birthday cutoff; they are printed  $\mu$  and  $\kappa$ , as for source 16 (section 1.3.1), so its  $\mu = 2^{<\theta}$  reads  $\mu = 2^{<\kappa}$ . Its  $\mathcal{P}_\kappa(X)$ ,  $\mathcal{I}_\kappa(X)$  are printed  $\mathcal{W}_\mu(X)$ ,  $\mathcal{I}_\mu(X)$ .

The phrase “the lexicographic ordering of all well-orders of surreal numbers” does not determine a single object. A well-ordering of a fixed set of surreal labels is a set relation. A well-ordering of the entire surreal class is a class relation. Its totality is not an ordinary class of objects in a two-sorted set/class theory. Moreover, a numerical increasing enumeration and an arbitrary enumeration carrying a different well-order are different constructions, even before these size issues arise.

The repository report [39] already develops much of this foundational separation, together with embedding spectra, birthday-cutoff phenomena, global support cores, and comparison of non-set-like class well-orders. The subsequent manuscript [52]<sup>11</sup> identifies the forward-prefix completion with injective words and asks, in its Question 13.1, whether an uncountable minimal-length permutation space of countable cofinality might nevertheless admit some other compatible complete metric. Incompleteness of the displayed prefix metric did not decide that question. Neither did the construction of a finer complete two-sided uniformity.

Here that question is answered negatively, by a stronger descriptive obstruction. Throughout the set-sized part, cardinals are identified with initial ordinals, and set choice is available. Write

$$\mathcal{W}_\mu(X) = \text{Bij}(\mu, X), \quad \mathcal{I}_\mu(X) = \text{Inj}(\mu, X), \quad |X| = \mu \geq \aleph_0.$$

The topology has basic neighborhoods prescribing a whole initial segment of length  $\alpha < \mu$ . It is not, at uncountable lengths, the usual finite-coordinate product topology.

**Main classification.** In the bounded-prefix topology,

$$\mathcal{W}_\mu(X) \text{ is metrizable} \iff \text{cf } \mu = \omega,$$

whereas

$$\mathcal{W}_\mu(X) \text{ is completely metrizable} \iff \mu = \omega.$$

If  $\mu > \omega$  and  $\text{cf } \mu = \omega$ , the stronger assertion

$$\mathcal{W}_\mu(X) \notin \text{Borel}(\mathcal{I}_\mu(X))$$

holds in ZFC, without a continuum-hypothesis assumption.

The classification concerns the *minimal* ordinal enumeration length  $\mu$ . It does not silently include all other ordinals of cardinality  $\mu$ . The distinction is essential: at a successor enumeration length, fixing the entire proper prefix can leave a unique final label and isolate a point.

The second main result is a closed-subspace universality theorem. If additionally  $\mu > \mathfrak{c} = 2^{\aleph_0}$ , then every zero-dimensional separable metrizable space is homeomorphic to a closed subspace of  $\mathcal{W}_\mu(X)$ . For any prescribed subset of binary Cantor space, the construction can preserve binary lexicographic order when  $X$  is linearly ordered. Thus the missing boundary is not a technical nuisance that disappears after a simple completeness repair; it can carry arbitrary Cantor traces.

The surreal specialization uses two different cardinal parameters. For an infinite birthday cutoff  $\kappa$ ,

$$\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}.$$

The relevant permutation length is  $\mu$ , not automatically  $\kappa$ . Every interval and ray of  $\mathbf{No}_{<\kappa}$  has size  $\mu$ , so its minimal enumeration order has the prefix topology as its order topology. In particular,  $\kappa = \beth_\omega$  supplies an unconditional example with  $\mu = \kappa > \mathfrak{c}$  and  $\text{cf } \mu = \omega$ .

<sup>11</sup>[merge] Source 24 of this report.

### 61.14.1 What is inherited, and what is developed here

The interpretation taxonomy, the basic set-alphabet embedding results, the support-core framework, direct common-prefix comparison, and the earlier forward-completion theorem belong to the preceding reports [39, 52]. Their elementary mechanisms are reproved below when they are needed for a self-contained argument. The main developments of this continuation are the compact omitted-label coding theorem, its non-Borel consequence, the resulting exact complete-metrizability classification, closed universality, and their simultaneous application to singular surreal birthday fragments. The bounded-image refinement in Section 69.7 gives a further localization of the obstruction.

This is a statement of dependence and of what the present manuscript contributes relative to the inspected drafts, not a claim that no related result exists elsewhere. Classical descriptive set theory supplies the perfect-set property for uncountable Borel subsets of Cantor space [57]. The newer singular-cardinal descriptive-set-theoretic framework of Dimonte and Motto Ros [58] is relevant context, but its generalized Borel hierarchies are not identified with the ordinary Borel sigma-algebra used in our theorem.

### 61.14.2 Conventions and logical scope

The set-sized proofs take place in ZFC. In the class-sized part, GB denotes Gödel–Bernays theory without an included choice principle, and  $GBC = GB + GC$ . We explicitly distinguish a theorem about two class relations from an assertion that a collection of all such relations is itself a class. A uniformly coded family means one admitted class relation whose sections are the family members; it never means an arbitrary set whose members are proper classes.

The words “well-order” and “well-founded” in a class model are internal unless an external viewpoint is stated. “Borel” throughout the main theorems means generated from open sets by *countable* unions and complements. “Closed-universal” means that the specified spaces occur as closed subspaces; it does not mean that all of them admit complete metrics. “Equimorphic” means mutually order-embeddable and does not mean order-isomorphic.

## 61.15 An atlas of the main interpretations

There is no finite list of every artificial coding one could call lexicographic. The following axes isolate the standard mathematically distinct meanings. Their combinations cover the principal interpretations of the question without imposing a misleading single answer.

| Interpretation             | Objects and comparison                                                                       | Foundational level                    |
|----------------------------|----------------------------------------------------------------------------------------------|---------------------------------------|
| Fixed labelled carrier     | Well-orders of a set $X \subseteq \mathbf{No}$ , compared through their ordinal enumerations | A set in ZFC                          |
| Fixed enumeration length   | Bijections $\lambda \rightarrow X$ ; especially the minimal slice $\lambda =  X $            | A set in ZFC                          |
| Varying set carriers       | Injective ordinal-length surreal words; termination must be ordered                          | A class of set codes                  |
| Inherited numerical order  | Strictly increasing words, or subsets well-ordered by $<_{\mathbf{No}}$                      | A class of set codes                  |
| Relation-code order        | Characteristic functions of well-order relations, along a separately chosen coordinate order | Set codes, or virtual class relations |
| Order-type quotient        | Forget labels and retain ordinal or class order type                                         | Different quotient problem            |
| Set-like global orders     | Class bijections $\text{Ord} \rightarrow \mathbf{No}$                                        | A predicate on class variables        |
| Unrestricted global orders | Class well-orders, possibly with proper-class predecessor segments                           | A predicate on class variables        |
| Higher-universe version    | The preceding objects interpreted inside a specified smaller universe or class model         | Sets externally; size shift explicit  |

### 61.15.1 Fixed sets: arbitrary well-orders are not numerical increasing orders

Let  $X$  be a set of surreal numbers, equipped with its numerical linear order. A different relation  $R \subseteq X^2$  can well-order  $X$ . Its unique ordinal enumeration is

$$e_R : \text{otp}(X, R) \longrightarrow X.$$

The values need not increase numerically. For instance,  $\langle 1, 0 \rangle$  is a legitimate enumeration of a two-element labelled well-order, but is not increasing in surreal numerical order. Numerical order is used to compare *different enumerations*, not to constrain each enumeration.

Two unequal enumerations of the same carrier cannot be related by proper initial-segment extension. If one exhausted  $X$  before the other ended, the latter would have no unused label for its next position. Hence their first disagreement occurs at a coordinate belonging to both domains, and its two labels determine their lexicographic comparison. No end marker is needed on a fixed exhaustive carrier.

For an infinite  $X$  of cardinality  $\mu$ , the possible order types are exactly the ordinals  $\lambda$  with  $\mu \leq \lambda < \mu^+$ . Thus

$$\mathcal{W}(X) = \bigcup_{\mu \leq \lambda < \mu^+} \mathcal{W}_\lambda(X), \quad \mathcal{W}_\lambda(X) = \text{Bij}(\lambda, X).$$

These slices need not be convex in the full lexicographic order. First labels can already place members of different lengths on either side of each other. Our principal topological theorem is about the minimal slice  $\mathcal{W}_\mu(X)$ , not an assertion about this entire union.

### 61.15.2 Variable carriers and termination conventions

If the carrier is allowed to vary, one word can be a proper prefix of another. For an ordinary class of all set-length injective words, the most familiar convention puts a proper prefix first. Equivalently, the termination symbol is below every genuine label. The empty word is then least. Putting termination above every label reverses the prefix comparison and makes the empty word greatest. Inserting termination at a chosen cut in the alphabet gives further inequivalent orders.

The standard sign presentation of a surreal uses the special convention

$$- < \mathbf{end} < +.$$

It therefore cannot be identified without proof with a surreal-labelled word order using termination below all letters. With the three binary words  $\langle - \rangle$ ,  $\langle \rangle$ ,  $\langle + \rangle$ , the central-termination order places the empty word in the middle, whereas prefix-first puts it before both nonempty words.

Allowing repeated labels changes injective words into arbitrary ordinal-length functions. Sorting by length before comparing labels is another construction, often called shortlex. A reverse-coordinate comparison at the *last* disagreement is not automatically defined for infinite words: two words may differ at all natural-number coordinates and have no last disagreement. These alternatives need their own definitions, rather than being treated as harmless notational changes.

### 61.15.3 Inherited well-orders, simplicity, and quotienting

A set  $A \subseteq \mathbf{No}$  well-ordered by numerical order has a unique increasing enumeration. This gives a natural restricted suborder of the varying-carrier word order. It excludes almost all arbitrary permutations. The numerical surreal order itself is not a well-order, as witnessed by

$$0 > -1 > -2 > -3 > \dots$$

The simplicity order on sign sequences, by proper prefix extension, is well-founded but partial. Birthday order is only a rank comparison, and equal birthdays require an additional tie-breaker to produce a linear well-order. None of these is the numerical lexicographic order on all well-order presentations.

Forgetting the labels removes the information on which enumeration lexicography depends. All bijections  $\mu \rightarrow X$  represent the same abstract order type  $\mu$ , although their first values can be arbitrary elements of  $X$ . Therefore the strict lex order does not descend to their order-isomorphism quotient. Ordering the resulting ordinal types in their usual ordinal order is legitimate, but it answers a different question. At the global level all proper-class set-like well-orders have type  $\mathbf{Ord}$ ; their type quotient collapses an especially rich labelled order to one type.

### 61.15.4 Relation-code lexicography

Choose a well-order  $q$  of the coordinate set  $X^2$ . A relation  $R$  is coded by its indicator function  $\chi_R : X^2 \rightarrow \{0, 1\}$ , and two relation codes can be compared at their  $q$ -least disagreement. Restricting to well-order relations produces another linear order. It depends on  $q$  and compares incidences, not next enumeration labels.

For example, let  $X = \{0, 1, 2\}$ , let  $R$  enumerate  $0, 1, 2$ , and let  $S$  enumerate  $1, 0, 2$ . Enumeration lexicography gives  $R < S$ . If the first differing coordinate in  $q$  is  $(0, 1)$  and  $0 < 1$  is the bit order, then  $\chi_R(0, 1) = 1$  and  $\chi_S(0, 1) = 0$ , so relation-code lexicography gives  $S < R$ . Both constructions are rigorous, but they are not the same construction.

The Kanovei–Shelah definable-hyperreal construction provides related motivation for lexicographically ordered presentations [16]. Its displayed indexing family consists of maps from the continuum into  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters, with lexicographic comparison of maps. That is not literally the set of all well-orders of  $\mathbb{R}$ . Repeated presentations there are part of a definability device; this distinction should be retained when transferring the idea to surreal labels.

### 61.16 Source 27: the question, its scope, and the audited starting point

[merge] Source 27’s Section 1, printed whole; its map of interpretations (Section 2) is printed in Part XII (section 80.2). Its birthday  $\ell(s)$  is printed  $\text{bd}(s)$  and its  $S_{<\theta}$  is printed  $\mathbf{No}_{<\theta}$ ; in its fixed-set paragraphs the cardinality  $\kappa$  of  $X$  is printed  $\mu$  and the cutoff  $\theta$  is printed  $\kappa$  (section 1.3.2). Its core  $\mathcal{D}_b$  (support codes) is printed  $\mathcal{P}_s(b)$  and its prefix tree  $\mathcal{T}$  of injective words is printed  $\mathcal{I}$ .

The numerical order on  $\mathbf{No}$  is not a well-order: already the finite surreals contain  $0 > -1 > -2 > \dots$ . A well-order of a set of surreal numbers therefore usually means an *additional* relation on those numbers, with their ordinary numerical order retained only for comparing labels. This distinction is the first decision in any rigorous formulation.

The second decision is the object to be compared. A well-order relation can be replaced by its increasing enumeration *in that well-order*, or by a binary table of its ordered pairs. These encodings carry different lexicographic orders. A third decision is size: a set of surreals is a set, the class of all surreals is a proper class, and a collection whose members are arbitrary proper-class relations is not an ordinary class of sets in two-sorted set theory.

The report in ProveIt already treats general-alphabet enumeration spectra, birthday cutoffs, set-length coding, direct comparison of class well-orders, bounded-support cores, represented cuts, several termination conventions, and nonextendible automorphisms of those cores [39, 1]. Re-proving the ordinal spectrum for a fixed set would not by itself constitute meaningful further study. Our main extensions instead concern information discarded by encodings and unlabelled prefix structure, sharper support optimization, and the difference between an ordinary class family and the whole raw order.

#### 61.16.1 What was actually inspected

We inspected the pinned surreal report’s `article.tex`, its README, the neighboring report on real well-orderings, and the Lean modules `SignSequence.lean`, `SignSequenceCut.lean`, `SignSequenceCutOperation.lean`, and `SizeObstructions.lean`. The merged surreal source has an important editorial limitation: it advertises a later Part IX, but its body at this pin contains Parts I–VIII.<sup>12</sup> Advertisements of the missing part are not used as proof evidence. Its dated October 3 status passages also supersede some questions retained from the October 2 question list.

This matters mathematically. The present source already contains the cardinal reserve theorem, the class-cut recognition theorem, and several word isomorphisms. We use those as established inputs of the research corpus, while supplying the arguments required to understand our extensions. We do not label them newly discovered. The exact least ordinal support bound is explicitly left unasserted there; relation-table scan dependence and exhaustive-branch failure for ordered prefix-tree automorphisms go beyond its recorded comparisons.

#### 61.16.2 Notation and conventions

A surreal number is represented by a sign sequence  $s : \alpha \rightarrow \{-, +\}$  for a set ordinal  $\alpha$ . Its birthday is  $\text{bd}(s) = \alpha$ . At the first disagreement we compare signs using  $- < \dagger < +$ , where  $\dagger$  means that the sequence has ended. Thus termination is distinct from both signs. We write

$$\mathbf{No}_{<\theta} = \{s \in \mathbf{No} : \text{bd}(s) < \theta\}.$$

This is a set for every set ordinal  $\theta$ .

<sup>12</sup>[merge] True at 7ff7736ec. Part IX was written before this addition, and the README is updated with it.

A word over a linearly ordered alphabet  $X$  is a function  $p : \alpha \rightarrow X$ . Unless stated otherwise, its length is a set ordinal. The notation  $p \smallfrown q$  denotes ordinal concatenation; ordinal multiplication in all scan and block lengths is genuine ordinal multiplication, not cardinal multiplication. The order  $L \times_{\text{lex}} M$  compares the  $L$  coordinate first. The sum  $\sum_{l \in L} M_l$  is the corresponding lexicographic sum. We write  $L \simeq_e M$  for mutual order embeddability; this never asserts  $L \cong M$ .

We use GB for Gödel–Bernays theory without a choice axiom, GBC for GB with Global Choice, and KM for Kelley–Morse class theory. A *uniformly class-indexed family*  $(R_i)_{i \in I}$  means that  $I$  is an ordinary class of sets and one class relation  $\mathcal{R}(i, x, y)$  gives every section  $R_i$ . The word “uniformly” is a mathematical requirement, not a convenience of notation.

### 61.17 Source 28: what is being continued

[merge] Source 28’s Sections 1 and 2, printed whole. Its  $\mathcal{P}_\mu(X)$ ,  $\mathcal{I}_\mu(X)$  are printed  $\mathcal{W}_\mu(X)$ ,  $\mathcal{I}_\mu(X)$ , and its  $\mathfrak{E}$  is  $\mathfrak{W}_{\text{sl}}$ ; its bounded ideal  $\mathcal{B}_\mu$  is printed  $\mathcal{J}_b$  and its coordinate ideals  $\mathcal{J}$  keep their letter, as in Part X (its macros `\BB` and `\JJ` name other objects here).

The phrase “the lexicographic ordering of all well-orders of surreal numbers” does not determine a unique mathematical structure. A well-order may have a fixed set of labels, a varying set of labels, or the entire surreal class as its carrier. It may be arbitrary or inherited from the numerical order. Labels may be retained or forgotten. A global well-order may or may not have only set-sized proper initial segments. A word may terminate, and the rule for comparing termination with a further label matters.

The ProveIt report [39] develops this taxonomy, set-sized embedding spectra, the singular birthday transition, global supported-permutation cores, and direct comparison of arbitrary class well-orders. The subsequent manuscript [52]<sup>13</sup> distinguishes forward prefix completion from Dedekind completion and derives exact continuity criteria for permutation operations. The present paper takes the latter criteria as a starting point, reproves them in a more general form, and answers several of that manuscript’s questions about the boundedness-preserving subgroup and natural topology refinements. It does not present the earlier completion or regular-cardinal Baire-space theorems as new discoveries.

#### 61.17.1 The principal developments

Let  $\mu$  be an infinite initial cardinal. On  $S_\mu = \text{Sym}(\mu)$ , let  $\tau_b$  be the topology of agreement on bounded initial segments. Put

$$H_\mu = \{p \in S_\mu : p \text{ and } p^{-1} \text{ send bounded sets to bounded sets}\}.$$

The following statements summarize the principal contributions relative to the inspected predecessor material.

**Main results.** For singular  $\mu$ :

- (i)  $H_\mu$  is the largest subgroup containing the finitary permutations on which the inherited  $\tau_b$  topology is a group topology. Its normalizer in  $S_\mu$  is exactly  $H_\mu$ .
- (ii) Its  $\tau_b$ -closure consists of all permutations preserving bounded sets in the forward direction. This closure is nowhere dense. Its forward uniform completion consists of all boundedness-preserving injections.
- (iii) The inverse-boundedness locus is dense and is an intersection of  $\text{cf } \mu$  explicit open sets in

<sup>13</sup>[merge] Source 24 of this report. Source 28 did not see source 26.



the injection completion. If  $\text{cf } \mu = \omega$ , it is a dense completely metrizable  $G_\delta$  subspace.

- (iv) The coarsest group topology containing  $\tau_b$  is  $\tau_{<\mu}$ , agreement on arbitrary subsets of  $\mu$  of cardinality less than  $\mu$ . Its character is

$$\chi(S_\mu, \tau_{<\mu}) = \text{cof}([\mu]^{<\mu}, \subseteq) > \mu.$$

- (v) The group  $(S_\mu, \tau_{<\mu})$  is complete for its two-sided group uniformity. This is a topology-changing repair, not a proof that the original lexicographic topology was already a group topology.

The numerical application uses  $\mathbf{No}_{<\kappa}$ , the surreals with sign-sequence length less than an infinite initial cardinal  $\kappa$ . Its cardinality is  $\mu = 2^{<\kappa}$ . The topology of its minimal exhaustive-enumeration slice  $\text{Bij}(\mu, \mathbf{No}_{<\kappa})$  is precisely the prefix topology. The singularity relevant to the main results is therefore the singularity of  $2^{<\kappa}$ , not merely that of  $\kappa$ .

### 61.17.2 Assumptions and limits

All set-sized results are proved in ZFC. We write GB for Gödel–Bernays class theory without building a choice principle into the abbreviation, and GBC for GB with Global Choice. Our global constructions use GBC and a specified baseline enumeration. An assertion about a family of proper classes is stated using one uniform class relation, or explicitly in an external metatheory. Neither GBC nor KM makes a proper class an element of another class.

The remaining complete-metrizability question for the *full* singular forward permutation space is not settled by the dense complete subspace constructed here. We state that question precisely in Section 113.22.<sup>14</sup> Likewise, self-normalization in the natural permutation action is not a classification of all abstract automorphisms of  $H_\mu$ .

## 61.18 The interpretations that must not be identified

### 61.18.1 Fixed sets of labels

Let  $(X, <)$  be a linearly ordered set. A labelled well-ordering  $R$  of  $X$  has a unique exhaustive ordinal enumeration

$$f_R : \alpha_R \longrightarrow X,$$

which is an isomorphism from the ordinal  $\alpha_R$  to  $(X, R)$ . There are two orders here:  $R$  tells us when a label is listed, whereas  $<$  tells us which differing label is numerically smaller. The enumeration need not be increasing for  $<$ .

Two different exhaustive enumerations of the same carrier cannot be proper prefixes of one another. A completed enumeration has already used every available label. Hence, for  $R \neq T$ , a first disagreement  $\delta$  exists and we define

$$R <_{\text{lex}} T \iff f_R(\delta) < f_T(\delta).$$

The standard least-disagreement argument proves a linear order. For example, in a comparison of three words, the smaller of the first-disagreement indices for the first two comparisons determines

<sup>14</sup>[merge] Answered negatively, in ZFC, by source 26, which source 28 did not see: for uncountable  $\mu$  of countable cofinality,  $\mathcal{W}_\mu(X)$  is not completely metrizable, and no nonempty open subspace of it is (theorem 69.18); see note 70.13 and the status of source 28's question 1 in section 113.12. The statement that source 28's own dense complete subspace does not settle it stays true.

the outer comparison; when those indices agree, transitivity of the alphabet order applies. No end-marker convention is required.

We distinguish the full order

$$\mathcal{W}(X) = \{R : R \text{ is a well-order on } X\}$$

from a fixed-length slice  $\mathcal{W}_\alpha(X) = \text{Bij}(\alpha, X)$ , and especially from the minimal slice

$$\mathcal{W}_\mu(X) = \text{Bij}(\mu, X), \quad \mu = |X| \text{ initial.}$$

The full carrier is a set: it is a subclass of the set  $\mathcal{P}(X \times X)$ . The apparent union over ordinal lengths does not make it a proper class. The minimal slice, however, has a uniform supply of unused labels after each proper prefix, which need not hold for other slices.

### 61.18.2 Varying set carriers and termination

A labelled well-order on some set of surreals is represented by an injective set-length word  $p : \alpha \rightarrow \mathbf{No}$ . Such words themselves have set codes, so the collection of all of them is an ordinary class. Now proper-prefix comparisons occur. End-first dictionary order places a word below every proper extension; end-last places it above every proper extension. In particular, the empty word is respectively least or greatest.

A central termination convention puts an end symbol at a chosen cut of the alphabet. On nonzero surreal labels, for example, one may place termination between the negative and positive labels. For fresh labels this gives

$$p \frown \langle -1 \rangle < p < p \frown \langle 1 \rangle.$$

The sign representation of an individual surreal uses the central convention  $- < \text{termination} < +$  on the *two-sign alphabet*. It is not the same as end-first dictionary order on *surreal-valued* words. Allowing repeated labels makes a word space rather than a space of labelled well-orders. Comparing length first makes a shortlex-type order rather than first-disagreement lexicography. These are legitimate alternatives, not interchangeable notations.

### 61.18.3 Inherited numerical well-orders

A different question asks for subsets that are already well-ordered by the numerical order of  $\mathbf{No}$ . Their increasing enumerations form a restricted collection of injective words. The full numerical surreal order is not well-founded, since it contains

$$0 > -1 > -2 > \dots$$

Even a numerically bounded class well-order can fail to be set-like. For nonnegative surreals  $u < v$ , ordered-field arithmetic gives

$$\frac{v}{v+1} - \frac{u}{u+1} = \frac{v-u}{(u+1)(v+1)} > 0.$$

Thus  $\{\alpha/(\alpha+1) : \alpha \in \text{Ord}\}$  has increasing type  $\text{Ord}$  and lies below 1. Appending 1 gives type  $\text{Ord} + 1$ ; the last point has a proper class of predecessors. Numerical boundedness does not imply set-likeness.

#### 61.18.4 Global class orders, codes, and quotients

A class order is *set-like* when every element has a set of predecessors. A set-like well-order of a proper class has an ordinal-class enumeration. For the whole surreal class, this produces the virtual domain of class bijections  $\text{Ord} \rightarrow \mathbf{No}$ . Arbitrary class well-orders can have proper-class initial segments and types such as  $\text{Ord} + 1$  or  $\text{Ord} + \text{Ord}$ ; ordinal-class words do not exhaust them.

There are also representation-dependent alternatives. After choosing an enumeration of pairs of labels, one can compare the characteristic functions of the well-order relations themselves. This compares the first differing *relation bit*, not the first differing label in an exhaustive list. Section 71.4 gives an explicit finite discrepancy.

Finally, forgetting labels and taking order-isomorphism types changes the problem drastically. All set-like global well-orders have type  $\text{Ord}$ , so their unlabelled quotient has one element. The rich lexicographic structure belongs to presentations. Lexicographic order generally does not descend to the quotient: differently labelled presentations of the same ordinal may occur on both sides of another presentation.

| Interpretation         | Objects                                               | Essential qualification                        |
|------------------------|-------------------------------------------------------|------------------------------------------------|
| Fixed set $X$          | Well-order relations or exhaustive ordinal lists      | No proper-prefix ambiguity.                    |
| Minimal slice          | Bijections $\mu \rightarrow X$ , $\mu =  X $          | The subject of the topology and group results. |
| Varying set carriers   | Injective set-length surreal words                    | A termination rule is needed.                  |
| Inherited order        | Numerically increasing enumerations                   | Not arbitrary labelled well-orders.            |
| Set-like global orders | Class bijections $\text{Ord} \rightarrow \mathbf{No}$ | A virtual higher-order domain.                 |
| All class well-orders  | Possibly non-set-like relations                       | Common prefixes can be proper classes.         |
| Relation-bit codes     | Characteristic functions of relations                 | Depends on a coordinate enumeration.           |
| Order-type quotient    | Unlabelled isomorphism types                          | Numerical lexicography is not inherited.       |

#### 61.19 Source 29: the question, the audit, and the new results

[merge] Source 29's Sections 1 and 3, printed whole except Subsection 3.3; its interpretations (Section 2) and Subsection 3.3 are printed in Part XII (sections 80.3 and 80.3.5). Its  $\mathbf{T}$  (the shorter-first order of all set-length surreal words, the type  $\mathbf{T}_{\min}$  of source 16's theorem 36.2) is printed  $\mathbf{T}_{\min}$  and its  $\mathbf{P} = \mathbf{T} \setminus \{\emptyset\}$  is printed  $\mathbf{T}_{\min}^+$ ; its  $\otimes_{\text{lex}}$  is printed  $\times_{\text{lex}}$ . Its internal labels  $\mathbf{gl}$ : stay under its own prefix ( $\mathbf{swo:bw:gl}$ ).

Lexicographic comparison combines two orders with different jobs. An auxiliary well-order says which label is visited first, second, and so on. The numerical order on the labels decides which of two visits comes earlier at their first disagreement. There is no reason for the resulting order on presentations to be a well-order. When the labels are surreal numbers, a further issue intervenes: all surreal numbers form a proper class. The well-orders of a fixed set of surreals and the well-orders of the entire surreal class therefore have fundamentally different logical sizes.

These observations do not prevent a substantial theory. They tell us which carriers, parameters, and axioms must be written into a theorem. The present continuation develops that theory in several directions that are not settled by the inspected merged article. Its main theme is the

amount of information that a chosen order or presentation keeps. The numerical birthday alphabet remembers its exact cutoff; some permutation-stratum embedding invariants forget it. An ordered prefix tree remembers every local numerical comparison but can still forget whether a class branch visits every label. A class condensation history has a finite description at each individual point, while its existence for all class well-orders has nontrivial foundational strength.

### 61.19.1 What was actually inspected

The source audit used the GitHub repository at a fixed commit, not a moving branch name. Sixteen source files were retrieved and their bytes checked against the Git blob hashes returned by GitHub. The central research file is

```
Algebra/SurrealNumbers/docs/foundations-and-computation/
surreal-well-orders/article.tex.
```

At the pin, its blob is `f3f69663cd63682ad2e1db808bcb3c6e8fd940ce` and its size is 903,287 bytes. The research README still describes an earlier 116-page, four-source merge.<sup>15</sup> The actual article already incorporates substantial additions from six further manuscripts, called sources 13–18 in its editorial apparatus. Its Parts VI–VIII supersede several entries in the older research-question list. The introduction also mentions an intended Part IX that is absent from the pinned body ([merge] Part IX was written before this addition). Accordingly we use actual theorem statements and proofs, together with explicitly identified earlier manuscripts, rather than infer established results from a roadmap or README [39, 47, 49, 48].<sup>16</sup>

This distinction changes what counts as further study. The pinned article already proves the general set-alphabet ordinal cutoff, substantial binary-cube spectra, fixed-stratum density and cellularity, minimal-stratum gap spectra, word-termination classifications, a raw comparison of arbitrary class well-orders, and a detailed represented-cut theory for a bounded-support core. Those results are recalled as background. We do not present them again as new discoveries.

### 61.19.2 A guide to the extensions

The following results organize the continuation.

1. **Exact numerical birthday spectra.** For every pair of set ordinals  $\beta, \theta$ ,

$$\mathbb{B}_\beta \hookrightarrow \mathbf{No}_{<\theta} \iff \beta < \theta.$$

This gives the exact numerical embeddability hierarchy and its least binary coding lengths, including noninitial ordinal cutoffs.

2. **Residual strata at noninitial birthdays.** We extend the earlier variable-residual analysis to every noninitial birthday cutoff. In particular, without the continuum hypothesis, all countable noninitial birthday alphabets have the same fixed-stratum equimorphism type at each common admissible enumeration length. This is a statement about mutual embeddings, not order isomorphism.
3. **An exact centered-word product.** Increasing words split at their termination cut. At a numerical threshold this yields  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ , and a small finite-fiber calculation classifies

<sup>15</sup>[merge] True at `7ff7736ec`; the README is updated with this addition.

<sup>16</sup>[merge] The three earlier manuscripts are sources 19, 21 and 20 of this report; source 29's manifest (`data/29-birthdays-source_manifest.json`) records their SHA-256, which equal the delivered texts. Source 29 did not see sources 22–28 or 30.

the zero conventions up to isomorphism. Translation disposes of thresholds at other actual surreal numbers; nonprincipal class cuts remain genuine parameters.

4. **New behavior at proper-class carriers.** Numerically well-ordered proper subclasses can have unattained surreal suprema. Exactly these create adjacent pairs under shorter-first comparison. This destroys a symmetry present for all set-length words and obstructs extending the set-prefix tree isomorphism to full class carriers.
5. **Topology and Borel structure after finite tails.** Over a countable dense alphabet, the types  $\omega$ ,  $\omega + 1$ ,  $\omega + 2$ , and  $\omega + n$  for  $n \geq 3$  exhibit different relations between native lexicographic topology and coordinate topology. The resulting distinction is measurable as well as topological.
6. **Prefix-tree and foundational limitations.** Uniform tree automorphisms can exchange exhaustive and nonexhaustive class branches. Separately, canonical terminating condensation histories are equivalent over GBC to strong class-well-order comparability. Neither statement conflates the existence of a class object with the existence of a class containing all such objects.

### 61.19.3 The Kanovei–Shelah connection

The historical motivation needs a precise formulation. In the preprint of Kanovei and Shelah, the relevant index set consists of maps

$$a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N}) \quad \text{whose ranges are ultrafilters on } \mathbb{N}.$$

The maps are compared at their first unequal coordinate, using the characteristic-function order on  $\mathcal{P}(\mathbb{N})$ . Repeated values are allowed. These are redundant ultrafilter presentations, not literally all well-orders of the real line. The index order is linear but is not itself required to be well-founded. Its role is in the definable nonstandard-real construction [16, §1 of the preprint]. What transfers here is the method of comparing sufficiently rich presentations lexicographically. No identification of the present surreal orders with that index order is assumed.

### 61.20 Source 29: the fixed-set baseline and what it does not identify

[merge] Source 29’s Section 3, printed whole. Its fixed set has cardinality  $\kappa$ ; following [section 1.3.1](#) it is printed  $\mu$  in the first two subsections (the birthday-fragment formulas keep  $\kappa$  as a cutoff). Its Theorem 3.1 is [theorem 6.1](#) and is recorded as a note.

#### 61.20.1 Exhaustive enumerations

For a linearly ordered set  $(X, <)$ , let

$$\mathcal{W}_\alpha(X) = \{f : \alpha \longrightarrow X : f \text{ is a bijection}\}, \quad \mathcal{W}(X) = \bigcup_{\alpha \in \text{Ord}} \mathcal{W}_\alpha(X).$$

If  $|X| = \mu$  is infinite, only  $\mu \leq \alpha < \mu^+$  can occur, so this union is a set. No exhaustive enumeration can be a proper prefix of another. Thus for distinct  $f, g$  there is a least coordinate at which both are defined and differ, and

$$f <_{\text{lex}} g \iff f(\delta) < g(\delta) \quad \text{at that coordinate } \delta.$$

The termination convention is immaterial here. It becomes relevant as soon as incomplete words or different carriers are admitted.

*Note 61.4* (Source 29, Theorem 3.1, “Inherited fixed-set spectrum”). *For every infinite linearly ordered set  $X$  of cardinality  $\mu$ ,*

$$|\mathcal{W}(X)| = 2^\mu, \quad \beta \hookrightarrow \mathcal{W}(X) \iff \beta < \mu^+$$

*for set ordinals  $\beta$ . Every linear order of size at most  $\mu$  embeds in the minimal stratum  $\mathcal{W}_\mu(X)$ . The reverse of every ordinal below  $\mu^+$  embeds as well, while neither direction admits a monotone sequence of length  $\mu^+$ .* This is [theorem 6.1](#) (the reals report’s `two:thm:ordinal-spectrum` for every infinite alphabet), as source 29 says: “This is established in the repository [39, 1]; we include the main mechanism because it underlies the later distinctions.” Its summary of the mechanism follows; it is source 09’s stabilization route ([theorem 6.1](#) and the note after it), not a new proof.

Partition  $X$  into pairs  $\{a_\xi, b_\xi\}$ , with  $a_\xi < b_\xi$ , indexed by  $\xi < \mu$ . A binary word chooses whether each pair is listed as  $(a_\xi, b_\xi)$  or  $(b_\xi, a_\xi)$ . Their concatenation has length  $2 \cdot \mu = \mu$  and preserves first-difference comparison. Hence  $\mathbb{B}_\mu \hookrightarrow \mathcal{W}_\mu(X)$ .

If  $L$  is any linear order of size at most  $\mu$ , enumerate its points among  $\mu$  coordinates and map  $x \in L$  to the bits recording whether the corresponding point is strictly below  $x$ . Unused coordinates are fixed at zero. For  $x < y$  all changing bits move from zero to one, and the coordinate naming  $x$  changes. This embeds  $L$  into  $\mathbb{B}_\mu$ . It also gives the lower cardinality bound  $2^\mu$  for the minimal stratum. The upper bound follows because well-orders are relations on  $X$  and there are only  $2^\mu$  such relations.

For the upper ordinal bound, suppose a strictly increasing sequence  $(f_i)_{i < \mu^+}$  existed. The first coordinates form a nondecreasing sequence in a set of size  $\mu$  and therefore are eventually constant. Continue by transfinite induction: after a common prefix of length  $\gamma < \mu^+$  has been fixed on a tail, the next coordinates are eventually constant on that tail. At a limit, regularity of  $\mu^+$  allows the at most  $\mu$  preceding tail bounds to be combined. The chosen labels are distinct. If a common prefix exhausts  $X$ , all subsequent  $f_i$  are identical, a contradiction. If none does, the construction chooses  $\mu^+$  distinct elements of  $X$ , also a contradiction. Reverse all comparisons for descending sequences. The lower ordinal assertion follows by embedding any order of cardinality at most  $\mu$ .

### 61.20.2 Why cardinality and abstract types are insufficient

The same theorem applies to many numerically different alphabets. For birthday fragments,

$$|\mathbf{No}_{<\kappa}| = 2^{<\kappa} \quad (\kappa \text{ infinite initial}), \quad |\mathbf{No}_{<\theta}| = 2^\kappa \quad (\kappa < \theta < \kappa^+).$$

The second formula follows because the full sign level of length  $\kappa$  is present and there are at most  $\kappa$  levels, each of size at most  $2^\kappa$ . It does not imply that the numerical alphabets are equimorphic. The next section proves exactly the contrary ([\[merge\] section 76](#), Part XI).

Nor does enumeration lexicography descend to abstract well-order types. Two presentations of the same type can fall on opposite sides of a third presentation after a change in the early labels. Ordering the possible types themselves by ordinal comparison gives the interval of types  $\mu \leq \alpha < \mu^+$ , whose order type is  $\mu^+$ . By [Note 61.4](#), that type cannot embed in the labelled order. For set-like global well-orders, quotienting by abstract class-order isomorphism collapses all objects to the single type  $\text{Ord}$ .

[\[merge\]](#) Source 29’s Subsection 3.3, “Relation scans and inverse ranks”, is printed in Part XII with the scans ([section 80.3.5](#)).

### 61.21 Source 30: scope, results, and the audited starting point

[merge] Source 30's Section 1, printed whole; its interpretations (Section 2) are printed in Part XII (section 80.4). Its  $\mathfrak{W}_{\text{sl}}$ ,  $\mathfrak{W}_{\text{all}}$  (macros `\SL`, `\ALL`) are  $\mathfrak{W}_{\text{sl}}$ ,  $\mathfrak{W}_{\text{all}}$ ; its  $X_\kappa$  is printed  $\mathbf{No}_{<\kappa}$  and its countable strata  $P_n = \mathcal{W}_{\omega+n}(\mathbb{Q})$  are printed  $P_{\omega+n}$ , as in source 29 (section 1.3.2).

Two choices precede every substantive question in this subject: *what objects are being ordered*, and *which representation is scanned*. A well-order of a fixed set can be represented by its ordinal enumeration, its relation table, or the ranks of labels in a previously fixed list. These representations give different comparisons. On the full surreal class, an individual well-order is itself a class relation, so the collection of all such relations is not an ordinary class of sets.

This article continues the merged ProveIt report *Lexicographic Orders of Well-Orderings of the Surreal Numbers* [39], inspected at the fixed revision

7ff7736ecf3a0adc8536b2d083cf819e1ba39ace.

Its location is

Algebra/SurrealNumbers/docs/foundations-and-computation/  
surreal-well-orders/article.tex.

We also inspected its guide and the four foundational Lean files listed in section 111.47. Some questions retained from the first wave are answered elsewhere in the merged text or in subsequent reports. Appearance in an old research list is therefore not, by itself, evidence that a problem remains open.

#### 61.21.1 The inherited baseline

The previous work develops fixed-set cardinalities and ordinal spectra; the singular strong-limit birthday transition; set-length words and termination conventions; the equivalence of global choice with the existence of any class well-order of  $\mathbf{No}$ ; direct comparison through proper-class common prefixes; and a bounded-support core, isomorphic to  $\mathbf{No}$  under global choice, together with cut recognition and automorphism-extension obstructions. Finite residual permutation blocks, non-monotone restrictions, class-model absoluteness, and countable external classifications have also already been studied. We do not present these as new findings.

#### 61.21.2 Four continuations

Throughout,  $\simeq_e$  means *equimorphism*: order embeddings exist in both directions. It does not mean isomorphism.

**Infinite remainders.** For an infinite cardinal  $\kappa$ , put

$$\mu = 2^{<\kappa}, \quad \rho_\kappa = \begin{cases} \kappa \cdot \kappa, & \kappa \text{ singular strong limit,} \\ \mu, & \text{otherwise.} \end{cases}$$

For every infinite regular  $\nu < \mu$ , every  $0 < \gamma < \mu^+$  and  $0 < \delta < \nu^+$ , and every finite  $n$ , we prove

$$\mathcal{W}_{\mu \cdot \gamma + \nu \cdot \delta + n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\rho_\kappa \cdot \gamma + \nu \cdot \delta} \times n!. \quad (\text{S30.1.1})$$

Here  $n!$  is a finite chain; products of orders are lexicographic; subscripts use ordinal arithmetic. In particular  $\nu = \omega$  handles every countably infinite remainder. The argument iterates through



finite descending sequences of residual cardinals. A singular strong-limit block of size  $\nu$  instead contributes  $\nu \cdot \nu$  per block. Mixing these two regimes gives a complete fixed-type equimorphism classification under GCH (note 64.18). The remaining unconditional boundary is a singular residual cardinal at most  $\kappa$  that is not strong limit. Residual cardinals above  $\kappa$  also have uninflated codes, even when singular. In particular, every fixed type is classified in ZFC whenever  $\kappa < \aleph_\omega$ , including  $\kappa = \omega_1$  without CH (corollary 64.20).

[merge] Source 19 proves the same classification independently and more generally (cube spectrum also at singular non-strong-limit layers): theorems 64.7 and 64.11. Source 30's explicit ZFC corollary for  $\kappa < \aleph_\omega$  is the one statement source 19 does not spell out (source 19 writes  $\mu \leq \aleph_\omega$ ). Neither source saw the other.

**Topological and measurable transitions.** For  $P_{\omega+n} = \mathcal{W}_{\omega+n}(\mathbb{Q})$ , the order topology is Polish and homeomorphic to Baire space at  $n = 0$ . At  $n = 1$  it is still second countable and metrizable but is meagre in itself. At  $n = 2$  it is separable and first countable, but nonmetrizable, and its Borel structure is not standard. At  $n \geq 3$  it has continuum many isolated points forming a dense open set and becomes a Baire space again. The maps between relation tables and enumerations have different, sharp measurability thresholds. Since  $\mathbf{No}_{<\omega}$  is a countable dense order without endpoints, the same results hold at the first infinite surreal stage  $\mathbf{No}_{<\omega}$ .

**Partial relation data and global type.** A set-sized partial table extends to a well-order of *any supplied proper-class well-order type* precisely when its forced orientations are well-founded, together with diagonal consistency. Each admitted global type is dense for set-supported relation observations. Finite observations approximate all linear orders; countable observations already detect ill-foundedness, but cannot detect set-likeness or the global class order type.

**Recovery of the class universe.** For GBC realizations of the same set model, inclusion of the admitted labelled global enumeration families is equivalent to inclusion of their whole class collections. Adjacent pair swaps along one common baseline already encode every class. The pure external order loses this information; the evaluation relation is essential.

### 61.21.3 The precise Kanovei–Shelah connection

Kanovei and Shelah lexicographically order maps  $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters [16, Section 2] ([merge] Section 2 of the published paper; Section 1, page 2, of the preprint). Such maps can repeat values. Their index order is not literally the set of all bijective well-orderings of  $\mathbb{R}$ , or all class well-orders of  $\mathbf{No}$ . The common idea is a well-ordered coordinate domain and a linearly ordered alphabet. Keeping the actual indexing object straight prevents a misleading identification at the start of the surreal generalization.

*Note 61.5* ([merge]; what the twelve maps repeat). All twelve separate the carriers of the batch-80 and batch-81 sources (sections 2 and 4.3 and note 34.2): a fixed set with all types or one type, varying set carriers (injective words, with a termination convention), words with repetitions, numerically well-ordered subsets and subclasses, set-like and arbitrary class well-orders, relation codes, abstract types, and, in sources 20, 21, 26, 27, 29 and 30, the universe- or model-relative versions.

- *Fixed carriers.* That two exhaustive enumerations of one set are never proper prefixes of one another, so that no termination convention is needed there, is argued again by sources 20, 21, 22, 24, 25, 26, 27, 28 and 30, and stated as source 23's note 80.1; it is the prefix-freeness of section 6 and source 13's lemma 34.1.

- *Termination.* Sources 20–28 and 30 name the three placements of a terminator (first, last, inside the alphabet) and length-first order as a further choice; sources 21, 24, 25, 26, 28 and 30 repeat source 14’s warning that the Conway sign order  $- < \mathbf{end} < +$  is not prefix-first dictionary order (section 5). Source 19 and source 27 describe the centered terminator  $\dagger$  of source 18 (section 36.5).
- *Increasing subclasses.* Sources 19, 20, 21, 24, 27, 28 and 30 show that a numerically well-ordered subclass need not be set-like by a copy of Ord below a point, as source 12 did with  $\{-1/(\alpha + 1) : \alpha \in \text{Ord}\} \cup \{0\}$  (section 18); sources 21 and 30 use that example, sources 24 and 28 use  $\{\alpha/(\alpha + 1)\} \cup \{1\}$ , source 19  $\{1 - 1/(\alpha + 2)\} \cup \{1\}$ , source 20 the sign sequences  $s_\alpha$  below 0, and source 27 any increasing copy of Ord inside  $(0, 1)$  followed by 1; sources 23 and 29 state it.
- *Relation codes and inverse ranks.* The finite examples of sources 21, 26 and 30 make the point of source 12’s shipped example and of proposition 17.1; source 15 proves theorems about these orders (theorem 38.2 and corollary 38.3), and sources 27 and 30 prove further ones (Part XII). Source 30’s three scans of six objects are checked by its finite program (section 111).
- *Abstract types.* That enumeration lexicography does not descend to abstract order types (all twelve; source 19’s  $R < S < T$  and source 21’s example 61.1 are explicit non-convexity examples) is printed with source 15’s example in section 38.
- *Kanovei–Shelah.* All twelve make source 08’s point (section 4.4): the index there is a lexicographically ordered family of redundant ultrafilter codes, not a space of well-orders. Ten do so in their opening sections (printed here, or in Part XII for source 27); sources 23 and 28 do so in their audit and foundations sections (section 111 and Part X). Sources 19 and 30 cite Section 2 of the published paper; sources 20, 21, 25, 27 and 29 cite Section 1 of the preprint, and source 23 its second page, as this report’s bibliography does; sources 22, 24, 26 and 28 give no section.
- *Class levels.* All twelve fix GB without choice and  $\text{GBC} = \text{GB} + \text{GC}$  as in section 1.3, read “the class of all class well-orders” as a predicate on class variables, and require uniform families to be coded by one class relation; their foundations sections are printed in section 111.

## 62 Thick alphabets, prefix cylinders and the prefix topology (source 24; sources 26, 28, 21, 22 and 19)

[merge] Source 24’s Sections 3 and 4 are printed, with its re-proofs of results of this report as notes. Sources 26, 28, 21 and 22 open their set-sized parts with the same toolkit; their versions follow in notes, except source 26’s statement that every cylinder is homeomorphic to the whole space and source 21’s remark on descending chains, which are printed. Source 24’s  $\mathcal{P}_\mu(X)$  and  $\mathcal{T}_\mu(X)$  are printed  $\mathcal{W}_\mu(X)$  and  $\mathcal{I}_{<\mu}(X)$  (part X). The word *thick* is source 24’s; source 22 says  $\mu$ -rich, source 21 *uniformly large intervals*, for the same hypothesis.

A surreal can be represented by a set-length sign sequence; see also Ehrlich [15] for the broader arithmetic and foundational setting. Write  $\text{bd}(x)$  for its length, and for an infinite initial cardinal  $\kappa$  put

$$\mathbf{No}_{<\kappa} = \{x : \text{bd}(x) < \kappa\}, \quad \mu = 2^{<\kappa} = \sup_{\alpha < \kappa} 2^{|\alpha|}.$$

All cardinal arithmetic in this section is external ZFC arithmetic. There is no assumption that  $\kappa$  is regular, that  $\mu = \kappa$ , or that a birthday cutoff is a subfield.

**Definition 62.1** (Thickness). A linear order  $X$  of infinite cardinality  $\mu$  is  $\mu$ -thick if it is dense without endpoints and every nonempty open interval and every ray has cardinality  $\mu$ .

*Note 62.2* (Source 24, Lemma 3.2, “Cutoff size and thickness”). For every infinite initial cardinal  $\kappa$ , the alphabet  $\mathbf{No}_{<\kappa}$  has cardinality  $2^{<\kappa}$  and is  $2^{<\kappa}$ -thick. Removing fewer than  $\mu$  points from a  $\mu$ -thick alphabet leaves a  $\mu$ -thick alphabet of the same cardinality. This is lemma 29.2 (source 17), by the same sign-cone argument. Source 24 makes the cone explicit in three cases: if neither of  $a < b$  is a prefix of the other, or if  $b$  is a proper prefix of  $a$ , every extension of  $r = a \frown (+)$  lies in  $(a, b)$ ; if  $a$  is a proper prefix of  $b$ , the next sign of  $b$  is plus and  $r = a \frown (+) \frown (-)^{\text{bd}(b)+1}$  works, since “either a later plus of  $b$  gives the first strict comparison, or the negative tail extends past the end of  $b$ ”. It stresses that “finite ordinal sums of ordinals below an infinite initial cardinal remain below it; regularity is not needed here”, and obtains the deletion clause by applying the cardinality count inside each old interval and ray. Source 15’s three-case construction (note 29.8) uses  $b \frown -$  in the second case.

*Remark 62.3* (What the arithmetic parameters do and do not control). The previous cutoff study (source 17; lemma 29.3 and theorem 30.2) distinguishes  $\text{cf}(2^{<\kappa})$ , which controls several alphabet-side characters, from  $\text{cf}(2^{<\kappa})$ , which controls prefix-neighborhood characters in the minimal enumeration slice. The topology below uses  $\mu = |X|$  as its domain length. Replacing it by the birthday bound  $\kappa$  without checking equality would change the problem.

The proof also explains why the relevant formalization should begin with sign sequences and a linear-order interface, not with the whole surreal field. Addition, multiplication, normal forms, and real closedness are unnecessary for the main results.

## 62.1 The minimal slice: prefixes and elementary spectra (source 24)

[merge] Source 24’s Section 4. Its prefix lemmas hold for every infinite alphabet and, under thickness, for the order topology; for  $X = \mathbf{No}_{<\kappa}$  and the minimal slice they are lemma 30.1 (source 17). New are the injection space  $\mathcal{I}_\mu(X)$  beside  $\mathcal{W}_\mu(X)$ , the metrizable criterion and the relabelling corollary.

For an infinite  $\mu$  and a set  $X$  of size  $\mu$ , let

$$\mathcal{I}_{<\mu}(X) = \bigcup_{\alpha < \mu} \text{Inj}(\alpha, X).$$

For  $p \in \mathcal{I}_{<\mu}(X)$ , define its cylinders in the two spaces by

$$[p]_P = \{f \in \mathcal{W}_\mu(X) : p \subseteq f\}, \quad [p]_I = \{h \in \mathcal{I}_\mu(X) : p \subseteq h\}.$$

The subscript will be omitted when the ambient space is clear.

**Lemma 62.4** (Small-prefix extension). Every  $p : \alpha \rightarrow X$  in  $\mathcal{I}_{<\mu}(X)$  extends to a member of  $\mathcal{W}_\mu(X)$ . Moreover, if  $z \notin \text{ran } p$ , it extends to a bijection  $\mu \rightarrow X \setminus \{z\}$ , viewed as a nonexhaustive member of  $\mathcal{I}_\mu(X)$ .

*Proof.* The unused labels have cardinality  $\mu$ , even after the one additional deletion. The ordinal tail  $[\alpha, \mu)$  has order type  $\mu$ : its cardinality is  $\mu$ , its type is at most  $\mu$ , and  $\mu$  is initial. Choose a bijection between the tail and the required unused labels and append it to  $p$ .  $\square$

**Lemma 62.5** (Cylinders and order topology). Suppose  $X$  is  $\mu$ -thick. In both  $\mathcal{W}_\mu(X)$  and  $\mathcal{I}_\mu(X)$ , cylinders are nonempty convex clopen sets without endpoints and form a base for the lexicographic order topology. Both orders are dense without endpoints.

*Proof.* If a word lies lexicographically between two words sharing  $p$ , it shares  $p$ ; otherwise its first disagreement with  $p$  puts it on the same exterior side of both words. Thus cylinders are convex. The complement of a length- $\alpha$  cylinder is the union of the other length- $\alpha$  cylinders.

Let  $f$  extend  $p$ , where  $\text{dom } p = \alpha$ . Fewer than  $\mu$  labels have been used. Choose unused labels  $u < f(\alpha) < v$ , and complete  $p \frown (u)$  and  $p \frown (v)$  using Lemma 62.4. These give points of  $[p]$  on both sides of  $f$ . Hence  $[p]$  is open and has no endpoints; its complement is open as well.

If  $g < f < h$ , take a prefix of  $f$  extending beyond its first disagreements with  $g$  and  $h$ . Its cylinder is contained in  $(g, h)$ . Rays are handled using one disagreement. This proves the base assertion. Finally, at the first differing coordinate of any two words, an unused label strictly between their two labels gives an intermediate exhaustive completion. The same construction gives points above or below any given word.  $\square$

*Note 62.6* (Source 24, Proposition 4.3, “Cardinality and exact ordinal spectrum”). *For any infinite linearly ordered alphabet  $X$  of size  $\mu$ ,  $|\mathcal{W}_\mu(X)| = 2^\mu$ . Every linear order of cardinality at most  $\mu$  embeds into  $\mathcal{W}_\mu(X)$ . An ordinal  $\beta$  embeds into this lexicographic order exactly when  $\beta < \mu^+$ ; the same bound holds for reverse ordinal orders.* This is the set baseline (theorem 6.1; reals report `lwo:prop:cardinality`, `lwo:thm:ordinal-spectrum`) on the minimal slice, by the pair-orientation, initial-segment and coordinate-stabilization argument of section 6; it is note 30.4 without its interval clause. Source 24 adds: “For comparison, the predecessor report proves that the full fixed-carrier order  $\mathcal{W}(X)$  also has cardinality  $2^\mu$  and exact ordinal spectrum  $\{\beta : \beta < \mu^+\}$ . The positive embeddings already occur in the minimal slice. The upper-bound argument for the full varying-length space is a separate result, not an automatic consequence of the fixed-coordinate proof above. Equal cardinality and ordinal spectrum do not identify their topologies or their adjacency structure.”

[merge] The character assertion of the next proposition is theorem 30.2 (point character  $q = \text{cf } \mu$ ) for an arbitrary alphabet; the metrizable criterion is source 24’s.

**Proposition 62.7** (Local character and metrizable). *In the prefix topology on  $\mathcal{W}_\mu(X)$ , each point has neighborhood character  $\text{cf } \mu$ . The space is metrizable exactly when  $\text{cf } \mu = \omega$ . The same holds for  $\mathcal{I}_\mu(X)$ .*

*Proof.* Restrictions to any cofinal sequence of prefix lengths give a base of size  $\text{cf } \mu$ . Conversely, for every member  $U$  of a local base at  $f$ , choose a cylinder  $[f \upharpoonright \alpha_U] \subseteq U$ . If there were fewer than  $\text{cf } \mu$  members, the lengths would be bounded by some  $\gamma < \mu$ . Every such  $U$  would contain  $[f \upharpoonright \gamma]$ . No one of them could be contained in the strictly smaller neighborhood  $[f \upharpoonright (\gamma + 1)]$ , a contradiction. The strictness follows by changing coordinate  $\gamma$  and completing the resulting prefix.

When  $\text{cf } \mu = \omega$ , fix increasing cofinal lengths  $\alpha_n < \mu$ . Give two distinct words distance  $2^{-n}$ , where  $n$  is the least index at which their restrictions to  $\alpha_n$  differ. This is a compatible ultrametric. If  $\text{cf } \mu > \omega$ , the space is not first countable and hence not metrizable.  $\square$

**Corollary 62.8** (The topology forgets the numerical labels). *A bijection  $a : X \rightarrow Y$  induces a homeomorphism  $f \mapsto a \circ f$  of the corresponding prefix spaces. If both alphabets are  $\mu$ -thick, this is a homeomorphism of their lexicographic order topologies. It is an order isomorphism precisely when  $a$  is increasing.*

*Proof.* Relabelling carries cylinders to cylinders. An increasing relabelling preserves first disagreement. Conversely, use exhaustive enumerations with prescribed distinct first labels to see that an order-preserving relabelling must preserve the alphabet order.  $\square$

This is the first indication that a topological classification and an order classification answer different questions. Numerical comparison is indispensable for the order, but under thickness it is invisible to the topology of the minimal slice.

## 62.2 The same toolkit in sources 26, 28, 21, 22 and 19

[merge] The other sources' versions of the toolkit, in notes; source 26's Proposition 3.2 and source 21's remark on descending chains are printed.

*Note 62.9* (Source 26, Section 3: Lemma 3.1, Definition 3.3 and Theorem 3.4). Source 26 writes  $N_p$  for the cylinder  $[p]$  and works with  $|X| = \mu$  (its  $\kappa$ ). Its Lemma 3.1 (“Extension of a short prefix”): *every admissible short prefix extends to a bijection  $\mu \rightarrow X$ ; every cylinder  $N_p$  is nonempty and contains two extensions whose values at the next coordinate are any prescribed distinct unused labels*. Its Definition 3.3 is thickness, and its Theorem 3.4 (“When the two topologies agree”): *if  $X$  is  $\mu$ -thick, the bounded-prefix topology on both  $\mathcal{W}_\mu(X)$  and  $\mathcal{I}_\mu(X)$  is their lexicographic order topology; these orders have no endpoints and are dense*. These are [lemmas 62.4](#) and [62.5](#), by the same arguments; source 26 also verifies that least-disagreement comparison is a linear order (“the earlier coordinate decides the comparison of  $f, h$ , or the two coordinates coincide and transitivity in  $X$  applies. Every cylinder is convex”). It adds the reason why the tail has the right type: “The ordinal tail  $[\alpha, \mu)$  has order type  $\mu$ : otherwise its order type and  $\alpha$  would both have cardinality below  $\mu$ , contradicting  $\alpha + \text{otp}[\alpha, \mu) = \mu$ ”, and the warning: “A prefix of length  $\alpha$  leaves, after increasing reindexing, another full  $\mu$ -word problem over a remaining alphabet of size  $\mu$ . This is not a license to replace an arbitrary nonminimal ordinal length by its cardinality.”

[merge] Source 26's Proposition 3.2 adds that every cylinder is homeomorphic to the whole space; its complete-metrizability classification uses this ([theorem 69.18](#)).

**Proposition 62.10** (Basic topology). *The cylinders form a clopen base for a Hausdorff zero-dimensional topology on  $\mathcal{W}_\mu(X)$  and on  $\mathcal{I}_\mu(X)$ . The former is dense in the latter. Every permutation cylinder is homeomorphic to  $\mathcal{W}_\mu(X)$ , after an arbitrary bijective relabelling of its remaining alphabet.*

*Proof.* Two prefixes are either incompatible or one extends the other, so cylinder intersections have the required base property. The complement of  $N_p$  is a union of cylinders witnessing a disagreement before  $\text{dom } p$ . Distinct functions have a coordinate of disagreement and disjoint longer cylinders. Density is [note 62.9](#). For the final statement, remove the prescribed prefix, increasingly reindex the ordinal tail, and bijectively relabel  $X \setminus \text{ran } p$  by  $X$ . Both this map and its inverse carry bounded-prefix neighborhoods to bounded-prefix neighborhoods.  $\square$

*Note 62.11* (Source 26, Section 3.1, “Embedding information inherited from the repository”). Source 26 records the set baseline of this report ([theorem 6.1](#)) for the full order  $\mathcal{W}(X)$ , calling these “earlier written-proof results . . . not new claims of this continuation. The upper bound on increasing ordinal chains is a genuine order-theoretic assertion, not a consequence of the cardinality  $2^\mu$  alone. We do not use that upper bound in the proofs of our main theorems.” It then gives the pair-orientation embedding  $(2^\mu, <_{\text{lex}}) \hookrightarrow (\mathcal{W}_\mu(X), <_{\text{lex}})$  directly (pairs  $\{a_\xi < b_\xi\}$  listed  $a_\xi, b_\xi$  for bit 0 and  $b_\xi, a_\xi$  for bit 1; “the pair order of type  $2 \cdot \mu$  is order-isomorphic to  $\mu$  for infinite initial  $\mu$ ”), and notes that “the finite executable checks in this package verify the first-disagreement mechanism, not the transfinite cardinal assertions”.

*Note 62.12* (Source 28, Section 3: Definition 3.1, Lemmas 3.2 and 3.3, Propositions 3.4 and 3.5). Source 28 uses source 24's thickness (its Definition 3.1) and proves again: Lemma 3.2 (“Size and thickness of birthday cutoffs”) = [note 62.2](#), with the same three-case sign prefix; Lemma 3.3 (“Short-prefix extension”), *a short injective prefix extends to a bijection  $\mu \rightarrow X$ , and also to an injection omitting any specified label outside its range* = [lemma 62.4](#); Proposition 3.4 (“The prefix topology is the numerical order topology”) = [lemma 62.5](#); Proposition 3.5 (“Basic size and embedding spectra, recalled with proof”) = [note 62.6](#); each by the same argument. It uses “the sign-sequence order of



surreals, not their full arithmetic”, citing Ehrlich [15] and the repository’s universe-indexed sign sequences [10]. It adds a warning about the full order: its topology “must not be inferred from” the minimal slice’s. “For example, take a common prefix enumerating  $X \setminus \{a, b\}$ , where  $a < b$ . The two exhaustive continuations ending in  $(a, b)$  and  $(b, a)$  are adjacent: any word between them must share the prefix, and only those two residual enumerations are possible. No such finite residual tail occurs in the minimal infinite slice” (this report’s [theorem 16.2](#)).

*Note 62.13* (Source 21, Section 3: Proposition 3.1, Definition 3.3 and Proposition 3.4). Source 21’s Proposition 3.1 (“Cardinality and a uniform lower embedding”): *for every  $\alpha$  of cardinality  $\mu$ ,  $|\mathcal{W}_\alpha(X)| = |\mathcal{W}(X)| = 2^\mu$ ; the binary cube  $\mathbb{B}_\mu$  order-embeds into  $\mathcal{W}_\mu(X)$ ; consequently every linear order of cardinality at most  $\mu$  embeds into  $\mathcal{W}_\mu(X)$* . Its Definition 3.3:  $X$  has *uniformly large intervals* if it has no endpoints and  $|(a, b)_X| = |X| = \mu$  for  $a < b$ . Its Proposition 3.4: *if  $X$  has uniformly large intervals, the order topology of  $\mathcal{W}_\mu(X)$  equals its topology of agreement on bounded prefixes; at every point its left and right characters are both  $\text{cf}(\mu)$* . These are [note 62.6](#), [lemma 62.5](#) and [theorem 30.2](#), by the same arguments. Source 21 writes  $\mathcal{I}_\mu(X)$  for the injective  $\mu$ -words from the start: “An injective word need not be exhaustive. This small-looking difference will control both the fiber decomposition and the automorphism obstruction.” It adds: that the topology of agreement on bounded initial segments is, “at  $\mu = \omega \dots$  the usual product topology when  $X$  has the discrete topology; at an uncountable  $\mu$  it is generally finer than the ordinary product topology, which only fixes finitely many coordinates”; that “cardinality  $2^\mu$  does not imply an increasing copy of every ordinal of that cardinality” (the ordinal bound of [theorem 6.1](#) is “a structural restriction of the lexicographic presentation order. The corresponding theorem for arbitrary linear orders of cardinality  $2^\mu$  would be false”); that for birthday fragments “the cardinal parameter is  $\mu = 2^{<\kappa}$ , not necessarily  $\kappa$ ”, with the sharper equimorphism [theorem 8.3](#) (“we will not replace it by cardinal multiplication or use a regular-cardinal argument at a singular cutoff”); a direct check that every  $\mathbf{No}_{<\kappa}$  has uniformly large intervals (a sign cone between the endpoints, of size  $2^{<\kappa}$  because  $\gamma + \lambda < \kappa$ ), with the warning “Noncardinal cutoffs require a separate argument; they need not have uniformly large numerical intervals”; and “This explains why point characters depend on  $\text{cf}(\mu)$  even when the global cofinality is controlled by the much smaller birthday parameter.”

[merge] Source 21 also records, after its Proposition 3.1:

*Remark 62.14* (A well-order of presentations need not result). Even  $\mathbb{B}_\omega$  has an infinite descending sequence: let the  $n$ th word have a single 1 at coordinate  $n$ . The pair code above therefore exhibits descending chains in minimum-length permutation spaces. The objects being compared are well-orders; the lexicographic order on those objects is usually not a well-order.

*Note 62.15* (Source 22, Section 3: Definition 3.1, Lemmas 3.2 and 3.3, Theorem 3.4). Source 22 works in ZFC with  $\mathcal{W}_\alpha(X)$  for every ordinal  $\alpha$  with  $|\alpha| = |X|$ , and writes  $\mathcal{W}_0(\emptyset)$  for the one-element order containing the empty enumeration. It calls a thick alphabet  $\mu$ -rich: “dense without endpoints and every nonempty open interval has cardinality  $\mu$ . Its open rays then also have cardinality  $\mu \dots$  We do not assume that all cuts of size less than  $\mu$  are filled.” Its Lemma 3.2 (“Richness of cardinal birthday cutoffs”):  $\mathbf{No}_{<\kappa}$  is  $\mu$ -rich, and  $\text{cf } X = \text{ci } X = \text{cf } \kappa$ , which is [note 62.2](#) with [lemma 29.3](#); its proof takes the interval cone above a separator of birthday exactly  $\delta$ , “the sign-word cut lemma used in the repository; its complete-lattice proof and universe-relative formal statement occur in `exists_cut_separator_of_birthday_1t`” [8]. Its Lemma 3.3 (“Short prefixes remain flexible”): *if  $X$  is  $\mu$ -rich,  $|\alpha| = \mu$  and  $p : \beta \rightarrow X$  is injective with  $\beta < \mu$ , then  $p$  extends to an element of  $\mathcal{W}_\alpha(X)$ ; at any further coordinate below  $\mu$ , both a smaller and a larger unused label than a prescribed label are available; the cylinder of  $p$  is nonempty and open in the order topology*, which is [lemmas 62.4](#) and [62.5](#) for every length  $\alpha$  of cardinality  $\mu$ , with the warning: “The restriction  $\beta < \mu$  is essential. A prefix of length  $\mu$  can leave a finite, countable, or otherwise special residual alphabet. Its cylinder

is always convex but need not be open. This is the source of the transitions below.” Its Theorem 3.4 (“Size and embeddings”): *for every infinite linearly ordered  $X$  of cardinality  $\mu$  and every ordinal  $\alpha$  of cardinality  $\mu$ ,  $|\mathcal{W}_\alpha(X)| = 2^\mu$ ; every linear order of size at most  $\mu$  embeds in  $\mathcal{W}_\alpha(X)$ ; an ordinal  $\beta$  embeds in it if and only if  $\beta < \mu^+$ , which is [theorem 6.1](#) for each slice; writing  $\alpha = \mu + \nu$ , the lower embedding orients pairs of a set of size  $\mu$  and appends a fixed enumeration of length  $\nu$  of the remaining labels. “This recovers the fixed-stratum portion of the earlier general-alphabet theory. In particular, the topological changes below do not reflect a change in the ordinal embedding spectrum.”*

*Note 62.16* (Source 19, Section 3, “Elementary tools and the known spectrum”). Source 19 recalls “a small toolkit so that the later proofs are understandable without the earlier reports”, citing Gonshor [12], Ehrlich [15] and Hamkins’s class-universality formulation. Its Lemma 3.1 (“Surreal set cuts and avoidance”): *if  $A < B$  are sets of surreals, a surreal  $z$  with  $A < z < B$  exists and can be chosen outside any prescribed set  $F$ ; every nonempty open surreal interval is a proper class, which is [theorem 5.6](#) and [corollary 5.7](#). Its Lemma 3.2 (“A birthday-controlled insertion”): *every linear order of cardinality  $\lambda$ , an infinite initial cardinal, embeds into  $\mathbf{No}_{<\lambda}$ , without regularity of  $\lambda$* ; its proof enumerates the order as  $(x_i)_{i<\lambda}$  and maps  $x_i$  to the simplest surreal between the images of the earlier elements below and above it, so that “the image of  $x_j$  has birthday at most  $j$ ” and the stage- $i$  cut has a separator of birthday at most  $\sup_{j<i}(j+1) \leq i$ ; “it is an ordinary set recursion”. This is the forth construction of [proposition 5.13](#) with the birthday bound of [theorem 5.6](#). Its Lemma 3.3 (“Deletion and sign cones”) is [lemma 29.2](#), and its Proposition 3.4 (“Fixed-carrier baseline”) is [theorem 6.1](#); for the ordinal upper bound source 19 says “we take that upper bound as a cited input here” from this report’s general-alphabet spectrum theorem and the reals report, and “the new residual-coding and model theorems below have their own proofs and do not depend on an informal extension of that bound”.*

## 63 Fixed-length strata: residual cylinders, weight and point characters (source 22)

[merge] Source 22’s Sections 4–6, printed whole. They treat every stratum  $\mathcal{W}_\alpha(X)$ ,  $|\alpha| = \mu$ , of a thick (source 22:  $\mu$ -rich) alphabet. The density and cellularity column of its table is source 16’s [theorem 27.2](#) ([note 63.5](#)); new in source 22 are the residual-cylinder identity and the first-disagreement character formula, the weight column, the complete point-character classification of the finite tails  $\mu + n$  and of the tail  $\mu + \omega$ , and the one-step Cantor–Bendixson derivative. Source 22 writes  $\Delta = \mu^{<\mu}$  and  $\chi^\pm$  for the left and right characters. Source 30’s [corollary 63.12](#), which removes the richness hypothesis from the weight column, follows the table.

### 63.1 Residual cylinders and an exact character formula (source 22)

#### 63.1.1 The residual-fibre identity

For an injective prefix  $p : \beta \rightarrow X$  of a length- $\alpha$  enumeration, let

$$R_p = X \setminus \text{ran } p, \quad \gamma_p = \text{otp}([\beta, \alpha)).$$

The prefix is extendible exactly when  $|R_p| = |\gamma_p|$ .

**Theorem 63.1** (Residual-cylinder identity). *For every extendible prefix  $p$ , concatenation is an order isomorphism*

$$\mathcal{W}_{\gamma_p}(R_p) \cong \{e \in \mathcal{W}_\alpha(X) : e \restriction \beta = p\}.$$



The image is convex. In particular, if  $\alpha = \mu + \nu$  and  $R \subseteq X$  satisfies  $|R| = |\nu|$  and  $|X \setminus R| = \mu$ , every length- $\mu$  enumeration of  $X \setminus R$  determines a convex copy of  $\mathcal{W}_\nu(R)$ .

*Proof.* Restriction to the remaining coordinate interval and reindexing by its order type are inverse to concatenation. All initial coordinates are equal, so first disagreement occurs in the tail. An enumeration between two extensions of  $p$  cannot disagree with  $p$  earlier than  $\beta$ , since such a disagreement puts it on the same outside side of both extensions.  $\square$

The theorem is elementary, but its consequence is substantial: after  $\mu$  coordinates, the original birthday alphabet imposes no uniform residual-cardinality condition beyond the prescribed tail length. A proof valid only while fewer than  $\mu$  labels have been consumed cannot simply be reused.

### 63.1.2 Characters and disagreement fibres

For a point  $e$  in a linear order  $P$ , define

$$\chi^-(e; P) = \text{cf}(P_{<e}), \quad \chi^+(e; P) = \text{ci}(P_{>e}).$$

We assign character 0 to an empty side and character 1 to a side with a greatest, respectively least, element. Thus  $(1, 1)$  means an isolated non-endpoint.

For  $e \in \mathcal{W}_\alpha(X)$  put

$$X_\beta = X \setminus e[\beta], \quad A_e^- = \{\beta < \alpha : (X_\beta)_{<e(\beta)} \neq \emptyset\},$$

with  $A_e^+$  defined similarly. These are the coordinates at which a smaller or larger first disagreement is possible. Let  $H_\beta^-$  and  $H_\beta^+$  be the corresponding fibres.

**Theorem 63.2** (General first-disagreement formula). *For any infinite set alphabet and any enumeration length,*

$$P_{<e} \cong \sum_{\beta \in A_e^-} H_\beta^-,$$

where the index set has its increasing ordinal order. The upper side is the analogous sum in reverse coordinate order. Consequently:

- (i) If  $A_e^-$  is empty,  $\chi^-(e; P) = 0$ .
- (ii) If  $A_e^-$  has no greatest member,  $\chi^-(e; P) = \text{cf}(A_e^-)$ .
- (iii) If  $A_e^-$  has greatest member  $\beta_*$ , then  $\chi^-(e; P) = \text{cf}(H_{\beta_*}^-)$ .

The dual statements hold on the right. More explicitly, with  $\gamma = \text{otp}([\beta + 1, \alpha))$ ,

$$H_\beta^- \cong \sum_{a \in (X_\beta)_{<e(\beta)}} \mathcal{W}_\gamma(X_\beta \setminus \{a\}).$$

Thus, if the next-label index order has no greatest element, its cofinality is the cofinality of the fibre; if it has greatest element  $a_*$ , the last residual enumeration order supplies the cofinality.

*Proof.* Every  $f < e$  has one first-disagreement coordinate. If this coordinate is  $\beta < \delta$ , and  $g < e$  first disagrees at  $\delta$ , then  $f < g$ , because  $g(\beta) = e(\beta) > f(\beta)$ . For fixed  $\beta$ , varying the label  $a < e(\beta)$  gives the displayed lexicographic sum, and the remainder is exactly the residual cylinder. Every allowed label has a completion: deleting any one member of the same residual set leaves the required cardinality.

For a sum of nonempty orders indexed by a limit well-order, choosing one point on a cofinal collection of indices yields a cofinal set. Conversely, every cofinal set projects to cofinally many indices. If there is a last index, the last summand is cofinal. Applying these facts twice proves the formula. Reverse the order for the upper side.  $\square$

This is an exact reduction, not an assertion that arbitrary residual endpoint cofinalities already have a simple closed formula. Limit prefixes can fail to be extendible even when every shorter prefix is extendible. For example, the decreasing enumeration of  $\omega^*$  has length  $\omega$ ; its finite prefixes extend to length  $\omega \cdot 2$  enumerations of that alphabet, but their union exhausts the alphabet too early.

### 63.1.3 Endpoint replacement after the first cardinal block

**Theorem 63.3** (Character transfer). *Let  $X$  be  $\mu$ -rich,  $q = \text{cf } \mu$ , and  $\alpha = \mu + \nu$ . Write  $e = p \frown t$ , where  $p$  has length  $\mu$ ,  $R = X \setminus \text{ran } p$ , and  $t \in \mathcal{W}_\nu(R)$ . Then*

$$\chi^\pm(e; \mathcal{W}_\alpha(X)) = \begin{cases} \chi^\pm(t; \mathcal{W}_\nu(R)), & \text{if that intrinsic side is nonempty,} \\ q, & \text{if that intrinsic side is empty.} \end{cases}$$

*Proof.* A nonempty part of the cylinder below  $e$  is cofinal in the entire lower side of  $e$ : every point below  $e$  outside the cylinder lies below all its points. This proves the first case. If  $t$  is the cylinder minimum, all lower first disagreements occur before  $\mu$ . At every  $\beta < \mu$ , richness supplies an unused smaller label, so the active coordinate set is exactly  $\mu$ . Theorem 63.2 gives  $q$ . The right side is dual. The proof also covers  $\nu = 0$ , when the cylinder is a singleton.  $\square$

## 63.2 The finite-tail transition and all fixed-length cardinal invariants (source 22)

The order topology is always intended. Density  $\text{d } P$  is the least cardinality of a dense subset, weight  $\text{w } P$  the least cardinality of a base, and cellularity  $\text{cell } P$  the supremum of the sizes of families of pairwise disjoint nonempty open sets. A  $\pi$ -base need only contain a nonempty open subset of each nonempty open set; it need not provide neighborhoods of every individual point.

**Theorem 63.4** (Topological classification of every fixed-length stratum). *Let  $X$  be  $\mu$ -rich, let  $\Delta = \mu^{<\mu}$ , and let  $|\alpha| = \mu$ . Then*

| <i>enumeration length</i> | $\text{d } \mathcal{W}_\alpha(X)$ | $\text{cell } \mathcal{W}_\alpha(X)$ | $\text{w } \mathcal{W}_\alpha(X)$ |
|---------------------------|-----------------------------------|--------------------------------------|-----------------------------------|
| $\mu$ or $\mu + 1$        | $\Delta$                          | $\Delta$                             | $\Delta$                          |
| $\mu + 2$                 | $\Delta$                          | $\Delta$                             | $2^\mu$                           |
| $\alpha \geq \mu + 3$     | $2^\mu$                           | $2^\mu$                              | $2^\mu$                           |

*Thus this table applies to every ordinal length of cardinality  $\mu$ , not merely to finite extensions of  $\mu$ .*

*Proof.* *The common lower bound.* For each  $\beta < \mu$ , there are  $\mu^{|\beta|}$  injective prefixes of length  $\beta$ . To verify the lower estimate, partition  $X$  into  $|\beta|$  disjoint sets of size  $\mu$  and choose the  $\xi$ th entry from the  $\xi$ th set; the finite and empty cases are immediate. Distinct prefixes of that length have disjoint, nonempty open cylinders by note 62.15. Consequently

$$\text{cell } \mathcal{W}_\alpha(X) \geq \sup_{\beta < \mu} \mu^{|\beta|} = \Delta.$$

There are altogether  $\Delta$  short prefixes, since  $\Delta \geq \mu$ .

*The first three lengths.* Suppose  $\alpha = \mu + n$  with  $n \leq 2$ . If two distinct enumerations have the same length- $\mu$  prefix, they lie in a fibre of size  $n! \leq 2$ . Their open interval is therefore empty. Hence the endpoints  $e < f$  of any nonempty open interval must first disagree at some  $\beta < \mu$ . Choose an unused label

$$e(\beta) < a < f(\beta)$$

above the common prefix, and extend that prefix by  $a$ . Richness ensures that deleting the fewer than  $\mu$  earlier labels does not exhaust this interval. The resulting short cylinder is contained in  $(e, f)$ .

The orders have no endpoints, because their first labels can always be moved down or up. Thus every nonempty open set contains such an interval, and the short cylinders form a  $\pi$ -base. Choose one point from each cylinder. This gives a dense subset of size at most  $\Delta$ . Since disjoint nonempty open sets require different points of a dense set,

$$\Delta \leq \text{cell } P \leq \text{d } P \leq \Delta.$$

For  $n = 0, 1$ , any two distinct enumerations differ before  $\mu$ , so the same interpolation argument shows that  $P$  is dense as an order. In a dense order without endpoints, intervals with endpoints in a dense subset form a base. Therefore  $\text{w } P = \Delta$  in these two cases.

For  $n = 2$ , fix a two-element subset  $R \subset X$ . There are  $2^\mu$  bijections  $p : \mu \rightarrow X \setminus R$ . Each cylinder of such a  $p$  consists of two adjacent points, say  $a_p < b_p$ . These are pairwise disjoint jumps. For each jump, a base must contain an element  $B_p$  with

$$b_p \in B_p \subseteq (a_p, \infty).$$

Different jumps require different base elements: if the  $p$ -jump precedes the  $p'$ -jump, then  $b_p \leq a_{p'}$ , so  $B_{p'}$  cannot contain  $b_p$ . Thus  $\text{w } P \geq 2^\mu$ . Every order of size  $2^\mu$  has an interval base of size at most  $2^\mu$ , proving equality.

*Every remaining length.* Write  $\alpha = \mu + \nu$ , where  $\nu \geq 3$ . Choose  $R \subset X$  with  $|R| = |\nu|$  and  $|X \setminus R| = \mu$ . There are  $2^\mu$  length- $\mu$  enumerations of  $X \setminus R$ . Their cylinders are pairwise disjoint convex copies of  $\mathcal{W}_\nu(R)$  by [theorem 63.1](#). This residual order has at least three points. Choose  $u_p < v_p < w_p$  in each cylinder. Then  $(u_p, w_p)$  is a nonempty open interval, contained in that cylinder. We have obtained  $2^\mu$  pairwise disjoint nonempty open sets. The inequalities

$$2^\mu \leq \text{cell } P \leq \text{d } P \leq \text{w } P \leq |P|^2 = 2^\mu$$

finish the proof. □

*Note 63.5* ([merge]; credits for [theorem 63.4](#)). The density and cellularity columns of the table are [theorem 27.2](#) (source 16; source 18's route in [section 27.1](#)), proved there for every infinite alphabet without the richness hypothesis; source 22's proof of them is a second route, by the same short-prefix  $\pi$ -base and the same  $2^\mu$  disjoint residual intervals. The weight column is new in source 22: weight  $\Delta$  at lengths  $\mu$  and  $\mu + 1$  and  $2^\mu$  from  $\mu + 2$  on, so that weight changes at the second terminal coordinate and density and cellularity at the third. For  $X = \mathbb{R}$  the table is the reals report's `two:prop:finite-topology`; source 30's [corollary 63.12](#) below removes the richness hypothesis from the weight column. Source 22 states that “the fixed-length density/cellularity/weight table and the redundancy of cylinder heights are now treated as proved results rather than left open”; for density and cellularity this is source 16's theorem, placed in the repository before source 22 was written (commit f0cd7032d) and printed in Part VI. The cofinality clause of [corollary 63.7](#) is source 16's [proposition 28.2](#).

*Remark 63.6* (Why the distinction between a base and a  $\pi$ -base matters). At length  $\mu + 2$ , short cylinders still detect every nonempty open set, but they cannot refine a neighborhood of an upper jump point which excludes its lower partner. Treating the short-prefix  $\pi$ -base as a neighborhood base would give the false conclusion  $wP = \Delta$  whenever  $\Delta < 2^\mu$ . Conversely, a length- $\mu$  cylinder is convex, but its two-point endpoints prevent it from being open. Both distinctions are essential.

**Corollary 63.7** (Birthday-cutoff form). *For every infinite cardinal  $\kappa$ , put  $\mu = 2^{<\kappa}$  and  $\Delta = \mu^{<\mu}$ . The table in [theorem 63.4](#) holds for  $X = \mathbf{No}_{<\kappa}$  without assuming that  $\kappa$  or  $\mu$  is regular. Moreover, for every  $\alpha$  of cardinality  $\mu$ ,*

$$\text{cf } \mathcal{W}_\alpha(X) = \text{ci } \mathcal{W}_\alpha(X) = \text{cf } \kappa.$$

*Proof.* Apply [note 62.15](#). For the last assertion, the first-label projection expresses the order as a lexicographic sum of nonempty fibres indexed by  $X$ . Since  $X$  has neither endpoint, the cofinality and coinitality are those of the indexing order.  $\square$

### 63.3 Finite fibres and exact point characters

Let

$$Q_n(X) = \{p : \mu \hookrightarrow X : |X \setminus \text{ran } p| = n\},$$

with first-disagreement order. Sorting each finite residual set identifies its permutation order with the finite chain of size  $n!$ .

**Proposition 63.8** (Finite-tail normal form). *For each finite  $n$ , there is an order isomorphism*

$$\mathcal{W}_{\mu+n}(X) \cong Q_n(X) \times_{\text{lex}} n!.$$

*The order  $Q_n(X)$  is dense without endpoints, has cardinality  $2^\mu$ , and every point has character  $(\text{cf } \mu, \text{cf } \mu)$ . Its density, cellularity, and weight are  $\mu^{<\mu}$ .*

*Proof.* The residual-cylinder identity gives the displayed lexicographic sum and hence the isomorphism. Distinct elements of  $Q_n(X)$  differ at a coordinate below  $\mu$ . At that coordinate one can insert an unused label strictly between their labels, then complete a length- $\mu$  injection leaving exactly  $n$  labels. This proves density; changing the first label proves the absence of endpoints.

For a fixed  $p$ , smaller and larger first disagreements are possible at every  $\beta < \mu$ . The argument of [theorem 63.2](#), applied directly to these prefixes, gives characters  $(\text{cf } \mu, \text{cf } \mu)$ . The cardinality is  $2^\mu$  by fixing a residual set of size  $n$ . The short-cylinder counting and  $\pi$ -base argument from [theorem 63.4](#) applies to  $Q_n(X)$  as well. Density of its order then converts the dense subset into a base of the same size.  $\square$

**Theorem 63.9** (All finite-tail point characters). *Let  $q = \text{cf } \mu$ . The complete character classification for  $\mathcal{W}_{\mu+n}(X)$  is*

| $n$        | characters and positions within each $n!$ -point block                      |
|------------|-----------------------------------------------------------------------------|
| 0, 1       | $(q, q)$ at every point                                                     |
| 2          | $(q, 1)$ at the first point; $(1, q)$ at the last point                     |
| $n \geq 3$ | $(q, 1)$ first; $(1, 1)$ at the $n! - 2$ interior points;<br>$(1, q)$ last. |

*For  $n \geq 3$ , there are exactly  $2^\mu$  isolated points, and they form a dense subset.*

*Proof.* Inside a finite chain, each nonempty side has character one. At a block endpoint, the missing intrinsic side acquires character  $q$  by [theorem 63.3](#). Singleton blocks acquire  $q$  on both sides.

A point with both an immediate predecessor and an immediate successor is isolated. Hence exactly the interior points of the finite blocks are isolated. There are  $2^\mu$  blocks and a fixed positive finite number of interior points per block. To see density, take a nonempty interval. If it lies within a single block, it contains an interior block point. If its endpoints have different blocks, density of  $Q_n(X)$  supplies a whole intervening block, which has an interior point.  $\square$

**Corollary 63.10** (One-step derivative). *For  $n \geq 3$ , the first Cantor–Bendixson derivative of  $\mathcal{W}_{\mu+n}(X)$  consists exactly of the first and last points of each finite block. The derivative has no isolated points in its subspace topology, so the derivatives stabilize at this stage. For  $n \leq 2$ , the original space has no isolated points.*

*Proof.* The first assertion is the isolated-point classification. Between distinct blocks there are further blocks arbitrarily close in the relevant direction. Consequently each retained first point is accumulated from below by retained endpoints, and each retained last point is accumulated from above. No retained endpoint is isolated after the interior points are removed. For singleton blocks both sides have character  $q \geq \omega$ ; for two-point blocks one side still has that character.  $\square$

**Example 63.11** (A countable alphabet). Take  $\kappa = \omega$ , so  $\mu = \omega$ ,  $q = \omega$ , and  $\Delta = \omega$ . The orders at lengths  $\omega$  and  $\omega + 1$  are second-countable. At length  $\omega + 2$  the order is separable and first-countable but not second-countable: it has  $2^{\aleph_0}$  distinct jumps. At length  $\omega + 3$  it has  $2^{\aleph_0}$  isolated points, densely distributed in six-point blocks. Thus extending an exhaustive enumeration by only two or three ordinal positions changes its topology sharply, although the alphabet cardinality and ordinal embedding spectrum have not changed.

[merge] Source 30 proves the weight column for every infinite alphabet, with no density, regularity, saturation or endpoint hypothesis. Its “established” inputs are the finite-residual decomposition ([proposition 63.8](#), whose proof does not use richness) and the density theorem [theorem 27.2](#).

**Corollary 63.12** (Weight for arbitrary infinite alphabets). *Let  $X$  be any infinite linearly ordered set of cardinality  $\mu$ , and let  $n < \omega$ . Then*

$$w(\text{Bij}(\mu + n, X)) = \begin{cases} \mu^{<\mu}, & n = 0, 1, \\ 2^\mu, & n \geq 2. \end{cases}$$

*No density, regularity, saturation or endpoint hypothesis on  $X$  is needed.*

*Proof.* Use the established finite-residual decomposition and dense-quotient lemma for arbitrary infinite alphabets. For  $n = 0, 1$  the fibers have one point, so the order is dense. In an infinite dense linear order, a dense subset gives a base of intervals and endpoint rays; thus weight equals density. The established density calculation is  $\mu^{<\mu}$  in these two cases.

For  $n = 2$ , fix an omitted pair. There are  $2^\mu$  bijections from  $\mu$  to its complement, producing  $2^\mu$  distinct adjacent pairs in the full order. The base-set argument in the preceding proof gives weight at least  $2^\mu$ . For  $n \geq 3$ , the same argument with a fixed omitted  $n$ -element set gives  $2^\mu$  isolated interior points of finite fibers. In all cases the whole order has cardinality  $2^\mu$ , which bounds its weight above.  $\square$

This refines the density and cellularity threshold in the existing fixed-type analysis. Weight changes at the *second* terminal coordinate, whereas density and cellularity change at the third. The numerical values can coincide when  $\mu^{<\mu} = 2^\mu$ ; the finite-fiber geometry still distinguishes the mechanisms.

### 63.4 Infinite tails and the general adjacency mechanism (source 22)

[merge] [Lemma 63.13](#) is the fixed-stratum form of this report's adjacency theorems ([theorems 16.2](#) and [16.6](#)); source 22's proof is printed because the next lemma uses its tail argument.

#### 63.4.1 Why infinite residuals cannot make jumps

The finite-tail origin of adjacency was established in the earlier reports for more general enumeration comparisons ([theorem 16.2](#); source 17). The following fixed-stratum form is useful here.

**Lemma 63.13** (Adjacency forces a finite residual). *Let  $X$  be an infinite linearly ordered set and write an infinite ordinal  $\alpha$  uniquely as  $\alpha = \lambda + n$ , where  $\lambda$  is a nonzero limit ordinal and  $n < \omega$ . The adjacent pairs in  $\mathcal{W}_\alpha(X)$  are precisely the consecutive permutations in a common length- $\lambda$  prefix cylinder. Its maximal finite adjacency components are therefore the  $n!$ -point cylinders, with singleton components when  $n = 0$  or  $1$ .*

*Proof.* Suppose  $e < f$  are adjacent, and let  $\beta$  be their first disagreement. Fixing the prefix through  $e(\beta)$ , the remaining tail of  $e$  must be lexicographically greatest among all exhaustive permutations of its residual labels. Otherwise a larger such tail would produce a point strictly between  $e$  and  $f$ . Swapping entries at an increasing pair of tail positions shows that this greatest tail is numerically strictly decreasing. Dually, the tail of  $f$  after its first differing label must be numerically strictly increasing.

Let  $R$  be the common set of unused labels before coordinate  $\beta$ . The inherited numerical order on

$$R \setminus \{e(\beta), f(\beta)\}$$

is both well-ordered and reverse well-ordered. Such a set is finite: if infinite, its first  $\omega$  increasing elements have no greatest member, contradicting reverse well-ordering. Hence  $R$  is finite, and the remaining coordinate interval  $[\beta, \alpha]$  is finite. Therefore  $\beta \geq \lambda$ .

Conversely, the common length- $\lambda$  prefix cylinder is a finite convex chain of  $n!$  permutations. Consecutive points in it are adjacent in the whole order. No adjacency crosses between these cylinders, proving the component assertion.  $\square$

#### 63.4.2 The tail of order type $\omega$

**Lemma 63.14** (Countable residual enumeration orders). *For a countably infinite linearly ordered set  $R$ , the order  $\mathcal{W}_\omega(R)$  has no adjacent pair and is separable. Each nonempty side of a point has character  $\omega$ . A minimum exists exactly when the numerical order type of  $R$  is  $\omega$ , and a maximum exists exactly when its order type is  $\omega^*$ .*

*Proof.* There are no jumps by [lemma 63.13](#). There are countably many finite injective prefixes. Choose one exhaustive completion of each. Given a nonempty interval, choose an interior enumeration  $h$ . Its comparisons with the endpoints are both decided at finite coordinates. A sufficiently long finite prefix of  $h$  has its entire cylinder in the interval. Therefore the chosen completions form a countable dense set. A nonempty side has no extremal point, because there are no jumps; its cofinality or coinitality is at most  $\omega$  by separability and cannot be finite. It is therefore  $\omega$ .

If an enumeration is the minimum, every pair of its entries must be numerically increasing; otherwise swapping an inversion produces a smaller enumeration. Thus it is a strictly increasing exhaustive enumeration of length  $\omega$ , and  $R$  has order type  $\omega$ . Conversely, that enumeration always chooses the least unused label, so is lexicographically least. The maximum assertion is dual.  $\square$

**Theorem 63.15** (Exact  $\mu + \omega$  character spectrum). *For a  $\mu$ -rich alphabet  $X$  and  $q = \text{cf } \mu$ , the complete set of point characters in  $\mathcal{W}_{\mu+\omega}(X)$  is*

$$\{(q, \omega), (\omega, q), (\omega, \omega)\}.$$

*Repeated pairs are counted only once. The order is dense without endpoints, has no isolated points, and has density, cellularity, and weight  $2^\mu$ .*

*Proof.* The length- $\mu$  prefix leaves a countably infinite residual set  $R$ . Apply [lemma 63.14](#) and [theorem 63.3](#). A countably infinite order cannot have both an increasing and a decreasing exhaustive  $\omega$ -enumeration, so its permutation order cannot have both endpoints. This gives exactly the three displayed possibilities.

Each possibility occurs. A rich alphabet contains a countable increasing sequence of type  $\omega$ , a decreasing one of type  $\omega^*$ , and a countable dense suborder without endpoints. These can be chosen inside a fixed interval with an outside complement of size  $\mu$ . Leaving one of these sets as the residual alphabet realizes the first, second, and third cases, respectively. For the dense suborder, recursively satisfy all rational-order requirements inside the interval, using density to place each new point. This requires only countably many choices.

The length  $\mu + \omega$  is a limit ordinal, so there are no adjacent pairs by [lemma 63.13](#). There are no global endpoints, since the first label can be changed in either direction. The last assertion is [theorem 63.4](#).  $\square$

*Remark 63.16* (A sharp contrast even when cardinal invariants coincide). If  $\text{cf } \mu > \omega$ , the minimal stratum has character  $(\text{cf } \mu, \text{cf } \mu)$  everywhere, while the  $\mu + \omega$  stratum does not have that character anywhere. Thus local characters can distinguish these strata even in cardinal-arithmetic situations in which their densities and weights happen to coincide. At  $\mu = \omega$ , the  $\omega + \omega$  stratum is first-countable but has continuum many disjoint nonempty open intervals. Unlike the  $\omega + 3$  stratum, it has no isolated points.

### 63.4.3 What remains for longer infinite tails

For  $\alpha = \mu + \nu$  with general  $\nu$ , [theorem 63.4](#) settles the three cardinal invariants, while [theorems 63.2](#) and [63.3](#) reduce local characters to residual enumeration orders. At  $\nu = \mu$ , the residual can be any subset  $R \subseteq X$  of cardinality  $\mu$  with complement of cardinality  $\mu$ . Its order need not resemble the original birthday fragment. A further classification of point characters or cuts must therefore analyze the actual residual order and not just substitute  $\text{cf } \mu$  into the minimal-stratum formulas.

The same distinction controls ordinal-tail refinements. For example, at  $\alpha = \mu + \omega + n$ , finite adjacency blocks have size  $n!$ , but the geometry between them is governed by the preceding countable residual enumeration. The residual-cylinder identity provides a recursive description; it is not, by itself, an order-isomorphism classification of all such strata.

## 64 Fixed types with infinite residual tails: the exact binary-cube spectrum (source 19; route of source 30)

[merge] Source 19's Section 5, printed whole. It continues Part VI's block-type theorem ([theorem 31.2](#), source 15), whose boundary ([section 31.3](#)) left every type  $\mu \cdot \gamma + \zeta$  with an infinite remainder  $\zeta < \mu$  open, and it answers source 12's question on all fixed enumeration types ([research question 113.11](#)) for the embedded binary cubes in every case and for the coding length except at singular non-strong-limit residual layers at most  $\kappa$  ([section 72](#)). Source 19 read source 15 as an unplaced manuscript;



“the previous continuation” and “the prior theorem” below are source 15’s [theorem 31.2](#), and the display (5) is that theorem. Its entry code and main-prefix lemma extend source 15’s [lemma 31.1](#), [lemma 31.3](#) and [lemma 31.4](#) from finite to small deletions; its finite-capacity lemma is [lemma 31.5](#) and is recorded as a note. Source 19’s  $r_\kappa$  is printed  $\rho_\kappa$ , its  $D_{\rho,m}$  is printed  $\mathbb{B}_\rho \times_{\text{lex}} m$ , and its residual weights  $r(\lambda)$ ,  $u_\kappa(\lambda)$  and their sums  $R$ ,  $U$  are printed  $\text{wt}(\lambda)$ ,  $\text{wt}_\kappa^+(\lambda)$ ,  $\varrho_{\text{lo}}$ ,  $\varrho_{\text{up}}$  ([part X](#)). Source 30 proves the equality cases again by a different organization, block by block; its statements follow in [section 64.7](#).

The finite-tail theorem in the previous continuation leaves a specific obstruction: a long initial enumeration can leave a smaller, variably ordered alphabet for its final infinite tail. We now control that variation. The result is an exact binary-cube spectrum for every fixed enumeration type, together with exact equimorphisms for a broad family that includes every type at alphabet cardinality at most  $\aleph_\omega$ .

Throughout this section  $\kappa$  is an infinite initial cardinal and

$$X = \mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}, \quad \rho_\kappa = \begin{cases} \kappa \cdot \kappa, & \kappa \text{ is singular strong limit,} \\ \mu, & \text{otherwise.} \end{cases} \quad (4)$$

For a positive finite integer  $m$  put

$$(\mathbb{B}_\rho \times_{\text{lex}} m) = \mathbb{B}_\rho \times_{\text{lex}} \{0, \dots, m-1\}, \quad \mathbb{B}_\rho = (\{0, 1\}^\rho, <_{\text{lex}}).$$

The notation  $\simeq_e$  always means an order embedding in each direction. It does not mean order isomorphism. All arithmetic on enumeration types and binary lengths is ordinal arithmetic.

Write an admissible type as

$$\alpha = \mu \cdot \gamma + \zeta, \quad 0 < \gamma < \mu^+, \quad \zeta < \mu, \quad \sigma = \rho_\kappa \cdot \gamma.$$

The previously proved finite-tail result is

$$\mathcal{W}_{\mu \cdot \gamma + n}(X) \simeq_e (\mathbb{B}_\sigma \times_{\text{lex}} n!) \quad (n < \omega). \quad (5)$$

Its proof uses a sign-sequence code and, at singular strong limits, a first-varying-row scheduler (source 15, [lemma 31.3](#); [lemma 8.1](#)). We first record exactly which parts continue to apply when infinitely many labels are reserved.

## 64.1 Uniform prefix codes and deletion of a small alphabet

**Lemma 64.1** (A prefix-free code for any small alphabet). *Every linear order  $Y$  of infinite cardinality  $\lambda$  admits an order-preserving prefix-free binary entry code whose code lengths are positive and strictly less than  $\lambda$ . This assertion does not require regularity of  $\lambda$ .*

*Proof.* By [note 62.16](#), insert  $Y$  into  $\mathbf{No}_{<\lambda}$ , with the element inserted at stage  $\xi$  having birthday at most  $\xi < \lambda$ . Replace each minus sign by 00, each plus sign by 11, and append 01. The three blocks express exactly minus  $<$  termination  $<$  plus. Distinct completed codes cannot be proper prefixes of one another, and their lexicographic comparison agrees with the comparison of the surreal labels. The code length is  $2 \cdot \text{bd}(y) + 2 < \lambda$ . Padding any completed code with zeros therefore preserves its comparison with every other completed code.  $\square$

**Lemma 64.2** (The main prefix and its small-deletion counterpart). *Every lexicographically ordered family of injective prefixes  $p : \mu \cdot \gamma \rightarrow X$  embeds into  $\mathbb{B}_\sigma$ . Moreover, for every  $Y \subseteq X$  with  $|Y| < \mu$ ,*

$$\mathbb{B}_\sigma \hookrightarrow \mathcal{W}_{\mu \cdot \gamma}(X \setminus Y).$$

*Proof.* Use the prefix-free sign code of Lemma 64.1 for the labels of  $X$ , of length less than  $\kappa$ . If  $\mu > \kappa$ , pad each entry to length  $\kappa$ . A block of  $\mu$  entries then has length  $\kappa \cdot \mu = \mu$ . If  $\mu = \kappa$  is regular, concatenate the unpadding codes. Every proper partial sum within a  $\kappa$ -entry block is below  $\kappa$ , and its full sum is  $\kappa$ . If  $\mu = \kappa$  is singular, it is a strong limit: otherwise an attained equality  $2^\rho = \kappa$ ,  $\rho < \kappa$ , would also hold at  $\max(\rho, \text{cf } \kappa)$ , contradicting König's inequality. In this last case pad entries to length  $\kappa$ ; each  $\kappa$ -entry block has length  $\kappa \cdot \kappa$ . Applying the appropriate construction to the  $\gamma$  consecutive blocks gives length exactly  $\sigma$ . Exhaustiveness of the prefix is never used.

For the lower embedding in the nonexceptional cases, partition  $X \setminus Y$  into pairs indexed by  $\mu \cdot \gamma$ . Orient each pair by its corresponding bit. Since  $2 \cdot (\mu \cdot \gamma) = \mu \cdot \gamma$ , this has exactly the required enumeration type and preserves first disagreement.

Suppose now  $\mu = \kappa$  is singular strong limit. The sign cones with prefixes  $+\xi \frown -$ ,  $\xi < \kappa$ , are disjoint copies of  $X$ . At most  $|Y| < \kappa$  of them meet  $Y$ . Choose  $\kappa$  unaffected cones and index them by  $\kappa \cdot \gamma$ . Their complement in  $X \setminus Y$  still has size  $\kappa$ : it contains all but fewer than  $\kappa$  of the all-plus sequences. Thus the row scheduler used in the prior theorem still applies.

For completeness, its relevant property is the following. Inside a  $\kappa$ -row block, schedule row-column pairs  $(i, j)$  first by  $\max(i, j)$  and then lexicographically within each level. Every pair has fewer than  $\kappa$  predecessors, so the schedule has type  $\kappa$ , including when  $\kappa$  is singular. The first occurrence of row  $i$  is  $(i, 0)$ , and these occurrences increase with  $i$ . Choose the first entry of each row by an input parameter and enumerate the rest of that cone in a fixed way depending on that parameter. Include the complement as a fixed filler row in the first block. This embeds  $X_{\text{lex}}^{\kappa \cdot \gamma}$  in the required permutation order.

Finally let  $\delta = \text{cf } \kappa$  and choose positive lengths  $(\tau_j)_{j < \delta}$  with sum  $\kappa$ . Since  $\delta \cdot (\kappa \cdot \gamma) = \kappa \cdot \gamma$ , group the input coordinates into  $\kappa \cdot \gamma$  successive blocks of  $\delta$  coordinates. In each group restrict coordinate  $j$  to surreals of sign length exactly  $\tau_j$ . This realizes the cube of length  $\kappa \cdot \kappa \cdot \gamma = \sigma$  inside the product, completing the proof.  $\square$

The small-deletion assertion is necessary: the independent residual alphabet can have any cardinality below  $\mu$ . Removing it is not being treated as if only finitely many cones were affected.

## 64.2 Extracting an arbitrary tail factor from one residual fiber

Let  $P$  be the order of injective prefixes  $p : \mu \cdot \gamma \rightarrow X$  whose omitted set has cardinality  $|\zeta|$ . There is an actual lexicographic-sum isomorphism

$$\mathcal{W}_{\mu \cdot \gamma + \zeta}(X) \cong \sum_{p \in P} \mathcal{W}_{\zeta}(X \setminus \text{ran } p). \quad (6)$$

The next result shows why this decomposition controls embeddings that need not initially respect the displayed fibers.

**Lemma 64.3** (Fusion into one fiber). *If  $Q \hookrightarrow \mathbb{B}_\rho$  and  $L = \sum_{q \in Q} F_q$ , then for every nonempty linear order  $K$ ,*

$$\mathbb{B}_\rho \times_{\text{lex}} K \hookrightarrow L \implies K \hookrightarrow F_q \text{ for some } q \in Q.$$

*Proof.* Compose an asserted embedding with the nondecreasing projection to  $Q$  and its binary code. At stage  $\xi < \rho$ , retain a full source cylinder with its first  $\xi$  bits fixed and with the first  $\xi$  target bits fixed. The next target bit makes an ordered split. If the lower full source child lies entirely on the lower side, retain it. Otherwise it contains a point on the upper side, and the upper full source child lies entirely on the upper side; retain that child. At limits take unions of prefixes. The whole  $K$  factor survives. After  $\rho$  stages it has one target binary code and therefore lies inside a single  $F_q$ . Restrict the original injective map to that factor.  $\square$

**Theorem 64.4** (Exact residual embedding reduction). *For every nonempty linear order  $K$ ,*

$$\begin{aligned} \mathbb{B}_\sigma \times_{\text{lex}} K &\hookrightarrow \mathcal{W}_{\mu \cdot \gamma + \zeta}(X) \\ \iff \exists Y \subseteq X (|Y| = |\zeta| \text{ and } K &\hookrightarrow \mathcal{W}_\zeta(Y)). \end{aligned}$$

*In particular, replacing  $K$  by  $\mathbb{B}_\tau$  gives the exact test for whether  $\mathbb{B}_{\sigma+\tau}$  embeds into the total order.*

*Proof.* Forwards, combine the prefix code from Lemma 64.2, the sum (6), and Lemma 64.3. Conversely fix  $Y$  and apply the small-deletion embedding to  $X \setminus Y$ . Append the chosen  $K$ -indexed permutations of  $Y$ . Every main prefix exhausts the same complement, so the residual alphabet is independent of its binary parameter.  $\square$

**Proposition 64.5** (Which small orders can occur as residual alphabets). *For an infinite cardinal  $\lambda < \mu$ , every linear order of cardinality  $\lambda$  embeds numerically into  $X$  if and only if  $\lambda \leq \kappa$ .*

*Proof.* If  $\lambda \leq \kappa$ , note 62.16 embeds the order into  $\mathbf{No}_{<\lambda} \subseteq X$ . If  $\lambda > \kappa$ , the padded sign code embeds  $X$  into  $\mathbb{B}_\kappa$ . This cube has no increasing sequence of type  $\kappa^+$ : successive stabilization of its bits, with bounds taken at the regular cardinal  $\kappa^+$ , would make every coordinate constant on a tail of such a sequence. But the  $\lambda$ -sized ordinal  $\lambda$  contains a  $\kappa^+$ -sequence. It cannot embed into  $X$ .  $\square$

Thus residual alphabets at or below  $\kappa$  include all small linear orders, not just a special family of surreal order types. Above  $\kappa$ , the inherited global sign code is a real restriction.

### 64.3 Finite cardinal layers and the coding sandwich

Repeated left division by the current infinite cardinal gives

$$\zeta = \lambda_0 \cdot \eta_0 + \cdots + \lambda_{k-1} \cdot \eta_{k-1} + n, \quad (7)$$

where

$$\mu > \lambda_0 > \cdots > \lambda_{k-1} \geq \omega, \quad 0 < \eta_i < \lambda_i^+, \quad n < \omega.$$

There are only finitely many layers: otherwise the successive remainders would give an infinite strictly decreasing sequence of initial ordinals. Finite  $\zeta$  means  $k = 0$ .

Define two weights on infinite cardinals:

$$\text{wt}(\lambda) = \begin{cases} \lambda \cdot \lambda, & \lambda \text{ singular strong limit,} \\ \lambda, & \text{otherwise,} \end{cases} \quad (8)$$

$$\text{wt}_\kappa^+(\lambda) = \begin{cases} \lambda \cdot \lambda, & \lambda \leq \kappa \text{ and } \lambda \text{ singular,} \\ \lambda, & \text{otherwise.} \end{cases} \quad (9)$$

Finally put

$$\varrho_{\text{lo}} = \sigma + \sum_{i < k} \text{wt}(\lambda_i) \cdot \eta_i, \quad \varrho_{\text{up}} = \sigma + \sum_{i < k} \text{wt}_\kappa^+(\lambda_i) \cdot \eta_i. \quad (10)$$

Both are nonzero limit ordinals and  $\varrho_{\text{lo}} \leq \varrho_{\text{up}}$ .

**Lemma 64.6** (Regular-cardinal blocks over any alphabet). *Let  $\lambda$  be an infinite regular cardinal,  $|Y| = \lambda$ , and  $0 < \eta < \lambda^+$ . Injective prefixes of length  $\lambda \cdot \eta$  in  $Y$  admit an order embedding into  $\mathbb{B}_{\lambda \cdot \eta}$ . For every  $n < \omega$ ,*

$$\mathcal{W}_{\lambda \cdot \eta + n}(Y) \simeq_e (\mathbb{B}_{\lambda \cdot \eta} \times_{\text{lex}} n!).$$

*Proof.* Use Lemma 64.1. Regularity makes every proper partial sum of entry-code lengths within a  $\lambda$ -entry block less than  $\lambda$ , while positivity makes the full sum exactly  $\lambda$ . Code the  $\eta$  consecutive blocks separately. In an exhaustive enumeration the last  $n$  labels are determined by the preceding prefix, so append their finite permutation rank. This gives the upper embedding. For the lower embedding reserve a fixed  $n$ -element set, orient pairs of the complement indexed by  $\lambda \cdot \eta$ , and append its selected terminal permutation.  $\square$

**Theorem 64.7** (Uniform coding bounds for every fixed type). *With the parameters above,*

$$(\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} n!) \hookrightarrow \mathcal{W}_\alpha(X) \hookrightarrow (\mathbb{B}_{\varrho_{\text{up}}} \times_{\text{lex}} n!). \quad (11)$$

*In particular, if  $\varrho_{\text{lo}} = \varrho_{\text{up}}$ , then  $\mathcal{W}_\alpha(X) \simeq_e (\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} n!)$ . Equality holds whenever every residual cardinal is regular, singular strong limit, or strictly larger than  $\kappa$ .*

*Proof.* For the upper embedding, code the main prefix in length  $\sigma$ . At the beginning of residual layer  $i$ , the remaining alphabet has cardinality  $\lambda_i$ . If it is regular use Lemma 64.6. If it is singular and at most  $\kappa$ , pad the entry code from Lemma 64.1 to length  $\lambda_i$ . The next  $\lambda_i \cdot \eta_i$  entries then use length  $\lambda_i \cdot \lambda_i \cdot \eta_i$ . If  $\lambda_i > \kappa$ , use the fixed global  $\kappa$ -bit sign code instead; its length for this block is  $\kappa \cdot \lambda_i \cdot \eta_i = \lambda_i \cdot \eta_i$ .

Choose an entry code for each possible residual alphabet. Such choices are indexed by a set. Two enumerations that have agreed up to a layer boundary have the same residual alphabet and use the same code; if they disagree before that boundary, their comparison is already decided. Consequently these conditional codes combine to an order-preserving code of length  $\varrho_{\text{up}}$ . Append the finite permutation rank, obtaining the upper embedding.

For the lower embedding choose fixed, pairwise disjoint alphabets for the finitely many residual layers. A singular strong-limit  $\lambda < \mu$  necessarily satisfies  $\lambda < \kappa$ , for otherwise  $\mu = 2^{<\kappa} \leq 2^{<\lambda} = \lambda$ . At such a layer choose an ordered copy of  $\mathbf{No}_{<\lambda}$ , which has cardinality  $\lambda$ , inside a sign cone of  $X$ . Finitely many distinct finite prefixes, such as  $+^i \frown -$ , make these copies disjoint. The singular scheduler in Lemma 64.2, applied at cutoff  $\lambda$ , realizes the factor  $\mathbb{B}_{\lambda \cdot \lambda \cdot \eta}$  there. For all other layers choose arbitrary disjoint sets of the requisite cardinality and realize  $\mathbb{B}_{\lambda \cdot \eta}$  by pair orientation. Also reserve a fixed  $n$ -element terminal set.

Their union has cardinality less than  $\mu$ . Its complement realizes  $\mathbb{B}_\sigma$  by Lemma 64.2. Concatenate all the independent block enumerations and the selected terminal permutation. The first changed source factor supplies the first changed target block. This gives the lower embedding in (11).  $\square$

**Corollary 64.8** (All regular residual layers). *If every  $\lambda_i$  in (7) is regular, and  $\tau = \sum_{i < \kappa} \lambda_i \cdot \eta_i$ , then*

$$\mathcal{W}_{\mu \cdot \gamma + \tau + n}(X) \simeq_e (\mathbb{B}_{\sigma + \tau} \times_{\text{lex}} n!).$$

*In particular every countably infinite residual is classified. Every admissible enumeration type is classified whenever  $\mu \leq \aleph_\omega$ .*

*Proof.* The first assertion is  $\varrho_{\text{lo}} = \varrho_{\text{up}} = \sigma + \tau$ . A countable ordinal is of the form  $\omega \cdot \eta + n$ . If  $\mu \leq \aleph_\omega$ , every smaller infinite cardinal is one of the regular cardinals  $\aleph_j$ ,  $j < \omega$ .  $\square$

Even with a singular non-strong-limit residual, ordinal absorption inside a coefficient  $\eta_i$  can occasionally make  $\varrho_{\text{lo}} = \varrho_{\text{up}}$ . The criterion is the equality of the computed ordinals, not merely the absence of one label in a list of cardinal cases. When they differ, Theorem 64.7 leaves a binary coding-length problem. The next argument nevertheless determines all embedded binary cubes exactly.

#### 64.4 A cardinality fusion lemma and the complete cube spectrum

**Lemma 64.9** (A constant subcylinder from a small range). *Let  $\chi$  be an infinite cardinal and  $A$  a linear order with  $|A| < 2^\chi$ . A nondecreasing map from a binary cylinder with at least  $\chi$  free initial bits, followed by any nonempty factor, into  $A$  is constant on a full subcylinder obtained by fixing fewer than  $\chi$  of those bits.*

*Proof.* Fix all later bits and choose one point in the final factor. This gives  $2^\chi$  ordered test points. Some value has at least three test preimages  $x_0 < x_1 < x_2$ , since a union of  $|A|$  sets of at most two elements has cardinality at most  $|A|$ . Choose  $\delta < \chi$  strictly beyond the first disagreements of  $x_1$  with both outer points. The entire source cylinder determined by the first  $\delta$  bits of  $x_1$  lies strictly between  $x_0$  and  $x_2$ , regardless of its later bits and final factor. Monotonicity forces the same value throughout it.  $\square$

**Lemma 64.10** (The singular non-strong-limit block). *Suppose  $\lambda$  is singular and not strong limit, and  $0 < \eta < \lambda^+$ . Consider a putative order embedding whose target enumerations share their earlier prefix, leaving an alphabet of cardinality  $\lambda$ , and whose source has a free initial block  $\mathbb{B}_{\lambda \cdot \eta}$  followed by a nonempty factor  $K$ . One can fix the source binary block so that the next  $\lambda \cdot \eta$  target entries are constant on the surviving whole  $K$  factor.*

*Proof.* Choose an exponent  $\theta < \lambda$  with  $2^\theta \geq \lambda$ , and put  $\chi = \max(|\theta|, \text{cf } \lambda) < \lambda$ . Then  $2^\chi \geq \lambda$ , and equality would contradict  $\text{cf}(2^\chi) > \chi \geq \text{cf } \lambda$ . Thus  $2^\chi > \lambda$ .

Process target coordinates  $j < \lambda \cdot \eta$ . All earlier target coordinates have been fixed. The current coordinate is therefore a nondecreasing map on the remaining source cylinder into a  $\lambda$ -element order. Lemma 64.9 makes it constant after fewer than  $\chi$  additional source bits. Before coordinate  $j$ , the total bits fixed are at most  $\chi \cdot j$ . There are still  $\chi$  free bits available:

$$\chi \cdot (j+1) < \lambda \cdot \eta \quad (j+1 < \lambda \cdot \eta).$$

Indeed, writing  $j = \lambda \cdot \beta + \delta$  with  $\delta < \lambda$  gives  $\chi \cdot j = \lambda \cdot \beta + \chi \cdot \delta$ , and  $\chi \cdot (\delta+1) < \lambda$ . The ordinal  $\lambda \cdot \eta$  is a limit, so  $j+1$  stays below it at every stage.

At limits take unions of the chosen prefixes. The total cost at the end is at most

$$\chi \cdot (\lambda \cdot \eta) = \lambda \cdot \eta.$$

If fewer bits were used, fix arbitrary additional bits to the exact block boundary. All target entries in the stated block are now fixed, and the entire subsequent factor  $K$  remains.  $\square$

**Theorem 64.11** (Exact binary-cube spectrum for every fixed type). *For every positive finite integer  $m$ ,*

$$(\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} m) \hookrightarrow \mathcal{W}_\alpha(X) \iff m \leq n!. \quad (12)$$

*Consequently, for every ordinal  $\theta$ ,*

$$\boxed{\mathbb{B}_\theta \hookrightarrow \mathcal{W}_\alpha(X) \iff \theta \leq \varrho_{\text{lo}} + \lfloor \log_2(n!) \rfloor.} \quad (13)$$

*This includes all singular non-strong-limit residual layers.*

*Proof.* The lower embedding in Theorem 64.7 proves the forward constructions in (12). Conversely suppose an embedding of  $(\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} m)$  is given. Fuse through the target's main prefix code, using Lemma 64.3; this fixes  $\sigma$  source bits and the entire main target prefix while preserving the later source factor.

At each regular residual layer fuse through the prefix code of length  $\lambda_i \cdot \eta_i$ . At each singular strong-limit layer fuse through the padded code of length  $\lambda_i \cdot \lambda_i \cdot \eta_i$ . At a singular non-strong-limit layer use Lemma 64.10, whose cost is only  $\lambda_i \cdot \eta_i$ . These are exactly the successive summands defining  $\varrho_{\text{lo}}$ . After finitely many layers, all target positions before the last  $n$  are fixed. The surviving  $m$  source points must inject into the  $n!$  permutations of one fixed residual  $n$ -element set. Thus  $m \leq n!$ .

Put  $j = \lfloor \log_2(n!) \rfloor$ . The lower construction contains  $\mathbb{B}_{\varrho_{\text{lo}}+j} = (\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} 2^j)$ , whereas (12) excludes  $\mathbb{B}_{\varrho_{\text{lo}}+j+1} = (\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} 2^{j+1})$ . Every larger cube contains this forbidden cube, and every smaller cube embeds into  $\mathbb{B}_{\varrho_{\text{lo}}+j}$  by padding zeros.  $\square$

The small-range argument concerns a map on a full binary cylinder, not merely on the chosen test points. Fixing a prefix of the middle test point puts every extension between the two outer points. This is what permits the fusion to retain an arbitrary subsequent source factor and to pass through limit stages.

## 64.5 Exact coding lengths, comparisons, and examples

Define the binary coding length  $\ell_2(L)$  to be the least ordinal  $\rho$  for which  $L \hookrightarrow \mathbb{B}_\rho$ . The finite-capacity argument has the following familiar consequence, recalled with its obstruction.

*Note 64.12* (Source 19, Lemma 5.12, “Finite factors over a limit-length cube”). *For nonzero limit ordinals  $\rho, \tau$  and positive finite integers  $m, n$ ,  $(\mathbb{B}_\rho \times_{\text{lex}} m) \hookrightarrow (\mathbb{B}_\tau \times_{\text{lex}} n)$  if and only if  $\rho < \tau$  or  $(\rho = \tau \text{ and } m \leq n)$ ; and  $\ell_2(\mathbb{B}_\rho \times_{\text{lex}} m) = \rho + \lceil \log_2 m \rceil$ . This is source 15’s lemma 31.5, by the same equal-length fusion; source 19 adds that the one-bit fusion also gives  $\mathbb{B}_\beta \hookrightarrow \mathbb{B}_\xi$  exactly when  $\beta \leq \xi$  (“if  $\beta > \xi$ , fixing  $\xi$  source bits to match all target bits leaves at least two source points”), which is lemma 6.4. Source 19 calls it “the familiar consequence ... recalled with its obstruction”. Source 29 proves it a third time (its Lemma 4.1 and Corollary 4.2).*

**Corollary 64.13** (Coding bounds and exact classifications). *For every type in this section,*

$$\varrho_{\text{lo}} + \lceil \log_2(n!) \rceil \leq \ell_2(\mathcal{W}_\alpha(X)) \leq \varrho_{\text{up}} + \lceil \log_2(n!) \rceil. \quad (14)$$

*If  $\varrho_{\text{lo}} = \varrho_{\text{up}}$ , both inequalities are equalities. Two such classified orders with parameters  $(\varrho_{\text{lo}}, n!)$  and  $(\varrho'_{\text{lo}}, m!)$  embed in the indicated direction exactly when*

$$\varrho_{\text{lo}} < \varrho'_{\text{lo}} \quad \text{or} \quad (\varrho_{\text{lo}} = \varrho'_{\text{lo}} \text{ and } n! \leq m!).$$

*Proof.* Apply note 64.12 to the sandwich (11). When its two outer orders coincide up to notation, use the resulting equimorphisms on both sides of a proposed embedding.  $\square$

Under CH,  $\kappa = \mu = \omega_1$  and every admissible type has the form

$$\alpha = \omega_1 \cdot \gamma + \omega \cdot \eta + n.$$

It follows that

$$\begin{aligned} \mathcal{W}_\alpha(\mathbf{No}_{<\omega_1}) &\simeq_e (\mathbb{B}_{\omega_1 \cdot \gamma + \omega \cdot \eta} \times_{\text{lex}} n!), \\ \ell_2(\mathcal{W}_\alpha(\mathbf{No}_{<\omega_1})) &= \omega_1 \cdot \gamma + \omega \cdot \eta + \lceil \log_2(n!) \rceil. \end{aligned}$$

For  $n = 3$ , the largest embedded cube has length  $\omega_1 \cdot \gamma + \omega \cdot \eta + 2$ , while the least coding length is that limit part plus 3. The difference follows from finite-capacity nonembedding, not from cardinality.



For an example with an infinite singular residual, let  $\lambda$  be singular strong limit, put  $\kappa = \lambda^+$  and  $\mu = 2^\lambda$ . Then

$$\mathcal{W}_{\mu \cdot \gamma + \lambda \cdot \eta + n}(\mathbf{No}_{<\lambda^+}) \simeq_e (\mathbb{B}_{\mu \cdot \gamma + \lambda \cdot \lambda \cdot \eta} \times_{\text{lex}} n!).$$

The singular residual contributes an additional ordinal factor  $\lambda$ , despite having smaller cardinality than the main alphabet.

## 64.6 What is settled and what remains

The arbitrary variable residual has now been controlled well enough to determine the complete embedded binary-cube spectrum for every fixed type. Exact equimorphism and coding length are also determined whenever the two computable bounds  $\varrho_{\text{lo}}, \varrho_{\text{up}}$  agree. This includes every type when  $\mu \leq \aleph_\omega$ , every countable residual at any cutoff, and the singular strong-limit residuals described above.

The remaining coding question occurs when singular non-strong-limit residual layers at or below  $\kappa$  make  $\varrho_{\text{lo}} < \varrho_{\text{up}}$ . Does the lower coding bound in (14) always hold with equality? A stronger formulation asks whether the whole order embeds into  $(\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} n!)$ . The simplest general-alphabet instance asks whether

$$\mathcal{W}_\lambda(Y) \hookrightarrow \mathbb{B}_\lambda$$

for every ordered  $Y$  of singular non-strong-limit cardinality  $\lambda$ . The argument above proves that its embedded cube spectrum is exactly  $\theta \leq \lambda$ , but the direct entry code only gives an embedding into  $\mathbb{B}_{\lambda \cdot \lambda}$ . One must not infer a coding embedding from the absence of the next larger cube.

Further refinements concern isomorphism instead of equimorphism, residual gap spectra, and which fiber structures can be recovered from the pure order. The exact residual reduction Theorem 64.4 supplies a useful test: an arbitrary product embedding with the full main binary factor must already realize its remaining factor on a single allowed residual alphabet.

## 64.7 Source 30's route to the residual codes

[merge] Source 30's Sections 4 and 5 classify the same fixed types again, independently: source 30 read this report at pin 7ff7736ec and cites no batch-81 or batch-83 manuscript. It builds the classification block by block (one smaller regular block, one singular strong-limit block, a finite descent through both, residual cardinals above  $\kappa$ ), where source 19 proves the coding sandwich for every type at once and adds the cube spectrum at singular non-strong-limit layers. Every statement of source 30 in these sections is a case of theorem 64.7 with corollary 64.13, or of its equality clause, proved with the same entry codes, padded codes, schedulers and reserved alphabets; they are recorded in notes. Source 30 states one consequence that source 19 does not state, printed as corollary 64.20, and an unconditional mixed example. Its weight  $r(\nu)$  is printed  $\text{wt}(\nu)$  and its bl is  $\ell_2$ .

*Note 64.14* (Source 30, opening of Section 4; dated correction). Source 30 opens: “The repository’s cardinal-block theorem treats  $\mathcal{W}_{\mu \gamma + n}(\mathbf{No}_{<\kappa})$ , where  $\mu = 2^{<\kappa}$  and  $n < \omega$ , but explicitly leaves infinite remainders  $\zeta < \mu$  untreated. See source label `swo:cb:new:thm:block-types`, lines 9200–9367, and the boundary discussion `swo:vi:sub:cb-boundary`, lines 9561–9580, at the source pin recorded above. The first new instance proved here is  $\mu + \omega$ . In fact the argument handles every countable remainder, and more generally a block of any smaller regular cardinal followed by a finite remainder.” [merge] (3 October 2026.) This was true of this report at source 30’s pin 7ff7736ec (the cited line numbers are those of theorem 31.2 and section 31.3). Within batch 83, source 19, written earlier at pin d5bd4a67b, treats every infinite remainder (theorems 64.7 and 64.11); the two treatments are independent.



*Note 64.15* (Source 30, Lemmas 4.1–4.3: entry codes, the long prefix, small deletion). Lemma 4.1 (“Short codes for arbitrary regular-cardinality alphabets”): *let  $\nu$  be an infinite regular cardinal and  $Y$  an arbitrary linear order of cardinality  $\nu$ ; there is an order-preserving, prefix-free binary code  $c_Y$  for  $Y$  with  $0 < \text{lh}(c_Y(y)) < \nu$ ; consequently  $\mathcal{W}_{\nu\delta+n}(Y) \simeq_e \mathbb{B}_{\nu\delta} \times_{\text{lex}} n!$  for every  $0 < \delta < \nu^+$  and  $n < \omega$ ; for  $\nu = \omega$  the codes are finite binary words, and the conclusion holds “for every countable linear alphabet, regardless of density, endpoints, jumps, or scatteredness”.* This is lemma 64.1 with lemma 64.6, by the same stage-controlled insertion into  $\mathbf{No}_{<\nu}$  (“regularity is not needed for this embedding step”) and the same  $00 < 01 < 11$  sign code; for  $\nu = \omega$  source 30 notes the alternative embedding into the dyadic rationals. Lemma 4.2 (“Exact coding length of an arbitrary long prefix”): *for  $0 < \gamma < \mu^+$  there is an order embedding  $X_{\text{lex}}^{\mu\gamma} \rightarrow \mathbb{B}_{\rho_\kappa\gamma}$ , in particular for all injective prefixes of that length; no exhaustiveness hypothesis is used, which is the first half of lemma 64.2 and the upper construction of theorem 31.2 (“This proves the asserted order embedding even for functions with repeated entries”).* Lemma 4.3 (“Small deletion in the singular scheduler”): *let  $\kappa$  be singular strong limit and  $D \subseteq \mathbf{No}_{<\kappa}$  with  $|D| < \kappa$ ; for  $0 < \gamma < \kappa^+$  and  $\lambda = \kappa\gamma$ ,  $\mathbb{B}_{\kappa\lambda}$  order-embeds into  $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa} \setminus D)$ .* This is the second half of lemma 64.2: it “extends the repository’s finite-deletion lemma” lemma 31.4 “from finite deletion to deletion of cardinality below  $\kappa$ ”, and it prints the row scheduler of lemma 31.3 in full “to expose the order, rather than merely cardinal, requirements”: the sign cones  $+\xi \frown -$ ,  $\xi < \kappa$ , of which fewer than  $\kappa$  meet  $D$ ; a filler row for the unselected labels; rows split into blocks of  $\kappa$  and scheduled by  $\max(\xi, j)$ ; and the regrouping of coordinates into blocks of cf  $\kappa$  restricted to sign words of lengths  $\eta_j$  with  $\sum_j \eta_j = \kappa$ .

*Note 64.16* (Source 30, Theorem 4.4 and Corollaries 4.5, 4.6 and 4.8; Remark 4.7). Theorem 4.4 (“A smaller regular-cardinal block after the long prefix”): *for an infinite regular  $\nu < \mu$ ,  $0 < \gamma < \mu^+$ ,  $0 < \delta < \nu^+$  and  $n < \omega$ ,  $\mathcal{W}_{\mu\gamma+\nu\delta+n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\rho_\kappa\gamma+\nu\delta} \times_{\text{lex}} n!$ ; all constructions are set constructions in ZFC, with no global choice or class comprehension.* Corollary 4.5 (“Every countable remainder”): *if  $\mu > \omega$ , then  $\mathcal{W}_{\mu\gamma+\omega\delta+n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\rho_\kappa\gamma+\omega\delta} \times_{\text{lex}} n!$  for  $0 < \gamma < \mu^+$ ,  $0 < \delta < \omega_1$ ,  $n < \omega$ ;* “Together with the repository’s finite-remainder theorem, this covers every  $\alpha = \mu\gamma + \zeta$  with  $\zeta < \omega_1$ . For a countable alphabet itself, Lemma 4.1 already treats every admissible ordinal length.” Corollary 4.6 (“Exact coding and cube spectrum”): *with  $\sigma = \rho_\kappa\gamma + \nu\delta$ , the least binary coding length is  $\sigma + \lceil \log_2(n!) \rceil$ ,  $\mathbb{B}_\theta$  embeds exactly when  $\theta \leq \sigma + \lceil \log_2(n!) \rceil$ , and two instances embed exactly when  $\sigma < \sigma'$  or ( $\sigma = \sigma'$  and  $n! \leq m!$ ), by lemma 31.5.* Remark 4.7: “At  $\kappa = \omega_1$ , the countable-remainder corollary already classifies every fixed type under CH: divide  $\alpha$  by  $\omega_1$  and apply the corollary to the countable remainder. Corollary 5.6 below removes CH entirely.” Corollary 4.8 (“Finite descent through regular residual cardinals”): *for regular  $\mu > \nu_1 > \dots > \nu_t$ ,  $\mathcal{W}_{\mu\gamma+\zeta+n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\sigma \times_{\text{lex}} n!$  with  $\zeta = \nu_1\delta_1 + \dots + \nu_t\delta_t$  and  $\sigma = \rho_\kappa\gamma + \zeta$ , with the coding-rank, cube-spectrum and comparison conclusions of Corollary 4.6.* These are corollary 64.8 and corollary 64.13, by the same upper codes (“ordinary set choice suffices” to choose one code for each possible residual alphabet; “equal earlier prefixes determine the same code”) and the same reserved alphabets with pair orientations. Source 30’s closing paragraph of its Section 4, “The remaining boundary”, observes that every remainder has a finite decomposition into blocks of strictly decreasing infinite cardinalities followed by a finite term, that “regularity is the exact step used to make a block of short entry codes have precisely the cardinal’s ordinal length, and the residual alphabet need not contain all of  $\mathbf{No}_{<\nu}$ ”, and that after its Section 5 “the remaining uncontrolled regime is therefore singular, not strong limit, and at most  $\kappa$ ”, as for source 19.

*Note 64.17* (Source 30, Theorem 5.1 and Corollary 5.2). Theorem 5.1 (“A singular strong-limit residual block”): *if  $\nu < \mu$  is a singular strong-limit cardinal,  $0 < \gamma < \mu^+$ ,  $0 < \delta < \nu^+$  and  $n < \omega$ , then  $\mathcal{W}_{\mu\gamma+\nu\delta+n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\rho_\kappa\gamma+\nu\cdot\nu\cdot\delta} \times_{\text{lex}} n!$ .* Corollary 5.2 (“A strict obstruction to the uninflated residual formula”): *with  $n = 0$  and  $\delta = 1$ ,  $\ell_2(\mathcal{W}_{\mu\gamma+\nu}(\mathbf{No}_{<\kappa})) = \rho_\kappa\gamma + \nu \cdot \nu$ , so*

$\mathcal{W}_{\mu\gamma+\nu}(\mathbf{No}_{<\kappa}) \not\rightarrow \mathbb{B}_{\rho_\kappa\gamma+\nu}$ ; “Thus the smaller regular-cardinal formula cannot be extended to all singular residual cardinals with the same uninflated tail length.” These are the singular strong-limit layer of [theorem 64.7](#), where  $\text{wt}(\nu) = \text{wt}_\kappa^+(\nu) = \nu \cdot \nu$ , and its coding consequence [corollary 64.13](#); source 30’s proof is source 19’s:  $\nu < \kappa$  because  $\mu = 2^{<\kappa} \leq 2^{<\nu} = \nu$  otherwise, every residual alphabet of size  $\nu$  embeds into  $\mathbf{No}_{<\nu}$  with codes of length below  $\nu$ , padded to length  $\nu$ , and the lower embedding reserves the full smaller cutoff  $\mathbf{No}_{<\nu}$  and runs the singular scheduler at cutoff  $\nu$ . Source 30 adds: “The explicit strict contrast above uses  $\delta = 1$ ; for arbitrary  $\delta$ , ordinal absorption can make  $\nu \cdot \nu \cdot \delta = \nu \cdot \delta$ . The smaller-cutoff argument alone does not classify a residual singular cardinal which is not strong limit. The padded upper bound remains available, but the smaller full surreal cutoff may have more than  $\nu$  labels, so it cannot be reserved as a residual alphabet of size  $\nu$ .”

*Note 64.18* (Source 30, Corollaries 5.3 and 5.4). For an infinite cardinal  $\nu$  that is regular or singular strong limit, source 30 puts  $\text{wt}(\nu) = \nu$  or  $\nu \cdot \nu$  respectively. Corollary 5.3 (“Finite descent through both controlled cardinal regimes”): *if  $\mu > \nu_0 > \dots > \nu_k$  are infinite cardinals, each regular or singular strong limit, then  $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\sigma \times_{\text{lex}} n!$  for  $\alpha = \mu\gamma + \sum_{i \leq k} \nu_i \delta_i + n$  and  $\sigma = \rho_\kappa\gamma + \sum_{i \leq k} \text{wt}(\nu_i) \delta_i$ .* Corollary 5.4 (“All fixed types when smaller cardinals are controlled”): *if every infinite cardinal below  $\mu$  is regular or singular strong limit, every fixed type of cardinality  $\mu$  is covered by this formula; in particular this yields a complete isomorphism classification of all fixed types under GCH*, because under GCH every singular cardinal is a strong limit. Under GCH, with the canonical decomposition  $\alpha = \sum_{i \leq k} \kappa_i \delta_i + n$ ,  $\kappa = \kappa_0 > \kappa_1 > \dots > \kappa_k \geq \omega$ , the complete formula reads

$$\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\sum_{i \leq k} \text{wt}(\kappa_i) \cdot \delta_i} \times_{\text{lex}} n!,$$

and “every product and sum here is ordinal; replacing a term  $\kappa_i \cdot \kappa_i$  by its cardinality would destroy the claimed classification”. These are the equality clause of [theorem 64.7](#) in the regular and singular strong-limit cases; the GCH classification follows from that clause, and source 19 leaves it implicit. The one point source 30 checks separately is that the reserved alphabets of the singular strong-limit blocks (copies of  $\mathbf{No}_{<\nu_i}$  behind distinct prefixes  $+^j \frown -$ ) are disjoint, as in source 19’s lower embedding.

[merge] Source 30’s unconditional example, with a singular residual inflation, a subsequent countable block and a non-power-of-two finite factor in one type:

**An unconditional mixed example.** Let  $\nu = \beth_\omega$ , let  $\kappa = \nu^+$ , and put  $\mu = 2^\nu = 2^{<\kappa}$ . Then  $\nu$  is singular strong limit, whereas the successor cutoff  $\kappa$  is not singular strong limit, so  $\rho_\kappa = \mu$ . For

$$\alpha = \mu + \nu + \omega + 3, \quad \sigma = \mu + \nu \cdot \nu + \omega,$$

the mixed theorem gives

$$\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\sigma \times_{\text{lex}} 6.$$

Its exact binary coding length is  $\sigma + 3$ , while its largest embedded binary cube has length  $\sigma + 2$ . This example needs no continuum hypothesis: it displays a singular residual inflation, a subsequent countable block, and a non-power-of-two finite factor in one fixed enumeration type.

*Note 64.19* (Source 30, Proposition 5.5, “Residual cardinals above the birthday cutoff”). *If  $\kappa < \nu < \mu$  is any infinite cardinal, with no regularity assumption, then  $\mathcal{W}_{\mu\gamma+\nu\delta+n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\mu\gamma+\nu\delta} \times_{\text{lex}} n!$  for  $0 < \gamma < \mu^+$ ,  $0 < \delta < \nu^+$  and  $n < \omega$ .* This is the clause  $\text{wt}_\kappa^+(\nu) = \nu$  for  $\nu > \kappa$  of [theorem 64.7](#), by the same padded native code: “since  $\mu$  and  $\nu$  are initial cardinals strictly larger than  $\kappa$ ,  $\kappa \cdot \mu = \mu$ ,  $\kappa \cdot \nu = \nu$ ”, with no regularity of either. Source 30 then combines the three admitted regimes into one cost per block,  $\nu$  when  $\nu$  is regular or larger than  $\kappa$  and  $\nu \cdot \nu$  when  $\nu$  is singular strong

limit (“a singular strong-limit  $\nu < \mu$  is necessarily below  $\kappa$ ”), and concludes: “The remaining unresolved residual-cardinal regime is therefore *singular, not strong limit, and at most  $\kappa$* , with  $\nu < \mu$ . Singularity by itself is not an obstruction to an uninflated residual code: the available uniform birthday bound must also be taken into account.” Source 19 proves more in that regime: the cube spectrum (theorem 64.11) and the coding bounds corollary 64.13.

[merge] Source 30’s Corollary 5.6 is a consequence of the equality clause of theorem 64.7 that source 19 does not state: source 19 states the classification of every type when  $\mu \leq \aleph_\omega$  (corollary 64.8) and at  $\kappa = \omega_1$  only under CH. It is printed whole.

**Corollary 64.20** (The first uncountable cutoff needs no continuum hypothesis). *Put  $X = \mathbf{No}_{<\omega_1}$  and  $\mu = 2^{\aleph_0}$ . For every ordinal  $\alpha$  of cardinality  $\mu$ , write  $\alpha = \lambda + n$ , where  $\lambda$  is a nonzero limit ordinal and  $n < \omega$ . Then, in ZFC,*

$$\mathcal{W}_\alpha(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!.$$

*More generally the same conclusion holds for  $X = \mathbf{No}_{<\kappa}$  at every infinite initial cutoff  $\kappa < \aleph_\omega$ , with  $\mu = 2^{<\kappa}$  and  $|\alpha| = \mu$ .*

*Proof.* For such a  $\kappa$ , every infinite cardinal at most  $\kappa$  is regular. Any larger residual cardinal has the uninflated native code from note 64.19. Thus every cardinal in the canonical finite residual decomposition is covered and has cost equal to itself. The main cutoff is regular, so  $\rho_\kappa = \mu$  as well. The mixed formula therefore has binary length equal to the entire limit part  $\lambda$  of  $\alpha$ . For  $\kappa = \omega_1$ , its alphabet cardinal is  $2^{\aleph_0}$ , with no condition on the value of that cardinal. For  $\kappa = \omega$ , there are no smaller infinite residual cardinals, and the existing cardinal-block theorem supplies the same conclusion.  $\square$

## 65 Completions of the minimal slice (sources 24, 21 and 19)

[merge] Three different completions of the minimal slice are printed here: the Hausdorff completion for agreement on bounded initial segments (source 24’s Section 5), the closure of the minimal stratum inside the full order  $\mathcal{W}(X)$  (source 21’s Section 4), and normal forms for the Dedekind completion (source 19’s Section 7). Source 24’s own summary of the difference: “Forward stabilization preserves injectivity but can lose labels. Inverse stabilization prevents that loss. Dedekind completion is larger again: it also fills cylinder edges and numerical gaps.” Sources 26 and 28 prove the forward completion again; source 28 extends it to every admissible coordinate ideal (theorem 70.25).

### 65.1 Forward prefix completion and the missing-label boundary (source 24)

#### 65.1.1 The uniformity that remembers only initial segments

For  $\alpha < \mu$ , put

$$E_\alpha = \{(f, g) : f \restriction \alpha = g \restriction \alpha\}.$$

These equivalence relations form a base of entourages for a Hausdorff uniformity on either  $\mathcal{W}_\mu(X)$  or  $\mathcal{I}_\mu(X)$ . We call it the *forward prefix uniformity*. Its topology is the cylinder topology. No ordering of  $X$  is needed to define it.

A net is Cauchy for this uniformity when, for every  $\alpha < \mu$ , all sufficiently late terms have exactly the same restriction to  $\alpha$ . The requirement is stronger than merely having many pairwise agreements, and at a singular  $\mu$  it must still be stated for every bounded initial segment, not merely for arbitrary finite sets of coordinates.

**Theorem 65.1** (Exact forward completion). *The uniform space  $\mathcal{I}_\mu(X)$  is complete. The inclusion*

$$\mathcal{W}_\mu(X) \hookrightarrow \mathcal{I}_\mu(X)$$

*is a dense uniform embedding and exhibits  $\mathcal{I}_\mu(X)$  as the Hausdorff completion of  $\mathcal{W}_\mu(X)$ .*

*Proof.* Let  $\mathcal{F}$  be a Cauchy filter on  $\mathcal{I}_\mu(X)$ . For each  $\alpha < \mu$  it contains a nonempty  $E_\alpha$ -small set, hence contains a cylinder  $[p_\alpha]$  of length  $\alpha$ . It cannot contain two distinct cylinders of that length, since they are disjoint. These unique prefixes are coherent: if  $\alpha < \beta$ , the two cylinders in the filter intersect, so  $p_\alpha = p_\beta \upharpoonright \alpha$ .

Their union  $h = \bigcup_{\alpha < \mu} p_\alpha$  is a function with domain  $\mu$ . It is injective, because any two proposed equal values occur in a common sufficiently long injective prefix. The filter contains every cylinder neighborhood of  $h$  and therefore converges to  $h$ . This proves completeness.

The entourages on  $\mathcal{W}_\mu(X)$  are the restrictions of the same entourages on  $\mathcal{I}_\mu(X)$ . Every cylinder of the latter meets  $\mathcal{W}_\mu(X)$  by Lemma 62.4. Completeness, density, and Hausdorffness give the claimed completion, including its uniqueness up to the usual uniform isomorphism fixing the dense subspace.  $\square$

**Theorem 65.2** (Exhaustion is an intersection of dense open requirements). *For each  $x \in X$ , let*

$$U_x = \{h \in \mathcal{I}_\mu(X) : x \in \text{ran } h\}, \quad F_x = \mathcal{I}_\mu(X) \setminus U_x.$$

*Then  $U_x$  is open dense and  $F_x$  is closed nowhere dense. Moreover,*

$$\mathcal{W}_\mu(X) = \bigcap_{x \in X} U_x, \quad \mathcal{I}_\mu(X) \setminus \mathcal{W}_\mu(X) = \bigcup_{x \in X} F_x.$$

*Both  $\mathcal{W}_\mu(X)$  and its complement are dense in  $\mathcal{I}_\mu(X)$ .*

*Proof.* Once  $x$  occurs, a prefix extending past that occurrence witnesses membership in  $U_x$ , so  $U_x$  is open. Any prefix can be extended to include  $x$ , if it does not already include it; then complete the new prefix. Hence  $U_x$  is dense, and  $F_x$  has the stated properties. The displayed equalities express surjectivity exactly.

Density of  $\mathcal{W}_\mu(X)$  was proved above. Given any short prefix, choose an unused label  $z$  and extend to an enumeration of  $X \setminus \{z\}$ . This gives a nonexhaustive point in the cylinder and proves density of the complement.  $\square$

The last result is a generalized dense-intersection description, not automatically an ordinary  $G_\delta$  statement. It becomes a  $G_\delta$  statement when  $X$  is countable. At larger cardinalities, the number of exhaustion requirements matters.

### 65.1.2 A concrete failure of forward completeness

**Example 65.3** (A label escaping to later and later positions). On  $\mathbb{N}$ , define

$$\pi_n(k) = \begin{cases} k+1, & k < n, \\ 0, & k = n, \\ k, & k > n. \end{cases}$$

Each  $\pi_n$  is the cycle  $(0 \ 1 \ \cdots \ n)$ . For every fixed  $k$ , eventually  $\pi_n(k) = k+1$ , so the sequence is forward Cauchy and converges in  $\mathcal{I}_\omega(\mathbb{N})$  to  $h(k) = k+1$ . The limit omits 0. Equivalently,  $\pi_n^{-1}(0) = n$  never stabilizes. Forward stabilization records where each position goes, but not where each target label comes from.

For arbitrary infinite  $\mu$ , choose a nononto injection  $h : \mu \rightarrow X$  and, for every  $\alpha < \mu$ , an exhaustive extension  $f_\alpha$  of  $h \upharpoonright \alpha$ . The  $\mu$ -indexed net  $(f_\alpha)$  is forward Cauchy with limit  $h$  in  $\mathcal{I}_\mu(X)$  and no limit in  $\mathcal{W}_\mu(X)$ . Thus the forward uniformity is incomplete for every infinite  $\mu$ .

### 65.1.3 Why this is not Dedekind completion

Assume thickness. Let  $\hat{P}$  be the Dedekind completion of the dense order  $P = \mathcal{W}_\mu(X)$ , using nonempty proper lower sets without a greatest element as the cut convention. For  $h \in \mathcal{I}_\mu(X)$ , put

$$L_h = \{f \in P : f <_{\text{lex}} h\}.$$

**Proposition 65.4** (The injection space sits strictly inside order completion). *The map  $h \mapsto L_h$  is an order embedding of  $\mathcal{I}_\mu(X)$  into  $\hat{P}$  extending the canonical embedding of  $P$ . If  $h$  is nonexhaustive, its image is a nonprincipal cut of  $P$ . This map is not onto: even cylinder-edge cuts need not come from injective words.*

*Proof.* For  $h < k$  in  $\mathcal{I}_\mu(X)$ , choose an unused label strictly between their values at the first disagreement and complete the resulting prefix exhaustively. This gives  $f \in P$  with  $h < f < k$ . Repeating the same argument supplies exhaustive words on either side of any injection and between any injection and a strictly smaller or larger exhaustive word. It follows that  $L_h$  is a valid cut, that the map is strictly order preserving, and that a nonexhaustive  $h$  has neither a greatest exhaustive word below it nor a least exhaustive word above it. If its cut were the cut of an exhaustive  $g$ , the same separation argument between  $h$  and  $g$  would be a contradiction.

Fix  $x \in X$ . Consider the lower edge of the cylinder  $[\langle x \rangle]$ :

$$L = \{f \in P : f(0) < x\}.$$

This is a nonempty proper lower set with no greatest element. An injection beginning with a label below  $x$  is not cofinal in this lower side, by density of the alphabet. An injection beginning above  $x$  lies above the entire  $x$ -cylinder. An injection beginning exactly with  $x$  has exhaustive words with that same first label strictly below it. Its lower cut therefore contains words not in  $L$ . These three cases exhaust the possibilities, so  $L$  is not  $L_h$  for any injection  $h$ .  $\square$

Thus there are at least two qualitatively different boundaries. A *branch boundary* is an injective full-length word missing labels. A *prefix edge* ends a cylinder without selecting a full continuation. Alphabet gaps create further order-completion points. Earlier cut-classification results (sources 17 and 13; [section 30.2](#)) organize these possibilities more finely; [Theorem 65.1](#) identifies exactly the part visible to the forward uniformity.

*Note 65.5* (Source 26, [Theorem 4.1](#) and [Proposition 4.2](#)). Source 26's [Theorem 4.1](#) (“Forward completion; earlier result reproved”):  $\mathcal{I}_\mu(X)$  is complete for the forward-prefix uniformity, and the inclusion  $\mathcal{W}_\mu(X) \subseteq \mathcal{I}_\mu(X)$  exhibits it as the Hausdorff uniform completion of the permutation space. Its [Proposition 4.2](#) (“Exhaustion requirements”): each  $U_x = \{h \in \mathcal{I}_\mu(X) : x \in \text{ran } h\}$  is open dense,  $\mathcal{W}_\mu(X) = \bigcap_{x \in X} U_x$ , and the complement  $F_x$  is closed nowhere dense and nonempty. These are [theorems 65.1](#) and [65.2](#); source 26 says “This is the mechanism of the earlier completion theorem” (source 24), and stresses that at uncountable  $\mu$  “a Cauchy filter must eventually stabilize every bounded prefix; pointwise stabilization of isolated coordinates alone is not the definition”. It adds: “The missing points of the uniform completion are therefore precisely the nonexhaustive injections. This is not a Dedekind completion theorem. Order completion can add cylinder edges and gaps in the alphabet that are not represented by a full injective branch ... the non-Borel theorem below concerns the branch boundary already present in  $\mathcal{I}_\mu(X)$ ” ([theorem 69.17](#)).



## 65.2 First-cardinality fibres and the closure of the minimal stratum (source 21)

[merge] Source 21's Section 4, printed whole. Its first-cardinality decomposition is the set-level form of the fibre decomposition of class well-orders that source 16 proves in Part VIII (equation (S16.12)) and of source 22's residual-cylinder identity (theorem 63.1); the closure theorem is new. Source 21's  $\mathcal{I}_\mu(X)$  is this Part's.

The set-sized setting already contains a useful prototype of the proper-class residual-tail phenomenon. No higher-order foundation is needed for it.

**Theorem 65.6** (First-cardinality decomposition). *Let  $X$  be any infinite linearly ordered set of cardinality  $\mu$ . There is a canonical order isomorphism*

$$\mathcal{W}(X) \cong \sum_{f \in \mathcal{I}_\mu(X)} \mathcal{W}(X \setminus \text{ran}(f)), \quad (15)$$

where the index set has its numerical lexicographic order and the sum is a lexicographic sum. The fiber over an exhaustive  $f$  is a singleton; the set of these singleton points is exactly  $\mathcal{W}_\mu(X)$ .

*Proof.* Every well-order of  $X$  has an enumeration of ordinal length at least  $\mu$ . Take its first  $\mu$  entries, an injective word  $f$ . The rest of the relation is exactly a well-order of the unused set  $Y_f = X \setminus \text{ran}(f)$ . Conversely, concatenate  $f$  with the enumeration of any well-order of  $Y_f$ . This uses every label exactly once and produces a well-order of  $X$ . The two constructions are inverse.

If two first words differ, their first disagreement occurs below  $\mu$  and decides the comparison of all corresponding completions. If they agree, comparison reduces to the residual well-orders. This is precisely the lexicographic sum in (15). For  $Y_f = \emptyset$  there is one empty residual relation. For  $Y_f \neq \emptyset$ , the concatenation has length strictly greater than  $\mu$ , so it does not belong to the minimum stratum.  $\square$

One should not confuse a singleton fiber with an exhaustive word. A one-element residual set also has only one well-order. Thus the unlabelled size of a fiber cannot detect exhaustivity.

**Lemma 65.7** (Endpoints of a residual enumeration order). *For any set  $Y$  with a linear label order,  $\mathcal{W}(Y)$  has a least element if and only if the label order  $<$  well-orders  $Y$ . It then consists of listing  $Y$  in that order. It has a greatest element if and only if the reverse label order well-orders  $Y$ .*

*Proof.* If  $<$  is a well-order, its increasing enumeration chooses the least unused label at every stage. Every first disagreement in another enumeration is larger, so this is the lexicographic minimum. Conversely, if a lexicographic minimum ever chooses a label while a smaller unused label remains, interchange the chosen label with that later label. The resulting well-order is smaller. Hence a minimum always chooses the least unused label; its enumeration shows that the label order itself is a well-order. Reverse the inequalities for the greatest element.  $\square$

**Lemma 65.8** (Density of exhaustive branches in the index order). *If  $X$  has uniformly large intervals, the exhaustive words  $\mathcal{W}_\mu(X)$  are order-dense in  $\mathcal{I}_\mu(X)$ , and there are exhaustive words on both sides of every nonexhaustive word.*

*Proof.* For  $f < g$ , let  $\delta < \mu$  be their first disagreement. Fewer than  $\mu$  labels occur in the common prefix. Choose an unused label strictly between  $f(\delta)$  and  $g(\delta)$  and complete that prefix to an exhaustive  $\mu$ -word. It lies strictly between  $f$  and  $g$ . To obtain exhaustive words on both sides of a single branch, choose unused labels below and above its first label, respectively, and complete.  $\square$

**Theorem 65.9** (Closure theorem). *Assume  $X$  has uniformly large intervals. In the order topology of  $\mathcal{W}(X)$ , the closure of  $\mathcal{W}_\mu(X)$  consists of:*

1. every point of  $\mathcal{W}_\mu(X)$ ; and
2. the least and greatest point, when they exist, of each fiber over a nonexhaustive  $f \in \mathcal{I}_\mu(X)$ .

By [lemma 65.7](#), the added endpoints are exactly the orders obtained by following  $f$  with an increasing or decreasing numerical well-order of its unused set.

*Proof.* An interior point of a nonexhaustive fiber has a point of that fiber on each side. The open interval between those two points is disjoint from  $\mathcal{W}_\mu(X)$ , so the point is not in the closure.

Suppose  $R$  is the least point of its fiber over  $f$ . If  $A < R < B$ , then  $A$  cannot lie in the same fiber. Its first  $\mu$  entries form a word  $g < f$ . By [lemma 65.8](#) choose an exhaustive word  $h$  with  $g < h < f$ . Its singleton fiber gives a point of  $\mathcal{W}_\mu(X)$  between  $A$  and  $R$ , hence inside  $(A, B)$ . Thus every neighborhood of  $R$  meets the minimum stratum. The greatest-point case is symmetric. A singleton fiber satisfies both arguments.  $\square$

**Example 65.10** (Finite and infinite residuals). If  $f$  omits a finite set of size  $n$ , its fiber is an  $n!$ -element chain. For  $n = 1$  the unique point is in the closure; for  $n = 2$  both points are; for  $n \geq 3$  only the two extreme permutations are. If  $X = \mathbb{Q}$  and  $f : \omega \rightarrow \mathbb{Q} \setminus \mathbb{N}$  is a bijection, the residual  $\mathbb{N}$  has a numerical well-order but its reverse does not. Exactly the least point of this infinite fiber is in the closure. If the unused set has order type  $\mathbb{Q}$ , neither endpoint exists, and its entire fiber is disjoint from the closure.

The closure theorem describes the relative position of strata, not just their separate cardinalities. It also identifies the information lost by retaining only the first  $\mu$  entries: all residual well-order choices are compressed to a single index. At the class level the first Ord-block plays the analogous role, but its fibers are no longer automatically set-sized.

### 65.3 Normal forms for the Dedekind completion of the minimal slice (source 19)

[merge] Source 19's Section 7, printed whole. Its three kinds of cut are source 17's cut trichotomy ([lemma 30.8](#)): a branch, an edge at a last straddling prefix, a limit-prefix boundary. New in source 19 are the uniqueness of the codes, the order read directly from them, and the sign codes of the residual splits ([proposition 65.12](#)). "The gap spectrum in the preceding research" is [theorem 30.11](#) (source 17) with source 18's multiplicities [theorem 30.16](#). Its full-branch cuts  $L_f$  are source 24's trace cuts ([proposition 65.4](#)). Source 19's  $\text{Perm}_\mu(D)$  is printed  $\mathcal{W}_\mu(D)$ ; its alphabet  $D$  is thick.

The gap spectrum in the preceding research identifies possible characters and their multiplicities. The same prefix analysis gives more information: it describes the arrangement of the completion points through unique decorated-word codes. This construction is entirely set-sized and therefore takes place in ZFC.

Let  $D$  be a dense linear order without endpoints, of infinite cardinality  $\mu$ , and assume that every nonempty open interval of  $D$  has cardinality  $\mu$ . Put

$$P = \mathcal{W}_\mu(D) = \{e : \mu \longrightarrow D : e \text{ is bijective}\}.$$

The application is  $D = \mathbf{No}_{<\kappa}$  and  $\mu = 2^{<\kappa}$  for an infinite initial cardinal  $\kappa$ . For an injective prefix  $p : \delta \rightarrow D$ ,  $\delta < \mu$ , write

$$C_p = \{e \in P : e \restriction \delta = p\}, \quad D_p = D \setminus \text{ran}(p).$$

The cylinder  $C_p$  is nonempty and convex. Since fewer than  $\mu$  labels have been used,  $D_p$  still meets every nonempty interval and is unbounded in both directions. In particular, every cylinder has neither a first nor a last member.



Use the following convention for the Dedekind completion:  $\overline{P}$  consists of the nonempty proper initial subsets of  $P$  having no greatest element, ordered by inclusion. The point  $e \in P$  is represented by  $P_{<e}$ . The two external endpoints can be added separately when a compactified completion is wanted.

**Theorem 65.11** (Unique normal forms for the completion). *Every point of  $\overline{P}$  has exactly one code of one of the following three kinds.*

1. **A full branch.** An injection  $f : \mu \rightarrow D$ , representing

$$L_f = \{e \in P : e < f\}.$$

*If  $f$  is onto, this is the original point  $f$ ; otherwise it is an omitted-branch gap.*

2. **A numerical split.** A pair  $(p, I)$ , where  $p$  has length  $\delta < \mu$  and  $I$  is a nonempty proper initial subset of  $D_p$ , representing

$$L_{p,I} = \{e : e < C_p\} \cup \{e \in C_p : e(\delta) \in I\}.$$

*Here  $e < C_p$  means that  $e$  lies below every member of the cylinder. The set  $I$  may have a greatest member, and its complement may have a least member. This kind always represents a proper gap.*

3. **A limit-prefix boundary.** A pair  $(p, \varepsilon)$ , where  $p$  has nonzero limit length  $\delta < \mu$  and  $\varepsilon \in \{-, +\}$ , representing respectively

$$L_{p,-} = \{e : e < C_p\}, \quad L_{p,+} = \{e : e < C_p\} \cup C_p.$$

*Both are proper gaps.*

*The restrictions in this list eliminate all duplication. In particular, a successor-prefix boundary is already a numerical-split code at its predecessor; a principal point is represented only by its full surjective branch.*

*Proof.* Let  $L \in \overline{P}$ , and let  $K$  consist of the short injective prefixes whose cylinders meet both  $L$  and  $P \setminus L$ . Two incompatible cylinders are disjoint convex sets, so they cannot both straddle one cut. Thus  $K$  is a downward-closed chain under extension. It contains the empty prefix because both sides of the cut are nonempty.

If  $K$  has a greatest member  $p$ , no child cylinder straddles. Partition  $D_p$  according to whether its entire child cylinder is in  $L$  or in  $P \setminus L$ . Both parts are nonempty since  $C_p$  straddles. The lower digit part is an initial subset  $I$ , and the cut is precisely  $L_{p,I}$ . Both sides have no endpoint: if a digit side has an endpoint, its final or initial child cylinder still has no endpoint; otherwise the varying next digits themselves defeat an endpoint. The longest straddling prefix and its partition are intrinsic to  $L$ , proving uniqueness in this case.

Suppose that  $K$  has no greatest member and its lengths have supremum  $\delta < \mu$ . The union  $p = \bigcup K$  is injective and has nonzero limit length  $\delta$ . Its cylinder does not straddle, for then  $p$  would be a greatest member of  $K$ . If  $C_p \subseteq P \setminus L$ , the cut is the lower boundary of  $C_p$ : any point below  $C_p$  disagrees with  $p$  at an earlier coordinate, and a longer member of  $K$  has a point of  $L$  above it. Every point above or in  $C_p$  belongs to the upper side. The reverse argument gives the upper boundary when  $C_p \subseteq L$ . The union  $p$  and the side containing its cylinder are intrinsic.

In the remaining case the lengths in  $K$  are cofinal in  $\mu$ . Their union is an injection  $f : \mu \rightarrow D$ . Every permutation  $e$  different from  $f$  has a first disagreement with it. Looking at a longer straddling prefix shows that  $e \in L$  exactly when  $e < f$ . If  $f$  is onto, it belongs to  $P$  and cannot lie in  $L$ ,

because then it would be the greatest member of  $L$ . Consequently  $L = L_f$  in the onto case as well as the non-onto case. The word  $f$  is uniquely recovered as the union of  $K$ .

Conversely, the displayed codes have exactly their asserted straddling patterns. For a numerical split,  $p$  is the last straddling prefix. For a limit boundary, all proper prefixes of  $p$  straddle and  $p$  itself does not. To see straddling arbitrarily late, choose unused smaller and larger next digits and complete the resulting prefixes. For a full branch, every short prefix straddles by the same construction. The full-interval-size hypothesis ensures the requisite fresh digits, and each resulting short injection has a bijective completion. All displayed lower subsets are nonempty, proper, and have no greatest member. Their distinct patterns exclude duplication between the three kinds.  $\square$

### 65.3.1 How the normal forms are ordered

The order can be read directly from the codes; it need not be reconstructed from cofinality data. Compare actual entries up to their first disagreement. If there is such a disagreement, use the order of  $D$ . If one code ends at  $p$  while the other continues with next label  $x$ , a numerical-split token  $I$  lies above the continuing code exactly when  $x \in I$ , and below it exactly when  $x \notin I$ . A boundary token  $p^-$  lies below every code continuing  $p$ , and  $p^+$  lies above them all. Numerical-split tokens at the same prefix compare by inclusion of their initial subsets. At a limit prefix,  $p^-$  is below all numerical-split tokens and  $p^+$  is above all of them.

There is a necessary distinction at an available next letter  $x$ :

$$I_x^- = \{y \in D_p : y < x\}, \quad I_x^+ = \{y \in D_p : y \leq x\}.$$

These are different initial subsets and represent the two different boundaries of  $C_{p \smallfrown x}$ . Identifying them would collapse a whole nonempty interval of the completion. In contrast, a successor boundary must not be added again as a separate boundary code.

The theorem realizes the completion as an explicit decorated tree order. Every automorphism of  $P$  extends uniquely to it through its action on cuts. This alone does not say that the automorphism preserves the three kinds of code or the displayed prefix tree.

### 65.3.2 Surreal sign codes for the residual numerical splits

For the surreal cutoff, even the residual-cut parameter has a concrete sign description.

**Proposition 65.12** (The exact identifications after deleting labels). *For  $D = \mathbf{No}_{<\kappa}$ , initial subsets of  $D$ , including the two external ends and both principal cuts at a point, are in bijection with sign words of length  $\kappa$ . Initial subsets of  $D_p$  are obtained by restricting these cuts. This restriction identifies exactly the pair of principal cuts immediately below and immediately above each deleted point  $x \in \text{ran}(p)$ , and makes no other identifications.*

*Proof.* Given an initial subset  $A$  of  $D$ , recursively construct  $s : \kappa \rightarrow \{-, +\}$  by

$$s(\alpha) = + \iff s \restriction \alpha \in A.$$

Every proper prefix is a surreal in  $D$ . A first-disagreement comparison proves  $A = \{x \in D : x < s\}$ . If  $x$  is a proper prefix of  $s$ , its membership is exactly the defining condition at its length. At a genuine first disagreement, the membership of the common prefix and initiality of  $A$  give the same conclusion. Conversely, the cut induced by a length- $\kappa$  sign word reconstructs that word by this recursion, proving the bijection.

Restriction to  $D_p$  is surjective: for an initial subset  $I$  of  $D_p$ , its downward closure in  $D$  induces exactly  $I$ . Suppose  $A \subsetneq B$  are two initial subsets of  $D$  with the same restriction. If  $B \setminus A$  contained

$x < y$ , there would be  $z \in D_p$  strictly between them because  $D_p$  is dense in  $D$ . Initiality gives  $z \in B \setminus A$ , a contradiction. The difference therefore consists of one deleted point  $x$ . The two cuts must then be  $\{y : y < x\}$  and  $\{y : y \leq x\}$ . Each such pair plainly has equal restriction. This proves that there are no further identifications.  $\square$

Thus numerical-split tokens can be coded by length- $\kappa$  sign words modulo these precisely specified deleted-point pairs, with the two residual endpoints excluded. All branches, prefixes, quotients, and completion points in this construction are sets.

An unrestricted analogue for global surreal well-orders would require a collection of arbitrary class cuts. A higher sort, an external containing universe, or a supplied uniform family must be specified before forming it. Applying the same first-failure argument to individually supplied classes does not create a GB class whose elements are all such cuts, and does not turn the raw global well-order family into a completion of its core.

## 66 A generalized Baire model at every regular size (source 24)

[merge] Source 24's Section 6, printed whole. It determines the topology, not the order, of the minimal slice at every regular enumeration cardinal  $\mu$ : the slice is homeomorphic to the bounded-topology space  $(\mu^{<\mu})^\mu$ , with no hypothesis  $\mu^{<\mu} = \mu$  and no GCH. For  $X = \mathbf{No}_{<\kappa}$  this answers source 12's question "Classify the order topology" (research question 113.13) whenever  $\mu = 2^{<\kappa}$  is regular (section 72). Source 28 recalls the construction and its singular obstruction (section 70.7).

For a cardinal  $\lambda$  and an infinite cardinal  $\mu$ , the *bounded topology* on  $\lambda^\mu$  has basic open sets

$$N_s = \{a \in \lambda^\mu : s \subseteq a\}, \quad s \in \lambda^\alpha, \quad \alpha < \mu.$$

When  $\mu$  is uncountable this is usually finer than the ordinary product topology, which permits only finitely many coordinate restrictions. Generalized descriptive set theory systematically studies this distinction [61]. We keep the topology explicit rather than relying on an ambiguous use of "Baire space."

### 66.0.1 The exhaustion-adapted prefix tree

Fix a bijection  $e : \mu \rightarrow X$ . For a short injective prefix  $p$ , let

$$m(p) = \min\{\xi < \mu : e(\xi) \notin \text{ran } p\}.$$

This exists because  $|\text{dom } p| < \mu$ . A *child* of  $p$  is an extension in which the label  $e(m(p))$  first appears at the very end:

$$q = p \frown t \frown \langle e(m(p)) \rangle, \quad t \in \text{Inj}(\delta, X \setminus (\text{ran } p \cup \{e(m(p))\})), \quad \delta < \mu.$$

All these extensions are short. Different children are incompatible: neither is a proper prefix of the other, because the earlier one would already have used the distinguished label. Every exhaustive extension of  $p$  extends exactly one child, namely the prefix ending at its first occurrence of that label.

**Lemma 66.1** (Exact branching number). *Each node  $p$  has exactly  $\lambda = \mu^{<\mu}$  children.*

*Proof.* The available labels for  $t$  have size  $\mu$ . For each ordinal  $\delta < \mu$ , the number of injections from  $\delta$  into a set of size  $\mu$  is  $\mu^{|\delta|}$ . For nonzero  $\delta$ , the lower bound follows by coding an arbitrary function  $a : \delta \rightarrow \mu$  as the injection  $\xi \mapsto (\xi, a(\xi))$  and then using a fixed bijection from  $\delta \times \mu$  to  $\mu$ ; the upper bound is immediate. The zero-length case contributes one. Summing over  $\delta < \mu$  gives  $\mu^{<\mu}$ , since that cardinal is at least  $\mu$ .  $\square$

The rule uses a well-order of the *set of labels* only to decide which coverage requirement to enforce next. It is independent of their numerical order. This is why it can classify the topology without classifying the linear order.

**Theorem 66.2** (Regular-cardinal topological classification). *If  $\mu$  is infinite regular,  $|X| = \mu$ , and  $\lambda = \mu^{<\mu}$ , there is a homeomorphism*

$$\Phi : \lambda^\mu \longrightarrow \mathcal{W}_\mu(X)$$

*for the bounded topology and the prefix topology. If  $X$  is  $\mu$ -thick, the latter is its lexicographic order topology.*

*Proof.* Using Choice, label the children of every short injective prefix by  $\lambda$ . Define prefixes  $p_s$  for  $s \in \lambda^{<\mu}$  by recursion. Let  $p_\emptyset$  be empty. At a successor stage let  $p_{s \smallfrown \langle i \rangle}$  be the child of  $p_s$  labelled by  $i$ . At a limit  $\eta < \mu$ , define

$$p_s = \bigcup_{\xi < \eta} p_{s \upharpoonright \xi} \quad (\text{dom } s = \eta).$$

Regularity of  $\mu$  is used here: the supremum of fewer than  $\mu$  prefix lengths, each below  $\mu$ , is still below  $\mu$ . The union is injective and hence is a valid node.

For a branch  $a \in \lambda^\mu$ , put

$$\Phi(a) = \bigcup_{\xi < \mu} p_{a \upharpoonright \xi}.$$

Every successor step increases the prefix length. Thus these lengths are cofinal in  $\mu$ , and the union has domain  $\mu$ . It is injective. At stage  $\xi$ , the prefix contains every label  $e(\zeta)$  with  $\zeta < \xi$ : prove this by induction. At a successor, either  $e(\xi)$  is already present or it is exactly the least missing label and the child adds it. At a limit, take the union. Therefore  $\Phi(a)$  is exhaustive.

Conversely, an exhaustive  $f : \mu \rightarrow X$  chooses a unique child of the empty prefix, then a unique child of that child, and so on. At limit stages take the already-determined prefix union. All intermediate prefixes are short by regularity and remain restrictions of  $f$ . This defines a unique branch  $a$  with  $\Phi(a) = f$ .

The correspondence satisfies

$$\Phi[N_s] = [p_s]_P.$$

Indeed, a permutation extending  $p_s$  must have followed the unique distinguished-label choices along the path to  $s$ . Hence  $\Phi$  is an open bijection. To check continuity, take  $f = \Phi(a)$  and a prescribed prefix length  $\alpha < \mu$ . Some  $p_{a \upharpoonright \xi}$  has length at least  $\alpha$ , so its cylinder lies inside  $[f \upharpoonright \alpha]$ . The preimage of every cylinder is open. This proves the homeomorphism, and Lemma 62.5 supplies the final order-topological identification.  $\square$

**Corollary 66.3** (Size-matched regular cutoffs). *If  $\mu = 2^{<\kappa}$  is regular, then*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \cong_{\text{top}} (\mu^{<\mu})^\mu.$$

*In particular, if  $\mu^{<\mu} = \mu$ , this is the usual bounded-topology generalized Baire space  $\mu^\mu$ .*

No continuum hypothesis or generalized continuum hypothesis has been used. The branching cardinal is retained as  $\mu^{<\mu}$  rather than being silently replaced by  $\mu$ . The regularity hypothesis is on the *enumeration cardinal*  $\mu$ ; regularity of the birthday parameter  $\kappa$  alone is not the statement of the theorem.

### 66.0.2 Why the construction does not contradict forward incompleteness

The full branch space  $\lambda^\mu$  is complete for its own prefix uniformity: a Cauchy filter stabilizes a unique prefix at every height, and the union is a branch. The homeomorphism  $\Phi$  transports this complete uniformity to a compatible complete uniformity on  $\mathcal{W}_\mu(X)$ .

This is not the forward uniformity of Section 65.1. A height in the exhaustion-adapted tree means that a prescribed list of labels has already been covered. The amount of ordinary enumeration needed to achieve that height varies from branch to branch and is not uniformly bounded. In the countable example, the place at which 0 first occurs can be arbitrarily late. Homeomorphisms preserve topology, not a chosen uniformity or the completeness of every compatible metric.

### 66.0.3 The exact point where singularity intervenes

If  $\mu$  is singular, a chain of fewer than  $\mu$  short prefixes can have union of length  $\mu$ . It can therefore reach a full injective word before all  $\mu$  coverage requirements have been visited. There is no short node at that limit stage to which the same child construction can be applied. This is a concrete obstruction to *this proof*, not a proof that no other generalized Baire presentation exists.

The distinction will recur in Section 69.1: at a singular cardinal a small-cardinality set of coordinates may be unbounded. Arguments that control finitely or cardinally few positions do not automatically control one bounded initial segment.

## 67 The countable cutoff (source 24; routes of sources 19, 20, 21, 26, 29 and 30)

*Note 67.1* ([merge]; one theorem, seven proofs). For every countable dense alphabet  $X$  without endpoints, in particular for the finite-birthday surreals  $\mathbf{No}_{<\omega}$ , the minimal slice  $\mathcal{W}_\omega(X)$  is homeomorphic to Baire space  $\omega^\omega$ , its Dedekind completion is the real line, every interval of the completion contains continuum many omitted points, and the slice is not order-isomorphic to the irrationals. Seven batch-83 sources prove this independently: source 24 (theorems 67.2 and 67.4, lemma 67.3, and corollary 67.5, printed below as the base: it alone gives both a complete compatible metric for the original topology and the exact descriptive form of the complement); source 19 (theorem 67.13, which adds that every interval also contains continuum many original points); source 20 (theorem 67.11, which adds that the slice is not  $F_\sigma$  in its completion); source 21 (theorems 67.6 and 67.10, which add a second mechanism of omitted cuts, and its homogeneity theorem theorem 67.8); source 26 (proposition 69.11, complete metrizability only); source 29 (theorem 68.2, case  $n = 0$ , set beside the slices with finitely many omitted labels); and source 30 (proposition 68.4 and note 68.5). Five build the homeomorphism with  $\omega^\omega$  by the same least-missing-label tree (sources 19, 20, 24, 30, and source 29 by citation); source 21 uses an even-odd back-and-forth coding. The  $G_\delta$  description through cylinder intervals and label requirements is common to sources 19, 20 and 24; source 26 obtains complete metrizability from the  $G_\delta$  criterion proposition 69.8. In this report the case was the example example 30.19 (“a Baire, zero-dimensional order”, with the warning that these facts do not identify it with the real line). No source has priority over another. The routes are printed after source 24’s text, each whole, because each adds a clause or a different proof.

[merge] Source 24’s Section 7, “The countable cutoff: a complete metric but a different order”.

Let  $X$  be countable, densely ordered, and without endpoints. This includes  $\mathbf{No}_{<\omega} = \mathbf{No}_{<\omega}$ , the finite sign sequences. Every nonempty interval and ray of such an  $X$  is infinite, so the thickness hypothesis holds with  $\mu = \omega$ .

### 67.0.1 An explicit complete compatible metric

Fix  $e : \omega \rightarrow X$  and identify  $f \in \mathcal{W}_\omega(X)$  with the permutation  $\pi = e^{-1} \circ f$  of  $\omega$ . On injections or permutations of  $\omega$ , define

$$d_{\rightarrow}(\pi, \sigma) = \begin{cases} 0, & \pi = \sigma, \\ 2^{-\min\{k : \pi(k) \neq \sigma(k)\}}, & \pi \neq \sigma. \end{cases}$$

This familiar forward ultrametric is not complete on permutations, by Example 65.3. Instead put

$$D(\pi, \sigma) = \max\{d_{\rightarrow}(\pi, \sigma), d_{\rightarrow}(\pi^{-1}, \sigma^{-1})\}.$$

**Theorem 67.2** (Complete metrization at countable size). *The metric  $D$  is a complete compatible ultrametric on  $\mathcal{W}_\omega(X)$ . This space is homeomorphic to  $\omega^\omega$ , has no isolated points, and is nowhere locally compact.*

*Proof.* The maximum of the two ultrametrics is an ultrametric. To see compatibility, finitely many inverse requirements at a fixed  $\pi$  are enforced by finitely many forward requirements: to ensure  $\sigma^{-1}(j) = \pi^{-1}(j)$  for  $j < n$ , require  $\sigma$  to agree with  $\pi$  at all the finitely many positions  $\pi^{-1}(j)$ . One sufficiently long forward prefix does this. Hence adding inverse requirements does not change the topology.

For a  $D$ -Cauchy sequence  $(\pi_n)$ , each forward coordinate and each inverse coordinate eventually stabilizes. Let their limits be  $h$  and  $g$ . If  $h(k) = j$ , sufficiently late terms have  $\pi_n(k) = j$  and  $\pi_n^{-1}(j) = g(j)$  simultaneously. Thus  $g(j) = k$ . Conversely, if  $g(j) = k$ , the same argument gives  $h(k) = j$ . Therefore  $g = h^{-1}$  and  $h$  is a permutation. The original sequence converges to it in  $D$ .

Theorem 66.2 gives the homeomorphism with  $\omega^\omega$ , since  $\omega^{<\omega} = \omega$ . Every finite prefix admits at least two continuations, so there are no isolated points. Each cylinder is partitioned by its exhaustion-adapted children into countably infinitely many nonempty open cylinders. That cover has no finite subcover, so the original cylinder is not compact. If a point had a compact neighborhood, it would contain a cylinder, which is closed in the whole space and hence would be a compact closed subset of that neighborhood. This contradiction proves nowhere local compactness.  $\square$

### 67.0.2 The Dedekind completion is a real line

**Lemma 67.3** (A countable order-dense subset). *The order  $\mathcal{W}_\omega(X)$  is separable and dense without endpoints. Its Dedekind completion is order-isomorphic to  $\mathbb{R}$ .*

*Proof.* There are countably many finite injective prefixes. Choose one exhaustive extension for each. These chosen points meet every nonempty cylinder and hence every nonempty open interval, giving a countable order-dense subset. Density and absence of endpoints were proved in Lemma 62.5.

The completion is consequently a complete dense linear order without endpoints with a countable order-dense subset. For completeness of the identification, a countable order-dense subset has no endpoints and is itself densely ordered; a standard alternating insertion construction gives an order isomorphism with  $\mathbb{Q}$ . Each point of the complete order is uniquely determined by the rational cut below it, and every real cut is realized by a supremum. This extends the countable isomorphism to an isomorphism with  $\mathbb{R}$ .  $\square$

Fix such an identification of the completion with  $\mathbb{R}$ . For a nonempty finite prefix  $p$ , let  $J_p$  be the open interval between the lower and upper completion edges of its cylinder. Then

$$J_p \cap \mathcal{W}_\omega(X) = [p]_P.$$



Both edges exist because a nonempty cylinder is bounded by words with smaller or larger first labels. It has no endpoints and is clopen in the original order, so neither edge lies in that cylinder. For the empty prefix set  $J_\emptyset = \mathbb{R}$ .

**Theorem 67.4** (Exact descriptive form of the omitted points). *Inside its Dedekind completion  $\mathbb{R}$ , the space  $P = \mathcal{W}_\omega(X)$  is a dense  $G_\delta$  set. Its complement is a dense meager  $F_\sigma$  set of cardinality  $2^{\aleph_0}$ .*

*Proof.* For  $n < \omega$  and  $x \in X$ , set

$$V_n = \bigcup_{\substack{p \in \mathcal{I}_{<\omega}(X) \\ |p|=n}} J_p, \quad W_x = \bigcup_{\substack{p \in \mathcal{I}_{<\omega}(X) \\ x \in \text{ran } p}} J_p.$$

All these sets are open and contain  $P$ , hence are dense. We claim

$$P = \bigcap_{n < \omega} V_n \cap \bigcap_{x \in X} W_x. \quad (16)$$

The forward inclusion is immediate. For the reverse inclusion, a point  $z$  belonging to all  $V_n$  chooses a unique length- $n$  prefix  $p_n$  with  $z \in J_{p_n}$ . Distinct same-level intervals are disjoint: a nonempty intersection would be open and meet the dense original order, contradicting disjointness of the corresponding cylinders. The same argument shows that the chosen prefixes are coherent. Their union is an injective  $\omega$ -word  $h$ .

Membership in every  $W_x$  ensures that each label occurs in one of these prefixes. Indeed, if  $z \in J_p$  and  $x \in \text{ran } p$ , comparison with a chosen prefix of length at least  $|p|$  forces compatibility and hence  $p \subseteq h$ . Thus  $h$  is exhaustive. Finally,  $z$  is its completion point: if they differed, an original point strictly between them would have to belong to every sufficiently long cylinder of  $h$ , contrary to its first disagreement with  $h$ . This proves (16).

The complement is therefore  $F_\sigma$  and meager; each closed set complementary to one of the displayed dense open sets is nowhere dense. It is dense as well. Every completion interval contains an original cylinder interval, and every such cylinder contains the trace cut of an injection that extends the prefix but omits an unused label.

For its cardinality, fix a label  $z_0$  and partition  $X \setminus \{z_0\}$  into countably many pairs. Independently ordering each pair gives  $2^{\aleph_0}$  distinct injective words with range exactly  $X \setminus \{z_0\}$ . Proposition 65.4 gives distinct omitted completion points. The opposite bound follows from the cardinality of  $\mathbb{R}$ .  $\square$

**Corollary 67.5** (Homeomorphic to, but not ordered like, the irrationals). *The minimal enumeration order of the countable surreal cutoff is homeomorphic to the Baire space and hence to the usual irrational-number space. It is not order-isomorphic to the irrationals in their usual order.*

*Proof.* The homeomorphism with Baire space was constructed above. The familiar identification of Baire space with irrationals can also be obtained from infinite continued fractions, or from successive countable interval partitions; only the topological assertion is used here. An order isomorphism of dense orders extends uniquely to their Dedekind completions, by transporting lower cuts. The completion of the irrationals omits only the countably many rational points, whereas Theorem 67.4 gives continuum many omitted points for our order. An isomorphism cannot have these two different complements.  $\square$

The completion description gives a particularly concrete answer to the original question at the first infinite cutoff. It is not a strange topology: it is Baire topology. Its unusual feature is the *placement of that topology in a linear order*, with many more omitted order cuts than the standard irrational presentation. The phenomenon is already visible before any genuinely class-sized issues arise.

### 67.1 Source 21's route, with homogeneity and non-intrinsic cylinders

[merge] Source 21's Section 7, printed whole. Its [theorem 67.6](#) and [theorem 67.10](#) are the theorem above by different proofs: the slice is a  $G_\delta$  in the discrete product  $\mathbb{Q}^\omega$ , and an even-odd back-and-forth coding gives the homeomorphism; the completion theorem adds omitted cuts from numerical next-coordinate splits to those from nonexhaustive branches. Its [lemma 67.7](#), [theorem 67.8](#) and [corollary 67.9](#) are new: the countable minimal slice is strongly finitely homogeneous, has  $2^{\aleph_0}$  automorphisms, and its proper prefix cylinders are not definable in the pure order. They answer source 12's question on the "degrees of homogeneity" ([research question 113.13](#)) at  $\kappa = \omega$ , which Part VI left open ([section 33](#), item (b)); the corollary is the set-level counterpart of source 13's class-level prefix non-rigidity ([theorem 47.12](#)).

Let

$$P = \mathcal{W}_\omega(\mathbb{Q}).$$

The rationals here carry their usual numerical order; the points of  $P$  are all exhaustive enumerations of them. Every countable dense alphabet without endpoints gives the same presentation order, by relabelling. In particular, this treats  $\mathcal{W}_\omega(\mathbf{No}_{<\omega})$ , since the finite-birthday surreals form a countable dense order without endpoints. It does not assert that the finite-birthday surreals equal the rational surreals as sets.

#### 67.1.1 Order topology and an explicit Baire coding

**Theorem 67.6.** *The order topology of  $P$  equals its finite-prefix topology, equivalently the subspace topology inherited from the product of countably many copies of discrete  $\mathbb{Q}$ . It is a zero-dimensional Polish space, homeomorphic to Baire space  $\omega^\omega$ , and it is nowhere locally compact.*

*Proof.* The equality of topologies and the clopen-cylinder assertion follow from [note 62.13](#). In discrete  $\mathbb{Q}^\omega$ , the injective words form a closed subspace. The surjectivity requirement is

$$\bigcap_{q \in \mathbb{Q}} \bigcup_{n < \omega} \{e : e(n) = q\},$$

a countable intersection of open sets. Thus  $P$  is a  $G_\delta$  subspace of a Polish space and is Polish.

Here is also a direct homeomorphism with Baire space, so no topological classification theorem is necessary. Fix a bijection of the underlying set  $\mathbb{Q}$  with  $\omega$  and view a point as a permutation of  $\omega$ . Given  $c \in \omega^\omega$ , build finite partial bijections  $p_s$ . At even stage  $s$ , take the least unused domain element and send it to the  $c(s)$ th unused range element. At odd stage  $s$ , take the least unused range element and assign it to the  $c(s)$ th unused domain element. Unused elements are listed in their natural order.

Every stage adds one pair. The even stages exhaust the domain and the odd stages exhaust the range, so the union is a permutation. Conversely, for a given permutation each prescribed even or odd step has a unique rank among the unused choices, recovering a unique  $c$ . The forward map is continuous because each finite construction stage uses only a finite prefix of  $c$ . Its inverse is continuous because a finite construction trace inspects finitely many values of the permutation or its inverse, and

$$\pi^{-1}(r) = d \iff \pi(d) = r$$

is a clopen coordinate condition. This proves the homeomorphism.

Finally, a cylinder with a prefix of length  $n$  has a continuous surjection, by its next coordinate, onto the infinite discrete set of unused rational labels. It is therefore not compact. If a point had a neighborhood with compact closure, that neighborhood would contain a clopen cylinder. The cylinder would be a closed subset of the compact closure, a contradiction.  $\square$

The adjective “discrete” matters. If one uses the ordinary Euclidean topology on the rational coordinate alphabet, the topology of coordinate agreement is no longer the product topology being described here. At uncountable lengths the bounded-prefix topology requires its own formulation as well.

### 67.1.2 Absorption and interval self-similarity

**Lemma 67.7** (Countable sums are absorbed). *For every nonempty countable linear order  $J$ ,*

$$\sum_{j \in J} P \cong P.$$

*Every nonempty finite-prefix cylinder of  $P$  is also isomorphic to  $P$ .*

*Proof.* After a finite prefix, the unused alphabet is  $\mathbb{Q}$  minus finitely many points. It is again a countable dense order without endpoints and hence isomorphic to  $\mathbb{Q}$ . Reindex the remaining  $\omega$  positions and relabel by such an isomorphism. This proves the cylinder assertion. In particular, first-coordinate decomposition gives

$$P \cong \sum_{q \in \mathbb{Q}} P.$$

Therefore

$$\sum_{j \in J} P \cong \sum_{j \in J} \sum_{q \in \mathbb{Q}} P \cong \sum_{t \in \sum_J \mathbb{Q}} P.$$

The index order  $\sum_J \mathbb{Q}$  is countable, dense, and without endpoints: inside a fiber use density of  $\mathbb{Q}$ , and between distinct fibers use the absence of an endpoint in either relevant copy of  $\mathbb{Q}$ . It is isomorphic to  $\mathbb{Q}$ , completing the proof.  $\square$

**Theorem 67.8** (Strong finite homogeneity). *Every nonempty bounded open interval and every open ray of  $P$  is order-isomorphic to  $P$ . Every finite increasing partial map from  $P$  to itself extends to an order automorphism. Moreover,*

$$|\text{Aut}(P, <)| = 2^{\aleph_0}.$$

*Proof.* Fix  $e \in P$ . Partition the points below  $e$  according to their first disagreement position  $n$ . At a fixed  $n$ , the smaller unused labels form a countable dense order without endpoints, and each such next label has a completion cylinder isomorphic to  $P$ . Thus each part is a  $\mathbb{Q}$ -sum of copies of  $P$ , hence isomorphic to  $P$ . These parts occur in increasing order of  $n$ , giving

$$(-\infty, e)_P \cong \sum_{n < \omega} P \cong P.$$

For the upper ray the first-disagreement parts occur in order  $\omega^*$ , and [lemma 67.7](#) again applies.

For  $e < f$ , let  $n$  be their first disagreement. The interval  $(e, f)$  splits into the upper ray after  $e$  within its length- $(n+1)$  cylinder, the cylinders whose next label lies strictly between  $e(n)$  and  $f(n)$ , and the lower ray before  $f$  within its corresponding cylinder. These are respectively a copy of  $P$ , a  $\mathbb{Q}$ -sum of copies of  $P$ , and a copy of  $P$ . Their sum is isomorphic to  $P$ .

Given finite increasing tuples  $e_1 < \dots < e_k$  and  $f_1 < \dots < f_k$ , match the points and choose isomorphisms of their corresponding intervening intervals and two outer rays. Patching the maps gives an automorphism.

Transitivity on the continuum many points of  $P$  gives the lower bound for the automorphism group. Choose one exhaustive completion of each finite injective prefix. The resulting countable set is order-dense. An order automorphism is determined by its values on this set, so there are at most  $|P|^{\aleph_0} = 2^{\aleph_0}$  automorphisms.  $\square$

**Corollary 67.9** (Prefix cylinders are not intrinsic). *The family of proper positive-length prefix cylinders is not invariant under all automorphisms of  $(P, <)$ . No individual such cylinder is definable in the pure order with finitely many parameters.*

*Proof.* Choose two ordered points in one cylinder and two ordered target points with different first coordinates. By [theorem 67.8](#) there is an automorphism carrying the first pair to the second. The image of the cylinder cannot be a positive-length cylinder, since its two target points disagree at their first coordinate. It cannot be the whole space, since the original cylinder was proper.

For the stronger statement, fix finitely many parameters. The automorphisms fixing them pointwise act transitively on each open component of their complement, by the same finite homogeneity theorem. A proper cylinder has no least or greatest point, and its two boundary cuts are omitted. Near either boundary one can choose an inside point and an outside point in the same component of the complement of the finite parameter set. An automorphism fixing all parameters moves the first point to the second and changes cylinder membership. A definable set would have to be invariant under this automorphism.  $\square$

This supplies a negative answer to a concrete prefix-recovery question. The labels and the displayed tree add real structure. An abstract order automorphism has no obligation to respect that tree.

### 67.1.3 Completion: two mechanisms for missing cuts

**Theorem 67.10.** *The Dedekind completion of  $P$  is order-isomorphic to  $\mathbb{R}$ . Every nonempty prefix cylinder has continuum many omitted cuts arising from nonexhaustive branches and continuum many arising from numerical cuts in a next-coordinate alphabet. In particular,*

$$P \not\cong (\mathbb{R} \setminus \mathbb{Q}, <),$$

*although the two spaces are homeomorphic.*

*Proof.*  $P$  is dense, without endpoints, and has a countable order-dense set as in the preceding proof. Its Dedekind completion is consequently a complete separable dense order without endpoints, hence has type  $\mathbb{R}$ .

Fix  $q_0 \in \mathbb{Q}$  and a bijection  $v : \omega \rightarrow \mathbb{Q} \setminus \{q_0\}$ . Its first-disagreement comparison with every member of  $P$  is defined. Put

$$L_v = \{e \in P : e < v\}, \quad R_v = \{e \in P : v < e\}.$$

Both sides are nonempty. Each has no endpoint: after any given finite first disagreement, agree with  $v$  for longer and then choose a fresh smaller or larger rational label. No point of  $P$  realizes the cut, since any different branch can be separated from  $v$  at its first disagreement by a fresh intermediate label. Distinct  $v$  give distinct cuts by the same separation argument. There are continuum many such bijections. The argument works after any fixed finite prefix by omitting a still-unused label.

For a different mechanism, take any irrational real  $r$  and split  $P$  according to  $e(0) < r$  or  $e(0) > r$ . This is a missing cut because no first coordinate equals  $r$ ; both sides approach it through rational first coordinates. Distinct irrationals give distinct cuts. Within a prefix cylinder use the next unused rational coordinate instead. These cuts do not determine a full rational branch through that level, whereas the preceding omitted-range cuts do.

An order isomorphism extends uniquely to the Dedekind completions and preserves the subset of represented points. The irrational order omits only the countably many rational points of its

completion;  $P$  omits continuum many. Hence they are not order-isomorphic. The homeomorphism assertion follows from [theorem 67.6](#) and the standard Baire-space description of the irrationals, or from their usual continued-fraction coding.  $\square$

Thus separability, complete metrizability, interval self-similarity, finite homogeneity, and the size of the completion do not suffice to identify this order with the irrational line. The placement of its omitted cuts is an additional invariant.

## 67.2 Source 20's route: the slice is not $F_\sigma$

[merge] Source 20's Section 9, printed whole. Its proof of the Baire-space homeomorphism and of the  $G_\delta$  description is source 24's; new are the strict Borel classification (the slice is not  $F_\sigma$  in its completion, because Baire space is not  $\sigma$ -compact) and the omitted cuts from irrational first coordinates, which are source 21's second mechanism ([theorem 67.10](#)).

At the first infinite birthday cutoff, the space of enumerations admits an accessible description that also illustrates a limitation of topological classification. Put

$$X = \mathbf{No}_{<\omega}, \quad P = \mathcal{W}_\omega(X) = \{f : \omega \longrightarrow X : f \text{ is a bijection}\},$$

with numerical first-difference comparison. The numerical order on  $X$  is countable, dense, and without endpoints. Thus it is isomorphic to  $\mathbb{Q}$ ; fix one such identification when needed. Everything in this section is an ordinary set theorem in ZFC. In fact, the proofs apply to any countable dense alphabet without endpoints.

For a finite injective word  $s$  over  $X$ , write

$$[s] = \{f \in P : s \text{ is an initial segment of } f\}.$$

Every such cylinder is nonempty: enumerate its countably infinite remaining alphabet after the prescribed prefix.

**Theorem 67.11** (Topology, completion, and omitted cuts). *The order  $P$  has the following properties.*

1. *Its order topology equals the topology of pointwise convergence when the alphabet  $X$  is discrete. The cylinders  $[s]$  are a countable clopen basis.*
2. *Its Dedekind completion is order-isomorphic to  $\mathbb{R}$ . Under any such identification,  $P$  is a dense  $G_\delta$  subset of  $\mathbb{R}$  that is not  $F_\sigma$ .*
3. *As a topological space,  $P$  is homeomorphic to Baire space  $\omega^\omega$ . No point has a compact neighborhood.*
4. *Every nonempty open interval of the completion contains  $2^{\aleph_0}$  omitted points. Consequently,  $P$  is not order-isomorphic to the irrational numbers with their usual order.*

The last two conclusions are compatible. A homeomorphism of spaces need not preserve their displayed linear orders. Here the order remembers the omitted Dedekind cuts, while the topology alone does not remember their number.

### 67.2.1 The topology comes from finite prefixes

**Lemma 67.12.** *Every cylinder  $[s]$  is convex, has no first or last point, and is clopen in the order topology. Conversely, every neighborhood of  $f \in P$  contains a cylinder determined by an initial segment of  $f$ .*

*Proof.* Convexity follows immediately from first-difference comparison: a word that disagrees with  $s$  before the end of  $s$  lies on the same side of every member of  $[s]$ . Let  $n$  be the length of  $s$  and let  $f \in [s]$ . The residual alphabet  $X \setminus \text{ran } s$  remains dense and without endpoints. Choose unused labels  $a < f(n) < b$ . Complete  $s$  followed by  $a$ , and  $s$  followed by  $b$ , to permutations  $g, h$ . Then  $g < f < h$ , and convexity gives  $(g, h) \subseteq [s]$ . This proves openness and also rules out endpoints. The complement of  $[s]$  is the union of all other cylinders of length  $n$ , so  $[s]$  is closed as well.

For the converse, suppose  $g < f < h$ . Let  $i$  and  $j$  be the first positions at which  $f$  differs from  $g$  and  $h$ , respectively. If  $n > \max\{i, j\}$ , then every word sharing  $f$ 's first  $n$  entries remains between  $g$  and  $h$ . The same argument, using one comparison, handles an open ray. Hence the finite cylinders form a basis for the order topology, exactly as they do for pointwise convergence on the discrete alphabet.  $\square$

There are countably many finite prefixes. Choosing one completion of each produces a countable dense subset of  $P$ . Moreover,  $P$  itself is densely ordered: at the first difference between two permutations, choose an unused numerical label strictly between their unequal labels and then complete the resulting prefix. Varying the first label also shows that  $P$  has no endpoints.

For completeness, these facts identify its completion without a cardinality argument. A countable dense subset  $D \subseteq P$  is itself a dense order without endpoints, and the usual alternating extension of finite partial isomorphisms identifies  $D$  with  $\mathbb{Q}$ . Cuts of  $D$  then identify the Dedekind completion of  $P$  with  $\mathbb{R}$ . We henceforth use one such identification and regard  $P$  as a dense suborder of  $\mathbb{R}$ .

### 67.2.2 An explicit homeomorphism with Baire space

Fix a listing  $X = \{x_0, x_1, \dots\}$ . At a finite injective prefix  $s$ , let  $x_{m(s)}$  be the first label in this listing that has not yet appeared in  $s$ . Partition  $[s]$  according to the finite extension ending at the first subsequent occurrence of this required label. The parts are the cylinders  $[t]$  with

$$t = s \frown u \frown \langle x_{m(s)} \rangle,$$

where  $u$  is any finite injective word over  $X \setminus (\text{ran } s \cup \{x_{m(s)}\})$ , including the empty word. These prefixes are pairwise incomparable under extension: each ends precisely at that first occurrence. Their cylinders therefore form a disjoint partition of  $[s]$ . The partition is countably infinite, and we enumerate its members using a fixed coding of finite words by natural numbers.

Given  $z \in \omega^\omega$ , start with  $s_0$  empty and take  $s_{n+1}$  to be the prefix indexing the  $z(n)$ -th part of this partition at  $s_n$ . The union

$$\Phi(z) = \bigcup_{n < \omega} s_n$$

is an injective sequence of length  $\omega$ , because the lengths strictly increase. It is surjective as well: after  $n$  steps all of  $x_0, \dots, x_{n-1}$  have appeared, since every step inserts the least missing label. Thus  $\Phi(z) \in P$ .

Conversely, any  $f \in P$  selects a unique member of each successive partition and hence determines a unique  $z$ . The image of a basic finite-prefix neighborhood in  $\omega^\omega$  is exactly the cylinder  $[s_n]$  reached by that prefix. These are open, and their lengths tend to infinity along each branch. They consequently form a neighborhood basis at  $\Phi(z)$ . This proves directly that  $\Phi$  is a homeomorphism. No compactness theorem or classification theorem for Polish spaces is needed for this construction.

There is also a direct local compactness obstruction. For each  $s$ , the cylinders  $[s \frown \langle a \rangle]$ , as  $a$  ranges over the unused alphabet, are a countably infinite disjoint open cover of  $[s]$  with no finite subcover. Thus  $[s]$  is not compact. If a point had a compact neighborhood, a sufficiently small clopen cylinder inside that neighborhood would be a closed subset of a compact space and hence compact, a contradiction.

### 67.2.3 Recognizing the genuine permutations inside the completion

The  $G_\delta$  assertion has an explicit description in terms of finite prefixes and eventual occurrence of every label. For each nonempty  $s$ , let

$$I_s = (\inf[s], \sup[s]) \subseteq \mathbb{R}, \quad I_\emptyset = \mathbb{R}.$$

The endpoints for nonempty  $s$  are finite: comparison of the first entry with smaller and larger labels bounds the cylinder. By convexity and the absence of endpoints in  $[s]$ ,

$$I_s \cap P = [s]. \quad (17)$$

Intervals belonging to different prefixes of the same length are disjoint. Otherwise their open intersection would meet the dense set  $P$ , contradicting disjointness of the cylinders. More generally, if  $I_s$  and  $I_t$  intersect, their prefixes are comparable under extension.

Define open subsets of the completion by

$$U_n = \bigcup_{\substack{s \text{ injective} \\ |s|=n}} I_s \quad (n \geq 1), \quad V_a = \bigcup_{\substack{s \text{ finite injective} \\ a \in \text{ran } s}} I_s \quad (a \in X).$$

Every permutation belongs to all these open sets. We claim the exact identity

$$P = \bigcap_{n \geq 1} U_n \cap \bigcap_{a \in X} V_a. \quad (18)$$

Indeed, a real point in every  $U_n$  determines a unique prefix  $s_n$  of every length, and the intersection rule makes these prefixes compatible. Their union is an injective sequence  $f$ . Membership in every  $V_a$  forces every label to occur in  $f$ , so  $f \in P$ . The real point in question and  $f$  both lie in every  $I_{s_n}$ . If they were different, a member of the dense set  $P$  strictly between them would lie in every  $I_{s_n}$  and hence share all entries of  $f$ , an impossibility. This proves the identity.

The conditions in (18) have distinct purposes. The  $U_n$  require a genuine label at every position; the  $V_a$  impose exhaustion of the alphabet. An infinite injective word omitting a label is excluded by the second requirement. A cut that separates possible next labels without selecting one is excluded by the first. Since each displayed open set contains the dense set  $P$ , the complement of  $P$  is a countable union of closed nowhere dense sets.

The Borel classification is strict:  $P$  is not  $F_\sigma$  in its completion. First observe that  $\omega^\omega$  is not  $\sigma$ -compact. Given compact sets  $K_n \subseteq \omega^\omega$ , their projections onto coordinate  $n$  are finite; choosing the  $n$ -th entry outside that finite projection constructs a point outside  $\bigcup_n K_n$ . The homeomorphism above gives the same conclusion for  $P$ . But an  $F_\sigma$  subset of  $\mathbb{R}$  is  $\sigma$ -compact, by intersecting its closed pieces with the compact intervals  $[-m, m]$ . This contradiction proves the assertion.

### 67.2.4 Continuum many missing points in every interval

Identify the alphabet with  $\mathbb{Q}$ . For every irrational  $r \in \mathbb{R}$ , split the permutations according to

$$L_r = \{f \in P : f(0) < r\}, \quad R_r = \{f \in P : f(0) > r\}.$$

These two sets partition  $P$ , the first has no greatest member, and the second has no least member. Indeed, a rational first entry can always be chosen closer to  $r$  on its own side and completed to a permutation. Thus this cut is an omitted point of the completion. Distinct irrational numbers



give distinct cuts, because a rational first entry between them separates the cuts. The completion therefore has  $2^{\aleph_0}$  omitted points.

The same argument works inside every cylinder: its residual alphabet is again a countable dense order without endpoints. Every nonempty open interval of the completion contains a cylinder, by density of  $P$  and [lemma 67.12](#). Hence every such interval contains  $2^{\aleph_0}$  omitted points. In particular, the complement is both dense and meager.

Finally, an order isomorphism between two dense orders extends uniquely to an isomorphism of their Dedekind completions, by mapping cuts. If  $P$  were order-isomorphic to  $\mathbb{R} \setminus \mathbb{Q}$ , this extension would identify the omitted points of their completions. One complement has cardinality  $2^{\aleph_0}$  and the other is countable. This is impossible, and completes the proof of [theorem 67.11](#).

This example supplies a specific answer at the level of topology and descriptive complexity. It does not by itself classify the arrangement of all omitted cuts up to order isomorphism. Extending the analysis to uncountable birthday cutoffs requires explicit choices of generalized topology and of the permitted lengths of prefixes; the countable Polish-space conclusions do not transfer merely by replacing  $\omega$  with a larger cardinal.

### 67.3 Source 19's route

[merge] Source 19's Section 8, printed whole. Its  $G_\delta$  proof reads the omitted points off the normal forms of [theorem 65.11](#), and it counts the original as well as the omitted points of every interval.

The first infinite birthday cutoff already distinguishes order type from topology in a concrete way. Let  $D = \mathbf{No}_{<\omega}$  and  $P = \mathcal{W}_\omega(D)$ . The proof below applies to every countable dense alphabet without endpoints.

**Theorem 67.13** (A real completion with many omitted points). *The order  $P$  has the following properties:*

1. *its Dedekind completion is order-isomorphic to  $\mathbb{R}$ ;*
2. *its order topology is homeomorphic to Baire space  $\omega^\omega$ ;*
3. *in its real-line completion it is a dense  $G_\delta$  subset;*
4. *every nonempty open interval of the completion contains  $2^{\aleph_0}$  original points and  $2^{\aleph_0}$  omitted completion points;*
5. *it is not order-isomorphic to the irrational numbers.*

*Proof.* Finite prefix cylinders form a clopen basis for the order topology. For  $e \in C_p$ , choose an unused smaller next digit and an unused larger next digit at some coordinate at or after the end of  $p$ , and complete to obtain  $a < e < b$  inside  $C_p$ . Convexity gives  $(a, b) \subseteq C_p$ , proving openness. Its complement is a union of incompatible prefix cylinders and is also open. Conversely, an interval around  $e$  contains a sufficiently long finite prefix cylinder of  $e$ , since both endpoint comparisons are decided at finite coordinates.

Only countably many finite prefixes exist. Choosing one completion of each gives a countable order-dense subset. The order is dense and has no endpoints. Its Dedekind completion is consequently a complete dense order without endpoints and with a countable dense subset, hence isomorphic to  $\mathbb{R}$ : identify a countable dense skeleton with  $\mathbb{Q}$  and then identify the two completions through their cuts. This proves the first assertion.

For an explicit Baire-space homeomorphism, fix an enumeration  $d_0, d_1, \dots$  of  $D$ . At a finite injective prefix  $p$ , let  $m(p)$  be the least-indexed unused label. Split  $C_p$  according to the finite block  $q$  that starts immediately after  $p$  and ends at the first occurrence of  $m(p)$ . Thus  $q$  is nonempty and injective, uses only unused labels, ends with  $m(p)$ , and has no earlier occurrence of that label.

There are countably infinitely many such blocks; enumerate them by natural numbers using the fixed alphabet enumeration.

Their cylinders  $C_{p \smallfrown q}$  are nonempty, clopen, pairwise disjoint, and cover  $C_p$ . Two such blocks cannot be proper prefixes of one another because both end at their first occurrence of the same required label. Given  $s \in \omega^\omega$ , at stage  $n$  append the block numbered  $s(n)$ . The prefix length increases at every stage, and each stage inserts the least missing label. The union is therefore an injection of domain  $\omega$  using every label, hence a permutation. Conversely, every permutation splits uniquely into these successive blocks. These procedures are inverse. A finite Baire cylinder maps to exactly one of the corresponding permutation cylinders, and these cylinders form bases on both sides. This proves the homeomorphism without a separate classification theorem for Polish spaces.

For the  $G_\delta$  assertion, view  $\overline{P}$  as the real line and, for each nonempty finite prefix  $p$ , put

$$J_p = (\inf C_p, \sup C_p).$$

These cylinder boundaries exist in the completion. For  $n \geq 1$  and  $d \in D$ , define the open sets

$$U_n = \bigcup_{\text{length}(p)=n} J_p, \quad V_d = \bigcup_{d \in \text{ran}(p)} J_p,$$

where the second union is over finite injective prefixes. Then

$$P = \bigcap_{n \geq 1} U_n \cap \bigcap_{d \in D} V_d.$$

Indeed, a numerical-split code at a prefix of length  $k$  fails  $U_{k+1}$ . There are no nonzero short limit prefixes below  $\omega$ . A full branch omitting  $d$  fails  $V_d$ . Every surjective branch belongs to every displayed open set. Theorem 65.11 exhausts all cases. This is a countable intersection of open sets. Each contains the dense set  $P$ , so the representation is a dense  $G_\delta$ .

Every nonempty interval contains a finite prefix cylinder. Reserve one unused label  $d$ . There are  $2^{\aleph_0}$  full injective branches extending that prefix and omitting  $d$ , for example by permuting the countably infinite remaining alphabet with  $d$  removed. Distinct branches give distinct proper gaps: at their first unequal coordinate, choose a fresh label strictly between their two entries and complete to a permutation between them. Thus the original interval contains continuum many omitted completion points. The cylinder also contains continuum many surjective permutations. This proves the fourth assertion.

Finally, the irrational order has real-line completion with exactly countably many omitted points, namely the rationals. An order isomorphism extends uniquely to the completions and preserves their complements. Our continuum-sized omitted set therefore excludes an order isomorphism with the irrationals.  $\square$

The conclusion illustrates a limitation of topological classification: the explicit Baire-space topology coexists with a substantially different pattern of Dedekind omissions. The completion normal forms retain that order information. For uncountable  $\mu$ , the same full-size-interval hypothesis makes bounded-length prefix cylinders a clopen basis, but the countable Baire-space construction and the real-line completion conclusion must not be transferred without new hypotheses.

## 68 Countable strata of length $\omega + n$ (sources 29 and 30)

[merge] One section from source 29's Section 8 (the base) and source 30's Section 6, written independently: neither saw the other. They prove the same theorems on the strata  $\mathcal{W}_{\omega+n}(D)$  of a

countable dense alphabet  $D$  without endpoints (for example  $\mathbf{No}_{<\omega}$ ). Source 29 writes  $P_\alpha = \mathcal{W}_\alpha(D)$  for every countable  $\alpha$ , source 30 writes  $P_n$  for the stratum of length  $\omega + n$  and  $X$  for the alphabet; here source 29's indexing is used throughout, so source 30's  $P_n$  is printed  $P_{\omega+n}$ , its  $X$  is printed  $D$ , its set  $D_n$  of nonisolated points is printed  $S_n$  (source 29's  $S$ ), and its relation-table space  $\mathcal{T}_n$  is printed  $\text{Tab}_n$ ; source 29's real completion  $C$  is printed  $\hat{I}$ . The base is source 29 for the category theorem, the comparison of the native and coordinate topologies at every countable length, the measurable trichotomy and the two product  $\sigma$ -algebras; source 30 for the nonisolated set and its Cantor–Bendixson analysis, the relation-table presentation, the ranking maps and the single orientation predicate; its weight corollary for arbitrary alphabets is printed with source 22's table (corollary 63.12). The duplicates: source 29's theorem 68.2 ( $n \geq 1$ ) is source 30's proposition 68.4, in a different ambient space; its proposition 68.6 is the third item of lemma 68.3; its theorem 68.8 is theorem 68.9; its theorem 68.13 is note 68.14 (classical, Fremlin's 419L(b)); its theorem 68.15 is theorem 68.16 in another formulation; the second clause of proposition 68.11 is corollary 68.17. At  $\mu = \omega$  the finite-tail normal form, the isolated points, the weight and the one-step derivative are source 22's proposition 63.8, theorem 63.9, corollary 63.10, and example 63.11; the density threshold at three labels is theorem 27.2. New against this report are the category, metrizability and complete-metrizability distinctions, the Baire property for  $n \geq 3$ , the topology comparison and all Borel results. The case  $n = 0$  is the countable-cutoff theorem of section 67.

The first countable stage gives a particularly concrete way to distinguish interpretations that have the same cardinality and the same embedding spectra. Let  $D$  be a countable dense linear order without endpoints; the intended example is  $D = \mathbf{No}_{<\omega}$  with its numerical order. For every countably infinite ordinal  $\alpha$ , put

$$P_\alpha = \{f : \alpha \longrightarrow D : f \text{ is bijective}\}.$$

We compare these functions at their first unequal ordinal coordinate. There are two topologies on the same set that must be named separately:

$$\tau_{\text{lex}} = \text{the topology of the displayed linear order}, \quad \tau_{\text{coord}} = \text{the subspace topology from } D^\alpha,$$

where the alphabet  $D$  is discrete in the second definition. The latter topology specifies finitely many coordinate values, at arbitrary positions of  $\alpha$ . A fixed prefix extending through an infinite ordinal need not be an open coordinate condition.

The previous countable-cutoff result says that these topologies agree on  $P_\omega$  and give Baire space (sources 21 and 20; theorems 67.6 and 67.11). The repository also already determines density and cellularity for every fixed stratum: in the present case the first three strata have countable density and all later ones have density  $\mathfrak{c}$  (theorem 27.2). We use these as background and prove refinements that are not consequences of the cardinal invariants: the precise topology comparison, a category distinction at one extra label, and different Borel sigma-algebras at two extra labels.

## 68.1 A common real completion for finite omissions

Let

$$I = \{p : \omega \longrightarrow D : p \text{ is injective}\}, \quad E_n = \{p \in I : |D \setminus \text{ran } p| = n\} \quad (n < \omega).$$

Here  $E_0 = P_\omega$ . Both  $I$  and each  $E_n$  carry finite-coordinate first-difference comparison. For a finite injective word  $s$ , its cylinder consists of all extensions of  $s$  in the order under discussion.

**Lemma 68.1** (A common completion). *The order  $I$  is dense, has no endpoints, and has a countable clopen cylinder basis. Its Dedekind completion  $\hat{I}$  is a copy of  $\mathbb{R}$ . Every  $E_n$  is order-dense in  $I$  and in  $\hat{I}$ . The order topology of  $E_n$  is its subspace topology in  $\hat{I}$ , and is also its finite-prefix topology.*

*Proof.* After any finite prefix the unused alphabet is again countable, dense, and without endpoints. Given unequal entries at the first difference, choose an unused entry strictly between them and extend to an infinite injection. The same construction on either side of the next entry proves that each cylinder has no endpoints and is open. Cylinders of one fixed finite length partition the order, so they are also closed. Every order interval containing  $p$  contains a sufficiently long finite prefix cylinder of  $p$ . Thus the countably many cylinders form a basis. Choosing one point of each nonempty cylinder gives a countable dense subset, whose order is the rational order. Passing to cuts identifies the completion with  $\mathbb{R}$ .

Every finite injective prefix can be completed with exactly  $n$  omitted labels: reserve  $n$  labels outside the prefix and enumerate all the other unused labels. Consequently each  $E_n$  meets every cylinder. It is order-dense in  $I$  and  $\hat{I}$ . Density ensures that the order topology is the inherited topology; the cylinder proof also works directly in  $E_n$ .  $\square$

The real completion in this lemma is shared by all  $E_n$ . In particular, it gives an unambiguous setting for comparing the complexity of their realized points. No class of proper-class orders occurs in this argument.

For each nonempty finite injective  $s$ , let

$$J_s = (\inf[s], \sup[s]) \subseteq \hat{I}, \quad J_\emptyset = \hat{I},$$

where the cylinder is now taken in  $I$ . Its finite endpoints exist, and  $J_s \cap I = [s]$ . Put

$$U_k = \bigcup_{|s|=k} J_s \quad (k \geq 1), \quad V_a = \bigcup_{a \in \text{ran } s} J_s \quad (a \in D).$$

All these sets are open in  $\hat{I}$ . Distinct cylinders of a fixed length give disjoint open intervals: an intersection would meet the dense order  $I$ . More generally, intersecting intervals have compatible prefixes. It follows, exactly as in the exhaustive case, that

$$I = \bigcap_{k \geq 1} U_k. \quad (19)$$

Indeed, a point in the intersection determines a compatible infinite injective branch. The point equals that branch in the completion, because otherwise an intervening member of  $I$  would share all of its coordinates. For a branch  $p \in I$ , membership in  $V_a$  says exactly that the label  $a$  occurs in  $p$ .

Define, for  $n \geq 0$ ,

$$G_{\leq n} = I \cap \bigcap_{A \in [D]^{n+1}} \bigcup_{a \in A} V_a. \quad (20)$$

This is a dense  $G_\delta$  subset of  $\hat{I}$ : it contains  $E_0$ , which is dense, and the displayed intersections are countable. Its branches omit at most  $n$  labels. Therefore

$$E_0 = G_{\leq 0}, \quad E_n = G_{\leq n} \setminus G_{\leq n-1} \quad (n \geq 1). \quad (21)$$

**Theorem 68.2** (The category change at one omission). *The following statements hold in the common real completion  $\hat{I}$ .*

1.  $E_0$  is a dense  $G_\delta$ , homeomorphic to Baire space, and is not  $F_\sigma$ .
2. For every  $n \geq 1$ ,  $E_n$  is a difference of two dense  $G_\delta$  sets, and hence is  $\Delta_3^0$ . It is neither  $G_\delta$  nor  $F_\sigma$ .
3. For every  $n \geq 1$ ,  $E_n$  is meager in itself, has no compatible complete metric, and contains a closed copy of Baire space.

Every nonempty interval of  $\widehat{I}$  contains  $\mathfrak{c}$  points of every  $E_n$ . Thus equal cardinalities of realized and omitted points do not detect this category distinction.

*Proof.* The first assertion is the previous  $\omega$ -enumeration theorem, applied to  $D$ . The explicit least-missing-label block construction gives the homeomorphism with  $\omega^\omega$ . For completeness, this space is not sigma-compact: if  $K_i$  are compact subsets, the projection of  $K_i$  to coordinate  $i$  is finite, and a diagonal choice outside these finite projections avoids  $\bigcup_i K_i$ . Since an  $F_\sigma$  subset of  $\mathbb{R}$  is sigma-compact,  $E_0$  is not  $F_\sigma$ .

Fix  $n \geq 1$ . For  $F \in [D]^n$ , let

$$A_F = \{p \in E_n : D \setminus \text{ran } p = F\}.$$

Within  $E_n$  this is the closed condition that no member of  $F$  ever occurs. It has empty interior. A finite prefix of a point of  $A_F$  can be extended by a member of  $F$  and then completed with  $n$  different omissions. Such a completion lies outside  $A_F$  in the same prescribed prefix neighborhood. The countable family  $(A_F)_{F \in [D]^n}$  covers  $E_n$ , proving that  $E_n$  is meager in itself. Each  $A_F$ , with its inherited prefix topology, is exactly the permutation space of the countable dense alphabet  $D \setminus F$ , hence is homeomorphic to Baire space. The Baire category theorem rules out a compatible complete metric on the nonempty space  $E_n$ .

Equation (21) proves the asserted Borel upper bound. If  $E_n$  were  $G_\delta$  in  $\widehat{I}$ , it would be a dense  $G_\delta$  disjoint from the dense  $G_\delta$  set  $E_0$ , contradicting the Baire theorem for  $\mathbb{R}$ . If  $E_n$  were  $F_\sigma$  in  $\widehat{I}$ , it would be sigma-compact, and its relatively closed subspace  $A_F$  would be sigma-compact as well. This contradicts the preceding diagonal argument for Baire space.

Finally, inside any fixed finite prefix choose one permissible omitted set  $F$  and vary the permutation of the remaining countably infinite alphabet. There are  $\mathfrak{c}$  such permutations. Every nonempty interval contains a prefix cylinder, and no  $E_n$  has more than  $\mathfrak{c}$  points.  $\square$

The description  $\Delta_3^0$  here is an upper bound together with strict exclusion from both second-level classes. It does not assert  $\Sigma_3^0$ - or  $\Pi_3^0$ -completeness. The more precise information we use is the explicit difference-of- $G_\delta$  representation.

[merge] Source 30 states the case  $n \geq 1$  in the injection space  $I = \text{Inj}(\omega, D) \subseteq D^\omega$  rather than in the completion; its Section 6 begins:

The countable birthday stage already distinguishes several notions that cardinality and ordinal-embedding spectra do not distinguish. Throughout this section let  $D$  be a countable dense linear order without endpoints, such as  $\mathbb{Q}$  or  $\mathbf{No}_{<\omega}$  with its numerical order. For a nonnegative integer  $n$  put

$$P_{\omega+n} = \text{Bij}(\omega + n, D), \quad E_n = \{p : \omega \hookrightarrow D : |D \setminus \text{ran}(p)| = n\}.$$

Both sets carry lexicographic orders;  $P_{\omega+n}$  carries its order topology. The topology on  $E_n$  will shortly be identified with the subspace topology inherited from  $D^\omega$ , where  $D$  is *discrete*. This use of the discrete topology is an encoding convention, and does not change the numerical order used in lexicographic comparisons.

The additional results in this section concern the progression

$$\omega, \quad \omega + 1, \quad \omega + 2, \quad \omega + 3.$$

At the first added terminal coordinate, complete metrizability fails. At the second, metrizability and standard Borel structure fail. At the third, a dense set of isolated points appears and the ranking map from relation tables ceases to be Borel measurable. These are different thresholds.

### 68.1.1 The cofinite-prefix model

For  $S \in [D]^n$  write

$$E_{n,S} = \{p \in E_n : \text{ran}(p) = D \setminus S\}.$$

For a finite injective sequence  $s$  in  $D$ , write  $[s] = \{p \in E_n : s \subseteq p\}$ . Every such cylinder is nonempty: choose  $n$  unused labels to omit, and enumerate all remaining labels.

**Lemma 68.3** (Prefix topology and finite fibers). *For every finite  $n$  the following assertions hold.*

1. *The finite-prefix cylinders form a countable clopen base for the order topology of  $E_n$ . Consequently this topology is precisely its product subspace topology.*
2.  *$E_n$  is densely ordered, has no endpoints, and has cardinality  $\mathfrak{c}$ .*
3. *Restriction to the first  $\omega$  coordinates identifies  $P_{\omega+n}$  as an ordered set with*

$$E_n \xrightarrow{\times} \{1, \dots, n!\},$$

*where the displayed finite set has its usual order and the product is lexicographic. The projection  $\pi_n : P_{\omega+n} \rightarrow E_n$  is continuous.*

*Proof.* If  $p$  extends  $s$ , choose two unused labels, one below and one above  $p(|s|)$ , and complete the two resulting finite prefixes to elements of  $E_n$ . The interval between these two elements contains  $p$  and lies in  $[s]$ . Thus  $[s]$  is open. Its complement is a union of the other cylinders of the same length, so it is also closed. If  $a < p < b$ , the two first disagreements with  $p$  occur at finite coordinates. A sufficiently long finite prefix of  $p$  therefore has its entire cylinder in  $(a, b)$ . The same argument applies to rays. This proves the first assertion.

If  $p < q$  first differ at  $k$ , choose an unused label strictly between  $p(k)$  and  $q(k)$  and extend the resulting prefix. The extension lies strictly between  $p$  and  $q$ . Choosing a smaller or larger first label proves the absence of endpoints. The upper cardinal bound is  $|D^\omega| = \mathfrak{c}$ . For a fixed omitted set  $S$ , the permutations of  $D \setminus S$  already give  $\mathfrak{c}$  elements: independently switch countably many disjoint pairs.

Finally, a prefix  $p \in E_n$  leaves exactly  $n$  labels, and its continuations are their  $n!$  permutations in lexicographic order. Different prefixes are compared before any of these terminal coordinates. This gives the displayed order isomorphism. For example, for  $a < b$  in  $E_n$ ,

$$\pi_n^{-1}((a, b)) = ((a, n!), (b, 1)),$$

and the corresponding ray identities prove continuity. □

**Proposition 68.4** (The category and Borel class of cofinite prefixes). *The space  $E_0$  is Polish and Baire. For each  $n \geq 1$ ,  $E_n$  is metrizable and second countable, but meagre in itself and therefore not Polish. As a subset of the Polish space  $I = \text{Inj}(\omega, D) \subseteq D^\omega$ ,  $E_n$  belongs to  $\Delta_3^0$  and is neither  $F_\sigma$  nor  $G_\delta$  when  $n \geq 1$ .*

*Proof.* Injectivity is a closed condition in  $D^\omega$ . Surjectivity is the  $G_\delta$  condition

$$\bigcap_{x \in D} \bigcup_{k < \omega} \{p : p(k) = x\}.$$

Thus  $E_0$  is a  $G_\delta$  subspace of a Polish space, hence Polish and Baire.

For  $n \geq 1$  the countable family  $\{E_{n,S} : S \in [D]^n\}$  covers  $E_n$ . Each member is closed in  $E_n$ , since it is specified there by the condition that every label in  $S$  is omitted. Each has empty interior: a

finite prefix can always be extended while omitting a different  $n$ -element set. This proves meagreness. A nonempty Polish space is Baire, so  $E_n$  is not Polish.

The condition “at most  $n$  labels are omitted” is  $G_\delta$ : for each  $(n+1)$ -element subset of  $D$ , at least one of its labels must occur. The condition “at least  $n$  labels are omitted” is  $F_\sigma$ : some  $n$ -element set is avoided. Their intersection is  $E_n$ , so  $E_n \in \Delta_3^0$ . It cannot be  $G_\delta$ , since that would make it Polish.

To rule out  $F_\sigma$ , fix  $S \in [D]^n$  and let

$$I_S = \{p \in I : \text{ran}(p) \subseteq D \setminus S\}.$$

This is a closed Polish subspace of  $I$ , and  $E_n \cap I_S = E_{n,S}$  is a dense  $G_\delta$  in  $I_S$ . Its complement is also dense: any finite injective prefix can be extended while omitting one additional label. Consequently every closed subset of  $I_S$  contained in  $E_{n,S}$  is nowhere dense. If  $E_n$  were  $F_\sigma$  in  $I$ , then  $E_{n,S}$  would be a meagre dense  $G_\delta$  in the Baire space  $I_S$ , a contradiction.  $\square$

*Note 68.5* (Source 30, Corollary 6.3, “An explicit Baire-space presentation”). *The space  $P_\omega = E_0$  is homeomorphic to  $\omega^\omega$  with its usual product topology.* This is [theorem 67.2](#), by the same least-missing-label tree ([theorem 66.2](#) at  $\mu = \omega$ ): at a cylinder  $[s]$  of full permutations, the first label  $a_j$  of a fixed list absent from  $s$  splits  $[s]$  into the countably many cylinders  $[s \cap v \cap (a_j)]$ , “exactly the possible blocks ending at the first future occurrence of  $a_j$ ”.

The preceding argument pinpoints why appending one final label matters. An enumeration of type  $\omega$  must eventually display *every* label; an enumeration of type  $\omega+1$  must leave exactly one label undisplayed at all finite stages. The latter condition is a countable union of closed, nowhere-dense alternatives, one for each possible final label. In particular  $P_{\omega+1}$  is not homeomorphic to the Baire space  $P_\omega$ .

## 68.2 Finite tails as finite split orders

**Proposition 68.6** (The exact finite-tail reduction). *For each  $n < \omega$  there is an order isomorphism*

$$P_{\omega+n} \cong E_n \times_{\text{lex}} \{0, \dots, n! - 1\}. \quad (22)$$

*It sends an enumeration to its first  $\omega$  entries and to the lexicographic rank of its finite terminal permutation. For  $n = 0$  and  $n = 1$  the second factor is a singleton.*

*Proof.* The first  $\omega$  entries omit exactly the  $n$  labels used in the tail. For a fixed prefix  $p \in E_n$ , the  $n!$  possible tails form a finite chain in numerical lexicographic order. Different prefixes first differ at a finite coordinate, so their entire tail fibers are ordered by the prefix comparison. This proves bijectivity and order preservation.  $\square$

**Corollary 68.7** (All the finite successors have different order types). *The orders  $P_{\omega+n}$ , for  $n < \omega$ , are pairwise nonisomorphic. In particular,  $P_\omega$  and  $P_{\omega+1}$  are equimorphic but are not isomorphic.*

*Proof.* The latter two orders are respectively  $E_0$  and  $E_1$ , and their order topologies are respectively Baire and meager in themselves. An order isomorphism is a homeomorphism, so no such isomorphism exists. Their equimorphism follows from the existing countable binary-cube classification ([theorem 31.2](#)); it is not a new embedding-spectrum assertion.

For  $n \geq 2$ , the equivalence relation saying that the closed interval between two points is finite has exactly the fibers in (22) as its classes. A fiber has  $n!$  elements. Different fibers have infinitely many whole fibers between them because  $E_n$  is densely ordered. Thus the size  $n!$  of the maximal finite convex blocks is an intrinsic invariant, distinguishing every  $n \geq 2$  from every other such  $n$  and from  $n = 0, 1$ .  $\square$



**Theorem 68.8** (Finite-tail topological classification). *In its native order topology,  $P_{\omega+n}$  has the following properties.*

1. *For  $n = 0$  it is Baire space.*
2. *For  $n = 1$  it is separable and metrizable, but is meager in itself and is not completely metrizable.*
3. *For  $n = 2$  it is first countable and separable, is meager in itself, and is not metrizable.*
4. *For  $n \geq 3$  it is first countable and nonmetrizable, and has a dense open set of  $\mathfrak{c}$  isolated points. It is therefore a Baire space, although its density is  $\mathfrak{c}$ .*

*For  $n \geq 2$  its topological weight is  $\mathfrak{c}$ .*

*Proof.* The cases  $n = 0, 1$  have already been proved. Consider a finite split order  $L = E \times_{\text{lex}} m$ , where  $E = E_n$  and  $m = n! \geq 2$ . The projection  $\pi : L \rightarrow E$  is continuous. A countable dense subset of  $E$  gives countable sequences approaching every base point from both sides. Together with the immediate neighbors inside a finite fiber, these give a countable local base at every point of  $L$ .

Let  $S = E \times \{0, m-1\}$  be the first and last points of the fibers. Its complement consists of isolated interior points and is open; hence  $S$  is closed. The induced topology on  $S$  is precisely the order topology of  $E \times_{\text{lex}} 2$ . This can also be seen directly from neighborhoods of the first and last point of each fiber: the first is approached through earlier base fibers and the last through later ones. Removing only the interior of the finite fibers changes neither of these neighborhoods.

The double split of  $E$  is separable. If  $B$  is a countable dense subset of  $E$ , then  $B \times 2$  is dense, since every nonempty open set contains a point over a nonempty base interval. It is not second countable. For each  $x \in E$ , choose a basis element containing  $(x, 0)$  and contained in the open ray  $(-\infty, (x, 1))$ . The chosen basis elements must be distinct: if  $x < y$ , the one chosen for  $x$  cannot contain  $(y, 0)$ . Thus every basis has cardinality at least  $|E| = \mathfrak{c}$ . A separable metrizable space is second countable, so the double split is not metrizable. Since  $S$  is a subspace, neither is  $L$ . First countability and  $|L| = \mathfrak{c}$  give the matching upper bound for weight.

When  $m = 2$ , the inverse images of the relatively closed nowhere dense sets  $A_F \subseteq E_n$  remain nowhere dense. Every nonempty open set in the double split contains all fibers over a nonempty base interval, which cannot be contained in  $A_F$ . These countably many inverse images cover  $L$ , proving that it is meager in itself.

When  $m \geq 3$ , the interior points of every fiber are isolated. They form a dense open set: any neighborhood of a first or last point meets another whole fiber, because the base order is dense and has no endpoints. Every dense open subset must contain every isolated point. Consequently a countable intersection of dense open subsets contains this same dense set, which proves the Baire assertion. There are  $\mathfrak{c}$  isolated points, so density is  $\mathfrak{c}$ .  $\square$

The threshold for category differs from the threshold for metrizability. One omitted label destroys complete metrizability. Two terminal labels produce uncountably many jumps and destroy metrizability. Three terminal labels introduce a dense set of isolated points and restore the Baire property. The density transition at three labels is already in the repository; the category and measurable-structure refinements here explain additional distinctions it does not record.

[merge] Source 30's Theorem 6.4 proves the four regimes again and adds the nonisolated set  $S_n$  (its  $D_n$ ), which is order-homeomorphic to the doubled order  $\text{Split}(E_n)$  and closed nowhere dense for  $n \geq 3$ :

### 68.2.1 The four topological regimes

We first isolate the elementary topology of a doubled order. If  $E$  is densely ordered without endpoints, put

$$\text{Split}(E) = E \overrightarrow{\times} \{-, +\}.$$

The two points  $p^- < p^+$  over each  $p \in E$  are adjacent, but neither is isolated. The term “split” here specifies this exact two-point construction.

**Theorem 68.9** (Finite-tail topology). *The following properties hold for  $P_{\omega+n}$ .*

1.  $P_\omega$  is zero-dimensional, Polish and Baire.
2.  $P_{\omega+1}$  is zero-dimensional, second countable and metrizable, but meagre in itself and not Polish.
3.  $P_{\omega+2}$  is zero-dimensional, first countable and separable, but meagre in itself and nonmetrizable. Its weight is  $\mathfrak{c}$ .
4. For  $n \geq 3$ ,  $P_{\omega+n}$  is zero-dimensional and first countable. It has a dense open discrete set of  $\mathfrak{c}$  isolated points, and is therefore Baire. Its density, cellularity and weight are all  $\mathfrak{c}$ , and it is nonmetrizable.

For  $n \geq 2$ , the nonisolated-point subspace is exactly

$$S_n = \{(p, 1), (p, n!) : p \in E_n\}.$$

With its inherited topology,  $S_n$  is order-homeomorphic to  $\text{Split}(E_n)$ . It is separable, meagre in itself and nonmetrizable, and has no isolated points. For  $n \geq 3$  it is closed and nowhere dense in  $P_{\omega+n}$ ; moreover  $P'_{\omega+n} = S_n = S'_n$ .

*Proof.* The assertions about  $P_\omega$  and  $P_{\omega+1}$  follow from the singleton-fiber cases of Lemma 68.3 and Proposition 68.4.

Consider  $E_n$  with each point replaced by two points. Every nonempty open subset contains the full inverse image of some finite-prefix cylinder. Indeed, a nonempty basic interval cannot have both bounds in one two-point fiber; between their distinct prefixes choose an intermediate prefix and then a sufficiently small finite-prefix cylinder. Choosing one point in the lift of each of the countably many finite-prefix cylinders gives a countable dense subset. Moreover

$$\text{Split}(E_n) = \bigcup_{S \in [D]^n} \pi^{-1}(E_{n,S}) \quad (n \geq 1)$$

is a countable union of closed nowhere-dense sets. Closedness follows from continuity of the projection; the preceding finite-prefix observation proves empty interior. This proves separability and meagreness for all these split spaces.

There are  $\mathfrak{c}$  adjacent pairs in  $\text{Split}(E_n)$ . From any base choose, for each  $p$ , a basic set  $U_p$  with

$$p^- \in U_p \subseteq (-\infty, p^+).$$

These basic sets are distinct. If  $p < q$ , then  $q^- \geq p^+$ , so a set contained in  $(-\infty, p^+)$  cannot also contain  $q^-$ . Hence the weight is at least  $\mathfrak{c}$ ; the family of all order intervals and rays gives the reverse bound. A separable metric space has countable weight, so the split space is nonmetrizable.

For completeness, the local topology can be described explicitly. Let  $C_k(p)$  be the length- $k$  prefix cylinder about  $p$  in  $E_n$ , and put  $m = n!$ . At the first point of a fiber with  $m \geq 2$ , the sets

$$\pi_n^{-1}(C_k(p)) \cap (-\infty, (p, 2)) \quad (k < \omega)$$

form a clopen local base. At the last point use

$$\pi_n^{-1}(C_k(p)) \cap ((p, m-1), \infty).$$

Every interior point of a finite fiber is an open singleton. This proves first countability and zero-dimensionality.

There are no adjacent points in distinct fibers, because  $E_n$  is densely ordered. Thus the isolated points in  $P_{\omega+n}$  are precisely  $(p, j)$  with  $1 < j < n!$ . For  $n \geq 3$  these points form a set of cardinality  $\mathfrak{c}$ . They are dense: a neighborhood of a fiber endpoint contains points from other fibers on its nonadjacent side, and hence contains an entire intervening fiber and an interior point of that fiber. Every dense open set contains every isolated point. Any countable intersection of dense open sets therefore contains this fixed dense set, proving the Baire property.

Deleting the interior points leaves exactly the two endpoints of each fiber. The displayed local bases show directly that their inherited topology is the order topology of  $\text{Split}(E_n)$ . This subspace has no isolated points. It is closed, and for  $n \geq 3$  it is nowhere dense because the isolated points are dense. Consequently the Cantor–Bendixson derivative stabilizes after the first step. Its nonmetrizability also proves that  $P_{\omega+n}$  itself is nonmetrizable for  $n \geq 3$ . Finally, the open singletons give cellularity and density at least  $\mathfrak{c}$ , while the size of the entire space gives all upper bounds claimed.  $\square$

Thus the Baire property is lost at  $\omega + 1$  and recovered at  $\omega + 3$  for an entirely different reason. The recovered Baire space is not completely Baire: it contains the nonempty closed subspace  $S_n$ , which is meagre in itself. There is no conflict with the minimal-type result, and no implication that the spaces in the final regime are Polish. Nor does the theorem assert that all  $P_{\omega+n}$  with  $n \geq 3$  are mutually homeomorphic.

### 68.3 Comparing the two topologies at every countable length

**Theorem 68.10** (Exact comparison of native and coordinate topologies). *For a countably infinite ordinal  $\alpha$ , the coordinate topology on  $P_\alpha$  is always homeomorphic to Baire space. The two displayed topologies satisfy*

| $\alpha$                            | comparison                                                      |
|-------------------------------------|-----------------------------------------------------------------|
| $\omega$                            | $\tau_{\text{lex}} = \tau_{\text{coord}}$                       |
| $\omega + 1$                        | $\tau_{\text{lex}} \subsetneq \tau_{\text{coord}}$              |
| $\omega + 2 \leq \alpha < \omega_1$ | $\tau_{\text{lex}}$ and $\tau_{\text{coord}}$ are incomparable. |

*The native order topology is first countable for every such  $\alpha$ .*

*Proof.* Choose any bijection  $b : \omega \rightarrow \alpha$ . Reindexing by  $b$  identifies the coordinate permutation space with the coordinate space  $P_\omega$ , which is Baire space. This reindexing makes no claim to preserve the native lexicographic order.

For  $\alpha = \omega + 1$ , two distinct exhaustive enumerations must differ before  $\omega$ : if their first  $\omega$  entries agreed, the single remaining label would agree too. Consequently finite initial cylinders form the order-topology basis, making every native-open set coordinate-open. The final-coordinate condition  $f(\omega) = a$  is coordinate-open but has empty interior in the native order: any finite prefix can be completed with a different omitted label. The inclusion is strict.

We give witnesses valid for every  $\alpha > \omega$ , writing  $\alpha = \omega + \tau$ . First, the coordinate topology is never contained in the native topology. Choose a bounded numerical interval in  $D$  and a suborder  $Y$  of that interval of order type  $\tau$ . Such a suborder exists because every countable linear order embeds in a countable dense order. For finite  $\tau$  this just means choosing finitely many increasing

labels. Enumerate  $D \setminus Y$  in the first  $\omega$  positions and enumerate  $Y$  increasingly thereafter; call the resulting point  $f$ . It is the least point in its first- $\omega$  prefix fiber. If  $a = f(\omega)$ , the coordinate-open set

$$O_a = \{g : g(\omega) = a\}$$

is not a native neighborhood of  $f$ . Given  $h < f$ , the first difference between  $h$  and  $f$  occurs at a finite position. Preserve a longer finite prefix of  $f$ , choose the next entry smaller than the corresponding entry of  $f$ , and prescribe a value other than  $a$  at coordinate  $\omega$ . The finite partial injection extends to a bijection, producing  $h < g < f$  with  $g \notin O_a$ . Thus every native neighborhood of  $f$  leaves  $O_a$ .

Second, if  $\tau$  has at least two elements, the native topology is not contained in the coordinate topology. Keep a first- $\omega$  prefix enumerating  $D \setminus Y$ , where  $Y$  lies in a bounded interval and has cardinality  $|\tau|$ ; choose two completions  $f < h$  of this prefix. The native ray  $(-\infty, h)$  contains  $f$ . In every finite-coordinate neighborhood of  $f$ , select two unfixed finite positions  $i < j$  with  $f(i) < f(j)$  and transpose their values. These exist arbitrarily late because  $D \setminus Y$  is unbounded above and deleting finitely many entries does not change that. The modified enumeration  $g$  respects the prescribed finite coordinates, but its first difference from  $h$  is at  $i$ , with a larger value. Thus  $g > h$ . The native ray is not coordinate-open.

Finally, no increasing sequence of type  $\omega_1$  can occur in  $P_\alpha$ , nor can a decreasing one. To see this directly, suppose  $(f_\xi)_{\xi < \omega_1}$  is increasing. On a final segment its first coordinate stabilizes, since it is monotone in a countable alphabet. Recursively stabilize each coordinate after stabilizing all earlier ones. There are only countably many coordinates, so the supremum of the stabilization indices is still below  $\omega_1$ . The sequence is eventually constant, a contradiction. If the portion below any point had no countable cofinal subset, recursion would build an increasing  $\omega_1$ -sequence there. The dual argument applies above the point. Countable cofinal and coinital sets give the required countable local bases.  $\square$

**Proposition 68.11** (Low-complexity comparison is not low-complexity topology). *For every countably infinite  $\alpha$ , the relation*

$$\{(f, g) \in P_\alpha^2 : f <_{\text{lex}} g\}$$

*is both  $F_\sigma$  and  $G_\delta$  in the product coordinate topology. Nevertheless, for every  $\alpha \geq \omega + 3$  there are native-open subsets of  $P_\alpha$  that are not coordinate-Borel.*

*Proof.* For each  $\beta < \alpha$ , the condition

$$f \restriction \beta = g \restriction \beta, \quad f(\beta) < g(\beta)$$

is closed in the product coordinate topology: the equality is an intersection of closed coordinate equalities, and the last condition is clopen because the alphabet is discrete. The lexicographic relation is the countable union of these closed sets, hence is  $F_\sigma$ . Its complement is the reverse strict relation together with the closed diagonal, again  $F_\sigma$ . Thus the strict relation is also  $G_\delta$ .

For the second assertion, write  $\alpha = \omega + \tau$  and choose  $Y \subset D$  with  $|Y| = |\tau|$  and infinite complement. There are  $\mathfrak{c}$  prefixes of length  $\omega$  enumerating  $D \setminus Y$ . Each of their convex tail fibers has at least three points, since  $\tau \geq 3$ , and therefore contains a nonempty ambient open interval. These intervals are pairwise disjoint. This is the countable case of the existing cellularity argument (theorem 27.2). Their arbitrary unions yield  $2^{\mathfrak{c}}$  distinct native-open sets, while the coordinate Polish space has only  $\mathfrak{c}$  Borel sets [57]. For finite tails, arbitrary subsets of the  $\mathfrak{c}$  isolated interior permutations give the same conclusion.  $\square$

Thus a Borel comparison relation on a standard Borel code space does not make every open set of the corresponding order topology Borel. The order topology permits arbitrary unions; the Borel sigma-algebra permits only countable iterations of its generating operations.

[merge] Source 30's presentation of the strata by relation tables, which its measurable comparison below uses:

### 68.3.1 Relation tables as a standard Borel presentation

Let  $\text{Tab}_n$  be the set of strict well-order relations of type  $\omega + n$  on  $D$ , regarded as a subspace of the Cantor space  $2^{D \times D}$ . Its topology records finitely many answers to questions of the form “does  $xRy$ ?” There are mutually inverse set bijections

$$\Phi_n : P_{\omega+n} \longrightarrow \text{Tab}_n, \quad r_n : \text{Tab}_n \longrightarrow P_{\omega+n},$$

where  $\Phi_n(e)$  is the relation induced by  $e$  and  $r_n$  is its unique increasing enumeration. The order topology on  $P_{\omega+n}$  and the table topology on  $\text{Tab}_n$  should not be identified.

**Proposition 68.12** (The fixed finite-tail table domains). *For every finite  $n$ ,  $\text{Tab}_n$  is a  $\Pi_3^0$ -complete Borel subset of  $2^{D \times D}$  and hence is a standard Borel space. Each coordinate of  $r_n$ , regarded as a map into the product  $D^{\omega+n}$  with discrete  $D$ , is Borel measurable. In fact, each singleton coordinate fiber is  $F_\sigma$  relative to  $\text{Tab}_n$ .*

*Proof.* Let  $\text{FinPred}_x$  say that  $x$  has finitely many predecessors. This is  $F_\sigma$ : some finite set contains all predecessors. Let  $\text{Succ}_k(x)$  say that  $x$  has exactly  $k$  successors. This too is  $F_\sigma$ , by choosing the finite successor set and specifying all remaining comparison bits. The set  $\text{LO}(D)$  of strict linear orders is closed.

For  $n = 0$  the required definition is

$$\text{LO}(D) \cap \bigcap_{x \in D} \text{FinPred}_x.$$

For  $n \geq 1$  a  $\Pi_3^0$  definition is the conjunction of

1. being a strict linear order;
2. among any  $n + 1$  distinct elements, at least one has finitely many predecessors;
3. some element has exactly  $n - 1$  successors.

To verify this definition, condition (2) leaves at most  $n$  points with infinitely many predecessors. The remaining points form an infinite initial segment of type  $\omega$ . Hence the entire order has type  $\omega + k$  for some  $k \leq n$ . Condition (3) forces  $k = n$ . The definability bounds follow from the displayed  $F_\sigma$  descriptions.

The  $\Pi_3^0$ -completeness of the isomorphism class of  $\omega$  is classical; see Miller [60, §2, Example 5]. Appending a fixed  $n$ -point terminal order, after relabeling  $D$  into a cofinite subset, is a continuous reduction from  $\text{Tab}_0$  to  $\text{Tab}_n$ . Indeed, appending finitely many points preserves any failure of well-foundedness, and for ordinals  $\gamma + n = \omega + n$  implies  $\gamma = \omega$ . This proves completeness for every finite  $n$ . Miller's full calculation in [60, §3, Lemma 3.5] gives the same classification as part of the classification for all countable ordinals.

Finally,  $r_n(R)(k) = x$  for  $k < \omega$  means exactly that  $x$  has  $k$  predecessors. For  $j < n$ ,

$$r_n(R)(\omega + j) = x \iff x \text{ has exactly } n - 1 - j \text{ successors.}$$

These are  $F_\sigma$  conditions. They prove the coordinate claim and product Borel measurability. □

The last assertion does *not* imply Borel measurability into the lexicographic order topology. This distinction becomes decisive below.

There is a third natural topology to keep separate. For every fixed countable ordinal  $\alpha$ , the space  $\text{Bij}(\alpha, D)$  is Polish in the *full discrete product topology* inherited from  $D^\alpha$ : injectivity is closed and surjectivity is  $G_\delta$ . The map from this product presentation to relation tables is continuous, since the relative order of two labels is determined once their two coordinates have been specified. For  $\alpha = \omega + 1$ , observing the terminal coordinate is product-open but not lexicographically open; its lexicographic fiber is one of the closed nowhere-dense sets above. Thus even the two topologies on the same set of enumerations must be named explicitly. The pathologies proved here concern the *order topology*, and do not contradict the existence of a Polish product presentation.

## 68.4 The exact native Borel sigma-algebra of a double split

The next elementary split-order lemma provides a full measurable description at  $\omega + 2$ . The general split-interval phenomenon is classical: for the split interval the following formula is Fremlin's [59, 419L(b)]. Applying it to the finite terminal permutation fibers supplies the present refinement. We include the complete proof so that no classification theorem is needed.

**Theorem 68.13** (Borel sets in the double split). *Let  $E$  be a separable dense linear order without endpoints, identified with a dense suborder of  $\mathbb{R}$ , and let*

$$L = E \times_{\text{lex}} \{0, 1\}, \quad \pi : L \rightarrow E.$$

*With the native order topologies,*

$$\mathcal{B}(L) = \{\pi^{-1}(B) \triangle N : B \in \mathcal{B}(E), N \subseteq L \text{ countable}\}. \quad (23)$$

*In particular, a native-Borel set can contain exactly one point of a split pair at only countably many base points. If  $E$  is uncountable,  $\mathcal{B}(L)$  is not countably generated and is not a standard Borel structure.*

*Proof.* The projection is continuous. In particular, inverse images of Borel base sets are Borel, and all countable subsets of  $L$  are Borel because singletons are closed. This proves one inclusion in (23).

The family on its right-hand side is a sigma-algebra. Complements preserve the form; the union of countably many such sets differs from the inverse image of the union of their base sets at only countably many points. It remains to show that it contains every open subset  $U \subseteq L$ .

The order  $L$  is separable, by the dense-set argument in the preceding theorem. Decompose  $U$  into its maximal convex components. Every component is open and contains a point of a fixed countable dense set, so there are only countably many of them. The projection of a convex component is a convex subset of  $E$ , hence a Borel interval of  $E$ , allowing any endpoint convention. A convex component contains either both or neither points of every fiber except, possibly, at its two boundary fibers. Consequently  $U$  differs from  $\pi^{-1}(\pi[U])$  at only countably many points, and  $\pi[U]$  is a countable union of Borel base intervals. This proves the other inclusion.

Suppose, finally, that  $(B_i)_{i < \omega}$  generated  $\mathcal{B}(L)$ . Each  $B_i$  distinguishes the two points over at most countably many base points. Outside the union of these countable exceptional sets, the two points have the same membership in every generator, hence in every set in the generated sigma-algebra. But their singletons are Borel and distinguish them. For uncountable  $E$  this is impossible. A standard Borel structure is countably generated [59, 424B(a)], so it is excluded as well.  $\square$

The hypothesis that  $E$  sits densely in its real completion matters for this formulation. We are not claiming that every separable linear order embeds densely in  $\mathbb{R}$ ; orders with jumps need a different statement. Each  $E_n$  satisfies exactly the hypothesis used here.

*Note 68.14* (Source 30, Lemma 6.7, “Borel sets split only countably many fibers”). *Let  $E$  be a second-countable dense linear order without endpoints, and suppose  $\text{Split}(E)$  is separable. With  $\pi$  the projection, its Borel sets are precisely the sets  $\pi^{-1}(B) \triangle C$  with  $B$  Borel in  $E$  and  $C$  countable; in particular every Borel set splits only countably many fibers; if  $E$  is uncountable, this Borel  $\sigma$ -algebra is not countably generated and is not a standard Borel structure.* This is [theorem 68.13](#), by the same component argument (an open set has countably many maximal convex components, each splitting at most its two boundary fibers). Source 30 adds: “For example, the set of all lower endpoints is not Borel. Its failure is stronger than a failure to be open: it splits every fiber. For  $E = E_n$ , the split space has exactly  $\mathfrak{c}$  Borel sets  $\dots$ , despite the failure of countable generation.” Source 29 credits the formula to Fremlin [59], 419L(b), for the split interval.

**Theorem 68.15** (The measurable trichotomy for finite tails). *Let  $\mathcal{B}_{\text{lex}}$  and  $\mathcal{B}_{\text{coord}}$  be the Borel sigma-algebras of the two topologies on  $P_{\omega+n}$ . Then*

| $n$        | comparison of Borel sigma-algebras                                            |
|------------|-------------------------------------------------------------------------------|
| 0, 1       | $\mathcal{B}_{\text{lex}} = \mathcal{B}_{\text{coord}}$                       |
| 2          | $\mathcal{B}_{\text{lex}} \subsetneq \mathcal{B}_{\text{coord}}$              |
| $n \geq 3$ | $\mathcal{B}_{\text{lex}}$ and $\mathcal{B}_{\text{coord}}$ are incomparable. |

For  $n = 2$ , formula (23) with  $E = E_2$  describes the native Borel sigma-algebra exactly, and it is not countably generated. For  $n \geq 3$ , the native Borel sigma-algebra has cardinality  $2^{\mathfrak{c}}$ .

*Proof.* For  $n = 0$  the topologies agree. For  $n = 1$ , native-open sets are coordinate-open. Conversely, the extra coordinate events are already native-Borel:

$$\{f : f(\omega) = a\} = P_{\omega+1} \setminus \bigcup_{k < \omega} \{f : f(k) = a\}.$$

The events on the right are native-clopen. A countable coordinate basis therefore consists of native-Borel sets, proving equality of the two sigma-algebras.

For  $n = 2$ , use the double-split representation. The prefix map to  $E_2$  is continuous for the coordinate topology, and countable sets are coordinate-Borel. Formula (23) therefore gives  $\mathcal{B}_{\text{lex}} \subseteq \mathcal{B}_{\text{coord}}$ . Fix  $a < b$  in  $D$  and consider the coordinate-clopen event

$$O_a = \{f : f(\omega) = a\}.$$

For each prefix enumerating  $D \setminus \{a, b\}$ , the two possible tails are  $(a, b)$  and  $(b, a)$ , and  $O_a$  contains exactly one of them. There are  $\mathfrak{c}$  such prefixes. The split-pair criterion in the preceding theorem shows that  $O_a$  is not native-Borel. The inclusion is strict.

For  $n \geq 3$ , let  $m = n!$  and let  $S = E_n \times \{0, m-1\}$  be the closed endpoint subspace from Theorem 68.8. It is the double split of  $E_n$ . Choose a fixed  $n$ -element residual set  $F$  with minimum  $a$  and with a different maximum. The coordinate-clopen event  $O_a$  contains the first but not the last point over every prefix enumerating  $D \setminus F$ : the first tail is increasing and the last decreasing. Its trace on  $S$  distinguishes  $\mathfrak{c}$  endpoint pairs, so the double-split theorem shows that the trace is not Borel in  $S$ . If  $O_a$  were native-Borel, its trace on this closed subspace would be Borel, a contradiction.

In the other direction, the isolated interior points form a set of cardinality  $\mathfrak{c}$ , every subset of which is native-open. There are  $2^{\mathfrak{c}}$  such subsets and only  $\mathfrak{c}$  coordinate-Borel sets [57]. Some native-open subsets are therefore not coordinate-Borel. This also proves the asserted cardinality of the native Borel sigma-algebra.  $\square$



There is a slightly fuller normal form when  $n \geq 3$ . Put

$$J = E_n \times \{1, \dots, n! - 2\}, \quad S = E_n \times \{0, n! - 1\}.$$

A set is native-Borel exactly when its intersection with  $J$  is arbitrary and its intersection with  $S$  has the form

$$\pi_S^{-1}(B) \triangle N, \quad B \in \mathcal{B}(E_n), \quad N \subseteq S \text{ countable.}$$

Indeed  $J$  is open and discrete,  $S$  is closed, and the induced topology on  $S$  is the double split. This gives an explicit description of every native-Borel set, not just a cardinality argument.

For comparison, under the same finite-split identification the coordinate Borel sigma-algebra is exactly

$$\left\{ \bigcup_{j < n!} B_j \times \{j\} : B_j \in \mathcal{B}(E_n) \right\}. \quad (24)$$

To verify this without a Borel-isomorphism theorem, the prefix projection is coordinate-continuous and the finite tail rank is coordinate-continuous. Conversely, partition  $E_n$  into the countably many closed sets  $A_F$  of prescribed omissions. On  $A_F \times \{j\}$ , every tail value is fixed. Thus the inverse image of every coordinate cylinder is a countable union of Borel pieces in the displayed finite product.

[merge] Source 30 states the measurable comparison through the ranking map  $r_n$  and the table map  $\Phi_n$ ; by its [proposition 68.18](#), coordinate Borel sets and relation-table Borel sets agree, so its theorem and [theorem 68.15](#) are one result in two formulations. Source 30's theorem adds that for  $n \leq 1$  ranking is a pointwise limit of continuous maps, and that the preimage of every lexicographic ray is  $F_\sigma$  for every  $n$ .

#### 68.4.1 A sharp, two-direction coding threshold

**Theorem 68.16** (Ranking and relation-table maps). *With the topologies specified above:*

1. For  $n = 0, 1$ ,  $\Phi_n$  is continuous and  $r_n$  is Borel measurable. Thus the two presentations have the same standard Borel structure. Nevertheless,  $r_n$  is discontinuous at every point; it is a pointwise limit of continuous maps into  $P_{\omega+n}$ .
2. For  $n = 2$ ,  $r_2$  is Borel measurable, but  $\Phi_2$  is not Borel measurable. The order-topological Borel structure of  $P_{\omega+2}$  is strictly smaller than the relation-table Borel structure and is not standard Borel.
3. For every finite  $n \geq 3$ , neither  $r_n$  nor  $\Phi_n$  is Borel measurable. The order topology on  $P_{\omega+n}$  has  $2^{\mathfrak{c}}$  open sets, whereas  $\text{Tab}_n$  has only  $\mathfrak{c}$  Borel sets.

More precisely, for  $n \leq 2$  the inverse image under  $r_n$  of every open set is  $F_\sigma$  relative to  $\text{Tab}_n$ . For every finite  $n$ , the inverse image of each individual lexicographic ray is  $F_\sigma$ , including when ranking is not Borel measurable.

*Proof.* For  $n \leq 1$ , at least one of any two distinct labels  $a, b$  occurs at a finite coordinate. The condition that  $a$  precedes  $b$  is therefore

$$\exists k < \omega \quad [e(k) = a \text{ and } b \notin \{e(0), \dots, e(k)\}].$$

Both this condition and its complementary orientation are unions of finite-prefix cylinders. Every relation bit is consequently continuous, proving continuity of  $\Phi_n$ . The countable finite-prefix base for  $P_{\omega+n}$  and [Proposition 68.12](#) prove Borel measurability of  $r_n$ .

To see discontinuity of ranking, fix a relation  $R$  and a table neighborhood determined by finitely many labels  $F$ . Move a finite-rank label outside  $F$ , distinct from the original first label, to the first position and preserve the relative order of all other labels. The order type remains  $\omega + n$  and all specified comparison bits remain unchanged, but the first enumerated label changes. The first-coordinate cylinder is open in  $P_{\omega+n}$ , so  $r_n$  is discontinuous at  $R$ .

There is also an explicit sequence of continuous approximations when  $n \leq 1$ . Fix an auxiliary enumeration  $x_0, x_1, \dots$  of  $D$ . At stage  $m$ , sort  $\{x_0, \dots, x_{m-1}\}$  by  $R$  and complete the resulting finite prefix to an element of  $P_{\omega+n}$  by a fixed deterministic rule. This map depends on finitely many comparison bits and is continuous. Every true finite initial segment eventually occurs correctly in these finite sorts, because each of its elements has only finitely many predecessors. The approximations therefore converge in the finite-prefix topology of  $P_\omega$  or  $P_{\omega+1}$  to  $r_n(R)$ .

For  $n = 2$ , write a Borel set of  $P_{\omega+2} = \text{Split}(E_2)$  as  $\pi_2^{-1}(B) \triangle C$  using [Note 68.14](#). The map  $R \mapsto r_2(R)|_\omega$  is Borel into  $E_2$ , by its countable finite-prefix base and the coordinate descriptions above. The inverse image of the countable set  $C$  is countable and therefore Borel in  $\text{Tab}_2$ . Thus  $r_2$  is Borel measurable.

In the reverse direction, fix  $a < b$  in  $D$ . In every fiber whose omitted pair is  $\{a, b\}$ , the lower point has terminal order  $(a, b)$  and the upper point has terminal order  $(b, a)$ . Therefore the inverse image under  $\Phi_2$  of the clopen table condition  $aRb$  splits  $\mathfrak{c}$  fibers. It is not Borel by [Note 68.14](#). This proves all the assertions at  $n = 2$ , including strictness of the two Borel structures.

For  $n \geq 3$ , arbitrary subsets of the dense isolated-point set are open. There are  $2^{\mathfrak{c}}$  such subsets. Distinct open subsets have distinct inverse images under the bijection  $r_n$ , while a standard Borel space of size  $\mathfrak{c}$  has only  $\mathfrak{c}$  Borel subsets. Thus  $r_n$  cannot be Borel measurable.

Finally, restrict to the closed subspace  $S_n \cong \text{Split}(E_n)$ . Fix an omitted  $n$ -element set  $S$  containing  $a < b$ . For each of the  $\mathfrak{c}$  prefixes omitting exactly  $S$ , the first terminal permutation is the increasing one and the last is the decreasing one. The table condition  $aRb$  distinguishes these two endpoints. Its inverse image in  $S_n$  splits  $\mathfrak{c}$  fibers and is not Borel. The trace on a closed subspace of a Borel set would be Borel. Consequently  $\Phi_n$  is not Borel measurable in  $P_{\omega+n}$  either.

For the final assertions, fix  $e \in P_{\omega+n}$ . The condition  $r_n(R) < e$  is a countable union over finite first-disagreement coordinates  $k$ , together with a finite exceptional set of relations whose enumerations agree with  $e$  through all of  $\omega$  and differ only in the terminal permutation. Each finite-disagreement condition is  $F_\sigma$ : it specifies finitely many exact finite ranks, and then one finite rank carrying any of countably many labels below  $e(k)$ . The exceptional relation codes are closed singletons. The same proof treats the opposite ray.

Every open subset of a separable linear order is a countable union of basic intervals and rays with actual order points as endpoints. To check the assertion without assuming a countable base, first decompose it into countably many maximal convex open components. In each component choose countable coinital and cofinal subsets, adding a least or greatest point if present. Intervals with endpoints from these countable subsets cover the component except possibly its least and greatest points; one additional ambient basic neighborhood covers each such point. These neighborhoods exist because the component is open. Applying this observation to the separable spaces  $P_\omega, P_{\omega+1}, P_{\omega+2}$ , and using closure of  $F_\sigma$  sets under finite intersections and countable unions, proves the open-set assertion. This is a statement about the inverse images of open sets; no Baire-class terminology for maps into the nonmetrizable space  $P_{\omega+2}$  is required.  $\square$

The theorem gives a concrete warning for formalization. On a fixed countable carrier, relation tables and enumerations are unambiguously interconvertible as *sets*. Their natural topologies and measurable structures need not be interconvertible, even though every individual enumeration coordinate has a low Borel description in table codes. At  $\omega + 2$  there are too many adjacent pairs

for a countably generated Borel structure; at  $\omega + 3$  there are too many open sets for ranking to be measurable at all.

**Corollary 68.17** (Nonmeasurable ranking at every later countable type). *Let  $\alpha \geq \omega + 3$  be countable. Equip the well-orders of type  $\alpha$  on  $D$  with their relation-table standard Borel structure, and their increasing enumerations with the lexicographic order topology. The ranking bijection from the former to the latter is not Borel measurable.*

*Proof.* The fixed-type table class is Borel by the classical Borelness of the isomorphism class of a countable structure; for ordinals see Miller [60, §3]. The established fixed-type cellularity theorem supplies  $\mathfrak{c}$  pairwise disjoint nonempty open intervals in the enumeration order. Their arbitrary unions are  $2^{\mathfrak{c}}$  distinct open sets. The same counting argument as in Theorem 68.16 applies.  $\square$

**Proposition 68.18** (One orientation predicate restores the standard Borel structure). *For  $P_{\omega+2}$ , let*

$$H = \{e \in P_{\omega+2} : e(\omega) <_D e(\omega + 1)\},$$

*the set of lower endpoints of the adjacent fibers. Then*

$$\sigma(\text{Bor}_{\text{lex}}(P_{\omega+2}) \cup \{H\}) = \text{Bor}_{\text{prod}}(P_{\omega+2}) = \{r_2(B) : B \in \text{Bor}(\text{Tab}_2)\}.$$

*Thus one additional binary predicate suffices, and none suffices.*

*Proof.* The equality between the full-product and transported table Borel structures follows from the coordinate formulas for ranking and continuity of relation coding in the product topology. Put

$$C_{a,b} = \{e \in P_{\omega+2} : D \setminus \text{ran}(e|_{\omega}) = \{a, b\}\}.$$

This is lexicographically closed: it says that neither  $a$  nor  $b$  ever occurs at a finite coordinate. For each  $a \in D$ ,

$$\{e : e(\omega) = a\} = \bigcup_{b > a} (C_{a,b} \cap H) \cup \bigcup_{b < a} (C_{a,b} \setminus H).$$

There is an analogous formula for  $e(\omega + 1) = a$ . All finite-coordinate observations are already lexicographically continuous. These countable unions therefore show that the enlarged sigma-algebra contains every product coordinate cylinder and hence the full-product Borel structure. Conversely, the lexicographic Borel structure is contained in the product Borel structure, and  $H$  is product-clopen. This proves equality. Finally  $H$  itself is not lexicographically Borel, since it splits every adjacent fiber, so an enrichment is necessary.  $\square$

*Remark 68.19* (Sigma-algebra inclusion is not topology inclusion). The full-product topology on  $P_{\omega+2}$  is Polish, but is not a refinement of its lexicographic topology. The two topologies are incomparable. The predicate  $H$  is product-clopen but not lexicographically Borel. In the other direction, the lexicographic ray  $(-\infty, p^+)$  is not product-open at  $p^-$ . A product neighborhood fixes only finitely many finite coordinates, along with possibly the terminal ones. At a later finite coordinate one can increase the label, retain the same omitted pair and terminal order, and complete the enumeration. The resulting point still belongs to that product neighborhood but lies above  $p^+$ . The proposition compares the generated sigma-algebras, not an inclusion of the two topologies.

### 68.5 Two different products of the native Borel structure

At two terminal labels a further distinction appears. For a topology on  $L$ , the Borel sigma-algebra  $\mathcal{B}(L \times L)$  of its topological product need not equal the sigma-algebra  $\mathcal{B}(L) \otimes \mathcal{B}(L)$  generated by Borel rectangles. The equality familiar from Polish spaces fails for the native split orders. Related split-interval product phenomena are classical [59, 419X(g)]; the following proof uses only the explicit Borel normal form above.

**Theorem 68.20** (Native comparison is not measurable for the rectangle product). *Let  $E$  be an uncountable dense suborder of  $\mathbb{R}$  without endpoints and let  $L = E \times_{\text{lex}} m$  for a finite  $m \geq 2$ . Then*

$$\{(x, y) \in L^2 : x < y\} \in \mathcal{B}(L \times L) \quad \text{but} \quad \{(x, y) \in L^2 : x < y\} \notin \mathcal{B}(L) \otimes \mathcal{B}(L).$$

*Consequently the two product sigma-algebras are distinct. This applies to  $P_{\omega+n}$  for every finite  $n \geq 2$ .*

*Proof.* Strict order is open in the product topology of any linearly ordered topological space. If  $x < z < y$ , the product of the rays below and above  $z$  is a suitable neighborhood of  $(x, y)$ . If  $x < y$  is an adjacent pair, use  $(-\infty, y) \times (x, \infty)$  instead. Thus the displayed relation belongs to  $\mathcal{B}(L \times L)$ .

First let  $m = 2$ . Every member of a sigma-algebra generated by a family uses only countably many members of that family: the sets having such a countable supporting subfamily themselves form a sigma-algebra containing all generators. Hence any member of  $\mathcal{B}(L) \otimes \mathcal{B}(L)$  is generated by countably many rectangles  $B_k \times C_k$ .

By Theorem 68.13, each  $B_k$  and  $C_k$  distinguishes the two points of only countably many fibers. Choose  $p \in E$  outside the union of all these exceptional base sets. Every generating rectangle has the same membership value at

$$((p, 0), (p, 1)) \quad \text{and} \quad ((p, 1), (p, 0)).$$

Equality of membership is preserved by complements and countable unions, so every set generated by these rectangles has the same value at those two ordered pairs. Strict order takes opposite values there. It cannot belong to the rectangle-generated product sigma-algebra.

For  $m > 2$ , the closed endpoint subspace  $S = E \times \{0, m-1\}$  is a double split. Its inclusion in  $L$  is continuous, hence Borel measurable. Pulling a rectangle-product measurable strict order on  $L$  back along the product of these inclusions would give a rectangle-product measurable strict order on  $S$ , contrary to the case just proved.  $\square$

The coordinate coding has none of this product ambiguity: it is Polish, and its strict lexicographic relation is  $\Delta_2^0$  by Proposition 68.11. Thus three separate objects occur naturally: the coordinate Borel structure, the native Borel structure, and the Borel structure of a native topological product. Their distinctions already matter for an elementary question such as whether comparison of two coded well-orders is a measurable operation.

### 68.6 What the countable analysis says about formalization

All the underlying carriers in this section are ordinary sets, and all arguments take place in ZFC. Nonetheless, the choice of coding and topology changes even the measurable sets. It is therefore insufficient to say that a formalization uses “the Borel structure on well-orders” without naming the topology.

A proof-assistant development can make the distinction concrete as follows. The set of bijections  $\alpha \rightarrow D$  is one underlying type. Use two tagged copies of it: one with the induced product topology

and one with the topology of the first-difference linear order. Prove the finite-tail order isomorphism before transporting any topological structure. The prefix-omission sets  $E_n$  should retain the prefix topology, not silently acquire the finer topology obtained by exposing the finite residual labels as extra coordinates. At  $n = 2$ , the coordinate event  $f(\omega) = a$  is a particularly useful negative regression test: it must be coordinate-clopen and fail to be native Borel. Identifying the two Borel structures would contradict an explicit mathematical theorem.

The comparison relation itself remains low in the coordinate Borel hierarchy for every countable ordinal length. This makes that code space useful for descriptive-set-theoretic questions even when the native order topology has a much larger family of open or Borel sets. The native topology is appropriate for order completion and cuts; the coordinate topology is appropriate for finite observations of enumerations. Both are rigorous interpretations, and the results above state exactly where they coincide and where they diverge.

### 68.7 Further questions made precise by these results

1. Are the dense base orders  $E_n$  and  $E_m$  isomorphic for distinct positive integers  $m, n$ ? The finite-tail orders themselves are pairwise nonisomorphic, but their finite-condensation quotients might forget the exact positive number of omitted labels. The common category and Borel bounds established here do not settle this question.
2. What is the exact difference-hierarchy or Wadge degree of  $E_n$  in the common real completion for  $n \geq 1$ ? The explicit difference of  $G_\delta$  sets is a sharper starting point than an unqualified third-level Borel bound.
3. Classify the ordered pairs  $(C, E_n)$  and their order automorphism groups. Any automorphism preserving  $E_0$  must preserve a comeager realized set; for positive  $n$  it preserves a dense meager realized set. Determine which features of the omitted-label decomposition can be recovered from the ordered pair alone.
4. For  $\alpha \geq \omega \cdot 2$ , characterize the native Borel sigma-algebra in terms of the first- $\omega$  prefix and the varying infinite residual orders. A low-complexity coordinate comparison relation is already available, but the finite split-pair normal form no longer describes the fibers.
5. Determine which infinite-length native order topologies are Baire, and where compatible complete metrics can exist. First countability holds at every countable length, whereas the coordinate topology is always Polish; these facts do not settle native metrizability beyond the finite-tail cases.
6. Find a functorial formalization of the two tagged topologies and their Borel sigma-algebras. The exact inclusion at one omitted label and failure of Borel measurability at two labels give concrete milestones before moving to generalized topologies at uncountable birthday stages.

## 69 Singular cardinalities: continuity, complete metrizability and non-Borel boundaries (sources 24 and 26)

[merge] Source 24's Section 8 and source 26's Sections 4–10. At a singular enumeration cardinal  $\mu$  a set of fewer than  $\mu$  positions can be unbounded, and controlling a bounded initial segment is no longer controlling a small set of coordinates. Source 24 finds the exact continuity criteria for the permutation operations and isolates the boundedness-preserving subgroup  $H_\mu$ ; source 26 answers source 24's question whether the minimal slice is completely metrizable when  $\text{cf } \mu = \omega < \mu$ : it never is, it is not even Borel in its completion, and no nonempty open subspace is completely metrizable

(theorem 69.18). Source 26 writes  $\kappa$  for  $|X|$  and  $\theta$  for the birthday cutoff; they are printed  $\mu$  and  $\kappa$  (part X). Source 28's further results on these groups are in section 70.

### 69.1 The permutation-group obstruction (source 24)

A fixed enumeration of the alphabet identifies  $\mathcal{W}_\mu(X)$  as a set with  $\text{Sym}(\mu)$ . At countable or regular cardinalities, it is tempting to import the usual topological permutation-group picture. At singular cardinalities that import is invalid unless an additional boundedness condition is checked.

**Definition 69.1** (Preservation of bounded sets). A function  $u : \mu \rightarrow \mu$  is *boundedness preserving* if

$$\forall \alpha < \mu \exists \beta < \mu \quad u[\alpha] \subseteq \beta.$$

Equivalently, it takes every bounded subset of  $\mu$  to a bounded subset. Here “bounded” refers to the ordinal order on  $\mu$ , not to cardinality alone.

On  $\text{Sym}(\mu)$  let  $L_\tau(\pi) = \tau \circ \pi$  and  $R_\sigma(\pi) = \pi \circ \sigma$ . All topology in the next theorem is the forward bounded-prefix topology.

**Theorem 69.2** (Exact continuity criteria). *For every infinite  $\mu$ :*

- (i) *Every left translation  $L_\tau$  is a homeomorphism.*
- (ii) *Right translation  $R_\sigma$  is continuous at every point if  $\sigma$  is boundedness preserving, and is continuous at no point otherwise.*
- (iii) *Inversion  $\pi \mapsto \pi^{-1}$  is continuous at a particular  $\pi$  exactly when  $\pi^{-1}$  is boundedness preserving.*

*Proof.* Left translation is a relabelling of output values. Equality of any forward restrictions is preserved in both directions, proving (i).

For (ii), to determine  $(\pi \circ \sigma) \upharpoonright \alpha$ , it suffices to determine  $\pi$  on  $\sigma[\alpha]$ . If that image is contained in a bounded initial segment  $\beta$ , agreement with  $\pi$  below  $\beta$  enforces the desired output agreement. Conversely, suppose  $\sigma[\alpha]$  is unbounded. Given any input neighborhood  $[\pi \upharpoonright \beta]$ , choose  $u \in \sigma[\alpha] \setminus \beta$  and a different  $v \in \mu \setminus \beta$ . Precompose  $\pi$  with the transposition of  $u, v$ . The new permutation agrees with  $\pi$  below  $\beta$ , but its composition with  $\sigma$  changes at the coordinate below  $\alpha$  whose image under  $\sigma$  is  $u$ . Thus no input neighborhood controls that output prefix, at any  $\pi$ .

For (iii), determining  $\pi^{-1}$  below  $\alpha$  is equivalent to fixing the positions  $\pi^{-1}[\alpha]$  of those labels. If this set is bounded, a sufficiently long forward restriction of  $\pi$  fixes all these preimages. If it is unbounded, given  $\beta < \mu$  choose  $u \in \pi^{-1}[\alpha] \setminus \beta$  and a different  $v \notin \beta$ . Again precompose with their transposition. The forward prefix below  $\beta$  is unchanged, but the inverse position of the label  $\pi(u) < \alpha$  changes from  $u$  to  $v$ . This is precisely failure of continuity at  $\pi$ .  $\square$

**Corollary 69.3** (Regular and singular cases diverge). *If  $\mu$  is regular, the forward topology makes  $\text{Sym}(\mu)$  a topological group. If  $\mu$  is singular, it does not: there are right translations discontinuous everywhere, and inversion has points of discontinuity.*

*Proof.* For regular  $\mu$ , every set of size less than  $\mu$  is bounded, so every permutation and its inverse preserve bounded sets. Joint continuity of multiplication follows directly: to control  $\pi' \circ \sigma'$  below  $\alpha$ , first require  $\sigma' \upharpoonright \alpha = \sigma \upharpoonright \alpha$  and then require  $\pi'$  to agree with  $\pi$  on a prefix containing the bounded set  $\sigma[\alpha]$ .

For singular  $\mu$ , let  $q = \text{cf } \mu < \mu$ . Choose  $A = \{a_\xi : \xi < q\}$  bounded in  $\mu$  and a disjoint cofinal set  $B = \{b_\xi : \xi < q\}$ . The involution  $\sigma$  exchanging each  $a_\xi$  with  $b_\xi$  and fixing everything else takes the bounded set  $A$  to the unbounded set  $B$ . It and its inverse fail the criterion in Theorem 69.2.  $\square$

This is not a pathology of surreal multiplication. It already occurs for permutations of a naked singular cardinal, before any numerical alphabet is supplied. Nor does it contradict Corollary 62.8: left translation is relabelling, whereas right translation changes the *positions* on which a bounded prefix depends.

### 69.1.1 The subgroup where the usual picture survives

Define

$$H_\mu = \{\pi \in \text{Sym}(\mu) : \pi \text{ and } \pi^{-1} \text{ preserve bounded sets}\}.$$

It is a subgroup: boundedness preservation is closed under composition, and the definition is symmetric under inversion. At a regular cardinal  $H_\mu = \text{Sym}(\mu)$ ; at a singular cardinal the involution above shows that it is proper.

For  $\alpha < \mu$  define two-sided entourages

$$E_\alpha^{\leftrightarrow} = \{(\pi, \sigma) : \pi \restriction \alpha = \sigma \restriction \alpha \text{ and } \pi^{-1} \restriction \alpha = \sigma^{-1} \restriction \alpha\}.$$

They define a Hausdorff uniformity on the full permutation set, independently of continuity of the group operations.

**Theorem 69.4** (Two-sided completeness and the bounded subgroup). *The full space  $\text{Sym}(\mu)$  is complete for the two-sided uniformity. On  $H_\mu$ , that uniformity induces the forward topology and is complete; with this topology  $H_\mu$  is a topological group. On all of  $\text{Sym}(\mu)$  the two uniform topologies agree when  $\mu$  is regular and differ when  $\mu$  is singular.*

*Proof.* A two-sided Cauchy filter determines coherent stabilized forward prefixes and coherent stabilized inverse prefixes. Their unions are injections  $h, g : \mu \rightarrow \mu$ . If  $h(\xi) = \eta$ , choose a filter member on which the forward value at  $\xi$  and the inverse value at  $\eta$  have both stabilized. Every permutation in that nonempty member witnesses  $g(\eta) = \xi$ . The converse is identical. Hence  $g = h^{-1}$  and  $h$  is a permutation. Stabilization on each initial segment gives convergence in the two-sided uniformity.

At a point  $\pi \in H_\mu$ , an inverse-prefix requirement below  $\alpha$  is implied by a forward requirement on a prefix bounding  $\pi^{-1}[\alpha]$ . Thus the two topologies agree on  $H_\mu$ . If the original Cauchy filter consists of elements of  $H_\mu$ , its limiting forward prefix  $h \restriction \alpha$  is a restriction of one such element and therefore has bounded image. The same holds for  $g \restriction \alpha$ . Thus  $h \in H_\mu$ , proving completeness of the subgroup. The proof of joint continuity and the inversion criterion above now apply inside the subgroup.

For regular  $\mu$ , all permutations belong to  $H_\mu$ , proving equality of the topologies on the full space. For singular  $\mu$ , inverse stabilization defines a two-sided neighborhood at the involution in Corollary 69.3 that contains no forward neighborhood there. The topology is strictly finer.  $\square$

**Corollary 69.5** (A complete metric on the bounded subgroup). *If  $\text{cf } \mu = \omega$ , the forward topology on  $H_\mu$  has a complete compatible ultrametric. Choose a countable cofinal sequence of ordinal lengths and take the maximum of the resulting forward and inverse first-disagreement ultrametrics.*

*Proof.* The cofinal sequence gives a countable base for the complete two-sided uniformity. On  $H_\mu$  that uniformity induces the forward topology by Theorem 69.4. Its ultrametric is therefore complete and compatible, exactly as in the countable permutation proof.  $\square$

*Remark 69.6* (A finer complete uniformity is not a universal group repair). The two-sided uniformity exists and is complete on the full singular permutation set, but this by itself does not make all group



operations continuous in its topology. For the unboundedness-preserving  $\sigma$  above, right translation is still discontinuous at the identity: at the identity, a sufficiently long fixed forward prefix also fixes that inverse prefix, while  $\sigma[\alpha]$  remains unbounded. The subgroup  $H_\mu$  is the clean topological-group statement proved here.

### 69.1.2 What is established when cofinality is countable

If  $\text{cf } \mu = \omega$ , Proposition 62.7 provides a compatible forward ultrametric, and Theorem 65.1 identifies its uniform completion with injections. When  $\mu = \omega$ , Theorem 67.2 supplies a different complete compatible metric. At an uncountable singular  $\mu$  of countable cofinality, Theorem 69.4 provides a complete *finer* uniform topology on the full permutation space, but not a complete metric for its original topology.

The earlier dense-open fusion theorem (source 17, theorem 30.5) proves that these forward spaces are Baire. Its proof extends a short prefix at stage  $n$  to meet the next dense open set and to cover a growing block of labels. This does not automatically give the point-preserving choices needed for a stronger topological completeness argument. Therefore neither forward incompleteness nor the failure of inversion continuity settles complete metrizability. The exact singular question is retained as its Question 13.1 (note 69.7).

*Note 69.7* ([merge]; source 24's singular question, answered (3 October 2026)). Source 24 retains the “exact singular question” as its Question 13.1: *for singular  $\mu > \omega$  with  $\text{cf } \mu = \omega$ , is the forward-prefix space  $\mathcal{W}_\mu(X)$  completely metrizable? Equivalently, can the dense permutation subspace of its metrizable injection completion be characterized as a  $G_\delta$  subset? Determine whether the answer depends on cardinal arithmetic.* Source 26 answers it negatively in ZFC, with no cardinal-arithmetic hypothesis: the permutation space is not even Borel in  $\mathcal{I}_\mu(X)$ , and no nonempty open subspace is completely metrizable (theorems 69.17 and 69.18 and corollary 69.19). Source 24's own results do not settle it, as it says; source 28's dense complete locus does not either (note 70.13).

## 69.2 The complete-metric subspace criterion (source 24, Appendix C)

[merge] Source 24's Appendix C, used in the singular question above and in source 26's classification: “For completeness, the characterization used in Question 13.1 can be proved without a separability hypothesis.”

**Proposition 69.8** (Complete metric subspaces and  $G_\delta$  sets). *Let  $B$  be a completely metrizable space and  $A \subseteq B$ . Then  $A$  has a complete compatible metric if and only if it is  $G_\delta$  in  $B$ .*

*Proof.* Fix a complete metric  $d$  on  $B$ . First suppose  $\rho$  is a complete compatible metric on  $A$ . For each positive integer  $n$ , cover  $A$  by ambient open sets  $U$  that meet  $A$ , have  $d$ -diameter less than  $2^{-n}$ , and have  $\rho$ -diameter less than  $2^{-n}$  on  $U \cap A$ . Such sets exist by compatibility of the two subspace topologies. Let  $O_n$  be their union.

If  $z \in \bigcap_n O_n$ , choose  $U_n \ni z$  from the  $n$ th cover and  $a_n \in U_n \cap A$ . Then  $a_n \rightarrow z$  in  $d$ . For fixed  $N$ , the ambient neighborhood  $U_N$  of  $z$  contains all sufficiently late  $a_n$ . Their pairwise  $\rho$ -distances are less than  $2^{-N}$ . Thus  $(a_n)$  is  $\rho$ -Cauchy and converges in  $A$  to some  $a$ . Compatibility gives convergence to  $a$  in  $d$  as well; hence  $z = a \in A$ . Therefore  $A = \bigcap_n O_n$ .

Conversely, suppose  $A = \bigcap_{n \geq 1} O_n$  with each  $O_n$  open in  $B$ . Put  $F_n = B \setminus O_n$  and, on  $A$ , set  $t_n(x) = 1/d(x, F_n)$ , with  $t_n = 0$  if  $F_n$  is empty. The denominator is positive because  $x \in O_n$ . Define

$$\rho(x, y) = d(x, y) + \sum_{n \geq 1} 2^{-n} \min\{1, |t_n(x) - t_n(y)|\}.$$

The triangle inequality follows term by term, and positivity follows from  $d$ . Continuity of every  $t_n$ , followed by control of the summable tail, shows that  $\rho$  induces the subspace topology.

A  $\rho$ -Cauchy sequence is  $d$ -Cauchy and has a limit  $z \in B$ . For each fixed  $n$ , its  $t_n$ -values are Cauchy in the ordinary real metric: truncated-distance Cauchyness is ordinary Cauchyness once the required error is less than 1. They therefore have a finite limit. If  $z \in F_n$ , the distances to  $F_n$  would tend to zero and their reciprocals would diverge, a contradiction. Hence  $z \in A$ . Continuity of the  $t_n$  and the same summable-tail estimate give convergence in  $\rho$ , proving completeness.  $\square$

*Note 69.9* (Source 26, Appendix A, Theorem A.1, and source 28, Appendix B, Proposition B.1). Both sources prove [proposition 69.8](#) again, by the same two constructions (a  $G_\delta$  from covers of shrinking diameter in both metrics; the complete metric  $d + \sum_n 2^{-n} \min\{1, |t_n(x) - t_n(y)|\}$  with  $t_n = 1/d(\cdot, F_n)$ ), “without assuming separability”. Source 26: “The earlier manuscript also records this criterion; the proof is repeated to make the logical dependence visible ... Only the necessity direction is required to turn Theorem 6.2 into the main negative conclusion. Its proof does not use completeness of the ambient metric; it uses the complete compatible metric hypothesized on the subspace. The converse supplies the positive conclusions for  $\mu = \omega$  and for  $H_\mu$ .” Source 28 uses it for its dense locus ([theorem 70.11](#)) and its own Question 1. Source 29 cites it in the form “an  $F_\sigma$  subset of  $\mathbb{R}$  is  $\sigma$ -compact”; the criterion is classical (Alexandrov).

### 69.3 Local character, the countable exception and the Baire property (source 26)

*Note 69.10* (Source 26, Theorem 4.3, “Local character”). *At every point of  $\mathcal{W}_\mu(X)$  and  $\mathcal{I}_\mu(X)$ , the least cardinality of a neighborhood base is  $\text{cf } \mu$ . In particular, either space is metrizable if and only if  $\text{cf } \mu = \omega$ .* This is [proposition 62.7](#), by the same argument; source 26 adds that the cofinal-length ultrametric is complete on  $\mathcal{I}_\mu(X)$  by [theorem 65.1](#), which its classification uses.

[merge] Source 26’s proof of the countable case is a route through the  $G_\delta$  criterion; its second paragraph is source 24’s metric  $D$  ([theorem 67.2](#)):

**Proposition 69.11** (The countable exception). *For  $\mu = \omega$ , the space  $\mathcal{W}_\omega(X)$  is completely metrizable.*

*Proof.* Its injection completion is completely metrizable, and [note 65.5](#) expresses the permutation subspace as a countable intersection of open sets. The complete-metric subspace criterion, proved in [note 69.9](#), applies.

An explicit alternative is to fix a bijection  $X \cong \omega$  and combine the forward first-disagreement metric with the corresponding metric on inverses. Stabilization of both a permutation and its inverse produces mutually inverse limiting injections. At  $\omega$ , a finite inverse-prefix requirement is already implied, at a given permutation, by a sufficiently long finite forward prefix. Thus the combined complete metric has the original topology.  $\square$

The last compatibility argument fails for general singular permutations: a bounded set of labels can have an unbounded inverse image. A finer complete metric on forward and inverse prefixes is not evidence that the forward topology itself is completely metrizable.

[merge] Source 26’s Baire theorem is the fusion theorem [theorem 30.5](#) (source 17, for  $X = \mathbf{No}_{<\kappa}$ ) for an arbitrary alphabet of every infinite cardinality, by the same fusion with coverage blocks:

### 69.3.1 Baire category survives the failure of completeness

**Theorem 69.12** (Baire property). *For every infinite cardinal  $\mu$ ,  $\mathcal{W}_\mu(X)$  is a Baire space in its bounded-prefix topology.*

*Proof.* Let  $(O_n)_{n < \omega}$  be dense open subsets and start inside an arbitrary cylinder  $N_p$ . If  $\text{cf } \mu > \omega$ , successively extend the prefix so that its cylinder lies inside  $O_n$ . The union of the countably many prefix domains remains bounded in  $\mu$ , and its injective union extends to a permutation. This extension belongs to every chosen cylinder and hence to every  $O_n$ .

If  $\text{cf } \mu = \omega$ , choose increasing sets  $Y_n \subseteq X$  with  $|Y_n| < \mu$  and union  $X$ , and cofinal domain lengths  $\gamma_n < \mu$ . At stage  $n$ , first refine the current cylinder into  $O_n$ , using density and openness. Then append all labels of  $Y_n$  not already used, in some order, and lengthen the prefix to length at least  $\gamma_n$ . Both operations still use fewer than  $\mu$  coordinates: a finite sum of ordinals whose cardinalities are below the initial cardinal  $\mu$  remains below  $\mu$ . [Note 62.9](#) supplies any additional unused labels required. The new cylinder stays inside  $O_n$ .

The union of the resulting prefixes has domain  $\mu$ , is injective, and covers every  $Y_n$ . It is a permutation in the original cylinder and in every  $O_n$ . The proof includes  $\mu = \omega$ , using finite  $Y_n$  and finite lengths.  $\square$

This fusion argument controls both domain growth and eventual coverage. It is not a proof of complete metrizability: it produces one point meeting a prescribed countable family of dense open sets, rather than a complete compatible metric or a  $G_\delta$  description inside the completion.

## 69.4 Compact coding of omitted labels (source 26)

[merge] Source 26's Section 5, printed whole: the central construction of the singular theory.

This section supplies the central construction. It only uses the cardinality of the alphabet and the countable cofinality of the domain. Surreal arithmetic is not involved.

**Theorem 69.13** (Compact coding of omitted labels). *Suppose  $\mu > \omega$ ,  $\text{cf } \mu = \omega$ , and  $|X| = \mu$ . For every  $A \subseteq 2^\omega$  with  $|A| < \mu$ , there exist distinct labels  $(y_a)_{a \in A}$  and a continuous topological embedding*

$$F : 2^\omega \longrightarrow \mathcal{I}_\mu(X)$$

*with compact, closed image such that*

$$X \setminus \text{ran } F(z) = \begin{cases} \{y_z\}, & z \in A, \\ \emptyset, & z \notin A. \end{cases} \quad (25)$$

*In particular,*

$$F^{-1}(\mathcal{W}_\mu(X)) = 2^\omega \setminus A. \quad (26)$$

*If  $X$  is linearly ordered,  $F$  can also be chosen as an order embedding for the binary and  $X$ -valued lexicographic orders.*

*Proof. Step 1: large finite-stage reservoirs.* Because  $\mu$  is an uncountable cardinal of countable cofinality, it is a limit cardinal. Choose strictly increasing infinite cardinals

$$\max\{|A|, \aleph_0\} < \lambda_0 < \lambda_1 < \cdots < \mu, \quad \sup_{n < \omega} \lambda_n = \mu.$$

Put  $\lambda_{-1} = 0$  as an endpoint convention and define consecutive domain blocks

$$D_n = [\lambda_{n-1}, \lambda_n), \quad |D_n| = \lambda_n.$$

Every bounded initial segment of  $\mu$  meets only finitely many of these blocks.

Choose a disjoint partition

$$X = Y \sqcup \bigsqcup_{n < \omega} R_n, \quad Y = \{y_a : a \in A\}, \quad |R_n| = \lambda_n.$$

This is possible because removing fewer than  $\mu$  labels leaves cardinality  $\mu$ , and  $\sum_n \lambda_n = \mu$ .

**Step 2: release a label at its first mismatch.** For  $t \in 2^{n+1}$ , define

$$A_t = \{a \in A : a \upharpoonright n = t \upharpoonright n \text{ and } a(n) \neq t(n)\}. \quad (27)$$

Thus  $A_t$  consists exactly of the branches of  $A$  whose first disagreement with  $t$  occurs at coordinate  $n$ . For any infinite branch  $z$ , the sets  $A_{z \upharpoonright (n+1)}$  are pairwise disjoint and

$$\bigcup_{n < \omega} A_{z \upharpoonright (n+1)} = A \setminus \{z\}. \quad (28)$$

When  $z \notin A$ , the right side is just  $A$ . Equation (28) is the entire logical mechanism of the code: distinct infinite binary sequences have a unique finite first disagreement.

**Step 3: fill each block and mark its bit.** For every  $n$  choose distinct marker labels  $r_n^0, r_n^1 \in R_n$ . If  $X$  is ordered, choose them with  $r_n^0 < r_n^1$ . Let  $d_n$  and  $d_n + 1$  be the first two positions of  $D_n$ . For each  $t \in 2^{n+1}$  choose a bijection

$$\vartheta_t : D_n \longrightarrow R_n \sqcup \{y_a : a \in A_t\} \quad (29)$$

subject to

$$\vartheta_t(d_n) = r_n^{t(n)}, \quad \vartheta_t(d_n + 1) = r_n^{1-t(n)}.$$

The remaining domain and target both have cardinality  $\lambda_n$ :  $\lambda_n$  is infinite and absorbs the smaller released set as well as the removal of two markers. Hence these bijections exist. Only set-sized choice is used.

**Step 4: assemble the injection.** For  $z \in 2^\omega$ , define

$$F(z) \upharpoonright D_n = \vartheta_{z \upharpoonright (n+1)}. \quad (30)$$

Within a block the map is injective. Distinct blocks use disjoint reservoirs  $R_n$ , and by (28) each distinguished label  $y_a$  is released into at most one block. Therefore  $F(z)$  is injective on all of  $\mu$ . Every reservoir is exhausted, and the union of the released labels is precisely  $Y \setminus \{y_z\}$  when  $z \in A$ , or all of  $Y$  otherwise. This proves (25) and (26).

**Step 5: topology and order.** A prescribed output prefix of length  $\alpha < \mu$  is contained in finitely many blocks. It is therefore determined by finitely many input bits. This proves continuity of  $F$ . Its markers recover every bit: the value at  $d_n$  is  $r_n^{z(n)}$ . Thus  $F$  is injective. A continuous injection from compact Cantor space into the Hausdorff space  $\mathcal{I}_\mu(X)$  is a homeomorphism onto a compact closed image.

Finally, if  $z < w$  first differ at bit  $n$ , they have the same output on every block before  $D_n$ . At the very first position  $d_n$  of that block, their values are  $r_n^0 < r_n^1$ . This is the first output disagreement, proving  $F(z) <_{\text{lex}} F(w)$ .  $\square$

### 69.4.1 Why countable cofinality is the effective resource

The construction hides arbitrarily many labels in an early reservoir  $Y$ , but schedules their release according to a finite first-disagreement test. To make the resulting map continuous, every bounded output prefix must see only finitely many stages. The block decomposition by a countable cofinal sequence guarantees precisely this. Each stage has a reservoir large enough to absorb all currently released labels without changing its domain cardinality.

The markers have a separate role. Release scheduling alone establishes the desired range equation but need not separate all input branches. Two permanent marker positions in each infinite block make the code injective and, with ordered markers, order-preserving. Finite tests check the release partition and first-disagreement markers; they cannot check the infinite cardinal absorption used in (29).

*Remark 69.14* (A particularly small defect). Every point in the compact image of  $F$  is either exhaustive or omits exactly one label. Thus the complexity established below does not require injections with large or complicated omitted ranges. The zero-versus-one omission distinction already suffices.

*Remark 69.15* (The strict cardinal bound). The hypothesis  $|A| < \mu$  is used to choose a first reservoir size larger than  $|A|$ . The theorem does not assert the same construction for  $|A| = \mu$ . Likewise, a cofinality- $\omega$  decomposition is not replaced without argument by finitely many stages or by stages indexed by a larger regular cardinal. These are genuine boundaries of the present proof.

## 69.5 Non-Borelness and exact complete metrizability (source 26)

[merge] Source 26's Section 6, printed whole. Its classification answers source 24's Question 13.1 (note 69.7) and source 28's Question 1 (note 70.13); it also answers, for the minimal slices, source 17's question on complete metrizability beyond the proved fusion length (research question 113.72), and source 22's question on descriptive topology when cofinalities are countable (section 72). "The earlier *Prefix Completions* manuscript" is source 24.

**Lemma 69.16** (A small non-Borel set of branches). *In ZFC there is a non-Borel set  $A \subseteq 2^\omega$  of cardinality  $\aleph_1$ .*

*Proof.* We use the classical theorem that every uncountable Borel subset of Cantor space contains a nonempty perfect subset, and every such perfect subset has cardinality  $\mathfrak{c}$  [57].

If  $\mathfrak{c} > \aleph_1$ , take any subset  $A$  of cardinality  $\aleph_1$ . It is uncountable and cannot contain a perfect subset, so it is not Borel.

If  $\mathfrak{c} = \aleph_1$ , enumerate all nonempty perfect subsets of Cantor space as  $(K_\xi)_{\xi < \omega_1}$ . There are exactly  $\mathfrak{c}$  such sets: at most that many because closed subsets are coded by countable trees, and at least that many, for example by fixing one infinite subsequence of coordinates while varying its binary values. Recursively choose two distinct new points  $a_\xi, b_\xi \in K_\xi$ , avoiding all earlier chosen points. Each stage forbids only countably many points, whereas  $K_\xi$  has size  $\aleph_1$ . Put  $A = \{a_\xi : \xi < \omega_1\}$ . No  $b_\xi$  belongs to  $A$ , by the continued avoidance requirement. Thus every perfect set meets the complement of  $A$ , and  $A$  contains no perfect subset. Again it is uncountable and not Borel.  $\square$

**Theorem 69.17** (The singular non-Borel boundary). *Let  $\mu > \omega$  have  $\text{cf } \mu = \omega$ , and let  $|X| = \mu$ . Then  $\mathcal{W}_\mu(X)$  is not an ordinary Borel subset of  $\mathcal{I}_\mu(X)$  in their bounded-prefix topology. This is already witnessed on a compact subspace consisting of injections with at most one omitted label.*

*Proof.* An uncountable cardinal of countable cofinality is strictly larger than  $\aleph_1$ . Choose  $A$  from Lemma 69.16 and apply Theorem 69.13. If  $\mathcal{W}_\mu(X)$  were Borel in  $\mathcal{I}_\mu(X)$ , continuity of  $F$  would imply that

$$2^\omega \setminus A = F^{-1}(\mathcal{W}_\mu(X))$$

is Borel in Cantor space. Taking its complement would make  $A$  Borel, a contradiction. The last assertion is (25).  $\square$

No separability assumption on  $\mathcal{I}_\mu(X)$  is being smuggled into this argument. Continuous inverse images preserve the sigma-algebra generated by open sets in arbitrary topological spaces. The perfect-set theorem is used only inside the ordinary separable Cantor domain.

**Theorem 69.18** (Complete-metrizability classification). *For every infinite cardinal  $\mu$  and every set  $X$  with  $|X| = \mu$ ,*

$$\mathcal{W}_\mu(X) \text{ is completely metrizable in the prefix topology} \iff \mu = \omega.$$

*For  $\mu > \omega$  of countable cofinality, no nonempty open subspace of  $\mathcal{W}_\mu(X)$  is completely metrizable.*

*Proof.* The countable case is Proposition 69.11. If  $\text{cf } \mu > \omega$ , note 69.10 shows that the space is not even first countable, hence is not metrizable.

It remains to consider uncountable  $\mu$  of countable cofinality. The injection space has a complete compatible metric by note 69.10. A completely metrizable subspace of a metrizable space must be  $G_\delta$  there; note 69.9 gives a proof that does not require separability. Complete metrizability of  $\mathcal{W}_\mu(X)$  would therefore contradict Theorem 69.17.

Finally, any nonempty open subspace contains a cylinder. The cylinder is clopen in the full permutation space and is homeomorphic to that space by Proposition 62.10. If the open subspace admitted a compatible complete metric, its closed cylinder would also be completely metrizable, which is impossible.  $\square$

**Corollary 69.19** (Answer to the earlier question). *Question 13.1 of the earlier Prefix Completions manuscript (source 24; note 69.7) has a negative answer in ZFC. The answer does not depend on the value of the continuum or on additional cardinal-arithmetic hypotheses.*

### 69.5.1 Why several plausible positive arguments fail

First, the fact that all exhaustion requirements  $U_x$  are open dense gives a  $\mu$ -indexed intersection, not necessarily a countable one. Theorem 69.17 shows that there is no replacement by a countable Borel construction at all. Second, the forward metric being incomplete is weaker than the conclusion: another compatible metric can repair that problem at  $\omega$ , but not at the uncountable singular lengths considered here. Third, imposing inverse-prefix stabilization gives a complete uniformity, but in general changes the topology. Fourth, the Baire fusion proof of Theorem 69.12 does not imply that every closed subspace is Baire or that the original space is completely metrizable.

The precise hierarchy is therefore useful:

$$\text{complete compatible metric} \implies \text{Baire}, \quad \text{but the converse fails here even locally.}$$

For these spaces, “the completion is well understood” and “the original topology is completely metrizable” are sharply different assertions.



## 69.6 Closed universality and ordered Cantor traces (source 26)

**Theorem 69.20** (Every Cantor subset as a closed trace). *Suppose  $\text{cf } \mu = \omega$  and  $\mu > \mathfrak{c}$ . For every  $B \subseteq 2^\omega$ , there is a closed subspace of  $\mathcal{W}_\mu(X)$  homeomorphic to  $B$ . If  $X$  is linearly ordered, the homeomorphism can simultaneously be an order isomorphism from  $B$  with its inherited binary lexicographic order to the image with its inherited enumeration order.*

*Proof.* Set  $A = 2^\omega \setminus B$ . Its cardinality is at most  $\mathfrak{c} < \mu$ . Apply Theorem 69.13 and put  $K = F[2^\omega]$ . Then  $K$  is closed in the injection space and

$$K \cap \mathcal{W}_\mu(X) = F[B].$$

This is closed relative to  $\mathcal{W}_\mu(X)$ , and the restriction of the embedding  $F$  is a homeomorphism from  $B$  to  $F[B]$ . The ordered-marker version of  $F$  gives the order statement.  $\square$

The theorem does not say that  $F[B]$  is closed in  $K$  or in the ambient injection space. Such a claim would be false for a nonclosed  $B$ . It is closed in the *permutation subspace*, precisely because that subspace meets the compact family in the prescribed trace.

**Corollary 69.21** (Closed topological universality). *If  $\text{cf } \mu = \omega$  and  $\mu > \mathfrak{c}$ , every zero-dimensional separable metrizable space is homeomorphic to a closed subspace of  $\mathcal{W}_\mu(X)$ .*

*Proof.* A zero-dimensional separable metrizable space has a countable clopen base  $(V_n)_{n < \omega}$ . Indeed, take a countable ordinary base, refine each of its members by clopen neighborhoods, and use Lindelöfness to extract countably many such refinements. The characteristic map

$$x \mapsto (\mathbf{1}_{V_n}(x))_{n < \omega}$$

is a topological embedding into  $2^\omega$ : the base separates points and recovers the topology. Apply Theorem 69.20 to its image.  $\square$

**Corollary 69.22** (Closed rational subspaces). *Under the hypotheses of Corollary 69.21, the permutation space has a closed subspace homeomorphic to  $\mathbb{Q}$ . In particular it is not hereditarily Baire.*

*Proof.* The rational line is a zero-dimensional separable metrizable space: intervals with irrational endpoints give a countable clopen base after choosing a countable dense set of irrational endpoints. Apply the corollary. The space  $\mathbb{Q}$  is a countable union of nowhere dense singletons in itself and has no isolated points, so it is not Baire.  $\square$

This supplies another obstruction to complete metrizability when  $\mu > \mathfrak{c}$ , since closed subspaces of completely metrizable spaces are completely metrizable and hence Baire. The non-Borel proof is stronger in range: it also covers singular cardinals  $\mu \leq \mathfrak{c}$ . It is therefore important not to replace Theorem 69.17 with a closed-rational argument whose hypothesis is stronger.

### 69.6.1 How much of the order is retained

The ordered-marker construction gives a true order embedding, not merely an injection continuous for an unrelated topology. When the alphabet is thick, [note 62.9](#) identifies the ambient topology with the lexicographic order topology, so the closedness statements concern that actual ordered space. A subset is always being given its inherited topology for the topological assertion and its restricted order for the order assertion. The order topology intrinsic to an arbitrary suborder need



not be silently identified with every ambient subspace topology; no such general identification is required for our proof.

These results distinguish order-theoretic largeness from completeness. A linear order can have strong embedding properties, can contain arbitrary closed topological traces, and can still fail complete metrizability on every nonempty open piece. Mutual order embeddability with a familiar lexicographic cube would not prevent this: equimorphism preserves neither the missing-cut pattern nor all topological completeness properties.

### 69.7 Bounded images and bounded inverses (source 26)

[merge] Source 26's Section 8, printed whole. Its  $\mathcal{I}_\mu^b$  (its  $\mathcal{B}^+$ ) is source 28's  $\mathcal{I}_\mu^b$ , and  $H_\mu$  is source 24's boundedness-preserving subgroup. Part (ii) re-derives source 24's complete metric on  $H_\mu$  (the corollary to [theorem 69.4](#)) by a  $G_\delta$  route; source 28 proves the closure and completion statements for every infinite  $\mu$  ([theorems 70.10](#) and [70.26](#)) and the dense inverse locus ([theorem 70.11](#)).

Fix a bijection  $X \cong \mu$  so that boundedness of labels is meaningful. A subset of  $\mu$  is bounded when it is contained in some  $\beta < \mu$ . Define

$$\mathcal{I}_\mu^b = \{f \in \mathcal{I}_\mu(\mu) : f[\alpha] \text{ is bounded for every } \alpha < \mu\}$$

and

$$H_\mu = \{\pi \in \text{Sym}(\mu) : \pi[\alpha] \text{ and } \pi^{-1}[\alpha] \text{ are bounded for every } \alpha < \mu\}.$$

The boundedness-preserving subgroup  $H_\mu$  and its compatible two-sided completeness occur in the earlier manuscript (source 24, [theorem 69.4](#)). The following ambient description and coding refinement separate the two boundedness requirements directly.

**Theorem 69.23** (The bounded-image completion and its complete subgroup). *Suppose  $\mu > \omega$  and cf  $\mu = \omega$ .*

- (i)  $\mathcal{I}_\mu^b$  is closed in  $\mathcal{I}_\mu(\mu)$ , hence completely metrizable.
- (ii)  $H_\mu$  is dense and  $G_\delta$  in  $\mathcal{I}_\mu^b$ , hence completely metrizable in the forward-prefix topology.
- (iii)  $\mathcal{I}_\mu^b \cap \mathcal{W}_\mu(\mu)$  is not Borel in  $\mathcal{I}_\mu^b$  and is not completely metrizable.

*Proof.* A failure of bounded-image preservation is witnessed by a single prefix  $f \upharpoonright \alpha$  whose range is unbounded. Every injection in that cylinder has the same failure. The complement of  $\mathcal{I}_\mu^b$  is therefore open, proving (i).

Fix cofinal lengths  $\gamma_n < \mu$ . For each  $n$ , let

$$V_n = \{f \in \mathcal{I}_\mu^b : \exists \beta < \mu \ \gamma_n \subseteq \text{ran}(f \upharpoonright \beta)\}.$$

This is open: a witnessing prefix continues to witness membership throughout its cylinder. We claim

$$H_\mu = \bigcap_{n < \omega} V_n. \tag{31}$$

A member of the right side covers every label and is thus a permutation. For any bounded label set, choose  $\gamma_n$  containing it; its inverse image lies in a witnessing  $\beta$ , so the inverse preserves boundedness. The forward condition is built into  $\mathcal{I}_\mu^b$ . Conversely, a member of  $H_\mu$  has bounded inverse image of each  $\gamma_n$ , giving the required witness. This proves (31).

To see density, start with a prefix  $p = f \upharpoonright \alpha$  of a bounded-image injection. Both its domain and range are bounded. Choose an infinite initial cardinal  $\rho < \mu$  strictly above their bounds. The

complements of  $\text{dom } p$  and  $\text{ran } p$  inside  $\rho$  both have size  $\rho$ . Complete  $p$  to a permutation of  $\rho$  and extend by the identity outside  $\rho$ . This permutation belongs to  $H_\mu$  and extends  $p$ . Part (ii) follows from [note 69.9](#).

For (iii), repeat the proof of [Theorem 69.13](#) with the distinguished labels  $Y$  placed in an initial segment  $\eta = |A| < \lambda_0$ . Take

$$R_0 = [\eta, \lambda_0), \quad R_n = [\lambda_{n-1}, \lambda_n) \quad (n \geq 1).$$

Each block target is contained in  $\lambda_n$ , so

$$F(z)[\lambda_n] \subseteq \lambda_n \quad (n < \omega).$$

Consequently the entire compact coded family lies in  $\mathcal{I}_\mu^b$ . Choosing the non-Borel  $A$  of [Lemma 69.16](#) proves (iii) by the same continuous-preimage argument, now in this closed ambient space. Noncomplete metrizability follows from the  $G_\delta$  criterion.  $\square$

**Proposition 69.24** (What the inverse condition sees on the code). *For the localized construction in the proof of [Theorem 69.23](#),*

$$F^{-1}(H_\mu) = 2^\omega \setminus \overline{A}, \quad F^{-1}(\mathcal{I}_\mu^b \cap \mathcal{W}_\mu(\mu)) = 2^\omega \setminus A,$$

where  $\overline{A}$  is the ordinary Cantor-space closure.

*Proof.* The second equation is already known. If  $z \notin \overline{A}$ , some finite prefix  $z \upharpoonright N$  has no extension in  $A$ . Every distinguished label is then released in a block before  $N$ . The inverse images of  $Y$  are bounded. The inverse images of the finitely many reservoir blocks intersecting any prescribed bounded label segment are bounded as well. Thus  $F(z)$  is exhaustive and has bounded inverse images.

If  $z \in A$ , the code is not exhaustive. If  $z \in \overline{A} \setminus A$ , then for every  $N$  some  $a \in A$  agrees with  $z$  on its first  $N$  bits. This  $a$  is distinct from  $z$  and is released at a first disagreement at stage at least  $N$ . The preimages of the bounded set  $Y$  therefore meet arbitrarily late domain blocks and are unbounded in  $\mu$ . Hence  $F(z) \notin H_\mu$ .  $\square$

This is a useful conceptual refinement. On the coded family, surjectivity tests membership outside  $A$ , whereas bounded inverse behavior tests membership outside its closure. The forward condition alone does not remove the descriptive complexity; the inverse condition removes exactly the accumulation-branch phenomenon responsible for delayed coverage of a bounded reservoir.

The proof also explains the limit of the group analogy. At a regular cardinal every set of size less than the cardinal is bounded, so every permutation preserves boundedness in both directions. At a singular cardinal a permutation can send a bounded cofinality-sized set onto an unbounded set. Consequently the full permutation space with forward-prefix topology is not automatically a topological group. Nor does merely naming a two-sided uniformity prove continuity of every translation on the full singular permutation set. The subgroup  $H_\mu$  is the precise safe statement used here.

## 69.8 The minimal surreal slices (source 26)

[merge] Source 26's Section 9, printed whole. Its birthday cutoff  $\theta$  is printed  $\kappa$ , so that  $\mu = 2^{<\kappa}$  as elsewhere in this report. Its [lemma 69.25](#) is [lemma 29.2](#) with a different sign trunk; its  $\beth_\omega$  example extends this report's  $\beth_\omega$  examples ([example 30.21](#), [theorem 8.3](#)) to the topology.

We use the ordinal-length sign model of surreal numbers [12]. A surreal  $x$  is a function from an ordinal  $\text{bd}(x)$  to  $\{-, +\}$ . Numerical comparison uses the first disagreement after extending each sign function by the termination value 0, with  $- < 0 < +$ . The ordinal  $\text{bd}(x)$  is its birthday. The model separates numerical order from simplicity, which is proper prefix extension.

For an infinite initial cardinal  $\kappa$ , set

$$\mathbf{No}_{<\kappa} = \{x \in \mathbf{No} : \text{bd}(x) < \kappa\}, \quad \mu = 2^{<\kappa} = \sup_{\alpha < \kappa} 2^{|\alpha|}.$$

This is a set of labels. It is not the whole surreal class and is not, without additional closure hypotheses, being treated as a subfield.

**Lemma 69.25** (Cardinality and thickness). *For every infinite cardinal  $\kappa$ ,*

$$|\mathbf{No}_{<\kappa}| = 2^{<\kappa},$$

*and every nonempty numerical interval and each ray in  $\mathbf{No}_{<\kappa}$  has this same cardinality.*

*Proof.* There are  $2^{|\alpha|}$  sign sequences of each birthday  $\alpha < \kappa$ . Their union has cardinality at least the supremum  $\mu$ . Also  $\kappa \leq \mu$ : successor cardinal bounds below a limit cardinal, or Cantor's inequality at the predecessor cardinal in the successor case, give this, and the case  $\kappa = \omega$  is immediate. Thus the union has size at most  $\kappa \cdot \mu = \mu$ .

For thickness, we construct a sign trunk all of whose extensions lie between given  $x < y$ . Let  $\delta$  be their first disagreement, with termination included. If  $\delta < \text{bd}(x)$ , take  $t = x \frown \langle + \rangle$ . Every extension of  $t$  is above  $x$ , by the plus at position  $\text{bd}(x)$ , and below  $y$ , by the earlier disagreement at  $\delta$ .

The remaining case is  $\delta = \text{bd}(x)$ : the lower word terminates and the upper word has a plus there. Start with the common prefix followed by that plus, follow the subsequent minus signs of  $y$  until its first later plus or its end, and put a minus at that position. Call the resulting trunk  $t$ . It lies above  $x$  by its plus at  $\delta$  and below  $y$  by the final minus, compared with either a plus or termination. Both comparisons are witnessed strictly before the end of  $t$ , so they persist through every extension. The length of  $t$  is at most one more than one of the original birthdays, hence is below  $\kappa$ .

In either case, if the trunk length is  $\beta < \kappa$ , every sign tail of length  $\alpha < \kappa$  gives an element of the interval, since  $\beta + \alpha < \kappa$ . These extensions are distinct and give at least  $2^{|\alpha|}$  elements for each  $\alpha$ . Taking the supremum yields  $\mu$ , and the opposite bound is the cardinality of the whole cutoff.

For a ray, choose an ordinal sign sequence above the given birthday, or its all-minus counterpart below it, still with birthday below  $\kappa$ . An interval between that point and the original point lies in the ray. The interval result supplies the same cardinality.  $\square$

*Remark 69.26* (No hidden regularity). The bound  $\beta + \alpha < \kappa$  uses that  $\kappa$  is an infinite initial cardinal and that only two ordinals are being added. It does not require  $\kappa$  to be regular. This is why the thickness argument remains valid at the singular cutoff central to source 26.

**Theorem 69.27** (Topological classification of the minimal surreal slices). *Let  $\kappa$  be an infinite cardinal, let  $\mu = 2^{<\kappa}$ , and compare the bijections  $\mu \rightarrow \mathbf{No}_{<\kappa}$  lexicographically by numerical surreal labels. Their order topology is the bounded-prefix topology. In that topology,*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \text{ is metrizable} \iff \text{cf}(2^{<\kappa}) = \omega,$$

*and*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \text{ is completely metrizable} \iff \kappa = \omega.$$

*If  $\mu > \omega$  and  $\text{cf } \mu = \omega$ , it is non-Borel in its injection completion and has no nonempty completely metrizable open subspace.*

*Proof.* Lemma 69.25 supplies exactly the thickness interface needed by note 62.9. Apply note 69.10 and Theorems 69.17 and 69.18 with cardinal parameter  $\mu$ . For  $\kappa = \omega$ , the union of all finite sign levels is countably infinite. For  $\kappa > \omega$ , the level of birthday  $\omega$  already contains  $2^{\aleph_0}$  signs, so  $\mu > \omega$ . Thus  $\mu = \omega$  holds exactly when  $\kappa = \omega$ .  $\square$

### 69.8.1 Three instructive cutoffs

At  $\kappa = \omega$ , the labels are finite sign sequences, equivalently the finite-birthday surreal numbers. The minimal enumeration space is completely metrizable; the forward metric still has nonexhaustive injections as completion points, but the topology admits a different complete metric. The earlier manuscript gives a more detailed Baire-space model and a distinction from the order type of the irrational line (source 24, corollary 67.5).

At  $\kappa = \omega_1$ , one has  $\mu = 2^{\aleph_0} = \mathfrak{c}$ . König's cofinality inequality gives  $\text{cf}(2^{\aleph_0}) > \omega$ , so the minimal enumeration space is not even first countable. Indeed, countable cofinality of  $\mathfrak{c}$  would yield a sequence of smaller cardinals whose successor product is greater than  $\mathfrak{c}$  by König's theorem, but at most  $\mathfrak{c}^\omega = \mathfrak{c}$ , a contradiction. This shows why “the labels contain real-like objects” does not make the space metrizable.

At

$$\kappa = \beth_\omega = \sup_{n < \omega} \beth_n, \quad \beth_0 = \aleph_0, \quad \beth_{n+1} = 2^{\beth_n},$$

we have a singular strong limit cardinal of countable cofinality. For each  $\nu < \kappa$ , choose  $n$  with  $\nu \leq \beth_n$ ; then  $2^\nu \leq \beth_{n+1} < \kappa$ . Hence  $2^{<\kappa} = \kappa$ , and  $\kappa > \mathfrak{c}$  because  $\beth_2 > \beth_1 = \mathfrak{c}$ .

**Corollary 69.28** (The unconditional  $\beth_\omega$  example). *The lexicographic minimal enumeration order*

$$L = \mathcal{W}_{\beth_\omega}(\mathbf{No}_{<\beth_\omega})$$

*has all of the following properties in ZFC: it is a dense linear order without endpoints; its order topology is zero-dimensional and metrizable; it is Baire; no nonempty open subspace is completely metrizable; it is non-Borel in its completely metrizable injection completion; and it is closed-universal for zero-dimensional separable metrizable spaces. Every subset of binary Cantor space can be realized as a closed trace by an order-preserving coding.*

*Proof.* Combine Lemma 69.25 with note 62.9, Theorems 69.12, 69.17, 69.18, and 69.20, and Corollary 69.21.  $\square$

This corollary is about an explicitly defined *set* of surreal-labelled well-orders. It is not obtained by pretending that  $\beth_\omega$  is inaccessible or that its birthday fragment is the set universe of a full class-theory model.

### 69.8.2 Relation to the earlier singular order-spectrum transition

The repository report also studies the minimal slices at these birthday cutoffs and gives a different singular transition (theorem 8.3). In its notation, if  $\mu = 2^{<\kappa}$ , the minimal enumeration order is equimorphic with the binary lexicographic cube of length  $\kappa \cdot \kappa$  when  $\kappa$  is singular strong limit, and with the cube of length  $\mu$  otherwise. Here  $\kappa \cdot \kappa$  is *ordinal* multiplication. This earlier assertion is not used as an input to the proofs above.

The two transitions measure different structures. The repository transition concerns mutual order embeddings into cubes. The present classification concerns the order topology and the descriptive complexity of exhaustiveness inside a prefix completion. They agree that singular strong-limit stages

deserve attention, but neither assertion can be substituted for the other. Equimorphism does not identify the topology with that of a compact cube, and the equality  $\mu = \kappa$  at  $\beth_\omega$  does not collapse ordinal length  $\kappa \cdot \kappa$  to  $\kappa$ .

## 69.9 Nonminimal lengths and varying carriers (source 26)

The minimality of the ordinal domain is structural, not a convenience in notation. For  $\lambda$  of cardinality  $\mu$ , one can define  $\mathcal{W}_\lambda(X) = \text{Bij}(\lambda, X)$  and its proper-prefix topology, but the cardinalities and order types of residual tails vary with the prefix.

**Proposition 69.29** (Successor-length isolation). *If  $\lambda = \delta + 1$  and  $|\lambda| = |X|$ , the proper-prefix topology on  $\mathcal{W}_\lambda(X)$  is discrete.*

*Proof.* For a given bijection  $f$ , fix its prefix of length  $\delta$ . Exactly one label remains, so exactly one full bijection extends this prefix. Thus the corresponding cylinder is the singleton  $\{f\}$ .  $\square$

Even for a dense numerical alphabet, this prefix topology need not coincide with enumeration order topology at nonminimal lengths: the singleton argument has no two unused choices with which to bracket the last coordinate. This gives a concrete reason that Theorem 69.18 cannot simply be restated with “all ordinals of cardinality  $\mu$ ” in place of the initial cardinal.

More generally, if a prefix leaves  $m$  positions and exactly  $m$  labels for finite  $m$ , its cylinder is a finite residual permutation order. Its adjacency and endpoints are finite combinatorial data. If it leaves an infinite residual cardinal, cardinal size alone still does not identify the residual *ordinal length*. A systematic theory should retain both the order type of the domain tail and the numerical order on the residual labels. The previous repository reports already analyze several such residual and adjacency phenomena (theorem 16.2 and section 31); the present minimal-slice proofs intentionally use only the uniform residual situation guaranteed by initial-cardinal length.

For varying set carriers, write

$$\mathcal{I} = \{p : p \text{ is a set function, } \text{dom } p \in \text{Ord}, \text{ran } p \subseteq \mathbf{No}, p \text{ injective}\}.$$

This is an ordinary class of sets. Every individual range has bounded birthdays: take the supremum of its birthday successors. Thus a fixed word belongs to a suitable set-sized fragment, while the class of all words is not bounded in one such fragment. The increasing-word subclass is obtained by requiring  $p(\xi) <_{\mathbf{No}} p(\eta)$  whenever  $\xi < \eta$ .

One can compare words with an explicitly specified termination rule. A prefix-first order has a least empty word and extensions lying immediately on one designated side of every terminating prefix; the numerical surreal sign order instead uses central termination. These endpoint and prefix distinctions prevent an automatic identification with  $\mathbf{No}$  even when an order embedding into  $\mathbf{No}$  is available. The earlier reports supply explicit sign codings and analyze the relevant limit-length cuts (part VII). Our new compact coding applies to fixed exhaustive and injective  $\mu$ -words, and does not purport to classify every variable-length cut.

## 70 Permutation groups at singular cardinalities: maximality, self-normalization and the canonical group envelope (source 28)

[merge] Source 28’s Sections 4–10, printed whole; its Section 3 is recorded in section 62 and its Sections 11–12 in section 71. Source 28 continues source 24 and answers source 24’s question on the boundedness-preserving subgroup (“its closure properties in the two ambient topologies,

its normalizers, and the possible topological-group structures on larger natural subgroups. How much of the ordinal bounded-set ideal is recoverable from this group together with its topology?”) in the natural action, and its question on uniformities with an exhaustion interpretation in part (section 72). It works on  $S_\mu = \text{Sym}(\mu)$ , identified with  $\mathcal{W}_\mu(X)$  by a fixed label enumeration  $e : \mu \rightarrow X$  ( $f \mapsto e^{-1} \circ f$ ), with topologies  $\tau_{\mathcal{J}}$  of agreement on the members of an admissible coordinate ideal  $\mathcal{J}$ ; its bounded ideal  $\mathcal{B}_\mu$  is printed  $\mathcal{J}_b$ , its loci  $B_\mu^\pm$  are printed  $H_\mu^\pm$ , and its  $\delta = \text{cf } \mu$  is printed  $q$  (part X).

Several of its statements are known from sources 24 and 26 and are credited where they occur. Theorem 70.2 and corollary 70.3 generalize source 24’s theorem 69.2 and the group clause of theorem 69.4 from the bounded ideal to every admissible ideal, by the same transposition witnesses. Proposition 70.19 is source 24’s remark 69.6. Source 28’s Proposition 8.1 is proposition 62.7 (note 70.20). Theorem 70.25 generalizes source 24’s theorem 65.1 to ideals; theorem 70.26, and the equality in theorem 70.10, are the closedness and density parts of source 26’s theorem 69.23, for every infinite  $\mu$  (source 26 assumes  $\text{cf } \mu = \omega < \mu$ ); lemma 70.9 is the bounded case of the balanced reserve (theorem 49.7; source 24’s note 71.1); theorem 70.11 uses the same open sets as source 26’s proof of theorem 69.23(ii), for a different set; the Baire clause of corollary 70.12 is a case of theorem 69.12, and the complete metric on  $H_\mu$  is its third proof (after corollary 69.5 and theorem 69.23(ii)); proposition 70.24 is, for  $\tau_b$ , the case  $\alpha = \mu$  of theorem 27.2 and of source 22’s weight column (theorem 63.4). New in source 28 are the maximality and self-normalization theorems, the recovery of the bounded ideal from the action, the nowhere density of  $H_\mu^+$  and  $H_\mu$ , the dense inverse locus, the explicit one-sided permutations, the canonical group envelope  $\tau_{<\mu}$ , its character  $\text{cof}([\mu]^{<\mu}, \subseteq)$ , the two-sided completeness of the envelope and the filtration-invariance proposition.

### 70.1 Coordinate control by an ideal (source 28)

Fix  $S_\mu = \text{Sym}(\mu)$  and use composition  $(p \circ q)(\xi) = p(q(\xi))$ . A fixed label enumeration  $e : \mu \rightarrow X$  identifies  $\mathcal{W}_\mu(X)$  with  $S_\mu$  by  $f \mapsto e^{-1} \circ f$ . The prefix topology on  $S_\mu$  has basic neighborhoods

$$N_\alpha(p) = \{q : q \upharpoonright \alpha = p \upharpoonright \alpha\}, \quad \alpha < \mu.$$

The numerical order on  $X$  does not enter this topology; thickness was what identified it with the order topology.

**Definition 70.1** (Admissible coordinate ideal). An ideal  $\mathcal{J}$  on  $\mu$  is *admissible* here if it contains every finite set, is closed under subsets and finite unions, and satisfies  $\mathcal{J} \subseteq [\mu]^{<\mu}$ . Define  $\tau_{\mathcal{J}}$  by the basic neighborhoods

$$N_A(p) = \{q : q \upharpoonright A = p \upharpoonright A\} \quad (A \in \mathcal{J}).$$

Write  $\text{Fix}(A) = N_A(\text{id})$ . A map  $u : \mu \rightarrow \mu$  preserves  $\mathcal{J}$  if  $u[A] \in \mathcal{J}$  for every  $A \in \mathcal{J}$ .

The bounded ideal

$$\mathcal{J}_b = \{A \subseteq \mu : \exists \alpha < \mu (A \subseteq \alpha)\}$$

gives  $\tau_{\mathcal{J}_b} = \tau_b$ . The small-set ideal  $[\mu]^{<\mu}$  gives  $\tau_{<\mu}$ . They agree at regular  $\mu$  and differ at singular  $\mu$ . Admissibility ensures that outside every control set there remain  $\mu$  coordinates, so transpositions can witness the necessity of continuity conditions.

**Theorem 70.2** (Exact ideal continuity criteria). *In  $(S_\mu, \tau_{\mathcal{J}})$ :*

- (i) *Every left translation  $L_s(p) = s \circ p$  is a homeomorphism.*
- (ii) *Right translation  $R_s(p) = p \circ s$  is continuous everywhere if  $s$  preserves  $\mathcal{J}$ , and nowhere otherwise.*



(iii) *Inversion is continuous at  $p$  if and only if  $p^{-1}$  preserves  $\mathcal{J}$ .*

*Proof.* Equality of restrictions on a fixed domain is unchanged by applying the same injective map to their values. This proves (i), including continuity of the inverse left translation.

To control  $p \circ s$  on  $A$ , it suffices to control  $p$  on  $s[A]$ . This proves sufficiency in (ii). If  $s[A] \notin \mathcal{J}$ , then for every  $B \in \mathcal{J}$  some  $u \in s[A] \setminus B$  exists; otherwise downward closure would put  $s[A]$  in  $\mathcal{J}$ . Choose  $v \notin B \cup \{u\}$ . The permutation  $q = p \circ (u \ v)$  agrees with  $p$  on  $B$ , but  $q \circ s$  changes at the coordinate  $s^{-1}(u) \in A$ . Thus continuity fails at every  $p$ .

To fix  $p^{-1}$  on  $A$ , it suffices to fix  $p$  on  $p^{-1}[A]$ . If this set belongs to  $\mathcal{J}$ , this proves the required local implication. If it does not, use the same transposition construction with  $u \in p^{-1}[A] \setminus B$ . The inverse position of the label  $p(u) \in A$  changes from  $u$  to  $v$ , while the forward restriction on  $B$  stays fixed. This proves necessity in (iii).  $\square$

**Corollary 70.3** (The ideal-preserving subgroup). *The set*

$$H_{\mathcal{J}} = \{p \in S_{\mu} : p \text{ and } p^{-1} \text{ preserve } \mathcal{J}\}$$

*is a subgroup, and its inherited  $\tau_{\mathcal{J}}$  topology makes it a topological group.*

*Proof.* Ideal preservation is closed under composition, and the definition is symmetric under inversion. For joint continuity at  $(p, s)$ , to control  $p' \circ s'$  on  $A \in \mathcal{J}$ , require  $s' \restriction A = s \restriction A$  and  $p' \restriction s[A] = p \restriction s[A]$ . Both are valid control requirements when  $s \in H_{\mathcal{J}}$ . Inversion is continuous by Theorem 70.2.  $\square$

At a singular cardinal, a bounded set of size  $\text{cf } \mu$  can be sent onto a cofinal set by a permutation. Thus Theorem 70.2 reproduces the predecessor's obstruction (source 24, [corollary 69.3](#)): the original forward topology need not make the full permutation group a topological group. The ideal formulation also identifies exactly which part of the proof is general and which part depends on ordinal boundedness.

## 70.2 Maximality and self-normalization (source 28)

**Theorem 70.4** (Maximal inherited topological subgroup). *Let  $\mathcal{J}$  be admissible. Among subgroups  $G \leq S_{\mu}$  that contain every finite-support permutation and for which the inherited  $\tau_{\mathcal{J}}$  topology is a group topology,  $H_{\mathcal{J}}$  is the largest.*

*Proof.* Corollary 70.3 proves that  $H_{\mathcal{J}}$  is such a subgroup; finitary permutations preserve  $\mathcal{J}$  in both directions because they change a set by only finitely many points.

Suppose  $G$  has the stated properties and  $p \in G$ . If  $p^{-1}$  does not preserve  $\mathcal{J}$ , the transposition witnesses in the proof of Theorem 70.2(iii) all lie in  $G$ : each is  $p \circ (u \ v)$ , and  $(u \ v)$  is finitary. They show that inversion on the subspace  $G$  is not continuous at  $p$ . This contradicts the group-topology assumption. Thus  $p^{-1}$  preserves  $\mathcal{J}$ . Applying the same argument to  $p^{-1} \in G$  shows that  $p$  preserves  $\mathcal{J}$ . Hence  $G \subseteq H_{\mathcal{J}}$ .  $\square$

The finitary hypothesis matters. Discontinuity in an ambient space does not automatically remain a discontinuity after restriction to an arbitrary subgroup. It is precisely the supply of internal transposition witnesses that makes this maximality argument valid.

For the bounded ideal, let  $H_{\mu} = H_{\mathcal{J}_b}$ . The next result answers the normalizer part of the predecessor's subgroup question (source 24; [section 72](#)).

**Lemma 70.5** (A bounded part and a cofinal part). *Let  $\mu$  be singular and  $q = \text{cf } \mu$ . If  $C \subseteq \mu$  is unbounded and  $|C| > q$ , then  $C$  contains a bounded subset  $D$  of size  $q$  and, disjoint from  $D$ , a cofinal subset  $E$  of size  $q$ .*



*Proof.* Choose an increasing cofinal sequence  $(\gamma_i)_{i < q}$  in  $\mu$ . If every  $C \cap \gamma_i$  had size less than  $q$ , their union would have size at most  $q \cdot q = q$ , contrary to  $|C| > q$ . Some bounded intersection therefore contains a subset  $D$  of size  $q$ . The portion of  $C$  above a bound for  $D$  remains unbounded, so it has a cofinal subset  $E$  of size  $q$ .  $\square$

**Theorem 70.6** (Self-normalization at singular cardinalities). *If  $\mu$  is singular, then*

$$N_{S_\mu}(H_\mu) = H_\mu.$$

*Proof.* Every element of  $H_\mu$  normalizes  $H_\mu$ . Conversely, let  $s$  normalize  $H_\mu$ . Suppose that  $s$  fails to preserve bounded sets. Choose a bounded  $A_0$  with  $s[A_0]$  unbounded. Since a singular cardinal is a limit cardinal, there is a bounded set  $A \supseteq A_0$  with

$$\text{cf } \mu < |A| < \mu.$$

For example, enlarge a bounded initial segment containing  $A_0$  to one of cardinality greater than  $\text{cf } \mu$ . Put  $C = s[A]$ . It is unbounded and has size greater than  $\text{cf } \mu$ .

Every permutation supported on  $A$  belongs to  $H_\mu$ . Conjugation by  $s$  therefore implies that *every* permutation supported on  $C$  belongs to  $H_\mu$ . By Lemma 70.5, choose a bounded  $D \subseteq C$  and a disjoint cofinal  $E \subseteq C$ , both of size  $\text{cf } \mu$ . Exchange  $D$  and  $E$  bijectively and fix all remaining coordinates. The resulting involution is supported on  $C$  but sends the bounded set  $D$  to the unbounded set  $E$ . It is not in  $H_\mu$ , a contradiction.

Thus  $s$  preserves bounded sets. Its inverse also normalizes  $H_\mu$ , so the same argument shows that  $s^{-1}$  preserves bounded sets. Consequently  $s \in H_\mu$ .  $\square$

*Remark 70.7* (Why the cardinal enlargement is essential). An unbounded set of size exactly  $\text{cf } \mu$  may have too few points in every bounded interval for the exchange used above. It is not valid to assert that every unbounded support set carries a permutation violating boundedness. Enlarging  $A_0$  to cardinality greater than  $\text{cf } \mu$  is what makes the proof work.

**Proposition 70.8** (Recovering boundedness from the natural action). *For  $A \subseteq \mu$ , the pointwise stabilizer of  $A$  in  $H_\mu$  is open in the inherited prefix topology if and only if  $A$  is bounded.*

*Proof.* If  $A$  is bounded, a bounded initial segment containing it has an open pointwise stabilizer contained in its stabilizer. That larger subgroup is open as a union of cosets. If  $A$  is unbounded, then for every  $\beta < \mu$  choose  $u \in A \setminus \beta$  and  $v \notin \beta \cup \{u\}$ . The finitary transposition  $(u \ v)$  belongs to  $H_\mu$  and fixes  $\beta$ , but does not fix  $A$ . No identity neighborhood can lie inside the stabilizer of  $A$ .  $\square$

This reconstructs the bounded-set ideal from the *topological group together with its specified permutation action*. It does not prove that the bare abstract group alone determines the ideal, nor that every abstract automorphism is induced by a permutation of  $\mu$ .

### 70.3 One-sided boundedness: closure, density and a complete locus (source 28)

Write

$$\begin{aligned} H_\mu^+ &= \{p \in S_\mu : p[A] \text{ is bounded for every bounded } A\}, \\ H_\mu^- &= \{p \in S_\mu : p^{-1}[A] \text{ is bounded for every bounded } A\}. \end{aligned}$$

Then  $H_\mu = H_\mu^+ \cap H_\mu^-$  and inversion exchanges  $H_\mu^+$  with  $H_\mu^-$ . By Theorem 70.2,  $H_\mu^-$  is exactly the continuity locus of inversion in the full forward topology. At singular cardinals these two one-sided loci have strikingly different positions in the same space.

### 70.3.1 A bounded extension lemma

**Lemma 70.9** (Extending a bounded partial injection). *Let  $A, C \subseteq \mu$  be bounded and let  $q : A \rightarrow C$  be a bijection. There is a permutation  $h \in H_\mu$  extending  $q$ , with bounded support.*

*Proof.* Put  $D_0 = A \cup C$ . If  $D_0$  is finite, the two residual sets  $D_0 \setminus A$  and  $D_0 \setminus C$  have equal finite size, so choose a bijection between them and complete  $q$  to a permutation of  $D_0$ .

If  $D_0$  is infinite, put  $\nu = |D_0| < \mu$ . Choose a bounded ordinal block  $R$  of size  $\nu$  above  $D_0$  and set  $D = D_0 \cup R$ . Such a block remains below  $\mu$  because both its starting ordinal and  $\nu$  have cardinality less than the initial cardinal  $\mu$ . Both  $D \setminus A$  and  $D \setminus C$  contain  $R$  and have size  $\nu$ . Complete  $q$  by a bijection between these residual sets.

In either case, extend the resulting bounded permutation by the identity outside its carrier. A bounded-support permutation and its inverse send a bounded set into the union of that set and the bounded support. Hence the extension belongs to  $H_\mu$ .  $\square$

The reserve in the infinite case avoids a genuine pitfall: an infinite partial bijection need not extend to a permutation of the union of its domain and range, since the residual sets inside that union need not have equal cardinality.

**Theorem 70.10** (Exact forward closure). *In  $(S_\mu, \tau_b)$ ,*

$$\overline{H_\mu} = H_\mu^+.$$

*For singular  $\mu$ , the set  $H_\mu^+$  is closed nowhere dense, and therefore  $H_\mu$  is nowhere dense as well.*

*Proof.* If  $p \notin H_\mu^+$ , choose  $\alpha < \mu$  for which  $p[\alpha]$  is unbounded. Every permutation with the same restriction to  $\alpha$  also fails the forward condition. Thus  $H_\mu^+$  is closed, and its closure contains no additional point of  $S_\mu$ .

If  $p \in H_\mu^+$ , then  $p \restriction \alpha$  has bounded domain and bounded range for every  $\alpha < \mu$ . Lemma 70.9 extends this restriction to an element of  $H_\mu$ . Every prefix neighborhood of  $p$  consequently meets  $H_\mu$ , proving the equality.

Now assume  $\mu$  singular and put  $q = \text{cf } \mu$ . Given any short injective prefix, the unused labels form an unbounded set: fewer than  $\mu$  used labels cannot contain a tail of cardinality  $\mu$ . Choose a cofinal sequence of  $q$  distinct unused labels and append it to the prefix. The enlarged prefix still has length less than  $\mu$ , since a finite ordinal sum of ordinals of cardinality less than  $\mu$  remains below  $\mu$ . It has unbounded range on a bounded domain. Its nonempty cylinder, obtained by [note 62.12](#), is disjoint from  $H_\mu^+$ . Thus the closed set  $H_\mu^+$  has empty interior.  $\square$

### 70.3.2 A dense inverse-continuity locus

Fix an increasing cofinal sequence  $(\mu_i)_{i < q}$  in  $\mu$ , where  $q = \text{cf } \mu$ . On the injection space  $\mathcal{I}_\mu(\mu)$  with its forward prefix topology, define

$$O_i = \{h : \exists \alpha < \mu (\mu_i \subseteq \text{ran}(h \restriction \alpha))\}.$$

Here  $\mu_i$  denotes the set of all ordinals below that ordinal; the condition requires an entire bounded block of labels to have appeared by some bounded time.

**Theorem 70.11** (Inverse-boundedness as a dense intersection). *Each  $O_i$  is open dense, and as a subspace of  $\mathcal{I}_\mu(\mu)$ ,*

$$H_\mu^- = \bigcap_{i < q} O_i.$$

*This intersection is dense. If  $q = \omega$ , it is a dense completely metrizable  $G_q$  subspace of  $\mathcal{I}_\mu(\mu)$  and of  $S_\mu$ . Its complement is meager in both spaces.*

*Proof.* Once the labels below  $\mu_i$  have appeared in a prefix, every continuation satisfies the same condition, so  $O_i$  is open. Any short prefix can be extended to include the as-yet unused labels below  $\mu_i$ ; there are fewer than  $\mu$  of them. The extension remains short, proving density.

If an injection belongs to every  $O_i$ , it is onto, since these label blocks cover  $\mu$ . Further, the preimage of each  $\mu_i$  is bounded. Every bounded subset of  $\mu$  is contained in some  $\mu_i$ , so the inverse of the resulting permutation preserves boundedness. Conversely, inverse boundedness supplies the required witness  $\alpha$  for every  $i$ . This proves the displayed equality.

For density of the full intersection, start with any prescribed short prefix. At stage  $i < q$ , extend it to include every label below  $\mu_i$  and to have length at least  $\mu_i$ . At a limit stage  $j < q$ , take the union of the preceding prefixes. This union is still short, since fewer than  $\text{cf } \mu$  lengths below  $\mu$  have been used. At the final stage  $q$ , the union has domain  $\mu$ , is injective, and covers every label. Each block  $\mu_i$  was covered at a short intermediate stage, so the final permutation belongs to  $H_\mu^-$  and extends the original prefix.

If  $q = \omega$ , agreement along the cofinal lengths  $\mu_i$  defines a compatible ultrametric on the injection space. A Cauchy sequence stabilizes a coherent injective prefix at every such length, and their union is its limit. Thus the injection space is completely metrizable. A  $G_q$  subspace of a completely metrizable space is completely metrizable; a proof not requiring separability is recalled in [note 69.9](#). Since every  $O_i$  is open dense, the complement of the intersection is a countable union of closed nowhere dense sets. Intersecting with  $S_\mu$  gives the same conclusion there, because each  $O_i \cap S_\mu$  is open dense by the same prefix-extension argument.  $\square$

**Corollary 70.12** (A complete core and the Baire property). *If  $\text{cf } \mu = \omega$ , then  $S_\mu$  with the forward topology is a Baire space, and  $H_\mu$  has a complete compatible metric. At an uncountable singular such  $\mu$ , this completely metrizable subgroup is nevertheless nowhere dense in  $S_\mu$ .*

*Proof.* The dense completely metrizable subspace  $H_\mu^-$  is Baire. If  $U_n$  are open dense in  $S_\mu$ , their intersections with  $H_\mu^-$  are open dense there: every open set in  $S_\mu$  meeting  $H_\mu^-$  meets  $U_n$  in a nonempty open set, which again meets the dense subspace. Hence  $\bigcap_n U_n$  meets every nonempty open set of  $S_\mu$ . This proves the Baire property.

By Theorem 70.10,  $H_\mu^+$  is closed in  $S_\mu$ , so  $H_\mu = H_\mu^+ \cap H_\mu^-$  is closed in the completely metrizable space  $H_\mu^-$ . The final assertion is Theorem 70.10.  $\square$

Complete metrizability is not inherited upward from a dense subspace. Consequently this corollary does not settle whether  $S_\mu$  itself admits a complete compatible metric when  $\mu > \omega$  is singular of countable cofinality. It does supply a concrete large complete locus and another proof of the Baire property in that case.

*Note 70.13* ([merge]; source 28's Question 1, answered by source 26 (3 October 2026)). Source 28 calls the complete metrizability of the full singular forward space its remaining question: “The remaining complete-metrizability question for the *full* singular forward permutation space is not settled by the dense complete subspace constructed here” (its introduction); the paragraph above; its ledger (“Does not settle complete metrizability of the full forward space”); and its Question 1, “Complete metrizability of the full singular space”: *let  $\mu > \omega$  be singular with  $\text{cf } \mu = \omega$ ; is  $\text{Sym}(\mu)$ , with its bounded-prefix topology, completely metrizable? Equivalently, is it a  $G_\delta$  subspace of the complete forward injection space?* Source 28's own results do not settle it, as it says: complete metrizability is not inherited upward from the dense locus  $H_\mu^-$ . The question is answered negatively in ZFC by source 26's [theorem 69.18](#): under the identification of  $S_\mu$  with  $\mathcal{W}_\mu(X)$ ,  $S_\mu$  is not Borel, hence

not  $G_\delta$ , in  $\mathcal{I}_\mu(\mu)$ , and no nonempty open subspace of it is completely metrizable. Source 28 read source 24 but not source 26: both were written at the same repository pin 7ff7736ec, and source 26 arrived first (commit 0d7b2441d; source 28 in 79049d58c). Source 28's Question 3, on the descriptive complexity of the one-sided loci, is answered in part by [theorem 69.17](#) and [theorem 69.23\(iii\)](#).

### 70.3.3 The forward and inverse conditions are independent

**Proposition 70.14** (An explicit one-sided permutation). *At every singular  $\mu$ ,*

$$H_\mu \subsetneq H_\mu^+, \quad H_\mu \subsetneq H_\mu^-.$$

*In particular,  $H_\mu$  is not closed in the forward topology.*

*Proof.* Put  $q = \text{cf } \mu < \mu$ . Choose a bounded set

$$B = \{b_{i,n} : i < q, n < \omega\}$$

of distinct coordinates. Above  $B$ , choose disjoint countably infinite blocks

$$A_i = \{a_{i,n} : n < \omega\} \quad (i < q)$$

in successively higher bounded ordinal intervals, with their starting positions cofinal in  $\mu$ . The intervals can be chosen so that every bounded set meets only blocks with indices bounded in  $q$ .

Define  $p$  on these coordinates by

$$p(a_{i,0}) = b_{i,0}, \quad p(a_{i,n+1}) = a_{i,n}, \quad p(b_{i,n}) = b_{i,n+1},$$

and fix all other coordinates. On each  $i$ -chain this is a bilateral shift, so it defines a permutation. A bounded input set meets only boundedly many of the high blocks. Its image is contained in the union of that input set, the bounded set  $B$ , and a bounded initial collection of the high blocks. Therefore  $p \in H_\mu^+$ .

But  $p^{-1}[B]$  contains every  $a_{i,0}$  and is cofinal in  $\mu$ . Thus  $p \notin H_\mu^-$ . Its inverse lies in  $H_\mu^- \setminus H_\mu^+$ . The nonclosedness assertion follows from  $\overline{H_\mu} = H_\mu^+$ .  $\square$

**Corollary 70.15** (A discrete subgroup outside the bounded subgroup). *There is a subgroup of  $S_\mu$  not contained in  $H_\mu$  whose inherited forward topology is a group topology.*

*Proof.* For the permutation  $p$  in Proposition 70.14, the orbit of  $b_{0,0}$  under the integer powers of  $p$  has no repetitions. Thus the only element of the cyclic subgroup  $\langle p \rangle$  fixing  $b_{0,0}$  is the identity. A sufficiently long prefix condition isolates that identity, and left translations isolate every other group element. The inherited topology is discrete, hence is a group topology. But  $p \notin H_\mu$ . This subgroup does not contain all finitary permutations, exactly as Theorem 70.4 requires.  $\square$

This example explains why a single condition such as “the permutation respects small prefixes” is insufficient. Forward boundedness concerns where early positions go; inverse boundedness concerns how late one must search to find early labels. They are different constraints at singular cardinalities.

## 70.4 The canonical group envelope (source 28)

The two-sided bounded topology  $\tau_{\leftrightarrow}$  has basic neighborhoods at  $p$  requiring

$$q \restriction \alpha = p \restriction \alpha, \quad q^{-1} \restriction \alpha = p^{-1} \restriction \alpha \quad (\alpha < \mu).$$

The predecessor proved that its defining uniformity is complete (source 24, [theorem 69.4](#)). This does not make the full singular permutation set a topological group: conjugating bounded control sets can produce unbounded control sets. We now identify the exact coarsest refinement that does make all group operations continuous.

**Definition 70.16** (Symmetric closure of a coordinate ideal). For an admissible ideal  $\mathcal{J}$ , let  $\mathcal{J}^{\text{sym}}$  be the ideal generated by the sets  $s[A]$ , with  $A \in \mathcal{J}$  and  $s \in S_\mu$ . Thus its members are subsets of finite unions of such images. It remains contained in  $[\mu]^{<\mu}$  and is invariant under every permutation.

**Theorem 70.17** (Smallest group refinement for an ideal). *The topology  $\tau_{\mathcal{J}^{\text{sym}}}$  is the coarsest group topology on  $S_\mu$  finer than  $\tau_{\mathcal{J}}$ .*

*Proof.* Every permutation and its inverse preserve  $\mathcal{J}^{\text{sym}}$ . Corollary 70.3, or the direct multiplication argument there, makes  $\tau_{\mathcal{J}^{\text{sym}}}$  a group topology. It contains  $\tau_{\mathcal{J}}$ .

Let  $\tau$  be any group topology containing  $\tau_{\mathcal{J}}$ . Each  $\text{Fix}(A)$  for  $A \in \mathcal{J}$  is  $\tau$ -open, and conjugation gives the  $\tau$ -open subgroup

$$s \text{Fix}(A) s^{-1} = \text{Fix}(s[A]).$$

Finite intersections give open stabilizers of finite unions of these image sets. If  $C$  is a subset of such a union  $D$ , then  $\text{Fix}(C)$  contains the open subgroup  $\text{Fix}(D)$ , and is itself open as a union of its cosets. Thus every  $\text{Fix}(C)$  with  $C \in \mathcal{J}^{\text{sym}}$  is  $\tau$ -open. Translating these identity neighborhoods proves  $\tau_{\mathcal{J}^{\text{sym}}} \subseteq \tau$ .  $\square$

**Theorem 70.18** (The bounded-prefix group envelope). *For every infinite  $\mu$ , the coarsest group topology finer than  $\tau_b$  is*

$$\tau_{<\mu} = \tau_{[\mu]^{<\mu}}.$$

*For regular  $\mu$ , it equals  $\tau_b$ . For singular  $\mu$ ,*

$$\tau_b \subsetneq \tau_{\leftrightarrow} \subsetneq \tau_{<\mu}.$$

*Proof.* Every small set  $A \subseteq \mu$  is the image of a bounded initial ordinal under a permutation: take a bijection  $|A| \rightarrow A$  and extend it over the two complements, both of size  $\mu$ . Conversely, images of bounded sets have size less than  $\mu$ . Therefore

$$(\mathcal{J}_b)^{\text{sym}} = [\mu]^{<\mu},$$

and Theorem 70.17 applies. At regular  $\mu$ , small and bounded are equivalent.

For every  $\mu$ , the two-sided bounded topology contains the forward one. A two-sided bounded requirement at  $p$  is enforced by forward agreement on

$$\alpha \cup p^{-1}[\alpha],$$

which has size less than  $\mu$ . Thus  $\tau_{\leftrightarrow} \subseteq \tau_{<\mu}$ .

Suppose  $\mu$  singular. Choose an involution  $s$  exchanging a bounded set of size  $\text{cf } \mu$  with a disjoint cofinal set of the same size. At  $s$ , an inverse requirement on the bounded set has no forward-prefix neighborhood enforcing it, by the transposition proof of Theorem 70.2. Hence  $\tau_b \neq \tau_{\leftrightarrow}$ .

At the identity, however, fixing  $\alpha$  automatically fixes the inverse on  $\alpha$ . Thus the  $\tau_{\leftrightarrow}$  and  $\tau_b$  identity neighborhood bases coincide. Let  $C$  be an unbounded set of size  $\text{cf } \mu < \mu$ . Its stabilizer is  $\tau_{<\mu}$ -open, but contains no bounded-prefix identity neighborhood: beyond any prescribed  $\beta$  one can swap a point of  $C$  with another point. It is not a  $\tau_{\leftrightarrow}$  neighborhood of the identity. Hence  $\tau_{\leftrightarrow} \neq \tau_{<\mu}$ .  $\square$

**Proposition 70.19** (Why two-sided bounded control is not enough). *At singular  $\mu$ ,  $\tau_{\leftrightarrow}$  does not make  $S_\mu$  a topological group.*

*Proof.* Use the permutation  $s$  from Theorem 70.18, with  $s[\alpha]$  unbounded for some  $\alpha$ . At the identity, every two-sided bounded neighborhood contains  $\text{Fix}(\beta)$  for some  $\beta$ . Choose a transposition  $t$  supported outside  $\beta$  that moves a point of  $s[\alpha]$ . Then  $t$  belongs to that input neighborhood, but  $t \circ s$  disagrees with  $s$  below  $\alpha$ . Right translation by  $s$  is not continuous at the identity.  $\square$

The envelope theorem gives a canonical answer to a natural formulation of the problem: retain all original prefix-open sets, keep ordinary permutation composition, and add no more open sets than group continuity forces. Its answer is *not* the original order topology and is *not* merely inverse-prefix control. It is control on all sets of positions with cardinality less than  $\mu$ .

## 70.5 Exact character and the singular cardinal penalty (source 28)

For a topological space, the character at a point is the least size of a local neighborhood base. For a poset,  $\text{cof}$  denotes the least size of a cofinal subset. Neither should be confused with the ordinal cofinality of the enumeration cardinal, although all three occur here.

*Note 70.20* (Source 28, Proposition 8.1, “Character of the forward space”). *Every point of  $(S_\mu, \tau_b)$  has character  $\text{cf } \mu$ . The forward space is metrizable exactly when  $\text{cf } \mu = \omega$ . This is [proposition 62.7](#) and [note 69.10](#), by the same argument (“Strictness follows by swapping two coordinates outside  $\gamma$ , one of them  $\gamma$ ”).*

**Theorem 70.21** (Character of the canonical envelope). *For every infinite cardinal  $\mu$ ,*

$$\chi(S_\mu, \tau_{<\mu}) = \text{cof}([\mu]^{<\mu}, \subseteq).$$

*For singular  $\mu$ ,*

$$\mu^+ \leq \chi(S_\mu, \tau_{<\mu}) \leq \mu^{<\mu}.$$

*In particular, the canonical group envelope is not first countable at any singular  $\mu$ .*

*Proof.* A cofinal family  $\mathcal{A} \subseteq [\mu]^{<\mu}$  gives the identity base  $\{\text{Fix}(A) : A \in \mathcal{A}\}$ . For the reverse inequality, let  $\mathcal{U}$  be a local base at the identity, and choose  $A_U$  small with  $\text{Fix}(A_U) \subseteq U$ . Given any small  $A$ , choose  $U \subseteq \text{Fix}(A)$ . Then

$$\text{Fix}(A_U) \subseteq \text{Fix}(A) \implies A \subseteq A_U.$$

Indeed, a point of  $A \setminus A_U$  could be transposed with a different point outside  $A_U$ , contradicting the inclusion. Thus the  $A_U$  form a cofinal family. Homogeneity of the topological group gives the formula at every point.

The upper bound is the number of small subsets,  $\mu^{<\mu}$ . To prove the strict lower bound at singular  $\mu$ , put  $q = \text{cf } \mu$  and suppose a family  $(A_\xi)_{\xi < \mu}$  were cofinal. Choose increasing infinite cardinals  $\kappa_i < \mu$ ,  $i < q$ , cofinal in  $\mu$ . For each  $i$ , the union

$$T_i = \bigcup \{A_\xi : \xi < \kappa_i, |A_\xi| \leq \kappa_i\}$$

has size at most  $\kappa_i$ , so choose  $x_i \in \mu \setminus T_i$ . The set  $C = \{x_i : i < q\}$  has size less than  $\mu$ . For any fixed  $\xi < \mu$ , choose  $i$  sufficiently large that  $\xi < \kappa_i$  and  $|A_\xi| \leq \kappa_i$ . Then  $x_i \notin A_\xi$ , and hence  $C \not\subseteq A_\xi$ . This contradicts cofinality of the family. A shorter family could have been padded to length  $\mu$ , so the cofinality is strictly greater than  $\mu$ .  $\square$

**Example 70.22** (A countably based order with a very large group envelope). Take  $\kappa = \beth_\omega$  and  $X = \mathbf{No}_{<\kappa}$ . The cardinal  $\kappa$  is a singular strong limit of cofinality  $\omega$ , and  $2^{<\kappa} = \kappa$ . Thus the minimal enumeration order has a metrizable order topology with local character  $\omega$ . Its canonical group envelope has character at least  $\beth_\omega^+$ . No additional hypothesis such as the generalized continuum hypothesis is needed for this example: strong limit follows because every  $\lambda < \beth_\omega$  is below some  $\beth_n$ , and  $2^\lambda \leq \beth_{n+1} < \beth_\omega$ .

*Remark 70.23* (What the character bound does not say). Character is not monotone under arbitrary refinement of a topology. In particular, the discrete topology is a finer metrizable group topology and has character 1. The theorem says that the *coarsest* group refinement has large character. It does not exclude all other finer metrizable group topologies. Nor does it claim that a different, artificially transported group law could not be placed on the underlying space. The fixed operations throughout are ordinary permutation composition and inversion.

**Proposition 70.24** (Equal global invariants can conceal different local geometry). *Both  $(S_\mu, \tau_b)$  and  $(S_\mu, \tau_{<\mu})$  have density and weight equal to  $\mu^{<\mu}$ .*

*Proof.* There are  $\mu^{<\mu}$  short injective prefixes, giving an upper bound for the weight of  $\tau_b$ ; choosing one completion of each gives the same bound for density. There are also at most  $\mu^{<\mu}$  small-domain partial injections, giving both upper bounds for  $\tau_{<\mu}$ .

For every cardinal  $\lambda < \mu$ , all injections from a fixed initial segment of size  $\lambda$  into  $\mu$  give pairwise disjoint nonempty cylinders in either topology. There are  $\mu^\lambda$  such injections: an arbitrary function  $a : \lambda \rightarrow \mu$  can be coded injectively as  $\xi \mapsto (\xi, a(\xi))$ , followed by a fixed identification  $\lambda \times \mu \cong \mu$  when  $\lambda > 0$ . Every dense set must meet every cylinder. Taking the supremum over  $\lambda < \mu$  gives density at least  $\mu^{<\mu}$ , and weight is at least density.  $\square$

Thus density and weight alone cannot detect the topology change. The local character and the continuity of right translations do detect it.

## 70.6 Three uniform completions, kept separate (source 28)

Uniform completeness depends on a chosen family of agreement requirements, not only on the induced topology. This is already visible for permutations of  $\omega$ : permutations can stabilize at every forward coordinate while one label escapes to arbitrarily late positions. The following results organize the analogous phenomena at arbitrary cardinalities.

### 70.6.1 Forward completion for an arbitrary coordinate ideal

For admissible  $\mathcal{J}$ , put on  $\mathcal{I}_\mu(\mu)$  the uniformity with basic entourages

$$E_A = \{(h, k) : h \restriction A = k \restriction A\}, \quad A \in \mathcal{J}.$$

Its restriction to  $S_\mu$  is the forward  $\mathcal{J}$ -uniformity.

**Theorem 70.25** (The forward ideal completion). *The injection space is complete for this uniformity, and  $S_\mu$  is a dense uniform subspace. Hence  $\mathcal{I}_\mu(\mu)$  is its Hausdorff uniform completion.*



*Proof.* A Cauchy filter determines, for each  $A \in \mathcal{J}$ , a unique stabilized restriction  $q_A$  on  $A$ : an  $E_A$ -small nonempty filter member lies in the cylinder determined by one such restriction, and two different cylinders are disjoint. Taking control on  $A \cup B$  proves compatibility of  $q_A$  and  $q_B$ . Since singletons belong to  $\mathcal{J}$ , the union determines a function  $h : \mu \rightarrow \mu$ . Control on each two-element set proves injectivity. The filter contains the cylinder of  $h \upharpoonright A$  for every  $A \in \mathcal{J}$ , hence converges to  $h$ .

Every small-domain partial injection extends to a permutation, because both unused domain and unused range have cardinality  $\mu$ . Therefore every basic cylinder in the injection space meets  $S_\mu$ . The agreement entourage restricts exactly to those on  $S_\mu$ , proving density and uniform embedding. Completeness and Hausdorffness give the completion assertion.  $\square$

This proof uses stabilization of an entire allowed control set, not merely separate eventual stabilization at each coordinate. For uncountable control sets those are distinct conditions.

**Theorem 70.26** (Forward completion of the bounded subgroup). *For the bounded-prefix uniformity, the Hausdorff completion of  $H_\mu$  is*

$$\mathcal{I}_\mu^b = \{h \in \text{Inj}(\mu, \mu) : h[A] \text{ is bounded whenever } A \text{ is bounded}\}.$$

*For every infinite  $\mu$ , the subgroup  $H_\mu$  is not complete for this forward uniformity.*

*Proof.* In the full injection completion from Theorem 70.25, a failure of boundedness is witnessed by an entire prefix with unbounded range. Its cylinder excludes  $\mathcal{I}_\mu^b$ , so that subspace is closed and complete. Each prefix of a boundedness-preserving injection has bounded domain and bounded range; Lemma 70.9 completes it inside  $H_\mu$ . Conversely, no prefix with unbounded range can occur in an element of  $H_\mu$ . Hence the closure of  $H_\mu$  in the injection completion is exactly  $\mathcal{I}_\mu^b$ .

For a concrete missing point, send each natural number  $n$  to  $n + 1$  and fix every coordinate in  $\mu \setminus \omega$ . This is a boundedness-preserving injection omitting 0. When  $\mu = \omega$ , bounded sets are finite and the assertion is immediate; when  $\mu > \omega$ , the changed coordinates lie in the bounded set  $\omega$ . Thus the complete closure strictly contains  $H_\mu$ .  $\square$

There is no conflict between this theorem and complete metrizability of  $H_\mu$  when  $\text{cf } \mu = \omega$ . A complete compatible metric can induce the same topology as an incomplete forward metric while defining a different uniformity.

### 70.6.2 Two-sided small-set completeness

On  $S_\mu$ , define

$$E_A^{\leftrightarrow} = \{(p, q) : p \upharpoonright A = q \upharpoonright A \text{ and } p^{-1} \upharpoonright A = q^{-1} \upharpoonright A\}, \quad |A| < \mu.$$

**Theorem 70.27** (Completeness of the canonical group envelope). *The two-sided small-set uniformity is complete, induces  $\tau_{<\mu}$ , and is the two-sided group uniformity of  $(S_\mu, \tau_{<\mu})$ . Thus the canonical envelope is a complete topological group in the two-sided sense.*

*Proof.* A Cauchy filter determines coherent stabilized forward and inverse restrictions. As in Theorem 70.25, their unions are injections  $h, g : \mu \rightarrow \mu$ . If  $h(\xi) = \eta$ , take a nonempty filter member on which the forward value at  $\xi$  and the inverse value at  $\eta$  are both stabilized. Every permutation in that member witnesses  $g(\eta) = \xi$ . Conversely,  $g(\eta) = \xi$  implies  $h(\xi) = \eta$ . Hence  $g = h^{-1}$ , and  $h$  is a permutation. Stabilization on each whole small control set proves convergence in the stated uniformity.

At a fixed  $p$ , inverse agreement on  $A$  is enforced by forward agreement on  $p^{-1}[A]$ , which is small. Thus adding inverse requirements does not change the topology  $\tau_{<\mu}$ .

Finally, the two group-uniform conditions associated with  $\text{Fix}(A)$  are

$$p^{-1}q \in \text{Fix}(A), \quad qp^{-1} \in \text{Fix}(A).$$

The first is equivalent to forward agreement on  $A$ . The second is equivalent to agreement of the inverses on  $A$ : it says  $q(p^{-1}(a)) = a$  for every  $a \in A$ . Their intersection is  $E_A^{\leftrightarrow}$ . Finite unions combine different control sets, identifying the uniformities exactly.  $\square$

| Control convention                       | Completion or completeness                                            | Singular-cardinal warning                                        |
|------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------|
| Forward bounded prefixes                 | Full injection space; for $H_\mu$ , boundedness-preserving injections | Surjectivity can be lost.                                        |
| Forward arbitrary small sets             | Full injection space, with small-set control                          | Coordinatewise convergence alone is weaker.                      |
| Forward and inverse bounded prefixes     | Complete on the full permutation set                                  | The induced topology is still not a group topology.              |
| Forward and inverse arbitrary small sets | Complete two-sided group uniformity                                   | Its topology is $\tau_{<\mu}$ , not the original order topology. |
| Dedekind cuts of the numerical order     | Also fills cylinder edges and alphabet gaps                           | Not a uniform completion of either preceding kind.               |

For the last row, even a full injective word cannot represent the lower edge of a first-label cylinder. If  $X$  is thick and  $x \in X$ , consider the lower cut consisting of exhaustive words with first label below  $x$ . An injection starting below  $x$  lies below some words in that cut, one starting above  $x$  lies above the entire  $x$ -cylinder, and one starting at  $x$  has exhaustive words with that same first label below it. None gives exactly the cylinder-edge cut. This is the elementary reason the forward injection boundary is only part of the order-completion boundary (source 24, [proposition 65.4](#)).

## 70.7 What the group results say about surreal lexicography (source 28)

### 70.7.1 Changing labels versus changing positions

A bijection  $a : X \rightarrow Y$  acts on exhaustive enumerations by  $f \mapsto a \circ f$ . It takes prefix cylinders to prefix cylinders, and is therefore always a homeomorphism of the prefix spaces. If both alphabets are thick, it is a homeomorphism of their order topologies. It is an order isomorphism exactly when  $a$  is increasing: sufficiency follows from first disagreement, and necessity follows by choosing exhaustive words with prescribed distinct first labels.

By contrast, a permutation of positions acts by  $f \mapsto f \circ s$ . Theorem [70.2](#) shows that this is a forward homeomorphism precisely when  $s \in H_\mu$ . The distinction is conceptual: relabelling does not change which positions have been inspected, but reordering positions can move a bounded inspection set to an unbounded one.

**Proposition 70.28** (Exact filtration invariance). *The position changes that preserve the bounded-prefix topology in both directions are exactly the automorphisms of the bounded-set ideal, namely  $H_\mu$ . The topology  $\tau_{<\mu}$  is invariant under every position permutation. For singular  $\mu$ , it is the coarsest topology that contains the prefix topology and is a group topology for all position permutations.*

*Proof.* The first assertion is Theorem 70.2(ii) applied to  $s$  and  $s^{-1}$ . Every permutation preserves cardinality, so it preserves the small-set ideal and acts homeomorphically for  $\tau_{<\mu}$ . The final assertion is Theorem 70.18.  $\square$

### 70.7.2 The permutation structure is relative to a baseline

The set  $\text{Bij}(\mu, X)$  is not intrinsically a group under composition: its members have domain  $\mu$  and codomain  $X$ , so  $f \circ g$  need not even be typed. Fixing  $e : \mu \rightarrow X$  gives the group law

$$f *_e g = f \circ e^{-1} \circ g,$$

with identity  $e$ . The usual right action of  $\text{Sym}(\mu)$  on enumerations is intrinsic, but choosing an identity for a group law is auxiliary. It is useful to regard the enumeration space as a *torsor*, a space on which the permutation group acts freely and transitively, before selecting a baseline.

The subgroup  $H_\mu$  then corresponds to enumerations that send bounded position sets into bounded portions of the baseline label filtration  $e[\alpha]$ , and whose inverses have the same property. Its placement inside the surreal enumeration space depends on that label filtration. Self-normalization is a precise theorem about this specified action; it is not a claim that numerical order alone canonically chooses  $H_\mu$ .

### 70.7.3 Concrete cutoff consequences

For  $\mathbf{No}_{<\kappa}$  and  $\mu = 2^{<\kappa}$ , the following conclusions are immediate from the proved theorems. The minimal order is dense without endpoints, has size  $2^\mu$ , and contains every linear order of size at most  $\mu$ . Its ordinal and reverse-ordinal spectra stop exactly at  $\mu^+$ . Its local order-topological character is cf  $\mu$ . After choosing a baseline, the boundedness-preserving subgroup is maximal in the sense of Theorem 70.4, and, when  $\mu$  is singular, is self-normalizing and nowhere dense. The least group refinement has character  $\text{cof}([\mu]^{<\mu})$  and is two-sided complete.

These conclusions are insensitive to whether the birthday cutoff is a subfield. Arithmetic closure is not needed. Conversely, the statements do not automatically apply to an arbitrary sparse subset of  $\mathbf{No}$  with its numerical order. Prefix topology still makes sense there, but its agreement with the order topology required thickness.

### 70.7.4 Context from earlier topological classifications

At regular  $\mu$ , the predecessor manuscript constructs a homeomorphism

$$\mathcal{W}_\mu(X) \cong_{\text{top}} (\mu^{<\mu})^\mu$$

with the bounded topology (source 24, theorem 66.2). Its tree forces the least not-yet-used label to appear at each stage. Regularity keeps the union of fewer than  $\mu$  intermediate short prefixes short. At singular  $\mu$ , that specific construction can reach full length before all coverage requirements have been imposed. This is why a regular argument cannot simply be copied.

At the countable thick cutoff, the same predecessor proves that the topology is ordinary Baire-space topology, while the Dedekind completion is a real line with continuum many omitted points. It is therefore not order-isomorphic to the usual irrationals, whose real completion omits only countably many points. This is another warning that topology alone does not classify the numerical order. The bounded-topology setting is part of the broader generalized descriptive-set-theoretic landscape studied in [61]; the present ideal arguments neither assume nor establish a general classification of all such spaces.

## 71 The class analogue: global prefix completion and its limits (source 24; sources 26 and 28)

[merge] Source 24’s Sections 9–11, with source 26’s Sections 11–12 and source 28’s Sections 11–12. The class layer re-proves Part IV: the core  $\mathcal{P}_{\text{bd}}(b)$  (the three sources’  $\mathcal{C}_b$ ), its prefix completion, its density and universality, the diagonal theorem and direct comparison; these are notes. New are source 24’s small-net rigidity, its global branch-completion scheme and its two-sided global completion, the corresponding remarks of sources 26 and 28, and source 28’s observation that every set of ordinal coordinates is bounded, so that the singular obstruction of [section 69](#) does not occur globally. All statements are GBC schemes about supplied class parameters; source 24’s  $\mathfrak{E}$ ,  $\mathfrak{I}$  are printed  $\mathfrak{W}_{\text{sl}}$ ,  $\mathfrak{J}$ .

### 71.1 Global prefix completion (source 24)

We now work in GBC with the fixed baseline  $b : \text{Ord} \rightarrow \mathbf{No}$ . Besides the virtual domain  $\mathfrak{W}_{\text{sl}}$  of class bijections, write  $\mathfrak{J}$  for the virtual domain of class injections  $h : \text{Ord} \rightarrow \mathbf{No}$ . Both have a well-defined first-disagreement comparison of individual members: a nonempty class of ordinals has a least member, since any one witness bounds a nonempty set in which the least witness can be found.

A set-sized injective prefix is an actual set  $p : \alpha \rightarrow \mathbf{No}$ . The expression “[ $p$ ]” means the predicate that a given class word extends  $p$ . It does not introduce a class with those class words as its members. This distinction is harmless for ordinary pairwise proofs and indispensable for statements about all words.

#### 71.1.1 A genuinely class-sized core made from set codes

A permutation of  $\text{Ord}$  is *set supported* when its moved coordinates form a set. Such a permutation is the identity above some ordinal. Code it canonically by a permutation of the least initial segment containing its support, extended by the identity outside that segment. The identity has the empty code. The collection of these codes is an ordinary class of sets. Evaluating a code as  $b \circ \pi$  gives an exhaustive surreal enumeration. Denote the ordered class of canonical codes by  $\mathcal{P}_{\text{bd}}(b)$ .

*Note 71.1* (Source 24, Lemma 9.1 and Propositions 9.2 and 9.3). Lemma 9.1 (“Global prefix extension with a reserve”): *every set-sized injective prefix  $p : \alpha \rightarrow \mathbf{No}$  extends to the evaluation of a code in  $\mathcal{P}_{\text{bd}}(b)$ ; the extension can be selected uniformly from  $p$  and the fixed baseline and global-choice parameters.* This is [lemma 19.4](#), by a balanced reserve: with  $q = b^{-1} \circ p$  and  $S = \alpha \cup \text{ran } q$ , a disjoint block  $R$  of size  $\nu \geq |S|$  makes  $D \setminus \alpha$  and  $D \setminus \text{ran } q$  equinumerous for  $D = S \cup R$ . Source 24 comments: “The reserve is not cosmetic. A naive instruction to ‘close every partial-injection chain’ can fail for an infinite chain with unequal endpoint behavior . . . Earlier work optimizes the size of that reserve; optimality is unnecessary here” (source 17’s [theorem 49.7](#)). Proposition 9.2 (“Prefix density and numerical density”): *every set-prefix cylinder in  $\mathfrak{W}_{\text{sl}}$  or  $\mathfrak{J}$  contains a core evaluation; a core evaluation lies strictly between any two distinct injections, and on either side of each, which is [proposition 19.25](#) with the density in [theorem 19.10](#).* Source 24: “The existing report proves more: with its explicit uniform set-cut convention, this core realizes all set-sized separated cuts and is order-isomorphic to the surreal line” ([corollary 19.15](#)). “The results below need only the prefix extension and density just proved. In particular, the fact that a dense core is a surreal line should not be turned into the assertion that the entire virtual enumeration domain is an ordinary copy of  $\mathbf{No}$ .” Proposition 9.3 (“Ordinary class-order universality, recalled”): *in GBC, every linear order carried by a class of sets admits a class order embedding into  $\mathbf{No}$ , and hence into  $\mathcal{P}_{\text{bd}}(b)$  and the virtual domain  $\mathfrak{W}_{\text{sl}}$ ; the variable-carrier set-word orders also contain such an image, even when words are*

required to be numerically increasing, which is [propositions 5.13](#) and [19.20](#); source 24 uses the set-cut property, “not an assertion about arbitrary class cuts”, and a first-label embedding  $\mathbf{No} \rightarrow \mathcal{P}_{\text{bd}}(b)$ , and adds: “The qualifier ‘carried by a class of sets’ is essential. The virtual domain of *all* class enumerations is not such a carrier, so the proposition does not embed that whole higher-order domain into  $\mathbf{No}$ . This universality statement is part of the predecessor framework; it is included to connect the new completion theory to the original embedding question.”

### 71.1.2 Set-indexed nets cannot make nontrivial global progress

A *uniformly set-indexed family of class words* is given by one class relation  $F(i, \xi, x)$ , where  $i$  belongs to a fixed set  $I$ , and the  $i$ th section defines a class word  $f_i$ . A directed order on  $I$  makes it a net. Forward Cauchyness means

$$\forall \alpha \in \text{Ord} \exists i_0 \in I \forall i, j \geq i_0 \quad f_i \upharpoonright \alpha = f_j \upharpoonright \alpha.$$

All restrictions displayed here are sets. This is a formula about the one class relation  $F$ , not a quantification over a set of proper-class objects.

**Theorem 71.2** (Small-net rigidity). *Every uniformly set-indexed forward-Cauchy net in  $\mathfrak{J}$  is eventually constant. Every uniformly set-indexed net converging in the prefix sense to a specified class word is eventually equal to that word. In particular, both statements hold for  $\mathfrak{M}_{\text{sl}}$  and for core evaluations.*

*Proof.* For each pair  $i, j \in I$  whose sections differ, let  $\delta(i, j)$  be their least disagreement, and set  $\delta(i, j) = 0$  when the sections are equal. This is a class function on the set  $I \times I$ . Class Replacement gives a set of its values. Thus

$$\Gamma = \sup\{\delta(i, j) + 1 : i, j \in I\}$$

is an ordinal. Two sections in this family agreeing below  $\Gamma$  are identical. Cauchyness at  $\Gamma$  makes all sections in a tail identical, proving eventual constancy.

For convergence to a specified word  $h$ , apply the same argument to the set of first disagreements between  $h$  and the sections not equal to it. A bound on those disagreements, together with convergence, makes every sufficiently late section equal to  $h$ .  $\square$

**Corollary 71.3** (No set-sized accumulation pattern). *Relative to the prefix-neighborhood predicates, any uniformly set-indexed collection of distinct global words is discrete and has no accumulation point, even at a word outside the collection. Every set of core codes is a closed discrete subset of the core’s order topology.*

*Proof.* For a specified word  $h$ , bound the first disagreements with all indexed words other than  $h$ . The corresponding prefix neighborhood of  $h$  excludes them all. On the core, prefix cylinders form the order-topology base by the same argument as [Lemma 62.5](#), using [note 71.1](#) for completions.  $\square$

This is a foundationally meaningful difference from the countable case. A sequence indexed by ordinary  $\omega$ , or a net indexed by any set, cannot successively explore every ordinal-length prefix in the global problem. Nontrivial global approximation must have a proper-class index range or be interpreted externally in a larger universe.

### 71.1.3 The correct global branch-completion statement

**Definition 71.4** (A coherent prefix system). A coherent prefix system is one class relation specifying, for every ordinal  $\alpha$ , a set injection  $p_\alpha : \alpha \rightarrow \mathbf{No}$ , with

$$p_\beta \upharpoonright \alpha = p_\alpha \quad (\alpha < \beta).$$

**Theorem 71.5** (Global branch completion, stated as a scheme). *A coherent prefix system has a unique union  $h : \text{Ord} \rightarrow \mathbf{No}$  that is a class injection. Conversely, every class injection determines such a system. For each specified injection  $h$ , there is a uniformly Ord-indexed net of core codes whose evaluations converge forward to  $h$ . It has a limit in  $\mathfrak{W}_{\text{sl}}$  exactly when  $h$  is exhaustive.*

*Proof.* Elementary class comprehension forms the union relation. Coherence proves that it is a function on all ordinals; injectivity is witnessed in a single set prefix for any proposed pair of coordinates. The reverse construction is restriction of a class function to each set ordinal, with Class Replacement ensuring that each restriction is a set.

For each  $\alpha$ , apply the uniform extension procedure in [note 71.1](#) to  $h \upharpoonright \alpha$ . Let  $c_\alpha$  be the resulting core code and  $f_\alpha$  its evaluation. For every  $\beta$ , all  $\alpha \geq \beta$  satisfy  $f_\alpha \upharpoonright \beta = h \upharpoonright \beta$ . This is forward convergence. Any other forward limit must share every set prefix with  $h$ , hence must equal  $h$ . Therefore a limit in the exhaustive domain exists exactly when  $h$  itself is exhaustive.  $\square$

This theorem is the global analogue of [Theorem 65.1](#), but its formulation is deliberately different. It does not form the collection of all Cauchy filters on a nonexistent ordinary space of all class words. It proves a uniform reconstruction statement about each supplied class-coded system. In a larger-universe set model, it can be read as an ordinary completion theorem; within two-sorted class theory it is a scheme with class parameters.

### 71.1.4 Inverse stabilization restores exhaustiveness

For a global enumeration  $f$ , its inverse is a class function  $f^{-1} : \mathbf{No} \rightarrow \text{Ord}$ . Relative to  $b$ , a two-sided prefix comparison requires agreement of  $f, g$  below a set ordinal  $\alpha$  and agreement of  $f^{-1}, g^{-1}$  on  $b[\alpha]$ .

**Theorem 71.6** (Two-sided global completion). *Let  $(f_\gamma)_{\gamma \in \text{Ord}}$  be a uniformly class-coded net of exhaustive global enumerations. Suppose that for every set ordinal  $\alpha$ , both the forward restrictions below  $\alpha$  and the inverse restrictions on  $b[\alpha]$  are eventually pairwise equal. Then there is a unique exhaustive global enumeration  $f$  to which the net converges in both senses.*

*At each specified enumeration, forward and two-sided neighborhood predicates generate the same topology. No elementary transfinite recursion principle beyond GBC is needed for this reconstruction.*

*Proof.* For each ordinal  $\xi$ , define  $h(\xi)$  to be the unique eventual value of  $f_\gamma(\xi)$ . For each surreal  $x$ , define  $g(x)$  to be the unique eventual value of  $f_\gamma^{-1}(x)$ . The assumed stabilization makes these values exist; they are defined by an elementary formula using the class parameter for the net. Elementary class comprehension therefore gives both graphs.

If  $h(\xi) = x$ , choose a stage beyond the stabilization thresholds for that forward coordinate and that inverse coordinate. Since every row is a bijection,  $g(x) = \xi$ . The converse gives  $h(g(x)) = x$ . Thus  $h$  and  $g$  are inverse class bijections. Stabilization of entire set prefixes, not just individual coordinates, is part of the hypothesis and gives convergence in both senses. Uniqueness follows from agreement on all set prefixes.



For compatibility at a fixed  $f$ , the preimage  $f^{-1}[b[\alpha]]$  is a set of ordinals by Class Replacement, hence bounded by some ordinal  $\beta$ . Agreement with  $f$  below  $\beta$  fixes the inverse on  $b[\alpha]$ . This is exactly the forward-to-inverse neighborhood implication. The other implication is immediate.

The construction selected eventual values by a formula. It did not perform a recursion with proper-class previous values; consequently no appeal to ETR is hidden in this argument.  $\square$

The global theorem behaves like the regular-cardinal case: every *set* of ordinal positions is bounded. The singular set case differs because a set of cardinality less than  $\mu$  need not be bounded in  $\mu$ . This one quantifier distinction explains both the continuity obstruction and the apparent contrast between small-net rigidity and nontrivial external approximations.

## 71.2 Comparison beyond ordinal words, and the foundations ledger (source 24)

*Note 71.7* (Source 24, Section 10: Definition 10.1, Lemma 10.2, Theorem 10.3 and Example 10.4). Source 24 gives “the direct comparison theorem because it is the cleanest way to include the unrestricted interpretation without assuming an inappropriate class-order-type calculus”. Its common labelled initial segment  $I(R, S)$ , its maximality lemma and its Theorem 10.3 (“Elementary lexicographic comparison of all class well-orders”): *for distinct class well-orders  $R, S$  of a labelled class  $X$  with a fixed linear order  $<_X$ ,  $R <_{\text{lex}} S$  iff  $a <_X b$  for the first residual labels; this is a strict linear comparison, it agrees with first-disagreement comparison for ordinal-indexed enumerations, and it uses elementary class comprehension, not general class well-order comparability or ETR*, are [theorem 14.2](#) and [theorem 14.7](#), by the same proof. Source 24: “This theorem reprises a central construction in the earlier report, with the logical dependencies exposed. It is included here as a tool, not presented as a new resolution of an old open problem.” Its Example 10.4 (“A genuinely proper-class prefix”: a set-like well-order of  $\mathbf{No} \setminus \{u < v\}$  followed by  $u, v$  or by  $v, u$ ) is the family of [theorem 16.2](#); source 24 adds that “non-set-like orders have tail geometry invisible to the first Ord-long set-like part . . . Therefore our cylinder topology, completion, and regular-cardinal Baire theorems should not be transferred wholesale to the unrestricted order.”

[merge] Source 24’s Section 11, “Paradox-free foundations and the size ledger”.

### 71.2.1 Four different levels of collection

The safest practice is to attach a size declaration to every carrier and every family. The following ledger separates constructions that are often written with deceptively similar notation.

| Object                                                    | Legitimate status                                                      | What must not be inferred                                                |
|-----------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------|
| $\mathcal{W}_\mu(X), \mathcal{I}_\mu(X)$ for set $X, \mu$ | Sets in ZFC; ordinary topology, filters, and powersets apply.          | They need not be sets inside a smaller chosen universe.                  |
| All set-length surreal words                              | A definable class of set graphs.                                       | This class is not a set; its powerclass is not an ordinary class.        |
| The support core $\mathcal{P}_{\text{bd}}(b)$             | A class of canonical set codes with a uniform evaluation relation.     | Its dense image does not exhaust all global enumerations.                |
| All class enumerations or all class well-orders           | A virtual domain, a higher-sort domain, or a set in an external model. | Proper classes do not become members of a same-level class in GBC or KM. |

The first two levels are enough to prove many order embeddings. The third supplies a practical surrogate for global prefixes. The fourth is needed only when one insists on quantifying over, or making functions whose arguments are, arbitrary global enumerations as objects.



*Note 71.8* (Source 24, Theorem 11.1, “No uniform class of set codes exhausts the global words”). In GBC, let  $A$  be a class of sets and let one class relation uniformly assign a global enumeration  $f_a : \text{Ord} \rightarrow \mathbf{No}$  to each  $a \in A$ ; this family cannot exhaust  $\mathfrak{W}_{\text{sl}}$ . This is [theorem 19.37](#), by the same pair-swap diagonal on the blocks  $\{2 \cdot \alpha, 2 \cdot \alpha + 1\}$ . Source 24: “This gives a concrete reason that  $\mathcal{P}_{\text{bd}}(b)$  is a core rather than an exhaustive parametrization. Density is a local statement about each prefix or each interval. Exhaustive coding would be a global surjectivity statement, and the diagonal argument forbids it. There is no contradiction between a surreal-sized dense core and a larger virtual domain.” Its Example 11.2 below is the inconsistency of unrestricted cuts recorded in the foundations report (`found:prop:allcuts`, formalized as `noUnrestrictedCuts`; [section 1.5](#)).

### 71.2.2 Set-like recursion versus recursion with class-valued stages

Several constructions above are sometimes described informally as “transfinite recursion,” but they do not all require the same principle. Recursion of set-length along a set ordinal is ordinary set recursion. A uniform Ord-length construction whose stage values and previous histories are sets can be described by its unique set-length initial solutions, then assembled by elementary class comprehension. This is different from a recursion in which the value at each stage is itself an arbitrary proper class.

The latter is the setting of elementary transfinite recursion principles for classes. Their nontrivial strength is reflected, for example, in the equivalence between suitable clopen class-game determinacy and ETR studied by Gitman and Hamkins [22]. It is therefore not valid to invoke arbitrary class recursion while claiming to work only in GBC.

In source 24, the common-prefix formula is elementary; the supported extension procedure takes set parameters to set codes; coherent-prefix unions are given class relations; and eventual-value reconstruction is an elementary definition from a supplied net. None of these arguments assumes an unrestricted proper-class recursion theorem. A future construction of an automorphism by class-length back-and-forth should be audited anew, rather than inheriting this conclusion merely because it also mentions prefixes.

### 71.2.3 An external inaccessible-universe model

There is a transparent way to turn the virtual global picture into an ordinary set-sized one. Work externally in ZFC and assume a strongly inaccessible cardinal  $\Theta$ . Let

$$M = V_\Theta, \quad \mathcal{X} = \mathcal{P}(V_\Theta).$$

Interpret sets as members of  $M$  and classes as all external subsets of  $V_\Theta$ . This full class model satisfies the class-comprehension and replacement requirements of KM with Global Choice: regularity bounds the ranks of images of internal sets, and strong inaccessibility ensures that the required internal sets and powersets lie below  $\Theta$ .

Its ordinals are externally the ordinals below  $\Theta$ , and its surreal class is externally  $\mathbf{No}_{<\Theta} = \mathbf{No}_{<\Theta}$ , of size  $\Theta$ . Consequently its full set-like global enumeration domain is externally the ordinary set

$$\text{Bij}(\Theta, \mathbf{No}_{<\Theta}) = \mathcal{W}_\Theta(\mathbf{No}_{<\Theta}).$$

Theorem 66.2 applies to it. In this model  $\Theta^{<\Theta} = \Theta$ , so the external topological presentation is  $\Theta^\Theta$  with the bounded topology. The arbitrary internal class well-orders, because all subclasses are available, correspond externally to all genuine well-orders of the set  $\mathbf{No}_{<\Theta}$ , not merely its minimal-length slice.

There is no conflict with Theorem 71.2. An internal set-indexed net has an index set in  $V_\Theta$  and hence fewer than  $\Theta$  indices. An external  $\Theta$ -indexed net need not have an internal set of indices and can make nontrivial progress. The distinction is exactly the one the theorem requires.

The inaccessible hypothesis is a sufficient external modeling device, not an axiom claimed to follow from ZFC, and not a prerequisite for the elementary set-sized results. Other class models can be used, but their available classes and their external size must then be specified.

#### 71.2.4 Definable classes, class models, and higher universes

There are three complementary formalization routes.

**Schematic definable classes.** In a set-theoretic metatheory, work with explicitly supplied formulas and set parameters defining the relevant classes. Each theorem is a schema about such definitions. This is efficient for fixed examples and elementary comparison. It does not yield a single internal truth predicate for all formulas or an internal enumeration of all semantic classes. Syntax codes are not automatically an exhaustive semantic universe.

**A two-sorted class model.** Specify a model  $(M, \mathcal{X})$  of GBC or a stronger theory. Every assertion about “all global enumerations” means all such class relations belonging to  $\mathcal{X}$ . Enlarging  $\mathcal{X}$  while preserving  $M$  can add new branches and new enumerations. A set-coded core can remain unchanged, but this does not make every assertion about all classes absolute. The previous cut-reconstruction work addresses more refined preservation questions (sources 17 and 14; section 48); here we use only the explicit parameter-relative interpretation.

**A universe hierarchy.** In a type-theoretic or Grothendieck-universe setting, collect the small surreals at one level and their large relations at the next. Their functions and cuts can then be ordinary objects at a sufficiently high level. This is often the easiest route for a proof assistant. It remains essential not to identify a type with a type containing all of its own higher-universe relations. Moving to a higher universe resolves a size problem; erasing the universe distinction recreates it.

**Example 71.9** (Why all class cuts cannot be filled by surreals). Every surreal is below some ordinal surreal: compare its sign sequence with a longer all-plus sequence. Thus the lower class consisting of all ordinal surreals has no surreal upper bound. The separated class cut with that lower side and empty upper side has no surreal separator. The familiar set-cut interpolation property is therefore not a theorem about arbitrary class-indexed cuts. The word “set” or “small” in such a theorem is a mathematical hypothesis, not bookkeeping.

### 71.3 Global well-orders without a class of classes (source 26)

#### 71.3.1 Set-like versus unrestricted class well-orders

Work in GBC. A class relation  $R$  on a class  $C$  is a class well-order when it is a strict linear order and every nonempty admitted subclass has an  $R$ -least element. It is *set-like* when

$$\text{Pred}_R(x) = \{y \in C : yRx\}$$

is a set for each  $x \in C$ . For a proper class, a set-like class well-order has a unique ordinal-class enumeration: the position of  $x$  is the set ordinal  $\text{otp}(\text{Pred}_R(x), R)$ . Its range of positions is an unbounded initial segment of  $\text{Ord}$ , hence all of  $\text{Ord}$ .

Fix a baseline class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . Global choice supplies such a bijection. Set-like surreal well-orders can be represented by other class bijections  $e : \text{Ord} \rightarrow \mathbf{No}$ . Their labels at the first differing ordinal determine their comparison. In contrast, an unrestricted class well-order can have type  $\text{Ord} + 1$  or  $\text{Ord} + \text{Ord}$ , after a suitable class bijection of its carrier with  $\mathbf{No}$ . It cannot be represented by an exhaustive Ord-word: the tail beyond the initial proper-class segment would be lost.

The repository proves the equivalence, over its convention for GB without set choice, between the existence of a class well-order of  $\mathbf{No}$  and global choice ([theorem 13.1](#)). We do not repeat the full weak-choice coding proof. All global constructions below explicitly assume GBC, so no additional choice claim is concealed in the selection of  $b$ .

The expressions “all set-like class well-orders” and “all class well-orders” below denote predicates on class variables. In two-sorted GB or KM, a class has *sets* as members. Stronger class comprehension in KM does not turn classes into allowable members of another class. An actual collection whose elements are class relations needs a higher sort, external semantics, or a separately specified coding restriction.

*Note 71.10* (Source 26, Section 11: Definition 11.1, Lemma 11.2, Theorem 11.3, Example 11.4, Lemma 11.5, Proposition 11.6 and Theorem 11.7). Source 26 prints “a useful earlier result”, with a proof “to expose its precise logical strength”: the common labelled prefix  $D_{R,S}$ , its Lemma 11.2 and its Theorem 11.3 (“A virtual linear order of all labelled class well-orders”), which are [theorems 14.2](#) and [14.7](#) and source 24’s [note 71.7](#). Its Example 11.4 (“A genuine adjacent pair”: a set-like enumeration of  $\mathbf{No} \setminus \{a, b\}$  followed by  $a, b$  or by  $b, a$ , two unrestricted well-orders of type  $\text{Ord} + 2$  that are adjacent) is [theorem 16.2](#); the example “also demonstrates why a topology using only set-length initial prefixes cannot distinguish every unrestricted tail: both orders have the same prefix at every ordinal position, yet they are distinct.” Its Lemma 11.5 (“Prefix extension with an explicit reserve”) is source 24’s [note 71.1](#) ([lemma 19.4](#)); “the previous report proves considerably more about this core ... The reserve lemma is the only part needed for the approximation arguments below.” It adds “a direct simple embedding of the numerical surreal order into the code class”: swap baseline position 0 with position  $b^{-1}(x)$ , so that the enumeration starts with  $x$ . Its Proposition 11.6 (“Class-order universality, with the coding boundary explicit”) is [propositions 5.13](#) and [19.20](#), with the remark that “the recursion here is set-valued ... It is not an ETR assertion producing a new proper class at every stage”; “This is an order-theoretic universality statement, not a claim that  $\mathcal{P}_{\text{bd}}(b)$  exhausts all global well-orders.” Its Theorem 11.7 (“Diagonal obstruction”): *no uniformly Ord-indexed family of class bijections  $\text{Ord} \rightarrow \mathbf{No}$  contains every such bijection; consequently, in GBC, no ordinary class of set codes with a uniform evaluation relation codes all set-like global surreal well-orders*, which is [theorem 19.37](#), by the same pair swap; “The conclusion is not an inconsistency in global well-orders. It is a limitation on a proposed uniform coding at the wrong logical level.”

*Note 71.11* (Source 26, Theorem 12.1, “Small-net rigidity”). *In GBC, every uniformly set-indexed prefix-Cauchy net of class Ord-words is eventually constant as a family of class functions.* This is source 24’s [theorem 71.2](#), by the same Class Replacement bound on the set of first disagreements. Source 26 observes that stating the scheme needs no “topology whose points are classes ... as a third-sort object”.

Class-indexed approximation is different. Given a class injection  $h : \text{Ord} \rightarrow \mathbf{No}$ , for every ordinal  $\alpha$  apply [note 71.10](#) to  $h \upharpoonright \alpha$  and choose a support-code extension uniformly. This Ord-indexed family stabilizes to  $h$  on each set prefix. The limit can fail to be exhaustive. For example, relative to a baseline, shift the finite ordinal positions by one and leave positions at least  $\omega$  fixed; the resulting injection omits exactly the first baseline label.

If forward and inverse prefixes both stabilize for a uniformly coded family of bijections, their

unions are mutually inverse. Indeed, when a stabilized forward value is  $h(\xi) = x$ , take a common sufficiently late stage at which that coordinate and the inverse coordinate  $x$  have stabilized. The bijection at that stage witnesses  $g(x) = \xi$ . The converse is the same. This is a reconstruction scheme for admitted class graphs, not the creation of a completed class whose elements are all class graphs.

### 71.3.2 A fixed class relation versus an enlarged class part

A model of class theory is written  $(M, \mathcal{X})$ , where  $M$  supplies the sets and  $\mathcal{X}$  the admitted classes. The class part need not consist of every externally available subset of  $M$ . A definition using an external collection of all subsets can therefore change the meaning of a supposed internal theorem.

There is, however, an important preservation fact for an *old fixed relation*. In GBC, internal well-foundedness of a fixed strict class relation is equivalent to the absence of a set-coded descending  $\omega$ -sequence. One direction is immediate from the range of such a sequence. Conversely, from a nonempty class with no minimal element, global choice selects a predecessor at each point, and recursion on  $\omega$  followed by Class Replacement yields a descending sequence that is a set.

Consequently, when  $(M, \mathcal{X})$  and  $(M, \mathcal{Y})$  both satisfy the relevant GBC assumptions, have the same sets, and an old relation  $R$  belongs to both class parts, enlarging the class part does not change whether that old relation is internally well-founded. What can change is *which relations exist*, which uniform families and cuts are admitted, and which prospective limits are available. The repository's reviewed scope correction makes this distinction explicit (this report's README, review notes). External well-foundedness of a nonstandard model is a different question: an external descending sequence need not be a sequence coded in  $M$ .

### 71.3.3 Higher universes are a legitimate alternative, not a size eraser

A universe-relative treatment can fix a strongly inaccessible  $\Theta$  and view

$$(V_\Theta, \mathcal{P}(V_\Theta))$$

from an ambient universe. Its internally proper classes are externally sets. The internal surreal sign carrier is represented externally by signs of length below  $\Theta$ , and global relations become external relations on that set. It is then legitimate to take an external set of those relations. What is illegitimate is to move that set back down and call it an internal class whose members are internal proper classes.

In type theory, the analogous safeguard is explicit universe levels. A carrier for ordinal-length signs lives above the universe of its ordinal lengths, while every fixed birthday-bounded fragment can be small in that lower universe. A type of all predicates on the carrier lives at a higher level again. These level changes can formalize the objects without paradox, but they alter which families count as small and which cut-filling theorem is applicable.

A singular birthday fragment such as  $\mathbf{No}_{<\beth_\omega}$  is a perfectly sound set-level alphabet. It is not a replacement for the inaccessible model just described. Regularity, replacement closure, power sets, and externally full class parts must be checked for a proposed model rather than inferred from an evocative notation.

### 71.3.4 Where stronger recursion principles actually enter

The direct common-prefix comparison in [note 71.10](#) does not need elementary transfinite recursion. Ordinary set-valued recursion used to select one surreal at each stage is also distinct from a recursion that constructs an entire new class at each stage. The latter is the setting in which ETR becomes

a substantive additional principle. Gitman and Hamkins identify its strength over GBC through clopen class-game determinacy [22].

A formal development should therefore state the exact recursion rule and the sort of its values. Invoking KM for every global argument overstates some dependencies; treating arbitrary class-valued recursion as an automatic consequence of elementary comprehension understates others. The present manuscript confines itself to elementary comparison, set-valued recursive constructions, and explicitly uniform family schemes. No theorem here assumes that all higher-order cuts can be filled.

## 71.4 Source 28's global sections

[merge] Source 28's Sections 11 and 12 re-prove the global layer; their one new point is printed whole: every set of ordinal coordinates is bounded, so the singular obstruction of sections 69 and 70 has no global counterpart.

### 71.4.1 The virtual enumeration domain

Work in GBC, and fix a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . Such a baseline can be obtained by globally choosing well-orderings of the set-sized birthday levels and listing the new surreals level by level. Every proper initial part of this listing is a set. The merged repository report analyzes the precise equivalence with Global Choice over its stated weak base theory (theorem 13.1); the arguments here assume the baseline rather than silently selecting one in bare GB.

Let  $\mathfrak{W}_{\text{sl}}$  mean the virtual domain of all class bijections  $f : \text{Ord} \rightarrow \mathbf{No}$ . It is notation for the objects over which class variables range under a stated predicate, not a declaration that those proper classes are members of an ordinary class. For two specified distinct class words, the first disagreement exists: choose one ordinal witness, and find the least witness inside the set of ordinals at most that one. Their numerical lexicographic comparison is therefore an elementary class-theoretic assertion.

A uniformly indexed family is represented by a single relation

$$F(i, \xi, x),$$

whose  $i$ th section is the graph of the relevant word. Quantifying over the set or class index  $i$  does not turn each section into a set. This representation is sufficient for the global prefix and continuity arguments below.

### 71.4.2 Every set of ordinal coordinates is bounded

**Proposition 71.12** (Why the global case has no singular-coordinate obstruction). *For every specified class permutation  $s : \text{Ord} \rightarrow \text{Ord}$ , both  $s$  and  $s^{-1}$  send bounded sets of ordinals to bounded sets. Forward and two-sided set-prefix neighborhood predicates agree at each specified global enumeration. Left and right translations and inversion satisfy the corresponding continuity formulas.*

*Proof.* A bounded collection of ordinals is a set. By Class Replacement, its image and inverse image under a specified class permutation are sets of ordinals. Every such set has an ordinal bound. The proofs of Theorem 70.2 now apply to each allowed control set, with set-sized images and inverse images. In particular, to enforce inverse agreement on  $b[\alpha]$  at  $f$ , it suffices to enforce forward agreement below an ordinal bounding the set  $f^{-1}[b[\alpha]]$ .  $\square$

This is a statement about formulas and specified class parameters, not an ordinary topology carried by a class whose elements are all the global enumerations. The analogy with regular cardinals is exact at the relevant quantifier: every *set* of ordinal coordinates is bounded. At singular  $\mu$ , a subset of cardinality less than  $\mu$  may instead be unbounded in  $\mu$ .

*Note 71.13* (Source 28, Lemma 11.2, Proposition 11.3, Theorem 11.4, Proposition 11.5, Lemma 12.1 and Theorem 12.2). Source 28’s Lemma 11.2 (“Global prefix extension by a set-supported code”) is [lemma 19.4](#) and source 24’s [note 71.1](#); its Proposition 11.3 (“Universality for ordinary class carriers”; “every set ordinal and every ordinary class linear order, including class well-orders such as  $\text{Ord} + 1$  and  $\text{Ord} + \text{Ord}$ , is included”) is [propositions 5.13](#) and [19.20](#); its Theorem 11.4 (“The diagonal obstruction, explicitly typed”:  $g_C = b \circ s_C$  with  $s_C$  exchanging the two members of pair  $i$  exactly when  $i \in C$ , decoded by one elementary formula  $D$ , and the diagonal class  $C_* = \{i : \neg D(i, i)\}$ ; “No truth predicate for arbitrary formulas is used”) is [theorem 19.37](#); its Proposition 11.5 (“Set-indexed prefix-Cauchy nets are eventually constant”) is source 24’s [theorem 71.2](#), its third proof in the batch; its Lemma 12.1 and Theorem 12.2 (“Direct lexicography beyond set-like orders”) are [theorem 14.2](#) and [theorem 14.7](#). Source 28 adds that a dense core of set codes “must not be confused with an exhaustive parametrization of the whole virtual domain”; that the coherent-branch approximations are “the appropriate global analogues of forward completion; they do not require a class containing all Cauchy filters” on  $\mathfrak{W}_{\text{sl}}$ ; that a supplied class-indexed net stabilizing forward prefixes and inverse restrictions to  $b[\alpha]$  reconstructs mutually inverse class maps by the two-coordinate argument of [theorem 70.27](#), with no “general recursion with previously constructed proper classes as values”; and, of unrestricted orders, that “an order with type  $\text{Ord} + 1$  cannot be exhaustively represented by an order-preserving list with domain  $\text{Ord}$ ”, that two orders sharing a common initial segment with complement  $\{a, b\}$  are adjacent, that relation coding gives a different order already on two labels, and that “numerical lexicography is not compatible with merely quotienting presentations by their unlabelled order types”.

## 71.5 Foundations, models and paradox avoidance (source 28)

[merge] Source 28’s Section 13, printed whole; it states for the group results what sources 24 and 26 state above for theirs.

### 71.5.1 Set approximations are ordinary mathematics

For a fixed set alphabet  $X$ , well-order relations lie in  $\mathcal{P}(X \times X)$ , and the spaces  $\text{Bij}(\mu, X)$  and  $\text{Inj}(\mu, X)$  are sets. Their topologies, filters, completions, and permutation subgroups are standard set-sized objects. This is the cleanest setting for formalizing the new theorems. A birthday cutoff specifies both the carrier and the relevant cardinal parameters explicitly.

One should not replace  $\mu = 2^{<\kappa}$  by  $\kappa$  merely because  $\kappa$  is the birthday bound. One should also distinguish subsets of size less than  $\mu$  from subsets belonging to a particular inner model. The first is a cardinal condition in the ambient set theory; the second is a membership condition relative to a foundational model.

### 71.5.2 Two-sorted class theory

In a model  $(M, \mathcal{X})$  of class theory, the sets are the elements of  $M$  and the available classes are the elements of the external collection  $\mathcal{X}$ . Internally, a class is a predicate on sets, not an element that can be inserted into an ordinary set or class. A formula may assert that a class variable is a bijection  $\text{Ord}^M \rightarrow \mathbf{No}^M$ , and may compare two such variables. It may not silently collect every one of those class variables into another ordinary class.

Global Choice supplies useful baseline listings and uniform choices of set-coded extensions. It is not a license for an exhaustive class list of all class bijections: [Note 71.13](#) rules that out. Stronger comprehension in KM still does not erase the type distinction between sets and classes.



Elementary comprehension is enough for the common-prefix definition, the diagonal construction, and the eventual-value reconstruction of a supplied stabilized net. General recursions in which earlier values are themselves proper classes require separate analysis. Gitman and Hamkins [22] distinguish elementary transfinite recursion over well-founded class relations from the strength of GBC and relate it to clopen class-game determinacy. We do not use that stronger recursion principle in the proofs just listed.

### 71.5.3 Larger universes and full external collections

There is a second rigorous approach: place the smaller theory inside a larger set universe. Conditionally, let  $\Theta$  be strongly inaccessible and take  $M = V_\Theta$ , with the full external collection of subclasses of  $M$ . The objects called proper classes inside  $M$  are sets externally. One may then form, externally, the set of all internal class enumerations and put an ordinary topology on it.

This is a relative model construction with a stated additional hypothesis, not a proof in bare ZFC that such an inaccessible exists. The external indexing ordinal is  $\Theta$ , and the surreals built from sign sequences of length less than  $\Theta$  form an external set of size  $\Theta$ . Since  $\Theta$  is regular, the singular-coordinate obstruction disappears for external subsets of size less than  $\Theta$ . But a general external subset of size  $\Theta$  is not an internal set-sized control object, so no claim about arbitrary external cuts follows from internal set-cut interpolation.

A Grothendieck-universe or dependent-type-theoretic presentation makes the same size distinction syntactic: words small in one universe form a type in a larger universe. The larger type is not automatically all-surreal mathematics at the same universe level.

### 71.5.4 Changing the available classes

If  $(M, \mathcal{X}) \subseteq (M, \mathcal{Y})$  have the same sets but more available classes in  $\mathcal{Y}$ , two old ordinal-class words retain their first-disagreement comparison. Their graphs and numerical labels have not changed. New class words can nevertheless appear, along with new approximation systems and new uniform families. An old proof that a class relation is a well-order must also be assessed relative to the intended available subclasses; internal and unrestricted external well-foundedness are not interchangeable.

In an externally countable model, an external countable sequence may be cofinal in  $\text{Ord}^M$ , despite the absence of such an internal set sequence. Thus the small-net rigidity theorem does not prohibit external approximations indexed by a set that is not an internal set of indices with the required internal uniform coding. Its quantifiers were internal and explicit.

### 71.5.5 Definable orders and syntactic codes

One can restrict to class orders defined by specified formulas with set parameters, but “all definable classes” is not automatically an internally evaluable class-coded family. A uniform evaluator for arbitrary definitions would require an appropriate satisfaction predicate. A safer construction fixes a single formula  $\varphi(i, \xi, x)$  and varies the set parameter  $i$ , or works externally with formula codes and an explicitly provided satisfaction relation. The diagonal obstruction applies to any genuinely uniform exhaustive family once the required evaluations are available.

The same caution applies to definability claims motivated by the Kanovei–Shelah construction [16]. Their published index set consists of functions from the continuum cardinal to  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters, ordered lexicographically. These functions need not be injective. That construction motivates definable lexicographic indexing, but it is not literally the order of all well-order relations on the reals or on the surreals.



### 71.5.6 A short paradox checklist

Before using an assertion about “all” surreal well-orders, specify the carrier, its size level, whether labels are retained, the permitted enumeration domains, and the comparison rule. For a completion argument, specify the topology or uniformity and the index size of the approximation. For a class construction, identify one uniform relation representing any claimed family, and say whether the statement is internal or external. For a recursion, identify the type of its values and the size of its previous history.

The apparent paradoxes arise when one of these specifications is suppressed. In particular, a class can contain codes for some class functions without containing every class function as an element; realizing every set cut does not mean realizing every class cut; and a universe-polymorphic construction does not collapse all its universe levels into one.

## 72 What this Part answers

[merge] Dated notes, 3 October 2026. Each item names the question, where it is asked, and the result that answers it. Where the question index of [section 113](#) or the status notes of Part VI ([section 33](#)) say otherwise, these notes supersede them; questions of the batch-81 and batch-83 sources are named by their numbers in those sources.

- (a) *Source 12’s question on all fixed enumeration types* ([research question 113.11](#)), *source 08’s question 4*, and *Part VI’s open boundary* ([section 31.3](#); [section 33](#), item (e)): the embedded binary cubes are determined for every fixed type,  $\mathbb{B}_\theta \hookrightarrow \mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$  iff  $\theta \leq \varrho_{\text{lo}} + \lfloor \log_2(n!) \rfloor$  ([theorem 64.11](#)); the equimorphism type and the binary coding length are determined whenever the two computed weights agree, in particular when every residual cardinal is regular, singular strong limit or above  $\kappa$  ([theorem 64.7](#) and [corollary 64.13](#)), hence for every type when  $\mu \leq \aleph_\omega$  ([corollary 64.8](#)), at every cutoff  $\kappa < \aleph_\omega$  in ZFC ([corollary 64.20](#)) and for every type under GCH ([note 64.18](#)). **Open:** the least coding length when a singular non-strong-limit residual cardinal at most  $\kappa$  makes  $\varrho_{\text{lo}} < \varrho_{\text{up}}$ ; source 19 asks it as its Question 14.1 (“The remaining coding gap”), and sources 29 and 30 ask it again (their Questions 13.1 and 12.1). The same results answer the fixed-type questions of sources 15–18 ([research questions 113.46](#), [113.58](#), [113.69](#) and [113.83](#)) in the same extent.
- (b) *Source 12’s question on the order topology, isomorphism invariants and degrees of homogeneity of the minimal slice* ([research question 113.13](#); Part VI left the isomorphism type and homogeneity open, [section 33](#), item (b)): answered further. The topology is determined at every regular  $\mu = 2^{<\kappa}$ , as the bounded-topology space  $(\mu^{<\mu})^\mu$  ([corollary 66.3](#)); the minimal surreal slice is metrizable exactly when  $\text{cf } \mu = \omega$  and completely metrizable exactly when  $\kappa = \omega$  ([theorem 69.27](#)); at  $\kappa = \omega$  it is homeomorphic to Baire space, has completion  $\mathbb{R}$  with continuum many omitted points in every interval, is not isomorphic to the irrationals ([note 67.1](#)), is strongly finitely homogeneous with  $2^{\aleph_0}$  automorphisms, and its proper prefix cylinders are not definable in the pure order ([theorem 67.8](#) and [corollary 67.9](#)). The weights and all point characters of the finite-tail and  $\mu + \omega$  strata of every thick alphabet are determined ([theorems 63.4](#), [63.9](#) and [63.15](#)). **Open:** the order type of the minimal slice at uncountable  $\kappa$ , and its topology at singular  $\mu$  (source 24’s question “A singular replacement for the exhaustion tree”). Source 15’s question on the finer topology of bounded slices ([research question 113.52](#)) is answered in the same extent for local bases and the cardinal invariants; source 18’s question on the arrangement of gaps ([research question 113.82](#)) in part, by the normal forms of the completion with its realized points ([theorem 65.11](#)); source 17’s question on isomorphism after

- the invariant calculation (research question 113.71) is not settled, but at  $\kappa = \omega$  the placement of the omitted cuts is an invariant beyond the topology (corollary 67.5 and theorem 67.10).
- (c) *Part VI's disclaimer after the fusion theorem* (“The theorem does not claim a complete metrizability”, after theorem 30.5) and *source 17's question “Beyond the proved fusion length”* (research question 113.72: “When  $q = \omega$ , determine complete metrizability and more precise descriptive topological properties”): answered for the minimal slices. When  $q = \omega$  the slice is completely metrizable exactly when  $\mu = \omega$ ; at uncountable  $\mu$  of countable cofinality it is not Borel in its injection completion, no nonempty open subspace is completely metrizable, and for  $\mu > \mathfrak{c}$  every zero-dimensional separable metrizable space is a closed subspace (theorems 69.17 and 69.18 and corollary 69.21). The least size of a family of dense open sets with nondense intersection remains open (source 24's and source 26's questions “Dense-open intersection thresholds”).
  - (d) *Source 24's Question 13.1* (complete metrizability at uncountable countable cofinality) and *source 28's Question 1*: **answered** negatively in ZFC (notes 69.7 and 70.13). Source 28's Question 3 (descriptive complexity of the one-sided loci): in part (theorem 69.23 and proposition 69.24).
  - (e) *Source 24's question “The boundedness-preserving subgroup”* (its closure properties, normalizers, topological-group structures on larger natural subgroups, and recovery of the bounded-set ideal): answered in the natural permutation action by source 28:  $H_\mu$  is the largest subgroup containing the finitary permutations on which the prefix topology is a group topology (theorem 70.4), it is self-normalizing at singular  $\mu$  (theorem 70.6), its forward closure is  $H_\mu^+$ , nowhere dense at singular  $\mu$  (theorem 70.10), and the bounded ideal is recovered from the group with its action (proposition 70.8). Source 24's question “Uniformities with an exhaustion interpretation”: in part (theorems 70.18 and 70.27). Abstract automorphisms of  $H_\mu$  remain open (source 28's question “Abstract automorphisms versus the natural normalizer”).
  - (f) *Source 22's Question 12.4* (“Descriptive topology when cofinalities are countable”): answered for the minimal lengths (theorem 69.18) and for the countable lengths  $\omega + n$ : metrizable exactly when  $n \leq 1$ , completely metrizable exactly when  $n = 0$ , the split space at  $\omega + 2$  separable, first countable, meagre and not metrizable (theorems 68.8 and 68.9). Its Question 12.2 (when are  $Q_n(X)$  and  $Q_m(X)$  isomorphic) remains open at  $\mu = \omega$ , where  $Q_n(D) = E_n$ ; the orders  $P_{\omega+n}$  themselves are pairwise nonisomorphic (corollary 68.7).
  - (g) *The reals report's question “Other ground orders”* (lwo:sec:research; section 33, item (g)): answered further. For  $X = \mathbf{No}_{<\kappa}$  and  $\alpha = \lambda + n$  with  $\lambda$  a limit, outside the open case of item (a) the display  $\mathcal{W}_{\lambda+n}(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$  holds exactly when the weighted length  $\varrho_{\text{lo}}$  equals  $\lambda$  (theorem 64.7 and note 64.12); the weight of every finite-tail stratum  $\mathcal{W}_{\mu+n}(X)$  is determined for every infinite alphabet (corollary 63.12). The gap spectrum of the full order  $\mathcal{W}(X)$  stays open.

Sources 24, 26 and 28 present their singular theorems as answers to questions of the sources they read; items (c)–(e) record which. Source 22 says its table is “now treated as proved”; for density and cellularity that was source 16's result (note 63.5).

## Part XI

# Across birthday cutoffs: restrictions, coherent limits and birthday spectra

[merge] (3 October 2026.) This Part belongs to the batch-83 addition of twelve further manuscripts, sources 19–30; its notation is that of [sections 1.3](#) and [1.3.1](#) with the batch-83 addendum ([section 1.3.2](#)), and the conventions fixed below. The set-sized Parts (I, VI and X) hold the birthday cutoff fixed. This Part prints what sources 19, 20, 21 and 29 prove about *changing* the cutoff: how the enumeration orders at two cutoffs are related by maps, how the orders at all cutoffs fit together into class well-orders of **No**, and what the cutoff itself, initial cardinal or not, leaves visible in the alphabet and in its enumeration strata.

- Restriction to a smaller carrier is order preserving for the native order in exactly three degenerate cases (source 19, [section 73](#)).
- Restriction-coherent profiles along all ordinal cutoffs are exactly the class well-orders of **No**; two labels decide every later local comparison, so eventual local comparison is the native order on the inverse limit; local ranks and a club of strong-limit cutoffs detect set-likeness; and the append-tail maps form a coherent direct system whose limit is the bounded-support core (source 20, [section 74](#)).
- Within one cardinality band, a single baseline gives exactly composing fixed-slot embeddings of the minimal slices (source 21, [section 75](#)).
- The binary cube  $\mathbb{B}_\beta$  embeds in  $\mathbf{No}_{<\theta}$  exactly when  $\beta < \theta$ , so the birthday alphabets are strictly ordered by embeddability even where their cardinality is constant (source 29, [section 76](#)).
- At a cutoff  $\theta$  that is not an initial cardinal, the minimal slice has exact cardinal invariants, coding rank  $\mu = 2^{|\theta|}$ , global cofinality  $\text{cf } \lambda$  ( $\theta = \lambda + n$ ), and is not isomorphic to any interval or ray of itself (source 21, [section 77](#)); every enumeration length at such a cutoff has a coding sandwich, an exact cube spectrum and an exact birthday threshold, and on the countable band all lengths are classified up to equimorphism without CH (source 29, [section 78](#)).

[Section 79](#) records which questions the Part answers.

The four sources' sections printed here are complementary, so there is no base to choose; one clause is proved twice, the coding rank  $\mu$  of the minimal slice at a noninitial cutoff (source 21's [theorem 77.4](#), and source 29's [corollary 78.9](#) at  $\alpha = \mu$ ), by different arguments, and both are printed as parts of larger theorems. Sources 19 and 20 (pin d5bd4a67b) and source 21 (pin 109aaca15) were written independently of one another and before Parts VI–VIII were written; source 29 (pin 7ff7736ec) read the delivered manuscripts of sources 19, 21 and 20 (its “earlier residual analysis”, “earlier structure” and “earlier limits” manuscripts) and credits them. Where source 29's Section 5 re-proves source 19's residual-tail lemmas for initial cardinal cutoffs, printed in Part X ([section 64](#)), the shared statements are recorded in notes and only the noninitial statements are printed in full; its Corollary 4.2 is recorded as a note to Parts I and VI. Source 19's restriction theorem refines source 15's three-label theorem of Part VII ([theorem 38.5](#)); source 20's eventual comparison explains why that obstruction does not prevent a native order on the inverse limit.

**Conventions of this Part.** [merge] All sums and products of ordinals are ordinal operations. For every ordinal  $\alpha$ ,  $\mathbf{No}_{<\alpha} = \{x \in \mathbf{No} : \text{bd}(x) < \alpha\}$ ; source 20's  $X_\alpha$  and source 29's  $X_\theta$  are printed this way. When the cutoff is an infinite initial cardinal it is written  $\kappa$  and  $\mu = 2^{<\kappa}$ , as in Parts VI–X.

When the cutoff is an infinite ordinal  $\theta$  that is *not* an initial cardinal (sources 21 and 29), both sources put  $\kappa = |\theta|$  and  $\mu = |\mathbf{No}_{<\theta}| = 2^\kappa$ , and this Part keeps their letters in those sections. *False reading:* there,  $\kappa < \theta$  is not the cutoff, and  $\mu = 2^\kappa$ , not  $2^{<\kappa}$ . In this Part  $\theta$  is a cutoff ordinal, never the regular cardinal that [section 1.3.1](#) writes  $\theta$  for source 18. Renamed symbols: source 19's restriction map  $r_Y$  and source 20's  $\rho_{\beta\alpha}$  are printed  $\pi_Y$ ,  $\pi_{\beta\alpha}$  (since  $r = \text{cf } \kappa$  and  $\rho_\kappa$  is source 15's minimal-slice coding rank); source 20's local ranks  $r_\alpha(x)$  are printed  $\text{lr}_\alpha(x)$ ; source 20's class  $A_R$  of minimum-type cutoffs is printed  $\text{Min}_R$  (source 16 writes  $A_R$  for the largest set final segment, [corollary 52.7](#)); source 20's core  $\mathcal{C}_b$  is printed  $\mathcal{P}_{\text{bd}}(b)$ , since its codes are normalized permutations of an ordinal ([section 1.3](#)); source 21's  $d$ ,  $c$  and  $\text{br}$  are printed  $\text{d}$ ,  $\text{cell}$  and  $\ell_2$  (its “least ordinal length of a fixed-length binary lexicographic cube into which the order embeds” is exactly  $\ell_2$ ); source 29's  $D_{\rho,m}$  and  $\otimes_{\text{lex}}$  are printed  $\mathbb{B}_\rho \times_{\text{lex}} m$  and  $\times_{\text{lex}}$ , its  $\ell_2$  is  $\ell_2$ , its termination symbol  $\dagger$  is **end** (Conway sign order  $- < \text{end} < +$ ), its layer weights  $r(\lambda)$ ,  $u_\kappa(\lambda)$  are printed  $\text{wt}(\lambda)$ ,  $\text{wt}_\kappa^+(\lambda)$ , and its weighted lengths  $R$ ,  $U$  are printed  $\varrho_{\text{lo}}$ ,  $\varrho_{\text{up}}$  (the letters  $r$ ,  $R$  are taken by  $\text{cf } \kappa$  and by raw well-orders). Displayed equations keep their source numbers as (SNN.*n*). Labels: **swo:rs:** (19), **swo:lm:** (20), **swo:fu:** (21, keeping its internal **bs:**), **swo:bw:** (29, keeping its internal **bs:** and **ns:**); text written for this Part has labels **swo:xi:**.

### 73 Exactly when restriction preserves the native order (source 19)

[merge] Source 19's Section 6, printed whole. Its restriction map  $r_Y$  is printed  $\pi_Y$ . “The earlier manuscripts” of its first sentence are, at this point of the report, source 15's three-label theorem ([theorem 38.5](#)) and the reversal  $021 < 102$ ,  $21 > 12$  printed with it in Part VII; source 19 cites sources 15, 16 and 18 as unrefereed prior manuscripts, which are now sources of this report. The two theorems are consistent: source 15's concerns all three two-label restrictions of a three-label carrier at once, under any choice of order on each target, while [theorem 73.1](#) concerns one specified restriction between native orders, and its exceptional case  $\{a < z < b\} \rightarrow \{a, b\}$  is the one restriction of source 15's carrier that is monotone.

The earlier manuscripts exhibit reversals under carrier restriction and show that no linearization can make all two-label restrictions monotone. There is also an exact answer for one specified restriction with the native numerical lexicographic orders on its source and target.

**Theorem 73.1** (Classification of monotone restrictions). *Let  $(X, <)$  be a linearly ordered set and  $Y \subseteq X$ . Compare the well-orders of each fixed labelled carrier by their ordinal enumerations, using the inherited label order at the first disagreement. The map*

$$\pi_Y : \mathcal{W}(X) \longrightarrow \mathcal{W}(Y), \quad R \longmapsto R \upharpoonright Y,$$

*is nondecreasing if and only if one of the following holds:*

1.  $|Y| \leq 1$ ;
2.  $Y = X$ ;
3.  $X = \{a < z < b\}$  and  $Y = \{a, b\}$ .

*For an infinite carrier  $X$  of cardinality  $\mu$ , every proper restriction retaining at least two labels already has a reversal on the minimal slice  $\mathcal{W}_\mu(X)$ .*

*Proof.* Each finite word displayed below is followed by the same fixed well-order of the unused labels. Both relevant comparisons are decided inside the displayed words.

Suppose  $a < b$  belong to  $Y$  and  $z \notin Y$ . If  $z < a$ , use

$$R = (z, b, a), \quad S = (a, b, z).$$

Then  $R < S$ , while their restrictions start with  $(b, a)$  and  $(a, b)$ , respectively, so  $\pi_Y(R) > \pi_Y(S)$ . If  $b < z$ , the pair

$$R = (b, a, z), \quad S = (z, a, b)$$

has exactly the same property. Therefore monotonicity requires every omitted label to lie between every pair of retained labels.

If  $Y$  contains  $a_0 < a_1 < a_2$ , any omitted  $z$  violates this requirement. For  $z < a_1$ , use the pair  $a_1 < a_2$ ; for  $a_1 < z$ , use the pair  $a_0 < a_1$ . These alternatives are exhaustive because  $z$  is omitted. Consequently, a proper monotone restriction retaining at least two labels must have  $Y = \{a, b\}$ , with every omitted label in  $(a, b)$ .

If two labels  $u < v$  are omitted, then  $a < u < v < b$ . The four-label pair

$$R = (u, b, a, v), \quad S = (v, a, b, u)$$

satisfies  $R < S$  because  $u < v$ , whereas the restrictions again begin  $(b, a)$  and  $(a, b)$ . There can therefore be only one omitted label, and the only remaining proper case is  $X = \{a < z < b\}$ .

For this exceptional case the six enumerations, in increasing lexicographic order, are

$$(a, z, b), (a, b, z), (z, a, b), (z, b, a), (b, a, z), (b, z, a).$$

The first three restrict to  $(a, b)$  and the last three to  $(b, a)$ , so restriction is nondecreasing. The singleton and identity cases are immediate.

Finally, if  $X$  is infinite, choose the common unused tail to have type  $\mu$ . Every displayed finite prefix followed by this tail has type  $n + \mu = \mu$ . Thus all reversals occur already in  $\mathcal{W}_\mu(X)$ .  $\square$

The same finite gadgets work for a supplied proper-class carrier in GBC: append a fixed set-like enumeration of the unused labels. They produce set-like global witnesses. In that setting the statement is interpreted as a theorem about supplied classes and explicit witnesses, not as the assertion that all class well-orders form an ordinary class of objects.

This sharpens the obstruction to a backwards system of birthday cutoffs. Whenever  $\mathbf{No}_{<\kappa} \subsetneq \mathbf{No}_{<\lambda}$  are infinite, unrestricted carrier restriction fails to be monotone, even on the minimal source slice. Forward extension maps, or specially constrained families of enumerations, must supply any coherent order-preserving system.

*Remark 73.2* (Finite verification). The accompanying script `check_restrictions.py` checks every subset of each carrier of size  $0 \leq n \leq 7$  and every adjacent pair in the lexicographic list of its permutations. It checks 695,228 adjacent comparisons. All cases agree with [theorem 73.1](#); the respective numbers of monotone subsets are 1, 2, 4, 6, 6, 7, 8, 9. The JSON report records the counts and sample reversals. This finite regression check does not replace the proof for arbitrary carriers.

[merge] The script ships as `code/19-residual-check_restrictions.py` and its recorded output as `data/19-residual-restriction_checks.json`. The counts 1, 2, 4, 6, 6, 7, 8, 9 are those of  $n = 0, 1, \dots, 7$  and agree with the theorem: for  $n \geq 4$  the monotone subsets are the empty set, the  $n$  singletons and  $X$  itself, and for  $n = 3$  the extra subset  $\{a, b\}$  is added.

[merge] The “forward extension maps” of the last paragraph are what the next two sections construct: source 20’s append-tail maps ([theorem 74.8](#)) and source 21’s fixed-slot insertions ([theorem 75.1](#)). Source 20’s [theorem 74.3](#) shows that the reversals of [theorem 73.1](#) disappear in the limit: for a fixed pair of well-orders, every sufficiently large restriction compares them as the originals are compared. Source 19 also records the restriction theorem as an attractive first formalization target and states that it “does not forbid carefully designed enlargement embeddings” (its results ledger).

## 74 Birthday limits: gluing, stabilization, and two different completions (source 20)

[merge] Source 20’s Section 4, printed whole. It is the only source of the batch to treat all ordinal cutoffs at once. Renamed:  $X_\alpha$  is printed  $\mathbf{No}_{<\alpha}$ ; the restriction maps  $\rho_{\beta\alpha}$  are printed  $\pi_{\beta\alpha}$ ; the local ranks  $r_\alpha(x)$  are printed  $\text{lr}_\alpha(x)$ ; the class  $A_R$  of minimum-type cutoffs is printed  $\text{Min}_R$ ; the core  $\mathcal{C}_b$  is printed  $\mathcal{P}_{\text{bd}}(b)$ ; and the ordinal  $r = \text{otp}(R)$  in the proof of [theorem 74.8](#) is printed  $\zeta$ . Its citations of the repository are replaced by cross-references to the results meant. Its first paragraph is accurate at this report’s head: the correction of “coherent” to “uniform” is the one recorded in [remarks 13.4](#) and [113.15](#) (3 October 2026). The nonmonotone example of its second subsection is source 15’s ([section 38](#)).

The distinction between uniformly chosen local well-orders and restriction-coherent local well-orders is mathematically consequential. The current repository explicitly corrects an earlier use of “coherent” in its global-choice argument: the recursively constructed orders on  $V_\alpha$  are uniform, but need not extend one another ([remark 13.4](#)). Here we study genuine restriction coherence and obtain an exact local-to-global description. We also show that the native lexicographic comparison is eventually recoverable from every increasing exhaustion, even though individual restriction maps need not preserve it.

Write

$$\mathbf{No}_{<\alpha} = \{x \in \mathbf{No} : \text{bd}(x) < \alpha\}.$$

These are increasing sets, their union is  $\mathbf{No}$ , and every set of surreal labels is contained in some  $\mathbf{No}_{<\alpha}$ . The last fact follows by applying Replacement to birthdays and then taking a strict ordinal upper bound. For  $\alpha \leq \beta$ , restriction gives a set map

$$\pi_{\beta\alpha} : \mathcal{W}(\mathbf{No}_{<\beta}) \longrightarrow \mathcal{W}(\mathbf{No}_{<\alpha}), \quad R \longmapsto R \upharpoonright \mathbf{No}_{<\alpha}^2.$$

A *uniform coherent profile* is one class function  $\alpha \mapsto R_\alpha$ , with set-valued entries, such that  $R_\alpha \in \mathcal{W}(\mathbf{No}_{<\alpha})$  and  $\pi_{\beta\alpha}(R_\beta) = R_\alpha$  whenever  $\alpha \leq \beta$ . This is a class of set-coded local relations, not a class containing global class relations as members.

### 74.1 Local well-foundedness reflects to the whole class

**Theorem 74.1** (Coherent birthday gluing in GB). *Every uniform coherent profile determines a unique strict class well-order  $R$  of  $\mathbf{No}$  with  $R \upharpoonright \mathbf{No}_{<\alpha}^2 = R_\alpha$  for all  $\alpha$ . Conversely, the restrictions of every supplied class well-order form such a profile. These constructions are inverse, uniformly in the supplied class parameters. Set-likeness is not required.*

*Proof.* Define  $xRy$  when  $xR_\alpha y$  at any stage containing  $x, y$ . Coherence makes this independent of the stage. Every finite collection of labels is contained in a common stage, so irreflexivity, transitivity, and linearity follow from the corresponding local properties. Elementary class comprehension supplies  $R$ .

It remains to prove that every nonempty available subclass  $A$  has a least element. Suppose otherwise and fix  $x_0 \in A$ . Given  $x_n$ , let  $\beta_n$  be the least ordinal such that

$$A \cap \mathbf{No}_{<\beta_n} \cap \{y : yRx_n\} \neq \emptyset.$$

Such an ordinal exists because  $A$  has no least element. A nonempty subclass of the ordinals has a least member, by restricting to an ordinal containing one witness. The displayed intersection is a set



by separation with class parameters. Let  $x_{n+1}$  be its unique  $R_{\beta_n}$ -least member. This is a uniquely specified set-valued recursion along  $\omega$ , available in GB; no global choice of witnesses is needed.

Class Replacement makes  $\{x_n : n < \omega\}$  a set. Its birthdays are bounded, so the entire strictly descending sequence lies in one  $\mathbf{No}_{<\gamma}$ . This contradicts the well-foundedness of  $R_\gamma$ . Uniqueness follows because every pair of labels occurs at a stage. The converse is immediate, and local restrictions are sets by separation with the supplied relation as a class parameter.  $\square$

Thus the inverse-limit notation

$$\{(R_\alpha)_{\alpha \in \text{Ord}} : \pi_{\beta\alpha}(R_\beta) = R_\alpha\} \longleftrightarrow \{R : R \text{ is a class well-order of } \mathbf{No}\} \quad (74.1a)$$

expresses a correspondence between predicates on supplied classes. It does not assert that either side is an ordinary GB class of members. In a larger universe in which the relevant classes are objects, it becomes a literal inverse-limit bijection, subject to that universe's well-foundedness interpretation.

**Proposition 74.2** (The sharp set-filtration criterion). *Work in ZFC. Let  $\lambda > 0$  and let  $(Y_\alpha)_{\alpha < \lambda}$  be an increasing ordinal-indexed family of sets with union  $Y$ . The following are equivalent:*

1. *Every countable subset of  $Y$  is contained in some  $Y_\alpha$ .*
2. *Every coherent family of well-orders on the  $Y_\alpha$  has a well-ordered union.*

*In particular, (2) holds if  $\text{cf}(\lambda) > \omega$ . If  $\text{cf}(\lambda) = \omega$  and the family is not eventually constant, there is a coherent counterexample.*

*Proof.* For (1) $\Rightarrow$ (2), an ill-founded linear union would admit a descending  $\omega$ -sequence, whose countable range would lie in one well-ordered stage.

Suppose (1) fails. Choose a countable set with no bounded stage of containment. Its stage indices give a countable cofinal sequence in  $\lambda$ . Recursively choose increasing cofinal indices  $\alpha_n$  and distinct labels  $y_n \notin Y_{\alpha_n}$ , enlarging  $\alpha_{n+1}$  so that it contains all previously chosen labels. For each fixed  $\alpha$ , only finitely many  $y_n$  belong to  $Y_\alpha$ : eventually  $\alpha_n \geq \alpha$ . Order  $\{y_n : n < \omega\}$  by  $\dots R y_2 R y_1 R y_0$ , and place it before a fixed well-order of its complement in  $Y$ . Every restriction to  $Y_\alpha$  consists of a finite initial block followed by a well-order, so it is a well-order. The union contains the descending sequence  $y_0, y_1, \dots$

If  $\text{cf}(\lambda) > \omega$ , countably many stage indices are bounded. If  $\text{cf}(\lambda) = \omega$  and no stage already equals the union, the same recursive choice of new labels produces a countable unbounded subset. This proves the final assertions.  $\square$

The familiar reverse- $\omega$  union of finite orders is therefore a real obstruction, but it does not refute birthday gluing along all ordinals. In the class theorem every set-sized witness has bounded birthdays. The proof also explains why Class Replacement matters: an arbitrary external branch through a small model's stages need not be an admissible class of a GB realization.

## 74.2 Native lexicography survives as an eventual comparison

Individual restriction maps are not monotone for native enumeration lexicography. On  $0 < 1 < 2$ ,

$$021 <_{\text{lex}} 102, \quad 21 >_{\text{lex}} 12 \quad \text{after restriction to } \{1, 2\}.$$

Nevertheless, each comparison has an unexpectedly small persistent witness.



**Theorem 74.3** (Two labels determine every sufficiently large restriction). *Let  $R, S$  be distinct well-orders of the same linearly ordered set or class of labels. There are distinct labels  $a, b$  such that every subcarrier  $Y$  containing  $a, b$  satisfies*

$$R \upharpoonright Y^2 <_{\text{lex}} S \upharpoonright Y^2 \iff a < b \iff R <_{\text{lex}} S.$$

*The assertion includes proper-class common prefixes and is provable in GB for supplied class well-orders.*

*Proof.* Let  $C$  be the largest common labelled initial segment of  $R, S$ . The elementary common-prefix construction for class well-orders provides  $C$ ; it does not compare abstract class order types (lemma 14.1 and theorem 14.7). Put

$$a = \min_R(\mathbf{No} \setminus C), \quad b = \min_S(\mathbf{No} \setminus C),$$

with the indicated carrier substituted for  $\mathbf{No}$  in the general case. These labels differ, since otherwise  $C$  could be extended by that common next label. In the restrictions to  $Y$ , the set or class  $C \cap Y$  is still an initial segment of both orders, and their orders on it agree. Their next labels are still  $a, b$ , because those labels belong to  $Y$  and were the least available labels outside  $C$ . The first unequal labels of the restrictions therefore coincide with those of the original pair. This is exactly the displayed comparison.  $\square$

The number two is optimal: restrictions to a one-element carrier cannot differ. The theorem is a persistence statement for restrictions of a fixed pair. It does not say that knowing only the orientation of one pair determines the comparison of arbitrary other orders.

**Corollary 74.4** (Native order on the coherent inverse limit). *For two coherent birthday profiles with global unions  $R, S$ ,*

$$R <_{\text{lex}} S \iff \exists \alpha \forall \beta \geq \alpha \ R_\beta <_{\text{lex}} S_\beta.$$

*For distinct profiles, the local comparisons are eventually strict and constant. Consequently eventual local comparison gives exactly the native global lexicographic order, despite the failure of monotonicity of individual bonding maps.*

*Proof.* Choose a cutoff containing the two labels from theorem 74.3. Every later cutoff has the same comparison. Equality of the global orders is equivalent to equality at every stage. Applying the same argument to the opposite strict comparison establishes the converse.  $\square$

This result is independent of the birthday filtration. Any increasing exhaustion of the carrier eventually contains the two decisive labels. Thus changing the exhaustion leaves the eventual order unchanged. There is also a stronger finiteness statement.

**Theorem 74.5** (Only finitely many changes of decisive labels). *Fix two well-orders  $R, S$  and any uniform increasing ordinal-indexed family of subcarriers. Once their restrictions are distinct, the pair of first unequal labels changes only finitely many times. In particular, the native lexicographic signs change only finitely often. There is no uniform finite bound on the number of changes, even for birthday restrictions of well-orders of  $\mathbf{No}$ .*

*Proof.* For  $Y \subseteq Z$  on which the restrictions differ, write  $(a_Y, b_Y)$  and  $(a_Z, b_Z)$  for their first unequal labels. The pair on  $Y$  is reversed:

$$a_Y R b_Y, \quad b_Y S a_Y.$$

Neither label can lie in the common labelled prefix  $C_Z$ . For if one did, initiality in the order putting the other one first would force both into  $C_Z$ , contradicting agreement there. Consequently

$$a_Z \leq_R a_Y, \quad b_Z \leq_S b_Y.$$

Each coordinate can strictly decrease only finitely many times. Otherwise, recursively taking the least later index with a strict decrease would produce a descending  $\omega$ -sequence in one of the given well-orders. In the class setting this is again a set-valued recursion with ordinal choices and Class Replacement. The union of the two finite sets of change indices is finite.

To obtain any prescribed finite sequence of strict comparison signs, choose strictly increasing positive ordinals  $\gamma_i$  and take  $a_i = -\gamma_i < b_i = \gamma_i$ , for  $0 \leq i \leq n$ . Both labels in a pair have the same birthday  $\gamma_i$ . In both global orders put these pairs before all other labels, with the pair blocks in decreasing order of  $i$ . Within the  $i$ th pair choose opposite orientations for  $R, S$ , selecting the orientation in  $R$  to give the prescribed sign. Append the same set-like birthday-respecting baseline on the remaining labels. At a cutoff containing pairs  $0, \dots, i$  and no later pair, the first unequal labels are exactly the  $i$ th pair. Both global orders are well-orders. Arbitrarily long finite alternating patterns follow.  $\square$

### 74.3 Detecting set-likeness from local ordinal ranks

For a profile and a label  $x$ , at every sufficiently large stage put

$$\text{lr}_\alpha(x) = \text{otp}(\{y \in \mathbf{No}_{<\alpha} : yR_\alpha x\}, R_\alpha).$$

These are set ordinals, even if the global predecessor class is proper. Coherence makes  $\text{lr}_\alpha(x)$  nondecreasing, because an ordered subset of an ordinal has order type at most that ordinal.

**Theorem 74.6** (Rank stabilization is exactly set-likeness). *For the global union  $R$  of a coherent profile and each label  $x$ , the following are equivalent:*

1. *the global predecessor class  $P_x = \{y : yRx\}$  is a set;*
2.  *$P_x$  is contained in a birthday cutoff;*
3. *the ordinals  $\text{lr}_\alpha(x)$  are bounded by one set ordinal;*
4.  *$\text{lr}_\alpha(x)$  is eventually constant.*

*Thus  $R$  is set-like exactly when every local rank profile eventually stabilizes. The theorem holds in GB.*

*Proof.* The first two conditions are equivalent by boundedness of birthdays and separation. If  $P_x \subseteq \mathbf{No}_{<\delta}$ , then every later local predecessor set is exactly  $P_x$ , proving (2) $\Rightarrow$ (4). Monotonicity gives (4) $\Rightarrow$ (3).

For (3) $\Rightarrow$ (1), suppose  $P_x$  is proper and all ranks are at most  $\theta$ . Using the global well-order supplied by [theorem 74.1](#), recursively select the least unused predecessor through  $\theta + 1$  stages. Each earlier range is a set and cannot exhaust  $P_x$ , so the recursion continues. It gives an increasing sequence of type  $\theta + 1$ . Its range, together with  $x$ , lies in one birthday cutoff. That local predecessor order then has type at least  $\theta + 1$ , contradicting its asserted bound  $\theta$ . Only set-valued recursion along a set ordinal was used.  $\square$

Coherence alone therefore does not impose set-likeness. For example, take a birthday-respecting set-like baseline on  $\mathbf{No} \setminus \{0\}$  and place 0 last. Its type is  $\text{Ord} + 1$ . All birthday restrictions are well-orders and cohere, but  $\text{lr}_\alpha(0)$  is unbounded. Local ordinal ranks diagnose the proper predecessor class without treating its order type as an ordinal.

#### 74.4 A club dichotomy for minimum enumeration types

For an infinite strong-limit cardinal  $\kappa$ , the birthday carrier  $\mathbf{No}_{<\kappa}$  has cardinality  $\kappa$ . Indeed every earlier stage has cardinality below  $\kappa$ , their  $\kappa$ -indexed union has cardinality at most  $\kappa$ , and the ordinal labels below  $\kappa$  give the reverse inequality. Thus  $\kappa$  is the least possible well-order type of this carrier. Define the class

$$\text{Min}_R = \{\kappa : \kappa \text{ is an infinite strong-limit cardinal and } \text{otp}(R \upharpoonright \mathbf{No}_{<\kappa}^2) = \kappa\}.$$

A class  $C \subseteq \text{Ord}$  is *club* if it is unbounded and contains every ordinal limit point of itself. A class is *stationary* if it meets every available club class. These are predicates on classes, not collections of classes regarded as ordinary members.

**Theorem 74.7** (Club reflection detects set-likeness). *In GBC, every class well-order  $R$  of  $\mathbf{No}$  satisfies exactly one of the following alternatives.*

1.  $R$  is set-like, and a club of infinite strong-limit cardinals  $\kappa$  satisfies:  $\mathbf{No}_{<\kappa}$  is an  $R$ -initial segment and has order type  $\kappa$ . In particular,  $\text{Min}_R$  contains a club.
2.  $R$  is not set-like, and there are a fixed label  $x$  and a club of infinite strong-limit cardinals  $\kappa$  such that  $x \in \mathbf{No}_{<\kappa}$  and  $\text{lr}_\kappa(x) \geq \kappa$ . Consequently  $\text{otp}(R \upharpoonright \mathbf{No}_{<\kappa}^2) > \kappa$ , and  $\text{Min}_R$  misses that club.

Hence  $R$  is set-like if and only if  $\text{Min}_R$  is stationary, equivalently if and only if  $\text{Min}_R$  contains a club. In particular, minimum type at every infinite strong-limit cutoff forces set-likeness.

*Proof.* We first record an explicit closure construction. Let  $h : \text{Ord} \rightarrow \text{Ord}$  be nondecreasing, with  $h(\alpha) > \alpha$ . The class

$$C_h = \{\kappa : \kappa \text{ is a nonzero limit ordinal and } (\forall \alpha < \kappa) h(\alpha) < \kappa\}$$

is club. For unboundedness, start above any prescribed ordinal and iterate  $\alpha_{n+1} = h(\alpha_n)$ ; its supremum belongs to  $C_h$ . Class Replacement justifies this set-sized iteration. Closure follows by bounding each fixed  $h(\alpha)$  below a member of  $C_h$  preceding the proposed limit. If each  $h(\alpha)$  is an infinite cardinal and exceeds  $2^{|\alpha|}$ , every member of  $C_h$  is a strong-limit cardinal: cardinals are cofinal below it, and its smaller power-set cardinals are bounded strictly below it.

If  $R$  is set-like, let  $f : \text{Ord} \rightarrow \mathbf{No}$  be its enumeration. Define  $h(\alpha)$  to be the least infinite cardinal strictly exceeding

$$\alpha, \quad 2^{|\alpha|}, \quad \sup_{\xi < \alpha} (\text{bd}(f(\xi)) + 1), \quad \sup_{x \in \mathbf{No}_{<\alpha}} (f^{-1}(x) + 1).$$

All the suprema concern sets by Class Replacement, and  $h$  is nondecreasing. For  $\kappa \in C_h$ , forward birthday closure gives  $f[\kappa] \subseteq \mathbf{No}_{<\kappa}$ . Conversely, each  $x \in \mathbf{No}_{<\kappa}$  already lies in some  $\mathbf{No}_{<\alpha}$  with  $\alpha < \kappa$ , so inverse closure gives  $f^{-1}(x) < \kappa$ . Thus  $f[\kappa] = \mathbf{No}_{<\kappa}$ , proving the first alternative, including initiality.

If  $R$  is not set-like, choose  $x$  with proper predecessor class  $P_x$ . Its first Ord-long segment has an increasing enumeration  $f : \text{Ord} \rightarrow P_x$ . Concretely, each set-length initial enumeration is obtained by repeatedly selecting the least unused predecessor; uniqueness makes these set-valued constructions uniform. Define  $h$  as above, omitting the inverse term and adding the fixed bound  $\text{bd}(x) + 1$ . For  $\kappa \in C_h$ , the label  $x$  and the increasing sequence  $f[\kappa]$  lie in  $\mathbf{No}_{<\kappa}$ . This sequence has type  $\kappa$  and precedes  $x$ , so  $\text{lr}_\kappa(x) \geq \kappa$ , and the whole local order has strictly larger type.

Finally, two club classes intersect in a club: alternate their least points above successive bounds and take the supremum of the resulting  $\omega$ -sequence. Closure places that supremum in both classes. This proves the stationary reformulation.  $\square$

**Unboundedly many minimum types do not suffice.** The stationary qualification is sharp. Let

$$P = \{\beth_{\beta+1} : \beta \in \text{Ord}\}, \quad Q = \mathbf{No} \setminus (P \cup \{0\}),$$

and order the carrier by  $P$ , in increasing  $\beta$ -order, followed by 0, followed by a birthday-respecting well-order of  $Q$ . This class well-order  $R$  is not set-like at 0. For any ordinal  $\alpha$ , put  $\lambda = \alpha + \omega$  and  $\kappa = \beth_\lambda$ . Then  $\kappa$  is strong-limit and  $\lambda < \kappa$ . For infinite  $\alpha$ , the ordinal  $\alpha + \omega$  is not an initial cardinal, whereas  $\beth_\lambda \geq \lambda$  is one; for finite  $\alpha$ , use  $\beth_\omega > \omega$ . The intersection  $P \cap \mathbf{No}_{<\kappa}$  has exactly type  $\lambda$ . Removing these fewer than  $\kappa$  labels and 0 leaves  $|Q \cap \mathbf{No}_{<\kappa}| = \kappa$ . Its birthday-respecting order has type  $\kappa$ : each earlier birthday segment has type below  $\kappa$ , while their union has cardinality  $\kappa$ . Consequently

$$\text{otp}(R \upharpoonright \mathbf{No}_{<\kappa}^2) = \lambda + 1 + \kappa = \kappa.$$

The cardinals  $\beth_{\alpha+\omega}$  are unbounded, so  $\text{Min}_R$  is unbounded, yet [theorem 74.7](#) supplies a club disjoint from it.

## 74.5 Coherent append-tail maps and their direct limit

Work now in GBC and choose a baseline well-order  $B$  which orders labels first by birthday and then by a chosen well-order of each birthday layer. It is set-like. Let  $b : \text{Ord} \rightarrow \mathbf{No}$  be its unique enumeration. Every  $\mathbf{No}_{<\alpha}$  is an initial segment of  $B$ ; write  $t_\alpha = \text{otp}(B \upharpoonright \mathbf{No}_{<\alpha})$ , so  $b[t_\alpha] = \mathbf{No}_{<\alpha}$ .

For  $\alpha \leq \beta$  define

$$j_{\alpha\beta}(R) = R + (B \upharpoonright (\mathbf{No}_{<\beta} \setminus \mathbf{No}_{<\alpha})).$$

The plus sign means ordered concatenation on the disjoint carriers.

**Theorem 74.8** (The direct system is precisely the bounded-support core). *The maps  $j_{\alpha\beta}$  are native lexicographic order embeddings, satisfy*

$$j_{\beta\gamma}j_{\alpha\beta} = j_{\alpha\gamma}, \quad \pi_{\beta\alpha}j_{\alpha\beta} = \text{id},$$

*and their direct limit identifies with the bounded-support core  $\mathcal{P}_{\text{bd}}(b)$ : the global enumerations which equal  $b$  at every sufficiently large ordinal. The identification is induced by*

$$J_\alpha(R) = R + (B \upharpoonright (\mathbf{No} \setminus \mathbf{No}_{<\alpha})).$$

*Proof.* The first disagreement of two well-orders on the same carrier occurs before the appended tail. Neither exhaustive enumeration can be a proper prefix of the other. Thus appending the common tail preserves and reflects comparison. Birthday compatibility of  $B$  puts  $\mathbf{No}_{<\beta} \setminus \mathbf{No}_{<\alpha}$  before  $\mathbf{No}_{<\gamma} \setminus \mathbf{No}_{<\beta}$ , proving the composition identity. The retraction identity is immediate.

Let  $\zeta = \text{otp}(R)$ , and choose an infinite initial cardinal  $\kappa > \zeta, t_\alpha$ . If  $f$  enumerates  $J_\alpha(R)$ , then on the appended tail

$$f(\zeta + \delta) = b(t_\alpha + \delta).$$

For every  $\gamma \geq \kappa$  both  $\zeta + \gamma = \gamma$  and  $t_\alpha + \gamma = \gamma$ : write  $\gamma = \kappa + \varepsilon$  and use  $\zeta + \kappa = t_\alpha + \kappa = \kappa$ . Hence  $f(\gamma) = b(\gamma)$  for every  $\gamma \geq \kappa$ . Each  $J_\alpha(R)$  therefore belongs to  $\mathcal{P}_{\text{bd}}(b)$ .

Conversely, let  $f$  equal  $b$  beyond  $\kappa$ , and let  $R_f$  be its induced order. The two bijections necessarily have  $f[\kappa] = b[\kappa]$ . Choose  $\alpha$  with  $b[\kappa] \subseteq \mathbf{No}_{<\alpha}$ , so  $t_\alpha \geq \kappa$ . Then  $f[t_\alpha] = b[t_\alpha] = \mathbf{No}_{<\alpha}$ ; on the complementary tail,  $f$  agrees with  $b$ . Consequently

$$R_f = J_\alpha(R_f \upharpoonright \mathbf{No}_{<\alpha}^2).$$

Finally, two local objects have the same global extension exactly when their images agree at a common later stage. This proves the asserted direct-limit identification.  $\square$

There is no need to make a proper equivalence class of stage codes into a member of another class. One can use the core's unique set-supported permutation codes, or choose the least stage at which a direct-limit object is represented. The direct limit is consequently an ordinary class order. The repository's core theorem identifies it with **No** under the stated choice assumptions ([corollary 19.15](#) and [note 19.16](#)).

The same stage spaces thus support two sharply different constructions:

| maps                               | limit                                          | comparison                   |
|------------------------------------|------------------------------------------------|------------------------------|
| $\pi_{\beta\alpha}$ by restriction | all supplied class well-orders                 | eventual native comparison   |
| $j_{\alpha\beta}$ by appending $B$ | $\mathcal{P}_{\text{bd}}(b) \cong \mathbf{No}$ | compatible native embeddings |

The inverse-limit side includes non-set-like orders; the direct-limit side consists of eventual perturbations of one baseline. Each local order has a direct-limit extension, so every bounded observation of a global class well-order is already realized in the core.

*Note 74.9* ([merge]; the append-tail maps and the earlier remarks on extension). The maps  $j_{\alpha\beta}$  are the operation  $R \mapsto R + Q$  with a fixed well-order  $Q$  of the new labels, which source 15 records as an order embedding in [section 38](#) (subsection on restrictions and extensions), there for one pair of carriers. New in source 20 is the coherence: one birthday-respecting baseline makes all of them compose exactly along all ordinal cutoffs, with  $\pi_{\beta\alpha}$  as a left inverse, and the direct limit is identified with the core. Naively appended tails need not preserve the minimal enumeration type ([research question 113.19](#) warns of this); here no minimality is claimed: the stage spaces are the full spaces  $\mathcal{W}(\mathbf{No}_{<\alpha})$  of all lengths. The fixed-slot construction of [theorem 75.1](#) keeps the minimal type instead, at the price of a fixed cardinality band.

## 74.6 A monotone inverse system using relation tables

There is also a literal monotone inverse system, obtained by changing the comparison. In GBC, well-order the pair layer

$$L_\gamma = \{(x, y) : \max(\text{bd}(x), \text{bd}(y)) = \gamma\}$$

for every  $\gamma$ , and concatenate these layer orders in increasing  $\gamma$ . The resulting scan  $Q$  of  $\mathbf{No}^2$  is set-like. Its restriction to  $\mathbf{No}_{<\alpha}^2$  is an initial segment of some ordinal length  $s_\alpha$ . The lengths are continuous at limit cutoffs and cofinal in Ord.

**Proposition 74.10** (Birthday table quotients). *Order each  $\mathcal{W}(\mathbf{No}_{<\alpha})$  by the characteristic table of its strict relation, scanned by  $Q \upharpoonright \mathbf{No}_{<\alpha}^2$ , with  $0 < 1$ . Every  $\pi_{\beta\alpha}$  is then a weakly order-preserving surjection with convex fibers. The append-tail maps  $j_{\alpha\beta}$  are order embeddings for these table orders as well. The coherent inverse-limit comparison is exactly global first-difference table comparison under  $Q$ .*

*Proof.* Restriction removes a final segment of the table. Prefix restriction of binary lexicographic words is weakly monotone, and all words with a fixed prefix form a convex set. Intersecting with the allowed well-order tables preserves convexity. Surjectivity follows by appending the baseline on the complementary carrier. The same argument shows that an append-tail map preserves the first differing bit of its input tables. For two global tables, their first differing pair belongs to some  $\mathbf{No}_{<\alpha}^2$ ; every later restriction has the same first differing bit. Finally, [theorem 74.1](#) identifies the compatible local well-order tables with the supplied global well-orders.  $\square$

These table orders depend on the chosen scans inside pair layers. They are a separate interpretation of lexicography. Their monotone restriction property does not imply that they agree with

native enumeration comparison. Native comparison instead has the stronger intrinsic interpretation of [theorem 74.3](#) and [corollary 74.4](#), and requires no scan of pairs.

[merge] Relation-table orders are source 15’s and source 16’s scan orders ([section 38](#), its subsection on relation tables, and [theorem 38.2](#)); [proposition 74.10](#) is the first statement about how they behave under change of cutoff. It answers in part source 15’s question “Coherent restricted filtrations” (its Question 13.9), which asks for restricted families on which restriction is monotone: a change of comparison achieves it for all well-orders at once.

## 74.7 The existence of a branch is already a choice principle

Over GB with set-level AC, each  $\mathcal{W}(\mathbf{No}_{<\alpha})$  is nonempty and every restriction map is surjective. It still takes a uniform class to provide a full coherent profile. By [theorem 74.1](#) and the repository’s equivalence between global choice and a class well-order of **No** ([theorem 13.1](#)),

$$\text{a full coherent birthday profile exists} \iff \text{Global Choice.}$$

The forward implication uses the glued well-order; the reverse implication restricts a birthday-respecting baseline. Thus nonempty set stages and surjective bonding maps do not by themselves supply a uniform class branch under the weaker stated assumptions. This pinpoints an actual foundational requirement, rather than attributing the difficulty to an informal “size paradox.”

[merge] [Theorem 13.1](#) is proved over GB without set choice, so the displayed equivalence holds over GB itself, not only over GB + AC: the forward implication uses [theorem 74.1](#) (GB), and the reverse one a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ , which GC supplies over GB. Source 20’s finite diagnostics (code/20-limits-finite\_diagnostics.py, recorded run data/20-limits-finite\_diagnostics\_results.json) check [theorem 74.3](#) on every restricted carrier and the coordinatewise monotonicity in the proof of [theorem 74.5](#) on every nested pair of carriers, for all pairs of permutations of carriers of sizes two to five: 58,255 decisive-pair restriction checks and 609,862 nested-witness checks over 7,432 permutation pairs, with the nonmonotone example  $021 < 102$ ,  $21 > 12$  reproduced. In source 20’s words they are “finite operational diagnostics”; they establish no transfinite or class-theoretic statement of this section.

## 75 Coherent embeddings within a cardinality band (source 21)

[merge] Source 21’s Section 6, printed whole. Its “proposed direction in the repository” is [research question 113.19](#); the cited noncardinal theorem is [theorem 77.4](#) ([section 77](#) below) and the cited topology proposition is source 21’s [note 62.13](#), printed in Part X. Source 20’s append-tail system ([theorem 74.8](#)), written independently, is the other coherent system of this Part; the two are compared in [note 74.9](#) and [section 79](#). Displayed equation (4) of source 21 is (S21.4).

Another proposed direction in the repository is compatibility between birthday cutoffs ([research question 113.19](#)). Appending an arbitrary tail generally changes the enumeration type. Within a fixed cardinality band, one baseline on an upper alphabet gives embeddings that keep the minimal type and compose exactly.

**Theorem 75.1** (Coherent insertion into fixed slots). *Let  $(X_i)_{i \in I}$  be a set-indexed nested family of ordered subsets of an ordered set  $Y$ , with  $|X_i| = |Y| = \mu$  infinite. Fix a bijection  $b : \mu \rightarrow Y$ . There are order embeddings*

$$J_{ij} : \mathcal{W}_\mu(X_i) \hookrightarrow \mathcal{W}_\mu(X_j) \quad (X_i \subseteq X_j)$$

*such that  $J_{ii}$  is the identity and  $J_{jk} \circ J_{ij} = J_{ik}$  whenever  $X_i \subseteq X_j \subseteq X_k$ .*

Each map is a homeomorphism onto its image for the prefix topologies. Its image is closed, and is clopen when the inserted positions are bounded in  $\mu$ ; if those positions are unbounded, the image is nowhere dense. These are also assertions about the order topologies whenever the relevant prefix and order topologies coincide.

*Proof.* Put  $A_i = b^{-1}[X_i] \subseteq \mu$ . A subset of the initial ordinal  $\mu$  having cardinality  $\mu$  has order type exactly  $\mu$ : its order type is at most  $\mu$  and cannot be smaller. Let  $a_i : \mu \rightarrow A_i$  be its unique increasing enumeration. For  $X_i \subseteq X_j$ , define

$$u_{ij} = a_j^{-1} \circ a_i : \mu \longrightarrow \mu, \quad D_{ij} = \mu \setminus u_{ij}[\mu].$$

Thus  $u_{ij}$  is increasing and cofinal. Set

$$J_{ij}(f)(\eta) = \begin{cases} f(\xi), & \eta = u_{ij}(\xi), \\ b(a_j(\eta)), & \eta \in D_{ij}. \end{cases} \quad (\text{S21.4})$$

Every old label occurs once in an old slot, and every new label occurs once in its fixed baseline slot, so the result is a bijection onto  $X_j$ . If  $f, g$  first disagree at  $\xi$ , their images first disagree at  $u_{ij}(\xi)$ : earlier variable slots agree, all inserted slots are fixed, and the numerical comparison at that disagreement is unchanged. Hence  $J_{ij}$  is an order embedding.

The slot maps satisfy  $u_{jk} \circ u_{ij} = u_{ik}$ . Under either successive insertion or direct insertion, the oldest labels retain the same ambient slots in  $\mu$ , and each additional label has its value prescribed by  $b$ . This proves the required equality of functions, rather than merely an isomorphism between two choices of extension. The identity assertion is immediate.

For continuity, fix a target prefix length  $\beta < \mu$ . The source coordinates whose slots are below  $\beta$  form a proper initial segment  $\delta < \mu$ , because  $u_{ij}$  is increasing and cofinal. Agreement on that source prefix therefore implies agreement on the entire target prefix; the other entries there are fixed. Conversely, for  $\delta < \mu$ , knowledge of the target prefix of length  $u_{ij}(\delta)$  determines all source coordinates below  $\delta$ . Both directions use genuine bounds below  $\mu$ ; no regularity of  $\mu$  is required. The map and its inverse on its image are continuous for the prefix topologies.

The image consists exactly of the target permutations satisfying

$$g(\eta) = b(a_j(\eta)) \quad (\eta \in D_{ij}).$$

Indeed these fixed positions contain exactly  $X_j \setminus X_i$ , so bijectivity forces the remaining positions to enumerate  $X_i$ . A violation is witnessed by a single coordinate and hence by a prefix neighborhood; the image is closed. If  $D_{ij}$  is bounded, a sufficiently long prefix checks all its conditions, so the image is open as well.

If  $D_{ij}$  is unbounded, take any cylinder meeting the image. Beyond its prescribed prefix choose an inserted position and a distinct later position, and swap their entries. The result remains in the cylinder but violates a fixed-position condition. The image therefore has empty interior and, being closed, is nowhere dense.  $\square$

**Corollary 75.2** (A bounded surreal system). *Let  $\Theta$  be a set ordinal and let  $I$  be a set of cutoffs  $\theta \leq \Theta$  such that*

$$|\mathbf{No}_{<\theta}| = |\mathbf{No}_{<\Theta}| = \mu \quad (\theta \in I).$$

*A single baseline enumeration of  $\mathbf{No}_{<\Theta}$  gives the coherent minimal-length system of [theorem 75.1](#) for all these cutoffs.*



In particular, for a fixed infinite cardinal  $\kappa$ , every cutoff  $\kappa < \theta \leq \kappa^+$  has alphabet cardinality  $2^\kappa$ . Use the single upper alphabet  $Y = \mathbf{No}_{<\kappa^+}$ , whose cardinality is  $2^{<\kappa^+} = 2^\kappa$ . Thus the construction works throughout this entire set-sized band, including its upper endpoint. The topology statements are then statements about order topology by [theorem 77.4](#) and [note 62.13](#). The distinction between a clopen image and a nowhere dense image records the placement of the new labels in the chosen baseline; it is not determined by their numerical order alone.

The restriction to one set-sized upper alphabet and one cardinality band is substantive. The proof gives no assertion that a single baseline works simultaneously across changing minimal cardinal lengths or across all proper-class birthday cutoffs. Those compatibility questions require additional constructions and an explicit foundational level.

[merge] Here  $\kappa$  is a cardinal of the band, not a cutoff of the convention of Part VI; the cutoffs  $\theta$  of the band are noninitial except its upper endpoint  $\kappa^+$ , and at the endpoint “ $\mu$ ” is  $2^{<\kappa^+} = 2^\kappa$  in either convention. The two coherent systems of this Part are complementary. Source 20’s append-tail maps ([theorem 74.8](#)) run along all ordinal cutoffs and compose exactly, but act on the full spaces  $\mathcal{W}(\mathbf{No}_{<\alpha})$  and change the enumeration length; source 21’s fixed-slot maps keep the minimal length  $\mu$ , are homeomorphisms onto closed images, but live in one cardinality band below one set-sized upper alphabet. Which of these maps preserve finite condensation, coding ranks or canonical cylinders, the further properties that [research question 113.19](#) asks about, is not stated by either source.

## 76 The exact binary spectrum of a birthday alphabet (source 29)

[merge] Source 29’s Section 4, printed whole. Its  $X_\theta$  is printed  $\mathbf{No}_{<\theta}$ , its  $D_{\rho,m}$  is printed  $\mathbb{B}_\rho \times_{\text{lex}} m$  and its  $\otimes_{\text{lex}}$  is printed  $\times_{\text{lex}}$ , its  $\ell_2$  is  $\ell_2$ , and its termination symbol  $\dagger$  is printed **end**; its displayed equations (1)–(4) are (S29.1)–(S29.4). The fusion lemma ([lemma 76.1](#)) is the cylinder fusion mechanism of [lemmas 31.1](#) and [31.5](#) and of source 19’s [lemma 64.3](#), stated in its cleanest form; it is printed because every later proof of this Part’s last three sections uses it. Its Corollary 4.2 is recorded in [note 76.2](#). Everything from [proposition 76.3](#) on is new against the repository and the rest of batch 83: the repository and its sources had determined the cube spectra of the binary cubes themselves and of the enumeration strata, but not of the birthday alphabets  $\mathbf{No}_{<\theta}$ .

The numerical order used as the alphabet already has a rigid embedding hierarchy. This remains true when increasing the birthday bound does not change the cardinality of the alphabet. We first determine that hierarchy; the result will then distinguish information retained by the alphabet from information retained by the lexicographic order of its enumerations.

Throughout this section all lengths are set ordinals. Write

$$\mathbf{No}_{<\theta} = \bigcup_{\xi < \theta} \{-, +\}^\xi, \quad \mathbb{B}_\beta = (\{0, 1\}^\beta, <_{\text{lex}}).$$

The order on  $\mathbf{No}_{<\theta}$  is numerical surreal order: at the first disagreement, termination lies strictly between minus and plus. This is the standard sign-sequence description of the surreals [12]. The symbol  $\simeq_e$  denotes order embeddings in both directions, not order isomorphism. For a positive finite integer  $m$ , put

$$\mathbb{B}_\rho \times_{\text{lex}} m = \mathbb{B}_\rho \times_{\text{lex}} \{0, \dots, m-1\}.$$

Every addition or multiplication of lengths below is ordinal arithmetic.

### 76.1 Binary fusion and finite capacity

The following elementary fusion argument is useful both for the native alphabet and for spaces of its enumerations. A full cylinder consists of all extensions of a fixed initial binary word, with every

later coordinate and every displayed final factor retained.

**Lemma 76.1** (Equal-length fusion). *Let  $\rho$  be an ordinal, let  $K$  be a nonempty linear order, and let*

$$f : \mathbb{B}_\rho \times_{\text{lex}} K \longrightarrow \mathbb{B}_\rho$$

*be nondecreasing. There is an  $s \in \mathbb{B}_\rho$  such that  $f$  is constant on  $\{s\} \times K$ .*

*Proof.* At stage  $\xi \leq \rho$  retain a full source cylinder whose first  $\xi$  bits are fixed and on which the first  $\xi$  target bits are constant. Suppose  $\xi < \rho$ . With the earlier target bits fixed, the next target bit is nondecreasing on that cylinder. If the lower source child maps entirely to 0 at that bit, retain it. Otherwise some point of the lower child maps to 1, so the whole upper child maps to 1; retain the upper child. At limit stages take unions of the fixed prefixes. After  $\rho$  stages the entire factor  $K$  survives over a single source word, and every target bit is fixed.  $\square$

*Note 76.2* (Source 29, Corollary 4.2, “Cube hierarchy and finite factors”). *For ordinals  $\beta, \gamma, \rho$  and positive finite integers  $m, n$ ,*

$$\mathbb{B}_\beta \hookrightarrow \mathbb{B}_\gamma \iff \beta \leq \gamma, \quad (\text{S29.1})$$

$$\mathbb{B}_\rho \times_{\text{lex}} m \hookrightarrow \mathbb{B}_\rho \times_{\text{lex}} n \iff m \leq n. \quad (\text{S29.2})$$

*If  $\rho = 0$  or  $\rho$  is an infinite limit ordinal, the least ordinal length of a binary cube containing  $\mathbb{B}_\rho \times_{\text{lex}} m$  is  $\ell_2(\mathbb{B}_\rho \times_{\text{lex}} m) = \rho + \lceil \log_2 m \rceil$ .*

[merge] (S29.1) is [lemma 6.4](#) (reals report `two:thm:cube-strict`; source 15’s [lemma 31.1](#)); (S29.2) and the coding length are [lemma 31.5](#) for nonzero limit  $\rho$  (the case  $\rho = 0$  is the count of finite chains), and source 19’s [note 64.12](#) proves the same. Source 29 derives all three from [lemma 76.1](#): for (S29.1), if  $\beta > \gamma$ , split the source after its first  $\gamma$  bits; the remaining factor has at least two points, and fusion of the composite with the target word makes them equal. For (S29.2), fusion of the composite with the projection to  $\mathbb{B}_\rho$  leaves all  $m$  source points over one target binary word, so they inject into its  $n$ -point fibre. For the coding length, coding the finite factor in  $j = \lceil \log_2 m \rceil$  bits gives the upper bound; a target length below  $\rho$  is excluded by the  $\mathbb{B}_\rho$  suborder, and a target length  $\rho + i$ ,  $i < j$ , has fibres of only  $2^i < m$  points, excluded by (S29.2). The proof is the argument of [lemma 31.5](#). New in the statement is only that (S29.2) is stated for every ordinal  $\rho$ , not only for nonzero limits.

## 76.2 The finite part of a birthday bound

Every ordinal  $\theta$  has a unique expression

$$\theta = \lambda + n, \quad n < \omega,$$

where  $\lambda = 0$  or  $\lambda$  is an infinite limit ordinal. A finite tail has an exact effect on the native alphabet.

**Proposition 76.3** (Finite terminal trees). *If  $n \geq 1$  and  $\lambda = 0$  or  $\lambda$  is an infinite limit ordinal, then*

$$\mathbf{No}_{<\lambda+n} \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} (2^n - 1).$$

*Proof.* Consider the strings whose length is at least  $\lambda$ . Each is a length- $\lambda$  sign word followed by a finite sign word of length less than  $n$ . The latter words, with centered termination, form a finite linear order of size

$$1 + 2 + \cdots + 2^{n-1} = 2^n - 1.$$

Thus this subset of  $\mathbf{No}_{<\lambda+n}$  is actually isomorphic to  $\mathbb{B}_\lambda \times_{\text{lex}} (2^n - 1)$ , proving one embedding.

For  $\lambda = 0$  this is the whole alphabet. Suppose therefore that  $\lambda$  is an infinite limit. The relevant identities are

$$2 \cdot \lambda = \lambda, \quad 2 \cdot \xi + 2 < \lambda \quad (\xi < \lambda). \quad (\text{S29.3})$$

Indeed  $\lambda = \omega \cdot \delta$  for some nonzero ordinal  $\delta$ , and left multiplication by 2 fixes this product. Strict inequality for the proper partial lengths follows because  $\xi + 1 < \lambda$ . The factor 2 in (S29.3) is on the left; no cardinal multiplication is being substituted here.

Replace each minus sign by 00 and each plus sign by 11. If a string  $x$  has length  $\xi < \lambda$ , append the termination block 01, then pad with zeros to length  $\lambda$ , and assign it finite fiber rank 0. If  $x$  has length  $\lambda + j$ , with  $j < n$ , double its first  $\lambda$  signs to form its length- $\lambda$  binary block, and assign its finite tail its rank in the  $(2^n - 1)$ -point centered tail order.

The first binary blocks separate any two short strings and every short string from every long string. At a termination disagreement the decisive comparison is precisely

$$00 < 01 < 11.$$

Different long prefixes also compare correctly. Long strings sharing their first  $\lambda$  signs compare by their finite tail ranks. Consequently this is an order embedding of the entire alphabet into  $\mathbb{B}_\lambda \times_{\text{lex}} (2^n - 1)$ .  $\square$

The full sign level  $\lambda + n$  has  $2^n$  points over each length- $\lambda$  prefix, whereas the available finite subtree has only  $2^n - 1$ . Thus [note 76.2](#) immediately gives

$$\mathbb{B}_{\lambda+n} \not\hookrightarrow \mathbf{No}_{<\lambda+n}. \quad (\text{S29.4})$$

The limit case requires a different argument: there is no final finite fiber in which to count points.

### 76.3 Fusion at a limit birthday

**Proposition 76.4** (The missing full-height cube). *For every infinite limit ordinal  $\lambda$ ,*

$$\mathbf{No}_{<\lambda} \hookrightarrow \mathbb{B}_\lambda, \quad \mathbb{B}_\lambda \not\hookrightarrow \mathbf{No}_{<\lambda}.$$

*Proof.* The first embedding is the short-string code from [proposition 76.3](#): double signs, append 01, and pad with zeros to length  $\lambda$ .

Suppose for a contradiction that  $e : \mathbb{B}_\lambda \hookrightarrow \mathbf{No}_{<\lambda}$ . Recursively construct binary prefixes  $p_\xi$ , for  $\xi \leq \lambda$ , of length  $2 \cdot \xi$ . Require that the images of all extensions of  $p_\xi$  share  $\xi$  actual surreal signs; in particular, none of these common coordinates is a termination symbol.

At a stage  $\xi < \lambda$ , the earlier target signs are fixed. The target symbol at coordinate  $\xi$  is therefore a nondecreasing function on the retained source cylinder with values in

$$- < \text{end} < +.$$

A nondecreasing map to a three-point chain becomes constant on a full binary subcylinder after at most two source-bit choices. To see this precisely, process the two threshold functions indicating whether the symbol is at least **end** and whether it is  $+$ . For each threshold, the one-bit child rule in the proof of [lemma 76.1](#) makes its value constant while retaining a full child cylinder. If fewer than two choices suffice, fix additional arbitrary bits. The resulting prefix has length  $2 \cdot (\xi + 1)$ .

The constant target symbol cannot be **end**. Since the earlier actual signs are common, termination at  $\xi$  would make every image the same length- $\xi$  surreal. The retained source cylinder still has at

least two elements by (S29.3), contradicting injectivity. Its next common symbol is therefore minus or plus, as required.

At a limit stage  $\xi < \lambda$ , take the union of the preceding prefixes. Its length is

$$\sup_{\eta < \xi} 2 \cdot \eta = 2 \cdot \xi < \lambda,$$

and every earlier chosen target sign persists. At the final stage  $p_\lambda$  is a full source word because  $2 \cdot \lambda = \lambda$ . Its image has an actual sign at every coordinate below  $\lambda$ . That image has birthday at least  $\lambda$ , contrary to membership in  $\mathbf{No}_{<\lambda}$ .  $\square$

**Theorem 76.5** (Exact native birthday spectrum). *For all ordinals  $\beta, \theta$ ,*

$$\mathbb{B}_\beta \hookrightarrow \mathbf{No}_{<\theta} \iff \beta < \theta.$$

*Proof.* If  $\beta < \theta$ , the strings of exactly length  $\beta$  form the required copy of  $\mathbb{B}_\beta$ . The empty alphabet at  $\theta = 0$  contains no such cube. For positive  $\theta$ , nonembedding at  $\beta = \theta$  is (S29.4) or proposition 76.4. Every larger cube contains  $\mathbb{B}_\theta$  by fixing its later coordinates, so larger  $\beta$  are excluded as well.  $\square$

**Corollary 76.6** (A strict hierarchy on every birthday band). *For all ordinals  $\theta, \eta$ ,*

$$\mathbf{No}_{<\theta} \hookrightarrow \mathbf{No}_{<\eta} \iff \theta \leq \eta.$$

*Proof.* Inclusion proves sufficiency. If  $\theta > \eta$ , the exact sign level  $\eta$  embeds  $\mathbb{B}_\eta$  into  $\mathbf{No}_{<\theta}$ . An embedding of that alphabet into  $\mathbf{No}_{<\eta}$  would contradict theorem 76.5.  $\square$

For an infinite initial cardinal  $\kappa$ , every noninitial cutoff  $\kappa < \theta < \kappa^+$  has alphabet cardinality  $2^\kappa$ . Nevertheless, corollary 76.6 distinguishes all these cutoffs under order embeddability. This distinction will disappear from the equimorphism type of several corresponding enumeration strata.

## 76.4 Coding length versus embedded cubes

**Corollary 76.7** (Exact native binary coding lengths). *Let  $\ell_2(L)$  be the least ordinal  $\rho$  such that  $L \hookrightarrow \mathbb{B}_\rho$ . Then  $\ell_2(\mathbf{No}_{<0}) = 0$ . For every nonzero limit  $\lambda$ ,*

$$\ell_2(\mathbf{No}_{<\lambda}) = \lambda.$$

*For  $n \geq 1$  and  $\lambda = 0$  or an infinite limit,*

$$\ell_2(\mathbf{No}_{<\lambda+n}) = \lambda + \lceil \log_2(2^n - 1) \rceil = \begin{cases} \lambda, & n = 1, \\ \lambda + n, & n \geq 2. \end{cases}$$

*Proof.* At a limit, proposition 76.4 supplies the upper code. Every proposed shorter target  $\mathbb{B}_\rho$  is excluded because  $\mathbb{B}_{\rho+1} \hookrightarrow \mathbf{No}_{<\lambda}$  and  $\mathbb{B}_{\rho+1} \not\hookrightarrow \mathbb{B}_\rho$ . At a successor cutoff use proposition 76.3 and note 76.2.  $\square$

For example,  $\mathbf{No}_{<\omega}$  and  $\mathbf{No}_{<\omega+1}$  both have least binary coding length  $\omega$ , but only  $\mathbf{No}_{<\omega+1}$  contains  $\mathbb{B}_\omega$ . At a limit birthday, all shorter cubes occur and their supremum is not attained. At the next birthday that full cube appears without increasing the least coding length. The two numerical invariants therefore answer different questions.

| Birthday bound       | Native alphabet information                                                | Least binary code |
|----------------------|----------------------------------------------------------------------------|-------------------|
| $\omega$             | cubes exactly of length $< \omega$                                         | $\omega$          |
| $\omega + 1$         | $\mathbf{No}_{<\omega+1} \simeq_e \mathbb{B}_\omega$                       | $\omega$          |
| $\omega + 2$         | $\mathbf{No}_{<\omega+2} \simeq_e \mathbb{B}_\omega \times_{\text{lex}} 3$ | $\omega + 2$      |
| $\omega + 3$         | $\mathbf{No}_{<\omega+3} \simeq_e \mathbb{B}_\omega \times_{\text{lex}} 7$ | $\omega + 3$      |
| $\omega \cdot 2$     | cubes exactly of length $< \omega \cdot 2$                                 | $\omega \cdot 2$  |
| $\omega \cdot 2 + 1$ | $\mathbf{No}_{<\omega \cdot 2+1} \simeq_e \mathbb{B}_{\omega \cdot 2}$     | $\omega \cdot 2$  |

These arguments use only set-sized sign strings, ordinal recursion, and canonical choices between two binary children. The native spectrum and strict hierarchy can consequently be developed in ZF; no enumeration of an arbitrary unordered set is needed. This makes them particularly suitable as an initial formalization target before the choice-dependent permutation constructions in the next section.

[merge] “The next section” of source 29 is printed as [section 78](#). The proposition on finite terminal trees is printed with  $\mathbb{B}_\lambda \times_{\text{lex}} (2^n - 1)$  where source 29’s typeset text has a stray double comma in the subscript of  $D$ ; no change of meaning. At  $\theta = \omega$  the theorem gives the countable alphabet  $\mathbf{No}_{<\omega}$  (order-isomorphic to the dyadic rationals) all finite cubes and no infinite one, consistent with  $\ell_2(\mathbf{No}_{<\omega}) = \omega$ . The repository’s earlier statements about birthday alphabets concern initial cardinal cutoffs: their cardinality  $\mu = 2^{<\kappa}$  and their cuts ([section 29](#)). The cardinality  $|\mathbf{No}_{<\theta}| = 2^\kappa$  is constant on each band  $\kappa < \theta \leq \kappa^+$ , while [corollary 76.6](#) separates every two cutoffs of the band by embeddability.

## 77 Noncardinal birthday cutoffs (source 21)

[merge] Source 21’s Section 5, printed whole. It keeps its letters  $\kappa = |\theta|$  and  $\mu = |\mathbf{No}_{<\theta}|$  (conventions of this Part: here  $\mu = 2^\kappa$ , not  $2^{<\kappa}$ ); its  $d$ ,  $c$  and  $\text{br}$  are printed  $\text{d}$ ,  $\text{cell}$  and  $\ell_2$  (its definition of  $\text{br}$  is that of  $\ell_2$ ); its displayed equation (3) is (S21.3). The question it answers in part is [research question 113.12](#) (“the repository explicitly asks”), which Part VI left open for noncardinal cutoffs ([section 33](#), items (b) and (g)). Some clauses are instances of results printed earlier:  $|P| = 2^\mu$  is [theorem 6.1](#); the formula  $\text{d} = \text{cell} = \mu^{<\mu}$  of [theorems 77.1](#) and [77.4](#) is the case  $\alpha = \mu$  of source 16’s [theorem 27.2](#), which needs no hypothesis on  $X$ ; the point characters of  $\mu$  are those that source 17 computes for initial cardinal cutoffs ([theorem 30.2](#), where  $q = \text{cf } \mu$ ). New are the sufficient condition of [theorem 77.1](#), its verification at every noncardinal cutoff, the coding rank, the global cofinality of  $\lambda$  and [corollary 77.5](#). Source 29 obtains the coding rank again, for every enumeration length, in [corollary 78.9](#).

The repository explicitly asks what survives when the birthday bound is an arbitrary ordinal rather than an initial cardinal ([research question 113.12](#)). There is a useful simplification in that case: the alphabet becomes strictly larger than the length of any of its monotone chains. This determines several exact invariants without any additional cardinal-arithmetic hypothesis.

### 77.1 A general permutation lemma

For an infinite initial cardinal  $\mu$  and an ordered set  $X$  of cardinality  $\mu$ , write  $\mathcal{W}_\mu(X)$  for the bijections  $\mu \rightarrow X$  in enumeration lexicographic order. For an injection  $p : \delta \rightarrow X$ , where  $\delta < \mu$ , put

$$C_p = \{f \in \mathcal{W}_\mu(X) : f \upharpoonright \delta = p\}.$$

The *prefix topology* has these cylinders as a base. Throughout this section, cellularity means the supremum of the sizes of pairwise disjoint families of nonempty open sets; the supremum need not be attained.

**Theorem 77.1** (Short monotone chains in the alphabet). *Suppose that  $X$  contains neither an increasing nor a decreasing sequence of order type  $\mu$ . Then  $P = \mathcal{W}_\mu(X)$  is dense without endpoints, and its order topology equals its prefix topology. The cylinders are clopen, and*

$$d(P) = \text{cell}(P) = \mu^{<\mu}.$$

*Every  $f \in P$  has left cofinality and right coinitiality exactly  $\text{cf}(\mu)$ .*

*Proof.* Define the feasible downward and upward divergence positions by

$$\begin{aligned} D^-(f) &= \{i < \mu : \exists j > i [f(j) < f(i)]\}, \\ D^+(f) &= \{i < \mu : \exists j > i [f(j) > f(i)]\}. \end{aligned}$$

If  $D^-(f)$  were bounded, a final part of  $f$  would be strictly increasing. Every final interval of the initial ordinal  $\mu$  has order type  $\mu$ , so this contradicts the hypothesis. Thus  $D^-(f)$  is cofinal in  $\mu$ , and the same argument applies to  $D^+(f)$ . At any feasible position, swap  $f(i)$  with the corresponding later value. This produces a bijection whose first disagreement with  $f$  occurs at  $i$ , in the desired direction.

For  $f < g$ , an upward swap of  $f$  beyond their first disagreement gives  $f < h < g$ . The same construction in both directions shows that every point is internal to each of its prefix cylinders. Since cylinders are convex, they are open; their complements are unions of other cylinders of the same length, so they are closed. Conversely, a prefix longer than the first disagreements with the endpoints of an order neighborhood lies inside that neighborhood. The two topologies therefore agree.

Choose a cofinal subset of  $D^-(f)$  of size  $\text{cf}(\mu)$  and one downward swap at each of its positions. These permutations are cofinal below  $f$ , since a later first disagreement brings a point closer to  $f$  from below. Any family of fewer than  $\text{cf}(\mu)$  points below  $f$ , however, has bounded disagreement positions; a downward swap beyond that bound lies above the entire family. The left character is exactly  $\text{cf}(\mu)$ . Reverse inequalities for the right character.

There are  $\mu^{<\mu}$  prefixes, so one completion per prefix gives a dense set of at most that size. Every prefix extends: its range has size less than  $\mu$ , its complement has size  $\mu$ , and the remaining coordinate interval has order type  $\mu$ . At level  $\delta < \mu$  there are exactly  $\mu^{|\delta|}$  injective prefixes. For the lower bound, encode an arbitrary function  $h : \delta \rightarrow \mu$  as the injection  $i \mapsto \langle h(i), i \rangle$  using a fixed injection  $\mu \times \delta \rightarrow \mu$ . Their cylinders form a disjoint open family of that size. Taking the supremum over  $\delta$  and using  $\text{cell}(P) \leq d(P)$  proves the formula.  $\square$

*Remark 77.2.* The hypothesis is sufficient, not necessary. It fails for  $X = \mathbb{Q}$  and  $\mu = \omega$ , even though the conclusion holds there because deleting a finite prefix leaves a dense alphabet without endpoints. It should not be silently substituted for that different argument.

**Lemma 77.3** (Finite-alphabet obstruction). *If  $|\tau| = \kappa$  is infinite and  $A$  is a finite linear order, then the lexicographic order on  $A^\tau$  contains no increasing or decreasing sequence of length  $\kappa^+$ .*

*Proof.* Suppose  $(x_\eta)_{\eta < \kappa^+}$  is strictly increasing. Recursively make each coordinate constant on a final segment. Once all coordinates below  $\beta$  have stabilized, their stabilization indices have a common upper bound below  $\kappa^+$ , since  $|\beta| \leq \kappa$  and  $\kappa^+$  is regular. Beyond that bound, coordinate  $\beta$  is

nondecreasing and takes only finitely many values, so it too eventually stabilizes. After  $\tau$  steps a single final segment has every coordinate constant, a contradiction. Reverse the alphabet order for decreasing sequences.  $\square$

[merge] This lemma is the finite-alphabet case of the stabilization argument used for source 19's [proposition 64.5](#) (no increasing  $\kappa^+$ -sequence in  $\mathbb{B}_\kappa$ ); it is stated here for every finite alphabet and every index ordinal of cardinality  $\kappa$ .

## 77.2 Exact invariants at an arbitrary noninitial bound

**Theorem 77.4** (Noncardinal cutoff). *Let  $\theta$  be an infinite ordinal that is not an initial cardinal, and put*

$$\kappa = |\theta|, \quad X = \mathbf{No}_{<\theta}, \quad \mu = |X|.$$

*Then  $\mu = 2^\kappa > \kappa$ . For  $P = \mathcal{W}_\mu(X)$  one has*

$$|P| = 2^\mu, \quad \ell_2(P) = \mu, \quad d(P) = \text{cell}(P) = \mu^{<\mu},$$

*where  $\ell_2$  is the least ordinal length of a fixed-length binary lexicographic cube into which the order embeds. The order  $P$  is dense without endpoints, its order topology is its prefix topology, and every point has left and right character  $\text{cf}(\mu)$ .*

*Write uniquely  $\theta = \lambda + n$ , where  $\lambda$  is a nonzero limit ordinal and  $n < \omega$ . Then*

$$\text{cf}(P) = \text{ci}(P) = \text{cf}(\lambda).$$

*Proof.* Since  $\kappa < \theta$ , all sign sequences of length  $\kappa$  lie in  $X$ . Thus  $|X| \geq 2^\kappa$ . There are  $\kappa$  possible lengths below  $\theta$ , each contributing at most  $2^\kappa$  sequences, proving  $\mu = 2^\kappa$ . Pad a sign sequence to length  $\theta$  with a symbol 0, ordered  $- < 0 < +$ . This embeds the numerical order of  $X$  into the lexicographic order on  $\{-, 0, +\}^\theta$ : the padding symbol reproduces the centered termination convention for surreal signs. By [lemma 77.3](#),  $X$  has no monotone sequence of length  $\kappa^+$ . As  $\mu \geq \kappa^+$ , [theorem 77.1](#) applies.

The number of bijections of an infinite  $\mu$ -set is  $2^\mu$ . The upper bound is  $\mu^\mu = 2^\mu$ ; independent swaps in a partition into pairs give the lower bound. To obtain the binary-rank upper bound, encode the three padding symbols as 00, 01, 10, respectively. This embeds  $X$  into binary words of fixed length  $\tau = 2 \cdot \theta$ . Here  $|\tau| = \kappa < \mu$ , so  $\tau < \mu$  and  $\tau \cdot \mu = \mu$ . Concatenating the entry codes of a permutation therefore embeds  $P$  into  $2^\mu$  with its lexicographic order.

Conversely, partition  $X$  into  $\mu$  pairs, order each pair numerically, and let each bit choose one of the pair's two enumeration orders. Since  $2 \cdot \mu = \mu$ , this embeds the lexicographic cube  $2^\mu$  into  $P$ . That cube contains every linear order  $L$  of cardinality at most  $\mu$ : index its elements by distinct coordinates and send  $t \in L$  to the characteristic function of its strict lower set, assigning 0 elsewhere. For  $s < t$ , these characteristic functions are nested, differ at the coordinate for  $s$ , and compare in the correct direction at their first disagreement. Thus, for every infinite  $\alpha < \mu$ , the order  $P$  contains the ordinal  $|\alpha|^+$ , whereas  $2^\alpha$  does not by [lemma 77.3](#). Finite target lengths are excluded by cardinality. The binary rank is exactly  $\mu$ .

It remains to compute the global cofinalities. In fact this argument works for every infinite birthday bound. If  $n = 0$ , the all-plus words  $+^\beta$ , for  $\beta < \lambda$ , are cofinal in  $X$ . A family of fewer than  $\text{cf}(\lambda)$  words has birthdays bounded below  $\lambda$ , so a longer all-plus word dominates the family. Hence  $\text{cf}(X) = \text{cf}(\lambda)$ . Whenever an alphabet  $Y$  has no maximum,

$$\text{cf}(\mathcal{W}_\mu(Y)) = \text{cf}(Y), \tag{S21.3}$$



because cofinal first entries give cofinal permutations, and the first entries of a cofinal family of permutations must be cofinal in  $Y$ .

If  $n > 0$ , let  $B$  consist of the words in  $X$  extending  $+^\lambda$ . These form a finite final block of the numerical alphabet, of size

$$|B| = 1 + 2 + \cdots + 2^{n-1} = 2^n - 1.$$

Every word outside  $B$  terminates or first has a minus sign below  $\lambda$ , and hence lies below every member of  $B$ . The complement  $Y = X \setminus B$  has no maximum. The words  $+^\beta$ ,  $\beta < \lambda$ , are cofinal in it. For the converse bound assign to each  $y \in Y$  its first disagreement or termination position relative to  $+^\lambda$ . Fewer than  $\text{cf}(\lambda)$  such positions are bounded below  $\lambda$ ; a longer all-plus word then dominates the corresponding family. Thus  $\text{cf}(Y) = \text{cf}(\lambda)$ .

The cylinder beginning by listing  $B$  in decreasing numerical order is a final convex suborder of  $P$ . Deleting this finite prefix leaves coordinate order type  $\mu$ , so that cylinder is isomorphic to  $\mathcal{W}_\mu(Y)$ . Equation (S21.3) gives  $\text{cf}(P) = \text{cf}(\lambda)$ . Negating all signs proves the matching coinitality formula.  $\square$

**Corollary 77.5** (Failure of interval self-similarity). *At every noncardinal cutoff of theorem 77.4, the whole order  $P$  is not isomorphic to any nonempty bounded open interval of itself, nor to either proper open ray.*

*Proof.* König's theorem gives

$$\text{cf}(P) = \text{ci}(P) = \text{cf}(\lambda) \leq \kappa < \text{cf}(2^\kappa) = \text{cf}(\mu).$$

Every interval  $(a, b)$  has cofinality and coinitality  $\text{cf}(\mu)$ , by the point-character calculation. A proper open ray has one of these endpoint characters. The corresponding invariant of  $P$  is strictly smaller.  $\square$

For example, the cutoffs  $\omega + 1$ ,  $\omega \cdot 2$ , and  $\omega^2$  all have  $\mu = 2^{\aleph_0}$ . Their minimal permutation orders have global cofinalities  $\omega$  and point characters  $\text{cf}(2^{\aleph_0}) > \omega$ . Moving from cutoff  $\omega$  to cutoff  $\omega + 1$  already changes the point character from countable to uncountable. These calculations answer part of the repository's proposed question, while leaving full cut spectra, saturation, and isomorphism classifications open. No historical novelty claim is attached to these deductions.

[merge] Two remarks on the comparison with initial cardinal cutoffs. First, at an initial cardinal cutoff  $\kappa$  the global cofinality is  $r = \text{cf } \kappa$  and the point characters are  $(q, q)$ ,  $q = \text{cf } \mu$  (theorem 30.2), with  $r \leq q$  (note 29.9). The argument of corollary 77.5 therefore applies at an initial cardinal cutoff exactly when  $r < q$ ; at a noncardinal cutoff the corresponding inequality  $\text{cf } \lambda < \text{cf } \mu$  holds always, by König's theorem. Second, at an initial cardinal cutoff the coding rank is  $\rho_\kappa$ , which differs from  $\mu$  exactly at singular strong limits (theorem 8.3); at a noncardinal cutoff it is always  $\mu$ , because  $\mathbf{No}_{<\theta}$  codes into binary words of the fixed length  $2 \cdot \theta < \mu$ . Source 29's corollary 76.6 shows that the noncardinal cutoffs of one band are nevertheless pairwise distinguished by their alphabets, while theorem 77.4 sees of the cutoff only  $\kappa$  and  $\text{cf } \lambda$ .

## 78 All enumeration lengths at a noninitial birthday bound (source 29)

[merge] Source 29's Section 5, printed whole except for four lemmas that source 19 proves for initial cardinal cutoffs in its Section 5 (printed in Part X, section 64) and that hold verbatim, for any alphabet, at noninitial cutoffs; these are recorded in notes carrying source 29's labels (notes 78.3,

78.4, 78.6 and 78.7). Source 29 credits all four to source 19 (“the earlier residual analysis”). Its main-prefix lemma and residual reduction (lemma 78.1 and proposition 78.2) are source 19’s lemma 64.2 and theorem 64.4 transferred to noninitial cutoffs, where the proof is simpler (no singular scheduler is needed for the main prefix); they are printed in full because their statements are new. The coding sandwich, the exact cube and birthday capacity, the coding bounds and the countable-band classification (theorems 78.5 and 78.8 and corollaries 78.9 and 78.10) are new; at initial cardinal cutoffs the corresponding statements are source 19’s theorems 64.7 and 64.11 and corollary 64.13, except the native birthday threshold (S29.13), which is new at every cutoff. Renamed (conventions of this Part):  $D_{\rho,m}$  is  $\mathbb{B}_{\rho} \times_{\text{lex}} m$ ; the layer weights  $r(\lambda)$ ,  $u_{\kappa}(\lambda)$  are  $\text{wt}(\lambda)$ ,  $\text{wt}_{\kappa}^+(\lambda)$ ; the weighted lengths  $R$ ,  $U$  are  $\varrho_{\text{lo}}$ ,  $\varrho_{\text{up}}$ ;  $\ell_2$  is  $\ell_2$ . Displayed equations (5)–(13) are (S29.5)–(S29.13). Its first sentence’s “fixed-length results” of the repository are theorem 31.2 (Part VI).

The repository’s fixed-length results include the minimum stratum and finite tails after a whole-cardinality prefix (theorem 31.2). The earlier residual analysis goes further for initial birthday bounds: it treats arbitrary smaller residual alphabets and obtains an exact embedded-cube spectrum, while retaining a specific upper coding gap at singular non-strong-limit residual cardinals (source 19, section 64). We now extend that analysis to every noninitial birthday cutoff. The residual fusion and singular scheduler are inherited mechanisms; the work here is to verify their hypotheses for the new alphabets and draw the resulting sharp birthday-embedding consequences.

Work in ZFC. Fix an infinite ordinal  $\theta$  which is not an initial cardinal, and put

$$\kappa = |\theta|, \quad X = \mathbf{No}_{<\theta}, \quad \mu = |X| = 2^{\kappa}. \quad (\text{S29.5})$$

The last equality follows because every sign word of length  $\kappa$  belongs to  $X$ , and there are only  $\kappa$  possible lengths, each contributing at most  $2^{\kappa}$  words. In particular  $\kappa < \mu$ . An admissible enumeration type is an ordinal  $\alpha$  of cardinality  $\mu$ , and  $\mathcal{W}_{\alpha}(X)$  denotes the bijections  $\alpha \rightarrow X$  ordered by their first numerical disagreement.

## 78.1 Main prefixes and variable residual alphabets

Write

$$\alpha = \mu \cdot \gamma + \zeta, \quad 0 < \gamma < \mu^+, \quad \zeta < \mu.$$

The difference between an injection of length  $\mu \cdot \gamma$  and a bijection of that length is essential: a proper prefix of the full enumeration may leave a smaller alphabet, with numerical order depending on the prefix.

**Lemma 78.1** (Noninitial main-prefix package). *Every lexicographically ordered family of injective prefixes  $p : \mu \cdot \gamma \rightarrow X$  embeds into  $\mathbb{B}_{\mu \cdot \gamma}$ . For every  $Y \subseteq X$  with  $|Y| < \mu$ ,*

$$\mathbb{B}_{\mu \cdot \gamma} \hookrightarrow \mathcal{W}_{\mu \cdot \gamma}(X \setminus Y).$$

*Proof.* Pad every sign word to length  $\theta$  with a termination symbol ordered between minus and plus, and encode the three symbols by 00, 01, 11. This embeds  $X$  into  $\mathbb{B}_{\tau}$ , where  $\tau = 2 \cdot \theta$  has cardinality  $\kappa$ . Thus  $0 < \tau < \mu$  and  $\tau \cdot \mu = \mu$ . Concatenating the fixed entry codes over  $\gamma$  successive  $\mu$ -blocks yields an order-preserving code of length  $\mu \cdot \gamma$ . Exhaustiveness is never used.

The complement  $X \setminus Y$  has cardinality  $\mu$ . Partition it into pairs indexed by  $\mu \cdot \gamma$ , orient each pair numerically, and let its corresponding bit choose the order in which the two labels are listed. Every label appears exactly once. The resulting enumeration has length  $2 \cdot (\mu \cdot \gamma) = \mu \cdot \gamma$ , and its first changed bit gives its first changed label in the same order direction.  $\square$

**Proposition 78.2** (Exact residual reduction). *For every nonempty linear order  $K$ ,*

$$\begin{aligned} \mathbb{B}_{\mu \cdot \gamma} \times_{\text{lex}} K &\hookrightarrow \mathcal{W}_{\mu \cdot \gamma + \zeta}(X) \\ \iff \exists Y \subseteq X (|Y| = |\zeta| \text{ and } K &\hookrightarrow \mathcal{W}_{\zeta}(Y)). \end{aligned}$$

*Proof.* There is an actual lexicographic-sum decomposition

$$\mathcal{W}_{\mu \cdot \gamma + \zeta}(X) \cong \sum_p \mathcal{W}_{\zeta}(X \setminus \text{ran } p), \quad (\text{S29.6})$$

where  $p$  ranges over the injective main prefixes whose omitted set has cardinality  $|\zeta|$ . The projection onto this prefix order is nondecreasing, and [lemma 78.1](#) embeds its range into  $\mathbb{B}_{\mu \cdot \gamma}$ . Apply [lemma 76.1](#) to the composition with a proposed product embedding. The whole factor  $K$  survives in one target fiber, proving necessity.

Conversely, fix a suitable  $Y$ . Use [lemma 78.1](#) to enumerate the fixed complement  $X \setminus Y$  according to the main binary factor, then append the chosen  $K$ -indexed permutations of  $Y$ .  $\square$

This reduction applies to arbitrary product embeddings. It does not assume that an embedding respects the displayed fibers. In particular, it prevents a variable residual alphabet from being replaced silently by a preferred copy of a birthday fragment.

[merge] The decomposition (S29.6) is source 19's (6) and, for the minimal length, the set-level fibre identity of sources 21 and 22 printed in Part X; the class version is source 16's [equation \(S16.12\)](#).

## 78.2 The finitely many residual cardinal layers

Repeated left division of  $\zeta$  by its current infinite initial cardinal gives a finite expression

$$\zeta = \lambda_0 \cdot \eta_0 + \cdots + \lambda_{s-1} \cdot \eta_{s-1} + n, \quad (\text{S29.7})$$

where

$$\mu > \lambda_0 > \cdots > \lambda_{s-1} \geq \omega, \quad 0 < \eta_i < \lambda_i^+, \quad n < \omega.$$

There cannot be infinitely many layers, since their cardinals would form an infinite strictly decreasing sequence of ordinals. When  $\zeta$  is finite,  $s = 0$ .

Define the two weights

$$\text{wt}(\lambda) = \begin{cases} \lambda \cdot \lambda, & \lambda \text{ singular strong limit,} \\ \lambda, & \text{otherwise,} \end{cases} \quad (\text{S29.8})$$

$$\text{wt}_{\kappa}^+(\lambda) = \begin{cases} \lambda \cdot \lambda, & \lambda \leq \kappa \text{ and } \lambda \text{ singular,} \\ \lambda, & \text{otherwise.} \end{cases} \quad (\text{S29.9})$$

Finally set

$$\varrho_{\text{lo}} = \mu \cdot \gamma + \sum_{i < s} \text{wt}(\lambda_i) \cdot \eta_i, \quad \varrho_{\text{up}} = \mu \cdot \gamma + \sum_{i < s} \text{wt}_{\kappa}^+(\lambda_i) \cdot \eta_i. \quad (\text{S29.10})$$

Both are nonzero limit ordinals, and  $\varrho_{\text{lo}} \leq \varrho_{\text{up}}$ . Their difference, when present, is an ordinal difference in coding cost; their cardinalities are both  $\mu$ .

*Note 78.3* (Source 29, Lemma 5.3, “Entry codes and regular blocks”). *Every linear order of infinite cardinality  $\lambda$  admits a prefix-free, order-preserving binary entry code with positive code lengths below  $\lambda$ . If  $\lambda$  is regular, concatenating these codes over a block of  $\lambda \cdot \eta$  entries has length exactly  $\lambda \cdot \eta$ .* The first sentence is source 19's [lemma 64.1](#) and the second the coding half of its [lemma 64.6](#), by

the same argument: insert the alphabet, in an enumeration of length  $\lambda$ , into  $\mathbf{No}_{<\lambda}$  by simplest separators (the birthday bound for a surreal cut, [12]), double minus and plus to 00 and 11 and append 01; regularity keeps every proper partial sum of fewer than  $\lambda$  code lengths below  $\lambda$ , and positivity makes the full sum  $\lambda$ . The code is source 15's entry code (lemma 31.1) applied after the insertion.

*Note 78.4* (Source 29, Lemma 5.4, “Inherited singular strong-limit scheduler”). *If  $\lambda$  is singular strong limit and  $0 < \eta < \lambda^+$ , then  $\mathbb{B}_{\lambda \cdot \lambda \cdot \eta} \hookrightarrow \mathcal{W}_{\lambda \cdot \eta}(\mathbf{No}_{<\lambda})$ .* This is the lower half of theorem 31.2 at the cutoff  $\lambda$  (there  $\mu = \lambda$  and  $\rho_\lambda = \lambda \cdot \lambda$ ), and source 29 says “We recall the construction from the earlier residual analysis”: the row scheduler of lemma 31.3 in the form given in the proof of source 19's lemma 64.2. Its steps, as source 29 recalls them: the sign cones with prefixes  $+^i \frown -$ ,  $i < \lambda$ , are pairwise disjoint copies of  $\mathbf{No}_{<\lambda}$ , indexed by  $\lambda \cdot \eta$ , and their complement (containing the  $\lambda$  all-plus words) is a fixed filler row placed first ( $1 + \lambda = \lambda$ ); row-column pairs are scheduled first by  $\max(i, j)$ , then lexicographically, which has order type  $\lambda$  even for singular  $\lambda$  and lets the first entry of the first varying row decide the comparison; this embeds  $(\mathbf{No}_{<\lambda})_{\text{lex}}^{\lambda \cdot \eta}$ . Finally, with  $\delta = \text{cf } \lambda$  and positive lengths  $(\tau_j)_{j < \delta}$  summing to  $\lambda$ , a binary word of length  $\lambda$  is encoded by  $\delta$  consecutive entries of sign lengths exactly  $\tau_j$ , so that  $\delta \cdot (\lambda \cdot \eta) = \lambda \cdot \eta$  entries carry  $\lambda \cdot \lambda \cdot \eta$  bits.

**Theorem 78.5** (Noninitial coding sandwich). *With (S29.5)–(S29.10),*

$$\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} n! \hookrightarrow \mathcal{W}_\alpha(X) \hookrightarrow \mathbb{B}_{\varrho_{\text{up}}} \times_{\text{lex}} n!.$$

*Proof.* For the upper embedding, code the main prefix in length  $\mu \cdot \gamma$  using lemma 78.1. At the start of a residual layer its remaining alphabet has cardinality  $\lambda_i$ : the remaining coordinate interval has that cardinality, and the enumeration is bijective.

At a regular layer, note 78.3 gives cost  $\lambda_i \cdot \eta_i$ . At a singular layer with  $\lambda_i \leq \kappa$ , pad its entry codes to length  $\lambda_i$ , giving cost  $\lambda_i \cdot \lambda_i \cdot \eta_i$ . If  $\lambda_i > \kappa$ , use the fixed global code of length  $\tau = 2 \cdot \theta$ . Now  $\tau < \lambda_i$ , so  $\tau \cdot \lambda_i = \lambda_i$ , and the cost is only  $\lambda_i \cdot \eta_i$ .

Choose one entry code for each possible residual alphabet. These are set-indexed choices. Two enumerations agreeing through a layer boundary have the same residual alphabet and use the same code. If they disagree earlier, their comparison has already been decided. The conditional codes therefore concatenate to an order-preserving code of length  $\varrho_{\text{up}}$ . After that prefix, the last  $n$  labels are determined as a set; append the rank of their permutation among its  $n!$  possibilities.

For the lower embedding, reserve disjoint fixed alphabets for the residual layers and a fixed  $n$ -element terminal set. If a residual  $\lambda < \mu$  is singular strong limit, necessarily  $\lambda \leq \kappa$ : otherwise strong limit would imply  $2^\kappa < \lambda$ , contradicting  $\lambda < \mu = 2^\kappa$ . Thus a copy of  $\mathbf{No}_{<\lambda}$  lies in a finite-prefix cone of  $X$ . All its sign lengths stay below  $\lambda \leq \kappa < \theta$ . Finitely many distinct prefixes make the alphabets for different such layers disjoint. Apply note 78.4 at each of them.

For every other residual layer reserve an arbitrary set of the required cardinality and orient pairs to realize  $\mathbb{B}_{\lambda_i \cdot \eta_i}$ . The union of all reserved sets has cardinality below  $\mu$ , so lemma 78.1 realizes the main binary factor on its complement. Concatenate the independent block enumerations and the selected terminal permutation. The first changed source factor gives the first changed target block, establishing the lower embedding.  $\square$

### 78.3 The exact capacity despite a possible coding gap

The only discrepancy between  $\text{wt}$  and  $\text{wt}_\kappa^+$  can occur at a singular non-strong-limit residual cardinal at most  $\kappa$ . The following inherited fusion mechanism determines its contribution to embedded cubes even when a short upper code is unavailable.

*Note 78.6* (Source 29, Lemma 5.6, “Small-range fusion”). *Let  $\chi$  be an infinite cardinal, let  $A$  be a linear order of cardinality less than  $2^\chi$ , and let  $K$  be nonempty. A nondecreasing map from a full binary cylinder with at least  $\chi$  initial free bits, followed by  $K$ , into  $A$  is constant on a full subcylinder obtained by fixing fewer than  $\chi$  of those bits.* This is source 19’s lemma 64.9, with the same proof: fixing the later bits and one point of  $K$  leaves  $2^\chi$  ordered test points, so some value has three preimages  $x_0 < x_1 < x_2$ ; a prefix of  $x_1$  of length below  $\chi$  beyond its first disagreements with  $x_0$  and  $x_2$  traps its whole cylinder between them.

*Note 78.7* (Source 29, Lemma 5.7, “A singular non-strong-limit layer”). *Suppose  $\lambda$  is singular and not strong limit, and  $0 < \eta < \lambda^+$ . In an order embedding, suppose the target enumerations share their earlier prefix and the remaining alphabet has cardinality  $\lambda$ . If the surviving source begins with  $\mathbb{B}_{\lambda \cdot \eta}$  followed by a nonempty factor  $K$ , one can fix that source binary block so that the next  $\lambda \cdot \eta$  target entries are constant on the surviving whole factor  $K$ .* This is source 19’s lemma 64.10, with the same proof: choose  $\rho < \lambda$  with  $2^{|\rho|} \geq \lambda$  and  $\chi = \max(|\rho|, \text{cf } \lambda) < \lambda$ , so that  $2^\chi > \lambda$  by König’s inequality; fix the target entries one at a time with note 78.6, each at a cost of fewer than  $\chi$  source bits. Source 29 makes the ordinal bookkeeping explicit: before coordinate  $j = \lambda \cdot \beta + \delta$  ( $\delta < \lambda$ ) the cost is at most  $\chi \cdot (j + 1) = \lambda \cdot \beta + \chi \cdot (\delta + 1) < \lambda \cdot \eta$ , and the total cost is at most  $\chi \cdot (\lambda \cdot \eta) = \lambda \cdot \eta$ .

**Theorem 78.8** (Exact cube and birthday capacity). *For every positive finite integer  $m$ ,*

$$\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} m \hookrightarrow \mathcal{W}_\alpha(X) \iff m \leq n!. \quad (\text{S29.11})$$

*Consequently, for every ordinal  $\beta$ ,*

$$\boxed{\mathbb{B}_\beta \hookrightarrow \mathcal{W}_\alpha(X) \iff \beta \leq \varrho_{\text{lo}} + \lfloor \log_2(n!) \rfloor.} \quad (\text{S29.12})$$

*For every ordinal  $\nu$ , the exact native-alphabet threshold is*

$$\boxed{\mathbf{No}_{<\nu} \hookrightarrow \mathcal{W}_\alpha(X) \iff \nu \leq \varrho_{\text{lo}} + \lfloor \log_2(n! + 1) \rfloor.} \quad (\text{S29.13})$$

*Proof.* The positive direction of (S29.11) follows from theorem 78.5. Conversely, suppose an embedding of  $\mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} m$  is given. Compose with the main-prefix projection and its binary code. Equal-length fusion fixes the main target prefix while retaining the whole later source factor, as in proposition 78.2.

At a regular residual layer fuse through its code of length  $\lambda_i \cdot \eta_i$ . At a singular strong-limit layer fuse through the padded code of length  $\lambda_i \cdot \lambda_i \cdot \eta_i$ . A layer with  $\lambda_i > \kappa$  has its short global code and cannot be singular strong limit, as shown in the sandwich proof. At a singular non-strong-limit layer at most  $\kappa$ , apply note 78.7; its cost is only  $\lambda_i \cdot \eta_i$ . These are exactly the successive summands defining  $\varrho_{\text{lo}}$ .

After the finitely many layers all target positions before the last  $n$  are fixed. The surviving  $m$  points must inject into the  $n!$  permutations of a single fixed residual  $n$ -element set. Hence  $m \leq n!$ , proving (S29.11).

Put  $a = \lfloor \log_2(n!) \rfloor$ . The allowed finite binary factor has  $2^a$  points, whereas the next one has  $2^{a+1} > n!$ . Thus  $\mathbb{B}_{\varrho_{\text{lo}}+a}$  embeds and  $\mathbb{B}_{\varrho_{\text{lo}}+a+1}$  does not. The cube hierarchy proves (S29.12) for every ordinal  $\beta$ .

For (S29.13), put  $b = \lfloor \log_2(n! + 1) \rfloor \geq 1$ . Since  $\varrho_{\text{lo}}$  is a nonzero limit, proposition 76.3 gives

$$\mathbf{No}_{<\varrho_{\text{lo}}+b} \simeq_e \mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} (2^b - 1), \quad \mathbf{No}_{<\varrho_{\text{lo}}+b+1} \simeq_e \mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} (2^{b+1} - 1).$$

The first finite factor has at most  $n!$  points, and the second has more. Equation (S29.11) permits the first alphabet and excludes the second. Inclusion between birthday alphabets supplies all smaller cutoffs and excludes all larger ones.  $\square$

**Corollary 78.9** (Coding bounds and exact equimorphisms). *One has*

$$\varrho_{\text{lo}} + \lceil \log_2(n!) \rceil \leq \ell_2(\mathcal{W}_\alpha(X)) \leq \varrho_{\text{up}} + \lceil \log_2(n!) \rceil.$$

If  $\varrho_{\text{lo}} = \varrho_{\text{up}}$ , then

$$\mathcal{W}_\alpha(X) \simeq_e \mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} n!, \quad \ell_2(\mathcal{W}_\alpha(X)) = \varrho_{\text{lo}} + \lceil \log_2(n!) \rceil.$$

In particular equality holds whenever every residual cardinal is regular, singular strong limit, or greater than  $\kappa$ .

*Proof.* Apply [note 76.2](#) to both sides of the sandwich. In the listed cases every lower and upper layer weight agrees. Ordinal absorption can also give  $\varrho_{\text{lo}} = \varrho_{\text{up}}$  in additional cases; equality of the computed ordinals is the precise criterion.  $\square$

The distinction in [theorem 78.8](#) and [corollary 78.9](#) is deliberate. The embedded cube spectrum and the embedded native birthday spectrum are exact even when  $\varrho_{\text{lo}} < \varrho_{\text{up}}$ . They do not, by themselves, supply an embedding of the whole order into the shorter upper cube. The remaining general-alphabet question is still whether

$$\mathcal{W}_\lambda(Y) \hookrightarrow \mathbb{B}_\lambda$$

holds for every ordered  $Y$  of singular non-strong-limit cardinality  $\lambda$ . Neither the present argument nor the earlier residual analysis settles that question (source 19, [section 64](#)).

[merge] At  $\alpha = \mu$  ( $\gamma = 1$ ,  $\zeta = 0$ ) the corollary gives  $\varrho_{\text{lo}} = \varrho_{\text{up}} = \mu$  and  $\ell_2(\mathcal{W}_\mu(\mathbf{No}_{<\theta})) = \mu$ , which is the coding rank of source 21's [theorem 77.4](#), there proved by the finite-alphabet obstruction ([lemma 77.3](#)) and here by fusion. The remaining question is source 19's "remaining coding gap" ([research question 113.94](#)) and source 29's [research question 113.212](#).

#### 78.4 A complete classification on the countable birthday band

**Corollary 78.10** (Every admissible length, without CH). *Let  $\omega < \theta < \omega_1$ , put  $\mu = 2^{\aleph_0}$ , and let  $\alpha$  be any ordinal of cardinality  $\mu$ . Write uniquely*

$$\alpha = \Lambda + n, \quad \Lambda \text{ a nonzero limit ordinal, } n < \omega.$$

Then

$$\begin{aligned} \mathcal{W}_\alpha(\mathbf{No}_{<\theta}) &\simeq_e \mathbb{B}_\Lambda \times_{\text{lex}} n!, \\ \ell_2(\mathcal{W}_\alpha(\mathbf{No}_{<\theta})) &= \Lambda + \lceil \log_2(n!) \rceil. \end{aligned}$$

Its embedded cubes have exactly the lengths

$$\beta \leq \Lambda + \lceil \log_2(n!) \rceil,$$

and its embedded birthday alphabets have exactly the bounds

$$\nu \leq \Lambda + \lceil \log_2(n! + 1) \rceil.$$

The same statements hold for any noninitial  $\theta$  with  $|\theta| < \aleph_\omega$ , using  $\mu = 2^{|\theta|}$ .



*Proof.* For a countable noninitial cutoff,  $\kappa = \omega$ . There is no singular cardinal at or below  $\kappa$ . A singular strong-limit cardinal  $\lambda > \kappa$  cannot lie below  $\mu$ , since strong limit would give  $2^\kappa < \lambda$ . Every residual weight is therefore just  $\lambda_i$ , so  $\varrho_{\text{lo}} = \varrho_{\text{up}} = \Lambda$ . Apply [theorem 78.8](#) and [corollary 78.9](#).

More generally, every infinite cardinal at or below a fixed  $\kappa < \aleph_\omega$  is one of the regular cardinals  $\aleph_j$  for finite  $j$ . Again there is no singular layer at or below  $\kappa$ , while no singular strong-limit residual can occur above it. The identical computation gives  $\varrho_{\text{lo}} = \varrho_{\text{up}} = \Lambda$ .  $\square$

For a limit  $\alpha$ , the corollary simply says  $\mathcal{W}_\alpha(\mathbf{No}_{<\theta}) \simeq_e \mathbb{B}_\alpha$ . A final single label adds no choice, so type  $\Lambda + 1$  still gives  $\mathbb{B}_\Lambda$ . A final pair gives two choices and hence  $\mathbb{B}_{\Lambda+1}$ . With three terminal labels one obtains  $\mathbb{B}_\Lambda \times_{\text{lex}} 6$ : the largest embedded cube has length  $\Lambda + 2$ , whereas the least binary coding length is  $\Lambda + 3$ .

At the minimum length  $\mu$ , the sharp birthday threshold is  $\mu + 1$ . Thus this stratum contains an ordered copy of  $\mathbf{No}_{\leq \mu} = \mathbf{No}_{< \mu+1}$  and contains no larger birthday alphabet. The  $+1$  here is ordinal successor, not the successor cardinal  $\mu^+$ .

Finally, [corollary 76.6](#) shows that the native alphabets at  $\omega + 1$ ,  $\omega + 2$ ,  $\omega \cdot 2$ ,  $\omega^2$ , and every other distinct countable noninitial cutoff are strictly ordered under order embeddability. Nevertheless, at each common admissible enumeration length their permutation orders have the same equimorphism type. No continuum hypothesis is used. Whether these corresponding permutation orders are actually isomorphic is a finer question about their cuts and local order structure, not one decided by the spectra proved here.

[merge] Two comparisons. (1) The initial-cutoff counterpart is source 30's [corollary 64.20](#) (Part X): at  $\kappa = \omega_1$ , and at every initial cutoff  $\kappa < \aleph_\omega$ , every admissible stratum is  $\simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$  in ZFC, by the equality clause of source 19's [theorem 64.7](#) (source 19's own [corollary 64.8](#) states the classification under  $\mu \leq \aleph_\omega$ ). So the cutoffs  $\omega_1$  and every countable  $\theta > \omega$  give equimorphic strata of each common length, although their alphabets are strictly ordered by [corollary 76.6](#). (2) By [theorem 77.4](#) the minimal strata ( $\alpha = \mu$ ) at the countable noninitial cutoffs share coding rank, density, cellularity and point characters, and their global cofinality  $\text{cf } \lambda$  is  $\omega$  at each of them; so no invariant computed in this Part distinguishes the permutation orders of two countable noninitial cutoffs, while their alphabets are distinguished by [corollary 76.6](#). Source 29's [research question 113.213](#) asks whether they are isomorphic.

## 79 What this Part answers

[merge] Dated notes, 3 October 2026. Each item names the question, where it is asked, and the result that answers it. Where the question index of [section 113](#) (dated 2 October 2026) or an earlier Part's status section says otherwise, these notes supersede it. Questions of sources 13–18 are cited by their own numbers; those of sources 19–30 by their labels.

- (a) *Source 12's question on noncardinal birthday bounds* ([research question 113.12](#)): **answered in part**; the index row “Noncardinal cutoffs; away from inaccessible truncations” and Part VI's status ([section 33](#), items (b) and (g)) are superseded for the noncardinal half. The question asks which features of the ordinal tail of  $\theta$  survive “in its coding rank and local character”. Answer: the coding rank of the minimal slice is always  $\mu = 2^{|\theta|}$ , its point characters are  $\text{cf } \mu$ , its density and cellularity  $\mu^{<\mu}$ , and of the ordinal tail only  $\text{cf } \lambda$  ( $\theta = \lambda + n$ ) survives, as the global cofinality and coinitality; the minimal slice is not isomorphic to any interval or ray of itself ([theorem 77.4](#) and [corollary 77.5](#)). The alphabet itself remembers  $\theta$  exactly:  $\mathbb{B}_\beta \hookrightarrow \mathbf{No}_{<\theta}$  iff  $\beta < \theta$ , so the alphabets of one band are strictly ordered by embeddability ([theorem 76.5](#) and [corollary 76.6](#)). Every enumeration length at a noninitial cutoff has a coding



sandwich, an exact cube spectrum and an exact birthday threshold (theorems 78.5 and 78.8 and corollary 78.9), and on the bands  $|\theta| < \aleph_\omega$  every length is classified up to equimorphism in ZFC, which forgets  $\theta$  (corollary 78.10). Open: the order type and isomorphism classification, cut spectra and saturation (source 21’s own reservation, and its research question 113.122); whether two countable noninitial cutoffs give isomorphic strata (research question 113.213); the upper coding gap at singular non-strong-limit residual cardinals at most  $\kappa$  (research questions 113.94 and 113.212). The same question is asked by source 15 (its Question 13.2), source 17 (Question 15.2) and source 18 (Question 19.3), with the same status, and by source 19 (research question 113.97), which is answered in the same part. Source 20’s research question 113.116 asks for gap spectra and their arrangement at noninitial cutoffs: the results here bear on it only through embedding spectra, and it stays open.

- (b) *Source 12’s question on compatibility between birthday cutoffs* (research question 113.19): **answered in part**; the index row “Compatibility between birthday cutoffs” is superseded. Restriction is not the answer: it is monotone for the native orders only in three degenerate cases (theorem 73.1), although two labels decide every sufficiently large restriction, so that eventual local comparison is the native order on the inverse limit, which is exactly the class well-orders of **No** (theorems 74.1 and 74.3 and corollary 74.4). Two coherent forward systems exist: the append-tail maps along all ordinal cutoffs, on the full spaces  $\mathcal{W}(\mathbf{No}_{<\alpha})$ , composing exactly, with the bounded-support core as direct limit (theorem 74.8); and the fixed-slot maps of the minimal slices within one cardinality band, composing exactly, homeomorphisms onto closed (clopen or nowhere dense) images (theorem 75.1 and corollary 75.2). With relation-table orders, restriction itself becomes a monotone inverse system (proposition 74.10). Open: which choices preserve finite condensation, coding ranks or canonical cylinders, and coherent minimal-length systems across a change of cardinality (research questions 113.113 and 113.124). The same question is asked by source 14 (its Question 14.10), source 17 (Question 15.11), source 18 (Question 19.4) and source 19 (research question 113.105), with the same status.
- (c) *Source 15’s question “Coherent restricted filtrations”* (its Question 13.9): **answered in part**. Native restriction is classified exactly (theorem 73.1); it is eventually correct for each fixed pair (theorems 74.3 and 74.5, with finitely many sign changes and no uniform bound); and relation-table orders give a monotone inverse system with convex fibres (proposition 74.10).
- (d) *The reals report’s question “Other ground orders”* (lwo:sec:research; Part VI’s status, section 33, item (g), which left noncardinal cutoffs open): **answered further**. For  $X = \mathbf{No}_{<\theta}$  with  $\theta$  noninitial and  $|\theta| < \aleph_\omega$ , the display  $\mathcal{W}_{\lambda+n}(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$  holds for every limit  $\lambda$  with  $|\lambda| = |X|$  and every  $n < \omega$  (corollary 78.10). For every noninitial  $\theta$ ,  $\mathcal{W}_{\lambda+n}(X) \simeq_e \mathbb{B}_{\varrho_{\text{lo}}} \times_{\text{lex}} n!$  whenever  $\varrho_{\text{lo}} = \varrho_{\text{up}}$  (corollary 78.9), which is the display when moreover  $\varrho_{\text{lo}} = \lambda$  (for instance when no residual layer is a singular strong limit).
- (e) *Set-likeness from bounded data*. No question of the report asks it, but theorems 74.6 and 74.7 give two exact criteria for set-likeness of a class well-order in terms of its restrictions to birthday cutoffs: eventual stabilization of every local rank (GB), and the minimum type  $\kappa$  on a stationary, equivalently club, class of strong-limit cutoffs (GBC); unboundedly many such cutoffs do not suffice. Source 20’s research question 113.114 (exact birthday profiles modulo the club filter) stays open. A full coherent birthday profile exists exactly under global choice, over GB (end of section 74).

Source 21 says of its noncardinal calculations that they “answer part of the repository’s proposed question, while leaving full cut spectra, saturation, and isomorphism classifications open”, and of its coherent system that “the restriction to one set-sized upper alphabet and one cardinality band is

substantive”; source 29 says that whether the countable-band permutation orders are isomorphic “is a finer question” not decided by its spectra; items (a) and (b) record the same.

## Part XII

# Words, termination markers and comparison codes

[merge] (3 October 2026.) This Part prints what four batch-83 sources prove about word orders, termination markers and the other comparison codes of Part VII: source 23 (its Sections 2–7), source 27 (its Sections 2, 4 and 5), source 29 (its Sections 2 and 6, its Subsection 3.3 and its Section 7) and source 30 (its Sections 2, 7 and 8). It continues Part VII ([part VII](#)) and answers, in part, two of the questions that Part VII left open ([section 85](#)). It is organized by theme, not by source.

- [Section 80](#) prints the four sources’ maps of interpretations, in the manner of [section 34](#).
- [Section 81](#) prints source 23’s termination-marker criterion and point characters for an abstract set-cut saturated alphabet. Both are [theorems 36.1](#) and [36.2](#) of Part VII (source 16) in a general alphabet; they are printed as notes with a second route.
- [Section 82](#) treats strictly increasing words: source 23’s theorem that no placement of a marker makes them set-cut saturated ([theorem 82.1](#), new), source 27’s factorization at the terminator, its product formula for the centered order, its zero conventions, interval richness and characters at an arbitrary gap ([theorems 82.4](#) and [82.7](#), [corollary 82.5](#), and [propositions 82.6](#) and [82.8](#)), and source 29’s Section 6, which proves the same product independently; its duplicates are notes with second routes, and its threshold reduction ([proposition 82.14](#)) is printed as a proposition.
- [Section 83](#) prints source 29’s Section 7 on numerically well-ordered proper-class carriers.
- [Section 84](#) treats relation-table lexicography: source 27’s scan-independent spectrum and the scan dependence of its topology ([theorems 84.2](#) to [84.4](#)), the density of the set-like orders under a set-like scan ([theorem 84.5](#), with source 30’s [theorem 84.11](#)), and source 30’s extension, approximation and canonical-scan theorems.
- [Section 85](#) records, with dates, the questions this Part answers.

The batch-83 notation addendum ([section 1.3.2](#)), which continues [section 1.3.1](#), fixes the symbols. The renamings that matter most here: source 23’s marker  $e$ , source 27’s terminator at a cut and source 29’s  $\dagger$  at a cut are all printed  $\varepsilon$ , as source 16’s marker in [section 36](#); the symbol  $\dagger$  is kept for the centered terminator in the place of zero, with  $\dagger_-$  and  $\dagger_+$  as in [section 37](#). Source 27’s  $\mathcal{B}$  and source 29’s  $\mathbf{T}$ , the shorter-first order of all set-length surreal words, are source 16’s  $\mathbf{T}_{\min}$  of [theorem 36.2](#) and are printed so; their nonempty part  $(\mathcal{B} \setminus \{\emptyset\}, \mathbf{P})$  is  $\mathbf{T}_{\min}^+$ . Source 27’s centered order  $J$  and source 29’s  $\mathcal{C}$  are Part VII’s  $\mathcal{T}_c^{\text{inc}}$ . Source 27’s scan  $q$  is printed  $s$  (this report’s  $q$  is cf  $\mu$ ), its table bits  $b_R$  are printed  $t_R$  (this report’s  $b$  is a baseline), and its  $\kappa = |X|$  is printed  $\mu$ .

## 80 The four sources’ maps of interpretations

[merge] Each of sources 23, 27, 29 and 30 opens, as the six batch-81 sources did ([section 34](#)), with a map of the readings of “all well-orders of surreal numbers”. The four maps are printed whole,

in source order. They agree with each other and with the batch-81 maps; their statements are inherited and none is claimed as new.

- Prefix-freeness of exhaustive enumerations of one set (source 23’s [note 80.1](#), and the corresponding paragraphs of sources 27, 29 and 30) is `lwo:lem:prefix-free` of the reals report, used in [section 6](#) and restated by source 15 ([note 32.3](#)).
- The fixed-set spectrum (source 27’s display (S27.1) and source 29’s [note 61.4](#), which is printed in its own Part with source 29’s Section 3) is [theorem 6.1](#).
- The three lexicographies of the six well-orders of a three-element carrier (source 30’s Subsection 2.4) are source 15’s table in [section 38](#), with the same pair scan and the same three orders.
- That the baseline is least for inverse-rank lexicography, and that this endpoint separates it from enumeration lexicography (source 30’s Subsection 2.5), is [theorem 38.1](#).
- The non-convexity of order-type classes and the collapse of the set-like global orders to the single type `Ord` are made in the batch-81 maps ([section 34](#)) and in [section 38](#).
- The example  $\{-1/(\alpha + 1) : \alpha \in \text{Ord}\} \cup \{0\}$  of a numerically increasing class of type `Ord + 1` (sources 27, 29 and 30) is printed after [proposition 38.6](#) and in [section 38.8](#).

Each source’s Kanovei–Shelah paragraph agrees with the qualification of [section 34](#): the index set of Kanovei and Shelah consists of maps  $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters, it is not the order of all well-orders of  $\mathbb{R}$ , and it motivates the method only [16]. Source 27’s paragraph is printed below. Those of sources 29 and 30 are in their Sections 1 (scope and provenance), which this Part does not print. Source 30 places the index in “Section 2” of the paper; source 29 (“§1 of the preprint”), source 27 (“Section 1”) and this report’s bibliography entry place it in Section 1, page 2, of the preprint, and that location is kept.

### 80.1 Source 23: a taxonomy of what can be ordered

| Construction           | Objects                                                                                               | Logical size and main caution                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed carrier          | Well-orders on a fixed set $X \subseteq \mathbf{No}$ ; equivalently bijections $\alpha \rightarrow X$ | A set in ZFC. Complete enumerations of the same carrier cannot be proper prefixes of one another.                                           |
| Fixed order type       | Bijections $\alpha \rightarrow X$ for one ordinal $\alpha$                                            | A set; it is nonempty exactly when $ \alpha  =  X $ .                                                                                       |
| Variable set carrier   | Well-orders on arbitrary set subsets of $\mathbf{No}$                                                 | An ordinary class, represented by injective ordinal-length words. Termination affects the order.                                            |
| Words with repetition  | Arbitrary $p : \alpha \rightarrow \mathbf{No}$ , with $\alpha$ a set ordinal                          | An ordinary class of labelled sequences, but not literally well-orders on their ranges unless one keeps positions as distinct objects.      |
| Inherited well-orders  | Set subsets well-ordered by the numerical $<$                                                         | Strictly increasing words. The restriction on continuations is substantial.                                                                 |
| Set-like global orders | Well-orders of all $\mathbf{No}$ with set predecessor classes                                         | Equivalent to class bijections $\text{Ord} \rightarrow \mathbf{No}$ ; a predicate on classes, not an ordinary class of all such bijections. |
| Raw global orders      | All class well-orders of all $\mathbf{No}$ , with no set-likeness requirement                         | Can have proper-class predecessor segments. Comparison must not assume an ordinal-indexed first disagreement.                               |

| Construction             | Objects                                                                          | Logical size and main caution                                                                          |
|--------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Increasing class subsets | Class subsets well-ordered by numerical $<$                                      | May have non-set-like inherited order. A global increasing enumeration of all <b>No</b> is impossible. |
| Unlabelled types         | Well-orders modulo order isomorphism                                             | Numerical lexicography does not descend to this quotient.                                              |
| Relation codes           | Characteristic functions of ordering relations, after choosing an order of pairs | A different lexicographic order; its dependence on the pair enumeration must be retained.              |

### 80.1.1 Enumeration lexicography

For words  $p : \alpha \rightarrow A$  and  $q : \beta \rightarrow A$ , compare their first different letter if one occurs before either word ends. If one is a proper prefix of the other, compare  $\varepsilon$  with the next letter of the longer word. We write  $p <_\varepsilon q$  for this order. The termination marker is not an allowed letter, even if one reserves a numerical label to model it.

End-first dictionary order means  $\varepsilon < A$ . End-last order means  $A < \varepsilon$ . A marker at a genuine class gap need not correspond to any surreal number. A marker immediately before  $a \in A$  has right side with minimum  $a$ ; immediately after  $a$  has left side with maximum  $a$ .

For injective words the ordinal indexing and the labels determine a well-order on the range. Conversely, every well-order on a set of labels has a unique ordinal enumeration. Thus injective words are a representation of all labelled set well-orders, not an additional choice of enumerations for the same labelled ordering.

*Note 80.1* (Source 23, Lemma 2.1, “No prefix ambiguity for complete enumerations”). *If  $p : \alpha \rightarrow X$  and  $q : \beta \rightarrow X$  are bijections and  $p$  is a prefix of  $q$ , then  $p = q$ .* Source 23’s proof: if  $\alpha < \beta$ , the next value  $q(\alpha)$  already occurs in  $p$ , since  $p$  exhausts  $X$ ; this contradicts injectivity of  $q$ . This is `two:lem:prefix-free` (section 6; source 15’s `note 32.3`).

Consequently none of the termination issues of source 23’s Part I (its Sections 4–7, printed in sections 81 and 82) changes the lexicographic order of exhaustive well-orders of a *fixed* carrier. They arise when carriers, or word lengths without exhaustivity, vary.

### 80.1.2 Other comparisons really are different

Shortlex first compares lengths and only then letters. Reverse lexicography requires a greatest disagreement, which arbitrary infinite words need not have. It is meaningful on finite words or suitable finite-difference families, but is not automatically a total order on all ordinal words. End-last lexicography is also sometimes used in tree orders; changing that convention changes prefix comparisons, not the numerical order of labels.

For relation-code lexicography, choose a set-like enumeration of ordered pairs of labels and compare the characteristic functions of the two relations at the first differing pair. Global choice supplies such an enumeration for a class carrier. This works even when the relations are not set-like. It is not the direct enumeration comparison of source 23’s Section 9 (section 106.1). A change in the schedule of pairs can change which disagreement decides the answer.

Finally, singleton words  $(a)$  and  $(b)$  have the same unlabelled well-order type for all labels  $a, b$ , yet  $(a) < (b)$  whenever  $a < b$ . Lexicography cannot be a comparison on unlabelled types that respects these representatives. Choosing the ordinal itself as a canonical representative gives a different, well-defined type order, rather than repairing this failure of descent.

## 80.2 Source 27: a map of the principal interpretations

There are arbitrarily many artificial encodings of well-orders, so no finite survey can enumerate every possible convention. The following table covers the structural choices responsible for the main differences. Once these choices are fixed, a further encoding is a new construction to be analyzed on its own terms.

| Interpretation     | Objects                                                              | Comparison and size                                                                                                 |
|--------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Fixed carrier      | All well-orders of a fixed set $X \subseteq \mathbf{No}$             | Compare their ordinal enumerations at the first unequal numerical label. A set-sized linear order.                  |
| Fixed type         | Bijections $\alpha \rightarrow X$ , for a specified ordinal $\alpha$ | A stratum of the preceding order; its type can matter beyond $ \alpha $ .                                           |
| Variable carrier   | Injective words of every set ordinal length in $\mathbf{No}$         | Prefix termination must be specified. An ordinary class of set objects.                                             |
| Increasing subsets | Words strictly increasing in the numerical order                     | Equivalent to set-sized numerically well-ordered subsets, with an explicit termination convention.                  |
| Global enumeration | Class bijections $E : \text{Ord} \rightarrow \mathbf{No}$            | Exactly the set-like class well-orders; comparison is on class variables.                                           |
| Raw global order   | Arbitrary class well-orders on $\mathbf{No}$                         | May have proper-class predecessor segments. Use a common labelled prefix, not an assumed ordinal enumeration.       |
| Relation table     | A well-order $R$ viewed as a binary function on pairs                | Requires an independently fixed well-ordered scan of pairs. Its order need not agree with enumeration lexicography. |
| Unlabelled type    | Well-orders modulo order isomorphism                                 | For set relations the quotient is ordinal type. Numerical-label lexicography does not descend to this quotient.     |
| Model-relative     | The preceding objects inside $(M, \mathcal{X})$                      | The admitted classes $\mathcal{X}$ , internal well-foundedness, and external size must all be specified.            |

### 80.2.1 Fixed carriers and the absence of a termination ambiguity

For a set  $X$ , let  $\mathcal{W}(X)$  be all strict well-orders of  $X$ . Each  $R \in \mathcal{W}(X)$  has a unique ordinal type  $\alpha_R$  and an enumeration  $e_R : \alpha_R \rightarrow X$ . If  $R \neq S$ , neither enumeration can be a proper prefix of the other: a proper prefix that is already onto  $X$  leaves no unused label for a longer injective word. Consequently the common domain contains a least disagreement  $\delta$ , and

$$R <_{\text{en}} S \iff e_R(\delta) <_X e_S(\delta).$$

This is a strict linear order. The familiar proof of transitivity compares the two first-disagreement positions for three words; the earlier position decides, and equal positions use transitivity in  $X$ .

For an infinite set  $X$  of cardinality  $\mu$ , the repository proves (theorem 6.1; the reals report)

$$|\mathcal{W}(X)| = 2^\mu, \quad \alpha \hookrightarrow \mathcal{W}(X) \iff \alpha < \mu^+, \quad (\text{S27.1})$$

and the same ordinal spectrum holds in the reverse order. Every linear order of size at most  $\mu$  embeds. These statements do not classify  $\mathcal{W}(X)$  up to isomorphism, do not imply separability, and do not identify its relation-table version. Its finite residual tails create adjacent points and finite

blocks: after an agreed prefix leaves  $n$  labels, the possible completions form the finite lexicographic order on  $n!$  permutations.

For a fixed type  $\alpha$ , write  $\mathcal{W}_\alpha(X)$  for the bijections  $\alpha \rightarrow X$ . Some strata have much greater binary coding length than others even though their cardinalities coincide. Birthday truncation of  $X$  is another independent parameter. At an infinite cardinal  $\kappa$ ,  $|\mathbf{No}_{<\kappa}| = 2^{<\kappa}$ , but neither a singular birthday bound nor a noninitial ordinal cutoff should be replaced by its cardinality without proof. The earlier corpus studies these distinctions extensively; the new scan results of [section 84.1](#) illustrate another reason why cardinality alone is insufficient.

### 80.2.2 Variable words and distinct terminators

If the carrier is allowed to vary, one word can be a proper prefix of another. Three standard conventions insert a distinct terminal symbol  $\varepsilon$  below every letter, above every letter, or at a cut  $X = L + R$ . The last convention means

$$x < \varepsilon \quad (x \in L), \quad \varepsilon < y \quad (y \in R).$$

Using the numerical value 0 as an ordinary letter *and* as an undistinguished padding symbol identifies different words: the empty word and (0) already have the same zero extension. One must either exclude 0 from the alphabet, place a distinct terminator immediately before or after it, or explicitly pass to the quotient defined by zero padding. These are different definitions.

Every set-length word is still a set. Hence the class of all such words does not create a class-of-classes problem. By contrast, a word of length Ord is itself a class, and treating all such words as members of one ordinary class would cross a type boundary.

### 80.2.3 Numerical well-ordering is much more restrictive

Every set-sized subset  $A \subseteq \mathbf{No}$  whose inherited numerical order is a well-order has a unique numerically increasing ordinal enumeration. This is the increasing-word interpretation studied in [section 82.2](#). It does not include arbitrary permutations of  $A$ . For example, the set  $\{0, 1\}$  supplies two labelled well-orders but only one numerically increasing enumeration.

There is no numerically increasing class bijection  $\text{Ord} \rightarrow \mathbf{No}$ , because its range would make the numerical surreal order a well-order. There are increasing proper-class subclasses, such as the ordinals. Such subclasses need not be set-like in their inherited order: take an increasing copy of Ord inside  $(0, 1)$  and append the point 1. The final point then has a proper class of predecessors. Thus even in the increasing interpretation, “ordinal sequence” and “arbitrary class well-ordered subset” must be distinguished.

### 80.2.4 Isomorphism types discard the lexicographic information

On a set carrier, every well-order is isomorphic to an ordinal. Two distinct enumerations of the same order type can compare either way after relabelling. For instance,  $(a, b)$  and  $(b, a)$  on  $a < b$  have the same ordinal type 2, while the first is lexicographically smaller. Numerical-label lexicography therefore does not descend to the quotient by abstract order isomorphism. Ordering the quotient by ordinal type is perfectly legitimate, but it is an additional, different order.

### 80.2.5 The Kanovei–Shelah connection

The historical connection should be stated precisely. Kanovei and Shelah use maps  $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters, ordered lexicographically using characteristic-function order on  $\mathcal{P}(\mathbb{N})$



[16]. Injectivity is not part of that definition. Their construction motivates the use of ordinal-indexed lexicographic codes without choosing a well-order of the underlying reals. It does not literally define the order of all well-order relations on  $\mathbb{R}$ , still less a collection of all class well-orders of **No**.

### 80.3 Source 29: interpretations and a disciplined choice of foundations

[merge] Source 29's Section 2, printed whole: its sign conventions, its interpretation table, its size ledger with the definition of uniform families (definition 80.2), used throughout its later sections, and its operations on orders. Its  $X_\theta$  is printed **No** $_{<\theta}$  (for an arbitrary set ordinal  $\theta$ , as in source 29) and its product  $\otimes_{\text{lex}}$  is printed  $\times_{\text{lex}}$ . Its Subsection 3.3, on relation scans and inverse ranks, follows at the end; the rest of its Section 3 is printed with its birthday spectra.

#### 80.3.1 Signs, birthdays, and numerical comparison

We represent a surreal number by a sign sequence

$$x : \text{bd}(x) \longrightarrow \{-, +\}, \quad \text{bd}(x) \in \text{Ord}.$$

At the first unequal coordinate, including the possibility that one sequence has ended, compare with

$$- < \text{termination} < +.$$

This is the numerical order. For each set ordinal  $\theta$ , write

$$\mathbf{No}_{<\theta} = \{x : \text{bd}(x) < \theta\}.$$

It is a set. The union over all set ordinals is the proper class **No**. Simplicity is a different relation: one sign sequence is simpler than another when it is a prefix. It is a tree order and is not a global well-order scan of the numerical line [12, 14].

The basic interpolation principle is the *set-cut property* (definition 5.1): if  $L, R$  are sets of surreals and  $L < R$ , then some  $z \in \mathbf{No}$  satisfies  $L < z < R$ . Empty sides are permitted. Deleting a set of forbidden labels does not destroy this property (lemma 5.2). One can choose a separator with birthday larger than the birthdays of all the constraints and forbidden labels; the inspected Lean foundation develops precisely this bounded-birthday route [8, 9]. It follows that **No** has no endpoints and every numerical interval between two distinct points is a proper class. The principle applies to sets of constraints. It does not fill an arbitrary cut between two proper classes.

#### 80.3.2 The interpretation table

The following choices cover the natural readings relevant to this problem. Altering the coding or comparison rule within one row can produce still further orders; the parameterized constructions below make those alterations explicit.

| Interpretation         | Object being compared                                                                                          | Logical size and decisive choice           |
|------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Fixed set of labels    | All auxiliary well-order relations on a set $X \subseteq \mathbf{No}$ , represented by exhaustive enumerations | A set; numerical first disagreement.       |
| Fixed enumeration type | Bijections $\alpha \rightarrow X$ for a specified ordinal $\alpha$                                             | A set; a stratum $\mathcal{W}_\alpha(X)$ . |



| Interpretation                 | Object being compared                                                                   | Logical size and decisive choice                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Variable set carrier           | Injective ordinal-length surreal words                                                  | A class of sets; termination placement matters.                                          |
| Inherited numerical well-order | Increasing words or their numerical carriers                                            | Set carriers form a class of sets; proper-class carriers require higher-order treatment. |
| Set-like global well-order     | Class bijections $\text{Ord} \rightarrow \mathbf{No}$                                   | Each is a class; the whole family is not an ordinary class of GB.                        |
| Arbitrary global well-order    | Class well-orders on $\mathbf{No}$ , possibly with proper-class predecessor segments    | Direct labelled-prefix comparison, without assigning set ordinal types.                  |
| Variable class carrier         | Class well-orders on arbitrary subclasses of $\mathbf{No}$                              | A higher-order collection; proper-class termination and prefixes must be specified.      |
| Relation-table encoding        | Characteristic bits of a relation under a fixed well-ordered scan of ordered pairs      | Scan-dependent; differs from enumeration comparison.                                     |
| Inverse-rank encoding          | Ranks of labels read in a fixed background scan                                         | Requires ranks to exist in the selected codomain; reverses some enumeration comparisons. |
| Abstract order types           | Well-orders modulo isomorphism, or ordinal notations encoding them                      | Labels have been forgotten; numerical lexicography does not descend to this quotient.    |
| Universe-relative version      | Any of the set constructions inside a specified set universe or type-theoretic universe | A legitimate bounded model; changing the universe changes the object.                    |

Free words with repeated labels are useful auxiliary objects, but are not themselves exhaustive well-orderings of their ranges. Likewise a surreal sign sequence has a well-ordered domain; this fact does not make it a well-ordering of the surreal class. These auxiliary encodings become legitimate only after their comparison and intended interpretation are stated.

### 80.3.3 Theories and the size ledger

Our set-sized arguments use ZFC unless a weaker basis is explicitly given. The native sign-spectrum argument uses only ZF. Our class arguments use GBC: Gödel–Bernays class theory with set choice and Global Choice. All classes have sets as their members. An expression such as “for every class well-order  $R$ ” is a quantification in the two-sorted language; it does not assert that there is a class whose members are the class relations  $R$ .

**Definition 80.2** (Uniform families). A family of class relations  $(R_i)_{i \in I}$  is *uniform* if one class relation  $F(i, x, y)$  codes all its members. Here the index  $i$  is always a set; its index carrier  $I$  may be a set or an ordinary class. A uniformly set-indexed family has a set index carrier. A class of set-coded words is an ordinary class because each word is a set.

This definition prevents a hidden appeal to a class choice principle. Choosing a class independently for each of a proper class of indices is stronger than exhibiting one elementary formula defining all rows. The local back-and-forth constructions used below are uniform in the latter sense: a tuple belongs to the output exactly when it appears in the unique deterministic set-length history with the indicated parameters.

A class well-order need not be set-like. We call  $R$  *set-like* when every predecessor segment  $\{y : yRx\}$  is a set. Under GBC, a proper-class set-like well-order has a class enumeration of type

Ord. A class well-order of type  $\text{Ord} + 1$  has a last point with a proper class of predecessors and admits no such exhaustive enumeration. Elementary comparison of two labelled orders remains possible; it is not the abstract comparability principle studied later.

### 80.3.4 Operations on orders

Write  $A \hookrightarrow B$  for an order embedding and  $A \simeq_e B$  for embeddings in both directions. These notions are strictly weaker than isomorphism in general. Define

$$A \times_{\text{lex}} B = \sum_{a \in A} B,$$

so the first coordinate is decisive. A natural number in such a product denotes its finite chain. The full binary cube is

$$\mathbb{B}_\beta = (\{0, 1\}^\beta, <_{\text{lex}}),$$

with the least unequal coordinate decisive. All products and sums of *lengths* are ordinal operations. For example,  $2 \cdot \omega = \omega$ , whereas  $\omega \cdot 2 > \omega$ . This distinction is essential to every coding-length calculation.

### 80.3.5 Relation scans and inverse ranks (source 29's Subsection 3.3)

A fixed well-ordered scan  $\mathbf{s}$  of  $X \times X$  permits another lexicographic order: compare the characteristic bits of two relations at the first scanned pair on which they differ. This is a legitimate order, but it uses the scan and the chosen orientation of zero and one. A first differing relation bit need not reveal the first differing enumerated label. The repository supplies finite counterexamples and scan-dependent endpoint classifications ([section 38](#), [theorem 38.2](#), and [corollary 38.3](#)).

Likewise one may scan the labels and compare their ordinal ranks in the two auxiliary well-orders. Already on  $0 < 1 < 2$ , the enumerations  $(1, 0, 2)$  and  $(2, 0, 1)$  compare in that order, whereas their rank vectors  $(1, 0, 2)$  and  $(1, 2, 0)$  show that the decisive scanned rank need not be attached to the first differing enumerated label. A reversal is given by  $(1, 2, 0)$  and  $(2, 0, 1)$ : enumeration comparison places the former first, while the rank of label 0 is respectively 2 and 1. On arbitrary non-set-like class well-orders, predecessor segments need not have set ordinal order types, so even the proposed rank codomain requires a further foundational choice.

These encodings are best treated as named additional structures. There is no canonical rule converting every such code order into the native numerical first-disagreement order.

## 80.4 Source 30: the interpretations and their foundational levels

There are indefinitely many artificial encodings. The following matrix organizes the natural independent choices rather than claiming to enumerate every possible coding.

| Version                | Objects and comparison                                         | Size and distinction                                                                 |
|------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Fixed labelled set $X$ | Well-order relations on $X$ , compared by ordinal enumerations | A set in ZFC. Numerical order compares labels; it is not the order being enumerated. |
| Fixed type $\alpha$    | Bijections $\alpha \rightarrow X$ , by first unequal value     | A set; nonempty iff $ \alpha  =  X $ . Fixing type changes the topology.             |
| Variable set support   | Injective ordinal-length words in <b>No</b>                    | A class of set codes. Proper prefixes require a termination rule.                    |

| Version                 | Objects and comparison                                  | Size and distinction                                                                                    |
|-------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Increasing support      | Words or subclasses well-ordered by native surreal $<$  | A restriction on the numerical order, not arbitrary labelled well-orders.                               |
| Set-like global orders  | Class bijections $\text{Ord} \rightarrow \mathbf{No}$   | A predicate on class variables in GBC. All have class type $\text{Ord}$ .                               |
| Arbitrary global orders | Class well-order relations on $\mathbf{No}$             | May have proper-class predecessors and common prefixes.                                                 |
| Relation-table scan     | Characteristic bits in a chosen well-order of pairs     | Depends on the scan; has a different topology and measurable structure.                                 |
| Inverse-rank scan       | Ranks of fixed labels in a baseline list                | Directly defined only for set-like class orders; the baseline is least.                                 |
| Type-first or quotient  | Compare types first, or identify isomorphic well-orders | Type-first changes the primary comparison; quotienting erases labels. Class types need not be ordinals. |
| Universe-relative       | Orders of $\mathbf{No}^u$ in a larger universe          | A higher-universe set/type; smallness remains relative to $u$ .                                         |

#### 80.4.1 Fixed sets and enumeration lexicography

For a linearly ordered set  $(X, <)$  define

$$\mathcal{W}_\alpha(X) = \{f : \alpha \rightarrow X : f \text{ is a bijection}\}.$$

For distinct  $f, g$  of the same length, compare  $f(\xi)$  and  $g(\xi)$  at their first unequal coordinate. Every well-order relation on  $X$  has a unique ordinal enumeration. For infinite  $|X| = \mu$  the union of these representations over  $\alpha < \mu^+$ ,  $|\alpha| = \mu$ , is a set, denoted  $\mathcal{W}(X)$ .

No proper-prefix tie occurs between distinct bijective enumerations of the same set: the shorter one would already have exhausted  $X$ . Thus the comparison extends across types without a termination rule. An order-isomorphism  $h : X \rightarrow Y$  induces  $f \mapsto h \circ f$  on enumerations. Arbitrary relabellings need not preserve the numerical comparison at the first disagreement.

#### 80.4.2 Variable support and termination

When the range varies,  $(x)$  can be a proper prefix of  $(x, y)$ . Putting the shorter word first, putting it last, comparing lengths first, or placing termination inside the alphabet produces inequivalent orders. The usual surreal sign order uses  $- < \text{termination} < +$ . A termination marker below every surreal entry is not that convention. Identifying a marker with the actual entry 0 also destroys injectivity unless an additional distinction is made.

All set-length words form a legitimate class: every individual word is a set. The prior reports embed these orders into  $\mathbf{No}$  and classify several termination families ([theorem 9.1](#) and [section 36](#)). Those embeddings do not authorize collecting all proper-class words into another ordinary class.

#### 80.4.3 Native order, birthday, and simplicity

The native surreal order is not a well-order, as witnessed by  $0, -1, -2, \dots$ . Simplicity is the initial-segment partial order on sign sequences; birthday is an ordinal rank with many ties. Neither supplies a well-ordered scan of all labels.

Ordering first by birthday and then by a selected well-order of each layer works under global choice. The existence of each individual layer order in ZFC does not provide that global selection.

Numerically increasing proper subclasses need not be set-like:

$$\{-1/(\alpha + 1) : \alpha \in \text{Ord}\} \cup \{0\}$$

has numerical order type  $\text{Ord} + 1$ . Its final point has a proper class of predecessors. This example separates increasing support from set-like global enumeration.

#### 80.4.4 Three scans of the same six objects

For labels  $0 < 1 < 2$ , enumeration order is 012, 021, 102, 120, 201, 210. Scanning inverse ranks of labels 0, 1, 2 gives 012, 021, 102, 201, 120, 210. Scanning relation bits in the pair order  $(0, 1), (0, 2), (1, 0), (1, 2), (2, 0), (2, 1)$  with  $0 < 1$  gives 210, 120, 102, 201, 021, 012. All three are abstract six-point chains, but they put the labelled objects in different positions. Infinite examples below distinguish their abstract topological and measurable structures as well. The accompanying script reproduces these finite comparisons. ([merge] The script is `code/30-tables-verify_finite.py`; the three orders are those of source 15's table in [section 38](#).)

#### 80.4.5 Inverse ranks, type-first comparison, and quotienting

The inverse-rank interpretation deserves a separate definition. Fix a baseline list  $b : \mu \rightarrow X$  of a set  $X$ . A well-order  $R$  of  $X$  assigns each label its ordinal rank  $r_R(x) = \text{otp}(\{y : yRx\}, R)$ . Scan the values

$$(r_R(b(\xi)))_{\xi < \mu}$$

and compare them as ordinals at the first unequal coordinate. This is a linear order on the same set of labelled well-orders, including comparisons between different order types. It generally differs from comparing the values in their increasing enumerations.

For the whole surreal class the same definition works for *set-like* well-orders, using a supplied  $b : \text{Ord} \rightarrow \mathbf{No}$ . It cannot assign ordinal ranks to a label whose predecessor class is proper. Its comparison is an elementary formula on two supplied class graphs; no collection of all rank functions is required.

The baseline order is the least point for inverse-rank lexicography. Indeed, if  $R$  first disagrees with the baseline at label index  $\delta$ , every earlier label has its unchanged rank. The rank of  $b(\delta)$  cannot be below  $\delta$ , because that rank is already occupied by an earlier baseline label. It must exceed  $\delta$ . In contrast, numerical enumeration lexicography on the full surreal class has no least point: any proposed first label can be replaced by a smaller one and the rest enumerated using the supplied global choice. The endpoint distinction alone prevents an order isomorphism between these two global interpretations. ([merge] This is [theorem 38.1](#), source 15's theorem, which also gives the characters of the inverse-rank order.)

Another convention compares order type first and uses enumeration lexicography only within a fixed type. For a set  $X$  of cardinality  $\mu$ , this is the ordinary lexicographic sum

$$\sum_{\substack{\alpha < \mu^+ \\ |\alpha| = \mu}} \mathcal{W}_\alpha(X).$$

Every fixed-type stratum is convex in this new order. Direct first-disagreement comparison across types has no such priority: with two different types, the first labels can be chosen in either numerical order. Thus these strata need not be convex in the original comparison.

Quotienting by isomorphism is a different operation. On a fixed set it leaves precisely the ordinal types  $\{\alpha < \mu^+ : |\alpha| = \mu\}$ . Ordering those types by their ordinal order is legitimate, but

is not induced by the labelled lexicographic comparison: changing representatives can reverse the comparison of two distinct type classes. On the whole class, all set-like global orders have type  $\text{Ord}$ , so their isomorphism quotient has just one member. Arbitrary global well-orders include such types as  $\text{Ord}+1$  and  $\text{Ord}+\text{Ord}$ ; they cannot all be replaced by set ordinals. A type-first construction there needs an explicitly specified class-type representation and comparison principle in an appropriate metatheory. The direct common-prefix formula ([theorem 14.2](#); source 30's Proposition 3.2) avoids that requirement.

#### 80.4.6 Why a reverse scan is not automatically defined

On finite words one can compare at the *last* unequal coordinate. On an infinite well-ordered domain a disagreement set need not have a greatest member. Two enumerations of a countable carrier which differ by switching every consecutive pair disagree unboundedly in  $\omega$ , so there is no last coordinate to inspect. A reverse-lexicographic rule therefore needs an additional restriction or completion convention. Even eventual agreement is insufficient by itself: a bounded set of disagreements can have limit order type and no maximum.

These examples are why the article fixes the scan, the label order, the allowed domains, and the treatment of termination before proving a lexicographic theorem. Allowing repetitions adds another independent choice: arbitrary ordinal-length words are legitimate set codes, but their multiplicities no longer represent a well-order of their range.

### 81 Termination markers over a saturated alphabet (source 23; second routes to Part VII)

[merge] Source 23's Sections 3–6, printed whole. Its alphabet is an arbitrary set-cut saturated class linear order  $A$  ([definition 5.1](#); source 23 says “set-cut saturated”) with a marker cut  $A = L \sqcup R$ , and its marker  $e$  is printed  $\varepsilon$ , as source 16's marker in [section 36](#); its word classes  $\mathcal{W}_e(A)$  (all set-length words) and  $\mathcal{I}_e(A)$  (injective ones) are printed  $\mathcal{T}_\varepsilon(A)$  and  $\mathcal{I}_\varepsilon(A)$ , and its prefix relation  $\sqsubseteq$  is kept. In a character pair source 23 writes **On**, printed **Ord**; in its words (its Subsection 1.2), the symbol “means ‘no set-sized cofinal family’ or its dual. It is not an additional cardinal.”

What is printed here is not new against the repository:

- Source 23's termination classification ([note 81.9](#)) and its surreal corollary ([note 81.10](#)) are source 16's [theorem 36.1](#) with an abstract saturated alphabet in place of **No**. In GBC every proper-class saturated alphabet is isomorphic to **No** ([proposition 5.13](#)), so the two statements say the same; source 23's version needs no such isomorphism, and its sufficiency proof, like source 16's, uses **GB** and the set-cut property only. The two proofs are the same common-prefix search with the same two-letter detour at an endpoint of a threshold side; source 23 organizes it by a set chain of straddling prefixes ([lemma 81.7](#)), as does the proof of Part II's [theorem 11.1](#).
- Source 23's uniform point characters ([note 81.12](#)) are the character part of [theorem 36.2](#) (source 16), and its regular-cardinal family of failures ([section 81.4.2](#)) is source 16's **T<sub>κ</sub>** there. The isomorphism classification of [theorem 36.2](#) (no repetitions; arbitrary and injective words isomorphic at every placement) is not in source 23, whose Question 16.1 asks for it.
- Its fresh-separator lemmas ([notes 81.2](#) and [81.3](#)) are [lemma 5.2](#) for an abstract alphabet; [lemma 81.4](#) and the three cylinder lemmas of [section 81.2](#) isolate steps of the proofs of [theorems 11.1](#) and [36.1](#).

*Pinned provenance.* Source 23 read this report at 109aaca15, before the batch-81 placement

(f0cd7032d) and Parts VI–VIII; its delivery calls the classification and the characters additions, and its Subsection 6.1 says that its adjacent placements “settle the additional case asked” in source 13’s question “A classification of termination cuts”. Both statements were true at its pin. At this report’s text that question is answered by source 16 (section 39, item (c)); source 23’s is an independent second proof. Its new result on words is the theorem on increasing words (theorem 82.1), printed in section 82.

### 81.1 Small cuts and fresh labels

*Note 81.1* (Source 23, Definition 3.1). *A class linear order  $A$  is set-cut saturated if for all sets  $B, C \subseteq A$  satisfying  $B < C$ , there is  $x \in A$  with  $B < x < C$ . Either set may be empty. A subset of a class order is cofinal if every point is at most some member of that subset; coinital is the dual notion. This is the set-cut property  $\eta_{\text{Ord}}$  of definition 5.1.*

The surreal order has this property: a separated set cut has a surreal separator. In sign-sequence language, bounding the option birthdays first supplies a separator at a suitably larger birthday. The repository proves a universe-relative form in `SignSequenceCut.lean` [8]. We use only the order property, not surreal arithmetic.

*Note 81.2* (Source 23, Lemma 3.2, “Two cuts give a fresh separator”). *Let  $A$  be set-cut saturated. Given sets  $B < C$  and a forbidden set  $D \subseteq A$ , there exists  $x \notin D$  with  $B < x < C$ . Consequently  $A \setminus D$  is set-cut saturated. This is lemma 5.2, with the same proof, for an arbitrary saturated alphabet.*

*Second route (source 23).* Choose  $z$  with  $B < z < C$ . Split  $D$  into  $D_- = \{d \in D : d \leq z\}$  and  $D_+ = \{d \in D : z < d\}$ . The sets  $B \cup D_-$  and  $C \cup D_+$  are strictly separated. Any separator  $x$  of this second cut is above every point of  $D_-$  and below every point of  $D_+$ , so it is outside  $D$  and still separates  $B, C$ .

*Note 81.3* (Source 23, Lemma 3.3, “No small bounds”). *A set-cut saturated class has no set-sized cofinal or coinital subset. Every nonempty initial or final segment of it is a proper class. For **No** this is in lemma 5.2 and the remarks after definition 5.1.*

*Second route (source 23).* A set cofinal subset would have a strict upper bound by the empty-right-side instance of saturation, a contradiction. The dual is identical. If an initial segment containing  $a$  were a set, its subset  $\{x : x < a\}$  would be a set. Saturating the cut consisting of that subset on the left and  $\{a\}$  on the right is impossible. The final-segment case is dual.

**Lemma 81.4** (Deletion preserves the relevant side conditions). *Let  $L$  be a nonempty initial segment of  $A$  and let  $D$  be a set. If  $L$  has no greatest element and has set cofinality  $\theta$ , then  $L \setminus D$  also has cofinality  $\theta$ . If  $L$  has no set cofinal subset, neither does  $L \setminus D$ . If  $L$  has a greatest element  $a$ , then either  $a \notin D$  remains greatest, or  $L \setminus D$  has no set cofinal subset. The dual assertions hold for final segments.*

*Proof.* When  $L$  has no maximum,  $L \setminus D$  is cofinal in  $L$ : above any  $x \in L$ , first choose  $y \in L$  with  $x < y$ , then choose a fresh point between  $x$  and  $y$ . A cofinal set in  $L \setminus D$  is therefore cofinal in  $L$ . Conversely, a set cofinal family in  $L$  can be moved point by point to larger fresh points of  $L$ , preserving its cardinal bound. This proves the first two assertions.

If  $a = \max L$  is removed and  $B \subseteq L \setminus D$  is a set, use note 81.2 on  $B < \{a\}$  to find a fresh point between them. It belongs to  $L \setminus D$  and is above  $B$ .  $\square$

We apply these lemmas both to  $A = \mathbf{No}$  and to a surreal alphabet with a set of reserved labels removed. This avoids a hidden injectivity problem in word extensions: all labels already used by a prefix form a set.

## 81.2 Cylinders and bounded straddling chains

[merge] Source 23 opens here its Part I, “Termination and the geometry of set-length words” (its Sections 4–7). It says that a prefix *crosses* a cut when its cylinder meets both sides, i.e. straddles it in the word of [section 1.3.1](#); its heading “Cylinders and bounded crossing chains” is printed with that word, and its proofs keep its own.

Fix a marker cut  $A = L \sqcup R$ . Let  $\mathcal{T}_\varepsilon(A)$  be all set-length words, and  $\mathcal{I}_\varepsilon(A)$  their injective suborder. For a word  $p$ , its cylinder is

$$C_p = \{w : p \sqsubseteq w\}.$$

It includes  $p$  itself. Concatenation always means ordinal concatenation; a suffix after a prefix of length  $\alpha$  is reindexed by the order type of the corresponding ordinal interval.

**Lemma 81.5** (Convex cylinders). *Every  $C_p$  is convex. Incompatible prefixes have disjoint, ordered cylinders. For arbitrary words,  $C_p$  is isomorphic to  $\mathcal{T}_\varepsilon(A)$ . For injective words it is isomorphic to  $\mathcal{I}_\varepsilon(A \setminus \text{ran } p)$ , with the induced marker cut.*

*Proof.* A word outside  $C_p$  either disagrees with  $p$  before  $p$  ends, or is a proper prefix of  $p$ . Its comparison with every member of  $C_p$  is decided by the same letter or the same marker comparison. It is therefore entirely on one side of the cylinder. Deleting the fixed prefix gives the claimed isomorphisms. For injective words the suffix must avoid exactly the labels already used by  $p$ .  $\square$

**Lemma 81.6** (Cylinders have no small end cofinalities). *Suppose  $L, R$  are nonempty. In either word order every cylinder has no set-sized cofinal or coinital subset, irrespective of whether the termination criterion of [section 81.3](#) holds.*

*Proof.* It suffices to work in a residual alphabet  $A \setminus D$ . Given a set of suffixes, collect their first letters into a set  $F$ . Choose a label  $r \in R \setminus D$  and a fresh label above  $F \cup \{r\}$ . Its singleton word is above all the suffixes, including the empty one. For a lower bound choose  $l \in L \setminus D$  and a fresh label below  $F \cup \{l\}$ . Nonemptiness of both residual sides follows from [note 81.3](#). Reattach the prefix.  $\square$

**Lemma 81.7** (Crossing-prefix reduction). *Let  $B < C$  be nonempty sets of words in either word order. Define*

$$T = \{p : B \cap C_p \neq \emptyset \text{ and } C \cap C_p \neq \emptyset\}.$$

*Then  $T$  is a set and a chain of prefixes. Its union  $p_*$  is a legitimate set word. Every constraint outside  $C_{p_*}$  lies on the correct side of that cylinder. Either  $p_* \notin T$ , and at most one constraint side meets  $C_{p_*}$ , or  $p_* \in T$  and no proper one-letter extension of  $p_*$  belongs to  $T$ .*

*Proof.* All members of  $T$  are prefixes of words in the set  $B \cup C$ , so they form a set. Two incompatible crossing prefixes would give a lower constraint in the higher cylinder and an upper constraint in the lower one, contradicting  $B < C$ . Thus  $T$  is a chain. Its lengths are bounded by

$$\Lambda = \sup\{\text{dom}(w) + 1 : w \in B \cup C\},$$

so its union is a set word. In the injective case two repeated entries in the union would already occur together in some member of the chain.

If a constraint  $w$  lies outside  $C_{p_*}$ , choose a crossing prefix long enough to decide the comparison between  $w$  and  $C_{p_*}$ . Such a prefix exists by the union definition, also when  $w$  is a proper prefix of  $p_*$ . A constraint of the opposite side extending that crossing prefix forces  $w$  to be on the required side of the entire cylinder. The last two alternatives follow directly from the definition of  $T$  and its union.  $\square$



This reduction is particularly useful for formalization. Its tree is a set of prefixes of the supplied data. It is not a recursion that successively constructs arbitrary classes, and it cannot run for Ord stages.

### 81.3 An exact termination-marker criterion

**Definition 81.8.** An interior marker cut  $(L, R)$  is *admissible* if

$$\begin{aligned} L &\text{ has a greatest element or has no set cofinal subset,} \\ R &\text{ has a least element or has no set coinital subset.} \end{aligned} \tag{32}$$

Here “interior” means both  $L$  and  $R$  are nonempty.

These are conditions (1)–(3) of [theorem 36.1](#).

*Note 81.9* (Source 23, Theorem 5.2, “Termination classification”). *Let  $A$  be a set-cut saturated class linear order. The following are equivalent:*

- (i) *the marker cut is interior and admissible;*
- (ii)  $\mathcal{T}_\varepsilon(A)$  *is set-cut saturated;*
- (iii)  $\mathcal{I}_\varepsilon(A)$  *is set-cut saturated.*

*The proof is uniform for the two word orders.* For  $A = \mathbf{No}$  this is [theorem 36.1](#) (source 16).

*Second route (source 23). Necessity.* If  $L$  is empty, the empty word is the least element; if  $R$  is empty, it is the greatest element. Either violates saturation with an empty constraint side.

Suppose  $L$  has no greatest element but has a set cofinal family  $S$ . The singleton words  $(s)$ , for  $s \in S$ , all lie below the empty word. No word lies strictly between all of them and the empty word: such a word must be nonempty, its first letter must belong to  $L$ , and that first letter is exceeded by a member of  $S$ . This is already a cut in the injective suborder. The argument for a set coinital family in  $R$ , with no minimum, is dual. Thus either (ii) or (iii) implies (i).

*Sufficiency.* Fix separated sets of constraints  $B < C$ . When a side is empty, [lemma 81.6](#) at the empty prefix supplies a separator. Otherwise apply [lemma 81.7](#) and write  $p = p_*$ . For injective words replace  $A$  by  $A' = A \setminus \text{ran } p$ ; for arbitrary words put  $A' = A$ . [Note 81.2](#) and [lemma 81.4](#) show that  $A'$  remains saturated and its induced marker cut is still admissible.

If  $p \notin T$ , at most one constraint side meets  $C_p$ . Choose a point in that cylinder strictly above, or strictly below, the active set of constraints, as appropriate. [Lemma 81.6](#) permits this, and all outside constraints were already correctly oriented.

It remains to consider  $p \in T$ . Project each active constraint to its next symbol after  $p$ , using  $\varepsilon$  for the constraint  $p$  itself. Call the resulting lower and upper symbol sets  $S$  and  $U$ . They are strictly separated: equality at an actual letter would make the corresponding child cylinder cross, contradicting maximality of  $p$ ; equality at  $\varepsilon$  would put the same word on both sides.

If neither symbol set contains  $\varepsilon$ , saturate the cut  $S < U$  in  $A'$ , choose a strict separating letter  $a$ , and take  $p \frown (a)$ .

Suppose next that  $\varepsilon \in S$ . Set  $S_0 = S \setminus \{\varepsilon\}$ ;  $U$  consists of actual letters in the right side  $R'$ . First choose  $a_0 \in A'$  with  $S_0 < a_0 < U$ . If  $a_0 \in R'$ , the word  $p \frown (a_0)$  works. Otherwise  $a_0 \in L'$ , so every lower actual symbol is below  $\varepsilon$ .

If  $R'$  has no set coinital subset, some  $a \in R'$  lies strictly below all of  $U$ ; then  $p \frown (a)$  works. If  $R'$  has minimum  $r$ , either  $r \notin U$  and  $r < U$ , or  $r \in U$ . In the first case choose any point of  $C_{p \frown (r)}$ . In the second case choose a point of this cylinder strictly below all upper constraints inside it, using

**lemma 81.6.** All lower constraints have next symbol at most  $\varepsilon$ , and all upper constraints outside this child have next symbol larger than  $r$ , so both choices separate the full cut.

The case  $\varepsilon \in U$  is the order-dual argument. The marker cannot belong to both symbol sets. This exhausts the possibilities and proves saturation in both word orders.

*Note 81.10* (Source 23, Corollary 5.3, “Surreal order type”). In GBC, when the marker is admissible, both  $\mathcal{T}_\varepsilon(A)$  and  $\mathcal{I}_\varepsilon(A)$  are order-isomorphic to **No**. This is the last sentence of [theorem 36.1](#), by class universality ([proposition 5.13](#)).

*Second route* (source 23). Each is an ordinary class of set codes and has a class enumeration by Ord. Back-and-forth along such enumerations constructs an isomorphism between any two set-cut saturated proper-class orders. At each stage the previous partial map is a set, the induced cut is a set cut, and a fresh separator extends it. Global choice selects the least available extension in fixed set-like well-orders. These are set-valued stages, so the resulting class map is obtained in GBC. Apply the construction to **No**.

*Remark 81.11* (What the endpoint exception does). If  $\varepsilon$  is immediately before  $a \in A$ , then the augmented alphabet has an adjacent pair  $\varepsilon < a$ . The words do not inherit that adjacency: there are entire cylinders with first letter  $a$ , and within them points below any prescribed set of constraints. This is exactly the minimum case in the proof. Removing  $a$  because it was already used in an injective prefix changes that side into one with no set coinitial family, by [lemma 81.4](#). It does not create a new obstruction.

[merge] This is the two-letter detour of source 16’s proof ([theorem 36.1](#)), and the reason why its adjacent placements  $0^-$  and  $0^+$  give **No** in (S16.6) ([equation \(S16.6\)](#)).

## 81.4 Exact point characters and examples

For an interior marker define

$$\begin{aligned} \chi_- &= \begin{cases} \text{cf}(L), & L \text{ has no maximum and has a set cofinal subset,} \\ \text{Ord}, & \text{otherwise,} \end{cases} \\ \chi_+ &= \begin{cases} \text{ci}(R), & R \text{ has no minimum and has a set coinitial subset,} \\ \text{Ord}, & \text{otherwise.} \end{cases} \end{aligned} \tag{33}$$

When a value is a set cardinal, it is necessarily infinite and regular.

*Note 81.12* (Source 23, Theorem 6.1, “Uniform point characters”). Every point of  $\mathcal{T}_\varepsilon(A)$  and every point of  $\mathcal{I}_\varepsilon(A)$  has left character  $\chi_-$  and right character  $\chi_+$ . The whole orders have no set-sized cofinal or coinitial subset. At most one of  $\chi_-, \chi_+$  can be a set cardinal. For  $A = \mathbf{No}$  the characters are those computed in the proof of [theorem 36.2](#) (source 16), and the last sentence is its observation that both threshold sides cannot fail.

*Second route* (source 23). Fix a word  $p$ . Every element below  $p$  that does not extend  $p$  lies below every extension of  $p$  whose next letter is in  $L$ . Thus the cylinders  $C_{p \smallfrown (a)}$ , for available  $a \in L$ , are cofinal below  $p$ . The available alphabet is  $A' = A$  or  $A \setminus \text{ran } p$ .

If  $L' = L \cap A'$  has no maximum and has set cofinality  $\theta$ , choose a strictly cofinal family of letters of size  $\theta$ . Their one-letter extensions of  $p$  are cofinal below  $p$ . Conversely, the first appended letters of any set cofinal family below  $p$  must be cofinal in  $L'$ : otherwise a child with a larger first appended letter dominates the family. Hence the left character is exactly  $\theta$ .

If  $L'$  has no set cofinal subset, the same projection argument rules out any set cofinal family below  $p$ . If it has maximum  $a$ , its last child cylinder  $C_{p \smallfrown (a)}$  is cofinal below  $p$ , and that cylinder

has no set cofinal subset. [Lemma 81.4](#) identifies these alternatives with the cases in (33), including removal of the original maximum. The right-hand proof is dual.

The assertions about the whole orders are [lemma 81.6](#). If both characters were small, there would be a set cofinal family  $S \subseteq L$  and a set cointial family  $U \subseteq R$ , with neither side having its endpoint. A separator of  $S < U$  in  $A$  could belong to neither side, contradicting  $A = L \sqcup R$ .

#### 81.4.1 Principal placements and reserved labels

A marker immediately before or after any surreal  $a$  is admissible. The side not having its endpoint has no small approach to  $a$ , by surreal set-cut saturation. Thus the corresponding arbitrary-word and injective-word orders are both isomorphic to **No**.

The reserved-central convention uses alphabet  $\mathbf{No} \setminus \{0\}$  and marker 0. Neither side has an endpoint or a small approaching family, so it is another admissible case. This recovers the successful special case in source 13 ([theorem 36.4](#)), while the adjacent placements settle the additional case asked there. ([merge] Pinned provenance: at this report's text the adjacent placements are source 16's (S16.6) ([equation \(S16.6\)](#)), and source 13's question is answered by [theorems 36.1](#) and [36.2](#); see the opening note of this section.)

#### 81.4.2 A regular-cardinal family of failures

For any infinite regular cardinal  $\theta$ , let

$$L_\theta = \{x \in \mathbf{No} : \exists \alpha < \theta (x < \alpha)\}, \quad R_\theta = \mathbf{No} \setminus L_\theta.$$

The ordinals below  $\theta$  are cofinal in  $L_\theta$ . A smaller cofinal set would give, by choosing ordinal upper witnesses for its members, a smaller cofinal set in  $\theta$ . Therefore  $\text{cf } L_\theta = \theta$ . Neither side has an endpoint. For example, an endpoint in  $R_\theta$  would contradict filling the set cut between all  $\alpha < \theta$  and that endpoint. The right side has no set cointial subset by the last argument of [note 81.12](#).

Putting the marker at this cut gives point character  $(\theta, \text{Ord})$  everywhere. Reversing the construction gives  $(\text{Ord}, \theta)$ . Distinct regular cardinals yield nonisomorphic word orders. They remain mutually embeddable with **No** under global choice, as discussed in source 23's Section 8 ([section 111.16](#)). Point characters distinguish isomorphism, not bi-embeddability.

[merge] This is source 16's threshold  $(L_\kappa, R_\kappa)$ , with  $\kappa$  for the regular cardinal, and the word orders are  $\mathbf{T}_\theta$  and  $\mathbf{T}_\theta^{\text{op}}$  of [theorem 36.2](#). Source 16 also shows that every termination order is equimorphic with **No** without global choice (the paragraph after [theorem 36.2](#)).

#### 81.4.3 Genuine two-large-sided gaps

The criterion also permits a marker at a nonprincipal cut with no set cofinal family on either side. Such a cut need not be realized by a surreal. For a concrete example, fix an  $\text{Ord}$ -long sign pattern in which both signs occur beyond every ordinal, such as alternation in consecutive two-position blocks. Compare each set-length sign sequence with this class pattern at its first disagreement or its end. The resulting cut has both sides nonempty and has no set cofinal or cointial family: any set of sequences has bounded lengths, and a sufficiently long prefix ending just before a suitable sign lies closer to the pattern. Its marker therefore produces surreal-isomorphic word orders.

This is not unrestricted cut filling inside **No**. It inserts a new symbol in the *alphabet presentation*, then studies an ordinary class of finite- or transfinite-length set codes built from that alphabet. The isomorphism in [note 81.10](#) is a new order map, not an extension fixing every original numerical label.

## 82 Increasing words: no saturating marker, a product at the terminator, and the zero conventions (sources 23, 27 and 29)

[merge] Part VII left two questions about strictly increasing words open (section 39, items (b) and (d)): the isomorphism type of the centered order  $\mathcal{T}_c^{\text{inc}}$  (source 18’s own question), and increasing words with an interior terminator at a threshold other than zero (source 16’s question “Centered increasing words”). Three batch-83 sources take them up independently. Source 23 shows that no placement of a marker, at zero or elsewhere, makes the increasing-word order set-cut saturated (theorem 82.1); for the threshold at zero this was known from source 16’s remark and source 18’s cut classification (theorem 37.2). Source 27 factors every increasing-word order at its terminator (theorem 82.4), which reduces the centered order to a lexicographic product of the shorter-first order of all words with its reverse (corollary 82.5), classifies the zero conventions up to isomorphism (proposition 82.6), shows that no interval is a surreal interval (theorem 82.7) and computes the point characters at an arbitrary class gap (proposition 82.8). Source 29 proves the factorization, the product and the zero conventions again, without knowledge of source 27; its statements are notes below. Its reduction of every actual threshold, under each zero convention, and of every gap with two large sides (proposition 82.14) is new as a proved statement; source 27 states its two simplest cases in a remark. The three results on thresholds other than zero complement one another; what remains open is recorded in section 85.

In this section, for an alphabet  $D$  (a set or a class, linearly ordered),  $\mathcal{T}^{\text{inc}}(D)$  is the class of strictly increasing set-length words over  $D$ , including the empty word;  $\mathcal{T}_{\text{sf}}^{\text{inc}}(D)$  and  $\mathcal{T}_{\text{lf}}^{\text{inc}}(D)$  compare proper prefixes first and last; and for a cut  $D = L + R$ ,  $\mathcal{T}_\varepsilon^{\text{inc}}(D)$  puts a distinct marker  $L < \varepsilon < R$ . Source 27 writes  $\text{Inc}_\perp$ ,  $\text{Inc}_\top$ ,  $\text{Inc}_C$ ; source 29 writes  $\text{Inc}_{\text{sf}}$ ,  $\text{Inc}_{\text{lf}}$ ,  $\text{Inc}_\dagger$ ; source 23 writes  $\mathcal{U}_e(A)$ .  $\mathbf{T}_{\min}$  is the shorter-first order of all set-length surreal words, repetitions allowed and the empty word included (theorem 36.2; source 27’s  $\mathcal{B}$ , source 29’s  $\mathbf{T}$ ), and  $\mathbf{T}_{\min}^+ = \mathbf{T}_{\min} \setminus \{\emptyset\}$  (source 27’s  $D$ , source 29’s  $\mathbf{P}$ ). The centered order with zero excluded is  $\mathcal{T}_c^{\text{inc}}$  (source 27’s  $J$ , source 29’s  $\mathcal{C}$ ); with zero retained and a distinct terminator immediately below or above it, as in proposition 37.5, the orders are  $\mathcal{T}_{\dagger-}^{\text{inc}}$  (source 27’s  $J_-$ , source 29’s  $\mathcal{C}_-$ ) and  $\mathcal{T}_{\dagger+}^{\text{inc}}$  ( $J_+$ ,  $\mathcal{C}_+$ ); the zero-padding quotient of proposition 37.4 is  $\mathcal{T}_{\text{pad}}^{\text{inc}}(Q_0, \mathcal{C}_{\text{pad}})$ . *False reading:*  $\mathbf{T}_{\min}$  is an order of *all* words; it is not  $\mathcal{T}$  with a central marker, and it is not the order of a binary cube.

### 82.1 Source 23: naturally increasing words, a universal obstruction

Let  $\mathcal{T}_\varepsilon^{\text{inc}}(A)$  be the suborder of all strictly increasing set-length words, for a set-cut saturated class alphabet  $A$  and a marker cut  $A = L \sqcup R$ . It represents exactly the set subsets well-ordered by the inherited numerical order when  $A = \mathbf{No}$ .

**Theorem 82.1** (No fixed marker saturates increasing words). *For every placement of  $\varepsilon$  relative to a set-cut saturated class alphabet  $A$ , the order  $\mathcal{T}_\varepsilon^{\text{inc}}(A)$  is not set-cut saturated. The same conclusion holds after deleting the empty word.*

*Proof.* Suppose first that  $R$  is nonempty. Choose a strictly increasing sequence  $a_0 < a_1 < \dots$  in  $R$ , and let  $p = (a_n)_{n < \omega}$  and  $p_n = p \restriction n$  for  $n \geq 1$ . Then

$$p_1 < p_2 < \dots < p.$$

If a word  $q$  were above every  $p_n$  but below  $p$ , it could not disagree with  $p$  at any finite position: disagreement below would put it below a longer  $p_n$ , and disagreement above would put it above  $p$ . Nor could it end at a finite position. Thus it would properly extend  $p$ . But any new letter of

a strictly increasing extension is greater than  $a_0$ , hence lies in  $R$ . Such an extension lies *above*  $p$ , again a contradiction.

If  $R$  is empty, the marker is above every letter. Choose any strictly increasing sequence  $(a_n)$  in  $A$ . Now  $p < p_{n+1} < p_n$  for all  $n \geq 1$ . A point above  $p$  and below every  $p_n$  would have to properly extend  $p$ , but every proper extension is below  $p$  under the end-last convention. This is impossible. All displayed constraints are nonempty words.  $\square$

*Remark 82.2* (Why the arbitrary injective case escapes). For an admissible interior marker, an arbitrary injective extension can append a fresh letter on either side of the marker, independently of the numerical values of earlier letters. A strictly increasing extension cannot. The failure is therefore caused by the inherited-order condition, not by injectivity or by class size alone.

Every set ordinal is still realized by a strictly increasing surreal word: use its ordinal labels. Every set or class linear order embeds into  $\mathcal{T}_\varepsilon^{\text{inc}}(\mathbf{No})$  under global choice by first embedding it into  $\mathbf{No}$  ([proposition 5.13](#)) and then taking singleton words. Thus universality can coexist with the explicit countable failures of saturation just proved.

*Note 82.3* ([merge]; what is new in [theorem 82.1](#)). For the marker at zero (zero excluded,  $\mathcal{T}_c^{\text{inc}}$ ) the theorem is source 16’s remark at the end of its Subsection 8.1 (printed after (S16.7): “a positive increasing word of length  $\omega$  still has its finite prefixes cofinal below it”) and the obstruction of source 18’s [theorem 37.2](#), which classifies all unfilled set cuts there; the endpoint placements are (S16.5). The theorem is new for every other placement, in particular for the thresholds other than zero left open in [section 39](#), items (b) and (d): no such threshold gives a saturated order, hence (by the set-cut property of  $\mathbf{No}$ ) none gives an order isomorphic to  $\mathbf{No}$ . It does not classify the unfilled cuts at a general threshold; source 27’s [proposition 82.8](#) computes the point characters and source 29’s [proposition 82.14](#) the isomorphism type for gaps with two large sides. Source 23’s proof uses only an  $\omega$ -sequence in  $R$  (or in  $A$ ); it needs neither global choice nor a classification of the threshold.

## 82.2 Source 27: increasing words and an exact split at the terminator

[merge] Source 27’s Section 5, printed whole. Its notation is replaced as listed at the start of this section. Its reference to “the known local behavior” of the centered order is source 18’s [theorem 37.1](#), and its reference to the uniform residual isomorphisms of its Section 6 is [lemma 96.1](#).

The increasing-word interpretation remembers the numerical order much more directly than an arbitrary well-order enumeration. Its restriction also creates a useful factorization that arbitrary words do not possess.

### 82.2.1 A choice-free two-part factorization

For an ordered alphabet  $A$ , let  $\mathcal{T}_{\text{sf}}^{\text{inc}}(A)$  and  $\mathcal{T}_{\text{lf}}^{\text{inc}}(A)$  consist of the strictly increasing set-length words, including the empty word, with proper prefixes respectively smaller and larger. If  $A = L + R$ , write  $\mathcal{T}_\varepsilon^{\text{inc}}(A)$  for the version with a distinct terminator between  $L$  and  $R$ .

**Theorem 82.4** (Factorization at a terminal cut). *For every set or class alphabet  $A = L + R$ ,*

$$\mathcal{T}_\varepsilon^{\text{inc}}(A) \cong \mathcal{T}_{\text{lf}}^{\text{inc}}(L) \times_{\text{lex}} \mathcal{T}_{\text{sf}}^{\text{inc}}(R). \quad (34)$$

*The isomorphism is canonical and uses no choice principle.*

*Proof.* In an increasing word, the  $L$  entries form an initial segment of its ordinal domain, and all remaining entries lie in  $R$ . Thus it has a unique form  $l \frown r$ . Conversely, every such concatenation is increasing because all  $L$  letters precede all  $R$  letters numerically.

If  $l, l'$  differ at actual entries, that difference decides the full words. If  $l$  is a proper prefix of  $l'$ , the next token of  $l' \frown r'$  lies in  $L$ , whereas the corresponding token of  $l \frown r$  is the terminator or lies in  $R$ . Therefore the entire block indexed by  $l'$  precedes the entire block indexed by  $l$ , exactly as in  $\mathcal{T}_{\text{lf}}^{\text{inc}}(L)$ . At equal  $l$ , every possible tail letter is above the terminator, so the tails are compared by  $\mathcal{T}_{\text{sf}}^{\text{inc}}(R)$ . This proves both bijectivity and order preservation.  $\square$

The first coordinate in (34) is primary. The factorization would fail for unrestricted words, which may repeatedly return from  $R$  to  $L$ .

### 82.2.2 The centered surreal order

Let  $\mathbf{T}_{\min}$  be all set-length surreal words, with repetitions permitted and proper prefixes smaller. Let  $\mathcal{T}_c^{\text{inc}}$  be the increasing words over  $\mathbf{No} \setminus \{0\}$ , with negative letters below the terminator and positive letters above it.

The following standard construction, used in the repository (theorems 35.1 and 35.7), is worth making explicit. After any increasing positive prefix, its permissible next letters form the ray above a set of surreals. After any increasing negative prefix, its permissible negative letters form the interval above that set and below 0. Both fill every separated set cut and have no endpoints. With a fixed global enumeration, set-state back-and-forth identifies each such class order with  $\mathbf{No}$ , uniformly in the prefix. Recursively applying these child isomorphisms to a set-length word gives an isomorphism of the corresponding rooted word trees. Every individual recursion has set length; no class of chosen class isomorphisms is needed. This argument is justified in greater detail for residual alphabets in lemma 96.1.

**Corollary 82.5** (A product description of the centered order). *In GBC,*

$$\boxed{\mathcal{T}_c^{\text{inc}} \cong \mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}} \quad (35)$$

*Proof.* The tree construction identifies  $\mathcal{T}_{\text{sf}}^{\text{inc}}(\mathbf{No}_{>0})$  with  $\mathbf{T}_{\min}$ . It identifies  $\mathcal{T}_{\text{lf}}^{\text{inc}}(\mathbf{No}_{<0})$  with the proper-prefix-larger arbitrary-word order. Coordinatewise negation identifies the latter with  $\mathbf{T}_{\min}^{\text{op}}$ : it reverses actual letter comparisons, and taking the opposite reverses the prefix convention. Apply theorem 82.4.  $\square$

This is an explicit isomorphism type in terms of an already studied word order, not an identification with  $\mathbf{No}$ . It also applies to a genuine gap  $\mathbf{No} = L + R$  when  $L$  has no set cofinal subset and  $R$  has no set coinital subset: the same admissible-child orders fill all set cuts. A threshold at an actual surreal point, with that point excluded, is the simpler special case obtained by translation.

[merge] Corollary 82.5 answers the first open item of section 39 (b) in part: it reduces the isomorphism type of  $\mathcal{T}_c^{\text{inc}}$  to that of  $\mathbf{T}_{\min}$ , the endpoint type of source 16's classification (theorem 36.2). It is consistent with source 18's corollary 37.9: the fibre over an all-negative prefix is a copy of  $\mathbf{T}_{\min}$  and is the positive-tail part of that prefix's cone. It does not identify  $\mathcal{T}_c^{\text{inc}}$  with  $\mathbf{No}$  (corollary 37.3), and whether  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$  is isomorphic to  $\mathbf{T}_{\min}^+$  is source 27's Question 12.4.



### 82.2.3 What happens when zero is retained

Different ways of retaining zero need not give different abstract order types, even though they give different labelled words. Set  $\mathbf{T}_{\min}^+ = \mathbf{T}_{\min} \setminus \{\emptyset\}$ . Splitting at the first letter gives

$$\mathbf{T}_{\min}^+ \cong \mathbf{No} \times_{\text{lex}} \mathbf{T}_{\min}, \quad \mathbf{T}_{\min} \cong 1 + \mathbf{T}_{\min}^+.$$

Since  $\mathbf{No} + \mathbf{No}$  fills all set cuts and has no endpoints, class back-and-forth gives  $\mathbf{No} + \mathbf{No} \cong \mathbf{No}$ . Hence  $\mathbf{T}_{\min}^+ + \mathbf{T}_{\min}^+ \cong \mathbf{T}_{\min}^+$  and  $\mathbf{T}_{\min} + \mathbf{T}_{\min}^+ \cong \mathbf{T}_{\min}$ . Splitting  $\mathbf{No}$  at zero also gives  $\mathbf{T}_{\min}^+ \cong \mathbf{T}_{\min}^+ + \mathbf{T}_{\min} + \mathbf{T}_{\min}^+ \cong \mathbf{T}_{\min}^+ + \mathbf{T}_{\min}$ . In particular

$$\mathbf{T}_{\min}^+ + \mathbf{T}_{\min}^+ \cong \mathbf{T}_{\min}^+, \quad \mathbf{T}_{\min}^+ + \mathbf{T}_{\min} \cong \mathbf{T}_{\min}^+, \quad \mathbf{T}_{\min} + \mathbf{T}_{\min} \cong \mathbf{T}_{\min}. \quad (36)$$

Let  $\mathcal{T}_{\dagger+}^{\text{inc}}$  retain zero and place  $0 < \dagger_+$  below all positives. Let  $\mathcal{T}_{\dagger-}^{\text{inc}}$  instead place all negatives below  $\dagger_- < 0$ . Let  $\mathcal{T}_{\text{pad}}^{\text{inc}}$  be the quotient of increasing words by equality of their zero extensions. An equivalence class has a unique representative obtained by deleting a final zero if one is present; an internal zero before positive entries remains.

**Proposition 82.6** (Exact zero-convention types). *In GBC,*

$$\mathcal{T}_{\dagger+}^{\text{inc}} \cong \mathcal{T}_{\text{pad}}^{\text{inc}} \cong \mathcal{T}_c^{\text{inc}}, \quad \mathcal{T}_{\dagger-}^{\text{inc}} \cong \mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} (1 + \mathbf{T}_{\min}).$$

*The adjacent pairs in  $\mathcal{T}_{\dagger-}^{\text{inc}}$  are exactly  $p < p \frown (0)$  for all-negative prefixes  $p$ , including the empty one.*

*Proof.* For  $\mathcal{T}_{\dagger+}^{\text{inc}}$ , the negative-or-zero part is either an all-negative word  $p$  or  $p \frown (0)$ . In the upper-terminator comparison these form a consecutive pair  $p \frown (0) < p$ . Therefore the first factor has type  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} 2$  and the second has type  $\mathbf{T}_{\min}$ . Absorption of  $\mathbf{T}_{\min} + \mathbf{T}_{\min}$  gives the result.

For  $\mathcal{T}_{\text{pad}}^{\text{inc}}$ , fix an all-negative prefix. Its reduced tails occur in three successive forms: empty; zero followed by a nonempty positive word; a nonempty positive word. Their types are  $1, \mathbf{T}_{\min}^+, \mathbf{T}_{\min}^+$ , giving  $1 + \mathbf{T}_{\min}^+ + \mathbf{T}_{\min}^+ \cong \mathbf{T}_{\min}$  for each negative-prefix block.

For  $\mathcal{T}_{\dagger-}^{\text{inc}}$ , the nonnegative tail has three successive parts: the empty word, the cone beginning with 0, and the cones beginning positively. Their types are  $1, \mathbf{T}_{\min}, \mathbf{T}_{\min}^+$ , whose sum is  $1 + \mathbf{T}_{\min}$  by (36). The first two elements of this tail order are consecutive. All other comparisons admit interpolation by a fresh numerical letter, proving the exact adjacency assertion.  $\square$

These isomorphisms do not preserve the original labelled words or their prefix trees. In particular, a quotient that merges  $p$  with  $p \frown (0)$  can have the same abstract order type as a convention that keeps them distinct.

[merge] The adjacency clause is source 18's [corollary 37.7](#); the isomorphism types are new. They answer, up to isomorphism, the comparison of the zero conventions that [propositions 37.4](#) and [37.5](#) and [corollary 37.7](#) make at the level of cuts: three of the four conventions have one type, and the fourth differs from it exactly by the adjacent pairs.

### 82.2.4 Every interval is rich, but none is a surreal interval

For a point  $x$  of an order, its left character means the cofinality of the open initial segment below  $x$ ; its right character is the coinitality above it. “Proper” below means no set cofinal, or coinital, family exists. It is not a cardinal named Ord.

The known local behavior in  $\mathcal{T}_c^{\text{inc}}$  is simple: every right character is proper; the left character is  $\text{cf}(\alpha)$  when a word has positive entries and nonzero limit length  $\alpha$ , and is proper for the other words



(theorem 37.1). At a positive limit word, sufficiently long proper prefixes form a cofinal sequence. For a smaller family, disagreement coordinates are bounded, so it is not cofinal. At a successor word, the set-cut property supplies a fresh intermediate last letter above any specified set of lower competitors.

**Theorem 82.7** (Convex word cones in every interval). *Every nonempty open interval of  $\mathcal{T}_c^{\text{inc}}$  contains a convex cone isomorphic to  $\mathbf{T}_{\min}$ . Every such interval contains points of left character  $\lambda$  for every infinite regular set cardinal  $\lambda$ , and also points with proper left character. No nonempty open interval of  $\mathcal{T}_c^{\text{inc}}$  is isomorphic to  $\mathbf{No}$ , and  $\mathcal{T}_c^{\text{inc}}$  has no convex suborder isomorphic to  $\mathbf{No}$ .*

*Proof.* Given  $f < g$ , choose a prefix whose comparisons with  $f, g$  are already settled strictly. If their first difference is at actual entries, use an intermediate letter above the common prefix. If  $f$  is a proper prefix of  $g$ , the next letter of  $g$  is positive; use a smaller admissible positive letter. If  $g$  is a proper prefix of  $f$ , the next letter of  $f$  is negative; use a larger admissible negative letter. All these choices follow from the surreal set-cut property.

Append a positive letter if necessary, keeping the comparisons already settled. The resulting prefix  $p$  ends positively. Its entire extension cone is convex, lies inside  $(f, g)$ , and has only letters above the terminator available afterward. The rooted-tree construction identifies that cone with  $\mathbf{T}_{\min}$ .

Append to  $p$  a strictly increasing positive tail of length  $\lambda$ . The resulting length has cofinality  $\lambda$ , and the local-character calculation gives left character exactly  $\lambda$ . A successor-length extension gives proper left character. Taking  $\lambda = \omega$  gives a set cofinal below a point inside the interval; this is an unfilled strict set cut with that point as upper endpoint. Every surreal interval fills all separated set cuts, so no such interval is isomorphic to  $\mathbf{No}$ . Finally, a convex copy of  $\mathbf{No}$  would contain a nonempty ambient interval isomorphic to a surreal interval, which has just been excluded.  $\square$

Each convex cone above contains a copy of  $\mathbf{No}$  among its one-letter tails. Thus every interval contains surreal copies, but these copies are necessarily nonconvex. Universality and convex realization are different questions even at this very local level.

[merge] This strengthens source 18's remark after corollary 37.3 (every nonempty open interval contains a nonconvex copy of  $\mathbf{No}$ ); the convex cone of type  $\mathbf{T}_{\min}$  is the positive-prefix case of corollary 37.9.

### 82.2.5 Genuine gap thresholds produce additional behavior

The factorization in theorem 82.4 works for any class gap  $\mathbf{No} = L + R$ . A concise exact local classification records what changes. Assume both sides are nonempty. Let  $\lambda_L = \text{cf}(L)$  if  $L$  has no maximum and has a set cofinal subset, and put  $\lambda_L = \text{proper}$  otherwise. Let  $\lambda_R = \text{ci}(R)$ , allowing 1 for a minimum and proper when no set is coinital.

**Proposition 82.8** (Characters at a general terminal gap). *The order  $\mathcal{T}_\varepsilon^{\text{inc}}(\mathbf{No})$  has no endpoints, and its characters are as follows. In the last two rows the word has successor length.*

| Word                                                     | Left character      | Right character |
|----------------------------------------------------------|---------------------|-----------------|
| All letters in $L$ , including the empty word            | $\lambda_L$         | $\lambda_R$     |
| An $R$ letter occurs; length $\alpha$ is a nonzero limit | $\text{cf}(\alpha)$ | proper          |
| $p \frown (\min R)$ with $p$ entirely in $L$             | 1                   | proper          |
| Every other successor word with an $R$ letter            | proper              | proper          |

The only adjacent pairs are  $p < p \frown (\min R)$  when  $\min R$  exists.

*Proof.* Above an all- $L$  word, the first new letter must lie in  $R$ ; its coinital possibilities give exactly  $\lambda_R$ . A least such letter gives the stated immediate successor. After an  $R$  letter has occurred, a new letter can be chosen above the whole prefix and below every relevant letter of a given set of upper competitors. Thus the right character is proper.

Below an all- $L$  word, if  $L$  has no maximum and its range is not cofinal in  $L$ , the possible new  $L$  letters form a cofinal tail of  $L$ , giving character  $\lambda_L$ . If the range is cofinal, its length is a limit of cofinality  $\lambda_L$ . Increasing the common-prefix length and then replacing the next letter by a smaller admissible one gives a cofinal family of that size; bounded disagreement coordinates exclude a smaller one. If  $L$  has no set cofinal family, append an  $L$  letter above all relevant set constraints instead.

When  $L$  has a maximum  $m$ , a word not yet containing  $m$  can move closer from below through the cone beginning with  $m$  and then arbitrarily large  $R$  letters. No set is cofinal in that cone. For a word ending at  $m$ , replace its final letter by one below  $m$  but above all earlier and competing set constraints. Its left character is again proper, which explains why a maximum of  $L$  does not give character 1 in the first row.

After entering  $R$ , a limit-length word is approached by its long proper prefixes, giving  $\text{cf}(\alpha)$ . At a successor word the only obstruction to interpolation at its last letter is that this letter is  $\min R$  and the earlier prefix is all in  $L$ . This is exactly the adjacent pair already identified. Otherwise include an  $R$  letter below the last letter among the lower constraints and use the set-cut property. The same constructions give a point on each side of every word, proving the absence of endpoints.  $\square$

For a concrete gap let

$$L_0 = \{x : x < n \text{ for some } n < \omega\}, \quad R_0 = \{x : n \leq x \text{ for every } n < \omega\}.$$

Here  $\text{cf}(L_0) = \omega$  and  $R_0$  has no set coinital subset; a set of such elements would admit a smaller surreal still above all integers. Reversing and negating this gap gives a right side of coinitality  $\omega$ . The associated increasing-word order has points of right character  $\omega$ , and therefore cannot be isomorphic to the standard centered  $\mathcal{T}_c^{\text{inc}}$ , whose right characters are all proper. Changing only the character of the distinguished empty word would not by itself prove nonisomorphism of unmarked orders; the right-character argument uses the full point spectrum.

[merge] [Proposition 82.8](#) answers in part the second open item of [section 39](#) (b) and item (d): it gives the point characters, and with them the adjacent pairs, at every interior threshold; with [source 23's theorem 82.1](#) (no threshold saturates) and [source 29's proposition 82.14](#) (actual thresholds and gaps with two large sides give  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ ) it settles the isomorphism type except at gaps with a small cofinality or coinitality on one side. There, characters distinguish such an order from  $\mathcal{T}_c^{\text{inc}}$ , as just shown, but the general cut classification “was not proved” ([source 27's audit](#)), and [source 27's Question 12.5](#) asks for the isomorphism types. The threshold  $L_0$  is [source 16's](#) “natural numbers” threshold printed before [theorem 36.2](#), and its reverse is [source 23's](#)  $(L_\omega, R_\omega)$  of [section 81.4.2](#), there for all words.

### 82.3 Source 29: a canonical product and the zero conventions

[merge] [Source 29's Section 6](#), printed whole. It proves [source 27's](#) factorization, centered product and zero-convention types independently; those statements are notes quoting [source 29's](#) formulation, followed by its proofs as second routes. Its threshold reduction ([proposition 82.14](#)) and its explicit absorption identity  $\mathbf{T}_{\min}^+ + 1 + \mathbf{T}_{\min}^+ \cong \mathbf{T}_{\min}^+$  are new; the collapsing clause of [note 82.13](#), that identifying the adjacent pairs of  $\mathcal{T}_{+-}^{\text{inc}}$  gives  $\mathcal{T}_c^{\text{inc}}$ , is not stated by [source 27](#). The threshold reduction

is printed as a proposition, since source 27 only remarks on its two cases. Source 29's local isomorphisms  $\theta_p$  are printed  $h_p$ , as in the proof of [theorem 35.1](#); its product  $\otimes_{\text{lex}}$  is printed  $\times_{\text{lex}}$ .

All words in this section have *set ordinal length*. Their entries may range over the whole surreal class, but each individual word is a set graph. The resulting word spaces and their products are therefore ordinary classes of sets. They should be distinguished from collections whose individual points are proper-class well-orders.

The repository already identifies the shorter-first trees of free, injective, and numerically increasing surreal words, and determines the set-cut obstructions for centered increasing words ([theorems 35.1](#), [35.7](#) and [37.2](#)). We use those mechanisms to obtain an explicit product description and to compare the different treatments of the entry zero up to isomorphism.

### 82.3.1 Splitting an increasing word at its terminator

For a linearly ordered alphabet  $D$ , let  $\mathcal{T}^{\text{inc}}(D)$  be the set or class of all strictly increasing set-length  $D$ -words, including the empty word. The notations  $\mathcal{T}_{\text{sf}}^{\text{inc}}(D)$  and  $\mathcal{T}_{\text{lf}}^{\text{inc}}(D)$  mean that proper prefixes come first and last, respectively; first unequal entries always compare in the order of  $D$ . For orders  $A, B$ , write

$$A \times_{\text{lex}} B = \sum_{a \in A} B.$$

Thus the first coordinate is decisive. The notation records an *ordered* product and makes no commutativity assertion.

Suppose  $D = L + R$  is an initial/final partition, and put a distinct formal terminator  $\varepsilon$  between its parts:

$$L < \varepsilon < R.$$

The word order  $\mathcal{T}_{\varepsilon}^{\text{inc}}(D)$  compares the terminator against the next entry when one word is a proper prefix of another. Either side of the partition is allowed to be empty.

*Note 82.9* (Source 29, Theorem 6.1, “Canonical factorization”). *For any such ordered alphabet and terminator,*

$$\mathcal{T}_{\varepsilon}^{\text{inc}}(D) \cong \mathcal{T}_{\text{lf}}^{\text{inc}}(L) \times_{\text{lex}} \mathcal{T}_{\text{sf}}^{\text{inc}}(R).$$

*The isomorphism is the canonical decomposition into the maximal initial  $L$ -word and the remaining  $R$ -word. For a class alphabet, the statement uses only elementary class definitions and set operations; Global Choice is not required.* This is source 27's [theorem 82.4](#).

*Second route* (source 29). If  $f : \alpha \rightarrow D$  is increasing, the indices at which its entries belong to  $L$  form an initial segment of  $\alpha$ , say  $\beta$ . The remaining interval of indices has a unique ordinal order type  $\gamma$ . Thus  $f$  splits uniquely as

$$f = u \frown v, \quad u \in \mathcal{T}^{\text{inc}}(L), \quad v \in \mathcal{T}^{\text{inc}}(R).$$

Conversely, every such concatenation is increasing because  $L < R$ . The domains, restrictions, ordinal tail, and concatenation are sets. This proves the claimed bijection of carriers.

Consider two pairs  $(u, v)$  and  $(u', v')$ . If  $u, u'$  disagree at a common coordinate, their first unequal  $L$ -entries decide the comparison of both concatenated words, independently of their  $R$ -tails. If  $u$  is a proper prefix of  $u'$ , the next entry of  $u'$  belongs to  $L$ . At that position  $u \frown v$  either terminates or has an entry in  $R$ . In either case it is larger. Hence the entire  $u'$ -fiber lies below the entire  $u$ -fiber, exactly as required by longer-first comparison on the first factor. The reverse prefix case is symmetric.

Finally, when  $u = u'$ , the tails use only entries above  $\varepsilon$ . Their comparison is therefore shorter-first lexicography on  $R$ . These cases prove the order identity.

This factorization does not require a numerical bound for the entire alphabet. In particular, it remains valid when the two sides of the terminator are proper classes. The theorem applies to set-length words even when the partition itself cannot be specified by an actual surreal number.

### 82.3.2 The two surreal factors

Let  $\mathbf{T}_{\min}$  be the shorter-first order of all set-length surreal words, with repetitions allowed and the empty word included. Put

$$\mathbf{T}_{\min}^+ = \mathbf{T}_{\min} \setminus \{\emptyset\}.$$

The empty word is the minimum of  $\mathbf{T}_{\min}$ . We will use the following form of the repository's prefix-tree construction, including its uniformity requirement ([theorem 35.7](#)).

*Note 82.10* (Source 29, Lemma 6.2, “Uniform isomorphisms of admissible alphabets”). *Work in GBC with a fixed set-like global well-order. Suppose a class of permitted set words contains the empty word and is closed under restrictions and unions of coherent set-length prefix chains. Suppose it has, after each permitted prefix  $p$ , a proper class  $D_p$  of admissible next letters, giving exactly its permitted one-letter extensions. Assume these classes are uniformly defined from  $p$  and fixed class parameters, and each fills every separated set cut, including cuts with an empty side. Then its ordered prefix tree is isomorphic to the free surreal-word tree, preserving word lengths, prefixes, and comparison of first unequal entries. The same tree isomorphism preserves either shorter-first or longer-first comparison. This is the construction of [theorem 35.1](#) (source 16) and [theorem 35.7](#) (source 18), stated for an arbitrary uniformly given family of admissible alphabets; source 27 uses it in the proof of [corollary 82.5](#) and states the residual case as [lemma 96.1](#).*

*Second route (source 29).* For each prefix  $p$ , construct an increasing bijection  $h_p : \mathbf{No} \rightarrow D_p$  by deterministic back-and-forth. Restrict the fixed global well-order to the two alphabets. At each ordinal stage, map the least unused point on the source side to the least unused point in the cut required by the current partial map, and then do the reverse insertion. The partial map and all cut constraints at a set ordinal stage are sets. Set-cut filling supplies each insertion.

This construction is uniform: membership in  $h_p$  is expressed by existence of a set-length initial history of the deterministic construction. The parameters of that assertion are  $p$ , the fixed class definitions, and the global well-order. One elementary class relation therefore gives all  $h_p$  simultaneously. There is no independent choice of a class isomorphism for each prefix.

Given a free source word  $f$ , construct a permitted target word  $g$  by ordinary set recursion,

$$g(\xi) = h_{g \upharpoonright \xi}(f(\xi)).$$

The inverse uses  $h_{g \upharpoonright \xi}^{-1}$  at each coordinate. Thus this is a length-preserving bijection of the two word trees. A shared source prefix gives a shared target prefix, and the same increasing letter map acts at the first unequal coordinate. This preserves first-entry comparisons. Prefix comparisons are preserved for either of the two specified conventions.

The lemma applies to increasing words over each half of the surreal line. After a negative prefix  $p$ , the admissible negative alphabet is

$$D_p^- = \{x : \text{ran}(p) < x < 0\}.$$

For separated sets  $A < B$  in this alphabet, add  $\text{ran}(p)$  to the lower constraints and 0 to the upper constraints. The surreal set-cut property supplies another point of  $D_p^-$ . The same argument handles

an empty side. For positive prefixes the admissible alphabet is

$$D_p^+ = \{x : \{0\} \cup \text{ran}(p) < x\};$$

add  $\{0\} \cup \text{ran}(p)$  to the lower constraints. These classes are proper: if one were a set, its own elements could be placed on one side of a cut that would require a point beyond all of them in that same class.

*Note 82.11* (Source 29, Corollary 6.3, “The centered increasing-word product”). Let  $\mathcal{T}_c^{\text{inc}}$  be the order of strictly increasing set-length words over  $\mathbf{No} \setminus \{0\}$ , with

$$\mathbf{No}_{<0} < \dagger < \mathbf{No}_{>0}.$$

In GBC,  $\mathcal{T}_c^{\text{inc}} \cong \mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ . This is source 27’s [corollary 82.5](#).

*Second route (source 29).* By [note 82.10](#),  $\mathcal{T}_{\text{sf}}^{\text{inc}}(\mathbf{No}_{>0}) \cong \mathbf{T}_{\min}$ . The same lemma identifies longer-first increasing negative words with longer-first *free* surreal words. On free words, entrywise negation reverses comparison of unequal entries and converts a longer-first comparison into the reverse of a shorter-first comparison. Consequently

$$\mathcal{T}_{\text{lf}}^{\text{inc}}(\mathbf{No}_{<0}) \cong \mathbf{T}_{\min}^{\text{op}}.$$

Apply [note 82.9](#). Passing first to free words is essential: negating an increasing word directly gives a decreasing word.

The factorization explains the two kinds of cones already isolated in the repository ([corollary 37.9](#)). An all-negative prefix still allows both a negative part and a positive part. Once a positive entry has occurred, only the shorter-first positive-tail factor remains. The product is an explicit description in previously studied orders. It does not by itself assert that its displayed factors or fibers can be recovered from the pure linear order.

### 82.3.3 Absorption in the dictionary order

Some useful ordered-sum identities make the zero conventions much closer up to isomorphism than their definitions suggest.

*Note 82.12* (Source 29, Lemma 6.4, “Dictionary absorption”). In GBC,

$$\begin{aligned} \mathbf{T}_{\min}^+ &\cong \mathbf{No} \times_{\text{lex}} \mathbf{T}_{\min}, & \mathbf{T}_{\min}^+ + \mathbf{T}_{\min}^+ &\cong \mathbf{T}_{\min}^+, \\ \mathbf{T}_{\min}^+ + 1 + \mathbf{T}_{\min}^+ &\cong \mathbf{T}_{\min}^+, & \mathbf{T}_{\min} + \mathbf{T}_{\min} &\cong \mathbf{T}_{\min}. \end{aligned}$$

Three of the four identities are source 27’s ([equation \(36\)](#) and the display before it); the third is equivalent to source 27’s  $\mathbf{T}_{\min}^+ + \mathbf{T}_{\min} \cong \mathbf{T}_{\min}^+$ , since  $\mathbf{T}_{\min} = 1 + \mathbf{T}_{\min}^+$ .

*Second route (source 29).* Every nonempty free word splits into its first letter and its tail, so  $\mathbf{T}_{\min}^+ \cong \mathbf{No} \times_{\text{lex}} \mathbf{T}_{\min}$ . The two-copy sum  $\mathbf{No} + \mathbf{No}$  fills every set cut and has no endpoints. For example, if lower constraints enter the second copy, interpolate there; if upper constraints enter the first, interpolate there. In the remaining case all lower constraints are in the first copy and all upper constraints in the second, and one can choose in the second copy below its set of upper constraints. Empty sides cause no problem. Class back-and-forth in GBC therefore gives  $\mathbf{No} + \mathbf{No} \cong \mathbf{No}$ , and hence

$$\mathbf{T}_{\min}^+ + \mathbf{T}_{\min}^+ \cong (\mathbf{No} + \mathbf{No}) \times_{\text{lex}} \mathbf{T}_{\min} \cong \mathbf{T}_{\min}^+.$$

Now split  $\mathbf{T}_{\min}^+$  at the one-letter word  $q = \langle 0 \rangle$ . The part below  $q$  consists exactly of the words with negative first letter, and has type  $\mathbf{No}_{<0} \times_{\text{lex}} \mathbf{T}_{\min} \cong \mathbf{T}_{\min}^+$ . The part above  $q$  consists of

the nonempty tails in the zero cone, followed by the words with positive first letter. Each part has type  $\mathbf{T}_{\min}^+$ , so their sum has type  $\mathbf{T}_{\min}^+$  by the identity just proved. Splitting at  $q$  thus gives  $\mathbf{T}_{\min}^+ \cong \mathbf{T}_{\min}^+ + 1 + \mathbf{T}_{\min}^+$ . Finally  $\mathbf{T}_{\min} = 1 + \mathbf{T}_{\min}^+$ , whence

$$\mathbf{T}_{\min} + \mathbf{T}_{\min} = 1 + \mathbf{T}_{\min}^+ + 1 + \mathbf{T}_{\min}^+ \cong 1 + \mathbf{T}_{\min}^+ = \mathbf{T}_{\min}.$$

All products and ordered sums here have set-coded points.

In particular, the proper-class boundary between two copies of **No** does not contradict set-cut saturation: no set is cofinal in the first copy or cinitial in the second. The proof never treats those entire copies as set-sized cut constraints.

### 82.3.4 An exact comparison of the zero conventions

Retaining zero as an actual entry requires a choice in addition to saying that termination is centered. We compare the following four orders, always using strictly increasing set-length words:

$$\begin{aligned} \mathcal{T}_c^{\text{inc}} &: \text{zero excluded, with termination in its place;} \\ \mathcal{T}_{\dagger+}^{\text{inc}} &: \mathbf{No}_{<0} < 0 < \dagger_+ < \mathbf{No}_{>0}; \\ \mathcal{T}_{\dagger-}^{\text{inc}} &: \mathbf{No}_{<0} < \dagger_- < 0 < \mathbf{No}_{>0}; \\ \mathcal{T}_{\text{pad}}^{\text{inc}} &: \text{zero padding, with equal padded words identified.} \end{aligned}$$

The distinct terminators in the middle two cases are formal symbols, not additional numerical entries. In the last case every word is extended by zero at all later ordinal coordinates and comparison is at the first unequal coordinate. On the original words this is a preorder; taking the indicated quotient is necessary.

For increasing words, a canonical padded representative is obtained by removing a terminal zero when one is present. Such a zero can only follow negative entries. No other terminal zero tail is possible in a strictly increasing word. Equality after padding is exactly equality of these reduced representatives, so the quotient has an ordinary class of set representatives.

*Note 82.13* (Source 29, Theorem 6.5, “The zero-convention isomorphism types”). *In GBC*,

$$\mathcal{T}_c^{\text{inc}} \cong \mathcal{T}_{\dagger+}^{\text{inc}} \cong \mathcal{T}_{\text{pad}}^{\text{inc}} \cong \mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}, \quad \mathcal{T}_{\dagger-}^{\text{inc}} \cong \mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} (2 + \mathbf{T}_{\min}^+).$$

*The first three orders are dense without endpoints. The adjacent pairs of  $\mathcal{T}_{\dagger-}^{\text{inc}}$  are exactly*

$$u < u \frown \langle 0 \rangle, \quad \text{ran}(u) < 0.$$

*The empty word is allowed for  $u$ . Collapsing these two-point intervals and leaving every other point alone gives an order isomorphic to  $\mathcal{T}_c^{\text{inc}}$ .* The isomorphism types and the adjacent pairs are source 27’s [proposition 82.6](#) (with  $1 + \mathbf{T}_{\min} = 2 + \mathbf{T}_{\min}^+$ ); the adjacent pairs are also source 18’s [corollary 37.7](#). Density without endpoints and the collapsing clause are proved only here.

*Second route (source 29).* Put  $A = \mathcal{T}_{\text{lf}}^{\text{inc}}(\mathbf{No}_{<0}) \cong \mathbf{T}_{\min}^{\text{op}}$ . Nonempty increasing positive words have type  $\mathbf{T}_{\min}^+$ , because the prefix-tree isomorphism used in [note 82.11](#) fixes the empty root. We examine the fibers over a negative word  $u$ .

For  $\mathcal{T}_{\dagger+}^{\text{inc}}$ , the initial nonpositive word is uniquely either  $u \frown \langle 0 \rangle$  or  $u$ . Under longer-first comparison they occur in exactly that order. Each has an arbitrary increasing positive tail, including an empty tail, so each contributes a copy of  $\mathbf{T}_{\min}$ . The fiber over  $u$  consequently has type  $\mathbf{T}_{\min} + \mathbf{T}_{\min} \cong \mathbf{T}_{\min}$ . The order of different negative prefixes is  $A$ , by the same comparison as in [note 82.9](#). Thus  $\mathcal{T}_{\dagger+}^{\text{inc}} \cong A \times_{\text{lex}} \mathbf{T}_{\min}$ .



For  $\mathcal{T}_{\text{pad}}^{\text{inc}}$ , a reduced representative with negative prefix  $u$  either stops there, has zero followed by a nonempty positive word, or has a nonempty positive word immediately. These three parts occur in the displayed order: after  $u$ , zero agrees with padded termination, and the next positive entry makes the zero-containing word larger than the reduced root; an immediate positive entry is larger still at the earlier coordinate. The fiber therefore has type

$$1 + \mathbf{T}_{\min}^+ + \mathbf{T}_{\min}^+ \cong 1 + \mathbf{T}_{\min}^+ = \mathbf{T}_{\min}.$$

This proves the padded-quotient assertion.

For  $\mathcal{T}_{\text{t-}}^{\text{inc}}$ , the fiber over  $u$  consists, in order, of the root  $u$  itself; the entire cone beginning with  $u \frown \langle 0 \rangle$ ; and the nonempty tails whose first new entry is positive. The middle cone has type  $\mathbf{T}_{\min}$ , including its root. Thus the fiber is

$$1 + \mathbf{T}_{\min} + \mathbf{T}_{\min}^+ = 2 + \mathbf{T}_{\min}^+ + \mathbf{T}_{\min}^+ \cong 2 + \mathbf{T}_{\min}^+.$$

This proves the second product formula.

The order  $\mathbf{T}_{\min}$  is dense: between different first entries choose an intermediate entry, and between a proper prefix and an extension choose a first new entry smaller than that of the extension. It has no greatest point because every word has a proper extension. Its reverse has no least point. Hence  $A \times_{\text{lex}} \mathbf{T}_{\min}$  is dense without endpoints, including at comparisons involving different fibers.

In a  $\mathcal{T}_{\text{t-}}^{\text{inc}}$  fiber, the first two points are adjacent. There is no point between  $u$  and  $u \frown \langle 0 \rangle$ : an extension of  $u$  whose next entry is positive exceeds the latter, and an extension whose next entry is negative is below  $u$ . Any continuation after the zero also exceeds the latter. The remaining parts of the fiber are dense, and the second point is followed by a copy of  $\mathbf{T}_{\min}^+$  with no least point. Different fibers have infinitely many points between them, since their index order  $A$  is dense. There are consequently no other adjacent pairs and no larger finite-interval blocks.

Collapsing the initial two points of each fiber replaces  $2 + \mathbf{T}_{\min}^+$  by  $1 + \mathbf{T}_{\min}^+ = \mathbf{T}_{\min}$  and keeps the index  $A$ . The quotient is therefore  $A \times_{\text{lex}} \mathbf{T}_{\min} \cong \mathcal{T}_c^{\text{inc}}$ . One can implement it by retaining the smaller representative of each adjacent pair; no class of proper-class equivalence classes is introduced.

The original presentations thus give two types among these four conventions. The distinction caused by a terminator immediately below zero survives in the pure order through adjacency. In the other three cases, the different presentations can be absorbed by the ordered-sum identities of [note 82.12](#). The isomorphisms are not assertions that the original words, their entries, or their negative-prefix fibers are identified pointwise.

### 82.3.5 Numerical thresholds and genuine remaining parameters

**Proposition 82.14** (Threshold reductions). *For every actual surreal  $c$ , the conventions centered at  $c$  transfer to the corresponding conventions at zero by entrywise translation  $x \mapsto x - c$ . This includes deletion of the entry  $c$ , either adjacent distinct terminator, and padding by  $c$  with its quotient.*

*More generally, suppose  $\mathbf{No} = L + R$  is a partition into nonempty classes,  $L$  has neither a greatest member nor a set cofinal subset, and  $R$  has neither a least member nor a set coinital subset. The increasing-word order with terminator at this cut is, in GBC, isomorphic to  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ .*

*Proof.* Translation preserves and reflects numerical inequalities, preserves strict increase, and sends  $c$  to zero. It also sends padding by  $c$  to zero padding. All first-disagreement and termination comparisons are therefore preserved.

For the second assertion, each of  $L, R$  fills every separated set cut relative to itself. For instance, a cut in  $L$  with a nonempty upper side can be filled in  $\mathbf{No}$  and its filler lies in  $L$ ; when the upper side is empty, the absence of a set cofinal subset provides a point above the lower constraints. The



argument for  $R$  is dual. After a set increasing prefix in  $L$ , its remaining upper part is still proper and set-cut saturated, by adding that prefix to the lower constraints. The same holds in  $R$ . Apply [notes 82.9](#) and [82.10](#).  $\square$

An actual numerical threshold is therefore not an additional isomorphism parameter. Nor are the two large sides of a nonprincipal class gap, under the stated hypotheses. A nonprincipal cut with a one-sided *set* cofinality or cointinality is different: a set increasing prefix can then exhaust the available room on one side. The canonical factorization still holds, but the saturated-alphabet identifications used above need not apply. This is the remaining threshold issue; the proof does not transfer the repository's complete free-word termination classification to increasing words.

Finally,  $\mathcal{T}_c^{\text{inc}}$  is not isomorphic to **No**. The positive word  $\langle 1, 2, 3, \dots \rangle$  is the supremum of its finite prefixes in  $\mathcal{T}_c^{\text{inc}}$ : any smaller word disagrees earlier in the smaller direction or is one of its proper prefixes. Such a nontrivial set supremum cannot occur in **No**, since a surreal set cut inserts a point above the given set and below its proposed supremum. The product formula retains this obstruction rather than removing it.

[merge] The last paragraph is the proof of [corollary 37.3](#) (source 18). The second assertion of [proposition 82.14](#) is the case of [corollary 82.5](#) that source 27 states in the paragraph after it (“It also applies to a genuine gap . . .”); source 29 proves it, and extends the translation remark there to all four zero conventions.

## 83 Numerically well-ordered proper-class carriers (source 29)

[merge] Source 29's Section 7, printed whole; no other source proves its results. Its comparison of carriers by the least point of the symmetric difference ([note 83.1](#) and the display after it) is source 15's characteristic comparison  $<_\chi$ , printed before [proposition 38.6](#), where it is stated for sets and classes alike; source 29's longer-first order  $\prec_{\text{lf}}$  is  $<_\chi$ . The example of a numerically increasing class of type  $\text{Ord} + 1$  printed after [proposition 38.6](#) is the case  $b = 0$  of [example 83.3](#). New are the adjacency theorem, the two-point blocks, longer-first density, the cofinal representatives and the tree non-isomorphism ([theorems 83.2](#) and [83.5](#) to [83.7](#) and [corollary 83.4](#)). Source 29 cites source 19 (its “EarlierResidual”) for the mention of this interpretation; the batch-81 maps make it as well ([section 34](#)).

The preceding word constructions use set ordinal lengths. We now admit all available subclasses of **No** that are well-ordered by their inherited numerical order. Such a carrier need not be set-like, and no set ordinal enumeration is assumed. This interpretation is mentioned in the earlier project reports, but its proper-class adjacency behavior is different from the set-word case.

All assertions in this section are made in GBC about individual class parameters, uniformly coded families, or the external collection belonging to a specified class model. We do not form a GB class whose members are these proper-class carriers.

### 83.1 Comparison without an abstract class ordinal

*Note 83.1* (Source 29, Lemma 7.1). *The union of finitely many numerically well-ordered class carriers is numerically well-ordered. Distinct such carriers  $A, B$  have a least point  $d$  of  $A \triangle B$ , and*

$$C = A \cap (-\infty, d) = B \cap (-\infty, d)$$

*is their common initial segment.* The first two assertions are source 15's paragraph before [proposition 38.6](#).

*Second route (source 29).* For a nonempty subclass of a finite union, intersect it with each carrier for which the intersection is nonempty and take the smallest of their finitely many least points. This proves the first assertion and applies in particular to the symmetric difference. Below its least point the carriers agree. Their inherited numerical relations agree there automatically.

If both carriers have a point after  $C$ , compare their least such points numerically. If one carrier is exactly  $C$ , a prefix convention is necessary. Denote the shorter-first order by  $\prec_{\text{sf}}$  and the longer-first order by  $\prec_{\text{lf}}$ . The latter has the particularly simple expression

$$A \prec_{\text{lf}} B \iff \min(A \triangle B) \in A.$$

It is characteristic-function lexicography with the bit order  $1 < 0$ , but its scan is the well-ordered union of the carriers being compared, not the whole numerical surreal line. Transitivity follows by comparing first differences in the union of three carriers. The shorter-first version reverses proper-prefix pairs and uses the same genuine first-difference comparisons.

### 83.2 Exact adjacency at an unattained supremum

For a numerical carrier  $A$ , define its class of possible next labels by

$$U(A) = \{x \in \mathbf{No} : \forall a \in A (a < x)\}.$$

If  $A$  is a set,  $U(A)$  has no least point by the set-cut property. For a proper-class  $A$ , it can have one. ([merge] For a set  $A$  this is the class  $U(q)$  of admissible next letters in the proof of [theorem 35.1](#).)

**Theorem 83.2** (Proper-class adjacency). *Two numerical well-ordered carriers  $A \prec_{\text{sf}} B$  are adjacent if and only if*

$$B = A \cup \{b\}, \quad b = \min U(A).$$

*Equivalently,  $b$  is a numerical supremum of  $A$  that does not belong to  $A$ . In every such pair,  $A$  is a proper class and has no greatest member.*

*Proof.* Suppose first that there is a genuine mismatch after the shared initial segment  $C$ , with next points  $a < b$ . Numerical density gives  $a < c < b$ . The carrier  $C \cup \{c\}$  is numerically well-ordered and lies strictly between the two given carriers. This argument remains valid when  $C$  is a proper class: adjoining one point above all its members preserves well-ordering.

Thus an adjacent pair must have  $A$  as a proper initial segment of  $B$ . Let  $b$  be the least point of  $B \setminus A$ . If the remainder has more than one point,  $A \cup \{b\}$  lies strictly between  $A$  and  $B$ . If some  $c \in U(A)$  satisfies  $c < b$ , then  $A \cup \{c\}$  lies strictly between them as well. Necessity follows.

Conversely, suppose  $B = A \cup \{b\}$  with  $b = \min U(A)$ . A carrier strictly between  $A$  and  $B$  must extend  $A$ : an earlier numerical divergence puts it on the same side of both, and a proper prefix of  $A$  lies below  $A$ . Its first new label  $c$  belongs to  $U(A)$ . If  $c > b$ , it lies above  $B$ ; if  $c = b$ , it is  $B$  or extends  $B$  and again cannot lie strictly between. The case  $c < b$  is impossible.

If  $A$  had a greatest member  $a$ , density would supply an upper bound strictly between  $a$  and  $b$ , contrary to the minimality of  $b$ . Thus the least strict upper bound is exactly an unattained supremum. If  $A$  were a set, the cut  $(A, \{b\})$  would have a surreal filler, giving the same contradiction.  $\square$

**Example 83.3** (A jump over every surreal). For every  $b \in \mathbf{No}$ , put

$$A_b = \left\{ b - \frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\},$$

where the ordinal values are embedded in  $\mathbf{No}$  and the arithmetic is surreal arithmetic. This is an increasing class of numerical type  $\text{Ord}$ , and it is cofinal below  $b$ . Indeed, if  $x < b$ , choose an ordinal  $\alpha$  with  $\alpha + 1 > 1/(b - x)$ . Then  $x < b - 1/(\alpha + 1) < b$ . Consequently

$$A_b \prec_{\text{sf}} A_b \cup \{b\}$$

is an adjacent pair. The upper carrier has type  $\text{Ord} + 1$ . The example is uniform in the set parameter  $b$ .

**Corollary 83.4.** *Every nontrivial adjacency component in the shorter-first numerical class-carrier order has exactly two members. Contracting these pairs gives a dense order with a minimum, in an external collection or in any appropriate higher-order formulation.*

*Proof.* An upper member of a jump has a greatest numerical point. A lower member has none. Hence one carrier cannot be both the upper member of one jump and the lower member of another. If two different contracted blocks had no block between them, their facing endpoints would be adjacent and would have belonged to the same block. The empty carrier remains least.  $\square$

**Theorem 83.5** (Longer-first density). *The longer-first numerical class-carrier order is dense and has the empty carrier as its greatest element. Therefore the full shorter-first and longer-first orders are neither isomorphic nor anti-isomorphic.*

*Proof.* A genuine mismatch is interpolated exactly as in [theorem 83.2](#). Otherwise  $B$  is a proper initial segment of  $A$  and  $A \prec_{\text{lf}} B$ . Let  $a$  be the first point of  $A \setminus B$ . If the remainder contains more points, the carrier  $B \cup \{a\}$  lies strictly between. If it does not, choose any  $c > a$ . It is still above every member of  $B$ , and

$$A = B \cup \{a\} \prec_{\text{lf}} B \cup \{c\} \prec_{\text{lf}} B.$$

This proves density. The empty carrier is a prefix of all carriers and is greatest under longer-first comparison. Density is preserved by both isomorphism and reversal; the shorter-first jumps therefore exclude either identification.  $\square$

[merge] For set carriers, [proposition 38.6](#) (source 15) embeds the longer-first order into  $\mathbf{No}$ ; source 15 remarks that the construction does not admit proper-class supports. [Theorems 83.2](#) and [83.5](#) show what changes at proper-class carriers: the shorter-first order acquires adjacent pairs, which no set-length word order of [theorem 35.1](#) has, while the longer-first order stays dense.

### 83.3 Every final class segment can occur after a prefix

**Theorem 83.6** (Cofinal numerical representatives). *Let  $L$  be an initial subclass of the numerical surreal line and let  $U = \mathbf{No} \setminus L$ . In GBC there is a numerically well-ordered subclass  $A \subseteq L$ , cofinal in  $L$ , such that*

$$U(A) = U.$$

*The carrier  $A$  can be a set exactly when  $L$  has a set cofinal subset. If no such set exists,  $A$  can be chosen as the range of an increasing class map  $\text{Ord} \rightarrow L$ .*

*Proof.* The empty  $L$  is handled by  $A = \emptyset$ . Otherwise fix a class bijection  $e : \text{Ord} \rightarrow \mathbf{No}$ . Construct an increasing sequence, stopping permanently as soon as its earlier range is cofinal in  $L$ . At stage  $\alpha$ , its earlier range  $S$  is a set. If it is not cofinal, some point of  $L$  is strictly above every member of  $S$ . Choose, by the  $e$ -scan, the least candidate  $y \in L$  satisfying

$$S < y, \quad e(\alpha) \in L \implies e(\alpha) \leq y.$$

Such a candidate exists: take the larger of the available strict upper bound and  $e(\alpha)$  when the latter belongs to  $L$ . Each step is a set-valued elementary recursion. A permanent stop flag can be included in the state, and the class construction is defined by its set-length histories. No recursion with class-valued slices is involved.

If the process stops, its range is set-sized and cofinal. If it does not, the resulting increasing class sequence is cofinal because every point  $x = e(\alpha) \in L$  is passed at its scan stage. Its range is numerically well-ordered, being isomorphic to  $\text{Ord}$ . If  $L$  has a set cofinal subset  $C$ , the set of indices  $e^{-1}[C]$  is ordinal-bounded. After that bound all of  $C$  has been passed, so the construction stops. The converse is immediate.

Finally, every point outside  $L$  is strictly above  $A$ , while every point of  $L$  is at most some member of  $A$ . Thus  $U(A) = U$ .  $\square$

This theorem describes the precise failure of a set-prefix argument at class length. After a set increasing prefix, the next-letter order has the full set-cut property. After a proper-class prefix, it can be any final class segment: empty, with a least point, with a set coinital sequence, or with no such sequence. A uniform tree isomorphism that relied on the same saturated successor alphabet at every set node cannot simply be continued through every class node.

### 83.4 A direct obstruction to extending the set-tree isomorphism

**Theorem 83.7.** *In GBC, the virtual prefix tree of all auxiliary class well-orders on arbitrary surreal subclasses is not isomorphic to the corresponding tree of inherited numerical class well-orders. In particular, the isomorphism between injective and increasing set-length word trees (theorems 35.1 and 35.7) cannot extend to an isomorphism of these full class-carrier trees.*

*Proof.* An immediate extension in either tree appends exactly one unused label. Global Choice provides an auxiliary class well-order of  $\mathbf{No} \setminus \{z\}$ . As a node of the full auxiliary tree, it has exactly one immediate extension, obtained by appending  $z$ .

In the numerical tree the immediate extensions of  $A$  are indexed by  $U(A)$ . This class is either empty or a proper class: with any  $x \in U(A)$  it contains every larger surreal. Thus no numerical node has exactly one immediate extension. A tree isomorphism would preserve that property, a contradiction.  $\square$

The proof can be read as a two-sorted assertion about any proposed uniform map, or as an ordinary external set-theoretic argument in a specified class model. It does not require a GB class of all nodes.

[merge] Theorem 83.7 limits the word isomorphisms of Part VII in length, not in choice: it does not bear on the question of sources 16 and 18, recorded at the end of section 39, whether they survive without a global well-order.

## 84 Relation-table lexicography: spectra, scans, extensions and density (sources 27 and 30)

[merge] Relation-table lexicography compares the characteristic functions of two relations at the first scanned pair on which they differ. Before batch 83 the report knew three facts about it: at class level its order of all class well-orders is equimorphic with  $\mathbb{B}_{\text{Ord}}$  for every scan (proposition 17.1, source 08); its endpoints are artifacts of the scan, each of the patterns “minimum only”, “maximum only” and “neither” occurring and “both” never (theorem 38.2 and corollary 38.3, source 15); and

it is not enumeration lexicography (section 38.8). The isomorphism question for two scans stayed open (section 39, item (e)). Two batch-83 sources continue this line.

- Source 27's Section 4 (section 84.1) works first on a set carrier. Every scan gives the same ordinal and reverse-ordinal spectrum (theorem 84.2, the set analogue of proposition 17.1), yet two scans of the same countable carrier, of types  $\omega$  and  $\omega + 6$ , give orders that are not isomorphic: one embeds in  $\mathbb{R}$  and has no isolated points, the other has continuum many (theorems 84.3 and 84.4). This answers the set-level form of item (e) negatively. At class level, under a set-like scan of type Ord, the core meets every nonempty interval of the order of all class well-order tables (theorem 84.5): a natural set-coded dense core, as source 09's direction "Non-set-like well-order relations" asks. Its complete ambient order (proposition 84.6) uses the constraint closure of source 15's endpoint-transfer proof.
- Source 30's Section 7 (section 84.2) shows that a set partial table extends to a class well-order of any prescribed type exactly when its forced relation is well-founded (theorem 84.7), characterizes the relations approximable on finite, countable and set-sized supports (theorem 84.8), and deduces that no continuous invariant detects the order type (corollary 84.10). Its Section 8 (section 84.3) proves that the set-like part of every set-like scan order is dense (theorem 84.11, whose second clause is the set-like part of source 27's theorem 84.5) and computes the local characters of one canonical scan, where bounded inversion support has an intrinsic description (theorem 84.13 and corollary 84.14).

Source 27's scan  $q : \theta \rightarrow X \times X$  is printed  $s : \vartheta \rightarrow X \times X$  (this report's  $q$  is cf  $\mu$  and its  $\theta$  a regular cardinal), its table bits  $b_R$  are printed  $t_R$ , its  $\kappa = |X|$  is printed  $\mu$ , its realized prefix cylinders  $C(p)$  are printed  $N(p)$  (they are source 30's neighbourhoods  $N_D(R)$  for an initial segment  $D$  of the scan), its density and cellularity  $d, c$  are printed  $d, \text{cell}$ , and its core  $\mathcal{D}_b$  is printed  $\mathcal{P}_s(b)$  (codes by moved points). In source 30's Sections 7–8, the supplied class bijection  $e : \text{Ord} \rightarrow X$  is a baseline in the sense of section 1.3; its supplied type  $J$  is a class well-order (not  $\mathfrak{J}$ , and not source 27's centered order); its rank function  $\rho$  and order type  $\theta$  are printed  $\varrho$  and  $\beta$ ; its external set  $\mathcal{W}^{\mathfrak{M}}$  of admitted class well-orders is printed  $\mathfrak{W}_{\text{all}}^{\mathfrak{M}}$ ; and its inversion class  $I(W)$  is printed  $\text{Inv}(W)$ , since  $\mathcal{I}$  is the class of injective words.

## 84.1 Source 27: six delayed comparisons change the topology

Relation-table lexicography orders the same well-order relations using different data. Its dependence on the scan is much stronger than a finite relabelling effect. Throughout the set-sized part of this section we work in ZFC and let  $|X| = \mu$  be infinite.

### 84.1.1 The scan and the information it uses

Fix a bijection  $s : \vartheta \rightarrow X \times X$ , where  $\vartheta$  is an ordinal of cardinality  $\mu$ . For a relation  $R$  define

$$t_R(\xi) = \begin{cases} 1, & s(\xi) \in R, \\ 0, & s(\xi) \notin R. \end{cases}$$

Compare these binary words with  $0 < 1$ , at their first disagreement. Write  $\mathcal{W}_s(X)$  for the well-orders with this comparison and  $\mathcal{W}_{s,\alpha}(X)$  for the relations of ordinal type  $\alpha$ . The numerical order on  $X$  does not occur in this definition. It matters only if an additional rule uses it to select  $s$ .

Diagonal entries are always zero. For an unordered pair  $\{x, y\}$ , the two directed entries are complements. Only whichever appears first can decide a comparison. Deleting diagonal entries

and the later directed entry of each pair produces an *effective scan*: a well-ordered list of oriented unordered pairs with the same induced order on relations.

The discrepancy from enumeration lexicography is visible on three labels. Let  $a < b < c$ , and compare the enumerations  $(a, c, b)$  and  $(b, a, c)$ . The first is enumeration-smaller. If the table scan first inspects  $(a, b)$ , its bits are respectively 1 and 0, so the first is table-larger. There is no contradiction: the two definitions ask different questions at their first step.

### 84.1.2 An invariant spectrum shared by every scan

**Lemma 84.1** (Independent adjacent-pair coding). *For every scan  $\mathbf{s}$  and every ordinal  $\alpha$  with  $|\alpha| = \mu$ , the lexicographic binary cube  $\mathbb{B}_\mu = (2^\mu, <_{\text{lex}})$  embeds into  $\mathcal{W}_{\mathbf{s}, \alpha}(X)$ .*

*Proof.* Choose one labelled well-order of type  $\alpha$ . Its initial length- $\mu$  segment contains  $\mu$  disjoint adjacent pairs at positions  $2 \cdot \xi, 2 \cdot \xi + 1$ . Swapping either pair changes only the two directed comparisons internal to that pair and leaves the ordinal type unchanged.

The first effective coordinates of these pairs form a well-ordered set of cardinality  $\mu$ , of order type at least  $\mu$ . Select its first  $\mu$  pairs. For  $s \in 2^\mu$ , orient selected pair  $\xi$  so that its first scheduled bit equals  $s(\xi)$ . Leave all other comparisons fixed. The first disagreement of two inputs is precisely the first disagreement of their output tables, proving the embedding.  $\square$

**Theorem 84.2** (Exact scan-independent ordinal spectrum). *For any scan  $\mathbf{s}$  and any fixed type  $\alpha$  of cardinality  $\mu$ , every linear order of size at most  $\mu$  embeds into  $\mathcal{W}_{\mathbf{s}, \alpha}(X)$ . The ordinal and reverse-ordinal embedding spectra of both this slice and  $\mathcal{W}_{\mathbf{s}}(X)$  are exactly*

$$\{\beta : \beta < \mu^+\}.$$

*Both spaces have cardinality  $2^\mu$ .*

*Proof.* For a linear order  $L$  with  $|L| \leq \mu$ , index some cube coordinates by its elements and encode  $l \in L$  by the bits  $s_l(t) = 1$  exactly when  $t < l$ , fixing unused coordinates at zero. If  $l < l'$ , the code increases coordinatewise and changes at  $t = l$ , so it increases under any lexicographic coordinate order. Compose with lemma 84.1. This proves the positive spectrum and the lower cardinality bound; the number of tables gives the upper bound.

For the negative spectrum it suffices to exclude a strictly increasing  $\mu^+$ -sequence in  $2^\vartheta$ . Recursively through the coordinates of  $\vartheta$ , pass to a final segment of the sequence on which the current bit is constant. Once the earlier bits are constant, the current binary coordinate is nondecreasing and therefore eventually constant. At a limit coordinate take the supremum of the previously chosen starting indices. There are at most  $\mu$  such indices, so regularity of  $\mu^+$  keeps the supremum below  $\mu^+$ . After all coordinates, a final segment consists of the same word, a contradiction. The decreasing case reverses the two bits. Every ordinal of cardinality greater than  $\mu$  contains  $\mu^+$  as an initial segment.  $\square$

This spectrum cannot detect the topological distinction proved next. The argument also explains why agreement on the list of embeddable ordinals is a weak form of classification.

[merge] The spectrum is that of enumeration lexicography (theorem 6.1), and the proofs are the same pair-orientation coding and the same stabilization argument. At class level the corresponding statement is source 08's proposition 17.1, whose lower embedding also orients two-element blocks at their first scanned pair.



### 84.1.3 Scans of the least possible order type

Now suppose  $s : \mu \rightarrow X^2$ , with  $\mu$  the initial ordinal of its cardinality. All topology below is the *order topology* of the indicated linear order. Its density  $d$  is the minimum cardinality of a dense subset; its cellularity  $cell$  is the supremum of cardinalities of disjoint families of nonempty open sets.

A realized prefix cylinder is

$$N(p) = \{R \in \mathcal{W}_s(X) : t_R \upharpoonright \gamma = p\}, \quad \gamma < \mu.$$

Every order-neighborhood of  $R$  contains such a cylinder about  $R$ . For a neighborhood  $(U, V)$ , choose  $\gamma$  beyond the first disagreements of  $U, R$  and  $R, V$ ; agreement through  $\gamma$  settles both comparisons. A one-sided neighborhood needs only one bound. This says that cylinders refine neighborhoods. It does not claim that every cylinder is itself open.

**Theorem 84.3** (Dense slices and exact topology). *For a scan of type  $\mu$ , every slice  $\mathcal{W}_{s,\alpha}(X)$  with  $|\alpha| = \mu$  is dense in  $\mathcal{W}_s(X)$ . Neither the full order nor any such slice has isolated points. Moreover*

$$d(\mathcal{W}_s) = cell(\mathcal{W}_s) = d(\mathcal{W}_{s,\alpha}) = cell(\mathcal{W}_{s,\alpha}) = 2^{<\mu}. \quad (37)$$

*Proof.* A prefix of length  $\gamma < \mu$  involves a set  $C$  of fewer than  $\mu$  labels. If  $R$  witnesses its feasibility,  $R \upharpoonright C$  has ordinal type  $\beta < \mu$ . Put this restriction first, followed by any well-order of  $X \setminus C$  of type  $\alpha$ . Since  $\beta + \alpha = \alpha$ , the result belongs to the desired slice and matches the prefix. The complement admits  $2^\mu$  such orders, so every neighborhood contains many members of that slice. The same argument with neighborhood endpoints in a slice excludes isolated points in its own order topology.

There are  $\sum_{\gamma < \mu} 2^{|\gamma|} = 2^{<\mu}$  possible prefixes. Choosing one representative of each realized prefix gives a dense set in the full order, or in any fixed slice. Thus  $d \leq 2^{<\mu}$  and  $cell \leq d$ .

For the lower bound use the cube embedding in lemma 84.1. For each  $s \in 2^\gamma$ , choose three distinct increasing extensions  $u_s < v_s < w_s$  of length  $\mu$ . There are still at least two coordinates available after  $\gamma$ . The intervals between the images of  $u_s, w_s$  are nonempty, witnessed by the image of  $v_s$ , and are disjoint for different prefixes  $s$ . Hence  $cell \geq 2^{|\gamma|}$  for every  $\gamma < \mu$ . Taking the supremum proves (37).  $\square$

In the countable case the scan can have type  $\omega$ , and there is an explicit order embedding into the real line:

$$R \mapsto \sum_{n < \omega} \frac{2t_R(n)}{3^{n+1}}. \quad (S27.2)$$

The first different digit dominates the entire later tail. This is an order embedding; no identification of every intrinsic suborder topology with a product subspace topology is being assumed.

### 84.1.4 A finite terminal block creates maximal isolation

**Theorem 84.4** (Six delayed coordinates). *On the same infinite carrier  $X$  there is a scan of type  $\mu + 6$  for which  $\mathcal{W}_s(X)$  has  $2^\mu$  isolated points. Its density and cellularity both equal  $2^\mu$ .*

*Proof.* Reserve three labels  $Z \subset X$  and put  $Y = X \setminus Z$ . First scan all pairs except the six off-diagonal directed pairs inside  $Z$ , using type  $\mu$ . Put those six comparisons at the end.

For each well-order  $Q$  of  $Y$ , consider the six completions that place all of  $Y$  in order  $Q$ , then the three labels of  $Z$  in any order. These have the same initial  $\mu$  table bits. Conversely those bits force exactly this form: only the order inside  $Z$  is undetermined. The fiber is therefore a convex six-point chain. Its middle four points are isolated in the full order, because their immediate neighbors in the chain delimit singleton open intervals.



There are  $2^\mu$  choices of  $Q$ , with disjoint fibers. Hence there are  $2^\mu$  isolated points. Their singleton neighborhoods give cellularity at least  $2^\mu$ , and any dense set must contain them. The total cardinality supplies the matching upper bounds.  $\square$

For countable  $X$ , the distinction is absolute: a scan of type  $\omega$  is separable with no isolated points, while the displayed scan of type  $\omega + 6$  has continuum many isolated points and cannot embed into  $\mathbb{R}$ . An embedding into  $\mathbb{R}$  would turn the disjoint intervals around these isolated points into disjoint real intervals, each containing a different rational, which is impossible.

Only the placement of six relation entries has changed. For larger  $\mu$  it is possible that  $2^{<\mu} = 2^\mu$ , so density alone may not distinguish the two scans. Absence versus presence of isolated points always does. A scan's order type must therefore be specified; replacing it by an equicardinal ordinal is not innocuous.

[merge] Together, [theorems 84.3 and 84.4](#) answer the set-level form of [section 39](#), item (e), and of source 08's question 9 ("Classification of relation-code orders", [section 113.4](#)), negatively: on every infinite set carrier two scans give non-isomorphic orders, although by [theorem 84.2](#) their embedding spectra agree. Isolated points are an artifact of the scan, in the sense of source 08's question; whether two set-like scans of type Ord can give non-isomorphic orders on **No** is not decided by source 27 (but see [theorem 84.13](#)).

### 84.1.5 A reversal of the global density picture

**Theorem 84.5** (Set-like orders are table-dense). *In GBC, fix a set-like scan  $s : \text{Ord} \rightarrow \mathbf{No}^2$ . The ordinary class of bounded-support codes  $\mathcal{P}_s(b)$ , now compared by its relation tables, meets every nonempty open interval of raw class well-order tables. In particular, set-like class well-orders are dense there. The analogous assertion holds for one-sided neighborhoods. Neither the raw nor the set-like table comparison has isolated points.*

*Proof.* Given  $U < R < V$ , choose a set ordinal  $\delta$  beyond the two first disagreements. The pairs inspected below  $\delta$  involve a set  $C$  of labels. The restriction  $R \upharpoonright C$  is a set well-order, even when  $R$  has proper-class predecessor segments. Enumerate  $C$  in that order as a set prefix, then complete the prefix by a permutation of  $b$  with set support, using [theorem 101.1](#). Its relation table matches  $R$  through the required coordinates, hence lies between  $U, V$ . Appending differently ordered fresh labels before completion gives multiple core elements. One-sided neighborhoods are identical with one bound omitted.  $\square$

In the enumeration comparison, the set-like orders miss all interiors of proper-tail fibers. In the relation-table comparison under an Ord scan, they meet every nonempty interval. This is a direct structural disagreement between the two interpretations. Indeed the relation-table order has a dense uniform ordinary class of codes, whereas [theorem 107.8](#) excludes every such dense family for raw enumeration comparison. The objects agree; their comparison relations differ. No class containing all set-like orders is being constructed in either argument.

Placing a few comparisons "after all ordinal stages" instead would give them proper-class predecessor collections. An  $\text{Ord} + 6$  scan is not a set-like Ord scan. A finite-looking change at the end can therefore change the foundational hypothesis as well as the topology.

[merge] The convex fibres missed in enumeration comparison are those of [proposition 15.3](#). [Theorem 84.5](#) answers source 09's question "Does it contain a natural set-coded dense core?" (its direction "Non-set-like well-order relations", [section 113.3](#)) for every set-like scan: the core  $\mathcal{P}_s(b)$  is one. The rest of that direction (when cuts of the relation-code order are represented by class

well-orders) remains open. Source 30's [theorem 84.11](#) proves the set-like clause again (its second sentence), adds that the set-like part is dense in itself, and needs no completion theorem.

### 84.1.6 Product topology and a complete ambient order

For a set  $X$ , the familiar space of all linear orders  $\mathcal{L}(X) \subseteq 2^{X^2}$  is closed in the product topology: failures of the linear-order axioms are witnessed by finitely many entries. This is the classical relation-table space used, for example, in [62, Section 4]. Every infinite fixed well-order type of cardinality  $|X|$  is product-dense there, since finite constraints can be put first and the remaining labels well-ordered afterward.

Order-density for a long scan is different. Inspect  $(x_{n+1}, x_n)$  first for  $n < \omega$ . The all-one initial prefix forces an infinite descending chain and cannot extend to a well-order, although it extends to many linear orders. Three such extensions, differing on reserved labels, delimit a nonempty table-order interval containing no well-order. Thus finite-coordinate density cannot be substituted for lexicographic density.

There is nevertheless a useful completeness fact for the *set* space of all linear-order tables.

**Proposition 84.6** (Complete ambient table order). *For every set-length scan  $s$ , the linear order  $\mathcal{L}_s(X)$  has endpoints and every subset has a supremum and an infimum.*

*Proof.* Translate each prescribed off-diagonal table bit into a directed constraint. A feasible partial table is exactly a constraint set with no finite directed cycle. It extends to a linear order by orienting unassigned pairs successively: at least one orientation avoids a cycle, since two forbidden orientations would give an old cycle. At limits a new finite cycle cannot appear. Greedily preferring the smaller, or larger, available bit gives the least, or greatest, completion of any feasible prefix, including the empty prefix.

Let  $\mathcal{A}$  be a nonempty subset of  $\mathcal{L}_s(X)$ . Follow the smallest bit realized by some member matching the current prefix. If a full member survives, it is the minimum. Otherwise there is a first limit stage  $\lambda$  at which the accumulated prefix  $p$  has no matching member. Every finite part of  $p$  is matched, so it is acyclic and has completions. Let  $G$  be its greatest completion. Every member of  $\mathcal{A}$  first differs from  $p$  before  $\lambda$ , with bit 1 against 0, so  $G$  is a lower bound. A larger lower bound would first differ upward from  $p$  at some earlier coordinate; a member of  $\mathcal{A}$  still matches  $p$  through that coordinate and lies below the proposed lower bound. Thus  $G = \inf \mathcal{A}$ . Reverse the construction for suprema; empty subsets use the endpoints.  $\square$

This does not identify  $\mathcal{L}_s(X)$  as the Dedekind completion of  $\mathcal{W}_s(X)$ : density can fail, and adjacent points require care even when a dense inclusion is available. Nor is the proposition being extended to a class-length scan without a separate recursion analysis.

[merge] The extension of a feasible partial table is the constraint closure of the proof of [theorem 38.2](#); for well-orders, source 30's [theorem 84.7](#) gives the exact criterion (well-foundedness of the forced relation, not merely acyclicity).

## 84.2 Source 30: extension, local limits, and invisible order types

[merge] Source 30's Section 7, printed whole; its assembly comments (lines 2056–2059 of its L<sup>A</sup>T<sub>E</sub>X source, a “research fragment” header) are not printed. No other source proves its results; source 27's [proposition 84.6](#) contains the acyclic set-level case of the extension step.

Relation-table lexicography admits a useful local theory that does not require aligning the enumeration domains of non-set-like orders. The main result is stronger than density of the set-like orders: every supplied global well-order type realizes exactly the same set-sized partial tables.

Throughout this section work in GBC, and supply a proper class  $X$  and a class bijection  $e : \text{Ord} \rightarrow X$ . The application is  $X = \mathbf{No}$ . A *class well-order* is a strict linear order under which every nonempty admitted subclass has a least element; it need not be set-like. Statements about several orders or an indexed family mean statements about supplied class parameters or one uniformly represented class relation. They do not collect class relations into another class.

### 84.2.1 An exact extension theorem for partial tables

Let  $D \subseteq X^2$  be a set and let  $p : D \rightarrow \{0, 1\}$  prescribe relation bits. Write  $A$  for the set of labels occurring in  $D$ . A positive off-diagonal bit requires its indicated orientation; a negative one requires the opposite orientation, because the extension must be total. Thus define a set relation  $P_p$  on  $A$  by

$$uP_pv \iff u \neq v \text{ and } (((u, v) \in D \ \& \ p(u, v) = 1) \text{ or } ((v, u) \in D \ \& \ p(v, u) = 0)).$$

Well-foundedness of  $P_p$  means that every nonempty subset of  $A$  has a  $P_p$ -minimal element. Transitivity of  $P_p$  is not required.

**Theorem 84.7** (Extension with an arbitrarily prescribed global type). *Fix any supplied class well-order  $J$  of  $X$ . The partial table  $p$  extends to a class well-order  $W$  isomorphic to  $J$  if and only if*

1. *every prescribed diagonal bit is 0;*
2. *the forced relation  $P_p$  is well-founded.*

*In particular, the conditions for extension to a set-like well-order are the same as the conditions for extension to any other supplied global well-order type.*

*Proof.* Necessity follows because  $P_p \subseteq W$ : the  $W$ -least element of any nonempty subset of  $A$  is  $P_p$ -minimal.

For sufficiency, rank the set-sized well-founded relation:

$$\varrho(v) = \sup\{\varrho(u) + 1 : uP_pv\}.$$

Order  $A$  lexicographically by the pairs  $(\varrho(v), e^{-1}(v))$ . This gives a set well-order  $\sigma$  extending  $P_p$ . Let  $\beta$  be its order type.

The first  $\beta$  elements of  $J$  form a set initial segment  $L$  of type  $\beta$ . To see this without invoking class-length recursion, recursively choose the  $J$ -least label outside the previously chosen set, for  $\xi < \beta$ . Such a label exists because  $X$  is proper and  $J$  is a class well-order. This is ordinary set-valued recursion of set length with a class parameter; class replacement makes its range a set. No instance of elementary transfinite recursion along a proper class is being used.

Let  $h : (A, \sigma) \rightarrow (L, J)$  be the increasing bijection. Both  $X \setminus A$  and  $X \setminus L$ , in their inherited  $e$ -orders, have type  $\text{Ord}$ . Their increasing enumerations extend  $h$  to a class permutation  $\pi$  of  $X$ . Pull  $J$  back along  $\pi$ :

$$xWy \iff \pi(x)J\pi(y).$$

Then  $W \cong J$  and  $W \upharpoonright A = \sigma$ , so every bit of  $p$  is realized. Taking  $J$  to be the baseline  $e$ -order gives a set-like extension.  $\square$

For a finite table, condition (2) is just absence of a directed cycle. For an infinite table, acyclicity is insufficient: the requirements  $x_{n+1} < x_n$  for  $n < \omega$  are acyclic but cannot extend to a well-order. The theorem therefore gives the exact additional condition needed at infinite observation scales.

When no particular global type is prescribed, a simpler extension suffices: place  $(A, \sigma)$  first and then append the  $e$ -ordered complement. The transport argument is needed to preserve a specified possibly non-set-like type.

### 84.2.2 Finite and countable observations have different closures

For a supplied class relation  $R$  and a set  $D \subseteq X^2$ , write

$$N_D(R) = \{W : W \text{ agrees with } R \text{ on } D\}.$$

Internally this is a *neighborhood notation*: membership is a formula about supplied class relations. It does not assert the existence of a set or class of all its members.

**Theorem 84.8** (Exact local-approximation criteria). *Fix a supplied class well-order  $J$  of  $X$ , and let  $R$  be an arbitrary supplied class relation on  $X$ .*

1. *Every finite partial table of  $R$  has an extension isomorphic to  $J$  if and only if  $R$  is a strict linear order.*
2. *Every countable partial table of  $R$  has such an extension if and only if  $R$  is a class well-order.*
3. *The criterion in (2) is unchanged if “countable” is replaced by “set-sized”, or by “a bounded prefix of a fixed set-like scan of  $X^2$ ”.*

*Proof.* Irreflexivity, totality and transitivity are all tested on finitely many coordinates. Thus the finite extension condition implies that  $R$  is a strict linear order. Conversely, its restriction to the finitely many labels of a finite table is a finite well-order, and [theorem 84.7](#) applies.

If  $R$  is a class well-order, its restriction to the set of labels of any set-sized table is a set well-order. The same theorem again supplies an extension of the prescribed type.

For the remaining implication, countable extension first gives linearity. Suppose a nonempty admitted subclass  $C$  has no  $R$ -least element. Choose  $c_0$  by the baseline enumeration, and recursively let  $c_{n+1}$  be the  $e$ -least member of

$$\{x \in C : xRc_n\}.$$

Set-valued iteration and class replacement produce the set sequence  $(c_n)_{n < \omega}$ . Its consecutive positive bits form a countable partial table that no well-order can realize. This contradiction establishes (2). The argument uses the supplied global enumeration to make the choices; it does not silently appeal to a stronger class dependent-choice principle.

Finally, any set of scanned coordinates has a set of ordinal positions and is therefore bounded in the scan. Every bounded prefix is itself a set. This proves the last equivalence.  $\square$

Consequently the finite-support closure of *each* admitted global isomorphism type consists of all strict linear orders, whereas its countable-support and set-support closures consist exactly of all class well-orders. These are equalities of closure criteria; they do not identify the three neighborhood topologies.

The numerical surreal order illustrates the finite/countable distinction. Every finite restriction is a well-order, but the countable chain  $0 > -1 > -2 > \dots$  prevents approximation on every countable support. Conversely, set-likeness has no such local obstruction: every set-sized table belonging to a non-set-like well-order also belongs to a set-like one.

More explicitly, put  $A_\alpha = e^{\alpha}$  and, for a class well-order  $R$ , define

$$R_\alpha = (R \upharpoonright A_\alpha) + (e \upharpoonright (X \setminus A_\alpha)).$$

This is a uniform class family of set-like well-orders. It eventually agrees with  $R$  on every set of coordinates. For example, let  $R$  put one distinguished label last and  $e$ -order all remaining labels. Then  $R$  has type  $\text{Ord} + 1$ , although every  $R_\alpha$  has type  $\text{Ord}$  and has no last element.

### 84.2.3 An ordinary class tree of set approximations

There is a useful formalization of this local theory that keeps all nodes set-sized. Let the level  $\alpha$  of a class tree consist of the set well-orders of  $A_\alpha$ , ordered by restriction between levels. Every level is a set, and the entire node collection is an ordinary class of set-coded pairs  $(\alpha, \sigma)$ .

**Proposition 84.9** (Branches and stabilization). *Supplied class well-orders of  $X$  correspond exactly to supplied coherent class branches meeting every level of this tree. The well-order obtained from a branch is set-like if and only if, for each label  $x$ , its predecessor sets in the successive levels eventually stabilize.*

*Proof.* Restriction sends a class well-order to a coherent branch. Conversely, coherence defines a strict linear order  $R$  on  $X$ . Every set of labels lies in some  $A_\alpha$ , so every set restriction is well-ordered. The countable descending-chain argument from [theorem 84.8](#) makes  $R$  a class well-order.

For  $x = e(\xi)$  and  $\alpha > \xi$ , let  $P_\alpha(x)$  be its predecessor set at level  $\alpha$ . These are nested sets whose class union is the full predecessor class. If that class is a set, its  $e$ -indices are bounded, so  $P_\alpha(x)$  eventually becomes constant. Conversely, a constant final value is the entire predecessor class and is a set.  $\square$

The limit-level requirement matters. Coherence of finite well-orders alone can produce an ill-founded limit; requiring a well-order at every set stage excludes the countable descending witness. No collection of all branches is introduced by this description.

### 84.2.4 An external topology and the absence of continuous type detectors

To turn the neighborhood notation into an ordinary topological space, choose an external set-sized two-sorted realization  $\mathfrak{M} = (M, \mathcal{C})$  of the stated theory. Let  $\mathfrak{W}_{\text{all}}^{\mathfrak{M}}$  be the external set of its admitted class well-orders. Give it the topology generated by  $N_D(R)$  with supports  $D$  that *the model regards as sets*. These cylinders are clopen and separate distinct admitted relations, so the space is Hausdorff. Equivalently, one may use the internally bounded scan prefixes.

For an admitted  $J$ , let  $[J]_{\mathfrak{M}}$  denote the orders isomorphic to  $J$  by an admitted class isomorphism. This explicitly fixes the meaning of isomorphism when the external realization is not transitive.

**Corollary 84.10** (Dense type fibers and continuous invariants). *Every  $[J]_{\mathfrak{M}}$  is dense and codense in  $\mathfrak{W}_{\text{all}}^{\mathfrak{M}}$ . The set-like orders and the non-set-like orders are both dense. Every continuous isomorphism-invariant map from  $\mathfrak{W}_{\text{all}}^{\mathfrak{M}}$  to a  $T_1$  space is constant.*

*Proof.* The extension theorem, interpreted in  $\mathfrak{M}$ , puts a member of  $[J]_{\mathfrak{M}}$  in every nonempty basic neighborhood. There are at least two distinct types, Ord and Ord + 1. Thus every type fiber has a different dense fiber in its complement. Type Ord provides the set-like approximants, while type Ord + 1 provides non-set-like approximants.

If  $f$  is invariant and continuous, then it is constant, with value  $f(J)$ , on the dense fiber  $[J]_{\mathfrak{M}}$ . Since the target is  $T_1$ , the preimage of  $\{f(J)\}$  is closed. A closed set containing a dense fiber is the whole domain, so  $f$  is constant.  $\square$

In particular, the indicator of set-likeness, with values in the discrete two-point space, is discontinuous at every point. The same holds for membership in any one type fiber. The full table does determine its order type; the obstruction concerns recovering that type continuously from set-sized observations.

The phrase “model regards as sets” is essential. Externally the whole carrier is a set. If all external coordinate subsets were allowed, observing the entire table would be a permitted basic

neighborhood and the topology would be discrete. The preceding corollary makes no assertion about that different topology.

### 84.3 Source 30: the order topology of relation-table lexicography

Fix a set-like scan of  $X^2$ , and compare table bits with  $0 < 1$ . The first disagreement exists for any two supplied distinct tables. Support neighborhoods and order neighborhoods are related, but they need not generate the same topology.

#### 84.3.1 Density for every scan on the set-like part

We first recall the endpoint-transfer argument, which already appears in the repository ([theorem 38.2](#), source 15). Suppose a well-order  $W$  is claimed to be an extremum but a strict linear order  $L$  improves its table at the first disagreement. Let  $P$  be the set of oriented constraints from the earlier coordinates. If the desired improvement puts  $y$  before  $x$ , take the  $P$ -predecessor closure  $D$  of  $y$ , including  $y$ . It is a set and does not contain  $x$ , since  $L$  extends  $P$ . Moving  $D$  to the front of  $W$ , preserving the old orders inside  $D$  and its complement, respects every earlier constraint and makes the desired improvement. It remains a well-order and remains set-like if  $W$  was set-like. Hence minima and maxima among well-orders, or among set-like well-orders, are also extrema among all strict linear orders.

**Theorem 84.11** (Scan-independent density of the set-like part). *For every set-like scan, the relation-table order of the set-like global well-orders is dense. Moreover, set-like well-orders are topologically dense in the order topology of all admitted class well-orders.*

*Proof.* Let  $R < S$  be set-like and let  $\delta$  be their first disagreement. Start with the set of labels in the scan through  $\delta$ . Close it under predecessors in both  $R$  and  $S$ , iterating this operation  $\omega$  times. Class replacement and set-likeness ensure that the resulting  $A$  is a set, and it is an initial segment of both orders. Put  $B = X \setminus A$ .

If no set-like order lay between  $R$  and  $S$ , replacing their  $B$ -tails independently would imply that  $R \restriction B$  is greatest and  $S \restriction B$  is least among set-like orders of  $B$ , under the induced scan. The endpoint-transfer argument makes them extrema among all linear orders of  $B$ . Reversal complements every nonconstant table bit, so these extrema are reversals of one another. An order on an infinite carrier and its reverse cannot both be well-orders. This is a contradiction.

For the final assertion, a point  $T$  of any nonempty open order interval has a bounded table prefix fixing its comparisons with the interval endpoints. A set-like extension of that prefix still lies in the interval. The same argument treats open rays.  $\square$

The last assertion is topological density, not a claim that every pair of distinct non-set-like orders has an intermediate order for an arbitrary scan. That stronger question requires another argument. The next explicit scan does have the stronger property.

*Note 84.12* ([merge]; one density theorem, two proofs). The second sentence of [theorem 84.11](#) is the set-like clause of source 27's [theorem 84.5](#), which is stronger: the core  $\mathcal{P}_s(b)$  already meets every nonempty open interval, and neither order has isolated points. Source 27 completes a set prefix by a bounded-support permutation ([theorem 101.1](#)); source 30 extends the prefix by [theorem 84.7](#). The first sentence of [theorem 84.11](#), that two set-like orders always have a set-like order strictly between them (no adjacent pairs), is source 30's only; its proof is the reversal argument of [corollary 38.3](#), applied to a tail. Source 30 did not see source 27.



### 84.3.2 A canonical baseline scan with asymmetric local character

Identify labels with their  $e$ -indices. Scan the unordered pairs  $(\xi, \eta)$ ,  $\xi < \eta$ , in increasing order of  $\eta$  and then  $\xi$ , and use bit 1 when  $e(\xi)$  precedes  $e(\eta)$ . This is a set-like class scan. It can be converted to a full directed-pair scan by listing the reverse direction immediately after each pair and inserting the constant diagonal bits.

For a set well-order  $\sigma$  on  $A_\alpha$ , let  $C_\sigma$  be the orders whose restriction to  $A_\alpha$  is  $\sigma$ .

**Theorem 84.13** (Canonical table characters). *For this scan, both the set-like order and the order of all class well-orders have the following properties.*

1. *They are dense, have the baseline order as greatest element, and have no least element.*
2. *Every  $C_\sigma$  has a greatest element and a strictly decreasing countable coinital sequence of set-like elements.*
3. *No point has a uniformly represented set-sized cofinal family strictly below it.*
4. *A nongreatest point  $W$  has a uniformly represented set-sized coinital family strictly above it if and only if its inversion class*

$$\text{Inv}(W) = \{(\xi, \eta) : \xi < \eta \text{ and } e(\eta)We(\xi)\}$$

*is a set.*

*Proof.* The greatest element of  $C_\sigma$  is  $M_\sigma$ , which puts  $\sigma$  first and then appends the baseline tail: every still undetermined bit is 1.

For  $n \geq 1$ , let  $W_n$  put

$$e(\alpha + n - 1), \dots, e(\alpha + 1), e(\alpha)$$

first, then  $\sigma$ , then the remaining baseline tail. These set-like orders form a strictly decreasing sequence in  $C_\sigma$ . They are coinital there. Indeed, their successive new labels are always inserted before every previously scanned label, the least possible insertion. An order that were below or equal to every  $W_n$  would have to make all these least insertions and would contain the descending sequence  $e(\alpha) > e(\alpha + 1) > \dots$ . A class well-order cannot do so. Thus every member of  $C_\sigma$  has some  $W_n$  strictly below it.

Given  $R < S$ , choose  $\alpha$  large enough that agreement with  $S$  on  $A_\alpha$  fixes the comparison  $R < S$ . A member of the preceding sequence below  $S$  lies strictly between  $R$  and  $S$ . This proves density and absence of a minimum. The baseline has all bits 1 and is the maximum.

For a supplied set family  $V_i < W$ , bound all first disagreements with  $W$  using class replacement. In a sufficiently long matching cylinder, choose a set-like element strictly below  $W$ . It still lies above every  $V_i$ , proving (3).

If  $\text{Inv}(W)$  is proper, every matching cylinder  $C_{W|A_\alpha}$  has maximum strictly greater than  $W$ . Bounding the first disagreements of any supplied set family above  $W$  and taking such a later maximum gives a still closer point above  $W$ . Thus no such family is coinital.

If  $\text{Inv}(W)$  is a set, choose  $\gamma$  containing the endpoints of all inversions. Every  $U > W$  first differs on a pair lying in  $A_\gamma^2$ . Therefore  $\tau = U|A_\gamma$  is table-greater than  $\sigma = W|A_\gamma$ . There are set-many such set orders  $\tau$ . Each cylinder  $C_\tau$  lies wholly above  $W$  and has the displayed countable coinital sequence. Their union is a uniformly represented set-sized family coinital strictly above  $W$ , proving (4).  $\square$

**Corollary 84.14** (The eventual-equality core in the canonical table order). *A class well-order has set-sized inversion support for this scan if and only if it is set-like and its ordinal enumeration is eventually equal to  $e$ .*

*Proof.* If all inversion endpoints lie in  $A_\gamma$ , the order puts some set well-order  $\sigma$  of  $A_\gamma$  first and then the unchanged baseline tail. Write  $\beta = \text{otp}(\sigma)$ . Choose an additively indecomposable ordinal  $\lambda > \gamma, \beta$ . The identities  $\gamma + \lambda = \beta + \lambda = \lambda$  align the two enumerations beyond  $\lambda$ , proving eventual equality. Conversely, if an enumeration agrees with  $e$  from  $\lambda$  onward, its first  $\lambda$  entries are exactly  $e[\lambda]$ . All inversions are consequently contained in the set  $\lambda \times \lambda$ .  $\square$

Bounded inversion support thus has an intrinsic description in this particular table order: away from its greatest point, it is exactly the existence of a set-sized right coinital family. The left side has no analogous small character. These properties are different from numerical enumeration lexicography and should not be transferred between the two representations.

The support topology is strictly finer than the order topology for this scan. Order neighborhoods contain sufficiently long matching table cylinders. In the other direction, swap the first two baseline labels and keep the remaining baseline tail. The result is the nongreatest maximum of its two-label cylinder. Every order neighborhood of it contains larger points by density, and those points leave the cylinder. That cylinder is consequently not order-open, although it is clopen in the support topology.

[merge] The canonical scan realizes source 15’s pattern “maximum only” (corollary 38.3) with the baseline as maximum. With  $e$  as the baseline  $b$ , the orders of corollary 84.14 are exactly the enumerations  $b \circ \pi$  coded by the core  $\mathcal{P}_s(b)$  of theorem 84.5: a permutation of Ord with set support fixes every ordinal beyond its support, and conversely, as the proof shows, an enumeration eventually equal to  $b$  permutes an initial segment of  $b$ ’s positions. So the core has an intrinsic description in this table order: apart from the maximum, its members are exactly the points with a set-sized coinital family strictly above them. Theorem 84.13 is the first intrinsic, scan-dependent local invariant of a relation-table order on **No**: item (4) separates points by their right character. It does not decide whether two set-like scans of type Ord can give isomorphic orders.

## 85 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of section 39; where that section or the question index of section 113 says otherwise, these notes supersede it. Open items that remain open are re-scoped, not closed.

- (a) *The isomorphism type of the centered increasing order* (section 39, item (b); source 18’s own question): **answered in part**.  $\mathcal{T}_c^{\text{inc}} \cong \mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$  in GBC, where  $\mathbf{T}_{\min}$  is the endpoint type of theorem 36.2 (source 27’s corollary 82.5; source 29’s note 82.11); the zero conventions  $\dagger_+$  and zero padding give the same type and  $\dagger_-$  differs from it exactly by its adjacent pairs (proposition 82.6 and note 82.13). Open: an identification of  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$  with a simpler order, for instance with  $\mathbf{T}_{\min}^+$  (source 27’s Question 12.4), and an intrinsic description of its factors (source 29’s Question 13.4).
- (b) *Increasing words with an interior terminator at a threshold other than zero* (section 39, items (b) and (d); source 16’s question “Centered increasing words”): **answered in part**. No placement of a marker gives a set-cut saturated order, so none gives **No** (theorem 82.1, source 23). Every actual threshold, under each zero convention, and every gap whose sides have no set cofinal (resp. coinital) subset and no endpoint, gives the centered type of (a) (proposition 82.14, source 29). At every interior gap the point characters and the adjacent pairs are known (proposition 82.8, source 27); they show that a gap with a small cofinality or coinitality on one side gives a type different from  $\mathcal{T}_c^{\text{inc}}$ . Open: the isomorphism types at those

gaps and the cut classification at a general threshold (source 27’s Question 12.5, source 29’s Question 13.5; source 23’s Question 16.2 asks which restrictions repair saturation).

- (c) *The isomorphism question for two scans* (section 39, item (e)) and *source 08’s question 9* (“Classification of relation-code orders”, section 113.4): **answered negatively on set carriers, open on No**. On every infinite set carrier there are two scans whose relation-table orders of well-orders are not isomorphic (theorems 84.3 and 84.4), while the ordinal and reverse-ordinal spectra and the cardinality do not depend on the scan (theorem 84.2): isolated points, density and cellularity are artifacts of the scan, the spectrum is not. For set-like scans of  $\mathbf{No}^2$  of type Ord no pair of non-isomorphic orders is known; the canonical scan of theorem 84.13 supplies a scan-dependent local invariant (right characters separate the eventual-equality core), and corollary 84.10 shows that no continuous invariant of set-sized observations detects the order type. The open forms are source 27’s Questions 12.1–12.3, source 29’s Question 13.11 and source 30’s Questions 12.6–12.7.
- (d) *Source 09’s direction “Non-set-like well-order relations”* (section 113.3; its status after the merge, which says that the dense core and the representation of cuts remain open): **answered in part**. For every set-like scan the core  $\mathcal{P}_s(b)$ , compared by relation tables, meets every nonempty open interval of the order of all class well-orders (theorem 84.5, source 27), and the set-like part is dense in itself (theorem 84.11, source 30). Which order properties depend on the coordinate enumeration is answered for endpoints (corollary 38.3), for isolated points on set carriers (item (c)) and for local characters at one canonical scan (theorem 84.13). Open: which cuts of a relation-code order are represented by class well-orders.
- (e) *Source 13’s question “A classification of termination cuts”* (section 39, item (c)): unchanged; source 23’s notes 81.9 and 81.12 are a second proof for an abstract saturated alphabet (section 81). *Source 23’s Question 16.1*, whether all marker orders with characters  $(\theta, \text{Ord})$  are isomorphic and whether the answer changes between arbitrary and injective words, is **already answered** at this report’s text: all are  $\mathbf{T}_\theta$ , and arbitrary and injective words are isomorphic at every placement (theorem 36.2, source 16, transported along an isomorphism of the alphabet with **No** in GBC). Source 23 had read this report before Part VII was written.

Numerically well-ordered proper-class carriers (section 83) were not the subject of a recorded question; source 29’s Question 13.6 asks for the type of the dense quotient of corollary 83.4.

## Part XIII

# The core and its cylinders: symmetries, decorated completions and class realizations

[merge] (3 October 2026.) This Part of the batch-83 addition continues Part IV (part IV), which built the core, and Part VIII (part VIII), which classified its class cuts, recognized the cuts of global enumerations, and showed that the bare order of the core forgets which of its cuts are represented. Part VIII left four things open (section 50, items (c), (e) and (f)): the stabilizer of the exhaustive cuts and which extra predicates make the reconstruction unique; what the external order type of the admitted enumerations remembers about the class realization; and ordinal support bounds. Eight batch-83 sources take these up, and this Part prints what they prove.

- *The core and its cylinders* (base: source 25). Source 22 shows that unranked cylinder inclusion recovers prefix heights (theorem 87.1) and recognizes represented cuts without ranks

(theorem 87.4). Source 25 classifies the orbits of every set-indexed configuration of injective branches under automorphisms preserving the core order and every cylinder (theorem 90.1): there are exactly two single-branch orbits, the core and the rest. Hence exhaustiveness is invisible to the core with all its cylinders, even with set-many parameters fixed (theorems 91.1 and 91.2), and a cylinder-preserving core automorphism need not extend to  $\mathfrak{W}_{\text{sl}}$  (theorem 92.2), which strengthens theorem 47.7. One cross-node relation, equality of edge labels, makes the structure rigid (theorem 93.3). At a regular  $\kappa$  with  $\kappa^{<\kappa} = \kappa$  the set version has exactly  $2^\kappa$  automorphisms (theorem 94.2); a family of cf  $\kappa$  approximants can force a noncore branch, so the bound in the set version is sharp at regular  $\kappa$  and the method fails at singular  $\kappa$  (theorem 95.1).

- *Ordered prefix trees without the core* (sources 27 and 29): branch transitivity on injective and on increasing words, the exact local obstruction, and families that preserve exhaustion (section 96).
- *Automorphisms of the bare core* (sources 20 and 21): re-proofs of Part VIII's orbit classification and extension criterion, a failure localized inside any interval while fixing any set of core points (theorem 97.7), disjointness of relabelled cores (proposition 97.9) and an explicit cut with no crossing first-level cylinder (proposition 97.20).
- *Decorated completions* (source 19): the Dedekind completion of the external raw order is the core completion with every represented fibre replaced by its interval completion (theorem 98.2); a core automorphism extends to the raw order exactly when it preserves fibre types (theorem 98.5); over residual alphabets at most two initial words share a trace (proposition 99.2); and every countable model omits continuum many amenable branches (theorem 100.1).
- *The least ordinal support* of a completion of a partial injection (source 27, theorem 101.2).
- *Class realizations* (sources 22, 27, 21 and 30): the admitted labelled traces recover the whole class realization (theorem 102.3, proved three times); the external order of a countable model is  $\mathbb{Q}$  even with the core coloured and the baseline fixed, and does not detect ETR (propositions 102.19 and 102.20).

Section 105 records which questions the Part answers.

*Printed once.* Height recovery is printed as source 22's theorem 87.1 (source 25, which credits it, is a note); the label-coherence theorem as source 25's theorem 93.1 (source 27's Theorem 6.5 is a note); the faithful reconstruction as source 22's theorem 102.3 (sources 27 and 30 are notes with their proofs as routes). The extension criterion and the orbit classification are Part VIII's theorems 47.2 and 47.5; their re-proofs in sources 20, 21 and 25 are notes with their proofs, as are the sources' re-proofs of Part IV (section 86). Sources 19–22 were written against pins (d5bd4a67b, 109aaca15) that predate Parts VI–VIII; sources 25, 27, 29 and 30 against 7ff7736ec, which contains Parts I–VIII. Source 25 read source 22 and source 23; source 29 read sources 19, 20 and 21; sources 27 and 30 cite no batch-83 source.

*Notation* (renamings for this Part; the batch-83 notation addendum, section 1.3.2, lists them with the other Parts'). The core is printed  $\mathcal{P}_s(b)$  in the support coding of sources 22, 25, 27 and 30 (their  $\mathcal{D}_b$ ,  $\mathcal{D}$ ,  $\mathcal{D}_b$  and  $\mathcal{B}_b$ ) and  $\mathcal{P}_{\text{bd}}(b)$  in the normalized-permutation coding of sources 19, 20, 21 and 29 (their  $\mathcal{C}$ ,  $\mathcal{C}_b$ ,  $\mathcal{S}_b = \mathcal{P}_b$  and  $\mathcal{P}_e$ ); the two codings have the same evaluations (section 1.3). *False reading:* source 25's  $\mathcal{C}_b$  is not the core but the two-sorted core-and-cylinder structure, printed  $\mathbf{C}_b$ ; sources 24 and 26 write  $\mathcal{C}_b$  for the core itself. The class of injective set-length words,  $\mathcal{T}$  in sources 22, 25 and 27 and  $\mathcal{J}$  in source 29, is printed  $\mathcal{I}$ , and source 29's increasing words  $\mathcal{J}$  are printed  $\mathcal{T}^{\text{inc}}$ ;  $\mathcal{T} = \mathcal{T}$  is all words here. Cylinders are  $\mathcal{C}_p$  (sources 20 and 22 write  $\mathcal{C}(p)$ , source 21  $\mathcal{S}_p$ ). Sources 22 and 25 write  $\mathfrak{E}$ ,  $\mathfrak{J}$  for  $\mathfrak{W}_{\text{sl}}$ ,  $\mathfrak{J}$ ; source 19's script  $\mathcal{H}_{\text{all}}$  is printed  $\mathfrak{W}_{\text{all}}$ . Source 25's

direction sign  $\varepsilon(f, g)$  is printed  $\text{sgn}(f, g)$ , since  $\varepsilon$  is a termination symbol in Part VII; its set-sized version at a regular  $\kappa$  carries a superscript:  $\mathcal{P}_s^\kappa(b)$ ,  $\mathcal{I}^\kappa$ ,  $\mathfrak{J}^\kappa = \text{Inj}(\kappa, X)$ ,  $\mathbf{C}_b^\kappa$ , and its  $\mathfrak{E}_\kappa = \text{Bij}(\kappa, X)$  is  $\mathcal{W}_\kappa(X)$ . Source 20 writes  $\Omega$  for “no set-sized witness”, which Part VIII writes  $\text{Ord}$ ; it is kept. Class realizations over one set universe are  $\mathcal{X} \subseteq \mathcal{Y}$  (sources 22, 27, 21; source 30’s  $\mathcal{C}$ ,  $\mathcal{D}$  are renamed); their external collections of admitted set-like enumerations are written  $H_{\mathcal{X}}$  (sources 21, 27) and  $\mathcal{E}(\mathcal{X})$  (source 30). Displayed equations of source *NN* keep their numbers as (*SNN.n*). Labels: *swo:rs*: (19), *swo:lm*: (20), *swo:fu*: (21, keeping its internal *gl*: and *fnd*:), *swo:rg*: (22), *swo:ph*: (25), *swo:sc*: (27), *swo:bw*: (29, keeping *ta*:, *gl*:), *swo:rt*: (30, keeping *fnd*:); labels written for this Part begin *swo:xiii*:.

## 86 The core as re-proved in sources 25, 22, 21 and 29

[merge] Four sources re-prove the layer of Parts III and IV on which their new results rest, and each credits it to this report. Their sections are printed here whole, as re-proofs; none is new against the repository.

- *Fixed-set spectrum and class universality*: [theorem 6.1](#) and [proposition 5.13](#) (source 25’s [notes 86.1](#) and [86.2](#)).
- *Fresh separators*: [lemma 5.2](#) and source 13’s [lemma 42.1](#) ([notes 86.3](#) and [86.8](#)).
- *Prefix completion with a reserve*: [lemma 19.4](#) with [remark 19.7](#); [theorem 49.7](#) gives the least reserve and source 27’s [theorem 101.2](#) the least ordinal support ([notes 86.4](#), [86.7](#) and [86.13](#)).
- *Interpolation and the core isomorphic to **No***: [theorem 19.10](#) and [corollary 19.15](#) ([notes 86.5](#), [86.9](#), [86.14](#) and [86.18](#)).
- *Trace injectivity*: the uniqueness clause of [theorem 45.1](#) ([note 86.6](#)).
- *Raw comparison*: [theorem 14.2](#) and source 12’s [theorem 14.7](#) ([notes 86.10](#) and [86.17](#), and source 25’s [note 103.1](#) in [section 103](#)).
- *First block, fibres and set-interval blocks*: [theorem 15.1](#) and [proposition 15.3](#), and the block decomposition of sources 15, 16 and 18 printed in Part IX ([theorems 52.3](#), [52.6](#) and [52.11](#)) ([notes 86.11](#) and [86.12](#)); the core sees only the first block: [corollary 46.4](#) and source 14’s [theorem 54.6](#) ([note 86.15](#)).
- *The diagonal theorem*: [theorems 19.37](#) and [19.38](#) ([notes 86.16](#) and [86.19](#); source 25’s [note 104.1](#) in [section 104](#); source 22’s pair-swap form at the end of [section 102.1](#)).
- *Global choice is an assumption*: [theorem 13.1](#) (opening of source 21’s section).

The proofs differ in bookkeeping only. Sources 21, 22 and 29 complete a prefix inside an infinite initial cardinal, as Part IV does; source 25 adjoins a fresh reserve of the size of the prefix data, which is the point of [remark 19.7](#). Source 25’s diagonal ([note 104.1](#)) and source 29’s ([note 86.19](#)) use a guard step at odd positions in place of a pair swap.

### 86.1 Source 25: baseline embedding results and the set/class transition

The following arguments are part of the earlier general-alphabet theory ([theorem 6.1](#) and [proposition 5.13](#); the reals report). They are included because they sharply delimit what changes when one passes from a fixed birthday fragment to the whole surreal line.

*Note 86.1* (Source 25, Theorem 3.1, “Fixed-set size and ordinal spectrum”). *Let  $X$  be an infinite linearly ordered set of cardinality  $\mu$ . Then*

$$|\mathcal{W}(X)| = 2^\mu, \quad \beta \hookrightarrow \mathcal{W}(X) \iff \beta < \mu^+.$$

Every linear order of cardinality at most  $\mu$  embeds already into  $\mathcal{W}_\mu(X)$ . There is no increasing or decreasing sequence of type  $\mu^+$  in  $\mathcal{W}(X)$ . This is [theorem 6.1](#) (the reals report's general-alphabet theorem), with its ordinal spectrum; source 25 recalls it with its proof and does not present it as new.

*Source 25's proof.* Partition  $X$  into  $\mu$  pairs and write the members of the  $\xi$ th pair as  $x_\xi^- < x_\xi^+$ . For a binary word  $s \in 2^\mu$ , enumerate each pair increasingly when  $s(\xi) = 0$  and decreasingly when  $s(\xi) = 1$ , using consecutive positions  $2 \cdot \xi$  and  $2 \cdot \xi + 1$ . Ordinal multiplication is intended; these positions partition  $\mu$  for an infinite initial ordinal  $\mu$ . This gives an order embedding

$$(2^\mu, <_{\text{lex}}) \hookrightarrow \mathcal{W}_\mu(X).$$

If  $L$  has cardinality at most  $\mu$ , inject its elements into  $\mu$  and associate to  $x \in L$  the characteristic word of the elements strictly below  $x$ , with zero at unused coordinates. For  $x < y$  the first word is coordinatewise at most the second and differs at the coordinate of  $x$ , so the assignment is strictly lexicographically increasing. This proves the lower embedding assertion and the lower cardinal bound. The upper bound follows because each well-order is a subset of  $X^2$ .

Suppose, for a contradiction, that  $(e_\xi)_{\xi < \mu^+}$  is a strictly increasing sequence of exhaustive words on  $X$ . The indices whose words extend a fixed prefix form a convex set of indices. If this set is unbounded in  $\mu^+$ , it therefore contains a final segment.

Starting with the empty prefix, recursively construct a common prefix extended by a final segment of the sequence. At a successor stage there are at most  $\mu$  possible next labels. Termination cannot occur on an unbounded set of indices, since an exhaustive word cannot be a proper prefix of another exhaustive word. Regularity of  $\mu^+$  implies that one next-label cylinder is unbounded, and hence contains a final segment. At a limit stage below  $\mu^+$ , the intersection of the previously chosen final segments is still a final segment. The construction would give an injective word of length  $\mu^+$  in  $X$ , a contradiction. Reversing the sequence gives the decreasing version.

Every ordinal below  $\mu^+$  has cardinality at most  $\mu$ , and every larger ordinal contains  $\mu^+$  as an initial segment. The spectrum follows.  $\square$

For an infinite cardinal  $\kappa$ , let

$$\mathbf{No}_{<\kappa} = \{x \in \mathbf{No} : \text{bd}(x) < \kappa\}.$$

The sign-sequence representation gives  $|\mathbf{No}_{<\kappa}| = 2^{<\kappa}$ . Thus the preceding theorem applies with  $\mu = 2^{<\kappa}$ . In particular, the size parameter is not automatically  $\kappa$ . The singular-cutoff results in the repository show why suppressing that distinction can give wrong coding statements ([theorem 8.3](#); source 22).

*Note 86.2* (Source 25, Proposition 3.2, “Class-order universality, in GBC”). *Every supplied class linear order admits a class order embedding into  $\mathbf{No}$ . It consequently admits an embedding into  $\mathfrak{W}_{\text{sl}}$  and into the set-supported core  $\mathcal{P}_s(b)$ .* This is [proposition 5.13](#) with the core universality of [corollary 19.15](#); source 25's embedding into the core uses one-letter completions ([note 86.4](#)).

*Source 25's proof.* Use global choice to arrange the underlying class in a set-like construction order. When treating one element, only a set of earlier elements has been assigned images. Separate the images of its earlier predecessors from those of its earlier successors by a surreal number. The usual surreal small-cut property supplies such a separator. This set-valued recursion gives an order embedding into  $\mathbf{No}$ .

For each surreal  $x$ , choose a core completion of the one-letter prefix  $(x)$ ; the existence of this completion is proved in [Note 86.4](#). The resulting map  $x \mapsto d_x$  is increasing because the first letters already decide the comparison. Composition proves the assertion.  $\square$



This explains an important discontinuity. Each fixed set fragment has a bounded ordinal embedding spectrum, but the full global construction contains every set ordinal and every supplied ordinary class linear order. Neither statement asserts that a same-level class of *all* class orders has been collected or enumerated.

## 86.2 Source 25: foundational framework and the supported core

### 86.2.1 The background theory

Throughout the global arguments, GBC means Gödel–Bernays class theory with global choice. Sets and classes are distinct sorts; classes have sets as members. Elementary class comprehension allows class parameters but does not quantify over classes in its defining formula. A uniformly  $I$ -indexed family of class words means one supplied class relation whose rows, indexed by the set  $I$ , are the words in question. It does not mean a set containing proper classes.

We use a fixed set-like global well-order of the sets to make constructions deterministic when desired. A recursion along  $\text{Ord}$  whose stage values and histories at set stages are sets is available by ordinary set recursion with class parameters and elementary comprehension. The materially stronger principle ETR concerns recursions with class-sized stage values. That distinction is essential and is audited in [Section 104](#); see [\[63, 22, 24\]](#) for the background.

### 86.2.2 Fresh small-cut separation

*Note 86.3* (Source 25, Lemma 4.1, “A separator avoiding a prescribed set”). *Let  $L, R, S$  be sets of surreals with  $L < R$ . Then there is an  $x \in \mathbf{No} \setminus S$  with  $L < x < R$ . Either of  $L, R$  may be empty.* This is [lemma 5.2](#) and source 13’s [lemma 42.1](#), in the exact-birthday form.

*Source 25’s proof.* The surreal separator theorem permits a separator of any birthday strictly above all birthdays of the options. Choose an ordinal  $\gamma$  strictly above the birthdays of  $L \cup R \cup S$  and choose a separator of birthday exactly  $\gamma$ . It cannot belong to  $S$ . The exact-birthday form is supplied by the sign-sequence construction and by the repository declaration `exists_cut_separator_of_birthday_lt` [\[8\]](#).  $\square$

Only this order-theoretic separation property, not multiplication or analysis on surreals, is used in the new homogeneity argument. In particular, every nonempty numerical open interval has points outside any prescribed set. Removing a set from  $\mathbf{No}$  leaves an order with the same small-cut separation property.

### 86.2.3 Normalized codes rather than class objects

A normalized support code consists of a set  $S \subseteq \text{Ord}$  and a fixed-point-free permutation  $\pi_S : S \rightarrow S$ . Extend it by the identity outside  $S$  to obtain a class permutation  $\pi$  of  $\text{Ord}$ , and evaluate the code as the class word

$$d_\pi(\alpha) = b(\pi(\alpha)).$$

The empty support code evaluates to  $b$ . Requiring  $S$  to be the exact support ensures that distinct normalized codes have distinct evaluations. The collection  $\mathcal{P}_s(b)$  of these codes is an ordinary class of sets. Its lexicographic order is elementary in the parameter  $b$ .

All sets of ordinals are bounded, so a core branch agrees with  $b$  beyond some ordinal. Conversely, an *exhaustive* branch eventually agreeing with  $b$  belongs to the core: its transition permutation  $b^{-1} \circ d$  has bounded, hence set, support. Exhaustiveness is necessary in this converse. An injective word can omit a label while still agreeing with  $b$  on a final segment.

*Note 86.4* (Source 25, Lemma 4.2, “Supported prefix completion”). *Every  $p \in \mathcal{I}$ , of any set-ordinal length, extends to a core branch. Core completions can be selected uniformly from their set-coded prefixes.* This is [lemma 19.4](#). Source 25 adjoins a fresh reserve of the size of  $A$ , the device of [remark 19.7](#); [theorem 49.7](#) computes the least reserve and source 27’s [theorem 101.2](#) the least ordinal support.

*Source 25’s proof.* Write  $\alpha = \text{lh}(p)$  and let  $q = b^{-1} \circ p : \alpha \rightarrow \text{Ord}$ . This is a set partial injection. Put  $A = \alpha \cup q[\alpha]$ .

If  $A$  is finite, the sets  $A \setminus \alpha$  and  $A \setminus q[\alpha]$  have the same finite cardinality, so complete  $q$  to a permutation of  $A$ . If  $A$  is infinite, choose a fresh set  $R$  of ordinals, disjoint from  $A$ , with  $|R| = |A|$ . In  $U = A \cup R$ , both  $U \setminus \alpha$  and  $U \setminus q[\alpha]$  have cardinality  $|A|$ . Choose a bijection between them and adjoin it to  $q$ . This is a permutation of  $U$ , and extension by the identity outside  $U$  is a set-supported permutation of  $\text{Ord}$ .

Normalize its support. The associated core word extends  $p$ . Selecting the first valid normalized code in the fixed global well-order gives a uniform selection. All searches and restrictions here involve sets and supplied class parameters.  $\square$

The reservoir in the infinite case is not cosmetic. Two complements of equinumerous infinite subsets inside their union need not have the same cardinality. Adding a reserve on both sides avoids an invalid cancellation of infinite cardinals.

*Note 86.5* (Source 25, Lemma 4.3, “Core separation of full branches”). *If  $f, g \in \mathfrak{J}$  and  $f <_{\text{lex}} g$ , there exists  $d \in \mathcal{P}_s(b)$  with*

$$f <_{\text{lex}} d <_{\text{lex}} g.$$

*Every cylinder  $C_p$  is nonempty and has at least two distinct child cylinders.* The first assertion is the two-family case of [theorem 19.10](#), which is stated for injective words; the second is the nonemptiness of cylinders used throughout Part VIII.

*Source 25’s proof.* Let  $\delta = \delta(f, g)$  and  $p = f \upharpoonright \delta = g \upharpoonright \delta$ . Choose

$$f(\delta) < x < g(\delta), \quad x \notin \text{ran}(p),$$

using [Note 86.3](#). Complete  $p \frown (x)$  by [Note 86.4](#). The first disagreement with both words occurs at  $\delta$ . Nonemptiness and the existence of distinct children follow in the same way from fresh labels.  $\square$

*Note 86.6* (Source 25, Corollary 4.4, “Injectivity of the cut trace”). *For a supplied branch  $f \in \mathfrak{J}$ , define its lower core trace by*

$$L_f = \{d \in \mathcal{P}_s(b) : d <_{\text{lex}} f\}.$$

*Distinct branches have distinct traces.* This is the uniqueness clause of [theorem 45.1](#) and its routes ([note 46.1](#)).

*Source 25’s proof.* A core branch strictly between two distinct branches belongs to exactly one of their lower traces.  $\square$

The core is proper because every surreal occurs as the first letter of a core completion, while the image of a set under evaluation at zero would be a set. Therefore global choice supplies a class enumeration of its normalized codes. This enumeration will be used only to schedule the back-and-forth construction, not as an enumeration of  $\mathfrak{W}_{\text{sl}}$ .

### 86.3 Source 22: global enumerations and their set-coded core

From this section onward, unless otherwise stated, work in GBC: Gödel–Bernays set theory with Global Choice. The convention GB alone includes no choice axiom. A class function is represented by its class graph, and a uniformly indexed family of class functions is one class graph with an additional *set-valued* index coordinate.

#### 86.3.1 The virtual enumeration order

Fix a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . Write

$$\mathfrak{W}_{\text{sl}} = \{e : e \text{ is a class bijection } \text{Ord} \rightarrow \mathbf{No}\}$$

only as shorthand for a predicate on class variables. This display does not assert that  $\mathfrak{W}_{\text{sl}}$  is a set or an ordinary class. Distinct enumerations have a least disagreement ordinal: restrict a nonempty disagreement class to one ordinal witnessing a disagreement and use the well-ordering of that set. Comparing the two surreal labels there gives a strict linear order on the available enumerations.

The same comparison applies to injective class words  $h : \text{Ord} \hookrightarrow \mathbf{No}$  which need not be exhaustive. Every set-length restriction of such a word is a set by class replacement. This fact will be used repeatedly; a proper-class word itself is not a permissible element of a set-coded family.

#### 86.3.2 Canonical support codes

Let  $\mathcal{P}_s(b)$  consist of canonical set codes for permutations  $\pi : \text{Ord} \rightarrow \text{Ord}$  whose support

$$\text{supp } \pi = \{\xi : \pi(\xi) \neq \xi\}$$

is a set. One canonical code is the set of all moved ordered pairs  $(\xi, \pi(\xi))$ , with the identity recovered off its domain. The code for the identity is empty. Membership in the code class is elementary: the code is a set bijection from a set onto itself with no fixed points, extended by the identity outside that set. Such codes are unique, so  $\mathcal{P}_s(b)$  is an ordinary class of sets. Evaluate its code as the global enumeration  $b \circ \pi$ , and compare codes by their evaluations. We occasionally write a code as if it were its evaluation when no ambiguity results.

This distinction between codes and evaluations is indispensable:  $\mathcal{P}_s(b)$  is an ordinary class, whereas its evaluations are class functions. The evaluation relation is a single class with set-coded arguments.

*Note 86.7* (Source 22, Lemma 7.1, “Every set prefix has a supported completion”). *Every injective set word  $p : \alpha \hookrightarrow \mathbf{No}$  extends the evaluation of some code in  $\mathcal{P}_s(b)$ . Completions can be chosen uniformly from the set prefix code and the fixed global-choice parameters.* This is [lemma 19.4](#), completed inside an infinite initial cardinal as in Part IV.

*Source 22’s proof.* Transport  $p$  along  $b^{-1}$  to a set partial injection  $q : \alpha \rightarrow \text{Ord}$ . Choose an infinite initial cardinal  $\theta$  larger than  $\alpha$  and all members of  $\text{ran } q$ , with  $|\alpha| < \theta$ . Both complements

$$\theta \setminus \alpha, \quad \theta \setminus \text{ran } q$$

have cardinality  $\theta$ . Choose a bijection between them, join it to  $q$ , and extend the resulting permutation of  $\theta$  by the identity outside  $\theta$ . Its moved-pair graph is a set. This gives the required code. Fixing a global choice of the relevant set bijections makes the construction uniform. Every stage is a set construction with class parameters; no class-valued transfinite recursion is involved.  $\square$

*Note 86.8* (Source 22, Lemma 7.2, “Fresh separators”). *If  $a < b$  are surreal numbers and  $S$  is a set of forbidden surreal labels, there is  $z \in (a, b) \setminus S$ . There are also fresh labels on either side of any prescribed surreal.* This is lemma 5.2 and source 13’s lemma 42.1.

*Source 22’s proof.* Choose an ordinal birthday bound strictly above the birthdays of  $a, b$  and every member of  $S$ . The small-cut separator theorem at that exact birthday supplies  $z$  between  $a$  and  $b$ , and its birthday excludes membership in  $S$ . For rays, use a small cut with only one nonempty side and again prescribe a sufficiently large birthday. This is precisely the smallness boundary respected by the repository’s sign-sequence construction [7, 8].  $\square$

For an injective set prefix  $p$ , write

$$C_p = \{d \in \mathcal{P}_s(b) : \text{ev}_b(d) \text{ extends } p\}.$$

Every  $C_p$  is nonempty and convex. If  $h$  is an injective Ord-word extending  $p$ , then there are points of  $C_p$  both below and above  $h$ : choose at the next coordinate a fresh label on the desired side of  $h(\text{dom } p)$  and apply note 86.7.

*Note 86.9* (Source 22, Proposition 7.3, “Density in all injective global words”). *For any distinct injective Ord-words  $h < k$ , there is a code  $d \in \mathcal{P}_s(b)$  with  $h < d < k$ . In particular,  $\mathcal{P}_s(b)$  is dense without endpoints, and its comparisons separate distinct global words.* This is the two-family case of theorem 19.10.

*Source 22’s proof.* Let  $\beta$  be the first disagreement of  $h, k$ . Choose a fresh label strictly between  $h(\beta)$  and  $k(\beta)$ , avoiding the set of common earlier labels. Complete the resulting set prefix by note 86.7. Both comparisons are decided at  $\beta$ . The endpoint assertion follows by applying the one-sided construction to the empty prefix.  $\square$

The earlier report proves the stronger assertion that  $\mathcal{P}_s(b)$  fills every uniformly set-indexed separated cut of global enumerations, and consequently that  $\mathcal{P}_s(b) \cong \mathbf{No}$  in GBC (theorem 19.10 and corollary 19.15). The classical universality and homogeneity of the surreal order underlie that identification [15]. Our reconstruction arguments below use only the explicitly proved prefix-completion and separation properties, not an unproved inference from density to saturation.

## 86.4 Source 21: class well-orders, canonical blocks, and a genuine dense core

[merge] Source 21’s Section 8, printed whole. Every numbered statement in it is a result of Parts III and IV or of Part IX and is printed as a note; source 21 calls two of them “repository baseline” and says, of the others, that canonical block decompositions “also arose in the preceding research” and are included “rather than treating every restatement as a new result”. Its theorem 65.6 is its set-level fibre theorem, printed in Part X.

We now pass from set carriers to the entire surreal class. A class well-order means a strict linear class relation for which every nonempty admitted subclass has a least element. It is *set-like* if each predecessor class is a set. In GBC, a set-like well-order of a proper class has unique order type Ord: sending a point to the ordinal type of its predecessor set gives an initial subclass of Ord, and properness forces that subclass to be all of Ord.

Thus  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  can be described using class bijections  $\text{Ord} \rightarrow \mathbf{No}$ . The unrestricted predicate  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  is larger. For example, enumerate  $\mathbf{No} \setminus \{a, b\}$  in type Ord and then append  $a, b$ . Interchanging the final two labels gives a second well-order. The two agree throughout a proper-class prefix, so no set ordinal is the first disagreement position of the complete orders. The comparison nevertheless has a direct elementary definition.

Existence of a global baseline is a substantive choice assumption. The repository proves, over its choice-free GB convention, that existence of a class well-order of **No** is equivalent to global choice ([theorem 13.1](#); Theorem 13.1 at the pin). We use GBC openly rather than treating the baseline as a canonical choice-free enumeration.

#### 86.4.1 Direct comparison of labelled class relations

For a class well-order  $R$  on a fixed linearly ordered class  $X$ , write  $\text{Pred}_R(x) = \{y : yRx\}$ . Given  $R, S$ , define a class  $C(R, S)$  by the elementary condition

$$x \in C(R, S) \iff \begin{array}{l} \text{Pred}_R(x) = \text{Pred}_S(x), \\ R \text{ and } S \text{ agree on this common predecessor class.} \end{array}$$

Equality here expands into quantification over set elements of the classes. There is no quantification over a class of candidate prefixes.

*Note 86.10* (Source 21, Theorem 8.1, “Largest common labelled prefix; repository baseline”).  $C(R, S)$  is the largest common labelled initial segment of  $R$  and  $S$ . If  $R \neq S$ , the least labels  $a, b$  outside it, in  $R, S$  respectively, exist and are distinct. The rule

$$R <_{\text{lex}} S \iff a <_X b$$

defines a strict linear comparison on the class well-orders of  $X$ . It uses elementary comprehension and does not require comparing abstract class order types or invoking ETR. This is [theorem 14.2](#) (source 12’s [theorem 14.7](#)); the source calls it a repository baseline.

*Source 21’s proof.* If  $x \in C(R, S)$  and  $yRx$ , the two orders agree below  $x$ . Thus  $ySx$ , its predecessors agree in the two orders, and the restrictions below  $y$  agree; hence  $y \in C(R, S)$ . The symmetric argument gives initiality in  $S$ . Any two members have the same ordering in  $R, S$  by the predecessor equality for the later member. Every common labelled initial segment is contained in  $C(R, S)$  by its defining condition.

If  $C(R, S) = X$ , the predecessor equalities give  $R = S$ . Otherwise each well-order has a least element outside the prefix. If these least elements were the same  $x$ , both would have predecessor class  $C(R, S)$ , with equal restricted orders, so  $x$  would belong to  $C(R, S)$ , a contradiction.

Totality and asymmetry now follow from the label order. For transitivity, suppose  $R <_{\text{lex}} S <_{\text{lex}} T$ . Their two common prefixes are initial in the same order  $S$  and are therefore comparable by inclusion. If one is shorter, the earlier first disagreement also decides the comparison of  $R$  with  $T$ . If they coincide, the three next labels satisfy  $a <_X b <_X c$ , so again  $R <_{\text{lex}} T$ .  $\square$

This is the direct proof developed in the repository ([section 14](#); Section 14 at the pin). It also proves that completion cylinders of a fixed labelled prefix are convex. A completion that departs earlier compares with every extension of the prefix in the same direction. For ordinal enumerations the definition agrees with the usual first-coordinate comparison.

#### 86.4.2 The first ordinal-class block and all residual fibers

*Note 86.11* (Source 21, Theorem 8.2, “Canonical first block”). For  $R \in \mathfrak{W}_{\text{all}}(\mathbf{No})$ , put

$$I_R = \{x : \text{Pred}_R(x) \text{ is a set}\}.$$

Then  $I_R$  is an initial proper class, possibly all of **No**, whose induced order is set-like. It has a unique enumeration  $F_R : \text{Ord} \rightarrow I_R$ . The remaining labels carry the residual well-order, and  $R$  is their concatenation after  $F_R$ . This is [theorem 15.1](#); the paragraph after it is [proposition 15.3](#).

*Source 21's proof.* Initiality follows because an earlier predecessor class is a subset of a predecessor set. If  $I_R$  were a set, the least point  $x$  outside it would have predecessor class exactly  $I_R$ , contradicting its exclusion. Thus  $I_R$  is proper. Its inherited predecessor classes are sets, so their ordinal types yield its unique Ord enumeration.  $\square$

Conversely, every injective class word  $F : \text{Ord} \rightarrow \mathbf{No}$  has an extension to a class well-order of the whole carrier: append a chosen well-order of  $\mathbf{No} \setminus \text{ran}(F)$ . Its first block is exactly  $\text{ran}(F)$ , because all later points have that proper class as a subset of their predecessor class. The completions of  $F$  are a convex fiber with the order of all well-orders of the unused carrier. This is the class analogue of [theorem 65.6](#). It is a statement about given class parameters and their completions, not the assertion that the displayed fibers form an ordinary class-sized lexicographic sum.

### 86.4.3 All set-interval blocks at once

The canonical first block can be extracted without recursion. In fact, the same idea gives every such block simultaneously.

*Note 86.12* (Source 21, Theorem 8.3, “Canonical set-interval decomposition”). *For a class well-order  $R$  on  $\mathbf{No}$ , define*

$$x \sim_R y \iff \text{the closed } R\text{-interval between } x, y \text{ is a set.}$$

*This is an elementary convex equivalence relation. Each class has a least element and, in its induced order, is either a set ordinal or a copy of Ord. There is at most one set-sized block, and if it exists it is final. Therefore  $R$  has a canonical representation*

$$R \cong \left( \sum_{b \in B_R} \text{Ord} \right) + \theta_R, \tag{S21.5}$$

*where  $B_R$  is the class of least representatives of its proper blocks with the induced order and  $\theta_R$  is a set ordinal, zero when there is no final set block. The representation uses no ETR. This is the canonical block decomposition of sources 15, 16 and 18, printed in Part IX ([theorems 52.3, 52.6 and 52.11](#)); source 21 proves it independently.*

*Source 21's proof.* Reflexivity and symmetry are immediate. An interval between  $x, z$  is contained in the union of the intervals between  $x, y$  and  $y, z$ , which proves transitivity. Convexity follows by taking subintervals. The condition that an interval is a set is expressed as existence of a set with exactly its members, so elementary comprehension defines the relation.

A block is a nonempty admitted subclass, hence has an  $R$ -least member  $m$ . The predecessors of  $x$  inside its block form the set interval  $[m, x)_R$ . Thus its induced order is set-like, and its ordinal-rank map identifies it either with a set ordinal or with all of Ord.

If a block  $A$  were a set and not final, choose the least point  $y$  after it. With  $m$  its least point,  $[m, y]_R = A \cup \{y\}$  would be a set. Hence  $y \sim_R m$ , contradicting its being outside the block. This proves the assertion about the final set block.

The least representatives are specified directly by the elementary relation. Within a proper block, the point of any specified ordinal rank is unique and is found using a set interval. Consequently the map evaluating  $(b, \alpha)$  at rank  $\alpha$  in block  $b$  is a class function given by an elementary formula. Together with the ordinary enumeration of the optional final set block it yields (S21.5). No sequence of class-valued residual states is constructed.  $\square$



The index order  $B_R$  need not be set-like. For any class well-order  $J$ , the class order  $\sum_{j \in J} \text{Ord}$ , transferred to  $\mathbf{No}$  by global choice, has  $J$  as its proper-block index order. Thus the theorem is a canonical decomposition, not a reduction of every class well-order to a set ordinal. These set-interval blocks inside  $R$  are also different from the finite permutation blocks in the lexicographic order of all such  $R$ , studied later.

#### 86.4.4 Normalized support codes

Fix a baseline class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . A normalized code is a set permutation  $\sigma : \gamma \rightarrow \gamma$  satisfying

$$\gamma = \sup\{\xi + 1 : \xi < \gamma, \sigma(\xi) \neq \xi\}.$$

The empty permutation is permitted. Extend  $\sigma$  by the identity outside  $\gamma$ , and evaluate it as  $E_\sigma = b \circ \hat{\sigma}$ . Let  $\mathcal{P}_{\text{bd}}(b)$  be the ordinary class of these codes, ordered by their evaluations. Normalization makes evaluation injective. Its range is exactly the exhaustive enumerations eventually equal to  $b$ .

*Note 86.13* (Source 21, Lemma 8.4, “Uniform completion of a set prefix”). *Every injective set word  $p : \delta \rightarrow \mathbf{No}$  extends to a core evaluation. The completion can be chosen by one elementary class construction from  $p$  and the fixed baseline and global well-order.* This is [lemma 19.4](#); the warning after it is [remark 19.7](#).

*Source 21’s proof.* For each  $\xi < \delta$ , let  $a_\xi = b^{-1}(p(\xi))$ . The prescribed partial permutation is  $\xi \mapsto a_\xi$ . Choose an infinite initial cardinal  $\nu$  strictly larger than  $\delta$  and all the  $a_\xi$ . Both the unused domain and unused range inside  $\nu$  then have cardinality  $\nu$ , so the partial injection extends to a permutation of  $\nu$ . Choose a completion using the fixed global well-order, extend by the identity, and normalize its support. Its evaluation has prefix  $p$ .  $\square$

Enlarging to a strictly larger cardinal is useful here. A partial injection cannot in general be completed on the mere union of its domain and range: the map  $n \mapsto n + 1$  on  $\omega$  is a simple counterexample. The preceding completion is the version verified in the repository, and it works uniformly for all set prefixes.

*Note 86.14* (Source 21, Theorem 8.5, “Uniform set-cut interpolation; repository baseline”). *Let  $(A_i)_{i \in I}$  and  $(B_j)_{j \in J}$  be uniformly set-indexed families of injective Ord-words, with  $A_i <_{\text{lex}} B_j$  for every cross pair. There is an injective set prefix every extension of which lies strictly between the two families. In particular, a point of  $\mathcal{P}_{\text{bd}}(b)$  lies between them. Empty sides are allowed.* This is [theorem 19.10](#), which the source credits; the paragraphs after it are [corollary 19.15](#) and [proposition 5.13](#).

*Source 21’s proof.* When both families are nonempty, class replacement bounds their cross first-disagreement ordinals by a set ordinal  $\beta$ ; otherwise take  $\beta = 0$ . Build a prefix and call a row active when it still agrees with the prefix. At stage  $\alpha$ , the next active left and right values form sets  $L, R$ . Every member of  $L$  is at most every member of  $R$ , because active cross rows have agreed so far. Thus their intersection has at most one member.

If they share a value  $z$ , append  $z$  and retain exactly the rows with that next value. It is fresh since every retained row is injective. The discarded rows have their desired strict inequality permanently decided. If the sets are disjoint, choose a fresh value strictly between them, using the surreal set-cut property after removing the set of previously used labels, append it, and stop. Removing a set does not destroy set-cut interpolation: first obtain more than that set’s cardinality many distinct separators by set recursion, then take one outside it. This also covers empty sides.

At limit stages take the union of the previous prefixes. Both sides cannot remain active through length  $\beta$ , since that would give an active cross pair agreeing longer than its bounded first

disagreement. So the procedure stops with a set prefix. Apply [note 86.13](#). All recursion states and restrictions of the uniformly indexed rows used in the construction are sets.  $\square$

This proof records the uniformity and size facts behind the repository interpolation theorem ([theorem 19.10](#); Theorem 19.10 at the pin). In particular,  $\mathcal{P}_{\text{bd}}(b)$  is a proper-class  $\eta_{\text{Ord}}$  order without endpoints. A set-valued ordinal back-and-forth identifies it with **No**: enumerate the two genuine classes by Ord, alternately insert the least unused point from each, and use set-cut interpolation to extend the current set partial isomorphism. This is the same structural principle as the classical uniqueness theorem for the surreal order [15].

The same insertion argument embeds every given class linear order into  $\mathcal{P}_{\text{bd}}(b)$ , because the set of already placed points supplies only set-sized constraints at each stage. This gives class-order universality of the core, with the input order and a global choice function as parameters. It does not produce an ordinary class whose elements are all input class orders.

*Note 86.15* (Source 21, Lemma 8.6, “The core sees exactly the first ordinal block”). *For*  $R \in \mathfrak{W}_{\text{all}}(\mathbf{No})$ , *let*  $F_R$  *be its canonical word from* [note 86.11](#). *Then*

$$\{d \in \mathcal{P}_{\text{bd}}(b) : E_d <_{\text{lex}} R\} = \{d \in \mathcal{P}_{\text{bd}}(b) : E_d <_{\text{lex}} F_R\}.$$

*Consequently any cut of the core represented by a full class well-order is represented by an injective Ord-word.* This is [corollary 46.4](#) and source 14’s trace-fibre theorem [theorem 54.6](#) (Part IX).

*Source 21’s proof.* If  $E_d \neq F_R$ , they first disagree at an ordinal coordinate, before any residual tail of  $R$  begins. That first disagreement also compares the well-order represented by  $E_d$  with  $R$ . If  $E_d = F_R$ , the word is exhaustive, so  $R$  has no unused labels or residual tail and equals the core well-order itself. Both strict comparisons then fail.  $\square$

Thus a cut forgets the residual well-order. In particular, the core is dense in the set-like fragment, but it is not dense inside a nonsingleton residual fiber of the full fragment. A proof using density must specify which of these ambient orders it concerns.

#### 86.4.5 Diagonal size obstruction

*Note 86.16* (Source 21, Proposition 8.7, “No exhaustive uniform set-code family”). *There is no class relation with a class of set codes whose evaluations include every set-like global enumeration of No.* This is [theorem 19.37](#).

*Source 21’s proof.* Under global choice, a class of set codes can be listed by Ord (with repetitions if it is a nonempty set). Suppose its rows are  $E_\alpha$ ,  $\alpha \in \text{Ord}$ . Decode from each row a bit at coordinate  $\alpha$ , setting the bit to 1 exactly when  $E_\alpha(2 \cdot \alpha) = b(2 \cdot \alpha + 1)$ . Form a new exhaustive enumeration by swapping baseline pair  $\alpha$  exactly when this decoded bit is 0. The new enumeration differs from row  $E_\alpha$  on that pair for every  $\alpha$ . It is given by elementary comprehension from the proposed uniform family, contradicting exhaustiveness. An empty family is immediately nonexhaustive.  $\square$

This is the relevant Cantor diagonal argument. The core has set codes because its support is a set; arbitrary class support cannot be compressed into an exhaustive uniform family of set codes. Moving to a larger universe permits the larger collection as a set there, but does not invalidate the internal diagonal statement.

## 86.5 Source 29: global well-orders, represented cuts, and the higher-order boundary

[merge] Source 29’s Section 10, printed whole. Its three numbered statements are results of this report, which source 29 marks as inherited; its subsection on the core-cut theorem summarizes Part VIII.

### 86.5.1 Raw comparison is elementary

Let  $R, S$  be class well-orders of the same carrier  $\mathbf{No}$ . Define  $C$  to consist of those labels  $x$  whose predecessor carriers in  $R$  and  $S$  agree and on which the two predecessor orders agree:

$$x \in C \iff \begin{aligned} &\forall u (uRx \leftrightarrow uSx), \\ &\forall u, v [(uRx \wedge vRx) \implies (uRv \leftrightarrow uSv)]. \end{aligned}$$

This is an elementary formula with the supplied class parameters. The class  $C$  is an initial segment of both orders and their restrictions to it agree. If  $R \neq S$ , both complements have least points

$$a = \min_R(\mathbf{No} \setminus C), \quad b = \min_S(\mathbf{No} \setminus C).$$

They are distinct; otherwise the common next point would belong to  $C$ . Define  $R <_{\text{lex}} S$  exactly when  $a < b$  numerically.

*Note 86.17* (Source 29, Proposition 10.1, “Inherited direct comparison”). *This construction linearly orders the available class well-orders of  $\mathbf{No}$  in the virtual sense. It needs elementary comprehension, not the comparability of abstract class well-orders and not ETR. For set-like orders it agrees with first disagreement of their Ord-enumerations.* This is [theorem 14.2](#); source 29 calls it inherited.

*Source 29’s proof.* Totality and asymmetry follow from the two different next labels. For transitivity, the common initial segments for  $(R, S)$  and  $(S, T)$  are both initial in  $S$ , so are comparable by inclusion. At the shorter segment, two of the orders still agree and the third comparison is inherited from the given strict inequality. If the segments are equal, the three next numerical labels give transitivity directly. This is comparison of labelled initial segments, not construction of an isomorphism between unrelated class orders. In the set-like case each common initial segment has an ordinal position unless the orders agree everywhere, and the description becomes ordinary first-coordinate disagreement. The full argument is part of the repository baseline ([theorem 14.2](#)).  $\square$

The formula explains why comparing raw well-orders and comparing their abstract order types are different foundational tasks. The latter asks for a new class map witnessing an initial-segment isomorphism; that map is not supplied merely by numerical comparison of two labels. The distinction will be central to source 29’s condensation section ([section 108.3](#)).

### 86.5.2 Set-like enumerations and an ordinary class core

Fix a class bijection  $e : \text{Ord} \rightarrow \mathbf{No}$ . A set-like global well-order has an exhaustive enumeration of this form. Denote the virtual collection of these enumerations by  $\mathfrak{W}_{\text{sl}}$  and the larger collection of all class well-orders on  $\mathbf{No}$  by  $\mathfrak{W}_{\text{all}}$ .

The bounded-support core relative to  $e$  consists of the enumerations  $e \circ \pi$ , where  $\pi$  is a permutation of  $\text{Ord}$  that is the identity outside some set ordinal. Such a permutation has a set code. Using the least possible support bound removes duplicate codes. The core, denoted  $\mathcal{P}_{\text{bd}}(e)$ , is therefore an ordinary class of sets. It is a small-code substructure of a higher-order collection, not an identification of that entire collection with a class of sets.

*Note 86.18* (Source 29, Theorem 10.2, “Inherited core theorem”). In GBC,  $\mathcal{P}_{\text{bd}}(e)$  has the set-cut property and is order-isomorphic to **No**. It interpolates every strict cut specified by two uniformly set-indexed families of set-like global enumerations. In particular,  $\mathfrak{W}_{\text{sl}}$  is dense and has no endpoints in the corresponding virtual sense, and every ordinary class linear order embeds in its core. This is [theorem 19.10](#) and [corollary 19.15](#) with [proposition 5.13](#); the source states it without proof and summarizes the repository’s proof in the next paragraph.

The proof in the repository bounds the first disagreements relevant to the set-indexed constraints, chooses a fresh numerical separator, and completes the resulting bounded modification to a permutation. The last assertion uses the usual Ord-stage back-and-forth: every stage has a set of earlier constraints, so the set-cut property supplies the next value. The same method proves that any two ordinary class orders with the full set-cut property are isomorphic in GBC ([proposition 5.13](#)) [14].

This universality includes every set ordinal, reverse ordinal, set linear order, and supplied class linear order with set-valued points. It does not assert an embedding of every *higher-order collection* of classes into **No**.

### 86.5.3 What the core-cut theorem records

The inspected article has a detailed normal form for proper normalized cuts of  $\mathcal{P}_{\text{bd}}(e)$ . Its three mechanisms are a boundary of a prefix cylinder, a numerical cut inside a next-letter alphabet, and a coherent injective branch. A cut is represented by an Ord-length injection exactly when it crosses appropriate prefix cylinders at every set length; exhaustivity additionally requires that the resulting prefixes cover all surreal labels ([theorems 43.3](#) and [45.1](#)). The explicit branch-automorphism theorem of this article shows why that coverage condition cannot be discarded even if the full ordered prefix tree is named.

An arbitrary class well-order can begin with a proper-class set-like segment and continue with a residual tail. Distinct tails can share the same first-block trace on the core. A faithful completion must therefore retain residual-order information in addition to a numerical cut. Earlier project work develops such decorated completions (sources 19, 21 and 20; [theorem 98.2](#)).

The unrestricted order  $\mathfrak{W}_{\text{all}}$  is not dense. For distinct labels  $a < b$ , enumerate  $\mathbf{No} \setminus \{a, b\}$  first, then append  $(a, b)$  or  $(b, a)$ . These two class well-orders are adjacent: every intermediate presentation would have to share their entire initial class prefix, and there are only the two permutations of the residual labels. More generally, a fixed prefix with  $n$  unused labels leaves a finite convex block of  $n!$  presentations. These terminal blocks cannot occur in an exhaustive set-like enumeration of type Ord.

### 86.5.4 Why uniform set codes cannot enumerate everything

*Note 86.19* (Source 29, Proposition 10.3, “Diagonal boundary”). No uniformly class-indexed family of set-like global enumerations, indexed by an ordinary class of sets, exhausts all such enumerations available in GBC. This is [theorem 19.37](#), in the guard-step form of source 25’s [note 104.1](#).

*Source 29’s proof.* An empty proposed family omits the baseline enumeration. For a nonempty family, Global Choice permits a listing as  $(F_\alpha)_{\alpha \in \text{Ord}}$ , allowing repetition if its index carrier is a set. Construct a new enumeration  $G$  in pairs of stages. At  $2 \cdot \alpha$ , choose an unused label different from  $F_\alpha(2 \cdot \alpha)$ . At  $2 \cdot \alpha + 1$ , choose the least still unused label in the baseline  $e$ -scan. There are always fresh labels because only a set has been used. Set-valued recursion gives one class map  $G$ .

It is injective by construction. The odd stages make it exhaustive: by transfinite induction they eventually pass every fixed baseline index. It differs from  $F_\alpha$  at coordinate  $2 \cdot \alpha$  for each  $\alpha$ . Thus the proposed family omits  $G$ .  $\square$

The statement is about uniform coding. It does not quantify over an imaginary GB class of all enumerations. An even more concrete size comparison is available in a specified full model. If  $\kappa$  is strongly inaccessible and one works externally with  $(V_\kappa, \mathcal{P}(V_\kappa))$ , then its surreal carrier has external cardinality  $\kappa$ , whereas its set-like global enumerations have external cardinality  $2^\kappa$ . They cannot all be injected into that carrier. Using this full-model semantics is a conditional modeling choice; it is not a proof in ZFC that an inaccessible cardinal exists.

### 86.5.5 Class extensions and external countability

For a fixed set universe and a fixed available class relation, well-foundedness is absolute between GBC class expansions. Under Global Choice, a failure of class well-foundedness yields a set-coded descending  $\omega$ -sequence. The equivalence is established in the primary class-theory literature [27, §3]. Enlarging the class collection can change which relations or branches are available; it does not thereby change well-foundedness of an old fixed relation.

A countable class model has only countably many available global enumerations externally. Its set-like enumeration order is then a countable dense linear order without endpoints and is externally isomorphic to  $\mathbb{Q}$ . This does not make the class model itself recoverable from that bare order. Nor does internal class well-foundedness automatically imply external well-foundedness; that stronger correctness property belongs to a beta-model assumption. Internally finite terminal blocks can also be externally infinite in an  $\omega$ -nonstandard model [27] (sources 19 and 20).

[merge] The last paragraph's remark that a countable class model has set-like enumeration order  $\mathbb{Q}$  is source 13's [proposition 55.5](#) (Part IX), proved again in source 21's [theorem 102.13](#) below. Its adjacent pairs and  $n!$  blocks are [theorem 16.2](#).

## 87 Unranked cylinders recover prefix heights (source 22)

[merge] Source 22's Section 8, printed whole. Its [theorem 87.1](#) is printed here once; source 25 re-proves it as its Proposition 5.1 and credits source 22 ([note 88.1](#)). Its [theorem 87.4](#) is a rank-free form of the represented-cut theorem [theorem 45.1](#) ([note 46.1](#)).

The earlier continuation recognizes global branches when cylinder labels and levels are supplied (sources 13 and 17: [theorems 46.15](#) and [46.21](#)). It asks whether the level predicate is additional essential structure. The answer is no, provided the full cylinder incidence structure is retained.

### 87.1 A size-safe meaning of the cylinder family

Let  $\mathcal{I}$  be the ordinary class of injective set-length prefixes. Use the class incidence relation

$$I(p, d) \iff d \in C_p, \quad p \in \mathcal{I}, \quad d \in \mathcal{P}_s(b).$$

The nodes  $p$  are sets; the cylinder  $C_p$  is its class of incident core points. We never collect those class subsets as elements of an ordinary class. To say that the cylinders are *unranked* means that the node sort, the core order, and incidence are retained, but the domain-length map is not supplied as an additional predicate. Inclusion is still expressible by the elementary formula

$$C_q \subseteq C_p \iff \forall d \in \mathcal{P}_s(b) (I(q, d) \Rightarrow I(p, d)).$$

All quantified objects in this formula are sets.

**Theorem 87.1** (Cylinder inclusion recovers prefix and height). *For  $p, q \in \mathcal{I}$ ,*

$$C_q \subseteq C_p \iff p \text{ is a prefix of } q.$$

*In particular, distinct prefixes have distinct cylinders. The strict ancestors of  $C_p$  under reverse inclusion form a set well-order of order type  $\text{dom } p$ . Consequently, in the ambient class set theory, the ordinal height is definable from the unranked incidence structure, and any available class incidence isomorphism preserves it.*

*Proof.* The forward implication for genuine prefixes is immediate. If  $p, q$  are incompatible, their cylinders are disjoint and nonempty, so inclusion is impossible. If  $q$  is a proper prefix of  $p$ , choose a fresh label at coordinate  $\text{dom } q$  different from the corresponding label of  $p$ , and complete this alternative prefix. Its code lies in  $C_q$  but not in  $C_p$ . These cases exhaust all possibilities.

The strict ancestors of  $C_p$  are therefore precisely

$$C_{p \upharpoonright \beta}, \quad \beta < \text{dom } p.$$

Their node codes form a set by replacement. Ordered from the largest cylinder toward the smaller ones, their order type is exactly  $\text{dom } p$ . Equivalently, height is the unique ordinal order-isomorphic to the set of strict ancestors under reverse inclusion. This description uses only incidence and the ambient set theory. It is not a claim of first-order definability of arbitrary ordinal heights in the pure two-sorted incidence language alone. It is not an appeal to a recursively constructed class-valued rank. An available class incidence isomorphism preserves inclusion and restricts to a set bijection on each ancestor set. It therefore preserves the unique ordinal order type. This is an internal statement; an arbitrary external map over a nonstandard model is not silently asserted to be an available class isomorphism.  $\square$

*Remark 87.2* (The fresh-choice hypothesis is real). The analogous assertion can fail for a finite exhaustive permutation tree. If one unused label remains, appending that forced label does not change the cylinder, so two different prefix lengths have the same cylinder. Here the whole surreal alphabet always leaves fresh labels after a set prefix; [notes 86.7](#) and [86.8](#) rule out that collapse. The finite regression suite records both the positive fresh-reserve case and this counterexample.

## 87.2 A rank-free criterion for a represented cut

A *normalized cut* of  $\mathcal{P}_s(b)$  is an initial subclass  $L \subset \mathcal{P}_s(b)$  such that  $L$  and  $U = \mathcal{P}_s(b) \setminus L$  are nonempty and  $L$  has no greatest point. The upper class  $U$  is allowed to have a least point. This convention agrees with strict lower traces

$$L_h = \{d \in \mathcal{P}_s(b) : d < h\}.$$

Normalization matters when  $h$  itself is the evaluation of a core code: it prevents choosing the wrong side of that point.

Say that  $L$  *crosses*  $C_p$  when both  $L \cap C_p$  and  $U \cap C_p$  are nonempty. Let

$$B_L = \{p \in \mathcal{I} : L \text{ crosses } C_p\}.$$

This is an ordinary class of set nodes, defined by elementary comprehension from  $L$  and incidence.

**Lemma 87.3** (Crossed cylinders form an initial branch). *The class  $B_L$  is a chain in the prefix order and is closed under taking prefixes. At most one crossed node has a given height.*



*Proof.* Ancestor closure follows from cylinder inclusion. Two incompatible cylinders are disjoint convex subsets of a linear order, with one entirely before the other. They cannot both meet both sides of an initial cut: an upper-side point in the earlier cylinder and a lower-side point in the later cylinder would contradict initiality. Thus crossed nodes are compatible. Nodes of equal length which are compatible are equal.  $\square$

**Theorem 87.4** (Unranked branch-recognition theorem). *For a normalized cut  $L$  of  $\mathcal{P}_s(b)$ , the following are equivalent:*

- (i)  $L = L_h$  for an injective class word  $h : \text{Ord} \hookrightarrow \mathbf{No}$ .
- (ii) The crossed-node class  $B_L$  is proper: it is not contained in any set.
- (iii) The recovered heights of crossed cylinders are unbounded in  $\text{Ord}$ .
- (iv) For every ordinal  $\alpha$ , there is exactly one crossed node of height  $\alpha$ .

*The representing word is unique and is definable by elementary comprehension from  $L$  and the labelled node incidence. No ETR is needed.*

*Proof.* If  $L = L_h$ , then every prefix  $h \upharpoonright \alpha$  is crossed, by the two-sided completion construction preceding [note 86.9](#). Conversely, a prefix incompatible with  $h$  has its cylinder wholly on one side. Therefore the crossed nodes are exactly the prefixes of  $h$ , proving (i) $\Rightarrow$ (iv).

By [lemma 87.3](#), there is at most one crossed node of each height. If all heights are bounded by an ordinal  $\lambda$ , class replacement applied to the uniquely specified nodes of heights at most  $\lambda$  shows that  $B_L$  is a set. This proves (ii) $\Rightarrow$ (iii). Unbounded heights together with ancestor closure give exactly one node of every height, proving (iii) $\Rightarrow$ (iv). A set of nodes has bounded ordinal heights, so (iv) $\Rightarrow$ (ii).

Assume (iv). The unique crossed nodes are compatible and nested. Define

$$h(\xi) = x \iff \exists p \in B_L (\xi \in \text{dom } p \wedge p(\xi) = x).$$

This is an elementary class definition, with set quantifiers only. Existence at each coordinate follows from the crossed node of height  $\xi + 1$ ; uniqueness and injectivity follow from compatibility. Thus  $h$  is an injective  $\text{Ord}$ -word.

Fix a core point  $d$  whose evaluation differs from  $h$ , and let  $\xi$  be the first disagreement. The cylinder of  $h \upharpoonright (\xi + 1)$  lies wholly above  $d$  if  $d < h$ , and wholly below  $d$  if  $d > h$ . Since that cylinder is crossed, initiality of  $L$  forces  $d \in L$  in the first case and  $d \in U$  in the second. If  $d = h$  is a core point, it cannot belong to  $L$ : all larger points lie in  $U$ , so it would be the greatest member of  $L$ . Hence  $L = L_h$ . Distinct words have distinct traces by [note 86.9](#), proving uniqueness.  $\square$

The properness assertion in (ii) is itself elementary in the class parameter  $L$ :

$$\forall a \exists p (p \in B_L \wedge p \notin a),$$

where  $a, p$  range over sets. Thus “proper class” is not concealing quantification over a new sort of classes of cylinders.

**Corollary 87.5** (Exhaustiveness is label coverage). *The word reconstructed in [theorem 87.4](#) is exhaustive if and only if*

$$\forall x \in \mathbf{No} \exists p \in B_L \quad x \in \text{ran } p.$$

*Therefore unranked cylinder incidence, together with the label-occurrence relation on nodes, suffices to recognize traces of global well-orders.*

*Proof.* The reconstructed range is exactly the union of the ranges of the crossed prefixes. Equality of that range with  $\mathbf{No}$  is the displayed condition.  $\square$

### 87.3 What has, and has not, been reconstructed

The level predicate is redundant: it is recovered from the ancestor order. Exhaustiveness still uses labels, and the word itself uses the coordinate-value incidence of a prefix. We have not shown that this label information can be discarded. Nor have we reconstructed cylinders from the pure core order. The earlier continuation gives an automorphism showing that the bare linear order does not determine its original prefix cylinders (source 13, [theorem 47.12](#)). These statements are compatible: once an unranked cylinder family is supplied, its levels are forced; without that family, the order alone need not determine it.

The criterion also sharpens the logical form of the earlier branch description. A normalized cut which crosses only set-many nodes is not an injective full-length branch trace. A failure can arise when a next-label cut is not realized, or at a limit-prefix boundary, as analyzed in the earlier reports. The present theorem does not need to choose a first failure recursively; it detects all such failures by the boundedness of the crossed branch.

*Note 87.6* ([merge]; the represented-cut theorem in batch 83). Part VIII prints six proofs of the represented-cut theorem ([note 46.1](#)). Batch 83 adds three: source 20's Theorem 3.10 ([note 111.16](#)), source 21's [notes 97.10](#) and [97.11](#) ([section 97](#)), and source 22's [lemma 87.3](#) and [theorem 87.4](#) above. Sources 25, 27 and 30 use the theorem and credit it ([note 92.1](#), source 27's display (S27.17), [note 102.18](#)). Source 22's form is the only one stated without the ranks of the crossing prefixes: properness of the class of crossed nodes replaces “a crossing prefix at every length”. Its [corollary 87.5](#) is the coverage clause of [theorem 45.1](#).

Source 22's sentence “We have not shown that this label information can be discarded” is answered by source 25, negatively: the core with all its unranked cylinders does not recognize exhaustive branches, even after naming set-many parameters ([theorems 91.1](#) and [91.2](#)), and the single cross-node relation of [section 93](#) suffices.

## 88 What unranked cylinders remember (source 25)

[merge] Source 25's Section 5. Its Proposition 5.1 is source 22's [theorem 87.1](#) and is printed as a note.

### 88.1 The two-sorted reduct

Let  $\mathbf{C}_b$  have a core sort  $\mathcal{P}_s(b)$ , a node sort  $\mathcal{I}$ , the order  $<_{\text{lex}}$  on  $\mathcal{P}_s(b)$ , the prefix relation on  $\mathcal{I}$ , and the incidence relation

$$\text{Inc}(d, p) \iff p \sqsubseteq d.$$

One may equivalently recover prefix order from cylinder inclusion. The structure has *no* coordinate-value relation, no ambient membership relation on the codes, and no predicate identifying equal letters used at different nodes. Those are genuinely additional relations, not silently included.

The two sorts are ordinary classes of sets. A cylinder is represented by its node code, not used as a proper-class element of a new class. This is how the informal phrase “the family of all cylinders” is made size-safe.

*Note 88.1* (Source 25, Proposition 5.1, “Cylinder recovery”). For  $p, q \in \mathcal{I}$ ,

$$C_q \subseteq C_p \iff p \sqsubseteq q.$$

*Distinct nodes have distinct cylinders. The proper ancestors of a node of length  $\alpha$  form a set-sized chain of order type  $\alpha$ . Hence the unranked cylinder structure determines all prefix lengths.* This is source 22's [theorem 87.1](#), which source 25 credits in the next paragraph; the proof is the same.

*Source 25's proof.* A prefix extension gives the displayed inclusion. Incompatible prefixes have disjoint nonempty cylinders. If  $q$  is a proper prefix of  $p$ , choose at the next position a fresh letter different from that of  $p$  and complete the resulting word to the core. Thus  $C_p$  is a proper subset of  $C_q$ . These observations give the converse and injectivity. The proper ancestors of  $p$  are exactly  $p \upharpoonright \beta$ ,  $\beta < \alpha$ , and their prefix order has type  $\alpha$ .  $\square$

This reconstruction was established in the residual-geometry continuation (source 22, [theorem 87.1](#)). We use it to convert the new profile theorem into a result about *unranked* cylinders rather than an expansion with a gratuitously named length function.

## 88.2 Automorphisms and first-disagreement profiles

An automorphism of  $\mathbf{C}_b$  is a pair  $(F, T)$  of class bijections on its two sorts preserving the displayed relations. An order automorphism  $F$  of  $\mathcal{P}_s(b)$  carrying cylinders onto cylinders determines  $T$  uniquely by

$$F[C_p] = C_{T(p)}.$$

By [Note 88.1](#),  $T$  preserves lengths and prefixes.

For distinct full words define

$$\text{pr}(f, g) = (\delta(f, g), \text{sgn}(f, g)), \quad \text{sgn}(f, g) = \begin{cases} -1, & f <_{\text{lex}} g, \\ +1, & g <_{\text{lex}} f. \end{cases}$$

These profiles determine every pairwise prefix-equivalence relation:

$$f \upharpoonright \alpha = g \upharpoonright \alpha \iff \alpha \leq \delta(f, g) \quad (f \neq g).$$

The inequality includes equality: a disagreement at  $\alpha$  still permits agreement on the restriction to  $\alpha$ .

**Proposition 88.2** (Automatic profile preservation). *Every automorphism of  $\mathbf{C}_b$  preserves the first-disagreement ordinal between distinct core branches. Conversely, an order automorphism of  $\mathcal{P}_s(b)$  preserving these ordinals induces a unique automorphism of  $\mathbf{C}_b$ .*

*Proof.* The longest prefix common to distinct core branches is the unique deepest node incident with both. Its height is their first-disagreement ordinal and is recovered by [Note 88.1](#).

Conversely, for  $p$  of length  $\alpha$ , choose  $d \in C_p$  and set

$$T(p) = F(d) \upharpoonright \alpha.$$

For any other  $e \in C_p$ , either  $e = d$  or  $\delta(e, d) \geq \alpha$ . Profile preservation makes the formula independent of the choice. Applying the same argument to  $F^{-1}$  gives an inverse on nodes. The equality  $F[C_p] = C_{T(p)}$  follows in both directions, and the remaining structural relations are preserved.  $\square$

Every such  $T$  acts on a supplied  $f \in \mathfrak{J}$  by

$$\widehat{T}(f)(\alpha) = T(f \upharpoonright (\alpha + 1))(\alpha). \tag{S25.1}$$

The transformed prefixes cohere, so this is an injective full class word. Its inverse is obtained from  $T^{-1}$ . It preserves all first-disagreement ordinals and lexicographic comparisons and agrees with  $F$  on the core. This is an elementary transformation of supplied class parameters, not the graph of a same-level function whose arguments are proper classes.

## 89 The one-point profile extension theorem (source 25)

The central issue is not extending a partial permutation of labels. Different nodes may relabel the same numerical value differently. What must be preserved is the ordered tree geometry: the first place at which two branches separate and the direction of that separation.

**Lemma 89.1** (One-point extension into the core). *Let  $I$  be a set, and let  $(a_i)_{i \in I}$  and  $(a'_i)_{i \in I}$  be uniformly indexed families of pairwise distinct members of  $\mathfrak{J}$  such that*

$$\text{pr}(a_i, a_j) = \text{pr}(a'_i, a'_j) \quad (i \neq j).$$

*Given  $d \in \mathfrak{J}$  distinct from every  $a_i$ , there is  $c \in \mathcal{P}_s(b)$  such that*

$$\text{pr}(d, a_i) = \text{pr}(c, a'_i) \quad (i \in I).$$

*The witness can be selected uniformly from the supplied class parameters and the set of indices.*

*Proof.* For  $I = \emptyset$ , choose any core code. Otherwise let

$$\delta_i = \delta(d, a_i), \quad \lambda = \sup_{i \in I} (\delta_i + 1).$$

Class replacement makes the collection of these ordinals a set. We construct a set-length injective prefix  $r$  that already decides every required comparison. There are two cases; the second is essential at transfinite lengths.

**Case 1: the disagreement ordinals have a maximum  $\delta_*$ .** Let  $I_* = \{i \in I : \delta_i = \delta_*\}$ . All the source branches with indices in  $I_*$  agree with  $d$  below  $\delta_*$ . Hence their target branches share a common prefix  $r_0$  of length  $\delta_*$ . Partition their next labels according to the required side of the new label:

$$\begin{aligned} L &= \{a'_i(\delta_*) : i \in I_*, a_i(\delta_*) < d(\delta_*)\}, \\ R &= \{a'_i(\delta_*) : i \in I_*, d(\delta_*) < a_i(\delta_*)\}. \end{aligned}$$

If a label lies in  $L$  and another in  $R$ , their source branches first disagree at  $\delta_*$  and are in that increasing order. Profile preservation implies the same for their target labels. Therefore  $L < R$ . [Note 86.3](#) supplies

$$x \notin \text{ran } r_0, \quad L < x < R.$$

Put  $r = r_0 \frown (x)$ . It has the required profile against every top-level constraint.

For an index  $i$  with  $\delta_i < \delta_*$ , choose  $j \in I_*$ . The source branch  $a_j$  agrees with  $d$  through coordinate  $\delta_i$ . Consequently the comparison of  $a_j$  with  $a_i$  has exactly the first disagreement and direction required of  $d$  against  $a_i$ . The target profile transfers this statement to  $a'_j$  against  $a'_i$ . Since  $r$  agrees with  $a'_j$  through coordinate  $\delta_i$ , it has the required profile against this lower-level constraint as well.

**Case 2: the disagreement ordinals have no maximum.** Here  $\lambda$  is a limit ordinal and the  $\delta_i$  are cofinal in  $\lambda$ . For each  $\beta < \lambda$ , define

$$r(\beta) = a'_i(\beta) \quad \text{for any } i \text{ with } \delta_i > \beta.$$

This is independent of  $i$ . Indeed, any two such source branches agree with  $d$  through  $\beta$ , so their target branches agree through  $\beta$ . For  $\gamma < \beta < \lambda$ , choose one  $i$  with  $\delta_i > \beta$ ; both  $r(\gamma)$  and  $r(\beta)$  occur at the corresponding positions of the injective branch  $a'_i$ . They are distinct. Thus  $r$  is an injective word of length  $\lambda$ .

Fix  $i \in I$  and choose  $j$  with  $\delta_j > \delta_i$ . The source  $a_j$  agrees with  $d$  through  $\delta_i$ . The profile of  $a_j$  against  $a_i$  therefore equals that of  $d$  against  $a_i$ . Transferring this profile to the targets and using the definition of  $r$  gives

$$\delta(r, a'_i) = \delta_i$$

with the required direction. This notation concerns a disagreement already occurring within the domain of  $r$ , not a comparison using a termination convention.

In either case [Note 86.4](#) extends  $r$  to a core branch  $c$ . All prescribed first disagreements occur inside  $r$ , so completion changes none of them. For uniformity, choose the least successful core code in a fixed set-like global well-order. The existence just proved guarantees that this elementary selection is defined.  $\square$

*Remark 89.2* (Why no numerical equality pattern is required). The argument does not preserve equality between a letter occurring in one node and a letter occurring at another depth or on another branch. Its only injectivity obligation is within the new prefix  $r$ . In Case 1 this is handled by avoiding  $\text{ran } r_0$ ; in Case 2 it follows from coherence inside individual target branches. This is the precise source of the extra symmetries studied below.

**Example 89.3** (An elementary local distortion). Take the two one-step extensions  $(0, 1)$  and  $(2, 1)$  of different first-letter nodes. An ordered-tree transformation can send their final labels to two different values, provided each local map is increasing and avoids the label already used on its own prefix. Nothing in sibling order or common-prefix lengths identifies these two occurrences of the original label 1. [Section 93](#) isolates exactly the relation that prohibits such a distortion.

## 90 Exact configuration homogeneity (source 25)

**Theorem 90.1** (Classification of set-indexed branch configurations). *Let  $(f_i)_{i \in I}$  and  $(g_i)_{i \in I}$  be uniformly set-indexed families of pairwise distinct members of  $\mathfrak{J}$ . There exists an automorphism  $(F, T)$  of  $\mathbf{C}_b$  such that*

$$\widehat{T}(f_i) = g_i \quad (i \in I)$$

*if and only if the following conditions hold:*

$$f_i \in \mathcal{P}_s(b) \iff g_i \in \mathcal{P}_s(b), \tag{S25.2}$$

$$\delta(f_i, f_j) = \delta(g_i, g_j) \quad (i \neq j), \tag{S25.3}$$

$$f_i <_{\text{lex}} f_j \iff g_i <_{\text{lex}} g_j \quad (i \neq j). \tag{S25.4}$$

*The construction uses GBC, not ETR.*

*Proof.* Necessity follows from [Proposition 88.2](#), the induced branch action, and the fact that  $F$  maps the core onto itself.

For sufficiency, start with the prescribed correspondences between the core members among the  $f_i$  and  $g_i$ . Their unique normalized codes form sets by class replacement. The remaining  $f_i, g_i$  are retained as fixed external-to-the-core constraints. The word “external” here means outside  $\mathcal{P}_s(b)$ , not outside the model or its admitted classes.

Choose a class enumeration of all core codes, indexed by Ord. Construct a growing partial bijection  $H$  between core codes by back-and-forth. At each set stage  $\alpha$ , its graph  $H_\alpha$  is a set. Maintain the invariant that all prescribed correspondences, consisting of  $H_\alpha$  and the noncore markers, preserve the profiles of distinct branches.

At a forth step, take the next enumerated core code  $d$  not yet in the domain. It is distinct from every source constraint: the core constraints already lie in the domain, and the other markers are noncore. [Lemma 89.1](#), applied to the uniformly set-indexed current constraints, gives a core target  $c$  with exactly the required profiles. In particular,  $c$  differs from every current target, so adding  $(d, c)$  preserves injectivity and the invariant.

At a back step, take the next core code not in the range and apply the same lemma with sources and targets interchanged. At a limit set stage, take the union of the earlier partial maps. This is still a set, and every pairwise requirement appeared together at an earlier stage. Choose the least possible witness at each extension step. Thus ordinary set-valued recursion along  $\text{Ord}$  defines the whole construction in GBC.

The union  $F$  is a class bijection of  $\mathcal{P}_s(b)$  preserving every core profile. It is an order automorphism, and [Proposition 88.2](#) gives its unique node action  $T$ .

For a noncore marker  $f_i$ , fix an ordinal  $\alpha$  and choose a core completion  $d$  of  $f_i \restriction \alpha$ . Since  $d \neq f_i$ , their first disagreement is at least  $\alpha$ . The maintained constraint implies that  $F(d)$  agrees with  $g_i$  below  $\alpha$ . By the definition of the node action,

$$T(f_i \restriction \alpha) = F(d) \restriction \alpha = g_i \restriction \alpha.$$

This holds at every set-ordinal length, so  $\hat{T}(f_i) = g_i$ . The prescribed core markers were already mapped correctly by  $F$ . This proves sufficiency.  $\square$

**Corollary 90.2** (Exactly two single-branch orbits). *Under the induced action of automorphisms of  $\mathbf{C}_b$ , the virtual domain  $\mathfrak{J}$  has precisely two orbits:*

$$\mathcal{P}_s(b) \quad \text{and} \quad \mathfrak{J} \setminus \mathcal{P}_s(b).$$

*More explicitly, any two supplied core branches can be carried to one another, and any two supplied noncore injective branches can be carried to one another, but no core branch can be carried to a noncore branch.*

*Proof.* Apply [Theorem 90.1](#) to a one-element index set. Only the membership condition remains. Both cases are nonempty:  $b$  is a core branch, and an injection omitting a label is not.  $\square$

*Remark 90.3* (The logical meaning of an orbit). The corollary abbreviates a quantified assertion about supplied class words and class maps on set codes. It does not construct an ordinary class whose elements are all automorphisms, all branches, or their equivalence classes. In the regular set model of [Section 94](#), these become ordinary sets and an ordinary group action.

The theorem also determines all finite and set-indexed tuple orbits. For example, a pair of noncore branches is classified solely by its first-disagreement ordinal and its order orientation. A triple must satisfy the usual compatibility of first-disagreement data, but once both triples exist, no additional cross-coordinate label pattern is an invariant. Repeated entries can be admitted by first matching their equality pattern and then passing to distinct representatives; a set of representatives exists by set choice.

*Note 90.4* ([merge]; how this compares with Part VIII). Source 14's [theorem 47.2](#) classifies cuts of the bare core order by their characters; the cuts of all noncore injective words have character  $(\text{Ord}, \text{Ord})$  and lie in one orbit there with every other cut of that character, for example the edges of successor-length cylinders. An automorphism that also carries cylinders to cylinders preserves the crossing condition of [theorem 45.1](#), so it maps branch cuts to branch cuts; [theorem 90.1](#) shows that on branches the remaining invariants are exactly core membership, first-disagreement ordinals



and order. Source 27's [theorem 96.2](#) and source 29's [theorem 96.7](#) keep the ordered prefix tree but not the core, and obtain one orbit of all injective branches; [corollary 90.2](#) shows that retaining the core splits it into exactly two.

## 91 Exhaustiveness is invisible to unlabelled cylinders (source 25)

### 91.1 An explicit pair of branches

Let  $e$  be obtained from  $b$  by swapping each pair of consecutive positions

$$2 \cdot \xi, \quad 2 \cdot \xi + 1 \quad (\xi \in \text{Ord}).$$

The finite factor is on the *left* in ordinal multiplication. These pairs partition  $\text{Ord}$ . The word  $e$  is a bijection onto  $\mathbf{No}$  and differs from  $b$  at every position, so  $e \notin \mathcal{P}_s(b)$ .

Define another class word by

$$f(\alpha) = b(1 + \alpha).$$

The map  $\alpha \mapsto 1 + \alpha$  is an increasing bijection from  $\text{Ord}$  onto its nonzero ordinals. Thus  $f$  is injective and omits exactly  $b(0)$ . Moreover,  $1 + \alpha = \alpha$  for every infinite ordinal  $\alpha$ , so  $f$  eventually agrees with  $b$ . This illustrates why eventual agreement without exhaustiveness does not characterize the core.

**Theorem 91.1** (Cylinder preservation does not preserve coverage). *There is an automorphism  $(F, T)$  of  $\mathbf{C}_b$  and branches  $e \in \mathfrak{W}_{\text{sl}}$ ,  $f \in \mathfrak{J} \setminus \mathfrak{W}_{\text{sl}}$  such that*

$$\hat{T}(e) = f.$$

*It preserves the core order, every cylinder as a member of the cylinder family, all prefix lengths, and all first-disagreement ordinals. Nevertheless, it carries an exhaustive branch to one omitting a label.*

*Proof.* The preceding  $e$  and  $f$  are both noncore. Apply [Corollary 90.2](#). □

Here preserving the cylinder family means permuting its members, not fixing every cylinder pointwise. Fixing every cylinder would fix every core branch and every supplied full branch.

### 91.2 The obstruction survives set-many parameters

**Theorem 91.2** (Parameter-robust non-invariance). *Let  $A \subseteq \mathcal{P}_s(b)$  and  $P \subseteq \mathcal{I}$  be sets. There is an automorphism of  $\mathbf{C}_b$  fixing every member of  $A$  and every node of  $P$ , but carrying some exhaustive noncore branch to a nonexhaustive injective branch.*

*Proof.* For each  $p \in P$ , choose a core completion  $d_p$ . Add these codes to  $A$ , obtaining a set  $A'$ . Choose a surreal  $x$  outside the set of first labels  $\{d(0) : d \in A'\}$ , and choose a core branch  $d_0$  beginning with  $x$ .

Construct an exhaustive branch  $e$  beginning with  $x$  by leaving the first position of  $d_0$  fixed and swapping a proper class of disjoint consecutive pairs in its tail. It differs from  $d_0$ , and hence from  $b$ , on a proper class of positions, so it is noncore. Construct  $f$  by

$$f(0) = d_0(0), \quad f(\alpha) = d_0(1 + \alpha) \quad (\alpha > 0).$$

This word begins with  $x$ , is injective, and omits  $d_0(1)$ .

For each  $a \in A'$ , both  $e$  and  $f$  disagree with  $a$  at coordinate 0, and their comparison directions are identical because their first label is  $x$ . The two configurations consisting of all the fixed branches in  $A'$  followed by  $e$ , respectively  $f$ , therefore satisfy [Theorem 90.1](#). The resulting automorphism fixes  $A'$ . It fixes each  $p \in P$  because  $p$  is the prefix of the fixed  $d_p$  at its invariant length. □

**Corollary 91.3** (No invariant unlabelled recognition test). *Exhaustiveness is not invariantly recognizable from  $\mathbf{C}_b$  and any prescribed set of core and node parameters. In particular, an automorphism-invariant definition in this reduct cannot recognize precisely the exhaustive full branches.*

This is deliberately a statement about the specified reduct and admitted branch action. In an ambient set theory that retains the full coordinate-value relation, exhaustiveness is simply the condition that every surreal occurs. We have not made that condition undefinable in the ambient theory; we have shown that the reduct forgets information needed to recover it. The result also does not assert that every enrichment strictly weaker than the one in [Section 93](#) fails.

### 91.3 Other properties lost by forgetting labels

The branch  $\alpha \mapsto \alpha$ , viewing each ordinal as a surreal, is numerically increasing and nonexhaustive. It is therefore noncore. [Corollary 90.2](#) places it in the same orbit as every other noncore injective branch, including an exhaustive branch. Consequently numerical increasingness is also invisible in the unlabelled cylinder structure. It concerns comparisons between labels occurring at *different* coordinates, whereas the prefix geometry orders siblings at one coordinate after a common prefix.

Similarly, the number or order type of omitted labels is not an invariant on the noncore orbit. This statement applies only to omitted ranges realized by supplied injective class words; it does not posit a same-level collection of all class cardinalities or all class order types.

*Note 91.4* ([merge]; relation to the bare-order results). [Theorem 91.1](#) strengthens [corollary 47.3](#), [theorem 47.7](#) with [corollary 47.8](#), and [theorem 47.10](#): those move an exhaustive cut by an automorphism of the bare core, whereas here every cylinder is carried to a cylinder. [Theorem 91.2](#) is the cylinder-preserving counterpart of source 20's [theorem 97.7](#), which fixes a set of core points pointwise in the bare order (written before Part VIII; source 25 did not cite it). Source 13's [theorem 47.12](#) is a different theorem: there an automorphism of  $\mathfrak{W}_{sl}$  moves a cylinder to a non-cylinder.

## 92 Extension to the full enumeration order (source 25)

[merge] Source 25's Section 9. Its Proposition 9.1 is source 17's [theorem 47.5](#), which source 25 credits; it is printed as a note with source 25's proof.

The previous sections construct an automorphism of the small-coded core and its full injective prefix completion. Does it extend to an order automorphism of the exhaustive domain  $\mathfrak{W}_{sl}$ ? The answer can fail even when the core automorphism respects every prefix cylinder.

*Note 92.1* (Source 25, Proposition 9.1, “Represented-trace criterion”). *Let  $F$  be an order automorphism of  $\mathcal{P}_s(b)$ . It has an order-automorphism extension to  $\mathfrak{W}_{sl}$  exactly when it permutes, in both directions, the cuts*

$$L_e = \{d \in \mathcal{P}_s(b) : d <_{\text{lex}} e\} \quad (e \in \mathfrak{W}_{sl})$$

*that are represented by admitted exhaustive branches. If such an extension exists, it is unique. The statement about an extension is understood as a uniform transformation of supplied class words, or externally in a larger universe. This is [theorem 47.5](#) (source 17's Theorem 10.1); sources 20 and 21 prove it again as [notes 97.8](#) and [97.13](#).*

*Source 25's proof.* Any extension must carry  $L_e$  to  $L_{G(e)}$ . Conversely, assume both  $F$  and  $F^{-1}$  carry represented traces to represented traces. By [Note 86.6](#), the branch representing  $F[L_e]$  is unique. Define  $G(e)$  to be this branch. The corresponding construction for  $F^{-1}$  gives its inverse. Density of  $\mathcal{P}_s(b)$  in  $\mathfrak{J}$  shows that trace inclusion strictly reflects the order, so  $G$  is increasing and extends  $F$ .

There is no need to choose from a collection of classes to make the uniform statement precise. For a supplied trace  $L$ , the condition that a prefix cylinder crosses its lower and upper sides is elementary in  $L$  and the core codes. When  $L$  is represented by an injective full branch, these crossing cylinders are exactly its prefixes: an agreeing prefix has core completions on both sides, while an incompatible prefix has its entire cylinder on one side. The unique prefix of each length is therefore recovered by elementary comprehension, and its union defines the representing branch. Apply this decoder to  $F[L_e]$ .  $\square$

The criterion is a conceptual reduction rather than a local classification of the stabilizing automorphism group. The criterion was already proved in the second continuation (source 17's Theorem 10.1, [theorem 47.5](#)); it is re-established here to fix the uniformity convention. The next conclusion strengthens the known obstruction by retaining all cylinders.

**Theorem 92.2** (A cylinder-preserving automorphism with no exhaustive extension). *There exists an order automorphism  $F$  of  $\mathcal{P}_s(b)$  preserving all prefix cylinders that has no order-automorphism extension to  $\mathfrak{W}_{sl}$ , even if the prospective extension is not required to preserve cylinders.*

*Proof.* Choose  $F, T, e, f$  as in [Theorem 91.1](#). The induced order-preserving branch action gives

$$F[L_e] = L_f.$$

An order-automorphism extension  $G$  on  $\mathfrak{W}_{sl}$  would also give  $F[L_e] = L_{G(e)}$ . Trace injectivity on all of  $\mathfrak{J}$  would imply  $G(e) = f$ . This contradicts  $f \notin \mathfrak{W}_{sl}$ .  $\square$

For a cylinder-preserving  $F$ , the action  $\hat{T}$  on  $\mathfrak{J}$  is already determined. The precise condition for an extension to  $\mathfrak{W}_{sl}$  is therefore

$$\hat{T}[\mathfrak{W}_{sl}] = \mathfrak{W}_{sl}.$$

This is not automatic from preserving the core, the lexicographic order, or the entire unranked prefix tree. Exhaustive completion is an additional choice of represented branches inside the injective completion.

[merge] [Theorem 92.2](#) answers in part Part VIII's open item on the stabilizer of the exhaustive cuts ([section 50](#) (c)): not every cylinder-preserving automorphism of  $\mathcal{P}_s(b)$  lies in that subgroup of  $\text{Aut}(\mathcal{P}_s(b))$ . Source 19's [theorem 98.5](#) gives the exact criterion for extension to the unrestricted order in an external set-sized model.

## 93 Cross-node label coherence and rigidity (source 25)

[merge] Source 25's Section 10. Source 27 proves [theorem 93.1](#) independently as its Theorem 6.5, printed as a note in [section 96.1](#) ([note 96.5](#)).

### 93.1 A single additional relation

An extension edge is a pair  $(p, q)$  of nodes such that  $q = p \frown (x)$  for one unused label  $x$ . These are exactly the immediate-successor pairs in the prefix order. Add a relation  $\sim_{\text{lab}}$  on edges by

$$(p, p \frown (x)) \sim_{\text{lab}} (q, q \frown (y)) \iff x = y.$$

The relation connects occurrences of one label at different nodes. It does not initially name any label, any birthday, or any field operation.

Each equivalence class of edges has exactly one root edge, namely  $(\emptyset, (x))$ . Thus labels can be represented by these existing set-coded root edges. There is no quotient whose elements would have to be proper classes. The root edges inherit the numerical order from their cylinders, and this reconstructs the ordered alphabet.

For the next theorem, consider the ordered prefix tree by itself: its node sort, prefix order, and the order of immediate successors at each node. This sibling order can either be included directly or recovered from the core cylinders.

**Theorem 93.1** (Label-coherent tree automorphisms). *An automorphism  $T$  of the ordered injective prefix tree preserves  $\sim_{\text{lab}}$  if and only if there is an order automorphism  $h$  of  $(\mathbf{No}, <)$  such that*

$$T(p) = h \circ p \quad (p \in \mathcal{I}).$$

*The alphabet automorphism  $h$  is unique.*

*Proof.* The action on root edges defines a unique bijection  $h : \mathbf{No} \rightarrow \mathbf{No}$  by

$$T((x)) = (h(x)).$$

Preservation of their sibling order makes  $h$  increasing. If the edge from  $p$  to  $p \frown (x)$  is considered, it is label-equivalent to the root edge for  $x$ . Its image must be label-equivalent to the root edge for  $h(x)$ . Therefore

$$T(p \frown (x)) = T(p) \frown (h(x)).$$

Induction along the set-ordinal length of  $p$  proves  $T(p) = h \circ p$ . At a limit length, the node is the union of its proper prefixes, and their images determine it uniquely. Conversely, coordinatewise application of any order automorphism  $h$  preserves injectivity, prefix order, sibling order, and edge-label equality. Its inverse is coordinatewise application of  $h^{-1}$ .  $\square$

**Lemma 93.2** (No nontrivial set-supported alphabet automorphism). *If an order automorphism  $h$  of  $\mathbf{No}$  fixes all points outside a set, then  $h = \text{id}$ .*

*Proof.* Suppose  $x < h(x)$ . Choose, by [Note 86.3](#), a point  $y$  strictly between them and outside the support of  $h$ . Monotonicity gives  $h(x) < h(y) = y$ , contradicting  $y < h(x)$ . If  $h(x) < x$ , choose a fixed point  $y$  between  $h(x)$  and  $x$ ; monotonicity gives  $y = h(y) < h(x)$ , again a contradiction. Thus every point is fixed.  $\square$

**Theorem 93.3** (The label-coherent core structure is rigid). *Every automorphism  $(F, T)$  of  $\mathbf{C}_b$  preserving  $\sim_{\text{lab}}$  is the identity. No numerical labels need to be named individually.*

*Proof.* [Theorem 93.1](#) gives  $T(p) = h \circ p$  for an alphabet order automorphism  $h$ . Hence the induced action on the baseline is

$$F(b) = \hat{T}(b) = h \circ b.$$

Because  $F(b) \in \mathcal{P}_s(b)$ , there is a set-supported permutation  $\pi$  of  $\text{Ord}$  such that  $h(b(\alpha)) = b(\pi(\alpha))$ . If  $S$  is the set support of  $\pi$ , then  $h$  fixes all labels outside the set  $b[S]$ . [Lemma 93.2](#) implies  $h = \text{id}$ . Thus  $T = \text{id}$ , and the prefixes of every core branch force  $F = \text{id}$  as well.  $\square$

There is a sharp contrast here. Without cross-node label equality the core structure has two large branch orbits and is homogeneous for all the configurations in [Theorem 90.1](#). After adding just this one relation, it is rigid. For the ordered tree without the distinguished supported core, the same relation still permits all coordinatewise alphabet automorphisms. The rigidity is therefore a joint effect of label coherence and the chosen eventual-agreement core.

**Proposition 93.4** (Coverage from coherent edges). *In the edge-coherent expansion, a supplied full branch is exhaustive if and only if every root edge is label-equivalent to some extension edge on the branch.*

*Proof.* Root edges represent every surreal label exactly once. An edge on the branch represents exactly the label introduced at its corresponding coordinate. The condition is therefore equivalent to every surreal occurring in its range.  $\square$

[Theorem 91.1](#) and [Proposition 93.4](#) give a negative and a positive recognition result in explicit languages. They do not establish that  $\sim_{\text{lab}}$  is the weakest conceivable enrichment that recognizes exhaustiveness; its rigidity is stronger than recognition alone, and weaker intermediate expansions are a natural subject for further study.

## 94 Regular set-sized models and an exact group size (source 25)

[merge] Source 25's Section 11. Its set structures are written with a superscript  $\kappa$  ([part XIII](#)):  $\mathcal{P}_s^\kappa(b)$  is source 25's  $\mathcal{D}_{b,\kappa}$ ,  $\mathcal{I}^\kappa$  its  $\mathcal{T}_\kappa$ ,  $\mathfrak{J}^\kappa$  its  $\mathfrak{J}_\kappa$ ,  $\mathcal{W}_\kappa(X)$  its  $\mathfrak{E}_\kappa$ , and  $\mathbf{C}_b^\kappa$  its  $\mathcal{C}_{b,\kappa}$ .

The class argument has a useful set-sized analogue. This provides ordinary sets of branches and an ordinary automorphism group, making the size claims and the nondefinability obstruction accessible without a higher-order interpretation.

### 94.1 The hypotheses and their purpose

Let  $\kappa$  be an infinite regular cardinal such that  $\kappa^{<\kappa} = \kappa$ . Let  $X$  be a linearly ordered set of cardinality  $\kappa$  satisfying

$$L < R, \quad |L \cup R| < \kappa \quad \implies \quad \exists x \in X \ (L < x < R). \quad (\text{S25.5})$$

Empty sides are permitted. This includes density and absence of endpoints.

The analogous fresh-separator property follows. First choose  $z$  between  $L$  and  $R$ . For a forbidden set  $S$  of size less than  $\kappa$ , apply (S25.5) again to

$$L \cup \{z\} \quad \text{and} \quad R \cup \{s \in S : z < s\}.$$

The resulting point lies above  $z$  but below every member of  $S$  above  $z$ , so it avoids  $S$ .

Fix a bijection  $b : \kappa \rightarrow X$ . Define

$$\begin{aligned} \mathcal{P}_s^\kappa(b) &= \{b \circ \pi : \pi \in \text{Sym}(\kappa), \ |\text{supp } \pi| < \kappa\}, \\ \mathcal{I}^\kappa &= \bigcup_{\alpha < \kappa} \text{Inj}(\alpha, X), \\ \mathfrak{J}^\kappa &= \text{Inj}(\kappa, X), \quad \mathcal{W}_\kappa(X) = \text{Bij}(\kappa, X). \end{aligned}$$

These are ordinary sets. Regularity makes support of size less than  $\kappa$  equivalent to bounded support. The code normalization used for the class core may be retained, although here the functions themselves are sets.

Every prefix in  $\mathcal{I}^\kappa$  has a core completion: the reserve argument in [Note 86.4](#) uses fewer than  $\kappa$  positions. Also

$$|\mathcal{P}_s^\kappa(b)| = |\mathcal{I}^\kappa| = \kappa.$$

The upper bounds follow from  $\kappa^{<\kappa} = \kappa$ , and the lower bounds follow already from distinct one-letter prefixes and their completions.

**Theorem 94.1** (Regular-cardinal configuration theorem). *Let  $\mathbf{C}_b^\kappa$  be the set-sized core-and-cylinder structure just defined. For configurations indexed by a set  $I$  of cardinality less than  $\kappa$ , [Theorem 90.1](#) holds with  $\mathcal{P}_s^\kappa(b)$ ,  $\mathfrak{J}^\kappa$ , and first-disagreement ordinals below  $\kappa$  in place of their class counterparts.*

*In particular, its action on  $\mathfrak{J}^\kappa$  has exactly two orbits, the core and its complement. The exhaustive branches are not invariant. Adding cross-node edge-label equality makes  $\mathbf{C}_b^\kappa$  rigid.*

*Proof.* The one-point proof applies verbatim once its supremum  $\lambda$  is below  $\kappa$ . This follows from regularity because there are fewer than  $\kappa$  constraints. The maximum case uses the fresh-separator property above, and the no-maximum case gives a prefix of length less than  $\kappa$ . Complete it to the core.

Enumerate the core in  $\kappa$  steps. At each stage below  $\kappa$ , the partial map together with the fixed markers has size less than  $\kappa$ , so back-and-forth can continue. Its union gives the required automorphism and branch action.

For non-invariance, pair-swap  $b$  throughout  $\kappa$  to obtain a noncore bijection, and use  $f(\alpha) = b(1 + \alpha)$  to omit  $b(0)$ . The latter is defined on  $\kappa$  and has the indicated omitted range for every infinite initial ordinal  $\kappa$ . Both are noncore and hence conjugate under the branch action.

For rigidity, an edge-coherent automorphism is coordinatewise application of an order automorphism  $h$  of  $X$ . Preservation of the supported core forces  $h$  to have support of size less than  $\kappa$ . The fresh-separator property supplies a fixed point between  $x$  and  $h(x)$  whenever they differ, giving the contradiction in [Lemma 93.2](#).  $\square$

**Theorem 94.2** (Exact cardinality of the automorphism group). *Under the preceding hypotheses,*

$$|\text{Aut}(\mathbf{C}_b^\kappa)| = 2^\kappa.$$

*After adjoining  $\sim_{\text{lab}}$ , the automorphism group has cardinality 1.*

*Proof.* An automorphism is determined by its action on the core, which has cardinality  $\kappa$ . Thus the group has size at most  $\kappa^\kappa = 2^\kappa$ .

There are  $2^\kappa$  injective full words: the upper bound is  $|X^\kappa| = 2^\kappa$ , and independent pair swaps of  $b$  give the lower bound. Removing the core, which has size  $\kappa$ , leaves  $2^\kappa$  words. By [Theorem 94.1](#), these form one orbit. A group acting transitively on a set of that size has at least  $2^\kappa$  elements. The equality follows. The enriched assertion is the rigidity theorem.  $\square$

## 94.2 Surreal birthday fragments and the countable example

If  $\kappa$  is regular and  $\kappa^{<\kappa} = \kappa$ , the alphabet  $X = \mathbf{No}_{<\kappa}$  has cardinality  $\kappa$  and satisfies [\(S25.5\)](#). Indeed, fewer than  $\kappa$  birthdays are bounded strictly below  $\kappa$  by regularity, and the exact-birthday separator may be chosen at a still larger ordinal below  $\kappa$ . The equality  $|X| = \kappa$  uses  $2^{<\kappa} = \kappa$ , which follows from the stated cardinal-arithmetic hypothesis for infinite  $\kappa$ .

At  $\kappa = \omega$ , one may take  $X = \mathbb{Q}$ , or the dyadic rational alphabet  $\mathbf{No}_{<\omega}$ . The core consists of the finite-support permutations of a fixed enumeration. Its countable ordered cylinder structure has  $2^{\aleph_0}$  automorphisms, and every infinite injective word outside the core lies in one orbit of their induced action. Yet the edge-coherent expansion is rigid. No continuum hypothesis is used.

This countable example is a concrete testing ground for the group-theoretic questions below. Its infinitary branch space is still indispensable: a complete ordered injection tree over a *finite* linear alphabet is rigid, since every finite sibling order is rigid. One should not expect an exhaustive finite-alphabet model to exhibit the same symmetries.

[merge] The bare countable order  $\mathcal{W}_\omega(\mathbb{Q})$  also has  $2^{\aleph_0}$  automorphisms (source 21's homogeneity theorem, Part X); that is a different structure, without cylinders.



## 95 Cofinality obstructions and singular lengths (source 25)

[merge] Source 25's Section 12. Here  $X$  is any linearly ordered set of size  $\kappa$  with a bijection  $b : \kappa \rightarrow X$ ;  $\mathcal{P}_s^{\text{bd}}(b)$  and  $\mathcal{P}_s^{<\kappa}(b)$  are source 25's  $\mathcal{D}_{\text{bd}}$  and  $\mathcal{D}_{<\kappa}$ , its bounded-support and small-support cores. At a singular  $\kappa$  they differ, as source 14's support hierarchy (section 49.1) shows for the class core.

The regularity restriction in the preceding proof is not cosmetic. A family of approximants cofinal in the word length can determine a noncore full word completely, leaving no core point with its profile.

**Theorem 95.1** (Cofinal profiles force the missing branch). *Let  $\kappa$  be any infinite cardinal and let  $b : \kappa \rightarrow X$  be a bijection onto any linearly ordered alphabet of size  $\kappa$ . Let  $\mathcal{P}_s^{\text{bd}}(b)$  be its bounded-support core and  $\mathcal{P}_s^{<\kappa}(b)$  its support-of-size-less-than- $\kappa$  core. There are a bijection  $d : \kappa \rightarrow X$  outside both cores and a family  $(a_i)_{i < \text{cf } \kappa}$  lying in both cores such that the only full word  $z$  satisfying*

$$\delta(z, a_i) = \delta(d, a_i) \quad (i < \text{cf } \kappa)$$

*is  $z = d$ . In particular, no member of either core realizes that profile over the fixed family.*

*Proof.* Let  $d$  swap every consecutive pair of  $b$ , with pair blocks indexed by  $\xi < \kappa$ . Choose pair-boundary ordinals  $\beta_i = 2 \cdot \xi_i < \kappa$  strictly increasing and cofinal in  $\kappa$ , for  $i < \text{cf } \kappa$ . Such boundaries are cofinal because the ordinal map  $\xi \mapsto 2 \cdot \xi$  is continuous and cofinal in every infinite initial ordinal.

Define  $a_i$  by performing the same swaps below  $\beta_i$  and leaving all positions at or above  $\beta_i$  fixed. Its support is bounded and has size less than  $\kappa$ , and

$$\delta(d, a_i) = \beta_i.$$

If  $z$  has the indicated first-disagreement ordinals, then  $z \upharpoonright \beta_i = a_i \upharpoonright \beta_i = d \upharpoonright \beta_i$  for every  $i$ . Cofinality forces  $z = d$ . But  $d$  moves all  $\kappa$  positions, so it belongs to neither core.  $\square$

For regular  $\kappa$ , the obstruction occurs at  $\kappa$  constraints, immediately beyond the less-than- $\kappa$  extension theorem. This establishes sharpness of that bound for the one-point core-realization mechanism. It does not by itself classify every possible larger partial automorphism.

For singular  $\kappa$ , the obstruction already occurs at  $\text{cf } \kappa < \kappa$  constraints. Consequently the less-than- $\kappa$  one-point extension statement fails even if one replaces bounded support by support of cardinality less than  $\kappa$ . The failure is not repaired merely by adopting a larger small-support core.

**Remark 95.2** (A second singular obstruction: unbounded prefix images). At a singular  $\kappa$ , not every injective prefix of length less than  $\kappa$  extends to the bounded-support core. Choose an injection  $q : \text{cf } \kappa \rightarrow \kappa$  with cofinal range and put  $p = b \circ q$ . A bounded-support permutation extending  $q$  would have to move positions to cofinally many ordinals above its support bound, which is impossible. In contrast, the reserve construction does extend this  $q$  to a permutation of support cardinality less than  $\kappa$ . Thus bounded support, small support, and prefix completeness must be separated at singular lengths.

The global GBC theorem has no analogous set-sized obstruction: every set of disagreement ordinals has a strict ordinal bound. A uniformly proper-class family of approximants can, however, force an entire noncore branch by the same argument. The set-indexing hypothesis is exactly what prevents this cofinality failure in Lemma 89.1.

## 96 Ordered prefix trees without the core (sources 27 and 29)

[merge] Sources 27 and 29 keep the ordered tree of set-length words (lengths, prefixes and the numerical order of children) but not the core. Both prove that its automorphisms act transitively on supplied injective class branches ([theorem 96.2](#); source 29's [theorem 96.7](#), which also covers the increasing-word tree), so exhaustiveness is not invariant ([corollary 96.3](#) and [note 96.8](#)). This is weaker than source 25's [theorem 91.1](#), where the core and every cylinder are preserved as well; retaining the core is what splits the single orbit into two ([note 90.4](#)). The new contributions here are the local description of every tree automorphism and the exact obstruction to preserving exhaustion ([theorem 96.4](#)), two families that do preserve it (source 27's antichain transformations and source 29's [proposition 96.11](#)), source 29's [corollary 96.12](#) (no fixed set height detects the failure), and source 29's increasing tree, where numerical cofinality is not invariant either ([corollary 96.9](#)). Both sources were written after Part VIII; neither cites source 25.

### 96.1 Source 27: prefix symmetries, exhaustiveness, and repeated labels

The corpus shows that the bare order of the bounded-support core does not recognize exhaustive branch cuts ([corollary 47.3](#) and [theorem 47.7](#)). A natural response is to retain the prefix tree as additional structure. We show that even this response is insufficient unless the structure also remembers which labels at different nodes are the same surreal.

Work in GBC with a fixed baseline  $b : \text{Ord} \rightarrow \mathbf{No}$ . Let  $\mathcal{I}$  be the ordinary class of all injective set-length surreal words. Its tree order is initial segment, and the children of  $p$  are  $p \frown (x)$  for  $x \in \mathbf{No} \setminus \text{ran } p$ , ordered by numerical  $x$ . An automorphism of this ordered tree is a class bijection preserving prefixes and each sibling order. It automatically preserves ordinal height, since the proper predecessors of a node of length  $\alpha$  have order type  $\alpha$ .

Every class injection  $E : \text{Ord} \rightarrow \mathbf{No}$  determines a maximal branch consisting of all  $E \upharpoonright \alpha$ . It is maximal whether or not  $E$  is onto: a compatible set-length node must be the corresponding prefix of  $E$ . Thus maximality is not exhaustiveness.

#### 96.1.1 A uniform isomorphism lemma with only set states

**Lemma 96.1** (Uniform pointed residual isomorphisms). *There is one class graph, definable from  $b$ , which for every pair of sets  $S, T \subseteq \mathbf{No}$  and points  $a \notin S$ ,  $c \notin T$  gives an increasing bijection*

$$J(S, T; a, c) : \mathbf{No} \setminus S \longrightarrow \mathbf{No} \setminus T, \quad J(S, T; a, c)(a) = c.$$

*There is also an unpointed version. The graph has set tuples as its elements; it is not a class whose elements are class maps.*

*Proof.* Deleting a set from  $\mathbf{No}$  preserves its ability to fill every separated set cut. To avoid a deleted set of cardinality  $\mu$ , first construct  $\mu^+$  distinct points between the two sides, using the surreal set-cut property at each set stage. At least one remains outside the deleted set. The same argument permits avoidance of the values already used by a set partial map.

Start with the increasing partial map  $\{(a, c)\}$ , or the empty map in the unpointed version. At ordinal stage  $\beta$ , handle the source label  $b(\beta)$  if it is eligible and unused. Its image must lie between the images of its earlier lower and upper domain elements. Choose the feasible fresh target of least  $b$ -index. Next handle  $b(\beta)$  on the target side by the symmetric extension of the inverse. The preceding cut argument ensures both steps are possible. At limits take unions. Every stage is a set partial increasing bijection, and every eligible label on both sides is eventually handled.

For each set ordinal  $\delta$ , the run through  $\delta$  is a unique set sequence. A pair belongs to the final map exactly when some such set run contains it. This is an elementary formula with parameters  $b, S, T, a, c$ . Elementary class comprehension therefore defines one uniform graph for all allowed set parameters. No stage of this construction chooses or collects a proper-class object as a state.  $\square$

The same proof works for the permissible next-letter orders of increasing words used in [section 82.2](#): those orders also fill every set cut, and their restrictions of the baseline enumerate them. The distinction between set-state recursion and a recursion whose individual values are classes is essential to the axiom count.

### 96.1.2 Every tree automorphism is governed by local residual maps

For an ordered-tree automorphism  $A$ , define its local map by

$$A(p \frown (x)) = A(p) \frown (\lambda_p(x)). \quad (\text{S27.8})$$

Then  $\lambda_p$  is an increasing bijection from  $\mathbf{No} \setminus \text{ran } p$  onto  $\mathbf{No} \setminus \text{ran } A(p)$ , uniformly in  $p$ , and

$$A(p)(\xi) = \lambda_{p \restriction \xi}(p(\xi)).$$

Conversely, suppose one class rule  $\Lambda(p, q; x, y)$  gives an increasing residual bijection from  $\mathbf{No} \setminus \text{ran } p$  onto  $\mathbf{No} \setminus \text{ran } q$  for every equal-height pair  $p, q$ . Define the image  $q = A(p)$  by the set recursion

$$q(\xi) = \Lambda(p \restriction \xi, q \restriction \xi)(p(\xi)). \quad (\text{S27.9})$$

The prefixes cohere by uniqueness. Inverting the residual maps in a recursion on a target word constructs its unique preimage, so  $A$  is a bijection at every height. It preserves prefixes and sibling orders. Thus (S27.9) is a precise way to construct an ordered-tree automorphism without selecting unrelated class maps at each node.

**Theorem 96.2** (Transitivity on all injective ordinal branches). *Given any class injections  $E, F : \text{Ord} \rightarrow \mathbf{No}$ , there exists an ordered-prefix-tree automorphism  $A$  such that*

$$A(E \restriction \alpha) = F \restriction \alpha \quad (\alpha \in \text{Ord}). \quad (\text{S27.10})$$

*It preserves ordinal levels, all prefix incidences, and numerical order among the children of each node.*

*Proof.* For equal-height  $p, q$  of length  $\alpha$ , use the pointed residual map

$$J(\text{ran } p, \text{ran } q; E(\alpha), F(\alpha))$$

when  $p = E \restriction \alpha$  and  $q = F \restriction \alpha$ . Its distinguished points are fresh by injectivity. In every other case use the unpointed residual map. This is a single elementary rule with class parameters  $b, E, F$ . Apply (S27.9). Induction on each set ordinal forces the pointed case along the designated branch, which proves (S27.10).  $\square$

**Corollary 96.3** (Ordered prefixes do not recognize exhaustiveness). *Exhaustiveness of an Ord-branch is not invariant under automorphisms of the ordered prefix tree, even if levels and sibling numerical order are retained.*

*Proof.* Take  $E = b$  and define

$$F(n) = b(n+1) \quad (n < \omega), \quad F(\xi) = b(\xi) \quad (\xi \geq \omega).$$

The map  $F$  is injective and omits exactly  $b(0)$ . Apply [theorem 96.2](#). The exhaustive branch is sent to a nonexhaustive branch, so no property invariant under this structure's automorphisms can distinguish the two.  $\square$

The displayed branches even agree outside a bounded set of coordinates. Agreement outside a set is not the same as being related by a permutation supported on that set: the infinite one-sided shift loses a label. This is the same boundary phenomenon measured by the reserve in [section 101](#).

For a set alphabet  $X$ , the full injective-word tree has a different answer. Its terminal nodes are exactly the exhaustive words, and any tree automorphism preserves terminal nodes. Every maximal branch has such a terminal node, because an injective word has length below the Hartogs number of  $X$  and a missing label would allow extension. This argument does not apply to a truncated set tree that stops before the terminal height, nor to the full proper-class tree.

### 96.1.3 An exact obstruction and useful sufficient bounds

An automorphism acts on a supplied class branch by the elementary formula

$$A_*E(\xi) = A(E \upharpoonright (\xi + 1))(\xi).$$

It acts on exhaustive branches precisely when it preserves coverage of the alphabet. A local formulation makes the obstruction explicit. For a target label  $y$ , put

$$\mathcal{I}_y(A) = \{p : y \notin \text{ran } A(p)\}.$$

This is a downward-closed subtree. At a node in it, exactly one available input child is forbidden if the branch is to continue omitting  $y$ , namely  $\lambda_p^{-1}(y)$ . Therefore  $A_*$  sends every exhaustive branch to an exhaustive one exactly when none of these subtrees has an exhaustive Ord-branch. To obtain a bijection on exhaustive branches, impose the same condition on  $A^{-1}$ . This criterion is exact, but it deliberately leaves a substantial branch problem.

**Theorem 96.4** (Set-bounded local inverse images). *Suppose for every label  $y$  there is a set  $K_y \subseteq \mathbf{No}$  such that*

$$y \notin \text{ran } A(p) \implies \lambda_p^{-1}(y) \in K_y$$

*for every node  $p$ . Then  $A_*$  preserves exhaustive branches. If additionally for every input label  $x$  all possible local values  $\lambda_p(x)$  lie in a set  $L_x$ , then  $A_*$  is a bijection on the exhaustive branches.*

*Proof.* Let  $E$  be exhaustive. The elements of the set  $K_y$  all occur in  $E$  before some set ordinal  $\alpha$ , by Class Replacement applied to their unique occurrence times. If  $y$  were absent from  $A(E \upharpoonright \alpha)$ , its local inverse image would have to belong both to  $K_y$  and to the unused source alphabet. This is impossible. Thus every  $y$  occurs in  $A_*E$ . The second hypothesis is exactly the first hypothesis for the inverse automorphism, giving the converse.  $\square$

A useful equivalent formulation bounds the birthdays of all possible local inverse images of labels of birthday below  $\beta$  by one set ordinal  $\mu(\beta)$ . The bounded sign stage is a set. Conversely, per-label set bounds have ordinal birthday bounds, and Class Replacement over a bounded sign stage gives such a uniform modulus. The symmetric forward condition is analogous.

A uniformly given action of a *set* group of numerical automorphisms supplies examples: if each local map is the restriction of a group element, every possible inverse image lies in the set orbit of  $y$ , and every forward image lies in the set orbit of  $x$ . This is a sufficient criterion, not a classification of all permissible local dynamics.

### 96.1.4 Independent transformations after a chosen prefix

Let  $P$  be a uniformly presented prefix antichain. Suppose that for every  $p \in P$  an increasing class bijection  $\varphi_p : \mathbf{No} \rightarrow \mathbf{No}$  is given by one class graph, and  $\varphi_p[\text{ran } p] = \text{ran } p$ . Keep  $p$  fixed and apply  $\varphi_p$  to every later letter in its extension cone; fix words in no such cone. The antichain makes the choice unique, the used set is preserved, and the inverse uses  $\varphi_p^{-1}$ . This is an ordered-tree automorphism preserving exhaustion in both directions. A branch meeting a cone has its remaining alphabet bijectively relabelled, and a branch meeting no cone is fixed.

For example, fix a length- $\omega$  prefix enumerating the embedded ordinary integers  $\mathbb{Z}$ . Above that prefix add 1 to every later letter, and act identically elsewhere. Since  $\mathbb{Z} + 1 = \mathbb{Z}$ , this is a valid transformation. It fixes all first-level nodes but is not the identity, so it is not a single global relabelling.

The bounds in [theorem 96.4](#) are not necessary. Choose a transformation after each first letter  $z$ ; for  $z > 1$  use

$$\varphi_z(x) = zx - z(z - 1),$$

and for  $z \leq 1$  use the identity. The affine map is increasing and fixes  $z$ , so the antichain construction applies. It preserves exhaustiveness, but  $\varphi_z^{-1}(0) = z - 1$  ranges over a proper class. There is no single set  $K_0$  bounding all these local inverse images. Once a first letter is fixed the tail behavior is simple; uniformity over all first letters is much stronger.

### 96.1.5 The additional predicate that gives an exact classification

Enrich the ordered tree by equality of labels on different successor edges:

$$(p, p \frown (x)) \sim (q, q \frown (y)) \iff x = y. \quad (\text{S27.11})$$

This predicate does not name individual surreals. It remembers which occurrences at different nodes refer to the same label.

*Note 96.5* (Source 27, Theorem 6.5, “Automorphisms after retaining label equality”). *The automorphisms of the ordered prefix tree preserving (S27.11) are exactly*

$$A(p) = \varphi \circ p$$

*for increasing class bijections  $\varphi : \mathbf{No} \rightarrow \mathbf{No}$ . They all preserve exhaustive branches. If the alphabet is also fixed pointwise, the only such tree automorphism is the identity.* This is source 25’s [theorem 93.1](#), with the trivial case of [theorem 93.3](#) in which the alphabet is fixed pointwise; source 27’s proof is the same root-edge argument. Source 25 adds that preserving the core as well forces the identity ([theorem 93.3](#)) and that coherent edges recognize coverage ([proposition 93.4](#)).

*Source 27’s proof.* The local map at the root is an increasing bijection  $\varphi : \mathbf{No} \rightarrow \mathbf{No}$ . Every edge appending  $x$  is equivalent to the root edge appending  $x$ . Preservation of the edge equivalence forces every such edge to append  $\varphi(x)$  after transformation. Thus every local map is the restriction of  $\varphi$ , and induction on a set word gives the formula for  $A$ . Conversely, coordinatewise relabeling by  $\varphi$  preserves every indicated relation and is inverted by relabeling with  $\varphi^{-1}$ .  $\square$

This answers a precise version of the question “what extra structure restores the labels?” It does not claim that cross-node label equality is minimal among all possible enrichments. Between the arbitrary tree automorphisms of [theorem 96.2](#) and the global relabellings of [note 96.5](#) lies a substantial subgroup preserving exhaustiveness, of which the bounds and antichain constructions describe useful parts.

All these tree automorphisms preserve first-disagreement comparison on full Ord-branches, which never stand in a proper-prefix relation. On set words they preserve the prefix-first and prefix-last orders. An interior terminator would additionally require the local maps to preserve its cut; that condition is not automatic.

[merge] Lemma 96.1 is the uniform construction of source 18's theorem 35.7 and of source 13's family  $h_S$  (lemma 42.2), with one prescribed point matched; source 29's lemma 96.6 below allows any set partial match and the increasing alphabets as well. The “reserve in section 101” mentioned after corollary 96.3 is source 27's ordinal support theorem (section 101).

## 96.2 Source 29: ordered prefix trees do not determine their exhaustive class branches

[merge] Source 29's Section 9, printed whole. Its theorem 96.7 is, for the injection tree  $\mathcal{I}$ , source 27's theorem 96.2 (a second proof); its case of the increasing-word tree  $\mathcal{T}^{\text{inc}}$  is new. Its Corollary 9.3 is source 27's corollary 96.3 with another target branch and is printed as a note. Its proposition 96.10 is source 27's remark on set alphabets after corollary 96.3.

The ordered prefix tree contains much more information than the pure lexicographic order. It records the length of every set word, all extension relations, and the numerical order of the possible next entries. Even this structure does not recognize which proper-class branches exhaust the surreal labels. We prove this by constructing automorphisms that carry any one supplied branch to any other. The same method applies to increasing branches and can exchange a numerically cofinal range with a bounded one.

### 96.2.1 The structures and the required uniform isomorphisms

Let  $\mathcal{I}$  be the class of all injective functions  $p : \alpha \rightarrow \mathbf{No}$  with  $\alpha$  a set ordinal. Let  $\mathcal{T}^{\text{inc}}$  be its strictly numerically increasing subclass. Both include the empty word. Each carries word length, the prefix relation, and shorter-first lexicographic order. At a node  $p$ , the one-step extensions have the order of their next entries. The respective admissible alphabets are

$$\begin{aligned} D_p^{\mathcal{I}} &= \mathbf{No} \setminus \text{ran}(p), \\ D_p^{\mathcal{T}^{\text{inc}}} &= \{x \in \mathbf{No} : \text{ran}(p) < x\}. \end{aligned}$$

The inequalities in the second line mean that every entry of  $p$  is less than  $x$ .

These are structures on a genuine class of set-coded nodes. A proper-class branch will be supplied separately as a class function  $F : \text{Ord} \rightarrow \mathbf{No}$  all of whose set restrictions are nodes of the relevant tree. The assertion below quantifies over such class parameters; it does not form a class whose members are all branches.

**Lemma 96.6** (Residual alphabets and a prescribed match). *Work in GBC with a fixed set-like global well-order. Each of the alphabets  $D_p^{\mathcal{I}}$  and  $D_p^{\mathcal{T}^{\text{inc}}}$  is a proper class filling every separated set cut, including cuts with an empty side. For two nodes  $p, q$  of the same tree and any set partial order isomorphism between their residual alphabets, there is an increasing bijection*

$$h : D_p \longrightarrow D_q$$

*extending it. If the nodes and the initial partial match are specified uniformly from set parameters and fixed class parameters, these bijections can be given by one elementary class relation.*



*Proof.* First recall set avoidance in the surreal line. Given sets  $A < B$  and a forbidden set  $S$ , choose  $z$  with  $A < z < B$ . If  $z \notin S$ , it is already a suitable filler. Otherwise apply the set-cut property again to obtain

$$A \cup \{z\} < y < B \cup \{s \in S : z < s\}.$$

All constraints are consistent:  $z < B$ , and every newly imposed upper constraint is larger than  $z$ . The point  $y$  is larger than every forbidden element at most  $z$  and smaller than every forbidden element above  $z$ , so  $y \notin S$ . This proves set-cut filling in  $D_p^{\mathcal{I}}$ . It is proper because removing a set from **No** cannot leave a set.

For  $D_p^{\mathcal{T}^{\text{inc}}}$ , append  $\text{ran}(p)$  to the lower constraints of any set cut in the residual alphabet. Every upper constraint already exceeds the whole prefix. Surreal interpolation therefore produces a filler in  $D_p^{\mathcal{T}^{\text{inc}}}$ , including when either original side was empty. If this residual alphabet were a set, using all of it as lower constraints would require a strictly larger point within that same set, a contradiction. It is therefore proper.

For the extension assertion, start a deterministic back-and-forth with the supplied set partial isomorphism. At each stage take the first unmatched source point in the fixed global well-order, and place its image in the cut specified by the present partial isomorphism. The image can be chosen as the first eligible point in the fixed global well-order. Then perform the corresponding inverse insertion for the first unmatched target point. The strict cut constraints automatically exclude previously used points. At a limit set ordinal take the union of the preceding partial maps. The histories and partial maps are sets at every stage, and both directions are eventually exhausted.

For uniformity, membership in the resulting bijection is expressed by existence of a set-length initial history obeying these deterministic rules. The nodes, initial match, stage ordinal, and history are all set variables. Only the original alphabet definitions and global well-order are class parameters. The same elementary formula works for all the specified instances. As in the repository's uniform prefix constructions ([lemma 42.2](#) and [theorem 35.7](#)), this avoids choosing an arbitrary class isomorphism independently for each node. The recursion has set-valued stages and does not require ETR or Class Choice.  $\square$

The extension assertion allows a single point of the source alphabet to be matched with any single point of the target alphabet. This freedom, applied along one distinguished branch, is the key to the next theorem.

### 96.2.2 Transitivity on supplied class branches

**Theorem 96.7** (Branch transitivity of the ordered trees). *Work in GBC. For either  $\mathcal{K} = \mathcal{I}$  or  $\mathcal{K} = \mathcal{T}^{\text{inc}}$ , let  $F, G : \text{Ord} \rightarrow \mathbf{No}$  be two supplied class branches. There is a class automorphism  $A : \mathcal{K} \rightarrow \mathcal{K}$  preserving lengths, prefixes, and lexicographic order such that*

$$A(F \upharpoonright \alpha) = G \upharpoonright \alpha \quad (\alpha \in \text{Ord}).$$

*In particular, it preserves the order of the immediate successors of every node.*

*Proof.* For each pair of nodes  $p, q$  of the same length  $\alpha$ , use [lemma 96.6](#) to define an increasing bijection

$$h_{p,q} : D_p \longrightarrow D_q.$$

When  $p = F \upharpoonright \alpha$  and  $q = G \upharpoonright \alpha$ , begin the deterministic construction with the single prescribed match

$$F(\alpha) \longmapsto G(\alpha).$$

These are admissible unused entries because  $F, G$  are branches. For every other pair  $p, q$ , begin with the empty partial map. The condition selecting the seeded case is elementary in the set parameters  $p, q, \alpha$  and the class parameters  $F, G$ . Thus one class relation gives all the  $h_{p,q}$  simultaneously.

For a source word  $f$  of set length  $\alpha$ , define its image  $g$  by set recursion. Begin with the empty target prefix and set

$$g(\xi) = h_{f \upharpoonright \xi, g \upharpoonright \xi}(f(\xi)) \quad (\xi < \alpha). \quad (\text{S29.20})$$

Take unions at limit stages. Each chosen letter belongs to the target residual alphabet, so the output is injective in  $\mathcal{I}$  and increasing in  $\mathcal{T}^{\text{inc}}$ . Set Replacement supplies the output graph. Restrictions of a construction are the unique constructions for the corresponding shorter input words. The resulting class map  $A$  thus preserves lengths and prefixes.

It is also bijective. Given a target word  $g$ , reconstruct a source prefix  $p$  recursively. At position  $\xi$ , the target prefix  $q = g \upharpoonright \xi$  is known and the next source letter is uniquely

$$h_{p,q}^{-1}(g(\xi)).$$

This belongs to  $D_p$ , so the reconstruction remains in the required tree. It is inverse to (S29.20) at successors and at limits. Hence every target node has exactly one source node.

To check lexicographic order, proper-prefix comparisons are preserved by the already proved prefix and length assertions. Otherwise let  $\xi$  be the first unequal coordinate of two words. They have the same source prefix  $p$  and therefore the same target prefix  $q$  at that stage. Their next entries are transformed by the same increasing map  $h_{p,q}$ , which preserves the comparison. This proves the order assertion and also its restriction to immediate successors.

Finally, set induction on  $\alpha$  proves  $A(F \upharpoonright \alpha) = G \upharpoonright \alpha$ . At each successor stage the appropriate  $h_{p,q}$  contains the prescribed match, and at limits both sides are the unions of their shorter restrictions. The graph of  $A$  is defined from the unique set recursion histories, with the fixed class parameters already described. No family of class-valued outputs is collected into a class of classes.  $\square$

For any supplied branch  $H$ , its image under a tree automorphism is itself an admitted class branch, defined by

$$A_*H(\xi) = A(H \upharpoonright (\xi + 1))(\xi).$$

The prefixes are coherent, and elementary comprehension gives the graph. The inverse tree automorphism supplies the inverse branch operation. This formulation is available within the stated class framework without replacing the virtual collection of branches by an undeclared higher-sort object.

*Note 96.8* (Source 29, Corollary 9.3, “Exhaustivity is additional information”). *There is a length-, prefix-, and lexicographic-order-preserving class automorphism of  $\mathcal{I}$  carrying the prefixes of an exhaustive bijection  $b : \text{Ord} \rightarrow \mathbf{No}$  to those of a nonexhaustive injection. Consequently exhaustivity of proper-class branches is not an invariant of this ordered prefix-tree structure.* This is source 27’s corollary 96.3, here with target branch  $\alpha \mapsto \alpha$ . Source 25’s theorem 91.1 proves more: the core and all its cylinders can be preserved as well.

*Source 29’s proof.* Global Choice gives an exhaustive enumeration  $b$  of the surreal class. Apply theorem 96.7 to  $F = b$  and to  $G(\alpha) = \alpha$ , where ordinals are regarded as surreals. The latter branch omits, for example,  $-1$ .  $\square$

**Corollary 96.9** (Numerical cofinality is also additional information). *An automorphism of the ordered increasing-word tree  $\mathcal{T}^{\text{inc}}$  can carry a numerically cofinal proper-class branch to one whose range is bounded above by zero.*

*Proof.* Take

$$F(\alpha) = \alpha, \quad G(\alpha) = -\frac{1}{\alpha + 1}.$$

Both are strictly increasing class functions. Every surreal is smaller than some ordinal—its sign length gives an ordinal bound—so  $F$  is cofinal in the numerical surreal line. Ordered-field arithmetic makes  $G$  strictly increasing, and every one of its values is negative. Apply [theorem 96.7](#) for  $\mathcal{T}^{\text{inc}}$ .  $\square$

The structures in these corollaries name lengths, prefix extension, and successor orders. They do *not* name the common evaluation relation assigning a literal surreal label to each successor. In particular, equality of labels used at different nodes is not part of the stated reduct. The automorphisms can change that information. If label evaluation were named, it could express exhaustivity directly. Similarly, the ambient set membership relation on the codes is not being asserted invariant under these tree automorphisms.

The automorphism in [note 96.8](#) is an automorphism of the set-word tree, not an automorphism of  $\mathfrak{W}_{\text{sl}}$ : its induced branch map fails to preserve the domain of exhaustive enumerations. The conclusion therefore complements the repository’s core-automorphism extension obstructions ([theorem 47.7](#) and [corollary 47.3](#)), while making a distinct assertion about a richer prefix structure.

### 96.2.3 Why the full tree over a set carrier behaves differently

**Proposition 96.10** (Terminal nodes over a set alphabet). *For a fixed set alphabet  $X$ , consider the full tree of all injective ordinal-length  $X$ -words, with all possible set lengths. Its terminal nodes are exactly the exhaustive enumerations of  $X$ . Every prefix-tree automorphism therefore preserves exhaustivity.*

*Proof.* If a word omits a label  $x \in X$ , appending  $x$  is a permitted one-letter extension. If its range is  $X$ , no label is available for such an extension, and any proper longer word would have to supply one at the first new coordinate. This proves the characterization. Terminality is preserved by every automorphism of the prefix relation.  $\square$

For an infinite set alphabet of cardinality  $\mu$ , the entire tree is a set: every injective domain is below  $\mu^+$ , so its nodes occur in a set-sized collection of function spaces. A truncation changes the argument. For instance, a tree cut off at height  $\mu$  has terminal height- $\mu$  nodes that may omit labels. In the surreal class tree, every node has set range and therefore has unused surreal labels. There is no terminal set node at the end of an Ord-length branch. This is exactly where the set-carrier characterization ceases to apply.

### 96.2.4 Local descriptions and a family preserving exhaustion

Every length- and prefix-preserving automorphism  $A$  of  $\mathcal{I}$  that preserves lexicographic order determines increasing bijections

$$\begin{aligned} h_p : \mathbf{No} \setminus \text{ran}(p) &\longrightarrow \mathbf{No} \setminus \text{ran}(A(p)), \\ A(p \frown \langle x \rangle) &= A(p) \frown \langle h_p(x) \rangle. \end{aligned}$$

Surjectivity on the immediate successors follows from bijectivity of  $A$  and preservation of prefixes. Conversely, a uniform collection of increasing maps between the relevant source and target residual alphabets, used coherently as in ([S29.20](#)), determines such an automorphism. [Theorem 96.7](#) shows that this local description alone imposes no preservation of exhaustion.

There is nevertheless a broad concrete family for which exhaustion is preserved, allowing the relabelling of a tail to depend on its entire initial set prefix.

**Proposition 96.11** (Fixed-stage continuation by residual bijections). *Fix a set ordinal  $\delta$  and a length-, prefix-, and lexicographic-order-preserving automorphism  $B$  of the injection tree through height  $\delta$ , including that height. For every node  $p$  of length  $\delta$ , specify uniformly an increasing bijection*

$$k_p : \mathbf{No} \setminus \text{ran}(p) \longrightarrow \mathbf{No} \setminus \text{ran}(B(p)).$$

*There is a class automorphism  $A$  of the full injection tree agreeing with  $B$  through height  $\delta$  and continuing each tail after  $p$  by the same entrywise map  $k_p$ . Its induced operation on class branches preserves exhaustivity in both directions. Such uniform families  $k_p$  exist in GBC.*

*Proof.* For words of length at most  $\delta$ , use  $B$ . For a longer word  $f$ , put  $p = f \restriction \delta$ , take  $B(p)$  as the initial target word, and define

$$A(f)(\xi) = k_p(f(\xi)) \quad (\delta \leq \xi < \text{dom}(f)).$$

The original tail avoids  $\text{ran}(p)$ , and its image avoids  $\text{ran}(B(p))$ ; injectivity of the tail is preserved. Thus the resulting word is a node. Restrictions are compatible at and on both sides of  $\delta$ . An inverse is obtained from  $B^{-1}$  on the initial prefix and the appropriate  $k_p^{-1}$  on its tail, proving bijectivity.

A first disagreement before  $\delta$  is preserved by  $B$ . If two words agree through  $\delta$ , the same increasing  $k_p$  acts at any later first disagreement. Proper-prefix comparisons are also preserved. Therefore  $A$  has all the asserted tree and order properties.

Now let  $F$  be any supplied class injection, and let  $p = F \restriction \delta$ . Its tail takes values in  $\mathbf{No} \setminus \text{ran}(p)$ . The range of  $F$  is all of  $\mathbf{No}$  if and only if that tail exhausts this residual alphabet. Since  $k_p$  is a bijection onto the residual alphabet of  $B(p)$ , this holds if and only if the image tail exhausts its own residual alphabet, which is equivalent to exhaustivity of  $A_*F$ . The same argument proves the reverse direction.

Finally, apply [lemma 96.6](#) with empty initial matches to obtain the  $k_p$  uniformly from  $p, B$ , and a fixed global well-order. Their elementary class definition supplies the required family without a choice of class-valued stages.  $\square$

At  $\delta = 0$  this includes uniform entrywise increasing relabellings of the entire surreal alphabet. At positive set stages it permits different residual relabellings after different prefixes. The fixed bijection on an entire tail provides the coverage argument that is absent from unrestricted coordinate-dependent choices. The theorem is a sufficient construction, not a classification of every exhaustion-preserving automorphism.

**Corollary 96.12** (Agreement through any prescribed set stage). *For every automorphism  $A$  of the ordered injection tree and every set ordinal  $\delta$ , there is an exhaustion-preserving automorphism  $\tilde{A}$  agreeing with  $A$  on all nodes of length at most  $\delta$ . Nevertheless some automorphisms of the full tree do not preserve exhaustion.*

*Proof.* Restrict  $A$  to the tree through height  $\delta$  and apply [proposition 96.11](#). The second assertion is [note 96.8](#).  $\square$

Thus the failure to preserve exhaustive proper-class branches cannot be diagnosed from the action through any one fixed set height, even when that entire height and all earlier levels are retained. The corollary is stated as a uniform approximation theorem about supplied class maps; it does not presuppose an internal class of all class automorphisms.

[merge] [Proposition 96.11](#) and [corollary 96.12](#) and source 27's [theorem 96.4](#) and antichain transformations are different sufficient families for preserving exhaustion; none classifies the exhaustion-preserving subgroup (sources 25, 27 and 29 each ask for it: [research question 113.165](#); source 27's Question 12.6; source 29's Question 13.7).

## 97 Automorphisms of the core and the failure of extension (sources 20 and 21)

[merge] Sources 20 and 21 were written before Part VIII and prove again, each independently, most of Part VIII’s [section 47](#): the classification of cut orbits by characters (source 20’s Theorem 5.3 is source 14’s [theorem 47.2](#); source 21’s Theorem 9.8 is its (Ord, Ord) case and [lemma 47.6](#)), the extension criterion (source 20’s Theorem 5.8 and source 21’s Theorem 9.4 are source 17’s [theorem 47.5](#)), and the existence of non-extendable automorphisms (source 20’s Corollary 5.6 is [corollary 47.3](#); source 21’s Theorem 9.10 is [theorem 47.7](#), and its Theorem 9.12 follows from [corollary 47.3](#)). Those statements are printed as notes with the sources’ proofs. New here are source 20’s localized failure ([theorem 97.7](#): the automorphism can fix any set of core points and be supported in one interval), its disjointness of relabelled cores ([proposition 97.9](#), which sharpens [proposition 47.9](#)), and source 21’s explicit cut with no crossing first-level cylinder ([proposition 97.20](#)).

Both sources say they answer the repository’s extension question ([research question 113.4](#)) negatively. At the time this was so against their pins (d5bd4a67b and 109aaca15, which predate Parts VI–VIII); at this report’s present state Part VIII has already answered that question in part, negatively ([section 50 \(c\)](#)), and the two sections are second routes to it, except for the three new results named above. Source 20 writes  $\Omega$  for “no set-sized witness”, which Part VIII writes Ord.

### 97.1 Source 20: automorphisms of the core and the failure of extension

The bounded-support core has an exceptionally homogeneous abstract order, but its position among global well-orders carries additional information. We now separate these two structures. The principal new conclusion is that an automorphism of the core need not extend even when only one additional exhaustive enumeration is to be included. This failure can occur while any prescribed set of core points is fixed pointwise. It answers negatively the unrestricted extension question posed in the repository and the previous continuation ([research question 113.4](#); source 18).

Throughout this section we work in GBC and fix a baseline class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . Write  $\mathcal{P}_{\text{bd}}(b)$  for the ordinary class of normalized set codes for permutations of Ord with set support, and  $E_c$  for the global enumeration evaluated from  $c \in \mathcal{P}_{\text{bd}}(b)$ . Codes are ordered by their evaluations. We use the established core theorem:  $\mathcal{P}_{\text{bd}}(b)$  fills every cut specified by sets, including cuts with an empty side, and is class-order-isomorphic to  $\mathbf{No}$ . We also use its stronger uniform interpolation form, whose upper and lower input families may be set-indexed families of arbitrary exhaustive global enumerations. The back-and-forth principle underlying the core theorem is classical in the study of the surreal order [15]; the arguments below record the relevant class-theoretic details explicitly.

#### 97.1.1 Class cuts and their automorphism orbits

**Definition 97.1** (Strict cuts and set characters). A *strict class cut* of  $\mathcal{P}_{\text{bd}}(b)$  is a nonempty proper initial class  $L \subseteq \mathcal{P}_{\text{bd}}(b)$  having no greatest element. Its upper side is  $U = \mathcal{P}_{\text{bd}}(b) \setminus L$ . Define

$$\chi(L) = (\text{cf}_s(L), \text{ci}_s(U)),$$

where  $\text{cf}_s$  is the least cardinality of a set cofinal subset when one exists, and is the marker  $\Omega$  otherwise; define  $\text{ci}_s$  dually. The marker  $\Omega$  means only “no set-sized witness.” It is not a set cardinal and is not used as an object containing all ordinals.

If a class without a greatest element has a set cofinal subset, the usual set-theoretic cofinality argument, applied to that subset, gives a strictly increasing cofinal sequence of infinite regular

length. The dual statement applies to coinitality. The upper side of a strict cut can instead have a minimum, in which case its character is 1. In particular, the strict cut below a core point is included in our convention. A closed principal initial segment is excluded because it has a greatest element.

*Note 97.2* (Source 20, Lemma 5.2, “Saturated sides”). *Let  $C$  be a class linear order filling every set cut. A nonempty initial class  $A \subseteq C$  with no set cofinal subset also fills every set cut, including empty sides. Dually, a nonempty final class with no set coinital subset has the same property. Every such class is class-order-isomorphic to  $\mathbf{No}$  in GBC.* For the sides of a core gap this is the first part of source 17’s lemma 47.6 (and source 21’s note 97.16); source 20 states it for every class order filling set cuts, and its second paragraph is the back-and-forth of proposition 5.13.

*Source 20’s proof.* Let  $P, Q \subseteq A$  be sets with  $P < Q$ . If  $Q$  is nonempty, a point of  $C$  between them belongs to  $A$ , because it lies below a member of the initial class  $A$ . If  $Q$  is empty, the absence of a set cofinal subset gives a point of  $A$  strictly above all of  $P$ . This includes the case  $P = Q = \emptyset$ . Thus  $A$  has the full set-cut property. It cannot be a set: applying that property with lower side all of  $A$  would produce a point of  $A$  above itself. The final-class statement is dual.

For the isomorphism, Global Choice provides set-like enumerations of the two proper classes. Alternately extend a set-sized partial order isomorphism to include the next point of each enumeration. The previous lower and upper images form sets, so the required new point exists. Choose the least eligible point in a fixed global well-order. At each ordinal stage the partial map is a set, and at a set limit its earlier stages have a set union. The resulting class is the union of the unique compatible set computations. This is ordinary set-valued transfinite recursion with class parameters, rather than recursion with a class-valued state; no ETR or class-choice scheme is needed.  $\square$

Every nonempty open interval  $(a, b)$  of  $\mathcal{P}_{\text{bd}}(b)$  also fills every set cut: adjoin  $a$  to the lower constraints and  $b$  to the upper constraints. It is therefore another copy of  $\mathbf{No}$ . This observation will permit the later construction to be confined to a prescribed interval.

*Note 97.3* (Source 20, Theorem 5.3, “Complete cut-orbit classification”). *Two strict class cuts  $L, L'$  of  $\mathcal{P}_{\text{bd}}(b)$  are carried to one another by a class order automorphism of  $\mathcal{P}_{\text{bd}}(b)$  if and only if  $\chi(L) = \chi(L')$ . The possible pairs are exactly*

$$(\Omega, 1), \quad (\lambda, \Omega), \quad (\Omega, \lambda), \quad (\Omega, \Omega),$$

where  $\lambda$  ranges over infinite regular set cardinals. The first pair consists precisely of strict cuts below core points. This is source 14’s theorem 47.2, with the marker  $\Omega$  in place of source 14’s character symbol  $\text{Ord}$ . Source 20 realizes  $(\Omega, \Omega)$  by the repeated-label cut of note 97.5, source 14 by a successor-prefix cylinder edge.

*Source 20’s proof.* An automorphism and its inverse send sets to sets by Class Replacement. They therefore preserve cofinal and coinital set witnesses and their least possible cardinalities. This proves necessity.

At least one coordinate of  $\chi(L)$  is  $\Omega$ . Otherwise take a set  $A_0$  cofinal in  $L$  and a set  $B_0$  coinital in its upper side. Set-cut interpolation gives  $A_0 < z < B_0$ . If  $z \in L$ , cofinality of  $A_0$  contradicts the first inequality; if  $z \notin L$ , coinitality of  $B_0$  contradicts the second. The endpoint and regularity observations following definition 97.1 give the displayed list of possible forms.

To prove sufficiency, construct isomorphisms between the two lower sides and between the two upper sides, then join the maps. A side whose relevant character is  $\Omega$  is a copy of  $\mathbf{No}$  by note 97.2. Suppose instead that the initial sides  $L, L'$  have the same set cofinality  $\lambda$ . Choose increasing cofinal sequences  $(a_\xi)_{\xi < \lambda}$  and  $(a'_\xi)_{\xi < \lambda}$ . Begin back-and-forth with the entire set partial isomorphism

$$a_\xi \longmapsto a'_\xi \quad (\xi < \lambda).$$



For a new  $x \in L$ , the already mapped points below and above  $x$  give a set cut in the ambient target core. Some  $a_\xi$  is strictly above  $x$ , so its image  $a'_\xi$  is an upper constraint. Every realizer of the cut consequently lies in  $L'$ . The inverse step is identical. This produces the required side isomorphism. For final sides with equal set coinitiality, use the dual argument. If both upper sides have a minimum, map the minima to one another and apply [note 97.2](#) to the open final tails. Each stage again has only a set partial map. Joining the two side isomorphisms preserves all comparisons across the cut and gives a class automorphism.

Strict principal cuts realize  $(\Omega, 1)$ . For an infinite regular  $\lambda$ , choose an increasing sequence  $(a_\xi)_{\xi < \lambda}$  in  $\mathcal{P}_{\text{bd}}(b)$  and put

$$L = \{x \in \mathcal{P}_{\text{bd}}(b) : (\exists \xi < \lambda) x < a_\xi\}.$$

The sequence is cofinal in  $L$ . If fewer than  $\lambda$  points were cofinal, choose an index bound for each of them; regularity would bound all these indices below  $\lambda$ , a contradiction. Thus  $\text{cf}_s(L) = \lambda$ . Its complement has no set coinitial subset by the preceding interpolation obstruction. This gives  $(\lambda, \Omega)$ , and the dual construction gives  $(\Omega, \lambda)$ . Finally the explicit cut in [note 97.5](#) below realizes  $(\Omega, \Omega)$ .  $\square$

The theorem is about individually supplied class cuts and class automorphisms. It does not form an ordinary class whose elements are all cuts. It is nevertheless a complete classification of the action: any two allowed cut parameters having the same displayed characters are conjugate.

### 97.1.2 A represented cut and an unrepresented cut in one orbit

For an exhaustive enumeration  $e : \text{Ord} \rightarrow \mathbf{No}$ , write

$$L_e = \{c \in \mathcal{P}_{\text{bd}}(b) : E_c <_{\text{lex}} e\}.$$

If  $e$  is not evaluated by a core code, its two strict comparison sides partition  $\mathcal{P}_{\text{bd}}(b)$ .

*Note 97.4* (Source 20, Lemma 5.4, “The character of a noncore enumeration”). *If  $e$  is an exhaustive global enumeration outside the chosen core, then  $\chi(L_e) = (\Omega, \Omega)$ .* This is source 17’s [proposition 46.26](#) (source 14’s [theorem 44.1](#)) for exhaustive noncore enumerations.

*Source 20’s proof.* For a set  $A \subseteq L_e$ , uniform core interpolation applied to the rows indexed by  $A$  and the one upper row  $e$  gives a core row strictly above every member of  $A$  and strictly below  $e$ . Thus  $A$  is not cofinal. The dual argument excludes a set coinitial subset above  $e$ . Smaller and larger first labels give nonempty sides.  $\square$

Such enumerations exist in GBC. For example, compose  $b$  with the class permutation interchanging each consecutive pair  $2 \cdot \alpha, 2 \cdot \alpha + 1$  of ordinals. The support is a proper class, so the resulting enumeration has no code in the bounded-support core.

For an injective set prefix  $p : \delta \rightarrow \mathbf{No}$ , let

$$C_p = \{c \in \mathcal{P}_{\text{bd}}(b) : E_c \restriction \delta = p\}$$

be its convex cylinder. Every such cylinder is nonempty. A prefix *straddles* a cut when its cylinder meets both sides. The established prefix-recognition theorem says that a strict class cut is represented by an injective Ord-word precisely when it has a straddling cylinder at every set-ordinal length. That word is unique. It represents a set-like global well-order precisely when it is surjective. A raw class well-order has the same cut as the injective enumeration of its canonical initial Ord-block, so raw representability satisfies the same prefix criterion, without the surjectivity requirement (source 18, [theorem 46.28](#); [theorem 45.1](#) and [corollary 46.4](#)).

*Note 97.5* (Source 20, Lemma 5.5, “A local deleted-label cut”). *Let  $p$  be an injective set prefix and choose  $x \in \mathbf{No} \setminus \text{ran}(p)$ . Form the noninjective set word*

$$q = p \frown \langle x \rangle, \quad t = p \frown \langle x, x \rangle.$$

*Every core row differs from  $t$  before  $t$  ends, and hence*

$$D_t = \{c \in \mathcal{P}_{\text{bd}}(b) : E_c <_{\text{lex}} t\}$$

*is well-defined. It is a strict class cut of character  $(\Omega, \Omega)$ , lying inside  $C_p$ , and is represented by no injective Ord-word and by no raw class well-order of  $\mathbf{No}$ .* This is source 18’s [example 46.33](#) (source 15’s repeated-label case, [example 46.11](#)), with its character computed.

*Source 20’s proof.* A core row cannot agree with both occurrences of  $x$ , so ordinary first-difference comparison partitions the core into initial and final sides. All codes below  $C_p$  are on the lower side and all codes above it are on the upper side. Within  $C_q$ , the side is decided by whether the next label is less or greater than  $x$ . Fresh labels on either side, followed by prefix completion, show that both parts are nonempty.

Let  $A \subseteq D_t$  be a set. The members outside  $C_q$  lie below that cylinder. For the members inside, collect their next labels, all of which are less than  $x$ . Choose a surreal  $y < x$  strictly above every collected label and outside  $\text{ran}(q)$ . Such a fresh realizer exists: the realizers of any surreal set cut form a proper class, since a set of all realizers could itself be adjoined to the lower constraints to obtain a still larger realizer. A set of forbidden labels cannot exhaust that class. Complete  $q \frown \langle y \rangle$  to a core row. It remains in  $D_t$  and lies above all of  $A$ . Thus the lower character is  $\Omega$ . The upper side is dual.

If  $\delta = \text{dom}(p)$ , the cylinder  $C_q$  straddles  $D_t$  at length  $\delta + 1$ . At length  $\delta + 2$ , every injective prefix already differs from  $t$ , and all its completions compare with  $t$  in the same direction. There is therefore no straddling cylinder of that length. Prefix recognition excludes an injective representative. The canonical initial-block description excludes a raw representative as well. Equivalently, any putative representative would be forced to begin with  $p, x, x$ , contrary to injectivity.  $\square$

*Note 97.6* (Source 20, Corollary 5.6, “Representability is extra structure”). *The strict cut of every exhaustive noncore enumeration is carried by some class automorphism of  $\mathcal{P}_{\text{bd}}(b)$  to a cut represented by no raw class well-order. Thus the bare core order does not distinguish these two kinds of cuts.* This is [corollary 47.3](#) with a repeated-label cut as target; source 15’s [theorem 47.10](#) uses a first-label cut and source 17’s [theorem 47.7](#) the cut of a nonexhaustive word.

*Source 20’s proof.* Apply [note 97.3](#) to [notes 97.4](#) and [97.5](#).  $\square$

Omitted-range injective branches also have character  $(\Omega, \Omega)$ . For a set of core rows below such a branch, Class Replacement bounds all their first-disagreement coordinates. Follow the branch beyond that bound, choose a fresh smaller next label, and complete the prefix to a core row. This improves the entire set while remaining below the branch; the upper side is dual. Consequently one orbit contains exhaustive branches, branches with residual tails, and nonbranch cuts caused by a repeated label. The prefix-recognition criterion therefore describes the order *with its specified cylinder structure*; it cannot be recovered as an invariant of the bare ordered core.

### 97.1.3 Nonextension while fixing arbitrary set data

**Theorem 97.7** (Localized failure of one-point extension). *Let  $e : \text{Ord} \rightarrow \mathbf{No}$  be an exhaustive enumeration outside  $\mathcal{P}_{\text{bd}}(b)$ , and let  $S \subseteq \mathcal{P}_{\text{bd}}(b)$  be any set. There is a class order automorphism  $\varphi$  of  $\mathcal{P}_{\text{bd}}(b)$ , fixing  $S$  pointwise, for which there is no raw class well-order  $R$  of  $\mathbf{No}$  satisfying*

$$\{d \in \mathcal{P}_{\text{bd}}(b) : E_d <_{\text{lex}} R\} = \varphi[L_e].$$

*In particular,  $\varphi$  has no order-embedding extension from the one-point enlargement  $\mathcal{P}_{\text{bd}}(b) \cup \{e\}$  into the raw comparison structure, and no automorphism extension to either  $\mathfrak{W}_{\text{sl}}$  or  $\mathfrak{W}_{\text{all}}$ .*

*Proof.* Partition  $S$  according to its comparison with  $e$ . Uniform interpolation produces core points  $a, d$  such that

$$\{s \in S : E_s < e\} < a < e < d < \{s \in S : e < E_s\}.$$

Empty outer sides are allowed. The interval  $I = (a, d)$  contains no member of  $S$ . Choose a prefix  $p$  of  $e$  extending beyond the first disagreement of  $e$  with each endpoint. Then  $C_p \subseteq I$ . Choose a fresh  $x$  and let  $D_t$  be the deleted-label cut from [note 97.5](#).

The lower and upper sides of  $L_e$  within  $I$  have no set cofinal and cointial subsets, respectively. For example, given a set on the lower side, adjoin  $a$  and interpolate below  $e$ ; this produces a larger member still in  $I$ . The upper side is dual. The construction in [note 97.5](#) produces its improving points inside  $C_q \subseteq I$ , so the two sides of  $D_t$  within  $I$  have the same characters. By [note 97.2](#), all four sides are copies of  $\mathbf{No}$ . Join isomorphisms of corresponding sides to obtain an automorphism of  $I$  carrying its first cut to its second. Extend it by the identity outside  $I$ . The resulting  $\varphi$  fixes  $S$  and satisfies  $\varphi[L_e] = D_t$ .

If a raw order  $R$  were the image of  $e$  under an order-embedding extension, then for every  $c \in \mathcal{P}_{\text{bd}}(b)$ ,

$$E_{\varphi(c)} <_{\text{lex}} R \iff E_c <_{\text{lex}} e.$$

Its strict core cut would therefore be  $D_t$ , contradicting [note 97.5](#). The same contradiction excludes the displayed cut identity directly, without constructing an encompassing object of all raw orders.  $\square$

This is stronger than the existence of a single bad automorphism. Any set of prescribed core data can be retained, and the failure can be confined to a core interval containing the cut of  $e$ . Accordingly, representedness is not an isomorphism-invariant property of the core order even after an arbitrary set of core parameters has been named. The theorem also identifies the exact limitation of set-sized partial homogeneity: extending a partial isomorphism within the core does not control its behavior at proper-class cuts.

The one-point formulation is legitimate already in GBC. Replace the additional point by one new set marker and compare it with core codes using the class parameter  $e$ . The impossibility asserts that no individual class relation  $R$  can realize its required image comparisons. No ordinary class containing all class relations, and no function whose ordinary arguments are classes, is required.

### 97.1.4 Exactly what an extension must preserve

Write  $\text{Exh}_b(L)$  for the assertion that a strict cut  $L$  is the strict core cut of an exhaustive Ord-enumeration. This is a predicate of a *class variable*, not a subset of an ordinary class of all cuts. Prefix recognition gives the following equivalent formulation using only set quantifiers with  $L$  as a class parameter:

1. at every ordinal length there is a straddling injective set prefix;

2. every surreal belongs to the range of some such prefix.

The straddling prefix at any fixed length is unique, since two disjoint convex cylinders cannot both cross an initial cut.

*Note 97.8* (Source 20, Theorem 5.8, “Extension criterion for the set-like order”). *Fix a class automorphism  $\varphi$  of  $\mathcal{P}_{\text{bd}}(b)$ . It has an extension to the set-like global comparison structure, given by one elementary transformation of each supplied class enumeration, if and only if*

$$\text{Exh}_b(L) \iff \text{Exh}_b(\varphi[L])$$

*for every strict class cut  $L$ . If the extension exists, it is unique. Here the displayed statement quantifies over class parameters; the conclusion does not assert an ordinary class function having arbitrary classes as its elements.* This is source 17’s [theorem 47.5](#), with the predicate  $\text{Exh}_b$  in place of source 17’s  $\mathcal{R}_b$ ; the two predicates agree by [theorem 45.1](#).

*Source 20’s proof.* Necessity follows by comparing the image of an enumeration with the images of all core rows, and then applying the inverse extension.

For sufficiency, supply any exhaustive enumeration  $e$  as a class parameter. Form the class cut  $\varphi[L_e]$ . By the hypothesis it has a straddling prefix at every ordinal length and its uniquely recovered word is exhaustive. Define its image  $e'$  by

$$e'(\alpha) = x \iff \begin{array}{l} \text{there is an injective prefix } p : \alpha + 1 \rightarrow \mathbf{No} \\ \text{straddling } \varphi[L_e] \text{ with } p(\alpha) = x. \end{array}$$

All quantified variables in this definition are sets. Membership in the image cut is the elementary condition

$$d \in \varphi[L_e] \iff (\exists c \in \mathcal{P}_{\text{bd}}(b)) (\varphi(c) = d \wedge E_c <_{\text{lex}} e).$$

Elementary class comprehension therefore produces  $e'$  for every supplied  $e$ , by the same formula. The prefix criterion proves that it is the unique exhaustive enumeration with that cut.

Core density gives, for distinct exhaustive enumerations,

$$e < f \iff L_e \subsetneq L_f.$$

Indeed a core row strictly between them witnesses strict inclusion. The construction consequently preserves and reflects order. At a core row it agrees with  $\varphi$ , because the strict cut below  $c$  is carried to the strict cut below  $\varphi(c)$ . The analogous construction with  $\varphi^{-1}$  gives its inverse. Finally any extension must carry  $L_e$  to  $\varphi[L_e]$ , and density makes the exhaustive representative unique.  $\square$

For one supplied noncore  $e$ , the corresponding criterion is sharper still. An image in  $\mathfrak{W}_{\text{all}}$  exists exactly when  $\varphi[L_e]$  has a straddling prefix at every ordinal length; an image in  $\mathfrak{W}_{\text{sl}}$  exists exactly when the recovered word is also surjective. For the entire raw structure, preserving these cuts alone is insufficient: several raw orders can have the same canonical initial Ord-block. An extension must additionally identify the orders of their residual fibers, with the required uniformity. This is the remaining structural question, after the unrestricted extension assertion has been refuted.

**Proposition 97.9** (Numerical relabeling moves a chosen core). *Let  $h$  be a nonidentity increasing class automorphism of  $(\mathbf{No}, <)$ . The evaluated bounded-support cores based at  $b$  and  $h \circ b$  are disjoint. In particular, no nonidentity numerical relabeling stabilizes one chosen core.*

*Proof.* If a row belonged to both cores, it would differ from each baseline on only a set of coordinates. Thus  $b$  and  $h \circ b$  would differ on a set of coordinates, and Class Replacement would imply that  $h$  moves only a set of surreal labels. If  $x < h(x)$ , then every  $y$  with  $x < y < h(x)$  satisfies  $h(y) > h(x) > y$ . The proper-class interval  $(x, h(x))$  would consist entirely of moved labels, a contradiction. If  $h(x) < x$ , every  $y \in (h(x), x)$  satisfies  $h(y) < h(x) < y$ , giving the same contradiction.  $\square$

Numerical relabeling still gives a natural isomorphism between the two different cores and a natural transformation of global enumerations. The proposition explains why it does not account for the nontrivial internal automorphisms of one fixed core.

[merge] Source 20's last paragraph before [proposition 97.9](#) calls the extension question for the whole raw structure “the remaining structural question”. Source 19's [theorem 98.5](#) answers it in an external set-sized model: a core automorphism extends to the raw order exactly when it preserves the isomorphism types of all residual fibres. Source 25 asks the elementary question about two baselines that [proposition 97.9](#) answers for relabelled ones: two cores coincide exactly when their baselines eventually agree, and are otherwise disjoint (its Question 16.6, [research question 113.170](#)).

## 97.2 Source 21: core automorphisms, a concrete obstruction to extension

The bounded-support core carries much of the local order geometry of  $\mathfrak{W}_{\text{sl}}$ , but its abstract order does not determine which of its large cuts are represented by exhaustive enumerations. This distinction gives a concrete negative answer to one of the repository's extension questions: some automorphisms of the core do not extend even to order embeddings from  $\mathfrak{W}_{\text{sl}}$  into  $\mathfrak{W}_{\text{all}}$ . We first establish a cut reconstruction lemma, then give an exact extension criterion for the set-like fragment and two successive forms of the obstruction.

### 97.2.1 The core and its cuts

Work in GBC, and fix a baseline class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . Write  $\mathcal{P}_{\text{bd}}(b)$  for the genuine class of normalized set-support codes, ordered by the lexicographic comparison of their evaluations. If  $d \in \mathcal{P}_{\text{bd}}(b)$ , write  $E_d : \text{Ord} \rightarrow \mathbf{No}$  for the corresponding global enumeration. Concretely, one can code a permutation of an ordinal, extend it by the identity, and compose with  $b$ . Normalization removes the duplicate codes obtained by adjoining an identity segment. Thus distinct points of  $\mathcal{P}_{\text{bd}}(b)$  have distinct evaluations.

The following established core properties are used below (Part IV: [lemma 19.4](#), [theorem 19.10](#), and [corollary 19.15](#)): every injective set prefix extends to an  $E_d$ ;  $\mathcal{P}_{\text{bd}}(b)$  realizes every separated cut given by two sets of its points, with empty sides allowed; and  $\mathcal{P}_{\text{bd}}(b)$  is order-dense in  $\mathfrak{W}_{\text{sl}}$ . The same density argument applies to injective Ord-words whose ranges need not be all of  $\mathbf{No}$ . In particular,  $\mathcal{P}_{\text{bd}}(b)$  is an  $\eta_{\text{Ord}}$  class order without endpoints. The class version of the back-and-forth underlying this characterization is also central to the usual uniqueness theorem for the surreal order [15].

For an injective class function  $F : \text{Ord} \rightarrow \mathbf{No}$ , define

$$L_F = \{d \in \mathcal{P}_{\text{bd}}(b) : E_d <_{\text{lex}} F\}, \quad R_F = \mathcal{P}_{\text{bd}}(b) \setminus L_F. \quad (\text{S21.*})$$

These are ordinary classes of set codes. When  $F = E_d$ , the lower class is the strict lower cut of that core point, and the upper class includes the point itself. Otherwise both sides are strict. For an injective set word  $p$ , put

$$C_p = \{d \in \mathcal{P}_{\text{bd}}(b) : p \subseteq E_d\}.$$

We say that  $C_p$  crosses a class  $L \subseteq \mathcal{P}_{\text{bd}}(b)$  if it meets both  $L$  and  $\mathcal{P}_{\text{bd}}(b) \setminus L$ .

*Note 97.10* (Source 21, Lemma 9.1, “Elementary reconstruction from a core cut”). *Let  $F : \text{Ord} \rightarrow \mathbf{No}$  be injective. For every injective set word  $p$ ,*

$$C_p \text{ crosses } L_F \iff p \subseteq F.$$

*Consequently  $F$  is recovered from  $L_F$  and  $b$  by*

$$F(\alpha) = x \iff \exists p [\text{dom}(p) = \alpha + 1, p \text{ injective}, \\ p(\alpha) = x, C_p \text{ crosses } L_F].$$

*The defining formula has only set quantifiers and the displayed class parameters. In particular, two injective Ord-words with the same core cut are equal. With [note 97.11](#) this is the represented-cut theorem [theorem 45.1](#) ([note 87.6](#)); the lemma is its uniqueness and reconstruction clause.*

*Source 21’s proof.* If  $p \not\subseteq F$ , the first discrepancy between  $p$  and  $F$  occurs before the end of  $p$ . Every evaluation extending  $p$  has the same comparison with  $F$  at that discrepancy. Thus all of  $C_p$  lies on one side of  $L_F$ .

Conversely, suppose  $p = F \upharpoonright \beta$ . Choose unused labels  $x < F(\beta) < y$ . For example, choose  $x$  below, and  $y$  above, the set  $\text{ran}(p) \cup \{F(\beta)\}$ . Extend  $p \frown \langle x \rangle$  and  $p \frown \langle y \rangle$  to core enumerations. The first is below  $F$  and the second is above it, so  $C_p$  crosses the cut. Applying this equivalence to words of length  $\alpha + 1$  proves the formula and uniqueness.  $\square$

The lemma applies equally to exhaustive and nonexhaustive words. A nonexhaustive word can therefore never induce the cut of an exhaustive one. This is a stronger statement than the mere absence of a core point at the cut. The reconstruction also works for principal core cuts: arbitrarily long prefixes still have core extensions on both sides of the represented enumeration.

*Note 97.11* (Source 21, Theorem 9.2, “Which core cuts are represented by words?”). *Let  $L \subsetneq \mathcal{P}_{\text{bd}}(b)$  be a nonempty lower class with no greatest element. Then  $L = L_F$  for some injective  $F : \text{Ord} \rightarrow \mathbf{No}$  if and only if, for every ordinal  $\alpha$ , some length- $\alpha$  prefix cylinder  $C_p$  crosses  $L$ . When this holds, the crossing prefix at each level is unique, these prefixes are coherent, and their union is the unique representing word. The word is exhaustive if and only if every surreal occurs in some crossing prefix. All these tests and the reconstruction use elementary formulas with the class parameters  $L, b$ . This is [theorem 45.1](#), proved again ([notes 46.1](#) and [87.6](#)).*

*Source 21’s proof.* Necessity follows from [note 97.10](#). For sufficiency, first observe that two distinct cylinders of the same length cannot both cross a lower class. Their fixed prefixes have a first discrepancy, so every point of one cylinder precedes every point of the other. A point outside  $L$  in the earlier cylinder and a point inside  $L$  in the later cylinder would contradict downward closure.

Thus there is a unique crossing prefix  $p_\alpha$  at each level. Every restriction of a crossing prefix also crosses, since its cylinder contains the original cylinder. Consequently  $p_\beta \upharpoonright \alpha = p_\alpha$  whenever  $\alpha < \beta$ . Define  $F(\alpha) = p_{\alpha+1}(\alpha)$ . Existence and uniqueness of these set prefixes give an elementary definition of the graph; there is no choice of a class-valued stage. Coherence shows that  $F \upharpoonright \alpha = p_\alpha$ , and therefore  $F$  is injective.

Take a core point  $E_d \neq F$ , and let  $\delta$  be their first discrepancy. If  $E_d <_{\text{lex}} F$ , then  $E_d$  precedes every point of the crossing cylinder  $C_{p_{\delta+1}}$ . That cylinder has a point in  $L$ , so downward closure gives  $d \in L$ . If  $E_d >_{\text{lex}} F$ , the cylinder has a point outside  $L$  below  $E_d$ , so  $d \notin L$ .

There remains only the possibility  $E_d = F$ . The comparisons just proved show that, if  $d \in L$ , it would be the greatest point of  $L$ . The hypothesis excludes this. Thus  $L = L_F$  also in that case. Finally,  $\text{ran}(F)$  is the union of the ranges of the crossing prefixes, which proves the exhaustivity criterion.  $\square$



The no-greatest-element hypothesis is essential for this formulation. For a core point  $E_d$ , both  $\{c : E_c <_{\text{lex}} E_d\}$  and  $\{c : E_c \leq_{\text{lex}} E_d\}$  have the same crossing cylinders at every level. Reconstruction gives  $E_d$  in both cases, but only the first is its strict lower cut. The second has greatest element  $d$ .

*Note 97.12* (Source 21, Proposition 9.3, “The first failed level of a cut”). *Let  $L$  be a proper nonempty lower class of  $\mathcal{P}_{\text{bd}}(b)$  with no greatest element. Suppose its first level with no crossing cylinder is a limit ordinal  $\lambda$ . The coherent earlier prefixes have a union  $p$  of length  $\lambda$ . If  $C_p$  lies on the upper side, then  $L$  has set cofinality exactly  $\text{cf}(\lambda)$ . If  $C_p$  lies on the lower side, the upper side has set coinitality exactly  $\text{cf}(\lambda)$ . Consequently a cut with no set-sized cofinal lower subset and no set-sized coinital upper subset (“large on both sides,” as in [definition 97.15](#)) either comes from a full injective Ord-word, or first loses its crossing cylinder at a successor level. In the latter case it is a cut between the next-label cylinders inside one fixed set prefix. This is the limit case of the first-failure classification (source 13’s [theorem 46.18](#), source 15’s [theorem 46.9](#), source 16’s [theorem 46.3](#), source 18’s [theorem 46.32](#)) with source 14’s characters ([theorem 44.1](#)).*

*Source 21’s proof.* Write  $p_\beta$  for the unique crossing prefix at level  $\beta < \lambda$ . Their restrictions agree, so their union is an injective set word  $p$ . Its cylinder is nonempty and wholly on one side. Suppose  $C_p \subseteq \mathcal{P}_{\text{bd}}(b) \setminus L$ . Choose a cofinal sequence of  $\text{cf}(\lambda)$  many levels  $\beta < \lambda$  and at each such level a code  $d_\beta \in L \cap C_{p_\beta}$ . Global choice and class replacement make these codes a set. Every  $d \in L$  must disagree with  $p$  at some position  $\delta < \lambda$ , in the smaller direction. For  $\beta > \delta$  one has  $d < d_\beta$ . Thus the chosen set is cofinal.

Conversely, for a family of fewer than  $\text{cf}(\lambda)$  codes in  $L$ , their disagreement positions with  $p$  are bounded below  $\lambda$ . A lower-side point in a later crossing cylinder is strictly above the entire family. Hence the cofinality is exactly  $\text{cf}(\lambda)$ . The other case is the order dual.

If a large cut ever fails the crossing test, it has a least failing level. Level zero crosses, and the preceding argument excludes limit levels. At a successor level  $\alpha + 1$ , the unique length- $\alpha$  cylinder still crosses, but each of its next-label cylinders lies entirely on one side. Their numeric ordering gives precisely the stated next-label cut.  $\square$

### 97.2.2 The exact extension criterion

Let  $h$  be a class automorphism of  $\mathcal{P}_{\text{bd}}(b)$ . For a class  $L \subseteq \mathcal{P}_{\text{bd}}(b)$ , its image

$$h_*L = \{h(d) : d \in L\}$$

is an ordinary class. The expression

$$\mathcal{C}_{\text{exh}} = \{L_E : E \in \mathfrak{W}_{\text{sl}}\}$$

will be used only as virtual notation for the represented exhaustive cuts. It does not assert that this collection of classes is itself a class of GBC.

*Note 97.13* (Source 21, Theorem 9.4, “Extension criterion”). *A class automorphism  $h$  of  $\mathcal{P}_{\text{bd}}(b)$  extends to an order automorphism of  $\mathfrak{W}_{\text{sl}}$  if and only if  $h_*$  carries the represented exhaustive cuts onto themselves. Explicitly, the condition is*

$$\begin{aligned} \forall E \in \mathfrak{W}_{\text{sl}} \exists F \in \mathfrak{W}_{\text{sl}} \quad L_F &= h_*L_E, \\ \forall F \in \mathfrak{W}_{\text{sl}} \exists E \in \mathfrak{W}_{\text{sl}} \quad L_F &= h_*L_E. \end{aligned}$$

*The extension is unique. The first condition alone gives an order embedding of  $\mathfrak{W}_{\text{sl}}$  into itself extending  $h$ . In the two-sorted formulation, the extension is a uniform operator on class parameters.*

*Its output is elementarily reconstructed from  $h, b, E$ ; neither ETR nor Class Choice is required.* This is source 17's [theorem 47.5](#); source 21 adds that the first condition alone gives an extending self-embedding of  $\mathfrak{M}_{\text{sl}}$ .

*Source 21's proof.* Suppose  $\Phi$  is an extending order automorphism. For  $d \in \mathcal{P}_{\text{bd}}(b)$ ,

$$E_d <_{\text{lex}} E \iff E_{h(d)} <_{\text{lex}} \Phi(E).$$

Hence  $L_{\Phi(E)} = h_*L_E$ . Surjectivity supplies the second condition. The same cut equation is necessary for an extending order embedding.

Conversely, given  $E$ , form  $h_*L_E$  by elementary comprehension and apply the formula in [note 97.10](#). The first condition asserts that the reconstructed word is an exhaustive enumeration  $F$ , and the lemma makes it unique. Thus this construction defines  $\Phi(E)$  without making a choice from a collection of class witnesses.

If  $E <_{\text{lex}} E'$ , order-density of the core gives  $L_E \subsetneq L_{E'}$ . Applying  $h$  preserves this proper inclusion, so the represented output cuts satisfy  $L_{\Phi(E)} \subsetneq L_{\Phi(E')}$ , which implies  $\Phi(E) <_{\text{lex}} \Phi(E')$ . For a core point  $E_d$ , the image of its strict lower cut is the strict lower cut of  $E_{h(d)}$ ; therefore  $\Phi(E_d) = E_{h(d)}$ . The second condition gives surjectivity. Finally, every extension must satisfy the cut equation, and reconstruction makes its value at each input unique.  $\square$

[Note 97.11](#) makes the criterion explicit in the prefix language. For each input  $E$ , the transformed lower cut  $h_*L_E$  must have a crossing cylinder at every ordinal level, and every surreal must occur in some such crossing prefix. These tests characterize an extending embedding. Applying the same tests to  $h^{-1}$  as well characterizes an extending automorphism.

*Note 97.14 (Source 21, Corollary 9.5, “Restriction to the core”). In an enlarged universe where the displayed collections are sets, restriction identifies*

$$\text{Aut}(\mathfrak{M}_{\text{sl}}; \mathcal{P}_{\text{bd}}(b)) \cong \text{Stab}_{\text{Aut}(\mathcal{P}_{\text{bd}}(b))}(\mathcal{C}_{\text{exh}}).$$

*Here  $\text{Aut}(\mathfrak{M}_{\text{sl}}; \mathcal{P}_{\text{bd}}(b))$  denotes the automorphisms carrying the core onto itself. In GBC, the same assertion is understood as the corresponding criterion for uniform operators.* This restates [theorem 47.5](#) as an identification of groups in an enlarged universe.

The criterion is exact, but it deliberately retains the represented-cut family as additional data. The next result shows that this data cannot be discarded.

### 97.2.3 Homogeneity of large cuts

**Definition 97.15.** A class cut  $(L, R)$  of  $\mathcal{P}_{\text{bd}}(b)$  is *large on both sides* if  $L, R$  are nonempty, partition  $\mathcal{P}_{\text{bd}}(b)$ , satisfy  $L < R$ , and  $L$  has no set-sized cofinal subset while  $R$  has no set-sized coinital subset.

*Note 97.16 (Source 21, Lemma 9.7, “Each side is an  $\eta_{\text{Ord}}$  order”). If  $(L, R)$  is large on both sides, both  $L$  and  $R$ , with their induced orders, are proper-class  $\eta_{\text{Ord}}$  orders without endpoints.* This is the first part of source 17's [lemma 47.6](#) (source 20's [note 97.2](#)).

*Source 21's proof.* Neither side can be a set, since it would then be its own set-sized cofinal or coinital subset. Consider set constraints  $A < B$  in  $L$ . If  $B \neq \emptyset$ , interpolate them in  $\mathcal{P}_{\text{bd}}(b)$ . The resulting point lies below an element of  $L$ , and hence belongs to  $L$ . If  $B = \emptyset$ , the set  $A$  is not cofinal in  $L$ , so some point of  $L$  lies strictly above all of  $A$ . This also handles  $A = \emptyset$ . The proof for  $R$  is dual.  $\square$

*Note 97.17* (Source 21, Theorem 9.8, “Conjugacy of large class cuts”). *Any two class cuts of  $\mathcal{P}_{\text{bd}}(b)$  that are large on both sides are carried to one another by a class automorphism of  $\mathcal{P}_{\text{bd}}(b)$ .* This is the (Ord, Ord) case of source 14’s [theorem 47.2](#) and the second part of [lemma 47.6](#).

*Source 21’s proof.* Write the cuts as  $(L_0, R_0)$  and  $(L_1, R_1)$ . By [note 97.16](#), the two left sides are proper-class  $\eta_{\text{Ord}}$  orders, as are the two right sides. Global Choice supplies set-like enumerations of these genuine classes of set codes. We give the standard back-and-forth for the left sides.

At stage  $\alpha$ , the partial isomorphism is a set. Insert the first unused point of  $L_0$ , placing its image between the images of all already mapped points below and above it. These are set constraints, so  $\eta_{\text{Ord}}$ -interpolation supplies a point. The strict inequalities make it fresh. Then perform the reverse insertion for the first unused point of  $L_1$ . A fixed global well-order selects the interpolants. At a limit stage take the union of the preceding partial maps. Since the stage is an ordinal, its history and the resulting map are sets. The union over all ordinals is an isomorphism  $L_0 \cong L_1$ .

The same construction gives  $R_0 \cong R_1$ . Join the two maps. Their domains and ranges partition the core, and the cross inequalities are preserved because the left side precedes the right side in each cut. The result is the required automorphism.

Every stage of this recursion is set-valued. Its existence is the ordinary class-parameter recursion available in GBC, rather than an application of ETR.  $\square$

*Note 97.18* (Source 21, Lemma 9.9, “Cuts induced by nonskeleton words”). *If an injective  $F : \text{Ord} \rightarrow \mathbf{No}$  is not the evaluation of any point of  $\mathcal{P}_{\text{bd}}(b)$ , the cut  $(L_F, R_F)$  is large on both sides.* This is source 17’s [proposition 46.26](#) for noncore injective words (source 20’s [note 97.4](#) for exhaustive ones).

*Source 21’s proof.* Let  $T \subseteq L_F$  be a set of core codes. For each  $d \in T$ , the first discrepancy  $\Delta(E_d, F)$  is an ordinal. Class Replacement gives an ordinal  $\beta$  strictly above all these discrepancies. Follow  $F$  up to  $\beta$ , choose a fresh label smaller than  $F(\beta)$ , and extend this set prefix to a core enumeration. The result lies below  $F$  but above every  $E_d$  with  $d \in T$ : at the first discrepancy of  $E_d$  with  $F$ , the new enumeration still agrees with  $F$ . Hence  $T$  is not cofinal in  $L_F$ .

For a set of codes in  $R_F$ , choose a fresh label larger than  $F(\beta)$  instead. This produces a point above  $F$  and below all the specified upper points. Taking empty sets of constraints shows that both sides are nonempty. The hypothesis excludes an equality case at a core point, so the upper side has no principal boundary.  $\square$

*Note 97.19* (Source 21, Theorem 9.10, “Nonextendable core automorphisms”). *There is a class automorphism of  $\mathcal{P}_{\text{bd}}(b)$  that has no extension even to an order embedding  $\mathfrak{W}_{\text{sl}} \rightarrow \mathfrak{W}_{\text{sl}}$ . Consequently the stabilizer in [note 97.14](#) is a proper subgroup of  $\text{Aut}(\mathcal{P}_{\text{bd}}(b))$ .* This is source 17’s [theorem 47.7](#), with the same pair-swapped  $E$  and the target word  $\alpha \mapsto \alpha$ ; source 21 states the failure for order embeddings  $\mathfrak{W}_{\text{sl}} \rightarrow \mathfrak{W}_{\text{sl}}$ , and source 17’s argument gives that too, since it uses only the cut equation.

*Source 21’s proof.* Define an exhaustive enumeration  $E$  by swapping every adjacent pair of positions of  $b$ :

$$E(2 \cdot \alpha) = b(2 \cdot \alpha + 1), \quad E(2 \cdot \alpha + 1) = b(2 \cdot \alpha).$$

Here ordinal multiplication has 2 on the left. The two-element blocks partition Ord, so  $E$  is a bijection. Its support is all of Ord, and therefore  $E$  is not a core evaluation.

Let  $F(\alpha) = \alpha$ , regarding each ordinal as a surreal. This word is injective but not exhaustive; for example, it omits  $-1$ . It too is not a core evaluation. By [note 97.18](#), both  $(L_E, R_E)$  and  $(L_F, R_F)$  are large on both sides. By [note 97.17](#), choose  $h \in \text{Aut}(\mathcal{P}_{\text{bd}}(b))$  with  $h_*L_E = L_F$ .

An extending order embedding  $\Phi$  would satisfy  $L_{\Phi(E)} = h_* L_E = L_F$ . The reconstruction lemma would then force  $\Phi(E) = F$  coordinate by coordinate. This is impossible, because  $\Phi(E)$  must be exhaustive and  $F$  is not.  $\square$

Thus all large class cuts lie in one orbit of the abstract core automorphism group, whereas exhaustive represented cuts form a proper part of that orbit. The distinction between exhaustive and nonexhaustive branches is not recoverable from the core order alone. This obstruction already occurs in the set-like fragment; no residual tail of an unrestricted class well-order is needed.

The result is existential and the construction uses the stated class parameters and Global Choice. It does not classify the extendable subgroup, prove parameter-free definability of the automorphism, or assert that every automorphism of  $\mathfrak{W}_{\text{sl}}$  preserves a chosen core. In a full inaccessible-universe model it becomes an ordinary set theorem: a saturated core of size  $\kappa$  has conjugate  $(\kappa, \kappa)$ -cuts, some represented by bijections of length  $\kappa$  and some represented only by nonexhaustive injective words.

#### 97.2.4 An obstruction even to extension into the unrestricted order

A nonexhaustive word can still be the initial Ord-word of an unrestricted class well-order, by placing its omitted labels afterward. To rule out extension into  $\mathfrak{W}_{\text{all}}$ , one therefore needs a cut that is not represented even by an injective Ord-word. Such a cut has a particularly simple definition.

**Proposition 97.20** (A large cut with no crossing first-level cylinder). *The class*

$$K = \{d \in \mathcal{P}_{\text{bd}}(b) : E_d(0) < 0\}$$

*is a proper lower cut, large on both sides, and no one-letter prefix cylinder crosses it. Consequently  $K \neq L_F$  for every injective  $F : \text{Ord} \rightarrow \mathbf{No}$ .*

*Proof.* Lexicographic comparison makes  $K$  a lower class. A completion of  $\langle -1 \rangle$  belongs to it, and a completion of  $\langle 0 \rangle$  does not, so the cut is nonempty and proper.

Let  $A \subseteq K$  be a set of codes. The set of first entries  $\{E_d(0) : d \in A\}$  lies strictly below 0. Surreal set-cut interpolation gives a label  $z$  strictly above all those entries and strictly below 0. A core completion of  $\langle z \rangle$  lies in  $K$  above every point of  $A$ . Thus no set is cofinal in the lower side.

Now let  $B \subseteq \mathcal{P}_{\text{bd}}(b) \setminus K$  be a set. Form the set

$$U = \{E_d(1) : d \in B, E_d(0) = 0\}.$$

Choose  $x$  below every element of  $U \cup \{0\}$ , and complete the injective prefix  $\langle 0, x \rangle$  to a core point. It lies in the upper side. It precedes every row in  $B$  starting with a positive label by comparison at position 0, and every row in  $B$  starting with 0 by comparison at position 1. Hence no set is coinital in the upper side.

For any one-letter prefix  $\langle a \rangle$ , its entire cylinder lies in  $K$  if  $a < 0$ , and its entire cylinder lies outside  $K$  if  $a \geq 0$ . None crosses. The final assertion follows already from [note 97.10](#) at level 1.  $\square$

*Note 97.21* (Source 21, Theorem 9.12, “No extension even from  $\mathfrak{W}_{\text{sl}}$  into  $\mathfrak{W}_{\text{all}}$ ”). *There is a class automorphism  $h$  of  $\mathcal{P}_{\text{bd}}(b)$  with no extending order embedding  $\mathfrak{W}_{\text{sl}} \rightarrow \mathfrak{W}_{\text{all}}$ . In particular it has no extending order embedding  $\mathfrak{W}_{\text{all}} \rightarrow \mathfrak{W}_{\text{all}}$ .* This follows from source 14’s [corollary 47.3](#), whose target is a cylinder edge, the trace of no unrestricted class well-order; source 15’s [theorem 47.10](#) uses a first-label cut  $\{d : E_d(0) \in A\}$  with  $A \cong \mathbf{No}$ , of which [proposition 97.20](#) is the case  $A = \{x : x < 0\}$ .

*Source 21's proof.* Choose the exhaustive adjacent-pair-swap enumeration  $E$  used in [note 97.19](#). Its core cut is large by [note 97.18](#). The cut  $K$  of [proposition 97.20](#) is also large, so [note 97.17](#) gives a core automorphism  $h$  with  $h_*L_E = K$ .

Suppose  $\Phi : \mathfrak{W}_{\text{sl}} \rightarrow \mathfrak{W}_{\text{all}}$  were an extending order embedding. For every core code  $d$ , order preservation gives

$$E_d <_{\text{lex}} E \iff E_{h(d)} <_{\text{lex}} \Phi(E).$$

Thus the unrestricted class well-order  $\Phi(E)$  would induce exactly the cut  $K$  on the core. By [note 86.15](#), every unrestricted class well-order induces the same core cut as its canonical initial injective Ord-word. This contradicts [proposition 97.20](#). An extending embedding on all of  $\mathfrak{W}_{\text{all}}$  would restrict to one on  $\mathfrak{W}_{\text{sl}}$ , so it too is impossible.  $\square$

The two obstructions locate different boundaries. The cut of a nonexhaustive word fails the exhaustivity test in [note 97.11](#), but does have crossing cylinders at every ordinal level. The cut  $K$  fails already at the first level. The latter failure cannot be repaired by adding a residual tail to a class well-order. These examples do not say that every large cut is unrepresented:  $L_E$  itself is a large represented cut.

For the unrestricted order, preservation of initial-word cuts is a necessary extension condition. A sufficient general extension theorem would also have to control the residual fibers that those cuts do not distinguish. The exact set-like criterion in [note 97.13](#) avoids that additional issue because the core cut uniquely determines an exhaustive enumeration.

## 98 Residual data and the completion of the raw order (source 19)

[merge] Source 19's Sections 9–11, printed whole. The inherited facts it starts from, display (S19.14), are [proposition 15.3](#) for the fibres and the uniqueness clause of [theorem 45.1](#) with [corollary 46.4](#) and source 14's [theorem 54.6](#) (Part IX) for the kernel of the trace map. [Theorem 98.2](#) answers source 15's question "Completion with residual data" ([research question 113.50](#)) in an external set-sized formulation, and [theorem 98.5](#) source 15's "Automorphism extension" ([research question 113.51](#)), source 14's "Automorphisms of raw trace fibers" ([research question 113.39](#)), source 20's [research question 113.108](#) and source 21's [research question 113.120](#) in the same formulation. Source 19's [lemma 98.3](#) is the class version of source 21's set-level endpoint lemma ([lemma 65.7](#), Part X).

The first-block theorem does more than exhibit an obstruction to a naive completion: it specifies the data that a faithful completion must retain. We first recall the established fiber statement, then derive an exact reconstruction at a size level where all the objects involved are sets. The resulting description is exact, although computing the order types of all residual fibers remains a separate problem.

Fix a baseline bijection  $b : \text{Ord} \rightarrow \mathbf{No}$  and its class  $\mathcal{P}_{\text{bd}}(b)$  of normalized bounded-support permutation codes. Write  $E_c$  for the enumeration evaluated from  $c \in \mathcal{P}_{\text{bd}}(b)$ . For a raw class well-order  $R$ , let  $F_R : \text{Ord} \rightarrow \mathbf{No}$  enumerate its initial set-like part, and put

$$B_R = \mathbf{No} \setminus \text{ran}(F_R), \quad t(R) = \{c \in \mathcal{P}_{\text{bd}}(b) : E_c <_{\text{lex}} R\}.$$

The previously proved first-block and represented-cut theorems give

$$t(R) = t(S) \iff F_R = F_S, \quad t^{-1}(t(R)) \cong \mathcal{W}(B_R). \quad (\text{S19.14})$$

These are inherited results. Their second isomorphism is restriction to the residual alphabet; its inverse appends a residual well-order after the fixed word  $F_R$ . The fibers are convex. Distinct initial

words are separated by a core row: at their first ordinal disagreement their common prefix is a set, so a fresh numerical surreal can be inserted strictly between the two unequal labels and the resulting prefix completed to a core enumeration.

An exhaustive initial word has a singleton fiber, as does a word omitting one label. A word omitting  $n$  labels has a finite fiber with  $n!$  points. Only the exhaustive word in the first two cases is set-like. Moreover a principal core cut  $\mathcal{P}_{\text{bd}}(b)_{<c}$  has exactly the representative  $E_c$ , because that enumeration is exhaustive.

### 98.1 An external set-sized formulation

Fix an externally set-sized two-sorted model  $\mathcal{M} = (M, \mathcal{S})$  of GBC. The baseline is an admitted class of this model. Let  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  be the external set of its admitted raw class well-orders, and interpret  $\mathcal{P}_{\text{bd}}(b)$  externally as the set of its core codes. All comparisons are those computed in  $\mathcal{M}$ . No transitivity,  $\omega$ -standardness, or  $\beta$ -model assumption is needed in this subsection. Internal linearity and the internal density and absence of endpoints of  $\mathcal{P}_{\text{bd}}(b)$  yield actual external statements about these interpreted orders.

For a residual carrier  $B$ , the notation  $\mathcal{W}^{\mathcal{S}}(B)$  means the external set of admitted relations satisfying the model's class-well-order predicate. It need not be the collection of all externally well-founded relations on the external carrier. This distinction permits the theorem to cover nonstandard models without an implicit absoluteness assumption.

Let  $\mathcal{Q} = \text{DC}(\mathcal{P}_{\text{bd}}(b))$  be the ordinary external Dedekind completion of the dense order  $\mathcal{P}_{\text{bd}}(b)$ . Concretely its points are all external subsets  $q \subseteq \mathcal{P}_{\text{bd}}(b)$  which are nonempty, proper, initial, and have no greatest member, ordered by inclusion. A core point  $c$  is identified with  $\mathcal{P}_{\text{bd}}(b)_{<c}$ . Thus  $\mathcal{Q}$  is conditionally complete, has no endpoints, and contains  $\mathcal{P}_{\text{bd}}(b)$  densely. For  $q \in \mathcal{Q}$  set

$$H_q = \{R \in \mathfrak{W}_{\text{all}}^{\mathcal{M}} : t(R) = q\}.$$

Each  $H_q$  is empty or is a convex copy of  $\mathcal{W}^{\mathcal{S}}(B_q)$  for the uniquely recovered initial word. Different nonempty fibers compare exactly as their indices in  $\mathcal{Q}$ . In particular, this gives a lexicographic-sum description of  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  over its represented core cuts.

**Definition 98.1** (Completion convention for a nondense order). For a nonempty external linear order  $L$ , let  $\text{DC}(L)$  retain every old point and adjoin one point for each proper unfilled Dedekind gap

$$L = A \sqcup B, \quad A < B, \quad A, B \neq \emptyset, \quad A \text{ has no greatest point, } B \text{ has no least point.}$$

Let  $\text{IC}(L)$  additionally acquire a least point if  $L$  had none and a greatest point if  $L$  had none. Then  $\text{IC}(L)$  is a complete bounded chain. Existing adjacent pairs are retained; they are not split by new points.

Taking all downsets of a nondense order would not enforce this convention: it can separate distinctions which are already realized by an old point or an adjacent pair. The convention above is the gap-filling Dedekind completion relevant to the raw well-order space.

**Theorem 98.2** (Reconstruction by completed residual fibers). *For each  $q \in \mathcal{Q}$  define*

$$K_q = \begin{cases} \{*_q\}, & H_q = \emptyset, \\ \text{IC}(H_q), & H_q \neq \emptyset. \end{cases} \quad K = \sum_{q \in \mathcal{Q}} K_q,$$



where the sum is lexicographic. Embed  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  by retaining each of its points in its own fiber. Then  $K$  is canonically its Dedekind completion  $\text{DC}(\mathfrak{W}_{\text{all}}^{\mathcal{M}})$  relative to this embedding. Every old adjacent pair remains adjacent. For any  $x < y$  in  $K$ , either an old raw order lies strictly between them, or  $x, y$  themselves are an adjacent pair of old raw orders.

*Proof.* Convexity and the comparison of different fibers make the displayed map an order embedding. To prove conditional completeness, let  $A \subseteq K$  be nonempty and bounded above, let  $P$  be its projection to  $\mathcal{Q}$ , and put  $q_0 = \sup P$ . If  $q_0 \notin P$ , then

$$\sup A = (q_0, \min K_{q_0}).$$

Every member of  $A$  has smaller first coordinate, and any candidate with smaller coordinate is exceeded by a member of  $A$  by the definition of  $q_0$ . If  $q_0 \in P$ , the nonempty section  $A_{q_0}$  has a supremum  $z$  in the bounded complete chain  $K_{q_0}$ , and

$$\sup A = (q_0, z).$$

These arguments also establish minimality of the proposed upper bound. Infima are dual. The sum has no endpoints because  $\mathcal{Q}$  has none.

If  $x < y$  have different first coordinates  $q < r$ , choose a core index strictly between  $q$  and  $r$ . Its original core point lies strictly between  $x$  and  $y$ . If they belong to the same fiber, the ordinary interval-completion property supplies an old fiber point between them unless they are adjacent old points. Adding a missing fiber endpoint does not create adjacency, since the original fiber had no endpoint on that side.

It remains to check that the new points represent exactly old unfilled gaps. An internal gap point of  $\text{IC}(H_q)$  has old fiber points cofinal below and coinital above. An added lower endpoint has old fiber points coinital above, while core points with smaller indices are cofinal below; an added upper endpoint is dual. If  $H_q$  is empty, core points on either side of  $q$  approach the new singleton from both sides. Thus every new point has no greatest old predecessor and no least old successor. Conversely conditional completeness supplies a point in every old proper unfilled gap. Exactly the required gap points have therefore been added, giving the canonical completion isomorphism.  $\square$

The interval endpoints in this theorem are essential. For example, suppose  $H_q$  has no least point. Core points with indices below  $q$  must have a supremum in the full completion. Their supremum is the new lower endpoint of the completed fiber, not an arbitrarily selected residual well-order. The analogous statement holds at the upper endpoint.

## 98.2 Which boundary points are already present?

**Lemma 98.3** (Extrema of the residual well-order space). *Let  $B$  be an admitted linearly ordered carrier with  $\mathcal{W}^S(B) \neq \emptyset$ . This external well-order space has a least point if and only if the numerical order on  $B$  is itself an admitted class well-order. In that case this numerical order is the least point. It has a greatest point if and only if the reverse numerical order is an admitted class well-order, in which case that reverse order is the greatest point.*

*Proof.* Suppose  $R$  is least. If  $a < b$  numerically but  $b$  precedes  $a$  in  $R$ , conjugate  $R$  by the transposition interchanging these labels. This gives an admitted well-order which agrees with  $R$  before the old position of  $b$  and then has the smaller label  $a$ . It is smaller in the raw lexicographic comparison, a contradiction. Therefore  $R$  agrees with numerical order on every pair, so that order is an admitted well-order. Conversely, the numerical well-order always selects the least remaining numerical label. At its first disagreement with any other well-order it is therefore smaller. Reverse the comparisons for the greatest-point assertion.  $\square$

Consequently the lower boundary of a represented completed fiber is new exactly when its numerical residual alphabet is not a class well-order; the upper boundary is new exactly when the reverse numerical order fails to be a class well-order. These tests use the model's admitted predicate, while their conclusions concern actual external endpoints.

The empty, one-point, and standard finite residual cases require no boundary additions. An internally infinite residual numerical well-order has an old minimum and acquires a new upper endpoint; a reverse well-order acquires a new lower endpoint. When neither direction is a class well-order, both boundaries are added. Both directions can be class well-orders only when the carrier is internally finite: an infinite carrier would yield, by successive least-element selection, an  $\omega$ -sequence increasing in one direction and decreasing in the other. Without  $\omega$ -standardness, internal finiteness must not be silently replaced by external finiteness.

**Proposition 98.4** (Exactly which suborders embed into the core completion). *Let  $H' \subseteq \mathfrak{W}_{\text{all}}^{\mathcal{M}}$  contain the core. It has an order embedding into  $\mathcal{Q}$  fixing the specified core embedding if and only if it meets each  $H_q$  in at most one point. In that case the embedding is the trace map and is unique.*

*Proof.* A point of the dense completion  $\mathcal{Q}$  is uniquely determined by its comparisons with the core. Hence two raw points with the same trace cannot have distinct images. Conversely the trace map is an order embedding whenever it is injective, because different fibers compare by their cuts. These facts also prove uniqueness.  $\square$

Choosing one representative from each represented cut therefore gives a suborder whose completion is  $\mathcal{Q}$ . Appending the restriction of the baseline to each residual alphabet supplies a parameterwise selector. This is an ordinary external selector here; internally it does not form a class whose elements are arbitrary classes.

### 98.3 Automorphisms and the required decorations

For  $g \in \text{Aut}(\mathcal{P}_{\text{bd}}(b))$  let  $\bar{g} \in \text{Aut}(\mathcal{Q})$  be its unique extension, given on cuts by  $\bar{g}(q) = g[q]$ . Decorate each  $q$  by the order type of  $H_q$ , with the empty fiber counted as a separate decoration. Let  $G$  be the subgroup of core automorphisms preserving these decorations.

**Theorem 98.5** (The exact extension criterion). *A core automorphism  $g$  extends to an automorphism of  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  if and only if  $g \in G$ , equivalently if*

$$H_q \cong H_{\bar{g}(q)} \quad (q \in \mathcal{Q}).$$

Writing  $\text{Aut}(\mathfrak{W}_{\text{all}}^{\mathcal{M}}; \mathcal{P}_{\text{bd}}(b))$  for automorphisms preserving the core setwise, there is a short exact sequence

$$1 \longrightarrow \prod_{H_q \neq \emptyset} \text{Aut}(H_q) \longrightarrow \text{Aut}(\mathfrak{W}_{\text{all}}^{\mathcal{M}}; \mathcal{P}_{\text{bd}}(b)) \longrightarrow G \longrightarrow 1.$$

In the external set setting this sequence splits noncanonically. Every such raw-order automorphism extends uniquely to its Dedekind completion from [Theorem 98.2](#).

*Proof.* If  $h$  extends  $g$ , comparison with each core point gives

$$t(h(R)) = g[t(R)].$$

Thus  $h$  maps  $H_q$  isomorphically onto  $H_{\bar{g}(q)}$ , proving necessity. Conversely, ordinary external Choice selects isomorphisms between the corresponding nonempty fibers. Their union is an automorphism

of the lexicographic sum. At principal core cuts the maps are forced singleton maps, so the union restricts to  $g$ .

An automorphism fixing the core pointwise fixes every trace and hence acts independently within its fibers; every such independent family does give an automorphism. This identifies the kernel and proves exactness. To obtain a noncanonical splitting, choose one representative chain  $L_d$  for every occurring nonempty fiber order type and fixed isomorphisms  $\phi_q : H_q \rightarrow L_d$ . Lift  $g \in G$  on  $H_q$  by

$$\phi_{\bar{g}(q)}^{-1} \circ \phi_q.$$

The intermediate identifications cancel under composition, so these lifts form a homomorphic section. No canonical choice of the  $\phi_q$  is asserted. Extension to the completion is the usual unique extension of an automorphism to its filled Dedekind gaps.  $\square$

Merely preserving which cuts are represented is therefore weaker than the required criterion. The entire residual fiber type, including its adjacencies and endpoint behavior, is the necessary decoration. The external Choice used to select fiber isomorphisms is ordinary set Choice; the theorem does not infer an internal higher-order choice principle.

## 99 Why residual classification is not automatically self-similar (source 19)

[merge] Over **No** itself the trace of an injective word is injective ([theorem 45.1](#)); source 14's trace-fibre theorem ([theorem 54.6](#), Part IX) is that case. Over a general residual alphabet the next statements show that two initial words can share a trace, never three. Source 19 writes  $\mathcal{C}_X$  for a core of exhaustive enumerations of a proper linearly ordered class  $X$ ; it is not the core of **No**.

The initial-word theorem is exact, but an iteration must retain the numerical residual alphabet. Deleting a set from **No** preserves fresh separators. Deleting the entire proper range of an Ord-word need not do so. We can, however, describe the possible failure quite sharply. Work parameterwise in GBC, or with admitted words in one fixed class realization. For a proper linearly ordered class  $X$ , choose a core  $\mathcal{C}_X$  of exhaustive Ord-enumerations such that every injective set prefix has a core completion. A bounded-support core of a fixed baseline has this property. Write  $\tau_X(U)$  for the strict lower trace of an injective admitted Ord-word  $U$  on this core; the trace need not be a normalized cut for a general alphabet.

**Lemma 99.1** (Exact criterion for an unseparated pair). *Suppose  $U < V$  first differ at  $\alpha$ , with common prefix  $p$  and next labels  $a < b$ . There is no core enumeration strictly between  $U$  and  $V$  if and only if all three conditions hold:*

1. *No label of  $X \setminus \text{ran}(p)$  lies strictly between  $a$  and  $b$ .*
2. *For every  $\beta > \alpha$ ,  $U(\beta)$  is the greatest label of  $X \setminus \text{ran}(U \upharpoonright \beta)$ .*
3. *For every  $\beta > \alpha$ ,  $V(\beta)$  is the least label of  $X \setminus \text{ran}(V \upharpoonright \beta)$ .*

*When these hold, neither word is exhaustive, no injective admitted Ord-word lies strictly between them, and  $\tau_X(U) = \tau_X(V)$ . Thus they also characterize equality of the two strict lower traces.*

*Proof.* A label between  $a$  and  $b$ , appended to  $p$ , gives a core separator. If  $U$  later chooses a label below some unused label, complete its prefix through that larger alternative. The result exceeds  $U$  but remains below  $V$  because its label at  $\alpha$  is still  $a$ . A later failure to choose the least unused label in  $V$  gives the dual separator. This proves necessity.

Conversely a word strictly between them must share  $p$  and at  $\alpha$  select  $a$ ,  $b$ , or an intermediate label. The third option is excluded. A word selecting  $a$  is at most  $U$ , since  $U$  selects the greatest unused label at every later stage. A word selecting  $b$  is at least  $V$ . This proves the stronger absence of any injective admitted word strictly between them.

To handle the strict-trace endpoint issue, suppose  $U$  were exhaustive. Its tail after  $a$  would then be the reverse numerical class well-order of  $X \setminus (\text{ran}(p) \cup \{a\})$ . The later tail of  $V$  is numerically strictly increasing and has proper range. Start beyond the possible occurrence of  $a$  and take  $V(\beta_0 + n)$  for  $n \in \omega$ . Class Replacement makes this an available set sequence, descending in the reverse numerical well-order, a contradiction. The dual argument excludes exhaustive  $V$ . Consequently neither word equals a core enumeration. Absence of a strict core separator is therefore exactly equality of their strict lower traces.  $\square$

**Proposition 99.2** (At most two initial words over an arbitrary residual cut). *There cannot be three distinct admitted injective Ord-words over  $X$  with the same core trace. In an external interpretation, each nonempty raw trace fiber is therefore either*

$$\mathcal{W}^S(B_U) \quad \text{or} \quad \mathcal{W}^S(B_U) + \mathcal{W}^S(B_V),$$

where the two-word case is the ordered pair described in [Lemma 99.1](#). Residual well-order completions are supplied by GBC. This assertion is independent of the baseline among cores having the stated prefix-completion property.

*Proof.* For three words with equal trace, apply [Lemma 99.1](#) to the outer pair. The middle word contradicts its absence of intervening injective words. For raw orders the trace is the trace of their initial word, and the orders with one fixed initial word form exactly its residual well-order space. Distinct initial-word fibers compare in the order of their words. These facts give the displayed possibilities. The criterion mentions only the numerical alphabet and the two words, which proves baseline independence.  $\square$

**Proposition 99.3** (An alphabet criterion for failure of core separation). *Keep the hypotheses above:  $X$  is a proper linearly ordered class, all words and class isomorphisms are admitted in the fixed realization, and every injective set prefix extends to an exhaustive row of  $\mathcal{C}_X$ . The following are equivalent:*

1. *Two distinct admitted injective Ord-words over  $X$  have the same strict lower trace on  $\mathcal{C}_X$ .*
2. *For some set  $S \subseteq X$ , the numerical order of  $X \setminus S$  has an admitted initial segment of type Ord and an admitted final segment of type Ord\*, with admitted class isomorphisms witnessing these types.*

*Consequently all distinct admitted injective words are separated by a core row exactly when no such set deletion exists.*

*Proof.* For the first implication, let  $U < V$  be a pair from [Lemma 99.1](#), with common set prefix  $p$  and unequal labels  $a < b$ . Delete  $S = \text{ran}(p) \cup \{a, b\}$ . The range of the later greedy-minimum tail of  $V$ , with  $a$  removed if present, is an initial numerical segment of  $X \setminus S$  of type Ord. Indeed any smaller residual label must have been selected earlier in that tail. Removing at most one label preserves its proper, set-like well-order type. Dually the later greedy-maximum range of  $U$ , with  $b$  removed if present, is a final segment of type Ord\*. These ranges and their canonical enumerations are admitted class constructions from  $U, V, p$ .

Conversely let  $J$  be the initial Ord-segment and  $K$  the final Ord\*-segment of  $Y = X \setminus S$ . They are disjoint: a common point would place an increasing  $\omega$ -sequence from the remaining tail of  $J$

inside  $K$ , contradicting the well-ordering of  $K$  in the reverse numerical direction. Well-order  $S$  as a common set prefix  $p$ , and let  $a < b$  be the first two labels of  $J$ . They are consecutive in  $Y$ . After  $p$ , form  $U$  by choosing  $a$  and then enumerating  $K$  downward; form  $V$  by choosing  $b$  and then enumerating  $J \setminus \{b\}$  upward. Each concatenation has type  $\text{Ord}$  and is an admitted injective word. The first tail always selects the greatest unused label because  $K$  is final; the second always selects the least because  $J \setminus \{b\}$  is initial in  $Y \setminus \{b\}$ . [Lemma 99.1](#) therefore gives equal strict core traces.  $\square$

The following example realizes the two-word obstruction even if the comparison core is enlarged to all exhaustive set-like enumerations.

For  $\alpha \in \text{Ord}$  put

$$\ell_\alpha = 1 - \frac{1}{\alpha + 2}, \quad r_\alpha = 2 + \frac{1}{\alpha + 2}, \quad Y = \{\ell_\alpha : \alpha \in \text{Ord}\} \cup \{r_\alpha : \alpha \in \text{Ord}\}.$$

Here  $\alpha$  is viewed as its ordinal surreal, and every displayed arithmetic operation is a surreal field operation. The left labels increase in type  $\text{Ord}$  inside  $[1/2, 1)$ ; the right labels decrease in type  $\text{Ord}^*$  inside  $(2, 5/2]$ . Every left label lies below every right label, so the numerical order of  $Y$  has type  $\text{Ord} + \text{Ord}^*$ .

**Proposition 99.4** (Different initial words with the same residual core cut). *There are injective  $\text{Ord}$ -words  $U < V$  over  $Y$  with different ranges whose strict lower traces on all admitted exhaustive set-like enumerations of  $Y$  coincide. Each word is the initial word of a raw well-order of  $Y$ . Moreover  $Y$  occurs as a residual alphabet after one proper initial block in a raw well-order of  $\mathbf{No}$ .*

*Proof.* Take

$$U = (\ell_0, r_0, r_1, r_2, \dots), \quad V = (\ell_1, \ell_0, \ell_2, \ell_3, \dots).$$

The first expression means the unique  $\text{Ord}$ -collapse of the order consisting of  $\ell_0$  followed by the  $r_\alpha$  sequence; this is valid because  $1 + \text{Ord}$  has type  $\text{Ord}$ . The second expression is the increasing left sequence with its first two labels transposed. Both words omit proper subclasses of  $Y$ , and  $U < V$  at the first label.

Let  $E : \text{Ord} \rightarrow Y$  be any admitted exhaustive set-like enumeration. If  $E(0) = \ell_0$ , then  $E \neq U$  since  $U$  omits every  $\ell_\alpha$  with  $\alpha \geq 2$  (and also  $\ell_1$ ). At their first later disagreement,  $U$  chooses the greatest still-unused numerical label: it is sweeping the right block downward from its maximum. Consequently  $E < U$ .

If  $E(0) = \ell_1$ , then  $E \neq V$  since  $V$  omits the right block. At their first later disagreement,  $V$  chooses the least still-unused numerical label, so  $E > V$ . If  $E(0) > \ell_1$ , that inequality holds immediately. There is no label below  $\ell_0$  or strictly between  $\ell_0$  and  $\ell_1$ . Therefore

$$\{E : E <_{\text{lex}} U\} = \{E : E(0) = \ell_0\} = \{E : E <_{\text{lex}} V\},$$

where  $E$  ranges over exhaustive set-like enumerations. Appending an admitted residual well-order to either word gives its raw completion, with that word as initial set-like block.

Finally  $\mathbf{No} \setminus Y$  is proper: it contains all negative ordinals. Restricting the baseline and collapsing the resulting set-like order gives a bijection  $F : \text{Ord} \rightarrow \mathbf{No} \setminus Y$ . Its raw completions have residual alphabet exactly  $Y$ .  $\square$

**Proposition 99.5** (An explicit interior completion point). *In an external set-sized interpretation of the preceding example, the shared raw trace fiber of  $U, V$  is exactly*

$$\mathcal{W}^S(\{\ell_\alpha : \alpha \geq 1\}) + \mathcal{W}^S(\{r_\alpha : \alpha \in \text{Ord}\}).$$

*The junction between these summands is a proper unfilled Dedekind gap of  $\mathcal{W}^S(Y)$ . It supplies an interior new point of the completed residual fiber, in addition to any points added at its outer boundaries.*

*Proof.* The two residual alphabets are exactly the labels omitted by  $U$  and  $V$ , respectively. [Proposition 99.2](#) excludes a third initial word with their trace, so their raw trace fiber is the displayed convex sum. The first alphabet is numerically of type  $\text{Ord}$ , and the second of type  $\text{Ord}^*$ . By [Lemma 98.3](#), the first summand has no greatest point and the second has no least point.

The downward closure of the first summand and the upward closure of the second partition the full raw order of  $Y$ : the two summands together form a convex trace fiber, and there is no further raw point between them. The first closure has no greatest point and the second no least point, giving the claimed unfilled gap. Its completion point has raw orders on both sides. Since  $Y$  is realized as a residual alphabet of  $\mathbf{No}$ , the point also occurs inside the interval completion of an actual raw surreal-order fiber in [Theorem 98.2](#).  $\square$

Thus the original surreal theorem cannot simply be reapplied inside every residual fiber. On this residual alphabet, equality of traces on its own set-like core merges different initial  $\text{Ord}$ -words. The merger is limited to two words by [Proposition 99.2](#), but its two residual well-order spaces must both be retained. A recursive theory must preserve this prefix-and-alphabet structure; the set/proper-class distinction alone is insufficient to make the process self-similar.

## 100 Persistent external cuts absent from the class realization (source 19)

The full external completion in [Theorem 98.2](#) uses all external cuts of the core, whereas the internal represented-cut theorem takes an admitted cut as a class parameter. This is a strict distinction in every externally countable model, even for cuts whose every internally bounded prefix exists.

**Theorem 100.1** (Continuum many omitted amenable branches). *Let  $\mathcal{M} = (M, \mathcal{S})$  be any externally countable two-sorted model of GBC. Its external core completion contains  $2^{\aleph_0}$  distinct cuts which have an internally set-coded straddling prefix at every internal ordinal but are represented by no admitted raw class well-order. These cuts can be induced by external exhaustive bijections*

$$F : \text{Ord}^{\mathcal{M}} \longrightarrow \mathbf{No}^{\mathcal{M}}$$

*whose restriction to every internal ordinal is coded by a set of  $M$ . No  $\omega$ -standardness assumption is required.*

*Proof.* Externally enumerate all admitted subclasses of  $\text{Ord}$  as  $A_0, A_1, \dots$ , and enumerate all internal ordinals as  $\delta_0, \delta_1, \dots$ . Set  $\gamma_0 = 0$ . At stage  $n$  put

$$\eta_n = \max^{\mathcal{M}}(\gamma_n, \delta_n), \quad \gamma_{n+1} = \eta_n + 2.$$

These lengths are independent of the following free choices and are externally cofinal in the interpreted ordinal order.

For an external binary sequence  $s \in 2^\omega$ , build coherent internally set-coded binary functions  $p_n : \gamma_n \rightarrow \{0, 1\}$ . Start with the empty function. Extend  $p_n$  by zeros on  $[\gamma_n, \eta_n)$ , and set

$$p_{n+1}(\eta_n) = 1 - \mathbf{1}_{A_n}(\eta_n), \quad p_{n+1}(\eta_n + 1) = s(n).$$



Each step is a finitely specified internal set construction. Their coherent external union defines the characteristic function of an external subclass  $A_s \subseteq \text{Ord}^M$ . For every internal ordinal  $\alpha$ , choose  $n$  with  $\alpha \leq^M \gamma_n$ ; its restriction to  $\alpha$  is then the internal set  $p_n \upharpoonright \alpha$ . This is the claimed amenability to internally bounded initial segments. At  $\eta_n$  it differs from  $A_n$ , so  $A_s$  is not admitted. Distinct  $s$  give distinct classes at the free coordinates  $\eta_n + 1$ .

Use the internal partition of the ordinals into pairs  $(2 \cdot \alpha, 2 \cdot \alpha + 1)$ , with ordinal left multiplication by 2. In the baseline  $b$ , swap the two values of a pair exactly when  $\alpha \in A_s$ . This defines an external exhaustive bijection  $F_s : \text{Ord}^M \rightarrow \mathbf{No}^M$ : the coordinate permutation is an involution on disjoint pairs. Every internally bounded initial restriction is coded in  $M$ , because it uses only an internally bounded restriction of  $A_s$  and of the admitted baseline. If  $F_s$  were admitted, elementary class comprehension from it and  $b$  would recover  $A_s$ , a contradiction.

Although the model's ordinals may be externally ill-founded,  $F_s$  and an admitted core row  $E_c$  have a least disagreement. They cannot be equal, since  $F_s$  is not admitted. Choose one disagreement  $\delta$ . Both restrictions through  $\delta + 1$  are coded internal sets, so their nonempty disagreement subset has an internally least member. Every earlier internal ordinal lies in that restriction; hence this is also their least disagreement in the external interpreted ordinal order.

Define

$$q_s = \{c \in \mathcal{P}_{\text{bd}}(b) : E_c <_{\text{lex}} F_s\}.$$

First-difference comparison makes  $q_s$  initial. Fresh labels and core completion make it nonempty, proper, and without a greatest member. For any internal ordinal  $\alpha$ , the prefix  $F_s \upharpoonright \alpha$  is coded in  $M$ . Complete it once with a fresh label below  $F_s(\alpha)$  and once with a fresh label above. The two core points lie on opposite sides of  $q_s$  in its prefix cylinder. Thus its prefix criterion persists at every internal ordinal.

Suppose an admitted raw order  $R$  represented  $q_s$ . Its admitted initial word  $F_R$  cannot equal  $F_s$ . Their least disagreement exists by the same bounded-restriction argument. A fresh core separator strictly between the two unequal labels, after their common set prefix, distinguishes their lower core cuts. This contradicts  $t(R) = q_s$ . Thus  $q_s$  has empty raw fiber. It cannot itself be an admitted class cut, since the internal persistent-prefix theorem would then assemble an admitted representing word.

Finally, distinct free-bit paths give different  $F_s$ . Their least disagreement exists by the same argument, and a core separator makes their cuts distinct. There are  $2^{\aleph_0}$  paths, proving the claim.  $\square$

These missing cuts are not terminal cuts of the internal first-failure classification. Their crossing prefixes persist throughout the model's ordinal range; the obstruction is that their coherent global branch is absent from the class realization. Their singleton fibers in the external completion are therefore an additional, explicitly exhibited source of completion points.

Here amenability means that internally bounded restrictions, and hence intersections with internal sets of ordinals, are coded in the model. It does not assert that adjoining  $A_s$  or  $F_s$  to the class realization preserves GBC, Class Replacement, or the relevant comprehension closure.

## 101 The exact least ordinal support bound (source 27)

[merge] Source 27's Section 8, printed whole. Its first subsection re-derives source 17's exact reserve cardinal ([theorem 49.7](#); display (S27.13) is (S17.7)), which it credits, and is printed as text. Its [theorem 101.2](#) answers the ordinal half of source 17's question "Ordinal and birthday optima for supported completion" ([research question 113.78](#)) and Part VIII's "Ordinal support bounds ...

remain open” (section 50 (f)); birthday bounds and the forced-prefix length remain open (section 105). Its first example extends source 17’s example 49.8.

Completing a set prefix to a global enumeration is a recurring step in the preceding constructions. A generic proof takes a sufficiently large cardinal block and completes a partial permutation there. The repository has already optimized the *number* of additional moved positions (theorem 49.7). We now optimize their *ordinal location*. A support of two very high ordinals is small in cardinality and large in height, so these are genuinely different problems.

### 101.1 Forced motion and the cardinal reserve

Let  $A \subseteq \text{Ord}$  be a set and  $h : A \rightarrow \text{Ord}$  injective. Write

$$\begin{aligned} D &= \{a \in A : h(a) \neq a\}, & R &= h[D], & S &= D \cup R, \\ u &= |R \setminus D|, & v &= |D \setminus R|, \\ \tau(h) &= \begin{cases} 0, & u = v, \\ \max(\aleph_0, u, v), & u \neq v. \end{cases} \end{aligned} \tag{S27.12}$$

Cardinals are identified with initial ordinals. The prescribed fixed points  $A \setminus D$  are disjoint from  $R$ , by injectivity of  $h$ .

Every extending permutation  $\pi$  must move each point of  $S$ . This is immediate for  $D$ . If  $r \in R \setminus D$ , some distinct prescribed point already maps to  $r$ , so  $\pi(r) = r$  would violate injectivity. For an extension with set support, put  $T = \text{supp } \pi \setminus S$ . Removing the prescribed bijection  $h : D \rightarrow R$  leaves a bijection

$$(R \setminus D) \cup T \longrightarrow (D \setminus R) \cup T. \tag{S27.13}$$

Consequently  $u + |T| = v + |T|$ . If  $u \neq v$  are finite, finite cancellation forces  $T$  to be infinite. If  $u \neq v$  and one is infinite, the rule for addition of well-ordered infinite cardinals forces  $|T| \geq \max(u, v)$ . This proves  $|T| \geq \tau(h)$ .

Conversely, a fresh set  $T$  of size  $\tau(h)$  outside  $A \cup R$  makes the two sides of (S27.13) equipotent. Any bijection between them, combined with  $h$  on  $D$  and the identity elsewhere, completes the partial injection. This is the earlier exact cardinal reserve result. Because all sets here are well-orderable subsets of ordinals, its pointwise existence argument needs no choice for arbitrary sets; it is available in ZF. A uniform simultaneous choice of matchings is a separate assertion.

### 101.2 A necessary and sufficient cutoff criterion

**Theorem 101.1** (Completion below a specified ordinal). *Let  $\beta$  be an ordinal. There exists a permutation  $\pi : \text{Ord} \rightarrow \text{Ord}$  extending  $h$ , with  $\text{supp } \pi \subseteq \beta$ , if and only if*

$$S \subseteq \beta \quad \text{and} \quad |\beta \setminus (A \cup R)| \geq \tau(h). \tag{S27.14}$$

*The second condition is automatic when  $\tau(h) = 0$ .*

*Proof.* Suppose such a permutation exists. Its forced support contains  $S$ . The remaining moved set  $T$  misses  $R$  by definition and misses  $A \setminus D$  because those points are prescribed fixed. Thus  $T \subseteq \beta \setminus (A \cup R)$ . The reserve argument above gives  $|T| \geq \tau(h)$ , proving both necessary conditions.

Conversely, choose  $T \subseteq \beta \setminus (A \cup R)$  of cardinality  $\tau(h)$ . Choose a bijection in (S27.13); both sides are well-orderable and have equal cardinality. Its domain,  $D$ , and the complement of  $S \cup T$  partition  $\text{Ord}$ , and their proposed images partition  $\text{Ord}$  as well. The resulting piecewise map is

therefore a permutation. It extends  $h$ , fixes every point outside  $S \cup T$ , and has support contained in  $\beta$ . Some reserve points may happen to be fixed, but the lower bound forces the cardinality of the moved reserve to remain  $\tau(h)$  when that cardinal is nonzero.  $\square$

### 101.3 A closed formula for the least bound

Let  $f : \text{Ord} \rightarrow \text{Ord} \setminus (A \cup R)$  be the increasing enumeration of the fresh ordinals. It is canonically defined by

$$f(\xi) = \gamma \iff \gamma \notin A \cup R \text{ and } \text{otp}(\gamma \setminus (A \cup R)) = \xi.$$

The complement of a set of ordinals has type  $\text{Ord}$ , so this formula defines  $f$  without a class-valued recursion. Put

$$\delta_S = \sup_{s \in S} (s + 1), \quad \delta_T = \sup_{\xi < \tau(h)} (f(\xi) + 1), \quad (\text{S27.15})$$

with an empty supremum equal to 0.

**Theorem 101.2** (Exact ordinal support optimum). *The least ordinal  $\beta$  for which  $h$  extends to a permutation supported in  $\beta$  is*

$$\boxed{\beta_{\min}(h) = \max(\delta_S, \delta_T)}. \quad (\text{S27.16})$$

*This minimum is attained. The formula specifies a canonical optimum, not a canonical completing permutation.*

*Proof.* We have  $S \subseteq \beta$  exactly when  $\delta_S \leq \beta$ . Since  $\tau(h)$  is an initial cardinal, the fresh set below  $\beta$  has cardinality at least  $\tau(h)$  exactly when its order type is at least  $\tau(h)$ . That is equivalent to  $f(\xi) < \beta$  for every  $\xi < \tau(h)$ , and therefore to  $\delta_T \leq \beta$ . The cutoff criterion now says that the admissible  $\beta$  are exactly the ordinals above both displayed bounds. Their least member is their maximum, and the converse part of [theorem 101.1](#) attains it.  $\square$

**Example 101.3** (The one-sided shift). Prescribe  $h(n) = n + 1$  for  $n < \omega$ . Then  $S = \omega$ ,  $u = 0$ ,  $v = 1$ ,  $\tau(h) = \omega$ , and  $f(n) = \omega + n$ . Hence

$$\beta_{\min}(h) = \omega \cdot 2.$$

An optimum completion maps  $\omega$  to 0 and maps  $\omega + n + 1$  to  $\omega + n$ , while retaining the prescribed shift on  $\omega$  and fixing all ordinals at least  $\omega \cdot 2$ . No finite reserve beyond  $\omega$  can close this one-sided infinite chain.

**Example 101.4** (Prescribed fixed positions can delay the reserve). Add the requirements  $h(\xi) = \xi$  for  $\omega \leq \xi < \gamma$ , where  $\gamma \geq \omega$ . The forced moved set and the cardinal reserve are unchanged, but the first available fresh ordinals are now  $\gamma, \gamma + 1, \dots$ . Therefore

$$\beta_{\min}(h) = \gamma + \omega.$$

This example explains why  $S$  and  $\tau(h)$  alone do not determine the ordinal optimum: the fixed domain  $A \setminus D$  can block the reserve.

**Example 101.5** (High fixed points need not matter). If  $h$  is the identity on  $A$ , then  $S = \emptyset$  and  $\beta_{\min}(h) = 0$ , however high the elements of  $A$  lie. Adding only finitely many fixed points outside  $\omega$  to the one-sided shift still leaves  $\beta_{\min} = \omega \cdot 2$ , even if one of those fixed points is arbitrarily high. The bound measures motion, not the height of the input data.

**Example 101.6** (Balanced infinite motion). For  $A = \{2n : n < \omega\}$  and  $h(2n) = 2n + 1$ , both  $u$  and  $v$  are  $\aleph_0$ , so  $\tau(h) = 0$ . Transposing each even number with the next odd number gives the optimum  $\beta_{\min} = \omega$ . Equal infinite boundary cardinalities require no reserve.

### 101.4 Application and exact limits of the result

For a prescribed injective surreal prefix  $p : \alpha \rightarrow \mathbf{No}$ , put  $h = b^{-1} \circ p$  with  $A = \alpha$ . Then [theorem 101.2](#) gives the least baseline coordinate bound below which one can modify  $b$  so that  $b \circ \pi$  begins with  $p$ . This is an exact optimization for a fixed baseline and a fixed prefix.

It does not optimize the birthday of the set code for  $\pi$ , the birthday of an order-embedded surreal representing that code, or the numerical size of the moved surreal labels. Nor does it select the prefix that optimizes an interpolant between two given cuts. Those involve additional choices and remain distinct research problems. The strict inequality in “support contained in  $\beta$ ” is the reason for every successor  $s + 1$  and  $f(\xi) + 1$  in ([S27.15](#)).

## 102 Class realizations: what labelled and marked enumeration spaces remember (sources 22, 27, 21 and 30)

[merge] Part VIII proved that, over a fixed set universe and baseline, an old core cut is represented in a larger class realization exactly when it is in the smaller one, and that every proper extension of the class part adds a global enumeration and a represented cut ([theorems 48.1](#) and [48.4](#)); it left open what the external order type of the admitted enumerations remembers ([section 50](#) (e)). Four sources answer as follows.

- *Labelled availability recovers the whole class part.* Source 22’s [theorem 102.3](#), printed first: with a common baseline  $b$  and a common class bijection  $g : \text{Ord} \rightarrow M$ , inclusion of the admitted exhaustive traces, or only of the pair-swap traces, is equivalent to inclusion of the class realizations. Source 27 proves it with only  $b$  shared, as a criterion for order embeddings fixing the core ([note 102.5](#)), and source 30 with no baseline given in advance ([note 102.16](#)); both are printed as notes with their proofs. The pair-swap code is the one source 17 uses for [theorem 48.4](#).
- *External size.* The external collection of admitted enumerations has the cardinality of the class part (source 21’s [theorem 102.13](#); source 30’s [note 102.17](#)).
- *Countable models: the bare order forgets.* The external order is  $\mathbb{Q}$  (source 13’s [proposition 55.5](#), Part IX, proved again as [theorem 102.13](#)); it stays  $\mathbb{Q}$  with the core as a dense colour and the baseline fixed ([proposition 102.19](#)), and it does not detect ETR ([proposition 102.20](#)). Fixing the core pointwise recovers everything ([note 102.5](#) and [corollary 102.12](#)).
- *A common baseline exists in every GBC-realization of a countable model exactly when some  $p$  has  $V = \text{HOD}_p$  there* ([theorem 102.21](#) and [corollary 102.22](#), applications of Williams’s minimum-model theorem).

Source 21’s section also proves the well-foundedness facts the comparisons need ([note 102.7](#) is the scope statement already printed in [section 5.6](#)).

### 102.1 Source 22: recovering an available class realization

#### 102.1.1 The coding parameters and the model comparison

Let  $M$  be a fixed set universe, and let  $\mathcal{X}$  and  $\mathcal{Y}$  be class realizations over it such that  $(M, \mathcal{X})$  and  $(M, \mathcal{Y})$  satisfy GBC. Comparisons between these class collections are made in the metatheory. Assume they share a global enumeration

$$g : \text{Ord}^M \longrightarrow M$$

and a baseline enumeration  $b : \text{Ord}^M \rightarrow \mathbf{No}^M$ . Supplied parameters are part of the hypotheses, not claimed to be canonically definable in an arbitrary model.

We may arrange that the two labels in each consecutive pair of baseline positions are numerically ordered:

$$b(2 \cdot \xi) < b(2 \cdot \xi + 1).$$

Indeed, sort each pair of an existing baseline. Every ordinal is uniquely  $2 \cdot \xi + i$  for  $i < 2$ ; this is ordinal multiplication with the finite factor on the *left*. The sorting transformation is elementary and available in both realizations.

Because  $M$  and  $b$  are fixed, the set-coded core  $\mathcal{P}_s(b)^M$ , its ordering, all set-prefix nodes, and their incidence are identical in the two realizations. What may differ is which class cuts and which full class branches are available.

### 102.1.2 Encoding an arbitrary class by pair swaps

For an available class  $A \subseteq M$ , define its ordinal bit class by

$$a_A(\xi) = 1 \iff g(\xi) \in A.$$

Define  $e_A$  by leaving the  $\xi$ th baseline pair in its original order when  $a_A(\xi) = 0$ , and reversing it when  $a_A(\xi) = 1$ . Explicitly,

$$\begin{aligned} e_A(2 \cdot \xi) &= \begin{cases} b(2 \cdot \xi), & g(\xi) \notin A, \\ b(2 \cdot \xi + 1), & g(\xi) \in A, \end{cases} \\ e_A(2 \cdot \xi + 1) &= \begin{cases} b(2 \cdot \xi + 1), & g(\xi) \notin A, \\ b(2 \cdot \xi), & g(\xi) \in A. \end{cases} \end{aligned}$$

It is a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ , since each baseline label occurs exactly once.

**Proposition 102.1** (Pair-swap encoding). *The assignment  $A \mapsto e_A$  is injective and elementary in the supplied class parameter  $A$ . On ordinal bit classes it preserves their first-disagreement lexicographic order. Moreover,*

$$e_A \text{ is evaluated by a code in } \mathcal{P}_s(b) \iff A \text{ is a set.}$$

*Proof.* At the first bit disagreement  $\xi$ , the first label disagreement is at  $2 \cdot \xi$ , where the zero bit yields the smaller baseline label and the one bit the larger. This proves order preservation and injectivity.

The supporting permutation moves exactly the two positions of those pairs with bit one. Its support is a set exactly when the one-bit ordinal class is a set. Since  $g$  is a class bijection, class replacement in either direction identifies this condition with  $A$  being a set. Uniqueness of the permutation relative to  $b$  rules out an alternative supported representation of the same enumeration.  $\square$

**Theorem 102.2** (Elementary decoding from a lower trace). *For every available class  $A$ , its normalized lower trace*

$$T_A = \{d \in \mathcal{P}_s(b) : d < e_A\}$$

*is available. From  $T_A$ , the common set-coded labelled incidence structure, and  $b, g$ , the class  $A$  is recovered by an elementary formula. In particular, no satisfaction predicate or class-valued transfinite recursion is required.*

*Proof.* The trace is an elementary class definition from  $e_A$  and evaluation. It is normalized by [note 86.9](#). Apply [theorem 87.4](#) to reconstruct its unique word, which must be  $e_A$ . Then

$$g(\xi) \in A \iff e_A(2 \cdot \xi) = b(2 \cdot \xi + 1).$$

Substituting the elementary crossed-prefix formula for the reconstructed value gives the promised decoder using only set quantifiers and the supplied class parameters. The existence of a crossed node long enough to read that coordinate is part of [theorem 87.4](#), not an extra choice requirement.  $\square$

Call a normalized represented trace a *binary-swap trace* when its reconstructed branch sends each pair of positions to the corresponding baseline pair in one of the two possible orders. Such a branch is automatically exhaustive, and the preceding decoder yields its unique class  $A$ . Thus the class sort can be represented by this specially recognized collection of traces; the entire family of exhaustive traces is more than is needed.

### 102.1.3 Faithfulness for same-set class realizations

In the metatheory, let  $\mathcal{B}_\mathcal{X}$  be the collection of lower traces on the common core represented by exhaustive enumerations available in  $\mathcal{X}$ . Let  $\mathcal{S}_\mathcal{X}$  be its subcollection of binary-swap traces. These are metatheoretic collections, not objects asserted to exist inside  $(M, \mathcal{X})$ .

**Theorem 102.3** (Faithful reconstruction of the class realization). *With the fixed  $M, b, g$  above, the following are equivalent:*

$$\mathcal{X} \subseteq \mathcal{Y}, \quad \mathcal{B}_\mathcal{X} \subseteq \mathcal{B}_\mathcal{Y}, \quad \mathcal{S}_\mathcal{X} \subseteq \mathcal{S}_\mathcal{Y}.$$

*In particular, equality of the admitted exhaustive traces, or just of the binary-swap traces, is equivalent to equality of the available class realizations.*

*Proof.* Inclusion of class realizations preserves all class enumerations and their elementary lower traces, so it implies inclusion of each trace collection.

Conversely, suppose the exhaustive trace inclusion holds, and take  $A \in \mathcal{X}$ . Construct  $T_A$  in  $\mathcal{X}$ . By hypothesis it is the lower trace of an exhaustive enumeration available in  $\mathcal{Y}$ , and hence it is itself an available class in  $\mathcal{Y}$ . The elementary decoder of [theorem 102.2](#), performed in  $(M, \mathcal{Y})$  using the common incidence and parameters, recovers exactly  $A$ . The decoding formula has only set quantifiers. Its interpretation is identical in the two models because their set universe, trace, and coding parameters agree. Thus the decoded external subclass really is the same  $A$ , and  $A \in \mathcal{Y}$ .

If only binary-swap trace inclusion is assumed, the same argument applies because every  $T_A$  is a binary-swap trace. This proves all equivalences. Applying the inclusion assertions in both directions proves the equality assertion.  $\square$

**Corollary 102.4** (Membership reconstruction). *Over the fixed set sort  $M$ , the class-membership structure is recoverable from a sort of admitted binary-swap traces with core incidence and the parameters  $b, g$ . The decoder interprets  $x \in A$  from the corresponding trace and the unique ordinal  $\xi$  with  $g(\xi) = x$ .*

This is a relative reconstruction statement. It is stronger than merely detecting that a proper same-set class extension adds some traces, which was established in the earlier continuation (source 17, [theorem 48.4](#)). It is not a reconstruction of  $M$ , of  $b, g$ , or of a class realization from the bare order type of  $\mathfrak{W}_{\text{sl}}$ .



### 102.1.4 Why this is not a diagonal paradox

The input to  $A \mapsto T_A$  is itself a class variable. The theorem does not give a set code for every  $A$ , nor a class function whose set-valued arguments enumerate all classes. A third sort may contain all these traces as objects, but then the sort and its axioms must be explicitly introduced. Inside GBC, each encoding or decoding is a schema uniform in one supplied class parameter.

Indeed, an exhaustive uniform enumeration of all global enumerations is impossible. Suppose a single class graph  $F(\xi, \eta)$  gave global bijections  $F_\xi$  for every ordinal  $\xi$  and covered every available global bijection. Define a bit class by

$$a(\xi) = 1 \iff F_\xi(2 \cdot \xi) \neq b(2 \cdot \xi + 1).$$

The associated pair-swap enumeration  $e_a$  differs from  $F_\xi$  at position  $2 \cdot \xi$ : when  $a(\xi) = 1$  it outputs the unequal upper baseline label; when  $a(\xi) = 0$  it outputs the lower label instead of the equal upper one. Thus  $e_a$  is missing from the family. This is an elementary diagonal argument, and it coexists with [theorem 102.3](#) because the latter never supplies such a set-indexed exhaustive family.

[merge] The pair-swap enumeration  $e_A$  and the decoder of [theorem 102.2](#) are source 17's construction in the proof of [theorem 48.4](#); [theorem 102.3](#) turns that construction into a complete invariant. The diagonal paragraph above is [theorem 19.37](#) in pair-swap form.

## 102.2 Source 27: the marked core detects the admitted classes of a model

The same first-order set universe can support different collections of admitted classes. Their unmarked external order types may conceal that difference. Marking the common bounded-support core pointwise gives a much more rigid invariant.

### 102.2.1 Reconstruction from a trace uses only set quantifiers

For a class injection  $E : \text{Ord} \rightarrow \mathbf{No}$ , define its lower trace on the core by

$$L_E = \{d \in \mathcal{P}_s(b) : E_d <_{\text{lex}} E\}.$$

For an injective set prefix  $p$ , let  $C_p = \{d \in \mathcal{P}_s(b) : E_d \text{ extends } p\}$ . Call  $p$  *straddling* if  $C_p$  meets both  $L_E$  and its complement.

The prefix  $E \upharpoonright \alpha$  straddles at every set length  $\alpha$ : append fresh letters on both sides of  $E(\alpha)$  and complete to core codes. No other prefix at that length straddles, since the cylinders at a fixed level are linearly arranged and a lower cut cannot cross two of them. Consequently the graph of  $E$  is recovered by

$$E(\alpha) = x \iff \exists p [\text{dom } p = \alpha + 1, p(\alpha) = x, C_p \cap L_E \neq \emptyset, C_p \setminus L_E \neq \emptyset]. \quad (\text{S27.17})$$

Every quantified prefix and every witness code is a set. This is the elementary part of the repository's more general represented-cut recognition theorem ([theorem 45.1](#)). We need this formula, rather than just an abstract appeal to density, when classes from different models are compared externally.

### 102.2.2 An embedding criterion for two class realizations

Let  $(M, \mathcal{X})$  and  $(M, \mathcal{Y})$  be set-sized models of GBC with the same first-order set structure. They need not be transitive, and neither class part is assumed contained in the other. Suppose they share a baseline  $b : \text{Ord}^M \rightarrow \mathbf{No}^M$ . Let  $H_{\mathcal{X}}$  and  $H_{\mathcal{Y}}$  be their *external* orders of admitted set-like enumerations, and identify the evaluated canonical core codes as distinguished elements of both orders.

*Note 102.5* (Source 27, Theorem 9.1, “Embeddings fixing the core”). *There exists an external order embedding*

$$J : H_{\mathcal{X}} \longrightarrow H_{\mathcal{Y}}$$

*fixing every common core element if and only if  $\mathcal{X} \subseteq \mathcal{Y}$ . Whenever such an embedding exists, it is the literal inclusion on enumeration graphs. In particular, an isomorphism fixing the core pointwise exists exactly when  $\mathcal{X} = \mathcal{Y}$ . This is source 22’s [theorem 102.3](#) in another form. Source 27 shares only the baseline  $b$  (its decoder  $d : \mathbf{No} \rightarrow V$  uses Mostowski collapses instead of a class bijection  $g$ ), assumes no inclusion between the realizations, and states the criterion for order embeddings  $H_{\mathcal{X}} \rightarrow H_{\mathcal{Y}}$  fixing the core, which must then be the inclusion. Source 22 fixes  $b$  and  $g$  and states inclusions of trace collections.*

*Source 27’s proof.* If  $J$  fixes the core, then for every  $d \in \mathcal{P}_s(b)$  and  $E \in H_{\mathcal{X}}$ ,

$$E_d < E \iff E_d < J(E).$$

The two lower traces are the same external subset of the common set-code domain. Formula ([S27.17](#)), evaluated in the same first-order structure with the same predicates  $L_E, b$ , therefore reconstructs the same graph in both expansions. Hence  $J(E) = E$  and  $H_{\mathcal{X}} \subseteq H_{\mathcal{Y}}$ .

To recover arbitrary admitted classes of sets, use a fixed definable surjection  $d : \mathbf{No} \rightarrow V$ , available in set theory with Choice. Here is its coding construction. A surreal sign word can encode an ordinal  $\alpha$ , a distinguished  $i < \alpha$ , and a binary relation on  $\alpha$ ; unary sign headers and an ordinal-length binary table give such a code. If the relation is well-founded and extensional, return the image of  $i$  under its Mostowski collapse; otherwise return  $\emptyset$ . Every set has a code by well-ordering the transitive closure of its singleton. No simultaneous choice of a code for every set is made. The decoding map  $d$  is the same elementary definition in both class realizations.

For  $C \in \mathcal{X}$ , let  $Z_C = \{\alpha : d(b(\alpha)) \in C\}$ . Swap baseline positions  $2 \cdot \alpha$  and  $2 \cdot \alpha + 1$  exactly when  $\alpha \in Z_C$ . The resulting enumeration  $E_C$  belongs to  $H_{\mathcal{X}}$ , hence also to  $H_{\mathcal{Y}}$ . Its pair orientations, together with  $b$ , recover  $Z_C$  by elementary comprehension. Surjectivity of  $d \circ b$  then recovers

$$C = \{x : \exists \alpha [d(b(\alpha)) = x \text{ and } \alpha \in Z_C]\}.$$

Thus  $C \in \mathcal{Y}$ , proving  $\mathcal{X} \subseteq \mathcal{Y}$ .

Conversely, class inclusion includes each admitted enumeration, and the same elementary first-disagreement formula preserves its order. This inclusion fixes the common core. The first part of the proof already establishes its uniqueness.  $\square$

The use of ([S27.17](#)) is substantive. Two classes from incomparable realizations need not coexist in any common class model. If  $M$  is nonstandard, their external disagreement set need not have a first element in its internal ordinal order. The proof does not assume one: it transfers an elementary reconstruction formula with shared parameters instead.

If  $(M, \mathcal{X})$  is externally countable,  $H_{\mathcal{X}}$  is countable, dense, and without endpoints. The usual back-and-forth argument makes it isomorphic to  $\mathbb{Q}$ . Hence an unmarked countable order can forget the class realization entirely, while fixing the embedded core pointwise forces exactly class inclusion. The marking is additional mathematical data, not a property extracted from the pure order.

**Corollary 102.6** (The raw-order existence criterion). *Under the same hypotheses, a core-fixing external embedding of the raw global well-order comparisons exists exactly when  $\mathcal{X} \subseteq \mathcal{Y}$ . Such an embedding fixes every old set-like enumeration. No uniqueness is asserted on non-set-like fibers.*

*Proof.* A raw well-order has the same core trace as its initial Ord block. If that trace is the trace of an exhaustive enumeration  $E$ , formula (S27.17) makes its initial block equal to  $E$ ; no labels remain for a tail. Thus  $E$  is the unique raw realization of that trace. A core-fixing embedding therefore fixes every set-like  $E$ , and the pair-code argument gives class inclusion.

For the converse, a shared relation's internal well-foundedness is absolute between same-set GBC expansions: it is equivalent to absence of a set-coded descending  $\omega$ -sequence. Class inclusion therefore includes the old raw well-orders, and elementary labelled comparison preserves their order. It fixes the core.  $\square$

[merge] The remark before [corollary 102.6](#) that a countable  $H_{\mathcal{X}}$  is  $\mathbb{Q}$  is source 13's [proposition 55.5](#) (Part IX) and source 21's [theorem 102.13](#) below.

### 102.3 Source 21: model dependence and preservation

Changing the admitted classes can change the collection of global enumerations even when every set remains fixed. Nevertheless, the abstract external order can forget much of this change. In particular, every externally countable GBC model gives the same abstract set-like enumeration order, namely  $\mathbb{Q}$ . Naming a common dense core reveals a much finer distinction. We prove these statements after separating the relevant notions of well-foundedness and preservation.

#### 102.3.1 Shared relations and well-foundedness

We use two standard facts from second-order set theory: realizations with the same sets agree about well-foundedness of shared class relations, and well-foundedness is preserved from a transitive submodel with the same ordinals. See [35, Observations 4.12–4.13]. Their elementary proofs explain exactly why they apply here.

*Note 102.7* (Source 21, Lemma 11.1, “A set witness to ill-foundedness”). *In GBC, a class relation  $R$  is ill-founded if and only if there is a set function  $f : \omega \rightarrow \text{dom}(R)$  such that*

$$f(n+1) R f(n) \quad (n \in \omega).$$

*Consequently, well-foundedness of a fixed class relation is expressible with only set quantifiers and the class parameter  $R$ .* This is the scope statement printed in [section 5.6](#) (the Hilbert's-tenth review's patch) and in source 11's [section 20](#), and the equivalence of [note 57.5](#) (Part IX); source 25 states it in [section 104](#).

*Source 21's proof.* The range of a descending sequence is a nonempty class with no  $R$ -minimal element. Conversely, suppose that a nonempty class  $A$  has no  $R$ -minimal element. Fix  $a_0 \in A$ , and use a set-like global well-order to select an  $R$ -predecessor in  $A$  at each subsequent step. Every nonempty class of possible predecessors has a first point in that global order. Ordinary recursion with the displayed class parameters produces the sequence, and Class Replacement makes its restriction to  $\omega$  a set.  $\square$

**Proposition 102.8** (Changing classes while keeping the sets). *Let  $(M, \mathcal{X})$  and  $(M, \mathcal{Y})$  be GBC realizations of the same first-order model. They agree about well-foundedness of every relation in  $\mathcal{X} \cap \mathcal{Y}$ . They also agree about the labeled lexicographic comparison of any shared class well-orders of a fixed shared carrier.*

*Proof.* The first assertion follows from [note 102.7](#): the possible set sequences and their membership relations are identical. For the second, the largest common labeled prefix and the two first residual labels have elementary definitions in the shared carrier and relations. Those definitions quantify over the same set domain and therefore have the same interpretation. For set-like enumerations, one can equivalently use their least ordinal of disagreement.  $\square$

No transitivity or external well-foundedness assumption is needed for this agreement statement. Both models can agree about a relation that is externally ill-founded. The assertion concerns their common interpretation, not its correctness in a larger metatheory.

**Proposition 102.9** (Preservation with the same ordinals). *Suppose  $(M, \mathcal{X}) \subseteq (N, \mathcal{Y})$  are transitive GBC models with  $\text{Ord}^M = \text{Ord}^N$ , and old classes are retained as the same relations on old elements. Every old class relation that  $(M, \mathcal{X})$  regards as well-founded remains well-founded in  $(N, \mathcal{Y})$ .*

*On a fixed old carrier, inclusion therefore preserves the old labeled lexicographic well-order space and its comparisons.*

*Proof.* Suppose that  $N$  has a descending  $\omega$ -sequence through an old relation  $R$ . In  $N$ , the ranks of its entries are bounded by some ordinal  $\beta$ . This ordinal belongs to  $M$ , since the models have the same ordinals. All the entries lie in the ground set  $A = \text{dom}(R) \cap V_\beta^M$ , increasing  $\beta$  if necessary. By Class Separation,  $R \cap (A \times A)$  is a set relation in  $M$ . The ground model regards it as well-founded and constructs its rank function into an ordinal. Transitivity makes that same rank function a witness to well-foundedness in  $N$ , contradicting the descending sequence.

For an old fixed carrier, the numerical labels and the two relation graphs have not changed. Their common labeled prefix and first residual labels are consequently unchanged. Thus the same first discrepancy gives the same lexicographic comparison.  $\square$

This applies to ordinary set-forcing extensions and, more generally, to class-forcing extensions satisfying the stated hypotheses. It does not assert the preservation of every second-order axiom, the existence of a uniform forcing relation for arbitrary class forcing, or that every *new* abstract class-well-order type was already present. Those are separate questions with stronger hypotheses and genuine recursion issues [\[24, 27\]](#).

**Proposition 102.10** (A sufficient condition for external correctness). *Let  $(M, \mathcal{X})$  be a transitive GBC model, and let  $\theta = \text{Ord}^M$  be its ordinal height in the metatheory. If  $\text{cf}(\theta) > \omega$ , every internally well-founded class relation is externally well-founded.*

*Proof.* An external descending  $\omega$ -sequence would have its entries' ranks bounded below  $\theta$ , by the cofinality hypothesis. Restrict the relation to a ground rank segment containing those entries. This is a well-founded set relation in the transitive model  $M$ , so its ground ordinal rank function also witnesses external well-foundedness. This contradicts the sequence.  $\square$

The rank-cut case is the familiar sufficient condition in [\[35, Observation 1.14\]](#). For height of countable cofinality, the rank bound can fail. A transitive model is correct about its well-founded *set* relations, but transitivity alone does not ensure correctness about all admitted class relations. A transitive realization with this additional correctness property is called a  $\beta$ -model.

### 102.3.2 The carrier also changes under forcing

There is an essential qualification to any claimed inclusion between surreal well-order spaces in different universes. If a forcing extension adds a new subset  $r \subseteq \omega$ , it adds the new surreal sign sequence of length  $\omega$  whose positive signs occur exactly at positions in  $r$ . Hence  $\mathbf{No}^M \subsetneq \mathbf{No}^N$ . An

old enumeration  $E : \text{Ord} \rightarrow \mathbf{No}^M$  is no longer onto  $\mathbf{No}^N$ , so it is not literally a point of the new  $\mathfrak{W}_{\text{sl}}(\mathbf{No}^N)$ .

After choosing auxiliary enumerations, there are straightforward comparison embeddings. For unrestricted class well-orders, fix a class well-order  $T$  of  $D = \mathbf{No}^N \setminus \mathbf{No}^M$  and send  $R$  to  $R + T$ . Every old label precedes every new one, so the first labeled discrepancy of two old orders remains their first discrepancy after extension.

For set-like orders, this sum generally has a proper-class initial segment. Instead choose a set-like enumeration  $d$  of  $D$ , indexed by  $\text{Ord}$  when  $D$  is proper or by a set ordinal when it is a set. Interleave  $E(\alpha)$  at slot  $(\alpha, 0)$  and, where defined,  $d(\alpha)$  at slot  $(\alpha, 1)$ , with  $\alpha$  the primary coordinate. Every initial segment is a set, so the resulting well-order is set-like and has type  $\text{Ord}$ . At the first old discrepancy all preceding new-label slots agree; thus lexicographic comparisons are preserved. These constructions require the stated auxiliary data. They should not be mistaken for a canonical inclusion or a functorial choice independent of that data.

### 102.3.3 Proper expansions of the class realization

Now fix the sets. Let

$$(M, \mathcal{X}) \subseteq (M, \mathcal{Y}), \quad b \in \mathcal{X},$$

be GBC realizations, and use the same normalized core  $\mathcal{P}_{\text{bd}}(b)$  in both. Its codes and evaluations are identical because the set part and the baseline are identical. Write  $H_{\mathcal{X}}$  and  $H_{\mathcal{Y}}$  for the externally interpreted collections of admitted set-like global enumerations. For set-sized two-sorted models, these collections are ordinary sets in the metatheory.

**Theorem 102.11** (A class expansion as an extension by cuts). *The following hold.*

- (i)  $H_{\mathcal{X}}$  is an order-dense suborder of  $H_{\mathcal{Y}}$ .
- (ii)  $H_{\mathcal{X}} = H_{\mathcal{Y}}$  if and only if  $\mathcal{X} = \mathcal{Y}$ .
- (iii) For  $E \in H_{\mathcal{Y}}$ , its lower cut  $L_E$  in the common core belongs to  $\mathcal{X}$  if and only if its enumeration graph  $E$  belongs to  $\mathcal{X}$ .

*Proof.* Both orders contain  $\mathcal{P}_{\text{bd}}(b)$ , which is dense in  $H_{\mathcal{Y}}$ . Old comparisons are unchanged by [proposition 102.8](#); this proves (i).

For (ii), choose a class bijection  $v : \text{Ord}^M \rightarrow M$  in  $\mathcal{X}$ , using Global Choice. Given any  $A \in \mathcal{Y}$ , define a class subset of the internal ordinals by

$$B_A = \{\alpha : v(\alpha) \in A\}.$$

Encode  $B_A$  by swapping the  $\alpha$ -th adjacent pair of  $b$  exactly when  $\alpha \in B_A$ . Denote the resulting enumeration by  $E_A$ . This is an element of  $H_{\mathcal{Y}}$ . If its graph belonged to  $\mathcal{X}$ , the orientations of its pairs, together with  $b$  and  $v$ , would recover  $A$  by elementary comprehension. Thus a new class supplies a new admitted enumeration. The converse is immediate since enumeration graphs are classes.

For (iii), an admitted  $E$  defines  $L_E$  by elementary comprehension. Conversely, if  $L_E \in \mathcal{X}$ , apply the elementary reconstruction formula of [note 97.10](#) inside  $(M, \mathcal{X})$ . Its set quantifiers have the same domain as in  $(M, \mathcal{Y})$ , so its resulting class graph is exactly  $E$ . Hence  $E \in \mathcal{X}$ .  $\square$

**Corollary 102.12** (Rigidity when the core is fixed pointwise). *If  $\mathcal{X} \subsetneq \mathcal{Y}$ , there is no weakly increasing map  $H_{\mathcal{Y}} \rightarrow H_{\mathcal{X}}$  fixing every point of the common core. In particular there is no such isomorphism or monotone retraction.*

*Proof.* Suppose  $r$  is such a map. If  $r(E) > E$ , choose a core point  $E_d$  strictly between them. Monotonicity gives  $r(E) \leq r(E_d) = E_d$ , a contradiction. If  $r(E) < E$ , choose a core point between them and use  $r(E_d) \leq r(E)$ , again a contradiction. Thus  $r(E) = E$  for every input, which is impossible for a point in  $H_Y \setminus H_X$ .  $\square$

The core therefore detects a proper expansion through newly represented cuts, provided its individual points are identified. This does not imply that the abstract unmarked enumeration order detects the expansion.

### 102.3.4 The external countable order is always rational

**Theorem 102.13** (External size and countable classification). *For every set-sized two-sorted GBC model  $(M, \mathcal{X})$ , the metatheory has*

$$|H_{\mathcal{X}}| = |\mathcal{X}|.$$

*If the entire two-sorted model is externally countable, then*

$$(H_{\mathcal{X}}, <_{\text{lex}}) \cong (\mathbb{Q}, <).$$

*Neither assertion requires transitivity,  $\beta$ -correctness, or  $\omega$ -standardness.*

*Proof.* The inequality  $|H_{\mathcal{X}}| \leq |\mathcal{X}|$  follows because the enumeration graphs are members of  $\mathcal{X}$ . Fix internal bijections  $v : \text{Ord}^M \rightarrow M$  and  $b : \text{Ord}^M \rightarrow \mathbf{No}^M$ . The binary pair construction in [theorem 102.11](#) gives an external injection  $A \mapsto E_A$  from  $\mathcal{X}$  into  $H_{\mathcal{X}}$ , since the graph of  $E_A$  determines  $A$ . Cantor–Bernstein proves the cardinal equality.

In the countable case, the GBC theorems that  $\mathfrak{W}_{\text{sl}}$  is a linear order, is dense, and has no endpoints hold in the model. Their class quantifiers range over exactly the externally admitted class objects in  $\mathcal{X}$ . Thus they say precisely that the external relation on  $H_{\mathcal{X}}$  is a countable dense linear order without endpoints. The usual back-and-forth for countable orders identifies it with  $\mathbb{Q}$ .  $\square$

There is no conflict with the internal statement that no uniformly set-indexed family is cofinal in  $\mathfrak{W}_{\text{sl}}$ . An external countable listing of all its admitted points need not be encoded by a uniform family inside the model. Nor is there a conflict with the diagonal obstruction to an internal exhaustive class list: an external list is not an internal decoder.

This theorem gives a concrete limitation on attempts to recover truth resources or class-comprehension strength from the unmarked order. Whenever the relevant two-sorted models are externally countable, their set-like enumeration orders are all isomorphic to  $\mathbb{Q}$ , even though their admitted class collections can differ. The named dense core retains finer information through [theorem 102.11](#).

Finally,  $\omega$ -standardness plays a different role in the unrestricted order. The preceding rational classification transfers internally proved density directly to the admitted class objects. By contrast, an interval that the model regards as finite can have externally infinitely many points when its claimed size is a nonstandard natural number. Statements about external finite-tail condensation must therefore be read with their separate standardness hypotheses. The distinction between Global Choice and selection of class witnesses is likewise unaffected: our cut reconstructions are explicit and unique, and do not use a Class Choice principle [\[66\]](#).

*Note 102.14* ([\[merge\]](#); overlaps in source 21’s Section 11). [Proposition 102.8](#) combines [note 102.7](#) with the elementarity of labelled comparison used in [theorem 48.1](#). In [theorem 102.11](#), item (iii) is source 17’s [theorem 48.4](#) and item (ii) its last clause; item (i) and [corollary 102.12](#) are new. In [theorem 102.13](#), the second assertion is source 13’s [proposition 55.5](#) (Part IX); the first,  $|H_{\mathcal{X}}| = |\mathcal{X}|$ , is new, and source 30 proves it again ([note 102.17](#)).



## 102.4 Source 30: what labelled enumeration spaces remember about their class universe

[merge] Source 30's Section 9, printed whole; source 30 cites none of sources 13–29. Its class realizations  $\mathcal{C}, \mathcal{D}$  are printed  $\mathcal{X}, \mathcal{Y}$  and its core  $\mathcal{B}_b$  is printed  $\mathcal{P}_s(b)$  (as a family of enumeration graphs). Its Proposition 9.1, Theorem 9.2 and Corollaries 9.3 and 9.4 are results of sources 22 and 21 and are printed as notes with source 30's proofs. [Proposition 102.19](#) (1), [proposition 102.20](#), [theorem 102.21](#) and [corollary 102.22](#) are new against the repository; source 30 calls the last two “applications of known class-theoretic machinery”.

The repository proves that enlarging the available classes, while keeping the sets and a baseline fixed, necessarily adds a new global enumeration ([theorem 48.4](#)). There is a stronger converse formulation: inclusion of the available *labelled* enumeration families detects inclusion of the entire class collections. In fact, the much narrower family obtained by independently swapping fixed adjacent pairs already contains all this information. This observation gives a precise positive answer to the reconstruction question with labels retained. It also explains why reconstruction from the pure external linear order is a substantially different problem.

The arguments in this section use two distinct levels of language. Within GB, each subset of the ordinals and each enumeration is one class variable, and the constructions below are elementary formulas in those variables. When comparing class realizations, we work externally with a fixed *set-sized* model  $M$  and collections  $\mathcal{X}, \mathcal{Y} \subseteq \mathcal{P}(M)$ . The external collection of their class graphs is then an ordinary set in the metatheory. No internal class whose elements are proper classes is introduced.

### 102.4.1 An elementary coding by pair orientations

Fix a class bijection

$$b : \text{Ord} \longrightarrow \text{No.}$$

Its existence already entails Global Choice, by the repository's theorem over the choiceless GB base ([theorem 13.1](#)). For each ordinal  $\alpha$ , sort the two labels  $b(2 \cdot \alpha), b(2 \cdot \alpha + 1)$  numerically:

$$u_\alpha = \min\{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\}, \quad v_\alpha = \max\{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\}.$$

Thus  $u_\alpha < v_\alpha$ . Define the pair-normalized baseline  $c$  by

$$c(2 \cdot \alpha) = u_\alpha, \quad c(2 \cdot \alpha + 1) = v_\alpha.$$

Here multiplication is ordinal multiplication, with the finite ordinal on the left. The intervals  $[2 \cdot \alpha, 2 \cdot \alpha + 2)$  partition  $\text{Ord}$ . All of these definitions are elementary in the original baseline.

For a class  $A \subseteq \text{Ord}$ , let  $E_A$  be the enumeration obtained by reversing exactly the pairs indexed by  $A$ :

$$\begin{aligned} E_A(2 \cdot \alpha) &= \begin{cases} v_\alpha, & \alpha \in A, \\ u_\alpha, & \alpha \notin A, \end{cases} \\ E_A(2 \cdot \alpha + 1) &= \begin{cases} u_\alpha, & \alpha \in A, \\ v_\alpha, & \alpha \notin A. \end{cases} \end{aligned} \tag{S30.9.1}$$

Each pair is still used once and distinct pairs are disjoint, so every  $E_A$  is a class bijection. Conversely, the elementary predicate  $\text{Pair}_c(E)$  says that  $E$  is a function on the ordinals and, for every  $\alpha$ ,

$$(E(2 \cdot \alpha), E(2 \cdot \alpha + 1)) \in \{(u_\alpha, v_\alpha), (v_\alpha, u_\alpha)\}. \tag{S30.9.2}$$

This predicate itself guarantees that  $E$  is a global bijection. Its unique orientation class is

$$A_E = \{\alpha \in \text{Ord} : E(2 \cdot \alpha) = v_\alpha\}. \quad (\text{S30.9.3})$$

*Note 102.15* (Source 30, Proposition 9.1, “Pair coding with an elementary inverse”). *In GB with the displayed baseline, the transformations  $A \mapsto E_A$  and  $E \mapsto A_E$  are mutually inverse, uniformly elementary transformations between ordinal classes and classes satisfying  $\text{Pair}_c$ . If  $A \neq B$  and  $\delta = \min(A \triangle B)$ , then*

$$E_A <_{\text{lex}} E_B \iff \delta \in B.$$

This is source 22’s [proposition 102.1](#) with its decoder [theorem 102.2](#), with the baseline pair-normalized; the Boolean structure below is source 30’s.

*Source 30’s proof.* The inverse assertions follow directly from (S30.9.1)–(S30.9.3). A nonempty ordinal class has a least member: after choosing any witness, minimize within the set of ordinals below that witness. Before position  $2 \cdot \delta$ , the two enumerations agree. At that position the enumeration whose orientation bit is zero has value  $u_\delta$ , and the other has value  $v_\delta > u_\delta$ . This proves the comparison formula.  $\square$

The coding also retains Boolean operations. Associate to  $A$  the class permutation of surreal labels

$$T_A = E_A \circ c^{-1}.$$

It either fixes both labels of a pair or exchanges them. Therefore

$$T_A^2 = \text{id}, \quad T_A \circ T_B = T_{A \triangle B}, \quad T_{\text{Ord} \setminus A} = T_{\text{Ord}} \circ T_A. \quad (\text{S30.9.4})$$

Intersection is recovered by the elementary operation that exchanges a pair precisely when both input permutations exchange that pair. Thus the orientation family records the Boolean algebra of available ordinal classes, including symmetric difference as actual permutation composition. Boolean operations alone do not characterize which families form a model of class theory: comprehension and Class Replacement impose additional requirements.

#### 102.4.2 A faithful invariant of class realization

Let  $(M, \mathcal{X})$  and  $(M, \mathcal{Y})$  be models of GBC with exactly the same first-order structure. They need not be transitive, and neither class collection is initially assumed to contain the other. Write

$$\mathcal{E}(\mathcal{X}) = \{E \in \mathcal{X} : (M, \mathcal{X}) \models E : \text{Ord} \xrightarrow{\sim} \mathbf{No}\}.$$

This is external notation for a collection of graphs. Internal ordinal arithmetic, surreal labels, and coded ordered pairs are all interpreted in  $M$ . If a baseline  $c$  is common to the two realizations, write

$$\mathcal{K}_c(\mathcal{X}) = \{E \in \mathcal{X} : (M, \mathcal{X}) \models \text{Pair}_c(E)\}.$$

*Note 102.16* (Source 30, Theorem 9.2, “Reconstruction from labelled enumeration availability”). *For any two such realizations,*

$$\mathcal{X} \subseteq \mathcal{Y} \iff \mathcal{E}(\mathcal{X}) \subseteq \mathcal{E}(\mathcal{Y}). \quad (\text{S30.9.5})$$

*If a pair-normalized baseline  $c$  belongs to both realizations, these conditions are also equivalent to*

$$\mathcal{K}_c(\mathcal{X}) \subseteq \mathcal{K}_c(\mathcal{Y}).$$

*In particular, equality of either labelled availability family implies equality of the complete class collections.* This is source 22’s [theorem 102.3](#) (and source 27’s [note 102.5](#)). Source 30 needs no baseline given in advance: it takes one from the assumed inclusion of enumeration families, and the class bijection  $g$  from that baseline.

*Source 30’s proof.* The forward implications use elementary comprehension and the fact that being a bijection between the displayed classes is a first-order assertion with set quantifiers and the specified class graph. Its truth is unchanged when the first-order structure and graph are unchanged.

Suppose now that  $\mathcal{E}(\mathcal{X}) \subseteq \mathcal{E}(\mathcal{Y})$ . Choose  $b \in \mathcal{E}(\mathcal{X})$ . The assumption makes it a common baseline, and pair normalization gives a common  $c$ . The standard coding proof of the baseline–Global Choice equivalence supplies a class bijection

$$g : \text{Ord}^M \longrightarrow M$$

uniformly definable from  $b$ ; hence the same  $g$  is available in both realizations. One can instead use a common class surjection, obtained by decoding ordinal presentations of rooted well-founded extensional relations, since injectivity is not needed below.

Take any class  $X \in \mathcal{X}$ , and form its ordinal code

$$A_X = \{\alpha \in \text{Ord}^M : g(\alpha) \in X\}.$$

Elementary comprehension in  $(M, \mathcal{X})$  gives  $A_X$ , and then  $E_{A_X} \in \mathcal{K}_c(\mathcal{X})$ . The assumed enumeration inclusion puts  $E_{A_X}$  in  $\mathcal{Y}$ . Applying the elementary inverse ([S30.9.3](#)) in the latter model recovers  $A_X$ , and the formula

$$x \in X \iff \exists \alpha \in \text{Ord}^M (g(\alpha) = x \wedge E_{A_X}(2 \cdot \alpha) = v_\alpha) \quad (\text{S30.9.6})$$

recovers  $X$ . Consequently  $X \in \mathcal{Y}$ , proving  $\mathcal{X} \subseteq \mathcal{Y}$ .

If the common baseline is given at the outset, the same proof uses only the inclusion of  $\mathcal{K}_c$ . Every quantifier in the reconstruction formula ranges over sets of  $M$ ; no quantification over an exhaustive family of classes is hidden in the construction.  $\square$

*Note 102.17* (Source 30, Corollary 9.3, “External size”). *For any set-sized GBC model  $(M, \mathcal{X})$  and a baseline  $c \in \mathcal{X}$ , the external sets  $\mathcal{X}$ ,  $\mathcal{E}(\mathcal{X})$ , and  $\mathcal{K}_c(\mathcal{X})$  are equinumerous.* This is the first assertion of source 21’s [theorem 102.13](#), with  $\mathcal{K}_c(\mathcal{X})$  added.

*Source 30’s proof.* The map  $X \mapsto E_{A_X}$  in the preceding proof injects all of  $\mathcal{X}$  into  $\mathcal{K}_c(\mathcal{X})$ , while  $\mathcal{K}_c(\mathcal{X}) \subseteq \mathcal{E}(\mathcal{X}) \subseteq \mathcal{X}$ . Apply Cantor–Bernstein externally.  $\square$

The reconstruction conclusion of [Note 102.16](#) also holds when the recorded objects are the labelled set-like well-order relations rather than their enumeration graphs. The transformations in both directions are elementary: transport ordinal order along the enumeration, or assign to each label the ordinal order type of its predecessor set. It also holds for the availability family of *all* labelled class well-orders. For the latter statement one uses the standard first-order characterization of internal class well-foundedness; see [64, Section 2]. Indeed, a nonempty ill-founded subclass yields a nonempty ill-founded *set* of labels: start with one label, and at each finite stage adjoin all predecessors within the given ill-founded subclass of least possible rank; then take the union. This construction uses ordinary set-valued recursion. Therefore two GB realizations with the same sets agree about the well-foundedness of a class relation they share. This is an internal absoluteness statement, and does not assert external well-foundedness.

*Note 102.18* (Source 30, Corollary 9.4, “The represented-cut version”). *Fix a common baseline and its set-coded support core. If two realizations have the same available represented trace cuts of that core, then they have the same classes. Inclusion of those represented-cut families likewise reflects inclusion of the class realizations.* This is the trace form of source 22’s [theorem 102.3](#).

*Source 30’s proof.* The repository’s recognition theorem ([theorem 45.1](#)) constructs the unique enumeration from its represented core cut by an elementary formula. Conversely, the cut is elementary in the enumeration. These operations are absolute between realizations with the same sets and baseline. Apply [Note 102.16](#).  $\square$

Thus a formalization may safely use the pair predicate and its elementary decoder as a small interface to arbitrary classes. The statement is uniform in any individually supplied class, or in a supplied uniform family. It does not furnish a same-level class of all pair-swap enumerations: the repository’s diagonal theorem ([theorem 19.37](#)) still rules out an exhaustive class-indexed evaluator. The positive reconstruction and the negative enumeration theorem describe different operations.

### 102.4.3 Why the pure external linear order loses this information

For an externally countable model of GBC, its admitted enumeration family is externally countable. Its lexicographic order is dense without endpoints. To see density, take the first disagreement of two enumerations, retain their common set prefix, choose an unused numerical label strictly between the two disagreeing labels, and complete the extended prefix to a global enumeration. The surreal set-cut property supplies the fresh label, and the baseline supplies the completion. Choosing a smaller or larger first label gives the absence of endpoints. Cantor’s classification of countable dense linear orders therefore makes this external order isomorphic to  $(\mathbb{Q}, <)$ .

**Proposition 102.19** (A core preserved setwise and a core fixed pointwise). *Let  $(M, \mathcal{X})$  and  $(M, \mathcal{Y})$  be externally countable GBC models with the same sets and a common baseline  $b$ . Let  $\mathcal{P}_s(b)$  be the common family of its internally set-supported permutations, viewed as enumeration graphs.*

1. *The core  $\mathcal{P}_s(b)$  and its complement are dense in each enumeration order. There is an external order isomorphism  $F : \mathcal{E}(\mathcal{X}) \longrightarrow \mathcal{E}(\mathcal{Y})$  with  $F[\mathcal{P}_s(b)] = \mathcal{P}_s(b)$  and  $F(b) = b$ .*
2. *If such an order isomorphism fixes every member of  $\mathcal{P}_s(b)$  pointwise, then  $\mathcal{X} = \mathcal{Y}$ , and the isomorphism is the identity on enumeration graphs.*

*Proof.* Core density is the repository’s prefix-completion theorem ([lemma 19.4](#) and [theorem 19.10](#)). For complement density, choose a prefix cylinder inside any prescribed open interval, and a core enumeration  $q = b \circ \sigma$  in that cylinder. Choose an ordinal bound beyond the prefix length and the support of  $\sigma$ . Define a class permutation  $\tau$  that fixes positions below a sufficiently large such bound and exchanges every later adjacent pair. Then  $q \circ \tau$  still lies in the cylinder. Beyond the bound,  $\sigma$  is the identity and  $\sigma \circ \tau$  moves every point of a proper-class tail. Thus  $q \circ \tau \notin \mathcal{P}_s(b)$ . All these constructions are elementary from the baseline and set parameters and work in each realization.

Each enumeration order is therefore a countable dense linear order with two dense colors: membership and nonmembership in the core. The usual back-and-forth proof of Cantor’s theorem applies with colors preserved. Start with the one-point partial map  $b \mapsto b$ , and extend alternately through countable external lists of the two domains; every required open interval contains an unused point of the required color. This proves the first assertion.

For the second, let  $E \in \mathcal{E}(\mathcal{X})$ . An isomorphism fixing the core pointwise makes  $E$  and  $F(E)$  determine the same cut of the common core. The repository’s elementary uniqueness theorem for represented cuts ([theorem 45.1](#)) implies that their labelled enumeration graphs are identical. This

uniqueness also follows directly by taking a set prefix through a disagreement and placing a common core enumeration strictly between the resulting two prefixes. Consequently the two enumeration families coincide. [Note 102.16](#) yields  $\mathcal{X} = \mathcal{Y}$ .  $\square$

Thus even a distinguished baseline point and a unary predicate for the support core are insufficient to reconstruct a countable class realization. The stronger requirement that the embedding of that core be fixed pointwise retains precisely the represented-cut information. The same back-and-forth may begin with any finite common suborder of the core fixed pointwise, so naming finitely many additional core points does not remove this loss of information.

**Proposition 102.20** (Pure order does not detect truth or ETR). *Suppose  $(M, \mathcal{Y})$  is an externally countable model of  $\text{GBC} + \text{ETR}$ , and  $M$  has a parametrically definable global well-order. Put  $\mathcal{X} = \text{Def}(M)$ , allowing set parameters in definitions. Then*

$$(M, \mathcal{X}) \models \text{GBC} + \neg\text{ETR}, \quad \mathcal{E}(\mathcal{X}) \cong \mathcal{E}(\mathcal{Y}) \cong (\mathbb{Q}, <)$$

*as external pure linear orders, while  $\mathcal{E}(\mathcal{X}) \subsetneq \mathcal{E}(\mathcal{Y})$  as labelled graph families. Moreover, relative to a common definable baseline, the external isomorphism can preserve the support core setwise and fix the baseline point.*

*Proof.* The definable classes of a first-order model satisfy GB, and the definable global well-order supplies Global Choice. Every class in  $\text{Def}(M)$  belongs to  $\mathcal{Y}$  by elementary comprehension. Elementary transfinite recursion constructs a truth predicate for the set universe, including satisfaction with set assignments, by recursion on formula complexity; this is a standard consequence of **ETR** [22]. Tarski's undefinability theorem excludes such a full satisfaction predicate from  $\text{Def}(M)$ , even with a set parameter. Hence the smaller realization fails ETR and is properly contained in the larger one. [Note 102.16](#) gives strict inclusion of their labelled spectra. Both pure orders are countable, dense, and have no endpoints, giving their external isomorphisms with  $\mathbb{Q}$ . [Proposition 102.19](#) supplies the additional preservation assertions.  $\square$

This proposition is conditional on the displayed model hypothesis; it does not derive the consistency of  $\text{GBC} + \text{ETR}$  from ZFC. The separation between definable-class models and truth extensions is standard [65]; its application here pinpoints the information discarded by forgetting coordinates. No claim that the external isomorphism is available internally is needed.

#### 102.4.4 Exactly when a definable baseline exists

Global Choice and a definable global choice function are different requirements. Fix one set parameter  $p$ , and let  $\text{HOD}_p$  mean the sets whose transitive closures consist of sets definable from ordinal parameters and the *single named parameter*  $p$ . This convention is explicit because notation such as  $\text{HOD}(p)$  is also used with other parameter conventions.

**Theorem 102.21** (Definability threshold). *Over ZF, the following are equivalent for classes definable by first-order formulas from the fixed set parameter  $p$ :*

1. *there is such a class well-order of  $\mathbf{No}$ ;*
2. *there is such a set-like class well-order of  $\mathbf{No}$ ;*
3. *there is such a bijection  $\text{Ord} \rightarrow \mathbf{No}$ ;*
4.  $V = \text{HOD}_p$ .

*The formulation of existence of a definable class is metatheoretic; for any specified definition, the constructions produce specified definitions of the other objects. In particular, parameter-free definability has threshold  $V = \text{HOD}$ .*

*Proof.* Apply the repository's bare-GB choice proof (theorem 13.1) to the given definable relation. Its operations use elementary comprehension and ordinary set-valued recursion, so every resulting class remains definable from  $p$ . In particular it gives a  $p$ -definable global well-order of  $V$ ; refining by rank makes it set-like. Each set is then the unique object at some ordinal position of this order, and so is definable from that ordinal and  $p$ . Every set, including every member of its transitive closure, is consequently ordinal-definable from  $p$ . Thus  $V = \text{HOD}_p$ .

Conversely, the canonical ordinal-definition codes give a  $p$ -definable well-order of  $\text{HOD}_p$ . Concretely, a code records a rank  $\theta$ , a formula number, and a finite tuple of ordinals, and asserts that the formula defines one object in the set structure  $V_\theta$  with parameter  $p$ . Order the codes first by a bound containing their ranks and ordinal parameters, and then by canonical well-orders within that set of codes; assign each object its least defining code. Satisfaction for the set  $V_\theta$  is uniformly definable in ZF. This construction therefore uses no truth predicate for  $V$ . If  $V = \text{HOD}_p$ , it well-orders all sets; birthday or rank refinement and restriction then give the required set-like surreal well-order and enumeration. The remaining implications are the usual transport and collapse of a set-like well-order.  $\square$

The parameter-free surreal statement is explicitly discussed by Hamkins [18]. The parameter-relative formulation makes precise what is required when a formal development proposes to replace an explicit class baseline by a definition in the language of sets. For a set model  $M$ , it follows that

$$\mathcal{E}(\text{Def}(M)) \neq \emptyset \iff M \models \exists p [V = \text{HOD}_p]. \quad (\text{S30.9.7})$$

Allowing arbitrary set parameters in  $\text{Def}(M)$  is essential to the quantifier on the right.

#### 102.4.5 A common baseline across all class realizations

The preceding reconstruction transfers precise existence results about class realizations to availability of global surreal enumerations. Williams proves that a countable  $M \models \text{ZFC}$  has a *minimum* GBC-realization exactly when it has a parametrically definable global well-order [64, Main Theorem 1]. Here minimum means contained in every realization; it is stronger than merely having no smaller one.

**Corollary 102.22** (Universal availability is definable availability). *Let  $M \models \text{ZFC}$  be countable and GBC-realizable. With the intersection taken externally over all its GBC-realizations,*

$$\bigcap_{\mathcal{X}: (M, \mathcal{X}) \models \text{GBC}} \mathcal{E}(\mathcal{X}) = \mathcal{E}(\text{Def}(M)).$$

*Consequently there is a global surreal enumeration admitted by every realization exactly when  $M \models \exists p [V = \text{HOD}_p]$ .*

*Proof.* A definable enumeration belongs to every realization. Conversely, suppose that one enumeration  $b$  belongs to them all. The classes definable from  $b$  and set parameters form  $\text{Def}(M; b)$ . This is a GB-realization, since it is contained in a given GBC-realization and its defining operations and Class Replacement are inherited there. It has Global Choice because it contains  $b$ . Since every realization contains  $b$  and is closed under elementary definitions,  $\text{Def}(M; b)$  is contained in every realization. It is therefore a minimum GBC-realization.



Williams’s theorem implies that  $M$  has a parametrically definable global well-order. Hence  $\text{Def}(M)$  itself is a GBC-realization. The assumed universal availability of  $b$  now puts  $b$  in  $\text{Def}(M)$ , proving the reverse inclusion. Equation (S30.9.7) gives the final equivalence.  $\square$

The countability hypothesis enters through the cited minimum-realization theorem and should not be silently removed. This corollary and [Note 102.16](#) are applications of known class-theoretic machinery and the elementary pair construction; they are not claims to new consistency theorems. They sharpen the repository’s class-extension analysis ([theorems 48.1](#) and [48.4](#)) in two ways: even a narrow labelled subfamily detects the whole class collection, and the existence of a baseline common to all realizations has an exact definability criterion. Equivalently, every nondefinable global surreal enumeration admitted by one realization of a countable set model can be omitted by another GBC-realization of those same sets.

[merge] [Proposition 102.19](#) (1) refines source 13’s [proposition 55.5](#) and source 21’s [theorem 102.13](#) by a back-and-forth with two dense colours fixing the baseline; item (2) is the isomorphism case of source 27’s [note 102.5](#). Together with [proposition 102.20](#) it answers [research question 113.18](#) for externally countable models: the bare order, even with the core named as a set and the baseline as a point, detects neither the class realization nor ETR.

## 103 Unrestricted class well-orders and other readings (source 25)

[merge] Source 25’s Section 13, on the scope of its theorems. Its Proposition 13.1 is [theorem 14.2](#) and is printed as a note.

The main homogeneity theorem concerns injective Ord-words and their exhaustive subclass. It must not be silently applied to non-set-like well-orders, arbitrary repeated words, or numerically increasing words. We now state how those interpretations fit around it.

### 103.1 Direct comparison without ordinal enumeration

Let  $R, S$  be class well-orders of the same labelled class  $X \subseteq \mathbf{No}$ , where well-order means that every nonempty admitted subclass has a least element. Write  $\text{Pred}_R(x)$  for the class of strict  $R$ -predecessors of  $x$ . Define  $I(R, S)$  to consist of those  $x$  for which

$$\text{Pred}_R(x) = \text{Pred}_S(x)$$

and the restrictions of  $R, S$  to that common predecessor class agree. This is an elementary formula with the supplied class parameters. No quantification over possible initial-segment classes is needed.

*Note 103.1* (Source 25, Proposition 13.1, “Largest common labelled prefix”). *The class  $I(R, S)$  is initial in both orders, and  $R, S$  agree on it. If  $R \neq S$ , its complement has distinct least elements  $u$  under  $R$  and  $v$  under  $S$ . The rule*

$$R <_{\text{lex}} S \iff u < v$$

*using the numerical order of the labels defines a strict linear comparison on supplied class well-orders of  $X$ . For set-like global well-orders, it agrees with first-disagreement comparison of their Ord-enumerations. This is [theorem 14.2](#) (source 12’s [theorem 14.7](#)); source 25 recalls it with its proof.*

*Source 25’s proof.* If  $x \in I(R, S)$  and  $yRx$ , then  $y$  is in the common predecessor class of  $x$ . Agreement on that class shows that  $y$  has the same predecessors in the two orders and that their restrictions

agree there. Thus  $y \in I(R, S)$ . The same argument works for  $S$ , proving initiality. For two points in  $I(R, S)$ , comparison with their predecessor classes shows agreement of the orders.

If  $I(R, S) = X$ , the orders coincide. Otherwise the class well-order property gives the two least points outside it. They cannot be the same point  $w$ : then the predecessor class of  $w$  in both orders would be exactly  $I(R, S)$ , with identical restrictions, putting  $w$  in  $I(R, S)$  after all.

For transitivity, compare  $I(R, S)$  and  $I(S, T)$  as initial segments of  $S$ . They are nested. If the first is properly shorter,  $S$  and  $T$  still agree at the next point where  $R$  separates from  $S$ ; hence the comparison of  $R$  with  $T$  is that of  $R$  with  $S$ . The symmetric argument applies if the second is shorter. If they coincide and  $R <_{\text{lex}} S <_{\text{lex}} T$ , their next labels satisfy  $u < v < w$ . Thus the distinct next labels  $u, w$  decide  $R <_{\text{lex}} T$  as well. Trichotomy and asymmetry are immediate from distinct next labels.

For exhaustive set-length words or full Ord-enumerations, the labels before the first unequal coordinate form exactly this common prefix. Hence the two definitions agree.  $\square$

This direct comparison is part of the maintained report ([theorem 14.2](#)). Its important foundational feature is that a proper-class common prefix causes no recursion problem. One compares the supplied relations directly rather than first trying to assign them class ordinal types.

### 103.2 Set-like is a genuine restriction

A class well-order is *set-like* when each point has only a set of predecessors. A proper-class set-like well-order has type Ord in GBC; the rank of each point is the set ordinal determined by its predecessor set, and these ranks exhaust Ord. This gives the enumeration interpretation used for  $\mathfrak{W}_{\text{sl}}$ .

Unrestricted well-orders may have a proper-class initial block followed by further elements: Ord + 1, Ord + Ord, and more elaborate supplied class well-orders are examples of possible types. They have no enumeration by Ord. Global choice provides enough coding to transport such orders to the carrier **No**, but does not turn them into set-like orders.

Numerically well-ordered subclasses also need not be set-like. [Note 86.2](#) embeds a supplied class order of type Ord + Ord into the numerical surreal line. Its range, with inherited numerical order, has that same non-set-like type. A claim that every increasing proper-class word has a set-like range would therefore require an additional hypothesis; a full injective Ord-word has a set-like range in its *enumeration* order, which need not coincide with the numerical order.

### 103.3 Termination, arbitrary words, and increasing subsets

When comparing exhaustive well-orders on one fixed carrier, a word cannot be a proper prefix of another exhaustive word. The position of a termination marker is irrelevant there. It becomes important for varying ranges, arbitrary words, and injective words of different lengths. An end-first convention places a proper prefix before all its extensions; a central marker places some next letters below and others above termination. These can produce different local characters and saturation behavior, as studied in the earlier reports ([part VII](#); sources 13 and 23).

The prefix geometry used here avoids that ambiguity. Nodes are ordered by extension, siblings by their numerical next labels, and full branches are compared at a genuine first disagreement. The new theorems therefore do not depend on selecting a termination marker for the entire node set.

The family of numerically increasing words is not preserved by our automorphisms, as [Section 91](#) shows. Nor is the injective extension lemma automatically valid after imposing monotonicity on each word: the next label must then be above all earlier labels, which adds cross-coordinate restrictions absent from its proof. Arbitrary repeated words remove the injectivity restriction but no longer

literally enumerate well-orders of their ranges. Each variation warrants its own theorem rather than a notational substitution.

### 103.4 Relation codes, order types, and quotients

One can fix a well-ordering of label pairs and compare characteristic functions of relations at their first unequal bit. In a set-like class pair order, a nonempty difference class has a least coordinate. This yields a valid code-dependent comparison, but generally not the common-prefix comparison in [Note 103.1](#). It orders the *bits describing the relations*, not the next labels in their enumerations.

Passing to unlabelled isomorphism types changes the problem again. Set-sized well-order types have canonical representatives in Ord, but a type-then-label ordering deliberately prioritizes length or type and need not preserve enumeration lexicography. For class well-orders, one must not form an ordinary class of proper-class equivalence classes. Work with supplied representatives and a definable equivalence relation, a higher sort, or an explicitly larger universe. The label relation of [Section 93](#) avoids this issue by representing each label class with its unique root edge.

## 104 A size and recursion audit (source 25)

### 104.1 Four different levels that should not be collapsed

For clarity, the objects in this article occupy the following levels.

| Object                                        | Legitimate representation in the global argument                                           |
|-----------------------------------------------|--------------------------------------------------------------------------------------------|
| One prefix or support code                    | A set, hence an element of an ordinary class                                               |
| The core and prefix tree                      | Ordinary classes of such sets; their incidence is a class relation                         |
| One full enumeration or raw order             | A supplied class relation, not an element of the core unless represented by a support code |
| All full enumerations, automorphisms, or cuts | Predicates on class variables; a literal collection requires a higher-level interpretation |

Kelley–Morse theory strengthens class comprehension, but its classes still have sets as elements. It does not authorize a same-level class whose members are all classes. A higher sort or a universe shift is needed when such a collection is to become an object.

### 104.2 Why the back-and-forth does not need ETR

The homogeneity proof uses the following operations, all at the indicated sizes.

| Operation                     | Size and principle used                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------|
| Collect disagreement ordinals | Class replacement on a set of indices produces a set of ordinals                         |
| Build the forced prefix       | A set-length function; values are specified by elementary formulas with class parameters |
| Choose a core completion      | A set support code; global choice selects the least successful code                      |
| One back-and-forth history    | A set partial map between set codes at each set stage                                    |
| Take a limit history          | A union over a set ordinal of set partial maps                                           |
| Take the final automorphism   | Elementary comprehension of the union of the set-valued recursion                        |
| Transform one supplied branch | Formula ( <a href="#">S25.1</a> ), with the branch as a class parameter                  |

A set-valued recursion with an elementary class parameter can be expressed by saying that a value occurs in a set-sized correct initial history. Uniqueness of the histories follows by ordinary set

transfinite induction, and their existence by set recursion with the fixed parameters. Elementary comprehension then collects the compatible stages into a class. The stage value is never a new arbitrary class.

By contrast, a recursion assigning an entire class to each stage, such as an iterated truth predicate of class length, falls under materially stronger recursion principles. The class-game and class-forcing literature makes these distinctions precise [63, 22, 24]. The present proof does not infer a class-valued recursion theorem merely from the availability of  $\text{Ord}$  as an index class.

### 104.3 No uniform enumeration of all global enumerations

*Note 104.1* (Source 25, Proposition 14.1, “Diagonal obstruction to exhaustive class coding”). *No uniformly class-indexed family of global enumerations, with its index an ordinary class of sets, contains every member of  $\mathfrak{W}_{\text{sl}}$ .* This is [theorem 19.37](#) (source 12’s [theorem 19.38](#)), proved with a guard step at odd positions instead of a pair swap.

*Source 25’s proof.* By global choice it suffices to treat a uniform family  $(e_\alpha)_{\alpha \in \text{Ord}}$ , allowing repetitions and padding a smaller family if necessary. Construct an injective word  $d$  in two-position blocks. At position  $2 \cdot \alpha$ , choose an unused label different from  $e_\alpha(2 \cdot \alpha)$ . At position  $2 \cdot \alpha + 1$ , choose the least-indexed still unused label in the baseline enumeration  $b$ .

At each set stage the used labels form a set, so both choices exist. This is set-valued recursion. The second choice guarantees exhaustiveness. Indeed, for any  $\beta$ , the labels  $b(\gamma)$  with  $\gamma < \beta$  have all been used by some set-ordinal stage once they have each been used; a supremum of those set-many stages exists. At the next guard step,  $b(\beta)$  is used unless it has already appeared. Transfinite induction on  $\beta$  proves that every baseline label eventually occurs. Finally,  $d$  disagrees with  $e_\alpha$  at coordinate  $2 \cdot \alpha$  for each  $\alpha$ .  $\square$

This earlier diagonal mechanism ([theorem 19.37](#)) explains why the class of support codes can be dense without representing every global enumeration. The availability of a dense coded core is not an exhaustive coding theorem.

### 104.4 Cuts and the Burali–Forti boundary

The small-cut property used in [Note 86.3](#) allows only set-sized option families. An unrestricted rule filling every proper-class cut would already demand a surreal strictly greater than every surreal, by taking all of **No** on the left and nothing on the right. This is impossible. Similarly, the proper class  $\text{Ord}$  must not be treated as one of its own set ordinals. The proper-class scope of the enumeration domain is not a license to use it as a set-length prefix.

A cut of  $\mathcal{P}_s(b)$  is one supplied class of core codes. The totality of all such class cuts is not an ordinary class of sets. The trace decoder in [Note 92.1](#) acts on one supplied cut, and its uniform variants act on a supplied class-indexed family. This is enough for the results proved here.

### 104.5 Class realizations and external models

For a model  $(M, \mathcal{X})$  of GBC, the words and automorphisms in the global theorems are those admitted by its class realization  $\mathcal{X}$ . The homogeneity construction from admitted uniform parameters yields an admitted class map. Its statement does not refer to every externally imaginable subset of  $M$ .

With an inaccessible cardinal  $\kappa$  in an ambient set theory, the full-class interpretation  $(V_\kappa, \mathcal{P}(V_\kappa))$  makes the collection of internal classes an ordinary set in the ambient universe. All its internal full words can then be collected externally, and ordinary external maps between them can be discussed.

This is a legitimate larger-universe interpretation, not a proof that GBC internally has a class of all classes. Inaccessible cardinals are not required for the regular set theorem of [Section 94](#).

For fixed class relations in GBC, well-foundedness is equivalent to the absence of a set-coded descending  $\omega$ -sequence. In the nontrivial direction, a nonempty class with no minimal element lets global choice select a descending sequence by set recursion. Consequently, enlarging the class realization while keeping the set universe and a fixed relation unchanged does not change that relation's well-foundedness. What can change is which class relations and full branches are available. This scope distinction is also recorded in the repository's review notes (this report's README, "Other discrepancies", and [section 5.6](#)).

Finally, "all definable classes" requires an explicit metatheoretic or syntactic interpretation. Formula codes alone do not supply an unrestricted internal truth predicate. Restricting to a stated formula family, supplying the relevant satisfaction relation, or working externally over a fixed model avoids conflating definability with a same-level exhaustive class coding.

## 105 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 50](#); where Part VIII's notes or the question index of [section 113](#) say otherwise, these notes supersede them. Questions of sources 13–30 are cited by their labels in this report (`swo:sub:nnumber` for unlabelled ones).

- (a) *Which extra predicates make the reconstruction unique* (Part VIII's item (c) and Part IX's item (f), [section 60](#); source 11's [research question 113.4](#); source 14's "Minimal enrichment for exhaustiveness", [research question 113.34](#); source 13's "Minimal extra structure for recognizing branches", [research question 113.20](#); source 22's "Eliminating label data", [research question 113.135](#)): source 14's question is **answered**, the others **in part**. Unranked cylinder inclusion recovers every prefix height ([theorem 87.1](#)) and, with label occurrence, recognizes represented and exhaustive cuts ([theorem 87.4](#) and [corollary 87.5](#)). Without labels it does not: the core with all its cylinders and the order of children does not recognize exhaustive branches, even after naming any set of core points and nodes ([theorems 91.1](#) and [91.2](#) and [corollary 91.3](#)); nor does the ordered prefix tree without the core ([corollary 96.3](#) and [note 96.8](#)). Equality of labels on different extension edges suffices ([proposition 93.4](#)) and makes the core-and-cylinder structure rigid ([theorem 93.3](#); source 27's [note 96.5](#)). Open: whether a strictly weaker enrichment recognizes exhaustiveness while leaving nontrivial automorphisms (source 25's [research question 113.166](#)), and whether a support filtration recovers part of the labels ([research question 113.167](#)).
- (b) *The stabilizer of the exhaustive cuts* (Part VIII's item (c); source 14's "The exhaustive-trace stabilizer", [research question 113.33](#); source 17's "The extendible automorphism group", [research question 113.73](#); source 20's [research question 113.107](#); source 21's [research question 113.119](#)): **in part**. Not every cylinder-preserving automorphism belongs to it ([theorem 92.2](#)), even among those fixing a given set of core points and nodes ([theorem 91.2](#)); in the bare order, not every automorphism fixing a given set of core points and supported in one interval belongs to it ([theorem 97.7](#)); every automorphism of the ordered prefix tree agrees through any set height with one that preserves exhaustion ([corollary 96.12](#)), and explicit families preserve it ([theorem 96.4](#), source 27's antichain transformations, [proposition 96.11](#)). The structure of the stabilizer remains open (source 25's [research question 113.165](#)).
- (c) *Automorphism extension across residual fibres* (source 15's "Automorphism extension", [research question 113.51](#); source 14's "Automorphisms of raw trace fibers", [research question 113.39](#);

- source 20's [research question 113.108](#); source 21's [research question 113.120](#)): **answered** in an external set-sized model: a core automorphism extends to the raw order exactly when it preserves the isomorphism types of all residual fibres, with a split exact sequence ([theorem 98.5](#)). A uniform internal version is not given.
- (d) *Completion with residual data* (source 15's question, [research question 113.50](#)): **answered** in an external set-sized formulation ([theorem 98.2](#)); over residual alphabets traces merge at most two initial words ([proposition 99.2](#)), with an explicit example and interior completion point ([propositions 99.4](#) and [99.5](#)).
- (e) *Baselines* (source 11's [research question 113.5](#), Part VIII's item (d); source 17's "Changing baselines", [research question 113.74](#)): **further in part**. Relabelled cores are disjoint ([proposition 97.9](#)), and two cores coincide exactly when their baselines eventually agree, being otherwise disjoint (source 25's remark to [research question 113.170](#)). The general case remains open.
- (f) *Dependence on the class realization* (Part VIII's item (e) and Part IX's item (e); source 11's [research question 113.8](#); source 12's [research question 113.18](#); source 08's question 12; source 17's "Recovering a class realization", [research question 113.75](#); source 14's "Which new cuts can a class extension add?", [research question 113.43](#)): source 17's question is **answered**: the labelled core with its admitted represented cuts recovers the whole class realization ([theorem 102.3](#), with  $b$  and  $g$  fixed; [note 102.5](#), with  $b$  only; [note 102.16](#), with no baseline given), and pair-swap traces already suffice. Source 12's [research question 113.18](#) is **answered for externally countable models**: the external order is  $\mathbb{Q}$ , stays so with the core named and the baseline fixed, and detects neither the class part nor ETR ([theorem 102.13](#) and [propositions 102.19](#) and [102.20](#)); fixing the core pointwise recovers everything ([note 102.5](#) and [corollary 102.12](#)). In general the external order remembers at least the cardinality of the class part ([theorem 102.13](#)); what more it remembers for uncountable models is open. Source 22's "Reconstruction without fixed coding parameters" ([research question 113.136](#)) is answered by [notes 102.5](#) and [102.16](#); a definable common baseline exists exactly when  $V = \text{HOD}_p$  for some  $p$  ([theorem 102.21](#) and [corollary 102.22](#)). Source 18's "The expanded countable model" ([research question 113.88](#)) asks what depends on the countable class realization once the label evaluation is added: all of it, by [note 102.16](#); the classification of the expanded structure is open.
- (g) *Ordinal support bounds* (Part VIII's item (f); source 17's "Ordinal and birthday optima for supported completion", [research question 113.78](#); source 09's direction "Sharp birthday and support bounds" and source 08's question 13): the ordinal half is **answered**: the least ordinal  $\beta$  with a completing permutation supported in  $\beta$  is  $\max(\delta_S, \delta_T)$  ([theorem 101.2](#)). Birthday bounds and the forced-prefix length remain open, as source 27 says. Source 17's sentence after [corollary 49.10](#) ("It does not assert a minimum ordinal support bound") remains true of source 17's theorem; source 27 supplies the bound.
- (h) *Stale claims kept as provenance*. Source 20's opening of its Section 5 and source 21's opening of its Section 9 say they answer the repository's extension question negatively; at their pins that was new, and at this report's present state it is a second route to Part VIII's item (c) ([section 97](#)). Source 22's sentence "We have not shown that this label information can be discarded" is answered by source 25 ([note 87.6](#)).



## Part XIV

# Unrestricted class well-orders II: local size, condensation histories and external models

[merge] (3 October 2026.) The sequel to Part IX on unrestricted class well-orders of  $\mathbf{No}$ , and the last Part of the batch-83 addition. Part IX (part IX) printed what batch-81 sources 13–18 prove about them: direct comparison by the largest common labelled prefix, the canonical decomposition of one class well-order into Ord-blocks (theorem 52.3), explicit condensation histories of every prescribed class length (theorem 53.4), finite components (theorem 54.4), products, a  $\Sigma_1^1$  comprehension bound, and source 18’s classification of the full external order in countable  $\omega$ -standard models (theorem 58.6). This Part prints what seven batch-83 manuscripts add on the same objects. It is “II” relative to Part IX: every result printed there is cited by its Part IX label, and the batch-83 re-proofs of it are notes.

- [Section 106](#): source 23’s Sections 9–10, its re-proofs of direct comparison and of the canonical set-interval blocks, with the iteration of condensation by explicit witnesses (limits as unions of equivalences).
- [Section 107](#): local size inside the unrestricted order. Source 20’s Section 6 (the maximal final set tail, an exact local coding dichotomy, the convex blocks of locally set-sized intervals and their dense quotient with an unfilled countable cut, and the external cardinal form in a full model) and source 27’s Section 7 (every uniform class-indexed family misses a whole convex copy of  $\mathfrak{W}_{\text{all}}(\mathbf{No})$ ; the set analogue at every birthday cutoff).
- [Section 108](#): complete condensation histories. Source 20’s Section 7 (termination by the input order; exact heights  $\min\{\alpha : \theta \leq \kappa^\alpha\}$  in full models), source 23’s Section 11 (a second route to source 15’s polynomial histories, in the union-of-equivalences convention), source 29’s Section 11 (floor maps, finite normal forms, and the theorem that over GBC the existence of canonical condensation histories is equivalent to strong comparability of class well-orders), and source 23’s Section 12 (copies of polynomial orders of any prescribed hierarchy inside every interval between two distinct set-like global enumerations).
- [Section 109](#): countable class models. Source 18’s question on  $\omega$ -nonstandard models ([research question 113.89](#)) is answered three times independently, by source 19’s Section 12, source 20’s Section 8 and source 21’s Section 10. Source 19 is printed first and is the base: its [theorem 109.9](#) gives one complete theory for the pure order of *every* externally set-sized GBC model, the strongest of the three answers. Sources 20 and 21 follow as routes, each whole; what only they prove is printed as theorems (source 20’s external finite-condensation blocks and decidability of the common theory, source 21’s automorphism group of the nonstandard order and the first Cantor–Bendixson derivative). Source 30’s compactness observation closes the section.
- [Section 110](#): source 27’s Section 10 (foundations, full rank semantics, internal and external well-foundedness, and its ledger).
- [Section 112](#): what the Part answers.

The rules of [section 1.2](#) apply: a result proved in an earlier Part, or twice in this one, is printed once and every other copy is a numbered note quoting the statement, carrying the copy’s own label, followed by its own proof when the argument differs or the source adds a point.

*Provenance and what each source saw.* Sources 19 and 20 are pinned to d5bd4a67b, sources 21 and 23 to 109aaca15, sources 27, 29 and 30 to 7ff7736ec; none of these revisions contains Part IX.

Source 19 read batch-81 sources 15, 16 and 18 as supplied manuscripts, source 20 read source 18, and source 23 read drafts of sources 13, 14 and 17; source 29 read sources 19, 20 and 21; sources 21, 27 and 30 cite none of these manuscripts. Their overlaps with Part IX and with one another are therefore independent re-proofs unless a note says otherwise.

*Notation in this Part* (the batch-83 notation addendum, [section 1.3.2](#), lists every renaming).  $\mathfrak{W}_{\text{all}}(\mathbf{No})$ ,  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$  as in [section 1.3](#); for a set-sized two-sorted model  $\mathcal{M}$ ,  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  and  $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$  are the external sets of its admitted class well-orders and of its admitted set-like ones, as in Part IX (source 19's  $H_{\text{all}}^{\mathcal{M}}$ ,  $H_{\text{sl}}^{\mathcal{M}}$  and source 20's  $H^{\mathcal{M}}$ ; source 21's  $L^{\mathcal{M}}$  is kept, being defined in its text).  $\sigma(\mathcal{A})$  is the dense shuffle, as in [section 58](#);  $\mathcal{F} = \{n! : n \in \omega\}$  (source 19's  $\mathcal{D}$ ); the nonstandard interval type  $\omega + \sum_{q \in \mathbb{Q}} \mathbb{Z} + \omega^*$  is  $I_{\infty}$  in all three countable-model sources (source 20's symbol; source 19's  $K$  and source 21's  $H$  are renamed, since  $K_R$  and  $H$  name a final set tail and a history here).  $P(\Gamma)$  is the class of finite-support ordinal-coefficient polynomials compared at their greatest disagreement, source 15's  $P(J)$  of [definition 53.1](#) (source 23's  $P(\Gamma)$  and source 29's  $\mathcal{P}(\Gamma)$  are renamed;  $\mathcal{P}(V_{\kappa})$  remains a power set). Source 20's common-prefix carrier  $P(R, S)$  is printed  $C(R, S)$ , this report's common prefix. Source 20's  $K_R$  is the largest final set segment of  $R$ , source 16's  $A_R$  ([corollary 52.7](#)). *False reading*: it is not the  $K_R$  of source 18's “Uniform removal through ordinary ordinal stages” in Part IX, which is an initial set-like part of a block index order. Source 27's  $\kappa = |X|$  is printed  $\mu$  and its  $S_{<\theta}$  is  $\mathbf{No}_{<\theta}$ ; its core  $\mathcal{D}_b$  is  $\mathcal{P}_s(b)$ ; source 23's  $\mathfrak{E}$  and  $\mathfrak{W}$  are  $\mathfrak{W}_{\text{sl}}$  and  $\mathfrak{W}_{\text{all}}$ . Displayed equations keep their source numbers as (SNN.n).

## 106 Direct comparison and set-interval condensation once more (source 23)

[merge] Source 23's Sections 9–10 recall the direct comparison of class well-orders and the canonical decomposition into set-interval blocks before its new constructions ([section 108](#)). Both are earlier results, printed as notes; the text around them is kept, including the warnings on what the raw order adds to the set-like one, and the iteration of condensation by explicit set witnesses.

### 106.1 Source 23: direct comparison beyond ordinal-class enumerations

A set-like well-order on  $\mathbf{No}$  has a class enumeration of order type  $\text{Ord}$ . A raw class well-order need not: examples include copies of  $\text{Ord} + \text{Ord}$  and lexicographic products of ordinal classes. Simply asking for the first *ordinal coordinate* of disagreement can miss an entire proper-class common prefix.

The repository gives a direct comparison ([theorems 14.2](#) and [14.7](#)). Here is the mechanism, with its foundational boundary made explicit.

*Note 106.1* (Source 23, Proposition 9.1, “Largest common labelled initial part”). *For two class well-orders  $H, K$  of the same labelled class  $X$ , there is an elementarily definable largest common initial part on which the two relations agree. If  $H \neq K$ , let  $a$  and  $b$  be the first remaining labels in  $H$  and  $K$ . Then  $a \neq b$ , and the rule*

$$H <_{\text{lex}} K \iff a <_X b$$

*defines a strict linear comparison on raw well-orders, when  $X$  has a fixed linear label order. No elementary transfinite recursion axiom is required.* This is [theorem 14.2](#) (source 12's [theorem 14.7](#)) with the same proof; Part IX prints the five batch-81 proofs ([section 51](#)).

*Source 23's proof.* Declare  $x$  to belong to  $P$  when its predecessor classes in  $H$  and  $K$  are equal and the two relations agree on that predecessor class. This is an elementary formula with class

parameters: equality of predecessor classes and agreement of relations use only quantification over set objects. If  $x \in P$ , every predecessor of  $x$  belongs to  $P$ ; the relations agree on  $P$ . Any common labelled initial part has this property and is contained in  $P$ .

If  $P = X$ , the orders agree. Otherwise the nonempty complement has an  $H$ -least point  $a$  and a  $K$ -least point  $b$ . Were  $a = b$ , its predecessor class would be  $P$  in both orders and it would belong to  $P$ , a contradiction.

For transitivity, compare the common parts  $P_{H,K}$  and  $P_{K,M}$  as initial segments of  $K$ . They are comparable by inclusion. If one is properly smaller, the first disagreement with the third order is the same as the earlier disagreement. If they are equal, the three next labels satisfy the transitive numerical comparison. This proves transitivity; trichotomy and asymmetry follow directly.  $\square$

The proposition remains a predicate-level statement about a fixed finite number of class relations. It does not collect every such relation into an ordinary class.

### 106.1.1 Why the raw order has phenomena absent from the set-like part

A raw order can have a proper-class common prefix with only two labels  $a < b$  left over. The two orders ending respectively in  $(a, b)$  and  $(b, a)$  are adjacent in the outer lexicographic order: there is no third permutation of the remaining carrier. This construction uses a well-order of  $\mathbf{No} \setminus \{a, b\}$  as the common prefix. The repository gives the more general exact finite-tail adjacency analysis (theorem 16.2; source 17's theorem 54.4).

By contrast, if two exhaustive Ord-enumerations differ after a set prefix, fresh surreal labels between the differing values remain available. Their interval contains many further enumerations and, as Section 108.4 will show, raw orders of arbitrarily prescribed condensation complexity.

## 106.2 Source 23: set-interval condensation without classes as elements

The classical finite condensation of a linear order identifies points with a finite interval between them; see, for example, Brown and Suárez [68]. We use a size-relative analogue for class well-orders: replace “finite” by “a set.” These are not the same operation, and finite-condensation theorems are not invoked as proofs of the class claims below.

**Definition 106.2.** For a class well-order  $H = (X, <_H)$ , put

$$x \sim_H y \iff [\min_H(x, y), \max_H(x, y)]_H \text{ is a set.}$$

Let  $B_H$  be the subclass of points that are least in their  $\sim_H$ -class. The *set-interval condensation* of  $H$  is  $D(H) = H \upharpoonright B_H$ .

Saying that the displayed interval is a set means that there exists a set whose members are exactly the points satisfying the interval formula. No powerclass is required.

*Note 106.3* (Source 23, Theorem 10.2, “Canonical block decomposition”). *The relation  $\sim_H$  is a convex equivalence relation. Every block has a least point, and  $B_H$  represents the quotient as a class well-order. Every block is either a set ordinal order or a copy of Ord. Every nonfinal block is a copy of Ord; a set-sized block can occur only as the final block.* This is source 15's theorem 52.3 (Part IX), proved there and by source 16 (theorem 52.6 with corollary 52.7) and source 18 (theorem 52.11) by the same rank argument; source 29's note 108.9 below is a fifth copy.

*Source 23's proof.* Reflexivity and symmetry are immediate. If  $x \sim_H y$  and  $y \sim_H z$ , the interval from  $x$  to  $z$  is contained in the union of the two given set intervals; Separation makes it a set.

Convexity follows by the same argument. Each block is an admitted nonempty subclass and has an  $H$ -least point  $b$ .

For  $x$  in this block, the interval  $[b, x)_H$  is a set well-order. Assign to  $x$  its ordinal order type. This is a class order isomorphism from the block onto an initial segment of  $\text{Ord}$ . If the block is a set, the range is a set ordinal. If the block is proper, the range cannot be a set, by Class Replacement applied to the inverse; hence the range is all of  $\text{Ord}$ .

Suppose a block  $C$  is a set and some point lies after it. The class of points strictly after every member of  $C$  has a least element  $d$ . For  $c \in C$ , the interval  $[c, d]_H$  is contained in  $C \cup \{d\}$ : a point between them outside  $C$  would be strictly after the whole convex block and contradict minimality of  $d$ . Thus  $c \sim_H d$ , impossible. Therefore a set block must be final.

The representative of a block is an ordinary set object. Restricting  $H$  to these representatives supplies the quotient order without ever using a proper block as an element of a class.  $\square$

In particular, every raw well-order is canonically a lexicographic sum of  $\text{Ord}$  blocks, possibly with one final set-ordinal block. The index order of those blocks can itself be a complicated raw well-order. This is a structural decomposition, not a claim that every raw order has only a set number of blocks.

### 106.2.1 Iteration with explicit witnesses

To keep every stage on the original domain, start with equality  $E_0$ . Given a convex equivalence relation  $E$ , define its condensation successor by

$$x D(E) y \iff \exists S \text{ a set } \forall z \in [x, y]_H \exists s \in S (zEs), \quad (\text{S23.3})$$

where the interval is written with its endpoints in increasing order. Thus a set of representatives meets all  $E$ -classes in the interval. At a limit stage take the union of earlier equivalences.

The witness in (S23.3) can be converted into a set of least representatives using Class Replacement. Accordingly it says exactly that the interval in the represented quotient is a set. This formula is elementary once  $E$  is a class parameter.

For a general raw order, asserting a uniform hierarchy of class-valued relations along an arbitrary class well-order is a genuine recursion principle.  $\text{GBC} + \text{ETR}$  is a sufficient setting; elementary transfinite recursion has strength beyond  $\text{GBC}$  [22, 24]. We make no claim here that full  $\text{ETR}$  is necessary for this particular operator. The next section avoids the issue for a large explicit family by writing down every stage simultaneously.

[merge] Source 23's successor rule (S23.3) with unions at limits is the convention of equivalence relations; sources 15, 20 and 29 iterate nested classes of least representatives and intersect them at limits (sections 53.1 and 108.1 and definition 108.10). Source 29's corollary 108.14 shows that the two conventions define the same history. Whether  $\text{ETR}$  is needed for general inputs is settled in the form of an equivalence by source 29's theorem 108.19, not by  $\text{ETR}$  itself (section 112).

## 107 Local size of intervals in the unrestricted order (sources 20 and 27)

[merge] Source 11's diagonal theorem (theorem 19.37) says that no uniform class of set codes exhausts the set-like order. Source 20 localizes it to intervals of the unrestricted order and finds an exact dichotomy: an interval  $[R, S]$  has a set-indexed exhaustive coding exactly when the residual carrier  $Y(R, S)$  after the common prefix is a set, and otherwise no uniform family indexed by any

class of sets exhausts even the open interval. The resulting equivalence  $\equiv_{\text{set}}$  has convex blocks  $\mathcal{W}(K_R)$ , and its quotient is dense but not set-cut-filling. Source 27 proves the stronger omission: a uniform family misses a whole convex copy of  $\mathfrak{W}_{\text{all}}(\mathbf{No})$ , so no uniform family is dense in the raw comparison. Both are new against the repository.

### 107.1 Source 20: A local size dichotomy for the unrestricted lexicographic order

The global failure of exhaustive set coding has a sharper local form. An interval can have an ordinary set of codes, or it can defeat every uniform family indexed by a class of sets. The distinction is determined by exactly one geometric feature: how many labels remain after the endpoints' common prefix. Throughout this section we work in GBC. Several of the structural lemmas only use GB once the displayed well-orders have been supplied.

We use the elementary comparison of labelled class well-orders established in this report (theorems 14.2 and 14.7) and source 18 (theorem 51.16). If  $R \neq S$  are class well-orders of  $\mathbf{No}$ , their maximal common labelled prefix has carrier  $C(R, S)$ . The first labels after this prefix are distinct; their numerical comparison decides  $R <_{\text{lex}} S$ . We put  $C(R, R) = \mathbf{No}$  and write

$$Y(R, S) = \mathbf{No} \setminus C(R, S).$$

The common prefix carries a relation as well as a carrier. Thus equality of prefix carriers below always includes agreement of their induced orders when that agreement is needed.

A labelled class prefix  $p$  consists of a class  $A_p \subseteq \mathbf{No}$  with a specified class well-order. Its cylinder

$$\text{Cyl}(p) = \{R : R \text{ extends } p \text{ as a labelled initial segment}\}$$

is notation for a predicate on class relations. It is convex under the native lexicographic comparison. In particular, its members need not have set codes, and the braces do not assert that these members form an ordinary GB class. Whenever  $\mathbf{No} \setminus A_p$  is nonempty, a completion exists: append a fixed global well-order of the remaining labels.

#### 107.1.1 Final set tails and interior cylinders

For a class well-order  $R$  of  $\mathbf{No}$ , put

$$K_R = \{x \in \mathbf{No} : \{y \in \mathbf{No} : xRy\} \text{ is a set}\}.$$

This definition uses only set quantifiers with  $R$  as a class parameter.

*Note 107.1* (Source 20, Lemma 6.1, “The maximal final set tail”). *The class  $K_R$  is either empty or a set. It is final in  $R$  and is the unique maximal final subset of  $R$  that is a set. If it is nonempty, it is the final set block in the canonical set-interval decomposition of  $R$ .* This is source 16’s corollary 52.7 (Part IX), whose  $A_R$  is the present  $K_R$ , with the same argument; the uniqueness and maximality clauses are explicit here.

*Source 20’s proof.* Finality follows because a tail after a later point is contained in the tail after an earlier point. If  $K_R$  is nonempty, let  $k$  be its  $R$ -least member. Then

$$K_R = \{k\} \cup \{y : kRy\},$$

which is a set by the definition of  $k$ . Every final set  $T$  lies in  $K_R$ : for  $x \in T$ , the tail after  $x$  is contained in  $T$ .

For the block assertion, recall that two labels lie in one canonical block precisely when the closed interval between them is a set (theorem 52.3; source 18's theorem 52.11). All of  $K_R$  lies in one such block. If  $zRk$  belonged to that block, the union of  $[z, k]_R$  and the set tail after  $k$  would make the tail after  $z$  a set, contradicting the minimality of  $k$ . Conversely every point of a final set block has a set tail. Thus the two descriptions agree.  $\square$

The next lemma is the main local argument. It is useful because the unused numerical alphabet can be a very thin proper subclass of  $\mathbf{No}$ ; one cannot assume that it contains a numerical separator between two specified labels.

**Lemma 107.2** (An interior cylinder with a proper residual alphabet). *Suppose  $R <_{\text{lex}} S$  and  $Y(R, S)$  is a proper class. There exists a labelled class prefix  $p$  such that*

$$\emptyset \neq \text{Cyl}(p) \subseteq (R, S), \quad \mathbf{No} \setminus A_p \text{ is a proper class.}$$

*This conclusion uses only GB and the supplied class well-orders, apart from the use of Global Choice to choose an arbitrary completion of the resulting prefix.*

*Proof.* Let  $C = C(R, S)$ , let  $Y = \mathbf{No} \setminus C$ , and let  $a < b$  be the first labels of  $R, S$  respectively after  $C$ . Consider an ascent in the  $R$ -tail,

$$aRxRy, \quad x < y, \quad \{z : xRz\} \text{ a proper class.} \quad (\text{A})$$

If such a pair exists, transpose the labels  $x, y$  in  $R$  to obtain an order  $T$ . At the first changed position,  $R$  has label  $x$  and  $T$  has label  $y > x$ , so  $R < T$ . Both still start with  $a$  immediately after  $C$ , whereas  $S$  starts with  $b > a$ , so  $T < S$ .

Take for  $p$  the prefix whose labels are  $\{z : zRx\}$ , in their  $R$ -order, followed by  $y$ . Every completion of  $p$  has the same two strict comparisons just described. Its residual carrier is

$$\mathbf{No} \setminus A_p = \{x\} \cup (\{z : xRz\} \setminus \{y\}),$$

which is proper. This supplies the desired cylinder. Dually, a descent

$$bSxSy, \quad x > y, \quad \{z : xSz\} \text{ a proper class} \quad (\text{D})$$

gives a cylinder inside  $(R, S)$  by transposing  $x, y$  in  $S$ .

Suppose neither (A) nor (D) exists. By note 107.1, the class

$$Z = Y \setminus (K_R \cup K_S \cup \{a, b\})$$

is proper. If  $uRv$  are in  $Z$ , then  $aRu$ , the tail after  $u$  is proper, and failure of (A) implies  $u > v$  numerically. Hence  $R$  on  $Z$  is strictly numerically decreasing. Failure of (D) similarly implies that  $S$  on  $Z$  is strictly numerically increasing.

The restriction of each relation to  $Z$  is a class well-order. Recursively take the first  $\omega$  elements of  $(Z, S)$ : at every finite stage the complement of the finitely many selected elements is nonempty, and its  $S$ -least member is uniquely specified. Ordinary set-valued recursion with class parameters yields a set sequence  $(z_n)_{n < \omega}$ . Its entries strictly increase numerically, so  $z_{n+1}Rz_n$  for every  $n$ . Its range is a nonempty available subclass with no  $R$ -least element, a contradiction. Thus (A) or (D) exists, completing the proof.  $\square$

[merge] Source 18's lemma 58.4 (Part IX) finds, by the same ascent–descent alternative, a cylinder with *infinite* residual carrier inside an interval joining two finite-condensation blocks; the lemma above is its proper-class analogue, with the final set tails  $K_R, K_S$  in place of the maximal finite tails.



### 107.1.2 Uniform coding and its diagonal obstruction

For precision, a *uniform  $I$ -indexed family* of class well-orders means a single class relation  $F \subseteq I \times \mathbf{No} \times \mathbf{No}$ , where  $I$  is a class of sets and the section

$$F_i = \{(x, y) : (i, x, y) \in F\}$$

is a class well-order for each  $i \in I$ . A predicate on class well-orders has an exhaustive such coding if every available well-order satisfying it is one of these sections. This last condition is an assertion with class quantifiers; it is not an application of elementary comprehension to a collection of classes. Set-indexed coding means that  $I$  is a set.

**Lemma 107.3** (A diagonal inside every proper-residual cylinder). *If  $\mathbf{No} \setminus A_p$  is proper, no uniform family indexed by any class of sets exhausts  $\text{Cyl}(p)$ . More strongly, for every uniform class-indexed family  $(R_i)_{i \in I}$  of well-orders of  $\mathbf{No}$ , some member of  $\text{Cyl}(p)$  differs from every  $R_i$ .*

*Proof.* Put  $Y = \mathbf{No} \setminus A_p$ . Global Choice supplies a class bijection from  $\text{Ord} \times \{0, 1\}$  to  $Y$ . Thus  $Y$  is partitioned into disjoint pairs  $\{u_\xi, v_\xi\}$  for  $\xi \in \text{Ord}$ ; name their labels so that  $u_\xi < v_\xi$  numerically. Global Choice also supplies a class injection  $\sigma : I \rightarrow \text{Ord}$ .

Define an ordered pair at every ordinal  $\xi$ . If  $\xi = \sigma(i)$ , put  $u_\xi, v_\xi$  in the opposite order to their order in  $R_i$ . The index  $i$  is unique. At other coordinates use the order  $(u_\xi, v_\xi)$ . This rule is elementary in the given class parameters. Append these pairs, in increasing ordinal order, after the prefix  $p$ . The result  $T$  is a class well-order of  $\mathbf{No}$ : a nonempty subclass either has a least member in the prefix, or has a least pair index and then a least member of that finite pair. It belongs to  $\text{Cyl}(p)$ .

For each  $i$ , the two labels in pair  $\sigma(i)$  occur in the opposite relative order in  $T$  and  $R_i$ . Hence  $T \neq R_i$ . The construction has produced one ordinary class relation from the given uniform family, without enumerating any collection of classes.  $\square$

[merge] The pair-swap diagonal is the one of source 11's [theorem 19.37](#) and source 12's [theorem 19.38](#), run inside a cylinder; source 27's [theorem 107.8](#) ([section 107.2](#)) strengthens the conclusion from one omitted order to a whole omitted convex copy of  $\mathfrak{W}_{\text{all}}(\mathbf{No})$ .

**Theorem 107.4** (Exact local coding dichotomy). *For two distinct class well-orders  $R <_{\text{lex}} S$  of  $\mathbf{No}$ , the following are equivalent:*

1.  $Y(R, S)$  is a set;
2. the closed interval  $[R, S]$  has a uniform set-indexed exhaustive coding;
3. the open interval  $(R, S)$  has a uniform exhaustive coding indexed by a class of sets.

*If these conditions fail, every proposed uniform class-indexed family omits a well-order strictly between  $R$  and  $S$ .*

*Proof.* Suppose  $Y = Y(R, S)$  is a set and let  $p$  be the common labelled prefix. The cylinder of  $p$  consists precisely of the orders obtained by appending a well-order of  $Y$ . All such well-orders form the ordinary set  $\mathcal{W}(Y) \subseteq \mathcal{P}(Y \times Y)$ . Concatenation gives a uniform family of class relations indexed by this set. Every order in  $[R, S]$  extends  $p$  by prefix convexity. Filter the index set by the elementary native comparisons  $R \leq F_i \leq S$ . Separation with the supplied class parameters gives a set of indices coding exactly the closed interval. Filtering with strict comparisons gives the open interval, proving  $(1) \Rightarrow (2) \Rightarrow (3)$ .

If  $Y$  is proper, [lemma 107.2](#) gives a proper-residual cylinder inside  $(R, S)$ . [Lemma 107.3](#) constructs within that cylinder a well-order omitted by any proposed class-indexed family. This contradicts (3), and proves both the remaining implication and the last assertion.  $\square$

This is a sharper conclusion than an estimate of cardinality. In GB there is no ordinary cardinal assigned to the collection of all class well-orders. The theorem instead gives a precise dichotomy in terms of legitimate uniform presentations. In particular, for these intervals, allowing a proper class of set indices does not enlarge the family of intervals that can be exhaustively presented.

### 107.1.3 The convex blocks of locally set-sized intervals

Define

$$R \equiv_{\text{set}} S \iff Y(R, S) \text{ is a set.}$$

Equality satisfies this convention because  $Y(R, R) = \emptyset$ .

**Theorem 107.5** (Canonical local-size blocks). *The relation  $\equiv_{\text{set}}$  is a convex equivalence relation on the predicate of class well-orders of  $\mathbf{No}$ . The block containing  $R$  consists exactly of the orders obtained by keeping the labelled prefix on  $\mathbf{No} \setminus K_R$  and replacing the final well-order of the set  $K_R$  arbitrarily. Consequently its native lexicographic order is represented by the ordinary set order  $\mathcal{W}(K_R)$ .*

*Proof.* Symmetry and reflexivity are immediate. For transitivity, the common prefixes of  $(R, S)$  and  $(S, T)$  are initial segments of the same order  $S$ , so one is contained in the other. The shorter is a common labelled prefix of  $R, T$ . If both original residuals are sets, so is the residual after this shorter prefix, and hence so is  $Y(R, T)$ . Convexity follows from prefix convexity: an order between  $R, S$  extends their common prefix, so its common-prefix residual with either endpoint is a subset of  $Y(R, S)$ .

Suppose  $R \neq S$  and  $Y(R, S)$  is a set. Let  $a$  be the first  $R$ -label outside their common prefix. Its strict  $R$ -tail is a subset of  $Y(R, S)$ , so  $a \in K_R$ . Hence the common prefix already contains every label of  $\mathbf{No} \setminus K_R$ , in its original  $R$ -order. Conversely, if  $S$  keeps that prefix and appends any well-order of  $K_R$ , then  $R, S$  share that prefix and their first-disagreement residual, if any, is contained in the set  $K_R$ . This proves the block description. The numerical first difference inside the common set tail is exactly the usual comparison on  $\mathcal{W}(K_R)$ .  $\square$

If  $K_R = \emptyset$  or  $|K_R| = 1$ , the block is a singleton. If  $|K_R| = n < \omega$ , its size is  $n!$ . If  $K_R$  is infinite of cardinality  $\mu$ , its size is  $2^\mu$ : the upper bound follows from coding a relation by a subset of  $K_R^2$ , and independent orientations of  $\mu$  disjoint pairs give the lower bound. The ordered structure for an infinite tail depends on its numerical alphabet, so cardinality alone does not classify these blocks. Every set alphabet  $K \subseteq \mathbf{No}$  occurs as such a final tail: well-order  $\mathbf{No} \setminus K$  in type Ord and append a well-order of  $K$ .

We may discuss a quotient comparison using representatives and  $\equiv_{\text{set}}$ . The notation  $[R]$  in the next statement is only an abbreviation for this equivalence relation. It does not make an equivalence class of class relations into a set or an ordinary class element.

**Theorem 107.6** (A dense quotient that does not fill countable cuts). *The quotient comparison by  $\equiv_{\text{set}}$  is dense and has no endpoints. Nevertheless there are two uniformly coded countable families whose quotient classes form an unfilled cut of character  $(\omega, \omega)$ . Thus deleting all locally set-sized blocks does not produce a surreal-like set-cut-filling order.*

*Proof.* Let  $R < S$  belong to different blocks. Their residual is proper, so lemma 107.2 gives a proper-residual cylinder inside  $(R, S)$ . Choose three different labels from its residual, sort them numerically, and form completions  $T_0 < T_1 < T_2$  beginning with these respective labels after the

prefix. Their first disagreements leave the entire proper residual unused, so the three completions lie in three different blocks. Convexity of the blocks gives

$$[R] < [T_1] < [S].$$

To obtain blocks on both sides of any  $R$ , choose first labels numerically below and above the first label of  $R$ , and complete them. Their disagreement with  $R$  occurs before any common label, leaving the proper residual **No**.

For the countable gap, use the proper numerical alphabet

$$Y = (\mathbb{Q} \setminus \{0\}) \cup \{\alpha : \alpha \in \text{Ord}, \alpha \geq \omega\} \cup \{-\alpha : \alpha \in \text{Ord}, \alpha \geq \omega\} \subseteq \mathbf{No}.$$

Its complement is proper as well: for example, the labels  $\alpha + \frac{1}{2}$  for infinite ordinals  $\alpha$  are in that complement and are pairwise distinct. Choose one class well-order  $p$  of  $\mathbf{No} \setminus Y$  and use it as a prefix. For each positive integer  $n$ , let  $R_n$  extend  $p$  with first remaining label  $-1/n$ , and let  $S_n$  extend it with first remaining label  $1/n$ . Append a fixed global order of the unused labels in each case. These constructions are uniform in  $n$ , and

$$[R_n] < [R_{n+1}] < [S_{m+1}] < [S_m] \quad (n, m \geq 1).$$

All the displayed classes are distinct because different first labels after  $p$  leave the proper residual  $Y$ .

Suppose a class  $[T]$  lies above every  $[R_n]$  and below every  $[S_n]$ . In particular,  $R_1 < T < S_1$ , so prefix convexity forces  $T$  to extend  $p$ . Let  $y \in Y$  be its first remaining label. The inequalities imply

$$-1/n \leq y \leq 1/n \quad (n \geq 1).$$

No member of  $Y$  satisfies these inequalities: a nonzero rational fails one of them by the Archimedean property of  $\mathbb{Q}$ , and an ordinal or negative ordinal of magnitude at least  $\omega$  fails already for  $n = 1$ . This is a contradiction.

The downward closure of the  $[R_n]$  and the upward closure of the  $[S_n]$  therefore partition the quotient comparison. The first family is cofinal on the lower side, and the second is coinital on the upper side. Neither side has an endpoint because the displayed sequences are strictly monotone. No finite family can be cofinal or coinital on the respective side, proving the exact character  $(\omega, \omega)$ . These closures are again predicates on representatives; the proof does not collect quotient objects.  $\square$

**Corollary 107.7** (The external cardinal form of the local dichotomy). *Let  $\kappa$  be strongly inaccessible in an ambient ZFC universe and interpret classes using  $(V_\kappa, \mathcal{P}(V_\kappa))$ . The raw order of all internal class well-orders of the internal surreal carrier is then an ordinary ambient set order. For any two distinct members  $R < S$ , its open and closed intervals satisfy*

$$\begin{aligned} |(R, S)| &\leq |[R, S]| < \kappa, & \text{if } Y(R, S) \text{ is an internal set,} \\ |(R, S)| &= |[R, S]| = 2^\kappa, & \text{if } Y(R, S) \text{ is internally proper.} \end{aligned}$$

*Thus  $\equiv_{\text{set}}$  is exactly the external condensation relation defined by  $|[R, S]| < \kappa$ . Its quotient has  $2^\kappa$  points, and every open quotient interval with distinct bounds has  $2^\kappa$  points.*

*Proof.* The full-model size check, proved again in [corollary 108.3](#), identifies internal sets with ambient subsets of cardinality less than  $\kappa$ . The surreal carrier has ambient cardinality  $\kappa$ . If  $Y(R, S)$  is an internal set of cardinality  $\mu < \kappa$ , the common-prefix cylinder is coded by a subset of  $\mathcal{P}(Y(R, S)^2)$ . Its cardinality is less than  $\kappa$  by the strong-limit property, proving the first case.

If the residual is internally proper, its ambient size is  $\kappa$ . By lemma 107.2, the open interval contains a cylinder whose residual also has size  $\kappa$ . Independent orientations of  $\kappa$  disjoint pairs in that residual give  $2^\kappa$  distinct completions in the cylinder. All are available because this is the full-class model. Conversely every raw well-order is a relation on a carrier of size  $\kappa$ , so there are at most  $2^\kappa$  such orders. This proves the second case and the condensation characterization.

The whole raw order has  $2^\kappa$  elements, again by pair coding and the relation-coding upper bound. Each equivalence block has fewer than  $\kappa$  elements by theorem 107.5 and the strong-limit property. Hence there are  $2^\kappa$  blocks: fewer blocks, even allowing  $\kappa$  elements per block, could not cover  $2^\kappa$  elements. Finally, for  $[R] < [S]$  the raw interval  $(R, S)$  has  $2^\kappa$  elements. Discarding the two endpoint blocks removes fewer than  $\kappa$  elements. Grouping the remaining elements into blocks of size less than  $\kappa$  still leaves  $2^\kappa$  distinct blocks strictly between  $[R]$  and  $[S]$ .  $\square$

[merge] Source 20's Section 10 uses the same full model to re-prove source 11's nonembedding theorem 20.1, by counting  $2^\kappa$  disjoint six-point blocks against a dense subset of size  $\kappa$ ; that passage is printed with source 20's foundations.

Two condensation operations have now appeared and must be kept separate. In corollary 107.7 the points being condensed are *whole well-orders*, and the retained blocks describe local interval sizes in their lexicographic order. The next section condenses intervals *inside one supplied well-order*, whose points are surreal labels. The common set/class boundary does not identify these different orders or their quotients.

## 107.2 Source 27: uniform families miss whole convex copies of the raw order

The bounded-support core is a dense ordinary class of set codes for the set-like interpretation. Could a more elaborate ordinary class of codes be dense in the raw comparison? The answer is negative. A diagonal argument leaves enough unused labels to retain a convex copy of the entire raw order, rather than just producing one omitted object.

### 107.2.1 The diagonal construction

**Theorem 107.8** (An omitted convex copy). *Work in GBC. Let  $I$  be any ordinary class, and let  $\mathcal{R}(i, x, y)$  be a uniform family of binary relations on  $\mathbf{No}$ . There is a convex family of raw class well-orders, described by explicit operations on class variables, that is order-isomorphic to  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  and contains none of the relations  $R_i$ .*

*In particular, no uniformly class-indexed family of raw well-orders meets every nonempty open interval of the raw comparison.*

*Proof.* Reserve  $A = (0, 1) \subseteq \mathbf{No}$  and put  $B = \mathbf{No} \setminus A$ . The map that prefixes the signs  $(+, -)$  is a numerical order isomorphism  $\varphi : \mathbf{No} \rightarrow A$ . The complement  $B$  is a proper class. Global Choice provides a bijection  $c : \text{Ord} \rightarrow B$  and an injection  $j : I \rightarrow \text{Ord}$ .

The ordinal pairs

$$\{2 \cdot \alpha, 2 \cdot \alpha + 1\}, \quad \alpha \in \text{Ord},$$

partition  $\text{Ord}$ . Here  $2 \cdot \alpha$  is left ordinal multiplication. Write  $x_\alpha = c(2 \cdot \alpha)$  and  $y_\alpha = c(2 \cdot \alpha + 1)$ . List these pairs in order of  $\alpha$ . Within pair  $\alpha = j(i)$ , choose its orientation opposite to the truth value of  $R_i(x_\alpha, y_\alpha)$ : put  $x_\alpha$  before  $y_\alpha$  exactly when that truth value is false. There is at most one such  $i$  because  $j$  is injective. Orient all unassigned pairs by a fixed default. This elementary definition gives a class bijection  $h : \text{Ord} \rightarrow B$  and its induced well-order  $P_h$ .

For every class well-order  $Q$  of  $A$ , form  $h + Q$ , placing all of  $B$  first in the order  $P_h$ , then all of  $A$  in order  $Q$ . Its initial set-like block is exactly  $h$ , since every point of  $A$  has the proper class

$B$  before it. Thus these orders constitute the convex fiber (S27.2) (source 11's [proposition 15.3](#)). Restriction to  $A$  and concatenation with  $h$  identify its raw comparison with  $\mathfrak{W}_{\text{all}}(A)$ ; conjugation by  $\varphi$  identifies the latter with  $\mathfrak{W}_{\text{all}}(\mathbf{No})$ .

Every member of this fiber differs from  $R_i$  on the ordered pair  $(x_{j(i)}, y_{j(i)})$ . Therefore the whole fiber is omitted, even though the input sections were not assumed to be well-orders.

For the final claim choose  $a_0 < a_1 < a_2$  in  $A$  and well-orders  $Q_0, Q_1, Q_2$  of  $A$  beginning with these respective labels. Global Choice completes the residual carriers. Then  $h + Q_0 < h + Q_1 < h + Q_2$ . Convexity puts the nonempty ambient interval  $(h + Q_0, h + Q_2)$  inside the omitted fiber.  $\square$

The conclusion is stronger than failure of exhaustive coding. A uniform family might omit many points and still be dense; this theorem rules that out for the raw comparison. It does not rule out cofinal or cointial uniform families: varying the first numerical label through  $\mathbf{No}$  gives such families immediately. It also does not contradict density of  $\mathcal{P}_s(b)$  in  $\mathfrak{W}_{\text{sl}}$ , because the fiber used above has a proper-class tail and contains no set-like global well-order.

The reference to a “convex family” is shorthand for a predicate on class variables together with the transformations  $Q \mapsto h + Q$  and restriction to  $A$ . It is not a claim that this family is a class of sets. Every step in the proof is elementary comprehension with class parameters, supplemented by Global Choice for the two enumerations.

### 107.2.2 The set analogue at birthday cutoffs

The same mechanism has a useful set version, where ordinary counting can replace class diagonalization.

**Theorem 107.9** (Many disjoint convex copies). *Let  $X$  be an infinite linearly ordered set of cardinality  $\mu$ . Suppose a convex subset  $A \subseteq X$  is order-isomorphic to  $X$  and  $B = X \setminus A$  has cardinality  $\mu$ . Then  $\mathcal{W}(X)$ , with enumeration lexicography, contains  $2^\mu$  pairwise disjoint convex copies of itself. Every subset of size less than  $2^\mu$  misses one such copy, and its order-topological density and cellularity both equal  $2^\mu$ .*

*Proof.* There are  $2^\mu$  bijections  $h : \mu \rightarrow B$ : independent pair swaps give the lower bound, while  $\mu^\mu = 2^\mu$  gives the upper bound. For each such  $h$ , consider the completions  $h + Q$  with  $Q \in \mathcal{W}(A)$ . They are a prefix cylinder and therefore convex. Restriction identifies the cylinder with  $\mathcal{W}(A) \cong \mathcal{W}(X)$ . Distinct bijections have distinct length- $\mu$  prefixes, so their cylinders are disjoint.

Each cylinder contains three points in increasing order, hence a nonempty ambient open interval. Selecting one such interval per cylinder gives  $2^\mu$  disjoint nonempty open intervals. Every dense subset must meet all of them. Since  $|\mathcal{W}(X)| = 2^\mu$ , both bounds are exact. A set of fewer than  $2^\mu$  points meets fewer than that many disjoint cylinders, proving the stronger omitted-copy assertion.  $\square$

For every infinite ordinal  $\theta$ , this theorem applies to  $X = \mathbf{No}_{<\theta}$ . The cone  $A = (0, 1) \cap \mathbf{No}_{<\theta}$  consists precisely of  $(+, -) \frown s$  for  $s \in \mathbf{No}_{<\theta}$  and is isomorphic to  $X$ . Finite prefixing preserves every infinite birthday cutoff: for finite  $\alpha$  the new length is finite, while for infinite  $\alpha$  one has  $2 + \alpha = \alpha$ . The complement contains the whole negative-sign cone, also isomorphic to  $X$ , and therefore has the same cardinality. No inaccessible-cardinal assumption enters this statement about a fixed set of sign sequences.

The exact density formula overlaps the earlier corpus (source 16's [theorem 27.2](#), its “Moreover” clause). The point of the argument here is its stronger geometric form: small subsets miss an entire convex self-copy. The class theorem replaces the unavailable hypercardinality count by an explicit diagonal on a uniform family.

*Note 107.10* ([merge]; how the omission results compare). Four uniform-coding obstructions are now printed. Source 11's [theorem 19.37](#) and source 12's [theorem 19.38](#) produce one global enumeration omitted by a given uniform family; source 20's [lemma 107.3](#) produces one omitted order inside every cylinder with proper residual carrier, hence inside every interval whose residual  $Y(R, S)$  is proper ([theorem 107.4](#)); source 27's [theorem 107.8](#) omits an entire convex fibre isomorphic to  $\mathfrak{W}_{\text{all}}(\mathbf{No})$ , so a uniform family is not even dense in the raw comparison. Source 27's theorem does not decide the local question of source 20's dichotomy (its omitted fibre is chosen by the diagonal, not inside a prescribed interval), and it does not contradict density of the core in the set-like order: its fibre has a proper-class tail and contains no set-like order, as source 27 says.

## 108 Complete condensation histories (sources 20, 23 and 29)

[merge] Part IX printed source 15's explicit histories: for every class well-order  $J$ , the polynomial order  $P(J)$  has an elementary history indexed by  $J + 1$  whose stages are proper and whose appended last stage is a singleton ([theorem 53.4](#)). Three batch-83 sources go further. Source 20 proves that a *supplied* nested history of an arbitrary class well-order  $W$  terminates by the stage  $W + 1$ , names ETR as a sufficient existence principle, and computes the exact heights in full models. Source 23 re-proves source 15's polynomial histories in the equivalence-relation convention and places copies of them inside every interval between two distinct set-like global enumerations. Source 29 gives each point of a well-order with a terminating history a finite polynomial address, and proves over GBC that the existence of canonical histories for all inputs is equivalent to strong comparability of class well-orders. All three take the same nested-representative operator  $D_W$  and the same limit convention, up to source 29's [corollary 108.14](#), which identifies source 23's union of equivalences with the intersection of representatives used by sources 15, 20 and 29.

### 108.1 Source 20: complete condensation histories: a bound and an exact model

The canonical set-interval decomposition is elementary. A supplied uniform history of successive decompositions can also be analysed elementarily. The issue of obtaining that history for an arbitrary input is separate. The preceding continuation realized arbitrary class well-orders as first-level block-index orders, analysed removal of initial blocks through ordinary ordinal stages, and supplied a polynomial example isomorphic to its own first condensation (source 18; [theorem 52.14](#) and [proposition 52.15](#)). Those results do not by themselves supply a uniform history of iterated quotients for every input. Here we prove that a complete history always terminates when it is run long enough, specify one sufficient recursion hypothesis, and calculate its exact height in full-class models.

#### 108.1.1 A precise limit convention

Let  $(X, W)$  be a nonempty class well-order. For a nonempty subclass  $A \subseteq X$  containing  $m = \min_W X$ , define  $D_W(A)$  to be the least representatives of the equivalence relation

$$x \sim_A y \iff \{z \in A : \min_W(x, y) \leq_W z \leq_W \max_W(x, y)\} \text{ is a set.}$$

Thus  $D_W(A) \subseteq A$  and it contains  $m$ . This is an elementary operator on classes with  $W$  as parameter. In particular, if  $A \neq \{m\}$  and  $s$  is the least member of  $A \setminus \{m\}$ , then  $s \notin D_W(A)$ , because  $[m, s]_{W \upharpoonright A} = \{m, s\}$ .

A *nested representative history* uses  $A$  itself as the carrier at each stage, rather than replacing it by an unspecified isomorphic copy. At successor stages apply  $D_W$ , and at nonzero limit stages



intersect all earlier retained subclasses. The limit convention is essential: abstract order isomorphism types alone do not retain the information needed to form these intersections.

**Theorem 108.1** (Termination by the input order). *Work in GB. Suppose a single class relation uniformly codes a nested representative history  $(H_i)_{i \in (X, W) + 1}$ , with*

$$H_m = X,$$

*successor stages given by  $D_W$ , and nonzero limit stages given by intersection. Here one fresh final stage  $*$  has been appended to the index order  $(X, W)$ . Then*

$$H_* = \{m\}.$$

*More precisely, for every  $x \in X$ ,*

$$H_x \subseteq \{m\} \cup \{y \in X : x \leq_W y\}. \quad (\text{S20.1})$$

*For arbitrary  $W$ , elementary transfinite recursion along  $(X, W) + 1$  is a sufficient hypothesis for the required uniform history to exist. No necessity or reverse-mathematical equivalence is asserted.*

*Proof.* The recursion rules imply that the retained classes are decreasing and all contain  $m$ . Transfinite induction along  $W$  is available in GB for formulas with the supplied history as a class parameter: a nonempty class of counterexamples would have a least member. We prove (S20.1) by that induction.

At the first stage  $x = m$ , the assertion is immediate. Suppose  $x$  is the immediate successor of  $t$ . By induction, no member of  $H_t$  lies strictly below  $t$  except possibly  $m$ . If  $t \neq m$  and  $t \in H_t$ , it is therefore the immediate successor of  $m$  in the induced order on  $H_t$ , and is deleted by  $D_W$ . All earlier members other than  $m$  have already disappeared. If  $t = m$ , there are no earlier nonminimal members to delete. In either case  $H_x = D_W(H_t)$  contains no  $y <_W x$  except  $m$ , as required.

If  $x$  is a nonzero limit point of the index order, take  $m \neq y <_W x$ . There is  $t$  with  $y <_W t <_W x$ ; the inductive bound excludes  $y$  from  $H_t$ , hence from  $H_x = \bigcap_{t <_W x} H_t$ . This proves the bound at limit stages.

If  $W$  has a greatest element  $\ell$ , the bound at  $\ell$  gives  $H_\ell \subseteq \{m, \ell\}$ . Its condensation is  $\{m\}$ , which is the appended successor stage. If  $W$  has no greatest element, every  $y \neq m$  has some index  $x >_W y$ , and the bound at that stage excludes it from the final intersection. The final stage is again  $\{m\}$ .

Finally, the rule for  $H_i$  is elementary in the earlier part of the history: one tests whether the index is first, a successor, or a limit, and then applies the corresponding displayed rule. The principle of elementary transfinite recursion along  $(X, W) + 1$  therefore supplies a class coding the history. This uses the standard meaning of ETR, whose stages may themselves be classes; its distinction from ordinary set-valued recursion is discussed in the primary literature on class recursion [24, 22]. The termination proof itself needed only GB and the already supplied history.  $\square$

The theorem does not contradict an isomorphism  $W \cong D_W(X)$ . The retained subclasses decrease even when their abstract order types repeat. In particular, no nontrivial retained class is literally a fixed point of  $D_W$ : its second member is always deleted at the next stage. A purported nonterminating history that keeps replacing a quotient by an arbitrary isomorphic copy has changed the specified limit convention.

### 108.1.2 The exact ordinal calculation

An external set theorem gives a completely explicit calculation. Let  $\kappa$  be any infinite initial cardinal and let  $\theta > 0$  be an ordinal. On a suborder of  $\theta$ , condense intervals of cardinality less than  $\kappa$ , retain least representatives, and intersect retained carriers at limits. Write  $H_\alpha(\theta)$  for the resulting ordinary transfinite history. Exponentiation in the next theorem is *ordinal exponentiation*.

**Theorem 108.2** (Multiples of ordinal powers). *For every ordinal  $\alpha$ ,*

$$H_\alpha(\theta) = \{\kappa^\alpha \cdot \beta : \kappa^\alpha \cdot \beta < \theta\}. \quad (\text{S20.2})$$

*Consequently the first singleton stage has exact height*

$$h_\kappa(\theta) = \min\{\alpha : \theta \leq \kappa^\alpha\}. \quad (\text{S20.3})$$

*Proof.* First observe that, on an ordinal  $\gamma$ , condensation by intervals of cardinality less than  $\kappa$  keeps exactly the points  $\kappa \cdot \delta < \gamma$ . Indeed, ordinary left division by  $\kappa$  writes each point uniquely as  $\kappa \cdot \delta + \rho$ , with  $\rho < \kappa$ . Within one  $\kappa$ -segment every closed interval has size less than  $\kappa$ . An interval meeting two different segments contains a final segment of the earlier  $\kappa$ -segment and therefore has size at least  $\kappa$ . This argument does not require regularity of  $\kappa$ .

At  $\alpha = 0$ , equation (S20.2) simply says that the retained carrier is all of  $\theta$ . If it holds at  $\alpha$ , the map  $\beta \mapsto \kappa^\alpha \cdot \beta$  order-isomorphically identifies an initial ordinal segment with  $H_\alpha(\theta)$ . Applying the preceding division observation to this induced order keeps precisely the indices  $\beta = \kappa \cdot \delta$ . Their images are

$$\kappa^\alpha \cdot (\kappa \cdot \delta) = \kappa^{\alpha+1} \cdot \delta,$$

proving the successor step.

Let  $\lambda$  be a nonzero limit ordinal. Every multiple of  $\kappa^\lambda$  is a multiple of  $\kappa^\alpha$  for  $\alpha < \lambda$ : write  $\lambda = \alpha + \delta$  and use  $\kappa^\lambda = \kappa^\alpha \cdot \kappa^\delta$ . For the converse, suppose  $x$  is a multiple of every  $\kappa^\alpha$  for  $\alpha < \lambda$ . Divide it by  $\kappa^\lambda$ :

$$x = \kappa^\lambda \cdot q + r, \quad r < \kappa^\lambda.$$

Continuity of ordinal exponentiation gives an  $\alpha < \lambda$  with  $r < \kappa^\alpha$ . The term  $\kappa^\lambda \cdot q$  is itself a multiple of  $\kappa^\alpha$ . Thus the displayed expression is the left division of  $x$  by  $\kappa^\alpha$ , with remainder  $r$ . Since  $x$  was a multiple of that power, uniqueness of division implies  $r = 0$ . This proves that the intersection of all earlier retained classes consists exactly of the multiples of  $\kappa^\lambda$ , establishing the limit step.

The right side of (S20.2) always contains 0 and contains another point exactly when  $\kappa^\alpha < \theta$ . This proves (S20.3). The set in that minimum is nonempty, since  $\alpha \mapsto \kappa^\alpha$  is increasing and  $\kappa^\theta \geq \theta$ .  $\square$

For example,

$$h_\kappa(\kappa) = 1, \quad h_\kappa(\kappa + 1) = 2, \quad h_\kappa(\kappa^\omega) = \omega, \quad h_\kappa(\kappa^\omega + 1) = \omega + 1.$$

The last pair makes the role of an appended terminal point especially clear: the class-limit intersection may retain that point as well as the minimum, requiring one more condensation.

**Corollary 108.3** (Exact heights in a full-class model). *Let  $\kappa$  be strongly inaccessible in an ambient ZFC universe, and use the full two-sorted model  $(V_\kappa, \mathcal{P}(V_\kappa))$ . If an internal class well-order of its surreal carrier has ambient ordinal type  $\theta$ , then  $|\theta| = \kappa$ , and its canonical nested condensation history has ambient height  $h_\kappa(\theta)$  from (S20.3). The possible heights are exactly*

$$0 < \alpha < \kappa^+.$$

*There is no single internal class-well-order schedule that bounds the heights of all such input orders. The bound by the input order in theorem 108.1 can be attained.*

*Proof.* In this full model, a subclass of  $V_\kappa$  is an internal set exactly when its ambient cardinality is less than  $\kappa$ . Inaccessibility bounds the cardinality of every  $V_\beta$  for  $\beta < \kappa$ ; conversely, regularity bounds the ranks of fewer than  $\kappa$  many elements of  $V_\kappa$ . This is the standard full-model size check used in the earlier continuation (source 18; [section 57.5](#)). All ambient subsets are available as classes, so internal class well-orders are actual well-orders externally. The surreal carrier has ambient cardinality  $\kappa$ . Its order type therefore satisfies  $\kappa \leq \theta < \kappa^+$ , and the condensation is exactly the one computed in [theorem 108.2](#).

The bound  $h_\kappa(\theta) \leq \theta < \kappa^+$  gives one inclusion for the height spectrum. Conversely, if  $0 < \alpha < \kappa^+$ , the ordinal  $\theta = \kappa^\alpha$  has cardinality  $\kappa$  and has height exactly  $\alpha$ , because ordinal exponentiation with base  $\kappa > 1$  is strictly increasing. For completeness, the cardinality assertion follows by induction: ordinal multiplication at a successor stage preserves cardinality  $\kappa$ , and at a limit below  $\kappa^+$  the union has at most  $\kappa$  terms, each of size at most  $\kappa$ . A bijection transfers this ordinal order to the surreal carrier, giving an available internal class relation.

Any internal class well-order schedule has an ambient order type less than  $\kappa^+$ . The displayed spectrum contains a larger height, proving the absence of a universal schedule in this model. For attainment of the input-order bound, take an ordinal fixed point

$$\theta = \kappa^\theta$$

of cardinality  $\kappa$ . One exists by starting with  $\theta_0 = \kappa$ , setting  $\theta_{n+1} = \kappa^{\theta_n}$ , and taking  $\theta = \sup_{n < \omega} \theta_n$ . Each stage and the supremum have cardinality  $\kappa$ , and continuity gives the fixed point. Strict increase of exponentiation shows  $\kappa^\alpha < \theta$  for every  $\alpha < \theta$ , so  $h_\kappa(\theta) = \theta$ . Such an input retains more than one point at every earlier stage and first becomes a singleton at its appended final stage.  $\square$

The last corollary is an external classification in a specified full-class model. It is not a claim that a GB universe contains the ordinal  $\kappa$  naming its own height, nor that it can collect all class well-orders and their histories into one class-valued object. The remaining foundational question is sharp: which fragment, if any, of elementary transfinite recursion is necessary for every supplied class well-order to admit the nested history? The termination theorem answers what happens to a history once it exists; it does not settle that existence principle over GB or GBC.

*Note 108.4* ([merge]; the existence question is answered over GBC). Source 29 answers this question, which source 20 also asks as its [research question 113.111](#), over GBC: the principle CTH that every class well-order admits its history on the schedule  $W + 1$  is equivalent to strong comparability CWO of class well-orders ([theorem 108.19](#)). Since ETR implies CWO (Hamkins–Woodin), ETR remains sufficient; whether it is necessary is not settled: source 29 leaves Hamkins and Woodin’s Question 7, on converses among these principles, open. Source 29 does not treat the question over GB without global choice. Source 20’s [theorem 108.1](#) is source 29’s [note 108.11](#), and its height formula is recovered there from finite normal forms (“A concrete full-model check in [section 108.3](#)).

[merge] (4 October 2026, batch 96.) Answered over GB as well. Sources 34 and 35 (Part XVI) prove independently that CTH, CWO and the two polynomial forms are equivalent over GB with ZF sets and no form of choice ([theorem XVI.5.8](#) and [note XVI.5.19](#)); source 29’s only use of global choice, the well-foundedness of  $P(\Gamma)$  ([note 108.15](#)), is replaced by a choice-free proof ([theorem XVI.4.2](#) and [note XVI.4.4](#); [note XVI.5.10](#)). Whether ETR is necessary remains open.

*Note 108.5* ([merge]; the polynomial orders in four sources). Source 15’s  $P(J)$  ([definition 53.1](#)), source 18’s  $(\mathcal{P}, \triangleleft)$  for  $J = \omega$  ([proposition 52.15](#)), source 23’s  $P(\Gamma)$  and source 29’s  $\mathcal{P}(\Gamma)$  are one class: finite partial functions from the exponent order to the nonzero ordinals, compared at the greatest exponent of disagreement. They are printed  $P(\Gamma)$  here. Source 23 compares them by ([S23.4](#)); source 29 uses them as the codomain of its finite normal forms.

## 108.2 Source 23: ordinal-polynomial models of every prescribed hierarchy

Let  $\Gamma = (G, <_\Gamma)$  be a given set or class well-order. A member of  $P(\Gamma)$  is a finite partial function from  $G$  to the nonzero ordinals. Evaluation outside its finite domain is defined to be zero. Thus we may conveniently write  $p : G \rightarrow \text{Ord}$  with finite support, but *the object  $p$  is its finite graph*. For  $p \neq q$ , the finite union of their supports has a greatest point of disagreement  $\delta$ , and we define

$$p < q \iff p(\delta) < q(\delta). \quad (\text{S23.4})$$

This is reverse lexicography on finite supports, not on arbitrary infinite-support words.

*Note 108.6* (Source 23, Lemma 11.1, “Polynomial order is a class well-order”). *In GBC,  $P(\Gamma)$  with (S23.4) is a class well-order. It is proper whenever  $G$  is nonempty.* This is source 15’s lemma 53.2 (Part IX), and source 29’s note 108.15; the three proofs map a descending sequence to ordinals by a base larger than all coefficients (source 15 uses an additively indecomposable base, source 23 an infinite base, source 29 the order type  $\delta^\alpha$ ).

*Source 23’s proof.* The greatest-disagreement comparison is a linear order: in a comparison of three finite graphs, work in the finite union of their supports, where ordinary finite lexicography applies.

Suppose a nonempty admitted subclass had no least point. Global choice would recursively select a strictly descending sequence  $p_0 > p_1 > \dots$ . The sequence is a set, by set recursion and Class Replacement. The union of its supports is a set well-ordered by  $\Gamma$ , say of order type  $\alpha$ . Its coefficients form a set of ordinals, so choose an infinite ordinal base  $b$  larger than all of them. Reindex the support set by  $\alpha$ . The map

$$p_n \longmapsto \sum_{\xi \in \text{supp}(p_n)}^{\text{decreasing } \xi} b^\xi p_n(\xi)$$

is strictly order-preserving into the ordinals. Indeed, at the highest differing exponent  $\delta$ , every finite sum of lower terms is less than  $b^\delta$ ; this follows successively from coefficients being below  $b$  and the laws of ordinal exponentiation. A larger coefficient at  $\delta$  therefore dominates all lower terms. We obtain a descending sequence of ordinals, a contradiction.

For any fixed  $g \in G$ , varying the coefficient at  $g$  embeds  $\text{Ord}$ . Thus the carrier is proper when  $G$  is nonempty.  $\square$

*Note 108.7* (Source 23, Lemma 11.2, “The first block removes the least coordinate”). *Suppose  $G$  is nonempty with least point  $g_0$ . Two polynomials have a set interval between them if and only if they agree at every coordinate strictly above  $g_0$ . Consequently*

$$D(P(\Gamma)) \cong P(\Gamma \setminus \{g_0\}),$$

where the order on the remaining coordinates is inherited. This is source 15’s lemma 53.3 (Part IX), with the same proof.

*Source 23’s proof.* If they agree above  $g_0$ , all points between them agree there too. Only the coefficient at  $g_0$  varies between two fixed ordinals, giving a set interval.

Otherwise let  $\delta > g_0$  be their greatest disagreement, with  $p < q$ . Keep every coordinate of  $p$  above  $g_0$  fixed, and vary its coefficient at  $g_0$  through all ordinals greater than or equal to its original value. Every resulting polynomial lies between  $p$  and  $q$ , since the higher disagreement at  $\delta$  still places it below  $q$ . These give a proper class of points in the interval.

The least representative of a block has coefficient zero at  $g_0$ . Deleting that coordinate identifies the representative order with the stated polynomial order.  $\square$

Adjoin a formally tagged new greatest stage  $\top$  to  $\Gamma$  (using a disjoint tagged copy of its old carrier). For  $t \in G$ , let  $I_t = \{\gamma : \gamma <_\Gamma t\}$  and put  $I_\top = G$ . Define

$$pE_tq \iff \forall \gamma \in G \setminus I_t (p(\gamma) = q(\gamma)). \quad (\text{S23.5})$$

The quantifier ranges over set objects  $\gamma$ ;  $p, q$  are finite graphs. This is one elementary definition with parameter  $\Gamma$ , and defines a class family on the set-coded stages of  $\Gamma + 1$ .

*Note 108.8* (Source 23, Theorem 11.3, “Prescribed-hierarchy theorem”). *Let  $\Gamma$  be any given class well-order, including a set well-order. The family (S23.5) is its polynomial order’s set-interval condensation hierarchy. More precisely:*

- (i) *At the first stage,  $E_t$  is equality. Each  $E_t$  is a convex equivalence relation.*
- (ii) *Its represented quotient is  $P(\Gamma \setminus I_t)$ .*
- (iii) *If  $t$  has immediate predecessor  $s$  in the stage order, then  $E_t = D(E_s)$ .*
- (iv) *At every nonzero limit stage  $t$ , including a limit final stage,  $E_t = \bigcup_{s < t} E_s$ .*
- (v)  *$E_\top$  is universal, and every  $E_t$  for  $t \in G$  is nonuniversal. In particular, collapse occurs exactly at the final stage.*

*These particular hierarchies are available in GBC without ETR.* This is source 15’s [theorem 53.4](#) (Part IX), stated for the equivalence relations  $E_t$  with unions at limits instead of nested classes of representatives with intersections; the two conventions agree by source 29’s [corollary 108.14](#). Source 23’s form adds the explicit quotient  $P(\Gamma \setminus I_t)$  at every stage and allows  $\Gamma$  to be a set well-order. Source 29’s [note 108.15](#) is a third copy.

*Source 23’s proof.* For each initial segment  $I_t$ , equivalence means equality on the remaining tail. This is a convex equivalence, and its least representative is obtained by deleting all coefficients in  $I_t$ . Its quotient therefore has exactly the stated polynomial order.

At a successor stage the remaining tail loses its least coordinate. [Note 108.7](#) proves precisely the successor clause, including when  $I_s$  is a proper class. At the first stage no coordinate has been removed, so equality is obtained.

At a limit stage, if  $pE_tq$ , their finite set of differing coordinates is contained in  $I_t$ . A finite subset of this initial segment is already contained in some  $I_s$  with  $s < t$ : take a stage strictly above its greatest member, which exists because  $t$  is not a successor. Thus  $pE_sq$ . The reverse inclusion is immediate. If there are no differing coordinates, any earlier stage suffices. The same proof applies when the predecessor segment of  $t$  is proper.

At the final stage all coordinates are deleted. Before that stage, the zero polynomial and the polynomial with coefficient one at coordinate  $t$  remain inequivalent. No recursion theorem was used: the displayed formula supplies the family and the preceding arguments verify its recursive specification.  $\square$

### 108.2.1 Every set ordinal, and beyond

For  $\Gamma = \alpha$  a set ordinal, we may say that the collapse rank is exactly  $\alpha$ . Every positive set ordinal is therefore realized by a proper-class well-order. For finite  $n$ , the order is the usual finite lexicographic ordinal-class power, with the highest coordinate compared first. Its successive condensations delete one least coordinate at a time.

For  $\Gamma = \omega$ , the quotient at every finite stage  $n$  is isomorphic to  $P(\omega)$  because  $[n, \omega)$  has order type  $\omega$ . Nevertheless the union of the finite-stage equivalences is universal: two finite graphs have bounded support. This gives a particularly transparent warning against declaring a hierarchy stable

merely because its successive quotient order types are isomorphic. ([merge] Source 15’s [remark 53.5](#) and source 18’s [proposition 52.15](#) make the same point with the same order; [note 53.6](#).)

For  $\Gamma = \text{Ord}$ , the tail after any set ordinal stage is again order-isomorphic to  $\text{Ord}$ . Consequently every set-stage quotient is isomorphic to  $P(\text{Ord})$ , and no set stage collapses it. The final stage of  $\text{Ord} + 1$  does collapse it, by the same finite-support argument. The final stage has a proper class of predecessors, but its relation is explicitly defined, not obtained by an unlicensed recursion.

Taking, for example,  $\Gamma = \text{Ord} + \text{Ord}$  continues the hierarchy beyond that first class-length block. The theorem is always relativized to a *given* admitted  $\Gamma$ . It does not assert a same-level class containing the hierarchies of all class well-orders.

### 108.2.2 Transport to surreal labels

Under global choice, every proper ordinary class admits a set-like  $\text{Ord}$ -enumeration. Hence for nonempty  $\Gamma$  there is a class bijection between  $P(\Gamma)$  and  $\mathbf{No}$ , and its well-order may be transported to the surreal carrier. The transported ordering is generally unrelated to the numerical ordering of the labels.

Thus the theorem supplies actual raw well-orders of *all surreal numbers*, not just abstract orders on an unrelated large carrier. The choice of transport affects the outer lexicographic position but not the condensation hierarchy of the transported relation.

### 108.3 Source 29: the exact strength of complete condensation histories

The elementary comparison of two labelled class well-orders does not require an isomorphism between their unlabelled order types. The repository’s common-prefix construction makes this distinction explicit ([theorem 14.2](#)). Iterating the canonical set-interval decomposition of one class well-order is a different problem. The preceding continuation, source 20, proved termination conditional on a supplied uniform history, gave elementary transfinite recursion as a sufficient existence principle, and asked for the exact strength of that existence assertion ([theorem 108.1](#), [research question 113.111](#)).

We now identify it with *strong class-well-order comparability*. The central construction assigns a finite ordinal-coefficient address to every point of a well-order with a supplied terminating history. These addresses form an initial segment of a class of formal polynomials. A common history for two convex suborders therefore gives their initial-segment comparison. Conversely, comparability places any class well-order inside a polynomial order with an explicit condensation history.

All results in this section are over **GBC**, unless a weaker setting is explicitly stated. No collection of classes is treated as an ordinary class of members. A history, an embedding, or a family of maps is always one supplied class relation. Formal sums and appended endpoints use tagged copies of their carriers; an arbitrary class carrier might be the entire universe, so a “new endpoint” must not mean an untagged set outside that carrier.

#### 108.3.1 The operator, the schedule, and the existing termination bound

Let  $W = (X, <_W)$  be a nonempty class well-order and put  $m = \min_W X$ . Here “class well-order” means that every nonempty admitted subclass has a least element. Given a nonempty subclass  $A \subseteq X$ , define

$$x \sim_A y \iff \{z \in A : \min_W(x, y) \leq_W z \leq_W \max_W(x, y)\} \text{ is a set,} \quad x, y \in A. \quad (\text{S29.21})$$

Let  $D_W(A)$  be the subclass of least representatives of these equivalence classes. The assertion that a displayed interval is a set is expressed by the existence of a set with exactly its members. Thus both  $\sim_A$  and  $D_W(A)$  have elementary definitions from the supplied class parameters.



*Note 108.9* (Source 29, Lemma 11.1, “One-step block facts”). *The relation  $\sim_A$  is a convex equivalence relation. Each of its blocks has a least element and, in its induced order, is either a set ordinal or a copy of Ord. A set-sized block can only be final. In particular, a block followed by another least representative has type Ord.* This is source 15’s [theorem 52.3](#) (Part IX) for the induced order on  $A$ , and source 23’s [note 106.3](#), with the same proof.

*Source 29’s proof.* Reflexivity and symmetry are immediate. An interval between two points is contained in the union of the intervals from those points to a third point, which proves transitivity. Taking subintervals proves convexity. Each block is an admitted nonempty subclass and therefore has a least point  $a$ . For a point  $x$  in that block, its predecessors within the block are exactly the set interval  $[a, x)_W \cap A$ . Their ordinal order types give a canonical rank map whose range is an initial subclass of Ord.

If that range is bounded by a set ordinal, Class Replacement applied to the inverse rank map makes the block a set. Otherwise the range is all of Ord. This proves the order-type assertion without assigning a set ordinal to a proper class. Finally, if a set block were not final, let  $y$  be the least point after it. Adjoining  $y$  to the block would give a set interval from its minimum, contradicting the maximality of the equivalence block.  $\square$

This is the canonical set-interval decomposition already used in the project ([theorem 52.3](#); source 20’s [section 108.1](#)). Its simultaneous definition requires no history of class-valued states.

Let  $\Gamma = (I, <_\Gamma)$  be a nonempty class well-order, and append a final stage  $*$ . Every  $i \in I$  has an immediate successor in  $\Gamma + 1$ , denoted  $i^+$ . A nonzero limit stage means a stage other than the first which has no immediate predecessor; its predecessor class need not be a set.

**Definition 108.10** (Canonical history). A canonical condensation history for  $W$ , with schedule  $\Gamma + 1$ , is one class

$$H \subseteq (I \cup \{*\}) \times X$$

whose slices satisfy

$$\begin{aligned} H_{\min \Gamma} &= X, \\ H_{i^+} &= D_W(H_i) \quad (i \in I), \\ H_\lambda &= \bigcap_{i <_{\Gamma+1} \lambda} H_i \quad \text{at every nonzero limit stage } \lambda. \end{aligned}$$

The final stage is included in the successor or limit clause as appropriate. The history is *terminating* if  $H_* = \{m\}$ . The principle CTH asserts that every nonempty class well-order  $W$  has such a history with the specific schedule  $W + 1$ .

The slices are decreasing and all contain  $m$ , by elementary transfinite induction with  $H$  as a class parameter. The same kind of induction proves the following bound from the preceding continuation. We include the argument because the distinction between existence and termination is central to the equivalence.

*Note 108.11* (Source 29, Proposition 11.3, “Termination for the input-order schedule; earlier result”). *In GB, a supplied canonical history on the schedule  $W + 1$  satisfies*

$$H_x \subseteq \{m\} \cup \{y \in X : x \leq_W y\} \quad (x \in X), \tag{S29.22}$$

*and its final slice is  $\{m\}$ .* This is source 20’s [theorem 108.1](#), which source 29 credits; the proof is the same induction.

*Source 29's proof.* Induct along the supplied class well-order  $W$ . The first-stage bound is immediate. If  $x$  is the successor of  $t$ , the inductive bound has already excluded all points below  $t$  except  $m$ . If  $t \neq m$  remains in  $H_t$ , it is the immediate successor of  $m$  there. Its interval from  $m$  has two points, so  $D_W$  deletes  $t$ . This proves the bound at  $x$ ; the case  $t = m$  is immediate.

At a nonzero limit  $x$ , each  $m \neq y <_W x$  lies below some  $t <_W x$ . The inductive bound excludes  $y$  from  $H_t$  and hence from the intersection at  $x$ . If  $W$  has a greatest element  $\ell$ , the bound leaves at most  $\{m, \ell\}$  at stage  $\ell$ , and the appended successor condenses it to  $\{m\}$ . If  $W$  has no greatest element, every  $y \neq m$  is excluded at some later stage, so the final intersection is again  $\{m\}$ .  $\square$

An appropriate instance of ETR supplies the history because the displayed successor and limit rules are elementary in the earlier history. This is class-valued recursion, not merely a recursion whose values and earlier histories are sets. The standard distinction is developed in the class-recursion literature [22, 27]. The remaining arguments first work with an already supplied history.

### 108.3.2 Greatest retained predecessors and finite changes

The essential new observation concerns projections onto retained carriers. A decreasing family of arbitrary subclasses need not have the property below; the successor rule forces it.

**Lemma 108.12** (Canonical floor maps). *For every stage of a supplied canonical history and every  $x \in X$ , the greatest retained predecessor exists:*

$$p_i(x) = \max_W \{a \in H_i : a \leq_W x\}. \quad (\text{S29.23})$$

*These maps are uniformly elementary in  $H, W$ . For  $i \leq j$ ,*

$$p_j(x) \leq_W p_i(x), \quad p_j(p_i(x)) = p_j(x), \quad p_i(p_j(x)) = p_j(x). \quad (\text{S29.24})$$

*Each  $p_i$  is nondecreasing as a function of  $x$ . At every nonzero limit stage  $\lambda$ , the earlier floors of a fixed point stabilize:*

$$\exists i_0 < \lambda \forall i (i_0 \leq i < \lambda \implies p_i(x) = p_\lambda(x)). \quad (\text{S29.25})$$

*Proof.* Prove existence by elementary transfinite induction. At the first stage the floor of  $x$  is  $x$ . Given  $p_i(x) = a$ , let  $b$  be the least representative of the  $\sim_{H_i}$ -block containing  $a$ . A representative between  $b$  and  $x$  would have to lie between  $b$  and  $a$ , by the defining property of  $a$ . It would therefore lie in the same block as  $b$ , contrary to least-representative uniqueness. Thus  $p_{i+}(x) = b$ .

At a nonzero limit  $\lambda$ , the earlier floors are nonincreasing because the retained carriers are nested. The class of their values is nonempty and has a  $W$ -least value  $b$ , attained at some stage  $i_0 < \lambda$ . Every later floor below  $\lambda$  equals  $b$ . Also  $b \in H_{i_0}$ , so nesting places it in every earlier slice. It follows that  $b$  belongs to the limit intersection. Any point of that intersection below  $x$  is at most  $p_{i_0}(x) = b$ . This proves both existence and (S29.25).

The induction assertion uses only set quantifiers and the supplied classes. In particular, the class of earlier floor values is defined directly by the greatest-predecessor formula; no family of newly chosen class maps is assumed. Once existence is proved, the same formula defines all  $p_i$  in a single class relation.

Monotonicity in  $x$  is immediate. If  $i \leq j$ , every  $H_j$ -point at most  $x$  is an  $H_i$ -point and is at most  $p_i(x)$ . This proves the first composition identity. The second follows because  $p_j(x)$  itself lies in  $H_i$ .  $\square$

**Proposition 108.13** (Each point changes its floor only finitely often). *For every fixed  $x$ , only finitely many stages  $i \in I$  satisfy  $p_{i+}(x) \neq p_i(x)$ . No new floor value first appears at a limit stage.*

*Proof.* The limit assertion is (S29.25). If the class of successor changes were not finite, repeatedly taking its least next member would give increasing stages  $i_0 < i_1 < \dots$  with

$$p_{i_{n+1}}(x) \leq_W p_{i_n^+}(x) <_W p_{i_n}(x).$$

These are uniquely chosen set-valued steps. Ordinary recursion along  $\omega$ , with the supplied classes as parameters, and Class Replacement produce a set-coded descending sequence in  $W$ , a contradiction.  $\square$

The finiteness is pointwise, not a finite bound on the length of the whole history. It is also internal finiteness when the theorem is interpreted inside a nonstandard model of class theory.

There is a precise quotient interpretation of the limit convention. Put  $x E_i y$  when  $p_i(x) = p_i(y)$ . The maps  $p_i$  are monotone surjections onto  $H_i$ , so  $E_i$  is a convex equivalence relation.

**Corollary 108.14** (Continuous identification at limits). *At a nonzero limit stage,*

$$E_\lambda = \bigcup_{i < \lambda} E_i.$$

*Thus no pair of original points is first identified at a limit.*

*Proof.* An earlier identification persists by (S29.24). Conversely, if the two floors agree at  $\lambda$ , choose an earlier stage after both stabilization stages in (S29.25). Their floors already agree there.  $\square$

Consequently the limit quotient is the directed quotient limit of the earlier stages. This can be stated without collecting a category of all class orders: for any supplied class order  $Y$  and any uniform compatible family of monotone maps  $f_i : H_i \rightarrow Y$ , there is a unique map from  $H_\lambda$  through which they factor. Define it by restricting the first  $f_i$  to  $H_\lambda$ . Equation (S29.25) and compatibility verify the factorization; surjectivity of each projection gives uniqueness. This answers one natural version of the earlier question about alternative limit conventions (source 20's [research question 113.112](#)): retaining the canonical quotient maps gives exactly the nested-representative limit.

### 108.3.3 A class of formal ordinal-coefficient polynomials

For a class well-order  $\Gamma = (I, <_\Gamma)$ , let  $P(\Gamma)$  be the ordinary class of finite partial functions from  $I$  to nonzero set ordinals. Missing coefficients are read as zero. Order these functions by the coefficient at the *greatest* exponent where they differ. This is a finite-support comparison, not a claim that Ord is a set-valued base for an ordinary exponentiation operation. The empty function, denoted 0, is the least polynomial.

*Note 108.15* (Source 29, Lemma 11.7, “Polynomial well-orders and their explicit histories”). *The order  $P(\Gamma)$  is a class well-order in GBC. It has an elementary canonical terminating history on the schedule  $\Gamma + 1$ , namely*

$$K_i = \{q \in P(\Gamma) : \text{supp}(q) \subseteq \{j : i \leq_\Gamma j\}\}, \quad K_* = \{0\}. \quad (\text{S29.26})$$

This combines source 15's [lemma 53.2](#) and [theorem 53.4](#) (Part IX), and source 23's [notes 108.6](#) and [108.8](#); source 29's well-foundedness proof computes the set order type  $\delta^\alpha$  instead of mapping into an ordinal base.

*Source 29's proof.* If the polynomial order had a nonempty subclass without a least element, Global Choice would produce a descending  $\omega$ -sequence. The union of its finite exponent supports is a set  $A$ , whose induced  $\Gamma$ -order has a set ordinal type  $\alpha$ . Class Replacement also bounds all coefficients occurring in that sequence by a single set ordinal  $\delta \geq 2$ , chosen strictly above every coefficient.

The sequence would therefore lie in the set order of finite-support  $\delta$ -valued functions on  $\alpha$ , with the same greatest-difference comparison. This is a well-order of ordinary ordinal type  $\delta^\alpha$ : the successor step is an ordinal sum of  $\delta$  copies of the preceding order, and at a limit every finite support is bounded in an earlier exponent segment. Ordinary set transfinite induction proves the assertion. The descending sequence is impossible, so the class polynomial order is well-founded.

At stage  $i$ , the  $K_i$ -polynomials with the same coefficients strictly above  $i$  form exactly one small-interval block, varying their  $i$ -coefficient through Ord. Within such a block every bounded coefficient interval is a set. If two higher parts differ, arbitrarily increasing the  $i$ -coefficient of the smaller polynomial stays below the larger polynomial. Their interval therefore contains a proper class. The least representative of a block has coefficient zero at  $i$ , proving the successor clause in (S29.26). At a nonzero limit, retaining only exponents above all earlier thresholds is exactly the intersection rule. The final threshold leaves only the zero polynomial.  $\square$

All polynomial objects are sets: a polynomial has a finite domain of set-valued stage labels and finitely many set ordinal coefficients. The possibly proper-class size is in the collection of available codes and in its ordering, not in an individual code.

### 108.3.4 Finite normal forms from a supplied history

For a history with schedule  $\Gamma + 1$ , define

$$c_i(x) = \text{otp}(\{z \in H_i : p_{i+}(x) \leq_W z <_W p_i(x)\}, <_W). \quad (\text{S29.27})$$

The displayed interval is a set. Indeed, the successor construction in lemma 108.12 identifies  $p_{i+}(x)$  as the least representative of the small-interval block containing  $p_i(x)$ . The digit is zero exactly when these two floors coincide. By proposition 108.13,

$$\text{NF}_H(x) = \{(i, c_i(x)) : i \in I, c_i(x) > 0\} \in P(\Gamma). \quad (\text{S29.28})$$

This finite graph is a set, and its dependence on  $x$  is given by one elementary class formula.

**Theorem 108.16** (Canonical finite normal form). *For a supplied terminating history  $H$ , the map  $\text{NF}_H$  is a class isomorphism of  $W$  onto an initial segment of  $P(\Gamma)$ .*

*Proof.* We first record precisely how to insert a digit. Suppose  $b \in H_{i+}$ , and let  $z \in H_i$  be the point of ordinal rank  $\rho$  in the small-interval block beginning at  $b$ , whenever that point exists. Then

$$p_i(z) = z, \quad p_{i+}(z) = b.$$

For every later stage, the floors of  $z$  agree with the floors of  $b$ , by (S29.24). At every earlier stage  $j < i$ , both relevant floors of  $z$  equal  $z$ , because  $z \in H_i \subseteq H_{j+}$ . It follows that  $\text{NF}_H(z)$  has the same digits as  $\text{NF}_H(b)$  above  $i$ , has digit  $\rho$  at  $i$ , and has zero digits below  $i$ . The digits of  $b$  at and below  $i$  are themselves zero. This is an exact one-digit insertion rule.

Now fix an actual normal form. At the last stage the floor of its point is  $m$ . Read its finitely many nonzero digits in decreasing stage order, starting from that anchor. At each step the one-digit rule says to take the unique point of the indicated ordinal rank in the indicated block. For an actual code, this point exists by (S29.27). Limit stages introduce no additional changes, by (S29.25). The

successive anchors therefore reconstruct the original point. Uniqueness at each rank proves that  $\text{NF}_H$  is injective. These are finitely many uniquely specified choices of sets, not a choice of class isomorphisms.

It remains to prove order preservation and, crucially, that the image is an initial segment. Let  $q < \text{NF}_H(x)$  be any polynomial, and let  $i$  be their greatest exponent of disagreement. Their higher digits are identical. Reading those common higher digits reconstructs the same anchor

$$b = p_{i+}(x).$$

Write  $d = c_i(x)$  and  $c = q(i)$ , with a missing digit of  $q$  read as zero. The comparison says  $c < d$ . In the  $H_i$ -block beginning at  $b$ , the point  $p_i(x)$  has rank  $d$ . Hence there are points  $a, a' \in H_i$  of ranks  $c, c+1$ , respectively, with

$$a <_W a' \leq_W p_i(x) \leq_W x. \quad (\text{S29.29})$$

Taking  $a$  inserts the digit  $c$ , leaving all smaller digits zero.

Process the remaining nonzero digits of  $q$  in decreasing order. For a lower exponent  $j < i$ , both the current anchor and the fixed cap  $a'$  belong to  $H_{j+}$ . Initially this follows from  $a, a' \in H_i$ , and it remains true after each insertion because the exponents decrease. The cap is a later least representative. Therefore the  $H_j$ -block beginning at the current anchor is nonfinal and has type Ord, by [note 108.9](#). Every requested set ordinal digit is available inside that block, and its point is still strictly below  $a'$ . The one-digit insertion rule proves that the step changes exactly the requested digit and preserves the previously inserted higher digits. After the finitely many steps, the resulting point  $y$  satisfies

$$\text{NF}_H(y) = q, \quad y <_W a' \leq_W x.$$

Thus every polynomial below an actual code is itself an actual code, and its decoded point is smaller. When  $q$  is the normal form of another given point, injectivity identifies that point with the constructed  $y$ . The two linear orders therefore have exactly the same comparisons. The same argument for arbitrary  $q$  proves initiality of the image.  $\square$

The initiality conclusion is stronger than an order embedding. It is the step that will give *strong* comparability, rather than merely an embedding as an arbitrary suborder. The cap in [\(S29.29\)](#) guarantees that every finite collection of lower digits can be filled without leaving the input order.

The theorem also explains how a history can have very large height while every point has a finite address. The exponents themselves are labels in a supplied class well-order, and may have proper classes of predecessors. They have not been replaced by set ordinals. Only the coefficients and the finite address are sets.

### 108.3.5 Restricting a history to a convex suborder

To compare two input orders, we need their histories on the same schedule. The following restriction rule supplies them from a single history of their sum.

**Lemma 108.17** (Anchored convex restriction). *Let  $Y$  be a nonempty convex subclass of  $W$ , and put  $a = \min_W Y$ . From a canonical history  $H$  on  $W$  define, on the same schedule,*

$$J_i = \{a\} \cup \{y \in H_i \cap Y : a <_W y\}. \quad (\text{S29.30})$$

*Then  $J$  is the canonical history of the induced order on  $Y$ . If  $H$  terminates, so does  $J$ .*

*Proof.* The first-stage clause is immediate, and intersecting the displayed subclasses at a limit gives the same formula at that limit. For the successor clause, fix  $x \in H_i \cap Y$  with  $a <_W x$ , and put  $u = p_i(a)$ . By the greatest-predecessor property, the global and local intervals from their respective anchors are

$$\begin{aligned} [u, x]_W \cap H_i &= \{u\} \cup \{z \in H_i : a <_W z \leq_W x\}, \\ [a, x]_W \cap J_i &= \{a\} \cup \{z \in H_i : a <_W z \leq_W x\}. \end{aligned}$$

Convexity of  $Y$  ensures that the tails displayed on the two lines coincide. The two intervals are sets simultaneously.

A witness above  $a$  that  $x$  is not a least representative is a witness in both orders. A global witness below  $a$  gives a local set interval by restricting its tail and inserting  $a$ . Conversely, a local witness equal to  $a$  can be replaced by  $u$ , using the displayed equality of tails. Consequently the retained representatives strictly above  $a$  agree exactly. The point  $a$  is the local minimum and is always retained. This proves the successor clause.

At the final stage, a singleton global carrier contributes either the same minimum or no point above  $a$ . Formula (S29.30) therefore gives  $\{a\}$ .  $\square$

Bare intersection with a noninitial convex interval is insufficient: it can discard the interval's own minimum. The anchor in (S29.30) is the necessary correction. For an initial segment, the anchor already is the global minimum, so ordinary restriction is enough.

**Corollary 108.18** (The exact criterion for a fixed schedule). *For nonempty class well-orders  $W$  and  $\Gamma$ , the following are equivalent:*

1.  $W$  has a supplied canonical terminating history on  $\Gamma + 1$ ;
2.  $W$  is class-isomorphic to an initial segment of  $P(\Gamma)$ .

*Proof.* The forward implication is theorem 108.16. For the reverse implication, restrict the explicit history (S29.26) to the given initial segment and pull it back through the supplied isomorphism. Initiality and lemma 108.17 verify the recursion clauses. The zero polynomial belongs to every nonempty initial segment, so the pulled-back history terminates at its minimum.  $\square$

This criterion uses an actual initial-segment embedding as its witness. It does not invoke a class of order types or compare unspecified cardinalities of class histories.

### 108.3.6 Equivalence with strong class-well-order comparability

Write CWO for the assertion that, for every two class well-orders, one is class-isomorphic to an initial segment of the other, allowing the whole order as an initial segment.

**Theorem 108.19** (Exact strength of canonical histories). *Over GBC, the following principles are equivalent:*

1. CWO: strong class-well-order comparability;
2. CTH: every nonempty class well-order  $W$  admits its canonical history on the input-order schedule  $W + 1$ ;
3. every nonempty class well-order admits a uniformly coded canonical terminating history on some class-well-order schedule  $\Gamma + 1$ ;
4. every nonempty class well-order  $W$  is class-isomorphic to an initial segment of  $P(W)$ .



*Proof.* (2) *implies* (3). This is [note 108.11](#): the specified input-order history terminates at its appended last stage.

(3) *implies* (1). Let  $R, S$  be two nonempty class well-orders on disjoint tagged carriers, and form their ordered sum  $T = R + S$ . Apply (3) once, to  $T$ , obtaining a terminating history with some schedule  $\Gamma + 1$ . Its two anchored convex restrictions give terminating histories for  $R$  and  $S$  on that same schedule. By [theorem 108.16](#), both are isomorphic to initial subclasses of  $P(\Gamma)$ .

Two initial subclasses of the same linear order are comparable by inclusion: if neither included the other, choosing a point in each set difference would contradict initiality in one direction or the other. Composing the map onto the smaller polynomial range with the inverse map on the larger range gives the required initial-segment isomorphism. The maps and their ranges are elementary classes defined from the one supplied history. There is no choice of a class witness for each member of a proper collection. An empty input order is handled separately and trivially.

(1) *implies* (4). We give an explicit argument that avoids assigning ordinals to class well-orders. First observe that every class order embedding  $f : W \rightarrow W$  satisfies  $x \leq_W f(x)$  for all  $x$ . Otherwise choose the least counterexample  $x$ . Since  $f(x) <_W x$ , order preservation gives  $f(f(x)) <_W f(x)$ , making  $f(x)$  an earlier counterexample. Consequently there is no class order embedding  $W + 1 \rightarrow W$ : if  $a$  were the image of the appended last point, the restriction to  $W$  would give

$$a \leq_W f(a) <_W f(*) = a.$$

Let  $\Gamma = W + 1$ . The polynomial order  $P(\Gamma)$  is a class well-order. It contains an embedded copy of  $W + 1$ , sending each exponent  $t$  to its coefficient-one monomial  $e_t$ . Comparability between  $W$  and  $P(\Gamma)$  cannot give an initial embedding of the latter into  $W$ , since composition would embed  $W + 1$  into  $W$ . Thus it supplies an initial-segment embedding

$$f : W \longrightarrow P(\Gamma).$$

Its range lies strictly below  $e_*$ . If a polynomial at least  $e_*$  lay in the range, initiality would put  $e_*$  and every  $e_x$ , for  $x \in X$ , into the range. Applying  $f^{-1}$  to these monomials would again embed  $W + 1$  into  $W$ . The polynomials strictly below  $e_*$  are exactly those supported on the original  $W$ -exponents, namely  $P(W)$ . Hence  $f$  gives (4).

(4) *implies* (2). Apply [corollary 108.18](#) with  $\Gamma = W$ . Concretely, if  $f$  is the supplied initial-segment isomorphism, define

$$H_i = \{x \in X : \text{supp}(f(x)) \subseteq \{j : i \leq_W j\}\}, \quad H_* = f^{-1}[\{0\}].$$

This is one elementary class relation. The polynomial history, initial restriction, and pullback prove all the required clauses.  $\square$

**Corollary 108.20** (The surreal carrier alone suffices). *In [theorem 108.19](#), the history principle (2) is unchanged if it is asserted only for class well-orders of the whole surreal carrier **No**.*

*Proof.* Under Global Choice, a proper class with the restriction of a set-like global well-order has type Ord. Therefore every proper class carrier has a class bijection with **No**. Transporting a supplied order through that bijection, applying the surreal-carrier history principle, and transporting both the label and stage coordinates back gives the input-order history on the original proper carrier. For a nonempty set carrier, every interval is a set, so the first successor stage is already its singleton minimum and all later stages remain there. Thus set carriers add no further requirement.  $\square$

### 108.3.7 What the equivalence settles, and what it leaves separate

The result identifies the exact condensation-history existence principle with a familiar class-theoretic comparison principle:

$$\text{GBC} \vdash \text{CTH} \longleftrightarrow \text{CWO}, \quad \text{GBC} + \text{ETR} \vdash \text{CTH}.$$

Hamkins–Woodin prove that ETR implies strong class-well-order comparability and explicitly ask about converses among the relevant ranking and comparison principles [27, Theorem 6 and Question 7]. Their Question 7 is not settled here. In particular, the presence of class-valued stages does not by itself prove that ETR is necessary. Our theorem provides an exact equivalence to CWO, without claiming an equivalence to arbitrary elementary transfinite recursion.

It also preserves the distinction with native labelled lexicography. The largest common labelled prefix of two supplied relations is elementarily definable, and their next labels can be compared without CWO or a condensation history (theorem 14.2). The stronger principle arises when all the iterated quotient information is required uniformly through a complete schedule. These are different tasks.

**Uniqueness and changes of the class collection.** Two histories for the same input and schedule agree wherever both are defined. A least stage of disagreement is impossible: the first, successor, and limit clauses all determine the slice from earlier slices. This is elementary induction with the two histories supplied as parameters. Consequently an extension of a GBC class collection with the same sets may supply a previously unavailable history, but cannot change an already available one. The recursion clauses, set-interval tests, and finite normal forms are elementary in the shared classes and sets. This is a uniqueness statement about individual histories, not a class-valued choice function assigning a history to every well-order.

**A concrete full-model check.** In a full inaccessible-universe interpretation  $(V_\kappa, \mathcal{P}(V_\kappa))$ , the polynomial order  $P(\Gamma)$  has ambient ordinal type  $\kappa^\alpha$  when  $\Gamma$  has ambient type  $\alpha$ . Its coefficients range over the internal ordinals, externally  $\kappa$ , and its supports are finite. Thus corollary 108.18 becomes exactly

$$\theta \leq \kappa^\alpha,$$

where  $\theta$  is the ambient type of the input order. This recovers the earlier exact height formula  $\min\{\alpha : \theta \leq \kappa^\alpha\}$  for the relevant nontrivial histories (source 20’s theorem 108.2 and corollary 108.3). The present proof explains that calculation through canonical finite normal forms, while remaining meaningful in arbitrary GBC realizations where external ordinal types are not available internally.

**Formalization boundary.** A direct formalization can separate four interfaces: a class well-order on set labels; a supplied history relation; the set-sized interval-rank operation; and elementary comprehension and Class Replacement. The floor-map and normal-form theorems then require no new recursion axiom. The finite polynomial codes belong to the set sort, and their comparison uses only a finite support and the supplied stage order. The existence theorem is a separate equivalence between two second-order assertions, CTH and CWO.

In a universe-relative type-theoretic implementation, the same proof can be expressed with explicit smallness and an enclosing universe. In a formalized two-sorted model, all claims of well-foundedness, finiteness, and set-sized intervals must retain that model’s interpretation. Neither approach permits forming an ordinary class whose members are all class well-orders, all histories, or all of their isomorphism classes.

*Note 108.21* ([merge]; one height calculation, five times). In the full model  $(V_\kappa, \mathcal{P}(V_\kappa))$  at an inaccessible  $\kappa$ , internal set-interval condensation is condensation by intervals of ambient cardinality below  $\kappa$ . Source 15's full-universe check (section 53.1) and source 18's (note 53.6) identify one condensation with left division by  $\kappa$ ; source 20's theorem 108.2 and corollary 108.3 iterate it to the history  $\{\kappa^\alpha \cdot \beta < \theta\}$  and the height  $\min\{\alpha : \theta \leq \kappa^\alpha\}$ , with every height  $0 < \alpha < \kappa^+$  attained and no universal schedule; source 23's universes section (proposition 111.24) computes that  $P(\Gamma)$  has type  $\kappa^\beta$  when  $\Gamma$  has type  $\beta$ , and source 29's "concrete full-model check" above turns corollary 108.18 into  $\theta \leq \kappa^\alpha$ . These agree. Source 20's corollary is the only one to state the height spectrum and the attainment of the input-order bound.

[merge] Source 23's Section 12 transports these polynomial orders into the lexicographic order of all class well-orders of **No**. Its theorem 108.23 is new against the repository and against Part IX: it follows from the convexity of the fibres over a fixed set-like head (source 11's proposition 15.3) and source 15's polynomial histories, but neither is stated with it.

#### 108.4 Source 23: arbitrary raw complexity inside global lexicographic intervals

Let  $\mathfrak{W}_{\text{sl}}$  be the virtual order of exhaustive class bijections  $e : \text{Ord} \rightarrow \mathbf{No}$ , and let  $\mathfrak{W}_{\text{all}}$  be the virtual order of all raw well-orders on **No**, both compared by the numerical label order.

**Lemma 108.22** (A set prefix is absorbed). *For every set ordinal  $\sigma$  and every nonempty class well-order  $\Gamma$ ,*

$$\sigma + P(\Gamma) \cong P(\Gamma)$$

*as class well-orders.*

*Proof.* The first set-interval block of  $P(\Gamma)$  is  $\text{Ord}$ , obtained by varying only the least coordinate. The order  $\sigma + \text{Ord}$  is isomorphic to  $\text{Ord}$ : map the first block identically onto the ordinals below  $\sigma$  and the following ordinal  $\xi$  onto  $\sigma + \xi$ . The latter enumerates the ordinal tail beginning at  $\sigma$ . Replace the first block by this isomorphism and leave the rest unchanged.  $\square$

**Theorem 108.23** (Dense realization of prescribed raw hierarchies). *In GBC, let  $e < f$  be two distinct members of  $\mathfrak{W}_{\text{sl}}$ , and let  $\Gamma$  be any nonempty admitted class well-order. There exists a raw well-order  $H$  on **No** such that*

$$e <_{\text{lex}} H <_{\text{lex}} f, \quad H \cong P(\Gamma).$$

*In particular, every such interval realizes every positive set-ordinal collapse rank and every prescribed class-indexed hierarchy of Note 108.8.*

*Proof.* Let  $\delta$  be the first ordinal where  $e$  and  $f$  disagree, and let  $p = e \upharpoonright \delta = f \upharpoonright \delta$ . This is a set prefix. Choose a fresh surreal  $c$  with

$$e(\delta) < c < f(\delta), \quad c \notin \text{ran } p,$$

using Note 81.2. The remaining label class  $Y = \mathbf{No} \setminus (\text{ran } p \cup \{c\})$  is proper. Transport the well-order  $P(\Gamma)$  to  $Y$  by a class bijection and place it after the prefix  $p \frown (c)$ .

The first disagreement with each endpoint occurs at the label  $c$ , so  $e < H < f$  regardless of the order chosen on  $Y$ . Its internal order type is  $(\delta + 1) + P(\Gamma)$ , which is  $P(\Gamma)$  by Lemma 108.22. The hierarchy is invariant under this isomorphism.  $\square$

*Remark 108.24* (The precise density assertion). The theorem concerns intervals whose endpoints are set-like exhaustive enumerations. It does not assert the same statement for every nonempty interval of  $\mathfrak{W}_{\text{all}}$ : finite residual intervals and adjacent pairs are counterexamples. Nor does it identify the condensation rank as an invariant definable from the outer lexicographic order alone. The extra structure is the relation encoded by an outer-order point.

A useful consequence is that a research program restricted to the canonical first set-like head cannot capture all internal order types appearing locally in the raw enlargement. Beyond that head one encounters hierarchies of any prescribed admitted well-ordered length. Nevertheless, all the witnesses in the theorem are constructed by one fresh-letter choice followed by a specified tail; the proof does not require a simultaneous choice of tails for all possible class parameters.

## 109 Countable class models: two external order types and one theory (sources 19, 20, 21 and 30)

[merge] Part IX printed source 18's classification of the external order  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  of an externally countable  $\omega$ -standard model of GBC as the factorial shuffle ([theorem 58.6](#)), its remark that the classification fails without  $\omega$ -standardness ([remark 58.8](#)), and its question on the  $\omega$ -nonstandard case ([research question 113.89](#)). Sources 19, 20 and 21 answer that question independently and identically: the order is the shuffle  $\sigma(\mathcal{F} \cup \{I_\infty\})$ , not isomorphic to the standard one but elementarily equivalent to it. Source 19 is printed first and is the base, because its [theorem 109.9](#) proves that the pure order of every externally set-sized model of GBC has the theory of the standard shuffle; its [proposition 109.7](#), [corollary 109.11](#) and [theorem 109.12](#) are proved only there. Sources 20 and 21 follow whole as routes. Their shared lemmas (maximal finite tail, permutation blocks, density of every tail length, the nonstandard interval, the Ehrenfeucht–Fraïssé threshold  $2^r - 1$ ) are notes pointing to the first printed copy; what only they prove is printed as theorems: source 20's external finite-condensation blocks ([corollary 109.18](#)) and the decidability of the common theory ([corollary 109.22](#)), and source 21's automorphism group of the nonstandard order and the first derivative ([theorem 109.36](#)). The lemmas on terminal tails, permutation blocks and dense tail lengths were proved first by sources 17 and 18 in Part IX ([theorem 54.4](#), [lemmas 58.1](#) and [58.5](#) and [proposition 58.3](#)); the three batch-83 copies are notes to those. Source 30's compactness observation, which explains why external finiteness of the blocks is not a first-order property, closes the section.

### 109.1 Source 19: the exact external spectrum in countable class models

The distinction between internal and external finiteness yields a complete answer to the countable-model question. It also reveals a useful contrast: the pure order has only two isomorphism types among countable GBC realizations, but these types have the same first-order theory, and that theory itself has the largest possible number of countable models.

Let  $\mathcal{M} = (M, \mathcal{X})$  be an externally countable two-sorted model of GBC. Its classes are the members of its admitted class sort  $\mathcal{X}$ . Write  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  for the external set of admitted class relations which  $\mathcal{M}$  regards as well-orders of its whole surreal carrier, and  $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$  for the subset which it regards as set-like. Compare them by the labelled common-prefix lexicographic comparison computed in  $\mathcal{M}$ . This produces an actual external linear order. No external well-foundedness of the individual relations is assumed.

For a countable nonempty collection  $\mathcal{A}$  of nonempty order types, let  $\sigma(\mathcal{A})$  denote the shuffle obtained by densely colouring the rational order with those types and replacing each rational by the

corresponding chain. Put

$$\mathcal{F} = \{n! : n \in \mathbb{N}\} = \{1, 2, 6, 24, 120, \dots\}, \quad I_\infty = \omega + \sum_{q \in \mathbb{Q}} \mathbb{Z} + \omega^*.$$

Here  $\mathbb{N}$  is the external standard set of natural numbers, and  $\mathbb{Z}$  denotes its usual order. The order  $I_\infty$  has a first and a last element. Its external finite-distance components are an initial  $\omega$ , a final  $\omega^*$ , and copies of  $\mathbb{Z}$  densely indexed between them.

**Theorem 109.1** (Exactly two countable GBC order types). *For an externally countable model  $\mathcal{M}$  of GBC,*

$$\mathfrak{W}_{\text{all}}^{\mathcal{M}} \cong \begin{cases} \sigma(\mathcal{F}), & \mathcal{M} \text{ is } \omega\text{-standard}, \\ \sigma(\mathcal{F} \cup \{I_\infty\}), & \mathcal{M} \text{ is } \omega\text{-nonstandard}. \end{cases}$$

*In both cases  $\mathfrak{W}_{\text{sl}}^{\mathcal{M}} \cong \mathbb{Q}$ . The two unrestricted orders are nonisomorphic but elementarily equivalent in the pure language of linear orders.*

The assertion concerns two possible types among GBC realizations, conditional on the existence of the respective kinds of model. It does not say that their common first-order theory has only two countable models. We prove the classification first, and then give the stronger first-order comparison.

*Note 109.2* ([merge]; which clauses are earlier, and the two other proofs). Two clauses of [theorem 109.1](#) are in Part IX: the  $\omega$ -standard formula is source 18’s [theorem 58.6](#), and  $\mathfrak{W}_{\text{sl}}^{\mathcal{M}} \cong \mathbb{Q}$  is source 13’s [proposition 55.5](#) (the second clause of source 18’s theorem), both by the proofs repeated below. The  $\omega$ -nonstandard formula, the nonisomorphism and the elementary equivalence are new against the repository and answer source 18’s [research question 113.89](#). Source 20 proves the same three clauses ([note 109.13](#)) and source 21 as well ([notes 109.33](#) and [109.35](#)); the three proofs use the same nonstandard-interval lemma and the same finite-game threshold. Source 19 obtains the elementary equivalence from the stronger [theorem 109.9](#), which needs no countability.

### 109.1.1 The internal finite-tail partition

The next constructions are first carried out internally in GBC. They apply to an infinite linearly ordered class whenever well-order completions are available; for the surreal carrier, Global Choice supplies them. They are parameterwise constructions of individual classes, not an internal collection containing all class orders.

*Note 109.3* (Source 19, Lemma 12.2, “The maximal finite terminal tail”). *Every class well-order  $R$  on an infinite carrier  $X$  has a unique finite terminal segment  $T_R$  such that  $R$  restricted to  $X \setminus T_R$  has no greatest element. Every finite terminal segment of  $R$  is contained in  $T_R$ . Write  $n(R) = |T_R| \in \omega^{\mathcal{M}}$ . This is source 18’s [lemma 58.1](#) (Part IX), with the same proof; source 19 adds that in an  $\omega$ -nonstandard model the stopping value may be nonstandard (the paragraph after the proof). Sources 20 and 21 repeat it ([notes 109.14](#) and [109.24](#)).*

*Source 19’s proof.* If  $R$  has no greatest element, take the empty tail. Otherwise remove the greatest element and repeatedly remove the greatest remaining element while one exists. If the process did not stop at an internal finite stage, ordinary set-valued recursion with class parameter  $R$  would produce an internal descending  $\omega$ -sequence. Class Replacement makes its range a set, contradicting well-foundedness. At the stopping stage the selected finite set is terminal and its complement has no greatest element. Each last-element or predecessor step was unique, so the tail is unique, and every finite terminal segment is among its initial sequence of removals.  $\square$

In an  $\omega$ -nonstandard model the stopping value  $n(R)$  may be nonstandard. The lemma is an internal assertion and does not claim that an external iteration through the standard natural numbers stops.

Let  $B_R$  be the cylinder obtained by keeping  $R$  unchanged on  $X \setminus T_R$  and permuting its  $n(R)$  terminal labels. These cylinders partition the admitted orders: every member has the same maximal terminal set and the same preceding prefix. Each cylinder is convex and has first and last elements. Internally it is a finite chain of size  $n(R)!$ . More explicitly, the finite set of permutations of  $T_R$  belongs to  $M$ , and each permutation defines a class relation by elementary comprehension. Sorting the terminal labels in their ambient order identifies their lexicographic permutations with the internal finite linear order of size  $n(R)!$ . Every admitted order in the cylinder arises in this manner. *Note 109.4* (Source 19, Lemma 12.3, “Adjacency has finite residual carrier”). *If  $R < S$  are adjacent, their residual carrier after the first common labelled prefix is finite. In particular, an adjacency step cannot leave the cylinder  $B_R$ .* This is source 18’s lemma 58.2 (Part IX) with the same proof, and the finite-residual half of source 11’s adjacency criterion theorem 16.2.

*Source 19’s proof.* Let  $C$  be the common prefix,  $Y = X \setminus C$ , and let the first unequal labels be  $a < b$ . Keeping  $C$  and the next label  $a$  fixed, the rest of the  $R$ -order must be the largest possible completion: a larger completion would lie between  $R$  and  $S$ . Consequently each successive label is the greatest unused label, so the reverse ambient order well-orders  $Y \setminus \{a\}$ . Dually, the ambient order well-orders  $Y \setminus \{b\}$ . Thus on  $Z = Y \setminus \{a, b\}$  both an order and its reverse are well-orders. This forces  $Z$  to be finite. Otherwise successively taking the least unused label would give an internal ascending  $\omega$ -sequence in one order, hence a descending sequence in the other. Therefore  $Y$  is finite. It is a finite terminal segment of  $R$ , and so is contained in  $T_R$  by Note 109.3.  $\square$

*Note 109.5* (Source 19, Lemma 12.4, “Every prescribed internal tail occurs densely”). *Between two distinct cylinders there is a cylinder with  $n(R) = n$ , for every prescribed  $n \in \omega^M$ .* This is source 18’s lemma 58.4 followed by lemma 58.5 (Part IX), with the same two-change argument, now read internally so that  $n$  may be nonstandard; source 20’s note 109.16 and source 21’s note 109.27 are the other copies.

*Source 19’s proof.* Take  $R < S$  in different cylinders, with common prefix  $C$ , residual carrier  $Y$ , and first unequal labels  $a < b$ . The residual  $Y$  is infinite, since otherwise the two orders would belong to one finite-tail cylinder. We first find a whole cylinder inside  $(R, S)$  with infinite residual carrier.

One possible change is to keep an  $R$ -prefix through its label  $a$ , then, at a later position outside  $T_R$ , replace the next label  $x$  by a numerically larger unused label. Its cylinder lies above  $R$  and below  $S$ , and its residual is infinite. A second possible change does the dual operation in  $S$ : after retaining its label  $b$ , replace a later label outside  $T_S$  by a smaller unused label.

At least one change is possible. Otherwise, on the infinite class

$$Z = Y \setminus (\{a, b\} \cup T_R \cup T_S),$$

every earlier  $R$ -label would be numerically greater than every later one, and every earlier  $S$ -label would be numerically smaller than every later one. The ambient order on  $Z$  and its reverse would both be well-orders, contradicting the argument in Note 109.4. The new cylinder is different from both endpoint cylinders, because its first disagreements with them occur while the residual carrier is infinite.

In that residual carrier choose an  $n$ -element set  $F$ . Well-order the remaining carrier without a greatest element: use its infinite initial cardinal if it is a set, or a set-like well-order of type Ord if it is proper. Append this order after the chosen prefix and then append the labels of  $F$ . The maximal finite terminal tail is exactly  $F$ , as required.  $\square$



For the surreal carrier the full order has no endpoints: change the first label upward or downward and complete the remaining carrier. Every cylinder has endpoints, so its quotient has no endpoints either. [Note 109.5](#) makes this quotient dense. When the class sort is externally countable the quotient is consequently a copy of  $\mathbb{Q}$ .

### 109.1.2 What a nonstandard finite interval looks like externally

**Lemma 109.6** (The external shape of a nonstandard finite interval). *In an externally countable nonstandard model of arithmetic, let  $c$  be a nonstandard natural number. The external order of the internal interval  $[0, c]$  has type*

$$I_\infty = \omega + \sum_{q \in \mathbb{Q}} \mathbb{Z} + \omega^*.$$

*The arithmetic available in any GBC model suffices.*

*Proof.* Declare  $x \sim y$  when their arithmetic difference in absolute value is a standard natural number. This is precisely external finite distance in  $[0, c]$ . The component of 0 consists of its standard successors and has type  $\omega$ . The component of  $c$  consists of its standard predecessors and has type  $\omega^*$ . Every other component has arbitrary standard predecessors and successors, and all its members are obtained by those finitely iterated steps from any chosen one. It has type  $\mathbb{Z}$ .

The quotient has a first and a last component. It is dense. Indeed, if  $x < y$  belong to different components then  $y - x$  is nonstandard, and

$$z = x + \left\lfloor \frac{y - x}{2} \right\rfloor$$

is at nonstandard distance from both endpoints. A standard bound on either half would give a standard bound on  $y - x$ . Hence  $[x] < [z] < [y]$ . The open interval between the endpoint components is a countable dense order without endpoints, so is a copy of  $\mathbb{Q}$ . This gives the displayed decomposition.  $\square$

*Proof of the isomorphism assertions in [Theorem 109.1](#).* In an  $\omega$ -standard model every  $n(R)$  is standard, so every cylinder has actual finite size  $n(R)!$ . Their quotient is countable and dense without endpoints, and every factorial size occurs densely by [Note 109.5](#). Countable colour-preserving back-and-forth gives  $\sigma(\mathcal{F})$ .

In an  $\omega$ -nonstandard model the cylinders with standard tail length again have actual factorial size. If  $n(R)$  is nonstandard, then  $n(R)!$  is nonstandard, and [Lemma 109.6](#) gives external type  $I_\infty$ . These are all the possibilities. Prescribing one fixed nonstandard tail length shows that the  $I_\infty$  colour occurs densely, and each standard factorial colour is dense as before. The same coloured back-and-forth gives  $\sigma(\mathcal{F} \cup \{I_\infty\})$ .

For the set-like suborder, the first unequal ordinal position has a set-sized common prefix. Choose an unused surreal strictly between the unequal next labels and complete the prefix to a global enumeration. The surreal set-cut property permits avoidance of the set of previously used labels. This gives internal and therefore external density. Varying the first label gives no endpoints, and external countability gives  $\mathbb{Q}$ . This argument never assumes  $\omega$ -standardness.  $\square$

### 109.1.3 Recovering the cylinders from the pure order

The internal finite-tail partition is visible in the external pure order through a single first-order formula. This observation removes any dependence on a hidden choice of presentation in the comparison that follows.

Define

$$\begin{aligned} \text{Adj}(x, y) &\iff x < y \wedge \neg \exists z (x < z < y), \\ S(x) &\iff \neg \exists y \text{ Adj}(y, x) \wedge \neg \exists y \text{ Adj}(x, y), \\ E(x, y) &\iff \neg \exists z (((x < z < y) \vee (y < z < x)) \wedge S(z)). \end{aligned}$$

**Proposition 109.7** (Intrinsic cylinder equivalence). *In the external unrestricted order of any set-sized two-sorted GBC model,  $S$  picks out exactly the singleton cylinders, and  $E(x, y)$  holds exactly when  $x$  and  $y$  belong to the same finite-tail cylinder.*

*Proof.* A singleton cylinder has no immediate predecessor or successor, because the quotient is dense. In a larger internally finite cylinder every point has at least one immediate neighbour inside that cylinder, including its endpoints; the internal finite linear order gives these neighbours even when it is externally infinite. Thus these cylinders contain no  $S$ -points. Consequently  $E$  holds within a cylinder. Between distinct cylinders, [Note 109.5](#) with tail length zero gives an  $S$ -point, so  $E$  fails.  $\square$

In the standard case  $E$  is external finite condensation. In the nonstandard case a  $I_\infty$ -class contains many external finite-distance components, so  $E$  is strictly coarser. In particular the two orders in [Theorem 109.1](#) are not isomorphic: the second has infinite  $E$ -classes and the first does not. The sentence asserting that every  $E$ -class is finite can be written in  $L_{\omega_1, \omega}$ , using a countable disjunction of the finite size formulas. It is not a first-order sentence.

#### 109.1.4 One complete first-order theory for all GBC realizations

The first-order comparison does not need external countability.

**Lemma 109.8** (Finite-game threshold). *Let  $h_r = 2^r - 1$ . Two finite chains of sizes at least  $h_r$  are  $r$ -equivalent. The same conclusion holds if either chain is replaced by an infinite discrete linear order with first and last elements, in which every point other than an endpoint has its corresponding immediate predecessor and successor.*

*Proof.* Use the  $r$ -round Ehrenfeucht–Fraïssé game. With  $s$  rounds left, maintain the following invariant in each interval between matched points, including the two end intervals: either the two intervals have the same finite size below  $h_s$ , or both have at least  $h_s$  points. Initially the invariant holds.

For a short interval use its unique finite order isomorphism. In a large interval, if the newly selected point is within fewer than  $h_{s-1}$  points of one end, match that exact finite distance from the corresponding end. Otherwise choose a point leaving at least  $h_{s-1}$  points on each side. The identity  $h_s = 2h_{s-1} + 1$  guarantees enough room. Finite successor and predecessor steps are available because the intervals have endpoints and are discrete. These choices preserve the invariant with  $s - 1$  rounds left and therefore preserve the final finite order pattern.  $\square$

**Theorem 109.9** (Uniform complete theory). *For every externally set-sized two-sorted model  $\mathcal{M}$  of GBC, the pure external order  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  has the same complete first-order theory*

$$T = \text{Th}(\sigma(\mathcal{F})).$$

*In particular, the standard and nonstandard countable orders of [Theorem 109.1](#) are elementarily equivalent.*

*Proof.* Fix a quantifier rank  $r$ . Colour a cylinder by its exact finite factorial size if that size is below  $h_r$ , and merge all other cylinders into one large colour. Only finitely many colours remain. Each occurs densely in the quotient: for the large colour prescribe any standard factorial size at least  $h_r$ . The quotient has no endpoints.

In a finite game, match a newly visited quotient class to a class of the same colour in the appropriate interval between previously matched classes. Density and absence of endpoints always permit this. Corresponding small cylinders are identical finite chains. Corresponding large cylinders are  $r$ -equivalent by Lemma 109.8, since every internally finite cylinder is externally discrete with endpoints. Play a fixed  $r$ -round winning game inside each matched pair of cylinders. There are at most  $r$  moves inside any one pair, and the quotient matching preserves the order between pairs. This gives an  $r$ -round winning strategy for the whole orders. As  $r$  was arbitrary, their complete theories agree.  $\square$

**Corollary 109.10** (A proper elementary suborder and an omitted type). *In the nonstandard countable order, the union  $L_{\text{fin}}$  of its genuinely finite  $E$ -classes is a proper elementary suborder isomorphic to  $\sigma(\mathcal{F})$ . The partial type*

$$p(x) = \{\text{the } E\text{-class of } x \text{ has at least } m \text{ members} : 1 \leq m < \omega\}$$

*is nonprincipal over  $T$ . The standard shuffle omits it; every point in a  $I_\infty$ -class of the nonstandard shuffle realizes it.*

*Proof.* Deleting the  $I_\infty$ -coloured quotient points leaves a countable dense order without endpoints in which all finite factorial colours are dense. Thus  $L_{\text{fin}} \cong \sigma(\mathcal{F})$ . For elementarity fix a finite tuple from this suborder and a quantifier rank  $r$ . Repeat the preceding quotient matching while fixing its parameter classes, and use the identity within those finite parameter classes. The same finite-game argument works for every  $r$  with the tuple fixed, proving elementarity.

The type is finitely satisfiable because factorial sizes are unbounded, and its realization and omission are immediate from the class sizes. Were it isolated by a formula consistent with the complete theory  $T$ , that formula would have a realization in the standard shuffle and force a realization of  $p$  there, a contradiction.  $\square$

**Corollary 109.11** (The standard shuffle is the prime model). *The standard shuffle is a countable atomic, and hence prime, model of  $T$ .*

*Proof.* For a finite tuple, specify its partition into  $E$ -classes, the order of these classes, the finite size of each class, and the exact position of each tuple entry in its class. Each condition is first-order expressible. Their finite conjunction specifies a single automorphism orbit: apply coloured-quotient back-and-forth and the unique isomorphisms of the finite chains. Consequently it isolates the tuple's complete type, establishing atomicity.

For completeness, primeness follows by an elementary enumeration argument. Enumerate the standard shuffle and embed its elements one at a time into any model of  $T$ . At each step the isolated type of the next finite tuple is consistent with, and extends, the previously isolated tuple type; its existential realization therefore follows from that preceding type. The resulting union is an elementary embedding.  $\square$

### 109.1.5 The common theory has many other countable models

The GBC classification is more restrictive than elementary equivalence. One can make this precise without appealing to an abstract model-spectrum theorem.

**Theorem 109.12** (All countable orders occur inside countable models of  $T$ ). *For every countable linear order  $A$ , put*

$$J_A = \omega + \sum_{a \in A} \mathbb{Z} + \omega^*.$$

*There is a countable model  $L_A$  of  $T$  with exactly one infinite  $E$ -class, of type  $J_A$ . Moreover,*

$$L_A \cong L_{A'} \iff A \cong A'.$$

*Consequently  $T$  has  $2^{\aleph_0}$  pairwise nonisomorphic countable models.*

*Proof.* Start with a rational quotient densely coloured by the finite factorial sizes, choose one quotient point, and replace its chain by  $J_A$ . All other classes retain their prescribed finite chains. The order  $J_A$  is infinite, discrete, and has first and last elements. The threshold argument of [Lemma 109.8](#) therefore applies. After merging the large colours at any fixed rank, all quotient colours remain dense. The same finite-game proof gives  $L_A \models T$ .

The formula  $E$  recovers the displayed classes exactly, because the singleton classes remain dense between them. Thus an isomorphism must preserve the unique infinite class. Its external finite-distance quotient is  $1 + A + 1$ : the endpoint components are  $\omega$  and  $\omega^*$ , and the interior components are the indicated copies of  $\mathbb{Z}$ . Removing the endpoint components recovers  $A$ . Conversely an isomorphism  $A \cong A'$  extends to their  $J$ -classes, and a coloured rational back-and-forth fixing the distinguished quotient point gives  $L_A \cong L_{A'}$ .

There are  $2^{\aleph_0}$  nonisomorphic countable linear orders. For an explicit family, encode a binary sequence  $(a_n)$  by  $\sum_{n < \omega} (\mathbb{Z} + (2 + a_n))$ . Its finite-distance components alternate a copy of  $\mathbb{Z}$  and a finite chain of length 2 or 3; their quotient has type  $\omega$ , so it recovers the sequence. This gives the lower bound for models of  $T$ . The upper bound is the usual count of relations on a countable domain.  $\square$

The arithmetic midpoint in [Lemma 109.6](#) explains why these other models do not produce additional countable GBC realizations. Within an internally finite nonstandard interval, its external finite-distance quotient must be dense, forcing the interior index order to be  $\mathbb{Q}$  in a countable model. The prescribed-tail construction additionally forces such  $I_\infty$ -classes to occur densely whenever the model is nonstandard. These conditions go beyond what the pure first-order theory can express.

### 109.1.6 What the result does and does not forget

For countable GBC realizations, the pure unrestricted order remembers  $\omega$ -standardness up to isomorphism and forgets it at the level of first-order theory. Its theory is even independent of the external size of the class model. The theorem does not classify expansions by the set-like predicate, evaluation at surreal labels, birthday restrictions, named baselines, or coding and definability information. Those structures remain natural places to look for dependence on the underlying set universe and class realization.

All constructions above use elementary class comprehension, ordinary set-valued recursion with supplied class parameters, and Global Choice for completions. They require neither ETR nor an internal class of all class orders. The quotient and model-theoretic games are external set constructions, with their scope explicitly fixed at the start.

## 109.2 Source 20: two external order types for countable class models

The earlier continuation, source 18, classifies the raw lexicographic order of an externally countable,  $\omega$ -standard model of GBC as a shuffle of finite factorial chains ([theorem 58.6](#)). It asks what happens

when the model has nonstandard natural numbers ([research question 113.89](#)). We now give the exact classification in that case, including its external finite-condensation blocks and its first-order theory. The classification concerns the *pure external linear order*; it discards the labels, the class membership relation, and the predicate of set-likeness.

Let  $\mathcal{M} = (M, \mathcal{X}) \models \text{GBC}$ , with both sorts countable in the ambient metatheory. Let  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  be the external set of those  $R \in \mathcal{X}$  that  $\mathcal{M}$  regards as class well-orders of its surreal carrier. The comparison  $R < S$  means that  $\mathcal{M}$  satisfies the elementary labelled first-disagreement formula. Thus “well-order” and “finite” inside a construction below are initially *internal* notions. We never assume that  $R$  is externally well-founded. Internal linearity of the comparison nevertheless makes  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  an actual external linear order.

For a countable nonempty family  $\mathcal{A}$  of countable order types, write  $\sigma(\mathcal{A})$  for its *dense shuffle*: colour  $\mathbb{Q}$  so that every colour in  $\mathcal{A}$  is dense, and replace each rational by a copy of its colour. The usual colour-preserving back-and-forth proves uniqueness of the resulting order type. Put

$$\mathcal{F} = \{1, 2, 6, 24, 120, \dots\} = \{n! : n \in \omega\}, \quad I_{\infty} = \omega + \sum_{q \in \mathbb{Q}} \mathbb{Z} + \omega^*.$$

The sum in  $I_{\infty}$  means a rationally ordered family of copies of the integer order; it does not mean an ordinal product.

*Note 109.13* (Source 20, Theorem 8.1, “Countable-model dichotomy”). *For every externally countable  $\mathcal{M} \models \text{GBC}$ ,*

$$\mathfrak{W}_{\text{all}}^{\mathcal{M}} \cong \begin{cases} \sigma(\mathcal{F}), & \mathcal{M} \text{ is } \omega\text{-standard}, \\ \sigma(\mathcal{F} \cup \{I_{\infty}\}), & \mathcal{M} \text{ is } \omega\text{-nonstandard}. \end{cases}$$

*These two orders are not isomorphic, but they are elementarily equivalent in the language containing only  $<$ . In the nonstandard case, the union of the blocks of standard finite size in the displayed shuffle is an elementary suborder isomorphic to  $\sigma(\mathcal{F})$ .* This is source 19’s [theorem 109.1](#) without its set-like clause, together with the first clause of source 19’s [corollary 109.10](#); source 20’s proof follows in the lemmas below, the proof after [note 109.17](#) and [note 109.20](#).

The proof separates an internally finite terminal permutation block from an externally finite-distance block. Conflating these two equivalence relations would give an incorrect classification.

### 109.2.1 A coarse equivalence that remains valid internally

For each admitted  $R$ , there is an internally finite terminal set  $T_R$  such that  $R$  on  $\mathbf{No}^{\mathcal{M}} \setminus T_R$  has no greatest element. Write  $n_R = |T_R|^{\mathcal{M}} \in \omega^{\mathcal{M}}$ . The following argument recalls the terminal-tail result with the nonstandard interpretation made explicit.

*Note 109.14* (Source 20, Lemma 8.2, “Internal terminal tails”). *The set  $T_R$  exists uniquely and contains every internally finite terminal segment of  $R$ .* This is source 18’s [lemma 58.1](#) (Part IX) and source 19’s [note 109.3](#), with the same proof.

*Source 20’s proof.* Work inside  $\mathcal{M}$ . If there is no greatest label, take  $T_R = \emptyset$ . Otherwise start with the greatest label and successively take its immediate predecessor whenever there is one. If this does not stop at some  $n \in \omega$ , ordinary set recursion with the class parameter  $R$  gives an  $R$ -descending sequence of length  $\omega$ . Class Replacement makes its range a set, contradicting the class well-order property. At the stopping stage the selected finite terminal segment has a remainder with no greatest element. Every finite terminal segment is obtained by finitely many of the same uniquely determined predecessor steps, proving uniqueness and maximality. In a nonstandard model the

stopping value  $n$  can be nonstandard; the argument does not assert termination after a standard finite number of external steps.  $\square$

For unequal  $R, S$ , let  $C(R, S)$  be their largest common labelled prefix. Its definition is elementary in the class parameters: a label  $x$  belongs to it exactly when the two predecessor classes of  $x$  are equal and the two relations agree on that predecessor class. Define an external relation on  $\mathfrak{M}_{\text{all}}^{\mathcal{M}}$  by

$$R E_{\mathcal{M}} S \iff R = S \text{ or } \mathcal{M} \models \text{“}\mathbf{No} \setminus C(R, S) \text{ is a finite set.”}$$

The formula on the right quantifies over a set coding the residual carrier and a natural number. It does *not* quantify over a set of class relations or assert the existence of a finite sequence whose entries are classes.

*Note 109.15* (Source 20, Lemma 8.3, “The internal permutation blocks”).  $E_{\mathcal{M}}$  is a convex equivalence relation. Its block through  $R$  consists exactly of the admitted orders that keep the prefix  $R \upharpoonright (\mathbf{No}^{\mathcal{M}} \setminus T_R)$  fixed and permute the internally finite set  $T_R$ . As an external linear order this block is isomorphic to

$$[0, n_R!)^{\mathcal{M}} = \{i \in \omega^{\mathcal{M}} : \mathcal{M} \models i < n_R!\}$$

with the model’s natural-number order. This is the internal form of source 18’s [proposition 58.3](#) (Part IX), as in source 19’s paragraph after [note 109.3](#) and source 21’s [note 109.25](#). Source 20 adds the coding argument that identifies the external block with the model’s interval  $[0, n_R!)$ , including when  $T_R$  is externally infinite.

*Source 20’s proof.* If the first-disagreement residual is internally finite, it is a terminal segment in  $R$ , hence is contained in  $T_R$  by [note 109.14](#). The two orders therefore agree on the prefix outside  $T_R$ . Conversely, two orders with that fixed prefix can first disagree only inside  $T_R$ , so they are  $E_{\mathcal{M}}$ -equivalent. Permuting  $T_R$  leaves a remainder with no greatest element, so it does not change the maximal terminal set. This proves transitivity; reflexivity and symmetry are immediate. Convexity follows because a fixed labelled prefix defines a convex lexicographic cylinder.

There is also a precise coding justification for the asserted block size. Internally let  $P$  be the set of permutations of  $T_R$ . For  $p \in P$ , elementary class comprehension defines the order  $R_p$  obtained by concatenating the fixed prefix with  $p$ . The construction can be presented as one class of triples  $(p, x, y)$  whose  $p$ -slice is  $R_p$ . Its slices are classes, not elements of a set. Conversely, the restriction of any admitted order in this cylinder to the set  $T_R$  is a set relation, by class separation, and hence is an internal permutation code in  $P$ .

Inside the model, the lexicographic order on this finite set of permutations is a chain of size  $n_R!$ , and therefore has an order isomorphism with  $[0, n_R!)$ . The external interpretations of this set isomorphism and of the slice construction give the claimed external isomorphism. In particular, when  $T_R$  is externally infinite we are using the model’s internally coded permutations, not all externally existing permutations of its external carrier.  $\square$

*Note 109.16* (Source 20, Lemma 8.4, “Every internal tail size is dense in the coarse quotient”). *The quotient  $\mathfrak{M}_{\text{all}}^{\mathcal{M}}/E_{\mathcal{M}}$  is a countable dense linear order without endpoints. For every fixed  $n \in \omega^{\mathcal{M}}$ , the blocks whose maximal terminal tail has internal size  $n$  occur densely in this quotient.* This is source 18’s [lemmas 58.4](#) and [58.5](#) (Part IX) read internally, and source 19’s [note 109.5](#), with the same argument.

*Source 20’s proof.* All constructions in this paragraph are performed inside the model. Take  $R < S$  in different  $E_{\mathcal{M}}$ -blocks. Let  $C$  be their common prefix, let  $Y = \mathbf{No} \setminus C$ , and let  $a < b$  be their



first unequal numerical labels. The residual  $Y$  is internally infinite. We first find a prefix cylinder contained in  $(R, S)$  whose residual carrier is internally infinite.

One possible construction changes, in  $R$ , a label  $x \in Y \setminus (\{a\} \cup T_R)$  to a larger unused label while retaining the prefix before  $x$ . The resulting cylinder stays above  $R$  and below  $S$ : the original next label  $a$  is retained. Because  $x$  is outside  $T_R$ , the unused carrier at that position is internally infinite, and remains so after choosing the replacement. A second possible construction changes, in  $S$ , a label  $x \in Y \setminus (\{b\} \cup T_S)$  to a smaller unused label. It has the symmetric properties.

At least one construction is possible. Otherwise, on the internally infinite class

$$Z = Y \setminus (\{a, b\} \cup T_R \cup T_S),$$

the restriction of  $R$  would equal the reverse numerical order, whereas the restriction of  $S$  would equal the numerical order. Both would be class well-orders. An infinite class cannot be well-ordered in both directions: successively choosing the least unused label in one direction produces an  $\omega$ -sequence descending in the other, again with a set range. This contradiction establishes the required cylinder. The first disagreements with both original endpoints have infinite residual carriers, so its members lie in blocks distinct from the endpoint blocks.

Let  $P$  be its fixed prefix and  $Y'$  its infinite residual carrier. Given any  $n \in \omega$ , choose an  $n$ -element set  $F \subseteq Y'$ . Well-order  $Y' \setminus F$  without a greatest element. If this carrier is a set, use an infinite initial cardinal as order type; if it is a proper class, use a set-like well-order of type  $\text{Ord}$ , available from Global Choice. Append that order after  $P$ , and then append  $F$  in any order. Its maximal finite terminal tail is exactly  $F$ . This construction works equally for a nonstandard internal  $n$ .

Returning to the metatheory, these statements give density and density of every indicated colour. Varying the first surreal label below or above a given first label gives a different coarse block, since the first residual is the entire internally infinite carrier. Thus the quotient has no endpoints. It is countable because the admitted class sort is externally countable.  $\square$

### 109.2.2 What an internally finite nonstandard interval looks like

The arithmetic ingredient is classical; compare the decomposition of countable nonstandard arithmetic models in [67]. We give the bounded-interval proof, since it identifies exactly which external blocks occur here.

*Note 109.17* (Source 20, Lemma 8.5, “The external type of a nonstandard finite interval”). *If  $N \in \omega^{\mathcal{M}}$  is nonstandard, then externally*

$$[0, N)^{\mathcal{M}} \cong I_{\infty}.$$

This is source 19’s lemma 109.6 for the half-open interval  $[0, N)$ , with the same midpoint argument; source 21’s note 109.32 is the third copy.

*Source 20’s proof.* Two points of this interval have a finite external interval between them exactly when their internal difference is a standard natural number. The block containing 0 consists of the standard natural numbers and has order type  $\omega$ . The block containing  $N - 1$  consists of

$$N - 1, N - 2, N - 3, \dots$$

and has order type  $\omega^*$ . These blocks are disjoint because  $N$  is nonstandard. A point outside them has nonstandard distance from both endpoints; adding and subtracting standard integers gives its entire finite-distance block, of type  $\mathbb{Z}$ .

The quotient of the remaining blocks is countable, dense, and has no endpoints. If  $a < b$  lie in different such blocks,  $b - a$  is nonstandard, and the model's integer midpoint  $\lfloor (a + b)/2 \rfloor$  has nonstandard distance from both. It therefore gives a strictly intermediate block. Given an interior block represented by  $a$ , the points  $\lfloor a/2 \rfloor$  and  $\lfloor (a + N - 1)/2 \rfloor$  give interior blocks respectively below and above it. Finally  $\lfloor N/2 \rfloor$  gives at least one interior block. The quotient is consequently isomorphic to  $\mathbb{Q}$ , proving the formula.  $\square$

*Source 20's proof of the isomorphism assertions in note 109.13.* If the model is  $\omega$ -standard, note 109.15 gives precisely the finite chains of sizes  $n!$ . The colours  $n = 0$  and  $n = 1$  both give the singleton type and can be merged. By note 109.16, the quotient is a densely coloured copy of  $\mathbb{Q}$ , proving the first formula.

If the model is  $\omega$ -nonstandard, standard  $n$  still give these same finite factorial chains. For every nonstandard  $n$ , the number  $n!$  is nonstandard, and note 109.17 identifies its block with  $I_\infty$ . Merge all these nonstandard colours into the one colour  $I_\infty$ . This colour is dense: choosing any one nonstandard  $n$  already gives a dense subcolour by note 109.16. The standard factorial colours remain dense. The countable colour-preserving back-and-forth now yields  $\sigma(\mathcal{F} \cup \{I_\infty\})$ .

The standard shuffle has only finite finite-distance blocks, whereas  $I_\infty$  contains an infinite such block. Thus the two external orders are not isomorphic. This conclusion needs no assumption that a class well-order in the model is externally well-founded.  $\square$

**Corollary 109.18** (External finite condensation). *In the nonstandard case, the external finite-distance blocks of  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  have exactly the following order types:*

$$1, 2, 6, 24, 120, \dots, \quad \omega, \quad \omega^*, \quad \mathbb{Z}.$$

*The uncoloured finite-condensation quotient is  $\mathbb{Q}$ . However, its block-type colours are not all dense: each coarse  $I_\infty$ -block contributes a closed quotient interval whose first point has colour  $\omega$ , whose last point has colour  $\omega^*$ , and whose interior is a rational order entirely coloured  $\mathbb{Z}$ .*

*Proof.* Inside a coarse standard block, the entire finite chain is a single finite-distance block. Inside a coarse nonstandard block, the proof of note 109.17 gives its bottom  $\omega$ -block, its top  $\omega^*$ -block, and its rationally indexed  $\mathbb{Z}$ -blocks. No finite-distance block crosses between coarse blocks: between any two distinct coarse blocks there are infinitely many others, by note 109.16. These are therefore all the possibilities.

Any two distinct points of the finite-condensation quotient either come from different coarse blocks, when coarse density supplies an intermediate point, or come from the same  $I_\infty$ -block, when note 109.17 supplies one. There are no endpoints and the quotient is countable, so it is  $\mathbb{Q}$ . The colour description follows from the same decomposition. In particular, replacing the classification in note 109.13 by a dense shuffle of the displayed finite-distance block types would be wrong.  $\square$

### 109.2.3 Elementary equivalence and an explicit elementary suborder

The new infinite block cannot be detected by a single first-order sentence because a sufficiently long finite factorial chain has the same bounded-depth order behaviour. We make this argument precise, including parameters.

*Note 109.19* (Source 20, Lemma 8.7, “The finite-chain threshold strategy”). *Put  $t_r = 2^r - 1$ . A chain of at least  $t_r$  elements that is finite externally, and an internally finite chain of at least  $t_r$  elements in  $\mathcal{M}$ , satisfy the same  $r$ -round Ehrenfeucht–Fraïssé game conditions in the pure order language. More generally, the Duplicator wins whenever every corresponding gap between previously*

*matched points has either equal size below  $t_r$ , or size at least  $t_r$  on both sides.* This is source 19's lemma 109.8, with the same strategy; source 21 uses it inside the proof of note 109.35.

*Source 20's proof.* Every nonempty gap in either chain has a least and a greatest point; for the internally finite chain this assertion holds externally because the internal endpoints are actual elements. Its size below any fixed standard finite bound is also absolute. Maintain the stated gap condition with  $r$  remaining moves. The assertion for  $r = 0$  is just preservation of the order of the selected points.

Suppose the next move divides a gap. If its size is less than  $t_r$ , the corresponding gap has exactly the same finite size, and the same-position response leaves identical gaps. Otherwise both gaps have size at least

$$t_r = 2t_{r-1} + 1.$$

If fewer than  $t_{r-1}$  unselected points lie on one side of the chosen point, match that finite distance from the corresponding endpoint. The other remaining gap is large enough on both sides. The two small-side cases cannot both occur. If neither occurs, choose a point with at least  $t_{r-1}$  unselected points on each side in the corresponding gap. Such a point exists by the displayed inequality, for example by counting  $t_{r-1}$  points from its left endpoint. The condition with  $r - 1$  remaining moves follows. Induction gives the strategy.  $\square$

*Note 109.20* (Source 20, Theorem 8.8, “An elementary standard part of the external order”).  
*Suppose  $\mathcal{M}$  is  $\omega$ -nonstandard, and let*

$$\mathfrak{W}_{\text{allstd}}^{\mathcal{M}} = \{R \in \mathfrak{W}_{\text{all}}^{\mathcal{M}} : n_R \text{ is a standard natural number}\}.$$

*This is a union of coarse blocks, and*

$$\mathfrak{W}_{\text{allstd}}^{\mathcal{M}} \cong \sigma(\mathcal{F}), \quad \mathfrak{W}_{\text{allstd}}^{\mathcal{M}} \preceq \mathfrak{W}_{\text{all}}^{\mathcal{M}}$$

*as pure linear orders. Standardness in this definition is an external condition, not a claimed internal predicate.* This is the first clause of source 19's corollary 109.10 (its  $L_{\text{fin}}$  is the present  $\mathfrak{W}_{\text{allstd}}^{\mathcal{M}}$ ), with the same quotient-matching proof; with note 109.13 it gives the countable case of source 19's theorem 109.9.

*Source 20's proof.* The standard factorial colours are dense in the coarse quotient, so restricting to them gives a densely coloured rational order. This proves the isomorphism assertion.

Fix a finite tuple of parameters in  $\mathfrak{W}_{\text{allstd}}^{\mathcal{M}}$  and a standard number  $r$  of game rounds. Pair every coarse block containing a parameter with itself, matching the whole finite chain identically. During the game maintain an order-preserving matching between the finitely many coarse blocks visited so far. Each paired pair of blocks is either two identical-size finite factorial chains, or one  $I_\infty$ -block and a finite factorial chain of size at least  $t_r$ .

If a move visits a new coarse block, choose a fresh matching block in the corresponding open interval of the other quotient, or in the corresponding ray. Every standard factorial colour is dense and unbounded there. Match a finite block by its same size. Match an  $I_\infty$ -block by any factorial size at least  $t_r$ . Start the local strategy of note 109.19 in the selected pair of blocks. If the move visits an already paired block, continue its local strategy. At most  $r$  moves occur in any one pair, and moves in different blocks are compatible because the quotient matching preserves order.

This supplies a winning strategy for every finite  $r$ , with the given parameters fixed. The Ehrenfeucht–Fraïssé criterion therefore gives the elementary inclusion. Together with the previous isomorphism results, this proves the elementary-equivalence assertion of note 109.13.  $\square$

*Note 109.21* (Source 20, Corollary 8.9, “A type that distinguishes the two countable orders”). *Let*

$$\text{Succ}(x, y) \iff x < y \wedge \neg \exists z (x < z < y).$$

*The partial one-variable type asserting that  $x$  begins a chain of  $k$  successive immediate successors, for every standard  $k$ , is finitely satisfiable but omitted in  $\sigma(\mathcal{F})$  and is realized in  $\sigma(\mathcal{F} \cup \{I_\infty\})$ .* This is the phenomenon of the second clause of source 19’s [corollary 109.10](#), with a different nonprincipal type: source 19’s type bounds the size of the class of  $x$  under its formula  $E$ , source 20’s counts immediate successors; source 20 adds the dual type and the points realizing both.

*Source 20’s proof.* Arbitrarily long finite factorial chains witness finite satisfiability. In the standard shuffle every point belongs to a finite block, and successive immediate successors cannot cross a coarse boundary, so no point realizes the whole type. In an  $I_\infty$ -block, every point of its bottom  $\omega$ -block realizes it. Dually there is a type asserting arbitrarily many successive immediate predecessors, realized in the top  $\omega^*$ -block. Every point in one of the middle  $\mathbb{Z}$ -blocks realizes both types simultaneously.  $\square$

**Corollary 109.22** (Decidability of the common pure-order theory). *The common first-order theory of the two orders in [note 109.13](#), in the language containing only  $<$ , is decidable.*

*Proof.* Given a sentence  $\varphi$ , compute its quantifier rank  $r$  and put  $t_r = 2^r - 1$ . Let  $s_r$  be the least factorial in  $\mathcal{F}$  that is at least  $\max\{1, t_r\}$ , and form the finite, effectively computable set

$$\mathcal{F}_r = \{s \in \mathcal{F} : s < t_r\} \cup \{s_r\}.$$

The quotient-matching strategy used in [note 109.20](#), together with [note 109.19](#), shows that both original orders are  $r$ -equivalent to  $L_r = \sigma(\mathcal{F}_r)$ . Indeed, pair a newly visited block of size less than  $t_r$  with the identical finite size, and pair any larger block, including  $I_\infty$ , with size  $s_r$ . Conversely, every block size appearing in  $L_r$  is available densely in the original order. All paired blocks are either isomorphic or above the local  $r$ -round threshold. It therefore suffices to decide whether  $L_r \models \varphi$ .

For completeness, this last decision is effective as follows. Consider a rational dense linear order  $D_r$  with unary predicates  $P_s$ , for  $s \in \mathcal{F}_r$ , partitioning it into dense colours. Let  $K = \max \mathcal{F}_r$ . Represent  $L_r$  as the pairs  $(q, i)$  with  $0 \leq i < K$  and

$$D_i(q) := \bigvee_{\substack{s \in \mathcal{F}_r \\ i < s}} P_s(q),$$

ordered lexicographically, first by  $q$  and then by  $i$ . This is an effective finite-tag interpretation: for fixed tags, equality translates to equality of the rational coordinates and equality of the tags, and

$$(q, i) < (q', j) \iff q < q' \vee (q = q' \wedge i < j).$$

Translate an existential quantifier over a point into the finite disjunction  $\bigvee_{i < K} \exists q (D_i(q) \wedge \dots)$ , translating the subformula with that tag fixed. Boolean operations and universal quantifiers are handled recursively. This produces a sentence in the finite language of  $D_r$ .

The theory of a dense linear order without endpoints partitioned into finitely many dense colours has effective quantifier elimination. To eliminate one existential quantifier, put the quantifier-free formula into disjunctive normal form. In each conjunction, the quantified point either equals a parameter, lies in an interval between parameters, or lies in an outer ray. Test the equality cases directly. For a nonempty interval or ray, existence is equivalent to compatibility of the finitely many colour conditions, since every permitted colour is dense and unbounded. Inconsistent order or colour

conditions give false. These tests are finite Boolean combinations of comparisons and colours of the parameters. Repeated elimination reduces the translated sentence to true or false and decides  $\varphi$ . The procedure concerns the pure-order reduct only; it does not decide the class theory or an expansion by evaluation or set-likeness predicates.  $\square$

Thus the two pure countable order types record whether the model's natural numbers are standard, but their first-order theories do not. Within either side of the dichotomy, the bare order loses all further information about the particular model and its admitted classes. In particular, it cannot recover the model's arithmetic, its class realization, or which labelled well-orders were originally supplied. These conclusions concern the reduct to  $<$ ; an expansion retaining set-likeness, label evaluation, internal permutation ranks, or the internal natural-number sort can retain substantially more information. The expanded structures remain separate research questions. No claim of elementary equivalence for those expansions is made here.

### 109.3 Source 21: finite tails, intrinsic blocks, and countable realizations

The largest-common-prefix comparison and the exact adjacency criterion are baseline results of the audited repository ([theorems 14.2](#) and [16.2](#)). We now derive a more complete description of the resulting finite blocks, their automorphisms, and their behavior in countable models. The proofs are included; no claim of historical priority is needed. A crucial qualification is that an internally finite block need not be externally finite in a nonstandard model.

Throughout this section, the carrier is  $X = \mathbf{No}$ , although the local arguments apply to any infinite linearly ordered carrier admitting the required well-orders. Write  $L = \mathfrak{W}_{\text{all}}(\mathbf{No})$  when speaking virtually. For a two-sorted model  $\mathcal{M} = (M, \mathcal{X}) \models \text{GBC}$ , write  $L^{\mathcal{M}}$  for the *external* collection of its admitted class well-orders, equipped with the externally interpreted lexicographic formula. Assertions involving quotient sets, automorphism groups, or topological derivatives are made in such an external realization. Their finite-variable ingredients remain formulas of class theory.

#### 109.3.1 The canonical terminal tail

The blocks in this section are blocks *of the lexicographic order of well-orders*. They must not be confused with the set-interval blocks inside an individual non-set-like class well-order.

**Definition 109.23** (Maximal finite terminal tail). For an admitted class well-order  $R$  on  $X$ , let

$$T_R = \{x \in X : \{y \in X : x \leq_R y\} \text{ is a finite set}\}.$$

When working in a model, “finite” in this formula means finite in that model. Put  $C_R = X \setminus T_R$ , let  $P_R = R \upharpoonright C_R$ , and write  $n(R) = |T_R|$ .

*Note 109.24* (Source 21, Lemma 10.2, “Existence of the maximal finite tail”). *The class  $T_R$  is a finite final segment, possibly empty. Its complement  $C_R$  is nonempty and has no  $R$ -greatest element. Moreover,  $n(R) = 0$  exactly when  $R$  has no greatest element. This is source 18's [lemma 58.1](#) (Part IX); source 21 obtains  $T_R$  by one comprehension instead of the predecessor recursion.*

*Source 21's proof.* Elementary comprehension defines  $T_R$ . If it is nonempty, let  $x_0$  be its  $R$ -least element. The final segment beginning at  $x_0$  is finite by definition, and every point of that segment also has a finite final segment. Minimality of  $x_0$  shows that this segment is exactly  $T_R$ . Since  $X$  is infinite,  $C_R$  is nonempty. If it had greatest element  $c$ , the final segment beginning at  $c$  would consist of  $c$  together with  $T_R$ , contradicting  $c \notin T_R$ . Finally, a nonempty finite final segment has a greatest element, while the greatest element of  $R$ , if present, belongs to  $T_R$ .  $\square$

Let  $B(R)$  consist of the orders obtained by retaining  $P_R$  as an initial labeled prefix and appending an arbitrary permutation of  $T_R$ . This family requires no choice of a class for each permutation. If  $n = n(R)$  and  $\pi : n \rightarrow T_R$  is a set bijection, its member is given by the single elementary formula

$$\begin{aligned} xR_\pi y \quad &\Longleftrightarrow \quad (x, y \in C_R \text{ and } xRy) \\ &\vee (x \in C_R \text{ and } y \in T_R) \\ &\vee (x, y \in T_R \text{ and } \pi^{-1}(x) < \pi^{-1}(y)). \end{aligned}$$

The set of bijections  $n \rightarrow T_R$  together with this formula defines one uniform class relation encoding the whole family. Thus ordinary GBC comprehension suffices; no class-choice principle is being used.

*Note 109.25* (Source 21, Theorem 10.3, “Exact terminal permutation blocks”). *Let  $E(R, S)$  mean that  $R = S$ , or that the complement of the maximal common labeled prefix of  $R, S$  is finite. Then*

$$E(R, S) \quad \Longleftrightarrow \quad S \in B(R).$$

*In particular,  $E$  is an equivalence relation. Its classes are convex chains of size  $n(R)!$ , ordered as the ordinary finite lexicographic permutation order; here  $0! = 1$ . Each class has a least and a greatest member. This is source 18’s [proposition 58.3](#) together with source 17’s [theorem 54.4](#) (Part IX), in the internal form of source 20’s [note 109.15](#); source 21 adds the single uniform formula  $R_\pi$  coding the whole block.*

*Source 21’s proof.* If  $S$  permutes  $T_R$  after  $P_R$ , the common-prefix residual is contained in  $T_R$  and is finite. Conversely, suppose  $R, S$  share an initial labeled prefix  $P$  with finite residual carrier  $F$ . Every point of  $F$  has finitely many successors in both orders. A point of  $P$  has finitely many successors in  $R$  exactly when it does in  $S$ : the restrictions to  $P$  agree, and both remaining carriers equal the same finite set  $F$ . Consequently  $T_R = T_S$ , and the orders agree off this set. This proves the equivalence and transitivity.

A fixed-prefix cylinder is convex: an order that first departs from the prefix compares with every completion of the prefix in the same direction. Hence  $B(R)$  is convex. Its uniform set codes are the permutations of the  $n(R)$ -element set  $T_R$ , and lexicographic comparison of the codes is exactly comparison of their appended finite words. The least and greatest append the labels in increasing and decreasing native order, respectively.  $\square$

*Note 109.26* (Source 21, Corollary 10.4, “Intrinsic finite condensation”). *In an  $\omega$ -standard model, or in an external realization where these tails are genuinely finite,  $E(R, S)$  holds exactly when the closed lexicographic interval between  $R$  and  $S$  is finite. Thus the possible finite-condensation block sizes are*

$$1, 2, 6, 24, 120, \dots$$

*For a nonsingleton block its intrinsic size  $n!$  recovers  $n \geq 2$ . Singleton blocks combine the possibilities  $n = 0$  and  $n = 1$ . This is the external reading of source 17’s [theorem 54.4](#) and source 18’s [proposition 58.3](#) (Part IX), with the same proof.*

*Source 21’s proof.* An  $E$ -class is finite and convex. Conversely, list a finite interval as  $R = T_0 < T_1 < \dots < T_k = S$ . Each consecutive pair is adjacent in the full order. The baseline adjacency theorem says that its common-prefix residual is finite. Transitivity of  $E$  gives  $E(R, S)$ . In a model this argument uses  $\omega$ -standardness precisely to turn the internally finite residuals supplied by that theorem into externally finite sets.  $\square$



### 109.3.2 A dense quotient with every factorial color

The following result uses the explicit internal equivalence  $E$ . It is valid internally even when  $\omega$  is nonstandard. Statements about a virtual quotient are abbreviations for their formulations using a finite number of class variables.

*Note 109.27* (Source 21, Theorem 10.5, “Density of each tail length”). *The quotient by  $E$  is dense and has no endpoints. Between any two distinct  $E$ -classes, for every  $n \in \omega$ , there is a class represented by an order with maximal finite tail of length exactly  $n$ . In particular, the tail lengths 0 and 1 are separately dense, although both give singleton blocks.* This is source 18’s lemma 58.5 (Part IX) and source 19’s note 109.5. Source 21’s proof differs: it passes to the greatest and least members  $R^+$ ,  $S^-$  of the two boundary blocks and uses their non-adjacency, instead of the two-change argument.

*Source 21’s proof.* Suppose  $B(R) < B(S)$ . Let  $R^+$  be the greatest member of  $B(R)$  and  $S^-$  the least member of  $B(S)$ . They cannot be adjacent, because the adjacency theorem would put them in the same  $E$ -class. Choose

$$R^+ < T < S^-.$$

The class of  $T$  differs from both boundary classes. This already proves density of the quotient.

For the prescribed tail length, compare  $T$  with each boundary. The two common prefixes are nested initial segments of  $T$ . Let  $C$  be the larger one and let  $t$  be the first  $T$ -label after  $C$ . The residual  $X \setminus C$  is infinite: otherwise  $T$  would be  $E$ -equivalent to the corresponding boundary. Thus

$$Y = X \setminus (C \cup \{t\})$$

is infinite. Every completion of  $T \upharpoonright (C \cup \{t\})$  remains strictly between  $R^+$  and  $S^-$ , because this prefix includes the two first disagreements determining those inequalities.

Choose an  $n$ -element set  $F \subseteq Y$  and a countably infinite set  $Z \subseteq Y \setminus F$ . One can obtain these sets from successive least elements of the restricted order  $T$ : ordinary set recursion with the class parameter and class replacement suffice. Let  $K = Y \setminus (F \cup Z)$ . Complete the fixed prefix by

$$(T \upharpoonright K) + \text{an } \omega\text{-ordering of } Z + \text{an ordering of } F.$$

Concatenation of these finitely many class well-orders is again a class well-order. The displayed  $\omega$ -part has no greatest element, so the maximal finite terminal tail of the result is exactly  $F$ . This construction is uniform in its set parameters and does not choose from a family of nonempty collections of classes.

Finally, the full lexicographic order has no endpoints: replace an order’s first label by a smaller or larger surreal and put the remaining labels afterward. Since every  $E$ -class has endpoints, the quotient also has no endpoints.  $\square$

The proof deliberately does not presume that native surreal density survives an arbitrary proper-class prefix. Such a prefix may already have exhausted all labels in a native open interval. What is used is a later completion cylinder with infinite residual carrier.

**Definition 109.28** (Rational shuffle). For a countable list of nonempty countable order types  $A_i$ , let  $\sigma(\{A_i\})$  denote the lexicographic sum over a copy of  $\mathbb{Q}$  partitioned into dense color classes  $Q_i$ , with each point of  $Q_i$  replaced by a copy of  $A_i$ . Back-and-forth for countably colored dense orders shows that its isomorphism type does not depend on the chosen dense partition.

*Note 109.29* (Source 21, Corollary 10.7, “Countable standard realization”). *If  $\mathcal{M} \models \text{GBC}$  is externally countable and  $\omega$ -standard, then*

$$L^{\mathcal{M}} \cong \sigma(\{1, 2, 6, 24, 120, \dots\}).$$

This is the first clause of source 18’s [theorem 58.6](#) (Part IX), with the same proof.

*Source 21’s proof.* By [notes 109.26](#) and [109.27](#), the external condensation quotient is countable dense without endpoints, with every listed block size dense. The colored back-and-forth theorem identifies its quotient with the corresponding colored rational order, and the finite blocks then determine the lexicographic sum.  $\square$

### 109.3.3 Automorphism lifting and a failure of reconstruction

*Note 109.30* (Source 21, Theorem 10.8, “Automorphisms of finite blocks”). *In the setting of [note 109.26](#), let  $D$  be the external finite-condensation quotient and color  $d \in D$  by its block size  $c(d)$ . Then*

$$\text{Aut}(L, <_{\text{lex}}) \cong \text{Aut}(D, <, c).$$

*The induced map on the quotient determines the automorphism uniquely.* This is the automorphism clause of source 18’s [corollary 58.9](#) (Part IX), with the same proof; the orbit description in the next paragraph is its orbit clause.

*Source 21’s proof.* An increasing automorphism preserves finiteness of closed intervals, hence preserves the condensation equivalence and the size of each class. Conversely, a color-preserving quotient automorphism lifts by the unique increasing bijection between each finite block and its target. The union of these bijections is an increasing bijection of  $L$ . Uniqueness follows from rigidity of finite chains.  $\square$

In the countable standard realization, two points are in the same automorphism orbit exactly when their blocks have the same size and they have the same rank inside those blocks. Sufficiency follows by extending a one-point color-preserving map on the rational quotient; necessity follows from [note 109.30](#).

**Corollary 109.31** (A greatest element is not intrinsic). *In a countable  $\omega$ -standard realization, there is an automorphism of the pure lexicographic order taking a represented well-order with no greatest element to one with a greatest element. Such an automorphism cannot be induced by any fixed relabeling bijection of the surreal carrier.*

*Proof.* By [note 109.27](#), choose orders of tail lengths 0 and 1. Both are singleton blocks, and colored rational-order homogeneity maps one block to the other. Lift that map using [note 109.30](#). [Note 109.24](#) distinguishes existence of a greatest element in the represented orders. Relabeling preserves the abstract order type of every represented well-order and therefore cannot have this effect.  $\square$

[merge] The automorphism is the one used in the proof of source 18’s [corollary 58.10](#), which exchanges a set-like order with one of type  $\text{Ord} + 1$ ; the clause on relabelings is source 21’s.

This is a concrete limit on recovering enumeration structure from the pure lexicographic order. It also answers an automorphism question more strongly than merely exhibiting numerical relabelings. The countable homogeneity argument must not be silently transferred to arbitrary uncountable class realizations.

### 109.3.4 The nonstandard correction

External countability does not imply  $\omega$ -standardness. Consequently, an unconditional claim that every countable realization has the finite-factorial shuffle type would be false. The correction has an exact form.

*Note 109.32* (Source 21, Lemma 10.10, “An externally infinite finite interval”). *Let  $\mathcal{M}$  be externally countable, and let  $N \in \omega^M$  be nonstandard. The external order on the internal arithmetic interval*

$$I_N = \{i \in \omega^M : i <^M N\}$$

*has type*

$$I_\infty = \omega + \sum_{q \in \mathbb{Q}} \mathbb{Z} + \omega^*.$$

This is source 19’s [lemma 109.6](#) and source 20’s [note 109.17](#), with the same midpoint argument.

*Source 21’s proof.* Externally declare  $i \sim j$  if their internal arithmetic difference is at most some *standard* natural number. The class of 0 is the standard natural numbers and has type  $\omega$ . The class of  $N - 1$  consists of  $N - 1 - k$  for standard  $k$  and has type  $\omega^*$ . Every other class has type  $\mathbb{Z}$ : it consists of the elements  $i + k$  for standard integers  $k$ , and both distances to the endpoints are nonstandard.

The quotient of the interior  $\mathbb{Z}$ -classes is nonempty, countable, dense, and has no endpoints. If  $i < j$  represent distinct classes,  $j - i$  is nonstandard, so the internally computed midpoint  $\lfloor (i + j)/2 \rfloor$  represents a strictly intermediate class. For an interior  $i$ , the element  $\lfloor i/2 \rfloor$  gives a class strictly between the initial class and that of  $i$ ; the symmetric construction works near the upper endpoint. The interior quotient therefore has type  $\mathbb{Q}$ . Taking the ordered sum of its classes proves the formula.  $\square$

*Note 109.33* (Source 21, Theorem 10.11, “Exact countable nonstandard realization”). *If  $\mathcal{M} \models \text{GBC}$  is externally countable and  $\omega$ -nonstandard, then*

$$L^{\mathcal{M}} \cong \sigma(\{1, 2, 6, 24, 120, \dots, I_\infty\}), \quad I_\infty = \omega + \sum_{q \in \mathbb{Q}} \mathbb{Z} + \omega^*.$$

*In particular it is not isomorphic to the order in [note 109.29](#).* This is the  $\omega$ -nonstandard clause of source 19’s [theorem 109.1](#) and of source 20’s [note 109.13](#); the nonisomorphism is proved by the same finite-condensation invariant.

*Source 21’s proof.* Use the external interpretation of the internal equivalence  $E$ . [Note 109.25](#) states internally that each class is the finite permutation order on  $n$  labels for some  $n \in \omega^M$ . Its ordinary lexicographic rank gives an internal order isomorphism with  $I_{n!}$ . The uniformly coded family preceding [note 109.25](#) includes every such permutation. Conversely, a class order on an internally finite set is a set relation, hence corresponds to one of these permutation codes. Thus this internal isomorphism is also an external bijection between the entire  $E$ -class and  $I_{n!}$ .

For standard  $n$  the block is the actual finite chain of size  $n!$ . For nonstandard  $n$ , its internal factorial is nonstandard, and [note 109.32](#) gives the external type  $I_\infty$ . The external quotient by  $E$  is countable dense without endpoints by [note 109.27](#). Every standard factorial color is dense. Fix one nonstandard  $n$ ; the same internal theorem says that its prescribed tail length occurs between every two distinct classes, so the external color  $I_\infty$  is also dense. Colored back-and-forth proves the asserted shuffle representation.

The  $I_\infty$  blocks have externally infinite finite-condensation classes of types  $\omega$ ,  $\mathbb{Z}$ , and  $\omega^*$ . None joins a neighboring  $E$ -block: the endpoints of a maximal terminal permutation block have no immediate neighbors outside it, by the internal adjacency theorem. The finite-factorial shuffle has only finite condensation classes. This intrinsic distinction proves nonisomorphism.  $\square$

*Remark 109.34* (Exactly what is being transferred). The preceding theorem does not assume that every admitted class well-order is externally well-founded. Internal well-orderhood and the proved internal formulas supply the external linear order, its explicit  $E$ -blocks, and the dense-color witnesses. It is precisely the external notion of finite condensation that changes in a nonstandard model. Calling its  $I_\infty$  blocks “finite-condensation blocks” externally would therefore be incorrect.

*Note 109.35* (Source 21, Proposition 10.13, “Different orders, the same elementary theory”). *The two countable orders*

$$A = \sigma(\{1, 2, 6, 24, \dots\}), \quad B = \sigma(\{1, 2, 6, 24, \dots, I_\infty\})$$

are elementarily equivalent in the pure language  $\{<\}$ , although they are not isomorphic. This is the countable case of source 19’s [theorem 109.9](#) and the elementary-equivalence clause of source 20’s [note 109.13](#). Source 21’s strategy differs: it plays an  $I_\infty$ -block against a single finite factorial block of size at least  $2t_r + 1$  instead of merging large colours.

*Source 21’s proof.* We give a winning strategy for every finite Ehrenfeucht–Fraïssé game. The displayed shuffle blocks are an auxiliary device for the strategy; no block predicate or color is added to the language. With  $r$  rounds remaining, put  $t_r = 2^r - 1$ . Match the finitely many blocks containing selected points in quotient order. Inside every matched pair, require each corresponding gap between selected points, including the initial and final gaps, to have either the same size less than  $t_r$ , or at least  $t_r$  points on both sides.

The standard finite-chain splitting argument preserves this invariant. Indeed,  $t_r = 2t_{r-1} + 1$ . In equal short gaps copy the position exactly. In a pair of long gaps, if the new point leaves fewer than  $t_{r-1}$  points on one side, copy that distance from the corresponding endpoint; the other side still has at least  $t_{r-1}$  points. If both resulting sides are long, choose a point leaving at least  $t_{r-1}$  points on each side. This works in  $I_\infty$  as well: every nonempty gap determined by finitely many selected points has a least and a greatest element, and successor or predecessor steps realize every required finite endpoint distance.

When a move enters a previously unmatched finite block, choose a fresh block of exactly the same size in the other quotient interval. Such blocks occur densely, and one can copy the selected position. When a move enters a previously unmatched  $I_\infty$  block, with  $r$  rounds left after this move, choose a fresh finite factorial block of size at least  $2t_r + 1$ . Match the selected point by copying a side of size less than  $t_r$ , if present; otherwise leave at least  $t_r$  points on both sides. The dense factorial colors always supply the required block in the correct quotient interval. Existing matched gaps satisfy the weaker invariant for the new number of remaining rounds.

After the last round the matching preserves order and equality. Thus the strategy wins every finite game, proving elementary equivalence. Nonisomorphism was proved by the external finite-condensation invariant in [note 109.33](#).  $\square$

[merge] The following group description of the nonstandard order is source 21’s alone.

There is also a useful group description. In the nonstandard countable order, let  $F$  be the set of points having an externally finite condensation class. Its convex components are the standard factorial blocks, and the convex components of its complement are the  $I_\infty$  blocks. Indeed, between any two distinct  $E$ -classes there are blocks of both kinds. Thus this partition is intrinsic, even

though individual nonstandard tail lengths are forgotten. After choosing identifications of its  $I_\infty$  blocks, one obtains

$$\text{Aut}(L^{\mathcal{M}}) \cong \left( \prod_{q \text{ colored } I_\infty} \text{Aut}(I_\infty) \right) \rtimes \text{Aut}(\mathbb{Q}, <, \text{colors}).$$

The kernel and quotient are intrinsic; the displayed splitting uses the chosen identifications. Similarly,

$$\text{Aut}(I_\infty) \cong \left( \prod_{q \in \mathbb{Q}} \mathbb{Z} \right) \rtimes \text{Aut}(\mathbb{Q}, <).$$

To verify the last formula, the initial  $\omega$  and final  $\omega^*$  condensation classes are rigid. The interior quotient is  $\mathbb{Q}$  and each of its classes is  $\mathbb{Z}$ , whose increasing automorphisms are exactly integer translations. A quotient automorphism and one translation per class give all automorphisms.

### 109.3.5 One derivative removes all permutation interiors

Let  $L'$  denote the set of nonisolated points in the external order topology. The topology here belongs to the external order of admitted class relations; it does not require a class of classes inside GBC.

**Theorem 109.36** (The first derivative). *For an external model realization as above,  $L'$  consists exactly of the singleton  $E$ -blocks and the two endpoints of every nonsingleton  $E$ -block. It has no isolated points, so*

$$L'' = L'.$$

*The isolated points of  $L$  are dense, and  $L'$  is closed and nowhere dense. In either countable case,*

$$L' \cong \sigma(\{1, 2\}).$$

*Proof.* In a linear order without endpoints, a point is isolated exactly when it has both an immediate predecessor and an immediate successor. The internal finite permutation chain has these two neighbors at every rank other than its endpoints, including nonstandard internal ranks. These remain actual neighbors externally. At an endpoint there is no neighbor on the exterior side: such an adjacency would extend the maximal terminal block. Thus precisely the stated points survive.

The quotient by  $E$  is dense without endpoints. Consequently every neighborhood, within  $L'$ , of a retained endpoint or singleton contains retained points from other blocks. This proves  $L'' = L'$ . By the density of the color 6, every neighborhood of a retained point also contains an interior point of a six-element block, which is isolated in  $L$ . The isolated points are therefore dense. Their union is open, so  $L'$  is closed; the same density makes its interior empty.

Finally, in a countable realization the dense quotient contains densely many singleton blocks and densely many nonsingleton blocks. After the latter are replaced by their endpoint pairs, colored back-and-forth yields  $\sigma(\{1, 2\})$ .  $\square$

Thus the first derivative forgets the distinction between the two countable raw orders, although external finite condensation detects it. For uncountable realizations the principal remaining invariant is the colored quotient: its isomorphism type, saturation, and automorphism group need not be fixed by the preceding local arguments.

### 109.4 Source 30: A compactness observation about finite condensation

This observation concerns ordinary set-sized linear orders, for example the external countable orders in a countable class-model realization. It does not treat a virtual collection of classes as the domain of an ordinary first-order structure.

**Proposition 109.37** (Finite factorial blocks are not first-order protected). *Let  $L$  be a linear order containing finite adjacent chains of arbitrarily large length. Then the complete first-order theory  $\text{Th}(L, <)$  has a countable model with an infinite finite-condensation class. In particular, if every finite-condensation class of  $L$  is finite, that property is not preserved by elementary equivalence.*

*Proof.* Expand the language by constants  $c_i$  for  $i \in \mathbb{Z}$ , and add

$$c_i < c_{i+1} \quad \text{and} \quad \neg \exists z (c_i < z < c_{i+1}) \quad (i \in \mathbb{Z})$$

to  $\text{Th}(L, <)$ . Every finite subset of these axioms is realized by placing the finitely many relevant constants consecutively inside a sufficiently long adjacent chain of  $L$ . Compactness gives a model of the complete theory with all these constants. Downward Löwenheim–Skolem gives a countable such model. Every interval  $[c_i, c_j]$  for finite integers  $i < j$  consists of the finitely many named successive points. Hence all the  $c_i$  belong to one finite-condensation class, which is infinite.  $\square$

Factorial terminal blocks supply the hypothesis because  $n!$  is unbounded in  $\omega$ . Thus an external classification as a dense shuffle of finite factorial chains cannot by itself serve as a first-order characterization of lexicographic well-order spaces. It requires the external structural condition or a suitable infinitary language. This conclusion does not contradict an exact classification within a previously specified family of countable class-model realizations: compactness may produce orders outside that family.

[merge] Source 30’s proposition is classical compactness, and its source does not cite the other countable-model manuscripts. Source 19’s [theorem 109.12](#) constructs such models of the common theory explicitly, one for every countable order  $A$ , and the  $\omega$ -nonstandard GBC realizations of [theorem 109.1](#) are models of the same theory with infinite finite-condensation classes. Together they show why the external finiteness condition in source 18’s [theorem 58.6](#) cannot be replaced by a first-order axiom, as source 19’s paragraph after [proposition 109.7](#) also says.

## 110 Foundations: full rank semantics and well-foundedness (source 27)

[merge] Source 27’s foundations section, printed whole. Its four levels of discourse and its account of ETR agree with the foundational sections of Parts V and IX ([part V](#), [section 57](#)); new is [theorem 110.1](#), which states as an equivalence what this report used only in one direction: the full structure  $(V_\kappa, \mathcal{P}(V_\kappa))$  satisfies GB exactly when  $\kappa$  is strongly inaccessible, and then satisfies KM + GC (sufficiency is used by source 08’s model (S08.12) and source 09’s [theorem 22.1](#)). Both directions are standard; source 27 does not claim novelty for them. Its ledger refers to results of its other sections, which this addition prints in other Parts.

### 110.1 Source 27: foundations that avoid size paradoxes

#### 110.1.1 Four safe levels of discourse

**A fixed set.** For  $X = \mathbf{No}_{<\theta}$  or any other set of labels,  $\mathcal{W}(X)$  is a subset of  $\mathcal{P}(X^2)$ . All relation-table spaces, all their subsets, their order topologies, and their Dedekind cuts are ordinary sets. This



is the cleanest environment for cardinal spectra, topology, computation, and first formalization milestones. An infinite birthday cutoff does not require an inaccessible cardinal.

**Definable classes in set theory.** In ZFC, the sign-sequence description defines a proper class **No**. One can state a schema about each formula defining a class relation, prove uniform constructions from specified formula parameters, and quantify over set-coded words. One cannot quantify over all arbitrary classes as objects of ZFC. Syntax codes do not erase this distinction: a universal satisfaction predicate for the universe is not supplied by the set of formula codes.

**Two-sorted class theory.** In GBC, individual class relations are variables of the second sort. Formulas such as  $R <_{\text{raw}} S$  and statements quantifying over class well-orders are legitimate. A uniformly given family has one graph  $\mathcal{R}(i, x, y)$  with set indices  $i$  in an ordinary class. This setting supports the comparison, residual maps, prefix automorphisms, and diagonal theorem proved above.

**An external model or a higher type.** If  $(M, \mathcal{X})$  is a set-sized model in a surrounding theory, its admitted classes are elements of the external set  $\mathcal{X}$ . The external collection of admitted well-orders is then a set, with an ordinary topology and automorphism group. Alternatively, a higher-order theory can explicitly add a sort for collections of classes. Either route must specify its semantics and comprehension principles; neither follows merely from writing braces around a class-variable predicate.

Kelley–Morse theory strengthens the permitted comprehension formulas, but its classes still have sets as members. It does not automatically turn an arbitrary collection of classes into a class. Work on hyperclass forcing makes such higher-level distinctions explicit [28]; that literature is context for the additional type level, not an axiom needed by the theorems here.

### 110.1.2 Why Global Choice is visible

Global Choice provides an ordinal enumeration of every proper class, including **No**. Its standard equivalent formulations distinguish it from ordinary set Choice [19]. In this particular setting, the repository proves the stronger converse: over GB, the existence of any class well-order of **No** implies Global Choice (theorem 13.1). Thus the baseline is not innocent notation.

For perspective, the set-code surjection  $d : \mathbf{No} \rightarrow V$  used above explains the mechanism once set Choice is available. A well-order of **No** lets one choose the least code of each set and hence obtain a global choice procedure. The choiceless version in the repository also establishes the required set Choice by set-valued rank recursion. We do not reclassify that established equivalence as an open question.

Set-sized results in sections 84.1 and 101 need much less. The explicit increasing-word split needs no choice at all; its homogeneous class-order identifications use Global Choice. The support optimum is a pointwise theorem about well-orderable sets of ordinals; only its use as a uniform completion procedure requires a fixed selection mechanism.

### 110.1.3 Set-state recursion and genuinely class-valued recursion

An ordinal-length process whose state and earlier history are sets at every set stage is available by ordinary recursion with class parameters. One constructs each set-length run, proves uniqueness, and takes its elementarily definable union. The residual isomorphism lemma follows this pattern. The fact that the final graph is a proper class does not make each intermediate state a class.

Elementary transfinite recursion, ETR, addresses stronger situations in which individual stages can themselves be classes and later stages refer to entire earlier class slices. Truth-predicate and class-game examples illustrate this distinction in the primary literature [22, 27]. We use no inference from “a recursion has length Ord” to “it requires ETR.” Conversely, we do not justify a simultaneous

choice of class isomorphisms merely by calling it ordinary recursion. The uniform graph construction in [lemma 96.1](#) is the required justification.

#### 110.1.4 Full rank models: an exact inaccessible-cardinal criterion

An inaccessible cardinal is a convenient way to replace the whole two-sorted universe by a full rank model. Its precise role can be stated as an equivalence.

**Theorem 110.1** (Full rank semantics). *In ambient ZFC, the full two-sorted structure*

$$(V_\kappa, \mathcal{P}(V_\kappa))$$

*satisfies GB, with the usual set part including Infinity, if and only if  $\kappa$  is strongly inaccessible. In that case it also satisfies KM with Global Choice.*

*Proof.* Suppose the full structure satisfies GB. Its set part makes  $\kappa$  a limit ordinal above  $\omega$ . If  $\text{cf}(\kappa) < \kappa$ , an external cofinal map  $f : \text{cf}(\kappa) \rightarrow \kappa$  is an available full class, with domain a set of  $V_\kappa$ . Its range has unbounded rank below  $\kappa$  and is not a set of  $V_\kappa$ , contrary to Class Replacement. Hence  $\text{cf}(\kappa) = \kappa$ , so  $\kappa$  is a regular cardinal.

If  $2^\lambda \geq \kappa$  for some cardinal  $\lambda < \kappa$ , ambient Choice gives a surjection  $\mathcal{P}(\lambda) \rightarrow \kappa$ . Its graph is also a full class, and its domain belongs to  $V_\kappa$ . Class Replacement would put its range  $\kappa$  in  $V_\kappa$ , again impossible. Thus  $\kappa$  is an uncountable regular strong limit cardinal.

Conversely, at a strongly inaccessible  $\kappa$ , every set in  $V_\kappa$  has external cardinality below  $\kappa$ . Any full class function on such a set has fewer than  $\kappa$  values, whose ranks are bounded below  $\kappa$  by regularity. Its range is therefore in  $V_\kappa$ . Standard rank closure supplies the set axioms, and this argument supplies Class Replacement. Every subset is admitted as a class, so even impredicative class comprehension holds. An external well-order of  $V_\kappa$  supplies a Global Choice class.  $\square$

The theorem characterizes *full rank initial-segment models*. It does not say that all Henkin models of GBC require inaccessible cardinals, nor that a set of surreals cut off at an arbitrary birthday is invalid. A singular birthday fragment is a legitimate set order; it is generally not the set part of a full class-theoretic universe.

#### 110.1.5 Internal versus external well-foundedness

There are three separate assertions: a relation is well-founded on sets of a model, it is well-founded on that model's admitted classes, and it is well-founded externally. Under GBC the first two agree, by the descending-sequence argument, but they need not agree with the third. Even transitivity of the set part is not by itself the full correctness condition for class well-foundedness [[27](#), Section 3].

For shared relations in two same-set GBC expansions, their internal well-foundedness agrees because their set-coded descending sequences agree. Their admitted relations may still differ, which is precisely what [note 102.5](#) detects. The distinction prevents two common errors: assuming that adding classes must destroy old internal well-foundedness, and assuming that an internal ordinal enumeration can always be traversed as an externally well-founded sequence.

#### 110.1.6 An axiom and object ledger

| Construction or theorem                            | Sufficient setting and object type                                                             |
|----------------------------------------------------|------------------------------------------------------------------------------------------------|
| All well-orders on a fixed set $X$                 | ZFC; a set of set relations.                                                                   |
| Terminal-cut factorization                         | ZF or elementary class definitions; a map on set-coded words.                                  |
| Pointwise minimum ordinal support                  | ZF with definable-class notation; all matching sets are well-orderable.                        |
| Direct raw labelled comparison                     | GBC; an elementary formula on two class parameters. No hyperclass is formed.                   |
| Uniform residual isomorphisms and branch transport | GBC with a baseline; one graph on set tuples, assembled from set runs.                         |
| Omitted convex copy for every uniform family       | GBC; diagonalization on an ordinary class index, with a formula on residual class orders.      |
| Dense core in raw relation-table order             | GBC and a set-like scan; an ordinary class of set codes.                                       |
| Marked-model embedding criterion                   | An external theorem about two set-sized GBC models with common sets and baseline.              |
| All subsets and topology of an admitted order      | A surrounding universe where the admitted class part is a set, or an explicitly stronger type. |
| Full $(V_\kappa, \mathcal{P}(V_\kappa))$ semantics | Exactly an ambient strongly inaccessible $\kappa$ for the usual GB axioms.                     |

## 111 Foundations, formalization plans and ledgers of sources 19–30

[merge] (3 October 2026.) Each batch-83 source states its foundations, a formalization plan tied to the repository’s Lean files, a ledger of results and assumptions, and a record of its checks and audit. They are printed here, whole and in source order, together with the baseline sections that no Part prints: source 19’s core and raw comparison (its Sections 3.1 and 4), source 20’s size-safe toolkit, source 23’s embedding spectra and core, and the class-comparison sections of sources 27 and 30. Their re-proofs of Parts I–IV are printed as numbered notes with the source’s proof kept; nothing in them is new against this report. Five sources’ foundations sections are printed where their mathematics is: source 24’s, source 26’s and source 28’s in Part X, source 25’s size and recursion audit in Part XIII, source 27’s foundations (with its full-rank criterion) in Part XIV. The sources’ research questions are in [section 113.12](#); the shipped finite checks and how to rerun them are in [section 111.49](#).

Every Lean declaration that these sections name as existing was checked on 3 October 2026 against `Algebra/SurrealNumbers/Surreal/Foundations/`: `SignSequence`, `SignSequence.lt_iff` (proved by `Iff.rfl`), `birthdays_bounded`, `small_bounded`, `not_small`, `IsPrefix`, `ordinalOrderEmbedding`, `ofSigns`, `simpler_wellFounded` (`SignSequence.lean`); `small_cut_fillers`, `exists_cut_separator_of_birthday_lt` (`SignSequenceCut.lean`); `cut`, `cut_realizes`, `birthday_cut_le_iSup`, `exists_separator_of_small_sets` (`SignSequenceCutOperation.lean`; a variant `HasSmallCutFillers.exists_separator_of_small_sets` and `ofSmallSets` are in `SmallCutData.lean`); `SmallCutData`, `HasSmallCutFillers`, `noUnrestrictedCuts` (`SizeObstructions.lean`); `isPrefix_of_minimum_birthday`, `existsUnique_prefix_of_ordConnected`, `existsUnique_minimum_birthday_of_ordConnected`, `ordConnected_separators` (`SignSequenceSimplicity.lean`) and `existsUnique_simplest_separator`, whose existence hypothesis the sources describe correctly. All exist, at the blobs the sources quote. Every other module or declaration

name in these plans (for example `WellOrders/ExhaustionTree`, `GroupEnvelope`, `ProfileExtension`) is a proposal of its source, as each source says; none exists, and none of this report's statements is formalized (section 1.5).

*Note 111.1* ([merge]; the class-theoretic conventions the twelve repeat). All twelve sources fix the same conventions, which are this report's (sections 1.3 and 1.3.1): GB has a ZF set part and no choice,  $GBC = GB + GC$ , and set-sized results are proved in ZFC (sources 19–30, in the subsections printed here and in section 61); a collection of class relations is a predicate on class variables, not a class (all twelve); a uniform family is one class relation with set indices (sources 20, 21, 22, 24, 27, 28, 29 with definition 80.2, 30); recursion with set values and set-sized histories is available in GBC, while class-valued recursion is ETR and is not used (sources 19, 20, 21, 22, 23, 25, 27, 29, 30, citing [22, 24]); in GBC a class well-order is well-founded exactly when it has no set-coded descending  $\omega$ -sequence, so well-foundedness of a fixed relation is absolute between class realizations with the same sets (sources 20, 22, 27, 30; the Hilbert's-tenth correction recorded in section 5.6), while internal and external well-foundedness differ (all twelve); KM does not make classes members of classes (sources 19, 20, 21, 22, 28, 29, 30); and an inaccessible full model  $(V_\kappa, \mathcal{P}(V_\kappa))$  is an optional external illustration (sources 19, 20, 21, 22, 23, 24, 28, 29), as in theorem 22.1. Each source's own wording is kept below.

### 111.1 Source 19: set-like global enumerations and their core

[merge] Source 19's Section 3.1, printed whole (the rest of its Section 3 is a note in Part X). Its core  $C$  is this report's core in the support coding,  $\mathcal{P}_s(e)$ ; (S19.2) is corollary 19.15 with the separation property of theorem 19.10.

Under Global Choice fix a set-like enumeration  $e : \text{Ord} \rightarrow \mathbf{No}$ . A bounded-support modification permutes some initial set of positions and agrees with  $e$  thereafter. Choose one canonical set code for each such modification. These codes form an ordinary class  $C$  of sets, with an evaluation relation that produces the associated global order. The earlier reports prove

$$C \cong \mathbf{No}, \quad C \text{ separates any two distinct exhaustive set-like enumerations.} \quad (\text{S19.2})$$

They also prove the stronger parameterwise interpolation statement for every uniformly set-indexed cut of set-like enumerations [39, §19] ([merge] theorem 19.10). Uniformity means that the family is supplied by one class relation with a set index; it does not mean an arbitrary external set of class objects.

One elementary fact will be used repeatedly. Every injective set prefix  $p : \alpha \rightarrow \mathbf{No}$  extends to a member of  $C$ : choose a large initial segment of  $e$  containing all labels in  $p$ , put the prefix first, and reorder the remaining set of labels before returning to the unchanged tail. If a proper class word  $F : \text{Ord} \rightarrow \mathbf{No}$  is injective, then at every set stage there are members of  $C$  on both sides of  $F$  sharing that prefix. Choose their next labels slightly below and above  $F(\alpha)$  and avoid the used set.

### 111.2 Source 19: raw class comparison and canonical blocks

[merge] Source 19's Section 4, printed whole; its two propositions are results of Parts III and IX and are recorded as notes.

An admitted class well-order is a strict linear class order whose restriction to every nonempty admitted subclass has a least element. In GBC one may equivalently use the usual well-foundedness test on sets; this is a statement inside the class model, not an assertion about all external subsets [27, §3].

**Definition 111.2** (Common labelled initial segment). For class well-orders  $R, S$  of the same labelled carrier  $X$ , let  $C(R, S)$  consist of the labels  $x$  for which

$$\text{Pred}_R(x) = \text{Pred}_S(x), \quad R \upharpoonright \text{Pred}_R(x) = S \upharpoonright \text{Pred}_S(x).$$

Here  $\text{Pred}_R(x) = \{y : yRx\}$ . Each equality expands to a first-order formula with  $R, S$  as class parameters.

*Note 111.3* (Source 19, Proposition 4.2, “Elementary comparison; prior result”). If  $R \neq S$ , the class  $C = C(R, S)$  is a proper initial segment of both orders, the orders agree on it, and their first labels outside  $C$  are different. Define  $R <_{\text{lex}} S$  when the first remaining  $R$ -label is numerically smaller than the first remaining  $S$ -label. This is a linear order, expressed as a formula on class parameters in GB. This is [theorems 14.2](#) and [14.7](#); source 19 calls it a prior result.

*Source 19’s proof.* If a label satisfies the defining property, every predecessor does too. Thus  $C$  is initial and the restrictions agree. Every common labelled initial segment is contained in  $C$ . If  $C = X$ , the relations agree everywhere. Otherwise take the least remaining label of each order. The predecessor class of each is exactly  $C$ ; equality of the two labels would put that label in  $C$ , a contradiction.

For transitivity, the common prefixes of  $R, S$  and of  $S, T$  are initial segments of  $S$ , so one contains the other. If containment is proper, the earlier comparison decides  $R$  against  $T$  with the same sign. If the prefixes coincide, compare the three next numerical labels; the transitivity of the label order applies. Trichotomy and asymmetry follow directly. No recursive matching of different labels and no class of equivalence classes is used.

*Note 111.4* (Source 19, Proposition 4.3, “First block and full block decomposition; prior results”). Let  $R$  well-order a proper class  $X$ . Its labels with set-sized predecessor class form a proper-class initial segment, possibly all of  $X$ , of type Ord. More generally, declaring  $x \sim_R y$  when the closed  $R$ -interval between them is a set gives convex blocks of type Ord, with at most one final set-sized block. In GB,

$$R \cong (J_R \times \text{Ord})_{\text{lex}} + \theta_R,$$

where  $J_R$  is represented by the actual least labels of the proper blocks and  $\theta_R$  is a set ordinal. This is the first block of [theorem 15.1](#) and the canonical block decomposition of Part IX (source 15’s [theorem 52.3](#); also [theorems 52.6](#) and [52.11](#)); source 19 calls both prior results.

*Source 19’s proof.* The first claim follows by enumerating each set predecessor segment by its unique ordinal rank. This segment cannot stop after only a set of labels: the least remaining label would also have set predecessors. Every ordinal rank occurs, since a missing rank would yield a first missing stage after a set initial segment. This yields its class isomorphism with Ord by elementary comprehension.

For the full decomposition, set intervals are closed under finite unions, proving transitivity of  $\sim_R$ . Each block has a least label by class well-ordering. Within a block every predecessor segment from that least label is a set. Thus its type is a set ordinal or Ord. A set block followed by another label would unite with the next label to remain a set interval, contradicting maximality. Hence a set block can only be final. The ranks within each proper block and the least block labels give the displayed isomorphism. This simultaneous construction does not iterate a class-valued quotient operation.

The quotient  $J_R$  need not be simpler than  $R$  up to isomorphism. Earlier work gives fixed-point examples [\[44, 46\]](#) ([merge] sources 16 and 18, Part IX). Consequently this decomposition alone does not justify a terminating recursive classification.

*Remark 111.5* (Existence and choice). Over GB without set choice, the existence of any class well-order of **No** is equivalent to Global Choice [39, Theorem 13.1] ([merge] [theorem 13.1](#)). The forward implication uses its restrictions to the binary sign levels to well-order every  $\mathcal{P}(\alpha)$  uniformly, and then constructs rank-wise well-orders of the universe by set recursion. In this article the global constructions simply state GBC as their hypothesis.

### 111.3 Source 19: formalizations that preserve the intended subject

The foundational issue is not repaired merely by replacing the word “set” with “class.” A class theory has its own sorts and its own comprehension principles. The earlier sections illustrate three distinct tasks: comparing supplied class relations, constructing one class from supplied parameters, and studying a domain whose elements are arbitrary classes. They require different formal interfaces.

#### 111.3.1 A size ledger

| Object                   | Safe internal treatment                                | Safe external treatment                                           |
|--------------------------|--------------------------------------------------------|-------------------------------------------------------------------|
| A surreal sign sequence  | A set with an ordinal domain                           | An element of the model’s interpreted surreal carrier             |
| <b>No</b> itself         | A proper class                                         | A set in a containing universe when the entire model is a set     |
| A global well-order $R$  | A class relation; allowed as a class parameter         | One element of the external set of admitted relations             |
| All global well-orders   | A predicate on class variables                         | The actual external set $\mathfrak{M}_{\text{all}}^{\mathcal{M}}$ |
| The bounded-support core | A class of canonical set codes                         | A set of codes with an interpreted evaluation relation            |
| A supplied core cut      | A class subset of those set codes                      | A subset of the external core                                     |
| All external core cuts   | No claim of a same-level class of such objects         | A subset of the power set of the external core                    |
| The decorated completion | A parameterwise or explicitly higher-sort construction | An ordinary set-theoretic lexicographic sum                       |

A two-sorted model need not be transitive,  $\omega$ -standard, or a  $\beta$ -model. Its assertions of well-foundedness concern the objects and subclasses that it admits. An external descending sequence might be absent from its set sort. None of the external model theorems silently replaces admitted well-orders by relations that are actually well-founded in the containing universe.

*Note 111.6* (Source 19, Proposition 13.1, “The uniform coding obstruction”). *In GBC there is no class relation  $A \subseteq I \times \text{Ord}$ , with  $I$  a class of set indices, whose rows enumerate every admitted binary class on  $\text{Ord}$ .* This is source 11’s diagonal non-coding theorem ([theorem 19.37](#)), here for rows that are binary classes on  $\text{Ord}$ , by the same diagonal class  $D$  along a class surjection onto the index class.

*Source 19’s proof.* If  $I$  is empty the assertion is immediate. Otherwise Global Choice gives a class surjection  $b : \text{Ord} \rightarrow I$ : enumerate a proper  $I$  in type  $\text{Ord}$ , or enumerate a set  $I$  and repeat a fixed value after its enumeration ends. Elementary comprehension forms

$$D = \{\alpha \in \text{Ord} : (b(\alpha), \alpha) \notin A\}.$$

For every  $i \in I$ , choose  $\alpha$  with  $b(\alpha) = i$ . Then  $D$  differs from the  $i$ th row at  $\alpha$ , a contradiction.

Pair-swapping a fixed global enumeration transfers this obstruction to uniform exhaustive coding of all set-like global orders. It explains why the core is useful but cannot contain every global



enumeration. It also explains the model-relative missing branches of Theorem 100.1. A single externally specified family can exist as a set in a larger universe without being a uniformly admitted family inside the original model.

### 111.3.2 Why the elementary constructions do not require ETR

In ordinary set recursion, the object constructed at every ordinal stage is a set, and the history below a fixed set ordinal is a set. With a fixed class parameter, GB proves the appropriate set recursion and Class Replacement. Unique compatible initial solutions can then be assembled by an elementary formula: a pair belongs to the final class precisely when it appears in some set-sized initial solution. This applies to birthday-controlled insertion, deterministic back-and-forth on partial set maps, and the set-valued recursive choices used throughout the global core construction.

By contrast, an elementary transfinite recursion with class-valued stages asks for a class  $S \subseteq \Gamma \times V$  satisfying

$$S_\gamma = \{x : \varphi(x, S \upharpoonright \gamma, Z)\},$$

where each  $S_\gamma$  may already be a proper class. This is the principle  $\text{ETR}_\Gamma$ , not a consequence obtained merely by calling the operation a recursion. Already  $\omega$ -length class-valued recursions can construct truth predicates. The distinction and its connections with determinacy and class forcing are developed in [22, 24, 27, 35].

The common labelled prefix is elementary because equality of the two predecessor relations expands into quantification over labels. The simultaneous set-interval block decomposition is elementary because one uses actual least block labels and set intervals. Neither construction attempts to recursively find an arbitrary isomorphism between differently labelled class well-orders. Such abstract comparison is the subject of stronger principles; for example, Williams's Lemma 3.17 uses an ETR hypothesis [35].

The model-theoretic shuffle proofs likewise use only ordinary set-valued constructions internally. The use of an external finite Ehrenfeucht–Fraïssé game does not assert that the entire external family of plays or strategies belongs to the class model.

### 111.3.3 Four workable foundational routes

**Bounded ZFC mathematics.** Fix  $X$ ,  $\alpha$ , or a birthday bound. Then every enumeration, cut, automorphism, and completion is a set. The infinite-tail spectrum, restriction classification, and bounded completion normal forms can all be formalized here. No inaccessible cardinal or class theory is needed for these results. This is the most direct route for the first machine-checked development.

**GBC with supplied class parameters.** State results as formulas about  $R, S, F, C$  and supplied uniform families. Define the comparison, first block, set-interval quotient, and trace cuts with elementary comprehension. When a theorem mentions all class well-orders, it universally quantifies a class variable; it does not create a new class containing those relations as elements. This route closely matches the elementary global proofs.

**External models.** Fix a set-sized structure  $\mathcal{M} = (M, \mathcal{X})$  in a containing ZFC universe. Its class sort is now an external set, so  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$ , its automorphism group, its power set, and its Dedekind completion are ordinary mathematical objects. This is the route used by the decorated-completion and model-spectrum theorems. The resulting object is model-relative; it is not a replacement for the class of all actual surreals in the containing universe.

A useful full-class example is  $(V_\Omega, \mathcal{P}(V_\Omega))$  for a strongly inaccessible  $\Omega$  in an outer universe. It gives strong comprehension internally. Such an example is a sufficient semantic setting, not a hidden assumption behind the set-sized theorems. In this model the set-like global orders correspond externally to  $\Omega$ -length exhaustive enumerations of the bounded surreal carrier, while arbitrary admitted well-orders can have any external order type of cardinality  $\Omega$ .

**An explicit higher sort or universe hierarchy.** One may introduce a sort of collections of classes, or work in nested universes and state exactly which objects are small at which level. This permits a domain of global orders and families of their cuts directly. It changes the ambient language or universe, and the additional comprehension or size assumptions must be listed. KM permits stronger quantification over classes in comprehension; it does not make proper classes into elements of other classes.

These routes are complementary. A proof can first be established as an ordinary theorem about a set linear order, then interpreted in a larger universe, or instantiated with the external orders of a class model. What must be proved separately is that the interpretation has the intended admitted relations and the intended smallness predicates.

#### 111.3.4 A Lean development tied to inspected repository sources

The inspected revision contains `Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean` and `SignSequenceCutOperation.lean` [53]. The first defines

$$\text{SignSequence.}\{u\} : \text{Type } (u + 1)$$

and proves `birthdays_bounded`, `not_small`, and `small_bounded`. The last declaration makes a fixed birthday-bounded fragment small relative to  $u$ ; the first requires an explicit `Small.}\{u\}` hypothesis on the indexing type. The cut-operation file proves `exists_separator_of_small_sets`, with explicit `Small.}\{u\}` instances for both option subsets. These are exactly the size-sensitive facts needed by the elementary order theory. They do not yet certify the new results of this article.

Lean's universe hierarchy and impredicative propositions do not erase these distinctions [29]. A predicate on a large type may be perfectly well-formed without its extension being small in the smaller universe. In particular, writing a relation on `SignSequence.}\{u\}` does not prove that its predecessor types are `Small.}\{u\}`. Set-likeness should be recorded as that explicit condition. Likewise, an external Lean well-foundedness predicate on a type is not automatically the internal well-foundedness predicate of a chosen GBC model.

The following module boundaries are proposals; no modules with these names are claimed to exist in the inspected repository.

| Proposed layer             | Main statements                                                             | Why it comes here                               |
|----------------------------|-----------------------------------------------------------------------------|-------------------------------------------------|
| Labelled enumeration order | First disagreement, prefix cylinders, literal residual lexicographic sum    | Common set-level interface                      |
| Binary codes and fusion    | Prefix-free sign code; constant-subcylinder lemma; fusion into one fiber    | The main embedding mechanism                    |
| Fixed-type spectra         | Cardinal-layer division; deletion scheduler; lower and upper weighted ranks | New ZFC classification                          |
| Completion codes           | Full branches, numerical splits, limit boundaries; canonical comparison     | Exact bounded arrangement, beyond characters    |
| Restriction maps           | Three- and four-label witnesses; exact classification                       | Small, independently reusable theorem           |
| Class-model interface      | Two sorts, admitted comprehension and replacement, evaluation of formulas   | Internal/external interpretation boundary       |
| Raw comparison and core    | Common labelled prefix; bounded-support codes; first-block trace            | Parameterwise class results                     |
| External reconstruction    | Decorated completion and automorphism extension                             | Ordinary theorem on the external set order      |
| Model invariants           | Tail cylinders, nonstandard intervals, EF games                             | Exact external classification and common theory |

The finite restriction theorem is an attractive first target: it needs no surreal arithmetic and has short explicit witnesses. The next high-value target is the fusion lemma, because it isolates the main reason for strict nonembedding results. The cardinal-layer theorem then becomes a composition of independently checked encoders and fusion costs. Formalizing the entire GBC metatheory should be a separate project rather than an implicit side effect of introducing a large type.

An alternative formalization may use a set-theoretic universe such as a Tarski–Grothendieck model. Existing surreal formalization work gives relevant experience with foundations and Conway normal forms [37]. That work is methodological context here; the present proofs are order-theoretic and do not require a formalized normal-form theorem or surreal multiplication.

#### 111.4 Source 19: results, assumptions, and verification status

The results have different logical scopes. Keeping them separate is part of their content, not a qualification added after the fact.

| Result                                             | Foundation                                          | Remaining limitation                                                    |
|----------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------|
| Weighted fixed-type bounds and exact cube spectrum | ZFC                                                 | Least coding length may lie between different computed bounds           |
| Bounded completion normal forms                    | ZFC, dense full-size intervals                      | An explicit completion description is not a short isomorphism invariant |
| Exact monotone restrictions                        | ZFC; class version in GBC                           | Does not forbid carefully designed enlargement embeddings               |
| Common labelled class comparison                   | GB, supplied class parameters                       | Does not compare unlabelled class order types by isomorphism            |
| Decorated completion and automorphism extension    | External ZFC around a specified set-sized GBC model | Model-relative admitted fibers                                          |
| Missing persistent external branches               | Externally countable GBC model                      | The constructed branches need not be admitted classes                   |
| Two countable realization types                    | Externally countable GBC model                      | The common theory has many other countable models                       |
| Common pure first-order theory                     | Any externally set-sized GBC model                  | No classification of uncountable isomorphism types                      |

The repository and same-day continuation results are identified where used. The new arguments were developed with separate checks of the ordinal coding, residual reconstruction, and model-theoretic proofs. That is proof review within the preparation of this manuscript; it is not external peer review.

The included Python program exhaustively verifies the restriction classification on all carriers of size at most seven. It tests every subset and every adjacent pair in the lexicographically sorted permutation list; the recorded run contains 695,228 comparisons. This is meaningful evidence for the finite witness analysis. It does not certify any transfinite, class-theoretic, or cardinal-arithmetic statement. No new Lean proof was compiled, and no uncompiled Lean sketch is presented as a certificate.

The accompanying source is a standalone L<sup>A</sup>T<sub>E</sub>X document with an internal bibliography and no external figures.<sup>17</sup> The PDF was compiled from that source and checked by rendering its pages. The archive also contains a build command, the finite-check program and its output, and a concise source and proof-status record.

## 111.5 Source 20: a size-safe toolkit

### 111.5.1 Signs, birthdays, and set cuts

We use the sign-sequence presentation

$$\mathbf{No} = \bigcup_{\alpha \in \mathbf{Ord}} \{-, +\}^\alpha.$$

The birthday  $\text{bd}(x)$  is the domain length of  $x$ . At the first unequal position, compare  $- < \text{undefined} < +$ . The birthday fragment  $\mathbf{No}_{<\theta}$  is a set for every set ordinal  $\theta$ . Every set of surreal

<sup>17</sup>[merge] Source 19's source and PDF are in arrival commit `b63f0c852`. Its build command, finite-check program, output and status record are shipped as `code/19-residual-build.sh`, `code/19-residual-check_restrictions.py`, `data/19-residual-restriction_checks.json` and `19-residual-source_status.txt`; the checker writes its JSON beside itself (section 111.49).

numbers has bounded birthdays. For an infinite initial cardinal  $\kappa$ , put

$$\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}.$$

This is an equality of cardinals, and it is not a claim that every ordinal calculation involving  $\mu$  can be replaced by cardinal arithmetic.

The basic surreal separation property is: if  $A, B$  are sets and  $A < B$ , a surreal lies strictly between them. Empty sides are allowed. It implies the following strengthening used repeatedly.

*Note 111.7* (Source 20, Lemma 3.1, “A fresh separator”). *If  $A, B, U$  are sets of surreals with  $A < B$ , there is a surreal  $z \notin U$  such that  $A < z < B$ .* This is source 11’s lemma 5.2 (a set can be avoided), with the same one-line proof.

*Source 20’s proof.* Let  $U_0$  be the members of  $U$  lying strictly between  $A$  and  $B$ . Then  $A \cup U_0 < B$ . Choose  $z$  between these two sets. If  $z \in U$ , the inequalities would imply  $z \in U_0$ , contrary to  $U_0 < z$ .

Every nonempty open numerical interval of  $\mathbf{No}$  is a proper class. Otherwise apply note 111.7 with that interval as the forbidden set. The same argument shows that deleting a set preserves the set-cut property.

### 111.5.2 What the class theories permit

Here GB means Gödel–Bernays theory whose set part is ZF, without Global Choice built into the abbreviation; GBC adds Global Choice. All global baseline constructions in this article take place in GBC. A class well-order is a strict linear class relation for which every nonempty available subclass has a least member. It is *set-like* when every strict predecessor class is a set.

An individual member of  $\mathfrak{W}_{\text{sl}}$  or  $\mathfrak{W}_{\text{all}}$  is a class relation. Consequently “ $R \in \mathfrak{W}_{\text{all}}$ ” abbreviates a predicate on a class variable. It does not assert membership in a GB class containing other proper classes. A uniform family  $(R_i)_{i \in I}$  is one class relation

$$T(i, x, y) \quad \text{with row } R_i(x, y),$$

where the indices  $i$  are sets and  $I$  can itself be a set or a class. This is what “set-indexed” and “class-indexed” mean below.

In GBC, ordinary recursion with *set values* along  $\text{Ord}$  is available: each set-ordinal initial history is a set, and the compatible local histories define one class. It does not require full elementary transfinite recursion. In contrast, a recursion whose value at each stage is an entire class may require ETR, even when the stage order is small. The distinctions among  $\text{ETR}_\omega$ ,  $\text{ETR}_{\text{Ord}}$ , and full ETR must not be suppressed [22, 24].

*Note 111.8* (Source 20, Proposition 3.2, “Set-like global orders”). *In GBC, a proper-class set-like well-order is uniquely isomorphic to  $\text{Ord}$ . In particular, a set-like well-order of  $\mathbf{No}$  is equivalent to a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ .* This is the identification of set-like class well-orders with class bijections  $\text{Ord} \rightarrow \mathbf{No}$  used throughout this report (the reading of  $\mathfrak{W}_{\text{sl}}$  in section 1.3).

*Source 20’s proof.* Send each label  $x$  to the ordinal order type of its predecessor set. The range is an initial subclass of  $\text{Ord}$ . If it were bounded, the inverse map on a bounding ordinal would make the original carrier a set by Class Replacement. Hence the range is all of  $\text{Ord}$ .

The existence of a class well-order of all surreal labels already has choice strength. Over  $\text{GB} + \text{AC}$  it is equivalent to Global Choice: sets can be coded by well-founded extensional relations on ordinals, those relations by sets of ordinals, and those by surreal sign sequences. Choosing the least code in the supplied surreal well-order produces a global choice operation. The repository proves the stronger equivalence over GB without assuming set choice separately [39, Theorem 13.1] ([merge] theorem 13.1). The further results below do not need that strengthening.

### 111.5.3 The set-sized baseline

*Note 111.9* (Source 20, Theorem 3.3, “Fixed carrier spectrum; earlier result”). *Let  $X$  be an infinite linearly ordered set of cardinality  $\mu$ . Then  $|\mathcal{W}(X)| = 2^\mu$ , every linear order of cardinality at most  $\mu$  embeds in  $\mathcal{W}(X)$ , and*

$$\alpha \hookrightarrow \mathcal{W}(X) \iff \alpha < \mu^+ \quad (\alpha \text{ an ordinal}).$$

*The same upper bound holds for reverse well-ordered suborders.* This is [theorem 6.1](#) (the reals report’s spectrum theorem for every infinite alphabet); source 20 calls it an earlier result.

*Source 20’s proof.* There are at most  $2^\mu$  relations on  $X$ . Partition  $X$  into  $\mu$  two-element pieces and enumerate those pieces in a fixed ordinal order. Choosing independently which member of each pair appears first embeds the binary lexicographic cube  $2^\mu$  into  $\mathcal{W}(X)$  and proves the lower cardinality bound. If  $L$  has cardinality at most  $\mu$ , enumerate its elements by coordinates below  $\mu$ . The characteristic functions of its initial segments are nested, distinct, and lexicographically ordered as  $L$ ; the pair construction then embeds  $L$  into  $\mathcal{W}(X)$ .

Suppose a strictly increasing sequence of enumerations has length  $\mu^+$ . On a tail, its first entries stabilize: a monotone sequence in an alphabet of size  $\mu$  cannot keep changing through a regular  $\mu^+$ -sequence. Repeating this observation stabilizes each successive entry on a tail. At any limit stage below  $\mu^+$ , regularity supplies a common tail for all earlier requirements. The stabilized prefix cannot exhaust  $X$ , since then the strictly increasing tail would be constant. Continuing would therefore give an injection from  $\mu^+$  into  $X$ , a contradiction. The decreasing case is identical with inequalities reversed. An ordinal below  $\mu^+$  has size at most  $\mu$ , so the embedding direction has already been proved.

For  $\mathbf{No}_{<\kappa}$ , the cutoff is therefore  $(2^{<\kappa})^+$ , not automatically  $\kappa^+$ . The recent continuation also computes density, cellularity, point characters, and proper-gap spectra for the minimal stratum [46] ([merge] printed in Part VI); those more detailed formulas are not needed by the new class-level theorems here.

### 111.5.4 Variable set words are still a class of sets

*Note 111.10* (Source 20, Proposition 3.4, “Explicit compression of dictionary words”). *All set-length words over  $\mathbf{No}$ , with prefix-first dictionary order, admit an explicit order embedding into  $\mathbf{No}$ , without Global Choice. If  $w = (x_i)_{i<\alpha}$ , its code has birthday*

$$\sum_{i<\alpha} (1 + 2 \cdot \text{bd}(x_i) + 2).$$

*Here all sums and products inside the birthday expression are ordinal.* This is source 11’s explicit word code ([theorem 9.1](#)), with the same pair-doubling of signs.

*Source 20’s proof.* Replace each minus sign of  $x$  by the pair  $--$ , each plus sign by  $++$ , and append the pair  $-+$ . Call the resulting prefix-free sign word  $c(x)$ . Because  $-- < -+ < ++$ , this transformation preserves numerical comparison, including the case where one original sign sequence ends first. Code  $w$  by concatenating, for each entry, one plus sign followed by  $c(x_i)$ . Unequal entries are decided within their prefix-free entry codes. If one whole word ends first, the next sign of the longer code is a plus, so the shorter code is smaller. The displayed length follows by counting the signs in each block.

This compression includes injective words, hence all well-orders of variable set-sized surreal subsets. It does not encode arbitrary class relations by sets; every individual input word in the proposition is a set. Nor is it an isomorphism with  $\mathbf{No}$ , as the limit-prefix example in [section 61.4](#) shows.



### 111.5.5 Comparison of arbitrary class well-orders

For class well-orders  $R, S$  of the same labelled carrier  $X$ , let  $C(R, S)$  consist of those  $x$  such that their strict predecessor classes agree and the two relations agree on that predecessor class. Membership in  $C(R, S)$  is elementary with class parameters: all its quantifiers range over labels.

*Note 111.11* (Source 20, Lemma 3.5, “Largest common labelled prefix”). *The class  $C(R, S)$  is an initial segment of both orders, their restrictions to it agree, and it contains every common labelled initial segment. If  $R \neq S$ , it is proper in  $X$  and the labels*

$$a = \min_R(X \setminus C(R, S)), \quad b = \min_S(X \setminus C(R, S))$$

*are distinct.* This is the largest common labelled prefix of [theorem 14.2](#) (source 12’s [theorem 14.7](#)).

*Source 20’s proof.* If  $x \in C(R, S)$ , each predecessor  $y$  of  $x$  has the same predecessor class and inherited predecessor order in both relations. Thus  $y$  also belongs to  $C(R, S)$ . For two members, comparison can be read inside the predecessors of the later one, where the relations agree. Conversely, a common labelled initial segment has the same predecessors and relations at every one of its points, so it is contained in  $C(R, S)$ . If the two next labels were equal, their common predecessor class would be exactly  $C(R, S)$ , and that label would itself belong to the class, a contradiction.

*Note 111.12* (Source 20, Theorem 3.6, “Native class lexicography; earlier result”). *For  $R \neq S$  in  $\mathfrak{W}_{\text{all}}(X)$ , define  $R <_{\text{lex}} S$  when  $a < b$  numerically, with  $a, b$  from [note 111.11](#). This is a strict linear comparison and agrees with enumeration lexicography when both orders are set-like. It requires no comparison theorem for abstract class well-order types. This is [theorems 14.2](#) and [14.7](#); source 20 calls it an earlier result.*

*Source 20’s proof.* Trichotomy follows from  $a \neq b$ . For transitivity, the prefixes  $C(R, S)$  and  $C(S, T)$  are initial segments of the *same* well-order  $S$ , hence one contains the other. If one is shorter, the corresponding earlier divergence determines the comparison of  $R$  and  $T$  as well. If they are equal, compare the three next labels and use numerical transitivity. For set-like orders the common prefix is a set, and its ordinal length is exactly the first differing coordinate of the canonical enumerations.

Equality of predecessor *sets alone* would be insufficient: the enumerations  $(0, 1, 2)$  and  $(1, 0, 2)$  give the same predecessor set at 2 but disagree before it. Agreement of the inherited relations is essential.

### 111.5.6 The bounded-support core

Fix a baseline class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . A normalized core code is a set permutation  $\pi : \lambda \rightarrow \lambda$  such that

$$\lambda = \sup\{\xi + 1 : \pi(\xi) \neq \xi\}.$$

The empty permutation represents the baseline. Extend  $\pi$  by the identity above  $\lambda$  and evaluate the resulting enumeration as  $E_\pi = b \circ \hat{\pi}$ . The ordinary class of codes, ordered by their evaluations, is denoted  $\mathcal{P}_{\text{bd}}(b)$ . Equivalent terminology is the class of enumerations eventually equal to  $b$ , represented by their unique set codes.

*Note 111.13* (Source 20, Lemma 3.7, “Uniform completion”). *Every injective set prefix  $p : \delta \rightarrow \mathbf{No}$  extends to an enumeration represented in  $\mathcal{P}_{\text{bd}}(b)$ . One completion can be selected uniformly in  $p$ . This is source 11’s supported completion ([lemma 19.4](#)).*

*Source 20’s proof.* The set injection  $b^{-1} \circ p$  takes values below some ordinal. Choose an infinite initial cardinal  $\gamma$  larger than both that bound and  $\delta$ . The domain and range complements in  $\gamma$  both have size  $\gamma$ , so a bijection between them extends the injection to a permutation of  $\gamma$ . Use a fixed global well-order to choose one such set bijection, extend by the identity, and normalize.

*Note 111.14* (Source 20, Theorem 3.8, “Core interpolation and identification; earlier result”). *In GBC, the class order  $\mathcal{P}_{\text{bd}}(b)$  is isomorphic to  $\mathbf{No}$ . More strongly, between any two uniformly set-indexed separated families of injective class words  $\text{Ord} \rightarrow \mathbf{No}$  there is a core enumeration. Empty families are allowed on either side.* This is [theorem 19.10](#) and [corollary 19.15](#); source 20 calls it an earlier result.

*Source 20’s proof.* Call the two families  $(A_i)_{i \in I}$  and  $(B_j)_{j \in J}$ . When both are nonempty, Class Replacement bounds all cross first-disagreement ordinals by a set ordinal  $\beta$ . Build an injective prefix and keep only rows still agreeing with it active. At the current coordinate, the active left values form a set  $L$  and the active right values a set  $U$ , with  $L \leq U$  pointwise. If their intersection is empty, choose a fresh strict separator using [note 111.7](#), append it, and stop. If their intersection is the singleton  $\{z\}$ , append  $z$  and discard all rows with another value. The comparison invariant is preserved. Two common values would contradict the cross inequalities. At limits take unions. Both sides cannot stay active through  $\beta$ , so the procedure stops with a set prefix. If one side was empty, separation is possible at once. Complete the prefix using [note 111.13](#).

Thus every separated pair of sets of core codes has a strict interpolant. The core is a proper class, since transpositions involving arbitrarily large ordinals have distinct codes. Enumerate it and  $\mathbf{No}$  by  $\text{Ord}$  using Global Choice. Alternating extensions of a set-sized partial isomorphism give an isomorphism: at each step the required lower and upper images are sets, so interpolation supplies a fresh image. The histories remain sets at every set stage. This is the usual class back-and-forth argument, not an invocation of class-valued ETR.

We call a class order with this set-cut property an  $\eta_{\text{Ord}}$  order. The same back-and-forth proof shows that any two such proper class orders are isomorphic, that set-sized partial order isomorphisms extend to automorphisms, and that every class linear order embeds into one under Global Choice. The classical surreal universality and homogeneity background is discussed by Ehrlich and Hamkins [15, 33]. The new automorphism theorem later in this article concerns the much stronger demand of extending an automorphism to the ambient space of global well-orders.

For an injective set prefix  $p$ , write

$$\mathcal{P}_{\text{bd}}(b)(p) = \{c \in \mathcal{P}_{\text{bd}}(b) : E_c \restriction \text{dom}(p) = p\}.$$

These nonempty prefix cylinders are convex. The core is dense in  $\mathfrak{W}_{\text{sl}}$ , but it need not meet an interval between arbitrary non-set-like raw orders. A common proper prefix with a small residual carrier is one basic reason. This distinction drives the local-size results below.

### 111.5.7 Which core cuts encode class orders

The new automorphism results require a precise account of represented cuts. We record the relevant earlier criterion, with its proof, so that the dependence on enumeration structure is visible.

*Note 111.15* (Source 20, Lemma 3.9, “The first ordinal block”). *For  $R \in \mathfrak{W}_{\text{all}}(\mathbf{No})$ , the class*

$$B_R = \{x \in \mathbf{No} : \{y : yRx\} \text{ is a set}\}$$

*is a proper initial class and, with its induced order, has type  $\text{Ord}$ . Let  $f_R : \text{Ord} \rightarrow B_R$  be its canonical enumeration. For every core code  $c$ , comparison of  $E_c$  with  $R$  is the same as comparison of  $E_c$  with  $f_R$ . Thus the strict core cut of  $R$  depends only on  $f_R$ .* This is source 11’s canonical first block ([theorem 15.1](#)) with source 16’s observation that the core cut of a raw order is that of its first block ([corollary 46.4](#)).

*Source 20’s proof.* The class is initial and set-like by definition. If it were a set, its complement would be nonempty, and the least point of that complement would have exactly  $B_R$  as its predecessor

set. It would then belong to  $B_R$ , a contradiction. Apply [note 111.8](#). If  $E_c$  and  $f_R$  differ, their first differing ordinal coordinate already determines the raw comparison with  $R$ . If they agree everywhere,  $f_R$  is onto  $\mathbf{No}$  because  $E_c$  is, so  $R$  is exactly that core order.

*Note 111.16* (Source 20, Theorem 3.10, “Recognition by straddling prefixes; earlier result”). *Let  $L$  be a nonempty proper initial class of  $\mathcal{P}_{\text{bd}}(b)$  with no greatest element. The following are equivalent in GBC:*

1.  $L = \{c : E_c <_{\text{lex}} f\}$  for an injective class word  $f : \text{Ord} \rightarrow \mathbf{No}$ ;
2. for every set ordinal  $\delta$ , some injective prefix of length  $\delta$  has a cylinder meeting both  $L$  and its complement;
3.  $L$  is the strict core cut of a raw class well-order of  $\mathbf{No}$ .

*The injective word in (1) is unique. The cut has a set-like representative exactly when this word is onto. Recovery of the word uses only elementary comprehension with  $L$  as a class parameter. This is the represented-cut theorem of Part VIII, source 14’s [theorem 45.1](#) ([note 46.1](#) lists its six earlier proofs); clause (3) is source 16’s raw-trace corollary. Source 20 calls it “the relevant earlier criterion” and gives its proof, a seventh route by straddling prefixes; Part XIII uses it.*

*Second route (source 20).* At most one cylinder of any fixed length straddles  $L$ : distinct cylinders are disjoint and ordered, so two straddlers would place a point outside an initial class below a point inside it. If straddlers exist at all lengths, uniqueness forces their prefixes to cohere. Define  $f(\alpha)$  as the last value of the unique straddling prefix of length  $\alpha + 1$ . This is an elementary definition from uniquely specified sets. Coherence proves injectivity. At the first difference between a core row and  $f$ , the straddling cylinder one coordinate longer determines which side of  $L$  that row occupies. If  $E_c = f$ , the code  $c$  cannot lie in  $L$ : a larger member of  $L$  would give the opposite comparison. Hence (1) follows.

Conversely, for every set prefix of an injective class word, continue to its next value, choose fresh labels on either side of that value, and complete the two prefixes in the core. They lie on opposite sides of the word’s cut. Thus every length has a straddler and (1) implies (2). The same first-difference argument proves uniqueness of the word.

By [note 111.15](#), a raw representative gives an injective word, so (3) implies (1). Given an injective word, append a fixed well-order of its unused labels. Its initial word is still the first ordinal block: all later points, if present, have the proper range of  $f$  among their predecessors. The resulting raw order has the required cut. It is set-like exactly when there are no unused labels. If some other set-like representative existed, uniqueness of the word would force it to be the same nonexhaustive word, which is impossible.

The test quantifies over all ordinal *lengths*, each of which is a set. It does not choose one class at each stage. This is why recovery is elementary even though the recovered word is a class. The later symmetry theorem will show that the cylinder structure in this test contains information absent from the bare linear order of the core.

## 111.6 Source 20: formalization without paradoxes

### 111.6.1 Four workable foundations

There is no single formalization that silently turns all rows of the interpretation table into the same object. Four complementary approaches are useful, and their relationships explain several apparent paradoxes.

**Ordinary set theory at a fixed stage.** Choose a set alphabet  $X$  or a birthday cutoff  $\mathbf{No}_{<\theta}$ . Its relations, well-orders, enumeration strata, order topologies, and Dedekind cuts are ordinary sets. The fixed-carrier theorem, sharp filtration criterion, ordinal condensation formula, and countable topological example fit this framework directly. One can also work with a set-indexed family of such stages. A proper-class inverse limit is an additional statement; it is not automatically supplied by existence of all its individual stages.

**Two-sorted class theory.** In GB or GBC, the labels are sets, individual global well-orders are classes, and their comparisons are formulas with class parameters. A uniform family has one relation on set indices and set labels. This supports birthday gluing, the core construction, class automorphisms, local coding obstructions, and analysis of a supplied condensation history. One may quantify over class variables in an assertion without using those variables as members of a new class.

**Stronger comprehension and recursion.** KM permits stronger class comprehension, but its second sort still does not consist of objects allowed to be members of ordinary classes. It therefore does not turn “all class well-orders” into a GB-style class of elements. ETR supplies class-valued recursion histories along specified well-founded class orders. Our termination theorem identifies what happens to such a history once it exists; it does not prove that its existence has the exact strength of full ETR. The known distinction between set-valued recursion and class-valued ETR is essential here [22, 24].

**A containing universe or an explicit model.** In an outer universe, fix a set model  $(M, \mathcal{X})$  of a two-sorted class theory. The admitted classes are now external objects, so their well-order space and any desired quotient are external sets. The full model  $(V_\kappa, \mathcal{P}(V_\kappa))$  at an inaccessible cardinal provides particularly transparent cardinal calculations. Countable models provide the shuffle classifications proved above. These are two different external realizations, not inconsistent answers to one unqualified cardinality question.

### 111.6.2 What an inaccessible full model actually contains

Let  $\kappa$  be inaccessible in the ambient universe and consider  $(V_\kappa, \mathcal{P}(V_\kappa))$ . Its internal surreal carrier is, externally,  $\mathbf{No}_{<\kappa}$  of cardinality  $\kappa$ . All its ambient subsets are available as classes. A subclass of this carrier is an internal set exactly when its ambient cardinality is less than  $\kappa$ . Consequently the external full well-order space is

$$\mathcal{W}(\mathbf{No}_{<\kappa}),$$

while its internally set-like part is

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}).$$

Indeed an actual well-order of a carrier of size  $\kappa$  is internally set-like precisely when all its predecessor segments have size below  $\kappa$ , which forces its type to be exactly  $\kappa$ .

Both spaces have cardinality  $2^\kappa$ , but their geometry differs. The minimal stratum has a dense set of size  $\kappa$ : choose a completion for each injective prefix of length below  $\kappa$ . There are only  $\kappa^{<\kappa} = \kappa$  such prefixes, and every nonempty order interval contains one of their cylinders. Fresh separators exist because  $\mathbf{No}_{<\kappa}$  fills cuts of size below  $\kappa$ .

The full order has  $2^\kappa$  disjoint nonempty intervals. Reserve three labels and vary the enumeration of all other labels in  $2^\kappa$  ways. For each common exhaustive prefix, its six possible final permutations form a convex six-point block. An open interval between the first and last point of each block is nonempty, and the different blocks are disjoint. An embedding into the minimal stratum would give

$2^\kappa$  disjoint nonempty intervals in an order with a dense subset of size  $\kappa$ , which is impossible. This re-proves the earlier full-model nonembedding result [39] ([merge] theorem 20.1) and locates the new interval and condensation calculations in the same concrete model.

There is no conflict with class universality of **No**. Internally, universality concerns class orders *whose elements are sets*. Externally the full well-order space has  $2^\kappa$  points, whereas the internal surreal carrier has only  $\kappa$ . Similarly, the new one-point nonextension theorem does not deny that a new class order can embed into **No**: it denies an embedding satisfying all comparisons already fixed by a specified automorphism on an entire proper-class core.

### 111.6.3 Changing the class collection while keeping the sets

Suppose  $(M, \mathcal{X})$  and  $(M, \mathcal{Y})$  are GBC realizations of the same first-order set model. For a relation available in both class collections, well-foundedness has the same truth value in both models. In GBC a failure of class well-foundedness is witnessed by a set-coded descending  $\omega$ -sequence; the sets, and hence all such witnesses, are unchanged. This is the precise absoluteness used here [35, Observation 4.12]; related stronger results appear in [27, §3].

The common native comparisons are stable as well, because their definitions use only set quantifiers and the supplied class relations. With a common baseline  $b$ , the set-coded core and its evaluations are also unchanged. What can change is which additional relations, class cuts, automorphisms, or uniform histories are available. Thus class extensions may add new points to the surrounding well-order space; they do not make an old shared well-order ill-founded merely by adding more classes.

An external branch through the stages of a countable model need not be an admissible internal class. The gluing theorem uses Class Replacement to bound the birthdays of a descending sequence. Omitting admissibility would remove that step. The countable shuffle theorem respects this boundary: its back-and-forth is external and its external enumeration need not belong to the model.

### 111.6.4 A practical route in Lean

The pinned ProveIt sources contain useful universe-aware interfaces. They were inspected at the revision on the title page; this article did not run a Lean build or add a Lean theorem.

| Existing file                      | Inspected declarations                                                                                                            | Appropriate use                                                          |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <code>SignSequence.lean</code>     | <code>SignSequence</code> ,<br><code>birthdays_bounded</code> ,<br><code>not_small</code> ,<br><code>ordinalOrderEmbedding</code> | Carrier, relative smallness, birthday bounds, and ordinal labels         |
| <code>SignSequenceCut.lean</code>  | <code>exists_cut_separator_of_birthday_lt</code> ,<br><code>small_cut_fillers</code>                                              | Separators with explicit universe and birthday bounds                    |
| <code>SizeObstructions.lean</code> | <code>SmallCutData</code> ,<br><code>HasSmallCutFillers</code> ,<br><code>noUnrestrictedCuts</code>                               | Distinguishing small cut filling from the inconsistent unrestricted form |

The files lie under `Algebra/SurrealNumbers/Surreal/Foundations/` [7, 8, 31]. In particular, `SignSequence.{u}` lives in `Type (u+1)`, and `not_small` rules out its being small relative to universe  $u$ . The fact that it has a Lean type is not a proof that the entire carrier may be used as one of the small index types in a cut theorem.

The most direct formalization begins with a generic linearly ordered carrier and well-order relation codes. Prove common-prefix comparison, its transitivity, restriction reversal examples,

two-label persistence, and the finite-change theorem before specializing to surreals. These arguments use no surreal arithmetic. The next layer formalizes coherent filtrations and the countable-witness condition, followed by the set-sized ordinal condensation formula. This yields useful verified results without first constructing a full class-theory library.

For genuine class statements there are two honest routes. One can work relative to explicit universes and retain every smallness condition, or formalize a two-sorted model with a class-code type and an evaluation relation. In the second route, “internally well-ordered” must be expressed using the model’s interpretation. Substituting Lean’s external `WellFounded` for that notion would exclude the very nonstandard models studied in [section 109.2](#). In a full transitive model the two notions coincide, which is why the inaccessible calculations are simpler.

Core automorphisms then require a general set-cut interpolation interface, uniform set-valued recursion, and a precise coding of evaluations. The one-point extension obstruction should be formalized with one new marker adjoined to the class of core codes, as in [theorem 97.7](#). That formulation avoids constructing a type intended to mean “every proper class of the whole ambient universe.” Complete condensation histories should take the history as an input first; a separate theorem can derive its existence from the selected ETR axiom or semantic model.

### 111.6.5 The exact boundaries that prevent paradoxes

The usual contradictions disappear when the quantifiers are kept in their declared domains. A cut theorem realizes cuts with small sets of constraints, not the cut having the entire carrier as its lower side. A global enumeration enumerates surreal *labels*, not all class relations on those labels. A representative of a canonical block is an actual least label; the whole proper block is not thereby made a set. An external power set is formed in the containing universe, not in a smaller universe that would regard its domain as proper.

Finally, first-divergence comparison never identifies arbitrary abstract class well-orders with ordinals. It uses shared labels and compares two initial segments of one supplied order. This is weaker than the general problem of constructing isomorphisms between arbitrary class well-order types. The literature proves ETR implications for that comparability problem; those implications must not be promoted to equivalences without a separate proof [\[27, 35\]](#).

### 111.7 Source 20: results and verification status

| Result                                                 | Foundation used                                                 | Status in this article                                               |
|--------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
| Fixed-set spectrum; word compression; core recognition | ZFC for fixed sets; GB/GBC at the displayed class steps         | Earlier project results, reproved as the toolkit                     |
| Coherent birthday gluing and rank stabilization        | GB, supplied uniform profile                                    | New proofs in <a href="#">theorems 74.1</a> and <a href="#">74.6</a> |
| Two-label persistence and finite changes               | Supplied well-orders; uniform class filtration where applicable | New proofs in <a href="#">theorems 74.3</a> and <a href="#">74.5</a> |
| Coherent append-tail system and core limit             | GBC with a birthday baseline                                    | New proof in <a href="#">theorem 74.8</a>                            |
| Club reflection of minimum local types                 | GBC; available club classes                                     | New dichotomy and sharpness example in <a href="#">theorem 74.7</a>  |



| Result                                       | Foundation used                                                      | Status in this article                                                               |
|----------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Cut orbits and localized nonextension        | GBC; class cuts and class automorphisms                              | New proofs in <a href="#">note 97.3</a> and <a href="#">theorem 97.7</a>             |
| Local coding dichotomy and convex blocks     | GBC; uniform indexed families                                        | New proofs in <a href="#">theorems 107.4</a> and <a href="#">107.5</a>               |
| Termination of a supplied history            | GB plus the supplied history; ETR is sufficient for existence        | New conditional proof in <a href="#">theorem 108.1</a>                               |
| Exact condensation heights                   | ZFC externally; inaccessible full model for the class interpretation | New calculation in <a href="#">theorem 108.2</a> and <a href="#">corollary 108.3</a> |
| Two countable pure order types               | External countability of both sorts; internal GBC                    | New nonstandard extension in <a href="#">note 109.13</a>                             |
| Elementary inclusion and equivalence         | Ordinary external model theory                                       | New finite-game proof in <a href="#">note 109.20</a>                                 |
| Decidability of the common pure-order theory | Effective finite-game reduction and coloured dense-order elimination | New procedure in <a href="#">corollary 109.22</a>                                    |
| Countable topology and omitted cuts          | ZFC, countable dense alphabet                                        | New proofs in <a href="#">theorem 67.11</a>                                          |

The principal arguments were read independently with particular attention to class quantifiers, residual alphabets that need not be dense, internal finiteness, and recursion strength. That review found and corrected several wording issues, including open-interval qualifications and an overbroad attribution about earlier condensation histories. It did not provide a formal certificate or establish literature priority.

### 111.7.1 Finite diagnostics supplied with the article

The archive includes a standard-library Python program and its saved output. The program checks all unordered pairs of permutations on carriers of sizes two through five. It verifies the decisive-pair persistence statement on every relevant restricted carrier and the coordinatewise monotonicity of witnesses on every relevant nested pair of carriers. It also evaluates finite surreal signs as exact dyadic rationals to check the dictionary-word encoder independently of the encoder's own sign comparison.

The saved run passes 58,255 decisive-pair restriction checks and 609,862 nested-witness checks, across 7,432 permutation pairs. It also passes 28,920 exact dyadic comparisons for 241 words with at most two entries, each entry having at most three signs. The explicit nonmonotone restriction example is reproduced in the output.

These are finite operational diagnostics. They establish no class automorphism theorem, transfinite induction, cardinal calculation, nonstandard-model classification, or ETR implication. Those conclusions rest on the written proofs. No Lean build, additional axiom audit, or machine verification of this article is claimed.

### 111.7.2 Rebuilding and reusing the deliverable

The main LaTeX file is self-contained and uses a conventional TeX Live installation. It has no external figures, bibliography database, or generated source dependencies. Compile with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  surreal_orders_limits_symmetries_models.tex
```

Run the independent finite diagnostics with

```
python3 finite_diagnostics.py --output finite_diagnostics_results.json
```

[merge] Shipped here as `code/20-limits-finite_diagnostics.py` and `data/20-limits-finite_diagnostics_results.json`; the L<sup>A</sup>T<sub>E</sub>X source and PDF are not shipped (they are in arrival commit `b63f0c852`). Rerun on a copy with an explicit `-output` (section 111.49). The article’s definitions, proofs, and distinctions between internal and external statements are part of the deliverable. The software output is an auxiliary record of finite checks, not a replacement for them.

## 111.8 Source 21: a practical route to rigorous formalization

The mathematics can be formalized in several compatible ways. The best starting point is a small-carrier development that isolates labelled order comparison from surreal arithmetic. The genuinely class results should then be expressed in a chosen model of class theory or through an explicit universe-relative smallness interface.

### 111.8.1 An axiom and size ledger

The following ledger records sufficient foundations used by the proofs. It is not an assertion that every entry is optimal.

| Construction                                                              | Sufficient setting                                       | Essential restriction                                                                                     |
|---------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Set alphabet, fibers, noncardinal invariants, fixed-cardinality coherence | ZFC                                                      | The carrier and all indexing families in these sections are sets.                                         |
| Largest common labelled prefix                                            | GB, given the class well-orders                          | Elementary comprehension; equality of predecessor classes uses set quantifiers.                           |
| Canonical first block and set-interval blocks                             | GB, given the class well-order                           | Ranks of set intervals define all blocks simultaneously.                                                  |
| Normalized core and uniform prefix completion                             | GBC                                                      | One fixed baseline and explicitly coded set support.                                                      |
| Set-cut interpolation and large-cut conjugacy                             | GBC                                                      | Uniform set-indexed inputs; set-valued recursion, even when its length is Ord.                            |
| Represented-cut reconstruction and extension obstruction                  | GBC                                                      | Cuts are individual classes of core codes; the family of all cuts is virtual.                             |
| External countable classifications                                        | A metatheory containing a countable two-sorted GBC model | Its admitted class objects form a countable set externally; standardness is stated separately.            |
| Full inaccessible-universe interpretation                                 | An inaccessible $\Omega$ in the metatheory               | The specific realization is $(V_\Omega, \mathcal{P}(V_\Omega))$ ; this is an additional model assumption. |
| Uniform class-valued recursion                                            | An appropriate ETR principle                             | Not supplied by calling an ordinal-length construction “recursion”; state the value size.                 |
| Actual hypercollection of class well-orders                               | A larger universe or a third-order theory                | Comprehension and choice at the new level must be specified.                                              |

For example, the canonical block theorem does not need class-valued recursion because its equivalence relation is defined directly. Conversely, an argument that selects an arbitrary isomorphism

for each of a proper collection of residual fibers does not become uniform just because each individual isomorphism exists. A proof must either give an elementary unique construction, assume a suitable higher-level choice principle, or restrict its conclusion to individually specified inputs. The cut-reconstruction proof of the extension criterion takes the first route.

### 111.8.2 The small-carrier formal core

A first formalization can parameterize everything by a type  $X$  with a linear label order. A well-order presentation consists of a relation  $R : X \rightarrow X \rightarrow \text{Prop}$ , proofs that it is a strict linear order, and well-foundedness. The following dependency order avoids involving surreal addition, multiplication, or analytic constructions.

1. Define predecessor sets and the common labelled prefix. Prove initiality, maximality, and existence of the two first outside labels for distinct well-orders on the same carrier.
2. Define lexicographic comparison through these labels and prove trichotomy and transitivity. Relate it to first disagreement for ordinal-indexed exhaustive enumerations.
3. Prove convexity of prefix cylinders, partial-injection completion inside a sufficiently large cardinal, and the first-cardinality fiber decomposition. These results can be tested on finite carriers, but their formal proofs should not be replaced by finite tests.
4. Add the elementary interval-size or short-chain hypothesis and derive the agreement topology, point characters, density, and cellularity. The noncardinal birthday theorem then becomes an application of the three-symbol padding code and an ordinal product calculation.
5. Formalize the fixed-slot embeddings and prove their literal composition equation. This supplies a useful coherence result before introducing any proper-class model.

The countable homogeneity theorem is a separate manageable milestone. Its main ingredients are the uniqueness of a countable dense order without endpoints, order sums, finite-prefix cylinders, and a back-and-forth on countable structures. The Baire coding can be verified as two continuous inverse maps on countable permutations; it need not be derived from a broad descriptive-set-theoretic library.

### 111.8.3 Using the repository's universe-relative surreal interface

The audited repository already contains sign-sequence foundations for bounded birthdays, small cuts, and size obstructions [7, 8, 9, 31]. The relevant files distinguish smallness relative to a universe from unrestricted families. Their existence does not confer formal status on the new results of this paper.

The concrete universe parameters matter. In the pinned source, `Surreal.Foundations.SignSequence` lives in `Type (u + 1)`, while its birthday is an `Ordinal.{u}`. The declarations `SignSequence.birthdays_bounded`, `SignSequence.small_bounded`, and `SignSequence.not_small` distinguish small families and bounded birthday subtypes from the encompassing type. Numerical first disagreement is exposed by `SignSequence.lt_iff`. Small-cut interpolation is available through `SignSequence.small_cut_fillers` and `SignSequence.exists_separator_of_small_sets`; the latter requires explicit `Small.{u}` instances for both option sets. The constructor `SignSequence.cut` comes with `SignSequence.cut_realizes` and `SignSequence.birthday_cut_le_iSup`. Finally, `Surreal.Foundations.noUnrestrictedCuts` rules out unrestricted cut filling for any irreflexive relation by taking the whole carrier as the left option set. These are source-audited interfaces, not a new kernel check of the repository or of this manuscript.

One sound design exposes the following mathematical interface:

- a universe-relative ordinal type and sign sequences on small ordinal domains;
- a linear numerical order using centered termination;
- bounded birthdays for small families;
- a separator for every small separated pair of families;
- the impossibility of replacing “small” by arbitrary families of the encompassing surreal type.

There are then two different formal targets. The first is a theorem about all well-orders of a particular type in a sufficiently large universe. That is an ordinary type-theoretic theorem, with explicit universe parameters. The second is a theorem inside a formalized two-sorted set theory, whose class variables range over the classes admitted by that model. One should not identify these targets merely because both are written with an expression resembling “all well-orders.”

The main nonextension theorem is particularly suitable for a modular formalization. Its abstract input is a genuine class order  $S$  with small-cut interpolation, a prefix completion operation, and an elementary cut decoder. The output is an obstruction to extending an automorphism. Surreal field structure is irrelevant. A final module can instantiate the interface with the normalized surreal core.

#### 111.8.4 Definable codes are a restriction, not a substitute for all classes

A formula code with set parameters can describe a definable class well-order. Such codes are sets, but the evaluation of every formula uniformly requires an appropriate satisfaction relation. Over a fixed set model, satisfaction is an external set relation and this approach is straightforward. For the entire set universe, an unrestricted internal truth predicate cannot be inserted silently into GB. Restricting formula complexity, adding a suitable truth class, or working externally with a fixed model are distinct options.

Even when evaluation is available, different codes can define the same relation. Numerical lexicography then orders their evaluations, giving a preorder on codes until extensional duplicates are identified. Finally, a definable-code family need not exhaust the admitted class relations, and [note 86.16](#) prevents any internally uniform set-code family from exhausting all global enumerations. The code language and its interpretation are part of the formal object.

### 111.9 Source 21: source audit and verification record

The principal repository input was read through the selected GitHub connection at the immutable revision `109aaca1505c12d70ae169bdc2011f40bd476f7d`. The relevant directory is `Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders`. Its README describes a 116-page synthesis of four manuscripts and explicitly states that the synthesis is not formally verified.<sup>18</sup> We read its complete `article.tex`, its README, and the supplied research status records. Its source labels remain useful identifiers even when page numbers change in later revisions.

| Audited topic                 | Repository label or location                              | Use here                                       |
|-------------------------------|-----------------------------------------------------------|------------------------------------------------|
| Class comparison              | <code>swo:cf:lex</code> , <code>swo:st:thm:fulllex</code> | Restated with a direct proof.                  |
| Canonical initial class block | <code>swo:cf:core</code>                                  | Basis of the residual-fiber and cut arguments. |

<sup>18</sup>[merge] True at `109aaca15`; the report has since grown to thirty sources and is still not formalized.

| Audited topic                     | Repository label or location                    | Use here                                                          |
|-----------------------------------|-------------------------------------------------|-------------------------------------------------------------------|
| Uniform set-cut interpolation     | <code>swo:gl:interpolation</code>               | Proof retained with the uniformity and recursion bounds explicit. |
| Core isomorphic to <b>No</b>      | <code>swo:gl:pbd-no</code>                      | Background for large-cut conjugacy.                               |
| Exact adjacency                   | Section 16                                      | Input to intrinsic terminal-block condensation.                   |
| Prefix and automorphism questions | <code>swo:n16.4</code> , <code>swo:n16.5</code> | Negative recovery results and the extension obstruction.          |
| Noncardinal cutoffs               | <code>swo:st:n14.2</code>                       | Exact invariants and the self-similarity obstruction.             |
| Coherent stage embeddings         | <code>swo:st:n14.9</code>                       | Solved within a common cardinality band.                          |
| Changes of class realization      | <code>swo:n16.8</code>                          | Preservation, new represented cuts, and external classifications. |

The new mathematical arguments were checked for their cardinality assumptions, ordinal arithmetic, use of uniform class parameters, behavior at principal cuts, and distinction between internal and external finiteness. Separate checks covered the noncardinal formulas, fixed-slot composition law, countable homogeneity and topology, core automorphism obstruction, and nonstandard terminal-block analysis. These were written-proof checks, not exhaustive searches over infinite objects and not kernel verification.

The deliverable is a standalone LaTeX source with an internal bibliography ([merge] not shipped here; source 21 ships no code or data, and its two audits are `21-obstructions-SOURCE_AUDIT.txt` and `21-obstructions-LEAN_INTERFACE_AUDIT.txt`). It compiles with pdfLaTeX; the accompanying build command uses `latexmk` to settle references. The final PDF was checked for unresolved references, overfull material, and rendered-page layout. Repository files were read but no repository build was performed and no repository content was changed.

## 111.10 Source 22: foundations, raw class orders, and universe dependence

### 111.10.1 Global choice and the interpretation of a class well-order

A class relation  $R$  on a class  $A$  is *set-like* when every predecessor segment  $\{y \in A : yRx\}$  is a set. A set-like class well-order of a proper class has order type  $\text{Ord}$ , and its unique ordinal enumeration is available by elementary class definitions using its set predecessor segments. An unrestricted class well-order need not be set-like. Orders of types  $\text{Ord} + 1$  and  $\text{Ord} + \text{Ord}$  have points with proper-class predecessor segments and cannot be enumerated exhaustively by a single  $\text{Ord}$ -word.

Global Choice supplies a set-like global well-order and hence the baselines used above. The maintained report proves the stronger choice equivalence, over **GB** without assuming set choice: the existence of a class well-order of the whole surreal class is equivalent to Global Choice [39, 40] ([merge] theorem 13.1). We use this as a previously established result, not as a new theorem. The explicit choice parameters in sections 86.3 and 102.1 therefore have genuine foundational content.

When “well-order” is stated internally in a two-sorted model, it refers to the model’s available subsets or classes. Under **GBC**, a fixed class linear order is well-founded exactly when it has no set-coded descending  $\omega$ -sequence. For the nontrivial direction, if an available nonempty subclass has no least member, use Global Choice to select a predecessor at each finite step; recursion on

$\omega$  with class parameters and replacement yield a set-coded descending sequence. Conversely, the range of a descending sequence is an available subclass with no least member.

It follows that enlarging the class realization over the same set universe does not change well-foundedness of a *fixed already available relation*. It can change which relations exist. External well-foundedness of a nonstandard model is a separate matter: an external descending sequence need not be a set of that model.

### 111.10.2 Direct comparison without an ordinal enumeration

The direct common-head construction from the earlier report [39] avoids forcing a non-set-like class order into an Ord-enumeration. We spell it out because it explains why raw comparison itself does not require ETR.

For class well-orders  $R, S$  on the same labelled carrier  $A \subseteq \mathbf{No}$ , put

$$I(R, S) = \{x \in A : \text{Pred}_R(x) = \text{Pred}_S(x) \text{ and the two orders agree there}\}.$$

Equality and agreement here are expressed by quantifying over elements, not by collecting proper predecessor classes into a set.

*Note 111.17* (Source 22, Proposition 10.1, “Elementary raw lexicographic comparison”). *The class  $I(R, S)$  is the largest common labelled initial part of  $R$  and  $S$ . If  $R \neq S$ , the complement has an  $R$ -least element  $a$  and an  $S$ -least element  $b$ , with  $a \neq b$ . Declaring*

$$R <_{\text{lex}} S \iff a < b \text{ numerically}$$

*defines a strict linear comparison of the available class well-orders. The definition is elementary in  $R, S$  and agrees with enumeration lexicography in the set-like case. This is [theorems 14.2 and 14.7](#); source 22 credits it to the earlier report.*

*Source 22’s proof.* If  $x \in I(R, S)$  and  $yRx$ , then  $y$  belongs to the common predecessor class of  $x$ . The agreement of the restricted orders there shows that the predecessor classes of  $y$  agree and carry the same restricted order. Thus  $y \in I(R, S)$ ; the argument is symmetric for  $S$ . Any common labelled initial part is contained in  $I(R, S)$ , proving maximality.

If  $R \neq S$ , this common part is proper. Its complement is an available nonempty class, so it has least elements  $a$  and  $b$  in the respective well-orders. If  $a = b$ , both predecessor classes would be  $I(R, S)$  with agreeing orders, forcing  $a \in I(R, S)$ , a contradiction. Trichotomy and asymmetry follow from the numerical label order.

For transitivity, suppose  $R <_{\text{lex}} S <_{\text{lex}} T$ . The heads  $I(R, S)$  and  $I(S, T)$  are initial parts of  $S$ , so they are comparable by inclusion. If one is strictly shorter, the other comparison still agrees at the first disagreement of that shorter head, and that comparison decides  $R <_{\text{lex}} T$ . If the heads are equal, the three next labels have the form  $a < b < c$  numerically, so again  $R <_{\text{lex}} T$ . When ordinal enumerations exist, the common head is exactly their common prefix.

One application of this construction, or extraction of one canonical set-like initial head, must not be confused with an arbitrary iteration whose successive states are proper classes. The latter can call for stronger recursion principles.

### 111.10.3 Which orders embed?

For set alphabets, [note 62.15](#) gives the exact ordinal spectrum and universality for linear orders of size at most the alphabet cardinal. In the global setting, the ordinary-class universality statement is stronger but must be carefully typed.



*Note 111.18* (Source 22, Proposition 10.2, “Ordinary-class universality”). *In GBC, every linear order whose carrier is an ordinary class of sets embeds, by a class map, into  $\mathbf{No}$ , and hence into the set-coded core  $\mathcal{P}_s(b)$  and the virtual global enumeration order.* This is the class universality of [propositions 5.13](#) and [19.20](#); source 22 says it is the usual surreal universality mechanism.

*Source 22’s proof.* Use Global Choice to enumerate the carrier in a set-like order, with domain a set ordinal or  $\text{Ord}$ . At stage  $\alpha$ , the previously placed points form a set. Their images below and above the next element determine a separated set cut of  $\mathbf{No}$ . Choose a separator, for example by the canonical simplest-cut operation or by the fixed global choice, and place the next point there. Every initial recursion segment is set-valued, so ordinary transfinite recursion with class parameters gives the desired class embedding.

To embed  $\mathbf{No}$  into  $\mathcal{P}_s(b)$ , choose uniformly a supported completion of each singleton prefix  $(x)$ . First labels show that  $x < y$  implies the completion of  $(x)$  is below the completion of  $(y)$ . Compose the two embeddings.

This is the usual surreal universality mechanism, not a historical novelty [15, 39]. It does not quantify over a same-level class whose elements are arbitrary classes. In particular, it does not imply an embedding of the entire higher-order collection of raw well-orders into an ordinary class order.

#### 111.10.4 Why GBC is enough for the new reconstruction

Four superficially similar operations have different strengths:

- (i) Set transfinite recursion with already supplied class parameters produces set-valued stages along each set ordinal; the compatible initial solutions can be collected by elementary comprehension.
- (ii) The height of one cylinder is the order type of a *set* of ancestors. No global class rank needs to be recursively constructed.
- (iii) The branch reconstructed from a cut has a unique value at each coordinate, explicitly defined by the existence of a crossed *set* prefix. It is obtained by comprehension, not by repeated choices of new classes.
- (iv) A recursion with proper classes as successive states is a different assertion. Its solution is a class of pairs coding all stage memberships, and its existence is not automatically a theorem of GBC.

The distinction is substantive. Gitman and Hamkins connect elementary transfinite recursion for class relations with clopen determinacy for class games [22]. Gitman, Hamkins, Holy, Schlicht, and Williams identify the class forcing theorem with  $\text{ETR}_{\text{Ord}}$  over GBC [24]. None of these stronger principles is silently used in [theorems 87.1, 87.4, 102.2](#) and [102.3](#).

#### 111.10.5 Three safe frameworks

**Set fragments in ZFC.** Fix a birthday cutoff and an enumeration ordinal. Every carrier, function space, and relation space is a set. All results of [note 62.15](#) and [sections 63.1, 63.2](#) and [63.4](#) live at this level. Singular cutoffs are allowed; inaccessible cardinals are not assumed.

**Classes in a two-sorted theory.** In GBC, set prefixes and support codes form ordinary classes. Enumerations and raw relations are class variables. Incidence and evaluation are ordinary class relations with set-coded arguments. Assertions about all enumerations are formulas with class quantifiers, not definitions of an ordinary class of enumerations. Stronger comprehension in KM does not by itself turn classes into members of classes; it changes the allowed formulas and resulting class existence, not the basic sort distinction.

**An explicitly larger universe or additional sort.** In an external universe, the classes of a fixed set model may form a set, and their functions and relation spaces can then be ordinary set objects externally. Alternatively, a higher-order theory may introduce a sort of collections of classes. Both approaches are legitimate when the ambient theory, available collections, and permitted maps are stated. Their conclusions must not be read back as same-level class constructions in GBC.

A ZFC treatment by definable-class schemata is also possible. A supplied definition can be substituted into the elementary constructions. However, syntactic codes of formulas are not a uniform truth predicate, and ranging over all formula codes does not automatically produce a definable evaluator for all classes.

### 111.10.6 A full inaccessible-universe interpretation

Suppose, in an external metatheory, that  $\kappa$  is strongly inaccessible and consider

$$(V_\kappa, \mathcal{P}(V_\kappa)).$$

This is a full class model of the relevant class axioms. Its surreal carrier is externally the set  $\mathbf{No}_{<\kappa}$ , of size  $\kappa$ . Its set-like global well-orders correspond exactly to exhaustive length- $\kappa$  enumerations of that set. Indeed, set-likeness says that each predecessor class has size less than  $\kappa$ ; an external well-order of a  $\kappa$ -sized set with this property has order type  $\kappa$ . Unrestricted class well-orders correspond to all external well-orders on that labelled set, including types  $\kappa + 3$  and longer.

Since  $\kappa^{<\kappa} = \kappa$ , [theorem 63.4](#) gives density and cellularity  $\kappa$  for the set-like enumeration order, while its  $\kappa + 3$  raw stratum has  $2^\kappa$  pairwise disjoint open intervals. This yields another proof of the earlier nonembedding phenomenon [39] ([merge] [theorem 20.1](#)).

*Note 111.19* (Source 22, Corollary 10.3, “A stratum obstruction to raw-to-set-like embedding”). *In the full inaccessible-universe interpretation, the unrestricted raw enumeration order does not order-embed into the set-like enumeration order.* This is source 11’s full-model nonembedding theorem ([theorem 20.1](#)); source 22 obtains it again from its invariant table (Part X), a different route.

*Second route (source 22).* Choose in each of  $2^\kappa$  distinct length- $\kappa$  cylinders of the  $\kappa + 3$  stratum three increasing points. Under an order embedding, their outer-point intervals remain pairwise disjoint nonempty intervals, because each contains the image of the middle point. This would force cellularity at least  $2^\kappa$  in the target, whose cellularity is  $\kappa$ . No assertion about equality of a suborder’s order topology with its inherited topology is needed.

At the opposite external size extreme, if both sorts of a model  $(M, \mathcal{X})$  are externally countable, its available global enumeration order is externally a countable dense order without endpoints. An external back-and-forth gives an isomorphism with  $\mathbb{Q}$ . That map need not be available in the model. This explains why an external order-type statement alone cannot replace the labelled reconstruction of [theorem 102.3](#).

### 111.10.7 The forbidden constructions

A hypothetical surreal realizing the cut  $(\mathbf{No}, \emptyset)$  would be greater than every surreal, including itself. Therefore only set-sized, or suitably universe-small, cuts may be filled without contradiction. Similarly, the collection of all ordinal lengths is not a set, and a set of all support codes cannot have unbounded heights. The diagonal argument in [section 102.1](#) rules out an exhaustive uniform class parametrization of all global enumerations by set-valued indices. These are distinct size obstructions, and none is repaired by merely changing notation from “set” to “class”.

## 111.11 Source 22: a concrete, universe-explicit formalization program

### 111.11.1 Actual inspected interfaces

The ProveIt sign carrier uses a lower universe for ordinal lengths and a larger universe for the entire carrier. The following source interface was inspected at the pinned revision [7]:

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : forall i, signAt i != 0 <-> i < birthday
```

This display transliterates logical symbols for readability; the repository uses Lean’s usual Unicode syntax. Its actual namespace is `Surreal.Foundations`. Inside `SignSequence`, the inspected declarations include

| Existing declaration                             | Role in the present mathematics                                                |
|--------------------------------------------------|--------------------------------------------------------------------------------|
| <code>lt_iff</code>                              | First-disagreement characterization of numerical order.                        |
| <code>birthdays_bounded</code>                   | Bounds the birthdays of a lower-universe-small family.                         |
| <code>not_small</code>                           | The complete sign carrier is not small in the universe of its ordinal lengths. |
| <code>small_bounded</code>                       | A fixed birthday bound gives a small fragment.                                 |
| <code>IsPrefix, truncate</code>                  | Prefix and restriction operations, separate from numerical order.              |
| <code>exists_cut_separator_of_birthday_lt</code> | A separator at a prescribed birthday above all option birthdays.               |
| <code>small_cut_fillers</code>                   | Realization of separated small cuts.                                           |

The last two declarations occur in `SignSequenceCut.lean`, in the namespace

`Surreal.Foundations.SignSequence.`

See [8]. Their parameter types explicitly use `SmallCutData` and `HasSmallCutFillers`. The file constructs the separator through a complete lexicographic order of bounded Boolean words. It does not assume a field structure or an unrestricted cut operation.

These source declarations are existing inputs, not formalizations of the new theorems. No Lean compiler or repository build was run for this manuscript, and no new Lean proof file is supplied. The formalization ledger of the earlier report likewise does not claim that its own global enumeration results have been kernel-checked [40].

### 111.11.2 Set-sized modules first

A first implementation should be independent of surreal arithmetic. It should expose an abstract linear alphabet, ordinal coordinate intervals, and a bijective enumeration subtype. The first-disagreement comparison can reuse a lexicographic function-order interface. The mathematical dependencies suggest the following sequence.

**Prefix and residual module.** Define extendible prefixes, residual carriers, and the order type of the remaining coordinate interval. Prove concatenation and restriction inverse, then cylinder convexity. The theorem must retain the complement-cardinality hypothesis rather than assuming that an arbitrary long partial injection extends.

**Characters and finite blocks.** Express the lower and upper sides as ordered sums over first-disagreement coordinates. Formalize the cofinality of a sum with no last index and with a last index. Derive character transfer, then the finite  $n!$  block description and the adjacency lemma. This isolates the infinite reasoning from finite permutation enumeration.

**Topology and cardinal arithmetic.** State richness as an alphabet hypothesis. Count short injective prefixes; distinguish an open cylinder, a convex cylinder, a  $\pi$ -base, and a base. The two separate lower-bound constructions—many short cylinders and many late residual intervals—then prove [theorem 63.4](#). Instantiate richness for the bounded sign carrier only after the abstract result is complete.

**Unranked tree incidence.** Formalize an abstract node/core incidence relation with extension, incompatible-cylinder separation, and fresh strict refinement. Prove that inclusion recovers prefix, that each ancestor collection is small, and that its ordinal order type recovers height. A bounded-height set version should be checked before passing to a class-model interface.

### 111.11.3 Do not identify all function types with available classes

Both `Ordinal.{u}` and `SignSequence.{u}` live at a level capable of containing their lower-universe-small fragments. The type of all functions between those types is a perfectly legitimate object in a sufficiently large type universe. It is nevertheless not automatically the class sort of a specified GBC model. A model may admit only some such external functions as classes.

Accordingly, the global layer should use an explicit two-sorted model or an explicit predicate of admissible class graphs, together with comprehension and replacement interfaces. A uniform family is represented by one admissible relation with set-valued indices. The trace decoder takes an admissible class parameter and constructs another admissible class by a verified elementary formula. In a higher-universe full-model implementation, label the conclusions as external statements about that full collection.

Universe polymorphism prevents type-in-type errors; it does not automatically express all the subtleties of an arbitrary class realization. Conversely, no new large-cardinal hypothesis is needed merely to prove the abstract set-alphabet theorems in Lean. The inaccessible cardinal in the preceding section belongs to one particular model comparison, not to their proof dependencies.

### 111.11.4 Useful interfaces rather than unsupported code claims

The following is a proposed mathematical specification, not compiled Lean:

**ResidualCylinder:**

input: an extendible injective prefix  $p$  of an alpha-enumeration  
output: an order isomorphism from enumerations of the residual alphabet onto the cylinder of  $p$

**RecoverHeight:**

input: cylinder incidence with distinct nodes and fresh refinement  
output: the unique ordinal type of the set of strict ancestors

**DecodeTrace:**

input: a normalized class cut crossing a proper class of nodes  
output: its unique injective ordinal-class word, defined by crossed set prefixes, together with a coverage criterion

These specifications expose the actual proof obligations: extendibility, small ancestor sets, uniqueness, and admissibility of the defining formulas. Writing a declaration named after a theorem without proving these obligations would not constitute a formalization.

### 111.12 Source 22: conclusion

There are two independent sources of complexity. In set-sized enumeration spaces, a prefix of length below the alphabet cardinal remains flexible, whereas a full cardinal-length prefix may leave an arbitrary residual order. That elementary distinction produces a complete cardinal-topological classification of fixed-length strata, finite factorial blocks, and a sharp contrast between finite and countable tails.

In the global problem, the decisive distinction is between an ordinary class of set codes and the higher-order collection of available class words. Retaining cylinder incidence recovers the missing ordinal ranks; retaining labels allows exhaustive branches to be recognized; and retaining the admitted traces over fixed parameters recovers the available class predicates themselves. Each step has an elementary definition with an explicit size boundary. None requires a same-level class of all well-orders, an unrestricted surreal cut, or an implicit class-valued recursion.

### 111.13 Source 22: theorem and dependency ledger

| Result                                                                       | Framework                           | Status and principal input                                                     |
|------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------|
| Birthday richness, <a href="#">note 62.15</a>                                | ZFC                                 | Derived here from the established sign separator theorem.                      |
| Size and ordinal spectrum, <a href="#">note 62.15</a>                        | ZFC                                 | Earlier general-alphabet mechanism; proof included.                            |
| Residual identity and characters, <a href="#">theorems 63.1 to 63.3</a>      | ZFC                                 | Explicit general reduction and transfer proved here.                           |
| All-length invariant table, <a href="#">theorem 63.4</a>                     | ZFC                                 | Continuation result; needs only a rich alphabet.                               |
| Finite and $\omega$ tail characters, <a href="#">theorems 63.9 and 63.15</a> | ZFC                                 | Continuation results; finite residuals and countable-order analysis.           |
| Finite residual adjacency, <a href="#">lemma 63.13</a>                       | ZFC                                 | Earlier mechanism specialized and reproved.                                    |
| Supported prefix completion, <a href="#">note 86.7</a>                       | GBC                                 | Earlier core mechanism, explicitly reproved.                                   |
| Unranked height and branches, <a href="#">theorems 87.1 and 87.4</a>         | GBC                                 | Continuation results; set ancestors and elementary comprehension.              |
| Trace decoder, <a href="#">theorem 102.2</a>                                 | GBC                                 | Elementary definitions with fixed $b, g$ .                                     |
| Faithful realization recovery, <a href="#">theorem 102.3</a>                 | Metatheory of GBC models            | Same set universe and shared parameters; no internal collection of all traces. |
| Raw comparison, <a href="#">note 111.17</a>                                  | Class theory with given well-orders | Earlier direct construction; proof included.                                   |
| Full-model nonembedding                                                      | External inaccessible model         | Earlier phenomenon, obtained again from the invariant table.                   |

No row asserts a new Lean-checked theorem. The finite executable checks have a different scope, described next.

### 111.14 Source 22: finite regression checks and reproducibility

The accompanying `code/finite_checks.py` uses only the Python standard library. It checks finite first-disagreement comparisons, residual-cylinder convexity and factorial size, recovery of prefix inclusion and height in an injection tree with spare labels, pair-swap order preservation and trace decoding, and the forced-last-label counterexample when spare labels are absent. Its machine-readable output is `data/finite_checks.json`. The delivered run passed 1,388,197 assertions, including 533,417 ordered word-pair cases, 16,071 residual cylinders, and all 267,289 node pairs in a 517-node finite injection tree. The pair-swap decoder was checked on all 16 four-bit words inside an ambient order of 40,320 permutations.

These checks do not verify uncountable cardinal arithmetic, transfinite cofinalities, small-cut filling, class comprehension, or model reconstruction. They guard the finite combinatorial mechanisms used in the written proofs. Their code is deterministic, and the recorded assertion counts can be reproduced by running

```
python3 code/finite_checks.py --output data/finite_checks.json
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The LaTeX source has an internal bibliography and no external image or font-file dependencies. A working build script, source audit, status note, and checksum manifest are included in the archive.<sup>19</sup> Fonts installed in the build environment are not distributed.

### 111.15 Source 22: source audit and boundaries of the review

The inspected repository commit is

d5bd4a67b41b89688a079997574a94bcd855bb83.

The maintained report and guide are at

```
Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex
Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/README.md.
```

The guide was used for the scope, merger history, theorem inventory, review status, and formalization boundary. The specific existing Lean declarations listed in [section 111.11](#) were read directly in the following pinned source blobs:

```
Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean
Blob: e2c093860f29b0b9ea312d50099fc4aa89a76a99.
Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceCut.lean
Blob: c5c35afd98a5ad293687367ee9fb0244f94cc91c.
```

The Library copies of the two earlier continuation manuscripts were consulted for their results and proposed next questions. Both cite the earlier repository revision

0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8.

<sup>19</sup>[merge] Shipped as `code/22-unranked-finite_checks.py`, `data/22-unranked-finite_checks.json`, `data/22-unranked-pdf_validation.json`, `22-unranked-SOURCE_AUDIT.md`, `22-unranked-RESEARCH_STATUS.md` and `code/22-unranked-build.sh`. The build script changes into its own directory and calls `code/finite_checks.py` and `article.tex`, so it builds nothing in this layout; the checksum manifest (9/9 entries verified at placement), the source and the PDF are not shipped. Rerun on a copy with an explicit `-output` ([section 111.49](#)).



They are separately credited in the bibliography and are not assumed to have been merged into the pinned repository.<sup>20</sup>

The source audit is targeted, not a certification of every theorem in the 116-page maintained report ([merge] 116 pages at d5bd4a67b). The new theorems are proved in this article rather than justified solely by an earlier manuscript's assertion. The bibliography distinguishes repository material, unpublished user-supplied continuations, and primary published or arXiv sources. Bibliographic verification does not establish historical novelty, and finite checks do not substitute for proof review.

## 111.16 Source 23: embedding spectra and what the new results do not change

### 111.16.1 Fixed set carriers

For an infinite linear alphabet  $X$  of cardinality  $\mu$ , let  $\mathcal{W}(X)$  consist of complete ordinal enumerations of  $X$ , and  $\mathcal{W}_\mu(X)$  of those of length exactly the initial ordinal  $\mu$ . The following baseline is already in the repository and its companion reals report [39, 1]. We include a proof to isolate it from the new termination theorems.

*Note 111.20* (Source 23, Theorem 8.1, “Fixed-set baseline”). *Both orders have cardinality  $2^\mu$ . They contain an embedded copy of every linear order of cardinality at most  $\mu$ . Their increasing ordinal embedding spectrum, and their decreasing ordinal embedding spectrum, consist exactly of the ordinals  $\beta < \mu^+$ . This is theorem 6.1; source 23 calls it a baseline “already in the repository” and includes a proof.*

*Source 23's proof.* The upper cardinal bound follows by representing each well-order as a relation on  $X$ . Partition  $X$  into  $\mu$  two-element sets and write each as  $\{a_i, b_i\}$  with  $a_i < b_i$ . For a binary word of length  $\mu$ , enumerate the  $i$ th pair as  $(a_i, b_i)$  for bit zero and  $(b_i, a_i)$  for bit one. Concatenating the pairs has ordinal length  $2 \cdot \mu = \mu$ . The first different bit decides the first different label. This embeds the binary lexicographic cube  $2^\mu$  and proves the lower cardinal bound.

For a nonempty linear order  $J$  with  $|J| \leq \mu$ , list its elements, with repetitions if needed, as  $(x_i)_{i < \mu}$ . Associate to  $x \in J$  the binary word  $c_x(i) = 1$  exactly when  $x_i < x$ . If  $x < y$ , then  $c_x \leq c_y$  coordinatewise and the inequality is strict at an index listing  $x$ . Hence  $x \mapsto c_x$  is a lexicographic embedding. Compose with the pair coding. The empty order embeds trivially.

For the upper ordinal bound, suppose  $(f_\eta)_{\eta < \mu^+}$  were strictly increasing in  $\mathcal{W}(X)$ . Construct by recursion a common eventual prefix of length  $\alpha$  for every  $\alpha < \mu^+$ . At a successor stage the next values on a tail of the family are nondecreasing in  $X$ . A nondecreasing  $\mu^+$ -sequence with at most  $\mu$  distinct values is eventually constant: otherwise the change stages would be cofinal in  $\mu^+$ , contradicting its regularity. At a limit stage less than  $\mu^+$ , take the supremum of the earlier stabilization stages.

No eventual common prefix can already exhaust  $X$ , because by Lemma 80.1 all later enumerations would then coincide. Thus the recursion yields an injection  $\mu^+ \rightarrow X$ , a contradiction. The decreasing case uses nonincreasing next values. The lower spectrum bound follows from the universality already proved.

The global word orders discussed above are ordinary proper classes, not one of these fixed-set orders. Under global choice they all embed into **No**, whether or not they are saturated, by the usual forth construction along an Ord-enumeration. Conversely singleton words embed **No** into each of them. Thus they are equimorphic. Theorem 81.9 and Theorem 82.1 show why equimorphism alone loses essential local information.

<sup>20</sup>[merge] They are sources 17 and 13 of this report, placed in f0cd7032d after source 22's pin and printed in Parts VI–IX.

### 111.16.2 Birthday cutoffs

For an infinite cardinal  $\kappa$ , the sign sequences of length below  $\kappa$  form the set  $\mathbf{No}_{<\kappa}$  of cardinality  $2^{<\kappa}$ . Theorem 111.20 therefore applies with  $\mu = 2^{<\kappa}$ . If  $\kappa$  is regular, a family of fewer than  $\kappa$  options has birthdays bounded below  $\kappa$ , so the usual sign-cut construction gives the corresponding small-cut saturation of this stage. At singular stages the birthday bound needs separate care.

The repository develops a finer singular strong-limit dichotomy for binary coding length ([merge] theorem 8.3). We do not use that dichotomy or re-prove its optimality here. Neither its correctness nor any purported strengthening is needed for the termination or condensation theorems.

### 111.16.3 The set-coded core of the global enumeration order

The class of set words must not be confused with the virtual order  $\mathfrak{W}_{\text{sl}}$  of all class bijections  $\text{Ord} \rightarrow \mathbf{No}$ . Nevertheless, an ordinary class of set codes is already dense in the latter and interpolates every uniformly set-indexed cut. This is a prior result of the repository [39]; the following proof connects it to the bounded-prefix method used above.

Fix a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . A set-supported permutation of  $\text{Ord}$  is coded canonically by its set of moved pairs, with identity understood elsewhere. Let  $\mathcal{P}_s(b)$  (source 23's  $P_b$ ) be the ordinary class of these codes, ordered by comparing their evaluations  $b \circ \pi$ . Distinct codes have distinct evaluations.

*Note 111.21* (Source 23, Lemma 8.2, “Completing a set prefix”). *Every injective set word  $p : \alpha \rightarrow \mathbf{No}$  extends to  $b \circ \pi$  for a set-supported permutation  $\pi$  of  $\text{Ord}$ . This is source 11's supported completion (lemma 19.4), for source 23's support-graph codes.*

*Source 23's proof.* Put  $u = b^{-1} \circ p$ , an injection from  $\alpha$  into  $\text{Ord}$  with set graph. Choose an infinite cardinal  $\kappa$  strictly above all ordinals in its domain and range and with  $|\alpha| < \kappa$ . The two complements  $\kappa \setminus \alpha$  and  $\kappa \setminus \text{ran } u$  both have cardinality  $\kappa$ . Any bijection between them completes  $u$  to a permutation of  $\kappa$ . Extend that permutation by the identity above  $\kappa$ , then retain only its moved pairs as the code.

*Note 111.22* (Source 23, Theorem 8.3, “Uniform small cuts have core separators”). *Suppose  $I, J$  are sets and  $(e_i)_{i \in I}, (f_j)_{j \in J}$  are uniformly presented class bijections  $\text{Ord} \rightarrow \mathbf{No}$ , with  $e_i < f_j$  for every  $i, j$ . Some evaluation of a code in  $\mathcal{P}_s(b)$  lies strictly between the two families. Consequently  $\mathcal{P}_s(b)$  is set-cut saturated and isomorphic to  $\mathbf{No}$  in GBC; its evaluations are dense in  $\mathfrak{W}_{\text{sl}}$ . This is theorem 19.10 and corollary 19.15; source 23 calls it “a prior result of the repository”.*

*Source 23's proof.* Uniform presentation means that each indexed family is given by a class relation on index, ordinal position, and surreal value. It permits Class Replacement on set index domains. If both sides are nonempty, let  $\delta_{ij}$  be the first disagreement of  $e_i$  and  $f_j$ . Choose a set ordinal  $\beta$  above all  $\delta_{ij}$ . Every prefix cylinder meeting both families has length less than  $\beta$ . All such prefixes therefore belong to a set: restrict the uniformly given words to set positions below  $\beta$ , collect their set restrictions, and use Replacement.

As in Lemma 81.7, these crossing prefixes form a chain. Their union  $p$  is an injective set word, and all constraints outside its cylinder are correctly oriented. If its cylinder meets both families, its lower and upper sets of next labels are strictly separated: a shared next label would yield a longer crossing prefix. Choose a fresh separator  $a$  of those sets. If only one family meets the cylinder, choose  $a$  strictly above, or strictly below, its set of next labels as required. If neither meets it, choose any unused  $a$ . No input word terminates at  $p$ , since every input has domain  $\text{Ord}$ .

In every case all comparisons are decided by the prefix  $p \frown (a)$ . Complete it by Lemma 111.21; any completion preserves those decisions. When an index set is empty, an appropriate fresh first

label bounds the other family’s first labels, and the same completion works. When both are empty, the baseline itself suffices.

Applying this to sets of codes proves saturation of  $\mathcal{P}_s(b)$ . It is a proper class, since every possible first surreal label is attained by the completion lemma. The back-and-forth argument of Corollary 81.10 gives  $\mathcal{P}_s(b) \cong \mathbf{No}$ . Taking two singleton index sets proves density in the virtual enumeration order.

This theorem realizes *uniformly set-indexed* cuts, not arbitrary class-indexed cuts and not an external collection of unrelated classes. The index sets being small is what bounds the first-disagreement positions. Also, the ordinary class  $\mathcal{P}_s(b)$  is not an exhaustive coding of  $\mathfrak{W}_{sl}$ ; Section 111.17 gives the diagonal obstruction. There is no conflict between a surreal-sized dense core and a strictly higher-level domain of all admitted global enumerations.

## 111.17 Source 23: why the constructions do not create a class paradox

### 111.17.1 Three levels that must not be conflated

The word orders  $\mathcal{T}_\varepsilon(\mathbf{No})$ ,  $\mathcal{I}_\varepsilon(\mathbf{No})$ , and  $\mathcal{T}_\varepsilon^{\text{inc}}(\mathbf{No})$  are ordinary classes: each element is a set function on a set ordinal. The polynomial carrier  $P(\Gamma)$  is also an ordinary class, even when  $\Gamma$  is proper, because its elements are finite graphs. Class parameters are used to define membership, but they do not become members.

By contrast, an exhaustive Ord-enumeration of  $\mathbf{No}$  is a proper-class function. The expressions  $\mathfrak{W}_{sl}$  and  $\mathfrak{W}_{all}$  abbreviate predicates on such functions or relations. A map between all their members requires an explicit interpretation: a uniform formula transforming class parameters, a higher sort, or an external map over a set-sized model. These meanings are not interchangeable.

The condensation quotient presents a third common hazard. A block may be a proper class, so the expression “the class of all blocks” is not an ordinary class in GB. The least-representative construction in Theorem 106.3 removes this hazard. The successor operation (S23.3) likewise quantifies over a set transversal, not a set of proper-class objects.

### 111.17.2 Global choice and the scope of class universality

Global choice supplies a set-like well-order of the universe and hence Ord-enumerations of proper classes. It supports the forth and back-and-forth constructions used here. The repository proves the stronger equivalence that a class well-order of  $\mathbf{No}$  exists exactly when global choice holds, over its stated GB conventions [39]. We rely on the forward global-choice construction; no new proof of that choiceless equivalence is claimed.

Under global choice, every admitted class linear order embeds into  $\mathbf{No}$ : enumerate its domain and at each set stage fill the cut determined by images already chosen. This standard class-universality argument is explicitly discussed by Hamkins [21]. It does not say that there is one class of *all* class linear orders or a uniform map on such a collection. We also do not infer any converse choice principle from universality alone.

### 111.17.3 No exhaustive class of set codes for global enumerations

The following diagonal makes the relevant obstruction concrete. It is consistent with, and independent of the new local word constructions; it is also part of the earlier report’s foundational picture [39].

*Note 111.23 (Source 23, Proposition 13.1, “Uniform-code diagonal”). There cannot be an ordinary class  $D$  of set codes with a uniform class evaluation  $F(d, \alpha)$  such that every exhaustive class bijection*

$\text{Ord} \rightarrow \mathbf{No}$  equals  $F(d, -)$  for some  $d \in D$ . This is [theorem 19.37](#), by the same pair switch along a class surjection onto the codes.

*Source 23's proof.* Assume otherwise and choose a class surjection  $i \mapsto d_i$  from  $\text{Ord}$  onto  $D$ ; global choice supplies it, allowing repetition if  $D$  is a set. Fix a class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . Partition the ordinal positions into consecutive pairs  $(2 \cdot i, 2 \cdot i + 1)$ . On each pair put the two values  $b(2 \cdot i)$  and  $b(2 \cdot i + 1)$  in the order that makes the first one different from  $F(d_i, 2 \cdot i)$ .

This defines an exhaustive class bijection  $g$  by an elementary formula, with no recursive choices. It differs from  $F(d_i, -)$  at position  $2 \cdot i$  for every  $i$ . Since every code is listed,  $g$  has no code, a contradiction.

A bounded-support core can nevertheless be an ordinary class of set codes and be isomorphic to  $\mathbf{No}$ . Such a core is not an exhaustive coding of all class bijections. The termination theorem constructs other ordinary class orders isomorphic to  $\mathbf{No}$ ; it does not evade this diagonal.

#### 111.17.4 Set-valued construction versus class-valued recursion

The fresh-separator lemma uses two small cuts. The crossing-prefix proof uses a bounded set of supplied prefixes. Direct raw comparison uses an elementary common-part predicate. Polynomial condensation uses one explicit formula. None requires a general recursion whose values are arbitrary classes.

A generic iteration of ([S23.3](#)), a recursively chosen class at each stage, or a truth predicate for a class structure is different. The literature on elementary transfinite recursion makes this distinction precise [[22](#), [24](#)]. A proof that an explicit formula satisfies a recursive specification is not a proof of the unrestricted recursion scheme. Conversely, it is unnecessarily strong to assume ETR merely because a verified explicit example has a class-length indexing order.

### 111.18 Source 23: bounded universes and a concrete interpretation

#### 111.18.1 Ordinary ZFC approximations

A set cutoff avoids higher-order collections. Choose a set of surreal labels  $X$ , a set ordinal bound on word lengths, and any chosen relation codes. All resulting carriers and relation families are sets. The class theorems cannot simply be copied without their smallness margins: lengths may reach the bound, required fresh labels may leave  $X$ , and set-cut saturation becomes saturation only for families below an appropriate cardinal.

For example, the proof of the termination theorem needs the union of a small family of prefixes still to have allowed length. Regularity of a length bound is a natural sufficient hypothesis. The finite checks in the package deliberately do not replace this requirement by a finite analogue and then claim a transfinite conclusion.

#### 111.18.2 A full inaccessible-universe model

Let  $\kappa$  be strongly inaccessible and work externally with the full two-sorted structure whose sets are  $V_\kappa$  and whose classes are all subsets of  $V_\kappa$ . Internally, ordinals form a proper class; externally that class has order type  $\kappa$ . Internal set-sized families have external cardinality below  $\kappa$ , and every such family of elements of  $V_\kappa$  belongs to  $V_\kappa$ .

The internal surreal sign carrier has external size  $\kappa$ , and its set-cut property becomes saturation for families of size below  $\kappa$ . Its set words also have external size  $\kappa$ . The full global enumeration collection is now an external set of bijections  $\kappa \rightarrow \mathbf{No}^{V_\kappa}$ , with cardinality  $2^\kappa$ . It is not an internal

ordinary class enumerated by  $\kappa$  set codes. This is a concrete setting in which Proposition 111.23 has an ordinary cardinal shadow.

**Proposition 111.24** (Polynomial orders become ordinal powers). *In this full model, if an admitted class well-order  $\Gamma$  has external order type  $\beta$ , then  $P(\Gamma)$  has external order type  $\kappa^\beta$ , with ordinal exponentiation. Internal set-interval condensation is the operation of collapsing intervals of external cardinality below  $\kappa$ .*

*Proof.* After externally reindexing  $\Gamma$  by  $\beta$ , the polynomials are finite-support functions  $\beta \rightarrow \kappa$ . They form the usual ordinal base- $\kappa$  expansions below  $\kappa^\beta$ , compared at their highest different exponent. This can also be proved without an expansion algorithm: at a successor exponent the order is the sum of  $\kappa$  copies of the preceding order, and at a limit it is the union of the initial suborders with bounded finite support. These are exactly the successor and limit equations for ordinal exponentiation.

A subclass of  $V_\kappa$  is an internal set exactly when it has external size below  $\kappa$ . The forward direction follows from inaccessibility; for the converse, regularity bounds the ranks of fewer than  $\kappa$  objects and places their set in  $V_\kappa$ . Apply this to the intervals.  $\square$

For  $\Gamma = \omega$ , the concrete example is  $\kappa^\omega$ . Its finite successive condensations again have order type  $\kappa^\omega$ , yet their limit equivalence is universal. Thus the class construction has an entirely familiar ordinal realization once the size level is fixed. The inaccessible hypothesis is a sufficient external model for this explanation, not an axiom needed by the GBC proofs of the explicit constructions.

### 111.18.3 Model dependence

In a two-sorted model with fewer available classes, “all global enumerations” means only the available class functions. An arbitrary external enumeration or isomorphism need not be internally available. Likewise, external countability of a model does not make its internal proper classes into internal sets. Statements about external order types, internal uniform transformations, and a higher-order powerclass must be labelled separately. No conclusion about all such class realizations is being inferred from the full inaccessible case.

## 111.19 Source 23: a size-explicit formalization program

### 111.19.1 Interfaces that actually exist

The inspected repository includes the following relevant interfaces [8, 54, 39]. The statements here describe their source; no build was run in this investigation.

| Source interface                                                                                                                                                   | Role in the proposed development                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <code>SignSequence.lean</code> : universe-indexed sign carrier and bounded birthdays                                                                               | Separates the large carrier from lower-universe families.                      |
| <code>SignSequenceCut.lean</code> :<br><code>exists_cut_separator_of_birthday_lt</code>                                                                            | Supplies a separator at a prescribed birthday strictly above option birthdays. |
| <code>SignSequenceCut.lean</code> :<br><code>small_cut_fillers</code>                                                                                              | The abstract small-cut property needed for the termination proof.              |
| <code>SmallCutData.lean</code> :<br><code>ofSmallSets</code> , reindexing, and<br><code>HasSmallCutFillers</code> .<br><code>exists_separator_of_small_sets</code> | Converts predicate-based sets of options to explicitly small indexed data.     |

| Source interface                                                       | Role in the proposed development                                    |
|------------------------------------------------------------------------|---------------------------------------------------------------------|
| <code>SizeObstructions.lean:</code><br><code>noUnrestrictedCuts</code> | Prevents an accidental same-size<br>unrestricted cut-filling axiom. |

In particular, a Lean type named `Set X` does not mean that every such set is small at the cut-index universe. The explicit `Small` hypotheses are the mathematical content of the size boundary. An attempt to fill all cuts indexed at the carrier's own size would admit the cut with every element on the left and nothing on the right, an immediate contradiction.

### 111.19.2 The first formalization milestone

A practical first development should stay abstract. Take a linearly ordered carrier  $A$  in a larger universe, with small-cut fillers indexed in a lower universe. Represent a word by a lower-universe ordinal length and a map from its initial segment to  $A$ . Record or prove smallness of the index type. In the injective version add injectivity as a predicate, not as a new carrier with silently different size.

The proof dependency order is:

1. Prove the two-cut fresh-separator lemma and preservation under a small forbidden family.
2. Define an external marker cut, its extended letter comparison, word lexicography, concatenation, and convex cylinders.
3. Construct the small set of prefixes of a small family of input words. Prove uniqueness of a crossing prefix at each length and the bounded union lemma.
4. Formalize the maximal-prefix separator proof, retaining separately the endpoint-child case and the no-small-approach case.
5. Instantiate with the sign carrier, then prove the increasing-word counterexample using an explicit countable increasing family.

No surreal field arithmetic is needed. This produces a useful checked theorem before attempting virtual orders or internal models of set theory. The article's exact character theorem can follow afterward, since it needs cofinality definitions and cardinal minimality beyond the separator proof.

### 111.19.3 Polynomial and raw-order formalization

There are two complementary routes. In a universe-relative type theory, finite-support polynomials can be represented by finite maps into a large ordinal type. The set-interval predicate should be replaced by *lower-universe smallness* of interval types. Proving that a higher coordinate disagreement contains a copy of the large ordinal carrier then proves non-smallness. This is the type-theoretic version of Lemma 108.7.

For literal GB or GBC claims, formalize a two-sorted model and interpret the constructions inside it. The theorem about arbitrary class well-orders needs an admitted-subclass well-order predicate, not only the absence of a small descending sequence under unstated choice. The quotient should be the subtype of least representatives. The family  $E_t$  should be defined directly by tail agreement; proving its recursive laws is then a theorem, not an axiom asserting an iteration exists.

A higher-universe representation of all relations on a given type is also legitimate. It creates a larger type of codes and must not be identified with a same-level proper-class collection. Universe raising solves the typing problem by changing the size level, not by eliminating Cantor-style obstructions.



### 111.19.4 Verification boundary

No new Lean or Rocq source is supplied as verified code. The package’s Python program tests finite lexicographic comparisons, cylinder convexity, crossing chains, finite-support ordering, and tail-equivalence identities. It cannot certify class well-foundedness, small-cut saturation, the absence of a set cofinal family, or a transfinite limit step. The ordinary proofs in this article are the evidence offered for those assertions.

### 111.20 Source 23: conclusions and result ledger

The main distinction is not between a small order and an unimaginably large one. It is between different carriers, different continuation rules, and different logical levels. For arbitrary and injective set-length words, the termination cut controls all local point characters and exactly when the order becomes surreal-like in the strong saturation sense. For increasing words, the continuation restriction defeats every fixed marker. For raw class well-orders, a safe quotient construction reveals arbitrarily long explicit condensation hierarchies, and those structures occur inside every interval between distinct set-like enumerations.

| Result                                             | Role here                    | Setting                                     |
|----------------------------------------------------|------------------------------|---------------------------------------------|
| Fixed-set cardinality and ordinal spectra          | Prior result, proof included | ZFC                                         |
| Dense set-coded core and uniform cut interpolation | Prior result, proof included | GBC                                         |
| Fresh separators and crossing reduction            | Proof tools developed here   | Saturated class alphabet                    |
| Exact termination criterion                        | Main extension; full proof   | GBC suffices                                |
| Uniform point characters                           | Main extension; full proof   | GBC suffices                                |
| Increasing-word obstruction for every marker       | Main extension; full proof   | GBC suffices                                |
| Direct raw comparison                              | Prior mechanism, re-proved   | Elementary class definitions                |
| Canonical set-interval block quotient              | Structural development       | GBC suffices                                |
| Prescribed polynomial condensation hierarchy       | Main extension; full proof   | GBC, no ETR needed                          |
| Dense raw complexity between set-like endpoints    | Main extension; full proof   | GBC                                         |
| General arbitrary-order iteration and collapse     | Not settled                  | Questions in Section <a href="#">113.17</a> |
| Lean verification of the new theorems              | Not claimed                  | Implementation plan only                    |

### 111.21 Source 23: repository and literature audit

The pinned repository source is 109aaca1505c12d70ae169bdc2011f40bd476f7d. The targeted inspection included:

- the surreal collection guide and the guide to `foundations-and-computation/surreal-well-orders`;
- the existing report’s stated theorem coverage and verification boundary, including its distinction between set words, set-like global orders, raw orders, and relation codes;
- `SignSequenceCut.lean` and `SmallCutData.lean`, including the explicit universe-relative smallness requirements.

The recursive repository tree was too large to inspect exhaustively. This was a targeted source audit, not a full repository review or build. The later branch-tip query returned a merge commit having the pinned revision as a parent; the article intentionally keeps the earlier inspected revision fixed.

The earlier continuation manuscript [41] was consulted for its exact termination and residual-iteration questions ([merge] source 13 of this report; its termination question was answered by source 16, Part VII). Its research agenda, not an unverified imported theorem, is the dependency for the new results. We have not certified all assertions in that continuation or in the merged report. The principal new proofs were written to be independent of the earlier fine cut classifications and support formulas.

The Kanovei–Shelah paper was checked directly, including the displayed index definition on its second page [16]. Its actual construction orders maps from the continuum cardinal into  $\mathcal{P}(\mathbb{N})$  whose ranges are ultrafilters. It should not be quoted as literally proving every theorem about the order of all well-orderings of  $\mathbb{R}$ , much less all surreal well-orders. Its relevance is the useful technique of linearly ordering rich codes rather than demanding a canonical choice of one representative.

The current paper on finite condensation [68] supplies context for the word “condensation,” not a proof of our set-interval analogue. The class-recursion references [22, 24] justify keeping ordinary elementary class constructions separate from a general class-valued recursion scheme. No claim of exhaustive literature search or historical priority is made.

## 111.22 Source 23: reproducibility and finite diagnostics

The package contains the standalone source `article.tex`, its compiled PDF, a build script, an executable standard-library Python checker, its JSON output, a research-status note, and checksums.<sup>21</sup> Run

```
python3 code/finite_checks.py -output data/finite_checks.json
sh build.sh
```

from the package directory.

The checker uses finite alphabets and exact integer comparisons. It verifies lexicographic ordering for several termination positions, convexity of all tested prefix cylinders, uniqueness of crossing prefixes at each length for separated finite constraint families, and the ordering and nested tail-equivalence identities for finite coefficient vectors. The checks also distinguish increasing words from arbitrary injective words in the examples used by the article. Finite raw-order comparisons and the paired-permutation diagonal are also checked. The recorded run passes 547,480 exact checks, with seed 20261003.

There is no finite surrogate test asserting that an interval is a set rather than a proper class. In a finite model every interval is a set, so such a surrogate would destroy the point of Lemma 108.7.

<sup>21</sup>[merge] Shipped as `code/23-termination-finite_checks.py`, `data/23-termination-finite_checks.json`, `code/23-termination-build.sh` (it compiles an `article.tex` beside itself and builds nothing here), `23-termination-RESEARCH_STATUS.md` and `23-termination-SOURCE_AUDIT.md`; the source, the PDF and the checksum manifest (8/8 verified) are not shipped.

Likewise, the limit stages of the polynomial hierarchy and the absence of small cofinal families are proved in the text, not inferred from a long finite run.

The PDF was compiled with pdfL<sup>A</sup>T<sub>E</sub>X and inspected by rendering pages. No Lean environment was available for a kernel check. Generated compilation auxiliaries are not part of the distribution.

### 111.23 Source 24: a formalization plan tied to the inspected ProveIt sources

The source inspection was deliberately narrow; the concrete module described below is cited as source code [10], not as verification of the new results. The relevant report’s README identifies the existing well-order study as an unformalized research report. The current work was not inserted into the repository and the repository was not built. We inspected an actual foundation module rather than treating a prose claim that the surreal library exists as verification of the new theorems.

#### 111.23.1 A concrete existing interface

At the pinned revision, the file

Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean

imports the sign-option theory and proves, in namespace `Surreal.Foundations.SignSequence`, declarations including

```
isPrefix_of_minimum_birthday,
existsUnique_prefix_of_ordConnected,
existsUnique_minimum_birthday_of_ordConnected,
ordConnected_separators,
existsUnique_simplest_separator.
```

The last declaration has the essential shape

```
(c : SmallCutData.{u, u + 1} SignSequence.{u} (\x y => x < y))
(hc : exists x, c.IsRealizedBy x)
: exists unique x,
  c.IsRealizedBy x and
  forall y, c.IsRealizedBy y -> IsPrefix x y
```

This display is an English-like rendering of the inspected signature, not a compilable Lean replacement. Most importantly, `hc` is an *existence hypothesis*. The theorem gives a unique simplest separator once existence is supplied; it does not itself prove that every cut has a separator. The explicit small-cut universe parameters are also part of the interface, not removable decoration.

#### 111.23.2 Recommended separation of modules

A new development should begin with an arbitrary infinite alphabet and only later instantiate it with surreal birthday cutoffs. One possible module organization is the following; all names here are proposed, not claims about files already present.

| Proposed module                  | Main obligations                                                                                             |
|----------------------------------|--------------------------------------------------------------------------------------------------------------|
| WellOrders/Enumeration           | Labelled well-order equivalence, fixed-length words, absence of proper-prefix ambiguity for a fixed carrier. |
| WellOrders/PrefixExtension       | Complement cardinalities, ordinal-tail type, completion of short injections.                                 |
| WellOrders/PrefixTopology        | Convex clopen cylinders; equality of prefix and order topologies under thickness.                            |
| WellOrders/ForwardCompletion     | Prefix uniformity, Cauchy-filter stabilization, injective completion, exhaustion requirements.               |
| WellOrders/ExhaustionTree        | Child partition, exact branching, regular-limit unions, the homeomorphism of Theorem 66.2.                   |
| WellOrders/PermutationContinuity | Bounded-image predicate; continuity criteria; complete two-sided uniformity.                                 |
| WellOrders/CountableCompletion   | Complete metric, real-line completion, explicit dense $G_\delta$ representation and gap cardinality.         |
| Surreal/ThickCutoffs             | Sign-cone interval lemma and cardinality at all infinite birthday cutoffs.                                   |
| ClassOrders/CommonPrefix         | The predecessor formula and raw lexicographic transitivity with explicit class-model semantics.              |
| ClassOrders/UniformFamilies      | Uniform family coding, Class Replacement bounds, small-net rigidity, class completion schemes.               |

Here *ordinal tail type* is a real obligation: a prefix of length  $\alpha < \mu$  leaves a tail of type  $\mu$  because  $\mu$  is initial. The corresponding statement for arbitrary fixed ordinal lengths is false in that simple form. Likewise, the formalization of the regular tree must expose the use of regularity at every limit stage.

### 111.23.3 Suggested proof boundaries and audit points

The general alphabet layer needs no surreal arithmetic. It requires an infinite cardinal, an ordinal or chosen well-ordered index type of that cardinality, a set of labels, and ordinary cardinal comparison. For the order topology, add thickness. The surreal-specific layer then discharges exactly that alphabet interface through signs.

The finite executable model should remain separate from the theorem layer. Enumerating finite permutations can test that a child-cylinder partition is disjoint and exhaustive at finite levels, but it cannot establish regular-limit closure. Testing finite partial injections can catch a reversed composition or a mistaken inverse, but it cannot distinguish regular from singular cardinalities. The distinction belongs in quantified cardinal and ordinal lemmas.

For a class-model formalization, represent a class as a predicate or relation on the universe of the model and specify which predicates are admitted in its class part. Do not identify the full ambient function type with the internal class collection unless a full-class model is the intended semantics. If a higher-universe formalization is chosen instead, write every smallness assumption explicitly; a Lean `Set` of large objects is not automatically small in the universe used by a small-cut theorem.

A useful audit sequence is: first verify all fixed-set theorems; then verify the cutoff instantiation; then verify statements about a single uniformly coded class family; only afterwards address a higher-sort domain of all class relations. This order keeps the most delicate logical issue out of otherwise elementary topology proofs.

### 111.24 Source 24: conclusions

The natural lexicographic enumeration spaces have more than one completion, and the difference is mathematically informative. Forward stabilization preserves injectivity but can lose labels. Inverse stabilization prevents that loss. Dedekind completion is larger again: it also fills cylinder edges and numerical gaps. The three constructions agree on many finite tests while diverging sharply in transfinite settings.

For regular minimal enumeration cardinalities there is an explicit generalized Baire model, with branching  $\mu^{<\mu}$ . At the countable surreal cutoff the topology is completely metrizable and familiar, while the order has continuum many missing real-line completion points. At singular cardinalities the bounded-set ideal becomes visible through discontinuous right translations and inversion. These are structural results about the requested lexicographic orders, not consequences of pretending that the surreals form a set.

At the global level, the same subject is rigorous once the size level is kept explicit. A class of set codes can supply every short prefix but cannot enumerate all global well-orders. Set-indexed approximation is rigid; class-indexed approximation is expressive. Arbitrary class well-orders admit an elementary labelled comparison even when their common prefix is a proper class, but their tails require a theory beyond ordinal-class words. These distinctions are the central safeguards against paradoxes and the most useful guide to further formalization.

### 111.25 Source 24: source audit, provenance, and limits

The repository snapshot used for the inspection is

109aaca1505c12d70ae169bdc2011f40bd476f7d

The main provenance paths are:

Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/README.md  
 Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex  
 Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean

The report README was inspected, including its scope, provenance, and unformalized status. The full merged article was not independently re-audited theorem by theorem for this project. Relevant continuation claims and the explicit complete-metrizability question were checked in the available prior manuscript source. The foundation Lean file was read; its Git blob is

3bf0fb22cb5a5c50b05360fd8e5dbeafb104e959.

No new Lean kernel verification is asserted, and no repository-wide build or audit was performed.

The distinction between inherited and developed results is as follows. The interpretation taxonomy, the support-core framework, raw common-prefix comparison, and earlier spectra and cut classifications come from the predecessor reports. Their relevant elementary mechanisms are reproved where needed. The exact forward-completion theorem, the exhaustion-adapted regular homeomorphism, the countable completion-complement analysis, the singular continuity criteria, and the small-net/inverse-reconstruction comparison are the principal developments of this article. This is a scope statement, not a historical-priority claim.

The references to the three continuation manuscripts denote available research drafts dated 3 October 2026. They are not described as journal publications or assumed to have been merged into the pinned repository.<sup>22</sup> Public sources were used to verify the Kanovei–Shelah construction, the

<sup>22</sup>[merge] They are sources 17, 13 and 14 of this report.

generalized bounded-topology context, and the distinction between elementary comprehension and stronger class recursion. None of those external sources is being cited as having proved the new results of this manuscript.

### 111.26 Source 24: assumption and dependency ledger

| Result                          | Essential assumptions                                         | Important exclusion                                                            |
|---------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------|
| Forward completion              | ZFC, infinite $ X  = \mu$ , domain the initial cardinal $\mu$ | No numerical order or regularity is needed.                                    |
| Prefix/order topology agreement | $\mu$ -thick numerical alphabet                               | Not asserted for arbitrary sparse alphabets or the full variable-length order. |
| Regular Baire model             | Regular $\mu$ , Choice                                        | No assumption $\mu^{<\mu} = \mu$ ; no singular conclusion.                     |
| Countable real completion       | Countable dense alphabet without endpoints                    | Homeomorphism does not imply order isomorphism with irrationals.               |
| Singular continuity criteria    | Bounded initial-segment topology on $\text{Sym}(\mu)$         | Boundedness is not the same as cardinality less than $\mu$ .                   |
| Global prefix extension         | GBC, fixed $b$ , set-sized prefix                             | Not an exhaustive parametrization of class words.                              |
| Small-net rigidity              | Uniformly set-indexed class relation, Class Replacement       | Does not apply to externally class-sized index sets.                           |
| Global inverse reconstruction   | Supplied uniform class net and two-sided stabilization        | No collection of all class nets and no unrestricted class recursion is formed. |
| Raw class comparison            | Given class well-orders, elementary comprehension             | No assumption that every class well-order has type Ord.                        |
| Full external model             | An inaccessible $\Theta$ in the external metatheory           | This is a conditional model, not a consequence of the weaker base theories.    |

### 111.27 Source 24: finite regression checks and reproducibility

The accompanying Python program uses only the standard library. It checks finite analogues of the mechanisms most vulnerable to indexing mistakes: first-disagreement comparison, convexity of prefix cylinders, disjointness and coverage of the least-unused-label child partition, independent pair-block binary coding, finite partial-injection extension, and central-termination sign-cone comparisons. It also checks the first several instances of the escaping-label example and the identities relating forward and inverse permutations.

The recorded run passed on 5,913 finite permutations and 16,071 prefix cylinders; it checked 10,158 child partitions, 20,775 lexicographic comparisons, 3,618 partial-injection extensions, and 3,255 interval-cone extensions. Full counts and the precise test scope are in `data/finite_checks.json`.

These checks are deliberately described as regression tests. Every finite injection from a finite set to itself is already surjective, so a finite experiment cannot reproduce the missing-label completion boundary. Nor can a finite permutation distinguish regular from singular cardinality. The limiting



examples and the class results are established by the written proofs, not by extrapolation from the test data.

The package contains the standalone LaTeX source, the compiled PDF, the verification script and its JSON output, a build script, a source-audit note, and a checksum manifest.<sup>23</sup> The PDF uses ordinary installed TeX packages. A clean rebuild needs a LaTeX distribution providing `newtx`, `tcolorbox`, and the other packages listed in the preamble, together with Python 3 for the optional checks. No external bibliography database, downloaded image, font file, or network request is needed by the build.

## 111.28 Source 25: a concrete formalization program

### 111.28.1 Existing interfaces actually inspected

The pinned repository revision contains a universe-indexed sign-sequence carrier [7]. Its relevant declaration is

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : forall i, signAt i != 0 <-> i < birthday
```

This display transcribes the interface using ASCII arrows; it is not an independently compiled source file. Numerical order is lifted from the lexicographic order of zero-extended sign functions. The module includes `lt_iff`, `birthdays_bounded`, `small_bounded`, `not_small`, and a separate `IsPrefix` relation. In particular, numerical order and simplicity are not conflated.

The inspected cut module [8] supplies two relevant declarations:

```
exists_cut_separator_of_birthday_lt
small_cut_fillers
```

The former constructs a separator of a specified birthday strictly above all option birthdays. These are precisely the inputs needed for Lemma 86.3; the homogeneity theorem does not require surreal arithmetic.

The repository research article is not itself a Lean development of the results proved here. This task did not run the repository's Lean build, and no new theorem in the present article is claimed to have been kernel-checked.

### 111.28.2 The first target should be the regular set theorem

Theorem 94.1 is a good initial formalization target because its structures, branch spaces, and automorphism groups are ordinary types of set-sized objects. The class theorem should not be simulated by silently treating all large-universe objects as small.

A reusable development can abstract over an ordered alphabet  $X$  and a fresh-cut extension property. It should then separate six mathematical layers.

<sup>23</sup>[merge] Shipped as `code/24-completions-finite_checks.py`, `data/24-completions-finite_checks.json`, `data/24-completions-build_validation.json`, `code/24-completions-build.sh` and `24-completions-SOURCE_AUDIT.md`; the checker's default output is `data/finite_checks.json` beside `code/`, so always pass `-output`; the build script compiles an `article.tex` that is not shipped. The source, PDF and checksum manifest (8/8 verified) are in arrival commit `b63f0c852`.

| Proposed module                  | Principal obligation                                                                  |
|----------------------------------|---------------------------------------------------------------------------------------|
| <code>PrefixProfile</code>       | First disagreement, prefix equivalence, profile preservation, and induced node action |
| <code>SupportedCompletion</code> | Extension of partial injections using an explicit reserve and a support bound         |
| <code>ProfileExtension</code>    | The maximum and no-maximum cases of Lemma 89.1                                        |
| <code>CoreHomogeneity</code>     | Set-sized back-and-forth, partial-bijection invariants, and branch transport          |
| <code>LabelCoherence</code>      | Root-edge representatives, coordinatewise normal form, and small-support rigidity     |
| <code>ExhaustiveExtension</code> | Core trace injectivity, the represented-trace criterion, and the nonextension witness |

These are proposed module names, not declarations reported as already present. The most valuable early milestone is the no-maximum case of the one-point lemma. It is easy to test only successor divergences and accidentally omit the genuinely transfinite content.

### 111.28.3 Proof data worth exposing explicitly

A partial-configuration record should contain its domain and range, a bijection between them, the proof of pairwise first-disagreement equality, and the proof of order preservation. The fixed noncore markers should be treated as additional supplied constraints, not incorrectly inserted into the type of core codes.

The extension lemma should return a *prefix witness* first, with proofs of injectivity and all profile constraints. Supported completion should then be a separate operation. This factorization makes the two independent sources of existence clear: alphabet cut filling and permutation completion. At singular lengths either source can fail for different reasons.

The inverse branch action needs its own proof. A function preserving prefixes injectively need not be a tree automorphism; the construction must retain surjectivity on the core and derive an inverse node map. Similarly, the trace criterion requires preservation of represented cuts in both directions, not only that exhaustive cuts map into exhaustive cuts.

The edge-label relation can be formalized on the subtype of immediate-successor pairs. Its canonical representative is a root edge, so no quotient of proper classes or higher-universe equivalence classes is needed. The coordinatewise normal-form theorem is then a transfinite induction on the prefix length.

### 111.28.4 Universe discipline for the eventual class analogue

In the repository’s sign interface, ordinals of universe  $u$  and the full sign carrier inhabit the next universe, while a fixed birthday fragment is small relative to  $u$ . A theorem using `Small.{u}` I must not be generalized to an arbitrary higher-universe index type without a new proof. The small-cut property only accepts the intended lower-universe option families.

A universe-relative analogue of the main theorem would need an explicit notion of small support, a chosen enumeration or appropriate presentation of the alphabet, and the exact closure principle required by its back-and-forth. It is not enough to replace `Ord` with `Ordinal` in an informal proof. Lean’s universe hierarchy is a size-management mechanism, not automatically a model of every GBC class statement.

For a direct foundational formalization, one may instead work inside a formal model of set and class theory, with elementary comprehension and class replacement explicit. That approach is heavier but makes the “no ETR” claim an actual theorem about axioms. The regular set theorem and the class-theory theorem are complementary targets rather than interchangeable encodings.

### 111.29 Source 25: conclusions and theorem ledger

The ambiguity in the original question is productive when made explicit. Fixed set carriers yield ordinary set orders with bounded ordinal spectra. Full surreal enumerations form a higher-level domain, but a set-coded core and an ordinary class of prefix nodes make a large part of their geometry accessible in GBC. Raw class well-orders require direct relation comparison rather than a fictitious enumeration of every class type by Ord.

The new structural conclusion is that the supported core and its complete unranked prefix geometry are still not enough to identify exhaustive branches. Their symmetries have a precise configuration classification, and all noncore injective branches lie in a single orbit. Exhaustiveness becomes visible only after adding information that links labels across nodes. The natural equality-of-edge-labels relation does more than recover coverage: together with the supported core, it removes every automorphism.

| Result                                                      | Framework                       | Status in this article                                       |
|-------------------------------------------------------------|---------------------------------|--------------------------------------------------------------|
| Fixed-set spectrum, Theorem 86.1                            | ZFC                             | Earlier general-alphabet result, proof supplied              |
| Class-order universality, Proposition 86.2                  | GBC                             | Standard small-cut construction, recalled                    |
| Supported completion, Lemma 86.4                            | GBC                             | Earlier core mechanism, proof with reserve supplied          |
| Unranked height recovery, Proposition 88.1                  | GBC                             | Earlier continuation result, re-established                  |
| One-point profile extension, Lemma 89.1                     | GBC                             | Main extension lemma proved here, including limit case       |
| Configuration orbits, Theorem 90.1                          | GBC                             | New continuation result; set-valued recursion                |
| Invisible exhaustiveness, Theorems 91.1, 91.2               | GBC                             | New negative recognition result, robust under set parameters |
| Cylinder-preserving nonextension, Theorem 92.2              | GBC                             | Strengthening of earlier core extension obstruction          |
| Edge-coherent normal form and rigidity, Theorems 93.1, 93.3 | GBC                             | New positive reconstruction and rigidity results             |
| Regular set theorem and group size, Theorems 94.1, 94.2     | ZFC, stated cardinal hypotheses | Set analogues and exact cardinality proved here              |
| Cofinal profile obstruction, Theorem 95.1                   | ZFC                             | Explicit sharpness mechanism and singular warning            |
| Raw comparison and diagonal obstruction                     | GBC                             | Earlier mechanisms, re-established for scope and foundations |

No entry in this ledger is a claim of independent refereeing, bibliographic priority, or a new Lean proof. The mathematical evidence for the new entries is the sequence of ordinary proofs above; the finite checks below have a narrower role.

### 111.30 Source 25: repository and literature audit

#### 111.30.1 Snapshot and scope

The repository was inspected at revision

7ff7736ecf3a0adc8536b2d083cf819e1ba39ace.

The targeted sources were the maintained surreal-well-orders article and its README, and the foundational Lean files `SignSequence.lean` and `SignSequenceCut.lean`. The repository source includes continuation material in addition to the originally merged manuscripts; we do not infer a current compiled page count from the README’s historical description.

The audit was not a line-by-line review of the entire repository, and it did not build Lean, validate all prior manuscript claims, or modify any remote file. The principal dependencies needed here—fresh separation, supported completion, core density, prefix recovery, and the class-order comparison—are either explicitly identified with inspected interfaces or re-proved in the body of this article.

The Library continuations consulted for provenance were the cut-reconstruction manuscript, the second continuation on spectra and extension obstructions, the residual-geometry manuscript, and the termination-and-condensation manuscript listed below. Their titles and relevant research questions were retrieved to avoid presenting an already supplied result as a fresh contribution. We do not claim to have independently reviewed all proofs in those manuscripts.

### 111.30.2 Literature scope

The Kanovei–Shelah paper was checked for the actual index space used in its construction, which explains the correction in Section 2. The class-recursion references distinguish elementary set-valued recursion from ETR and related stronger principles. Ehrlich’s survey [15] provides historical context for the surreal continuum; the specific new prefix-homogeneity and rigidity arguments are not attributed to that survey.

Historical novelty has not been established by an exhaustive literature search. Theorems marked as continuation results are claimed relative to the inspected material, and their validity must be judged from their stated hypotheses and proofs.

### 111.31 Source 25: executable checks and reproducibility

The archive includes a deterministic Python program using only the standard library. It tests finite restrictions of ordered prefix recodings over rational labels, the maximum-attained one-point construction in both directions, the first-disagreement/prefix-equivalence identity, finite supported completions, and a cross-node label-coherence counterexample.

The recorded run used seed 20261003 and 2,500 trials. It passed 269,329 assertions, broken down as follows.

| Checked finite mechanism                         | Assertions |
|--------------------------------------------------|------------|
| Profile preservation in sampled prefix recodings | 36,847     |
| Fresh injective extension prefixes               | 2,500      |
| Forward one-point profile extension              | 13,587     |
| Backward prefix injectivity                      | 2,500      |
| Backward one-point profile extension             | 13,587     |
| Prefix equivalence from first disagreement       | 184,235    |
| Finite supported completions                     | 16,071     |
| Failure of cross-node label coherence            | 1          |
| Two-sided fresh-cut example                      | 1          |
| Total                                            | 269,329    |

A finite sampled recoding is not asserted to be an automorphism of the entire rational injection tree. It tests the local finite profile mechanism. In particular, these checks do *not* verify the no-maximum limit case, proper-class recursion, global homogeneity, the exhaustive-branch non-invariance theorem, cardinal arithmetic, or Lean kernel correctness. The JSON result file records these exclusions explicitly.

The supplied build script runs the checks and compiles the standalone LaTeX source. The principal commands are

```
python3 code/finite_checks.py --output data/finite_checks.json
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is internal to the source. There are no external figure dependencies. The archive includes the compiled PDF, this source, code and recorded results, a status note, an audit note, and a checksum manifest.<sup>24</sup> Installed font files are not distributed.

## 111.32 Source 26: a concrete formalization program

### 111.32.1 Existing inspected Lean interfaces

The pinned repository contains a universe-indexed sign carrier [7] in

`Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean`.

The inspected declaration has the shape

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff :
    forall i, signAt i != 0 <-> i < birthday
```

[merge] Transliterated to ASCII; the source prints Lean’s Unicode symbols. The carrier is the sign presentation, with termination represented by zero. Its inspected declarations include `lt_iff` for first-disagreement numerical comparison, `ofSigns`, `ordinalOrderEmbedding`, `birthdays_bounded`, `not_small`, and `small_bounded`. The same file distinguishes `IsPrefix` from numerical comparison and proves `simpler_wellFounded` for proper-prefix simplicity.

The repository README additionally records small-cut results in neighboring files. Those declarations were reported there, but their full proofs were not re-audited for this continuation. No theorem about the new non-Borel boundary is being represented as an existing Lean theorem. The article’s proposed module names below are plans, not claims that files with those names already exist.

### 111.32.2 Separate the general alphabet layer from surreal arithmetic

The principal construction can be formalized before any surreal theory. A minimal interface is an index type with a well-order and a cardinal/minimal-length condition, a label type of the same cardinality, and a prefix-agreement relation. At the level of schematic Lean definitions, the core data are as simple as

<sup>24</sup>[merge] Shipped as `code/25-symmetries-finite_checks.py`, `data/25-symmetries-finite_checks.json`, `code/25-symmetries-build.sh`, `25-symmetries-RESEARCH_STATUS.md` and `25-symmetries-SOURCE_AUDIT.md`; source, PDF and checksum manifest (8/8 verified) are in arrival commit `5ad44b1ed`.

```

universe u

def InjectiveWords (I X : Type u) :=
  {f : I -> X // Function.Injective f}

def PermutationWords (I X : Type u) := I ~- X

def PrefixAgree {I X : Type u} [LT I]
  (a : I) (f g : I -> X) : Prop :=
  forall i, i < a -> f i = g i

```

[merge] Transliterated to ASCII; the source prints Lean’s Unicode symbols. These type-level sketches do not implement the cardinal hypotheses or proofs and were not compiled for this article. The substantive obligations are not these wrappers. They are cardinal complement calculations, cofinal block decomposition, the topology generated by cylinders, and the proof that the omitted-label map is continuous.

A suitable development should avoid making “cardinal  $\mu$ ” both an arbitrary well-ordered type and an initial ordinal without an explicit bridge. The proof of Lemma 62.9 must retain the fact that every proper prefix has smaller cardinality and that its residual tail has order type  $\mu$ . At a nonminimal domain the corresponding statement is false, as Proposition 69.29 illustrates.

| Proposed module             | Main obligations                                                                                                          |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Words/MinimalPrefixes       | Initial-cardinal tail reindexing; extension of short injections; two unused-label extensions.                             |
| Words/PrefixTopology        | Clopen cylinder base, Hausdorffness, local character, thick-alphabet order-topology equivalence.                          |
| Words/PrefixCompletion      | Coherent stabilized prefixes, complete injection uniformity, density, open exhaustion requirements.                       |
| Words/OmittedLabelCode      | First-mismatch partition, disjoint reservoirs, marker bijections, exact omitted range, continuity and order preservation. |
| Words/SingularBoundary      | Small non-Borel Cantor set; continuous-preimage contradiction; complete-metric $G_\delta$ criterion.                      |
| Words/ClosedUniversality    | Compact image closedness and the closed-trace argument; countable clopen-base embedding.                                  |
| Words/BoundedImages         | Closed forward-bounded injection space, dense $G_\delta$ bounded subgroup, closure-trace equation.                        |
| Surreal/ThickCutoffs        | Sign-trunk construction and the two cardinal parameters $\kappa$ and $2^{<\kappa}$ .                                      |
| ClassOrders/CommonPrefix    | Admitted-class semantics, elementary prefix formula, raw comparison and transitivity.                                     |
| ClassOrders/UniformFamilies | Support reserves, diagonalization, Class Replacement bounds, set-indexed net rigidity.                                    |

### 111.32.3 The smallest trustworthy boundary proof

The first-mismatch identity (28) and the ordered markers are finite-coordinate combinatorics. Their transfinite use requires a separate cardinal block lemma. Once that lemma is available, the range calculation and continuity proof do not require a developed theory of surreal addition, multiplication, or analytic functions. The only specifically surreal input is thickness of the birthday alphabet.

For the final non-Borel step, a formalization can either use a library perfect-set theorem for Borel Cantor subsets, if the relevant version is available in the pinned environment, or isolate it as a named imported theorem whose proof must be supplied. It should not introduce an unproved



new axiom while presenting the resulting theorem as kernel-verified mathematics. The two cases in Lemma 69.16 also require the ordinary choice-based cardinal alternatives for the continuum and the Bernstein recursion in the equality case.

The complete-metric criterion is proved in Appendix 69.9 precisely to avoid a hidden Polish/separability assumption. Its necessity direction is sufficient for the main negative result. Formalizing that direction first would minimize dependencies while retaining the full conclusion.

For the global part, a sound first target is a class model represented by a carrier of sets, an explicit predicate selecting admitted class relations, and the relevant comprehension/replacement axioms. Alternatively, work one universe higher and label the theorem as an external, universe-relative statement. Using the full ambient function type as if it were automatically the internal class part would erase the distinction that the proof is meant to preserve.

### 111.33 Source 26: conclusions

The singular boundary of the minimal enumeration space is now visible on a very small and explicit test family. Compactly many injections, each omitting at most one label, already force non-Borelness of exhaustiveness. As a result, the complete-metrizability question has a cardinal-independent negative answer at every uncountable countable-cofinality length. Together with local character and the countable case, this yields an exact classification rather than merely an example of a bad metric.

The closed-trace theorem strengthens the conclusion when the enumeration cardinal exceeds the continuum. The  $\sqcup_\omega$  surreal cutoff satisfies all needed assumptions in ZFC and turns the abstract alphabet theorem into a statement about the numerical lexicographic order of labelled surreal well-orders. It is a Baire, metrizable, densely ordered space with extensive closed-subspace universality and no nonempty completely metrizable open part.

These advances coexist with, rather than replace, the earlier order spectra and class-theoretic constructions. Fixed carriers, variable carriers, numerical increasing words, nonminimal lengths, raw class well-orders, and quotient order types remain distinct objects. At the global level, uniform coding is a powerful rigorous substitute for a nonexistent ordinary class of all class relations, but diagonalization marks its limit. Keeping the alphabet, length, topology, and logical level explicit is both the main defense against paradoxes and the clearest route to further formal verification.

**111.34 Source 26: assumptions and dependency ledger**

| Result                        | Assumptions                                       | Main dependencies                              |
|-------------------------------|---------------------------------------------------|------------------------------------------------|
| Prefix extension and topology | ZFC, $ X  = \mu$ initial infinite                 | Complement cardinalities; ordinal tail type    |
| Order-topology equality       | Previous assumptions; thick linear alphabet       | Two-sided unused labels; first disagreements   |
| Forward completion            | Initial infinite $\mu$                            | Cauchy-filter stabilized prefixes              |
| Metric classification         | Initial infinite $\mu$                            | Local character of $\mu$                       |
| Compact coding                | cf $\mu = \omega$ , $\mu > \omega$ , $ A  < \mu$  | Reservoir absorption; first mismatch; markers  |
| Non-Borelness                 | ZFC, uncountable cf $\mu = \omega$                | Coding; classical Borel perfect-set property   |
| Closed universality           | Previous cofinality; $\mu > \mathfrak{c}$         | Arbitrary Cantor-complement coding             |
| Bounded-image refinement      | Fixed baseline $X \cong \mu$                      | Localized reservoirs; bounded inverse coverage |
| Surreal application           | Infinite birthday cardinal $\kappa$               | Sign trunks; $\mu = 2^{<\kappa}$ ; thickness   |
| Raw class comparison          | Given class well-orders; elementary comprehension | Common predecessor formula                     |
| Support core and diagonal     | GBC, fixed baseline                               | Set reserves; pair swaps; uniform evaluation   |
| Small-net rigidity            | Uniform set-indexed family; Class Replacement     | Set of least disagreement ordinals             |

No main set-sized theorem uses large cardinals,  $CH$ ,  $GCH$ , or a regularity assumption on the birthday cutoff. An inaccessible cardinal appears only in a separately labelled external class-model illustration. No result about all unrestricted class well-orders is deduced by treating them as exhaustive Ord-words.

**111.35 Source 26: source audit and reproducibility****111.35.1 Pinned repository inspection**

The GitHub default-branch reference was read during this work and pinned to

7ff7736ecf3a0adc8536b2d083cf819e1ba39ace

The relevant paths were:

Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/README.md

Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex

Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean

The report README was inspected in successive ranges, including its scope, source-status disclosures, formalization boundary, and review corrections. The beginning of the merged article was inspected for definitions, provenance, and the stated result inventory. Its full, very large synthesis was *not* independently re-audited theorem by theorem. The first 230 lines of the sign-sequence foundation file were read at the pinned revision, covering the declarations listed in Section 111.32.

The inspected Git blob identifiers are:

| File           | Blob identifier                          |
|----------------|------------------------------------------|
| Report README  | 4a60c55651c7fddba35441fe0283e6956f838551 |
| Merged article | f3f69663cd63682ad2e1db808bcb3c6e8fd940ce |
| Sign carrier   | e2c093860f29b0b9ea312d50099fc4aa89a76a99 |

The earlier *Prefix Completions* source was retrieved from the user’s available file library. Its explicit singular question, completion theorem, two-sided-topology limitations, and complete-metric appendix were read. It is cited as an available research draft, not a journal publication or an assumed part of the pinned merge ([merge] source 24 of this report).

The external references were checked against primary publisher or author/archive sources. The Kanovei–Shelah indexing definition was checked in the paper, including its rendered page. Publisher metadata for the Gonshor and Kechris books were verified, but neither book was newly read in full for this task. The Dimonte–Motto Ros reference supplies context from its abstract and declared scope, not a theorem imported without checking its hypotheses. There is no claim of an exhaustive bibliographic novelty search.

### 111.35.2 Executable checks and their exact scope

The companion Python program checks three finite mechanisms. It exhausts finite sets of binary branches through depth four and verifies disjoint first-mismatch release with exactly the matching branch unreleased. It verifies first-disagreement order preservation for the two-marker pair code. It also checks common-labelled-prefix comparison on finite permutation orders and convexity of finite prefix cylinders.

These tests can detect a reversed bit convention, a mistaken release stage, a duplicate release, or an incorrect finite comparison formula. They cannot prove infinite cardinal absorption, the perfect-set property, non-Borelness, complete metrizability, Class Replacement, or a proper-class theorem. The infinite results stand on the written arguments and their stated standard inputs. No Lean build, kernel verification, or repository modification was performed.

The recorded run passed 5,645,785 assertions, including 1,050,698 subset–branch configurations, 174,251 marker-code pairs, 15,018 finite order pairs, and 16,072 prefix cylinders. These counts measure regression coverage only.

The package contains the standalone source and compiled PDF, a build script, the finite-check program, its recorded results, a README, and a checksum manifest.<sup>25</sup> It needs no external images, bibliography database, or nonstandard font files. Rebuild and rerun with

```
python3 code/finite_checks.py --output data/finite_checks.json
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  surreal_lexicographic_boundaries.tex
```

## 111.36 Source 27: comparing global well-orders without a class of classes

### 111.36.1 Well-foundedness and set-likeness

A class relation  $R$  on  $X$  is a class well-order if it is a strict linear order and every nonempty admitted subclass has an  $R$ -least element. In GBC, equivalently, every nonempty set subclass has a least element. Indeed, if a nonempty class  $A$  had none, Global Choice would recursively choose a

<sup>25</sup>[merge] Shipped as `code/26-nonborel-finite_checks.py`, `data/26-nonborel-finite_checks.json`, `code/26-nonborel-build.sh` and `26-nonborel-RESEARCH_STATUS.md`; source, PDF, README and checksum manifest (7/7 verified) are in arrival commit 0d7b2441d.

descending  $\omega$ -sequence in  $A$ ; its range is a set, a contradiction. This equivalence concerns classes admitted by the model. It does not identify an internally well-founded relation in a nonstandard model with an externally well-founded one.

The relation is *set-like* if  $\text{Pred}_R(x) = \{y : yRx\}$  is a set for every  $x$ . A set-like class well-order on a proper class has order type  $\text{Ord}$ : each predecessor set has a set ordinal type, these ranks form an initial subclass of  $\text{Ord}$ , and a bounded range would make the carrier a set. Conversely, any class bijection  $\text{Ord} \rightarrow X$  transports ordinal order to a set-like class well-order.

An arbitrary class well-order need not be set-like. A well-order of type  $\text{Ord} + 1$ , realized on **No** using Global Choice, already supplies a counterexample. This explains why a naive instruction to compare the “first unequal ordinal entries” is incomplete in the raw interpretation.

### 111.36.2 An elementary common-prefix formula

The next construction is an established foundation of the repository [39]; we reproduce it because the new large-family theorem depends on its exact strength.

**Definition 111.25.** For class well-orders  $R, S$  on the same labelled class  $X$ , put

$$C(R, S) = \{x \in X : \text{Pred}_R(x) = \text{Pred}_S(x) \text{ and } R, S \text{ agree on } \text{Pred}_R(x)^2\}.$$

Here agreement means agreement of the two class predicates on all pairs in the common predecessor class. No power class is formed.

*Note 111.26* (Source 27, Theorem 3.2, “Direct labelled comparison”). *In GBC,  $C(R, S)$  is the largest common labelled initial segment on which  $R$  and  $S$  agree. If  $R \neq S$ , let  $a$  and  $b$  be the respective least elements outside it. Then  $a \neq b$ , and*

$$R <_{\text{raw}} S \iff a <_X b$$

*defines a strict linear comparison of class well-orders. Its formula uses only set quantifiers with  $R, S$  as class parameters. No ETR is needed. This is theorems 14.2 and 14.7; source 27 calls it “an established foundation of the repository”.*

*Source 27’s proof.* If  $x \in C(R, S)$  and  $yRx$ , then  $y$  belongs to the common predecessor class. Any predecessor of  $y$  lies in that class, and agreement there shows that  $\text{Pred}_R(y) = \text{Pred}_S(y)$  with the induced orders equal. Thus  $y \in C(R, S)$ , and the same argument applies to  $S$ . The orders agree on  $C(R, S)$  because  $uRv$  with  $v \in C(R, S)$  is equivalent to  $u \in \text{Pred}_R(v) = \text{Pred}_S(v)$ , and hence to  $uSv$ . Conversely, every common labelled initial segment with agreement is contained in  $C(R, S)$ .

If  $C(R, S) = X$ , the two orders are equal. Otherwise its complement has least members  $a, b$  in the two well-orders. Both predecessor classes are exactly  $C(R, S)$ . If  $a = b$ , the definition would put this point in  $C(R, S)$ , a contradiction.

For transitivity suppose  $R < S < T$ . The common prefixes  $C(R, S)$  and  $C(S, T)$  are initial segments of  $S$ , hence one contains the other. If the first is properly shorter, the first divergence of  $R$  from both  $S$  and  $T$  occurs there, with the same smaller label. If the second is shorter, use the symmetric argument. If the prefixes coincide, the three next labels satisfy  $a < b < c$ . In every case  $R < T$ . Totality and asymmetry follow directly from totality of  $<_X$ .

This theorem constructs a *formula*  $R <_{\text{raw}} S$  on class variables. It does not assert the existence of a class containing all such  $R$ . General isomorphism comparison of unrelated class well-orders is a different problem and can require stronger principles [27]. The labelled identity comparison avoids that task.

### 111.36.3 The first set-like block and its convex fibers

Let  $I_R$  be the elements whose  $R$ -predecessor class is a set. It is an initial segment. Its inherited order has a canonical enumeration  $c_R : \text{Ord} \rightarrow I_R$ : for every set ordinal one can perform the set-length recursion choosing the next element, and uniqueness assembles the prefixes. Equivalently  $c_R(\alpha) = x$  exactly when the set  $\text{Pred}_R(x)$  has order type  $\alpha$ . This also proves uniformity for a uniformly given family  $R_i$ .

For a fixed class injection  $h : \text{Ord} \rightarrow X$ , consider all raw well-orders with  $c_R = h$ . If  $Y = X \setminus \text{ran } h$ , these are exactly the orders

$$h + Q, \quad Q \text{ a class well-order of } Y. \quad (\text{S27.2})$$

Every point of  $Y$  has all of  $\text{ran } h$  before it, so none belongs to the initial set-like block. Two different blocks  $h, k$  first disagree at a set ordinal, and that disagreement decides comparison of *every* order in their respective fibers. Therefore each fiber is convex. Within one fiber, comparison is precisely raw comparison of the tails  $Q$ . These are formula-level statements, and remain meaningful when the fibers are collections of class variables.

The fiber may have one point, finitely many points, a set-sized order, or an entire raw class-order comparison, depending on  $Y$ . This residual information is why replacing every raw well-order by its first ordinal block loses much more than redundant notation.

### 111.36.4 A set-coded core for the set-like interpretation

Fix a baseline class bijection  $b : \text{Ord} \rightarrow \mathbf{No}$ . A permutation  $\pi$  of  $\text{Ord}$  with set support is completely represented by the set graph of its moved points. The associated enumeration is  $E_\pi = b \circ \pi$ . Let  $\mathcal{P}_s(b)$  denote the ordinary class of these canonical codes, compared by their enumerations. Distinct codes have distinct enumerations.

Every injective set prefix extends to such an enumeration: translate it by  $b^{-1}$  to a partial injection of ordinals and complete that injection to a permutation with set support. An explicit optimum is obtained in [section 101](#). Between two set-like enumerations, take their common set prefix and insert a fresh numerical label strictly between their first unequal labels, then complete. A set of used labels cannot exhaust a surreal interval, so a fresh label exists. Hence the core meets every nonempty interval of the set-like comparison.

More strongly, the repository proves that  $\mathcal{P}_s(b)$  fills every separated cut specified by a *uniform set-indexed* family of core elements, and in GBC it is isomorphic to  $\mathbf{No}$  [39]. This is a property of an ordinary class of set codes. Theorems below show why it cannot be promoted to a dense uniform family for the entire raw order.

## 111.37 Source 27: a concrete formalization route

The best first machine-checked development is a generic library of well-ordered words and relation scans. It should depend on an ordered alphabet and explicit size parameters, then specialize to the surreal carrier. This keeps the proofs reusable and makes the size distinctions visible in the interfaces.

### 111.37.1 What ProveIt already provides

At the pinned revision the independent sign carrier is defined as `SignSequence : Type (u + 1)`. It has an ordinal birthday in `Ordinal.{u}`, a sign at each such ordinal, and the condition that the

sign is nonzero exactly below the birthday. Its numerical order is lifted from lexicographic order on the zero-extended signs. The inspected declarations include the following [7, 8, 9, 31].

| Existing declaration                             | Role in a future development                                                |
|--------------------------------------------------|-----------------------------------------------------------------------------|
| <code>SignSequence.lt_iff</code>                 | First-disagreement characterization of the numerical order.                 |
| <code>ordinalOrderEmbedding</code>               | The all-plus embedding of ordinals into the sign carrier.                   |
| <code>birthdays_bounded</code>                   | Every small indexed family has a strict ordinal birthday bound.             |
| <code>small_bounded</code>                       | Each bounded-birthday subtype is small in the lower universe.               |
| <code>not_small</code>                           | The whole sign carrier cannot be resized to the lower universe $u$ .        |
| <code>small_cut_fillers</code>                   | Existence of a strict separator for small separated cuts.                   |
| <code>exists_separator_of_small_sets</code>      | A set-interface form of small-cut interpolation.                            |
| <code>exists_cut_separator_of_birthday_lt</code> | A separator at a prescribed birthday above all option birthdays.            |
| <code>noUnrestrictedCuts</code>                  | Formal obstruction to demanding strict separators for arbitrary large cuts. |

The first entries are in `SignSequence.lean`; the last four are in the cut and size modules named in [section 61.16](#). We inspected their source statements. This article does not claim to have rebuilt `ProveIt` or machine-checked its new lexicographic theorems.

### 111.37.2 Universe levels should remain part of the statement

Lean’s universe hierarchy provides a disciplined way to say that a carrier is too large for one universe while still being an object in a larger one [29]. It does not identify all universe levels or provide a type of every type. In the inspected development, both `Ordinal.{u}` and `SignSequence.{u}` live in `Type (u + 1)`. The lower universe  $u$  controls the small well-ordered representatives; `Small.{u}` asks whether a type can be resized to that lower universe. Predicate types and collections of predicates on a fixed carrier do not each automatically require a new universe. The crucial distinction is between a construction with explicit fixed universe parameters and an unbounded totality across all universe levels. Relation types, family indices, and the model in which a class interpretation is made should retain their actual universe parameters in every theorem interface. An order embedding of a bounded universe’s word type into a larger surreal type is compatible with all the negative results above. It does not establish an embedding of a literal collection of every proper-class relation into the same foundational level. Similarly, proving a small-cut property does not permit instantiating it with the whole carrier as a small set.

### 111.37.3 Recommended theorem interfaces and implementation order

**Stage 1: generic words.** Represent a word by a well-ordered domain and an entry function, or by an ordinal and entries on its initial segment. Use an explicit sum type for the terminator; do not use a distinguished alphabet element unless taking a stated quotient. Prove first-difference uniqueness, linearity, common-prefix rules, convexity of cylinders, and prefix-freeness for exhaustive



injections. The split map of [theorem 82.4](#) should be formalized here, as an exact order isomorphism independent of any surreal arithmetic.

**Stage 2: set relations and scans.** A scan is an order-indexed bijection onto ordered pairs; retain its ordinal type in the data. Define the characteristic table and its order embedding into a binary power. Formalize adjacent-pair coding, stabilization of a  $\mu^+$ -sequence of binary words, and the exact spectrum. Then prove the prefix-neighborhood lemma and the two scan-topology theorems. The six-coordinate construction is a small, concrete counterexample generator with a genuinely transfinite consequence.

**Stage 3: permutation completion.** Separate forced moved points from available reserve points. The main reusable equivalence is ([S27.14](#)). Formalize the finite and infinite cardinal cancellation cases, followed by increasing enumeration of fresh ordinals and ([S27.16](#)). Keep the existence of a matching separate from selecting one computationally. The exact finite optimizer in the accompanying diagnostics is a test of the specification, not a proof of the infinite reserve case.

**Stage 4: an explicit class model.** For the class theorems, formalize a two-sorted model or a fixed universe-bounded interpretation. Expose an interface for elementary comprehension, Class Replacement, and a set-like global enumeration. Prove the common labelled prefix formula and its linearity without introducing a class of class relations. Define a family as a single relation with a set index argument. The diagonal theorem can then be formalized directly.

**Stage 5: set-state recursion and prefix automorphisms.** Prove one lemma assembling compatible unique set-length runs. Apply it first to the residual back-and-forth, then to the local-map evaluator. This isolates the exact foundational step and prevents accidentally assuming a principle of class choice. The branch-transitivity theorem, the coverage bounds, and the cross-node label classification then follow with short structural arguments.

**Stage 6: external model comparison.** Express the trace reconstruction formula and prove its invariance for models sharing their first-order set structure and the displayed class predicates. The marked-model theorem should be an external theorem about these models, not an internal quantification over a universe's own entire semantic apparatus.

These stages form a dependency order, not a promise of an existing implementation. The source audit is deliberately explicit about which items are already Lean declarations and which are proposed new modules.

### 111.38 Source 27: finite diagnostics, proof status, and reproduction

The archive includes a Python program using only the standard library. It checks all effective oriented scans on three labels, the explicit enumeration-versus-table reversal, finite versions of the six-point fiber construction, the exact increasing-word factorization at every cut of a finite alphabet, and the finite support optimum by exhaustive permutation search. The program writes a JSON record of counts and results. Its checks remain active when Python optimization is enabled.

These checks have a limited and explicit role. They detect mistakes in orientation, prefix termination, partitioning, and support-height conventions. They do not prove regularity arguments at  $\mu^+$ , infinite reserve requirements, Global Choice, any class recursion claim, or absence of a uniform class family. Those conclusions rest on the mathematical proofs in the article. In particular, the finite support search always has balanced finite boundary cardinalities; the  $\omega$ -shift obstruction cannot be discovered by extrapolating that test alone.

The theorem statements and their foundational assumptions were reviewed independently during preparation. This is not external peer review and does not change their status to machine-checked mathematics. The stronger general-gap cut-classification proposal was not proved and is left in [section 113.21](#), rather than included among the theorems.

The LaTeX source is self-contained and uses an internal bibliography. Compile with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error
surreal_lexicographic_orders.tex
```

Run the finite diagnostics with

```
python3 finite_checks.py -output finite_checks.json
```

[merge] Shipped as `code/27-scans-finite_checks.py` and `data/27-scans-finite_checks.json`, with `27-scans-SOURCE_AUDIT.txt`; the source and PDF are in arrival commit 79049d58c. The distributed PDF was compiled from the distributed source, checked for unresolved references and citations, and rendered for visual inspection. The separate source-audit note records exactly which repository files were inspected and preserves the pinned revision.

## 111.39 Source 28: a formalization plan tied to the inspected repository

### 111.39.1 Separate the generic theorem from its surreal instance

The new algebra and topology use only a set of coordinates, an ideal of allowed control sets, and permutations. A Lean development should first formalize that generic layer. Surreal-specific code is needed only to supply a thick linearly ordered alphabet and to connect its numerical order topology with the generic prefix topology. This avoids importing arithmetic or real-closed-field results into proofs that do not use them.

A sensible dependency order is as follows. First implement injective partial maps, exhaustive completions, and prefix agreement. Then implement an admissible control ideal, its agreement topology and uniformity, and the exact translation and inversion criteria. Next prove the maximal-subgroup theorem and the symmetric-ideal envelope theorem. The bounded ideal then supplies the singular normalizer and closure results; the small-set ideal supplies the character and complete-group statements. Finally instantiate the order-topological interface at a birthday cutoff.

| Proposed module                    | Main obligations                          | Dependencies to keep explicit                               |
|------------------------------------|-------------------------------------------|-------------------------------------------------------------|
| <code>PartialInjection</code>      | Small and bounded extension lemmas        | Cardinality of both complements; reserve construction.      |
| <code>AgreementIdeal</code>        | Neighborhoods, uniformity, control ideal  | Finite unions, singleton coverage, complement size.         |
| <code>PermutationContinuity</code> | Exact criteria and maximal subgroup       | Internal transposition witnesses.                           |
| <code>GroupEnvelope</code>         | Symmetric ideal and minimality            | Conjugation of pointwise stabilizers.                       |
| <code>BoundedPermutation</code>    | Self-normalizer and one-sided closure     | Singular cofinality; enlarged bounded support.              |
| <code>SmallSetCharacter</code>     | Exact character and diagonal lower bound  | Cofinal small-subset families, not just ordinal cofinality. |
| <code>AgreementCompletion</code>   | Injection completion and two-sided limits | Stabilization on whole control sets.                        |
| <code>SurrealEnumeration</code>    | Thickness and order-topology instance     | Sign cones, universe-indexed alphabet size.                 |

These are proposed module names, not claims that such files already exist in ProveIt. No uncompiled code with admitted proofs is presented as a verified implementation.

### 111.39.2 What the inspected Lean file actually proves

The inspected source [10] is

```
Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean.
```

In the namespace `Surreal.Foundations.SignSequence`, it proves, among other declarations,

```
isPrefix_of_minimum_birthday,  
existsUnique_prefix_of_ordConnected,  
existsUnique_minimum_birthday_of_ordConnected,  
ordConnected_separators,  
existsUnique_simplest_separator.
```

The final theorem has an explicit existence premise for the separator of a `SmallCutData.{u, u + 1}` cut. It establishes a unique simplest separator *given existence*. It must not be used as an independent proof that every permitted small cut is filled. The minimum-birthday theorem similarly concerns a nonempty order-convex set.

There are also two unrelated prefix notions to keep separate: a surreal sign prefix, relevant to birthday and simplicity, and an enumeration prefix, relevant to the coordinate topology studied here. A theorem about one is not an instance of a theorem about the other without an explicit map and proof.

### 111.39.3 Universe discipline and class semantics

A Lean expression of the form `Set A` denotes predicates on the type `A`; it does not, merely by its name, impose the set-versus-proper-class distinction of GBC. A universe-indexed surreal carrier is a type at a specified level. To model global class statements, one should either formalize a two-sorted class model inside the type theory or state external comparison results for a smaller universe embedded in a larger one.

Set-level filter completeness can be formalized in the usual way once its carriers are genuine types. The global small-net and coherent-prefix results should be separate theorems about a model and a supplied class relation. Their hypotheses include Class Replacement or the corresponding semantic closure property. Treating a type of all class relations as an ordinary internal class would reproduce precisely the foundational error the manuscript avoids.

### 111.39.4 A feasible first verification target

The shortest useful verified chain is: conjugation of pointwise stabilizers; the ideal continuity criteria; maximality under the finitary-subgroup hypothesis; and the ideal-envelope theorem. These do not need ordinal arithmetic. A second chain formalizes the bounded extension lemma and the singular normalizer argument, with Lemma 70.5 as a separate cardinal-combinatorial result. The cofinality diagonal in Theorem 70.21 is an independent, compact target.

This modular sequence would provide checked mathematical advances before tackling the larger surreal and class-model interfaces. The current manuscript supplies the proofs and dependency map, not a completed new Lean development or a repository-wide verification certificate.

## 111.40 Source 28: conclusion

The lexicographic study becomes more informative, not less, when its competing interpretations are separated. For a fixed thick set of labels, the minimal enumeration order has a prefix topology. At singular enumeration size, that topology sees bounded positions rather than all small coordinate

sets. Its failure to support the full permutation-group structure is therefore a precise filtration phenomenon, not a defect in surreal arithmetic.

The boundedness-preserving subgroup has a strong algebraic position: it is maximal under the finitary hypothesis and self-normalizing. Its forward closure and completion are exactly describable, while the inverse-bounded locus provides a dense complete metric subspace at countable cofinality. The canonical group envelope replaces bounded control by arbitrary small-set control. Its large character is forced by a direct diagonal argument, and its two-sided completeness has an explicit inverse-stabilization proof.

At the global level, every set of ordinal coordinates is bounded, so the singular obstruction does not transfer unchanged. The real issue is logical size: a virtual domain of class enumerations is not a class of such enumerations. Uniform coding, supported set codes, model-relative class semantics, and the direct common-prefix comparison allow the subject to be studied without making that identification. These distinctions also determine a realistic route to formal verification and a focused list of remaining problems.

#### 111.41 Source 28: assumption and proof-dependency ledger

| Statement                       | Assumptions                                                                  | Important limitation                                                                                                                     |
|---------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Cutoff thickness                | Infinite initial $\kappa$ ;                                                  | No field closure or regularity of $\kappa$                                                                                               |
| Prefix/order topology agreement | sign-sequence order; ZFC<br>$ X  = \mu$ , $\mu$ -thick, minimal domain $\mu$ | assumed.<br>Not asserted for arbitrary sparse alphabets or every ordinal-length slice.                                                   |
| Ideal continuity criteria       | Admissible ideal contained in $[\mu]^{<\mu}$                                 | Necessity uses spare coordinates outside every control set.                                                                              |
| Maximal inherited group         | Subgroup contains all finitary permutations                                  | Ambient discontinuity alone would not suffice for arbitrary subgroups.                                                                   |
| Self-normalizer                 | Singular $\mu$                                                               | Uses an enlarged bounded set of size greater than cf $\mu$ .                                                                             |
| Forward closure and completion  | Bounded-image partial maps; reserve lemma                                    | Completeness of one uniformity is not completeness of every compatible metric.                                                           |
| Dense complete inverse locus    | cf $\mu = \omega$ for the metric assertion                                   | Does not settle complete metrizability of the full forward space ([merge] settled negatively by source 26: <a href="#">note 70.13</a> ). |
| Canonical group envelope        | Ordinary permutation operations; topology must contain $\tau_b$              | A different topology or group law is a different question.                                                                               |
| Character lower bound           | Singular $\mu$ , ZFC                                                         | Bounds the coarsest group refinement, not all finer topologies.                                                                          |
| Global supported extension      | GBC, specified baseline $b$                                                  | Produces a class of set codes, not all class words as elements.                                                                          |
| Global small-net rigidity       | Uniform family with an internal set of indices                               | Does not apply to arbitrary external sequences in a countable model.                                                                     |

| Statement                       | Assumptions                                               | Important limitation                                                        |
|---------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------|
| Unrestricted class lexicography | Given class well-orders and numerical label order         | Does not assert that every class well-order has type $\text{Ord}$ .         |
| Full external universe model    | External strongly inaccessible $\Theta$                   | Conditional modeling approach, not an existence theorem in the base theory. |
| Lean simplicity interface       | Nonempty convex set, or an explicit cut-existence premise | Not a standalone proof of universal small-cut filling.                      |

## 111.42 Source 28: source audit and provenance

The repository was inspected at the pinned revision

`7ff7736ecf3a0adc8536b2d083cf819e1ba39ace`.

The recursive tree response was too large to inspect exhaustively. The focused inspection read the merged-report README and the complete sign-sequence simplicity source. It did not recheck every theorem in the 116-page merged report and did not run a repository-wide build ([merge] the README said 116 pages at `7ff7736ec`, where the article already had Parts I–VIII). The two inspected paths are

`Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/README.md`  
`Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean`.

The Lean file’s inspected Git blob is

`3bf0fb22cb5a5c50b05360fd8e5dbeafb104e959`.

The README describes the merged report as AI-assisted, unrefereed, and not formalized, and records its source and review history. We preserve that distinction rather than upgrading a research report into a formal verification claim.

The available prior manuscript `Surreal_Lexicographic_Orders_Prefix_Completions.tex` was read in substantial source ranges covering its definitions, set-level completion and continuity proofs, global prefix arguments, and research questions. In particular, its questions about the bounded subgroup’s normalizer and the full singular space’s complete metrizability were checked directly. The former is answered here in the specified natural permutation action; the latter remains a question of this manuscript.<sup>26</sup> The prior draft `Surreal_Well_Orders_Further_Study.tex` supplies the cut-reconstruction and prefix-non-rigidity context [41]; no claim is made that every continuation draft has been merged into the pinned repository.

The interpretation taxonomy, basic spectra, cutoff thickness mechanism, bounded continuity criteria, distinction between forward and Dedekind completion, and global prefix framework are inherited or reproved background. The ideal formulation, maximality theorem, singular self-normalizer, one-sided closure/completion analysis, dense inverse locus, canonical group envelope, and exact character analysis are the principal developments presented here. This is a provenance distinction relative to inspected material, not a historical-priority claim.

Primary external sources were checked for the Kanovei–Shelah indexing construction, generalized descriptive-set-theoretic context, and the distinction between elementary comprehension and stronger

<sup>26</sup>[merge] Answered negatively in ZFC by source 26, which source 28 did not see (theorem 69.18; note 70.13). The two delivery files named here are sources 24 and 13 of this report.

class recursion. None is cited as a source of the new normalizer or group-envelope proofs. The bibliography is internal to the standalone LaTeX source, so rebuilding does not require an external database or network access.

### 111.43 Source 28: finite regression checks and reproducibility

The accompanying standard-library Python program checks finite mechanisms that are vulnerable to composition-order and indexing errors: convexity of prefix cylinders, the identity relating conjugated stabilizers to images of control sets, the two forward/inverse group-uniformity identities, extension of finite partial injections, the predecessor-based common-prefix definition, and the reversal between enumeration lexicography and relation-bit lexicography in a two-label example. It records counts and the precise tested size ranges in `data/finite_checks.json`.

The recorded run passed on 16,072 prefix cylinders, 470,323 conjugation identities, 940,646 group-uniformity identities, 1,799 partial-injection extensions, and 15,018 independent common-prefix comparisons, together with the relation-code counterexample. The prefix tests range over all permutations through size 7; the algebraic pair tests range through size 5.

These tests do not prove singular-cardinal or class-theoretic assertions. Every finite injection from a finite set to itself is surjective, and finite permutations cannot distinguish bounded control from arbitrary small-set control at a singular cardinal. The normalizer theorem, the diagonal character bound, the missing-label completion boundary, and the global size restrictions are established by the written arguments, not by extrapolating finite experiments.

The package contains `article.tex`, the compiled `article.pdf`, a build script, the test program and recorded output, a source-audit note, and a checksum manifest.<sup>27</sup> A rebuild requires a LaTeX installation with the packages in the preamble; Python 3 is needed only for the optional regression tests. No font files, generated auxiliary TeX files, or copies of external copyrighted papers are distributed.

### 111.44 Source 29: formalization without paradoxes

The choice of foundation is part of the statement of the problem. The preceding results suggest four workable approaches, each useful for a different portion of the theory.

#### 111.44.1 Four routes and their scope

| Framework                 | What it treats directly                                                                                             | Boundary that must remain explicit                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| ZFC with set bounds       | Birthday fragments, fixed strata, binary codes, countable topology, and every specified set-word construction       | A class is represented schematically by a formula; there is no internal quantifier over all definable classes or universal truth predicate.          |
| GBC with supplied classes | Global enumerations, numerical class carriers, uniform families, raw comparison, and set-valued class constructions | The full collection of class relations is not an ordinary class. A history supplied as a class parameter is different from a theorem that it exists. |

<sup>27</sup>[merge] Shipped as `code/28-envelope-finite_checks.py`, `data/28-envelope-finite_checks.json`, `code/28-envelope-build.sh` (it changes into `code/` and builds nothing there) and `28-envelope-SOURCE_AUDIT.md`; source, PDF and checksum manifest (7/7 verified) are in arrival commit 79049d58c.



| Framework                           | What it treats directly                                                                                                                    | Boundary that must remain explicit                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Higher-order class theory           | Collections whose members are classes, and maps or completions on these collections                                                        | The additional sort and its comprehension axioms must be specified. <b>KM</b> strengthens class comprehension but still does not make proper classes members of ordinary classes. |
| External or universe-relative model | All available objects of a specified model can be studied as external sets; type universes separate carriers from collections of relations | Internal size, external size, internal well-foundedness, and external correctness must not be identified. Enlarging the universe changes the object.                              |

For a ZFC-only development, the first milestone should be fully set-sized: fix  $\theta$  and  $\alpha$ , construct  $\mathbf{No}_{<\theta}$  and  $\mathcal{W}_\alpha(\mathbf{No}_{<\theta})$ , and prove the desired theorem uniformly in those parameters. Statements about all set ordinals then form ordinary first-order theorems. A statement about all *definable class* well-orders is instead a schema, unless one moves to a specified external model with a satisfaction relation available outside it. Coding formulas alone does not supply an internal satisfaction predicate for arbitrary formulas of the same universe.

For GBC, the most economical discipline is to keep all relation arguments explicit. The common-prefix comparison in [Note 86.17](#) uses two supplied classes. The branch automorphism in [Theorem 96.7](#) uses two supplied branches and one fixed global scan. The condensation normal form in [Theorem 108.16](#) uses a supplied history. Each can be described by one elementary class formula after its parameters are available.

By contrast, asserting a solution to every elementary recursion with class-valued slices along an arbitrary class well-order is the principle **ETR**, not a routine consequence of the ordinary set recursion theorem [\[22, 27\]](#). The new **CTH**  $\leftrightarrow$  **CWO** equivalence identifies exactly the existence principle for the particular complete condensation histories used here. It neither proves that arbitrary class recursion follows from comparability nor requires such recursion once a history has already been supplied.

#### 111.44.2 What the inspected Lean code already provides

The repository has genuine Lean files supporting the sign-sequence foundation. Their existence and statements were inspected at the pin; they were not rebuilt in this task. All five files below lie in

`Algebra/SurrealNumbers/Surreal/Foundations/.`

| File                                       | Relevant support actually present                                                                                                                                          |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>SignSequence.lean</code>             | A carrier in <b>Type</b> $(u+1)$ with birthdays in <b>Ordinal</b> . $\{u\}$ , numerical first-disagreement order, bounded birthdays, ordinal embedding, and non-smallness. |
| <code>SignSequenceCut.lean</code>          | Small-cut fillers and separators at a prescribed birthday above the birthdays of the constraints.                                                                          |
| <code>SignSequenceCutOperation.lean</code> | Canonical small-cut operations, birthday bounds, density, and absence of endpoints.                                                                                        |

| File                                     | Relevant support actually present                                                                                                            |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <code>SignSequenceSimplicity.lean</code> | The unique simplest separator <i>assuming an existing separator</i> ; the existence hypothesis is not an unrestricted cut-existence theorem. |
| <code>SizeObstructions.lean</code>       | A lower-universe indexing discipline for small cuts and impossibility of unrestricted cut filling.                                           |

In particular, the assertion named `existsUnique_simplest_separator` must not be used with its separator-existence premise omitted. The small-cut theory and the non-smallness theorem fit together: allowing every subset of the carrier as a lower cut would demand a point above the entire carrier. The obstruction is a formal mathematical statement, not a limitation to bypass with an unchecked universe coercion [53].

A Lean universe-relative surreal type is also not a paradoxical set of all surreals at its own universe. Its birthdays range over ordinals at a smaller universe, and the carrier lives at the next level. One may form relations or collections in a larger type universe, but then the resulting construction is explicitly relative to those universe parameters. No identification of a type with the totality of all types is involved.

### 111.44.3 A practical sequence of formal milestones

The most useful initial formalization is a generic first-difference library rather than a large class-theory encoding. The following milestones have exact mathematical specifications.

1. **Set ordinal words and explicit termination.** Represent a word with its ordinal domain and a function on that domain. Prove existence of the first disagreement, strict-order transitivity, prefix-free comparison of bijections, and the three orders obtained from termination below, between, or above labels. Keep zero padding as a preorder until the stated quotient or reduced representatives have been constructed.
2. **The native birthday spectrum.** Formalize [Lemma 76.1](#) and [propositions 76.3](#) and [76.4](#). The crucial invariant is that the target prefix contains actual signs, not padded termination symbols. At a limit, take the union of the source prefixes and verify the ordinal identity  $2 \cdot \lambda = \lambda$ . This proves the full spectrum in [Theorem 76.5](#) without any class quantification.
3. **The canonical word factorization.** Formalize [Note 82.9](#) over an arbitrary ordered alphabet split as  $L + R$ . The proof only needs the ordinal initial segment of lower-side entries and its residual tail. The proper-class corollary  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$  can then depend explicitly on supplied local alphabet isomorphisms.
4. **Residual fibers and finite tails.** Prove the exact decomposition over an injective prefix into well-orders of its unused labels, before introducing any cardinal coding. For countable alphabets, formalize  $P_{\omega+n} \cong E_n \times_{\text{lex}} n!$  and the adjacent endpoint pairs. This separates the combinatorial core from the descriptive set theory.
5. **Two named topologies and two named Borel structures.** The discrete coordinate topology and the order topology must be distinct definitions. The theorem that comparison is  $\Delta_2^0$  in one coding must not rewrite the Borel structure of the other. The same discipline applies to the topological product Borel structure and the tensor sigma-algebra generated by measurable rectangles.
6. **Condensation from a supplied history.** Prove the floor identities, finite support, and initiality of the normal-form image with the history as a parameter. Only after these lemmas

are complete should an existence axiom such as CWO or ETR be connected to the construction. This order of development makes the exact logical dependency visible in the theorem statements.

These are proposed modules and proof targets, not newly compiled Lean developments. Their advantage is that most of the difficult mathematics can be checked in set-sized or universe-relative form before the final class-theoretic packaging is attempted.

#### 111.44.4 The restrictions that prevent paradoxes

The restrictions are concrete. A surreal’s sign sequence has a set ordinal domain. A Conway normal form has set support. A member of a GB class is a set. A comparison between two class relations does not create a set of all class relations. A uniformly indexed family has one specified class code. A power collection of classes belongs to a higher sort or to an external model. A class well-order with a proper-class predecessor segment is not assigned a set ordinal rank at that point. Finally, a nonprincipal cut between proper classes is not filled merely because every set-sized cut is filled.

With these conditions retained, the phenomena proved here are compatible rather than contradictory. The same set-prefix tree can support very different class branches, and an ordinary class core can be universal for all ordinary class linear orders while failing to contain the entire higher-order family of global enumerations.

#### 111.45 Source 29: results, dependencies, and verification status

The table records the logical basis and the kind of conclusion. “Extension” means a result established here beyond the inspected source treatment; it is not a claim of historical priority.

| Result                                                     | Basis                            | Status and scope                                                                                                                                        |
|------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed-set cardinality and ordinal cutoff                   | ZFC                              | Inherited; reproved in <a href="#">Note 61.4</a> .                                                                                                      |
| Exact native birthday cube spectrum and hierarchy          | ZF                               | Extension; <a href="#">Theorem 76.5</a> and <a href="#">corollary 76.6</a> .                                                                            |
| Noninitial residual strata and exact capacity              | ZFC                              | Extension of the explicitly credited residual machinery; <a href="#">Theorems 78.5</a> and <a href="#">78.8</a> . A specified upper-coding gap remains. |
| Countable birthday-band formula at every admissible length | ZFC                              | Exact equimorphism; no CH and no isomorphism claim; <a href="#">Corollary 78.10</a> .                                                                   |
| Canonical terminator factorization                         | Elementary set/class formulation | Extension; <a href="#">Note 82.9</a> .                                                                                                                  |
| Centered product and zero-convention isomorphisms          | GBC                              | Extension using inherited uniform alphabet normalization; <a href="#">Notes 82.11</a> and <a href="#">82.13</a> .                                       |
| Numerical proper-class adjacency and full-tree obstruction | GBC                              | Extension; <a href="#">Theorems 83.2</a> and <a href="#">83.7</a> .                                                                                     |
| Finite-tail topology and Borel distinctions                | ZFC                              | New application and classification for these strata; classical split-space antecedent credited to Fremlin.                                              |

| Result                                                          | Basis | Status and scope                                                                                                          |
|-----------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------|
| Ordered-prefix branch transitivity                              | GBC   | Extension; <a href="#">Theorem 96.7</a> ; literal labels are not preserved.                                               |
| Raw global comparison and the bounded-support core              | GBC   | Inherited and recalled; <a href="#">Notes 86.17</a> and <a href="#">86.18</a> .                                           |
| Finite normal forms and $\text{CTH} \leftrightarrow \text{CWO}$ | GBC   | Extension; <a href="#">Theorems 108.16</a> and <a href="#">108.19</a> . No $\text{CWO} \leftrightarrow \text{ETR}$ claim. |

The principal proof arguments underwent independent internal mathematical review, including the ordinal fusion at limit birthdays, the finite residual capacities, the termination-factorization signs, the proper-class common-prefix cases, the split-space Borel argument, the uniform tree-automorphism construction, and the initiality step in the condensation normal form. This review is evidence of scrutiny, not a substitute for refereeing or a proof-assistant certificate.

The delivered source is a single self-contained LaTeX file with an embedded bibliography. The PDF is compiled from that source. The package also contains build instructions, an audit of the repository and earlier project sources, and a machine-readable provenance manifest.<sup>28</sup> No remote repository was modified, and no new Lean theorem is represented as compiled or verified.

## 111.46 Source 30: small cuts, class comparison, and the inherited tools

### 111.46.1 Sign sequences and set-sized cuts

Write  $\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha$  as a class definition, not a set union. The birthday  $\text{bd}(x)$  is its sign length. For an infinite cardinal  $\kappa$ ,

$$|\mathbf{No}_{<\kappa}| = \sum_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}.$$

Cardinals are identified with initial ordinals when they serve as coordinate lengths. Ordinal and cardinal multiplication remain different operations.

We use the standard small-cut lemma: for sets  $L < R$  of surreal numbers, there is a simplest separator, whose birthday is at most  $\sup\{\text{bd}(z) + 1 : z \in L \cup R\}$ . The usual sign construction proves this by meeting each lower and upper requirement before or at that bound. It includes empty option sets and gives upper and lower bounds for every set of surreals. These facts and birthday estimates are implemented on the inspected sign carrier [53, 15].

The word *set* is indispensable. Taking all of  $\mathbf{No}$  as lower options would require a surreal larger than itself. A small-cut theorem cannot be generalized to arbitrary class options.

### 111.46.2 Class theory and recursion

We take GB to have ZF as its set part and no choice axiom;  $\text{GBC} = \text{GB} + \text{GC}$  includes set choice. A class has sets as elements. Although the language has class variables, elementary class comprehension permits only set quantifiers in the defining formula, with class parameters.

Ordinary recursion with set values and set-sized initial histories is available with fixed class parameters: solve on each set-sized domain and assemble by uniqueness. A recursion assigning a

<sup>28</sup>[merge] Shipped as 29-birthdays-SOURCE\_AUDIT.txt, 29-birthdays-PROOF\_STATUS.txt, data/29-birthdays-source\_manifest.json and data/29-birthdays-verification.json; source 29 has no executable checks. The manifest also contains three opaque library-file identifiers of the authoring tool for its earlier manuscripts (sources 19–21); they are identifiers, not credentials. Source and PDF are in arrival commit 51b0a69d7.

proper class at each stage is different. ETR supplies additional elementary class-valued recursions; it is not a synonym for ordinary transfinite recursion [22, 24].

A *class well-order* is a strict linear class relation for which every nonempty admitted subclass has a least member. It is *set-like* when each predecessor class is a set. In GBC, for linear class relations, well-foundedness is equivalent to absence of a descending  $\omega$ -sequence. From a nonminimal subclass, choose successively its baseline-least smaller member; ordinary recursion produces such a sequence. Equivalently every nonempty set subset has a least member. This formulation also gives absoluteness for an existing relation between class realizations with the same sets.

### 111.46.3 The choice requirement is exact

*Note 111.27* (Source 30, Theorem 3.1, “Inherited choice equivalence”). *Over GB, the following are equivalent: global choice; a class well-order of  $\mathbf{No}$ ; a set-like class well-order of  $\mathbf{No}$ ; and a class bijection  $\text{Ord} \rightarrow \mathbf{No}$ .* This is source 11’s [theorem 13.1](#) (over GB without set choice); source 30 recalls its argument. ([merge] “Uniformly”, not “coherently”: the orders  $w_\alpha$  need not extend one another, [remark 13.4](#).)

*Source 30’s proof.* We recall the choiceless argument of [39, statement labelled `swo:cf:choice`] ([theorem 13.1](#)). A supplied well-order  $W$  of sign sequences recursively gives well-orders  $w_\alpha$  of  $V_\alpha$ . At successors, the unique ordinal enumeration of the already ordered  $V_\alpha$  identifies its subsets with sign sequences of one fixed length; transport  $W$  to order its power set. At limits compare rank first, then use the existing order on that rank. All stages are sets. Selecting the least element of a nonempty set in a sufficiently large  $w_\alpha$  gives global choice.

Conversely global choice selects a well-order of each birthday layer. Their birthday-ordered concatenation is set-like. The ordinal ranks of a set-like well-order of the proper class  $\mathbf{No}$  form an initial subclass of  $\text{Ord}$ . This range cannot be a set, by class replacement, so it is all of  $\text{Ord}$ . Its inverse is the required bijection. A bijection conversely transports the ordinal order to a set-like one.

This is existence in the class sort, not first-order definability. We sharpen the distinction later. Background and classical choice equivalences appear in [18, 19].

### 111.46.4 Comparing arbitrary class well-orders directly

For class well-orders  $R, S$  on the same labelled class  $X$ , let  $P_R(x) = \{y : yRx\}$  and define

$$x \in C(R, S) \iff P_R(x) = P_S(x) \text{ and } R \restriction P_R(x) = S \restriction P_R(x). \quad (\text{S30.3.1})$$

All equalities expand into quantification over set elements. Elementary comprehension therefore forms this class.

*Note 111.28* (Source 30, Proposition 3.2, “Inherited common-prefix comparison”). *For  $R \neq S$ ,  $C(R, S)$  is their largest common labelled initial segment. Its complement has different least points  $a$  in  $R$  and  $b$  in  $S$ . Comparing  $a$  and  $b$  in the native label order defines a strict linear comparison of class well-orders.* This is [theorems 14.2](#) and [14.7](#).

*Source 30’s proof.* If  $x \in C(R, S)$ , all its predecessors also belong, with the same induced order. Any common labelled initial segment satisfies the formula, proving maximality. Equal residual minima would themselves satisfy the formula, a contradiction. For transitivity,  $C(R, S)$  and  $C(S, T)$  are nested as initial segments of  $S$ . If one is shorter, that earlier disagreement controls the third comparison. If they coincide, use transitivity on the next labels.

No ordinal is assigned to a proper-class prefix. In particular the construction applies to transported orders of type  $\text{Ord} + \text{Ord}$  and their tail rearrangements. This is a comparison formula on class variables; it is not a class containing all the relations as elements.

### 111.46.5 Embedding benchmarks

Write  $\mathbb{B}_\theta = (\{0, 1\}^\theta, <_{\text{lex}})$ . For infinite  $|X| = \mu$ , pair orientations give  $\mathbb{B}_\mu \hookrightarrow \mathcal{W}_\mu(X)$ . Every linear order of cardinality at most  $\mu$  embeds into  $\mathbb{B}_\mu$  by coding its initial segments on a well-ordered list of its labels.

The preceding reports prove the complete fixed-set benchmark ([merge] theorem 6.1):  $|\mathcal{W}(X)| = 2^\mu$ ; all linear orders of size at most  $\mu$  embed; and an ordinal embeds exactly when it is below  $\mu^+$ . At birthday stages the minimal stratum satisfies

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\rho_\kappa}. \quad (\text{S30.3.2})$$

The next section states the precise prefix-coding ingredients needed from this theorem and proves the new residual extensions. An embedding classification does not determine order topology: equimorphic orders may have different endpoints, jumps, and completions. This distinction is central to the later results.

### 111.47 Source 30: formalization without size paradoxes

The distinctions in the theorems suggest a practical formalization architecture. Its purpose is to make a theorem's domain and quantifiers visible in its type, so that an innocent-looking change of representation cannot silently change the mathematics.

#### 111.47.1 Three sound routes, serving different questions

**Set stages in ZFC.** All of the residual coding, countable topology, and Borel results are ordinary set mathematics. Fix  $\mathbf{No}_{<\kappa}$ , an ordinal domain  $\alpha$ , and the relevant set of bijections or relations. Power sets, products, spaces of codes, and quantification over all those codes are legitimate. No inaccessible cardinal, global choice, or class theory is needed. Ordinary choice is used to select well-orderings and codes for a set-indexed collection of residual alphabets.

For this part, a theorem should take an ordinal-indexed domain as data. A bare bijection between that domain and its cardinal does not preserve the first-disagreement order. The singular scheduler and the regular code-sum lemma are precisely the places where erasing the ordinal structure would invalidate the proof.

**Two-sorted GB/GBC.** To study the whole surreal class, express a global enumeration by a formula about one class graph. Express its lexicographic comparison with a second graph by another formula. For a uniform family use one class relation  $F(i, \xi, x)$ , with the index  $i$  ranging over sets. The partial-table extension theorem has the safe shape

$$\forall J \forall p (J \text{ is a class well-order and } p \text{ is a valid set table} \implies \exists W \text{ desired properties}).$$

The proof supplies  $W$  by elementary definitions from its parameters. It never creates a class containing  $J$  or  $W$  as an element.

This route is adequate for the elementary reconstruction and approximation theorems. The present arguments use set-valued recursions with set-sized initial histories. More general class recursions, whether their values or dependency histories are proper classes, require separately stated principles such as ETR. Class forcing is another area where strength must be specified: the unrestricted class forcing theorem is equivalent over GBC to  $\text{ETR}_{\text{Ord}}$  [24]. That theorem is background for a future forcing analysis, not an assumption hidden in our extension arguments.



**A containing universe or an external model.** An actual topology on a collection of classes, an automorphism group of that collection, or an intersection over all class realizations is cleanly formulated in a metatheory in which the model  $(M, \mathcal{C})$  is a set. Then  $\mathcal{C}$  and its subcollections are ordinary sets there. State explicitly whether a support, sequence, or isomorphism is internal to  $M$  or merely exists externally.

For example, a model-internal set may be externally countable without belonging to any prescribed small internal rank. Conversely the whole internally proper surreal class is a set externally. Allowing its whole relation table as an external support makes the observation topology discrete, destroying the density result. This is a change of topology, not a contradiction.

One may alternatively posit a third sort of collections of classes, with explicitly stated axioms. Merely passing from GB to Kelley–Morse does not do that: KM strengthens class comprehension but still does not make proper classes elements of ordinary classes. A higher sort must be introduced as such. There is no need for it in the set theorems or in the elementary class formulas proved here.

### 111.47.2 Why an exhaustive same-level evaluator is impossible

The pair construction gives an immediate diagnostic for an incorrect size claim. Suppose one class relation  $F(\alpha, \xi, x)$  uniformly listed every global enumeration as  $E_\alpha$  for  $\alpha \in \text{Ord}$ . Use the normalized pairs  $u_\alpha < v_\alpha$  from [note 102.15](#), and form by elementary comprehension

$$A = \{\alpha : E_\alpha(2 \cdot \alpha) = u_\alpha\}.$$

At position  $2 \cdot \alpha$ , the pair enumeration  $E_A$  is  $v_\alpha$  if  $E_\alpha$  is  $u_\alpha$ , and is  $u_\alpha$  otherwise. In either case they differ. Thus  $E_A$  is omitted.

Under global choice a purported exhaustive class of set indices can be reindexed by  $\text{Ord}$  (with repetition if the index domain is a set), giving the same contradiction. There is therefore no uniform exhaustive same-level evaluator. This familiar diagonal argument is entirely compatible with [note 102.16](#): that theorem recovers each individually supplied class from a corresponding enumeration; it does not list all classes.

Likewise, enumerating formulas is not an automatic escape. For a fixed formula, satisfaction in a given set structure is available. A uniform predicate evaluating every formula in the whole universe supplies additional truth strength; Tarski’s theorem prevents it from being definable in the same language. Stronger class theories can supply truth for a fixed language, without thereby listing all available classes. A metatheoretic schema of separate definitions does not itself supply one uniform internal evaluator.

### 111.47.3 The Lean interface actually present in ProveIt

The inspected files lie under

Algebra/SurrealNumbers/Surreal/Foundations/.

Their source, rather than a proposed module name in a report, is the basis for the following inventory.

| Source file                    | Existing declaration and mathematical role                                                                                                                                                                                                    |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>SignSequence.lean</code> | <code>SignSequence.Type(u+1)</code> has a birthday in <code>Ordinal.{u}</code> and a zero-extended sign function. <code>birthdays_bounded</code> bounds small families; <code>not_small</code> proves the carrier is not small at level $u$ . |

| Source file                                | Existing declaration and mathematical role                                                                                                                                        |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>SignSequenceCut.lean</code>          | <code>exists_cut_separator_of_birthday_lt</code> supplies a separator of a prescribed birthday above all option birthdays; <code>small_cut_fillers</code> realizes small cuts.    |
| <code>SignSequenceCutOperation.lean</code> | <code>birthday_cut_le_iSup</code> controls the simplest cut; <code>exists_separator_of_small_sets</code> requires explicit <code>Small.{u}</code> witnesses for both option sets. |
| <code>SizeObstructions.lean</code>         | <code>SmallCutData</code> and <code>HasSmallCutFillers</code> express the smallness boundary; <code>noUnrestrictedCuts</code> rules out unrestricted cut realization.             |

These declarations are in the namespace `Surreal.Foundations`, with carrier-specific results in `Surreal.Foundations.SignSequence`. Their universe-relative form is mathematically essential. The fact that all surreals at one level can be treated as a type at a higher level does not make that type small at the original level. A proof of small-cut filling must retain its smallness hypotheses.

We inspected these declarations but did not rebuild the repository or extend its Lean development. No theorem in the new residual, topology, or reconstruction sections is being reported as Lean-verified. The following is an implementation plan, with proof obligations, rather than claimed existing code.

#### 111.47.4 A dependency-respecting implementation plan

1. **Separate the representations.** Define different types for a well-order relation, a bijective ordinal enumeration, a partial injection, and a relation table with its scan. Define conversion maps explicitly. The first-disagreement theorem should say which order each conversion preserves. It should not install one undifferentiated order instance on every code.
2. **Formalize prefix-free binary coding.** Prove the sign code  $- \mapsto 00$ ,  $+ \mapsto 11$ , termination  $\mapsto 01$  is prefix-free and order-preserving. State the concatenation theorem with ordinal sums of code lengths. Its regular-cardinal specialization and its constant-length padding specialization should be separate lemmas.
3. **Verify the singular scheduler.** Give the max-coordinate schedule its actual ordinal order type and prove that the first occurrence of a row precedes every larger-index row. Establish exhaustiveness, injectivity, and first-disagreement preservation separately. This is the most delicate set-theoretic component of the embedding proof.
4. **Use residual alphabets as dependent data.** After a prefix  $p$ , the remaining alphabet is  $X \setminus \text{ran}(p)$ . A selected code depends on that set. The proof that equal prefixes select equal codes is as important as the existence of the codes. The upper and lower embeddings can then be assembled block by block.
5. **Name all three topologies.** On fixed countable enumerations, distinguish the full discrete-product topology, the lexicographic order topology, and the pulled-back relation-table topology. Their Borel sigma-algebras agree in some cases and diverge in others. Using the wrong ambient topology would turn a false continuity statement into an apparently routine proof goal.
6. **Formalize the finite-tail decomposition first.** The order equivalence  $P_{\omega+n} \cong E_n \times n!$  is a small interface from permutation combinatorics to topology. Build the split-order Borel lemma and its countable-fiber exception formula above this interface.

7. **Develop class results through set nodes.** Formalize the partial-table rank extension for sets. Then use a supplied model of class theory, or an explicit universe-relative surrogate, to prove the branch and stabilization statements. Pair coding and its decoder are elementary and make a suitable first reconstruction theorem.

This order of work provides useful certified results before requiring a full formal theory of arbitrary class realizations. Finite testing of a scheduler or a code checks indexing conventions, but cannot replace the proofs of regularity, class replacement, or absence of exhaustive evaluators.

#### 111.47.5 Assumption ledger

| Result or construction                                   | Foundation used                                                                                                               |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Fixed-stage residual embeddings and finite-tail topology | ZFC; ordinary choice for set-indexed code selections. Regularity and strong-limit hypotheses occur exactly where stated.      |
| Every fixed type at $\kappa < \aleph_\omega$             | ZFC, without CH or GCH. All smaller infinite cardinals are regular, and larger residual alphabets use the native padded code. |
| The GCH classification of every fixed type               | ZFC+GCH. It is a conditional classification, not a ZFC assertion that all singular residual cardinals are strong limit.       |
| Any class well-order of <b>No</b> iff global choice      | GB without prior set choice; inherited theorem with explicit set-valued rank recursion.                                       |
| Common-prefix class comparison                           | Elementary class comprehension with supplied well-orders; no class ordinal or ETR is introduced.                              |
| Partial-table extension, closure, and canonical scan     | GBC with a supplied baseline. All recursive stages used here are sets.                                                        |
| Labelled realization reconstruction                      | Elementary coding internally; an external set metatheory for comparison of realization collections.                           |
| Universal baseline across all realizations               | Countable set model, plus Williams's minimum-realization theorem.                                                             |
| Pure-order invisibility of ETR                           | Conditional on the stated countable GBC+ETR realization; uses a full truth predicate and Tarski's theorem.                    |
| Actual topologies on admitted classes                    | External set-sized models, with the allowed supports specified as internal.                                                   |

#### 111.48 Source 30: verification and the scope of the conclusions

The mathematical review was organized around independent failure modes: ordinal length rather than cardinal length; the change of residual alphabet after a prefix; topology versus a particular measurable presentation; class-valued versus set-valued recursion; and internal versus external collections. The final regular, singular strong-limit, mixed-block, finite-tail, table-extension, and reconstruction arguments received separate proof review. Review is additional scrutiny, not a substitute for a formal proof or external peer review.

The archive includes a small executable check of the finite mechanisms: the three scans of six labelled orders, prefix-free sign coding, pair orientations, and the finite partial-table extension criterion. It exhaustively compares acyclicity of partial orientations with existence of an extending

permutation on carriers through size five. The resulting record is included.<sup>29</sup> These checks validate finite conventions only; no finite test establishes the transfinite scheduler, Borel nonmeasurability, class replacement, or the model-theoretic conclusions.

The classifications proved here concern precisely the representations and hypotheses stated. GCH gives a full fixed-type embedding classification at infinite cardinal birthday cutoffs. The unconditional theorem already covers every fixed type at each infinite  $\kappa < \aleph_\omega$ , and every residual block above the birthday cutoff. Only the regime of singular non-strong-limit residual cardinals at most  $\kappa$  remains outside the stated exact formulas. Even a complete embedding classification leaves the topological distinctions at  $\omega + 1$  and  $\omega + 2$  visible. At the global level, relation tables can make all types locally indistinguishable while labelled enumeration availability recovers the entire class universe. Both phenomena are compatible because they retain different information.

### 111.49 The finite checks of sources 19–30 as shipped

[merge] Ten of the twelve sources ship a standard-library Python program and its recorded output; sources 21 and 29 ship none (source 29 ships a provenance manifest and build metadata instead). All check finite mechanisms only; none verifies a transfinite, class-theoretic or cardinal statement, as each source says. At placement (2e06337d5) every program was rerun on a copy with Python 3.14.4 on Windows and reproduced its record exactly apart from line endings: the programs write CRLF on Windows and the records have LF, so compare with `diff -strip-trailing-cr`. The delivered checksum manifests of sources 22, 23, 24, 25, 26 and 28 were verified and are not shipped; source 29’s manifest hashes (its sixteen repository files, the three earlier manuscripts and its six deliverables) were verified as well.

| Src. | Program and record                                                                                              | Recorded checks                                                                                                                                          | Output                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 19   | <code>code/19-residual-check_restrictions.py</code> ; <code>data/19-residual-restriction_checks.json</code>     | restriction classification on all carriers of size at most 7; 695,228 adjacent comparisons                                                               | always writes <code>restriction_checks.json</code> beside itself               |
| 20   | <code>code/20-limits-finite_diagnostics.py</code> ; <code>data/20-limits-finite_diagnostics_results.json</code> | 58,255 decisive-pair and 609,862 nested-witness checks over 7,432 permutation pairs; 28,920 dyadic word comparisons; the nonmonotone restriction example | the file named by <code>-output</code> , else standard output                  |
| 22   | <code>code/22-unranked-finite_checks.py</code> ; <code>data/22-unranked-finite_checks.json</code>               | 1,388,197 assertions (word pairs, residual cylinders, an injection tree, pair-swap decoding)                                                             | default <code>data/finite_checks.json</code> relative to the working directory |
| 23   | <code>code/23-termination-finite_checks.py</code> ; <code>data/23-termination-finite_checks.json</code>         | 547,480 checks, seed 20261003                                                                                                                            | default <code>data/finite_checks.json</code> relative to the working directory |
| 24   | <code>code/24-completions-finite_checks.py</code> ; <code>data/24-completions-finite_checks.json</code>         | 5,913 permutations, 16,071 cylinders, 10,158 child partitions, 20,775 comparisons, 3,618 extensions, 3,255 cone extensions                               | default <code>data/finite_checks.json</code> beside <code>code/</code>         |

<sup>29</sup>[merge] Shipped as `code/30-tables-verify_finite.py` and `data/30-tables-verification_results.json`, with `code/30-tables-build.sh` (it compiles the unshipped `surreal_well_orders.tex` beside itself), `30-tables-source_audit.txt` and `30-tables-proof_review.txt`. The checker writes `verification_results.json` beside itself unless `-output` is passed (section 111.49).

| Src. | Program and record                                                         | Recorded checks                                                                                                                                 | Output                                          |
|------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 25   | code/25-symmetries-finite_checks.py; data/25-symmetries-finite_checks.json | 269,329 assertions, seed 20261003, 2,500 trials                                                                                                 | default relative data/finite_checks.json        |
| 26   | code/26-nonborel-finite_checks.py; data/26-nonborel-finite_checks.json     | 5,645,785 assertions                                                                                                                            | default relative data/finite_checks.json        |
| 27   | code/27-scans-finite_checks.py; data/27-scans-finite_checks.json           | 48 scans (36 orders), 720 reductions, 28,672 increasing-word pairs, 15,126 partial injections, 103,772 support-bound checks                     | -output, else standard output                   |
| 28   | code/28-envelope-finite_checks.py; data/28-envelope-finite_checks.json     | 16,072 cylinders, 470,323 conjugation and 940,646 group-uniformity identities, 1,799 extensions, 15,018 comparisons, one relation-code reversal | -output; also prints the record                 |
| 30   | code/30-tables-verify_finite.py; data/30-tables-verification_results.json  | 59,810 partial orientation tables, 110 boundary cases, 3,969 sign-word comparisons, pair swaps, the three scans of six objects                  | default verification_results.json beside itself |

Run them on copies, never in the report directory, and always name the output file. From the report directory, with an empty scratch directory `$$`:

```
cp code/19-residual-check_restrictions.py "$$/ " && python "$$/19-residual-
check_restrictions.py"
python code/20-limits-finite_diagnostics.py --output "$$/20.json"
python code/22-unranked-finite_checks.py --output "$$/22.json"
python code/23-termination-finite_checks.py --output "$$/23.json"
python code/24-completions-finite_checks.py --output "$$/24.json"
python code/25-symmetries-finite_checks.py --output "$$/25.json"
python code/26-nonborel-finite_checks.py --output "$$/26.json"
python code/27-scans-finite_checks.py --output "$$/27.json"
python code/28-envelope-finite_checks.py --output "$$/28.json"
python code/30-tables-verify_finite.py --output "$$/30.json"
```

and compare with the records in `data/` (source 19's run writes `$$/restriction_checks.json`). The delivered build scripts `code/19-residual-build.sh`, `code/22-unranked-build.sh`, `code/23-termination-build.sh`, `code/24-completions-build.sh`, `code/25-symmetries-build.sh`, `code/26-nonborel-build.sh`, `code/28-envelope-build.sh` and `code/30-tables-build.sh` compile L<sup>A</sup>T<sub>E</sub>X sources that are not shipped (and some of them run their checker with a relative output path first); they build nothing in this layout and are kept as delivered.

## 112 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 50](#); where the question index of [section 113](#) (dated 2 October 2026) or an earlier status section says otherwise, these notes supersede it.

- (a) *Source 18's research question 113.89,  $\omega$ -nonstandard class models:* **answered**, three times independently. The external order is  $\sigma(\mathcal{F} \cup \{I_\infty\})$  (theorem 109.1; note 109.13; note 109.33); its external finite-condensation blocks have the types  $1, 2, 6, \dots, \omega, \omega^*, \mathbb{Z}$  in the arrangement of corollary 109.18; it has the first-order theory of the standard factorial shuffle, as has the external order of every set-sized GBC model (theorem 109.9); it realizes types the standard shuffle omits (corollary 109.10, note 109.21); the common theory is decidable (corollary 109.22) and has  $2^{\aleph_0}$  countable models, of which the standard shuffle is prime (corollary 109.11 and theorem 109.12).
- (b) *Source 11's research question 113.3, exact recursion strength of iterated residual decompositions, and the related questions of sources 13, 15, 16 and 18 (research questions 113.30, 113.48, 113.64 and 113.86), of source 20 (research question 113.111), of source 23 (research question 113.147) and of source 19 (research question 113.104):* answered in part, in the form of canonical condensation histories with the nested-representative limit. A supplied history on the schedule  $W + 1$  always terminates (theorem 108.1); over GBC the existence of such histories for every input is equivalent to strong comparability of class well-orders (theorem 108.19), and every point then has a finite polynomial address (theorem 108.16), the canonical notation that research question 113.30 asks for, for the orders that have a terminating history. The limit convention is canonical (corollary 108.14); no nontrivial retained class is a fixed point of the operator (the remark after theorem 108.1). Open: whether ETR is necessary (equivalently for this operator, whether CWO implies ETR; Hamkins and Woodin's Question 7 is not settled), the strength over GB without global choice, and source 15's first question whether GB codes all finite condensations uniformly. The block-index half of research question 113.3 was answered by Part IX. *Dated note, 4 October 2026 (batch 96):* the strength over GB without global choice is answered: the equivalence with CWO holds over GB (section XVI.12 (a)).
- (c) *Source 23's research question 113.148, which orders eventually collapse:* answered for supplied histories (each collapses at the stage  $W + 1$ , theorem 108.1); in a full model the exact heights are  $\min\{\alpha : \theta \leq \kappa^\alpha\}$ , every height below  $\kappa^+$  occurs, and no single schedule bounds them all (corollary 108.3).
- (d) *Source 20's research question 113.112, alternative limits:* answered in part: keeping the canonical quotient maps gives exactly the nested-representative limit, and source 23's union of equivalences is the same history (corollary 108.14).
- (e) *Source 12's research question 113.17, automorphisms and finite condensation:* answered in part, for countable models. In the  $\omega$ -standard case the automorphism group is that of the coloured rational quotient (source 18's corollary 58.9; source 21's note 109.30); in the nonstandard case it is a semidirect product over the  $I_\infty$ -blocks (source 21's group description after note 109.35). Automorphisms of the pure order act on enumeration structure beyond relabelings: one moves an order without a greatest element to one with a greatest element (corollary 109.31). Uncountable realizations remain open (source 21's research question 113.125).
- (f) *Source 11's research question 113.4, what the bare order defines:* a further part. In the external order of every set-sized GBC model the singleton blocks and the finite-tail permutation blocks are first-order definable (proposition 109.7); existence of a greatest element is not invariant (corollary 109.31).
- (g) *Source 11's research question 113.8, source 12's research question 113.18, and the open item "what the unlabelled external order type remembers" of section 50 (e):* answered in part for the pure order. Among countable models it remembers exactly whether  $\omega$  is standard (theorem 109.1); its first-order theory is the same for all set-sized models (theorem 109.9). Expansions by the set-like predicate, labels or baselines are not classified here (source 20's



research questions 113.109 and 113.110, source 19’s research question 113.102).

- (h) *Source 17’s research question 113.77, the condensed unrestricted order*: answered in part, for countable models: its finite-condensation quotient is  $\mathbb{Q}$ , with the colour arrangement of corollary 109.18 in the nonstandard case. For the coarser condensation by set-sized residuals, source 20’s quotient is dense without endpoints but has an unfilled cut of character  $(\omega, \omega)$  (theorem 107.6).
- (i) *Uniform coding of the unrestricted order* (no earlier question): an interval  $[R, S]$  is exhaustively coded by a uniform set-indexed family exactly when its residual  $Y(R, S)$  is a set (theorem 107.4), and no uniform class-indexed family is dense in the raw comparison (theorem 107.8).

## Part XV

# Countable global choice and surreal coding over Zermelo set theory

[merge] (3 October 2026, batch 89.) This Part prints source 31, batch-89 manuscript 01 (section 1.10), whole and in its own order. It read this report at 2b7b388ba, with Parts I–XIV, and starts from three of its labels: the equivalence of global choice with well-orders of **No** over GB (theorem 13.1), source 12’s full-class model at an inaccessible cardinal (theorem 21.1), and source 28’s question on separating Class Replacement from uniform set choice (research question 113.209). Its two halves have different logical settings, as its introduction says.

- Sections XV.2 to XV.5: first-order Zermelo set theory with Choice, ZC (and ZC<sub>F</sub> with Foundation), with a selector  $c$  and Separation for formulas that mention it. A selector on  $D$ -small sets and uniform injections into  $D$  are interconvertible by uniform definitions (theorem XV.3.3); bounded trace compression needs no Collection (theorem XV.4.1); selection from nonempty countable sets is not conservative over ZC or ZC<sub>F</sub> (theorem XV.5.1), while without expanded Separation it is (proposition XV.5.5).
- Sections XV.6 to XV.11: in an external ZFC universe, the full-class rank models  $\mathcal{M}_\lambda = (V_\lambda, \mathcal{P}(V_\lambda))$  at every limit  $\lambda > \omega$ : exact thresholds for ordinal and birthday enumeration of the sign carrier, for universal set coding and ordinal abstraction, and for Class Replacement on each set domain; the cardinalities of internal proper classes; separating heights. This extends Part V’s inaccessible model (section 21) to every limit height.
- Section XV.12: consequences for this report’s questions and a formalization architecture; sections XV.13 and XV.14: the axiom ledger and source audit (source 31’s Appendices A and B); section XV.15: what the Part answers.

Its twelve research questions are printed with the other sources’ in the questions part (section 113.26), its title page and abstract in Appendix C; appendix D maps its numbered statements.

*The dependency on Glazer’s countermodels.* Theorem XV.5.1 derives a contradiction from Glazer’s obstruction sentence through an abstract interface (section XV.5.1) and takes from Glazer’s paper that the base theory does not refute that sentence. That the interface matches the obstruction of Glazer’s Propositions 3.8 and 4.11 exactly has not been reviewed here; source 31 itself asks for expert review (section XV.14). The theorem is printed as claimed, conditional on that interface.

*Numbering.* The sections of this Part are numbered XV.1–XV.15, so that the questions part and the appendices keep their numbers (Sections 113–115, Appendices A–D). Sections XV.1–XV.12

are source 31's Sections 1–12, and its statement  $k.m$  is printed as  $XV.k.m$ ; the [merge] paragraphs inside them add no numbered statement. Its equations (1)–(10) are tagged (S31.1)–(S31.10).

*Notation* (section 1.3.3 lists every renaming).  $\mathbf{Z}$  is Zermelo set theory and  $\mathbf{ZC}$  adds Choice; neither has classes or Replacement.  $\mathbf{GC}_{\leq\omega}$  is a selector on nonempty countable sets, not this report's  $\mathbf{GC}$ .  $X \mapsto D$  means that an injection exists (source 31's  $X \hookrightarrow D$ ;  $\hookrightarrow$  keeps its meaning of order embedding).  $\mathcal{M}_\lambda = (V_\lambda, \mathcal{P}(V_\lambda))$  has all external subclasses as classes;  $\kappa = |\lambda|$ , which is the cutoff only when  $\lambda$  is a cardinal;  $S_\lambda = \bigcup_{\alpha < \lambda} \alpha^2$  is the sign carrier, the set  $\mathbf{No}_{<\lambda}$  without its field structure. The threads of lemma XV.3.1 form the set  $\text{Thr}_X$  (source 31's  $\mathcal{T}_X$ ).

## XV.1 The problem and the contributions

Global choice makes a single selection simultaneously throughout the universe. Ordinary Choice supplies selections for each set-sized family. With Replacement available, these principles often have the same consequences about sets. Removing Replacement changes the problem: a family described by a formula and indexed by  $\omega$  need not itself be a set. A uniform selection from that family can then produce information that ordinary Choice cannot assemble.

Glazer proved that global choice with expanded Separation is not conservative over  $\mathbf{ZC}$ , both with and without Foundation [70]. We use the countable-family incidence obstruction in his Propositions 3.8 and 4.11; we do not reconstruct his symmetric-extension countermodels. Our contribution to that argument is to show that a selector on countable inputs already supplies the uniform presentations needed for its contradiction.

The relevant ProveIt source is the report on surreal well-orders [69]. Its Theorem 13.1, label `swo:cf:choice` (theorem 13.1), gives over  $\mathbf{GB}$  the equivalence between global choice, class well-orders of  $\mathbf{No}$ , set-like such well-orders, and an Ord-enumeration of  $\mathbf{No}$ . Its Question `swo:ge:n10` (research question 113.209) asks how image bounding and uniform choice separate over weaker class theories. We analyze particular mechanisms in that question; the question's other reconstruction and extension schemes remain open.

The two parts of the article are related, but their logical settings differ. The countable-selection theorem is a theorem about a first-order theory with an additional operation symbol. The rank-model results are semantic classifications of specific two-sorted structures with all external subclasses admitted. No identification of these two settings is assumed.

### XV.1.1 The localized theorem

Write  $\mathbf{GC}_{\leq\omega}$  for a total operation  $c$  satisfying

$$X \neq \emptyset \wedge X \mapsto \omega \implies c(X) \in X.$$

Separation must be allowed to mention  $c$ . The main refinement is

$$\boxed{\mathbf{ZC} + \mathbf{GC}_{\leq\omega} + \mathbf{Sep}(c) \text{ is not conservative over } \mathbf{ZC}.}$$

The corresponding statement with Foundation also holds. Moreover, replacing “at most countable” by “countably infinite” does not restore conservativity. The proof is in section XV.5.

There is a genuine obstacle to the naive proof. Repeatedly deleting the selected element for  $\omega$  steps need not exhaust a countable set. A selector can list one infinite part first and leave another part untouched. Nor may we simply continue through all countable ordinals: the relevant ordinal types need not exist in Zermelo set theory. We instead collect all compatible partial well-orders inside a fixed power set and then normalize the resulting order against a single set of order codes.

### XV.1.2 The exact classification

For a limit ordinal  $\lambda > \omega$ , put

$$S_\lambda = \bigcup_{\alpha < \lambda} {}^\alpha 2, \quad \text{bd}(s) = \text{dom}(s).$$

Here  $\text{bd}(s)$  is the ordinal length, or birthday, of a binary sign sequence. The following table previews the classification. Every cardinal condition is evaluated in the external ZFC universe.

| Principle in $\mathcal{M}_\lambda$                                                                                     | Exact condition on the height                               |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Global choice; a birthday-monotone set-like well-order of $S_\lambda$                                                  | Every limit $\lambda > \omega$                              |
| A class bijection $\text{Ord} \longleftrightarrow S_\lambda$                                                           | $\lambda = \kappa$ is a cardinal and $2^{<\kappa} = \kappa$ |
| Such a bijection with nondecreasing birthdays                                                                          | $\lambda = \kappa$ is a strong-limit cardinal               |
| A class injection $V_\lambda \hookrightarrow S_\lambda$ ; equivalently, every set well-order has an ordinal order type | $\lambda = \kappa = \beth_\kappa$                           |
| Class Replacement on every ordinal domain                                                                              | $\lambda$ is an uncountable regular cardinal                |
| Full Class Replacement                                                                                                 | $\lambda$ is strongly inaccessible                          |

The fourth row is an equivalence *within this family of full-class rank models*. It is not asserted over every weak class theory. The table also concerns the sign-sequence carrier; at arbitrary cutoffs we do not assert closure under the surreal field operations.

### XV.1.3 What is new, what is inherited, and what remains open

The bounded proof of the well-ordering theorem used below is classical in method. Its application to a uniform presentation theorem, and thence to the countable-input refinement, is the main additional deduction. Targeted searches did not find that refinement explicitly stated; this is not a proof of priority. Restricted global-choice principles have an established literature, including Enayat's treatment of global finite choice [71].

The distinction between well-orders and ordinal order types without Replacement is also established. Hamkins discusses ordinal abstraction and the example  $V_{\omega_1}$  in his account of Rin's work [63, 72]. Carneiro's discussion addresses set-like class orders without Replacement [74]. The contribution of our second part is the simultaneous exact classification of the displayed coding and replacement properties, including their different thresholds and explicit separating heights. The usual global-choice equivalences over GB are collected in [19]; the beth-fixed-point background also has prior discussion [75].

## XV.2 Axioms, classes, and bounded constructions

### XV.2.1 The first-order base theory

Following Glazer's convention, Z comprises Extensionality, Empty Set, Pairing, Union, Power Set, Infinity, and the Separation schema. Including Empty Set explicitly is harmless; the other usual formulations derive it. Set Choice is added to obtain ZC. We write  $\text{ZC}_F$  when Foundation, in the form that every nonempty set has an  $\in$ -minimal member, is added. No form of Replacement or Collection is part of these base theories.

All occurrences of  $X \mapsto D$  mean that there exists a *set graph* of an injection. A set well-order is a strict linear order on a set such that every nonempty subset has a least member. We do not replace its underlying set by an ordinal unless that replacement has been justified.

When a class operation  $c$  is named,  $\text{Sep}(c)$  means Separation for every formula of the language containing  $c$ , with set parameters allowed. A formula defining a unique set-valued operation gives a definitional expansion: formulas using the new operation can be translated into the original expanded language. Consequently, such a definition transports Separation, but does not introduce Replacement.

### XV.2.2 Fixed bounds

Fix a well-ordered set  $(D, <_D)$ , and define

$$\text{Small}_D(X) \iff X \mapsto D.$$

A  $D$ -restricted selector chooses from every nonempty  $D$ -small set. For definiteness it returns  $\emptyset$  elsewhere. Its restriction to the subsets of a fixed  $D$ -small set  $X$  can be represented by a set:

$$\{(A, x) \in \mathcal{P}(X) \times X : A \neq \emptyset \text{ and } c(A) = x\}.$$

This follows by Separation. The restriction's being a set does not imply that the family of all such restrictions, as  $X$  varies, is a set.

We shall also use the set of codes

$$\mathcal{C}_D = \mathcal{P}(D) \times \mathcal{P}(D \times D). \quad (\text{S31.1})$$

A code records both a carrier and a relation. Recording the carrier is essential: the empty relation alone cannot distinguish different singleton orders. Ordinary Choice supplies one fixed well-order  $<_D$  of  $\mathcal{C}_D$ . No choice of a code for each  $X$  has yet been made.

### XV.2.3 A bounded-existence convention

Every set introduced in the first part will have an explicit containing set. For example, a proposed graph from  $X$  to  $D$  is separated from  $X \times D$ ; a family of relations on  $X$  is separated from  $\mathcal{P}(X \times X)$ . Cartesian products themselves exist by the ordinary Pairing, Union, Power Set, and Separation construction.

These bounds are more than presentation details. A statement of the form  $\forall n \in \omega \exists! X \varphi(n, X)$  need not provide a set containing all the values  $X$ . On the other hand, if a containing set  $B$  is already known, then its graph is a subset of  $\omega \times B$ , and Separation suffices. Our argument repeatedly uses the second situation while avoiding the first.

## XV.3 Uniform presentations from restricted selection

### XV.3.1 A selector-compatible well-order

**Lemma XV.3.1** (Bounded well-order construction). *Let  $X$  be a set and suppose  $c(A) \in A$  for every nonempty  $A \subseteq X$ . In  $Z + \text{Sep}(c)$  there is a unique well-order  $W_X$  of  $X$  satisfying*

$$a = c(X \setminus \text{Pred}_{W_X}(a)) \quad (a \in X). \quad (\text{S31.2})$$

*The relation  $W_X$  is uniformly definable from  $X$  and  $c$ . No ordinal collapse, Replacement, or Foundation is used.*

*Proof.* Call  $(A, R)$  a *thread on  $X$*  if  $A \subseteq X$ ,  $R$  well-orders  $A$ , and

$$a = c(X \setminus \text{Pred}_R(a)) \quad (a \in A).$$

The ambient set in this equation is always  $X$ , even for a proper partial domain  $A$ . The set  $\text{Thr}_X$  of threads is separated from  $\mathcal{P}(X) \times \mathcal{P}(X \times X)$ .

We first prove that any two threads  $(A, R)$ ,  $(B, T)$  are comparable by initial segment. Let  $C \subseteq A \cap B$  consist of those  $x$  whose predecessor sets in the two orders are equal and on whose predecessor set the two orders agree. If  $x \in C$  and  $yRx$ , then  $yTx$ , and agreement below  $x$  implies agreement of the predecessor sets and orders below  $y$ . Thus  $y \in C$ . The symmetric statement holds, so  $C$  is an initial segment of both orders, with the same order on it.

If both threads continue beyond  $C$ , let  $a$  and  $b$  be the respective first points beyond it. Their predecessor sets are exactly  $C$ , whence

$$a = c(X \setminus C) = b.$$

The predecessor orders agree, so this common point belongs to  $C$ , a contradiction. One thread is therefore an initial segment of the other.

Separate from  $X$  the union  $A_*$  of all thread domains, and from  $X \times X$  the union  $R_*$  of their relations. Comparability makes  $R_*$  a linear order on  $A_*$ . To verify well-ordering, let  $Y \subseteq A_*$  be nonempty. Choose  $y \in Y$  and a thread containing  $y$ . Let  $z$  be the least element of the nonempty intersection of its domain with  $Y$ . Any other thread is an initial segment or an extension of this one; its new points cannot lie below  $z$ . Thus  $z$  is least in  $Y$  for  $R_*$ .

Every thread containing a point  $a$  has the same predecessors of  $a$  as the union. Consequently  $(A_*, R_*)$  is itself a thread. If  $A_* \neq X$ , append  $c(X \setminus A_*)$  to its end. The resulting well-order is a thread with a larger domain, contradicting the definition of  $A_*$ . Hence  $A_* = X$ , proving existence. Two full-domain threads must coincide by initial-segment comparability, proving uniqueness.  $\square$

*Remark XV.3.2* (Why ordinary iteration is insufficient). Let  $X = \omega \times \{0, 1\}$ . A selector on subsets of  $X$  can select the least available  $(n, 0)$  whenever any such point remains. Successive finite deletions then visit only  $\omega \times \{0\}$ . The compatible well-order can have type  $\omega + \omega$ , so its first  $\omega$  positions need not enumerate  $X$ . The next theorem produces an injection into  $\omega$  without asserting that this particular well-order has a countable ordinal representative in the weak theory.

### XV.3.2 Normalizing against one set of order codes

**Theorem XV.3.3** (Uniform presentation theorem). *Fix  $(D, <_D)$  and a well-order  $\prec_D$  of  $\mathcal{C}_D$ . Over  $\mathbf{Z}$  with Separation in the relevant expanded language, the following existence principles are interconvertible by uniform definitions:*

- (i) *a selector  $c$  for every nonempty  $D$ -small set;*
- (ii) *a class assignment  $X \mapsto i_X$  which supplies a set injection  $i_X : X \rightarrow D$  for every  $D$ -small set  $X$ .*

*In  $\mathbf{ZC}$ , the needed order  $\prec_D$  exists by ordinary Choice.*

*Proof.* Assume (i). Every subset of a  $D$ -small  $X$  is  $D$ -small, so lemma XV.3.1 constructs  $W_X$ . Form by Separation

$$\mathcal{A}_X = \{(A, R) \in \mathcal{C}_D : R \text{ well-orders } A \text{ and } (X, W_X) \cong (A, R)\}.$$

This set is nonempty. Indeed, an existing injection  $u : X \rightarrow D$  transports  $W_X$  to its image; both the image and the transported relation are separated from  $D$  and  $D \times D$ . This argument uses the existence of a particular  $u$  only to prove nonemptiness. It does not choose such a  $u$  uniformly.

Let  $(A_X, R_X)$  be the  $\prec_D$ -least code in  $\mathcal{A}_X$ . There is a unique order isomorphism

$$i_X : (X, W_X) \longrightarrow (A_X, R_X),$$

because a well-order has no nonidentity order automorphism. For completeness, two isomorphisms would have a least point of disagreement; their values there have the same predecessor set and therefore are equal, a contradiction. The isomorphism graph is a subset of  $X \times D$ . The assertions defining the least code and its unique isomorphism constitute a uniform definition of  $i_X$ . Set its value to  $\emptyset$  on nonsmall inputs.

Conversely, assume (ii). For nonempty  $D$ -small  $X$ , the image  $i_X[X]$  is a nonempty subset of  $D$ . Choose the unique element of  $X$  mapped to its  $\prec_D$ -least member. This uniformly defines a selector. All image and inverse graphs here have displayed set bounds.

In either direction, the new operation has a unique set value specified by a formula in the old expanded language. Substitution of this definition translates each instance of Separation involving the new operation into an available instance of Separation. This verifies the assertion about the whole expanded theories, not just about an isolated operation.  $\square$

*Remark XV.3.4* (Uniformity is not compatibility under restriction). The theorem does not say that  $i_Y = i_X \upharpoonright Y$  when  $Y \subseteq X$ . Nor does it assert a definitional equivalence that recovers an original selector after a round trip. It gives uniform constructions establishing equivalence of the two existence principles, with a fixed set of code-ordering parameters.

**Corollary XV.3.5** (Uniform countable enumeration). *Under  $ZC + GC_{\leq \omega} + Sep(c)$ , a single class definition assigns to every nonempty countable set  $X$  a set surjection  $e_X : \omega \rightarrow X$ .*

*Proof.* Apply [theorem XV.3.3](#) with  $D = \omega$ . Let  $m_X = \min i_X[X]$ , and define

$$e_X(n) = \begin{cases} i_X^{-1}(n), & n \in i_X[X], \\ i_X^{-1}(m_X), & n \notin i_X[X]. \end{cases}$$

The graph is a subset of  $\omega \times X$ , and its range is  $X$ .  $\square$

## XV.4 Incidence coding without Collection

The next theorem captures what a uniform presentation makes possible. Its hypothesis is weaker than requiring a different incidence trace for every label: equal traces need only determine the same target value.

**Theorem XV.4.1** (Bounded trace compression). *Work in  $Z$  with Separation for all the named class parameters. Let  $I, D, T$  be sets. Suppose:*

- (a)  $i \mapsto X_i$ , for  $i \in I$ , is a class assignment of sets, and a uniform class definition gives set injections  $u_i : X_i \rightarrow D$ ;
- (b)  $A$  is a class,  $E(x, a)$  a class relation, and  $F : A \rightarrow T$  a class surjection;
- (c) whenever  $a, b \in A$  satisfy

$$\forall i \in I \forall x \in X_i (E(x, a) \leftrightarrow E(x, b)),$$

one has  $F(a) = F(b)$ .



Then there is a set partial surjection

$$Q : \mathcal{P}(I \times D) \rightarrow T. \quad (\text{S31.3})$$

If  $\mathcal{P}(I \times D)$  is well-orderable, then  $T \hookrightarrow \mathcal{P}(I \times D)$ . In particular this last conclusion holds in ZC.

*Proof.* For  $a \in A$ , define its trace by Separation from  $I \times D$ :

$$H_a = \{(i, d) \in I \times D : \exists x \in X_i [u_i(x) = d \wedge E(x, a)]\}. \quad (\text{S31.4})$$

For each  $i$  and  $x \in X_i$ , injectivity gives

$$E(x, a) \iff (i, u_i(x)) \in H_a.$$

Thus  $H_a = H_b$  implies the hypothesis in (c) and hence  $F(a) = F(b)$ . Now separate from  $\mathcal{P}(I \times D) \times T$  the relation

$$Q = \{(H, t) : \exists a \in A [H = H_a \wedge F(a) = t]\}. \quad (\text{S31.5})$$

It is single-valued by the preceding observation and surjective onto  $T$  because  $F$  is. To get an injection in the other direction, well-order  $\mathcal{P}(I \times D)$ , and assign to each  $t$  its least  $Q$ -preimage. The resulting graph is again a subset of a fixed set product.  $\square$

**Corollary XV.4.2** (Compression from a restricted selector). *In ZC +  $\text{GC}_{\leq D}$  +  $\text{Sep}(c)$ , the same conclusion holds when (a) is replaced by the assumption that every  $X_i$  injects into  $D$ .*

*Proof.* Use [theorem XV.3.3](#) to define  $u_i = i_{X_i}$ .  $\square$

#### XV.4.1 The logical point of the compression theorem

Neither  $\{X_i : i \in I\}$ , nor  $\bigcup_{i \in I} X_i$ , nor the collection of traces  $\{H_a : a \in A\}$  was obtained by Replacement. The first two may fail to be sets. The third can, when needed, be separated from  $\mathcal{P}(I \times D)$ . The proof does not first collect the class of labels and then take its image. It directly defines a bounded relation whose fibres are determined by the trace.

Empty rows  $X_i = \emptyset$  cause no problem: their injection has empty graph and contributes no coordinate. This is a small advantage of formulating the theorem using injections instead of surjections onto the rows.

The target-separation condition also permits quotient labels. A class may have many labels describing the same trace, provided their target values agree. Thus the method applies to incidence presentations with redundant codes, not just to extensional representations of individual objects.

#### XV.4.2 What ordinary set Choice already provides

If the family of rows is an actual set, ordinary Choice supplies simultaneous injections. To see the bounded reason, let  $U$  be the union of that family. Every candidate injection into  $D$  is a subset of  $U \times D$ . The sets of candidate injections therefore form a subfamily of  $\mathcal{P}(\mathcal{P}(U \times D))$ . Choice on this set-sized family provides a selection. Consequently, the new phenomenon concerns class assignments whose values have no common set bound. It is not a failure of ordinary countable choice inside ZC.

## XV.5 Countable global selection is nonconservative

### XV.5.1 The external input from Glazer

We use the following precise interface to Glazer’s countermodel theorem [70, Propositions 3.8 and 4.11]. There is a pure set-theoretic obstruction sentence  $\Theta$ , with separate versions for  $\mathbf{ZC}$  and  $\mathbf{ZC}_F$ , realized in his respective models. Its definable data include countable sets  $X_n$ , an extensional incidence class on their class union, and a class surjection onto a set carrier  $T$  satisfying

$$T \not\rightarrow \mathcal{P}(\omega).$$

Extensionality of incidence means that equal traces imply equal labels and hence equal target values. The inequality is the part of his cardinal obstruction used here.

This formulation suppresses unnecessary ordinal representatives. If a target is given by a prewell-order on a set  $B$ , its quotient can be represented as a subset of  $\mathcal{P}(B)$ : separate the equivalence classes from that power set. The argument therefore does not assume that a von Neumann  $\omega_2$  exists internally. The published construction, rather than any new forcing assertion in this article, supplies the countermodels.

[merge] *Status of the interface.* The two paragraphs above are source 31’s statement of what it uses from [70, Propositions 3.8 and 4.11]. This write has not checked them against Glazer’s paper: whether his obstruction supplies countable rows  $X_n$ , an extensional incidence class and a class surjection onto a set  $T$  with  $T \not\rightarrow \mathcal{P}(\omega)$  in exactly this form is unreviewed, and source 31 names expert review of the localized theorem as the next step (section XV.14). Theorem XV.5.1 and its corollaries are printed as claimed, conditional on this interface.

### XV.5.2 The refinement

**Theorem XV.5.1** (Countable-input nonconservativity). *Let  $\mathbf{T}$  be either  $\mathbf{ZC}$  or  $\mathbf{ZC}_F$ . Add a total unary operation  $c$ , allow Separation to mention it, and require only*

$$\forall X [X \neq \emptyset \wedge X \rightarrow \omega \rightarrow c(X) \in X].$$

*This extension is not conservative over  $\mathbf{T}$  for sentences of the original language  $\{\in\}$ . In particular, it proves the negation of the appropriate Glazer obstruction sentence  $\Theta$ , while  $\mathbf{T}$  does not.*

*Proof.* Assume  $\Theta$  in the expanded theory. Apply theorem XV.3.3 with  $D = \omega$  to obtain, uniformly in  $n$ , injections  $u_n : X_n \rightarrow \omega$ . The countable rows and incidence data supplied by  $\Theta$  satisfy theorem XV.4.1. Thus

$$T \rightarrow \mathcal{P}(\omega \times \omega).$$

The standard pairing bijection  $\omega \times \omega \longleftrightarrow \omega$  induces a bijection of their power sets. We obtain  $T \rightarrow \mathcal{P}(\omega)$ , contradicting  $\Theta$ .

The base theory has a model satisfying  $\Theta$  by the cited countermodel theorem, so it cannot prove  $\neg\Theta$ . No step of the additional uniformization or compression argument used Foundation; applying the relevant version of the input proves both cases.  $\square$

*Remark XV.5.2* (Metatheory). The positive implication  $\Theta \rightarrow \perp$  is a derivation in the specified expanded weak theory. The nonderivability statement uses the published countermodel result. Glazer states that his model-construction theorems are established in  $\mathbf{ZF}$ ; our refinement inherits that input. This is not an assertion that  $\mathbf{ZC}$  proves its own consistency. Readers preferring conditional consistency language may assume  $\text{Con}(\mathbf{ZF})$ ; no claim of an optimal metatheoretic bound is needed here.

[merge] (4 October 2026, batch 90.) Glazer’s Oberwolfach abstract *A hypertalk* (Oberwolfach Reports 22 (2025), Report No. 2/2025, pp. 106–107) reports a parallel phenomenon without Foundation: the classification of connected one-dimensional manifolds, not necessarily second countable or set-sized, is not a theorem of ZFC without Foundation but becomes provable when Global Choice is added. He asks whether  $\text{ZFC} - \text{Fnd} + \text{GC}$  proves that every topological manifold is a set. Part XII of *polish-models-of-omnific-arithmetic* (its source 18, batch-90 manuscript 03; AI-assisted, unrefereed, not formalized) prints that question for connected Hausdorff real class manifolds of dimension at least two (its Question 165.6, `pma:gcm:q:glazer`), and proves, without Global Choice, that a connected real class manifold is a set when its carrier has an ordinal exhaustion, a compatible real class metric, or a class injection into or surjection from some  $\mathbf{No}^m$ ; the question itself stays open. Neither that part nor source 31 uses the other’s theorems.

**Corollary XV.5.3** (A single infinite input cardinality suffices). *The nonconservativity conclusion remains true if the selector is required to choose only from sets in bijection with  $\omega$ .*

*Proof.* Let  $c_\infty$  select from countably infinite sets. If  $\emptyset \neq X \rightarrow \omega$ , then  $X \times \omega$  is countably infinite. Indeed, it injects into  $\omega \times \omega$ , and after fixing one element of  $X$ , it contains a copy of  $\omega$ . The ordinary set Cantor–Bernstein theorem, provable by the bounded finite-path construction, gives the bijection. Define

$$c(X) = \pi_1(c_\infty(X \times \omega))$$

for these inputs, and  $c(X) = \emptyset$  otherwise. This is a countable-set selector. Separation for it translates into Separation for  $c_\infty$ , so [theorem XV.5.1](#) applies.  $\square$

**Corollary XV.5.4** (Any fixed infinite bound). *Suppose a fixed well-ordered  $D$  contains a set copy of  $\omega$ . A selector on all nonempty sets embeddable into  $D$ , with expanded Separation, already implies the countable-input conclusion.*

*Proof.* Every countable set embeds into  $D$ . Restrict the selector’s specified behavior to those inputs and apply [theorem XV.5.1](#).  $\square$

### XV.5.3 The exact role of expanded Separation

**Proposition XV.5.5** (An uninteracting selector is conservative). *If the Separation schema is not extended to the new symbol, adjoining the selector requirement alone is conservative over  $\text{ZC}$ , and likewise over  $\text{ZC}_F$ .*

*Proof.* Use the ordinary external ZFC metatheory for this model-theoretic argument. Every set-sized model  $M$  of the base theory can be expanded by selecting an  $\in^M$ -member of each internally nonempty set. External Choice makes these selections on the external set  $M$ . The resulting operation satisfies even unrestricted global selection in the expanded structure. All original axioms still hold because their schemas do not mention the new operation. Thus every base model has an expansion, and completeness gives conservativity for original-language sentences.  $\square$

Expanded Separation is therefore part of the mathematical content of [theorem XV.5.1](#). Merely writing a choice operation into a formal language is not the same assertion.

## XV.6 Full-class rank models and the sign carrier

From this section onward, work in an external universe satisfying ZFC. Fix a limit ordinal  $\lambda > \omega$ . The set sort of  $\mathcal{M}_\lambda$  is  $V_\lambda$ , and its class sort consists of every external subset of  $V_\lambda$ . Ordered pairs and finite tuples are encoded in the usual way. Since  $\lambda$  is a limit, any finite rank increase remains below  $\lambda$ ; hence every relation or function on this domain can be represented by a class of coded tuples.

**Proposition XV.6.1** (Available axioms).  *$\mathcal{M}_\lambda$  satisfies  $\text{ZC}_F$  on its set sort, elementary class comprehension, Separation with arbitrary class parameters, and global choice. Its internal ordinals are precisely the ordinals below  $\lambda$ .*

*Proof.* Transitivity and limit height give Extensionality, Pairing, Union, Power Set, Infinity, and Foundation. Every external subset of a set  $a \in V_\lambda$  again belongs to  $V_\lambda$ , proving Separation even with arbitrary class parameters. External Choice gives a choice function on a set-sized family  $a$ ; its graph has rank bounded by the ranks of  $a$  and its union, plus a finite amount, so belongs to  $V_\lambda$ . Thus internal set Choice holds.

Every extension of a formula on the domain is an admitted class, giving class comprehension. External Choice also selects one member of every nonempty element of the external set  $V_\lambda$ ; its graph is an admitted class and witnesses global choice. Transitivity and the actual well-foundedness of membership identify the internal ordinals with  $\lambda$ .  $\square$

### XV.6.1 Sign sequences and bounded birthdays

Each binary function on an ordinal  $\alpha < \lambda$  has rank between  $\alpha$  and  $\max(\omega, \alpha) + m$  for an absolute finite coding constant  $m$ , after harmless adjustment for finite  $\alpha$ . It therefore belongs to  $V_\lambda$ . Conversely, the domain of any internally represented ordinal-length binary function is an ordinal below  $\lambda$ . The internal sign-sequence class is exactly  $S_\lambda$ .

The following boundedness criterion will be used repeatedly:

$$B \subseteq S_\lambda \text{ is an internal set} \iff \sup\{\text{bd}(s) : s \in B\} < \lambda. \quad (\text{S31.6})$$

For the reverse implication, uniformly bounded lengths give a fixed rank bound on all the coded functions and on their collection. For the forward implication, the rank of the internal set bounds the ranks, and therefore the lengths, of its members. In particular,  $S_\lambda$  is an internal proper class.

This identification is about the sign presentation and its birthday filtration. For example, at a noncardinal cutoff, operations on two sign sequences may produce a sequence whose length reaches the cutoff. Accordingly, no theorem below treats every  $S_\lambda$  as a real-closed field or as a model of all surreal arithmetic.

### XV.6.2 Set-like well-orders at every height

**Proposition XV.6.2** (Choice orders without ordinal enumeration). *For every limit  $\lambda > \omega$ ,  $V_\lambda$  has a rank-monotone set-like class well-order, and  $S_\lambda$  has a birthday-monotone set-like class well-order. These orders give a least element to every nonempty admitted class, not only to every internal nonempty set.*

*Proof.* Externally choose a well-order of each rank layer of  $V_\lambda$ , and order first by rank, then by the selected layer order. Every nonempty subclass has a least occurring rank and then a least element on that layer. The predecessors of a given point are contained in a fixed rank segment, so form an

internal set. The construction for  $S_\lambda$  is identical, with birthday layers  ${}^a2$ . Its predecessors have bounded birthday, and equation (S31.6) applies.  $\square$

One must not infer that either of these orders has type  $\text{Ord}^{\mathcal{M}_\lambda}$ . Such an inference requires both the availability of ordinal representatives for predecessor orders and a justification that the resulting enumeration has the appropriate range. We determine exactly when the corresponding statements hold.

[merge] At a strongly inaccessible  $\kappa$ ,  $\mathcal{M}_\kappa$  is source 12's model  $\mathcal{M}$  of section 21, and the identification of the internal sign-sequence class with  $S_\kappa = \mathbf{No}_{<\kappa}$  is clause (i) of theorem 21.1; source 09's universe enlargement (section 22) and the universe sections of Part IX (section 57) also work at an inaccessible height. This section and the next five drop inaccessibility. Proposition XV.6.2 gives a set-like class well-order of  $S_\lambda$  at every height, which by theorem XV.7.2 need not have type  $\text{Ord}$ ; over GB, which has class Replacement, the two are equivalent (theorem 13.1 and the dated note after it).

## XV.7 Ordinal enumeration and birthday order

For an infinite cardinal  $\kappa$ , write

$$2^{<\kappa} = \sup_{\mu < \kappa, \mu \text{ a cardinal}} 2^\mu.$$

Our use of cardinal equalities in this and the following sections is entirely external.

**Lemma XV.7.1** (Size of the sign carrier). *Let  $\lambda > \omega$  be a limit ordinal and  $\kappa = |\lambda|$ . Then*

$$|S_\lambda| = \begin{cases} 2^{<\kappa}, & \lambda = \kappa, \\ 2^\kappa, & \kappa < \lambda. \end{cases} \quad (\text{S31.7})$$

*Proof.* The layers have size  $2^{|\alpha|}$ , and distinct layers are disjoint. For  $\lambda = \kappa$ , their cardinal sum is  $\kappa \cdot 2^{<\kappa} = 2^{<\kappa}$ . Here  $2^{<\kappa} \geq \kappa$ : at a successor cardinal this follows from Cantor's theorem at the predecessor cardinal; at a limit cardinal it follows by taking the supremum of the successor cardinals below it.

If  $\kappa < \lambda$ , the birthday  $\kappa$  layer is present and has size  $2^\kappa$ . Every layer has at most this size, and there are  $\kappa$  layers, giving the matching upper bound.  $\square$

[merge] The case  $\lambda = \kappa$  is the count  $|\mathbf{No}_{<\kappa}| = 2^{<\kappa}$  of section 7 (before corollary 7.1), proved there by the same sum.

**Theorem XV.7.2** (Bare ordinal enumeration). *There is an admitted class bijection  $\text{Ord}^{\mathcal{M}_\lambda} \longleftrightarrow S_\lambda$  if and only if  $\lambda$  is a cardinal  $\kappa$  satisfying  $2^{<\kappa} = \kappa$ . The same condition characterizes a class injection  $S_\lambda \hookrightarrow \text{Ord}^{\mathcal{M}_\lambda}$ .*

*Proof.* All external functions on the relevant carriers are admitted class functions. The assertion is therefore cardinal equality with  $|\lambda|$ , or the corresponding cardinal inequality. At a noninitial  $\lambda$ , lemma XV.7.1 and Cantor's theorem rule it out. At an initial  $\kappa$ , the same lemma gives exactly the displayed condition. An injection of  $\lambda$  into  $S_\lambda$  is always available by taking the all-zero sequence at each birthday, so injection and bijection have the same threshold.  $\square$

**Definition XV.7.3.** An ordinal birthday enumeration is a class bijection  $e : \lambda \rightarrow S_\lambda$  satisfying  $\xi < \eta \Rightarrow \text{bd}(e(\xi)) \leq \text{bd}(e(\eta))$ .

**Theorem XV.7.4** (Birthday enumeration threshold). *An ordinal birthday enumeration exists if and only if  $\lambda$  is a strong-limit cardinal.*

*Proof.* Such an enumeration is in particular a bare enumeration, so [theorem XV.7.2](#) implies  $\lambda = \kappa$  is a cardinal. Fix  $\alpha < \kappa$  and choose a sign sequence of birthday  $\alpha + 1$ . It occurs at some position  $\xi < \kappa$ . Monotonicity places every sequence of birthday  $\alpha$  before that position. Hence

$$2^{|\alpha|} \leq |\xi| < \kappa.$$

Thus  $\kappa$  is strong limit.

Conversely, suppose  $\kappa$  is strong limit. For each  $\alpha < \kappa$ , externally well-order  ${}^\alpha 2$  in its initial cardinal order type  $\theta_\alpha = 2^{|\alpha|} < \kappa$ . Concatenate these orders in increasing birthday. For every  $\beta < \kappa$ , the ordinal sum of the layers up through  $\beta$  has cardinal at most

$$\max(\aleph_0, |\beta|, 2^{|\beta|}) < \kappa.$$

As  $\kappa$  is an initial ordinal, each such partial sum is below  $\kappa$ . Their supremum is therefore at most  $\kappa$ . It is at least  $\kappa$ , since there is a nonempty layer for every birthday. The concatenated order has type exactly  $\kappa$ , giving the required enumeration. No regularity of  $\kappa$  was used.  $\square$

*Remark XV.7.5.* A birthday-monotone *set-like well-order* exists at every limit height by [proposition XV.6.2](#); the extra demand in [theorem XV.7.4](#) is that its order type be the internal Ord. The difference is visible even at  $\omega_1$  under CH, as shown in [section XV.11](#).

## XV.8 Universal set coding and ordinal abstraction

Recall the external beth hierarchy:

$$\beth_0 = \aleph_0, \quad \beth_{\alpha+1} = 2^{\beth_\alpha}, \quad \beth_\delta = \sup_{\alpha < \delta} \beth_\alpha \quad (\delta \text{ limit}).$$

For an uncountable initial cardinal  $\kappa$ ,

$$|V_\kappa| = \beth_\kappa. \tag{S31.8}$$

Indeed,  $|V_{\omega+\alpha}| = \beth_\alpha$  by induction, and  $\omega + \kappa = \kappa$ . In particular, one must not substitute this formula without adjustment at  $\lambda = \omega + \omega$ , where the cardinality is  $\beth_\omega$ .

**Lemma XV.8.1** (Growth facts). *For an uncountable cardinal  $\kappa$ ,  $\beth_\kappa \geq \kappa$ , and  $\beth_\kappa$  is a strong-limit cardinal. If  $\beth_\kappa = \kappa$ , then  $|V_\alpha| < \kappa$  for every  $\alpha < \kappa$ .*

*Proof.* The beth sequence is strictly increasing and continuous. At the limit index  $\kappa$  its supremum has cardinal at least  $\kappa$ . If  $\mu < \beth_\kappa$ , some  $\beth_\beta$  exceeds  $\mu$ , with  $\beta + 1 < \kappa$ ; hence  $2^\mu \leq \beth_{\beta+1} < \beth_\kappa$ . This proves the strong-limit assertion. In the fixed-point case, the rank-cardinality formula, or its monotone upper bounds for small ranks, puts every  $|V_\alpha|$  below  $\beth_\kappa = \kappa$ .  $\square$

**Theorem XV.8.2** (Universal coding threshold). *For a limit ordinal  $\lambda > \omega$ , the following are equivalent:*

- (i) *there is a class injection  $V_\lambda \hookrightarrow S_\lambda$ ;*
- (ii) *there is a class bijection  $V_\lambda \longleftrightarrow S_\lambda$ ;*
- (iii) *there is a class bijection  $V_\lambda \longleftrightarrow \lambda$ ;*
- (iv)  $\lambda = \kappa = \beth_\kappa$ .



*Proof.* All the functions in question are external functions between sets, represented internally as classes. Because  $S_\lambda \subseteq V_\lambda$ , (i) and (ii) are equivalent by external Cantor–Bernstein.

First suppose  $\lambda$  is not an initial ordinal, and put  $\kappa = |\lambda| < \lambda$ . If  $\kappa = \omega$ , then  $\lambda \geq \omega + \omega$ , so

$$|V_\lambda| \geq \beth_\omega > 2^\omega = |S_\lambda|.$$

If  $\kappa > \omega$ , then  $\kappa + 2 < \lambda$ , because  $\lambda$  is a limit. By equation (S31.8),

$$|V_\lambda| \geq |V_{\kappa+2}| = 2^{2^{\beth_\kappa}} > 2^\kappa = |S_\lambda|.$$

Thus neither (i) nor (iii) can hold at a noninitial height.

Now let  $\lambda = \kappa$ . If  $\beth_\kappa > \kappa$ , the strong-limit property in lemma XV.8.1 gives

$$|S_\kappa| = 2^{<\kappa} \leq 2^\kappa < \beth_\kappa = |V_\kappa|.$$

Both (i) and (iii) fail. If  $\beth_\kappa = \kappa$ , then  $\kappa$  is strong limit, so  $|S_\kappa| = \kappa = |V_\kappa|$ . All three coding statements hold.  $\square$

**Definition XV.8.3** (Ordinal abstraction). OA asserts that every internal set well-order is isomorphic, by an internal set graph, to an internal von Neumann ordinal.

**Theorem XV.8.4** (Ordinal abstraction threshold).  $\mathcal{M}_\lambda \models \text{OA}$  if and only if  $\lambda = \kappa = \beth_\kappa$ .

*Proof.* Assume first that  $\lambda = \kappa = \beth_\kappa$ . Every internal set  $a$  has size below  $\kappa$ , by lemma XV.8.1. An internal well-order on  $a$  is externally a genuine well-order: every external subset of  $a$  is internal. Let its external order type be  $\alpha$ . Since  $|\alpha| = |a| < \kappa$  and  $\kappa$  is initial,  $\alpha < \kappa$ . Its isomorphism graph has bounded rank and belongs to  $V_\kappa$ . Thus the collapse is internal.

Conversely, suppose first that  $\lambda$  is not initial. The ordinal  $\kappa = |\lambda|$  belongs to  $V_\lambda$ . Transport the order  $\lambda$  to the carrier  $\kappa$  by an external bijection. The resulting relation is a subset of  $\kappa \times \kappa$ , so has rank below  $\lambda$ . It is an internal well-order of external type  $\lambda$ , which has no internal ordinal representative. This refutes OA.

Finally, let  $\lambda = \kappa$  and suppose  $\beth_\kappa > \kappa$ . Continuity of the beth hierarchy gives some  $\beta < \kappa$  with  $\beth_\beta > \kappa$ ; otherwise its supremum would be at most  $\kappa$ . A rank segment below  $\kappa$  therefore contains an internal set of size at least  $\kappa$ . Choose an external subset  $a$  of that set of size  $\kappa$ , and transport the order  $\kappa$  to  $a$ . Both  $a$  and the relation are internal by rank boundedness. This well-order has type  $\kappa$ , again with no internal ordinal representative.  $\square$

### XV.8.1 Why this does not prove Replacement

At a singular beth fixed point, every internal set well-order has an ordinal type and the entire set universe has a class bijection with Ord. Nevertheless, a class function on a small ordinal can have unbounded-rank values. An external class bijection does not imply that the images of all of its set-sized domain restrictions are internal sets. The next section makes this obstruction exact.

## XV.9 The exact domain size for Class Replacement

**Definition XV.9.1.** For an internal set  $a$ , let  $\text{CR}(a)$  assert that the range of every admitted class function  $F : a \rightarrow V_\lambda$  is an internal set. Let  $\text{CR}(\text{Ord})$  assert  $\text{CR}(\alpha)$  for every internal ordinal  $\alpha$ . Full CR asserts  $\text{CR}(a)$  for every internal set.

**Lemma XV.9.2** (The domain-size criterion). *For every  $a \in V_\lambda$ ,*

$$\mathcal{M}_\lambda \models \text{CR}(a) \iff |a| < \text{cf}(\lambda). \quad (\text{S31.9})$$

*The same condition characterizes full-class Collection on the domain  $a$ .*

*Proof.* If  $|a| < \text{cf}(\lambda)$ , the ranks of the values of any function  $F : a \rightarrow V_\lambda$  form an externally bounded subset of  $\lambda$ . Its range therefore has rank below  $\lambda$ , so is internal.

If  $|a| \geq \text{cf}(\lambda)$ , choose externally a surjection  $p : a \rightarrow \text{cf}(\lambda)$  and a cofinal map  $g : \text{cf}(\lambda) \rightarrow \lambda$ . The admitted class function  $g \circ p$  has an unbounded set of ordinal values. That range is not an internal set, refuting  $\text{CR}(a)$ .

For Collection, a total class relation on  $a$  has an external witness selection by ambient Choice. When  $|a| < \text{cf}(\lambda)$ , these selected witnesses have bounded rank and give a collecting set. In the other direction, the functional relation just constructed has unique witnesses and no internal collecting set.  $\square$

**Theorem XV.9.3** (Replacement thresholds). *For every limit ordinal  $\lambda > \omega$ ,*

$$\begin{aligned} \mathcal{M}_\lambda \models \text{CR}(\omega) &\iff \text{cf}(\lambda) > \omega, \\ \mathcal{M}_\lambda \models \text{CR}(\text{Ord}) &\iff \lambda \text{ is an uncountable regular cardinal,} \\ \mathcal{M}_\lambda \models \text{CR} &\iff \lambda \text{ is strongly inaccessible.} \end{aligned}$$

*Proof.* The first assertion is [lemma XV.9.2](#) with  $a = \omega$ .

For the second, if  $\lambda$  is regular initial, every  $\alpha < \lambda$  has size below  $\text{cf}(\lambda) = \lambda$ . Conversely, if  $\text{cf}(\lambda) < \lambda$ , then  $\text{cf}(\lambda)$  is itself an internal ordinal and its domain violates [lemma XV.9.2](#). The only remaining possibility is  $\text{cf}(\lambda) = \lambda$ , which says that  $\lambda$  is an initial regular cardinal.

For full Replacement, the second assertion first forces regular cardinal height. If that regular cardinal  $\kappa$  is not strong limit, choose an infinite cardinal  $\mu < \kappa$  with  $2^\mu \geq \kappa$ . The internal set  $\mathcal{P}(\mu)$  has size at least  $\text{cf}(\kappa) = \kappa$ , so violates [lemma XV.9.2](#). Conversely, at a strongly inaccessible  $\kappa$ , induction on  $\alpha < \kappa$  gives  $|V_\alpha| < \kappa$ : the successor step uses strong limit and the limit step uses regularity. Every internal set has size below  $\text{cf}(\kappa)$ , and [lemma XV.9.2](#) proves full Class Replacement.  $\square$

[merge] The inaccessible case of the third equivalence is the classical fact that source 12’s model uses (“regularity and the strong-limit property ensure the required closure of set operations and class Replacement”, [section 21](#)). The first equivalence makes exact that section’s warning that  $V_{\beth_\omega}$  “lacks the regularity used in Replacement”: there  $\text{cf} \lambda = \omega$ , so Class Replacement fails already on the domain  $\omega$  ([section XV.11](#)).

**Corollary XV.9.4** (Separating the two requirements). *Within the structures  $\mathcal{M}_\lambda$ , the conjunction of ordinal-domain Class Replacement and ordinal birthday enumeration is equivalent to full Class Replacement. The same is true with ordinal birthday enumeration replaced by ordinal abstraction or universal set coding.*

*Proof.* The first conjunction says exactly that the height is regular and strong limit. For the second, a beth fixed point is strong limit, and regularity again gives strong inaccessibility. Every inaccessible cardinal is a beth fixed point: induction gives  $\beth_\alpha < \kappa$  for  $\alpha < \kappa$ , and continuity gives  $\beth_\kappa = \kappa$ .  $\square$

**Remark XV.9.5** (Ordinal domains are not all well-orderable domains). The distinction in [theorem XV.9.3](#) concerns *ordinal domains*. It must not be confused with Replacement on every

well-orderable set. Freire and Hamkins prove that the latter implies full Replacement over Zermelo set theory with Foundation, without assuming Choice [73]. An internal well-ordered carrier may have no internal ordinal representative, precisely as [theorem XV.8.4](#) shows.

*Remark XV.9.6* (Full classes versus definable classes). The inaccessible threshold characterizes Replacement for *all external class functions* admitted in  $\mathcal{M}_\lambda$ . It is not a classification of heights satisfying the first-order Replacement schema for formulas of the pure membership language. Such heights are usually discussed through worldly rank segments; the class parameter restriction changes the problem. No result about first-order Replacement at all singular heights is asserted here.

## XV.10 The sizes of internal proper classes

The available classes form an external set, but many are proper classes relative to the internal set sort. This distinction is useful rather than paradoxical.

**Theorem XV.10.1** (Proper-class cardinal spectrum). *A class  $C \subseteq V_\lambda$  is an internal proper class exactly when it is unbounded in rank. Its possible external cardinalities are precisely*

$$\{\theta : \theta \text{ is a cardinal and } \text{cf}(\lambda) \leq \theta \leq |V_\lambda|\}. \quad (\text{S31.10})$$

*Proof.* A rank-bounded subclass is an external subset of some  $V_\alpha$  and therefore belongs to  $V_\lambda$ . Conversely, an element of  $V_\lambda$  bounds the ranks of all of its members. An unbounded subclass must have at least  $\text{cf}(\lambda)$  elements. A cofinal sequence of ordinals below  $\lambda$  realizes that lower bound.

For any cardinal  $\theta$  in the displayed interval, choose an external subset  $B \subseteq V_\lambda$  of size  $\theta$ , and adjoin a fixed cofinal set of ordinals of size  $\text{cf}(\lambda)$ . The union is rank-unbounded and still has size  $\theta$ . It is an admitted proper class of the required cardinality.  $\square$

**Corollary XV.10.2** (Two different limitation-of-size conditions). *In  $\mathcal{M}_\lambda$ :*

- (i) *every proper class admits an injection from Ord if and only if  $\text{cf}(\lambda) = |\lambda|$ ;*
- (ii) *every proper class is class-bijective with Ord if and only if  $\lambda$  is strongly inaccessible.*

*Proof.* For (i), compare the least possible size  $\text{cf}(\lambda)$  with  $|\text{Ord}^{\mathcal{M}_\lambda}| = |\lambda|$ . For (ii), the universal class itself first forces  $|V_\lambda| = |\lambda|$ , which by [theorem XV.8.2](#) forces a beth-fixed-point cardinal height  $\kappa$ . The minimum proper-class size then forces  $\text{cf}(\kappa) = \kappa$ , hence strong inaccessibility. At an inaccessible height both the minimum and maximum sizes equal  $\kappa$ , so all proper classes have the required bijections.  $\square$

At a noninitial height, condition (i) is not the same as regular cardinal height. For example  $\lambda = \omega_1 \cdot 2$  has  $\text{cf}(\lambda) = |\lambda| = \omega_1$ , but fails ordinal-domain Class Replacement since  $\omega_1$  is an internal ordinal domain. This example separates cardinal comparability from internal image bounding.

## XV.11 Explicit separating models

### XV.11.1 Global choice without any ordinal sign enumeration

Take  $\lambda = \omega + \omega$ . Then

$$|\text{Ord}^{\mathcal{M}_\lambda}| = \aleph_0, \quad |S_\lambda| = 2^{\aleph_0}, \quad |V_\lambda| = \beth_\omega.$$

Global choice and birthday-monotone set-like well-orders exist, but even a class injection  $S_\lambda \rightarrow \text{Ord}$  is impossible. Class Replacement on  $\omega$  fails, witnessed by the class function  $n \mapsto \omega + n$ , whose range

is unbounded in  $\lambda$ . Thus a global selector, including its countable restriction, need not supply any countable image-bounding principle.

### XV.11.2 Ordinal sign enumeration without birthday enumeration

Assume CH and take  $\lambda = \omega_1$ . Then  $2^{<\omega_1} = 2^{\aleph_0} = \omega_1$ , so a class bijection  $\omega_1 \longleftrightarrow S_{\omega_1}$  exists. But there are  $\omega_1$  sequences of birthday  $\omega$ . A birthday-monotone enumeration would have to place all of them before any sequence of birthday  $\omega + 1$ . No position below  $\omega_1$  has that many predecessors.

Ordinal-domain Class Replacement holds by regularity, but full Class Replacement and ordinal abstraction fail. This refines the role of the established  $V_{\omega_1}$  example by separating arbitrary sign enumeration from enumeration that respects the birthday filtration.

### XV.11.3 Birthday enumeration without universal set coding

Take  $\kappa = \beth_\omega$ . This is a singular strong-limit cardinal of cofinality  $\omega$ . Hence ordinal birthday enumeration of  $S_\kappa$  exists. But

$$|V_\kappa| = \beth_\kappa > \beth_\omega = \kappa = |S_\kappa|.$$

Universal set coding and ordinal abstraction fail. This example requires no continuum hypothesis and no inaccessible cardinal.

### XV.11.4 Universal coding and ordinal abstraction without Replacement

Define externally

$$\kappa_0 = \omega, \quad \kappa_{n+1} = \beth_{\kappa_n}, \quad \kappa = \sup_{n < \omega} \kappa_n.$$

The sequence is strictly increasing, so  $\text{cf}(\kappa) = \omega$ . Continuity gives  $\beth_\kappa = \sup_n \beth_{\kappa_n} = \sup_n \kappa_{n+1} = \kappa$ . This is the first beth fixed point reached above the starting value. Consequently

$$V_\kappa \longleftrightarrow S_\kappa \longleftrightarrow \kappa$$

by class bijections, and every internal set well-order has an internal ordinal type. Nevertheless the admitted class function  $n \mapsto \kappa_n$  has unbounded range. Even  $\text{CR}(\omega)$  fails.

No first-order definability assertion about this particular sequence is needed for the example: every external class function is admitted in the structure. This makes its failure of *Class* Replacement immediate and avoids confusing the two replacement schemas.

### XV.11.5 The inaccessible case

If a strongly inaccessible  $\kappa$  exists, all properties in the main classification table hold in  $\mathcal{M}_\kappa$ . In particular, every internal proper class has external cardinality  $\kappa$ , and its class functions on internal sets have internally bounded images. This is the setting in which the rank-model examples in the inspected ProveIt report are placed. ([merge] This report: [sections 21, 22 and 57.](#))

| Height                         | GC  | bare sign<br>enumeration | birthday<br>enumeration | universal<br>set coding | full CR |
|--------------------------------|-----|--------------------------|-------------------------|-------------------------|---------|
| $\omega + \omega$              | yes | no                       | no                      | no                      | no      |
| $\omega_1$ , assuming CH       | yes | yes                      | no                      | no                      | no      |
| $\beth_\omega$                 | yes | yes                      | yes                     | no                      | no      |
| First beth fixed point         | yes | yes                      | yes                     | yes                     | no      |
| Strongly inaccessible $\kappa$ | yes | yes                      | yes                     | yes                     | yes     |

## XV.12 Consequences for surreal foundations and formalization

### XV.12.1 A precise extension of the ProveIt questions

The classification explains why a proof formulated over GB cannot simply be read over ZC with class comprehension. Three jobs must be separated:

1. *Selection*: choose a local presentation or local order.
2. *Collapse and coding*: represent that presentation by an available ordinal or sign sequence.
3. *Image bounding*: ensure that the values obtained over a set domain still form an internal set.

The first part of this article shows that selection restricted to countable inputs already has non-conservative consequences when combined with expanded Separation. The second shows that even unrestricted selection leaves collapse and image bounding at distinct thresholds in full-class rank models.

These are concrete answers about the mechanisms named in the repository’s weak foundations agenda ([merge] [research question 113.209](#); the questions [research questions 113.161](#) and [113.185](#) of sources 24 and 26 ask about the same resources). They do not prove that its supported-prefix extension, small-net rigidity, or inverse-reconstruction schemes are each equivalent to a particular row of our table. Establishing those equivalences would require encoding the exact input and output types of each scheme in a common weak base.

### XV.12.2 A formalization architecture with an explicit trust boundary

A proof assistant development can formalize the first part without implementing all the symmetric-extension machinery. The natural target is a theorem about models of a specified first-order theory, with Glazer’s obstruction isolated as an input interface. Its parameters would be formulas defining the class family, incidence relation, label domain, and target map, together with their required properties. The formal theorem would show that no expanded model can satisfy both that interface and countable selection with the relevant Separation instances.

The thread construction should be represented by a bounded object:  $\text{Thr}_X \subseteq \mathcal{P}(X) \times \mathcal{P}(X^2)$ . The main lemmas are initial-segment comparability, well-foundedness of its union, maximality, rigidity of well-orders, and correctness of least-code normalization. The compression theorem then needs only injectivity of each row presentation and target separation by traces. This organization makes accidental appeals to Replacement visible.

For the second part, use a universe separation: the metatheory has an actual set  $V_\lambda$ , and its class sort is another set  $\mathcal{P}(V_\lambda)$  in the larger universe. Internal “set” and external “set” are distinct predicates. No term denoting a set of all sets or a set of all actual surreal numbers is required. The sign carrier is explicitly the set  $\bigcup_{\alpha < \lambda} {}^\alpha 2$  in the metatheory.

A first machine-checked milestone should be the domain-size criterion [lemma XV.9.2](#), followed by the sign-cardinality formula and the two enumeration thresholds. These use ordinary external cardinal arithmetic and rank bounds. The ordinal-abstraction theorem adds the distinction between an internal well-order relation and its external order type.

### XV.12.3 Why no experimental computation is offered as proof

Finite experiments with selectors can illustrate the thread construction, but cannot verify non-conservativity, lack of ordinal collapse, or Class Replacement. The mathematical verification here consists of displayed set bounds and independent checks of the proofs. The supplied source is an

ordinary LaTeX article, not a proof-assistant certificate. A later formal development should report separately the checked uniformization theorem and any assumed countermodel construction.

## XV.13 Axiom ledger and proof dependencies

[merge] Source 31's Appendix A.

| Construction or result           | Existence bound or main argument                                    | Logical setting                         |
|----------------------------------|---------------------------------------------------------------------|-----------------------------------------|
| Thread collection $\text{Thr}_X$ | Subset of $\mathcal{P}(X) \times \mathcal{P}(X^2)$                  | $\mathbf{Z} + \mathbf{Sep}(c)$          |
| Union thread $W_X$               | Domain subset of $X$ , relation subset of $X^2$                     | Initial-segment comparison; no collapse |
| Uniform injection $i_X$          | Least code in $\mathcal{C}_D$ ; graph subset of $X \times D$        | One fixed code well-order               |
| Trace $H_a$                      | Subset of $I \times D$                                              | Expanded Separation                     |
| Partial surjection $Q$           | Subset of $\mathcal{P}(I \times D) \times T$                        | Target determined by trace              |
| Countable nonconservativity      | Uniformization and compression contradict the published obstruction | Glazer countermodel input               |
| Sign enumeration thresholds      | Cardinal count and ordinal sums of birthday layers                  | External ZFC, full classes              |
| Universal set coding and OA      | $ V_\kappa  = \beth_\kappa$ , bounded rank of internal relations    | External ZFC, full classes              |
| Replacement on $a$               | All $ a $ -sized rank sets bounded iff $ a  < \text{cf}(\lambda)$   | External ZFC, full classes              |
| Proper-class size spectrum       | Rank-unbounded subsets and cofinal sets                             | External ZFC, full classes              |

The main derivation has the following dependency order. First prove the bounded thread lemma. Then prove least-code uniformization. Use the uniform injections to form bounded traces and the partial surjection. Finally instantiate the published obstruction. The rank-model part is independent of that nonconservativity argument; its common inputs are the rank-boundedness criterion and external cardinal arithmetic.

## XV.14 Source audit and research status

[merge] Source 31's Appendix B.

### XV.14.1 The inspected repository snapshot

The report was inspected at commit

2b7b388ba81a3355b19a7c2d2fe797e92f572355.

Its path is

Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex.



The corresponding Git blob is 05807d292fbed8b68aa5744b502fece703057c74. The source is a merged research report. We inspected its relevant foundations statements and proof passages; no repository-wide correctness audit is claimed. No repository files were modified. The exact replacement-free classification developed here was not found in the inspected passages and targeted searches.

[merge] The commit 2b7b388ba is “Catalogue batches 81 to 86”; the blob 05807d292 is this article as written by 07ee2b30c (batch 83), with Parts I–XIV, unchanged until this write.

### XV.14.2 Status of the principal assertions

**Proved refinement.** The countable-input and exactly-countably-infinite-input nonconservativity statements follow from the fully displayed uniformization argument and Glazer’s published countermodels.

**Proved structural results.** The full-class rank-model spectra, domain-size criterion, and separation examples are proved here in external ZFC. Several ingredients and motivating examples are classical or already discussed in the cited literature.

**Not claimed.** There is no claimed new solution of the entire surreal transfer problem, no classification for arbitrary definable-class realizations, no proof of the finite-input conservativity boundary, and no machine-checked formalization.

**Novelty qualification.** The article provides a concrete additional theorem and an exact synthesis. Targeted literature and repository checks support its relevance but do not establish that every formulation is historically new. Expert review of the localized nonconservativity statement would be the next scholarly step.

## XV.15 What this Part answers

[merge] Dated notes, 3 October 2026, in the form of [section 50](#).

- (a) *Source 28’s research question 113.209, the weakest class-theoretic assumptions for the supported-prefix extension, small-net rigidity and inverse-reconstruction schemes, and whether their uses of Class Replacement and uniform set choice separate sharply:* answered in part. Source 31 separates three mechanisms: selection, collapse and coding, and image bounding ([section XV.12](#)). Selection restricted to countable inputs, with Separation for the selector, is already nonconservative over ZC and ZC<sub>F</sub> ([theorem XV.5.1](#), through the Glazer interface; countably infinite inputs suffice, [corollary XV.5.3](#)), and conservative without expanded Separation ([proposition XV.5.5](#)). In the full-class rank models global choice holds at every height, while ordinal coding and image bounding have different exact thresholds ([theorems XV.7.2, XV.7.4, XV.8.2, XV.8.4 and XV.9.3](#)), Class Replacement on a set holds exactly below the cofinality of the height ([lemma XV.9.2](#)), and ordinal-domain Class Replacement together with birthday enumeration, ordinal abstraction or universal coding is full Class Replacement ([corollary XV.9.4](#)). Open: the minimal assumptions of each of the three schemes, which source 31 restates as [research question 113.242](#); it proves no equivalence of a scheme with a row of its table. The sibling questions [research questions 113.161 and 113.185](#) of sources 24 and 26 ask about the same resources and stay open.
- (b) *Source 12’s full-class model* ([theorem 21.1](#), a theorem, not a question): extended from inaccessible  $\kappa$  to every limit height for global choice, ordinal and birthday enumeration of the

sign carrier, universal set coding, ordinal abstraction, Class Replacement on each set domain and the sizes of proper classes ([theorem XV.10.1](#) and [corollary XV.10.2](#)); the warning about  $V_{\sqcup_\omega}$  in [section 21](#) is made exact. The external identification of the set-like and unrestricted orders in clauses (ii)–(iv) of [theorem 21.1](#) is not extended ([research question 113.241](#)).

- (c) *The base theory of [theorem 13.1](#):* in  $\mathcal{M}_\lambda$ , global choice and set-like class well-orders of the sign carrier exist at every height, but an Ord-enumeration exists only when  $\lambda = \kappa$  with  $2^{<\kappa} = \kappa$  ([proposition XV.6.2](#) and [theorem XV.7.2](#)); the equivalence of [theorem 13.1](#) uses the class Replacement of GB (the dated note after it). No earlier question asked this.
- (d) *Not settled:* Hamkins’s question whether universality of the surreal line is a weak global choice principle ([section 1.7](#) (f)); Hamkins’s surreal transfer question [76], on which source 31 asks [research question 113.244](#); the questions listed as not affected in [section 113.25](#) (q).

[merge] *New against the repository* (source 31 claims no priority for any formulation): the uniform presentation theorem and uniform countable enumeration ([theorem XV.3.3](#) and [corollary XV.3.5](#)), bounded trace compression ([theorem XV.4.1](#) and [corollary XV.4.2](#)), the countable-input nonconservativity theorem and its corollaries ([theorem XV.5.1](#) and [corollaries XV.5.3](#) and [XV.5.4](#), conditional on the Glazer interface), the conservativity of an uninteracting selector ([proposition XV.5.5](#)), choice orders without ordinal enumeration at every height ([proposition XV.6.2](#)), the bare, birthday, universal-coding and ordinal-abstraction thresholds ([theorems XV.7.2](#), [XV.7.4](#), [XV.8.2](#) and [XV.8.4](#)), the domain-size criterion and the Replacement thresholds ([lemma XV.9.2](#), [theorem XV.9.3](#), and [corollary XV.9.4](#)), the proper-class spectrum ([theorem XV.10.1](#) and [corollary XV.10.2](#)) and the separating heights of [section XV.11](#). *Inherited or re-proved:* the method of [lemma XV.3.1](#) (a bounded proof of the well-ordering theorem), the available axioms ([proposition XV.6.1](#)), the case  $\lambda = \kappa$  of [lemma XV.7.1](#) ([section 7](#)), the beth facts ([lemma XV.8.1](#)), the inaccessible case of [theorem XV.9.3](#) ([section 21](#)), ordinal abstraction and the example  $V_{\omega_1}$  (Hamkins, Rin), set-like orders without Replacement (Carneiro) and well-ordered Replacement (Freire and Hamkins, [remark XV.9.5](#)).

## Part XVI

# Definable class well-orders beyond Ord: arithmetic, histories over GB, fixed points and definability ceilings

[merge] (4 October 2026, batch 96.) This Part merges four manuscripts of batch 96 ([section 1.11](#)), written independently on 4 and 5 October 2026 in answer to one request: how far definable class well-orders reach beyond Ord, what their arithmetic is, how they can be named, and how the subject can be formalized. They are this report’s sources 32–35, batch-96 manuscripts 08–11: source 32 (“Beyond Ord”, definability ceilings and truth jumps), source 33 (“Beyond Ord”, definability heights), source 34 (“Beyond Ord”, canonical arithmetic and the limits of notation; the base) and source 35 (fixed-point completion and the limits of uniform definability). All four read this report with Parts I–XV, exactly as it stood before this write ([section 1.11](#)); their subject continues Part IX’s canonical blocks and histories ([sections 52](#) and [53](#)) and Part XIV’s complete condensation histories ([section 108](#)).

*The question answered.* Source 29 proved over GBC that strong comparability CWO of class well-orders and the existence CTH of canonical condensation histories are equivalent ([theorem 108.19](#)), and

this report listed the same question over GB without global choice as open (note 108.4, section 112 (b), research question 113.111). Sources 34 and 35 answer it independently: the equivalence holds over GB with ZF sets and no form of choice (theorem XVI.5.8 and note XVI.5.19). The only use of global choice in source 29’s argument was the well-foundedness of the polynomial order  $P(\Gamma)$  (note 108.15); both sources prove it without choice (theorem XVI.4.2 and note XVI.4.4). Neither proves CWO or CTH in GB, and whether CWO implies ETR (Hamkins and Woodin’s Question 7) stays open. The places that listed the question carry dated notes of this write.

*Arrangement.* Sections XVI.1–XVI.10 are source 34’s Sections 1–10, in its order and whole. Each is followed by subsections that print the corresponding material of source 35, then source 32, then source 33, each source’s sections whole and in their own order where possible. Source 34’s research agenda (its Section 11) is printed with the other three sources’ questions in section 113.28; section XVI.11 collects the four sources’ ledgers, checks and source audits (source 34’s Appendix A and the appendices of the others), and section XVI.12 records what the Part answers. A statement proved again by the same argument is a numbered *Note* that gives the source’s statement, number and differences, with the statement it repeats; a proof that differs is printed as a marked route (“Source 35’s proof”); results of one source are printed in full. Where a source re-proves a result of Parts I–XV the Note names the earlier label. Every result, proof, example, remark, question and limitation of the four sources is printed; text written for the merge is marked [merge].

*Three stale or answered items* are kept on record with their proofs (Vladimir’s standing rule of 4 October 2026). Source 32’s claim to answer this report’s termination question repeats source 20’s theorem 108.1 (remark XVI.1.1). Source 33’s Remark 5.5 and its research question 12.7 leave open whether its finite-digit map has initial image; it always has, because the map is the normal-form map of Part XIV (remark XVI.5.29). Sources 32 and 33 ask for the strength of tower existence (their questions 12.1 and 12.6); it is CWO, over GBC by source 29 and over GB by sources 34 and 35, so only  $\text{CWO} \Rightarrow \text{ETR}$  remains (section 113.28). Source 35’s question 10.6, a choice-free completion theorem, is answered here positively from source 34’s choice-free set-subset criterion (theorem XVI.6.21).

*Numbering.* The sections of this Part are numbered XVI.1–XVI.12, so that the questions part and the appendices keep their numbers (Sections 113–115, Appendices A–D), as in Part XV. Numbered statements are XVI. $k.m$ ; appendix D maps every numbered statement of the four sources to its place here. Source 34’s displayed equation ( $k$ ) is tagged (S34. $k$ ), source 35’s (S35. $k$ ), source 32’s (S32. $k$ ), and source 33’s equation ( $s.k$ ) (S33. $s.k$ ).

*Notation* (section 1.3.4 lists every renaming).  $\Omega$  is the presentation (Ord,  $<$ ), never a set ordinal or a surreal.  $A^{[B]}$  is the finite-support power (source 34’s macro `\Pow`, printed `\fsp`; sources 32 and 33 write  $A^B$ , printed  $A^{[B]}$  where the operands are class orders), and  $P(\Gamma) = \Omega^{[\Gamma]}$  is the polynomial order of Parts IX and XIV.  $\mathbf{E}$  is the hereditary term order (source 34’s and source 35’s  $\mathcal{E}$ ; this report’s  $\mathcal{E}_\kappa$  and  $\mathcal{E}$  are different objects) and  $\mathbf{E}(A)$  its version over a base  $A$ .  $A \hookrightarrow B$  is an order embedding,  $A \preceq_{\text{init}} B$  an initial one,  $A \not\hookrightarrow B$  the absence of an embedding.

## XVI.1 The question and the contribution

There are two quite different senses in which an ordering can be longer than all ordinals. An ordinary set cannot carry a well-order longer than every ordinal: its order type is itself an ordinal. A proper class can carry such an ordering, and can carry many different ones. For example, tagging each ordinal with 0 and putting a new tagged endpoint above all of them gives  $\Omega + 1$ . Putting one entire tagged copy of the ordinals after another gives  $\Omega + \Omega$ . The endpoint or the first point of the second copy has a proper class of predecessors. That is the precise structural change.

The purpose of this paper is not to postulate an absolute ordinal above all ordinals. It is to organize the well-orders already expressible as class relations, determine which operations are legitimate, and identify what limits their notation. A formula defining a relation, a proof that it is well-ordered, an isomorphism to another presentation, and a uniform procedure ranging over all definitions are different mathematical objects. Much of the difficulty lies in keeping these objects separate.

### XVI.1.1 Relation to the inspected ProveIt material

We inspected the ProveIt repository at commit `e1d2f3048b947801873a6e9ddc4e03736234b328`. The principal source is the consolidated report *Lexicographic Orders of Well-Orderings of the Surreal Numbers*, especially its treatment of canonical blocks, polynomial presentations, and Source 29’s canonical-history equivalence [77]. Its status ledger explicitly leaves the theory over GB open after establishing the corresponding equivalence over GBC. The present paper supplies the missing choice-free polynomial argument and checks the remaining dependencies. The theorem applies to arbitrary supplied class well-orders, not only to a surreal carrier.

[merge] Source 34 read this report at commit `e1d2f3048`, where `article.tex` (blob `72eee5cc8`) is the text of Parts I–XV that this write extends. Its “Source 29’s canonical-history equivalence” is [theorem 108.19](#), and “its status ledger explicitly leaves the theory over GB open” refers to [note 108.4](#), to item (b) of [section 112](#) and to [research question 113.111](#). The third row of the table below is [theorem XVI.5.8](#).

The repository also contains work on omnific notation and definable surreal numbers. Those directions concern additional algebraic structures or definability of individual numbers. Here the objects are order presentations. In particular, a formal symbol  $\Omega$  in this paper does not designate a surreal larger than every ordinal surreal. There is no such surreal in the usual class of surreal numbers, since each surreal is set-coded and has an ordinal birthday.

| Result                                              | Status in this paper                                            | Foundational scope                                                    |
|-----------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------|
| Set-like boundary and one-step division by $\Omega$ | Established structure; included with proofs                     | GB                                                                    |
| Finite-support power is a class well-order          | Choice-free refinement of the repository proof                  | GB; full subclass-minimum definition                                  |
| Canonical histories $\iff$ CWO                      | Repository equivalence lowered from GBC to GB                   | Arbitrary class carriers                                              |
| Hereditary $\Omega$ normal forms                    | Proper-class development of a standard ordinal-notation pattern | Explicit finite syntax; GB suffices                                   |
| Uniform-family diagonal and truth-relative growth   | Proved synthesis; priority not asserted                         | Supplied uniform family; full truth required for truth specialization |
| Definable spectral cofinality                       | Proved external model-theoretic synthesis                       | Transitive $\beta$ -model; named global well-order                    |

The most direct answer to a repository open question is the change of base theory in the third row. The other developments provide a systematic framework for constructing, interpreting, and bounding long notations. The much broader problem of whether strong class-well-order comparability implies ETR is not solved here. Hamkins and Woodin formulate that question explicitly; the sources checked during preparation did not provide a verified later resolution [27, Theorem 6 and Question 7]. This is a dated literature-status statement, not a proof that no resolution exists.

### XVI.1.2 What “farther” will mean

For class well-orders  $A, B$ , write  $A \hookrightarrow B$  if there is a class order embedding and  $A \preccurlyeq_{\text{init}} B$  if an embedding has an initial range, allowing the whole range. We use  $A \cong B$  for a class isomorphism. The claim that  $B$  is strictly longer than  $A$  is strongest when  $A$  embeds as a proper initial segment of  $B$ . A construction that contains  $A + 1$  as a suborder also proves  $B \not\hookrightarrow A$ , even when initial-segment comparability is unavailable. We specify which conclusion is proved rather than silently identifying the two relations.

In a sufficiently large external universe, a class model can itself be set-sized. Its internally proper class orders then have ordinary external ordinal lengths. This supplies precise numerical calibrations, but those external ordinals are not new ordinals of the original model.

### XVI.1.3 Source 35: scope, results, and their provenance

There is a coherent arithmetic of well-ordered classes that continue past the order  $\text{Ord}$ . Its objects should be understood as ordered presentations, and its equality should be understood as the existence of an appropriate class isomorphism. This immediately permits  $\text{Ord} + 1$  without asserting the existence of a set ordinal larger than every ordinal. The new final point is a tagged set in a class carrier; its predecessor class is proper. This distinction is the starting point for everything that follows.

The question “how far beyond  $\text{Ord}$ ?” has several answers. Explicit finite syntax reaches sums, products, class powers, and an exponential fixed point containing every finite tower of  $\text{Ord}$ -powers. Given a fixed point, one can construct another strictly above it. A uniformly coded family with a supplied well-ordered index class always has a strict upper bound for ordinary suborder embeddability, so no such family is cofinal among all class well-orders. In ambient ZFC, for strongly inaccessible  $\kappa$ , the full two-sorted model based on  $V_\kappa$  has external upper boundary  $\kappa^+$ , while a family restricted to first-order definitions occupies a bounded part of that spectrum. None of these statements identifies a last class ordinal.

The main results fall into three groups. First, we give an explicit arithmetic with proofs requiring only predicative class comprehension and class Replacement. Second, we connect the repository’s canonical condensation machinery with fixed-point completion. Third, we quantify the obstruction to one all-purpose definable notation system. The axioms used in the second and third groups are specified separately: an explicit constructor being elementary is a much weaker assertion than every class well-order having a comparable canonical presentation.

#### XVI.1.3.1 What is inherited, and what is developed here

The relevant ProveIt source is the maintained *surreal-well-orders* article, inspected at commit `f158f27b46bd1efb5a771ffc3c07e64eea970501`, dated 4 October 2026 [78]. Its polynomial orders and canonical histories are directly relevant even though the surrounding report also studies lexicographic orders of labelled well-orders. This paper studies the individual well-order presentations themselves.

| Topic                | Inspected prior development                                          | Role in this article                                     |
|----------------------|----------------------------------------------------------------------|----------------------------------------------------------|
| Set-interval blocks  | Repository Theorem 52.3 gives an elementary canonical decomposition. | Reproved briefly and expressed as division by $\Omega$ . |
| Polynomial histories | Repository Theorem 53.4 gives explicit histories for $P(\Gamma)$ .   | Used as the arithmetic model for normal forms.           |



| Topic                       | Inspected prior development                                                                                   | Role in this article                                                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Finite normal forms         | Repository Theorem 108.16 obtains an initial polynomial image from a supplied history.                        | Proof checked and included, with its choice assumptions isolated.                                                      |
| Comparability and histories | Repository Theorem 108.19 proves $\text{CWO} \leftrightarrow \text{CTH}$ over $\text{GBC}$ .                  | Strengthened to $\text{GB}$ using the new direct minimization proof.                                                   |
| Class epsilon notation      | Class-order notation for $\varepsilon_{\text{Ord}+1}$ occurs in Rathjen’s work, as recorded by Jeon [89, 88]. | A hereditary $\Omega$ -polynomial presentation is developed explicitly, without a priority claim for epsilon notation. |
| Fixed-point completion      | Not located in the inspected repository development.                                                          | A local equivalence and the global principle $\text{FPE}$ are proved here.                                             |
| Uniform definability        | General truth and coding obstructions are classical.                                                          | A strict successor-sum bound and a fixed-complexity separation theorem are proved here.                                |

The repository identifies its relevant additions as unrefereed and not machine formalized. We therefore supply proofs of the machinery needed for the extensions, rather than treating a manuscript label as a proof certificate. The results called extensions are new relative to the inspected development; this statement is deliberately narrower than a claim of first discovery in the literature. In particular, the article does not claim a solution to the general reverse-mathematical problem asking whether  $\text{CWO}$  implies  $\text{ETR}$ .

### XVI.1.3.2 A precise reading of the choice-free improvement

The repository explicitly retains the question of the strength of canonical histories over  $\text{GB}$  without Global Choice. The improvement proved below addresses this boundary: the same four-way characterization already holds over  $\text{GB}$ . This identifies the history principle with strong comparability in that weaker base. It does *not* prove either principle outright in  $\text{GB}$ , give a complete independence classification, or remove Choice from every theorem about orders of the entire surreal carrier. Transporting arbitrary proper carriers to one fixed carrier is a separate use of Global Choice.

Over  $\text{GBC}$ , the key local result concerning fixed points can be read independently of any universal comparison principle. For a supplied well-order  $W$ , one specific initial embedding  $W \preceq_{\text{init}} P(W)$  is enough to construct a fixed-point extension of  $W$ . Conversely, a fixed-point extension provides this embedding. Thus a question about completing one order has a concrete structural certificate.

[merge] Source 35 read this report at commit `f158f27b4`, with the same `article.tex` (blob `72eee5cc8`) and `README` as source 34. Its “repository Theorems 52.3, 53.4, 108.16 and 108.19” are [theorems 52.3](#), [53.4](#), [108.16](#) and [108.19](#), with these numbers; “its Part XIV explicitly leaves the corresponding question over  $\text{GB}$  without Global Choice open” refers to [note 108.4](#) and [section 112](#) (b). Its attribution of class epsilon notation to Rathjen through Jeon’s note is recorded as source 35 states it; the status of its 2026 citations is in [section 1.11](#).

### XVI.1.4 Source 32: the question and the scope of the answer

The expression “longer than  $\text{Ord}$ ” becomes meaningful as soon as one stops requiring every predecessor segment of a well-order to be a set. For example, take two disjoint copies of  $\text{Ord}$ , place the first entirely before the second, and compare ordinals normally within each copy. Every nonempty admitted subclass has a least point, but the first point of the second copy has a proper class of predecessors. This is a legitimate class well-order, not a new von Neumann ordinal.



We write

$$\Omega = (\text{Ord}, <)$$

for this *order presentation*. An expression such as  $\Omega^{[\Omega]}$  will denote a specifically defined class of sets with a specifically defined ordering. It will never mean that  $\text{Ord}$  has become an element of the set universe. Likewise, a variable  $W$  denotes a supplied class relation and its domain, not an element of a class of all proper-class order types.

There are three different questions behind the user’s problem. Which long orders have explicit definitions in a weak class theory? Which recursive operations can that theory perform along those orders? And, relative to a specified model and a specified meaning of “definable”, which external ordinal lengths can its class presentations realize? These questions have different answers. A large presentation need not support strong recursion, and an externally available isomorphism need not be an admitted class.

#### XVI.1.4.1 Main results and their provenance

The constructions of sums, products, and finite-support powers in [section XVI.3.4](#) are presentation-level extensions of ordinary ordinal arithmetic. The set-like boundary and the distinction between arbitrary and initial embeddings are foundational facts, not proposed discoveries. The repository already develops least-representative set-interval blocks and explicit polynomial condensation examples [\[79, 80\]](#). We reprove the ingredients needed here and use them to sharpen the structural picture.

The principal additional results proved in this manuscript are the uniform diagonal theorem ([note XVI.7.14](#)), the conditional termination bound for general condensation ([note XVI.5.20](#)), and the definability-ceiling and exact truth-presentation theorems ([theorems XVI.8.9](#) and [XVI.8.12](#)). These are mathematical contributions of this treatment in the sense that complete arguments are supplied here. Their historical novelty has not been established. The condensation theorem answers the repository’s termination question under an explicit recursion hypothesis, while leaving the exact strength of hierarchy existence open.

*Remark XVI.1.1* ([merge]; source 32’s termination theorem is source 20’s). The sentence of source 32 above, “The condensation theorem answers the repository’s termination question under an explicit recursion hypothesis, while leaving the exact strength of hierarchy existence open”, is stale on both counts, and is kept on record with the following correction. This report proved the termination theorem before source 32 was written: source 20’s [theorem 108.1](#) (Part XIV) shows over GB that a supplied nested history of  $W$  on the schedule  $W + 1$  ends in the singleton of the least point, and names elementary transfinite recursion along  $W + 1$  as a sufficient existence hypothesis; source 29 proves it again ([note 108.11](#)), and so do sources 34 and 35 ([lemma XVI.5.2](#) and [note XVI.5.13](#)). Source 32’s [note XVI.5.20](#) is the same theorem for its hierarchies of equivalence relations, which define exactly the same stages as nested histories ([remark XVI.5.24](#)). The question source 32 read (23-termination-RESEARCH\_STATUS.md, source 23’s status file, unchanged since it was staged) predates Part XIV; source 32 did not read Part XIV, whose status note records it as answered. The “exact strength of hierarchy existence” is answered as well: over GBC it is strong comparability CWO (source 29, [theorem 108.19](#)), and over GB likewise ([theorem XVI.5.8](#) and [note XVI.5.19](#)). Only whether CWO implies ETR remains open ([section 113.28](#)). Source 32 itself asserts no priority (its research status and [section XVI.11.3](#)).

The repository inspection was targeted rather than exhaustive. The relevant research-status file explicitly distinguishes its GBC polynomial examples from a general termination theorem. The merged report identifies the source as manuscript 23, with its condensation material incorporated

into the larger surreal well-order study [79, 80]. We do not certify the other manuscripts in that collection.

#### XVI.1.4.2 Conventions about foundations

Our default theory is GBC: Gödel–Bernays class theory with Global Choice and a ZFC set part. Every member of a class is a set. A class relation can therefore be coded by a class of ordered pairs or tuples of sets. Class parameters in elementary comprehension are allowed, but quantification over classes inside that comprehension is not.

When a statement is made in a model  $(M, \mathcal{X})$ , “class” means an element of the admitted class collection  $\mathcal{X}$ . This is especially important for well-foundedness and isomorphisms. Sections [section XVI.8.6](#) and [section XVI.8.7](#) explicitly change viewpoint to an external ZFC metatheory with a strongly inaccessible cardinal. Their ordinal spectra are external sets of ordinals; they are not same-level class quotients in GBC.

### XVI.1.5 Source 33: the problem and the principal results

#### XVI.1.5.1 What does “longer than Ord” mean?

Write

$$\Omega = (\text{Ord}, \in)$$

for the class well-order of all set ordinals. The symbol  $\Omega$  is an abbreviation for an order presentation, *not* a new set and not an ordinal of the same universe. The tagged disjoint union  $\Omega + 1$  is a perfectly legitimate class well-order. Its last point has a proper class of predecessors. This is precisely why the usual theorem assigning an ordinal to every *set* well-order does not collapse  $\Omega + 1$  to a set ordinal or to  $\Omega$ .

There are several different questions hidden in the request to go beyond Ord:

- (i) Which longer orders can be constructed by explicit formulas?
- (ii) Which arithmetic operations preserve class well-foundedness?
- (iii) When can two arbitrary class well-orders be compared by an initial embedding?
- (iv) How does the answer depend on allowed parameters, truth predicates, and the surrounding class model?
- (v) Can all relevant orders be organized into one uniform notation system?

The answers are not interchangeable. In particular, constructing  $\Omega^{[\Omega]}$  is much easier than producing a uniform comparison procedure for arbitrary class well-orders. The literature on class recursion makes this separation explicit [27, 22, 84].

#### XVI.1.5.2 A map of the results proved here

The article’s main statements are the following.

##### Structural and definability results.

**Canonical division.** Every class well-order  $W$  has a canonical presentation

$$W \cong \Omega \cdot Q + \alpha,$$

where  $Q$  is a class well-order and  $\alpha$  is a set ordinal ([note XVI.3.11](#)).

**Finite digits.** An exhausting, uniformly supplied set-interval condensation tower indexed by  $\Gamma + 1$  yields an order embedding

$$W \longrightarrow \Omega^{[\Gamma]}$$

whose values have finite support ([theorem XVI.5.27](#)).

**Definability diagonal.** At every fixed finite syntactic level  $n$ , a parameter-free-in- $(\in, G)$  order  $D_n$  dominates every set-parameter  $\Sigma_n(G)$  presentation, and  $D_n$  embeds into none of those orders ([theorem XVI.7.20](#)).

**External height.** Over  $V_\kappa$  with  $\kappa$  inaccessible, the supremum  $\delta_G$  of the  $G$ -definable well-order types satisfies

$$\kappa < \delta_G < \kappa^+, \quad \text{cf}(\delta_G) = \omega, \quad \delta_{G,T} > \delta_G$$

when  $T$  is truth for  $(V_\kappa, \in, G)$  ([note XVI.8.16](#) and [theorem XVI.8.19](#)).

The most important common mechanism is *uniformity*. A class-indexed family of class relations is usable when one class of tuples gives all its sections. A metatheoretic collection of individually definable relations does not automatically have such a code. The difference accounts both for positive constructions and for diagonal obstructions.

### XVI.1.5.3 Relationship with ProveIt and the literature

The targeted repository review used the main README, the surreal research catalogue, and the guide to `Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/` [82, 83, 80]. The latter guide describes twenty-three merged manuscripts and an 860-page report. It explicitly separates unrefereed research text from formalized Lean results. Its later parts already contain class-order and transfinite condensation material. Therefore neither the existence of orders beyond Ord nor the general idea of an arbitrarily long condensation hierarchy is claimed here as a new contribution to the repository.

This manuscript takes a different organizing route: intrinsic class-order arithmetic, canonical division, finite digits, and the relationship between syntactic complexity and attainable order lengths. Its proofs are self-contained except for clearly cited foundational background. The repository review was targeted, not a line-by-line audit of the full merged report, and it does not establish absence of overlapping arguments elsewhere in the repository.

Three literature interfaces matter particularly. First, class comparison is part of the reverse mathematics of elementary transfinite recursion [27]. Second, truth predicates and their iterations are genuine class-existence principles rather than harmless uniform notation [22, 24]. Third, unrolling and class-theoretic potentialism supply ways to study external order heights while retaining the distinction between sets and classes [35, 84, 28]. The results below are not presented as solutions of the published comparison/recursion problems.

[merge] (4 October 2026.) Source 33's description of the guide was accurate at its reading (README blob d2e24b58, unchanged from e50dde15b to this write); with this Part the report merges twenty-seven manuscripts. The finite-digit embedding announced in the box above is in fact initial: it is the normal-form map of Part XIV ([remark XVI.5.29](#)).

## XVI.2 Foundations: classes, definability, and well-foundedness

### XVI.2.1 The base theories and the object language

We use the two-sorted formulation of Gödel–Bernays theory. Sets are members of classes; proper classes are not members of other classes. Elementary comprehension permits formulas with set quantifiers and supplied class parameters. Class Replacement says that the image of a set under a class function is a set. The notation **GB** here includes the usual ZF set axioms but does not assume Global Choice; none of the choice-free proofs below needs a choice axiom for arbitrary sets. **GBC** adds a global choice function, equivalently in this setting a set-like global well-order. The letter  $G$  will denote such a well-order when one is used.

A class relation on  $X$  is represented by a class of coded pairs of sets. One relation suffices to code a family: its section at a set index  $i$  is a class. This does not turn the family into a class whose members are themselves proper classes. Appended endpoints and sums use tagged sets, such as  $(0, x)$  and  $(1, \emptyset)$ . An arbitrary carrier can already equal the whole universe, so an untagged “new set outside the carrier” is not a legitimate general construction.

**Definition XVI.2.1** (Class well-order). A class well-order is a strict linear order  $<_W$  on a class  $X$  such that every nonempty admitted subclass of  $X$  has a least member. A relation is *set-like* if the predecessors of each individual point form a set.

The word “admitted” refers to the class part of the model. An internal class well-order is not automatically externally well-founded when the model is set-sized. A transitive model of class theory is a  $\beta$ -model if its assertions of class well-foundedness are externally correct. We invoke this extra hypothesis when taking external order types.

Over **GBC**, class well-foundedness is equivalent to the absence of a set-coded infinite descending sequence. Given a subclass without a least point, use  $G$  to select a predecessor repeatedly; ordinary recursion on  $\omega$  and Class Replacement produce a descending set sequence. Conversely, the range of a descending sequence has no least member. This standard distinction is discussed in [22, 27]. Over **GB** alone we retain the subclass-minimum definition and do not use the reverse implication without an additional selection argument.

**Lemma XVI.2.2** (A first-order criterion for linear orders). *In **GB**, a class linear order is a class well-order if and only if every nonempty set subset of its domain has a least point. Consequently a common class linear order has absolute well-foundedness between two **GB** realizations with the same sets.*

*Proof.* Only the reverse implication needs proof. Suppose a nonempty admitted subclass  $C$  has no least point. Below any  $x \in C$ , take the least ordinal  $\rho$  such that  $C_{<x} \cap V_\rho$  is nonempty, and then take the order-least point of that set. The hypothesis supplies this latter minimum, so the selected predecessor is unique. Starting from any one point of  $C$ , ordinary set-valued recursion yields an infinite descending sequence. Its range is a set subset with no least point, a contradiction. The criterion quantifies only over sets and the given relation; hence it is unchanged in a same-set class extension.  $\square$

This lemma is specific to linear orders: their set-sized restrictions provide unique choices. It does not silently identify the set-subset criterion with absence of descending sequences in arbitrary choiceless settings.

## XVI.2.2 Four meanings of definable

The formulas defining a class order can have no parameters, set parameters, ordinal parameters, or specified class predicates. These give different spectra. An order defined from a truth class is not thereby definable in the pure language of set theory. Nor does the fact that each member of an external list is definable give one formula that evaluates every definition in that list.

The principal elementary constructions preserve their input parameters. If  $A$  and  $B$  are defined using  $p$ , their tagged sum, product, and finite-support power are defined using  $p$ . Our hereditary  $\Omega$  system uses no extra predicate: its terms are finite trees with ordinal labels. The diagonal theorem later uses a uniform relation already supplied, while its truth specialization explicitly enlarges the language.

For a ZFC-only presentation, each such construction can be read as a metatheoretic scheme about specified defining formulas. Statements quantifying over all classes, such as CWO, require a class theory or an external interpretation. A universal internal predicate that evaluates every pure set-theoretic definition cannot be assumed.

## XVI.2.3 Induction is not the same as class-valued recursion

Induction along a supplied class well-order is elementary: if the class of failures of an elementary property were nonempty, its least member would contradict the induction step. This proves uniqueness and many structural assertions. It does not create an unknown class-valued history. A history satisfying

$$S_i = \{x : \varphi(x, S \restriction i, A)\}$$

is a new class whose sections may all be proper classes. The principle ETR asserts the existence of such solutions along class well-orders. Even a length- $\omega$  class-valued recursion can require more than GBC, as the recursive construction of a full truth predicate shows. In contrast, recursion on a finite tree, or on  $\omega$  with uniquely specified set values, is available in the elementary arguments below.

## XVI.2.4 Source 35: foundations and the meaning of length

### XVI.2.4.1 The base theory and class parameters

Throughout the choice-free results, GB means the two-sorted Gödel–Bernays theory with ZF for sets, elementary class comprehension allowing class parameters, and class Replacement. No form of Choice is built into this notation. We write GBC for its extension by Global Choice. This convention is stated because authors differ on whether the abbreviation GB includes set Choice. KM denotes Kelley–Morse class theory; any additional choice or collection principle will be named when used.

A *class presentation*  $W = (X, <_W)$  consists of a class of sets and a class relation on those sets. In a model  $(M, \mathcal{X})$ , all the classes mentioned belong to  $\mathcal{X}$ . A class is never silently an element of another class. Expressions involving families of classes mean one class of tuples with set-valued indices.

*Note XVI.2.3* (Source 35, Definition 2.1, “Class well-order”). A *class presentation*  $W = (X, <_W)$  is a *class well-order* if  $<_W$  is a strict linear order and every nonempty admitted subclass of  $X$  has a least element. It is *set-like* if, for every  $x \in X$ , the predecessor class  $W_{<x} = \{y \in X : y <_W x\}$  is a set. This is source 34’s [definition XVI.2.1](#); source 35 calls the pair  $(X, <_W)$  a *class presentation*, a class of sets with a class relation.

The word “admitted” matters in model-theoretic discussions. The internal assertion is interpreted using the model’s sets and classes. In an external full model the quantification includes every subset

of the carrier. For fixed relations on a fixed set universe satisfying the usual choice assumptions, well-foundedness has a first-order description by absence of a set-coded descending sequence. Consequently one should not claim that changing only the class collection in two same-set GBC realizations automatically changes well-foundedness. The later formalization section explains the distinct issue of a transitive set model being externally wrong because it is missing a descending sequence as a *set*.

*Note XVI.2.4* (Source 35, Proposition 2.2, “The exact boundary of set-likeness”). *In GB, a set-like class well-order is isomorphic to a set ordinal if its carrier is a set, and to Ord if its carrier is proper. Every proper-class well-order has an initial subclass isomorphic to Ord. If it is not set-like, it has an initial subclass isomorphic to Ord + 1.* This is source 34’s [theorem XVI.3.1](#) (Section XVI.3), by the same proof: the rank  $x \mapsto \text{otp}(W_{<x})$  maps onto an initial subclass of Ord, which is all of Ord for a proper carrier by Class Replacement applied to the inverse; the points with set-sized predecessor classes form the initial copy of Ord, and its first successor caps a copy of Ord + 1. Source 35 adds that the rank function is elementary because the isomorphism certifying each value is a set. Sources 32 and 33 prove it as [notes XVI.2.9](#) and [XVI.2.14](#); in this report it is the class-carrier form of source 11’s canonical initial part ([theorem 15.1](#)) and of the first block of [theorem 52.3](#).

#### XVI.2.4.2 Two notions of comparison

An order embedding is an injective class map preserving and reflecting the strict order. Write  $A \hookrightarrow B$  if one exists. Write  $A \preceq_{\text{init}} B$  if an embedding exists whose image is an initial subclass of  $B$ ; the entire target is allowed. Write  $A \prec_{\text{init}} B$  if its image is proper. When we say that one order is *longer* without further qualification, we mean this proper-initial-segment comparison.

The existence of an arbitrary suborder embedding is weaker as a *witness*: it need not itself identify an initial segment. Under strong class-well-order comparability, the two usual comparisons can be related as for set ordinals. We do not assume that principle while constructing elementary examples. A class isomorphism, denoted  $\cong$ , must be a class available in the chosen formalization.

**Lemma XVI.2.5** (Rigidity and the successor obstruction). *In GB, every order embedding  $f : W \rightarrow W$  of a class well-order into itself is inflationary:  $x \leq_W f(x)$ . There is no embedding  $W + 1 \hookrightarrow W$ . Between two class well-orders there is at most one isomorphism onto an initial subclass.*

*Proof.* If the class of counterexamples to inflationarity is nonempty, let  $x$  be its least element. From  $f(x) < x$  and order preservation,  $f(f(x)) < f(x)$ , contradicting minimality. If an embedding of  $W + 1$  into  $W$  maps its final point to  $a$ , its restriction  $f$  to  $W$  satisfies  $a \leq f(a) < a$ , again impossible. For uniqueness of initial isomorphisms, a least point of disagreement would have two different images with exactly the same predecessor class in the target; strict linearity forbids this.  $\square$

This elementary obstruction replaces several tempting but unjustified appeals to a universal ordinal rank for class well-orders.

[merge] The first two assertions of the lemma are source 34’s [lemma XVI.3.2](#) (Section XVI.3), proved in the same way, and sources 32 and 33’s [notes XVI.2.10](#), [XVI.2.15](#) and [XVI.2.16](#); source 35’s [note XVI.7.7](#) is a second proof of the second. The third, uniqueness of an initial-segment isomorphism, is stated as a lemma only here; source 34 uses it after [corollary XVI.6.8](#) and source 32 notes it after [note XVI.2.10](#). It is printed whole because [theorem XVI.6.17](#) uses it.



### XVI.2.4.3 The symbol $\Omega$ and the absence of a paradox

We use  $\Omega$  as a typographical symbol for the *presentation*  $(\text{Ord}, \in)$  when it occurs in class-order arithmetic. It is not a member of  $\text{Ord}$ . For example, a carrier for  $\Omega + 1$  is

$$(\{0\} \times \text{Ord}) \cup \{(1, 0)\},$$

with every point of the first tagged copy below the final point. The pair  $(1, 0)$  is a set. Its predecessor class is proper. A carrier for  $\Omega \cdot \Omega$  is  $\text{Ord} \times \text{Ord}$ , ordered by the second coordinate first and by the first coordinate within each block. Neither construction asks for a set containing all ordinals.

Similarly,  $\Omega^{[\Gamma]}$  below is a class of *finite set codes*. An exponent label  $i \in \Gamma$  is a set, even when its predecessor class in  $\Gamma$  is proper. The exponent label need not be an ordinal. Finite support is what makes the object a legal set code.

*Remark XVI.2.6* (Set-valued and class-valued recursion). A recursion on  $\omega$  whose next value is a uniquely specified set can be carried out in GB with supplied class parameters. One takes finite approximations and uses class Replacement to obtain the set-coded sequence. A recursion on  $\omega$  whose stages are newly constructed *classes* is different: GB does not in general supply one class coding the entire sequence. The Tarskian recursion for truth is the familiar warning. The finite-trace constructions in this paper repeatedly use the former fact and never assume the latter without a stated principle [24].

### XVI.2.5 Source 32: well-ordering, set-likeness, and comparison

*Note XVI.2.7* (Source 32, Definition 2.1). A class well-order is a strict linear order  $<_W$  on a class  $\text{dom } W$  such that every nonempty admitted subclass of  $\text{dom } W$  has a least element; it is set-like when  $W_{<x} = \{y \in \text{dom } W : y <_W x\}$  is a set for every  $x \in \text{dom } W$ . An embedding preserves and reflects the strict order; an initial embedding additionally has an initial-segment image. One writes  $A \hookrightarrow B$  and  $A \preceq_{\text{init}} B$  only when the witnessing map is an admitted class. This is [definition XVI.2.1](#) with the two comparison relations of source 34's Section 1.2 (XVI.1.2). Source 32 writes  $D_W$  for the domain of  $W$ , printed  $\text{dom } W$  here, because  $D_W$  is the condensation operator of Parts IX and XIV and of Section XVI.5; it writes  $\preceq_e$  and  $\preceq_i$  for  $\hookrightarrow$  and  $\preceq_{\text{init}}$ .

In GBC the apparently second-order well-order condition has an elementary test in the supplied class parameters. The following familiar equivalence is also used in the class-order literature [27].

**Lemma XVI.2.8** (The descending-sequence test). *For a strict linear class order  $W$ , the following are equivalent: every nonempty admitted subclass has a least point; every nonempty set subset has a least point; and there is no set function  $f : \omega \rightarrow \text{dom } W$  with  $f(n+1) <_W f(n)$  for every  $n$ .*

*Proof.* The first assertion implies the second. A descending sequence has a set range with no least point, so the second implies the third. Conversely, let  $S$  be a nonempty admitted subclass with no least point. Fix a set-like global well-order of the universe. Choose its least point in  $S$  as  $f(0)$ ; after choosing  $f(n)$ , choose the globally least element of  $S$  below  $f(n)$  in the  $W$  order. Recursion of length  $\omega$  with set values and class parameters, followed by Class Replacement, gives a set function  $f$ . It is descending. This is ordinary set-valued recursion, not class-valued elementary transfinite recursion.  $\square$

[merge] Source 32's [lemma XVI.2.8](#) is the GBC equivalence already printed in Part IX (source 15's paragraph "Exactly where recursion principles enter" in [section 57.4](#)) and stated by source 33 as [note XVI.2.12](#); source 35 recalls it in Section XVI.8 (its Section 9.3) and source 34 in [theorem XVI.7.4](#)'s preamble. Over GB without global choice, source 34's [lemma XVI.2.2](#) gives the

equivalence of the first two clauses for linear orders; the third clause, absence of descending set sequences, is not claimed there to be equivalent without choice.

Consequently, validity of a supplied class-order presentation can be tested by a first-order formula with its domain and relation as class parameters. This fact will later let us filter invalid fibers of a uniformly presented family without impredicative class comprehension.

*Note XVI.2.9* (Source 32, Theorem 2.3, “The exact set-like boundary”). *Every set-like class well-order is isomorphic either to a set ordinal or to  $\Omega$ . Moreover, every proper-class well-order has a canonical initial segment isomorphic to  $\Omega$ , namely  $I_W = \{x \in \text{dom } W : W_{<x} \text{ is a set}\}$ . This is source 34’s [theorem XVI.3.1](#) and source 35’s [note XVI.2.4](#), by the same rank collapse and the same least point outside  $I_W$ ; source 32 works in GBC, although the proof uses no choice.*

This gives the precise answer to the first threshold question: longer orders are possible exactly because they are not set-like. No additional element of Ord, and no inconsistency, is involved.

*Note XVI.2.10* (Source 32, Lemma 2.4, “No self-compression”). *If  $W$  is a class well-order, no admitted embedding  $W + 1 \rightarrow W$  exists; every admitted self-embedding  $f : W \rightarrow W$  satisfies  $x \leq_W f(x)$ . This is source 34’s [lemma XVI.3.2](#). Source 32’s proof takes the least  $x$  with  $f(x) <_W x$ , puts  $y = f(x)$ , and derives  $y \leq_W f(y)$  from minimality and  $f(y) <_W f(x) = y$  from order preservation; for  $W + 1$ , the image  $a$  of the new point satisfies  $g(a) <_W a$  for the restriction  $g$ .*

Uniqueness of an initial-segment isomorphism follows from the same least-disagreement argument. Existence is another matter: GBC does not permit one simply to collect all class partial isomorphisms and take their union. That proposal uses quantification over classes inside the proposed defining formula. Under ETR, class well-orders are comparable by initial embeddings [27]; our explicit constructions below do not assume this comparison principle.

## XVI.2.6 Source 33: foundations, well-foundedness, and size

### XVI.2.6.1 The base theory and the meaning of a class presentation

Our default base theory is GBC: Gödel–Bernays set theory with global choice, including the usual set axioms of ZFC. Set variables range over sets; class variables range over the available classes. Elementary class comprehension permits formulas with set quantifiers and named class parameters. Class replacement says that the image of a set under a class function is a set.

A *class order presentation* is a pair  $(D, R)$  of classes, with  $R \subseteq D \times D$ . Elements of  $D$  are always sets. A function between presentations is represented by a class graph of ordered pairs of sets. An isomorphism type will be discussed externally or by comparing presentations; we do not form an internal class whose elements are proper-class relations.

Global choice supplies a set-like global well-order  $G$  of  $V$ , which we fix when needed. Its restriction to any nonempty class has a least point. This last assertion follows by choosing one point, intersecting the class with its set-sized  $G$ -initial segment, and applying ordinary set well-ordering.

*Note XVI.2.11* (Source 33, Definition 2.1). *A class well-order is a linear class order for which every nonempty available subclass has a least element; it is set-like if every proper principal initial segment  $D_{<x} = \{y \in D : yRx\}$  is a set. This is [definition XVI.2.1](#); source 33 writes a presentation as a pair  $(D, R)$  of classes.*

There is an important first-order test for this definition. The equivalence below is standard in the class-recursion literature [27, §3]; the proof records exactly the use of choice and replacement needed later.

*Note XVI.2.12* (Source 33, Lemma 2.2, “The descending-sequence test”). *For a linear class order  $R$  on  $D$ , the following are equivalent in GBC: (a) every nonempty available subclass of  $D$  has a least*

point; (b) every nonempty set subset of  $D$  has a least point; (c) there is no set function  $f : \omega \rightarrow D$  with  $f(n+1)Rf(n)$  for all  $n$ . Consequently well-orderhood is expressible by a first-order formula with  $D, R$  as class predicates. This is source 32's lemma XVI.2.8, by the same argument: from a subclass  $C$  without least point, choose  $x_0 \in C$  and then the  $G$ -least member of  $\{y \in C : yRx_n\}$ ; finite approximations define the  $n$ th value by an elementary formula, and Class Replacement on  $\omega$  collects the values into a set function. Source 33 stresses that only an ordinary set-length sequence of sets is constructed, with no class-valued stages. Over GB, (a) $\Leftrightarrow$ (b) is source 34's lemma XVI.2.2. Remark XVI.2.13 (Internal versus external well-foundedness). The word “set” in note XVI.2.12 refers to the sets of the model under consideration. An external descending sequence need not belong to that model. Thus a general transitive class model can misjudge well-foundedness. The inaccessible-rank models used in section XVI.8.8 are verified separately to avoid this problem. We do not silently identify internal well-foundedness with external well-foundedness.

### XVI.2.6.2 Set-like orders stop at Ord

Note XVI.2.14 (Source 33, Proposition 2.4). *A set-like class well-order is isomorphic either to a set ordinal or to  $\Omega$ ; in particular a proper-class set-like well-order has type  $\Omega$ .* This is source 34's theorem XVI.3.1, by the ordinal collapse  $\rho(x) = \text{otp}(D_{<x})$ ; if its range is a proper initial segment of Ord, its least omitted ordinal bounds it, and Class Replacement applied to the inverse makes  $D$  a set.

An order beyond  $\Omega$  therefore needs at least one point with a proper class of predecessors. That requirement is not an inconsistency: it simply places the order outside the set-like case of note XVI.2.14.

### XVI.2.6.3 Embeddings, initial embeddings, and rigidity

Write  $A \hookrightarrow B$  if an available class order embedding  $A \rightarrow B$  exists, and  $A \preceq_{\text{init}} B$  if one exists with initial image. The first relation is enough for our diagonal results. We do not assume, in GBC, that arbitrary class well-orders admit an initial comparison.

Note XVI.2.15 (Source 33, Lemma 2.5, “Inflation of a self-embedding”). *If  $f : W \rightarrow W$  is an order embedding of a class well-order into itself, then  $x \leq_W f(x)$  for every  $x \in W$ .* This is the first assertion of source 34's lemma XVI.3.2, with the same least-counterexample proof.

Note XVI.2.16 (Source 33, Corollary 2.6, “The successor obstruction”). *For every class well-order  $W$ ,  $W \hookrightarrow W + 1$  and  $W + 1 \not\hookrightarrow W$ ; in particular, there is no greatest class well-order under embeddability.* The nonembedding is the second assertion of source 34's lemma XVI.3.2; source 33's proof restricts  $h : W + 1 \rightarrow W$  to  $W$  and applies note XVI.2.15 at  $a = h(\text{top})$ , and the empty  $W$  is immediate.

This proof is useful precisely because it needs neither a universal comparison theorem nor a class recursion along  $W$ .

## XVI.3 The boundary at $\Omega$ and elementary arithmetic

**Theorem XVI.3.1** (The set-like boundary). *In GB, every set-like class well-order is isomorphic either to a set ordinal or to  $\Omega$ . Every proper class well-order  $W$  has a canonical initial copy of  $\Omega$ , consisting exactly of its points with set-sized predecessor classes. If  $W$  is not set-like, it has an initial copy of  $\Omega + 1$ .*

*Proof.* For a set-like order define  $r(x) = \text{otp}(W_{<x})$ . Each value is a set ordinal, and  $r$  is an isomorphism onto an initial subclass of  $\text{Ord}$ . A bounded initial subclass is an ordinal; Class Replacement applied to the inverse map then makes the original carrier a set. An unbounded initial subclass is all of  $\text{Ord}$ .

For general  $W$ , let  $S$  consist of the points with set predecessor classes. It is an initial subclass and is set-like. If  $W$  is proper and  $S$  were a set, the complement would be nonempty and have a least point  $b$ . Its predecessors would be exactly  $S$ , contradicting  $b \notin S$ . Thus  $S$  is proper and has type  $\Omega$ . If its complement is nonempty, its first point caps the claimed initial copy of  $\Omega + 1$ .  $\square$

This theorem excludes a set-valued increasing rank map from a genuinely longer class well-order into  $\text{Ord}$ . It does not exclude class rankings into another long order. In particular, assigning every point the set ordinal “number of its predecessors” would simply assume away the phenomenon being studied.

**Lemma XVI.3.2** (No backward self-embedding). *For a class well-order  $W$ , every class order embedding  $f : W \rightarrow W$  satisfies  $x \leq_W f(x)$ . There is no class order embedding  $W + 1 \rightarrow W$ .*

*Proof.* If  $f(x) <_W x$ , choose the least such  $x$ . Then  $f(f(x)) <_W f(x)$  makes  $f(x)$  an earlier counterexample. For an embedding of  $W + 1$ , let  $a$  be the image of its endpoint. The restriction to  $W$  would imply  $a \leq_W f(a) <_W a$ . The empty order is immediate.  $\square$

### XVI.3.1 Sums and products

Define  $A + B$  by tagged concatenation. Define  $A \cdot B$  on pairs  $(b, a)$ , comparing  $b$  first and then  $a$ ; thus it is  $B$  copies of  $A$ . More generally, a uniformly given family  $(A_b)_{b \in B}$  has a sum  $\sum_{b \in B} A_b$ . For a nonempty subclass of the sum, first choose the least occupied index and then the least point in that fiber. This proves well-ordering in GB without class choice. The coordinate maps prove, with explicit isomorphisms,

$$(A + B) + C \cong A + (B + C), \quad (A \cdot B) \cdot C \cong A \cdot (B \cdot C),$$

$$A \cdot (B + C) \cong A \cdot B + A \cdot C.$$

Commutativity and distribution of  $(A + B) \cdot C$  over its left summands are not asserted; set-ordinal examples refute them.

| Presentation          | Concrete model                                                | Distinguishing feature                                                           |
|-----------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------|
| $\Omega + 1$          | Ordinals followed by a tagged endpoint                        | A proper-class predecessor segment with a final point                            |
| $\Omega \cdot 2$      | Two consecutive ordinal blocks                                | The second block has no last point                                               |
| $\Omega \cdot \omega$ | Countably many ordinal blocks                                 | Has a set-coded countable cofinal sequence                                       |
| $\Omega^2$            | Pairs of ordinals, outer coordinate first                     | Proper-class many ordinal blocks                                                 |
| $\Omega^{[\Omega]}$   | Finite ordinal-coefficient polynomials with ordinal exponents | Contains every $\Omega^\alpha$ , $\alpha \in \text{Ord}$ , as an initial segment |

In the last two lines the superscripts mean the finite-support construction defined below. A cofinal set in  $\Omega \cdot \omega$  does not collapse its carrier to a set: the initial segments bounded by that set can themselves be proper classes.

### XVI.3.2 Canonical division by $\Omega$

**Theorem XVI.3.3** (One-step block decomposition). *Every class well-order  $W$  has a canonical decomposition*

$$W \cong \Omega \cdot Q + \delta,$$

where  $Q$  is a class well-order and  $\delta$  is a set ordinal, possibly zero. The representatives and the isomorphism are elementary in  $W$ , and the construction is valid in GB.

*Proof.* Declare  $x \sim y$  when the closed interval between them is a set. The relation is an elementary, convex equivalence relation. Every block has a least point  $r$ , and each of its points has a set predecessor interval within the block. Its local rank  $\text{otp}([r, x))$  identifies the block with a set ordinal or with  $\Omega$ , by [Theorem XVI.3.1](#).

If a block were a set and not final, the class of points following it would have a least point  $z$ . The interval from the block's minimum to  $z$  would be that set with one point adjoined. This contradicts distinctness of the blocks. Let  $Q$  be the class of least representatives of all proper blocks, with the inherited order. There is at most one set block, and it is final; take its order type as  $\delta$ , or take  $\delta = 0$  if it does not exist. Local ranks give the displayed isomorphism uniformly.  $\square$

This is a canonical decomposition of a supplied order, not an automatic transfinite iteration of decompositions. The question whether all successive quotients can be supplied as one history is the subject of [Section XVI.5](#). The one-step construction and the polynomial examples already appear in [ProveIt \[77\]](#); they are included here to make the refinement self-contained.

[merge] [Theorem XVI.3.1](#) is source 35's [note XVI.2.4](#), source 32's [note XVI.2.9](#) and source 33's [note XVI.2.14](#); [lemma XVI.3.2](#) is source 35's [lemma XVI.2.5](#) (first two clauses), source 32's [note XVI.2.10](#) and source 33's [notes XVI.2.15](#) and [XVI.2.16](#). [Theorem XVI.3.3](#) is the canonical block decomposition of Part IX, proved there three times ([note 52.1](#): source 15's [theorem 52.3](#), source 16's [theorem 52.6](#), source 18's [theorem 52.11](#)), and here again by sources 35, 32 and 33 ([\(S35.8\)](#), [theorem XVI.3.6](#) and [note XVI.3.11](#)); source 34's remainder  $\delta$  is Part IX's  $\theta_R$  and source 35's  $\rho_W$ .

### XVI.3.3 Source 35: arithmetic of presented class well-orders; sums and products

The operations in this section act on *presentations*: a class of set-valued labels together with a class ordering relation. They do not turn proper classes into elements of a larger same-level class. Write  $\Omega$  for the presentation  $(\text{Ord}, <)$ . An expression such as  $\Omega \cdot \Omega$  denotes a specified class order; an equation between such expressions means a class order-isomorphism. All the direct constructions and their well-ordering proofs below work in GB, with the convention that a class well-order gives a least element to every nonempty available subclass. Global Choice is unnecessary for these results.

The repository already develops the ordinal-coefficient polynomial construction  $P(\Gamma)$  and uses it to analyze condensation histories [\[78\]](#). Here we place that construction inside a general arithmetic, allow arbitrary class well-ordered coefficients, and give a choice-free proof of well-foundedness. Class expressions of the form  $\text{Ord}^\Gamma$  also occur in the literature on elementary transfinite recursion [\[24, Section 9\]](#). The finite-support definition makes the meaning and the required resources explicit.

#### XVI.3.3.1 Sums and products

For class well-orders  $A$  and  $B$ , their sum  $A + B$  has the disjointly tagged domain

$$(\{0\} \times A) \cup (\{1\} \times B),$$

with every point of the first copy before every point of the second. Their product  $A \cdot B$  has domain  $A \times B$  and comparison

$$(a, b) < (a', b') \iff b <_B b' \text{ or } (b = b' \text{ and } a <_A a').$$

Thus  $A \cdot B$  consists of copies of  $A$  indexed by  $B$ . A nonempty subclass of the product has a least occupied  $B$ -coordinate, and its fiber there has a least  $A$ -coordinate. This proves the well-ordering claim without a recursion or any choice principle. The sum is even more immediate.

The same argument handles a *supplied uniform family*. More precisely, let  $I$  be a class well-order, let  $D \subseteq I \times V$  be a class, and let a class relation specify a well-order  $A_i$  on each fiber  $D_i$ . The tagged domain  $D$ , with the  $I$ -coordinate compared first, is the sum  $\sum_{i \in I} A_i$ . There is one class of pairs and one class comparison relation. No class whose members are the classes  $A_i$  is being formed. Choosing the least occupied index and then the least point of its fiber proves that this sum is well-ordered.

Reassociation and retagging give explicit isomorphisms

$$(A + B) + C \cong A + (B + C), \quad (\text{S35.1})$$

$$(A \cdot B) \cdot C \cong A \cdot (B \cdot C), \quad (\text{S35.2})$$

$$A \cdot (B + C) \cong A \cdot B + A \cdot C. \quad (\text{S35.3})$$

For example, the product reassociation sends  $((a, b), c)$  to  $(a, (b, c))$ ; both comparisons prioritize  $c$ , then  $b$ , then  $a$ . These verifications use only the displayed definitions, so there is no hidden comparison theorem for arbitrary class orders.

The direction in (S35.3) matters. The other distributive law already fails for set ordinals:

$$(1 + 1) \cdot \omega \cong \omega, \quad 1 \cdot \omega + 1 \cdot \omega \cong \omega + \omega.$$

The familiar noncommutativity of ordinal arithmetic persists as well. It is helpful to record the order of the factors before simplifying an expression.

### XVI.3.4 Source 32: ordinal arithmetic with explicit class presentations; uniform sums and products

#### XVI.3.4.1 Uniform sums and products

A family  $(W_i)_{i \in I}$  of class orders will always be *uniformly coded*: there are classes  $D$  and  $R$  such that

$$\text{dom } W_i = \{x : (i, x) \in D\}, \quad x <_{W_i} y \iff (i, x, y) \in R.$$

This is not a class function whose values are proper classes. Its elements are ordinary set tuples, and it is a legitimate object of GBC.

**Proposition XVI.3.4** (Uniform sum theorem). *If  $I$  is a class well-order and  $(W_i)_{i \in I}$  is a uniformly coded family of class well-orders, then*

$$\sum_{i \in I} W_i$$

*is a class well-order on the pairs  $(i, x)$  with  $x \in \text{dom } W_i$ , ordered first by  $i$  and then within the fiber.*

*Proof.* For a nonempty subclass of the sum, its occupied indices form an admitted nonempty subclass of  $I$ . Take the least occupied index and then the least point in that fiber. Both choices use elementary comprehension and the well-order hypotheses.  $\square$



The binary sum  $A + B$  is the two-fiber special case. The ordinal product  $A \cdot B$  is the sum of copies of  $A$  indexed by  $B$ ; on  $\text{dom } A \times \text{dom } B$ , the *second coordinate has priority*. Rearranging finite tags and tuples gives explicit admitted isomorphisms

$$(A + B) + C \cong A + (B + C), \quad (A \cdot B) \cdot C \cong A \cdot (B \cdot C),$$

$$A \cdot (B + C) \cong A \cdot B + A \cdot C.$$

Commutativity and the other distributive law are not asserted. These are ordinal, not cardinal, operations.

[merge] Proposition XVI.3.4 is the sum argument of source 34's section XVI.3 and of source 35's Section 3.1 (in section XVI.3.3), printed as a numbered statement because sources 32 and 33 cite it so; source 33's note XVI.3.9 is the same proposition.

### XVI.3.5 Source 32: canonical division by $\Omega$ and class cofinality

#### XVI.3.5.1 Set-interval blocks

For  $x, y \in W$ , let  $[x, y]_W$  denote the closed interval between them, regardless of which is smaller. Define

$$x \sim_W y \iff [x, y]_W \text{ is a set.} \quad (\text{S32.5})$$

The predicate on the right is elementary: it asserts the existence of a set with exactly the specified members. In particular, it does not quantify over a collection of proper classes.

*Note XVI.3.5* (Source 32, Lemma 5.1). *The relation  $\sim_W$  is a convex equivalence relation. Each block has a least point, and the induced order inside each block is set-like. Every nonfinal block is proper. Consequently there can be at most one set-sized block, and if it exists it is final.* This is the block lemma of Part IX and Part XIV (source 15's theorem 52.3, source 18's lemma 52.10, source 29's note 108.9), proved inside source 34's theorem XVI.3.3 and as source 35's note XVI.5.11 and source 33's note XVI.3.10, by the same argument: transitivity from the union of two set intervals, set-likeness because the predecessors of  $x$  in its block lie in the set interval from the block's least point  $r$ , and, if a nonfinal block  $B$  were a set, the least point  $z$  after it would satisfy  $[r, z]_W = B \cup \{z\}$ , a set.

Least representatives solve the size problem in forming the quotient. Instead of putting proper-class blocks into another class as elements, form the class of their least points. An original point determines its representative by an elementary formula. The resulting quotient is an ordinary class relation on sets. This is the block-representation method already used in the repository [79]; the following division formulation makes its consequence explicit.

**Theorem XVI.3.6** (Canonical  $\Omega$ -division). *For every class well-order  $W$  there are a canonically presented class well-order  $Q$  and a set ordinal  $\alpha$  such that*

$$W \cong \Omega \cdot Q + \alpha. \quad (\text{S32.6})$$

*Here  $Q$  is the order of least representatives of the proper set-interval blocks;  $\alpha$  is the type of the final set block, or 0 when there is none. The pair is unique up to an admitted isomorphism of  $Q$  and equality of  $\alpha$ .*

*Proof.* By note XVI.3.5, every proper block is set-like and therefore has type  $\Omega$  by note XVI.2.9. The rank of a point  $x$  within its block is the ordinal type of the set of block predecessors of  $x$ . The

formula defining this rank is uniform in the block representative and  $x$ . If a final set block exists, apply the ordinary set-order collapse to it and call its type  $\alpha$ .

Assigning each point its block representative and its local rank gives the required admitted isomorphism. For the empty order take  $Q = 0$  and  $\alpha = 0$ . For a nonempty set order there is just one set block, so  $Q = 0$  and  $\alpha = \text{otp}(W)$ .

To verify uniqueness, observe that in any presentation  $\Omega \cdot Q + \alpha$  the copies of  $\Omega$  are exactly the proper set-interval blocks. Points from distinct copies have a proper interval between them, because it contains a final part of the earlier copy of  $\Omega$ . All intervals within a copy are sets, and the final ordinal  $\alpha$ , when nonzero, is the unique set block. An isomorphism preserves set-sized intervals: images of sets under an admitted class function are sets, and the inverse gives the converse. It therefore induces an isomorphism of the proper-block representative orders and an isomorphism of the final set blocks. Ordinary ordinal uniqueness gives equality of the remainders.  $\square$

[merge] Theorem XVI.3.6 is source 34's theorem XVI.3.3 with the uniqueness clause of source 18's proposition 52.12 (Part IX), proved in the same way: an admitted isomorphism and its inverse preserve set-sized intervals by Class Replacement. Source 33's note XVI.3.11 states the same canonicity. In proposition XVI.3.7, the description of the least cofinal size is source 18's corollary 52.13 in other words; the first clause, an increasing cofinal map  $\Omega \rightarrow W$  when no set is cofinal, is proved here only by source 32 (its recursion along  $\text{Ord}$  uses a set-like global well-order).

The theorem is not a global Cantor normal form theorem. In particular,  $Q$  may be another non-set-like proper-class order, and iterating the quotient construction introduces a separate recursion problem. Nevertheless, it provides a canonical first structural invariant for every supplied  $W$ , with no ETR assumption and no global comparison theorem.

### XVI.3.5.2 Cofinality need not measure class size

A subclass  $C \subseteq W$  is cofinal if every point of  $W$  lies at or below some member of  $C$ . Proper-class well-orders can have a last point, a countable cofinal set, or no set cofinal subset at all. For example,

$$\Omega + 1 \text{ has cofinality } 1, \quad \Omega \cdot \omega \text{ has cofinality } \omega,$$

whereas  $\Omega$  has no set cofinal subset. The hereditary order  $E(\Omega)$  has a countable cofinal set by note XVI.6.24.

**Proposition XVI.3.7** (Set cofinality or an  $\Omega$ -sequence). *If a class well-order  $W$  has no set cofinal subset, it has an admitted increasing cofinal map  $\Omega \rightarrow W$ . If it does have a set cofinal subset, its least possible set cardinality is 1 or an infinite regular cardinal.*

*Proof.* In the first case  $W$  is proper. A set-like global well-order restricted to its domain gives an enumeration  $e : \text{Ord} \rightarrow \text{dom } W$ . Recursively choose  $f(\xi)$  to be the  $W$ -least point strictly above  $e(\xi)$  and all earlier values  $f(\eta)$ ,  $\eta < \xi$ . The collection to be bounded is a set. Since no set is cofinal, a strict upper bound exists. This is set-valued recursion with set histories along  $\text{Ord}$ , available in GBC. The result is increasing and is cofinal because it bounds every value of  $e$ .

In the second case, choose a set cofinal subset  $C$ . Its induced order has a set ordinal type  $\lambda$ . If  $W$  has a greatest point, one point is enough. Otherwise the cofinality of  $W$  equals the ordinary cofinality of  $\lambda$ : a cofinal subset of  $\lambda$  gives one in  $W$ , and any set cofinal in  $W$  maps cofinally into  $C$  by taking least upper bounds in  $C$  point by point. Ordinary ordinal cofinality is an infinite regular cardinal.  $\square$

The first clause does not say that  $W$  has order type  $\Omega$ . A cofinal suborder is not necessarily an initial suborder. In  $\Omega^{[\Omega]}$ , for example, the monomials indexed by ordinal exponents are cofinal, although the whole order is much longer than its initial set-like segment.

### XVI.3.6 Source 33: an explicit arithmetic of class well-orders; uniformly indexed sums

Disjoint tags are implicit in all constructions. Define  $A + B$  by placing every point of  $A$  before every point of  $B$ . Define  $A \cdot B$  on pairs  $(b, a)$ , with  $b \in B$ ,  $a \in A$ , by comparing the  $B$ -coordinate first. Thus  $A \cdot B$  consists of  $B$ -many consecutive copies of  $A$ .

*Note XVI.3.8* (Source 33, Definition 3.1, “Uniform family”). *A family  $(A_i)_{i \in I}$  of class order presentations is uniform if there are classes  $U$  and  $S$  with  $\text{dom}(A_i) = \{x : U(i, x)\}$  and  $x <_{A_i} y \leftrightarrow S(i, x, y)$ ; the index collection  $I$  is itself a class of sets, whose members are codes or labels, not proper classes.* This is source 34’s [definition XVI.7.1](#) (Section XVI.7) and source 32’s uniformly coded families before [proposition XVI.3.4](#).

*Note XVI.3.9* (Source 33, Proposition 3.2). *If  $I$  is a class well-order and  $(A_i)_{i \in I}$  is a uniform family of class well-orders, then the lexicographic sum  $\sum_{i \in I} A_i$  is a class well-order, definable from the given data in GBC.* This is source 32’s [proposition XVI.3.4](#), by the same least occupied index and least point of its fibre; the argument uses no choice (source 34’s [section XVI.3](#)).

The statement includes a subtle restriction: merely being told, for each index, that some class order exists does not supply the uniform pair of predicates  $U, S$ . Even a set-indexed collection of class-valued choices must not be treated as an ordinary set function.

### XVI.3.7 Source 33: canonical division by the order of all ordinals

#### XVI.3.7.1 Set intervals and their blocks

For points  $x, y$  of a class linear order  $W$ , let  $[x, y]_W$  denote the closed interval between them, regardless of their order. Define

$$x \sim_W y \iff [x, y]_W \text{ is a set.} \quad (\text{S33.4.1})$$

Being a set is expressed by the existence of a set with exactly the specified members. Thus [\(S33.4.1\)](#) is an elementary definition from  $W$ .

*Note XVI.3.10* (Source 33, Lemma 4.1). *For a class well-order  $W$ , the relation  $\sim_W$  is a convex equivalence relation. Each block has a least point and is a set-like well-order. Any block that is a set must be the final block of  $W$ .* This is source 32’s [note XVI.3.5](#), with the same proof.

#### XVI.3.7.2 The division theorem

*Note XVI.3.11* (Source 33, Theorem 4.2, “Canonical  $\Omega$ -division”). *Every class well-order  $W$  has a decomposition*

$$W \cong \Omega \cdot Q + \alpha, \quad (\text{S33.4.2})$$

*where  $Q$  is a class well-order and  $\alpha$  a set ordinal. The factors are canonical:  $Q$  is the order of the least points of the proper-class set-interval blocks, and  $\alpha$  the type of the final set-sized block, if there is one, or zero. The decomposition and the displayed isomorphism exist in GBC.* This is source 34’s [theorem XVI.3.3](#) and source 32’s [theorem XVI.3.6](#), including its canonicity: source 33 sends  $x$  to the pair  $(r(x), \rho(x))$  of its block minimum and  $\rho(x) = \text{otp}\{y : r(x) \leq_W y <_W x\}$ , and proves canonicity

by observing that in any presentation  $\Omega \cdot Q + \alpha$  the displayed blocks are exactly the set-interval blocks. Its displayed equation (4.2) is this decomposition.

**Example XVI.3.12.** The canonical decompositions of  $\Omega + \omega$  and  $\Omega^2 + 17$  have  $(Q, \alpha) = (1, \omega)$  and  $(\Omega, 17)$ , respectively. A nonempty set ordinal has  $Q = 0$  and is its own remainder. The empty order has both factors zero.

The theorem is a genuine structural statement about arbitrary class well-orders. It does not assume initial comparability of two unrelated orders, nor does it recursively construct classes along  $W$ . Every point's coordinate is the type of a *set* interval.

### XVI.3.7.3 Why repeated quotients need coherent data

One may iterate the operation of identifying set-interval blocks. At a limit stage, however, knowing only the isomorphism types of the previous quotients is insufficient: one needs to know *which points have already been identified*. The next section makes this precise with an explicit family. It also avoids any implication that the existence of individually described quotients automatically supplies a uniformly coded transfinite iteration.

## XVI.4 Finite-support exponentiation without Global Choice

**Definition XVI.4.1** (Finite-support power). Let  $A$  be a nonempty class well-order with least point  $0_A$ , and let  $B$  be a class well-order. The carrier of  $A^{[B]}$  is the class of finite partial functions  $f : B \rightarrow A \setminus \{0_A\}$ . Missing values mean  $0_A$ . For distinct  $f, g$ , take the greatest point  $b$  in their finite disagreement set and put

$$f < g \iff f(b) <_A g(b).$$

Write  $P(B) = \Omega^{[B]}$ . Its elements are formal polynomials  $\Omega^{b_0}a_0 + \cdots + \Omega^{b_{k-1}}a_{k-1}$ , where  $b_0 >_B \cdots >_B b_{k-1}$  and  $0 < a_i \in \text{Ord}$ .

The superscript  $b_i$  is a point of the exponent order; when interpreting the monomial's predecessor length, it represents the initial segment  $B_{<b_i}$ . A polynomial is a finite set, even when both available coordinates and available coefficients range over proper classes. No class-valued function is being made an element of the carrier. The greatest-disagreement rule is a strict linear order by the ordinary finite lexicographic argument.

**Theorem XVI.4.2** (Choice-free power theorem). *GB proves that  $A^{[B]}$  is a class well-order whenever  $A$  and  $B$  are class well-orders and  $A$  is nonempty.*

*Proof.* Let  $C$  be a nonempty subclass of the power. We construct its least member directly. Regard each polynomial as a finite list of exponent-coefficient pairs in decreasing exponent order. For a chosen prefix  $p$ , let  $C_p$  consist of those members of  $C$  whose lists begin with  $p$ . Maintain two invariants:  $C_p$  is nonempty, and every member of  $C \setminus C_p$  is larger than every member of  $C_p$ . The empty prefix satisfies these conditions.

If  $p$  itself is in  $C$ , it is least in  $C_p$ , since each proper extension has a nonzero next coefficient. It is therefore least in  $C$ , and the construction stops. Otherwise each member of  $C_p$  has a next pair. Among the next exponents choose the  $B$ -least one, say  $b$ . Among the corresponding nonzero coefficients choose the  $A$ -least one, say  $a$ . These are uniquely defined minima of admitted subclasses. Append  $(b, a)$  to  $p$ . The new  $C_p$  is nonempty. A member rejected at this step has a larger next exponent, or the same exponent with larger coefficient; in either case it is larger than every retained member. Thus both invariants persist.

If this process did not stop, ordinary recursion on  $\omega$  would produce a sequence of uniquely defined finite prefixes. All values are sets, so Class Replacement supplies the sequence as a set. Its successive selected exponents strictly decrease in  $B$ , because every retained polynomial lists its support in decreasing order. The range of this sequence would be a nonempty subclass of  $B$  without a least member, contrary to the hypothesis. Hence the process stops and yields the required least member of  $C$ . No choices from unrelated subclasses are made.  $\square$

The difference from the repository proof is exact. A general subclass without a minimum cannot in bare GB simply be turned into a descending sequence by invoking Global Choice. The proof above uses the already supplied well-orders of the coordinates and digits to choose unique minima. This is the dependency removed in the canonical-history theorem.

There is also an independent short route to the same result. For a nonempty set of finite graphs, collect its exponent supports and its digits into sets, including the zero digit. Their induced well-orders have set ordinal types. The given graphs therefore sit inside an ordinary set ordinal power, so that set of graphs has a minimum. [Lemma XVI.2.2](#) upgrades this set-subset assertion to the full class well-order assertion in GB. The explicit greedy proof records a minimum-selection procedure and exposes the same absence of choice directly.

#### XVI.4.1 Arithmetic laws with concrete witnesses

The finite-support construction agrees with ordinary ordinal exponentiation on set ordinals. For a successor exponent, the last coefficient arranges copies of the preceding power; at a limit, every finite support lies below some earlier exponent. These observations give the usual set transfinite proof.

For class well-orders, explicit splitting and flattening of finite graphs prove the following laws without class-valued recursion:

$$A^{[0]} \cong 1, \quad A^{[1]} \cong A, \quad (\text{S34.1})$$

$$A^{[(B+C)]} \cong A^{[B]} \cdot A^{[C]}, \quad (\text{S34.2})$$

$$(A^{[B]})^{[C]} \cong A^{[(B \cdot C)]}. \quad (\text{S34.3})$$

In (S34.2), restrict a finite function to the two tagged pieces; the  $C$  piece is the more significant coordinate. In (S34.3), a finitely supported function with finitely supported values flattens to a finite function on pairs  $(c, b)$ . The greatest differing  $c$ , followed by the greatest differing  $b$ , is exactly the product order. Empty inner functions represent the zero digit. The inverse maps group the finite pairs into their fibers, so these are isomorphisms, not just embeddings.

An initial embedding  $j : B \rightarrow C$  induces an initial embedding  $A^{[B]} \rightarrow A^{[C]}$  by relabelling supports. If  $A$  has at least two points and the image of  $j$  is proper, the latter embedding is proper: the first missing exponent supports a new coefficient-one monomial. This gives a reliable strict-growth rule.

**Proposition XVI.4.3** (The exponent is not a power class). *For every set ordinal  $2 \leq a$ , one has*

$$a^{[\Omega]} \cong \Omega, \quad a^{[(\Omega+1)]} \cong \Omega \cdot a.$$

*By contrast,  $P(\Omega)$  has a proper initial segment of type  $\Omega$  and is strictly longer than it.*

*Proof.* For set  $a$ , any polynomial has a greatest support exponent  $\beta$  unless it is zero. All smaller polynomials have support in  $\beta + 1$  and digits below  $a$ . They therefore form a set. The whole power

is proper, since there are monomials at every ordinal exponent. [Theorem XVI.3.1](#) gives its type  $\Omega$ . Equation (S34.2) gives the second formula. For base  $\Omega$ , the constant polynomials form an initial copy of  $\Omega$ , and the monomial at exponent 1 follows all of them.  $\square$

Thus  $2^{[\Omega]}$  and  $\omega^{[\Omega]}$  do not leap beyond  $\Omega$  under this definition. A notation that presents them as larger has either changed the exponentiation operation or confused order exponentiation with a collection of subsets. Full functions from a proper class are not set-coded elements of an ordinary class; functions of unrestricted support require a different foundation and a separately checked order.

## XVI.4.2 A presentation language and a useful normal-form rule

A safe presentation language begins with  $0, 1$ , set ordinal constants, and  $\Omega$ , and closes under tagged sum, product, finite-support power, and explicitly supplied uniform sums. It can also include the hereditary constructor in the next development. Each expression denotes a concrete class relation. Equalities between expressions are witnessed by class isomorphisms, and the language need not choose a representative of every possible isomorphism type.

For  $f \in P(B)$  with decreasing support  $b_0, \dots, b_{k-1}$ , the strict predecessor class has the canonical presentation

$$(P(B))_{<f} \cong \sum_{i < k} \Omega^{[B_{<b_i}]} \cdot f(b_i). \quad (\text{S34.4})$$

Indeed, partition the predecessors by the first, most significant coordinate at which they fall below  $f$ . At coordinate  $b_i$  the smaller coefficient has  $f(b_i)$  choices in ordinal order, and the lower coordinates form  $\Omega^{[B_{<b_i}]}$ . Concatenating these pieces in decreasing significance proves the formula. It explains exactly what the polynomial notation means as a description of an initial segment.

For example, a point written  $\Omega^{\omega+2} \cdot 3 + \Omega^7 \cdot \omega + 5$  names the initial segment with those three successive terms. It does not claim that the ambient proper class order is a von Neumann ordinal. Constants and finite trees give systematic descriptions, but the comparison of arbitrary formula-defined presentations is not replaced by a universal syntactic normalizer. The diagonal results below show why that stronger objective has a real obstruction.

[merge] [Theorem XVI.4.2](#) removes the one use of global choice in this report's proofs that polynomial orders are well-ordered: source 15's [lemma 53.2](#), source 23's [note 108.6](#) and source 29's [note 108.15](#) each turn a subclass without a least element into a descending sequence by global choice. Source 35 proves the same theorem with a different termination argument ([note XVI.4.4](#) and its proof below); sources 32 and 33 prove it in GBC by the set-support localization of source 34's second route ([notes XVI.4.10](#) and [XVI.4.12](#)). The laws (S34.1)–(S34.3) are source 35's (S35.5) and (S35.6), source 32's [note XVI.4.11](#) and source 33's [note XVI.4.13](#); the initial-embedding rule is source 35's [lemma XVI.4.5](#); the first identity of [proposition XVI.4.3](#) is the third of source 35's [proposition XVI.4.6](#) and source 33's [proposition XVI.4.14](#). Formula (S34.4) is source 35's [note XVI.6.22](#).

## XVI.4.3 Source 35: exponentiation by finite support, cofinality and continuity

Let  $A$  be a nonempty class well-order, with least point  $0_A$ , and let  $B$  be any class well-order. Define  $A^{[B]}$  to be the class of *finite partial functions*

$$p : B \rightarrow A \setminus \{0_A\}.$$

Every such function is a set: its graph contains finitely many ordered pairs of sets. Missing values are read as  $0_A$ . Its finite domain is written  $\text{supp}(p)$ . For distinct  $p, q$ , the finite nonempty set

$$\{b \in \text{supp}(p) \cup \text{supp}(q) : p(b) \neq q(b)\}$$



has a greatest element  $b_*$ . Put

$$p < q \iff p(b_*) <_A q(b_*). \quad (\text{S35.4})$$

The empty graph is the least element. The brackets emphasize this order-theoretic exponentiation; it is not the collection of all class functions  $B \rightarrow A$ , or cardinal exponentiation.

*Note XVI.4.4* (Source 35, Theorem 3.1, “Choice-free finite-support theorem”). *If  $A$  is a nonempty class well-order and  $B$  is a class well-order, then  $A^{[B]}$  is a class well-order. This is provable in GB.* This is source 34’s [theorem XVI.4.2](#), proved independently. Both proofs build the least member of a nonempty subclass  $C$  greedily, appending at each step the least next exponent and then the least next coefficient, and both keep every discarded polynomial above every retained one. They differ in the termination argument: source 34 collects the uniquely specified prefixes into a set sequence by recursion on  $\omega$  and Class Replacement, and derives a descending sequence of exponents; source 35 avoids the  $\omega$ -sequence and considers the class of last exponents of all finite histories obeying the rule, whose least element would have to be followed by a smaller one. Its proof is printed as a second route.

*Source 35’s proof.* The comparison is linear. One way to check transitivity is to put the supports of three polynomials into one finite decreasing list of exponents and compare their coefficient lists lexicographically, starting at the greatest exponent. This is an ordinary finite lexicographic comparison.

Let  $C \subseteq A^{[B]}$  be a nonempty class. Write each polynomial as its finite list of nonzero terms, with exponents strictly decreasing. We construct the beginning of a least member of  $C$ . Start with the empty prefix  $s$ . At every step the survivor class

$$C_s = \{p \in C : p \text{ begins with the list } s\}$$

is nonempty. If  $s$  itself is a polynomial in  $C_s$ , stop. It is the least survivor: every proper extension adds a positive coefficient at a smaller exponent.

Otherwise every survivor has a next term. The class of their next exponents is a nonempty subclass of  $B$ , so choose its *least* element  $b$ . Among survivors having next exponent  $b$ , choose the least next coefficient  $a$  in  $A$ . Append  $(b, a)$  to  $s$ . The new survivor class is nonempty. Moreover, each discarded survivor is greater than every retained survivor: at the first new place of disagreement it has either a larger exponent, or the same exponent and a larger coefficient.

This process must stop after finitely many steps. To see this without making an infinite sequence of choices, consider all *finite histories* obeying the rule. They are set codes, and the condition of being such a history is an elementary formula with the supplied classes as parameters. If no history terminated, the class of last exponents of nonempty histories would be a nonempty subclass of  $B$ . Let  $b$  be its least element and take a history ending in  $b$ . Its next extension exists by the rule, and its new exponent is strictly below  $b$ , since exponents in a polynomial decrease. This contradicts the choice of  $b$ .

At termination, the resulting finite list is a member of  $C$  and is below every polynomial discarded at an earlier step. It is therefore the least member of  $C$ . All selections used designated least elements; Global Choice and elementary transfinite recursion were not used.  $\square$

For set ordinals  $\alpha \geq 1$  and  $\beta$ , this construction has ordinary ordinal type  $\alpha^\beta$ . Indeed, a last exponent splits the order into  $\alpha$  successive copies of the preceding power, and at a nonzero limit every finite support is contained in an earlier exponent segment. These are the usual successor and limit equations, starting with  $\alpha^{[0]} = 1$ . Consequently the construction genuinely extends ordinal exponentiation. The empty base can be treated separately by the usual conventions  $0^0 = 1$  and  $0^\beta = 0$  for  $\beta > 0$ ; it is not needed in the results below.

**XVI.4.3.1 The exponent laws are explicit isomorphisms**

For nonempty  $A$ , splitting a finite support between the two tagged parts of  $B + C$  gives

$$A^{[B+C]} \cong A^{[B]} \cdot A^{[C]}. \quad (\text{S35.5})$$

The right-hand coordinate records the restriction to  $C$ . A difference there is more significant than every difference in  $B$ , exactly as required by the product convention. Joining the two tagged graphs is the inverse map. In particular,

$$A^{[0]} \cong 1, \quad A^{[1]} \cong A, \quad A^{[B+1]} \cong A^{[B]} \cdot A.$$

There is also a flattening isomorphism

$$(A^{[B]})^{[C]} \cong A^{[B \cdot C]}. \quad (\text{S35.6})$$

An element on the left assigns a nonzero finite  $B$ -polynomial  $p_c$  to finitely many  $c \in C$ . Send it to the finite graph on  $B \times C$  whose value at  $(b, c)$  is  $p_c(b)$ . Conversely, slice a finite graph on  $B \times C$  at its finitely many occupied  $C$ -coordinates. Empty slices are omitted. The greatest different  $C$ -coordinate determines the first comparison; inside that slice, the greatest different  $B$ -coordinate determines the second. This is precisely greatest-difference comparison on  $B \cdot C$ . Thus both the bijection and its compatibility with the order are verified directly.

These identities are available even when several operands and their initial segments are proper classes. They require no recursion whose stages would themselves have to be classes. They also do not assert the generally false identity  $(A \cdot B)^{[C]} \cong A^{[C]} \cdot B^{[C]}$ .

**Lemma XVI.4.5** (Preservation of initial embeddings). *An order embedding  $f : B \rightarrow C$  induces an order embedding*

$$A^{[f]} : A^{[B]} \longrightarrow A^{[C]}, \quad A^{[f]}(p)(f(b)) = p(b).$$

*If the image of  $f$  is initial, the induced image is initial. If that image is proper and  $A$  has at least two elements, the induced image is proper as well.*

*Proof.* The map transports a finite support and its coefficients; an embedding preserves the greatest point of a finite difference set. Hence it preserves and reflects comparisons. Suppose the image of  $f$  is initial. Every exponent outside it is above every exponent inside it. A polynomial with an occupied outside exponent is therefore greater than every polynomial supported in the image. The latter polynomials consequently form an initial subclass, and each has a unique finite inverse image under the displayed map. If the exponent image is proper, let  $c_*$  be its least omitted point and let  $1_A$  be the least point of  $A$  above  $0_A$ . The monomial supported at  $c_*$  with coefficient  $1_A$  is the least polynomial outside the induced image. This also proves properness explicitly.  $\square$

The polynomial operation used in the repository is exactly

$$P(B) := \Omega^{[B]}.$$

Its coefficients are the nonzero set ordinals. The lemma shows that  $P$  transports initial embeddings, including their properness. Identity maps and compositions are preserved literally by support transport. This functorial behavior is what makes the fixed-point constructions in the next part possible.

Changing the *base* deserves separate attention. An initial embedding  $A \rightarrow A'$  also gives a coefficientwise order embedding  $A^{[B]} \rightarrow (A')^{[B]}$ , but that particular map need not be initial. Already

for  $2 \subset 3$  and exponent 2, the image contains the monomial at exponent 1 while omitting the constant coefficient 2 below it. One must not infer initiality from coefficientwise inclusion. For set ordinals another initial embedding exists by ordinary ordinal comparison; applying an analogous comparison to arbitrary class orders requires its own justification.

### XVI.4.3.2 What collapses to Ord, and what passes it

A set-like well-order on a proper class has its canonical ordinal-rank isomorphism with  $\Omega$ : each point has a set of predecessors, its rank is a set ordinal, and the range is an initial subclass of Ord that is a proper class. It must therefore be all of Ord. This observation gives useful simplification rules.

**Proposition XVI.4.6** (Set-sized operands at the class boundary). *For set ordinals  $\alpha$  and  $\beta$ ,*

$$\alpha + \Omega \cong \Omega, \quad \alpha \cdot \Omega \cong \Omega \quad (\alpha > 0), \quad \beta^{[\Omega]} \cong \Omega \quad (\beta \geq 2).$$

*Proof.* In  $\alpha + \Omega$ , the predecessor set of a point in the second copy has ordinary type  $\alpha + \gamma$  for some set ordinal  $\gamma$ . In  $\alpha \cdot \Omega$ , the predecessors of  $(a, \gamma)$  have ordinary type  $\alpha \cdot \gamma + a$ . Thus both displayed orders are set-like, and their domains are proper classes under the stated nonemptiness assumption.

For  $\beta^{[\Omega]}$ , a polynomial and all its predecessors have their supports in some set ordinal  $\gamma$ . There are only set-many finite  $\beta$ -valued graphs on  $\gamma$ , so its predecessor class is a set. The monomials with coefficient 1 at successive ordinal exponents show that the whole domain is proper. The same set-like classification applies. More explicitly, the power on exponents below  $\gamma$  has type  $\beta^\gamma$ , and these ordinary ordinals are cofinal in Ord. The resulting identification agrees with ordinary base- $\beta$  evaluation on each finite polynomial.  $\square$

[merge] The third identity is source 34's [proposition XVI.4.3](#); the first two are source 32's remark after [note XVI.4.11](#) and source 33's examples after [note XVI.4.13](#). Source 35's continuity principle (S35.7) is the last clause of source 32's [note XVI.4.11](#). [Proposition XVI.4.7](#) and [lemma XVI.4.8](#) are proved here only by source 35; the second holds over GB as well ([theorem XVI.6.21](#)).

The reversed expressions behave differently. The new endpoint of  $\Omega + 1$  has a proper class of predecessors, and the first point of the second block of  $\Omega \cdot 2$  has the same property. Neither order is isomorphic to  $\Omega$ . Their displayed first copies of  $\Omega$  are proper initial subclasses, so the sense in which they are longer is witnessed directly.

In  $\Omega^{[\Omega]}$ , the constant polynomials form an initial copy of  $\Omega$ . The monomial with coefficient 1 at exponent 1 lies immediately after that whole copy. Moreover,

$$\Omega^{[2]} \cong \Omega \cdot \Omega$$

is a proper initial subclass of  $\Omega^{[\Omega]}$ , by [Lemma XVI.4.5](#). In particular,

$$\beta^{[\Omega]} \cong \Omega \quad \text{for every set } \beta \geq 2, \quad \Omega^{[\Omega]} \not\cong \Omega.$$

The size of the coefficient collection changes the construction: allowing all set ordinals as coefficients introduces proper-class intervals. There is no legitimate inference that a proper-class base can be replaced by a sufficiently large set base for the *whole* order.

### XVI.4.3.3 Cofinality is a separate invariant

A long class well-order may still have a set of cofinal points. The exponent construction gives exact elementary examples of both possibilities.

**Proposition XVI.4.7** (Cofinality of a limit exponent). *Suppose  $A$  has at least two elements and  $B$  is nonempty with no greatest point. Then  $A^{[B]}$  has a set-sized cofinal subclass if and only if  $B$  does. Whenever these cofinalities are set cardinals, their least possible cardinalities agree.*

*Proof.* Let  $1_A$  be the least nonzero coefficient. The map sending  $b \in B$  to the monomial with coefficient  $1_A$  at exponent  $b$  is strictly increasing and cofinal. Indeed, choose an exponent above the greatest exponent of any given nonzero polynomial; its monomial is larger. A cofinal set in  $B$  therefore gives a cofinal set of the same size in the power.

Conversely, let  $Q$  be a cofinal set of polynomials. Take the greatest support exponent of each nonzero member of  $Q$ . These exponents form a set by Class Replacement. They are cofinal in  $B$ : if all lay below some  $b$ , the monomial at  $b$  would exceed every member of  $Q$ . Their number is at most  $|Q|$ . The two constructions prove both assertions.  $\square$

In particular,  $\Omega^{[\Omega]}$  has no set-sized cofinal subclass, whereas  $\Omega^{[\omega]}$  has cofinality  $\omega$ . The latter is nevertheless longer than every finite power  $\Omega^{[n]}$ : these powers are proper initial subclasses, and the monomials at the finite exponents give a cofinal sequence. If the exponent instead has a greatest point,  $\Omega^{[B]}$  has no cofinal set. For any proposed set of polynomials, their coefficients at that point have a common ordinal bound; a larger coefficient there produces a polynomial above the entire set. Thus passing from  $\Omega^{[\omega]}$  to  $\Omega^{[\omega+1]}$  changes the availability of a set-sized cofinal sequence. Cofinality and initial-segment length measure different features.

#### XVI.4.3.4 Continuity and bounded interpretations

Continuity is meaningful for a given uniform system. Suppose  $(B_i)_{i \in I}$  is uniformly presented,  $I$  is a nonempty class well-order, the domains are nested, and each earlier order is an initial subclass of every later one. Then

$$B = \bigcup_{i \in I} B_i, \quad A^{[B]} = \bigcup_{i \in I} A^{[B_i]}. \quad (\text{S35.7})$$

The first union is well-ordered: given a nonempty subclass, take one of its points in a stage and take its least point inside that stage. All points outside the stage are later. For the second equality, the finitely many exponents of a polynomial lie together in a single stage. This proves continuity directly from finite support.

In particular, if  $B$  is nonempty and has no greatest point,

$$A^{[B]} = \bigcup_{b \in B} A^{[B_{<b}]}$$

Some of the orders  $B_{<b}$  may be proper classes; nevertheless this family is elementary and uniform. These assertions do not supply an unspecified family of class stages, or a supremum of every class well-order. The existence and coding of the diagram precede the union.

There is another useful verification principle when a notation system consists of finite trees carrying ordinal labels.

**Lemma XVI.4.8** (Bounded-specialization principle). *Work in GBC. Let  $T$  be an elementary class of finite ordinal-labelled terms with a class linear order. Suppose that, for arbitrarily large set ordinals  $\kappa$ , all terms whose labels are below  $\kappa$  admit a common strictly order-preserving interpretation into a set ordinal. Then  $T$  is a class well-order.*

*Proof.* If a nonempty subclass had no least element, Global Choice would produce a descending  $\omega$ -sequence in it. Its terms use only a set of ordinal labels, whose union is bounded by some set ordinal. Choose a larger  $\kappa$  with the asserted interpretation. The entire sequence then maps to a descending sequence in one set ordinal, a contradiction.  $\square$

For nested ordinal-coefficient polynomials, the relevant interpretation substitutes an ordinary ordinal  $\kappa$  for the formal symbol  $\Omega$ , with every coefficient below  $\kappa$ . Ordinary base- $\kappa$  comparison verifies order preservation on the bounded terms. This principle explains how set-ordinal mathematics can verify a class notation system locally without collapsing its global order type to a set ordinal. For the finite-support operation itself, [Note XVI.4.4](#) is stronger in its axiomatic economy, since its direct least-element argument already works without Global Choice.

All comparisons made in this section have explicit witnesses. No step assumes that arbitrary class well-orders are comparable by initial embeddings. That additional issue belongs to the foundational analysis, rather than to the elementary support calculations.

#### XVI.4.4 Source 32: finite-support exponentiation and familiar expressions

Let  $A$  have a least point  $0_A$  and at least two points. Define  $A^{[B]}$  to be the class of finite partial functions

$$f : \text{dom } B \rightarrow \text{dom } A \setminus \{0_A\},$$

read as total functions with default value  $0_A$ . For distinct  $f, g$ , their finite set of differing coordinates has a greatest point  $b$ . Declare

$$f < g \iff f(b) <_A g(b). \quad (\text{S32.1})$$

The empty function is the zero point. Only finitely many nonzero digits are stored, so every element is a set even when both parameters have proper-class domains.

**Lemma XVI.4.9** (Set-sized digit representation). *For ordinary ordinals  $a \geq 2$  and  $b$ , finite-support functions  $b \rightarrow a$  with the order (S32.1) have order type  $a^b$  in ordinary ordinal arithmetic.*

*Proof.* Use ordinary transfinite induction on the set ordinal  $b$ . At zero there is one function. At  $b = c + 1$ , grouping by the digit at  $c$  gives  $a$  successive copies of the order at  $c$ , hence type  $a^c \cdot a$ . At a nonzero limit, every finite support lies below some  $c < b$ , and these smaller orders are nested initial segments. Their union has type  $\sup_{c < b} a^c = a^b$ . This also proves that the digit order is a well-order and that the displayed embeddings at smaller exponent bounds are initial. The same induction identifies the rank of a digit function with the finite sum  $\sum a^{\beta_i} d_i$  in decreasing order of its occupied exponents.  $\square$

*Note XVI.4.10* (Source 32, Theorem 3.3, “Explicit class exponentiation in GBC”). *For class well-orders  $A$  and  $B$ , with  $A$  having at least two points, the finite-support presentation  $A^{[B]}$  is a class well-order. Its construction and proof require no ETR axiom.* This is source 34’s [theorem XVI.4.2](#), in GBC. Source 32’s proof is source 34’s second route with descending sequences in place of set subsets: a descending set sequence uses a set of exponents and a set of digits, whose collapses to ordinals translate it into a descending sequence in an ordinary ordinal power ([lemma XVI.4.9](#)), and [lemma XVI.2.8](#) converts this into class well-foundedness. That conversion is where global choice enters; over GB, [lemma XVI.2.2](#) replaces it.

*Source 32’s proof.* Linearity and transitivity can be checked on the finite union of the supports of any three functions, using the corresponding finite lexicographic comparison. Suppose there were a descending set sequence of functions. The union of its finite supports is a set  $S \subseteq \text{dom } B$ , and the set of all its coefficient values together with  $0_A$  is a set  $T \subseteq \text{dom } A$ . The induced well-orders on  $S$  and  $T$  have ordinary ordinal types  $b$  and  $a$ . If  $T$  contains only zero, there cannot be a strict descent. Otherwise  $a \geq 2$ , and replacing coordinates and digits by their ranks translates the sequence into a descending sequence in the set well-order from [lemma XVI.4.9](#). This is impossible. Apply [lemma XVI.2.8](#).  $\square$

The same *set-support localization* argument is useful more generally: a purported countable obstruction to a finite-coded class construction often uses only a set of labels. One can then reduce the obstruction to ordinary set mathematics without collapsing the entire proper-class order.

*Note XVI.4.11* (Source 32, Theorem 3.4, “Exponent laws by explicit maps”). *For the preceding presentations,  $A^{[0]} \cong 1$ ,  $A^{[1]} \cong A$ ,*

$$A^{[B+C]} \cong A^{[B]} \cdot A^{[C]}, \quad A^{[B \cdot C]} \cong (A^{[B]})^{[C]}. \quad (\text{S32.2})$$

*If  $j : B \rightarrow C$  is an initial embedding, pushing supports forward along  $j$  gives an initial embedding  $A^{[B]} \rightarrow A^{[C]}$ , proper when  $j$  is proper. If  $B$  has no greatest point,  $A^{[B]}$  is the union of its initial suborders supported in  $B_{<b}$ , for  $b \in B$ . These are source 34’s (S34.1)–(S34.3) and the initial-embedding paragraph after them, source 35’s lemma XVI.4.5 and (S35.7), and source 33’s note XVI.4.13, proved by the same splitting, grouping and support transport; source 32 assumes  $A$  has at least two points.*

We deliberately avoid the assertion that initial embeddings in the *base* always induce initial embeddings of powers. Even for ordinary finite ordinals that assertion is false: the digit subset for  $2^2$  inside the base-3 presentation of  $3^2$  is not an initial segment.

#### XVI.4.4.1 Which familiar expressions really get longer?

For every set ordinal  $\alpha$  and every nonzero set ordinal  $a$ ,

$$\alpha + \Omega \cong \Omega, \quad a \cdot \Omega \cong \Omega.$$

Indeed, the orders on the left are proper and set-like, so note XVI.2.9 applies. By contrast,  $\Omega + 1$  is not set-like, and  $\Omega \cdot 2$  consists of two genuinely successive proper-class blocks.

For every set ordinal  $a \geq 2$ ,

$$a^{[\Omega]} \cong \Omega. \quad (\text{S32.3})$$

To see set-likeness, a predecessor of a given finite-support function can use no exponent above its greatest occupied exponent. The available exponents are therefore bounded by a set ordinal, and the coefficient alphabet is a set. There are only a set of possible predecessors. The order is proper because there is a distinct monomial at each ordinal exponent. Again note XVI.2.9 applies.

This absorption fails for a proper-class base. In  $\Omega^{[\Omega]}$ , the monomial at exponent 1 has all ordinal constants as predecessors. Thus  $\Omega^{[\Omega]}$  is not set-like. The constructions provide, among many others,

$$\Omega, \quad \Omega + 1, \quad \Omega \cdot 2, \quad \Omega^2, \quad \Omega^{[\Omega]}, \quad \Omega^{[\Omega^{[\Omega]}]}, \quad \dots$$

The displayed tower steps are connected by explicit proper initial embeddings. No universal comparison theorem is needed to certify their increase.

#### XVI.4.5 Source 33: finite-support exponentiation and first landmarks beyond Ord

Suppose  $A$  is a class well-order with least point  $0_A$ . Define  $A^{[B]}$  to be the class of finite partial functions

$$p : B \rightarrow A \setminus \{0_A\}.$$

Missing values are read as  $0_A$ . The empty function is zero. For distinct  $p, q$ , let  $b$  be the *greatest* exponent at which they differ and set

$$p < q \iff p(b) <_A q(b).$$



The greatest differing exponent exists because the union of the two supports is finite. This is reverse lexicographic comparison on the exponent positions, not comparison at the first difference.

*Note XVI.4.12* (Source 33, Theorem 3.3, “Arithmetic closure”). *If  $A, B$  are class well-orders and  $A$  is nonempty, then the finite-support presentation  $A^{[B]}$  is a class well-order. For set ordinal bases and exponents, it is the usual ordinal power. No elementary transfinite recursion principle is needed for its construction or well-foundedness.* This is source 34’s [theorem XVI.4.2](#) in GBC, proved as source 32’s [note XVI.4.10](#): a descending set sequence uses a set  $S$  of exponents and a set  $C$  of coefficients, whose collapses to ordinals  $\beta$  and  $\alpha$  send it to a descending sequence in the finite-support presentation of  $\alpha^\beta$ ; if  $C$  contains only zero all the polynomials are equal. The last assertion is source 32’s [lemma XVI.4.9](#). Source 33 adds that for a one-point base the construction has one point and that zero bases follow the conventions  $0^0 = 1$ ,  $0^B = 0$  for nonempty  $B$ .

For a one-point base, the construction always has one point. We use the usual separate conventions  $0^0 = 1$  and  $0^{[B]} = 0$  for nonempty  $B$  when zero bases occur. All substantial uses below have base at least two.

*Note XVI.4.13* (Source 33, Proposition 3.4, “Explicit arithmetic laws”). *The following isomorphisms are supplied by definable coordinate maps:  $(A + B) + C \cong A + (B + C)$ ,  $(A \cdot B) \cdot C \cong A \cdot (B \cdot C)$ ,  $A \cdot (B + C) \cong A \cdot B + A \cdot C$ , and, for nonempty  $A$ ,  $A^{[B+C]} \cong A^{[B]} \cdot A^{[C]}$  and  $(A^{[B]})^{[C]} \cong A^{[B \cdot C]}$ .* These are the identities of source 34’s [section XVI.3](#) and (S34.2) and (S34.3), source 35’s (S35.3), (S35.5) and (S35.6) and source 32’s [note XVI.4.11](#), by the same retagging, splitting and flattening.

There is no general commutativity and no general distributive law on the other side. For instance,  $1 + \Omega \cong \Omega$  but  $\Omega + 1 \not\cong \Omega$ . Also  $(1 + 1) \cdot \Omega \cong \Omega$ , whereas  $1 \cdot \Omega + 1 \cdot \Omega \cong \Omega \cdot 2$ .

#### XVI.4.5.1 First landmarks beyond Ord

**Proposition XVI.4.14.** *For every set ordinal  $a \geq 2$ ,*

$$a^{[\Omega]} \cong \Omega.$$

*Consequently,*

$$2^{[\Omega+1]} \cong \Omega \cdot 2, \quad 2^{[\Omega \cdot 2]} \cong \Omega^2.$$

*Proof.* A polynomial below a given element of  $a^{[\Omega]}$  cannot use an exponent greater than the largest exponent of that element. All such polynomials therefore belong to a set of finite functions from one bounded ordinal into the set  $a$ . Hence  $a^{[\Omega]}$  is set-like. It is a proper class because every ordinal can occur as the exponent of a monomial with coefficient one. Apply [note XVI.2.14](#). The remaining identities follow from [note XVI.4.13](#).  $\square$

[merge] The first identity of [proposition XVI.4.14](#) is source 34’s [proposition XVI.4.3](#) and source 35’s [proposition XVI.4.6](#); the two consequences  $2^{[\Omega+1]} \cong \Omega \cdot 2$  and  $2^{[\Omega \cdot 2]} \cong \Omega^2$  are printed only by source 33 (the first is the case  $a = 2$  of source 34’s  $a^{[\Omega+1]} \cong \Omega \cdot a$ ).

Thus the symbols  $\Omega + 1$ ,  $\Omega \cdot 2$ ,  $\Omega^2$ ,  $\Omega^\omega$ , and  $\Omega^{[\Omega]}$  admit completely explicit meanings. Later we construct a coherent fixed point above finite towers of powers of  $\Omega$ . None of these expressions turns  $\Omega$  into a set ordinal.

*Remark XVI.4.15* (Not cardinal exponentiation). The class  $A^{[B]}$  just defined contains *finite* function codes. It is not the collection of all class functions  $B \rightarrow A$ . The latter would generally be a higher-order collection, not a class of sets in GBC. Likewise, arbitrary infinite-support reverse lexicographic orders need not be well-founded; a support convention is part of a definition, not a cosmetic choice.

## XVI.5 The canonical-history theorem over GB

The repository already proves a substantial theorem over GBC: canonical terminating condensation histories, strong comparability of class well-orders, and certain initial-segment polynomial representations are equivalent [77]. Its proof constructs canonical floor maps and finite normal forms. We retain that construction and its attribution. The contribution here is a sharper foundation: the choice-free polynomial well-order theorem proved above removes the only use of Global Choice in the arbitrary-carrier argument. Consequently the equivalence holds over GB, including when the set universe does not satisfy Choice. This answers the repository's remaining question about the GB version for the precise limit convention below. It does not establish the converse from comparability to ETR discussed by Hamkins and Woodin [27].

Throughout this section a class well-order means that every nonempty admitted subclass has a least element. We do not substitute the potentially weaker absence of a set-coded descending  $\omega$ -sequence over a choiceless base. All constructions use elementary comprehension and Class Replacement. In particular, a supplied history is a single class relation, not a collection whose members are classes.

### XVI.5.1 Histories and their termination bound

Let  $W = (X, <_W)$  be a nonempty class well-order with minimum  $m$ . For a nonempty subclass  $A \subseteq X$  containing  $m$ , let  $D_W(A)$  be the class of least representatives for the equivalence relation

$$x \sim_A y \iff [\min_W(x, y), \max_W(x, y)]_W \cap A \text{ is a set.}$$

The preceding block decomposition applies to the induced order on  $A$ . In particular, a block followed by another block has type Ord.

Let  $\Gamma = (I, <_\Gamma)$  be another nonempty class well-order. Form  $\Gamma + 1$  with a tagged new final stage  $*$ . Write  $i^+$  for the immediate successor of  $i \in I$  in this extended order. A nonzero limit stage is a noninitial stage without an immediate predecessor; it may have a proper class of predecessors.

**Definition XVI.5.1** (Canonical history). A canonical history of  $W$  on  $\Gamma + 1$  is one class relation  $H$  with slices

$$H_{\min \Gamma} = X, \quad H_{i^+} = D_W(H_i), \quad H_\lambda = \bigcap_{i < \lambda} H_i \quad \text{at nonzero limits } \lambda.$$

It is terminating if  $H_* = \{m\}$ . The principle CTH asserts that every nonempty class well-order  $W$  has a canonical history on its own input-order schedule  $W + 1$ .

The displayed intersection is elementary: membership says that every earlier set-valued stage label satisfies the appropriate slice predicate. Elementary induction along the supplied schedule shows that the slices are decreasing and contain  $m$ . The history's existence, however, is not supplied by that induction.

**Lemma XVI.5.2** (Termination on the input schedule). *In GB, any supplied canonical history on  $W + 1$  terminates. More precisely,*

$$H_x \subseteq \{m\} \cup \{y \in X : x \leq_W y\} \quad (x \in X).$$

*Proof.* Induct along  $W$ , using the given history as a class parameter. The first-stage claim is immediate. At the successor  $x = t^+$ , the inductive bound has removed all points below  $t$  except  $m$ .

If  $t \neq m$  remains, it is the second point of  $H_t$ , so its two-point interval with  $m$  causes its deletion by  $D_W$ . This proves the successor bound. At a nonzero limit  $x$ , each  $m \neq y < x$  lies below some  $t < x$ , and is absent from  $H_t$  by induction.

If  $W$  has a last point  $\ell$ , its slice is contained in  $\{m, \ell\}$ ; the final successor leaves only  $m$ . Otherwise every nonminimal point is excluded by some later stage, hence from the final intersection.  $\square$

This is the repository's supplied-history termination argument [77]. It is consistent with  $W$  being isomorphic to its first condensation: the slices are actual nested subclasses, so repeated abstract order types do not prevent the specified limit intersection from shrinking.

## XVI.5.2 Canonical floors and finite addresses

**Lemma XVI.5.3** (Floors at all stages). *For each stage  $i$  and point  $x \in X$ , there is a greatest retained point at most  $x$ :*

$$p_i(x) = \max_W \{a \in H_i : a \leq_W x\}.$$

*The maps are uniformly elementary in  $H, W$ , nondecreasing in  $x$ , and for  $i \leq j$  satisfy*

$$p_j(x) \leq_W p_i(x), \quad p_j(p_i(x)) = p_j(x), \quad p_i(p_j(x)) = p_j(x).$$

*At every nonzero limit  $\lambda$ , the floors of each fixed  $x$  stabilize before  $\lambda$  at  $p_\lambda(x)$ .*

*Proof.* At the first stage the floor is  $x$ . Suppose  $p_i(x) = a$ , and let  $b$  be the least representative of  $a$ 's block in  $H_i$ . A representative strictly between  $b$  and  $x$  would be at most  $a$ , by the floor property, and would belong to the same block as  $b$ . This is impossible. Thus  $p_{i+}(x) = b$ .

At a nonzero limit  $\lambda$ , the earlier floor values form a nonempty admitted subclass of  $W$ . Take its least value  $b$ , attained at some  $i_0 < \lambda$ . Because the floors decrease, every subsequent floor before  $\lambda$  equals  $b$ . Nesting places  $b$  in every earlier slice as well, so  $b \in H_\lambda$ . Every point of  $H_\lambda$  at most  $x$  is at most  $p_{i_0}(x) = b$ . This proves existence and stabilization.

This is elementary induction: the greatest-predecessor formula defines all floors from the supplied history without selecting class maps. Nesting also proves the two composition identities, and monotonicity in  $x$  is immediate.  $\square$

**Lemma XVI.5.4** (Finitely many changes). *For each fixed  $x$ , only finitely many stages  $i \in I$  have  $p_{i+}(x) \neq p_i(x)$ . No floor value first appears at a limit stage.*

*Proof.* The limit assertion follows from Lemma XVI.5.3. If the class of successor changes were not finite, taking its least remaining member would uniquely define increasing stages  $i_0 < i_1 < \dots$  with strictly decreasing floors. Ordinary set-valued recursion along  $\omega$ , followed by Class Replacement, gives a set-coded descending sequence in  $W$ , a contradiction. These are unique minimum choices, so neither form of Choice is used.  $\square$

Define the ordinal digit

$$c_i(x) = \text{otp}(\{z \in H_i : p_{i+}(x) \leq_W z <_W p_i(x)\}, <_W).$$

This interval is a set because its endpoints lie in one set-interval block. The digit is nonzero exactly at a floor change. Consequently

$$\text{NF}_H(x) = \{(i, c_i(x)) : c_i(x) > 0\} \in P(\Gamma)$$

is a finite set graph. Its dependence on  $x$  is elementary. Finiteness here, like the theorem itself, is interpreted internally in a nonstandard model.

**Theorem XVI.5.5** (Finite normal form with initial image). *If  $H$  terminates on  $\Gamma + 1$ , then  $\text{NF}_H$  is a class order isomorphism of  $W$  onto an initial segment of  $P(\Gamma)$ .*

*Proof.* First establish a digit-insertion rule. Let  $b \in H_{i+}$ , and let  $z$  be the point of ordinal rank  $\rho$  in the  $H_i$  block beginning at  $b$ , when such a point exists. Then  $p_i(z) = z$  and  $p_{i+}(z) = b$ . At later stages, the floors of  $z$  and  $b$  agree by Lemma XVI.5.3. At every earlier stage both floors determining a digit equal  $z$ , since  $z \in H_i$ . Thus insertion adds the digit  $\rho$  at  $i$ , keeps the higher digits of  $b$ , and makes all lower digits zero.

An actual code reconstructs its point uniquely. Start from the final anchor  $m$  and read its finitely many nonzero digits in decreasing stage order, using the insertion rule. Each requested rank exists for an actual code. Lemma XVI.5.4 ensures there are no other floor changes, including at limits. The resulting point is therefore  $x$ . This proves injectivity.

To prove initiality, fix  $q < \text{NF}_H(x)$  and let  $i$  be the greatest exponent of disagreement. Decoding the common higher digits gives the anchor  $b = p_{i+}(x)$ . Write  $d = c_i(x)$  and  $c = q(i)$ , so  $c < d$ . The  $H_i$  block at  $b$  contains points  $a, a'$  of ranks  $c, c + 1$ , with

$$a <_W a' \leq_W p_i(x) \leq_W x.$$

Insert digit  $c$ , obtaining anchor  $a$ , and keep  $a'$  as a fixed upper bound.

Process the remaining nonzero digits of  $q$  in decreasing exponent order. For each lower exponent  $j < i$ , the current anchor and  $a'$  both belong to  $H_{j+}$ . Initially this follows from membership in  $H_i$ ; after an insertion at  $k > j$ , the new anchor lies in  $H_k \subseteq H_{j+}$ . They are distinct retained representatives. Therefore the  $H_j$  block beginning at the current anchor is nonfinal and has type Ord. Every requested ordinal digit exists inside it, strictly below  $a'$ . Insert that digit and continue.

After finitely many steps the resulting  $y$  satisfies  $\text{NF}_H(y) = q$  and  $y < a' \leq x$ . Thus every polynomial below an actual code occurs with a smaller preimage. Applied to two actual codes, this also gives exact order preservation, using injectivity and linearity. The image is initial, as required.  $\square$

The fixed upper bound in this proof is essential: it guarantees complete Ord blocks for all lower digits. Mere injectivity into a polynomial order would not yield strong comparability.

### XVI.5.3 Restrictions and prescribed schedules

**Lemma XVI.5.6** (Anchored convex restriction). *Let  $Y$  be a nonempty convex subclass of  $W$ , with  $a = \min_W Y$ . If  $H$  is a canonical history on  $W$ , then on the same schedule*

$$J_i = \{a\} \cup \{y \in H_i \cap Y : a <_W y\}$$

*is the canonical history of the induced order on  $Y$ . It terminates whenever  $H$  terminates.*

*Proof.* The initial clause and the limit intersections follow directly. For a successor stage fix  $x \in H_i \cap Y$  with  $a < x$ , and let  $u = p_i(a)$ . There are no  $H_i$  points in  $(u, a]$ . Convexity gives

$$\begin{aligned} [u, x]_W \cap H_i &= \{u\} \cup \{z \in H_i : a < z \leq x\}, \\ [a, x]_W \cap J_i &= \{a\} \cup \{z \in H_i : a < z \leq x\}. \end{aligned}$$

These intervals are sets simultaneously. A global witness below  $x$  showing that  $x$  is not a least representative is also a local witness if it is above  $a$ . If it is at or below  $a$ , restriction of its interval and insertion of  $a$  gives a local set interval. Conversely, a local witness above  $a$  is global too,

whereas the local witness  $a$  can be replaced by  $u$  using the displayed identities. Hence the retained representatives strictly above  $a$  coincide. The local minimum  $a$  is always retained, proving the successor clause. A final global singleton contributes no point above  $a$ , so the local final slice is  $\{a\}$ .  $\square$

Bare intersection with  $Y$  would sometimes discard its minimum; the anchor cannot be omitted. The earlier polynomial construction supplies, in GB, the explicit history

$$K_i = \{q \in P(\Gamma) : \text{supp } q \subseteq I_{\geq i}\}, \quad K_* = \{0\}.$$

Condensation deletes the least available exponent; the limit intersection retains exactly the exponents above every earlier threshold.

**Corollary XVI.5.7** (Exact fixed-schedule criterion). *For nonempty class well-orders  $W, \Gamma$ , a canonical terminating history of  $W$  on  $\Gamma + 1$  exists if and only if  $W$  is class-isomorphic to an initial segment of  $P(\Gamma)$ .*

*Proof.* The forward implication is [Theorem XVI.5.5](#). Conversely, restrict the explicit polynomial history to the supplied nonempty initial range and pull it back through the class isomorphism. [Lemma XVI.5.6](#) verifies the recursion clauses; the zero polynomial gives the final singleton.  $\square$

#### XVI.5.4 The equivalence and its exact scope

**Theorem XVI.5.8** (Canonical histories over GB). *Over GB, the following principles are equivalent:*

1. CWO: *of any two class well-orders, one is class-isomorphic to an initial segment of the other, allowing the whole order.*
2. CTH: *every nonempty  $W$  has a canonical history on  $W + 1$ .*
3. *Every nonempty  $W$  has a canonical terminating history on some class-well-order schedule  $\Gamma + 1$ .*
4. *Every nonempty  $W$  is class-isomorphic to an initial segment of  $P(W)$ .*

*Proof.* The implication  $(2) \Rightarrow (3)$  is [Lemma XVI.5.2](#).

For  $(3) \Rightarrow (1)$ , form the tagged sum  $T = R + S$  of two nonempty orders. Apply (3) once to obtain one history on a schedule  $\Gamma + 1$ . Anchored convex restriction gives histories for  $R$  and  $S$  on that same schedule. Their normal forms have initial ranges in the same order  $P(\Gamma)$ . Two initial subclasses of a linear order are comparable by inclusion. Compose the map onto the smaller range with the inverse map on the larger range to obtain the required class initial-segment isomorphism. Empty inputs are immediate. There is no simultaneous choice of histories for many orders.

For  $(1) \Rightarrow (4)$ , first observe that any class order embedding  $f : W \rightarrow W$  satisfies  $x \leq_W f(x)$ . Otherwise the least counterexample  $x$  has  $f(x) < x$  and  $f(f(x)) < f(x)$ , giving an earlier counterexample. It follows that  $W + 1$  cannot embed into  $W$ : if the last point mapped to  $a$ , the restricted embedding would give  $a \leq f(a) < f(*) = a$ .

Now put  $\Gamma = W + 1$ . The choice-free polynomial theorem makes  $P(\Gamma)$  a class well-order. The coefficient-one monomials  $e_t$  embed  $W + 1$  into it. Comparability with  $W$  must therefore give an initial embedding  $f : W \rightarrow P(\Gamma)$ ; the opposite direction would embed  $W + 1$  into  $W$ . Its range lies strictly below  $e_*$ , since otherwise initiality would put  $e_*$  and every  $e_x$  in the range, again embedding  $W + 1$  into  $W$  through  $f^{-1}$ . The polynomials below  $e_*$  are exactly  $P(W)$ , which proves (4).

Finally,  $(4) \Rightarrow (2)$  follows from [Corollary XVI.5.7](#) with  $\Gamma = W$ . Explicitly, pull the support-threshold polynomial history back through the supplied initial-segment isomorphism.  $\square$

**Proposition XVI.5.9** (Uniqueness with a fixed schedule). *Two canonical histories for the same input order and schedule agree at every stage. Suppose the sets, input order, and schedule remain fixed while the collection of available classes is enlarged, and both orders remain class well-orders in the larger realization. Then an already available history and its normal forms retain their values and agree with any newly available canonical history on that schedule.*

*Proof.* The class of disagreement stages is elementary in the two supplied histories. If nonempty, it has a least member. That member cannot be the first stage, whose slice is prescribed. At a successor the equal preceding slices have equal condensations, and at a limit the equal earlier slices have equal intersections. Each possibility contradicts disagreement. For the second claim, the set-interval tests and recursion clauses quantify only over the unchanged sets with the given classes as parameters. The old history still satisfies those clauses, and the assumed well-ordering of the schedule in the larger realization permits uniqueness to identify it with any new candidate. The ordinal interval ranks defining the normal forms are unchanged as well.  $\square$

Thus a stronger class theory can supply a previously absent history without changing an old one. This distinction concerns existence of one class witness; it does not provide a class-valued operation selecting histories for a totality of class well-orders. Over the choiceless base, preservation of an old relation's class-well-order property under enlargement of the class collection is not being assumed without a hypothesis.

The repository's four-way equivalence and the preceding normal-form machinery are the mathematical starting point [77]. Our strengthening consists of the GB polynomial theorem and the checked elimination of Choice from the remaining dependencies. It gives neither a proof of CWO in GB nor a proof that CWO yields arbitrary elementary transfinite recursion. Suitable ETR supplies the required histories; the reverse comparison remains separate [27].

There is also a necessary limitation concerning carriers. The repository reduces the history principle to well-orders of the entire surreal class by transporting arbitrary proper class carriers through a global bijection. That corollary retains its Global Choice hypothesis. The choice-free theorem above quantifies directly over the given class carriers and does not assume that every proper class is bijective with Ord or with the surreals.

*Note XVI.5.10* ([merge]; this section re-proves Part XIV over GB). Every statement of this section except [theorem XVI.5.8](#)'s base theory is a re-proof, with the same arguments, of source 29's results in Part XIV, which source 34 credits throughout: [definition XVI.5.1](#) is [definition 108.10](#); [lemma XVI.5.2](#) is source 20's [theorem 108.1](#) and source 29's [note 108.11](#); [lemma XVI.5.3](#) is [lemma 108.12](#); [lemma XVI.5.4](#) is [proposition 108.13](#); the digits  $c_i(x)$  are (S29.27); [theorem XVI.5.5](#) is [theorem 108.16](#); [lemma XVI.5.6](#) is [lemma 108.17](#); the explicit history  $K_i$  is (S29.26); [corollary XVI.5.7](#) is [corollary 108.18](#); [theorem XVI.5.8](#) is [theorem 108.19](#), there over GBC; and [proposition XVI.5.9](#) is source 29's paragraph "Uniqueness and changes of the class collection" after [corollary 108.20](#). Source 29's argument used global choice only through the well-foundedness of  $P(\Gamma)$  ([note 108.15](#)); source 34 checked that floors, finite changes, normal forms, restriction and the four implications use only uniquely specified minima, and replaced that one step by [theorem XVI.4.2](#). This report agrees with that dependency analysis: the proofs of [lemmas 108.12](#) and [108.17](#), [proposition 108.13](#), [theorem 108.16](#), and [corollary 108.18](#) as printed in Part XIV take least elements of admitted classes and collect uniquely specified  $\omega$ -sequences, and nothing else. Source 35 proves the same equivalence over GB independently ([note XVI.5.19](#)). Source 29's reduction to the surreal carrier ([corollary 108.20](#)) keeps global choice, as both sources say. The equivalence answers over GB the question of [note 108.4](#), of item (b) of [section 112](#) and of source 20's [research question 113.111](#); each now carries a dated note.



### XVI.5.5 Source 35: condensation, finite addresses, and comparison without Choice

This section strengthens a result recorded in the ProveIt manuscript [78]. That manuscript proves, over GBC, an equivalence between complete canonical condensation histories and strong comparison of class well-orders; its Part XIV explicitly leaves the corresponding question over GB without Global Choice open. We prove the equivalence over the choice-free base used here: the sets satisfy ZF, and the class axioms provide elementary comprehension and Class Replacement. Neither set Choice nor Global Choice occurs in the proof. The principal repair is the choice-free finite-support theorem, [Note XVI.4.4](#); the remaining argument is established below with its class-existence requirements explicit. The normal-form construction itself, and the equivalence over GBC, are credited to the repository.

Throughout, a class well-order means an order in which every nonempty admitted subclass has a least element. In particular, we do not replace this hypothesis by the absence of a descending  $\omega$ -sequence. An initial segment may be the whole order. Class carriers always consist of sets; a block or history slice can be a proper class, but never becomes an element of another class.

#### XVI.5.5.1 One condensation is elementary

Let  $W = (X, <_W)$  be nonempty. For a nonempty subclass  $A \subseteq X$ , with its induced order, define

$$x \sim_A y \iff [\min_W(x, y), \max_W(x, y)]_W \cap A \text{ is a set.}$$

The right side is elementary: it asserts that there exists a set whose members are exactly the displayed interval. Let  $D_W(A)$  consist of the least elements of the  $\sim_A$ -blocks.

*Note XVI.5.11* (Source 35, Lemma 4.1, “Set-interval blocks”). *The relation  $\sim_A$  is a convex equivalence relation. Its blocks have canonical ordinal rank maps, and each block is isomorphic either to a set ordinal or to Ord. A set-sized block, if present, is final. In particular, a block followed by another block has type  $\Omega$ .* This is source 29’s [note 108.9](#) for the induced order on  $A$ , with the same proof, and source 32’s and source 33’s [notes XVI.3.5](#) and [XVI.3.10](#); source 35 notes that the rank of a point is the order type of a set that already carries the supplied well-order, so no well-ordering of a set is chosen.

This gives the repository’s canonical division theorem. Let  $B_W$  be the ordered class of minima of the proper blocks of  $W$ , and let  $\rho_W$  be the ordinal type of the final set block, or zero if there is none. Then

$$W \cong \Omega \cdot B_W + \rho_W. \tag{S35.8}$$

Here the product is a  $B_W$ -indexed sum of copies of  $\Omega$ . The isomorphism sends a point in a proper block to its block minimum and its ordinal rank within that block; it sends the final set block to  $\rho_W$ . This is one elementary class map. Thus the word “canonical” does not conceal a choice of one isomorphism for each block. When  $W$  is a set,  $B_W$  is empty and  $\rho_W = \text{otp}(W)$ . For a proper carrier,  $B_W$  is nonempty. Formula (S35.8) and its proof are already in [78]; the quotient can be as complicated as an arbitrary class well-order, so this single division does not finish the analysis.

#### XVI.5.5.2 What a complete history asserts

Let  $\Gamma = (I, <_\Gamma)$  be a nonempty class well-order and append a tagged final point  $*$ . Each  $i \in I$  has an immediate successor  $i^+$  in  $\Gamma + 1$ . A nonzero limit stage has no immediate predecessor; its predecessor class need not be a set.

*Note XVI.5.12* (Source 35, Definition 4.2, “Canonical condensation history”). *A history for  $W$  on  $\Gamma + 1$  is a single class relation  $H \subseteq (I \cup \{*\}) \times X$  with  $H_{\min \Gamma} = X$ ,  $H_{i+} = D_W(H_i)$  for  $i \in I$ , and  $H_\lambda = \bigcap_{i <_{\Gamma+1} \lambda} H_i$  at nonzero limits (source 35’s displays (9)–(11)). It terminates if  $H_* = \{m\}$ ,  $m = \min_W X$ ; CTH says that every nonempty  $W$  has such a history on  $W + 1$ . This is [definition XVI.5.1](#) and [definition 108.10](#).*

The recursion clauses are elementary in the earlier history. Their solutions, however, are class-valued histories; elementary comprehension alone does not assert that a solution exists. Once  $H$  is supplied, elementary induction along its schedule proves that the slices decrease and all contain  $m$ . It also proves uniqueness for fixed  $W$  and schedule: a least stage at which two histories differ is incompatible with the corresponding recursion clause.

*Note XVI.5.13* (Source 35, Proposition 4.3, “Input-order termination”). *Every supplied history on  $W + 1$  terminates; more precisely  $H_x \subseteq \{m\} \cup \{y \in X : x \leq_W y\}$  for  $x \in X$ . This is [lemma XVI.5.2](#), source 20’s [theorem 108.1](#) and source 29’s [note 108.11](#), by the same induction along  $W$ .*

### XVI.5.5.3 Floors and the finiteness of every individual address

*Note XVI.5.14* (Source 35, Lemma 4.4, “Canonical floors”). *For every stage  $i$  of a supplied history and every  $x \in X$ , the maximum  $p_i(x) = \max_W(H_i \cap (-\infty, x]_W)$  exists; the graph of all these maps is elementary in  $H, W$ ; for  $i \leq j$ ,*

$$p_j(x) \leq_W p_i(x), \quad p_j(p_i(x)) = p_j(x), \quad p_i(p_j(x)) = p_j(x); \quad (\text{S35.12})$$

*and at each nonzero limit  $\lambda$  the earlier floors of a fixed point are eventually constant, with value  $p_\lambda(x)$ . This is [lemma XVI.5.3](#) and source 29’s [lemma 108.12](#) with (S29.24), by the same induction; source 35 stresses that no new class is defined at an induction step, the value class being defined by one formula from the single supplied  $H$ .*

*Note XVI.5.15* (Source 35, Lemma 4.5, “Finite changes”). *For a fixed  $x$ , the class  $\{i \in I : p_{i+}(x) \neq p_i(x)\}$  is finite. No new floor value first occurs at a limit stage. This is [lemma XVI.5.4](#) and [proposition 108.13](#): an infinite change class would yield, by a uniquely determined recursion collected by Class Replacement (“not an application of Dependent Choice”), increasing stages with strictly decreasing floors.*

The limit rule also has an intrinsic quotient interpretation. Put  $x E_i y$  if  $p_i(x) = p_i(y)$ . The projection identities and eventual stabilization give

$$E_\lambda = \bigcup_{i < \lambda} E_i$$

at every nonzero limit: if two limit floors agree, choose an earlier stage after the stabilization stage of each floor. Thus an identification of original points never first appears at a limit.

[merge] The remark above is source 29’s [corollary 108.14](#). The explicit history (S35.13) is source 34’s  $K_i$  after [lemma XVI.5.6](#) and source 29’s (S29.26); source 35’s paragraph after it is the fullest verification of its successor clause printed here.

### XVI.5.5.4 Polynomial histories and canonical finite normal forms

Write  $P(\Gamma)$  for the class of finite partial functions from  $I$  to nonzero set ordinals, compared at the greatest exponent where their coefficients differ. Missing coefficients are zero. By [Note XVI.4.4](#), this is a class well-order in the choice-free base. Its explicit history on  $\Gamma + 1$  is

$$K_i = \{q : \text{supp}(q) \subseteq \{j : i \leq_\Gamma j\}\}, \quad K_* = \{0\}. \quad (\text{S35.13})$$

Indeed, fixing all coefficients strictly above  $i$  gives one set-interval block of  $K_i$ : its  $i$ -coefficient varies through  $\text{Ord}$ . Bounded intervals within this fiber are sets. Between different higher parts, the interval contains arbitrarily large  $i$ -coefficients on the lower higher part and is proper. Removing the  $i$ -coefficient therefore selects exactly the block minima. At a limit, intersection of the support thresholds is the required threshold; at the final stage only the empty polynomial remains. The formula is uniform and elementary, so this particular history needs no recursion axiom.

Now let  $H$  be a supplied terminating history for  $W$  on  $\Gamma + 1$ . Define

$$c_i(x) = \text{otp}([p_{i+}(x), p_i(x))_W \cap H_i, <_W). \quad (\text{S35.14})$$

The interval is a set because its endpoints lie in the same  $\sim_{H_i}$ -block. Its rank is zero exactly when the two floors coincide. Consequently

$$\text{NF}_H(x) = \{(i, c_i(x)) : i \in I, c_i(x) > 0\}$$

is a finite polynomial. This is a set-valued elementary class map.

*Note XVI.5.16* (Source 35, Theorem 4.6, “Finite normal form from a supplied history”). *The map  $\text{NF}_H$  is an isomorphism from  $W$  onto an initial segment of  $P(\Gamma)$ .* This is [theorem XVI.5.5](#) and source 29’s [theorem 108.16](#), with the same proof: one insertion adds the digit  $\rho$  at stage  $i$  and zeros below it; reading the finitely many nonzero digits from the final floor  $m$  reconstructs  $x$ ; for  $q$  below an actual code, the points  $a <_W a'$  of ranks  $c$  and  $c + 1$  in the block at  $p_{i+}(x)$  satisfy

$$a <_W a' \leq_W p_i(x) \leq_W x \quad (\text{S35.15})$$

(source 29’s [\(S29.29\)](#)), and the fixed upper bound  $a'$  keeps every later block nonfinal, so every requested lower digit can be inserted below  $a'$ .

The upper bound  $a'$  in [\(S35.15\)](#) is essential. It ensures that all subsequently requested smaller digits can be inserted inside nonfinal blocks, even when the input order has a truncated last block. Consequently the theorem gives an initial-segment embedding, which is stronger than an embedding into an arbitrary suborder.

#### XVI.5.5.5 Restriction to a convex interval

*Note XVI.5.17* (Source 35, Lemma 4.7, “Anchored restriction”). *If  $Y$  is a nonempty convex subclass of  $W$  and  $a = \min_W Y$ , then*

$$J_i = \{a\} \cup \{y \in H_i \cap Y : a <_W y\} \quad (\text{S35.16})$$

*is the canonical history of  $Y$  on the same schedule, and it terminates whenever  $H$  does.* This is [lemma XVI.5.6](#) and [lemma 108.17](#), with the same two displayed intervals.

Intersecting with  $Y$  without restoring its minimum would generally fail. For an initial segment the anchor is the global minimum, and ordinary intersection suffices.

*Note XVI.5.18* (Source 35, Theorem 4.8, “Exact criterion for a fixed schedule”). *For nonempty class well-orders  $W$  and  $\Gamma$ , the following are equivalent in GB:  $W$  has a canonical terminating history on  $\Gamma + 1$ ;  $W$  is class-isomorphic to an initial segment of  $P(\Gamma)$ .* This is [corollary XVI.5.7](#) and [corollary 108.18](#), by the same restriction and pullback; every nonempty initial image contains 0.

### XVI.5.5.6 The equivalence over the choice-free base

Let CWO assert that any two class well-orders are comparable by a class isomorphism of one onto an initial segment of the other.

*Note XVI.5.19* (Source 35, Theorem 4.9, “Choice-free history–comparison equivalence”). *Over GB with ZF sets and no form of Choice assumed, the following are equivalent: (1) CWO; (2) CTH: every nonempty  $W$  has a history on  $W + 1$ ; (3) every nonempty  $W$  has a terminating history on some class-well-order schedule  $\Gamma + 1$ ; (4) every nonempty  $W$  is class-isomorphic to an initial segment of  $P(W)$ .* This is source 34’s [theorem XVI.5.8](#), proved independently with the same four implications: (2) $\Rightarrow$ (3) by termination; (3) $\Rightarrow$ (1) from one history of the tagged sum  $R + S$  and its two anchored restrictions; (1) $\Rightarrow$ (4) by comparing  $W$  with  $P(W + 1)$ , which contains  $W + 1$  as the monomials  $e_i$ ; (4) $\Rightarrow$ (2) by the explicit relation  $H_i = \{x \in X : \text{supp}(f(x)) \subseteq \{j : i \leq_W j\}\}$ ,  $H_* = f^{-1}[\{0\}]$ . Source 29’s [theorem 108.19](#) is the same theorem over GBC. Source 35’s choice-free input is its [note XVI.4.4](#).

This proves the characterization over GB requested in the repository’s open-item list. It does not determine whether CWO follows from GB, and it does not establish  $\text{CWO} \Rightarrow \text{ETR}$ . Elementary transfinite recursion supplies the displayed history equations, so the proof also gives

$$\text{GB} + \text{ETR} \vdash \text{CTH} \wedge \text{CWO}.$$

The converse from comparison to unrestricted elementary transfinite recursion remains a separate question.

The axiom distinction matters for scope. The theorem applies to every *supplied* class well-order on a class of set labels. Transporting all proper carriers to a single preferred carrier, such as the whole surreal class, can require Global Choice and is not part of this strengthening. Similarly, the conclusion supplies an individual history for each individual input; it does not produce a same-level class whose members are all class histories.

Finally, finite addresses do not impose a finite or ordinal bound on the length of an order. Their exponents belong to the supplied class schedule and can themselves have proper classes of predecessors. What the theorem controls is the finite amount of changing information attached to each individual point. The existence of a complete uniform history, and hence of these addresses for every input order, has exactly the comparison content expressed in [Note XVI.5.19](#).

### XVI.5.6 Source 32: condensation: general termination and sharp examples

The repository asks whether iterating set-interval condensation eventually collapses every raw class well-order, while distinguishing its explicit polynomial examples from arbitrary recursion [79]. There are really two issues: existence of an admitted hierarchy and termination of a hierarchy once available. The second has a general answer.

[merge] The question source 32 quotes is the one of source 23’s status file; this report had answered it before source 32 was written ([remark XVI.1.1](#)).

#### XVI.5.6.1 An elementary condensation operator

Let  $E$  be a convex equivalence relation on  $W$ . Write  $r_E(x)$  for the least point in the  $E$ -class of  $x$ , and let  $Q_E$  be the class of these least representatives in their inherited order. Define

$$x \mathcal{C}_W(E) y \iff [r_E(x), r_E(y)]_{Q_E} \text{ is a set.} \quad (\text{S32.7})$$

This is again a convex equivalence relation, by [note XVI.3.5](#) applied to  $Q_E$  and then pulled back to  $W$ . It is coarser than  $E$ . All of [\(S32.7\)](#) is elementary in  $W, E$ : least representatives and set-sized intervals are expressed by quantifying only over sets.

For a supplied class well-order of stages, a *condensation hierarchy* begins with equality, applies  $\mathcal{C}_W$  at successors, and takes unions of its previous equivalence relations at nonzero limits. Unions are along an increasing chain, so they are convex equivalence relations. Stage domains and relations must be coded by one admitted class; an external assertion that each individual stage exists is not enough.

### XVI.5.6.2 Termination by the order's own length

*Note XVI.5.20* (Source 32, Theorem 6.1, “Conditional general termination”). *Let  $W$  be a nonempty class well-order, and append a new greatest stage  $\infty$  to form  $W+1$ . Suppose an admitted condensation hierarchy  $(E_t)_{t \in W+1}$  on  $W$  exists, using equality at the least stage,  $\mathcal{C}_W$  at successors, and unions at nonzero limits. Then  $E_\infty$  is universal. In particular,  $\text{GBC} + \text{ETR}_{W+1}$  supplies a hierarchy that collapses  $W$  by its own order length.* This is source 20's [theorem 108.1](#), source 29's [note 108.11](#) and [lemma XVI.5.2](#), for hierarchies of equivalence relations; such hierarchies are the nested histories of those statements ([remark XVI.5.24](#)). Source 32's proof follows; its invariant [\(S32.8\)](#) is the retention bound [\(S20.1\)](#) read through that correspondence.

*Source 32's proof.* Let  $w_0$  be the least point of  $W$ . We prove by class well-order induction on the stages  $t \in W+1$  that

$$x \in W \text{ and } x <_W t \implies x E_t w_0, \quad (\text{S32.8})$$

where all points of  $W$  precede  $\infty$ . At the first stage the assertion is vacuous.

Suppose  $t$  is the immediate successor of  $s$ . If  $s = w_0$ , the assertion needed is just reflexivity. Otherwise, the induction hypothesis says that all points below  $s$  lie in the one  $E_s$ -block containing  $w_0$ . The quotient interval from that block to the block containing  $s$  has at most two points: a point below  $s$  already belongs to the first block, and there are no other points before  $s$  to create intermediate blocks. A two-point or one-point interval is a set. Applying  $\mathcal{C}_W$  therefore merges  $s$  and all its predecessors with  $w_0$ , proving [\(S32.8\)](#) at  $t$ .

If  $t$  is a nonzero limit, take  $x < t$ . There is a stage  $s$  with  $x < s < t$ , since otherwise  $x$  would be the immediate predecessor of  $t$ . The induction hypothesis gives  $x E_s w_0$ , and the union rule gives  $x E_t w_0$ . This also handles  $t = \infty$  when  $W$  has no greatest point. If  $W$  has a greatest point, the successor argument handles  $\infty$ . Thus every point is equivalent to  $w_0$  at the final stage.

For existence under ETR, code the history by a class of triples  $(t, x, y)$ . The first-stage, successor, and limit clauses above form an elementary recursion rule with parameters  $W$  and its stage order. Formula [\(S32.7\)](#) is elementary, so  $\text{ETR}_{W+1}$  gives the required class. The induction establishing that its sections are increasing convex equivalence relations uses the operator properties already proved.  $\square$

**Status.** The termination implication itself is a theorem of GBC about any admitted hierarchy satisfying the stated clauses. ETR is a sufficient existence hypothesis. This is not an argument that bare GBC admits every such hierarchy, nor that this particular recursion operator has the full strength of ETR.

The invariant [\(S32.8\)](#) avoids a potentially misleading strategy: it does not try to assign a decreasing external ordinal size to the successive quotients. A quotient may be isomorphic to the previous quotient for a very long time while the quotient *maps* continue to identify new points.

### XVI.5.6.3 An explicit family attaining arbitrarily long bounds

Here we re-establish the polynomial mechanism already present in the repository, so that the sharpness argument is self-contained. Fix an admitted class well-order  $\Gamma$  and put

$$P(\Gamma) = \Omega^{[\Gamma]}.$$

For a stage  $c \in \Gamma + 1$ , define directly

$$p E_c q \iff p(d) = q(d) \text{ for every } d \in \Gamma \text{ with } c \leq d. \quad (\text{S32.9})$$

At the added last stage there are no such  $d$ , so the relation is universal. The formula defines the entire hierarchy at once, without ETR.

*Note XVI.5.21* (Source 32, Theorem 6.2, “Exact polynomial condensation length”). *The relations (S32.9) form a condensation hierarchy on  $P(\Gamma)$ . For nonempty  $\Gamma$ , no stage belonging to  $\Gamma$  is universal, while the added final stage is universal. Thus its first collapse occurs at the stage of length  $\Gamma$ .* This is source 23’s [note 108.8](#) (Part XIV), source 15’s [theorem 53.4](#) and the explicit history  $K_i$  of [lemma XVI.5.6](#)’s sequel, and source 33’s [note XVI.5.25](#); the proof is the same: the least representative of an  $E_c$ -class clears the coefficients below  $c$ ; at a successor two representatives agreeing above  $s$  span a set interval, while different higher parts leave a proper class of coefficients at  $s$  between them; at a limit a finite set of differing exponents is bounded below an earlier stage; and the monomial at  $c$  separates zero from the next stage.

**Corollary XVI.5.22** (Isomorphic quotients can hide real progress). *For  $P(\Omega) = \Omega^{[\Omega]}$ , the quotient by  $E_\alpha$  is isomorphic to  $P(\Omega)$  for every set ordinal  $\alpha$ , but the quotient at the class-limit stage  $\Omega$  is a singleton. In particular, no set-ordinal bound on condensation lengths can work for all explicit class well-orders.*

*Proof.* At stage  $\alpha$ , the retained exponent tail is  $[\alpha, \text{Ord})$ , which is isomorphic to  $\text{Ord}$  by the ordinary ordinal-tail correspondence. Transporting finite supports gives the asserted quotient isomorphism. Nevertheless the equivalence relations strictly increase: the monomial at coordinate  $\alpha$  survives  $E_\alpha$  and is identified with zero at  $E_{\alpha+1}$ . Formula (S32.9) is universal only at the final class-limit stage. For each proposed set bound, this same example survives beyond it.  $\square$

[merge] [Corollary XVI.5.22](#) is the self-similarity of source 18’s [proposition 52.15](#) (Part IX) at every set stage, for  $P(\Omega)$  instead of  $P(\omega)$ , and the remark after source 20’s [theorem 108.1](#) that repeated quotient types do not stop a history; source 33’s [example XVI.5.26](#) gives the case  $\Gamma = \omega$ .

This is a particularly important warning for a formalization: equality of quotient order types is not equality of equivalence relations, and it is not a reason to stop an iteration. The bonding maps contain indispensable information.

## XVI.5.7 Source 33: polynomial condensation and finite digits

### XVI.5.7.1 Ordinal-coefficient polynomials

For a class well-order  $\Gamma$ , let

$$P(\Gamma) = \Omega^{[\Gamma]}$$

be the finite-support order from [section XVI.4.5](#). Its elements may be written formally as

$$p = \Omega^{\gamma_0} a_0 + \Omega^{\gamma_1} a_1 + \cdots + \Omega^{\gamma_{m-1}} a_{m-1}, \quad \gamma_0 >_\Gamma \cdots >_\Gamma \gamma_{m-1}, \quad 0 < a_i \in \text{Ord}. \quad (\text{S33.5.1})$$



This is a notation for a finite list of exponent–coefficient pairs. It does not presuppose that a point  $\gamma_i$  of  $\Gamma$  is a set ordinal. Repeated terms are combined only according to the chosen coding convention; in the canonical presentation each exponent occurs once.

Write  $\Gamma^+ = \Gamma + 1$  and denote its new greatest point by  $\infty$ . For  $b \in \Gamma^+$ , define

$$p E_b q \iff p(\gamma) = q(\gamma) \text{ for every } \gamma \in \Gamma \text{ with } \gamma \geq b. \quad (\text{S33.5.2})$$

At  $b = \infty$  the condition is empty. At the least stage it is equality. The family  $E_b$  is a single elementary class relation in  $b, p, q$ .

**Definition XVI.5.23** (Exact condensation tower). Let  $W$  be a nonempty class well-order. An *exact condensation tower* indexed by  $\Gamma^+$  is a uniform family of convex equivalence relations  $F_b$  on  $W$  satisfying:

- (i)  $F_b$  is equality at the least stage;
- (ii) at a successor stage  $a^+$ , two points are  $F_{a^+}$ -equivalent exactly when the interval between their  $F_a$ -classes in  $W/F_a$  is a set;
- (iii) at a nonzero limit stage  $b$ ,  $F_b = \bigcup_{a < b} F_a$ .

Quotients are always represented by the minima of their classes. The tower is *exhausting* if  $F_\infty$  is universal.

For any  $a \in \Gamma$ , its immediate successor exists in  $\Gamma^+$ : either there is a next point of  $\Gamma$ , or the next point is  $\infty$ . A nonzero limit means a point with no immediate predecessor. These descriptions require no external ordinal numbering of the stages.

*Remark XVI.5.24* ([merge]; towers of equivalence relations are nested histories). Let  $W = (X, <_W)$  be nonempty with least point  $m$ , let  $\Gamma$  be nonempty, and let  $(F_b)_{b \in \Gamma^+}$  be an exact condensation tower in the sense of [definition XVI.5.23](#). Source 32's hierarchies ([note XVI.5.20](#)) are the same objects: its operator ([S32.7](#)) is clause (ii). Put

$$H_b = \{x \in X : x = \min_W[x]_{F_b}\},$$

the class of least points of the  $F_b$ -classes, elementary in the tower. Then  $H$  is a canonical history of  $W$  on  $\Gamma + 1$  ([definition XVI.5.1](#)), its floors are  $p_b(x) = \min_W[x]_{F_b}$ , and the tower is exhausting exactly when  $H$  terminates. Conversely every canonical history  $H$  arises in this way from exactly one exact tower, namely  $x F_b y \iff p_b(x) = p_b(y)$ , which is source 35's equivalence  $E_b$  after [note XVI.5.15](#).

*Proof.* Each  $F_b$ -class is convex, so its least point  $r$  is the greatest point of  $H_b$  below any of its members  $x$ : a point  $z$  of  $H_b$  with  $r <_W z \leq_W x$  would lie in the class of  $x$  by convexity and so would not be a least point. Hence  $p_b(x) = \min_W[x]_{F_b}$ . At the least stage  $F$  is equality and  $H = X$ . At a successor  $a^+$ , clause (ii) says that  $x$  and  $y$  are  $F_{a^+}$ -equivalent exactly when the  $F_a$ -classes between theirs form a set; these classes are represented by their least points, which are exactly the points of  $H_a$  between  $p_a(x)$  and  $p_a(y)$ , so the condition is  $p_a(x) \sim_{H_a} p_a(y)$  for the relation  $\sim_A$  that defines  $D_W$  before [definition XVI.5.1](#). Every  $F_{a^+}$ -class is therefore the union of the  $F_a$ -classes whose least points form one  $\sim_{H_a}$ -block, and its least point is the least point of that block:  $H_{a^+} = D_W(H_a)$ . At a nonzero limit  $\lambda$ ,  $F_\lambda = \bigcup_{a < \lambda} F_a$ , so  $x$  is least in its  $F_\lambda$ -class exactly when no  $y <_W x$  is  $F_a$ -equivalent to  $x$  for some  $a < \lambda$ , that is, exactly when  $x \in H_a$  for every  $a < \lambda$ :  $H_\lambda = \bigcap_{a < \lambda} H_a$ .  $F_\infty$  is universal exactly when its one class has least point  $m$  and no other, that is  $H_\infty = \{m\}$ . Conversely, for a history  $H$  the relations  $p_b(x) = p_b(y)$  are convex by the monotonicity of floors, satisfy (i) and (ii) by the computation just made, and satisfy (iii) by the stabilization of floors at limits ([lemma XVI.5.3](#); source 29's [corollary 108.14](#)). The two passages are inverse because  $p_b(x) = \min_W[x]_{F_b}$ .  $\square$

Consequently source 33's [note XVI.5.32](#) and source 32's [note XVI.5.20](#) are source 20's [theorem 108.1](#), and source 33's digit map is a normal-form map ([remark XVI.5.29](#)).

[merge] [corrected after independent check, 4 October 2026] The hypothesis that  $\Gamma$  is nonempty was missing; [definition XVI.5.1](#) requires it. Source 33's [definition XVI.5.23](#) does not: for  $\Gamma = \emptyset$  its tower is the single relation  $F_\infty$ , equality, which is exhausting exactly when  $W$  has one point, while no history is defined.

*Note XVI.5.25* (Source 33, Theorem 5.2, “Exact polynomial condensation”). *The family (S33.5.2) is an exhausting exact condensation tower for  $P(\Gamma)$ . For every stage  $b$ ,*

$$P(\Gamma)/E_b \cong P(\Gamma_{\geq b}), \quad (\text{S33.5.3})$$

*with the quotient represented by truncating all exponents below  $b$ . No stage  $b < \infty$  is universal. All these assertions hold in GBC.* This is source 32's [note XVI.5.21](#) and source 23's [note 108.8](#), whose clause (ii) is the same quotient formula; source 33's proof is the same truncation argument.

**Example XVI.5.26** (Stationary quotient types, nonstationary identifications). For  $\Gamma = \omega$ , every finite-stage quotient  $P(\omega)/E_n$  is isomorphic to  $P(\omega)$ , since the tail  $[n, \omega)$  has type  $\omega$ . Nevertheless  $E_n$  is nonuniversal for every finite  $n$  and  $E_\omega$  is universal. Replacing a coherent tower by a list of quotient *types* therefore loses essential information.

[merge] [Example XVI.5.26](#) is the case  $\Gamma = \omega$  of source 32's [corollary XVI.5.22](#) and source 18's [proposition 52.15](#).

The general phenomenon of prescribed long condensation is related to the repository's existing work. The contribution of [note XVI.5.25](#) here is the explicit arithmetic presentation, uniform formula, and exact quotient calculation used in the next theorem; priority for the broad realization principle is not claimed.

### XVI.5.7.2 A finite-digit theorem for arbitrary supplied towers

For an exhausting tower  $(F_b)$  on  $W$  and  $a \in \Gamma$ , the  $F_a$ -classes inside the  $F_{a+}$ -class of  $x$  form one set-interval block of  $W/F_a$ . That block is set-like by [note XVI.3.10](#). Consequently the position of the  $F_a$ -class of  $x$  within it is a set ordinal. Denote this ordinal by

$$r_a(x) = \text{otp}\{C : C \text{ is an } F_a\text{-class in } [x]_{F_{a+}}, C < [x]_{F_a}\}. \quad (\text{S33.5.4})$$

In (S33.5.4), the notation  $C$  is shorthand for its least representative, so the displayed collection is genuinely a set of sets.

**Theorem XVI.5.27** (Finite-digit embedding). *Suppose  $W$  is nonempty and has a supplied uniform exhausting exact condensation tower  $(F_b)_{b \in \Gamma^+}$ . Then*

$$\text{supp}_W(x) = \{a \in \Gamma : r_a(x) \neq 0\}$$

*is finite for every  $x \in W$ , and the map*

$$\Phi(x) = \{(a, r_a(x)) : r_a(x) \neq 0\} \quad \text{is an order embedding } W \longrightarrow P(\Gamma). \quad (\text{S33.5.5})$$

*The theorem is provable in GBC from the supplied tower; it does not assert that GBC constructs every such tower.*

*Proof. Step 1: every digit is an ordinal.* The set-likeness argument preceding the theorem proves that each predecessor collection in (S33.5.4) is a set well-order. Its ordinary ordinal collapse is unique, so all digits are specified by one elementary formula with the tower as a parameter.

*Step 2: nonzero digits occur at only finitely many stages.* Fix  $x$ . Start with  $\sigma = \infty$  and with  $z$  the least point of  $[x]_{F_\sigma} = W$ . More generally maintain that  $z$  is the least point of  $[x]_{F_\sigma}$ . If  $x = z$ , every digit at a stage below  $\sigma$  is zero:  $x$  occupies the first  $F_a$ -class in its  $F_{a+}$ -class for each such  $a$ .

Otherwise, let  $\tau \leq \sigma$  be the least stage at which  $x$  and  $z$  are equivalent. It is not the first stage, since  $x \neq z$ . It is not a limit, since the limit rule would give equivalence at an earlier stage. Therefore  $\tau = a^+$  for some  $a \in \Gamma$ . At stage  $a$ , the two points lie in different  $F_a$ -classes within one  $F_{a+}$ -class. Since  $z < x$ , we have  $r_a(x) > 0$ .

For  $a < b < \sigma$ , the points  $z, x$  are already  $F_b$ -equivalent, and  $z$  is still least in the larger  $F_{b+}$ -class containing them. Hence  $r_b(x) = 0$ . Replace  $\sigma$  by  $a$  and  $z$  by the least point of  $[x]_{F_a}$ . Repeat the argument inside that class.

If the process did not terminate, the successive selected exponents would form a strictly descending  $\omega$ -sequence in  $\Gamma$ . The next selected exponent and the new minimum are uniquely specified by elementary formulas. As in [note XVI.2.12](#), ordinary set recursion and class replacement would collect this into a forbidden set sequence. The process therefore terminates after finitely many steps. The selected exponents are precisely all stages with nonzero digit. Thus (S33.5.5) is a finite set code.

*Step 3: comparison is preserved.* Let  $x < y$  and take the least stage at which  $x, y$  become equivalent. Again it has the form  $a^+$ . At all stages  $b > a$  their digits agree, because their  $F_b$ -classes are the same. At  $a$ , their distinct  $F_a$ -classes lie in one set-like block, and the class of  $x$  precedes the class of  $y$ . Therefore

$$r_a(x) < r_a(y).$$

The greatest differing digit is exactly  $a$ , which is precisely the rule for  $\Phi(x) < \Phi(y)$  in  $P(\Gamma)$ . In particular, the map is injective.  $\square$

*Remark XVI.5.28* (What the theorem does not claim). The conclusion is an order embedding. No claim is made here that this specific digit map always has initial image. Converting arbitrary suborders to initial segments is a separate comparison issue. Under a theory with class well-order comparability, an initial embedding of  $W$  into  $P(\Gamma)$  can be obtained, but that assertion is not used in the proof above.

*Remark XVI.5.29* ([merge]; the digit map always has initial image). [Remark XVI.5.28](#) makes no claim that  $\Phi$  has initial image, and source 33's research question [research question 113.263](#) asks for which supplied towers it is initial. It always is; the remark and the first part of the question are stale, and stay on record with this proof. Let  $(F_b)_{b \in \Gamma^+}$  be a supplied uniform exhausting exact tower on  $W$ . If  $\Gamma$  is empty, exhaustion of equality forces the nonempty  $W$  to be a singleton; then  $P(\Gamma) = 1$  and  $\Phi$  is the unique isomorphism, hence initial. Assume henceforth that  $\Gamma$  is nonempty, and let  $H$  be the terminating history of [remark XVI.5.24](#), with floors  $p_b(x) = \min_W [x]_{F_b}$ . Fix  $a \in \Gamma$  and  $x \in W$ . The  $F_a$ -classes  $C$  contained in  $[x]_{F_{a+}}$  with  $C < [x]_{F_a}$  are represented by their least points; these lie in  $H_a$ , are at least  $\min_W [x]_{F_{a+}} = p_{a+}(x)$  and are below  $\min_W [x]_{F_a} = p_a(x)$ ; conversely, a point  $z \in H_a$  with  $p_{a+}(x) \leq_W z <_W p_a(x)$  lies in  $[x]_{F_{a+}}$  by convexity and is the least point of an  $F_a$ -class below that of  $x$ . Hence

$$r_a(x) = \text{otp}(\{z \in H_a : p_{a+}(x) \leq_W z <_W p_a(x)\}, <_W) = c_a(x),$$

the digit of source 34's normal form ([theorem XVI.5.5](#); source 29's (S29.27)), and  $\Phi = \text{NF}_H$ . By [theorem XVI.5.5](#) (over GB; source 35's [note XVI.5.16](#); source 29's [theorem 108.16](#) over GBC),  $\Phi$  is an

isomorphism of  $W$  onto an initial segment of  $P(\Gamma)$ . So  $\Phi$  is initial for every supplied tower, without any comparison principle; initiality is the capped reconstruction in the proof of [theorem XVI.5.5](#), which [theorem XVI.5.27](#)'s proof does not attempt. The last sentence of the question, whether a different canonical compression map has a weaker axiom bound, has two readings. As a question about the base theory for  $\Phi$ , it is answered by the same identification:  $\Phi$  needs only GB and the supplied tower. As a question about existence for every  $W$ , any map with initial image costs exactly CWO over GB ([corollary XVI.5.7](#) and [theorem XVI.5.8](#)), while mere embeddings  $x \mapsto e_x$  into  $P(W)$  are free.

[merge] [corrected after independent check, 4 October 2026] The text of 4 October 2026 said that this part of the question “is answered by the same identification:  $\Phi$  needs only GB and the supplied tower”; that holds only for the first reading, and the two readings are now separated. A further bounded review found that the earlier proof immediately invoked a terminating history for every supplied tower. For  $W = 1$  and  $\Gamma = 0$  the exhausting tower exists but the printed canonical-history definition requires nonempty  $\Gamma$ ; the singleton case above repairs that invocation without restricting the initiality conclusion. The old domain claim is retained here with this counterexample.

### XVI.5.7.3 Recursion gives a converse

The principle  $\text{ETR}_\Lambda$  says that every elementary class recursion along the specified class well-order  $\Lambda$  has a solution. It is more than a notational license to write a recursively defined family.

**Proposition XVI.5.30** (Conditional converse). *Assume  $\text{GBC} + \text{ETR}_{\Gamma^+}$ . If a nonempty class well-order  $W$  embeds into  $P(\Gamma)$ , then its exact condensation tower indexed by  $\Gamma^+$  exists and is exhausting. Consequently, in this theory,*

$$W \hookrightarrow P(\Gamma) \iff W \text{ has an exhausting exact } \Gamma^+ \text{-tower.}$$

*Proof.* The successor operation in [definition XVI.5.23](#), represented using block minima, is elementary in the previous equivalence relation. The limit operation uses an existential set quantifier over earlier stage labels. Thus the stated recursion principle produces the exact tower  $(F_b)$  of  $W$ . Transport  $W$  to the image of its embedding in  $P(\Gamma)$ .

We prove by class well-order induction that, for points of  $W$ ,

$$E_b \subseteq F_b.$$

The first stage is immediate. At a successor, two points in a common  $E_{a^+}$ -class meet only a set of ambient  $E_a$ -classes between them. Within the interval of  $W$ , an occupied  $E_a$ -class lies inside a single  $F_a$ -class by induction. Select the least point of its intersection with that interval, then the least representative of the resulting  $F_a$ -class. Class replacement on the set of ambient classes shows that the intervening  $F_a$ -classes form a set. Hence the two points are  $F_{a^+}$ -equivalent.

At a limit the inclusion follows from the union rules. Since  $E_\infty$  is universal, so is  $F_\infty$ . The reverse implication is [theorem XVI.5.27](#).  $\square$

*Remark XVI.5.31* ([merge]; under  $\text{ETR}_{\Gamma+1}$  an embedding into  $P(\Gamma)$  becomes initial). In  $\text{GBC} + \text{ETR}_{\Gamma^+}$ , suppose a class well-order  $W$  embeds into  $P(\Gamma)$ . If  $W$  is empty, its unique embedding is initial. If  $W$  is nonempty and  $\Gamma$  is empty, then  $P(\Gamma) = 1$  forces  $W$  to be a singleton, again giving an initial embedding. For nonempty  $W$  and  $\Gamma$ , combine [proposition XVI.5.30](#) with [remark XVI.5.29](#): its exact tower exists and is exhausting, so its normal-form map has initial image. Thus in that theory, for every  $W$  and  $\Gamma$ ,  $W \hookrightarrow P(\Gamma) \Leftrightarrow W \preceq_{\text{init}} P(\Gamma)$ , and, for nonempty  $W$  and  $\Gamma$ , by [corollary XVI.5.7](#)

both say that  $W$  has a terminating history on  $\Gamma + 1$ . Source 33 states only the embedding form; the upgrade is this write's, by the two cited proofs.

[merge] [corrected after independent check, 4 October 2026] The hypothesis that  $W$  and  $\Gamma$  are nonempty, which [corollary XVI.5.7](#) requires, was missing from the last clause. For  $W = 1$  and  $\Gamma = 0$  the initial embedding exists but the stated canonical history is undefined, as in [remark XVI.5.24](#). The source's [proposition XVI.5.30](#) and this remark also said that the exact tower exists for “a class well-order  $W$ ”, without the nonemptiness required by [definition XVI.5.23](#). For  $W = 0$  and  $\Gamma = 1$  the embedding premise holds but that tower lies outside its definition. The proposition now states its nonempty domain, and the separate empty cases above prove the broader embedding/initial-embedding equivalence. These counterexamples retain both earlier omissions; no tower convention is silently extended.

*Note XVI.5.32* (Source 33, Proposition 5.7, “Exhaustion under a sufficient recursion principle”). *If  $W$  is a nonempty class well-order and  $\text{ETR}_{W+1}$  holds, then the exact condensation tower of  $W$  indexed by  $W + 1$  is exhausting.* This is source 20's [theorem 108.1](#) with its existence clause, and source 32's [note XVI.5.20](#), through [remark XVI.5.24](#). Source 33's proof inducts on stages  $x \in W + 1$ , showing that all points strictly below  $x$  lie in one  $F_x$ -class: at  $x = y^+$  the point  $y$  lies in that class or the next, and one condensation step merges the prefix through  $y$ ; at a limit any two points of the prefix lie below an earlier stage. It calls this “a sufficient axiom bound, not an optimality theorem”. The existence of exhausting towers for all nonempty class well-orders is CTH by [remark XVI.5.24](#), hence equivalent to CWO ([theorem XVI.5.8](#)). The earlier wording “all class well-orders” included  $W = 0$ , which lies outside [definition XVI.5.23](#); that domain counterexample is retained here, and the nonempty qualification is explicit.

This is a sufficient axiom bound, not an optimality theorem. The explicit polynomial towers of [note XVI.5.25](#) require no such axiom because their entire family is given by the single truncation formula ([S33.5.2](#)).

## XVI.6 Hereditary normal forms and a first fixed point

Finite-support exponentiation admits an explicit fixed-point construction that goes beyond every finite tower with base  $\text{Ord}$ . This section works in GB. We write  $\Omega$  as a typographical symbol for the class order  $\text{Ord}$ , not as a new set ordinal. The construction uses the familiar hereditary normal-form pattern from ordinal analysis: for example, hereditary base- $\omega_1$  normal forms below  $\varepsilon_{\omega_1+1}$  already appear in Weiermann's discussion [85]. Finite supports and preservation of initial segments also belong to the established framework of normal dilators [86]. Our purpose is to give the proper-class construction, its foundational justification, and its universal property directly. We do not claim to have invented hereditary ordinal notation.

[merge] The hereditary system is new to this report; the four sources construct it independently. Source 35's  $\mathcal{E}$  ([note XVI.6.11](#)) is this  $\text{E}$ , with the same terms and comparison, proved well-ordered over GBC ([note XVI.6.13](#)); source 32's  $\mathcal{H}(A)$ , printed  $\text{E}(A)$ , is the general base of [corollary XVI.6.10](#), over GBC ([note XVI.6.24](#)); source 33 reaches the same fixed point as the direct limit  $\mathbb{T}$  of the finite towers ([theorem XVI.6.26](#)). In this section  $\text{E}^{\leq n}$  is source 34's  $E_n$ , the terms of height at most  $n$ , renamed because  $E_i$  is a floor equivalence in Section XVI.5.

### XVI.6.1 A class of finite codes

**Definition XVI.6.1** (Hereditary base- $\Omega$  terms). The raw expressions are finite rooted ordered trees with ordinal coefficients. We write the empty tree as 0, and a nonempty tree as

$$t = [(t_0, a_0), \dots, (t_{k-1}, a_{k-1})], \quad k > 0,$$

where the  $t_i$  are its exponent subtrees and the  $a_i$  are nonzero set ordinals. Compare two raw expressions recursively by lexicographically comparing their lists of exponent-coefficient pairs: compare exponents first and coefficients second; after an identical prefix, the shorter list is smaller. A term is *valid* when all its exponent subterms are valid and

$$t_0 > t_1 > \dots > t_{k-1}.$$

Let  $\mathbf{E}$  be the class of valid terms, with this order. Its intended display notation is

$$t = \Omega^{t_0} a_0 + \dots + \Omega^{t_{k-1}} a_{k-1}.$$

Every expression is a set. The recursive comparison terminates because each recursive call concerns smaller finite trees. Validity, comparison, and the finite set of subterms therefore have uniform first-order definitions, using finite computation witnesses if desired. In particular, the definition does not invoke a recursion of length  $\omega$  whose values are classes. This distinction matters: arbitrary class-valued recursion can require additional axioms even when its indexing order is small.

For an infinite initial cardinal  $\kappa$ , let  $\mathbf{E}[\kappa]$  consist of the terms all of whose coefficients are below  $\kappa$ . It is a set of finite codes. Define an ordinal interpretation on this set by

$$v_\kappa(0) = 0, \quad v_\kappa(t) = \kappa^{v_\kappa(t_0)} a_0 + \dots + \kappa^{v_\kappa(t_{k-1})} a_{k-1}. \quad (\text{S34.5})$$

This is ordinary evaluation of a finite tree in set ordinal arithmetic.

**Lemma XVI.6.2** (Bounded-coefficient comparison). *For  $s, t \in \mathbf{E}[\kappa]$ ,*

$$s < t \iff v_\kappa(s) < v_\kappa(t).$$

*Proof.* Induct on the total size of the two trees. By the induction hypothesis, the interpreted exponent lists remain strictly decreasing, and comparison of their exponents agrees with comparison of the subterms. All coefficients lie below  $\kappa$ . The usual comparison theorem for finite-support base- $\kappa$  representations therefore compares the two ordinal values at exactly the first exponent-coefficient difference of their lists. If one list ends after an identical prefix, its remaining value is zero and is smaller than the nonempty remaining sum of the other list. These are exactly the defining cases of term comparison.  $\square$

**Theorem XVI.6.3** (Well-foundedness in GB). *The definable relation on  $\mathbf{E}$  is a class well-order.*

*Proof.* Strict linearity follows from the comparison lemma: the finitely many coefficients in any two or three terms have a common cardinal bound, so irreflexivity, trichotomy, and transitivity transfer from ordinal comparison. Now let  $S$  be any nonempty set of terms. The coefficients appearing anywhere in  $S$  form a set of ordinals. Choose an infinite initial cardinal  $\kappa$  above all of them; arbitrarily large well-orderable initial cardinals exist in ZF. The comparison lemma embeds  $S$  into the ordinals, so  $S$  has a least term. Every nonempty set subset therefore has a minimum. [Lemma XVI.2.2](#) implies that every nonempty admitted subclass has a minimum as well. This last step requires neither set Choice nor Global Choice.  $\square$



This proof uses a different ordinal interpretation for different bounded collections of terms. It never attempts an ordinal-valued order embedding of the whole proper class  $\mathbf{E}$ .

**Example XVI.6.4** (Reading and comparing terms). Consider

$$s = \Omega^{\Omega+3}7 + \Omega^{\Omega+1}\omega^2 + 19, \quad t = \Omega^{\Omega+3}7 + \Omega^{\Omega+1}(\omega^2 + 1).$$

These are finite codes with ordinary ordinal coefficient labels. The leading exponent and coefficient agree. At the next common exponent,  $\omega^2 < \omega^2 + 1$ , so  $s < t$ ; the final coefficient 19 cannot change the outcome. Under any sufficiently large cardinal interpretation, the same comparison is an ordinary ordinal calculation. No object called the ordinal  $\Omega + 3$  is required inside the original universe: its occurrence is itself an exponent subtree.

The class and its order are parameter-free definable, although individual terms carry arbitrary ordinal data. A fixed finite syntax with constants from a set supplies only a set of terms. Allowing all set ordinal coefficients is what makes this notation system a proper class. This is compatible with ordinary definability: every particular term is constructed from a finite tuple of ordinal parameters and a finite tree shape. It does not entail a truth predicate deciding which ordinals are definable without parameters. Thus the presentation keeps separate the definability of an entire class order, the parameters describing one of its elements, and the availability of a short written name for a coefficient.

## XVI.6.2 Height, initial segments, and finite towers

Define the syntactic height by

$$h(0) = 0, \quad h(t) = 1 + \max_{i < k} h(t_i) \quad (t \neq 0),$$

and put  $\mathbf{E}^{\leq n} = \{t \in \mathbf{E} : h(t) \leq n\}$ . Thus  $\mathbf{E}^{\leq 0} = \{0\}$ , while  $\mathbf{E}^{\leq 1}$  consists of zero and the constants  $\Omega^0 a$ , and is canonically isomorphic to  $\text{Ord}$ . Write 1 for the constant with coefficient 1, and define the finite sequence of constructions

$$\tau_0 = 1, \quad \tau_{n+1} = \Omega^{\tau_n}. \quad (\text{S34.6})$$

Here the latter expression is the singleton root list  $[(\tau_n, 1)]$ . Each  $\tau_n$  is a finite term; the resulting  $\omega$ -sequence is set-coded.

**Theorem XVI.6.5** (Exact height cuts). *For every  $n < \omega$ ,*

$$\mathbf{E}^{\leq n} = \mathbf{E}_{< \tau_n}, \quad \mathbf{E}^{\leq n+1} \cong \text{Ord}^{[\mathbf{E}^{\leq n}]}$$

*Consequently the height stages are proper initial segments with order types*

$$1, \quad \Omega, \quad \Omega^{[\Omega]}, \quad \Omega^{[\Omega^{[\Omega]}]}, \quad \dots,$$

*and  $\mathbf{E} = \bigcup_{n < \omega} \mathbf{E}^{\leq n}$  has cofinality  $\omega$ .*

*Proof.* Only zero lies below the least positive term 1, proving the initial-cut claim for  $n = 0$ . Removing the root identifies  $\mathbf{E}^{\leq n+1}$  with finite graphs whose exponent support lies in  $\mathbf{E}^{\leq n}$ , with arbitrary nonzero ordinal coefficients. This proves the displayed isomorphism. Suppose  $\mathbf{E}^{\leq n} = \mathbf{E}_{< \tau_n}$ . A term lies below the monomial  $\Omega^{\tau_n}$  exactly when all its exponent support lies below  $\tau_n$ : a greater

exponent makes it larger, while an equal greatest exponent requires coefficient at least 1 and also makes it at least the monomial. Thus  $E^{\leq n+1} = E^{< \tau_{n+1}}$ .

The term  $\tau_n$  has height  $n+1$ , so it belongs to  $E^{\leq n+1}$  and not to  $E^{\leq n}$ . The cuts are therefore proper and the sequence  $(\tau_n)$  is strictly increasing. Every finite tree has finite height, so the cuts exhaust  $E$  and this sequence is cofinal. There is no largest term; a finite subset cannot be cofinal. Hence the least possible cofinality is exactly  $\omega$ .  $\square$

A proper class well-order can have set cofinality because it need not be setlike. Here the predecessor class of the term  $\Omega$  already contains a copy of every set ordinal. The theorem also shows that every exponent subterm of a nonzero term is smaller than the whole term: its height lies below that of the term, and the height levels are initial.

### XVI.6.3 The fixed-point isomorphism and leastness

**Theorem XVI.6.6** (A definable fixed point). *There is a canonical definable class isomorphism*

$$\text{Ord}^{[E]} \cong E.$$

*Proof.* Sort the finite support of a graph in decreasing  $E$ -order and make its exponent-coefficient pairs the children of a new root. This produces a valid term. Conversely, remove the root of a term to recover the finite graph. The constructions are inverse, and both orders compare the greatest exponent at which a coefficient differs.  $\square$

Leastness has a precise formulation which requires neither a collection of all class order types nor a general comparability theorem.

**Theorem XVI.6.7** (Least initial pre-fixed point). *Let  $X$  be a class well-order, and suppose that*

$$j : \text{Ord}^{[X]} \longrightarrow X$$

*is an initial order embedding. There is a unique finite-term evaluation  $e : E \longrightarrow X$  satisfying*

$$e(0) = j(\emptyset), \quad e([(t_i, a_i)]_{i < k}) = j(\{(e(t_i), a_i) : i < k\}). \quad (\text{S34.7})$$

*This map is an initial order embedding. In particular,  $E$  embeds initially into every class well-order equipped with an isomorphism  $\text{Ord}^{[X]} \cong X$ .*

*Proof.* Define  $e(t) = x$  by the existence of a finite assignment of elements of  $X$  to all subterms of  $t$ , satisfying (S34.7) at every node and assigning  $x$  to the root. Finite-tree induction proves existence and uniqueness, simultaneously proving preservation and reflection of comparison. Indeed, already evaluated exponent subterms have the same strict order as before, so the graph at the next root is well-defined; finite-support comparison followed by the order embedding  $j$  gives exactly the desired comparison. Elementary comprehension supplies the class graph of  $e$ . All witnesses in this definition are finite sets.

It remains to prove initiality. Put  $I_n = e(E^{\leq n})$ . Since  $j$  is initial and the empty graph is least,  $I_0 = \{j(\emptyset)\}$  is an initial segment of  $X$ . If  $I_n$  is initial in  $X$ , the finite graphs supported on  $I_n$  form an initial segment of  $\text{Ord}^{[X]}$ . The evaluation equation identifies their image under  $j$  with  $I_{n+1}$ . Hence  $I_{n+1}$  is initial. Their increasing union  $e(E)$  is therefore initial as well. Uniqueness follows from finite-tree induction on the evaluation equation.  $\square$

The argument defines one evaluation class by finite witnesses and then studies its restrictions. It does not recursively assemble an  $\omega$ -sequence of classes. In this concrete sense,  $\mathbf{E}$  is the least fixed point of the base-Ord power constructor, already in  $\mathbf{GB}$ . The assertion neither supplies  $\mathbf{ETR}$  along  $\mathbf{E}$  nor makes  $\mathbf{E}$  a largest definable class well-order.

**Corollary XVI.6.8** (An explicit long condensation history). *The order  $\mathbf{E}$  has an elementary canonical history on  $\mathbf{E} + 1$ . Its first singleton slice occurs at the appended final stage.*

*Proof.* Transport the explicit history of  $P(\mathbf{E})$  through the root isomorphism. At a stage  $t \in \mathbf{E}$ , the retained terms are precisely those whose root exponents are all at least  $t$ , together with zero. This is the polynomial history used in Section XVI.5. The monomial  $\Omega^t$  is a nonzero retained point, so that slice is not a singleton. The final slice is  $\{0\}$ .  $\square$

The universal property also makes the meaning of “least” independent of our chosen tree coding. A different presentation with the same fixed-point property and the same initial universal property receives an initial embedding from  $\mathbf{E}$  and supplies one in the reverse direction. Their composites are initial self-embeddings of class well-orders and hence are identities: a least point of disagreement would contradict initiality. The presentations are therefore uniquely isomorphic. This is a statement about specified class maps, not a quotient whose members would themselves be proper classes.

There is also a precise distinction between reaching the fixed-point order and naming it as one of its elements. Every term lies below some  $\tau_n$ , while the sequence of these terms is cofinal in  $\mathbf{E}$ . Consequently no term denotes an initial segment of type  $\mathbf{E}$  itself. The fixed-point isomorphism concerns the whole class presentation. To continue beyond it one may use a new class-order construction, such as  $\mathbf{E} + 1$  or the general-base construction below; the finite terms already defined do not silently acquire an additional terminal element.

#### XVI.6.4 Exact calibration in larger universes

**Theorem XVI.6.9** (Bounded-coefficient order type). *For an infinite initial cardinal  $\kappa$ , the map  $v_\kappa$  is an isomorphism of  $\mathbf{E}[\kappa]$  onto the least fixed point of ordinary ordinal exponentiation  $\xi \mapsto \kappa^\xi$ . This is the least epsilon number strictly greater than  $\kappa$ . In particular,*

$$\text{otp}(\mathbf{E}[\omega]) = \varepsilon_0, \quad \text{otp}(\mathbf{E}[\kappa]) = \varepsilon_{\kappa+1} \quad \text{if } \kappa \text{ is uncountable.}$$

*Proof.* The coefficient-bounded height stages have ordinal types  $c_0 = 1$  and  $c_{n+1} = \kappa^{c_n}$ , by root removal and ordinary finite-support exponentiation. They exhaust  $\mathbf{E}[\kappa]$ , so its ordinal type is  $\lambda = \sup_{n < \omega} c_n$ . Continuity gives  $\kappa^\lambda = \lambda$ . Any fixed point bounds all  $c_n$ , by induction, and therefore bounds  $\lambda$ .

If  $\kappa^\xi = \xi$ , then  $\xi \leq \omega^\xi \leq \kappa^\xi = \xi$ , so  $\xi$  is an epsilon number. Also  $\xi > \kappa$ : for  $2 \leq \xi \leq \kappa$  one has  $\kappa^\xi \geq \kappa^2 > \kappa$ , while  $\xi = 0, 1$  are not fixed. Conversely, let  $\varepsilon > \kappa$  be an epsilon number. Choose  $0 < \delta < \varepsilon$  with  $\kappa \leq \omega^\delta$ . Since  $\delta\varepsilon = \varepsilon$ ,

$$\varepsilon = \omega^\varepsilon \leq \kappa^\varepsilon \leq (\omega^\delta)^\varepsilon = \omega^{\delta\varepsilon} = \varepsilon.$$

For completeness,  $\delta\varepsilon = \varepsilon$  follows by bounding  $\delta$  by  $\omega^\eta$  for some  $\eta < \varepsilon$  and using  $\eta + \varepsilon = \varepsilon$ .

Finally, every uncountable initial cardinal satisfies  $\varepsilon_\kappa = \kappa$ . Indeed, for  $\alpha < \kappa$  the usual construction of the epsilon enumeration gives  $|\varepsilon_\alpha| \leq \max(\aleph_0, |\alpha|) < \kappa$ ; continuity, together with  $\varepsilon_\alpha \geq \alpha$ , now gives  $\sup_{\alpha < \kappa} \varepsilon_\alpha = \kappa$ . The next epsilon number is consequently  $\varepsilon_{\kappa+1}$ .  $\square$

The bounded-coefficient sets must not be confused with the height cuts. For example, the constant term  $\kappa$  is excluded from  $E[\kappa]$ , whereas the larger term  $\Omega$  is included. Thus  $E[\kappa]$  is not initial in the full  $E$ . In a full class model  $(V_\kappa, \mathcal{P}(V_\kappa))$  with  $\kappa$  inaccessible, all the finite codes in question belong to  $V_\kappa$ . Its internally defined  $E$  therefore has external order type exactly  $\varepsilon_{\kappa+1}$ . This supplies an exact external measurement without turning that ordinal into a set of the smaller universe.

[merge] Source 34's general-base system  $H(A)$  is printed  $E(A)$ ;  $E = E(\Omega)$  up to the identification of  $\Omega$  with its nonzero coefficients.

**Corollary XVI.6.10** (Other class bases). *For every class well-order  $A$  with at least two elements, the analogous hereditary finite normal forms  $E(A)$  form a class well-order in GB. They satisfy  $A^{[E(A)]} \cong E(A)$  and the same least initial pre-fixed-point property. Their height stages are the finite base- $A$  towers and are cofinal with order type  $\omega$ .*

*Proof.* Use nonzero elements of  $A$  as coefficients. In any nonempty set  $S$  of terms, all coefficients, together with the least element of  $A$ , form a set. Collapse their induced well-order to an ordinal, choose an infinite initial cardinal above that ordinal, and interpret all terms of  $S$  at that cardinal with the collapsed coefficients. This preserves comparison and supplies a least element of  $S$ . Finite triples establish linearity, and Lemma XVI.2.2 gives class well-foundedness in GB. Root removal, the least nonzero coefficient, and finite evaluation prove the remaining assertions exactly as above.  $\square$

## XVI.6.5 Source 35: hereditary notation and fixed-point completion

The polynomial operator admits a systematic extension beyond every fixed finite tower of powers. Its first fixed point has finite descriptions, and its existence does not require elementary transfinite recursion. The more delicate question is whether an *arbitrary supplied* class well-order can be placed inside such a fixed point. We prove an exact criterion for that extension, and thereby obtain another characterization of the repository's condensation-history principle.

Class notation for epsilon-like orders above Ord is established proof-theoretic practice. Jeon's note explicitly recalls Rathjen's class order  $E_0 = \varepsilon_{\text{Ord}+1}$  and its role in induction principles [88, Proposition 1.5]; the cited source is [89]. The finite hereditary construction below belongs to that tradition. We use our own notation  $E$  so that its precise meaning is fixed by a class-order construction, without presupposing a universe of generalized ordinal objects. The local extension criterion in Theorem XVI.6.18 is an additional result developed here; its priority outside the sources examined has not been independently established.

### XVI.6.5.1 Finite hereditary $\Omega$ -terms

*Note XVI.6.11* (Source 35, Definition 5.1, "Normalized hereditary terms"). *An  $\Omega$ -term is a finite tree: the empty list is the term 0, and a nonzero term is a list  $t = ((t_0, \alpha_0), \dots, (t_{r-1}, \alpha_{r-1}))$  with  $t_0 > \dots > t_{r-1}$  and  $0 < \alpha_i \in \text{Ord}$ , whose exponent entries are themselves terms, displayed as  $\Omega^{t_0} \alpha_0 + \dots + \Omega^{t_{r-1}} \alpha_{r-1}$ . Lists are compared lexicographically from the first pair, comparing exponents first and coefficients second; a proper common-prefix extension is larger, and exponent comparisons use the same rule recursively.  $E$  is the class of normalized terms. This is source 34's definition XVI.6.1, with the same order. Source 35 writes  $\mathcal{E}$ , printed  $E$  (section 1.3.4).*

The definition concerns finite trees labelled by set ordinals. It does not place a proper class inside another class. Recursive comparisons involve smaller finite trees, so normalization and comparison can be expressed by the existence of a finite set-coded computation trace. Consequently both predicates are elementary definitions in GB. In a nonstandard model, "finite" has that model's internal meaning, as throughout this article.

Write  $c(0) = 0$  and  $c(\alpha) = ((0, \alpha))$  for  $\alpha > 0$ . These are the constant terms. To avoid confusing syntax with an actual proper-class element, write

$$u = ((c(1), 1)), \quad M(t) = ((t, 1)).$$

Thus  $u$  is the term displayed as  $\Omega$ , while  $M(t)$  is displayed as  $\Omega^t$ . The symbol  $\Omega$  in the display is a constructor symbol;  $u$  itself is an ordinary set code.

**Lemma XVI.6.12** (Bounded specialization). *Let  $S$  be any set of terms, closed under taking exponent subterms. There is an additively indecomposable set ordinal  $\Lambda > 1$  such that the evaluation*

$$v_\Lambda(0) = 0, \quad v_\Lambda(t) = \sum_{i < r} \Lambda^{v_\Lambda(t_i)} \alpha_i$$

*is strictly order preserving on  $S$ .*

*Proof.* Choose  $\Lambda$  strictly above every coefficient occurring in  $S$ ; a sufficiently large power of  $\omega$  suffices. The displayed evaluation is ordinary set recursion on finite trees. Simultaneous induction on the sizes of two trees proves preservation of comparison. For the induction step, exponent comparisons have already been preserved. If  $\gamma < \delta$  and  $a < \Lambda$ , then

$$\Lambda^\gamma a < \Lambda^{\gamma+1} \leq \Lambda^\delta.$$

For  $\delta > 0$ , the ordinal  $\Lambda^\delta$  is additively indecomposable, so a finite tail of lower-exponent summands remains below it. At exponent zero there is no lower tail. Therefore the largest exponent of disagreement, followed by its coefficient, decides both the syntactic and the evaluated comparison. This is exactly the comparison rule in [Note XVI.6.11](#).  $\square$

*Note XVI.6.13* (Source 35, Theorem 5.3, “The first explicit polynomial fixed point”). *In GBC,  $\mathbf{E}$  is a proper class well-order, and assembling a polynomial into one new root gives an elementary class isomorphism  $\rho : P(\mathbf{E}) \rightarrow \mathbf{E}$ . This is source 34’s [theorem XVI.6.3](#), which needs only GB, and [theorem XVI.6.6](#). Source 35’s well-ordering proof applies [lemma XVI.6.12](#) to the subterms of a set-coded descending sequence, which global choice produces from a subclass without least element (first minimizing set-theoretic rank, then using the global choice function); source 34 applies its [lemma XVI.6.2](#) to an arbitrary nonempty set of terms and concludes by [lemma XVI.2.2](#). The root map is the same.*

*Note XVI.6.14* (Source 35, Proposition 5.4, “Exact tower cuts and cofinality”). *Put  $b_0 = u$  and  $b_{n+1} = M(b_n)$ . Then  $\mathbf{E}_{<b_0} \cong \text{Ord}$ ,  $\mathbf{E}_{<b_{n+1}} \cong P(\mathbf{E}_{<b_n})$  and  $\mathbf{E}_{<b_n} \cong P^n(\text{Ord})$ . The sequence  $(b_n)_{n < \omega}$  is strictly increasing and cofinal in  $\mathbf{E}$ . In particular,  $\mathbf{E}$  has cofinality  $\omega$ . This is source 34’s [theorem XVI.6.5](#) with the index shifted:  $b_n = \tau_{n+1}$  and  $\mathbf{E}_{<b_n} = \mathbf{E}^{\leq n+1}$ ; the proof is the same (the cut below  $M(b)$  consists of the terms whose root exponents lie below  $b$ ). Source 35 adds that the sequence is obtained by set-valued recursion on  $\omega$ , so its range is a set.*

The apparent tension between a proper-class carrier and a countable cofinal set is useful. It is the *set-like* proper well-order  $\text{Ord}$  that has no set cofinal subset. An order with proper-class initial segments can have set cofinality. Here each  $b_n$  is a set code whose predecessors form the indicated, generally proper, class.

*Note XVI.6.15* (Source 35, Theorem 5.5, “Leastness with an explicit witness”). *Suppose  $F$  is a class well-order and  $\varphi : P(F) \cong F$  is a supplied class isomorphism. There is a unique class map  $v : \mathbf{E} \rightarrow F$  with  $v(0) = \varphi(\emptyset)$  and  $v(t) = \varphi(\{(v(t_i), \alpha_i) : i < r\})$ ; it is an isomorphism onto an initial segment of  $F$ . This is the case of source 34’s [theorem XVI.6.7](#) in which the supplied initial*

embedding  $j : \text{Ord}^{[F]} \rightarrow F$  is an isomorphism; source 34's hypothesis, an initial pre-fixed point, is weaker. The evaluation is the same finite-trace definition. For initiality source 34 inducts on the height levels  $I_n = e(\mathbf{E}^{\leq n})$ , source 35 by structural induction on  $t$ ; its proof is printed as a second route.

*Source 35's proof.* Evaluate each finite tree from its leaves upward. This defines one elementary class map by finite traces, with  $\varphi$  as a parameter. Simultaneous induction on two trees shows that distinct exponent subterms have distinct images, so the displayed finite graph is a polynomial, and that comparison is preserved. Uniqueness follows from the same finite recursion.

We prove initiality by structural induction on  $t$ . There is nothing below  $v(0)$ . For nonzero  $t$ , let  $s$  be its largest exponent subterm, and take  $y <_F v(t)$ . Every exponent  $a$  in  $q = \varphi^{-1}(y)$  satisfies  $a \leq_F v(s)$ , since  $q < \varphi^{-1}(v(t))$  in polynomial order. If  $a < v(s)$ , the inductive assertion for  $s$  supplies its preimage under  $v$ ; equality has preimage  $s$ . Decode these finitely many exponents and retain their ordinal coefficients. They form a normalized term  $t'$  with  $v(t') = y$ . Thus everything below an image point belongs to the image. No comparison principle for arbitrary class well-orders was used.  $\square$

This theorem gives a precise sense in which  $\mathbf{E}$  is the first fixed point of  $A \mapsto P(A)$ . The existence of the isomorphism is part of the fixed-point assertion. An external equality of order types in an enclosing universe would not supply the class parameter needed by the proof in an arbitrary class model.

The use of bounded specialization also has a precise limitation. Bounding every coefficient by a set ordinal produces a set of finite trees, but generally does not produce an initial subclass of the full term order. A new, larger constant coefficient still denotes a term below  $u$ . Specialization is therefore a device for ruling out a particular set-coded descending sequence. The actual initial subclasses used to measure the tower are the exponent cuts in [Note XVI.6.14](#), not arbitrary coefficient cutoffs.

**An enclosing-universe check.** In the full model  $(V_\kappa, \mathcal{P}(V_\kappa))$  at a strongly inaccessible  $\kappa$ , the internal coefficients range externally over the ordinals below  $\kappa$ . Evaluate terms at the ordinary ordinal base  $\kappa$ . The evaluation identifies  $\mathbf{E}$  with the least positive solution of  $\kappa^\lambda = \lambda$ . Indeed, the successive tower cuts have types  $\kappa, \kappa^\kappa, \kappa^{\kappa^\kappa}, \dots$ , and ordinary base- $\kappa$  normal forms show that their union contains every ordinal below the supremum of this tower. The supremum is the first fixed point by continuity and strict increase of ordinal exponentiation.

This is also the least ordinary epsilon number strictly above  $\kappa$ . If  $\lambda = \kappa^\lambda$ , then  $\omega^\lambda \leq \lambda \leq \omega^\lambda$ , so  $\lambda$  is an epsilon number; the fixed point is larger than  $\kappa$ . Conversely, the first epsilon number  $\epsilon > \kappa$  satisfies

$$\kappa^\epsilon = (\omega^\kappa)^\epsilon = \omega^{\kappa \cdot \epsilon} = \omega^\epsilon = \epsilon.$$

Here  $\omega^\kappa = \kappa$  and  $\kappa \cdot \epsilon = \epsilon$ . Since an uncountable initial ordinal  $\kappa$  also satisfies  $\varepsilon_\kappa = \kappa$ , this ambient type is  $\varepsilon_{\kappa+1}$ . The calculation explains the familiar epsilon notation while retaining the necessary distinction: internally  $\mathbf{E}$  is a proper class order, whereas externally the displayed epsilon number is a set ordinal.

[merge] The enclosing-universe check is a second proof of source 34's [theorem XVI.6.9](#) for uncountable inaccessible  $\kappa$ , and source 33 states the tower  $t_{n+1} = \kappa^{t_n}$  in its Section 9.2 (Section XVI.8).

### XVI.6.5.2 Completion from an initial polynomial embedding

The next construction permits an arbitrary seed order, provided its initial embedding into its polynomial order is supplied. This hypothesis isolates exactly the potentially missing comparison



information.

*Note XVI.6.16* (Source 35, Lemma 5.6, “Finite-support preservation”). *If  $f : A \rightarrow B$  is an initial embedding of class well-orders, mapping each exponent through  $f$  gives an initial embedding  $P(f) : P(A) \rightarrow P(B)$ . This operation preserves identities and composition, and preserves properness of initial embeddings.* This is source 35’s own lemma XVI.4.5 for the base  $\Omega$ , with functoriality added, and source 34’s rule after (S34.3); the proof transports finite supports, and an exponent  $b$  omitted by  $f$  gives the omitted monomial at  $b$ .

**Theorem XVI.6.17** (Explicit fixed-point completion). *In GBC, suppose that  $j : A \rightarrow P(A)$  is an initial embedding of class well-orders. There is a class well-order  $C(A, j)$ , an initial embedding  $\iota : A \rightarrow C(A, j)$ , and a class isomorphism*

$$\Phi : P(C(A, j)) \longrightarrow C(A, j),$$

*all elementary in the supplied parameters. If  $j$  is proper,  $\iota$  is proper and  $C(A, j)$  has a definable increasing cofinal  $\omega$ -sequence. Moreover,  $C(A, j)$  initially embeds into every polynomial fixed point into which  $A$  initially embeds.*

*Proof.* For notation, write

$$A_n = P^n(A), \quad j_n = P^n(j) : A_n \longrightarrow A_{n+1}.$$

These symbols do not assume an arbitrary class recursion. A member of  $A_n$  is a finite nested polynomial code, with at most  $n$  polynomial levels and leaves labelled in  $A$ . Empty polynomials can terminate a branch earlier. Tags distinguish leaves from polynomial nodes. Membership and comparison are verified on a finite tree. The map  $j_n$  replaces every bottom-level  $A$ -label  $a$  by the finite polynomial  $j(a)$  and retains the outer nodes. This too is checked by a finite trace. Thus one elementary formula defines all stages and all maps uniformly in  $n$ .

Each  $A_n$  is a class well-order and each  $j_n$  is initial. Uniform finite induction is legitimate here: in GBC, well-foundedness can be expressed as the absence of a set-coded descending  $\omega$ -sequence, an elementary assertion about the uniformly defined stages. Alternatively, a proposed descending sequence at one stage uses only a set of bottom labels and coefficients; their set well-order types and ordinary finite polynomial constructions give a direct contradiction. Initiality follows from Note XVI.6.16.

For  $n \leq m$ , let  $j_{n,m}$  be the finite composite from  $A_n$  to  $A_m$ , with  $j_{n,n}$  the identity. On tagged pairs  $(n, a)$ , put

$$(n, a) \sim (m, b) \iff j_{n,k}(a) = j_{m,k}(b), \quad k = \max(n, m).$$

Each equivalence class has a unique representative at its least stage. Use those representatives as the carrier of  $C(A, j)$ . Concretely, all pairs at stage zero are retained; a pair  $(n, a)$ ,  $n > 0$ , is retained exactly when  $a$  is not in the range of  $j_{n-1}$ . This is a class of set codes, not a class whose members are equivalence classes. Order representatives by comparison of their images in a common finite stage. Injectivity and compatibility of the connecting maps prove that equality and order are well defined.

Let  $\iota_n : A_n \rightarrow C(A, j)$  send a point to its least-stage representative. The maps are initial embeddings. Indeed, a point below  $\iota_n(a)$  can be compared with it in a common later stage; initiality of  $j_{n,k}$  supplies a unique preimage in  $A_n$ . Every point belongs to some  $\iota_n[A_n]$ . Given a nonempty subclass of the colimit, choose one of its points and a stage containing that point. Its intersection

with that stage has a least element. Initiality shows that no point outside the stage can be smaller. Thus the colimit is a class well-order.

To construct  $\Phi$ , take a polynomial  $p$  on the colimit. Its finite support is contained in one  $\iota_n[A_n]$ : take the maximum of the finitely many least-stage numbers of its exponents. Decode the support into  $A_n$ , retaining the coefficients, to obtain  $p_n \in P(A_n) = A_{n+1}$ , and put

$$\Phi(p) = \iota_{n+1}(p_n).$$

For the empty support choose  $n = 0$ . The result is independent of the chosen stage because  $j_{n+1,m+1} = P(j_{n,m})$ . Choosing a common stage for two supports proves that  $\Phi$  preserves order strictly. It is onto: a point represented at a positive stage already is a polynomial on the preceding stage; a point at stage zero can first be moved to stage one using  $j$ . This proves the fixed-point assertion.

If  $j$  is proper, so is every  $j_n$ . Let  $d_n$  be the least point of  $A_{n+1}$  omitted by  $j_n$ . The points  $\iota_{n+1}(d_n)$  increase strictly, and the  $n$ th one lies above the whole image of  $A_n$ . They are therefore cofinal. Their definition is uniform and set valued, so Class Replacement supplies the corresponding set sequence.

Finally suppose  $h : A \rightarrow F$  is initial and  $\varphi : P(F) \cong F$ . Finite evaluation gives initial maps

$$h_0 = h, \quad h_{n+1} = \varphi \circ P(h_n) : A_{n+1} \rightarrow F.$$

Two initial embeddings from one class well-order into another must agree: at a least disagreement, the smaller image would lie below the other and hence would be the latter map's image of an earlier point, contradicting injectivity and earlier agreement. Applying this uniqueness to  $h_{n+1} \circ j_n$  and  $h_n$  proves compatibility. The maps therefore induce an initial embedding of the colimit into  $F$ . All evaluations and identifications above are finite set computations relative to the supplied classes; no  $\text{ETR}_\omega$  instance has been invoked.  $\square$

For  $A = \text{Ord}$ , take  $j$  to send an ordinal coefficient to the constant polynomial at the least exponent. This is a proper initial embedding. Its completion is exactly the hereditary system above: a finite stage has finitely nested polynomial roots, and identifying constant expansions gives the normalized hereditary trees. The first point missing from the zeroth-stage image is  $b_0 = u$ ; the first point missing from the next-stage image is  $b_1 = \Omega^u$ ; the subsequent missing points are  $b_2, b_3, \dots$ . Thus the general colimit construction and the explicitly indexed tower cuts give the same order and the same cofinal sequence.

### XVI.6.5.3 The exact local criterion

**Theorem XVI.6.18** (Local fixed-point extension criterion). *For any class well-order  $W$ , the following are equivalent over GBC:*

1. *there is a class initial embedding  $W \rightarrow P(W)$ ;*
2. *there are a class well-order  $F$ , a class initial embedding  $W \rightarrow F$ , and a class isomorphism  $P(F) \cong F$ .*

*For nonempty  $W$ , these are also equivalent to the existence of its canonical terminating condensation history on the schedule  $W + 1$ .*

*Proof.* The forward implication is [Theorem XVI.6.17](#). For the converse, identify  $W$  with its initial image  $A \subseteq F$  and write  $\varphi : P(F) \cong F$ . The monomial map

$$g(b) = \varphi(\{(b, 1)\})$$

is an order embedding of  $F$  into itself. Every such embedding is inflationary. Otherwise its least counterexample  $b$  would satisfy  $g(g(b)) < g(b) < b$ , making  $g(b)$  an earlier counterexample.

Take  $a \in A$  and  $q = \varphi^{-1}(a)$ . If  $b \in \text{supp}(q)$ , then the coefficient-one monomial at  $b$  is at most  $q$ . Hence

$$b \leq_F g(b) \leq_F a.$$

Initiality gives  $b \in A$ . Thus  $\varphi^{-1}$  maps  $A$  into  $P(A)$ . Its range there is initial: a polynomial below  $\varphi^{-1}(a)$  has image under  $\varphi$  below  $a$ , which belongs to  $A$ . Transport back through the original embedding to obtain  $W \rightarrow P(W)$ . The last equivalence is exactly the repository's fixed-schedule normal-form criterion proved in the preceding section.  $\square$

**Corollary XVI.6.19** (A fixed-point characterization of comparability). *Let FPE assert that every class well-order initially embeds in a class well-order  $F$  with a supplied isomorphism  $P(F) \cong F$ . Then*

$$\text{GBC} \vdash \text{FPE} \longleftrightarrow \text{CWO} \longleftrightarrow \text{CTH}.$$

*Requiring the initial extension in FPE to be proper gives the same global principle.*

*Proof.* Apply [Theorem XVI.6.18](#) to each input and use the repository equivalence between CWO, CTH, and the assertion that every  $W$  initially embeds in  $P(W)$ . For a proper extension, apply that assertion to the tagged order  $W + 1$  before taking its completion. The converse is immediate.  $\square$

This conclusion calibrates *extending every input*, even though one explicit fixed point already exists in GBC. It also avoids an invalid inference from a countable display of class-valued stages to a need for  $\text{ETR}_\omega$ . The special operator used here is finitary and has uniform finite codes; arbitrary elementary class recursions need not have that property. No equivalence with full ETR follows from this argument.

There is no hidden choice of a class witness for every member of a proper collection. The local construction takes one order and one embedding as supplied parameters and returns one explicitly defined completion. The global principle quantifies over inputs and asserts existence separately for each one. It does not assert a single class whose entries are all those completions, nor does it provide a uniform satisfaction predicate for arbitrary formulas defining orders. This distinction is essential when translating the result into a theory whose classes are precisely the definable classes of a fixed set-theoretic model.

#### XVI.6.5.4 A next fixed point, with no maximal endpoint

**Corollary XVI.6.20** (Successor fixed-point extension). *Given  $B$  and a class isomorphism  $\varphi : P(B) \cong B$ , GBC constructs a polynomial fixed point strictly extending  $B$  initially. It is least among polynomial fixed points properly extending  $B$ , in the sense of supplied initial embeddings.*

*Proof.* Let  $A = B + 1$ , with appended point  $*$ . Define an initial embedding  $j : A \rightarrow P(A)$  by

$$j(b) = \varphi^{-1}(b) \quad (b \in B), \quad j(*) = \{(*, 1)\}.$$

The first part fills exactly  $P(B)$ , and the second point is the first polynomial after that initial subclass. Thus  $j$  is initial and proper. Its completion contains  $B + 1$  initially and is a fixed point, by [Theorem XVI.6.17](#). Any fixed point properly extending  $B$  contains an initial copy of  $B + 1$ , by adjoining its least point outside the image of  $B$ . The universal property of the completion therefore proves leastness.  $\square$

Consequently the first hereditary fixed point is not a final boundary. It has a next fixed point, and every supplied fixed point has another one beyond it. Each externally specified finite repetition is available from these explicit constructions. Organizing all such extensions along a prescribed arbitrary class index is a further uniformity problem; it is not supplied merely by writing an indexed epsilon or Veblen symbol. A rigorous notation system must specify its set codes, its comparison rule, the admitted embeddings, and the existence principle used to assemble any genuinely transfinite family.

[merge] Theorems XVI.6.17 and XVI.6.18 and corollaries XVI.6.19 and XVI.6.20 are proved here only by source 35; none of sources 32–34 and nothing earlier in this report constructs a fixed-point completion. Source 35 keeps them over GBC and asks whether they hold over GB (its question 10.6, research question 113.270); they do (theorem XVI.6.21). The “repository equivalence” used in the proof of corollary XVI.6.19 is theorem 108.19 over GBC, and theorem XVI.5.8 over GB.

**Theorem XVI.6.21** ([merge]; completion without choice). *The following hold over GB, with ZF sets and no form of set choice or global choice.*

- (1) Source 35’s bounded-specialization principle, lemma XVI.4.8.
- (2)  $E$  is a class well-order and root formation is an isomorphism  $P(E) \cong E$  (note XVI.6.13).
- (3) The fixed-point completion theorem XVI.6.17: for every initial embedding  $j : A \rightarrow P(A)$  of class well-orders there are  $C(A, j)$ ,  $\iota$  and  $\Phi$  with all the properties stated there.
- (4) The local criterion theorem XVI.6.18; the characterization corollary XVI.6.19 in the form  $GB \vdash FPE \leftrightarrow CWO \leftrightarrow CTH$ , with its proper-extension variant; and the successor fixed point corollary XVI.6.20.

*Proof.* Apart from citing the GBC form of the history–comparison equivalence in the proof of corollary XVI.6.19 (replaced in (4) by theorem XVI.5.8), source 35’s proofs use global choice only to infer that a linear order is a class well-order from the absence of descending set sequences. Over GB use instead source 34’s lemma XVI.2.2: a class linear order is a class well-order exactly when every nonempty set subset has a least element. That condition quantifies only over sets, so it is elementary in the order.

(1) Let  $S$  be a nonempty set of terms of the class  $T$  of lemma XVI.4.8. The labels occurring in  $S$  form a set of ordinals, by Replacement, and so have a strict upper bound. By hypothesis there is a larger  $\kappa$  for which all terms with labels below  $\kappa$ , in particular all members of  $S$ , have a strictly order-preserving interpretation into a set ordinal. The member of  $S$  with the least interpretation is least in  $S$ , since  $S$  is linearly ordered. Thus every nonempty set subset of  $T$  has a least element, and lemma XVI.2.2 applies.

(2) Source 35’s  $\mathcal{E}$  and source 34’s  $E$  are the same class with the same order (note XVI.6.11); this is theorems XVI.6.3 and XVI.6.6.

(3) In the proof of theorem XVI.6.17 the stages  $A_n = P^n(A)$  and the maps  $j_n = P^n(j)$  are defined by one elementary formula uniformly in  $n \in \omega$ , with  $A$  and  $j$  as class parameters, through finite nested codes, and  $A_{n+1}$  is literally  $P(A_n)$  under this coding. Let  $\varphi(n)$  say:  $<_{A_n}$  is a strict linear order on  $A_n$ , every nonempty set subset of  $A_n$  has a  $<_{A_n}$ -least element, and  $j_n$  is an order embedding of  $A_n$  onto an initial subclass of  $A_{n+1}$ . Every quantifier of  $\varphi$  ranges over sets, so  $\varphi$  is elementary, and  $\{n \in \omega : \neg\varphi(n)\}$  is a set; if it were nonempty it would have a least element. Hence internal induction on  $n$  is available for  $\varphi$ .  $\varphi(0)$  holds because  $A$  is a class well-order,  $P(A)$  is linearly ordered and  $j$  is initial. Suppose  $\varphi(n)$ . By lemma XVI.2.2,  $A_n$  is a class well-order; then  $A_{n+1} = P(A_n)$  is a class well-order by theorem XVI.4.2, so it is linear and its nonempty set subsets have least elements;  $A_{n+2} = P(A_{n+1})$  is linear by the finite lexicographic comparison; and  $j_{n+1} = P(j_n)$  is initial by note XVI.6.16, whose proof only transports finite supports. So  $\varphi(n+1)$

holds, and  $\varphi(n)$  holds for every  $n \in \omega$ . By [lemma XVI.2.2](#) once more, applied with  $n$  as a parameter, each  $A_n$  is a class well-order. This replaces the paragraph of source 35’s proof that invokes global choice (“Uniform finite induction is legitimate here: in GBC, . . .”); its alternative route through a proposed descending sequence also needs choice to produce the sequence.

The rest of source 35’s proof makes no choice. The carrier of  $C(A, j)$  consists of least-stage representatives, defined by an elementary formula, and its order compares images in a common finite stage. A nonempty subclass of the colimit has a point in some  $\iota_n[A_n]$ ; the intersection of the subclass with that stage has a least element because  $A_n$  is a class well-order, and initiality makes it least overall. The isomorphism  $\Phi$  decodes finite supports. If  $j$  is proper,  $d_n$  is the least element of the nonempty class  $A_{n+1} \setminus j_n[A_n]$ , uniquely specified, and Class Replacement collects  $(d_n)_{n < \omega}$ . For the universal property, the maps  $h_n$  are defined by finite evaluation uniformly in  $n$ ; “ $h_n$  is an initial embedding of  $A_n$  into  $F$ ” is elementary, and induction on  $n$  proves it by [note XVI.6.16](#) and the isomorphism  $\varphi : P(F) \cong F$ ;  $h_{n+1} \circ j_n = h_n$  because two initial embeddings between class well-orders agree ([lemma XVI.2.5](#), whose least-disagreement proof is choice-free).

(4) In [theorem XVI.6.18](#) the forward implication is (3). The converse uses that the monomial map  $g$  is an order embedding of a class well-order into itself, hence inflationary ([lemma XVI.2.5](#)), and initiality; the final equivalence is [note XVI.5.18](#) with  $\Gamma = W$ , which holds over GB. For [corollary XVI.6.19](#), apply the local criterion to each input and replace the GBC equivalence [theorem 108.19](#) by its GB form [theorem XVI.5.8](#); for the proper variant, apply it to  $W + 1$ . [Corollary XVI.6.20](#) uses only (3) and the universal property.  $\square$

[merge] This answers source 35’s question 10.6 ([research question 113.270](#)) positively. The point source 35 raises there, the “internal uniformity of the iterated stages”, is exactly the induction on  $\varphi$ : [lemma XVI.2.2](#) makes “ $A_n$  is a class well-order” equivalent to an elementary property of  $n$ , so the induction is internal to the model and needs no metatheoretic iteration and no class-valued recursion. Source 35’s leastness ([note XVI.6.15](#)), tower cuts ([note XVI.6.14](#)) and bounded specialization at an indecomposable base ([lemma XVI.6.12](#)) use no choice as printed. The proof is this write’s; no source states it. It had not been reviewed when written; an independent adversarial check of 4 October 2026 ([section XVI.12](#)) found it valid, with no mathematical gap, after the two editorial corrections recorded next.

**Review remark H4 (retained dependency correction).** [merge] [corrected after independent check, 4 October 2026] In (1) the class of terms of [lemma XVI.4.8](#), printed  $T$  there, was misprinted  $T_m$  twice. The first sentence of the proof said that source 35’s proofs use global choice only for that inference; its proof of [corollary XVI.6.19](#) also cites the GBC equivalence [theorem 108.19](#), which item (4) already replaces by [theorem XVI.5.8](#), so the argument was complete and only the sentence is reworded. The check also notes that in (3) the finite-trace definitions of membership and order in  $A_n$  agree with  $P$  applied to  $A_{n-1}$ ; that agreement holds by construction and can be proved by the same induction, folded into  $\varphi(n)$ .

## XVI.6.6 Source 35: a systematic notation discipline

The purpose of a notation is to make a specified family of orders usable. A notation should identify its carrier, its comparison rule, the axioms proving well-ordering, and the meaning of its equality. There is no need to pretend that every class order has one canonical set-valued name. The preceding results instead suggest several compatible levels of notation.

### XVI.6.6.1 Order constructors and labelled normal forms

At the first level, a term is an instruction for constructing a presentation. Constants name the empty order, the singleton order, set ordinals,  $\Omega$ , and explicitly supplied class well-orders. Binary constructors form  $A + B$ ,  $A \cdot B$ , and the finite-support power  $A^{[B]}$ . A family constructor  $\sum_{i \in I} W_i$  is permitted only when its input is one uniform class relation. An initial direct limit constructor includes its actual connecting maps as data. Consequently every constructor has a finite, well-typed specification.

At the second level, once an exponent order  $\Gamma$  is fixed, a polynomial code is

$$q = ((i_0, \alpha_0), \dots, (i_{r-1}, \alpha_{r-1})), \quad i_0 >_\Gamma \dots >_\Gamma i_{r-1}, \quad 0 < \alpha_j \in \text{Ord}.$$

The code is a set. Its displayed notation may be written

$$\sum_{j < r} \Omega^{\langle i_j \rangle_\Gamma} \alpha_j,$$

where  $\langle i \rangle_\Gamma$  means the initial segment  $\Gamma_{<i}$ , not an unexplained ordinal assigned to  $i$ . This superscript convention is optional when  $\Gamma$  is the ordinary ordinal class, but indispensable for an arbitrary class exponent order.

*Note XVI.6.22* (Source 35, Proposition 6.1, “Meaning of a polynomial as a cut”). *For the preceding code  $q \in P(\Gamma)$ , its strict initial segment has the explicit order decomposition  $P(\Gamma)_{<q} \cong \sum_{j < r} P(\Gamma_{<i_j}) \cdot \alpha_j$ . Thus the displayed polynomial is an actual finite sum of class well-orders. The assertion holds in GB.* This is source 34’s (S34.4), by the same partition of the predecessors of  $q$  by the greatest exponent at which they fall below it.

This proposition separates two objects that a compact notation often conflates:  $q$  is one set code, whereas the order denoted by the cut below  $q$  can be a proper class. For example, the code with coefficient 1 at exponent 1 and zero elsewhere is a set, but its predecessor order in  $P(\text{Ord})$  is  $\Omega$ . A notation system for orders beyond  $\text{Ord}$  is therefore possible without any individual notation being a proper-class object.

### XVI.6.6.2 Worked comparisons and arithmetic

Finite polynomials with ordinal exponents provide the following strict comparisons, all witnessed inside the same polynomial order:

$$\Omega + 1 < \Omega \cdot 2 < \Omega \cdot \omega < \Omega^2 < \Omega^2 + \Omega \cdot \omega < \Omega^3.$$

Here and in worked displays within one notation order,  $<$  abbreviates the proper initial-segment comparison of the represented cuts. Compare the greatest exponent of disagreement; if it agrees, compare its ordinal coefficient. No search through all lower exponents is needed.

For instance,

$$q = \Omega^3 \cdot 2 + \Omega \cdot \omega + 7, \quad r = \Omega^3 \cdot 2 + \Omega \cdot (\omega + 1)$$

satisfy  $q < r$ . Their exponent-3 coefficients agree. At exponent 1, the coefficient  $\omega$  is smaller than  $\omega + 1$ , so the finite tail 7 does not affect the comparison.

Ordinal addition is visibly noncommutative. The identities

$$(\Omega^2 + \Omega \cdot 3 + 7) + (\Omega \cdot 4 + 5) = \Omega^2 + \Omega \cdot 7 + 5$$

and

$$(\Omega \cdot 4 + 5) + (\Omega^2 + \Omega \cdot 3 + 7) = \Omega^2 + \Omega \cdot 3 + 7$$



hold as presentation isomorphisms. In the first calculation the later  $\Omega$ -block absorbs the earlier finite tail. In the second it absorbs the entire earlier summand. More generally, when adding a nonzero right-hand polynomial, retain the left-hand exponents greater than its leading exponent, add the two coefficients at that exponent in the *ordinal* sense, and retain the right-hand lower tail. Each coefficient sum is a set ordinal, so there is no carry into the class base  $\Omega$ .

For products, the direction of the factors must remain explicit:

$$(\Omega + 1) \cdot \omega = \Omega \cdot \omega, \quad \omega \cdot (\Omega + 1) = \Omega + \omega.$$

The first has  $\omega$  successive proper blocks. In the second, the initial  $\omega \cdot \Omega$  is set-like and hence has type  $\Omega$ , followed by one copy of  $\omega$ . The equality

$$\Omega^{[\Omega+1]} \cong \Omega^{[\Omega]} \cdot \Omega$$

comes directly from splitting the final exponent coordinate. It does not depend on a recursive definition taking a supremum over a proper collection of class objects.

The absorption laws for a set base are particularly instructive:

$$2^{[\Omega]} \cong \omega^{[\Omega]} \cong \Omega, \quad \Omega^{[\Omega]} \not\cong \Omega.$$

The first presentations have set-sized predecessors and proper carriers. In the last one, the predecessor class below the monomial at exponent 1 is already proper, so the order is not set-like. This is a structural proof of the difference, rather than a symbolic rule about an imagined largest ordinal.

### XVI.6.6.3 Hereditary terms and the first exponential closure

In the hereditary system of the preceding section, the exponents themselves are terms. Comparison descends through finitely many subtrees until it reaches ordinal coefficients. The following is a typical legal term:

$$\Omega^{\Omega^2 + \Omega \cdot 3 + 5} \cdot \omega + \Omega^{\Omega + 7} \cdot 4 + 9.$$

Its meaning is a cut in the class of hereditary terms. The term's finite tree is a set even though its first exponent already denotes a proper-class order.

Let  $\mathbf{E}$  be the complete hereditary term order. The finite tower markers give a strictly increasing cofinal sequence in  $\mathbf{E}$ ; every single term lies below some such marker. Hence  $\mathbf{E}$  is longer than every finite tower stage and has a specified isomorphism  $P(\mathbf{E}) \cong \mathbf{E}$ . The notation  $\mathbf{E}$  names the whole order, while a term  $t$  names one of its proper initial cuts. This avoids representing the entire fixed point by one of its own elements.

There are two useful ways of continuing. One may introduce a new named constant for the order  $\mathbf{E}$  and construct its successor, products, and powers. Or one may apply the fixed-point completion construction to  $\mathbf{E} + 1$ , obtaining a larger exponential fixed point. Repeating a fixed, explicitly specified finite construction is legitimate. Claiming a universal class-valued recursion through all class well-orders requires a further principle and is not built into this syntax.

### XVI.6.6.4 What is effective, and what is only definable

There are four distinct levels of computational content.

1. A *mathematical class presentation* permits arbitrary set ordinal coefficients. These coefficients have exact mathematical order, but they are not all finite strings.

2. A *relative comparison procedure* compares hereditary terms using a supplied comparison procedure for their coefficient notations. Finite structural recursion then suffices.
3. An *effective notation language* restricts coefficients to a chosen recursively presented ordinal system. It has countably many expressions and names only the corresponding selected cuts.
4. A *proof-carrying presentation* stores, with each new constructor, a proof of well-order preservation or an initial embedding certificate. This is the appropriate interface for a formal mathematics library.

The small executable artifact accompanying this article implements the finite-tree comparison idea with nonnegative integer coefficients. It also checks the finite-support split and flatten maps against ordinary finite ordered sets. Its task is to make the conventions reproducible; it is not a decision procedure for arbitrary definable class well-orders or a formal proof of the theorems above.

[merge] The executable artifact mentioned above is shipped as `code/35-completion-finite_notation_checks.py`, with its recorded output `data/35-completion-finite_notation_results.json` ([appendix A.13](#)).

### XVI.6.6.5 An interface for extensible notations

A useful formal specification associates four items with each presentation: a predicate for legal codes, a comparison relation, a stated well-order theorem, and a semantics relating codes to initial cuts. A constructor additionally records how it acts on class maps. For example, the exponent-order constructor  $P$  acts on an initial embedding by replacing each exponent label by its image. This preservation law is what makes the completion theorem possible.

A notation package should therefore expose more than a pretty printer. It should expose the explicit initial embedding of one stage into the next, the inverse on its image, and the map that interprets a finite code. Such witnesses permit comparisons and direct limits in weak class theory even where unrestricted class well-order comparability is unavailable. The definability theorem in the next section shows why this disciplined extensibility is preferable to promising one complete language for all class orders.

### XVI.6.7 Source 32: finite terms beyond every finite exponential tower

The finite-support construction can itself be iterated inside finite syntax. This gives an exponential fixed-point presentation without performing a class-valued recursion of unbounded length.

*Note XVI.6.23* (Source 32, Definition 4.1, “Hereditary digit terms”). *Fix a class well-order  $A$  with least points  $0_A < 1_A$ . A hereditary  $A$ -term is either 0, or a finite list  $t = [(t_0, a_0), \dots, (t_{m-1}, a_{m-1})]$  where the  $t_i$  are hereditary terms in strictly decreasing order and  $a_i \in A \setminus \{0_A\}$ . Compare nonzero terms by their leading exponents, then their leading coefficients, and then their remaining lists; an exhausted list is smaller than a nonexhausted one. The term 0 is least.* Source 32 writes  $\mathcal{H}(A)$  for this presentation, printed  $\mathbf{E}(A)$ ; it is source 34’s general base of [corollary XVI.6.10](#), and for  $A = \Omega$  it is  $\mathbf{E}$  up to identifying coefficients.

This definition is a finite-tree definition, not a recursive specification along a proper-class well-order. Membership and comparison are computed by recursion on the finite heights of the input trees (or the maximum of the heights of two trees). The requirement that the exponent list decrease is checked using comparisons of strictly smaller trees. Terms are finite sets of labels and finite-tree structure.

*Note XVI.6.24* (Source 32, Theorem 4.2, “A hereditary fixed-point presentation”). In GBC,  $E(A)$  is a class well-order and there is an explicit isomorphism

$$E(A) \cong A^{[E(A)]}. \quad (\text{S32.4})$$

The constant terms form a proper initial copy of  $A$ . There is a set-coded increasing  $\omega$ -sequence cofinal in  $E(A)$ . This is source 34’s [corollary XVI.6.10](#) (over GB), with the cofinality of [theorem XVI.6.5](#). Source 32’s proof is source 34’s: evaluate the terms occurring in a set of terms at the ordinal type  $a \geq 2$  of their coefficient labels,  $\text{val}(t) = \sum_{i < m} a^{\text{val}(t_i)} \rho(a_i)$ , which preserves and reflects comparison by [lemma XVI.4.9](#); a descending sequence would give a descending sequence of ordinals, and [lemma XVI.2.8](#) concludes in GBC. The constants are the segment below  $[(1, 1_A)]$  with  $1 = [(0, 1_A)]$ , and the cofinal sequence is  $v_0 = 1$ ,  $v_{n+1} = [(v_n, 1_A)]$ , source 34’s  $\tau_n$ .

For  $A = \Omega$ , these terms provide an explicit proper-class order above all finite towers based at  $\Omega$ , with an isomorphism to its own  $\Omega$ -power. More precisely, the segment below  $v_n$  has the corresponding finite-tower presentation: root decomposition identifies the segment below  $v_{n+1}$  with  $A^{(E(A))^{<v_n}}$ , starting with the one-point segment below  $v_0 = 1$ .

This is a useful answer to “how far can one stretch?” but it is not an endpoint. Adjoining a last point produces a strictly larger order by [note XVI.2.10](#). One can also repeat finite-term constructions over the new base. The significance is that these examples are already definable by finite syntax in GBC; their existence is not evidence that GBC supports arbitrary recursion along them.

## XVI.6.8 Source 33: coherent towers of powers and a fixed point

### XVI.6.8.1 An explicit tower, not an arbitrary class recursion

Define presentations

$$T_0 = \Omega, \quad T_{n+1} = \Omega^{[T_n]} \quad (n < \omega).$$

These can be coded uniformly by finite rooted trees: leaves carry ordinal labels, a node carries a finite descending list of exponent subtrees and nonzero ordinal coefficients, and a level label specifies the required nesting depth. Validity and comparison are obtained by finite bottom-up evaluation on the code tree. In particular, a set records the entire finite evaluation at every node. No truth predicate for the whole universe is required.

This description supplies one elementary definition of the family of carriers and orders. First-order induction on  $n$ , using [note XVI.4.12](#), verifies that every stage is a class well-order. This is different from asserting that *every* arbitrary class-valued recursion of length  $\omega$  has a solution.

Embed  $T_0$  into  $T_1$  by sending  $\alpha$  to the constant polynomial with coefficient  $\alpha$  at the least exponent, with zero sent to the empty polynomial. Call this  $i_0$ . Inductively, let  $i_{n+1}$  replace every exponent by its image under  $i_n$ , leaving coefficients unchanged. These maps also have uniform finite-tree definitions.

**Lemma XVI.6.25.** *Every  $i_n : T_n \rightarrow T_{n+1}$  is a proper initial embedding.*

*Proof.* The constants form a proper initial segment of  $T_1$ . More generally, if  $j : A \rightarrow B$  has initial image, the induced exponent map  $\Omega^{[A]} \rightarrow \Omega^{[B]}$  has initial image: any polynomial below an image polynomial cannot have a nonzero coefficient at an exponent outside the initial image. If  $j$  is proper, a monomial at the least omitted exponent is not in that image. Apply induction.  $\square$

### XVI.6.8.2 The direct limit

Take the disjoint union of all pairs  $(n, t)$  with  $t \in T_n$ . Identify  $(n, t)$  and  $(m, s)$  when their images agree at a common later finite stage. Each equivalence class has a unique representative at its least level. Use these least-level representatives as a class carrier  $\mathbb{T}$ . Comparison is computed by lifting two representatives to the larger of their levels. Thus both equality and order on  $\mathbb{T}$  are elementary from the finite-tree coding.

**Theorem XVI.6.26** (A definable fixed point beyond finite towers). *The direct limit  $\mathbb{T}$  is a class well-order, every  $T_n$  is a proper initial suborder of  $\mathbb{T}$ , and there is a definable order isomorphism*

$$\Omega^{[\mathbb{T}]} \cong \mathbb{T}.$$

*This construction is available in GBC.*

*Proof.* The coherent initial embeddings make each  $T_n$  an initial suborder of the limit. For a nonempty subclass  $S \subseteq \mathbb{T}$ , choose  $x \in S$  and a stage  $T_n$  containing it. The intersection of  $S$  with that stage has a least point. Every point of  $\mathbb{T}$  below  $x$  already belongs to that same initial stage, so this least point is least in  $S$ . Thus  $\mathbb{T}$  is well-ordered. Properness follows from [lemma XVI.6.25](#).

An element of  $\Omega^{[\mathbb{T}]}$  uses only finitely many exponent points of  $\mathbb{T}$ . All those points lie in a common finite stage  $T_n$ . Interpret the same polynomial as a point of  $T_{n+1}$ , hence of  $\mathbb{T}$ . A different sufficiently large choice of stage gives the same limit point, by the definition of  $i_{n+1}$ .

This map is order preserving because any two finite supports can be put into one stage. It is onto because every point of a stage  $T_{n+1}$  already is a polynomial over  $T_n$ , and every point of  $T_0$  is represented among the constants of  $T_1$ . It is therefore the required isomorphism.  $\square$

The fixed point is not a ceiling:  $\mathbb{T} + 1$  again strictly exceeds  $\mathbb{T}$  under embeddability. Further explicit notation systems can be built, but their uniform definition and well-foundedness must be checked rather than inferred from suggestive ordinal symbols. The diagonal theorem below rules out one uniform system containing every definable possibility in its own language.

[merge] Source 33's direct limit, which it writes  $E$ , is printed  $\mathbb{T}$ :  $E$  is a floor equivalence in Section XVI.5 and  $E$  is the hereditary order.  $\mathbb{T}$  is  $E$  up to a canonical isomorphism: source 33's embedding  $i_0$  of  $T_0 = \Omega$  into  $T_1$  is source 35's seed embedding  $j$  for  $A = \text{Ord}$ , its maps  $i_{n+1}$  are  $P(i_n)$ , so  $\mathbb{T}$  is source 35's completion  $C(\text{Ord}, j)$ , which source 35 identifies with the hereditary system (the paragraph after [theorem XVI.6.17](#)). Directly: by [theorem XVI.6.7](#), applied to the isomorphism  $\Omega^{[\mathbb{T}]} \cong \mathbb{T}$  of [theorem XVI.6.26](#),  $E$  embeds initially into  $\mathbb{T}$ ; by the universal property in [theorem XVI.6.17](#), applied to root formation  $P(E) \cong E$  and the initial copy of  $\text{Ord}$  formed by the constant terms,  $\mathbb{T}$  embeds initially into  $E$ ; both composites are initial self-embeddings of class well-orders, hence identities ([lemma XVI.2.5](#)), so both maps are isomorphisms. Source 33 constructs  $\mathbb{T}$  in GBC; the construction works in GB by [theorem XVI.6.21](#) (3). The finite tower  $T_n$  is source 34's  $E^{\leq n+1} = E_{<\tau_{n+1}}$  ([theorem XVI.6.5](#)).

## XVI.7 Uniform catalogs and truth jumps

The preceding notation systems describe substantial families of orders. A different question is whether one uniform notation system can encompass every available class well-order. There is a direct obstruction, which also leads to an explicit construction showing how an added truth predicate increases the definable order spectrum. The results in this section are proved as structural consequences; no claim of historical priority is intended.

Write  $C \hookrightarrow D$  when there is a class order embedding of  $C$  into  $D$ . This notation requires neither a cofinal image nor an initial image. In particular, we will not substitute the general class comparison principle for an embedding argument.

### XVI.7.1 A diagonal sum for uniform families

**Definition XVI.7.1** (Uniform class family). A uniform family  $(C_i)_{i \in I}$  consists of class relations  $D(i, x)$  and  $R(i, x, y)$ : the domain of  $C_i$  is  $\{x : D(i, x)\}$ , and its strict order is the corresponding slice of  $R$ . Thus the family is represented by classes of *set tuples*. An external list of formulas is not, by itself, such a uniformly interpreted family.

**Theorem XVI.7.2** (Uniform-family diagonal). *Work in GB, and let  $(I, <_I)$  be an available class well-order. If  $(C_i)_{i \in I}$  is a uniform family of class well-orders, then*

$$\Delta = \sum_{i \in I} (C_i + 1)$$

*is a class well-order. For every  $i \in I$ , the order  $C_i + 1$  embeds as a convex block of  $\Delta$ , while*

$$\Delta \not\hookrightarrow C_i.$$

*If  $I$  is proper, then  $\Delta$  is proper.*

*Proof.* Use tagged set codes  $\langle i, 0, x \rangle$  for  $x \in C_i$  and  $\langle i, 1 \rangle$  for the last point of its block. Order these codes first by  $<_I$  and then by the order of  $C_i + 1$ . Elementary comprehension supplies both the domain and the relation.

Let  $S$  be a nonempty subclass of this order. The class of indices of blocks meeting  $S$  has a least member  $i$ . The intersection of  $S$  with that block has a least member, since  $C_i + 1$  is a class well-order. That member is least in  $S$ , proving well-foundedness.

The inclusion of the  $i$ th block is explicit. If an embedding  $f : \Delta \rightarrow C_i$  existed, its restriction to that block would embed  $C_i + 1$  into  $C_i$ , contrary to the successor obstruction [lemma XVI.3.2](#). This contradiction excludes *every* available class embedding, including one using additional class parameters. Finally,  $i \mapsto \langle i, 1 \rangle$  injects  $I$  into  $\Delta$ ; if the latter were a set, class replacement applied to the inverse would make  $I$  a set.  $\square$

**Corollary XVI.7.3** (No uniform cofinal catalog). *In GBC, no uniform class-indexed family of class well-orders is cofinal for order embedding among all class well-orders. In particular, no such family represents every class well-order up to class isomorphism.*

*Proof.* Restrict a global well-order to the index class and apply [Theorem XVI.7.2](#). Its diagonal sum embeds into no catalog member.  $\square$

This remains an obstruction when the proposed catalog is redundant or ineffective. What matters is uniform interpretation as actual class relations. If the family and its index order are definable from fixed parameters, the diagonal sum is definable from the same parameters. Consequently such a family cannot even be cofinal among all well-orders definable from those parameters. This does not prohibit partial notation systems, bounded fragments, or interpretation using stronger parameters. It identifies why simply enumerating formulas does not produce a universal notation system: their intended interpretations have not thereby been collected into one class relation.

### XVI.7.2 A well-order supplied by one truth predicate

Fix a set-like global well-order  $G$  of  $V$  and a class parameter  $A$ ; finitely many class predicates can be combined into  $A$  by tags. We do not assume that  $G$  is definable from  $A$ . Let  $T$  be a full elementary satisfaction class for

$$(V, \in, A, G).$$

Here “full” means that  $T$  evaluates all first-order formula codes with finite set-valued assignments and satisfies the compositional Tarskian clauses. It is an additional assumed class, not something being defined in the language whose truth it expresses. The established connection between truth predicates and elementary transfinite recursion provides the foundational context [22].

The key to a uniform catalog at this stronger level is that class well-foundedness has a first-order test in GBC. Namely, a class relation is well-founded exactly when there is no set-coded descending  $\omega$ -sequence. In the converse direction, if a nonempty class has no minimal member,  $G$  selects a predecessor of each of its points; ordinary set-valued recursion on  $\omega$  yields a descending sequence. This is not class-valued ETR. The sharper GB + DC version is proved in [22, Section 4].

Index pairs of domain and relation formulas, together with a packed set parameter, by

$$i = (e, f, p), \quad \varphi_e(x, p; A, G), \quad \psi_f(x, y, p; A, G).$$

Restrict to valid syntactic codes of these arities, and order the resulting proper index class  $I$  by  $G$ . Packing finite tuples into  $p$  includes every presentation with finitely many set parameters.

From  $e, f$  one effectively obtains the code of a first-order formula  $\text{WO}_{e,f}(p; A, G)$ . It asserts that  $\psi_f$  is a strict linear order on the  $\varphi_e$ -domain, and that no set function  $s$  on  $\omega$  has all its values in that domain with

$$\psi_f(s(n+1), s(n), p; A, G) \quad \text{for every } n \in \omega.$$

Consequently

$$\text{Valid}(i) \iff T(\ulcorner \text{WO}_{e,f} \urcorner, p)$$

is a uniform elementary condition using  $T$ . Define  $W_i$  by interpreting  $\varphi_e, \psi_f$  through  $T$  when  $i$  is valid, and by taking the empty order otherwise. Explicitly, its domain is given by

$$x \in W_i \iff \text{Valid}(i) \wedge T(e, x, p),$$

with  $x <_{W_i} y$  defined by  $x, y \in W_i$  and  $T(f, x, y, p)$ . By the compositional clauses and the preceding test, every fiber is a class well-order. Invalid formulas cannot introduce an ill-founded summand.

**Theorem XVI.7.4** (Relative truth jump). *Under the preceding GBC assumptions, let*

$$J(T; A, G) = \sum_{i \in I}^G (W_i + 1).$$

*This is a proper class well-order definable from  $T, A, G$ . For every class well-order  $C$  whose domain and order are first-order definable from  $A, G$  and finitely many set parameters,*

$$C + 1 \hookrightarrow J(T; A, G), \quad J(T; A, G) \nrightarrow C.$$

*The first embedding identifies  $C + 1$  with a convex block. In particular,  $J(T; A, G)$  is not class-isomorphic to any such  $C$ .*



*Proof.* The satisfaction class uniformly interprets the fiber family, so [Theorem XVI.7.2](#) applies. A presentation of  $C$  occurs at the index formed by its formula codes and packed parameters. Its fiber is exactly  $C$ , which gives the asserted block and rules out every reverse embedding. Each index contributes its last point, ensuring properness.  $\square$

This gives a precise sense in which truth increases the order spectrum: one newly definable order dominates every earlier definable presentation and cannot embed back into any of them. It cannot itself have an  $A, G$ -definable presentation with set parameters, since that presentation would be one of the fibers to which the theorem applies. The obstruction concerns order types, not only the definability of a particular isomorphism.

In bare GBC these conclusions concern order embeddings. If the ambient theory additionally proves comparison, for example in  $\text{GBC} + \text{ETR}$ , then each  $C$  is isomorphic to a proper initial segment of  $J(T; A, G)$ : comparison in the other direction is excluded by the reverse-embedding obstruction. The argument needs no class choice or second-order comprehension to gather the fibers. Their simultaneous interpretation comes from the supplied  $T$ .

### XVI.7.3 Iteration and the scope of the notation

Let  $\Gamma$  be a class well-order, include its domain and relation among the base predicates  $A$ , and suppose a  $\Gamma$ -iterated truth predicate  $U$  is available. Its slice  $U_\gamma$  is a truth predicate for

$$(V, \in, A, G, U_{<\gamma}).$$

The earlier aggregate is represented by set pairs  $\langle \delta, z \rangle$ ; it is not a class whose elements are classes. The construction above is uniform, so it defines a uniform family

$$J_\gamma = J(U_\gamma; A \oplus U_{<\gamma}, G),$$

where  $\oplus$  denotes the tagged join of predicates. Here the fixed syntactic translation between separate predicates and their tagged join is understood when applying  $U_\gamma$ .

**Corollary XVI.7.5** (Growth along iterated truth). *For  $\delta <_\Gamma \gamma$ ,*

$$J_\delta + 1 \hookrightarrow J_\gamma, \quad J_\gamma \not\hookrightarrow J_\delta.$$

*Proof.* From  $U_{<\gamma}$  and the set parameter  $\delta$ , one recovers the slice  $U_\delta$  and the restriction  $U_{<\delta}$ . Thus the fixed formulas constructing  $J_\delta$  can be expressed using  $A, G, U_{<\gamma}$  and parameter  $\delta$ . Apply [Theorem XVI.7.4](#) at stage  $\gamma$ .  $\square$

This conditional statement handles successor and limit stages alike. It assumes the whole truth iteration as one class; it does not infer that iteration from the separate existence of its slices. Arbitrary available class well-order lengths of iterated truth belong to the established ETR framework [22], while extensions obtained by adjoining truth predicates are studied in class-theoretic potentialism [84].

The symbols  $J_\gamma$  therefore provide systematic relative notation for a strictly increasing family in the stated embedding sense. They do not denote canonical absolute class ordinals: the ordering of formula codes and parameters through  $G$  affects the sum. Their robust content is the domination property and the definability environment in which they appear. Determining whether weaker hypotheses suffice for such dominating orders, or whether an appropriate domination principle recovers truth, remains a separate question; no converse follows from the construction.

**Corollary XVI.7.6** (Bounded-fragment jumps). *For each fixed natural number  $n$ , GBC admits an  $A, G$ -definable class well-order  $K_n$  dominating every class well-order whose domain and relation have  $\Sigma_n$  or  $\Pi_n$  definitions in the Lévy hierarchy of  $\mathcal{L}(\in, A, G)$ , with arbitrary finite tuples of set parameters. Each such  $C$  satisfies*

$$C + 1 \hookrightarrow K_n, \quad K_n \nrightarrow C.$$

*There is no single  $A, G$ -set-parameter-definable uniform family of class well-orders having these domination properties simultaneously for all  $n$ .*

*Proof.* For a fixed  $n$ , partial satisfaction for syntactically normalized  $\Sigma_n$  and  $\Pi_n$  formulas is elementary definable. Bounded matrices can be evaluated in a transitive set containing their assignments, with the predicates restricted to that set; the fixed number of unbounded quantifier blocks is then interpreted directly.

Use this partial evaluator to interpret the candidate domains and relations. Apply the first-order linearity and descending-sequence test to those interpreted relations, replacing invalid fibers by the empty order. Their uniform diagonal sum gives  $K_n$  by [Theorem XVI.7.2](#); a full truth predicate is unnecessary for this fixed fragment.

If all the resulting domination properties belonged to one definable uniform family, that family would be cofinal among all  $A, G$ -set-parameter definable class well-orders: every fixed defining formula has some finite Lévy complexity. Its definable diagonal sum would contradict that cofinality by [Theorem XVI.7.2](#).  $\square$

Thus separate constructions for every fixed complexity bound must be distinguished from an internally uniform sequence of those constructions. No assertion of separation at each pair of adjacent complexity levels is being made.

[merge] This report's earlier diagonal theorems work at the level of relations: no uniform class of codes lists every class well-order of **No** (source 15's [theorem 57.4](#); source 08's [note 19.41](#); [theorems 19.37](#) and [19.38](#)), because a diagonal enumeration differs from each row on one pair. [Theorem XVI.7.2](#) works at the level of order types: the diagonal sum embeds into no row at all. It is proved independently by all four sources ([note XVI.7.8](#), [note XVI.7.14](#), [note XVI.7.17](#)); [corollary XVI.7.3](#) is [corollary XVI.7.9](#) and [notes XVI.7.16](#) and [XVI.7.18](#). [Corollary XVI.7.6](#) is the bounded-complexity diagonal of source 33's [theorem XVI.7.20](#), printed in full below because source 33's corollaries use its parameter-free form, and of source 35's [note XVI.7.13](#); source 35's [theorem XVI.7.11](#) is the pure-language form on Ord.

#### XVI.7.4 Source 35: definability, uniform bounds, and complexity separation

An explicit notation can describe a very long class well-order without describing all such orders. The distinction becomes sharp when the interpretation of the notations is itself one class relation. We prove that every uniformly interpreted family has a strict upper bound, even for the comparatively weak relation of suborder embeddability. Combining this observation with partial satisfaction gives a separation theorem for the complexity of presentations of class order types.

The separation below is a consequence of elementary coding and well-foundedness; no claim of historical priority is made. Its role here is to strengthen the repository's obstruction to exhaustive lists of labelled orders. That earlier obstruction can omit another presentation of an already represented type. The obstruction below escapes every represented type, and even every type embedding into one of the represented orders.

### XVI.7.4.1 A uniform family is always bounded

Throughout this section an embedding is an order-preserving injection whose graph is an available class. The notation  $A \hookrightarrow B$  permits an arbitrary suborder as image. It does not assert the stronger initial-segment relation  $A \preceq_{\text{init}} B$ .

*Note XVI.7.7* (Source 35, Lemma 7.1, “An appended point cannot be absorbed”). *In GB, if  $W$  is a class well-order, then  $\neg(W+1 \hookrightarrow W)$ .* This is the second assertion of source 34’s lemma XVI.3.2 and of source 35’s own lemma XVI.2.5. The proof differs: instead of the least point moved downward, it iterates the restricted map from the image of the appended point and obtains a descending set sequence. It is printed as a second route.

*Source 35’s proof.* Suppose that  $f : W+1 \rightarrow W$  is an embedding, and let  $*$  denote the tagged appended point. Identify the original copy of  $W$  with  $W$  only for purposes of this proof, and put  $g(x) = f(x)$  on that copy. Set  $a_0 = f(*)$ . Since every original point precedes  $*$ , we have  $g(a_0) <_W a_0$ . Define  $a_{k+1} = g(a_k)$  for  $k < \omega$ . Order preservation gives  $a_{k+2} <_W a_{k+1}$  whenever  $a_{k+1} <_W a_k$ . Ordinary set-valued recursion and Class Replacement produce this sequence as a set. Its range has no  $W$ -least element, contradicting well-foundedness. The tags ensure that no set outside the original carrier has been tacitly assumed to exist.  $\square$

*Note XVI.7.8* (Source 35, Theorem 7.2, “Strict bound for a uniform family”). *Work in GB. Let  $I$  carry a supplied class well-order  $\triangleleft$ , and let  $(W_i)_{i \in I}$  be a uniform family of class well-orders. There is a class well-order  $U$ , elementary definable from these data, such that for every  $i \in I$ ,  $W_i \hookrightarrow U$  and  $\neg(U \hookrightarrow W_i)$ .* *In GBC, the conclusion applies to a family indexed by any class of sets, since Global Choice supplies the required order of  $I$ .* This is source 34’s theorem XVI.7.2 with  $U = \sum_{i \in (I, \triangleleft)} (W_i + 1)$  (source 34’s  $\Delta$ ), and its last clause is corollary XVI.7.3’s proof; the proof is the same: least occupied row, then least point in the row, and a row inclusion composed with  $U \hookrightarrow W_i$  would embed  $W_i + 1$  into  $W_i$  (note XVI.7.7).

**Corollary XVI.7.9** (No cofinal uniform notation system). *In GBC, no uniformly class-coded family of class well-orders is cofinal among all class well-orders under suborder embeddability. In particular, no such family represents every class well-order type. If strong comparability CWO is additionally assumed, the bound in Note XVI.7.8 is also a strict upper bound under initial-segment comparison.*

*Proof.* Apply the theorem to the proposed family. Its output  $U$  embeds into none of its members, which contradicts cofinality. Under CWO, comparison of  $W_i$  and  $U$  must give an initial-segment embedding  $W_i \preceq_{\text{init}} U$ : the reverse alternative would in particular give  $U \hookrightarrow W_i$ . Equality of types is excluded by the same argument.  $\square$

[merge] The first two sentences of corollary XVI.7.9 are source 34’s corollary XVI.7.3; the clause under CWO is source 34’s remark after theorem XVI.7.4 for its truth jump. Lemma XVI.7.10 is the ordinal-carrier case of the equivalence of lemma XVI.2.8, proved without choice by least-ordinal selection; source 34’s lemma XVI.2.2 gives, over GB, the set-subset form for every linear carrier.

The hypothesis concerns a semantic interpreter, not just a collection of strings. A syntax class  $T$  together with one class relation decoding the carrier and comparison relation of every term in  $T$  is covered by the corollary. Enlarging the alphabet, allowing ordinal parameters, or admitting terms of set size does not remove the obstruction if the result still has such an interpreter. The construction even gives an explicit next order: sum the successors of all currently interpreted orders. A proposed system cannot both interpret that entire sum and remain exhaustive.

This does not prohibit useful notation systems for large fragments, nor a metatheoretic scheme assigning a different defining formula to each notation. The latter need not supply a single internal

evaluation class. It also does not turn arbitrary suborder embeddings into initial-segment embeddings in GB; the final assertion of the corollary deliberately exposes that extra premise.

#### XVI.7.4.2 A definable test for well-foundedness on ordinal carriers

The fixed-complexity argument needs to select the well-orders among uniformly evaluated relations. This selection does not require a class quantifier.

**Lemma XVI.7.10** (Ordinal-carrier well-foundedness). *In GB, a strict linear class order  $R$  on Ord is a class well-order if and only if there is no set function  $f : \omega \rightarrow \text{Ord}$  satisfying*

$$f(k+1) R f(k) \quad (k < \omega).$$

*Consequently this property is elementary in the class parameter  $R$ .*

*Proof.* A descending sequence has a nonempty range without an  $R$ -least member. Conversely, suppose that an available nonempty subclass  $B \subseteq \text{Ord}$  has no  $R$ -least member. Choose its least member in the usual ordinal order as  $f(0)$ . Given  $f(k)$ , take the usual-ordinal-least member of

$$\{\beta \in B : \beta R f(k)\}$$

as  $f(k+1)$ . This class is nonempty by hypothesis, and the choice is unique. Set-valued recursion along  $\omega$ , followed by Class Replacement, gives the prohibited set function. No choice of a class witness is involved.  $\square$

For arbitrary class carriers, the corresponding equivalence is established in the standard class-recursion literature over GB + DC, and hence over GBC [22, discussion preceding Definition 6]. The ordinal carrier permits the direct least-ordinal selection above. In a nonstandard or externally ill-founded model, this is an internal criterion: it refers to the model's own set functions and available classes. External well-foundedness must be stated separately.

#### XVI.7.4.3 Strict separation at every fixed definability level

Fix a standard finite  $n \geq 1$ . A  $\Sigma_n$  presentation on Ord means a strict order relation given by a formula of the set-theoretic Lévy hierarchy, allowing a finite tuple of ordinal parameters. We use prenex representatives; over the present set theory the customary versions of the hierarchy give the same class of definable relations.

For this fixed  $n$  there is a first-order partial satisfaction formula  $\text{Sat}_{\Sigma_n}$  that evaluates all such formula codes. Its construction is a metatheoretic finite induction, not a uniform truth definition for arbitrary  $n$ . The distinction is explicit in the standard development: bounded truth is  $\Delta_1$ -definable, and the fixed  $\Sigma_n$  truth predicate is  $\Sigma_n$ -definable for  $n \geq 1$  [87, Lemma 2.13 and Definitions 2.14–2.15].

**Theorem XVI.7.11** (Fixed-complexity separation of class order types). *For every fixed standard  $n \geq 1$ , there is a parameter-free first-order definition of a class well-order  $U_n$  on Ord with the following properties, provably in GB. If  $R$  is a class well-order on Ord admitting a  $\Sigma_n$  definition with ordinal parameters, then*

$$R \hookrightarrow U_n, \quad \neg(U_n \hookrightarrow R).$$

*In particular,  $U_n$  has no  $\Sigma_n$  ordinal-parameter presentation up to available class isomorphism. Restricted to definable classes and definable maps, these statements are also theorems of ZFC understood as a scheme.*

*Proof.* A potential presentation is coded by  $c = (e, \vec{\alpha})$ , where  $e$  is a code for a  $\Sigma_n$  formula with the appropriate free variables and  $\vec{\alpha}$  is a finite tuple of ordinals. Define

$$\xi R_c \eta \iff \text{Sat}_{\Sigma_n}(e, \langle \xi, \eta, \vec{\alpha} \rangle).$$

Let  $I_n$  consist of exactly those codes for which  $R_c$  is a strict linear order on Ord and has no descending set function of length  $\omega$ . This is an elementary class definition: the displayed satisfaction formula has already been fixed, and [Lemma XVI.7.10](#) expresses the remaining condition using only set quantifiers. Every section indexed by  $I_n$  is therefore a class well-order.

The codes can be treated as nonempty finite tuples of ordinals. There is a parameter-free set-like well-order of all such tuples: compare their maximum coordinates first, their finite lengths second, and then use the lexicographic order at a fixed length. Every nonempty subclass has a least maximum, a least length at that maximum, and finally a least tuple in the indicated finite lexicographic order. The predecessors of a given tuple form a set, since their entries are bounded by its maximum coordinate and they belong to the set of finite sequences from that bound. Restrict this order to  $I_n$ .

[Note XVI.7.8](#) now constructs

$$S_n = \sum_{c \in I_n} (R_c + 1).$$

Every input  $R$  in the theorem is one of its sections, by correctness of the fixed partial satisfaction predicate. Thus  $R \hookrightarrow S_n$ , whereas an embedding  $S_n \hookrightarrow R$  would contradict [Note XVI.7.7](#).

It remains to put the carrier on Ord without adding a definable global choice assumption. The tagged carrier of  $S_n$  is itself parameter-free definably coded by finite ordinal tuples. Use the set-like tuple order just described on this carrier, and assign to each point the ordinal order type of its predecessor set. The assignment is definable and is an isomorphism with an initial subclass of Ord. The carrier is proper:  $I_n$  includes a code for the usual ordinal order, whose row already has a proper class of points. Hence the image cannot be a set ordinal, since Class Replacement applied to the inverse map would then make the carrier a set. The image is all of Ord.

Transport  $S_n$  across this canonical bijection and call the result  $U_n$ . All definitions used only the fixed natural number  $n$ , which can be built into the defining formula; there are no set parameters. Transport preserves the embedding conclusions. Finally, if a  $\Sigma_n$  ordinal-parameter presentation were isomorphic to  $U_n$ , that isomorphism would in particular embed  $U_n$  into an input order, a contradiction.  $\square$

The theorem separates order types, rather than merely the complexity of different names for one relation. Its negative conclusion allows any admitted embedding, even one defined using additional parameters or available as a nondefinable class. It does not claim separation of two consecutive syntactic levels: no optimal upper complexity for the output formula has been asserted. The same construction can be made for any fixed syntactic fragment having a definable partial evaluator, provided that its codes and permitted parameters have a supplied well-order.

**Corollary XVI.7.12** (The separation family has no uniform ordinal-parameter evaluator). *There is no first-order formula  $H(n, \xi, \eta, \vec{\gamma})$ , with one fixed finite tuple of ordinal parameters  $\vec{\gamma}$ , whose sections  $H_n$ , for every standard  $n \geq 1$ , are class well-orders on Ord satisfying the negative conclusion of [Theorem XVI.7.11](#) at level  $n$ .*

*Proof.* The fixed formula  $H$  is equivalent to a  $\Sigma_m$  formula for some standard finite  $m \geq 1$ . Its  $m$ -th section, with the additional ordinal parameter  $m$ , is itself a  $\Sigma_m$  ordinal-parameter presentation. The stipulated negative property for that section would therefore prohibit its identity embedding into itself.  $\square$

This is why the assertion *for each fixed  $n$*  cannot be silently rewritten as the existence of one definable sequence of the resulting class relations. The corollary has the stated ordinal-parameter scope. It does not, by this argument alone, exclude a uniform definition using an arbitrary non-ordinal- definable set parameter or a new truth predicate.

*Note XVI.7.13* (Source 35, Proposition 7.7, “Version with arbitrary set parameters”). *Fix an available global well-order  $G$  of the universe. In the language  $\{\in, G\}$ , and for each fixed standard  $n \geq 1$ , there is a  $G$ -definable class well-order strictly bounding all  $\Sigma_n(\in, G)$ -definable class well-orders with arbitrary set parameters, under suborder embeddability. The carrier and relation of an input presentation may both be defined by such formulas. No single formula in this language, even with additional fixed set parameters, uniformly provides such bounding orders for every standard  $n$ . This is source 33’s [theorem XVI.7.20](#) with its [corollary XVI.7.22](#), and source 34’s [corollary XVI.7.6](#) for  $A$  empty; source 35’s proof evaluates the two formulas by the fixed partial satisfaction predicate, filters by the GBC well-foundedness test, orders the codes by a set-like global well-order definable from  $G$ , applies [note XVI.7.8](#), and obtains the nonuniformity as in [corollary XVI.7.12](#), all fixed set parameters of a proposed uniform formula being among those allowed in its own offending section.*

Naming  $G$  is substantive. Global Choice as a class axiom does not assert that a global well-order is definable in the pure membership language. The proposition consequently gives a relative definability theorem, whereas [Theorem XVI.7.11](#) on ordinal carriers gives the unconditional pure-language version.

### XVI.7.5 Source 32: no uniformly definable family reaches the end

There is no largest class well-order, since  $W + 1$  always defeats  $W$ . A stronger statement excludes one uniformly coded family from being cofinal among all admitted class well-orders, even for the weaker arbitrary-embedding relation.

*Note XVI.7.14* (Source 32, Theorem 7.1, “Uniform diagonal domination”). *Let  $I$  be a nonempty class well-order and  $(W_i)_{i \in I}$  a uniformly coded family of class well-orders. Define*

$$D = \sum_{i \in I} (W_i + 1). \quad (\text{S32.10})$$

*Then  $D$  is a class well-order, every  $W_i$  embeds into  $D$ , and  $D$  embeds into none of the  $W_i$ . The theorem is provable in GBC without class-order comparability or ETR. This is source 34’s [theorem XVI.7.2](#), with the same proof by [proposition XVI.3.4](#) and [note XVI.2.10](#); source 34 shows it in GB. Source 32’s diagonal  $D$  is source 34’s  $\Delta$ ; it is not the condensation operator  $D_W$ .*

Notice what the theorem does *not* assert: the copy of  $W_i$  used in the sum need not be initial. No initial comparison is required for the contradiction. In an external setting where these presentations are ordinary well-orders, the usual ordinal comparison converts the conclusion into the strict inequality of order types.

**Corollary XVI.7.15** (A fixed finite syntax has a definable bound). *Fix finitely many first-order formula pairs for a domain and an order relation, with arbitrary set parameters and fixed admitted class parameters. There is one class well-order, definable from the same class parameters and a fixed global well-order, that does not embed into any valid well-order presentation defined by those formulas.*

*Proof.* Encode each finite tuple of set parameters by one set. A finite disjunction of the given formulas provides a uniform family indexed by a finite formula tag and the parameter code. By [lemma XVI.2.8](#), being a valid well-order is an elementary predicate on each fiber. Replace invalid



fibers by the empty order. Global Choice well-orders the index class, and [note XVI.7.14](#) supplies the desired bound. All definitions involve only the given finitely many formulas, so no truth predicate for arbitrary formulas is being used.  $\square$

[merge] [Corollary XVI.7.15](#) is the finite-list case of source 34’s [corollary XVI.7.6](#); source 32 uses it in [theorem XVI.8.9](#).

*Note XVI.7.16* (Source 32, Corollary 7.3, “No universal admitted list”). *There is no admitted uniform class-coded family that contains, up to admitted isomorphism, every admitted class well-order. More strongly, no such family is cofinal for the relation  $\hookrightarrow$  among all admitted class well-orders.* This is source 34’s [corollary XVI.7.3](#), by restricting a global well-order to the index class; an empty family is plainly not cofinal.

An external list of formulas does not contradict this result. It is easy externally to enumerate finite strings that might define classes. The missing internal object is a uniform satisfaction relation evaluating all those formulas. Confusing an external enumeration of syntax with an admitted family of semantic evaluations is precisely the mistake that the next two sections exploit constructively.

## XVI.7.6 Source 33: a diagonal theorem for uniform families

### XVI.7.6.1 Strict domination without a comparison theorem

*Note XVI.7.17* (Source 33, Theorem 7.1, “Uniform-family diagonal”). *Let  $(W_i)_{i \in I}$  be a uniform family of class well-orders, where  $I$  is a nonempty class well-order. Define*

$$D = \sum_{i \in I} (W_i + 1). \quad (\text{S33.7.1})$$

*Then  $D$  is a class well-order and, for every  $i \in I$ ,*

$$W_i \hookrightarrow D, \quad D \not\hookrightarrow W_i. \quad (\text{S33.7.2})$$

*The construction and proof are available in GBC.* This is source 34’s [theorem XVI.7.2](#) and source 32’s [note XVI.7.14](#), with the same proof ([notes XVI.2.16](#) and [XVI.3.9](#)).

The theorem is stronger than a statement that a family has no maximum. It produces one order that strictly dominates *every member at once*, without choosing isomorphisms between members or comparing their lengths. It also works when some fibers are empty.

*Note XVI.7.18* (Source 33, Corollary 7.2, “No uniform catalogue”). *There is no class-coded family, indexed by sets, that contains up to isomorphism every available class well-order. More strongly, no such family is cofinal under embeddability among all available class well-orders.* This is source 34’s [corollary XVI.7.3](#) and source 32’s [note XVI.7.16](#).

This is an appropriate class-order version of the obstruction behind Burali–Forti, but its proof does not illegally collect all classes as objects. It quantifies over an already supplied uniform family and constructs one new relation from it.

### XVI.7.6.2 What remains possible

A language can have a set of finite expressions and a definable evaluator for a useful fragment: sums, products, finite powers, or the coherent tower of [section XVI.6.8](#). The obstruction says that such an evaluator cannot be cofinal among *all* well-orders definable in the same language if its evaluation is

itself definable there. One must increase complexity, add a truth predicate, weaken the coverage claim, or move to a stronger semantic setting.

In a theory with initial comparability, (S33.7.2) can be read as strict inequality of the usual external ordinal-like lengths. In the base theory it is deliberately stated using ordinary embeddings, where the proof is unconditional.

### XVI.7.7 Source 33: definability levels and parameter-free cofinality

[merge] Theorem XVI.7.20 is source 34's corollary XVI.7.6 in the language  $\{\in, G\}$ , with the diagonal order shown free of set parameters; source 35's note XVI.7.13 is the same statement, and its theorem XVI.7.11 removes  $G$  for orders on Ord with ordinal parameters. Lemma XVI.7.19 is the partial evaluator of source 34's proof of corollary XVI.7.6 and of source 35's paragraph before theorem XVI.7.11. Corollary XVI.7.21 is the external cofinality behind source 34's theorem XVI.8.3, reached by a different route (note XVI.8.16).

#### XVI.7.7.1 Fix the language before asking what is definable

Fix a named set-like global well-order  $G$  and the first-order language

$$\mathcal{L}_G = \{\in, G\}.$$

A presentation is *G-definable with set parameters* if its domain and relation are defined by two formulas of this language with a finite tuple of set parameters. It is *set-parameter-free* if no such tuple is needed. The named class  $G$  is still allowed.

Allowing an arbitrary new class parameter for each presentation would make this distinction meaningless: every relation is definable from itself. Likewise, allowing a truth predicate changes the language and must be recorded explicitly.

Choose an ordinary effective coding of formulas and finite parameter tuples. For each fixed standard natural number  $n$ , let  $\Sigma_n(G)$  denote a standard bounded level of the first-order hierarchy. It is harmless to pad formulas with unused quantifiers, so every particular first-order formula belongs to  $\Sigma_n(G)$  for some sufficiently large finite  $n$ . We require both the domain formula and the relation formula of a  $\Sigma_n(G)$  presentation to lie at that level.

**Lemma XVI.7.19** (Partial evaluation at a fixed level). *For each fixed standard finite  $n$ , there is an elementary formula  $\text{Sat}_n(e, \vec{a}; G)$  that correctly evaluates  $\Sigma_n(G)$  formulas coded by  $e$  at assignments  $\vec{a}$ . Its construction is a metatheoretic schema in  $n$ , not an assertion that all the predicates  $\text{Sat}_n$  form one  $G$ -definable class uniformly in  $n$ .*

*Proof.* Bounded formulas can be evaluated inside a set containing the transitive closures of the parameter values. Restrict the named class predicate  $G$  to the relevant set of pairs; elementary class separation supplies this restriction as a set. Ordinary satisfaction for that set structure evaluates the bounded formula.

For a fixed number of unbounded quantifier blocks, place those blocks in front of the bounded evaluation formula and handle their finite tuples of variables by set coding. This gives a finite first-order formula for the chosen level. The passage from level  $n$  to the next is a construction of formulas outside the object theory; it does not supply truth for all formulas in one predicate.  $\square$

#### XVI.7.7.2 The bounded-complexity diagonal

**Theorem XVI.7.20** (Bounded definability cannot bound class-order length). *For each fixed standard finite  $n$ , there is a class well-order  $D_n$  with a set-parameter-free definition in  $\mathcal{L}_G$  such that:*

- (a) every class well-order having a  $\Sigma_n(G)$  presentation with arbitrary set parameters embeds into  $D_n$ ;
- (b)  $D_n$  embeds into no class well-order having such a presentation;
- (c) in particular,  $D_n$  has no isomorphic  $\Sigma_n(G)$  presentation, even one using set parameters.

The quantification over  $n$  is a schema. Each instance is provable in GBC. Nonembedding in (b) holds for all embeddings available in the ambient class model, not merely for definable embeddings of bounded complexity.

*Proof.* Coding all candidate parameters. Let the index class be all sets in the order  $G$ . A valid index is a code

$$i = (e_D, e_R, \vec{p})$$

for a pair of  $\Sigma_n(G)$  formulas and a finite set-parameter tuple. Invalid codes will designate the empty order. Using the fixed evaluator from lemma XVI.7.19, define the candidate domain and relation jointly in  $i, x, y$ .

*Selecting only well-orders.* For a valid index, test whether the candidate relation, restricted to its domain, is a linear order and has no descending set function of domain  $\omega$ . This is a first-order property of  $i$  in the language with  $G$ : replace each occurrence of the candidate domain and relation by its  $\text{Sat}_n$  definition. By note XVI.2.12, the test is equivalent to being a class well-order in the ambient model. If the test fails, replace the candidate by the empty order.

Thus we have one  $G$ -definable uniform family  $(W_i)_{i \in V}$  of genuine class well-orders, containing every  $\Sigma_n(G)$  presentation with any set parameters. No truth predicate beyond the chosen partial level was used. The well-orderhood test can increase formula complexity; it is not claimed to remain  $\Sigma_n(G)$ .

*Diagonalization.* Put

$$D_n = \sum_{i \in (V, G)} (W_i + 1).$$

The index  $n$  has been fixed externally and is represented by a numeral in the resulting formula. All set parameters of all candidates are now variables ranging through the index class. Consequently this presentation of  $D_n$  has no set parameters.

Apply note XVI.7.17. It gives (a) and (b). If (c) failed, choose an index  $i$  coding an isomorphic  $\Sigma_n(G)$  presentation. That isomorphism would itself be an embedding of  $D_n$  into  $W_i$ , contradicting (b).  $\square$

**Corollary XVI.7.21** (Parameter-free presentations are externally cofinal). *Every  $G$ -definable class well-order with set parameters embeds into a set-parameter-free  $G$ -definable class well-order. Externally, the countable family of presentations*

$$D_0, D_1, D_2, \dots$$

*is cofinal under embeddability among all  $G$ -definable class well-orders with set parameters.*

*Proof.* The two formulas defining a given presentation have some fixed finite complexity bound  $n$ . Apply theorem XVI.7.20(a).  $\square$

This is a cofinality statement, not pointwise elimination of parameters. It does not say that every parameter-definable order has an isomorphic parameter-free presentation. The replacement order can be substantially longer, as the diagonal construction intentionally ensures.

**Corollary XVI.7.22** (No definable uniform cofinal family). *There is no  $G$ -definable uniform family of class well-orders, indexed by a  $G$ -definable class of sets and allowing a fixed finite tuple of set parameters in its definition, which is cofinal among all  $G$ -definable class well-orders with set parameters.*

*Proof.* Order the indices by the restriction of  $G$ . The diagonal of that family is again  $G$ -definable with the same fixed parameters. Cofinality would place its order type below a member under embeddability, contradicting [note XVI.7.17](#).  $\square$

In particular, the external list of formulas defining the  $D_n$  is not a uniform  $G$ -definable evaluation of all their order relations. The schema and the missing uniform evaluator coexist without contradiction. This is one place where the analogy with truth hierarchies is exact in its logical shape: every fixed level is available, while their complete uniform assembly requires a change of resources.

### XVI.7.7.3 Removing the named global order in special universes

**Corollary XVI.7.23.** *Suppose the universe has a parameter-free first-order definable set-like global well-order. Then every set-parameter-definable class well-order embeds into a parameter-free first-order definable class well-order. There is an external countable cofinal family of the latter presentations.*

*Proof.* Use the definable global order as  $G$  in [corollary XVI.7.21](#), then replace occurrences of  $G$  in each resulting formula by its fixed parameter-free definition. The substitution may increase finite complexity but introduces no parameters.  $\square$

The constructible universe provides the standard intended example through its canonical global ordering. Without a definable global well-order, this proof does not remove the named class parameter. It is important not to replace “relative to  $G$ ” by an unconditional parameter-free claim about every model of ZFC.

### XVI.7.7.4 A precise account of what has been resolved

The preceding theorems answer three concrete structural questions: *Can one fixed definability level be cofinal?* No. *Do arbitrary set parameters increase the external supremum beyond parameter-free presentations, once  $G$  is fixed?* No. *Can the resulting countable cofinal construction itself be uniformly evaluated in the same language?* No.

These are proved statements with explicit hypotheses. They are not claimed to settle a historically recognized open problem. Establishing whether the same arguments or consequences occur in the existing literature requires a broader bibliographic and expert review than the targeted search recorded here.

## XVI.8 External spectra and the effect of definability

The question “how far beyond Ord?” acquires a precise numerical answer after a model and its available classes have been fixed. The answer belongs to the external metatheory: a proper class of a set model is itself an ordinary set externally. Its genuinely well-founded orderings therefore have ordinary external ordinal ordertypes. This viewpoint does not turn an internal proper class into an internal set. It compares two different levels of discourse.

Throughout this section the metatheory is ZFC. A *transitive  $\beta$ -model*  $(M, \mathcal{X})$  is a transitive model of class theory which is correct about the well-foundedness of its class relations. Transitivity

alone does not imply this correctness: an external descending sequence through a class relation need not be a set of  $M$ . The distinction is standard in the model theory of second-order set theories [35]. Our spectra below count genuinely externally well-founded orders. The  $\beta$ -hypothesis will allow an internal definability test to select exactly those orders.

### XVI.8.1 The full spectrum over an inaccessible universe

**Theorem XVI.8.1** (Full spectrum). *Let  $\kappa$  be an inaccessible cardinal in the external universe. In the full class model*

$$\mathfrak{V}_\kappa = (V_\kappa, \mathcal{P}(V_\kappa)),$$

*the external ordertypes of class well-orders are exactly the ordinals  $\alpha < \kappa^+$ . Those whose domains are internally proper classes have exactly the ordertypes*

$$\kappa \leq \alpha < \kappa^+.$$

*Here  $\text{Ord}^{\mathfrak{V}_\kappa}$  has external ordertype  $\kappa$ .*

*Proof.* Inaccessibility gives  $|V_\kappa| = \kappa$ . Every class domain is an external subset of  $V_\kappa$ , so any well-order on it has cardinality at most  $\kappa$  and hence ordertype below  $\kappa^+$ . Conversely, every  $\alpha < \kappa^+$  has cardinality at most  $\kappa$ , so an injection into  $V_\kappa$  transports its ordering to a subclass of  $V_\kappa$ . Every such relation is available in the full class collection. Finite tuple coding stays inside  $V_\kappa$ , so binary relations can be represented by unary class predicates.

Every element of  $V_\kappa$  has cardinality below  $\kappa$ . Conversely, an external subset  $D \subseteq V_\kappa$  of cardinality below  $\kappa$  has bounded element ranks by regularity; consequently  $D \in V_\kappa$ . Thus a subclass is internally proper precisely when its external cardinality is  $\kappa$ . The second assertion follows.  $\square$

The use of inaccessibility also makes the full class model a legitimate model of the usual strong class axioms: arbitrary class comprehension is available, and regularity ensures that the image of an internal set under a class function remains an internal set. It is a  $\beta$ -model. Indeed, an external countable sequence of elements of  $V_\kappa$  has bounded ranks and belongs to  $V_\kappa$ . The same argument shows  $\beta$ -correctness for any GBC realization over this  $V_\kappa$ .

**Theorem XVI.8.1** supplies an exact ceiling, but only relative to this particular model and the full external class collection. The ceiling  $\kappa^+$  is not itself represented by one of its class well-orders. It is neither an internal successor cardinal of a set  $\text{Ord}$  nor a largest class ordertype. Restricting the available definitions gives a much smaller external ceiling.

### XVI.8.2 Parameter spectra are intervals; parameter-free spectra need not be

Fix a transitive set model  $M$  with a finite language  $L$  expanding membership by named class predicates. Assume these predicates occur in a GBC realization, so first-order definitions and their set restrictions are legitimate available classes. Put  $\lambda = |M|$ , externally. Let  $\mathcal{S}_{\text{par}}^L$  be the external ordertypes of well-orders on subclasses of  $M$  whose domains and relations are first-order  $L$ -definable using finite tuples of set parameters from  $M$ . Write  $\mathcal{S}_0^L$  for the parameter-free version. A named predicate is part of the language; “parameter-free” here excludes additional set parameters, not the named predicates themselves.

**Proposition XVI.8.2** (The parameter spectrum). *There is a limit ordinal  $\Theta_L < \lambda^+$  such that*

$$\mathcal{S}_{\text{par}}^L = [0, \Theta_L).$$

*The set  $\mathcal{S}_0^L$  is countable, is closed under successors, and has no greatest member. Its supremum has cofinality  $\omega$ . In general  $\mathcal{S}_0^L$  need not be an ordinal interval.*

*Proof.* There are countably many formulas and only  $\lambda$  finite parameter tuples, so  $|\mathcal{S}_{\text{par}}^L| \leq \lambda$ . If an order has external ordertype  $\alpha$  and  $\beta < \alpha$ , let  $x \in M$  be its element at position  $\beta$ . The initial segment below  $x$  is definable with  $x$  as one more set parameter and has ordertype  $\beta$ . Therefore the parameter spectrum is downward closed.

Tagged addition of one last element turns any definable presentation of  $\alpha$  into a definable presentation of  $\alpha + 1$ , using the same parameters. Thus this downward closed set has no largest member. It is an ordinal interval  $[0, \Theta_L)$  for a limit  $\Theta_L$ , and its cardinality bound implies  $\Theta_L < \lambda^+$ .

There are only countably many parameter-free definitions. Successor addition is still parameter-free, so their ordertypes have no largest member. Their supremum consequently has cofinality exactly  $\omega$ . The initial-segment argument does not establish downward closure here, because the cutoff point need not be parameter-free definable.  $\square$

In fact, for a parameter-free definable *set* well-order, its unique ordinal collapse is parameter-free definable. Thus, over  $V_\kappa$  for uncountable inaccessible  $\kappa$ , the countable parameter-free spectrum omits many ordinals below  $\kappa$ , whereas the definable class  $\text{Ord}$  itself supplies the external type  $\kappa$ . This gives a concrete reason the parameter-free spectrum is not downward closed in that setting.

For  $M = V_\kappa$ , the parameter spectrum contains  $\kappa$  and its successor, and [Proposition XVI.8.2](#) gives the strict bounds

$$\kappa < \Theta_L < \kappa^+.$$

These are external statements about definitions over the fixed structure. They do not require the isomorphism between two presentations of the same external type to be definable over that structure. The external initial-segment theorem therefore does not settle the internal class well-order comparability problem.

### XVI.8.3 Parameter elimination by a definable sum

We now prove a more informative relation between the two spectra. It uses explicit class sums to remove set parameters from an upper bound on a given definable ordertype. We present it as a synthesis theorem with a full proof, without a claim of historical priority. Naming the predicates is essential. The broader study of expansions by truth predicates and additional classes provides the appropriate surrounding context [\[84\]](#).

**Theorem XVI.8.3** (Parameter-free cofinality, relative to a global order). *Let  $(M, \mathcal{X})$  be a transitive  $\beta$ -model of GBC. Fix finitely many class predicates  $\vec{A} \in \mathcal{X}$  and a global well-order  $G \in \mathcal{X}$ . In the finite language  $L = \{\in, \vec{A}, G\}$ , the parameter-free spectrum is cofinal in the set-parameter spectrum. Consequently,*

$$\sup \mathcal{S}_0^L = \Theta_L, \quad \text{cf}(\Theta_L) = \omega.$$

*If  $G$  is parameter-free definable from the other named predicates, it need not be added as a separate symbol.*

*Proof.* Fix one pair of ordinary finite  $L$ -formulas  $\delta(x, \vec{p})$  and  $\rho(x, y, \vec{p})$ , with a fixed finite number of parameter positions. For every tuple  $\vec{q}$  of the corresponding length from  $M$ , let  $D_{\vec{q}}$  and  $R_{\vec{q}}$  be the domain and relation obtained by substituting  $\vec{q}$ . Let  $I_{\delta, \rho}$  be the class of those tuples for which  $M$  judges  $R_{\vec{q}}$  to be a class well-order on  $D_{\vec{q}}$ .

This filtering condition is first-order and uses no set parameters. Substitute the two fixed formulas into the requirements for a linear order and into the assertion that no descending *set* sequence of length  $\omega$  exists. There is no variable formula code to evaluate, so no satisfaction predicate is needed. Elementary comprehension supplies  $I_{\delta, \rho}$ , and the restriction of  $G$  well-orders its tuple codes.



Form the lexicographic sum

$$U_{\delta,\rho} = \sum_{\vec{q} \in (I_{\delta,\rho}, G)} (D_{\vec{q}}, R_{\vec{q}}).$$

Its elements are set pairs  $(\vec{q}, x)$  with  $\vec{q} \in I_{\delta,\rho}$  and  $x \in D_{\vec{q}}$ . Distinct blocks are compared by  $G$ , and elements of one block by  $R_{\vec{q}}$ . The domain and ordering are  $L$ -definable without set parameters, with the fixed formulas substituted into their definitions. Internally this is a class well-order: a nonempty subclass has a least occupied index and then a least element in that block. It is therefore externally well-founded by  $\beta$ -correctness. The same correctness ensures that the filter admitted no externally ill-founded candidates.

Every genuinely well-founded instance of this formula pair, with any particular parameter tuple  $\vec{p}$ , passes the internal test and occurs as a block of  $U_{\delta,\rho}$ . Its external ordertype is therefore at most  $\text{otp}(U_{\delta,\rho})$ . This proves that every member of  $\mathcal{S}_{\text{par}}^L$  is bounded by a member of  $\mathcal{S}_0^L$ .

Externally there are countably many ordinary finite formula pairs. Enumerating them gives a countable cofinal family of the separately defined orders  $U_{\delta,\rho}$ . The two spectra have the same supremum, and successor closure excludes a largest member. Hence  $\text{cf}(\Theta_L) = \omega$ .  $\square$

The hypotheses are deliberately explicit. The realization  $\text{Def}(M; \vec{A}, G)$ , allowing set parameters, inherits class replacement and satisfies GBC. It is also a  $\beta$ -model: both realizations have the same sets, and in each the well-foundedness of a common class relation is equivalent to the absence of a descending  $M$ -set  $\omega$ -sequence. They therefore agree about well-foundedness, and the larger realization is  $\beta$ -correct. No assumption that every possible external class is available is needed. Conversely, mere transitivity would not justify the filter: a candidate with an external descending sequence missing from  $M$  could be admitted. Naming  $G$  does not assert a parameter-free definable global well-order in the original pure membership language.

**Corollary XVI.8.4.** *For an inaccessible  $\kappa$  and fixed named predicates  $\vec{A}, G$  in the full class model over  $V_\kappa$ , with  $G$  a global well-order,*

$$\kappa < \Theta_L < \kappa^+, \quad \text{cf}(\Theta_L) = \omega.$$

*Parameter-free  $L$ -definitions are cofinal in this interval of parameter-definable types, although they realize only countably many of its members.*

#### XVI.8.4 Why the cofinal family does not create a largest notation

Choose an external enumeration of the finite formula pairs, and denote the resulting orders by  $U_n$ . For each standard  $n$  its definition is a finite formula, but the proof supplies no single  $L$ -formula that evaluates all these definitions uniformly. Such a uniform,  $L$ -definable cofinal family is impossible: its  $\omega$ -indexed lexicographic sum followed by one new last point would again be  $L$ -definable and would have ordertype larger than every member of the supposedly cofinal family.

A stronger class collection may contain a uniform code for the family. For example, the full collection  $\mathcal{P}(V_\kappa)$  does. There is no contradiction: using that additional class as a predicate enlarges the language of permitted definitions. The obstruction concerns a uniform code definable in the *same fixed language*. One could instead organize the construction using separate bounded-complexity satisfaction predicates. They likewise do not assemble into a definable full truth predicate for that language; the fixed-formula proof shows that this additional machinery is unnecessary here.

Thus external countable cofinality is compatible with the impossibility of internally listing a cofinal family by one definition. A research question suggested by these results is to classify the ordinals  $\Theta_L$  obtainable from a fixed  $V_\kappa$  as the named class predicates vary, subject to amenability

and the desired recursion axioms. Another is to determine which restricted notation calculi are cofinal in particular definability spectra. The present theorems give necessary bounds and an explicit external construction; they do not identify one absolute endpoint beyond Ord.

[merge] [Theorem XVI.8.1](#) is source 12’s full-class model [theorem 21.1](#) (iv) (Part V; extended to every limit height by source 31, [theorem XV.9.3](#)), proved again by sources 35, 32 and 33 (notes [XVI.8.5](#), [XVI.8.8](#) and [XVI.8.15](#)). The definable ceilings of the four sources are one invariant. Over  $V_\kappa$  with  $\kappa$  inaccessible, source 32’s  $\delta_A$  is  $\Theta_L$  for  $L = \{\in, G, A\}$  ([theorem XVI.8.9](#): its spectrum is the interval  $[0, \delta_A)$ ), source 33’s  $\delta_G$  and  $\delta_G^0$  are  $\Theta_L$  and  $\sup \mathcal{S}_0^L$  for  $L = \{\in, G\}$  ([note XVI.8.16](#)), and source 35’s  $\delta_{\text{set}}$  is  $\Theta_L$  for the pure language  $L = \{\in\}$ , by [proposition XVI.8.2](#) applied in the full model ([theorem XVI.8.6](#)). [Theorem XVI.8.3](#) is the most general form, for any transitive  $\beta$ -model of GBC with a named global well-order; sources 32 and 33 prove the case  $V_\kappa$ , and source 32 adds  $\kappa^{\delta_A} = \delta_A$ , source 33 the closure under ordinal arithmetic. Together they answer in part source 31’s research question 113.239 (section 113.29).

### XVI.8.5 Source 35: external lengths: the full-model spectrum and definability-sensitive cutoffs

The constructions above are meaningful internally, without assigning a set ordinal to a proper class order. An ambient set universe can nevertheless measure their lengths when the entire class model is a set there. Keeping these two viewpoints separate gives a precise answer to “how far beyond Ord?” and prevents an external cardinal bound from becoming an internal largest notation.

#### XVI.8.5.1 The exact full-model spectrum

Work in an ambient universe satisfying ZFC, and let  $\kappa$  be strongly inaccessible. Consider the two-sorted structure

$$\mathcal{M}_\kappa = (V_\kappa, \mathcal{P}(V_\kappa)).$$

Its sets are the elements of  $V_\kappa$ ; its classes are *all ambient subsets* of  $V_\kappa$ , with the appropriate tagged copies if the two sorts are formally disjoint. This notation does not assert that  $\mathcal{P}(V_\kappa)$  is a set or a class *inside*  $\mathcal{M}_\kappa$ . Relations and functions use the usual set codes for ordered pairs.

This structure satisfies KM with Global Choice. Full comprehension holds because every subset of the first-order universe is available. For Class Replacement, if  $a \in V_\kappa$  and a class function has domain  $a$ , its image has cardinality less than  $\kappa$ . Regularity bounds the ranks of its elements below  $\kappa$ , so the image belongs to  $V_\kappa$ . A bijection between  $\kappa$  and  $V_\kappa$  in the ambient universe provides Global Choice. Here the identity  $|V_\kappa| = \kappa$  uses strong inaccessibility.

*Note XVI.8.5* (Source 35, Theorem 9.1, “Full external spectrum”). *The external order types of class well-orders in  $\mathcal{M}_\kappa$  are exactly the ordinals  $\theta < \kappa^+$ : those of sets are the  $\theta < \kappa$ , those of proper classes the  $\theta$  with  $\kappa \leq \theta < \kappa^+$ . The same proper-class spectrum is obtained if the carrier is required to be exactly  $\text{Ord}^{\mathcal{M}_\kappa} = \kappa$ .* This is source 34’s [theorem XVI.8.1](#) and source 12’s [theorem 21.1](#) (iv), by the same cardinality and regularity count; source 35 adds the last sentence (transport to the carrier  $\kappa$  by an ambient bijection) and that internal and external well-foundedness agree because every external witness of ill-foundedness is an available class.

In this example  $\Omega$  has external length  $\kappa$ , while  $\Omega + 1$ ,  $\Omega \cdot 2$ , and  $\Omega^{[\Omega]}$  have the corresponding ordinary ordinal lengths  $\kappa + 1$ ,  $\kappa \cdot 2$ , and  $\kappa^\kappa$ . There is no conflict with the absence of a largest internal class well-order. The external ordinal  $\kappa^+$  is not represented by a class well-order in  $\mathcal{M}_\kappa$ , and the model does not possess the collection of all its classes as a same-level object. This full spectrum is background already present in the repository; the next refinement records the effect of restricting definitions.

### XVI.8.5.2 Definability changes the external ceiling

Use the fixed countable language  $\{\in\}$ . A presentation consists of formulas for its carrier and relation, evaluated over  $V_\kappa$ . Let  $S_0$ ,  $S_{\text{ord}}$ , and  $S_{\text{set}}$  be the external sets of order types of genuine well-orders with presentations using, respectively, no parameters, finitely many ordinal parameters, and finitely many arbitrary set parameters from  $V_\kappa$ . Put

$$\delta_0 = \sup S_0, \quad \delta_{\text{ord}} = \sup S_{\text{ord}}, \quad \delta_{\text{set}} = \sup S_{\text{set}}.$$

These are definitions in the ambient universe, which has the satisfaction relation of the set structure  $V_\kappa$ . They are not proposed as uniform truth definitions inside  $V_\kappa$ .

[merge] In [theorem XVI.8.6](#),  $\delta_{\text{set}}$  is  $\Theta_L$  for  $L = \{\in\}$  and the set-parameter spectrum is the interval  $[0, \delta_{\text{set}})$ , by source 34's [proposition XVI.8.2](#) applied to the full model, which is a GBC realization containing the predicate;  $\text{cf}(\delta_0) = \omega$  is also there. Whether the inequalities  $\delta_0 \leq \delta_{\text{ord}} \leq \delta_{\text{set}}$  are strict in the pure language, and  $\text{cf}(\delta_{\text{set}})$ , stay open; with a named global well-order the analogue of  $\delta_0 = \delta_{\text{set}}$  holds ([theorem XVI.8.3](#); [note XVI.8.16](#)). The reflection-tree example below is the standard mechanism of Hamkins's answer to Glazer's MathOverflow question, as source 35 says; it is not new.

**Theorem XVI.8.6** (Definability-sensitive spectrum bounds). *With these conventions,*

$$\kappa < \delta_0 \leq \delta_{\text{ord}} \leq \delta_{\text{set}} < \kappa^+, \quad \text{cf}(\delta_0) = \omega.$$

*None of the three spectra has a maximum. Each of the other two suprema has cofinality at most  $\kappa$ . Allowing arbitrary class parameters instead recovers the full spectrum of [Note XVI.8.5](#).*

*Proof.* There are countably many parameter-free formula pairs, and at most  $\kappa$  formula pairs with finite parameters from  $V_\kappa$ . The same bound holds for ordinal parameters. Thus

$$|S_0| \leq \aleph_0, \quad |S_{\text{ord}}| \leq \kappa, \quad |S_{\text{set}}| \leq \kappa.$$

Every member lies below  $\kappa^+$  by [Note XVI.8.5](#). The ambient successor cardinal  $\kappa^+$  is regular, so the supremum of at most  $\kappa$  of its members is still below  $\kappa^+$ .

Given any presentation of  $A$ , form the tagged carrier  $(\{0\} \times A) \cup \{(1, 0)\}$ , retaining the old order in the first part and putting the new point last. This defines  $A + 1$  with exactly the same permitted parameters. Its external type is strictly larger. Hence no spectrum has a maximum. A countable nonempty set of ordinals without a largest member has a supremum of cofinality  $\omega$ . The parameter-free presentation of  $\Omega$  and its successors gives  $\delta_0 > \kappa$ . Inclusion of the permitted parameter families gives the remaining inequalities and cardinal bounds on cofinality.

If class parameters are unrestricted, use a carrier and its ordering relation themselves as parameters. Every class presentation is then definable, so there is no remaining restriction.  $\square$

The same suprema result if one restricts to proper-class carriers: given any definable set order, append a copy of  $\Omega$  to obtain a definable proper-class order of greater external type. We make no claim that the inequalities between the three definability suprema are strict. Their exact values require additional information about  $V_\kappa$ . In particular, the counting argument does not assign an exact cofinality to  $\delta_{\text{set}}$ .

It is also important not to identify these spectra with an automatic GBC model  $(V_\kappa, \text{Def}(V_\kappa))$ . The definable classes always give the expected elementary comprehension and replacement, but a global well-order definable from a set parameter need not exist. The spectrum theorem takes place inside the full model and merely restricts which presentations are counted.

### XVI.8.5.3 Internal well-foundedness and external well-foundedness

For a class model  $(M, \mathcal{X})$ , internal well-foundedness quantifies over the subclasses in  $\mathcal{X}$ . A transitive first-order universe  $M$  does not by itself make this equivalent to external well-foundedness. The relevant additional correctness condition is the usual  $\beta$ -model condition: relations declared well-founded internally must be genuinely well-founded externally.

In GBC, a class relation is well-founded if and only if it has no descending  $\omega$ -sequence coded by a set. For the nontrivial direction, Global Choice recursively chooses a smaller member of any nonempty class with no minimal member; Class Replacement collects the resulting  $\omega$ -sequence. Thus two GBC realizations with the same first-order universe agree about well-foundedness of every relation they share. This invariance concerns their *internal* judgment, and does not imply that judgment is externally correct.

Here is the standard reflection-tree mechanism giving a transitive counterexample; compare Hamkins’s answer in [90]. Assume a countable transitive model of ZFC exists, and choose  $M = L_\alpha$  of least possible height. Fix a primitive recursive increasing sequence  $k_i$  large enough to cover the first  $i$  axioms in a recursive enumeration of ZFC. Let  $T$  consist of finite increasing ordinal sequences  $(\beta_i)$  such that  $L_{\beta_i}$  satisfies those axioms and

$$L_{\beta_i} \prec_{\Sigma_{k_i}} L_{\beta_{i+1}} \quad \text{whenever both entries occur.}$$

This is definable using satisfaction for set structures. Reflection produces an external infinite branch. An  $M$ -set branch would have  $\lambda = \sup_i \beta_i < \alpha$ ; eventual elementarity for every formula would give  $L_\lambda \models \text{ZFC}$ , contradicting minimality. The Kleene–Brouwer order of  $T$ , with extensions preceding their prefixes, is therefore internally well-founded but externally ill-founded in  $(M, \text{Def}(M))$ . Internally, a descending sequence would yield a branch by successive coordinate stabilization. Externally, the branch itself yields descending finite prefixes.

The inaccessible-height situation is much better. Every ambient  $\omega$ -sequence of elements of  $V_\kappa$  belongs to  $V_\kappa$ : regularity bounds its coordinate ranks, and the graph then has rank below  $\kappa$ . Consequently *any* GBC realization with first-order universe  $V_\kappa$  is correct about well-foundedness of its admitted relations. This gives a second proof of the correctness used in [Note XVI.8.5](#), and explains exactly why that external calculation is safe.

### XVI.8.6 Source 32: external spectra and the ceiling of definability

We now give a precise model-relative answer to “how much farther than Ord?” This section takes place in an external ZFC metatheory and assumes a strongly inaccessible cardinal  $\kappa$ . This is a sufficient setting for the results, not a claim that such a cardinal follows from ZFC or is necessary for every related phenomenon.

Put  $M = V_\kappa$ . Fix a class predicate  $G$  coding a bijection  $\kappa \rightarrow V_\kappa$ , and let  $A \subseteq V_\kappa$  be an arbitrary additional predicate. Tuples and relations may be coded as unary predicates using ordinary set pairing. Define

$$\mathcal{X}_A = \text{Def}(V_\kappa; \in, G, A), \tag{S32.11}$$

where *all set parameters from  $V_\kappa$  are allowed*. The parameters  $G, A$  are fixed; they are not varied separately for each definition within this one class realization.

The distinction between different admitted class collections over one set universe is central in work on class-theoretic potentialism [84]. Here it lets us isolate a numerical invariant of definability rather than assigning an absolute length to “the definable classes” without specifying a language or a model.

### XVI.8.6.1 Why this setting has honest external well-orders

**Lemma XVI.8.7** (The inaccessible-rank model). *Both  $(V_\kappa, \mathcal{X}_A)$  and  $(V_\kappa, \mathcal{P}(V_\kappa))$  satisfy GBC. For every class relation on  $V_\kappa$ , its internal class well-foundedness in either model is equivalent to external well-foundedness.*

*Proof.* The set structure  $V_\kappa$  satisfies ZFC. A subset of any  $x \in V_\kappa$  belongs to  $V_\kappa$ , so separation with arbitrary class predicates holds. If a class function is evaluated on a set  $x \in V_\kappa$ , then  $|x| < \kappa$ . Its values form fewer than  $\kappa$  many sets of rank below  $\kappa$ . Regularity of  $\kappa$  bounds all their ranks below  $\kappa$ , so the image belongs to  $V_\kappa$ . This verifies Class Replacement even for arbitrary external class predicates. The predicate  $G$  gives Global Choice. For  $\mathcal{X}_A$ , elementary comprehension holds by substituting the definitions of its finitely many class parameters; for the full power set it is automatic.

Every external sequence  $f : \omega \rightarrow V_\kappa$  belongs to  $V_\kappa$ : the ranks of its countably many values have supremum below  $\kappa$ , and its graph then has rank below  $\kappa$ . Consequently an external descending  $\omega$ -sequence is an internal set witness to failure of well-foundedness. Conversely an internal descending sequence is certainly external. External dependent choice and lemma XVI.2.8 give the equivalence.  $\square$

The last assertion is not valid for arbitrary countable or nonstandard models. Such models may admit internally well-founded presentations that are externally ill-founded. We use the inaccessible setting precisely to rule out that ambiguity, rather than silently identifying internal and external order types everywhere.

*Note XVI.8.8* (Source 32, Theorem 8.2, “The full-class spectrum”). *In  $(V_\kappa, \mathcal{P}(V_\kappa))$ , the external order types of class well-orders are exactly the ordinals below  $\kappa^+$ , and those of proper-class well-orders exactly*

$$[\kappa, \kappa^+). \quad (\text{S32.12})$$

*The set-like proper-class type is  $\kappa$ , the external type of the internal  $\Omega$ .* This is source 34’s theorem XVI.8.1 and source 12’s theorem 21.1 (iv), with the same proof.

Thus full second-order semantics produces every length below the next external cardinal, not merely finite expressions in  $\kappa$ . It does not produce a greatest type:  $\kappa^+$  is a supremum outside the range, not an extra class ordinal in that model. Restricting to definable classes produces a much smaller, structured ceiling.

### XVI.8.6.2 The definability ceiling theorem

Let  $S_A$  be the external set of ordinal types of the  $\mathcal{X}_A$ -admitted class well-orders. There are only a set of these presentations externally, so the following definition is ordinary metatheoretic set formation:

$$\delta_A = \sup\{\beta + 1 : \beta \in S_A\}. \quad (\text{S32.13})$$

**Theorem XVI.8.9** (Definability ceiling). *The realized spectrum is exactly the initial segment*

$$S_A = \{\beta : \beta < \delta_A\}. \quad (\text{S32.14})$$

Moreover,

$$\kappa < \delta_A < \kappa^+, \quad \text{cf}(\delta_A) = \omega, \quad \kappa^{\delta_A} = \delta_A, \quad (\text{S32.15})$$

where the exponentiation and cofinality in (S32.15) are external ordinal operations. In particular,  $\delta_A$  is not itself represented by any class well-order in  $\mathcal{X}_A$ .

*Proof. Initiality and no last type.* Suppose  $\beta \in S_A$ , represented by  $W$ , and  $\gamma < \beta$ . The point of  $W$  at external position  $\gamma$  is an element  $x \in V_\kappa$ . Restricting  $W$  below  $x$  is definable with the additional allowed set parameter  $x$  and has type  $\gamma$ . Thus  $S_A$  is downward closed. It is closed under successor because  $W + 1$  is a definable class presentation. Therefore it has no largest member, and it is exactly the initial segment in (S32.14). Its supremum is not represented. The internal ordinal class has external type  $\kappa$ , so  $\kappa \in S_A$  and  $\kappa < \delta_A$ .

*A countable cofinal sequence of bounds.* Externally enumerate all first-order formula pairs for domains and binary relations in the fixed finite language with  $G, A$ :

$$(\varphi_0, \psi_0), (\varphi_1, \psi_1), \dots$$

Each formula pair is arranged to use one code for its finite tuple of set parameters. For each ordinary natural number  $n$ , apply [corollary XVI.7.15](#) to the first  $n + 1$  pairs, with all such set parameters. Let  $D_n$  be the resulting definable diagonal well-order and put

$$\theta_n = \text{otp}(D_n).$$

Each  $\theta_n$  belongs to  $S_A$ , so  $\theta_n < \delta_A$ . Every definable well-order  $W$  uses some formula pair, say the  $j$ th. For all  $n \geq j$ , the diagonal theorem says that  $W$  embeds into  $D_n$  and that  $D_n$  does not embed into  $W$ . These are actual external well-orders by [lemma XVI.8.7](#). Ordinary ordinal comparison therefore gives

$$\text{otp}(W) < \theta_n.$$

Hence  $(\theta_n)_{n < \omega}$  is externally cofinal in  $\delta_A$ .

Each  $\theta_n < \kappa^+$  by [note XVI.8.8](#). Since  $\kappa^+$  is regular and uncountable, the supremum of this countable sequence is strictly below  $\kappa^+$ . The ordinal  $\delta_A$  is a nonzero limit with no largest represented type, so a finite subset cannot be cofinal. Its cofinality is therefore exactly  $\omega$ .

*Closure and the fixed-point equation.* If  $\alpha, \beta \in S_A$  and  $\alpha \geq 2$ , choose definable class presentations of those types. Their finite-support power is definable in the same structure and is a well-order by [note XVI.4.10](#). Externally its type is the ordinary ordinal  $\alpha^\beta$ , by [lemma XVI.4.9](#) applied in the external universe. Thus  $S_A$  is closed under this exponentiation, as well as under sums and products.

In particular, since  $\kappa \in S_A$ , for every  $\beta < \delta_A$  we have  $\kappa^\beta < \delta_A$ . Continuity of ordinary exponentiation at the limit  $\delta_A$  gives

$$\kappa^{\delta_A} = \sup_{\beta < \delta_A} \kappa^\beta \leq \delta_A.$$

For every ordinal  $\beta$  and base at least 2, ordinary induction gives  $\beta \leq \kappa^\beta$ . Taking suprema yields the reverse inequality. This proves all of ([S32.15](#)).  $\square$

[merge] [Theorem XVI.8.9](#) is source 34's [proposition XVI.8.2](#) and [corollary XVI.8.4](#) at  $V_\kappa$  for the language  $\{\in, G, A\}$ , by a different route: source 34 bounds each formula pair by its own parameter-free sum, source 32 by the diagonals of finite lists of formula pairs ([corollary XVI.7.15](#)). The equation  $\kappa^{\delta_A} = \delta_A$  is proved here only by source 32; source 33's [corollary XVI.8.17](#) adds  $\omega^\delta = \delta$  and closure under ordinal arithmetic. [Corollary XVI.8.11](#) is proved only by source 32.

*Remark XVI.8.10* (What the ceiling does and does not mean). The spectrum statement uses set parameters essentially: they permit restriction below an arbitrary point of a presentation. It is not a theorem about parameter-free definability. Likewise, “definable with arbitrary class parameters” would mean something different if the class parameter were allowed to change separately for each presentation. Here  $A$  and  $G$  are fixed before  $\delta_A$  is formed.



The countable sequence  $(D_n)$  above is at this point an *external sequence of individually admitted classes*. It is not asserted to be a single  $\mathcal{X}_A$ -admitted family. If it were such a family, its sum could already reach the forbidden ceiling. The distinction is not bookkeeping: it is exactly where truth enters.

### XVI.8.6.3 Further structural consequences

**Corollary XVI.8.11.** *For fixed  $\kappa$  and  $G$ , the ordinals  $\delta_A$ , as  $A \subseteq V_\kappa$  varies, are unbounded in  $\kappa^+$ . More precisely,*

$$\sup_{A \subseteq V_\kappa} \delta_A = \kappa^+.$$

*Proof.* Given  $\theta < \kappa^+$ , transfer a well-order of type  $\theta$  to a subset of  $V_\kappa$  and code its domain and relation in one unary predicate  $A$ . That order is then definable from  $A$ , so  $\theta \in S_A$  and  $\theta < \delta_A$ . The reverse bound follows from [theorem XVI.8.9](#) for each  $A$ .  $\square$

The necessary conditions  $\text{cf}(\delta_A) = \omega$  and  $\kappa^{\delta_A} = \delta_A$  are not a classification of possible ceilings. In fact the spectrum is closed under more than ordinary exponentiation: it is closed under every fixed finite presentation constructor developed here, including  $A \mapsto E(A)$ . Determining which such closure properties are sufficient, or how these ordinals are related to other definability invariants, remains a substantive question.

A useful schematic comparison is therefore

| admitted presentations over $V_\kappa$   | external spectrum               |
|------------------------------------------|---------------------------------|
| set-like class well-orders               | $\{\beta : \beta \leq \kappa\}$ |
| $\text{Def}(V_\kappa; G, A)$ well-orders | $\{\beta : \beta < \delta_A\}$  |
| all class well-orders                    | $\{\beta : \beta < \kappa^+\}.$ |

In the first row the set-order types are below  $\kappa$ , and the single proper set-like type is  $\kappa$ . In the other rows many different non-set-like types appear. None of these external spectra is being asserted to be an admitted universal family in the model it describes.

### XVI.8.7 Source 32: a truth predicate reaches the old ceiling exactly

Let  $T_A$  be the full satisfaction predicate for the *set-sized external structure*  $(V_\kappa, \in, G, A)$ , with finite assignments coded in  $V_\kappa$ . Such a predicate exists externally and is an element of  $\mathcal{P}(V_\kappa)$ . It is not definable in that same structure with set parameters, by the usual diagonal argument for truth. Form the enlarged definable class realization

$$\mathcal{X}_{T_A} = \text{Def}(V_\kappa; \in, G, T_A).$$

The predicate  $A$  is recoverable from the truth values of its atomic formulas, so  $\mathcal{X}_A \subseteq \mathcal{X}_{T_A}$ .

**Theorem XVI.8.12** (Exact truth presentation and strict jump). *There is an  $\mathcal{X}_{T_A}$ -definable class well-order whose external type is exactly  $\delta_A$ . Consequently,*

$$\delta_A < \delta_{T_A}. \tag{S32.16}$$

*Proof.* Use the fixed external effective enumeration of formula pairs from the proof of [theorem XVI.8.9](#). In the language with  $T_A$ , evaluation of these pairs is uniform in their natural-number codes and set parameters. Validity of each evaluated fiber is still an elementary condition, by [lemma XVI.2.8](#); its

external correctness follows from [lemma XVI.8.7](#). Fix once and for all the global ordering of the formula-tag and parameter-code index class obtained from  $G$ .

The diagonal construction for the first  $n + 1$  formula pairs can now be carried out uniformly in  $n$ . It gives one admitted family  $(D_n)_{n < \omega}$  in  $\mathcal{X}_{T_A}$ . Its  $n$ th fiber is exactly the individually definable order used in the ceiling proof: for a fixed standard  $n$ , the finitely many evaluations can be replaced by their actual formulas, eliminating the truth predicate from that one fiber.

Define the class order

$$L = \sum_{n < \omega} (D_n + 1). \quad (\text{S32.17})$$

It is definable from  $G, T_A$  and is a class well-order by [proposition XVI.3.4](#). Every finite partial sum of (S32.17) belongs to  $\mathcal{X}_A$ , because it involves only finitely many of the individually definable  $D_n$ . Thus each finite partial sum has external type strictly below  $\delta_A$ . Since these are nested initial segments whose union is  $L$ , continuity of the external ordinal sum gives

$$\text{otp}(L) \leq \delta_A.$$

On the other hand, every  $D_n$  embeds into  $L$ . The ordinals  $\theta_n = \text{otp}(D_n)$  are cofinal in  $\delta_A$ , so  $\text{otp}(L) \geq \delta_A$ . Hence  $\text{otp}(L) = \delta_A$  exactly.

The new spectrum therefore contains  $\delta_A$ . It is successor closed and is the initial segment below its own ceiling, by [theorem XVI.8.9](#) applied with  $T_A$  as the extra predicate. This implies (S32.16).  $\square$

This theorem gives a precise interpretation of one step beyond a definability boundary. Truth does not merely yield some unrelated longer order. It assembles the formerly nonuniform finite-syntax bounds into a presentation of the old supremum itself. The construction is parallel in spirit to class-theoretic truth hierarchies, but its conclusion here follows from the explicit diagonal argument rather than from an assumed global iterated-truth principle.

[merge] [Theorem XVI.8.12](#) is the strongest of the four truth results: source 34's [theorem XVI.7.4](#) produces, internally and from an assumed full satisfaction class, an order dominating every definable one; source 33's [theorem XVI.8.19](#) shows at  $V_\kappa$  that truth raises the height, with the bounds of the new height; source 32 shows that truth represents the old height exactly. Its [corollary XVI.8.13](#) gives one family with no uniform code, as source 33's [corollary XVI.7.22](#) and source 34's [corollary XVI.7.6](#) do for theirs.

**Corollary XVI.8.13** (The old bounds cannot have been uniformly admitted). *The particular family  $(D_n)_{n < \omega}$  used above has no uniform code in  $\mathcal{X}_A$ .*

*Proof.* Such a code would make (S32.17) an  $\mathcal{X}_A$ -admitted class well-order of type  $\delta_A$ , contradicting [theorem XVI.8.9](#).  $\square$

One can externally repeat the construction: set  $A_0 = A$  and let  $A_{n+1}$  code the full truth predicate for the structure with  $A_n$ . Then

$$\delta_{A_0} < \delta_{A_1} < \delta_{A_2} < \cdots < \kappa^+.$$

Each individual step is covered by [theorem XVI.8.12](#). This assertion does not automatically supply one admitted internal hierarchy of all these truth predicates; combining them is again a uniformity question. The same caution applies at limit stages.

## XVI.8.8 Source 33: external spectra, definability heights, and truth

### XVI.8.8.1 A full-class model at an inaccessible cutoff

In this section the surrounding metatheory is ordinary external ZFC. Assume  $\kappa$  is strongly inaccessible and let

$$M = V_\kappa, \quad \mathcal{X} = \mathcal{P}(M).$$

Relations and tuples are coded as elements of  $M$ , so writing all subclasses as  $\mathcal{P}(M)$  covers binary relations as well. The set structure  $M$  satisfies ZFC. Every subset of  $M$  is an available class in  $(M, \mathcal{X})$ . Full class comprehension holds externally. Class replacement holds because a set of  $M$  has external cardinality below  $\kappa$ , and regularity bounds the ranks of fewer than  $\kappa$  many output values. A global ordering can be chosen externally and belongs to  $\mathcal{X}$ .

*Note XVI.8.14* (Source 33, Lemma 9.1, “Correct well-foundedness at the cutoff”). *Every class well-order of  $(V_\kappa, \mathcal{P}(V_\kappa))$  is externally well-founded, and conversely every externally well-founded class order is internally a well-order.* This is source 32’s lemma XVI.8.7 for the full model and source 34’s  $\beta$ -model remark after theorem XVI.8.1: an external descending sequence of elements of  $V_\kappa$  has ranks bounded below  $\kappa$  by regularity and uncountability, so it belongs to  $V_\kappa$ .

*Note XVI.8.15* (Source 33, Theorem 9.2, “Exact full-class spectrum”). *In the full-class model, the external order types of class well-orders are exactly*

$$\{\alpha : \alpha < \kappa^+\}; \tag{S33.9.1}$$

*those with proper-class carriers exactly the  $\alpha$  with  $\kappa \leq \alpha < \kappa^+$ ; and the set-like proper-class order  $\Omega^M$  has external type  $\kappa$ .* This is source 34’s theorem XVI.8.1, with the same proof.

The endpoint  $\kappa^+$  is an external bound, not a last represented class order. For each  $\alpha < \kappa^+$ , its successor still lies below  $\kappa^+$ . This reconciles note XVI.8.15 with the internal no-greatest-order theorem.

### XVI.8.8.2 Arithmetic viewed from outside

If a class exponent order  $\Gamma$  has external type  $\gamma$ , the finite-support construction has external type

$$\text{otp}(P(\Gamma)) = \kappa^\gamma, \tag{S33.9.2}$$

where the exponent on the right is *ordinal* exponentiation. Indeed, all exponent points and ordinal coefficients below  $\kappa$  are available as elements of  $M$ , and every finite code on them belongs to  $M$ . The external collapse is the ordinary finite-support presentation of that ordinal power.

All powers in (S33.9.2) still lie below  $\kappa^+$  when  $\gamma < \kappa^+$ : their finite-support presentations have cardinality at most  $\kappa$ . This gives a concrete sense in which very long ordinal expressions can have the same underlying class size.

For the fixed point of theorem XVI.6.26, set

$$t_0 = \kappa, \quad t_{n+1} = \kappa^{t_n}.$$

Its external type is  $\sup_{n < \omega} t_n$ . Continuity of ordinal exponentiation shows that this is the least fixed point of  $\xi \mapsto \kappa^\xi$  above  $\kappa$ . It is still below  $\kappa^+$ .

[merge] The external type (S33.9.2) is source 29’s full-model check after corollary 108.20 and source 34’s substitution rule in section XVI.10; the type  $\sup_n t_n$  of the fixed point is source 34’s theorem XVI.6.9,  $\varepsilon_{\kappa+1}$ .

### XVI.8.8.3 The smaller definable spectrum

Fix a set-like global ordering  $G$  in the full class model and let  $\text{Def}(M; G)$  consist of all first-order definable subclasses using  $G$  and finite tuples of parameters from  $M$ . This is a legitimate GBC class realization: comprehension is built into the definition, and replacement follows from the same rank and cardinality bound used above. Its well-orders are externally correct by the proof of [note XVI.8.14](#).

Define, externally,

$$\delta_G = \sup\{\text{otp}(W) : W \text{ has a first-order } G\text{-definable presentation with set parameters in } M\}. \quad (\text{S33.9.3})$$

Let  $\delta_G^0$  be the same supremum with no set parameters. This supremum is a set ordinal in the external metatheory. We make no claim that it is an ordinal or a represented well-order of the definable class model itself.

*Note XVI.8.16* (Source 33, Theorem 9.3, “Countably cofinal definability height”). *Under the inaccessible-rank hypotheses,*

$$\delta_G^0 = \delta_G, \quad \kappa < \delta_G < \kappa^+, \quad \text{cf}(\delta_G) = \omega. \quad (\text{S33.9.4})$$

*No  $G$ -definable class well-order has type  $\delta_G$ . The full class model does have a class well-order of that external type.* This is source 34’s [theorem XVI.8.3](#) and [corollary XVI.8.4](#) for  $L = \{\in, G\}$ , with  $\delta_G = \Theta_L$  and  $\delta_G^0 = \sup \mathcal{S}_0^L$ . The proof differs: source 34 bounds the instances of one formula pair by one parameter-free sum; source 33 uses the diagonals  $D_n$  of its [theorem XVI.7.20](#), one for each complexity level. It is printed as a second route. The last sentence is source 34’s [theorem XVI.8.1](#).

*Source 33’s proof.* For each external standard  $n$ , interpret the parameter-free formula for  $D_n$  from [theorem XVI.7.20](#) in  $(M, \in, G)$ , and let  $d_n = \text{otp}(D_n)$ . The order is externally well-founded. Every  $G$ -definable presentation has a fixed standard finite formula complexity, so it embeds into some  $D_n$ . Conversely every  $D_n$  is itself  $G$ -definable without set parameters. Therefore

$$\delta_G = \delta_G^0 = \sup_{n < \omega} d_n.$$

Every  $d_n < \kappa^+$  by [note XVI.8.15](#). The successor cardinal  $\kappa^+$  is regular in the external metatheory, so the supremum of this countable sequence is still strictly below  $\kappa^+$ .

The definable spectrum has no largest type: taking a tagged successor is a definable operation and strictly increases external ordinal type. Thus its supremum is not attained and has no finite cofinal subset. The countable cofinal family above gives  $\text{cf}(\delta_G) = \omega$ . The definable orders  $\Omega + 1$  and their further successors show that  $\delta_G > \kappa$ .

Nonattainment follows again from successor closure. Since  $\delta_G < \kappa^+$ , the full-class spectrum theorem supplies a full class well-order with that type, though not a  $G$ -definable one.  $\square$

[merge] [Corollary XVI.8.17](#) extends source 32’s  $\kappa^{\delta_A} = \delta_A$  ([theorem XVI.8.9](#)) to closure under ordinal sum, product and power and to  $\omega^{\delta_G} = \delta_G$ . The same argument gives  $\omega^{\Theta_L} = \Theta_L$ , and  $\theta^{\Theta_L} = \Theta_L$  for the external type  $\theta$  of  $\text{Ord}^M$ , for every transitive  $\beta$ -model and language of [proposition XVI.8.2](#): the parameter spectrum is the interval  $[0, \Theta_L)$ ; finite-support powers of  $L$ -definable presentations are  $L$ -definable with the joined parameters; and their external types are the ordinal powers, because the model contains every finite partial function between its sets and is correct about well-foundedness. [Theorem XVI.8.19](#) is weaker than source 32’s [theorem XVI.8.12](#) on exactness and adds the upper bound and cofinality of the new height.

**Corollary XVI.8.17** (Arithmetic closure of the definability ceiling). *For all external ordinals  $\alpha, \beta < \delta_G$ ,*

$$\alpha + \beta < \delta_G, \quad \alpha \cdot \beta < \delta_G, \quad \alpha^\beta < \delta_G.$$

Moreover,

$$\omega^{\delta_G} = \kappa^{\delta_G} = \delta_G. \quad (\text{S33.9.5})$$

*In particular,  $\delta_G$  is an external epsilon number, not just an unspecified ordinal of countable cofinality.*

*Proof.* Let  $S$  be the spectrum whose supremum is  $\delta_G$ . Given  $\alpha, \beta < \delta_G$ , choose  $a, b \in S$  with  $a \geq \max(2, \alpha)$  and  $b \geq \beta$ . The explicit sum, product, and power constructions preserve first-order definability, allowing the two original parameter tuples together. Their external types are  $a + b$ ,  $a \cdot b$ , and  $a^b$ , respectively. Each belongs to  $S$  and is strictly below its unattained supremum. Monotonicity of ordinary ordinal arithmetic gives the first three inequalities; zero and one cause no difficulty.

For  $\lambda = \omega$  or  $\kappa$ , the base order of type  $\lambda$  is definable, and  $\lambda^a \in S$  whenever  $a \in S$ . Continuity in the exponent and cofinality of  $S$  give

$$\lambda^{\delta_G} = \sup_{a \in S} \lambda^a \leq \delta_G.$$

Ordinal exponentiation with base at least two satisfies  $\lambda^\xi \geq \xi$ , as follows by transfinite induction. Hence equality holds. The case  $\lambda = \omega$  is the epsilon-number assertion.  $\square$

*Remark XVI.8.18* (Two different cofinalities). The equality  $\text{cf}(\delta_G) = \omega$  is entirely external. It does not say that the internal class  $\text{Ord}^M$  has a countable cofinal sequence. Indeed  $\text{Ord}^M$  has external type  $\kappa$ , which is regular and uncountable. Nor does it provide a  $G$ -definable uniform sequence of relations cofinal in the definable spectrum; [corollary XVI.7.22](#) forbids exactly that uniformization.

#### XVI.8.8.4 Truth strictly raises the definability height

A full first-order truth predicate for  $(M, \in, G)$  is an external subset of the class of formula-assignment codes, so it belongs to the full class part. Call it  $T$ . It evaluates formulas of  $\mathcal{L}_G$ ; it does *not* evaluate formulas of the expanded language containing  $T$  itself.

**Theorem XVI.8.19** (A strict truth jump in order height). *Let  $\delta_{G,T}$  be the definability height in the fixed expanded language  $\{\in, G, T\}$ , allowing set parameters. Then*

$$\delta_G < \delta_{G,T} < \kappa^+, \quad \text{cf}(\delta_{G,T}) = \omega. \quad (\text{S33.9.6})$$

*The two class realizations have exactly the same set universe and the same set ordinals.*

*Proof.* Using  $T$ , all  $G$ -formula pairs together with all finite parameter tuples can now be evaluated in one uniform family, with no bounded-complexity restriction. Apply the first-order well-orderhood test from [note XVI.2.12](#) to each decoded relation and replace failures by the empty order. The resulting family is  $G, T$ -definable and contains every  $G$ -definable class well-order.

Let  $K$  be its diagonal sum of successor fibers. Every  $G$ -definable well-order embeds into  $K$ , so externally  $\text{otp}(K) \geq \delta_G$ . The  $G, T$ -definable successor  $K + 1$  has type strictly greater than  $\delta_G$ . Hence  $\delta_{G,T} > \delta_G$ .

The proof of [note XVI.8.16](#) applies verbatim to any fixed finite collection of named predicates whose class realization satisfies GBC. At  $V_\kappa$  the rank argument supplies this requirement for  $G, T$ . It gives the remaining bound and cofinality in [\(S33.9.6\)](#). No sets were added; only the allowed class definitions were enlarged.  $\square$

This result makes precise a form of “farther than Ord” that cannot be captured merely by listing longer ordinal expressions. The height of attainable *definable* order types can increase while the set ordinals, all sets, and even the ambient full-class spectrum remain fixed. The role of truth expansion is consistent with the broader potentialist view of changing class resources over a fixed set model [84].

#### XVI.8.8.5 What fails outside the full inaccessible model

For any set-sized external model with carrier  $M$ , genuinely well-founded orders on subsets of  $M$  have external type below  $|M|^+$ . Equality with the full interval of types need not follow. The class part may omit many relations, the set model may not be externally correct about well-foundedness, and small external subsets of  $M$  need not be sets of  $M$ .

Accordingly, neither the exact interval (S33.9.1) nor the specific cutoff  $\kappa$  should be asserted for arbitrary models of GBC or KM. The inaccessible construction is an explicit semantic laboratory, not an additional hypothesis smuggled into the elementary GBC theorems of the earlier sections.

### XVI.9 How far the constructions reach, and what requires more axioms

The preceding results give several answers to “how much farther than Ord?” They answer different versions of the question and should not be compressed into a single purported largest class ordinal.

First, elementary presentations already go beyond every fixed finite combination of ordinal blocks. If  $\alpha$  is a set ordinal, the power  $\Omega^{[\alpha]}$  is a definable class well-order, and these are proper initial segments of  $\Omega^{[\Omega]}$ . Iterating the same finite-support constructor inside hereditary syntax gives

$$\Omega, \quad \Omega^{[\Omega]}, \quad \Omega^{[\Omega^{[\Omega]}]}, \quad \dots$$

as proper initial segments of  $\mathbf{E}$ . Appending an endpoint or a new block to  $\mathbf{E}$  immediately gives further orders. None of these explicit constructions requires the existence of an unknown class-valued transfinite history.

Second, no single uniform family is cofinal among all class well-orders under order embedding. This is stronger than saying that there is no largest element: it blocks a catalog indexed by sets with one uniform evaluation relation. A syntax can describe an enormous domain and still have a definable extension outside its reach.

Third, definability imposes a relative ceiling in an external model. For a fixed transitive  $\beta$ -model and a fixed finite collection of named predicates, the set-parameter definable order types form an initial segment  $[0, \Theta)$ . Naming a suitable truth predicate strictly enlarges the range available in that language. Under a named global well-order,  $\Theta$  has external cofinality  $\omega$ . This does not make  $\Theta$  countable, nor does it make the external cofinal sequence an admitted uniform class sequence.

Fourth, under the full interpretation over  $V_\kappa$ , the precise external ceiling for all class presentations is  $\kappa^+$ . This bound is relative to the chosen universe. It is not an absolute object that can be appended to Ord inside the original theory. The difference between explicit definability and arbitrary available classes is responsible for the difference between  $\Theta$  and  $\kappa^+$ .

#### XVI.9.1 The comparison and recursion boundary

The established class-recursion literature supplies the implications

$$\text{GBC} + \text{ETR} \vdash \text{CWO}, \quad \text{GBC} + \text{ETR} \vdash \text{CTH}.$$



The first follows by recursively matching the least unmatched points of two orders; the second also follows from the defining elementary recursion for the history. Gitman and Hamkins identify full ETR with clopen class-game determinacy and, in its full all-length formulation, with iterated elementary truth. Gitman, Hamkins, Holy, Schlicht, and Williams identify  $\text{ETR}_{\text{Ord}}$  with the class forcing theorem [22, 24]. These are theorems about stronger existence principles, not prerequisites for the finite graph manipulations used here.

For fixed small lengths, iteration conventions matter. For example,  $\text{ETR}_1$  is just an elementary comprehension instance; the existence of one full truth predicate is already a stronger assertion. We therefore do not replace the all-length equivalence by the unqualified formula “ $\text{ETR}_\Gamma$  equals  $\Gamma$ -iterated truth” for every  $\Gamma$ .

There is also a useful boundary between arbitrary rankings and continuous rankings. Hamkins and Woodin prove that every well-founded class relation has a ranking into some class well-order over  $\text{GBC}^-$ , while ETR supplies continuous rankings. Their comparison question asks whether appropriate converse implications hold [27, Theorems 4–6 and Question 7]. The present GB refinement identifies the strength of the particular condensation-history principle with CWO; it does not convert all elementary class recursions into finite normal forms.

[merge] The implications displayed above are recorded by all four sources and by this report (theorem 108.19 and the paragraph after it). Source 34’s last sentence identifies the strength of the condensation-history principle with CWO over GB (theorem XVI.5.8); with source 35’s corollary XVI.6.19 and theorem XVI.6.21, the fixed-point extension principle FPE is a further equivalent over GB. Whether CWO implies ETR is Hamkins and Woodin’s Question 7, which no source settles (section 113.28).

## XVI.9.2 Source 35: the boundary between comparison and class recursion

The preceding constructions use elementary comprehension, set-valued recursion, and Class Replacement. They do not solve arbitrary recursions whose individual stages are classes. For an elementary formula  $\varphi(x, i, F, A)$  and a class well-order  $\Gamma$ , a class recursion asks for a single class  $F$  satisfying

$$F_i = \{x : \varphi(x, i, F \restriction \Gamma_{<i}, A)\} \quad (i \in \Gamma).$$

The earlier history on the right may be a proper class, even when  $i$  has a set of predecessors. This is the content of elementary transfinite recursion, ETR, rather than the ordinary set-valued recursion theorem.

Over GBC, ETR is equivalent to the existence, for every class parameter and every class well-order, of an iterated first-order truth predicate along that order. It is also equivalent to determinacy of clopen games on classes [22, Theorems 8–9]. This supplies a precise reason to track the formalism when writing an iteration beyond Ord: the length and the type of the values are separate resources.

For a fixed  $\Gamma$ ,  $\text{GBC} + \text{ETR}_\Gamma$  proves that  $\Gamma$  is comparable with every class well-order by initial-segment embeddings. One recursively matches each point of  $\Gamma$  with the least unmatched point of the other order; if that construction stops, its inverse supplies the opposite comparison [35, Lemma 3.17]. Finite concatenations of the same recursion length add no strength, because the solution on one piece can be used as a parameter for the next. This observation does not justify a uniform iteration over all finite pieces.

The distinction is already visible at the important fragment  $\text{ETR}_{\text{Ord}}$ . Over GBC, it is equivalent to the class forcing theorem and to the existence of Ord-iterated truth predicates, among several other formulations [24, Main Theorem]. These are established calibration theorems. They do not identify arbitrary well-order comparability with full ETR.

The name “elementary transfinite recursion” also occurs in a different weak-set-theory setting. Frittaion and Genovesi prove that their first-order theory **C**, which includes Axiom Beta, proves  $\Delta_0$  transfinite recursion and generates relativized constructible hierarchies [91]. Their base theory and recursion scheme differ from the class-valued scheme displayed above. Applying that result to the present comparison problem would require an explicit interpretation and a proof that the relevant class solutions are obtained; the shared terminology alone supplies no such implication.

For clarity, the established implication chain over **GBC**, and over the weaker **GBC**<sup>−</sup> base used in the cited paper, is

$$\begin{aligned} \text{ETR} &\implies \text{continuous rankings for all well-founded class relations} \\ &\implies \text{CWO} \\ &\implies \text{weak class-well-order comparability.} \end{aligned}$$

Here a continuous ranking assigns to a point the least available rank strictly exceeding every predecessor rank; weak comparability permits arbitrary suborder embeddings. Hamkins and Woodin establish the first two implications in their Theorems 5–6 and ask in Question 7 whether the displayed principles are equivalent [27]. The present article neither proves those converses nor treats the repository’s condensation characterization of **CWO** as a proof of them.

Another independent distinction concerns the ambient model. Two externally isomorphic orders need not have their isomorphism available in a specified class realization. Glazer’s question about definable class well-orders explicitly separates genuine external well-foundedness from internal recognition, and separates external isomorphism from definability of the isomorphism [90]. Its discussion includes a truth-encoding proposal at length  $\text{Ord}^\omega$ ; we do not require that proposal as a lemma here. The bounds and separation theorems above are proved directly for admitted classes and maps. In a  $\beta$ -correct realization, their selected well-orders are also genuine external well-orders. Without that correctness hypothesis an internally valid construction may have an externally ill-founded interpretation.

These qualifications suggest a workable formalization policy. Store a notation, its carrier, its comparison relation, and its well-foundedness proof as separate data. State whether an interpreter is one available class or an external formula scheme. Record initial-segment embeddings separately from arbitrary embeddings, and require an explicit history when class-valued iteration is used. With these distinctions, long notations remain useful while their limits are mathematical theorems rather than hidden changes of universe or truth resources.

[merge] The citations of Frittaion and Genovesi’s 2026 preprint, of the correction to the countdown argument recorded in the revised class-forcing paper, and of Glazer’s MathOverflow question are source 35’s; their checking status is recorded in [section 1.11](#).

### XVI.9.3 Source 32: order length is not recursion strength

#### XVI.9.3.1 Three different uses of recursion

An elementary transfinite recursion along a supplied class well-order  $\Gamma$  asks for a class  $S \subseteq \Gamma \times V$  whose sections satisfy a rule of the form

$$S_\gamma = \{x : \varphi(x, S \upharpoonright \Gamma_{<\gamma}, Z)\}, \tag{S32.18}$$

where  $\varphi$  has only set quantifiers and  $Z$  is a fixed class parameter. The sections  $S_\gamma$  are classes, not necessarily sets. ETR asserts the existence of solutions for the relevant class well-orders. The distinction between class-valued recursion and finite or set-valued recursion is standard in the literature on second-order set theory [35, 84].

The proofs in this manuscript use three quite different mechanisms. Finite syntax allows recursive evaluation of a single finite tree. Set-valued recursion with a set history constructs, for example, an  $\omega$ -sequence of finite term codes or the cofinal map in [proposition XVI.3.7](#). Class-valued recursion constructs all the equivalence relations in [note XVI.5.20](#), with previous relations themselves as class data. Only the third is an invocation of ETR.

Even a recursion whose stage order is the set  $\omega$  can be class-valued: its stages might be increasingly complicated satisfaction classes. Therefore “the index is only countable” is not a reason to replace a class-valued recursion by the ordinary recursion theorem of ZFC. Conversely, the existence of  $E(\Omega)$  is not an ETR theorem, since its points are finite codes and well-foundedness was proved by support localization.

### XVI.9.3.2 Comparison principles and what is left open

Hamkins and Woodin prove that ETR implies initial-segment comparability of class well-orders and discuss reverse implications relating ETR, continuous rankings, and weaker comparison principles [\[27\]](#). We use their theorem only where explicitly stated. No claimed solution of those reverse-implication questions is part of the present work.

It is tempting to argue that two class well-orders can always be compared by recursively matching their least unmatched points. The difficulty appears when the matched initial history is proper: that history is no longer a set on which the ordinary set recursion theorem operates. It is equally tempting to take the union of all partial class matches, but this invokes a collection of class objects outside elementary class comprehension. Both arguments need additional justification.

Our diagonal theorem was designed to avoid this issue. Its contradiction is local to a single known block  $W_i + 1$ . Likewise, the division theorem obtains canonical isomorphisms from set-sized local ranks. These examples illustrate how to obtain substantial structural results without solving the general comparison problem.

In the external spectrum theorems, ordinary order types can of course be compared in the metatheory. That comparison is not being pulled back as an admitted isomorphism in  $\mathcal{X}_A$ . For example, the assertion that a definable suborder has an external ordinal type below another does not, by itself, prove that its initial-segment isomorphism is definable over  $(V_\kappa, \in, G, A)$ .

### XVI.9.3.3 Why there is no universal final ordinal

Three rigorous negatives replace a vague search for “the biggest ordinal beyond Ord”. First,  $W + 1$  is always a strictly larger presentation. Second, [note XVI.7.14](#) defeats any one admitted uniform family. Third, the exact externally realized spectrum changes with the admitted class collection: it is bounded by  $\delta_A$  for one definable realization and by  $\kappa^+$  for the full realization, with truth strictly increasing the former.

None of this contradicts the fact that every ordinary set well-order is isomorphic to a unique ordinal. The problematic step would be to apply the set collapse theorem to a relation whose predecessor segments are proper. The theorem’s set-likeness hypothesis is doing real work.

## XVI.9.4 Source 32: conclusion

The ordinal class is a boundary for set-like orderings, not for definable well-order presentations. Finite tuples, finite supports, and finite hereditary trees already produce a rich arithmetic beyond it. Every class well-order has a canonical first division by  $\Omega$ , and no admitted uniform family can be cofinal among all admitted class well-orders.

The strongest distinctions arise when uniformity is made explicit. A general condensation history, once admitted, must terminate by the order's own length, but existence of that history is a separate recursion principle. At an inaccessible set-universe cutoff, all definable class well-orders have a sharply structured external ceiling of countable cofinality, yet a truth predicate assembles a presentation of exactly that previously unavailable ceiling. Full classes extend farther still, through every external type below  $\kappa^+$ .

Thus the rigorous answer to “how far beyond Ord?” is not one terminal ordinal. It is an arithmetic of presentations, a hierarchy of admissible constructions, and a model-relative spectrum whose boundary moves when the available definitions become uniform.

### **XVI.9.5 Source 33: conclusion and status of the contribution**

The order of all set ordinals is a boundary for set-like orders, not a boundary for class well-orders. Explicit sums, products, finite-support powers, and coherent fixed points pass beyond it in elementary class set theory. The price is not inconsistency but proper-class initial segments and careful tracking of what is uniformly coded.

The structural analysis is quite concrete. Set intervals divide every class well-order canonically into  $\Omega$ -blocks and a final set remainder. An exhausting condensation tower gives finite ordinal digits and an embedding into a polynomial class order. The important existential issue is whether the tower is supplied by an explicit formula or requires a recursion principle.

The definability analysis has a different conclusion: there is no final uniform catalogue. Every fixed complexity bound can be diagonally exceeded, even when arbitrary set parameters are allowed. At an inaccessible rank model this yields a countably cofinal, arithmetically closed external ceiling strictly above the internal Ord, and truth strictly raises that ceiling without changing any set ordinal.

These are proved structural answers and a research program. The article is not a claim to have settled the general published comparison problem, not a report of a completed Lean formalization, and not evidence of historical priority. The most valuable next checks are an independent mathematical review of the bounded-definability construction and a formal proof of the supplied-tower digit theorem with every size assertion made explicit.

## **XVI.10 Rigorous formalization routes**

### **XVI.10.1 Two-sorted class theory**

For the reverse-mathematical result, a two-sorted formalization is the most faithful route. Use a type of sets, a type of admitted classes, membership of sets in classes, and codes for ordered pairs. Class relations are classes of pair codes. The elementary comprehension and Class Replacement schemes must be explicit assumptions of the formalized object theory. A class well-order quantifies over the admitted class sort, exactly as in the mathematical definition.

In this setting, a supplied history is one class of stage–point pairs. Its slices are defined by comprehension; they are not members of a newly created class of all slices. The polynomial carrier consists of finite graphs of sets. The greedy proof uses only a least point in a supplied well-order and ordinary recursion on a set. The resulting theorem can therefore track the absence of Choice at the level of the object theory.

A proof assistant's unrestricted collection of predicates on the set type should not silently replace the admitted class sort. That would usually provide full external semantics and could make

comparability automatic. Such a proof may be correct for that strong interpretation while failing to establish a theorem over arbitrary Henkin models of GB.

### XVI.10.2 Universe levels and ordinary order types

A second route treats  $\Omega$  as all ordinals small relative to one universe. In Lean’s ordinary ordinal development, `Ordinal.{u}` lives in `Type (u+1)` and represents order types of `Type u` carriers. Its complete order can be assigned an order type at the larger universe level. A tagged endpoint, finite product, or finite-support function space then produces an ordinary order in that larger universe. Mathlib’s ordinal library explicitly distinguishes these levels and their lifts [30].

This route is well suited to validating algebraic laws and the hereditary tree construction. It also makes clear why  $\Omega + 1$  is harmless: it lies beyond the smaller universe’s ordinals, not beyond the ordinals at every level. Moving all objects back to the smaller universe would require a smallness theorem that fails at exactly the proper-class boundary.

For a theorem about the exact strength over GB, the universe route is a semantic test rather than a replacement for a formal object-theory proof. Its strong ambient supply of subsets and functions cannot be used as evidence that every weak class realization contains the required isomorphism or history.

### XVI.10.3 Full inaccessible-universe semantics

Assume an inaccessible  $\kappa$  in an external ZFC universe and interpret sets by  $V_\kappa$  and classes by all its external subsets. Every point and every finite code belongs to  $V_\kappa$ . Class relations become external sets, so ordinary ordinal arithmetic computes their lengths. This interpretation gives the exact  $\kappa^+$  spectrum and the substitution rule

$$\Omega \rightsquigarrow \kappa, \quad P(B) \rightsquigarrow \kappa^{\text{otp}(B)}, \quad E \rightsquigarrow \varepsilon_{\kappa+1}.$$

It provides a strong consistency check on indexing and orientation of products. It does not prove a universal theorem over weak class theories simply by being one model of them.

### XVI.10.4 Unrolling and higher sorts

There is a more structural way to pass beyond the original sets. The unrolling construction uses well-founded extensional class graphs with a distinguished top to represent virtual sets. A class well-order with an appended top then represents an ordinal of the unrolled structure. Williams develops the resulting interpretations and explains how their strength depends on the ground class theory [35, Chapter 2]. This is a natural framework if one wants actual ordinal-like objects for class-order types, rather than working presentation by presentation.

One must prove the required graph comparisons and the axioms of the unrolled structure; they are not consequences of having written down a quotient notation. Similarly, a third sort of hyperclasses can collect class presentations, but its comprehension and replacement principles must be stated. Adding a sort moves the collection problem upward. It does not supply an absolute class of all orders across every higher level.

### XVI.10.5 A concrete formalization plan for ProveIt

The proof dependencies suggest the following small modules. The names are proposed organization, not claims of existing checked files.

| Module                 | Main content                                                                    | Required boundary                              |
|------------------------|---------------------------------------------------------------------------------|------------------------------------------------|
| Class orders           | Tagged carriers, admitted-subclass minima, initial embeddings, anti-embedding   | GB                                             |
| Finite-support powers  | Finite list comparison, greedy minimum, split and flatten isomorphisms          | GB; no Choice instance                         |
| Canonical blocks       | Set-interval predicate, local ordinal ranks, least representatives              | GB                                             |
| Canonical histories    | Supplied histories, floors, finite changes, normal-form initiality, equivalence | GB; existence as a named principle             |
| Hereditary syntax      | Finite tree validation, structural comparison, specialization, fixed-point map  | GB via the set-subset criterion                |
| Definability and truth | Formula codes, validity compiler, uniform diagonal, external spectra            | Truth and model assumptions tracked separately |

The most valuable first target is the choice-free power theorem. Its invariant concerns an actual nonempty subclass and the prefix selected from it; an implementation must not quietly replace this by a finite sample. Next formalize the fixed-cap step in normal-form initiality: it is what upgrades an arbitrary embedding to an initial embedding. These two points carry the mathematical content of the base-theory refinement.

For the hereditary syntax, define comparison by recursion on the sizes of two finite trees before asserting that the resulting order is well-founded. Define the height-bounded stages syntactically. This avoids accidentally using  $\text{ETR}_\omega$  to construct a sequence of proper-class stages. The ordinary specialization proof then gives a useful independent semantic specification.

No natural-number program can literally contain an arbitrary ordinal coefficient. A computational implementation should therefore state its coefficient interface: natural numbers, a specified notation system for set ordinals, or an abstract well-order with a comparison oracle. Claims of decidability apply to the chosen interface, not to every set ordinal or every formula-defined class presentation.

[merge] No module of the plan above exists in the repository; no statement of this Part is formalized (section 1.5). The ProveIt files the sources name are `SetTheory/ZF/Lean/ZF/Zf.lean` (source 32's Section 11.6, printed below), whose declarations `Sep_form`, `Repl_form`, `ZFax` and `ZFprov` and import `FirstOrder.Calculus` exist as source 32 says. Source 32 also reports `Ordinal.univ` and `Ordinal.type_lt_ordinal` in `Mathlib/SetTheory/Ordinal/Univ.lean`, with `Ordinal.univ_id` deprecated in favour of the latter ("since 2026-03-20"). These remain attributed API/deprecation claims; this review has not rechecked their external history. Both repository manifests pin `v4.32.0`, revision `81a5d257c8e410db227a6665ed08f64fea08e997`.

## XVI.10.6 Source 35: five formalization routes and a proof-assistant design

**First-order set theory with definable classes.** In a ZFC development, write each class by a formula and regard theorems involving arbitrary definitions as schemes. Carrier membership, comparison, finite supports, finite trees, and the explicit witnesses constructed in this article can all be handled in this way. A theorem for each fixed complexity level is legitimate; substituting a single purported truth predicate for all levels is not. This route is economical when the intended output is a specific notation system with explicitly defined operations.

**GB and GBC.** A two-sorted first-order theory makes class quantification explicit. Elementary comprehension permits fixed class parameters but forbids class quantifiers in the defining formula.



Finite-support normal forms and finite-tree evaluation remain ordinary classes of sets. A family of classes is represented by one binary class relation with sections, not by treating its proper-class sections as members. Global Choice permits arbitrary carriers to be recoded uniformly and supplies the general descending-sequence argument; constructions using only ordinal codes often need less choice.

**GBC + ETR.** Elementary transfinite recursion supplies a class-coded solution when each stage may itself be a proper class and the recursion follows an arbitrary admitted class well-order. It justifies general recursively defined comparison and iteration constructions. Its equivalence with clopen class-game determinacy, and the related equivalences for  $\text{ETR}_{\text{Ord}}$ , explain why this is a substantive additional axiom rather than a routine use of set recursion [22, 24]. The finite hereditary constructions in this article were arranged to avoid assuming it. Producing one explicit very long class order does not supply arbitrary recursions along that order.

**KM and unrolling.** Impredicative comprehension allows classes defined with class quantifiers. Unrolling instead represents larger objects by class membership codes and compares those codes by class isomorphisms. The exact axiom statement matters: Williams proves that KM plus Class Collection is bi-interpretable with  $\text{ZFC}_I^-$ , which includes Separation and Collection and has a largest inaccessible cardinal. Here inaccessibility includes the required powersets below that cardinal. For  $\text{GBC} + \text{ETR}$ , the corresponding theory is the weaker  $\text{wZFC}_I^-(0) + \Sigma_0\text{-Transfinite Recursion}$ , not full  $\text{ZFC}^-$  with Collection [35, Definition 2.1, Theorem 2.2, Corollary 2.32]. An externally well-founded collapse additionally requires the relevant  $\beta$ -correctness. An internal interpretation alone does not turn every represented order into an externally genuine ordinal.

**Higher-order theories and universe hierarchies.** One may allow hyperclasses whose elements are classes, or work in a larger universe in which the previous universe is a set. Such a move makes a previous collection of class orders available as an object at the new level. The level must be recorded: its ordinal operations are ordinary operations in the larger setting, while its lower-level interpretation may be a proper class order. Repeating this construction never produces a same-level object containing every order at every level. It produces a hierarchy of formally distinct domains.

#### XVI.10.6.1 A proof-assistant design with explicit universes

A direct Lean development should begin with a model of sets and an explicit collection of admitted predicates and relations. Its class-well-order structure should record a carrier predicate, a strict linear order on that carrier, and minimality for every admitted nonempty subclass. Set-likeness must be a separate field or predicate. Requiring it in the definition of every class well-order would exclude the main examples by construction.

The initial implementation can consist of the following independent interfaces:

1. Classes and class functions over a fixed set model, together with elementary comprehension and Class Replacement.
2. General class well-orders, arbitrary order embeddings, and initial embeddings, with each distinction retained in theorem statements.
3. Finite-support polynomials and their greatest-disagreement order, followed by explicit sum and product isomorphisms.
4. Finite hereditary syntax, its canonical comparison, and the compatible finite-stage embeddings defining the fixed-point order.

5. A separate semantics in an ambient inaccessible-height model, including the spectrum and definability theorems above.

An alternative begins with universe-indexed types. The type of ordinals of one universe lives in a larger universe; its lexicographic square and finite hereditary notation types can be formed there. This is an effective way to verify explicit order algebra. It does not by itself formalize the distinction between an arbitrary subset and an admitted class of a particular GB model. That distinction requires the model interface just described.

There are likewise two different uses of well-founded recursion. A proof assistant's recursion on a genuinely well-founded type can evaluate finite syntax directly. Applying the same mechanism to a model's *internally* well-founded class relation would silently add external correctness unless its hypotheses are translated and proved. Keeping the syntax layer, internal class semantics, and external realization separate exposes this obligation. The discussion here is a formalization design; it is not a claim that a Lean implementation has been compiled or kernel-checked.

### XVI.10.7 Source 32: five rigorous formalization routes

The right formalization depends on whether the goal is to check the arithmetic, to establish its minimal axioms, or to compare models with different admitted classes. No one encoding makes these three tasks identical.

#### XVI.10.7.1 ZFC with definable-class schemas

In a purely first-order treatment, replace each class symbol by a formula with a fixed finite tuple of parameters. A theorem about a fixed finite number of classes becomes a schema of first-order theorems. The constructions of sums, products, finite-support powers, and hereditary terms are syntactic transformations of formulas. Their proofs can be represented in an ordinary first-order proof system.

This route is especially faithful to the phrase “definable class well-order”. It also exposes the limit of the approach: a variable ranging over formula codes does not come equipped with an internal semantic evaluator. One may manipulate codes syntactically, prove substitution lemmas, and handle each fixed finite collection of formulas; one may not introduce a full satisfaction predicate for the ambient universe without an additional assumption.

A global well-order needs attention. ZFC alone does not assert a parameter-definable global well-order of its universe. One should either work over a structure with an explicitly supplied global predicate, assume a definable global-order hypothesis, or weaken the claims so that they do not require Global Choice. The GBC results above should not be translated into parameter-free ZFC assertions by simply erasing the class parameters.

#### XVI.10.7.2 A two-sorted admitted-class model

For questions about GBC itself, use a structure with a set sort  $M$ , a class sort  $\mathcal{X}$ , an internal set-membership relation, and an interpretation relation  $x \in C$  for  $x \in M$ ,  $C \in \mathcal{X}$ . Tuples and graphs are internally encoded in  $M$ . An embedding is required to have a graph in  $\mathcal{X}$ .

The well-order predicate must quantify over admitted subclasses:

$$\text{WO}_{\mathcal{X}}(D, R) \implies \forall S \in \mathcal{X} (\emptyset \neq S \subseteq D \implies S \text{ has an } R\text{-least point}).$$

Alternatively, after formalizing lemma XVI.2.8, use the elementary internal condition forbidding descending functions whose codes are *sets of the model*. These are not automatically all external functions from an external  $\omega$ .

This matters in Lean. A bare declaration  $R : M \rightarrow M \rightarrow \text{Prop}$  followed by ambient `WellFounded`  $R$  usually means well-foundedness for the proof assistant’s full external predicate and sequence universe. It is stronger than the internal condition of an arbitrary Henkin class model. Similarly, setting the class sort equal to all predicates  $M \rightarrow \text{Prop}$  chooses full external classes, not a general GBC realization. To study  $\mathcal{X}_A$ , encode the admissible predicates and admissible graphs explicitly, together with closure under elementary definitions.

For the inaccessible-rank models of [lemma XVI.8.7](#), a separate absoluteness theorem then relates internal well-ordering to external well-foundedness. That theorem is a bridge to be proved, not a convention to apply to all models.

### XVI.10.7.3 GBC with explicit recursion packages, or stronger class theory

An ETR interface should take a stage presentation, an elementary formula code, and admitted parameters, and return an admitted history satisfying ([S32.18](#)). The condensation development can then be split cleanly in two: the termination invariant is proved from a supplied history; a separate theorem invokes the ETR interface to obtain that history. A stronger theory such as Morse–Kelley class theory can also supply the requisite recursion, but that should not obscure the weaker conditional theorem [[35](#)].

This separation gives an auditable dependency boundary. The explicit polynomial hierarchy needs no recursion package at all: formula ([S32.9](#)) supplies its code directly. Its verification consists of successor and limit identities. By contrast, an arbitrary  $W$  enters the existence theorem only after the ETR assumption has been made.

### XVI.10.7.4 Higher universes and ordinary ordinal libraries

A practical route to checking the arithmetic is to view the ordinals of one Lean universe as a type in a larger universe. Mathlib’s current ordinal-universe interface provides `Ordinal.univ` and `Ordinal.type_lt_ordinal`; the latter identifies the order type of the smaller-universe ordinals with the corresponding larger-universe universal ordinal. The old name `Ordinal.univ_id` is marked deprecated in the documentation inspected on 4 October 2026 [[92](#)].

This makes the presentation  $\Omega$  into an ordinary ordinal in a higher universe, after which the usual ordinal library can verify powers and initial embeddings. It is an excellent arithmetic test bed. It does not identify the predicates admitted by an arbitrary  $(M, \mathcal{X})$ , and it does not establish that a construction was provable in bare GBC. A universe-level smallness notion is also not automatically the model’s internal “is a set” predicate. The inaccessible-universe interpretation should never be read as a proof in ordinary ZFC that a strongly inaccessible set cardinal exists.

A useful implementation architecture therefore has an abstract presentation layer and a separate realization functor into higher-universe ordinals. The functor certifies arithmetic identities once their maps are defined, while the abstract layer tracks which graphs and histories are admitted by the chosen class theory.

### XVI.10.7.5 Unrolling into virtual sets

Another approach represents virtual sets by well-founded extensional class membership codes. A class well-order, with an added top point, can serve as a code for an ordinal in an unrolled structure. This is a change of interpretation, not a collapse to an actual ordinal in the original set universe. Williams develops the unrolling approach and the dependence of its properties on the underlying second-order theory [[35](#)].

For the present project, the attraction is that arithmetic on long orders can be compared with ordinary arithmetic on the corresponding virtual ordinals. The danger is assuming all the axioms of the target theory before proving them. In particular, existence of suitable code comparisons and set-forming operations needs its own theorem. An unrolled ordinal representing  $\Omega + \Omega$  is not a counterexample to the original model's assertion that its von Neumann ordinals form the class `Ord`.

### XVI.10.7.6 Connection to the inspected ProveIt foundation

The inspected file

`SetTheory/ZF/Lean/ZF/Zf.lean`

works with first-order formulas and explicitly coded ZF axioms. Among the declarations inspected are `Sep_form`, `Repl_form`, `ZFax`, and `ZFprov`; the file imports `FirstOrder.Calculus`. Its documented internal-recursion development concerns finite recursion and closure for definable set-like relations [81]. These are relevant foundations, but they do not already constitute a formalization of non-set-like class recursion.

A compatible extension would first add syntax for finitely many class predicate parameters and prove substitution and satisfaction bridges. A second layer would package admitted classes and their elementary-comprehension closure. The arithmetic could then be developed with no ETR import. Condensation existence would live in a module whose assumptions explicitly include the recursion package, while termination from a supplied hierarchy would remain in the weak layer.

Within a proposed `ClassOrders` namespace, useful modules would be `Basic`, `FiniteSupport`, `Arithmetic`, `Condensation`, and `DefinabilitySpectrum`. These names describe a proposed organization, not files created in the repository. The first machine-checked milestones should be least-representative quotients, no self-compression, and the finite-support comparison laws. The model-relative ceiling theorem should follow only after the syntax/semantics boundary has been formalized.

**Status.** No Lean or Rocq proof was produced or checked for this article. No repository build or modification was performed. The formalization discussion is a source-informed implementation plan, not a verification claim.

## XVI.10.8 Source 33: formalization approaches and their tradeoffs

### XVI.10.8.1 ZFC with definable-class schemas

One can express each explicit construction as a theorem schema in ordinary ZFC. Fix formulas defining the domain, order, and any needed global ordering, substitute those formulas into the construction, and prove the resulting first-order assertion. This is particularly well suited to finite-support exponentiation and the bounded-complexity diagonal.

The limitation is quantification. A schema over formulas is not a single internal quantifier over all definable classes. The diagonal proof makes this limitation visible rather than hiding it: the evaluator exists at each fixed level, but a uniform evaluator for every level is not thereby supplied. For formalization, formula substitution and the distinction between a metatheoretic natural number and an object-theoretic variable must be explicit.

The use of a named  $G$  can be handled either by assuming its defining formula has the required global-order properties or by working in a language with an additional class predicate and the corresponding separation/replacement instances. Pure ZFC without a definable global order does not automatically satisfy the parameter-free formulation of [corollary XVI.7.23](#).

### XVI.10.8.2 Two-sorted GB, GBC, and NBG

A two-sorted theory is the most faithful framework for the core article. The set sort contains every point, finite code, and ordinal coordinate. The class sort contains domains, relations, embeddings, and uniform towers. Elementary comprehension creates the needed classes, and replacement recognizes when their set-indexed images are sets.

Terminology varies: some formulations of NBG include global choice as part of the default theory and some distinguish it. Here GBC always includes a set-like global choice order. No argument should be transferred to a choiceless or Power-Set-free version merely by changing the theory's name.

The central size invariant is a predicate saying that a specified subclass has a set with exactly its members. For example,

$$\text{SmallInterval}(x, y) \iff \exists z \forall t [t \in z \leftrightarrow t \in [x, y]_W].$$

It must not be replaced by external cardinality or by the fact that an implementation stores a finite approximation. Quotient elements should be least representatives. These design choices let the core statements remain inside the original two sorts.

### XVI.10.8.3 Elementary transfinite recursion and comparison

A general recursion of the form

$$S_\gamma = \{x : \varphi(x, S \upharpoonright \gamma, A)\}, \quad \gamma \in \Gamma,$$

with  $\varphi$  first-order and  $A$  a class parameter, is what  $\text{ETR}_\Gamma$  supplies. The stages  $S_\gamma$  are classes, so ordinary set recursion does not suffice. Even recursion of length  $\omega$  with class-valued stages can encode a full satisfaction predicate [22].

The known implication

$$\text{GBC} + \text{ETR} \implies \text{initial comparability of class well-orders}$$

is established in Hamkins–Woodin [27, Theorem 6]. This supplies a conventional ordinal-style comparison theory when it is needed. The finite-support operations, canonical division, supplied-tower digit map, and diagonal theorems do not use that implication.

One should also distinguish  $\text{ETR}_\Omega$  from full ETR over every class well-order. The exact strength of the class forcing theorem identifies it with the former, among several equivalent principles [24]. A notation such as  $\text{ETR}_{\Omega[\Omega]}$  therefore carries real information about the permitted recursion lengths.

### XVI.10.8.4 KM, stronger recursion, and unrolling

Kelley–Morse theory allows impredicative class comprehension, where the defining formulas may quantify over classes. The corresponding second-order transfinite recursion principle is equivalent to KM over GBC [26]. This is a stronger setting than the elementary recursions used in the conditional condensation results.

Stronger class comprehension still does not make proper classes ordinary members of classes. To treat class well-orders as ordinal-like objects in a larger first-order universe, one may instead *unroll*: use well-founded extensional class membership codes and identify codes by isomorphism. Williams describes, in particular, the unrolling of KM plus Class Collection to a model of  $\text{ZFC}^-$  with a largest cardinal, and analyzes weaker theories as well [35]. Antos–Friedman use a related coding framework for beta-models of an explicitly strengthened Morse–Kelley theory [28].

These are interpretive constructions with hypotheses, not permission to write “the class of all class ordinals” inside ordinary GBC. Collection, well-foundedness correctness, and the exact set theory obtained by unrolling must be tracked separately. This manuscript uses unrolling as a formalization option, not as an unproved shortcut in its elementary proofs.

### XVI.10.8.5 Inaccessible universes and full semantics

The  $V_\kappa$  model of [section XVI.8.8](#) translates the class orders into ordinary external ordinals. It is particularly effective for testing arithmetic formulas and explaining the spectrum endpoint  $\kappa^+$ . It also supports all full subclasses, so it is stronger than a minimal definable-class realization.

A full powerset of classes is not harmless over an arbitrary set model. For example, if a transitive model is externally countable, its full class part includes an external countable cofinal enumeration of its ordinals. Treating that enumeration as a class function on its  $\omega$  contradicts class replacement. The regular inaccessible cutoff avoids exactly this failure. This is why the hypothesis is proved relevant, rather than being used only as an informal picture of a “large enough universe.”

### XVI.10.8.6 Lean universes: relative largeness rather than an absolute class

Mathlib’s ordinal type is universe indexed. In its documented interface, `Ordinal.{u}` lives in `Type (u+1)`, and `Ordinal.type` assigns an ordinal to a well-order on a type of the chosen universe [30]. Thus the order on all `Ordinal.{u}` can itself receive an order type one universe higher. Abstractly, define

$$\Omega_u = \text{type}(\text{Ordinal}^{\{u\}}, <) \in \text{Ordinal}^{\{u+1\}}.$$

Its successor, square, and higher powers then use ordinary ordinal arithmetic at that higher level.

This approach is excellent for concrete relative-universe models. It does not identify the smaller-universe ordinals with a set ordinal in the same smaller universe. Nor does Lean provide a single type of ordinals over *all* universes. Universe lifting is part of the result, not a technicality to suppress.

There are two distinct formalization projects:

- (a) **Ambient type-theoretic mathematics.** Use ordinary well-orders on large types, smallness predicates for predecessor types, and larger-universe ordinals. This is the shorter route to arithmetic and representation theorems.
- (b) **A theorem about class set theory.** Define a two-sorted model interface, an explicit available-class sort, elementary comprehension, class replacement, and a named global order. Prove each implication using only that interface. This is the route that can certify a stated GBC versus ETR axiom bound.

A structure that simply interprets “class” as *every* predicate  $U \rightarrow \text{Prop}$  gives full external semantics. It must not be used to claim an elementary GBC proof when the argument secretly quantifies over predicates unavailable in a weaker realization. The explicit available-class sort prevents this mistake.

The current article provides interfaces and proof obligations, not a Lean implementation. No displayed mathematical interface is advertised as compiler-tested code.

### XVI.10.8.7 Transport to surreal carriers

Under global choice, every proper class of sets can be enumerated in a set-like order of type  $\Omega$  by restricting a global order. Hence proper class carriers of the orders studied here can be put in class



bijection with the surreal numbers **No**, and their order relations transported to that carrier. This connects the article with the repository’s surreal well-order projects.

The transported order is not the natural ordered-field relation on **No**. That relation is not a well-order; for example, the negative integers descend indefinitely. Nor is the symbol  $\Omega$  a surreal number larger than every set ordinal. Each surreal is represented by a set-length sign sequence, and an ordinal longer than its birthday bounds it above. There is no surreal with proper-class birthday in the ordinary construction.

Finally, ordinal arithmetic is not surreal field arithmetic. Already  $1 + \omega = \omega$  in ordinal addition, whereas adding the surreal number 1 to the surreal  $\omega$  gives a strictly larger number. Reusing symbols must not erase which operations or which ordering are being studied.

## XVI.10.9 Source 33: a formalization and verification plan for ProveIt

### XVI.10.9.1 Suggested module boundary

A proposed new development could live under `SetTheory/ClassWellOrders/`; this is a suggested location, not a claim that the directory already exists. Its mathematical dependency graph should separate the following components.

| Component     | Main specification                                                                                         | Trust boundary                                            |
|---------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Presentations | Domains, class relations, embeddings, initial images; least-subclass and descending-sequence formulations. | Explicit class-model interface.                           |
| FiniteSupport | Finite maps with a greatest-difference comparison; split, group, and flatten maps.                         | Finite combinatorics plus ordinary set well-orders.       |
| Arithmetic    | Sums, products, powers, and uniform lexicographic sums.                                                    | Elementary comprehension and replacement.                 |
| SetIntervals  | Convexity, least representatives, set-like blocks, and canonical $\Omega$ -division.                       | Internal sethood, not external size.                      |
| Condensation  | Supplied towers, polynomial truncations, ordinal digits, termination and order preservation.               | Separate supplied-tower theorem from existence under ETR. |
| Definability  | Formula codes, each partial satisfaction predicate, the well-orderhood test, and the diagonal compiler.    | Schema-level indexing must remain explicit.               |
| RankModels    | Correctness of $V_\kappa$ , full spectrum, definability height, and truth jump.                            | External inaccessible hypothesis.                         |

The successor obstruction should be proved early and imported by every diagonal argument. It is short, robust, and avoids the temptation to assume initial comparability as an unrecorded global fact.

### XVI.10.9.2 The most important proof obligations

For the structural side, the critical obligations are that interval sethood is internal, quotient representatives are sets, the ordinal coordinate in (S33.5.4) exists, and the digit-extraction process terminates by well-foundedness of the stage order. For the definability side, they are that bounded

satisfaction is genuinely available, the well-orderhood test has only set quantifiers, and the family of all parameter tuples is jointly definable before the diagonal sum is formed.

A formalization of [theorem XVI.7.20](#) should not package all the evaluators into a predicate assumed to satisfy unrestricted truth. A syntactic compiler that takes a *metatheoretic* numeral  $n$  to the formula defining  $D_n$  is appropriate. A purported internal evaluator for every resulting formula would need a separate existence theorem and would contradict cofinality if it were definable in the unchanged language.

### XVI.10.9.3 Executable checks included with this article

The archive includes a small Python verification program for finite positional models. It checks greatest-difference comparison, splitting an exponent sum, flattening nested powers, and the partition into fibers of truncation. These tests can catch a reversed product convention or an incorrect significance order in the finite coding layer.

They do *not* verify that a proper-class order is well-founded, that a class relation is a set, that partial truth exists, or that a class recursion is licensed. In a genuinely finite order every interval is a set, so replacing set-interval condensation by naive finite testing would collapse everything at the first step. The tests intentionally avoid that false analogy.

A useful next implementation would prove the finite checker correct, then prove that the class construction uses the verified combinatorial maps. Until that bridge and the foundational arguments are formalized, the Python output remains a consistency check on conventions, not a mathematical certificate for the class-theoretic theorems.

[merge] The program described above is shipped as `code/33-heights-finite_checks.py`, its recorded run as `data/33-heights-verification_results.json` ([appendix A.13](#)).

## XVI.11 Ledgers, checks, and source audits of sources 32–35

[merge] The four sources' proof ledgers, verification records and source audits, as delivered (source 34's Appendix A; source 35's Appendix A; source 32's Appendices A–C; source 33's Appendices A–C). Their file names are the delivery names; the shipped names are in [appendix A.13](#) and the README. The finite checks were rerun for this write on copies, with the recorded results ([appendix A.13](#)).

### XVI.11.1 Source 34: proof and provenance ledger

The proofs in this report were independently checked during preparation for the following specific risks: an illicit proper-class element, an unacknowledged use of choice, a class-valued recursion disguised as finite recursion, a noninitial embedding used as an initial one, and an external well-order assertion without the required model hypothesis. These checks are mathematical review within this preparation process, not external peer review or a kernel check.

The exact repository dependency is Source 29 of [77], especially the source labels `swo:bw:ch:floors`, `swo:bw:ch:finite-changes`, `swo:bw:ch:normal-form-theorem`, `swo:bw:ch:convex-restriction`, and `swo:bw:ch:equivalence`. The repository's polynomial lemma invokes Global Choice; [Theorem XVI.4.2](#) replaces that invocation. The remaining arguments are re-established here with their weaker dependencies. The endpoint-transport corollary for a universal carrier is not included in the choice-free strengthening.

The hereditary finite-tree pattern is standard in ordinal notation. Its appearance in the BIRS problem collection concerns a set-sized  $\Omega$ , commonly  $\omega_1$  [85]. The present development supplies

proper-class semantics, finite-code well-foundedness, and the corresponding fixed-point interpretation. It does not claim to have invented hereditary Cantor normal form.

The uniform-sum and parameter-spectrum arguments are proved in full and presented as a synthesis. No bibliographic search can certify their historical novelty, and no such certification is asserted. The explicit resolution claimed here is limited to the base-theory question in the inspected ProveIt report.

The accompanying source is a single self-contained LaTeX file with an internal bibliography. Rebuild it with two runs of `pdflatex -interaction=nonstopmode -halt-on-error beyond_ord.tex`, or with `latexmk -pdf beyond_ord.tex`. The PDF was compiled and inspected for layout. No Lean or Rocq proof is included, and the mathematical result is not represented as a formalization.

[merge] The five labels source 34 names are this report’s [lemmas 108.12](#) and [108.17](#), [proposition 108.13](#), and [theorems 108.16](#) and [108.19](#); the “polynomial lemma” is [note 108.15](#), and the endpoint-transport corollary [corollary 108.20](#). Source 34’s single file and PDF are not shipped; Part XVI is its text.

### XVI.11.2 Source 35: proof status and reproducibility

The article supplies mathematical proofs, independently checked during its preparation. It is not a refereed publication and does not contain a Lean-verified proof. The following ledger records what the principal claims actually require.

| Claim                                 | Base or setting                     | Principal verification obligation                                                                                           |
|---------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Finite-support powers are well-orders | GB                                  | The minimization uses uniquely determined finite prefixes, with no selection of a class sequence.                           |
| Arithmetic identities                 | GB                                  | Explicit support splitting, flattening, and inverse maps.                                                                   |
| Canonical history criterion           | GB                                  | Floors at limits, finite changes, and capped reconstruction of every smaller polynomial.                                    |
| CWO $\leftrightarrow$ CTH             | GB                                  | Common-schedule convex restriction and the successor obstruction. Neither principle is assumed to be a theorem of the base. |
| Hereditary fixed point                | GBC                                 | Bounded coefficient specialization rules out descending sequences; root formation is an isomorphism.                        |
| Local fixed-point completion          | GBC                                 | Uniform finite traces, least-stage representatives, preservation of initial embeddings, and finite-support continuity.      |
| Uniform-family strict bound           | GB plus a supplied index well-order | A successor copy of each input is a block of the sum.                                                                       |
| Fixed-complexity separation           | Fixed- $n$ scheme                   | Partial truth at one fixed complexity and a definable well-order test on ordinal carriers.                                  |
| Full and definable spectra            | Ambient ZFC, inaccessible $\kappa$  | External cardinality and regularity; the full class part is specified.                                                      |

The accompanying program `artifacts/finite_notation_checks.py` performs exhaustive finite checks of the split and flatten maps for bases 2,3 and small finite exponent sets. It also compares a collection of hereditary terms against exact numerical interpretations at bases 5,7,11,

each larger than every coefficient used. It writes its counts and ordered examples to `artifacts/finite_notation_results.json`. These checks detect coding-convention errors such as reversing the product order or flattening the two exponent coordinates in the wrong direction. They do not establish infinite or class well-foundedness; that conclusion comes from the proofs in the article.

The archive contains a self-contained LaTeX source, the compiled PDF, the illustrative program and its recorded output, and a README with build instructions. The LaTeX source uses standard mathematical packages and an inline bibliography, so it requires no bibliography database, external images, or network connection. Run `pdflatex` until the cross-references and contents stabilize (normally three runs from a clean directory); `latexmk -pdf` automates these passes.

Repository statements were tied to the explicit commit in the bibliography. The external literature was consulted on 4–5 October 2026. An unverified priority claim, a numerical experiment, and a theorem proved from stated assumptions are different kinds of evidence; the article keeps them separate. Future work should begin by formalizing the new choice-free proof and the fixed-point completion theorem, then compare the exact results with a broader literature search.

[merge] Source 35’s “fixed-point construction and local criterion” are now proved over GB as well (theorem XVI.6.21). Its program is shipped as `code/35-completion-finite_notation_checks.py`, and its recorded output as `data/35-completion-finite_notation_results.json`. The program always writes `finite_notation_results.json` beside itself, here in `code/`: rerun it on a copy (appendix A.13).

### XVI.11.3 Source 32: dependency and claim audit

| Result                                                     | Sufficient setting                     | Scope of the claim                                                                                 |
|------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------|
| Set-like boundary,<br><a href="#">note XVI.2.9</a>         | GBC                                    | Canonical local-rank proof; standard foundational result.                                          |
| Finite-support powers,<br><a href="#">note XVI.4.10</a>    | GBC                                    | Explicit presentation;<br>countable-obstruction localization to set orders.                        |
| Hereditary fixed point,<br><a href="#">note XVI.6.24</a>   | GBC                                    | Finite syntax, not general class recursion; no novelty priority asserted.                          |
| $\Omega$ -division, <a href="#">theorem XVI.3.6</a>        | GBC                                    | Sharp formulation from canonical set-interval blocks; block antecedent credited to the repository. |
| General termination,<br><a href="#">note XVI.5.20</a>      | GBC plus a supplied hierarchy          | Termination invariant is unconditional on further recursion axioms once the history exists.        |
| Hierarchy existence                                        | GBC + $\text{ETR}_{W+1}$               | Sufficient, not proved necessary.                                                                  |
| Exact polynomial heights,<br><a href="#">note XVI.5.21</a> | GBC                                    | Direct formula; prior repository mechanism re-established.                                         |
| Uniform diagonal,<br><a href="#">note XVI.7.14</a>         | GBC                                    | Defeats arbitrary embeddings from the diagonal into any fiber; no comparability assumption.        |
| Full spectrum, <a href="#">note XVI.8.8</a>                | External ZFC,<br>inaccessible $\kappa$ | Full class realization; ordinary external order types.                                             |

| Result                                                        | Sufficient setting                        | Scope of the claim                                                                       |
|---------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------|
| Definability ceiling,<br><a href="#">theorem XVI.8.9</a>      | Same external setting;<br>fixed $G, A$    | All set parameters allowed;<br>internal/external correctness supplied<br>by $V_\kappa$ . |
| Exact truth presentation,<br><a href="#">theorem XVI.8.12</a> | Same setting, enlarged<br>predicate $T_A$ | Represents $\delta_A$ itself; does not<br>compute $\delta_{T_A}$ .                       |
| General initial comparison                                    | GBC + ETR                                 | Literature result, not a new theorem<br>of this article.                                 |

The most important proof checks concern size rather than algebraic manipulation. A quotient uses least representatives instead of class-valued elements. A uniformly coded family is one relation on set tuples. A limit condensation stage is one admitted section of one admitted history. A finite list of formulas is evaluated by finitely many substitutions, whereas evaluation at a variable formula code requires truth. Finally, external ordinal spectra are formed only after moving to a set-sized external model.

No proof above deduces a new large-cardinal inconsistency, a same-level set of all ordinals, full ETR from mere presentation closure, or a resolution of general class-order comparability over bare GBC. These would be different claims requiring different arguments.

#### XVI.11.4 Source 32: computational checks and reproducibility

The accompanying Python script checks finite digit comparisons, support splitting, grouping of nested exponents, initial support embeddings, and hereditary base representations of ordinary natural numbers. It also checks the algebra of tail-deletion equivalence relations on bounded digit arrays. The run completed 689,808 exact assertions without failure; the category counts are recorded in `finite_checks.json`.

These computations are regression tests for the concrete coding conventions. They do not prove the infinitary results. In particular, every interval in a finite order is a set, so an actual finite test of the distinction between a set interval and a proper-class interval would be vacuous. The tail-deletion tests therefore verify only the stated finite mask formulas; they do not claim to simulate class condensation. Likewise, no finite program here verifies GBC, ETR, an inaccessible cardinal, or an external truth predicate.

The source is self-contained LaTeX with an internal bibliography. Build it with a standard TeX distribution, for example by running `pdflatex beyond_ord.tex` repeatedly until references stabilize. No nonstandard font files or external bibliography database are required. The separate verification-status file records the compilation and rendering checks performed for the delivered PDF.

[merge] The “accompanying Python script” is `code/32-ceilings-finite_checks.py`, `finite_checks.json` is `data/32-ceilings-finite_checks.json`, and the “separate verification-status file” is `32-ceilings-PROOF_STATUS.md`. The source and its PDF are not shipped.

#### XVI.11.5 Source 32: source inspection record

The repository was read on 4 October 2026. Because it was active during inspection, a root-tree read and later file reads need not represent one atomic snapshot. The inspected root tree reported commit

8f7d4a5c8cb87e66534d88ec6698f6f46d32a152.

The relevant research-status file reported blob

5333d792f02b0d578b7a9907a30d06b5f154e14a,

and the inspected first-order Lean foundation reported blob

b33db57cfb71e00c50fa3c57ea355ee4a9a4d523.

The status file was read in full; the Lean inspection covered its header and initial 150 lines, including the displayed first-order axiom declarations. The merged report’s README was inspected for scope and attribution. No claim is made to have audited all of the merged mathematical proofs or all of the repository’s formal code.

The literature search focused on primary sources concerning class well-orders, elementary transfinite recursion, class realizations, and unrolling. It supports the attributions made in the article but is not an exhaustive priority search for every theorem proved here. The exposed exact interfaces in the Mathlib documentation were checked separately rather than inferred from an older memory of their names.

[merge] The “root tree” 8f7d4a5c8 is a commit (2026-10-04 16:32 PDT, tree 08039c482), at which this article is blob 72eee5cc8, the text with Parts I–XV; the two blobs it records are the files it names, both unchanged at the time of this write.

### XVI.11.6 Source 33: axiom and proof-status ledger

| Statement                                         | Sufficient assumptions                       | Status here                                                                                          |
|---------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------|
| Descending-sequence test; set-like classification | GBC                                          | Standard background, with proofs and conventions.                                                    |
| Successor obstruction; uniform-family diagonal    | GBC                                          | Complete written proofs; no comparison axiom.                                                        |
| Finite-support power and arithmetic laws          | GBC                                          | Complete written proofs; finite convention checks supplied.                                          |
| Canonical $\Omega$ -division                      | GBC                                          | Complete written proof with least-representative quotient.                                           |
| Polynomial condensation                           | GBC                                          | Explicit uniform formula and proof; related long-condensation work already exists in the repository. |
| Finite-digit embedding                            | GBC plus a supplied uniform exhausting tower | Complete written proof; tower existence is not assumed free.                                         |
| Converse from a polynomial embedding              | GBC + $\text{ETR}_{\Gamma+}$                 | Conditional written theorem.                                                                         |
| Exhaustion of arbitrary $W$ along $W + 1$         | GBC + $\text{ETR}_{W+1}$                     | Sufficient bound proved; optimality not established.                                                 |
| Explicit finite-power fixed point                 | GBC                                          | Finite-tree coding and direct limit proof; not arbitrary class recursion.                            |



| Statement                                    | Sufficient assumptions                     | Status here                                                         |
|----------------------------------------------|--------------------------------------------|---------------------------------------------------------------------|
| Bounded-definability diagonal and cofinality | GBC, fixed named $G$ ; standard finite $n$ | Schema of complete written proofs; no uniform full truth predicate. |
| Exact full-class spectrum                    | External ZFC and inaccessible $\kappa$     | Complete proof for $(V_\kappa, \mathcal{P}(V_\kappa))$ only.        |
| Definability height and arithmetic closure   | Same cutoff, fixed $G$                     | External supremum theorem; not an internal cofinal sequence in Ord. |
| Strict truth-height jump                     | Same cutoff, truth $T$ available           | Complete written proof; language changes explicitly.                |
| General class initial comparison from ETR    | GBC + ETR                                  | Established literature, cited rather than claimed new.              |
| Unrolling and stronger recursion             | Theory-dependent extra principles          | Literature interface only; no new equivalence claimed.              |

None of the class-theoretic proofs in this manuscript has been checked by Lean, Rocq, or another proof assistant as part of this delivery. The finite Python checks establish only the enumerated finite identities. No axiom ledger entry should be upgraded on the basis of those tests.

### XVI.11.7 Source 33: notation and error-prevention guide

| Notation                    | Meaning                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------|
| $\Omega$                    | The class order $(\text{Ord}, \in)$ , not a set ordinal.                                       |
| $A \cong B$                 | An available class order isomorphism, or its explicit external interpretation where specified. |
| $A \hookrightarrow B$       | An available order embedding; its image need not be initial.                                   |
| $A \preceq_{\text{init}} B$ | An order embedding whose image is an initial segment.                                          |
| $A \cdot B$                 | $B$ -many ordered copies of $A$ , with $B$ the significant coordinate.                         |
| $A^{[B]}$                   | Finite-support ordinal exponentiation; not all functions $B \rightarrow A$ .                   |
| $P(\Gamma)$                 | The class of finite Ord-coefficient polynomials on exponent order $\Gamma$ .                   |
| $\Gamma^+$                  | The order $\Gamma + 1$ with a new top stage; not a successor cardinal.                         |
| $\kappa^+$                  | The external successor cardinal of the inaccessible $\kappa$ in the spectrum analysis.         |
| $F_b$                       | A section of one uniform condensation-tower relation.                                          |
| $D_n$                       | A separately defined diagonal presentation for each fixed standard finite syntax level $n$ .   |
| $\delta_G$                  | The external supremum of types definable with $G$ and set parameters over $V_\kappa$ .         |
| $T$                         | Truth for the language with $G$ , not truth for its own expanded language.                     |

The most common invalid shortcuts are collecting proper classes as members, using a full truth evaluator in a bounded-truth argument, confusing a schema with one uniform class, treating internal

well-foundedness as automatically external, and using an arbitrary class recursion without recording ETR. The constructions above were organized to avoid these shortcuts.

[merge] In this report source 33's  $\Omega$  is printed  $\Omega$ , its  $A^B$  is  $A^{[B]}$ , its  $P_\Gamma$  is  $P(\Gamma)$ , and its direct limit  $E$  is  $\mathbb{T}$  (section 1.3.4).

### XVI.11.8 Source 33: repository and source audit

The repository paths reviewed were the main README, the surreal report catalogue, and the surreal-well-orders guide, accessed on 4 October 2026. The catalogue blob returned by the connector was `b7b69377ff172577b642b4fb78bfb2b09989dcd0`; the well-orders guide blob was `d2e24b58617bbbe56aadbdeb11c2018da51ae79`. These are file-content identifiers, not claimed commit identifiers. The guide's declared scope and status were used for continuity and to avoid misrepresenting existing work as new. The full 860-page report was not reviewed line by line.

Primary external sources were consulted in their arXiv abstract or HTML versions, the author's exposition for stronger recursion, and the official mathlib documentation. Bibliographic years identify the cited version or publication; they do not assert that an older open-question statement is still unresolved as of the delivery date. A targeted search found the Hamkins–Woodin comparison question but did not establish a later definitive resolution.

The source audit and proof-status file supplied in the archive give a machine-readable-path companion to this appendix. The article has no external figure or font-file dependency; its bibliography is included in the LaTeX source.

[merge] The two blobs are the current `Algebra/SurrealNumbers/docs/README.md` (introduced in `c14a866bc`) and this report's README (as of `e50dde15b`); source 33 gives no commit, so its reading lies between `c14a866bc` and the arrival `e3839ad2c`, and in that range this article is the text with Parts I–XV. “The source audit and proof-status file” are `33-heights-SOURCE_AUDIT.md` and `33-heights-PROOF_STATUS.md`; the PDF and `MANIFEST.sha256` are not shipped.

## XVI.12 What this Part answers

[merge] Dated notes, 4 October 2026, in the form of section 112; where an earlier status section or question index says otherwise, these notes supersede it, and the places concerned carry dated notes of their own.

- (a) *The strength of complete condensation histories over GB without global choice* (note 108.4; item (b) of section 112; source 20's research question 113.111; the GBC theorem is source 29's theorem 108.19): **answered**, twice independently. Over GB with ZF sets and no form of choice, CWO, CTH, terminating histories on some schedule, and initial embeddings  $W \preceq_{\text{init}} P(W)$  are equivalent (theorem XVI.5.8 and note XVI.5.19), and so is the fixed-point extension principle FPE (corollary XVI.6.19 and theorem XVI.6.21). Not answered: whether GB proves CWO (no source claims it), and whether CWO implies ETR (Hamkins and Woodin's Question 7; source 29's research question 113.219).
- (b) *The strength of tower or hierarchy existence* (source 32's research question 113.258, source 33's research question 113.259, and the ledgers' “optimality not established” in sections XVI.11.3 and XVI.11.6): **answered** by item (a) and, over GBC, already by source 29, because towers and hierarchies of equivalence relations are nested histories (remark XVI.5.24); only CWO  $\Rightarrow$  ETR remains. Source 32's claim to answer the termination question repeats source 20 (remark XVI.1.1).

- (c) *Source 33's remark XVI.5.28 and research question research question 113.263, initial image of the digit map: answered:* the map is always initial (remark XVI.5.29); under  $\text{ETR}_{\Gamma+1}$  an arbitrary embedding into  $P(\Gamma)$  yields an initial one (remark XVI.5.31).
- (d) *Source 35's research question 113.270, a choice-free completion theorem: answered* positively by this write (theorem XVI.6.21).
- (e) *Source 31's research question 113.239, definable-class versions of the spectra:* answered in part, for the external order types at a strongly inaccessible height. For the classes definable over  $V_\kappa$  in a fixed finite language with set parameters, the realized types form an interval  $[0, \Theta_L)$  with  $\kappa < \Theta_L < \kappa^+$  (proposition XVI.8.2); with a named global well-order  $\text{cf}(\Theta_L) = \omega$  and parameter-free definitions are cofinal (theorem XVI.8.3 and corollary XVI.8.4),  $\kappa^{\Theta_L} = \omega^{\Theta_L} = \Theta_L$  (theorem XVI.8.9 and corollary XVI.8.17), the ceilings exhaust  $\kappa^+$  as the predicate varies (corollary XVI.8.11), and a truth predicate represents the old ceiling exactly and raises it (theorems XVI.8.12 and XVI.8.19); in the pure language,  $\kappa < \delta_0 \leq \delta_{\text{ord}} \leq \delta_{\text{set}} < \kappa^+$  with  $\text{cf}(\delta_0) = \omega$  (theorem XVI.8.6). The rest of research question 113.239 stays open: the analogues of source 31's thresholds for global choice, enumeration, coding and Class Replacement, heights  $\lambda$  that are not inaccessible, and the exact values of the ceilings.
- (f) *Source 12's full-class model (theorem 21.1 (iv), a theorem):* proved again by all four sources (theorem XVI.8.1 and notes XVI.8.5, XVI.8.8 and XVI.8.15); its hereditary calibration  $\text{otp}(\mathbf{E}) = \varepsilon_{\kappa+1}$  is new (theorem XVI.6.9).
- (g) *Not settled:* Hamkins and Woodin's Question 7; source 15's research question 113.48, whether GB codes all finite condensations uniformly, which source 35 asks again (research question 113.260); and the questions of section 113.28.

[merge] *New against the repository* (checked against the passages of Parts I–XV cited in this Part; none of the four sources claims historical priority): the choice-free power theorem (theorem XVI.4.2 and note XVI.4.4) and source 34's set-subset criterion (lemma XVI.2.2); the equivalence over GB; the cofinality of a limit exponent (proposition XVI.4.7) and bounded specialization (lemma XVI.4.8); the increasing cofinal map  $\Omega \rightarrow W$  (proposition XVI.3.7); the identities  $2^{[\Omega+1]} \cong \Omega \cdot 2$ ,  $2^{[\Omega \cdot 2]} \cong \Omega^2$  (proposition XVI.4.14); the hereditary order, its fixed point, leastness, height cuts, calibration and general base (theorems XVI.6.3, XVI.6.5 to XVI.6.7 and XVI.6.9 and corollary XVI.6.10, with the routes of sources 35, 32 and 33); the fixed-point completion, the local criterion and FPE (theorems XVI.6.17 and XVI.6.18 and corollaries XVI.6.19 and XVI.6.20); the type-level uniform-family diagonal (theorem XVI.7.2 and its three copies); the truth jumps (theorems XVI.7.4, XVI.8.12 and XVI.8.19 and corollary XVI.7.5); the bounded-complexity diagonals and separations (corollaries XVI.7.6 and XVI.7.12 and theorems XVI.7.11 and XVI.7.20); the definability ceilings (proposition XVI.8.2, theorems XVI.8.3, XVI.8.6 and XVI.8.9, note XVI.8.16, and corollaries XVI.8.11 and XVI.8.17); source 33's conditional converse (proposition XVI.5.30); and this write's theorem XVI.6.21 and remarks XVI.5.29 and XVI.5.31. *Inherited or re-proved:* the set-like boundary, the successor obstruction, uniform sums, division by  $\Omega$  and its uniqueness (sources 15, 16 and 18), the block lemma, the GBC descending-sequence test (source 15), polynomial histories (sources 15, 23 and 29), termination (source 20), floors, finite changes, normal forms, anchored restriction and the fixed-schedule criterion (source 29), the full spectrum (source 12), and self-similar quotients (source 18; corollary XVI.5.22 at every set stage). The reflection-tree example, the epsilon notation and the class-recursion calibrations are credited by the sources to Hamkins, Rathjen (through Jeon), Gitman, Hamkins, Holy, Schlicht and Williams.

[merge] *Reviews.* The Hilbert's-tenth research tree reviewed the four archives in a bounded intake (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/revie

w\_beyond\_ord\_e3839ad2c.md, commit fcc4098ec): it read 2,468 archive-member lines, including 1,665 manuscript lines, checked source 34's choice-free power theorem and set-subset criterion in full (theorem XVI.4.2 and lemma XVI.2.2) and "found no defect", confirmed that this report states the GBC equivalence and leaves the GB base open at the sources' pins, and found no demonstrably wrong computational-interface claim; it did not audit the floor, finite-change, reconstruction and restriction dependencies of the GB equivalence, nor the fixed-point, condensation and spectrum proofs, and certifies none of the four manuscripts. A second note there (review\_beyond\_ord\_placement\_111c38012.md, commit b16786caf) authenticates the placement byte for byte and adds no mathematical review. A third (ordinal\_two\_type\_effectivity\_boundary.md, commit c6082d075) proves that no uniform effective procedure decides, for computable well-orders of  $\mathbb{N}$  promised to have type  $\omega$  or  $\omega + 1$ , which type occurs; it is consistent with the sources' explicit restrictions of their notations to supplied coefficient orders and selected effective systems (source 34's section XVI.10, source 35's Section 6.4 in section XVI.6.6) and refutes no claim. None of these covers the text written for this Part. The claims most in need of review are the full dependency analysis of the base change from GBC to GB (repeated by this write against Part XIV, note XVI.5.10), source 32's exact truth presentation (theorem XVI.8.12), source 35's separation on Ord (theorem XVI.7.11), and this write's theorem XVI.6.21.

[merge] *Later bounded publication review.* The subsequent review\_beyond\_ord\_write\_62b16914e.md and separate review\_beyond\_ord\_gb\_62b16914e.md in the same research tree examine the actual Part XVI publication. The selected GB history, floors, finite changes, normal forms, anchored restriction and completion arguments pass, including the internal finite-code induction in theorem XVI.6.21. This is neither a full audit nor a formalization. Truth presentation, separation and spectrum proofs remain outside that scope. Finite notation relative to supplied coefficient orders does not provide a paid fixed-arity ordinary-integer computation history.

**Review remark R1 (coverage counterevidence).** The earlier phrase "2,468 selected lines of the four manuscripts" counted non-manuscript material as manuscript reading. The pinned intake receipt instead counts  $392 + 318 + 584 + 371 = 1,665$  TeX lines and 803 other archive-member lines, totaling 2,468. Its 446 context lines are separate. The corrected wording above does not enlarge the earlier proof coverage.

**Review remark R2 (dependency-pin counterevidence).** The earlier claim that this repository pins v4.31.0 is false at 62b16914e: both the root and surreal lake-manifest.json give inputRev: v4.32.0, with revision 81a5d257c8e410db227a6665ed08f64fea08e997. This corrects the repository pin, not any separately consulted documentation. The named API/deprecation claims remain attributed to source 32, without a new audit of Mathlib history.

**Review remark R3 (output-path counterevidence).** The early guide said source 35 writes artifacts/finite\_notation\_results.json relative to its working directory. Line 161 of the placed program instead constructs Path(\_\_file\_\_).with\_name("finite\_notation\_results.json"), which writes beside the script. The later delivery instructions already state that convention. This is an inert source observation; no supplied program was run.

[merge] *Independent check (dated note, 4 October 2026).* An adversarial verification of the four statements this write added to Part XVI, theorem XVI.6.21 and remarks XVI.5.24, XVI.5.29 and XVI.5.31, was made for the batch-97 write (a working note of that session, not shipped). It rechecked source 34's set-subset criterion, every use of global choice in source 35's proofs of its 3.5, 5.3, 5.7, 5.8, 5.9 and 5.10, the internal induction in item (3) of theorem XVI.6.21, the identification of

towers with histories, the identification of the digit map with the normal-form map, and source 33's converse, and found no mathematical gap and no counterexample. [Theorem XVI.6.21](#) is valid after two editorial fixes; the three remarks are valid after adding a missing nonemptiness hypothesis ([remarks XVI.5.24](#) and [XVI.5.31](#)) or separating two readings of a question ([remark XVI.5.29](#)). The fixes are made, each with a dated note. The check is a careful reading by the intake, not a formal verification or an external review.

**Review remark H5 (qualification of earlier review passes).** The preceding working-note report said that it “found no mathematical gap and no counterexample” and that the remarks were valid after the listed fixes. The frozen Hilbert’s-tenth reviews of [62b16914e](#) likewise passed the tower/history and digit/converse chain without its empty-domain qualifications. A bounded review of [a21a42d8c](#), recorded in `review_horizons_xvi_corrections_a21a42d8c.md` in that research tree, found the residual cases now repaired in [remarks XVI.5.29](#) and [XVI.5.31](#):  $W = 1, \Gamma = 0$  has an exhausting tower but no canonical history under the printed definition;  $W = 0, \Gamma = 1$  has an initial embedding but is outside the exact-tower definition. The earlier unqualified passes remain historical evidence and are qualified by these counterexamples. The nonempty arguments, the embedding endpoints with their new empty cases, and the choice-free completion theorem remain valid within this bounded review. No full review of Part XVI is asserted.

## Part XVII

# Finite-support arithmetic and definability horizons beyond Ord

[merge] (4 October 2026, batch 97.) This Part prints source 36, batch-97 manuscript 01 ([section 1.12](#)), whole and in its own order, as Part XV prints source 31. Source 36 is a fifth independent answer to the request that produced sources 32–35: how far definable class well-orders reach beyond Ord, what their arithmetic is, and how they can be named and formalized. It read this report at commit [e1d2f3048](#), the pin of source 34, with Parts I–XV. It did not see Part XVI, which prints sources 32–35 and was written on 4 October 2026, and it arrived fifteen minutes after their placement, too late to join their merge.

*What it adds to Part XVI.* Explicit normal-form addition and multiplication on the points of  $E$ , representing sums and products of their initial segments; the addition rule also applies to points of every  $\Omega^{[B]}$  ([theorems XVII.5.1](#) and [XVII.5.2](#)); the exact predecessor decomposition of  $A^{[B]}$  for an arbitrary base and a complete criterion for its set-likeness ([theorems XVII.3.5](#) and [XVII.3.6](#)), with  $a^{[\Omega \cdot B]} \cong \Omega^{[B]}$  for set ordinals  $a \geq 2$  ([corollary XVII.3.7](#)); a choice-free presentation of every proper class well-order on the carrier Ord ([proposition XVII.2.4](#)); a choice-free transfer theorem for finite-support term operators ([theorem XVII.6.2](#)); the epsilon-term functor  $\text{Ep}$  on class well-orders ([theorem XVII.6.4](#)); over every transitive model of ZFC of height  $\kappa$ , the calibrations  $\text{otp}(E) = \varepsilon_{\kappa+1}$  and  $\text{otp}(\text{Ep}(E)) = \varepsilon_{\varepsilon_{\kappa+1}}$  and a parameter-free order of type  $\zeta_{\kappa+1}$  ([theorems XVII.6.6](#) and [XVII.6.7](#)); the definable spectrum of every transitive model, with its definability horizon an  $\varepsilon$ -fixed point above  $\zeta_{\kappa+1}$ ,  $\varepsilon_{\mathfrak{H}(M)} = \mathfrak{H}(M)$  ([theorem XVII.6.9](#) and [proposition XVII.6.11](#)), stronger than the closure properties proved in Part XVI; and the horizons  $\mathfrak{H}_G$  over  $V_\kappa$  are unbounded in  $\kappa^+$  as the named enumeration  $G$  itself varies, with no further predicate ([theorem XVII.7.11](#)).

*What it proves again.* Its set-localization lemma, its power calculus, its hereditary order with its fixed point and towers, its uniform diagonal, its  $V_\kappa$  horizon theorem, its bounded-complexity

diagonal and its truth promotion are theorems of sources 32–35, proved independently on the same day. Each is printed as delivered and followed by a numbered credit Note that names the statements of Part XVI (notes XVII.2.A, XVII.3.A, XVII.4.A, XVII.7.B, XVII.7.D, XVII.7.E and XVII.7.G and the others of this Part). Its least-fixed-point theorem (theorem XVII.4.5) states leastness only against isomorphisms  $\Omega^{[X]} \cong X$ ; its proof gives source 34’s stronger form, against initial pre-fixed points, which note XVII.4.B states with credit to source 34. The sentences in which source 36 presents its horizon theorems as “proved results of this investigation” carry dated credit notes (after Section XVII.1.5 and in Appendix C); none is changed. No statement of source 36 was found false.

*Numbering.* The sections of this Part are numbered XVII.1–XVII.11, so that the questions part and the appendices keep their numbers (Sections 113–115, Appendices A–D), as in Parts XV and XVI. Sections XVII.1–XVII.8 are source 36’s Sections 1–8, and its statement  $k.m$  is printed as XVII. $k.m$ ; its equation  $(k.m)$  is tagged (S36. $k.m$ ), and its starred equation  $(*)$  is tagged (S36.\*). Section XVII.9 is its Section 9, whose ten research questions are printed with the other sources’ questions (section 113.30); Section XVII.10 is its Appendix A (proof status, source audit and reproducibility); Section XVII.11 records what the Part answers. So that source 36’s numbers are kept, the credit Notes written for the merge are numbered XVII. $k.A$ , XVII. $k.B$ , ... within each section, and the [merge] paragraphs add no numbered statement. Appendix D.28 maps every numbered statement.

*Notation* (section 1.3.5 lists every renaming). As in Part XVI,  $\Omega$  is the presentation  $(\text{Ord}, \in)$  (source 36’s bold  $\Omega$ );  $A^{[B]}$  is the finite-support power, source 36’s  $A^B$ , printed  $A^{[B]}$  where the operands are class orders (in a term display  $\Omega^t \alpha$  the superscript is a term, and  $\Omega^n$  for finite  $n$  abbreviates  $\Omega^{[n]}$ ); and  $\mathbf{E}$  is the hereditary term order, source 36’s sans-serif  $\mathsf{T}_\Omega$ , whose height stages  $T_n$  are printed  $\mathsf{E}^{\leq n+1}$ . Source 36’s sans-serif  $\mathbf{E}$  for the epsilon-term functor is printed  $\mathsf{Ep}$ , because  $\mathbf{E} = \mathbf{E}$  is the hereditary order here. Its horizons  $H(M)$ ,  $H_0(M)$ ,  $H_p(M)$ ,  $H_G$ ,  $H_{G,T}$  are printed  $\mathfrak{H}(M)$ ,  $\mathfrak{H}_0(M)$ ,  $\mathfrak{H}_p(M)$ ,  $\mathfrak{H}_G$ ,  $\mathfrak{H}_{G,T}$ , because  $H$  and  $H_i$  are histories here;  $H_{\kappa+}$  in proposition XVII.8.1 keeps its usual meaning. Its truth predicate keeps the letter  $T$ , as source 33’s does in Part XVI.  $X \preccurlyeq_{\text{init}} Y$  and  $X \prec_{\text{init}} Y$  are initial and proper initial embeddings, and  $L_{<x}$  is the predecessor class  $L_{<x}$ .

*Status.* Source 36 calls itself a research manuscript with conventional written proofs and “independent review during preparation”. It claims no publication priority, no resolution of a recognized open problem, no referee and no proof-assistant certification (section XVII.10). The Hilbert’s-tenth research tree reviewed its archive in a bounded intake (commit 39bb02ece) and found no defect in the proofs it read; what that review covers is recorded in section XVII.11.

## XVII.1 The question and its three different answers

The usual well-order of all ordinals is not the end of definable well-ordering. It is the end of *set-like* proper class well-ordering. Once an element is allowed to have a proper class of predecessors, there are elementary presentations of

$$\text{Ord} + 1, \quad \text{Ord} + \text{Ord}, \quad \text{Ord}^2, \quad \text{Ord}^{[\text{Ord}]},$$

and much longer orders. This article makes these expressions precise, calculates with them, and explains the reach and limits of systematic notations.

This is closely aligned with Reshetnikov’s 2021 question about definable well-ordered class relations [95]. His examples already identify the essential phenomenon. Modern second-order set theory explicitly uses orders beyond  $\text{Ord}$  and arithmetic on them [84]. The ProveIt corpus contains a substantial neighboring development of labelled class orders, set-like cores, block decomposition,



and condensation histories [93]. Our emphasis is on the lengths of class orders and the definability resources needed to present them.

### XVII.1.1 An elementary example on the same carrier

Let the carrier be  $\text{Ord}$  itself. Keep the positive ordinals in their usual order and put 0 after all of them:

$$\alpha R \beta \iff (0 < \alpha, \beta = 0) \vee (0 < \alpha, 0 < \beta, \alpha < \beta).$$

Every nonempty admitted subclass has a least  $R$ -element: use its least positive ordinal if one exists, and otherwise use 0. The resulting order has type  $\text{Ord} + 1$ . Its last point has a proper class of predecessors.

There is no new set ordinal here. A relation has been defined on a class of ordinary sets. Trying to perform the usual von Neumann collapse at the last point would require all its predecessors to become one set. The set-sized collapse theorem does not permit that step.

Throughout the article,  $\Omega$  denotes the presentation  $(\text{Ord}, \in)$ . It is a symbol for an ordered class, not an extra member adjoined to  $\text{Ord}$ . Its operations are ordinal order operations, not cardinal arithmetic and not surreal field arithmetic.

### XVII.1.2 Three interpretations of length

A complete answer needs three levels of interpretation.

1. **Individual presentations.** Given specified class orders, can we define their sum, product, power, or a fixed-point notation system? This is a question about explicit elementary class constructions.
2. **An external horizon over a fixed model.** If  $M$  is a transitive set model, every externally well-founded class presentation over  $M$  has an ordinary order type in the ambient universe. We can take the supremum of the definable ones there.
3. **A universal internal notation system.** Can one class of set codes, with one available evaluator, name or dominate all class well-orders of the same universe? A diagonal argument gives a negative answer.

The first question has remarkably long constructive answers. The second has exact structural bounds but depends on the model and on which predicates may be used in definitions. The third fails even when domination is weakened to arbitrary increasing embeddability.

### XVII.1.3 Main results and a map of the notation scale

The elementary part is developed over  $\text{GB}$ , with Power Set but without a tacit choice axiom. It constructs finite-support powers, proves their algebraic laws and exact predecessor decompositions, and gives a complete set-likeness criterion. The notation class  $\mathbf{E}$  consists of finite hereditary  $\Omega$ -normal forms. It satisfies

$$\Omega^{[\mathbf{E}]} \cong \mathbf{E},$$

and is the least such fixed point under initial embedding. All recursion needed for this construction operates on finite set codes.

A second operator,  $\text{Ep}$ , is the classical epsilon-term functor [97], extended to class orders by localization. If its input has external type  $\beta$ , its output has type  $\varepsilon_\beta$ . Here  $\varepsilon_\varepsilon$  is the increasing

continuous enumeration of the fixed points of  $\alpha \mapsto \omega^\alpha$ . Write  $\zeta_\xi$  for the increasing continuous enumeration of the fixed points of  $\xi \mapsto \varepsilon_\xi$ .

For external calibration, fix a transitive  $M \models \text{ZFC}$  and put  $\kappa = M \cap \text{Ord}$ . The right column below uses ordinary external *ordinal* arithmetic.

| Presentation                                  | Construction                   | External order type over $M$           |
|-----------------------------------------------|--------------------------------|----------------------------------------|
| $\Omega$                                      | Usual ordinal order            | $\kappa$                               |
| $\Omega + 1$                                  | Append a greatest point        | $\kappa + 1$                           |
| $\Omega \cdot 2$                              | Two consecutive copies         | $\kappa \cdot 2$                       |
| $\Omega^2$                                    | Pairs, last coordinate major   | $\kappa^2$                             |
| $\Omega^{[\Omega]}$                           | Finite ordinal digit strings   | $\kappa^\kappa$                        |
| $\mathbf{E}$                                  | Hereditary finite digit trees  | $\varepsilon_{\kappa+1}$               |
| $\mathbf{Ep}(\mathbf{E})$                     | Epsilon terms over those trees | $\varepsilon_{\varepsilon_{\kappa+1}}$ |
| $\sum_{n < \omega} \mathbf{Ep}^n(\mathbf{E})$ | Uniform nested finite syntax   | $\zeta_{\kappa+1}$                     |

The exact subscripts use the closure of a transitive ZFC model under the ordinary epsilon and zeta functions: externally  $\varepsilon_\kappa = \zeta_\kappa = \kappa$ . This is proved below; it is not an assumption that  $\kappa$  is a cardinal in the ambient universe.

The trees are set codes. A point displayed as  $\Omega^\Omega$  is a code whose *predecessor order* has that length. It is not a proper class inserted as an element of another class. This distinction makes the notation usable in ordinary class theory.

#### XVII.1.4 The definability horizon

Let  $\mathfrak{H}(M)$  be the external supremum of order types of externally well-founded orders definable over  $M$ , allowing finitely many set parameters. The article proves

$$\text{Spec}(M) = [0, \mathfrak{H}(M)), \quad \zeta_{\kappa+1} < \mathfrak{H}(M) < |\kappa|^+, \quad \varepsilon_{\mathfrak{H}(M)} = \mathfrak{H}(M).$$

The proper-class presentations realize exactly  $[\kappa, \mathfrak{H}(M))$ ; arbitrary proper definable carriers give the same spectrum as the carrier  $\text{Ord}^M$ .

A sharper result holds at an inaccessible height. Work externally in ZFC, let  $M = V_\kappa$  for strongly inaccessible  $\kappa$ , and name a predicate  $G$  coding a bijection  $\kappa \rightarrow V_\kappa$ . For the corresponding horizon  $\mathfrak{H}_G$ ,

$$\kappa < \mathfrak{H}_G < \kappa^+, \quad \text{cf}(\mathfrak{H}_G) = \omega, \quad \varepsilon_{\mathfrak{H}_G} = \mathfrak{H}_G.$$

Presentations with *no set parameters* are already cofinal in  $\mathfrak{H}_G$ , although  $G$  remains in the language. The proof constructs one dominating class order for each fixed level of formula complexity. These form an effective external schema of formulas, not a single available semantic evaluator.

Two further results explain how horizons can be crossed:

$$\mathfrak{H}_G < \mathfrak{H}_{G,T} \quad \text{when } T \text{ names full truth for } (V_\kappa, \in, G), \quad \sup_G \mathfrak{H}_G = \kappa^+.$$

Naming  $T$  gives a presentation of  $\mathfrak{H}_G$  itself. If every class is available, the external spectrum is exactly  $[0, \kappa^+)$ . Thus a fixed definability regime and the full class collection over the same sets have different precise cutoffs.

### XVII.1.5 Status and the role of ProveIt

Long class orders, basic class arithmetic, and the epsilon operator are established background. We give complete proofs of the particular constructions and structural statements used here. The integrated horizon theorem, its parameter-free cofinality conclusion, truth promotion, and variation with  $G$  are presented as proved results of this investigation. The source search does not establish publication priority for these combinations.

No recognized major open problem is claimed solved. In particular, the converse from general class-well-order comparability to elementary transfinite recursion is not established here. It is posed in the cited literature [35, 27]; the targeted current search located no verified later resolution. Our explicit constructions do not require that converse.

The repository audit records inspected sources and a pinned commit. It does not turn the repository’s research manuscripts into imported axioms. None of the principal proofs depends on accepting an unreviewed theorem from those manuscripts. The formalization roadmap explains how this work could connect to ProveIt’s ordinal-code development [94].

[merge] (4 October 2026.) Source 36 did not see Part XVI (part XVII). Its “integrated horizon theorem” and its “parameter-free cofinality conclusion” (theorem XVII.7.5), its bounded-complexity diagonal (proposition XVII.7.6) and its truth promotion (theorem XVII.7.9) are theorems that sources 32, 33 and 34 proved independently on 4 October 2026; notes XVII.7.D, XVII.7.E and XVII.7.G give the correspondences. Its variation with  $G$  (theorem XVII.7.11) refines source 32’s corollary XVI.8.11, which varies an additional predicate with  $G$  fixed. What is new against Part XVI is listed in section XVII.11. The sentences above are printed as delivered; the same credit applies to the “Main result” of source 36’s title page (Appendix C).

## XVII.2 Class well-orders and the set boundary

Our base theory in this part is GB with the axioms of ZF, including Power Set, for its sets. Neither the Axiom of Choice for sets nor Global Choice is assumed. Classes have sets as their members. Elementary class comprehension permits fixed class parameters, and Class Replacement takes the image of a set under a class function to be a set. These conventions matter: the constructions below concern relations on classes of sets, not objects having proper classes as members.

**Definition XVII.2.1.** A *class well-order* is a class  $L$ , together with a strict linear ordering  $<_L$ , such that every nonempty available subclass of  $L$  has a least element. Its predecessor class at  $x \in L$  is

$$L_{<x} = \{y \in L : y <_L x\}.$$

The order is *setlike* if every  $L_{<x}$  is a set. An *initial embedding*  $j : L \rightarrow K$  is an order embedding whose image is downward closed in  $K$ ; it is *proper* when it is not onto.

We write  $L \preceq_{\text{init}} K$  when such an initial embedding exists, and  $L \prec_{\text{init}} K$  when a proper one exists. The latter assertion includes an actual class map. It does not presuppose that arbitrary class well-orders are comparable. Likewise,  $\Omega$  denotes the particular presentation  $(\text{Ord}, \in)$ . An expression involving the order type of a proper class is an abbreviation for a presentation considered up to class isomorphism. We do not form a class of equivalence classes whose members would themselves be proper classes.

### XVII.2.1 Localization and internal well-foundedness

**Lemma XVII.2.2** (Set localization). *A class linear order  $L$  is a class well-order if and only if every nonempty set  $s \subseteq L$  has a least element.*

*Proof.* The forward implication is immediate. For the converse, suppose that a nonempty class  $C \subseteq L$  has no least element. For each  $x \in C$ , there is  $y \in C$  with  $y <_L x$ . The least ordinal  $\beta$  such that a suitable  $y$  belongs to  $V_\beta$  is uniquely determined by  $x$ . For a fixed ordinal  $\alpha$ , Class Replacement bounds these least ordinals as  $x$  ranges over the set  $C \cap V_\alpha$ . Consequently there is an ordinal  $F(\alpha) > \alpha$  such that

$$x \in C \cap V_\alpha \implies \exists y \in C \cap V_{F(\alpha)} (y <_L x).$$

Choose  $F(\alpha)$  to be the least such ordinal, obtaining a set-valued class function.

Take  $c \in C$ , choose  $\alpha_0$  with  $c \in V_{\alpha_0}$ , and define  $\alpha_{n+1} = F(\alpha_n)$ . This is ordinary recursion producing a sequence of sets, with fixed class parameters. It is available in GB; it is not a recursion producing a sequence of classes. Put  $\theta = \sup_{n < \omega} \alpha_n$ . The set  $C \cap V_\theta$  is nonempty. If  $x$  belongs to it, then  $x \in V_{\alpha_n}$  for some  $n$ , and there is a smaller member of  $C \cap V_{\alpha_{n+1}}$ . Thus  $C \cap V_\theta$  has no least element, contrary to the hypothesis.  $\square$

*Note XVII.2.A* ([merge]; source 34's set-subset criterion). [Lemma XVII.2.2](#) is source 34's [lemma XVI.2.2](#) (Part XVI), proved independently, with the same hypotheses: GB with ZF sets, a class *linear* order, no form of choice. The proofs differ in the last step. Source 34 selects a unique predecessor below each point (least rank, then least in that set) and builds a descending set sequence; source 36 closes an ordinal under the rank bound  $F$  and exhibits the set  $C \cap V_\theta$  with no least element, so it needs no order-least selection. Source 36's proof of [theorem XVII.6.2](#) repeats the argument with the sets  $L(x)$  of least-rank lower points. The caveat after [lemma XVI.2.2](#) applies here too: the criterion is for linear orders. Part XVI's completion theorem without choice ([theorem XVI.6.21](#)) rests on this lemma.

This lemma allows well-foundedness to be checked using only set quantifiers, with the ordering as a class parameter. In particular, two GB-expansions with the same sets agree about a common class relation's internal well-foundedness. This useful distinction is emphasized in the class-theoretic literature [84].

Internal well-foundedness must nevertheless be distinguished from well-foundedness in an external metatheory. If  $(M, \mathcal{X}) \models \text{GB}$ , the sets and classes quantified over above are those of that structure. An externally visible descending sequence need not be a set of  $M$ . Later sections explicitly impose external well-foundedness when taking external ordinal order types. No external correctness assumption is needed for the internal theorems of this section.

### XVII.2.2 The canonical beginning and a standard carrier

**Theorem XVII.2.3** (The ordinal initial segment). *Every proper class well-order  $L$  has a canonical initial segment isomorphic to  $\Omega$ . It consists exactly of the points with set predecessor classes. In particular, every setlike proper class well-order is isomorphic to  $\Omega$ .*

*Proof.* Let

$$S = \{x \in L : L_{<x} \text{ is a set}\}.$$

This is a class by elementary comprehension. It is initial: if  $y < x$  and  $L_{<x}$  is a set, then  $L_{<y}$  is its subclass and hence a set. For  $x \in S$ , let  $\rho(x)$  be the unique ordinal order type of the set well-order  $L_{<x}$ . The ordinary order-type theorem uses no choice. Its uniqueness makes  $\rho$  a class function.

The map  $\rho$  is an order embedding. Moreover, every ordinal below  $\rho(x)$  is  $\rho(y)$  for a unique  $y < x$ . Therefore its range is an initial subclass of  $\text{Ord}$ . If this range were not all of  $\text{Ord}$ , the least missing ordinal would show that it is a set ordinal  $\lambda$ . Applying Class Replacement to the inverse map on  $\lambda$  would then make  $S$  a set. Since  $L$  is proper, the class  $L \setminus S$  would have a least member  $p$ . Initiality of  $S$  would give  $L_{<p} = S$ , a set, contradicting  $p \notin S$ . Thus  $\rho[S] = \text{Ord}$ , as required.  $\square$

Accordingly, passing beyond  $\text{Ord}$  means allowing points with proper class predecessor classes. It does not mean finding a set ordinal larger than every ordinal. The first new point in  $\Omega + 1$ , for example, has all of  $\Omega$  below it.

*Note XVII.2.B* ([merge]; the set-like boundary). [Theorem XVII.2.3](#) is the set-like boundary of Part XVI, source 34's [theorem XVI.3.1](#) (with source 35's [note XVI.2.4](#), source 32's [note XVI.2.9](#) and source 33's [note XVI.2.14](#)), by the same proof; for well-orders of the surreal carrier it is source 11's [theorem 15.1](#). Source 34 adds that a proper class well-order which is not set-like has an initial copy of  $\Omega + 1$ .

**Proposition XVII.2.4** (Reduction to the carrier  $\text{Ord}$ ). *Every proper class well-order has an isomorphic presentation on the carrier  $\text{Ord}$ . The transport is definable from the given presentation and requires no additional choice principle.*

*Proof.* On the carrier  $L$ , define  $x \triangleleft y$  when either  $\text{rank}(x) < \text{rank}(y)$ , or their ranks agree and  $x <_L y$ . A nonempty subclass first has a least occurring rank and then a least member at that rank under  $<_L$ . Thus  $\triangleleft$  is a class well-order. The predecessors of  $y$  are contained in  $L \cap V_{\text{rank}(y)+1}$ , so it is setlike. By [Theorem XVII.2.3](#), there is a canonical isomorphism  $e : \Omega \rightarrow (L, \triangleleft)$ . Transport the original relation  $<_L$ , rather than  $\triangleleft$ , along  $e$ .  $\square$

The supplied well-order orders each rank level, which is why this argument does not need Global Choice. It is not an argument that an arbitrary proper class can be well-ordered. Related rank-first constructions appear in the standard analysis of Global Choice [19]. If the input presentation is first-order definable with specified parameters, both the enumeration just constructed and the transported relation are definable with those parameters.

[merge] [Proposition XVII.2.4](#) is new to this report in this form. It complements source 29's [corollary 108.20](#), whose transfer to the surreal carrier **No** uses global choice, which a well-order of **No** amounts to over **GB** ([theorem 13.1](#)). Nothing contradicts: the carrier  $\text{Ord}$  needs no global choice because the supplied order breaks the ties inside each rank level, as the paragraph above says, and source 34's reduction to the surreal carrier keeps global choice for that reason ([section 1.4](#), addendum for sources 32–35).

## XVII.3 Finite-support ordinal arithmetic

Arithmetic on class well-orders is established mathematical practice [84]. We give the constructions explicitly because they remain meaningful when predecessor classes are proper. The finite-support description agrees with familiar set ordinal exponentiation and with its formal development in Isabelle [98].

The sum  $A + B$  is the tagged disjoint union, with the  $A$ -copy first. The product  $A \cdot B$  has carrier  $A \times B$ , ordered by the second coordinate first:

$$(a, b) < (a', b') \iff b <_B b' \text{ or } (b = b' \text{ and } a <_A a').$$

Thus it comprises a  $B$ -ordered sequence of copies of  $A$ . The empty and singleton presentations are denoted by 0 and 1. Both constructions preserve class well-orders: for products, minimize the second coordinate occurring in a nonempty subclass and then the first.

### XVII.3.1 Powers and their structural laws

**Definition XVII.3.1** (Finite-support power). Let  $A \neq 0$  be a class well-order, with least element  $0_A$ . For a class well-order  $B$ , the carrier of  $A^{[B]}$  consists of finite set graphs  $f$  whose domain is a finite subset of  $B$ , and whose values belong to  $A \setminus \{0_A\}$ . Missing entries are interpreted as  $0_A$ . For distinct  $f, g$ , let  $b$  be the largest exponent where their interpreted values differ, and put

$$f < g \iff f(b) <_A g(b).$$

Every code is a set, even when  $A$  and  $B$  are proper classes. In particular, this definition does not quantify over a collection of all class functions from  $B$  to  $A$ .

**Theorem XVII.3.2** (Power calculus). *Finite-support powers are class well-orders. There are explicit class isomorphisms*

$$A^{[0]} \cong 1, \quad A^{[1]} \cong A, \quad 1^{[B]} \cong 1, \quad (\text{S36.3.1})$$

$$A^{[B+C]} \cong A^{[B]} \cdot A^{[C]}, \quad (\text{S36.3.2})$$

$$(A^{[B]})^{[C]} \cong A^{[B \cdot C]}. \quad (\text{S36.3.3})$$

*The sum and product are associative, and*

$$A \cdot (B + C) \cong A \cdot B + A \cdot C.$$

*All these assertions are valid for the indicated class presentations in GB.*

*Proof.* The power ordering is linear because a nonempty finite disagreement set has a largest member. For transitivity, take the larger of the determining exponents in two successive comparisons; the coefficient comparison there gives the required comparison of the first and third codes. To prove well-foundedness, take a nonempty set  $Q$  of codes. Let  $S_B$  be the union of their supports, and let  $S_A$  consist of  $0_A$  and all their nonzero coefficients. These are sets with induced well-orders. Collapse them to set ordinals  $b$  and  $a$ , respectively. Transporting the codes embeds  $Q$  into the ordinary finite-support presentation of the set ordinal  $a^b$ . Therefore  $Q$  has a least element. Lemma XVII.2.2 proves the class assertion.

For (S36.3.2), split a finite support into its  $B$ -part and its  $C$ -part. A disagreement in the  $C$ -part dominates every disagreement in the  $B$ -part, exactly as in the product on the right. For (S36.3.3), an outer code assigns to finitely many  $c \in C$  a nonzero finite code on  $B$ . Flatten its entries to pairs  $(b, c)$ . Conversely, group a finite graph on  $B \times C$  by its second coordinate. The largest outer disagreement and then the largest inner disagreement give exactly the product ordering on exponents. The unit laws are immediate. Associativity of sums and products, and the displayed distributive law, follow by retagging the relevant tuples. All maps and their inverses act on sets.  $\square$

For an empty base one may separately stipulate  $0^{[0]} = 1$  and  $0^{[B]} = 0$  for nonempty  $B$ . We keep the main power statements at nonempty bases. Familiar distinctions from commutative arithmetic remain: for example  $1 + \Omega \cong \Omega$ , whereas  $\Omega + 1 \not\cong \Omega$ :  $\Omega + 1$  has a greatest point and  $\Omega$  does not.

**Lemma XVII.3.3** (Transport of initial segments). *An initial embedding  $j : B \rightarrow C$  induces an initial embedding  $A^{[B]} \rightarrow A^{[C]}$  by transporting supports. If  $A$  has at least two elements and  $j$  is proper, the induced embedding is proper.*



*Proof.* Any nonzero digit outside  $j[B]$  occurs above all exponents in that initial segment. A code with such a digit is therefore above every code supported inside  $j[B]$ . This proves initiality. In the proper case, let  $c$  be the least point outside  $j[B]$ , and let  $1_A$  be the least member of  $A \setminus \{0_A\}$ . The singleton code with value  $1_A$  at  $c$  has exactly the transported  $A^{[B]}$  as its predecessor class. Indeed a smaller code must be supported below  $c$ , and every such code is smaller.  $\square$

*Note XVII.3.A* ([merge]; the power theorem of Part XVI). [Definition XVII.3.1](#) is source 34's [definition XVI.4.1](#). [Theorem XVII.3.2](#) is source 34's choice-free power theorem [theorem XVI.4.2](#) with its laws (S34.1), (S34.2) and (S34.3), also proved by source 35 ([note XVI.4.4](#), over GB; its laws (S35.5) and (S35.6)) and, over GBC, by sources 32 and 33 ([notes XVI.4.10 to XVI.4.13](#)); for the base  $\Omega$  it is source 29's [note 108.15](#) over GBC. Source 36's well-foundedness proof is the collapse route: a set of codes uses a set of exponents and a set of coefficients, whose collapses send it into the finite-support presentation of an ordinary ordinal power, and [lemma XVII.2.2](#) lifts the result to classes. That is source 34's second proof, and the route of sources 32 and 33 with set subsets in place of descending sequences; source 34's first proof and source 35's are greedy minimizations. [Lemma XVII.3.3](#) is source 35's [lemma XVI.4.5](#) for initial embeddings and the paragraph before [proposition XVI.4.3](#); source 33's conventions  $0^0 = 1$ ,  $0^B = 0$  are those of the paragraph after [theorem XVII.3.2](#).

**Proposition XVII.3.4** (Continuity of the presented power). *If  $B$  is nonempty and has no greatest point, the canonical initial copies of  $A^{[B_{<b}]}$ , for  $b \in B$ , exhaust  $A^{[B]}$ . If  $b$  is the greatest point of  $B$ , then*

$$A^{[B]} \cong A^{[B_{<b}]} \cdot A.$$

*Proof.* In the first case, a nonzero code has a largest support exponent  $c$ . Choose  $b > c$ . Its support is then contained in  $B_{<b}$ . The zero code is in every copy, and the copies are initial by [Lemma XVII.3.3](#). In the second case,  $B \cong B_{<b} + 1$ , so the assertion follows from (S36.3.2).  $\square$

The first statement is a presentation-level continuity law. It can be written as the elementary class condition  $\text{supp}(f) \subseteq B_{<b}$  on pairs  $(b, f)$ . It does not require a collection whose members are the proper classes  $A^{[B_{<b}]}$ , nor an unrestricted supremum operation on all class order types.

[merge] The first clause of [proposition XVII.3.4](#) is the case  $B_b = B_{<b}$  of source 35's continuity law (S35.7), which source 35 also states in this form; the second clause,  $A^{[B]} \cong A^{[B_{<b}]} \cdot A$  at a greatest point, is new in this form, and is (S34.2) with  $C = 1$ . Source 35's [proposition XVI.4.7](#) computes the cofinality of the first case. The uniform sums of the next paragraph are source 34's sums ([section XVI.3](#)) and source 32's [proposition XVI.3.4](#).

The same discipline permits uniform ordered sums. Suppose one class relation specifies fibers  $A_b$ , for  $b \in B$ , and another class relation specifies a well-order on each fiber. The carrier

$$\{(a, b) : b \in B, a \in A_b\}$$

ordered by  $b$  first and by its fiber order second is a class well-order. A nonempty subclass has a least occurring index and then a least point in that fiber. This observation allows arbitrarily long, already specified sums. The hypothesis of a single supplied definition of the fibers is essential: writing a formal family of class names does not by itself supply the class relation needed for the construction.

### XVII.3.2 Predecessor normal forms and set-likeness

**Theorem XVII.3.5** (Exact predecessor decomposition). *If  $f \in A^{[B]}$  has support  $b_0 > \dots > b_{n-1}$  and values  $a_i$ , then*

$$(A^{[B]})_{<f} \cong \sum_{i < n} A^{[B_{<b_i}]} \cdot A_{<a_i}. \quad (\text{S36.*})$$

*The empty support gives the empty sum.*

*Proof.* For  $g < f$ , its largest disagreement with  $f$  must occur at a point  $b_i$  of the support of  $f$ . At a position where  $f$  has value zero, a disagreement could only make  $g$  larger. Fixing this index  $i$ , the code  $g$  agrees with  $f$  above  $b_i$ , takes an arbitrary value below  $a_i$  at  $b_i$ , and has an arbitrary finite tail below  $b_i$ . These choices give precisely  $A^{[B_{<b_i}]} \cdot A_{<a_i}$ , with the coefficient as major coordinate. The blocks are ordered by increasing  $i$ : agreeing with one more leading digit places a code above all codes that dropped earlier. This identifies both the carrier and its ordering.  $\square$

For the base  $\Omega$ , the coefficients are positive set ordinals, so (S36.\*) reads

$$(\Omega^{[B]})_{<f} \cong \sum_{i < n} \Omega^{[B_{<b_i}]} \alpha_i. \quad (\text{S36.3.4})$$

This licenses the notation  $\Omega^{b_0} \alpha_0 + \cdots + \Omega^{b_{n-1}} \alpha_{n-1}$  for a *point code*. In its order-type interpretation, the exponent is  $B_{<b_i}$ , not a class somehow made into a member of  $B$ . The distinction remains essential when  $B_{<b_i}$  is proper.

For a concrete example, let  $B = \Omega + 1$ , and name its final point  $\top$ . The code with coefficient 1 at  $\top$ , coefficient 2 at the ordinal point 3, and coefficient 7 at 0 has predecessor presentation

$$\Omega^{[\Omega]} + \Omega^3 \cdot 2 + 7.$$

The first exponent is  $\Omega$  because the predecessors of  $\top$  are the entire first copy of  $\text{Ord}$ . The other two predecessor classes have the ordinary set order types 3 and 0. The formula therefore handles a proper exponent cut and set exponent cuts in one normal form, with no change of foundational meaning.

[merge] For the base  $\Omega$ , (S36.3.4) is source 34's predecessor formula (S34.4), with the same proof, and the polynomial reading of Parts IX and XIV. Theorem XVII.3.5 for an arbitrary class base  $A$ , where the blocks are  $A^{[B_{<b_i}]} \cdot A_{<a_i}$ , is new to this report.

**Theorem XVII.3.6** (Complete set-likeness criterion). *Suppose  $A$  has at least two elements and  $B \neq 0$ . Then  $A^{[B]}$  is setlike if and only if at least one of the following holds:*

1.  $A$  is a set and  $B$  is setlike;
2.  $B$  is a singleton and  $A$  is setlike.

*Proof.* Digits supported at the least exponent, together with the zero code, form an initial copy of  $A$ . Thus set-likeness of  $A^{[B]}$  implies set-likeness of  $A$ . If some  $b \in B$  has a proper predecessor class, the singleton code with coefficient  $1_A$  at  $b$  has  $A^{[B_{<b}]}$  below it. This contains a copy of  $B_{<b}$ , obtained by putting coefficient  $1_A$  at one exponent at a time. That copy is proper by Class Replacement applied to the inverse map if its image were a set. Hence  $B$  must be setlike.

If  $A$  is proper and  $B$  has a second point, the singleton code at that second point has an entire copy of  $A$  below it. This rules out set-likeness, proving necessity of the alternatives.

Conversely, suppose  $A$  is a set and  $B$  is setlike. Below a nonzero code with largest exponent  $b$ , every support is contained in the set  $B_{<b} \cup \{b\}$ . All such finite graphs belong to the power set of  $(B_{<b} \cup \{b\}) \times A$ . Its appropriate subclass is a set, so the predecessor class is a set. The zero code has no predecessors. The singleton-exponent alternative follows from  $A^{[1]} \cong A$ .  $\square$

Thus  $\Omega^2$  already passes the setlike boundary. More generally,  $\Omega^{[B]}$  is setlike precisely when  $B$  has at most one point. A proper class carrier alone, however, does not force such a passage.

**Corollary XVII.3.7** (Collapse of set bases). *For every set ordinal  $a \geq 2$  and every class well-order  $B$ ,*

$$a^{[\Omega]} \cong \Omega, \quad a^{[\Omega \cdot B]} \cong \Omega^{[B]}.$$

*Proof.* The first power is setlike by Theorem XVII.3.6 and proper because singleton supports give an injection from Ord. Apply Theorem XVII.2.3. The second assertion follows from  $(a^{[\Omega]})^{[B]} \cong a^{[\Omega \cdot B]}$ .  $\square$

For example,

$$2^{[\Omega]} \cong \omega^{[\Omega]} \cong \Omega, \quad \omega^{[\Omega+1]} \cong \Omega \cdot \omega, \quad \omega^{[\Omega \cdot \Omega]} \cong \Omega^{[\Omega]}.$$

These are identities of ordered presentations, not cardinal arithmetic.

[merge] Theorem XVII.3.6 is new to this report; Part XVI shows set-likeness and its failure only in examples (the proofs of propositions XVI.4.3 and XVI.4.14 and source 32's remark after note XVI.4.11 that  $\Omega^{[\Omega]}$  is not set-like). Corollary XVII.3.7 generalizes source 34's proposition XVI.4.3 ( $a^{[\Omega]} \cong \Omega$ ,  $a^{[\Omega+1]} \cong \Omega \cdot a$ ), source 35's proposition XVI.4.6 ( $\beta^{[\Omega]} \cong \Omega$ ) and source 33's proposition XVI.4.14 ( $2^{[\Omega+1]} \cong \Omega \cdot 2$ ,  $2^{[\Omega \cdot 2]} \cong \Omega^2$ ) to every exponent of the form  $\Omega \cdot B$ .

## XVII.4 Hereditary $\Omega$ -normal forms

Finite supports can themselves have finite-support exponents. The right way to iterate this observation is to give a single finite syntax. This avoids a purported recursion that would collect a new proper class at each stage.

**Definition XVII.4.1** (The term class). A raw term is a finite rooted ordered tree with a positive ordinal coefficient on each edge. Write an empty node as 0, and a nonempty node as

$$[(t_0, \alpha_0), \dots, (t_{k-1}, \alpha_{k-1})].$$

Compare raw terms recursively by lexicographic comparison of these lists: compare exponent subtrees first, their coefficients second, and then the remaining entries. A proper prefix is smaller. A term is *normal* when all its subterms are normal and  $t_0 > \dots > t_{k-1}$ . The class of normal terms is  $\mathbf{E}$ .

The recursion here is on finite syntax trees. More explicitly, one can code a tree by a finite prefix-closed subset of  $\omega^{<\omega}$ , with consecutive child indices at each node, and a finite ordinal labeling of its edges. Comparisons are computed from pairs of proper subtrees. Normality is then checked at its finitely many nodes. These are uniform first-order definitions; neither a truth predicate nor class-valued transfinite recursion occurs.

The constant  $\bar{\alpha}$ , for  $\alpha > 0$ , is the single-entry term  $[(0, \alpha)]$ , and  $\bar{0} = 0$ . There are no separate constant atoms duplicating these terms. We also write a node suggestively as

$$\Omega^{t_0} \alpha_0 + \dots + \Omega^{t_{k-1}} \alpha_{k-1}.$$

This is syntax with the predecessor semantics of Theorem XVII.3.5.

The word *finite* describes the number of nodes and labels, not the size of an ordinal label. There are proper class many possible coefficients. A tree is a set carrying finitely many of them, so the whole notation domain can be a proper class even though each tree has finite shape. A parameter-free definition of this domain is compatible with individual terms having arbitrary ordinal labels. For computational use one must additionally specify how those labels are represented and compared; a finite alphabet of executable labels describes a smaller notation domain.

### XVII.4.1 Well-foundedness from set ordinal evaluation

**Theorem XVII.4.2** (Localization of hereditary normal forms). *The syntactic ordering on  $\mathbf{E}$  is a class well-order. For every set ordinal  $\kappa \geq 2$ , terms whose coefficients are below  $\kappa$  admit an injective order-preserving evaluation into the ordinary ordinals, given by*

$$|0|_\kappa = 0, \quad |[(t_i, \alpha_i)]_{i < \kappa}|_\kappa = \sum_{i < \kappa} \kappa^{|t_i|_\kappa} \alpha_i. \quad (\text{S36.4.1})$$

*Evaluation also reflects strict order and equality.*

*Proof.* We use the ordinary ordinal radix comparison principle at base  $\kappa$ . In a finite expansion with decreasing exponents and coefficients below  $\kappa$ , a tail supported below  $\gamma$  has value below  $\kappa^\gamma$ . To see the relevant bound, if the largest exponent is  $\delta$  and its coefficient is  $c < \kappa$ , induct on the number of lower terms to obtain

$$\kappa^\delta c + \text{tail} < \kappa^\delta (c + 1) \leq \kappa^{\delta+1}.$$

For exponents below  $\gamma$ , this is at most  $\kappa^\gamma$ . The successor and limit laws for ordinary ordinal exponentiation justify the last comparison. Hence a larger leading exponent dominates the whole smaller expansion, and a larger first unequal coefficient dominates its lower tail. Common leading parts can be removed from a comparison because ordinal addition is strictly increasing in its right argument. This proves the radix comparison principle. The general base expansion, rather than only base  $\omega$ , is also developed in Mathlib [99].

Induction on the maximum of the two finite tree heights now shows that the evaluated exponent subtrees have exactly their syntactic ordering. The radix comparison principle therefore applies at the roots. It proves preservation and reflection of comparison and equality, as well as the required decreasing-exponent condition at each root.

Finally let  $Q \subseteq \mathbf{E}$  be a nonempty set. The ordinal coefficients occurring anywhere in its trees form a set. Choose one ordinal  $\kappa \geq 2$  above them all. Evaluation embeds  $Q$  into the ordinals, and its image is a set by Replacement. It follows that  $Q$  has a least member. Lemma XVII.2.2 finishes the proof.  $\square$

No special closure property of  $\kappa$  is needed for this localization: it need only bound the coefficients. The theorem is an internal proof in the stated base theory, not an appeal to a larger universe.

**Proposition XVII.4.3** (Root decomposition). *Removing the root identifies*

$$\mathbf{E} \cong \Omega^{[\mathbf{E}]}.$$

*Proof.* The root of a normal term is exactly a finite-support map from normal exponent terms to positive ordinal coefficients. Conversely, attach a new root to any such finite family. A finite family of finite trees is finite, and the two constructions are inverse. Their comparisons agree because both inspect the largest exponent of disagreement first.  $\square$

### XVII.4.2 Strict towers and the least fixed point

Let the height of 0 be zero, and let a nonzero tree's height be one plus the maximum height of its exponent subtrees. For  $n < \omega$ , put

$$\mathbf{E}^{\leq n+1} = \{t \in \mathbf{E} : \text{height}(t) \leq n+1\}.$$

These are uniform restrictions of one previously defined class. Define actual term codes

$$w_0 = [(\bar{1}, 1)], \quad w_{n+1} = [(w_n, 1)].$$

**Theorem XVII.4.4** (Canonical strict towers). *Each  $E^{\leq n+1}$  is initial in  $E$ , and*

$$E^{\leq 1} \cong \Omega, \quad E^{\leq n+2} \cong \Omega^{[E^{\leq n+1}]}, \quad E^{\leq n+1} = E_{<w_n}.$$

*In particular,*

$$\Omega \prec_{\text{init}} \Omega^{[\Omega]} \prec_{\text{init}} \Omega^{[\Omega^{[\Omega]}]} \prec_{\text{init}} \cdots \prec_{\text{init}} E.$$

*Proof.* Height at most one gives exactly the ordinal constants. Every nonconstant normal term has a positive leading exponent, and is therefore above every constant. The least such term is  $w_0$ , since  $\bar{1}$  is the least positive term and 1 is the least positive coefficient. This proves the assertions for  $E^{\leq 1}$ .

Root decomposition identifies  $E^{\leq n+2}$  with  $\Omega^{[E^{\leq n+1}]}$ . If  $E^{\leq n+1} = E_{<w_n}$ , transport of this initial segment through the power construction and Proposition XVII.4.3 identifies  $E^{\leq n+2}$  with the predecessors of the singleton digit at  $w_n$ . That singleton is  $w_{n+1}$ , by Lemma XVII.3.3. Induction proves all the equalities. Moreover  $w_n$  has height  $n+2$ , so it belongs to  $E^{\leq n+2} \setminus E^{\leq n+1}$ ; each inclusion is proper.  $\square$

The code  $w_0$  is a point of  $E$  whose predecessor presentation has type  $\Omega$ . The entire class Ord has not become a member of a class. The same distinction applies to every tower boundary.

*Note XVII.4.A* ([merge]; the hereditary order of Part XVI). Source 36's term class (its sans-serif  $T_\Omega$ , printed  $E$ ) is source 34's  $E$  (definition XVI.6.1): the same finite trees with nonzero set-ordinal coefficients, the same comparison (exponents first, coefficients second, a proper prefix smaller) and the same normality condition; it is also source 35's  $\mathcal{E}$  (note XVI.6.11) and, up to isomorphism, source 33's direct limit  $\mathbb{T}$  (theorem XVI.6.26). Theorem XVII.4.2 is source 34's lemma XVI.6.2 and theorem XVI.6.3, by the same argument; source 34 evaluates at an infinite initial cardinal above the coefficients, source 36 at any ordinal  $\kappa \geq 2$  above them, which the radix comparison principle allows. Proposition XVII.4.3 is theorem XVI.6.6. Theorem XVII.4.4 is theorem XVI.6.5 with the index shifted by one: source 36's height stage  $T_n$ , printed  $E^{\leq n+1}$ , is the cut below  $w_n$ , and  $w_n$  is source 34's term  $\tau_{n+1}$  (S34.6); it is also source 35's note XVI.6.14 (with  $b_n = w_n$ ) and source 33's tower (lemma XVI.6.25). Source 36, like source 34, proves all of this in GB without choice; sources 33 and 35 work in GBC, and Part XVI lowers source 35's version to GB (theorem XVI.6.21).

**Theorem XVII.4.5** (Least  $\Omega$ -exponential fixed point). *The class  $E$  is isomorphic to  $\Omega^{[E]}$ . If a class well-order  $X$  admits a class order isomorphism  $\phi : \Omega^{[X]} \rightarrow X$ , then  $E \prec_{\text{init}} X$ .*

*Proof.* The fixed-point assertion is Proposition XVII.4.3. For leastness, evaluate each finite tree through the given isomorphism:

$$h(0) = \phi(\emptyset), \quad h([(t_i, \alpha_i)]_{i < k}) = \phi(\{(h(t_i), \alpha_i) : i < k\}). \quad (\text{S36.4.2})$$

A valuation on the finite set of nodes is a set. Its equations use only the fixed class parameter  $\phi$ . Thus, once existence and uniqueness are checked, elementary comprehension defines  $h$  as a class map, without a recursion along  $E$ .

These checks can be made simultaneously by induction on finite height. The value  $h(0)$  is the least point of  $X$ . The images of the constants are  $\phi$  applied to the zero code and to singleton digits at that least point. They form an initial copy of  $\Omega$  in  $X$ . Suppose the evaluation on  $E^{\leq n+1}$  is an initial embedding. Transport of finite supports makes its extension on  $E^{\leq n+2} \cong \Omega^{[E^{\leq n+1}]}$  an initial embedding into  $\Omega^{[X]}$ . Composition with  $\phi$  makes it initial in  $X$ . In particular, distinct exponent subtrees remain distinct and correctly ordered, so the graph in (S36.4.2) is a legitimate finite support. The same induction gives unique values at every node.

Every term lies in some  $E^{\leq n+1}$ , and these evaluations agree because they satisfy the same finite equations. If  $x < h(t)$ , choose  $n$  with  $t \in E^{\leq n+1}$ . Initiality at that stage gives  $x = h(s)$  for some  $s \in E^{\leq n+1}$ . Therefore the whole image of  $h$  is initial, proving the assertion.  $\square$

Leastness here does not invoke comparability of arbitrary class well-orders. It compares this explicit term class with each supplied fixed point. Nor is a fixed point a final boundary for class order types. The power law gives

$$\Omega^{[E+1]} \cong \Omega^{[E]} \cdot \Omega \cong E \cdot \Omega,$$

which contains its first  $E$ -copy as a proper initial segment. The simpler successor presentation  $E + 1$  also passes it.

*Note XVII.4.B* ([merge]; leastness against initial pre-fixed points). [Theorem XVII.4.5](#) states leastness only against fixed points given by an isomorphism  $\phi : \Omega^{[X]} \cong X$ ; in that form it is source 35's [note XVI.6.15](#). Its proof gives more, with no change: *if a class well-order  $X$  admits an initial order embedding  $\phi : \Omega^{[X]} \rightarrow X$  (an initial pre-fixed point), then the evaluation  $h$  of (S36.4.2) is a unique initial embedding  $E \rightarrow X$ ; in particular  $E \preccurlyeq_{\text{init}} X$ .* The proof uses of  $\phi$  only that it is an order embedding with initial image:  $h(0) = \phi(\emptyset)$  is the least point of  $X$  because the image of  $\phi$  is initial and contains  $\phi(\emptyset)$  as its least point; the constants go to an initial copy of  $\Omega$  for the same reason; and in the induction a composite of initial embeddings is initial. This stronger statement is source 34's [theorem XVI.6.7](#) (Part XVI), which is credited for it; source 36 did not see it. The phrase “the least such fixed point under initial embedding” in source 36's introduction (Section XVII.1.3) holds in this stronger sense. The remark after the theorem, that leastness invokes no comparability of arbitrary class well-orders, is source 34's remark after [theorem XVI.6.7](#).

## XVII.5 Arithmetic of the represented initial segments

Write  $\|t\| = E_{<t}$ , interpreting this notation as a class presentation. It is useful to give actual algorithms for adding and multiplying these presentations. The algorithms make finitely many structural decisions, together with operations on ordinal coefficients. They are therefore relative procedures on supplied ordinal labels, not a recursive finite-alphabet notation for all ordinals.

### XVII.5.1 Addition by a final-segment isomorphism

**Theorem XVII.5.1** (Normal-form addition). *There is a finite normal-form operation  $f \boxplus g$  on  $E$  such that*

$$\|f \boxplus g\| \cong \|f\| + \|g\|.$$

*The same rule applies to the points of every  $\Omega^{[B]}$ .*

*Proof.* If  $g = 0$ , return  $f$ . Otherwise let  $b$  be the leading exponent of  $g$ . Keep the digits of  $f$  above  $b$ , add its coefficient at  $b$  to that of  $g$  by ordinary ordinal addition, discard its digits below  $b$ , and append the lower tail of  $g$ . A missing coefficient of  $f$  is zero. The resulting coefficient is a positive ordinal, and the exponents remain strictly decreasing.

Fix  $f$ . The map  $R_f(g) = f \boxplus g$  is strictly increasing. If two inputs have different leading exponents, at the larger one the output for that input strictly increases the corresponding coefficient of  $f$ , whereas the other output leaves it unchanged. If their leading exponents agree, unequal leading coefficients are compared by strict monotonicity of ordinary ordinal addition in the right argument. When the leading coefficients also agree, their lower tails are unchanged by the rule and decide the comparison.

Its range is exactly the final segment starting at  $f$ . Certainly  $R_f(0) = f$ . For  $h > f$ , let  $b$  be their largest disagreement. Then  $h(b) > f(b)$ . There is a unique positive ordinal  $\delta$  with  $f(b) + \delta = h(b)$ :



it is the order type of the ordinary ordinal interval from  $f(b)$  to  $h(b)$ . Take  $g$  to have coefficient  $\delta$  at  $b$  and the lower tail of  $h$ . The displayed rule gives  $R_f(g) = h$ .

Consequently  $R_f$  is an isomorphism onto that final segment. Restricting it to the predecessors of  $g$  identifies them with the interval  $[f, f \boxplus g)$ . Prepending the predecessors of  $f$  proves the required sum identity. The argument depends only on the ordered exponent carrier and ordinal coefficients, so applies to  $\Omega^{[B]}$  as asserted.  $\square$

For every point-cut  $D$  of  $\Omega^{[B]}$ , the same proof gives  $D + \Omega^{[B]} \cong \Omega^{[B]}$ . Thus the absorption used in normal-form arithmetic has been established by an explicit class isomorphism.

### XVII.5.2 Multiplication without class-valued recursion

**Theorem XVII.5.2** (Normal-form multiplication). *There is a finite normal-form operation  $f \boxtimes g$  on  $\mathbf{E}$  satisfying*

$$\|f \boxtimes g\| \cong \|f\| \cdot \|g\|.$$

*Proof.* Zero factors give zero. Write a nonzero first factor in root notation as

$$f = \Omega^a \alpha + r,$$

where  $\alpha > 0$  is an ordinal and every exponent in  $r$  is below  $a$ . Put  $A = \|a\|$  and  $D = \|f\|$ . The predecessor theorem gives  $D \cong \Omega^{[A]} \alpha + \|r\|$ .

For each set ordinal  $\beta$ , define a term  $F_\beta$  directly:

$$F_0 = 0, \quad F_\beta = \begin{cases} \Omega^a(\alpha\beta) + r, & \beta > 0 \text{ is a successor,} \\ \Omega^a(\alpha\beta), & \beta > 0 \text{ is a limit.} \end{cases} \quad (\text{S36.5.1})$$

Normal-form addition gives  $F_\beta \boxplus f = F_{\beta+1}$ . Hence  $[F_\beta, F_{\beta+1})$  is a copy of  $D$ . The points  $F_\beta$  increase strictly. They are continuous at limit indices in the following concrete sense: every point below  $F_\lambda$  lies below some  $F_\beta$ ,  $\beta < \lambda$ . This follows from ordinary ordinal continuity  $\alpha\lambda = \sup_{\beta < \lambda} \alpha\beta$ ; at a limit  $F_\lambda$  has no lower tail.

These intervals partition the point-cut  $\Omega^{a \boxplus 1}$ , whose order type is  $\Omega^{[A+1]}$ . Indeed its points have leading exponent at most  $a$  and an ordinal coefficient at  $a$ . Some  $\alpha\beta$  exceeds that coefficient, so the  $F_\beta$  are cofinal. Given a point, choose the least  $\gamma$  with that point below  $F_\gamma$ . Continuity excludes limit  $\gamma$ , placing it in a unique interval  $[F_\beta, F_{\beta+1})$ . Thus

$$(x, \beta) \mapsto F_\beta \boxplus x$$

is an explicit isomorphism  $D \cdot \Omega \cong \Omega^{[A+1]}$ . Restricting the index to any ordinal  $\beta$  also proves  $D \cdot \beta \cong \|F_\beta\|$ . These maps use the single formula (S36.5.1); no sequence of class isomorphisms was recursively selected.

Now let  $b > 0$  be an exponent term and put  $B = \|b\|$ . Remove the least point of  $B$ , writing  $B \cong 1 + B'$ . The power laws and product associativity give

$$D \cdot \Omega^{[B]} \cong (D \cdot \Omega) \cdot \Omega^{[B']} \cong \Omega^{[A+1]} \cdot \Omega^{[B']} \cong \Omega^{[A+B]}.$$

By Theorem XVII.5.1, the last presentation is the point-cut denoted by  $\Omega^{a \boxplus b}$ .

The algorithm is now explicit. For each summand  $\Omega^{b_i} \beta_i$  of  $g$  with  $b_i > 0$ , output  $\Omega^{a \boxplus b_i} \beta_i$ . If  $g$  has an ordinal constant tail  $\beta > 0$ , append  $F_\beta$ . Right distributivity and the identities just proved identify this output with  $D \cdot \|g\|$ . The positive exponents remain strictly decreasing because  $b \mapsto a \boxplus b$  is strictly increasing. They all exceed  $a$ , so an appended  $F_\beta$  preserves normality.  $\square$

For example, the algorithms give

$$(\Omega + 1)\omega = \Omega\omega, \quad (\Omega + 1)(\omega + 1) = \Omega(\omega + 1) + 1,$$

and

$$(\Omega^2 + \Omega + 7)(\Omega^3 \cdot 2 + \omega + 1) = \Omega^5 \cdot 2 + \Omega^2(\omega + 1) + \Omega + 7.$$

Here equality abbreviates the proved isomorphism of predecessor presentations. Arbitrary powers of such presentations are already available through Definition XVII.3.1. We have not proved in this section a normalization algorithm placing every such power back inside  $\mathbf{E}$ ; addition and multiplication closure should not be mistaken for that further assertion.

[merge] (4 October 2026.) Theorems XVII.5.1 and XVII.5.2 are new to this report. They answer the addition and multiplication part of source 34’s agenda item research question 113.275, which asks for algorithms on the initial segments represented by finite trees “proved against class-order isomorphisms”: both are proved by explicit class isomorphisms of the predecessor orders, and the absorption  $D + \Omega^{[B]} \cong \Omega^{[B]}$  after theorem XVII.5.1 extends source 35’s  $\alpha + \Omega \cong \Omega$  (proposition XVI.4.6) to every point-cut. Exponentiation is not answered: source 36 says so above and asks for it as its sixth question (research question 113.300); the dated note at research question 113.275 records the status. The algorithms are relative to the ordinal arithmetic of the coefficients ( $\alpha\beta$ ,  $f(b) + \delta$ , successor or limit  $\beta$ ); the Hilbert’s-tenth review of the archive (commit 39bb02ece) read both proofs, found no defect, and records that these procedures retain ordinal-label primitives and are not an implementation on a finite alphabet. The three displayed products were recomputed by hand at the placement (6571ee1af).

## XVII.6 Epsilon notation and exact external spectra

Finite hereditary powers provide one substantial interval of notation, but they do not exhaust the useful finite constructions. We next extend a standard epsilon-term operator to class well-orders, then calculate exact external order types. The distinction between a concrete presentation and an externally calculated ordinal remains essential throughout.

### XVII.6.1 A transfer principle for finite terms

The method behind Lemma XVII.2.2 applies to many established ordinal notation systems. Here is a formulation that isolates the required uniformity.

**Definition XVII.6.1.** A *finite-support term operator*  $F$  assigns a class of finite, set-coded terms to a linear order  $B$ . Each term  $t$  has a finite set  $\text{supp}_B(t)$  of labels from  $B$ . Validity and comparison are given by fixed first-order definitions, allowing the relation of  $B$  as a class parameter. We require that whenever  $S \subseteq B$  contains the labels of the terms under consideration, validity and comparison in  $F(B)$  agree with validity and comparison in  $F(B \upharpoonright S)$ .

This definition includes an actual restriction property, rather than just the informal assertion that expressions are finite. In particular, a rule that consults the whole predecessor class of an input label would require separate justification.

**Theorem XVII.6.2** (Finite-support transfer). *Work in GB. Suppose  $F$  is a finite-support term operator, its terms on a set input form a set, and  $F$  sends every set well-order to a set well-order. Then  $F$  sends every class well-order to a class well-order. No choice principle is needed.*

*Proof.* Let  $A$  be any nonempty set of terms of  $F(B)$ . Replacement and Union give the set

$$S = \bigcup_{t \in A} \text{supp}_B(t) \subseteq B.$$

The induced order  $B \upharpoonright S$  is a set well-order. By hypothesis  $F(B \upharpoonright S)$  is a set well-order, and its restriction to  $A$  agrees with the original comparison. Thus  $A$  has a least element. The same argument on finite sets of terms proves linearity, including transitivity and antisymmetry.

For completeness, the passage from least elements for sets to least elements for classes does not require dependent choice. If a nonempty class  $C$  had no least element, then for each  $x \in C$  let  $L(x)$  be the nonempty set of lower elements of  $C$  having the least possible rank. This set exists by class separation on an appropriate rank level. Starting with  $A_0 = \{x_0\} \subseteq C$ , form

$$A_{n+1} = \bigcup_{x \in A_n} L(x), \quad A = \bigcup_{n < \omega} A_n.$$

These are sets by ordinary set recursion and Replacement with the fixed class parameters. Every member of  $A$  has a smaller member in  $A$ , contradicting the result for nonempty sets. Hence every nonempty class also has a least element.  $\square$

The theorem supplies a controlled route to further notation systems: specify a finite grammar, prove the restriction property, and prove the ordinary set-order interpretation. It does not authorize substituting classes into an arbitrary ordinal function merely because that function is normal.

[merge] [Theorem XVII.6.2](#) is new to this report as a general method. Its first paragraph is source 34's set-subset criterion ([lemma XVI.2.2](#), here [lemma XVII.2.2](#)) applied to the set of labels of a set of terms; source 35's bounded-specialization principle ([lemma XVI.4.8](#), over GB by [theorem XVI.6.21](#) (1)) is the same transfer for ordinal-labelled terms with an ordinal interpretation, and source 34's and source 36's well-foundedness proofs for  $\mathbf{E}$  ([theorems XVII.4.2](#) and [XVI.6.3](#)) are instances. Its second paragraph is a further proof of [lemma XVII.2.2](#), through the sets of least-rank lower points ([note XVII.2.A](#)).

## XVII.6.2 The epsilon-term functor

The ordinary notation operator  $B \mapsto \varepsilon_B$  is established in the literature; see the explicit syntax in [97, Definition 2.3]. We use a compressed Cantor-normal-form presentation and supply its set and class proofs. The resulting construction will be denoted by  $\text{Ep}(B)$ .

**Definition XVII.6.3** (Normal epsilon terms). For a linear order  $B$ , normal terms are finite tagged trees of the following forms:

1. the zero term 0;
2. a constant  $e_b$ , for  $b \in B$ ;
3. a nonempty list

$$[(t_0, n_0), \dots, (t_{k-1}, n_{k-1})], \quad t_0 > \dots > t_{k-1}, \quad 0 < n_i < \omega,$$

whose intended reading is  $\omega^{t_0} n_0 + \dots + \omega^{t_{k-1}} n_{k-1}$ .

In the third clause all exponent terms must themselves be normal. The singleton list  $[(e_b, 1)]$  is excluded: it would duplicate  $e_b$ .

Here normality and comparison are defined together by finite recursion. Zero is least; constants satisfy  $e_b < e_c$  exactly when  $b <_B c$ . Two list terms are compared lexicographically, first comparing their exponents and then their positive natural coefficients. A proper prefix is smaller. For a list term  $s$  with leading exponent  $t_0$ , the mixed comparison is

$$s < e_b \iff t_0 < e_b, \quad e_b < s \iff e_b \leq t_0. \quad (\text{S36.6.1})$$

If  $t_0 = e_b$ , either the leading coefficient exceeds 1 or a nonempty tail follows, so  $s$  is strictly larger than  $e_b$ . This explains the normal-form exclusion.

Each recursive comparison replaces at least one list by a proper subterm; comparisons between constants consult  $B$  directly. Consequently the combined finite size of the two displayed syntax trees is a termination measure. Labels are leaves: their transitive closures are not traversed. One can formalize validity and comparison by a finite computation record over the nodes of the input trees. There is no recursion through the order  $B$ , and no evaluation of arbitrary set-theoretic formulas.

**Theorem XVII.6.4** (Exact epsilon interpretation and class extension). *If  $B$  is a set well-order of order type  $\beta$ , then*

$$\text{otp}(\text{Ep}(B)) = \varepsilon_\beta.$$

*For an arbitrary class well-order  $B$ , the same finite syntax defines a class well-order  $\text{Ep}(B)$  in GB. The construction preserves the parameters used to define  $B$ .*

*Proof.* First suppose  $r : B \rightarrow \beta$  is the ordinal collapse of a set well-order. Evaluate the terms by

$$\llbracket 0 \rrbracket = 0, \quad \llbracket e_b \rrbracket = \varepsilon_{r(b)}, \quad \llbracket [(t_i, n_i)]_{i < k} \rrbracket = \sum_{i < k} \omega^{\llbracket t_i \rrbracket} n_i.$$

We use the convention that  $\varepsilon_0$  is the first fixed point of  $\xi \mapsto \omega^\xi$ .

Induction on finite term size shows that every value is below  $\varepsilon_\beta$ . Constants have this property because their indices are below  $\beta$ . A finite sum of powers with exponents below an epsilon number is again below that epsilon number. For the empty input  $B$ , there are no constants, and this argument gives values below  $\varepsilon_0$ .

A simultaneous induction on pairs of terms proves that comparison is preserved and reflected. For two lists this is the uniqueness and comparison rule of ordinary Cantor normal form. For a list and a constant of value  $\lambda = \varepsilon_{r(b)}$ , a leading exponent below  $\lambda$  makes the whole list value smaller than  $\lambda$ . A leading exponent above  $\lambda$  makes it larger. Equality of the leading exponent to  $\lambda$  also makes it larger, because the only list that could evaluate to  $\lambda$  is the excluded singleton. The recursive comparison of exponents is justified by the induction hypothesis. Thus evaluation is an order embedding.

To prove surjectivity, use induction on  $\alpha < \varepsilon_\beta$ . Zero is represented. If  $\alpha$  is an epsilon number, then  $\alpha = \varepsilon_\gamma$  for a unique  $\gamma < \beta$ , and the corresponding constant represents it. Otherwise write its nonzero Cantor normal form

$$\alpha = \omega^{\alpha_0} n_0 + \cdots + \omega^{\alpha_{k-1}} n_{k-1}.$$

Every exponent  $\alpha_i$  is strictly below  $\alpha$ . Indeed,  $\omega^\xi \geq \xi$  always, and an exponent equal to  $\alpha$  could occur only in the singleton expression  $\alpha = \omega^\alpha$ , which is the epsilon case already handled. Represent the exponents inductively and form the list. Its exponents are correctly ordered by the embedding property, and it cannot be the excluded singleton because  $\alpha$  is not an epsilon number. This proves the exact set-case order type.

For the class case, the support of a term consists of its finitely many labels  $b$ . Both normality and comparison use only the inherited order between labels, so the restriction property holds. On a set input the collection of finite terms is a set. Theorem XVII.6.2 now applies. Finally, all definitions are fixed finite-syntax definitions relative to the relation  $B$ , so no new parameter is introduced.  $\square$

An order embedding  $B \rightarrow C$  acts on terms by replacing each label with its image. Structural induction shows that this is an embedding  $\text{Ep}(B) \rightarrow \text{Ep}(C)$ . This explains the functor terminology at the level needed here; no collection of all class orders is being formed. For example,  $\text{Ep}(\Omega) \cong \Omega$ : evaluate  $e_\alpha$  as the ordinary set ordinal  $\varepsilon_\alpha$  and use ordinary arithmetic. All such values are sets, and every set ordinal is represented. Applying the operator to a longer presentation can have a very different effect.

[merge] The epsilon-term functor  $\text{Ep}$  on class well-orders ([definition XVII.6.3](#) and [theorem XVII.6.4](#)) is new to this report; source 36 credits its syntax to Marcone and Montalbán [97], and Part XVI's sources credit class epsilon notation to Rathjen through Jeon ([88]; [section 1.4](#), addendum for sources 32–35). It is a first step of the Veblen-style hierarchy of class-order constructors that source 34's [research question 113.272](#) and source 35's [research question 113.273](#) ask for (dated note at [research question 113.272](#)). Source 36's sans-serif  $\mathsf{E}$  is printed  $\text{Ep}$  because  $\mathsf{E}$  is the hereditary order here;  $\text{Ep}(\Omega) \cong \Omega$ , whereas  $\mathsf{E}$  is far longer.

### XVII.6.3 Exact calibration over a transitive set model

From this point through the spectrum calculations, the ambient metatheory is ZFC. Fix a transitive set model

$$M \models \text{ZFC}, \quad \kappa = M \cap \text{Ord}.$$

The symbol  $\kappa$  denotes an ordinary ordinal of the ambient universe, although it is not an element of  $M$ . All displayed order types in this subsection are external order types. Transitivity makes finite syntax absolute, but it does not make arbitrary class well-foundedness absolute.

**Lemma XVII.6.5.** *Let  $\zeta_\alpha$  enumerate the fixed points of  $\xi \mapsto \varepsilon_\xi$ , starting with the least such fixed point. Then*

$$\varepsilon_\kappa = \kappa = \zeta_\kappa.$$

*Proof.* Ordinal arithmetic and the ordinary recursive definitions of the epsilon and zeta enumerations are absolute between  $M$  and the ambient universe. For every  $\alpha < \kappa$ , the model constructs  $\varepsilon_\alpha$  and  $\zeta_\alpha$ , which therefore lie below  $\kappa$ . Both normal enumerations dominate their arguments. Continuity at the limit ordinal  $\kappa$  gives the equalities by taking suprema over  $\alpha < \kappa$ .  $\square$

**Theorem XVII.6.6** (The first hereditary power boundary). *For the presentation  $\mathsf{E}$  of Theorem XVII.4.5,*

$$\text{otp}(\mathsf{E}^M) = \varepsilon_{\kappa+1}, \quad \text{otp}(\text{Ep}(\mathsf{E})^M) = \varepsilon_{\varepsilon_{\kappa+1}}.$$

*Proof.* Every finite tree whose ordinal coefficients lie below  $\kappa$  belongs to  $M$ , by closure under finite coding. Conversely, every tree recognized by  $M$  is such an external finite tree, since transitive models have the standard natural numbers. Thus  $\mathsf{E}^M$  is exactly the hereditary base- $\kappa$  normal-form system, with its usual recursive comparison.

Restricting root decomposition to coefficients below  $\kappa$  shows that the bounded-height stages have types  $\tau_0 = \kappa$  and  $\tau_{n+1} = \kappa^{\tau_n}$ . All finite trees form their increasing union, whose type is  $\sup_{n < \omega} \tau_n$ , the least fixed point of  $\xi \mapsto \kappa^\xi$ . The fixed points of this function are exactly the epsilon numbers strictly above  $\kappa$ . To check this explicitly, use  $\kappa = \omega^\kappa$ . If  $\xi = \kappa^\xi$ , then

$$\xi = \omega^{\kappa \cdot \xi} \geq \kappa \cdot \xi \geq \xi,$$

so  $\kappa \cdot \xi = \xi$  and  $\omega^\xi = \xi$ . Moreover  $\xi > \kappa$ , since  $\xi \geq \kappa$  and  $\kappa^\kappa > \kappa$ . Conversely, an epsilon number  $\xi > \kappa$  satisfies  $\kappa \cdot \xi = \xi$ , hence  $\kappa^\xi = \omega^{\kappa \cdot \xi} = \xi$ . Lemma XVII.6.5 identifies the least such number as  $\varepsilon_{\kappa+1}$ .

Finite syntax is also absolute for the epsilon operator. Apply Theorem XVII.6.4 externally to the set order  $E^M$  to obtain the second equality.  $\square$

The first equality is an exact calibration, not a claim that  $\varepsilon_{\kappa+1}$  is an ordinal available to  $M$ . It also shows that these explicit presentations depend only on  $\kappa$  when compared between transitive models of equal ordinal height.

*Note XVII.6.A* ([merge]; source 34's calibration at every transitive height). The first equality of theorem XVII.6.6 is source 34's theorem XVI.6.9 ( $\text{otp}(E[\kappa]) = \varepsilon_{\kappa+1}$  for an uncountable initial cardinal  $\kappa$ , by the same computation of the least fixed point of  $\xi \mapsto \kappa^\xi$ ), carried to the height  $\kappa = M \cap \text{Ord}$  of every transitive model  $M$  of ZFC, which need not be a cardinal:  $E^M$  is  $E[\kappa]$ , and lemma XVII.6.5 supplies  $\varepsilon_\kappa = \kappa$  where source 34 uses that  $\kappa$  is an uncountable cardinal. Source 34 states the case of a full class model over an inaccessible  $\kappa$ , and source 33 computes the same type as  $\sup_n t_n$  there (section XVI.8.8). The second equality, lemma XVII.6.5 and theorem XVII.6.7 are new to this report. In the proof above,  $\tau_n$  are ordinals (the types of the height stages), not source 34's terms  $\tau_n$  of (S34.6).

**Theorem XVII.6.7** (An explicit zeta boundary). *There is a parameter-free class well-order  $W$ , defined by finite nested syntax and an ordered sum over  $\omega$ , such that for every transitive  $M \models \text{ZFC}$  as above,*

$$\text{otp}(W^M) = \zeta_{\kappa+1}.$$

*Proof.* Put  $A_0 = E$  and  $A_{n+1} = \text{Ep}(A_n)$ , and define

$$W = \sum_{n < \omega} A_n.$$

The uniform definition of this family deserves attention. Attach a natural-number level to a finite term code. At level 0 use the grammar for  $E$ . At level  $n + 1$ , use the epsilon grammar, with each epsilon constant carrying a valid level- $n$  term as its label. Exponent subterms remain at the same level but are proper subtrees. Validity and comparison are consequently determined by a single finite recursive calculation on the complete nested code. The recursion follows the displayed syntax tree, treating base ordinal coefficients as leaves, rather than exploring their transitive closures. Equivalently, one can require a finite computation record verifying all its nodes and comparisons. This defines the family uniformly; it is not an evaluation map for arbitrary formulas defining classes.

Finite induction using Theorem XVII.6.4 proves that every  $A_n$  is a class well-order. The ordered sum is a class well-order: a nonempty subclass has a least occupied natural-number index, and a least element in that summand.

Externally write

$$a_0 = \varepsilon_{\kappa+1}, \quad a_{n+1} = \varepsilon_{a_n}.$$

These are the exact types of the summands. Let  $\lambda$  be the least fixed point of epsilon enumeration above  $\kappa$ . Then  $\kappa < a_0 < \lambda$ , and normality proves that the  $a_n$  increase strictly while remaining below  $\lambda$ . Their supremum is itself a fixed point by continuity and exceeds  $\kappa$ , so it equals  $\lambda$ . Each  $a_n$  is an epsilon number; therefore addition absorbs all earlier summands and

$$\sum_{n < \omega} a_n = \sup_{n < \omega} a_n = \lambda.$$

Finally Lemma XVII.6.5 gives  $\lambda = \zeta_{\kappa+1}$ .  $\square$



### XVII.6.4 The spectrum of definable presentations

The next definition makes precise the external version of the question posed in the earlier discussion of definable class orders [93, 100].

**Definition XVII.6.8.** Let  $\text{Spec}(M)$  consist of the external order types of externally well-founded well-orders on subclasses of  $M$ , where both the carrier and the order are first-order definable over  $M$  with finitely many set parameters. Define the *definability horizon*

$$\mathfrak{H}(M) = \sup \text{Spec}(M).$$

A carrier is *internally proper* if it is not equal to any set belonging to  $M$ .

The ambient universe can form this spectrum using satisfaction for the set structure  $M$ . The definition does not posit a satisfaction predicate for  $M$  inside  $M$ , nor does it turn all classes of the original universe into a class.

**Theorem XVII.6.9** (Spectrum and carrier theorem). *For every transitive set model  $M \models \text{ZFC}$  of height  $\kappa$ ,*

$$\text{Spec}(M) = [0, \mathfrak{H}(M)).$$

*The types with internally proper carriers are exactly*

$$[\kappa, \mathfrak{H}(M)).$$

*Moreover every such proper presentation can be transferred definably to the carrier  $\text{Ord}^M$ , without adding parameters or assuming a definable global choice function.*

*Proof.* If a presentation has type  $\gamma$ , then for each  $\beta < \gamma$  its restriction below the point of index  $\beta$  has type  $\beta$ . That point belongs to  $M$  and can be added as a parameter. Thus the spectrum is downward closed. Taking a tagged copy of the carrier and appending one new tagged point gives a definable successor presentation. The spectrum has no maximum, proving the first formula.

If a definable carrier equals  $a \in M$ , Separation inside  $M$  forms its order as a set relation. External well-foundedness implies that  $M$  regards this set relation as a well-order. Its ordinal collapse belongs to  $M$ , and transitivity makes that collapse externally correct. Its type is therefore below  $\kappa$ .

Now let  $R$  well-order an internally proper carrier  $D$ . Consider the definable initial part

$$P = \{x \in D : \{y \in D : yRx\} \text{ is a set belonging to } M\}.$$

It is initial because a predecessor segment below a predecessor is a definable subset of a set. Its inherited order is setlike internally. If  $P$  were a set in  $M$ , the least element  $x$  of  $D \setminus P$  would have exactly  $P$  as its predecessors. Then  $x \in P$ , a contradiction. Hence  $P$  is internally proper.

Map each  $x \in P$  to the ordinal type of its set of predecessors. This is a definable order isomorphism onto an initial subclass of  $\text{Ord}^M$ . If its image were bounded below  $\kappa$ , it would be a set in  $M$ ; Replacement applied to the inverse would make  $P$  a set in  $M$ . Thus its image is all of  $\text{Ord}^M$ . The original order therefore begins with an initial copy of  $\kappa$ . Together with the first formula and the set-carrier case, this proves the asserted proper spectrum.

Finally, re-order  $D$  by rank, retaining  $R$  within each rank:

$$xEy \iff \text{rank}(x) < \text{rank}(y) \vee (\text{rank}(x) = \text{rank}(y) \wedge xRy).$$

This is a well-order, and every predecessor segment is contained in an internal rank level and is therefore a set in  $M$ . Its canonical setlike collapse gives a definable bijection  $D \rightarrow \text{Ord}^M$ . Transport  $R$  along this bijection. All definitions use the original parameters only; the supplied order on  $D$  provides the tie-breaking that a global choice function would otherwise supply.  $\square$

Allowing all finite set parameters is essential for the interval statement: taking an arbitrary initial segment normally introduces its endpoint as a parameter. The carrier-transfer statement itself does preserve parameter-free definability.

**Proposition XVII.6.10** (Cardinality bound). *Let  $\mu = |\kappa|$ , calculated externally. Then*

$$|M| = \mu, \quad \mathfrak{H}(M) < \mu^+.$$

*Proof.* For each  $\alpha < \kappa$ , the internal rank level  $V_\alpha^M$  is a set of  $M$ . Choice in  $M$  provides a bijection with some ordinal below  $\kappa$ , so externally its cardinality is at most  $\mu$ . Since  $M = \bigcup_{\alpha < \kappa} V_\alpha^M$ , ambient choice gives  $|M| \leq \mu$ ; the reverse inequality follows from  $\kappa \subseteq M$ .

There are at most  $\mu$  formulas with finite tuples of parameters from  $M$ . Every definable carrier has external cardinality at most  $\mu$ , so each realized order type is below  $\mu^+$ . The supremum of at most  $\mu$  such ordinals is still below  $\mu^+$ , proving the strict bound.  $\square$

*Note XVII.6.B* ([merge]; the parameter spectrum of Part XVI). The first formula of [theorem XVII.6.9](#) and [proposition XVII.6.10](#) are source 34's [proposition XVI.8.2](#) for the pure language  $L = \{\in\}$ :  $\mathcal{S}_{\text{par}}^L = [0, \Theta_L)$  with  $\Theta_L < \lambda^+$ ,  $\lambda = |M|$ , so that  $\mathfrak{H}(M) = \Theta_{\{\in\}}$ ; the proofs are the same (downward closure through the endpoint as a new parameter, successor closure, counting). Source 36 adds  $|M| = |\kappa|$ . For  $M = V_\kappa$  they are source 35's  $\delta_{\text{set}}$  ([theorem XVI.8.6](#)), and with a named enumeration  $G$  source 32's and source 33's ceilings ([note XVII.7.D](#)). Like source 34, source 36 counts externally well-founded orders over an arbitrary transitive model, with no  $\beta$ -hypothesis. New to this report: the proper-carrier spectrum  $[\kappa, \mathfrak{H}(M))$  and the parameter-preserving transfer of every internally proper presentation to the carrier  $\text{Ord}^M$ , which is [proposition XVII.2.4](#) carried out inside  $M$ .

### XVII.6.5 Closure and parameter budgets

**Proposition XVII.6.11** (Closure and cofinality of horizons). *The spectrum  $\text{Spec}(M)$  is closed under ordinary ordinal addition, multiplication, and exponentiation, and under  $\alpha \mapsto \varepsilon_\alpha$ . Consequently*

$$\varepsilon_{\mathfrak{H}(M)} = \mathfrak{H}(M).$$

*Let  $\mathfrak{H}_0(M)$  be the horizon for parameter-free presentations and  $\mathfrak{H}_p(M)$  the horizon for presentations definable from one fixed finite tuple  $p$ . Then*

$$\text{cf}(\mathfrak{H}_0(M)) = \text{cf}(\mathfrak{H}_p(M)) = \omega, \quad \varepsilon_{\mathfrak{H}_0(M)} = \mathfrak{H}_0(M), \quad \varepsilon_{\mathfrak{H}_p(M)} = \mathfrak{H}_p(M).$$

*Furthermore,*

$$\zeta_{\kappa+1} < \mathfrak{H}_0(M) \leq \mathfrak{H}_p(M) \leq \mathfrak{H}(M) < |\kappa|^+.$$

*Proof.* Given two definable presentations, combine their finite tuples of parameters and apply the explicit sums, products, and finite-support powers of the preceding section. External transitivity ensures that all finite codes occur in  $M$ , so these constructions have the ordinary ordinal sum, product, and power as their external types. The conventional cases with base 0 or 1 use the empty or singleton presentations. [Theorem XVII.6.4](#) similarly gives a presentation of type  $\varepsilon_\alpha$  from one of type  $\alpha$ .

Write  $\mathfrak{H} = \mathfrak{H}(M)$ . Successor closure makes  $\mathfrak{H}$  a limit ordinal, and realized types are cofinal in  $\mathfrak{H}$ . Normality of epsilon enumeration gives

$$\varepsilon_{\mathfrak{H}} = \sup_{\alpha \in \text{Spec}(M)} \varepsilon_\alpha \leq \mathfrak{H} \leq \varepsilon_{\mathfrak{H}}.$$

The middle inequality uses that each epsilon value is again realized.

For a fixed parameter tuple there are only countably many formulas. Their realized types are cofinal in the corresponding horizon and have no maximum, because successor construction preserves that tuple. The horizon therefore has cofinality exactly  $\omega$ . The epsilon construction also preserves the tuple, so the same continuity argument applies. No downward-closure assertion for this restricted spectrum is needed. Finally  $W + 1$ , with  $W$  from Theorem XVII.6.7, is parameter-free and has type  $\zeta_{\kappa+1} + 1$ . The remaining inequalities follow by inclusion of the permitted definitions and Proposition XVII.6.10.  $\square$

The same epsilon-closure argument applies when a fixed finite list of class predicates is named in the language: it is enough that the relevant definitions and finite coding are available. The next section exploits this observation for a named global enumeration. For unrestricted set parameters over an arbitrary  $M$ , the elementary cardinality argument alone gives  $\text{cf}(\mathfrak{H}(M)) \leq |\kappa|$ , not  $\text{cf}(\mathfrak{H}(M)) = \omega$ .

[merge] (4 October 2026.) Proposition XVII.6.11 is new against Part XVI in two respects. First,  $\varepsilon_{\mathfrak{H}} = \mathfrak{H}$  is stronger than the closure properties proved there for the ceilings over  $V_\kappa$ : source 32's  $\kappa^{\delta_A} = \delta_A$  (theorem XVI.8.9), source 33's  $\omega^{\delta_G} = \kappa^{\delta_G} = \delta_G$  (corollary XVI.8.17), and their extension to transitive  $\beta$ -models in the [merge] paragraph before corollary XVI.8.17. An epsilon number  $\delta > \kappa$  need not satisfy  $\varepsilon_\delta = \delta$  (for example  $\varepsilon_{\kappa+1}$ ), while  $\varepsilon_\delta = \delta > \kappa$  gives  $\omega^\delta = \kappa^\delta = \delta$  by the computation in the proof of theorem XVII.6.6. Second, it holds for every transitive  $M \models \text{ZFC}$ , with no  $\beta$ -hypothesis and no inaccessibility, and it adds the lower bound  $\zeta_{\kappa+1} < \mathfrak{H}_0(M)$  through the parameter-free order of theorem XVII.6.7. It proves source 32's remark after corollary XVI.8.11 that “closure under hereditary finite term formation gives further necessary restrictions” (also in source 32's question research question 113.284), in this precise form. The cofinality  $\text{cf}(\mathfrak{H}_0(M)) = \omega$  is source 34's statement on  $\mathcal{S}_0^L$  in proposition XVI.8.2. Dated notes at research questions 113.284 and 113.285 record the advance.

## XVII.6.6 Full classes and the admissible upper bound

**Proposition XVII.6.12** (The full-class calibration). *Let  $\kappa$  be inaccessible in the ambient universe. In the full class realization*

$$(V_\kappa, \mathcal{P}(V_\kappa)),$$

*the external order types of internally proper class well-orders are exactly  $[\kappa, \kappa^+)$ .*

*Proof.* Here  $|V_\kappa| = \kappa$ . Regularity ensures that a class function on an internal set has a set-sized, rank-bounded image in  $V_\kappa$ ; all subclasses are available. Thus the usual class-theoretic arguments apply, including the initial-Ord argument of Theorem XVII.6.9. A proper class well-order has type at least  $\kappa$  and cardinality at most  $\kappa$ , hence type below  $\kappa^+$ .

Conversely, if  $\kappa \leq \alpha < \kappa^+$ , choose an external bijection  $V_\kappa \rightarrow \alpha$ . Pulling back the ordinal order gives a relation on  $V_\kappa$ . Its ordered-pair codes belong to  $V_\kappa$ , so the relation is an admitted class in the full realization. Its carrier is proper and its type is exactly  $\alpha$ .  $\square$

This cutoff is an external ordinal. It is not a new member of  $V_\kappa$  or a set of all its class order types. The comparison with  $\mathfrak{H}(V_\kappa) < \kappa^+$  also shows that definable classes and all classes have different spectra even over the same set universe.

*Note XVII.6.C* ([merge]; the full-class spectrum). Proposition XVII.6.12 is source 12's theorem 21.1 (iv) (Part V; at every limit height, source 31's theorem XV.9.3) and source 34's theorem XVI.8.1, proved again by sources 35, 32 and 33 (notes XVI.8.5, XVI.8.8 and XVI.8.15); the proof is the same.

There is a finer historical bound. If  $A$  is an admissible transitive set containing  $M$  as an element, then

$$\mathfrak{H}(M) \leq \text{Ord} \cap A. \quad (\text{S36.6.2})$$

Indeed, satisfaction for the *set structure*  $M$  is available in  $A$  by admissible recursion. Each specified definable carrier and relation is therefore an element of  $A$ . An externally well-founded one has its ordinal collapse in  $A$ , by the well-founded recursion theorem for admissible sets. Each individual type is thus below the height of  $A$ ; taking the external supremum proves (S36.6.2). Applying this to the least admissible set above  $M$  gives the bound discussed by Blass [100].

Sharpness requires care. The cited discussions distinguish closure conditions giving a strict bound from examples giving equality; Taranovsky supplies equality examples in [101]. We do not claim a new resolution of that earlier existence question. Nor do we identify  $\mathfrak{H}(M)$  with a supremum over arbitrary computable presentations of  $M$ , whose formulation requires a precise choice of quantifiers and coding. The results proved here use explicit class presentations, ordinary external order types, and stated parameter budgets throughout.

[merge] (4 October 2026.) The admissible bound (S36.6.2) rests on the theorem that a well-founded relation which is an element of an admissible set has its rank below the height of that set; source 36 sketches it and credits Blass. With proposition XVII.8.1 (the unrolling height  $\kappa^+$  of the full model) it is the first comparison of a definability cutoff with admissible and unrolling heights that source 34's agenda item research question 113.283 asks for; the identification of  $\mathfrak{H}(M)$  stays open (source 36's research question 113.302). The MathOverflow discussions cited here were not checked by this write.

## XVII.7 Uniform notation systems and definability horizons

The preceding constructions supply long, explicit orders. We now ask how their lengths compare with *all* presentations available in a specified language. There are two complementary answers. Internally, a uniform family can always be surpassed by another class well-order. Externally, over a fixed set universe, definable presentations have an exact ordinal cutoff. At an inaccessible-height universe with a named global enumeration, this cutoff has countable cofinality, and allowing arbitrary set parameters does not increase the supremum already approached by parameter-free presentations.

The dependence on the available classes is a central feature of second-order set theory and class-theoretic potentialism [35, 84]. Here we isolate it in explicit constructions and prove the resulting bounds. The arguments do not require a converse to the known implication from elementary transfinite recursion to comparison of arbitrary class well-orders.

### XVII.7.1 The obstruction to a uniform universal notation system

An order embedding throughout this subsection is simply a strictly increasing injection; its image need not be an initial segment. This weaker notion makes the obstruction stronger.

**Lemma XVII.7.1** (A well-order cannot absorb an appended point). *In GB, if  $A$  is a class well-order, there is no order embedding  $A + 1 \rightarrow A$ .*

*Proof.* First observe that every increasing class map  $f : A \rightarrow A$  satisfies  $a \leq_A f(a)$  for every  $a \in A$ . Otherwise the class of counterexamples has a least member  $a$ . Put  $b = f(a) <_A a$ . Leastness gives  $b \leq_A f(b)$ , whereas monotonicity gives

$$f(b) <_A f(a) = b,$$

a contradiction.

Now suppose  $f : A + 1 \rightarrow A$  were an embedding, and let  $*$  be the appended point. Its image  $b = f(*)$  belongs to  $A$ . Applied to the restriction of  $f$  to  $A$ , the observation gives  $b \leq_A f(b)$ . But  $b <_{A+1} *$  gives  $f(b) <_A f(*) = b$ . These inequalities contradict each other. The same argument includes the empty case, in which  $f(*) \in A$  is already impossible.  $\square$

*Note XVII.7.A* ([merge]; the successor obstruction). [Lemma XVII.7.1](#) is source 34's [lemma XVI.3.2](#), with the same proof (an increasing self-map is inflationary), as are source 35's [lemma XVI.2.5](#) and [note XVI.7.7](#), source 32's [note XVI.2.10](#) and source 33's [notes XVI.2.15](#) and [XVI.2.16](#).

**Theorem XVII.7.2** (Uniform diagonal obstruction). *Work in GB. Let  $I$  be a class well-order, and let  $(A_i)_{i \in I}$  be a uniformly class-coded family of class well-orders. Then*

$$\mathcal{D} = \sum_{i \in I} (A_i + 1) \tag{S36.7.1}$$

*is a class well-order which does not order-embed into any  $A_i$ . Consequently, in GBC, no uniformly class-coded family indexed by an arbitrary class is cofinal in all class well-orders under order embeddability.*

*Proof.* Uniform coding means that one class specifies all pairs  $(i, x)$  with  $x \in A_i$ , and another specifies all triples  $(i, x, y)$  satisfying  $x <_{A_i} y$ . Thus tagged pairs and elementary comprehension construct  $\mathcal{D}$ . To find the least element of a nonempty subclass of  $\mathcal{D}$ , first find the least index at which it meets a block, and then its least element in that block. Hence  $\mathcal{D}$  is well-ordered.

If  $e : \mathcal{D} \rightarrow A_i$  were an embedding, its restriction to the  $i$ -th block would embed  $A_i + 1$  into  $A_i$ , contradicting [Lemma XVII.7.1](#).

For the last assertion, global choice places any class of indices in a set-like class well-order: restrict a global enumeration of the universe to that class. Apply the first assertion using this index order. If the family were cofinal, some member would have to receive an embedding of  $\mathcal{D}$ , which has just been excluded.  $\square$

The theorem addresses a precise form of notation system. Suppose the valid notations form an available class  $N$  of sets, and one available class relation uniformly evaluates their carriers and orders. If every evaluation is a well-order, the represented orders cannot be cofinal in all class well-orders. Neither quotienting by isomorphism nor allowing several notations for one order avoids the argument.

The hypotheses also identify the scope of the conclusion. An external enumeration of formulas is not automatically a class evaluator for their meanings. If proposed notations include ill-founded presentations, a uniform class of the valid codes is an additional requirement. Collections of class orders formed in a stronger surrounding universe are different objects from the same-level coded families considered here. Thus the obstruction rules out a universal evaluator at one level, while leaving systematic restricted calculi and extensions of the ambient language entirely available.

*Note XVII.7.B* ([merge]; the uniform-family diagonal). [Theorem XVII.7.2](#) is source 34's [theorem XVI.7.2](#) and [corollary XVI.7.3](#) (over GB, and GBC for the catalogue clause), with the same proof; it is also source 35's [note XVI.7.8](#) and [corollary XVI.7.9](#), source 32's [notes XVI.7.14](#) and [XVI.7.16](#) and source 33's [notes XVI.7.17](#) and [XVI.7.18](#), so it is now proved five times independently. Source 34 adds that  $\mathcal{D}$  is proper when  $I$  is. The earlier diagonal theorems of this report work at the level of relations ([theorems 19.37](#) and [19.38](#)); see the [merge] paragraph after [corollary XVI.7.6](#).

### XVII.7.2 An inaccessible-height universe with a named enumeration

For the remainder of this section, work in an external universe satisfying ZFC. Fix a strongly inaccessible cardinal  $\kappa$ , put  $M = V_\kappa$ , and let  $G$  be a predicate coding a bijection

$$G : \kappa \longrightarrow V_\kappa.$$

The graph of this bijection is coded by a subset of  $M$ . In the structure  $(M, \in, G)$ , its domain is the internal proper class  $\text{Ord}^M$ . Accordingly,  $\kappa$  is an external ordinal, not a set of  $M$ .

**Definition XVII.7.3.** Let  $\text{Spec}_G$  be the set of external ordinal order types of externally well-founded linear orders whose domains and relations are first-order definable in  $(M, \in, G)$ , allowing finite tuples of set parameters from  $M$ . Let  $\text{Spec}_G^0$  impose the additional requirement that no set parameters occur. Define the *definability horizon*

$$\mathfrak{H}_G = \sup \text{Spec}_G.$$

The word *parameter-free* here always refers to the expanded language with the named predicate  $G$ .

These are legitimate external sets of ordinals: there is a set of formulas and parameter tuples, and each interpreted carrier is a subset of the set  $M$ . We have not formed a class, inside  $M$ , whose members are proper classes or their isomorphism types.

**Lemma XVII.7.4** (Closure and correctness of the ambient model). *The structure  $(M, \text{Def}(M; G))$  satisfies GBC, and every class relation on  $M$  is externally well-founded if and only if it has no descending  $\omega$ -sequence coded by a set of  $M$ .*

*Proof.* Strong inaccessibility gives  $|V_\kappa| = \kappa$ , and if  $a \in V_\kappa$ , then  $|a| < \kappa$ . Every subset of  $a$  belongs to  $V_\kappa$ , so separation remains valid with  $G$  in the language. The image of  $a$  under any class function has fewer than  $\kappa$  members. Regularity bounds their ranks below  $\kappa$ , so that image also belongs to  $V_\kappa$ . This proves Class Replacement. Definable classes are closed under elementary comprehension, and  $G$  supplies global choice.

Every external countable sequence of elements of  $V_\kappa$  is a set in  $V_\kappa$ : regularity bounds the ranks of its entries below  $\kappa$ , and coding the sequence still gives rank below  $\kappa$ . In the external universe, an ill-founded relation on a set admits a descending  $\omega$ -sequence by dependent choice. Such a sequence for a relation on  $M$  therefore belongs to  $M$ . Conversely, a sequence in the transitive structure  $M$  witnessing descent witnesses actual external descent.  $\square$

The last assertion is what will make a definable validity filter sound. Transitivity alone does not suffice for this purpose: a countable transitive model can omit an external descending sequence. Our proofs below use the stated inaccessible-height hypotheses rather than silently treating an internal well-foundedness assertion as an external one.

*Note XVII.7.C* ([merge]; the inaccessible-rank model). [Lemma XVII.7.4](#) is source 32's [lemma XVI.8.7](#) (for  $\text{Def}(V_\kappa; \in, G, A)$  and for the full model), source 33's [note XVI.8.14](#) and source 34's  $\beta$ -model remark after [theorem XVI.8.1](#), with the same proof. Source 36's named predicate  $G$ , a bijection  $\kappa \rightarrow V_\kappa$ , is source 32's  $G$  and source 33's set-like global well-order  $G$  (the order induced by a bijection with  $\text{Ord}^M$ ).

**Theorem XVII.7.5** (The definable horizon). *Under these hypotheses,*

$$\text{Spec}_G = [0, \mathfrak{H}_G), \quad \kappa < \mathfrak{H}_G < \kappa^+, \quad \varepsilon_{\mathfrak{H}_G} = \mathfrak{H}_G. \quad (\text{S36.7.2})$$



Moreover,

$$\text{cf}(\mathfrak{H}_G) = \omega \quad \text{and} \quad \sup \text{Spec}_G^0 = \mathfrak{H}_G. \quad (\text{S36.7.3})$$

All ordinals and cofinalities in these displays are external.

We first verify (S36.7.2); the following subsections prove the more distinctive assertions (S36.7.3).

*Proof of the interval, bounds, and fixed-point assertion.* The parameter-preserving arguments of Theorem XVII.6.9 apply to the expanded language. In particular, an initial segment is defined using its endpoint as an additional set parameter, and appending a point shows that the spectrum has no maximum. Thus it is exactly  $[0, \mathfrak{H}_G)$ .

There are  $\kappa$  possible formula-and-parameter tuples. Every carrier has size at most  $\kappa$ , so every represented type is below the external successor cardinal  $\kappa^+$ . Its regularity implies  $\mathfrak{H}_G < \kappa^+$ . The definable presentation  $\text{Ord}^M + 1$  has type  $\kappa + 1$ , giving the strict lower bound.

The epsilon-term construction of Theorem XVII.6.4 preserves the allowed predicates and parameters. Therefore  $\alpha < \mathfrak{H}_G$  implies  $\varepsilon_\alpha < \mathfrak{H}_G$ . Since  $\mathfrak{H}_G$  is a limit, continuity of epsilon enumeration gives

$$\varepsilon_{\mathfrak{H}_G} = \sup_{\alpha < \mathfrak{H}_G} \varepsilon_\alpha \leq \mathfrak{H}_G.$$

The reverse inequality follows from  $\alpha \leq \varepsilon_\alpha$ , proving equality. The same constructions also give closure of  $[0, \mathfrak{H}_G)$  under ordinal sums, products, and powers. In particular  $\omega^{\mathfrak{H}_G} = \mathfrak{H}_G$ , and finite sums of ordinals below  $\mathfrak{H}_G$  remain below it.  $\square$

*Note XVII.7.D* ([merge]; the  $V_\kappa$  horizon theorem is a theorem of sources 32–34). Theorem XVII.7.5 is, apart from the clause  $\varepsilon_{\mathfrak{H}_G} = \mathfrak{H}_G$ , the definability-ceiling theorem that sources 32, 33 and 34 proved independently on 4 October 2026 and that Part XVI prints: source 32's theorem XVI.8.9 ( $S_A = [0, \delta_A)$ ,  $\kappa < \delta_A < \kappa^+$ ,  $\text{cf}(\delta_A) = \omega$ ,  $\kappa^{\delta_A} = \delta_A$ , for the language  $\{\in, G, A\}$ ; with  $A$  trivial,  $\delta_A = \mathfrak{H}_G$ ), source 33's note XVI.8.16 ( $\delta_G^0 = \delta_G$ ,  $\kappa < \delta_G < \kappa^+$ ,  $\text{cf}(\delta_G) = \omega$ ;  $\delta_G = \mathfrak{H}_G$  and  $\delta_G^0 = \sup \text{Spec}_G^0$ ) and source 34's proposition XVI.8.2, theorem XVI.8.3, and corollary XVI.8.4 ( $\Theta_L$  for  $L = \{\in, G\}$ , in every transitive  $\beta$ -model of GBC with a named global well-order). Source 36's is the fourth independent proof at  $V_\kappa$ , and source 35's theorem XVI.8.6 is the pure-language companion. Its route to the cofinality is source 33's: the bounded-complexity diagonals  $D_n$  (proposition XVII.7.6; theorem XVI.7.20); source 34 bounds each formula pair by its own parameter-free sum, source 32 uses diagonals of finite lists of formula pairs. New against Part XVI is  $\varepsilon_{\mathfrak{H}_G} = \mathfrak{H}_G$ , obtained from the epsilon-term functor, which strengthens source 32's  $\kappa^{\delta_A} = \delta_A$  and source 33's  $\omega^{\delta_G} = \delta_G$  as recorded after proposition XVII.6.11. Source 36's novelty sentences (Section XVII.1.5 and its title page) carry this credit in dated notes.

### XVII.7.3 A sound filter at every fixed formula complexity

Fix a standard natural number  $n$ , externally. Choose a nested syntactic complexity convention for first-order formulas, for example normalized Lévy complexity allowing both  $\Sigma_n$  and  $\Pi_n$  formulas. Every formula belongs to some finite level. At this one fixed level there is a first-order partial satisfaction formula  $\text{Sat}_n$  for  $(M, \in, G)$ . The subscript indexes a schema of formulas; no full uniform truth predicate is being assumed.

For completeness, partial truth here can be obtained by ordinary syntactic recursion at the metatheoretic level. Truth for bounded formulas is evaluated in a transitive set containing the parameters, with  $G$  restricted to that set. The restriction exists by amenability, and bounded truth is absolute between this set structure and  $M$ . Its satisfaction relation is a set constructed by ordinary set-valued recursion. For a normalized formula with a fixed finite number of unbounded quantifier

blocks, place the corresponding quantifiers over  $M$  in front of this bounded evaluation. Finite tuples encode a block containing several variables. Combining the two possible initial quantifier polarities yields the required  $\text{Sat}_n$ .

This construction is effective for each standard  $n$ , so the code of its output formula can be generated from  $n$ . It does not give one formula which correctly evaluates all levels at once: the unbounded quantifiers used at successive levels occur in the output formulas themselves. This distinction will be used both in the complexity hierarchy and in the truth-adjunction argument below.

For explicit coding, represent a class presentation by a unary predicate: entries  $\langle 0, x \rangle$  mark its domain, and entries  $\langle 1, x, y \rangle$  mark its strict order. An index  $u \in M$  codes a formula together with a finite parameter tuple. Partial satisfaction gives uniform candidate classes

$$\widehat{D}_{n,u}, \quad \widehat{R}_{n,u} \subseteq \widehat{D}_{n,u}^2.$$

Malformed syntactic codes are assigned empty candidates. Every presentation of the chosen complexity, with every allowed parameter tuple, occurs among these candidates.

Define  $\text{Valid}_n(u)$  to say that the candidate is a strict linear order and that

$$\neg \exists f \left[ f \text{ is a function with domain } \omega, \quad \text{ran}(f) \subseteq \widehat{D}_{n,u}, \right. \\ \left. \forall k \in \omega \quad f(k+1) \widehat{R}_{n,u} f(k) \right]. \quad (\text{S36.7.4})$$

All quantified objects in this formula are sets of  $M$ . Occurrences of the candidate predicates are replaced by the fixed formula  $\text{Sat}_n$ . Thus  $\text{Valid}_n$  is first-order in  $\{\in, G\}$ , although usually of greater complexity than the formulas being tested.

By Lemma XVII.7.4, the filter accepts precisely the externally well-founded candidates. Put

$$A_{n,u} = \begin{cases} (\widehat{D}_{n,u}, \widehat{R}_{n,u}), & \text{Valid}_n(u), \\ 0, & \text{otherwise.} \end{cases}$$

This is one uniformly definable family of genuine well-orders for the fixed  $n$ . Invalid formulas cause no problem because their entire block has been replaced by the empty order.

**Proposition XVII.7.6** (Bounded-complexity domination). *For each standard  $n$ , there is a parameter-free  $\{\in, G\}$ -definable class well-order*

$$D_n = \sum_{\xi \in \text{Ord}^M} (A_{n,G(\xi)} + 1) \quad (\text{S36.7.5})$$

whose external type  $\delta_n$  is strictly larger than every type presented at complexity level  $n$ , even when arbitrary finite set parameters are allowed in those presentations. Nevertheless  $\delta_n < \mathfrak{H}_G$ .

*Proof.* The preceding filter and the named enumeration  $G$  make the displayed sum a first-order definable presentation. Its index order and all its blocks are externally well-founded, so the sum is a well-order. A given level- $n$  presentation with its parameter tuple has some code  $u \in M$ , and  $u = G(\xi)$  for a unique  $\xi < \kappa$ . If its type is  $\alpha$ , the corresponding block has type  $\alpha + 1$ . Consequently  $\alpha < \delta_n$ , either by ordinary external ordinal comparison or by the obstruction in Theorem XVII.7.2.

The definition of  $D_n$  uses no set parameter: the standard  $n$  is compiled into the chosen partial satisfaction formula, and the remaining coding conventions are fixed. Thus  $\delta_n \in \text{Spec}_G^0 \subseteq \text{Spec}_G = [0, \mathfrak{H}_G)$ .  $\square$

This proposition explains why bounding the *number of set parameters* does not capture the decisive resource. Even a single fixed formula can range through arbitrarily many parameter values. Partial truth lets one collect all those values at a specified formula complexity; a modest increase in complexity then constructs an order exceeding all their types.

*Note XVII.7.E* ([merge]; the bounded-complexity diagonal is a theorem of sources 33–35). [Proposition XVII.7.6](#) is source 33’s [theorem XVI.7.20](#), by the same construction: a partial satisfaction formula for one fixed level (source 33’s [lemma XVI.7.19](#)), a first-order validity filter that is sound because the model is correct about well-foundedness, invalid candidates replaced by the empty order, and the tagged sum along  $G$ , with no set parameter. It is also source 34’s [corollary XVI.7.6](#) (internally, in GBC with named predicates) and source 35’s [note XVI.7.13](#); source 35’s [theorem XVI.7.11](#) is the pure-language form for orders on Ord with ordinal parameters. Source 33 states the domination by embeddings; source 36 by external order types, which at  $V_\kappa$  is the same by [lemma XVII.7.4](#).

#### XVII.7.4 Countable cofinality and the role of parameter-free syntax

Let

$$h_n = \sup\{\alpha + 1 : \alpha \text{ has a level-}n \text{ presentation with set parameters}\}. \quad (\text{S36.7.6})$$

The levels are nested, so  $h_n$  is nondecreasing. Every first-order definition has finite complexity, giving

$$\sup_{n < \omega} h_n = \mathfrak{H}_G.$$

[Proposition XVII.7.6](#) gives the sharper inequalities

$$h_n \leq \delta_n < \mathfrak{H}_G \quad (n < \omega). \quad (\text{S36.7.7})$$

Hence  $(\delta_n)_{n < \omega}$  is cofinal in  $\mathfrak{H}_G$ . Because  $\mathfrak{H}_G$  is a nonzero limit ordinal, this proves  $\text{cf}(\mathfrak{H}_G) = \omega$ . Since each  $\delta_n$  belongs to  $\text{Spec}_G^0$ , it simultaneously proves  $\sup \text{Spec}_G^0 = \mathfrak{H}_G$ , completing [Theorem XVII.7.5](#).

The parameter-free spectrum itself is not an interval. It has only countably many members externally, while it contains the type  $\kappa$  of  $\text{Ord}^M$ . It therefore omits many smaller external ordinals. Set parameters fill these gaps by naming endpoints; they do not raise the supremum beyond the cofinal parameter-free sequence just constructed.

**Corollary XVII.7.7** (An effective strict complexity hierarchy). *There is an effective function  $f : \omega \rightarrow \omega$ , with  $f(n) > n$ , such that*

$$h_n < h_{f(n)}.$$

*Thus an effective sequence of increasing finite complexity levels has strictly increasing spectrum bounds cofinal in  $\mathfrak{H}_G$ .*

*Proof.* The construction of the partial satisfaction formula, the validity test, and the tagged sum is an effective transformation of syntax. Read off a finite complexity bound for the resulting formula defining  $D_n$ , and increase it if necessary to obtain  $f(n) > n$ . Then  $\delta_n$  occurs at level  $f(n)$ , so

$$h_n \leq \delta_n < \delta_n + 1 \leq h_{f(n)}.$$

Starting at any level and iterating  $f$  gives a strictly increasing, unbounded sequence of natural-number levels, hence a cofinal sequence of their bounds.  $\square$

This statement avoids attaching significance to a numerical syntax overhead. With a fixed conventional prenex coding, the construction uses bounded truth followed by fixed formula templates, so one can audit a finite shift bound. The effective function formulation is enough for every mathematical conclusion above and is independent of such implementation choices.

[merge] [Corollary XVII.7.7](#) is new to this report in this form. Source 35's [corollary XVI.7.12](#) and [theorem XVI.7.11](#) separate definability levels for orders on  $\text{Ord}$  in the pure language, and source 35 claims no separation of two consecutive levels; source 36 claims none either, since  $f$  is effective but not computed. The height  $h_n$  is a supremum of  $\alpha + 1$ , which gives the strict inequality used in the proof (as the Hilbert's-tenth review notes); it is not source 20's height function  $h_\kappa$  of [corollary 108.3](#).

**Proposition XVII.7.8** (No uniform evaluator for the cofinal schema). *The sequence of definitions  $D_n$  can be generated effectively as syntax, but no first-order formula over  $(M, \in, G)$ , even with finitely many set parameters, uniformly evaluates presentations of all the  $D_n$ .*

*Proof.* Such an evaluator would make  $\sum_{n < \omega} (D_n + 1)$  a single definable class well-order. Its external type is at least  $\sup_n \delta_n = \mathfrak{H}_G$ , because it contains a copy of every summand. This contradicts  $\text{Spec}_G = [0, \mathfrak{H}_G)$ .  $\square$

There is therefore no conflict between the external countable cofinality and the internal diagonal obstruction. The external sequence lists definitions and their interpreted types. The class universe  $\text{Def}(M; G)$  does not contain a uniform semantic evaluation of that sequence. In particular, the countable cofinality of  $\mathfrak{H}_G$  does not supply an internal set sequence of ordinals approaching  $\text{Ord}^M$ : the entries  $\delta_n$  are external types beyond the internal ordinal range.

*Note XVII.7.F* ([merge]; no uniform evaluator). [Proposition XVII.7.8](#) is source 32's [corollary XVI.8.13](#), source 33's [corollary XVI.7.22](#) and the subsection of source 34 after [corollary XVI.8.4](#) (“Why the cofinal family does not create a largest notation”), with the same argument; the paragraph after it is source 33's [remark XVI.8.18](#).

### XVII.7.5 Truth makes the old horizon presentable

Let  $T$  be a full first-order satisfaction predicate for the structure  $(M, \in, G)$ , including assignments of finite tuples from  $M$ . It exists as an external subset of  $M$ , and hence among the full classes. Write  $\mathfrak{H}_{G,T}$  for the horizon obtained by allowing  $T$  as an additional named predicate. All the preceding arguments apply to this expanded language:  $G$  still supplies the enumeration, and the ambient closure properties are unchanged.

**Theorem XVII.7.9** (Truth promotion). *There is a parameter-free  $\{\in, G, T\}$ -definable class well-order of external type exactly  $\mathfrak{H}_G$ . Consequently*

$$\kappa < \mathfrak{H}_G < \mathfrak{H}_{G,T} < \kappa^+. \quad (\text{S36.7.8})$$

*Proof.* Choose the partial satisfaction formulas effectively, so that there is a recursive map  $n \mapsto d(n)$  giving the code of the formula defining the tagged presentation  $D_n$ . The natural numbers of  $M = V_\kappa$  are standard. Thus this syntactic map has its usual arithmetical definition inside  $M$ , and  $T$  evaluates the formula with code  $d(n)$  on any given assignment. The entire family  $(D_n)_{n < \omega}$  is now uniformly first-order definable using the predicate  $T$ .

Form

$$E_G = \sum_{n < \omega} (D_n + 1). \quad (\text{S36.7.9})$$

Its external type is

$$\lambda = \sup_{m < \omega} \sum_{n < m} (\delta_n + 1).$$

Every finite partial sum is below  $\mathfrak{H}_G$ , by closure of  $\text{Spec}_G$  under successor and finite addition. Hence  $\lambda \leq \mathfrak{H}_G$ . Conversely the sum contains every  $D_n$ , so  $\lambda \geq \sup_n \delta_n = \mathfrak{H}_G$ . Therefore  $\lambda = \mathfrak{H}_G$ .

The definition uses only the named predicates  $G, T$ ; no set parameter is required. It follows that  $\mathfrak{H}_G \in \text{Spec}_{G,T}$ , and the initial-segment description of that spectrum gives  $\mathfrak{H}_G < \mathfrak{H}_{G,T}$ . Its upper bound is the same cardinal counting argument used in Theorem XVII.7.5.  $\square$

This is a precise way in which additional definitional resources extend length. A truth predicate converts the former cutoff into one presented order, after which successor and the other constructors continue beyond it. The theorem also gives another proof that such a full  $T$  is not definable from  $G$  and finitely many set parameters: otherwise the two horizons would coincide.

*Note XVII.7.G* ([merge]; truth promotion is source 32's exact truth presentation). Theorem XVII.7.9 is source 32's theorem XVI.8.12, proved independently on the same day: a full satisfaction predicate makes the diagonal family uniform, and the  $\omega$ -sum (S36.7.9) (source 32's  $L$ , (S32.17)) has type exactly the old ceiling. Source 32's  $T_A$  is the truth predicate for  $(V_\kappa, \in, G, A)$ , which is source 36's  $T$  when  $A$  is trivial; its diagonals are those of finite lists of formula pairs, source 36's those of the complexity levels. Source 33's theorem XVI.8.19 gives the strict jump with  $\text{cf}(\delta_{G,T}) = \omega$  but not the exact presentation, and source 34's theorem XVI.7.4 gives, internally and from an assumed full satisfaction class, an order dominating every definable one. Source 36's order  $E_G$  is a sum of diagonal orders; it is not a stage equivalence  $E_i$  of Section XVI.5. Source 36's closing remark, that  $T$  is not definable from  $G$  with set parameters, is also made by source 32. The size of the jump stays open (research question 113.298).

### XVII.7.6 The horizons are unbounded when the enumeration varies

The upper bound  $\kappa^+$  is independent of  $G$ , but no smaller external ordinal bounds all the  $\mathfrak{H}_G$ . The following coding argument shows this without any appeal to class recursion.

**Lemma XVII.7.10** (Encoding a predicate by adjacent swaps). *For every external  $A \subseteq \kappa$ , there is a bijection  $G_A : \kappa \rightarrow V_\kappa$  from which  $A$  is first-order definable without set parameters.*

*Proof.* For each  $\xi < \kappa$ , reserve input positions  $4 \cdot \xi, 4 \cdot \xi + 1$  and output ordinals  $2 \cdot \xi, 2 \cdot \xi + 1$ . Here the finite factor is on the *left* in ordinal multiplication. Define

|                |                    |                        |
|----------------|--------------------|------------------------|
|                | $G_A(4 \cdot \xi)$ | $G_A(4 \cdot \xi + 1)$ |
| $\xi \notin A$ | $2 \cdot \xi$      | $2 \cdot \xi + 1$      |
| $\xi \in A$    | $2 \cdot \xi + 1$  | $2 \cdot \xi$          |

Ordinal division by a positive finite left factor uniquely partitions  $\kappa$  into these finite blocks. Thus the specified positions map bijectively onto  $\text{Ord} \cap V_\kappa = \kappa$ .

The remaining positions  $\{4 \cdot \xi + 2, 4 \cdot \xi + 3 : \xi < \kappa\}$  have cardinality  $\kappa$ . So does  $V_\kappa \setminus \text{Ord}$ : for example,  $\xi \mapsto \{\xi\}$ , for  $0 < \xi < \kappa$ , gives  $\kappa$  distinct nonordinals. Fill the remaining positions by any external bijection onto this complement. This completes  $G_A$  to the required bijection.

The decoding formula is simply

$$\xi \in A \iff G_A(4 \cdot \xi) = 2 \cdot \xi + 1.$$

All ordinal operations appearing here are definable in  $M$ , and no additional parameter occurs.  $\square$

**Theorem XVII.7.11** (Unbounded variation of the horizon). *As  $G$  ranges over all external bijections  $\kappa \rightarrow V_\kappa$ ,*

$$\sup_G \mathfrak{H}_G = \kappa^+.$$

*Indeed, every  $\alpha$  with  $\kappa \leq \alpha < \kappa^+$  is the type of a parameter-free  $G$ -definable class well-order for some such  $G$ .*

*Proof.* Fix  $\alpha$  in the stated interval. Its cardinality is  $\kappa$ , so there is an external well-order  $R$  on  $\kappa$  of type  $\alpha$ . Choose the standard parameter-free ordinal pairing  $\pi : \text{Ord}^2 \rightarrow \text{Ord}$ , obtained by ordering pairs first by their maximum coordinate and then lexicographically, and assigning to each pair the type of its predecessor set. Its restriction maps  $\kappa^2$  bijectively onto  $\kappa$ . To check this last point, predecessor sets of pairs below  $\kappa$  have cardinality below  $\kappa$ , while a pair with a coordinate at least  $\kappa$  has at least  $\kappa$  predecessors. The same pairing is computed correctly in the transitive structure  $V_\kappa$ .

Encode  $R$  by

$$A = \{\pi(\xi, \eta) : \xi R \eta\} \subseteq \kappa,$$

and apply Lemma XVII.7.10. In the language with  $G_A$ , the carrier  $\text{Ord}^M$  and the relation

$$\xi R \eta \iff G_A(4 \cdot \pi(\xi, \eta)) = 2 \cdot \pi(\xi, \eta) + 1$$

are parameter-free definable. Hence  $\alpha \in \text{Spec}_{G_A}^0$ , so  $\alpha < \mathfrak{H}_{G_A}$ . Such  $\alpha$ 's are cofinal in  $\kappa^+$ , and every individual  $\mathfrak{H}_G$  is below  $\kappa^+$  by Theorem XVII.7.5.  $\square$

Thus the fixed-point and countable-cofinality restrictions on an individual horizon coexist with an unbounded range of possible horizons. A named global enumeration need not be mathematically neutral: its ordering of the universe can encode substantial additional information.

[merge] (4 October 2026.) Lemma XVII.7.10 and theorem XVII.7.11 are new to this report. Source 32's corollary XVI.8.11 proves  $\sup_A \delta_A = \kappa^+$  by keeping  $G$  fixed and adding a predicate  $A$  that codes a well-order of the required type; source 36 varies the named enumeration itself, with no further predicate and no set parameter, by coding any  $A \subseteq \kappa$  into  $G$  through adjacent swaps. This answers source 33's research question research question 113.286, which asks how much  $\delta_G$  can vary as  $G$  ranges over set-like global well-orders of one  $V_\kappa$  and what information such a  $G$  can carry:  $\delta_G = \mathfrak{H}_G$  is unbounded in  $\kappa^+$ , every subset of  $\kappa$  can be coded into  $G$ , and every  $\alpha \in [\kappa, \kappa^+)$  is the type of a parameter-free  $G$ -definable order for some  $G$ ; which values  $\mathfrak{H}_G$  itself takes stays open (research question 113.295). The  $G$  here is a bijection  $\kappa \rightarrow V_\kappa$ , which induces a set-like global well-order of type  $\text{Ord}$  as source 33 requires. The dated note at research question 113.286 records this.

### XVII.7.7 Comparison with the full-class cutoff

Proposition XVII.6.12 gives the exact full-class spectrum of proper carriers,  $[\kappa, \kappa^+)$ . Including set carriers adds precisely the types below  $\kappa$ , so the spectrum of all class presentations is  $\text{Spec}_{\text{full}} = [0, \kappa^+)$ .

The set universe has remained  $V_\kappa$  throughout. With definable classes and a fixed  $G$ , the exact cutoff is  $\mathfrak{H}_G < \kappa^+$ ; adding truth presents that old cutoff and strictly raises it; allowing all classes reaches every type below  $\kappa^+$ . None of these cutoffs is itself an ordinal of  $V_\kappa$ . They are external measurements of the lengths carried by its class presentations, which is the distinction needed to speak rigorously about how far those presentations stretch beyond  $\text{Ord}^M$ .



## XVII.8 Rigorous formalization choices

Different formalizations answer different questions. A useful foundation must specify what its classes are, which maps are available, and whether a claimed well-order is internal or externally well-founded. Increasing the number of sorts or the universe level changes what counts as a set; it does not create a contradiction with the original set–class boundary.

### XVII.8.1 First-order ZFC and formula schemas

For a fixed first-order definition, ZFC is often sufficient. A presentation consists of formulas  $D(x, p)$  and  $R(x, y, p)$ . A construction such as sum, product, or finite-support power produces new formulas effectively from these particular inputs. A proof that every set subset of the defined order has a least element is an ordinary first-order proof. Instances involving another definable subclass are obtained by substituting its defining formula.

This framework is suitable for an executable library of individual constructions. It does not contain one object that is the collection of all proper-class order types. A natural number coding a formula is a set, but its semantic interpretation over the entire universe is not uniformly first-order definable. Formula syntax, semantic truth, and proof certificates must be represented separately.

Set parameters must also be part of the specification. A definition using an arbitrary set parameter is different from a parameter-free definition, and a named class predicate is a stronger resource than either. If a predicate is added to the first-order language, Separation and Replacement in that expanded language must be justified. At the inaccessible-height models used in the horizon theorem, this justification was explicit.

### XVII.8.2 GB, NBG, and admitted class maps

A two-sorted class theory is the most economical setting for the elementary calculus. Its objects are sets and classes; members of classes are sets. A function between class carriers is represented by a class graph of set-coded pairs. These data support all the constructions in the arithmetic and epsilon sections.

The notations GB, NBG, and GBC occur with different choice conventions in the literature. Here GB includes Power Set and Class Replacement, but does not include Choice by abbreviation. GBC adds Global Choice. This prevents a notation convention from silently strengthening a theorem.

There is an additional issue with order types. Two abstract class presentations can be externally isomorphic even when the isomorphism is not an admitted class in the chosen model. Accordingly, a formal interface should store an explicit comparison map when a theorem uses one. An external equality of order types is not automatically a usable internal class isomorphism.

The elementary constructors have a convenient implementation contract: the output domain, order, and each displayed isomorphism are given by a formula relative to the supplied input classes. They do not invoke an unspecified choice of representatives of class-isomorphism types.

### XVII.8.3 What elementary transfinite recursion adds

Elementary transfinite recursion concerns a family whose stages are *classes*. For a class well-order  $\Gamma$ , a typical instance asks for a class  $F$  satisfying

$$F_\gamma = \{x : \varphi(x, \gamma, F \upharpoonright \Gamma_{<\gamma}, A)\}, \quad \gamma \in D_\Gamma,$$

where  $\varphi$  has only set quantifiers and  $A$  is a fixed class parameter. The history supplied to a stage can itself be a proper class.

This is substantially different from evaluating a finite tree. Even an  $\omega$ -length class-valued recursion can assemble truth by formula depth; bare GBC does not supply a full first-order truth predicate. The explicit finite syntax used in this article avoids that extra requirement.

Over GBC, full ETR is equivalent to determinacy for clopen class games [22]. This connection helps locate class recursion within the broader landscape of principles for class constructions.

A useful established theorem is that  $\text{GBC} + \text{ETR}_\Gamma$  compares the fixed  $\Gamma$  with every class well-order [35, Lemma 3.17]. The recursion tries to map each point of  $\Gamma$  to the least target point not used earlier. If it never runs out of target points, it gives an initial embedding of  $\Gamma$ . If it does run out, its inverse on the completed domain gives an initial embedding in the other direction. Thus full ETR implies class-well-order comparability.

The known implication chain includes

$\text{ETR} \implies \text{continuous rankings} \implies \text{initial comparability} \implies \text{weak embeddability comparability}.$

The question of converses is explicitly posed by Hamkins and Woodin [27, Question 7]. We do not identify these principles here. The more specific principle  $\text{ETR}_{\text{Ord}}$  is the exact strength of the class forcing theorem [24]; having recursion at that length should not be silently treated as recursion along every longer class well-order.

[merge] Hamkins and Woodin’s Question 7 is source 29’s [research question 113.219](#) and the second theme of [section 113.28](#); over GB, comparability CWO is equivalent to canonical histories and to source 35’s FPE (theorems XVI.5.8 and XVI.6.21 and corollary XVI.6.19). Source 36 asks it again as its tenth question ([research question 113.304](#)).

| Framework                 | Appropriate role                                                                                                                |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Fixed ZFC formulas        | Individual definable carriers, finite syntax, and formula-by-formula theorems.                                                  |
| GB                        | Elementary class arithmetic, set localization, explicit class isomorphisms, and the finite-term systems of this article.        |
| GBC                       | The same constructions with a supplied global enumeration for uniform indexing of arbitrary set codes.                          |
| GBC +ETR                  | General elementary class-valued recursion and comparison of arbitrary class well-orders.                                        |
| KM                        | Impredicative comprehension and stronger class constructions; still no ordinary class with arbitrary proper classes as members. |
| KM + Class Collection     | The stronger unrolling theorem described next.                                                                                  |
| External transitive model | Ordinary ordinal measurements of class presentations, with external well-foundedness explicitly checked.                        |
| Higher type universe      | Set-sized implementations of lower-universe class presentations and their order types.                                          |

## XVII.8.4 Unrolling and a concrete larger universe

Unrolling treats suitable well-founded extensional class relations as membership codes for new objects. This is a principled way for a proper class of one universe to become a set in another.

The strong classical result uses KM together with Class Collection. In this statement, KM has ZFC, including Choice, as its set theory, following the convention of the cited source. As stated in the modern account of Freire and Williams [96, Theorem 2.1.1], this theory is bi-interpretable with ZFC without Power Set, with Collection in the weak-set-theory axiomatization, Choice expressed as the assertion that every set can be well-ordered, and a largest cardinal that is inaccessible. The Class Collection hypothesis and the Collection convention in the first-order theory are material; simply deleting Power Set from a Replacement-only axiom list does not specify the same theory.

For the full rank model, the construction has a particularly transparent external description.

**Proposition XVII.8.1** (The full unrolling at an inaccessible height). *Let  $\kappa$  be strongly inaccessible. The externally interpreted unrolling of  $(V_\kappa, \mathcal{P}(V_\kappa))$  is  $H_{\kappa^+}$ , the sets whose transitive closures have cardinal at most  $\kappa$ . Its ordinal height is  $\kappa^+$ .*

*Proof.* A class membership code has its nodes in  $V_\kappa$ , so it has at most  $\kappa$  nodes externally. A well-founded extensional code has its ordinary external Mostowski collapse. The transitive closure of the object represented by its designated root has cardinal at most  $\kappa$ ; hence the object belongs to  $H_{\kappa^+}$ .

Conversely, if  $x \in H_{\kappa^+}$ , the membership relation on the transitive closure of  $\{x\}$  has at most  $\kappa$  nodes. Transport it to a subset of  $V_\kappa$ , designating the node corresponding to  $x$ . All the resulting relations are available in the full class collection. The external collapse recovers  $x$ . The usual identification of isomorphic codes therefore gives exactly  $H_{\kappa^+}$ . Its ordinal members are precisely the ordinals below  $\kappa^+$ .  $\square$

This explains the full-class cutoff from another direction. A class well-order of external type  $\alpha$ , with  $\kappa \leq \alpha < \kappa^+$ , becomes an ordinary ordinal in the unrolled universe. It was never an ordinal set of the original  $V_\kappa$ . Power Set fails at  $\kappa$  in the unrolled universe: each subset of  $\kappa$  is present, but the collection of all of them is too large to be one of its sets.

[merge] Proposition XVII.8.1 is a standard computation, new to this report as a stated proposition; source 34 discusses unrolling without it (section XVI.10), and source 33 asks about unrolled order codes (research question 113.276). The account of Freire and Williams [96] is cited as source 36 states it and was not checked by this write.

### XVII.8.5 Universes and proof assistants

A universe hierarchy gives another rigorous implementation. In a Lean-style notation, the type of  $u$ -small ordinals lives one universe level higher:

$$\text{Ordinal}_u : \text{Type}_{u+1}.$$

Viewed from the next level, its natural well-order has an ordinary order type

$$\Omega_u = \text{otp}(\text{Ordinal}_u, <) \quad \text{in} \quad \text{Ordinal}_{u+1}.$$

The sums, products, finite-support powers, and finite trees can then be implemented as ordinary types at the larger level. Increasing a universe level makes the previously proper collection small relative to the new level. It does not provide a single type of ordinals containing every universe's ordinals.

[merge] Here  $\Omega_u$  is an ordinal of the next universe level, the order type of  $\text{Ordinal}_u$ ; it is not the presentation  $\Omega = (\text{Ord}, \in)$  of this Part (source 36 writes the latter in bold and  $\Omega_u$  in light face). Mathlib's corresponding declarations are discussed in Section XVI.10 (source 32's section XVI.10.7).

There are two distinct verification targets.

1. **External typed orders.** Implement the finite syntax and interpret it in a larger ordinal type. Prove that the comparator agrees with that interpretation. This gives genuine well-foundedness in the ambient proof assistant.
2. **Internal class theory.** Formalize a first-order set model  $M$ , a class realization  $\mathcal{X}$ , and the evaluation relation between them. Then prove the GB theorems about admitted presentations and admitted maps. Internal well-foundedness must use that interpretation, rather than being silently replaced by the proof assistant's external well-foundedness predicate.

The first target is considerably smaller and provides a practical first milestone. The second is required for the exact foundational-strength claims and the comparison of different class realizations.

The following proposed modules make the dependencies explicit. These are design targets, not assertions that corresponding files have already been added to ProveIt.

| Proposed module                     | Mathematical responsibility                                                                                    |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>ClassOrderPresentation</code> | Carrier and relation data, set-likeness, admitted embeddings, and the set-subset well-foundedness formulation. |
| <code>FiniteSupportPower</code>     | Finite support representation, comparison, power laws, predecessor decomposition, and set-likeness criterion.  |
| <code>OmegaNormalForm</code>        | Hereditary syntax, validity, comparison, local radix semantics, tower cuts, addition, and multiplication.      |
| <code>EpsilonTerms</code>           | Epsilon atoms, mixed comparison, exact set semantics, and finite-support transfer to classes.                  |
| <code>DefinabilityHorizon</code>    | Transitive-model spectra, partial satisfaction, bounded-complexity diagonalization, and truth promotion.       |

ProveIt's existing ListCoding documentation reports a common hereditary Cantor-normal-form coding below  $\varepsilon_0$ , with Lean denotation theorems and separately scoped Rocq syntactic results [94]. That is a natural source of engineering patterns. Its current theorem statements are not automatically theorems about proper classes. Mathlib's general-base Cantor normal form is a directly relevant set-level component [99].

[merge] (4 October 2026.) None of the five proposed modules exists in the repository, and no statement of Part XVII is formalized (section 1.5). The ListCoding documentation is `Logi c/PeanoArithmetic/ListCoding/README.md`; it describes the hereditary codes below  $\varepsilon_0$ , the Lean denotation theorems and the separately scoped Rocq results as source 36 says. Mathlib's `Mathlib/SetTheory/Ordinal/CantorNormalForm.lean` is in the Mathlib version this repository pins. The formalization plans of Part XVI's sources (sections XVI.10 and XVI.10.9) propose comparable first milestones.

### XVII.8.6 A contract for systematic notations

A reusable class-order notation should state five items.

1. Its input signature: ordinal labels, set parameters, named class predicates, or supplied class orders.
2. The set codes representing its points, together with a normality test.
3. The comparison procedure and the hypotheses needed for well-foundedness.
4. Its semantic coverage: a precise initial segment or a family of presentations, not an unspecified claim to represent every long ordinal.
5. Its map discipline: which isomorphisms, embeddings, or normalization functions are explicitly definable from the inputs.

Our  $A^{[B]}$ ,  $E$ , and  $\text{Ep}(B)$  satisfy this contract. The polynomial display for a point of  $A^{[B]}$  denotes its predecessor presentation. The whole notation class is a separate object. A proposed Veblen or collapsing-function extension should follow the same contract before its printed expressions are used as if they were established class ordinal arithmetic.

## XVII.9 Further research questions and topics

[merge] Source 36's introduction to its questions and its ten research questions (its lines 2519–2635) are printed with the other sources' questions in [section 113.30](#), each with its status after this write; the questions of this report that it answers or advances are in [section 113.31](#). Its closing subsection follows.

### XVII.9.1 A concrete next formalization project

A manageable first project is to formalize the hereditary term comparator, its local radix theorem, and root decomposition. The next stage adds the epsilon grammar and support transfer. These results would certify a long, explicit class-notation family without first formalizing all of KM or settling general class comparison.

The horizon theorem is a separate model-theoretic project. Its first formal milestone should be the external theorem for  $V_\kappa$  with a named  $G$ : downward closure of the spectrum, the cardinal bound, the sound partial-truth filter, and the diagonal sequence. This keeps the genuine semantic difficulty visible instead of burying it in a notation parser.

## XVII.10 Proof status, source audit, and reproducibility

### XVII.10.1 What was checked mathematically

The main statements were developed with separate reviews of class arithmetic, finite syntax, external spectra, and the bounded-complexity argument. A further independent review checked the inaccessible-height horizon theorem, truth promotion, and the epsilon and zeta extensions. The final article distinguishes assertions about admitted class maps from equalities of external ordinal order types.

Several hypotheses are especially important to the proofs:

- The choice-free localization argument is a GB argument with Power Set. It uses rank sets and Class Replacement.
- The ordinal labels in hereditary terms are leaves of a finite syntax tree. Recursion is not performed on their possibly infinite transitive closures.
- An arbitrary transitive model need not recognize external ill-foundedness of its class relations. Every  $\mathfrak{H}(M)$  spectrum in the article uses external well-foundedness.
- The bounded-complexity validity filter is externally sound at  $V_\kappa$  because  $\kappa$  is inaccessible. This hypothesis is not silently dropped.
- Parameter-free cofinality in the  $G$ -theorem means no *set* parameters in the language already naming  $G$ . It does not mean that every represented type has a parameter-free presentation.
- The diagonal formulas form an effective syntactic schema. Their interpretations are not supplied by one formula before truth is added.
- General class powers are constructed, but a complete power-normalization algorithm on hereditary point-cuts is not claimed.

These checks are conventional mathematical review, not external refereeing or proof-assistant certification. The code checks described next have a strictly finite scope.

### XVII.10.2 The executable notation demonstrator

The file `code/notation_demo.py` implements the natural-coefficient fragment of the hereditary  $\Omega$ -term syntax. Terms are immutable tuples of exponent–coefficient pairs. The empty tuple represents zero. The program validates normal forms, compares them, prints them, and builds the tower boundary terms. It does not silently sort malformed input.

Only positive natural numbers are implemented as coefficients. Infinite set-ordinal coefficients, general class-order presentations, and arbitrary class arithmetic are outside this program’s scope. A complete finite alphabet naming every set ordinal would be a different and impossible interpretation of the mathematical grammar.

The verification report records the following run:

| Check                                                    | Count or value |
|----------------------------------------------------------|----------------|
| Distinct bounded normal terms                            | 994            |
| Unordered pairs checked against an independent evaluator | 493,521        |
| Additional ordered pair comparisons                      | 2,000          |
| Reflexive comparisons                                    | 994            |
| Malformed inputs rejected                                | 16             |
| Tower boundaries checked structurally                    | 8              |
| Common finite base in the independent evaluator          | 7              |
| Largest integer produced by that evaluator               | 92 bits        |

For the independent check, the program substitutes one ordinary natural base 7, greater than every coefficient in the bounded sample, and evaluates each finite tree by exact integer exponentiation and addition. The evaluator calls neither the validator nor the structural comparator. All 994 values are distinct and their ordering agrees with the structural comparison. A size bound prevents the test from trying to evaluate enormous towers.

Substitution of a finite base is a comparison test, not the interpretation of  $\Omega$  as Ord. In particular, the finite tests do not establish class well-foundedness, which is proved separately in the article.

To run the examples and reproduce the recorded check:

```
python3 code/notation_demo.py --examples-only
python3 code/notation_demo.py --report code/verification.json
```

[merge] (4 October 2026.) The program is shipped as `code/36-horizons-notation_demo.py` and its report as `data/36-horizons-verification.json`; the commands above use the delivery names, and the second overwrites the recorded report in place, so it must be run on a copy ([appendix A.14](#)). On Windows the program needs `PYTHONUTF8=1` (or `py -X utf8`): with the default code page it stops with `UnicodeEncodeError` on the letter  $\Omega$ , and the report run then leaves a 0-byte file. Rerun on a copy with `PYTHONUTF8=1` on 5 October 2026 (Python 3.14.4): status passed, with the counts of the table above; the regenerated report equals the delivered one except for `python_version` and line endings. The program implements natural coefficients only; it is not a formalization, and the theorems of this Part do not depend on it.

### XVII.10.3 The ProveIt snapshot

The targeted repository audit used the main-branch commit

e1d2f3048b947801873a6e9ddc4e03736234b328,



whose recorded commit time is 4 October 2026, 23:34:43 UTC. The code-search index initially reported an older snapshot; selected documents were therefore fetched at the pinned commit.

The inspected materials were:

1. The repository README, establishing the organization and the distinction between formal developments and research reports.
2. The catalogue at `Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/README.md`. Its relevant entries describe the canonical ordinal block structure, finite normal forms from condensation, and the comparison-specific scope of certain conclusions.
3. The accompanying `29-birthdays-PROOF_STATUS.txt`, which explicitly separates the condensation/comparability result from an unproved converse to ETR.
4. `Logic/PeanoArithmetic/ListCoding/README.md`, particularly the documented hereditary ordinal codes below  $\varepsilon_0$  and the separate Lean and Rocq theorem scopes.

The approximately 3.3-MB consolidated surreal-well-orders LaTeX manuscript was not inspected end to end. The targeted audit used its detailed catalogue and proof-status documents; it is not an exhaustive duplicate search or an independent verification of that corpus. This limits claims about novelty relative to every repository passage. It does not affect the proofs in the present article, which are supplied independently.

No repository files were modified, and no Lean or Rocq build of ProveIt was performed. The package is an article and a finite notation demonstrator, with a proposed formalization route.

[merge] The commit `e1d2f3048` (4 October 2026, 16:34:43 PDT) is the pin of source 34; this report’s `article.tex` there is the text with Parts I–XV, unchanged until Part XVI was written. The older index commit `555ee36fb` also exists. The four inspected paths exist at the current revision, and their contents are as described; “the approximately 3.3-MB” manuscript is this report’s `article.tex`, 3,289,057 bytes at the pin. The source audit is shipped as `36-horizons-SOURCE_AUDIT.txt` ([appendix A.14](#)).

## XVII.10.4 How to rebuild the document

The main `beyond_ord.tex` in the final package is self-contained: the section source and bibliography are included in that one file. No external bibliography database or downloaded image is required. A standard TeX Live installation with the listed packages suffices.

To rebuild, run:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error beyond_ord.tex
```

The accompanying `build.sh` performs the same build. The delivered PDF was compiled from this source and checked for unresolved references and layout defects. The separate `DOCUMENT_CHECKS.json` records the final page count, compilation diagnostics, and rendering review.

[merge] The manuscript `beyond_ord.tex`, its 40-page PDF and the delivery README are not shipped; they survive in the arrival commit’s archive ([appendix A.14](#)). `build.sh` is shipped as `code/36-horizons-build.sh` and builds a file that is not shipped; `DOCUMENT_CHECKS.json` is shipped as `data/36-horizons-DOCUMENT_CHECKS.json`, and its four SHA-256 values match the delivered files. This Part is built as part of this report.

### XVII.10.5 Research status in one table

| Item                                             | Status                                                                                           |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Long class orders and basic arithmetic           | Established background, with explicit constructions and proofs supplied.                         |
| Hereditary $\Omega$ -terms and epsilon extension | Fully specified and proved here; epsilon syntax has stated classical prior art.                  |
| External spectra and the $G$ -horizon theorem    | Conventional proofs included; hypotheses and parameter regimes explicit.                         |
| Truth promotion and unbounded variation with $G$ | Proved by the displayed uniformization and coding constructions.                                 |
| Historical priority                              | Not established by the targeted source search.                                                   |
| General comparability versus ETR                 | Not resolved; cited literature and search limitation stated.                                     |
| Proof-assistant verification                     | A roadmap is given; no such certification is claimed for this article.                           |
| Finite software checks                           | Reproducible comparator checks only; no inference from finite testing to class well-foundedness. |

[merge] (4 October 2026.) The rows “External spectra and the  $G$ -horizon theorem” and “Truth promotion” concern theorems proved independently on the same day by sources 32–34 (notes XVII.7.D and XVII.7.G); “Historical priority: not established” applies to them in that sense too.

### XVII.11 What this Part answers

[merge] Dated notes, 4 October 2026, in the form of section XVI.12; the questions concerned carry dated notes of their own (sections 113.28 and 113.31).

- (a) *Source 33’s research question research question 113.286, how much the definability ceiling depends on the global-order predicate:* **answered**. The ceilings  $\delta_G = \mathfrak{H}_G$  are unbounded in  $\kappa^+$  as  $G$  varies over the bijections  $\kappa \rightarrow V_\kappa$ , every  $A \subseteq \kappa$  can be coded into some  $G$ , and every  $\alpha \in [\kappa, \kappa^+)$  is the type of a parameter-free  $G$ -definable order for some  $G$  (lemma XVII.7.10 and theorem XVII.7.11).
- (b) *Source 34’s agenda item research question 113.275, arithmetic of the hereditary notation:* **answered for addition and multiplication** (theorems XVII.5.1 and XVII.5.2); exponentiation open (research question 113.300).
- (c) *Source 32’s research question 113.284 and source 33’s research question 113.285, the definability ceiling:* advanced by the necessary conditions  $\varepsilon_\delta = \delta$  and  $\delta > \zeta_{\kappa+1}$ , for every transitive model (proposition XVII.6.11) and at  $V_\kappa$  with a named enumeration (theorem XVII.7.5).
- (d) *Source 34’s research question 113.272 and source 35’s research question 113.273, Veblen-style hierarchies of class-order constructors:* advanced by the finite-support transfer theorem and the epsilon-term functor (theorems XVII.6.2 and XVII.6.4).
- (e) *Source 34’s research question 113.283, comparison of the ceiling with admissible and unrolling heights:* advanced by (S36.6.2) and proposition XVII.8.1.
- (f) *Source 31’s research question 113.239:* advanced for external order types at every transitive height (section 113.31 (f)).
- (g) *Not settled:* Hamkins and Woodin’s Question 7, and source 36’s ten questions (section 113.30); its second question is answered in part by source 34’s theorem XVI.8.3.

[merge] *New against the repository* (checked against Parts I–XVI; source 36 claims no historical priority): normal-form addition and multiplication with the absorption  $D + \Omega^{[B]} \cong \Omega^{[B]}$  (theorems XVII.5.1 and XVII.5.2); the predecessor decomposition for an arbitrary base, the set-likeness criterion and  $a^{\Omega \cdot B} \cong \Omega^{[B]}$  (theorems XVII.3.5 and XVII.3.6 and corollary XVII.3.7); the second clause of proposition XVII.3.4; the carrier Ord without choice (proposition XVII.2.4); the finite-support transfer theorem (theorem XVII.6.2); the epsilon-term functor on class orders and its calibrations,  $\varepsilon_\kappa = \zeta_\kappa = \kappa$  at transitive heights, and the order of type  $\zeta_{\kappa+1}$  (theorems XVII.6.4, XVII.6.6 and XVII.6.7 and lemma XVII.6.5); the proper-carrier spectrum, the transfer to  $\text{Ord}^M$  and  $\varepsilon_{\mathfrak{H}(M)} = \mathfrak{H}(M)$  with  $\zeta_{\kappa+1} < \mathfrak{H}_0(M)$  (theorem XVII.6.9 and proposition XVII.6.11); the clause  $\varepsilon_{\mathfrak{H}_G} = \mathfrak{H}_G$  of theorem XVII.7.5; the effective hierarchy corollary XVII.7.7; the swap coding and the variation of  $G$  (lemma XVII.7.10 and theorem XVII.7.11); and the unrolling computation (proposition XVII.8.1). *Proved again* (credit Notes): set localization (source 34), the set-like boundary (sources 11 and 32–35), the power theorem and its laws (sources 29 and 32–35), the hereditary order, its fixed point, towers and leastness (sources 32–35), the calibration at cardinals (source 34), the parameter spectrum (source 34), the full-class spectrum (sources 12 and 32–35), the successor obstruction and uniform diagonal (sources 32–35), the inaccessible-rank model (sources 32–34), the  $V_\kappa$  horizon theorem (sources 32–34), the bounded-complexity diagonal (sources 33–35), the missing uniform evaluator (sources 32–34) and truth promotion (source 32). *Inherited and credited by source 36*: class arithmetic and long class orders (Barton and Williams), the epsilon-term syntax (Marcone and Montalbán), the admissible bound (Blass) and its equality examples (Taranovsky), the unrolling theorem for KM with Class Collection (Freire and Williams), and the comparison theorem GBC + ETR<sub>F</sub> (Williams).

[merge] *Review*. The Hilbert’s-tenth research tree reviewed the archive in a bounded intake (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review\_beyond\_ord\_package\_d7cf7d554.md, commit 39bb02ece): it read 1,879 lines of the manuscript; checked the set-localization lemma, the ordinal initial segment and carrier reduction, the power calculus with its transport, continuity, predecessor and set-likeness results, the hereditary comparison, root decomposition, towers and the least fixed point through a supplied isomorphism, the normal-form addition and multiplication, and the transfer theorem; read the uniform diagonal and the setting, validity filter and complexity hierarchy of the horizon theorem; and found no defect. It records that the arithmetic procedures keep ordinal-label primitives and that the demonstrator’s interface has no addition or multiplication. It did not read the proofs of the epsilon interpretation, the epsilon and zeta calibrations, the general spectrum theorem, truth promotion, the variation of  $G$ , the admissible bound or the unrolling proposition, and it ran no delivered program. The placement check (6571ee1af) read the main proofs, recomputed the three worked products and found no false statement. A later bounded publication review at a21a42d8c, recorded in review\_horizons\_write\_a21a42d8c.md in the same research tree, reads the whole 557-line guide diff and 1,698 selected article lines. It authenticates the source archive, all five ancillary placements and all 70 source-label routes. The selected arithmetic proofs, the initial-embedding leastness of note XVII.4.B, and the fixed-standard-complexity syntax hierarchy pass within their hypotheses. Three editorial arithmetic summaries are corrected below. This review does not certify the epsilon interpretation, the epsilon/zeta calibrations, general transitive-model spectrum, truth promotion, variation of  $G$ , admissible bound or unrolling proof; downstream uses retain those dependencies. No supplied program was run and no paid fixed-arity integer compiler follows from the ordinal-relative arithmetic.

**Review remark H1 (multiplication scope).** The guide originally extended “addition and multiplication” to every  $\Omega^{[B]}$ ; the notation table said “points of  $\mathbf{E}$  or  $\Omega^{[B]}$ ”, and this Part’s introduction made the same extension. For  $B = 2$ , the point-cut  $\Omega$  belongs to  $\Omega^{[2]}$ , but its product with itself is the whole order  $\Omega^2 = \Omega^{[2]}$ , which cannot be a proper initial point-cut of itself. Only the addition rule is asserted on every such power; both operations are defined on  $\mathbf{E}$ . The printed theorems retain that scope.

**Review remark H2 (set-likeness hypotheses).** The guide omitted “ $A$  has at least two elements and  $B$  is nonempty” from [theorem XVII.3.6](#). For  $A = 1$ ,  $B = \text{Ord} + 1$ , the power is the set-like singleton, but neither stated alternative holds. Independently,  $A = \text{Ord}$ ,  $B = 0$  again gives the set-like singleton without either alternative. Both restrictions are now explicit in the guide; the theorem is unchanged.

**Review remark H3 (set-base collapse).** The guide stated  $a^{[\Omega \cdot B]} \cong \Omega^{[B]}$  without the assumption that  $a$  is a set ordinal at least two. Taking  $a = 1$  and  $B = 1$  would assert  $1 \cong \text{Ord}$ . The guide now repeats the premise of [corollary XVII.3.7](#); its proof is unchanged.

# Questions, conclusions and status

## 113 Research questions

[merge] The four sources' questions are printed per source in the following subsections, as proposed continuations, not as a census of open problems. The index below groups the duplicates, credits every source that asks them, and records the status after the merge (dated 2 October 2026). Questions of source 09 and source 08 are numbered by the order of their subsections and paragraphs in [sections 113.3](#) and [113.4](#). [merge] (3 October 2026.) The status column below carries dated updates for the batch-81 addition; the questions of sources 13–18 have their own index in [section 113.5](#). [merge] (3 October 2026, batch 83.) Entries marked *batch 83*: give the status after Parts X–XIV, with the item of [section 113.25](#) that records it; the questions of sources 19–30 are indexed in [section 113.12](#).

| Topic                                                   | Asked by                                                                                                        | Status                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Weak-choice strength of a well-order of <b>No</b>       | 12: <a href="#">remark 113.15</a>                                                                               | answered by <a href="#">theorem 13.1</a>                                                                                                                                                                                                                                                                                                                                                                                                        |
| Comparison of non-set-like class well-orders            | 08: question 11                                                                                                 | answered by <a href="#">theorems 14.2</a> and <a href="#">14.7</a>                                                                                                                                                                                                                                                                                                                                                                              |
| Coding lengths at singular cutoffs                      | 08: question 3                                                                                                  | answered in part by <a href="#">theorem 8.3</a> ; <i>3 Oct.</i> : cut half answered ( <a href="#">section 33</a> )                                                                                                                                                                                                                                                                                                                              |
| Invariants, cut spectra and isomorphism type at cutoffs | 11: <a href="#">research question 113.7</a> ; 12: <a href="#">research question 113.13</a> ; 08: questions 1, 2 | open; regular case partly by <a href="#">proposition 12.1</a> ; <i>3 Oct.</i> : <a href="#">research question 113.7</a> and question 2 answered, <a href="#">research question 113.13</a> except the isomorphism type, question 1 open ( <a href="#">section 33</a> ); <i>batch 83</i> : homogeneity, topology and completion at $\mu = \omega$ , complete metrizability only at $\mu = \omega$ , weights ( <a href="#">section 113.25</a> (c)) |
| Fixed noninitial enumeration types                      | 12: <a href="#">research question 113.11</a> ; 08: question 4                                                   | open (reals report: <b>lwo:sec:research</b> ); <i>3 Oct.</i> : answered for $\mu \cdot \gamma + n$ ( <a href="#">section 33</a> ); <i>batch 83</i> : cube spectrum of every type, coding length except at singular non-strong-limit residual cardinals ( <a href="#">section 113.25</a> (a))                                                                                                                                                    |

| Topic                                                      | Asked by                                                                                                                                                    | Status                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Noncardinal cutoffs; away from inaccessible truncations    | 12: <a href="#">research question 113.12</a> ; 09: direction 7                                                                                              | open; <i>3 Oct.</i> : direction 7 answered for the minimal slice ( <a href="#">section 33</a> ); <i>batch 83</i> : in part ( <a href="#">section 113.25</a> (b))                                                                                                                       |
| Represented cuts, exhaustive branches and higher-size cuts | 11: <a href="#">research question 113.1</a> ; 09: direction 3; 08: question 8                                                                               | open; <i>3 Oct.</i> : <b>answered</b> ; question 8 for cuts of the core ( <a href="#">section 50</a> )                                                                                                                                                                                 |
| Full virtual lexicographic products                        | 08: question 7                                                                                                                                              | open; <i>3 Oct.</i> : <b>answered</b> ( <a href="#">section 60</a> )                                                                                                                                                                                                                   |
| Weakest comprehension and minimal axioms                   | 11: <a href="#">research question 113.2</a> ; 12: <a href="#">research question 113.16</a> ; 09: direction 2; 08: question 10                               | open; <i>3 Oct.</i> : $\Sigma_1^1$ sufficiency for <a href="#">research question 113.2</a> ( <a href="#">section 60</a> )                                                                                                                                                              |
| Iterated residual decomposition; proper-class blocks       | 11: <a href="#">research question 113.3</a> ; 12: <a href="#">research question 113.14</a>                                                                  | open; <i>3 Oct.</i> : block-index orders answered, iteration strength open ( <a href="#">section 60</a> ); <i>batch 83</i> : canonical histories $\Leftrightarrow$ CWO over GBC ( <a href="#">section 113.25</a> (g)); <i>batch 96</i> : over GB ( <a href="#">section 113.29</a> (b)) |
| Automorphisms; recovering prefix structure; baselines      | 11: <a href="#">research questions 113.4</a> and <a href="#">113.5</a> ; 12: <a href="#">research question 113.17</a> ; 09: direction 4; 08: questions 5, 6 | open; <i>3 Oct.</i> : in part, mostly negative ( <a href="#">sections 50</a> and <a href="#">60</a> ); <i>batch 83</i> : label-edge equality suffices, unlabelled cylinders do not; <a href="#">research question 113.17</a> in part ( <a href="#">section 113.25</a> (d), (h))        |
| Birthday and support bounds                                | 11: <a href="#">research question 113.6</a> ; 09: direction 6; 08: question 13                                                                              | open; <i>3 Oct.</i> : support in part ( <a href="#">section 50</a> ); birthday bounds open; <i>batch 83</i> : least ordinal support <b>answered</b> ( <a href="#">section 113.25</a> (i))                                                                                              |
| Dependence on the class realization                        | 11: <a href="#">research question 113.8</a> ; 12: <a href="#">research question 113.18</a> ; 08: question 12                                                | open; <i>3 Oct.</i> : in part ( <a href="#">sections 50</a> and <a href="#">60</a> ); <i>batch 83</i> : labelled enumerations recover the class part; the pure order of countable models classified ( <a href="#">section 113.25</a> (e))                                              |
| A first verified milestone                                 | 11: <a href="#">research question 113.9</a> ; 09: direction 1; 08: question 13                                                                              | proposals                                                                                                                                                                                                                                                                              |
| Uniform constructions indexed by codes (Kanovei–Shelah)    | 11: <a href="#">research question 113.10</a> ; 09: direction 9; 08: question 15                                                                             | open                                                                                                                                                                                                                                                                                   |



| Topic                                  | Asked by                                     | Status                                                                                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relation-code comparisons              | 09: direction 5; 08: question 9              | open; equimorphism type by <a href="#">proposition 17.1</a> ; <i>3 Oct.</i> : endpoint patterns ( <a href="#">section 39</a> ); <i>batch 83</i> : scan dependence at set level; dense core <b>answered</b> ( <a href="#">section 113.25 (k)</a> )                   |
| Termination conventions for set words  | 09: direction 8; 08: question 14             | open; <i>3 Oct.</i> : direction 8 <b>answered</b> , question 14 except one type ( <a href="#">section 39</a> ); <i>batch 83</i> : the centered type is $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ ( <a href="#">section 113.25 (j)</a> ) |
| Compatibility between birthday cutoffs | 12: <a href="#">research question 113.19</a> | open; <i>batch 83</i> : in part ( <a href="#">section 113.25 (f)</a> )                                                                                                                                                                                              |

### 113.1 Source 11's research questions and topics

The questions below are proposed extensions of the present development. Except where a source is explicitly named, they are not asserted to be established open problems in the literature. Several ask for a stronger classification than the proved results supply.

#### 113.1.1 A precise description of the represented cuts

Fix the dense ordinary class  $\mathcal{P}_{\text{bd}} \cong \mathbf{No}$ . Every set-like global enumeration determines a lower cut of  $\mathcal{P}_{\text{bd}}$ , and so does every injective nonexhaustive Ord-word. The omitted-range example shows that the first collection does not fill every cut generated by the second.

**Research question 113.1.** Characterize, using the prefix tree and its numerical labels, exactly which class cuts of  $\mathcal{P}_{\text{bd}}$  arise from injective Ord-words, and which of these words are exhaustive.

A useful first target is a necessary-and-sufficient compatibility condition on the intersections of a cut with prefix cylinders. The challenge is to separate missing labels, impossible repeated labels, and cuts that escape every coherent branch. This should first be stated for a fixed uniform class cut, before quantifying over an entire hyperclass of cuts.

#### 113.1.2 The embedding relationship between the two global orders

The set-like fragment and binary class words are equimorphic, while the unrestricted order has additional convex fibers. Theorem [20.1](#) rules out a reverse embedding in the full universe interpretation. Theorem [20.2](#) also rules out a definable uniform reverse embedding in  $\mathbf{KM} + \mathbf{GC}$ .

**Research question 113.2.** What is the weakest comprehension scheme over GBC that proves the uniform nonembedding result? Can the inverse-evaluation step be replaced by an elementary argument, or is a genuine strengthening necessary?

A precise project should specify the syntactic complexity of the proposed transformation and of the comprehension used to invert its code assignment. In the bounded setting one can instead

compare the density and cellularity of all well-orders of  $\mathbf{No}_{<\kappa}$  with their minimum-length stratum, especially when  $\kappa$  is not inaccessible. The full-universe case is already settled above; the remaining questions concern strength and wider hypotheses.

### 113.1.3 Iterated residual decomposition

One extraction of the canonical initial Ord-part requires only elementary class comprehension. A residual proper class has another such part, suggesting a hierarchy of successive blocks.

**Research question 113.3.** What is the exact recursion strength needed to organize iterated residual decompositions uniformly, and what class order types can occur as their block-index orders?

An initial project could analyze finitely many blocks and specified set-length constructions, recording where the recursion state becomes a class. The general case should be compared with known ETR results on class well-order comparability, rather than assumed to follow from the one-step theorem.

### 113.1.4 Recovering the prefix structure from the bare order

The definitions exhibit a labeled tree of prefixes. An automorphism of the bare lexicographic order need not obviously preserve its displayed labels or cylinders.

**Research question 113.4.** Which prefix cones, canonical-core fibers, and finite permutation blocks are definable from the linear order alone? What extra predicates make the reconstruction unique?

Adjacency directly detects finite stretches, but a singleton fiber can be exhaustive or can omit one label. Thus local block size alone cannot recover set-likeness. An analysis of the automorphism group may clarify what the order remembers about the original enumeration.

### 113.1.5 The bounded-support choice of baseline

Every baseline enumeration gives an ordinary class  $\mathcal{P}_{\text{bd}}$  abstractly isomorphic to  $\mathbf{No}$ . If two baselines differ on a proper class of positions, their bounded-support subfamilies can be disjoint.

**Research question 113.5.** How do the two dense classes sit inside the same  $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ , and when does an automorphism of that higher-order order carry one onto the other?

The abstract isomorphism of the two dense orders does not automatically extend to all global enumerations. The represented-cut problem is likely relevant. A restricted first case is a baseline change obtained by an order automorphism of the numerical surreal line.

### 113.1.6 Sharper birthday bounds and minimal codes

The explicit code gives a usable bound

$$\text{bd}(E(f)) = \sum_{i < \text{dom } f} (1 + 2 \cdot \text{bd}(f(i)) + 2).$$

It is designed for a transparent proof, not asserted to be optimal.

**Research question 113.6.** Which birthday bounds are optimal for uniform embeddings of injective surreal words? Can shorter prefix-free coefficient codes preserve useful simplicity relations?

Regular cutoffs permit same-stage compression; singular cutoffs show why a cardinal bound alone is insufficient. A sharpened result should specify whether it preserves only numerical order, prefix extension, simplicity, or all of these.

### 113.1.7 Saturation of more general bounded alphabets

If  $X$  is an  $\eta_\kappa$  order of cardinality  $\kappa$ , its minimal-length permutation order inherits  $\eta_\kappa$ . Birthday fragments often have cardinality  $2^{<\kappa}$ , which may exceed  $\kappa$ .

**Research question 113.7.** Determine the exact cut spectrum and point characters of  $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa})$ , where  $\lambda = 2^{<\kappa}$ , without assuming  $2^{<\kappa} = \kappa$ .

The issue combines the size of a used prefix with the number and birthdays of possible numerical separators. The real-alphabet formulas in the precursor report cannot settle this case by substitution.

### 113.1.8 Class extensions and available global orders

Two models may have the same sets and hence the same individual surreal numbers, but different available classes and different global well-orders.

**Research question 113.8.** Which features of the lexicographic theory are invariant under extensions of the class part, and how does the represented-cut collection change?

Uniform formulas such as first divergence remain the same for the existing relations. Existence of additional enumerations changes the larger totality. This offers a way to distinguish properties of the labeled surreal order from properties of the surrounding class theory.

### 113.1.9 A verified first milestone for ProveIt

**Research question 113.9.** Can the explicit word compression and the direct predecessor comparison be formalized first, with a small, reusable interface independent of surreal arithmetic?

These are concrete, finite-dependency targets: ordinal concatenation, numerical first difference, injectivity, and well-order predecessor sets. The next milestone is the general successor-cardinal obstruction, followed by bounded-support completion and the small-family interpolation theorem. This ordering keeps the central algorithms separate from a formalization of an entire higher-order set theory.

### 113.1.10 Uniform mathematical constructions indexed by well-orders

The Kanovei–Shelah method suggests using a definable linear order of codes to index constructions without selecting one privileged code. The surreal setting raises additional smallness conditions.

**Research question 113.10.** Which finite-support or bounded-support constructions can be indexed by a uniform class family of global well-orders, and which require a genuine higher-order index object?

A first investigation should specify its support sets, evaluation maps, and quantifier domains before adding algebra or model theory. The diagonal theorem prevents an exhaustive class-code shortcut, but leaves many useful uniform families available.

[merge] Status of source 11’s questions after the merge: all remain open. Source 12’s dichotomy (theorem 8.3) computes the binary coding length of  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  in every case, but not the cut spectrum or the point characters that research question 113.7 asks for.

[merge] Dated note, 3 October 2026 (batch 81). Research questions 113.1 and 113.7 are answered (sections 33 and 50); research questions 113.2 to 113.5 and 113.8 are answered in part (sections 50 and 60); research questions 113.6, 113.9 and 113.10 stay open, the explicit codes and support bounds of sources 13–17 bearing on research question 113.6 (sections 39 and 50).

### 113.2 Source 12’s research questions and prioritized program

The following are proposed continuations. Except where an older question is explicitly attributed, they are not claimed to be recognized open problems in the literature.

**Research question 113.11** (All fixed enumeration types). Determine  $\ell_2(\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}))$  for every ordinal  $\alpha$  of cardinality  $2^{<\kappa}$ , not only the minimal type. Can the answer be expressed by a precise ordinal-sum invariant of admissible entry-code lengths, together with the finite terminal permutation factor?

The singular theorem identifies the first obstruction that any such formula must capture. A useful initial target is  $\alpha = \mu \cdot 2$  and then  $\mu \cdot \gamma + n$  for finite  $n$ .

**Research question 113.12** (Noncardinal birthday bounds). For an arbitrary infinite ordinal  $\theta$ , classify the minimal permutation order of  $\mathbf{No}_{<\theta}$ . Which features of the ordinal tail of  $\theta$ , beyond  $|\theta|$  and  $|\mathbf{No}_{<\theta}|$ , survive in its coding rank and local character?

The cone argument for an initial cardinal relies on closure under  $i + 1 + \alpha$  below the cutoff. At a noncardinal cutoff that closure can fail and should be replaced by a stratified family of cones.

**Research question 113.13** (Isomorphism, not merely equimorphism). Classify the order topology and isomorphism invariants of  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ . At regular cutoffs, and at singular strong limits, what are its density, cellularity, cut-character spectra, and degrees of homogeneity? Which distinguish it from other orders of the same binary rank?

The binary cube itself cannot be the literal answer: it has endpoints and adjacent pairs, while the surreal permutation order has different local behavior. A dense suborder equimorphic with a cube can retain substantial information not measured by rank.

**Research question 113.14** (The hierarchy of proper-class blocks). Study the subpredicates of  $\mathfrak{W}_{\text{all}}(\mathbf{No})$  with types  $\text{Ord} \cdot \gamma + \alpha$  for fixed set ordinals  $\gamma, \alpha$ . Give explicit, size-safe embedding and nonembedding theorems between these subpredicates and the set-like order.

The inaccessible-model calculation provides concrete bounded targets and shows that the full order is strictly more complex than the set-like order there. Extending binary fusion to uniform transformations on class predicates may require stronger comprehension; that dependence should be proved, not assumed away.

*Remark 113.15* (Source 12’s question “The exact weak-choice strength”, answered). *Source 12’s text*: Remove set-level AC from the base theory in [theorem 13.2](#). What principle is equivalent to a class well-order of the sign-sequence surreals then? Compare it with coherent well-ordering of power sets of ordinals and with uniform ordinal presentations of transitive closures.

[merge] *Answered* (2 October 2026). Source 11’s [theorem 13.1](#) proves the equivalence over GB with no set-level choice: a class well-order of the sign-sequence surreals is equivalent to global choice. Its proof is precisely a uniform well-ordering of the power sets  $V_{\beta+1}$ , obtained by set-valued recursion from the given well-order on the sign sequences of each length  $\theta$ ; see [remark 13.4](#). (Corrected on 3 October 2026 after the Hilbert’s-tenth review [review\\_surreal\\_synthesis\\_0be9b9134.md](#): “uniform” replaces “coherent”, since these orders need not extend one another; [remark 13.4](#) gives a finite example.)

This is separate from the question whether surreal universality for class linear orders itself implies global choice, raised by Hamkins [21]. The present equivalence concerns a well-order of all surreal labels and does not settle that universality question.

**Research question 113.16** (Elementary common-prefix theory). Formalize [theorem 14.7](#) over a precisely chosen weak class theory. How little comprehension and class well-foundedness are needed? Can one characterize the same relation directly by a formula avoiding explicit mention of the common-prefix class?

The definition already expands to elementary quantifiers. A mechanized proof would make the separation from abstract class comparability particularly transparent and yield a reusable theorem for other carriers.

**Research question 113.17** (Automorphisms and finite condensation). Determine natural automorphism groups of the full lexicographic order, and of the order after collapsing finite residual blocks. Which operations arise from numerical automorphisms or permutations of the underlying surreal carrier, and which genuinely act on enumeration structure?

Finite-tail blocks are now characterized intrinsically by adjacency. That makes condensation a plausible first invariant, but no complete classification of its quotient is claimed here.

**Research question 113.18** (Dependence on the class realization). Fix a transitive set universe  $M$  and vary its admissible class collection  $\mathcal{X}$ . Which external order invariants of the admitted global well-orders detect this change? Can the order reconstruct useful information about available truth predicates or class choice functions?

The diagonal obstruction persists internally, while external cardinality can vary. This makes the distinction between full and Henkin semantics a source of mathematical questions rather than merely a formal warning.

**Research question 113.19** (Compatibility between birthday cutoffs). Construct and analyze coherent embeddings from the well-order spaces at smaller birthday cutoffs into those at larger cutoffs. Which choices of reserve labels and tail enumerations are functorial, and which preserve finite condensation, coding ranks, or canonical cylinders?

Naively extending every enumeration by an arbitrary tail can change minimal length, and fixed choices need not be compatible across three cutoffs. A coherence theorem would make universe-lifting arguments much more useful in a proof assistant.

[merge] Status of source 12's questions after the merge: [remark 113.15](#) is answered (it is printed above as a remark). [Research question 113.11](#) continues the reals report's question "Other ground orders" for the surreal cutoffs, whose minimal type [theorem 8.3](#) settles. The others remain open.

[merge] Dated note, 3 October 2026 (batch 81). [Research question 113.11](#) is answered for the types  $\mu \cdot \gamma + n$ , and [research question 113.13](#) except for the isomorphism type and homogeneity ([section 33](#)); [research questions 113.14](#), [113.17](#) and [113.18](#) are answered in part ([sections 50](#) and [60](#)); [research questions 113.12](#), [113.16](#) and [113.19](#) stay open, and sources 13–18 ask them again ([section 113.5](#)).

### 113.3 Source 09's research directions

The following are proposed problems emerging from the development, not a claim that all are already catalogued open problems. Each is stated with a foundational level so that its intended meaning is clear.

### 113.3.1 Formal proof of the surreal core theorem

**Framework:** universe-indexed Lean with explicit smallness and a baseline parameter. Complete the prefix-extension and interpolation modules, then prove the back-and-forth isomorphism. The difficult part is expected to be the bookkeeping of small histories and variable ordinal prefixes, not the finite first-disagreement arguments. A useful first milestone is the inaccessible or regular-cardinal set-sized version, parameterized by an abstract saturated alphabet.

### 113.3.2 Exact axiomatic strength of the global conclusions

**Framework:** GB with ordinary set choice but without global choice. [Proposition 13.7](#) settles the baseline question: a set-like global well-order of **No** is equivalent to global choice. Determine the exact strength of alternative universality and homogeneity assertions that do not already assume such a baseline. In particular, which conclusions can be stated and proved for individually definable class orders without supplying a single uniform global enumeration? Distinguish uniform existence assertions from a schema treating one fixed definition at a time; the latter is not automatically a uniform class construction.

### 113.3.3 Which cuts of the core represent exhaustive branches?

**Framework:** class cut parameters over  $\mathcal{P}_s(b)$  or an explicitly higher-sorted completion. Every global enumeration determines a cut in the dense core, and density makes this representation injective. Characterize its image intrinsically. The missing-branch construction shows that injectivity of a class word is insufficient: exhaustiveness must be detected. A promising formulation separates coherence of set prefixes, realizability of each next-coordinate cut, and eventual coverage of every surreal value.

### 113.3.4 Automorphisms beyond set-sized partial maps

**Framework:** GBC for the core; higher-sort or universe-bounded versions for the full comparison object. Set-sized partial isomorphisms extend in the core, but proper-class partial maps need not. Classify which preserve enough prefix or support structure to extend. Also distinguish automorphisms of the bare order from those preserving the preferred filtration by support bounds. These are different classification problems even though the bare core is always **No**.

### 113.3.5 Non-set-like well-order relations

**Framework:** GBC or GBC + ETR, with a fixed relation-code coordinate order. Study the alternative binary comparison of all global well-order relations. Which order properties depend on the coordinate enumeration? Does it contain a natural set-coded dense core, and when are its cuts represented by actual class well-orders? No conclusion about that different comparison follows merely from the enumeration results proved here.

### 113.3.6 Sharp birthday and support bounds

**Framework:** ZFC on bounded fragments. Given bounds on the birthdays and first-disagreement coordinates of a set family, determine optimal bounds for an interpolating prefix and its supported completion. The present proof bounds the forced prefix length by  $\sup\{\Delta(E_i, F_j) + 1 : i \in I, j \in J\} + 1$  and uses an oversized cardinal for support. That bound is safe but not generally minimal. This direction could yield concrete complexity estimates for symbolic surreal order calculations.



### 113.3.7 What survives away from inaccessible truncations?

**Framework:** ZFC with specified cardinal arithmetic. Replace the inaccessible  $\kappa$  by a regular cardinal and a set alphabet with appropriate cut saturation. Separate the roles of regularity,  $\kappa^{<\kappa} = \kappa$ , and closure under the required cut constructor. At singular cardinals, the first-disagreement bound can reach the ambient length even for relatively small families; identify the resulting interpolation spectrum and construct sharp counterexamples.

### 113.3.8 Other termination conventions for set words

**Framework:** GBC or bounded ZFC. Our dictionary order makes proper prefixes smaller and therefore creates the limit-prefix suprema of [note 11.4](#). Study a distinguished internal termination value, restricted alphabets, or prefix-free codes of set well-orders. Determine exactly which variants have the set-cut property and hence are isomorphic to **No**. The convention must be part of the statement, not an implicit analogy with surreal signs.

### 113.3.9 Uniform infinitary constructions indexed by redundant codes

**Framework:** bounded universes first. Kanovei–Shelah-style redundant indexing suggests avoiding a choice of one representative for each well-order or auxiliary object. Investigate constructions using the support core or bounded strata as indices, while checking that the resulting products, functions, and quotient relations remain at the declared level. The diagonal theorem forbids a uniform same-level list of all global enumerations, but does not preclude purpose-built redundant families adequate for a particular theorem.

[merge] Status of source 09's directions after the merge. Direction 2: the baseline question it already settles over  $\text{GB} + \text{AC}$  needs no set choice ([theorem 13.1](#)); the strength of the other universality and homogeneity assertions remains open. Direction 3 is source 11's [research question 113.1](#). Direction 5: source 08's [proposition 17.1](#) gives the equimorphism type of a relation-code order; the dense core and the representation of its cuts remain open. Direction 7: source 12's dichotomy treats coding lengths at singular cutoffs, not the interpolation spectrum the direction asks for. The others remain open.

[merge] Dated note, 3 October 2026 (batch 81). Direction 3 is answered ([section 50](#)), direction 7 for the minimal slice and direction 8 for unrestricted and injective words ([sections 33](#) and [39](#)); directions 4, 5 and 6 are answered in part ([sections 39](#) and [50](#)); directions 1, 2 and 9 stay open, with source 16's  $\Sigma_1^1$  bound bearing on direction 2 ([section 60](#)).

## 113.4 Source 08's research questions and topics

The following are directions not resolved by the present manuscript. They are not blanket assertions that no relevant answer exists in the literature. Each should begin with a targeted literature search and a precise choice of size level.

### 113.4.1 Set-sized enumeration orders

**1. Isomorphism classification beyond the common spectrum.** The ordinal spectrum and cardinality of  $\mathcal{W}(X)$  depend only on  $|X|$ , while its local cylinder structure remembers the numerical order of residual alphabets. Which additional invariants of  $X$  can be recovered from  $\mathcal{W}(X)$ ? Determine sufficient and necessary conditions for  $\mathcal{W}(X) \cong \mathcal{W}(Y)$ , rather than merely embeddings in both directions. Dense alphabets, well-orders, and mixed sums provide natural test cases.

**2. Exact invariants at regular cutoffs ([merge] partly answered; answered 3 October 2026; see the notes after this list).** For regular  $\kappa$  with  $2^{<\kappa} = \kappa$ , compute the density, cellularity, point characters, and cut cofinalities of  $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ . The binary equimorphism in (S08.6) does not by itself determine its order-topological isomorphism type. Investigate whether the support skeleton construction has a natural  $\kappa$ -sized dense realization inside this set order under weaker assumptions than strong inaccessibility.

**3. Singular birthdays and code lengths ([merge] answered in part; see the notes after this list).** At singular  $\kappa$ , exactly which cuts of size below  $\kappa$  fail to be realized in  $\mathbf{No}_{<\kappa}$  or in its fixed-length enumeration order? Determine the least binary coding lengths for these orders and how they depend on cofinality and cardinal arithmetic. The three specific uses of regularity in note 7.6 identify where a new argument is needed.

**4. Prescribed noninitial lengths.** For  $\alpha$  of cardinality  $\kappa$ , classify  $\mathcal{W}_\alpha(\mathbf{No}_{<\lambda})$ . Finite tails are understood through note 6.25, but noninitial limit lengths may preserve finer lexicographic complexity. Compare this problem with the fixed-type analysis proposed in the continuum manuscript [2], checking its hypotheses and proofs independently for nonseparable alphabets.

#### 113.4.2 The skeleton and the virtual ambient order

**5. Definable choices of the surreal isomorphism.** The back-and-forth isomorphism  $\mathcal{P}_s(b) \cong \mathbf{No}$  depends on witness choices. With a specified global well-order, can it be chosen to respect useful support or birthday bounds? What additional normalization gives uniqueness, if any? Distinguish definability from parameters, parameter-free definability, and invariance under a change of baseline.

**6. Automorphisms and extension from a skeleton.** An order automorphism  $\varphi$  of  $\mathbf{No}$  acts on  $\mathcal{W}_{\text{sl}}$  by  $e \mapsto \varphi \circ e$  and sends  $O_b$  onto  $O_{\varphi \circ b}$ . Which automorphisms of one skeleton extend to uniform automorphisms of the full virtual order? Can order alone recognize eventual-agreement orbits, or are these additional structure supplied by the coordinate presentation?

**7. Full virtual lexicographic products ([merge] answered 3 October 2026; see the notes after this list).** For an ordinary class carrier, every set-indexed lexicographic power of  $\mathbf{No}$  embeds into a skeleton. The *full virtual* product  $\mathcal{W}_{\text{sl}} \times_{\text{lex}} \mathcal{W}_{\text{sl}}$  is a different object. Does it admit a uniform order embedding into  $\mathcal{W}_{\text{sl}}$ ? More generally, determine an absorption spectrum for uniform virtual powers. Simply concatenating two Ord-long words has domain  $\text{Ord} + \text{Ord}$ , not a set-like Ord domain, so the set-sized concatenation argument is unavailable.

**8. Higher-size cuts and completion ([merge] answered 3 October 2026 for cuts of the core; see the notes after this list).** The set-cut theorem deliberately leaves out cuts specified by proper class families of enumeration codes or by higher collections of classes. Classify which such cuts are realized once a precise higher-type language is chosen. A proposed Dedekind completion must identify the type of its cut objects and cannot be treated as another ordinary class without a size proof.

**9. Classification of relation-code orders.** For two global coordinate enumerations  $q, q'$  of  $\mathbf{No}^2$ , the relation-code orders have the same binary equimorphism type. Are there useful hypotheses yielding an isomorphism? Determine which invariants are artifacts of the coordinate order and which reflect genuine properties of class well-orders.

### 113.4.3 Foundations and formalization

**10. Minimal axioms, stated theorem by theorem.** Separate the strength of a baseline  $\text{Ord} \cong \mathbf{No}$ , uniform prefix completion, orbit cut filling, and the global back-and-forth theorem. Over variants of GB with set Choice, which global selection principles are necessary for their uniform formulations? The present proofs establish sufficiency of GBC, not an exact reverse-mathematical classification.

**11. Comparison of non-set-like class well-orders ([merge] answered; see the note after this list).** Formulate an explicit alignment or comparison principle for arbitrary class well-orders and locate its recursion requirements. Which restricted domains, such as finite sums of  $\text{Ord}$  and set ordinals, admit direct elementary algorithms? A theorem about all class domains should not be extrapolated from these examples without proof.

**12. Dependence on the class collection.** For fixed  $M$  and  $\mathcal{C} \subseteq \mathcal{D}$ , compare the virtual orders of global enumerations in  $(M, \mathcal{C})$  and  $(M, \mathcal{D})$ . The same set-coded skeleton relative to a shared baseline remains available, but new class rows may appear. Study which higher-size cut properties, operator embeddings, and definability properties change.

**13. Quantitative and certified prefix extension.** Formalize the support bound in [note 19.8](#), the forced-prefix length in [note 19.13](#), and the birthday of its stopping entry. This would make the central theorem a sequence of explicit bounded lemmas rather than one monolithic existence proof. It is a strong first target for the repository's existing small-cut interface.

**14. Increasing-subset and alternate-termination orders ([merge] answered 3 October 2026 except one type; see the notes after this list).** Study lexicographic orders of numerically increasing set sequences, as opposed to arbitrary enumerations, and compare prefix-small, prefix-large, and centered-termination conventions. The limit-prefix counterexample shows that these choices affect local cofinalities even when the alphabet is  $\mathbf{No}$ .

**15. An order-theoretic interface for class nonstandard models.** Determine which part of a Kanovei–Shelah-style indexing construction can use a set-coded skeleton and which genuinely needs a larger collection. Keep the order embedding problem separate from the existence of a class ultrafilter, satisfaction relation, or transfer principle. A successful construction should include an explicit uniform-family and smallness audit at every step.

[merge] Status of source 08's questions after the merge, dated 2 October 2026. Question 11 is answered: the labelled comparisons of [theorem 14.2](#) and [theorem 14.7](#) compare arbitrary class well-orders of a labelled class with elementary comprehension, without ETR (see the note after [proposition 17.1](#)). Question 3 is answered in part: the least binary coding length of the fixed-length enumeration order at a singular cutoff is  $\kappa \cdot \kappa$  at a strong limit and  $2^{<\kappa}$  otherwise ([theorem 8.3](#)); which cuts of size below  $\kappa$  fail to be realized remains open. Question 2 is answered in part by [proposition 12.1](#) (the order is  $\eta_\kappa$ , and all its global and local cofinalities and coinitalities equal  $\kappa$ ). Question 4 is source 12's [research question 113.11](#). The others remain open.

[merge] Dated note, 3 October 2026 (batch 81). Questions 2 and 7 are answered ([sections 33](#) and [60](#)); question 3 also in its cut half, in character terms ([section 33](#)); question 8 for class cuts of the core ([section 50](#)); question 14 except the isomorphism type of the centered increasing order ([section 39](#)); questions 4, 6, 9, 12 and 13 in part ([sections 33](#), [39](#), [50](#) and [60](#)); questions 1, 5, 10 and 15 stay open.

### 113.5 The questions of sources 13–18: index and status (3 October 2026)

[merge] Sources 13–18 ask 74 questions (13, 13, 12, 11, 13 and 12), printed per source in the following subsections as proposed continuations, not as a census of open problems. Many repeat each other or a question above; the table groups them, names the earlier question each continues, and records the status after the whole batch-81 addition. A question answered by a sibling source is marked in its source’s subsection. The status of the questions of sources 11, 12, 09 and 08 after the addition is recorded in the dated notes after their subsections and in [sections 33, 39, 50 and 60](#).

| Topic                                                                           | Asked by                                                                                                                                                                                                                                                                                   | Status, 3 October 2026                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local geometry and gap spectrum of the minimal slice                            | 14: <a href="#">research question 113.41</a> ; 16: <a href="#">research question 113.59</a>                                                                                                                                                                                                | <b>answered</b> by source 17 ( <a href="#">theorem 30.11</a> ; Part VI)                                                                                                                                                                                                                                                |
| Isomorphism type, arrangement of gaps, finer topology of set strata             | 15: <a href="#">research question 113.52</a> ; 17: <a href="#">research questions 113.71 and 113.72</a> ; 18: <a href="#">research question 113.82</a> (continuing 12: <a href="#">research question 113.13</a> )                                                                          | open; characters, density, gap spectrum and multiplicities of the minimal slice are known ( <a href="#">section 33</a> ); <i>batch 83</i> : <a href="#">research question 113.72 answered</a> ; topology at $\mu = \omega$ ( <a href="#">section 113.25</a> (c), (m))                                                  |
| Longer and nonminimal enumeration types                                         | 15: <a href="#">research question 113.46</a> ; 16: <a href="#">research question 113.58</a> ; 17: <a href="#">research question 113.69</a> ; 18: <a href="#">research question 113.83</a> (continuing 12: <a href="#">research question 113.11</a> )                                       | coding length and cube spectrum for $\mu \cdot \gamma + n$ ( <a href="#">theorem 31.2</a> ), density of every stratum ( <a href="#">theorem 27.2</a> ); open for infinite remainders and for gap spectra beyond the minimal slice; <i>batch 83</i> : infinite remainders covered ( <a href="#">section 113.25</a> (a)) |
| Noncardinal birthday cutoffs                                                    | 15: <a href="#">research question 113.47</a> ; 17: <a href="#">research question 113.70</a> ; 18: <a href="#">research question 113.84</a> (= 12: <a href="#">research question 113.12</a> )                                                                                               | open; <i>batch 83</i> : in part ( <a href="#">section 113.25</a> (b))                                                                                                                                                                                                                                                  |
| Coherent embeddings across cutoffs; quantitative compatibility                  | 14: <a href="#">research question 113.42</a> ; 15: <a href="#">research question 113.54</a> ; 16: <a href="#">research question 113.68</a> ; 17: <a href="#">research question 113.79</a> ; 18: <a href="#">research question 113.85</a> (= 12: <a href="#">research question 113.19</a> ) | open; <i>batch 83</i> : in part ( <a href="#">section 113.25</a> (f))                                                                                                                                                                                                                                                  |
| Normal forms for class cuts; cutoff analogues of first failure                  | 13: <a href="#">research questions 113.22 and 113.23</a>                                                                                                                                                                                                                                   | in part: unique-data trichotomy ( <a href="#">theorem 43.3</a> ); minimal-slice cuts ( <a href="#">theorem 30.11</a> )                                                                                                                                                                                                 |
| Recovering prefix structure, exhaustiveness or set-likeness from the bare order | 13: <a href="#">research question 113.20</a> ; 14: <a href="#">research question 113.34</a> ; 15: <a href="#">research question 113.49</a> ; 16: <a href="#">research question 113.61</a> (continuing 11: <a href="#">research question 113.4</a> )                                        | negative half answered ( <a href="#">sections 50 and 60</a> ); which enrichments suffice is open; <i>batch 83</i> : label-edge equality suffices, unlabelled cylinders do not ( <a href="#">section 113.25</a> (h))                                                                                                    |

| Topic                                                                        | Asked by                                                                                                                                                                                                                                            | Status, 3 October 2026                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Automorphisms preserving a core or its represented cuts; changes of baseline | 13: research question 113.21; 14: research question 113.33; 15: research question 113.51; 16: research question 113.62; 17: research questions 113.73 and 113.74; 18: research question 113.87 (continuing 11: research question 113.5)             | extension criterion (theorem 47.5), non-extension (theorem 47.7), relabelling (proposition 47.9); stabilizers and general baselines open; <i>batch 83</i> : partial families; extension across residual fibres externally (section 113.25 (h), (n))                                                                                                         |
| Completion with residual data                                                | 15: research question 113.50                                                                                                                                                                                                                        | open; <i>batch 83</i> : <b>answered</b> externally by source 19 (section 113.25 (n))                                                                                                                                                                                                                                                                        |
| Products and splits                                                          | 13: research questions 113.24 and 113.25                                                                                                                                                                                                            | in part: KM + GC suffices (theorem 55.4); hierarchy open                                                                                                                                                                                                                                                                                                    |
| Class realizations                                                           | 13: research questions 113.26 and 113.27; 14: research question 113.43; 15: research question 113.56; 16: research question 113.66; 17: research question 113.75; 18: research questions 113.88 and 113.89 (continuing 11: research question 113.8) | in part (section 50 (e), section 60 (e)); <i>batch 83</i> : research question 113.89 and research question 113.75 <b>answered</b> (section 113.25 (e), (l))                                                                                                                                                                                                 |
| Termination conventions                                                      | 13: research question 113.28; 16: research question 113.60; 18: research question 113.91                                                                                                                                                            | 13's <b>answered</b> by source 16; 16's answered by source 18 at the zero threshold; 18's open (section 39); <i>batch 83</i> : 18's in part (section 113.25 (j))                                                                                                                                                                                            |
| Birthday, support and code bounds                                            | 13: research question 113.29; 14: research questions 113.35 to 113.37; 17: research question 113.78; 18: research question 113.90 (continuing 11: research question 113.6)                                                                          | support: hierarchy and exact reserve (theorems 49.3 and 49.7); the rest open; <i>batch 83</i> : least ordinal support <b>answered</b> (section 113.25 (i))                                                                                                                                                                                                  |
| Iterated condensation, residual blocks, raw fibres                           | 13: research question 113.30; 14: research questions 113.38 and 113.39; 15: research question 113.48; 16: research question 113.64; 17: research questions 113.76 and 113.77; 18: research question 113.86 (continuing 11: research question 113.3) | block indices answered (note 52.1); 17's research question 113.77 answered externally for countable $\omega$ -standard models (theorem 58.6); iteration strength open; <i>batch 83</i> : canonical histories over GBC (section 113.25 (g)); research question 113.77 further (section 113.25 (o)); <i>batch 96</i> : histories over GB (section 113.29 (b)) |
| Relation codes and other scans                                               | 13: research question 113.31; 14: research question 113.40; 15: research questions 113.53 and 113.55                                                                                                                                                | in part: endpoint patterns (theorem 38.2); <i>batch 83</i> : scan dependence at set level, dense core (section 113.25 (k))                                                                                                                                                                                                                                  |

| Topic                                        | Asked by                                                                                                                                                                                                                                                                                | Status, 3 October 2026                                                                                                                                       |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Comprehension and choice strength            | 14: <a href="#">research question 113.44</a> ; 16: <a href="#">research questions 113.63 and 113.65</a> ; 18: <a href="#">research question 113.92</a> (continuing 11: <a href="#">research question 113.2</a> )                                                                        | in part: $\Sigma_1^1$ sufficiency ( <a href="#">theorem 56.2</a> ); one set-like parameter suffices for the word isomorphisms ( <a href="#">section 39</a> ) |
| First formal milestones                      | 13: <a href="#">research question 113.32</a> ; 14: <a href="#">research question 113.45</a> ; 15: <a href="#">research question 113.57</a> ; 16: <a href="#">research question 113.67</a> ; 17: <a href="#">research question 113.80</a> ; 18: <a href="#">research question 113.93</a> | proposals; nothing is formalized                                                                                                                             |
| Typed constructions indexed by presentations | 17: <a href="#">research question 113.81</a> (continuing 11: <a href="#">research question 113.10</a> )                                                                                                                                                                                 | open                                                                                                                                                         |

### 113.6 Source 13: further research questions and proposed projects

These questions are proposed continuations of the present development, not a claim that each is a previously published open problem. The prefix-trace criterion, the first-failure classification, and the negative pure-order cylinder result settle the versions stated here; the questions below ask for stronger or different conclusions.

#### 113.6.1 Cuts, trees, and enriched structure

**Research question 113.20** (Minimal extra structure for recognizing branches). What is the weakest expansion of the pure core order  $\mathcal{P}_s(b) \cong \mathbf{No}$  that makes its branch traces recognizable? Is the family of all cylinders enough as an unranked family, or must one retain the prefix-length relation? Can a support filtration recover the necessary levels without explicitly naming prefixes?

[Theorem 47.12](#) shows that the pure order alone cannot canonically recover the original cylinder family. The positive target is therefore an enriched reconstruction theorem, not a repetition of the disproved pure-order assertion.

**Research question 113.21** (Automorphisms preserving a chosen core). Classify the uniform order automorphisms of  $\mathfrak{W}_{sl}$  that preserve the specific eventual-agreement core  $\mathcal{P}_s(b)$ . Which preserve its support filtration, and which preserve the set of exhaustive traces after an isomorphism between two cores? Does adding birthday or simplicity information force a substantial rigidity theorem?

The finite-block automorphisms need not preserve the baseline orbit. A useful initial project is to characterize tail alphabet transformations that do preserve it.

**Research question 113.22** (A normal form for class cuts). Can the branch/horizontal/limit-boundary trichotomy be organized into a canonical recursive notation for every available normalized cut? For horizontal cuts, how much of the next-label cut can be encoded by surreal simplicity data? What uniqueness survives when changes of prefix presentation are allowed?

A complete class-cut collection is not an ordinary class object in GBC. The first implementation should concern one supplied class cut or one uniform class-indexed family, before introducing a higher sort.



**Research question 113.23** (Sharp cutoff analogues). For a birthday fragment  $X_\kappa$ , determine which part of the first-failure classification survives when prefix extension can fail through exhaustion, complement-cardinality mismatch, or a missing label separator. Can these three obstructions be combined with the singular strong-limit transition into a practical classification of cut characters in fixed-length strata?

This asks for more than substituting  $X_\kappa$  into the global theorem; the hypotheses that must be replaced are explicitly identified in [Section 32.2](#).

### 113.6.2 Products, comprehension, and model dependence

**Research question 113.24** (The weakest uniform split obstruction). Over GBC, what additional comprehension principle suffices to rule out a fixed-formula uniform embedding  $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} 2 \rightarrow \mathfrak{W}_{\text{sl}}$ ? Can the inverse-evaluator argument be reduced to a restricted existential-class comprehension schema? For more restrictive formulas defining the proposed embedding, is there an elementary diagonal argument avoiding that comprehension altogether?

[Theorem 55.4](#) gives a sufficient theory and identifies the point to optimize. It is not a proof that full KM is necessary.

**Research question 113.25** (A hierarchy beyond one split). In a full inaccessible universe, classify the mutual embeddability of  $\mathfrak{W}_{\text{sl}}^n$  for finite  $n \geq 2$ , of  $\mathfrak{W}_{\text{sl}} \times_{\text{lex}} K$  for various right factors  $K$ , and of iterated point-splittings. Which invariants distinguish these orders after the first dense-core obstruction has saturated at cardinality  $2^\kappa$ ?

All these orders can have the same cardinality, so a useful invariant must record more than the number of elements. Families of disjoint intervals, patterns of adjacent pairs, and iterated condensation are natural candidates.

**Research question 113.26** (Class realizations with intermediate behavior). For a fixed set model  $M$ , how do different collections of available classes  $\mathcal{C}$  affect the external density, cellularity, and product-embedding behavior of its enumeration order? Which intermediate behaviors lie between the externally countable rational case and the full inaccessible-universe obstruction?

A precise answer should state whether a map is arbitrary external, definable over the two-sorted model, or coded in an additional sort. Those are distinct classification problems.

**Research question 113.27** (Uniformity versus external isomorphism). For an externally countable model, can the complexity of an external isomorphism  $\mathfrak{W}_{\text{sl}} \cong \mathbb{Q}$  be measured relative to presentations of its set and class sorts? What information about the model is necessary to compute, define, or select such an isomorphism?

This connects the order-theoretic collapse to computability and model presentation without mistaking an external construction for a class available internally.

### 113.6.3 Termination, bounds, and residual orders

**Research question 113.28** (A classification of termination cuts). For a class alphabet order-isomorphic to **No**, characterize the placements of an extra terminal marker for which arbitrary set-word or injective set-word lexicography is set-cut saturated. What happens when the marker is placed immediately on one side of an existing label, rather than reserving that label? Can the answer be expressed in terms of the two characters of the marker cut?

The end-first and end-last failures and the reserved-central success are settled here. They need not exhaust all intermediate behavior.

**Research question 113.29** (Optimal birthday and support bounds). For prescribed bounds on input birthdays, word length, and baseline indices, sharpen the prefix-length and support-cardinality bounds in [Theorem 42.4](#). Is the explicit central-code birthday sum optimal within a useful class of prefix-free local codes? Which bounds are compatible with embeddings across birthday cutoffs?

The current support completion deliberately chooses a large cardinal to equalize complements. It proves existence, not optimality.

**Research question 113.30** (Iterating raw residual decomposition). The ordinal-head fiber theorem removes one set-like initial part. How far can residual decomposition be iterated with set-valued codes, and what class-recursion principles are needed when an iteration has class-valued stages? Is there a canonical rank or notation for a substantial family of raw class well-orders beyond the first head?

This is where ETR or a stronger scheme could become genuinely relevant. The elementary one-head construction alone does not justify an arbitrary iteration.

**Research question 113.31** (Relation-code dependence). For two set-like enumerations of pairs, determine which transformations relate their relation-bit orders on raw well-orders. When do the two induced orders become isomorphic, merely equimorphic, or distinct under definability restrictions? Which code orders can be characterized by simple invariants of the pair enumeration?

#### 113.6.4 A concrete formalization milestone

**Research question 113.32** (First checked extension). Formalize the split-order density lemma and the crossing-prefix uniqueness/reconstruction argument in a size-explicit abstract setting. Then instantiate the small-cut and fresh-value interfaces using `SignSequenceCut.lean`. Can the central-word saturation theorem be completed without introducing surreal field arithmetic?

The answer to the last design question should be positive at the mathematical level: the proof uses only order, the reserved marker, smallness, and fresh set-cut separators. Establishing this dependency boundary in Lean would be a useful substantial first checked result.

[merge] Status of source 13's questions, 3 October 2026. [Research question 113.28](#) is **answered** by source 16 ([theorems 36.1](#) and [36.2](#), transported along an isomorphism of the alphabet with **No**; [section 39](#) (c)). [Research question 113.22](#) is answered in part by source 14's unique-data trichotomy ([theorem 43.3](#)) and source 15's finer successor cases; [research question 113.23](#) in part, for the minimal slice, by source 17's gap spectrum ([theorem 30.11](#)). [Research question 113.21](#) bears on source 17's extension criterion ([theorem 47.5](#)), [research question 113.26](#) on source 18's countable models ([theorem 58.6](#)), [research question 113.29](#) on source 17's exact reserve ([theorem 49.7](#)), [research question 113.30](#) on the block decomposition ([note 52.1](#)), [research question 113.31](#) on source 15's endpoint transfer ([theorem 38.2](#)), and [research question 113.24](#) on source 16's  $\Sigma_1^1$  bound ([note 56.4](#)). The others remain open.

#### 113.7 Source 14: further research questions and priorities

The following are proposed continuations of the present work. They are not all established named open problems in the literature. In particular, the prefix-structured exhaustive-cut question and the support-cardinal threshold are now answered by the theorems above; the questions here state stronger or different targets rather than restating those results.

### 113.7.1 Cut recognition beyond the named prefix structure

**Research question 113.33** (The exhaustive-trace stabilizer). Characterize the subgroup of class order automorphisms of  $\mathcal{P}_s(b)$  that preserves the exhaustive-trace collection. How does it compare with the groups preserving all prefix cylinders, or all label incidences? Can it be generated by explicit transformations of enumeration structure?

[Theorem 47.2](#) classifies the bare-order cut orbits and [Corollary 47.3](#) proves that this stabilizer is proper. What remains is its structure, not the now-answered yes/no question whether every core automorphism preserves exhaustive representability.

**Research question 113.34** (Minimal enrichment for exhaustiveness). Can the coverage predicate be recovered from cylinder inclusion and numerical order of children without naming the label evaluation itself? What additional incidence relation records that the same surreal label appears at different possible coordinates?

Coherence is essentially a branch property of the cylinder tree, whereas exhaustiveness compares labels along that branch with the entire alphabet. Separating these logical resources should clarify exactly what the recognition theorem uses.

### 113.7.2 The support hierarchy beyond one saturation invariant

**Research question 113.35** (Isomorphism of support layers). For infinite cardinals  $\kappa, \lambda$  with  $\text{cf } \kappa = \text{cf } \lambda$ , when are  $\mathcal{P}_{<\kappa}(b)$  and  $\mathcal{P}_{<\lambda}(b)$  isomorphic as bare class linear orders? Does some cut invariant recover more than the cofinality of the support threshold?

[Theorem 49.3](#) separates layers with different cofinalities but leaves this equal-cofinality case untouched. Their shared embedded copy of **No** is too weak to decide it. The attained minimum support cost is an invariant of the *support-enriched* structure, not yet an invariant of its bare order.

**Research question 113.36** (Support ideals). Replace the cardinality condition by an ideal of permitted set supports. Under what closure and permutation-invariance hypotheses is saturation controlled by the ideal's additivity? Can a failure of union closure always be converted into a coherent prefix configuration forcing a support outside the ideal?

The compression lemma bounds cardinality, not membership in an arbitrary ideal: it can adjoin images of moved coordinates and a buffer. Therefore a general ideal theorem needs explicit closure under those operations. It would be incorrect simply to replace  $< \kappa$  by “belongs to the ideal” in the existing proof.

**Research question 113.37** (Optimal simultaneous support cost). Given a specific separated set family, characterize the least support cardinality of a separator by its forced-prefix tree, rather than only by the union of all input supports. Can the finite version be computed without enumerating all possible separator permutations?

The sharp examples attain the threshold, but many input supports contain coordinates irrelevant to their comparisons. The first-outside-support argument suggests an algorithmic pruning principle. Finite optimization and the transfinite cardinal minimum should be studied separately.

### 113.7.3 Raw tails, condensation, and alternative comparison orders

**Research question 113.38** (Iterated initial-branch quotients). After quotienting raw well-orders by their first Ord-long branch, can one iterate a similarly explicit trace construction to recover successive proper-class blocks? What can be proved for prescribed block patterns without assuming a canonical class ordinal for every well-order?

The trace-fiber theorem completely describes the first loss of information. Iterating it along class-valued histories may require stronger recursion principles. The exact point at which elementary comprehension ceases to suffice should be isolated as a theorem.

**Research question 113.39** (Automorphisms of raw trace fibers). Which automorphisms of the branch order extend to the unrestricted well-order order, and what compatibility conditions do the residual alphabets impose? Which automorphisms preserve finite residual blocks but fail to preserve the core-trace quotient?

The singleton omitted-label fiber shows that being a singleton is not enough to recognize the exhaustive suborder. Any extension theorem must also account for residual alphabets with more than one label.

**Research question 113.40** (Relation-code comparison). For a specified coordinate schedule on pairs of surreal labels, classify the class cuts represented by genuine well-order relations. Is there a useful analogue of crossing cylinders that detects transitivity, linearity, and well-foundedness from a coherent branch of relation bits?

Coherent binary bits always define a relation, but well-foundedness is a global constraint. The strong reconstruction phenomenon for exhaustive enumerations may not carry over. A schedule-independent theorem must explicitly quantify which coordinate changes it tolerates.

### 113.7.4 Birthday cutoffs and universe coherence

**Research question 113.41** (Exact local geometry of set slices). For  $\mu = 2^{<\kappa}$ , determine the complete cut-character spectrum of  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ , especially when  $\mu > \kappa$  and at singular cutoffs. Which parts of the crossing-cylinder proof survive when deleting a long used prefix can exhaust intervals of the alphabet?

This continues an unfilled target of the predecessor. The global proof uses the fact that deleting *any set* leaves plentiful fresh surreals; a fixed birthday fragment lacks that unrestricted property. The inherited binary equimorphism does not decide the local characters.

**Research question 113.42** (Coherent embeddings across cutoffs). Construct explicit embeddings between enumeration spaces for increasing birthday bounds that compose coherently. Which preserve the minimal allowed enumeration type, the prefix cylinders, or the relevant support filtration?

Appending an arbitrary tail preserves a first disagreement but may change the ordinal type. A functorial construction should record both the alphabet inclusion and the position schedule, rather than silently identify cardinally equivalent domains.

### 113.7.5 Class realizations and proof strength

**Research question 113.43** (Which new cuts can a class extension add?). For a fixed set model and old baseline, characterize the additional exhaustive trace cuts admitted by specified class extensions. Can the new-cut collection distinguish natural classes of realizations even when their old-cut realizability relations agree?

[Theorem 48.1](#) supplies a rigidity constraint: no old cut changes status. It does not classify the new parameters. The tail fibers also suggest studying extensions that add raw orders with an old trace but a new tail.

**Research question 113.44** (Axiomatic economy). What is the weakest class theory proving the crossing-chain trichotomy and its reconstruction theorem once a baseline and fresh-separator property are supplied? Can the proof of the support threshold be formalized in a set-only regular-cardinal model before passing to classes?

The existing proofs visibly use elementary comprehension and class Replacement, plus set Choice for extensions. Identifying which are essential is more informative than merely observing that a stronger class theory also proves the theorem.

**Research question 113.45** (First kernel-verified milestone). Can the common-prefix comparison and the support-compression lemma be formalized as small reusable Lean modules, then independently connected to the pinned surreal sign-sequence library?

These targets avoid the largest foundational infrastructure while capturing the most error-prone finite and cardinal bookkeeping. A useful deliverable would include separate tests for fixed-position preservation, complement cardinalities, and the distinction between strict right character and the character of a normalized cut complement.

[merge] Status of source 14's questions, 3 October 2026. [Research question 113.41](#) is **answered** by source 17 ([theorems 30.2](#) and [30.11](#), with source 18's multiplicities, [theorem 30.16](#)), at every cutoff, including  $\mu > \kappa$  and singular  $\kappa$ . [Research question 113.43](#) is answered in part by source 17's [theorem 48.4](#) (every proper same-set extension adds new represented cuts), [research question 113.38](#) in part by the block decomposition ([note 52.1](#)), and [research question 113.33](#) bears on source 17's extension criterion ([theorem 47.5](#)), which identifies the stabilizer of the represented-cut predicate abstractly. [Research question 113.42](#) is source 12's [research question 113.19](#). The others remain open.

## 113.8 Source 15: further research questions and a prioritized program

**Research question 113.46** (All fixed enumeration types). Write  $\alpha = \mu \cdot \gamma + \zeta$  with  $\zeta < \mu$  infinite. Classify the possible residual numerical suborders left after the long prefix, and use them to compute  $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$ . The theorem here handles exactly finite  $\zeta$ . A useful first family is  $\zeta = \omega$ , followed by regular and singular infinite remainders.

**Research question 113.47** (Noncardinal birthday bounds). For an arbitrary ordinal  $\theta$ , determine the order structure of  $\mathbf{No}_{<\theta}$  and its permutation strata. The cone and ordinal absorption arguments for an initial cardinal need not survive unchanged. Identify which ordinal invariants of  $\theta$  replace the present regular/singular dichotomy.

**Research question 113.48** (The exact strength of arbitrary-input condensation). Does GB prove a uniform class coding all finite condensations of every class well-order? If not, determine a sharp sufficient and necessary principle. Then study longer histories with specified limit intersections. The explicit polynomial examples show that existence for particular long histories alone is insufficient to calibrate the general scheme.

**Research question 113.49** (Intrinsic information in the full raw order). Which prefix cones, first-block fibres, and finite residual blocks are definable from the bare lexicographic order of raw well-orders? [Theorem 47.10](#) shows that the abstract core does not determine its represented cuts.

Determine how much extra structure is needed for reconstruction, and whether the full raw order restores any of the lost distinctions.

**Research question 113.50** (Completion with residual data). Construct and characterize a stratified completion that contains all terminal cuts while recording the residual well-order fibres over persistent branches. Determine its universal property and its minimum required size level. Ordinary Dedekind completion alone loses the residual data.

**Research question 113.51** (Automorphism extension). Characterize those automorphisms of a bounded-support core that preserve persistent cuts, preserve exhaustive cuts, or extend to native global enumerations. Compare cores built from baselines differing on a proper class of positions. The abstract isomorphism of two cores does not decide their placement inside the surrounding order.

**Research question 113.52** (Finer topology of bounded slices). After the density, cellularity, and point-character calculations, classify the exact gap spectra, local bases, and isomorphism types of the bounded permutation spaces. Equimorphism with a binary product does not determine all these invariants. Study which distinctions survive when cardinal arithmetic makes displayed density values coincide.

**Research question 113.53** (Inverse ranks and prescribed relation scans). Classify the cuts and point characters of ordinal-rank lexicography beyond the points  $M_\lambda$ . For table lexicography, use endpoint transfer to characterize scans whose extremal linear-order table is well-founded or set-like. Then seek invariants beyond the three possible endpoint patterns.

**Research question 113.54** (Coherent restricted filtrations). The three-label theorem rules out monotonicity for every restriction. Which nested chains of carriers, or which restricted families of orders, admit a useful monotone restriction system? Can birthday cutoffs be linked by canonical order embeddings rather than by unrestricted deletion?

**Research question 113.55** (Naturally well-ordered supports). Characterize the image and omitted cuts of  $A \mapsto -\sum_{a \in A} \omega^{-a}$ . Compare shorter-first and longer-first orders on increasing set words, and develop a higher-level analogue that treats proper-class supports without calling the result a single surreal normal form.

**Research question 113.56** (Class-model dependence). For a fixed set universe  $M$ , study changes in the lexicographic theory as the class realization grows from  $\mathcal{C}$  to  $\mathcal{C}'$ . Which statements about existing orders are absolute, which require well-foundedness preservation, and which detect newly available cuts or enumerations? Keep full-class and Henkin interpretations separate.

**Research question 113.57** (First formal milestones). Formalize the set-prefix common-disagreement formula and the elementary set-interval quotient before the entire class theory. Next verify the binary scheduler and finite-fibre fusion; then the persistent-cut reconstruction with explicit smallness. These have concise interfaces and clear failure cases, making them useful independent additions to ProveIt.

The most concrete next mathematical projects are the smaller infinite-tail classification, exact bounded gap spectra, and the automorphism-extension criterion. The most concrete foundational project is the arbitrary-input condensation scheme. Each now has a precise point where the proved construction stops, rather than an undifferentiated request to study “all well-orders.”

[merge] Status of source 15’s questions, 3 October 2026. [Research question 113.51](#) is answered in its extension clause by source 17: a core automorphism extends to the global enumerations



exactly when it preserves the represented cuts (theorem 47.5), and some do not (theorem 47.7); the comparison of cores from baselines differing on a proper class stays open. Research question 113.52 is answered in part for the minimal slice (gap spectrum and multiplicities, theorems 30.11 and 30.16); local bases and isomorphism types stay open. Research question 113.49 and research question 113.56 are answered in part as recorded in section 60 (e)–(f); for research question 113.56 see also note 57.5. Research questions 113.46 and 113.47 are source 12’s research questions 113.11 and 113.12 beyond the cases answered in Part VI. The others remain open.

### 113.9 Source 16: further research questions, with the resolved parts removed

The following are proposed continuation problems, not a census of currently open problems in the literature. In particular, the exact termination family, represented-cut recognition with the labelled prefix tree, simultaneous residual blocks, and the stated regular-stage topological invariants have been settled under the assumptions of this article. They should not be relisted as wholly open.

#### 113.9.1 Highest-priority structural problems

**Research question 113.58** (Longer-stratum binary coding ranks). For  $\mathbf{No}_{<\kappa}$  and  $\mu = 2^{<\kappa}$ , determine the least ordinal  $\rho$  for which  $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}) \hookrightarrow 2_{\text{lex}}^\rho$  for every  $|\alpha| = \mu$ , and determine the reverse cube embeddings.

The topology is now known for all these strata, but it does not determine binary coding rank. The singular strong-limit transition (S16.2) and the finite residual permutation blocks give concrete starting cases. The first targets should be  $\alpha = \mu + 1$ ,  $\mu + 2$ , and  $\mu + \omega$ , with separate treatment of singular strong-limit  $\kappa$ .

**Research question 113.59** (Singular birthday gaps). Determine the unfilled cut-character spectrum of  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  when  $\kappa$  is singular, and when  $\kappa$  is regular but  $\mu > \kappa$ .

The global characters  $\text{cf}(\kappa)$  and the exact density  $\mu^{<\mu}$  are known here. The missing information is local: which numerical gaps in the truncated alphabet combine with exhaustion and prefix edges to produce each pair of cofinalities? The bounded first-failure analysis should be formulated before attempting a general classification by cardinal arithmetic alone.

**Research question 113.60** (Centered increasing words). Classify the order of numerically increasing set-length words when the termination symbol is placed at a fixed interior numerical threshold.

Theorem 36.2 covers unrestricted and injective words. The increasing case has prefix-dependent available alphabets and already fails the centered saturation theorem. A useful first problem is to determine all point characters for the  $0^-$  and  $0^+$  placements, including words that cross the threshold only at a limit stage.

**Research question 113.61** (Recovering prefix structure from order alone). Which features of the labelled prefix tree can be reconstructed from the bare order  $\mathcal{P}_{\text{bd}}(b)$ , from  $\mathfrak{W}_{\text{sl}}$  as a definable comparison, or from  $\mathfrak{W}_{\text{all}}$ ?

Our cut recognition uses the family of cylinders and numerical labels as extra predicates. A bare core is isomorphic to  $\mathbf{No}$ , and singleton raw fibers can arise either from an exhaustive branch or from a branch omitting one label. Thus neither the core order nor the size of a fiber immediately identifies exhaustion. One can ask for the least additional predicates that make the tree and exhaustion relation definable.

**Research question 113.62** (Changing baselines and extending automorphisms). For two baselines  $b, c$ , characterize when an automorphism of the set-like global comparison carries  $\mathcal{P}_{\text{bd}}(b)$  onto  $\mathcal{P}_{\text{bd}}(c)$ .

An increasing numerical automorphism  $h$  gives the explicit example  $E \mapsto h \circ E$ , carrying  $\mathcal{P}_{\text{bd}}(b)$  onto  $\mathcal{P}_{\text{bd}}(h \circ b)$ . General abstract isomorphisms between the two cores need not preserve the distinguished set of exhaustive branch cuts. [Theorem 46.2](#) suggests an extension criterion stated in terms of precisely those cuts.

### 113.9.2 Foundational problems with explicit targets

**Research question 113.63** (Exact strength of uniform nonembedding). Can the conclusion of [Theorem 56.2](#) for elementary transformation formulas be proved in GBC alone? If not, what is its reverse-mathematical strength?

The missing step is sharply localized: forming the inverse evaluator ([S16.16](#)). A lower bound would need a model in which a specified elementary transformation exists or a reduction from a comprehension principle. The mere presence of an existential class quantifier in the present proof does not establish necessity.

**Research question 113.64** (Repeated condensation of the quotient). Determine the strength and possible stabilization behavior of the transfinite tower obtained by repeatedly taking least representatives of set-interval blocks and intersecting at limits.

[Theorem 52.6](#) has removed the need for ETR to extract all blocks of one order. The new question concerns repeated condensation of the resulting index orders. Suitable ETR is a sufficient bound; whether this particular operator needs it, and which class orders can remain nontrivial at prescribed stages, are separate questions.

**Research question 113.65** (Choice needed for the word isomorphisms). How far can [Theorems 35.1](#) and [36.2](#) be proved without a global well-order?

Their explicit embeddings and the termination saturation criterion use less than the GBC back-and-forth isomorphisms. Investigate whether set choice, a restricted class-enumeration principle, or a canonical sign construction suffices for particular cases. This does not settle the broader historical question about the choice strength of universal embeddability into **No**.

**Research question 113.66** (Class extensions and canonical data). For a fixed set universe with two admitted-class expansions, describe which new branch cuts, global well-orders, and uniform transformations become available.

Shared relations retain the same internal well-foundedness under GBC, but the available class graphs can change. It is useful to distinguish absoluteness of the elementary formulas for blocks and recognition from existence of their class parameters or of the transformations being considered.

### 113.9.3 Formalization and quantitative coding

**Research question 113.67** (A first verified theorem package). Implement the set-alphabet ordinal spectrum and the  $\mu, \mu + 1, \mu + 2$  versus later-strata density theorem, then instantiate them at the small birthday carriers already in ProveIt.

This package would verify both the combinatorial mechanism and its smallness boundaries without first formalizing arbitrary class models. The two-point-fiber case is a particularly useful regression target: counting jumps as disjoint open intervals would produce a false proof.

**Research question 113.68** (Quantitative compatibility of the constructions). Find sharp birthday and support bounds for prefix completion, termination encodings, and the tree isomorphisms; determine which maps commute with increasing birthday cutoffs.

Equation (S16.7) is explicit, while the global back-and-forth isomorphisms generally are not birthday-minimal. Compatibility should mean equality or a specified coherent commuting diagram of maps, not only the existence of unrelated embeddings at each stage.

[merge] Status of source 16's questions, 3 October 2026. [Research question 113.59](#) is **answered** by source 17 ([theorem 30.11](#); [section 33](#)). [Research question 113.60](#) is answered by source 18 for the threshold at zero ([section 37](#); [section 39 \(d\)](#)); other interior thresholds are open. [Research question 113.58](#) is answered for  $\alpha = \mu \cdot \gamma + n$ , which includes the first targets  $\mu + 1$  and  $\mu + 2$ , by source 15's [theorem 31.2](#);  $\alpha = \mu + \omega$  and the reverse cube embeddings beyond those types are open. [Research question 113.62](#) is answered in part by source 17's extension criterion and relabelling ([theorem 47.5](#) and [proposition 47.9](#)), which is the criterion the commentary anticipates. [Research question 113.65](#): source 18's proof of the word isomorphisms uses one set-like global well-order as a parameter ([section 39](#)). [Research question 113.61](#) and [research question 113.66](#) are answered in part as recorded in [sections 50](#) and [60](#). The others remain open.

## 113.10 Source 17: further research questions and a prioritized program

These questions are continuations of this article, not claims that each is a historically catalogued open problem. The results above narrow their scope: questions about minimal-slice characters, the displayed gap spectrum, or mere recognition with the full labelled prefix structure are no longer left open within this development.

### 113.10.1 Set-sized order geometry

**Research question 113.69** (Nonminimal enumeration types). For  $|\alpha| = \mu = 2^{<\kappa}$ , classify the characters, gap spectrum, density, and binary coding length of  $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$ . Begin with  $\alpha = \mu \cdot 2$  and  $\mu + n$ .

A prefix can now use  $\mu$  labels before exhausting its coordinate domain, so residual alphabets can be small, have endpoints, or be finite. The proof of [Theorem 30.11](#) suggests replacing the single residual-alphabet interface by a tree whose nodes carry their actual residual order types. A theorem expressed only in  $|\alpha|$  cannot capture all such effects.

**Research question 113.70** (Noncardinal birthday cutoffs). Replace the initial cardinal  $\kappa$  by an arbitrary infinite ordinal  $\theta$ . Which local invariants depend on the ordinal tail of  $\theta$ , not merely on  $|\theta|$  and  $|\mathbf{No}_{<\theta}|$ ?

Our cone argument uses  $\delta + \gamma < \kappa$  for smaller ordinals. That property fails at many noncardinal cutoffs. A useful first target is a cutoff of the form  $\kappa + \omega$ , followed by  $\kappa \cdot 2$ , with the contributions of each birthday layer kept separate.

**Research question 113.71** (Isomorphism after the invariant calculation). When do two minimal permutation orders with the same values of  $r, q, \mu^{<\mu}$  and the same gap spectrum have the same order type? Which additional invariants distinguish them?

The explicit gap classification provides more information than binary equimorphism, but still need not determine an order. Candidate invariants include the arrangement of cylinder-boundary cuts, densities of particular convex suborders, and the interaction of branch cuts with a chosen prefix filtration. No general isomorphism conclusion is asserted here.

**Research question 113.72** (Beyond the proved fusion length). Determine the least size of a family of dense open sets whose intersection fails to be dense. When  $q = \omega$ , determine complete metrizability and more precise descriptive topological properties directly from the permutation presentation.

[Theorem 30.5](#) supplies a lower bound: every family of size at most  $q$  succeeds. For longer constructions, intermediate prefixes can already have length  $\mu$ , so the simple small-history argument no longer applies. This is an obstruction to that proof, not a proof that longer fusion always fails.

### 113.10.2 Global branches, automorphisms, and class realizations

**Research question 113.73** (The extendible automorphism group). Describe the stabilizer of the represented-cut predicate in  $\text{Aut}(\mathcal{P}_s(b))$ . Can one characterize it by preservation of fewer predicates than the entire labelled cylinder and coverage structure?

[Theorem 47.5](#) identifies this stabilizer abstractly, and [Theorem 47.7](#) shows it is proper. The remaining task is a useful structural description. Numerical relabellings give a natural family, but need not exhaust it.

**Research question 113.74** (Changing baselines). For two baselines  $b, b'$ , when does an automorphism of  $\mathfrak{W}_{\text{sl}}$  carry  $\mathcal{P}_s(b)$  onto  $\mathcal{P}_s(b')$ ? How does the answer depend on their eventual-agreement classes and on their represented-cut predicates?

Relabelling by an increasing numerical bijection is sufficient when it carries one baseline to the other. Abstract order-isomorphism of the two cores is not sufficient by itself; the obstruction proved here explains why the represented cuts must enter the analysis.

**Research question 113.75** (Recovering a class realization). What information about  $\mathcal{X}$  can be reconstructed from the labelled core together with its admitted represented cuts? What is lost if the baseline and cylinder predicates are forgotten?

[Theorem 48.4](#) detects every proper same-set extension through new represented cuts. Its pair encoding recovers an individual ordinal subclass from a suitable enumeration. A stronger reconstruction should specify the decoding language and distinguish existence of individual codes from a universal truth or satisfaction predicate.

**Research question 113.76** (Iterated proper-class blocks). Determine the precise recursion strength needed to organize successive canonical Ord-blocks of a raw well-order along a prescribed index order. Compare finite iteration, set-length iteration, and class-length iteration.

One extraction is elementary; iterating a procedure whose state is a class can require more. The distinctions involving ETR in [22] provide the appropriate setting for a reverse- mathematical analysis. Our direct comparison and branch recognition do not by themselves solve this recursion problem.

**Research question 113.77** (The condensed unrestricted order). Classify the dense comparison obtained by collapsing the finite components of [Theorem 54.4](#). Which residual fibres remain detectable from its bare order, and what embeds into it?

Finite adjacency is now fully described. The quotient can still contain nontrivial fibres with infinite or proper-class residual carrier. It would be incorrect to conclude from density alone that it is the set-like order or an ordinary surreal-sized class.

### 113.10.3 Effective bounds, coherence, and verification

**Research question 113.78** (Ordinal and birthday optima for supported completion). Refine the cardinal reserve formula to optimal ordinal support bounds and optimal birthdays of canonical support codes. Which optima are uniform without an additional global choice parameter?

Two fresh reserves of the same cardinality can occupy very different ordinal intervals. [Theorem 49.7](#) isolates the unavoidable cardinality; ordinal placement and coding complexity remain separate. A constructive version should specify the representation of partial injections and the operation used to choose a complement bijection.

**Research question 113.79** (Coherent cutoff embeddings). Construct embeddings between the orders at different birthday cutoffs which compose exactly and preserve a useful prefix filtration. Determine which can also preserve cut types or finite condensation.

Appending an arbitrary tail preserves some comparisons but can change the required enumeration type. A coherent reserve mechanism should be chosen once for the whole system rather than separately at each pair of cutoffs. The support theorem may help isolate its minimal size costs.

**Research question 113.80** (First checked milestones). Formalize the reserve theorem, elementary common-prefix comparison, and set-sized cut trichotomy before attempting the class automorphism theorem. Can the abstract alphabet interface support other saturated or non-Archimedean ordered alphabets without changes to the proofs?

These milestones have explicit input and output types and no dependency on a formal universe of all class well-orders. Their value is not merely in reproducing finite tests, but in certifying the limit and cardinal arguments that the tests cannot inspect.

**Research question 113.81** (Typed constructions indexed by presentations). Which algebraic or model-theoretic constructions can use the set orders  $\mathcal{W}_\alpha(X)$  or uniformly coded global subfamilies as index orders, while keeping every support and evaluation map within a specified size level?

A promising first investigation should use a bounded index set, verify that the output is independent of the chosen presentation in the intended sense, and only then consider a uniform class version. The analogy with Kanovei–Shelah is a guide to construction, not permission to use an uncoded higher-order totality as a set.

[merge] Status of source 17's questions, 3 October 2026. [Research question 113.77](#) is answered externally for countable  $\omega$ -standard models by source 18: there the collapsed order is the rational order, densely coloured by the factorial block sizes ([theorem 58.6](#)); its internal classification is open. [Research question 113.69](#) is answered for the binary coding length and the density of the strata  $\mu \cdot \gamma + n$  by sources 15 and 16 ([theorems 27.2](#) and [31.2](#)); characters and gap spectra beyond the minimal slice are open. [Research question 113.76](#) is answered in its block clause by the block decomposition ([note 52.1](#)). [Research questions 113.70](#) and [113.79](#) are source 12's [research questions 113.12](#) and [113.19](#). Source 17's own table of the earlier questions it addresses (in [section 4](#)) is corrected there. The others remain open.

## 113.11 Source 18: revised research questions and priorities

The following are proposed directions left after the theorems above. They are not presented as a list of previously catalogued open problems. Each states the extra information that is still needed.

### 113.11.1 Bounded order types and their completions

**Research question 113.82** (The arrangement of gaps). For fixed  $\kappa$ , classify  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  up to order isomorphism, or classify its completion together with the subset of realized points. Which features go beyond the complete character spectrum and the multiplicities proved here?

The new spectrum and counting formulas turn this into an arrangement problem. In particular, a large family of gaps of one character may have a highly nontrivial induced order. A first target is  $\kappa = \omega$ , followed by  $\omega_1$  under specified continuum arithmetic. Equimorphism with a binary cube does not settle this.

**Research question 113.83** (Longer fixed enumeration types). Determine gap spectra and binary coding lengths for  $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$  when  $\alpha > \mu + 1$ , especially  $\alpha = \mu \cdot 2$  and  $\mu \cdot \gamma + n$ .

Density and cellularity are now known for every  $\alpha$  of cardinality  $\mu$ , and finite-tail point characters are explicit. Later infinite blocks can use up an entire  $\mu$ -sized residual region, so the minimal-type proof cannot simply be repeated.

**Research question 113.84** (Noncardinal birthday bounds). For a noninitial ordinal  $\theta$ , determine the spectrum of  $\mathcal{W}_{|\mathbf{No}_{<\theta}|}(\mathbf{No}_{<\theta})$ . Which ordinal features of  $\theta$  survive in interval sizes and cut characters?

The present proof uses the closure  $\beta + \gamma < \kappa$  for  $\beta, \gamma < \kappa$ , which may fail for  $\theta$ . The appropriate replacement is a stratified description of sign cones, not a substitution of  $|\theta|$  for  $\theta$ .

**Research question 113.85** (Coherent enlargement of cutoffs). Construct embeddings between order spaces for increasing birthday cutoffs that compose exactly, and specify whether they preserve minimal enumeration length, prefix cylinders, finite condensation, or selected cut characters.

Appending a fixed tail gives an embedding of full well-order spaces, but need not preserve minimum-length strata or compose with a second independently chosen tail. A global rank-layer schedule gives a candidate for coherent full spaces; the sharper minimal-stratum requirements remain separate.

### 113.11.2 Class cuts, blocks, and automorphisms

**Research question 113.86** (Iterating the quotient rather than removing blocks). Choose a precise limit rule and determine the strength of iterating  $R \mapsto B_R$  along specified class well-orders. Which fixed points, periodic behaviors, or invariants occur?

The single decomposition and ordinary ordinal-indexed block removal are elementary. The finite-support example is already a fixed point up to isomorphism. Any claim that repeated condensation automatically reduces complexity must confront that example. The axiomatic strength of the genuinely iterated operation is not determined here.

**Research question 113.87** (Order structure on represented class cuts). Study the class cuts satisfying the straddling-prefix criterion, with predicates marking exhaustive branches and residual tail types. Which natural automorphisms of the bounded-support core preserve that distinguished collection of cuts?

The criterion gives a recognition theorem. It does not classify the automorphism action on all class cuts, or show that every automorphism of the core extends to the global order. Baseline changes differing on a proper class of positions are a concrete first case.



**Research question 113.88** (The expanded countable model). Classify  $(\mathfrak{W}_{\text{all}}^{\mathcal{M}}, \mathfrak{W}_{\text{sl}}^{\mathcal{M}})$ , and then add the evaluation relation to the surreal labels. Which features depend on the countable class realization once the pure order has become the same factorial shuffle?

The pure order cannot define set-likeness even with finite parameters. An added predicate can therefore retain information that the shuffle classification necessarily discards.

**Research question 113.89** ( $\omega$ -nonstandard class models). Describe the external finite-condensation blocks and the elementary theory of  $\mathfrak{W}_{\text{all}}$  for countable  $\omega$ -nonstandard GBC models. Do these orders share the first-order theory of the standard factorial shuffle, while realizing additional types?

Internally finite permutation cylinders become externally infinite discrete intervals. Standard finite adjacency chains inside them already invalidate the standard classification. Determining the possible global assembly of these intervals is a concrete model-theoretic problem; no positive answer to the question is claimed here.

### 113.11.3 Words, encoding, and formalization

**Research question 113.90** (Sharp word-code bounds). Optimize the birthday of explicit order embeddings of injective and increasing words, recording separately preservation of numerical order, prefix extension, and simplicity. Can the centered encodings be made comparably efficient at singular stages?

The duplicated-sign delimiters provide transparent proofs, rather than optimal codes. Uniform tree isomorphisms obtained by back-and-forth supply little direct birthday control.

**Research question 113.91** (The centered increasing order itself). Classify the entire centered increasing-word order up to isomorphism and determine the intrinsic definability of its all-negative cones and its positive-limit points.

Its local characters, its complete set-cut obstruction, and the two types of rooted cones are known here. The contrast between that intrinsic information and the displayed entry signs makes the automorphism problem meaningful.

**Research question 113.92** (Weak choice and uniform constructions). Which word-tree isomorphisms and class-core identifications can be proved without Global Choice? For a specified operation, distinguish existence, a uniform definable implementation, and an effective implementation on concrete codes.

The explicit sign embeddings and many cut arguments need no Global Choice. Our class isomorphisms use deterministic global back-and-forth. These are different strength claims; a uniform family of class maps should not be selected without proof.

**Research question 113.93** (A first machine-checked milestone). Formalize the elementary common-prefix relation and the set-interval decomposition for a supplied class-model relation, alongside the set-level ordinal spectrum and minimal permutation cut trichotomy. Can one order-theoretic interface serve all four?

This is a realistic modular target because it avoids almost all surreal arithmetic. The initial deliverables should expose their size assumptions in the theorem signatures and make the conventional proof-to-formal-statement correspondence reviewable.

[merge] Status of source 18's questions, 3 October 2026. Source 18 was written before sources 13–17 arrived. [Research question 113.83](#) is answered in part by source 15 (binary coding lengths of  $\mu \cdot \gamma + n$ , [theorem 31.2](#)); [research question 113.87](#) in part by source 17's extension criterion ([theorem 47.5](#)), which shows that a core automorphism extends to the global order exactly when it preserves the represented cuts; [research question 113.92](#) bears on source 16's [research question 113.65](#). [Research questions 113.84](#) and [113.85](#) are source 12's [research questions 113.12](#) and [113.19](#); [research question 113.90](#) is source 11's [research question 113.6](#). The others remain open.

### 113.12 The questions of sources 19–30: index and status (3 October 2026)

[merge] Sources 19–30 ask 139 questions (13, 12, 12, 12, 10, 12, 11, 12, 12, 12, 12 and 9), printed per source in the following subsections as proposed continuations, not as a census of open problems; every source says so. Many repeat each other or an earlier question; the table groups them, names the earlier question each continues, and records the status after the whole batch-83 addition. A question answered by another batch-83 source is marked in the dated note after its source's subsection. The status of the earlier questions that batch 83 answers fully or in part is recorded in [section 113.25](#).

| Topic                                                          | Asked by                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Status, 3 October 2026                                                                                                             |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Coding length at singular non-strong-limit residual cardinals  | 19: <a href="#">research question 113.94</a> ; 29: <a href="#">research question 113.212</a> ; 30: <a href="#">research questions 113.224</a> and <a href="#">113.225</a> (continuing 12: <a href="#">research question 113.11</a> )                                                                                                                                                                                                                                                                                                                                                                            | open; cube spectrum settled ( <a href="#">theorem 64.11</a> )                                                                      |
| Isomorphism type of fixed strata; gap arrangements             | 19: <a href="#">research questions 113.95</a> and <a href="#">113.96</a> ; 20: <a href="#">research questions 113.115</a> and <a href="#">113.116</a> ; 21: <a href="#">research questions 113.123</a> and <a href="#">113.127</a> ; 22: <a href="#">research questions 113.131</a> to <a href="#">113.133</a> ; 24: <a href="#">research questions 113.156</a> and <a href="#">113.157</a> ; 26: <a href="#">research questions 113.180</a> to <a href="#">113.182</a> ; 28: <a href="#">research questions 113.207</a> and <a href="#">113.208</a> (continuing 12: <a href="#">research question 113.13</a> ) | open; in part at the countable alphabet and for $\mu + \omega$ (notes after each source)                                           |
| Noncardinal cutoffs and the countable birthday band            | 19: <a href="#">research question 113.97</a> ; 21: <a href="#">research question 113.122</a> ; 22: <a href="#">research question 113.138</a> ; 29: <a href="#">research question 113.213</a> (= 12: <a href="#">research question 113.12</a> )                                                                                                                                                                                                                                                                                                                                                                  | in part (Part XI); order types open                                                                                                |
| Coherence across cutoffs and lengths                           | 19: <a href="#">research question 113.105</a> ; 20: <a href="#">research question 113.113</a> ; 21: <a href="#">research question 113.124</a> ; 22: <a href="#">research question 113.139</a> ; 27: <a href="#">research question 113.198</a> (= 12: <a href="#">research question 113.19</a> )                                                                                                                                                                                                                                                                                                                 | in part ( <a href="#">theorems 74.8</a> and <a href="#">75.1</a> ); across a change of cardinality open                            |
| Singular topology of the minimal slice; complete metrizability | 24: <a href="#">research questions 113.153</a> to <a href="#">113.155</a> ; 26: <a href="#">research questions 113.176</a> to <a href="#">113.178</a> and <a href="#">113.183</a> ; 28: <a href="#">research questions 113.200</a> and <a href="#">113.202</a> (continuing 17: <a href="#">research question 113.72</a> )                                                                                                                                                                                                                                                                                       | complete metrizability <b>answered</b> negatively by source 26 ( <a href="#">theorem 69.18</a> ); finer descriptive questions open |

| Topic                                                      | Asked by                                                                                                                                                                                                                                                                                                                                                    | Status, 3 October 2026                                                                                                    |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Permutation groups and uniformities of the minimal slice   | 24: research questions 113.158 to 113.160; 26: research question 113.179; 28: research questions 113.201 and 113.203 to 113.206                                                                                                                                                                                                                             | 24's subgroup question <b>answered</b> by source 28; the rest open or in part                                             |
| Countable strata $\omega + n$ and later countable types    | 29: research question 113.214; 30: research questions 113.226 to 113.228; 22: research question 113.134                                                                                                                                                                                                                                                     | finite tails classified (theorems 68.8 and 68.9); later types open                                                        |
| Words, terminators and the centered product                | 23: research questions 113.143 and 113.144; 25: research question 113.173; 27: research questions 113.191 and 113.192; 29: research questions 113.215 to 113.217 (continuing 18: research question 113.91)                                                                                                                                                  | 23's first <b>answered</b> by source 16; the others open or in part                                                       |
| Relation-table scans                                       | 21: research question 113.129; 27: research questions 113.188 to 113.190; 29: research question 113.222; 30: research questions 113.229 and 113.230 (continuing 08: question 9)                                                                                                                                                                             | in part (theorems 84.4 and 84.13); open on <b>No</b> in general                                                           |
| The exhaustive-branch stabilizer; extendible automorphisms | 19: research question 113.100; 20: research question 113.107; 21: research questions 113.119 and 113.121; 22: research question 113.137; 25: research questions 113.165, 113.167 to 113.169 and 113.171; 27: research question 113.193; 29: research question 113.218                                                                                       | open; partial families (theorem 96.4 and corollary 96.12)                                                                 |
| Extension across residual fibres                           | 20: research question 113.108; 21: research question 113.120; 27: research question 113.195; 19: research question 113.99                                                                                                                                                                                                                                   | <b>answered</b> externally by source 19 (theorem 98.5); fibre types open                                                  |
| Which enrichments recognize exhaustion or the realization  | 19: research question 113.101; 20: research questions 113.109 and 113.110; 21: research question 113.126; 22: research questions 113.135 and 113.136; 23: research question 113.145; 25: research questions 113.166, 113.170, 113.172 and 113.174; 27: research questions 113.194 and 113.196; 29: research question 113.221; 30: research question 113.231 | 22's label question <b>answered</b> by source 25; labelled traces suffice, setwise cores do not; minimal enrichments open |
| Changes of the class realization                           | 19: research questions 113.102 and 113.103; 20: research question 113.117; 21: research question 113.128; 24: research question 113.162; 26: research question 113.186; 30: research question 113.232 (continuing 11: research question 113.8)                                                                                                              | in part (theorem 102.3 and note 102.16); uncountable models open                                                          |

| Topic                                              | Asked by                                                                                                                                                                                                                                                                                                                                            | Status, 3 October 2026                                                                                                                                               |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Condensation histories and their strength          | 19: research question 113.104; 20: research questions 113.111, 113.112 and 113.114; 21: research question 113.125; 22: research question 113.140; 23: research questions 113.147 to 113.150; 29: research questions 113.219 and 113.220 (continuing 11: research question 113.3)                                                                    | 20's research question 113.111 <b>answered</b> over GBC by source 29 (theorem 108.19); relation to ETR open; <i>batch 96</i> : answered over GB (section 113.29 (a)) |
| Residual class alphabets and the first-block trace | 19: research question 113.98                                                                                                                                                                                                                                                                                                                        | open                                                                                                                                                                 |
| Unrestricted tails and their completions           | 24: research questions 113.161 and 113.163; 26: research questions 113.184 and 113.185; 28: research questions 113.209 and 113.210                                                                                                                                                                                                                  | open; <i>batch 89</i> : research question 113.209 in part (section 113.27)                                                                                           |
| Bounds and choice strength                         | 23: research questions 113.146 and 113.151; 27: research question 113.197                                                                                                                                                                                                                                                                           | open                                                                                                                                                                 |
| First formal milestones                            | 19: research question 113.106; 20: research question 113.118; 21: research question 113.130; 22: research question 113.142; 23: research question 113.152; 24: research question 113.164; 25: research question 113.175; 26: research question 113.187; 27: research question 113.199; 28: research question 113.211; 29: research question 113.223 | proposals; nothing is formalized                                                                                                                                     |
| Long strata coding lengths                         | 22: research question 113.141                                                                                                                                                                                                                                                                                                                       | <b>answered</b> by source 19 except one regime                                                                                                                       |

### 113.13 Source 19: further research questions after the new results

These questions are proposed continuations of the present work. They are not asserted to be a catalogue of historically established open problems. The distinction matters because a question may already have an answer in the wider literature even when the project has not yet identified it.

#### 113.13.1 Fixed enumeration types and bounded completions

**Research question 113.94** (The remaining coding gap). When a singular non-strong-limit residual cardinal at most  $\kappa$  produces different lower and upper weighted ranks  $\varrho_{\text{lo}} < \varrho_{\text{up}}$ , determine the least binary coding length of  $\mathcal{W}_\alpha(\mathbf{No}_{<\kappa})$ . Is the lower bound always attained? A stronger and especially useful target is an embedding into  $\mathbb{B}_{\varrho_{\text{lo}}} \times n!$ .

The embedded-cube spectrum is already settled by the constant-cylinder argument. The unresolved issue is an upper coding construction, not the existence of a still larger binary cube inside the order. The generic subproblem is whether  $\mathcal{W}_\lambda(Y)$  embeds into  $\mathbb{B}_\lambda$  for every alphabet  $Y$  of singular non-strong-limit size  $\lambda$ . Birthday-controlled insertion shows that all such alphabets occur in the relevant residual range when  $\lambda \leq \kappa$ .

**Research question 113.95** (From equimorphism to isomorphism). Classify the actual order types of the fixed strata whose binary coding ranks and cube spectra agree. Which additional arrangements of numerical gaps, residual alphabets, and finite cylinders distinguish equimorphic examples?

The first-cutoff theorem gives a warning: two familiar topological models can have the same ordered completion  $\mathbb{R}$  and different complements inside it. The canonical decorated-word completion is a more informative object than a list of cardinal invariants.

**Research question 113.96** (Gap arrangements in longer strata). Extend the bounded normal forms from minimum-length permutations to every fixed enumeration type, using the residual lexicographic sum and the cardinal-layer decomposition. Compute the order of the gap loci, together with their point characters and multiplicities.

Infinite tails introduce several successive residual alphabets. The completion should record those transitions; merely adding the minimal-stratum spectra cannot determine how the gaps are arranged.

**Research question 113.97** (Noninitial birthday bounds). For an arbitrary ordinal  $\theta$ , study  $\mathcal{W}_{|\mathbf{No}_{<\theta}|}(\mathbf{No}_{<\theta})$  and its longer strata. Which features depend on  $\theta$  as an ordinal rather than just on its cardinality?

The full-interval-size and sign-cone arguments above use the closure of an infinite initial cardinal under sums of two smaller ordinals. A noninitial bound requires a finer decomposition of its sign tree.

### 113.13.2 Residual class alphabets and automorphisms

**Research question 113.98** (Stability of first-block separation). Analyze the set-deletion criterion of Proposition 99.3 in residual surreal alphabets defined by sign, birthday, or algebraic conditions. Which natural embeddings, convex restrictions, or extensions of the class realization preserve injectivity of the initial-word trace map?

The surreal fresh-label property is sufficient: at a first difference  $a < b$  after a set prefix, an unused label strictly between them gives an exhaustive separator. Requiring every open interval of  $Y$  to remain nonempty after deleting any set guarantees this step. The exact obstruction is now an initial Ord-segment and a final Ord\*-segment exposed by deleting a set, with admitted witnesses. The Ord + Ord\* counterexample shows that cardinality of the carrier alone is insufficient. Applying the criterion systematically, and tracking the admission of its witnesses under a change of class realization, would make recursive residual analysis more effective.

**Research question 113.99** (Residual fiber isomorphism types). Determine which orders  $\mathcal{W}^{\mathcal{M}}(Y)$  occur as fibers over core cuts and when two such fibers are isomorphic. Can useful invariants be extracted from the numerical order on  $Y$  without classifying its entire family of well-orders?

This is the remaining mathematical content in the automorphism extension criterion. The decorated reconstruction is exact, but a decoration by the full fiber order type is not by itself a simple classification of those fibers.

**Research question 113.100** (Which core automorphisms extend?). Compute the decoration-preserving subgroup of  $\text{Aut}(C)$  in specified external class models. How does it depend on the chosen baseline enumeration and on a change of baseline with proper-class support?

Preservation of represented versus unrepresented cuts is necessary but may omit the residual order type. The exact criterion requires preserving that full type. It would be valuable to replace it with more concrete combinatorial invariants in important models.

**Research question 113.101** (Enrichments that remember the class theory). Expand the pure global order by its set-like suborder, a selected core, its prefix tree, and an evaluation relation. Which fragments of the underlying class theory become interpretable in these successively richer structures?

The common pure first-order theory cannot distinguish even  $\omega$ -standardness. This makes the question of the smallest informative enrichment precise. Retaining the numerical alphabet and the prefix-evaluation relation is a natural first experiment.

### 113.13.3 External models and foundational strength

**Research question 113.102** (Uncountable external isomorphism spectra). Classify  $\mathfrak{W}_{\text{all}}^{\mathcal{M}}$  and  $\mathfrak{W}_{\text{sl}}^{\mathcal{M}}$  for specified uncountable class models. What additional conditions on their quotients and nonstandard internally finite cylinders are forced by saturation or by a full-class realization?

The common elementary theory is already fixed. Uncountable isomorphism types must retain information beyond finite games, including cut cofinalities and the external geometry of internally finite intervals. The full-class model at an inaccessible cardinal is a concrete first case.

**Research question 113.103** (Admitted versus amenable branches). Study the order and descriptive complexity of the persistent external cuts whose bounded prefixes are all internally coded but whose full branches are not admitted. How does this family change when the class realization is enlarged while the set universe is kept fixed?

Theorem 100.1 produces continuum many such cuts in every countable realization. The next issue is how these cuts are arranged and which extensions admit selected branches without adding sets.

**Research question 113.104** (Iterating set-interval condensation). Specify a limit-stage rule for iterating  $R \mapsto J_R$  and determine the exact recursion or comprehension principles needed for the resulting construction. Which fixed points and strictly changing orbits occur?

The one-step quotient is elementary and may preserve isomorphism type. Any iteration theorem must therefore prove its own structural or well-foundedness claim rather than infer descent from the word “quotient.”

### 113.13.4 Coherence and formal proof development

**Research question 113.105** (Coherent enlargement maps). Construct a directed system of embeddings between birthday-cutoff well-order spaces, with explicitly stated preservation of prefixes, finite-tail cylinders, or selected completion data.

The restriction theorem proves that arbitrary carrier restriction cannot provide the required monotone maps. A construction should instead specify how new labels are inserted and how the schedules compose. Full well-order spaces and minimum-length strata impose different scheduling requirements.



**Research question 113.106** (A small certified core of the theory). Formalize the restriction classification, first-disagreement comparison, prefix-code concatenation, and cylinder fusion as general set-level lemmas, then derive the weighted fixed-type spectrum from them.

This would convert the most reusable part of the present analysis into a compact foundation for later work. It also provides a clear test of which cardinal-arithmetic lemmas and ordinal-sum bounds are actually needed by the proof.

[merge] Status of source 19's questions, 3 October 2026. [Research question 113.94](#) is open; sources 29 and 30 ask it again ([research questions 113.212](#) and [113.224](#)). [Research question 113.95](#) is open; at the countable alphabet the finite-tail strata are pairwise nonisomorphic ([corollary 68.7](#)). [Research question 113.96](#) is open beyond the characters of  $\mathcal{W}_{\mu+\omega}$  ([theorem 63.15](#)). [Research question 113.97](#) is answered in part by source 21 ([theorem 77.4](#)) and source 29 ([theorem 76.5](#) and [corollary 78.10](#)); the order type is open. [Research questions 113.98](#) to [113.100](#), [113.102](#) and [113.103](#) are open. [Research question 113.101](#) is answered in part: the labelled global enumerations of a realization recover its whole class part (sources 22, 27, 30: [theorem 102.3](#) and [notes 102.5](#) and [102.16](#)), while a setwise-named core does not ([proposition 102.19](#)). [Research question 113.104](#) is answered over GBC for canonical histories: they exist for all class well-orders exactly when strong class comparability holds (source 29, [theorem 108.19](#)), and a supplied history terminates (source 20, [theorem 108.1](#)); the limit convention is [corollary 108.14](#). [Research question 113.105](#) is answered in part by source 20's append-tail system ([theorem 74.8](#)) and source 21's fixed-slot embeddings ([theorem 75.1](#)). [Research question 113.106](#) is a proposal.

### 113.14 Source 20: a revised research agenda

The following questions concern extensions not established by this article. They are project proposals, except where their background is explicitly cited. The previous broad question whether *every* core automorphism extends has a negative answer in [theorem 97.7](#); it should no longer appear as unresolved.

**Research question 113.107** (The extendible subgroup). Give useful generators or intrinsic descriptions of the subgroup of core automorphisms preserving exhaustive represented cuts. How does it depend on the chosen baseline and the displayed prefix structure?

[Note 97.8](#) supplies an exact criterion, but it does not compute that subgroup. A concrete starting point is the automorphism group of the labelled prefix structure together with its exhaustion predicate. Numerical relabelling alone does not stabilize a fixed core, by [proposition 97.9](#).

**Research question 113.108** (Extensions across raw residual fibers). Which core automorphisms preserving all injective represented cuts extend to the entire raw comparison structure? Formulate necessary and sufficient conditions including the order types of residual fibers and the uniformity of the chosen fiber identifications.

Different raw well-orders can have the same first ordinal block and hence the same core cut. A cut-preservation criterion alone cannot choose their relative images. The local-size blocks of [theorem 107.5](#) offer a first manageable family of fibers.

**Research question 113.109** (The expanded countable structure). Classify the external pairs  $(\mathfrak{M}_{\text{all}}^{\mathcal{M}}, \mathfrak{M}_{\text{sl}}^{\mathcal{M}})$ , or expand further by prefix evaluation. Which distinctions between class realizations survive after the pure order has collapsed to the two types in [note 109.13](#)?

The pure order forgets almost all class-theoretic information. Set-likeness is a natural first predicate to restore, but its distribution inside singleton or larger cut fibers can depend on admitted branches. A full answer should specify whether the underlying set model is fixed.

**Research question 113.110** (Effective theory with additional predicates). Determine the theory and decision complexity after adding a named set-like fragment, selected prefix predicates, or a baseline evaluation relation to the pure countable order.

The pure order theory is decidable by [corollary 109.22](#); adding the enumeration data can reintroduce much more information. It is useful to identify the smallest expansion that detects a change of class realization, rather than assuming that the pure order already does so.

**Research question 113.111** (The strength of complete condensation histories). Over GB or GBC, what is the exact strength of the assertion that every supplied class well-order admits its nested least-representative condensation history through the input order plus one stage?

[Theorem 108.1](#) proves the terminal behavior conditional on the history and gives ETR as a sufficient existence principle. A reversal would require encoding a genuinely class-valued recursion into this particular elementary operator. Merely observing its transfinite appearance would not establish such a reversal.

[merge] (4 October 2026, batch 96.) Answered over GBC by source 29 and over GB by sources 34 and 35: the assertion is equivalent to strong comparability CWO ([theorems XVI.5.8](#) and [108.19](#) and [note XVI.5.19](#)); whether it implies ETR is open ([research question 113.219](#)).

**Research question 113.112** (Alternative limits for quotient iteration). Which other mathematically natural limit rules give canonical condensation histories, and how do their heights compare with  $h_\kappa(\theta)$  in the full model?

The nested-carrier convention retains embedding information at limits. An isomorphism-type-only process discards that information. One should first define any replacement rule as an actual operator and examine the self-similar examples before seeking a general height theorem.

**Research question 113.113** (Embeddings that retain more than order). Can birthday-stage embeddings preserve designated cylinders, finite condensation blocks, or chosen cut characters while composing exactly? What is possible for specified minimal enumeration strata?

The append-tail maps in [theorem 74.8](#) solve coherence for all enumeration types. They need not stay in a prescribed minimal stratum. The local-rank criterion identifies the obstruction to set-likeness; preserving exact local ordinal types is a stricter requirement.

**Research question 113.114** (Exact birthday profiles and club diagnostics). Classify the attainable functions  $\kappa \mapsto \text{otp}(R \upharpoonright \mathbf{No}_{<\kappa}^2)$  and the associated classes  $\text{Min}_R$  of strong-limit cutoffs at which the minimum type is attained. Which finer invariants remain after passing to equivalence modulo the club filter?

[Theorem 74.7](#) detects set-likeness:  $\text{Min}_R$  contains a club in the set-like case and misses a club otherwise (source 20 writes  $A_R$ ). Unboundedness alone does not suffice, as the sparse-beth example shows. A natural next step is to apply analogous diagnostics to the successive canonical ordinal blocks and ask whether their combined profiles recover more of a raw class well-order.

**Research question 113.115** (Gap arrangements beyond spectra). Classify the ordered pair consisting of the Dedekind completion of  $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$  and its distinguished realized subset. At the countable cutoff, how much of the prefix tree is recoverable from that pair?

[Theorem 67.11](#) identifies the countable completion and the topological complexity of the realized points. Equal gap counts and equal topology do not by themselves identify the arrangement of the missing cuts. At uncountable stages a generalized topology must first be specified.

**Research question 113.116** (Longer strata and noninitial birthdays). Determine the full proper-gap spectra and their arrangement for  $\mathcal{W}_{\mu,2}(\mathbf{No}_{<\kappa})$  and longer strata, and for birthday cutoffs which are not initial cardinals.

The known density transition and binary coding ranks do not determine these arrangements. At noninitial cutoffs, ordinal tails can destroy the closure of the cones used in the cardinal-cutoff proofs. A first target is a single additional infinite enumeration block with a fixed finite terminal tail.

**Research question 113.117** (Robustness under class extensions). For extensions of the class collection with the same sets, characterize which new represented cuts, automorphisms, and condensation histories can appear while all old comparisons remain fixed.

The absoluteness result in [section 111.6](#) rules out changing the well-foundedness of an old common relation. The remaining freedom lies in new relations and in new uniform witnesses. This sharper formulation avoids a false source of model dependence.

**Research question 113.118** (A verified generic first-divergence library). Formalize common-prefix comparison, two-label persistence, finite changes along restrictions, and the sharp coherent-gluing criterion for arbitrary ordered carriers. Then instantiate at the inspected ProveIt sign carrier.

This is a concrete first milestone: its dependencies are relation restriction, set-valued recursion, ordinal first difference, and well-foundedness. It would verify several central mechanisms before the more demanding task of formalizing the class automorphism and nonstandard-model arguments.

[merge] Status of source 20's questions, 3 October 2026. [Research question 113.107](#) is open; it is asked again by sources 21, 25, 27 and 29 ([research questions 113.119](#), [113.165](#), [113.193](#) and [113.218](#)), with partial results in [theorem 96.4](#) and [corollary 96.12](#). [Research question 113.108](#) is **answered** in the external formulation by source 19, which source 20 did not see: a core automorphism extends to the admitted raw orders of a set-sized model exactly when it preserves the isomorphism types of the residual fibres ([theorem 98.5](#)). [Research question 113.109](#) is answered in part for countable models by source 30 ([proposition 102.19](#)) and, for labelled data, by [theorem 102.3](#) and [note 102.16](#). [Research question 113.110](#) is open. [Research question 113.111](#) is **answered** over GBC by source 29: the assertion is equivalent to strong comparability of class well-orders ([theorem 108.19](#)); its relation to ETR is open ([research question 113.219](#)). *Dated note, 4 October 2026 (batch 96)*: answered over GB as well ([section 113.29](#) (a)). [Research question 113.112](#) is answered in part ([corollary 108.14](#): under the nested rule the limit equivalence is the union of the earlier ones). [Research question 113.113](#) is answered within a cardinality band by source 21 ([theorem 75.1](#)); across cardinalities it is open ([research question 113.124](#)). [Research questions 113.114](#) and [113.115](#) are open; the countable completion is described further by sources 21, 24 and 29 ([theorems 67.2](#), [67.10](#) and [68.2](#)). [Research question 113.116](#) is answered in part: characters at  $\mu + \omega$  ([theorem 63.15](#)) and the spectrum level at noncardinal cutoffs (Part XI); the gap spectra are open. [Research question 113.117](#) is answered in part by the reconstruction theorems ([theorem 102.3](#) and [notes 102.5](#) and [102.16](#)). [Research question 113.118](#) is a proposal.

### 113.15 Source 21: further research questions and concrete targets

The following are proposed continuations. They are not a claim that each question has already appeared as an established open problem in the literature. Each is formulated after the results

proved here, so it does not ask again whether core extension always works or whether noncardinal cutoffs have computable basic invariants.

**Research question 113.119** (Classify the extendable subgroup). For a fixed baseline  $b$ , describe  $\text{Stab}_{\text{Aut}(\mathcal{P}_{\text{bd}}(b))}(\mathcal{C}_{\text{exh}})$  by useful generators, definability conditions, or an intrinsic expansion of the core. Which such descriptions avoid explicitly naming every exhaustive represented cut?

The stabilizer criterion is exact, and the subgroup is proper. The remaining task is a structural description. Cofinalities alone cannot suffice: exhaustive, nonexhaustive, and vertical large cuts are conjugate in the unexpanded core order.

**Research question 113.120** (Extension to the full class-well-order space). Which core automorphisms preserve the represented injective cuts and the order types of their residual fibers, and when can their induced fiber isomorphisms be made uniform?

The full extension problem requires more than the set-like criterion: two orders with the same initial word may have different residual well-orders. A sufficient theorem should construct the fiber maps explicitly or state the additional choice principle used to assemble them. The vertical-cut obstruction already rules out arbitrary extension.

**Research question 113.121** (Prefix-preserving automorphisms). Classify automorphisms of the displayed prefix tree that preserve its exhaustive branches. Which numerical relabelings and coordinate-dependent successor-order isomorphisms generate them?

The countable theorem shows that bare-order automorphisms can destroy the cylinder family. Even a tree automorphism need not visibly preserve the global requirement that every label eventually occurs. This asks for the precise extra condition, rather than assuming that local compatibility implies exhaustivity.

**Research question 113.122** (Noncardinal cut spectra and isomorphism types). Determine the complete cofinality spectrum of missing cuts in  $\mathcal{W}_{2|\theta|}(\mathbf{No}_{<\theta})$  for noncardinal  $\theta$ . When do two such spaces with equal basic invariants have the same order type?

For example, the cutoffs  $\omega + 1$ ,  $\omega \cdot 2$ , and  $\omega^2$  have the same cardinality, density, global cofinalities, point characters, and binary coding rank calculated here. These equalities do not prove an isomorphism. The finite numerical end blocks and the different sign-prefix structures suggest concrete cut configurations to compare first.

**Research question 113.123** (Fixed nonminimal enumeration types). Extend fixed-type cardinal and character formulas to full isomorphism and embedding classifications for  $\mathcal{W}_\alpha(\mathbf{No}_{<\theta})$ , with  $|\alpha| = |\mathbf{No}_{<\theta}|$ , and describe the relative closures of these strata.

Existing finite-terminal and density calculations are not by themselves a classification. The first-cardinality decomposition and closure theorem give a starting point that tracks the unused alphabet, not only the ordinal length. A useful next case is to compare consecutive finite terminal lengths while retaining the prefix-fiber structure.

**Research question 113.124** (Coherence across a change of cardinality). Can the fixed-slot construction be replaced by embeddings between minimum-length spaces when their cardinalities change, with strict composition identities along a specified chain of birthday cutoffs?

Within a fixed band, [theorem 75.1](#) supplies such a system. Across a cardinal jump the coordinate domains change; across a proper class of cutoffs, the uniform family itself requires a size audit. Specify whether the embeddings should preserve the order topology, prefix structure, or both.

**Research question 113.125** (The uncountable colored quotient). For uncountable external realizations, classify the quotient of  $\mathfrak{W}_{\text{all}}$  by its canonical terminal permutation blocks, with the factorial-size colors retained. Which saturation and homogeneity properties follow from the underlying class realization?

The countable standard and nonstandard types are completely described here. Their local block analysis remains available in larger models, but dense colors no longer imply uniqueness of the quotient order. This is a natural setting for separating cardinal invariants from actual order type.

**Research question 113.126** (How much model structure does a marked core retain?). When are two pairs  $(H_{\mathcal{X}}, \mathcal{P}_{\text{bd}}(b))$  and  $(H_{\mathcal{Y}}, \mathcal{P}_{\text{bd}}(c))$  isomorphic if the core is preserved setwise but its individual codes are allowed to move?

Pointwise identification of a common core detects proper class expansions through represented cuts, while the unmarked countable orders are always rational. The intermediate classification should be expressed through the action of core automorphisms on represented-cut families. It could distinguish model-theoretic information that neither extreme viewpoint captures.

**Research question 113.127** (Automorphism and descriptive complexity of the countable-alphabet space). Give an explicit description of the subgroup of  $\text{Aut}(\mathbb{R}, <)$  preserving the embedded set  $P = \mathcal{W}_{\omega}(\mathbb{Q})$  in its Dedekind completion. Classify its orbits on omitted cuts and determine the complexity of the branch-exhaustivity condition in a chosen completion code.

The Baire homeomorphism, finite homogeneity, and continuum many missing cuts are already proved. A sharper description would separate vertical numerical cuts, omitted-range branch cuts, and any further cut types by explicit invariants.

**Research question 113.128** (Presentation orders under forcing). Under hypotheses preserving old class well-order types, which lexicographic presentation orders acquire new cuts or new automorphism obstructions? What changes after identifying old and new carriers through specified interleavings?

Preservation of old relations is proved above; preservation of all new abstract types is a different issue studied under stronger hypotheses in [27]. New sign sequences can enlarge the carrier even when ordinals are unchanged. A useful first target is a forcing extension adding one real, with the old carrier named explicitly.

**Research question 113.129** (Relation-table comparison with extra structure). Which choices of the coordinate enumeration of  $\mathbf{No}^2$  give isomorphic expanded relation-table orders, and which preserve a specified set-coded dense family? Can any of them recover the numerical first-prefix comparison after adding a small amount of structure?

Binary equimorphism does not settle these questions. The two-label example already rules out identifying the comparisons on the nose.

**Research question 113.130** (Formal verification of the obstruction). Formalize first the small-carrier prefix comparison and fixed-slot coherence theorem, then the abstract cut-reconstruction and nonextension argument over a universe-relative class interface.

The theorem dependencies are short and arithmetic-free. A verified result should include an axiom report and an explicit smallness boundary. In particular, the final nonextension theorem should be checked as a statement about permitted uniform operators or about a specified larger-universe collection, not an undeclared type of all classes.



[merge] Status of source 21's questions, 3 October 2026. Research question 113.119 is open (with source 20's research question 113.107). Research question 113.120 is **answered** in the external formulation by source 19 (theorem 98.5); the uniformity of internal fibre isomorphisms is open. Research question 113.121 is answered in part: tree automorphisms preserving the core and every cylinder act on branches with exactly two orbits (theorem 90.1); label-coherent ones are coordinatewise (theorem 93.1, and source 27's note 96.5); exhaustion-preserving families are in theorem 96.4 and proposition 96.11. Research questions 113.122, 113.124, 113.125 and 113.128 are open. Research question 113.123 is answered in part (embedding spectra by sources 19 and 30, theorem 64.11 and note 64.18; cardinal invariants by source 22, theorem 63.4); isomorphism is open. Research question 113.126 is answered for countable models by source 30: the two pairs are always isomorphic, by an isomorphism fixing the baseline (proposition 102.19). Research question 113.127 is answered in part (the omitted cuts form a dense meagre  $F_\sigma$  of size  $\mathfrak{c}$ , theorem 67.4; complexity of the  $E_n$ , theorem 68.2). Research question 113.129 is answered in part at set level by source 27 (theorems 84.2 to 84.4). Research question 113.130 is a proposal.

### 113.16 Source 22: further research questions and proposed projects

The following questions are continuations of the present program. They are not advertised as a list of historically established famous open problems. In particular, the fixed-length density/cellularity/weight table and the redundancy of cylinder heights are now treated as proved results rather than left open.

#### 113.16.1 Residual orders and topology

**Research question 113.131** (Nonminimal cut spectra). Classify the nonprincipal cut-character spectra of  $\mathcal{W}_{\mu+n}(\mathbf{No}_{<\kappa})$ ,  $\mathcal{W}_{\mu+\omega}(\mathbf{No}_{<\kappa})$ , and  $\mathcal{W}_{\mu\cdot 2}(\mathbf{No}_{<\kappa})$ . Which cuts arise inside a residual fibre, and which occur at a boundary where a limiting prefix loses extendibility?

The general first-disagreement formula settles point characters only after the relevant residual orders are understood. A cut may follow a coherent prefix chain whose union exhausts too many labels. Such a failure is not detected by the three cardinal topological invariants.

**Research question 113.132** (The dense quotient orders  $Q_n(X)$ ). When are  $Q_n(X)$  and  $Q_m(X)$  isomorphic for distinct finite  $n, m$ ? Does the answer depend only on the numerical order type of  $X$ , or on additional saturation properties not implied by richness?

They share the invariants computed in proposition 63.8, but equality of those invariants is not an isomorphism theorem. An isomorphism would sharpen the finite-tail normal forms and compare their nonisolated derivatives.

**Research question 113.133** (Long-tail local characters). Describe the possible point characters for  $\alpha = \mu \cdot 2$  by a tractable invariant of the residual alphabet. Can the iterated first-disagreement reduction be organized as a useful normal form rather than an arbitrary tree of residual orders?

The main topological table already applies. The remaining issue is local order structure, including endpoints of residual permutation orders and cofinalities of their extremal fibres.

**Research question 113.134** (Descriptive topology when cofinalities are countable). For which alphabets and lengths are these first-countable spaces metrizable, completely metrizable, or representable by familiar ordered spaces? In particular, analyze the split space at  $\mu + 2$  without mistaking separability for second countability.



The countable-alphabet examples provide controlled test cases:  $\omega + 2$  has many jumps,  $\omega + 3$  has a dense set of isolated points, and  $\omega + \omega$  has no jumps but continuum cellularity.

### 113.16.2 What structure must be retained?

**Research question 113.135** (Eliminating label data). After unranked cylinder incidence is supplied, what is the weakest additional relation that recognizes exhaustive branches? Can label occurrence be replaced by a coarser invariant, or can two incidence presentations with identical unlabelled data disagree on the exhaustive-trace predicate?

Height is no longer an unknown ingredient: [theorem 87.1](#) recovers it. The open issue is label coverage. No claim that it is eliminable, or provably indispensable in every presentation, is made here.

**Research question 113.136** (Reconstruction without fixed coding parameters). How much of [theorem 102.3](#) survives when  $b$  or  $g$  is not named? Can one formulate a functorial reconstruction up to isomorphism of two-sorted models, rather than inclusion over an identical set universe with shared parameters?

The elementary decoder is exact once the parameters are fixed. Removing them introduces an automorphism and definability problem, not a failure of the existing decoder.

**Research question 113.137** (Automorphisms preserving admitted traces). Characterize the automorphisms of the unranked incidence structure which preserve exhaustive traces in a given class realization. Which core-order automorphisms extend to this incidence structure, and which preserve the binary-swap trace substructure after changing the baseline?

The earlier reports supply extension obstructions for pure core automorphisms. The present reconstruction suggests that preservation of admitted traces can constrain the induced action on the entire class sort.

### 113.16.3 Bounds, foundations, and verification

**Research question 113.138** (Noncardinal birthday cutoffs). For an ordinal cutoff such as  $\kappa + \omega$  or  $\kappa \cdot 2$ , determine exactly which intervals retain full alphabet cardinality. Extend the topological classification to fragments where richness fails or varies from interval to interval.

The proof of [note 62.15](#) uses  $\delta + \eta < \kappa$  for smaller ordinals at an initial-cardinal cutoff. That mechanism must be replaced, not assumed, for arbitrary ordinal cutoffs.

**Research question 113.139** (Coherent length and cutoff embeddings). Construct embeddings among different birthday fragments and enumeration strata which compose exactly and preserve a useful prefix filtration. Determine the optimal additional support or reserve needed for this coherence.

An arbitrary appended tail preserves a comparison but may change the prescribed order type or destroy compatibility with another embedding. The earlier optimal support-cardinality theorem is a natural input; optimal ordinal placement remains separate.

**Research question 113.140** (Iterated raw heads and recursion strength). Determine the exact class-recursion strength needed to iterate the canonical common-head or ordinal-head decomposition through set and class lengths. Which bounded iterations are elementary, and which imply fragments of ETR?

The elementary direct comparison in [note 111.17](#) does not settle this. A reverse-mathematical treatment should specify whether a stage is a set object or a whole class relation.

**Research question 113.141** (Binary coding lengths for long strata). Extend the earlier binary equimorphism and singular strong-limit coding-length results to nonminimal lengths. How do finite jumps, infinite residual orders, and failures of residual extendibility affect the least binary-word length admitting an order embedding?

The constant ordinal embedding spectrum does not answer this finer question. Nor does cardinality  $2^\mu$  alone determine a binary lexicographic coding length.

**Research question 113.142** (First kernel-checked milestones). Formalize the residual-cylinder identity, the first-disagreement character formula, and height recovery from unranked incidence. Then instantiate the richness and fresh-separator interfaces using the repository's sign carrier.

These milestones are small enough to isolate precise size hypotheses but substantial enough to verify the limit arguments which finite regression checks cannot reach. Only after they are checked should the class-realization decoder be encoded in an explicit model interface.

[merge] Status of source 22's questions, 3 October 2026. [Research questions 113.131 to 113.133](#) are open (for  $\mu = \omega$  the finite-tail spaces are classified by sources 29 and 30, [theorems 68.8](#) and [68.9](#)). [Research question 113.134](#) is answered for the minimal lengths by source 26 (completely metrizable exactly at  $\mu = \omega$ , [theorem 69.18](#)) and for the lengths  $\omega + n$  of a countable alphabet by sources 29 and 30 (same references). [Research question 113.135](#) is **answered** by source 25: unranked cylinders, the core order and any set of parameters do not recognize exhaustive branches ([theorems 91.1](#) and [91.2](#)), and equality of labels across edges does ([theorem 93.1](#)); a minimal relation is open ([research question 113.166](#)). [Research question 113.136](#) is answered in part: source 27 needs only the baseline ([note 102.5](#)), source 30 needs no common baseline in advance ([note 102.16](#) and [corollary 102.22](#)). [Research question 113.137](#) is answered in part by source 25 ([theorems 90.1](#) and [93.3](#)). [Research question 113.138](#) is answered in part (Part XI). [Research question 113.139](#) is answered in part ([theorems 74.8](#) and [75.1](#)). [Research question 113.140](#) is answered for canonical histories ([theorem 108.19](#)). [Research question 113.141](#) is **answered** by source 19 except at singular non-strong-limit residual cardinals at most  $\kappa$  ([theorem 64.11](#); source 30's route, [note 64.18](#)). [Research question 113.142](#) is a proposal.

## 113.17 Source 23: further research questions and proposed projects

The termination question for the two unrestricted set-word carriers is settled by [Theorem 81.9](#); the questions below begin beyond that result. The explicit polynomial hierarchies likewise answer an existence problem, not the full classification or proof-strength problem for arbitrary raw orders.

### 113.17.1 Word orders and enriched lexicographic structure

**Research question 113.143** (Classification of the nonsaturated marker orders). For a fixed infinite regular  $\theta$ , are all word orders arising from marker cuts of type  $(\theta, \text{Ord})$  isomorphic? Does the answer change between arbitrary and injective words? Which cut spectra, beyond point characters, decide the answer?

The character computation gives necessary invariants but not a complete classification. A focused first step is a cut reconstruction theorem for the nonsaturated word orders, keeping their named cylinders.

**Research question 113.144** (Restricted increasing-word repairs). Which length restrictions or endpoint decorations make an increasing-word order saturated at a prescribed small cardinal? Can one characterize the obstruction by cofinal subsets of the terminally available alphabet at each prefix, rather than by one global marker?

Theorem 82.1 rules out every fixed marker for all set lengths. Prefix-dependent markers or extra continuation labels are substantive new structures, not exceptions to that theorem.

**Research question 113.145** (Intrinsic versus named cylinders). For which expansions of the outer linear order are prefix cylinders, termination points, or inherited monotonicity recoverable? Can one find a minimal additional relation that distinguishes these presentations without naming every prefix individually?

The previous work's prefix non-rigidity makes a claim of recovery from bare order alone unsafe. The new admissible-marker isomorphisms supply further examples in which the same linear order has different alphabet presentations.

**Research question 113.146** (Quantitative birthday bounds). For a small input cut of words, bound the separator's length and the birthdays of its letters in terms of the input and forbidden-label bounds. Can the two-cut fresh-separator construction be arranged to use one prescribed extra birthday level?

The crossing chain already gives a set length bound from the input. Optimizing the label bound would connect the abstract proof directly to `exists_cut_separator_of_birthday_lt`.

### 113.17.2 Raw orders and condensation

**Research question 113.147** (Proof strength of general condensation iteration). Over GBC, what is the exact strength of the assertion that every raw class well-order admits its set-interval condensation hierarchy along a given class well-order? Is a restricted form of ETR sufficient and necessary, or does this particular operator admit a weaker construction?

Theorem 108.8 shows that long explicit examples by themselves cannot establish a lower bound on the recursion principle. A reversal would need to encode genuinely arbitrary recursive information.

**Research question 113.148** (Which orders eventually collapse?). In a theory providing the required hierarchy, does every raw class well-order collapse at some admitted class-well-order stage? How should collapse length be represented without an illegitimate order type object for every proper-class well-order?

The polynomial theorem realizes every prescribed admitted length. It does not give a universal bound or prove eventual collapse for all raw orders.

**Research question 113.149** (Lexicographic distribution of internal invariants). Can Theorem 108.23 be sharpened from existence in every set-like interval to a uniform embedding of a specified parameter order, with its points carrying prescribed condensation hierarchies? Which such families remain definable from a fixed baseline enumeration?

The quantifier order matters: a single class parameter  $\Gamma$  is allowed in the proved theorem. A uniform family ranging over all class well-orders needs an explicitly higher-order interpretation.

**Research question 113.150** (Condensation versus finite residual adjacency). How do set-interval blocks inside a raw well-order interact with finite adjacency components in the outer lexicographic order? Can an enriched outer order describe the distinction between an exhausted finite residual tail and a proper residual class?

These are different condensations on different orders. Treating them as one operation would erase the key carrier distinction.

### 113.17.3 Foundations and implementation

**Research question 113.151** (Choice strength of the new classification). What is the weakest class theory proving the separator part of Theorem 81.9, with a supplied saturated alphabet and marker cut? Can its order-isomorphism conclusion be proved without a supplied global enumeration, for the special surreal presentations?

Most local proofs use only set-sized constructions with class parameters. The back-and-forth conclusion deliberately uses global choice. These should be optimized separately.

**Research question 113.152** (A checked pair of contrasting theorems). Formalize the admissible-marker theorem for the injective sign-carrier word order and, in the same framework, the failure for increasing words. Then formalize the explicit  $\omega$  polynomial hierarchy and its non-stabilizing limit, before addressing arbitrary  $\Gamma$ .

This yields a manageable development with positive and negative tests. The finite diagnostics supplied here can guide examples, but the kernel proof must retain the universe and cofinality distinctions.

[merge] Status of source 23's questions, 3 October 2026. [Research question 113.143](#) was **answered** by source 16 before source 23 was placed: all marker orders of type  $(\theta, \text{Ord})$  are  $\mathbf{T}_\theta$ , and arbitrary and injective words give isomorphic orders ([theorem 36.2](#)). [Research questions 113.144, 113.146 and 113.149 to 113.151](#) are open. [Research question 113.145](#) is answered in part, negatively: cylinders are not intrinsic ([theorem 47.12](#) and [corollary 67.9](#)) and not enough to recognize exhaustion ([theorem 91.1](#)). [Research question 113.147](#) is answered for canonical histories over GBC by source 29 ([theorem 108.19](#)); whether ETR is necessary is open. [Research question 113.148](#) is answered for supplied histories by source 20: a history along  $W + 1$  terminates ([theorem 108.1](#)), with exact heights in the full model ([corollary 108.3](#)). [Research question 113.152](#) is a proposal.

## 113.18 Source 24: further research questions and proposed projects

These are proposed continuations of the results proved here. They are not all claimed to be previously published open problems. Questions about structured cut recognition, prefix non-rigidity, optimal support reserves, and some automorphism extensions have already been addressed in the earlier drafts; a new project should use those results instead of restating them as untouched.

### 113.18.1 Singular topology and completeness

**Research question 113.153** (Complete metrizability at uncountable countable cofinality). Let  $\mu > \omega$  be singular with  $\text{cf } \mu = \omega$ . Is the forward-prefix space  $\mathcal{W}_\mu(X)$  completely metrizable? Equivalently, can the dense permutation subspace of its metrizable injection completion be characterized as a  $G_\delta$  subset? Determine whether the answer depends on cardinal arithmetic.

The equivalence is justified directly in Appendix 69.2; its ambient injection space is completely metrizable by Theorem 65.1 and Proposition 62.7. The question concerns the original topology, not the finer two-sided topology. The shift-type counterexample proves only the incompleteness of one uniformity. A constructive positive answer would need to enforce uncountably many coverage requirements while keeping the appropriate neighborhoods; a negative answer could be obtained by showing that no countable open intersection in the injection space characterizes exhaustiveness.

**Research question 113.154** (A singular replacement for the exhaustion tree). Classify  $\mathcal{W}_\mu(X)$  up to homeomorphism when  $\mu$  is singular. Can it be represented as the body of a tree with an explicit acceptance condition at prematurely full prefixes, rather than as the unrestricted body of a regular-height tree?

The exact obstruction in Section 66 suggests retaining both domain-length growth and coverage growth as separate coordinates. A useful result would identify an invariant that distinguishes singular spaces with the same cofinality and weight.

**Research question 113.155** (Dense-open intersection thresholds). Determine the least cardinality of a family of dense open subsets of  $\mathcal{W}_\mu(X)$  whose intersection is not dense. Compare that number with  $\text{cf } \mu$ , the exhaustion family in  $\mathcal{I}_\mu(X)$ , and any stronger point-preserving fusion properties.

The existing fusion theorem is a lower bound, not a complete classification. An intersection theorem in the permutation space and one in the injection completion are different statements: the latter has all the closed missing-label boundaries available.

### 113.18.2 Order structure beyond the topological classification

**Research question 113.156** (The embedded real-line model at countable size). Describe the complement  $\mathbb{R} \setminus \mathcal{W}_\omega(X)$ , up to increasing homeomorphism of  $\mathbb{R}$ , using its cylinder-edge, numerical-gap, and nonexhaustive-branch parts. Which of these parts are recoverable from the bare linear order and which require named prefix predicates?

The present paper gives its Borel class, category, density, and cardinality, but not a complete orbit classification of its placement in  $\mathbb{R}$ . The earlier non-rigidity results warn that a prefix-based description need not be intrinsic to the bare order.

**Research question 113.157** (Regular homeomorphisms versus order embeddings). Theorem 66.2 forgets the numerical alphabet. Which additional data on the exhaustion tree recover lexicographic order, and what order-embedding or order-isomorphism criteria can be proved from those data?

Even the countable case shows that recognizing Baire topology does not identify the order with its standard irrational model. Quantitative versions could track birthday bounds and support costs during the tree coding.

**Research question 113.158** (Completion-preserving automorphisms). Characterize those order or topological automorphisms of the permutation space that extend to its forward injection completion, its two-sided completion, or its Dedekind completion while preserving specified boundary types.

Every order isomorphism extends to the Dedekind completion, but preservation of the branch subspace is an extra condition. Earlier represented-cut criteria should be combined with the uniform-continuity obstruction, rather than replaced by a blanket extension claim.

### 113.18.3 Permutation algebra at singular cardinals

**Research question 113.159** (The boundedness-preserving subgroup). Analyze  $H_\mu$  intrinsically at singular  $\mu$ : its closure properties in the two ambient topologies, its normalizers, and the possible topological-group structures on larger natural subgroups. How much of the ordinal bounded-set ideal is recoverable from this group together with its topology?

Theorem 69.2 is an exact local criterion, but it does not classify the subgroup algebraically. The explicit transposition between a bounded cofinality-sized set and a cofinal set provides a basic test witness for any proposed extension.

**Research question 113.160** (Uniformities with an exhaustion interpretation). Classify compatible complete uniformities on the regular permutation space that admit a direct label-coverage interpretation. At singular sizes, determine which inverse-control conditions can be added without changing the original topology on a specified subspace.

The forward, two-sided, and exhaustion-tree uniformities already separate three natural notions. A useful classification should distinguish equality of topology, uniform equivalence, and equality of completion boundary.

### 113.18.4 Class theory, models, and formal verification

**Research question 113.161** (An axiomatization of virtual completion). Give a reusable formal theory of prefix-completion schemes for class-coded structures, with the exact hypotheses needed for coherent unions, small-net rigidity, and inverse reconstruction. Can the whole theory be developed over a fragment weaker than the full assumptions used here?

The small-net argument isolates Class Replacement as the source of the uniform disagreement bound. An appropriate reverse-mathematical analysis would test that dependency, rather than assuming all class-theoretic principles have the same strength.

**Research question 113.162** (Changing the class part without changing the sets). For models  $(M, \mathcal{X}) \subseteq (M, \mathcal{Y})$ , characterize which newly available injective branches are exhaustive and which new boundary traces they induce. Relate this to the existing preservation results for old, prefix-structured cuts.

A new class may be a new net, a new limit, or a new parameter defining a cut. Those are different mechanisms. Any forcing interpretation should separately verify the relevant class-forcing theorem and preservation hypotheses; none is assumed in this article.

**Research question 113.163** (Unrestricted tails and their completion theory). Develop analogues of the present completion constructions for arbitrary non-set-like labelled class well-orders. What prefix index structures are appropriate after a proper-class common segment, and how does finite-tail adjacency interact with those structures?

An Ord-word model cannot merely be reused: it forgets the very tail that makes such an order non-set-like. The elementary common-prefix relation is a starting point that does not presuppose a comparison theorem for all class order types.

**Research question 113.164** (A small trusted Lean core). Formalize, without admitted new mathematical axioms, the short-prefix extension lemma, the cylinder-base theorem, forward completion, and the regular exhaustion-tree homeomorphism. Then instantiate them at  $\mathbf{No}_{<\kappa}$  and separately formalize the class-model comparison theorem.



This would convert the main set-sized advances into checked mathematics while preserving the explicit boundary around global class semantics. The inspected simplicity interface can be reused when simplest separators are needed, but its existence hypothesis must still be discharged by an independent small-cut theorem.

[merge] Status of source 24's questions, 3 October 2026. [Research question 113.153](#) is **answered negatively** in ZFC by source 26: for uncountable  $\mu$  of countable cofinality  $\mathcal{W}_\mu(X)$  is not Borel in  $\mathcal{I}_\mu(X)$  and has no compatible complete metric, on any nonempty open subspace ([theorems 69.17](#) and [69.18](#)); the answer does not depend on cardinal arithmetic ([note 69.7](#)). [Research question 113.154](#) is answered in part by the same theorems and by closed universality ([theorem 69.20](#)); the homeomorphism type is open. [Research question 113.155](#) is open (asked again as [research question 113.183](#)). [Research questions 113.156](#) to [113.158](#), [113.161](#) and [113.163](#) are open. [Research question 113.159](#) is **answered** in the natural action by source 28:  $H_\mu$  is the largest subgroup containing the finitary permutations on which the prefix topology is a group topology, it is self-normalizing at singular  $\mu$ , the bounded ideal is recovered from it with its action, and its closure and completion are described ([theorems 70.4](#), [70.6](#), [70.10](#) and [70.26](#) and [proposition 70.8](#)). [Research question 113.160](#) is answered in part by source 28's group envelope and its two-sided completeness ([theorems 70.18](#) and [70.27](#)). [Research question 113.162](#) is answered in part by the reconstruction theorems ([theorem 102.3](#) and [note 102.16](#)). [Research question 113.164](#) is a proposal.

### 113.19 Source 25: further research questions and proposed projects

The questions below continue beyond the results proved here. They are proposed research problems arising from this investigation, not assertions that each is a previously published open problem.

**Research question 113.165** (The exhaustive-branch stabilizer). Characterize the subgroup of cylinder-preserving core automorphisms whose induced action permutes  $\mathfrak{W}_{\text{sl}}$ . Can it be described by local compatibility conditions on the increasing maps between residual alphabets, rather than the global requirement that every branch still cover every label?

The represented-trace criterion is exact but global. [Theorem 92.2](#) shows that the entire cylinder group is too large. The countable rational-alphabet case is already a substantial, concrete version of this classification problem.

**Research question 113.166** (Recognition without full rigidity). Find natural enrichments strictly weaker than edge-label equality that recognize exhaustive branches while still admitting nontrivial automorphisms. Which relations are necessary to connect label occurrences across different depths, and which merely overdetermine that information?

Our negative result rules out recognition from the unlabelled core and any set of parameters. Our positive enrichment is sufficient and rigid. It does not establish a minimal language between those two extremes.

**Research question 113.167** (Preserving a support filtration). Let  $\mathcal{P}_s(b)_\alpha$  consist of core branches whose transition permutation is supported below  $\alpha$ . What remains of the automorphism group after requiring preservation of every  $\mathcal{P}_s(b)_\alpha$ , or a cofinal subsequence of these levels? Does a support filtration recover any part of edge-label coherence?

The present back-and-forth preserves membership in the whole core but does not control the support of each image. The filtration is additional structure, distinct from the prefix-height relation, which is already recovered by cylinders.

**Research question 113.168** (Larger configurations and exact obstruction patterns). Which uniformly class-indexed configurations admit a simultaneous profile-preserving automorphism? Is there a useful criterion combining boundedness of divergence constraints at each stage with the avoidance of already determined noncore limit branches?

Theorem 95.1 supplies one necessary obstruction. A full characterization would need to distinguish failure of one-point realization from failure of completing an otherwise compatible partial automorphism.

**Research question 113.169** (Singular-cardinal automorphism groups). For singular  $\kappa$ , compare the cylinder structures of bounded-support and less-than- $\kappa$ -support cores. Determine their orbit invariants, first-disagreement realization spectra, and automorphism-group cardinalities for saturated fragments or specified surreal birthday alphabets.

One must keep track of  $|\mathbf{No}_{<\kappa}| = 2^{<\kappa}$  as well as cf  $\kappa$ . Remark 95.2 shows that even the underlying prefix families can differ. The regular formula  $2^\kappa$  cannot simply be transferred by notation.

**Research question 113.170** (Two baselines at once). Given global enumerations  $b, c$ , study the structure carrying both cores  $\mathcal{P}_s(b)$  and  $\mathcal{P}_s(b)_c$  and their common prefix tree. How does the transition permutation  $b^{-1} \circ c$  affect the joint automorphism group and recognition of exhaustive branches?

The elementary intersection question has an immediate answer: the two cores coincide exactly when  $b$  and  $c$  eventually agree, and otherwise they are disjoint. Indeed, a common core member would express both baselines as set-supported modifications of that member, making their transition permutation set-supported. A second, disjointly positioned core may constrain the symmetries of the first without naming every label; that joint structure is the unresolved part of the question.

**Research question 113.171** (The bare order of exhaustive enumerations). Classify order automorphisms of  $\mathfrak{W}_{\text{sl}}$  without assuming that they preserve a chosen core or its cylinder family. Which invariants of the enriched structures survive after both are forgotten? How do answers depend on the admitted class realization and on permitted uniformity of maps?

The earlier pure-order nonrigidity results and the present cylinder-preserving nonextension theorem constrain different directions. Neither is a classification of the full bare-order automorphism problem.

**Research question 113.172** (Raw tails and outer-order invariants). Can structural properties of the non-set-like tail of a raw class well-order be recognized from the outer lexicographic order, perhaps after adding a natural incidence relation? Which convex-fibre invariants distinguish raw well-orders that share the same initial set-like head?

Earlier condensation constructions produce complicated encoded tail orders [51]. The complication of an order encoded by a point is not automatically an invariant definable from the outer order of points. An explicit reconstruction or non-invariance argument is needed.

**Research question 113.173** (The increasing-word analogue). For the subtree of numerically increasing set-length words, determine the exact one-point extension property and its obstructions. Can an appropriate language of cross-coordinate inequalities yield an orbit classification analogous to Theorem 90.1?

The present automorphism group does not preserve this subtree. A successful analogue must retain more than sibling order and divergence, but perhaps less than full equality of labels across all edges.

**Research question 113.174** (Class-realization reconstruction after forgetting labels). Earlier continuation work reconstructs an admitted class realization from exhaustive traces over a labelled core. What part of that reconstruction survives under unlabelled isomorphism of core-and-cylinder structures? Which additional relations recover the lost distinction between exhaustive and nonexhaustive traces?

Theorem 91.1 prevents a direct label-free transfer of the coverage test. The question is now to identify surviving invariants, not to assume that the original decoder remains canonical in the reduct.

**Research question 113.175** (A checked minimal development). Formalize the regular set theorem, the countable two-orbit case, and the edge-coherent rigidity theorem with a common abstract alphabet interface. Can the maximum/no-maximum extension lemma be isolated as a reusable ordered-tree theorem independent of surreal numbers?

The suggested first deliverable is a kernel-checked one-point lemma plus supported completion. These are the two nontrivial inputs on which the longer back-and-forth proof depends.

[merge] Status of source 25's questions, 3 October 2026. [Research question 113.165](#) is answered in part by source 27's exact obstruction and sufficient bounds ([theorem 96.4](#)) and source 29's fixed-stage family ([proposition 96.11](#) and [corollary 96.12](#)). [Research questions 113.166 to 113.169](#) and [113.171](#) are open (source 27 asks [research question 113.166](#) again, [research question 113.194](#)). [Research question 113.170](#): its intersection part is settled in its own commentary; source 20 shows that an increasing relabelling makes the two cores disjoint ([proposition 97.9](#)); the joint structure is open. [Research question 113.172](#) is open; source 23 shows that every interval between distinct set-like enumerations contains raw orders of any prescribed condensation complexity ([theorem 108.23](#)). [Research question 113.173](#) is answered in part by source 29's transitivity on the increasing tree ([theorem 96.7](#)). [Research question 113.174](#) is answered in part: the pure order and a setwise-named core lose the realization ([propositions 102.19](#) and [102.20](#)). [Research question 113.175](#) is a proposal.

## 113.20 Source 26: further research questions and proposed projects

The following questions are proposed continuations. They are not all claimed to be previously published open problems. The singular complete-metrizability question of the earlier draft is no longer on this list: it is answered by Theorem 69.18. Likewise, the existence of direct comparison for non-set-like orders and the basic support-core coding are established background, not unresolved proposals.

### 113.20.1 Descriptive complexity and closed subspaces

**Research question 113.176** (Generalized Borel complexity). Determine the sharp complexity of  $\mathcal{W}_\mu(X)$  in appropriate generalized Borel and projective hierarchies when  $\mu > \omega$  and  $\text{cf } \mu = \omega$ . Which hierarchy matches the actual weight of the prefix space, rather than merely the cardinality of its index set?

The ordinary non-Borel theorem is a lower bound, not a complete answer to a generalized hierarchy question. The exhaustion formula is a  $\mu$ -indexed intersection of opens, and should be compared with the definitions and weight assumptions in singular generalized descriptive set theory [58].

**Research question 113.177** (Closed universality without  $\mu > \mathfrak{c}$ ). For an uncountable cardinal of countable cofinality with  $\mu \leq \mathfrak{c}$ , characterize the separable metrizable spaces occurring as closed subspaces of  $\mathcal{W}_\mu(X)$ . Must a closed copy of  $\mathbb{Q}$  exist in ZFC?

The present coding handles complements indexed by fewer than  $\mu$  branches. It settles noncomplete metrizability in this regime but does not supply arbitrary Cantor traces. Improving the reservoir scheme or finding a genuinely different closed-subspace construction would be meaningful progress.

**Research question 113.178** (Higher cofinality coding). For cf  $\mu = \tau > \omega$ , is there an analogue of Theorem 69.13 with a bounded-topology binary space of height  $\tau$ ? What compactness, closed-image, and perfect-set substitutes remain valid?

A transfinite release schedule retains the least-disagreement combinatorics, but its topology is no longer ordinary compact metrizable Cantor space. The continuity and closed-image arguments must be replaced, not merely relabelled.

**Research question 113.179** (The exact role of bounded inverse images). Classify the topological and descriptive complexity of the intermediate spaces between  $H_\mu$  and  $\mathcal{I}_\mu^b \cap \mathcal{W}_\mu(\mu)$  (source 26's  $\mathcal{B}^+ \cap \mathcal{P}_\kappa(\kappa)$ ). Can the trace calculation involving  $A$  and  $\bar{A}$  be extended to higher closure or derived-set operations by additional inverse-control conditions?

Proposition 69.24 provides a concrete starting model: the inverse condition transforms arbitrary exclusion of  $A$  into exclusion of its topological closure. Any proposed hierarchy should state the controlled family of label sets exactly.

### 113.20.2 Enumeration lengths and order structure

**Research question 113.180** (A classification across nonminimal strata). For every ordinal  $\lambda$  with  $|\lambda| = |X|$ , classify the prefix topology of  $\mathcal{W}_\lambda(X)$  and determine when it agrees with the enumeration order topology. Which residual-tail invariants govern complete metrizability?

Successor lengths are prefix-discrete, whereas initial infinite cardinal lengths have the classification proved here. Limit lengths with long finite or small-cardinal residual configurations form the next natural family. A statement depending only on  $|\lambda|$  cannot suffice.

**Research question 113.181** (Closed order embeddings versus topological embeddings). For thick surreal alphabets, characterize which linear orders admit embeddings with closed image in the minimal enumeration space, and distinguish this from merely embedding their order topologies. Can prescribed Cantor traces be required to be convex or to respect named cylinder relations?

The marker construction gives closed images relative to the permutation space, but does not generally give convex images. Earlier prefix non-rigidity results caution that bare order automorphisms need not preserve an externally named prefix structure.

**Research question 113.182** (Embedding spectra refined by topology). Refine the earlier cardinal and ordinal embedding spectra by asking for continuous, closed, or prefix-preserving embeddings. Which of these spectra distinguish equimorphic surreal enumeration slices?

The new obstruction shows that mutual order embeddings alone leave substantial topology unresolved. The unconditional  $\beth_\omega$  example is a useful test case because it does not depend on additional cardinal arithmetic.

**Research question 113.183** (Dense-open intersection thresholds). Determine the least cardinality of a family of dense open subsets of  $\mathcal{W}_\mu(X)$  whose intersection is not dense, and compare it with the corresponding invariant of  $\mathcal{I}_\mu(X)$  and of  $H_\mu$ .

Theorem 69.12 settles the countable lower bound. The injection space has the explicit missing-label closed sets, while the bounded subgroup has the countable uniform-exhaustion description (31). These different structures may lead to different thresholds.

### 113.20.3 Foundations, global orders, and verification

**Research question 113.184** (A topology that retains unrestricted class tails). Develop a precise higher-sort or model-relative topology for arbitrary labelled class well-orders that distinguishes tails after a common proper-class prefix. How do the finite residual adjacent pairs sit in its completion theory?

Set-length prefix agreement alone is not Hausdorff on the unrestricted domain. The common-labelled-prefix formula gives an elementary comparison, but not by itself a full higher-order topology.

**Research question 113.185** (Reverse foundations of uniform approximation). What fragments of Class Replacement and choice suffice for small-net rigidity, the reserve extension theorem, and the diagonal noncoding theorem? Which implications can be reversed over a weak two-sorted base theory?

The proofs isolate different resources: set collection of disagreement bounds, completion of set partial permutations, and an elementary pair-swap diagonal. They need not all have the same axiomatic strength.

**Research question 113.186** (Changing admitted class families). For class-model extensions with the same sets, describe which new injective or exhaustive branches become available and how their traces interact with old uniformly coded cuts. What preservation principles keep the relevant prefix-completion schemes intact?

The problem concerns new relations and new uniform families, not a change in the internal well-foundedness of an old fixed relation under the GBC hypotheses explained in Section 71.3.

**Research question 113.187** (A minimal kernel-checked continuation). Formalize the compact omitted-label theorem, the non-Borel corollary, and the complete-metrizability classification on a general alphabet, then instantiate the result at  $\mathbf{No}_{<\beth_\omega}$ . How small can the trusted interface to classical descriptive set theory be made?

A useful milestone is a machine-checked exact range equation for the code, before any Borel hierarchy formalization. That theorem captures the main construction and can be reused independently of the final topological obstruction.

[merge] Status of source 26's questions, 3 October 2026. [Research questions 113.176 to 113.178](#) and [113.181 to 113.185](#) are open. [Research question 113.179](#) is answered in part by source 28's one-sided loci ([theorems 70.10 and 70.11](#)). [Research question 113.180](#) is answered in part for the lengths  $\omega + n$  of a countable alphabet ([theorems 68.8 to 68.10](#)). [Research question 113.186](#) is answered in part by the reconstruction theorems ([theorem 102.3](#) and [note 102.16](#)). [Research question 113.187](#) is a proposal.

### 113.21 Source 27: further research questions and priorities

The following questions are continuations motivated by the proved results. They are not asserted to be a catalogue of open problems recognized in the published literature.

**Research question 113.188** (Classify intermediate scan types). For a set carrier of cardinality  $\mu$ , classify density, cellularity, point characters, and finite convex blocks in  $\mathcal{W}_s(X)$  when the scan has type  $\vartheta$  with  $|\vartheta| = \mu$ . The cases  $\vartheta = \mu$  and a carefully chosen  $\mu + 6$  are settled here; the arrangement of dependencies between scanned pairs, not only  $\theta$ , may matter. A first target is the complete classification of countable scans of type  $\omega + n$ .

**Research question 113.189** (Recover the effective scan). Which oriented pair priorities are determined by the labelled order on well-orders? Swapping a single adjacent pair recovers its first orientation; swapping two disjoint adjacent pairs recovers their relative priority. Priorities of pairs sharing a label can be redundant because transitivity constrains their bits. Classify effective scans modulo equality of their induced labelled orders. The three-label diagnostic gives 36 orders from 48 scans.

**Research question 113.190** (The table-ordered bounded-support core). The same set-coded core is dense in the raw table order under a set-like scan, but its order now depends on that scan. Determine when it has endpoints, which uniform set cuts it fills, and which scan invariants control its isomorphism type. Its numerical-enumeration order is isomorphic to **No**; that fact cannot simply be transferred to its table order.

**Research question 113.191** (A sharper type for centered increasing words). Is

$$\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min} \cong \mathbf{T}_{\min}^+?$$

The product theorem answers the factorization problem, and (36) proves several sum identities, but neither settles this isomorphism. Determine a distinguishing invariant or construct an isomorphism. Equality of known local character spectra would not alone suffice.

**Research question 113.192** (Orders at genuine surreal gaps). Classify  $\mathcal{T}_{\varepsilon}^{\text{inc}}(\mathbf{No})$  (source 27's  $\text{Inc}_C(\mathbf{No})$ ) for cuts  $\mathbf{No} = L + R$  with one side of set cofinality or coinitality. [Proposition 82.8](#) computes point characters, and a gap with a set-coinitial right side is separated from the standard centered order. Do those characters classify all such constructions? A stronger classification of every unfilled set cut has not been proved here and should be treated as a new theorem to establish, not inferred from the point table.

**Research question 113.193** (The subgroup preserving exhaustive branches). Classify the ordered-prefix-tree automorphisms for which both the map and its inverse preserve exhaustive Ord-branches. The exact obstruction is an exhaustive branch through a subtree avoiding one output label. Uniform birthday bounds are sufficient but not necessary, and antichain tail transformations give additional examples. An ordinal-valued progress condition for the varying forbidden labels is a concrete next target.

**Research question 113.194** (Weaker enrichments than label equality). Cross-node equality of appended labels yields exactly global numerical relabellings. Which strictly weaker predicates already force preservation of exhaustiveness? Possible data include birthday strata, equality of birthdays, and incidence with designated bounded label sets. These enrichments should be tested separately; preserving the unlabelled tree and sibling orders is now known to be insufficient.

**Research question 113.195** (Raw extensions of core automorphisms). For a class part fixed externally, characterize automorphisms of a chosen core that extend to the full raw order, taking account of the complete tail fiber over each represented nonexhaustive branch. The set-like extension criterion concerns exhaustive traces; raw extensions must also preserve the residual orders.

**Research question 113.196** (Marked model reconstruction with fewer constants). [Note 102.5](#) fixes every core element pointwise. What smaller collection of predicates or distinguished subsets suffices to recover the class part, perhaps up to an automorphism of the set universe? Investigate expanded countable models before claiming an intrinsic invariant of the unmarked order.



**Research question 113.197** (Optimize the interpolant, not just a fixed prefix). The ordinal support optimum is exact for a prescribed partial injection. For a separated cut, many prefixes may yield an interpolant. Minimize the optimum over those prefixes, then compare ordinal support height with sign-code birthday. This may produce sharper constructive bounds for the dense-core realization theorem.

**Research question 113.198** (Controlled bounded approximations). Determine which class constructions admit coherent versions across birthday stages, especially singular and noninitial cutoffs. The set self-copy theorem here works at every infinite cutoff, but the homogeneous back-and-forth used for word and prefix isomorphisms requires additional closure at a fixed stage. Preserve the difference between full set-word trees and trees truncated before their terminal level.

**Research question 113.199** (First machine-checked theorem package). Implement the generic terminal-cut factorization, the six-coordinate scan theorem, and the support cutoff criterion first. These give small, independently useful milestones with explicit set-level statements. The next milestone should be the elementary labelled class-prefix comparison inside an explicit model, followed by the uniform diagonal theorem. This ordering resolves concrete foundational obligations before attempting the larger automorphism constructions.

### 113.21.1 What the continuation has resolved

The new results locate several boundaries more sharply than an ordinal spectrum alone can. A scan can retain that spectrum while changing isolation and density. An ordered prefix tree can retain every local numerical comparison while losing exhaustive coverage. An ordinary class core can be dense for relation tables while no uniform ordinary class family is dense for raw enumeration comparison. A support bound can be optimal in cardinality while far from optimal in ordinal height. The surviving questions ask which additional structure recovers the information lost at each boundary.

[merge] Status of source 27's questions, 3 October 2026. [Research questions 113.188, 113.189, 113.191 and 113.197](#) are open. [Research question 113.190](#) is answered in part by source 30's canonical max-index scan ([theorem 84.13](#)) and its density theorem ([theorem 84.11](#)). [Research question 113.192](#) is answered in part: actual thresholds and gaps with two large sides by source 29 ([proposition 82.14](#)); no marker makes increasing words saturated ([theorem 82.1](#)); gaps with one small side are open ([research question 113.216](#)). [Research question 113.193](#) is answered in part ([proposition 96.11](#) and [corollary 96.12](#)). [Research question 113.194](#) is open ([research question 113.166](#)). [Research question 113.195](#) is **answered** in the external formulation by source 19 ([theorem 98.5](#)). [Research question 113.196](#) is answered in part: a core named setwise with the baseline does not suffice in countable models ([proposition 102.19](#)). [Research question 113.198](#) is answered in part ([theorems 74.8 and 75.1](#)). [Research question 113.199](#) is a proposal.

## 113.22 Source 28: further research questions and topics

### 113.22.1 Singular topology and descriptive complexity

**Research question 113.200** (Complete metrizability of the full singular space). Let  $\mu > \omega$  be singular with  $\text{cf } \mu = \omega$ . Is  $\text{Sym}(\mu)$ , with its bounded-prefix topology, completely metrizable? Equivalently, is it a  $G_\delta$  subspace of the complete forward injection space?

The dense complete locus  $B_\mu^-$  and the Baire property do not decide this question. Nor does the incompleteness of the forward metric. A positive proof must organize all label-coverage requirements

into countably many ambient open requirements, or construct a complete compatible metric directly. A negative proof must rule out all compatible complete metrics, not just the chosen forward one. Appendix 69.9 supplies the precise equivalence.

**Research question 113.201** (The singular envelope’s exact character). Determine  $\text{cof}([\mu]^{<\mu}, \subseteq)$  for significant singular cardinals, and isolate which additional cardinal-arithmetic hypotheses yield an exact value between  $\mu^+$  and  $\mu^{<\mu}$ .

The elementary diagonal gives a uniform strict lower bound, not an exact formula. Further work should compare the cofinality of the whole small-subset poset with fixed-size subset cofinalities and with appropriate possible-cofinality methods, without replacing one invariant by another on the basis of notation alone.

**Research question 113.202** (Descriptive complexity of the one-sided loci). For uncountable cf  $\mu$ , determine the natural generalized Borel complexity of  $B_\mu^+$ ,  $B_\mu^-$ , and  $H_\mu$  in the forward injection and permutation spaces. For countable cofinality, classify  $B_\mu^-$  as a complete nonseparable metric space and its placement inside  $S_\mu$ .

The formulas in Section 70.3 are exact starting points. A classification should keep track of the actual weight  $\mu^{<\mu}$ , which can exceed  $\mu$ . The singular descriptive-set-theoretic framework of Dimonte and Motto Ros [58] is relevant background, but applying its results requires matching the space’s weight and topology rather than only its indexing cardinal.

### 113.22.2 Algebra and reconstruction

**Research question 113.203** (Abstract automorphisms versus the natural normalizer). Is every abstract or continuous automorphism of  $H_\mu$  induced by a permutation of its natural carrier? Under what hypotheses can the action, or the bounded-set ideal, be reconstructed intrinsically?

Theorem 70.6 only classifies conjugations by carrier permutations that preserve the subgroup. Proposition 70.8 reconstructs the ideal when the action is already named. These are useful constraints, but neither is an automatic solution of the abstract automorphism problem.

**Research question 113.204** (Subgroups without all finitary permutations). Classify natural subgroups not contained in  $H_\mu$  and not containing all finitary permutations on which the inherited forward topology is a group topology. Which non-discrete examples exist?

The maximality theorem identifies its exact hypothesis rather than forbidding every alternative subgroup. Such a classification should determine when internal transposition witnesses are absent and whether that absence reflects a useful structural constraint or merely a very small group.

**Research question 113.205** (Intermediate coordinate ideals). For admissible ideals  $\mathcal{J}$ , classify  $H_{\mathcal{J}}$ , the closure of  $H_{\mathcal{J}}$  in  $\tau_{\mathcal{J}}$ , and the gap between  $\mathcal{J}$  and  $\mathcal{J}^{\text{sym}}$ . Which ideals lead to self-normalizing ideal-preserving subgroups?

The general continuity and envelope theorems already apply. The self-normalizer proof for boundedness uses a special “bounded part plus cofinal part” lemma, so its extension should isolate an analogue of that lemma rather than assume every ideal behaves like the bounded ideal.

### 113.22.3 Numerical order and nonminimal enumerations

**Research question 113.206** (Order automorphisms and the coordinate bornology). Which order automorphisms of a thick minimal enumeration order preserve its forward uniformity, its injective completion, or the small-set group envelope? What additional named structure is needed to recover prefix cylinders or the chosen label filtration?

The earlier continuation [41] exhibits prefix non-rigidity, so a universal reconstruction claim from the pure order alone is not the right target. The question is to identify the appropriate expanded structure or the exact extension criterion.

**Research question 113.207** (Nonminimal ordinal lengths and residual tails). For  $\text{Bij}(\alpha, X)$  with  $|\alpha| = |X|$  but  $\alpha$  not the initial cardinal, develop analogues of the ideal-continuity and completion results. How do finite residual tails and limit positions change local character, adjacency, and the supply of extendible prefixes?

The initial-cardinal hypothesis in Lemma 62.12 should be replaced by exact residual-domain and residual-range conditions. This is likely a more informative approach than trying to transfer the minimal-slice theorems wholesale to the full fixed-carrier order.

**Research question 113.208** (Order-type classification beyond topology). Which numerical features of two thick alphabets force their minimal enumeration orders to be isomorphic, equimorphic, or nonisomorphic? Which invariants survive after their homeomorphic prefix spaces have forgotten the alphabet order?

The countable comparison with the irrationals already shows that topological classification is insufficient. Dedekind boundary types, interval decompositions, and the interaction with label order provide concrete candidates, whereas cardinality and ordinal spectrum alone are too coarse.

### 113.22.4 Class foundations and verification

**Research question 113.209** (Global control schemes over weaker theories). What are the weakest class-theoretic assumptions sufficient for the supported-prefix extension, small-net rigidity, and inverse-reconstruction schemes? Can their uses of Class Replacement and uniform set choice be separated sharply?

These proofs have different logical mechanisms: image bounding, choosing residual set bijections, and elementary eventual-value comprehension. A reverse-mathematical analysis should examine those mechanisms individually rather than assign the same strength to every global completion assertion.

[merge] (3 October 2026, batch 89.) Answered in part by source 31 (Part XV), which names this question. It separates three mechanisms, selection, collapse and coding, and image bounding (section XV.12). Over Zermelo set theory with Choice, selection restricted to countable inputs, with Separation for the selector, is already nonconservative (theorem XV.5.1, through an unreviewed interface to Glazer’s countermodels); in the full-class rank models global choice holds at every height, while ordinal coding and image bounding have different exact thresholds, and Class Replacement on a set holds exactly when its size is below the cofinality of the height (lemma XV.9.2, theorem XV.9.3, and corollary XV.9.4). The weakest assumptions for each of the three named schemes are not determined; source 31 restates that remainder as its research question 113.242 (section 113.27).

**Research question 113.210** (Non-set-like tails and class completion). Construct a completion theory for arbitrary class well-orders that uses the direct common-prefix relation and continues through proper-class initial segments. Which kinds of index orders, or external universe levels, are necessary to retain the tails?

An Ord-word completion cannot see a tail following a full Ord prefix. The finite-tail examples give mandatory tests for any proposed theory, and Theorem 71.13 supplies a comparison relation that does not presuppose a class-order-type calculus.

**Research question 113.211** (A verified ideal-to-surreal pipeline). Formalize the generic ideal-envelope and maximal-subgroup results, the singular self-normalizer, and the character diagonal, then instantiate them at a rigorously sized surreal cutoff. Develop the global class-model statements as a separate semantic layer.

This target directly extends the inspected source without conflating its simplicity theorem with cut existence. A successful implementation should record theorem-to-file mappings, universe levels, and the distinction between compiled proofs, imported assumptions, and model-theoretic hypotheses.

[merge] Status of source 28's questions, 3 October 2026. [Research question 113.200](#) is **answered negatively** in ZFC by source 26, which source 28 did not see: for uncountable  $\mu$  of countable cofinality,  $\text{Sym}(\mu) \cong \mathcal{W}_\mu(\mu)$  with the bounded-prefix topology is not a  $G_\delta$  in the forward injection space, so it is not completely metrizable, and no nonempty open subspace is ([theorem 69.18](#)). Source 28's statement that its own dense complete locus does not settle the question stays true ([note 70.13](#)). [Research questions 113.201, 113.203 to 113.205, 113.209 and 113.210](#) are open. [Research question 113.202](#) is answered in part at countable cofinality by source 26 ([theorem 69.23](#)). [Research question 113.206](#) is answered in part at  $\mu = \omega$ : prefix cylinders are not invariant ([corollary 67.9](#)). [Research question 113.207](#) is answered in part ([proposition 69.29](#); the lengths  $\omega + n$ , [theorem 68.8](#)). [Research question 113.208](#) is answered in part at  $\mu = \omega$  ([theorems 67.2 and 67.8](#)). [Research question 113.211](#) is a proposal. [merge] (3 October 2026, batch 89.) [Research question 113.209](#) is now answered in part by source 31 ([section 113.27 \(a\)](#)); the other questions listed above as open are not affected by it.

### 113.23 Source 29: further research questions after these results

The agenda below is revised in light of both the pinned source and the present proofs. Several questions in older lists have acquired answers: the centered increasing-word type has a product description; actual numerical thresholds reduce by translation; all noninitial strata have the stated residual analysis; and complete condensation histories have an exact equivalence with strong class comparability. The following questions concern what is still missing.

**Research question 113.212** (The remaining singular upper code). Let  $\lambda$  be singular and not strong limit. Under what hypotheses on a  $\lambda$ -sized ordered residual alphabet  $Y$  does

$$\mathcal{W}_\lambda(Y) \hookrightarrow \mathbb{B}_\lambda$$

hold? In particular, settle the residual cases  $\lambda \leq \kappa$  in [Corollary 78.9](#) for which the ordinal bounds  $\varrho_{\text{lo}} < \varrho_{\text{up}}$  remain different after absorption.

The exact embedded-cube spectrum does not answer this question. A proof must construct an upper code, or exhibit an obstruction stronger than the absence of a larger binary cube. The countable and finite-aleph birthday bands avoid the unresolved residual case and supply useful fully classified test instances.

**Research question 113.213** (Isomorphism on a constant-cardinality birthday band). For fixed  $\alpha$  of cardinality  $\mathfrak{c}$ , determine when

$$\mathcal{W}_\alpha(\mathbf{No}_{<\theta}) \cong \mathcal{W}_\alpha(\mathbf{No}_{<\eta}), \quad \omega < \theta, \eta < \omega_1.$$

Do some of these orders distinguish  $\theta$  and  $\eta$  even though all have the same equimorphism type given by [Corollary 78.10](#)?

The native alphabets themselves form a strict embeddability chain. The problem is to identify which of that information survives in the pure permutation order. Gap arrangement, local completeness, and the distribution of residual order types are possible invariants; the finite-tail topology illustrates why cardinal spectra alone can miss a distinction.

**Research question 113.214** (Infinite tails at the countable alphabet). Classify the native topology and Borel structure of  $\mathcal{W}_\alpha(\mathbb{Q})$  for countable  $\alpha \geq \omega \cdot 2$ . Determine their exact relation to the coordinate Borel structure, and identify natural standard Borel subspaces or quotients.

The article compares the coordinate and native topologies at every countable length, but gives the complete measurable classification only for finite tails after  $\omega$ . For longer types, the tail is an infinite order varying with the omitted alphabet. It cannot be replaced by a constant finite split. The cellularity argument in [Proposition 68.11](#) already implies that all these full native Borel structures are non-countably-generated and hence nonstandard: they contain  $2^\mathfrak{c}$  distinct open sets, whereas a countably generated sigma-algebra has at most  $\mathfrak{c}$  members.

**Research question 113.215** (Intrinsic geometry of the centered product). What are the order automorphisms of  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ ? Are its displayed lower-prefix fibers recognizable from the pure order, perhaps after naming a small amount of structure? Is there a simpler canonical presentation that explains its full cut arrangement?

The factorization is an explicit isomorphism theorem. It does not assert that its fibers are definable from the resulting bare order. The earlier failures of prefix recovery make this a real additional problem rather than a notational reformulation.

**Research question 113.216** (Nonprincipal termination cuts). Classify increasing-word orders whose distinct terminator lies at a nonprincipal class cut with a set cofinal family on the lower side or a set cointial family on the upper side.

The factorization in [Note 82.9](#) reduces the question to increasing words over the two cut sides. Actual surreal thresholds and nonprincipal cuts whose two sides both retain the full set-cut property are already covered by [Proposition 82.14](#). The remaining cases should be parameterized by genuine cofinality and endpoint data, not by the numerical value of a threshold that translation removes.

**Research question 113.217** (The dense quotient of numerical class carriers). Classify the shorter-first numerical class-carrier order after contracting the adjacent pairs of [Theorem 83.2](#). Which properties of the upper cones  $U(A)$  survive in this quotient? Can the range of available final-segment orders be encoded as a canonical family of decorations on it?

[Theorem 83.6](#) supplies every possible numerical final class segment as an available-letter order after an increasing prefix. A classification must therefore address arbitrary class cuts, rather than only bounded-support or set-prefix data.

**Research question 113.218** (The exhaustive-branch stabilizer). Characterize the uniform systems of local increasing residual bijections that preserve every exhaustive class branch of the injection tree. Are there useful generators or closure operations for the resulting subgroup?

[Proposition 96.11](#) gives a substantial explicit family of such automorphisms. [Theorem 96.7](#) and [corollary 96.12](#) show that prefix coherence and every single fixed set-height restriction are insufficient tests. The condition must express global coverage of labels, not just local successor-order behavior.

**Research question 113.219** (Comparability versus stronger recursion). Investigate the relation of CWO, and therefore CTH, to ETR, continuous rankings, and weak comparability by arbitrary suborder embeddings.

Hamkins–Woodin formulate these comparisons in their Question 7 [27]. The present article supplies a new equivalent presentation of the strong comparability principle; it does not settle the converses involving stronger recursion. Finite normal forms from a history may make that boundary easier to study, because they identify what a condensation-specific construction can provide without a universal truth recursion.

[merge] (4 October 2026, batch 96.) Not settled by Part XVI. Over GB, sources 34 and 35 add equivalent forms of CWO: canonical histories and initial embeddings into  $P(W)$  ([theorem XVI.5.8](#) and [note XVI.5.19](#)), and source 35’s fixed-point extension principle FPE ([corollary XVI.6.19](#), over GB by [theorem XVI.6.21](#)).

**Research question 113.220** (Alternative condensation limits). How do different limit conventions compare with the nested least-representative rule used here? Which retain finite normal forms or the equivalence with strong comparability?

For the present rule, [Corollary 108.14](#) proves that the induced equivalence relation at a limit is the union of the earlier equivalences. This removes one ambiguity. Conventions that choose new representatives or form an external quotient without retained anchors can have different existence requirements.

**Research question 113.221** (Recovering a class realization from an expanded order). Which additional predicates on the global enumeration order, its core, or its ordered prefix tree determine the available class collection up to an appropriate isomorphism?

The bare external set-like enumeration order of a countable two-sorted class model can be  $\mathbb{Q}$ . Naming heights and successor orders still permits the branch changes of [Theorem 96.7](#). Literal label evaluation is sufficient to rule out those particular changes, but whether a substantially smaller invariant family of predicates recovers the intended realization remains open in this analysis.

**Research question 113.222** (Scan equivalence for relation encodings). Classify the order types obtained from different well-ordered scans of relation tables on a fixed surreal carrier or birthday set. Identify natural hypotheses under which two scans induce isomorphic, equimorphic, or provably different orders of well-orders.

The existing endpoint examples already rule out scan-independence in general. A useful next theorem should specify a structural class of scans and compare their effects without silently replacing relation-bit comparison by enumeration comparison.

**Research question 113.223** (A certified mathematical core). Implement the set-sized first-difference, binary fusion, birthday spectrum, and canonical word-factorization results in Lean on top of the inspected sign-sequence foundation. Then formalize the normal-form theorem from a supplied history in an explicitly chosen class or universe-relative framework.



The native spectrum is a particularly focused first target: its statement is exact at every ordinal cutoff, its proof is independent of Global Choice, and its delicate limit invariant is explicit. Success there would provide a reusable library for the later permutation and class-level constructions.

[merge] Status of source 29's questions, 3 October 2026. [Research question 113.212](#) is open (with [research questions 113.94](#) and [113.224](#)). [Research questions 113.213](#) to [113.215](#), [113.217](#), [113.219](#) and [113.220](#) are open. [Research question 113.216](#) is open; source 27 computes the point characters ([proposition 82.8](#)). [Research question 113.218](#) is answered in part ([theorem 96.4](#)). [Research question 113.221](#) is answered in part: literal label evaluation suffices ([theorem 102.3](#) and [notes 102.5](#) and [102.16](#)), and label-edge equality makes the core-and-cylinder structure rigid ([theorem 93.3](#)). [Research question 113.222](#) is answered in part at set level ([theorems 84.2](#) and [84.4](#)) and for one scan on **No** ([theorem 84.13](#)). [Research question 113.223](#) is a proposal.

### 113.24 Source 30: research questions after the new results

The following are proposed research tasks for this program. Where a problem is inherited, its remaining scope is narrowed by the theorems above. The list is not a claim that every item is an independently established open problem throughout the literature.

#### 113.24.1 Residual coding beyond the classified cases

**Research question 113.224** (Singular residual cardinals that are not strong limit). Suppose  $\nu \leq \kappa$ ,  $\nu < 2^{<\kappa}$ , is singular and not strong limit. Determine the exact equimorphism type of  $\mathcal{W}_{\mu\gamma+\nu\delta+n}(\mathbf{No}_{<\kappa})$  in ZFC.

The present proof gives a concrete interval of possibilities: pair orientations give a lower binary factor of length  $\nu\delta$ , while padding the entry codes gives an upper factor of length  $\nu^2\delta$ , appended after  $\rho_\kappa\gamma$ . The strong-limit lower construction cannot simply be reused because  $\mathbf{No}_{<\nu}$  may contain more than  $\nu$  labels. A productive approach is to identify which  $\nu$ -sized residual alphabets can replace its disjoint sign cones. The aim is a coding-capacity theorem for arbitrary residual alphabets, not just a cardinality estimate.

**Research question 113.225** (Dependence on cardinal arithmetic). Which additional cardinal-arithmetic hypotheses, weaker than GCH, guarantee the full fixed-type classification? Can two models with the same relevant cardinalities have different residual binary coding lengths?

The mixed-block theorem already gives one sufficient condition: every singular cardinal at most  $\kappa$  actually encountered in the finite cardinal descent is strong limit; larger residual cardinals are covered by the native padded code. No hypothesis about unrelated cardinals is needed. The next step is to decide whether this condition is necessary or merely sufficient.

#### 113.24.2 Topological classification beyond invariants

**Research question 113.226** (The positive finite-prefix spaces). Are  $E_n$  and  $E_m$  homeomorphic for all positive finite  $n, m$ ? What about their doubled orders and the spaces  $P_n$  for  $n \geq 3$ ?

All  $E_n$  for  $n > 0$  share the proper  $\Delta_3^0$  description and meagreness proved here. These invariants do not classify their homeomorphism types. ([merge] Source 30's  $P_n$  is  $P_{\omega+n}$  here.) The doubled-space Borel formula adds a precise constraint: any candidate homeomorphism has to respect the countable-exception structure of measurable separation of fibers.

**Research question 113.227** (Later countable ordinal types). For arbitrary countable  $\alpha > \omega + 2$ , determine the Cantor–Bendixson derivative sequence and Baire-category behavior of  $\mathcal{W}_\alpha(\mathbb{Q})$ .

The non-Borel ranking obstruction is already known from [corollary 68.17](#); that should not be left as the main unresolved question. What remains is a structural analysis of the order topology itself, especially at limit remainders, where finite fibers no longer describe the whole local geometry.

**Research question 113.228** (Reverse coding at later countable types). For which countable  $\alpha \geq \omega + \omega$  is the map from lexicographic enumerations to relation tables Borel measurable? What structural feature of the terminal order determines its failure or success?

The opposite-direction ranking map already fails to be Borel at every  $\alpha \geq \omega + 3$ . The reverse direction requires a separate analysis. At  $\omega + 2$ , one binary endpoint-orientation predicate repairs the measurable presentation exactly ([proposition 68.18](#)); that case is settled. For  $n \geq 3$ , the lexicographic Borel structure contains too many sets to be contained in any standard Borel structure on the same carrier. An appropriate standard presentation must therefore change which sets are measurable, rather than simply add predicates to all existing order-open sets.

### 113.24.3 Global scans and reconstruction

**Research question 113.229** (Arbitrary scans and non-set-like jumps). Is relation-table lexicography on all class well-orders dense for every set-like scan? If not, characterize a scan and a pair of non-set-like orders producing a jump.

Theorem [84.11](#) settles the set-like suborder, and Theorem [84.13](#) settles the explicit max-index scan even on all class orders. Neither proof alone settles every scan on all orders. A counterexample would have to exploit proper-class predecessor structure that cannot be closed into the common initial set used in the set-like proof.

**Research question 113.230** (Topologies determined by scans). Classify when a scan’s support topology and its order topology agree. Determine which changes of scan preserve the intrinsic local-character description of bounded inversion support.

Our canonical scan already separates these topologies and identifies one-sided set coinitality with bounded inversion support. A general classification should explain how much of this asymmetry is a scan artifact and how much is forced by well-foundedness.

**Research question 113.231** (How much named core suffices?). Find intermediate named structures between a unary predicate for the support core and its pointwise embedding that recover the class realization.

The endpoints are now precise: [proposition 102.19](#) shows that even a baseline point and the whole core named setwise can lose ETR, while fixing the core pointwise recovers all labels through its represented cuts. Candidates include prefix cylinders, the evaluation map on a structured subfamily, or the pair-swap Boolean operations. The distinction between a countable external language and an internal uniform family must be preserved.

**Research question 113.232** (Uncountable realizations and common baselines). Which analogues of the universal-baseline theorem hold for uncountable set models? Which external order invariants of their admitted enumeration spaces detect more than the cardinality of the class collection?

The countable argument uses both Cantor back-and-forth and Williams’s countable minimum-realization theorem. Removing countability is not a routine restatement. The labelled inclusion theorem itself does not need it and provides a useful comparison standard.

#### 113.24.4 Foundational and formal priorities

The most concrete next formalization target is the prefix-free code together with the regular block-length lemma. It is self-contained, uses already available small-cut foundations, and supports every regular residual case. The next difficult target is the singular scheduler. The most compact class-theoretic target is pair coding and its decoder; the most informative local-to-global target is partial-table extension.

A broader foundational question asks how much choice follows from universality of the surreal *numerical* order for all class linear orders, without assuming a global well-order of its labels. That is a different question from [note 111.27](#); it was posed explicitly by Hamkins [21]. Nothing in the present proof settles that question, and the global baselines used here cannot be assumed in an argument attempting to derive global choice.

[merge] Status of source 30's questions, 3 October 2026. [Research question 113.224](#) is open (with [research questions 113.94](#) and [113.212](#)); the cube spectrum of these strata is settled by [theorem 64.11](#). [Research questions 113.225](#) to [113.230](#) and [113.232](#) are open; source 29's [corollary 68.7](#) bears on [research question 113.226](#). [Research question 113.231](#) is answered in part: a core fixed pointwise recovers the class part ([note 102.5](#)). Hamkins's question on universality and choice [21] is not settled ([section 1.7](#) (f)).

#### 113.25 Earlier questions answered by sources 19–30 (dated notes, 3 October 2026)

[merge] Dated notes in the form of [section 50](#); where the question indexes of [section 113](#) and [section 113.5](#) say otherwise, these notes supersede them. The “What this Part answers” sections of Parts X–XIV give the same status with the theorems in context.

- (a) *Source 12's [research question 113.11](#), all fixed enumeration types* (with source 08's question 4, Part VI status (e) and [section 31.3](#), and sources 15–18's [research questions 113.46](#), [113.58](#), [113.69](#) and [113.83](#)): answered in part. For every type  $\alpha = \mu\gamma + \sum_{i < k} \lambda_i \eta_i + n$  of  $\mathbf{No}_{<\kappa}$  the embedded binary-cube spectrum is exact, and the least coding length is determined whenever the two weighted bounds agree, in particular for every type when every residual cardinal is regular, a singular strong limit or above  $\kappa$  (source 19, [theorems 64.7](#) and [64.11](#)); hence every type is classified when  $\kappa < \aleph_\omega$  in ZFC, and every type under GCH (source 30, [corollary 64.20](#) and [note 64.18](#)). Open: singular non-strong-limit residual cardinals at most  $\kappa$  ([research questions 113.94](#), [113.212](#) and [113.224](#)). The infinite-remainder case that Part VI left open ([section 31.3](#)) is covered.
- (b) *Source 12's [research question 113.12](#), noncardinal birthday bounds* (with sources 15, 17 and 18's [research questions 113.47](#), [113.70](#) and [113.84](#)): answered in part. The native alphabet remembers its cutoff exactly:  $\mathbb{B}_\beta \hookrightarrow \mathbf{No}_{<\theta}$  iff  $\beta < \theta$  (source 29, [theorem 76.5](#) and [corollary 76.6](#)); the minimal slice at a noncardinal cutoff has coding rank  $\mu$ , characters and cofinalities computed, and is isomorphic to no bounded interval or ray of itself (source 21, [theorem 77.4](#)); on the countable band the permutation strata forget  $\theta$  up to equimorphism ([corollary 78.10](#)). The order type remains open.
- (c) *Source 12's [research question 113.13](#), isomorphism invariants and homogeneity*, and source 08's question 1: answered in part. At the countable cutoff the minimal slice is homeomorphic to Baire space, has the real line as Dedekind completion with a dense meagre  $F_\sigma$  of omitted points, every open interval is isomorphic to the whole order, finite partial maps extend, and prefix cylinders are not intrinsic (sources 21 and 24, [theorems 67.2](#), [67.4](#) and [67.8](#) and [corollary 67.9](#)); at regular  $\mu$  it is homeomorphic to  $(\mu^{<\mu})^\mu$  ([theorem 66.2](#)); it is completely metrizable only

- at  $\mu = \omega$  (theorem 69.18); weights and finite-tail characters of every stratum are known (theorems 63.4 and 63.9). The isomorphism type at uncountable cutoffs is open.
- (d) *Source 12's research question 113.17, automorphisms and finite condensation*: answered in part for countable models (source 21, note 109.30 and corollary 109.31); the finite-tail blocks are first-order definable in the pure order of every set-sized model (source 19, proposition 109.7).
  - (e) *Source 11's research question 113.8, source 12's research question 113.18, source 08's question 12, and Part VIII's item (e) "What the unlabelled external order type remembers"*: answered further. The admitted labelled global enumerations recover the whole class part over the same sets (sources 22, 27 and 30, theorem 102.3 and notes 102.5 and 102.16); the external pure order of a countable model is one of two shuffles, according as the model is  $\omega$ -standard or not, and all set-sized models have the same complete theory (source 19, theorems 109.1 and 109.9); naming the core setwise and the baseline still loses the realization, and ETR is invisible to the pure order (source 30, propositions 102.19 and 102.20). Uncountable models are open. Source 17's research question 113.75 (recovering a class realization) is **answered** with fixed coding parameters by source 22, and source 14's research question 113.43 in part.
  - (f) *Source 12's research question 113.19, compatibility between cutoffs* (with sources 14, 15, 17 and 18's research questions 113.42, 113.54, 113.79 and 113.85): answered in part. Restriction is monotone only in three degenerate cases (source 19, theorem 73.1); append-tail maps form a coherent direct system whose limit is the bounded-support core (source 20, theorem 74.8); fixed-slot embeddings compose exactly within a cardinality band (source 21, theorem 75.1). Preservation of condensation, coding ranks or cylinders, and coherence across a change of cardinality, remain open.
  - (g) *Source 11's research question 113.3, iterated residual decomposition* (with sources 13, 15, 16 and 18's research questions 113.30, 113.48, 113.64 and 113.86): answered for the canonical-history formulation over GBC: every class well-order has its canonical terminating condensation history exactly when strong comparability of class well-orders holds (source 29, theorem 108.19); a supplied history terminates (source 20, theorem 108.1); limits are unions (corollary 108.14). Whether ETR is necessary (Hamkins–Woodin's Question 7) and the theory over GB remain open. *Dated note, 4 October 2026 (batch 96)*: the theory over GB is answered by sources 34 and 35 (section 113.29 (b)); Question 7 remains open.
  - (h) *Source 11's research question 113.4 and research question 113.5; Part VIII's item (c)*: answered further. Exhaustive branches are not recognized by the core with all its unranked cylinders and any set of parameters (source 25, theorems 91.1 and 91.2), nor by the ordered prefix tree (theorems 96.2 and 96.7); equality of labels across extension edges suffices and makes the structure rigid (theorems 93.1 and 93.3 and note 96.5); this answers the "extra predicates" item and source 14's research question 113.34. An increasing relabelling makes two cores disjoint (proposition 97.9). The stabilizer of the exhaustive cuts is still undescribed, with partial families (theorem 96.4 and corollary 96.12).
  - (i) *Part VIII's item (f), ordinal support bounds*, and source 17's research question 113.78: the ordinal half is **answered**: a partial injection completes with support below  $\beta$  exactly when the stated cardinal criterion holds, and the least such  $\beta$  is given in closed form (source 27, theorems 101.1 and 101.2). Birthday bounds and the forced-prefix length remain open.
  - (j) *Part VII's open items on centered increasing words and interior terminators*, and source 18's research question 113.91: answered in part. The centered order is  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ , with all zero conventions identified (sources 27 and 29, corollary 82.5 and notes 82.11 and 82.13); no fixed marker makes increasing words set-cut saturated (source 23, theorem 82.1); actual thresholds

reduce to zero and gaps with two large sides are classified ([proposition 82.14](#)); characters at a general gap are computed ([proposition 82.8](#)). Gaps with a small side and a cut classification remain open.

- (k) *Part VII's item on two scans, source 08's question 9 and source 09's fifth direction*: the scan question is answered negatively at set level: two scans of one countable carrier can give a real-embeddable order and an order with continuum many disjoint intervals, with the same ordinal spectrum (source 27, [theorems 84.2 to 84.4](#)). Source 09's "natural set-coded dense core" exists: the bounded-support core is dense in the relation-table order for every set-like scan (source 27, [theorem 84.5](#); source 30, [theorem 84.11](#)). The representation of its cuts, and scans on all of **No**, remain open.
- (l) *Source 18's research question 113.89,  $\omega$ -nonstandard class models*: **answered**, three times independently: the external order is the shuffle of the factorial colours together with one further block type, and it has the same complete theory as the standard shuffle, which is decidable (sources 19, 20 and 21, [theorem 109.1](#), [notes 109.13](#) and [109.33](#), and [corollary 109.22](#)).
- (m) *Source 17's research question 113.72, complete metrizable when  $q = \omega$* : **answered**: the minimal slice is completely metrizable exactly when  $\mu = \omega$  (source 24, [theorem 67.2](#); source 26, [theorem 69.18](#)). The dense-open threshold is open ([research questions 113.155](#) and [113.183](#)).
- (n) *Source 15's research questions 113.50 and 113.51* (completion with residual data; automorphisms across residual fibres) *and source 14's research question 113.39*: **answered** in the external formulation by source 19 ([theorems 98.2](#) and [98.5](#)).
- (o) *Source 17's research question 113.77, the condensed unrestricted order*: answered in part (source 20's local-size blocks, [theorem 107.5](#); the countable models of item (l)).
- (p) *Source 13's research question 113.28* (termination cuts): already answered by source 16; source 23's criterion is a second route ([theorem 36.1](#)). *Source 13's research question 113.20* (extra structure for recognizing branches): answered as in item (h).
- (q) *Questions not affected*: [research questions 113.2](#), [113.14](#) and [113.16](#) (source 23's [theorem 108.23](#) bears on [research question 113.14](#) but does not answer it), [research questions 113.9](#) and [113.10](#), and Hamkins's question on universality and choice.

### 113.26 Source 31's research questions (batch 89)

[merge] (3 October 2026.) Source 31's Section 13, "Further research questions", printed whole; its questions 1–12 are numbered here in this section's sequence and labelled `swo:gcz:n1–swo:gcz:n12`.

The following questions are proposed directions. Except where explicitly identified as repository questions, they are formulated here and are not being advertised as a catalog of established open problems. Each distinguishes a specific missing step from results proved above.

**Research question 113.233** (Global selection from finite sets). Is ZC with a selector on all nonempty finite sets and expanded Separation nonconservative over ZC? What changes with Foundation?

The present proof uses selection on every nonempty subset of a countable carrier. Finite-input selection does not supply that operation. Enayat's global finite choice principles provide relevant background, but do not by themselves settle this Zermelo conservativity question.

**Research question 113.234** (Pairs and bounded finite cardinalities). For a fixed integer  $m \geq 2$ , what pure set-theoretic consequences follow from a class selector on  $m$ -element sets, with expanded Separation? Can these consequences distinguish different  $m$ ?



A proof would require either a different obstruction model or a new reduction from finite-input choices to a useful uniform presentation. Arbitrary countable-set enumeration cannot simply be assumed.

**Research question 113.235** (The Separation complexity needed by the refinement). Which finite collection, or syntactic fragment, of Separation instances involving the selector suffices for [theorem XV.5.1](#)?

The proof gives explicit containing sets, but its defining formulas include well-order tests and unbounded quantification over class labels. Boundedness of the output is not a claim that every defining formula is  $\Delta_0$ . Tracking the actual formulas is a feasible formalization project.

**Research question 113.236** (Reducing ordinary Choice). How much of ordinary AC can be removed from the localized nonconservativity theorem?

The positive uniformization proof only asks for well-orders of  $D$  and  $\mathcal{C}_D$ ; for  $D = \omega$ , the latter can be coded by a fixed set of reals. A complete weakening must also recheck the base countermodel and the final cardinal obstruction, rather than dropping Choice from the proof statement alone.

**Research question 113.237** (Fragments that restore conservativity). Which Replacement or Collection fragments suffice to restore conservativity for countable-input global selection with expanded Separation?

Collecting every definable  $\omega$ -family destroys the particular class-family obstruction used here. That observation does not establish a general conservativity theorem, which would require an expansion or forcing argument for arbitrary models of the chosen base.

**Research question 113.238** (Selectors beyond the countable threshold). For well-ordered sets  $D$  and  $E$ , when do restricted global selection on  $D$ -small and on  $E$ -small sets have the same original-language consequences over a fixed weak base?

Every bound containing  $\omega$  already gives our nonconservativity witness. The uniform presentation theorem supplies a common method of comparison but does not collapse the full hierarchy of consequences.

**Research question 113.239** (Definable-class versions of the spectra). What replaces the rank-model classification when the admitted classes are  $\text{Def}(V_\lambda, p)$ , rather than all external subclasses?

External cardinal equality then need not produce an admitted bijection, and a cofinal witness to Replacement failure need not be definable from the permitted parameter. The full-class answers provide bounds and test cases, not an answer to this definability problem.

[merge] (4 October 2026, batch 96.) Answered in part, for the external order types at a strongly inaccessible height: for the classes definable over  $V_\kappa$  in a fixed finite language with set parameters the realized types form an interval  $[0, \Theta_L)$  with  $\kappa < \Theta_L < \kappa^+$ , of countable cofinality under a named global well-order, closed under ordinal arithmetic, and raised by a truth predicate, which represents the old ceiling exactly (sources 32–35; [section XVI.12](#) (e)). The analogues of the thresholds of Part XV for definable classes remain open.

**Research question 113.240** (Abstract weak-theory equivalences). Find a natural two-sorted extension of ZC over which particular surreal coding principles characterize ordinal abstraction, an image-bounding scheme, or a restricted class-choice principle.



Our exact equivalences are semantic statements about  $\mathcal{M}_\lambda$ . Turning one into an abstract equivalence requires a proof for arbitrary models, including models without the full external rank hierarchy or all subclasses.

**Research question 113.241** (All set-like order types at a cutoff). Classify the external order types of every internally set-like well-order of  $S_\lambda$ , especially at singular and non-strong-limit heights.

We classify whether some such order has type  $\lambda$ , and whether it can respect birthdays. A classification of all possible types requires control of the birthday bounds on every proper initial segment and may contain more information than cardinality alone.

**Research question 113.242** (The repository’s global control schemes). For each precise supported-prefix extension, small-net rigidity, and inverse-reconstruction scheme in ProveIt, identify the minimum selection, collapse, and image-bounding assumptions.

This is the remaining program behind `swo:ge:n10` ([research question 113.209](#)). The domain-size criterion can test candidate image-bounding claims, while bounded trace compression may test claims that a uniform presentation is available. No equivalence for all three schemes has been proved here.

**Research question 113.243** (A verified bounded uniformization theorem). Can a proof assistant verify [theorems XV.3.3](#) and [XV.4.1](#) over an explicit Zermelo axiom interface, with no imported theorem using Replacement?

This is a concrete near-term target. Every newly formed set can carry its containing-set witness as part of the formal construction. A separate module can then instantiate the theorem with the Glazer obstruction interface.

**Research question 113.244** (Surreal transfer and parameter-sensitive choice). Can the distinctions between uniform choice, ordinal coding, and admitted definability be used to sharpen the obstruction to definable transfer expansions of **No**?

Hamkins’s surreal transfer question and Glazer’s contribution to its discussion are a direct further intersection [76]. Our results concern weak foundations and full-class rank models. They do not settle transfer in ZFC with arbitrary set parameters; a solution would require a separate definability or forcing argument.

[merge] Status of source 31’s questions, 3 October 2026. All twelve are proposed continuations. [Research question 113.241](#) is answered at strongly inaccessible heights by source 12’s [theorem 21.1](#) (ii): there every internally set-like global well-order of  $\mathbf{No}_{<\kappa} = S_\kappa$  has external type  $\kappa$ ; elsewhere it is open. [Research question 113.242](#) restates the open remainder of source 28’s [research question 113.209](#), item (a) below. [Research question 113.243](#) is a formalization proposal. [Research question 113.244](#) concerns Hamkins’s surreal transfer question [76], not the universality question of [section 1.7](#) (f). The others are open; this write found no answer to them elsewhere in this report.

## 113.27 Earlier questions answered by source 31 (dated notes, 3 October 2026)

[merge] Dated notes in the form of [section 113.25](#); where the question indexes of [sections 113](#) and [113.12](#) say otherwise, these notes supersede them. [Section XV.15](#) gives the same status with the theorems in context.

- (a) *Source 28's research question 113.209, separating Class Replacement from uniform set choice:* answered in part, for the mechanisms rather than for the three named schemes. Countable selection with expanded Separation is nonconservative over  $ZC$  and  $ZC_F$ , conditional on the Glazer interface (theorem XV.5.1 and section XV.5.1), and conservative without expanded Separation (proposition XV.5.5); in the full-class rank models, global choice, ordinal coding and image bounding have different exact thresholds (theorems XV.7.2, XV.7.4, XV.8.2, XV.8.4 and XV.9.3, lemma XV.9.2, and corollary XV.9.4). Open: the weakest assumptions for the supported-prefix extension, small-net rigidity and inverse-reconstruction schemes themselves (research question 113.242). The sibling questions of sources 24 and 26 on the same resources, research questions 113.161 and 113.185, stay open.
- (b) *Questions not affected:* those listed in section 113.25 (q), including Hamkins's question on universality and choice.

### 113.28 The questions of sources 32–35, merged by theme (batch 96)

[merge] (4 October 2026.) The research questions of the four sources of Part XVI, with their commentary, merged into one list: source 34's agenda items R1–R12 (its Section 11), source 35's questions 10.1–10.13 (its Section 10), source 32's questions 12.1–12.11 (its Section 12) and source 33's research questions 12.1–12.14 (its Section 12). Each is printed whole, under the theme it belongs to, with its source and number in its heading; a dated [merge] paragraph at the head of each theme says which questions coincide, which are answered and by what, and what remains open. Answered questions stay on record (Vladimir's standing rule of 4 October 2026). The four introductions follow, then the themes; earlier questions of this report that sources 32–35 answer are in section 113.29.

**Source 34's introduction to its agenda.** The following agenda distinguishes a localized extension from a published foundational question. The first three items are especially suitable continuations because their input objects and proof obligations are already explicit.

**Source 35's introduction to its program.** The results give several directions with different levels of risk. The first group concerns formal verification and relatively local extensions of the proofs. The second asks reverse-mathematical questions that remain substantially harder. The third concerns systematic notation design. An answer in one group should not be advertised as settling a stronger question in another.

**Source 32's introduction to its questions.** The following questions distinguish repository continuations from broader published questions. A question described as published was posed in the cited source; that description does not assert an exhaustive verification that its status has remained unchanged through every later publication.

**Source 33's introduction to its questions.** The questions below are a proposed program. Except where an explicit published source is named, “question” means a direction not settled by this manuscript, not a claim that the problem is known to be open worldwide.

#### 113.28.1 Verifying the choice-free core and proof-carrying notation

[merge] Source 34's R1, source 35's questions 10.1 and 10.2, source 32's question 12.11 and source 33's question 12.13 ask for the same first formalization: the two-sorted class interface, the choice-free power theorem and the capped normal-form reconstruction; source 34's R12 and source 33's

question 12.12 ask for proof-carrying notation. All open: no statement of Part XVI is formalized (section 1.5). This report's earlier formalization plans for the class orders are in section 59 and section 111.

**Research question 113.245** (Source 34, R1: kernel-check the GB refinement). Formalize [Theorem XVI.4.2](#) and the canonical-history equivalence in a two-sorted encoding that records comprehension instances. The key deliverable is an axiom audit showing that no Choice or ETR axiom entered either proof. A full-model ordinal calculation alone would not meet this objective.

### Verify the choice-free core first

**Research question 113.246** (Source 35, Question 10.1: Formalize the finite-support minimization theorem). Can the greedy proof of [note XVI.4.4](#) be formalized in a two-sorted theory whose set axioms are precisely ZF, with all uses of class comprehension and Replacement recorded?

This is the most immediate verification target. A successful formalization should use the original subclass of candidate polynomials and a finite prefix as parameters, rather than postulating a sequence of newly formed tail classes. Its termination argument should explicitly construct the uniquely specified sequence of exponent labels in the nonterminating case. Checking this point also provides a reusable pattern for other finite-support functors.

**Research question 113.247** (Source 35, Question 10.2: Verify the complete normal-form argument). Can one give a machine-checked proof of the GB fixed-schedule criterion, including the capped reconstruction establishing that the polynomial image is an *initial* subclass?

An ordinary order embedding would not suffice for the equivalence with strong comparability. The finite reconstruction with a retained upper cap is the most important part to preserve. A formal library should then separate the choice-free result for arbitrary carriers from the Global-Choice-dependent reduction to the whole surreal carrier. The inspected repository has not already supplied a machine verification of this later manuscript development.

**Research question 113.248** (Source 32, Question 12.11: A verified presentation library). Can the weak presentation theorems, the ETR-dependent existence theorem, and the external spectrum theorems be formalized in one library without identifying their notions of well-foundedness or smallness?

A successful implementation should expose a theorem's class-admissibility assumptions in its type, and should make it impossible to import an external well-foundedness proof into an arbitrary internal model without an explicit absoluteness bridge. This is a substantive design constraint, not merely a namespace convention.

**Research question 113.249** (Source 33, Research question 12.13: Formal reverse mathematics). Implement the two-sorted class-model interface and verify the distinction between the GBC theorems and the ETR-dependent existence theorems. A particularly focused first milestone is a complete formal proof of the uniform-family diagonal, followed by a bounded-satisfaction compiler for one fixed complexity level.

**Research question 113.250** (Source 34, R12: a proof-carrying notation interface). Design a syntax whose constructors carry certificates of class well-ordering and whose conversions carry explicit initial embeddings. The goal is an extensible and checkable family, not a final catalog. The uniform diagonal theorem proves that the latter goal is too strong.

**Research question 113.251** (Source 33, Research question 12.12: Proof-carrying notation systems). Construct useful finite-expression languages with verified evaluation and well-foundedness proofs for long fragments beyond  $\Omega$ . Which fragments have decidable equality or comparison? The no-uniform-cofinal-family theorem gives a precise reason not to promise one self-contained system covering all definable well-orders in its own language.

### 113.28.2 Comparison, recursion, and the strength of histories

[merge] Source 34's R5, source 35's question 10.3, source 32's question 12.8 and source 33's question 12.10 are Hamkins and Woodin's Question 7, which this report records as source 29's [research question 113.219](#): **open**. Over GB it is now a question about any of the equivalent principles CWO, CTH and FPE ([theorems XVI.5.8](#) and [XVI.6.21](#) and [corollary XVI.6.19](#)). Source 34's R6 and source 35's question 10.5 ask the weak-versus-strong part of the same question: **open**. Source 32's question 12.1 and source 33's question 12.6 ask for the strength of the existence of condensation hierarchies or towers: **answered**, and kept on record: it is CWO, over GBC by source 29 ([theorem 108.19](#)) and over GB by sources 34 and 35, because hierarchies and towers are nested histories ([remark XVI.5.24](#)); only  $\text{CWO} \Rightarrow \text{ETR}$  remains open. Source 35's question 10.4 is source 15's [research question 113.48](#): **open**. Source 34's R7 is **open**. Source 32's question 12.10, on removing global choice, is answered for the arguments it names: the descending-sequence test is replaced over GB by [lemma XVI.2.2](#) for linear orders, support localization and least-representative division need no choice ([theorems XVI.3.3](#) and [XVI.4.2](#)), and the diagonal of an already well-ordered uniform family is a GB theorem ([theorem XVI.7.2](#)); global choice remains for ordering arbitrary index classes ([corollary XVI.7.3](#)), and how much set theory each argument needs is **open**.

**Research question 113.252** (Source 34, R5: comparability versus elementary transfinite recursion). Investigate the converse to  $\text{ETR} \Rightarrow \text{CWO}$ , and its relationship with continuous ranking. This is Hamkins–Woodin's published Question 7, not resolved by the present history equivalence [27]. The GB refinement isolates an additional equivalent formulation that may be useful for building separating models or extracting recursion.

### The remaining recursion and comparison boundary

**Research question 113.253** (Source 35, Question 10.3: Does comparability yield elementary transfinite recursion?). Determine whether strong class-well-order comparability implies ETR over the usual base theories, or construct a separating model.

This is a classical open-question direction, not a problem solved by the choice-free improvement. The results here add an equivalent fixed-point extension formulation over GBC: every class well-order has an initial extension admitting an isomorphism with its own  $\Omega$ -power. One may therefore try to recover iterated truth from the completion witnesses, or show that those witnesses can exist without all elementary recursion solutions. A bare order isomorphism need not provide a truth predicate; the missing reduction must be proved rather than inferred from the height of the order.

**Research question 113.254** (Source 32, Question 12.8: Comparison and recursion). What are the reverse implications between ETR, existence of continuous class rankings, initial class-order comparability, and arbitrary-suborder comparability?

This is the family of comparison questions posed by Hamkins and Woodin [27]. Our diagonal theorem uses neither comparison principle and therefore does not answer those reversals. Condensation existence may provide a new intermediate principle to test, rather than a presumed equivalence.

## Connections to published problems and formal systems

**Research question 113.255** (Source 33, Research question 12.10: Comparison versus recursion: a published interface). Hamkins–Woodin ask whether ETR, continuous rankings, initial comparison, and weak comparison of class well-orders are equivalent over  $\text{GBC}^-$  [27, Question 7]. Can the diagonal and condensation methods isolate new implications or separating model constructions? This article proves no reversal. The targeted search located the published question but did not establish a complete up-to-date resolution of its status; a project pursuing it should begin with a renewed specialist literature check.

**Research question 113.256** (Source 34, R6: weak versus strong comparison). Determine the exact strength of arbitrary-suborder comparability relative to initial-segment comparability. The diagonal theorem does not settle this question. A reversal must produce an initial range or a genuine countermodel, rather than appeal to set-order intuition.

**Research question 113.257** (Source 35, Question 10.5: Weak comparison and ranking principles). Which of the implications among ordinary suborder comparability, initial-segment comparability, continuous class rankings, and ETR reverse over GB or GBC?

The successor-sum bound proved here uses ordinary suborder embeddings and needs no universal comparison principle. By contrast, fixed-point completion uses an initial embedding certificate. This distinction may help formulate intermediate principles. The relevant established implications and the original Hamkins–Woodin question should remain explicit when constructing models or proposed reductions [27].

**Research question 113.258** (Source 32, Question 12.1: Exact strength of condensation existence). Over GBC, what is the strength of the assertion that every class well-order  $W$  admits the set-interval condensation hierarchy along  $W + 1$ ? Is it weaker than full ETR, equivalent to a recognizable fragment, or equivalent to a natural class-order comparison principle?

The termination part is settled by [note XVI.5.20](#); hierarchy existence is the remaining issue. A productive separation test would construct a GBC realization closed under this one elementary operator but not under arbitrary elementary recursions.

## Condensation, ranks, and recursion

**Research question 113.259** (Source 33, Research question 12.6: The exact recursion strength of exhaustion). Determine the weakest theory proving the existence and exhaustion of the condensation tower for every class well-order. [Note XVI.5.32](#) gives a sufficient order-dependent ETR assumption. Does an appropriately uniform exhaustion principle recover a substantial recursion principle?

**Research question 113.260** (Source 35, Question 10.4: Uniform finite condensation). For every supplied class well-order  $W$ , does GB prove the existence of one class coding all finite iterates of its set-interval condensation operator?

Each externally fixed finite iterate is elementary. The uniform assertion is different, as the repository correctly emphasizes. The finite-trace proof for repeated  $P$  does not answer this question:  $P$  acts by local transformations of finite supports, whereas set-interval condensation contains a global set-existence test. A promising approach is to analyze whether a single input order can make its successive condensations encode increasingly complex partial truth. A negative result must construct such an input and recover the truth information explicitly.

**Research question 113.261** (Source 34, R7: restricted history principles). Study histories only for orders of a specified definability complexity, only on a specified family of code carriers, or only on specified schedules. Track the existence of a well-order on each carrier explicitly, so a restriction cannot become vacuous when Choice fails. Determine when these fragments recover full comparability and when they measure a weaker principle. The whole-surreal-carrier reduction in the repository uses its separate Global Choice analysis and is not a new open problem here.

**Research question 113.262** (Source 32, Question 12.10: Reducing choice and background strength). Which arguments can be carried out without Global Choice, given well-ordered input domains? How much set theory is needed for support localization, least-representative division, and diagonal domination of an already well-ordered uniform family?

The direct least-element proof for sums is choice-free once the orders are supplied. Global Choice was used in the descending-sequence test and in ordering arbitrary parameter-index classes. Separating these uses may yield versions suited to choiceless GB or weaker set fragments.

### 113.28.3 Condensation lengths, quotients, and normal forms

[merge] Source 33's question 12.7 is **answered** for its first part, and kept on record: the digit map of every supplied tower has initial image, and needs only GB and the tower ([remark XVI.5.29](#)). Source 32's question 12.2 and source 33's question 12.8 are answered in part in full inaccessible models by source 20's exact heights  $\min\{\alpha : \theta \leq \kappa^\alpha\}$  ([corollary 108.3](#)); the internal classification is open. Source 35's questions 10.12 and 10.13 continue source 18's [proposition 52.15](#) and source 20's [research question 113.112](#), which is answered in part by [corollary 108.14](#); open. Source 32's question 12.3 and source 33's question 12.9 are open.

**Research question 113.263** (Source 33, Research question 12.7: Sharper canonical digit representations). For which supplied towers is the canonical map  $\Phi$  of [Theorem XVI.5.27](#) already initial? Can one give an intrinsic criterion on tower fibers that proves initiality without invoking general class comparison? A separate question is whether a different canonical compression map has a weaker axiom bound.

**Research question 113.264** (Source 32, Question 12.2: Optimal bounds for individual orders). Given a supplied hierarchy, can its first universal stage be characterized from intrinsic structural data of  $W$ ? Which invariants yield better bounds than  $W$  itself, while distinguishing the genuinely changing quotient maps in [corollary XVI.5.22](#)?

The explicit examples realize every supplied class length as an exact collapse length. A classification therefore has to retain more information than the external cardinality of the domain or the repeated quotient's isomorphism type.

**Research question 113.265** (Source 33, Research question 12.8: Minimal condensation lengths). Under which hypotheses can one assign a minimal exhaustion length to an arbitrary class well-order, compare these lengths, and do so uniformly across a supplied family? The coherent-relation requirement must be kept; the repeated quotient-type example  $P(\omega)/E_n \cong P(\omega)$  rules out a classification using quotient types alone.

**Research question 113.266** (Source 32, Question 12.3: Normal forms after repeated  $\Omega$ -division). For which presentations does iteration of [theorem XVI.3.6](#) yield a useful canonical normal form? Can one recognize the finite-support powers and hereditary term orders directly from their block structure? What hypotheses ensure uniqueness of more elaborate normal-form expansions?



The one-step division theorem is unconditional in GBC. Neither a universal Cantor normal form for all class well-orders nor a uniformly admitted iteration through all possible class lengths has been established here.

### Condensation beyond finite polynomial forms

**Research question 113.267** (Source 35, Question 10.12: Classify isomorphism-level condensation fixed points). Which class well-orders are isomorphic to their set-interval quotient, and which can occur in nontrivial isomorphism-level cycles?

The example  $P(\omega)$  already shows that one-step condensation can preserve the order type. It does not show that the actual retained subclass is unchanged. Nor does it prevent a specified transfinite history from terminating. Any classification should retain the quotient map, the literal retained representatives, and the isomorphism type as distinct data. This avoids the false inference that every quotient must have a strictly smaller abstract length.

**Research question 113.268** (Source 35, Question 10.13: Alternative limits with an explicit universal property). What genuinely different class-order theories arise when limits of condensation systems are defined by other universal properties or additional completion operations?

The nested-representative limit and the union of its induced equivalence relations already describe the same canonical process. A new theory must change something substantive, such as the admitted maps, the identifications made at limit stages, or the ambient type level. The benefit should be an exact theorem or a preserved invariant, rather than an alternative spelling of the same construction.

**Research question 113.269** (Source 33, Research question 12.9: Other smallness ideals). Replace “is a set” by a specified smallness ideal or by internal cardinality below a cutoff. Which analogues of canonical division and finite-digit representation survive? Singular cutoffs require particular care because the rank-bounding and replacement arguments used at an inaccessible  $\kappa$  may fail.

### 113.28.4 Notation systems, fixed points, and support conditions

[merge] Source 35’s question 10.6 is **answered** positively by this write ([theorem XVI.6.21](#)), and kept on record. Source 34’s R4 and source 35’s question 10.8 ask for the same Veblen-style hierarchy of class-order fixed points; with source 35’s question 10.7 (completion for other functors), source 34’s R2 and R3, and source 33’s question 12.14, they are open.

### Fixed-point completions and their functorial behavior

**Research question 113.270** (Source 35, Question 10.6: A choice-free completion theorem). Can the local fixed-point completion theorem itself be lowered from GBC to GB in full generality, with a completely uniform proof of well-ordering for the finite-trace direct limit?

The explicit finite-support constructor is now known to preserve well-ordering without Choice. The remaining point is the internal uniformity of the iterated stages for an arbitrary supplied seed embedding. Proving each externally fixed stage is not by itself a proof of the needed uniform assertion in a weak base. This article keeps the stated GBC hypothesis for the completion theorem instead of using a metatheoretic iteration as an unstated class-existence axiom.

**Research question 113.271** (Source 35, Question 10.7: Completion as a universal construction). For which class-order functors  $F$  does a supplied initial embedding  $W \preceq_{\text{init}} F(W)$  yield a least fixed-point extension by an elementary finite-trace construction?

For  $F = P$ , finite support guarantees preservation of the direct limit, while preservation of initial embeddings guarantees that every predecessor of a point lies in a single stage. These two properties suggest sufficient hypotheses. They should be phrased for one uniformly supplied functor on codes and maps, without quantifying over a nonexistent class containing all classes as elements. Normal dilators are a natural source of examples, but a claim about every dilator must specify its coding and axioms separately.

**Research question 113.272** (Source 34, R4: iterated fixed-point notation). Develop a Veblen-style hierarchy of class-order constructors with finite support and local certificates. State a well-foundedness theorem for the complete proposed syntax before assigning it a large symbolic ordinal name. Compare the resulting theory with the support-based constructions used in ordinal analysis [86].

[merge] (4 October 2026, batch 97.) Advanced by source 36: its finite-support transfer theorem turns a finite-support term operator that preserves set well-orders into one that preserves class well-orders, over GB without choice ([theorem XVII.6.2](#)), and the epsilon-term functor  $\mathbf{Ep}$  is such an operator, with set-case type  $\varepsilon_\beta$  ([theorem XVII.6.4](#)) and an order of type  $\zeta_{\kappa+1}$  from its iterates ([theorem XVII.6.7](#)). The hierarchy itself, here and in source 35's [research question 113.273](#), stays open (source 36's [research question 113.299](#); [section 113.31](#) (d)).

**Research question 113.273** (Source 35, Question 10.8: Successive fixed points and Veblen-style systems). Construct explicit hereditary notation systems for successive  $P$ -fixed points, their common fixed points with other constructors, and suitable Veblen-style closures. Identify the exact comparison and well-ordering proofs for each system.

The classical use of class epsilon notation shows that such growth is a legitimate proof-theoretic direction. The contribution needed here is a coherent class-presentation semantics and verified constructor laws. A symbol resembling a familiar ordinal notation is not enough to establish a class well-order. Bounded specialization and finite-support direct limits offer two complementary proof methods.

**Research question 113.274** (Source 34, R2: support conditions beyond finite trees). Determine which families of set-sized supports admit a coherent replacement for greatest-disagreement comparison while preserving class well-foundedness. An unrestricted infinite support need not have a greatest differing coordinate. The finite-support proofs identify exactly where such a replacement must supply additional structure; a proposed extension should include counterexamples showing why its support restrictions are needed.

**Research question 113.275** (Source 34, R3: normalization of arithmetic inside the hereditary system). Construct certified algorithms for addition, multiplication, and exponentiation of the initial-segment notations represented by finite trees, with an explicit interface for ordinal coefficients. Prove the algorithms against class-order isomorphisms, not only against a few specializations  $\Omega \mapsto \kappa$ . Determine precisely which operations preserve finite normal forms.

[merge] (4 October 2026, batch 97.) **Answered for addition and multiplication** by source 36's [theorems XVII.5.1](#) and [XVII.5.2](#), proved against class isomorphisms of the represented initial segments, relative to the ordinal arithmetic of the coefficients ([section 113.31](#) (b)). Exponentiation, and which operations preserve finite normal forms in general, stay open (source 36's [research question 113.300](#)).

**Research question 113.276** (Source 33, Research question 12.14: Higher-order extension without a final universe). Which useful ordinal operations and comparison principles can be expressed on unrolled or hyperclass order codes, and what are their exact collection and recursion requirements? A satisfactory extension should reproduce the appropriate diagonal obstruction at its own level rather than postulate an absolute final ordinal of every universe.

### 113.28.5 Diagonals and definability complexity

[merge] Source 35's question 10.9, source 32's question 12.6 and source 33's question 12.1 ask the same thing: the least complexity of an order escaping a fixed definability level; open. Source 35's question 10.11 is open. Source 33's question 12.2 asks to remove the named global order: for orders on Ord with ordinal parameters source 35 removes it ([theorem XVI.7.11](#)); for arbitrary carriers and set parameters it is open. Source 34's R8 is the same distinction between ordinal and set parameters, which source 35's  $\delta_{\text{ord}} \leq \delta_{\text{set}}$  measures ([theorem XVI.8.6](#)); open.

### Definability spectra and uniform bounds

**Research question 113.277** (Source 35, Question 10.9: Quantitative complexity of escape orders). For a fixed finite  $n$ , what is the smallest definability complexity of a class well-order that strictly dominates all  $\Sigma_n$  ordinal- parameter presentations in the sense of the successor-sum theorem?

Our construction proves an escape at some finite higher complexity. It does not identify the least possible increase, and it does not prove that every adjacent level of a chosen syntactic hierarchy has a particular canonical representative. Establishing sharp upper and lower bounds would turn a qualitative nonuniformity theorem into a finer ordinal-analysis tool.

**Research question 113.278** (Source 32, Question 12.6: Syntactic complexity of diagonal bounds). Given formulas of fixed quantifier complexity, how complicated must a formula defining a diagonal upper bound be? Can the valid-fiber filter and the global index order be optimized to produce a sharp complexity hierarchy?

The proof uses only a fixed finite list and an elementary well-foundedness test. Making the syntactic transformations explicit would both sharpen the mathematics and provide a suitable finite verification target for the repository's formula calculus.

### Definability and order-height spectra

**Research question 113.279** (Source 33, Research question 12.1: Sharp complexity of the diagonal). For a fixed syntax convention, what is the best explicit upper bound on the complexity of the formulas defining  $D_n$ ? [Theorem XVI.7.20](#) proves that no isomorphic  $\Sigma_n(G)$  presentation exists. Can a matching small additive increase in the Levy hierarchy be obtained, and can optimality be shown for natural coding schemes?

**Research question 113.280** (Source 35, Question 10.11: Optimal uniform-family bounding). Given a uniform family, can one construct a more canonical or smaller strict bound than its successor-sum, while staying in the same weak theory and preserving useful functorial properties?

The successor-sum depends on an ordering of the index class and makes no claim to be a least upper bound. Removing that dependence may require comparison witnesses or higher-order selection. The problem is therefore a useful test case for the difference between explicit bounding and canonical classification.

**Research question 113.281** (Source 33, Research question 12.2: The role of a definable global order). Which weaker hypotheses suffice for parameter-free cofinality in the pure language  $\{\in\}$ ? Can the global-order parameter be removed in models without a parameter-free definable global well-order, perhaps for selected families of carriers or selected formula complexities?

**Research question 113.282** (Source 34, R8: ordinal parameters versus arbitrary set parameters). The spectrum theorem permits all set parameters. Characterize models in which ordinal parameters already give the same spectrum, and models in which the two differ. Under an appropriate definable global coding of sets by ordinals the distinction disappears; without such coding it should be treated as a real restriction.

### 113.28.6 Definability ceilings and truth

[merge] Source 34's R9, source 32's question 12.4, source 33's questions 12.3 and 12.4, and source 35's question 10.10 ask for the same invariant, the definable ceiling  $\Theta_L$  (source 32's  $\delta_A$ , source 33's  $\delta_G$ , source 35's  $\delta_{\text{set}}$ ; [section XVI.12](#) (e)): open beyond the necessary conditions  $\kappa < \Theta_L < \kappa^+$ ,  $\text{cf} = \omega$  under a named global order, and closure under ordinal arithmetic, including  $\kappa^{\Theta_L} = \omega^{\Theta_L} = \Theta_L$ . Source 32's question 12.7 is answered in part: with a named global order, parameter-free definitions are cofinal ([theorem XVI.8.3](#) and [corollary XVI.7.21](#)); in the pure language source 35 bounds the three suprema ([theorem XVI.8.6](#)); initiality of the parameter-free spectrum fails at  $V_\kappa$  (the remark after [proposition XVI.8.2](#)). Source 34's R10, source 32's question 12.5 and source 33's question 12.5 ask how large a truth jump is: source 32 shows that truth represents the old ceiling exactly ([theorem XVI.8.12](#)); the size of the new ceiling is open.

**Research question 113.283** (Source 34, R9: the invariant  $\Theta_{A,G}$ ). Analyze the dependence of the external definable-order cutoff on the model and named predicates. Countable cofinality is now fixed under the stated hypotheses, but the exact cutoff need not be. Compare it with admissible structures, unrolling heights, and definable closure. A theorem should specify whether it concerns an external ordinal or an internally definable object.

[merge] (4 October 2026, batch 97.) Advanced by source 36: the cutoff over any transitive model of ZFC is a fixed point of  $\xi \mapsto \varepsilon_\xi$  above  $\zeta_{\kappa+1}$  ([proposition XVII.6.11](#)); it is bounded by the height of every admissible set containing the model ([S36.6.2](#)); and the full model over  $V_\kappa$  unrolls to  $H_{\kappa^+}$ , of height  $\kappa^+$  ([proposition XVII.8.1](#)). The exact cutoff stays open (source 36's [research questions 113.295](#) and [113.302](#); [section 113.31](#) (e)).

**Research question 113.284** (Source 32, Question 12.4: Classifying definability ceilings). For fixed inaccessible  $\kappa$  and global predicate  $G$ , which ordinals occur as  $\delta_A$ ? Are there natural sufficient conditions supplementing

$$\kappa < \delta < \kappa^+, \quad \text{cf}(\delta) = \omega, \quad \kappa^\delta = \delta?$$

Closure under hereditary finite term formation gives further necessary restrictions. It would be valuable to compare the ceiling with heights arising from admissibility, definable well-founded relations, or an appropriate unrolling, but no identification of these invariants is assumed in the proofs above.

[merge] (4 October 2026, batch 97.) Advanced, not answered: source 36 proves that these ceilings satisfy  $\varepsilon_\delta = \delta$  and  $\delta > \zeta_{\kappa+1}$  ([theorem XVII.7.5](#) and [proposition XVII.6.11](#)), which implies  $\kappa^\delta = \delta$  and makes the remark above on hereditary term formation precise; it bounds them by admissible heights ([S36.6.2](#)); and varying  $G$  instead of  $A$  gives the same supremum  $\kappa^+$  ([theorem XVII.7.11](#)). Which ordinals occur is open (source 36's [research question 113.295](#); [section 113.31](#) (c)).

**Research question 113.285** (Source 33, Research question 12.3: Identifying the external ceiling). Can  $\delta_G$  be characterized in admissibility, fine-structural, or unrolling terms for canonical choices of  $G$  over  $V_\kappa$  or constructible rank models? The manuscript proves countable cofinality, arithmetic closure, and the epsilon-number property, but not an exact ordinal notation for this ceiling.

[merge] (4 October 2026, batch 97.) Advanced, not answered:  $\delta_G$  is a fixed point of  $\xi \mapsto \varepsilon_\xi$  above  $\zeta_{\kappa+1}$ , not only an epsilon number (source 36's [theorem XVII.7.5](#) and [proposition XVII.6.11](#)), and it is bounded by the height of every admissible set containing  $V_\kappa$  (S36.6.2); no exact characterization is known (source 36's [research questions 113.295](#) and [113.302](#); [section 113.31](#) (c)).

**Research question 113.286** (Source 33, Research question 12.4: Dependence on the global-order predicate). How much can  $\delta_G$  vary as  $G$  ranges over set-like global well-orders of one fixed  $V_\kappa$ ? What information can be coded into such a  $G$  while preserving its order-theoretic role, and what external heights can occur? Distinguish a change of predicate from a change of the set model.

[merge] (4 October 2026, batch 97.) **Answered** by source 36 ([lemma XVII.7.10](#) and [theorem XVII.7.11](#); [section 113.31](#) (a)). As  $G$  ranges over the bijections  $\kappa \rightarrow V_\kappa$ ,  $\delta_G = \mathfrak{H}_G$  is unbounded in  $\kappa^+$ ; any  $A \subseteq \kappa$  can be coded into  $G$  by adjacent swaps so as to be parameter-free  $G$ -definable; and every  $\alpha \in [\kappa, \kappa^+)$  is the type of a parameter-free  $G$ -definable class well-order for some  $G$ . The set model is fixed throughout. Which ordinals occur is source 36's [research question 113.295](#).

**Research question 113.287** (Source 35, Question 10.10: Characterize the definability-sensitive cutoffs). For inaccessible  $\kappa$ , analyze the external suprema of the well-order types definable over  $V_\kappa$  with no parameters, ordinal parameters, real parameters, or arbitrary set parameters. Which equalities between these suprema are forced by structural properties of  $V_\kappa$ ?

The counting argument gives strict bounds below  $\kappa^+$ , and successor closure gives no maximum. It does not calculate the suprema. These are external invariants of a specified structure and language. They should not be presented as internally definable ordinals of  $V_\kappa$ , since many lie above its entire ordinal class. Models with a definable global well-order may behave differently from models where arbitrary set parameters carry information unavailable from ordinal parameters.

**Research question 113.288** (Source 32, Question 12.7: Parameter-free and restricted-parameter spectra). What replaces [theorem XVI.8.9](#) when set parameters are forbidden, restricted to ordinals, or restricted to a specified set of parameters? Which closure and initiality properties survive?

Downward closure in the proof used the arbitrary point parameter  $x$  of a presentation. Without that parameter, the spectrum need not be justified as an initial segment by the same argument. The first task is to specify the parameter class before comparing ceilings.

**Research question 113.289** (Source 34, R10: how large is a single truth jump?). Our sum construction gives an explicit upper bound beyond every old definable order. Find sharper external bounds on its order type and determine how much the indexing well-order affects it. There is no reason to assume that the chosen diagonal sum is the least possible representative of the new definability gap.

**Research question 113.290** (Source 32, Question 12.5: Quantitative truth jumps). Can  $\delta_{T_A}$  be bounded or characterized in terms of  $\delta_A$  and a natural additional invariant of  $(V_\kappa, G, A)$ ? How do successive truth predicates and uniformly admitted limit truth predicates affect the spectrum?

The present exact result is that the first truth extension represents the old ceiling, not a formula for the new one. Distinguishing a finite tower of individual predicates from one admitted sequence of all stages remains essential.

**Research question 113.291** (Source 33, Research question 12.5: Iterated truth growth). How large is the gap  $\delta_{G,T} - \delta_G$  in an appropriate ordinal sense, and how does the hierarchy behave under finite, set-length, or class-length iterated truth predicates? [Theorem XVI.8.19](#) provides strictness for one truth expansion, not a quantitative classification of the iterated spectrum.

### 113.28.7 Changes of class realization and forcing

[merge] Source 34's R11, source 32's question 12.9 and source 33's question 12.11 ask how ceilings and notation systems behave under changes of the class realization and under forcing; open. This report's questions on changes of class realization are indexed in [section 113.12](#) (row "Changes of the class realization").

**Research question 113.292** (Source 34, R11: forcing and new class-order types). Hamkins and Woodin establish preservation results under specified forcing hypotheses and formulate more general questions [27]. Study preservation of the explicit notation systems and of  $\Theta_{A,G}$  separately from preservation of all class-order types. Old orders can remain well-founded while new isomorphisms or new presentations become available.

**Research question 113.293** (Source 32, Question 12.9: Changes of class realization and forcing). How does the definability ceiling change when one enlarges the admitted predicates while retaining the same set universe? Which analogous invariants behave well when the set universe also changes by forcing?

The first situation includes the truth extension proved here. The second must be kept distinct: preservation results for class well-orders and ETR in the forcing literature impose their own hypotheses [27]. A useful starting point is to formulate comparison maps between two explicitly specified spectra instead of presuming that all externally identical ordinal types have admitted isomorphisms.

**Research question 113.294** (Source 33, Research question 12.11: Preservation of definability heights). How do  $\delta_G$  and the truth-expanded heights behave under forcing or changes of class realization? Known class-order preservation results [27] concern carefully specified hypotheses. Equality of set ordinals does not, by itself, settle preservation of a *definability* spectrum in a changed language.

**Source 34's closing remark and source 35's practical sequence.** The immediate mathematical deliverable is the choice-free history theorem. The fixed-point and definability results supply two distinct ways to extend the project: one builds increasingly expressive finite syntax, while the other measures how the available class language changes the range of order types.

**A practical sequence of next steps** A sensible implementation sequence starts with the choice-free finite-support lemma and explicit map laws, then verifies the normal-form theorem, and finally formalizes fixed-point completion. These steps have short dependencies and precise acceptance criteria. The full reverse-mathematical separation problems should be pursued after these constructive foundations are checked, because they rely on subtler model existence and truth considerations.

For the repository, the immediate mathematical changes would be to replace the polynomial well-order proof, lower the assumptions of the specific canonical-history theorems, add the local fixed-point criterion, and attach an explicit status note to each theorem. A separate module should



contain the notation interface and its constructor certificates. This article and its archive are a standalone deliverable; no repository mutation or publication is part of the present work.

The resulting picture is precise: class-order arithmetic has many explicit, arbitrarily extensible constructions; the canonical histories available for all inputs are governed by strong comparability; and uniform definability imposes its own barriers even when individual presentations are elementary. These three facts give distinct, testable directions for further work.

### 113.29 Earlier questions answered by sources 32–35 (dated notes, 4 October 2026)

[merge] Dated notes in the form of [section 113.27](#); where the question indexes of [sections 113](#), [113.5](#) and [113.12](#) say otherwise, these notes supersede them, and the cells concerned carry dated notes. [Section XVI.12](#) gives the same status with the theorems in context.

- (a) *Source 20's research question 113.111, the strength of complete condensation histories “over GB or GBC”*: answered over GBC by source 29 ([theorem 108.19](#)) and now over GB by sources 34 and 35 ([theorem XVI.5.8](#) and [note XVI.5.19](#)): the existence of the histories for all inputs is equivalent to CWO, and to source 35's FPE ([corollary XVI.6.19](#) and [theorem XVI.6.21](#)). Whether CWO implies ETR stays open ([research question 113.219](#)).
- (b) *Source 11's research question 113.3 and the related questions of sources 13, 15, 16 and 18 (research questions 113.30, 113.48, 113.64 and 113.86)*, whose status after Part XIV left “the theory over GB” open ([section 113.25](#) (g)): the canonical-history formulation is now answered over GB as in (a). Source 15's first question, whether GB codes all finite condensations uniformly ([research question 113.48](#)), stays open; source 35 asks it again ([research question 113.260](#)).
- (c) *Source 31's research question 113.239, definable-class versions of the spectra*: answered in part, for external order types at an inaccessible height ([section XVI.12](#) (e)); its classification of global choice, enumeration, coding and Class Replacement for  $\text{Def}(V_\lambda, p)$  stays open.
- (d) *Source 29's research question 113.219*: not answered; sources 34 and 35 add equivalent forms of CWO over GB ([theorem XVI.5.8](#) and [corollary XVI.6.19](#)).

### 113.30 The questions of source 36 (batch 97)

[merge] (4 October 2026.) Source 36's ten research questions (its Section 9; Section XVII.9), printed whole with its commentary after its introduction to them, with the renamings of Part XVII (horizons  $\mathfrak{H}$ , the hereditary order  $E$ ). Each question is followed by a dated [merge] paragraph naming the questions of [section 113.28](#) it coincides with and its status. All ten are open; the second is answered in part by Part XVI. Questions of this report that source 36 answers or advances are in [section 113.31](#).

The first group below consists of questions suggested by the proved structure. They are an agenda for further investigation, not claims that each is a newly identified open problem in the literature. The final comparison question has an explicit published antecedent.

**Research question 113.295** (Source 36, Research question 1: Which horizons are realizable?). For fixed inaccessible  $\kappa$ , characterize

$$\mathcal{H}_\kappa = \{\mathfrak{H}_G : G \text{ codes a bijection } \kappa \rightarrow V_\kappa\}.$$

The article proves this set is unbounded in  $\kappa^+$ . Every member has cardinality  $\kappa$ , cofinality  $\omega$ , is fixed by epsilon enumeration, and exceeds the explicit zeta bound. These are necessary conditions, not a sufficient characterization. Further finitary notation operators may force additional closure

conditions. A useful first objective is to identify a nontrivial sufficient condition, or to exclude a candidate satisfying all conditions established here.

[merge] Open. It is the question of the theme “Definability ceilings and truth” of [section 113.28](#) (source 34’s [research question 113.283](#), source 32’s [research question 113.284](#), source 33’s [research questions 113.285](#) and [113.286](#), source 35’s [research question 113.287](#)) with the predicate  $G$  varying instead of an additional predicate  $A$ . The necessary conditions now known for one  $\mathfrak{H}_G$ :  $\kappa < \zeta_{\kappa+1} < \mathfrak{H}_G < \kappa^+$ ,  $\text{cf} = \omega$ ,  $\varepsilon_{\mathfrak{H}_G} = \mathfrak{H}_G$  ([theorem XVII.7.5](#) and [proposition XVII.6.11](#)), closure under the finite-support constructors and  $\text{Ep}$ ; unboundedness of the set is [theorem XVII.7.11](#). The sentence “Further finitary notation operators may force additional closure conditions” is a conjecture of source 36, not proved.

**Research question 113.296** (Source 36, Research question 2: Parameter-free versus parameter-allowing horizons). For which transitive set models  $M$  does  $\mathfrak{H}_0(M) = \mathfrak{H}(M)$ , and when can there be a strict gap?

The equality is proved here in the inaccessible-height setting after a global enumeration is named. The unrestricted transitive-model case lacks the same uniformly sound well-order filter and may lack a definable global enumeration. A classification should identify separately the roles of these two issues. Merely observing that there are countably many parameter-free formulas does not settle the question.

[merge] Answered in part by Part XVI. Source 34’s [theorem XVI.8.3](#) proves  $\sup \mathcal{S}_0^L = \Theta_L$  for every transitive  $\beta$ -model of GBC whose language names a global well-order  $G$ , and its last sentence removes  $G$  from the language when  $G$  is parameter-free definable from the other predicates; so  $\mathfrak{H}_0(M) = \mathfrak{H}(M)$  whenever  $(M, \text{Def}(M))$  is a  $\beta$ -model with a parameter-free definable global well-order (source 33’s [corollary XVI.7.23](#) is the internal form of the same elimination, by embeddings). Open without the  $\beta$ -hypothesis or without such a well-order: the two issues source 36 separates. The question is source 32’s [research question 113.288](#) and source 33’s [research question 113.281](#) for the pure language, and source 34’s [research question 113.282](#) concerns the related comparison of ordinal and set parameters.

**Research question 113.297** (Source 36, Research question 3: The complexity profile). For a fixed nested hierarchy of first-order formulas, calculate the bounded-complexity heights  $h_n$  and sharpen the effective bound  $h_n < h_{f(n)}$ .

The proof gives an effective increase without optimizing quantifier alternations. An audited coding might yield a small constant shift. This is a concrete syntactic project: fix the representation of domains, relations, partial truth, and validity, then count the actual complexity of the diagonal formula. Numerical shifts should be conclusions of that audit, not guesses.

[merge] Open. It is the question of the theme “Diagonals and definability complexity” of [section 113.28](#): source 35’s [research question 113.277](#), source 32’s [research question 113.278](#), source 33’s [research question 113.279](#). “An audited coding might yield a small constant shift” is a conjecture of source 36, not proved.

**Research question 113.298** (Source 36, Research question 4: The size of truth promotion). How much information about  $\mathfrak{H}_{G,T}$  is determined by  $\mathfrak{H}_G$ , when  $T$  is truth for  $(V_\kappa, \in, G)$ ?

We prove that  $\mathfrak{H}_G$  acquires a presentation and that the horizon strictly increases. We do not calculate the new endpoint from the old one. One can first examine finite iterates of truth in a fixed external full model, then ask which features depend on the predicate’s content rather than merely the previous horizon.

[merge] Open. It is source 34's [research question 113.289](#), source 32's [research question 113.290](#) and source 33's [research question 113.291](#). Known about  $\mathfrak{H}_{G,T}$ : it exceeds  $\mathfrak{H}_G$  and is presented exactly at  $\mathfrak{H}_G$  ([theorem XVII.7.9](#); source 32's [theorem XVI.8.12](#)), it has cofinality  $\omega$  (source 33's [theorem XVI.8.19](#)), and, since [theorem XVII.7.5](#) applies to the expanded language as source 36 says before [theorem XVII.7.9](#),  $\varepsilon_{\mathfrak{H}_{G,T}} = \mathfrak{H}_{G,T}$ .

**Research question 113.299** (Source 36, Research question 5: Higher finitary notation functors). Develop and compare class versions of Veblen and other well-order preserving finite-term operators, with exact predecessor semantics.

The support-localization theorem gives a method once a valid set-order operator and its support invariance are supplied. The task is to provide those data and prove the resulting class theorems. The Veblen constructions of [97] are a precise starting point. One should not infer that every ordinal function admits such a finite-support realization.

[merge] Open. It is source 34's [research question 113.272](#) and source 35's [research question 113.273](#), with source 35's [research question 113.271](#) for functors with a supplied initial embedding. Source 36's transfer theorem ([theorem XVII.6.2](#)) and epsilon-term functor ([theorem XVII.6.4](#)) are its first step (dated note at [research question 113.272](#)).

**Research question 113.300** (Source 36, Research question 6: Normalizing general powers of point-cuts). Extend the explicit arithmetic on hereditary  $\Omega$ -terms to a fully proved power-normalization algorithm.

The article proves direct addition and multiplication of represented point-cuts and supplies arbitrary class exponentiation as a presentation. It does not prove an algorithm converting every such power back into a term of E. A complete treatment should handle zero, successors, limit tails, and the exact distinction between a point and its predecessor class. This is a proof and implementation objective, rather than a claim that the corresponding set-ordinal algorithms are unknown.

[merge] Open. It is the exponentiation part of source 34's [research question 113.275](#), whose addition and multiplication part source 36 answers ([theorems XVII.5.1](#) and [XVII.5.2](#); dated note at [research question 113.275](#)).

**Research question 113.301** (Source 36, Research question 7: Weaker set theories). Which localization and finite-term transfer results survive without Power Set or with weaker forms of Collection?

The choice-free localization proof uses rank sets. A global enumeration can sometimes supply alternate set bounds. The appropriate question is which bounding and replacement principles are sufficient for each construction, not whether the displayed GB proof can be copied unchanged into a weaker theory.

[merge] Open. It is the set-theoretic half of source 32's [research question 113.262](#) ("how much set theory is needed for support localization . . ."), whose choice half is answered over GB ([section 113.28](#), theme "Comparison, recursion, and the strength of histories"). Part XV studies Zermelo set theory without Replacement, a related weakening ([part XV](#)).

**Research question 113.302** (Source 36, Research question 8: Admissible and definability bounds). Give sharper model-sensitive bounds for  $\mathfrak{H}(M)$ , and identify when the admissible upper bound is attained.

The existence of equality examples is already discussed in the historical sources [100, 101]; it should not be posed again as an unanswered existence question. A more informative objective is a classification by closure properties, class well-foundedness correctness, and the available parameter regime.

[merge] Open. It is the comparison with admissible structures asked by source 34’s [research question 113.283](#), by source 32 after [research question 113.284](#) and by source 33’s [research question 113.285](#); source 36’s bound (S36.6.2) and [proposition XVII.8.1](#) are first steps (dated note at [research question 113.283](#)). The MathOverflow discussions cited were not checked by this write.

**Research question 113.303** (Source 36, Research question 9: Stability under extensions). How do the spectra and horizons change under forcing, under changes of class realization, and under expansions by specific predicates?

Changing a class realization with the same sets preserves well-foundedness of a shared relation under the relevant base axioms, but can add comparison maps and new presentations. The truth and coding theorems give two sharply controlled expansion mechanisms. A forcing analysis must account for both new predicates and potentially new externally well-founded presentations. The broader forcing questions about ETR have a separate literature [27].

[merge] Open. It is the theme “Changes of class realization and forcing” of [section 113.28](#): source 34’s [research question 113.292](#), source 32’s [research question 113.293](#), source 33’s [research question 113.294](#).

**Research question 113.304** (Source 36, Research question 10: Class comparison and recursion strength). What are the exact reverse implications among ETR, continuous rankings, initial comparability, and weak class-order comparability over the relevant weak class theories?

This is the question posed in [35, 27]. No solution is claimed here, and no later resolution was verified by the targeted search. The explicit finite-code orders offer a controlled test family: their comparison maps can be constructed directly, which helps separate the strength of specific notation systems from the strength of comparison for every class well-order.

[merge] Open. It is Hamkins and Woodin’s Question 7, source 29’s [research question 113.219](#), and the theme “Comparison, recursion, and the strength of histories” of [section 113.28](#) (source 34’s [research questions 113.252](#) and [113.256](#), source 35’s [research questions 113.253](#) and [113.257](#), source 32’s [research question 113.254](#), source 33’s [research question 113.255](#)).

### 113.31 Earlier questions answered or advanced by source 36 (dated notes, 4 October 2026)

[merge] Dated notes in the form of [section 113.29](#); the questions concerned carry dated notes of their own in [section 113.28](#), and [section XVII.11](#) gives the same status with the theorems in context.

- (a) *Source 33’s [research question 113.286](#), dependence of  $\delta_G$  on the global-order predicate: answered.* As  $G$  ranges over the bijections  $\kappa \rightarrow V_\kappa$ , the ceilings  $\delta_G = \mathfrak{H}_G$  are unbounded in  $\kappa^+$ ; every  $A \subseteq \kappa$  can be coded into some  $G$  so that  $A$  is parameter-free  $G$ -definable; and every  $\alpha \in [\kappa, \kappa^+)$  is the type of a parameter-free  $G$ -definable class well-order for some  $G$  ([lemma XVII.7.10](#) and [theorem XVII.7.11](#)). The change is of the predicate only, with the set model fixed, as the question asks to distinguish. Which ordinals occur as  $\mathfrak{H}_G$  is source 36’s [research question 113.295](#).
- (b) *Source 34’s [research question 113.275](#), normalization of arithmetic inside the hereditary system: answered for addition and multiplication by [theorems XVII.5.1](#) and [XVII.5.2](#), proved*

- against class isomorphisms of the represented initial segments. Exponentiation, and the question which operations preserve finite normal forms, stay open ([research question 113.300](#)).
- (c) *Source 32's research question 113.284 and source 33's research question 113.285, classifying or identifying the definability ceiling*: advanced, not answered. New necessary conditions:  $\varepsilon_\delta = \delta$  and  $\delta > \zeta_{\kappa+1}$  ([proposition XVII.6.11](#) and [theorem XVII.7.5](#)), which proves source 32's remark that closure under hereditary finite term formation gives further necessary restrictions.
  - (d) *Source 34's research question 113.272 (and source 35's research question 113.273), a Veblen-style hierarchy of class-order constructors*: advanced by the finite-support transfer theorem and the epsilon-term functor on class well-orders ([theorems XVII.6.2, XVII.6.4 and XVII.6.7](#)); the hierarchy itself is open ([research question 113.299](#)).
  - (e) *Source 34's research question 113.283, the invariant  $\Theta_{A,G}$  and its comparison with admissible structures and unrolling heights*: advanced by the admissible bound ([S36.6.2](#)), the unrolling height  $\kappa^+$  of the full model ([proposition XVII.8.1](#)) and the closure properties of (c); open ([research question 113.302](#)).
  - (f) *Source 31's research question 113.239, definable-class versions of the spectra*: advanced for the external order types at heights that are not inaccessible: for every transitive  $M \models \text{ZFC}$  the definable spectrum is an interval  $[0, \mathfrak{H}(M))$  with  $\mathfrak{H}(M) < |\kappa|^+$ , its proper-carrier part is  $[\kappa, \mathfrak{H}(M))$ , and  $\varepsilon_{\mathfrak{H}(M)} = \mathfrak{H}(M)$  ([theorem XVII.6.9](#) and [propositions XVII.6.10 and XVII.6.11](#)). The thresholds for global choice, enumeration, coding and Class Replacement stay open, as in [section 113.29](#) (c).
  - (g) *Not answered*: Hamkins and Woodin's Question 7 (source 29's [research question 113.219](#); source 36's [research question 113.304](#)).

## 114 The sources' conclusions

[merge] Source 11 has no conclusion section; its summary is [section 115](#). The conclusions of sources 12, 09 and 08 are printed whole.

### 114.1 Source 12's conclusion

There is a rigorous full lexicographic order of all global surreal well-orders, provided it is understood as a relation on class objects, not as an ordinary class containing those objects. Its comparison needs no universal enumeration of abstract class order types: the largest common *labelled* prefix is elementary and sufficient.

The set-like suborder has a particularly uniform geometry. It fills set-indexed cuts, contains every class linear order even locally, and is equimorphic with binary classes on Ord. The unrestricted order adds new proper-class prefix phenomena, including finite adjacent blocks. The diagonal theorem prevents either construction from being exhausted by a same-level class of set codes.

At set-sized birthday bounds the theory becomes quantitative. The full well-order space of a carrier of cardinality  $\mu$  has exact ordinal spectrum below  $\mu^+$  and binary rank  $\mu^+$ . The minimal surreal slice has rank  $\mu$ , except at singular strong-limit cutoffs, where it has rank  $\kappa \cdot \kappa$ . This is an order-theoretic effect of cofinal blocks of code lengths, invisible to cardinal counting alone.

The inaccessible-rank interpretation shows exactly how these facts can coexist: internal classes and the external collection of all such classes occupy different levels. Respecting those levels removes the apparent paradox without weakening the embedding and universality questions into trivialities.



## 114.2 Source 09’s conclusions

The lexicographic well-ordering problem has three different answers, depending on its carrier. On a fixed set of surreals of size  $\kappa$ , the exact ordinal spectrum stops at  $\kappa^+$ . Allowing arbitrary set-length words gives a class-universal dictionary order, but prefix limits give it local order properties different from those of **No**. For set-like global well-orders, first disagreement again defines a linear comparison. Existence of a global enumeration is equivalent to global choice over **GB** with set choice, and all global enumerations cannot be collected into an ordinary class of set codes.

The eventual-agreement core resolves the constructive side of the global problem: it is a genuine class, it is isomorphic to the surreal line, it realizes every uniformly set-sized cut of global enumerations, and it is dense in every global interval. Bounded reflection explains why set-indexed configurations can always be studied in a suitable bounded permutation stage. The explicit missing branch and binary diagonal identify two independent boundaries: arbitrary class-sized cuts need not have solutions, and no single uniform class-indexed coding covers all global enumerations.

These boundaries are not paradoxes. They are the information needed to state theorems at the correct logical level.

## 114.3 Source 08’s conclusion

The global enumeration problem has a particularly clean answer once the types of objects are separated. Set-like well-orders of **No** are represented by class bijections from **Ord**, and numerical first-disagreement gives a rigorous virtual linear order. That order is large enough to encode every class binary word, yet its set-cut geometry is already witnessed by a single class of set-supported perturbations. This code class is another copy of the surreal line.

At a fixed set carrier, the structure is different: its exact ordinal spectrum stops at the successor of the carrier cardinal, and finite permutation tails account for every jump. At varying set carriers, termination introduces new local cofinalities. For non-set-like well-orders, a further index convention or relation code is necessary. None of these distinctions is optional bookkeeping: each changes a mathematical conclusion.

The principal proofs are elementary in their stated foundations. Their formalization should begin with bounded objects and the repository’s existing sign-sequence cut theorem, and only then pass to the uniform class formulation. The dense-skeleton theorem provides a concrete target: it combines first-disagreement combinatorics, controlled set support, small-cut filling, and universe-aware back-and-forth without requiring a paradoxical collection of all classes.

[merge] (3 October 2026.) The conclusions of sources 13, 14 and 17 follow; source 13’s includes its result ledger. Sources 15, 16 and 18 end with status tables instead, printed with the ledgers in the next section. In source 13’s ledger, the row “No full two-point split embedding” is a further statement whose proof is source 11’s ([note 55.6](#)).

## 114.4 Source 13: conclusions and result ledger

The strongest organizing principle is not that a single enormous well-order has been discovered. It is that different notions of “all” and different representations produce distinct linear orders, while a small collection of explicit tools connects them safely.

The fixed-set layer has its own cardinal and ordinal spectra. The ordinary class of set words depends decisively on termination. The full set-like enumeration domain has a surreal-sized dense core but no exhaustive uniform class of set codes. The raw order domain expands individual branch fibers, not the entire cut space of that core. A higher-level product can exceed the dense core’s



capacity even though ordinary class-indexed sums are absorbed. Finally, pure order isomorphisms can change the prefix presentation itself.

| Result                                               | Status in this article                                                              | Stated setting                                     |
|------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------|
| Fixed-alphabet spectra and singular cutoff dichotomy | Credited prior written results                                                      | ZFC                                                |
| Direct raw comparison and head fibers                | Prior mechanism, re-proved for use                                                  | GBC suffices here                                  |
| Core completion, set interpolation, diagonal         | Prior foundation, self-contained proofs                                             | GBC                                                |
| Crossing criterion and label coverage                | Further result, <a href="#">Theorem 46.15</a>                                       | GBC                                                |
| First-failure trichotomy and character formulas      | Further result, <a href="#">Theorem 46.18</a> and <a href="#">Proposition 46.19</a> | GBC                                                |
| <b>No</b> -left-factor absorption                    | Further formulation and proof, <a href="#">Theorem 55.1</a>                         | Uniform GBC maps                                   |
| No full two-point split embedding                    | Further result, <a href="#">Theorem 55.3–55.4</a>                                   | Full inaccessible universe; uniform KM + GC schema |
| External rational presentation                       | Semantic contrast, <a href="#">Proposition 55.5</a>                                 | Externally countable model                         |
| Cylinder non-rigidity                                | Further result, <a href="#">Theorem 47.12</a>                                       | Uniform GBC automorphism                           |
| Central-word surreal isomorphism                     | Further result, <a href="#">Theorem 36.4</a>                                        | GBC, reserved zero marker                          |
| New Lean proofs                                      | Not supplied                                                                        | Formalization plan only                            |

The follow-up therefore gives explicit answers to several concrete questions raised by the repository while leaving stronger classification and minimal-axiom problems clearly identified. The proofs and their stated size hypotheses are the basis on which these conclusions should be evaluated.

### 114.5 Source 14: conclusion

The strongest organizing distinction is not between large and small cardinalities alone. It is between set-coded presentations, class-valued branches, and raw class relations with further tails. Prefix cylinders make that distinction mathematically visible.

Every normalized class cut of the set-supported core either stops at a cylinder edge, stops at a genuine numerical gap, or follows an injective branch through all ordinals. Exhaustive well-orders are exactly the branches that cover the alphabet. Because this criterion reconstructs a graph using only set quantifiers and old parameters, it yields a no-new-realization theorem for old cuts. Raw well-orders have the same trace precisely when their canonical initial branches agree, leaving an explicit residual well-ordering problem in each fiber.

Support restrictions provide a different axis of complexity. Allowing support of cardinality  $< \kappa$  gives exactly the cut-filling threshold of  $\kappa$ , and the failure beyond it has an attained support cost. All these layers still contain a copy of the full surreal order. Thus class universality, density in an ambient comparison order, and saturation are genuinely different properties.

The resulting theory remains paradox-free because each “all” has a specified sort, each prefix has a specified size, and each use of surreal cut filling has set option families. The automorphism-orbit theorem sharpens this distinction: bare-order symmetries can exchange exhaustive traces with nonbranch boundaries. The principal open targets are now the exhaustive-trace stabilizer, finer support invariants, and the interaction between raw tail structure and higher class realizations.

## 114.6 Source 17: conclusion

The lexicographic ordering problem separates into several mathematically substantial structures. On the minimal birthday-cutoff slice, the relevant parameters are the cardinality  $\mu = 2^{<\kappa}$  and two different cofinalities  $r = \text{cf } \kappa$ ,  $q = \text{cf } \mu$ . Point characters are  $(q, q)$ , cylinder cofinalities are  $r$ , and three distinct cut mechanisms give the exact gap spectrum. Exhaustiveness is also the key issue in fusion: a coherent full-length injection need not be a permutation.

The same distinction persists at class scale. A single lower class cut of a supported core determines an injective global word exactly when its labelled cylinders straddle at every ordinal length. A separate coverage condition distinguishes exhaustive words. Reconstruction is elementary, but exhaustion is invisible to the bare core order: a core automorphism can carry an exhaustive branch gap to a nonexhaustive one. The represented- cut predicate is therefore genuine additional structure, and same-set class extensions change it while leaving the code carrier unchanged.

For unrestricted class well-orders, a proper-class prefix can leave a finite residual alphabet. The resulting maximal adjacency components are exactly the factorial blocks of a finite final segment; their condensation is dense without becoming an ordinary class of all class orders. Across all these conclusions, typing the objects and bounding the recursion histories are part of the mathematics, not merely notational cautions.

[merge] (3 October 2026, batch 89.) Source 31 has no conclusion section; its closing section, on the consequences for this report’s questions and for formalization, is printed in Part XV ([section XV.12](#)).

## 115 Results, assumptions, and verification status

| Result                                                                   | Foundation used here                                              |
|--------------------------------------------------------------------------|-------------------------------------------------------------------|
| Fixed-alphabet cardinality and ordinal spectrum                          | ZFC; existing repository theorem                                  |
| Explicit set-length word compression                                     | ZF, as a definable class construction                             |
| Local word characters and set-cut arguments                              | Surreal set cuts; ZFC for cardinal-valued characters              |
| Any class well-order of <b>No</b> implies global choice                  | GB with ZF set part                                               |
| Raw comparison, canonical initial part, finite-jump criterion            | GB, with the given class orders                                   |
| Bounded-support completion and interpolation                             | GBC, with a fixed global enumeration                              |
| $\mathcal{P}_{\text{bd}} \cong \mathbf{No}$ and class-order universality | GBC, by set-valued back-and-forth                                 |
| Binary class coding and no exhaustive uniform evaluator                  | GBC, interpreted with class parameters                            |
| Unrestricted order does not embed into its set-like fragment             | Full inaccessible universe; or KM + GC for definable uniform maps |
| All higher-order collections treated as actual objects                   | An explicitly chosen higher-order or containing-universe setting  |

The proof dependencies are short. Set cuts yield fresh coefficients after deleting a set. Fresh coefficients and bounded-support completion yield strong interpolation. Strong interpolation yields density, local cofinality statements, the **No**-isomorphism, and cut coding. Independently, the

predecessor comparison yields convex cones, the canonical initial-part decomposition, and the finite-residual jump theorem. Pair swaps supply both positive binary embeddings and the diagonal obstruction.

The bundled Python script checks finite instances of the two comparison rules and of the sign encoding. It checks 465 coefficient-code comparisons, 79,800 word-code comparisons, and 266,272 pairs of finite permutations, with the predecessor rule checked in both directions and the adjacency criterion compared with the finite lexicographic order. These checks help detect a reversed delimiter or an incorrect finite comparison. They do not verify an ordinal induction, a class comprehension argument, or a cardinal nonembedding theorem.

The article contains full mathematical proofs for the claims it promotes to theorems. It does not claim that the new results have been checked in Lean, that the whole repository has been audited, or that every research question listed here is a previously published open problem.

[merge] The other sources' ledgers follow; each records the hypotheses its source used, not that each is necessary.

### 115.1 Source 12's proof-dependency and assumption ledger

| Result                                             | Stated setting                         | Essential input                                                                                     |
|----------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------|
| Common labelled prefix and full lexical comparison | GBC; comparison is elementary          | Existing well-orders; class minima; ambient linear order.                                           |
| Global-choice equivalence                          | GB + set AC                            | Ordinal coding of transitive closures; restriction of a surreal well-order to a set of codes.       |
| No exhaustive class-coded family                   | GBC                                    | A baseline Ord-bijection; elementary diagonal pair orientation.                                     |
| Set-like cut interpolation                         | GBC                                    | Set-sized discrepancy bound; surreal small-cut filling; prefix extension.                           |
| Local class universality                           | GBC                                    | Set-valued forth and back-and-forth recursions; global enumeration.                                 |
| Finite-tail adjacency                              | GBC                                    | Label transpositions; finite residual cylinders; no infinite order well-ordered in both directions. |
| Set-sized full spectra                             | ZFC                                    | Pair coding; regularity of $\mu^+$ ; cylinder-fusion nonembedding.                                  |
| Minimal-slice singular transition                  | ZFC                                    | Cone partition; scheduler; ordinal block sums; regularity or singular cofinality as appropriate.    |
| Full-class rank model                              | Ambient ZFC + an inaccessible $\kappa$ | Closure of $V_\kappa$ ; all external classes; regularity for descending sequences.                  |

### 115.2 Source 09's theorem-status ledger

| Result           | Logical setting and proof input                | Status here                                 |
|------------------|------------------------------------------------|---------------------------------------------|
| Surreal set cuts | Set options; standard sign-sequence simplicity | Classical; repository primitives inspected. |

| Result                                                      | Logical setting and proof input                       | Status here                                    |
|-------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|
| Class universality and back-and-forth                       | GBC; set-valued recursion                             | Classical mechanism, proved in full.           |
| Global baseline $\Leftrightarrow$ GC                        | GB plus set choice; well-founded ordinal codes        | Standard coding argument, proved here.         |
| Fixed-set cardinality and ordinal spectrum                  | ZFC; pair switches and regular-cardinal stabilization | Generalization/reproof of the set-sized route. |
| Limit-prefix character of $\mathcal{I}$                     | ZFC schema or GBC class statement                     | Written proof.                                 |
| Bounded-support extension                                   | GBC with fixed $b$ ; set cardinal arithmetic          | Written proof; finite mechanism tested.        |
| Core interpolation and $\mathcal{P}_s(b) \cong \mathbf{No}$ | GBC; uniform families, small cuts, extension          | Main construction; written proof only.         |
| Bounded reflection                                          | GBC; Replacement for a uniform set family             | Written proof.                                 |
| Missing (Ord, Ord) branch                                   | GBC; ordinal prefixes and uniform completions         | Explicit counterexample with proof.            |
| Uniform no-coding                                           | GBC; pair switches and diagonal class                 | Written proof; finite diagonal tested.         |
| Inaccessible model                                          | ZFC plus a strongly inaccessible cardinal             | Conditional set-sized theorem, proved here.    |
| Proposed Lean modules                                       | Explicit universes, smallness, and baseline parameter | Plan only; not compiled or verified.           |

### 115.3 Source 08's assumption and proof-dependency ledger

| Result                                      | Assumptions                                  | Main dependencies                                         |
|---------------------------------------------|----------------------------------------------|-----------------------------------------------------------|
| Fixed-carrier cardinality and universality  | ZFC, $ X  = \kappa$ infinite                 | Pair blocks and lower-cut bits                            |
| Exact ordinal spectrum                      | Same                                         | Regularity of $\kappa^+$ , not of $\kappa$                |
| Adjacency classification                    | Same                                         | Extreme residual enumerations, two-sided well-foundedness |
| Regular cutoff theorem                      | $\kappa$ regular, $2^{<\kappa} = \kappa$     | Birthday bounds, terminal coding, active-row cuts         |
| Pairwise global comparison                  | Class parameters, elementary comprehension   | A least disagreement below one witness                    |
| Orbit prefix extension                      | Baseline $b$ , class replacement, set Choice | Enlargement to a bigger set cardinal                      |
| Orbit interpolation                         | Baseline and uniform set families            | Cross-disagreement bound, fresh entry, prefix extension   |
| Skeleton isomorphism and class universality | GBC                                          | Global enumerations, uniform set-valued recursion         |
| Virtual binary equimorphism                 | Baseline, elementary class operations        | Pair blocks and ordinal concatenation                     |
| No exhaustive class-uniform list            | GBC for arbitrary class indices              | Pairwise diagonalization                                  |
| Full-class model example                    | External strongly inaccessible $\kappa$      | Relative sets versus external sets                        |

The ledger records the hypotheses actually used in the supplied proofs. It does not claim that every hypothesis is logically necessary. In particular, the foundational optimization questions in [section 113.4](#) remain distinct from the sufficiency results.

[merge] (3 October 2026.) The ledgers and status tables of sources 14, 15, 16, 17 and 18 follow (source 13's is with its conclusions in [section 114](#)). Each records what its source proved in writing

and what it inherited; “new” there means new relative to this report at the source’s pin, not new relative to the sibling sources, which proved several of the same theorems independently (section 4.5). Source 18’s table is printed whole, as is its remark that its review corrected the arbitrary-alphabet density construction to include isolated global endpoints (the correction is in the printed theorem, theorem 27.9). Source 16’s last subsection describes its delivered package; the shipped names are listed in appendix B.

#### 115.4 Source 14: result ledger and verification status

| Result                                                       | Status in this package                                                                                                                  |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Fixed-set cardinality, ordinal spectrum, adjacency           | Inherited from the pinned ProveIt report; stated with source, not newly established here.                                               |
| Birthday minimal-slice dichotomy                             | Inherited; ordinal multiplication and mutual embeddability explicitly retained.                                                         |
| Core cylinders and prefix completion                         | Reproved as prerequisites; basic constructions already present in the predecessor.                                                      |
| Boundary trichotomy, Theorem 43.3                            | Main continuation result; complete written proof, relative to named cylinders.                                                          |
| Character table, Theorem 44.1                                | Written derivation from the trichotomy, including the principal-cut correction.                                                         |
| Exhaustive recognition, Theorem 45.1                         | Answers the prefix-enriched version of a predecessor research question.                                                                 |
| Class-model absoluteness, Theorem 48.1                       | Written proof using the elementary reconstruction formula and fixed old parameters.                                                     |
| Cut automorphism orbits and nonrecognizability, Theorem 47.2 | Written proof by set homogeneity and two-sided back-and-forth; distinguishes order-invariant classification from prefix classification. |
| Support compression and exact threshold, Theorem 49.3        | Main continuation result; finite kernel checked computationally, infinite/cardinal proof supplied in text.                              |
| Core isomorphism with <b>No</b> ; diagonal non-coding        | Inherited statements; proofs or recovery arguments included for context.                                                                |
| Direct raw comparison and canonical branch                   | Inherited constructions; short proofs given to support the trace analysis.                                                              |
| Trace fibers, Theorem 54.6                                   | New synthesis of the canonical branch with cut recognition; written proof.                                                              |
| Historical priority of continuation results                  | Not established; no claim of a first result in the published literature.                                                                |
| Lean verification of continuation results                    | Not performed. Proposed module names and pseudocode are not compiled artifacts.                                                         |

The entire manuscript is an unrefereed research contribution. The ledger separates what is proved in writing, what is inherited, and what remains unverified; a finite diagnostic is never substituted for a class theorem.

#### 115.5 Source 15: what is settled here and what remains

| Topic                       | Result and boundary                                                                                                               | Foundation                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Set-sized spectra           | Cardinality and ordinal universality are inherited background, with proof reproduced                                              | ZFC                                  |
| Native class comparison     | Direct first unequal labels; no abstract class-type comparison required                                                           | GB with given class well-orders      |
| Canonical block quotient    | All first-level blocks defined simultaneously; any class index can occur                                                          | GB for quotient; GBC for realization |
| Long condensation histories | Explicit polynomial histories for every given class well-order; no arbitrary-input equivalence to ETR                             | GBC                                  |
| Represented cuts            | Persistence at every set prefix level is necessary and sufficient; all first failures classified                                  | GBC with fixed core                  |
| Abstract reconstruction     | An automorphism of the core can destroy representability                                                                          | GBC                                  |
| Bounded local geometry      | Local character $\text{cf}(2^{<\kappa})$ , global cofinality $\text{cf } \kappa$ , and exact $\eta_{\text{cf } \kappa}$ threshold | ZFC                                  |
| Longer fixed types          | Exact cube and finite-factor classification for $\mu \cdot \gamma + n$ ; not all smaller infinite residual tails                  | ZFC                                  |
| Alternative codes           | Rank endpoints, table endpoint classification, quotient and restriction obstructions                                              | ZFC or GBC as stated                 |
| Formal verification         | Concrete interfaces proposed; finite examples checked; new transfinite proofs not machine-verified                                | Explicit status, not an added axiom  |

The research questions below are proposed extensions of this development. Except for a cited historical question, they are not claims that a named problem has an independently verified current open status in the literature.

### 115.6 Source 16: proof status, assumptions, and reproducibility

| Result                                                                    | Assumptions                                            | Status relative to the inspected report                   |
|---------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------|
| Set-alphabet cardinality and ordinal spectrum, <a href="#">Note 32.1</a>  | ZFC                                                    | Existing result, reproved.                                |
| All-stratum density and cellularity; countable incompleteness             | ZFC                                                    | New extension here, with full proofs.                     |
| Exact regular-stage gap spectrum                                          | ZFC; regular $\mu$ ; $\eta_\mu$ alphabet of size $\mu$ | New extension here.                                       |
| Endpoint word-tree isomorphisms; complete termination type classification | GBC                                                    | New extension here; explicit embeddings need less choice. |



| Result                                                             | Assumptions                               | Status relative to the inspected report                        |
|--------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------|
| Direct raw comparison and GC equivalence                           | GB; given class well-orders               | Existing result, explained and reproved.                       |
| Simultaneous set-interval block decomposition and largest set tail | GB                                        | New extension here; no ETR.                                    |
| Bounded-support completion and interpolation                       | GBC; uniform families                     | Existing result, reproved.                                     |
| Represented-cut recognition and first-failure classification       | GBC; one available lower cut              | New extension here; elementary formulas.                       |
| Elementary uniform nonembedding                                    | GBC plus $\Sigma_1^1$ class comprehension | Sharper sufficient bound; no reversal claimed.                 |
| Full-model identification and nonembedding                         | Ambient ZFC plus an inaccessible cardinal | Existing result, rederived with additional stratum comparison. |
| Proposed Lean architecture                                         | Explicit universe and model choices       | Design only; no new Lean verification claimed.                 |

### 115.6.1 What was checked

The source audit read the merged surreal well-order article and README, the reals report README, the project agent instructions, and the specific sign-sequence and size-obstruction declarations listed above, at revision 491ed8411cccf7bd57916eb7666bbba762e92f96. The audit was targeted, rather than a proof that no overlapping statement occurs anywhere else in the repository. The new arguments were developed and checked independently within this preparation, including the set-theoretic scopes of the cut, block, topology, and termination proofs. This is mathematical review, not a kernel proof or external peer review.

The accompanying Python script checks finite relation-prefix comparison, the centered and prefix-first sign encodings, and the distinction between relation-table and enumeration comparison. It evaluates finite surreal sign strings as exact dyadic rationals, so the numerical side of the encoding check is independent of the encoded lexicographic comparator. Its successful finite tests do not establish class comprehension, transfinite well-foundedness, cardinal arithmetic, or saturation.

The PDF was compiled from the included standalone LaTeX source and checked for missing references, compilation errors, and layout defects. The archive includes the source, PDF, finite diagnostic script and its results, a short build guide, and a source/status note. A standard TeX installation with the packages named in the preamble is sufficient:

```
latexmk -pdf -interaction=nonstopmode surreal_well_orders_further.tex
python3 finite_checks.py
```

No network access is needed to compile the article or run the finite checks. The bibliography gives pinned repository locations and primary literature links for independent investigation.

## 115.7 Source 17: theorem and dependency ledger

| Result                                                    | Main dependency                                        | Foundation                            | Status here              |
|-----------------------------------------------------------|--------------------------------------------------------|---------------------------------------|--------------------------|
| Alphabet gaps,<br><a href="#">Theorem 29.4</a>            | Sign paths, set recursion, small deletion              | ZFC                                   | Proved                   |
| Minimal-slice invariants,<br><a href="#">Theorem 30.2</a> | Cylinder base, divergence bounds                       | ZFC                                   | Proved                   |
| Fusion, <a href="#">Theorem 30.5</a>                      | Dense refinement, coverage schedule                    | ZFC                                   | Proved                   |
| Gap spectrum,<br><a href="#">Theorem 30.11</a>            | Alphabet gaps, cut trichotomy, character calculation   | ZFC                                   | Proved                   |
| Exact reserve,<br><a href="#">Theorem 49.7</a>            | Cardinal deficiency balance, complement bijection      | Set choice; class notation elementary | Proved                   |
| Core interpolation and isomorphism                        | Surreal set cuts, supported completion, set histories  | GBC                                   | Source results re-proved |
| Recognition,<br><a href="#">Theorem 46.21</a>             | Unique straddlers, elementary class comprehension      | GBC                                   | Proved                   |
| Automorphism extension and failure                        | Recognition, large-gap homogeneity                     | GBC                                   | Proved                   |
| Class-extension detection                                 | Recognition, pair coding, common baseline              | Same-set GBC models                   | Proved                   |
| Finite condensation                                       | Source adjacency, no descending $\omega$ -chain        | GBC; comparison-level statement       | Proved                   |
| All-type or noncardinal classification                    | Residual alphabets not controlled by current interface | Not fixed                             | Proposed questions       |

“Proved” means that a written argument is supplied in this article. It does not mean external proof review, historical priority, or formal verification. The targeted source audit read the merged report, its README, its open-question sections, and the relevant source proofs at the pinned revision. It was not an exhaustive audit of the repository or the literature.

## 115.8 Source 18: assumptions, provenance, and verification status

| Result family                                      | Framework                                          | Status in this deliverable                                |
|----------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------|
| Set ordinal spectra                                | ZFC                                                | Repository result, proof supplied                         |
| Bounded cardinal invariants and exact gaps         | ZFC                                                | Proved here; independent mathematical audit               |
| Elementary labelled class comparison               | GB with supplied well-orders                       | Repository method, proof supplied                         |
| Canonical set-interval decomposition               | GB with supplied well-order                        | Proved here; no ETR used                                  |
| Block-index realizations on <b>No</b>              | GBC                                                | Proved by explicit construction and transport             |
| Core interpolation and exhaustive-code obstruction | GBC                                                | Repository methods, proofs supplied                       |
| Represented-cut criterion                          | GB with a fixed baseline                           | Proved here; surjectivity distinguishes the set-like case |
| Countable factorial shuffle                        | Externally countable, $\omega$ -standard GBC model | Proved here; external classification                      |
| Centered word cut theorems                         | GB/definable-class formulation                     | Proved here; isomorphisms additionally use GBC            |
| Full-class nonembedding model                      | Ambient ZFC with inaccessible cardinal             | External interpretation and cardinal proof supplied       |
| Lean realization of new results                    | Future work                                        | Not claimed or built                                      |

The repository audit was targeted to the surreal well-order report, the reals precursor, their source-status documents, and four relevant sign-sequence foundation files. It was not an audit of the whole repository. We fixed a revision so that subsequent repository changes cannot silently change the basis of this article. The mathematical proof review checked the cutoff hypotheses, singular-cardinal steps, class replacement and comprehension uses, finite terminal fibers, and external-model assumptions. It also corrected the arbitrary-alphabet density construction to include possible isolated global endpoints.

The complete source is self-contained LaTeX with an embedded bibliography. The PDF was compiled from that source and its rendered pages inspected. Those checks establish document consistency and layout; the mathematical validation remains the conventional proofs and their independent review. The attached package makes no claim of a formal kernel certificate or a literature-priority result.

[merge] (3 October 2026, batch 89.) Source 31’s axiom ledger and its status of the principal assertions close Part XV ([sections XV.13](#) and [XV.14](#)).

## A Source crosswalk and reproducibility

The immutable sources used in the targeted audit are recorded in the accompanying source audit. For convenient verification, the most relevant mappings are:

| Source | Relevant content                                                                                                                                                                                                                               |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [1]    | Labels <code>lwo:lem:prefix-free</code> , <code>lwo:lem:cylinder</code> , <code>lwo:prop:cardinality</code> , <code>lwo:thm:no-long</code> , <code>lwo:thm:ordinal-spectrum</code> ; bounded sign orders in <code>lwo:sp:thm:surreals</code> . |
| [3]    | Set-coded surreals and bounded birthdays; unrestricted cut obstruction; class-quantification and universe distinctions; set-valued recursion.                                                                                                  |
| [7]    | Actual sign carrier, ordinal embedding, smallness obstruction, and bounded fragments.                                                                                                                                                          |
| [9]    | Small cut operation, separator properties, and birthday bounds.                                                                                                                                                                                |
| [5]    | A nearby use of a set-like global well-order and ordinary class-parameter recursion.                                                                                                                                                           |

To compile the delivered standalone source, run `latexmk -pdf surreal_well_orders.tex` from its directory. All article sections and the bibliography are included in that single file; no separate bibliography processor is needed. Run `python3 finite_checks.py` for the finite checks. The supplied PDF was compiled from the included sources and inspected after rendering.

[merge] *Delivery names.* The paragraph above describes source 11’s delivered package. Its `surreal_well_orders.tex` is shipped as `article.tex`, which this merged text replaces; its PDF is not shipped (`article.pdf` is a build of this text); its `finite_checks.py` and `finite_checks_results.txt` are `code/11-raw-orders-finite_checks.py` and `data/11-raw-orders-finite_checks_results.txt`; the “accompanying source audit” is `11-raw-orders-repository_audit.md`. Build this report with `latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex`, and rerun the checks as in [appendix B](#). The same applies to the reproducibility records of the other sources that follow.

## A.1 Source 12: reproducibility and inspection record

The package contains `article.tex`, the compiled `article.pdf`, a build script, an exact finite-check program, its recorded JSON output, and a research-status and source-audit file. The Python program uses only the standard library. The recorded run passed all 180,905 finite comparison cases. Its comparisons are exact; there are no floating-point tolerances or inferred transfinite results. The PDF was compiled and rendered for visual inspection.

The repository audit directly inspected the merged real-well-order README and the sign-sequence simplicity Lean source at the stated commit. It did not independently compile the repository or verify the predecessor report’s proofs. The needed predecessor patterns have therefore been reproved in the present manuscript. Literature checking was focused on surreal foundations, Kanovei–Shelah indexing, and class recursion and well-orders; it was not sufficient to certify historical originality.

## A.2 Source 09: repository and source audit

The repository was inspected through its GitHub connector. All cited repository files are pinned to

63bb8b1d99359e7188abf0a33fb152e121858150.

A later search surfaced a newer commit; the cited foundational files were fetched at the earlier pinned snapshot rather than mixing versions. No repository file was changed.

The targeted inspection covered the merged real well-ordering report’s README and the relevant returned source of the sign-sequence and cut-operation modules. The real report identifies itself as unrefereed and not formalized. Its README describes a merged 54-page report and separates its results from nearby Lean developments. That report is used as research context, not as a kernel-certified dependency.

The sign-sequence file provides actual Lean declarations for the universe-indexed carrier, order, ordinal embedding, and smallness properties; the cut file provides the small-cut operation and its birthday bound. Their source statements were read, but this project did not rebuild the repository or audit all transitive dependencies. No claim of independent kernel verification is made.

The primary literature checked includes Kanovei–Shelah’s original index definition, Rangel–Mariano’s explicit set/class framework, and the class-recursion distinctions in Gitman–Hamkins–Holy–Schlicht–Williams. The Kanovei–Shelah index page was also visually inspected. The literature search was targeted rather than exhaustive. General universality and back-and-forth arguments are classical; the support-core organization and the accompanying obstruction statements are proved here without a priority claim.

### A.3 Source 09: verification and reproducibility

The accompanying `code/finite_checks.py` uses only Python’s standard library. It checks finite versions of the pair-switch lexicographic embedding, partial-prefix permutation completion, the forced-prefix separator algorithm, support extension preserving first disagreements, and the diagonal switch construction. Exact rational arithmetic is used for the separator tests. A deterministic random seed makes the sampled checks reproducible; other suites exhaust their specified finite ranges.

**These checks do not verify any transfinite theorem.** Their role is to detect indexing, orientation, and extension mistakes in the finite combinatorial mechanisms. Proofs of the infinite statements are the written arguments in the article. The recorded run passed 164,560 assertions, including 1,101 separator-family cases. The test report is `data/finite_checks.json`. The archive also contains a build script and a theorem-status ledger.

To compile the article, run `bash build.sh` in the archive’s root directory. It makes three passes with `pdflatex`, resolves the internal bibliography and cross-references, and writes build products into `build/` before copying the finished PDF to the root.

### A.4 Source 08: finite regression checks

The package includes `verify_finite.py`, requiring Python 3.10 or later and only its standard library. Running it writes `verification_results.json`. The supplied run passed:

| Check                                                          | Instances |
|----------------------------------------------------------------|-----------|
| Consecutive permutation pairs, sizes 1 through 7               | 5,906     |
| All increasing pairs of distinct permutations, sizes through 5 | 7,432     |
| Pair-code order comparisons, bit lengths 0 through 7           | 21,845    |
| Sign words of lengths 0 through 6                              | 127       |
| All sign-code order and prefix-free comparisons                | 16,129    |

The permutation test checks both directions of the adjacency rule on all manageable pairs, not just a few positive examples. The sign test compares the numerical convention  $- < 0 < +$  with the

binary blocks  $00 < 01 < 11$ , including pairs in which one sign word is a proper prefix. The pair-code test checks that every output is a permutation of the correct finite alphabet and that comparisons agree with the input binary order.

These tests can detect a reversed pair orientation, a wrong terminal block, or a mistaken finite-tail rule. They cannot prove regularity arguments, transfinite stabilization, set/class separation, or the existence of a global isomorphism. Those depend on the mathematical proofs in the text.

## A.5 Source 08: source audit and reproducibility

Two mathematical source files were inspected. One is `SignSequence.lean`; the other is `SignSequenceCut.lean`. Both were read at the pinned ProveIt commit. The project’s README was used only for its stated organization and audit policy. No build was run, and no claim was imported merely because a research report’s title suggested it. The source files have blob hashes

|                              |                                                       |
|------------------------------|-------------------------------------------------------|
| <code>SignSequence</code>    | <code>e2c093860f29b0b9ea312d50099fc4aa89a76a99</code> |
| <code>SignSequenceCut</code> | <code>c5c35afd98a5ad293687367ee9fb0244f94cc91c</code> |

The Kanovei–Shelah preprint was inspected directly, including the page defining its index. The class-recursion references were consulted through the author’s research summaries. Classical book and journal references provide background attribution; no claim of an exhaustive literature or priority search is made. The prior continuum manuscript was identified in the user’s file library and used as a contextual predecessor, not as a formally certified dependency.

The package contains a self-contained L<sup>A</sup>T<sub>E</sub>X source with an embedded bibliography, the compiled PDF, a build script, the finite verification script and its output, and a README. No external figure or private font file is needed. Compile with a standard TeX Live installation using `latexmk -pdf surreal_well_orders.tex`, or run `pdflatex` sufficiently many times to resolve cross-references.

[merge] (3 October 2026.) The audit and reproducibility records of sources 13, 14, 15 and 17 follow; source 16’s is the last subsection of its status section and source 18’s is part of its status table (both printed with the ledgers). They describe the delivered packages, whose layout differs from this directory: the shipped names of every delivered file are in [appendix B](#), and none of the six manuscripts or their PDFs is shipped. Their checksum manifests (sources 13, 14, 16, 17) and source 13’s build logs are not shipped either. Source 15 ships no checksums; source 18 ships no code or data.

## A.6 Source 13: repository and literature audit

### A.6.1 Pinned source and scope of inspection

The current main branch was resolved through the connected GitHub interface to

`0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8`

with commit timestamp 3 October 2026, 17:15:49 UTC. The working tree was not modified. The audit was targeted, not an exhaustive review of every proof in the repository or a rebuild of the Lean project.

The merged surreal report’s introduction, guide, selected dense-core and cylinder proofs, raw omitted-range and model sections, formalization plan, and research-question index were inspected. Retrieval targets included source ranges around 5000–5600, 5730–6205, and 6800–7120; these describe the targeted portions, not a claim that the entire 115-page report was independently certified. The real well-ordering report’s guide was inspected for provenance and conventions.



| Source                | Path or Git blob identity                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| Merged surreal report | <code>Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/article.tex</code>                   |
| Report blob           | <code>f20837b76f8d981cb1dce3da79f633e280edad41</code>                                                                  |
| Sign carrier          | <code>Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean</code>                                              |
| Carrier blob          | <code>e2c093860f29b0b9ea312d50099fc4aa89a76a99</code>                                                                  |
| Cut constructor       | <code>Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceCut.lean</code>                                           |
| Cut blob              | <code>c5c35afd98a5ad293687367ee9fb0244f94cc91c</code>                                                                  |
| Real report guide     | <code>SetTheory/Cardinals/docs/reports/ordinals-and-order-types/lexicographic-well-orderings-of-reals/README.md</code> |

The mathematical claims newly developed here were checked by reviewing their proofs and by finite tests of the code-level constructions. No independent referee report or machine verification is implied. The literature search was targeted to primary sources on the surreal continuum, class well-orders and recursion, higher-class foundations, and the Kanovei–Shelah index. It was not a complete priority search for every abstract order-theoretic lemma.

### A.6.2 Specific dependency choices

The fresh-value lemma uses an already constructed small-cut interface. The dense-core interpolation argument is written out rather than attributed only to a theorem name. The global product obstruction explicitly reuses the earlier report’s density and inverse-evaluator method. The cut and automorphism theorems do not require its singular-cardinal classification, and the central-word theorem does not require surreal arithmetic. Thus a later correction in a nonessential prior theorem would not automatically affect the main new proofs.

## A.7 Source 13: reproducibility and finite checks

The archive contains the standalone `article.tex`, its compiled PDF, a build script, a standard-library Python checker, its recorded JSON output, and a status note. No external image or private font file is required. The bibliography is internal to the TeX source.

The finite checks cover the following distinct mechanisms. First, they build well-order relations on finite alphabets and compare the common-predecessor formula with direct permutation lexicography. Second, they test convexity of every finite prefix cylinder and the uniqueness and coherence of crossing prefixes at all finite cut positions. Third, they test finite prefix completion. Fourth, they compare the central-word sign code with independent rational values of finite surreal sign sequences.

The test alphabets for the last group are  $-2, -1, -\frac{1}{2}, \frac{1}{2}, 1, 2$ , with all words of length at most three, including repeated-letter words. Finite signs are evaluated as dyadic rationals, not merely re-compared by a second copy of the same lexicographic function. The reserved-marker condition is also checked by exhibiting the collision that would result from admitting a trailing zero as a letter.

These tests do not establish class comprehension, ordinal-limit coherence, global choice, the back-and-forth theorem, or a transfinite birthday bound. They are reproducible checks of finite instances and independent representations, with the logical trust boundary stated explicitly.

```
# Run from the extracted archive directory.
python3 code/finite_checks.py --output data/finite_results.json
bash build.sh
```

The build script runs three LaTeX passes to settle cross-references and the table of contents. A PDF inspection record and checksums are included in the delivery’s status files.

## A.8 Source 14: finite diagnostics and reproducibility

The accompanying `verification.py` uses only Python’s standard library. It checks finite instances of the following kernels: relation-based common prefixes versus enumeration prefixes; convexity of prefix cylinders; the chain property of crossing prefixes at every finite cut; extension of partial injections on a finite support; preservation of all comparisons under support compression; finite self-delimiting sign and word codes; and pair-orientation coding. The exact tested ranges, case counts, and pass/fail results are written to `verification_results.json` by the delivered execution. The delivered run passed every diagnostic. Its principal counts are:

| Kernel                                             | Tested cases or comparisons |
|----------------------------------------------------|-----------------------------|
| Labelled common-prefix comparison                  | 15,018                      |
| Prefix cylinders / proper finite cuts              | 2,371 / 867                 |
| Finite partial injections                          | 15,126                      |
| Compressed candidates / retained comparisons       | 3,868 / 66,873              |
| Numerical sign-code pairs                          | 16,129                      |
| Outer word-code pairs, two termination conventions | 320,000                     |
| Pair-orientation comparisons                       | 21,844                      |

Finite alphabets have endpoints and adjacent labels, unlike the residual surreal alphabet. Therefore the finite crossing-prefix test checks the chain lemma, not the full boundary trichotomy. The buffer argument for infinite partial injections, cofinality at singular cardinals, Global Choice, and class-model absoluteness are not experimentally verified. Those assertions stand on their written proofs and remain appropriate subjects for independent mathematical and formal review.

To reproduce the package locally:

```
python3 verification.py
bash build.sh
```

The build script compiles the standalone source twice with `pdflatex`; the bibliography is internal, so no external database or network access is needed for compilation. Standard TeX packages named in the preamble must be installed. `SHA256SUMS.txt` records the delivered source, PDF, scripts, status files, and diagnostic output. The archive does not include fonts, repository clones, or LaTeX auxiliary files. The repository itself has not been modified.

## A.9 Source 14: source audit and bibliographic scope

The primary repository inputs were read through the GitHub connector at the pinned revision below. The targeted mathematical ranges were the opening status and abstract, the core/interpolation and binary-coding sections, the universe and formalization discussion, and the continuation questions. The actual Lean simplicity file was separately inspected. The metadata in the README distinguishes four source manuscripts from their merged report. We did not treat their statements as kernel-verified results or as established published literature.

External sources were used to check the Kanovei–Shelah indexing construction, the distinction between elementary comparison and class-valued recursion, and standard surreal background. The consulted version of the class-recursion paper is explicitly dated in the bibliography. No unresolved question quoted from it is asserted here to remain open as of the present date. The new arguments in this article are self-contained relative to the stated surreal set-cut theorem, GBC, and explicitly marked inherited set-spectrum results.

### A.10 Source 15: source provenance and reproducibility

The repository inputs were read through the GitHub connection at the pinned revision stated in [Section 4.8](#). The main paths are the surreal well-orders report, the real well-orderings report, and `Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean`. The new results are deductions in this article, not changes committed to the repository. No claim of source-wide novelty follows from this targeted audit.

The primary literature was checked for the precise Kanovei–Shelah index, surreal order versus simplicity, global choice, class well-foundedness and ETR, and the standard interval-condensation mechanism. In particular the countable condensation studied by Brown and Suárez is related methodology, not the same as the model-relative set-interval condensation used here [34]. The final journal and arXiv versions of a source may have different pagination; theorem references specify the version where needed.

The accompanying `examples.py` regenerates the three-label table, checks the explicit restriction reversal and the Boolean restriction bound, and checks the finite predecessor-comparison formula. These calculations are reproducibility aids for finite examples only. They do not validate cardinal arithmetic, class comprehension, transfinite recursion, or the new proper-class theorems.

The source is a standalone L<sup>A</sup>T<sub>E</sub>X document with an internal bibliography and no external figures. Build it with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error
surreal_lexicographic_orders.tex
```

The PDF was compiled and inspected for unresolved references, overflowing formulas, and page-layout defects. Mathematical review and layout review are separate from formal proof checking.

### A.11 Source 17: finite checks and reproducibility

The package contains `finite_checks.py`, which requires only the Python standard library. Its deterministic recorded run is in `finite_checks.json`. The run passed all of the following checks:

| Finite mechanism checked                                              | Instances |
|-----------------------------------------------------------------------|-----------|
| Predecessor-signature comparison against tuple lexicography           | 533,418   |
| Exact adjacency condition against consecutive finite permutations     | 266,272   |
| Convexity of prefix cylinders, carriers of size at most 7             | 16,072    |
| Uniqueness and coherence of straddlers at finite cut levels           | 852       |
| Finite partial-injection extension on its necessary support           | 15,126    |
| Prefix-free coefficient-code comparison                               | 961       |
| Variable-length word-code comparison (400 words, repetitions allowed) | 160,000   |

The empty carrier and empty word are included where appropriate. The coefficient tests use all sign sequences of length at most four. Word tests use an alphabet of seven sign sequences of length

at most two, and words of length at most three. Finite partial injections are tested on carriers of size at most six, including partial maps with fixed points.

These tests do not verify singular cardinals, limits of prefix chains, class comprehension, large-gap homogeneity, or the necessity of an infinite reserve. They are consistency checks on the finite mechanisms used by the proofs, not experimental evidence for an unproved theorem.

To reproduce the checks and build the PDF from a TeX Live installation, run the supplied `build.sh`. The bibliography is internal to the LaTeX source, so no external bibliography database is required. The ZIP includes a README and a research-status note, with SHA-256 checksums of its payload. It does not bundle fonts or claim to reproduce the separate source report.

## A.12 Source 31: delivery and reproducibility

[merge] (3 October 2026.) Source 31 was delivered as `global_choice_surreal_coding.zip` (arrival `7c0f2d9f9`) with three files: the manuscript `global_choice_surreal_coding.tex` (1,547 lines, 70,676 bytes), its 21-page PDF (400,246 bytes) and `README.txt` (2,590 bytes). None is shipped; the placement `23adb85f9` staged nothing, and all three are recoverable with `git show 7c0f2d9f9:docs/incoming/global_choice_surreal_coding.zip`. Source 31 ships no program and no data, by design (section XV.12.3). Its README lists the main results, builds the article with `latexmk -pdf -interaction=nonstopmode -halt-on-error global_choice_surreal_coding.tex` (no images, bibliography database, download or shell escape), says that the supplied PDF was built with pdfTeX, checked for unresolved references and layout warnings and visually reviewed, records the inspected commit and path of this report, and states that “No repository files were modified”. Rebuilt from a fresh extraction with MiKTeX on 3 October 2026, the delivered manuscript gives 21 pages with no warning. Part XV is that manuscript with the changes listed in section 1.10.

## A.13 Sources 32–35: delivery and reproducibility

[merge] (4 October 2026.) The four archives (arrival `e3839ad2c`) were placed in `111c38012`. Staged, byte-identical to the delivery (13 files, 39,802 bytes, LF only): source 32’s `code/32-ceilings-Makefile`, `code/32-ceilings-finite_checks.py`, `data/32-ceilings-finite_checks.json` and `32-ceilings-PROOF_STATUS.md`; source 33’s `code/33-heights-build.sh`, `code/33-heights-finite_checks.py`, `data/33-heights-verification_results.json`, `data/33-heights-BUILD_REPORT.json`, `33-heights-PROOF_STATUS.md` and `33-heights-SOURCE_AUDIT.md`; source 34’s `34-choicefree-PROOF_STATUS.txt`; source 35’s `code/35-completion-finite_notation_check.s.py` and `data/35-completion-finite_notation_results.json`. Not shipped (they survive in the arrival commit): the four manuscripts and PDFs, the four delivery READMEs, source 32’s `SHA256SUMS` (7 of 7 hashes verified at placement) and source 33’s `MANIFEST.sha256` (9 of 9). Part XVI is the text of the manuscripts.

*The checks, rerun for this write on copies* (Python 3.14): source 32’s script with `-output` passes 689,808 assertions (seed 20261004); source 33’s passes 155,530; source 35’s passes 1,825 splitting, 21,993 flattening and 2,700 comparison cases (30 terms, bases 5, 7 and 11). Each output equals the shipped record as JSON and byte for byte up to line endings (Windows writes CRLF). All three check finite coding conventions only, as each says.

*Delivery names.* The shipped build files name unshipped sources: `code/32-ceilings-Makefile` compiles `beyond_ord.tex` and writes `finite_checks.json`; `code/33-heights-build.sh` compiles `article.tex` into `.build/` and copies `article.pdf`; both build nothing here. Source 35’s script writes `finite_notation_results.json` beside itself (in `code/`), and source 32’s program defaults to `finite_checks.json` in the working directory; source 32’s status file names `finite_checks.json`,

source 33's names `article.tex` and its “generated package manifest”; the printed appendices name the delivery files and are annotated in Section XVI.11. The archives can be recovered with `git show e3839ad2c:docs/incoming/<archive>.zip`. The README gives rerun commands that write only to a copy.

### A.14 Source 36: delivery and reproducibility

[merge] (4 October 2026.) The archive `Beyond_Ord_Research_Package.zip` (715,002 bytes, arrival `d7cf7d554`) holds nine files under `Beyond_Ord/`. The placement `6571ee1af` staged five of them, byte-identical to the delivery (24,944 bytes, LF only): `36-horizons-SOURCE_AUDIT.txt` (from `SOURCE_AUDIT.txt`), `code/36-horizons-build.sh` (`build.sh`), `code/36-horizons-notation_demo.py` (`code/notation_demo.py`), `data/36-horizons-verification.json` (`code/verification.json`) and `data/36-horizons-DOCUMENT_CHECKS.json` (`DOCUMENT_CHECKS.json`). Not shipped (they survive in the arrival commit): the manuscript `beyond_ord.tex`, its 40-page PDF, the delivery `README.txt` and the demonstrator's `code/README.md`, whose scope and limitations are reproduced in this report's README. Part XVII is the text of the manuscript. The four SHA-256 values in `DOCUMENT_CHECKS.json` (manuscript, PDF, demonstrator, report) match the delivered files.

*The demonstrator, rerun for this write on a copy* (Python 3.14.4, 4 October 2026, with `PYTHONUTF8=1`): `-examples-only` prints the examples, and `-report` passes with 994 distinct terms, 493,521 unordered and 2,000 ordered comparisons against the independent base-7 evaluator, 994 reflexive checks, 16 malformed inputs rejected and 8 tower boundaries, as recorded; the regenerated report equals `data/36-horizons-verification.json` except for `python_version` (3.14.4 against the delivered 3.12.14) and its CRLF line endings. Without `PYTHONUTF8=1` the program stops with `UnicodeEncodeError` under the Windows code page, and the report run leaves a 0-byte file; this is disclosed, not patched.

*Delivery names.* `code/36-horizons-build.sh` compiles `beyond_ord.tex`, which is not shipped, and builds nothing here. The source audit and `DOCUMENT_CHECKS.json` name `beyond_ord.tex`, `beyond_ord.pdf` and `code/...` delivery paths; Section XVII.10 is annotated with the shipped names. The demonstrator's documented command `-report code/verification.json` overwrites its argument; the README gives a rerun that writes only to a copy. The archive can be recovered with `git show d7cf7d554:docs/incoming/Beyond_Ord_Research_Package.zip`.

## B The finite checks as shipped

[merge] All four programs use only the Python standard library and check finite instances; they verify no transfinite, class-theoretic or cardinal statement. Rerun on 2 October 2026 on copies, with Python 3.14.4 on Windows, each reproduced its recorded output exactly apart from line endings: on Windows all four write CRLF, while the records have LF, so compare with `diff -strip-trailing-cr`.

| Source | Program and record                                                                                           | Checks                                                                                                                                                    | Output                    |
|--------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 11     | <code>code/11-raw-orders-finite_checks.py</code> ; <code>data/11-raw-orders-finite_checks_results.txt</code> | 465 sign-code pairs; 79,800 word-code pairs; predecessor comparison and adjacency on all 266,272 pairs of permutations of at most 6 points (867 adjacent) | prints to standard output |

| Source                                                                                 | Program and record                                                                            | Checks                                                                                                                                                                                                                                                                  | Output                                                                                                                                                            |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12                                                                                     | <code>code/12-singular-finite_checks.py; data/12-singular-finite_checks.json</code>           | 180,905 cases: sign code, pair orientation, common prefix and adjacency, larger adjacent pairs, row scheduler                                                                                                                                                           | writes the file named by <code>-output</code>                                                                                                                     |
| 09                                                                                     | <code>code/09-core-finite_checks.py; data/09-core-finite_checks.json</code>                   | 164,560 assertions, 1,101 separator families, seed 20261002: pair switch, prefix completion, forced separator, bounded reflection, diagonal switch                                                                                                                      | default <code>-output</code> is <code>data/finite_checks.json</code> beside <code>code/</code> , an unprefixed new file; always pass <code>-output</code>         |
| 08                                                                                     | <code>code/08-skeleton-verify_finite.py; data/08-skeleton-verification_results.json</code>    | 5,906 consecutive and 7,432 ordered pairs of permutations; 21,845 pair-code comparisons; 16,129 sign-code comparisons on 127 words                                                                                                                                      | writes <code>verification_results.json</code> beside itself                                                                                                       |
| <i>[merge] Batch 81 (sources 13–17; source 18 ships no code), added 3 October 2026</i> |                                                                                               |                                                                                                                                                                                                                                                                         |                                                                                                                                                                   |
| 13                                                                                     | <code>code/13-products-finite_checks.py; data/13-products-finite_results.json</code>          | 533,417 relation-versus-lexicographic pairs of permutations of at most 6 points; 2,371 convex cylinders and prefix completions; 5,885 cut-level uniqueness checks; 867 successor-failure partitions; 2,044 transitivity triples; 67,081 central-word pairs on 259 words | default <code>-output</code> is <code>data/finite_results.json</code> relative to the working directory, an unprefixed new file; always pass <code>-output</code> |
| 14                                                                                     | <code>code/14-trichotomy-verification.py; data/14-trichotomy-verification_results.json</code> | 15,018 labelled common-prefix pairs; 2,371 cylinders and 867 finite cuts; 15,126 partial injections; 3,868 compressed candidates with 66,873 comparisons; 16,129 sign-code and 320,000 word-code pairs; 21,844 pair orientations                                        | writes <code>verification_results.json</code> beside itself (no option); run a copy                                                                               |
| 15                                                                                     | <code>code/15-blocks-examples.py; data/15-blocks-finite_examples.txt</code>                   | the three-label table, restriction reversal and relabelling, all eight orientations of the pair-restriction spaces, and the raw predecessor formula against enumeration lexicography for all permutations of 0–5 points                                                 | prints to standard output                                                                                                                                         |



| Source | Program and record                                                         | Checks                                                                                                                                                                                           | Output                                                                                                               |
|--------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 16     | code/16-strata-finite_checks.py; data/16-strata-finite_checks_results.json | four sign-code schemes on 241 words (232,324 word comparisons, exact dyadic values); the raw predecessor formula on 15,018 relation pairs of at most 5 points; the relation-table counterexample | default -output is finite_check_results.json beside the script, an unprefixed new file in code/; always pass -output |
| 17     | code/17-gaps-finite_checks.py; data/17-gaps-finite_checks.json             | 533,418 predecessor comparisons; 266,272 adjacency checks; 16,072 cylinders; 852 straddling checks; 15,126 partial injections; 961 coefficient-code and 160,000 word-code comparisons            | default -output is finite_checks.json in the working directory; always pass -output                                  |

Run them on copies, never in the report directory. From the report directory, with an empty scratch directory `$$`:

```
cp code/08-skeleton-verify_finite.py "$S/"
python "$S/08-skeleton-verify_finite.py"          # writes $$/verification_results.json
python code/09-core-finite_checks.py --output "$S/09.json"
python code/11-raw-orders-finite_checks.py > "$S/11.txt"
python code/12-singular-finite_checks.py --output "$S/12.json"
```

and compare with the four records in `data/`. The delivered build scripts build nothing in this layout: `code/08-skeleton-build.sh` runs `verify_finite.py` and compiles `surreal_well_orders.tex` beside itself, `code/09-core-build.sh` compiles an `article.tex` in `code/`, and `code/12-singular-build.sh` changes into `code/` and then calls `code/finite_checks.py`. They are kept as delivered.

[merge] (3 October 2026, batch 81.) The five programs of sources 13–17 also use only the Python standard library and check finite instances only. Rerun on 3 October 2026 on copies, with Python 3.14.4 on Windows, each reproduced its recorded output apart from line endings and, for sources 13 and 14, the recorded `python_version` (3.13.5). From the report directory, with an empty scratch directory `$$`:

```
python code/13-products-finite_checks.py --output "$S/13.json"
cp code/14-trichotomy-verification.py "$S/"
python "$S/14-trichotomy-verification.py"          # writes $$/verification_results.json
python code/15-blocks-examples.py > "$S/15.txt"
python code/16-strata-finite_checks.py --output "$S/16.json"
python code/17-gaps-finite_checks.py --output "$S/17.json"
```

and compare with the records in `data/` using `diff -strip-trailing-cr`. The delivered build scripts build nothing in this layout: `code/13-products-build.sh` and `code/14-trichotomy-build.sh` compile an `article.tex` in `code/` (the first writes `build-pass-1.log` there before failing), `code/15-blocks-build.sh` and `code/18-shuffle-build.sh` run `latexmk` on the unshipped manuscripts `surreal_lexicographic_orders.tex` and `surreal_lexicographic_orders_further_study.tex`, and `code/17-gaps-build.sh` first runs `python3 finite_checks.py`, absent under that name. They are kept as delivered.

*Delivery names.* The reproducibility records printed in [appendix A](#) and the ledgers name the delivered files. Source 13's `code/finite_checks.py`, `data/finite_results.json`, `data/pdf_inspection.json`, `build.sh` and `RESEARCH_STATUS.md` are `code/13-products-finite_checks.py`, `data/13-products-finite_results.json`, `data/13-products-pdf_inspection.json` (it records the layout check of source 13's 33-page PDF, which is not shipped), `code/13-products-build.sh` and `13-products-RESEARCH_STATUS.md`; source 14's `verification.py`, `verification_results.json`, `build.sh` and `RESEARCH_STATUS.md` are `code/14-trichotomy-verification.py`, `data/14-trichotomy-verification_results.json`, `code/14-trichotomy-build.sh` and `14-trichotomy-RESEARCH_STATUS.md`; source 15's `examples.py`, `finite_examples.txt` and `build.sh` are `code/15-blocks-examples.py`, `data/15-blocks-finite_examples.txt` and `code/15-blocks-build.sh`; source 16's `finite_checks.py`, `finite_check_results.json` and `SOURCE_STATUS.txt` are `code/16-strata-finite_checks.py`, `data/16-strata-finite_check_results.json` and `16-strata-SOURCE_STATUS.txt`; source 17's `finite_checks.py`, `finite_checks.json`, `build.sh` and `research_status.md` are `code/17-gaps-finite_checks.py`, `data/17-gaps-finite_checks.json`, `code/17-gaps-build.sh` and `17-gaps-research_status.md`; source 18's `build.sh` is `code/18-shuffle-build.sh`. The six manuscripts, their PDFs, their delivery READMEs and the checksum manifests are not shipped; they remain in the arrival commits ([section 1.8](#)).

[merge] (3 October 2026, batch 89.) Source 31 ships no program ([appendix A.12](#)).

[merge] (4 October 2026, batch 97.) Source 36's natural-coefficient demonstrator is `code/36-horizons-notation_demo.py`, with its recorded report `data/36-horizons-verification.json`; its scope and the rerun for this write are in Section XVII.10 and [appendix A.14](#).

## C The other sources' title pages and abstracts

[merge] The results these summaries announce are printed in the body; the summaries are kept for the record.

### Source 12

**Title-page summary. Main distinction.** An arbitrary class well-order of **No** need not have order type  $\text{Ord}$ . Nevertheless, two such well-orders have an *elementarily definable largest common labelled initial segment*, possibly a proper class. This supplies a lexicographic comparison in GBC without invoking general class well-order comparability or ETR.

**Principal quantitative result.** Put  $\mathbf{No}_{<\kappa} = \{s : \alpha \rightarrow \{-, +\} : \alpha < \kappa\}$  and  $\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}$ , for an infinite initial cardinal  $\kappa$ . The minimal-length permutation order satisfies

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \mu = \kappa \text{ and } \text{cf } \kappa < \kappa, \\ \mathbb{B}_\mu, & \text{otherwise,} \end{cases}$$

where  $\mathbb{B}_\gamma = (2^\gamma, <_{\text{lex}})$  and  $\simeq_e$  means mutual order-embeddability, not isomorphism.

**Status.** This is an AI-assisted, unrefereed research manuscript. Proofs are supplied for the stated results; historical priority is not claimed. The accompanying exact finite tests do not verify the transfinite arguments. No new Lean formalization is claimed.

**Abstract.** We study the lexicographic order of well-orderings of the surreal numbers, carefully distinguishing bounded sets of surreals, set-like global well-orders, and arbitrary global class well-orders. For the full class problem, ordinal-indexed enumerations are insufficient. We construct the largest common labelled initial segment of two well-orders directly from their relations, prove that comparison of their next elements defines a linear order, and show that this construction needs only elementary class comprehension once the well-orders exist. Over GB with set choice, the existence of any class well-order of the surreals is equivalent to global choice. The collection of all these well-orders is not itself an ordinary class of GBC or Kelley–Morse theory; a diagonal theorem rules out any uniform exhaustive coding by a class of sets.

For set-like global well-orders we prove density, interpolation for set-indexed cuts, local universality for class linear orders, and mutual embeddability with the lexicographic comparison of binary classes on  $\text{Ord}$ . For unrestricted global well-orders we classify adjacent pairs: they arise precisely from finite residual permutations after a common, necessarily proper-class, prefix. At a birthday cutoff  $\kappa$ , we obtain an exact regular/singular dichotomy for the binary coding rank of minimal-length enumerations. We also establish the exact ordinal and binary-cube embedding spectra for the full set-sized order. Finally, a full-class model based on an inaccessible rank explains exactly how the bounded and global theories fit together, and provides a rigorous example in which the full order and the set-like order have the same cardinality and ordinal spectrum but different coding ranks. Explicit proof-assistant interfaces and further research questions are included.

## Source 09

**Abstract.** The phrase “the lexicographic order of all well-orders of the surreal numbers” has several mathematically different interpretations. We develop three: exhaustive enumerations of a fixed set of surreals; all set-length injective surreal words; and class bijections from the ordinals onto the surreal numbers, representing set-like global well-orders. For an infinite set  $X$  of cardinality  $\kappa$ , we give a self-contained proof that its lexicographic well-ordering space has cardinality  $2^\kappa$ , contains every linear order of size at most  $\kappa$ , and contains an ordinal precisely when that ordinal is below  $\kappa^+$ .

For the global problem, we work in Gödel–Bernays set theory with global choice and do not pretend that a collection of proper classes is itself an ordinary class. Fix a global enumeration  $b$ . Its eventually-equal perturbations admit canonical set codes. We prove that their lexicographic order is isomorphic to the surreal line, meets every set-prefix cylinder, and interpolates every uniformly set-indexed separated pair of families of arbitrary global enumerations. Thus every class linear order embeds into this genuine class core. We also prove bounded reflection for set-indexed families, construct an explicit unfilled  $(\text{Ord}, \text{Ord})$ -configuration, and diagonalize against every uniform class-indexed list of all global enumerations.

A strongly inaccessible truncation gives a fully set-sized model of the size distinction: the core has cardinality  $\kappa$ , whereas all length  $\kappa$  enumerations have cardinality  $2^\kappa$ . The article includes foundational and Lean formalization plans, a repository audit, finite verification code, and a focused research agenda.

**Title-page status.** Status: AI-assisted research manuscript; complete written arguments are provided for the stated results. Not peer-reviewed or Lean-verified. Classical ingredients and repository results are distinguished from the constructions developed here. No claim of historical priority is made.

## Source 08

**Abstract.** There is no single unqualified object called “the lexicographic order of all well-orders of the surreal numbers.” We distinguish well-orderings of a fixed set of surreals, well-orderings of varying set-sized subsets, set-like well-orderings of the whole surreal class, and arbitrary class well-orderings. For an infinite ordered set  $X$  of cardinality  $\kappa$ , the exhaustive-enumeration order  $\mathcal{W}(X)$  has size  $2^\kappa$ , contains every linear order of size at most  $\kappa$ , and contains an ordinal  $\beta$  exactly when  $\beta < \kappa^+$ . Its adjacent pairs arise precisely from finite permutation tails.

For the whole surreal class, work in Gödel–Bernays set theory with Global Choice. The numerical first-disagreement comparison of class bijections  $\text{Ord} \rightarrow \mathbf{No}$  is meaningful without treating classes as elements of classes. Fix one such bijection  $b$ . We prove that every set-sized separated family of global enumerations has a strict separator obtained from  $b$  by a permutation of set support. The class of canonical codes for these perturbations is therefore order-isomorphic to  $\mathbf{No}$ . It is a dense, set-cut-realizing skeleton inside the much larger virtual enumeration order, and every class linear order embeds into it. The full virtual order is equimorphic with the lexicographic order of class binary sequences, but admits no class-uniform exhaustive parametrization. We supply proofs, cutoff versions, counterexamples to tempting identifications, a universe-sensitive Lean implementation plan tied to inspected ProveIt sources, and a research agenda.

### Status box.

**Status and scope.** The results below are mathematical derivations with explicit proofs, not claims of historical priority. The manuscript has not been independently refereed, and its new results have not been checked by a proof assistant. Repository declarations are identified separately from proposed extensions. The accompanying executable checks cover finite coding and adjacency examples only. All uses of “class” are relative to the stated foundational model.

[merge] (3 October 2026.) The title pages of sources 13–18 follow, with their abstracts and status statements; layout commands are omitted.

## Source 13

**Title page.** “Further research report”. *Lexicographic Orders of Surreal Well-Orderings: Cut reconstruction, asymmetric products, prefix non-rigidity, and size-safe foundations* (PDF metadata: “Further Structure”). Research manuscript prepared for Vladimir Reshetnikov, 3 October 2026.

**Abstract.** We further investigate the lexicographic comparison of well-orderings of the surreal numbers, separating set-sized alphabets, set-length words, set-like global enumerations, unrestricted class well-orders, and code-dependent alternatives. The central new result is an exact prefix-cylinder criterion for a normalized class cut of a surreal-sized dense core to be represented by an injective ordinal-class word. Exhaustive words are characterized by an additional label-coverage condition. Cuts that fail the criterion have a first failure of one of two precisely described kinds: a next-label cut or a limit-prefix boundary.

We prove an asymmetric product phenomenon. The global enumeration order absorbs a surreal left factor, and more generally admits every ordinary class-indexed lexicographic sum of copies of itself. Its two-point split, however, does not embed back in the full inaccessible-universe interpretation, nor by a definable uniform transformation in Kelley–Morse theory with global choice. Externally countable models give the opposite answer for unrestricted external maps. A finite-block recoding automorphism shows that the pure linear order does not recover its prefix

cylinders. Finally, replacing end-first dictionary comparison by a central termination convention on the nonzero alphabet makes both arbitrary and injective set-length word orders isomorphic to the surreal line.

The proofs specify their logical size levels. An implementation plan uses the repository’s actual universe-indexed sign-sequence and small-cut interfaces, without claiming a new Lean formalization.

**Title-page status. Research status.** AI-assisted, unrefereed mathematical manuscript. The proofs are presented for scrutiny; historical priority and kernel verification are not claimed. This is a follow-up to the repository’s merged report, not a replacement or a certification of all its results. The inspected repository snapshot is pinned in [Appendix A.6](#). Finite checks accompany the article, but do not establish transfinite or class-theoretic theorems.

## Source 14

**Title page.** “Research continuation”. *Lexicographic Well-Orders of the Surreals II: Cut classification, support thresholds, and class-model absoluteness*. Prepared for Vladimir Reshetnikov, 3 October 2026.

**Abstract.** The earlier ProveIt reports identify a set-coded dense skeleton of the lexicographic order of global surreal enumerations, but leave open which cuts of that skeleton come from exhaustive enumerations. We answer the prefix-structured version of that question. Every proper normalized class cut of the skeleton has exactly one of three forms: an edge of a set-prefix cylinder, a nonprincipal numerical gap after a set prefix, or a coherent injective branch of length  $\text{Ord}$ . The last form is realized by a global well-order of set-like type exactly when its branch covers every surreal. Reconstruction is elementary; a separate orbit theorem shows that bare-order automorphisms can destroy exhaustive representability.

This yields an absoluteness theorem: enlarging the class part of a model while retaining its sets and a fixed baseline enumeration cannot newly realize an old skeleton cut by a global enumeration. New enumerations have genuinely new trace cuts. For unrestricted labelled class well-orders, the trace map forgets exactly the tail after the canonical  $\text{Ord}$ -long initial segment; its fibers are described explicitly.

A separate support analysis gives a sharp hierarchy. If  $\kappa$  is an infinite set cardinal, the permutations of a baseline with support of cardinality  $< \kappa$  form an  $\eta_{\text{cf } \kappa}$ -order, but not an  $\eta_{(\text{cf } \kappa)^+}$ -order. A separated pair of  $\text{cf } \kappa$ -long families forces at least  $\kappa$  moved coordinates, and this support cost is attained in the full skeleton. Thus universality for ordinary class linear orders can coexist with sharply limited saturation.

We distinguish fixed carriers, variable set domains, increasing subclasses, global set-like orders, unrestricted class relations, relation-code lexicography, and order-type quotients. The exposition includes proofs, exact quantifier conventions, a repository audit, universe-relative and Lean formalization plans, finite regression tests, and a focused research agenda.

**Title-page status. Research status.** The new results below are proved in this manuscript, not machine-verified. They are unrefereed and no claim of historical priority is made. The cut classification is relative to the named prefix cylinders; it is not a classification in the language of bare linear order alone. The repository is unchanged.

## Source 15

**Title page.** *Lexicographic Orders of Surreal Well-Orders: Interpretations, canonical blocks, represented cuts, and fixed-type spectra.* “A continuation of the ProveIt research report”. Research article prepared for Vladimir Reshetnikov, 3 October 2026.

**Abstract.** There is no single unqualified object called the lexicographic order of all well-orders of the surreal numbers. The carrier, representation, comparison alphabet, indexing order, prefix convention, and foundational size level all affect the answer. We give a systematic account of the principal interpretations and develop three extensions of the existing ProveIt report. First, the relation that a closed interval in a class well-order is a set produces its entire canonical block decomposition by an elementary formula; every class well-order can occur as the block-index order. Explicit ordinal polynomials realize arbitrarily long class-indexed condensation histories. Second, a cut of the bounded-support surreal core is represented by an injective ordinal-class word exactly when every set-length level has a cylinder meeting both sides. First failure gives an exhaustive description of terminal cuts, including repeated-label obstructions and limit-prefix boundaries. The lower-core-cut map forgets exactly the residual well-order after the first proper block. Third, for birthday cutoffs we extend the minimal-slice dichotomy to enumeration types consisting of any admissible set number of full cardinal blocks followed by a finite tail, with exact binary coding lengths. Exact local characters and the saturation threshold are computed for every cardinal birthday cutoff. Complete proofs, counterexamples, foundational assumptions, source provenance, and a prioritized research program are included.

**Title-page status. Status.** This is an AI-assisted, unrefereed research exposition. “Extension” means a deduction beyond the audited repository report, not a claim of historical priority. Its new theorems have written proofs and independent mathematical review within this task. They have not been checked by Lean or another proof assistant. Finite calculations check only the explicit finite examples.

Repository revision inspected: 0ecd158fd163bc4aca0ee40ccf6479ca6cf5f2a8.

## Source 16

**Title page.** “Order theory and foundations”. *Lexicographic Orders of Surreal Well-Orders: Cuts, Canonical Blocks, and the Limits of Set-Sized Coding.* “A further study of the ProveIt research program”. Research article prepared for Vladimir Reshetnikov, 3 October 2026.

**Abstract.** The phrase “the lexicographic ordering of all well-orders of the surreal numbers” describes several substantially different objects. We separate well-orders on a fixed set, ordinal-length words, numerically increasing subsets, set-like global well-orders, arbitrary class well-orders, and relation-table encodings. Building on the ProveIt report, we establish several extensions. For every infinite alphabet we compute the density and cellularity of every fixed enumeration stratum, including a transition at three terminal positions, and prove that every stratum fails conditional Dedekind completeness. At regular saturated stages we determine the complete cofinality spectrum of unfilled cuts. For global enumerations we give an exact prefix-cylinder criterion for the cuts represented by injective ordinal-class branches, with an additional criterion for exhaustiveness. Every failure has a first coordinate or limit-prefix obstruction. For an arbitrary class well-order, the elementary relation “the interval between these points is a set” simultaneously extracts all its canonical ordinal-class blocks, without elementary transfinite recursion. We also classify alternative



termination conventions, sharpen a comprehension bound for uniform nonembedding, and give a formalization plan that distinguishes smallness, Lean universes, and internal class semantics. All new results are presented with ordinary mathematical proofs; no claim of historical priority or machine-checked verification is made.

**Title-page status. Repository snapshot.** 491ed8411cccf7bd57916eb7666bbba762e92f96.

The source audit is targeted. The article is independent of later changes to the repository, and its new theorems are not additions to the Lean library.

## Source 17

**Title page.** *Lexicographic Well-Orders of the Surreal Numbers, II: Exact cut spectra, branch recognition, optimal support reserves, and automorphism-extension obstructions.* Research manuscript prepared for Vladimir Reshetnikov, 3 October 2026.

**Abstract.** We continue the ProveIt report on lexicographic well-orders of surreal numbers, separating set-sized enumeration spaces, set-length words, set-like global well-orders, unrestricted class well-orders, and relation-code orders. For every infinite cardinal  $\kappa$ , put  $X = \mathbf{No}_{<\kappa}$ ,  $\mu = 2^{<\kappa}$ ,  $r = \text{cf } \kappa$ , and  $q = \text{cf } \mu$ . For the minimal-length enumeration order  $P = \mathcal{W}_\mu(X)$  we prove that every point has two-sided character  $(q, q)$ , that  $\text{cf } P = \text{ci } P = r$ , and that density, weight, and cellularity are  $\mu^{<\mu}$ . Its nonprincipal cut-character spectrum is exactly

$$\{(r, r), (q, q)\} \cup \{(\theta, r), (r, \theta) : \theta \in \text{Reg}, \theta < \mu\}.$$

Here  $\text{Reg}$  denotes the infinite regular cardinals. These statements require neither regularity of  $\kappa$  nor  $2^{<\kappa} = \kappa$ . We also establish an exact saturation threshold and a fusion theorem for dense open sets.

In *GBC*, we give a necessary-and-sufficient cylinder criterion for a class cut of a bounded-support surreal core to represent an injective Ord-word; a separate coverage condition characterizes exhaustive words. The reconstruction is elementary and needs no ETR. This yields an exact criterion for extending core automorphisms to the full global enumeration order and an obstruction: some core automorphisms do not extend. Proper extensions of the available classes necessarily add new represented cuts while leaving the set-coded core unchanged. A cardinal deficiency formula gives the optimal number of additional moved positions needed to extend a set partial injection to a supported permutation. We also identify maximal finite adjacency components in the unrestricted order and show that their quotient is dense. All higher-order statements are formulated without a same-level class of classes.

**Title-page status. Status.** The new assertions have ordinary mathematical proofs in this manuscript. They are not claimed to be historically first, peer-reviewed, or Lean-verified. The repository was read at the pinned revision stated in [Section 4.10](#); it was not modified or built. Finite checks supplied with this article test finite mechanisms, not the transfinite theorems. The remaining questions are explicitly distinguished from the results proved here.

## Source 18

**Title page.** *Lexicographic Well-Orders of the Surreal Numbers: Cuts, canonical class blocks, and changes of interpretation.* Research article prepared for Vladimir Reshetnikov, 3 October 2026.

**Abstract.** The phrase “the lexicographic ordering of all well-orders of surreal numbers” describes several different objects. We give a common framework for well-orders of fixed sets, fixed enumeration types, set-length words, numerically increasing words, ordinal-indexed global enumerations, and arbitrary class well-orders. We retain the repository’s elementary comparison by common labelled prefixes and develop its unresolved topological and foundational questions.

For  $X = \mathbf{No}_{<\kappa}$  and  $\mu = 2^{<\kappa}$  we compute the density, cellularity, point characters, saturation threshold, and complete Dedekind-gap character spectrum, including exact multiplicities, of the minimal permutation order  $\mathcal{W}_\mu(X)$ . A third terminal position causes its density and cellularity to reach  $2^\mu$ . For arbitrary class well-orders, an elementary set-interval equivalence yields a canonical decomposition into Ord blocks and a final set ordinal, without ETR. The quotient can be isomorphic to the original order. An exact prefix criterion recognizes which class cuts of a dense core are represented by global well-orders; a repeated-label cut gives an explicit obstruction. We also classify the external full lexicographic order of every countable  $\omega$ -standard GBC model as a dense shuffle of finite chains of factorial sizes, and show that its pure order does not define the set-like fragment. Finally, termination conventions distinguish word orders isomorphic to  $\mathbf{No}$  from orders with limit-prefix obstructions and even adjacent pairs.

All new claims are supplied with conventional proofs. Their status is relative to a pinned repository snapshot, without a claim of literature priority or machine verification. Explicit size boundaries and a staged formalization plan accompany a revised research agenda.

**Title-page status. Repository basis.** <https://github.com/VladimirReshetnikov/ProveIt> Snapshot: 83befe707f2840c2b53e0606701d8a2b28598e47. This is an AI-assisted research manuscript with mathematical proof review; it is not a refereed paper or a Lean certificate.

## Sources 19–30

[merge] The title pages and abstracts of the twelve batch-83 sources, with the status statements on their title pages. The results they announce are printed in Parts X–XIV; the summaries are kept for the record, with this report’s notation (section 1.3.2). Each title page names Vladimir Reshetnikov as the person the manuscript was prepared for and is dated 3 October 2026.

**Source 19.** *Lexicographic Orders of Surreal Well-Orders: Residual spectra, decorated completions, and model invariants.*

There is no single object called the lexicographic order of all well-orders of the surreal numbers. The answer changes with the carrier, enumeration type, termination rule, coding convention, and foundational universe. We give a common framework for the principal interpretations and continue the repository’s analysis beyond its set-sized spectra, canonical class blocks, and dense set-like core.

The main developments concern infinite residual tails, explicit completion codes, and the structure visible in an external model. We determine the embedded binary-cube spectrum for every fixed enumeration type, and the least coding length whenever two explicit ordinal bounds coincide. A cylinder-fusion argument reduces long fixed-type embeddings to residual alphabets. We describe the Dedekind completion of the global order by retaining the entire residual well-order fiber over each core cut, and obtain an exact automorphism-extension criterion. For general residual alphabets, at most two initial ordinal words can have the same trace, and an exact greedy-tail criterion detects this ambiguity; an equivalent numerical criterion uses deletion of a set. For countable models, nonstandard finite tails produce a second, exactly determined shuffle order; nevertheless the pure orders arising from all externally set-sized models have the same first-order theory. We also classify

all monotone restriction maps and give a canonical bounded completion description, including the topology at the first infinite birthday cutoff.

The proofs separate ordinary set recursion from elementary transfinite recursion with class-valued stages, and distinguish admitted classes from external subsets. A practical formalization plan and a revised research agenda record both the results obtained and the remaining problems.

**Repository basis.** <https://github.com/VladimirReshetnikov/ProveIt>; pinned revision d5bd4a67b41b89688a079997574a94bcd855bb83. Conventional mathematical proofs, with independent checks during preparation. This AI-assisted manuscript is not a refereed publication or a machine-checked proof certificate.

**Source 20.** *Lexicographic Orders of Surreal Well-Orders: Coherent Limits, Nonextendible Symmetries, and Countable Models.*

The phrase “the lexicographic ordering of all well-orders of the surreal numbers” combines several different mathematical objects. We separate fixed set carriers, variable domains, ordinal enumeration strata, numerical well-ordered subclasses, set-like global enumerations, arbitrary class well-orders, and characteristic-relation codes. A common toolkit makes their size boundaries and comparison rules explicit.

Building on the ProveIt investigation and its recent continuations, we develop further results in five directions. Restriction-coherent birthday orders glue to class well-orders; their native lexicographic comparisons stabilize despite nonmonotone restriction maps, and minimum local types on a club of birthday cutoffs detect set-likeness. Automorphisms of the surreal-sized bounded-support core can destroy representability of cuts, even while fixing any prescribed set of core points. A local coding dichotomy identifies exactly when intervals of the full order admit uniform set or class codes. Iterated condensation has a termination bound under an explicitly stated recursion principle. Finally, the full order in a countable model with nonstandard natural numbers has a universal shuffle description involving both finite factorial chains and one additional infinite interval type. The two countable orders are elementarily equivalent and their common pure-order theory is decidable. A countable-alphabet example connects the theory with Baire space and Dedekind completion.

The article supplies conventional proofs, counterexamples, a theorem dependency ledger, and a revised research agenda. The results are research claims with internal mathematical review, not machine-checked theorems or claims of priority in the literature.

**Repository basis.** <https://github.com/VladimirReshetnikov/ProveIt>; pinned revision d5bd4a67b41b89688a079997574a94bcd855bb83. The repository was read through its GitHub interface. This is an independent AI-assisted continuation, not a modification of the repository.

**Source 21.** *Lexicographic Orders of Surreal Well-Orders: Further Structure, Automorphism Obstructions, and Model Dependence.*

There is no single unqualified object called the lexicographic order of all well-orders of the surreal numbers. A well-order of a fixed set of surreals, an ordinal-length injective word, a numerically well-ordered subset, a set-like global enumeration, and an arbitrary class well-order live at different logical levels and have different order-theoretic properties. We organize these interpretations and continue the corresponding ProveIt research program with explicit proofs and a careful size discipline.

The principal further result concerns automorphisms. A fixed global enumeration has an ordinary class of bounded-support modifications, order-isomorphic to the surreal line. Some order automorphisms of this dense core cannot extend even to order embeddings from the set-like enumeration space into the space of all class well-orders. They can move the cut of an exhaustive enumeration to a cut represented by no class well-order at all. A weaker obstruction moves an

exhaustive cut to the cut of an injective branch that omits labels. Within the set-like space, we characterize extendable automorphisms exactly by preservation of exhaustive represented cuts.

Several complementary results make the distinction concrete. For a countable dense alphabet, the minimum-length permutation order is homeomorphic to Baire space, every open interval is isomorphic to the whole order, and finite increasing partial maps extend to automorphisms; its prefix cylinders are nevertheless not intrinsic to its pure order. Its Dedekind completion is the real line, with continuum many omitted cuts, so it is not order-isomorphic to the irrational numbers. For a general set alphabet we identify the convex fibers over its first cardinality-many coordinates and compute the closure of the minimum-length stratum. We also treat noncardinal birthday cutoffs, coherent embeddings within a fixed cardinality, intrinsic finite permutation blocks, and the distinct external classifications arising from countable standard and nonstandard models.

The foundational discussion separates elementary class comprehension, ordinary set-valued recursion, elementary transfinite recursion with class-valued slices, higher-order collections, and universe-relative formalizations. A final research agenda records exactly what has been settled here and what still needs stronger classification or verification.

**Proof and provenance status.** This is an AI-assisted mathematical research manuscript with written proofs. Its principal arguments received independent internal proof checks during preparation; that is neither refereeing nor proof-assistant verification. No new Lean theorem is claimed. The ProveIt source is pinned to commit `109aaca1505c12d70ae169bdc2011f40bd476f7d`, observed on 3 October 2026. Results called extensions are extensions developed and proved in this manuscript, not claims of historical priority. Classical background and results already in the repository are identified explicitly.

**Source 22.** *Lexicographic Orders of Surreal Well-Orderings: Residual geometry, the finite-tail transition, unranked cylinders, and reconstruction of class realizations.*

The phrase “all well-orders of surreal numbers” ranges from ordinary sets of labelled enumerations to predicates on proper-class relations. We separate these interpretations and continue the ProveIt investigation in two directions.

For an infinite cardinal  $\kappa$ , let  $X = \mathbf{No}_{<\kappa}$ ,  $\mu = 2^{<\kappa}$ , and  $\Delta = \mu^{<\mu}$ . We determine the density, cellularity, and weight of *every fixed enumeration-length stratum*  $\mathcal{W}_\alpha(X)$ ,  $|\alpha| = \mu$ . At lengths  $\mu$  and  $\mu + 1$  all three invariants are  $\Delta$ . At  $\mu + 2$ , density and cellularity remain  $\Delta$ , but weight is  $2^\mu$ . At every length  $\alpha \geq \mu + 3$ , all three are  $2^\mu$ . We give exact point characters for finite tails and the tail  $\omega$ , and a general residual-fibre formula for arbitrary lengths. The ordinal embedding spectrum stays unchanged throughout.

For global enumerations, the inclusion order of the unranked prefix cylinders recovers their ordinal heights. A normalized cut represents an injective Ord-word exactly when the cylinders it crosses form a proper class; a label-coverage condition recognizes exhaustive words. Finally, a uniform pair-swap encoding and an elementary decoder show that admitted exhaustive traces, over a fixed labelled core and coding parameters, recover the entire available class realization. These arguments require no class-valued transfinite recursion and never form a same-level class of all classes.

**Status.** AI-assisted mathematical research manuscript, with complete ordinary proofs of the stated extensions. Historical priority, independent refereeing, and Lean verification are not claimed. The repository and two earlier continuation manuscripts are credited explicitly. Executable checks cover finite mechanisms only. No repository files were modified.

**Source 23.** *Lexicographic Surreal Well-Orders: Termination markers, increasing-word obstructions, and transfinite condensation.*

The expression “all well-orders of surreal numbers” encompasses several inequivalent constructions. We separate fixed-carrier enumerations, set-length words, naturally increasing well-ordered subsets, set-like global enumerations, arbitrary class well-orders, and orders on relation codes. Two extensions of the existing research are developed in detail.

First, we classify exactly which placements of a fixed termination marker make the order of all set-length words, or all injective set-length words, set-cut saturated. An interior marker works precisely when each side of its alphabet cut either has the appropriate endpoint or has no small cofinal or coinital subset. In particular, a marker immediately adjacent to an existing surreal label works: an adjacent pair in the alphabet need not produce an adjacent pair of words. We compute every point’s two-sided character. By contrast, no fixed termination marker makes the order of strictly increasing set-length surreal words set-cut saturated.

Second, we introduce set-interval condensation of raw class well-orders, using least representatives rather than proper classes as quotient elements. For every given class well-order  $\Gamma$ , a finite-support ordinal-polynomial construction supplies a class well-order with an explicitly defined condensation hierarchy indexed by  $\Gamma + 1$ , collapsing exactly at its last stage. These particular hierarchies require no elementary transfinite recursion axiom. Every interval between distinct set-like global enumerations contains raw well-orders realizing every such prescribed hierarchy. We explain why this does not classify the bare outer lexicographic order, and we give a size-explicit formalization program and further questions.

**Status.** Ordinary mathematical proofs are provided for the new statements. This is an AI-assisted, unrefereed research manuscript; historical priority and Lean verification are not claimed. The accompanying executable checks concern finite mechanisms only. The inspected repository revision and the distinction between prior results, new arguments, and open problems are recorded explicitly.

**Source 24.** *Lexicographic Well-Orders of the Surreals: Prefix completions, generalized Baire models, and the set-class boundary.*

We continue the study of lexicographic orders of labelled well-orderings of surreal numbers. The main advance is a distinction between order completion and completion by stabilization of initial segments. For an infinite alphabet of cardinality  $\mu$ , the latter completion of the minimal-length permutation space is exactly the space of injective  $\mu$ -words; failure of exhaustiveness is its entire missing boundary. For regular  $\mu$  we construct an explicit homeomorphism from the permutation space to the bounded-topology space  $(\mu^{<\mu})^\mu$ , without assuming  $\mu^{<\mu} = \mu$ .

At the countable birthday cutoff this gives a complete compatible metric and a Baire-space presentation. Nevertheless, the resulting linear order is not isomorphic to the irrationals: its Dedekind completion is the real line, but it omits continuum many points. At singular cardinalities we identify exactly when right translation and inversion are continuous in the prefix topology, and isolate the boundedness-preserving subgroup on which the usual permutation-group picture remains valid.

For global surreal enumerations, we formulate corresponding results in Gödel–Bernays class theory without forming a class of classes. Every uniformly set-indexed prefix-Cauchy net is eventually constant, whereas class-indexed approximation recovers all injective ordinal-class words. Inverse stabilization precisely repairs the loss of exhaustiveness. We also distinguish fixed and varying carriers, inherited numerical well-orders, arbitrary non-set-like class well-orders, relation-code orders, and order-type quotients. A concrete Lean plan is tied to inspected repository declarations, and the remaining singular and foundational questions are stated separately.

**Status.** This is an AI-assisted, unrefereed proof manuscript. The results are supported by

the mathematical arguments given here, not by a new Lean verification or a claim of historical priority. Finite regression checks are included but do not prove the transfinite theorems. The repository was inspected at the pinned revision recorded in Appendix 111.25; it was not changed.

**Source 25.** *Lexicographic Surreal Well-Orders: Prefix homogeneity, invisible exhaustiveness, and label rigidity.*

We continue the investigation of lexicographically ordered well-orderings of the surreal numbers, distinguishing fixed set carriers, variable-length words, naturally increasing words, set-like global enumerations, arbitrary class well-orders, and relation codes.

The principal extension concerns the set-coded, set-supported core of global enumerations. We classify the orbits of every uniformly set-indexed configuration of injective ordinal-class words under automorphisms of the core that preserve its order and all unranked prefix cylinders. The complete invariants are core membership, pairwise first-disagreement ordinals, and pairwise lexicographic comparisons. In particular, there are exactly two single-branch orbits: core branches and all other injective branches.

Consequently, even the order together with every prefix cylinder does not recognize exhaustive branches. A cylinder-preserving core automorphism can carry the trace of a global well-order to that of an injection omitting a surreal label, and therefore need not extend to the full enumeration order. This remains possible while fixing any prescribed set of core and prefix parameters. Conversely, one cross-node relation saying that two extension edges use the same label makes the core-and-cylinder structure rigid. Label coherence without the core yields exactly coordinatewise order automorphisms of the surreal alphabet.

We prove regular-cardinal set versions, including an exact automorphism-group cardinality, and a cofinality obstruction at singular lengths. All class constructions are formulated in GBC using set-valued recursion, with no same-level collection of classes and no appeal to elementary transfinite recursion.

**Status.** AI-assisted, unrefereed research manuscript. Complete ordinary proofs are supplied for the stated extensions; historical priority and Lean verification are not claimed. Earlier results are credited separately. The accompanying executable checks concern finite mechanisms, not the transfinite theorems.

**Source 26.** *Lexicographic Well-Orders of the Surreals: Non-Borel boundaries, closed universality, and safe foundations.*

We continue the investigation of lexicographic orders of labelled well-orderings of surreal numbers. The principal result answers the singular complete-metrizability question in the earlier *Prefix Completions* manuscript. For every uncountable cardinal  $\mu$  of countable cofinality, the space of bijections  $\mu \rightarrow X$ , with bounded-prefix topology, is not Borel in its completely metrizable space of injective words. Consequently it has no compatible complete metric. Together with the local-character calculation and the countable case, this gives an exact classification: for infinite minimal enumeration lengths, complete metrizability holds precisely at  $\mu = \omega$ .

The proof is an explicit omitted-label coding construction. A compact Cantor family of injections can encode any set of fewer than  $\mu$  binary branches through the omission of exactly one label. If  $\mu > 2^{\aleph_0}$ , every zero-dimensional separable metrizable space occurs as a closed subspace of the permutation space. Ordered marker labels make the construction an order embedding as well. At the unconditional birthday cutoff  $\beth_\omega$ , these results apply to the actual lexicographic order topology, yielding a metrizable Baire order with no nonempty completely metrizable open subspace.

We also distinguish fixed and varying carriers, minimal and nonminimal enumeration lengths, inherited numerical well-orders, termination conventions, relation-code lexicography, order-type



quotients, set-like global enumerations, and unrestricted class well-orders. Direct labelled comparison, bounded-support coding, diagonalization, and class-indexed approximation are formulated without a class of classes. A concrete universe-aware Lean plan, a source audit, finite regression checks, and a proposed research program complete the report.

**Status.** AI-assisted and unrefereed. The main advances are supplied with written proofs; no new Lean verification or historical-priority claim is made. Earlier repository results are identified as such. The repository was inspected at a pinned revision and was not modified.

**Source 27.** *Lexicographic Orders of Surreal Well-Orders: Further Structure from Encodings, Prefix Trees, and Support Bounds.*

There is no single order determined by the phrase “the lexicographic ordering of all well-orders of surreal numbers.” A fixed set of surreal labels, all set-length words, numerically increasing words, ordinal enumerations of the whole surreal class, arbitrary class well-orders, and well-ordered scans of relation tables produce different objects. Their differences remain substantial after their common elementary theory has been understood.

Building on the current ProveIt research corpus, we develop five further directions. First, relation-table orders on the same countable carrier can be real-embeddable or can have continuum many disjoint nonempty intervals, depending on whether the scan has order type  $\omega$  or  $\omega + 6$ ; their ordinal embedding spectra nevertheless agree. Second, an increasing word with an interior terminator splits exactly into a left part and a right part, giving a lexicographic-product description of the centered surreal construction. Third, ordered prefix trees admit automorphisms taking exhaustive ordinal branches to nonexhaustive ones: preserving levels, extensions, and the numerical ordering of siblings does not recover exhaustiveness. Fourth, every uniformly class-indexed family of arbitrary global well-orders misses a convex copy of the entire raw comparison order. A related rigidity theorem shows that a set-like enumeration order with its canonical core fixed pointwise detects the admitted class part of a model. Finally, we sharpen the existing optimum for the cardinality of permutation support to an exact least ordinal support bound.

All constructions are given with explicit size and uniformity conditions. We explain their implementation in set theory, two-sorted class theory, set models, and universe-polymorphic type theory. The proofs are conventional mathematical proofs; the accompanying finite checks test finite mechanisms, not transfinite or class-theoretic assertions.

**Status and provenance.** The repository snapshot is `7ff7736ecf3a0adc8536b2d083cf819e1ba39ace`, committed at 21:47:46 UTC on 3 October 2026. The present article is a continuation, not a replacement for the much larger merged report. Statements identified as background come from that report or the cited primary literature. The additional theorems are proved here, with no claim of historical priority or external peer review. No new Lean proof certificate is supplied.

**Source 28.** *Lexicographic Well-Orders of the Surreals: Singular permutation geometry and the canonical group envelope.*

We continue the study of lexicographic orders of labelled well-orderings of surreal numbers, with particular attention to a distinction that appears at singular enumeration cardinalities: controlling a bounded initial segment is not the same as controlling an arbitrary small set of positions. For the minimal enumeration slice, this distinction separates the numerical order topology from a canonical topology compatible with all permutation operations.

We prove that the subgroup preserving bounded sets in both directions is maximal among subgroups containing the finitary permutations and carrying the inherited topological-group structure. At every singular cardinal it is self-normalizing. We determine its forward closure and completion,

describe a dense inverse-continuity locus, and construct explicit permutations separating the one-sided boundedness conditions. The smallest group topology finer than the prefix topology is exactly agreement on arbitrary sets of size less than the enumeration cardinal. Its character is the cofinality of the small-subset poset, which is strictly greater than that cardinal in the singular case. It is complete for its two-sided group uniformity.

These results apply to surreal birthday cutoffs after verifying their thickness. We also distinguish fixed and varying carriers, inherited numerical well-orders, set-like and unrestricted global class well-orders, relation codes, and order-type quotients. Class arguments are formulated without a class of classes, and a formalization plan separates universe-indexed Lean mathematics from model-relative class semantics.

**Research status.** AI-assisted, unrefereed mathematical manuscript. The new assertions are supported by the proofs below, not by a claim of historical priority or a new Lean kernel verification. Finite regression checks accompany the text but do not prove its infinitary statements. The inspected repository revision is pinned in Appendix 111.42; the repository was not modified.

**Source 29.** *Lexicographic Orders of Surreal Well-Orders: Exact Birthday Spectra, Word Products, Topological Transitions, and Class Comparability.*

The phrase “the lexicographic ordering of all well-orders of the surreal numbers” refers to several different mathematical constructions. Some are sets, some are classes of set codes, and some are higher-order collections of class relations. We give a common framework for these interpretations and extend the corresponding ProveIt research program.

Our first result recovers the exact birthday cutoff from the numerical order: the binary lexicographic cube  $\mathbb{B}_\beta$  embeds in  $\mathbf{No}_{<\theta}$  exactly when  $\beta < \theta$ . Consequently the birthday fragments form a strictly increasing hierarchy under order embeddability, even throughout a band on which their cardinality is constant. In contrast, for every countable  $\omega < \theta < \omega_1$  and every enumeration length  $\alpha = \Lambda + n$  of cardinality the continuum, with  $\Lambda$  limit and  $n$  finite, the well-ordering stratum is equimorphic to  $\mathbb{B}_\Lambda \times_{\text{lex}} n!$ . We extend the corresponding residual-cardinal analysis to arbitrary noninitial birthday bounds and state the remaining singular upper-coding gap explicitly.

For increasing set-length words, splitting at the termination cut gives a canonical factorization. The centered surreal case has order type  $\mathbf{T}_{\min}^{\text{op}} \times_{\text{lex}} \mathbf{T}_{\min}$ , where  $\mathbf{T}_{\min}$  is the shorter-first order of all set-length surreal words. This identifies three previously separate zero conventions and describes the fourth by adjacent-pair expansion. If proper-class numerical carriers are admitted, shorter-first adjacency instead occurs exactly when a carrier acquires its unattained numerical supremum. The longer-first class-carrier order remains dense.

At the countable alphabet  $\mathbf{No}_{<\omega} \cong \mathbb{Q}$ , finite terminal tails produce a sharp progression in topology and measurable structure. The native lexicographic and coordinate Borel structures agree at  $\omega + 1$ , differ at  $\omega + 2$ , and are incomparable at every finite successor  $\omega + n$  with  $n \geq 3$ . We also prove that the entire ordered set-prefix tree fails to recognize exhaustive or numerically cofinal class branches. Finally, over GBC, the existence of complete terminating canonical set-interval condensation histories is equivalent to strong comparability of class well-orders. The proof supplies finite ordinal-coefficient normal forms from any such history.

Every assertion distinguishes isomorphism from equimorphism, supplied classes from collections of classes, and elementary definitions from class-recursion existence principles. The article ends with concrete formalization targets and a revised research agenda.

**Proof and provenance status.** This is an AI-assisted research manuscript with written proofs and independent internal mathematical review. It is not a refereed paper, and no new Lean verification or publication-priority claim is made. The repository audit is pinned to

commit 7ff7736ecf3a0adc8536b2d083cf819e1ba39ace, observed on 3 October 2026. Earlier project manuscripts are distinguished from the results developed here. A source manifest and a proof-status guide accompany the article.

**Source 30.** *Lexicographic Well-Orders of the Surreals: Residual Codes, Topological Transitions, and Class Reconstruction.*

The lexicographic ordering of all well-orders of the surreal numbers has several mathematically inequivalent interpretations. After separating them, we develop four continuations of the existing ProveIt reports. We determine binary-cube equimorphism types for new fixed strata with infinite remainders, including the previously isolated case  $\mu + \omega$ . Finite descent through regular and singular strong-limit cardinal blocks gives a complete fixed-type equimorphism classification under GCH. An additional bounded-birthday argument classifies every fixed type in ZFC at each infinite cutoff  $\kappa < \aleph_\omega$ . We prove a sharp topological and measurable transition between the countable strata of types  $\omega$ ,  $\omega + 1$ ,  $\omega + 2$ , and  $\omega + n$  for  $n \geq 3$ ; relation tables and enumeration codes cease to be measurably interchangeable already at  $\omega + 2$ . An exact partial-table extension theorem shows that every admitted proper-class well-order type is locally indistinguishable from every other by set-sized relation data. Finally, the admitted *labelled* global enumerations recover an entire class realization, despite the severe information loss of its external pure order. Sets, classes, and metatheoretic collections are kept distinct throughout. The article includes proofs, source and assumption audits, a formalization plan, and a prioritized research agenda.

**Status.** Conventional mathematical arguments with independent AI-assisted proof review. This is not a machine-checked formalization or an assertion of historical priority. Inherited results are identified separately from the continuations proved here.

## Source 31

[merge] The title page, abstract and scope statement of source 31 (batch-89 manuscript 01). The results it announces are printed in Part XV; the summary is kept for the record, with this report's notation (section 1.3.3). The title page reads “Research article prepared for Vladimir Reshetnikov” and is dated 4 October 2026.

**Source 31.** *Countable Global Choice and Surreal Coding over Zermelo Set Theory: A localized nonconservativity theorem and exact rank-model spectra.*

We investigate an intersection of Elliot Glazer's work on weak choice principles and ProveIt's research on surreal numbers and class well-orders. The first result is a refinement of Glazer's nonconservativity theorem: a global selector required to choose only from nonempty countable sets already produces new set-theoretic consequences over Zermelo set theory with Choice, provided Separation is extended to the selector. Selection on exactly countably infinite sets suffices as well. The additional argument is a bounded uniformization theorem. It converts a selector on sets embeddable into a fixed well-ordered set into a uniform assignment of injections into that set, without Replacement or ordinal collapse. A general incidence-coding theorem then connects this uniformization to Glazer's published countermodel.

The second part studies the full-class structures  $\mathcal{M}_\lambda = (V_\lambda, \mathcal{P}(V_\lambda))$ , for limit ordinals  $\lambda > \omega$ . Global choice and set-like well-orders exist at every such height. Nevertheless, ordinal enumeration of binary sign sequences, enumeration in birthday order, universal coding of sets by sign sequences, and full Class Replacement have different exact thresholds: respectively  $2^{<\kappa} = \kappa$ , strong-limit cardinality, a beth fixed point, and strong inaccessibility. We also determine replacement on each individual set domain and the external cardinalities of internal proper classes. These results isolate several

mechanisms behind the usual global-choice equivalences. Full proofs, strict separation examples, an axiom audit, and twelve further research questions are included. The article claims a proved refinement and a structural synthesis; it makes no claim of resolving unrestricted surreal transfer or of establishing historical priority for every result.

**Scope.** Conventional mathematical proofs. No Lean or Rocq verification is claimed. The nonconservativity conclusion uses Glazer’s countermodel theorem as an explicitly cited input. The rank-model classification is proved in an external ZFC metatheory.

## Sources 32–35

[merge] The title pages, abstracts and status statements of sources 32–35 (batch-96 manuscripts 08–11). The results they announce are printed in Part XVI; the summaries are kept for the record, with this report’s notation ([section 1.3.4](#)).

**Source 34 (base).** *Beyond Ord: Definable Class Well-Orders, Canonical Arithmetic, and the Limits of Notation.* “Research report prepared for Vladimir Reshetnikov”, dated 4 October 2026.

The class of all ordinals is the endpoint of set-sized ordinal notation, but it is not the endpoint of definable well-ordering. Tagged sums, lexicographic products, and finite-support powers produce well-orders such as  $\text{Ord} + 1$ ,  $\text{Ord} \cdot \text{Ord}$ , and  $\text{Ord}^{[\text{Ord}]}$  without creating a new set ordinal. We develop these constructions as relations on sets, separate their explicit arithmetic from stronger class-recursion principles, and give systematic notation by finite labelled trees.

The main refinement of the ProveIt program is a choice-free proof that finite-support powers of class well-orders are well-ordered. It lowers the repository’s equivalence between canonical terminating condensation histories and strong class-well-order comparability from GBC to GB, for arbitrary class carriers and the full subclass-minimum definition of well-ordering. This resolves the corresponding base-theory question in the inspected repository; it does not decide whether comparability implies elementary transfinite recursion. Further results give a hereditary fixed-point notation system, a uniform-family diagonal obstruction, a truth-relative construction of longer definable orders, and an external cofinality theorem for definable order spectra. In a full inaccessible-universe interpretation, all class well-order types have external lengths below  $\kappa^+$ , while the hereditary system has the exact length  $\varepsilon_{\kappa+1}$ .

Every substantive theorem is proved or explicitly attributed. The paper distinguishes repository refinements and new combinations of arguments from established literature, makes no claim of historical priority, and records a formalization plan and a focused research agenda. The proofs are mathematical proofs; no kernel-checked formalization is claimed.

**Notation convention.**  $\Omega$  denotes a presentation of  $(\text{Ord}, \in)$ , never an ordinal that belongs to the same universe.  $A^{[B]}$  denotes finite-support order exponentiation, not a power class.

**Source 35.** *Definable Class Well-Orders Beyond Ord: Arithmetic, canonical notation, fixed-point completion, and the limits of uniform definability.* “Research report prepared for Vladimir Reshetnikov”, dated 5 October 2026.

The ordinal class is the end of the set ordinals, but it is not the end of definable well-order presentations whose elements are sets. A proper-class well-order may have points with proper classes of predecessors, giving familiar-looking expressions such as  $\text{Ord} + 1$ ,  $\text{Ord} \cdot \text{Ord}$ , and  $\text{Ord}^{\text{Ord}}$  rigorous meanings. We develop their arithmetic using tagged sums, lexicographic products, and finite-support exponentiation, while keeping class presentations separate from nonexistent set ordinals beyond  $\text{Ord}$ .

The first contribution is a direct, choice-free proof that finite-support exponentiation preserves class well-ordering. It lowers the ProveIt manuscript’s equivalence between strong class-well-order comparability and complete canonical condensation histories from GBC to GB, with no use of set Choice. A second contribution, over GBC, characterizes precisely which orders have a fixed-point completion for  $P(X) = \text{Ord}^X$ : an order  $W$  initially embeds into such a fixed point if and only if  $W$  initially embeds into  $P(W)$ . For nonempty  $W$ , this is also equivalent to its canonical terminating history on  $W + 1$ . The proof uses finite traces and an explicitly coded direct limit, rather than arbitrary class-valued recursion. We also prove a strict bounding theorem for every uniformly coded family with a supplied well-ordered index class, and derive a hierarchy of definability complexities of their presentations up to admitted isomorphism.

Hereditary ordinal-coefficient terms supply a concrete exponential fixed point and a transparent comparison procedure. Full-model and definability-sensitive spectra explain the model-relative sense in which such orders can stretch farther than Ord. The final sections compare set-theoretic and type-theoretic formalizations and propose a research program. Existing polynomial and condensation results are credited to the inspected repository; classical class-epsilon notation is also credited. The extensions are proved here, but no claim of literature-wide priority or resolution of CWO versus ETR is made.

**Keywords.** Class well-order; ordinal arithmetic; elementary transfinite recursion; definability; canonical condensation; fixed point; finite-support notation; Gödel–Bernays set theory.

**Source 32.** *Beyond Ord: Definable Class Well-Orders. Explicit arithmetic, condensation, definability ceilings, and truth jumps*; headed “Foundations and order theory”. “Research manuscript prepared for Vladimir Reshetnikov”, dated 4 October 2026.

The ordinal class is the largest set-like well-order, but not the largest class well-order. We develop an explicit arithmetic of non-set-like well-orders without treating proper classes as elements of a same-level universe. Finite-support exponentiation and hereditary finite normal forms give orders far beyond Ord already in Gödel–Bernays set theory with Global Choice. A canonical division theorem expresses every class well-order as  $\Omega \cdot Q + \alpha$ , where  $\Omega = (\text{Ord}, <)$ ,  $Q$  is a class well-order presentation, and  $\alpha$  is a set ordinal.

Three further results organize the limitations of this arithmetic. First, a diagonal sum dominates every member of any one uniformly class-coded family, even without class-order comparability. Second, whenever a set-interval condensation hierarchy exists along  $W + 1$ , it collapses  $W$  by its final stage; elementary transfinite recursion supplies such a hierarchy. Explicit polynomial examples show that no set-ordinal bound suffices. Third, for an inaccessible  $\kappa$  and the classes definable over  $(V_\kappa, \in, G, A)$  with set parameters, the externally realized order types form an ordinal  $\delta_A$  satisfying

$$\kappa < \delta_A < \kappa^+, \quad \text{cf}(\delta_A) = \omega, \quad \kappa^{\delta_A} = \delta_A.$$

A full truth predicate for that structure makes  $\delta_A$  itself representable. We give proofs, isolate the relevant axioms, relate the constructions to the ProveIt repository, and propose a formalization and research program.

**Research status.** AI-assisted, unrefereed manuscript. Theorems below have written proofs; no historical-priority, peer-review, or machine-verification claim is made. Standard results and repository antecedents are identified separately. In particular, the condensation result is conditional on existence of the indicated hierarchy, not a proof of its existence in bare GBC.

**Source 33.** *Beyond Ord: Definable Class Well-Orders, Arithmetic, Condensation, and Definability Heights*; headed “Foundations / research article”. “Research manuscript prepared for Vladimir Reshetnikov”, dated 4 October 2026.

Proper-class well-orders can extend beyond the order of all set ordinals without creating new set ordinals. We develop a size-safe calculus for these orders and separate explicit constructions in Gödel–Bernays set theory with global choice from principles requiring elementary transfinite recursion. The structural results include canonical division by the order of all ordinals, an explicit condensation hierarchy for finite ordinal-coefficient polynomials, and a finite-digit embedding theorem for orders equipped with an exhausting condensation tower.

The main definability result is a uniform-family diagonal theorem. For each fixed finite formula-complexity bound, it produces a class well-order definable without set parameters, relative to a named global order, into which every well-order of that complexity with arbitrary set parameters embeds, while the constructed order embeds into none of them. Consequently, set-parameter-free presentations are externally cofinal among all set-parameter-definable presentations, but no definable uniform family can witness that cofinality. At an inaccessible rank model  $V_\kappa$ , all full-class order types are exactly the ordinals below  $\kappa^+$ ; the smaller definable spectrum has a supremum strictly between  $\kappa$  and  $\kappa^+$  of external cofinality  $\omega$ ; this supremum is an epsilon number closed under ordinal arithmetic. Adding a truth predicate strictly increases that definability height.

The article supplies proofs, an axiom ledger, repository integration guidance, and further research problems. Historical priority for the derived results is not established, and no machine verification of the class-theoretic proofs is claimed.

**Status.** AI-assisted, unrefereed research manuscript. The established literature, arguments proved here, conditional results, and open directions are distinguished throughout. The basic existence of long class well-orders and earlier condensation constructions are not advertised as new discoveries.

## Source 36

[merge] The title page, abstract and scope statement of source 36 (batch-97 manuscript 01). The results it announces are printed in Part XVII, with this report’s notation ([section 1.3.5](#)).

**Source 36.** *Beyond Ord: Definable Class Well-Orders, Finite-Support Arithmetic, and Definability Horizons*, subtitled “A constructive notation calculus with exact model-relative bounds”; “An investigation prompted by the ProveIt repository”, October 2026; no author line.

**Main result.** With a named global enumeration of an inaccessible-height universe, all first-order definable class well-orders have an external definability horizon of countable cofinality, fixed by epsilon enumeration. Presentations without set parameters already approach this horizon. Adjoining a truth predicate makes the horizon itself presentable and produces a strictly larger one.

[merge] (4 October 2026.) Apart from “fixed by epsilon enumeration”, this main result is the definability-ceiling and truth-presentation theorem that sources 32, 33 and 34 proved independently on 4 October 2026 ([notes XVII.7.D](#) and [XVII.7.G](#)); source 36 did not see Part XVI. The statement is printed as delivered.

**Scope and status.** This article gives conventional mathematical proofs, explicit constructions, source attribution, and a research agenda. Long class well-orders, their arithmetic, and the epsilon notation functor have established antecedents. The structural results are proved here as a synthesis; publication priority and a breakthrough resolution of a recognized open problem are not asserted. The accompanying finite computations check a notation demonstrator; they are not proof-assistant certification.

The class of all ordinals is the longest set-like proper class well-order, but it is far from the longest definable class well-order. We develop a systematic account with explicit carriers, arithmetic, and normal forms. Finite-support reverse-lexicographic powers define class ordinal arithmetic in



Gödel–Bernays set theory. We prove their laws, exact predecessor decompositions, and a complete criterion for set-likeness. Finite hereditary  $\Omega$ -normal forms yield a parameter-free class  $\mathbf{E} \cong \Omega^{[\mathbf{E}]}$ , with canonical strict initial segments  $\Omega, \Omega^{[\Omega]}, \Omega^{[\Omega^{[\Omega]}]}, \dots$ . Localization to set ordinals proves well-foundedness without class-valued recursion. The standard epsilon-term functor extends the notation scale to fixed points of epsilon enumeration.

For a transitive set model  $M \models \text{ZFC}$  of ordinal height  $\kappa$ , externally well-founded definable presentations with finite set parameters realize precisely  $[0, \mathfrak{H}(M))$ , where  $\mathfrak{H}(M) < |\kappa|^+$  and  $\varepsilon_{\mathfrak{H}(M)} = \mathfrak{H}(M)$ . Explicit parameter-free syntax already passes the first fixed point of epsilon enumeration above  $\kappa$ . For  $M = V_\kappa$ , with  $\kappa$  inaccessible and a named global enumeration  $G$ , we prove  $\text{cf}(\mathfrak{H}_G) = \omega$  and that presentations without set parameters are cofinal in the parameter-allowing spectrum. Each fixed formula complexity admits a definable order dominating all presentations of that complexity. Full truth makes  $\mathfrak{H}_G$  itself presentable; varying  $G$  makes the horizons unbounded in  $\kappa^+$ . With all classes available, the exact external cutoff is  $\kappa^+$ .

We distinguish formula schemas, admitted classes, external order types, unrolling, and higher-universe formalizations. A uniform diagonal theorem rules out a same-level notation system cofinal in all class well-orders. The known class-comparability and recursion-strength questions remain separate from the explicit constructions proved here.

## D Crosswalk of the sources' numbered statements

[merge] Each row gives a numbered statement of a source (numbered as in that source's own PDF), and where it is printed or recorded here. A note records a statement proved here by the same argument; a section labelled “route” prints a different argument. Labels are the source's own with its prefix, or `swo:sub:number` for unlabelled statements.

### D.1 Source 11 (base)

| No.  | Statement in source 11                           | Printed here as         |
|------|--------------------------------------------------|-------------------------|
| 3.1  | Definition                                       | Definition 5.1, p. 60   |
| 3.2  | Lemma (Deleting a set)                           | Lemma 5.2, p. 60        |
| 3.3  | Proposition                                      | Proposition 5.13, p. 67 |
| 4.1  | Theorem (Cardinality and exact ordinal spectrum) | Theorem 6.1, p. 69      |
| 4.2  | Proposition (Exactly where jumps occur)          | Proposition 6.2, p. 70  |
| 5.1  | Corollary                                        | Corollary 7.1, p. 76    |
| 5.2  | Lemma                                            | Lemma 7.2, p. 76        |
| 6.1  | Theorem (Explicit sign coding)                   | Theorem 9.1, p. 81      |
| 6.2  | Proposition (Birthday bounds)                    | Proposition 9.2, p. 82  |
| 7.1  | Theorem (Exact local character)                  | Theorem 10.1, p. 83     |
| 7.2  | Corollary                                        | Corollary 10.2, p. 83   |
| 8.1  | Theorem (Word-cut classification)                | Theorem 11.1, p. 84     |
| 8.2  | Corollary (Successor words)                      | Corollary 11.2, p. 84   |
| 9.1  | Proposition                                      | Proposition 12.1, p. 86 |
| 10.1 | Theorem (Existence and global choice)            | Theorem 13.1, p. 87     |
| 11.1 | Lemma (Unique first disagreement)                | Lemma 14.1, p. 92       |
| 11.2 | Theorem (Raw lexicographic order)                | Theorem 14.2, p. 92     |
| 11.3 | Proposition (Convex prefix cones)                | Proposition 14.3, p. 93 |
| 12.1 | Theorem (Canonical initial part)                 | Theorem 15.1, p. 95     |
| 12.2 | Example (Two proper blocks)                      | Example 15.2, p. 95     |
| 12.3 | Proposition (Convex fibers)                      | Proposition 15.3, p. 96 |

| No.   | Statement in source 11                                           | Printed here as                  |
|-------|------------------------------------------------------------------|----------------------------------|
| 12.4  | Remark (Singleton fibers and iteration)                          | Remark 15.4, p. 96               |
| 13.1  | Lemma (Extreme enumerations)                                     | Lemma 16.1, p. 96                |
| 13.2  | Theorem (Exact adjacency criterion)                              | Theorem 16.2, p. 97              |
| 13.3  | Example (An adjacent pair of type $\text{Ord} + 2$ )             | Example 16.3, p. 97              |
| 14.1  | Definition                                                       | Definition 19.1, p. 101          |
| 14.2  | Definition                                                       | Definition 19.3, p. 102          |
| 14.3  | Lemma (Completion of an arbitrary set prefix)                    | Lemma 19.4, p. 103               |
| 14.4  | Theorem (Prefix interpolation)                                   | Theorem 19.10, p. 105            |
| 14.5  | Corollary                                                        | Corollary 19.15, p. 106          |
| 14.6  | Corollary (The class of realizations of a set cut)               | Corollary 19.18, p. 107          |
| 14.7  | Proposition                                                      | Proposition 19.20, p. 108        |
| 14.8  | Proposition (Prefix cylinders)                                   | Proposition 19.25, p. 109        |
| 14.9  | Proposition (Approximants at every point)                        | Proposition 19.27, p. 110        |
| 14.10 | Corollary (Set saturation is not Dedekind completeness)          | Corollary 19.29, p. 110          |
| 14.11 | Theorem                                                          | Theorem 19.32, p. 112            |
| 14.12 | Corollary                                                        | Corollary 19.36, p. 113          |
| 14.13 | Theorem (No exhaustive uniform class of codes)                   | Theorem 19.37, p. 114            |
| 14.14 | Theorem (An omitted-range cut)                                   | Theorem 19.42, p. 116            |
| 14.15 | Example (Two omitted points produce adjacent raw orders)         | Example 19.45, p. 117            |
| 14.16 | Corollary (Exactly which raw set cuts admit bounded completions) | Corollary 19.46, p. 118          |
| 15.1  | Theorem (Nonembedding in a full universe)                        | Theorem 20.1, p. 120             |
| 15.2  | Theorem                                                          | Theorem 20.2, p. 120             |
| 16.1  | Research question                                                | Research question 113.1, p. 892  |
| 16.2  | Research question                                                | Research question 113.2, p. 892  |
| 16.3  | Research question                                                | Research question 113.3, p. 893  |
| 16.4  | Research question                                                | Research question 113.4, p. 893  |
| 16.5  | Research question                                                | Research question 113.5, p. 893  |
| 16.6  | Research question                                                | Research question 113.6, p. 893  |
| 16.7  | Research question                                                | Research question 113.7, p. 894  |
| 16.8  | Research question                                                | Research question 113.8, p. 894  |
| 16.9  | Research question                                                | Research question 113.9, p. 894  |
| 16.10 | Research question                                                | Research question 113.10, p. 894 |

## D.2 Source 12

| No. | Statement in source 12                                          | Printed here as        |
|-----|-----------------------------------------------------------------|------------------------|
| 3.1 | Definition (Common labelled prefix)                             | Definition 14.4, p. 93 |
| 3.2 | Lemma (Maximal-prefix lemma)                                    | Lemma 14.5, p. 94      |
| 3.3 | Definition (Full lexicographic comparison)                      | Definition 14.6, p. 94 |
| 3.4 | Theorem (Full class lexicographic order)                        | Theorem 14.7, p. 94    |
| 3.5 | Remark (Why this does not prove class order-type comparability) | Remark 14.8, p. 94     |
| 3.6 | Example (A disagreement after every ordinary ordinal)           | Example 14.9, p. 94    |
| 3.7 | Corollary (Agreement with enumeration lexicography)             | Corollary 14.10, p. 95 |
| 4.1 | Theorem (Choice equivalence)                                    | Theorem 13.2, p. 88    |

| No.  | Statement in source 12                                               | Printed here as                  |
|------|----------------------------------------------------------------------|----------------------------------|
| 4.2  | Remark (The explicit set-choice hypothesis)                          | Remark 13.3, p. 89               |
| 5.1  | Theorem (No exhaustive class-coded family)                           | Theorem 19.38, p. 115            |
| 5.2  | Corollary (No uniform set-code solution)                             | Corollary 19.39, p. 115          |
| 6.1  | Lemma (Extending a set prefix)                                       | Lemma 19.5, p. 103               |
| 6.2  | Lemma (Avoiding a forbidden set)                                     | Lemma 5.3, p. 60                 |
| 6.3  | Theorem (Set-indexed interpolation)                                  | Note 19.11, p. 106               |
| 6.4  | Proposition (Cofinality and local character)                         | Note 19.28, p. 110               |
| 6.5  | Lemma (Surreal universality, with the size of the recursion exposed) | Lemma 5.14, p. 67                |
| 6.6  | Lemma (Deleting a set does not change the numerical class order)     | Lemma 5.15, p. 67                |
| 6.7  | Theorem (Local class universality and cylinders)                     | Note 19.21, p. 108               |
| 7.1  | Lemma (A self-delimiting numerical code)                             | Lemma 19.33, p. 113              |
| 7.2  | Theorem (Binary-class equimorphism)                                  | Theorem 19.34, p. 113            |
| 8.1  | Definition (A prefix cylinder in the full order)                     | Definition 16.4, p. 98           |
| 8.2  | Proposition (Residual-order decomposition)                           | Proposition 16.5, p. 98          |
| 8.3  | Theorem (Exact adjacency criterion)                                  | Theorem 16.6, p. 98              |
| 8.4  | Corollary (Proper-class prefixes are responsible)                    | Corollary 16.7, p. 99            |
| 9.1  | Lemma (Prefix-freeness of exhaustive enumerations)                   | Lemma 6.3, p. 72                 |
| 9.2  | Lemma (Strict binary hierarchy)                                      | Lemma 6.4, p. 72                 |
| 9.3  | Lemma (Characteristic-cut coding)                                    | Lemma 6.5, p. 72                 |
| 9.4  | Lemma (Pair orientation coding)                                      | Lemma 6.6, p. 72                 |
| 9.5  | Lemma (No $\mu^+$ -long monotone chain)                              | Lemma 6.7, p. 72                 |
| 9.6  | Theorem (Cardinality and exact spectra)                              | Theorem 6.8, p. 73               |
| 9.7  | Remark (What the rank does not say)                                  | Remark 6.9, p. 73                |
| 10.1 | Lemma (Row scheduler)                                                | Lemma 8.1, p. 79                 |
| 10.2 | Lemma (Surreal cone decomposition)                                   | Lemma 8.2, p. 79                 |
| 10.3 | Theorem (Exact minimal-slice dichotomy)                              | Theorem 8.3, p. 80               |
| 10.4 | Corollary (A complete cube test for the minimal slice)               | Corollary 8.4, p. 80             |
| 10.5 | Example (An unconditional singular example)                          | Example 8.5, p. 80               |
| 10.6 | Example (Regular cutoffs)                                            | Example 8.6, p. 80               |
| 11.1 | Theorem (Exact external identification)                              | Theorem 21.1, p. 123             |
| 14.1 | Research question (All fixed enumeration types)                      | Research question 113.11, p. 895 |
| 14.2 | Research question (Noncardinal birthday bounds)                      | Research question 113.12, p. 895 |
| 14.3 | Research question (Isomorphism, not merely equimorphism)             | Research question 113.13, p. 895 |
| 14.4 | Research question (The hierarchy of proper-class blocks)             | Research question 113.14, p. 895 |
| 14.5 | Research question (The exact weak-choice strength)                   | Remark 113.15, p. 895            |
| 14.6 | Research question (Elementary common-prefix theory)                  | Research question 113.16, p. 896 |
| 14.7 | Research question (Automorphisms and finite condensation)            | Research question 113.17, p. 896 |
| 14.8 | Research question (Dependence on the class realization)              | Research question 113.18, p. 896 |
| 14.9 | Research question (Compatibility between birthday cutoffs)           | Research question 113.19, p. 896 |

### D.3 Source 09

| No.  | Statement in source 09                                                   | Printed here as           |
|------|--------------------------------------------------------------------------|---------------------------|
| 2.1  | Definition                                                               | Definition 5.9, p. 65     |
| 2.2  | Lemma (Avoiding a prescribed set)                                        | Lemma 5.10, p. 65         |
| 3.1  | Theorem (Class universality)                                             | Theorem 5.16, p. 68       |
| 3.2  | Theorem (Uniqueness and set-homogeneity)                                 | Theorem 5.17, p. 68       |
| 3.3  | Corollary                                                                | Corollary 5.18, p. 68     |
| 4.1  | Definition                                                               | Note 6.10, p. 73          |
| 4.2  | Lemma (Exhaustiveness removes the prefix ambiguity)                      | Note 6.11, p. 74          |
| 4.3  | Theorem (Cardinality and universality)                                   | Note 6.12, p. 74          |
| 4.4  | Lemma (Eventual stabilization)                                           | Lemma 6.13, p. 74         |
| 4.5  | Theorem (Exact ordinal spectrum)                                         | Theorem 6.14, p. 74       |
| 4.6  | Remark                                                                   | Remark 6.15, p. 74        |
| 4.7  | Proposition                                                              | Note 6.16, p. 74          |
| 4.8  | Proposition (Finite residuals are exactly the adjacency mechanism)       | Note 6.17, p. 75          |
| 5.1  | Proposition                                                              | Proposition 7.3, p. 77    |
| 6.1  | Proposition                                                              | Proposition 11.3, p. 85   |
| 6.2  | Proposition (Limit prefixes create small one-sided character)            | Note 11.4, p. 85          |
| 7.1  | Definition                                                               | Definition 13.5, p. 90    |
| 7.2  | Proposition                                                              | Proposition 13.6, p. 90   |
| 7.3  | Proposition (The baseline already entails global choice)                 | Proposition 13.7, p. 90   |
| 8.1  | Lemma (Bounded-support prefix completion)                                | Lemma 19.6, p. 104        |
| 8.2  | Remark (Why extra unused positions matter)                               | Remark 19.7, p. 104       |
| 9.1  | Theorem (Uniform set-indexed interpolation)                              | Note 19.12, p. 106        |
| 9.2  | Corollary (The surreal core theorem)                                     | Note 19.16, p. 107        |
| 9.3  | Corollary (Dense copies in every interval)                               | Note 19.19, p. 108        |
| 10.1 | Theorem (Bounded reflection)                                             | Theorem 19.30, p. 111     |
| 10.2 | Theorem                                                                  | Note 19.22, p. 108        |
| 10.3 | Proposition (Representable suborders do not exceed surreal universality) | Proposition 19.24, p. 109 |
| 10.4 | Proposition (Cylinder self-similarity)                                   | Note 19.26, p. 109        |
| 11.1 | Theorem (An explicit unfilled (Ord, Ord) configuration)                  | Theorem 19.43, p. 117     |
| 11.2 | Remark (A general missing-branch mechanism)                              | Remark 19.44, p. 117      |
| 12.1 | Theorem (Uniform no-coding theorem)                                      | Note 19.40, p. 115        |
| 13.1 | Theorem (Inaccessible set-sized model)                                   | Theorem 22.1, p. 124      |

## D.4 Source 08

| No. | Statement in source 08                    | Printed here as        |
|-----|-------------------------------------------|------------------------|
| 2.1 | Proposition                               | Proposition 5.4, p. 61 |
| 2.2 | Lemma                                     | Lemma 5.5, p. 61       |
| 2.3 | Theorem (Birthday-controlled cut filling) | Theorem 5.6, p. 61     |
| 2.4 | Corollary (Fresh separators)              | Corollary 5.7, p. 62   |
| 2.5 | Definition                                | Definition 5.8, p. 62  |
| 3.1 | Lemma (No prefix ambiguity)               | Note 6.18, p. 75       |
| 3.2 | Lemma (Pair coding)                       | Note 6.19, p. 75       |
| 3.3 | Lemma                                     | Note 6.20, p. 75       |

| No.  | Statement in source 08                         | Printed here as           |
|------|------------------------------------------------|---------------------------|
| 3.4  | Theorem                                        | Note 6.21, p. 75          |
| 4.1  | Theorem (Prefix stabilization)                 | Note 6.22, p. 75          |
| 4.2  | Corollary (Exact spectra)                      | Note 6.23, p. 75          |
| 4.3  | Proposition (A uniform binary upper coding)    | Note 6.24, p. 76          |
| 5.1  | Theorem (Adjacency classification)             | Note 6.25, p. 76          |
| 5.2  | Corollary (Finite condensation blocks)         | Note 6.26, p. 76          |
| 6.1  | Proposition                                    | Note 7.4, p. 78           |
| 6.2  | Lemma                                          | Note 7.5, p. 78           |
| 6.3  | Theorem (The initial-length cutoff order)      | Note 7.6, p. 78           |
| 6.4  | Remark                                         | Note 7.7, p. 78           |
| 7.1  | Proposition                                    | Note 11.5, p. 86          |
| 8.1  | Definition                                     | Definition 5.11, p. 66    |
| 8.2  | Theorem                                        | Note 5.12, p. 66          |
| 9.1  | Proposition                                    | Note 19.2, p. 102         |
| 10.1 | Lemma (Prefix extension inside one orbit)      | Note 19.8, p. 105         |
| 10.2 | Corollary                                      | Note 19.9, p. 105         |
| 11.1 | Theorem (Orbit interpolation)                  | Note 19.13, p. 106        |
| 11.2 | Corollary                                      | Note 19.14, p. 106        |
| 12.1 | Theorem (Uniqueness of a class saturated line) | Note 5.19, p. 68          |
| 12.2 | Theorem (Surreal-sized skeleton)               | Note 19.17, p. 107        |
| 12.3 | Theorem (Universality for class linear orders) | Note 5.20, p. 69          |
| 12.4 | Corollary                                      | Note 19.23, p. 108        |
| 12.5 | Proposition                                    | Proposition 19.31, p. 111 |
| 13.1 | Theorem (Virtual binary equimorphism)          | Note 19.35, p. 113        |
| 13.2 | Theorem (No class-uniform exhaustive list)     | Note 19.41, p. 115        |
| 14.1 | Proposition                                    | Proposition 17.1, p. 100  |

[merge] (3 October 2026.) Sources 13–18 follow, numbered as in their own PDFs. Each is printed in full where it is first proved, or recorded in a numbered note where it repeats a printed result (the batch-80 practice).

## D.5 Source 13

| No. | Statement in source 13                               | Printed here as          |
|-----|------------------------------------------------------|--------------------------|
| 2.1 | Lemma (Prefix-freeness of exhaustive enumerations)   | Lemma 34.1, p. 167       |
| 3.1 | Lemma (Fresh set-cut separator)                      | Lemma 42.1, p. 206       |
| 3.2 | Lemma (Class universality and uniqueness in GBC)     | Lemma 42.2, p. 207       |
| 4.1 | Theorem (Fixed-alphabet results from the repository) | Note 32.4, p. 163        |
| 5.1 | Proposition (Limit-prefix character)                 | Note 35.3, p. 178        |
| 6.1 | Proposition (Elementary labeled comparison)          | Proposition 51.7, p. 256 |
| 6.2 | Proposition (Canonical head; prior structural lemma) | Proposition 51.8, p. 257 |
| 7.1 | Lemma (Set-prefix completion)                        | Lemma 42.3, p. 208       |
| 7.2 | Theorem (Uniform set-cut interpolation)              | Theorem 42.4, p. 209     |
| 7.3 | Corollary (The core and its cylinders)               | Corollary 42.5, p. 209   |
| 7.4 | Lemma (Two-sided ordinal approximants)               | Lemma 42.6, p. 209       |
| 7.5 | Theorem (No exhaustive uniform evaluator)            | Theorem 42.7, p. 210     |
| 8.1 | Definition                                           | Definition 46.13, p. 227 |

| No.   | Statement in source 13                                               | Printed here as                  |
|-------|----------------------------------------------------------------------|----------------------------------|
| 8.2   | Lemma (Uniqueness and coherence of crossing prefixes)                | Lemma 46.14, p. 228              |
| 8.3   | Theorem (Exact branch representation)                                | Theorem 46.15, p. 228            |
| 8.4   | Corollary (Trace image and fibers of raw orders)                     | Corollary 46.16, p. 229          |
| 8.5   | Example (A branch that fails only exhaustion)                        | Example 46.17, p. 229            |
| 9.1   | Theorem (Branch, horizontal cut, or limit boundary)                  | Theorem 46.18, p. 229            |
| 9.2   | Proposition (Characters at first failure)                            | Proposition 46.19, p. 230        |
| 9.3   | Example (A horizontal (Ord, Ord) cut)                                | Example 46.20, p. 231            |
| 10.1  | Theorem (Left-factor absorption)                                     | Theorem 55.1, p. 278             |
| 10.2  | Lemma (Split-order density bound)                                    | Lemma 55.2, p. 279               |
| 10.3  | Theorem (No full split in an inaccessible universe)                  | Theorem 55.3, p. 279             |
| 10.4  | Theorem (No definable uniform split embedding)                       | Theorem 55.4, p. 280             |
| 10.5  | Proposition (External countable collapse of the order)               | Proposition 55.5, p. 280         |
| 11.1  | Lemma (Finite tuples as a surreal class order)                       | Lemma 47.11, p. 242              |
| 11.2  | Theorem (Cylinder non-rigidity)                                      | Theorem 47.12, p. 242            |
| 12.1  | Theorem (Central-word isomorphism)                                   | Theorem 36.4, p. 186             |
| 12.2  | Proposition (Direct central-word coding)                             | Proposition 36.5, p. 187         |
| 15.1  | Research question (Minimal extra structure for recognizing branches) | Research question 113.20, p. 903 |
| 15.2  | Research question (Automorphisms preserving a chosen core)           | Research question 113.21, p. 903 |
| 15.3  | Research question (A normal form for class cuts)                     | Research question 113.22, p. 903 |
| 15.4  | Research question (Sharp cutoff analogues)                           | Research question 113.23, p. 904 |
| 15.5  | Research question (The weakest uniform split obstruction)            | Research question 113.24, p. 904 |
| 15.6  | Research question (A hierarchy beyond one split)                     | Research question 113.25, p. 904 |
| 15.7  | Research question (Class realizations with intermediate behavior)    | Research question 113.26, p. 904 |
| 15.8  | Research question (Uniformity versus external isomorphism)           | Research question 113.27, p. 904 |
| 15.9  | Research question (A classification of termination cuts)             | Research question 113.28, p. 904 |
| 15.10 | Research question (Optimal birthday and support bounds)              | Research question 113.29, p. 905 |
| 15.11 | Research question (Iterating raw residual decomposition)             | Research question 113.30, p. 905 |
| 15.12 | Research question (Relation-code dependence)                         | Research question 113.31, p. 905 |
| 15.13 | Research question (First checked extension)                          | Research question 113.32, p. 905 |

## D.6 Source 14

| No. | Statement in source 14                      | Printed here as          |
|-----|---------------------------------------------|--------------------------|
| 2.1 | Theorem (Set-carrier baseline, inherited)   | Note 32.5, p. 164        |
| 4.1 | Lemma (Fresh separators)                    | Lemma 41.1, p. 204       |
| 4.2 | Lemma (Completion of a set prefix)          | Lemma 41.2, p. 205       |
| 4.3 | Definition (Cylinder and residual alphabet) | Definition 41.3, p. 205  |
| 4.4 | Proposition (Cylinder geometry)             | Proposition 41.4, p. 205 |
| 4.5 | Lemma (Density around injective branches)   | Lemma 41.5, p. 205       |



| No.   | Statement in source 14                                         | Printed here as                  |
|-------|----------------------------------------------------------------|----------------------------------|
| 5.1   | Definition                                                     | Definition 43.1, p. 217          |
| 5.2   | Lemma (The crossing-chain lemma)                               | Lemma 43.2, p. 217               |
| 5.3   | Theorem (Boundary trichotomy)                                  | Theorem 43.3, p. 217             |
| 5.4   | Remark (What the theorem does not construct)                   | Remark 43.4, p. 218              |
| 6.1   | Theorem (Cut-character calculation)                            | Theorem 44.1, p. 219             |
| 6.2   | Corollary (Recovery of the core saturation theorem)            | Corollary 44.2, p. 219           |
| 6.3   | Corollary (The known surreal order type)                       | Corollary 44.3, p. 220           |
| 7.1   | Theorem (Exhaustive-cut recognition)                           | Theorem 45.1, p. 220             |
| 7.2   | Theorem (No new realization of an old trace cut)               | Theorem 48.1, p. 243             |
| 7.3   | Corollary                                                      | Corollary 48.2, p. 243           |
| 7.4   | Remark (Scope of absoluteness)                                 | Remark 48.3, p. 244              |
| 8.1   | Lemma (Set homogeneity of the core)                            | Lemma 47.1, p. 238               |
| 8.2   | Theorem (Complete character invariant for cut orbits)          | Theorem 47.2, p. 238             |
| 8.3   | Corollary (Exhaustiveness is not a bare-order invariant)       | Corollary 47.3, p. 238           |
| 9.1   | Lemma (Small-support extension of a partial injection)         | Lemma 49.1, p. 245               |
| 9.2   | Lemma (Support compression)                                    | Lemma 49.2, p. 245               |
| 9.3   | Theorem (Exact support saturation and an attained obstruction) | Theorem 49.3, p. 246             |
| 9.4   | Proposition (Universality without high saturation)             | Proposition 49.4, p. 246         |
| 9.5   | Remark (Cardinality versus location of support)                | Remark 49.5, p. 247              |
| 10.1  | Lemma (Initial set-like branch)                                | Lemma 54.5, p. 277               |
| 10.2  | Theorem (Trace fibers and their exact loss of information)     | Theorem 54.6, p. 277             |
| 10.3  | Example (Singleton fibers need not be set-like)                | Example 54.7, p. 278             |
| 10.4  | Corollary (Raw orders do not fill every core boundary)         | Corollary 54.8, p. 278           |
| 11.1  | Proposition (Prefix-first word coding)                         | Proposition 35.6, p. 181         |
| 11.2  | Theorem (No exhaustive uniform class of codes, inherited)      | Theorem 56.3, p. 283             |
| 14.1  | Research question (The exhaustive-trace stabilizer)            | Research question 113.33, p. 906 |
| 14.2  | Research question (Minimal enrichment for exhaustiveness)      | Research question 113.34, p. 906 |
| 14.3  | Research question (Isomorphism of support layers)              | Research question 113.35, p. 906 |
| 14.4  | Research question (Support ideals)                             | Research question 113.36, p. 906 |
| 14.5  | Research question (Optimal simultaneous support cost)          | Research question 113.37, p. 906 |
| 14.6  | Research question (Iterated initial-branch quotients)          | Research question 113.38, p. 907 |
| 14.7  | Research question (Automorphisms of raw trace fibers)          | Research question 113.39, p. 907 |
| 14.8  | Research question (Relation-code comparison)                   | Research question 113.40, p. 907 |
| 14.9  | Research question (Exact local geometry of set slices)         | Research question 113.41, p. 907 |
| 14.10 | Research question (Coherent embeddings across cut-offs)        | Research question 113.42, p. 907 |
| 14.11 | Research question (Which new cuts can a class extension add?)  | Research question 113.43, p. 907 |
| 14.12 | Research question (Axiomatic economy)                          | Research question 113.44, p. 908 |
| 14.13 | Research question (First kernel-verified milestone)            | Research question 113.45, p. 908 |

## D.7 Source 15

| No. | Statement in source 15                                              | Printed here as           |
|-----|---------------------------------------------------------------------|---------------------------|
| 3.1 | Lemma (Fresh separators)                                            | Lemma 51.2, p. 252        |
| 3.2 | Lemma (Exhaustiveness removes prefix ambiguity)                     | Note 32.3, p. 162         |
| 3.3 | Theorem (Set-sized baseline, from the prior reports)                | Note 32.3, p. 162         |
| 4.1 | Theorem (First unequal labels without class ordinals)               | Theorem 51.9, p. 258      |
| 4.2 | Proposition (Prefix convexity and initial-word fibres)              | Proposition 51.10, p. 259 |
| 5.1 | Definition (Condensation by set intervals)                          | Definition 52.2, p. 264   |
| 5.2 | Theorem (Elementary canonical block decomposition)                  | Theorem 52.3, p. 264      |
| 5.3 | Corollary (Allowed block profiles)                                  | Corollary 52.4, p. 265    |
| 5.4 | Theorem (Every class well-order occurs as a block index)            | Theorem 52.5, p. 265      |
| 6.1 | Definition (Finite ordinal polynomials over a class order)          | Definition 53.1, p. 271   |
| 6.2 | Lemma (The polynomial order is a class well-order)                  | Lemma 53.2, p. 272        |
| 6.3 | Lemma (One condensation removes the least exponent)                 | Lemma 53.3, p. 272        |
| 6.4 | Theorem (Every class well-order is an explicit condensation length) | Theorem 53.4, p. 272      |
| 6.5 | Remark (A proper quotient can have the same abstract order type)    | Remark 53.5, p. 273       |
| 7.1 | Lemma (Every injective set prefix has a core completion)            | Lemma 42.8, p. 210        |
| 7.2 | Theorem (Exact criterion for representation)                        | Theorem 46.6, p. 224      |
| 7.3 | Remark (The normalization is necessary)                             | Remark 46.7, p. 225       |
| 7.4 | Corollary (Exhaustiveness and the kernel of the raw cut map)        | Corollary 46.8, p. 225    |
| 7.5 | Theorem (Terminal-cut classification)                               | Theorem 46.9, p. 225      |
| 7.6 | Example (A cut not filled even by arbitrary class well-orders)      | Example 46.10, p. 226     |
| 7.7 | Example (Repeated label, numerical gap, and limit boundary)         | Example 46.11, p. 226     |
| 7.8 | Proposition (Cofinality read from first failure)                    | Proposition 46.12, p. 227 |
| 7.9 | Theorem (An automorphism destroys prefix representability)          | Theorem 47.10, p. 241     |
| 8.1 | Lemma (An explicit entry code and the strict cube hierarchy)        | Lemma 31.1, p. 156        |
| 8.2 | Theorem (Cardinal blocks and a finite terminal permutation)         | Theorem 31.2, p. 157      |
| 8.3 | Lemma (Blockwise row scheduling)                                    | Lemma 31.3, p. 157        |
| 8.4 | Lemma (Finite deletion does not obstruct the singular scheduler)    | Lemma 31.4, p. 157        |
| 8.5 | Lemma (Finite fibers over a binary cube)                            | Lemma 31.5, p. 158        |
| 8.6 | Corollary (The complete invariant on these block types)             | Corollary 31.6, p. 159    |
| 8.7 | Corollary (Density and cellularity of the block types)              | Corollary 31.7, p. 160    |
| 9.1 | Lemma (Full-size numerical intervals)                               | Note 29.8, p. 141         |
| 9.2 | Theorem (Local and global cofinalities)                             | Note 30.7, p. 144         |

| No.   | Statement in source 15                                                 | Printed here as                  |
|-------|------------------------------------------------------------------------|----------------------------------|
| 9.3   | Lemma (Cofinality of the alphabet cardinal)                            | Note 29.9, p. 142                |
| 9.4   | Theorem (Exact saturation threshold for the minimal slice)             | Theorem 30.23, p. 152            |
| 10.1  | Theorem (Endpoint and character of inverse ranks)                      | Theorem 38.1, p. 196             |
| 10.2  | Theorem (Endpoint transfer for relation tables)                        | Theorem 38.2, p. 197             |
| 10.3  | Corollary (Exactly three possible endpoint patterns)                   | Corollary 38.3, p. 197           |
| 10.4  | Example (Failure of a numerical pair scan)                             | Example 38.4, p. 197             |
| 10.5  | Theorem (No total linearization respects every restriction)            | Theorem 38.5, p. 198             |
| 10.6  | Proposition (Coefficient-one normal forms realize set supports)        | Proposition 38.6, p. 199         |
| 11.1  | Theorem (No exhaustive uniform class of set codes)                     | Theorem 57.4, p. 289             |
| 13.1  | Research question (All fixed enumeration types)                        | Research question 113.46, p. 908 |
| 13.2  | Research question (Noncardinal birthday bounds)                        | Research question 113.47, p. 908 |
| 13.3  | Research question (The exact strength of arbitrary-input condensation) | Research question 113.48, p. 908 |
| 13.4  | Research question (Intrinsic information in the full raw order)        | Research question 113.49, p. 908 |
| 13.5  | Research question (Completion with residual data)                      | Research question 113.50, p. 909 |
| 13.6  | Research question (Automorphism extension)                             | Research question 113.51, p. 909 |
| 13.7  | Research question (Finer topology of bounded slices)                   | Research question 113.52, p. 909 |
| 13.8  | Research question (Inverse ranks and prescribed relation scans)        | Research question 113.53, p. 909 |
| 13.9  | Research question (Coherent restricted filtrations)                    | Research question 113.54, p. 909 |
| 13.10 | Research question (Naturally well-ordered supports)                    | Research question 113.55, p. 909 |
| 13.11 | Research question (Class-model dependence)                             | Research question 113.56, p. 909 |
| 13.12 | Research question (First formal milestones)                            | Research question 113.57, p. 909 |

## D.8 Source 16

| No. | Statement in source 16                                     | Printed here as           |
|-----|------------------------------------------------------------|---------------------------|
| 3.1 | Lemma (Fresh separators)                                   | Lemma 51.3, p. 253        |
| 3.2 | Proposition (Universality and uniqueness)                  | Proposition 51.4, p. 253  |
| 4.1 | Theorem (Basic set-alphabet results)                       | Note 32.1, p. 161         |
| 5.1 | Lemma (No jump between distinct short-prefix branches)     | Lemma 27.1, p. 134        |
| 5.2 | Theorem (Exact density and cellularity)                    | Theorem 27.2, p. 134      |
| 5.3 | Remark (Why two terminal labels differ from three)         | Remark 27.3, p. 135       |
| 5.4 | Corollary (Birthday-stage topology)                        | Corollary 27.4, p. 135    |
| 6.1 | Theorem (Every infinite stratum is incomplete)             | Theorem 28.1, p. 138      |
| 6.2 | Proposition (Global characters from the first label)       | Proposition 28.2, p. 139  |
| 6.3 | Theorem (Complete gap-character spectrum)                  | Theorem 30.14, p. 147     |
| 7.1 | Theorem (Three interpretations with the same prefix order) | Theorem 35.1, p. 177      |
| 8.1 | Theorem (Termination criterion)                            | Theorem 36.1, p. 182      |
| 8.2 | Theorem (All termination types)                            | Theorem 36.2, p. 184      |
| 9.1 | Proposition (Existence has a precise choice strength)      | Proposition 51.11, p. 259 |
| 9.2 | Lemma (A common labelled prefix without class ordinals)    | Lemma 51.12, p. 260       |

| No.   | Statement in source 16                                              | Printed here as                  |
|-------|---------------------------------------------------------------------|----------------------------------|
| 9.3   | Proposition (Convexity and finite residual jumps)                   | Proposition 51.13, p. 260        |
| 10.1  | Theorem (Canonical block decomposition)                             | Theorem 52.6, p. 266             |
| 10.2  | Corollary (Largest set final segment)                               | Corollary 52.7, p. 267           |
| 10.3  | Theorem (All block-index orders occur)                              | Theorem 52.8, p. 267             |
| 11.1  | Lemma (Uniform completion of every set prefix)                      | Lemma 42.9, p. 211               |
| 11.2  | Theorem (Uniform set-cut interpolation)                             | Theorem 42.10, p. 211            |
| 11.3  | Corollary (A surreal-sized dense core)                              | Corollary 42.11, p. 212          |
| 12.1  | Theorem (Prefix-cylinder recognition)                               | Theorem 46.2, p. 222             |
| 12.2  | Theorem (Complete classification of the first failure)              | Theorem 46.3, p. 223             |
| 12.3  | Corollary (Exactly what raw tails add)                              | Corollary 46.4, p. 224           |
| 13.1  | Theorem (No exhaustive uniform class of set codes)                  | Theorem 56.1, p. 281             |
| 13.2  | Theorem (Elementary transformations and $\Sigma_1^1$ comprehension) | Theorem 56.2, p. 282             |
| 14.1  | Theorem (Exact external identification)                             | Theorem 57.2, p. 284             |
| 14.2  | Theorem (Strict embedding separation in the full model)             | Theorem 57.3, p. 284             |
| 16.1  | Research question (Longer-stratum binary coding ranks)              | Research question 113.58, p. 910 |
| 16.2  | Research question (Singular birthday gaps)                          | Research question 113.59, p. 910 |
| 16.3  | Research question (Centered increasing words)                       | Research question 113.60, p. 910 |
| 16.4  | Research question (Recovering prefix structure from order alone)    | Research question 113.61, p. 910 |
| 16.5  | Research question (Changing baselines and extending automorphisms)  | Research question 113.62, p. 911 |
| 16.6  | Research question (Exact strength of uniform nonembedding)          | Research question 113.63, p. 911 |
| 16.7  | Research question (Repeated condensation of the quotient)           | Research question 113.64, p. 911 |
| 16.8  | Research question (Choice needed for the word isomorphisms)         | Research question 113.65, p. 911 |
| 16.9  | Research question (Class extensions and canonical data)             | Research question 113.66, p. 911 |
| 16.10 | Research question (A first verified theorem package)                | Research question 113.67, p. 911 |
| 16.11 | Research question (Quantitative compatibility of the constructions) | Research question 113.68, p. 912 |

## D.9 Source 17

| No. | Statement in source 17                                          | Printed here as          |
|-----|-----------------------------------------------------------------|--------------------------|
| 3.1 | Proposition (Direct labelled comparison; background with proof) | Proposition 51.6, p. 255 |
| 4.1 | Definition (Characters and genuine gaps)                        | Definition 29.1, p. 140  |
| 4.2 | Lemma                                                           | Lemma 29.2, p. 140       |
| 4.3 | Lemma                                                           | Lemma 29.3, p. 140       |
| 4.4 | Theorem (Cut spectrum of the birthday alphabet)                 | Theorem 29.4, p. 140     |
| 4.5 | Example (A warning about equal-character guesses)               | Example 29.5, p. 141     |
| 4.6 | Lemma                                                           | Lemma 29.6, p. 141       |
| 5.1 | Lemma (The cylinder base)                                       | Lemma 30.1, p. 143       |
| 5.2 | Theorem (Characters and size invariants)                        | Theorem 30.2, p. 143     |
| 5.3 | Remark                                                          | Remark 30.3, p. 143      |

| No.  | Statement in source 17                                               | Printed here as                  |
|------|----------------------------------------------------------------------|----------------------------------|
| 5.4  | Proposition (Set-alphabet background, with a direct proof)           | Note 30.4, p. 143                |
| 5.5  | Theorem (A dense-open fusion theorem)                                | Theorem 30.5, p. 144             |
| 6.1  | Lemma (Cut trichotomy)                                               | Lemma 30.8, p. 145               |
| 6.2  | Lemma (Characters in the three cases)                                | Lemma 30.9, p. 145               |
| 6.3  | Theorem (Exact gap spectrum)                                         | Theorem 30.11, p. 146            |
| 6.4  | Definition                                                           | Definition 30.17, p. 150         |
| 6.5  | Corollary (Exact saturation threshold)                               | Corollary 30.18, p. 150          |
| 6.6  | Example (The countable birthday cutoff)                              | Example 30.19, p. 151            |
| 6.7  | Example (Regular size-matched cutoffs)                               | Example 30.20, p. 151            |
| 6.8  | Example (A singular strong-limit cutoff)                             | Example 30.21, p. 151            |
| 6.9  | Remark (What has not been classified)                                | Remark 30.22, p. 151             |
| 7.1  | Theorem (Exact reserve cardinal)                                     | Theorem 49.7, p. 247             |
| 7.2  | Example (Why a finite reserve can never repair a one-sided shift)    | Example 49.8, p. 248             |
| 7.3  | Example (Finite partial injections require no reserve)               | Example 49.9, p. 248             |
| 7.4  | Corollary (Application to prescribed surreal prefixes)               | Corollary 49.10, p. 248          |
| 8.1  | Lemma (Avoiding a set of labels)                                     | Lemma 42.12, p. 213              |
| 8.2  | Theorem (Core interpolation; source result)                          | Theorem 42.13, p. 213            |
| 8.3  | Corollary (A dense surreal order core; source result)                | Corollary 42.14, p. 213          |
| 8.4  | Corollary (Universality for ordinary class linear orders)            | Corollary 42.15, p. 214          |
| 8.5  | Theorem (Uniform diagonal obstruction; source result)                | Theorem 42.16, p. 214            |
| 9.1  | Theorem (Recognition of injective branches)                          | Theorem 46.21, p. 232            |
| 9.2  | Remark (Why every ordinal length and the endpoint convention matter) | Remark 46.22, p. 232             |
| 9.3  | Example (A coordinate cut that does not follow a branch)             | Example 46.23, p. 233            |
| 9.4  | Example (A limit boundary that defeats finite-depth testing)         | Example 46.24, p. 233            |
| 9.5  | Example (A coherent branch which is not exhaustive)                  | Example 46.25, p. 233            |
| 9.6  | Proposition (Characters of branch cuts outside the core)             | Proposition 46.26, p. 233        |
| 9.7  | Proposition (Raw lifts and loss of residual information)             | Proposition 46.27, p. 233        |
| 10.1 | Theorem (Extension criterion)                                        | Theorem 47.5, p. 239             |
| 10.2 | Lemma (Homogeneity of large two-sided core gaps)                     | Lemma 47.6, p. 240               |
| 10.3 | Theorem (Some core automorphisms do not extend)                      | Theorem 47.7, p. 240             |
| 10.4 | Corollary (A limit to pure-order recognition)                        | Corollary 47.8, p. 240           |
| 10.5 | Proposition (Relabelling by numerical automorphisms)                 | Proposition 47.9, p. 241         |
| 11.1 | Theorem (New enumerations are exactly new represented core cuts)     | Theorem 48.4, p. 244             |
| 12.1 | Theorem (Adjacency after a common prefix; source result)             | Theorem 54.2, p. 275             |
| 12.2 | Definition (Finite-adjacency components)                             | Definition 54.3, p. 275          |
| 12.3 | Theorem (Exact maximal finite components)                            | Theorem 54.4, p. 275             |
| 15.1 | Research question (Nonminimal enumeration types)                     | Research question 113.69, p. 912 |

| No.   | Statement in source 17                                                   | Printed here as                                   |
|-------|--------------------------------------------------------------------------|---------------------------------------------------|
| 15.2  | Research question (Noncardinal birthday cutoffs)                         | <a href="#">Research question 113.70</a> , p. 912 |
| 15.3  | Research question (Isomorphism after the invariant calculation)          | <a href="#">Research question 113.71</a> , p. 912 |
| 15.4  | Research question (Beyond the proved fusion length)                      | <a href="#">Research question 113.72</a> , p. 913 |
| 15.5  | Research question (The extendible automorphism group)                    | <a href="#">Research question 113.73</a> , p. 913 |
| 15.6  | Research question (Changing baselines)                                   | <a href="#">Research question 113.74</a> , p. 913 |
| 15.7  | Research question (Recovering a class realization)                       | <a href="#">Research question 113.75</a> , p. 913 |
| 15.8  | Research question (Iterated proper-class blocks)                         | <a href="#">Research question 113.76</a> , p. 913 |
| 15.9  | Research question (The condensed unrestricted order)                     | <a href="#">Research question 113.77</a> , p. 913 |
| 15.10 | Research question (Ordinal and birthday optima for supported completion) | <a href="#">Research question 113.78</a> , p. 914 |
| 15.11 | Research question (Coherent cutoff embeddings)                           | <a href="#">Research question 113.79</a> , p. 914 |
| 15.12 | Research question (First checked milestones)                             | <a href="#">Research question 113.80</a> , p. 914 |
| 15.13 | Research question (Typed constructions indexed by presentations)         | <a href="#">Research question 113.81</a> , p. 914 |

## D.10 Source 18

| No. | Statement in source 18                                             | Printed here as                          |
|-----|--------------------------------------------------------------------|------------------------------------------|
| 2.1 | Lemma (Surreal order tools)                                        | <a href="#">Lemma 51.5</a> , p. 254      |
| 3.1 | Theorem (Set-sized universality and the exact ordinal spectrum)    | <a href="#">Note 32.2</a> , p. 161       |
| 4.1 | Lemma (The dense order of cofinite prefixes)                       | <a href="#">Lemma 27.7</a> , p. 137      |
| 4.2 | Theorem (Exact density and cellularity at the minimal type)        | <a href="#">Theorem 27.8</a> , p. 137    |
| 4.3 | Theorem (Density and cellularity for every fixed type)             | <a href="#">Theorem 27.9</a> , p. 137    |
| 4.4 | Remark                                                             | <a href="#">Remark 27.10</a> , p. 138    |
| 4.5 | Lemma (Large intervals and unchanged cut sides after deletion)     | <a href="#">Note 29.8</a> , p. 141       |
| 4.6 | Lemma (The two cofinalities)                                       | <a href="#">Note 29.9</a> , p. 142       |
| 4.7 | Theorem (Complete local and global characters at the minimal type) | <a href="#">Theorem 30.24</a> , p. 153   |
| 4.8 | Theorem (Exact guaranteed interpolation)                           | <a href="#">Theorem 30.25</a> , p. 154   |
| 4.9 | Corollary (Two especially transparent regimes)                     | <a href="#">Corollary 30.26</a> , p. 155 |
| 5.1 | Theorem (Exact gap spectrum)                                       | <a href="#">Note 30.13</a> , p. 147      |
| 5.2 | Lemma (Alphabet gap spectrum)                                      | <a href="#">Note 29.10</a> , p. 142      |
| 5.3 | Lemma (Small deletions preserve order density)                     | <a href="#">Note 29.8</a> , p. 141       |
| 5.4 | Lemma (Characters of cylinders and their limit boundaries)         | <a href="#">Note 30.13</a> , p. 147      |
| 5.5 | Theorem (Gap spectrum under permutation expansion)                 | <a href="#">Theorem 30.10</a> , p. 145   |
| 5.6 | Theorem (Exact multiplicities)                                     | <a href="#">Theorem 30.16</a> , p. 149   |
| 6.1 | Proposition (Explicit order coding; repository result)             | <a href="#">Note 35.4</a> , p. 179       |
| 6.2 | Proposition (The prefix-supremum obstruction)                      | <a href="#">Note 35.5</a> , p. 180       |
| 7.1 | Theorem (Centered arbitrary and injective words)                   | <a href="#">Theorem 36.6</a> , p. 188    |



| No.   | Statement in source 18                                            | Printed here as                  |
|-------|-------------------------------------------------------------------|----------------------------------|
| 7.2   | Proposition (An explicit centered sign embedding)                 | Proposition 36.7, p. 189         |
| 8.1   | Theorem (Local characters of centered increasing words)           | Theorem 37.1, p. 190             |
| 8.2   | Theorem (All set-cut obstructions)                                | Theorem 37.2, p. 191             |
| 8.3   | Corollary                                                         | Corollary 37.3, p. 191           |
| 9.1   | Proposition (The zero-padding quotient)                           | Proposition 37.4, p. 192         |
| 9.2   | Proposition                                                       | Proposition 37.5, p. 193         |
| 9.3   | Lemma (Adjacency test for increasing words)                       | Lemma 37.6, p. 193               |
| 9.4   | Corollary (Exactly where the zero jump appears)                   | Corollary 37.7, p. 193           |
| 10.1  | Theorem (A prefix-preserving tree isomorphism)                    | Theorem 35.7, p. 181             |
| 10.2  | Corollary (Two kinds of centered increasing cones)                | Corollary 37.9, p. 194           |
| 12.1  | Theorem (Existence of a global well-order; repository result)     | Theorem 51.14, p. 261            |
| 12.2  | Lemma (Largest common labelled initial segment)                   | Lemma 51.15, p. 261              |
| 12.3  | Theorem (Elementary full lexicographic order)                     | Theorem 51.16, p. 262            |
| 12.4  | Theorem (Adjacency is finite residual adjacency)                  | Theorem 51.17, p. 262            |
| 13.1  | Definition (Set-interval equivalence)                             | Definition 52.9, p. 268          |
| 13.2  | Lemma (Block structure)                                           | Lemma 52.10, p. 268              |
| 13.3  | Theorem (Canonical division by the ordinal class)                 | Theorem 52.11, p. 269            |
| 13.4  | Proposition (Uniqueness and functoriality)                        | Proposition 52.12, p. 269        |
| 13.5  | Corollary (Set-sized cofinality from the blocks)                  | Corollary 52.13, p. 269          |
| 13.6  | Theorem (Exact spectrum of block index orders)                    | Theorem 52.14, p. 270            |
| 13.7  | Proposition (Self-similar block quotient)                         | Proposition 52.15, p. 270        |
| 14.1  | Lemma (Completion of every set prefix)                            | Lemma 42.17, p. 215              |
| 14.2  | Theorem (Uniform set-cut interpolation)                           | Theorem 42.18, p. 215            |
| 14.3  | Theorem (No ordinary class of codes for all global well-orders)   | Theorem 42.19, p. 215            |
| 15.1  | Theorem (The prefix criterion for represented class cuts)         | Theorem 46.28, p. 234            |
| 15.2  | Remark (Why the strict-cut convention matters)                    | Remark 46.29, p. 235             |
| 15.3  | Corollary (Surjectivity is the exact remaining obstruction)       | Corollary 46.30, p. 235          |
| 15.4  | Corollary (Elementary recognition in uniform families)            | Corollary 46.31, p. 236          |
| 15.5  | Theorem (First-failure classification)                            | Theorem 46.32, p. 236            |
| 15.6  | Example (A deleted-label cut caused by repetition)                | Example 46.33, p. 237            |
| 15.7  | Remark (Class character does not decide representability)         | Remark 46.34, p. 237             |
| 16.1  | Lemma (The maximal finite terminal tail)                          | Lemma 58.1, p. 293               |
| 16.2  | Lemma (Adjacency forces a finite residual carrier)                | Lemma 58.2, p. 294               |
| 16.3  | Proposition (Factorial blocks)                                    | Proposition 58.3, p. 294         |
| 16.4  | Lemma                                                             | Lemma 58.4, p. 294               |
| 16.5  | Lemma (Prescribing the finite tail)                               | Lemma 58.5, p. 295               |
| 16.6  | Theorem (Countable-model factorial-shuffle theorem)               | Theorem 58.6, p. 295             |
| 16.7  | Remark (An explicit computable presentation)                      | Remark 58.7, p. 296              |
| 16.8  | Remark (The hypothesis cannot simply be dropped)                  | Remark 58.8, p. 296              |
| 16.9  | Corollary (External embeddings and intrinsic automorphism orbits) | Corollary 58.9, p. 296           |
| 16.10 | Corollary (Set-likeness is not definable from the external order) | Corollary 58.10, p. 296          |
| 19.1  | Research question (The arrangement of gaps)                       | Research question 113.82, p. 915 |

| No.   | Statement in source 18                                                 | Printed here as                                   |
|-------|------------------------------------------------------------------------|---------------------------------------------------|
| 19.2  | Research question (Longer fixed enumeration types)                     | <a href="#">Research question 113.83</a> , p. 915 |
| 19.3  | Research question (Noncardinal birthday bounds)                        | <a href="#">Research question 113.84</a> , p. 915 |
| 19.4  | Research question (Coherent enlargement of cutoffs)                    | <a href="#">Research question 113.85</a> , p. 915 |
| 19.5  | Research question (Iterating the quotient rather than removing blocks) | <a href="#">Research question 113.86</a> , p. 915 |
| 19.6  | Research question (Order structure on represented class cuts)          | <a href="#">Research question 113.87</a> , p. 915 |
| 19.7  | Research question (The expanded countable model)                       | <a href="#">Research question 113.88</a> , p. 916 |
| 19.8  | Research question ( $\omega$ -nonstandard class models)                | <a href="#">Research question 113.89</a> , p. 916 |
| 19.9  | Research question (Sharp word-code bounds)                             | <a href="#">Research question 113.90</a> , p. 916 |
| 19.10 | Research question (The centered increasing order itself)               | <a href="#">Research question 113.91</a> , p. 916 |
| 19.11 | Research question (Weak choice and uniform constructions)              | <a href="#">Research question 113.92</a> , p. 916 |
| 19.12 | Research question (A first machine-checked milestone)                  | <a href="#">Research question 113.93</a> , p. 916 |

[merge] (3 October 2026.) Sources 19–30 follow, with the same conventions; their statements are numbered as in their own PDFs (source 26 numbers its research questions within sections, source 28 its questions consecutively). A row whose target is a note records a statement printed there as a duplicate of an earlier result; the note names the result and the proof that is printed.

## D.11 Source 19

| No.  | Statement in source 19                                                | Printed here as                           |
|------|-----------------------------------------------------------------------|-------------------------------------------|
| 3.1  | Lemma (Surreal set cuts and avoidance)                                | <a href="#">Note 62.16</a> , p. 354       |
| 3.2  | Lemma (A birthday-controlled insertion)                               | <a href="#">Note 62.16</a> , p. 354       |
| 3.3  | Lemma (Deletion and sign cones)                                       | <a href="#">Note 62.16</a> , p. 354       |
| 3.4  | Proposition (Fixed-carrier baseline)                                  | <a href="#">Note 62.16</a> , p. 354       |
| 4.1  | Definition (Common labelled initial segment)                          | <a href="#">Definition 111.2</a> , p. 650 |
| 4.2  | Proposition (Elementary comparison; prior result)                     | <a href="#">Note 111.3</a> , p. 650       |
| 4.3  | Proposition (First block and full block decomposition; prior results) | <a href="#">Note 111.4</a> , p. 650       |
| 4.4  | Remark (Existence and choice)                                         | <a href="#">Remark 111.5</a> , p. 651     |
| 5.1  | Lemma (A prefix-free code for any small alphabet)                     | <a href="#">Lemma 64.1</a> , p. 362       |
| 5.2  | Lemma (The main prefix and its small-deletion counterpart)            | <a href="#">Lemma 64.2</a> , p. 362       |
| 5.3  | Lemma (Fusion into one fiber)                                         | <a href="#">Lemma 64.3</a> , p. 363       |
| 5.4  | Theorem (Exact residual embedding reduction)                          | <a href="#">Theorem 64.4</a> , p. 364     |
| 5.5  | Proposition (Which small orders can occur as residual alphabets)      | <a href="#">Proposition 64.5</a> , p. 364 |
| 5.6  | Lemma (Regular-cardinal blocks over any alphabet)                     | <a href="#">Lemma 64.6</a> , p. 364       |
| 5.7  | Theorem (Uniform coding bounds for every fixed type)                  | <a href="#">Theorem 64.7</a> , p. 365     |
| 5.8  | Corollary (All regular residual layers)                               | <a href="#">Corollary 64.8</a> , p. 365   |
| 5.9  | Lemma (A constant subcylinder from a small range)                     | <a href="#">Lemma 64.9</a> , p. 366       |
| 5.10 | Lemma (The singular non-strong-limit block)                           | <a href="#">Lemma 64.10</a> , p. 366      |
| 5.11 | Theorem (Exact binary-cube spectrum for every fixed type)             | <a href="#">Theorem 64.11</a> , p. 366    |
| 5.12 | Lemma (Finite factors over a limit-length cube)                       | <a href="#">Note 64.12</a> , p. 367       |

| No.   | Statement in source 19                                                 | Printed here as                  |
|-------|------------------------------------------------------------------------|----------------------------------|
| 5.13  | Corollary (Coding bounds and exact classifications)                    | Corollary 64.13, p. 367          |
| 6.1   | Theorem (Classification of monotone restrictions)                      | Theorem 73.1, p. 448             |
| 6.2   | Remark (Finite verification)                                           | Remark 73.2, p. 449              |
| 7.1   | Theorem (Unique normal forms for the completion)                       | Theorem 65.11, p. 376            |
| 7.2   | Proposition (The exact identifications after deleting labels)          | Proposition 65.12, p. 377        |
| 8.1   | Theorem (A real completion with many omitted points)                   | Theorem 67.13, p. 389            |
| 9.1   | Definition (Completion convention for a nondense order)                | Definition 98.1, p. 567          |
| 9.2   | Theorem (Reconstruction by completed residual fibers)                  | Theorem 98.2, p. 567             |
| 9.3   | Lemma (Extrema of the residual well-order space)                       | Lemma 98.3, p. 568               |
| 9.4   | Proposition (Exactly which suborders embed into the core completion)   | Proposition 98.4, p. 569         |
| 9.5   | Theorem (The exact extension criterion)                                | Theorem 98.5, p. 569             |
| 10.1  | Lemma (Exact criterion for an unseparated pair)                        | Lemma 99.1, p. 570               |
| 10.2  | Proposition (At most two initial words over an arbitrary residual cut) | Proposition 99.2, p. 571         |
| 10.3  | Proposition (An alphabet criterion for failure of core separation)     | Proposition 99.3, p. 571         |
| 10.4  | Proposition (Different initial words with the same residual core cut)  | Proposition 99.4, p. 572         |
| 10.5  | Proposition (An explicit interior completion point)                    | Proposition 99.5, p. 572         |
| 11.1  | Theorem (Continuum many omitted amenable branches)                     | Theorem 100.1, p. 573            |
| 12.1  | Theorem (Exactly two countable GBC order types)                        | Theorem 109.1, p. 626            |
| 12.2  | Lemma (The maximal finite terminal tail)                               | Note 109.3, p. 626               |
| 12.3  | Lemma (Adjacency has finite residual carrier)                          | Note 109.4, p. 627               |
| 12.4  | Lemma (Every prescribed internal tail occurs densely)                  | Note 109.5, p. 627               |
| 12.5  | Lemma (The external shape of a nonstandard finite interval)            | Lemma 109.6, p. 628              |
| 12.6  | Proposition (Intrinsic cylinder equivalence)                           | Proposition 109.7, p. 629        |
| 12.7  | Lemma (Finite-game threshold)                                          | Lemma 109.8, p. 629              |
| 12.8  | Theorem (Uniform complete theory)                                      | Theorem 109.9, p. 629            |
| 12.9  | Corollary (A proper elementary suborder and an omitted type)           | Corollary 109.10, p. 630         |
| 12.10 | Corollary (The standard shuffle is the prime model)                    | Corollary 109.11, p. 630         |
| 12.11 | Theorem (All countable orders occur inside countable models of $T$ )   | Theorem 109.12, p. 631           |
| 13.1  | Proposition (The uniform coding obstruction)                           | Note 111.6, p. 651               |
| 14.1  | Research question (The remaining coding gap)                           | Research question 113.94, p. 919 |
| 14.2  | Research question (From equimorphism to isomorphism)                   | Research question 113.95, p. 920 |
| 14.3  | Research question (Gap arrangements in longer strata)                  | Research question 113.96, p. 920 |
| 14.4  | Research question (Noninitial birthday bounds)                         | Research question 113.97, p. 920 |
| 14.5  | Research question (Stability of first-block separation)                | Research question 113.98, p. 920 |
| 14.6  | Research question (Residual fiber isomorphism types)                   | Research question 113.99, p. 920 |

| No.   | Statement in source 19                                         | Printed here as                   |
|-------|----------------------------------------------------------------|-----------------------------------|
| 14.7  | Research question (Which core automorphisms extend?)           | Research question 113.100, p. 920 |
| 14.8  | Research question (Enrichments that remember the class theory) | Research question 113.101, p. 921 |
| 14.9  | Research question (Uncountable external isomorphism spectra)   | Research question 113.102, p. 921 |
| 14.10 | Research question (Admitted versus amenable branches)          | Research question 113.103, p. 921 |
| 14.11 | Research question (Iterating set-interval condensation)        | Research question 113.104, p. 921 |
| 14.12 | Research question (Coherent enlargement maps)                  | Research question 113.105, p. 921 |
| 14.13 | Research question (A small certified core of the theory)       | Research question 113.106, p. 922 |

## D.12 Source 20

| No.  | Statement in source 20                                              | Printed here as           |
|------|---------------------------------------------------------------------|---------------------------|
| 3.1  | Lemma (A fresh separator)                                           | Note 111.7, p. 656        |
| 3.2  | Proposition (Set-like global orders)                                | Note 111.8, p. 656        |
| 3.3  | Theorem (Fixed carrier spectrum; earlier result)                    | Note 111.9, p. 657        |
| 3.4  | Proposition (Explicit compression of dictionary words)              | Note 111.10, p. 657       |
| 3.5  | Lemma (Largest common labelled prefix)                              | Note 111.11, p. 658       |
| 3.6  | Theorem (Native class lexicography; earlier result)                 | Note 111.12, p. 658       |
| 3.7  | Lemma (Uniform completion)                                          | Note 111.13, p. 658       |
| 3.8  | Theorem (Core interpolation and identification; earlier result)     | Note 111.14, p. 659       |
| 3.9  | Lemma (The first ordinal block)                                     | Note 111.15, p. 659       |
| 3.10 | Theorem (Recognition by straddling prefixes; earlier result)        | Note 111.16, p. 660       |
| 4.1  | Theorem (Coherent birthday gluing in GB)                            | Theorem 74.1, p. 450      |
| 4.2  | Proposition (The sharp set-filtration criterion)                    | Proposition 74.2, p. 451  |
| 4.3  | Theorem (Two labels determine every sufficiently large restriction) | Theorem 74.3, p. 452      |
| 4.4  | Corollary (Native order on the coherent inverse limit)              | Corollary 74.4, p. 452    |
| 4.5  | Theorem (Only finitely many changes of decisive labels)             | Theorem 74.5, p. 452      |
| 4.6  | Theorem (Rank stabilization is exactly set-likeness)                | Theorem 74.6, p. 453      |
| 4.7  | Theorem (Club reflection detects set-likeness)                      | Theorem 74.7, p. 454      |
| 4.8  | Theorem (The direct system is precisely the bounded-support core)   | Theorem 74.8, p. 455      |
| 4.9  | Proposition (Birthday table quotients)                              | Proposition 74.10, p. 456 |
| 5.1  | Definition (Strict cuts and set characters)                         | Definition 97.1, p. 554   |
| 5.2  | Lemma (Saturated sides)                                             | Note 97.2, p. 555         |
| 5.3  | Theorem (Complete cut-orbit classification)                         | Note 97.3, p. 555         |
| 5.4  | Lemma (The character of a noncore enumeration)                      | Note 97.4, p. 556         |
| 5.5  | Lemma (A local deleted-label cut)                                   | Note 97.5, p. 557         |
| 5.6  | Corollary (Representability is extra structure)                     | Note 97.6, p. 557         |
| 5.7  | Theorem (Localized failure of one-point extension)                  | Theorem 97.7, p. 558      |

| No.   | Statement in source 20                                           | Printed here as                                    |
|-------|------------------------------------------------------------------|----------------------------------------------------|
| 5.8   | Theorem (Extension criterion for the set-like order)             | <a href="#">Note 97.8</a> , p. 559                 |
| 5.9   | Proposition (Numerical relabeling moves a chosen core)           | <a href="#">Proposition 97.9</a> , p. 559          |
| 6.1   | Lemma (The maximal final set tail)                               | <a href="#">Note 107.1</a> , p. 602                |
| 6.2   | Lemma (An interior cylinder with a proper residual alphabet)     | <a href="#">Lemma 107.2</a> , p. 603               |
| 6.3   | Lemma (A diagonal inside every proper-residual cylinder)         | <a href="#">Lemma 107.3</a> , p. 604               |
| 6.4   | Theorem (Exact local coding dichotomy)                           | <a href="#">Theorem 107.4</a> , p. 604             |
| 6.5   | Theorem (Canonical local-size blocks)                            | <a href="#">Theorem 107.5</a> , p. 605             |
| 6.6   | Theorem (A dense quotient that does not fill countable cuts)     | <a href="#">Theorem 107.6</a> , p. 605             |
| 6.7   | Corollary (The external cardinal form of the local dichotomy)    | <a href="#">Corollary 107.7</a> , p. 606           |
| 7.1   | Theorem (Termination by the input order)                         | <a href="#">Theorem 108.1</a> , p. 610             |
| 7.2   | Theorem (Multiples of ordinal powers)                            | <a href="#">Theorem 108.2</a> , p. 611             |
| 7.3   | Corollary (Exact heights in a full-class model)                  | <a href="#">Corollary 108.3</a> , p. 611           |
| 8.1   | Theorem (Countable-model dichotomy)                              | <a href="#">Note 109.13</a> , p. 632               |
| 8.2   | Lemma (Internal terminal tails)                                  | <a href="#">Note 109.14</a> , p. 632               |
| 8.3   | Lemma (The internal permutation blocks)                          | <a href="#">Note 109.15</a> , p. 633               |
| 8.4   | Lemma (Every internal tail size is dense in the coarse quotient) | <a href="#">Note 109.16</a> , p. 633               |
| 8.5   | Lemma (The external type of a nonstandard finite interval)       | <a href="#">Note 109.17</a> , p. 634               |
| 8.6   | Corollary (External finite condensation)                         | <a href="#">Corollary 109.18</a> , p. 635          |
| 8.7   | Lemma (The finite-chain threshold strategy)                      | <a href="#">Note 109.19</a> , p. 635               |
| 8.8   | Theorem (An elementary standard part of the external order)      | <a href="#">Note 109.20</a> , p. 636               |
| 8.9   | Corollary (A type that distinguishes the two countable orders)   | <a href="#">Note 109.21</a> , p. 637               |
| 8.10  | Corollary (Decidability of the common pure-order theory)         | <a href="#">Corollary 109.22</a> , p. 637          |
| 9.1   | Theorem (Topology, completion, and omitted cuts)                 | <a href="#">Theorem 67.11</a> , p. 386             |
| 9.2   | Lemma                                                            | <a href="#">Lemma 67.12</a> , p. 386               |
| 11.1  | Question (The extendible subgroup)                               | <a href="#">Research question 113.107</a> , p. 922 |
| 11.2  | Question (Extensions across raw residual fibers)                 | <a href="#">Research question 113.108</a> , p. 922 |
| 11.3  | Question (The expanded countable structure)                      | <a href="#">Research question 113.109</a> , p. 922 |
| 11.4  | Question (Effective theory with additional predicates)           | <a href="#">Research question 113.110</a> , p. 923 |
| 11.5  | Question (The strength of complete condensation histories)       | <a href="#">Research question 113.111</a> , p. 923 |
| 11.6  | Question (Alternative limits for quotient iteration)             | <a href="#">Research question 113.112</a> , p. 923 |
| 11.7  | Question (Embeddings that retain more than order)                | <a href="#">Research question 113.113</a> , p. 923 |
| 11.8  | Question (Exact birthday profiles and club diagnostics)          | <a href="#">Research question 113.114</a> , p. 923 |
| 11.9  | Question (Gap arrangements beyond spectra)                       | <a href="#">Research question 113.115</a> , p. 923 |
| 11.10 | Question (Longer strata and noninitial birthdays)                | <a href="#">Research question 113.116</a> , p. 924 |
| 11.11 | Question (Robustness under class extensions)                     | <a href="#">Research question 113.117</a> , p. 924 |
| 11.12 | Question (A verified generic first-divergence library)           | <a href="#">Research question 113.118</a> , p. 924 |

## D.13 Source 21

| No.  | Statement in source 21                                                                        | Printed here as           |
|------|-----------------------------------------------------------------------------------------------|---------------------------|
| 2.1  | Example (The unlabelled quotient is incompatible with lexicography)                           | Example 61.1, p. 321      |
| 3.1  | Proposition (Cardinality and a uniform lower embedding)                                       | Note 62.13, p. 353        |
| 3.2  | Remark (A well-order of presentations need not result)                                        | Remark 62.14, p. 353      |
| 3.3  | Definition                                                                                    | Note 62.13, p. 353        |
| 3.4  | Proposition                                                                                   | Note 62.13, p. 353        |
| 4.1  | Theorem (First-cardinality decomposition)                                                     | Theorem 65.6, p. 374      |
| 4.2  | Lemma (Endpoints of a residual enumeration order)                                             | Lemma 65.7, p. 374        |
| 4.3  | Lemma (Density of exhaustive branches in the index order)                                     | Lemma 65.8, p. 374        |
| 4.4  | Theorem (Closure theorem)                                                                     | Theorem 65.9, p. 374      |
| 4.5  | Example (Finite and infinite residuals)                                                       | Example 65.10, p. 375     |
| 5.1  | Theorem (Short monotone chains in the alphabet)                                               | Theorem 77.1, p. 464      |
| 5.2  | Remark                                                                                        | Remark 77.2, p. 464       |
| 5.3  | Lemma (Finite-alphabet obstruction)                                                           | Lemma 77.3, p. 464        |
| 5.4  | Theorem (Noncardinal cutoff)                                                                  | Theorem 77.4, p. 465      |
| 5.5  | Corollary (Failure of interval self-similarity)                                               | Corollary 77.5, p. 466    |
| 6.1  | Theorem (Coherent insertion into fixed slots)                                                 | Theorem 75.1, p. 457      |
| 6.2  | Corollary (A bounded surreal system)                                                          | Corollary 75.2, p. 458    |
| 7.1  | Theorem                                                                                       | Theorem 67.6, p. 383      |
| 7.2  | Lemma (Countable sums are absorbed)                                                           | Lemma 67.7, p. 384        |
| 7.3  | Theorem (Strong finite homogeneity)                                                           | Theorem 67.8, p. 384      |
| 7.4  | Corollary (Prefix cylinders are not intrinsic)                                                | Corollary 67.9, p. 385    |
| 7.5  | Theorem                                                                                       | Theorem 67.10, p. 385     |
| 8.1  | Theorem (Largest common labelled prefix; repository baseline)                                 | Note 86.10, p. 524        |
| 8.2  | Theorem (Canonical first block)                                                               | Note 86.11, p. 524        |
| 8.3  | Theorem (Canonical set-interval decomposition)                                                | Note 86.12, p. 525        |
| 8.4  | Lemma (Uniform completion of a set prefix)                                                    | Note 86.13, p. 526        |
| 8.5  | Theorem (Uniform set-cut interpolation; repository baseline)                                  | Note 86.14, p. 526        |
| 8.6  | Lemma (The core sees exactly the first ordinal block)                                         | Note 86.15, p. 527        |
| 8.7  | Proposition (No exhaustive uniform set-code family)                                           | Note 86.16, p. 527        |
| 9.1  | Lemma (Elementary reconstruction from a core cut)                                             | Note 97.10, p. 561        |
| 9.2  | Theorem (Which core cuts are represented by words?)                                           | Note 97.11, p. 561        |
| 9.3  | Proposition (The first failed level of a cut)                                                 | Note 97.12, p. 562        |
| 9.4  | Theorem (Extension criterion)                                                                 | Note 97.13, p. 562        |
| 9.5  | Corollary (Restriction to the core)                                                           | Note 97.14, p. 563        |
| 9.6  | Definition                                                                                    | Definition 97.15, p. 563  |
| 9.7  | Lemma (Each side is an $\eta_{\text{Ord}}$ order)                                             | Note 97.16, p. 563        |
| 9.8  | Theorem (Conjugacy of large class cuts)                                                       | Note 97.17, p. 564        |
| 9.9  | Lemma (Cuts induced by nonskeleton words)                                                     | Note 97.18, p. 564        |
| 9.10 | Theorem (Nonextendable core automorphisms)                                                    | Note 97.19, p. 564        |
| 9.11 | Proposition (A large cut with no crossing first-level cylinder)                               | Proposition 97.20, p. 565 |
| 9.12 | Theorem (No extension even from $\mathfrak{W}_{\text{sl}}$ into $\mathfrak{W}_{\text{all}}$ ) | Note 97.21, p. 565        |
| 10.1 | Definition (Maximal finite terminal tail)                                                     | Definition 109.23, p. 638 |
| 10.2 | Lemma (Existence of the maximal finite tail)                                                  | Note 109.24, p. 638       |



| No.   | Statement in source 21                                                                      | Printed here as                                   |
|-------|---------------------------------------------------------------------------------------------|---------------------------------------------------|
| 10.3  | Theorem (Exact terminal permutation blocks)                                                 | <a href="#">Note 109.25, p. 639</a>               |
| 10.4  | Corollary (Intrinsic finite condensation)                                                   | <a href="#">Note 109.26, p. 639</a>               |
| 10.5  | Theorem (Density of each tail length)                                                       | <a href="#">Note 109.27, p. 640</a>               |
| 10.6  | Definition (Rational shuffle)                                                               | <a href="#">Definition 109.28, p. 640</a>         |
| 10.7  | Corollary (Countable standard realization)                                                  | <a href="#">Note 109.29, p. 641</a>               |
| 10.8  | Theorem (Automorphisms of finite blocks)                                                    | <a href="#">Note 109.30, p. 641</a>               |
| 10.9  | Corollary (A greatest element is not intrinsic)                                             | <a href="#">Corollary 109.31, p. 641</a>          |
| 10.10 | Lemma (An externally infinite finite interval)                                              | <a href="#">Note 109.32, p. 642</a>               |
| 10.11 | Theorem (Exact countable nonstandard realization)                                           | <a href="#">Note 109.33, p. 642</a>               |
| 10.12 | Remark (Exactly what is being transferred)                                                  | <a href="#">Remark 109.34, p. 643</a>             |
| 10.13 | Proposition (Different orders, the same elementary theory)                                  | <a href="#">Note 109.35, p. 643</a>               |
| 10.14 | Theorem (The first derivative)                                                              | <a href="#">Theorem 109.36, p. 644</a>            |
| 11.1  | Lemma (A set witness to ill-foundedness)                                                    | <a href="#">Note 102.7, p. 582</a>                |
| 11.2  | Proposition (Changing classes while keeping the sets)                                       | <a href="#">Proposition 102.8, p. 582</a>         |
| 11.3  | Proposition (Preservation with the same ordinals)                                           | <a href="#">Proposition 102.9, p. 583</a>         |
| 11.4  | Proposition (A sufficient condition for external correctness)                               | <a href="#">Proposition 102.10, p. 583</a>        |
| 11.5  | Theorem (A class expansion as an extension by cuts)                                         | <a href="#">Theorem 102.11, p. 584</a>            |
| 11.6  | Corollary (Rigidity when the core is fixed pointwise)                                       | <a href="#">Corollary 102.12, p. 584</a>          |
| 11.7  | Theorem (External size and countable classification)                                        | <a href="#">Theorem 102.13, p. 585</a>            |
| 13.1  | Research question (Classify the extendable subgroup)                                        | <a href="#">Research question 113.119, p. 925</a> |
| 13.2  | Research question (Extension to the full class-well-order space)                            | <a href="#">Research question 113.120, p. 925</a> |
| 13.3  | Research question (Prefix-preserving automorphisms)                                         | <a href="#">Research question 113.121, p. 925</a> |
| 13.4  | Research question (Noncardinal cut spectra and isomorphism types)                           | <a href="#">Research question 113.122, p. 925</a> |
| 13.5  | Research question (Fixed nonminimal enumeration types)                                      | <a href="#">Research question 113.123, p. 925</a> |
| 13.6  | Research question (Coherence across a change of cardinality)                                | <a href="#">Research question 113.124, p. 925</a> |
| 13.7  | Research question (The uncountable colored quotient)                                        | <a href="#">Research question 113.125, p. 926</a> |
| 13.8  | Research question (How much model structure does a marked core retain?)                     | <a href="#">Research question 113.126, p. 926</a> |
| 13.9  | Research question (Automorphism and descriptive complexity of the countable-alphabet space) | <a href="#">Research question 113.127, p. 926</a> |
| 13.10 | Research question (Presentation orders under forcing)                                       | <a href="#">Research question 113.128, p. 926</a> |
| 13.11 | Research question (Relation-table comparison with extra structure)                          | <a href="#">Research question 113.129, p. 926</a> |
| 13.12 | Research question (Formal verification of the obstruction)                                  | <a href="#">Research question 113.130, p. 926</a> |

## D.14 Source 22

| No.  | Statement in source 22                                                | Printed here as                   |
|------|-----------------------------------------------------------------------|-----------------------------------|
| 3.1  | Definition                                                            | Note 62.15, p. 353                |
| 3.2  | Lemma (Richness of cardinal birthday cutoffs)                         | Note 62.15, p. 353                |
| 3.3  | Lemma (Short prefixes remain flexible)                                | Note 62.15, p. 353                |
| 3.4  | Theorem (Size and embeddings)                                         | Note 62.15, p. 353                |
| 4.1  | Theorem (Residual-cylinder identity)                                  | Theorem 63.1, p. 354              |
| 4.2  | Theorem (General first-disagreement formula)                          | Theorem 63.2, p. 355              |
| 4.3  | Theorem (Character transfer)                                          | Theorem 63.3, p. 356              |
| 5.1  | Theorem (Topological classification of every fixed-length stratum)    | Theorem 63.4, p. 356              |
| 5.2  | Remark (Why the distinction between a base and a $\pi$ -base matters) | Remark 63.6, p. 358               |
| 5.3  | Corollary (Birthday-cutoff form)                                      | Corollary 63.7, p. 358            |
| 5.4  | Proposition (Finite-tail normal form)                                 | Proposition 63.8, p. 358          |
| 5.5  | Theorem (All finite-tail point characters)                            | Theorem 63.9, p. 358              |
| 5.6  | Corollary (One-step derivative)                                       | Corollary 63.10, p. 359           |
| 5.7  | Example (A countable alphabet)                                        | Example 63.11, p. 359             |
| 6.1  | Lemma (Adjacency forces a finite residual)                            | Lemma 63.13, p. 360               |
| 6.2  | Lemma (Countable residual enumeration orders)                         | Lemma 63.14, p. 360               |
| 6.3  | Theorem (Exact $\mu + \omega$ character spectrum)                     | Theorem 63.15, p. 361             |
| 6.4  | Remark (A sharp contrast even when cardinal invariants coincide)      | Remark 63.16, p. 361              |
| 7.1  | Lemma (Every set prefix has a supported completion)                   | Note 86.7, p. 522                 |
| 7.2  | Lemma (Fresh separators)                                              | Note 86.8, p. 523                 |
| 7.3  | Proposition (Density in all injective global words)                   | Note 86.9, p. 523                 |
| 8.1  | Theorem (Cylinder inclusion recovers prefix and height)               | Theorem 87.1, p. 531              |
| 8.2  | Remark (The fresh-choice hypothesis is real)                          | Remark 87.2, p. 531               |
| 8.3  | Lemma (Crossed cylinders form an initial branch)                      | Lemma 87.3, p. 531                |
| 8.4  | Theorem (Unranked branch-recognition theorem)                         | Theorem 87.4, p. 532              |
| 8.5  | Corollary (Exhaustiveness is label coverage)                          | Corollary 87.5, p. 532            |
| 9.1  | Proposition (Pair-swap encoding)                                      | Proposition 102.1, p. 578         |
| 9.2  | Theorem (Elementary decoding from a lower trace)                      | Theorem 102.2, p. 578             |
| 9.3  | Theorem (Faithful reconstruction of the class realization)            | Theorem 102.3, p. 579             |
| 9.4  | Corollary (Membership reconstruction)                                 | Corollary 102.4, p. 579           |
| 10.1 | Proposition (Elementary raw lexicographic comparison)                 | Note 111.17, p. 669               |
| 10.2 | Proposition (Ordinary-class universality)                             | Note 111.18, p. 670               |
| 10.3 | Corollary (A stratum obstruction to raw-to-set-like embedding)        | Note 111.19, p. 671               |
| 12.1 | Question (Nonminimal cut spectra)                                     | Research question 113.131, p. 927 |
| 12.2 | Question (The dense quotient orders $Q_n(X)$ )                        | Research question 113.132, p. 927 |
| 12.3 | Question (Long-tail local characters)                                 | Research question 113.133, p. 927 |
| 12.4 | Question (Descriptive topology when cofinalities are countable)       | Research question 113.134, p. 927 |
| 12.5 | Question (Eliminating label data)                                     | Research question 113.135, p. 928 |
| 12.6 | Question (Reconstruction without fixed coding parameters)             | Research question 113.136, p. 928 |
| 12.7 | Question (Automorphisms preserving admitted traces)                   | Research question 113.137, p. 928 |
| 12.8 | Question (Noncardinal birthday cutoffs)                               | Research question 113.138, p. 928 |

| No.   | Statement in source 22                               | Printed here as                                    |
|-------|------------------------------------------------------|----------------------------------------------------|
| 12.9  | Question (Coherent length and cutoff embeddings)     | <a href="#">Research question 113.139</a> , p. 928 |
| 12.10 | Question (Iterated raw heads and recursion strength) | <a href="#">Research question 113.140</a> , p. 928 |
| 12.11 | Question (Binary coding lengths for long strata)     | <a href="#">Research question 113.141</a> , p. 929 |
| 12.12 | Question (First kernel-checked milestones)           | <a href="#">Research question 113.142</a> , p. 929 |

## D.15 Source 23

| No.  | Statement in source 23                                               | Printed here as                                    |
|------|----------------------------------------------------------------------|----------------------------------------------------|
| 2.1  | Lemma (No prefix ambiguity for complete enumerations)                | <a href="#">Note 80.1</a> , p. 476                 |
| 3.1  | Definition                                                           | <a href="#">Note 81.1</a> , p. 485                 |
| 3.2  | Lemma (Two cuts give a fresh separator)                              | <a href="#">Note 81.2</a> , p. 485                 |
| 3.3  | Lemma (No small bounds)                                              | <a href="#">Note 81.3</a> , p. 485                 |
| 3.4  | Lemma (Deletion preserves the relevant side conditions)              | <a href="#">Lemma 81.4</a> , p. 485                |
| 4.1  | Lemma (Convex cylinders)                                             | <a href="#">Lemma 81.5</a> , p. 486                |
| 4.2  | Lemma (Cylinders have no small end cofinalities)                     | <a href="#">Lemma 81.6</a> , p. 486                |
| 4.3  | Lemma (Crossing-prefix reduction)                                    | <a href="#">Lemma 81.7</a> , p. 486                |
| 5.1  | Definition                                                           | <a href="#">Equation (32)</a> , p. 487             |
| 5.2  | Theorem (Termination classification)                                 | <a href="#">Note 81.9</a> , p. 487                 |
| 5.3  | Corollary (Surreal order type)                                       | <a href="#">Note 81.10</a> , p. 488                |
| 5.4  | Remark (What the endpoint exception does)                            | <a href="#">Remark 81.11</a> , p. 488              |
| 6.1  | Theorem (Uniform point characters)                                   | <a href="#">Note 81.12</a> , p. 488                |
| 7.1  | Theorem (No fixed marker saturates increasing words)                 | <a href="#">Theorem 82.1</a> , p. 490              |
| 7.2  | Remark (Why the arbitrary injective case escapes)                    | <a href="#">Remark 82.2</a> , p. 491               |
| 8.1  | Theorem (Fixed-set baseline)                                         | <a href="#">Note 111.20</a> , p. 676               |
| 8.2  | Lemma (Completing a set prefix)                                      | <a href="#">Note 111.21</a> , p. 677               |
| 8.3  | Theorem (Uniform small cuts have core separators)                    | <a href="#">Note 111.22</a> , p. 677               |
| 9.1  | Proposition (Largest common labelled initial part)                   | <a href="#">Note 106.1</a> , p. 599                |
| 10.1 | Definition                                                           | <a href="#">Definition 106.2</a> , p. 600          |
| 10.2 | Theorem (Canonical block decomposition)                              | <a href="#">Note 106.3</a> , p. 600                |
| 11.1 | Lemma (Polynomial order is a class well-order)                       | <a href="#">Note 108.6</a> , p. 613                |
| 11.2 | Lemma (The first block removes the least coordinate)                 | <a href="#">Note 108.7</a> , p. 613                |
| 11.3 | Theorem (Prescribed-hierarchy theorem)                               | <a href="#">Note 108.8</a> , p. 614                |
| 12.1 | Lemma (A set prefix is absorbed)                                     | <a href="#">Lemma 108.22</a> , p. 624              |
| 12.2 | Theorem (Dense realization of prescribed raw hierarchies)            | <a href="#">Theorem 108.23</a> , p. 624            |
| 12.3 | Remark (The precise density assertion)                               | <a href="#">Remark 108.24</a> , p. 625             |
| 13.1 | Proposition (Uniform-code diagonal)                                  | <a href="#">Note 111.23</a> , p. 678               |
| 14.1 | Proposition (Polynomial orders become ordinal powers)                | <a href="#">Proposition 111.24</a> , p. 680        |
| 16.1 | Research question (Classification of the nonsaturated marker orders) | <a href="#">Research question 113.143</a> , p. 929 |
| 16.2 | Research question (Restricted increasing-word repairs)               | <a href="#">Research question 113.144</a> , p. 930 |
| 16.3 | Research question (Intrinsic versus named cylinders)                 | <a href="#">Research question 113.145</a> , p. 930 |
| 16.4 | Research question (Quantitative birthday bounds)                     | <a href="#">Research question 113.146</a> , p. 930 |

| No.   | Statement in source 23                                                | Printed here as                                    |
|-------|-----------------------------------------------------------------------|----------------------------------------------------|
| 16.5  | Research question (Proof strength of general condensation iteration)  | <a href="#">Research question 113.147</a> , p. 930 |
| 16.6  | Research question (Which orders eventually collapse?)                 | <a href="#">Research question 113.148</a> , p. 930 |
| 16.7  | Research question (Lexicographic distribution of internal invariants) | <a href="#">Research question 113.149</a> , p. 930 |
| 16.8  | Research question (Condensation versus finite residual adjacency)     | <a href="#">Research question 113.150</a> , p. 931 |
| 16.9  | Research question (Choice strength of the new classification)         | <a href="#">Research question 113.151</a> , p. 931 |
| 16.10 | Research question (A checked pair of contrasting theorems)            | <a href="#">Research question 113.152</a> , p. 931 |

## D.16 Source 24

| No. | Statement in source 24                                                  | Printed here as                           |
|-----|-------------------------------------------------------------------------|-------------------------------------------|
| 1.1 | Theorem (Main results, with hypotheses explicit)                        | <a href="#">Theorem 61.2</a> , p. 328     |
| 3.1 | Definition (Thickness)                                                  | <a href="#">Definition 62.1</a> , p. 349  |
| 3.2 | Lemma (Cutoff size and thickness)                                       | <a href="#">Note 62.2</a> , p. 350        |
| 3.3 | Remark (What the arithmetic parameters do and do not control)           | <a href="#">Remark 62.3</a> , p. 350      |
| 4.1 | Lemma (Small-prefix extension)                                          | <a href="#">Lemma 62.4</a> , p. 350       |
| 4.2 | Lemma (Cylinders and order topology)                                    | <a href="#">Lemma 62.5</a> , p. 350       |
| 4.3 | Proposition (Cardinality and exact ordinal spectrum)                    | <a href="#">Note 62.6</a> , p. 351        |
| 4.4 | Proposition (Local character and metrizability)                         | <a href="#">Proposition 62.7</a> , p. 351 |
| 4.5 | Corollary (The topology forgets the numerical labels)                   | <a href="#">Corollary 62.8</a> , p. 351   |
| 5.1 | Theorem (Exact forward completion)                                      | <a href="#">Theorem 65.1</a> , p. 372     |
| 5.2 | Theorem (Exhaustion is an intersection of dense open requirements)      | <a href="#">Theorem 65.2</a> , p. 372     |
| 5.3 | Example (A label escaping to later and later positions)                 | <a href="#">Example 65.3</a> , p. 372     |
| 5.4 | Proposition (The injection space sits strictly inside order completion) | <a href="#">Proposition 65.4</a> , p. 373 |
| 6.1 | Lemma (Exact branching number)                                          | <a href="#">Lemma 66.1</a> , p. 378       |
| 6.2 | Theorem (Regular-cardinal topological classification)                   | <a href="#">Theorem 66.2</a> , p. 379     |
| 6.3 | Corollary (Size-matched regular cutoffs)                                | <a href="#">Corollary 66.3</a> , p. 379   |
| 7.1 | Theorem (Complete metrizability at countable size)                      | <a href="#">Theorem 67.2</a> , p. 381     |
| 7.2 | Lemma (A countable order-dense subset)                                  | <a href="#">Lemma 67.3</a> , p. 381       |
| 7.3 | Theorem (Exact descriptive form of the omitted points)                  | <a href="#">Theorem 67.4</a> , p. 382     |
| 7.4 | Corollary (Homeomorphic to, but not ordered like, the irrationals)      | <a href="#">Corollary 67.5</a> , p. 382   |
| 8.1 | Definition (Preservation of bounded sets)                               | <a href="#">Definition 69.1</a> , p. 408  |
| 8.2 | Theorem (Exact continuity criteria)                                     | <a href="#">Theorem 69.2</a> , p. 408     |
| 8.3 | Corollary (Regular and singular cases diverge)                          | <a href="#">Corollary 69.3</a> , p. 408   |
| 8.4 | Theorem (Two-sided completeness and the bounded subgroup)               | <a href="#">Theorem 69.4</a> , p. 409     |

| No.   | Statement in source 24                                                 | Printed here as                   |
|-------|------------------------------------------------------------------------|-----------------------------------|
| 8.5   | Corollary (A complete metric on the bounded subgroup)                  | Corollary 69.5, p. 409            |
| 8.6   | Remark (A finer complete uniformity is not a universal group repair)   | Remark 69.6, p. 409               |
| 9.1   | Lemma (Global prefix extension with a reserve)                         | Note 71.1, p. 434                 |
| 9.2   | Proposition (Prefix density and numerical density)                     | Note 71.1, p. 434                 |
| 9.3   | Proposition (Ordinary class-order universality, recalled)              | Note 71.1, p. 434                 |
| 9.4   | Theorem (Small-net rigidity)                                           | Theorem 71.2, p. 435              |
| 9.5   | Corollary (No set-sized accumulation pattern)                          | Corollary 71.3, p. 435            |
| 9.6   | Definition (A coherent prefix system)                                  | Definition 71.4, p. 436           |
| 9.7   | Theorem (Global branch completion, stated as a scheme)                 | Theorem 71.5, p. 436              |
| 9.8   | Theorem (Two-sided global completion)                                  | Theorem 71.6, p. 436              |
| 10.1  | Definition (The common labelled initial segment)                       | Note 71.7, p. 437                 |
| 10.2  | Lemma (Maximality of the common segment)                               | Note 71.7, p. 437                 |
| 10.3  | Theorem (Elementary lexicographic comparison of all class well-orders) | Note 71.7, p. 437                 |
| 10.4  | Example (A genuinely proper-class prefix)                              | Note 71.7, p. 437                 |
| 11.1  | Theorem (No uniform class of set codes exhausts the global words)      | Note 71.8, p. 438                 |
| 11.2  | Example (Why all class cuts cannot be filled by surreals)              | Example 71.9, p. 439              |
| 13.1  | Question (Complete metrizability at uncountable countable cofinality)  | Research question 113.153, p. 931 |
| 13.2  | Question (A singular replacement for the exhaustion tree)              | Research question 113.154, p. 932 |
| 13.3  | Question (Dense-open intersection thresholds)                          | Research question 113.155, p. 932 |
| 13.4  | Question (The embedded real-line model at countable size)              | Research question 113.156, p. 932 |
| 13.5  | Question (Regular homeomorphisms versus order embeddings)              | Research question 113.157, p. 932 |
| 13.6  | Question (Completion-preserving automorphisms)                         | Research question 113.158, p. 932 |
| 13.7  | Question (The boundedness-preserving subgroup)                         | Research question 113.159, p. 933 |
| 13.8  | Question (Uniformities with an exhaustion interpretation)              | Research question 113.160, p. 933 |
| 13.9  | Question (An axiomatization of virtual completion)                     | Research question 113.161, p. 933 |
| 13.10 | Question (Changing the class part without changing the sets)           | Research question 113.162, p. 933 |
| 13.11 | Question (Unrestricted tails and their completion theory)              | Research question 113.163, p. 933 |
| 13.12 | Question (A small trusted Lean core)                                   | Research question 113.164, p. 933 |
| C.1   | Proposition (Complete metric subspaces and $G_\delta$ sets)            | Proposition 69.8, p. 410          |

## D.17 Source 25

| No. | Statement in source 25                                                | Printed here as      |
|-----|-----------------------------------------------------------------------|----------------------|
| 1.1 | Theorem (Configuration classification; precise version in Section 90) | Theorem 61.3, p. 332 |

| No.  | Statement in source 25                                                    | Printed here as                   |
|------|---------------------------------------------------------------------------|-----------------------------------|
| 3.1  | Theorem (Fixed-set size and ordinal spectrum)                             | Note 86.1, p. 518                 |
| 3.2  | Proposition (Class-order universality, in GBC)                            | Note 86.2, p. 519                 |
| 4.1  | Lemma (A separator avoiding a prescribed set)                             | Note 86.3, p. 520                 |
| 4.2  | Lemma (Supported prefix completion)                                       | Note 86.4, p. 521                 |
| 4.3  | Lemma (Core separation of full branches)                                  | Note 86.5, p. 521                 |
| 4.4  | Corollary (Injectivity of the cut trace)                                  | Note 86.6, p. 521                 |
| 5.1  | Proposition (Cylinder recovery)                                           | Note 88.1, p. 533                 |
| 5.2  | Proposition (Automatic profile preservation)                              | Proposition 88.2, p. 534          |
| 6.1  | Lemma (One-point extension into the core)                                 | Lemma 89.1, p. 535                |
| 6.2  | Remark (Why no numerical equality pattern is required)                    | Remark 89.2, p. 536               |
| 6.3  | Example (An elementary local distortion)                                  | Example 89.3, p. 536              |
| 7.1  | Theorem (Classification of set-indexed branch configurations)             | Theorem 90.1, p. 536              |
| 7.2  | Corollary (Exactly two single-branch orbits)                              | Corollary 90.2, p. 537            |
| 7.3  | Remark (The logical meaning of an orbit)                                  | Remark 90.3, p. 537               |
| 8.1  | Theorem (Cylinder preservation does not preserve coverage)                | Theorem 91.1, p. 538              |
| 8.2  | Theorem (Parameter-robust non-invariance)                                 | Theorem 91.2, p. 538              |
| 8.3  | Corollary (No invariant unlabelled recognition test)                      | Corollary 91.3, p. 539            |
| 9.1  | Proposition (Represented-trace criterion)                                 | Note 92.1, p. 539                 |
| 9.2  | Theorem (A cylinder-preserving automorphism with no exhaustive extension) | Theorem 92.2, p. 540              |
| 10.1 | Theorem (Label-coherent tree automorphisms)                               | Theorem 93.1, p. 541              |
| 10.2 | Lemma (No nontrivial set-supported alphabet automorphism)                 | Lemma 93.2, p. 541                |
| 10.3 | Theorem (The label-coherent core structure is rigid)                      | Theorem 93.3, p. 541              |
| 10.4 | Proposition (Coverage from coherent edges)                                | Proposition 93.4, p. 542          |
| 11.1 | Theorem (Regular-cardinal configuration theorem)                          | Theorem 94.1, p. 543              |
| 11.2 | Theorem (Exact cardinality of the automorphism group)                     | Theorem 94.2, p. 543              |
| 12.1 | Theorem (Cofinal profiles force the missing branch)                       | Theorem 95.1, p. 544              |
| 12.2 | Remark (A second singular obstruction: unbounded prefix images)           | Remark 95.2, p. 544               |
| 13.1 | Proposition (Largest common labelled prefix)                              | Note 103.1, p. 592                |
| 14.1 | Proposition (Diagonal obstruction to exhaustive class coding)             | Note 104.1, p. 595                |
| 16.1 | Research question (The exhaustive-branch stabilizer)                      | Research question 113.165, p. 934 |
| 16.2 | Research question (Recognition without full rigidity)                     | Research question 113.166, p. 934 |
| 16.3 | Research question (Preserving a support filtration)                       | Research question 113.167, p. 934 |
| 16.4 | Research question (Larger configurations and exact obstruction patterns)  | Research question 113.168, p. 935 |
| 16.5 | Research question (Singular-cardinal automorphism groups)                 | Research question 113.169, p. 935 |
| 16.6 | Research question (Two baselines at once)                                 | Research question 113.170, p. 935 |
| 16.7 | Research question (The bare order of exhaustive enumerations)             | Research question 113.171, p. 935 |
| 16.8 | Research question (Raw tails and outer-order invariants)                  | Research question 113.172, p. 935 |
| 16.9 | Research question (The increasing-word analogue)                          | Research question 113.173, p. 935 |



| No.   | Statement in source 25                                                       | Printed here as                   |
|-------|------------------------------------------------------------------------------|-----------------------------------|
| 16.10 | Research question (Class-realization reconstruction after forgetting labels) | Research question 113.174, p. 936 |
| 16.11 | Research question (A checked minimal development)                            | Research question 113.175, p. 936 |

## D.18 Source 26

| No.  | Statement in source 26                                                    | Printed here as                   |
|------|---------------------------------------------------------------------------|-----------------------------------|
| 3.1  | Lemma (Extension of a short prefix)                                       | Note 62.9, p. 352                 |
| 3.2  | Proposition (Basic topology)                                              | Proposition 62.10, p. 352         |
| 3.3  | Definition                                                                | Note 62.9, p. 352                 |
| 3.4  | Theorem (When the two topologies agree)                                   | Note 62.9, p. 352                 |
| 4.1  | Theorem (Forward completion; earlier result re-proved)                    | Note 65.5, p. 373                 |
| 4.2  | Proposition (Exhaustion requirements)                                     | Note 65.5, p. 373                 |
| 4.3  | Theorem (Local character)                                                 | Note 69.10, p. 411                |
| 4.4  | Proposition (The countable exception)                                     | Proposition 69.11, p. 411         |
| 4.5  | Theorem (Baire property)                                                  | Theorem 69.12, p. 412             |
| 5.1  | Theorem (Compact coding of omitted labels)                                | Theorem 69.13, p. 412             |
| 5.2  | Remark (A particularly small defect)                                      | Remark 69.14, p. 414              |
| 5.3  | Remark (The strict cardinal bound)                                        | Remark 69.15, p. 414              |
| 6.1  | Lemma (A small non-Borel set of branches)                                 | Lemma 69.16, p. 414               |
| 6.2  | Theorem (The singular non-Borel boundary)                                 | Theorem 69.17, p. 414             |
| 6.3  | Theorem (Complete-metrizability classification)                           | Theorem 69.18, p. 415             |
| 6.4  | Corollary (Answer to the earlier question)                                | Corollary 69.19, p. 415           |
| 7.1  | Theorem (Every Cantor subset as a closed trace)                           | Theorem 69.20, p. 416             |
| 7.2  | Corollary (Closed topological universality)                               | Corollary 69.21, p. 416           |
| 7.3  | Corollary (Closed rational subspaces)                                     | Corollary 69.22, p. 416           |
| 8.1  | Theorem (The bounded-image completion and its complete subgroup)          | Theorem 69.23, p. 417             |
| 8.2  | Proposition (What the inverse condition sees on the code)                 | Proposition 69.24, p. 418         |
| 9.1  | Lemma (Cardinality and thickness)                                         | Lemma 69.25, p. 419               |
| 9.2  | Remark (No hidden regularity)                                             | Remark 69.26, p. 419              |
| 9.3  | Theorem (Topological classification of the minimal surreal slices)        | Theorem 69.27, p. 419             |
| 9.4  | Corollary (The unconditional $\beth_\omega$ example)                      | Corollary 69.28, p. 420           |
| 10.1 | Proposition (Successor-length isolation)                                  | Proposition 69.29, p. 421         |
| 11.1 | Definition (Common labelled prefix)                                       | Note 71.10, p. 440                |
| 11.2 | Lemma                                                                     | Note 71.10, p. 440                |
| 11.3 | Theorem (A virtual linear order of all labelled class well-orders)        | Note 71.10, p. 440                |
| 11.4 | Example (A genuine adjacent pair)                                         | Note 71.10, p. 440                |
| 11.5 | Lemma (Prefix extension with an explicit reserve)                         | Note 71.10, p. 440                |
| 11.6 | Proposition (Class-order universality, with the coding boundary explicit) | Note 71.10, p. 440                |
| 11.7 | Theorem (Diagonal obstruction)                                            | Note 71.10, p. 440                |
| 12.1 | Theorem (Small-net rigidity)                                              | Note 71.11, p. 440                |
| 14.1 | Research question (Generalized Borel complexity)                          | Research question 113.176, p. 936 |

| No.   | Statement in source 26                                                    | Printed here as                                    |
|-------|---------------------------------------------------------------------------|----------------------------------------------------|
| 14.2  | Research question (Closed universality without $\mu > \mathfrak{c}$ )     | <a href="#">Research question 113.177</a> , p. 936 |
| 14.3  | Research question (Higher cofinality coding)                              | <a href="#">Research question 113.178</a> , p. 937 |
| 14.4  | Research question (The exact role of bounded inverse images)              | <a href="#">Research question 113.179</a> , p. 937 |
| 14.5  | Research question (A classification across nonminimal strata)             | <a href="#">Research question 113.180</a> , p. 937 |
| 14.6  | Research question (Closed order embeddings versus topological embeddings) | <a href="#">Research question 113.181</a> , p. 937 |
| 14.7  | Research question (Embedding spectra refined by topology)                 | <a href="#">Research question 113.182</a> , p. 937 |
| 14.8  | Research question (Dense-open intersection thresholds)                    | <a href="#">Research question 113.183</a> , p. 937 |
| 14.9  | Research question (A topology that retains unrestricted class tails)      | <a href="#">Research question 113.184</a> , p. 938 |
| 14.10 | Research question (Reverse foundations of uniform approximation)          | <a href="#">Research question 113.185</a> , p. 938 |
| 14.11 | Research question (Changing admitted class families)                      | <a href="#">Research question 113.186</a> , p. 938 |
| 14.12 | Research question (A minimal kernel-checked continuation)                 | <a href="#">Research question 113.187</a> , p. 938 |
| A.1   | Theorem                                                                   | <a href="#">Note 69.9</a> , p. 411                 |

## D.19 Source 27

| No. | Statement in source 27                                       | Printed here as                            |
|-----|--------------------------------------------------------------|--------------------------------------------|
| 3.1 | Definition                                                   | <a href="#">Definition 111.25</a> , p. 697 |
| 3.2 | Theorem (Direct labelled comparison)                         | <a href="#">Note 111.26</a> , p. 697       |
| 4.1 | Lemma (Independent adjacent-pair coding)                     | <a href="#">Lemma 84.1</a> , p. 506        |
| 4.2 | Theorem (Exact scan-independent ordinal spectrum)            | <a href="#">Theorem 84.2</a> , p. 506      |
| 4.3 | Theorem (Dense slices and exact topology)                    | <a href="#">Theorem 84.3</a> , p. 507      |
| 4.4 | Theorem (Six delayed coordinates)                            | <a href="#">Theorem 84.4</a> , p. 507      |
| 4.5 | Theorem (Set-like orders are table-dense)                    | <a href="#">Theorem 84.5</a> , p. 508      |
| 4.6 | Proposition (Complete ambient table order)                   | <a href="#">Proposition 84.6</a> , p. 509  |
| 5.1 | Theorem (Factorization at a terminal cut)                    | <a href="#">Theorem 82.4</a> , p. 491      |
| 5.2 | Corollary (A product description of the centered order)      | <a href="#">Corollary 82.5</a> , p. 492    |
| 5.3 | Proposition (Exact zero-convention types)                    | <a href="#">Proposition 82.6</a> , p. 493  |
| 5.4 | Theorem (Convex word cones in every interval)                | <a href="#">Theorem 82.7</a> , p. 494      |
| 5.5 | Proposition (Characters at a general terminal gap)           | <a href="#">Proposition 82.8</a> , p. 494  |
| 6.1 | Lemma (Uniform pointed residual isomorphisms)                | <a href="#">Lemma 96.1</a> , p. 545        |
| 6.2 | Theorem (Transitivity on all injective ordinal branches)     | <a href="#">Theorem 96.2</a> , p. 546      |
| 6.3 | Corollary (Ordered prefixes do not recognize exhaustiveness) | <a href="#">Corollary 96.3</a> , p. 546    |
| 6.4 | Theorem (Set-bounded local inverse images)                   | <a href="#">Theorem 96.4</a> , p. 547      |
| 6.5 | Theorem (Automorphisms after retaining label equality)       | <a href="#">Note 96.5</a> , p. 548         |
| 7.1 | Theorem (An omitted convex copy)                             | <a href="#">Theorem 107.8</a> , p. 607     |

| No.   | Statement in source 27                                                | Printed here as                                    |
|-------|-----------------------------------------------------------------------|----------------------------------------------------|
| 7.2   | Theorem (Many disjoint convex copies)                                 | <a href="#">Theorem 107.9</a> , p. 608             |
| 8.1   | Theorem (Completion below a specified ordinal)                        | <a href="#">Theorem 101.1</a> , p. 575             |
| 8.2   | Theorem (Exact ordinal support optimum)                               | <a href="#">Theorem 101.2</a> , p. 576             |
| 8.3   | Example (The one-sided shift)                                         | <a href="#">Example 101.3</a> , p. 576             |
| 8.4   | Example (Prescribed fixed positions can delay the reserve)            | <a href="#">Example 101.4</a> , p. 576             |
| 8.5   | Example (High fixed points need not matter)                           | <a href="#">Example 101.5</a> , p. 576             |
| 8.6   | Example (Balanced infinite motion)                                    | <a href="#">Example 101.6</a> , p. 576             |
| 9.1   | Theorem (Embeddings fixing the core)                                  | <a href="#">Note 102.5</a> , p. 581                |
| 9.2   | Corollary (The raw-order existence criterion)                         | <a href="#">Corollary 102.6</a> , p. 581           |
| 10.1  | Theorem (Full rank semantics)                                         | <a href="#">Theorem 110.1</a> , p. 647             |
| 12.1  | Research question (Classify intermediate scan types)                  | <a href="#">Research question 113.188</a> , p. 938 |
| 12.2  | Research question (Recover the effective scan)                        | <a href="#">Research question 113.189</a> , p. 939 |
| 12.3  | Research question (The table-ordered bounded-support core)            | <a href="#">Research question 113.190</a> , p. 939 |
| 12.4  | Research question (A sharper type for centered increasing words)      | <a href="#">Research question 113.191</a> , p. 939 |
| 12.5  | Research question (Orders at genuine surreal gaps)                    | <a href="#">Research question 113.192</a> , p. 939 |
| 12.6  | Research question (The subgroup preserving exhaustive branches)       | <a href="#">Research question 113.193</a> , p. 939 |
| 12.7  | Research question (Weaker enrichments than label equality)            | <a href="#">Research question 113.194</a> , p. 939 |
| 12.8  | Research question (Raw extensions of core automorphisms)              | <a href="#">Research question 113.195</a> , p. 939 |
| 12.9  | Research question (Marked model reconstruction with fewer constants)  | <a href="#">Research question 113.196</a> , p. 939 |
| 12.10 | Research question (Optimize the interpolant, not just a fixed prefix) | <a href="#">Research question 113.197</a> , p. 940 |
| 12.11 | Research question (Controlled bounded approximations)                 | <a href="#">Research question 113.198</a> , p. 940 |
| 12.12 | Research question (First machine-checked theorem package)             | <a href="#">Research question 113.199</a> , p. 940 |

## D.20 Source 28

| No. | Statement in source 28                                              | Printed here as                          |
|-----|---------------------------------------------------------------------|------------------------------------------|
| 3.1 | Definition (Thickness)                                              | <a href="#">Note 62.12</a> , p. 352      |
| 3.2 | Lemma (Size and thickness of birthday cutoffs)                      | <a href="#">Note 62.12</a> , p. 352      |
| 3.3 | Lemma (Short-prefix extension)                                      | <a href="#">Note 62.12</a> , p. 352      |
| 3.4 | Proposition (The prefix topology is the numerical order topology)   | <a href="#">Note 62.12</a> , p. 352      |
| 3.5 | Proposition (Basic size and embedding spectra, recalled with proof) | <a href="#">Note 62.12</a> , p. 352      |
| 4.1 | Definition (Admissible coordinate ideal)                            | <a href="#">Definition 70.1</a> , p. 422 |
| 4.2 | Theorem (Exact ideal continuity criteria)                           | <a href="#">Theorem 70.2</a> , p. 422    |
| 4.3 | Corollary (The ideal-preserving subgroup)                           | <a href="#">Corollary 70.3</a> , p. 423  |
| 5.1 | Theorem (Maximal inherited topological subgroup)                    | <a href="#">Theorem 70.4</a> , p. 423    |
| 5.2 | Lemma (A bounded part and a cofinal part)                           | <a href="#">Lemma 70.5</a> , p. 423      |
| 5.3 | Theorem (Self-normalization at singular cardinalities)              | <a href="#">Theorem 70.6</a> , p. 424    |

| No.  | Statement in source 28                                                     | Printed here as                   |
|------|----------------------------------------------------------------------------|-----------------------------------|
| 5.4  | Remark (Why the cardinal enlargement is essential)                         | Remark 70.7, p. 424               |
| 5.5  | Proposition (Recovering boundedness from the natural action)               | Proposition 70.8, p. 424          |
| 6.1  | Lemma (Extending a bounded partial injection)                              | Lemma 70.9, p. 425                |
| 6.2  | Theorem (Exact forward closure)                                            | Theorem 70.10, p. 425             |
| 6.3  | Theorem (Inverse-boundedness as a dense intersection)                      | Theorem 70.11, p. 425             |
| 6.4  | Corollary (A complete core and the Baire property)                         | Corollary 70.12, p. 426           |
| 6.5  | Proposition (An explicit one-sided permutation)                            | Proposition 70.14, p. 427         |
| 6.6  | Corollary (A discrete subgroup outside the bounded subgroup)               | Corollary 70.15, p. 427           |
| 7.1  | Definition (Symmetric closure of a coordinate ideal)                       | Definition 70.16, p. 428          |
| 7.2  | Theorem (Smallest group refinement for an ideal)                           | Theorem 70.17, p. 428             |
| 7.3  | Theorem (The bounded-prefix group envelope)                                | Theorem 70.18, p. 428             |
| 7.4  | Proposition (Why two-sided bounded control is not enough)                  | Proposition 70.19, p. 429         |
| 8.1  | Proposition (Character of the forward space)                               | Note 70.20, p. 429                |
| 8.2  | Theorem (Character of the canonical envelope)                              | Theorem 70.21, p. 429             |
| 8.3  | Example (A countably based order with a very large group envelope)         | Example 70.22, p. 430             |
| 8.4  | Remark (What the character bound does not say)                             | Remark 70.23, p. 430              |
| 8.5  | Proposition (Equal global invariants can conceal different local geometry) | Proposition 70.24, p. 430         |
| 9.1  | Theorem (The forward ideal completion)                                     | Theorem 70.25, p. 430             |
| 9.2  | Theorem (Forward completion of the bounded subgroup)                       | Theorem 70.26, p. 431             |
| 9.3  | Theorem (Completeness of the canonical group envelope)                     | Theorem 70.27, p. 431             |
| 10.1 | Proposition (Exact filtration invariance)                                  | Proposition 70.28, p. 432         |
| 11.1 | Proposition (Why the global case has no singular-coordinate obstruction)   | Proposition 71.12, p. 442         |
| 11.2 | Lemma (Global prefix extension by a set-supported code)                    | Note 71.13, p. 443                |
| 11.3 | Proposition (Universality for ordinary class carriers)                     | Note 71.13, p. 443                |
| 11.4 | Theorem (The diagonal obstruction, explicitly typed)                       | Note 71.13, p. 443                |
| 11.5 | Proposition (Set-indexed prefix-Cauchy nets are eventually constant)       | Note 71.13, p. 443                |
| 12.1 | Lemma (Common-prefix lemma)                                                | Note 71.13, p. 443                |
| 12.2 | Theorem (Direct lexicography beyond set-like orders)                       | Note 71.13, p. 443                |
| 1    | Question (Complete metrizability of the full singular space)               | Research question 113.200, p. 940 |
| 2    | Question (The singular envelope's exact character)                         | Research question 113.201, p. 941 |
| 3    | Question (Descriptive complexity of the one-sided loci)                    | Research question 113.202, p. 941 |
| 4    | Question (Abstract automorphisms versus the natural normalizer)            | Research question 113.203, p. 941 |
| 5    | Question (Subgroups without all finitary permutations)                     | Research question 113.204, p. 941 |
| 6    | Question (Intermediate coordinate ideals)                                  | Research question 113.205, p. 941 |

| No. | Statement in source 28                                              | Printed here as                   |
|-----|---------------------------------------------------------------------|-----------------------------------|
| 7   | Question (Order automorphisms and the coordinate bornology)         | Research question 113.206, p. 942 |
| 8   | Question (Nonminimal ordinal lengths and residual tails)            | Research question 113.207, p. 942 |
| 9   | Question (Order-type classification beyond topology)                | Research question 113.208, p. 942 |
| 10  | Question (Global control schemes over weaker theories)              | Research question 113.209, p. 942 |
| 11  | Question (Non-set-like tails and class completion)                  | Research question 113.210, p. 943 |
| 12  | Question (A verified ideal-to-surreal pipeline)                     | Research question 113.211, p. 943 |
| B.1 | Proposition (Complete metric subspaces and ambient $G_\delta$ sets) | Note 69.9, p. 411                 |

## D.21 Source 29

| No.  | Statement in source 29                                | Printed here as           |
|------|-------------------------------------------------------|---------------------------|
| 2.1  | Definition (Uniform families)                         | Definition 80.2, p. 480   |
| 3.1  | Theorem (Inherited fixed-set spectrum)                | Note 61.4, p. 346         |
| 4.1  | Lemma (Equal-length fusion)                           | Lemma 76.1, p. 460        |
| 4.2  | Corollary (Cube hierarchy and finite factors)         | Note 76.2, p. 460         |
| 4.3  | Proposition (Finite terminal trees)                   | Proposition 76.3, p. 460  |
| 4.4  | Proposition (The missing full-height cube)            | Proposition 76.4, p. 461  |
| 4.5  | Theorem (Exact native birthday spectrum)              | Theorem 76.5, p. 462      |
| 4.6  | Corollary (A strict hierarchy on every birthday band) | Corollary 76.6, p. 462    |
| 4.7  | Corollary (Exact native binary coding lengths)        | Corollary 76.7, p. 462    |
| 5.1  | Lemma (Noninitial main-prefix package)                | Lemma 78.1, p. 467        |
| 5.2  | Proposition (Exact residual reduction)                | Proposition 78.2, p. 468  |
| 5.3  | Lemma (Entry codes and regular blocks)                | Note 78.3, p. 468         |
| 5.4  | Lemma (Inherited singular strong-limit scheduler)     | Note 78.4, p. 469         |
| 5.5  | Theorem (Noninitial coding sandwich)                  | Theorem 78.5, p. 469      |
| 5.6  | Lemma (Small-range fusion)                            | Note 78.6, p. 470         |
| 5.7  | Lemma (A singular non-strong-limit layer)             | Note 78.7, p. 470         |
| 5.8  | Theorem (Exact cube and birthday capacity)            | Theorem 78.8, p. 470      |
| 5.9  | Corollary (Coding bounds and exact equimorphisms)     | Corollary 78.9, p. 471    |
| 5.10 | Corollary (Every admissible length, without CH)       | Corollary 78.10, p. 471   |
| 6.1  | Theorem (Canonical factorization)                     | Note 82.9, p. 496         |
| 6.2  | Lemma (Uniform isomorphisms of admissible alphabets)  | Note 82.10, p. 497        |
| 6.3  | Corollary (The centered increasing-word product)      | Note 82.11, p. 498        |
| 6.4  | Lemma (Dictionary absorption)                         | Note 82.12, p. 498        |
| 6.5  | Theorem (The zero-convention isomorphism types)       | Note 82.13, p. 499        |
| 6.6  | Proposition (Threshold reductions)                    | Proposition 82.14, p. 500 |
| 7.1  | Lemma                                                 | Note 83.1, p. 501         |
| 7.2  | Theorem (Proper-class adjacency)                      | Theorem 83.2, p. 502      |
| 7.3  | Example (A jump over every surreal)                   | Example 83.3, p. 502      |
| 7.4  | Corollary                                             | Corollary 83.4, p. 503    |
| 7.5  | Theorem (Longer-first density)                        | Theorem 83.5, p. 503      |
| 7.6  | Theorem (Cofinal numerical representatives)           | Theorem 83.6, p. 503      |

| No.   | Statement in source 29                                                  | Printed here as                                    |
|-------|-------------------------------------------------------------------------|----------------------------------------------------|
| 7.7   | Theorem                                                                 | <a href="#">Theorem 83.7</a> , p. 504              |
| 8.1   | Lemma (A common completion)                                             | <a href="#">Lemma 68.1</a> , p. 391                |
| 8.2   | Theorem (The category change at one omission)                           | <a href="#">Theorem 68.2</a> , p. 392              |
| 8.3   | Proposition (The exact finite-tail reduction)                           | <a href="#">Proposition 68.6</a> , p. 395          |
| 8.4   | Corollary (All the finite successors have different order types)        | <a href="#">Corollary 68.7</a> , p. 395            |
| 8.5   | Theorem (Finite-tail topological classification)                        | <a href="#">Theorem 68.8</a> , p. 396              |
| 8.6   | Theorem (Exact comparison of native and coordinate topologies)          | <a href="#">Theorem 68.10</a> , p. 398             |
| 8.7   | Proposition (Low-complexity comparison is not low-complexity topology)  | <a href="#">Proposition 68.11</a> , p. 399         |
| 8.8   | Theorem (Borel sets in the double split)                                | <a href="#">Theorem 68.13</a> , p. 401             |
| 8.9   | Theorem (The measurable trichotomy for finite tails)                    | <a href="#">Theorem 68.15</a> , p. 402             |
| 8.10  | Theorem (Native comparison is not measurable for the rectangle product) | <a href="#">Theorem 68.20</a> , p. 406             |
| 9.1   | Lemma (Residual alphabets and a prescribed match)                       | <a href="#">Lemma 96.6</a> , p. 549                |
| 9.2   | Theorem (Branch transitivity of the ordered trees)                      | <a href="#">Theorem 96.7</a> , p. 550              |
| 9.3   | Corollary (Exhaustivity is additional information)                      | <a href="#">Note 96.8</a> , p. 551                 |
| 9.4   | Corollary (Numerical cofinality is also additional information)         | <a href="#">Corollary 96.9</a> , p. 551            |
| 9.5   | Proposition (Terminal nodes over a set alphabet)                        | <a href="#">Proposition 96.10</a> , p. 552         |
| 9.6   | Proposition (Fixed-stage continuation by residual bijections)           | <a href="#">Proposition 96.11</a> , p. 553         |
| 9.7   | Corollary (Agreement through any prescribed set stage)                  | <a href="#">Corollary 96.12</a> , p. 553           |
| 10.1  | Proposition (Inherited direct comparison)                               | <a href="#">Note 86.17</a> , p. 528                |
| 10.2  | Theorem (Inherited core theorem)                                        | <a href="#">Note 86.18</a> , p. 529                |
| 10.3  | Proposition (Diagonal boundary)                                         | <a href="#">Note 86.19</a> , p. 529                |
| 11.1  | Lemma (One-step block facts)                                            | <a href="#">Note 108.9</a> , p. 616                |
| 11.2  | Definition (Canonical history)                                          | <a href="#">Definition 108.10</a> , p. 616         |
| 11.3  | Proposition (Termination for the input-order schedule; earlier result)  | <a href="#">Note 108.11</a> , p. 616               |
| 11.4  | Lemma (Canonical floor maps)                                            | <a href="#">Lemma 108.12</a> , p. 617              |
| 11.5  | Proposition (Each point changes its floor only finitely often)          | <a href="#">Proposition 108.13</a> , p. 617        |
| 11.6  | Corollary (Continuous identification at limits)                         | <a href="#">Corollary 108.14</a> , p. 618          |
| 11.7  | Lemma (Polynomial well-orders and their explicit histories)             | <a href="#">Note 108.15</a> , p. 618               |
| 11.8  | Theorem (Canonical finite normal form)                                  | <a href="#">Theorem 108.16</a> , p. 619            |
| 11.9  | Lemma (Anchored convex restriction)                                     | <a href="#">Lemma 108.17</a> , p. 620              |
| 11.10 | Corollary (The exact criterion for a fixed schedule)                    | <a href="#">Corollary 108.18</a> , p. 621          |
| 11.11 | Theorem (Exact strength of canonical histories)                         | <a href="#">Theorem 108.19</a> , p. 621            |
| 11.12 | Corollary (The surreal carrier alone suffices)                          | <a href="#">Corollary 108.20</a> , p. 622          |
| 13.1  | Research question (The remaining singular upper code)                   | <a href="#">Research question 113.212</a> , p. 943 |
| 13.2  | Research question (Isomorphism on a constant-cardinality birthday band) | <a href="#">Research question 113.213</a> , p. 944 |
| 13.3  | Research question (Infinite tails at the countable alphabet)            | <a href="#">Research question 113.214</a> , p. 944 |



| No.   | Statement in source 29                                                    | Printed here as                                    |
|-------|---------------------------------------------------------------------------|----------------------------------------------------|
| 13.4  | Research question (Intrinsic geometry of the centered product)            | <a href="#">Research question 113.215</a> , p. 944 |
| 13.5  | Research question (Nonprincipal termination cuts)                         | <a href="#">Research question 113.216</a> , p. 944 |
| 13.6  | Research question (The dense quotient of numerical class carriers)        | <a href="#">Research question 113.217</a> , p. 944 |
| 13.7  | Research question (The exhaustive-branch stabilizer)                      | <a href="#">Research question 113.218</a> , p. 945 |
| 13.8  | Research question (Comparability versus stronger recursion)               | <a href="#">Research question 113.219</a> , p. 945 |
| 13.9  | Research question (Alternative condensation limits)                       | <a href="#">Research question 113.220</a> , p. 945 |
| 13.10 | Research question (Recovering a class realization from an expanded order) | <a href="#">Research question 113.221</a> , p. 945 |
| 13.11 | Research question (Scan equivalence for relation encodings)               | <a href="#">Research question 113.222</a> , p. 945 |
| 13.12 | Research question (A certified mathematical core)                         | <a href="#">Research question 113.223</a> , p. 945 |

## D.22 Source 30

| No. | Statement in source 30                                                 | Printed here as                           |
|-----|------------------------------------------------------------------------|-------------------------------------------|
| 3.1 | Theorem (Inherited choice equivalence)                                 | <a href="#">Note 111.27</a> , p. 710      |
| 3.2 | Proposition (Inherited common-prefix comparison)                       | <a href="#">Note 111.28</a> , p. 710      |
| 4.1 | Lemma (Short codes for arbitrary regular-cardinality alphabets)        | <a href="#">Note 64.15</a> , p. 369       |
| 4.2 | Lemma (Exact coding length of an arbitrary long prefix)                | <a href="#">Note 64.15</a> , p. 369       |
| 4.3 | Lemma (Small deletion in the singular scheduler)                       | <a href="#">Note 64.15</a> , p. 369       |
| 4.4 | Theorem (A smaller regular-cardinal block after the long prefix)       | <a href="#">Note 64.16</a> , p. 369       |
| 4.5 | Corollary (Every countable remainder)                                  | <a href="#">Note 64.16</a> , p. 369       |
| 4.6 | Corollary (Exact coding and cube spectrum)                             | <a href="#">Note 64.16</a> , p. 369       |
| 4.7 | Remark                                                                 | <a href="#">Note 64.16</a> , p. 369       |
| 4.8 | Corollary (Finite descent through regular residual cardinals)          | <a href="#">Note 64.16</a> , p. 369       |
| 5.1 | Theorem (A singular strong-limit residual block)                       | <a href="#">Note 64.17</a> , p. 369       |
| 5.2 | Corollary (A strict obstruction to the uninflated residual formula)    | <a href="#">Note 64.17</a> , p. 369       |
| 5.3 | Corollary (Finite descent through both controlled cardinal regimes)    | <a href="#">Note 64.18</a> , p. 370       |
| 5.4 | Corollary (All fixed types when smaller cardinals are controlled)      | <a href="#">Note 64.18</a> , p. 370       |
| 5.5 | Proposition (Residual cardinals above the birthday cutoff)             | <a href="#">Note 64.19</a> , p. 370       |
| 5.6 | Corollary (The first uncountable cutoff needs no continuum hypothesis) | <a href="#">Corollary 64.20</a> , p. 371  |
| 6.1 | Lemma (Prefix topology and finite fibers)                              | <a href="#">Lemma 68.3</a> , p. 394       |
| 6.2 | Proposition (The category and Borel class of cofinite prefixes)        | <a href="#">Proposition 68.4</a> , p. 394 |
| 6.3 | Corollary (An explicit Baire-space presentation)                       | <a href="#">Note 68.5</a> , p. 395        |
| 6.4 | Theorem (Finite-tail topology)                                         | <a href="#">Theorem 68.9</a> , p. 397     |
| 6.5 | Corollary (Weight for arbitrary infinite alphabets)                    | <a href="#">Corollary 63.12</a> , p. 359  |

| No.  | Statement in source 30                                                        | Printed here as                   |
|------|-------------------------------------------------------------------------------|-----------------------------------|
| 6.6  | Proposition (The fixed finite-tail table domains)                             | Proposition 68.12, p. 400         |
| 6.7  | Lemma (Borel sets split only countably many fibers)                           | Note 68.14, p. 402                |
| 6.8  | Theorem (Ranking and relation-table maps)                                     | Theorem 68.16, p. 403             |
| 6.9  | Corollary (Nonmeasurable ranking at every later countable type)               | Corollary 68.17, p. 405           |
| 6.10 | Proposition (One orientation predicate restores the standard Borel structure) | Proposition 68.18, p. 405         |
| 6.11 | Remark (Sigma-algebra inclusion is not topology inclusion)                    | Remark 68.19, p. 405              |
| 7.1  | Theorem (Extension with an arbitrarily prescribed global type)                | Theorem 84.7, p. 510              |
| 7.2  | Theorem (Exact local-approximation criteria)                                  | Theorem 84.8, p. 511              |
| 7.3  | Proposition (Branches and stabilization)                                      | Proposition 84.9, p. 512          |
| 7.4  | Corollary (Dense type fibers and continuous invariants)                       | Corollary 84.10, p. 512           |
| 8.1  | Theorem (Scan-independent density of the set-like part)                       | Theorem 84.11, p. 513             |
| 8.2  | Theorem (Canonical table characters)                                          | Theorem 84.13, p. 514             |
| 8.3  | Corollary (The eventual-equality core in the canonical table order)           | Corollary 84.14, p. 514           |
| 9.1  | Proposition (Pair coding with an elementary inverse)                          | Note 102.15, p. 587               |
| 9.2  | Theorem (Reconstruction from labelled enumeration availability)               | Note 102.16, p. 587               |
| 9.3  | Corollary (External size)                                                     | Note 102.17, p. 588               |
| 9.4  | Corollary (The represented-cut version)                                       | Note 102.18, p. 589               |
| 9.5  | Proposition (A core preserved setwise and a core fixed pointwise)             | Proposition 102.19, p. 589        |
| 9.6  | Proposition (Pure order does not detect truth or ETR)                         | Proposition 102.20, p. 590        |
| 9.7  | Theorem (Definability threshold)                                              | Theorem 102.21, p. 590            |
| 9.8  | Corollary (Universal availability is definable availability)                  | Corollary 102.22, p. 591          |
| 10.1 | Proposition (Finite factorial blocks are not first-order protected)           | Proposition 109.37, p. 645        |
| 12.1 | Question (Singular residual cardinals that are not strong limit)              | Research question 113.224, p. 946 |
| 12.2 | Question (Dependence on cardinal arithmetic)                                  | Research question 113.225, p. 946 |
| 12.3 | Question (The positive finite-prefix spaces)                                  | Research question 113.226, p. 946 |
| 12.4 | Question (Later countable ordinal types)                                      | Research question 113.227, p. 947 |
| 12.5 | Question (Reverse coding at later countable types)                            | Research question 113.228, p. 947 |
| 12.6 | Question (Arbitrary scans and non-set-like jumps)                             | Research question 113.229, p. 947 |
| 12.7 | Question (Topologies determined by scans)                                     | Research question 113.230, p. 947 |
| 12.8 | Question (How much named core suffices?)                                      | Research question 113.231, p. 947 |
| 12.9 | Question (Uncountable realizations and common baselines)                      | Research question 113.232, p. 947 |

[merge] (3 October 2026, batch 89.) Source 31 follows, with the same conventions. Its statement  $k.m$  is printed as XV. $k.m$  (part XV); its research questions, numbered 1–12 in its own PDF, are printed in section 113.26.

**D.23 Source 31**

| No.  | Statement in source 31                                        | Printed here as                   |
|------|---------------------------------------------------------------|-----------------------------------|
| 3.1  | Lemma (Bounded well-order construction)                       | Lemma XV.3.1, p. 721              |
| 3.2  | Remark (Why ordinary iteration is insufficient)               | Remark XV.3.2, p. 722             |
| 3.3  | Theorem (Uniform presentation theorem)                        | Theorem XV.3.3, p. 722            |
| 3.4  | Remark (Uniformity is not compatibility under restriction)    | Remark XV.3.4, p. 723             |
| 3.5  | Corollary (Uniform countable enumeration)                     | Corollary XV.3.5, p. 723          |
| 4.1  | Theorem (Bounded trace compression)                           | Theorem XV.4.1, p. 723            |
| 4.2  | Corollary (Compression from a restricted selector)            | Corollary XV.4.2, p. 724          |
| 5.1  | Theorem (Countable-input nonconservativity)                   | Theorem XV.5.1, p. 725            |
| 5.2  | Remark (Metatheory)                                           | Remark XV.5.2, p. 725             |
| 5.3  | Corollary (A single infinite input cardinality suffices)      | Corollary XV.5.3, p. 726          |
| 5.4  | Corollary (Any fixed infinite bound)                          | Corollary XV.5.4, p. 726          |
| 5.5  | Proposition (An uninteracting selector is conservative)       | Proposition XV.5.5, p. 726        |
| 6.1  | Proposition (Available axioms)                                | Proposition XV.6.1, p. 727        |
| 6.2  | Proposition (Choice orders without ordinal enumeration)       | Proposition XV.6.2, p. 727        |
| 7.1  | Lemma (Size of the sign carrier)                              | Lemma XV.7.1, p. 728              |
| 7.2  | Theorem (Bare ordinal enumeration)                            | Theorem XV.7.2, p. 728            |
| 7.3  | Definition                                                    | Definition XV.7.3, p. 728         |
| 7.4  | Theorem (Birthday enumeration threshold)                      | Theorem XV.7.4, p. 729            |
| 7.5  | Remark                                                        | Remark XV.7.5, p. 729             |
| 8.1  | Lemma (Growth facts)                                          | Lemma XV.8.1, p. 729              |
| 8.2  | Theorem (Universal coding threshold)                          | Theorem XV.8.2, p. 729            |
| 8.3  | Definition (Ordinal abstraction)                              | Definition XV.8.3, p. 730         |
| 8.4  | Theorem (Ordinal abstraction threshold)                       | Theorem XV.8.4, p. 730            |
| 9.1  | Definition                                                    | Definition XV.9.1, p. 730         |
| 9.2  | Lemma (The domain-size criterion)                             | Lemma XV.9.2, p. 731              |
| 9.3  | Theorem (Replacement thresholds)                              | Theorem XV.9.3, p. 731            |
| 9.4  | Corollary (Separating the two requirements)                   | Corollary XV.9.4, p. 731          |
| 9.5  | Remark (Ordinal domains are not all well-orderable domains)   | Remark XV.9.5, p. 731             |
| 9.6  | Remark (Full classes versus definable classes)                | Remark XV.9.6, p. 732             |
| 10.1 | Theorem (Proper-class cardinal spectrum)                      | Theorem XV.10.1, p. 732           |
| 10.2 | Corollary (Two different limitation-of-size conditions)       | Corollary XV.10.2, p. 732         |
| 1    | Question (Global selection from finite sets)                  | Research question 113.233, p. 950 |
| 2    | Question (Pairs and bounded finite cardinalities)             | Research question 113.234, p. 950 |
| 3    | Question (The Separation complexity needed by the refinement) | Research question 113.235, p. 951 |
| 4    | Question (Reducing ordinary Choice)                           | Research question 113.236, p. 951 |
| 5    | Question (Fragments that restore conservativity)              | Research question 113.237, p. 951 |
| 6    | Question (Selectors beyond the countable threshold)           | Research question 113.238, p. 951 |
| 7    | Question (Definable-class versions of the spectra)            | Research question 113.239, p. 951 |
| 8    | Question (Abstract weak-theory equivalences)                  | Research question 113.240, p. 951 |
| 9    | Question (All set-like order types at a cutoff)               | Research question 113.241, p. 952 |
| 10   | Question (The repository's global control schemes)            | Research question 113.242, p. 952 |
| 11   | Question (A verified bounded uniformization theorem)          | Research question 113.243, p. 952 |

| No. | Statement in source 31                                     | Printed here as                                    |
|-----|------------------------------------------------------------|----------------------------------------------------|
| 12  | Question (Surreal transfer and parameter-sensitive choice) | <a href="#">Research question 113.244</a> , p. 952 |

## D.24 Source 34 (base of Part XVI)

| No.  | Statement in source 34                                                 | Printed here as                                    |
|------|------------------------------------------------------------------------|----------------------------------------------------|
| 2.1  | Definition (Class well-order)                                          | <a href="#">Definition XVI.2.1</a> , p. 745        |
| 2.2  | Lemma (A first-order criterion for linear orders)                      | <a href="#">Lemma XVI.2.2</a> , p. 745             |
| 3.1  | Theorem (The set-like boundary)                                        | <a href="#">Theorem XVI.3.1</a> , p. 750           |
| 3.2  | Lemma (No backward self-embedding)                                     | <a href="#">Lemma XVI.3.2</a> , p. 751             |
| 3.3  | Theorem (One-step block decomposition)                                 | <a href="#">Theorem XVI.3.3</a> , p. 752           |
| 4.1  | Definition (Finite-support power)                                      | <a href="#">Definition XVI.4.1</a> , p. 757        |
| 4.2  | Theorem (Choice-free power theorem)                                    | <a href="#">Theorem XVI.4.2</a> , p. 757           |
| 4.3  | Proposition (The exponent is not a power class)                        | <a href="#">Proposition XVI.4.3</a> , p. 758       |
| 5.1  | Definition (Canonical history)                                         | <a href="#">Definition XVI.5.1</a> , p. 767        |
| 5.2  | Lemma (Termination on the input schedule)                              | <a href="#">Lemma XVI.5.2</a> , p. 767             |
| 5.3  | Lemma (Floors at all stages)                                           | <a href="#">Lemma XVI.5.3</a> , p. 768             |
| 5.4  | Lemma (Finitely many changes)                                          | <a href="#">Lemma XVI.5.4</a> , p. 768             |
| 5.5  | Theorem (Finite normal form with initial image)                        | <a href="#">Theorem XVI.5.5</a> , p. 769           |
| 5.6  | Lemma (Anchored convex restriction)                                    | <a href="#">Lemma XVI.5.6</a> , p. 769             |
| 5.7  | Corollary (Exact fixed-schedule criterion)                             | <a href="#">Corollary XVI.5.7</a> , p. 770         |
| 5.8  | Theorem (Canonical histories over GB)                                  | <a href="#">Theorem XVI.5.8</a> , p. 770           |
| 5.9  | Proposition (Uniqueness with a fixed schedule)                         | <a href="#">Proposition XVI.5.9</a> , p. 771       |
| 6.1  | Definition (Hereditary base- $\Omega$ terms)                           | <a href="#">Definition XVI.6.1</a> , p. 783        |
| 6.2  | Lemma (Bounded-coefficient comparison)                                 | <a href="#">Lemma XVI.6.2</a> , p. 783             |
| 6.3  | Theorem (Well-foundedness in GB)                                       | <a href="#">Theorem XVI.6.3</a> , p. 783           |
| 6.4  | Example (Reading and comparing terms)                                  | <a href="#">Example XVI.6.4</a> , p. 784           |
| 6.5  | Theorem (Exact height cuts)                                            | <a href="#">Theorem XVI.6.5</a> , p. 784           |
| 6.6  | Theorem (A definable fixed point)                                      | <a href="#">Theorem XVI.6.6</a> , p. 785           |
| 6.7  | Theorem (Least initial pre-fixed point)                                | <a href="#">Theorem XVI.6.7</a> , p. 785           |
| 6.8  | Corollary (An explicit long condensation history)                      | <a href="#">Corollary XVI.6.8</a> , p. 786         |
| 6.9  | Theorem (Bounded-coefficient order type)                               | <a href="#">Theorem XVI.6.9</a> , p. 786           |
| 6.10 | Corollary (Other class bases)                                          | <a href="#">Corollary XVI.6.10</a> , p. 787        |
| 7.1  | Definition (Uniform class family)                                      | <a href="#">Definition XVI.7.1</a> , p. 800        |
| 7.2  | Theorem (Uniform-family diagonal)                                      | <a href="#">Theorem XVI.7.2</a> , p. 800           |
| 7.3  | Corollary (No uniform cofinal catalog)                                 | <a href="#">Corollary XVI.7.3</a> , p. 800         |
| 7.4  | Theorem (Relative truth jump)                                          | <a href="#">Theorem XVI.7.4</a> , p. 801           |
| 7.5  | Corollary (Growth along iterated truth)                                | <a href="#">Corollary XVI.7.5</a> , p. 802         |
| 7.6  | Corollary (Bounded-fragment jumps)                                     | <a href="#">Corollary XVI.7.6</a> , p. 803         |
| 8.1  | Theorem (Full spectrum)                                                | <a href="#">Theorem XVI.8.1</a> , p. 812           |
| 8.2  | Proposition (The parameter spectrum)                                   | <a href="#">Proposition XVI.8.2</a> , p. 812       |
| 8.3  | Theorem (Parameter-free cofinality, relative to a global order)        | <a href="#">Theorem XVI.8.3</a> , p. 813           |
| 8.4  | Corollary                                                              | <a href="#">Corollary XVI.8.4</a> , p. 814         |
| R1   | Agenda item (Kernel-check the GB refinement)                           | <a href="#">Research question 113.245</a> , p. 954 |
| R2   | Agenda item (Support conditions beyond finite trees)                   | <a href="#">Research question 113.274</a> , p. 959 |
| R3   | Agenda item (Normalization of arithmetic inside the hereditary system) | <a href="#">Research question 113.275</a> , p. 959 |
| R4   | Agenda item (Iterated fixed-point notation)                            | <a href="#">Research question 113.272</a> , p. 959 |

| No. | Statement in source 34                                              | Printed here as                   |
|-----|---------------------------------------------------------------------|-----------------------------------|
| R5  | Agenda item (Comparability versus elementary transfinite recursion) | Research question 113.252, p. 955 |
| R6  | Agenda item (Weak versus strong comparison)                         | Research question 113.256, p. 956 |
| R7  | Agenda item (Restricted history principles)                         | Research question 113.261, p. 957 |
| R8  | Agenda item (Ordinal parameters versus arbitrary set parameters)    | Research question 113.282, p. 961 |
| R9  | Agenda item (The invariant $\Theta_{A,G}$ )                         | Research question 113.283, p. 961 |
| R10 | Agenda item (How large is a single truth jump?)                     | Research question 113.289, p. 962 |
| R11 | Agenda item (Forcing and new class-order types)                     | Research question 113.292, p. 963 |
| R12 | Agenda item (A proof-carrying notation interface)                   | Research question 113.250, p. 954 |

## D.25 Source 35

| No.  | Statement in source 35                                      | Printed here as             |
|------|-------------------------------------------------------------|-----------------------------|
| 2.1  | Definition (Class well-order)                               | Note XVI.2.3, p. 746        |
| 2.2  | Proposition (The exact boundary of set-likeness)            | Note XVI.2.4, p. 747        |
| 2.3  | Lemma (Rigidity and the successor obstruction)              | Lemma XVI.2.5, p. 747       |
| 2.4  | Remark (Set-valued and class-valued recursion)              | Remark XVI.2.6, p. 748      |
| 3.1  | Theorem (Choice-free finite-support theorem)                | Note XVI.4.4, p. 760        |
| 3.2  | Lemma (Preservation of initial embeddings)                  | Lemma XVI.4.5, p. 761       |
| 3.3  | Proposition (Set-sized operands at the class boundary)      | Proposition XVI.4.6, p. 762 |
| 3.4  | Proposition (Cofinality of a limit exponent)                | Proposition XVI.4.7, p. 763 |
| 3.5  | Lemma (Bounded-specialization principle)                    | Lemma XVI.4.8, p. 763       |
| 4.1  | Lemma (Set-interval blocks)                                 | Note XVI.5.11, p. 772       |
| 4.2  | Definition (Canonical condensation history)                 | Note XVI.5.12, p. 773       |
| 4.3  | Proposition (Input-order termination)                       | Note XVI.5.13, p. 773       |
| 4.4  | Lemma (Canonical floors)                                    | Note XVI.5.14, p. 773       |
| 4.5  | Lemma (Finite changes)                                      | Note XVI.5.15, p. 773       |
| 4.6  | Theorem (Finite normal form from a supplied history)        | Note XVI.5.16, p. 774       |
| 4.7  | Lemma (Anchored restriction)                                | Note XVI.5.17, p. 774       |
| 4.8  | Theorem (Exact criterion for a fixed schedule)              | Note XVI.5.18, p. 774       |
| 4.9  | Theorem (Choice-free history-comparison equivalence)        | Note XVI.5.19, p. 775       |
| 5.1  | Definition (Normalized hereditary terms)                    | Note XVI.6.11, p. 787       |
| 5.2  | Lemma (Bounded specialization)                              | Lemma XVI.6.12, p. 788      |
| 5.3  | Theorem (The first explicit polynomial fixed point)         | Note XVI.6.13, p. 788       |
| 5.4  | Proposition (Exact tower cuts and cofinality)               | Note XVI.6.14, p. 788       |
| 5.5  | Theorem (Leastness with an explicit witness)                | Note XVI.6.15, p. 788       |
| 5.6  | Lemma (Finite-support preservation)                         | Note XVI.6.16, p. 790       |
| 5.7  | Theorem (Explicit fixed-point completion)                   | Theorem XVI.6.17, p. 790    |
| 5.8  | Theorem (Local fixed-point extension criterion)             | Theorem XVI.6.18, p. 791    |
| 5.9  | Corollary (A fixed-point characterization of comparability) | Corollary XVI.6.19, p. 792  |
| 5.10 | Corollary (Successor fixed-point extension)                 | Corollary XVI.6.20, p. 792  |
| 6.1  | Proposition (Meaning of a polynomial as a cut)              | Note XVI.6.22, p. 795       |
| 7.1  | Lemma (An appended point cannot be absorbed)                | Note XVI.7.7, p. 804        |
| 7.2  | Theorem (Strict bound for a uniform family)                 | Note XVI.7.8, p. 804        |
| 7.3  | Corollary (No cofinal uniform notation system)              | Corollary XVI.7.9, p. 804   |

| No.   | Statement in source 35                                                       | Printed here as                   |
|-------|------------------------------------------------------------------------------|-----------------------------------|
| 7.4   | Lemma (Ordinal-carrier well-foundedness)                                     | Lemma XVI.7.10, p. 805            |
| 7.5   | Theorem (Fixed-complexity separation of class order types)                   | Theorem XVI.7.11, p. 805          |
| 7.6   | Corollary (The separation family has no uniform ordinal-parameter evaluator) | Corollary XVI.7.12, p. 806        |
| 7.7   | Proposition (Version with arbitrary set parameters)                          | Note XVI.7.13, p. 807             |
| 9.1   | Theorem (Full external spectrum)                                             | Note XVI.8.5, p. 815              |
| 9.2   | Theorem (Definability-sensitive spectrum bounds)                             | Theorem XVI.8.6, p. 816           |
| 10.1  | Question (Formalize the finite-support minimization theorem)                 | Research question 113.246, p. 954 |
| 10.2  | Question (Verify the complete normal-form argument)                          | Research question 113.247, p. 954 |
| 10.3  | Question (Does comparability yield elementary transfinite recursion?)        | Research question 113.253, p. 955 |
| 10.4  | Question (Uniform finite condensation)                                       | Research question 113.260, p. 956 |
| 10.5  | Question (Weak comparison and ranking principles)                            | Research question 113.257, p. 956 |
| 10.6  | Question (A choice-free completion theorem)                                  | Research question 113.270, p. 958 |
| 10.7  | Question (Completion as a universal construction)                            | Research question 113.271, p. 958 |
| 10.8  | Question (Successive fixed points and Veblen-style systems)                  | Research question 113.273, p. 959 |
| 10.9  | Question (Quantitative complexity of escape orders)                          | Research question 113.277, p. 960 |
| 10.10 | Question (Characterize the definability-sensitive cutoffs)                   | Research question 113.287, p. 962 |
| 10.11 | Question (Optimal uniform-family bounding)                                   | Research question 113.280, p. 960 |
| 10.12 | Question (Classify isomorphism-level condensation fixed points)              | Research question 113.267, p. 958 |
| 10.13 | Question (Alternative limits with an explicit universal property)            | Research question 113.268, p. 958 |

## D.26 Source 32

| No. | Statement in source 32                                | Printed here as             |
|-----|-------------------------------------------------------|-----------------------------|
| 2.1 | Definition                                            | Note XVI.2.7, p. 748        |
| 2.2 | Lemma (The descending-sequence test)                  | Lemma XVI.2.8, p. 748       |
| 2.3 | Theorem (The exact set-like boundary)                 | Note XVI.2.9, p. 749        |
| 2.4 | Lemma (No self-compression)                           | Note XVI.2.10, p. 749       |
| 3.1 | Proposition (Uniform sum theorem)                     | Proposition XVI.3.4, p. 753 |
| 3.2 | Lemma (Set-sized digit representation)                | Lemma XVI.4.9, p. 764       |
| 3.3 | Theorem (Explicit class exponentiation in GBC)        | Note XVI.4.10, p. 764       |
| 3.4 | Theorem (Exponent laws by explicit maps)              | Note XVI.4.11, p. 765       |
| 4.1 | Definition (Hereditary digit terms)                   | Note XVI.6.23, p. 797       |
| 4.2 | Theorem (A hereditary fixed-point presentation)       | Note XVI.6.24, p. 797       |
| 5.1 | Lemma                                                 | Note XVI.3.5, p. 754        |
| 5.2 | Theorem (Canonical $\Omega$ -division)                | Theorem XVI.3.6, p. 754     |
| 5.3 | Proposition (Set cofinality or an $\Omega$ -sequence) | Proposition XVI.3.7, p. 755 |
| 6.1 | Theorem (Conditional general termination)             | Note XVI.5.20, p. 776       |
| 6.2 | Theorem (Exact polynomial condensation length)        | Note XVI.5.21, p. 777       |



| No.   | Statement in source 32                                         | Printed here as                   |
|-------|----------------------------------------------------------------|-----------------------------------|
| 6.3   | Corollary (Isomorphic quotients can hide real progress)        | Corollary XVI.5.22, p. 777        |
| 7.1   | Theorem (Uniform diagonal domination)                          | Note XVI.7.14, p. 807             |
| 7.2   | Corollary (A fixed finite syntax has a definable bound)        | Corollary XVI.7.15, p. 807        |
| 7.3   | Corollary (No universal admitted list)                         | Note XVI.7.16, p. 808             |
| 8.1   | Lemma (The inaccessible-rank model)                            | Lemma XVI.8.7, p. 818             |
| 8.2   | Theorem (The full-class spectrum)                              | Note XVI.8.8, p. 818              |
| 8.3   | Theorem (Definability ceiling)                                 | Theorem XVI.8.9, p. 818           |
| 8.4   | Remark (What the ceiling does and does not mean)               | Remark XVI.8.10, p. 819           |
| 8.5   | Corollary                                                      | Corollary XVI.8.11, p. 820        |
| 9.1   | Theorem (Exact truth presentation and strict jump)             | Theorem XVI.8.12, p. 820          |
| 9.2   | Corollary (The old bounds cannot have been uniformly admitted) | Corollary XVI.8.13, p. 821        |
| 12.1  | Question (Exact strength of condensation existence)            | Research question 113.258, p. 956 |
| 12.2  | Question (Optimal bounds for individual orders)                | Research question 113.264, p. 957 |
| 12.3  | Question (Normal forms after repeated $\Omega$ -division)      | Research question 113.266, p. 957 |
| 12.4  | Question (Classifying definability ceilings)                   | Research question 113.284, p. 961 |
| 12.5  | Question (Quantitative truth jumps)                            | Research question 113.290, p. 962 |
| 12.6  | Question (Syntactic complexity of diagonal bounds)             | Research question 113.278, p. 960 |
| 12.7  | Question (Parameter-free and restricted-parameter spectra)     | Research question 113.288, p. 962 |
| 12.8  | Question (Comparison and recursion)                            | Research question 113.254, p. 955 |
| 12.9  | Question (Changes of class realization and forcing)            | Research question 113.293, p. 963 |
| 12.10 | Question (Reducing choice and background strength)             | Research question 113.262, p. 957 |
| 12.11 | Question (A verified presentation library)                     | Research question 113.248, p. 954 |

## D.27 Source 33

| No. | Statement in source 33                             | Printed here as              |
|-----|----------------------------------------------------|------------------------------|
| 2.1 | Definition                                         | Note XVI.2.11, p. 749        |
| 2.2 | Lemma (The descending-sequence test)               | Note XVI.2.12, p. 749        |
| 2.3 | Remark (Internal versus external well-foundedness) | Remark XVI.2.13, p. 750      |
| 2.4 | Proposition                                        | Note XVI.2.14, p. 750        |
| 2.5 | Lemma (Inflation of a self-embedding)              | Note XVI.2.15, p. 750        |
| 2.6 | Corollary (The successor obstruction)              | Note XVI.2.16, p. 750        |
| 3.1 | Definition (Uniform family)                        | Note XVI.3.8, p. 756         |
| 3.2 | Proposition                                        | Note XVI.3.9, p. 756         |
| 3.3 | Theorem (Arithmetic closure)                       | Note XVI.4.12, p. 766        |
| 3.4 | Proposition (Explicit arithmetic laws)             | Note XVI.4.13, p. 766        |
| 3.5 | Proposition                                        | Proposition XVI.4.14, p. 766 |
| 3.6 | Remark (Not cardinal exponentiation)               | Remark XVI.4.15, p. 766      |
| 4.1 | Lemma                                              | Note XVI.3.10, p. 756        |
| 4.2 | Theorem (Canonical $\Omega$ -division)             | Note XVI.3.11, p. 756        |
| 4.3 | Example                                            | Example XVI.3.12, p. 757     |
| 5.1 | Definition (Exact condensation tower)              | Definition XVI.5.23, p. 778  |

| No.   | Statement in source 33                                                 | Printed here as                   |
|-------|------------------------------------------------------------------------|-----------------------------------|
| 5.2   | Theorem (Exact polynomial condensation)                                | Note XVI.5.25, p. 779             |
| 5.3   | Example (Stationary quotient types, nonstationary identifications)     | Example XVI.5.26, p. 779          |
| 5.4   | Theorem (Finite-digit embedding)                                       | Theorem XVI.5.27, p. 779          |
| 5.5   | Remark (What the theorem does not claim)                               | Remark XVI.5.28, p. 780           |
| 5.6   | Proposition (Conditional converse)                                     | Proposition XVI.5.30, p. 781      |
| 5.7   | Proposition (Exhaustion under a sufficient recursion principle)        | Note XVI.5.32, p. 782             |
| 6.1   | Lemma                                                                  | Lemma XVI.6.25, p. 798            |
| 6.2   | Theorem (A definable fixed point beyond finite towers)                 | Theorem XVI.6.26, p. 799          |
| 7.1   | Theorem (Uniform-family diagonal)                                      | Note XVI.7.17, p. 808             |
| 7.2   | Corollary (No uniform catalogue)                                       | Note XVI.7.18, p. 808             |
| 8.1   | Lemma (Partial evaluation at a fixed level)                            | Lemma XVI.7.19, p. 809            |
| 8.2   | Theorem (Bounded definability cannot bound class-order length)         | Theorem XVI.7.20, p. 809          |
| 8.3   | Corollary (Parameter-free presentations are externally cofinal)        | Corollary XVI.7.21, p. 810        |
| 8.4   | Corollary (No definable uniform cofinal family)                        | Corollary XVI.7.22, p. 811        |
| 8.5   | Corollary                                                              | Corollary XVI.7.23, p. 811        |
| 9.1   | Lemma (Correct well-foundedness at the cutoff)                         | Note XVI.8.14, p. 822             |
| 9.2   | Theorem (Exact full-class spectrum)                                    | Note XVI.8.15, p. 822             |
| 9.3   | Theorem (Countably cofinal definability height)                        | Note XVI.8.16, p. 823             |
| 9.4   | Corollary (Arithmetic closure of the definability ceiling)             | Corollary XVI.8.17, p. 824        |
| 9.5   | Remark (Two different cofinalities)                                    | Remark XVI.8.18, p. 824           |
| 9.6   | Theorem (A strict truth jump in order height)                          | Theorem XVI.8.19, p. 824          |
| 12.1  | Research question (Sharp complexity of the diagonal)                   | Research question 113.279, p. 960 |
| 12.2  | Research question (The role of a definable global order)               | Research question 113.281, p. 961 |
| 12.3  | Research question (Identifying the external ceiling)                   | Research question 113.285, p. 962 |
| 12.4  | Research question (Dependence on the global-order predicate)           | Research question 113.286, p. 962 |
| 12.5  | Research question (Iterated truth growth)                              | Research question 113.291, p. 963 |
| 12.6  | Research question (The exact recursion strength of exhaustion)         | Research question 113.259, p. 956 |
| 12.7  | Research question (Sharper canonical digit representations)            | Research question 113.263, p. 957 |
| 12.8  | Research question (Minimal condensation lengths)                       | Research question 113.265, p. 957 |
| 12.9  | Research question (Other smallness ideals)                             | Research question 113.269, p. 958 |
| 12.10 | Research question (Comparison versus recursion: a published interface) | Research question 113.255, p. 956 |
| 12.11 | Research question (Preservation of definability heights)               | Research question 113.294, p. 963 |
| 12.12 | Research question (Proof-carrying notation systems)                    | Research question 113.251, p. 955 |
| 12.13 | Research question (Formal reverse mathematics)                         | Research question 113.249, p. 954 |
| 12.14 | Research question (Higher-order extension without a final universe)    | Research question 113.276, p. 960 |

**D.28 Source 36 (Part XVII)**

[merge] Source 36's statement  $k.m$  is printed as XVII. $k.m$ ; its research questions are printed in [section 113.30](#). The credit notes XVII. $k.A$ , ... of Part XVII are not source statements and are not listed.

| No.  | Statement in source 36                                     | Printed here as                                |
|------|------------------------------------------------------------|------------------------------------------------|
| 2.1  | Definition                                                 | <a href="#">Definition XVII.2.1</a> , p. 852   |
| 2.2  | Lemma (Set localization)                                   | <a href="#">Lemma XVII.2.2</a> , p. 853        |
| 2.3  | Theorem (The ordinal initial segment)                      | <a href="#">Theorem XVII.2.3</a> , p. 853      |
| 2.4  | Proposition (Reduction to the carrier Ord)                 | <a href="#">Proposition XVII.2.4</a> , p. 854  |
| 3.1  | Definition (Finite-support power)                          | <a href="#">Definition XVII.3.1</a> , p. 855   |
| 3.2  | Theorem (Power calculus)                                   | <a href="#">Theorem XVII.3.2</a> , p. 855      |
| 3.3  | Lemma (Transport of initial segments)                      | <a href="#">Lemma XVII.3.3</a> , p. 855        |
| 3.4  | Proposition (Continuity of the presented power)            | <a href="#">Proposition XVII.3.4</a> , p. 856  |
| 3.5  | Theorem (Exact predecessor decomposition)                  | <a href="#">Theorem XVII.3.5</a> , p. 856      |
| 3.6  | Theorem (Complete set-likeness criterion)                  | <a href="#">Theorem XVII.3.6</a> , p. 857      |
| 3.7  | Corollary (Collapse of set bases)                          | <a href="#">Corollary XVII.3.7</a> , p. 858    |
| 4.1  | Definition (The term class)                                | <a href="#">Definition XVII.4.1</a> , p. 858   |
| 4.2  | Theorem (Localization of hereditary normal forms)          | <a href="#">Theorem XVII.4.2</a> , p. 859      |
| 4.3  | Proposition (Root decomposition)                           | <a href="#">Proposition XVII.4.3</a> , p. 859  |
| 4.4  | Theorem (Canonical strict towers)                          | <a href="#">Theorem XVII.4.4</a> , p. 860      |
| 4.5  | Theorem (Least $\Omega$ -exponential fixed point)          | <a href="#">Theorem XVII.4.5</a> , p. 860      |
| 5.1  | Theorem (Normal-form addition)                             | <a href="#">Theorem XVII.5.1</a> , p. 861      |
| 5.2  | Theorem (Normal-form multiplication)                       | <a href="#">Theorem XVII.5.2</a> , p. 862      |
| 6.1  | Definition                                                 | <a href="#">Definition XVII.6.1</a> , p. 863   |
| 6.2  | Theorem (Finite-support transfer)                          | <a href="#">Theorem XVII.6.2</a> , p. 863      |
| 6.3  | Definition (Normal epsilon terms)                          | <a href="#">Definition XVII.6.3</a> , p. 864   |
| 6.4  | Theorem (Exact epsilon interpretation and class extension) | <a href="#">Theorem XVII.6.4</a> , p. 865      |
| 6.5  | Lemma                                                      | <a href="#">Lemma XVII.6.5</a> , p. 866        |
| 6.6  | Theorem (The first hereditary power boundary)              | <a href="#">Theorem XVII.6.6</a> , p. 866      |
| 6.7  | Theorem (An explicit zeta boundary)                        | <a href="#">Theorem XVII.6.7</a> , p. 867      |
| 6.8  | Definition                                                 | <a href="#">Definition XVII.6.8</a> , p. 868   |
| 6.9  | Theorem (Spectrum and carrier theorem)                     | <a href="#">Theorem XVII.6.9</a> , p. 868      |
| 6.10 | Proposition (Cardinality bound)                            | <a href="#">Proposition XVII.6.10</a> , p. 869 |
| 6.11 | Proposition (Closure and cofinality of horizons)           | <a href="#">Proposition XVII.6.11</a> , p. 869 |
| 6.12 | Proposition (The full-class calibration)                   | <a href="#">Proposition XVII.6.12</a> , p. 870 |
| 7.1  | Lemma (A well-order cannot absorb an appended point)       | <a href="#">Lemma XVII.7.1</a> , p. 871        |
| 7.2  | Theorem (Uniform diagonal obstruction)                     | <a href="#">Theorem XVII.7.2</a> , p. 872      |
| 7.3  | Definition                                                 | <a href="#">Definition XVII.7.3</a> , p. 873   |
| 7.4  | Lemma (Closure and correctness of the ambient model)       | <a href="#">Lemma XVII.7.4</a> , p. 873        |
| 7.5  | Theorem (The definable horizon)                            | <a href="#">Theorem XVII.7.5</a> , p. 873      |
| 7.6  | Proposition (Bounded-complexity domination)                | <a href="#">Proposition XVII.7.6</a> , p. 875  |
| 7.7  | Corollary (An effective strict complexity hierarchy)       | <a href="#">Corollary XVII.7.7</a> , p. 876    |
| 7.8  | Proposition (No uniform evaluator for the cofinal schema)  | <a href="#">Proposition XVII.7.8</a> , p. 877  |
| 7.9  | Theorem (Truth promotion)                                  | <a href="#">Theorem XVII.7.9</a> , p. 877      |
| 7.10 | Lemma (Encoding a predicate by adjacent swaps)             | <a href="#">Lemma XVII.7.10</a> , p. 878       |
| 7.11 | Theorem (Unbounded variation of the horizon)               | <a href="#">Theorem XVII.7.11</a> , p. 879     |
| 8.1  | Proposition (The full unrolling at an inaccessible height) | <a href="#">Proposition XVII.8.1</a> , p. 882  |

| No. | Statement in source 36                                                | Printed here as                                    |
|-----|-----------------------------------------------------------------------|----------------------------------------------------|
| 1   | Research question (Which horizons are realizable?)                    | <a href="#">Research question 113.295</a> , p. 964 |
| 2   | Research question (Parameter-free versus parameter-allowing horizons) | <a href="#">Research question 113.296</a> , p. 965 |
| 3   | Research question (The complexity profile)                            | <a href="#">Research question 113.297</a> , p. 965 |
| 4   | Research question (The size of truth promotion)                       | <a href="#">Research question 113.298</a> , p. 965 |
| 5   | Research question (Higher finitary notation functors)                 | <a href="#">Research question 113.299</a> , p. 966 |
| 6   | Research question (Normalizing general powers of point-cuts)          | <a href="#">Research question 113.300</a> , p. 966 |
| 7   | Research question (Weaker set theories)                               | <a href="#">Research question 113.301</a> , p. 966 |
| 8   | Research question (Admissible and definability bounds)                | <a href="#">Research question 113.302</a> , p. 966 |
| 9   | Research question (Stability under extensions)                        | <a href="#">Research question 113.303</a> , p. 967 |
| 10  | Research question (Class comparison and recursion strength)           | <a href="#">Research question 113.304</a> , p. 967 |

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